



```
In [1]: from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

Thesis Project by Group CLS

THESIS 1-3

Thesis Project

Group Members:

- Celles, Aaron Kent M.
- Leviste, Lee Ryan A.
- Santeco, Enrique S.

Section 1. Introduction

1.1 Problem Overview

Large-scale global optimization (LSGO) presents a critical challenge in computational intelligence, where the "curse of dimensionality" leads to an explosive increase in local optima and a high risk of premature convergence. While foundational algorithms like Differential Evolution (DE) are robust, their dependence on crossover and random mutation may restrict guided exploration — reducing particle diversity over time and leading to premature convergence.

This notebook implements and evaluates the proposed **Diversity-Preserving Level-Based Learning Differential Evolution (DP-LLDE)** algorithm, tested against the **CEC 2017 Large-Scale Global Optimization Benchmark** across 29 benchmark functions (F1-F30, excluding deprecated F2) at **10 dimensions**.

1.2 Research Objectives

1. Design and implement DP-LLDE, integrating sequential relative particle mutation and dynamic level-based learning into the Differential Evolution framework.

2. Evaluate DP-LLDE on the CEC 2017 benchmark suite (F1–F30) at 10 dimensions.
3. Examine convergence behavior, solution quality, and the effect of diversity-preserving strategies across unimodal, multimodal, hybrid, and composition functions.

1.3 Algorithm Overview - DP-LLDE

DP-LLDE extends LBLDE (Qiao et al., 2022) with three core contributions:

1. **Sequential Level Assignment** — each individual is assigned a learning source level drawn randomly from $\{2, \dots, NL\}$ on the first generation, decaying toward the elite level (level 1) in subsequent generations.
2. **Diversity-Aware Exemplar Selection** — if two exemplars e_1, e_2 fall below a diversity threshold, one is replaced with the most distant candidate in that level, preventing exemplar collapse.
3. **Normalized Mutation Pressure (Eq. 10):**

$$V_i = e_1 + F_i \times (e_2 - X_i) \times \frac{NL - k}{NL}$$

The factor $(NL-k)/NL$ decays to zero as $k \rightarrow NL$, naturally reducing mutation pressure as learning quality improves.

Parameter	Value
Population Size (NP)	100
Number of Levels (NL)	4
Initial μ_{CR}	0.35
Max FES	$10,000 \times D$
Dimensions (D)	10
Seed	42

Section 2. Benchmark Function Description

2.1 Overview

The **CEC 2017 Large-Scale Global Optimization Benchmark** (Awad et al., 2017) is a standardized suite of 30 test functions designed to evaluate optimization algorithms on complex, high-dimensional search spaces. The benchmark tests an algorithm's ability to avoid false local optima, convergence speed, and solution accuracy.

All functions are defined over the search space $[-100, 100]^D$ with their respective global optima offset by a function-specific bias value.

Library used: `cec2017` by Duncan Tilley — provides the official shift vectors, rotation matrices, and shuffle permutations from the CEC 2017 specification.

2.2 Unimodal Function Overview

Unimodal functions contain a **single global optimum and no local optima**. They are used to test the convergence speed and accuracy of an algorithm in a straightforward landscape.

No.	Function Name	Global Optimum
F1	Shifted and Rotated Bent Cigar	100
F2	Shifted and Rotated Sum of Different Power (<i>Deprecated</i>)	200
F3	Shifted and Rotated Zakharov	300

Note: F2 is officially deprecated from the CEC 2017 benchmark but is included here for completeness as it remains implemented in the `cec2017` library.

2.3 Multimodal Function Overview

Multimodal functions contain **multiple local optima**, making them significantly harder to solve. They are used to evaluate an algorithm's capacity to avoid premature convergence and escape false local optima.

No.	Function Name	Global Optimum
F4	Shifted and Rotated Rosenbrock's	400
F5	Shifted and Rotated Rastrigin's	500

No.	Function Name	Global Optimum
F6	Shifted and Rotated Expanded Schaffer's F6	600
F7	Shifted and Rotated Lunacek Bi-Rastrigin	700
F8	Shifted and Rotated Non-Continuous Rastrigin's	800
F9	Shifted and Rotated Levy	900
F10	Shifted and Rotated Schwefel's	1000

2.4 Hybrid Function Overview

Hybrid functions divide the search space into several regions, each governed by a **different basic function**. They assess an algorithm's adaptability to heterogeneous landscapes and mixed search behaviors.

No.	Component Functions	Global Optimum
F11	Zakharov; Rosenbrock; Rastrigin	1100
F12	Elliptic; Schwefel; Bent Cigar	1200
F13	Bent Cigar; Rosenbrock; Bi-Rastrigin	1300
F14	Elliptic; Ackley; Schaffer F7; Rastrigin	1400
F15	Bent Cigar; HGBat; Rastrigin; Rosenbrock	1500
F16	Schaffer F6; HGBat; Rosenbrock; Schwefel	1600
F17	Katsuura; Ackley; Griewank+Rosenbrock; Schwefel; Rastrigin	1700
F18	Elliptic; Ackley; Rastrigin; HGBat; Discus	1800
F19	Bent Cigar; Rastrigin; Griewank+Rosenbrock; Weierstrass; Schaffer F6	1900
F20	HappyCat; Katsuura; Ackley; Rastrigin; Schwefel; Schaffer F7	2000

2.5 Composite Function Overview

Composition functions **combine multiple shifted and rotated basic functions** into a single search space, creating highly complex and deceptive optimization landscapes. They are the most challenging category in the CEC 2017 suite and significantly test an algorithm's overall robustness and global search capability.

No.	Component Functions	Global Optimum
F21	Rosenbrock; Elliptic; Rastrigin	2100
F22	Rastrigin; Griewank; Schwefel	2200
F23	Rosenbrock; Ackley; Schwefel; Rastrigin	2300
F24	Ackley; Elliptic; Griewank; Rastrigin	2400
F25	Rastrigin; HappyCat; Ackley; Discus; Rosenbrock	2500
F26	Schaffer F6; Schwefel; Griewank; Rosenbrock; Rastrigin	2600
F27	HGBat; Rastrigin; Schwefel; Bent Cigar; Elliptic; Schaffer F6	2700
F28	Ackley; Griewank; Discus; Rosenbrock; HappyCat; Schaffer F6	2800
F29	F15; F16; F17	2900
F30	F15; F18; F19	3000

Section 3. List of Required Libraries

3.1 List of Libraries and Versions

The following libraries and modules were used for optimization, benchmarking, and visualization in this project:

Library / Module	Purpose	Version
<code>python</code>	Core programming language	3.10+
<code>numpy</code>	Numerical computing and array operations	2.0.2
<code>matplotlib</code>	Plotting and saving convergence graphs	3.10.0
<code>cec2017</code>	CEC 2017 benchmark functions (simple, hybrid, composition)	Latest (pip)
<code>cec2017.simple</code>	Simple benchmark functions F1-F10	—
<code>cec2017.hybrid</code>	Hybrid benchmark functions F11-F20	—
<code>cec2017.composition</code>	Composition benchmark functions F21-F30	—
<code>dplldc</code>	DP-LLDE optimization algorithm (uploaded locally)	Custom
<code>google.colab.files</code>	File upload utility for loading <code>dplldc.py</code> into the Colab environment	Built-in
<code>sys</code>	Clears cached module before re-importing <code>dplldc.py</code>	Built-in

Section 4. Environment Setup and Validation

4.1 Installation

Install `cec2017`

```
In [ ]: !pip install cec2017 --quiet
```

ERROR: Could not find a version that satisfies the requirement cec2017 (from versions: none)
ERROR: No matching distribution found for cec2017

4.2 File Validation

Import `simple.py`, `hybrid.py`, and `composition.py` python files for benchmark functions

```
In [2]: from cec2017 import simple, hybrid, composition
print('cec2017 loaded!')
```

cec2017 loaded!

Import `sys` control over warning messages for verification that `dpllde.py` exists

```
In [3]: from google.colab import files
import sys

uploaded = files.upload()

if 'dpllde.py' in uploaded:
    if 'dpllde' in sys.modules:
        del sys.modules['dpllde']
    import dpllde
    from dpllde import DPLLDE
    print('dpllde.py uploaded and imported successfully!')
else:
    print('ERROR: Please upload a file named dpllde.py')
```

Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

Saving `dpllde.py` to `dpllde.py`

`dpllde.py` uploaded and imported successfully!

```
In [4]: import dpllde
        from dpllde import DPLLDE
```

Section 5. Configuration and Helper Functions

5.1 Library Imports

Import libraries `numpy`, `matplotlib`, and `cec2017` for numerical computation, visualization, and CEC 2017 benchmark functions.

```
In [5]: import numpy as np
import matplotlib.pyplot as plt
from cec2017 import simple, hybrid, composition
```

5.2 Global Parameters

Define global parameters

```
In [6]: D = 100 # Change this when testing different dimensions
```

```
In [7]: bounds = np.array([[-100.0, 100.0]] * D)
MAX_FES = 10000 * D
NP, NL, MU_CR = 100, 4, 0.35
SEED = 42
```

Store all results for summary table under `all_results` dict

```
In [8]: all_results = {}
```

```
In [10]: import pickle
import os
from google.colab import drive

# Save functions
def save_results_checkpoint(results_dict, filename='all_results_checkpoint.pkl'):
    """Save all results to Google Drive."""
    drive_path = '/content/drive/My Drive/thesis_results'
    os.makedirs(drive_path, exist_ok=True)
    filepath = os.path.join(drive_path, filename)

    with open(filepath, 'wb') as f:
        pickle.dump(results_dict, f)
    print(f'✓ Checkpoint saved to: {filepath}')

def save_plots_to_drive(plot_filename):
    """Copy a single plot to Google Drive."""
```

```

drive_path = '/content/drive/My Drive/thesis_results/plots'
os.makedirs(drive_path, exist_ok=True)

if os.path.exists(plot_filename):
    import shutil
    dest = os.path.join(drive_path, plot_filename)
    shutil.copy(plot_filename, dest)
    print(f'✓ Plot saved to: {dest}')

def load_results_checkpoint(filename='all_results_checkpoint.pkl'):
    """Load previous results from Google Drive if they exist."""
    drive_path = '/content/drive/My Drive/thesis_results'
    filepath = os.path.join(drive_path, filename)

    if os.path.exists(filepath):
        with open(filepath, 'rb') as f:
            return pickle.load(f)
    return {}

```

5.3 Function Definitions

Define `make_wrapper` to adapt CEC 2017 functions to accept a 1D input array from DP-LLDE, returning a large penalty value if the output is invalid (nan/inf).

```

In [11]: def make_wrapper(fn):
    """Wrap a cec2017 function (expects (1,D) input) to accept (D,) input."""
    def wrapper(x):
        val = fn(x.reshape(1, -1))
        if hasattr(val, '__len__'):
            val = float(val[0])
        else:
            val = float(val)
        return val if np.isfinite(val) else 1e18
    return wrapper

```

Define `run_and_plot` to execute a single DP-LLDE optimization run on a given benchmark function, print the results, store them for the summary table, and save the convergence graph.

```

In [12]: def run_and_plot(fn_name, fn_eval, f_global, fn_label):
    """Run DP-LLDE on a function and plot convergence + diversity graphs."""
    print('=' * 80)
    print(f'DP-LLDE on CEC 2017 {fn_name} – Single Run')
    print('=' * 80)
    print(f'Function      : {fn_label} ({D}D)')
    print(f'Bounds        : [-100, 100]')
    print(f'Global optimum : {f_global}')

    opt = DP-LLDE(
        objective_func=fn_eval,

```



```

        bounds=bounds,
        NP=NP, NL=NL, mu_CR_ini=MU_CR,
        max_fes=MAX_FES,
        seed=SEED
    )
    # ← Unpack 4 values now
    best_sol, best_fit, history, diversity_history = opt.optimize(verbose=True)

    error = abs(best_fit - f_global)
    print(f'\n{"="*80}')
    print('Results:')
    print(f'    Best Fitness           : {best_fit:.6e}')
    print(f'    Global Optimum          : {f_global:.6e}')
    print(f'    Error (|best-optimum|) : {error:.6e}')
    print(f'    Best Solution (5 dim)   : {best_sol[:5]}')
    print(f'    D^(0) (init. diversity): {opt.D_0:.4e}')
    print(f'\n{"="*80}')

    # Store for summary
    all_results[fn_name] = {
        'label': fn_label,
        'best_fit': best_fit,
        'f_global': f_global,
        'error': error
    }

    save_results_checkpoint(all_results)

    fes_axis = np.linspace(0, MAX_FES, len(history))
    gen_axis = np.arange(len(diversity_history)) # Generations (0, 1, 2, ...)

    # — Two-panel figure —
    fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(10, 10))

    # Convergence curve (top) – FEs on x-axis
    ax1.plot(fes_axis, history, color='steelblue', linewidth=1.5)
    ax1.set_ylabel('Best Fitness', fontsize=12)
    ax1.set_xlabel('Function Evaluations (FEs)', fontsize=12)
    ax1.set_title(
        f'DP-LLDE – CEC 2017 {fn_name}: {fn_label} ({D}D)',
        fontsize=12
    )

    ax1.grid(True, which='both', linestyle='--', alpha=0.6)

    # Diversity curve (bottom) – Generations on x-axis
    ax2.plot(gen_axis, diversity_history, color='darkorange', linewidth=1.5)
    ax2.set_xlabel('Generation', fontsize=12)
    ax2.set_ylabel('Population Diversity\n(Mean Std Dev)', fontsize=12)
    ax2.set_title(f'Diversity Across Generations – {fn_name}', fontsize=12)
    ax2.grid(True, which='both', linestyle='--', alpha=0.6)

    plt.tight_layout()
    fname = f'convergence_{fn_name}.png'

```

```
plt.savefig(fname, dpi=150)
plt.show()
print(f'Graph saved as {fname}\n')

save_plots_to_drive(fname)
```

```
In [13]: def load_results_checkpoint(filename='all_results_checkpoint.pkl'):
        """Load previous results from local Colab storage."""
        filepath = f'/content/{filename}'

        if os.path.exists(filepath):
            with open(filepath, 'rb') as f:
                return pickle.load(f)
        return {}
```

```
In [14]: all_results = load_results_checkpoint()
        if all_results:
            print(f'✓ Loaded {len(all_results)} previous results')
```

✓ Loaded 27 previous results

```
In [15]: all_results = load_results_checkpoint()

        if all_results:
            print(f'✓ Loaded {len(all_results)} previous results')
            print('Already completed:')
            for fn_name in sorted(all_results.keys()):
                print(f'✓ {fn_name}')
            completed = set(all_results.keys())
        else:
            print('No checkpoint found')
            completed = set()
```

✓ Loaded 27 previous results

Already completed:

- ✓ F1
- ✓ F10
- ✓ F11
- ✓ F12
- ✓ F13
- ✓ F14
- ✓ F15
- ✓ F16
- ✓ F17
- ✓ F18
- ✓ F19
- ✓ F2
- ✓ F20
- ✓ F21
- ✓ F22
- ✓ F23
- ✓ F24
- ✓ F25
- ✓ F26
- ✓ F27
- ✓ F3
- ✓ F4
- ✓ F5
- ✓ F6
- ✓ F7
- ✓ F8
- ✓ F9

Section 6. CEC2017 Benchmark Functions

6.1 Unimodal Functions (F1-F3)

```
In [ ]: # =====  
# F1 – Shifted and Rotated Bent Cigar  
# =====  
run_and_plot('F1', make_wrapper(simple.f1), 100.0, 'Shifted and Rotated Bent C
```

=====

=

DP-LLDE on CEC 2017 F1 – Single Run

=====

=

Function : Shifted and Rotated Bent Cigar (100D)

Bounds : [-100, 100]

Global optimum : 100.0

Initial $D^{(0)} = 5.7220e+01$ | Diversity threshold = 5.7220e+00

Generation 0: Best Fitness = 6.191709e+12

Generation 10: Best = 2.375406e+12 FES = 1100 $\mu_F = 0.542$ $\mu_{CR} = 0.353$

Diversity = 2.6289e+01 $k_t = [1, 1, 1, 1]$

Generation 20: Best = 1.374961e+12 FES = 2100 $\mu_F = 0.588$ $\mu_{CR} = 0.355$

Diversity = 1.4751e+01 $k_t = [2, 3, 2, 3]$

Generation 30: Best = 1.031836e+12 FES = 3100 $\mu_F = 0.621$ $\mu_{CR} = 0.358$

Diversity = 9.6337e+00 $k_t = [1, 1, 1, 1]$

Generation 40: Best = 8.063596e+11 FES = 4100 $\mu_F = 0.659$ $\mu_{CR} = 0.362$

Diversity = 6.8515e+00 $k_t = [3, 2, 3, 3]$

Generation 50: Best = 6.344313e+11 FES = 5100 $\mu_F = 0.673$ $\mu_{CR} = 0.365$

Diversity = 6.0810e+00 $k_t = [1, 1, 1, 1]$

Generation 60: Best = 4.717910e+11 FES = 6100 $\mu_F = 0.692$ $\mu_{CR} = 0.364$

Diversity = 4.6773e+00 $k_t = [3, 4, 4, 3]$

Generation 70: Best = 4.195450e+11 FES = 7100 $\mu_F = 0.719$ $\mu_{CR} = 0.372$

Diversity = 4.5307e+00 $k_t = [1, 1, 1, 1]$

Generation 80: Best = 3.372732e+11 FES = 8100 $\mu_F = 0.731$ $\mu_{CR} = 0.374$

Diversity = 3.9570e+00 $k_t = [2, 4, 4, 3]$

Generation 90: Best = 2.574591e+11 FES = 9100 $\mu_F = 0.745$ $\mu_{CR} = 0.371$

Diversity = 4.0826e+00 $k_t = [1, 1, 1, 1]$

Generation 100: Best = 1.984105e+11 FES = 10100 $\mu_F = 0.754$ $\mu_{CR} = 0.373$

Diversity = 3.8628e+00 $k_t = [4, 4, 2, 4]$

Generation 110: Best = 1.645739e+11 FES = 11100 $\mu_F = 0.754$ $\mu_{CR} = 0.385$

Diversity = 3.1301e+00 $k_t = [1, 1, 1, 1]$

Generation 120: Best = 1.348857e+11 FES = 12100 $\mu_F = 0.761$ $\mu_{CR} = 0.390$

Diversity = 2.8239e+00 $k_t = [3, 3, 2, 4]$

Generation 130: Best = 1.131632e+11 FES = 13100 $\mu_F = 0.764$ $\mu_{CR} = 0.397$

Diversity = 2.4753e+00 $k_t = [1, 1, 1, 1]$

Generation 140: Best = 8.837354e+10 FES = 14100 $\mu_F = 0.770$ $\mu_{CR} = 0.398$

Diversity = 2.3512e+00 $k_t = [2, 4, 2, 3]$

Generation 150: Best = 6.856760e+10 FES = 15100 $\mu_F = 0.767$ $\mu_{CR} = 0.398$

Diversity = 2.3850e+00 $k_t = [1, 1, 1, 1]$

Generation 160: Best = 5.782279e+10 FES = 16100 $\mu_F = 0.759$ $\mu_{CR} = 0.398$

Diversity = 1.9144e+00 $k_t = [4, 3, 2, 2]$

Generation 170: Best = 4.588447e+10 FES = 17100 $\mu_F = 0.753$ $\mu_{CR} = 0.404$

Diversity = 1.8297e+00 $k_t = [1, 1, 1, 1]$

Generation 180: Best = 3.627879e+10 FES = 18100 $\mu_F = 0.753$ $\mu_{CR} = 0.397$

Diversity = 1.6468e+00 $k_t = [3, 4, 4, 3]$

Generation 190: Best = 2.726994e+10 FES = 19100 $\mu_F = 0.750$ $\mu_{CR} = 0.402$

Diversity = 1.4904e+00 $k_t = [1, 1, 1, 1]$

Generation 200: Best = 2.061382e+10 FES = 20100 $\mu_F = 0.757$ $\mu_{CR} = 0.400$

Diversity = 1.3609e+00 $k_t = [2, 3, 3, 3]$

Generation 210: Best = 1.605854e+10 FES = 21100 $\mu_F = 0.760$ $\mu_{CR} = 0.411$

Diversity = 1.2873e+00 $k_t = [1, 1, 1, 1]$

Generation 220: Best = 1.306558e+10 FES = 22100 $\mu_F = 0.751$ $\mu_{CR} = 0.408$

Diversity = 1.0365e+00 $k_t = [4, 2, 2, 3]$

Generation	230:	Best = 1.042228e+10	FES = 23100	$\mu_F = 0.742$	$\mu_{CR} = 0.406$
Diversity	= 8.9157e-01	k_t = [1, 1, 1, 1]			
Generation	240:	Best = 8.364558e+09	FES = 24100	$\mu_F = 0.743$	$\mu_{CR} = 0.414$
Diversity	= 7.2132e-01	k_t = [4, 2, 4, 2]			
Generation	250:	Best = 6.547795e+09	FES = 25100	$\mu_F = 0.750$	$\mu_{CR} = 0.418$
Diversity	= 6.8699e-01	k_t = [1, 1, 1, 1]			
Generation	260:	Best = 5.196653e+09	FES = 26100	$\mu_F = 0.750$	$\mu_{CR} = 0.418$
Diversity	= 5.7349e-01	k_t = [2, 4, 4, 3]			
Generation	270:	Best = 4.135379e+09	FES = 27100	$\mu_F = 0.756$	$\mu_{CR} = 0.415$
Diversity	= 5.6994e-01	k_t = [1, 1, 1, 1]			
Generation	280:	Best = 3.246476e+09	FES = 28100	$\mu_F = 0.752$	$\mu_{CR} = 0.416$
Diversity	= 4.9183e-01	k_t = [4, 2, 2, 4]			
Generation	290:	Best = 2.712486e+09	FES = 29100	$\mu_F = 0.750$	$\mu_{CR} = 0.424$
Diversity	= 4.3694e-01	k_t = [1, 1, 1, 1]			
Generation	300:	Best = 2.336912e+09	FES = 30100	$\mu_F = 0.733$	$\mu_{CR} = 0.423$
Diversity	= 3.5717e-01	k_t = [3, 3, 2, 4]			
Generation	310:	Best = 2.056445e+09	FES = 31100	$\mu_F = 0.699$	$\mu_{CR} = 0.420$
Diversity	= 3.2608e-01	k_t = [1, 1, 1, 1]			
Generation	320:	Best = 1.813920e+09	FES = 32100	$\mu_F = 0.682$	$\mu_{CR} = 0.414$
Diversity	= 2.6898e-01	k_t = [4, 2, 4, 3]			
Generation	330:	Best = 1.737389e+09	FES = 33100	$\mu_F = 0.670$	$\mu_{CR} = 0.417$
Diversity	= 2.2427e-01	k_t = [1, 1, 1, 1]			
Generation	340:	Best = 1.611586e+09	FES = 34100	$\mu_F = 0.676$	$\mu_{CR} = 0.408$
Diversity	= 1.8335e-01	k_t = [4, 3, 4, 2]			
Generation	350:	Best = 1.529059e+09	FES = 35100	$\mu_F = 0.684$	$\mu_{CR} = 0.411$
Diversity	= 1.4839e-01	k_t = [1, 1, 1, 1]			
Generation	360:	Best = 1.476352e+09	FES = 36100	$\mu_F = 0.681$	$\mu_{CR} = 0.410$
Diversity	= 1.2557e-01	k_t = [2, 4, 3, 4]			
Generation	370:	Best = 1.411404e+09	FES = 37100	$\mu_F = 0.687$	$\mu_{CR} = 0.405$
Diversity	= 1.2283e-01	k_t = [1, 1, 1, 1]			
Generation	380:	Best = 1.368296e+09	FES = 38100	$\mu_F = 0.679$	$\mu_{CR} = 0.399$
Diversity	= 1.0558e-01	k_t = [2, 3, 4, 2]			
Generation	390:	Best = 1.331869e+09	FES = 39100	$\mu_F = 0.668$	$\mu_{CR} = 0.395$
Diversity	= 9.4629e-02	k_t = [1, 1, 1, 1]			
Generation	400:	Best = 1.286936e+09	FES = 40100	$\mu_F = 0.678$	$\mu_{CR} = 0.386$
Diversity	= 8.1799e-02	k_t = [3, 4, 2, 3]			
Generation	410:	Best = 1.259159e+09	FES = 41100	$\mu_F = 0.702$	$\mu_{CR} = 0.390$
Diversity	= 8.6166e-02	k_t = [1, 1, 1, 1]			
Generation	420:	Best = 1.232073e+09	FES = 42100	$\mu_F = 0.700$	$\mu_{CR} = 0.384$
Diversity	= 7.8262e-02	k_t = [3, 2, 4, 3]			
Generation	430:	Best = 1.204624e+09	FES = 43100	$\mu_F = 0.700$	$\mu_{CR} = 0.382$
Diversity	= 8.4792e-02	k_t = [1, 1, 1, 1]			
Generation	440:	Best = 1.181117e+09	FES = 44100	$\mu_F = 0.707$	$\mu_{CR} = 0.383$
Diversity	= 7.7938e-02	k_t = [4, 4, 3, 2]			
Generation	450:	Best = 1.161220e+09	FES = 45100	$\mu_F = 0.700$	$\mu_{CR} = 0.379$
Diversity	= 7.1370e-02	k_t = [1, 1, 1, 1]			
Generation	460:	Best = 1.144193e+09	FES = 46100	$\mu_F = 0.699$	$\mu_{CR} = 0.380$
Diversity	= 6.0951e-02	k_t = [2, 3, 3, 4]			
Generation	470:	Best = 1.134319e+09	FES = 47100	$\mu_F = 0.706$	$\mu_{CR} = 0.376$
Diversity	= 5.6631e-02	k_t = [1, 1, 1, 1]			
Generation	480:	Best = 1.119373e+09	FES = 48100	$\mu_F = 0.708$	$\mu_{CR} = 0.375$
Diversity	= 5.3762e-02	k_t = [2, 2, 3, 4]			
Generation	490:	Best = 1.113487e+09	FES = 49100	$\mu_F = 0.700$	$\mu_{CR} = 0.363$
Diversity	= 5.2606e-02	k_t = [1, 1, 1, 1]			

Generation 500: Best = 1.103486e+09	FES = 50100	$\mu_F = 0.693$	$\mu_{CR} = 0.361$
Diversity = 4.9525e-02	$k_t = [3, 4, 2, 2]$		
Generation 510: Best = 1.097439e+09	FES = 51100	$\mu_F = 0.693$	$\mu_{CR} = 0.348$
Diversity = 4.9879e-02	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 1.090324e+09	FES = 52100	$\mu_F = 0.691$	$\mu_{CR} = 0.331$
Diversity = 4.4698e-02	$k_t = [3, 3, 3, 2]$		
Generation 530: Best = 1.084036e+09	FES = 53100	$\mu_F = 0.694$	$\mu_{CR} = 0.324$
Diversity = 4.2821e-02	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 1.078453e+09	FES = 54100	$\mu_F = 0.691$	$\mu_{CR} = 0.320$
Diversity = 4.0497e-02	$k_t = [4, 4, 2, 2]$		
Generation 550: Best = 1.071292e+09	FES = 55100	$\mu_F = 0.702$	$\mu_{CR} = 0.328$
Diversity = 4.1762e-02	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 1.062937e+09	FES = 56100	$\mu_F = 0.710$	$\mu_{CR} = 0.317$
Diversity = 3.9365e-02	$k_t = [3, 4, 4, 4]$		
Generation 570: Best = 1.058654e+09	FES = 57100	$\mu_F = 0.718$	$\mu_{CR} = 0.319$
Diversity = 4.0219e-02	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 1.048958e+09	FES = 58100	$\mu_F = 0.719$	$\mu_{CR} = 0.318$
Diversity = 4.1812e-02	$k_t = [2, 3, 3, 2]$		
Generation 590: Best = 1.037915e+09	FES = 59100	$\mu_F = 0.725$	$\mu_{CR} = 0.317$
Diversity = 4.6993e-02	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 1.022328e+09	FES = 60100	$\mu_F = 0.737$	$\mu_{CR} = 0.315$
Diversity = 5.0635e-02	$k_t = [2, 2, 3, 4]$		
Generation 610: Best = 1.004053e+09	FES = 61100	$\mu_F = 0.751$	$\mu_{CR} = 0.303$
Diversity = 5.8632e-02	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 9.812654e+08	FES = 62100	$\mu_F = 0.757$	$\mu_{CR} = 0.306$
Diversity = 6.8374e-02	$k_t = [4, 3, 2, 2]$		
Generation 630: Best = 9.688585e+08	FES = 63100	$\mu_F = 0.751$	$\mu_{CR} = 0.301$
Diversity = 6.8299e-02	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 9.557453e+08	FES = 64100	$\mu_F = 0.739$	$\mu_{CR} = 0.301$
Diversity = 5.9750e-02	$k_t = [3, 3, 3, 4]$		
Generation 650: Best = 9.386751e+08	FES = 65100	$\mu_F = 0.734$	$\mu_{CR} = 0.302$
Diversity = 6.2701e-02	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 9.250770e+08	FES = 66100	$\mu_F = 0.739$	$\mu_{CR} = 0.287$
Diversity = 6.2325e-02	$k_t = [3, 2, 4, 4]$		
Generation 670: Best = 9.115703e+08	FES = 67100	$\mu_F = 0.728$	$\mu_{CR} = 0.284$
Diversity = 6.1433e-02	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 8.907462e+08	FES = 68100	$\mu_F = 0.722$	$\mu_{CR} = 0.276$
Diversity = 5.7399e-02	$k_t = [2, 3, 3, 4]$		
Generation 690: Best = 8.823172e+08	FES = 69100	$\mu_F = 0.729$	$\mu_{CR} = 0.273$
Diversity = 5.9324e-02	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 8.666170e+08	FES = 70100	$\mu_F = 0.738$	$\mu_{CR} = 0.259$
Diversity = 6.4021e-02	$k_t = [4, 2, 4, 2]$		
Generation 710: Best = 8.624741e+08	FES = 71100	$\mu_F = 0.735$	$\mu_{CR} = 0.255$
Diversity = 6.3164e-02	$k_t = [1, 1, 1, 1]$		
Generation 720: Best = 8.534579e+08	FES = 72100	$\mu_F = 0.731$	$\mu_{CR} = 0.259$
Diversity = 5.5392e-02	$k_t = [3, 3, 4, 3]$		
Generation 730: Best = 8.475307e+08	FES = 73100	$\mu_F = 0.727$	$\mu_{CR} = 0.255$
Diversity = 5.2044e-02	$k_t = [1, 1, 1, 1]$		
Generation 740: Best = 8.453991e+08	FES = 74100	$\mu_F = 0.716$	$\mu_{CR} = 0.253$
Diversity = 4.5458e-02	$k_t = [3, 3, 3, 4]$		
Generation 750: Best = 8.397718e+08	FES = 75100	$\mu_F = 0.722$	$\mu_{CR} = 0.250$
Diversity = 4.1356e-02	$k_t = [1, 1, 1, 1]$		
Generation 760: Best = 8.355742e+08	FES = 76100	$\mu_F = 0.711$	$\mu_{CR} = 0.245$
Diversity = 3.5769e-02	$k_t = [2, 4, 2, 4]$		

Generation 770: Best = 8.333079e+08	FES = 77100	$\mu_F = 0.708$	$\mu_{CR} = 0.234$
Diversity = 3.4687e-02	$k_t = [1, 1, 1, 1]$		
Generation 780: Best = 8.282760e+08	FES = 78100	$\mu_F = 0.721$	$\mu_{CR} = 0.232$
Diversity = 3.3249e-02	$k_t = [4, 3, 3, 4]$		
Generation 790: Best = 8.245820e+08	FES = 79100	$\mu_F = 0.734$	$\mu_{CR} = 0.247$
Diversity = 3.5570e-02	$k_t = [1, 1, 1, 1]$		
Generation 800: Best = 8.190403e+08	FES = 80100	$\mu_F = 0.725$	$\mu_{CR} = 0.247$
Diversity = 3.5753e-02	$k_t = [2, 2, 2, 3]$		
Generation 810: Best = 8.151595e+08	FES = 81100	$\mu_F = 0.728$	$\mu_{CR} = 0.253$
Diversity = 3.9732e-02	$k_t = [1, 1, 1, 1]$		
Generation 820: Best = 8.103532e+08	FES = 82100	$\mu_F = 0.733$	$\mu_{CR} = 0.246$
Diversity = 3.9144e-02	$k_t = [3, 2, 2, 4]$		
Generation 830: Best = 8.055108e+08	FES = 83100	$\mu_F = 0.740$	$\mu_{CR} = 0.239$
Diversity = 3.9555e-02	$k_t = [1, 1, 1, 1]$		
Generation 840: Best = 7.990082e+08	FES = 84100	$\mu_F = 0.738$	$\mu_{CR} = 0.235$
Diversity = 4.2882e-02	$k_t = [3, 3, 2, 2]$		
Generation 850: Best = 7.940607e+08	FES = 85100	$\mu_F = 0.734$	$\mu_{CR} = 0.235$
Diversity = 4.5872e-02	$k_t = [1, 1, 1, 1]$		
Generation 860: Best = 7.860312e+08	FES = 86100	$\mu_F = 0.738$	$\mu_{CR} = 0.237$
Diversity = 4.3185e-02	$k_t = [4, 3, 3, 4]$		
Generation 870: Best = 7.777428e+08	FES = 87100	$\mu_F = 0.735$	$\mu_{CR} = 0.248$
Diversity = 4.1020e-02	$k_t = [1, 1, 1, 1]$		
Generation 880: Best = 7.694798e+08	FES = 88100	$\mu_F = 0.745$	$\mu_{CR} = 0.251$
Diversity = 4.3907e-02	$k_t = [4, 3, 4, 3]$		
Generation 890: Best = 7.617916e+08	FES = 89100	$\mu_F = 0.742$	$\mu_{CR} = 0.256$
Diversity = 4.7551e-02	$k_t = [1, 1, 1, 1]$		
Generation 900: Best = 7.429089e+08	FES = 90100	$\mu_F = 0.742$	$\mu_{CR} = 0.254$
Diversity = 4.8532e-02	$k_t = [2, 2, 4, 4]$		
Generation 910: Best = 7.275222e+08	FES = 91100	$\mu_F = 0.749$	$\mu_{CR} = 0.257$
Diversity = 5.5550e-02	$k_t = [1, 1, 1, 1]$		
Generation 920: Best = 7.180005e+08	FES = 92100	$\mu_F = 0.752$	$\mu_{CR} = 0.264$
Diversity = 5.8343e-02	$k_t = [4, 3, 3, 3]$		
Generation 930: Best = 7.058855e+08	FES = 93100	$\mu_F = 0.741$	$\mu_{CR} = 0.263$
Diversity = 5.9239e-02	$k_t = [1, 1, 1, 1]$		
Generation 940: Best = 6.915479e+08	FES = 94100	$\mu_F = 0.739$	$\mu_{CR} = 0.268$
Diversity = 5.5372e-02	$k_t = [2, 4, 3, 3]$		
Generation 950: Best = 6.402027e+08	FES = 95100	$\mu_F = 0.759$	$\mu_{CR} = 0.271$
Diversity = 6.7590e-02	$k_t = [1, 1, 1, 1]$		
Generation 960: Best = 6.052985e+08	FES = 96100	$\mu_F = 0.752$	$\mu_{CR} = 0.280$
Diversity = 7.5489e-02	$k_t = [4, 3, 2, 3]$		
Generation 970: Best = 5.902538e+08	FES = 97100	$\mu_F = 0.742$	$\mu_{CR} = 0.279$
Diversity = 7.3734e-02	$k_t = [1, 1, 1, 1]$		
Generation 980: Best = 5.826612e+08	FES = 98100	$\mu_F = 0.741$	$\mu_{CR} = 0.272$
Diversity = 6.4386e-02	$k_t = [4, 4, 3, 4]$		
Generation 990: Best = 5.679795e+08	FES = 99100	$\mu_F = 0.743$	$\mu_{CR} = 0.276$
Diversity = 6.2114e-02	$k_t = [1, 1, 1, 1]$		
Generation 1000: Best = 5.539050e+08	FES = 100100	$\mu_F = 0.743$	$\mu_{CR} = 0.273$
Diversity = 6.3901e-02	$k_t = [4, 4, 2, 4]$		
Generation 1010: Best = 5.457478e+08	FES = 101100	$\mu_F = 0.736$	$\mu_{CR} = 0.268$
Diversity = 6.3020e-02	$k_t = [1, 1, 1, 1]$		
Generation 1020: Best = 5.371458e+08	FES = 102100	$\mu_F = 0.723$	$\mu_{CR} = 0.266$
Diversity = 5.6278e-02	$k_t = [4, 2, 4, 2]$		
Generation 1030: Best = 5.323548e+08	FES = 103100	$\mu_F = 0.709$	$\mu_{CR} = 0.261$
Diversity = 5.2684e-02	$k_t = [1, 1, 1, 1]$		

Generation 1040: Best = 5.234985e+08 FES = 104100 μ_F = 0.699 μ_{CR} = 0.260
 Diversity = 4.5528e-02 k_t = [2, 3, 2, 4]
 Generation 1050: Best = 5.165550e+08 FES = 105100 μ_F = 0.692 μ_{CR} = 0.264
 Diversity = 4.6255e-02 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 5.105560e+08 FES = 106100 μ_F = 0.694 μ_{CR} = 0.266
 Diversity = 4.1676e-02 k_t = [3, 4, 2, 4]
 Generation 1070: Best = 5.047341e+08 FES = 107100 μ_F = 0.726 μ_{CR} = 0.256
 Diversity = 4.3688e-02 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 4.956574e+08 FES = 108100 μ_F = 0.730 μ_{CR} = 0.246
 Diversity = 4.5981e-02 k_t = [2, 4, 3, 2]
 Generation 1090: Best = 4.864639e+08 FES = 109100 μ_F = 0.715 μ_{CR} = 0.256
 Diversity = 5.0434e-02 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 4.769644e+08 FES = 110100 μ_F = 0.737 μ_{CR} = 0.267
 Diversity = 5.2739e-02 k_t = [2, 4, 4, 2]
 Generation 1110: Best = 4.715321e+08 FES = 111100 μ_F = 0.741 μ_{CR} = 0.265
 Diversity = 5.3760e-02 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 4.627300e+08 FES = 112100 μ_F = 0.732 μ_{CR} = 0.254
 Diversity = 5.0957e-02 k_t = [3, 2, 3, 2]
 Generation 1130: Best = 4.567947e+08 FES = 113100 μ_F = 0.716 μ_{CR} = 0.243
 Diversity = 5.2099e-02 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 4.508025e+08 FES = 114100 μ_F = 0.704 μ_{CR} = 0.246
 Diversity = 4.9457e-02 k_t = [3, 2, 2, 4]
 Generation 1150: Best = 4.450650e+08 FES = 115100 μ_F = 0.700 μ_{CR} = 0.246
 Diversity = 4.3342e-02 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 4.426811e+08 FES = 116100 μ_F = 0.695 μ_{CR} = 0.243
 Diversity = 3.8420e-02 k_t = [2, 3, 2, 3]
 Generation 1170: Best = 4.408095e+08 FES = 117100 μ_F = 0.684 μ_{CR} = 0.236
 Diversity = 3.5432e-02 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 4.351309e+08 FES = 118100 μ_F = 0.690 μ_{CR} = 0.241
 Diversity = 3.3367e-02 k_t = [3, 4, 4, 3]
 Generation 1190: Best = 4.282536e+08 FES = 119100 μ_F = 0.698 μ_{CR} = 0.250
 Diversity = 3.4002e-02 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 4.235053e+08 FES = 120100 μ_F = 0.720 μ_{CR} = 0.254
 Diversity = 3.6758e-02 k_t = [2, 2, 3, 4]
 Generation 1210: Best = 4.163621e+08 FES = 121100 μ_F = 0.732 μ_{CR} = 0.256
 Diversity = 3.9430e-02 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 4.108613e+08 FES = 122100 μ_F = 0.748 μ_{CR} = 0.261
 Diversity = 4.0459e-02 k_t = [3, 2, 2, 2]
 Generation 1230: Best = 4.043851e+08 FES = 123100 μ_F = 0.745 μ_{CR} = 0.260
 Diversity = 4.1168e-02 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 3.993357e+08 FES = 124100 μ_F = 0.750 μ_{CR} = 0.255
 Diversity = 4.1403e-02 k_t = [3, 3, 4, 3]
 Generation 1250: Best = 3.922471e+08 FES = 125100 μ_F = 0.739 μ_{CR} = 0.256
 Diversity = 4.2924e-02 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 3.886346e+08 FES = 126100 μ_F = 0.735 μ_{CR} = 0.258
 Diversity = 4.1976e-02 k_t = [3, 2, 4, 3]
 Generation 1270: Best = 3.855644e+08 FES = 127100 μ_F = 0.735 μ_{CR} = 0.256
 Diversity = 3.8911e-02 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 3.826949e+08 FES = 128100 μ_F = 0.712 μ_{CR} = 0.253
 Diversity = 3.1952e-02 k_t = [3, 4, 4, 4]
 Generation 1290: Best = 3.810987e+08 FES = 129100 μ_F = 0.715 μ_{CR} = 0.255
 Diversity = 2.9089e-02 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 3.787745e+08 FES = 130100 μ_F = 0.707 μ_{CR} = 0.254
 Diversity = 2.7053e-02 k_t = [2, 2, 4, 2]

Generation 1310: Best = 3.778218e+08 FES = 131100 μ_F = 0.701 μ_{CR} = 0.249
 Diversity = 2.5244e-02 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 3.765393e+08 FES = 132100 μ_F = 0.697 μ_{CR} = 0.249
 Diversity = 2.2908e-02 k_t = [4, 3, 3, 2]
 Generation 1330: Best = 3.752273e+08 FES = 133100 μ_F = 0.706 μ_{CR} = 0.250
 Diversity = 2.2019e-02 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 3.738148e+08 FES = 134100 μ_F = 0.710 μ_{CR} = 0.258
 Diversity = 1.9966e-02 k_t = [4, 4, 3, 4]
 Generation 1350: Best = 3.728130e+08 FES = 135100 μ_F = 0.719 μ_{CR} = 0.270
 Diversity = 1.8076e-02 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 3.721602e+08 FES = 136100 μ_F = 0.727 μ_{CR} = 0.270
 Diversity = 1.7236e-02 k_t = [2, 3, 4, 4]
 Generation 1370: Best = 3.715034e+08 FES = 137100 μ_F = 0.726 μ_{CR} = 0.270
 Diversity = 1.7940e-02 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 3.704807e+08 FES = 138100 μ_F = 0.723 μ_{CR} = 0.280
 Diversity = 1.7887e-02 k_t = [2, 3, 4, 2]
 Generation 1390: Best = 3.695176e+08 FES = 139100 μ_F = 0.711 μ_{CR} = 0.275
 Diversity = 1.8343e-02 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 3.684289e+08 FES = 140100 μ_F = 0.707 μ_{CR} = 0.275
 Diversity = 1.7237e-02 k_t = [3, 3, 3, 3]
 Generation 1410: Best = 3.679770e+08 FES = 141100 μ_F = 0.704 μ_{CR} = 0.281
 Diversity = 1.6524e-02 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 3.674350e+08 FES = 142100 μ_F = 0.697 μ_{CR} = 0.273
 Diversity = 1.4422e-02 k_t = [3, 4, 4, 4]
 Generation 1430: Best = 3.667927e+08 FES = 143100 μ_F = 0.699 μ_{CR} = 0.274
 Diversity = 1.3338e-02 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 3.664458e+08 FES = 144100 μ_F = 0.695 μ_{CR} = 0.277
 Diversity = 1.2156e-02 k_t = [3, 4, 4, 4]
 Generation 1450: Best = 3.660260e+08 FES = 145100 μ_F = 0.691 μ_{CR} = 0.269
 Diversity = 1.1062e-02 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 3.656017e+08 FES = 146100 μ_F = 0.697 μ_{CR} = 0.275
 Diversity = 1.0311e-02 k_t = [2, 2, 4, 3]
 Generation 1470: Best = 3.652869e+08 FES = 147100 μ_F = 0.706 μ_{CR} = 0.274
 Diversity = 1.0548e-02 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 3.647363e+08 FES = 148100 μ_F = 0.725 μ_{CR} = 0.266
 Diversity = 1.0939e-02 k_t = [3, 3, 3, 3]
 Generation 1490: Best = 3.641573e+08 FES = 149100 μ_F = 0.729 μ_{CR} = 0.263
 Diversity = 1.2362e-02 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 3.632913e+08 FES = 150100 μ_F = 0.737 μ_{CR} = 0.265
 Diversity = 1.2892e-02 k_t = [2, 3, 4, 3]
 Generation 1510: Best = 3.627554e+08 FES = 151100 μ_F = 0.742 μ_{CR} = 0.267
 Diversity = 1.4260e-02 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 3.622148e+08 FES = 152100 μ_F = 0.733 μ_{CR} = 0.270
 Diversity = 1.3613e-02 k_t = [4, 2, 2, 2]
 Generation 1530: Best = 3.619344e+08 FES = 153100 μ_F = 0.738 μ_{CR} = 0.268
 Diversity = 1.3662e-02 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 3.614561e+08 FES = 154100 μ_F = 0.727 μ_{CR} = 0.259
 Diversity = 1.1700e-02 k_t = [2, 4, 4, 4]
 Generation 1550: Best = 3.612241e+08 FES = 155100 μ_F = 0.714 μ_{CR} = 0.260
 Diversity = 1.0429e-02 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 3.610260e+08 FES = 156100 μ_F = 0.714 μ_{CR} = 0.255
 Diversity = 9.6313e-03 k_t = [2, 3, 3, 3]
 Generation 1570: Best = 3.607024e+08 FES = 157100 μ_F = 0.708 μ_{CR} = 0.260
 Diversity = 9.6828e-03 k_t = [1, 1, 1, 1]

Generation 1580: Best = 3.604512e+08 FES = 158100 μ_F = 0.713 μ_{CR} = 0.255
 Diversity = 9.8845e-03 k_t = [2, 2, 2, 2]
 Generation 1590: Best = 3.601105e+08 FES = 159100 μ_F = 0.713 μ_{CR} = 0.252
 Diversity = 9.8673e-03 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 3.598288e+08 FES = 160100 μ_F = 0.722 μ_{CR} = 0.246
 Diversity = 9.1217e-03 k_t = [2, 2, 4, 4]
 Generation 1610: Best = 3.594725e+08 FES = 161100 μ_F = 0.713 μ_{CR} = 0.242
 Diversity = 8.4085e-03 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 3.591900e+08 FES = 162100 μ_F = 0.708 μ_{CR} = 0.234
 Diversity = 8.3960e-03 k_t = [2, 2, 3, 2]
 Generation 1630: Best = 3.589044e+08 FES = 163100 μ_F = 0.723 μ_{CR} = 0.233
 Diversity = 9.9670e-03 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 3.585587e+08 FES = 164100 μ_F = 0.724 μ_{CR} = 0.234
 Diversity = 1.0039e-02 k_t = [2, 3, 3, 4]
 Generation 1650: Best = 3.578665e+08 FES = 165100 μ_F = 0.729 μ_{CR} = 0.231
 Diversity = 1.1527e-02 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 3.573521e+08 FES = 166100 μ_F = 0.734 μ_{CR} = 0.232
 Diversity = 1.2412e-02 k_t = [3, 4, 2, 4]
 Generation 1670: Best = 3.569260e+08 FES = 167100 μ_F = 0.726 μ_{CR} = 0.232
 Diversity = 1.2550e-02 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 3.563033e+08 FES = 168100 μ_F = 0.719 μ_{CR} = 0.240
 Diversity = 1.1314e-02 k_t = [3, 3, 3, 4]
 Generation 1690: Best = 3.556574e+08 FES = 169100 μ_F = 0.729 μ_{CR} = 0.239
 Diversity = 1.2094e-02 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 3.553398e+08 FES = 170100 μ_F = 0.738 μ_{CR} = 0.238
 Diversity = 1.1698e-02 k_t = [4, 3, 3, 2]
 Generation 1710: Best = 3.551788e+08 FES = 171100 μ_F = 0.732 μ_{CR} = 0.235
 Diversity = 1.1137e-02 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 3.547207e+08 FES = 172100 μ_F = 0.725 μ_{CR} = 0.228
 Diversity = 1.0579e-02 k_t = [3, 2, 3, 2]
 Generation 1730: Best = 3.543801e+08 FES = 173100 μ_F = 0.719 μ_{CR} = 0.230
 Diversity = 1.0856e-02 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 3.540607e+08 FES = 174100 μ_F = 0.718 μ_{CR} = 0.237
 Diversity = 1.0073e-02 k_t = [3, 3, 4, 3]
 Generation 1750: Best = 3.538603e+08 FES = 175100 μ_F = 0.723 μ_{CR} = 0.235
 Diversity = 9.9238e-03 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 3.535589e+08 FES = 176100 μ_F = 0.737 μ_{CR} = 0.239
 Diversity = 9.6902e-03 k_t = [2, 3, 4, 4]
 Generation 1770: Best = 3.532678e+08 FES = 177100 μ_F = 0.727 μ_{CR} = 0.250
 Diversity = 1.0203e-02 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 3.529749e+08 FES = 178100 μ_F = 0.719 μ_{CR} = 0.256
 Diversity = 9.8868e-03 k_t = [2, 3, 2, 4]
 Generation 1790: Best = 3.526321e+08 FES = 179100 μ_F = 0.712 μ_{CR} = 0.267
 Diversity = 9.7158e-03 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 3.523809e+08 FES = 180100 μ_F = 0.708 μ_{CR} = 0.266
 Diversity = 9.4541e-03 k_t = [2, 2, 3, 4]
 Generation 1810: Best = 3.521275e+08 FES = 181100 μ_F = 0.714 μ_{CR} = 0.270
 Diversity = 9.6780e-03 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 3.517158e+08 FES = 182100 μ_F = 0.714 μ_{CR} = 0.274
 Diversity = 9.3149e-03 k_t = [4, 4, 3, 3]
 Generation 1830: Best = 3.516211e+08 FES = 183100 μ_F = 0.712 μ_{CR} = 0.278
 Diversity = 9.6498e-03 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 3.512437e+08 FES = 184100 μ_F = 0.702 μ_{CR} = 0.279
 Diversity = 8.9695e-03 k_t = [2, 4, 2, 3]

Generation 1850: Best = 3.508455e+08	FES = 185100	$\mu_F = 0.710$	$\mu_{CR} = 0.279$
Diversity = 9.9851e-03	k_t = [1, 1, 1, 1]		
Generation 1860: Best = 3.502519e+08	FES = 186100	$\mu_F = 0.719$	$\mu_{CR} = 0.279$
Diversity = 9.7170e-03	k_t = [4, 2, 2, 3]		
Generation 1870: Best = 3.500435e+08	FES = 187100	$\mu_F = 0.715$	$\mu_{CR} = 0.291$
Diversity = 9.4996e-03	k_t = [1, 1, 1, 1]		
Generation 1880: Best = 3.498862e+08	FES = 188100	$\mu_F = 0.710$	$\mu_{CR} = 0.293$
Diversity = 8.6843e-03	k_t = [3, 3, 4, 2]		
Generation 1890: Best = 3.496363e+08	FES = 189100	$\mu_F = 0.712$	$\mu_{CR} = 0.280$
Diversity = 9.0246e-03	k_t = [1, 1, 1, 1]		
Generation 1900: Best = 3.492918e+08	FES = 190100	$\mu_F = 0.722$	$\mu_{CR} = 0.283$
Diversity = 9.4803e-03	k_t = [4, 4, 4, 2]		
Generation 1910: Best = 3.488950e+08	FES = 191100	$\mu_F = 0.706$	$\mu_{CR} = 0.286$
Diversity = 9.1859e-03	k_t = [1, 1, 1, 1]		
Generation 1920: Best = 3.486113e+08	FES = 192100	$\mu_F = 0.713$	$\mu_{CR} = 0.286$
Diversity = 9.3876e-03	k_t = [4, 4, 3, 2]		
Generation 1930: Best = 3.482756e+08	FES = 193100	$\mu_F = 0.718$	$\mu_{CR} = 0.285$
Diversity = 9.3460e-03	k_t = [1, 1, 1, 1]		
Generation 1940: Best = 3.478687e+08	FES = 194100	$\mu_F = 0.718$	$\mu_{CR} = 0.288$
Diversity = 9.3638e-03	k_t = [3, 4, 4, 3]		
Generation 1950: Best = 3.476366e+08	FES = 195100	$\mu_F = 0.726$	$\mu_{CR} = 0.295$
Diversity = 9.0457e-03	k_t = [1, 1, 1, 1]		
Generation 1960: Best = 3.473649e+08	FES = 196100	$\mu_F = 0.727$	$\mu_{CR} = 0.289$
Diversity = 8.6999e-03	k_t = [4, 3, 4, 2]		
Generation 1970: Best = 3.470062e+08	FES = 197100	$\mu_F = 0.729$	$\mu_{CR} = 0.290$
Diversity = 9.4674e-03	k_t = [1, 1, 1, 1]		
Generation 1980: Best = 3.466043e+08	FES = 198100	$\mu_F = 0.733$	$\mu_{CR} = 0.288$
Diversity = 9.7651e-03	k_t = [4, 4, 2, 4]		
Generation 1990: Best = 3.464295e+08	FES = 199100	$\mu_F = 0.744$	$\mu_{CR} = 0.276$
Diversity = 9.5428e-03	k_t = [1, 1, 1, 1]		
Generation 2000: Best = 3.460074e+08	FES = 200100	$\mu_F = 0.731$	$\mu_{CR} = 0.262$
Diversity = 8.8441e-03	k_t = [3, 3, 4, 4]		
Generation 2010: Best = 3.457356e+08	FES = 201100	$\mu_F = 0.733$	$\mu_{CR} = 0.264$
Diversity = 9.8542e-03	k_t = [1, 1, 1, 1]		
Generation 2020: Best = 3.455013e+08	FES = 202100	$\mu_F = 0.734$	$\mu_{CR} = 0.260$
Diversity = 9.4793e-03	k_t = [3, 2, 2, 4]		
Generation 2030: Best = 3.453129e+08	FES = 203100	$\mu_F = 0.726$	$\mu_{CR} = 0.255$
Diversity = 9.0303e-03	k_t = [1, 1, 1, 1]		
Generation 2040: Best = 3.451883e+08	FES = 204100	$\mu_F = 0.727$	$\mu_{CR} = 0.238$
Diversity = 7.9501e-03	k_t = [4, 2, 3, 3]		
Generation 2050: Best = 3.450371e+08	FES = 205100	$\mu_F = 0.709$	$\mu_{CR} = 0.240$
Diversity = 7.3188e-03	k_t = [1, 1, 1, 1]		
Generation 2060: Best = 3.449254e+08	FES = 206100	$\mu_F = 0.711$	$\mu_{CR} = 0.241$
Diversity = 6.8269e-03	k_t = [3, 3, 3, 2]		
Generation 2070: Best = 3.447924e+08	FES = 207100	$\mu_F = 0.714$	$\mu_{CR} = 0.247$
Diversity = 6.7585e-03	k_t = [1, 1, 1, 1]		
Generation 2080: Best = 3.447228e+08	FES = 208100	$\mu_F = 0.714$	$\mu_{CR} = 0.241$
Diversity = 6.1540e-03	k_t = [4, 3, 4, 2]		
Generation 2090: Best = 3.446102e+08	FES = 209100	$\mu_F = 0.714$	$\mu_{CR} = 0.242$
Diversity = 5.7527e-03	k_t = [1, 1, 1, 1]		
Generation 2100: Best = 3.444796e+08	FES = 210100	$\mu_F = 0.718$	$\mu_{CR} = 0.242$
Diversity = 5.6840e-03	k_t = [4, 4, 4, 3]		
Generation 2110: Best = 3.443708e+08	FES = 211100	$\mu_F = 0.718$	$\mu_{CR} = 0.252$
Diversity = 5.6477e-03	k_t = [1, 1, 1, 1]		

Generation 2120: Best = 3.441671e+08 FES = 212100 μ_F = 0.725 μ_{CR} = 0.254
 Diversity = 6.2126e-03 k_t = [3, 2, 4, 2]
 Generation 2130: Best = 3.440585e+08 FES = 213100 μ_F = 0.729 μ_{CR} = 0.257
 Diversity = 6.7902e-03 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 3.439579e+08 FES = 214100 μ_F = 0.729 μ_{CR} = 0.260
 Diversity = 6.1886e-03 k_t = [4, 2, 4, 2]
 Generation 2150: Best = 3.439001e+08 FES = 215100 μ_F = 0.726 μ_{CR} = 0.260
 Diversity = 5.6879e-03 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 3.438626e+08 FES = 216100 μ_F = 0.712 μ_{CR} = 0.269
 Diversity = 4.7872e-03 k_t = [4, 4, 2, 3]
 Generation 2170: Best = 3.438394e+08 FES = 217100 μ_F = 0.720 μ_{CR} = 0.269
 Diversity = 4.5910e-03 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 3.437918e+08 FES = 218100 μ_F = 0.722 μ_{CR} = 0.255
 Diversity = 4.2591e-03 k_t = [4, 2, 2, 2]
 Generation 2190: Best = 3.437439e+08 FES = 219100 μ_F = 0.711 μ_{CR} = 0.263
 Diversity = 4.4019e-03 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 3.437046e+08 FES = 220100 μ_F = 0.690 μ_{CR} = 0.255
 Diversity = 3.9817e-03 k_t = [4, 4, 4, 2]
 Generation 2210: Best = 3.436874e+08 FES = 221100 μ_F = 0.683 μ_{CR} = 0.263
 Diversity = 3.6915e-03 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 3.436588e+08 FES = 222100 μ_F = 0.680 μ_{CR} = 0.263
 Diversity = 3.1715e-03 k_t = [3, 4, 3, 3]
 Generation 2230: Best = 3.436387e+08 FES = 223100 μ_F = 0.673 μ_{CR} = 0.254
 Diversity = 2.7848e-03 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 3.436145e+08 FES = 224100 μ_F = 0.678 μ_{CR} = 0.261
 Diversity = 2.5050e-03 k_t = [2, 2, 2, 2]
 Generation 2250: Best = 3.436069e+08 FES = 225100 μ_F = 0.685 μ_{CR} = 0.245
 Diversity = 2.4488e-03 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 3.435741e+08 FES = 226100 μ_F = 0.700 μ_{CR} = 0.245
 Diversity = 2.3851e-03 k_t = [4, 2, 3, 2]
 Generation 2270: Best = 3.435611e+08 FES = 227100 μ_F = 0.702 μ_{CR} = 0.239
 Diversity = 2.3653e-03 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 3.435468e+08 FES = 228100 μ_F = 0.711 μ_{CR} = 0.242
 Diversity = 2.1801e-03 k_t = [4, 2, 4, 4]
 Generation 2290: Best = 3.435283e+08 FES = 229100 μ_F = 0.720 μ_{CR} = 0.242
 Diversity = 2.1756e-03 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 3.435166e+08 FES = 230100 μ_F = 0.728 μ_{CR} = 0.239
 Diversity = 2.2276e-03 k_t = [4, 3, 2, 4]
 Generation 2310: Best = 3.435033e+08 FES = 231100 μ_F = 0.729 μ_{CR} = 0.236
 Diversity = 2.1827e-03 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 3.434906e+08 FES = 232100 μ_F = 0.716 μ_{CR} = 0.235
 Diversity = 2.0423e-03 k_t = [3, 3, 4, 3]
 Generation 2330: Best = 3.434860e+08 FES = 233100 μ_F = 0.711 μ_{CR} = 0.232
 Diversity = 2.0593e-03 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 3.434756e+08 FES = 234100 μ_F = 0.713 μ_{CR} = 0.230
 Diversity = 1.7993e-03 k_t = [2, 2, 4, 3]
 Generation 2350: Best = 3.434668e+08 FES = 235100 μ_F = 0.725 μ_{CR} = 0.226
 Diversity = 1.8769e-03 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 3.434554e+08 FES = 236100 μ_F = 0.735 μ_{CR} = 0.219
 Diversity = 1.9324e-03 k_t = [2, 3, 4, 4]
 Generation 2370: Best = 3.434473e+08 FES = 237100 μ_F = 0.736 μ_{CR} = 0.221
 Diversity = 1.9936e-03 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 3.434406e+08 FES = 238100 μ_F = 0.736 μ_{CR} = 0.219
 Diversity = 1.8420e-03 k_t = [3, 2, 4, 4]

Generation 2390: Best = 3.434322e+08 FES = 239100 μ_F = 0.746 μ_{CR} = 0.223
 Diversity = 1.8084e-03 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 3.434244e+08 FES = 240100 μ_F = 0.741 μ_{CR} = 0.220
 Diversity = 1.7232e-03 k_t = [4, 3, 2, 3]
 Generation 2410: Best = 3.434192e+08 FES = 241100 μ_F = 0.730 μ_{CR} = 0.221
 Diversity = 1.6259e-03 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 3.434132e+08 FES = 242100 μ_F = 0.731 μ_{CR} = 0.224
 Diversity = 1.4954e-03 k_t = [4, 2, 3, 4]
 Generation 2430: Best = 3.434054e+08 FES = 243100 μ_F = 0.739 μ_{CR} = 0.227
 Diversity = 1.5454e-03 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 3.433995e+08 FES = 244100 μ_F = 0.734 μ_{CR} = 0.222
 Diversity = 1.4944e-03 k_t = [4, 3, 4, 3]
 Generation 2450: Best = 3.433946e+08 FES = 245100 μ_F = 0.736 μ_{CR} = 0.218
 Diversity = 1.5076e-03 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 3.433860e+08 FES = 246100 μ_F = 0.742 μ_{CR} = 0.217
 Diversity = 1.4441e-03 k_t = [2, 3, 4, 4]
 Generation 2470: Best = 3.433781e+08 FES = 247100 μ_F = 0.744 μ_{CR} = 0.210
 Diversity = 1.4805e-03 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 3.433691e+08 FES = 248100 μ_F = 0.741 μ_{CR} = 0.219
 Diversity = 1.6047e-03 k_t = [4, 2, 4, 2]
 Generation 2490: Best = 3.433578e+08 FES = 249100 μ_F = 0.733 μ_{CR} = 0.224
 Diversity = 1.6061e-03 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 3.433556e+08 FES = 250100 μ_F = 0.733 μ_{CR} = 0.223
 Diversity = 1.5475e-03 k_t = [3, 3, 4, 2]
 Generation 2510: Best = 3.433443e+08 FES = 251100 μ_F = 0.721 μ_{CR} = 0.225
 Diversity = 1.6723e-03 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 3.433374e+08 FES = 252100 μ_F = 0.721 μ_{CR} = 0.237
 Diversity = 1.7444e-03 k_t = [3, 4, 4, 2]
 Generation 2530: Best = 3.433293e+08 FES = 253100 μ_F = 0.721 μ_{CR} = 0.230
 Diversity = 1.6601e-03 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 3.433252e+08 FES = 254100 μ_F = 0.720 μ_{CR} = 0.233
 Diversity = 1.6511e-03 k_t = [2, 3, 4, 3]
 Generation 2550: Best = 3.433207e+08 FES = 255100 μ_F = 0.712 μ_{CR} = 0.228
 Diversity = 1.5433e-03 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 3.433124e+08 FES = 256100 μ_F = 0.714 μ_{CR} = 0.224
 Diversity = 1.3782e-03 k_t = [4, 3, 4, 4]
 Generation 2570: Best = 3.433074e+08 FES = 257100 μ_F = 0.711 μ_{CR} = 0.221
 Diversity = 1.4058e-03 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 3.433033e+08 FES = 258100 μ_F = 0.702 μ_{CR} = 0.219
 Diversity = 1.2546e-03 k_t = [4, 2, 2, 4]
 Generation 2590: Best = 3.432986e+08 FES = 259100 μ_F = 0.695 μ_{CR} = 0.234
 Diversity = 1.2106e-03 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 3.432933e+08 FES = 260100 μ_F = 0.700 μ_{CR} = 0.228
 Diversity = 1.1406e-03 k_t = [4, 3, 3, 4]
 Generation 2610: Best = 3.432871e+08 FES = 261100 μ_F = 0.707 μ_{CR} = 0.224
 Diversity = 1.1706e-03 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 3.432826e+08 FES = 262100 μ_F = 0.716 μ_{CR} = 0.220
 Diversity = 1.2034e-03 k_t = [3, 2, 2, 3]
 Generation 2630: Best = 3.432793e+08 FES = 263100 μ_F = 0.719 μ_{CR} = 0.217
 Diversity = 1.2637e-03 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 3.432750e+08 FES = 264100 μ_F = 0.725 μ_{CR} = 0.209
 Diversity = 1.2178e-03 k_t = [2, 3, 4, 2]
 Generation 2650: Best = 3.432709e+08 FES = 265100 μ_F = 0.730 μ_{CR} = 0.211
 Diversity = 1.3238e-03 k_t = [1, 1, 1, 1]

Generation 2660: Best = 3.432629e+08 FES = 266100 μ_F = 0.735 μ_{CR} = 0.211
 Diversity = 1.4437e-03 k_t = [3, 4, 4, 2]
 Generation 2670: Best = 3.432524e+08 FES = 267100 μ_F = 0.736 μ_{CR} = 0.215
 Diversity = 1.5406e-03 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 3.432418e+08 FES = 268100 μ_F = 0.732 μ_{CR} = 0.221
 Diversity = 1.6818e-03 k_t = [3, 3, 3, 3]
 Generation 2690: Best = 3.432323e+08 FES = 269100 μ_F = 0.745 μ_{CR} = 0.217
 Diversity = 1.8469e-03 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 3.432057e+08 FES = 270100 μ_F = 0.738 μ_{CR} = 0.221
 Diversity = 2.0314e-03 k_t = [2, 4, 3, 2]
 Generation 2710: Best = 3.431807e+08 FES = 271100 μ_F = 0.736 μ_{CR} = 0.215
 Diversity = 2.5423e-03 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 3.431574e+08 FES = 272100 μ_F = 0.738 μ_{CR} = 0.222
 Diversity = 2.7964e-03 k_t = [3, 4, 3, 2]
 Generation 2730: Best = 3.431016e+08 FES = 273100 μ_F = 0.743 μ_{CR} = 0.233
 Diversity = 3.3044e-03 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 3.430824e+08 FES = 274100 μ_F = 0.735 μ_{CR} = 0.233
 Diversity = 3.2033e-03 k_t = [2, 3, 2, 3]
 Generation 2750: Best = 3.430434e+08 FES = 275100 μ_F = 0.733 μ_{CR} = 0.240
 Diversity = 3.4185e-03 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 3.430081e+08 FES = 276100 μ_F = 0.737 μ_{CR} = 0.247
 Diversity = 3.2458e-03 k_t = [2, 3, 2, 4]
 Generation 2770: Best = 3.429857e+08 FES = 277100 μ_F = 0.726 μ_{CR} = 0.248
 Diversity = 3.1658e-03 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 3.429302e+08 FES = 278100 μ_F = 0.732 μ_{CR} = 0.241
 Diversity = 3.3917e-03 k_t = [2, 3, 4, 2]
 Generation 2790: Best = 3.429183e+08 FES = 279100 μ_F = 0.732 μ_{CR} = 0.241
 Diversity = 3.5691e-03 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 3.428960e+08 FES = 280100 μ_F = 0.719 μ_{CR} = 0.229
 Diversity = 3.3735e-03 k_t = [4, 2, 3, 4]
 Generation 2810: Best = 3.428826e+08 FES = 281100 μ_F = 0.716 μ_{CR} = 0.233
 Diversity = 2.9497e-03 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 3.428740e+08 FES = 282100 μ_F = 0.707 μ_{CR} = 0.233
 Diversity = 2.4192e-03 k_t = [3, 3, 2, 4]
 Generation 2830: Best = 3.428620e+08 FES = 283100 μ_F = 0.697 μ_{CR} = 0.234
 Diversity = 2.0461e-03 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 3.428552e+08 FES = 284100 μ_F = 0.697 μ_{CR} = 0.246
 Diversity = 1.8579e-03 k_t = [4, 2, 3, 3]
 Generation 2850: Best = 3.428494e+08 FES = 285100 μ_F = 0.702 μ_{CR} = 0.250
 Diversity = 1.7120e-03 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 3.428446e+08 FES = 286100 μ_F = 0.714 μ_{CR} = 0.247
 Diversity = 1.5743e-03 k_t = [3, 3, 4, 2]
 Generation 2870: Best = 3.428366e+08 FES = 287100 μ_F = 0.724 μ_{CR} = 0.239
 Diversity = 1.6096e-03 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 3.428250e+08 FES = 288100 μ_F = 0.714 μ_{CR} = 0.243
 Diversity = 1.6245e-03 k_t = [3, 3, 3, 3]
 Generation 2890: Best = 3.428216e+08 FES = 289100 μ_F = 0.717 μ_{CR} = 0.234
 Diversity = 1.6712e-03 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 3.428153e+08 FES = 290100 μ_F = 0.716 μ_{CR} = 0.243
 Diversity = 1.5797e-03 k_t = [3, 3, 3, 3]
 Generation 2910: Best = 3.428095e+08 FES = 291100 μ_F = 0.717 μ_{CR} = 0.240
 Diversity = 1.5687e-03 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 3.428041e+08 FES = 292100 μ_F = 0.709 μ_{CR} = 0.240
 Diversity = 1.3517e-03 k_t = [4, 2, 2, 4]

Generation 2930: Best = 3.428014e+08 FES = 293100 μ_F = 0.697 μ_{CR} = 0.237
 Diversity = 1.1837e-03 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 3.427997e+08 FES = 294100 μ_F = 0.705 μ_{CR} = 0.237
 Diversity = 1.1135e-03 k_t = [3, 2, 3, 2]
 Generation 2950: Best = 3.427989e+08 FES = 295100 μ_F = 0.687 μ_{CR} = 0.243
 Diversity = 9.9970e-04 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 3.427964e+08 FES = 296100 μ_F = 0.690 μ_{CR} = 0.233
 Diversity = 8.7455e-04 k_t = [3, 4, 3, 4]
 Generation 2970: Best = 3.427938e+08 FES = 297100 μ_F = 0.693 μ_{CR} = 0.231
 Diversity = 8.6887e-04 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 3.427913e+08 FES = 298100 μ_F = 0.703 μ_{CR} = 0.230
 Diversity = 8.5076e-04 k_t = [2, 3, 2, 4]
 Generation 2990: Best = 3.427892e+08 FES = 299100 μ_F = 0.708 μ_{CR} = 0.233
 Diversity = 8.6623e-04 k_t = [1, 1, 1, 1]
 Generation 3000: Best = 3.427864e+08 FES = 300100 μ_F = 0.712 μ_{CR} = 0.239
 Diversity = 8.9025e-04 k_t = [4, 2, 4, 2]
 Generation 3010: Best = 3.427840e+08 FES = 301100 μ_F = 0.715 μ_{CR} = 0.235
 Diversity = 9.0260e-04 k_t = [1, 1, 1, 1]
 Generation 3020: Best = 3.427804e+08 FES = 302100 μ_F = 0.720 μ_{CR} = 0.244
 Diversity = 8.6065e-04 k_t = [3, 3, 3, 3]
 Generation 3030: Best = 3.427793e+08 FES = 303100 μ_F = 0.717 μ_{CR} = 0.247
 Diversity = 9.0986e-04 k_t = [1, 1, 1, 1]
 Generation 3040: Best = 3.427757e+08 FES = 304100 μ_F = 0.717 μ_{CR} = 0.252
 Diversity = 8.6686e-04 k_t = [4, 3, 2, 3]
 Generation 3050: Best = 3.427740e+08 FES = 305100 μ_F = 0.710 μ_{CR} = 0.253
 Diversity = 8.7189e-04 k_t = [1, 1, 1, 1]
 Generation 3060: Best = 3.427729e+08 FES = 306100 μ_F = 0.702 μ_{CR} = 0.256
 Diversity = 7.5261e-04 k_t = [4, 3, 2, 3]
 Generation 3070: Best = 3.427722e+08 FES = 307100 μ_F = 0.695 μ_{CR} = 0.254
 Diversity = 6.6156e-04 k_t = [1, 1, 1, 1]
 Generation 3080: Best = 3.427715e+08 FES = 308100 μ_F = 0.697 μ_{CR} = 0.236
 Diversity = 6.0341e-04 k_t = [3, 2, 3, 2]
 Generation 3090: Best = 3.427709e+08 FES = 309100 μ_F = 0.683 μ_{CR} = 0.237
 Diversity = 5.8648e-04 k_t = [1, 1, 1, 1]
 Generation 3100: Best = 3.427704e+08 FES = 310100 μ_F = 0.690 μ_{CR} = 0.247
 Diversity = 5.0992e-04 k_t = [3, 3, 4, 3]
 Generation 3110: Best = 3.427696e+08 FES = 311100 μ_F = 0.686 μ_{CR} = 0.248
 Diversity = 4.8504e-04 k_t = [1, 1, 1, 1]
 Generation 3120: Best = 3.427693e+08 FES = 312100 μ_F = 0.689 μ_{CR} = 0.249
 Diversity = 4.3978e-04 k_t = [3, 3, 4, 3]
 Generation 3130: Best = 3.427689e+08 FES = 313100 μ_F = 0.689 μ_{CR} = 0.249
 Diversity = 4.0549e-04 k_t = [1, 1, 1, 1]
 Generation 3140: Best = 3.427685e+08 FES = 314100 μ_F = 0.700 μ_{CR} = 0.242
 Diversity = 3.6980e-04 k_t = [4, 2, 4, 4]
 Generation 3150: Best = 3.427683e+08 FES = 315100 μ_F = 0.697 μ_{CR} = 0.241
 Diversity = 3.5023e-04 k_t = [1, 1, 1, 1]
 Generation 3160: Best = 3.427679e+08 FES = 316100 μ_F = 0.703 μ_{CR} = 0.240
 Diversity = 3.4091e-04 k_t = [2, 4, 3, 2]
 Generation 3170: Best = 3.427676e+08 FES = 317100 μ_F = 0.706 μ_{CR} = 0.245
 Diversity = 3.5668e-04 k_t = [1, 1, 1, 1]
 Generation 3180: Best = 3.427674e+08 FES = 318100 μ_F = 0.693 μ_{CR} = 0.250
 Diversity = 3.2475e-04 k_t = [2, 4, 4, 3]
 Generation 3190: Best = 3.427670e+08 FES = 319100 μ_F = 0.691 μ_{CR} = 0.253
 Diversity = 3.0421e-04 k_t = [1, 1, 1, 1]

Generation 3200: Best = 3.427667e+08 FES = 320100 μ_F = 0.687 μ_{CR} = 0.253
 Diversity = 2.9225e-04 k_t = [4, 3, 4, 3]
 Generation 3210: Best = 3.427666e+08 FES = 321100 μ_F = 0.694 μ_{CR} = 0.253
 Diversity = 2.8793e-04 k_t = [1, 1, 1, 1]
 Generation 3220: Best = 3.427663e+08 FES = 322100 μ_F = 0.702 μ_{CR} = 0.252
 Diversity = 2.7520e-04 k_t = [4, 4, 4, 2]
 Generation 3230: Best = 3.427661e+08 FES = 323100 μ_F = 0.706 μ_{CR} = 0.254
 Diversity = 2.6022e-04 k_t = [1, 1, 1, 1]
 Generation 3240: Best = 3.427658e+08 FES = 324100 μ_F = 0.716 μ_{CR} = 0.259
 Diversity = 2.5113e-04 k_t = [4, 4, 4, 3]
 Generation 3250: Best = 3.427656e+08 FES = 325100 μ_F = 0.731 μ_{CR} = 0.253
 Diversity = 2.5602e-04 k_t = [1, 1, 1, 1]
 Generation 3260: Best = 3.427653e+08 FES = 326100 μ_F = 0.742 μ_{CR} = 0.249
 Diversity = 2.7633e-04 k_t = [2, 3, 2, 4]
 Generation 3270: Best = 3.427649e+08 FES = 327100 μ_F = 0.744 μ_{CR} = 0.244
 Diversity = 3.0631e-04 k_t = [1, 1, 1, 1]
 Generation 3280: Best = 3.427646e+08 FES = 328100 μ_F = 0.734 μ_{CR} = 0.245
 Diversity = 3.3792e-04 k_t = [4, 3, 2, 2]
 Generation 3290: Best = 3.427643e+08 FES = 329100 μ_F = 0.729 μ_{CR} = 0.250
 Diversity = 3.4644e-04 k_t = [1, 1, 1, 1]
 Generation 3300: Best = 3.427640e+08 FES = 330100 μ_F = 0.735 μ_{CR} = 0.250
 Diversity = 3.4596e-04 k_t = [4, 4, 2, 2]
 Generation 3310: Best = 3.427635e+08 FES = 331100 μ_F = 0.735 μ_{CR} = 0.257
 Diversity = 3.2860e-04 k_t = [1, 1, 1, 1]
 Generation 3320: Best = 3.427633e+08 FES = 332100 μ_F = 0.725 μ_{CR} = 0.266
 Diversity = 3.2969e-04 k_t = [2, 4, 2, 2]
 Generation 3330: Best = 3.427627e+08 FES = 333100 μ_F = 0.731 μ_{CR} = 0.273
 Diversity = 3.3902e-04 k_t = [1, 1, 1, 1]
 Generation 3340: Best = 3.427624e+08 FES = 334100 μ_F = 0.735 μ_{CR} = 0.272
 Diversity = 3.4467e-04 k_t = [3, 3, 3, 2]
 Generation 3350: Best = 3.427623e+08 FES = 335100 μ_F = 0.726 μ_{CR} = 0.269
 Diversity = 3.4357e-04 k_t = [1, 1, 1, 1]
 Generation 3360: Best = 3.427619e+08 FES = 336100 μ_F = 0.709 μ_{CR} = 0.276
 Diversity = 3.1841e-04 k_t = [2, 3, 4, 2]
 Generation 3370: Best = 3.427616e+08 FES = 337100 μ_F = 0.706 μ_{CR} = 0.269
 Diversity = 3.3253e-04 k_t = [1, 1, 1, 1]
 Generation 3380: Best = 3.427614e+08 FES = 338100 μ_F = 0.714 μ_{CR} = 0.268
 Diversity = 3.0240e-04 k_t = [2, 2, 4, 3]
 Generation 3390: Best = 3.427612e+08 FES = 339100 μ_F = 0.717 μ_{CR} = 0.276
 Diversity = 3.0314e-04 k_t = [1, 1, 1, 1]
 Generation 3400: Best = 3.427610e+08 FES = 340100 μ_F = 0.710 μ_{CR} = 0.277
 Diversity = 2.8201e-04 k_t = [4, 3, 4, 2]
 Generation 3410: Best = 3.427609e+08 FES = 341100 μ_F = 0.699 μ_{CR} = 0.276
 Diversity = 2.5634e-04 k_t = [1, 1, 1, 1]
 Generation 3420: Best = 3.427607e+08 FES = 342100 μ_F = 0.702 μ_{CR} = 0.275
 Diversity = 2.3959e-04 k_t = [4, 2, 2, 3]
 Generation 3430: Best = 3.427606e+08 FES = 343100 μ_F = 0.701 μ_{CR} = 0.265
 Diversity = 2.3577e-04 k_t = [1, 1, 1, 1]
 Generation 3440: Best = 3.427604e+08 FES = 344100 μ_F = 0.700 μ_{CR} = 0.254
 Diversity = 2.1514e-04 k_t = [3, 3, 4, 4]
 Generation 3450: Best = 3.427603e+08 FES = 345100 μ_F = 0.705 μ_{CR} = 0.262
 Diversity = 2.1696e-04 k_t = [1, 1, 1, 1]
 Generation 3460: Best = 3.427602e+08 FES = 346100 μ_F = 0.707 μ_{CR} = 0.253
 Diversity = 2.0257e-04 k_t = [3, 2, 2, 3]

Generation 3470: Best = 3.427601e+08 FES = 347100 μ_F = 0.711 μ_{CR} = 0.253
 Diversity = 1.9441e-04 k_t = [1, 1, 1, 1]
 Generation 3480: Best = 3.427600e+08 FES = 348100 μ_F = 0.704 μ_{CR} = 0.251
 Diversity = 1.7964e-04 k_t = [3, 2, 2, 3]
 Generation 3490: Best = 3.427599e+08 FES = 349100 μ_F = 0.714 μ_{CR} = 0.251
 Diversity = 1.9839e-04 k_t = [1, 1, 1, 1]
 Generation 3500: Best = 3.427598e+08 FES = 350100 μ_F = 0.702 μ_{CR} = 0.251
 Diversity = 1.7972e-04 k_t = [2, 3, 4, 4]
 Generation 3510: Best = 3.427597e+08 FES = 351100 μ_F = 0.702 μ_{CR} = 0.248
 Diversity = 1.8263e-04 k_t = [1, 1, 1, 1]
 Generation 3520: Best = 3.427597e+08 FES = 352100 μ_F = 0.702 μ_{CR} = 0.251
 Diversity = 1.6353e-04 k_t = [4, 4, 4, 4]
 Generation 3530: Best = 3.427596e+08 FES = 353100 μ_F = 0.704 μ_{CR} = 0.254
 Diversity = 1.3905e-04 k_t = [1, 1, 1, 1]
 Generation 3540: Best = 3.427595e+08 FES = 354100 μ_F = 0.703 μ_{CR} = 0.247
 Diversity = 1.2315e-04 k_t = [4, 4, 4, 2]
 Generation 3550: Best = 3.427595e+08 FES = 355100 μ_F = 0.705 μ_{CR} = 0.247
 Diversity = 1.2354e-04 k_t = [1, 1, 1, 1]
 Generation 3560: Best = 3.427594e+08 FES = 356100 μ_F = 0.715 μ_{CR} = 0.241
 Diversity = 1.3101e-04 k_t = [2, 3, 4, 2]
 Generation 3570: Best = 3.427593e+08 FES = 357100 μ_F = 0.720 μ_{CR} = 0.238
 Diversity = 1.5130e-04 k_t = [1, 1, 1, 1]
 Generation 3580: Best = 3.427592e+08 FES = 358100 μ_F = 0.721 μ_{CR} = 0.242
 Diversity = 1.4065e-04 k_t = [4, 4, 4, 4]
 Generation 3590: Best = 3.427591e+08 FES = 359100 μ_F = 0.747 μ_{CR} = 0.249
 Diversity = 1.5531e-04 k_t = [1, 1, 1, 1]
 Generation 3600: Best = 3.427590e+08 FES = 360100 μ_F = 0.761 μ_{CR} = 0.243
 Diversity = 1.6466e-04 k_t = [4, 3, 4, 4]
 Generation 3610: Best = 3.427590e+08 FES = 361100 μ_F = 0.745 μ_{CR} = 0.246
 Diversity = 1.6998e-04 k_t = [1, 1, 1, 1]
 Generation 3620: Best = 3.427589e+08 FES = 362100 μ_F = 0.745 μ_{CR} = 0.241
 Diversity = 1.6283e-04 k_t = [4, 3, 3, 4]
 Generation 3630: Best = 3.427588e+08 FES = 363100 μ_F = 0.744 μ_{CR} = 0.238
 Diversity = 1.6735e-04 k_t = [1, 1, 1, 1]
 Generation 3640: Best = 3.427587e+08 FES = 364100 μ_F = 0.738 μ_{CR} = 0.242
 Diversity = 1.7240e-04 k_t = [3, 3, 4, 3]
 Generation 3650: Best = 3.427586e+08 FES = 365100 μ_F = 0.735 μ_{CR} = 0.254
 Diversity = 1.7827e-04 k_t = [1, 1, 1, 1]
 Generation 3660: Best = 3.427586e+08 FES = 366100 μ_F = 0.714 μ_{CR} = 0.252
 Diversity = 1.5817e-04 k_t = [2, 4, 3, 4]
 Generation 3670: Best = 3.427585e+08 FES = 367100 μ_F = 0.712 μ_{CR} = 0.246
 Diversity = 1.4396e-04 k_t = [1, 1, 1, 1]
 Generation 3680: Best = 3.427585e+08 FES = 368100 μ_F = 0.704 μ_{CR} = 0.244
 Diversity = 1.3400e-04 k_t = [2, 4, 4, 2]
 Generation 3690: Best = 3.427585e+08 FES = 369100 μ_F = 0.701 μ_{CR} = 0.236
 Diversity = 1.2575e-04 k_t = [1, 1, 1, 1]
 Generation 3700: Best = 3.427584e+08 FES = 370100 μ_F = 0.692 μ_{CR} = 0.227
 Diversity = 1.0921e-04 k_t = [2, 3, 3, 3]
 Generation 3710: Best = 3.427584e+08 FES = 371100 μ_F = 0.695 μ_{CR} = 0.229
 Diversity = 1.0879e-04 k_t = [1, 1, 1, 1]
 Generation 3720: Best = 3.427584e+08 FES = 372100 μ_F = 0.689 μ_{CR} = 0.234
 Diversity = 9.8019e-05 k_t = [3, 4, 2, 4]
 Generation 3730: Best = 3.427583e+08 FES = 373100 μ_F = 0.696 μ_{CR} = 0.237
 Diversity = 9.1752e-05 k_t = [1, 1, 1, 1]

Generation 3740: Best = 3.427583e+08 FES = 374100 μ_F = 0.720 μ_{CR} = 0.257
 Diversity = 1.0101e-04 k_t = [4, 4, 3, 2]
 Generation 3750: Best = 3.427582e+08 FES = 375100 μ_F = 0.722 μ_{CR} = 0.263
 Diversity = 1.1192e-04 k_t = [1, 1, 1, 1]
 Generation 3760: Best = 3.427582e+08 FES = 376100 μ_F = 0.731 μ_{CR} = 0.263
 Diversity = 1.1868e-04 k_t = [3, 4, 3, 3]
 Generation 3770: Best = 3.427582e+08 FES = 377100 μ_F = 0.724 μ_{CR} = 0.256
 Diversity = 1.1084e-04 k_t = [1, 1, 1, 1]
 Generation 3780: Best = 3.427581e+08 FES = 378100 μ_F = 0.716 μ_{CR} = 0.245
 Diversity = 1.0330e-04 k_t = [3, 2, 2, 4]
 Generation 3790: Best = 3.427581e+08 FES = 379100 μ_F = 0.713 μ_{CR} = 0.243
 Diversity = 9.0340e-05 k_t = [1, 1, 1, 1]
 Generation 3800: Best = 3.427581e+08 FES = 380100 μ_F = 0.718 μ_{CR} = 0.246
 Diversity = 8.6680e-05 k_t = [2, 4, 2, 3]
 Generation 3810: Best = 3.427581e+08 FES = 381100 μ_F = 0.704 μ_{CR} = 0.246
 Diversity = 8.9291e-05 k_t = [1, 1, 1, 1]
 Generation 3820: Best = 3.427580e+08 FES = 382100 μ_F = 0.704 μ_{CR} = 0.246
 Diversity = 8.4156e-05 k_t = [3, 4, 3, 4]
 Generation 3830: Best = 3.427580e+08 FES = 383100 μ_F = 0.709 μ_{CR} = 0.244
 Diversity = 8.8837e-05 k_t = [1, 1, 1, 1]
 Generation 3840: Best = 3.427580e+08 FES = 384100 μ_F = 0.706 μ_{CR} = 0.243
 Diversity = 8.6636e-05 k_t = [4, 2, 2, 2]
 Generation 3850: Best = 3.427580e+08 FES = 385100 μ_F = 0.711 μ_{CR} = 0.236
 Diversity = 8.6858e-05 k_t = [1, 1, 1, 1]
 Generation 3860: Best = 3.427580e+08 FES = 386100 μ_F = 0.721 μ_{CR} = 0.238
 Diversity = 7.9462e-05 k_t = [2, 2, 4, 4]
 Generation 3870: Best = 3.427579e+08 FES = 387100 μ_F = 0.720 μ_{CR} = 0.247
 Diversity = 7.7269e-05 k_t = [1, 1, 1, 1]
 Generation 3880: Best = 3.427579e+08 FES = 388100 μ_F = 0.709 μ_{CR} = 0.243
 Diversity = 7.6115e-05 k_t = [4, 2, 4, 2]
 Generation 3890: Best = 3.427579e+08 FES = 389100 μ_F = 0.717 μ_{CR} = 0.248
 Diversity = 7.3569e-05 k_t = [1, 1, 1, 1]
 Generation 3900: Best = 3.427579e+08 FES = 390100 μ_F = 0.725 μ_{CR} = 0.246
 Diversity = 7.1883e-05 k_t = [2, 4, 2, 3]
 Generation 3910: Best = 3.427579e+08 FES = 391100 μ_F = 0.727 μ_{CR} = 0.253
 Diversity = 7.8830e-05 k_t = [1, 1, 1, 1]
 Generation 3920: Best = 3.427578e+08 FES = 392100 μ_F = 0.728 μ_{CR} = 0.248
 Diversity = 7.6702e-05 k_t = [2, 3, 2, 4]
 Generation 3930: Best = 3.427578e+08 FES = 393100 μ_F = 0.720 μ_{CR} = 0.252
 Diversity = 7.8817e-05 k_t = [1, 1, 1, 1]
 Generation 3940: Best = 3.427578e+08 FES = 394100 μ_F = 0.712 μ_{CR} = 0.253
 Diversity = 7.4891e-05 k_t = [2, 2, 2, 3]
 Generation 3950: Best = 3.427578e+08 FES = 395100 μ_F = 0.699 μ_{CR} = 0.255
 Diversity = 7.4434e-05 k_t = [1, 1, 1, 1]
 Generation 3960: Best = 3.427578e+08 FES = 396100 μ_F = 0.701 μ_{CR} = 0.262
 Diversity = 6.8613e-05 k_t = [2, 4, 3, 3]
 Generation 3970: Best = 3.427578e+08 FES = 397100 μ_F = 0.701 μ_{CR} = 0.270
 Diversity = 6.5868e-05 k_t = [1, 1, 1, 1]
 Generation 3980: Best = 3.427577e+08 FES = 398100 μ_F = 0.701 μ_{CR} = 0.273
 Diversity = 6.5386e-05 k_t = [3, 4, 2, 4]
 Generation 3990: Best = 3.427577e+08 FES = 399100 μ_F = 0.705 μ_{CR} = 0.278
 Diversity = 6.5221e-05 k_t = [1, 1, 1, 1]
 Generation 4000: Best = 3.427577e+08 FES = 400100 μ_F = 0.725 μ_{CR} = 0.293
 Diversity = 6.7311e-05 k_t = [3, 2, 4, 3]

Generation 4010: Best = 3.427576e+08 FES = 401100 μ_F = 0.721 μ_{CR} = 0.293
 Diversity = 8.2585e-05 k_t = [1, 1, 1, 1]
 Generation 4020: Best = 3.427576e+08 FES = 402100 μ_F = 0.718 μ_{CR} = 0.291
 Diversity = 7.8918e-05 k_t = [2, 3, 4, 2]
 Generation 4030: Best = 3.427576e+08 FES = 403100 μ_F = 0.724 μ_{CR} = 0.286
 Diversity = 7.6794e-05 k_t = [1, 1, 1, 1]
 Generation 4040: Best = 3.427575e+08 FES = 404100 μ_F = 0.736 μ_{CR} = 0.296
 Diversity = 8.6740e-05 k_t = [3, 2, 3, 4]
 Generation 4050: Best = 3.427575e+08 FES = 405100 μ_F = 0.718 μ_{CR} = 0.301
 Diversity = 9.5252e-05 k_t = [1, 1, 1, 1]
 Generation 4060: Best = 3.427575e+08 FES = 406100 μ_F = 0.707 μ_{CR} = 0.298
 Diversity = 8.0388e-05 k_t = [4, 4, 3, 3]
 Generation 4070: Best = 3.427575e+08 FES = 407100 μ_F = 0.708 μ_{CR} = 0.287
 Diversity = 7.9457e-05 k_t = [1, 1, 1, 1]
 Generation 4080: Best = 3.427574e+08 FES = 408100 μ_F = 0.707 μ_{CR} = 0.292
 Diversity = 7.4440e-05 k_t = [3, 3, 2, 3]
 Generation 4090: Best = 3.427574e+08 FES = 409100 μ_F = 0.695 μ_{CR} = 0.291
 Diversity = 6.8563e-05 k_t = [1, 1, 1, 1]
 Generation 4100: Best = 3.427574e+08 FES = 410100 μ_F = 0.705 μ_{CR} = 0.294
 Diversity = 6.6225e-05 k_t = [2, 3, 3, 2]
 Generation 4110: Best = 3.427574e+08 FES = 411100 μ_F = 0.700 μ_{CR} = 0.275
 Diversity = 6.3620e-05 k_t = [1, 1, 1, 1]
 Generation 4120: Best = 3.427574e+08 FES = 412100 μ_F = 0.699 μ_{CR} = 0.269
 Diversity = 6.2622e-05 k_t = [2, 2, 3, 2]
 Generation 4130: Best = 3.427574e+08 FES = 413100 μ_F = 0.686 μ_{CR} = 0.280
 Diversity = 6.6125e-05 k_t = [1, 1, 1, 1]
 Generation 4140: Best = 3.427573e+08 FES = 414100 μ_F = 0.700 μ_{CR} = 0.281
 Diversity = 6.2861e-05 k_t = [2, 2, 2, 4]
 Generation 4150: Best = 3.427573e+08 FES = 415100 μ_F = 0.690 μ_{CR} = 0.282
 Diversity = 5.6683e-05 k_t = [1, 1, 1, 1]
 Generation 4160: Best = 3.427573e+08 FES = 416100 μ_F = 0.699 μ_{CR} = 0.286
 Diversity = 5.0740e-05 k_t = [3, 4, 3, 2]
 Generation 4170: Best = 3.427573e+08 FES = 417100 μ_F = 0.705 μ_{CR} = 0.294
 Diversity = 5.0714e-05 k_t = [1, 1, 1, 1]
 Generation 4180: Best = 3.427573e+08 FES = 418100 μ_F = 0.689 μ_{CR} = 0.301
 Diversity = 4.3664e-05 k_t = [3, 2, 2, 4]
 Generation 4190: Best = 3.427573e+08 FES = 419100 μ_F = 0.680 μ_{CR} = 0.302
 Diversity = 4.0348e-05 k_t = [1, 1, 1, 1]
 Generation 4200: Best = 3.427573e+08 FES = 420100 μ_F = 0.686 μ_{CR} = 0.296
 Diversity = 3.5238e-05 k_t = [3, 3, 2, 3]
 Generation 4210: Best = 3.427573e+08 FES = 421100 μ_F = 0.688 μ_{CR} = 0.301
 Diversity = 3.5252e-05 k_t = [1, 1, 1, 1]
 Generation 4220: Best = 3.427573e+08 FES = 422100 μ_F = 0.687 μ_{CR} = 0.298
 Diversity = 3.0797e-05 k_t = [3, 4, 3, 3]
 Generation 4230: Best = 3.427573e+08 FES = 423100 μ_F = 0.698 μ_{CR} = 0.305
 Diversity = 3.0180e-05 k_t = [1, 1, 1, 1]
 Generation 4240: Best = 3.427573e+08 FES = 424100 μ_F = 0.713 μ_{CR} = 0.300
 Diversity = 3.0582e-05 k_t = [3, 4, 2, 2]
 Generation 4250: Best = 3.427573e+08 FES = 425100 μ_F = 0.719 μ_{CR} = 0.310
 Diversity = 3.0062e-05 k_t = [1, 1, 1, 1]
 Generation 4260: Best = 3.427573e+08 FES = 426100 μ_F = 0.708 μ_{CR} = 0.318
 Diversity = 2.8438e-05 k_t = [2, 3, 2, 2]
 Generation 4270: Best = 3.427573e+08 FES = 427100 μ_F = 0.702 μ_{CR} = 0.312
 Diversity = 2.8656e-05 k_t = [1, 1, 1, 1]

Generation 4280: Best = 3.427573e+08 FES = 428100 μ_F = 0.711 μ_{CR} = 0.312
 Diversity = 2.7588e-05 k_t = [3, 2, 2, 2]
 Generation 4290: Best = 3.427573e+08 FES = 429100 μ_F = 0.691 μ_{CR} = 0.317
 Diversity = 2.6820e-05 k_t = [1, 1, 1, 1]
 Generation 4300: Best = 3.427573e+08 FES = 430100 μ_F = 0.696 μ_{CR} = 0.317
 Diversity = 2.3584e-05 k_t = [4, 4, 3, 2]
 Generation 4310: Best = 3.427573e+08 FES = 431100 μ_F = 0.688 μ_{CR} = 0.312
 Diversity = 2.1922e-05 k_t = [1, 1, 1, 1]
 Generation 4320: Best = 3.427573e+08 FES = 432100 μ_F = 0.699 μ_{CR} = 0.308
 Diversity = 1.9670e-05 k_t = [2, 3, 3, 4]
 Generation 4330: Best = 3.427573e+08 FES = 433100 μ_F = 0.706 μ_{CR} = 0.304
 Diversity = 1.9583e-05 k_t = [1, 1, 1, 1]
 Generation 4340: Best = 3.427573e+08 FES = 434100 μ_F = 0.712 μ_{CR} = 0.298
 Diversity = 1.8442e-05 k_t = [2, 3, 2, 3]
 Generation 4350: Best = 3.427573e+08 FES = 435100 μ_F = 0.719 μ_{CR} = 0.293
 Diversity = 1.9124e-05 k_t = [1, 1, 1, 1]
 Generation 4360: Best = 3.427573e+08 FES = 436100 μ_F = 0.711 μ_{CR} = 0.297
 Diversity = 1.9677e-05 k_t = [2, 2, 2, 3]
 Generation 4370: Best = 3.427573e+08 FES = 437100 μ_F = 0.712 μ_{CR} = 0.284
 Diversity = 1.9887e-05 k_t = [1, 1, 1, 1]
 Generation 4380: Best = 3.427573e+08 FES = 438100 μ_F = 0.707 μ_{CR} = 0.283
 Diversity = 1.8484e-05 k_t = [2, 4, 4, 3]
 Generation 4390: Best = 3.427573e+08 FES = 439100 μ_F = 0.711 μ_{CR} = 0.284
 Diversity = 1.9867e-05 k_t = [1, 1, 1, 1]
 Generation 4400: Best = 3.427573e+08 FES = 440100 μ_F = 0.706 μ_{CR} = 0.284
 Diversity = 1.8325e-05 k_t = [3, 4, 2, 2]
 Generation 4410: Best = 3.427573e+08 FES = 441100 μ_F = 0.718 μ_{CR} = 0.283
 Diversity = 2.0377e-05 k_t = [1, 1, 1, 1]
 Generation 4420: Best = 3.427573e+08 FES = 442100 μ_F = 0.708 μ_{CR} = 0.281
 Diversity = 1.9823e-05 k_t = [2, 4, 2, 3]
 Generation 4430: Best = 3.427573e+08 FES = 443100 μ_F = 0.689 μ_{CR} = 0.270
 Diversity = 1.8358e-05 k_t = [1, 1, 1, 1]
 Generation 4440: Best = 3.427573e+08 FES = 444100 μ_F = 0.687 μ_{CR} = 0.276
 Diversity = 1.6337e-05 k_t = [2, 3, 3, 4]
 Generation 4450: Best = 3.427573e+08 FES = 445100 μ_F = 0.689 μ_{CR} = 0.273
 Diversity = 1.5182e-05 k_t = [1, 1, 1, 1]
 Generation 4460: Best = 3.427573e+08 FES = 446100 μ_F = 0.689 μ_{CR} = 0.276
 Diversity = 1.5023e-05 k_t = [2, 2, 3, 3]
 Generation 4470: Best = 3.427573e+08 FES = 447100 μ_F = 0.690 μ_{CR} = 0.275
 Diversity = 1.5438e-05 k_t = [1, 1, 1, 1]
 Generation 4480: Best = 3.427573e+08 FES = 448100 μ_F = 0.707 μ_{CR} = 0.269
 Diversity = 1.6159e-05 k_t = [4, 2, 3, 4]
 Generation 4490: Best = 3.427573e+08 FES = 449100 μ_F = 0.717 μ_{CR} = 0.273
 Diversity = 1.7945e-05 k_t = [1, 1, 1, 1]
 Generation 4500: Best = 3.427573e+08 FES = 450100 μ_F = 0.723 μ_{CR} = 0.281
 Diversity = 1.8612e-05 k_t = [2, 4, 3, 3]
 Generation 4510: Best = 3.427573e+08 FES = 451100 μ_F = 0.722 μ_{CR} = 0.268
 Diversity = 2.0771e-05 k_t = [1, 1, 1, 1]
 Generation 4520: Best = 3.427573e+08 FES = 452100 μ_F = 0.732 μ_{CR} = 0.278
 Diversity = 2.0766e-05 k_t = [4, 2, 2, 3]
 Generation 4530: Best = 3.427572e+08 FES = 453100 μ_F = 0.742 μ_{CR} = 0.282
 Diversity = 2.5230e-05 k_t = [1, 1, 1, 1]
 Generation 4540: Best = 3.427572e+08 FES = 454100 μ_F = 0.736 μ_{CR} = 0.284
 Diversity = 2.6375e-05 k_t = [2, 3, 4, 3]

Generation 4550: Best = 3.427572e+08 FES = 455100 μ_F = 0.748 μ_{CR} = 0.281
 Diversity = 3.0564e-05 k_t = [1, 1, 1, 1]
 Generation 4560: Best = 3.427572e+08 FES = 456100 μ_F = 0.747 μ_{CR} = 0.285
 Diversity = 3.4494e-05 k_t = [2, 4, 2, 4]
 Generation 4570: Best = 3.427572e+08 FES = 457100 μ_F = 0.739 μ_{CR} = 0.277
 Diversity = 3.6824e-05 k_t = [1, 1, 1, 1]
 Generation 4580: Best = 3.427572e+08 FES = 458100 μ_F = 0.746 μ_{CR} = 0.264
 Diversity = 4.1622e-05 k_t = [4, 3, 2, 3]
 Generation 4590: Best = 3.427572e+08 FES = 459100 μ_F = 0.757 μ_{CR} = 0.261
 Diversity = 4.9804e-05 k_t = [1, 1, 1, 1]
 Generation 4600: Best = 3.427572e+08 FES = 460100 μ_F = 0.747 μ_{CR} = 0.262
 Diversity = 5.0210e-05 k_t = [4, 4, 2, 2]
 Generation 4610: Best = 3.427572e+08 FES = 461100 μ_F = 0.744 μ_{CR} = 0.255
 Diversity = 4.9813e-05 k_t = [1, 1, 1, 1]
 Generation 4620: Best = 3.427572e+08 FES = 462100 μ_F = 0.723 μ_{CR} = 0.245
 Diversity = 4.3056e-05 k_t = [4, 3, 4, 3]
 Generation 4630: Best = 3.427572e+08 FES = 463100 μ_F = 0.715 μ_{CR} = 0.247
 Diversity = 4.1828e-05 k_t = [1, 1, 1, 1]
 Generation 4640: Best = 3.427572e+08 FES = 464100 μ_F = 0.706 μ_{CR} = 0.246
 Diversity = 3.6648e-05 k_t = [4, 3, 2, 4]
 Generation 4650: Best = 3.427572e+08 FES = 465100 μ_F = 0.722 μ_{CR} = 0.251
 Diversity = 3.5642e-05 k_t = [1, 1, 1, 1]
 Generation 4660: Best = 3.427572e+08 FES = 466100 μ_F = 0.716 μ_{CR} = 0.251
 Diversity = 3.1943e-05 k_t = [4, 4, 4, 2]
 Generation 4670: Best = 3.427572e+08 FES = 467100 μ_F = 0.729 μ_{CR} = 0.241
 Diversity = 2.8764e-05 k_t = [1, 1, 1, 1]
 Generation 4680: Best = 3.427572e+08 FES = 468100 μ_F = 0.724 μ_{CR} = 0.237
 Diversity = 2.6095e-05 k_t = [4, 3, 2, 4]
 Generation 4690: Best = 3.427572e+08 FES = 469100 μ_F = 0.725 μ_{CR} = 0.239
 Diversity = 2.4974e-05 k_t = [1, 1, 1, 1]
 Generation 4700: Best = 3.427572e+08 FES = 470100 μ_F = 0.714 μ_{CR} = 0.245
 Diversity = 2.2502e-05 k_t = [3, 2, 4, 3]
 Generation 4710: Best = 3.427572e+08 FES = 471100 μ_F = 0.722 μ_{CR} = 0.246
 Diversity = 2.1856e-05 k_t = [1, 1, 1, 1]
 Generation 4720: Best = 3.427572e+08 FES = 472100 μ_F = 0.711 μ_{CR} = 0.248
 Diversity = 1.9357e-05 k_t = [4, 3, 4, 4]
 Generation 4730: Best = 3.427572e+08 FES = 473100 μ_F = 0.704 μ_{CR} = 0.251
 Diversity = 1.7866e-05 k_t = [1, 1, 1, 1]
 Generation 4740: Best = 3.427572e+08 FES = 474100 μ_F = 0.711 μ_{CR} = 0.248
 Diversity = 1.6518e-05 k_t = [3, 3, 2, 3]
 Generation 4750: Best = 3.427572e+08 FES = 475100 μ_F = 0.723 μ_{CR} = 0.241
 Diversity = 1.8242e-05 k_t = [1, 1, 1, 1]
 Generation 4760: Best = 3.427572e+08 FES = 476100 μ_F = 0.723 μ_{CR} = 0.237
 Diversity = 1.9593e-05 k_t = [4, 2, 2, 2]
 Generation 4770: Best = 3.427572e+08 FES = 477100 μ_F = 0.710 μ_{CR} = 0.238
 Diversity = 1.8750e-05 k_t = [1, 1, 1, 1]
 Generation 4780: Best = 3.427572e+08 FES = 478100 μ_F = 0.709 μ_{CR} = 0.236
 Diversity = 1.6342e-05 k_t = [3, 3, 3, 2]
 Generation 4790: Best = 3.427572e+08 FES = 479100 μ_F = 0.716 μ_{CR} = 0.236
 Diversity = 1.7065e-05 k_t = [1, 1, 1, 1]
 Generation 4800: Best = 3.427572e+08 FES = 480100 μ_F = 0.705 μ_{CR} = 0.231
 Diversity = 1.5663e-05 k_t = [2, 4, 3, 3]
 Generation 4810: Best = 3.427572e+08 FES = 481100 μ_F = 0.707 μ_{CR} = 0.231
 Diversity = 1.6441e-05 k_t = [1, 1, 1, 1]

Generation 4820: Best = 3.427572e+08 FES = 482100 μ_F = 0.709 μ_{CR} = 0.232
 Diversity = 1.5984e-05 k_t = [2, 3, 4, 3]
 Generation 4830: Best = 3.427572e+08 FES = 483100 μ_F = 0.724 μ_{CR} = 0.234
 Diversity = 1.7847e-05 k_t = [1, 1, 1, 1]
 Generation 4840: Best = 3.427572e+08 FES = 484100 μ_F = 0.734 μ_{CR} = 0.230
 Diversity = 1.7279e-05 k_t = [2, 3, 2, 4]
 Generation 4850: Best = 3.427572e+08 FES = 485100 μ_F = 0.728 μ_{CR} = 0.230
 Diversity = 1.8020e-05 k_t = [1, 1, 1, 1]
 Generation 4860: Best = 3.427572e+08 FES = 486100 μ_F = 0.729 μ_{CR} = 0.232
 Diversity = 1.6743e-05 k_t = [2, 4, 3, 3]
 Generation 4870: Best = 3.427572e+08 FES = 487100 μ_F = 0.724 μ_{CR} = 0.231
 Diversity = 1.6313e-05 k_t = [1, 1, 1, 1]
 Generation 4880: Best = 3.427572e+08 FES = 488100 μ_F = 0.723 μ_{CR} = 0.242
 Diversity = 1.4957e-05 k_t = [4, 3, 3, 4]
 Generation 4890: Best = 3.427572e+08 FES = 489100 μ_F = 0.719 μ_{CR} = 0.244
 Diversity = 1.3959e-05 k_t = [1, 1, 1, 1]
 Generation 4900: Best = 3.427572e+08 FES = 490100 μ_F = 0.712 μ_{CR} = 0.238
 Diversity = 1.2685e-05 k_t = [2, 4, 2, 3]
 Generation 4910: Best = 3.427572e+08 FES = 491100 μ_F = 0.704 μ_{CR} = 0.239
 Diversity = 1.2541e-05 k_t = [1, 1, 1, 1]
 Generation 4920: Best = 3.427572e+08 FES = 492100 μ_F = 0.713 μ_{CR} = 0.235
 Diversity = 1.1063e-05 k_t = [2, 4, 2, 4]
 Generation 4930: Best = 3.427572e+08 FES = 493100 μ_F = 0.709 μ_{CR} = 0.230
 Diversity = 1.0148e-05 k_t = [1, 1, 1, 1]
 Generation 4940: Best = 3.427571e+08 FES = 494100 μ_F = 0.735 μ_{CR} = 0.227
 Diversity = 9.7436e-06 k_t = [4, 2, 3, 4]
 Generation 4950: Best = 3.427571e+08 FES = 495100 μ_F = 0.736 μ_{CR} = 0.235
 Diversity = 1.0551e-05 k_t = [1, 1, 1, 1]
 Generation 4960: Best = 3.427571e+08 FES = 496100 μ_F = 0.735 μ_{CR} = 0.242
 Diversity = 1.1387e-05 k_t = [3, 4, 2, 2]
 Generation 4970: Best = 3.427571e+08 FES = 497100 μ_F = 0.736 μ_{CR} = 0.239
 Diversity = 1.1874e-05 k_t = [1, 1, 1, 1]
 Generation 4980: Best = 3.427571e+08 FES = 498100 μ_F = 0.732 μ_{CR} = 0.240
 Diversity = 1.3369e-05 k_t = [3, 3, 4, 2]
 Generation 4990: Best = 3.427571e+08 FES = 499100 μ_F = 0.727 μ_{CR} = 0.253
 Diversity = 1.4250e-05 k_t = [1, 1, 1, 1]
 Generation 5000: Best = 3.427571e+08 FES = 500100 μ_F = 0.714 μ_{CR} = 0.256
 Diversity = 1.3840e-05 k_t = [2, 2, 4, 3]
 Generation 5010: Best = 3.427571e+08 FES = 501100 μ_F = 0.732 μ_{CR} = 0.270
 Diversity = 1.6120e-05 k_t = [1, 1, 1, 1]
 Generation 5020: Best = 3.427571e+08 FES = 502100 μ_F = 0.729 μ_{CR} = 0.266
 Diversity = 1.7545e-05 k_t = [4, 3, 3, 3]
 Generation 5030: Best = 3.427571e+08 FES = 503100 μ_F = 0.737 μ_{CR} = 0.275
 Diversity = 1.9046e-05 k_t = [1, 1, 1, 1]
 Generation 5040: Best = 3.427571e+08 FES = 504100 μ_F = 0.742 μ_{CR} = 0.267
 Diversity = 1.9829e-05 k_t = [4, 2, 2, 3]
 Generation 5050: Best = 3.427571e+08 FES = 505100 μ_F = 0.745 μ_{CR} = 0.277
 Diversity = 2.6213e-05 k_t = [1, 1, 1, 1]
 Generation 5060: Best = 3.427571e+08 FES = 506100 μ_F = 0.764 μ_{CR} = 0.286
 Diversity = 3.0930e-05 k_t = [3, 4, 4, 3]
 Generation 5070: Best = 3.427571e+08 FES = 507100 μ_F = 0.758 μ_{CR} = 0.289
 Diversity = 3.2423e-05 k_t = [1, 1, 1, 1]
 Generation 5080: Best = 3.427571e+08 FES = 508100 μ_F = 0.753 μ_{CR} = 0.291
 Diversity = 3.2299e-05 k_t = [3, 2, 4, 3]

Generation 5090: Best = 3.427571e+08 FES = 509100 μ_F = 0.752 μ_{CR} = 0.287
 Diversity = 3.4815e-05 k_t = [1, 1, 1, 1]
 Generation 5100: Best = 3.427571e+08 FES = 510100 μ_F = 0.736 μ_{CR} = 0.299
 Diversity = 3.2213e-05 k_t = [3, 2, 3, 2]
 Generation 5110: Best = 3.427571e+08 FES = 511100 μ_F = 0.725 μ_{CR} = 0.285
 Diversity = 3.1913e-05 k_t = [1, 1, 1, 1]
 Generation 5120: Best = 3.427571e+08 FES = 512100 μ_F = 0.737 μ_{CR} = 0.283
 Diversity = 3.2700e-05 k_t = [4, 2, 3, 2]
 Generation 5130: Best = 3.427571e+08 FES = 513100 μ_F = 0.734 μ_{CR} = 0.281
 Diversity = 3.4252e-05 k_t = [1, 1, 1, 1]
 Generation 5140: Best = 3.427571e+08 FES = 514100 μ_F = 0.730 μ_{CR} = 0.276
 Diversity = 3.2805e-05 k_t = [2, 3, 3, 4]
 Generation 5150: Best = 3.427571e+08 FES = 515100 μ_F = 0.743 μ_{CR} = 0.275
 Diversity = 3.2972e-05 k_t = [1, 1, 1, 1]
 Generation 5160: Best = 3.427571e+08 FES = 516100 μ_F = 0.738 μ_{CR} = 0.273
 Diversity = 3.3986e-05 k_t = [4, 4, 2, 3]
 Generation 5170: Best = 3.427571e+08 FES = 517100 μ_F = 0.735 μ_{CR} = 0.265
 Diversity = 3.5750e-05 k_t = [1, 1, 1, 1]
 Generation 5180: Best = 3.427571e+08 FES = 518100 μ_F = 0.730 μ_{CR} = 0.270
 Diversity = 3.3501e-05 k_t = [4, 4, 4, 4]
 Generation 5190: Best = 3.427571e+08 FES = 519100 μ_F = 0.729 μ_{CR} = 0.275
 Diversity = 3.0645e-05 k_t = [1, 1, 1, 1]
 Generation 5200: Best = 3.427571e+08 FES = 520100 μ_F = 0.722 μ_{CR} = 0.275
 Diversity = 2.9054e-05 k_t = [2, 3, 3, 4]
 Generation 5210: Best = 3.427571e+08 FES = 521100 μ_F = 0.712 μ_{CR} = 0.274
 Diversity = 2.9307e-05 k_t = [1, 1, 1, 1]
 Generation 5220: Best = 3.427571e+08 FES = 522100 μ_F = 0.703 μ_{CR} = 0.276
 Diversity = 2.5924e-05 k_t = [3, 2, 4, 4]
 Generation 5230: Best = 3.427571e+08 FES = 523100 μ_F = 0.688 μ_{CR} = 0.270
 Diversity = 2.6543e-05 k_t = [1, 1, 1, 1]
 Generation 5240: Best = 3.427571e+08 FES = 524100 μ_F = 0.688 μ_{CR} = 0.282
 Diversity = 2.2841e-05 k_t = [3, 4, 3, 4]
 Generation 5250: Best = 3.427571e+08 FES = 525100 μ_F = 0.695 μ_{CR} = 0.278
 Diversity = 2.1467e-05 k_t = [1, 1, 1, 1]
 Generation 5260: Best = 3.427571e+08 FES = 526100 μ_F = 0.687 μ_{CR} = 0.272
 Diversity = 1.9651e-05 k_t = [2, 2, 3, 3]
 Generation 5270: Best = 3.427571e+08 FES = 527100 μ_F = 0.701 μ_{CR} = 0.269
 Diversity = 2.0546e-05 k_t = [1, 1, 1, 1]
 Generation 5280: Best = 3.427571e+08 FES = 528100 μ_F = 0.716 μ_{CR} = 0.289
 Diversity = 2.1764e-05 k_t = [3, 3, 3, 2]
 Generation 5290: Best = 3.427571e+08 FES = 529100 μ_F = 0.717 μ_{CR} = 0.290
 Diversity = 2.1490e-05 k_t = [1, 1, 1, 1]
 Generation 5300: Best = 3.427571e+08 FES = 530100 μ_F = 0.717 μ_{CR} = 0.300
 Diversity = 2.2064e-05 k_t = [4, 2, 4, 3]
 Generation 5310: Best = 3.427571e+08 FES = 531100 μ_F = 0.720 μ_{CR} = 0.301
 Diversity = 2.2384e-05 k_t = [1, 1, 1, 1]
 Generation 5320: Best = 3.427571e+08 FES = 532100 μ_F = 0.739 μ_{CR} = 0.306
 Diversity = 2.3238e-05 k_t = [2, 3, 4, 3]
 Generation 5330: Best = 3.427571e+08 FES = 533100 μ_F = 0.748 μ_{CR} = 0.311
 Diversity = 2.4628e-05 k_t = [1, 1, 1, 1]
 Generation 5340: Best = 3.427571e+08 FES = 534100 μ_F = 0.753 μ_{CR} = 0.320
 Diversity = 2.9923e-05 k_t = [2, 3, 2, 3]
 Generation 5350: Best = 3.427570e+08 FES = 535100 μ_F = 0.749 μ_{CR} = 0.315
 Diversity = 3.5763e-05 k_t = [1, 1, 1, 1]

Generation 5360: Best = 3.427570e+08 FES = 536100 μ_F = 0.739 μ_{CR} = 0.312
 Diversity = 3.8296e-05 k_t = [2, 4, 2, 2]
 Generation 5370: Best = 3.427570e+08 FES = 537100 μ_F = 0.735 μ_{CR} = 0.309
 Diversity = 4.3228e-05 k_t = [1, 1, 1, 1]
 Generation 5380: Best = 3.427570e+08 FES = 538100 μ_F = 0.725 μ_{CR} = 0.307
 Diversity = 3.9016e-05 k_t = [4, 3, 3, 2]
 Generation 5390: Best = 3.427570e+08 FES = 539100 μ_F = 0.716 μ_{CR} = 0.301
 Diversity = 3.5981e-05 k_t = [1, 1, 1, 1]
 Generation 5400: Best = 3.427570e+08 FES = 540100 μ_F = 0.722 μ_{CR} = 0.296
 Diversity = 3.3474e-05 k_t = [3, 3, 4, 3]
 Generation 5410: Best = 3.427570e+08 FES = 541100 μ_F = 0.712 μ_{CR} = 0.305
 Diversity = 3.3906e-05 k_t = [1, 1, 1, 1]
 Generation 5420: Best = 3.427570e+08 FES = 542100 μ_F = 0.719 μ_{CR} = 0.294
 Diversity = 3.3997e-05 k_t = [3, 4, 2, 4]
 Generation 5430: Best = 3.427570e+08 FES = 543100 μ_F = 0.699 μ_{CR} = 0.296
 Diversity = 2.8432e-05 k_t = [1, 1, 1, 1]
 Generation 5440: Best = 3.427570e+08 FES = 544100 μ_F = 0.695 μ_{CR} = 0.296
 Diversity = 2.4818e-05 k_t = [4, 4, 4, 4]
 Generation 5450: Best = 3.427570e+08 FES = 545100 μ_F = 0.690 μ_{CR} = 0.294
 Diversity = 2.1583e-05 k_t = [1, 1, 1, 1]
 Generation 5460: Best = 3.427570e+08 FES = 546100 μ_F = 0.689 μ_{CR} = 0.287
 Diversity = 1.9415e-05 k_t = [4, 3, 2, 2]
 Generation 5470: Best = 3.427570e+08 FES = 547100 μ_F = 0.696 μ_{CR} = 0.283
 Diversity = 1.9125e-05 k_t = [1, 1, 1, 1]
 Generation 5480: Best = 3.427570e+08 FES = 548100 μ_F = 0.697 μ_{CR} = 0.283
 Diversity = 1.7873e-05 k_t = [2, 2, 4, 3]
 Generation 5490: Best = 3.427570e+08 FES = 549100 μ_F = 0.681 μ_{CR} = 0.276
 Diversity = 1.6963e-05 k_t = [1, 1, 1, 1]
 Generation 5500: Best = 3.427570e+08 FES = 550100 μ_F = 0.692 μ_{CR} = 0.276
 Diversity = 1.5488e-05 k_t = [2, 2, 2, 3]
 Generation 5510: Best = 3.427570e+08 FES = 551100 μ_F = 0.690 μ_{CR} = 0.269
 Diversity = 1.4197e-05 k_t = [1, 1, 1, 1]
 Generation 5520: Best = 3.427570e+08 FES = 552100 μ_F = 0.692 μ_{CR} = 0.265
 Diversity = 1.2582e-05 k_t = [3, 2, 3, 2]
 Generation 5530: Best = 3.427570e+08 FES = 553100 μ_F = 0.685 μ_{CR} = 0.273
 Diversity = 1.2198e-05 k_t = [1, 1, 1, 1]
 Generation 5540: Best = 3.427570e+08 FES = 554100 μ_F = 0.688 μ_{CR} = 0.272
 Diversity = 1.1539e-05 k_t = [2, 3, 2, 4]
 Generation 5550: Best = 3.427570e+08 FES = 555100 μ_F = 0.697 μ_{CR} = 0.265
 Diversity = 1.1019e-05 k_t = [1, 1, 1, 1]
 Generation 5560: Best = 3.427570e+08 FES = 556100 μ_F = 0.685 μ_{CR} = 0.266
 Diversity = 9.9480e-06 k_t = [2, 2, 4, 2]
 Generation 5570: Best = 3.427570e+08 FES = 557100 μ_F = 0.698 μ_{CR} = 0.268
 Diversity = 1.0102e-05 k_t = [1, 1, 1, 1]
 Generation 5580: Best = 3.427570e+08 FES = 558100 μ_F = 0.706 μ_{CR} = 0.265
 Diversity = 9.6242e-06 k_t = [2, 3, 2, 4]
 Generation 5590: Best = 3.427570e+08 FES = 559100 μ_F = 0.700 μ_{CR} = 0.263
 Diversity = 9.3888e-06 k_t = [1, 1, 1, 1]
 Generation 5600: Best = 3.427570e+08 FES = 560100 μ_F = 0.707 μ_{CR} = 0.257
 Diversity = 9.4456e-06 k_t = [2, 3, 2, 2]
 Generation 5610: Best = 3.427570e+08 FES = 561100 μ_F = 0.690 μ_{CR} = 0.249
 Diversity = 9.5821e-06 k_t = [1, 1, 1, 1]
 Generation 5620: Best = 3.427570e+08 FES = 562100 μ_F = 0.697 μ_{CR} = 0.255
 Diversity = 9.2710e-06 k_t = [4, 3, 4, 4]

Generation 5630: Best = 3.427570e+08 FES = 563100 μ_F = 0.711 μ_{CR} = 0.254
 Diversity = 9.2959e-06 k_t = [1, 1, 1, 1]
 Generation 5640: Best = 3.427570e+08 FES = 564100 μ_F = 0.707 μ_{CR} = 0.246
 Diversity = 8.2386e-06 k_t = [2, 2, 2, 3]
 Generation 5650: Best = 3.427570e+08 FES = 565100 μ_F = 0.712 μ_{CR} = 0.247
 Diversity = 8.4751e-06 k_t = [1, 1, 1, 1]
 Generation 5660: Best = 3.427570e+08 FES = 566100 μ_F = 0.719 μ_{CR} = 0.240
 Diversity = 8.0328e-06 k_t = [3, 2, 3, 4]
 Generation 5670: Best = 3.427570e+08 FES = 567100 μ_F = 0.708 μ_{CR} = 0.238
 Diversity = 7.6866e-06 k_t = [1, 1, 1, 1]
 Generation 5680: Best = 3.427570e+08 FES = 568100 μ_F = 0.718 μ_{CR} = 0.243
 Diversity = 7.5655e-06 k_t = [3, 4, 4, 2]
 Generation 5690: Best = 3.427570e+08 FES = 569100 μ_F = 0.711 μ_{CR} = 0.249
 Diversity = 7.0142e-06 k_t = [1, 1, 1, 1]
 Generation 5700: Best = 3.427570e+08 FES = 570100 μ_F = 0.710 μ_{CR} = 0.251
 Diversity = 6.1986e-06 k_t = [4, 4, 2, 3]
 Generation 5710: Best = 3.427570e+08 FES = 571100 μ_F = 0.721 μ_{CR} = 0.250
 Diversity = 5.9190e-06 k_t = [1, 1, 1, 1]
 Generation 5720: Best = 3.427570e+08 FES = 572100 μ_F = 0.712 μ_{CR} = 0.252
 Diversity = 5.3598e-06 k_t = [4, 2, 3, 4]
 Generation 5730: Best = 3.427570e+08 FES = 573100 μ_F = 0.713 μ_{CR} = 0.249
 Diversity = 5.3253e-06 k_t = [1, 1, 1, 1]
 Generation 5740: Best = 3.427570e+08 FES = 574100 μ_F = 0.713 μ_{CR} = 0.243
 Diversity = 5.0875e-06 k_t = [2, 2, 4, 3]
 Generation 5750: Best = 3.427570e+08 FES = 575100 μ_F = 0.709 μ_{CR} = 0.249
 Diversity = 5.4417e-06 k_t = [1, 1, 1, 1]
 Generation 5760: Best = 3.427570e+08 FES = 576100 μ_F = 0.705 μ_{CR} = 0.253
 Diversity = 5.0777e-06 k_t = [2, 2, 2, 3]
 Generation 5770: Best = 3.427570e+08 FES = 577100 μ_F = 0.713 μ_{CR} = 0.252
 Diversity = 5.2633e-06 k_t = [1, 1, 1, 1]
 Generation 5780: Best = 3.427570e+08 FES = 578100 μ_F = 0.732 μ_{CR} = 0.257
 Diversity = 5.4473e-06 k_t = [3, 4, 2, 4]
 Generation 5790: Best = 3.427570e+08 FES = 579100 μ_F = 0.719 μ_{CR} = 0.258
 Diversity = 5.1868e-06 k_t = [1, 1, 1, 1]
 Generation 5800: Best = 3.427570e+08 FES = 580100 μ_F = 0.726 μ_{CR} = 0.254
 Diversity = 4.7419e-06 k_t = [3, 3, 2, 3]
 Generation 5810: Best = 3.427570e+08 FES = 581100 μ_F = 0.725 μ_{CR} = 0.249
 Diversity = 4.5583e-06 k_t = [1, 1, 1, 1]
 Generation 5820: Best = 3.427570e+08 FES = 582100 μ_F = 0.737 μ_{CR} = 0.249
 Diversity = 4.4784e-06 k_t = [4, 4, 3, 3]
 Generation 5830: Best = 3.427570e+08 FES = 583100 μ_F = 0.749 μ_{CR} = 0.249
 Diversity = 4.5808e-06 k_t = [1, 1, 1, 1]
 Generation 5840: Best = 3.427570e+08 FES = 584100 μ_F = 0.742 μ_{CR} = 0.252
 Diversity = 4.2821e-06 k_t = [2, 4, 3, 3]
 Generation 5850: Best = 3.427570e+08 FES = 585100 μ_F = 0.736 μ_{CR} = 0.252
 Diversity = 4.8852e-06 k_t = [1, 1, 1, 1]
 Generation 5860: Best = 3.427570e+08 FES = 586100 μ_F = 0.739 μ_{CR} = 0.252
 Diversity = 4.9782e-06 k_t = [3, 3, 4, 3]
 Generation 5870: Best = 3.427570e+08 FES = 587100 μ_F = 0.741 μ_{CR} = 0.255
 Diversity = 6.0940e-06 k_t = [1, 1, 1, 1]
 Generation 5880: Best = 3.427570e+08 FES = 588100 μ_F = 0.751 μ_{CR} = 0.246
 Diversity = 7.0829e-06 k_t = [2, 3, 3, 2]
 Generation 5890: Best = 3.427570e+08 FES = 589100 μ_F = 0.738 μ_{CR} = 0.245
 Diversity = 7.5193e-06 k_t = [1, 1, 1, 1]

Generation 5900: Best = 3.427570e+08 FES = 590100 μ_F = 0.738 μ_{CR} = 0.247
 Diversity = 7.4741e-06 k_t = [4, 2, 2, 3]
 Generation 5910: Best = 3.427570e+08 FES = 591100 μ_F = 0.743 μ_{CR} = 0.245
 Diversity = 7.9994e-06 k_t = [1, 1, 1, 1]
 Generation 5920: Best = 3.427570e+08 FES = 592100 μ_F = 0.751 μ_{CR} = 0.254
 Diversity = 7.7429e-06 k_t = [3, 4, 4, 3]
 Generation 5930: Best = 3.427570e+08 FES = 593100 μ_F = 0.741 μ_{CR} = 0.252
 Diversity = 7.7429e-06 k_t = [1, 1, 1, 1]
 Generation 5940: Best = 3.427570e+08 FES = 594100 μ_F = 0.746 μ_{CR} = 0.251
 Diversity = 7.7770e-06 k_t = [2, 2, 2, 4]
 Generation 5950: Best = 3.427570e+08 FES = 595100 μ_F = 0.736 μ_{CR} = 0.251
 Diversity = 8.0757e-06 k_t = [1, 1, 1, 1]
 Generation 5960: Best = 3.427570e+08 FES = 596100 μ_F = 0.723 μ_{CR} = 0.258
 Diversity = 8.1116e-06 k_t = [2, 2, 2, 2]
 Generation 5970: Best = 3.427570e+08 FES = 597100 μ_F = 0.723 μ_{CR} = 0.257
 Diversity = 9.3253e-06 k_t = [1, 1, 1, 1]
 Generation 5980: Best = 3.427570e+08 FES = 598100 μ_F = 0.723 μ_{CR} = 0.254
 Diversity = 1.0081e-05 k_t = [2, 3, 2, 3]
 Generation 5990: Best = 3.427570e+08 FES = 599100 μ_F = 0.721 μ_{CR} = 0.242
 Diversity = 1.0702e-05 k_t = [1, 1, 1, 1]
 Generation 6000: Best = 3.427570e+08 FES = 600100 μ_F = 0.734 μ_{CR} = 0.237
 Diversity = 1.0646e-05 k_t = [3, 4, 4, 3]
 Generation 6010: Best = 3.427570e+08 FES = 601100 μ_F = 0.737 μ_{CR} = 0.241
 Diversity = 1.1214e-05 k_t = [1, 1, 1, 1]
 Generation 6020: Best = 3.427570e+08 FES = 602100 μ_F = 0.744 μ_{CR} = 0.249
 Diversity = 1.1562e-05 k_t = [2, 2, 4, 4]
 Generation 6030: Best = 3.427570e+08 FES = 603100 μ_F = 0.751 μ_{CR} = 0.246
 Diversity = 1.3289e-05 k_t = [1, 1, 1, 1]
 Generation 6040: Best = 3.427570e+08 FES = 604100 μ_F = 0.760 μ_{CR} = 0.256
 Diversity = 1.5189e-05 k_t = [4, 3, 3, 2]
 Generation 6050: Best = 3.427570e+08 FES = 605100 μ_F = 0.742 μ_{CR} = 0.245
 Diversity = 1.5422e-05 k_t = [1, 1, 1, 1]
 Generation 6060: Best = 3.427570e+08 FES = 606100 μ_F = 0.738 μ_{CR} = 0.239
 Diversity = 1.4862e-05 k_t = [2, 2, 2, 2]
 Generation 6070: Best = 3.427570e+08 FES = 607100 μ_F = 0.734 μ_{CR} = 0.245
 Diversity = 1.7157e-05 k_t = [1, 1, 1, 1]
 Generation 6080: Best = 3.427570e+08 FES = 608100 μ_F = 0.728 μ_{CR} = 0.248
 Diversity = 1.5459e-05 k_t = [4, 4, 4, 3]
 Generation 6090: Best = 3.427570e+08 FES = 609100 μ_F = 0.737 μ_{CR} = 0.252
 Diversity = 1.6006e-05 k_t = [1, 1, 1, 1]
 Generation 6100: Best = 3.427570e+08 FES = 610100 μ_F = 0.730 μ_{CR} = 0.258
 Diversity = 1.4525e-05 k_t = [4, 4, 2, 3]
 Generation 6110: Best = 3.427570e+08 FES = 611100 μ_F = 0.735 μ_{CR} = 0.253
 Diversity = 1.4125e-05 k_t = [1, 1, 1, 1]
 Generation 6120: Best = 3.427570e+08 FES = 612100 μ_F = 0.721 μ_{CR} = 0.251
 Diversity = 1.2228e-05 k_t = [4, 3, 2, 2]
 Generation 6130: Best = 3.427570e+08 FES = 613100 μ_F = 0.706 μ_{CR} = 0.261
 Diversity = 1.0758e-05 k_t = [1, 1, 1, 1]
 Generation 6140: Best = 3.427570e+08 FES = 614100 μ_F = 0.707 μ_{CR} = 0.261
 Diversity = 9.8086e-06 k_t = [4, 4, 2, 3]
 Generation 6150: Best = 3.427570e+08 FES = 615100 μ_F = 0.709 μ_{CR} = 0.260
 Diversity = 1.0394e-05 k_t = [1, 1, 1, 1]
 Generation 6160: Best = 3.427570e+08 FES = 616100 μ_F = 0.711 μ_{CR} = 0.253
 Diversity = 1.0483e-05 k_t = [2, 2, 2, 3]

Generation 6170: Best = 3.427570e+08	FES = 617100	$\mu_F = 0.720$	$\mu_{CR} = 0.255$
Diversity = 1.1443e-05	k_t = [1, 1, 1, 1]		
Generation 6180: Best = 3.427570e+08	FES = 618100	$\mu_F = 0.716$	$\mu_{CR} = 0.265$
Diversity = 1.1690e-05	k_t = [3, 2, 3, 2]		
Generation 6190: Best = 3.427570e+08	FES = 619100	$\mu_F = 0.721$	$\mu_{CR} = 0.263$
Diversity = 1.1924e-05	k_t = [1, 1, 1, 1]		
Generation 6200: Best = 3.427570e+08	FES = 620100	$\mu_F = 0.706$	$\mu_{CR} = 0.262$
Diversity = 9.8153e-06	k_t = [3, 4, 4, 3]		
Generation 6210: Best = 3.427570e+08	FES = 621100	$\mu_F = 0.705$	$\mu_{CR} = 0.262$
Diversity = 9.1690e-06	k_t = [1, 1, 1, 1]		
Generation 6220: Best = 3.427570e+08	FES = 622100	$\mu_F = 0.696$	$\mu_{CR} = 0.253$
Diversity = 7.5678e-06	k_t = [3, 4, 2, 3]		
Generation 6230: Best = 3.427570e+08	FES = 623100	$\mu_F = 0.692$	$\mu_{CR} = 0.232$
Diversity = 6.8325e-06	k_t = [1, 1, 1, 1]		
Generation 6240: Best = 3.427570e+08	FES = 624100	$\mu_F = 0.696$	$\mu_{CR} = 0.225$
Diversity = 6.3302e-06	k_t = [3, 4, 2, 2]		
Generation 6250: Best = 3.427570e+08	FES = 625100	$\mu_F = 0.695$	$\mu_{CR} = 0.219$
Diversity = 6.4714e-06	k_t = [1, 1, 1, 1]		
Generation 6260: Best = 3.427570e+08	FES = 626100	$\mu_F = 0.708$	$\mu_{CR} = 0.220$
Diversity = 7.4472e-06	k_t = [2, 4, 2, 2]		
Generation 6270: Best = 3.427570e+08	FES = 627100	$\mu_F = 0.713$	$\mu_{CR} = 0.211$
Diversity = 8.3379e-06	k_t = [1, 1, 1, 1]		
Generation 6280: Best = 3.427570e+08	FES = 628100	$\mu_F = 0.722$	$\mu_{CR} = 0.209$
Diversity = 9.1204e-06	k_t = [2, 3, 4, 4]		
Generation 6290: Best = 3.427570e+08	FES = 629100	$\mu_F = 0.743$	$\mu_{CR} = 0.195$
Diversity = 8.7736e-06	k_t = [1, 1, 1, 1]		
Generation 6300: Best = 3.427570e+08	FES = 630100	$\mu_F = 0.743$	$\mu_{CR} = 0.191$
Diversity = 8.3148e-06	k_t = [3, 2, 2, 3]		
Generation 6310: Best = 3.427570e+08	FES = 631100	$\mu_F = 0.752$	$\mu_{CR} = 0.182$
Diversity = 8.8514e-06	k_t = [1, 1, 1, 1]		
Generation 6320: Best = 3.427570e+08	FES = 632100	$\mu_F = 0.735$	$\mu_{CR} = 0.177$
Diversity = 8.2953e-06	k_t = [3, 4, 3, 4]		
Generation 6330: Best = 3.427570e+08	FES = 633100	$\mu_F = 0.734$	$\mu_{CR} = 0.157$
Diversity = 7.6158e-06	k_t = [1, 1, 1, 1]		
Generation 6340: Best = 3.427570e+08	FES = 634100	$\mu_F = 0.726$	$\mu_{CR} = 0.146$
Diversity = 7.7178e-06	k_t = [4, 3, 4, 4]		
Generation 6350: Best = 3.427570e+08	FES = 635100	$\mu_F = 0.732$	$\mu_{CR} = 0.139$
Diversity = 7.2499e-06	k_t = [1, 1, 1, 1]		
Generation 6360: Best = 3.427570e+08	FES = 636100	$\mu_F = 0.744$	$\mu_{CR} = 0.137$
Diversity = 7.9059e-06	k_t = [3, 2, 3, 2]		
Generation 6370: Best = 3.427570e+08	FES = 637100	$\mu_F = 0.752$	$\mu_{CR} = 0.142$
Diversity = 9.0962e-06	k_t = [1, 1, 1, 1]		
Generation 6380: Best = 3.427570e+08	FES = 638100	$\mu_F = 0.754$	$\mu_{CR} = 0.134$
Diversity = 9.8906e-06	k_t = [4, 4, 4, 2]		
Generation 6390: Best = 3.427570e+08	FES = 639100	$\mu_F = 0.748$	$\mu_{CR} = 0.125$
Diversity = 1.2510e-05	k_t = [1, 1, 1, 1]		
Generation 6400: Best = 3.427570e+08	FES = 640100	$\mu_F = 0.757$	$\mu_{CR} = 0.117$
Diversity = 1.3422e-05	k_t = [3, 4, 4, 4]		
Generation 6410: Best = 3.427570e+08	FES = 641100	$\mu_F = 0.764$	$\mu_{CR} = 0.116$
Diversity = 1.2998e-05	k_t = [1, 1, 1, 1]		
Generation 6420: Best = 3.427570e+08	FES = 642100	$\mu_F = 0.775$	$\mu_{CR} = 0.116$
Diversity = 1.2139e-05	k_t = [3, 4, 3, 4]		
Generation 6430: Best = 3.427570e+08	FES = 643100	$\mu_F = 0.777$	$\mu_{CR} = 0.114$
Diversity = 1.2119e-05	k_t = [1, 1, 1, 1]		

Generation 6440: Best = 3.427570e+08 FES = 644100 μ_F = 0.767 μ_{CR} = 0.114
 Diversity = 1.4353e-05 k_t = [3, 2, 3, 2]
 Generation 6450: Best = 3.427570e+08 FES = 645100 μ_F = 0.767 μ_{CR} = 0.111
 Diversity = 1.7060e-05 k_t = [1, 1, 1, 1]
 Generation 6460: Best = 3.427570e+08 FES = 646100 μ_F = 0.771 μ_{CR} = 0.104
 Diversity = 1.9597e-05 k_t = [3, 2, 3, 4]
 Generation 6470: Best = 3.427570e+08 FES = 647100 μ_F = 0.771 μ_{CR} = 0.101
 Diversity = 2.0697e-05 k_t = [1, 1, 1, 1]
 Generation 6480: Best = 3.427570e+08 FES = 648100 μ_F = 0.788 μ_{CR} = 0.101
 Diversity = 1.9896e-05 k_t = [4, 3, 2, 3]
 Generation 6490: Best = 3.427570e+08 FES = 649100 μ_F = 0.778 μ_{CR} = 0.105
 Diversity = 1.8213e-05 k_t = [1, 1, 1, 1]
 Generation 6500: Best = 3.427570e+08 FES = 650100 μ_F = 0.795 μ_{CR} = 0.101
 Diversity = 1.7234e-05 k_t = [3, 3, 4, 3]
 Generation 6510: Best = 3.427570e+08 FES = 651100 μ_F = 0.786 μ_{CR} = 0.104
 Diversity = 1.9368e-05 k_t = [1, 1, 1, 1]
 Generation 6520: Best = 3.427570e+08 FES = 652100 μ_F = 0.782 μ_{CR} = 0.107
 Diversity = 1.9967e-05 k_t = [2, 3, 2, 3]
 Generation 6530: Best = 3.427570e+08 FES = 653100 μ_F = 0.787 μ_{CR} = 0.097
 Diversity = 2.0897e-05 k_t = [1, 1, 1, 1]
 Generation 6540: Best = 3.427570e+08 FES = 654100 μ_F = 0.784 μ_{CR} = 0.098
 Diversity = 2.2672e-05 k_t = [2, 4, 4, 4]
 Generation 6550: Best = 3.427570e+08 FES = 655100 μ_F = 0.760 μ_{CR} = 0.090
 Diversity = 2.1379e-05 k_t = [1, 1, 1, 1]
 Generation 6560: Best = 3.427570e+08 FES = 656100 μ_F = 0.767 μ_{CR} = 0.087
 Diversity = 2.2829e-05 k_t = [2, 3, 3, 3]
 Generation 6570: Best = 3.427570e+08 FES = 657100 μ_F = 0.750 μ_{CR} = 0.086
 Diversity = 2.2838e-05 k_t = [1, 1, 1, 1]
 Generation 6580: Best = 3.427570e+08 FES = 658100 μ_F = 0.738 μ_{CR} = 0.097
 Diversity = 2.2330e-05 k_t = [4, 2, 2, 3]
 Generation 6590: Best = 3.427570e+08 FES = 659100 μ_F = 0.753 μ_{CR} = 0.101
 Diversity = 2.1575e-05 k_t = [1, 1, 1, 1]
 Generation 6600: Best = 3.427570e+08 FES = 660100 μ_F = 0.753 μ_{CR} = 0.093
 Diversity = 2.0365e-05 k_t = [3, 2, 3, 4]
 Generation 6610: Best = 3.427570e+08 FES = 661100 μ_F = 0.750 μ_{CR} = 0.087
 Diversity = 1.8405e-05 k_t = [1, 1, 1, 1]
 Generation 6620: Best = 3.427570e+08 FES = 662100 μ_F = 0.764 μ_{CR} = 0.084
 Diversity = 1.7082e-05 k_t = [3, 2, 2, 3]
 Generation 6630: Best = 3.427570e+08 FES = 663100 μ_F = 0.767 μ_{CR} = 0.084
 Diversity = 1.7354e-05 k_t = [1, 1, 1, 1]
 Generation 6640: Best = 3.427570e+08 FES = 664100 μ_F = 0.781 μ_{CR} = 0.094
 Diversity = 1.6669e-05 k_t = [3, 2, 4, 4]
 Generation 6650: Best = 3.427570e+08 FES = 665100 μ_F = 0.779 μ_{CR} = 0.100
 Diversity = 2.0363e-05 k_t = [1, 1, 1, 1]
 Generation 6660: Best = 3.427570e+08 FES = 666100 μ_F = 0.778 μ_{CR} = 0.088
 Diversity = 2.2840e-05 k_t = [2, 2, 3, 3]
 Generation 6670: Best = 3.427570e+08 FES = 667100 μ_F = 0.768 μ_{CR} = 0.094
 Diversity = 2.7077e-05 k_t = [1, 1, 1, 1]
 Generation 6680: Best = 3.427570e+08 FES = 668100 μ_F = 0.768 μ_{CR} = 0.087
 Diversity = 2.5591e-05 k_t = [3, 4, 4, 3]
 Generation 6690: Best = 3.427570e+08 FES = 669100 μ_F = 0.773 μ_{CR} = 0.082
 Diversity = 2.9385e-05 k_t = [1, 1, 1, 1]
 Generation 6700: Best = 3.427570e+08 FES = 670100 μ_F = 0.769 μ_{CR} = 0.087
 Diversity = 3.1367e-05 k_t = [3, 4, 2, 4]

Generation 6710: Best = 3.427570e+08 FES = 671100 μ_F = 0.777 μ_{CR} = 0.085
 Diversity = 3.7107e-05 k_t = [1, 1, 1, 1]
 Generation 6720: Best = 3.427570e+08 FES = 672100 μ_F = 0.786 μ_{CR} = 0.090
 Diversity = 4.2202e-05 k_t = [4, 2, 3, 4]
 Generation 6730: Best = 3.427570e+08 FES = 673100 μ_F = 0.793 μ_{CR} = 0.085
 Diversity = 6.0562e-05 k_t = [1, 1, 1, 1]
 Generation 6740: Best = 3.427570e+08 FES = 674100 μ_F = 0.818 μ_{CR} = 0.081
 Diversity = 7.5884e-05 k_t = [3, 2, 3, 4]
 Generation 6750: Best = 3.427570e+08 FES = 675100 μ_F = 0.801 μ_{CR} = 0.089
 Diversity = 8.3316e-05 k_t = [1, 1, 1, 1]
 Generation 6760: Best = 3.427570e+08 FES = 676100 μ_F = 0.800 μ_{CR} = 0.091
 Diversity = 7.9796e-05 k_t = [2, 4, 3, 2]
 Generation 6770: Best = 3.427570e+08 FES = 677100 μ_F = 0.788 μ_{CR} = 0.085
 Diversity = 7.5933e-05 k_t = [1, 1, 1, 1]
 Generation 6780: Best = 3.427570e+08 FES = 678100 μ_F = 0.793 μ_{CR} = 0.086
 Diversity = 6.7457e-05 k_t = [2, 2, 3, 3]
 Generation 6790: Best = 3.427570e+08 FES = 679100 μ_F = 0.783 μ_{CR} = 0.079
 Diversity = 6.3412e-05 k_t = [1, 1, 1, 1]
 Generation 6800: Best = 3.427570e+08 FES = 680100 μ_F = 0.788 μ_{CR} = 0.083
 Diversity = 5.2158e-05 k_t = [4, 2, 3, 3]
 Generation 6810: Best = 3.427570e+08 FES = 681100 μ_F = 0.789 μ_{CR} = 0.079
 Diversity = 6.3310e-05 k_t = [1, 1, 1, 1]
 Generation 6820: Best = 3.427570e+08 FES = 682100 μ_F = 0.780 μ_{CR} = 0.077
 Diversity = 5.8860e-05 k_t = [2, 3, 3, 4]
 Generation 6830: Best = 3.427570e+08 FES = 683100 μ_F = 0.776 μ_{CR} = 0.081
 Diversity = 7.1959e-05 k_t = [1, 1, 1, 1]
 Generation 6840: Best = 3.427570e+08 FES = 684100 μ_F = 0.777 μ_{CR} = 0.079
 Diversity = 7.6118e-05 k_t = [4, 2, 4, 3]
 Generation 6850: Best = 3.427570e+08 FES = 685100 μ_F = 0.793 μ_{CR} = 0.079
 Diversity = 8.5508e-05 k_t = [1, 1, 1, 1]
 Generation 6860: Best = 3.427570e+08 FES = 686100 μ_F = 0.777 μ_{CR} = 0.081
 Diversity = 8.4947e-05 k_t = [3, 4, 4, 2]
 Generation 6870: Best = 3.427570e+08 FES = 687100 μ_F = 0.760 μ_{CR} = 0.092
 Diversity = 9.6822e-05 k_t = [1, 1, 1, 1]
 Generation 6880: Best = 3.427570e+08 FES = 688100 μ_F = 0.775 μ_{CR} = 0.090
 Diversity = 8.6441e-05 k_t = [4, 2, 3, 4]
 Generation 6890: Best = 3.427570e+08 FES = 689100 μ_F = 0.760 μ_{CR} = 0.085
 Diversity = 8.0859e-05 k_t = [1, 1, 1, 1]
 Generation 6900: Best = 3.427570e+08 FES = 690100 μ_F = 0.760 μ_{CR} = 0.087
 Diversity = 8.6555e-05 k_t = [4, 3, 4, 3]
 Generation 6910: Best = 3.427570e+08 FES = 691100 μ_F = 0.765 μ_{CR} = 0.082
 Diversity = 9.6076e-05 k_t = [1, 1, 1, 1]
 Generation 6920: Best = 3.427570e+08 FES = 692100 μ_F = 0.764 μ_{CR} = 0.084
 Diversity = 1.0166e-04 k_t = [2, 4, 2, 2]
 Generation 6930: Best = 3.427570e+08 FES = 693100 μ_F = 0.781 μ_{CR} = 0.082
 Diversity = 1.1605e-04 k_t = [1, 1, 1, 1]
 Generation 6940: Best = 3.427570e+08 FES = 694100 μ_F = 0.787 μ_{CR} = 0.079
 Diversity = 1.3442e-04 k_t = [2, 4, 4, 3]
 Generation 6950: Best = 3.427570e+08 FES = 695100 μ_F = 0.800 μ_{CR} = 0.076
 Diversity = 1.5445e-04 k_t = [1, 1, 1, 1]
 Generation 6960: Best = 3.427570e+08 FES = 696100 μ_F = 0.788 μ_{CR} = 0.069
 Diversity = 1.9055e-04 k_t = [4, 4, 3, 2]
 Generation 6970: Best = 3.427570e+08 FES = 697100 μ_F = 0.779 μ_{CR} = 0.066
 Diversity = 2.1303e-04 k_t = [1, 1, 1, 1]

Generation 6980: Best = 3.427569e+08 FES = 698100 μ_F = 0.802 μ_{CR} = 0.065
 Diversity = 2.7516e-04 k_t = [2, 4, 3, 3]
 Generation 6990: Best = 3.427569e+08 FES = 699100 μ_F = 0.811 μ_{CR} = 0.065
 Diversity = 3.1515e-04 k_t = [1, 1, 1, 1]
 Generation 7000: Best = 3.427569e+08 FES = 700100 μ_F = 0.789 μ_{CR} = 0.064
 Diversity = 2.9996e-04 k_t = [3, 2, 2, 3]
 Generation 7010: Best = 3.427569e+08 FES = 701100 μ_F = 0.767 μ_{CR} = 0.068
 Diversity = 3.0475e-04 k_t = [1, 1, 1, 1]
 Generation 7020: Best = 3.427569e+08 FES = 702100 μ_F = 0.783 μ_{CR} = 0.066
 Diversity = 2.6476e-04 k_t = [2, 4, 3, 4]
 Generation 7030: Best = 3.427569e+08 FES = 703100 μ_F = 0.773 μ_{CR} = 0.062
 Diversity = 2.9147e-04 k_t = [1, 1, 1, 1]
 Generation 7040: Best = 3.427569e+08 FES = 704100 μ_F = 0.780 μ_{CR} = 0.065
 Diversity = 3.3228e-04 k_t = [2, 2, 4, 4]
 Generation 7050: Best = 3.427569e+08 FES = 705100 μ_F = 0.792 μ_{CR} = 0.062
 Diversity = 3.2754e-04 k_t = [1, 1, 1, 1]
 Generation 7060: Best = 3.427569e+08 FES = 706100 μ_F = 0.802 μ_{CR} = 0.058
 Diversity = 3.4321e-04 k_t = [4, 2, 4, 2]
 Generation 7070: Best = 3.427569e+08 FES = 707100 μ_F = 0.799 μ_{CR} = 0.063
 Diversity = 3.5967e-04 k_t = [1, 1, 1, 1]
 Generation 7080: Best = 3.427569e+08 FES = 708100 μ_F = 0.792 μ_{CR} = 0.065
 Diversity = 3.8862e-04 k_t = [2, 2, 2, 2]
 Generation 7090: Best = 3.427569e+08 FES = 709100 μ_F = 0.780 μ_{CR} = 0.057
 Diversity = 4.0114e-04 k_t = [1, 1, 1, 1]
 Generation 7100: Best = 3.427569e+08 FES = 710100 μ_F = 0.788 μ_{CR} = 0.058
 Diversity = 3.2263e-04 k_t = [3, 2, 3, 4]
 Generation 7110: Best = 3.427569e+08 FES = 711100 μ_F = 0.776 μ_{CR} = 0.058
 Diversity = 2.4946e-04 k_t = [1, 1, 1, 1]
 Generation 7120: Best = 3.427569e+08 FES = 712100 μ_F = 0.790 μ_{CR} = 0.065
 Diversity = 2.9581e-04 k_t = [4, 3, 2, 2]
 Generation 7130: Best = 3.427569e+08 FES = 713100 μ_F = 0.778 μ_{CR} = 0.072
 Diversity = 2.6944e-04 k_t = [1, 1, 1, 1]
 Generation 7140: Best = 3.427569e+08 FES = 714100 μ_F = 0.783 μ_{CR} = 0.066
 Diversity = 2.6609e-04 k_t = [3, 2, 2, 3]
 Generation 7150: Best = 3.427569e+08 FES = 715100 μ_F = 0.772 μ_{CR} = 0.070
 Diversity = 3.0003e-04 k_t = [1, 1, 1, 1]
 Generation 7160: Best = 3.427569e+08 FES = 716100 μ_F = 0.779 μ_{CR} = 0.068
 Diversity = 2.5336e-04 k_t = [4, 2, 4, 3]
 Generation 7170: Best = 3.427569e+08 FES = 717100 μ_F = 0.761 μ_{CR} = 0.072
 Diversity = 2.8077e-04 k_t = [1, 1, 1, 1]
 Generation 7180: Best = 3.427569e+08 FES = 718100 μ_F = 0.769 μ_{CR} = 0.073
 Diversity = 2.5988e-04 k_t = [3, 4, 3, 2]
 Generation 7190: Best = 3.427569e+08 FES = 719100 μ_F = 0.781 μ_{CR} = 0.062
 Diversity = 2.3752e-04 k_t = [1, 1, 1, 1]
 Generation 7200: Best = 3.427569e+08 FES = 720100 μ_F = 0.779 μ_{CR} = 0.064
 Diversity = 2.1625e-04 k_t = [3, 2, 2, 3]
 Generation 7210: Best = 3.427569e+08 FES = 721100 μ_F = 0.784 μ_{CR} = 0.056
 Diversity = 2.1877e-04 k_t = [1, 1, 1, 1]
 Generation 7220: Best = 3.427569e+08 FES = 722100 μ_F = 0.774 μ_{CR} = 0.060
 Diversity = 2.6107e-04 k_t = [3, 4, 2, 3]
 Generation 7230: Best = 3.427569e+08 FES = 723100 μ_F = 0.786 μ_{CR} = 0.062
 Diversity = 2.4820e-04 k_t = [1, 1, 1, 1]
 Generation 7240: Best = 3.427569e+08 FES = 724100 μ_F = 0.794 μ_{CR} = 0.061
 Diversity = 2.6939e-04 k_t = [2, 3, 3, 3]

Generation 7250: Best = 3.427569e+08 FES = 725100 μ_F = 0.776 μ_{CR} = 0.065
 Diversity = 2.2640e-04 k_t = [1, 1, 1, 1]
 Generation 7260: Best = 3.427569e+08 FES = 726100 μ_F = 0.787 μ_{CR} = 0.062
 Diversity = 2.2759e-04 k_t = [3, 3, 3, 2]
 Generation 7270: Best = 3.427569e+08 FES = 727100 μ_F = 0.781 μ_{CR} = 0.066
 Diversity = 2.2407e-04 k_t = [1, 1, 1, 1]
 Generation 7280: Best = 3.427569e+08 FES = 728100 μ_F = 0.790 μ_{CR} = 0.066
 Diversity = 2.1700e-04 k_t = [3, 4, 2, 2]
 Generation 7290: Best = 3.427569e+08 FES = 729100 μ_F = 0.796 μ_{CR} = 0.068
 Diversity = 2.4515e-04 k_t = [1, 1, 1, 1]
 Generation 7300: Best = 3.427569e+08 FES = 730100 μ_F = 0.804 μ_{CR} = 0.069
 Diversity = 2.3797e-04 k_t = [4, 4, 3, 4]
 Generation 7310: Best = 3.427569e+08 FES = 731100 μ_F = 0.796 μ_{CR} = 0.071
 Diversity = 2.4922e-04 k_t = [1, 1, 1, 1]
 Generation 7320: Best = 3.427569e+08 FES = 732100 μ_F = 0.804 μ_{CR} = 0.068
 Diversity = 2.8511e-04 k_t = [2, 3, 2, 2]
 Generation 7330: Best = 3.427569e+08 FES = 733100 μ_F = 0.804 μ_{CR} = 0.066
 Diversity = 3.3758e-04 k_t = [1, 1, 1, 1]
 Generation 7340: Best = 3.427569e+08 FES = 734100 μ_F = 0.809 μ_{CR} = 0.067
 Diversity = 4.0406e-04 k_t = [3, 2, 2, 3]
 Generation 7350: Best = 3.427569e+08 FES = 735100 μ_F = 0.815 μ_{CR} = 0.060
 Diversity = 5.1397e-04 k_t = [1, 1, 1, 1]
 Generation 7360: Best = 3.427569e+08 FES = 736100 μ_F = 0.811 μ_{CR} = 0.063
 Diversity = 4.7744e-04 k_t = [3, 4, 2, 4]
 Generation 7370: Best = 3.427569e+08 FES = 737100 μ_F = 0.810 μ_{CR} = 0.063
 Diversity = 5.5091e-04 k_t = [1, 1, 1, 1]
 Generation 7380: Best = 3.427569e+08 FES = 738100 μ_F = 0.821 μ_{CR} = 0.067
 Diversity = 5.9064e-04 k_t = [4, 4, 2, 2]
 Generation 7390: Best = 3.427569e+08 FES = 739100 μ_F = 0.808 μ_{CR} = 0.073
 Diversity = 5.9077e-04 k_t = [1, 1, 1, 1]
 Generation 7400: Best = 3.427569e+08 FES = 740100 μ_F = 0.813 μ_{CR} = 0.065
 Diversity = 6.2020e-04 k_t = [3, 2, 2, 3]
 Generation 7410: Best = 3.427569e+08 FES = 741100 μ_F = 0.806 μ_{CR} = 0.066
 Diversity = 6.7076e-04 k_t = [1, 1, 1, 1]
 Generation 7420: Best = 3.427569e+08 FES = 742100 μ_F = 0.804 μ_{CR} = 0.068
 Diversity = 7.5413e-04 k_t = [4, 3, 2, 4]
 Generation 7430: Best = 3.427569e+08 FES = 743100 μ_F = 0.796 μ_{CR} = 0.066
 Diversity = 7.9298e-04 k_t = [1, 1, 1, 1]
 Generation 7440: Best = 3.427569e+08 FES = 744100 μ_F = 0.810 μ_{CR} = 0.059
 Diversity = 8.0858e-04 k_t = [3, 3, 3, 4]
 Generation 7450: Best = 3.427569e+08 FES = 745100 μ_F = 0.788 μ_{CR} = 0.060
 Diversity = 7.5730e-04 k_t = [1, 1, 1, 1]
 Generation 7460: Best = 3.427569e+08 FES = 746100 μ_F = 0.787 μ_{CR} = 0.058
 Diversity = 7.2426e-04 k_t = [3, 4, 4, 3]
 Generation 7470: Best = 3.427569e+08 FES = 747100 μ_F = 0.784 μ_{CR} = 0.063
 Diversity = 7.0220e-04 k_t = [1, 1, 1, 1]
 Generation 7480: Best = 3.427569e+08 FES = 748100 μ_F = 0.795 μ_{CR} = 0.065
 Diversity = 5.9735e-04 k_t = [2, 3, 4, 2]
 Generation 7490: Best = 3.427569e+08 FES = 749100 μ_F = 0.798 μ_{CR} = 0.066
 Diversity = 6.1549e-04 k_t = [1, 1, 1, 1]
 Generation 7500: Best = 3.427569e+08 FES = 750100 μ_F = 0.796 μ_{CR} = 0.065
 Diversity = 7.8970e-04 k_t = [3, 2, 4, 2]
 Generation 7510: Best = 3.427569e+08 FES = 751100 μ_F = 0.789 μ_{CR} = 0.061
 Diversity = 8.3676e-04 k_t = [1, 1, 1, 1]

Generation 7520: Best = 3.427569e+08 FES = 752100 μ_F = 0.797 μ_{CR} = 0.063
 Diversity = 9.4748e-04 k_t = [2, 4, 2, 4]
 Generation 7530: Best = 3.427569e+08 FES = 753100 μ_F = 0.797 μ_{CR} = 0.059
 Diversity = 1.2229e-03 k_t = [1, 1, 1, 1]
 Generation 7540: Best = 3.427569e+08 FES = 754100 μ_F = 0.805 μ_{CR} = 0.060
 Diversity = 1.1152e-03 k_t = [2, 3, 2, 3]
 Generation 7550: Best = 3.427569e+08 FES = 755100 μ_F = 0.807 μ_{CR} = 0.062
 Diversity = 1.1611e-03 k_t = [1, 1, 1, 1]
 Generation 7560: Best = 3.427569e+08 FES = 756100 μ_F = 0.814 μ_{CR} = 0.061
 Diversity = 1.3485e-03 k_t = [3, 4, 2, 2]
 Generation 7570: Best = 3.427569e+08 FES = 757100 μ_F = 0.812 μ_{CR} = 0.063
 Diversity = 1.6800e-03 k_t = [1, 1, 1, 1]
 Generation 7580: Best = 3.427569e+08 FES = 758100 μ_F = 0.810 μ_{CR} = 0.065
 Diversity = 1.6622e-03 k_t = [3, 4, 2, 4]
 Generation 7590: Best = 3.427569e+08 FES = 759100 μ_F = 0.796 μ_{CR} = 0.060
 Diversity = 1.6915e-03 k_t = [1, 1, 1, 1]
 Generation 7600: Best = 3.427569e+08 FES = 760100 μ_F = 0.780 μ_{CR} = 0.067
 Diversity = 1.9574e-03 k_t = [3, 3, 2, 2]
 Generation 7610: Best = 3.427569e+08 FES = 761100 μ_F = 0.810 μ_{CR} = 0.060
 Diversity = 2.4380e-03 k_t = [1, 1, 1, 1]
 Generation 7620: Best = 3.427569e+08 FES = 762100 μ_F = 0.812 μ_{CR} = 0.065
 Diversity = 2.0555e-03 k_t = [2, 4, 3, 4]
 Generation 7630: Best = 3.427569e+08 FES = 763100 μ_F = 0.821 μ_{CR} = 0.057
 Diversity = 1.9001e-03 k_t = [1, 1, 1, 1]
 Generation 7640: Best = 3.427569e+08 FES = 764100 μ_F = 0.808 μ_{CR} = 0.061
 Diversity = 1.7620e-03 k_t = [2, 3, 3, 2]
 Generation 7650: Best = 3.427569e+08 FES = 765100 μ_F = 0.808 μ_{CR} = 0.060
 Diversity = 2.6792e-03 k_t = [1, 1, 1, 1]
 Generation 7660: Best = 3.427569e+08 FES = 766100 μ_F = 0.781 μ_{CR} = 0.062
 Diversity = 2.6260e-03 k_t = [2, 4, 2, 3]
 Generation 7670: Best = 3.427568e+08 FES = 767100 μ_F = 0.785 μ_{CR} = 0.056
 Diversity = 3.5618e-03 k_t = [1, 1, 1, 1]
 Generation 7680: Best = 3.427568e+08 FES = 768100 μ_F = 0.772 μ_{CR} = 0.058
 Diversity = 3.4327e-03 k_t = [4, 3, 3, 2]
 Generation 7690: Best = 3.427568e+08 FES = 769100 μ_F = 0.764 μ_{CR} = 0.056
 Diversity = 3.6755e-03 k_t = [1, 1, 1, 1]
 Generation 7700: Best = 3.427568e+08 FES = 770100 μ_F = 0.778 μ_{CR} = 0.061
 Diversity = 3.9690e-03 k_t = [4, 3, 2, 4]
 Generation 7710: Best = 3.427568e+08 FES = 771100 μ_F = 0.780 μ_{CR} = 0.062
 Diversity = 4.2107e-03 k_t = [1, 1, 1, 1]
 Generation 7720: Best = 3.427568e+08 FES = 772100 μ_F = 0.771 μ_{CR} = 0.060
 Diversity = 4.0659e-03 k_t = [4, 2, 4, 4]
 Generation 7730: Best = 3.427568e+08 FES = 773100 μ_F = 0.777 μ_{CR} = 0.057
 Diversity = 4.5513e-03 k_t = [1, 1, 1, 1]
 Generation 7740: Best = 3.427568e+08 FES = 774100 μ_F = 0.773 μ_{CR} = 0.059
 Diversity = 4.9163e-03 k_t = [3, 4, 3, 2]
 Generation 7750: Best = 3.427568e+08 FES = 775100 μ_F = 0.796 μ_{CR} = 0.052
 Diversity = 5.0267e-03 k_t = [1, 1, 1, 1]
 Generation 7760: Best = 3.427568e+08 FES = 776100 μ_F = 0.803 μ_{CR} = 0.053
 Diversity = 4.6766e-03 k_t = [2, 4, 4, 4]
 Generation 7770: Best = 3.427568e+08 FES = 777100 μ_F = 0.794 μ_{CR} = 0.060
 Diversity = 4.6580e-03 k_t = [1, 1, 1, 1]
 Generation 7780: Best = 3.427568e+08 FES = 778100 μ_F = 0.784 μ_{CR} = 0.056
 Diversity = 4.2542e-03 k_t = [3, 3, 2, 4]

Generation 7790: Best = 3.427568e+08 FES = 779100 μ_F = 0.782 μ_{CR} = 0.052
 Diversity = 4.3085e-03 k_t = [1, 1, 1, 1]
 Generation 7800: Best = 3.427568e+08 FES = 780100 μ_F = 0.806 μ_{CR} = 0.050
 Diversity = 3.6416e-03 k_t = [3, 4, 3, 2]
 Generation 7810: Best = 3.427568e+08 FES = 781100 μ_F = 0.807 μ_{CR} = 0.062
 Diversity = 4.2524e-03 k_t = [1, 1, 1, 1]
 Generation 7820: Best = 3.427568e+08 FES = 782100 μ_F = 0.802 μ_{CR} = 0.058
 Diversity = 4.5314e-03 k_t = [4, 3, 3, 4]
 Generation 7830: Best = 3.427568e+08 FES = 783100 μ_F = 0.796 μ_{CR} = 0.054
 Diversity = 4.3306e-03 k_t = [1, 1, 1, 1]
 Generation 7840: Best = 3.427568e+08 FES = 784100 μ_F = 0.769 μ_{CR} = 0.060
 Diversity = 4.1381e-03 k_t = [3, 4, 4, 4]
 Generation 7850: Best = 3.427568e+08 FES = 785100 μ_F = 0.773 μ_{CR} = 0.061
 Diversity = 3.6571e-03 k_t = [1, 1, 1, 1]
 Generation 7860: Best = 3.427568e+08 FES = 786100 μ_F = 0.779 μ_{CR} = 0.061
 Diversity = 3.1457e-03 k_t = [2, 4, 3, 3]
 Generation 7870: Best = 3.427568e+08 FES = 787100 μ_F = 0.788 μ_{CR} = 0.065
 Diversity = 3.5124e-03 k_t = [1, 1, 1, 1]
 Generation 7880: Best = 3.427568e+08 FES = 788100 μ_F = 0.807 μ_{CR} = 0.070
 Diversity = 3.7713e-03 k_t = [3, 3, 4, 4]
 Generation 7890: Best = 3.427568e+08 FES = 789100 μ_F = 0.794 μ_{CR} = 0.064
 Diversity = 5.4502e-03 k_t = [1, 1, 1, 1]
 Generation 7900: Best = 3.427568e+08 FES = 790100 μ_F = 0.789 μ_{CR} = 0.067
 Diversity = 6.2551e-03 k_t = [3, 3, 2, 4]
 Generation 7910: Best = 3.427567e+08 FES = 791100 μ_F = 0.787 μ_{CR} = 0.061
 Diversity = 6.8754e-03 k_t = [1, 1, 1, 1]
 Generation 7920: Best = 3.427567e+08 FES = 792100 μ_F = 0.787 μ_{CR} = 0.060
 Diversity = 6.4935e-03 k_t = [3, 3, 3, 2]
 Generation 7930: Best = 3.427567e+08 FES = 793100 μ_F = 0.785 μ_{CR} = 0.059
 Diversity = 7.4490e-03 k_t = [1, 1, 1, 1]
 Generation 7940: Best = 3.427567e+08 FES = 794100 μ_F = 0.786 μ_{CR} = 0.055
 Diversity = 9.5394e-03 k_t = [2, 3, 4, 3]
 Generation 7950: Best = 3.427567e+08 FES = 795100 μ_F = 0.799 μ_{CR} = 0.052
 Diversity = 1.1758e-02 k_t = [1, 1, 1, 1]
 Generation 7960: Best = 3.427567e+08 FES = 796100 μ_F = 0.800 μ_{CR} = 0.055
 Diversity = 1.4494e-02 k_t = [3, 2, 3, 4]
 Generation 7970: Best = 3.427567e+08 FES = 797100 μ_F = 0.803 μ_{CR} = 0.051
 Diversity = 1.3226e-02 k_t = [1, 1, 1, 1]
 Generation 7980: Best = 3.427567e+08 FES = 798100 μ_F = 0.814 μ_{CR} = 0.054
 Diversity = 1.2439e-02 k_t = [4, 4, 4, 3]
 Generation 7990: Best = 3.427567e+08 FES = 799100 μ_F = 0.819 μ_{CR} = 0.049
 Diversity = 1.3956e-02 k_t = [1, 1, 1, 1]
 Generation 8000: Best = 3.427566e+08 FES = 800100 μ_F = 0.828 μ_{CR} = 0.055
 Diversity = 1.3065e-02 k_t = [4, 2, 2, 4]
 Generation 8010: Best = 3.427566e+08 FES = 801100 μ_F = 0.821 μ_{CR} = 0.054
 Diversity = 1.2729e-02 k_t = [1, 1, 1, 1]
 Generation 8020: Best = 3.427566e+08 FES = 802100 μ_F = 0.824 μ_{CR} = 0.056
 Diversity = 1.2584e-02 k_t = [3, 2, 3, 3]
 Generation 8030: Best = 3.427566e+08 FES = 803100 μ_F = 0.816 μ_{CR} = 0.052
 Diversity = 1.2219e-02 k_t = [1, 1, 1, 1]
 Generation 8040: Best = 3.427566e+08 FES = 804100 μ_F = 0.822 μ_{CR} = 0.055
 Diversity = 1.1416e-02 k_t = [4, 4, 3, 2]
 Generation 8050: Best = 3.427566e+08 FES = 805100 μ_F = 0.809 μ_{CR} = 0.051
 Diversity = 1.1911e-02 k_t = [1, 1, 1, 1]

Generation 8060: Best = 3.427566e+08 FES = 806100 μ_F = 0.800 μ_{CR} = 0.059
 Diversity = 1.1284e-02 k_t = [2, 3, 4, 2]
 Generation 8070: Best = 3.427566e+08 FES = 807100 μ_F = 0.813 μ_{CR} = 0.056
 Diversity = 1.1780e-02 k_t = [1, 1, 1, 1]
 Generation 8080: Best = 3.427566e+08 FES = 808100 μ_F = 0.830 μ_{CR} = 0.053
 Diversity = 1.3884e-02 k_t = [2, 3, 3, 4]
 Generation 8090: Best = 3.427566e+08 FES = 809100 μ_F = 0.807 μ_{CR} = 0.057
 Diversity = 1.3121e-02 k_t = [1, 1, 1, 1]
 Generation 8100: Best = 3.427566e+08 FES = 810100 μ_F = 0.803 μ_{CR} = 0.053
 Diversity = 1.2172e-02 k_t = [3, 4, 2, 2]
 Generation 8110: Best = 3.427566e+08 FES = 811100 μ_F = 0.799 μ_{CR} = 0.052
 Diversity = 1.2978e-02 k_t = [1, 1, 1, 1]
 Generation 8120: Best = 3.427566e+08 FES = 812100 μ_F = 0.789 μ_{CR} = 0.055
 Diversity = 1.4287e-02 k_t = [2, 3, 4, 3]
 Generation 8130: Best = 3.427566e+08 FES = 813100 μ_F = 0.797 μ_{CR} = 0.059
 Diversity = 1.6032e-02 k_t = [1, 1, 1, 1]
 Generation 8140: Best = 3.427566e+08 FES = 814100 μ_F = 0.797 μ_{CR} = 0.060
 Diversity = 1.4317e-02 k_t = [3, 4, 3, 3]
 Generation 8150: Best = 3.427565e+08 FES = 815100 μ_F = 0.813 μ_{CR} = 0.067
 Diversity = 1.5109e-02 k_t = [1, 1, 1, 1]
 Generation 8160: Best = 3.427565e+08 FES = 816100 μ_F = 0.811 μ_{CR} = 0.068
 Diversity = 1.5102e-02 k_t = [2, 2, 3, 3]
 Generation 8170: Best = 3.427565e+08 FES = 817100 μ_F = 0.812 μ_{CR} = 0.068
 Diversity = 1.5712e-02 k_t = [1, 1, 1, 1]
 Generation 8180: Best = 3.427565e+08 FES = 818100 μ_F = 0.805 μ_{CR} = 0.059
 Diversity = 1.5066e-02 k_t = [4, 2, 2, 4]
 Generation 8190: Best = 3.427565e+08 FES = 819100 μ_F = 0.794 μ_{CR} = 0.054
 Diversity = 1.4682e-02 k_t = [1, 1, 1, 1]
 Generation 8200: Best = 3.427565e+08 FES = 820100 μ_F = 0.790 μ_{CR} = 0.055
 Diversity = 1.4049e-02 k_t = [2, 4, 4, 3]
 Generation 8210: Best = 3.427565e+08 FES = 821100 μ_F = 0.784 μ_{CR} = 0.051
 Diversity = 1.4706e-02 k_t = [1, 1, 1, 1]
 Generation 8220: Best = 3.427565e+08 FES = 822100 μ_F = 0.774 μ_{CR} = 0.060
 Diversity = 1.4971e-02 k_t = [2, 3, 3, 2]
 Generation 8230: Best = 3.427565e+08 FES = 823100 μ_F = 0.769 μ_{CR} = 0.058
 Diversity = 1.4059e-02 k_t = [1, 1, 1, 1]
 Generation 8240: Best = 3.427565e+08 FES = 824100 μ_F = 0.774 μ_{CR} = 0.060
 Diversity = 1.4693e-02 k_t = [3, 4, 2, 2]
 Generation 8250: Best = 3.427564e+08 FES = 825100 μ_F = 0.773 μ_{CR} = 0.059
 Diversity = 1.5615e-02 k_t = [1, 1, 1, 1]
 Generation 8260: Best = 3.427564e+08 FES = 826100 μ_F = 0.785 μ_{CR} = 0.058
 Diversity = 1.4255e-02 k_t = [2, 2, 3, 3]
 Generation 8270: Best = 3.427564e+08 FES = 827100 μ_F = 0.781 μ_{CR} = 0.064
 Diversity = 1.7331e-02 k_t = [1, 1, 1, 1]
 Generation 8280: Best = 3.427564e+08 FES = 828100 μ_F = 0.796 μ_{CR} = 0.057
 Diversity = 1.5969e-02 k_t = [4, 2, 2, 4]
 Generation 8290: Best = 3.427564e+08 FES = 829100 μ_F = 0.803 μ_{CR} = 0.052
 Diversity = 1.7901e-02 k_t = [1, 1, 1, 1]
 Generation 8300: Best = 3.427564e+08 FES = 830100 μ_F = 0.793 μ_{CR} = 0.053
 Diversity = 1.9074e-02 k_t = [3, 4, 2, 3]
 Generation 8310: Best = 3.427563e+08 FES = 831100 μ_F = 0.800 μ_{CR} = 0.057
 Diversity = 1.9840e-02 k_t = [1, 1, 1, 1]
 Generation 8320: Best = 3.427563e+08 FES = 832100 μ_F = 0.789 μ_{CR} = 0.052
 Diversity = 2.3960e-02 k_t = [3, 2, 2, 3]

Generation 8330: Best = 3.427563e+08 FES = 833100 μ_F = 0.780 μ_{CR} = 0.049
 Diversity = 2.5853e-02 k_t = [1, 1, 1, 1]
 Generation 8340: Best = 3.427563e+08 FES = 834100 μ_F = 0.779 μ_{CR} = 0.059
 Diversity = 2.2669e-02 k_t = [4, 2, 3, 3]
 Generation 8350: Best = 3.427563e+08 FES = 835100 μ_F = 0.782 μ_{CR} = 0.056
 Diversity = 2.1519e-02 k_t = [1, 1, 1, 1]
 Generation 8360: Best = 3.427563e+08 FES = 836100 μ_F = 0.793 μ_{CR} = 0.058
 Diversity = 2.1499e-02 k_t = [2, 3, 3, 3]
 Generation 8370: Best = 3.427563e+08 FES = 837100 μ_F = 0.790 μ_{CR} = 0.061
 Diversity = 1.9091e-02 k_t = [1, 1, 1, 1]
 Generation 8380: Best = 3.427563e+08 FES = 838100 μ_F = 0.783 μ_{CR} = 0.056
 Diversity = 1.8074e-02 k_t = [3, 3, 4, 2]
 Generation 8390: Best = 3.427563e+08 FES = 839100 μ_F = 0.777 μ_{CR} = 0.055
 Diversity = 1.6410e-02 k_t = [1, 1, 1, 1]
 Generation 8400: Best = 3.427563e+08 FES = 840100 μ_F = 0.783 μ_{CR} = 0.059
 Diversity = 1.4738e-02 k_t = [2, 3, 3, 2]
 Generation 8410: Best = 3.427563e+08 FES = 841100 μ_F = 0.784 μ_{CR} = 0.057
 Diversity = 1.5818e-02 k_t = [1, 1, 1, 1]
 Generation 8420: Best = 3.427563e+08 FES = 842100 μ_F = 0.797 μ_{CR} = 0.059
 Diversity = 1.4558e-02 k_t = [4, 2, 2, 4]
 Generation 8430: Best = 3.427563e+08 FES = 843100 μ_F = 0.783 μ_{CR} = 0.060
 Diversity = 1.5849e-02 k_t = [1, 1, 1, 1]
 Generation 8440: Best = 3.427562e+08 FES = 844100 μ_F = 0.784 μ_{CR} = 0.060
 Diversity = 1.4458e-02 k_t = [3, 3, 4, 4]
 Generation 8450: Best = 3.427562e+08 FES = 845100 μ_F = 0.798 μ_{CR} = 0.058
 Diversity = 2.0975e-02 k_t = [1, 1, 1, 1]
 Generation 8460: Best = 3.427562e+08 FES = 846100 μ_F = 0.806 μ_{CR} = 0.059
 Diversity = 2.4588e-02 k_t = [4, 2, 2, 3]
 Generation 8470: Best = 3.427562e+08 FES = 847100 μ_F = 0.783 μ_{CR} = 0.062
 Diversity = 2.2351e-02 k_t = [1, 1, 1, 1]
 Generation 8480: Best = 3.427562e+08 FES = 848100 μ_F = 0.780 μ_{CR} = 0.057
 Diversity = 1.9782e-02 k_t = [4, 4, 4, 4]
 Generation 8490: Best = 3.427561e+08 FES = 849100 μ_F = 0.777 μ_{CR} = 0.061
 Diversity = 2.3346e-02 k_t = [1, 1, 1, 1]
 Generation 8500: Best = 3.427561e+08 FES = 850100 μ_F = 0.769 μ_{CR} = 0.064
 Diversity = 2.1441e-02 k_t = [3, 3, 4, 2]
 Generation 8510: Best = 3.427561e+08 FES = 851100 μ_F = 0.761 μ_{CR} = 0.062
 Diversity = 2.4251e-02 k_t = [1, 1, 1, 1]
 Generation 8520: Best = 3.427560e+08 FES = 852100 μ_F = 0.775 μ_{CR} = 0.062
 Diversity = 3.0073e-02 k_t = [4, 4, 2, 3]
 Generation 8530: Best = 3.427560e+08 FES = 853100 μ_F = 0.775 μ_{CR} = 0.062
 Diversity = 3.9881e-02 k_t = [1, 1, 1, 1]
 Generation 8540: Best = 3.427560e+08 FES = 854100 μ_F = 0.793 μ_{CR} = 0.055
 Diversity = 3.3102e-02 k_t = [3, 4, 4, 3]
 Generation 8550: Best = 3.427560e+08 FES = 855100 μ_F = 0.805 μ_{CR} = 0.055
 Diversity = 3.2756e-02 k_t = [1, 1, 1, 1]
 Generation 8560: Best = 3.427560e+08 FES = 856100 μ_F = 0.822 μ_{CR} = 0.056
 Diversity = 3.8824e-02 k_t = [3, 2, 4, 2]
 Generation 8570: Best = 3.427559e+08 FES = 857100 μ_F = 0.820 μ_{CR} = 0.059
 Diversity = 3.5848e-02 k_t = [1, 1, 1, 1]
 Generation 8580: Best = 3.427559e+08 FES = 858100 μ_F = 0.827 μ_{CR} = 0.061
 Diversity = 3.6071e-02 k_t = [3, 4, 2, 2]
 Generation 8590: Best = 3.427559e+08 FES = 859100 μ_F = 0.802 μ_{CR} = 0.058
 Diversity = 3.3548e-02 k_t = [1, 1, 1, 1]

Generation 8600: Best = 3.427559e+08 FES = 860100 μ_F = 0.808 μ_{CR} = 0.060
 Diversity = 3.2102e-02 k_t = [2, 4, 3, 4]
 Generation 8610: Best = 3.427559e+08 FES = 861100 μ_F = 0.794 μ_{CR} = 0.055
 Diversity = 3.1958e-02 k_t = [1, 1, 1, 1]
 Generation 8620: Best = 3.427558e+08 FES = 862100 μ_F = 0.799 μ_{CR} = 0.055
 Diversity = 3.2512e-02 k_t = [3, 2, 4, 2]
 Generation 8630: Best = 3.427558e+08 FES = 863100 μ_F = 0.797 μ_{CR} = 0.058
 Diversity = 3.5730e-02 k_t = [1, 1, 1, 1]
 Generation 8640: Best = 3.427558e+08 FES = 864100 μ_F = 0.821 μ_{CR} = 0.059
 Diversity = 3.5769e-02 k_t = [3, 3, 3, 3]
 Generation 8650: Best = 3.427558e+08 FES = 865100 μ_F = 0.810 μ_{CR} = 0.063
 Diversity = 3.7604e-02 k_t = [1, 1, 1, 1]
 Generation 8660: Best = 3.427558e+08 FES = 866100 μ_F = 0.805 μ_{CR} = 0.057
 Diversity = 2.7037e-02 k_t = [4, 2, 3, 4]
 Generation 8670: Best = 3.427558e+08 FES = 867100 μ_F = 0.821 μ_{CR} = 0.056
 Diversity = 2.1232e-02 k_t = [1, 1, 1, 1]
 Generation 8680: Best = 3.427558e+08 FES = 868100 μ_F = 0.816 μ_{CR} = 0.054
 Diversity = 1.7812e-02 k_t = [3, 2, 4, 4]
 Generation 8690: Best = 3.427558e+08 FES = 869100 μ_F = 0.815 μ_{CR} = 0.054
 Diversity = 2.0866e-02 k_t = [1, 1, 1, 1]
 Generation 8700: Best = 3.427558e+08 FES = 870100 μ_F = 0.809 μ_{CR} = 0.053
 Diversity = 2.2473e-02 k_t = [2, 2, 4, 2]
 Generation 8710: Best = 3.427558e+08 FES = 871100 μ_F = 0.823 μ_{CR} = 0.059
 Diversity = 2.3054e-02 k_t = [1, 1, 1, 1]
 Generation 8720: Best = 3.427558e+08 FES = 872100 μ_F = 0.830 μ_{CR} = 0.063
 Diversity = 2.4735e-02 k_t = [4, 3, 3, 3]
 Generation 8730: Best = 3.427558e+08 FES = 873100 μ_F = 0.822 μ_{CR} = 0.060
 Diversity = 2.5522e-02 k_t = [1, 1, 1, 1]
 Generation 8740: Best = 3.427558e+08 FES = 874100 μ_F = 0.819 μ_{CR} = 0.060
 Diversity = 2.6581e-02 k_t = [2, 4, 3, 3]
 Generation 8750: Best = 3.427558e+08 FES = 875100 μ_F = 0.811 μ_{CR} = 0.056
 Diversity = 2.4707e-02 k_t = [1, 1, 1, 1]
 Generation 8760: Best = 3.427558e+08 FES = 876100 μ_F = 0.811 μ_{CR} = 0.056
 Diversity = 2.2642e-02 k_t = [4, 2, 2, 2]
 Generation 8770: Best = 3.427557e+08 FES = 877100 μ_F = 0.804 μ_{CR} = 0.056
 Diversity = 2.1282e-02 k_t = [1, 1, 1, 1]
 Generation 8780: Best = 3.427557e+08 FES = 878100 μ_F = 0.783 μ_{CR} = 0.057
 Diversity = 1.4819e-02 k_t = [3, 2, 3, 4]
 Generation 8790: Best = 3.427557e+08 FES = 879100 μ_F = 0.767 μ_{CR} = 0.055
 Diversity = 1.3990e-02 k_t = [1, 1, 1, 1]
 Generation 8800: Best = 3.427557e+08 FES = 880100 μ_F = 0.792 μ_{CR} = 0.058
 Diversity = 1.5060e-02 k_t = [3, 4, 4, 3]
 Generation 8810: Best = 3.427557e+08 FES = 881100 μ_F = 0.796 μ_{CR} = 0.064
 Diversity = 1.6882e-02 k_t = [1, 1, 1, 1]
 Generation 8820: Best = 3.427557e+08 FES = 882100 μ_F = 0.787 μ_{CR} = 0.054
 Diversity = 1.5587e-02 k_t = [4, 4, 3, 4]
 Generation 8830: Best = 3.427557e+08 FES = 883100 μ_F = 0.787 μ_{CR} = 0.055
 Diversity = 1.5274e-02 k_t = [1, 1, 1, 1]
 Generation 8840: Best = 3.427557e+08 FES = 884100 μ_F = 0.794 μ_{CR} = 0.057
 Diversity = 1.6577e-02 k_t = [4, 3, 4, 4]
 Generation 8850: Best = 3.427557e+08 FES = 885100 μ_F = 0.802 μ_{CR} = 0.052
 Diversity = 1.4702e-02 k_t = [1, 1, 1, 1]
 Generation 8860: Best = 3.427557e+08 FES = 886100 μ_F = 0.813 μ_{CR} = 0.060
 Diversity = 1.3309e-02 k_t = [4, 4, 4, 3]

Generation 8870: Best = 3.427557e+08 FES = 887100 μ_F = 0.801 μ_{CR} = 0.063
 Diversity = 1.6145e-02 k_t = [1, 1, 1, 1]
 Generation 8880: Best = 3.427557e+08 FES = 888100 μ_F = 0.787 μ_{CR} = 0.057
 Diversity = 1.6959e-02 k_t = [2, 4, 2, 4]
 Generation 8890: Best = 3.427557e+08 FES = 889100 μ_F = 0.794 μ_{CR} = 0.062
 Diversity = 1.6707e-02 k_t = [1, 1, 1, 1]
 Generation 8900: Best = 3.427556e+08 FES = 890100 μ_F = 0.800 μ_{CR} = 0.058
 Diversity = 1.4336e-02 k_t = [2, 3, 2, 4]
 Generation 8910: Best = 3.427556e+08 FES = 891100 μ_F = 0.785 μ_{CR} = 0.056
 Diversity = 1.7174e-02 k_t = [1, 1, 1, 1]
 Generation 8920: Best = 3.427556e+08 FES = 892100 μ_F = 0.806 μ_{CR} = 0.064
 Diversity = 2.5389e-02 k_t = [3, 2, 4, 2]
 Generation 8930: Best = 3.427556e+08 FES = 893100 μ_F = 0.826 μ_{CR} = 0.065
 Diversity = 2.9619e-02 k_t = [1, 1, 1, 1]
 Generation 8940: Best = 3.427556e+08 FES = 894100 μ_F = 0.823 μ_{CR} = 0.058
 Diversity = 2.7835e-02 k_t = [2, 4, 2, 2]
 Generation 8950: Best = 3.427556e+08 FES = 895100 μ_F = 0.820 μ_{CR} = 0.056
 Diversity = 2.8237e-02 k_t = [1, 1, 1, 1]
 Generation 8960: Best = 3.427556e+08 FES = 896100 μ_F = 0.824 μ_{CR} = 0.056
 Diversity = 2.8051e-02 k_t = [3, 2, 3, 2]
 Generation 8970: Best = 3.427556e+08 FES = 897100 μ_F = 0.833 μ_{CR} = 0.058
 Diversity = 2.8480e-02 k_t = [1, 1, 1, 1]
 Generation 8980: Best = 3.427556e+08 FES = 898100 μ_F = 0.810 μ_{CR} = 0.064
 Diversity = 2.0336e-02 k_t = [4, 2, 4, 2]
 Generation 8990: Best = 3.427556e+08 FES = 899100 μ_F = 0.793 μ_{CR} = 0.060
 Diversity = 1.9553e-02 k_t = [1, 1, 1, 1]
 Generation 9000: Best = 3.427556e+08 FES = 900100 μ_F = 0.807 μ_{CR} = 0.063
 Diversity = 1.8139e-02 k_t = [3, 2, 2, 3]
 Generation 9010: Best = 3.427556e+08 FES = 901100 μ_F = 0.814 μ_{CR} = 0.061
 Diversity = 1.8638e-02 k_t = [1, 1, 1, 1]
 Generation 9020: Best = 3.427556e+08 FES = 902100 μ_F = 0.813 μ_{CR} = 0.062
 Diversity = 1.8089e-02 k_t = [4, 2, 4, 3]
 Generation 9030: Best = 3.427556e+08 FES = 903100 μ_F = 0.792 μ_{CR} = 0.050
 Diversity = 1.7188e-02 k_t = [1, 1, 1, 1]
 Generation 9040: Best = 3.427556e+08 FES = 904100 μ_F = 0.793 μ_{CR} = 0.056
 Diversity = 1.6741e-02 k_t = [3, 3, 3, 4]
 Generation 9050: Best = 3.427556e+08 FES = 905100 μ_F = 0.800 μ_{CR} = 0.059
 Diversity = 1.8432e-02 k_t = [1, 1, 1, 1]
 Generation 9060: Best = 3.427556e+08 FES = 906100 μ_F = 0.782 μ_{CR} = 0.058
 Diversity = 1.5303e-02 k_t = [4, 2, 2, 2]
 Generation 9070: Best = 3.427556e+08 FES = 907100 μ_F = 0.776 μ_{CR} = 0.059
 Diversity = 1.8516e-02 k_t = [1, 1, 1, 1]
 Generation 9080: Best = 3.427556e+08 FES = 908100 μ_F = 0.788 μ_{CR} = 0.056
 Diversity = 2.3916e-02 k_t = [4, 4, 3, 3]
 Generation 9090: Best = 3.427556e+08 FES = 909100 μ_F = 0.786 μ_{CR} = 0.055
 Diversity = 2.2576e-02 k_t = [1, 1, 1, 1]
 Generation 9100: Best = 3.427556e+08 FES = 910100 μ_F = 0.804 μ_{CR} = 0.053
 Diversity = 2.0387e-02 k_t = [2, 4, 2, 3]
 Generation 9110: Best = 3.427556e+08 FES = 911100 μ_F = 0.814 μ_{CR} = 0.051
 Diversity = 2.1569e-02 k_t = [1, 1, 1, 1]
 Generation 9120: Best = 3.427556e+08 FES = 912100 μ_F = 0.824 μ_{CR} = 0.053
 Diversity = 1.7545e-02 k_t = [4, 4, 3, 2]
 Generation 9130: Best = 3.427555e+08 FES = 913100 μ_F = 0.804 μ_{CR} = 0.061
 Diversity = 1.5325e-02 k_t = [1, 1, 1, 1]

Generation 9140: Best = 3.427555e+08 FES = 914100 μ_F = 0.803 μ_{CR} = 0.063
 Diversity = 1.3237e-02 k_t = [4, 4, 3, 4]
 Generation 9150: Best = 3.427555e+08 FES = 915100 μ_F = 0.797 μ_{CR} = 0.061
 Diversity = 2.0533e-02 k_t = [1, 1, 1, 1]
 Generation 9160: Best = 3.427555e+08 FES = 916100 μ_F = 0.804 μ_{CR} = 0.061
 Diversity = 2.4152e-02 k_t = [3, 3, 2, 2]
 Generation 9170: Best = 3.427555e+08 FES = 917100 μ_F = 0.807 μ_{CR} = 0.057
 Diversity = 2.9231e-02 k_t = [1, 1, 1, 1]
 Generation 9180: Best = 3.427555e+08 FES = 918100 μ_F = 0.816 μ_{CR} = 0.060
 Diversity = 2.7070e-02 k_t = [2, 2, 3, 3]
 Generation 9190: Best = 3.427555e+08 FES = 919100 μ_F = 0.808 μ_{CR} = 0.061
 Diversity = 2.8160e-02 k_t = [1, 1, 1, 1]
 Generation 9200: Best = 3.427555e+08 FES = 920100 μ_F = 0.814 μ_{CR} = 0.056
 Diversity = 2.8393e-02 k_t = [2, 2, 3, 2]
 Generation 9210: Best = 3.427555e+08 FES = 921100 μ_F = 0.807 μ_{CR} = 0.057
 Diversity = 2.5145e-02 k_t = [1, 1, 1, 1]
 Generation 9220: Best = 3.427555e+08 FES = 922100 μ_F = 0.814 μ_{CR} = 0.060
 Diversity = 2.2032e-02 k_t = [3, 3, 4, 3]
 Generation 9230: Best = 3.427555e+08 FES = 923100 μ_F = 0.805 μ_{CR} = 0.063
 Diversity = 2.1171e-02 k_t = [1, 1, 1, 1]
 Generation 9240: Best = 3.427555e+08 FES = 924100 μ_F = 0.807 μ_{CR} = 0.064
 Diversity = 2.1660e-02 k_t = [4, 3, 3, 3]
 Generation 9250: Best = 3.427555e+08 FES = 925100 μ_F = 0.816 μ_{CR} = 0.067
 Diversity = 2.0956e-02 k_t = [1, 1, 1, 1]
 Generation 9260: Best = 3.427555e+08 FES = 926100 μ_F = 0.816 μ_{CR} = 0.062
 Diversity = 2.1387e-02 k_t = [4, 4, 2, 4]
 Generation 9270: Best = 3.427555e+08 FES = 927100 μ_F = 0.829 μ_{CR} = 0.057
 Diversity = 2.7016e-02 k_t = [1, 1, 1, 1]
 Generation 9280: Best = 3.427555e+08 FES = 928100 μ_F = 0.837 μ_{CR} = 0.051
 Diversity = 2.6848e-02 k_t = [4, 3, 2, 2]
 Generation 9290: Best = 3.427555e+08 FES = 929100 μ_F = 0.831 μ_{CR} = 0.051
 Diversity = 2.4748e-02 k_t = [1, 1, 1, 1]
 Generation 9300: Best = 3.427555e+08 FES = 930100 μ_F = 0.818 μ_{CR} = 0.058
 Diversity = 1.9184e-02 k_t = [3, 4, 4, 3]
 Generation 9310: Best = 3.427555e+08 FES = 931100 μ_F = 0.809 μ_{CR} = 0.058
 Diversity = 1.8507e-02 k_t = [1, 1, 1, 1]
 Generation 9320: Best = 3.427555e+08 FES = 932100 μ_F = 0.827 μ_{CR} = 0.052
 Diversity = 1.8040e-02 k_t = [4, 2, 4, 2]
 Generation 9330: Best = 3.427555e+08 FES = 933100 μ_F = 0.832 μ_{CR} = 0.054
 Diversity = 1.7339e-02 k_t = [1, 1, 1, 1]
 Generation 9340: Best = 3.427555e+08 FES = 934100 μ_F = 0.832 μ_{CR} = 0.059
 Diversity = 1.5692e-02 k_t = [3, 2, 4, 2]
 Generation 9350: Best = 3.427555e+08 FES = 935100 μ_F = 0.830 μ_{CR} = 0.058
 Diversity = 1.5344e-02 k_t = [1, 1, 1, 1]
 Generation 9360: Best = 3.427555e+08 FES = 936100 μ_F = 0.839 μ_{CR} = 0.058
 Diversity = 1.6570e-02 k_t = [3, 3, 4, 2]
 Generation 9370: Best = 3.427555e+08 FES = 937100 μ_F = 0.845 μ_{CR} = 0.054
 Diversity = 1.3135e-02 k_t = [1, 1, 1, 1]
 Generation 9380: Best = 3.427555e+08 FES = 938100 μ_F = 0.839 μ_{CR} = 0.058
 Diversity = 1.3102e-02 k_t = [3, 3, 4, 3]
 Generation 9390: Best = 3.427555e+08 FES = 939100 μ_F = 0.827 μ_{CR} = 0.057
 Diversity = 1.3994e-02 k_t = [1, 1, 1, 1]
 Generation 9400: Best = 3.427555e+08 FES = 940100 μ_F = 0.832 μ_{CR} = 0.058
 Diversity = 1.2825e-02 k_t = [4, 4, 2, 4]

Generation 9410: Best = 3.427555e+08 FES = 941100 μ_F = 0.840 μ_{CR} = 0.064
 Diversity = 1.5406e-02 k_t = [1, 1, 1, 1]
 Generation 9420: Best = 3.427554e+08 FES = 942100 μ_F = 0.821 μ_{CR} = 0.057
 Diversity = 1.6484e-02 k_t = [2, 4, 2, 2]
 Generation 9430: Best = 3.427554e+08 FES = 943100 μ_F = 0.837 μ_{CR} = 0.054
 Diversity = 1.6743e-02 k_t = [1, 1, 1, 1]
 Generation 9440: Best = 3.427554e+08 FES = 944100 μ_F = 0.811 μ_{CR} = 0.062
 Diversity = 1.3496e-02 k_t = [2, 3, 4, 3]
 Generation 9450: Best = 3.427554e+08 FES = 945100 μ_F = 0.816 μ_{CR} = 0.066
 Diversity = 1.5003e-02 k_t = [1, 1, 1, 1]
 Generation 9460: Best = 3.427554e+08 FES = 946100 μ_F = 0.814 μ_{CR} = 0.066
 Diversity = 1.5907e-02 k_t = [3, 4, 3, 2]
 Generation 9470: Best = 3.427554e+08 FES = 947100 μ_F = 0.809 μ_{CR} = 0.059
 Diversity = 1.4914e-02 k_t = [1, 1, 1, 1]
 Generation 9480: Best = 3.427554e+08 FES = 948100 μ_F = 0.796 μ_{CR} = 0.057
 Diversity = 1.2158e-02 k_t = [3, 3, 3, 4]
 Generation 9490: Best = 3.427554e+08 FES = 949100 μ_F = 0.792 μ_{CR} = 0.056
 Diversity = 1.0270e-02 k_t = [1, 1, 1, 1]
 Generation 9500: Best = 3.427554e+08 FES = 950100 μ_F = 0.808 μ_{CR} = 0.061
 Diversity = 1.0272e-02 k_t = [4, 2, 3, 2]
 Generation 9510: Best = 3.427554e+08 FES = 951100 μ_F = 0.792 μ_{CR} = 0.059
 Diversity = 1.0356e-02 k_t = [1, 1, 1, 1]
 Generation 9520: Best = 3.427554e+08 FES = 952100 μ_F = 0.789 μ_{CR} = 0.058
 Diversity = 1.0672e-02 k_t = [3, 2, 2, 3]
 Generation 9530: Best = 3.427554e+08 FES = 953100 μ_F = 0.791 μ_{CR} = 0.052
 Diversity = 1.1403e-02 k_t = [1, 1, 1, 1]
 Generation 9540: Best = 3.427554e+08 FES = 954100 μ_F = 0.806 μ_{CR} = 0.061
 Diversity = 1.0024e-02 k_t = [2, 3, 2, 4]
 Generation 9550: Best = 3.427554e+08 FES = 955100 μ_F = 0.805 μ_{CR} = 0.056
 Diversity = 1.0457e-02 k_t = [1, 1, 1, 1]
 Generation 9560: Best = 3.427554e+08 FES = 956100 μ_F = 0.796 μ_{CR} = 0.064
 Diversity = 1.1197e-02 k_t = [3, 3, 4, 3]
 Generation 9570: Best = 3.427554e+08 FES = 957100 μ_F = 0.800 μ_{CR} = 0.057
 Diversity = 1.0882e-02 k_t = [1, 1, 1, 1]
 Generation 9580: Best = 3.427554e+08 FES = 958100 μ_F = 0.808 μ_{CR} = 0.056
 Diversity = 1.1028e-02 k_t = [2, 3, 3, 2]
 Generation 9590: Best = 3.427554e+08 FES = 959100 μ_F = 0.798 μ_{CR} = 0.059
 Diversity = 9.6126e-03 k_t = [1, 1, 1, 1]
 Generation 9600: Best = 3.427554e+08 FES = 960100 μ_F = 0.797 μ_{CR} = 0.060
 Diversity = 1.0439e-02 k_t = [3, 2, 3, 4]
 Generation 9610: Best = 3.427554e+08 FES = 961100 μ_F = 0.804 μ_{CR} = 0.056
 Diversity = 9.4425e-03 k_t = [1, 1, 1, 1]
 Generation 9620: Best = 3.427554e+08 FES = 962100 μ_F = 0.783 μ_{CR} = 0.063
 Diversity = 9.7128e-03 k_t = [4, 3, 2, 4]
 Generation 9630: Best = 3.427554e+08 FES = 963100 μ_F = 0.799 μ_{CR} = 0.060
 Diversity = 1.1058e-02 k_t = [1, 1, 1, 1]
 Generation 9640: Best = 3.427554e+08 FES = 964100 μ_F = 0.793 μ_{CR} = 0.057
 Diversity = 1.0977e-02 k_t = [2, 4, 2, 3]
 Generation 9650: Best = 3.427554e+08 FES = 965100 μ_F = 0.808 μ_{CR} = 0.059
 Diversity = 1.0970e-02 k_t = [1, 1, 1, 1]
 Generation 9660: Best = 3.427554e+08 FES = 966100 μ_F = 0.809 μ_{CR} = 0.061
 Diversity = 1.0994e-02 k_t = [4, 4, 2, 2]
 Generation 9670: Best = 3.427554e+08 FES = 967100 μ_F = 0.818 μ_{CR} = 0.061
 Diversity = 1.0775e-02 k_t = [1, 1, 1, 1]

Generation 9680: Best = 3.427554e+08 FES = 968100 μ_F = 0.831 μ_{CR} = 0.055
 Diversity = 1.1263e-02 k_t = [2, 2, 4, 4]
 Generation 9690: Best = 3.427554e+08 FES = 969100 μ_F = 0.832 μ_{CR} = 0.055
 Diversity = 9.3119e-03 k_t = [1, 1, 1, 1]
 Generation 9700: Best = 3.427554e+08 FES = 970100 μ_F = 0.823 μ_{CR} = 0.064
 Diversity = 9.4710e-03 k_t = [4, 2, 3, 2]
 Generation 9710: Best = 3.427554e+08 FES = 971100 μ_F = 0.827 μ_{CR} = 0.063
 Diversity = 8.6108e-03 k_t = [1, 1, 1, 1]
 Generation 9720: Best = 3.427554e+08 FES = 972100 μ_F = 0.807 μ_{CR} = 0.065
 Diversity = 7.9171e-03 k_t = [3, 4, 4, 3]
 Generation 9730: Best = 3.427554e+08 FES = 973100 μ_F = 0.786 μ_{CR} = 0.060
 Diversity = 7.9772e-03 k_t = [1, 1, 1, 1]
 Generation 9740: Best = 3.427554e+08 FES = 974100 μ_F = 0.785 μ_{CR} = 0.061
 Diversity = 8.1606e-03 k_t = [4, 2, 3, 3]
 Generation 9750: Best = 3.427554e+08 FES = 975100 μ_F = 0.784 μ_{CR} = 0.060
 Diversity = 7.9312e-03 k_t = [1, 1, 1, 1]
 Generation 9760: Best = 3.427554e+08 FES = 976100 μ_F = 0.784 μ_{CR} = 0.060
 Diversity = 7.4142e-03 k_t = [2, 4, 2, 3]
 Generation 9770: Best = 3.427554e+08 FES = 977100 μ_F = 0.799 μ_{CR} = 0.061
 Diversity = 8.2040e-03 k_t = [1, 1, 1, 1]
 Generation 9780: Best = 3.427554e+08 FES = 978100 μ_F = 0.803 μ_{CR} = 0.056
 Diversity = 8.8592e-03 k_t = [4, 3, 3, 3]
 Generation 9790: Best = 3.427554e+08 FES = 979100 μ_F = 0.794 μ_{CR} = 0.052
 Diversity = 9.3815e-03 k_t = [1, 1, 1, 1]
 Generation 9800: Best = 3.427554e+08 FES = 980100 μ_F = 0.802 μ_{CR} = 0.055
 Diversity = 8.3582e-03 k_t = [2, 4, 3, 4]
 Generation 9810: Best = 3.427554e+08 FES = 981100 μ_F = 0.786 μ_{CR} = 0.058
 Diversity = 8.7691e-03 k_t = [1, 1, 1, 1]
 Generation 9820: Best = 3.427554e+08 FES = 982100 μ_F = 0.805 μ_{CR} = 0.057
 Diversity = 8.5697e-03 k_t = [4, 3, 3, 4]
 Generation 9830: Best = 3.427554e+08 FES = 983100 μ_F = 0.797 μ_{CR} = 0.056
 Diversity = 7.3970e-03 k_t = [1, 1, 1, 1]
 Generation 9840: Best = 3.427554e+08 FES = 984100 μ_F = 0.797 μ_{CR} = 0.062
 Diversity = 7.3431e-03 k_t = [2, 3, 3, 3]
 Generation 9850: Best = 3.427554e+08 FES = 985100 μ_F = 0.787 μ_{CR} = 0.059
 Diversity = 7.2625e-03 k_t = [1, 1, 1, 1]
 Generation 9860: Best = 3.427554e+08 FES = 986100 μ_F = 0.809 μ_{CR} = 0.062
 Diversity = 8.2799e-03 k_t = [3, 2, 3, 2]
 Generation 9870: Best = 3.427554e+08 FES = 987100 μ_F = 0.819 μ_{CR} = 0.062
 Diversity = 8.1144e-03 k_t = [1, 1, 1, 1]
 Generation 9880: Best = 3.427554e+08 FES = 988100 μ_F = 0.819 μ_{CR} = 0.061
 Diversity = 8.9783e-03 k_t = [3, 2, 3, 2]
 Generation 9890: Best = 3.427554e+08 FES = 989100 μ_F = 0.815 μ_{CR} = 0.061
 Diversity = 8.3609e-03 k_t = [1, 1, 1, 1]
 Generation 9900: Best = 3.427554e+08 FES = 990100 μ_F = 0.822 μ_{CR} = 0.063
 Diversity = 8.5461e-03 k_t = [4, 2, 3, 2]
 Generation 9910: Best = 3.427554e+08 FES = 991100 μ_F = 0.836 μ_{CR} = 0.058
 Diversity = 7.7836e-03 k_t = [1, 1, 1, 1]
 Generation 9920: Best = 3.427554e+08 FES = 992100 μ_F = 0.824 μ_{CR} = 0.055
 Diversity = 7.8419e-03 k_t = [2, 3, 3, 3]
 Generation 9930: Best = 3.427554e+08 FES = 993100 μ_F = 0.816 μ_{CR} = 0.060
 Diversity = 6.5692e-03 k_t = [1, 1, 1, 1]
 Generation 9940: Best = 3.427554e+08 FES = 994100 μ_F = 0.792 μ_{CR} = 0.069
 Diversity = 6.1544e-03 k_t = [3, 3, 2, 3]

Generation 9950: Best = 3.427554e+08 FES = 995100 μ_F = 0.793 μ_{CR} = 0.064
 Diversity = 6.3619e-03 k_t = [1, 1, 1, 1]
 Generation 9960: Best = 3.427554e+08 FES = 996100 μ_F = 0.802 μ_{CR} = 0.062
 Diversity = 7.4171e-03 k_t = [3, 4, 4, 2]
 Generation 9970: Best = 3.427554e+08 FES = 997100 μ_F = 0.795 μ_{CR} = 0.064
 Diversity = 7.9937e-03 k_t = [1, 1, 1, 1]
 Generation 9980: Best = 3.427554e+08 FES = 998100 μ_F = 0.792 μ_{CR} = 0.065
 Diversity = 6.9705e-03 k_t = [2, 4, 2, 2]
 Generation 9990: Best = 3.427554e+08 FES = 999100 μ_F = 0.788 μ_{CR} = 0.064
 Diversity = 7.2982e-03 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 3.427554e+08

Total Function Evals = 1000000

Total Successful Updates = 357258

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Results:

Best Fitness : 3.427554e+08

Global Optimum : 1.000000e+02

Error (|best-optimum|) : 3.427553e+08

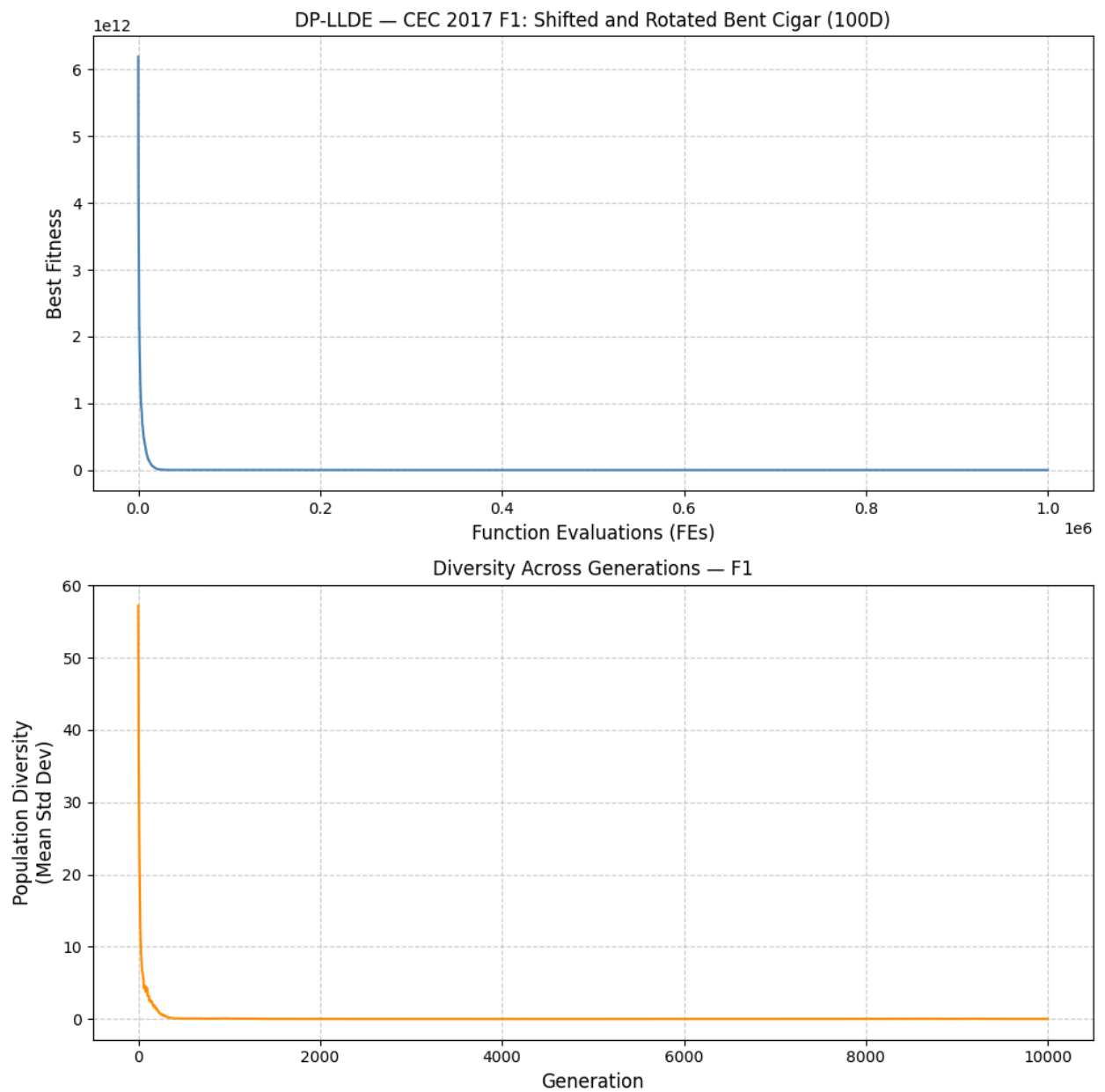
Best Solution (5 dim) : [-55.2538685 -70.42952094 -29.40750933 -58.32666564
22.08969344]

D^(0) (init. diversity): 5.7220e+01

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✓ Checkpoint saved to: /content/drive/My Drive/thesis_results/all_results_checkpoint.pkl



Graph saved as convergence_F1.png

✓ Plot saved to: /content/drive/My Drive/thesis_results/plots/convergence_F1.png

```
In [ ]: # =====
# F2 – Shifted and Rotated Sum of Different Power
# =====
run_and_plot('F2', make_wrapper(simple.f2), 200.0, 'Shifted and Rotated Sum of
```

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DP-LLDE on CEC 2017 F2 – Single Run

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Function : Shifted and Rotated Sum of Different Power (Deprecated) (100D)
Bounds : [-100, 100]
Global optimum : 200.0
WARNING: f2 has been deprecated from the CEC 2017 benchmark suite
Initial $D^*(0) = 5.7220e+01$ | Diversity threshold = 5.7220e+00
Generation 0: Best Fitness = 1.048956e+189
Generation 10: Best = 5.831948e+166 FES = 1100 $\mu_F = 0.552$ $\mu_{CR} = 0.343$
Diversity = 3.6851e+01 $k_t = [1, 1, 1, 1]$
Generation 20: Best = 1.936649e+152 FES = 2100 $\mu_F = 0.585$ $\mu_{CR} = 0.326$
Diversity = 2.5294e+01 $k_t = [2, 3, 2, 3]$
Generation 30: Best = 3.677410e+139 FES = 3100 $\mu_F = 0.596$ $\mu_{CR} = 0.342$
Diversity = 1.9521e+01 $k_t = [1, 1, 1, 1]$
Generation 40: Best = 8.935641e+135 FES = 4100 $\mu_F = 0.598$ $\mu_{CR} = 0.341$
Diversity = 1.4739e+01 $k_t = [3, 2, 3, 3]$
Generation 50: Best = 1.230261e+129 FES = 5100 $\mu_F = 0.589$ $\mu_{CR} = 0.339$
Diversity = 1.1775e+01 $k_t = [1, 1, 1, 1]$
Generation 60: Best = 6.646984e+124 FES = 6100 $\mu_F = 0.598$ $\mu_{CR} = 0.332$
Diversity = 8.8646e+00 $k_t = [3, 4, 4, 3]$
Generation 70: Best = 2.124276e+121 FES = 7100 $\mu_F = 0.605$ $\mu_{CR} = 0.338$
Diversity = 7.0031e+00 $k_t = [1, 1, 1, 1]$
Generation 80: Best = 3.654992e+117 FES = 8100 $\mu_F = 0.614$ $\mu_{CR} = 0.332$
Diversity = 5.4699e+00 $k_t = [4, 2, 2, 4]$
Generation 90: Best = 1.729022e+116 FES = 9100 $\mu_F = 0.640$ $\mu_{CR} = 0.328$
Diversity = 4.1078e+00 $k_t = [1, 1, 1, 1]$
Generation 100: Best = 2.014218e+114 FES = 10100 $\mu_F = 0.649$ $\mu_{CR} = 0.323$
Diversity = 3.2586e+00 $k_t = [4, 2, 2, 3]$
Generation 110: Best = 7.718435e+112 FES = 11100 $\mu_F = 0.662$ $\mu_{CR} = 0.312$
Diversity = 2.7689e+00 $k_t = [1, 1, 1, 1]$
Generation 120: Best = 6.721396e+111 FES = 12100 $\mu_F = 0.675$ $\mu_{CR} = 0.308$
Diversity = 2.4355e+00 $k_t = [3, 4, 4, 3]$
Generation 130: Best = 9.173418e+110 FES = 13100 $\mu_F = 0.686$ $\mu_{CR} = 0.305$
Diversity = 2.2431e+00 $k_t = [1, 1, 1, 1]$
Generation 140: Best = 1.227596e+110 FES = 14100 $\mu_F = 0.696$ $\mu_{CR} = 0.304$
Diversity = 2.3042e+00 $k_t = [2, 2, 3, 2]$
Generation 150: Best = 8.905298e+108 FES = 15100 $\mu_F = 0.713$ $\mu_{CR} = 0.308$
Diversity = 2.2760e+00 $k_t = [1, 1, 1, 1]$
Generation 160: Best = 1.653382e+108 FES = 16100 $\mu_F = 0.714$ $\mu_{CR} = 0.307$
Diversity = 2.3266e+00 $k_t = [2, 3, 3, 2]$
Generation 170: Best = 2.365897e+107 FES = 17100 $\mu_F = 0.716$ $\mu_{CR} = 0.299$
Diversity = 2.4143e+00 $k_t = [1, 1, 1, 1]$
Generation 180: Best = 4.408481e+106 FES = 18100 $\mu_F = 0.717$ $\mu_{CR} = 0.302$
Diversity = 2.4459e+00 $k_t = [2, 4, 2, 3]$
Generation 190: Best = 8.034259e+105 FES = 19100 $\mu_F = 0.718$ $\mu_{CR} = 0.295$
Diversity = 2.4419e+00 $k_t = [1, 1, 1, 1]$
Generation 200: Best = 2.537995e+105 FES = 20100 $\mu_F = 0.706$ $\mu_{CR} = 0.284$
Diversity = 2.2148e+00 $k_t = [4, 2, 2, 4]$
Generation 210: Best = 2.891943e+104 FES = 21100 $\mu_F = 0.712$ $\mu_{CR} = 0.273$
Diversity = 2.2253e+00 $k_t = [1, 1, 1, 1]$
Generation 220: Best = 6.434350e+103 FES = 22100 $\mu_F = 0.724$ $\mu_{CR} = 0.260$

Diversity = 2.2113e+00 k_t = [2, 4, 4, 3]
 Generation 230: Best = 1.157052e+103 FES = 23100 μ_F = 0.724 μ_{CR} = 0.253
 Diversity = 2.1899e+00 k_t = [1, 1, 1, 1]
 Generation 240: Best = 2.179406e+102 FES = 24100 μ_F = 0.710 μ_{CR} = 0.248
 Diversity = 2.0726e+00 k_t = [2, 3, 4, 3]
 Generation 250: Best = 4.323720e+101 FES = 25100 μ_F = 0.713 μ_{CR} = 0.250
 Diversity = 2.1368e+00 k_t = [1, 1, 1, 1]
 Generation 260: Best = 2.819207e+101 FES = 26100 μ_F = 0.714 μ_{CR} = 0.251
 Diversity = 1.8936e+00 k_t = [4, 4, 4, 4]
 Generation 270: Best = 9.226685e+100 FES = 27100 μ_F = 0.725 μ_{CR} = 0.255
 Diversity = 1.7950e+00 k_t = [1, 1, 1, 1]
 Generation 280: Best = 2.396391e+100 FES = 28100 μ_F = 0.728 μ_{CR} = 0.253
 Diversity = 1.6587e+00 k_t = [4, 2, 4, 4]
 Generation 290: Best = 1.860067e+100 FES = 29100 μ_F = 0.736 μ_{CR} = 0.254
 Diversity = 1.6033e+00 k_t = [1, 1, 1, 1]
 Generation 300: Best = 5.977058e+99 FES = 30100 μ_F = 0.753 μ_{CR} = 0.255
 Diversity = 1.8324e+00 k_t = [4, 3, 2, 2]
 Generation 310: Best = 1.894698e+99 FES = 31100 μ_F = 0.748 μ_{CR} = 0.251
 Diversity = 1.9770e+00 k_t = [1, 1, 1, 1]
 Generation 320: Best = 9.818873e+98 FES = 32100 μ_F = 0.733 μ_{CR} = 0.245
 Diversity = 1.8409e+00 k_t = [3, 3, 4, 3]
 Generation 330: Best = 3.490960e+98 FES = 33100 μ_F = 0.741 μ_{CR} = 0.241
 Diversity = 1.8796e+00 k_t = [1, 1, 1, 1]
 Generation 340: Best = 1.559396e+98 FES = 34100 μ_F = 0.741 μ_{CR} = 0.248
 Diversity = 1.7258e+00 k_t = [4, 2, 2, 3]
 Generation 350: Best = 4.772368e+97 FES = 35100 μ_F = 0.763 μ_{CR} = 0.242
 Diversity = 2.0769e+00 k_t = [1, 1, 1, 1]
 Generation 360: Best = 2.115147e+97 FES = 36100 μ_F = 0.758 μ_{CR} = 0.252
 Diversity = 2.2775e+00 k_t = [2, 4, 3, 3]
 Generation 370: Best = 1.183698e+97 FES = 37100 μ_F = 0.751 μ_{CR} = 0.248
 Diversity = 2.4309e+00 k_t = [1, 1, 1, 1]
 Generation 380: Best = 4.642747e+96 FES = 38100 μ_F = 0.727 μ_{CR} = 0.248
 Diversity = 2.4303e+00 k_t = [2, 3, 4, 2]
 Generation 390: Best = 2.553356e+96 FES = 39100 μ_F = 0.735 μ_{CR} = 0.234
 Diversity = 2.4913e+00 k_t = [1, 1, 1, 1]
 Generation 400: Best = 1.245300e+96 FES = 40100 μ_F = 0.738 μ_{CR} = 0.218
 Diversity = 2.4553e+00 k_t = [3, 4, 2, 3]
 Generation 410: Best = 6.182129e+95 FES = 41100 μ_F = 0.731 μ_{CR} = 0.203
 Diversity = 2.4108e+00 k_t = [1, 1, 1, 1]
 Generation 420: Best = 3.347530e+95 FES = 42100 μ_F = 0.734 μ_{CR} = 0.200
 Diversity = 2.2012e+00 k_t = [2, 3, 3, 3]
 Generation 430: Best = 1.994732e+95 FES = 43100 μ_F = 0.743 μ_{CR} = 0.206
 Diversity = 2.2515e+00 k_t = [1, 1, 1, 1]
 Generation 440: Best = 1.181730e+95 FES = 44100 μ_F = 0.749 μ_{CR} = 0.203
 Diversity = 2.2349e+00 k_t = [2, 2, 3, 2]
 Generation 450: Best = 7.706221e+94 FES = 45100 μ_F = 0.749 μ_{CR} = 0.207
 Diversity = 2.4011e+00 k_t = [1, 1, 1, 1]
 Generation 460: Best = 6.074874e+94 FES = 46100 μ_F = 0.739 μ_{CR} = 0.199
 Diversity = 2.2479e+00 k_t = [2, 4, 3, 2]
 Generation 470: Best = 3.595042e+94 FES = 47100 μ_F = 0.737 μ_{CR} = 0.200
 Diversity = 2.3542e+00 k_t = [1, 1, 1, 1]
 Generation 480: Best = 2.681376e+94 FES = 48100 μ_F = 0.740 μ_{CR} = 0.201
 Diversity = 2.3572e+00 k_t = [4, 2, 3, 4]
 Generation 490: Best = 2.077375e+94 FES = 49100 μ_F = 0.740 μ_{CR} = 0.203

Diversity = 2.4377e+00	k_t = [1, 1, 1, 1]					
Generation 500: Best = 1.490623e+94	FES = 50100	$\mu_F = 0.746$	$\mu_{CR} = 0.203$			
Diversity = 2.4639e+00	k_t = [4, 4, 2, 2]					
Generation 510: Best = 8.904113e+93	FES = 51100	$\mu_F = 0.740$	$\mu_{CR} = 0.199$			
Diversity = 2.5666e+00	k_t = [1, 1, 1, 1]					
Generation 520: Best = 7.122475e+93	FES = 52100	$\mu_F = 0.750$	$\mu_{CR} = 0.195$			
Diversity = 2.5652e+00	k_t = [2, 4, 2, 3]					
Generation 530: Best = 5.358564e+93	FES = 53100	$\mu_F = 0.747$	$\mu_{CR} = 0.193$			
Diversity = 2.5177e+00	k_t = [1, 1, 1, 1]					
Generation 540: Best = 4.511032e+93	FES = 54100	$\mu_F = 0.755$	$\mu_{CR} = 0.192$			
Diversity = 2.6336e+00	k_t = [3, 4, 3, 4]					
Generation 550: Best = 3.490658e+93	FES = 55100	$\mu_F = 0.763$	$\mu_{CR} = 0.193$			
Diversity = 2.6294e+00	k_t = [1, 1, 1, 1]					
Generation 560: Best = 2.756559e+93	FES = 56100	$\mu_F = 0.765$	$\mu_{CR} = 0.185$			
Diversity = 2.7461e+00	k_t = [3, 4, 3, 2]					
Generation 570: Best = 2.386944e+93	FES = 57100	$\mu_F = 0.749$	$\mu_{CR} = 0.182$			
Diversity = 3.0190e+00	k_t = [1, 1, 1, 1]					
Generation 580: Best = 1.927916e+93	FES = 58100	$\mu_F = 0.751$	$\mu_{CR} = 0.176$			
Diversity = 2.8395e+00	k_t = [3, 4, 3, 2]					
Generation 590: Best = 1.556064e+93	FES = 59100	$\mu_F = 0.763$	$\mu_{CR} = 0.170$			
Diversity = 3.1745e+00	k_t = [1, 1, 1, 1]					
Generation 600: Best = 1.426866e+93	FES = 60100	$\mu_F = 0.752$	$\mu_{CR} = 0.166$			
Diversity = 2.9945e+00	k_t = [2, 2, 3, 4]					
Generation 610: Best = 1.301455e+93	FES = 61100	$\mu_F = 0.748$	$\mu_{CR} = 0.172$			
Diversity = 2.8401e+00	k_t = [1, 1, 1, 1]					
Generation 620: Best = 1.127080e+93	FES = 62100	$\mu_F = 0.745$	$\mu_{CR} = 0.175$			
Diversity = 2.8435e+00	k_t = [2, 2, 4, 3]					
Generation 630: Best = 9.825766e+92	FES = 63100	$\mu_F = 0.750$	$\mu_{CR} = 0.176$			
Diversity = 3.0903e+00	k_t = [1, 1, 1, 1]					
Generation 640: Best = 8.693070e+92	FES = 64100	$\mu_F = 0.750$	$\mu_{CR} = 0.182$			
Diversity = 3.1534e+00	k_t = [4, 3, 4, 3]					
Generation 650: Best = 7.629920e+92	FES = 65100	$\mu_F = 0.744$	$\mu_{CR} = 0.181$			
Diversity = 3.1500e+00	k_t = [1, 1, 1, 1]					
Generation 660: Best = 6.643734e+92	FES = 66100	$\mu_F = 0.744$	$\mu_{CR} = 0.187$			
Diversity = 3.2344e+00	k_t = [3, 3, 2, 3]					
Generation 670: Best = 5.424450e+92	FES = 67100	$\mu_F = 0.737$	$\mu_{CR} = 0.184$			
Diversity = 3.1101e+00	k_t = [1, 1, 1, 1]					
Generation 680: Best = 4.742909e+92	FES = 68100	$\mu_F = 0.746$	$\mu_{CR} = 0.181$			
Diversity = 2.9898e+00	k_t = [4, 4, 4, 4]					
Generation 690: Best = 3.737953e+92	FES = 69100	$\mu_F = 0.753$	$\mu_{CR} = 0.191$			
Diversity = 3.0681e+00	k_t = [1, 1, 1, 1]					
Generation 700: Best = 2.871476e+92	FES = 70100	$\mu_F = 0.763$	$\mu_{CR} = 0.196$			
Diversity = 3.0340e+00	k_t = [3, 4, 3, 4]					
Generation 710: Best = 2.327515e+92	FES = 71100	$\mu_F = 0.762$	$\mu_{CR} = 0.196$			
Diversity = 3.1746e+00	k_t = [1, 1, 1, 1]					
Generation 720: Best = 2.048475e+92	FES = 72100	$\mu_F = 0.754$	$\mu_{CR} = 0.205$			
Diversity = 3.1964e+00	k_t = [2, 2, 3, 2]					
Generation 730: Best = 1.483965e+92	FES = 73100	$\mu_F = 0.747$	$\mu_{CR} = 0.202$			
Diversity = 3.3454e+00	k_t = [1, 1, 1, 1]					
Generation 740: Best = 1.124909e+92	FES = 74100	$\mu_F = 0.743$	$\mu_{CR} = 0.208$			
Diversity = 3.4346e+00	k_t = [4, 4, 4, 2]					
Generation 750: Best = 9.023935e+91	FES = 75100	$\mu_F = 0.738$	$\mu_{CR} = 0.204$			
Diversity = 3.6029e+00	k_t = [1, 1, 1, 1]					
Generation 760: Best = 7.050949e+91	FES = 76100	$\mu_F = 0.735$	$\mu_{CR} = 0.205$			

Diversity = 3.5735e+00	k_t = [2, 2, 2, 3]				
Generation 770: Best = 5.837700e+91	FES = 77100	$\mu_F = 0.726$	$\mu_{CR} = 0.211$		
Diversity = 3.5260e+00	k_t = [1, 1, 1, 1]				
Generation 780: Best = 4.750939e+91	FES = 78100	$\mu_F = 0.718$	$\mu_{CR} = 0.202$		
Diversity = 3.4653e+00	k_t = [2, 3, 3, 2]				
Generation 790: Best = 3.747340e+91	FES = 79100	$\mu_F = 0.716$	$\mu_{CR} = 0.196$		
Diversity = 3.4607e+00	k_t = [1, 1, 1, 1]				
Generation 800: Best = 3.124442e+91	FES = 80100	$\mu_F = 0.718$	$\mu_{CR} = 0.196$		
Diversity = 3.4293e+00	k_t = [4, 2, 2, 2]				
Generation 810: Best = 2.713298e+91	FES = 81100	$\mu_F = 0.724$	$\mu_{CR} = 0.194$		
Diversity = 3.4004e+00	k_t = [1, 1, 1, 1]				
Generation 820: Best = 2.357531e+91	FES = 82100	$\mu_F = 0.725$	$\mu_{CR} = 0.200$		
Diversity = 3.3800e+00	k_t = [4, 2, 2, 4]				
Generation 830: Best = 2.001981e+91	FES = 83100	$\mu_F = 0.732$	$\mu_{CR} = 0.203$		
Diversity = 3.3013e+00	k_t = [1, 1, 1, 1]				
Generation 840: Best = 1.628923e+91	FES = 84100	$\mu_F = 0.741$	$\mu_{CR} = 0.214$		
Diversity = 3.3331e+00	k_t = [4, 2, 3, 2]				
Generation 850: Best = 1.402050e+91	FES = 85100	$\mu_F = 0.741$	$\mu_{CR} = 0.218$		
Diversity = 3.4413e+00	k_t = [1, 1, 1, 1]				
Generation 860: Best = 1.155043e+91	FES = 86100	$\mu_F = 0.740$	$\mu_{CR} = 0.215$		
Diversity = 3.2056e+00	k_t = [4, 4, 3, 3]				
Generation 870: Best = 9.847361e+90	FES = 87100	$\mu_F = 0.747$	$\mu_{CR} = 0.206$		
Diversity = 3.3378e+00	k_t = [1, 1, 1, 1]				
Generation 880: Best = 8.018175e+90	FES = 88100	$\mu_F = 0.753$	$\mu_{CR} = 0.209$		
Diversity = 3.1866e+00	k_t = [3, 3, 3, 3]				
Generation 890: Best = 6.066568e+90	FES = 89100	$\mu_F = 0.747$	$\mu_{CR} = 0.217$		
Diversity = 3.4831e+00	k_t = [1, 1, 1, 1]				
Generation 900: Best = 4.946645e+90	FES = 90100	$\mu_F = 0.753$	$\mu_{CR} = 0.206$		
Diversity = 3.5477e+00	k_t = [3, 3, 4, 4]				
Generation 910: Best = 4.098954e+90	FES = 91100	$\mu_F = 0.752$	$\mu_{CR} = 0.210$		
Diversity = 3.4987e+00	k_t = [1, 1, 1, 1]				
Generation 920: Best = 3.275225e+90	FES = 92100	$\mu_F = 0.743$	$\mu_{CR} = 0.210$		
Diversity = 3.5085e+00	k_t = [3, 3, 4, 2]				
Generation 930: Best = 2.763666e+90	FES = 93100	$\mu_F = 0.744$	$\mu_{CR} = 0.213$		
Diversity = 3.4920e+00	k_t = [1, 1, 1, 1]				
Generation 940: Best = 2.325424e+90	FES = 94100	$\mu_F = 0.744$	$\mu_{CR} = 0.220$		
Diversity = 3.4789e+00	k_t = [3, 2, 2, 3]				
Generation 950: Best = 2.120679e+90	FES = 95100	$\mu_F = 0.733$	$\mu_{CR} = 0.219$		
Diversity = 3.5095e+00	k_t = [1, 1, 1, 1]				
Generation 960: Best = 1.838029e+90	FES = 96100	$\mu_F = 0.737$	$\mu_{CR} = 0.212$		
Diversity = 3.2747e+00	k_t = [3, 3, 4, 3]				
Generation 970: Best = 1.606819e+90	FES = 97100	$\mu_F = 0.737$	$\mu_{CR} = 0.213$		
Diversity = 3.2779e+00	k_t = [1, 1, 1, 1]				
Generation 980: Best = 1.324194e+90	FES = 98100	$\mu_F = 0.749$	$\mu_{CR} = 0.219$		
Diversity = 3.2515e+00	k_t = [3, 3, 2, 4]				
Generation 990: Best = 1.171664e+90	FES = 99100	$\mu_F = 0.746$	$\mu_{CR} = 0.220$		
Diversity = 3.2584e+00	k_t = [1, 1, 1, 1]				
Generation 1000: Best = 9.807610e+89	FES = 100100	$\mu_F = 0.754$	$\mu_{CR} = 0.219$		
Diversity = 3.4014e+00	k_t = [3, 4, 3, 2]				
Generation 1010: Best = 7.625804e+89	FES = 101100	$\mu_F = 0.758$	$\mu_{CR} = 0.218$		
Diversity = 3.5130e+00	k_t = [1, 1, 1, 1]				
Generation 1020: Best = 6.605475e+89	FES = 102100	$\mu_F = 0.758$	$\mu_{CR} = 0.213$		
Diversity = 3.4050e+00	k_t = [4, 3, 3, 2]				
Generation 1030: Best = 5.427397e+89	FES = 103100	$\mu_F = 0.761$	$\mu_{CR} = 0.213$		

Diversity = 3.6620e+00	k_t = [1, 1, 1, 1]			
Generation 1040: Best = 4.501399e+89	FES = 104100	$\mu_F = 0.738$	$\mu_{CR} = 0.200$	
Diversity = 3.5803e+00	k_t = [4, 4, 4, 2]			
Generation 1050: Best = 4.025017e+89	FES = 105100	$\mu_F = 0.725$	$\mu_{CR} = 0.198$	
Diversity = 3.7316e+00	k_t = [1, 1, 1, 1]			
Generation 1060: Best = 3.559483e+89	FES = 106100	$\mu_F = 0.715$	$\mu_{CR} = 0.198$	
Diversity = 3.3641e+00	k_t = [4, 3, 3, 3]			
Generation 1070: Best = 3.227615e+89	FES = 107100	$\mu_F = 0.727$	$\mu_{CR} = 0.204$	
Diversity = 3.2433e+00	k_t = [1, 1, 1, 1]			
Generation 1080: Best = 2.722131e+89	FES = 108100	$\mu_F = 0.737$	$\mu_{CR} = 0.201$	
Diversity = 2.9882e+00	k_t = [2, 2, 4, 2]			
Generation 1090: Best = 2.514868e+89	FES = 109100	$\mu_F = 0.728$	$\mu_{CR} = 0.203$	
Diversity = 2.8751e+00	k_t = [1, 1, 1, 1]			
Generation 1100: Best = 2.232123e+89	FES = 110100	$\mu_F = 0.727$	$\mu_{CR} = 0.207$	
Diversity = 2.8106e+00	k_t = [3, 4, 2, 2]			
Generation 1110: Best = 1.986071e+89	FES = 111100	$\mu_F = 0.740$	$\mu_{CR} = 0.211$	
Diversity = 2.9242e+00	k_t = [1, 1, 1, 1]			
Generation 1120: Best = 1.789478e+89	FES = 112100	$\mu_F = 0.743$	$\mu_{CR} = 0.214$	
Diversity = 2.8593e+00	k_t = [2, 4, 3, 3]			
Generation 1130: Best = 1.653135e+89	FES = 113100	$\mu_F = 0.749$	$\mu_{CR} = 0.217$	
Diversity = 3.0193e+00	k_t = [1, 1, 1, 1]			
Generation 1140: Best = 1.438358e+89	FES = 114100	$\mu_F = 0.754$	$\mu_{CR} = 0.223$	
Diversity = 2.7656e+00	k_t = [3, 3, 4, 3]			
Generation 1150: Best = 1.273613e+89	FES = 115100	$\mu_F = 0.754$	$\mu_{CR} = 0.229$	
Diversity = 2.8923e+00	k_t = [1, 1, 1, 1]			
Generation 1160: Best = 1.152375e+89	FES = 116100	$\mu_F = 0.756$	$\mu_{CR} = 0.234$	
Diversity = 2.9166e+00	k_t = [3, 2, 4, 3]			
Generation 1170: Best = 1.038271e+89	FES = 117100	$\mu_F = 0.759$	$\mu_{CR} = 0.234$	
Diversity = 2.9610e+00	k_t = [1, 1, 1, 1]			
Generation 1180: Best = 8.804600e+88	FES = 118100	$\mu_F = 0.751$	$\mu_{CR} = 0.231$	
Diversity = 3.0399e+00	k_t = [2, 3, 3, 3]			
Generation 1190: Best = 7.765814e+88	FES = 119100	$\mu_F = 0.757$	$\mu_{CR} = 0.230$	
Diversity = 3.1383e+00	k_t = [1, 1, 1, 1]			
Generation 1200: Best = 6.689696e+88	FES = 120100	$\mu_F = 0.747$	$\mu_{CR} = 0.227$	
Diversity = 3.1385e+00	k_t = [2, 4, 2, 2]			
Generation 1210: Best = 5.944625e+88	FES = 121100	$\mu_F = 0.747$	$\mu_{CR} = 0.221$	
Diversity = 3.1728e+00	k_t = [1, 1, 1, 1]			
Generation 1220: Best = 5.036909e+88	FES = 122100	$\mu_F = 0.752$	$\mu_{CR} = 0.225$	
Diversity = 3.1749e+00	k_t = [3, 3, 3, 2]			
Generation 1230: Best = 4.169513e+88	FES = 123100	$\mu_F = 0.751$	$\mu_{CR} = 0.214$	
Diversity = 3.4157e+00	k_t = [1, 1, 1, 1]			
Generation 1240: Best = 3.420054e+88	FES = 124100	$\mu_F = 0.744$	$\mu_{CR} = 0.208$	
Diversity = 3.4354e+00	k_t = [3, 3, 2, 2]			
Generation 1250: Best = 2.998731e+88	FES = 125100	$\mu_F = 0.742$	$\mu_{CR} = 0.201$	
Diversity = 3.4751e+00	k_t = [1, 1, 1, 1]			
Generation 1260: Best = 2.516386e+88	FES = 126100	$\mu_F = 0.745$	$\mu_{CR} = 0.200$	
Diversity = 3.4582e+00	k_t = [3, 3, 2, 3]			
Generation 1270: Best = 2.254799e+88	FES = 127100	$\mu_F = 0.734$	$\mu_{CR} = 0.198$	
Diversity = 3.3340e+00	k_t = [1, 1, 1, 1]			
Generation 1280: Best = 2.078105e+88	FES = 128100	$\mu_F = 0.729$	$\mu_{CR} = 0.198$	
Diversity = 3.0543e+00	k_t = [2, 3, 4, 4]			
Generation 1290: Best = 1.940013e+88	FES = 129100	$\mu_F = 0.744$	$\mu_{CR} = 0.197$	
Diversity = 3.2064e+00	k_t = [1, 1, 1, 1]			
Generation 1300: Best = 1.798263e+88	FES = 130100	$\mu_F = 0.747$	$\mu_{CR} = 0.195$	

Diversity = 3.1550e+00	k_t = [3, 4, 3, 2]			
Generation 1310: Best = 1.675195e+88	FES = 131100	$\mu_F = 0.740$	$\mu_{CR} = 0.193$	
Diversity = 3.3232e+00	k_t = [1, 1, 1, 1]			
Generation 1320: Best = 1.512280e+88	FES = 132100	$\mu_F = 0.738$	$\mu_{CR} = 0.189$	
Diversity = 3.1760e+00	k_t = [4, 2, 2, 2]			
Generation 1330: Best = 1.365955e+88	FES = 133100	$\mu_F = 0.750$	$\mu_{CR} = 0.193$	
Diversity = 3.1310e+00	k_t = [1, 1, 1, 1]			
Generation 1340: Best = 1.209076e+88	FES = 134100	$\mu_F = 0.749$	$\mu_{CR} = 0.193$	
Diversity = 2.9478e+00	k_t = [4, 4, 2, 4]			
Generation 1350: Best = 1.154198e+88	FES = 135100	$\mu_F = 0.757$	$\mu_{CR} = 0.209$	
Diversity = 2.9829e+00	k_t = [1, 1, 1, 1]			
Generation 1360: Best = 1.043817e+88	FES = 136100	$\mu_F = 0.763$	$\mu_{CR} = 0.217$	
Diversity = 3.0113e+00	k_t = [3, 3, 2, 4]			
Generation 1370: Best = 9.298216e+87	FES = 137100	$\mu_F = 0.751$	$\mu_{CR} = 0.221$	
Diversity = 3.2263e+00	k_t = [1, 1, 1, 1]			
Generation 1380: Best = 8.774806e+87	FES = 138100	$\mu_F = 0.750$	$\mu_{CR} = 0.215$	
Diversity = 3.0987e+00	k_t = [2, 4, 4, 4]			
Generation 1390: Best = 8.116452e+87	FES = 139100	$\mu_F = 0.757$	$\mu_{CR} = 0.219$	
Diversity = 3.3435e+00	k_t = [1, 1, 1, 1]			
Generation 1400: Best = 7.404223e+87	FES = 140100	$\mu_F = 0.756$	$\mu_{CR} = 0.212$	
Diversity = 3.3338e+00	k_t = [4, 3, 2, 3]			
Generation 1410: Best = 7.066120e+87	FES = 141100	$\mu_F = 0.759$	$\mu_{CR} = 0.212$	
Diversity = 3.6415e+00	k_t = [1, 1, 1, 1]			
Generation 1420: Best = 6.411126e+87	FES = 142100	$\mu_F = 0.767$	$\mu_{CR} = 0.212$	
Diversity = 3.7294e+00	k_t = [2, 4, 2, 4]			
Generation 1430: Best = 5.763673e+87	FES = 143100	$\mu_F = 0.770$	$\mu_{CR} = 0.213$	
Diversity = 3.6646e+00	k_t = [1, 1, 1, 1]			
Generation 1440: Best = 5.142671e+87	FES = 144100	$\mu_F = 0.763$	$\mu_{CR} = 0.212$	
Diversity = 3.8042e+00	k_t = [2, 3, 2, 4]			
Generation 1450: Best = 4.626077e+87	FES = 145100	$\mu_F = 0.755$	$\mu_{CR} = 0.206$	
Diversity = 3.8157e+00	k_t = [1, 1, 1, 1]			
Generation 1460: Best = 4.126548e+87	FES = 146100	$\mu_F = 0.756$	$\mu_{CR} = 0.207$	
Diversity = 3.6411e+00	k_t = [2, 4, 3, 2]			
Generation 1470: Best = 3.685943e+87	FES = 147100	$\mu_F = 0.748$	$\mu_{CR} = 0.216$	
Diversity = 3.6428e+00	k_t = [1, 1, 1, 1]			
Generation 1480: Best = 3.493532e+87	FES = 148100	$\mu_F = 0.741$	$\mu_{CR} = 0.213$	
Diversity = 3.6529e+00	k_t = [2, 2, 4, 2]			
Generation 1490: Best = 3.076859e+87	FES = 149100	$\mu_F = 0.733$	$\mu_{CR} = 0.209$	
Diversity = 3.6990e+00	k_t = [1, 1, 1, 1]			
Generation 1500: Best = 2.974701e+87	FES = 150100	$\mu_F = 0.727$	$\mu_{CR} = 0.201$	
Diversity = 3.6730e+00	k_t = [4, 2, 4, 4]			
Generation 1510: Best = 2.781883e+87	FES = 151100	$\mu_F = 0.735$	$\mu_{CR} = 0.197$	
Diversity = 3.5911e+00	k_t = [1, 1, 1, 1]			
Generation 1520: Best = 2.478136e+87	FES = 152100	$\mu_F = 0.746$	$\mu_{CR} = 0.202$	
Diversity = 3.5316e+00	k_t = [2, 3, 3, 3]			
Generation 1530: Best = 2.401923e+87	FES = 153100	$\mu_F = 0.742$	$\mu_{CR} = 0.198$	
Diversity = 3.4548e+00	k_t = [1, 1, 1, 1]			
Generation 1540: Best = 2.247717e+87	FES = 154100	$\mu_F = 0.747$	$\mu_{CR} = 0.195$	
Diversity = 3.4418e+00	k_t = [3, 4, 3, 3]			
Generation 1550: Best = 2.134352e+87	FES = 155100	$\mu_F = 0.748$	$\mu_{CR} = 0.210$	
Diversity = 3.5353e+00	k_t = [1, 1, 1, 1]			
Generation 1560: Best = 1.995643e+87	FES = 156100	$\mu_F = 0.739$	$\mu_{CR} = 0.200$	
Diversity = 3.1696e+00	k_t = [3, 2, 4, 4]			
Generation 1570: Best = 1.867363e+87	FES = 157100	$\mu_F = 0.745$	$\mu_{CR} = 0.202$	

Diversity = 3.2480e+00	k_t = [1, 1, 1, 1]			
Generation 1580: Best = 1.706469e+87	FES = 158100	$\mu_F = 0.751$	$\mu_{CR} = 0.201$	
Diversity = 3.3703e+00	k_t = [3, 3, 4, 2]			
Generation 1590: Best = 1.571111e+87	FES = 159100	$\mu_F = 0.744$	$\mu_{CR} = 0.195$	
Diversity = 3.4658e+00	k_t = [1, 1, 1, 1]			
Generation 1600: Best = 1.432427e+87	FES = 160100	$\mu_F = 0.749$	$\mu_{CR} = 0.196$	
Diversity = 3.1626e+00	k_t = [4, 3, 3, 4]			
Generation 1610: Best = 1.382781e+87	FES = 161100	$\mu_F = 0.751$	$\mu_{CR} = 0.204$	
Diversity = 3.2217e+00	k_t = [1, 1, 1, 1]			
Generation 1620: Best = 1.328630e+87	FES = 162100	$\mu_F = 0.745$	$\mu_{CR} = 0.201$	
Diversity = 3.1795e+00	k_t = [2, 2, 2, 3]			
Generation 1630: Best = 1.228422e+87	FES = 163100	$\mu_F = 0.741$	$\mu_{CR} = 0.206$	
Diversity = 3.2273e+00	k_t = [1, 1, 1, 1]			
Generation 1640: Best = 1.159740e+87	FES = 164100	$\mu_F = 0.742$	$\mu_{CR} = 0.206$	
Diversity = 3.2099e+00	k_t = [3, 4, 2, 4]			
Generation 1650: Best = 1.041645e+87	FES = 165100	$\mu_F = 0.751$	$\mu_{CR} = 0.201$	
Diversity = 3.3394e+00	k_t = [1, 1, 1, 1]			
Generation 1660: Best = 9.305191e+86	FES = 166100	$\mu_F = 0.738$	$\mu_{CR} = 0.200$	
Diversity = 3.3047e+00	k_t = [4, 3, 3, 2]			
Generation 1670: Best = 8.206151e+86	FES = 167100	$\mu_F = 0.737$	$\mu_{CR} = 0.198$	
Diversity = 3.4165e+00	k_t = [1, 1, 1, 1]			
Generation 1680: Best = 7.360718e+86	FES = 168100	$\mu_F = 0.750$	$\mu_{CR} = 0.201$	
Diversity = 3.3822e+00	k_t = [3, 4, 4, 3]			
Generation 1690: Best = 6.558446e+86	FES = 169100	$\mu_F = 0.757$	$\mu_{CR} = 0.198$	
Diversity = 3.5839e+00	k_t = [1, 1, 1, 1]			
Generation 1700: Best = 5.679562e+86	FES = 170100	$\mu_F = 0.751$	$\mu_{CR} = 0.203$	
Diversity = 3.5064e+00	k_t = [2, 4, 3, 4]			
Generation 1710: Best = 4.658181e+86	FES = 171100	$\mu_F = 0.743$	$\mu_{CR} = 0.201$	
Diversity = 3.6128e+00	k_t = [1, 1, 1, 1]			
Generation 1720: Best = 4.113333e+86	FES = 172100	$\mu_F = 0.745$	$\mu_{CR} = 0.198$	
Diversity = 3.5074e+00	k_t = [2, 4, 2, 3]			
Generation 1730: Best = 3.521437e+86	FES = 173100	$\mu_F = 0.755$	$\mu_{CR} = 0.198$	
Diversity = 3.7021e+00	k_t = [1, 1, 1, 1]			
Generation 1740: Best = 3.093320e+86	FES = 174100	$\mu_F = 0.748$	$\mu_{CR} = 0.206$	
Diversity = 3.6614e+00	k_t = [4, 4, 4, 3]			
Generation 1750: Best = 2.719565e+86	FES = 175100	$\mu_F = 0.752$	$\mu_{CR} = 0.197$	
Diversity = 3.6762e+00	k_t = [1, 1, 1, 1]			
Generation 1760: Best = 2.398574e+86	FES = 176100	$\mu_F = 0.746$	$\mu_{CR} = 0.186$	
Diversity = 3.6149e+00	k_t = [2, 2, 4, 4]			
Generation 1770: Best = 2.070931e+86	FES = 177100	$\mu_F = 0.743$	$\mu_{CR} = 0.191$	
Diversity = 3.5849e+00	k_t = [1, 1, 1, 1]			
Generation 1780: Best = 1.345191e+86	FES = 178100	$\mu_F = 0.746$	$\mu_{CR} = 0.193$	
Diversity = 3.4771e+00	k_t = [4, 2, 2, 2]			
Generation 1790: Best = 1.155627e+86	FES = 179100	$\mu_F = 0.748$	$\mu_{CR} = 0.191$	
Diversity = 3.6453e+00	k_t = [1, 1, 1, 1]			
Generation 1800: Best = 8.567742e+85	FES = 180100	$\mu_F = 0.746$	$\mu_{CR} = 0.186$	
Diversity = 3.6123e+00	k_t = [2, 2, 3, 3]			
Generation 1810: Best = 6.546900e+85	FES = 181100	$\mu_F = 0.747$	$\mu_{CR} = 0.184$	
Diversity = 3.5935e+00	k_t = [1, 1, 1, 1]			
Generation 1820: Best = 5.188390e+85	FES = 182100	$\mu_F = 0.743$	$\mu_{CR} = 0.185$	
Diversity = 3.5442e+00	k_t = [3, 3, 4, 3]			
Generation 1830: Best = 4.241804e+85	FES = 183100	$\mu_F = 0.740$	$\mu_{CR} = 0.188$	
Diversity = 3.5623e+00	k_t = [1, 1, 1, 1]			
Generation 1840: Best = 3.891300e+85	FES = 184100	$\mu_F = 0.733$	$\mu_{CR} = 0.188$	

Diversity = 3.3819e+00	k_t = [4, 2, 4, 3]			
Generation 1850: Best = 3.312444e+85	FES = 185100	$\mu_F = 0.739$	$\mu_{CR} = 0.181$	
Diversity = 3.5261e+00	k_t = [1, 1, 1, 1]			
Generation 1860: Best = 2.827320e+85	FES = 186100	$\mu_F = 0.726$	$\mu_{CR} = 0.195$	
Diversity = 3.3442e+00	k_t = [4, 3, 3, 3]			
Generation 1870: Best = 2.303784e+85	FES = 187100	$\mu_F = 0.726$	$\mu_{CR} = 0.197$	
Diversity = 3.2942e+00	k_t = [1, 1, 1, 1]			
Generation 1880: Best = 1.800823e+85	FES = 188100	$\mu_F = 0.738$	$\mu_{CR} = 0.194$	
Diversity = 3.1797e+00	k_t = [3, 2, 3, 4]			
Generation 1890: Best = 1.485390e+85	FES = 189100	$\mu_F = 0.751$	$\mu_{CR} = 0.187$	
Diversity = 3.1653e+00	k_t = [1, 1, 1, 1]			
Generation 1900: Best = 1.122880e+85	FES = 190100	$\mu_F = 0.747$	$\mu_{CR} = 0.190$	
Diversity = 3.0516e+00	k_t = [2, 3, 4, 3]			
Generation 1910: Best = 8.073775e+84	FES = 191100	$\mu_F = 0.749$	$\mu_{CR} = 0.192$	
Diversity = 3.3062e+00	k_t = [1, 1, 1, 1]			
Generation 1920: Best = 7.000121e+84	FES = 192100	$\mu_F = 0.748$	$\mu_{CR} = 0.192$	
Diversity = 3.1834e+00	k_t = [2, 3, 4, 3]			
Generation 1930: Best = 6.010193e+84	FES = 193100	$\mu_F = 0.750$	$\mu_{CR} = 0.186$	
Diversity = 3.3264e+00	k_t = [1, 1, 1, 1]			
Generation 1940: Best = 4.947155e+84	FES = 194100	$\mu_F = 0.746$	$\mu_{CR} = 0.187$	
Diversity = 3.3511e+00	k_t = [4, 3, 3, 2]			
Generation 1950: Best = 4.208028e+84	FES = 195100	$\mu_F = 0.729$	$\mu_{CR} = 0.192$	
Diversity = 3.3498e+00	k_t = [1, 1, 1, 1]			
Generation 1960: Best = 3.574671e+84	FES = 196100	$\mu_F = 0.741$	$\mu_{CR} = 0.187$	
Diversity = 3.3836e+00	k_t = [2, 4, 3, 3]			
Generation 1970: Best = 3.130588e+84	FES = 197100	$\mu_F = 0.734$	$\mu_{CR} = 0.178$	
Diversity = 3.4323e+00	k_t = [1, 1, 1, 1]			
Generation 1980: Best = 2.575892e+84	FES = 198100	$\mu_F = 0.726$	$\mu_{CR} = 0.184$	
Diversity = 3.1605e+00	k_t = [4, 2, 2, 4]			
Generation 1990: Best = 2.190661e+84	FES = 199100	$\mu_F = 0.736$	$\mu_{CR} = 0.192$	
Diversity = 3.0369e+00	k_t = [1, 1, 1, 1]			
Generation 2000: Best = 1.873751e+84	FES = 200100	$\mu_F = 0.748$	$\mu_{CR} = 0.193$	
Diversity = 3.1601e+00	k_t = [2, 3, 2, 4]			
Generation 2010: Best = 1.662286e+84	FES = 201100	$\mu_F = 0.747$	$\mu_{CR} = 0.182$	
Diversity = 3.3178e+00	k_t = [1, 1, 1, 1]			
Generation 2020: Best = 1.486180e+84	FES = 202100	$\mu_F = 0.750$	$\mu_{CR} = 0.189$	
Diversity = 3.2212e+00	k_t = [4, 4, 2, 3]			
Generation 2030: Best = 1.311602e+84	FES = 203100	$\mu_F = 0.744$	$\mu_{CR} = 0.187$	
Diversity = 2.9972e+00	k_t = [1, 1, 1, 1]			
Generation 2040: Best = 1.184966e+84	FES = 204100	$\mu_F = 0.754$	$\mu_{CR} = 0.195$	
Diversity = 3.0539e+00	k_t = [2, 2, 4, 4]			
Generation 2050: Best = 1.027112e+84	FES = 205100	$\mu_F = 0.754$	$\mu_{CR} = 0.197$	
Diversity = 3.1169e+00	k_t = [1, 1, 1, 1]			
Generation 2060: Best = 8.371424e+83	FES = 206100	$\mu_F = 0.735$	$\mu_{CR} = 0.204$	
Diversity = 2.9023e+00	k_t = [3, 3, 4, 3]			
Generation 2070: Best = 7.407878e+83	FES = 207100	$\mu_F = 0.745$	$\mu_{CR} = 0.203$	
Diversity = 3.0140e+00	k_t = [1, 1, 1, 1]			
Generation 2080: Best = 5.713600e+83	FES = 208100	$\mu_F = 0.741$	$\mu_{CR} = 0.208$	
Diversity = 3.1640e+00	k_t = [2, 4, 4, 2]			
Generation 2090: Best = 4.859266e+83	FES = 209100	$\mu_F = 0.739$	$\mu_{CR} = 0.207$	
Diversity = 3.2014e+00	k_t = [1, 1, 1, 1]			
Generation 2100: Best = 4.427937e+83	FES = 210100	$\mu_F = 0.744$	$\mu_{CR} = 0.199$	
Diversity = 2.9826e+00	k_t = [3, 4, 3, 3]			
Generation 2110: Best = 3.969086e+83	FES = 211100	$\mu_F = 0.750$	$\mu_{CR} = 0.201$	

Diversity = 2.9455e+00	k_t = [1, 1, 1, 1]			
Generation 2120: Best = 3.705796e+83	FES = 212100	$\mu_F = 0.744$	$\mu_{CR} = 0.197$	
Diversity = 3.0231e+00	k_t = [2, 2, 2, 2]			
Generation 2130: Best = 3.395740e+83	FES = 213100	$\mu_F = 0.736$	$\mu_{CR} = 0.196$	
Diversity = 2.9824e+00	k_t = [1, 1, 1, 1]			
Generation 2140: Best = 3.088831e+83	FES = 214100	$\mu_F = 0.724$	$\mu_{CR} = 0.204$	
Diversity = 2.7826e+00	k_t = [3, 2, 4, 2]			
Generation 2150: Best = 2.805725e+83	FES = 215100	$\mu_F = 0.729$	$\mu_{CR} = 0.196$	
Diversity = 2.8430e+00	k_t = [1, 1, 1, 1]			
Generation 2160: Best = 2.563754e+83	FES = 216100	$\mu_F = 0.743$	$\mu_{CR} = 0.194$	
Diversity = 2.7166e+00	k_t = [3, 2, 3, 4]			
Generation 2170: Best = 2.289979e+83	FES = 217100	$\mu_F = 0.742$	$\mu_{CR} = 0.207$	
Diversity = 2.7416e+00	k_t = [1, 1, 1, 1]			
Generation 2180: Best = 2.006835e+83	FES = 218100	$\mu_F = 0.755$	$\mu_{CR} = 0.210$	
Diversity = 2.8741e+00	k_t = [3, 3, 3, 3]			
Generation 2190: Best = 1.804105e+83	FES = 219100	$\mu_F = 0.748$	$\mu_{CR} = 0.205$	
Diversity = 2.8419e+00	k_t = [1, 1, 1, 1]			
Generation 2200: Best = 1.615581e+83	FES = 220100	$\mu_F = 0.746$	$\mu_{CR} = 0.201$	
Diversity = 2.9529e+00	k_t = [4, 2, 2, 3]			
Generation 2210: Best = 1.480234e+83	FES = 221100	$\mu_F = 0.751$	$\mu_{CR} = 0.194$	
Diversity = 3.0678e+00	k_t = [1, 1, 1, 1]			
Generation 2220: Best = 1.284114e+83	FES = 222100	$\mu_F = 0.753$	$\mu_{CR} = 0.194$	
Diversity = 3.1802e+00	k_t = [2, 3, 2, 2]			
Generation 2230: Best = 1.163251e+83	FES = 223100	$\mu_F = 0.750$	$\mu_{CR} = 0.208$	
Diversity = 3.2261e+00	k_t = [1, 1, 1, 1]			
Generation 2240: Best = 1.048306e+83	FES = 224100	$\mu_F = 0.748$	$\mu_{CR} = 0.210$	
Diversity = 3.1960e+00	k_t = [3, 3, 4, 2]			
Generation 2250: Best = 8.974272e+82	FES = 225100	$\mu_F = 0.751$	$\mu_{CR} = 0.207$	
Diversity = 3.2373e+00	k_t = [1, 1, 1, 1]			
Generation 2260: Best = 6.907172e+82	FES = 226100	$\mu_F = 0.747$	$\mu_{CR} = 0.207$	
Diversity = 3.2115e+00	k_t = [4, 3, 4, 2]			
Generation 2270: Best = 6.101723e+82	FES = 227100	$\mu_F = 0.729$	$\mu_{CR} = 0.208$	
Diversity = 3.2588e+00	k_t = [1, 1, 1, 1]			
Generation 2280: Best = 4.486744e+82	FES = 228100	$\mu_F = 0.739$	$\mu_{CR} = 0.209$	
Diversity = 3.0225e+00	k_t = [3, 4, 2, 4]			
Generation 2290: Best = 3.260072e+82	FES = 229100	$\mu_F = 0.754$	$\mu_{CR} = 0.213$	
Diversity = 3.1872e+00	k_t = [1, 1, 1, 1]			
Generation 2300: Best = 2.450555e+82	FES = 230100	$\mu_F = 0.748$	$\mu_{CR} = 0.218$	
Diversity = 3.0719e+00	k_t = [2, 2, 4, 4]			
Generation 2310: Best = 1.550795e+82	FES = 231100	$\mu_F = 0.757$	$\mu_{CR} = 0.214$	
Diversity = 3.1857e+00	k_t = [1, 1, 1, 1]			
Generation 2320: Best = 1.098211e+82	FES = 232100	$\mu_F = 0.760$	$\mu_{CR} = 0.195$	
Diversity = 3.2250e+00	k_t = [3, 2, 2, 3]			
Generation 2330: Best = 7.312018e+81	FES = 233100	$\mu_F = 0.757$	$\mu_{CR} = 0.199$	
Diversity = 3.3687e+00	k_t = [1, 1, 1, 1]			
Generation 2340: Best = 5.148753e+81	FES = 234100	$\mu_F = 0.751$	$\mu_{CR} = 0.194$	
Diversity = 3.0220e+00	k_t = [4, 3, 3, 4]			
Generation 2350: Best = 3.391574e+81	FES = 235100	$\mu_F = 0.741$	$\mu_{CR} = 0.188$	
Diversity = 2.8900e+00	k_t = [1, 1, 1, 1]			
Generation 2360: Best = 2.303367e+81	FES = 236100	$\mu_F = 0.736$	$\mu_{CR} = 0.187$	
Diversity = 2.8364e+00	k_t = [4, 2, 2, 3]			
Generation 2370: Best = 1.625527e+81	FES = 237100	$\mu_F = 0.729$	$\mu_{CR} = 0.185$	
Diversity = 2.8460e+00	k_t = [1, 1, 1, 1]			
Generation 2380: Best = 1.269106e+81	FES = 238100	$\mu_F = 0.728$	$\mu_{CR} = 0.183$	

Diversity = 2.8124e+00	k_t = [3, 2, 3, 3]			
Generation 2390: Best = 9.965081e+80	FES = 239100	$\mu_F = 0.737$	$\mu_{CR} = 0.184$	
Diversity = 3.0363e+00	k_t = [1, 1, 1, 1]			
Generation 2400: Best = 8.975511e+80	FES = 240100	$\mu_F = 0.731$	$\mu_{CR} = 0.185$	
Diversity = 3.0574e+00	k_t = [2, 4, 4, 2]			
Generation 2410: Best = 7.011606e+80	FES = 241100	$\mu_F = 0.740$	$\mu_{CR} = 0.192$	
Diversity = 3.1998e+00	k_t = [1, 1, 1, 1]			
Generation 2420: Best = 5.462745e+80	FES = 242100	$\mu_F = 0.734$	$\mu_{CR} = 0.186$	
Diversity = 3.0449e+00	k_t = [3, 4, 2, 4]			
Generation 2430: Best = 5.048836e+80	FES = 243100	$\mu_F = 0.744$	$\mu_{CR} = 0.184$	
Diversity = 2.9389e+00	k_t = [1, 1, 1, 1]			
Generation 2440: Best = 4.161495e+80	FES = 244100	$\mu_F = 0.747$	$\mu_{CR} = 0.179$	
Diversity = 2.8962e+00	k_t = [2, 2, 3, 3]			
Generation 2450: Best = 3.565651e+80	FES = 245100	$\mu_F = 0.739$	$\mu_{CR} = 0.176$	
Diversity = 2.9068e+00	k_t = [1, 1, 1, 1]			
Generation 2460: Best = 3.094229e+80	FES = 246100	$\mu_F = 0.734$	$\mu_{CR} = 0.174$	
Diversity = 2.8161e+00	k_t = [4, 2, 3, 4]			
Generation 2470: Best = 2.663552e+80	FES = 247100	$\mu_F = 0.736$	$\mu_{CR} = 0.177$	
Diversity = 2.7571e+00	k_t = [1, 1, 1, 1]			
Generation 2480: Best = 2.394411e+80	FES = 248100	$\mu_F = 0.738$	$\mu_{CR} = 0.178$	
Diversity = 2.6148e+00	k_t = [2, 4, 2, 3]			
Generation 2490: Best = 2.085307e+80	FES = 249100	$\mu_F = 0.760$	$\mu_{CR} = 0.176$	
Diversity = 2.6963e+00	k_t = [1, 1, 1, 1]			
Generation 2500: Best = 1.756058e+80	FES = 250100	$\mu_F = 0.761$	$\mu_{CR} = 0.178$	
Diversity = 2.7743e+00	k_t = [4, 3, 4, 4]			
Generation 2510: Best = 1.589282e+80	FES = 251100	$\mu_F = 0.766$	$\mu_{CR} = 0.186$	
Diversity = 2.8783e+00	k_t = [1, 1, 1, 1]			
Generation 2520: Best = 1.342772e+80	FES = 252100	$\mu_F = 0.760$	$\mu_{CR} = 0.184$	
Diversity = 2.7218e+00	k_t = [4, 4, 3, 3]			
Generation 2530: Best = 1.250290e+80	FES = 253100	$\mu_F = 0.746$	$\mu_{CR} = 0.183$	
Diversity = 2.7616e+00	k_t = [1, 1, 1, 1]			
Generation 2540: Best = 1.037615e+80	FES = 254100	$\mu_F = 0.740$	$\mu_{CR} = 0.175$	
Diversity = 2.6985e+00	k_t = [3, 4, 3, 4]			
Generation 2550: Best = 8.741150e+79	FES = 255100	$\mu_F = 0.734$	$\mu_{CR} = 0.186$	
Diversity = 2.7089e+00	k_t = [1, 1, 1, 1]			
Generation 2560: Best = 7.462822e+79	FES = 256100	$\mu_F = 0.742$	$\mu_{CR} = 0.189$	
Diversity = 2.9188e+00	k_t = [4, 3, 2, 4]			
Generation 2570: Best = 5.894181e+79	FES = 257100	$\mu_F = 0.749$	$\mu_{CR} = 0.189$	
Diversity = 3.0164e+00	k_t = [1, 1, 1, 1]			
Generation 2580: Best = 5.015805e+79	FES = 258100	$\mu_F = 0.741$	$\mu_{CR} = 0.187$	
Diversity = 3.0043e+00	k_t = [3, 2, 2, 3]			
Generation 2590: Best = 4.199654e+79	FES = 259100	$\mu_F = 0.734$	$\mu_{CR} = 0.176$	
Diversity = 2.8369e+00	k_t = [1, 1, 1, 1]			
Generation 2600: Best = 3.770567e+79	FES = 260100	$\mu_F = 0.740$	$\mu_{CR} = 0.187$	
Diversity = 2.9656e+00	k_t = [3, 2, 3, 2]			
Generation 2610: Best = 3.304024e+79	FES = 261100	$\mu_F = 0.724$	$\mu_{CR} = 0.188$	
Diversity = 2.8398e+00	k_t = [1, 1, 1, 1]			
Generation 2620: Best = 2.752228e+79	FES = 262100	$\mu_F = 0.725$	$\mu_{CR} = 0.182$	
Diversity = 2.8168e+00	k_t = [4, 3, 4, 3]			
Generation 2630: Best = 2.279725e+79	FES = 263100	$\mu_F = 0.730$	$\mu_{CR} = 0.191$	
Diversity = 2.7754e+00	k_t = [1, 1, 1, 1]			
Generation 2640: Best = 1.992813e+79	FES = 264100	$\mu_F = 0.728$	$\mu_{CR} = 0.194$	
Diversity = 2.5653e+00	k_t = [2, 4, 2, 4]			
Generation 2650: Best = 1.765329e+79	FES = 265100	$\mu_F = 0.737$	$\mu_{CR} = 0.195$	

Diversity = 2.5240e+00	k_t = [1, 1, 1, 1]			
Generation 2660: Best = 1.546991e+79	FES = 266100	$\mu_F = 0.744$	$\mu_{CR} = 0.197$	
Diversity = 2.7290e+00	k_t = [4, 2, 2, 2]			
Generation 2670: Best = 1.349229e+79	FES = 267100	$\mu_F = 0.741$	$\mu_{CR} = 0.196$	
Diversity = 2.8295e+00	k_t = [1, 1, 1, 1]			
Generation 2680: Best = 1.226328e+79	FES = 268100	$\mu_F = 0.743$	$\mu_{CR} = 0.197$	
Diversity = 2.8696e+00	k_t = [2, 3, 3, 2]			
Generation 2690: Best = 1.185468e+79	FES = 269100	$\mu_F = 0.729$	$\mu_{CR} = 0.196$	
Diversity = 2.9407e+00	k_t = [1, 1, 1, 1]			
Generation 2700: Best = 1.054307e+79	FES = 270100	$\mu_F = 0.733$	$\mu_{CR} = 0.194$	
Diversity = 2.6716e+00	k_t = [4, 2, 3, 3]			
Generation 2710: Best = 9.929555e+78	FES = 271100	$\mu_F = 0.729$	$\mu_{CR} = 0.204$	
Diversity = 2.7139e+00	k_t = [1, 1, 1, 1]			
Generation 2720: Best = 9.350091e+78	FES = 272100	$\mu_F = 0.740$	$\mu_{CR} = 0.206$	
Diversity = 2.7042e+00	k_t = [4, 3, 2, 4]			
Generation 2730: Best = 8.751245e+78	FES = 273100	$\mu_F = 0.756$	$\mu_{CR} = 0.203$	
Diversity = 2.8794e+00	k_t = [1, 1, 1, 1]			
Generation 2740: Best = 8.179153e+78	FES = 274100	$\mu_F = 0.759$	$\mu_{CR} = 0.201$	
Diversity = 2.9054e+00	k_t = [3, 4, 3, 2]			
Generation 2750: Best = 7.540448e+78	FES = 275100	$\mu_F = 0.753$	$\mu_{CR} = 0.198$	
Diversity = 2.8859e+00	k_t = [1, 1, 1, 1]			
Generation 2760: Best = 6.813605e+78	FES = 276100	$\mu_F = 0.750$	$\mu_{CR} = 0.201$	
Diversity = 2.8480e+00	k_t = [4, 2, 3, 2]			
Generation 2770: Best = 6.478950e+78	FES = 277100	$\mu_F = 0.751$	$\mu_{CR} = 0.197$	
Diversity = 2.9726e+00	k_t = [1, 1, 1, 1]			
Generation 2780: Best = 5.865692e+78	FES = 278100	$\mu_F = 0.751$	$\mu_{CR} = 0.186$	
Diversity = 3.0478e+00	k_t = [2, 4, 3, 3]			
Generation 2790: Best = 5.603014e+78	FES = 279100	$\mu_F = 0.747$	$\mu_{CR} = 0.188$	
Diversity = 3.0650e+00	k_t = [1, 1, 1, 1]			
Generation 2800: Best = 5.230556e+78	FES = 280100	$\mu_F = 0.745$	$\mu_{CR} = 0.194$	
Diversity = 2.8980e+00	k_t = [2, 4, 3, 3]			
Generation 2810: Best = 4.899375e+78	FES = 281100	$\mu_F = 0.738$	$\mu_{CR} = 0.198$	
Diversity = 2.9976e+00	k_t = [1, 1, 1, 1]			
Generation 2820: Best = 4.632904e+78	FES = 282100	$\mu_F = 0.735$	$\mu_{CR} = 0.191$	
Diversity = 2.7470e+00	k_t = [3, 2, 2, 2]			
Generation 2830: Best = 4.215473e+78	FES = 283100	$\mu_F = 0.745$	$\mu_{CR} = 0.189$	
Diversity = 2.9165e+00	k_t = [1, 1, 1, 1]			
Generation 2840: Best = 4.145071e+78	FES = 284100	$\mu_F = 0.725$	$\mu_{CR} = 0.179$	
Diversity = 2.9157e+00	k_t = [3, 2, 4, 2]			
Generation 2850: Best = 3.954732e+78	FES = 285100	$\mu_F = 0.715$	$\mu_{CR} = 0.178$	
Diversity = 2.8814e+00	k_t = [1, 1, 1, 1]			
Generation 2860: Best = 3.812134e+78	FES = 286100	$\mu_F = 0.724$	$\mu_{CR} = 0.176$	
Diversity = 2.6230e+00	k_t = [4, 2, 2, 3]			
Generation 2870: Best = 3.647963e+78	FES = 287100	$\mu_F = 0.727$	$\mu_{CR} = 0.175$	
Diversity = 2.6592e+00	k_t = [1, 1, 1, 1]			
Generation 2880: Best = 3.502206e+78	FES = 288100	$\mu_F = 0.723$	$\mu_{CR} = 0.183$	
Diversity = 2.6119e+00	k_t = [2, 2, 4, 4]			
Generation 2890: Best = 3.365917e+78	FES = 289100	$\mu_F = 0.723$	$\mu_{CR} = 0.188$	
Diversity = 2.7061e+00	k_t = [1, 1, 1, 1]			
Generation 2900: Best = 3.258031e+78	FES = 290100	$\mu_F = 0.729$	$\mu_{CR} = 0.194$	
Diversity = 2.5984e+00	k_t = [4, 3, 3, 4]			
Generation 2910: Best = 3.141190e+78	FES = 291100	$\mu_F = 0.735$	$\mu_{CR} = 0.199$	
Diversity = 2.5875e+00	k_t = [1, 1, 1, 1]			
Generation 2920: Best = 3.040581e+78	FES = 292100	$\mu_F = 0.741$	$\mu_{CR} = 0.202$	

Diversity = 2.5629e+00	k_t = [4, 4, 2, 4]			
Generation 2930: Best = 2.856265e+78	FES = 293100	$\mu_F = 0.749$	$\mu_{CR} = 0.209$	
Diversity = 2.6062e+00	k_t = [1, 1, 1, 1]			
Generation 2940: Best = 2.681565e+78	FES = 294100	$\mu_F = 0.758$	$\mu_{CR} = 0.215$	
Diversity = 2.7251e+00	k_t = [3, 4, 2, 4]			
Generation 2950: Best = 2.464424e+78	FES = 295100	$\mu_F = 0.758$	$\mu_{CR} = 0.209$	
Diversity = 2.7178e+00	k_t = [1, 1, 1, 1]			
Generation 2960: Best = 2.328040e+78	FES = 296100	$\mu_F = 0.759$	$\mu_{CR} = 0.205$	
Diversity = 2.8082e+00	k_t = [3, 4, 4, 2]			
Generation 2970: Best = 2.101397e+78	FES = 297100	$\mu_F = 0.742$	$\mu_{CR} = 0.202$	
Diversity = 2.8605e+00	k_t = [1, 1, 1, 1]			
Generation 2980: Best = 2.022293e+78	FES = 298100	$\mu_F = 0.738$	$\mu_{CR} = 0.192$	
Diversity = 2.5729e+00	k_t = [4, 2, 2, 4]			
Generation 2990: Best = 1.825383e+78	FES = 299100	$\mu_F = 0.750$	$\mu_{CR} = 0.196$	
Diversity = 2.6047e+00	k_t = [1, 1, 1, 1]			
Generation 3000: Best = 1.623766e+78	FES = 300100	$\mu_F = 0.750$	$\mu_{CR} = 0.202$	
Diversity = 2.5350e+00	k_t = [2, 3, 3, 4]			
Generation 3010: Best = 1.487958e+78	FES = 301100	$\mu_F = 0.762$	$\mu_{CR} = 0.203$	
Diversity = 2.8431e+00	k_t = [1, 1, 1, 1]			
Generation 3020: Best = 1.373147e+78	FES = 302100	$\mu_F = 0.751$	$\mu_{CR} = 0.202$	
Diversity = 2.8804e+00	k_t = [2, 2, 3, 4]			
Generation 3030: Best = 1.194492e+78	FES = 303100	$\mu_F = 0.754$	$\mu_{CR} = 0.201$	
Diversity = 2.8475e+00	k_t = [1, 1, 1, 1]			
Generation 3040: Best = 1.102387e+78	FES = 304100	$\mu_F = 0.747$	$\mu_{CR} = 0.187$	
Diversity = 2.7122e+00	k_t = [4, 3, 4, 3]			
Generation 3050: Best = 1.020000e+78	FES = 305100	$\mu_F = 0.749$	$\mu_{CR} = 0.176$	
Diversity = 2.7608e+00	k_t = [1, 1, 1, 1]			
Generation 3060: Best = 8.340215e+77	FES = 306100	$\mu_F = 0.748$	$\mu_{CR} = 0.176$	
Diversity = 2.8762e+00	k_t = [3, 2, 3, 4]			
Generation 3070: Best = 7.705987e+77	FES = 307100	$\mu_F = 0.743$	$\mu_{CR} = 0.177$	
Diversity = 2.9116e+00	k_t = [1, 1, 1, 1]			
Generation 3080: Best = 7.054474e+77	FES = 308100	$\mu_F = 0.743$	$\mu_{CR} = 0.181$	
Diversity = 2.8995e+00	k_t = [2, 3, 2, 4]			
Generation 3090: Best = 6.077434e+77	FES = 309100	$\mu_F = 0.748$	$\mu_{CR} = 0.188$	
Diversity = 2.9122e+00	k_t = [1, 1, 1, 1]			
Generation 3100: Best = 5.403978e+77	FES = 310100	$\mu_F = 0.747$	$\mu_{CR} = 0.191$	
Diversity = 2.9298e+00	k_t = [4, 4, 4, 4]			
Generation 3110: Best = 4.841612e+77	FES = 311100	$\mu_F = 0.735$	$\mu_{CR} = 0.188$	
Diversity = 2.9777e+00	k_t = [1, 1, 1, 1]			
Generation 3120: Best = 4.406314e+77	FES = 312100	$\mu_F = 0.744$	$\mu_{CR} = 0.188$	
Diversity = 3.0009e+00	k_t = [4, 2, 2, 2]			
Generation 3130: Best = 3.908772e+77	FES = 313100	$\mu_F = 0.748$	$\mu_{CR} = 0.190$	
Diversity = 3.1098e+00	k_t = [1, 1, 1, 1]			
Generation 3140: Best = 3.502776e+77	FES = 314100	$\mu_F = 0.744$	$\mu_{CR} = 0.197$	
Diversity = 3.1053e+00	k_t = [3, 2, 4, 2]			
Generation 3150: Best = 3.238812e+77	FES = 315100	$\mu_F = 0.740$	$\mu_{CR} = 0.201$	
Diversity = 3.2941e+00	k_t = [1, 1, 1, 1]			
Generation 3160: Best = 2.916500e+77	FES = 316100	$\mu_F = 0.731$	$\mu_{CR} = 0.200$	
Diversity = 3.1882e+00	k_t = [2, 2, 4, 2]			
Generation 3170: Best = 2.616795e+77	FES = 317100	$\mu_F = 0.729$	$\mu_{CR} = 0.195$	
Diversity = 3.0507e+00	k_t = [1, 1, 1, 1]			
Generation 3180: Best = 2.524506e+77	FES = 318100	$\mu_F = 0.727$	$\mu_{CR} = 0.193$	
Diversity = 2.8909e+00	k_t = [4, 4, 4, 3]			
Generation 3190: Best = 2.389652e+77	FES = 319100	$\mu_F = 0.736$	$\mu_{CR} = 0.191$	

Diversity = 2.8812e+00	k_t = [1, 1, 1, 1]			
Generation 3200: Best = 2.223121e+77	FES = 320100	$\mu_F = 0.744$	$\mu_{CR} = 0.194$	
Diversity = 2.6105e+00	k_t = [4, 2, 4, 4]			
Generation 3210: Best = 2.117384e+77	FES = 321100	$\mu_F = 0.748$	$\mu_{CR} = 0.201$	
Diversity = 2.6819e+00	k_t = [1, 1, 1, 1]			
Generation 3220: Best = 1.983118e+77	FES = 322100	$\mu_F = 0.756$	$\mu_{CR} = 0.195$	
Diversity = 2.6524e+00	k_t = [4, 2, 2, 4]			
Generation 3230: Best = 1.759068e+77	FES = 323100	$\mu_F = 0.767$	$\mu_{CR} = 0.199$	
Diversity = 2.7069e+00	k_t = [1, 1, 1, 1]			
Generation 3240: Best = 1.661889e+77	FES = 324100	$\mu_F = 0.756$	$\mu_{CR} = 0.211$	
Diversity = 2.7139e+00	k_t = [4, 3, 3, 3]			
Generation 3250: Best = 1.562777e+77	FES = 325100	$\mu_F = 0.757$	$\mu_{CR} = 0.208$	
Diversity = 2.8618e+00	k_t = [1, 1, 1, 1]			
Generation 3260: Best = 1.457099e+77	FES = 326100	$\mu_F = 0.740$	$\mu_{CR} = 0.209$	
Diversity = 2.8755e+00	k_t = [4, 2, 2, 2]			
Generation 3270: Best = 1.368705e+77	FES = 327100	$\mu_F = 0.742$	$\mu_{CR} = 0.206$	
Diversity = 2.8010e+00	k_t = [1, 1, 1, 1]			
Generation 3280: Best = 1.288080e+77	FES = 328100	$\mu_F = 0.748$	$\mu_{CR} = 0.212$	
Diversity = 2.8519e+00	k_t = [2, 4, 2, 2]			
Generation 3290: Best = 1.221283e+77	FES = 329100	$\mu_F = 0.742$	$\mu_{CR} = 0.213$	
Diversity = 2.8941e+00	k_t = [1, 1, 1, 1]			
Generation 3300: Best = 1.135749e+77	FES = 330100	$\mu_F = 0.739$	$\mu_{CR} = 0.210$	
Diversity = 2.7733e+00	k_t = [4, 3, 2, 4]			
Generation 3310: Best = 1.092726e+77	FES = 331100	$\mu_F = 0.752$	$\mu_{CR} = 0.207$	
Diversity = 2.7749e+00	k_t = [1, 1, 1, 1]			
Generation 3320: Best = 1.050876e+77	FES = 332100	$\mu_F = 0.757$	$\mu_{CR} = 0.207$	
Diversity = 3.0990e+00	k_t = [3, 3, 2, 2]			
Generation 3330: Best = 1.010591e+77	FES = 333100	$\mu_F = 0.738$	$\mu_{CR} = 0.202$	
Diversity = 3.0473e+00	k_t = [1, 1, 1, 1]			
Generation 3340: Best = 9.661288e+76	FES = 334100	$\mu_F = 0.735$	$\mu_{CR} = 0.190$	
Diversity = 3.0276e+00	k_t = [4, 3, 2, 2]			
Generation 3350: Best = 9.252909e+76	FES = 335100	$\mu_F = 0.730$	$\mu_{CR} = 0.186$	
Diversity = 2.8847e+00	k_t = [1, 1, 1, 1]			
Generation 3360: Best = 9.014697e+76	FES = 336100	$\mu_F = 0.732$	$\mu_{CR} = 0.189$	
Diversity = 2.6963e+00	k_t = [4, 2, 3, 3]			
Generation 3370: Best = 8.543893e+76	FES = 337100	$\mu_F = 0.733$	$\mu_{CR} = 0.188$	
Diversity = 2.6341e+00	k_t = [1, 1, 1, 1]			
Generation 3380: Best = 8.408006e+76	FES = 338100	$\mu_F = 0.731$	$\mu_{CR} = 0.187$	
Diversity = 2.5194e+00	k_t = [4, 4, 3, 4]			
Generation 3390: Best = 8.141378e+76	FES = 339100	$\mu_F = 0.731$	$\mu_{CR} = 0.187$	
Diversity = 2.4592e+00	k_t = [1, 1, 1, 1]			
Generation 3400: Best = 7.773289e+76	FES = 340100	$\mu_F = 0.744$	$\mu_{CR} = 0.196$	
Diversity = 2.4734e+00	k_t = [4, 4, 2, 3]			
Generation 3410: Best = 7.450955e+76	FES = 341100	$\mu_F = 0.755$	$\mu_{CR} = 0.194$	
Diversity = 2.6535e+00	k_t = [1, 1, 1, 1]			
Generation 3420: Best = 7.040342e+76	FES = 342100	$\mu_F = 0.757$	$\mu_{CR} = 0.200$	
Diversity = 2.5043e+00	k_t = [4, 3, 2, 3]			
Generation 3430: Best = 6.766445e+76	FES = 343100	$\mu_F = 0.763$	$\mu_{CR} = 0.188$	
Diversity = 2.8116e+00	k_t = [1, 1, 1, 1]			
Generation 3440: Best = 6.446553e+76	FES = 344100	$\mu_F = 0.757$	$\mu_{CR} = 0.184$	
Diversity = 2.8280e+00	k_t = [2, 3, 2, 4]			
Generation 3450: Best = 6.215508e+76	FES = 345100	$\mu_F = 0.753$	$\mu_{CR} = 0.189$	
Diversity = 2.8307e+00	k_t = [1, 1, 1, 1]			
Generation 3460: Best = 6.019960e+76	FES = 346100	$\mu_F = 0.743$	$\mu_{CR} = 0.195$	

Diversity = 2.6936e+00	k_t = [2, 2, 4, 3]			
Generation 3470: Best = 5.609073e+76	FES = 347100	$\mu_F = 0.737$	$\mu_{CR} = 0.203$	
Diversity = 2.7368e+00	k_t = [1, 1, 1, 1]			
Generation 3480: Best = 5.410500e+76	FES = 348100	$\mu_F = 0.732$	$\mu_{CR} = 0.206$	
Diversity = 2.7301e+00	k_t = [4, 4, 4, 4]			
Generation 3490: Best = 5.223200e+76	FES = 349100	$\mu_F = 0.731$	$\mu_{CR} = 0.220$	
Diversity = 2.7147e+00	k_t = [1, 1, 1, 1]			
Generation 3500: Best = 5.062634e+76	FES = 350100	$\mu_F = 0.732$	$\mu_{CR} = 0.218$	
Diversity = 2.6821e+00	k_t = [3, 3, 2, 2]			
Generation 3510: Best = 4.870732e+76	FES = 351100	$\mu_F = 0.728$	$\mu_{CR} = 0.218$	
Diversity = 2.7125e+00	k_t = [1, 1, 1, 1]			
Generation 3520: Best = 4.630874e+76	FES = 352100	$\mu_F = 0.727$	$\mu_{CR} = 0.219$	
Diversity = 2.5765e+00	k_t = [4, 4, 3, 4]			
Generation 3530: Best = 4.459183e+76	FES = 353100	$\mu_F = 0.737$	$\mu_{CR} = 0.223$	
Diversity = 2.6604e+00	k_t = [1, 1, 1, 1]			
Generation 3540: Best = 4.348417e+76	FES = 354100	$\mu_F = 0.732$	$\mu_{CR} = 0.217$	
Diversity = 2.7290e+00	k_t = [3, 3, 2, 2]			
Generation 3550: Best = 4.247009e+76	FES = 355100	$\mu_F = 0.730$	$\mu_{CR} = 0.212$	
Diversity = 2.7707e+00	k_t = [1, 1, 1, 1]			
Generation 3560: Best = 4.142650e+76	FES = 356100	$\mu_F = 0.732$	$\mu_{CR} = 0.202$	
Diversity = 2.6663e+00	k_t = [3, 4, 2, 3]			
Generation 3570: Best = 4.045120e+76	FES = 357100	$\mu_F = 0.733$	$\mu_{CR} = 0.203$	
Diversity = 2.8091e+00	k_t = [1, 1, 1, 1]			
Generation 3580: Best = 3.945187e+76	FES = 358100	$\mu_F = 0.733$	$\mu_{CR} = 0.197$	
Diversity = 2.7058e+00	k_t = [3, 4, 2, 3]			
Generation 3590: Best = 3.813255e+76	FES = 359100	$\mu_F = 0.739$	$\mu_{CR} = 0.200$	
Diversity = 2.7602e+00	k_t = [1, 1, 1, 1]			
Generation 3600: Best = 3.651199e+76	FES = 360100	$\mu_F = 0.753$	$\mu_{CR} = 0.203$	
Diversity = 2.6835e+00	k_t = [2, 2, 2, 4]			
Generation 3610: Best = 3.478093e+76	FES = 361100	$\mu_F = 0.744$	$\mu_{CR} = 0.215$	
Diversity = 2.7039e+00	k_t = [1, 1, 1, 1]			
Generation 3620: Best = 3.374411e+76	FES = 362100	$\mu_F = 0.756$	$\mu_{CR} = 0.219$	
Diversity = 2.6964e+00	k_t = [2, 2, 4, 3]			
Generation 3630: Best = 3.272779e+76	FES = 363100	$\mu_F = 0.753$	$\mu_{CR} = 0.220$	
Diversity = 2.8606e+00	k_t = [1, 1, 1, 1]			
Generation 3640: Best = 3.178449e+76	FES = 364100	$\mu_F = 0.754$	$\mu_{CR} = 0.219$	
Diversity = 2.7635e+00	k_t = [4, 3, 3, 3]			
Generation 3650: Best = 3.124842e+76	FES = 365100	$\mu_F = 0.756$	$\mu_{CR} = 0.209$	
Diversity = 2.9182e+00	k_t = [1, 1, 1, 1]			
Generation 3660: Best = 3.006076e+76	FES = 366100	$\mu_F = 0.757$	$\mu_{CR} = 0.202$	
Diversity = 3.0386e+00	k_t = [4, 3, 2, 2]			
Generation 3670: Best = 2.941018e+76	FES = 367100	$\mu_F = 0.743$	$\mu_{CR} = 0.198$	
Diversity = 2.9906e+00	k_t = [1, 1, 1, 1]			
Generation 3680: Best = 2.859830e+76	FES = 368100	$\mu_F = 0.746$	$\mu_{CR} = 0.196$	
Diversity = 2.9057e+00	k_t = [2, 2, 2, 4]			
Generation 3690: Best = 2.766516e+76	FES = 369100	$\mu_F = 0.739$	$\mu_{CR} = 0.192$	
Diversity = 2.9319e+00	k_t = [1, 1, 1, 1]			
Generation 3700: Best = 2.643662e+76	FES = 370100	$\mu_F = 0.740$	$\mu_{CR} = 0.199$	
Diversity = 2.8606e+00	k_t = [2, 3, 4, 3]			
Generation 3710: Best = 2.555006e+76	FES = 371100	$\mu_F = 0.726$	$\mu_{CR} = 0.203$	
Diversity = 2.8826e+00	k_t = [1, 1, 1, 1]			
Generation 3720: Best = 2.420084e+76	FES = 372100	$\mu_F = 0.725$	$\mu_{CR} = 0.208$	
Diversity = 2.6895e+00	k_t = [2, 4, 2, 2]			
Generation 3730: Best = 2.357632e+76	FES = 373100	$\mu_F = 0.720$	$\mu_{CR} = 0.206$	

Diversity = 2.8237e+00	k_t = [1, 1, 1, 1]			
Generation 3740: Best = 2.303417e+76	FES = 374100	$\mu_F = 0.709$	$\mu_{CR} = 0.198$	
Diversity = 2.6831e+00	k_t = [3, 2, 3, 3]			
Generation 3750: Best = 2.202095e+76	FES = 375100	$\mu_F = 0.719$	$\mu_{CR} = 0.206$	
Diversity = 2.8575e+00	k_t = [1, 1, 1, 1]			
Generation 3760: Best = 1.995182e+76	FES = 376100	$\mu_F = 0.730$	$\mu_{CR} = 0.201$	
Diversity = 2.8171e+00	k_t = [3, 2, 2, 3]			
Generation 3770: Best = 1.949624e+76	FES = 377100	$\mu_F = 0.743$	$\mu_{CR} = 0.191$	
Diversity = 2.9046e+00	k_t = [1, 1, 1, 1]			
Generation 3780: Best = 1.882733e+76	FES = 378100	$\mu_F = 0.732$	$\mu_{CR} = 0.182$	
Diversity = 2.8939e+00	k_t = [4, 3, 3, 2]			
Generation 3790: Best = 1.858123e+76	FES = 379100	$\mu_F = 0.728$	$\mu_{CR} = 0.186$	
Diversity = 2.9904e+00	k_t = [1, 1, 1, 1]			
Generation 3800: Best = 1.803089e+76	FES = 380100	$\mu_F = 0.722$	$\mu_{CR} = 0.187$	
Diversity = 3.0204e+00	k_t = [2, 3, 2, 2]			
Generation 3810: Best = 1.759530e+76	FES = 381100	$\mu_F = 0.723$	$\mu_{CR} = 0.197$	
Diversity = 3.1099e+00	k_t = [1, 1, 1, 1]			
Generation 3820: Best = 1.695110e+76	FES = 382100	$\mu_F = 0.722$	$\mu_{CR} = 0.200$	
Diversity = 2.7597e+00	k_t = [4, 4, 2, 3]			
Generation 3830: Best = 1.663109e+76	FES = 383100	$\mu_F = 0.743$	$\mu_{CR} = 0.204$	
Diversity = 2.8647e+00	k_t = [1, 1, 1, 1]			
Generation 3840: Best = 1.627828e+76	FES = 384100	$\mu_F = 0.744$	$\mu_{CR} = 0.190$	
Diversity = 2.7865e+00	k_t = [2, 2, 4, 3]			
Generation 3850: Best = 1.589094e+76	FES = 385100	$\mu_F = 0.733$	$\mu_{CR} = 0.184$	
Diversity = 2.7612e+00	k_t = [1, 1, 1, 1]			
Generation 3860: Best = 1.530273e+76	FES = 386100	$\mu_F = 0.729$	$\mu_{CR} = 0.175$	
Diversity = 2.7344e+00	k_t = [4, 3, 3, 4]			
Generation 3870: Best = 1.481207e+76	FES = 387100	$\mu_F = 0.731$	$\mu_{CR} = 0.178$	
Diversity = 2.4775e+00	k_t = [1, 1, 1, 1]			
Generation 3880: Best = 1.413236e+76	FES = 388100	$\mu_F = 0.746$	$\mu_{CR} = 0.186$	
Diversity = 2.5386e+00	k_t = [3, 3, 3, 3]			
Generation 3890: Best = 1.357905e+76	FES = 389100	$\mu_F = 0.757$	$\mu_{CR} = 0.193$	
Diversity = 2.7897e+00	k_t = [1, 1, 1, 1]			
Generation 3900: Best = 1.290715e+76	FES = 390100	$\mu_F = 0.759$	$\mu_{CR} = 0.198$	
Diversity = 2.7391e+00	k_t = [2, 4, 3, 4]			
Generation 3910: Best = 1.241263e+76	FES = 391100	$\mu_F = 0.753$	$\mu_{CR} = 0.208$	
Diversity = 2.7432e+00	k_t = [1, 1, 1, 1]			
Generation 3920: Best = 1.186783e+76	FES = 392100	$\mu_F = 0.757$	$\mu_{CR} = 0.213$	
Diversity = 2.8041e+00	k_t = [4, 2, 2, 4]			
Generation 3930: Best = 1.164786e+76	FES = 393100	$\mu_F = 0.750$	$\mu_{CR} = 0.204$	
Diversity = 2.7683e+00	k_t = [1, 1, 1, 1]			
Generation 3940: Best = 1.125740e+76	FES = 394100	$\mu_F = 0.748$	$\mu_{CR} = 0.198$	
Diversity = 2.7896e+00	k_t = [4, 2, 2, 2]			
Generation 3950: Best = 1.079746e+76	FES = 395100	$\mu_F = 0.734$	$\mu_{CR} = 0.189$	
Diversity = 2.9082e+00	k_t = [1, 1, 1, 1]			
Generation 3960: Best = 1.055908e+76	FES = 396100	$\mu_F = 0.729$	$\mu_{CR} = 0.193$	
Diversity = 2.7600e+00	k_t = [3, 4, 2, 2]			
Generation 3970: Best = 1.036941e+76	FES = 397100	$\mu_F = 0.733$	$\mu_{CR} = 0.197$	
Diversity = 2.8044e+00	k_t = [1, 1, 1, 1]			
Generation 3980: Best = 1.006708e+76	FES = 398100	$\mu_F = 0.736$	$\mu_{CR} = 0.207$	
Diversity = 2.7837e+00	k_t = [3, 4, 3, 3]			
Generation 3990: Best = 9.742120e+75	FES = 399100	$\mu_F = 0.739$	$\mu_{CR} = 0.205$	
Diversity = 2.8024e+00	k_t = [1, 1, 1, 1]			
Generation 4000: Best = 9.540370e+75	FES = 400100	$\mu_F = 0.741$	$\mu_{CR} = 0.204$	

Diversity = 2.7152e+00	k_t = [2, 3, 2, 4]			
Generation 4010: Best = 9.273364e+75	FES = 401100	$\mu_F = 0.748$	$\mu_{CR} = 0.195$	
Diversity = 2.7945e+00	k_t = [1, 1, 1, 1]			
Generation 4020: Best = 9.073339e+75	FES = 402100	$\mu_F = 0.750$	$\mu_{CR} = 0.204$	
Diversity = 2.6965e+00	k_t = [4, 2, 3, 4]			
Generation 4030: Best = 8.741579e+75	FES = 403100	$\mu_F = 0.760$	$\mu_{CR} = 0.205$	
Diversity = 2.8622e+00	k_t = [1, 1, 1, 1]			
Generation 4040: Best = 8.470217e+75	FES = 404100	$\mu_F = 0.767$	$\mu_{CR} = 0.201$	
Diversity = 2.9686e+00	k_t = [4, 4, 3, 4]			
Generation 4050: Best = 8.264971e+75	FES = 405100	$\mu_F = 0.753$	$\mu_{CR} = 0.198$	
Diversity = 2.9373e+00	k_t = [1, 1, 1, 1]			
Generation 4060: Best = 7.945051e+75	FES = 406100	$\mu_F = 0.748$	$\mu_{CR} = 0.185$	
Diversity = 2.9716e+00	k_t = [3, 4, 2, 3]			
Generation 4070: Best = 7.724949e+75	FES = 407100	$\mu_F = 0.747$	$\mu_{CR} = 0.181$	
Diversity = 3.0627e+00	k_t = [1, 1, 1, 1]			
Generation 4080: Best = 7.516456e+75	FES = 408100	$\mu_F = 0.748$	$\mu_{CR} = 0.177$	
Diversity = 2.9389e+00	k_t = [3, 4, 2, 3]			
Generation 4090: Best = 7.343334e+75	FES = 409100	$\mu_F = 0.750$	$\mu_{CR} = 0.171$	
Diversity = 3.0805e+00	k_t = [1, 1, 1, 1]			
Generation 4100: Best = 7.130165e+75	FES = 410100	$\mu_F = 0.738$	$\mu_{CR} = 0.180$	
Diversity = 2.9042e+00	k_t = [4, 4, 4, 2]			
Generation 4110: Best = 6.765880e+75	FES = 411100	$\mu_F = 0.738$	$\mu_{CR} = 0.186$	
Diversity = 3.0337e+00	k_t = [1, 1, 1, 1]			
Generation 4120: Best = 6.577355e+75	FES = 412100	$\mu_F = 0.744$	$\mu_{CR} = 0.184$	
Diversity = 3.0742e+00	k_t = [4, 3, 4, 3]			
Generation 4130: Best = 6.362301e+75	FES = 413100	$\mu_F = 0.736$	$\mu_{CR} = 0.181$	
Diversity = 3.0105e+00	k_t = [1, 1, 1, 1]			
Generation 4140: Best = 6.202010e+75	FES = 414100	$\mu_F = 0.729$	$\mu_{CR} = 0.179$	
Diversity = 2.8877e+00	k_t = [2, 4, 3, 2]			
Generation 4150: Best = 6.049500e+75	FES = 415100	$\mu_F = 0.732$	$\mu_{CR} = 0.199$	
Diversity = 2.8631e+00	k_t = [1, 1, 1, 1]			
Generation 4160: Best = 5.847916e+75	FES = 416100	$\mu_F = 0.733$	$\mu_{CR} = 0.198$	
Diversity = 2.7718e+00	k_t = [2, 2, 3, 2]			
Generation 4170: Best = 5.617421e+75	FES = 417100	$\mu_F = 0.730$	$\mu_{CR} = 0.195$	
Diversity = 2.8207e+00	k_t = [1, 1, 1, 1]			
Generation 4180: Best = 5.460457e+75	FES = 418100	$\mu_F = 0.728$	$\mu_{CR} = 0.198$	
Diversity = 2.6885e+00	k_t = [2, 3, 2, 3]			
Generation 4190: Best = 5.261616e+75	FES = 419100	$\mu_F = 0.721$	$\mu_{CR} = 0.202$	
Diversity = 2.8695e+00	k_t = [1, 1, 1, 1]			
Generation 4200: Best = 4.949425e+75	FES = 420100	$\mu_F = 0.733$	$\mu_{CR} = 0.196$	
Diversity = 2.7239e+00	k_t = [3, 4, 3, 3]			
Generation 4210: Best = 4.804521e+75	FES = 421100	$\mu_F = 0.736$	$\mu_{CR} = 0.208$	
Diversity = 2.8754e+00	k_t = [1, 1, 1, 1]			
Generation 4220: Best = 4.609843e+75	FES = 422100	$\mu_F = 0.753$	$\mu_{CR} = 0.211$	
Diversity = 2.6781e+00	k_t = [2, 4, 3, 4]			
Generation 4230: Best = 4.318048e+75	FES = 423100	$\mu_F = 0.761$	$\mu_{CR} = 0.213$	
Diversity = 2.7357e+00	k_t = [1, 1, 1, 1]			
Generation 4240: Best = 4.158861e+75	FES = 424100	$\mu_F = 0.767$	$\mu_{CR} = 0.209$	
Diversity = 2.7680e+00	k_t = [2, 2, 3, 4]			
Generation 4250: Best = 3.947688e+75	FES = 425100	$\mu_F = 0.761$	$\mu_{CR} = 0.207$	
Diversity = 2.8035e+00	k_t = [1, 1, 1, 1]			
Generation 4260: Best = 3.790551e+75	FES = 426100	$\mu_F = 0.765$	$\mu_{CR} = 0.202$	
Diversity = 2.9039e+00	k_t = [4, 3, 3, 4]			
Generation 4270: Best = 3.629665e+75	FES = 427100	$\mu_F = 0.772$	$\mu_{CR} = 0.197$	

Diversity = 3.1597e+00	k_t = [1, 1, 1, 1]			
Generation 4280: Best = 3.455403e+75	FES = 428100	$\mu_F = 0.770$	$\mu_{CR} = 0.189$	
Diversity = 2.8640e+00	k_t = [4, 2, 3, 4]			
Generation 4290: Best = 3.261560e+75	FES = 429100	$\mu_F = 0.779$	$\mu_{CR} = 0.190$	
Diversity = 3.0309e+00	k_t = [1, 1, 1, 1]			
Generation 4300: Best = 3.091489e+75	FES = 430100	$\mu_F = 0.774$	$\mu_{CR} = 0.184$	
Diversity = 2.8498e+00	k_t = [3, 4, 3, 4]			
Generation 4310: Best = 2.873098e+75	FES = 431100	$\mu_F = 0.761$	$\mu_{CR} = 0.187$	
Diversity = 2.9550e+00	k_t = [1, 1, 1, 1]			
Generation 4320: Best = 2.617704e+75	FES = 432100	$\mu_F = 0.750$	$\mu_{CR} = 0.188$	
Diversity = 2.7834e+00	k_t = [3, 2, 3, 3]			
Generation 4330: Best = 2.431367e+75	FES = 433100	$\mu_F = 0.744$	$\mu_{CR} = 0.189$	
Diversity = 2.8567e+00	k_t = [1, 1, 1, 1]			
Generation 4340: Best = 2.330208e+75	FES = 434100	$\mu_F = 0.744$	$\mu_{CR} = 0.186$	
Diversity = 2.8443e+00	k_t = [3, 2, 3, 4]			
Generation 4350: Best = 2.253432e+75	FES = 435100	$\mu_F = 0.754$	$\mu_{CR} = 0.178$	
Diversity = 2.8914e+00	k_t = [1, 1, 1, 1]			
Generation 4360: Best = 2.165156e+75	FES = 436100	$\mu_F = 0.745$	$\mu_{CR} = 0.174$	
Diversity = 2.9435e+00	k_t = [3, 3, 2, 3]			
Generation 4370: Best = 2.078895e+75	FES = 437100	$\mu_F = 0.745$	$\mu_{CR} = 0.180$	
Diversity = 2.8965e+00	k_t = [1, 1, 1, 1]			
Generation 4380: Best = 1.976877e+75	FES = 438100	$\mu_F = 0.753$	$\mu_{CR} = 0.184$	
Diversity = 2.8005e+00	k_t = [4, 2, 4, 4]			
Generation 4390: Best = 1.901148e+75	FES = 439100	$\mu_F = 0.773$	$\mu_{CR} = 0.186$	
Diversity = 2.8005e+00	k_t = [1, 1, 1, 1]			
Generation 4400: Best = 1.826697e+75	FES = 440100	$\mu_F = 0.776$	$\mu_{CR} = 0.186$	
Diversity = 2.8940e+00	k_t = [3, 4, 2, 4]			
Generation 4410: Best = 1.751966e+75	FES = 441100	$\mu_F = 0.767$	$\mu_{CR} = 0.186$	
Diversity = 2.9719e+00	k_t = [1, 1, 1, 1]			
Generation 4420: Best = 1.685330e+75	FES = 442100	$\mu_F = 0.758$	$\mu_{CR} = 0.197$	
Diversity = 3.0628e+00	k_t = [2, 3, 2, 3]			
Generation 4430: Best = 1.610854e+75	FES = 443100	$\mu_F = 0.757$	$\mu_{CR} = 0.190$	
Diversity = 3.0475e+00	k_t = [1, 1, 1, 1]			
Generation 4440: Best = 1.501678e+75	FES = 444100	$\mu_F = 0.754$	$\mu_{CR} = 0.194$	
Diversity = 3.0684e+00	k_t = [3, 2, 3, 2]			
Generation 4450: Best = 1.447168e+75	FES = 445100	$\mu_F = 0.740$	$\mu_{CR} = 0.200$	
Diversity = 3.1987e+00	k_t = [1, 1, 1, 1]			
Generation 4460: Best = 1.353559e+75	FES = 446100	$\mu_F = 0.728$	$\mu_{CR} = 0.199$	
Diversity = 3.1174e+00	k_t = [4, 3, 2, 3]			
Generation 4470: Best = 1.299505e+75	FES = 447100	$\mu_F = 0.731$	$\mu_{CR} = 0.199$	
Diversity = 2.9640e+00	k_t = [1, 1, 1, 1]			
Generation 4480: Best = 1.202880e+75	FES = 448100	$\mu_F = 0.718$	$\mu_{CR} = 0.190$	
Diversity = 2.7949e+00	k_t = [2, 3, 4, 2]			
Generation 4490: Best = 1.136735e+75	FES = 449100	$\mu_F = 0.714$	$\mu_{CR} = 0.192$	
Diversity = 2.7360e+00	k_t = [1, 1, 1, 1]			
Generation 4500: Best = 1.072363e+75	FES = 450100	$\mu_F = 0.720$	$\mu_{CR} = 0.185$	
Diversity = 2.7361e+00	k_t = [4, 3, 2, 3]			
Generation 4510: Best = 9.965690e+74	FES = 451100	$\mu_F = 0.721$	$\mu_{CR} = 0.196$	
Diversity = 2.6639e+00	k_t = [1, 1, 1, 1]			
Generation 4520: Best = 9.116651e+74	FES = 452100	$\mu_F = 0.723$	$\mu_{CR} = 0.199$	
Diversity = 2.6096e+00	k_t = [2, 4, 4, 4]			
Generation 4530: Best = 8.639689e+74	FES = 453100	$\mu_F = 0.738$	$\mu_{CR} = 0.197$	
Diversity = 2.5267e+00	k_t = [1, 1, 1, 1]			
Generation 4540: Best = 8.191519e+74	FES = 454100	$\mu_F = 0.742$	$\mu_{CR} = 0.196$	

Diversity = 2.5844e+00	k_t = [4, 3, 2, 2]			
Generation 4550: Best = 7.568677e+74	FES = 455100	$\mu_F = 0.738$	$\mu_{CR} = 0.200$	
Diversity = 2.7915e+00	k_t = [1, 1, 1, 1]			
Generation 4560: Best = 7.209491e+74	FES = 456100	$\mu_F = 0.728$	$\mu_{CR} = 0.198$	
Diversity = 2.7984e+00	k_t = [3, 2, 3, 2]			
Generation 4570: Best = 6.698030e+74	FES = 457100	$\mu_F = 0.724$	$\mu_{CR} = 0.206$	
Diversity = 2.9461e+00	k_t = [1, 1, 1, 1]			
Generation 4580: Best = 6.457399e+74	FES = 458100	$\mu_F = 0.719$	$\mu_{CR} = 0.201$	
Diversity = 2.7379e+00	k_t = [4, 3, 2, 4]			
Generation 4590: Best = 5.873409e+74	FES = 459100	$\mu_F = 0.725$	$\mu_{CR} = 0.197$	
Diversity = 2.7101e+00	k_t = [1, 1, 1, 1]			
Generation 4600: Best = 5.465339e+74	FES = 460100	$\mu_F = 0.717$	$\mu_{CR} = 0.208$	
Diversity = 2.5462e+00	k_t = [2, 4, 3, 3]			
Generation 4610: Best = 5.132747e+74	FES = 461100	$\mu_F = 0.732$	$\mu_{CR} = 0.212$	
Diversity = 2.7589e+00	k_t = [1, 1, 1, 1]			
Generation 4620: Best = 4.691434e+74	FES = 462100	$\mu_F = 0.730$	$\mu_{CR} = 0.216$	
Diversity = 2.6167e+00	k_t = [4, 4, 4, 3]			
Generation 4630: Best = 4.166719e+74	FES = 463100	$\mu_F = 0.733$	$\mu_{CR} = 0.222$	
Diversity = 2.5210e+00	k_t = [1, 1, 1, 1]			
Generation 4640: Best = 3.833178e+74	FES = 464100	$\mu_F = 0.742$	$\mu_{CR} = 0.219$	
Diversity = 2.6037e+00	k_t = [4, 4, 3, 4]			
Generation 4650: Best = 3.500162e+74	FES = 465100	$\mu_F = 0.746$	$\mu_{CR} = 0.224$	
Diversity = 2.4978e+00	k_t = [1, 1, 1, 1]			
Generation 4660: Best = 3.034168e+74	FES = 466100	$\mu_F = 0.740$	$\mu_{CR} = 0.229$	
Diversity = 2.4760e+00	k_t = [3, 4, 2, 3]			
Generation 4670: Best = 2.748376e+74	FES = 467100	$\mu_F = 0.734$	$\mu_{CR} = 0.237$	
Diversity = 2.7143e+00	k_t = [1, 1, 1, 1]			
Generation 4680: Best = 2.518117e+74	FES = 468100	$\mu_F = 0.741$	$\mu_{CR} = 0.226$	
Diversity = 2.6976e+00	k_t = [3, 3, 2, 4]			
Generation 4690: Best = 2.405798e+74	FES = 469100	$\mu_F = 0.730$	$\mu_{CR} = 0.226$	
Diversity = 2.6447e+00	k_t = [1, 1, 1, 1]			
Generation 4700: Best = 2.274510e+74	FES = 470100	$\mu_F = 0.731$	$\mu_{CR} = 0.219$	
Diversity = 2.5607e+00	k_t = [4, 3, 4, 2]			
Generation 4710: Best = 2.170020e+74	FES = 471100	$\mu_F = 0.729$	$\mu_{CR} = 0.216$	
Diversity = 2.5481e+00	k_t = [1, 1, 1, 1]			
Generation 4720: Best = 2.056325e+74	FES = 472100	$\mu_F = 0.731$	$\mu_{CR} = 0.222$	
Diversity = 2.4622e+00	k_t = [3, 3, 2, 3]			
Generation 4730: Best = 1.974104e+74	FES = 473100	$\mu_F = 0.740$	$\mu_{CR} = 0.215$	
Diversity = 2.6522e+00	k_t = [1, 1, 1, 1]			
Generation 4740: Best = 1.885664e+74	FES = 474100	$\mu_F = 0.742$	$\mu_{CR} = 0.223$	
Diversity = 2.7641e+00	k_t = [3, 3, 3, 2]			
Generation 4750: Best = 1.808589e+74	FES = 475100	$\mu_F = 0.737$	$\mu_{CR} = 0.213$	
Diversity = 2.7680e+00	k_t = [1, 1, 1, 1]			
Generation 4760: Best = 1.739736e+74	FES = 476100	$\mu_F = 0.722$	$\mu_{CR} = 0.211$	
Diversity = 2.7878e+00	k_t = [2, 4, 3, 3]			
Generation 4770: Best = 1.663591e+74	FES = 477100	$\mu_F = 0.725$	$\mu_{CR} = 0.221$	
Diversity = 2.8325e+00	k_t = [1, 1, 1, 1]			
Generation 4780: Best = 1.595440e+74	FES = 478100	$\mu_F = 0.724$	$\mu_{CR} = 0.226$	
Diversity = 2.5947e+00	k_t = [4, 4, 3, 3]			
Generation 4790: Best = 1.464248e+74	FES = 479100	$\mu_F = 0.728$	$\mu_{CR} = 0.226$	
Diversity = 2.6369e+00	k_t = [1, 1, 1, 1]			
Generation 4800: Best = 1.401174e+74	FES = 480100	$\mu_F = 0.727$	$\mu_{CR} = 0.218$	
Diversity = 2.7024e+00	k_t = [2, 2, 2, 2]			
Generation 4810: Best = 1.311913e+74	FES = 481100	$\mu_F = 0.717$	$\mu_{CR} = 0.216$	

Diversity = 2.7027e+00	k_t = [1, 1, 1, 1]			
Generation 4820: Best = 1.240192e+74	FES = 482100	$\mu_F = 0.740$	$\mu_{CR} = 0.200$	
Diversity = 2.6322e+00	k_t = [4, 2, 4, 4]			
Generation 4830: Best = 1.184436e+74	FES = 483100	$\mu_F = 0.735$	$\mu_{CR} = 0.204$	
Diversity = 2.7825e+00	k_t = [1, 1, 1, 1]			
Generation 4840: Best = 1.110992e+74	FES = 484100	$\mu_F = 0.733$	$\mu_{CR} = 0.204$	
Diversity = 2.7388e+00	k_t = [3, 3, 2, 3]			
Generation 4850: Best = 1.054604e+74	FES = 485100	$\mu_F = 0.732$	$\mu_{CR} = 0.208$	
Diversity = 2.7707e+00	k_t = [1, 1, 1, 1]			
Generation 4860: Best = 9.831065e+73	FES = 486100	$\mu_F = 0.732$	$\mu_{CR} = 0.213$	
Diversity = 2.6635e+00	k_t = [3, 4, 3, 4]			
Generation 4870: Best = 9.401976e+73	FES = 487100	$\mu_F = 0.747$	$\mu_{CR} = 0.223$	
Diversity = 2.6677e+00	k_t = [1, 1, 1, 1]			
Generation 4880: Best = 8.988208e+73	FES = 488100	$\mu_F = 0.737$	$\mu_{CR} = 0.226$	
Diversity = 2.5469e+00	k_t = [2, 4, 2, 4]			
Generation 4890: Best = 8.158750e+73	FES = 489100	$\mu_F = 0.752$	$\mu_{CR} = 0.242$	
Diversity = 2.6574e+00	k_t = [1, 1, 1, 1]			
Generation 4900: Best = 7.927107e+73	FES = 490100	$\mu_F = 0.746$	$\mu_{CR} = 0.243$	
Diversity = 2.7339e+00	k_t = [2, 2, 3, 2]			
Generation 4910: Best = 7.490191e+73	FES = 491100	$\mu_F = 0.737$	$\mu_{CR} = 0.232$	
Diversity = 2.7280e+00	k_t = [1, 1, 1, 1]			
Generation 4920: Best = 7.016481e+73	FES = 492100	$\mu_F = 0.741$	$\mu_{CR} = 0.240$	
Diversity = 2.7203e+00	k_t = [3, 4, 2, 2]			
Generation 4930: Best = 6.634716e+73	FES = 493100	$\mu_F = 0.740$	$\mu_{CR} = 0.235$	
Diversity = 2.8294e+00	k_t = [1, 1, 1, 1]			
Generation 4940: Best = 6.311965e+73	FES = 494100	$\mu_F = 0.725$	$\mu_{CR} = 0.234$	
Diversity = 2.8472e+00	k_t = [3, 2, 3, 2]			
Generation 4950: Best = 6.104761e+73	FES = 495100	$\mu_F = 0.724$	$\mu_{CR} = 0.218$	
Diversity = 2.7930e+00	k_t = [1, 1, 1, 1]			
Generation 4960: Best = 5.776066e+73	FES = 496100	$\mu_F = 0.718$	$\mu_{CR} = 0.214$	
Diversity = 2.6307e+00	k_t = [2, 4, 3, 4]			
Generation 4970: Best = 5.501217e+73	FES = 497100	$\mu_F = 0.722$	$\mu_{CR} = 0.219$	
Diversity = 2.5253e+00	k_t = [1, 1, 1, 1]			
Generation 4980: Best = 5.169859e+73	FES = 498100	$\mu_F = 0.726$	$\mu_{CR} = 0.235$	
Diversity = 2.3282e+00	k_t = [3, 3, 4, 3]			
Generation 4990: Best = 4.709794e+73	FES = 499100	$\mu_F = 0.727$	$\mu_{CR} = 0.235$	
Diversity = 2.4313e+00	k_t = [1, 1, 1, 1]			
Generation 5000: Best = 4.309089e+73	FES = 500100	$\mu_F = 0.734$	$\mu_{CR} = 0.241$	
Diversity = 2.2901e+00	k_t = [4, 4, 2, 4]			
Generation 5010: Best = 3.859212e+73	FES = 501100	$\mu_F = 0.754$	$\mu_{CR} = 0.237$	
Diversity = 2.3787e+00	k_t = [1, 1, 1, 1]			
Generation 5020: Best = 3.453861e+73	FES = 502100	$\mu_F = 0.747$	$\mu_{CR} = 0.243$	
Diversity = 2.5434e+00	k_t = [3, 2, 3, 2]			
Generation 5030: Best = 2.965423e+73	FES = 503100	$\mu_F = 0.733$	$\mu_{CR} = 0.243$	
Diversity = 2.5593e+00	k_t = [1, 1, 1, 1]			
Generation 5040: Best = 2.636414e+73	FES = 504100	$\mu_F = 0.731$	$\mu_{CR} = 0.246$	
Diversity = 2.4451e+00	k_t = [2, 2, 2, 4]			
Generation 5050: Best = 2.258645e+73	FES = 505100	$\mu_F = 0.733$	$\mu_{CR} = 0.240$	
Diversity = 2.5301e+00	k_t = [1, 1, 1, 1]			
Generation 5060: Best = 1.942300e+73	FES = 506100	$\mu_F = 0.747$	$\mu_{CR} = 0.239$	
Diversity = 2.4866e+00	k_t = [4, 2, 4, 4]			
Generation 5070: Best = 1.783139e+73	FES = 507100	$\mu_F = 0.749$	$\mu_{CR} = 0.235$	
Diversity = 2.4820e+00	k_t = [1, 1, 1, 1]			
Generation 5080: Best = 1.623305e+73	FES = 508100	$\mu_F = 0.740$	$\mu_{CR} = 0.236$	

Diversity = 2.3346e+00	k_t = [4, 3, 2, 3]			
Generation 5090: Best = 1.472926e+73	FES = 509100	$\mu_F = 0.733$	$\mu_{CR} = 0.236$	
Diversity = 2.3883e+00	k_t = [1, 1, 1, 1]			
Generation 5100: Best = 1.340012e+73	FES = 510100	$\mu_F = 0.727$	$\mu_{CR} = 0.236$	
Diversity = 2.3504e+00	k_t = [3, 4, 2, 3]			
Generation 5110: Best = 1.262056e+73	FES = 511100	$\mu_F = 0.729$	$\mu_{CR} = 0.239$	
Diversity = 2.4202e+00	k_t = [1, 1, 1, 1]			
Generation 5120: Best = 1.172246e+73	FES = 512100	$\mu_F = 0.727$	$\mu_{CR} = 0.228$	
Diversity = 2.4019e+00	k_t = [4, 3, 2, 2]			
Generation 5130: Best = 1.052220e+73	FES = 513100	$\mu_F = 0.732$	$\mu_{CR} = 0.225$	
Diversity = 2.4735e+00	k_t = [1, 1, 1, 1]			
Generation 5140: Best = 9.384512e+72	FES = 514100	$\mu_F = 0.747$	$\mu_{CR} = 0.226$	
Diversity = 2.5058e+00	k_t = [3, 3, 3, 2]			
Generation 5150: Best = 8.681996e+72	FES = 515100	$\mu_F = 0.750$	$\mu_{CR} = 0.223$	
Diversity = 2.5240e+00	k_t = [1, 1, 1, 1]			
Generation 5160: Best = 7.774780e+72	FES = 516100	$\mu_F = 0.744$	$\mu_{CR} = 0.227$	
Diversity = 2.4307e+00	k_t = [3, 2, 4, 3]			
Generation 5170: Best = 7.209661e+72	FES = 517100	$\mu_F = 0.739$	$\mu_{CR} = 0.223$	
Diversity = 2.4426e+00	k_t = [1, 1, 1, 1]			
Generation 5180: Best = 6.395526e+72	FES = 518100	$\mu_F = 0.734$	$\mu_{CR} = 0.230$	
Diversity = 2.4011e+00	k_t = [2, 3, 3, 4]			
Generation 5190: Best = 5.776557e+72	FES = 519100	$\mu_F = 0.735$	$\mu_{CR} = 0.233$	
Diversity = 2.4281e+00	k_t = [1, 1, 1, 1]			
Generation 5200: Best = 5.320174e+72	FES = 520100	$\mu_F = 0.729$	$\mu_{CR} = 0.223$	
Diversity = 2.4055e+00	k_t = [2, 2, 4, 4]			
Generation 5210: Best = 4.713365e+72	FES = 521100	$\mu_F = 0.746$	$\mu_{CR} = 0.215$	
Diversity = 2.5224e+00	k_t = [1, 1, 1, 1]			
Generation 5220: Best = 4.382590e+72	FES = 522100	$\mu_F = 0.741$	$\mu_{CR} = 0.217$	
Diversity = 2.4397e+00	k_t = [2, 3, 2, 3]			
Generation 5230: Best = 3.952347e+72	FES = 523100	$\mu_F = 0.722$	$\mu_{CR} = 0.223$	
Diversity = 2.4260e+00	k_t = [1, 1, 1, 1]			
Generation 5240: Best = 3.534011e+72	FES = 524100	$\mu_F = 0.727$	$\mu_{CR} = 0.216$	
Diversity = 2.3593e+00	k_t = [3, 3, 2, 4]			
Generation 5250: Best = 3.084959e+72	FES = 525100	$\mu_F = 0.745$	$\mu_{CR} = 0.210$	
Diversity = 2.4945e+00	k_t = [1, 1, 1, 1]			
Generation 5260: Best = 2.612326e+72	FES = 526100	$\mu_F = 0.746$	$\mu_{CR} = 0.217$	
Diversity = 2.4190e+00	k_t = [3, 4, 4, 4]			
Generation 5270: Best = 2.158450e+72	FES = 527100	$\mu_F = 0.739$	$\mu_{CR} = 0.218$	
Diversity = 2.3733e+00	k_t = [1, 1, 1, 1]			
Generation 5280: Best = 1.905647e+72	FES = 528100	$\mu_F = 0.736$	$\mu_{CR} = 0.214$	
Diversity = 2.3769e+00	k_t = [3, 2, 3, 3]			
Generation 5290: Best = 1.643373e+72	FES = 529100	$\mu_F = 0.735$	$\mu_{CR} = 0.212$	
Diversity = 2.3752e+00	k_t = [1, 1, 1, 1]			
Generation 5300: Best = 1.342649e+72	FES = 530100	$\mu_F = 0.743$	$\mu_{CR} = 0.209$	
Diversity = 2.4074e+00	k_t = [3, 3, 4, 2]			
Generation 5310: Best = 1.162756e+72	FES = 531100	$\mu_F = 0.747$	$\mu_{CR} = 0.204$	
Diversity = 2.5445e+00	k_t = [1, 1, 1, 1]			
Generation 5320: Best = 9.574077e+71	FES = 532100	$\mu_F = 0.740$	$\mu_{CR} = 0.205$	
Diversity = 2.4708e+00	k_t = [4, 2, 4, 2]			
Generation 5330: Best = 7.452571e+71	FES = 533100	$\mu_F = 0.757$	$\mu_{CR} = 0.212$	
Diversity = 2.6045e+00	k_t = [1, 1, 1, 1]			
Generation 5340: Best = 5.586187e+71	FES = 534100	$\mu_F = 0.741$	$\mu_{CR} = 0.208$	
Diversity = 2.4430e+00	k_t = [4, 4, 4, 4]			
Generation 5350: Best = 3.584358e+71	FES = 535100	$\mu_F = 0.736$	$\mu_{CR} = 0.209$	

Diversity = 2.4061e+00	k_t = [1, 1, 1, 1]			
Generation 5360: Best = 2.481741e+71	FES = 536100	$\mu_F = 0.734$	$\mu_{CR} = 0.202$	
Diversity = 2.4321e+00	k_t = [4, 3, 2, 3]			
Generation 5370: Best = 1.675139e+71	FES = 537100	$\mu_F = 0.738$	$\mu_{CR} = 0.193$	
Diversity = 2.3165e+00	k_t = [1, 1, 1, 1]			
Generation 5380: Best = 1.297015e+71	FES = 538100	$\mu_F = 0.742$	$\mu_{CR} = 0.199$	
Diversity = 2.2459e+00	k_t = [4, 2, 2, 4]			
Generation 5390: Best = 6.664223e+70	FES = 539100	$\mu_F = 0.739$	$\mu_{CR} = 0.204$	
Diversity = 2.3091e+00	k_t = [1, 1, 1, 1]			
Generation 5400: Best = 4.614060e+70	FES = 540100	$\mu_F = 0.736$	$\mu_{CR} = 0.210$	
Diversity = 2.2568e+00	k_t = [3, 2, 2, 3]			
Generation 5410: Best = 3.257679e+70	FES = 541100	$\mu_F = 0.741$	$\mu_{CR} = 0.219$	
Diversity = 2.3581e+00	k_t = [1, 1, 1, 1]			
Generation 5420: Best = 2.511844e+70	FES = 542100	$\mu_F = 0.730$	$\mu_{CR} = 0.213$	
Diversity = 2.2257e+00	k_t = [2, 4, 2, 3]			
Generation 5430: Best = 1.832348e+70	FES = 543100	$\mu_F = 0.725$	$\mu_{CR} = 0.214$	
Diversity = 2.0963e+00	k_t = [1, 1, 1, 1]			
Generation 5440: Best = 1.382812e+70	FES = 544100	$\mu_F = 0.729$	$\mu_{CR} = 0.202$	
Diversity = 2.0226e+00	k_t = [3, 4, 4, 4]			
Generation 5450: Best = 8.591792e+69	FES = 545100	$\mu_F = 0.742$	$\mu_{CR} = 0.193$	
Diversity = 2.0852e+00	k_t = [1, 1, 1, 1]			
Generation 5460: Best = 5.812945e+69	FES = 546100	$\mu_F = 0.754$	$\mu_{CR} = 0.186$	
Diversity = 2.0787e+00	k_t = [2, 3, 4, 4]			
Generation 5470: Best = 2.975648e+69	FES = 547100	$\mu_F = 0.760$	$\mu_{CR} = 0.182$	
Diversity = 2.1485e+00	k_t = [1, 1, 1, 1]			
Generation 5480: Best = 2.049997e+69	FES = 548100	$\mu_F = 0.749$	$\mu_{CR} = 0.186$	
Diversity = 2.1614e+00	k_t = [3, 4, 2, 2]			
Generation 5490: Best = 1.363635e+69	FES = 549100	$\mu_F = 0.748$	$\mu_{CR} = 0.183$	
Diversity = 2.1334e+00	k_t = [1, 1, 1, 1]			
Generation 5500: Best = 1.094013e+69	FES = 550100	$\mu_F = 0.749$	$\mu_{CR} = 0.192$	
Diversity = 2.1537e+00	k_t = [2, 2, 3, 3]			
Generation 5510: Best = 7.508687e+68	FES = 551100	$\mu_F = 0.745$	$\mu_{CR} = 0.191$	
Diversity = 2.2488e+00	k_t = [1, 1, 1, 1]			
Generation 5520: Best = 5.627431e+68	FES = 552100	$\mu_F = 0.733$	$\mu_{CR} = 0.185$	
Diversity = 2.2323e+00	k_t = [4, 4, 4, 3]			
Generation 5530: Best = 4.509992e+68	FES = 553100	$\mu_F = 0.727$	$\mu_{CR} = 0.184$	
Diversity = 2.2281e+00	k_t = [1, 1, 1, 1]			
Generation 5540: Best = 3.799580e+68	FES = 554100	$\mu_F = 0.719$	$\mu_{CR} = 0.181$	
Diversity = 2.1351e+00	k_t = [4, 3, 3, 2]			
Generation 5550: Best = 3.566415e+68	FES = 555100	$\mu_F = 0.714$	$\mu_{CR} = 0.177$	
Diversity = 2.0719e+00	k_t = [1, 1, 1, 1]			
Generation 5560: Best = 3.104142e+68	FES = 556100	$\mu_F = 0.707$	$\mu_{CR} = 0.173$	
Diversity = 2.0564e+00	k_t = [3, 2, 3, 2]			
Generation 5570: Best = 2.692447e+68	FES = 557100	$\mu_F = 0.705$	$\mu_{CR} = 0.171$	
Diversity = 2.0321e+00	k_t = [1, 1, 1, 1]			
Generation 5580: Best = 2.225149e+68	FES = 558100	$\mu_F = 0.707$	$\mu_{CR} = 0.171$	
Diversity = 1.9234e+00	k_t = [4, 4, 2, 2]			
Generation 5590: Best = 1.916475e+68	FES = 559100	$\mu_F = 0.704$	$\mu_{CR} = 0.173$	
Diversity = 1.8545e+00	k_t = [1, 1, 1, 1]			
Generation 5600: Best = 1.563232e+68	FES = 560100	$\mu_F = 0.717$	$\mu_{CR} = 0.174$	
Diversity = 1.7588e+00	k_t = [3, 2, 4, 3]			
Generation 5610: Best = 1.153080e+68	FES = 561100	$\mu_F = 0.732$	$\mu_{CR} = 0.172$	
Diversity = 1.8807e+00	k_t = [1, 1, 1, 1]			
Generation 5620: Best = 1.038466e+68	FES = 562100	$\mu_F = 0.728$	$\mu_{CR} = 0.179$	

Diversity = 1.7457e+00	k_t = [3, 3, 4, 4]			
Generation 5630: Best = 8.509037e+67	FES = 563100	$\mu_F = 0.749$	$\mu_{CR} = 0.183$	
Diversity = 1.8319e+00	k_t = [1, 1, 1, 1]			
Generation 5640: Best = 6.719990e+67	FES = 564100	$\mu_F = 0.763$	$\mu_{CR} = 0.183$	
Diversity = 1.9819e+00	k_t = [3, 4, 4, 2]			
Generation 5650: Best = 5.741935e+67	FES = 565100	$\mu_F = 0.777$	$\mu_{CR} = 0.172$	
Diversity = 2.1836e+00	k_t = [1, 1, 1, 1]			
Generation 5660: Best = 5.176850e+67	FES = 566100	$\mu_F = 0.764$	$\mu_{CR} = 0.167$	
Diversity = 2.1445e+00	k_t = [4, 3, 4, 3]			
Generation 5670: Best = 4.775209e+67	FES = 567100	$\mu_F = 0.750$	$\mu_{CR} = 0.165$	
Diversity = 2.1388e+00	k_t = [1, 1, 1, 1]			
Generation 5680: Best = 4.216241e+67	FES = 568100	$\mu_F = 0.743$	$\mu_{CR} = 0.164$	
Diversity = 2.1319e+00	k_t = [3, 2, 4, 2]			
Generation 5690: Best = 3.774406e+67	FES = 569100	$\mu_F = 0.739$	$\mu_{CR} = 0.167$	
Diversity = 2.1343e+00	k_t = [1, 1, 1, 1]			
Generation 5700: Best = 3.307704e+67	FES = 570100	$\mu_F = 0.726$	$\mu_{CR} = 0.165$	
Diversity = 2.0396e+00	k_t = [3, 3, 3, 2]			
Generation 5710: Best = 2.828142e+67	FES = 571100	$\mu_F = 0.732$	$\mu_{CR} = 0.162$	
Diversity = 2.0401e+00	k_t = [1, 1, 1, 1]			
Generation 5720: Best = 2.474838e+67	FES = 572100	$\mu_F = 0.744$	$\mu_{CR} = 0.164$	
Diversity = 2.0795e+00	k_t = [2, 3, 4, 3]			
Generation 5730: Best = 2.258640e+67	FES = 573100	$\mu_F = 0.750$	$\mu_{CR} = 0.173$	
Diversity = 2.1152e+00	k_t = [1, 1, 1, 1]			
Generation 5740: Best = 2.011674e+67	FES = 574100	$\mu_F = 0.749$	$\mu_{CR} = 0.173$	
Diversity = 1.9849e+00	k_t = [2, 2, 4, 3]			
Generation 5750: Best = 1.774746e+67	FES = 575100	$\mu_F = 0.755$	$\mu_{CR} = 0.174$	
Diversity = 2.0430e+00	k_t = [1, 1, 1, 1]			
Generation 5760: Best = 1.588095e+67	FES = 576100	$\mu_F = 0.753$	$\mu_{CR} = 0.173$	
Diversity = 2.0551e+00	k_t = [4, 2, 4, 2]			
Generation 5770: Best = 1.414987e+67	FES = 577100	$\mu_F = 0.742$	$\mu_{CR} = 0.172$	
Diversity = 2.0855e+00	k_t = [1, 1, 1, 1]			
Generation 5780: Best = 1.225974e+67	FES = 578100	$\mu_F = 0.750$	$\mu_{CR} = 0.182$	
Diversity = 2.0738e+00	k_t = [2, 3, 3, 2]			
Generation 5790: Best = 1.063773e+67	FES = 579100	$\mu_F = 0.750$	$\mu_{CR} = 0.183$	
Diversity = 2.2463e+00	k_t = [1, 1, 1, 1]			
Generation 5800: Best = 8.897926e+66	FES = 580100	$\mu_F = 0.755$	$\mu_{CR} = 0.182$	
Diversity = 2.1667e+00	k_t = [4, 4, 3, 3]			
Generation 5810: Best = 7.443246e+66	FES = 581100	$\mu_F = 0.743$	$\mu_{CR} = 0.181$	
Diversity = 2.1439e+00	k_t = [1, 1, 1, 1]			
Generation 5820: Best = 5.548483e+66	FES = 582100	$\mu_F = 0.750$	$\mu_{CR} = 0.177$	
Diversity = 2.1451e+00	k_t = [2, 3, 4, 4]			
Generation 5830: Best = 3.897359e+66	FES = 583100	$\mu_F = 0.762$	$\mu_{CR} = 0.182$	
Diversity = 2.0548e+00	k_t = [1, 1, 1, 1]			
Generation 5840: Best = 2.920370e+66	FES = 584100	$\mu_F = 0.753$	$\mu_{CR} = 0.186$	
Diversity = 2.1374e+00	k_t = [2, 2, 3, 3]			
Generation 5850: Best = 2.357789e+66	FES = 585100	$\mu_F = 0.740$	$\mu_{CR} = 0.185$	
Diversity = 2.1985e+00	k_t = [1, 1, 1, 1]			
Generation 5860: Best = 2.030729e+66	FES = 586100	$\mu_F = 0.749$	$\mu_{CR} = 0.195$	
Diversity = 2.1757e+00	k_t = [4, 4, 2, 2]			
Generation 5870: Best = 1.702655e+66	FES = 587100	$\mu_F = 0.730$	$\mu_{CR} = 0.190$	
Diversity = 2.1299e+00	k_t = [1, 1, 1, 1]			
Generation 5880: Best = 1.324741e+66	FES = 588100	$\mu_F = 0.735$	$\mu_{CR} = 0.183$	
Diversity = 2.0307e+00	k_t = [4, 3, 2, 3]			
Generation 5890: Best = 9.522481e+65	FES = 589100	$\mu_F = 0.738$	$\mu_{CR} = 0.192$	

Diversity = 2.0672e+00	k_t = [1, 1, 1, 1]			
Generation 5900: Best = 7.154273e+65	FES = 590100	$\mu_F = 0.729$	$\mu_{CR} = 0.192$	
Diversity = 1.9647e+00	k_t = [2, 4, 4, 4]			
Generation 5910: Best = 5.751643e+65	FES = 591100	$\mu_F = 0.743$	$\mu_{CR} = 0.200$	
Diversity = 2.0117e+00	k_t = [1, 1, 1, 1]			
Generation 5920: Best = 4.567586e+65	FES = 592100	$\mu_F = 0.736$	$\mu_{CR} = 0.194$	
Diversity = 1.9247e+00	k_t = [3, 2, 4, 4]			
Generation 5930: Best = 3.846049e+65	FES = 593100	$\mu_F = 0.745$	$\mu_{CR} = 0.198$	
Diversity = 1.9511e+00	k_t = [1, 1, 1, 1]			
Generation 5940: Best = 2.833440e+65	FES = 594100	$\mu_F = 0.735$	$\mu_{CR} = 0.192$	
Diversity = 1.9839e+00	k_t = [2, 2, 4, 4]			
Generation 5950: Best = 2.181751e+65	FES = 595100	$\mu_F = 0.727$	$\mu_{CR} = 0.196$	
Diversity = 1.9511e+00	k_t = [1, 1, 1, 1]			
Generation 5960: Best = 1.676888e+65	FES = 596100	$\mu_F = 0.720$	$\mu_{CR} = 0.197$	
Diversity = 1.9296e+00	k_t = [2, 4, 3, 2]			
Generation 5970: Best = 1.313994e+65	FES = 597100	$\mu_F = 0.716$	$\mu_{CR} = 0.197$	
Diversity = 1.9005e+00	k_t = [1, 1, 1, 1]			
Generation 5980: Best = 1.004633e+65	FES = 598100	$\mu_F = 0.730$	$\mu_{CR} = 0.194$	
Diversity = 1.8193e+00	k_t = [4, 2, 2, 3]			
Generation 5990: Best = 8.233869e+64	FES = 599100	$\mu_F = 0.743$	$\mu_{CR} = 0.194$	
Diversity = 1.9393e+00	k_t = [1, 1, 1, 1]			
Generation 6000: Best = 6.236654e+64	FES = 600100	$\mu_F = 0.757$	$\mu_{CR} = 0.182$	
Diversity = 1.9546e+00	k_t = [2, 4, 4, 2]			
Generation 6010: Best = 3.784713e+64	FES = 601100	$\mu_F = 0.763$	$\mu_{CR} = 0.181$	
Diversity = 2.0789e+00	k_t = [1, 1, 1, 1]			
Generation 6020: Best = 2.803324e+64	FES = 602100	$\mu_F = 0.771$	$\mu_{CR} = 0.173$	
Diversity = 2.0745e+00	k_t = [4, 2, 3, 3]			
Generation 6030: Best = 2.023186e+64	FES = 603100	$\mu_F = 0.764$	$\mu_{CR} = 0.177$	
Diversity = 2.0766e+00	k_t = [1, 1, 1, 1]			
Generation 6040: Best = 1.161461e+64	FES = 604100	$\mu_F = 0.741$	$\mu_{CR} = 0.182$	
Diversity = 2.0034e+00	k_t = [3, 3, 2, 3]			
Generation 6050: Best = 6.277223e+63	FES = 605100	$\mu_F = 0.748$	$\mu_{CR} = 0.186$	
Diversity = 1.9884e+00	k_t = [1, 1, 1, 1]			
Generation 6060: Best = 3.437296e+63	FES = 606100	$\mu_F = 0.739$	$\mu_{CR} = 0.188$	
Diversity = 1.8862e+00	k_t = [3, 2, 3, 3]			
Generation 6070: Best = 1.599876e+63	FES = 607100	$\mu_F = 0.739$	$\mu_{CR} = 0.184$	
Diversity = 1.8847e+00	k_t = [1, 1, 1, 1]			
Generation 6080: Best = 9.324230e+62	FES = 608100	$\mu_F = 0.738$	$\mu_{CR} = 0.192$	
Diversity = 1.8070e+00	k_t = [4, 2, 4, 3]			
Generation 6090: Best = 5.230162e+62	FES = 609100	$\mu_F = 0.739$	$\mu_{CR} = 0.196$	
Diversity = 1.7691e+00	k_t = [1, 1, 1, 1]			
Generation 6100: Best = 2.154348e+62	FES = 610100	$\mu_F = 0.750$	$\mu_{CR} = 0.193$	
Diversity = 1.7000e+00	k_t = [3, 3, 2, 3]			
Generation 6110: Best = 1.097277e+62	FES = 611100	$\mu_F = 0.751$	$\mu_{CR} = 0.183$	
Diversity = 1.6666e+00	k_t = [1, 1, 1, 1]			
Generation 6120: Best = 5.073841e+61	FES = 612100	$\mu_F = 0.761$	$\mu_{CR} = 0.183$	
Diversity = 1.5789e+00	k_t = [2, 2, 4, 3]			
Generation 6130: Best = 2.175116e+61	FES = 613100	$\mu_F = 0.755$	$\mu_{CR} = 0.183$	
Diversity = 1.6584e+00	k_t = [1, 1, 1, 1]			
Generation 6140: Best = 1.059350e+61	FES = 614100	$\mu_F = 0.754$	$\mu_{CR} = 0.180$	
Diversity = 1.5990e+00	k_t = [3, 4, 3, 3]			
Generation 6150: Best = 6.407511e+60	FES = 615100	$\mu_F = 0.754$	$\mu_{CR} = 0.184$	
Diversity = 1.6538e+00	k_t = [1, 1, 1, 1]			
Generation 6160: Best = 3.259373e+60	FES = 616100	$\mu_F = 0.751$	$\mu_{CR} = 0.192$	

Diversity = 1.5515e+00	k_t = [4, 3, 2, 2]			
Generation 6170: Best = 1.117354e+60	FES = 617100	$\mu_F = 0.746$	$\mu_{CR} = 0.191$	
Diversity = 1.5377e+00	k_t = [1, 1, 1, 1]			
Generation 6180: Best = 6.915417e+59	FES = 618100	$\mu_F = 0.744$	$\mu_{CR} = 0.185$	
Diversity = 1.4956e+00	k_t = [3, 3, 4, 2]			
Generation 6190: Best = 4.099170e+59	FES = 619100	$\mu_F = 0.735$	$\mu_{CR} = 0.183$	
Diversity = 1.4676e+00	k_t = [1, 1, 1, 1]			
Generation 6200: Best = 2.325980e+59	FES = 620100	$\mu_F = 0.736$	$\mu_{CR} = 0.177$	
Diversity = 1.3644e+00	k_t = [2, 4, 3, 4]			
Generation 6210: Best = 1.307453e+59	FES = 621100	$\mu_F = 0.750$	$\mu_{CR} = 0.178$	
Diversity = 1.3706e+00	k_t = [1, 1, 1, 1]			
Generation 6220: Best = 6.148185e+58	FES = 622100	$\mu_F = 0.753$	$\mu_{CR} = 0.187$	
Diversity = 1.3721e+00	k_t = [3, 2, 3, 2]			
Generation 6230: Best = 2.899948e+58	FES = 623100	$\mu_F = 0.753$	$\mu_{CR} = 0.187$	
Diversity = 1.3603e+00	k_t = [1, 1, 1, 1]			
Generation 6240: Best = 1.505005e+58	FES = 624100	$\mu_F = 0.753$	$\mu_{CR} = 0.185$	
Diversity = 1.2623e+00	k_t = [3, 2, 4, 4]			
Generation 6250: Best = 5.936010e+57	FES = 625100	$\mu_F = 0.750$	$\mu_{CR} = 0.191$	
Diversity = 1.2294e+00	k_t = [1, 1, 1, 1]			
Generation 6260: Best = 3.708488e+57	FES = 626100	$\mu_F = 0.748$	$\mu_{CR} = 0.191$	
Diversity = 1.2096e+00	k_t = [4, 3, 2, 3]			
Generation 6270: Best = 2.173762e+57	FES = 627100	$\mu_F = 0.741$	$\mu_{CR} = 0.188$	
Diversity = 1.1752e+00	k_t = [1, 1, 1, 1]			
Generation 6280: Best = 1.454916e+57	FES = 628100	$\mu_F = 0.740$	$\mu_{CR} = 0.192$	
Diversity = 1.0551e+00	k_t = [4, 4, 2, 2]			
Generation 6290: Best = 8.868613e+56	FES = 629100	$\mu_F = 0.734$	$\mu_{CR} = 0.199$	
Diversity = 1.0115e+00	k_t = [1, 1, 1, 1]			
Generation 6300: Best = 5.083197e+56	FES = 630100	$\mu_F = 0.742$	$\mu_{CR} = 0.188$	
Diversity = 9.9523e-01	k_t = [2, 3, 4, 4]			
Generation 6310: Best = 3.837291e+56	FES = 631100	$\mu_F = 0.750$	$\mu_{CR} = 0.183$	
Diversity = 1.0526e+00	k_t = [1, 1, 1, 1]			
Generation 6320: Best = 2.565013e+56	FES = 632100	$\mu_F = 0.750$	$\mu_{CR} = 0.185$	
Diversity = 1.0707e+00	k_t = [3, 3, 2, 4]			
Generation 6330: Best = 1.761364e+56	FES = 633100	$\mu_F = 0.747$	$\mu_{CR} = 0.190$	
Diversity = 1.0785e+00	k_t = [1, 1, 1, 1]			
Generation 6340: Best = 1.183853e+56	FES = 634100	$\mu_F = 0.743$	$\mu_{CR} = 0.187$	
Diversity = 1.0785e+00	k_t = [4, 4, 4, 3]			
Generation 6350: Best = 8.993800e+55	FES = 635100	$\mu_F = 0.747$	$\mu_{CR} = 0.185$	
Diversity = 1.0724e+00	k_t = [1, 1, 1, 1]			
Generation 6360: Best = 6.659091e+55	FES = 636100	$\mu_F = 0.749$	$\mu_{CR} = 0.189$	
Diversity = 1.0532e+00	k_t = [3, 2, 3, 2]			
Generation 6370: Best = 4.210927e+55	FES = 637100	$\mu_F = 0.754$	$\mu_{CR} = 0.185$	
Diversity = 1.0646e+00	k_t = [1, 1, 1, 1]			
Generation 6380: Best = 2.895650e+55	FES = 638100	$\mu_F = 0.749$	$\mu_{CR} = 0.188$	
Diversity = 1.0992e+00	k_t = [3, 4, 2, 4]			
Generation 6390: Best = 1.721312e+55	FES = 639100	$\mu_F = 0.752$	$\mu_{CR} = 0.184$	
Diversity = 1.1145e+00	k_t = [1, 1, 1, 1]			
Generation 6400: Best = 1.155187e+55	FES = 640100	$\mu_F = 0.756$	$\mu_{CR} = 0.183$	
Diversity = 1.1069e+00	k_t = [2, 3, 3, 3]			
Generation 6410: Best = 7.665962e+54	FES = 641100	$\mu_F = 0.762$	$\mu_{CR} = 0.185$	
Diversity = 1.1845e+00	k_t = [1, 1, 1, 1]			
Generation 6420: Best = 4.316989e+54	FES = 642100	$\mu_F = 0.743$	$\mu_{CR} = 0.186$	
Diversity = 1.1933e+00	k_t = [3, 3, 2, 2]			
Generation 6430: Best = 3.050379e+54	FES = 643100	$\mu_F = 0.745$	$\mu_{CR} = 0.186$	

Diversity = 1.2126e+00	k_t = [1, 1, 1, 1]			
Generation 6440: Best = 2.118797e+54	FES = 644100	$\mu_F = 0.733$	$\mu_{CR} = 0.190$	
Diversity = 1.1897e+00	k_t = [3, 4, 3, 2]			
Generation 6450: Best = 1.538756e+54	FES = 645100	$\mu_F = 0.730$	$\mu_{CR} = 0.198$	
Diversity = 1.1172e+00	k_t = [1, 1, 1, 1]			
Generation 6460: Best = 1.089243e+54	FES = 646100	$\mu_F = 0.743$	$\mu_{CR} = 0.199$	
Diversity = 1.0877e+00	k_t = [3, 3, 4, 2]			
Generation 6470: Best = 7.337505e+53	FES = 647100	$\mu_F = 0.748$	$\mu_{CR} = 0.191$	
Diversity = 1.1296e+00	k_t = [1, 1, 1, 1]			
Generation 6480: Best = 5.494455e+53	FES = 648100	$\mu_F = 0.750$	$\mu_{CR} = 0.191$	
Diversity = 1.1067e+00	k_t = [2, 4, 4, 3]			
Generation 6490: Best = 3.604901e+53	FES = 649100	$\mu_F = 0.746$	$\mu_{CR} = 0.185$	
Diversity = 1.1349e+00	k_t = [1, 1, 1, 1]			
Generation 6500: Best = 1.979535e+53	FES = 650100	$\mu_F = 0.740$	$\mu_{CR} = 0.190$	
Diversity = 1.1043e+00	k_t = [2, 2, 3, 3]			
Generation 6510: Best = 1.692314e+53	FES = 651100	$\mu_F = 0.737$	$\mu_{CR} = 0.194$	
Diversity = 1.1163e+00	k_t = [1, 1, 1, 1]			
Generation 6520: Best = 8.817242e+52	FES = 652100	$\mu_F = 0.730$	$\mu_{CR} = 0.197$	
Diversity = 1.1066e+00	k_t = [2, 3, 3, 2]			
Generation 6530: Best = 6.343469e+52	FES = 653100	$\mu_F = 0.724$	$\mu_{CR} = 0.200$	
Diversity = 1.1275e+00	k_t = [1, 1, 1, 1]			
Generation 6540: Best = 4.878078e+52	FES = 654100	$\mu_F = 0.722$	$\mu_{CR} = 0.206$	
Diversity = 1.0237e+00	k_t = [4, 4, 2, 4]			
Generation 6550: Best = 3.008125e+52	FES = 655100	$\mu_F = 0.738$	$\mu_{CR} = 0.209$	
Diversity = 1.0188e+00	k_t = [1, 1, 1, 1]			
Generation 6560: Best = 2.117541e+52	FES = 656100	$\mu_F = 0.740$	$\mu_{CR} = 0.218$	
Diversity = 1.0485e+00	k_t = [3, 3, 2, 2]			
Generation 6570: Best = 1.636489e+52	FES = 657100	$\mu_F = 0.749$	$\mu_{CR} = 0.214$	
Diversity = 1.0684e+00	k_t = [1, 1, 1, 1]			
Generation 6580: Best = 1.054871e+52	FES = 658100	$\mu_F = 0.744$	$\mu_{CR} = 0.218$	
Diversity = 1.0490e+00	k_t = [4, 4, 3, 3]			
Generation 6590: Best = 8.356992e+51	FES = 659100	$\mu_F = 0.738$	$\mu_{CR} = 0.209$	
Diversity = 1.0251e+00	k_t = [1, 1, 1, 1]			
Generation 6600: Best = 5.229998e+51	FES = 660100	$\mu_F = 0.738$	$\mu_{CR} = 0.212$	
Diversity = 1.0190e+00	k_t = [3, 4, 4, 3]			
Generation 6610: Best = 4.315304e+51	FES = 661100	$\mu_F = 0.741$	$\mu_{CR} = 0.210$	
Diversity = 1.0193e+00	k_t = [1, 1, 1, 1]			
Generation 6620: Best = 2.967817e+51	FES = 662100	$\mu_F = 0.752$	$\mu_{CR} = 0.207$	
Diversity = 9.0884e-01	k_t = [3, 3, 4, 4]			
Generation 6630: Best = 2.069519e+51	FES = 663100	$\mu_F = 0.741$	$\mu_{CR} = 0.212$	
Diversity = 9.1174e-01	k_t = [1, 1, 1, 1]			
Generation 6640: Best = 1.707784e+51	FES = 664100	$\mu_F = 0.754$	$\mu_{CR} = 0.208$	
Diversity = 8.4040e-01	k_t = [3, 4, 3, 4]			
Generation 6650: Best = 1.350540e+51	FES = 665100	$\mu_F = 0.747$	$\mu_{CR} = 0.207$	
Diversity = 8.3992e-01	k_t = [1, 1, 1, 1]			
Generation 6660: Best = 1.100744e+51	FES = 666100	$\mu_F = 0.746$	$\mu_{CR} = 0.210$	
Diversity = 8.0578e-01	k_t = [4, 4, 4, 4]			
Generation 6670: Best = 8.608858e+50	FES = 667100	$\mu_F = 0.747$	$\mu_{CR} = 0.216$	
Diversity = 7.9375e-01	k_t = [1, 1, 1, 1]			
Generation 6680: Best = 7.450949e+50	FES = 668100	$\mu_F = 0.730$	$\mu_{CR} = 0.223$	
Diversity = 7.4408e-01	k_t = [4, 3, 3, 4]			
Generation 6690: Best = 6.487980e+50	FES = 669100	$\mu_F = 0.736$	$\mu_{CR} = 0.212$	
Diversity = 7.5157e-01	k_t = [1, 1, 1, 1]			
Generation 6700: Best = 5.545012e+50	FES = 670100	$\mu_F = 0.736$	$\mu_{CR} = 0.203$	

Diversity = 7.9359e-01	k_t = [4, 4, 4, 3]			
Generation 6710: Best = 4.782099e+50	FES = 671100	$\mu_F = 0.730$	$\mu_{CR} = 0.209$	
Diversity = 7.6277e-01	k_t = [1, 1, 1, 1]			
Generation 6720: Best = 3.966583e+50	FES = 672100	$\mu_F = 0.737$	$\mu_{CR} = 0.211$	
Diversity = 7.5790e-01	k_t = [3, 4, 3, 2]			
Generation 6730: Best = 3.502813e+50	FES = 673100	$\mu_F = 0.735$	$\mu_{CR} = 0.221$	
Diversity = 7.8522e-01	k_t = [1, 1, 1, 1]			
Generation 6740: Best = 2.818518e+50	FES = 674100	$\mu_F = 0.733$	$\mu_{CR} = 0.228$	
Diversity = 7.6236e-01	k_t = [2, 3, 2, 3]			
Generation 6750: Best = 2.262363e+50	FES = 675100	$\mu_F = 0.741$	$\mu_{CR} = 0.230$	
Diversity = 7.8496e-01	k_t = [1, 1, 1, 1]			
Generation 6760: Best = 1.808061e+50	FES = 676100	$\mu_F = 0.748$	$\mu_{CR} = 0.227$	
Diversity = 8.1576e-01	k_t = [3, 2, 3, 2]			
Generation 6770: Best = 1.404324e+50	FES = 677100	$\mu_F = 0.738$	$\mu_{CR} = 0.237$	
Diversity = 8.3072e-01	k_t = [1, 1, 1, 1]			
Generation 6780: Best = 1.148461e+50	FES = 678100	$\mu_F = 0.737$	$\mu_{CR} = 0.230$	
Diversity = 8.1112e-01	k_t = [3, 4, 3, 2]			
Generation 6790: Best = 8.984684e+49	FES = 679100	$\mu_F = 0.739$	$\mu_{CR} = 0.225$	
Diversity = 8.4086e-01	k_t = [1, 1, 1, 1]			
Generation 6800: Best = 7.263408e+49	FES = 680100	$\mu_F = 0.735$	$\mu_{CR} = 0.225$	
Diversity = 8.2530e-01	k_t = [4, 3, 4, 2]			
Generation 6810: Best = 6.158209e+49	FES = 681100	$\mu_F = 0.742$	$\mu_{CR} = 0.220$	
Diversity = 8.2463e-01	k_t = [1, 1, 1, 1]			
Generation 6820: Best = 4.611307e+49	FES = 682100	$\mu_F = 0.742$	$\mu_{CR} = 0.221$	
Diversity = 8.1828e-01	k_t = [4, 3, 4, 4]			
Generation 6830: Best = 3.767469e+49	FES = 683100	$\mu_F = 0.748$	$\mu_{CR} = 0.222$	
Diversity = 8.7140e-01	k_t = [1, 1, 1, 1]			
Generation 6840: Best = 3.043563e+49	FES = 684100	$\mu_F = 0.745$	$\mu_{CR} = 0.222$	
Diversity = 8.3243e-01	k_t = [3, 3, 3, 4]			
Generation 6850: Best = 2.447181e+49	FES = 685100	$\mu_F = 0.762$	$\mu_{CR} = 0.224$	
Diversity = 8.4767e-01	k_t = [1, 1, 1, 1]			
Generation 6860: Best = 1.714096e+49	FES = 686100	$\mu_F = 0.762$	$\mu_{CR} = 0.217$	
Diversity = 8.6261e-01	k_t = [4, 2, 4, 3]			
Generation 6870: Best = 1.347259e+49	FES = 687100	$\mu_F = 0.757$	$\mu_{CR} = 0.210$	
Diversity = 9.2560e-01	k_t = [1, 1, 1, 1]			
Generation 6880: Best = 1.061299e+49	FES = 688100	$\mu_F = 0.744$	$\mu_{CR} = 0.201$	
Diversity = 9.2148e-01	k_t = [2, 2, 4, 2]			
Generation 6890: Best = 7.913953e+48	FES = 689100	$\mu_F = 0.739$	$\mu_{CR} = 0.199$	
Diversity = 9.2446e-01	k_t = [1, 1, 1, 1]			
Generation 6900: Best = 6.356695e+48	FES = 690100	$\mu_F = 0.738$	$\mu_{CR} = 0.195$	
Diversity = 8.5704e-01	k_t = [3, 3, 3, 3]			
Generation 6910: Best = 4.927590e+48	FES = 691100	$\mu_F = 0.748$	$\mu_{CR} = 0.188$	
Diversity = 8.6219e-01	k_t = [1, 1, 1, 1]			
Generation 6920: Best = 3.841474e+48	FES = 692100	$\mu_F = 0.766$	$\mu_{CR} = 0.190$	
Diversity = 8.3884e-01	k_t = [4, 2, 2, 4]			
Generation 6930: Best = 2.936343e+48	FES = 693100	$\mu_F = 0.756$	$\mu_{CR} = 0.192$	
Diversity = 8.6686e-01	k_t = [1, 1, 1, 1]			
Generation 6940: Best = 2.154134e+48	FES = 694100	$\mu_F = 0.748$	$\mu_{CR} = 0.196$	
Diversity = 8.9793e-01	k_t = [4, 2, 4, 3]			
Generation 6950: Best = 1.754114e+48	FES = 695100	$\mu_F = 0.740$	$\mu_{CR} = 0.201$	
Diversity = 8.3609e-01	k_t = [1, 1, 1, 1]			
Generation 6960: Best = 1.395397e+48	FES = 696100	$\mu_F = 0.734$	$\mu_{CR} = 0.202$	
Diversity = 7.8214e-01	k_t = [4, 3, 4, 2]			
Generation 6970: Best = 9.644641e+47	FES = 697100	$\mu_F = 0.743$	$\mu_{CR} = 0.205$	

Diversity = 8.5364e-01	k_t = [1, 1, 1, 1]			
Generation 6980: Best = 8.544153e+47	FES = 698100	$\mu_F = 0.740$	$\mu_{CR} = 0.204$	
Diversity = 8.1754e-01	k_t = [4, 3, 3, 4]			
Generation 6990: Best = 6.241169e+47	FES = 699100	$\mu_F = 0.753$	$\mu_{CR} = 0.201$	
Diversity = 8.1808e-01	k_t = [1, 1, 1, 1]			
Generation 7000: Best = 5.358442e+47	FES = 700100	$\mu_F = 0.748$	$\mu_{CR} = 0.201$	
Diversity = 8.1180e-01	k_t = [3, 4, 3, 3]			
Generation 7010: Best = 4.015617e+47	FES = 701100	$\mu_F = 0.751$	$\mu_{CR} = 0.207$	
Diversity = 8.3612e-01	k_t = [1, 1, 1, 1]			
Generation 7020: Best = 2.699039e+47	FES = 702100	$\mu_F = 0.754$	$\mu_{CR} = 0.206$	
Diversity = 8.2213e-01	k_t = [4, 3, 4, 3]			
Generation 7030: Best = 2.309555e+47	FES = 703100	$\mu_F = 0.747$	$\mu_{CR} = 0.210$	
Diversity = 8.0534e-01	k_t = [1, 1, 1, 1]			
Generation 7040: Best = 1.792728e+47	FES = 704100	$\mu_F = 0.749$	$\mu_{CR} = 0.211$	
Diversity = 7.5924e-01	k_t = [4, 3, 4, 2]			
Generation 7050: Best = 1.372311e+47	FES = 705100	$\mu_F = 0.751$	$\mu_{CR} = 0.209$	
Diversity = 7.7625e-01	k_t = [1, 1, 1, 1]			
Generation 7060: Best = 1.076647e+47	FES = 706100	$\mu_F = 0.743$	$\mu_{CR} = 0.206$	
Diversity = 7.5159e-01	k_t = [4, 2, 3, 4]			
Generation 7070: Best = 8.064722e+46	FES = 707100	$\mu_F = 0.737$	$\mu_{CR} = 0.207$	
Diversity = 7.5407e-01	k_t = [1, 1, 1, 1]			
Generation 7080: Best = 6.709000e+46	FES = 708100	$\mu_F = 0.733$	$\mu_{CR} = 0.210$	
Diversity = 7.4340e-01	k_t = [2, 2, 2, 3]			
Generation 7090: Best = 5.686103e+46	FES = 709100	$\mu_F = 0.733$	$\mu_{CR} = 0.219$	
Diversity = 7.6507e-01	k_t = [1, 1, 1, 1]			
Generation 7100: Best = 4.607049e+46	FES = 710100	$\mu_F = 0.737$	$\mu_{CR} = 0.219$	
Diversity = 7.3068e-01	k_t = [2, 3, 3, 2]			
Generation 7110: Best = 3.600152e+46	FES = 711100	$\mu_F = 0.731$	$\mu_{CR} = 0.223$	
Diversity = 7.6375e-01	k_t = [1, 1, 1, 1]			
Generation 7120: Best = 2.984050e+46	FES = 712100	$\mu_F = 0.747$	$\mu_{CR} = 0.215$	
Diversity = 7.2090e-01	k_t = [4, 4, 2, 4]			
Generation 7130: Best = 2.092376e+46	FES = 713100	$\mu_F = 0.753$	$\mu_{CR} = 0.212$	
Diversity = 7.6482e-01	k_t = [1, 1, 1, 1]			
Generation 7140: Best = 1.645094e+46	FES = 714100	$\mu_F = 0.752$	$\mu_{CR} = 0.213$	
Diversity = 7.5677e-01	k_t = [3, 4, 3, 2]			
Generation 7150: Best = 1.351199e+46	FES = 715100	$\mu_F = 0.742$	$\mu_{CR} = 0.208$	
Diversity = 7.8942e-01	k_t = [1, 1, 1, 1]			
Generation 7160: Best = 1.082560e+46	FES = 716100	$\mu_F = 0.739$	$\mu_{CR} = 0.213$	
Diversity = 7.3167e-01	k_t = [3, 2, 2, 3]			
Generation 7170: Best = 9.053672e+45	FES = 717100	$\mu_F = 0.738$	$\mu_{CR} = 0.208$	
Diversity = 7.1770e-01	k_t = [1, 1, 1, 1]			
Generation 7180: Best = 7.405450e+45	FES = 718100	$\mu_F = 0.733$	$\mu_{CR} = 0.201$	
Diversity = 7.1809e-01	k_t = [2, 4, 2, 3]			
Generation 7190: Best = 6.873751e+45	FES = 719100	$\mu_F = 0.732$	$\mu_{CR} = 0.203$	
Diversity = 7.5559e-01	k_t = [1, 1, 1, 1]			
Generation 7200: Best = 5.942811e+45	FES = 720100	$\mu_F = 0.737$	$\mu_{CR} = 0.197$	
Diversity = 7.0094e-01	k_t = [3, 3, 3, 4]			
Generation 7210: Best = 5.053105e+45	FES = 721100	$\mu_F = 0.747$	$\mu_{CR} = 0.190$	
Diversity = 7.2932e-01	k_t = [1, 1, 1, 1]			
Generation 7220: Best = 4.288235e+45	FES = 722100	$\mu_F = 0.754$	$\mu_{CR} = 0.193$	
Diversity = 7.3251e-01	k_t = [4, 3, 4, 4]			
Generation 7230: Best = 3.681094e+45	FES = 723100	$\mu_F = 0.755$	$\mu_{CR} = 0.201$	
Diversity = 7.1187e-01	k_t = [1, 1, 1, 1]			
Generation 7240: Best = 3.437642e+45	FES = 724100	$\mu_F = 0.758$	$\mu_{CR} = 0.208$	

Diversity = 6.9041e-01	k_t = [3, 4, 4, 3]			
Generation 7250: Best = 3.060558e+45	FES = 725100	$\mu_F = 0.756$	$\mu_{CR} = 0.210$	
Diversity = 7.0938e-01	k_t = [1, 1, 1, 1]			
Generation 7260: Best = 2.841004e+45	FES = 726100	$\mu_F = 0.743$	$\mu_{CR} = 0.207$	
Diversity = 6.8191e-01	k_t = [2, 2, 4, 3]			
Generation 7270: Best = 2.567349e+45	FES = 727100	$\mu_F = 0.747$	$\mu_{CR} = 0.205$	
Diversity = 6.8553e-01	k_t = [1, 1, 1, 1]			
Generation 7280: Best = 2.407594e+45	FES = 728100	$\mu_F = 0.743$	$\mu_{CR} = 0.208$	
Diversity = 6.6486e-01	k_t = [2, 4, 3, 3]			
Generation 7290: Best = 2.106234e+45	FES = 729100	$\mu_F = 0.743$	$\mu_{CR} = 0.213$	
Diversity = 6.7157e-01	k_t = [1, 1, 1, 1]			
Generation 7300: Best = 2.016978e+45	FES = 730100	$\mu_F = 0.737$	$\mu_{CR} = 0.208$	
Diversity = 6.3590e-01	k_t = [4, 4, 2, 4]			
Generation 7310: Best = 1.910900e+45	FES = 731100	$\mu_F = 0.750$	$\mu_{CR} = 0.211$	
Diversity = 6.2664e-01	k_t = [1, 1, 1, 1]			
Generation 7320: Best = 1.807449e+45	FES = 732100	$\mu_F = 0.758$	$\mu_{CR} = 0.207$	
Diversity = 6.2642e-01	k_t = [2, 4, 2, 3]			
Generation 7330: Best = 1.740783e+45	FES = 733100	$\mu_F = 0.751$	$\mu_{CR} = 0.207$	
Diversity = 6.5703e-01	k_t = [1, 1, 1, 1]			
Generation 7340: Best = 1.652360e+45	FES = 734100	$\mu_F = 0.749$	$\mu_{CR} = 0.208$	
Diversity = 6.6382e-01	k_t = [2, 3, 2, 2]			
Generation 7350: Best = 1.617540e+45	FES = 735100	$\mu_F = 0.735$	$\mu_{CR} = 0.209$	
Diversity = 6.6962e-01	k_t = [1, 1, 1, 1]			
Generation 7360: Best = 1.547473e+45	FES = 736100	$\mu_F = 0.733$	$\mu_{CR} = 0.214$	
Diversity = 6.6041e-01	k_t = [2, 2, 3, 2]			
Generation 7370: Best = 1.476572e+45	FES = 737100	$\mu_F = 0.736$	$\mu_{CR} = 0.225$	
Diversity = 6.6538e-01	k_t = [1, 1, 1, 1]			
Generation 7380: Best = 1.402038e+45	FES = 738100	$\mu_F = 0.746$	$\mu_{CR} = 0.228$	
Diversity = 6.1416e-01	k_t = [3, 4, 2, 4]			
Generation 7390: Best = 1.355421e+45	FES = 739100	$\mu_F = 0.752$	$\mu_{CR} = 0.226$	
Diversity = 6.3164e-01	k_t = [1, 1, 1, 1]			
Generation 7400: Best = 1.308820e+45	FES = 740100	$\mu_F = 0.752$	$\mu_{CR} = 0.220$	
Diversity = 6.4943e-01	k_t = [3, 2, 3, 2]			
Generation 7410: Best = 1.239694e+45	FES = 741100	$\mu_F = 0.762$	$\mu_{CR} = 0.214$	
Diversity = 7.2453e-01	k_t = [1, 1, 1, 1]			
Generation 7420: Best = 1.187993e+45	FES = 742100	$\mu_F = 0.749$	$\mu_{CR} = 0.204$	
Diversity = 7.2909e-01	k_t = [2, 2, 4, 2]			
Generation 7430: Best = 1.141164e+45	FES = 743100	$\mu_F = 0.744$	$\mu_{CR} = 0.210$	
Diversity = 7.5157e-01	k_t = [1, 1, 1, 1]			
Generation 7440: Best = 1.085257e+45	FES = 744100	$\mu_F = 0.748$	$\mu_{CR} = 0.205$	
Diversity = 7.3991e-01	k_t = [4, 2, 4, 3]			
Generation 7450: Best = 1.040573e+45	FES = 745100	$\mu_F = 0.753$	$\mu_{CR} = 0.209$	
Diversity = 7.5266e-01	k_t = [1, 1, 1, 1]			
Generation 7460: Best = 9.792374e+44	FES = 746100	$\mu_F = 0.760$	$\mu_{CR} = 0.212$	
Diversity = 7.3507e-01	k_t = [2, 2, 2, 4]			
Generation 7470: Best = 9.348632e+44	FES = 747100	$\mu_F = 0.769$	$\mu_{CR} = 0.201$	
Diversity = 7.9456e-01	k_t = [1, 1, 1, 1]			
Generation 7480: Best = 8.894033e+44	FES = 748100	$\mu_F = 0.760$	$\mu_{CR} = 0.201$	
Diversity = 8.3057e-01	k_t = [3, 4, 3, 2]			
Generation 7490: Best = 8.450739e+44	FES = 749100	$\mu_F = 0.755$	$\mu_{CR} = 0.203$	
Diversity = 8.1661e-01	k_t = [1, 1, 1, 1]			
Generation 7500: Best = 7.829556e+44	FES = 750100	$\mu_F = 0.753$	$\mu_{CR} = 0.214$	
Diversity = 8.5152e-01	k_t = [2, 3, 2, 4]			
Generation 7510: Best = 7.485901e+44	FES = 751100	$\mu_F = 0.751$	$\mu_{CR} = 0.202$	

Diversity = 8.3460e-01	k_t = [1, 1, 1, 1]			
Generation 7520: Best = 6.992032e+44	FES = 752100	$\mu_F = 0.748$	$\mu_{CR} = 0.195$	
Diversity = 8.1276e-01	k_t = [4, 3, 3, 3]			
Generation 7530: Best = 6.651640e+44	FES = 753100	$\mu_F = 0.745$	$\mu_{CR} = 0.194$	
Diversity = 8.3842e-01	k_t = [1, 1, 1, 1]			
Generation 7540: Best = 6.316262e+44	FES = 754100	$\mu_F = 0.747$	$\mu_{CR} = 0.195$	
Diversity = 8.3075e-01	k_t = [3, 4, 3, 3]			
Generation 7550: Best = 5.985127e+44	FES = 755100	$\mu_F = 0.748$	$\mu_{CR} = 0.199$	
Diversity = 8.4123e-01	k_t = [1, 1, 1, 1]			
Generation 7560: Best = 5.721064e+44	FES = 756100	$\mu_F = 0.744$	$\mu_{CR} = 0.203$	
Diversity = 8.1198e-01	k_t = [3, 3, 2, 2]			
Generation 7570: Best = 5.402782e+44	FES = 757100	$\mu_F = 0.747$	$\mu_{CR} = 0.203$	
Diversity = 8.5298e-01	k_t = [1, 1, 1, 1]			
Generation 7580: Best = 4.954916e+44	FES = 758100	$\mu_F = 0.756$	$\mu_{CR} = 0.204$	
Diversity = 8.2057e-01	k_t = [4, 4, 2, 3]			
Generation 7590: Best = 4.612796e+44	FES = 759100	$\mu_F = 0.763$	$\mu_{CR} = 0.209$	
Diversity = 8.7563e-01	k_t = [1, 1, 1, 1]			
Generation 7600: Best = 4.316093e+44	FES = 760100	$\mu_F = 0.759$	$\mu_{CR} = 0.201$	
Diversity = 8.9981e-01	k_t = [4, 2, 3, 3]			
Generation 7610: Best = 3.884628e+44	FES = 761100	$\mu_F = 0.754$	$\mu_{CR} = 0.199$	
Diversity = 9.5510e-01	k_t = [1, 1, 1, 1]			
Generation 7620: Best = 3.654058e+44	FES = 762100	$\mu_F = 0.746$	$\mu_{CR} = 0.198$	
Diversity = 8.9833e-01	k_t = [4, 4, 2, 4]			
Generation 7630: Best = 3.387497e+44	FES = 763100	$\mu_F = 0.765$	$\mu_{CR} = 0.191$	
Diversity = 8.7128e-01	k_t = [1, 1, 1, 1]			
Generation 7640: Best = 3.192657e+44	FES = 764100	$\mu_F = 0.765$	$\mu_{CR} = 0.185$	
Diversity = 8.8305e-01	k_t = [2, 2, 3, 3]			
Generation 7650: Best = 3.053280e+44	FES = 765100	$\mu_F = 0.754$	$\mu_{CR} = 0.192$	
Diversity = 8.7537e-01	k_t = [1, 1, 1, 1]			
Generation 7660: Best = 2.861147e+44	FES = 766100	$\mu_F = 0.758$	$\mu_{CR} = 0.190$	
Diversity = 8.7538e-01	k_t = [4, 3, 4, 2]			
Generation 7670: Best = 2.646650e+44	FES = 767100	$\mu_F = 0.764$	$\mu_{CR} = 0.192$	
Diversity = 9.0883e-01	k_t = [1, 1, 1, 1]			
Generation 7680: Best = 2.500311e+44	FES = 768100	$\mu_F = 0.759$	$\mu_{CR} = 0.189$	
Diversity = 9.0643e-01	k_t = [3, 3, 4, 2]			
Generation 7690: Best = 2.373966e+44	FES = 769100	$\mu_F = 0.752$	$\mu_{CR} = 0.188$	
Diversity = 9.1329e-01	k_t = [1, 1, 1, 1]			
Generation 7700: Best = 2.282557e+44	FES = 770100	$\mu_F = 0.747$	$\mu_{CR} = 0.186$	
Diversity = 8.7255e-01	k_t = [2, 4, 4, 3]			
Generation 7710: Best = 2.150037e+44	FES = 771100	$\mu_F = 0.744$	$\mu_{CR} = 0.187$	
Diversity = 8.7309e-01	k_t = [1, 1, 1, 1]			
Generation 7720: Best = 2.060057e+44	FES = 772100	$\mu_F = 0.758$	$\mu_{CR} = 0.180$	
Diversity = 9.0333e-01	k_t = [4, 2, 4, 3]			
Generation 7730: Best = 1.924724e+44	FES = 773100	$\mu_F = 0.748$	$\mu_{CR} = 0.182$	
Diversity = 9.8389e-01	k_t = [1, 1, 1, 1]			
Generation 7740: Best = 1.851327e+44	FES = 774100	$\mu_F = 0.739$	$\mu_{CR} = 0.183$	
Diversity = 8.8837e-01	k_t = [3, 3, 4, 3]			
Generation 7750: Best = 1.679849e+44	FES = 775100	$\mu_F = 0.747$	$\mu_{CR} = 0.188$	
Diversity = 9.0082e-01	k_t = [1, 1, 1, 1]			
Generation 7760: Best = 1.566955e+44	FES = 776100	$\mu_F = 0.755$	$\mu_{CR} = 0.187$	
Diversity = 8.6054e-01	k_t = [3, 2, 3, 4]			
Generation 7770: Best = 1.480438e+44	FES = 777100	$\mu_F = 0.764$	$\mu_{CR} = 0.193$	
Diversity = 9.0453e-01	k_t = [1, 1, 1, 1]			
Generation 7780: Best = 1.395305e+44	FES = 778100	$\mu_F = 0.756$	$\mu_{CR} = 0.190$	

Diversity = 8.7762e-01	k_t = [4, 4, 4, 4]			
Generation 7790: Best = 1.305828e+44	FES = 779100	$\mu_F = 0.762$	$\mu_{CR} = 0.187$	
Diversity = 8.7015e-01	k_t = [1, 1, 1, 1]			
Generation 7800: Best = 1.273376e+44	FES = 780100	$\mu_F = 0.761$	$\mu_{CR} = 0.186$	
Diversity = 8.3903e-01	k_t = [4, 3, 3, 3]			
Generation 7810: Best = 1.197958e+44	FES = 781100	$\mu_F = 0.769$	$\mu_{CR} = 0.200$	
Diversity = 8.7046e-01	k_t = [1, 1, 1, 1]			
Generation 7820: Best = 1.141566e+44	FES = 782100	$\mu_F = 0.765$	$\mu_{CR} = 0.188$	
Diversity = 8.8972e-01	k_t = [2, 4, 3, 3]			
Generation 7830: Best = 1.078002e+44	FES = 783100	$\mu_F = 0.772$	$\mu_{CR} = 0.184$	
Diversity = 9.0939e-01	k_t = [1, 1, 1, 1]			
Generation 7840: Best = 1.017657e+44	FES = 784100	$\mu_F = 0.768$	$\mu_{CR} = 0.182$	
Diversity = 9.3563e-01	k_t = [4, 4, 4, 2]			
Generation 7850: Best = 9.599438e+43	FES = 785100	$\mu_F = 0.763$	$\mu_{CR} = 0.183$	
Diversity = 9.6621e-01	k_t = [1, 1, 1, 1]			
Generation 7860: Best = 9.013483e+43	FES = 786100	$\mu_F = 0.758$	$\mu_{CR} = 0.188$	
Diversity = 8.7519e-01	k_t = [4, 3, 3, 3]			
Generation 7870: Best = 8.369389e+43	FES = 787100	$\mu_F = 0.758$	$\mu_{CR} = 0.181$	
Diversity = 9.2829e-01	k_t = [1, 1, 1, 1]			
Generation 7880: Best = 8.039771e+43	FES = 788100	$\mu_F = 0.744$	$\mu_{CR} = 0.173$	
Diversity = 9.2934e-01	k_t = [3, 4, 2, 2]			
Generation 7890: Best = 7.711662e+43	FES = 789100	$\mu_F = 0.736$	$\mu_{CR} = 0.172$	
Diversity = 9.4301e-01	k_t = [1, 1, 1, 1]			
Generation 7900: Best = 7.382572e+43	FES = 790100	$\mu_F = 0.738$	$\mu_{CR} = 0.171$	
Diversity = 9.1619e-01	k_t = [4, 4, 4, 4]			
Generation 7910: Best = 7.195629e+43	FES = 791100	$\mu_F = 0.742$	$\mu_{CR} = 0.174$	
Diversity = 8.7270e-01	k_t = [1, 1, 1, 1]			
Generation 7920: Best = 6.819605e+43	FES = 792100	$\mu_F = 0.739$	$\mu_{CR} = 0.176$	
Diversity = 8.8030e-01	k_t = [4, 3, 4, 3]			
Generation 7930: Best = 6.541106e+43	FES = 793100	$\mu_F = 0.740$	$\mu_{CR} = 0.180$	
Diversity = 9.0765e-01	k_t = [1, 1, 1, 1]			
Generation 7940: Best = 6.346255e+43	FES = 794100	$\mu_F = 0.721$	$\mu_{CR} = 0.177$	
Diversity = 8.9405e-01	k_t = [2, 2, 2, 2]			
Generation 7950: Best = 6.087209e+43	FES = 795100	$\mu_F = 0.727$	$\mu_{CR} = 0.172$	
Diversity = 9.0086e-01	k_t = [1, 1, 1, 1]			
Generation 7960: Best = 5.863199e+43	FES = 796100	$\mu_F = 0.726$	$\mu_{CR} = 0.177$	
Diversity = 8.3540e-01	k_t = [3, 4, 3, 4]			
Generation 7970: Best = 5.561364e+43	FES = 797100	$\mu_F = 0.739$	$\mu_{CR} = 0.194$	
Diversity = 8.1160e-01	k_t = [1, 1, 1, 1]			
Generation 7980: Best = 5.367149e+43	FES = 798100	$\mu_F = 0.736$	$\mu_{CR} = 0.197$	
Diversity = 8.1531e-01	k_t = [4, 4, 2, 2]			
Generation 7990: Best = 5.201077e+43	FES = 799100	$\mu_F = 0.745$	$\mu_{CR} = 0.210$	
Diversity = 8.6166e-01	k_t = [1, 1, 1, 1]			
Generation 8000: Best = 4.934441e+43	FES = 800100	$\mu_F = 0.746$	$\mu_{CR} = 0.213$	
Diversity = 8.3500e-01	k_t = [4, 4, 3, 3]			
Generation 8010: Best = 4.758619e+43	FES = 801100	$\mu_F = 0.753$	$\mu_{CR} = 0.206$	
Diversity = 9.1676e-01	k_t = [1, 1, 1, 1]			
Generation 8020: Best = 4.568018e+43	FES = 802100	$\mu_F = 0.740$	$\mu_{CR} = 0.210$	
Diversity = 8.9992e-01	k_t = [3, 2, 2, 2]			
Generation 8030: Best = 4.399342e+43	FES = 803100	$\mu_F = 0.732$	$\mu_{CR} = 0.212$	
Diversity = 9.0872e-01	k_t = [1, 1, 1, 1]			
Generation 8040: Best = 4.216909e+43	FES = 804100	$\mu_F = 0.745$	$\mu_{CR} = 0.209$	
Diversity = 8.6321e-01	k_t = [3, 4, 4, 3]			
Generation 8050: Best = 3.972142e+43	FES = 805100	$\mu_F = 0.745$	$\mu_{CR} = 0.211$	

Diversity = 8.8713e-01	k_t = [1, 1, 1, 1]			
Generation 8060: Best = 3.725135e+43	FES = 806100	$\mu_F = 0.737$	$\mu_{CR} = 0.208$	
Diversity = 8.6559e-01	k_t = [2, 3, 2, 2]			
Generation 8070: Best = 3.585082e+43	FES = 807100	$\mu_F = 0.727$	$\mu_{CR} = 0.211$	
Diversity = 8.9883e-01	k_t = [1, 1, 1, 1]			
Generation 8080: Best = 3.441900e+43	FES = 808100	$\mu_F = 0.719$	$\mu_{CR} = 0.204$	
Diversity = 8.2330e-01	k_t = [4, 2, 3, 3]			
Generation 8090: Best = 3.246482e+43	FES = 809100	$\mu_F = 0.729$	$\mu_{CR} = 0.214$	
Diversity = 8.4260e-01	k_t = [1, 1, 1, 1]			
Generation 8100: Best = 3.102293e+43	FES = 810100	$\mu_F = 0.744$	$\mu_{CR} = 0.214$	
Diversity = 8.6817e-01	k_t = [2, 3, 3, 2]			
Generation 8110: Best = 2.927015e+43	FES = 811100	$\mu_F = 0.740$	$\mu_{CR} = 0.210$	
Diversity = 9.2837e-01	k_t = [1, 1, 1, 1]			
Generation 8120: Best = 2.794595e+43	FES = 812100	$\mu_F = 0.732$	$\mu_{CR} = 0.210$	
Diversity = 9.3069e-01	k_t = [2, 3, 4, 2]			
Generation 8130: Best = 2.688036e+43	FES = 813100	$\mu_F = 0.734$	$\mu_{CR} = 0.206$	
Diversity = 8.9030e-01	k_t = [1, 1, 1, 1]			
Generation 8140: Best = 2.598665e+43	FES = 814100	$\mu_F = 0.726$	$\mu_{CR} = 0.202$	
Diversity = 8.3605e-01	k_t = [2, 2, 3, 3]			
Generation 8150: Best = 2.436952e+43	FES = 815100	$\mu_F = 0.731$	$\mu_{CR} = 0.197$	
Diversity = 8.4671e-01	k_t = [1, 1, 1, 1]			
Generation 8160: Best = 2.318613e+43	FES = 816100	$\mu_F = 0.738$	$\mu_{CR} = 0.208$	
Diversity = 8.2192e-01	k_t = [4, 3, 3, 4]			
Generation 8170: Best = 2.201048e+43	FES = 817100	$\mu_F = 0.758$	$\mu_{CR} = 0.213$	
Diversity = 8.5271e-01	k_t = [1, 1, 1, 1]			
Generation 8180: Best = 1.987984e+43	FES = 818100	$\mu_F = 0.762$	$\mu_{CR} = 0.219$	
Diversity = 8.1286e-01	k_t = [2, 4, 3, 4]			
Generation 8190: Best = 1.877615e+43	FES = 819100	$\mu_F = 0.763$	$\mu_{CR} = 0.217$	
Diversity = 8.7771e-01	k_t = [1, 1, 1, 1]			
Generation 8200: Best = 1.765461e+43	FES = 820100	$\mu_F = 0.745$	$\mu_{CR} = 0.217$	
Diversity = 8.7558e-01	k_t = [3, 4, 3, 3]			
Generation 8210: Best = 1.670559e+43	FES = 821100	$\mu_F = 0.752$	$\mu_{CR} = 0.221$	
Diversity = 9.0156e-01	k_t = [1, 1, 1, 1]			
Generation 8220: Best = 1.518856e+43	FES = 822100	$\mu_F = 0.739$	$\mu_{CR} = 0.214$	
Diversity = 8.5790e-01	k_t = [2, 2, 4, 4]			
Generation 8230: Best = 1.424510e+43	FES = 823100	$\mu_F = 0.744$	$\mu_{CR} = 0.210$	
Diversity = 8.5028e-01	k_t = [1, 1, 1, 1]			
Generation 8240: Best = 1.333438e+43	FES = 824100	$\mu_F = 0.749$	$\mu_{CR} = 0.206$	
Diversity = 8.1521e-01	k_t = [4, 3, 3, 2]			
Generation 8250: Best = 1.218122e+43	FES = 825100	$\mu_F = 0.739$	$\mu_{CR} = 0.212$	
Diversity = 8.3810e-01	k_t = [1, 1, 1, 1]			
Generation 8260: Best = 1.104260e+43	FES = 826100	$\mu_F = 0.733$	$\mu_{CR} = 0.213$	
Diversity = 7.9072e-01	k_t = [3, 4, 3, 4]			
Generation 8270: Best = 1.033188e+43	FES = 827100	$\mu_F = 0.742$	$\mu_{CR} = 0.218$	
Diversity = 8.0288e-01	k_t = [1, 1, 1, 1]			
Generation 8280: Best = 9.452577e+42	FES = 828100	$\mu_F = 0.754$	$\mu_{CR} = 0.222$	
Diversity = 8.3739e-01	k_t = [3, 4, 2, 2]			
Generation 8290: Best = 8.775070e+42	FES = 829100	$\mu_F = 0.754$	$\mu_{CR} = 0.220$	
Diversity = 9.0541e-01	k_t = [1, 1, 1, 1]			
Generation 8300: Best = 7.648858e+42	FES = 830100	$\mu_F = 0.755$	$\mu_{CR} = 0.224$	
Diversity = 8.9347e-01	k_t = [4, 4, 3, 4]			
Generation 8310: Best = 6.758850e+42	FES = 831100	$\mu_F = 0.762$	$\mu_{CR} = 0.220$	
Diversity = 9.1646e-01	k_t = [1, 1, 1, 1]			
Generation 8320: Best = 6.225558e+42	FES = 832100	$\mu_F = 0.761$	$\mu_{CR} = 0.221$	

Diversity = 9.4852e-01 k_t = [2, 4, 3, 4]
 Generation 8330: Best = 5.854737e+42 FES = 833100 μ_F = 0.759 μ_{CR} = 0.214
 Diversity = 9.5326e-01 k_t = [1, 1, 1, 1]
 Generation 8340: Best = 5.339894e+42 FES = 834100 μ_F = 0.758 μ_{CR} = 0.209
 Diversity = 9.0877e-01 k_t = [2, 3, 3, 4]
 Generation 8350: Best = 4.931489e+42 FES = 835100 μ_F = 0.757 μ_{CR} = 0.204
 Diversity = 9.3009e-01 k_t = [1, 1, 1, 1]
 Generation 8360: Best = 4.467107e+42 FES = 836100 μ_F = 0.765 μ_{CR} = 0.196
 Diversity = 9.4187e-01 k_t = [2, 4, 2, 3]
 Generation 8370: Best = 3.867734e+42 FES = 837100 μ_F = 0.766 μ_{CR} = 0.198
 Diversity = 1.0183e+00 k_t = [1, 1, 1, 1]
 Generation 8380: Best = 3.729283e+42 FES = 838100 μ_F = 0.758 μ_{CR} = 0.198
 Diversity = 9.9443e-01 k_t = [4, 2, 2, 2]
 Generation 8390: Best = 3.384210e+42 FES = 839100 μ_F = 0.753 μ_{CR} = 0.192
 Diversity = 9.9380e-01 k_t = [1, 1, 1, 1]
 Generation 8400: Best = 2.867366e+42 FES = 840100 μ_F = 0.754 μ_{CR} = 0.197
 Diversity = 9.1347e-01 k_t = [2, 4, 3, 4]
 Generation 8410: Best = 2.630065e+42 FES = 841100 μ_F = 0.751 μ_{CR} = 0.200
 Diversity = 9.1061e-01 k_t = [1, 1, 1, 1]
 Generation 8420: Best = 2.331117e+42 FES = 842100 μ_F = 0.743 μ_{CR} = 0.207
 Diversity = 9.2153e-01 k_t = [4, 2, 4, 2]
 Generation 8430: Best = 2.280320e+42 FES = 843100 μ_F = 0.741 μ_{CR} = 0.212
 Diversity = 9.1325e-01 k_t = [1, 1, 1, 1]
 Generation 8440: Best = 1.949394e+42 FES = 844100 μ_F = 0.739 μ_{CR} = 0.220
 Diversity = 8.8768e-01 k_t = [3, 2, 2, 2]
 Generation 8450: Best = 1.860440e+42 FES = 845100 μ_F = 0.733 μ_{CR} = 0.217
 Diversity = 9.1998e-01 k_t = [1, 1, 1, 1]
 Generation 8460: Best = 1.726259e+42 FES = 846100 μ_F = 0.736 μ_{CR} = 0.216
 Diversity = 8.9192e-01 k_t = [2, 4, 2, 2]
 Generation 8470: Best = 1.588849e+42 FES = 847100 μ_F = 0.740 μ_{CR} = 0.205
 Diversity = 8.8967e-01 k_t = [1, 1, 1, 1]
 Generation 8480: Best = 1.447455e+42 FES = 848100 μ_F = 0.740 μ_{CR} = 0.194
 Diversity = 8.3323e-01 k_t = [3, 2, 2, 4]
 Generation 8490: Best = 1.343594e+42 FES = 849100 μ_F = 0.748 μ_{CR} = 0.196
 Diversity = 8.3006e-01 k_t = [1, 1, 1, 1]
 Generation 8500: Best = 1.196758e+42 FES = 850100 μ_F = 0.747 μ_{CR} = 0.196
 Diversity = 8.2539e-01 k_t = [4, 4, 2, 2]
 Generation 8510: Best = 1.143517e+42 FES = 851100 μ_F = 0.749 μ_{CR} = 0.198
 Diversity = 8.3389e-01 k_t = [1, 1, 1, 1]
 Generation 8520: Best = 1.043862e+42 FES = 852100 μ_F = 0.753 μ_{CR} = 0.197
 Diversity = 8.4689e-01 k_t = [4, 3, 4, 3]
 Generation 8530: Best = 9.854729e+41 FES = 853100 μ_F = 0.748 μ_{CR} = 0.205
 Diversity = 8.8021e-01 k_t = [1, 1, 1, 1]
 Generation 8540: Best = 9.488634e+41 FES = 854100 μ_F = 0.735 μ_{CR} = 0.203
 Diversity = 8.5181e-01 k_t = [3, 4, 2, 2]
 Generation 8550: Best = 8.793681e+41 FES = 855100 μ_F = 0.752 μ_{CR} = 0.204
 Diversity = 8.4658e-01 k_t = [1, 1, 1, 1]
 Generation 8560: Best = 8.110228e+41 FES = 856100 μ_F = 0.743 μ_{CR} = 0.211
 Diversity = 8.5175e-01 k_t = [3, 4, 2, 4]
 Generation 8570: Best = 7.704252e+41 FES = 857100 μ_F = 0.755 μ_{CR} = 0.221
 Diversity = 8.3481e-01 k_t = [1, 1, 1, 1]
 Generation 8580: Best = 7.199902e+41 FES = 858100 μ_F = 0.762 μ_{CR} = 0.214
 Diversity = 8.7397e-01 k_t = [3, 4, 2, 2]
 Generation 8590: Best = 6.669778e+41 FES = 859100 μ_F = 0.752 μ_{CR} = 0.213

Diversity = 9.0131e-01	k_t = [1, 1, 1, 1]			
Generation 8600: Best = 6.062613e+41	FES = 860100	$\mu_F = 0.758$	$\mu_{CR} = 0.203$	
Diversity = 9.2457e-01	k_t = [3, 3, 3, 2]			
Generation 8610: Best = 5.597398e+41	FES = 861100	$\mu_F = 0.750$	$\mu_{CR} = 0.202$	
Diversity = 9.6317e-01	k_t = [1, 1, 1, 1]			
Generation 8620: Best = 5.278330e+41	FES = 862100	$\mu_F = 0.739$	$\mu_{CR} = 0.197$	
Diversity = 9.4200e-01	k_t = [4, 4, 2, 2]			
Generation 8630: Best = 4.825789e+41	FES = 863100	$\mu_F = 0.733$	$\mu_{CR} = 0.204$	
Diversity = 9.4663e-01	k_t = [1, 1, 1, 1]			
Generation 8640: Best = 4.315827e+41	FES = 864100	$\mu_F = 0.725$	$\mu_{CR} = 0.203$	
Diversity = 8.5881e-01	k_t = [3, 3, 2, 3]			
Generation 8650: Best = 4.093292e+41	FES = 865100	$\mu_F = 0.730$	$\mu_{CR} = 0.205$	
Diversity = 8.5551e-01	k_t = [1, 1, 1, 1]			
Generation 8660: Best = 3.811389e+41	FES = 866100	$\mu_F = 0.736$	$\mu_{CR} = 0.200$	
Diversity = 8.6945e-01	k_t = [3, 3, 4, 2]			
Generation 8670: Best = 3.550182e+41	FES = 867100	$\mu_F = 0.738$	$\mu_{CR} = 0.193$	
Diversity = 8.3831e-01	k_t = [1, 1, 1, 1]			
Generation 8680: Best = 3.339246e+41	FES = 868100	$\mu_F = 0.748$	$\mu_{CR} = 0.189$	
Diversity = 8.2080e-01	k_t = [4, 3, 3, 3]			
Generation 8690: Best = 3.123392e+41	FES = 869100	$\mu_F = 0.753$	$\mu_{CR} = 0.190$	
Diversity = 8.1411e-01	k_t = [1, 1, 1, 1]			
Generation 8700: Best = 2.984619e+41	FES = 870100	$\mu_F = 0.758$	$\mu_{CR} = 0.190$	
Diversity = 8.1058e-01	k_t = [4, 3, 2, 3]			
Generation 8710: Best = 2.816136e+41	FES = 871100	$\mu_F = 0.741$	$\mu_{CR} = 0.188$	
Diversity = 8.8719e-01	k_t = [1, 1, 1, 1]			
Generation 8720: Best = 2.692558e+41	FES = 872100	$\mu_F = 0.731$	$\mu_{CR} = 0.187$	
Diversity = 8.3787e-01	k_t = [4, 2, 2, 3]			
Generation 8730: Best = 2.549786e+41	FES = 873100	$\mu_F = 0.742$	$\mu_{CR} = 0.193$	
Diversity = 8.8148e-01	k_t = [1, 1, 1, 1]			
Generation 8740: Best = 2.404194e+41	FES = 874100	$\mu_F = 0.747$	$\mu_{CR} = 0.196$	
Diversity = 8.8755e-01	k_t = [4, 2, 2, 4]			
Generation 8750: Best = 2.212298e+41	FES = 875100	$\mu_F = 0.749$	$\mu_{CR} = 0.190$	
Diversity = 9.2013e-01	k_t = [1, 1, 1, 1]			
Generation 8760: Best = 2.145486e+41	FES = 876100	$\mu_F = 0.750$	$\mu_{CR} = 0.181$	
Diversity = 9.2006e-01	k_t = [2, 2, 2, 3]			
Generation 8770: Best = 2.055664e+41	FES = 877100	$\mu_F = 0.733$	$\mu_{CR} = 0.187$	
Diversity = 9.1186e-01	k_t = [1, 1, 1, 1]			
Generation 8780: Best = 1.986268e+41	FES = 878100	$\mu_F = 0.726$	$\mu_{CR} = 0.187$	
Diversity = 8.8347e-01	k_t = [4, 2, 3, 4]			
Generation 8790: Best = 1.897759e+41	FES = 879100	$\mu_F = 0.731$	$\mu_{CR} = 0.189$	
Diversity = 8.4697e-01	k_t = [1, 1, 1, 1]			
Generation 8800: Best = 1.794703e+41	FES = 880100	$\mu_F = 0.744$	$\mu_{CR} = 0.199$	
Diversity = 8.6863e-01	k_t = [2, 3, 4, 3]			
Generation 8810: Best = 1.699689e+41	FES = 881100	$\mu_F = 0.742$	$\mu_{CR} = 0.201$	
Diversity = 8.9455e-01	k_t = [1, 1, 1, 1]			
Generation 8820: Best = 1.625498e+41	FES = 882100	$\mu_F = 0.742$	$\mu_{CR} = 0.191$	
Diversity = 8.5953e-01	k_t = [3, 3, 3, 4]			
Generation 8830: Best = 1.547439e+41	FES = 883100	$\mu_F = 0.741$	$\mu_{CR} = 0.199$	
Diversity = 8.7211e-01	k_t = [1, 1, 1, 1]			
Generation 8840: Best = 1.463720e+41	FES = 884100	$\mu_F = 0.744$	$\mu_{CR} = 0.207$	
Diversity = 8.7586e-01	k_t = [3, 4, 2, 2]			
Generation 8850: Best = 1.395326e+41	FES = 885100	$\mu_F = 0.737$	$\mu_{CR} = 0.209$	
Diversity = 8.9334e-01	k_t = [1, 1, 1, 1]			
Generation 8860: Best = 1.319183e+41	FES = 886100	$\mu_F = 0.723$	$\mu_{CR} = 0.212$	

Diversity = 8.7575e-01	k_t = [2, 2, 4, 2]			
Generation 8870: Best = 1.214266e+41	FES = 887100	$\mu_F = 0.730$	$\mu_{CR} = 0.217$	
Diversity = 8.8177e-01	k_t = [1, 1, 1, 1]			
Generation 8880: Best = 1.152393e+41	FES = 888100	$\mu_F = 0.732$	$\mu_{CR} = 0.212$	
Diversity = 8.7700e-01	k_t = [3, 3, 2, 3]			
Generation 8890: Best = 1.109757e+41	FES = 889100	$\mu_F = 0.733$	$\mu_{CR} = 0.203$	
Diversity = 8.7672e-01	k_t = [1, 1, 1, 1]			
Generation 8900: Best = 1.071045e+41	FES = 890100	$\mu_F = 0.733$	$\mu_{CR} = 0.203$	
Diversity = 8.2955e-01	k_t = [3, 4, 3, 4]			
Generation 8910: Best = 1.029669e+41	FES = 891100	$\mu_F = 0.739$	$\mu_{CR} = 0.208$	
Diversity = 8.2868e-01	k_t = [1, 1, 1, 1]			
Generation 8920: Best = 9.995742e+40	FES = 892100	$\mu_F = 0.739$	$\mu_{CR} = 0.209$	
Diversity = 8.3275e-01	k_t = [2, 3, 3, 3]			
Generation 8930: Best = 9.612198e+40	FES = 893100	$\mu_F = 0.737$	$\mu_{CR} = 0.203$	
Diversity = 8.4809e-01	k_t = [1, 1, 1, 1]			
Generation 8940: Best = 8.839616e+40	FES = 894100	$\mu_F = 0.730$	$\mu_{CR} = 0.199$	
Diversity = 8.2620e-01	k_t = [4, 2, 3, 2]			
Generation 8950: Best = 8.627573e+40	FES = 895100	$\mu_F = 0.744$	$\mu_{CR} = 0.201$	
Diversity = 8.3091e-01	k_t = [1, 1, 1, 1]			
Generation 8960: Best = 8.285895e+40	FES = 896100	$\mu_F = 0.741$	$\mu_{CR} = 0.204$	
Diversity = 7.9108e-01	k_t = [4, 3, 3, 3]			
Generation 8970: Best = 7.857269e+40	FES = 897100	$\mu_F = 0.750$	$\mu_{CR} = 0.216$	
Diversity = 8.0480e-01	k_t = [1, 1, 1, 1]			
Generation 8980: Best = 7.413335e+40	FES = 898100	$\mu_F = 0.754$	$\mu_{CR} = 0.222$	
Diversity = 8.4938e-01	k_t = [4, 2, 3, 4]			
Generation 8990: Best = 7.234765e+40	FES = 899100	$\mu_F = 0.758$	$\mu_{CR} = 0.220$	
Diversity = 8.5352e-01	k_t = [1, 1, 1, 1]			
Generation 9000: Best = 6.803484e+40	FES = 900100	$\mu_F = 0.751$	$\mu_{CR} = 0.227$	
Diversity = 8.3580e-01	k_t = [3, 4, 3, 3]			
Generation 9010: Best = 6.642362e+40	FES = 901100	$\mu_F = 0.743$	$\mu_{CR} = 0.231$	
Diversity = 8.5052e-01	k_t = [1, 1, 1, 1]			
Generation 9020: Best = 6.351517e+40	FES = 902100	$\mu_F = 0.743$	$\mu_{CR} = 0.230$	
Diversity = 8.3967e-01	k_t = [4, 3, 2, 2]			
Generation 9030: Best = 6.066265e+40	FES = 903100	$\mu_F = 0.747$	$\mu_{CR} = 0.224$	
Diversity = 8.5497e-01	k_t = [1, 1, 1, 1]			
Generation 9040: Best = 5.818681e+40	FES = 904100	$\mu_F = 0.745$	$\mu_{CR} = 0.238$	
Diversity = 8.4477e-01	k_t = [2, 3, 3, 3]			
Generation 9050: Best = 5.599812e+40	FES = 905100	$\mu_F = 0.731$	$\mu_{CR} = 0.240$	
Diversity = 8.8782e-01	k_t = [1, 1, 1, 1]			
Generation 9060: Best = 5.405061e+40	FES = 906100	$\mu_F = 0.735$	$\mu_{CR} = 0.235$	
Diversity = 8.2489e-01	k_t = [3, 4, 3, 3]			
Generation 9070: Best = 5.190868e+40	FES = 907100	$\mu_F = 0.735$	$\mu_{CR} = 0.238$	
Diversity = 8.2229e-01	k_t = [1, 1, 1, 1]			
Generation 9080: Best = 4.985910e+40	FES = 908100	$\mu_F = 0.730$	$\mu_{CR} = 0.237$	
Diversity = 7.6597e-01	k_t = [4, 4, 4, 2]			
Generation 9090: Best = 4.797710e+40	FES = 909100	$\mu_F = 0.730$	$\mu_{CR} = 0.236$	
Diversity = 7.9158e-01	k_t = [1, 1, 1, 1]			
Generation 9100: Best = 4.563409e+40	FES = 910100	$\mu_F = 0.732$	$\mu_{CR} = 0.232$	
Diversity = 7.6583e-01	k_t = [4, 2, 2, 4]			
Generation 9110: Best = 4.381539e+40	FES = 911100	$\mu_F = 0.734$	$\mu_{CR} = 0.237$	
Diversity = 7.5585e-01	k_t = [1, 1, 1, 1]			
Generation 9120: Best = 4.104496e+40	FES = 912100	$\mu_F = 0.739$	$\mu_{CR} = 0.237$	
Diversity = 7.6774e-01	k_t = [4, 4, 4, 3]			
Generation 9130: Best = 4.013252e+40	FES = 913100	$\mu_F = 0.748$	$\mu_{CR} = 0.242$	

Diversity = 8.3098e-01	k_t = [1, 1, 1, 1]			
Generation 9140: Best = 3.897909e+40	FES = 914100	$\mu_F = 0.739$	$\mu_{CR} = 0.241$	
Diversity = 8.6237e-01	k_t = [2, 4, 4, 3]			
Generation 9150: Best = 3.746519e+40	FES = 915100	$\mu_F = 0.739$	$\mu_{CR} = 0.247$	
Diversity = 8.5083e-01	k_t = [1, 1, 1, 1]			
Generation 9160: Best = 3.637699e+40	FES = 916100	$\mu_F = 0.734$	$\mu_{CR} = 0.245$	
Diversity = 8.2328e-01	k_t = [2, 2, 3, 4]			
Generation 9170: Best = 3.477931e+40	FES = 917100	$\mu_F = 0.744$	$\mu_{CR} = 0.241$	
Diversity = 8.5617e-01	k_t = [1, 1, 1, 1]			
Generation 9180: Best = 3.359771e+40	FES = 918100	$\mu_F = 0.736$	$\mu_{CR} = 0.235$	
Diversity = 8.4453e-01	k_t = [4, 3, 2, 2]			
Generation 9190: Best = 3.221335e+40	FES = 919100	$\mu_F = 0.745$	$\mu_{CR} = 0.235$	
Diversity = 8.7675e-01	k_t = [1, 1, 1, 1]			
Generation 9200: Best = 3.091408e+40	FES = 920100	$\mu_F = 0.748$	$\mu_{CR} = 0.238$	
Diversity = 8.4245e-01	k_t = [4, 4, 2, 4]			
Generation 9210: Best = 2.910133e+40	FES = 921100	$\mu_F = 0.749$	$\mu_{CR} = 0.247$	
Diversity = 8.2781e-01	k_t = [1, 1, 1, 1]			
Generation 9220: Best = 2.739848e+40	FES = 922100	$\mu_F = 0.736$	$\mu_{CR} = 0.248$	
Diversity = 7.8579e-01	k_t = [3, 2, 4, 3]			
Generation 9230: Best = 2.601677e+40	FES = 923100	$\mu_F = 0.747$	$\mu_{CR} = 0.245$	
Diversity = 8.5163e-01	k_t = [1, 1, 1, 1]			
Generation 9240: Best = 2.528373e+40	FES = 924100	$\mu_F = 0.743$	$\mu_{CR} = 0.242$	
Diversity = 8.6001e-01	k_t = [3, 2, 2, 4]			
Generation 9250: Best = 2.455169e+40	FES = 925100	$\mu_F = 0.749$	$\mu_{CR} = 0.240$	
Diversity = 8.6778e-01	k_t = [1, 1, 1, 1]			
Generation 9260: Best = 2.343940e+40	FES = 926100	$\mu_F = 0.734$	$\mu_{CR} = 0.242$	
Diversity = 8.5771e-01	k_t = [3, 3, 2, 3]			
Generation 9270: Best = 2.257693e+40	FES = 927100	$\mu_F = 0.746$	$\mu_{CR} = 0.237$	
Diversity = 9.1278e-01	k_t = [1, 1, 1, 1]			
Generation 9280: Best = 2.190115e+40	FES = 928100	$\mu_F = 0.742$	$\mu_{CR} = 0.238$	
Diversity = 8.7880e-01	k_t = [3, 4, 3, 3]			
Generation 9290: Best = 2.127839e+40	FES = 929100	$\mu_F = 0.747$	$\mu_{CR} = 0.234$	
Diversity = 9.1425e-01	k_t = [1, 1, 1, 1]			
Generation 9300: Best = 2.072108e+40	FES = 930100	$\mu_F = 0.739$	$\mu_{CR} = 0.232$	
Diversity = 8.5725e-01	k_t = [3, 3, 2, 4]			
Generation 9310: Best = 2.011388e+40	FES = 931100	$\mu_F = 0.733$	$\mu_{CR} = 0.227$	
Diversity = 8.6279e-01	k_t = [1, 1, 1, 1]			
Generation 9320: Best = 1.891403e+40	FES = 932100	$\mu_F = 0.737$	$\mu_{CR} = 0.220$	
Diversity = 8.2847e-01	k_t = [2, 2, 2, 3]			
Generation 9330: Best = 1.810455e+40	FES = 933100	$\mu_F = 0.743$	$\mu_{CR} = 0.226$	
Diversity = 8.8050e-01	k_t = [1, 1, 1, 1]			
Generation 9340: Best = 1.752807e+40	FES = 934100	$\mu_F = 0.743$	$\mu_{CR} = 0.223$	
Diversity = 8.3368e-01	k_t = [4, 4, 3, 4]			
Generation 9350: Best = 1.665673e+40	FES = 935100	$\mu_F = 0.743$	$\mu_{CR} = 0.224$	
Diversity = 8.3432e-01	k_t = [1, 1, 1, 1]			
Generation 9360: Best = 1.595953e+40	FES = 936100	$\mu_F = 0.743$	$\mu_{CR} = 0.217$	
Diversity = 7.9003e-01	k_t = [3, 3, 2, 4]			
Generation 9370: Best = 1.540745e+40	FES = 937100	$\mu_F = 0.751$	$\mu_{CR} = 0.223$	
Diversity = 8.1255e-01	k_t = [1, 1, 1, 1]			
Generation 9380: Best = 1.488692e+40	FES = 938100	$\mu_F = 0.750$	$\mu_{CR} = 0.221$	
Diversity = 7.9347e-01	k_t = [2, 3, 3, 4]			
Generation 9390: Best = 1.440147e+40	FES = 939100	$\mu_F = 0.753$	$\mu_{CR} = 0.222$	
Diversity = 8.2447e-01	k_t = [1, 1, 1, 1]			
Generation 9400: Best = 1.393832e+40	FES = 940100	$\mu_F = 0.750$	$\mu_{CR} = 0.221$	

Diversity = 7.9878e-01	k_t = [2, 2, 3, 4]			
Generation 9410: Best = 1.346652e+40	FES = 941100	$\mu_F = 0.759$	$\mu_{CR} = 0.224$	
Diversity = 8.1092e-01	k_t = [1, 1, 1, 1]			
Generation 9420: Best = 1.314679e+40	FES = 942100	$\mu_F = 0.737$	$\mu_{CR} = 0.223$	
Diversity = 7.8388e-01	k_t = [3, 3, 4, 3]			
Generation 9430: Best = 1.285943e+40	FES = 943100	$\mu_F = 0.736$	$\mu_{CR} = 0.215$	
Diversity = 8.0130e-01	k_t = [1, 1, 1, 1]			
Generation 9440: Best = 1.224235e+40	FES = 944100	$\mu_F = 0.745$	$\mu_{CR} = 0.209$	
Diversity = 8.4898e-01	k_t = [3, 2, 2, 2]			
Generation 9450: Best = 1.195698e+40	FES = 945100	$\mu_F = 0.743$	$\mu_{CR} = 0.216$	
Diversity = 8.4888e-01	k_t = [1, 1, 1, 1]			
Generation 9460: Best = 1.145507e+40	FES = 946100	$\mu_F = 0.742$	$\mu_{CR} = 0.220$	
Diversity = 8.7183e-01	k_t = [3, 3, 4, 2]			
Generation 9470: Best = 1.102782e+40	FES = 947100	$\mu_F = 0.754$	$\mu_{CR} = 0.218$	
Diversity = 8.9893e-01	k_t = [1, 1, 1, 1]			
Generation 9480: Best = 1.063686e+40	FES = 948100	$\mu_F = 0.745$	$\mu_{CR} = 0.214$	
Diversity = 8.4183e-01	k_t = [2, 3, 3, 4]			
Generation 9490: Best = 1.022799e+40	FES = 949100	$\mu_F = 0.754$	$\mu_{CR} = 0.205$	
Diversity = 8.9902e-01	k_t = [1, 1, 1, 1]			
Generation 9500: Best = 9.693080e+39	FES = 950100	$\mu_F = 0.755$	$\mu_{CR} = 0.190$	
Diversity = 9.0805e-01	k_t = [2, 2, 2, 2]			
Generation 9510: Best = 9.446758e+39	FES = 951100	$\mu_F = 0.746$	$\mu_{CR} = 0.189$	
Diversity = 9.4070e-01	k_t = [1, 1, 1, 1]			
Generation 9520: Best = 9.147277e+39	FES = 952100	$\mu_F = 0.751$	$\mu_{CR} = 0.186$	
Diversity = 9.1445e-01	k_t = [3, 3, 4, 3]			
Generation 9530: Best = 8.785827e+39	FES = 953100	$\mu_F = 0.753$	$\mu_{CR} = 0.180$	
Diversity = 9.2195e-01	k_t = [1, 1, 1, 1]			
Generation 9540: Best = 8.530435e+39	FES = 954100	$\mu_F = 0.757$	$\mu_{CR} = 0.182$	
Diversity = 8.5271e-01	k_t = [3, 2, 4, 4]			
Generation 9550: Best = 8.171767e+39	FES = 955100	$\mu_F = 0.754$	$\mu_{CR} = 0.184$	
Diversity = 8.8903e-01	k_t = [1, 1, 1, 1]			
Generation 9560: Best = 7.975656e+39	FES = 956100	$\mu_F = 0.749$	$\mu_{CR} = 0.188$	
Diversity = 9.0851e-01	k_t = [2, 2, 4, 3]			
Generation 9570: Best = 7.671306e+39	FES = 957100	$\mu_F = 0.753$	$\mu_{CR} = 0.190$	
Diversity = 9.5638e-01	k_t = [1, 1, 1, 1]			
Generation 9580: Best = 7.413280e+39	FES = 958100	$\mu_F = 0.741$	$\mu_{CR} = 0.179$	
Diversity = 8.8897e-01	k_t = [2, 3, 2, 4]			
Generation 9590: Best = 7.272088e+39	FES = 959100	$\mu_F = 0.743$	$\mu_{CR} = 0.181$	
Diversity = 8.9876e-01	k_t = [1, 1, 1, 1]			
Generation 9600: Best = 7.038981e+39	FES = 960100	$\mu_F = 0.749$	$\mu_{CR} = 0.186$	
Diversity = 8.9525e-01	k_t = [3, 3, 3, 2]			
Generation 9610: Best = 6.935285e+39	FES = 961100	$\mu_F = 0.749$	$\mu_{CR} = 0.194$	
Diversity = 9.0891e-01	k_t = [1, 1, 1, 1]			
Generation 9620: Best = 6.696997e+39	FES = 962100	$\mu_F = 0.754$	$\mu_{CR} = 0.188$	
Diversity = 8.9557e-01	k_t = [2, 3, 2, 3]			
Generation 9630: Best = 6.596120e+39	FES = 963100	$\mu_F = 0.749$	$\mu_{CR} = 0.188$	
Diversity = 9.2778e-01	k_t = [1, 1, 1, 1]			
Generation 9640: Best = 6.420682e+39	FES = 964100	$\mu_F = 0.756$	$\mu_{CR} = 0.184$	
Diversity = 9.5414e-01	k_t = [2, 4, 4, 3]			
Generation 9650: Best = 6.057870e+39	FES = 965100	$\mu_F = 0.758$	$\mu_{CR} = 0.190$	
Diversity = 9.6534e-01	k_t = [1, 1, 1, 1]			
Generation 9660: Best = 5.868862e+39	FES = 966100	$\mu_F = 0.745$	$\mu_{CR} = 0.186$	
Diversity = 9.6680e-01	k_t = [2, 4, 4, 2]			
Generation 9670: Best = 5.684870e+39	FES = 967100	$\mu_F = 0.753$	$\mu_{CR} = 0.181$	

Diversity = 9.8162e-01	k_t = [1, 1, 1, 1]			
Generation 9680: Best = 5.500518e+39	FES = 968100	$\mu_F = 0.734$	$\mu_{CR} = 0.181$	
Diversity = 9.3574e-01	k_t = [3, 3, 2, 3]			
Generation 9690: Best = 5.352604e+39	FES = 969100	$\mu_F = 0.736$	$\mu_{CR} = 0.192$	
Diversity = 9.1774e-01	k_t = [1, 1, 1, 1]			
Generation 9700: Best = 5.178509e+39	FES = 970100	$\mu_F = 0.729$	$\mu_{CR} = 0.191$	
Diversity = 8.9437e-01	k_t = [2, 4, 2, 3]			
Generation 9710: Best = 5.023500e+39	FES = 971100	$\mu_F = 0.728$	$\mu_{CR} = 0.200$	
Diversity = 8.8056e-01	k_t = [1, 1, 1, 1]			
Generation 9720: Best = 4.911000e+39	FES = 972100	$\mu_F = 0.737$	$\mu_{CR} = 0.202$	
Diversity = 9.2182e-01	k_t = [4, 2, 3, 2]			
Generation 9730: Best = 4.732233e+39	FES = 973100	$\mu_F = 0.728$	$\mu_{CR} = 0.200$	
Diversity = 9.0774e-01	k_t = [1, 1, 1, 1]			
Generation 9740: Best = 4.425636e+39	FES = 974100	$\mu_F = 0.727$	$\mu_{CR} = 0.198$	
Diversity = 8.2601e-01	k_t = [2, 4, 2, 3]			
Generation 9750: Best = 4.299081e+39	FES = 975100	$\mu_F = 0.737$	$\mu_{CR} = 0.206$	
Diversity = 8.5024e-01	k_t = [1, 1, 1, 1]			
Generation 9760: Best = 4.103972e+39	FES = 976100	$\mu_F = 0.737$	$\mu_{CR} = 0.211$	
Diversity = 8.1189e-01	k_t = [3, 3, 4, 4]			
Generation 9770: Best = 3.893854e+39	FES = 977100	$\mu_F = 0.748$	$\mu_{CR} = 0.213$	
Diversity = 8.2172e-01	k_t = [1, 1, 1, 1]			
Generation 9780: Best = 3.805468e+39	FES = 978100	$\mu_F = 0.757$	$\mu_{CR} = 0.214$	
Diversity = 7.8510e-01	k_t = [4, 3, 4, 4]			
Generation 9790: Best = 3.705546e+39	FES = 979100	$\mu_F = 0.764$	$\mu_{CR} = 0.214$	
Diversity = 8.3406e-01	k_t = [1, 1, 1, 1]			
Generation 9800: Best = 3.587458e+39	FES = 980100	$\mu_F = 0.750$	$\mu_{CR} = 0.225$	
Diversity = 8.4332e-01	k_t = [4, 4, 4, 3]			
Generation 9810: Best = 3.511307e+39	FES = 981100	$\mu_F = 0.751$	$\mu_{CR} = 0.230$	
Diversity = 8.4449e-01	k_t = [1, 1, 1, 1]			
Generation 9820: Best = 3.440276e+39	FES = 982100	$\mu_F = 0.754$	$\mu_{CR} = 0.236$	
Diversity = 8.2920e-01	k_t = [2, 4, 3, 4]			
Generation 9830: Best = 3.321650e+39	FES = 983100	$\mu_F = 0.769$	$\mu_{CR} = 0.240$	
Diversity = 8.9185e-01	k_t = [1, 1, 1, 1]			
Generation 9840: Best = 3.236441e+39	FES = 984100	$\mu_F = 0.758$	$\mu_{CR} = 0.229$	
Diversity = 8.9445e-01	k_t = [2, 2, 3, 3]			
Generation 9850: Best = 3.185863e+39	FES = 985100	$\mu_F = 0.754$	$\mu_{CR} = 0.226$	
Diversity = 8.8566e-01	k_t = [1, 1, 1, 1]			
Generation 9860: Best = 3.129696e+39	FES = 986100	$\mu_F = 0.754$	$\mu_{CR} = 0.226$	
Diversity = 8.9516e-01	k_t = [2, 4, 3, 2]			
Generation 9870: Best = 3.057915e+39	FES = 987100	$\mu_F = 0.760$	$\mu_{CR} = 0.209$	
Diversity = 9.3053e-01	k_t = [1, 1, 1, 1]			
Generation 9880: Best = 2.989827e+39	FES = 988100	$\mu_F = 0.754$	$\mu_{CR} = 0.210$	
Diversity = 9.2485e-01	k_t = [2, 2, 3, 2]			
Generation 9890: Best = 2.901175e+39	FES = 989100	$\mu_F = 0.755$	$\mu_{CR} = 0.206$	
Diversity = 9.5211e-01	k_t = [1, 1, 1, 1]			
Generation 9900: Best = 2.855102e+39	FES = 990100	$\mu_F = 0.745$	$\mu_{CR} = 0.207$	
Diversity = 8.2657e-01	k_t = [3, 2, 4, 4]			
Generation 9910: Best = 2.720848e+39	FES = 991100	$\mu_F = 0.747$	$\mu_{CR} = 0.203$	
Diversity = 8.6963e-01	k_t = [1, 1, 1, 1]			
Generation 9920: Best = 2.585408e+39	FES = 992100	$\mu_F = 0.741$	$\mu_{CR} = 0.203$	
Diversity = 8.4085e-01	k_t = [4, 3, 4, 4]			
Generation 9930: Best = 2.479175e+39	FES = 993100	$\mu_F = 0.740$	$\mu_{CR} = 0.220$	
Diversity = 8.9396e-01	k_t = [1, 1, 1, 1]			
Generation 9940: Best = 2.416599e+39	FES = 994100	$\mu_F = 0.746$	$\mu_{CR} = 0.225$	

```

Diversity = 8.6648e-01 k_t = [2, 4, 4, 3]
Generation 9950: Best = 2.309210e+39 FES = 995100  $\mu_F$  = 0.745  $\mu_{CR}$  = 0.216
Diversity = 9.0373e-01 k_t = [1, 1, 1, 1]
Generation 9960: Best = 2.252281e+39 FES = 996100  $\mu_F$  = 0.744  $\mu_{CR}$  = 0.216
Diversity = 9.1738e-01 k_t = [2, 2, 2, 2]
Generation 9970: Best = 2.162260e+39 FES = 997100  $\mu_F$  = 0.738  $\mu_{CR}$  = 0.214
Diversity = 9.4760e-01 k_t = [1, 1, 1, 1]
Generation 9980: Best = 2.109277e+39 FES = 998100  $\mu_F$  = 0.735  $\mu_{CR}$  = 0.215
Diversity = 8.7265e-01 k_t = [3, 3, 4, 4]
Generation 9990: Best = 2.061209e+39 FES = 999100  $\mu_F$  = 0.741  $\mu_{CR}$  = 0.222
Diversity = 9.0038e-01 k_t = [1, 1, 1, 1]

```

Optimization complete.

```

Best Fitness          = 2.018874e+39
Total Function Evals  = 1000000
Total Successful Updates = 431534

```

```

=====
=

```

Results:

```

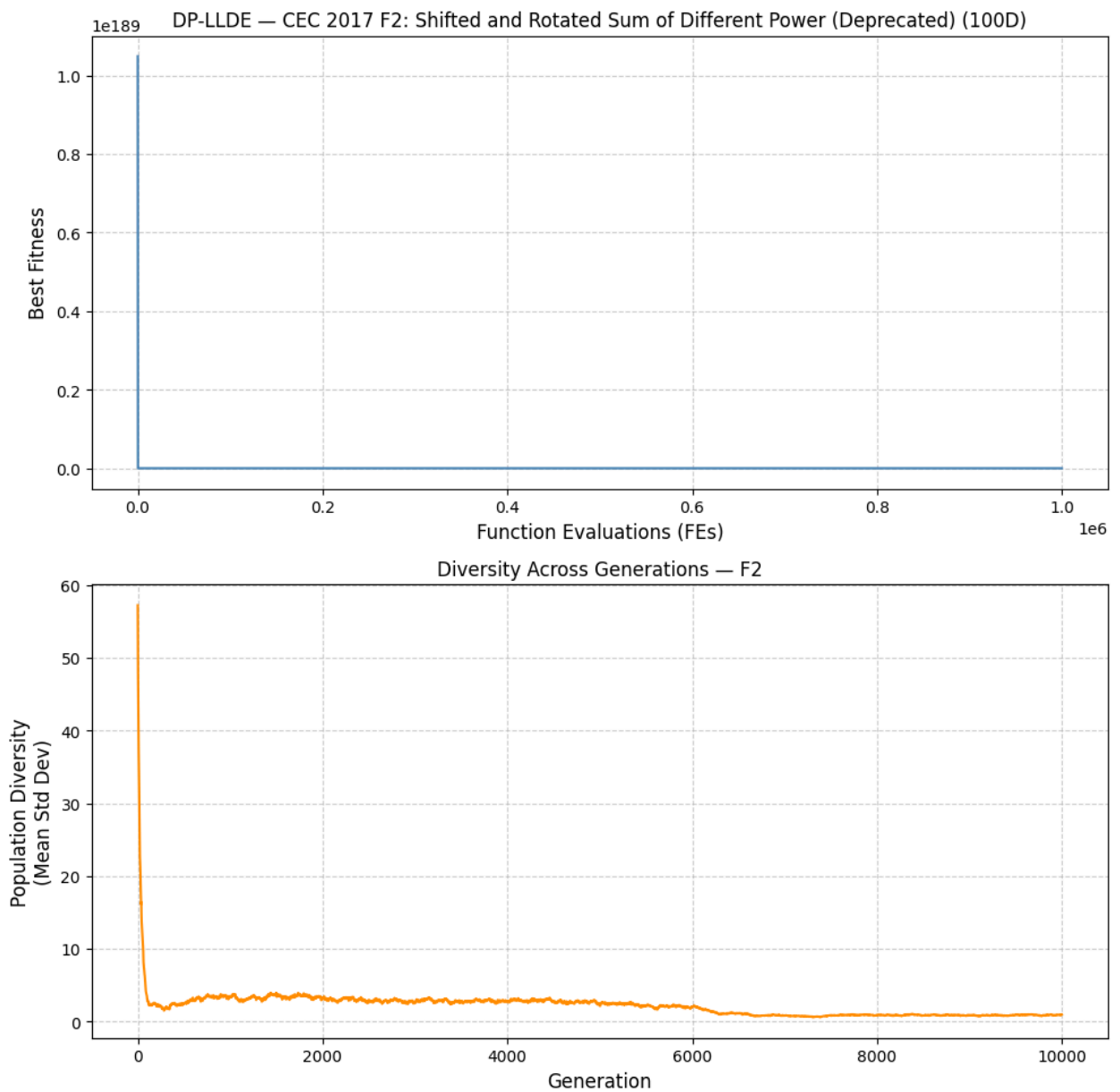
Best Fitness          : 2.018874e+39
Global Optimum        : 2.000000e+02
Error (|best-optimum|) : 2.018874e+39
Best Solution (5 dim) : [-95.60139409 -47.08249977 -56.73845887  2.1558729
6.50729657]
D^(0) (init. diversity): 5.7220e+01

```

```

=====
=
✓ Checkpoint saved to: /content/drive/My Drive/thesis_results/all_results_checkpoint.pkl

```

Graph saved as convergence_F2.png

✓ Plot saved to: /content/drive/My Drive/thesis_results/plots/convergence_F2.png

```
In [ ]: # =====
# F3 — Shifted and Rotated Zakharov
# =====
run_and_plot('F3', make_wrapper(simple.f3), 300.0, 'Shifted and Rotated Zakharov')
```

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DP-LLDE on CEC 2017 F3 – Single Run

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Function : Shifted and Rotated Zakharov (100D)

Bounds : [-100, 100]

Global optimum : 300.0

Initial $D^*(0) = 5.7220e+01$ | Diversity threshold = 5.7220e+00

Generation 0: Best Fitness = 7.602487e+06

Generation 10: Best = 6.479446e+05 FES = 1100 $\mu_F = 0.562$ $\mu_{CR} = 0.345$

Diversity = 4.5978e+01 $k_t = [1, 1, 1, 1]$

Generation 20: Best = 5.356490e+05 FES = 2100 $\mu_F = 0.577$ $\mu_{CR} = 0.351$

Diversity = 3.9790e+01 $k_t = [2, 3, 2, 3]$

Generation 30: Best = 4.494429e+05 FES = 3100 $\mu_F = 0.579$ $\mu_{CR} = 0.313$

Diversity = 3.2680e+01 $k_t = [1, 1, 1, 1]$

Generation 40: Best = 3.649203e+05 FES = 4100 $\mu_F = 0.594$ $\mu_{CR} = 0.298$

Diversity = 2.1734e+01 $k_t = [3, 2, 3, 3]$

Generation 50: Best = 3.627213e+05 FES = 5100 $\mu_F = 0.609$ $\mu_{CR} = 0.301$

Diversity = 1.3663e+01 $k_t = [1, 1, 1, 1]$

Generation 60: Best = 3.450727e+05 FES = 6100 $\mu_F = 0.649$ $\mu_{CR} = 0.287$

Diversity = 9.1411e+00 $k_t = [3, 4, 4, 3]$

Generation 70: Best = 3.421989e+05 FES = 7100 $\mu_F = 0.668$ $\mu_{CR} = 0.281$

Diversity = 6.7787e+00 $k_t = [1, 1, 1, 1]$

Generation 80: Best = 3.376511e+05 FES = 8100 $\mu_F = 0.671$ $\mu_{CR} = 0.282$

Diversity = 5.3172e+00 $k_t = [3, 4, 2, 4]$

Generation 90: Best = 3.337805e+05 FES = 9100 $\mu_F = 0.705$ $\mu_{CR} = 0.317$

Diversity = 4.1162e+00 $k_t = [1, 1, 1, 1]$

Generation 100: Best = 3.275465e+05 FES = 10100 $\mu_F = 0.725$ $\mu_{CR} = 0.333$

Diversity = 3.3707e+00 $k_t = [4, 4, 4, 4]$

Generation 110: Best = 3.255659e+05 FES = 11100 $\mu_F = 0.756$ $\mu_{CR} = 0.315$

Diversity = 3.0959e+00 $k_t = [1, 1, 1, 1]$

Generation 120: Best = 3.233532e+05 FES = 12100 $\mu_F = 0.768$ $\mu_{CR} = 0.312$

Diversity = 2.6536e+00 $k_t = [2, 3, 4, 4]$

Generation 130: Best = 3.207742e+05 FES = 13100 $\mu_F = 0.769$ $\mu_{CR} = 0.302$

Diversity = 2.1525e+00 $k_t = [1, 1, 1, 1]$

Generation 140: Best = 3.207742e+05 FES = 14100 $\mu_F = 0.777$ $\mu_{CR} = 0.315$

Diversity = 2.1347e+00 $k_t = [2, 2, 3, 2]$

Generation 150: Best = 3.155027e+05 FES = 15100 $\mu_F = 0.795$ $\mu_{CR} = 0.326$

Diversity = 2.1029e+00 $k_t = [1, 1, 1, 1]$

Generation 160: Best = 3.111964e+05 FES = 16100 $\mu_F = 0.790$ $\mu_{CR} = 0.325$

Diversity = 2.1879e+00 $k_t = [3, 4, 2, 2]$

Generation 170: Best = 3.105230e+05 FES = 17100 $\mu_F = 0.799$ $\mu_{CR} = 0.345$

Diversity = 2.3861e+00 $k_t = [1, 1, 1, 1]$

Generation 180: Best = 3.090752e+05 FES = 18100 $\mu_F = 0.799$ $\mu_{CR} = 0.333$

Diversity = 2.2828e+00 $k_t = [4, 3, 3, 2]$

Generation 190: Best = 3.058818e+05 FES = 19100 $\mu_F = 0.803$ $\mu_{CR} = 0.327$

Diversity = 2.4505e+00 $k_t = [1, 1, 1, 1]$

Generation 200: Best = 3.045136e+05 FES = 20100 $\mu_F = 0.795$ $\mu_{CR} = 0.357$

Diversity = 2.2003e+00 $k_t = [3, 2, 2, 2]$

Generation 210: Best = 2.990113e+05 FES = 21100 $\mu_F = 0.805$ $\mu_{CR} = 0.364$

Diversity = 2.2213e+00 $k_t = [1, 1, 1, 1]$

Generation 220: Best = 2.969938e+05 FES = 22100 $\mu_F = 0.828$ $\mu_{CR} = 0.377$

Diversity = 2.9527e+00 $k_t = [3, 3, 2, 4]$

Generation 230: Best = 2.961590e+05	FES = 23100	$\mu_F = 0.821$	$\mu_{CR} = 0.365$
Diversity = 2.5371e+00	k_t = [1, 1, 1, 1]		
Generation 240: Best = 2.944118e+05	FES = 24100	$\mu_F = 0.821$	$\mu_{CR} = 0.364$
Diversity = 2.0444e+00	k_t = [4, 4, 2, 4]		
Generation 250: Best = 2.938063e+05	FES = 25100	$\mu_F = 0.809$	$\mu_{CR} = 0.350$
Diversity = 1.8839e+00	k_t = [1, 1, 1, 1]		
Generation 260: Best = 2.928881e+05	FES = 26100	$\mu_F = 0.808$	$\mu_{CR} = 0.349$
Diversity = 1.5467e+00	k_t = [2, 4, 3, 4]		
Generation 270: Best = 2.919701e+05	FES = 27100	$\mu_F = 0.827$	$\mu_{CR} = 0.341$
Diversity = 1.4400e+00	k_t = [1, 1, 1, 1]		
Generation 280: Best = 2.905535e+05	FES = 28100	$\mu_F = 0.821$	$\mu_{CR} = 0.351$
Diversity = 1.3070e+00	k_t = [4, 3, 2, 4]		
Generation 290: Best = 2.898790e+05	FES = 29100	$\mu_F = 0.823$	$\mu_{CR} = 0.384$
Diversity = 1.4926e+00	k_t = [1, 1, 1, 1]		
Generation 300: Best = 2.880107e+05	FES = 30100	$\mu_F = 0.823$	$\mu_{CR} = 0.410$
Diversity = 1.4253e+00	k_t = [3, 3, 4, 3]		
Generation 310: Best = 2.861418e+05	FES = 31100	$\mu_F = 0.823$	$\mu_{CR} = 0.390$
Diversity = 1.6440e+00	k_t = [1, 1, 1, 1]		
Generation 320: Best = 2.851811e+05	FES = 32100	$\mu_F = 0.823$	$\mu_{CR} = 0.391$
Diversity = 1.7799e+00	k_t = [4, 4, 4, 2]		
Generation 330: Best = 2.825887e+05	FES = 33100	$\mu_F = 0.818$	$\mu_{CR} = 0.366$
Diversity = 1.9786e+00	k_t = [1, 1, 1, 1]		
Generation 340: Best = 2.824433e+05	FES = 34100	$\mu_F = 0.817$	$\mu_{CR} = 0.366$
Diversity = 1.9935e+00	k_t = [2, 4, 4, 3]		
Generation 350: Best = 2.805995e+05	FES = 35100	$\mu_F = 0.806$	$\mu_{CR} = 0.358$
Diversity = 2.0322e+00	k_t = [1, 1, 1, 1]		
Generation 360: Best = 2.800956e+05	FES = 36100	$\mu_F = 0.786$	$\mu_{CR} = 0.375$
Diversity = 1.7478e+00	k_t = [2, 4, 4, 3]		
Generation 370: Best = 2.783495e+05	FES = 37100	$\mu_F = 0.776$	$\mu_{CR} = 0.341$
Diversity = 1.6273e+00	k_t = [1, 1, 1, 1]		
Generation 380: Best = 2.757307e+05	FES = 38100	$\mu_F = 0.777$	$\mu_{CR} = 0.369$
Diversity = 1.5763e+00	k_t = [4, 2, 4, 3]		
Generation 390: Best = 2.757307e+05	FES = 39100	$\mu_F = 0.794$	$\mu_{CR} = 0.369$
Diversity = 1.8967e+00	k_t = [1, 1, 1, 1]		
Generation 400: Best = 2.753423e+05	FES = 40100	$\mu_F = 0.808$	$\mu_{CR} = 0.355$
Diversity = 1.6855e+00	k_t = [4, 4, 4, 4]		
Generation 410: Best = 2.740529e+05	FES = 41100	$\mu_F = 0.824$	$\mu_{CR} = 0.359$
Diversity = 1.5127e+00	k_t = [1, 1, 1, 1]		
Generation 420: Best = 2.732710e+05	FES = 42100	$\mu_F = 0.822$	$\mu_{CR} = 0.346$
Diversity = 1.5091e+00	k_t = [3, 2, 3, 2]		
Generation 430: Best = 2.720549e+05	FES = 43100	$\mu_F = 0.795$	$\mu_{CR} = 0.333$
Diversity = 1.4723e+00	k_t = [1, 1, 1, 1]		
Generation 440: Best = 2.706498e+05	FES = 44100	$\mu_F = 0.806$	$\mu_{CR} = 0.321$
Diversity = 1.3836e+00	k_t = [4, 3, 4, 4]		
Generation 450: Best = 2.701151e+05	FES = 45100	$\mu_F = 0.808$	$\mu_{CR} = 0.326$
Diversity = 1.4085e+00	k_t = [1, 1, 1, 1]		
Generation 460: Best = 2.690326e+05	FES = 46100	$\mu_F = 0.812$	$\mu_{CR} = 0.302$
Diversity = 1.4351e+00	k_t = [3, 2, 4, 3]		
Generation 470: Best = 2.682212e+05	FES = 47100	$\mu_F = 0.802$	$\mu_{CR} = 0.282$
Diversity = 1.5171e+00	k_t = [1, 1, 1, 1]		
Generation 480: Best = 2.682212e+05	FES = 48100	$\mu_F = 0.791$	$\mu_{CR} = 0.289$
Diversity = 1.3595e+00	k_t = [2, 4, 2, 4]		
Generation 490: Best = 2.672550e+05	FES = 49100	$\mu_F = 0.808$	$\mu_{CR} = 0.249$
Diversity = 1.3612e+00	k_t = [1, 1, 1, 1]		

Generation 500: Best = 2.661554e+05	FES = 50100	$\mu_F = 0.806$	$\mu_{CR} = 0.249$
Diversity = 1.1913e+00	$k_t = [2, 4, 2, 4]$		
Generation 510: Best = 2.638668e+05	FES = 51100	$\mu_F = 0.828$	$\mu_{CR} = 0.278$
Diversity = 1.3551e+00	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 2.638668e+05	FES = 52100	$\mu_F = 0.818$	$\mu_{CR} = 0.296$
Diversity = 1.4862e+00	$k_t = [2, 2, 2, 3]$		
Generation 530: Best = 2.609493e+05	FES = 53100	$\mu_F = 0.819$	$\mu_{CR} = 0.328$
Diversity = 1.7591e+00	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 2.596220e+05	FES = 54100	$\mu_F = 0.808$	$\mu_{CR} = 0.322$
Diversity = 1.8890e+00	$k_t = [3, 2, 4, 2]$		
Generation 550: Best = 2.590929e+05	FES = 55100	$\mu_F = 0.803$	$\mu_{CR} = 0.366$
Diversity = 1.9177e+00	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 2.576192e+05	FES = 56100	$\mu_F = 0.791$	$\mu_{CR} = 0.380$
Diversity = 1.5874e+00	$k_t = [2, 4, 2, 3]$		
Generation 570: Best = 2.564608e+05	FES = 57100	$\mu_F = 0.794$	$\mu_{CR} = 0.369$
Diversity = 1.6390e+00	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 2.547568e+05	FES = 58100	$\mu_F = 0.818$	$\mu_{CR} = 0.380$
Diversity = 1.8384e+00	$k_t = [2, 3, 3, 4]$		
Generation 590: Best = 2.543049e+05	FES = 59100	$\mu_F = 0.816$	$\mu_{CR} = 0.334$
Diversity = 1.7236e+00	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 2.532279e+05	FES = 60100	$\mu_F = 0.819$	$\mu_{CR} = 0.326$
Diversity = 1.6788e+00	$k_t = [2, 2, 3, 4]$		
Generation 610: Best = 2.525843e+05	FES = 61100	$\mu_F = 0.822$	$\mu_{CR} = 0.340$
Diversity = 1.4813e+00	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 2.508122e+05	FES = 62100	$\mu_F = 0.813$	$\mu_{CR} = 0.352$
Diversity = 1.4350e+00	$k_t = [2, 3, 3, 3]$		
Generation 630: Best = 2.500177e+05	FES = 63100	$\mu_F = 0.820$	$\mu_{CR} = 0.340$
Diversity = 1.5492e+00	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 2.487206e+05	FES = 64100	$\mu_F = 0.810$	$\mu_{CR} = 0.332$
Diversity = 1.7600e+00	$k_t = [4, 2, 4, 4]$		
Generation 650: Best = 2.466782e+05	FES = 65100	$\mu_F = 0.810$	$\mu_{CR} = 0.322$
Diversity = 1.7242e+00	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 2.464042e+05	FES = 66100	$\mu_F = 0.811$	$\mu_{CR} = 0.317$
Diversity = 1.7551e+00	$k_t = [3, 2, 2, 4]$		
Generation 670: Best = 2.440996e+05	FES = 67100	$\mu_F = 0.815$	$\mu_{CR} = 0.339$
Diversity = 1.7514e+00	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 2.425562e+05	FES = 68100	$\mu_F = 0.827$	$\mu_{CR} = 0.346$
Diversity = 1.9386e+00	$k_t = [2, 4, 4, 3]$		
Generation 690: Best = 2.400145e+05	FES = 69100	$\mu_F = 0.824$	$\mu_{CR} = 0.369$
Diversity = 2.0160e+00	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 2.393237e+05	FES = 70100	$\mu_F = 0.831$	$\mu_{CR} = 0.377$
Diversity = 2.1818e+00	$k_t = [2, 2, 2, 2]$		
Generation 710: Best = 2.375175e+05	FES = 71100	$\mu_F = 0.831$	$\mu_{CR} = 0.389$
Diversity = 2.3634e+00	$k_t = [1, 1, 1, 1]$		
Generation 720: Best = 2.362406e+05	FES = 72100	$\mu_F = 0.817$	$\mu_{CR} = 0.364$
Diversity = 2.1000e+00	$k_t = [2, 2, 3, 3]$		
Generation 730: Best = 2.351488e+05	FES = 73100	$\mu_F = 0.808$	$\mu_{CR} = 0.381$
Diversity = 2.1192e+00	$k_t = [1, 1, 1, 1]$		
Generation 740: Best = 2.332537e+05	FES = 74100	$\mu_F = 0.795$	$\mu_{CR} = 0.363$
Diversity = 1.8638e+00	$k_t = [4, 2, 2, 2]$		
Generation 750: Best = 2.320663e+05	FES = 75100	$\mu_F = 0.780$	$\mu_{CR} = 0.351$
Diversity = 1.8775e+00	$k_t = [1, 1, 1, 1]$		
Generation 760: Best = 2.302000e+05	FES = 76100	$\mu_F = 0.780$	$\mu_{CR} = 0.352$
Diversity = 1.8828e+00	$k_t = [2, 3, 2, 2]$		

Generation 770:	Best = 2.300597e+05	FES = 77100	$\mu_F = 0.790$	$\mu_{CR} = 0.351$
Diversity = 1.9727e+00	$k_t = [1, 1, 1, 1]$			
Generation 780:	Best = 2.295204e+05	FES = 78100	$\mu_F = 0.789$	$\mu_{CR} = 0.348$
Diversity = 1.7114e+00	$k_t = [2, 2, 3, 3]$			
Generation 790:	Best = 2.281626e+05	FES = 79100	$\mu_F = 0.801$	$\mu_{CR} = 0.326$
Diversity = 1.6974e+00	$k_t = [1, 1, 1, 1]$			
Generation 800:	Best = 2.281224e+05	FES = 80100	$\mu_F = 0.815$	$\mu_{CR} = 0.301$
Diversity = 1.7230e+00	$k_t = [2, 2, 3, 2]$			
Generation 810:	Best = 2.253342e+05	FES = 81100	$\mu_F = 0.811$	$\mu_{CR} = 0.332$
Diversity = 1.7418e+00	$k_t = [1, 1, 1, 1]$			
Generation 820:	Best = 2.253342e+05	FES = 82100	$\mu_F = 0.815$	$\mu_{CR} = 0.331$
Diversity = 2.0547e+00	$k_t = [2, 4, 4, 2]$			
Generation 830:	Best = 2.246651e+05	FES = 83100	$\mu_F = 0.816$	$\mu_{CR} = 0.308$
Diversity = 1.9702e+00	$k_t = [1, 1, 1, 1]$			
Generation 840:	Best = 2.234212e+05	FES = 84100	$\mu_F = 0.811$	$\mu_{CR} = 0.326$
Diversity = 1.7521e+00	$k_t = [3, 2, 3, 3]$			
Generation 850:	Best = 2.206789e+05	FES = 85100	$\mu_F = 0.802$	$\mu_{CR} = 0.327$
Diversity = 1.7680e+00	$k_t = [1, 1, 1, 1]$			
Generation 860:	Best = 2.197751e+05	FES = 86100	$\mu_F = 0.828$	$\mu_{CR} = 0.326$
Diversity = 1.9690e+00	$k_t = [4, 3, 2, 2]$			
Generation 870:	Best = 2.177154e+05	FES = 87100	$\mu_F = 0.816$	$\mu_{CR} = 0.339$
Diversity = 1.8908e+00	$k_t = [1, 1, 1, 1]$			
Generation 880:	Best = 2.158516e+05	FES = 88100	$\mu_F = 0.827$	$\mu_{CR} = 0.329$
Diversity = 1.8941e+00	$k_t = [2, 3, 2, 2]$			
Generation 890:	Best = 2.151607e+05	FES = 89100	$\mu_F = 0.842$	$\mu_{CR} = 0.309$
Diversity = 2.0723e+00	$k_t = [1, 1, 1, 1]$			
Generation 900:	Best = 2.125318e+05	FES = 90100	$\mu_F = 0.844$	$\mu_{CR} = 0.296$
Diversity = 2.0769e+00	$k_t = [3, 3, 2, 4]$			
Generation 910:	Best = 2.115598e+05	FES = 91100	$\mu_F = 0.839$	$\mu_{CR} = 0.284$
Diversity = 2.3300e+00	$k_t = [1, 1, 1, 1]$			
Generation 920:	Best = 2.105606e+05	FES = 92100	$\mu_F = 0.844$	$\mu_{CR} = 0.327$
Diversity = 2.1412e+00	$k_t = [2, 3, 3, 3]$			
Generation 930:	Best = 2.069867e+05	FES = 93100	$\mu_F = 0.838$	$\mu_{CR} = 0.367$
Diversity = 2.4687e+00	$k_t = [1, 1, 1, 1]$			
Generation 940:	Best = 2.060541e+05	FES = 94100	$\mu_F = 0.832$	$\mu_{CR} = 0.375$
Diversity = 2.4248e+00	$k_t = [3, 3, 3, 4]$			
Generation 950:	Best = 2.042839e+05	FES = 95100	$\mu_F = 0.817$	$\mu_{CR} = 0.390$
Diversity = 2.1475e+00	$k_t = [1, 1, 1, 1]$			
Generation 960:	Best = 2.027963e+05	FES = 96100	$\mu_F = 0.817$	$\mu_{CR} = 0.373$
Diversity = 2.0754e+00	$k_t = [4, 2, 2, 4]$			
Generation 970:	Best = 2.019475e+05	FES = 97100	$\mu_F = 0.823$	$\mu_{CR} = 0.373$
Diversity = 2.1092e+00	$k_t = [1, 1, 1, 1]$			
Generation 980:	Best = 1.988738e+05	FES = 98100	$\mu_F = 0.813$	$\mu_{CR} = 0.371$
Diversity = 2.1559e+00	$k_t = [3, 3, 4, 2]$			
Generation 990:	Best = 1.988738e+05	FES = 99100	$\mu_F = 0.816$	$\mu_{CR} = 0.371$
Diversity = 2.2302e+00	$k_t = [1, 1, 1, 1]$			
Generation 1000:	Best = 1.988738e+05	FES = 100100	$\mu_F = 0.814$	$\mu_{CR} = 0.373$
Diversity = 1.9604e+00	$k_t = [4, 4, 4, 4]$			
Generation 1010:	Best = 1.981071e+05	FES = 101100	$\mu_F = 0.811$	$\mu_{CR} = 0.372$
Diversity = 1.7393e+00	$k_t = [1, 1, 1, 1]$			
Generation 1020:	Best = 1.971679e+05	FES = 102100	$\mu_F = 0.814$	$\mu_{CR} = 0.352$
Diversity = 1.6362e+00	$k_t = [3, 4, 2, 3]$			
Generation 1030:	Best = 1.956111e+05	FES = 103100	$\mu_F = 0.819$	$\mu_{CR} = 0.380$
Diversity = 1.6882e+00	$k_t = [1, 1, 1, 1]$			

Generation 1040: Best = 1.955703e+05 FES = 104100 μ_F = 0.821 μ_{CR} = 0.379
 Diversity = 1.8905e+00 k_t = [3, 3, 4, 3]
 Generation 1050: Best = 1.950285e+05 FES = 105100 μ_F = 0.818 μ_{CR} = 0.353
 Diversity = 1.6173e+00 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 1.944300e+05 FES = 106100 μ_F = 0.803 μ_{CR} = 0.333
 Diversity = 1.3809e+00 k_t = [4, 2, 4, 2]
 Generation 1070: Best = 1.934034e+05 FES = 107100 μ_F = 0.819 μ_{CR} = 0.323
 Diversity = 1.3248e+00 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 1.921236e+05 FES = 108100 μ_F = 0.830 μ_{CR} = 0.317
 Diversity = 1.5505e+00 k_t = [2, 2, 4, 2]
 Generation 1090: Best = 1.912727e+05 FES = 109100 μ_F = 0.831 μ_{CR} = 0.316
 Diversity = 1.7477e+00 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 1.900402e+05 FES = 110100 μ_F = 0.837 μ_{CR} = 0.355
 Diversity = 1.9008e+00 k_t = [3, 4, 3, 3]
 Generation 1110: Best = 1.877897e+05 FES = 111100 μ_F = 0.824 μ_{CR} = 0.370
 Diversity = 1.8246e+00 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 1.871526e+05 FES = 112100 μ_F = 0.822 μ_{CR} = 0.391
 Diversity = 1.9586e+00 k_t = [2, 4, 3, 4]
 Generation 1130: Best = 1.859887e+05 FES = 113100 μ_F = 0.823 μ_{CR} = 0.411
 Diversity = 1.8726e+00 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 1.853942e+05 FES = 114100 μ_F = 0.823 μ_{CR} = 0.397
 Diversity = 1.9103e+00 k_t = [4, 3, 4, 3]
 Generation 1150: Best = 1.840797e+05 FES = 115100 μ_F = 0.816 μ_{CR} = 0.376
 Diversity = 2.0553e+00 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 1.836453e+05 FES = 116100 μ_F = 0.807 μ_{CR} = 0.370
 Diversity = 1.8051e+00 k_t = [2, 3, 4, 4]
 Generation 1170: Best = 1.830098e+05 FES = 117100 μ_F = 0.805 μ_{CR} = 0.377
 Diversity = 1.5706e+00 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 1.818263e+05 FES = 118100 μ_F = 0.794 μ_{CR} = 0.379
 Diversity = 1.5789e+00 k_t = [2, 3, 3, 3]
 Generation 1190: Best = 1.810182e+05 FES = 119100 μ_F = 0.784 μ_{CR} = 0.395
 Diversity = 1.5605e+00 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 1.801599e+05 FES = 120100 μ_F = 0.782 μ_{CR} = 0.367
 Diversity = 1.5130e+00 k_t = [3, 3, 4, 3]
 Generation 1210: Best = 1.795929e+05 FES = 121100 μ_F = 0.785 μ_{CR} = 0.349
 Diversity = 1.3130e+00 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 1.768819e+05 FES = 122100 μ_F = 0.786 μ_{CR} = 0.382
 Diversity = 1.3232e+00 k_t = [4, 2, 4, 3]
 Generation 1230: Best = 1.754310e+05 FES = 123100 μ_F = 0.805 μ_{CR} = 0.406
 Diversity = 1.6610e+00 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 1.721534e+05 FES = 124100 μ_F = 0.806 μ_{CR} = 0.367
 Diversity = 1.6746e+00 k_t = [2, 4, 2, 3]
 Generation 1250: Best = 1.721534e+05 FES = 125100 μ_F = 0.795 μ_{CR} = 0.371
 Diversity = 1.7199e+00 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 1.720728e+05 FES = 126100 μ_F = 0.805 μ_{CR} = 0.349
 Diversity = 1.5637e+00 k_t = [4, 2, 2, 3]
 Generation 1270: Best = 1.716642e+05 FES = 127100 μ_F = 0.794 μ_{CR} = 0.390
 Diversity = 1.4986e+00 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 1.709160e+05 FES = 128100 μ_F = 0.799 μ_{CR} = 0.408
 Diversity = 1.4152e+00 k_t = [3, 2, 4, 4]
 Generation 1290: Best = 1.684010e+05 FES = 129100 μ_F = 0.816 μ_{CR} = 0.418
 Diversity = 1.4826e+00 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 1.684010e+05 FES = 130100 μ_F = 0.816 μ_{CR} = 0.475
 Diversity = 1.5000e+00 k_t = [3, 4, 2, 3]

Generation 1310: Best = 1.665589e+05 FES = 131100 μ_F = 0.817 μ_{CR} = 0.483
 Diversity = 1.5736e+00 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 1.653105e+05 FES = 132100 μ_F = 0.819 μ_{CR} = 0.475
 Diversity = 1.4144e+00 k_t = [4, 3, 2, 4]
 Generation 1330: Best = 1.645379e+05 FES = 133100 μ_F = 0.830 μ_{CR} = 0.430
 Diversity = 1.3901e+00 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 1.640220e+05 FES = 134100 μ_F = 0.827 μ_{CR} = 0.469
 Diversity = 1.3680e+00 k_t = [2, 4, 4, 4]
 Generation 1350: Best = 1.629097e+05 FES = 135100 μ_F = 0.811 μ_{CR} = 0.445
 Diversity = 1.3770e+00 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 1.613557e+05 FES = 136100 μ_F = 0.810 μ_{CR} = 0.458
 Diversity = 1.4960e+00 k_t = [4, 3, 2, 4]
 Generation 1370: Best = 1.609147e+05 FES = 137100 μ_F = 0.814 μ_{CR} = 0.463
 Diversity = 1.4967e+00 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 1.594258e+05 FES = 138100 μ_F = 0.818 μ_{CR} = 0.499
 Diversity = 1.3523e+00 k_t = [3, 4, 3, 4]
 Generation 1390: Best = 1.587729e+05 FES = 139100 μ_F = 0.820 μ_{CR} = 0.519
 Diversity = 1.4790e+00 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 1.575284e+05 FES = 140100 μ_F = 0.822 μ_{CR} = 0.525
 Diversity = 1.6532e+00 k_t = [2, 3, 2, 2]
 Generation 1410: Best = 1.572818e+05 FES = 141100 μ_F = 0.812 μ_{CR} = 0.498
 Diversity = 1.5697e+00 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 1.556515e+05 FES = 142100 μ_F = 0.819 μ_{CR} = 0.494
 Diversity = 1.4936e+00 k_t = [4, 3, 3, 3]
 Generation 1430: Best = 1.538492e+05 FES = 143100 μ_F = 0.820 μ_{CR} = 0.488
 Diversity = 1.6756e+00 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 1.533335e+05 FES = 144100 μ_F = 0.838 μ_{CR} = 0.501
 Diversity = 1.9623e+00 k_t = [2, 4, 2, 3]
 Generation 1450: Best = 1.522930e+05 FES = 145100 μ_F = 0.840 μ_{CR} = 0.514
 Diversity = 2.0848e+00 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 1.514344e+05 FES = 146100 μ_F = 0.831 μ_{CR} = 0.514
 Diversity = 1.9326e+00 k_t = [3, 4, 4, 3]
 Generation 1470: Best = 1.513267e+05 FES = 147100 μ_F = 0.823 μ_{CR} = 0.495
 Diversity = 1.7220e+00 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 1.497703e+05 FES = 148100 μ_F = 0.824 μ_{CR} = 0.500
 Diversity = 1.6114e+00 k_t = [4, 3, 4, 2]
 Generation 1490: Best = 1.485938e+05 FES = 149100 μ_F = 0.823 μ_{CR} = 0.526
 Diversity = 1.7433e+00 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 1.481414e+05 FES = 150100 μ_F = 0.806 μ_{CR} = 0.505
 Diversity = 1.7996e+00 k_t = [2, 2, 4, 3]
 Generation 1510: Best = 1.462789e+05 FES = 151100 μ_F = 0.819 μ_{CR} = 0.525
 Diversity = 1.3429e+00 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 1.453169e+05 FES = 152100 μ_F = 0.805 μ_{CR} = 0.520
 Diversity = 1.3077e+00 k_t = [3, 3, 2, 4]
 Generation 1530: Best = 1.446884e+05 FES = 153100 μ_F = 0.824 μ_{CR} = 0.602
 Diversity = 1.4243e+00 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 1.433310e+05 FES = 154100 μ_F = 0.823 μ_{CR} = 0.590
 Diversity = 1.3087e+00 k_t = [2, 3, 4, 3]
 Generation 1550: Best = 1.427305e+05 FES = 155100 μ_F = 0.807 μ_{CR} = 0.576
 Diversity = 1.3130e+00 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 1.419181e+05 FES = 156100 μ_F = 0.820 μ_{CR} = 0.565
 Diversity = 1.2797e+00 k_t = [4, 2, 3, 4]
 Generation 1570: Best = 1.413475e+05 FES = 157100 μ_F = 0.813 μ_{CR} = 0.569
 Diversity = 1.4024e+00 k_t = [1, 1, 1, 1]

Generation 1580: Best = 1.407316e+05 FES = 158100 μ_F = 0.804 μ_{CR} = 0.571
 Diversity = 1.2455e+00 k_t = [4, 3, 2, 3]
 Generation 1590: Best = 1.403706e+05 FES = 159100 μ_F = 0.789 μ_{CR} = 0.552
 Diversity = 1.0838e+00 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 1.402270e+05 FES = 160100 μ_F = 0.792 μ_{CR} = 0.543
 Diversity = 1.0645e+00 k_t = [2, 4, 3, 3]
 Generation 1610: Best = 1.394297e+05 FES = 161100 μ_F = 0.798 μ_{CR} = 0.530
 Diversity = 1.0642e+00 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 1.390110e+05 FES = 162100 μ_F = 0.782 μ_{CR} = 0.492
 Diversity = 1.0694e+00 k_t = [4, 4, 2, 2]
 Generation 1630: Best = 1.384591e+05 FES = 163100 μ_F = 0.790 μ_{CR} = 0.523
 Diversity = 1.0742e+00 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 1.380335e+05 FES = 164100 μ_F = 0.790 μ_{CR} = 0.529
 Diversity = 1.0495e+00 k_t = [4, 2, 4, 4]
 Generation 1650: Best = 1.377484e+05 FES = 165100 μ_F = 0.811 μ_{CR} = 0.524
 Diversity = 1.0961e+00 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 1.370469e+05 FES = 166100 μ_F = 0.820 μ_{CR} = 0.576
 Diversity = 1.2009e+00 k_t = [3, 4, 2, 3]
 Generation 1670: Best = 1.354287e+05 FES = 167100 μ_F = 0.810 μ_{CR} = 0.663
 Diversity = 1.1790e+00 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 1.354287e+05 FES = 168100 μ_F = 0.804 μ_{CR} = 0.672
 Diversity = 1.1387e+00 k_t = [4, 2, 4, 4]
 Generation 1690: Best = 1.351129e+05 FES = 169100 μ_F = 0.810 μ_{CR} = 0.682
 Diversity = 1.1553e+00 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 1.346569e+05 FES = 170100 μ_F = 0.802 μ_{CR} = 0.747
 Diversity = 1.0444e+00 k_t = [2, 2, 4, 2]
 Generation 1710: Best = 1.344705e+05 FES = 171100 μ_F = 0.806 μ_{CR} = 0.844
 Diversity = 1.0261e+00 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 1.337235e+05 FES = 172100 μ_F = 0.795 μ_{CR} = 0.940
 Diversity = 9.6544e-01 k_t = [3, 4, 2, 3]
 Generation 1730: Best = 1.323075e+05 FES = 173100 μ_F = 0.808 μ_{CR} = 0.974
 Diversity = 8.2106e-01 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 1.310930e+05 FES = 174100 μ_F = 0.814 μ_{CR} = 0.987
 Diversity = 7.7609e-01 k_t = [2, 4, 2, 3]
 Generation 1750: Best = 1.306506e+05 FES = 175100 μ_F = 0.811 μ_{CR} = 0.992
 Diversity = 6.1421e-01 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 1.299594e+05 FES = 176100 μ_F = 0.828 μ_{CR} = 0.993
 Diversity = 5.5526e-01 k_t = [4, 3, 2, 4]
 Generation 1770: Best = 1.293194e+05 FES = 177100 μ_F = 0.827 μ_{CR} = 0.995
 Diversity = 5.3521e-01 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 1.284455e+05 FES = 178100 μ_F = 0.835 μ_{CR} = 0.996
 Diversity = 5.8821e-01 k_t = [4, 2, 4, 3]
 Generation 1790: Best = 1.276098e+05 FES = 179100 μ_F = 0.837 μ_{CR} = 0.995
 Diversity = 5.9282e-01 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 1.270481e+05 FES = 180100 μ_F = 0.840 μ_{CR} = 0.996
 Diversity = 5.8451e-01 k_t = [2, 3, 3, 4]
 Generation 1810: Best = 1.259945e+05 FES = 181100 μ_F = 0.842 μ_{CR} = 0.994
 Diversity = 7.1847e-01 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 1.252177e+05 FES = 182100 μ_F = 0.841 μ_{CR} = 0.995
 Diversity = 7.0017e-01 k_t = [4, 4, 4, 2]
 Generation 1830: Best = 1.248193e+05 FES = 183100 μ_F = 0.833 μ_{CR} = 0.997
 Diversity = 5.7631e-01 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 1.242339e+05 FES = 184100 μ_F = 0.840 μ_{CR} = 0.996
 Diversity = 6.1110e-01 k_t = [2, 4, 2, 3]

Generation 1850: Best = 1.237138e+05 FES = 185100 μ_F = 0.838 μ_{CR} = 0.997
 Diversity = 5.9497e-01 k_t = [1, 1, 1, 1]
 Generation 1860: Best = 1.234528e+05 FES = 186100 μ_F = 0.828 μ_{CR} = 0.997
 Diversity = 4.2201e-01 k_t = [2, 4, 3, 4]
 Generation 1870: Best = 1.228991e+05 FES = 187100 μ_F = 0.820 μ_{CR} = 0.996
 Diversity = 6.0842e-01 k_t = [1, 1, 1, 1]
 Generation 1880: Best = 1.221278e+05 FES = 188100 μ_F = 0.825 μ_{CR} = 0.997
 Diversity = 5.0932e-01 k_t = [2, 3, 2, 3]
 Generation 1890: Best = 1.211813e+05 FES = 189100 μ_F = 0.840 μ_{CR} = 0.996
 Diversity = 6.8035e-01 k_t = [1, 1, 1, 1]
 Generation 1900: Best = 1.204131e+05 FES = 190100 μ_F = 0.844 μ_{CR} = 0.996
 Diversity = 7.2181e-01 k_t = [3, 3, 4, 2]
 Generation 1910: Best = 1.191255e+05 FES = 191100 μ_F = 0.838 μ_{CR} = 0.997
 Diversity = 7.1586e-01 k_t = [1, 1, 1, 1]
 Generation 1920: Best = 1.180197e+05 FES = 192100 μ_F = 0.844 μ_{CR} = 0.997
 Diversity = 7.6731e-01 k_t = [4, 2, 3, 2]
 Generation 1930: Best = 1.171580e+05 FES = 193100 μ_F = 0.839 μ_{CR} = 0.996
 Diversity = 6.8470e-01 k_t = [1, 1, 1, 1]
 Generation 1940: Best = 1.162318e+05 FES = 194100 μ_F = 0.847 μ_{CR} = 0.996
 Diversity = 6.1436e-01 k_t = [4, 3, 4, 4]
 Generation 1950: Best = 1.158837e+05 FES = 195100 μ_F = 0.845 μ_{CR} = 0.995
 Diversity = 5.3334e-01 k_t = [1, 1, 1, 1]
 Generation 1960: Best = 1.155744e+05 FES = 196100 μ_F = 0.841 μ_{CR} = 0.995
 Diversity = 4.4168e-01 k_t = [3, 4, 4, 4]
 Generation 1970: Best = 1.154894e+05 FES = 197100 μ_F = 0.832 μ_{CR} = 0.994
 Diversity = 3.4555e-01 k_t = [1, 1, 1, 1]
 Generation 1980: Best = 1.152361e+05 FES = 198100 μ_F = 0.828 μ_{CR} = 0.994
 Diversity = 4.8943e-01 k_t = [3, 4, 2, 4]
 Generation 1990: Best = 1.149765e+05 FES = 199100 μ_F = 0.834 μ_{CR} = 0.994
 Diversity = 4.2159e-01 k_t = [1, 1, 1, 1]
 Generation 2000: Best = 1.143545e+05 FES = 200100 μ_F = 0.828 μ_{CR} = 0.995
 Diversity = 5.9444e-01 k_t = [3, 2, 2, 2]
 Generation 2010: Best = 1.135578e+05 FES = 201100 μ_F = 0.837 μ_{CR} = 0.996
 Diversity = 5.6462e-01 k_t = [1, 1, 1, 1]
 Generation 2020: Best = 1.128855e+05 FES = 202100 μ_F = 0.847 μ_{CR} = 0.996
 Diversity = 5.4565e-01 k_t = [4, 4, 3, 2]
 Generation 2030: Best = 1.124474e+05 FES = 203100 μ_F = 0.843 μ_{CR} = 0.996
 Diversity = 5.5621e-01 k_t = [1, 1, 1, 1]
 Generation 2040: Best = 1.119309e+05 FES = 204100 μ_F = 0.842 μ_{CR} = 0.994
 Diversity = 5.2555e-01 k_t = [2, 4, 3, 2]
 Generation 2050: Best = 1.111198e+05 FES = 205100 μ_F = 0.844 μ_{CR} = 0.995
 Diversity = 7.4988e-01 k_t = [1, 1, 1, 1]
 Generation 2060: Best = 1.104888e+05 FES = 206100 μ_F = 0.847 μ_{CR} = 0.996
 Diversity = 6.2630e-01 k_t = [2, 2, 2, 3]
 Generation 2070: Best = 1.096069e+05 FES = 207100 μ_F = 0.839 μ_{CR} = 0.994
 Diversity = 6.5565e-01 k_t = [1, 1, 1, 1]
 Generation 2080: Best = 1.088520e+05 FES = 208100 μ_F = 0.834 μ_{CR} = 0.994
 Diversity = 6.5222e-01 k_t = [4, 2, 2, 2]
 Generation 2090: Best = 1.073376e+05 FES = 209100 μ_F = 0.830 μ_{CR} = 0.993
 Diversity = 6.9538e-01 k_t = [1, 1, 1, 1]
 Generation 2100: Best = 1.061713e+05 FES = 210100 μ_F = 0.826 μ_{CR} = 0.993
 Diversity = 6.4093e-01 k_t = [2, 2, 4, 2]
 Generation 2110: Best = 1.051064e+05 FES = 211100 μ_F = 0.818 μ_{CR} = 0.994
 Diversity = 6.2015e-01 k_t = [1, 1, 1, 1]

Generation 2120: Best = 1.048072e+05 FES = 212100 μ_F = 0.822 μ_{CR} = 0.995
 Diversity = 4.4293e-01 k_t = [2, 2, 4, 4]
 Generation 2130: Best = 1.034062e+05 FES = 213100 μ_F = 0.826 μ_{CR} = 0.996
 Diversity = 7.3593e-01 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 1.027379e+05 FES = 214100 μ_F = 0.828 μ_{CR} = 0.997
 Diversity = 5.4670e-01 k_t = [3, 3, 2, 4]
 Generation 2150: Best = 1.022545e+05 FES = 215100 μ_F = 0.826 μ_{CR} = 0.996
 Diversity = 4.6932e-01 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 1.015705e+05 FES = 216100 μ_F = 0.824 μ_{CR} = 0.995
 Diversity = 4.7719e-01 k_t = [3, 3, 3, 4]
 Generation 2170: Best = 1.010124e+05 FES = 217100 μ_F = 0.825 μ_{CR} = 0.993
 Diversity = 5.4189e-01 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 1.006258e+05 FES = 218100 μ_F = 0.832 μ_{CR} = 0.995
 Diversity = 4.4827e-01 k_t = [4, 3, 2, 4]
 Generation 2190: Best = 9.986181e+04 FES = 219100 μ_F = 0.833 μ_{CR} = 0.993
 Diversity = 6.0338e-01 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 9.922420e+04 FES = 220100 μ_F = 0.835 μ_{CR} = 0.995
 Diversity = 4.6731e-01 k_t = [2, 4, 3, 4]
 Generation 2210: Best = 9.797947e+04 FES = 221100 μ_F = 0.841 μ_{CR} = 0.994
 Diversity = 6.9541e-01 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 9.717242e+04 FES = 222100 μ_F = 0.840 μ_{CR} = 0.995
 Diversity = 5.4448e-01 k_t = [2, 2, 3, 2]
 Generation 2230: Best = 9.625466e+04 FES = 223100 μ_F = 0.835 μ_{CR} = 0.993
 Diversity = 5.5409e-01 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 9.524081e+04 FES = 224100 μ_F = 0.846 μ_{CR} = 0.993
 Diversity = 6.7417e-01 k_t = [4, 3, 2, 4]
 Generation 2250: Best = 9.444658e+04 FES = 225100 μ_F = 0.846 μ_{CR} = 0.995
 Diversity = 6.1133e-01 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 9.364400e+04 FES = 226100 μ_F = 0.836 μ_{CR} = 0.995
 Diversity = 5.5652e-01 k_t = [4, 2, 4, 3]
 Generation 2270: Best = 9.324255e+04 FES = 227100 μ_F = 0.829 μ_{CR} = 0.995
 Diversity = 4.7889e-01 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 9.295429e+04 FES = 228100 μ_F = 0.825 μ_{CR} = 0.995
 Diversity = 4.1218e-01 k_t = [2, 2, 2, 2]
 Generation 2290: Best = 9.242778e+04 FES = 229100 μ_F = 0.827 μ_{CR} = 0.994
 Diversity = 5.6451e-01 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 9.161320e+04 FES = 230100 μ_F = 0.828 μ_{CR} = 0.994
 Diversity = 4.6999e-01 k_t = [4, 4, 2, 2]
 Generation 2310: Best = 9.031613e+04 FES = 231100 μ_F = 0.836 μ_{CR} = 0.994
 Diversity = 6.9917e-01 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 8.937054e+04 FES = 232100 μ_F = 0.839 μ_{CR} = 0.995
 Diversity = 5.4818e-01 k_t = [3, 3, 3, 3]
 Generation 2330: Best = 8.847958e+04 FES = 233100 μ_F = 0.834 μ_{CR} = 0.994
 Diversity = 5.9206e-01 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 8.809803e+04 FES = 234100 μ_F = 0.838 μ_{CR} = 0.995
 Diversity = 4.3070e-01 k_t = [3, 4, 2, 4]
 Generation 2350: Best = 8.755779e+04 FES = 235100 μ_F = 0.833 μ_{CR} = 0.994
 Diversity = 3.9524e-01 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 8.706998e+04 FES = 236100 μ_F = 0.841 μ_{CR} = 0.994
 Diversity = 4.5178e-01 k_t = [3, 4, 3, 3]
 Generation 2370: Best = 8.654319e+04 FES = 237100 μ_F = 0.841 μ_{CR} = 0.994
 Diversity = 4.9652e-01 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 8.623433e+04 FES = 238100 μ_F = 0.842 μ_{CR} = 0.995
 Diversity = 4.2881e-01 k_t = [2, 3, 2, 4]

Generation 2390: Best = 8.506023e+04 FES = 239100 μ_F = 0.847 μ_{CR} = 0.993
 Diversity = 6.7621e-01 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 8.341647e+04 FES = 240100 μ_F = 0.851 μ_{CR} = 0.994
 Diversity = 7.0162e-01 k_t = [2, 2, 2, 2]
 Generation 2410: Best = 8.215424e+04 FES = 241100 μ_F = 0.848 μ_{CR} = 0.994
 Diversity = 7.3669e-01 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 8.172804e+04 FES = 242100 μ_F = 0.845 μ_{CR} = 0.994
 Diversity = 4.5826e-01 k_t = [3, 3, 2, 2]
 Generation 2430: Best = 8.104685e+04 FES = 243100 μ_F = 0.842 μ_{CR} = 0.992
 Diversity = 5.8368e-01 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 8.009406e+04 FES = 244100 μ_F = 0.846 μ_{CR} = 0.994
 Diversity = 5.6147e-01 k_t = [2, 4, 4, 4]
 Generation 2450: Best = 7.982740e+04 FES = 245100 μ_F = 0.839 μ_{CR} = 0.993
 Diversity = 4.5926e-01 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 7.942441e+04 FES = 246100 μ_F = 0.831 μ_{CR} = 0.995
 Diversity = 4.2012e-01 k_t = [2, 3, 2, 3]
 Generation 2470: Best = 7.893997e+04 FES = 247100 μ_F = 0.838 μ_{CR} = 0.995
 Diversity = 5.7764e-01 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 7.855085e+04 FES = 248100 μ_F = 0.838 μ_{CR} = 0.994
 Diversity = 5.0046e-01 k_t = [3, 3, 4, 2]
 Generation 2490: Best = 7.816754e+04 FES = 249100 μ_F = 0.833 μ_{CR} = 0.994
 Diversity = 4.5342e-01 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 7.775255e+04 FES = 250100 μ_F = 0.836 μ_{CR} = 0.994
 Diversity = 4.6954e-01 k_t = [4, 4, 4, 2]
 Generation 2510: Best = 7.754203e+04 FES = 251100 μ_F = 0.832 μ_{CR} = 0.995
 Diversity = 4.1337e-01 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 7.738410e+04 FES = 252100 μ_F = 0.823 μ_{CR} = 0.994
 Diversity = 3.3042e-01 k_t = [2, 3, 2, 2]
 Generation 2530: Best = 7.724062e+04 FES = 253100 μ_F = 0.820 μ_{CR} = 0.993
 Diversity = 4.0235e-01 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 7.708134e+04 FES = 254100 μ_F = 0.820 μ_{CR} = 0.995
 Diversity = 3.4340e-01 k_t = [3, 3, 3, 2]
 Generation 2550: Best = 7.673523e+04 FES = 255100 μ_F = 0.811 μ_{CR} = 0.994
 Diversity = 4.7483e-01 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 7.644443e+04 FES = 256100 μ_F = 0.829 μ_{CR} = 0.996
 Diversity = 3.5944e-01 k_t = [4, 2, 4, 4]
 Generation 2570: Best = 7.631963e+04 FES = 257100 μ_F = 0.818 μ_{CR} = 0.994
 Diversity = 3.5294e-01 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 7.610584e+04 FES = 258100 μ_F = 0.815 μ_{CR} = 0.993
 Diversity = 4.0989e-01 k_t = [2, 3, 3, 2]
 Generation 2590: Best = 7.599733e+04 FES = 259100 μ_F = 0.806 μ_{CR} = 0.991
 Diversity = 3.4250e-01 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 7.575900e+04 FES = 260100 μ_F = 0.797 μ_{CR} = 0.993
 Diversity = 3.6351e-01 k_t = [3, 3, 2, 2]
 Generation 2610: Best = 7.558179e+04 FES = 261100 μ_F = 0.786 μ_{CR} = 0.994
 Diversity = 3.8437e-01 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 7.537929e+04 FES = 262100 μ_F = 0.786 μ_{CR} = 0.994
 Diversity = 3.2093e-01 k_t = [4, 2, 4, 2]
 Generation 2630: Best = 7.526847e+04 FES = 263100 μ_F = 0.798 μ_{CR} = 0.992
 Diversity = 3.3121e-01 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 7.518355e+04 FES = 264100 μ_F = 0.814 μ_{CR} = 0.992
 Diversity = 2.6788e-01 k_t = [2, 3, 2, 4]
 Generation 2650: Best = 7.507343e+04 FES = 265100 μ_F = 0.812 μ_{CR} = 0.991
 Diversity = 3.0394e-01 k_t = [1, 1, 1, 1]

Generation 2660: Best = 7.494622e+04 FES = 266100 μ_F = 0.808 μ_{CR} = 0.992
 Diversity = 3.1700e-01 k_t = [2, 3, 2, 3]
 Generation 2670: Best = 7.483416e+04 FES = 267100 μ_F = 0.790 μ_{CR} = 0.991
 Diversity = 2.8029e-01 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 7.471672e+04 FES = 268100 μ_F = 0.801 μ_{CR} = 0.992
 Diversity = 2.4951e-01 k_t = [3, 4, 4, 4]
 Generation 2690: Best = 7.465839e+04 FES = 269100 μ_F = 0.788 μ_{CR} = 0.992
 Diversity = 2.6316e-01 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 7.436427e+04 FES = 270100 μ_F = 0.786 μ_{CR} = 0.989
 Diversity = 3.1487e-01 k_t = [2, 4, 2, 3]
 Generation 2710: Best = 7.420284e+04 FES = 271100 μ_F = 0.771 μ_{CR} = 0.991
 Diversity = 3.1106e-01 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 7.413423e+04 FES = 272100 μ_F = 0.778 μ_{CR} = 0.992
 Diversity = 2.4959e-01 k_t = [3, 2, 3, 4]
 Generation 2730: Best = 7.410225e+04 FES = 273100 μ_F = 0.760 μ_{CR} = 0.993
 Diversity = 1.8370e-01 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 7.402846e+04 FES = 274100 μ_F = 0.762 μ_{CR} = 0.991
 Diversity = 1.9999e-01 k_t = [4, 3, 2, 3]
 Generation 2750: Best = 7.381993e+04 FES = 275100 μ_F = 0.780 μ_{CR} = 0.990
 Diversity = 2.8739e-01 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 7.356830e+04 FES = 276100 μ_F = 0.789 μ_{CR} = 0.991
 Diversity = 3.2307e-01 k_t = [2, 3, 4, 4]
 Generation 2770: Best = 7.327082e+04 FES = 277100 μ_F = 0.799 μ_{CR} = 0.990
 Diversity = 3.6736e-01 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 7.300076e+04 FES = 278100 μ_F = 0.799 μ_{CR} = 0.993
 Diversity = 3.4350e-01 k_t = [2, 3, 4, 2]
 Generation 2790: Best = 7.272906e+04 FES = 279100 μ_F = 0.805 μ_{CR} = 0.993
 Diversity = 4.0431e-01 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 7.251855e+04 FES = 280100 μ_F = 0.800 μ_{CR} = 0.994
 Diversity = 3.2058e-01 k_t = [2, 3, 2, 2]
 Generation 2810: Best = 7.218539e+04 FES = 281100 μ_F = 0.807 μ_{CR} = 0.992
 Diversity = 3.6071e-01 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 7.194301e+04 FES = 282100 μ_F = 0.810 μ_{CR} = 0.994
 Diversity = 3.3528e-01 k_t = [3, 3, 3, 4]
 Generation 2830: Best = 7.153152e+04 FES = 283100 μ_F = 0.812 μ_{CR} = 0.993
 Diversity = 4.4794e-01 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 7.122597e+04 FES = 284100 μ_F = 0.802 μ_{CR} = 0.995
 Diversity = 3.8381e-01 k_t = [2, 3, 2, 3]
 Generation 2850: Best = 7.097180e+04 FES = 285100 μ_F = 0.803 μ_{CR} = 0.996
 Diversity = 3.7601e-01 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 7.049159e+04 FES = 286100 μ_F = 0.802 μ_{CR} = 0.995
 Diversity = 4.4741e-01 k_t = [2, 3, 2, 2]
 Generation 2870: Best = 7.019589e+04 FES = 287100 μ_F = 0.808 μ_{CR} = 0.996
 Diversity = 4.1758e-01 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 6.963167e+04 FES = 288100 μ_F = 0.811 μ_{CR} = 0.995
 Diversity = 5.6579e-01 k_t = [2, 2, 4, 2]
 Generation 2890: Best = 6.946575e+04 FES = 289100 μ_F = 0.823 μ_{CR} = 0.995
 Diversity = 4.1607e-01 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 6.885087e+04 FES = 290100 μ_F = 0.833 μ_{CR} = 0.996
 Diversity = 5.5770e-01 k_t = [3, 2, 4, 4]
 Generation 2910: Best = 6.862364e+04 FES = 291100 μ_F = 0.834 μ_{CR} = 0.994
 Diversity = 4.3751e-01 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 6.817736e+04 FES = 292100 μ_F = 0.842 μ_{CR} = 0.994
 Diversity = 4.0638e-01 k_t = [4, 3, 3, 4]

Generation 2930: Best = 6.784581e+04 FES = 293100 μ_F = 0.836 μ_{CR} = 0.994
 Diversity = 4.3487e-01 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 6.738216e+04 FES = 294100 μ_F = 0.841 μ_{CR} = 0.995
 Diversity = 4.9103e-01 k_t = [2, 2, 2, 4]
 Generation 2950: Best = 6.710745e+04 FES = 295100 μ_F = 0.834 μ_{CR} = 0.993
 Diversity = 4.7191e-01 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 6.689094e+04 FES = 296100 μ_F = 0.821 μ_{CR} = 0.993
 Diversity = 3.5167e-01 k_t = [4, 2, 3, 3]
 Generation 2970: Best = 6.684530e+04 FES = 297100 μ_F = 0.793 μ_{CR} = 0.992
 Diversity = 2.6626e-01 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 6.670216e+04 FES = 298100 μ_F = 0.790 μ_{CR} = 0.990
 Diversity = 2.2416e-01 k_t = [2, 3, 4, 3]
 Generation 2990: Best = 6.647514e+04 FES = 299100 μ_F = 0.794 μ_{CR} = 0.991
 Diversity = 3.1289e-01 k_t = [1, 1, 1, 1]
 Generation 3000: Best = 6.606724e+04 FES = 300100 μ_F = 0.796 μ_{CR} = 0.991
 Diversity = 4.0700e-01 k_t = [4, 3, 3, 4]
 Generation 3010: Best = 6.555430e+04 FES = 301100 μ_F = 0.813 μ_{CR} = 0.991
 Diversity = 5.2120e-01 k_t = [1, 1, 1, 1]
 Generation 3020: Best = 6.522179e+04 FES = 302100 μ_F = 0.822 μ_{CR} = 0.993
 Diversity = 4.1333e-01 k_t = [4, 3, 2, 4]
 Generation 3030: Best = 6.506889e+04 FES = 303100 μ_F = 0.808 μ_{CR} = 0.993
 Diversity = 3.8141e-01 k_t = [1, 1, 1, 1]
 Generation 3040: Best = 6.475817e+04 FES = 304100 μ_F = 0.813 μ_{CR} = 0.994
 Diversity = 4.2963e-01 k_t = [3, 4, 2, 2]
 Generation 3050: Best = 6.432105e+04 FES = 305100 μ_F = 0.818 μ_{CR} = 0.995
 Diversity = 3.9924e-01 k_t = [1, 1, 1, 1]
 Generation 3060: Best = 6.395414e+04 FES = 306100 μ_F = 0.827 μ_{CR} = 0.994
 Diversity = 4.3311e-01 k_t = [2, 3, 3, 4]
 Generation 3070: Best = 6.363299e+04 FES = 307100 μ_F = 0.825 μ_{CR} = 0.993
 Diversity = 4.2716e-01 k_t = [1, 1, 1, 1]
 Generation 3080: Best = 6.337769e+04 FES = 308100 μ_F = 0.836 μ_{CR} = 0.995
 Diversity = 4.0166e-01 k_t = [4, 2, 2, 4]
 Generation 3090: Best = 6.318206e+04 FES = 309100 μ_F = 0.823 μ_{CR} = 0.994
 Diversity = 3.5960e-01 k_t = [1, 1, 1, 1]
 Generation 3100: Best = 6.305326e+04 FES = 310100 μ_F = 0.822 μ_{CR} = 0.996
 Diversity = 3.6953e-01 k_t = [3, 4, 2, 2]
 Generation 3110: Best = 6.281132e+04 FES = 311100 μ_F = 0.805 μ_{CR} = 0.994
 Diversity = 3.9752e-01 k_t = [1, 1, 1, 1]
 Generation 3120: Best = 6.258998e+04 FES = 312100 μ_F = 0.816 μ_{CR} = 0.994
 Diversity = 4.2319e-01 k_t = [4, 3, 3, 3]
 Generation 3130: Best = 6.234004e+04 FES = 313100 μ_F = 0.810 μ_{CR} = 0.992
 Diversity = 3.8562e-01 k_t = [1, 1, 1, 1]
 Generation 3140: Best = 6.197118e+04 FES = 314100 μ_F = 0.820 μ_{CR} = 0.994
 Diversity = 4.4210e-01 k_t = [4, 3, 4, 2]
 Generation 3150: Best = 6.179668e+04 FES = 315100 μ_F = 0.819 μ_{CR} = 0.993
 Diversity = 3.4604e-01 k_t = [1, 1, 1, 1]
 Generation 3160: Best = 6.166681e+04 FES = 316100 μ_F = 0.798 μ_{CR} = 0.994
 Diversity = 3.2488e-01 k_t = [3, 3, 3, 2]
 Generation 3170: Best = 6.155708e+04 FES = 317100 μ_F = 0.785 μ_{CR} = 0.991
 Diversity = 3.3161e-01 k_t = [1, 1, 1, 1]
 Generation 3180: Best = 6.149265e+04 FES = 318100 μ_F = 0.779 μ_{CR} = 0.990
 Diversity = 3.0065e-01 k_t = [3, 2, 4, 2]
 Generation 3190: Best = 6.134350e+04 FES = 319100 μ_F = 0.776 μ_{CR} = 0.991
 Diversity = 3.2692e-01 k_t = [1, 1, 1, 1]

Generation 3200: Best = 6.119685e+04 FES = 320100 μ_F = 0.780 μ_{CR} = 0.992
 Diversity = 3.2840e-01 k_t = [2, 2, 4, 4]
 Generation 3210: Best = 6.108216e+04 FES = 321100 μ_F = 0.787 μ_{CR} = 0.987
 Diversity = 2.7143e-01 k_t = [1, 1, 1, 1]
 Generation 3220: Best = 6.087656e+04 FES = 322100 μ_F = 0.801 μ_{CR} = 0.990
 Diversity = 3.2734e-01 k_t = [2, 2, 4, 4]
 Generation 3230: Best = 6.075840e+04 FES = 323100 μ_F = 0.812 μ_{CR} = 0.993
 Diversity = 3.4327e-01 k_t = [1, 1, 1, 1]
 Generation 3240: Best = 6.053503e+04 FES = 324100 μ_F = 0.801 μ_{CR} = 0.991
 Diversity = 3.8573e-01 k_t = [2, 2, 2, 4]
 Generation 3250: Best = 5.996480e+04 FES = 325100 μ_F = 0.814 μ_{CR} = 0.990
 Diversity = 5.0139e-01 k_t = [1, 1, 1, 1]
 Generation 3260: Best = 5.970755e+04 FES = 326100 μ_F = 0.815 μ_{CR} = 0.991
 Diversity = 4.0094e-01 k_t = [4, 3, 2, 2]
 Generation 3270: Best = 5.951721e+04 FES = 327100 μ_F = 0.801 μ_{CR} = 0.993
 Diversity = 3.4877e-01 k_t = [1, 1, 1, 1]
 Generation 3280: Best = 5.923400e+04 FES = 328100 μ_F = 0.808 μ_{CR} = 0.992
 Diversity = 4.1070e-01 k_t = [4, 4, 2, 2]
 Generation 3290: Best = 5.897859e+04 FES = 329100 μ_F = 0.809 μ_{CR} = 0.993
 Diversity = 4.8883e-01 k_t = [1, 1, 1, 1]
 Generation 3300: Best = 5.875730e+04 FES = 330100 μ_F = 0.801 μ_{CR} = 0.995
 Diversity = 3.5127e-01 k_t = [4, 2, 2, 3]
 Generation 3310: Best = 5.843400e+04 FES = 331100 μ_F = 0.808 μ_{CR} = 0.991
 Diversity = 3.8807e-01 k_t = [1, 1, 1, 1]
 Generation 3320: Best = 5.803918e+04 FES = 332100 μ_F = 0.816 μ_{CR} = 0.993
 Diversity = 4.3495e-01 k_t = [3, 4, 3, 2]
 Generation 3330: Best = 5.775895e+04 FES = 333100 μ_F = 0.813 μ_{CR} = 0.991
 Diversity = 3.8401e-01 k_t = [1, 1, 1, 1]
 Generation 3340: Best = 5.734365e+04 FES = 334100 μ_F = 0.819 μ_{CR} = 0.995
 Diversity = 5.0739e-01 k_t = [3, 3, 2, 2]
 Generation 3350: Best = 5.713143e+04 FES = 335100 μ_F = 0.819 μ_{CR} = 0.993
 Diversity = 4.3530e-01 k_t = [1, 1, 1, 1]
 Generation 3360: Best = 5.678448e+04 FES = 336100 μ_F = 0.821 μ_{CR} = 0.993
 Diversity = 4.6971e-01 k_t = [2, 4, 3, 2]
 Generation 3370: Best = 5.660486e+04 FES = 337100 μ_F = 0.819 μ_{CR} = 0.993
 Diversity = 3.8464e-01 k_t = [1, 1, 1, 1]
 Generation 3380: Best = 5.613703e+04 FES = 338100 μ_F = 0.817 μ_{CR} = 0.991
 Diversity = 4.0554e-01 k_t = [2, 3, 4, 2]
 Generation 3390: Best = 5.596215e+04 FES = 339100 μ_F = 0.810 μ_{CR} = 0.993
 Diversity = 4.4744e-01 k_t = [1, 1, 1, 1]
 Generation 3400: Best = 5.582176e+04 FES = 340100 μ_F = 0.810 μ_{CR} = 0.995
 Diversity = 3.3665e-01 k_t = [4, 4, 3, 2]
 Generation 3410: Best = 5.560071e+04 FES = 341100 μ_F = 0.816 μ_{CR} = 0.991
 Diversity = 3.6889e-01 k_t = [1, 1, 1, 1]
 Generation 3420: Best = 5.540977e+04 FES = 342100 μ_F = 0.818 μ_{CR} = 0.994
 Diversity = 3.3626e-01 k_t = [4, 2, 3, 4]
 Generation 3430: Best = 5.527677e+04 FES = 343100 μ_F = 0.821 μ_{CR} = 0.994
 Diversity = 3.1683e-01 k_t = [1, 1, 1, 1]
 Generation 3440: Best = 5.505402e+04 FES = 344100 μ_F = 0.826 μ_{CR} = 0.994
 Diversity = 3.5858e-01 k_t = [2, 3, 3, 2]
 Generation 3450: Best = 5.494355e+04 FES = 345100 μ_F = 0.813 μ_{CR} = 0.995
 Diversity = 3.4890e-01 k_t = [1, 1, 1, 1]
 Generation 3460: Best = 5.472231e+04 FES = 346100 μ_F = 0.812 μ_{CR} = 0.995
 Diversity = 3.6609e-01 k_t = [2, 3, 4, 3]

Generation 3470: Best = 5.460269e+04 FES = 347100 μ_F = 0.805 μ_{CR} = 0.994
 Diversity = 3.4051e-01 k_t = [1, 1, 1, 1]
 Generation 3480: Best = 5.442775e+04 FES = 348100 μ_F = 0.796 μ_{CR} = 0.994
 Diversity = 2.3959e-01 k_t = [4, 4, 3, 3]
 Generation 3490: Best = 5.436187e+04 FES = 349100 μ_F = 0.787 μ_{CR} = 0.989
 Diversity = 2.1904e-01 k_t = [1, 1, 1, 1]
 Generation 3500: Best = 5.409318e+04 FES = 350100 μ_F = 0.793 μ_{CR} = 0.987
 Diversity = 3.1869e-01 k_t = [4, 2, 3, 3]
 Generation 3510: Best = 5.385069e+04 FES = 351100 μ_F = 0.802 μ_{CR} = 0.990
 Diversity = 3.7688e-01 k_t = [1, 1, 1, 1]
 Generation 3520: Best = 5.353143e+04 FES = 352100 μ_F = 0.811 μ_{CR} = 0.992
 Diversity = 3.6369e-01 k_t = [3, 4, 3, 3]
 Generation 3530: Best = 5.321803e+04 FES = 353100 μ_F = 0.799 μ_{CR} = 0.993
 Diversity = 3.7591e-01 k_t = [1, 1, 1, 1]
 Generation 3540: Best = 5.296444e+04 FES = 354100 μ_F = 0.805 μ_{CR} = 0.993
 Diversity = 3.3261e-01 k_t = [3, 4, 3, 4]
 Generation 3550: Best = 5.288280e+04 FES = 355100 μ_F = 0.787 μ_{CR} = 0.993
 Diversity = 2.5896e-01 k_t = [1, 1, 1, 1]
 Generation 3560: Best = 5.282876e+04 FES = 356100 μ_F = 0.774 μ_{CR} = 0.991
 Diversity = 2.4137e-01 k_t = [4, 2, 4, 3]
 Generation 3570: Best = 5.274295e+04 FES = 357100 μ_F = 0.762 μ_{CR} = 0.987
 Diversity = 2.0800e-01 k_t = [1, 1, 1, 1]
 Generation 3580: Best = 5.266547e+04 FES = 358100 μ_F = 0.773 μ_{CR} = 0.988
 Diversity = 1.6688e-01 k_t = [2, 3, 4, 4]
 Generation 3590: Best = 5.262269e+04 FES = 359100 μ_F = 0.768 μ_{CR} = 0.988
 Diversity = 1.6047e-01 k_t = [1, 1, 1, 1]
 Generation 3600: Best = 5.260995e+04 FES = 360100 μ_F = 0.740 μ_{CR} = 0.986
 Diversity = 1.1317e-01 k_t = [2, 2, 4, 3]
 Generation 3610: Best = 5.259217e+04 FES = 361100 μ_F = 0.726 μ_{CR} = 0.988
 Diversity = 1.0604e-01 k_t = [1, 1, 1, 1]
 Generation 3620: Best = 5.256499e+04 FES = 362100 μ_F = 0.748 μ_{CR} = 0.980
 Diversity = 1.0256e-01 k_t = [4, 4, 4, 4]
 Generation 3630: Best = 5.253182e+04 FES = 363100 μ_F = 0.763 μ_{CR} = 0.983
 Diversity = 1.0136e-01 k_t = [1, 1, 1, 1]
 Generation 3640: Best = 5.249196e+04 FES = 364100 μ_F = 0.764 μ_{CR} = 0.980
 Diversity = 6.4281e-02 k_t = [3, 3, 3, 4]
 Generation 3650: Best = 5.247913e+04 FES = 365100 μ_F = 0.763 μ_{CR} = 0.977
 Diversity = 9.2873e-02 k_t = [1, 1, 1, 1]
 Generation 3660: Best = 5.245250e+04 FES = 366100 μ_F = 0.743 μ_{CR} = 0.978
 Diversity = 8.3147e-02 k_t = [4, 3, 3, 2]
 Generation 3670: Best = 5.242701e+04 FES = 367100 μ_F = 0.749 μ_{CR} = 0.976
 Diversity = 6.9186e-02 k_t = [1, 1, 1, 1]
 Generation 3680: Best = 5.239816e+04 FES = 368100 μ_F = 0.757 μ_{CR} = 0.974
 Diversity = 6.4414e-02 k_t = [4, 3, 2, 3]
 Generation 3690: Best = 5.238685e+04 FES = 369100 μ_F = 0.756 μ_{CR} = 0.977
 Diversity = 5.1787e-02 k_t = [1, 1, 1, 1]
 Generation 3700: Best = 5.235718e+04 FES = 370100 μ_F = 0.765 μ_{CR} = 0.966
 Diversity = 4.7035e-02 k_t = [4, 2, 3, 2]
 Generation 3710: Best = 5.233288e+04 FES = 371100 μ_F = 0.771 μ_{CR} = 0.971
 Diversity = 5.4790e-02 k_t = [1, 1, 1, 1]
 Generation 3720: Best = 5.228646e+04 FES = 372100 μ_F = 0.771 μ_{CR} = 0.970
 Diversity = 7.5578e-02 k_t = [4, 2, 4, 2]
 Generation 3730: Best = 5.224655e+04 FES = 373100 μ_F = 0.775 μ_{CR} = 0.971
 Diversity = 7.7153e-02 k_t = [1, 1, 1, 1]

Generation 3740: Best = 5.220711e+04 FES = 374100 μ_F = 0.771 μ_{CR} = 0.971
 Diversity = 7.6128e-02 k_t = [4, 4, 3, 3]
 Generation 3750: Best = 5.214702e+04 FES = 375100 μ_F = 0.768 μ_{CR} = 0.973
 Diversity = 8.1564e-02 k_t = [1, 1, 1, 1]
 Generation 3760: Best = 5.210007e+04 FES = 376100 μ_F = 0.773 μ_{CR} = 0.982
 Diversity = 1.1560e-01 k_t = [3, 3, 2, 3]
 Generation 3770: Best = 5.155498e+04 FES = 377100 μ_F = 0.785 μ_{CR} = 0.981
 Diversity = 2.5800e-01 k_t = [1, 1, 1, 1]
 Generation 3780: Best = 5.142848e+04 FES = 378100 μ_F = 0.792 μ_{CR} = 0.982
 Diversity = 1.5190e-01 k_t = [3, 4, 2, 3]
 Generation 3790: Best = 5.129047e+04 FES = 379100 μ_F = 0.779 μ_{CR} = 0.981
 Diversity = 1.6270e-01 k_t = [1, 1, 1, 1]
 Generation 3800: Best = 5.118989e+04 FES = 380100 μ_F = 0.786 μ_{CR} = 0.986
 Diversity = 2.0747e-01 k_t = [4, 4, 2, 2]
 Generation 3810: Best = 5.107548e+04 FES = 381100 μ_F = 0.778 μ_{CR} = 0.986
 Diversity = 2.2804e-01 k_t = [1, 1, 1, 1]
 Generation 3820: Best = 5.096424e+04 FES = 382100 μ_F = 0.784 μ_{CR} = 0.985
 Diversity = 2.0774e-01 k_t = [4, 2, 2, 4]
 Generation 3830: Best = 5.073202e+04 FES = 383100 μ_F = 0.770 μ_{CR} = 0.982
 Diversity = 1.8351e-01 k_t = [1, 1, 1, 1]
 Generation 3840: Best = 5.056455e+04 FES = 384100 μ_F = 0.785 μ_{CR} = 0.980
 Diversity = 2.1613e-01 k_t = [3, 2, 3, 4]
 Generation 3850: Best = 5.049029e+04 FES = 385100 μ_F = 0.780 μ_{CR} = 0.985
 Diversity = 1.9948e-01 k_t = [1, 1, 1, 1]
 Generation 3860: Best = 5.043581e+04 FES = 386100 μ_F = 0.771 μ_{CR} = 0.985
 Diversity = 1.4864e-01 k_t = [4, 4, 2, 4]
 Generation 3870: Best = 5.040567e+04 FES = 387100 μ_F = 0.768 μ_{CR} = 0.985
 Diversity = 1.0896e-01 k_t = [1, 1, 1, 1]
 Generation 3880: Best = 5.037407e+04 FES = 388100 μ_F = 0.761 μ_{CR} = 0.980
 Diversity = 9.5177e-02 k_t = [2, 3, 2, 3]
 Generation 3890: Best = 5.029892e+04 FES = 389100 μ_F = 0.780 μ_{CR} = 0.975
 Diversity = 1.2141e-01 k_t = [1, 1, 1, 1]
 Generation 3900: Best = 5.028317e+04 FES = 390100 μ_F = 0.760 μ_{CR} = 0.975
 Diversity = 7.5578e-02 k_t = [2, 4, 2, 4]
 Generation 3910: Best = 5.024439e+04 FES = 391100 μ_F = 0.758 μ_{CR} = 0.974
 Diversity = 8.8712e-02 k_t = [1, 1, 1, 1]
 Generation 3920: Best = 5.019988e+04 FES = 392100 μ_F = 0.758 μ_{CR} = 0.975
 Diversity = 1.0736e-01 k_t = [3, 2, 2, 4]
 Generation 3930: Best = 5.016068e+04 FES = 393100 μ_F = 0.766 μ_{CR} = 0.980
 Diversity = 1.2750e-01 k_t = [1, 1, 1, 1]
 Generation 3940: Best = 5.014434e+04 FES = 394100 μ_F = 0.748 μ_{CR} = 0.981
 Diversity = 1.0927e-01 k_t = [3, 2, 2, 2]
 Generation 3950: Best = 5.010392e+04 FES = 395100 μ_F = 0.757 μ_{CR} = 0.981
 Diversity = 1.0436e-01 k_t = [1, 1, 1, 1]
 Generation 3960: Best = 5.009474e+04 FES = 396100 μ_F = 0.760 μ_{CR} = 0.983
 Diversity = 7.8687e-02 k_t = [3, 4, 2, 3]
 Generation 3970: Best = 5.008321e+04 FES = 397100 μ_F = 0.756 μ_{CR} = 0.977
 Diversity = 5.7458e-02 k_t = [1, 1, 1, 1]
 Generation 3980: Best = 5.007145e+04 FES = 398100 μ_F = 0.733 μ_{CR} = 0.974
 Diversity = 5.6647e-02 k_t = [4, 3, 4, 2]
 Generation 3990: Best = 5.005212e+04 FES = 399100 μ_F = 0.740 μ_{CR} = 0.971
 Diversity = 5.9117e-02 k_t = [1, 1, 1, 1]
 Generation 4000: Best = 5.003855e+04 FES = 400100 μ_F = 0.752 μ_{CR} = 0.967
 Diversity = 4.4874e-02 k_t = [3, 4, 4, 4]

Generation 4010: Best = 5.003188e+04 FES = 401100 μ_F = 0.757 μ_{CR} = 0.969
 Diversity = 4.3194e-02 k_t = [1, 1, 1, 1]
 Generation 4020: Best = 5.000944e+04 FES = 402100 μ_F = 0.740 μ_{CR} = 0.967
 Diversity = 4.7469e-02 k_t = [4, 4, 3, 2]
 Generation 4030: Best = 5.000325e+04 FES = 403100 μ_F = 0.748 μ_{CR} = 0.975
 Diversity = 5.2796e-02 k_t = [1, 1, 1, 1]
 Generation 4040: Best = 4.999554e+04 FES = 404100 μ_F = 0.744 μ_{CR} = 0.968
 Diversity = 4.1283e-02 k_t = [4, 2, 2, 4]
 Generation 4050: Best = 4.997889e+04 FES = 405100 μ_F = 0.744 μ_{CR} = 0.971
 Diversity = 5.3161e-02 k_t = [1, 1, 1, 1]
 Generation 4060: Best = 4.997694e+04 FES = 406100 μ_F = 0.730 μ_{CR} = 0.973
 Diversity = 4.0410e-02 k_t = [3, 4, 2, 4]
 Generation 4070: Best = 4.997078e+04 FES = 407100 μ_F = 0.749 μ_{CR} = 0.968
 Diversity = 2.5853e-02 k_t = [1, 1, 1, 1]
 Generation 4080: Best = 4.995573e+04 FES = 408100 μ_F = 0.749 μ_{CR} = 0.966
 Diversity = 3.0570e-02 k_t = [4, 2, 2, 3]
 Generation 4090: Best = 4.993720e+04 FES = 409100 μ_F = 0.763 μ_{CR} = 0.971
 Diversity = 5.1495e-02 k_t = [1, 1, 1, 1]
 Generation 4100: Best = 4.992558e+04 FES = 410100 μ_F = 0.774 μ_{CR} = 0.973
 Diversity = 3.3259e-02 k_t = [3, 2, 4, 3]
 Generation 4110: Best = 4.991534e+04 FES = 411100 μ_F = 0.770 μ_{CR} = 0.965
 Diversity = 3.5712e-02 k_t = [1, 1, 1, 1]
 Generation 4120: Best = 4.991109e+04 FES = 412100 μ_F = 0.761 μ_{CR} = 0.966
 Diversity = 2.1889e-02 k_t = [3, 4, 4, 4]
 Generation 4130: Best = 4.990238e+04 FES = 413100 μ_F = 0.773 μ_{CR} = 0.966
 Diversity = 2.5075e-02 k_t = [1, 1, 1, 1]
 Generation 4140: Best = 4.989969e+04 FES = 414100 μ_F = 0.788 μ_{CR} = 0.959
 Diversity = 1.8473e-02 k_t = [2, 2, 4, 3]
 Generation 4150: Best = 4.989613e+04 FES = 415100 μ_F = 0.771 μ_{CR} = 0.953
 Diversity = 1.8196e-02 k_t = [1, 1, 1, 1]
 Generation 4160: Best = 4.988504e+04 FES = 416100 μ_F = 0.774 μ_{CR} = 0.958
 Diversity = 2.5490e-02 k_t = [2, 3, 4, 3]
 Generation 4170: Best = 4.987974e+04 FES = 417100 μ_F = 0.771 μ_{CR} = 0.958
 Diversity = 2.3304e-02 k_t = [1, 1, 1, 1]
 Generation 4180: Best = 4.986596e+04 FES = 418100 μ_F = 0.763 μ_{CR} = 0.961
 Diversity = 2.6359e-02 k_t = [2, 3, 3, 4]
 Generation 4190: Best = 4.985980e+04 FES = 419100 μ_F = 0.762 μ_{CR} = 0.966
 Diversity = 3.7672e-02 k_t = [1, 1, 1, 1]
 Generation 4200: Best = 4.984521e+04 FES = 420100 μ_F = 0.750 μ_{CR} = 0.966
 Diversity = 4.1088e-02 k_t = [3, 3, 3, 4]
 Generation 4210: Best = 4.983960e+04 FES = 421100 μ_F = 0.754 μ_{CR} = 0.968
 Diversity = 3.6154e-02 k_t = [1, 1, 1, 1]
 Generation 4220: Best = 4.982767e+04 FES = 422100 μ_F = 0.748 μ_{CR} = 0.968
 Diversity = 2.9269e-02 k_t = [3, 4, 4, 4]
 Generation 4230: Best = 4.981839e+04 FES = 423100 μ_F = 0.747 μ_{CR} = 0.969
 Diversity = 2.6552e-02 k_t = [1, 1, 1, 1]
 Generation 4240: Best = 4.981578e+04 FES = 424100 μ_F = 0.758 μ_{CR} = 0.966
 Diversity = 2.5089e-02 k_t = [2, 4, 3, 4]
 Generation 4250: Best = 4.979398e+04 FES = 425100 μ_F = 0.775 μ_{CR} = 0.964
 Diversity = 3.4784e-02 k_t = [1, 1, 1, 1]
 Generation 4260: Best = 4.976938e+04 FES = 426100 μ_F = 0.776 μ_{CR} = 0.963
 Diversity = 4.0283e-02 k_t = [4, 2, 3, 2]
 Generation 4270: Best = 4.975789e+04 FES = 427100 μ_F = 0.772 μ_{CR} = 0.954
 Diversity = 3.4843e-02 k_t = [1, 1, 1, 1]

Generation 4280: Best = 4.974274e+04 FES = 428100 μ_F = 0.772 μ_{CR} = 0.963
 Diversity = 2.6819e-02 k_t = [2, 2, 3, 4]
 Generation 4290: Best = 4.973150e+04 FES = 429100 μ_F = 0.781 μ_{CR} = 0.966
 Diversity = 3.5305e-02 k_t = [1, 1, 1, 1]
 Generation 4300: Best = 4.970763e+04 FES = 430100 μ_F = 0.778 μ_{CR} = 0.969
 Diversity = 3.2623e-02 k_t = [3, 2, 4, 4]
 Generation 4310: Best = 4.969895e+04 FES = 431100 μ_F = 0.779 μ_{CR} = 0.957
 Diversity = 3.8906e-02 k_t = [1, 1, 1, 1]
 Generation 4320: Best = 4.968849e+04 FES = 432100 μ_F = 0.779 μ_{CR} = 0.963
 Diversity = 2.6067e-02 k_t = [3, 3, 4, 4]
 Generation 4330: Best = 4.966970e+04 FES = 433100 μ_F = 0.776 μ_{CR} = 0.967
 Diversity = 4.6573e-02 k_t = [1, 1, 1, 1]
 Generation 4340: Best = 4.965127e+04 FES = 434100 μ_F = 0.772 μ_{CR} = 0.970
 Diversity = 4.4715e-02 k_t = [3, 4, 2, 2]
 Generation 4350: Best = 4.964105e+04 FES = 435100 μ_F = 0.769 μ_{CR} = 0.978
 Diversity = 5.2633e-02 k_t = [1, 1, 1, 1]
 Generation 4360: Best = 4.962041e+04 FES = 436100 μ_F = 0.761 μ_{CR} = 0.978
 Diversity = 5.8463e-02 k_t = [4, 2, 4, 2]
 Generation 4370: Best = 4.959383e+04 FES = 437100 μ_F = 0.763 μ_{CR} = 0.970
 Diversity = 5.0304e-02 k_t = [1, 1, 1, 1]
 Generation 4380: Best = 4.959274e+04 FES = 438100 μ_F = 0.757 μ_{CR} = 0.972
 Diversity = 3.4968e-02 k_t = [2, 3, 3, 3]
 Generation 4390: Best = 4.958465e+04 FES = 439100 μ_F = 0.761 μ_{CR} = 0.974
 Diversity = 3.6147e-02 k_t = [1, 1, 1, 1]
 Generation 4400: Best = 4.956325e+04 FES = 440100 μ_F = 0.763 μ_{CR} = 0.974
 Diversity = 3.5602e-02 k_t = [4, 3, 4, 3]
 Generation 4410: Best = 4.955564e+04 FES = 441100 μ_F = 0.755 μ_{CR} = 0.969
 Diversity = 3.6744e-02 k_t = [1, 1, 1, 1]
 Generation 4420: Best = 4.954598e+04 FES = 442100 μ_F = 0.757 μ_{CR} = 0.968
 Diversity = 2.7305e-02 k_t = [2, 4, 3, 4]
 Generation 4430: Best = 4.954093e+04 FES = 443100 μ_F = 0.763 μ_{CR} = 0.971
 Diversity = 3.9482e-02 k_t = [1, 1, 1, 1]
 Generation 4440: Best = 4.952722e+04 FES = 444100 μ_F = 0.764 μ_{CR} = 0.973
 Diversity = 3.7079e-02 k_t = [3, 2, 3, 3]
 Generation 4450: Best = 4.952057e+04 FES = 445100 μ_F = 0.764 μ_{CR} = 0.973
 Diversity = 4.3344e-02 k_t = [1, 1, 1, 1]
 Generation 4460: Best = 4.950333e+04 FES = 446100 μ_F = 0.767 μ_{CR} = 0.977
 Diversity = 4.6997e-02 k_t = [3, 3, 4, 3]
 Generation 4470: Best = 4.949210e+04 FES = 447100 μ_F = 0.789 μ_{CR} = 0.979
 Diversity = 3.9395e-02 k_t = [1, 1, 1, 1]
 Generation 4480: Best = 4.947775e+04 FES = 448100 μ_F = 0.783 μ_{CR} = 0.978
 Diversity = 4.6361e-02 k_t = [2, 3, 3, 2]
 Generation 4490: Best = 4.946804e+04 FES = 449100 μ_F = 0.790 μ_{CR} = 0.973
 Diversity = 5.5973e-02 k_t = [1, 1, 1, 1]
 Generation 4500: Best = 4.945408e+04 FES = 450100 μ_F = 0.792 μ_{CR} = 0.975
 Diversity = 5.0126e-02 k_t = [4, 2, 4, 3]
 Generation 4510: Best = 4.944437e+04 FES = 451100 μ_F = 0.798 μ_{CR} = 0.971
 Diversity = 4.4652e-02 k_t = [1, 1, 1, 1]
 Generation 4520: Best = 4.942259e+04 FES = 452100 μ_F = 0.784 μ_{CR} = 0.975
 Diversity = 4.0527e-02 k_t = [4, 3, 4, 4]
 Generation 4530: Best = 4.941445e+04 FES = 453100 μ_F = 0.782 μ_{CR} = 0.972
 Diversity = 3.5960e-02 k_t = [1, 1, 1, 1]
 Generation 4540: Best = 4.940845e+04 FES = 454100 μ_F = 0.776 μ_{CR} = 0.974
 Diversity = 3.8725e-02 k_t = [3, 3, 3, 2]

Generation 4550: Best = 4.940397e+04 FES = 455100 μ_F = 0.762 μ_{CR} = 0.970
 Diversity = 3.8396e-02 k_t = [1, 1, 1, 1]
 Generation 4560: Best = 4.939816e+04 FES = 456100 μ_F = 0.750 μ_{CR} = 0.974
 Diversity = 3.8608e-02 k_t = [4, 2, 3, 2]
 Generation 4570: Best = 4.938966e+04 FES = 457100 μ_F = 0.734 μ_{CR} = 0.975
 Diversity = 3.2412e-02 k_t = [1, 1, 1, 1]
 Generation 4580: Best = 4.938511e+04 FES = 458100 μ_F = 0.747 μ_{CR} = 0.972
 Diversity = 3.4626e-02 k_t = [4, 2, 2, 3]
 Generation 4590: Best = 4.937404e+04 FES = 459100 μ_F = 0.751 μ_{CR} = 0.971
 Diversity = 3.1635e-02 k_t = [1, 1, 1, 1]
 Generation 4600: Best = 4.936362e+04 FES = 460100 μ_F = 0.761 μ_{CR} = 0.965
 Diversity = 4.1678e-02 k_t = [3, 2, 3, 3]
 Generation 4610: Best = 4.935078e+04 FES = 461100 μ_F = 0.769 μ_{CR} = 0.967
 Diversity = 4.2311e-02 k_t = [1, 1, 1, 1]
 Generation 4620: Best = 4.934416e+04 FES = 462100 μ_F = 0.767 μ_{CR} = 0.966
 Diversity = 3.3145e-02 k_t = [4, 4, 2, 2]
 Generation 4630: Best = 4.933796e+04 FES = 463100 μ_F = 0.765 μ_{CR} = 0.963
 Diversity = 3.4687e-02 k_t = [1, 1, 1, 1]
 Generation 4640: Best = 4.932825e+04 FES = 464100 μ_F = 0.755 μ_{CR} = 0.962
 Diversity = 3.6329e-02 k_t = [4, 3, 2, 2]
 Generation 4650: Best = 4.932477e+04 FES = 465100 μ_F = 0.750 μ_{CR} = 0.962
 Diversity = 3.2043e-02 k_t = [1, 1, 1, 1]
 Generation 4660: Best = 4.932135e+04 FES = 466100 μ_F = 0.745 μ_{CR} = 0.960
 Diversity = 2.1887e-02 k_t = [2, 2, 4, 4]
 Generation 4670: Best = 4.931937e+04 FES = 467100 μ_F = 0.751 μ_{CR} = 0.960
 Diversity = 1.8673e-02 k_t = [1, 1, 1, 1]
 Generation 4680: Best = 4.931478e+04 FES = 468100 μ_F = 0.740 μ_{CR} = 0.955
 Diversity = 2.1308e-02 k_t = [3, 2, 4, 2]
 Generation 4690: Best = 4.930745e+04 FES = 469100 μ_F = 0.750 μ_{CR} = 0.957
 Diversity = 2.6959e-02 k_t = [1, 1, 1, 1]
 Generation 4700: Best = 4.930220e+04 FES = 470100 μ_F = 0.736 μ_{CR} = 0.960
 Diversity = 2.6075e-02 k_t = [4, 3, 3, 3]
 Generation 4710: Best = 4.929737e+04 FES = 471100 μ_F = 0.747 μ_{CR} = 0.956
 Diversity = 2.3522e-02 k_t = [1, 1, 1, 1]
 Generation 4720: Best = 4.929346e+04 FES = 472100 μ_F = 0.750 μ_{CR} = 0.951
 Diversity = 1.9337e-02 k_t = [3, 3, 2, 4]
 Generation 4730: Best = 4.929021e+04 FES = 473100 μ_F = 0.734 μ_{CR} = 0.951
 Diversity = 2.2229e-02 k_t = [1, 1, 1, 1]
 Generation 4740: Best = 4.928194e+04 FES = 474100 μ_F = 0.732 μ_{CR} = 0.951
 Diversity = 2.3460e-02 k_t = [4, 3, 2, 4]
 Generation 4750: Best = 4.927721e+04 FES = 475100 μ_F = 0.730 μ_{CR} = 0.949
 Diversity = 1.9033e-02 k_t = [1, 1, 1, 1]
 Generation 4760: Best = 4.926839e+04 FES = 476100 μ_F = 0.739 μ_{CR} = 0.947
 Diversity = 1.8208e-02 k_t = [2, 3, 4, 4]
 Generation 4770: Best = 4.926180e+04 FES = 477100 μ_F = 0.739 μ_{CR} = 0.942
 Diversity = 1.7128e-02 k_t = [1, 1, 1, 1]
 Generation 4780: Best = 4.925831e+04 FES = 478100 μ_F = 0.742 μ_{CR} = 0.937
 Diversity = 1.2212e-02 k_t = [3, 2, 3, 4]
 Generation 4790: Best = 4.924967e+04 FES = 479100 μ_F = 0.756 μ_{CR} = 0.929
 Diversity = 1.8634e-02 k_t = [1, 1, 1, 1]
 Generation 4800: Best = 4.924348e+04 FES = 480100 μ_F = 0.757 μ_{CR} = 0.929
 Diversity = 1.3517e-02 k_t = [4, 2, 4, 4]
 Generation 4810: Best = 4.924005e+04 FES = 481100 μ_F = 0.751 μ_{CR} = 0.928
 Diversity = 1.2044e-02 k_t = [1, 1, 1, 1]

Generation 4820: Best = 4.923632e+04 FES = 482100 μ_F = 0.731 μ_{CR} = 0.940
 Diversity = 1.2274e-02 k_t = [4, 4, 3, 2]
 Generation 4830: Best = 4.923392e+04 FES = 483100 μ_F = 0.741 μ_{CR} = 0.936
 Diversity = 1.1444e-02 k_t = [1, 1, 1, 1]
 Generation 4840: Best = 4.922892e+04 FES = 484100 μ_F = 0.751 μ_{CR} = 0.934
 Diversity = 1.1582e-02 k_t = [2, 4, 3, 3]
 Generation 4850: Best = 4.922484e+04 FES = 485100 μ_F = 0.764 μ_{CR} = 0.934
 Diversity = 1.4859e-02 k_t = [1, 1, 1, 1]
 Generation 4860: Best = 4.921985e+04 FES = 486100 μ_F = 0.755 μ_{CR} = 0.940
 Diversity = 1.5814e-02 k_t = [3, 3, 4, 2]
 Generation 4870: Best = 4.921585e+04 FES = 487100 μ_F = 0.746 μ_{CR} = 0.933
 Diversity = 1.5079e-02 k_t = [1, 1, 1, 1]
 Generation 4880: Best = 4.921011e+04 FES = 488100 μ_F = 0.752 μ_{CR} = 0.925
 Diversity = 1.2287e-02 k_t = [4, 2, 4, 4]
 Generation 4890: Best = 4.920699e+04 FES = 489100 μ_F = 0.756 μ_{CR} = 0.920
 Diversity = 1.1785e-02 k_t = [1, 1, 1, 1]
 Generation 4900: Best = 4.920388e+04 FES = 490100 μ_F = 0.762 μ_{CR} = 0.916
 Diversity = 1.0261e-02 k_t = [4, 4, 3, 4]
 Generation 4910: Best = 4.919729e+04 FES = 491100 μ_F = 0.771 μ_{CR} = 0.920
 Diversity = 1.8975e-02 k_t = [1, 1, 1, 1]
 Generation 4920: Best = 4.919235e+04 FES = 492100 μ_F = 0.766 μ_{CR} = 0.931
 Diversity = 1.9731e-02 k_t = [3, 2, 2, 4]
 Generation 4930: Best = 4.918858e+04 FES = 493100 μ_F = 0.750 μ_{CR} = 0.936
 Diversity = 2.4118e-02 k_t = [1, 1, 1, 1]
 Generation 4940: Best = 4.918678e+04 FES = 494100 μ_F = 0.753 μ_{CR} = 0.940
 Diversity = 2.0635e-02 k_t = [3, 3, 2, 2]
 Generation 4950: Best = 4.918276e+04 FES = 495100 μ_F = 0.735 μ_{CR} = 0.944
 Diversity = 2.3537e-02 k_t = [1, 1, 1, 1]
 Generation 4960: Best = 4.917967e+04 FES = 496100 μ_F = 0.731 μ_{CR} = 0.946
 Diversity = 1.5644e-02 k_t = [3, 3, 4, 2]
 Generation 4970: Best = 4.917617e+04 FES = 497100 μ_F = 0.731 μ_{CR} = 0.946
 Diversity = 1.9227e-02 k_t = [1, 1, 1, 1]
 Generation 4980: Best = 4.917241e+04 FES = 498100 μ_F = 0.733 μ_{CR} = 0.952
 Diversity = 1.7529e-02 k_t = [3, 3, 2, 2]
 Generation 4990: Best = 4.916924e+04 FES = 499100 μ_F = 0.727 μ_{CR} = 0.943
 Diversity = 1.6139e-02 k_t = [1, 1, 1, 1]
 Generation 5000: Best = 4.916409e+04 FES = 500100 μ_F = 0.739 μ_{CR} = 0.940
 Diversity = 1.6861e-02 k_t = [2, 3, 3, 4]
 Generation 5010: Best = 4.915976e+04 FES = 501100 μ_F = 0.746 μ_{CR} = 0.942
 Diversity = 1.6968e-02 k_t = [1, 1, 1, 1]
 Generation 5020: Best = 4.915589e+04 FES = 502100 μ_F = 0.747 μ_{CR} = 0.945
 Diversity = 1.6622e-02 k_t = [2, 2, 2, 4]
 Generation 5030: Best = 4.915295e+04 FES = 503100 μ_F = 0.739 μ_{CR} = 0.948
 Diversity = 1.8384e-02 k_t = [1, 1, 1, 1]
 Generation 5040: Best = 4.915183e+04 FES = 504100 μ_F = 0.729 μ_{CR} = 0.952
 Diversity = 1.2419e-02 k_t = [2, 3, 3, 4]
 Generation 5050: Best = 4.915010e+04 FES = 505100 μ_F = 0.734 μ_{CR} = 0.950
 Diversity = 1.2567e-02 k_t = [1, 1, 1, 1]
 Generation 5060: Best = 4.914893e+04 FES = 506100 μ_F = 0.723 μ_{CR} = 0.945
 Diversity = 9.3506e-03 k_t = [2, 2, 3, 4]
 Generation 5070: Best = 4.914780e+04 FES = 507100 μ_F = 0.737 μ_{CR} = 0.941
 Diversity = 1.0117e-02 k_t = [1, 1, 1, 1]
 Generation 5080: Best = 4.914472e+04 FES = 508100 μ_F = 0.733 μ_{CR} = 0.942
 Diversity = 1.2228e-02 k_t = [2, 4, 2, 3]

Generation 5090: Best = 4.914254e+04 FES = 509100 μ_F = 0.735 μ_{CR} = 0.941
 Diversity = 1.3421e-02 k_t = [1, 1, 1, 1]
 Generation 5100: Best = 4.913982e+04 FES = 510100 μ_F = 0.737 μ_{CR} = 0.939
 Diversity = 1.4780e-02 k_t = [2, 4, 3, 4]
 Generation 5110: Best = 4.913779e+04 FES = 511100 μ_F = 0.736 μ_{CR} = 0.946
 Diversity = 1.3229e-02 k_t = [1, 1, 1, 1]
 Generation 5120: Best = 4.913435e+04 FES = 512100 μ_F = 0.746 μ_{CR} = 0.946
 Diversity = 1.1286e-02 k_t = [4, 3, 4, 4]
 Generation 5130: Best = 4.913229e+04 FES = 513100 μ_F = 0.763 μ_{CR} = 0.940
 Diversity = 1.3317e-02 k_t = [1, 1, 1, 1]
 Generation 5140: Best = 4.912808e+04 FES = 514100 μ_F = 0.755 μ_{CR} = 0.939
 Diversity = 1.2740e-02 k_t = [2, 4, 4, 2]
 Generation 5150: Best = 4.912469e+04 FES = 515100 μ_F = 0.756 μ_{CR} = 0.934
 Diversity = 1.6140e-02 k_t = [1, 1, 1, 1]
 Generation 5160: Best = 4.912130e+04 FES = 516100 μ_F = 0.746 μ_{CR} = 0.930
 Diversity = 1.0436e-02 k_t = [4, 4, 4, 4]
 Generation 5170: Best = 4.911954e+04 FES = 517100 μ_F = 0.747 μ_{CR} = 0.925
 Diversity = 1.0955e-02 k_t = [1, 1, 1, 1]
 Generation 5180: Best = 4.911833e+04 FES = 518100 μ_F = 0.750 μ_{CR} = 0.925
 Diversity = 1.0726e-02 k_t = [2, 2, 3, 4]
 Generation 5190: Best = 4.911356e+04 FES = 519100 μ_F = 0.753 μ_{CR} = 0.920
 Diversity = 1.3550e-02 k_t = [1, 1, 1, 1]
 Generation 5200: Best = 4.911252e+04 FES = 520100 μ_F = 0.768 μ_{CR} = 0.925
 Diversity = 1.1791e-02 k_t = [3, 2, 3, 4]
 Generation 5210: Best = 4.910980e+04 FES = 521100 μ_F = 0.769 μ_{CR} = 0.921
 Diversity = 1.1214e-02 k_t = [1, 1, 1, 1]
 Generation 5220: Best = 4.910775e+04 FES = 522100 μ_F = 0.772 μ_{CR} = 0.926
 Diversity = 1.0842e-02 k_t = [4, 2, 4, 3]
 Generation 5230: Best = 4.910434e+04 FES = 523100 μ_F = 0.766 μ_{CR} = 0.933
 Diversity = 1.4344e-02 k_t = [1, 1, 1, 1]
 Generation 5240: Best = 4.910111e+04 FES = 524100 μ_F = 0.777 μ_{CR} = 0.945
 Diversity = 1.1634e-02 k_t = [4, 4, 4, 3]
 Generation 5250: Best = 4.909937e+04 FES = 525100 μ_F = 0.772 μ_{CR} = 0.940
 Diversity = 9.4014e-03 k_t = [1, 1, 1, 1]
 Generation 5260: Best = 4.909799e+04 FES = 526100 μ_F = 0.773 μ_{CR} = 0.936
 Diversity = 9.2693e-03 k_t = [3, 4, 2, 4]
 Generation 5270: Best = 4.909661e+04 FES = 527100 μ_F = 0.782 μ_{CR} = 0.934
 Diversity = 1.0883e-02 k_t = [1, 1, 1, 1]
 Generation 5280: Best = 4.909334e+04 FES = 528100 μ_F = 0.783 μ_{CR} = 0.930
 Diversity = 1.1791e-02 k_t = [4, 4, 4, 3]
 Generation 5290: Best = 4.909247e+04 FES = 529100 μ_F = 0.789 μ_{CR} = 0.925
 Diversity = 1.0884e-02 k_t = [1, 1, 1, 1]
 Generation 5300: Best = 4.909073e+04 FES = 530100 μ_F = 0.777 μ_{CR} = 0.925
 Diversity = 1.0354e-02 k_t = [2, 4, 2, 2]
 Generation 5310: Best = 4.908967e+04 FES = 531100 μ_F = 0.780 μ_{CR} = 0.927
 Diversity = 1.1027e-02 k_t = [1, 1, 1, 1]
 Generation 5320: Best = 4.908756e+04 FES = 532100 μ_F = 0.768 μ_{CR} = 0.918
 Diversity = 1.1309e-02 k_t = [2, 2, 2, 2]
 Generation 5330: Best = 4.908473e+04 FES = 533100 μ_F = 0.757 μ_{CR} = 0.922
 Diversity = 1.5151e-02 k_t = [1, 1, 1, 1]
 Generation 5340: Best = 4.908233e+04 FES = 534100 μ_F = 0.751 μ_{CR} = 0.924
 Diversity = 1.3762e-02 k_t = [2, 3, 4, 3]
 Generation 5350: Best = 4.907961e+04 FES = 535100 μ_F = 0.750 μ_{CR} = 0.935
 Diversity = 1.6369e-02 k_t = [1, 1, 1, 1]

Generation 5360: Best = 4.907780e+04 FES = 536100 μ_F = 0.743 μ_{CR} = 0.940
 Diversity = 1.1732e-02 k_t = [4, 2, 4, 4]
 Generation 5370: Best = 4.907630e+04 FES = 537100 μ_F = 0.737 μ_{CR} = 0.944
 Diversity = 1.2858e-02 k_t = [1, 1, 1, 1]
 Generation 5380: Best = 4.907483e+04 FES = 538100 μ_F = 0.744 μ_{CR} = 0.944
 Diversity = 9.8994e-03 k_t = [3, 2, 3, 4]
 Generation 5390: Best = 4.907310e+04 FES = 539100 μ_F = 0.741 μ_{CR} = 0.946
 Diversity = 1.2092e-02 k_t = [1, 1, 1, 1]
 Generation 5400: Best = 4.907102e+04 FES = 540100 μ_F = 0.743 μ_{CR} = 0.950
 Diversity = 9.5354e-03 k_t = [4, 4, 4, 3]
 Generation 5410: Best = 4.906401e+04 FES = 541100 μ_F = 0.756 μ_{CR} = 0.945
 Diversity = 2.0548e-02 k_t = [1, 1, 1, 1]
 Generation 5420: Best = 4.905663e+04 FES = 542100 μ_F = 0.763 μ_{CR} = 0.948
 Diversity = 1.9189e-02 k_t = [3, 4, 4, 4]
 Generation 5430: Best = 4.905038e+04 FES = 543100 μ_F = 0.770 μ_{CR} = 0.943
 Diversity = 1.8358e-02 k_t = [1, 1, 1, 1]
 Generation 5440: Best = 4.904174e+04 FES = 544100 μ_F = 0.780 μ_{CR} = 0.941
 Diversity = 1.7941e-02 k_t = [4, 3, 2, 4]
 Generation 5450: Best = 4.903071e+04 FES = 545100 μ_F = 0.787 μ_{CR} = 0.938
 Diversity = 2.2299e-02 k_t = [1, 1, 1, 1]
 Generation 5460: Best = 4.901997e+04 FES = 546100 μ_F = 0.792 μ_{CR} = 0.932
 Diversity = 2.0159e-02 k_t = [2, 3, 2, 4]
 Generation 5470: Best = 4.901113e+04 FES = 547100 μ_F = 0.789 μ_{CR} = 0.936
 Diversity = 2.4846e-02 k_t = [1, 1, 1, 1]
 Generation 5480: Best = 4.900310e+04 FES = 548100 μ_F = 0.785 μ_{CR} = 0.938
 Diversity = 2.6989e-02 k_t = [2, 3, 4, 2]
 Generation 5490: Best = 4.898960e+04 FES = 549100 μ_F = 0.781 μ_{CR} = 0.946
 Diversity = 3.1832e-02 k_t = [1, 1, 1, 1]
 Generation 5500: Best = 4.896854e+04 FES = 550100 μ_F = 0.781 μ_{CR} = 0.954
 Diversity = 3.9494e-02 k_t = [2, 2, 4, 4]
 Generation 5510: Best = 4.895528e+04 FES = 551100 μ_F = 0.777 μ_{CR} = 0.959
 Diversity = 2.8728e-02 k_t = [1, 1, 1, 1]
 Generation 5520: Best = 4.894376e+04 FES = 552100 μ_F = 0.783 μ_{CR} = 0.968
 Diversity = 3.1209e-02 k_t = [2, 3, 2, 3]
 Generation 5530: Best = 4.893495e+04 FES = 553100 μ_F = 0.782 μ_{CR} = 0.962
 Diversity = 3.3624e-02 k_t = [1, 1, 1, 1]
 Generation 5540: Best = 4.892625e+04 FES = 554100 μ_F = 0.764 μ_{CR} = 0.958
 Diversity = 2.7829e-02 k_t = [3, 2, 2, 2]
 Generation 5550: Best = 4.888252e+04 FES = 555100 μ_F = 0.772 μ_{CR} = 0.955
 Diversity = 4.5886e-02 k_t = [1, 1, 1, 1]
 Generation 5560: Best = 4.887166e+04 FES = 556100 μ_F = 0.770 μ_{CR} = 0.957
 Diversity = 3.2512e-02 k_t = [4, 3, 3, 3]
 Generation 5570: Best = 4.886124e+04 FES = 557100 μ_F = 0.757 μ_{CR} = 0.959
 Diversity = 2.2943e-02 k_t = [1, 1, 1, 1]
 Generation 5580: Best = 4.884488e+04 FES = 558100 μ_F = 0.779 μ_{CR} = 0.953
 Diversity = 2.0396e-02 k_t = [2, 2, 3, 4]
 Generation 5590: Best = 4.883237e+04 FES = 559100 μ_F = 0.769 μ_{CR} = 0.945
 Diversity = 2.2898e-02 k_t = [1, 1, 1, 1]
 Generation 5600: Best = 4.881705e+04 FES = 560100 μ_F = 0.785 μ_{CR} = 0.947
 Diversity = 2.9624e-02 k_t = [2, 2, 4, 3]
 Generation 5610: Best = 4.880696e+04 FES = 561100 μ_F = 0.781 μ_{CR} = 0.952
 Diversity = 2.9743e-02 k_t = [1, 1, 1, 1]
 Generation 5620: Best = 4.878792e+04 FES = 562100 μ_F = 0.785 μ_{CR} = 0.948
 Diversity = 3.7342e-02 k_t = [4, 4, 2, 3]

Generation 5630: Best = 4.877535e+04 FES = 563100 μ_F = 0.788 μ_{CR} = 0.952
 Diversity = 2.7042e-02 k_t = [1, 1, 1, 1]
 Generation 5640: Best = 4.876740e+04 FES = 564100 μ_F = 0.778 μ_{CR} = 0.956
 Diversity = 2.1370e-02 k_t = [4, 3, 2, 3]
 Generation 5650: Best = 4.876129e+04 FES = 565100 μ_F = 0.778 μ_{CR} = 0.949
 Diversity = 1.9302e-02 k_t = [1, 1, 1, 1]
 Generation 5660: Best = 4.875338e+04 FES = 566100 μ_F = 0.756 μ_{CR} = 0.945
 Diversity = 1.8328e-02 k_t = [3, 2, 3, 3]
 Generation 5670: Best = 4.873635e+04 FES = 567100 μ_F = 0.776 μ_{CR} = 0.946
 Diversity = 2.5124e-02 k_t = [1, 1, 1, 1]
 Generation 5680: Best = 4.871835e+04 FES = 568100 μ_F = 0.779 μ_{CR} = 0.952
 Diversity = 2.4313e-02 k_t = [3, 2, 2, 4]
 Generation 5690: Best = 4.869820e+04 FES = 569100 μ_F = 0.772 μ_{CR} = 0.951
 Diversity = 3.0568e-02 k_t = [1, 1, 1, 1]
 Generation 5700: Best = 4.868443e+04 FES = 570100 μ_F = 0.768 μ_{CR} = 0.948
 Diversity = 2.2927e-02 k_t = [2, 2, 2, 4]
 Generation 5710: Best = 4.867401e+04 FES = 571100 μ_F = 0.766 μ_{CR} = 0.950
 Diversity = 1.9125e-02 k_t = [1, 1, 1, 1]
 Generation 5720: Best = 4.865792e+04 FES = 572100 μ_F = 0.768 μ_{CR} = 0.949
 Diversity = 1.9167e-02 k_t = [4, 3, 3, 3]
 Generation 5730: Best = 4.865140e+04 FES = 573100 μ_F = 0.779 μ_{CR} = 0.948
 Diversity = 1.6956e-02 k_t = [1, 1, 1, 1]
 Generation 5740: Best = 4.864757e+04 FES = 574100 μ_F = 0.750 μ_{CR} = 0.951
 Diversity = 1.6600e-02 k_t = [3, 2, 3, 2]
 Generation 5750: Best = 4.864290e+04 FES = 575100 μ_F = 0.756 μ_{CR} = 0.946
 Diversity = 1.8725e-02 k_t = [1, 1, 1, 1]
 Generation 5760: Best = 4.863825e+04 FES = 576100 μ_F = 0.756 μ_{CR} = 0.929
 Diversity = 1.3393e-02 k_t = [3, 2, 3, 4]
 Generation 5770: Best = 4.863374e+04 FES = 577100 μ_F = 0.756 μ_{CR} = 0.939
 Diversity = 1.3551e-02 k_t = [1, 1, 1, 1]
 Generation 5780: Best = 4.862843e+04 FES = 578100 μ_F = 0.765 μ_{CR} = 0.940
 Diversity = 1.2322e-02 k_t = [2, 4, 2, 2]
 Generation 5790: Best = 4.861776e+04 FES = 579100 μ_F = 0.764 μ_{CR} = 0.938
 Diversity = 1.7732e-02 k_t = [1, 1, 1, 1]
 Generation 5800: Best = 4.860609e+04 FES = 580100 μ_F = 0.765 μ_{CR} = 0.943
 Diversity = 1.8719e-02 k_t = [4, 4, 4, 2]
 Generation 5810: Best = 4.859826e+04 FES = 581100 μ_F = 0.772 μ_{CR} = 0.944
 Diversity = 1.8725e-02 k_t = [1, 1, 1, 1]
 Generation 5820: Best = 4.858957e+04 FES = 582100 μ_F = 0.777 μ_{CR} = 0.947
 Diversity = 2.2686e-02 k_t = [4, 2, 2, 2]
 Generation 5830: Best = 4.857737e+04 FES = 583100 μ_F = 0.781 μ_{CR} = 0.942
 Diversity = 2.2833e-02 k_t = [1, 1, 1, 1]
 Generation 5840: Best = 4.855773e+04 FES = 584100 μ_F = 0.789 μ_{CR} = 0.943
 Diversity = 2.7253e-02 k_t = [3, 3, 3, 3]
 Generation 5850: Best = 4.853667e+04 FES = 585100 μ_F = 0.789 μ_{CR} = 0.945
 Diversity = 2.7199e-02 k_t = [1, 1, 1, 1]
 Generation 5860: Best = 4.851972e+04 FES = 586100 μ_F = 0.786 μ_{CR} = 0.951
 Diversity = 2.3382e-02 k_t = [4, 3, 2, 4]
 Generation 5870: Best = 4.850841e+04 FES = 587100 μ_F = 0.772 μ_{CR} = 0.949
 Diversity = 2.7738e-02 k_t = [1, 1, 1, 1]
 Generation 5880: Best = 4.848003e+04 FES = 588100 μ_F = 0.773 μ_{CR} = 0.947
 Diversity = 2.8171e-02 k_t = [4, 4, 2, 3]
 Generation 5890: Best = 4.847075e+04 FES = 589100 μ_F = 0.778 μ_{CR} = 0.941
 Diversity = 1.9212e-02 k_t = [1, 1, 1, 1]

Generation 5900: Best = 4.845648e+04 FES = 590100 μ_F = 0.773 μ_{CR} = 0.942
 Diversity = 1.8473e-02 k_t = [4, 4, 4, 4]
 Generation 5910: Best = 4.844791e+04 FES = 591100 μ_F = 0.778 μ_{CR} = 0.930
 Diversity = 1.7521e-02 k_t = [1, 1, 1, 1]
 Generation 5920: Best = 4.844184e+04 FES = 592100 μ_F = 0.777 μ_{CR} = 0.934
 Diversity = 1.1582e-02 k_t = [4, 2, 2, 4]
 Generation 5930: Best = 4.843907e+04 FES = 593100 μ_F = 0.782 μ_{CR} = 0.940
 Diversity = 1.3571e-02 k_t = [1, 1, 1, 1]
 Generation 5940: Best = 4.843351e+04 FES = 594100 μ_F = 0.772 μ_{CR} = 0.937
 Diversity = 1.2918e-02 k_t = [4, 2, 2, 4]
 Generation 5950: Best = 4.842824e+04 FES = 595100 μ_F = 0.769 μ_{CR} = 0.928
 Diversity = 1.2940e-02 k_t = [1, 1, 1, 1]
 Generation 5960: Best = 4.842553e+04 FES = 596100 μ_F = 0.774 μ_{CR} = 0.922
 Diversity = 1.1052e-02 k_t = [3, 3, 4, 3]
 Generation 5970: Best = 4.842260e+04 FES = 597100 μ_F = 0.765 μ_{CR} = 0.929
 Diversity = 1.2952e-02 k_t = [1, 1, 1, 1]
 Generation 5980: Best = 4.841471e+04 FES = 598100 μ_F = 0.777 μ_{CR} = 0.926
 Diversity = 1.6522e-02 k_t = [3, 2, 3, 3]
 Generation 5990: Best = 4.841223e+04 FES = 599100 μ_F = 0.776 μ_{CR} = 0.943
 Diversity = 1.6011e-02 k_t = [1, 1, 1, 1]
 Generation 6000: Best = 4.840821e+04 FES = 600100 μ_F = 0.790 μ_{CR} = 0.943
 Diversity = 1.3394e-02 k_t = [3, 3, 2, 4]
 Generation 6010: Best = 4.840134e+04 FES = 601100 μ_F = 0.782 μ_{CR} = 0.951
 Diversity = 2.1962e-02 k_t = [1, 1, 1, 1]
 Generation 6020: Best = 4.839783e+04 FES = 602100 μ_F = 0.781 μ_{CR} = 0.957
 Diversity = 2.0581e-02 k_t = [2, 2, 3, 2]
 Generation 6030: Best = 4.839641e+04 FES = 603100 μ_F = 0.765 μ_{CR} = 0.951
 Diversity = 1.7295e-02 k_t = [1, 1, 1, 1]
 Generation 6040: Best = 4.839382e+04 FES = 604100 μ_F = 0.753 μ_{CR} = 0.950
 Diversity = 1.1403e-02 k_t = [2, 4, 3, 3]
 Generation 6050: Best = 4.839233e+04 FES = 605100 μ_F = 0.747 μ_{CR} = 0.945
 Diversity = 1.0938e-02 k_t = [1, 1, 1, 1]
 Generation 6060: Best = 4.839036e+04 FES = 606100 μ_F = 0.742 μ_{CR} = 0.934
 Diversity = 1.2800e-02 k_t = [2, 2, 2, 3]
 Generation 6070: Best = 4.838746e+04 FES = 607100 μ_F = 0.735 μ_{CR} = 0.930
 Diversity = 1.3983e-02 k_t = [1, 1, 1, 1]
 Generation 6080: Best = 4.838491e+04 FES = 608100 μ_F = 0.748 μ_{CR} = 0.934
 Diversity = 1.1246e-02 k_t = [3, 4, 3, 4]
 Generation 6090: Best = 4.838313e+04 FES = 609100 μ_F = 0.739 μ_{CR} = 0.940
 Diversity = 1.2564e-02 k_t = [1, 1, 1, 1]
 Generation 6100: Best = 4.838131e+04 FES = 610100 μ_F = 0.743 μ_{CR} = 0.947
 Diversity = 1.1454e-02 k_t = [3, 2, 4, 4]
 Generation 6110: Best = 4.837861e+04 FES = 611100 μ_F = 0.748 μ_{CR} = 0.951
 Diversity = 1.3497e-02 k_t = [1, 1, 1, 1]
 Generation 6120: Best = 4.837718e+04 FES = 612100 μ_F = 0.748 μ_{CR} = 0.941
 Diversity = 1.1031e-02 k_t = [3, 3, 3, 2]
 Generation 6130: Best = 4.837466e+04 FES = 613100 μ_F = 0.760 μ_{CR} = 0.944
 Diversity = 1.1860e-02 k_t = [1, 1, 1, 1]
 Generation 6140: Best = 4.837249e+04 FES = 614100 μ_F = 0.769 μ_{CR} = 0.949
 Diversity = 1.0132e-02 k_t = [3, 3, 4, 3]
 Generation 6150: Best = 4.836977e+04 FES = 615100 μ_F = 0.770 μ_{CR} = 0.942
 Diversity = 1.4798e-02 k_t = [1, 1, 1, 1]
 Generation 6160: Best = 4.836529e+04 FES = 616100 μ_F = 0.769 μ_{CR} = 0.946
 Diversity = 1.8339e-02 k_t = [4, 3, 3, 4]

Generation 6170: Best = 4.835958e+04 FES = 617100 μ_F = 0.764 μ_{CR} = 0.943
 Diversity = 2.1990e-02 k_t = [1, 1, 1, 1]
 Generation 6180: Best = 4.835376e+04 FES = 618100 μ_F = 0.764 μ_{CR} = 0.949
 Diversity = 1.8368e-02 k_t = [3, 4, 3, 3]
 Generation 6190: Best = 4.834971e+04 FES = 619100 μ_F = 0.763 μ_{CR} = 0.954
 Diversity = 2.0688e-02 k_t = [1, 1, 1, 1]
 Generation 6200: Best = 4.834559e+04 FES = 620100 μ_F = 0.758 μ_{CR} = 0.956
 Diversity = 1.8618e-02 k_t = [3, 2, 4, 3]
 Generation 6210: Best = 4.834042e+04 FES = 621100 μ_F = 0.760 μ_{CR} = 0.948
 Diversity = 1.7999e-02 k_t = [1, 1, 1, 1]
 Generation 6220: Best = 4.833465e+04 FES = 622100 μ_F = 0.753 μ_{CR} = 0.951
 Diversity = 1.7459e-02 k_t = [3, 4, 4, 3]
 Generation 6230: Best = 4.832951e+04 FES = 623100 μ_F = 0.765 μ_{CR} = 0.950
 Diversity = 2.0391e-02 k_t = [1, 1, 1, 1]
 Generation 6240: Best = 4.832498e+04 FES = 624100 μ_F = 0.763 μ_{CR} = 0.950
 Diversity = 1.7908e-02 k_t = [4, 2, 4, 3]
 Generation 6250: Best = 4.832236e+04 FES = 625100 μ_F = 0.761 μ_{CR} = 0.941
 Diversity = 1.5775e-02 k_t = [1, 1, 1, 1]
 Generation 6260: Best = 4.831901e+04 FES = 626100 μ_F = 0.745 μ_{CR} = 0.941
 Diversity = 1.4381e-02 k_t = [3, 3, 2, 4]
 Generation 6270: Best = 4.831678e+04 FES = 627100 μ_F = 0.732 μ_{CR} = 0.944
 Diversity = 1.2669e-02 k_t = [1, 1, 1, 1]
 Generation 6280: Best = 4.831207e+04 FES = 628100 μ_F = 0.742 μ_{CR} = 0.941
 Diversity = 1.2016e-02 k_t = [4, 3, 4, 2]
 Generation 6290: Best = 4.831002e+04 FES = 629100 μ_F = 0.744 μ_{CR} = 0.942
 Diversity = 1.1211e-02 k_t = [1, 1, 1, 1]
 Generation 6300: Best = 4.830635e+04 FES = 630100 μ_F = 0.746 μ_{CR} = 0.945
 Diversity = 8.1562e-03 k_t = [3, 3, 4, 4]
 Generation 6310: Best = 4.830491e+04 FES = 631100 μ_F = 0.751 μ_{CR} = 0.934
 Diversity = 1.0067e-02 k_t = [1, 1, 1, 1]
 Generation 6320: Best = 4.830138e+04 FES = 632100 μ_F = 0.751 μ_{CR} = 0.938
 Diversity = 1.0324e-02 k_t = [3, 3, 3, 2]
 Generation 6330: Best = 4.829681e+04 FES = 633100 μ_F = 0.756 μ_{CR} = 0.948
 Diversity = 1.5220e-02 k_t = [1, 1, 1, 1]
 Generation 6340: Best = 4.829019e+04 FES = 634100 μ_F = 0.769 μ_{CR} = 0.949
 Diversity = 1.3909e-02 k_t = [4, 4, 3, 3]
 Generation 6350: Best = 4.828630e+04 FES = 635100 μ_F = 0.770 μ_{CR} = 0.953
 Diversity = 1.4606e-02 k_t = [1, 1, 1, 1]
 Generation 6360: Best = 4.828185e+04 FES = 636100 μ_F = 0.752 μ_{CR} = 0.940
 Diversity = 1.5621e-02 k_t = [2, 2, 2, 2]
 Generation 6370: Best = 4.827813e+04 FES = 637100 μ_F = 0.759 μ_{CR} = 0.943
 Diversity = 1.5662e-02 k_t = [1, 1, 1, 1]
 Generation 6380: Best = 4.827353e+04 FES = 638100 μ_F = 0.749 μ_{CR} = 0.951
 Diversity = 1.5791e-02 k_t = [3, 2, 2, 4]
 Generation 6390: Best = 4.826893e+04 FES = 639100 μ_F = 0.753 μ_{CR} = 0.950
 Diversity = 2.0752e-02 k_t = [1, 1, 1, 1]
 Generation 6400: Best = 4.826074e+04 FES = 640100 μ_F = 0.761 μ_{CR} = 0.952
 Diversity = 1.7292e-02 k_t = [3, 3, 2, 4]
 Generation 6410: Best = 4.825771e+04 FES = 641100 μ_F = 0.764 μ_{CR} = 0.961
 Diversity = 2.2472e-02 k_t = [1, 1, 1, 1]
 Generation 6420: Best = 4.825193e+04 FES = 642100 μ_F = 0.754 μ_{CR} = 0.964
 Diversity = 2.0292e-02 k_t = [2, 3, 4, 4]
 Generation 6430: Best = 4.824528e+04 FES = 643100 μ_F = 0.745 μ_{CR} = 0.967
 Diversity = 2.2354e-02 k_t = [1, 1, 1, 1]

Generation 6440: Best = 4.823898e+04 FES = 644100 μ_F = 0.759 μ_{CR} = 0.965
 Diversity = 2.3481e-02 k_t = [3, 4, 3, 3]
 Generation 6450: Best = 4.822813e+04 FES = 645100 μ_F = 0.758 μ_{CR} = 0.964
 Diversity = 2.6128e-02 k_t = [1, 1, 1, 1]
 Generation 6460: Best = 4.821399e+04 FES = 646100 μ_F = 0.763 μ_{CR} = 0.962
 Diversity = 2.9531e-02 k_t = [2, 2, 4, 4]
 Generation 6470: Best = 4.820377e+04 FES = 647100 μ_F = 0.764 μ_{CR} = 0.966
 Diversity = 3.4111e-02 k_t = [1, 1, 1, 1]
 Generation 6480: Best = 4.819383e+04 FES = 648100 μ_F = 0.753 μ_{CR} = 0.965
 Diversity = 2.2908e-02 k_t = [2, 2, 2, 4]
 Generation 6490: Best = 4.817815e+04 FES = 649100 μ_F = 0.766 μ_{CR} = 0.962
 Diversity = 2.8965e-02 k_t = [1, 1, 1, 1]
 Generation 6500: Best = 4.816898e+04 FES = 650100 μ_F = 0.753 μ_{CR} = 0.956
 Diversity = 2.3265e-02 k_t = [3, 4, 4, 3]
 Generation 6510: Best = 4.815446e+04 FES = 651100 μ_F = 0.764 μ_{CR} = 0.959
 Diversity = 3.1094e-02 k_t = [1, 1, 1, 1]
 Generation 6520: Best = 4.814244e+04 FES = 652100 μ_F = 0.769 μ_{CR} = 0.963
 Diversity = 2.9592e-02 k_t = [2, 2, 4, 2]
 Generation 6530: Best = 4.812796e+04 FES = 653100 μ_F = 0.764 μ_{CR} = 0.967
 Diversity = 2.9823e-02 k_t = [1, 1, 1, 1]
 Generation 6540: Best = 4.812499e+04 FES = 654100 μ_F = 0.772 μ_{CR} = 0.971
 Diversity = 2.9503e-02 k_t = [3, 3, 3, 2]
 Generation 6550: Best = 4.810815e+04 FES = 655100 μ_F = 0.755 μ_{CR} = 0.972
 Diversity = 3.9177e-02 k_t = [1, 1, 1, 1]
 Generation 6560: Best = 4.810214e+04 FES = 656100 μ_F = 0.758 μ_{CR} = 0.973
 Diversity = 3.0740e-02 k_t = [4, 3, 3, 3]
 Generation 6570: Best = 4.809177e+04 FES = 657100 μ_F = 0.765 μ_{CR} = 0.976
 Diversity = 3.0947e-02 k_t = [1, 1, 1, 1]
 Generation 6580: Best = 4.808313e+04 FES = 658100 μ_F = 0.750 μ_{CR} = 0.972
 Diversity = 2.3364e-02 k_t = [2, 2, 3, 4]
 Generation 6590: Best = 4.807512e+04 FES = 659100 μ_F = 0.749 μ_{CR} = 0.971
 Diversity = 1.8295e-02 k_t = [1, 1, 1, 1]
 Generation 6600: Best = 4.805100e+04 FES = 660100 μ_F = 0.773 μ_{CR} = 0.963
 Diversity = 2.0196e-02 k_t = [3, 4, 3, 4]
 Generation 6610: Best = 4.802599e+04 FES = 661100 μ_F = 0.789 μ_{CR} = 0.959
 Diversity = 3.4853e-02 k_t = [1, 1, 1, 1]
 Generation 6620: Best = 4.801655e+04 FES = 662100 μ_F = 0.788 μ_{CR} = 0.967
 Diversity = 2.9836e-02 k_t = [4, 3, 4, 2]
 Generation 6630: Best = 4.799619e+04 FES = 663100 μ_F = 0.793 μ_{CR} = 0.961
 Diversity = 3.3366e-02 k_t = [1, 1, 1, 1]
 Generation 6640: Best = 4.798168e+04 FES = 664100 μ_F = 0.796 μ_{CR} = 0.962
 Diversity = 3.4428e-02 k_t = [4, 2, 2, 4]
 Generation 6650: Best = 4.795444e+04 FES = 665100 μ_F = 0.805 μ_{CR} = 0.969
 Diversity = 6.0424e-02 k_t = [1, 1, 1, 1]
 Generation 6660: Best = 4.793286e+04 FES = 666100 μ_F = 0.777 μ_{CR} = 0.974
 Diversity = 6.5925e-02 k_t = [3, 4, 2, 2]
 Generation 6670: Best = 4.791663e+04 FES = 667100 μ_F = 0.777 μ_{CR} = 0.977
 Diversity = 5.3529e-02 k_t = [1, 1, 1, 1]
 Generation 6680: Best = 4.786170e+04 FES = 668100 μ_F = 0.767 μ_{CR} = 0.976
 Diversity = 6.8827e-02 k_t = [3, 3, 4, 4]
 Generation 6690: Best = 4.782764e+04 FES = 669100 μ_F = 0.776 μ_{CR} = 0.976
 Diversity = 6.9837e-02 k_t = [1, 1, 1, 1]
 Generation 6700: Best = 4.779715e+04 FES = 670100 μ_F = 0.776 μ_{CR} = 0.976
 Diversity = 6.6525e-02 k_t = [2, 4, 2, 2]

Generation 6710: Best = 4.778655e+04	FES = 671100	μ_F = 0.767	μ_{CR} = 0.976
Diversity = 5.0844e-02	k_t = [1, 1, 1, 1]		
Generation 6720: Best = 4.776246e+04	FES = 672100	μ_F = 0.753	μ_{CR} = 0.975
Diversity = 4.7536e-02	k_t = [3, 2, 2, 2]		
Generation 6730: Best = 4.773878e+04	FES = 673100	μ_F = 0.757	μ_{CR} = 0.974
Diversity = 5.5083e-02	k_t = [1, 1, 1, 1]		
Generation 6740: Best = 4.772035e+04	FES = 674100	μ_F = 0.767	μ_{CR} = 0.975
Diversity = 3.9726e-02	k_t = [2, 3, 2, 4]		
Generation 6750: Best = 4.768086e+04	FES = 675100	μ_F = 0.764	μ_{CR} = 0.971
Diversity = 4.3076e-02	k_t = [1, 1, 1, 1]		
Generation 6760: Best = 4.766898e+04	FES = 676100	μ_F = 0.760	μ_{CR} = 0.972
Diversity = 4.4401e-02	k_t = [4, 2, 3, 3]		
Generation 6770: Best = 4.765215e+04	FES = 677100	μ_F = 0.757	μ_{CR} = 0.969
Diversity = 2.8528e-02	k_t = [1, 1, 1, 1]		
Generation 6780: Best = 4.764362e+04	FES = 678100	μ_F = 0.757	μ_{CR} = 0.972
Diversity = 3.5230e-02	k_t = [4, 3, 3, 4]		
Generation 6790: Best = 4.763111e+04	FES = 679100	μ_F = 0.754	μ_{CR} = 0.974
Diversity = 4.0702e-02	k_t = [1, 1, 1, 1]		
Generation 6800: Best = 4.761900e+04	FES = 680100	μ_F = 0.753	μ_{CR} = 0.974
Diversity = 3.0707e-02	k_t = [2, 4, 4, 3]		
Generation 6810: Best = 4.761020e+04	FES = 681100	μ_F = 0.769	μ_{CR} = 0.966
Diversity = 2.4906e-02	k_t = [1, 1, 1, 1]		
Generation 6820: Best = 4.760549e+04	FES = 682100	μ_F = 0.759	μ_{CR} = 0.964
Diversity = 1.8099e-02	k_t = [3, 2, 4, 2]		
Generation 6830: Best = 4.759907e+04	FES = 683100	μ_F = 0.749	μ_{CR} = 0.962
Diversity = 2.1720e-02	k_t = [1, 1, 1, 1]		
Generation 6840: Best = 4.758737e+04	FES = 684100	μ_F = 0.755	μ_{CR} = 0.964
Diversity = 2.0724e-02	k_t = [3, 3, 3, 4]		
Generation 6850: Best = 4.758366e+04	FES = 685100	μ_F = 0.749	μ_{CR} = 0.962
Diversity = 1.4672e-02	k_t = [1, 1, 1, 1]		
Generation 6860: Best = 4.757787e+04	FES = 686100	μ_F = 0.763	μ_{CR} = 0.965
Diversity = 1.5964e-02	k_t = [2, 2, 3, 3]		
Generation 6870: Best = 4.757335e+04	FES = 687100	μ_F = 0.759	μ_{CR} = 0.962
Diversity = 1.6673e-02	k_t = [1, 1, 1, 1]		
Generation 6880: Best = 4.756706e+04	FES = 688100	μ_F = 0.742	μ_{CR} = 0.959
Diversity = 1.5150e-02	k_t = [4, 2, 2, 3]		
Generation 6890: Best = 4.756485e+04	FES = 689100	μ_F = 0.721	μ_{CR} = 0.951
Diversity = 1.2168e-02	k_t = [1, 1, 1, 1]		
Generation 6900: Best = 4.756108e+04	FES = 690100	μ_F = 0.735	μ_{CR} = 0.951
Diversity = 1.0382e-02	k_t = [2, 4, 3, 4]		
Generation 6910: Best = 4.755863e+04	FES = 691100	μ_F = 0.741	μ_{CR} = 0.945
Diversity = 1.0615e-02	k_t = [1, 1, 1, 1]		
Generation 6920: Best = 4.755430e+04	FES = 692100	μ_F = 0.743	μ_{CR} = 0.955
Diversity = 1.2090e-02	k_t = [2, 3, 2, 2]		
Generation 6930: Best = 4.755115e+04	FES = 693100	μ_F = 0.744	μ_{CR} = 0.960
Diversity = 1.3789e-02	k_t = [1, 1, 1, 1]		
Generation 6940: Best = 4.754820e+04	FES = 694100	μ_F = 0.742	μ_{CR} = 0.956
Diversity = 9.5583e-03	k_t = [4, 4, 3, 3]		
Generation 6950: Best = 4.754575e+04	FES = 695100	μ_F = 0.755	μ_{CR} = 0.958
Diversity = 1.0125e-02	k_t = [1, 1, 1, 1]		
Generation 6960: Best = 4.754404e+04	FES = 696100	μ_F = 0.776	μ_{CR} = 0.960
Diversity = 8.9587e-03	k_t = [4, 4, 2, 3]		
Generation 6970: Best = 4.754116e+04	FES = 697100	μ_F = 0.781	μ_{CR} = 0.957
Diversity = 1.0781e-02	k_t = [1, 1, 1, 1]		

Generation 6980: Best = 4.753980e+04 FES = 698100 μ_F = 0.780 μ_{CR} = 0.957
 Diversity = 8.4610e-03 k_t = [4, 4, 2, 3]
 Generation 6990: Best = 4.753835e+04 FES = 699100 μ_F = 0.780 μ_{CR} = 0.950
 Diversity = 9.4380e-03 k_t = [1, 1, 1, 1]
 Generation 7000: Best = 4.753470e+04 FES = 700100 μ_F = 0.779 μ_{CR} = 0.951
 Diversity = 1.1512e-02 k_t = [2, 3, 2, 4]
 Generation 7010: Best = 4.753205e+04 FES = 701100 μ_F = 0.776 μ_{CR} = 0.952
 Diversity = 1.3788e-02 k_t = [1, 1, 1, 1]
 Generation 7020: Best = 4.752853e+04 FES = 702100 μ_F = 0.769 μ_{CR} = 0.950
 Diversity = 1.4416e-02 k_t = [3, 2, 3, 4]
 Generation 7030: Best = 4.752489e+04 FES = 703100 μ_F = 0.758 μ_{CR} = 0.959
 Diversity = 1.4492e-02 k_t = [1, 1, 1, 1]
 Generation 7040: Best = 4.752143e+04 FES = 704100 μ_F = 0.760 μ_{CR} = 0.958
 Diversity = 1.2996e-02 k_t = [3, 4, 3, 4]
 Generation 7050: Best = 4.751871e+04 FES = 705100 μ_F = 0.757 μ_{CR} = 0.961
 Diversity = 1.3379e-02 k_t = [1, 1, 1, 1]
 Generation 7060: Best = 4.751380e+04 FES = 706100 μ_F = 0.759 μ_{CR} = 0.951
 Diversity = 1.5989e-02 k_t = [3, 2, 2, 3]
 Generation 7070: Best = 4.750786e+04 FES = 707100 μ_F = 0.751 μ_{CR} = 0.950
 Diversity = 2.0655e-02 k_t = [1, 1, 1, 1]
 Generation 7080: Best = 4.749901e+04 FES = 708100 μ_F = 0.757 μ_{CR} = 0.957
 Diversity = 3.0008e-02 k_t = [3, 3, 4, 4]
 Generation 7090: Best = 4.748820e+04 FES = 709100 μ_F = 0.743 μ_{CR} = 0.950
 Diversity = 2.3881e-02 k_t = [1, 1, 1, 1]
 Generation 7100: Best = 4.748572e+04 FES = 710100 μ_F = 0.748 μ_{CR} = 0.951
 Diversity = 1.6103e-02 k_t = [2, 2, 4, 3]
 Generation 7110: Best = 4.748150e+04 FES = 711100 μ_F = 0.755 μ_{CR} = 0.952
 Diversity = 1.7401e-02 k_t = [1, 1, 1, 1]
 Generation 7120: Best = 4.747666e+04 FES = 712100 μ_F = 0.760 μ_{CR} = 0.954
 Diversity = 1.5632e-02 k_t = [3, 2, 3, 4]
 Generation 7130: Best = 4.746807e+04 FES = 713100 μ_F = 0.764 μ_{CR} = 0.940
 Diversity = 1.7715e-02 k_t = [1, 1, 1, 1]
 Generation 7140: Best = 4.746200e+04 FES = 714100 μ_F = 0.766 μ_{CR} = 0.934
 Diversity = 1.1386e-02 k_t = [4, 4, 3, 4]
 Generation 7150: Best = 4.745531e+04 FES = 715100 μ_F = 0.770 μ_{CR} = 0.934
 Diversity = 1.4004e-02 k_t = [1, 1, 1, 1]
 Generation 7160: Best = 4.745058e+04 FES = 716100 μ_F = 0.781 μ_{CR} = 0.936
 Diversity = 1.4249e-02 k_t = [3, 4, 3, 2]
 Generation 7170: Best = 4.744451e+04 FES = 717100 μ_F = 0.784 μ_{CR} = 0.938
 Diversity = 1.5961e-02 k_t = [1, 1, 1, 1]
 Generation 7180: Best = 4.744252e+04 FES = 718100 μ_F = 0.782 μ_{CR} = 0.944
 Diversity = 1.0217e-02 k_t = [4, 3, 4, 3]
 Generation 7190: Best = 4.743727e+04 FES = 719100 μ_F = 0.772 μ_{CR} = 0.943
 Diversity = 1.5073e-02 k_t = [1, 1, 1, 1]
 Generation 7200: Best = 4.743373e+04 FES = 720100 μ_F = 0.775 μ_{CR} = 0.939
 Diversity = 1.4758e-02 k_t = [3, 3, 3, 3]
 Generation 7210: Best = 4.743161e+04 FES = 721100 μ_F = 0.760 μ_{CR} = 0.927
 Diversity = 1.7149e-02 k_t = [1, 1, 1, 1]
 Generation 7220: Best = 4.742738e+04 FES = 722100 μ_F = 0.744 μ_{CR} = 0.927
 Diversity = 1.6868e-02 k_t = [3, 2, 4, 2]
 Generation 7230: Best = 4.742255e+04 FES = 723100 μ_F = 0.734 μ_{CR} = 0.927
 Diversity = 1.3908e-02 k_t = [1, 1, 1, 1]
 Generation 7240: Best = 4.741163e+04 FES = 724100 μ_F = 0.738 μ_{CR} = 0.927
 Diversity = 2.4796e-02 k_t = [2, 2, 2, 3]

Generation 7250: Best = 4.740498e+04 FES = 725100 μ_F = 0.759 μ_{CR} = 0.936
 Diversity = 2.3650e-02 k_t = [1, 1, 1, 1]
 Generation 7260: Best = 4.739763e+04 FES = 726100 μ_F = 0.774 μ_{CR} = 0.945
 Diversity = 2.8932e-02 k_t = [3, 2, 2, 4]
 Generation 7270: Best = 4.738822e+04 FES = 727100 μ_F = 0.763 μ_{CR} = 0.947
 Diversity = 2.7302e-02 k_t = [1, 1, 1, 1]
 Generation 7280: Best = 4.738066e+04 FES = 728100 μ_F = 0.757 μ_{CR} = 0.953
 Diversity = 2.6129e-02 k_t = [2, 2, 3, 2]
 Generation 7290: Best = 4.737222e+04 FES = 729100 μ_F = 0.762 μ_{CR} = 0.947
 Diversity = 2.5782e-02 k_t = [1, 1, 1, 1]
 Generation 7300: Best = 4.736363e+04 FES = 730100 μ_F = 0.763 μ_{CR} = 0.947
 Diversity = 2.6339e-02 k_t = [4, 4, 4, 2]
 Generation 7310: Best = 4.735622e+04 FES = 731100 μ_F = 0.775 μ_{CR} = 0.943
 Diversity = 2.6967e-02 k_t = [1, 1, 1, 1]
 Generation 7320: Best = 4.734083e+04 FES = 732100 μ_F = 0.769 μ_{CR} = 0.942
 Diversity = 2.6305e-02 k_t = [2, 3, 3, 4]
 Generation 7330: Best = 4.733593e+04 FES = 733100 μ_F = 0.768 μ_{CR} = 0.949
 Diversity = 2.7131e-02 k_t = [1, 1, 1, 1]
 Generation 7340: Best = 4.732209e+04 FES = 734100 μ_F = 0.774 μ_{CR} = 0.949
 Diversity = 3.0718e-02 k_t = [4, 3, 4, 2]
 Generation 7350: Best = 4.731384e+04 FES = 735100 μ_F = 0.760 μ_{CR} = 0.937
 Diversity = 1.9099e-02 k_t = [1, 1, 1, 1]
 Generation 7360: Best = 4.730971e+04 FES = 736100 μ_F = 0.756 μ_{CR} = 0.937
 Diversity = 1.6733e-02 k_t = [4, 3, 2, 2]
 Generation 7370: Best = 4.730619e+04 FES = 737100 μ_F = 0.765 μ_{CR} = 0.919
 Diversity = 1.2930e-02 k_t = [1, 1, 1, 1]
 Generation 7380: Best = 4.730343e+04 FES = 738100 μ_F = 0.767 μ_{CR} = 0.917
 Diversity = 9.8414e-03 k_t = [4, 3, 2, 2]
 Generation 7390: Best = 4.729801e+04 FES = 739100 μ_F = 0.773 μ_{CR} = 0.921
 Diversity = 1.2150e-02 k_t = [1, 1, 1, 1]
 Generation 7400: Best = 4.729196e+04 FES = 740100 μ_F = 0.765 μ_{CR} = 0.915
 Diversity = 1.3808e-02 k_t = [2, 2, 2, 2]
 Generation 7410: Best = 4.728736e+04 FES = 741100 μ_F = 0.759 μ_{CR} = 0.906
 Diversity = 1.5152e-02 k_t = [1, 1, 1, 1]
 Generation 7420: Best = 4.728404e+04 FES = 742100 μ_F = 0.758 μ_{CR} = 0.917
 Diversity = 1.3796e-02 k_t = [3, 4, 4, 3]
 Generation 7430: Best = 4.727942e+04 FES = 743100 μ_F = 0.767 μ_{CR} = 0.924
 Diversity = 1.7536e-02 k_t = [1, 1, 1, 1]
 Generation 7440: Best = 4.727456e+04 FES = 744100 μ_F = 0.763 μ_{CR} = 0.931
 Diversity = 1.5924e-02 k_t = [3, 2, 3, 3]
 Generation 7450: Best = 4.727066e+04 FES = 745100 μ_F = 0.759 μ_{CR} = 0.941
 Diversity = 1.8599e-02 k_t = [1, 1, 1, 1]
 Generation 7460: Best = 4.726511e+04 FES = 746100 μ_F = 0.752 μ_{CR} = 0.951
 Diversity = 2.1735e-02 k_t = [3, 3, 2, 3]
 Generation 7470: Best = 4.726230e+04 FES = 747100 μ_F = 0.752 μ_{CR} = 0.955
 Diversity = 2.0413e-02 k_t = [1, 1, 1, 1]
 Generation 7480: Best = 4.725742e+04 FES = 748100 μ_F = 0.762 μ_{CR} = 0.960
 Diversity = 2.2939e-02 k_t = [3, 2, 2, 3]
 Generation 7490: Best = 4.725029e+04 FES = 749100 μ_F = 0.770 μ_{CR} = 0.956
 Diversity = 2.5379e-02 k_t = [1, 1, 1, 1]
 Generation 7500: Best = 4.724500e+04 FES = 750100 μ_F = 0.765 μ_{CR} = 0.958
 Diversity = 2.5784e-02 k_t = [2, 3, 2, 4]
 Generation 7510: Best = 4.724119e+04 FES = 751100 μ_F = 0.762 μ_{CR} = 0.953
 Diversity = 2.1676e-02 k_t = [1, 1, 1, 1]

Generation 7520: Best = 4.723776e+04 FES = 752100 μ_F = 0.757 μ_{CR} = 0.953
 Diversity = 1.9367e-02 k_t = [3, 4, 2, 4]
 Generation 7530: Best = 4.723643e+04 FES = 753100 μ_F = 0.762 μ_{CR} = 0.958
 Diversity = 1.4613e-02 k_t = [1, 1, 1, 1]
 Generation 7540: Best = 4.723389e+04 FES = 754100 μ_F = 0.758 μ_{CR} = 0.956
 Diversity = 1.5021e-02 k_t = [2, 4, 4, 3]
 Generation 7550: Best = 4.722992e+04 FES = 755100 μ_F = 0.755 μ_{CR} = 0.956
 Diversity = 1.9684e-02 k_t = [1, 1, 1, 1]
 Generation 7560: Best = 4.722682e+04 FES = 756100 μ_F = 0.755 μ_{CR} = 0.962
 Diversity = 1.9076e-02 k_t = [3, 4, 3, 3]
 Generation 7570: Best = 4.722499e+04 FES = 757100 μ_F = 0.744 μ_{CR} = 0.968
 Diversity = 1.5053e-02 k_t = [1, 1, 1, 1]
 Generation 7580: Best = 4.722308e+04 FES = 758100 μ_F = 0.734 μ_{CR} = 0.959
 Diversity = 1.2873e-02 k_t = [2, 4, 2, 3]
 Generation 7590: Best = 4.721906e+04 FES = 759100 μ_F = 0.735 μ_{CR} = 0.949
 Diversity = 1.5945e-02 k_t = [1, 1, 1, 1]
 Generation 7600: Best = 4.721734e+04 FES = 760100 μ_F = 0.723 μ_{CR} = 0.948
 Diversity = 1.2969e-02 k_t = [2, 3, 4, 2]
 Generation 7610: Best = 4.721348e+04 FES = 761100 μ_F = 0.738 μ_{CR} = 0.954
 Diversity = 1.5924e-02 k_t = [1, 1, 1, 1]
 Generation 7620: Best = 4.720943e+04 FES = 762100 μ_F = 0.735 μ_{CR} = 0.953
 Diversity = 1.2287e-02 k_t = [3, 4, 3, 3]
 Generation 7630: Best = 4.720660e+04 FES = 763100 μ_F = 0.728 μ_{CR} = 0.944
 Diversity = 1.1022e-02 k_t = [1, 1, 1, 1]
 Generation 7640: Best = 4.720120e+04 FES = 764100 μ_F = 0.741 μ_{CR} = 0.951
 Diversity = 1.5607e-02 k_t = [2, 3, 3, 4]
 Generation 7650: Best = 4.719896e+04 FES = 765100 μ_F = 0.739 μ_{CR} = 0.961
 Diversity = 1.7609e-02 k_t = [1, 1, 1, 1]
 Generation 7660: Best = 4.719439e+04 FES = 766100 μ_F = 0.741 μ_{CR} = 0.956
 Diversity = 1.4737e-02 k_t = [4, 3, 4, 3]
 Generation 7670: Best = 4.719084e+04 FES = 767100 μ_F = 0.755 μ_{CR} = 0.959
 Diversity = 1.7215e-02 k_t = [1, 1, 1, 1]
 Generation 7680: Best = 4.718926e+04 FES = 768100 μ_F = 0.760 μ_{CR} = 0.961
 Diversity = 1.4274e-02 k_t = [2, 3, 2, 2]
 Generation 7690: Best = 4.718784e+04 FES = 769100 μ_F = 0.750 μ_{CR} = 0.964
 Diversity = 1.5115e-02 k_t = [1, 1, 1, 1]
 Generation 7700: Best = 4.718551e+04 FES = 770100 μ_F = 0.749 μ_{CR} = 0.964
 Diversity = 1.2084e-02 k_t = [2, 4, 2, 4]
 Generation 7710: Best = 4.718239e+04 FES = 771100 μ_F = 0.752 μ_{CR} = 0.959
 Diversity = 1.6630e-02 k_t = [1, 1, 1, 1]
 Generation 7720: Best = 4.717734e+04 FES = 772100 μ_F = 0.752 μ_{CR} = 0.959
 Diversity = 1.7005e-02 k_t = [3, 4, 2, 3]
 Generation 7730: Best = 4.717537e+04 FES = 773100 μ_F = 0.744 μ_{CR} = 0.948
 Diversity = 1.3405e-02 k_t = [1, 1, 1, 1]
 Generation 7740: Best = 4.717131e+04 FES = 774100 μ_F = 0.743 μ_{CR} = 0.950
 Diversity = 1.3159e-02 k_t = [4, 3, 2, 4]
 Generation 7750: Best = 4.716388e+04 FES = 775100 μ_F = 0.750 μ_{CR} = 0.954
 Diversity = 1.9649e-02 k_t = [1, 1, 1, 1]
 Generation 7760: Best = 4.715783e+04 FES = 776100 μ_F = 0.751 μ_{CR} = 0.951
 Diversity = 2.4110e-02 k_t = [3, 4, 3, 3]
 Generation 7770: Best = 4.715280e+04 FES = 777100 μ_F = 0.760 μ_{CR} = 0.952
 Diversity = 2.5626e-02 k_t = [1, 1, 1, 1]
 Generation 7780: Best = 4.714607e+04 FES = 778100 μ_F = 0.767 μ_{CR} = 0.949
 Diversity = 2.1833e-02 k_t = [4, 2, 3, 2]

Generation 7790: Best = 4.714111e+04 FES = 779100 μ_F = 0.767 μ_{CR} = 0.951
 Diversity = 2.0964e-02 k_t = [1, 1, 1, 1]
 Generation 7800: Best = 4.713590e+04 FES = 780100 μ_F = 0.768 μ_{CR} = 0.957
 Diversity = 1.8173e-02 k_t = [3, 3, 2, 4]
 Generation 7810: Best = 4.713285e+04 FES = 781100 μ_F = 0.769 μ_{CR} = 0.951
 Diversity = 2.3042e-02 k_t = [1, 1, 1, 1]
 Generation 7820: Best = 4.712939e+04 FES = 782100 μ_F = 0.776 μ_{CR} = 0.952
 Diversity = 1.4528e-02 k_t = [4, 2, 2, 4]
 Generation 7830: Best = 4.712812e+04 FES = 783100 μ_F = 0.779 μ_{CR} = 0.948
 Diversity = 1.0818e-02 k_t = [1, 1, 1, 1]
 Generation 7840: Best = 4.712521e+04 FES = 784100 μ_F = 0.773 μ_{CR} = 0.947
 Diversity = 1.4302e-02 k_t = [3, 4, 3, 2]
 Generation 7850: Best = 4.712308e+04 FES = 785100 μ_F = 0.761 μ_{CR} = 0.950
 Diversity = 1.2040e-02 k_t = [1, 1, 1, 1]
 Generation 7860: Best = 4.711846e+04 FES = 786100 μ_F = 0.777 μ_{CR} = 0.943
 Diversity = 1.5341e-02 k_t = [3, 2, 2, 4]
 Generation 7870: Best = 4.711452e+04 FES = 787100 μ_F = 0.785 μ_{CR} = 0.949
 Diversity = 2.2531e-02 k_t = [1, 1, 1, 1]
 Generation 7880: Best = 4.710826e+04 FES = 788100 μ_F = 0.784 μ_{CR} = 0.955
 Diversity = 2.2711e-02 k_t = [3, 2, 2, 2]
 Generation 7890: Best = 4.710424e+04 FES = 789100 μ_F = 0.785 μ_{CR} = 0.957
 Diversity = 2.4159e-02 k_t = [1, 1, 1, 1]
 Generation 7900: Best = 4.709395e+04 FES = 790100 μ_F = 0.767 μ_{CR} = 0.958
 Diversity = 2.3153e-02 k_t = [3, 2, 3, 3]
 Generation 7910: Best = 4.709116e+04 FES = 791100 μ_F = 0.760 μ_{CR} = 0.962
 Diversity = 2.3971e-02 k_t = [1, 1, 1, 1]
 Generation 7920: Best = 4.708455e+04 FES = 792100 μ_F = 0.751 μ_{CR} = 0.967
 Diversity = 2.3477e-02 k_t = [2, 3, 2, 3]
 Generation 7930: Best = 4.706178e+04 FES = 793100 μ_F = 0.743 μ_{CR} = 0.969
 Diversity = 3.7773e-02 k_t = [1, 1, 1, 1]
 Generation 7940: Best = 4.705947e+04 FES = 794100 μ_F = 0.756 μ_{CR} = 0.964
 Diversity = 2.3678e-02 k_t = [2, 4, 4, 3]
 Generation 7950: Best = 4.705593e+04 FES = 795100 μ_F = 0.767 μ_{CR} = 0.961
 Diversity = 2.0428e-02 k_t = [1, 1, 1, 1]
 Generation 7960: Best = 4.705164e+04 FES = 796100 μ_F = 0.754 μ_{CR} = 0.960
 Diversity = 2.0793e-02 k_t = [2, 4, 4, 2]
 Generation 7970: Best = 4.704830e+04 FES = 797100 μ_F = 0.765 μ_{CR} = 0.957
 Diversity = 1.8970e-02 k_t = [1, 1, 1, 1]
 Generation 7980: Best = 4.704509e+04 FES = 798100 μ_F = 0.768 μ_{CR} = 0.964
 Diversity = 1.5682e-02 k_t = [4, 2, 2, 2]
 Generation 7990: Best = 4.704342e+04 FES = 799100 μ_F = 0.770 μ_{CR} = 0.963
 Diversity = 1.3554e-02 k_t = [1, 1, 1, 1]
 Generation 8000: Best = 4.704154e+04 FES = 800100 μ_F = 0.767 μ_{CR} = 0.963
 Diversity = 1.1368e-02 k_t = [3, 2, 2, 4]
 Generation 8010: Best = 4.704001e+04 FES = 801100 μ_F = 0.769 μ_{CR} = 0.957
 Diversity = 1.1204e-02 k_t = [1, 1, 1, 1]
 Generation 8020: Best = 4.703727e+04 FES = 802100 μ_F = 0.765 μ_{CR} = 0.956
 Diversity = 1.6169e-02 k_t = [4, 2, 2, 3]
 Generation 8030: Best = 4.703593e+04 FES = 803100 μ_F = 0.755 μ_{CR} = 0.959
 Diversity = 1.4593e-02 k_t = [1, 1, 1, 1]
 Generation 8040: Best = 4.703437e+04 FES = 804100 μ_F = 0.766 μ_{CR} = 0.961
 Diversity = 1.1359e-02 k_t = [2, 4, 3, 3]
 Generation 8050: Best = 4.703243e+04 FES = 805100 μ_F = 0.760 μ_{CR} = 0.956
 Diversity = 1.3706e-02 k_t = [1, 1, 1, 1]

Generation 8060: Best = 4.703138e+04 FES = 806100 μ_F = 0.758 μ_{CR} = 0.953
 Diversity = 1.3026e-02 k_t = [2, 2, 2, 3]
 Generation 8070: Best = 4.703028e+04 FES = 807100 μ_F = 0.757 μ_{CR} = 0.949
 Diversity = 1.4756e-02 k_t = [1, 1, 1, 1]
 Generation 8080: Best = 4.702927e+04 FES = 808100 μ_F = 0.754 μ_{CR} = 0.948
 Diversity = 1.3436e-02 k_t = [3, 3, 4, 2]
 Generation 8090: Best = 4.702583e+04 FES = 809100 μ_F = 0.748 μ_{CR} = 0.952
 Diversity = 1.7574e-02 k_t = [1, 1, 1, 1]
 Generation 8100: Best = 4.702572e+04 FES = 810100 μ_F = 0.751 μ_{CR} = 0.957
 Diversity = 1.3674e-02 k_t = [4, 3, 2, 2]
 Generation 8110: Best = 4.702472e+04 FES = 811100 μ_F = 0.744 μ_{CR} = 0.958
 Diversity = 1.0896e-02 k_t = [1, 1, 1, 1]
 Generation 8120: Best = 4.702365e+04 FES = 812100 μ_F = 0.709 μ_{CR} = 0.952
 Diversity = 8.3975e-03 k_t = [2, 2, 2, 3]
 Generation 8130: Best = 4.702249e+04 FES = 813100 μ_F = 0.722 μ_{CR} = 0.948
 Diversity = 8.4713e-03 k_t = [1, 1, 1, 1]
 Generation 8140: Best = 4.702034e+04 FES = 814100 μ_F = 0.731 μ_{CR} = 0.952
 Diversity = 1.1625e-02 k_t = [2, 2, 3, 4]
 Generation 8150: Best = 4.701705e+04 FES = 815100 μ_F = 0.746 μ_{CR} = 0.950
 Diversity = 1.6409e-02 k_t = [1, 1, 1, 1]
 Generation 8160: Best = 4.701424e+04 FES = 816100 μ_F = 0.744 μ_{CR} = 0.956
 Diversity = 1.8040e-02 k_t = [4, 4, 3, 2]
 Generation 8170: Best = 4.701192e+04 FES = 817100 μ_F = 0.755 μ_{CR} = 0.959
 Diversity = 1.3689e-02 k_t = [1, 1, 1, 1]
 Generation 8180: Best = 4.700990e+04 FES = 818100 μ_F = 0.732 μ_{CR} = 0.959
 Diversity = 1.2712e-02 k_t = [2, 4, 3, 4]
 Generation 8190: Best = 4.700824e+04 FES = 819100 μ_F = 0.734 μ_{CR} = 0.957
 Diversity = 1.3470e-02 k_t = [1, 1, 1, 1]
 Generation 8200: Best = 4.700723e+04 FES = 820100 μ_F = 0.746 μ_{CR} = 0.950
 Diversity = 1.1306e-02 k_t = [4, 3, 4, 2]
 Generation 8210: Best = 4.700646e+04 FES = 821100 μ_F = 0.761 μ_{CR} = 0.934
 Diversity = 8.6968e-03 k_t = [1, 1, 1, 1]
 Generation 8220: Best = 4.700550e+04 FES = 822100 μ_F = 0.761 μ_{CR} = 0.941
 Diversity = 1.0349e-02 k_t = [2, 2, 4, 3]
 Generation 8230: Best = 4.700494e+04 FES = 823100 μ_F = 0.738 μ_{CR} = 0.941
 Diversity = 9.1905e-03 k_t = [1, 1, 1, 1]
 Generation 8240: Best = 4.700327e+04 FES = 824100 μ_F = 0.743 μ_{CR} = 0.942
 Diversity = 1.1262e-02 k_t = [3, 3, 3, 2]
 Generation 8250: Best = 4.700320e+04 FES = 825100 μ_F = 0.736 μ_{CR} = 0.946
 Diversity = 8.1852e-03 k_t = [1, 1, 1, 1]
 Generation 8260: Best = 4.700253e+04 FES = 826100 μ_F = 0.738 μ_{CR} = 0.949
 Diversity = 6.8903e-03 k_t = [3, 4, 2, 4]
 Generation 8270: Best = 4.700178e+04 FES = 827100 μ_F = 0.738 μ_{CR} = 0.944
 Diversity = 6.5159e-03 k_t = [1, 1, 1, 1]
 Generation 8280: Best = 4.700155e+04 FES = 828100 μ_F = 0.727 μ_{CR} = 0.941
 Diversity = 4.6969e-03 k_t = [4, 3, 3, 3]
 Generation 8290: Best = 4.700131e+04 FES = 829100 μ_F = 0.747 μ_{CR} = 0.939
 Diversity = 3.9461e-03 k_t = [1, 1, 1, 1]
 Generation 8300: Best = 4.700118e+04 FES = 830100 μ_F = 0.754 μ_{CR} = 0.938
 Diversity = 2.6130e-03 k_t = [4, 4, 2, 4]
 Generation 8310: Best = 4.700086e+04 FES = 831100 μ_F = 0.769 μ_{CR} = 0.937
 Diversity = 2.8564e-03 k_t = [1, 1, 1, 1]
 Generation 8320: Best = 4.700050e+04 FES = 832100 μ_F = 0.787 μ_{CR} = 0.942
 Diversity = 2.9359e-03 k_t = [2, 4, 4, 4]

Generation 8330: Best = 4.700020e+04 FES = 833100 μ_F = 0.790 μ_{CR} = 0.940
 Diversity = 3.3410e-03 k_t = [1, 1, 1, 1]
 Generation 8340: Best = 4.699999e+04 FES = 834100 μ_F = 0.791 μ_{CR} = 0.935
 Diversity = 2.6596e-03 k_t = [4, 2, 2, 3]
 Generation 8350: Best = 4.699956e+04 FES = 835100 μ_F = 0.793 μ_{CR} = 0.931
 Diversity = 3.3476e-03 k_t = [1, 1, 1, 1]
 Generation 8360: Best = 4.699933e+04 FES = 836100 μ_F = 0.789 μ_{CR} = 0.922
 Diversity = 3.0732e-03 k_t = [2, 4, 4, 4]
 Generation 8370: Best = 4.699902e+04 FES = 837100 μ_F = 0.792 μ_{CR} = 0.925
 Diversity = 3.0435e-03 k_t = [1, 1, 1, 1]
 Generation 8380: Best = 4.699877e+04 FES = 838100 μ_F = 0.781 μ_{CR} = 0.927
 Diversity = 2.5345e-03 k_t = [2, 3, 3, 4]
 Generation 8390: Best = 4.699845e+04 FES = 839100 μ_F = 0.783 μ_{CR} = 0.921
 Diversity = 3.5022e-03 k_t = [1, 1, 1, 1]
 Generation 8400: Best = 4.699805e+04 FES = 840100 μ_F = 0.791 μ_{CR} = 0.931
 Diversity = 3.0713e-03 k_t = [3, 2, 3, 4]
 Generation 8410: Best = 4.699748e+04 FES = 841100 μ_F = 0.796 μ_{CR} = 0.938
 Diversity = 4.8880e-03 k_t = [1, 1, 1, 1]
 Generation 8420: Best = 4.699696e+04 FES = 842100 μ_F = 0.781 μ_{CR} = 0.945
 Diversity = 6.3316e-03 k_t = [2, 2, 3, 3]
 Generation 8430: Best = 4.699682e+04 FES = 843100 μ_F = 0.777 μ_{CR} = 0.938
 Diversity = 4.9451e-03 k_t = [1, 1, 1, 1]
 Generation 8440: Best = 4.699590e+04 FES = 844100 μ_F = 0.779 μ_{CR} = 0.942
 Diversity = 5.3684e-03 k_t = [3, 4, 3, 2]
 Generation 8450: Best = 4.699519e+04 FES = 845100 μ_F = 0.783 μ_{CR} = 0.946
 Diversity = 5.7229e-03 k_t = [1, 1, 1, 1]
 Generation 8460: Best = 4.699449e+04 FES = 846100 μ_F = 0.779 μ_{CR} = 0.951
 Diversity = 4.7836e-03 k_t = [3, 4, 4, 3]
 Generation 8470: Best = 4.699390e+04 FES = 847100 μ_F = 0.769 μ_{CR} = 0.947
 Diversity = 4.2361e-03 k_t = [1, 1, 1, 1]
 Generation 8480: Best = 4.699334e+04 FES = 848100 μ_F = 0.773 μ_{CR} = 0.944
 Diversity = 4.7210e-03 k_t = [4, 2, 4, 2]
 Generation 8490: Best = 4.699216e+04 FES = 849100 μ_F = 0.780 μ_{CR} = 0.941
 Diversity = 6.6330e-03 k_t = [1, 1, 1, 1]
 Generation 8500: Best = 4.699115e+04 FES = 850100 μ_F = 0.779 μ_{CR} = 0.942
 Diversity = 6.0062e-03 k_t = [2, 4, 4, 4]
 Generation 8510: Best = 4.698978e+04 FES = 851100 μ_F = 0.795 μ_{CR} = 0.941
 Diversity = 7.1195e-03 k_t = [1, 1, 1, 1]
 Generation 8520: Best = 4.698781e+04 FES = 852100 μ_F = 0.797 μ_{CR} = 0.933
 Diversity = 7.7086e-03 k_t = [3, 2, 2, 3]
 Generation 8530: Best = 4.698673e+04 FES = 853100 μ_F = 0.787 μ_{CR} = 0.937
 Diversity = 7.6922e-03 k_t = [1, 1, 1, 1]
 Generation 8540: Best = 4.698442e+04 FES = 854100 μ_F = 0.793 μ_{CR} = 0.941
 Diversity = 8.6415e-03 k_t = [2, 4, 2, 2]
 Generation 8550: Best = 4.698179e+04 FES = 855100 μ_F = 0.796 μ_{CR} = 0.940
 Diversity = 9.0884e-03 k_t = [1, 1, 1, 1]
 Generation 8560: Best = 4.698045e+04 FES = 856100 μ_F = 0.802 μ_{CR} = 0.940
 Diversity = 8.1069e-03 k_t = [3, 4, 3, 2]
 Generation 8570: Best = 4.697966e+04 FES = 857100 μ_F = 0.802 μ_{CR} = 0.946
 Diversity = 8.6394e-03 k_t = [1, 1, 1, 1]
 Generation 8580: Best = 4.697862e+04 FES = 858100 μ_F = 0.787 μ_{CR} = 0.940
 Diversity = 6.5097e-03 k_t = [2, 3, 3, 3]
 Generation 8590: Best = 4.697725e+04 FES = 859100 μ_F = 0.781 μ_{CR} = 0.948
 Diversity = 1.0664e-02 k_t = [1, 1, 1, 1]

Generation 8600: Best = 4.697575e+04 FES = 860100 μ_F = 0.766 μ_{CR} = 0.943
 Diversity = 8.9225e-03 k_t = [3, 4, 3, 2]
 Generation 8610: Best = 4.697382e+04 FES = 861100 μ_F = 0.776 μ_{CR} = 0.946
 Diversity = 1.1823e-02 k_t = [1, 1, 1, 1]
 Generation 8620: Best = 4.697191e+04 FES = 862100 μ_F = 0.765 μ_{CR} = 0.954
 Diversity = 1.0786e-02 k_t = [2, 3, 4, 4]
 Generation 8630: Best = 4.696989e+04 FES = 863100 μ_F = 0.756 μ_{CR} = 0.956
 Diversity = 1.3873e-02 k_t = [1, 1, 1, 1]
 Generation 8640: Best = 4.696822e+04 FES = 864100 μ_F = 0.747 μ_{CR} = 0.957
 Diversity = 1.1175e-02 k_t = [2, 2, 4, 3]
 Generation 8650: Best = 4.696690e+04 FES = 865100 μ_F = 0.735 μ_{CR} = 0.958
 Diversity = 1.1176e-02 k_t = [1, 1, 1, 1]
 Generation 8660: Best = 4.696491e+04 FES = 866100 μ_F = 0.746 μ_{CR} = 0.951
 Diversity = 7.6480e-03 k_t = [2, 4, 4, 2]
 Generation 8670: Best = 4.696432e+04 FES = 867100 μ_F = 0.751 μ_{CR} = 0.956
 Diversity = 7.0416e-03 k_t = [1, 1, 1, 1]
 Generation 8680: Best = 4.695983e+04 FES = 868100 μ_F = 0.763 μ_{CR} = 0.958
 Diversity = 8.4681e-03 k_t = [3, 2, 4, 4]
 Generation 8690: Best = 4.695903e+04 FES = 869100 μ_F = 0.771 μ_{CR} = 0.955
 Diversity = 6.8162e-03 k_t = [1, 1, 1, 1]
 Generation 8700: Best = 4.695807e+04 FES = 870100 μ_F = 0.774 μ_{CR} = 0.945
 Diversity = 5.6790e-03 k_t = [3, 3, 4, 3]
 Generation 8710: Best = 4.695644e+04 FES = 871100 μ_F = 0.780 μ_{CR} = 0.945
 Diversity = 7.2236e-03 k_t = [1, 1, 1, 1]
 Generation 8720: Best = 4.695585e+04 FES = 872100 μ_F = 0.777 μ_{CR} = 0.948
 Diversity = 6.5651e-03 k_t = [2, 4, 3, 4]
 Generation 8730: Best = 4.695408e+04 FES = 873100 μ_F = 0.765 μ_{CR} = 0.951
 Diversity = 1.2916e-02 k_t = [1, 1, 1, 1]
 Generation 8740: Best = 4.694935e+04 FES = 874100 μ_F = 0.766 μ_{CR} = 0.956
 Diversity = 1.8787e-02 k_t = [3, 4, 3, 2]
 Generation 8750: Best = 4.694748e+04 FES = 875100 μ_F = 0.765 μ_{CR} = 0.958
 Diversity = 1.8362e-02 k_t = [1, 1, 1, 1]
 Generation 8760: Best = 4.694507e+04 FES = 876100 μ_F = 0.760 μ_{CR} = 0.964
 Diversity = 1.7045e-02 k_t = [3, 4, 3, 3]
 Generation 8770: Best = 4.694175e+04 FES = 877100 μ_F = 0.754 μ_{CR} = 0.968
 Diversity = 1.9972e-02 k_t = [1, 1, 1, 1]
 Generation 8780: Best = 4.693900e+04 FES = 878100 μ_F = 0.767 μ_{CR} = 0.967
 Diversity = 1.4337e-02 k_t = [4, 3, 4, 2]
 Generation 8790: Best = 4.693789e+04 FES = 879100 μ_F = 0.779 μ_{CR} = 0.963
 Diversity = 1.3468e-02 k_t = [1, 1, 1, 1]
 Generation 8800: Best = 4.693717e+04 FES = 880100 μ_F = 0.766 μ_{CR} = 0.959
 Diversity = 9.4001e-03 k_t = [3, 3, 4, 2]
 Generation 8810: Best = 4.693589e+04 FES = 881100 μ_F = 0.789 μ_{CR} = 0.952
 Diversity = 9.9290e-03 k_t = [1, 1, 1, 1]
 Generation 8820: Best = 4.693492e+04 FES = 882100 μ_F = 0.781 μ_{CR} = 0.952
 Diversity = 8.9454e-03 k_t = [2, 3, 2, 2]
 Generation 8830: Best = 4.693374e+04 FES = 883100 μ_F = 0.789 μ_{CR} = 0.955
 Diversity = 1.1005e-02 k_t = [1, 1, 1, 1]
 Generation 8840: Best = 4.692965e+04 FES = 884100 μ_F = 0.795 μ_{CR} = 0.959
 Diversity = 1.4258e-02 k_t = [2, 3, 4, 4]
 Generation 8850: Best = 4.692822e+04 FES = 885100 μ_F = 0.778 μ_{CR} = 0.960
 Diversity = 1.3304e-02 k_t = [1, 1, 1, 1]
 Generation 8860: Best = 4.692622e+04 FES = 886100 μ_F = 0.774 μ_{CR} = 0.958
 Diversity = 1.0401e-02 k_t = [2, 4, 2, 2]

Generation 8870: Best = 4.692549e+04	FES = 887100	$\mu_F = 0.747$	$\mu_{CR} = 0.953$
Diversity = 9.0112e-03	k_t = [1, 1, 1, 1]		
Generation 8880: Best = 4.692433e+04	FES = 888100	$\mu_F = 0.751$	$\mu_{CR} = 0.954$
Diversity = 6.4331e-03	k_t = [2, 2, 2, 4]		
Generation 8890: Best = 4.692412e+04	FES = 889100	$\mu_F = 0.759$	$\mu_{CR} = 0.953$
Diversity = 6.0752e-03	k_t = [1, 1, 1, 1]		
Generation 8900: Best = 4.692303e+04	FES = 890100	$\mu_F = 0.772$	$\mu_{CR} = 0.947$
Diversity = 4.9704e-03	k_t = [4, 2, 2, 2]		
Generation 8910: Best = 4.692239e+04	FES = 891100	$\mu_F = 0.773$	$\mu_{CR} = 0.946$
Diversity = 6.0911e-03	k_t = [1, 1, 1, 1]		
Generation 8920: Best = 4.692118e+04	FES = 892100	$\mu_F = 0.788$	$\mu_{CR} = 0.941$
Diversity = 6.4173e-03	k_t = [3, 4, 2, 2]		
Generation 8930: Best = 4.691983e+04	FES = 893100	$\mu_F = 0.783$	$\mu_{CR} = 0.940$
Diversity = 7.2041e-03	k_t = [1, 1, 1, 1]		
Generation 8940: Best = 4.691893e+04	FES = 894100	$\mu_F = 0.759$	$\mu_{CR} = 0.936$
Diversity = 6.2485e-03	k_t = [3, 4, 4, 2]		
Generation 8950: Best = 4.691760e+04	FES = 895100	$\mu_F = 0.758$	$\mu_{CR} = 0.940$
Diversity = 6.6330e-03	k_t = [1, 1, 1, 1]		
Generation 8960: Best = 4.691721e+04	FES = 896100	$\mu_F = 0.757$	$\mu_{CR} = 0.938$
Diversity = 4.4326e-03	k_t = [3, 2, 3, 3]		
Generation 8970: Best = 4.691616e+04	FES = 897100	$\mu_F = 0.749$	$\mu_{CR} = 0.940$
Diversity = 6.3562e-03	k_t = [1, 1, 1, 1]		
Generation 8980: Best = 4.691527e+04	FES = 898100	$\mu_F = 0.749$	$\mu_{CR} = 0.942$
Diversity = 7.3089e-03	k_t = [4, 3, 4, 4]		
Generation 8990: Best = 4.691439e+04	FES = 899100	$\mu_F = 0.754$	$\mu_{CR} = 0.941$
Diversity = 6.2476e-03	k_t = [1, 1, 1, 1]		
Generation 9000: Best = 4.691377e+04	FES = 900100	$\mu_F = 0.753$	$\mu_{CR} = 0.945$
Diversity = 4.8948e-03	k_t = [4, 2, 3, 4]		
Generation 9010: Best = 4.691296e+04	FES = 901100	$\mu_F = 0.755$	$\mu_{CR} = 0.941$
Diversity = 7.1295e-03	k_t = [1, 1, 1, 1]		
Generation 9020: Best = 4.691218e+04	FES = 902100	$\mu_F = 0.756$	$\mu_{CR} = 0.934$
Diversity = 6.9265e-03	k_t = [2, 3, 4, 4]		
Generation 9030: Best = 4.691070e+04	FES = 903100	$\mu_F = 0.765$	$\mu_{CR} = 0.924$
Diversity = 1.0112e-02	k_t = [1, 1, 1, 1]		
Generation 9040: Best = 4.690926e+04	FES = 904100	$\mu_F = 0.757$	$\mu_{CR} = 0.934$
Diversity = 9.5010e-03	k_t = [4, 3, 2, 3]		
Generation 9050: Best = 4.690579e+04	FES = 905100	$\mu_F = 0.753$	$\mu_{CR} = 0.937$
Diversity = 1.2768e-02	k_t = [1, 1, 1, 1]		
Generation 9060: Best = 4.690444e+04	FES = 906100	$\mu_F = 0.750$	$\mu_{CR} = 0.935$
Diversity = 8.3557e-03	k_t = [3, 3, 4, 3]		
Generation 9070: Best = 4.690171e+04	FES = 907100	$\mu_F = 0.763$	$\mu_{CR} = 0.940$
Diversity = 1.0781e-02	k_t = [1, 1, 1, 1]		
Generation 9080: Best = 4.689997e+04	FES = 908100	$\mu_F = 0.764$	$\mu_{CR} = 0.942$
Diversity = 9.3796e-03	k_t = [3, 3, 4, 4]		
Generation 9090: Best = 4.689805e+04	FES = 909100	$\mu_F = 0.782$	$\mu_{CR} = 0.947$
Diversity = 8.8763e-03	k_t = [1, 1, 1, 1]		
Generation 9100: Best = 4.689492e+04	FES = 910100	$\mu_F = 0.780$	$\mu_{CR} = 0.936$
Diversity = 8.4120e-03	k_t = [2, 3, 4, 4]		
Generation 9110: Best = 4.689457e+04	FES = 911100	$\mu_F = 0.786$	$\mu_{CR} = 0.936$
Diversity = 6.8689e-03	k_t = [1, 1, 1, 1]		
Generation 9120: Best = 4.689383e+04	FES = 912100	$\mu_F = 0.771$	$\mu_{CR} = 0.934$
Diversity = 5.1325e-03	k_t = [2, 3, 3, 3]		
Generation 9130: Best = 4.689285e+04	FES = 913100	$\mu_F = 0.782$	$\mu_{CR} = 0.937$
Diversity = 5.7260e-03	k_t = [1, 1, 1, 1]		

Generation 9140: Best = 4.689184e+04 FES = 914100 μ_F = 0.796 μ_{CR} = 0.943
 Diversity = 6.3601e-03 k_t = [3, 2, 4, 2]
 Generation 9150: Best = 4.689023e+04 FES = 915100 μ_F = 0.788 μ_{CR} = 0.942
 Diversity = 8.7138e-03 k_t = [1, 1, 1, 1]
 Generation 9160: Best = 4.688912e+04 FES = 916100 μ_F = 0.770 μ_{CR} = 0.932
 Diversity = 6.7137e-03 k_t = [3, 3, 3, 2]
 Generation 9170: Best = 4.688804e+04 FES = 917100 μ_F = 0.786 μ_{CR} = 0.932
 Diversity = 7.6559e-03 k_t = [1, 1, 1, 1]
 Generation 9180: Best = 4.688630e+04 FES = 918100 μ_F = 0.762 μ_{CR} = 0.933
 Diversity = 7.5304e-03 k_t = [3, 2, 2, 4]
 Generation 9190: Best = 4.688466e+04 FES = 919100 μ_F = 0.767 μ_{CR} = 0.932
 Diversity = 7.6267e-03 k_t = [1, 1, 1, 1]
 Generation 9200: Best = 4.688205e+04 FES = 920100 μ_F = 0.753 μ_{CR} = 0.927
 Diversity = 8.8902e-03 k_t = [2, 2, 4, 2]
 Generation 9210: Best = 4.688092e+04 FES = 921100 μ_F = 0.761 μ_{CR} = 0.931
 Diversity = 8.4165e-03 k_t = [1, 1, 1, 1]
 Generation 9220: Best = 4.687779e+04 FES = 922100 μ_F = 0.780 μ_{CR} = 0.935
 Diversity = 9.8553e-03 k_t = [2, 2, 3, 4]
 Generation 9230: Best = 4.687523e+04 FES = 923100 μ_F = 0.768 μ_{CR} = 0.937
 Diversity = 9.2857e-03 k_t = [1, 1, 1, 1]
 Generation 9240: Best = 4.687255e+04 FES = 924100 μ_F = 0.768 μ_{CR} = 0.939
 Diversity = 7.4288e-03 k_t = [3, 3, 4, 4]
 Generation 9250: Best = 4.687097e+04 FES = 925100 μ_F = 0.778 μ_{CR} = 0.933
 Diversity = 9.4203e-03 k_t = [1, 1, 1, 1]
 Generation 9260: Best = 4.686962e+04 FES = 926100 μ_F = 0.768 μ_{CR} = 0.923
 Diversity = 8.2473e-03 k_t = [3, 3, 2, 4]
 Generation 9270: Best = 4.686794e+04 FES = 927100 μ_F = 0.770 μ_{CR} = 0.923
 Diversity = 8.7213e-03 k_t = [1, 1, 1, 1]
 Generation 9280: Best = 4.686688e+04 FES = 928100 μ_F = 0.761 μ_{CR} = 0.924
 Diversity = 6.6376e-03 k_t = [3, 3, 2, 3]
 Generation 9290: Best = 4.686421e+04 FES = 929100 μ_F = 0.776 μ_{CR} = 0.928
 Diversity = 9.7297e-03 k_t = [1, 1, 1, 1]
 Generation 9300: Best = 4.686272e+04 FES = 930100 μ_F = 0.782 μ_{CR} = 0.926
 Diversity = 7.4127e-03 k_t = [4, 4, 3, 3]
 Generation 9310: Best = 4.686142e+04 FES = 931100 μ_F = 0.790 μ_{CR} = 0.926
 Diversity = 7.2106e-03 k_t = [1, 1, 1, 1]
 Generation 9320: Best = 4.685968e+04 FES = 932100 μ_F = 0.791 μ_{CR} = 0.927
 Diversity = 6.5316e-03 k_t = [3, 3, 3, 4]
 Generation 9330: Best = 4.685818e+04 FES = 933100 μ_F = 0.793 μ_{CR} = 0.925
 Diversity = 8.5929e-03 k_t = [1, 1, 1, 1]
 Generation 9340: Best = 4.685664e+04 FES = 934100 μ_F = 0.792 μ_{CR} = 0.931
 Diversity = 5.9081e-03 k_t = [2, 3, 4, 2]
 Generation 9350: Best = 4.685558e+04 FES = 935100 μ_F = 0.787 μ_{CR} = 0.925
 Diversity = 7.4531e-03 k_t = [1, 1, 1, 1]
 Generation 9360: Best = 4.685467e+04 FES = 936100 μ_F = 0.771 μ_{CR} = 0.922
 Diversity = 6.6796e-03 k_t = [4, 4, 2, 2]
 Generation 9370: Best = 4.685385e+04 FES = 937100 μ_F = 0.777 μ_{CR} = 0.921
 Diversity = 7.5624e-03 k_t = [1, 1, 1, 1]
 Generation 9380: Best = 4.685257e+04 FES = 938100 μ_F = 0.777 μ_{CR} = 0.925
 Diversity = 5.4232e-03 k_t = [4, 3, 4, 4]
 Generation 9390: Best = 4.685153e+04 FES = 939100 μ_F = 0.790 μ_{CR} = 0.916
 Diversity = 6.7159e-03 k_t = [1, 1, 1, 1]
 Generation 9400: Best = 4.684973e+04 FES = 940100 μ_F = 0.784 μ_{CR} = 0.906
 Diversity = 6.6727e-03 k_t = [4, 3, 3, 3]

Generation 9410: Best = 4.684737e+04 FES = 941100 μ_F = 0.768 μ_{CR} = 0.917
 Diversity = 6.9831e-03 k_t = [1, 1, 1, 1]
 Generation 9420: Best = 4.684430e+04 FES = 942100 μ_F = 0.774 μ_{CR} = 0.924
 Diversity = 8.5154e-03 k_t = [3, 3, 4, 3]
 Generation 9430: Best = 4.684371e+04 FES = 943100 μ_F = 0.770 μ_{CR} = 0.923
 Diversity = 6.3286e-03 k_t = [1, 1, 1, 1]
 Generation 9440: Best = 4.684271e+04 FES = 944100 μ_F = 0.775 μ_{CR} = 0.927
 Diversity = 5.0291e-03 k_t = [4, 3, 3, 4]
 Generation 9450: Best = 4.684088e+04 FES = 945100 μ_F = 0.791 μ_{CR} = 0.931
 Diversity = 7.9315e-03 k_t = [1, 1, 1, 1]
 Generation 9460: Best = 4.683822e+04 FES = 946100 μ_F = 0.764 μ_{CR} = 0.926
 Diversity = 7.0644e-03 k_t = [2, 2, 2, 3]
 Generation 9470: Best = 4.683669e+04 FES = 947100 μ_F = 0.768 μ_{CR} = 0.922
 Diversity = 7.9055e-03 k_t = [1, 1, 1, 1]
 Generation 9480: Best = 4.683496e+04 FES = 948100 μ_F = 0.773 μ_{CR} = 0.924
 Diversity = 6.7219e-03 k_t = [4, 3, 3, 4]
 Generation 9490: Best = 4.683358e+04 FES = 949100 μ_F = 0.774 μ_{CR} = 0.916
 Diversity = 6.7056e-03 k_t = [1, 1, 1, 1]
 Generation 9500: Best = 4.683163e+04 FES = 950100 μ_F = 0.783 μ_{CR} = 0.916
 Diversity = 6.0780e-03 k_t = [3, 4, 3, 3]
 Generation 9510: Best = 4.683071e+04 FES = 951100 μ_F = 0.790 μ_{CR} = 0.918
 Diversity = 6.0903e-03 k_t = [1, 1, 1, 1]
 Generation 9520: Best = 4.682970e+04 FES = 952100 μ_F = 0.797 μ_{CR} = 0.927
 Diversity = 5.5230e-03 k_t = [2, 4, 2, 3]
 Generation 9530: Best = 4.682897e+04 FES = 953100 μ_F = 0.777 μ_{CR} = 0.925
 Diversity = 5.1209e-03 k_t = [1, 1, 1, 1]
 Generation 9540: Best = 4.682861e+04 FES = 954100 μ_F = 0.757 μ_{CR} = 0.928
 Diversity = 5.0521e-03 k_t = [3, 3, 3, 2]
 Generation 9550: Best = 4.682783e+04 FES = 955100 μ_F = 0.762 μ_{CR} = 0.925
 Diversity = 4.7264e-03 k_t = [1, 1, 1, 1]
 Generation 9560: Best = 4.682698e+04 FES = 956100 μ_F = 0.770 μ_{CR} = 0.930
 Diversity = 4.4935e-03 k_t = [3, 3, 2, 3]
 Generation 9570: Best = 4.682613e+04 FES = 957100 μ_F = 0.777 μ_{CR} = 0.928
 Diversity = 5.6386e-03 k_t = [1, 1, 1, 1]
 Generation 9580: Best = 4.682518e+04 FES = 958100 μ_F = 0.784 μ_{CR} = 0.931
 Diversity = 5.0046e-03 k_t = [3, 4, 4, 3]
 Generation 9590: Best = 4.682316e+04 FES = 959100 μ_F = 0.795 μ_{CR} = 0.927
 Diversity = 8.0186e-03 k_t = [1, 1, 1, 1]
 Generation 9600: Best = 4.682149e+04 FES = 960100 μ_F = 0.793 μ_{CR} = 0.929
 Diversity = 7.6977e-03 k_t = [3, 3, 3, 3]
 Generation 9610: Best = 4.681981e+04 FES = 961100 μ_F = 0.788 μ_{CR} = 0.931
 Diversity = 8.0302e-03 k_t = [1, 1, 1, 1]
 Generation 9620: Best = 4.681817e+04 FES = 962100 μ_F = 0.791 μ_{CR} = 0.937
 Diversity = 8.3483e-03 k_t = [3, 3, 3, 4]
 Generation 9630: Best = 4.681665e+04 FES = 963100 μ_F = 0.792 μ_{CR} = 0.937
 Diversity = 1.0664e-02 k_t = [1, 1, 1, 1]
 Generation 9640: Best = 4.681042e+04 FES = 964100 μ_F = 0.782 μ_{CR} = 0.943
 Diversity = 1.6833e-02 k_t = [4, 3, 4, 3]
 Generation 9650: Best = 4.680697e+04 FES = 965100 μ_F = 0.785 μ_{CR} = 0.949
 Diversity = 1.9345e-02 k_t = [1, 1, 1, 1]
 Generation 9660: Best = 4.680317e+04 FES = 966100 μ_F = 0.794 μ_{CR} = 0.956
 Diversity = 1.5173e-02 k_t = [4, 4, 4, 3]
 Generation 9670: Best = 4.679791e+04 FES = 967100 μ_F = 0.801 μ_{CR} = 0.950
 Diversity = 1.7780e-02 k_t = [1, 1, 1, 1]

Generation 9680: Best = 4.679449e+04 FES = 968100 μ_F = 0.796 μ_{CR} = 0.947
 Diversity = 1.6753e-02 k_t = [2, 4, 3, 3]
 Generation 9690: Best = 4.679086e+04 FES = 969100 μ_F = 0.783 μ_{CR} = 0.955
 Diversity = 1.6739e-02 k_t = [1, 1, 1, 1]
 Generation 9700: Best = 4.678824e+04 FES = 970100 μ_F = 0.795 μ_{CR} = 0.962
 Diversity = 1.9433e-02 k_t = [2, 4, 4, 3]
 Generation 9710: Best = 4.678225e+04 FES = 971100 μ_F = 0.770 μ_{CR} = 0.964
 Diversity = 2.1921e-02 k_t = [1, 1, 1, 1]
 Generation 9720: Best = 4.677234e+04 FES = 972100 μ_F = 0.775 μ_{CR} = 0.967
 Diversity = 3.3633e-02 k_t = [4, 3, 2, 4]
 Generation 9730: Best = 4.676584e+04 FES = 973100 μ_F = 0.784 μ_{CR} = 0.966
 Diversity = 2.6706e-02 k_t = [1, 1, 1, 1]
 Generation 9740: Best = 4.675852e+04 FES = 974100 μ_F = 0.770 μ_{CR} = 0.970
 Diversity = 2.6989e-02 k_t = [3, 3, 4, 3]
 Generation 9750: Best = 4.675469e+04 FES = 975100 μ_F = 0.761 μ_{CR} = 0.967
 Diversity = 2.3097e-02 k_t = [1, 1, 1, 1]
 Generation 9760: Best = 4.675036e+04 FES = 976100 μ_F = 0.759 μ_{CR} = 0.959
 Diversity = 2.3029e-02 k_t = [3, 2, 4, 2]
 Generation 9770: Best = 4.674079e+04 FES = 977100 μ_F = 0.753 μ_{CR} = 0.965
 Diversity = 3.3285e-02 k_t = [1, 1, 1, 1]
 Generation 9780: Best = 4.673748e+04 FES = 978100 μ_F = 0.746 μ_{CR} = 0.969
 Diversity = 2.6983e-02 k_t = [3, 3, 3, 2]
 Generation 9790: Best = 4.673263e+04 FES = 979100 μ_F = 0.731 μ_{CR} = 0.962
 Diversity = 2.4590e-02 k_t = [1, 1, 1, 1]
 Generation 9800: Best = 4.673137e+04 FES = 980100 μ_F = 0.746 μ_{CR} = 0.967
 Diversity = 1.8111e-02 k_t = [2, 4, 4, 2]
 Generation 9810: Best = 4.672907e+04 FES = 981100 μ_F = 0.741 μ_{CR} = 0.970
 Diversity = 2.0006e-02 k_t = [1, 1, 1, 1]
 Generation 9820: Best = 4.672774e+04 FES = 982100 μ_F = 0.737 μ_{CR} = 0.964
 Diversity = 1.5008e-02 k_t = [3, 3, 3, 3]
 Generation 9830: Best = 4.671928e+04 FES = 983100 μ_F = 0.740 μ_{CR} = 0.961
 Diversity = 2.3386e-02 k_t = [1, 1, 1, 1]
 Generation 9840: Best = 4.671062e+04 FES = 984100 μ_F = 0.753 μ_{CR} = 0.963
 Diversity = 2.9284e-02 k_t = [2, 3, 4, 2]
 Generation 9850: Best = 4.670624e+04 FES = 985100 μ_F = 0.752 μ_{CR} = 0.960
 Diversity = 2.5740e-02 k_t = [1, 1, 1, 1]
 Generation 9860: Best = 4.670268e+04 FES = 986100 μ_F = 0.760 μ_{CR} = 0.956
 Diversity = 1.4667e-02 k_t = [4, 4, 4, 4]
 Generation 9870: Best = 4.669883e+04 FES = 987100 μ_F = 0.761 μ_{CR} = 0.956
 Diversity = 1.6339e-02 k_t = [1, 1, 1, 1]
 Generation 9880: Best = 4.669475e+04 FES = 988100 μ_F = 0.759 μ_{CR} = 0.962
 Diversity = 1.8745e-02 k_t = [2, 3, 4, 3]
 Generation 9890: Best = 4.669296e+04 FES = 989100 μ_F = 0.734 μ_{CR} = 0.954
 Diversity = 1.4951e-02 k_t = [1, 1, 1, 1]
 Generation 9900: Best = 4.668783e+04 FES = 990100 μ_F = 0.741 μ_{CR} = 0.952
 Diversity = 1.7068e-02 k_t = [2, 4, 2, 3]
 Generation 9910: Best = 4.668722e+04 FES = 991100 μ_F = 0.747 μ_{CR} = 0.961
 Diversity = 1.4143e-02 k_t = [1, 1, 1, 1]
 Generation 9920: Best = 4.668428e+04 FES = 992100 μ_F = 0.746 μ_{CR} = 0.958
 Diversity = 1.3914e-02 k_t = [4, 2, 4, 2]
 Generation 9930: Best = 4.668098e+04 FES = 993100 μ_F = 0.735 μ_{CR} = 0.959
 Diversity = 1.3373e-02 k_t = [1, 1, 1, 1]
 Generation 9940: Best = 4.667966e+04 FES = 994100 μ_F = 0.735 μ_{CR} = 0.959
 Diversity = 1.4818e-02 k_t = [2, 3, 3, 2]

Generation 9950: Best = 4.667496e+04 FES = 995100 μ_F = 0.739 μ_{CR} = 0.962
 Diversity = 2.1494e-02 k_t = [1, 1, 1, 1]
 Generation 9960: Best = 4.667284e+04 FES = 996100 μ_F = 0.729 μ_{CR} = 0.961
 Diversity = 1.2374e-02 k_t = [2, 3, 2, 4]
 Generation 9970: Best = 4.667053e+04 FES = 997100 μ_F = 0.726 μ_{CR} = 0.959
 Diversity = 1.0529e-02 k_t = [1, 1, 1, 1]
 Generation 9980: Best = 4.666661e+04 FES = 998100 μ_F = 0.734 μ_{CR} = 0.952
 Diversity = 1.3051e-02 k_t = [4, 3, 3, 4]
 Generation 9990: Best = 4.666440e+04 FES = 999100 μ_F = 0.738 μ_{CR} = 0.948
 Diversity = 1.1891e-02 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 4.666286e+04

Total Function Evals = 1000000

Total Successful Updates = 338249

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Results:

Best Fitness : 4.666286e+04

Global Optimum : 3.000000e+02

Error (|best-optimum|) : 4.636286e+04

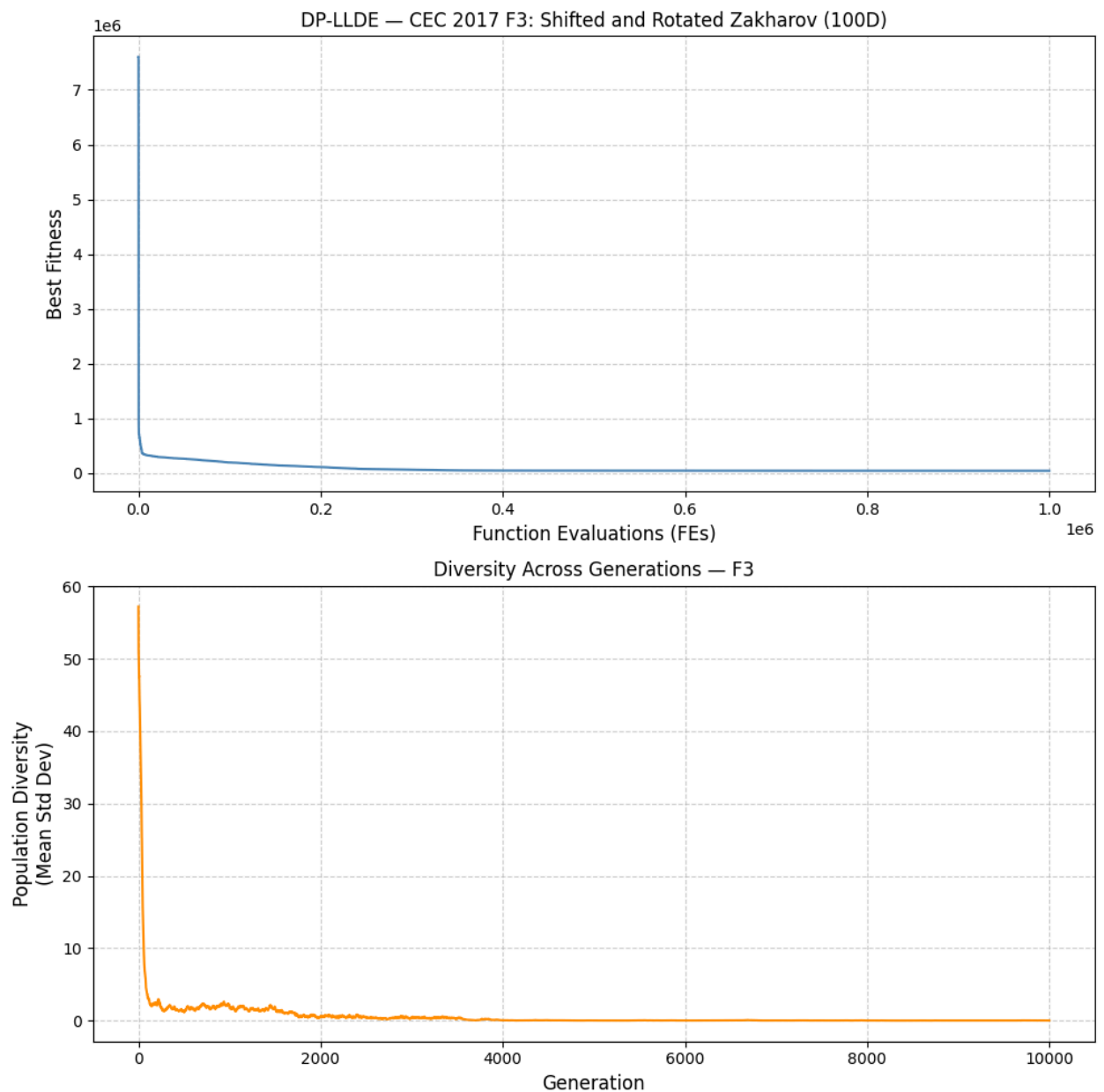
Best Solution (5 dim) : [-32.92703988 -1.23134861 32.43795311 0.89160719
-20.67733074]

D^(0) (init. diversity): 5.7220e+01

=====

=

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results/all_results_checkpoint.pkl



Graph saved as convergence_F3.png

✓ Plot saved to: /content/drive/My Drive/thesis_results/plots/convergence_F3.png

6.2 Multimodal Functions (F4-F10)

```
In [ ]: # =====
# F4 – Shifted and Rotated Rosenbrock
# =====
run_and_plot('F4', make_wrapper(simple.f4), 400.0, "Shifted and Rotated Rosent
```



```

=====
=
DP-LLDE on CEC 2017 F4 – Single Run
=====
=
Function      : Shifted and Rotated Rosenbrock's (100D)
Bounds       : [-100, 100]
Global optimum : 400.0
Initial D^(0) = 5.7220e+01 | Diversity threshold = 5.7220e+00
Generation 0: Best Fitness = 2.335375e+05
Generation 10: Best = 6.448679e+04 FES = 1100  $\mu_F$  = 0.556  $\mu_{CR}$  = 0.351
Diversity = 3.0946e+01 k_t = [1, 1, 1, 1]
Generation 20: Best = 3.028908e+04 FES = 2100  $\mu_F$  = 0.586  $\mu_{CR}$  = 0.350
Diversity = 1.7575e+01 k_t = [2, 3, 2, 3]
Generation 30: Best = 1.553869e+04 FES = 3100  $\mu_F$  = 0.610  $\mu_{CR}$  = 0.352
Diversity = 1.1493e+01 k_t = [1, 1, 1, 1]
Generation 40: Best = 1.173428e+04 FES = 4100  $\mu_F$  = 0.645  $\mu_{CR}$  = 0.361
Diversity = 7.9388e+00 k_t = [3, 2, 3, 3]
Generation 50: Best = 9.195586e+03 FES = 5100  $\mu_F$  = 0.657  $\mu_{CR}$  = 0.367
Diversity = 6.1616e+00 k_t = [1, 1, 1, 1]
Generation 60: Best = 7.327133e+03 FES = 6100  $\mu_F$  = 0.677  $\mu_{CR}$  = 0.365
Diversity = 4.9043e+00 k_t = [3, 4, 4, 3]
Generation 70: Best = 6.080730e+03 FES = 7100  $\mu_F$  = 0.703  $\mu_{CR}$  = 0.379
Diversity = 4.1385e+00 k_t = [1, 1, 1, 1]
Generation 80: Best = 5.093245e+03 FES = 8100  $\mu_F$  = 0.714  $\mu_{CR}$  = 0.381
Diversity = 3.7595e+00 k_t = [2, 4, 4, 3]
Generation 90: Best = 4.298254e+03 FES = 9100  $\mu_F$  = 0.724  $\mu_{CR}$  = 0.377
Diversity = 3.6050e+00 k_t = [1, 1, 1, 1]
Generation 100: Best = 3.484488e+03 FES = 10100  $\mu_F$  = 0.733  $\mu_{CR}$  = 0.382
Diversity = 3.3232e+00 k_t = [4, 4, 4, 2]
Generation 110: Best = 3.042339e+03 FES = 11100  $\mu_F$  = 0.742  $\mu_{CR}$  = 0.394
Diversity = 3.1381e+00 k_t = [1, 1, 1, 1]
Generation 120: Best = 2.687407e+03 FES = 12100  $\mu_F$  = 0.743  $\mu_{CR}$  = 0.403
Diversity = 2.6010e+00 k_t = [3, 3, 2, 4]
Generation 130: Best = 2.400414e+03 FES = 13100  $\mu_F$  = 0.752  $\mu_{CR}$  = 0.408
Diversity = 2.3305e+00 k_t = [1, 1, 1, 1]
Generation 140: Best = 2.156889e+03 FES = 14100  $\mu_F$  = 0.754  $\mu_{CR}$  = 0.410
Diversity = 2.1150e+00 k_t = [2, 4, 2, 3]
Generation 150: Best = 1.953984e+03 FES = 15100  $\mu_F$  = 0.760  $\mu_{CR}$  = 0.414
Diversity = 2.1175e+00 k_t = [1, 1, 1, 1]
Generation 160: Best = 1.777397e+03 FES = 16100  $\mu_F$  = 0.749  $\mu_{CR}$  = 0.420
Diversity = 1.8859e+00 k_t = [4, 3, 2, 2]
Generation 170: Best = 1.625577e+03 FES = 17100  $\mu_F$  = 0.742  $\mu_{CR}$  = 0.418
Diversity = 1.8285e+00 k_t = [1, 1, 1, 1]
Generation 180: Best = 1.518951e+03 FES = 18100  $\mu_F$  = 0.740  $\mu_{CR}$  = 0.423
Diversity = 1.4495e+00 k_t = [3, 4, 2, 3]
Generation 190: Best = 1.441344e+03 FES = 19100  $\mu_F$  = 0.735  $\mu_{CR}$  = 0.427
Diversity = 1.3065e+00 k_t = [1, 1, 1, 1]
Generation 200: Best = 1.377888e+03 FES = 20100  $\mu_F$  = 0.740  $\mu_{CR}$  = 0.426
Diversity = 1.1996e+00 k_t = [2, 3, 3, 3]
Generation 210: Best = 1.321421e+03 FES = 21100  $\mu_F$  = 0.742  $\mu_{CR}$  = 0.435
Diversity = 1.0973e+00 k_t = [1, 1, 1, 1]
Generation 220: Best = 1.281611e+03 FES = 22100  $\mu_F$  = 0.739  $\mu_{CR}$  = 0.442
Diversity = 8.6267e-01 k_t = [4, 2, 2, 3]

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Generation 230: Best = 1.235717e+03	FES = 23100	$\mu_F = 0.752$	$\mu_{CR} = 0.445$
Diversity = 8.5337e-01	k_t = [1, 1, 1, 1]		
Generation 240: Best = 1.203201e+03	FES = 24100	$\mu_F = 0.752$	$\mu_{CR} = 0.450$
Diversity = 8.7042e-01	k_t = [4, 2, 4, 2]		
Generation 250: Best = 1.163597e+03	FES = 25100	$\mu_F = 0.759$	$\mu_{CR} = 0.449$
Diversity = 8.2778e-01	k_t = [1, 1, 1, 1]		
Generation 260: Best = 1.136042e+03	FES = 26100	$\mu_F = 0.759$	$\mu_{CR} = 0.450$
Diversity = 7.5358e-01	k_t = [2, 4, 4, 3]		
Generation 270: Best = 1.111444e+03	FES = 27100	$\mu_F = 0.756$	$\mu_{CR} = 0.450$
Diversity = 7.6722e-01	k_t = [1, 1, 1, 1]		
Generation 280: Best = 1.084988e+03	FES = 28100	$\mu_F = 0.755$	$\mu_{CR} = 0.451$
Diversity = 6.9488e-01	k_t = [4, 2, 2, 4]		
Generation 290: Best = 1.067078e+03	FES = 29100	$\mu_F = 0.761$	$\mu_{CR} = 0.466$
Diversity = 6.7550e-01	k_t = [1, 1, 1, 1]		
Generation 300: Best = 1.048245e+03	FES = 30100	$\mu_F = 0.751$	$\mu_{CR} = 0.465$
Diversity = 5.3672e-01	k_t = [3, 3, 2, 4]		
Generation 310: Best = 1.036731e+03	FES = 31100	$\mu_F = 0.763$	$\mu_{CR} = 0.455$
Diversity = 5.3787e-01	k_t = [1, 1, 1, 1]		
Generation 320: Best = 1.022266e+03	FES = 32100	$\mu_F = 0.765$	$\mu_{CR} = 0.450$
Diversity = 5.0404e-01	k_t = [4, 2, 4, 3]		
Generation 330: Best = 1.009074e+03	FES = 33100	$\mu_F = 0.752$	$\mu_{CR} = 0.454$
Diversity = 4.9355e-01	k_t = [1, 1, 1, 1]		
Generation 340: Best = 9.971079e+02	FES = 34100	$\mu_F = 0.757$	$\mu_{CR} = 0.452$
Diversity = 4.8800e-01	k_t = [4, 3, 4, 2]		
Generation 350: Best = 9.863949e+02	FES = 35100	$\mu_F = 0.757$	$\mu_{CR} = 0.459$
Diversity = 5.1591e-01	k_t = [1, 1, 1, 1]		
Generation 360: Best = 9.751933e+02	FES = 36100	$\mu_F = 0.758$	$\mu_{CR} = 0.457$
Diversity = 4.6607e-01	k_t = [2, 4, 3, 3]		
Generation 370: Best = 9.647871e+02	FES = 37100	$\mu_F = 0.756$	$\mu_{CR} = 0.457$
Diversity = 4.8661e-01	k_t = [1, 1, 1, 1]		
Generation 380: Best = 9.507560e+02	FES = 38100	$\mu_F = 0.747$	$\mu_{CR} = 0.461$
Diversity = 5.1598e-01	k_t = [2, 3, 4, 2]		
Generation 390: Best = 9.374659e+02	FES = 39100	$\mu_F = 0.750$	$\mu_{CR} = 0.465$
Diversity = 5.4221e-01	k_t = [1, 1, 1, 1]		
Generation 400: Best = 9.244892e+02	FES = 40100	$\mu_F = 0.757$	$\mu_{CR} = 0.470$
Diversity = 4.8927e-01	k_t = [3, 4, 2, 3]		
Generation 410: Best = 9.139695e+02	FES = 41100	$\mu_F = 0.767$	$\mu_{CR} = 0.469$
Diversity = 4.8870e-01	k_t = [1, 1, 1, 1]		
Generation 420: Best = 9.044608e+02	FES = 42100	$\mu_F = 0.765$	$\mu_{CR} = 0.472$
Diversity = 4.7008e-01	k_t = [3, 2, 4, 3]		
Generation 430: Best = 8.944467e+02	FES = 43100	$\mu_F = 0.759$	$\mu_{CR} = 0.468$
Diversity = 5.1185e-01	k_t = [1, 1, 1, 1]		
Generation 440: Best = 8.849252e+02	FES = 44100	$\mu_F = 0.756$	$\mu_{CR} = 0.471$
Diversity = 4.5495e-01	k_t = [4, 4, 3, 2]		
Generation 450: Best = 8.766344e+02	FES = 45100	$\mu_F = 0.750$	$\mu_{CR} = 0.477$
Diversity = 4.3742e-01	k_t = [1, 1, 1, 1]		
Generation 460: Best = 8.690128e+02	FES = 46100	$\mu_F = 0.748$	$\mu_{CR} = 0.481$
Diversity = 3.8784e-01	k_t = [2, 3, 3, 4]		
Generation 470: Best = 8.641822e+02	FES = 47100	$\mu_F = 0.741$	$\mu_{CR} = 0.480$
Diversity = 3.8452e-01	k_t = [1, 1, 1, 1]		
Generation 480: Best = 8.581800e+02	FES = 48100	$\mu_F = 0.738$	$\mu_{CR} = 0.480$
Diversity = 3.4949e-01	k_t = [2, 4, 3, 3]		
Generation 490: Best = 8.515383e+02	FES = 49100	$\mu_F = 0.744$	$\mu_{CR} = 0.477$
Diversity = 4.2112e-01	k_t = [1, 1, 1, 1]		

Generation 500: Best = 8.442882e+02	FES = 50100	$\mu_F = 0.742$	$\mu_{CR} = 0.476$
Diversity = 4.0317e-01	$k_t = [2, 3, 3, 2]$		
Generation 510: Best = 8.375430e+02	FES = 51100	$\mu_F = 0.732$	$\mu_{CR} = 0.477$
Diversity = 4.3002e-01	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 8.315091e+02	FES = 52100	$\mu_F = 0.720$	$\mu_{CR} = 0.479$
Diversity = 3.5994e-01	$k_t = [3, 2, 2, 2]$		
Generation 530: Best = 8.265976e+02	FES = 53100	$\mu_F = 0.697$	$\mu_{CR} = 0.471$
Diversity = 3.6124e-01	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 8.226857e+02	FES = 54100	$\mu_F = 0.697$	$\mu_{CR} = 0.457$
Diversity = 3.2090e-01	$k_t = [3, 3, 3, 2]$		
Generation 550: Best = 8.175076e+02	FES = 55100	$\mu_F = 0.692$	$\mu_{CR} = 0.457$
Diversity = 2.9853e-01	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 8.137532e+02	FES = 56100	$\mu_F = 0.706$	$\mu_{CR} = 0.461$
Diversity = 3.0014e-01	$k_t = [2, 3, 2, 2]$		
Generation 570: Best = 8.105900e+02	FES = 57100	$\mu_F = 0.693$	$\mu_{CR} = 0.460$
Diversity = 2.8891e-01	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 8.080853e+02	FES = 58100	$\mu_F = 0.694$	$\mu_{CR} = 0.447$
Diversity = 2.4438e-01	$k_t = [2, 3, 3, 3]$		
Generation 590: Best = 8.056982e+02	FES = 59100	$\mu_F = 0.696$	$\mu_{CR} = 0.442$
Diversity = 2.5383e-01	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 8.033573e+02	FES = 60100	$\mu_F = 0.694$	$\mu_{CR} = 0.441$
Diversity = 2.3071e-01	$k_t = [4, 3, 2, 3]$		
Generation 610: Best = 8.005750e+02	FES = 61100	$\mu_F = 0.701$	$\mu_{CR} = 0.436$
Diversity = 2.2233e-01	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 7.985260e+02	FES = 62100	$\mu_F = 0.693$	$\mu_{CR} = 0.448$
Diversity = 1.8613e-01	$k_t = [2, 2, 4, 3]$		
Generation 630: Best = 7.968518e+02	FES = 63100	$\mu_F = 0.689$	$\mu_{CR} = 0.448$
Diversity = 1.7722e-01	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 7.955785e+02	FES = 64100	$\mu_F = 0.705$	$\mu_{CR} = 0.445$
Diversity = 1.4162e-01	$k_t = [2, 4, 4, 4]$		
Generation 650: Best = 7.936665e+02	FES = 65100	$\mu_F = 0.718$	$\mu_{CR} = 0.438$
Diversity = 1.5682e-01	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 7.918937e+02	FES = 66100	$\mu_F = 0.734$	$\mu_{CR} = 0.436$
Diversity = 1.5688e-01	$k_t = [3, 2, 2, 4]$		
Generation 670: Best = 7.906317e+02	FES = 67100	$\mu_F = 0.732$	$\mu_{CR} = 0.430$
Diversity = 1.6309e-01	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 7.890291e+02	FES = 68100	$\mu_F = 0.736$	$\mu_{CR} = 0.428$
Diversity = 1.5112e-01	$k_t = [3, 3, 4, 3]$		
Generation 690: Best = 7.877147e+02	FES = 69100	$\mu_F = 0.745$	$\mu_{CR} = 0.437$
Diversity = 1.6134e-01	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 7.854528e+02	FES = 70100	$\mu_F = 0.747$	$\mu_{CR} = 0.435$
Diversity = 1.7354e-01	$k_t = [4, 2, 3, 3]$		
Generation 710: Best = 7.837282e+02	FES = 71100	$\mu_F = 0.759$	$\mu_{CR} = 0.433$
Diversity = 1.8825e-01	$k_t = [1, 1, 1, 1]$		
Generation 720: Best = 7.818655e+02	FES = 72100	$\mu_F = 0.757$	$\mu_{CR} = 0.429$
Diversity = 1.7796e-01	$k_t = [4, 4, 3, 4]$		
Generation 730: Best = 7.803205e+02	FES = 73100	$\mu_F = 0.754$	$\mu_{CR} = 0.428$
Diversity = 1.7189e-01	$k_t = [1, 1, 1, 1]$		
Generation 740: Best = 7.785767e+02	FES = 74100	$\mu_F = 0.739$	$\mu_{CR} = 0.429$
Diversity = 1.7220e-01	$k_t = [4, 3, 3, 3]$		
Generation 750: Best = 7.765608e+02	FES = 75100	$\mu_F = 0.751$	$\mu_{CR} = 0.429$
Diversity = 1.7457e-01	$k_t = [1, 1, 1, 1]$		
Generation 760: Best = 7.742355e+02	FES = 76100	$\mu_F = 0.748$	$\mu_{CR} = 0.424$
Diversity = 1.6837e-01	$k_t = [3, 3, 3, 3]$		

Generation 770: Best = 7.727386e+02	FES = 77100	$\mu_F = 0.742$	$\mu_{CR} = 0.418$
Diversity = 1.7952e-01	$k_t = [1, 1, 1, 1]$		
Generation 780: Best = 7.712562e+02	FES = 78100	$\mu_F = 0.739$	$\mu_{CR} = 0.417$
Diversity = 1.7333e-01	$k_t = [3, 4, 2, 3]$		
Generation 790: Best = 7.697461e+02	FES = 79100	$\mu_F = 0.739$	$\mu_{CR} = 0.408$
Diversity = 1.7356e-01	$k_t = [1, 1, 1, 1]$		
Generation 800: Best = 7.670392e+02	FES = 80100	$\mu_F = 0.739$	$\mu_{CR} = 0.406$
Diversity = 1.9420e-01	$k_t = [2, 4, 2, 2]$		
Generation 810: Best = 7.649896e+02	FES = 81100	$\mu_F = 0.728$	$\mu_{CR} = 0.407$
Diversity = 2.0944e-01	$k_t = [1, 1, 1, 1]$		
Generation 820: Best = 7.631347e+02	FES = 82100	$\mu_F = 0.732$	$\mu_{CR} = 0.400$
Diversity = 2.0355e-01	$k_t = [4, 2, 2, 2]$		
Generation 830: Best = 7.610336e+02	FES = 83100	$\mu_F = 0.737$	$\mu_{CR} = 0.407$
Diversity = 1.8556e-01	$k_t = [1, 1, 1, 1]$		
Generation 840: Best = 7.596724e+02	FES = 84100	$\mu_F = 0.733$	$\mu_{CR} = 0.404$
Diversity = 1.8011e-01	$k_t = [3, 2, 3, 2]$		
Generation 850: Best = 7.590290e+02	FES = 85100	$\mu_F = 0.726$	$\mu_{CR} = 0.388$
Diversity = 1.7977e-01	$k_t = [1, 1, 1, 1]$		
Generation 860: Best = 7.576109e+02	FES = 86100	$\mu_F = 0.737$	$\mu_{CR} = 0.392$
Diversity = 1.5392e-01	$k_t = [2, 3, 4, 3]$		
Generation 870: Best = 7.564279e+02	FES = 87100	$\mu_F = 0.733$	$\mu_{CR} = 0.384$
Diversity = 1.6090e-01	$k_t = [1, 1, 1, 1]$		
Generation 880: Best = 7.555070e+02	FES = 88100	$\mu_F = 0.736$	$\mu_{CR} = 0.371$
Diversity = 1.3977e-01	$k_t = [4, 4, 3, 3]$		
Generation 890: Best = 7.547927e+02	FES = 89100	$\mu_F = 0.740$	$\mu_{CR} = 0.377$
Diversity = 1.2684e-01	$k_t = [1, 1, 1, 1]$		
Generation 900: Best = 7.535311e+02	FES = 90100	$\mu_F = 0.754$	$\mu_{CR} = 0.369$
Diversity = 1.3487e-01	$k_t = [4, 3, 3, 3]$		
Generation 910: Best = 7.526126e+02	FES = 91100	$\mu_F = 0.764$	$\mu_{CR} = 0.367$
Diversity = 1.5481e-01	$k_t = [1, 1, 1, 1]$		
Generation 920: Best = 7.510086e+02	FES = 92100	$\mu_F = 0.750$	$\mu_{CR} = 0.369$
Diversity = 1.5307e-01	$k_t = [2, 2, 4, 3]$		
Generation 930: Best = 7.496565e+02	FES = 93100	$\mu_F = 0.746$	$\mu_{CR} = 0.372$
Diversity = 1.8328e-01	$k_t = [1, 1, 1, 1]$		
Generation 940: Best = 7.478637e+02	FES = 94100	$\mu_F = 0.747$	$\mu_{CR} = 0.370$
Diversity = 1.8393e-01	$k_t = [2, 4, 3, 3]$		
Generation 950: Best = 7.464790e+02	FES = 95100	$\mu_F = 0.746$	$\mu_{CR} = 0.368$
Diversity = 1.9088e-01	$k_t = [1, 1, 1, 1]$		
Generation 960: Best = 7.444267e+02	FES = 96100	$\mu_F = 0.749$	$\mu_{CR} = 0.355$
Diversity = 1.8946e-01	$k_t = [3, 3, 3, 3]$		
Generation 970: Best = 7.432616e+02	FES = 97100	$\mu_F = 0.738$	$\mu_{CR} = 0.346$
Diversity = 1.9468e-01	$k_t = [1, 1, 1, 1]$		
Generation 980: Best = 7.415982e+02	FES = 98100	$\mu_F = 0.739$	$\mu_{CR} = 0.335$
Diversity = 1.9291e-01	$k_t = [4, 4, 4, 2]$		
Generation 990: Best = 7.406203e+02	FES = 99100	$\mu_F = 0.745$	$\mu_{CR} = 0.337$
Diversity = 1.7758e-01	$k_t = [1, 1, 1, 1]$		
Generation 1000: Best = 7.386717e+02	FES = 100100	$\mu_F = 0.741$	$\mu_{CR} = 0.336$
Diversity = 1.6223e-01	$k_t = [2, 3, 3, 4]$		
Generation 1010: Best = 7.374542e+02	FES = 101100	$\mu_F = 0.741$	$\mu_{CR} = 0.329$
Diversity = 1.7065e-01	$k_t = [1, 1, 1, 1]$		
Generation 1020: Best = 7.358492e+02	FES = 102100	$\mu_F = 0.760$	$\mu_{CR} = 0.329$
Diversity = 1.6936e-01	$k_t = [3, 4, 2, 3]$		
Generation 1030: Best = 7.345128e+02	FES = 103100	$\mu_F = 0.768$	$\mu_{CR} = 0.330$
Diversity = 1.7738e-01	$k_t = [1, 1, 1, 1]$		

Generation 1040: Best = 7.328132e+02 FES = 104100 μ_F = 0.781 μ_{CR} = 0.324
 Diversity = 1.7883e-01 k_t = [4, 4, 2, 3]
 Generation 1050: Best = 7.309620e+02 FES = 105100 μ_F = 0.770 μ_{CR} = 0.324
 Diversity = 1.9230e-01 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 7.283801e+02 FES = 106100 μ_F = 0.765 μ_{CR} = 0.318
 Diversity = 2.0775e-01 k_t = [4, 3, 2, 4]
 Generation 1070: Best = 7.261417e+02 FES = 107100 μ_F = 0.757 μ_{CR} = 0.312
 Diversity = 2.0857e-01 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 7.234965e+02 FES = 108100 μ_F = 0.754 μ_{CR} = 0.324
 Diversity = 2.1974e-01 k_t = [3, 3, 3, 2]
 Generation 1090: Best = 7.216684e+02 FES = 109100 μ_F = 0.745 μ_{CR} = 0.313
 Diversity = 2.3902e-01 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 7.202626e+02 FES = 110100 μ_F = 0.746 μ_{CR} = 0.312
 Diversity = 2.1593e-01 k_t = [3, 2, 2, 2]
 Generation 1110: Best = 7.181287e+02 FES = 111100 μ_F = 0.752 μ_{CR} = 0.301
 Diversity = 2.2531e-01 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 7.165886e+02 FES = 112100 μ_F = 0.746 μ_{CR} = 0.286
 Diversity = 1.9333e-01 k_t = [4, 4, 4, 4]
 Generation 1130: Best = 7.150703e+02 FES = 113100 μ_F = 0.742 μ_{CR} = 0.294
 Diversity = 1.8975e-01 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 7.139000e+02 FES = 114100 μ_F = 0.741 μ_{CR} = 0.287
 Diversity = 1.7765e-01 k_t = [4, 3, 3, 3]
 Generation 1150: Best = 7.129943e+02 FES = 115100 μ_F = 0.737 μ_{CR} = 0.285
 Diversity = 1.8023e-01 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 7.112015e+02 FES = 116100 μ_F = 0.744 μ_{CR} = 0.283
 Diversity = 1.6926e-01 k_t = [3, 3, 2, 4]
 Generation 1170: Best = 7.101727e+02 FES = 117100 μ_F = 0.739 μ_{CR} = 0.282
 Diversity = 1.7632e-01 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 7.091546e+02 FES = 118100 μ_F = 0.721 μ_{CR} = 0.280
 Diversity = 1.5819e-01 k_t = [3, 2, 4, 3]
 Generation 1190: Best = 7.075656e+02 FES = 119100 μ_F = 0.731 μ_{CR} = 0.272
 Diversity = 1.6138e-01 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 7.066714e+02 FES = 120100 μ_F = 0.733 μ_{CR} = 0.274
 Diversity = 1.5850e-01 k_t = [4, 4, 4, 2]
 Generation 1210: Best = 7.053812e+02 FES = 121100 μ_F = 0.738 μ_{CR} = 0.284
 Diversity = 1.5882e-01 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 7.043294e+02 FES = 122100 μ_F = 0.741 μ_{CR} = 0.280
 Diversity = 1.6327e-01 k_t = [3, 4, 2, 4]
 Generation 1230: Best = 7.031536e+02 FES = 123100 μ_F = 0.749 μ_{CR} = 0.275
 Diversity = 1.5920e-01 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 7.016130e+02 FES = 124100 μ_F = 0.761 μ_{CR} = 0.266
 Diversity = 1.7123e-01 k_t = [3, 4, 4, 4]
 Generation 1250: Best = 7.001739e+02 FES = 125100 μ_F = 0.767 μ_{CR} = 0.274
 Diversity = 1.9519e-01 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 6.986274e+02 FES = 126100 μ_F = 0.759 μ_{CR} = 0.277
 Diversity = 1.9499e-01 k_t = [2, 2, 2, 4]
 Generation 1270: Best = 6.976853e+02 FES = 127100 μ_F = 0.741 μ_{CR} = 0.272
 Diversity = 2.0502e-01 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 6.965132e+02 FES = 128100 μ_F = 0.746 μ_{CR} = 0.275
 Diversity = 1.9300e-01 k_t = [3, 4, 3, 3]
 Generation 1290: Best = 6.953519e+02 FES = 129100 μ_F = 0.739 μ_{CR} = 0.278
 Diversity = 1.9243e-01 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 6.940909e+02 FES = 130100 μ_F = 0.755 μ_{CR} = 0.276
 Diversity = 1.9284e-01 k_t = [2, 4, 2, 2]

Generation 1310: Best = 6.922233e+02 FES = 131100 μ_F = 0.751 μ_{CR} = 0.278
 Diversity = 1.9815e-01 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 6.909489e+02 FES = 132100 μ_F = 0.748 μ_{CR} = 0.276
 Diversity = 2.0929e-01 k_t = [4, 2, 3, 2]
 Generation 1330: Best = 6.890397e+02 FES = 133100 μ_F = 0.746 μ_{CR} = 0.279
 Diversity = 2.2736e-01 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 6.877204e+02 FES = 134100 μ_F = 0.739 μ_{CR} = 0.277
 Diversity = 2.0630e-01 k_t = [4, 3, 3, 2]
 Generation 1350: Best = 6.867040e+02 FES = 135100 μ_F = 0.747 μ_{CR} = 0.281
 Diversity = 1.9538e-01 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 6.853892e+02 FES = 136100 μ_F = 0.750 μ_{CR} = 0.280
 Diversity = 1.9154e-01 k_t = [4, 3, 4, 2]
 Generation 1370: Best = 6.843155e+02 FES = 137100 μ_F = 0.757 μ_{CR} = 0.288
 Diversity = 2.0188e-01 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 6.826031e+02 FES = 138100 μ_F = 0.755 μ_{CR} = 0.283
 Diversity = 1.8750e-01 k_t = [3, 3, 3, 3]
 Generation 1390: Best = 6.808628e+02 FES = 139100 μ_F = 0.755 μ_{CR} = 0.290
 Diversity = 2.0369e-01 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 6.798348e+02 FES = 140100 μ_F = 0.749 μ_{CR} = 0.291
 Diversity = 1.9111e-01 k_t = [2, 3, 2, 2]
 Generation 1410: Best = 6.785518e+02 FES = 141100 μ_F = 0.752 μ_{CR} = 0.292
 Diversity = 1.8871e-01 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 6.770818e+02 FES = 142100 μ_F = 0.742 μ_{CR} = 0.305
 Diversity = 1.8433e-01 k_t = [4, 3, 2, 2]
 Generation 1430: Best = 6.760953e+02 FES = 143100 μ_F = 0.742 μ_{CR} = 0.313
 Diversity = 1.7551e-01 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 6.754066e+02 FES = 144100 μ_F = 0.742 μ_{CR} = 0.310
 Diversity = 1.6535e-01 k_t = [2, 3, 2, 3]
 Generation 1450: Best = 6.743339e+02 FES = 145100 μ_F = 0.717 μ_{CR} = 0.310
 Diversity = 1.6081e-01 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 6.729636e+02 FES = 146100 μ_F = 0.720 μ_{CR} = 0.309
 Diversity = 1.6089e-01 k_t = [4, 2, 4, 4]
 Generation 1470: Best = 6.716169e+02 FES = 147100 μ_F = 0.727 μ_{CR} = 0.305
 Diversity = 1.6380e-01 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 6.709297e+02 FES = 148100 μ_F = 0.743 μ_{CR} = 0.307
 Diversity = 1.4380e-01 k_t = [3, 2, 4, 4]
 Generation 1490: Best = 6.698337e+02 FES = 149100 μ_F = 0.751 μ_{CR} = 0.312
 Diversity = 1.3603e-01 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 6.692563e+02 FES = 150100 μ_F = 0.765 μ_{CR} = 0.319
 Diversity = 1.3675e-01 k_t = [4, 3, 3, 3]
 Generation 1510: Best = 6.685442e+02 FES = 151100 μ_F = 0.761 μ_{CR} = 0.318
 Diversity = 1.3336e-01 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 6.674884e+02 FES = 152100 μ_F = 0.757 μ_{CR} = 0.322
 Diversity = 1.3457e-01 k_t = [4, 2, 3, 3]
 Generation 1530: Best = 6.668215e+02 FES = 153100 μ_F = 0.758 μ_{CR} = 0.319
 Diversity = 1.4508e-01 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 6.652612e+02 FES = 154100 μ_F = 0.763 μ_{CR} = 0.325
 Diversity = 1.4848e-01 k_t = [2, 4, 3, 4]
 Generation 1550: Best = 6.642940e+02 FES = 155100 μ_F = 0.771 μ_{CR} = 0.309
 Diversity = 1.7751e-01 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 6.628693e+02 FES = 156100 μ_F = 0.768 μ_{CR} = 0.306
 Diversity = 1.8786e-01 k_t = [2, 4, 3, 2]
 Generation 1570: Best = 6.618650e+02 FES = 157100 μ_F = 0.762 μ_{CR} = 0.317
 Diversity = 1.9721e-01 k_t = [1, 1, 1, 1]

Generation 1580: Best = 6.602201e+02 FES = 158100 μ_F = 0.753 μ_{CR} = 0.308
 Diversity = 1.7320e-01 k_t = [3, 3, 4, 3]
 Generation 1590: Best = 6.594142e+02 FES = 159100 μ_F = 0.752 μ_{CR} = 0.302
 Diversity = 1.6855e-01 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 6.587161e+02 FES = 160100 μ_F = 0.753 μ_{CR} = 0.295
 Diversity = 1.6342e-01 k_t = [2, 4, 2, 3]
 Generation 1610: Best = 6.576968e+02 FES = 161100 μ_F = 0.748 μ_{CR} = 0.300
 Diversity = 1.6170e-01 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 6.565211e+02 FES = 162100 μ_F = 0.751 μ_{CR} = 0.293
 Diversity = 1.6034e-01 k_t = [4, 4, 2, 2]
 Generation 1630: Best = 6.557513e+02 FES = 163100 μ_F = 0.759 μ_{CR} = 0.286
 Diversity = 1.5554e-01 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 6.548032e+02 FES = 164100 μ_F = 0.757 μ_{CR} = 0.279
 Diversity = 1.4326e-01 k_t = [4, 2, 4, 4]
 Generation 1650: Best = 6.539018e+02 FES = 165100 μ_F = 0.764 μ_{CR} = 0.278
 Diversity = 1.4718e-01 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 6.525853e+02 FES = 166100 μ_F = 0.773 μ_{CR} = 0.282
 Diversity = 1.5029e-01 k_t = [2, 2, 4, 4]
 Generation 1670: Best = 6.515078e+02 FES = 167100 μ_F = 0.774 μ_{CR} = 0.283
 Diversity = 1.7345e-01 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 6.508745e+02 FES = 168100 μ_F = 0.774 μ_{CR} = 0.274
 Diversity = 1.6841e-01 k_t = [3, 4, 4, 3]
 Generation 1690: Best = 6.500430e+02 FES = 169100 μ_F = 0.766 μ_{CR} = 0.267
 Diversity = 1.6421e-01 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 6.491861e+02 FES = 170100 μ_F = 0.767 μ_{CR} = 0.263
 Diversity = 1.6467e-01 k_t = [2, 2, 4, 2]
 Generation 1710: Best = 6.481451e+02 FES = 171100 μ_F = 0.770 μ_{CR} = 0.267
 Diversity = 1.6975e-01 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 6.473769e+02 FES = 172100 μ_F = 0.751 μ_{CR} = 0.267
 Diversity = 1.5564e-01 k_t = [3, 4, 2, 3]
 Generation 1730: Best = 6.465211e+02 FES = 173100 μ_F = 0.743 μ_{CR} = 0.276
 Diversity = 1.5143e-01 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 6.458422e+02 FES = 174100 μ_F = 0.731 μ_{CR} = 0.273
 Diversity = 1.3648e-01 k_t = [3, 4, 2, 3]
 Generation 1750: Best = 6.451590e+02 FES = 175100 μ_F = 0.735 μ_{CR} = 0.278
 Diversity = 1.3908e-01 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 6.442489e+02 FES = 176100 μ_F = 0.734 μ_{CR} = 0.275
 Diversity = 1.2577e-01 k_t = [3, 3, 4, 3]
 Generation 1770: Best = 6.436206e+02 FES = 177100 μ_F = 0.733 μ_{CR} = 0.279
 Diversity = 1.2960e-01 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 6.430587e+02 FES = 178100 μ_F = 0.749 μ_{CR} = 0.274
 Diversity = 1.2543e-01 k_t = [3, 4, 4, 3]
 Generation 1790: Best = 6.425977e+02 FES = 179100 μ_F = 0.754 μ_{CR} = 0.267
 Diversity = 1.2514e-01 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 6.416179e+02 FES = 180100 μ_F = 0.755 μ_{CR} = 0.260
 Diversity = 1.2500e-01 k_t = [4, 2, 3, 2]
 Generation 1810: Best = 6.412287e+02 FES = 181100 μ_F = 0.748 μ_{CR} = 0.254
 Diversity = 1.1802e-01 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 6.407292e+02 FES = 182100 μ_F = 0.744 μ_{CR} = 0.251
 Diversity = 1.1211e-01 k_t = [2, 2, 2, 3]
 Generation 1830: Best = 6.403090e+02 FES = 183100 μ_F = 0.755 μ_{CR} = 0.248
 Diversity = 1.1894e-01 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 6.398085e+02 FES = 184100 μ_F = 0.742 μ_{CR} = 0.244
 Diversity = 1.1528e-01 k_t = [4, 4, 4, 4]

Generation 1850: Best = 6.393619e+02	FES = 185100	$\mu_F = 0.741$	$\mu_{CR} = 0.244$
Diversity = 1.1074e-01	k_t = [1, 1, 1, 1]		
Generation 1860: Best = 6.389133e+02	FES = 186100	$\mu_F = 0.753$	$\mu_{CR} = 0.235$
Diversity = 1.0948e-01	k_t = [3, 3, 3, 2]		
Generation 1870: Best = 6.385422e+02	FES = 187100	$\mu_F = 0.759$	$\mu_{CR} = 0.239$
Diversity = 1.1322e-01	k_t = [1, 1, 1, 1]		
Generation 1880: Best = 6.379416e+02	FES = 188100	$\mu_F = 0.758$	$\mu_{CR} = 0.250$
Diversity = 1.1102e-01	k_t = [3, 4, 3, 3]		
Generation 1890: Best = 6.376794e+02	FES = 189100	$\mu_F = 0.746$	$\mu_{CR} = 0.249$
Diversity = 1.1648e-01	k_t = [1, 1, 1, 1]		
Generation 1900: Best = 6.371630e+02	FES = 190100	$\mu_F = 0.757$	$\mu_{CR} = 0.244$
Diversity = 1.1624e-01	k_t = [2, 3, 3, 3]		
Generation 1910: Best = 6.369535e+02	FES = 191100	$\mu_F = 0.743$	$\mu_{CR} = 0.230$
Diversity = 1.1456e-01	k_t = [1, 1, 1, 1]		
Generation 1920: Best = 6.366723e+02	FES = 192100	$\mu_F = 0.736$	$\mu_{CR} = 0.223$
Diversity = 1.0155e-01	k_t = [2, 3, 2, 4]		
Generation 1930: Best = 6.359603e+02	FES = 193100	$\mu_F = 0.749$	$\mu_{CR} = 0.226$
Diversity = 1.1555e-01	k_t = [1, 1, 1, 1]		
Generation 1940: Best = 6.353918e+02	FES = 194100	$\mu_F = 0.757$	$\mu_{CR} = 0.236$
Diversity = 1.1520e-01	k_t = [4, 3, 4, 2]		
Generation 1950: Best = 6.350919e+02	FES = 195100	$\mu_F = 0.755$	$\mu_{CR} = 0.240$
Diversity = 1.1993e-01	k_t = [1, 1, 1, 1]		
Generation 1960: Best = 6.346455e+02	FES = 196100	$\mu_F = 0.765$	$\mu_{CR} = 0.240$
Diversity = 1.1920e-01	k_t = [2, 4, 3, 3]		
Generation 1970: Best = 6.342357e+02	FES = 197100	$\mu_F = 0.770$	$\mu_{CR} = 0.237$
Diversity = 1.2089e-01	k_t = [1, 1, 1, 1]		
Generation 1980: Best = 6.338472e+02	FES = 198100	$\mu_F = 0.776$	$\mu_{CR} = 0.228$
Diversity = 1.2394e-01	k_t = [4, 2, 2, 2]		
Generation 1990: Best = 6.334830e+02	FES = 199100	$\mu_F = 0.769$	$\mu_{CR} = 0.230$
Diversity = 1.2959e-01	k_t = [1, 1, 1, 1]		
Generation 2000: Best = 6.329138e+02	FES = 200100	$\mu_F = 0.764$	$\mu_{CR} = 0.234$
Diversity = 1.1696e-01	k_t = [3, 3, 2, 2]		
Generation 2010: Best = 6.323873e+02	FES = 201100	$\mu_F = 0.762$	$\mu_{CR} = 0.233$
Diversity = 1.2318e-01	k_t = [1, 1, 1, 1]		
Generation 2020: Best = 6.318552e+02	FES = 202100	$\mu_F = 0.765$	$\mu_{CR} = 0.234$
Diversity = 1.2407e-01	k_t = [4, 3, 4, 3]		
Generation 2030: Best = 6.315283e+02	FES = 203100	$\mu_F = 0.742$	$\mu_{CR} = 0.233$
Diversity = 1.1704e-01	k_t = [1, 1, 1, 1]		
Generation 2040: Best = 6.311733e+02	FES = 204100	$\mu_F = 0.749$	$\mu_{CR} = 0.220$
Diversity = 1.1463e-01	k_t = [3, 2, 2, 3]		
Generation 2050: Best = 6.307955e+02	FES = 205100	$\mu_F = 0.744$	$\mu_{CR} = 0.212$
Diversity = 1.1654e-01	k_t = [1, 1, 1, 1]		
Generation 2060: Best = 6.302812e+02	FES = 206100	$\mu_F = 0.749$	$\mu_{CR} = 0.214$
Diversity = 1.1540e-01	k_t = [3, 3, 3, 2]		
Generation 2070: Best = 6.298475e+02	FES = 207100	$\mu_F = 0.761$	$\mu_{CR} = 0.219$
Diversity = 1.2234e-01	k_t = [1, 1, 1, 1]		
Generation 2080: Best = 6.296523e+02	FES = 208100	$\mu_F = 0.758$	$\mu_{CR} = 0.219$
Diversity = 1.1557e-01	k_t = [4, 3, 4, 2]		
Generation 2090: Best = 6.292515e+02	FES = 209100	$\mu_F = 0.749$	$\mu_{CR} = 0.214$
Diversity = 1.1359e-01	k_t = [1, 1, 1, 1]		
Generation 2100: Best = 6.289588e+02	FES = 210100	$\mu_F = 0.747$	$\mu_{CR} = 0.217$
Diversity = 9.9940e-02	k_t = [4, 4, 4, 3]		
Generation 2110: Best = 6.285888e+02	FES = 211100	$\mu_F = 0.755$	$\mu_{CR} = 0.224$
Diversity = 9.8094e-02	k_t = [1, 1, 1, 1]		

Generation 2120: Best = 6.283372e+02 FES = 212100 μ_F = 0.756 μ_{CR} = 0.232
 Diversity = 9.3794e-02 k_t = [3, 4, 2, 3]
 Generation 2130: Best = 6.281073e+02 FES = 213100 μ_F = 0.760 μ_{CR} = 0.219
 Diversity = 8.9346e-02 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 6.279208e+02 FES = 214100 μ_F = 0.775 μ_{CR} = 0.218
 Diversity = 9.0255e-02 k_t = [3, 2, 2, 3]
 Generation 2150: Best = 6.276586e+02 FES = 215100 μ_F = 0.774 μ_{CR} = 0.220
 Diversity = 8.7840e-02 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 6.275319e+02 FES = 216100 μ_F = 0.766 μ_{CR} = 0.222
 Diversity = 8.4366e-02 k_t = [4, 4, 4, 2]
 Generation 2170: Best = 6.272654e+02 FES = 217100 μ_F = 0.761 μ_{CR} = 0.231
 Diversity = 8.3125e-02 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 6.270834e+02 FES = 218100 μ_F = 0.750 μ_{CR} = 0.231
 Diversity = 7.9654e-02 k_t = [3, 2, 3, 3]
 Generation 2190: Best = 6.269549e+02 FES = 219100 μ_F = 0.745 μ_{CR} = 0.240
 Diversity = 8.0458e-02 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 6.267430e+02 FES = 220100 μ_F = 0.737 μ_{CR} = 0.229
 Diversity = 7.3276e-02 k_t = [4, 2, 4, 4]
 Generation 2210: Best = 6.266183e+02 FES = 221100 μ_F = 0.739 μ_{CR} = 0.238
 Diversity = 7.1341e-02 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 6.264001e+02 FES = 222100 μ_F = 0.741 μ_{CR} = 0.233
 Diversity = 7.4802e-02 k_t = [2, 2, 3, 3]
 Generation 2230: Best = 6.262343e+02 FES = 223100 μ_F = 0.743 μ_{CR} = 0.234
 Diversity = 7.8315e-02 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 6.260899e+02 FES = 224100 μ_F = 0.747 μ_{CR} = 0.227
 Diversity = 7.1395e-02 k_t = [2, 3, 4, 3]
 Generation 2250: Best = 6.259515e+02 FES = 225100 μ_F = 0.746 μ_{CR} = 0.223
 Diversity = 7.0943e-02 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 6.258020e+02 FES = 226100 μ_F = 0.744 μ_{CR} = 0.226
 Diversity = 6.5147e-02 k_t = [3, 4, 4, 3]
 Generation 2270: Best = 6.256884e+02 FES = 227100 μ_F = 0.756 μ_{CR} = 0.221
 Diversity = 6.9168e-02 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 6.255076e+02 FES = 228100 μ_F = 0.752 μ_{CR} = 0.214
 Diversity = 7.0384e-02 k_t = [3, 2, 2, 2]
 Generation 2290: Best = 6.253242e+02 FES = 229100 μ_F = 0.738 μ_{CR} = 0.212
 Diversity = 7.7666e-02 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 6.252426e+02 FES = 230100 μ_F = 0.732 μ_{CR} = 0.210
 Diversity = 6.8563e-02 k_t = [4, 3, 2, 4]
 Generation 2310: Best = 6.250923e+02 FES = 231100 μ_F = 0.736 μ_{CR} = 0.214
 Diversity = 6.8629e-02 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 6.249746e+02 FES = 232100 μ_F = 0.743 μ_{CR} = 0.216
 Diversity = 6.4772e-02 k_t = [2, 3, 4, 4]
 Generation 2330: Best = 6.247915e+02 FES = 233100 μ_F = 0.746 μ_{CR} = 0.219
 Diversity = 6.3963e-02 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 6.247142e+02 FES = 234100 μ_F = 0.755 μ_{CR} = 0.216
 Diversity = 6.1452e-02 k_t = [2, 3, 4, 4]
 Generation 2350: Best = 6.245372e+02 FES = 235100 μ_F = 0.763 μ_{CR} = 0.213
 Diversity = 6.8790e-02 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 6.243675e+02 FES = 236100 μ_F = 0.769 μ_{CR} = 0.219
 Diversity = 7.2059e-02 k_t = [4, 2, 2, 2]
 Generation 2370: Best = 6.241699e+02 FES = 237100 μ_F = 0.775 μ_{CR} = 0.215
 Diversity = 7.8101e-02 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 6.240616e+02 FES = 238100 μ_F = 0.770 μ_{CR} = 0.230
 Diversity = 6.8974e-02 k_t = [3, 3, 3, 4]

Generation 2390: Best = 6.239086e+02 FES = 239100 μ_F = 0.756 μ_{CR} = 0.230
 Diversity = 6.8957e-02 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 6.237193e+02 FES = 240100 μ_F = 0.766 μ_{CR} = 0.225
 Diversity = 6.6976e-02 k_t = [4, 4, 4, 3]
 Generation 2410: Best = 6.235331e+02 FES = 241100 μ_F = 0.771 μ_{CR} = 0.219
 Diversity = 6.7501e-02 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 6.232747e+02 FES = 242100 μ_F = 0.778 μ_{CR} = 0.217
 Diversity = 7.4663e-02 k_t = [2, 4, 3, 2]
 Generation 2430: Best = 6.229856e+02 FES = 243100 μ_F = 0.767 μ_{CR} = 0.217
 Diversity = 8.3530e-02 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 6.227987e+02 FES = 244100 μ_F = 0.766 μ_{CR} = 0.225
 Diversity = 7.8648e-02 k_t = [4, 3, 2, 4]
 Generation 2450: Best = 6.226588e+02 FES = 245100 μ_F = 0.755 μ_{CR} = 0.230
 Diversity = 7.5586e-02 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 6.224888e+02 FES = 246100 μ_F = 0.762 μ_{CR} = 0.231
 Diversity = 7.0110e-02 k_t = [2, 4, 4, 4]
 Generation 2470: Best = 6.223885e+02 FES = 247100 μ_F = 0.766 μ_{CR} = 0.228
 Diversity = 6.6348e-02 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 6.222728e+02 FES = 248100 μ_F = 0.763 μ_{CR} = 0.224
 Diversity = 6.5117e-02 k_t = [4, 2, 4, 3]
 Generation 2490: Best = 6.221514e+02 FES = 249100 μ_F = 0.768 μ_{CR} = 0.229
 Diversity = 6.5675e-02 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 6.220430e+02 FES = 250100 μ_F = 0.767 μ_{CR} = 0.234
 Diversity = 6.2018e-02 k_t = [3, 4, 4, 4]
 Generation 2510: Best = 6.219418e+02 FES = 251100 μ_F = 0.749 μ_{CR} = 0.222
 Diversity = 5.8869e-02 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 6.218523e+02 FES = 252100 μ_F = 0.740 μ_{CR} = 0.228
 Diversity = 5.8040e-02 k_t = [3, 4, 2, 2]
 Generation 2530: Best = 6.217420e+02 FES = 253100 μ_F = 0.730 μ_{CR} = 0.230
 Diversity = 5.5850e-02 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 6.216339e+02 FES = 254100 μ_F = 0.738 μ_{CR} = 0.233
 Diversity = 5.2466e-02 k_t = [4, 2, 3, 2]
 Generation 2550: Best = 6.215667e+02 FES = 255100 μ_F = 0.752 μ_{CR} = 0.241
 Diversity = 5.1616e-02 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 6.214595e+02 FES = 256100 μ_F = 0.748 μ_{CR} = 0.233
 Diversity = 4.8069e-02 k_t = [3, 2, 4, 2]
 Generation 2570: Best = 6.213999e+02 FES = 257100 μ_F = 0.746 μ_{CR} = 0.237
 Diversity = 4.7874e-02 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 6.213380e+02 FES = 258100 μ_F = 0.753 μ_{CR} = 0.239
 Diversity = 4.4935e-02 k_t = [4, 4, 2, 4]
 Generation 2590: Best = 6.212666e+02 FES = 259100 μ_F = 0.764 μ_{CR} = 0.242
 Diversity = 4.2983e-02 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 6.211900e+02 FES = 260100 μ_F = 0.752 μ_{CR} = 0.244
 Diversity = 4.3935e-02 k_t = [3, 4, 2, 4]
 Generation 2610: Best = 6.211154e+02 FES = 261100 μ_F = 0.749 μ_{CR} = 0.249
 Diversity = 4.6329e-02 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 6.210420e+02 FES = 262100 μ_F = 0.761 μ_{CR} = 0.244
 Diversity = 4.3382e-02 k_t = [4, 2, 4, 3]
 Generation 2630: Best = 6.209921e+02 FES = 263100 μ_F = 0.762 μ_{CR} = 0.229
 Diversity = 4.3633e-02 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 6.209274e+02 FES = 264100 μ_F = 0.757 μ_{CR} = 0.227
 Diversity = 4.3476e-02 k_t = [3, 2, 2, 4]
 Generation 2650: Best = 6.208518e+02 FES = 265100 μ_F = 0.756 μ_{CR} = 0.229
 Diversity = 4.6965e-02 k_t = [1, 1, 1, 1]

Generation 2660: Best = 6.207719e+02 FES = 266100 μ_F = 0.765 μ_{CR} = 0.233
 Diversity = 4.3740e-02 k_t = [2, 3, 4, 4]
 Generation 2670: Best = 6.206829e+02 FES = 267100 μ_F = 0.772 μ_{CR} = 0.234
 Diversity = 4.8345e-02 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 6.206258e+02 FES = 268100 μ_F = 0.776 μ_{CR} = 0.228
 Diversity = 4.7482e-02 k_t = [2, 2, 4, 4]
 Generation 2690: Best = 6.205136e+02 FES = 269100 μ_F = 0.775 μ_{CR} = 0.226
 Diversity = 4.7887e-02 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 6.204435e+02 FES = 270100 μ_F = 0.777 μ_{CR} = 0.218
 Diversity = 4.9008e-02 k_t = [2, 3, 2, 4]
 Generation 2710: Best = 6.203220e+02 FES = 271100 μ_F = 0.771 μ_{CR} = 0.210
 Diversity = 4.8429e-02 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 6.202786e+02 FES = 272100 μ_F = 0.766 μ_{CR} = 0.209
 Diversity = 4.7382e-02 k_t = [4, 3, 3, 4]
 Generation 2730: Best = 6.202147e+02 FES = 273100 μ_F = 0.765 μ_{CR} = 0.218
 Diversity = 4.7094e-02 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 6.200819e+02 FES = 274100 μ_F = 0.766 μ_{CR} = 0.221
 Diversity = 4.5910e-02 k_t = [4, 4, 4, 2]
 Generation 2750: Best = 6.199990e+02 FES = 275100 μ_F = 0.767 μ_{CR} = 0.223
 Diversity = 4.6747e-02 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 6.199266e+02 FES = 276100 μ_F = 0.759 μ_{CR} = 0.221
 Diversity = 4.4173e-02 k_t = [2, 3, 2, 3]
 Generation 2770: Best = 6.198639e+02 FES = 277100 μ_F = 0.761 μ_{CR} = 0.215
 Diversity = 4.6358e-02 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 6.197968e+02 FES = 278100 μ_F = 0.747 μ_{CR} = 0.220
 Diversity = 4.1550e-02 k_t = [2, 4, 2, 4]
 Generation 2790: Best = 6.197138e+02 FES = 279100 μ_F = 0.752 μ_{CR} = 0.217
 Diversity = 4.3791e-02 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 6.196756e+02 FES = 280100 μ_F = 0.753 μ_{CR} = 0.222
 Diversity = 4.1792e-02 k_t = [4, 2, 3, 2]
 Generation 2810: Best = 6.196142e+02 FES = 281100 μ_F = 0.756 μ_{CR} = 0.217
 Diversity = 4.0709e-02 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 6.195541e+02 FES = 282100 μ_F = 0.751 μ_{CR} = 0.222
 Diversity = 3.5761e-02 k_t = [4, 2, 4, 3]
 Generation 2830: Best = 6.195057e+02 FES = 283100 μ_F = 0.760 μ_{CR} = 0.218
 Diversity = 3.7812e-02 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 6.194363e+02 FES = 284100 μ_F = 0.764 μ_{CR} = 0.220
 Diversity = 3.7752e-02 k_t = [4, 3, 2, 3]
 Generation 2850: Best = 6.193716e+02 FES = 285100 μ_F = 0.766 μ_{CR} = 0.222
 Diversity = 4.0622e-02 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 6.192723e+02 FES = 286100 μ_F = 0.771 μ_{CR} = 0.218
 Diversity = 4.1012e-02 k_t = [4, 3, 2, 3]
 Generation 2870: Best = 6.191949e+02 FES = 287100 μ_F = 0.782 μ_{CR} = 0.221
 Diversity = 4.2031e-02 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 6.191068e+02 FES = 288100 μ_F = 0.790 μ_{CR} = 0.225
 Diversity = 4.4995e-02 k_t = [3, 3, 2, 4]
 Generation 2890: Best = 6.190371e+02 FES = 289100 μ_F = 0.777 μ_{CR} = 0.223
 Diversity = 4.5753e-02 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 6.189803e+02 FES = 290100 μ_F = 0.768 μ_{CR} = 0.220
 Diversity = 4.3894e-02 k_t = [4, 4, 4, 3]
 Generation 2910: Best = 6.189126e+02 FES = 291100 μ_F = 0.761 μ_{CR} = 0.216
 Diversity = 4.2081e-02 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 6.188430e+02 FES = 292100 μ_F = 0.762 μ_{CR} = 0.217
 Diversity = 4.1265e-02 k_t = [2, 3, 2, 3]

Generation 2930: Best = 6.187819e+02 FES = 293100 μ_F = 0.762 μ_{CR} = 0.217
 Diversity = 4.1966e-02 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 6.187505e+02 FES = 294100 μ_F = 0.766 μ_{CR} = 0.223
 Diversity = 3.9808e-02 k_t = [2, 3, 2, 4]
 Generation 2950: Best = 6.187167e+02 FES = 295100 μ_F = 0.761 μ_{CR} = 0.218
 Diversity = 3.9020e-02 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 6.186497e+02 FES = 296100 μ_F = 0.757 μ_{CR} = 0.218
 Diversity = 3.7963e-02 k_t = [3, 4, 2, 3]
 Generation 2970: Best = 6.186006e+02 FES = 297100 μ_F = 0.753 μ_{CR} = 0.225
 Diversity = 4.1928e-02 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 6.184883e+02 FES = 298100 μ_F = 0.759 μ_{CR} = 0.238
 Diversity = 4.2511e-02 k_t = [3, 3, 4, 2]
 Generation 2990: Best = 6.184266e+02 FES = 299100 μ_F = 0.760 μ_{CR} = 0.223
 Diversity = 4.4759e-02 k_t = [1, 1, 1, 1]
 Generation 3000: Best = 6.183754e+02 FES = 300100 μ_F = 0.760 μ_{CR} = 0.218
 Diversity = 4.2477e-02 k_t = [3, 3, 3, 4]
 Generation 3010: Best = 6.183181e+02 FES = 301100 μ_F = 0.762 μ_{CR} = 0.220
 Diversity = 4.1738e-02 k_t = [1, 1, 1, 1]
 Generation 3020: Best = 6.182549e+02 FES = 302100 μ_F = 0.758 μ_{CR} = 0.222
 Diversity = 4.0241e-02 k_t = [2, 4, 4, 3]
 Generation 3030: Best = 6.182189e+02 FES = 303100 μ_F = 0.757 μ_{CR} = 0.221
 Diversity = 3.8358e-02 k_t = [1, 1, 1, 1]
 Generation 3040: Best = 6.181701e+02 FES = 304100 μ_F = 0.756 μ_{CR} = 0.216
 Diversity = 3.8881e-02 k_t = [2, 2, 4, 2]
 Generation 3050: Best = 6.181288e+02 FES = 305100 μ_F = 0.751 μ_{CR} = 0.221
 Diversity = 4.0185e-02 k_t = [1, 1, 1, 1]
 Generation 3060: Best = 6.180847e+02 FES = 306100 μ_F = 0.755 μ_{CR} = 0.223
 Diversity = 3.5810e-02 k_t = [4, 3, 3, 4]
 Generation 3070: Best = 6.180438e+02 FES = 307100 μ_F = 0.744 μ_{CR} = 0.224
 Diversity = 3.4050e-02 k_t = [1, 1, 1, 1]
 Generation 3080: Best = 6.180162e+02 FES = 308100 μ_F = 0.746 μ_{CR} = 0.208
 Diversity = 3.2818e-02 k_t = [2, 2, 3, 3]
 Generation 3090: Best = 6.179723e+02 FES = 309100 μ_F = 0.758 μ_{CR} = 0.197
 Diversity = 3.3618e-02 k_t = [1, 1, 1, 1]
 Generation 3100: Best = 6.179259e+02 FES = 310100 μ_F = 0.767 μ_{CR} = 0.203
 Diversity = 3.4017e-02 k_t = [4, 2, 3, 3]
 Generation 3110: Best = 6.178659e+02 FES = 311100 μ_F = 0.774 μ_{CR} = 0.202
 Diversity = 3.6873e-02 k_t = [1, 1, 1, 1]
 Generation 3120: Best = 6.178387e+02 FES = 312100 μ_F = 0.775 μ_{CR} = 0.202
 Diversity = 3.7146e-02 k_t = [2, 4, 4, 2]
 Generation 3130: Best = 6.178036e+02 FES = 313100 μ_F = 0.777 μ_{CR} = 0.212
 Diversity = 3.7516e-02 k_t = [1, 1, 1, 1]
 Generation 3140: Best = 6.177560e+02 FES = 314100 μ_F = 0.768 μ_{CR} = 0.201
 Diversity = 3.3885e-02 k_t = [3, 2, 2, 3]
 Generation 3150: Best = 6.177310e+02 FES = 315100 μ_F = 0.756 μ_{CR} = 0.202
 Diversity = 3.5652e-02 k_t = [1, 1, 1, 1]
 Generation 3160: Best = 6.176793e+02 FES = 316100 μ_F = 0.758 μ_{CR} = 0.200
 Diversity = 3.6228e-02 k_t = [4, 3, 4, 2]
 Generation 3170: Best = 6.176546e+02 FES = 317100 μ_F = 0.760 μ_{CR} = 0.201
 Diversity = 3.6369e-02 k_t = [1, 1, 1, 1]
 Generation 3180: Best = 6.176276e+02 FES = 318100 μ_F = 0.757 μ_{CR} = 0.202
 Diversity = 3.3728e-02 k_t = [2, 3, 4, 3]
 Generation 3190: Best = 6.175970e+02 FES = 319100 μ_F = 0.749 μ_{CR} = 0.203
 Diversity = 3.3883e-02 k_t = [1, 1, 1, 1]

Generation 3200: Best = 6.175733e+02 FES = 320100 μ_F = 0.748 μ_{CR} = 0.206
 Diversity = 3.1163e-02 k_t = [4, 3, 2, 3]
 Generation 3210: Best = 6.175558e+02 FES = 321100 μ_F = 0.746 μ_{CR} = 0.206
 Diversity = 2.6867e-02 k_t = [1, 1, 1, 1]
 Generation 3220: Best = 6.175272e+02 FES = 322100 μ_F = 0.745 μ_{CR} = 0.209
 Diversity = 2.6043e-02 k_t = [2, 3, 3, 2]
 Generation 3230: Best = 6.175125e+02 FES = 323100 μ_F = 0.743 μ_{CR} = 0.210
 Diversity = 2.6603e-02 k_t = [1, 1, 1, 1]
 Generation 3240: Best = 6.174850e+02 FES = 324100 μ_F = 0.744 μ_{CR} = 0.206
 Diversity = 2.6209e-02 k_t = [4, 4, 3, 3]
 Generation 3250: Best = 6.174636e+02 FES = 325100 μ_F = 0.745 μ_{CR} = 0.213
 Diversity = 2.5642e-02 k_t = [1, 1, 1, 1]
 Generation 3260: Best = 6.174370e+02 FES = 326100 μ_F = 0.749 μ_{CR} = 0.206
 Diversity = 2.4532e-02 k_t = [4, 3, 4, 2]
 Generation 3270: Best = 6.174261e+02 FES = 327100 μ_F = 0.749 μ_{CR} = 0.199
 Diversity = 2.5395e-02 k_t = [1, 1, 1, 1]
 Generation 3280: Best = 6.173931e+02 FES = 328100 μ_F = 0.748 μ_{CR} = 0.202
 Diversity = 2.3940e-02 k_t = [2, 3, 4, 3]
 Generation 3290: Best = 6.173690e+02 FES = 329100 μ_F = 0.747 μ_{CR} = 0.211
 Diversity = 2.4988e-02 k_t = [1, 1, 1, 1]
 Generation 3300: Best = 6.173460e+02 FES = 330100 μ_F = 0.755 μ_{CR} = 0.208
 Diversity = 2.4919e-02 k_t = [4, 4, 2, 2]
 Generation 3310: Best = 6.173125e+02 FES = 331100 μ_F = 0.767 μ_{CR} = 0.215
 Diversity = 2.6968e-02 k_t = [1, 1, 1, 1]
 Generation 3320: Best = 6.172913e+02 FES = 332100 μ_F = 0.778 μ_{CR} = 0.210
 Diversity = 2.6075e-02 k_t = [2, 3, 2, 2]
 Generation 3330: Best = 6.172646e+02 FES = 333100 μ_F = 0.770 μ_{CR} = 0.207
 Diversity = 2.6426e-02 k_t = [1, 1, 1, 1]
 Generation 3340: Best = 6.172430e+02 FES = 334100 μ_F = 0.765 μ_{CR} = 0.215
 Diversity = 2.6616e-02 k_t = [3, 2, 3, 2]
 Generation 3350: Best = 6.172195e+02 FES = 335100 μ_F = 0.772 μ_{CR} = 0.217
 Diversity = 2.6992e-02 k_t = [1, 1, 1, 1]
 Generation 3360: Best = 6.171932e+02 FES = 336100 μ_F = 0.771 μ_{CR} = 0.220
 Diversity = 2.5688e-02 k_t = [3, 3, 4, 4]
 Generation 3370: Best = 6.171742e+02 FES = 337100 μ_F = 0.766 μ_{CR} = 0.226
 Diversity = 2.6632e-02 k_t = [1, 1, 1, 1]
 Generation 3380: Best = 6.171438e+02 FES = 338100 μ_F = 0.765 μ_{CR} = 0.220
 Diversity = 2.7890e-02 k_t = [2, 4, 3, 2]
 Generation 3390: Best = 6.171351e+02 FES = 339100 μ_F = 0.766 μ_{CR} = 0.223
 Diversity = 2.7924e-02 k_t = [1, 1, 1, 1]
 Generation 3400: Best = 6.171126e+02 FES = 340100 μ_F = 0.758 μ_{CR} = 0.223
 Diversity = 2.6070e-02 k_t = [4, 3, 3, 4]
 Generation 3410: Best = 6.170960e+02 FES = 341100 μ_F = 0.761 μ_{CR} = 0.229
 Diversity = 2.4924e-02 k_t = [1, 1, 1, 1]
 Generation 3420: Best = 6.170804e+02 FES = 342100 μ_F = 0.743 μ_{CR} = 0.233
 Diversity = 2.2906e-02 k_t = [2, 2, 2, 2]
 Generation 3430: Best = 6.170647e+02 FES = 343100 μ_F = 0.740 μ_{CR} = 0.237
 Diversity = 2.2397e-02 k_t = [1, 1, 1, 1]
 Generation 3440: Best = 6.170506e+02 FES = 344100 μ_F = 0.742 μ_{CR} = 0.236
 Diversity = 2.0593e-02 k_t = [4, 4, 2, 2]
 Generation 3450: Best = 6.170377e+02 FES = 345100 μ_F = 0.736 μ_{CR} = 0.245
 Diversity = 1.9490e-02 k_t = [1, 1, 1, 1]
 Generation 3460: Best = 6.170208e+02 FES = 346100 μ_F = 0.743 μ_{CR} = 0.242
 Diversity = 2.0034e-02 k_t = [4, 2, 2, 2]

Generation 3470: Best = 6.170119e+02 FES = 347100 μ_F = 0.751 μ_{CR} = 0.249
 Diversity = 2.0233e-02 k_t = [1, 1, 1, 1]
 Generation 3480: Best = 6.170010e+02 FES = 348100 μ_F = 0.748 μ_{CR} = 0.242
 Diversity = 1.8134e-02 k_t = [4, 4, 4, 3]
 Generation 3490: Best = 6.169865e+02 FES = 349100 μ_F = 0.754 μ_{CR} = 0.242
 Diversity = 1.8374e-02 k_t = [1, 1, 1, 1]
 Generation 3500: Best = 6.169753e+02 FES = 350100 μ_F = 0.743 μ_{CR} = 0.250
 Diversity = 1.7462e-02 k_t = [2, 2, 3, 4]
 Generation 3510: Best = 6.169677e+02 FES = 351100 μ_F = 0.739 μ_{CR} = 0.249
 Diversity = 1.6714e-02 k_t = [1, 1, 1, 1]
 Generation 3520: Best = 6.169587e+02 FES = 352100 μ_F = 0.739 μ_{CR} = 0.240
 Diversity = 1.6304e-02 k_t = [3, 4, 3, 2]
 Generation 3530: Best = 6.169451e+02 FES = 353100 μ_F = 0.744 μ_{CR} = 0.234
 Diversity = 1.7547e-02 k_t = [1, 1, 1, 1]
 Generation 3540: Best = 6.169325e+02 FES = 354100 μ_F = 0.740 μ_{CR} = 0.235
 Diversity = 1.5930e-02 k_t = [3, 4, 2, 4]
 Generation 3550: Best = 6.169258e+02 FES = 355100 μ_F = 0.746 μ_{CR} = 0.225
 Diversity = 1.5631e-02 k_t = [1, 1, 1, 1]
 Generation 3560: Best = 6.169171e+02 FES = 356100 μ_F = 0.748 μ_{CR} = 0.231
 Diversity = 1.5707e-02 k_t = [4, 2, 3, 3]
 Generation 3570: Best = 6.169079e+02 FES = 357100 μ_F = 0.749 μ_{CR} = 0.228
 Diversity = 1.5282e-02 k_t = [1, 1, 1, 1]
 Generation 3580: Best = 6.168974e+02 FES = 358100 μ_F = 0.757 μ_{CR} = 0.224
 Diversity = 1.4921e-02 k_t = [2, 4, 3, 3]
 Generation 3590: Best = 6.168884e+02 FES = 359100 μ_F = 0.758 μ_{CR} = 0.233
 Diversity = 1.5874e-02 k_t = [1, 1, 1, 1]
 Generation 3600: Best = 6.168821e+02 FES = 360100 μ_F = 0.754 μ_{CR} = 0.233
 Diversity = 1.5440e-02 k_t = [3, 2, 2, 2]
 Generation 3610: Best = 6.168756e+02 FES = 361100 μ_F = 0.753 μ_{CR} = 0.220
 Diversity = 1.4545e-02 k_t = [1, 1, 1, 1]
 Generation 3620: Best = 6.168645e+02 FES = 362100 μ_F = 0.757 μ_{CR} = 0.219
 Diversity = 1.4774e-02 k_t = [2, 2, 4, 4]
 Generation 3630: Best = 6.168571e+02 FES = 363100 μ_F = 0.753 μ_{CR} = 0.209
 Diversity = 1.5398e-02 k_t = [1, 1, 1, 1]
 Generation 3640: Best = 6.168477e+02 FES = 364100 μ_F = 0.762 μ_{CR} = 0.203
 Diversity = 1.5612e-02 k_t = [3, 4, 3, 3]
 Generation 3650: Best = 6.168412e+02 FES = 365100 μ_F = 0.757 μ_{CR} = 0.201
 Diversity = 1.5866e-02 k_t = [1, 1, 1, 1]
 Generation 3660: Best = 6.168284e+02 FES = 366100 μ_F = 0.773 μ_{CR} = 0.203
 Diversity = 1.5845e-02 k_t = [3, 3, 2, 3]
 Generation 3670: Best = 6.168238e+02 FES = 367100 μ_F = 0.796 μ_{CR} = 0.210
 Diversity = 1.6767e-02 k_t = [1, 1, 1, 1]
 Generation 3680: Best = 6.168136e+02 FES = 368100 μ_F = 0.798 μ_{CR} = 0.203
 Diversity = 1.6292e-02 k_t = [3, 3, 4, 4]
 Generation 3690: Best = 6.168092e+02 FES = 369100 μ_F = 0.798 μ_{CR} = 0.207
 Diversity = 1.6604e-02 k_t = [1, 1, 1, 1]
 Generation 3700: Best = 6.168004e+02 FES = 370100 μ_F = 0.790 μ_{CR} = 0.210
 Diversity = 1.5854e-02 k_t = [4, 2, 4, 4]
 Generation 3710: Best = 6.167935e+02 FES = 371100 μ_F = 0.785 μ_{CR} = 0.201
 Diversity = 1.5467e-02 k_t = [1, 1, 1, 1]
 Generation 3720: Best = 6.167844e+02 FES = 372100 μ_F = 0.792 μ_{CR} = 0.204
 Diversity = 1.6280e-02 k_t = [3, 4, 4, 3]
 Generation 3730: Best = 6.167766e+02 FES = 373100 μ_F = 0.790 μ_{CR} = 0.205
 Diversity = 1.7018e-02 k_t = [1, 1, 1, 1]

Generation 3740: Best = 6.167706e+02 FES = 374100 μ_F = 0.785 μ_{CR} = 0.207
 Diversity = 1.6824e-02 k_t = [2, 2, 2, 4]
 Generation 3750: Best = 6.167625e+02 FES = 375100 μ_F = 0.780 μ_{CR} = 0.195
 Diversity = 1.7630e-02 k_t = [1, 1, 1, 1]
 Generation 3760: Best = 6.167538e+02 FES = 376100 μ_F = 0.763 μ_{CR} = 0.195
 Diversity = 1.6277e-02 k_t = [2, 4, 2, 4]
 Generation 3770: Best = 6.167472e+02 FES = 377100 μ_F = 0.766 μ_{CR} = 0.200
 Diversity = 1.6307e-02 k_t = [1, 1, 1, 1]
 Generation 3780: Best = 6.167430e+02 FES = 378100 μ_F = 0.762 μ_{CR} = 0.209
 Diversity = 1.5622e-02 k_t = [2, 4, 4, 3]
 Generation 3790: Best = 6.167322e+02 FES = 379100 μ_F = 0.751 μ_{CR} = 0.217
 Diversity = 1.5707e-02 k_t = [1, 1, 1, 1]
 Generation 3800: Best = 6.167266e+02 FES = 380100 μ_F = 0.743 μ_{CR} = 0.217
 Diversity = 1.4660e-02 k_t = [4, 4, 4, 2]
 Generation 3810: Best = 6.167214e+02 FES = 381100 μ_F = 0.753 μ_{CR} = 0.218
 Diversity = 1.3775e-02 k_t = [1, 1, 1, 1]
 Generation 3820: Best = 6.167131e+02 FES = 382100 μ_F = 0.764 μ_{CR} = 0.211
 Diversity = 1.3262e-02 k_t = [3, 3, 4, 4]
 Generation 3830: Best = 6.167063e+02 FES = 383100 μ_F = 0.766 μ_{CR} = 0.209
 Diversity = 1.3364e-02 k_t = [1, 1, 1, 1]
 Generation 3840: Best = 6.166979e+02 FES = 384100 μ_F = 0.765 μ_{CR} = 0.212
 Diversity = 1.2437e-02 k_t = [4, 3, 4, 2]
 Generation 3850: Best = 6.166934e+02 FES = 385100 μ_F = 0.770 μ_{CR} = 0.207
 Diversity = 1.3790e-02 k_t = [1, 1, 1, 1]
 Generation 3860: Best = 6.166851e+02 FES = 386100 μ_F = 0.777 μ_{CR} = 0.203
 Diversity = 1.3768e-02 k_t = [2, 2, 4, 4]
 Generation 3870: Best = 6.166759e+02 FES = 387100 μ_F = 0.782 μ_{CR} = 0.206
 Diversity = 1.6130e-02 k_t = [1, 1, 1, 1]
 Generation 3880: Best = 6.166677e+02 FES = 388100 μ_F = 0.784 μ_{CR} = 0.213
 Diversity = 1.6009e-02 k_t = [2, 4, 2, 3]
 Generation 3890: Best = 6.166589e+02 FES = 389100 μ_F = 0.775 μ_{CR} = 0.198
 Diversity = 1.7674e-02 k_t = [1, 1, 1, 1]
 Generation 3900: Best = 6.166534e+02 FES = 390100 μ_F = 0.782 μ_{CR} = 0.197
 Diversity = 1.5619e-02 k_t = [4, 2, 4, 4]
 Generation 3910: Best = 6.166452e+02 FES = 391100 μ_F = 0.779 μ_{CR} = 0.198
 Diversity = 1.4791e-02 k_t = [1, 1, 1, 1]
 Generation 3920: Best = 6.166401e+02 FES = 392100 μ_F = 0.775 μ_{CR} = 0.200
 Diversity = 1.4249e-02 k_t = [3, 3, 4, 4]
 Generation 3930: Best = 6.166363e+02 FES = 393100 μ_F = 0.768 μ_{CR} = 0.211
 Diversity = 1.4058e-02 k_t = [1, 1, 1, 1]
 Generation 3940: Best = 6.166277e+02 FES = 394100 μ_F = 0.768 μ_{CR} = 0.208
 Diversity = 1.3958e-02 k_t = [3, 2, 3, 2]
 Generation 3950: Best = 6.166246e+02 FES = 395100 μ_F = 0.753 μ_{CR} = 0.209
 Diversity = 1.4231e-02 k_t = [1, 1, 1, 1]
 Generation 3960: Best = 6.166177e+02 FES = 396100 μ_F = 0.745 μ_{CR} = 0.207
 Diversity = 1.3027e-02 k_t = [4, 2, 3, 3]
 Generation 3970: Best = 6.166130e+02 FES = 397100 μ_F = 0.745 μ_{CR} = 0.207
 Diversity = 1.3039e-02 k_t = [1, 1, 1, 1]
 Generation 3980: Best = 6.166081e+02 FES = 398100 μ_F = 0.745 μ_{CR} = 0.211
 Diversity = 1.1883e-02 k_t = [2, 2, 3, 3]
 Generation 3990: Best = 6.166038e+02 FES = 399100 μ_F = 0.748 μ_{CR} = 0.211
 Diversity = 1.2211e-02 k_t = [1, 1, 1, 1]
 Generation 4000: Best = 6.165978e+02 FES = 400100 μ_F = 0.748 μ_{CR} = 0.203
 Diversity = 1.1830e-02 k_t = [4, 4, 3, 2]

Generation 4010: Best = 6.165945e+02 FES = 401100 μ_F = 0.737 μ_{CR} = 0.203
 Diversity = 1.1421e-02 k_t = [1, 1, 1, 1]
 Generation 4020: Best = 6.165885e+02 FES = 402100 μ_F = 0.744 μ_{CR} = 0.205
 Diversity = 1.0804e-02 k_t = [3, 2, 2, 4]
 Generation 4030: Best = 6.165842e+02 FES = 403100 μ_F = 0.751 μ_{CR} = 0.215
 Diversity = 1.1842e-02 k_t = [1, 1, 1, 1]
 Generation 4040: Best = 6.165779e+02 FES = 404100 μ_F = 0.760 μ_{CR} = 0.200
 Diversity = 1.2175e-02 k_t = [3, 4, 3, 2]
 Generation 4050: Best = 6.165733e+02 FES = 405100 μ_F = 0.761 μ_{CR} = 0.201
 Diversity = 1.2292e-02 k_t = [1, 1, 1, 1]
 Generation 4060: Best = 6.165673e+02 FES = 406100 μ_F = 0.770 μ_{CR} = 0.198
 Diversity = 1.1798e-02 k_t = [4, 4, 3, 3]
 Generation 4070: Best = 6.165616e+02 FES = 407100 μ_F = 0.775 μ_{CR} = 0.199
 Diversity = 1.2115e-02 k_t = [1, 1, 1, 1]
 Generation 4080: Best = 6.165591e+02 FES = 408100 μ_F = 0.774 μ_{CR} = 0.197
 Diversity = 1.1418e-02 k_t = [4, 3, 3, 3]
 Generation 4090: Best = 6.165555e+02 FES = 409100 μ_F = 0.758 μ_{CR} = 0.197
 Diversity = 1.0724e-02 k_t = [1, 1, 1, 1]
 Generation 4100: Best = 6.165512e+02 FES = 410100 μ_F = 0.753 μ_{CR} = 0.195
 Diversity = 9.6008e-03 k_t = [4, 2, 2, 4]
 Generation 4110: Best = 6.165476e+02 FES = 411100 μ_F = 0.764 μ_{CR} = 0.198
 Diversity = 9.4231e-03 k_t = [1, 1, 1, 1]
 Generation 4120: Best = 6.165447e+02 FES = 412100 μ_F = 0.773 μ_{CR} = 0.196
 Diversity = 1.0384e-02 k_t = [3, 3, 4, 2]
 Generation 4130: Best = 6.165374e+02 FES = 413100 μ_F = 0.777 μ_{CR} = 0.197
 Diversity = 1.1512e-02 k_t = [1, 1, 1, 1]
 Generation 4140: Best = 6.165358e+02 FES = 414100 μ_F = 0.774 μ_{CR} = 0.194
 Diversity = 1.2181e-02 k_t = [2, 2, 2, 2]
 Generation 4150: Best = 6.165319e+02 FES = 415100 μ_F = 0.770 μ_{CR} = 0.199
 Diversity = 1.2005e-02 k_t = [1, 1, 1, 1]
 Generation 4160: Best = 6.165270e+02 FES = 416100 μ_F = 0.771 μ_{CR} = 0.203
 Diversity = 1.1450e-02 k_t = [2, 3, 3, 4]
 Generation 4170: Best = 6.165246e+02 FES = 417100 μ_F = 0.764 μ_{CR} = 0.196
 Diversity = 1.1346e-02 k_t = [1, 1, 1, 1]
 Generation 4180: Best = 6.165197e+02 FES = 418100 μ_F = 0.773 μ_{CR} = 0.192
 Diversity = 1.1559e-02 k_t = [2, 3, 3, 3]
 Generation 4190: Best = 6.165145e+02 FES = 419100 μ_F = 0.763 μ_{CR} = 0.194
 Diversity = 1.1547e-02 k_t = [1, 1, 1, 1]
 Generation 4200: Best = 6.165098e+02 FES = 420100 μ_F = 0.756 μ_{CR} = 0.204
 Diversity = 1.1688e-02 k_t = [2, 3, 3, 4]
 Generation 4210: Best = 6.165055e+02 FES = 421100 μ_F = 0.754 μ_{CR} = 0.194
 Diversity = 1.1978e-02 k_t = [1, 1, 1, 1]
 Generation 4220: Best = 6.165007e+02 FES = 422100 μ_F = 0.754 μ_{CR} = 0.191
 Diversity = 1.1274e-02 k_t = [3, 2, 3, 4]
 Generation 4230: Best = 6.164951e+02 FES = 423100 μ_F = 0.762 μ_{CR} = 0.198
 Diversity = 1.1553e-02 k_t = [1, 1, 1, 1]
 Generation 4240: Best = 6.164902e+02 FES = 424100 μ_F = 0.771 μ_{CR} = 0.192
 Diversity = 1.1761e-02 k_t = [3, 2, 2, 4]
 Generation 4250: Best = 6.164863e+02 FES = 425100 μ_F = 0.767 μ_{CR} = 0.192
 Diversity = 1.1562e-02 k_t = [1, 1, 1, 1]
 Generation 4260: Best = 6.164832e+02 FES = 426100 μ_F = 0.772 μ_{CR} = 0.189
 Diversity = 1.1477e-02 k_t = [2, 3, 2, 2]
 Generation 4270: Best = 6.164800e+02 FES = 427100 μ_F = 0.773 μ_{CR} = 0.195
 Diversity = 1.1909e-02 k_t = [1, 1, 1, 1]

Generation 4280: Best = 6.164749e+02 FES = 428100 μ_F = 0.756 μ_{CR} = 0.196
 Diversity = 1.1860e-02 k_t = [3, 2, 4, 2]
 Generation 4290: Best = 6.164703e+02 FES = 429100 μ_F = 0.744 μ_{CR} = 0.200
 Diversity = 1.2474e-02 k_t = [1, 1, 1, 1]
 Generation 4300: Best = 6.164648e+02 FES = 430100 μ_F = 0.732 μ_{CR} = 0.205
 Diversity = 1.1200e-02 k_t = [4, 4, 3, 3]
 Generation 4310: Best = 6.164618e+02 FES = 431100 μ_F = 0.746 μ_{CR} = 0.218
 Diversity = 1.0522e-02 k_t = [1, 1, 1, 1]
 Generation 4320: Best = 6.164591e+02 FES = 432100 μ_F = 0.752 μ_{CR} = 0.218
 Diversity = 9.7306e-03 k_t = [4, 2, 3, 4]
 Generation 4330: Best = 6.164562e+02 FES = 433100 μ_F = 0.757 μ_{CR} = 0.210
 Diversity = 9.3323e-03 k_t = [1, 1, 1, 1]
 Generation 4340: Best = 6.164519e+02 FES = 434100 μ_F = 0.762 μ_{CR} = 0.197
 Diversity = 9.4136e-03 k_t = [3, 2, 3, 4]
 Generation 4350: Best = 6.164479e+02 FES = 435100 μ_F = 0.769 μ_{CR} = 0.205
 Diversity = 9.3839e-03 k_t = [1, 1, 1, 1]
 Generation 4360: Best = 6.164449e+02 FES = 436100 μ_F = 0.763 μ_{CR} = 0.201
 Diversity = 9.5043e-03 k_t = [4, 4, 2, 3]
 Generation 4370: Best = 6.164427e+02 FES = 437100 μ_F = 0.768 μ_{CR} = 0.214
 Diversity = 1.0229e-02 k_t = [1, 1, 1, 1]
 Generation 4380: Best = 6.164377e+02 FES = 438100 μ_F = 0.767 μ_{CR} = 0.206
 Diversity = 1.0512e-02 k_t = [4, 4, 2, 4]
 Generation 4390: Best = 6.164353e+02 FES = 439100 μ_F = 0.759 μ_{CR} = 0.197
 Diversity = 1.0257e-02 k_t = [1, 1, 1, 1]
 Generation 4400: Best = 6.164320e+02 FES = 440100 μ_F = 0.751 μ_{CR} = 0.191
 Diversity = 1.0085e-02 k_t = [2, 2, 2, 2]
 Generation 4410: Best = 6.164285e+02 FES = 441100 μ_F = 0.762 μ_{CR} = 0.186
 Diversity = 1.0250e-02 k_t = [1, 1, 1, 1]
 Generation 4420: Best = 6.164254e+02 FES = 442100 μ_F = 0.760 μ_{CR} = 0.176
 Diversity = 9.5722e-03 k_t = [3, 3, 2, 3]
 Generation 4430: Best = 6.164227e+02 FES = 443100 μ_F = 0.757 μ_{CR} = 0.178
 Diversity = 9.9203e-03 k_t = [1, 1, 1, 1]
 Generation 4440: Best = 6.164200e+02 FES = 444100 μ_F = 0.758 μ_{CR} = 0.175
 Diversity = 9.4678e-03 k_t = [2, 3, 3, 2]
 Generation 4450: Best = 6.164181e+02 FES = 445100 μ_F = 0.756 μ_{CR} = 0.176
 Diversity = 9.8072e-03 k_t = [1, 1, 1, 1]
 Generation 4460: Best = 6.164156e+02 FES = 446100 μ_F = 0.755 μ_{CR} = 0.183
 Diversity = 8.7860e-03 k_t = [3, 3, 4, 4]
 Generation 4470: Best = 6.164135e+02 FES = 447100 μ_F = 0.757 μ_{CR} = 0.181
 Diversity = 8.2602e-03 k_t = [1, 1, 1, 1]
 Generation 4480: Best = 6.164100e+02 FES = 448100 μ_F = 0.764 μ_{CR} = 0.183
 Diversity = 8.2313e-03 k_t = [2, 3, 3, 2]
 Generation 4490: Best = 6.164079e+02 FES = 449100 μ_F = 0.761 μ_{CR} = 0.185
 Diversity = 8.3054e-03 k_t = [1, 1, 1, 1]
 Generation 4500: Best = 6.164050e+02 FES = 450100 μ_F = 0.768 μ_{CR} = 0.198
 Diversity = 8.2528e-03 k_t = [4, 2, 2, 2]
 Generation 4510: Best = 6.164021e+02 FES = 451100 μ_F = 0.771 μ_{CR} = 0.192
 Diversity = 8.6120e-03 k_t = [1, 1, 1, 1]
 Generation 4520: Best = 6.164007e+02 FES = 452100 μ_F = 0.779 μ_{CR} = 0.190
 Diversity = 8.7737e-03 k_t = [3, 3, 2, 3]
 Generation 4530: Best = 6.163980e+02 FES = 453100 μ_F = 0.761 μ_{CR} = 0.191
 Diversity = 8.9114e-03 k_t = [1, 1, 1, 1]
 Generation 4540: Best = 6.163950e+02 FES = 454100 μ_F = 0.756 μ_{CR} = 0.198
 Diversity = 8.9429e-03 k_t = [2, 3, 2, 3]

Generation 4550: Best = 6.163902e+02 FES = 455100 μ_F = 0.761 μ_{CR} = 0.196
 Diversity = 9.3435e-03 k_t = [1, 1, 1, 1]
 Generation 4560: Best = 6.163843e+02 FES = 456100 μ_F = 0.764 μ_{CR} = 0.207
 Diversity = 1.0553e-02 k_t = [3, 3, 3, 2]
 Generation 4570: Best = 6.163810e+02 FES = 457100 μ_F = 0.767 μ_{CR} = 0.204
 Diversity = 1.1124e-02 k_t = [1, 1, 1, 1]
 Generation 4580: Best = 6.163770e+02 FES = 458100 μ_F = 0.768 μ_{CR} = 0.202
 Diversity = 1.0567e-02 k_t = [4, 4, 3, 3]
 Generation 4590: Best = 6.163744e+02 FES = 459100 μ_F = 0.768 μ_{CR} = 0.196
 Diversity = 9.9629e-03 k_t = [1, 1, 1, 1]
 Generation 4600: Best = 6.163714e+02 FES = 460100 μ_F = 0.764 μ_{CR} = 0.208
 Diversity = 9.7409e-03 k_t = [3, 2, 3, 2]
 Generation 4610: Best = 6.163685e+02 FES = 461100 μ_F = 0.771 μ_{CR} = 0.204
 Diversity = 9.3839e-03 k_t = [1, 1, 1, 1]
 Generation 4620: Best = 6.163633e+02 FES = 462100 μ_F = 0.773 μ_{CR} = 0.200
 Diversity = 9.3919e-03 k_t = [4, 3, 4, 3]
 Generation 4630: Best = 6.163602e+02 FES = 463100 μ_F = 0.774 μ_{CR} = 0.200
 Diversity = 9.5267e-03 k_t = [1, 1, 1, 1]
 Generation 4640: Best = 6.163593e+02 FES = 464100 μ_F = 0.765 μ_{CR} = 0.205
 Diversity = 8.9748e-03 k_t = [3, 3, 2, 2]
 Generation 4650: Best = 6.163563e+02 FES = 465100 μ_F = 0.758 μ_{CR} = 0.197
 Diversity = 9.0531e-03 k_t = [1, 1, 1, 1]
 Generation 4660: Best = 6.163531e+02 FES = 466100 μ_F = 0.763 μ_{CR} = 0.207
 Diversity = 9.1957e-03 k_t = [3, 2, 2, 3]
 Generation 4670: Best = 6.163492e+02 FES = 467100 μ_F = 0.761 μ_{CR} = 0.202
 Diversity = 9.4037e-03 k_t = [1, 1, 1, 1]
 Generation 4680: Best = 6.163446e+02 FES = 468100 μ_F = 0.748 μ_{CR} = 0.205
 Diversity = 9.1965e-03 k_t = [2, 4, 3, 3]
 Generation 4690: Best = 6.163422e+02 FES = 469100 μ_F = 0.747 μ_{CR} = 0.202
 Diversity = 9.4529e-03 k_t = [1, 1, 1, 1]
 Generation 4700: Best = 6.163394e+02 FES = 470100 μ_F = 0.746 μ_{CR} = 0.203
 Diversity = 9.1890e-03 k_t = [2, 2, 3, 2]
 Generation 4710: Best = 6.163368e+02 FES = 471100 μ_F = 0.748 μ_{CR} = 0.213
 Diversity = 8.7523e-03 k_t = [1, 1, 1, 1]
 Generation 4720: Best = 6.163333e+02 FES = 472100 μ_F = 0.748 μ_{CR} = 0.222
 Diversity = 8.1805e-03 k_t = [2, 2, 3, 3]
 Generation 4730: Best = 6.163317e+02 FES = 473100 μ_F = 0.750 μ_{CR} = 0.218
 Diversity = 8.2600e-03 k_t = [1, 1, 1, 1]
 Generation 4740: Best = 6.163270e+02 FES = 474100 μ_F = 0.763 μ_{CR} = 0.226
 Diversity = 8.6483e-03 k_t = [4, 4, 4, 2]
 Generation 4750: Best = 6.163250e+02 FES = 475100 μ_F = 0.757 μ_{CR} = 0.226
 Diversity = 8.8622e-03 k_t = [1, 1, 1, 1]
 Generation 4760: Best = 6.163207e+02 FES = 476100 μ_F = 0.764 μ_{CR} = 0.220
 Diversity = 8.7839e-03 k_t = [4, 3, 3, 4]
 Generation 4770: Best = 6.163181e+02 FES = 477100 μ_F = 0.768 μ_{CR} = 0.227
 Diversity = 9.3215e-03 k_t = [1, 1, 1, 1]
 Generation 4780: Best = 6.163147e+02 FES = 478100 μ_F = 0.770 μ_{CR} = 0.221
 Diversity = 9.5073e-03 k_t = [3, 3, 3, 2]
 Generation 4790: Best = 6.163095e+02 FES = 479100 μ_F = 0.774 μ_{CR} = 0.225
 Diversity = 1.0538e-02 k_t = [1, 1, 1, 1]
 Generation 4800: Best = 6.163058e+02 FES = 480100 μ_F = 0.772 μ_{CR} = 0.225
 Diversity = 1.0892e-02 k_t = [3, 2, 3, 2]
 Generation 4810: Best = 6.162992e+02 FES = 481100 μ_F = 0.768 μ_{CR} = 0.228
 Diversity = 1.1418e-02 k_t = [1, 1, 1, 1]

Generation 4820: Best = 6.162960e+02 FES = 482100 μ_F = 0.775 μ_{CR} = 0.221
 Diversity = 1.1474e-02 k_t = [3, 3, 4, 2]
 Generation 4830: Best = 6.162928e+02 FES = 483100 μ_F = 0.778 μ_{CR} = 0.221
 Diversity = 1.1423e-02 k_t = [1, 1, 1, 1]
 Generation 4840: Best = 6.162903e+02 FES = 484100 μ_F = 0.781 μ_{CR} = 0.216
 Diversity = 1.0819e-02 k_t = [4, 2, 4, 2]
 Generation 4850: Best = 6.162858e+02 FES = 485100 μ_F = 0.757 μ_{CR} = 0.207
 Diversity = 1.0536e-02 k_t = [1, 1, 1, 1]
 Generation 4860: Best = 6.162835e+02 FES = 486100 μ_F = 0.752 μ_{CR} = 0.203
 Diversity = 9.6413e-03 k_t = [3, 2, 3, 4]
 Generation 4870: Best = 6.162819e+02 FES = 487100 μ_F = 0.744 μ_{CR} = 0.201
 Diversity = 9.0915e-03 k_t = [1, 1, 1, 1]
 Generation 4880: Best = 6.162791e+02 FES = 488100 μ_F = 0.744 μ_{CR} = 0.209
 Diversity = 9.0520e-03 k_t = [3, 3, 3, 2]
 Generation 4890: Best = 6.162752e+02 FES = 489100 μ_F = 0.755 μ_{CR} = 0.205
 Diversity = 9.2996e-03 k_t = [1, 1, 1, 1]
 Generation 4900: Best = 6.162729e+02 FES = 490100 μ_F = 0.764 μ_{CR} = 0.218
 Diversity = 9.5688e-03 k_t = [4, 4, 3, 2]
 Generation 4910: Best = 6.162698e+02 FES = 491100 μ_F = 0.777 μ_{CR} = 0.212
 Diversity = 9.4303e-03 k_t = [1, 1, 1, 1]
 Generation 4920: Best = 6.162680e+02 FES = 492100 μ_F = 0.769 μ_{CR} = 0.208
 Diversity = 9.2716e-03 k_t = [4, 3, 3, 4]
 Generation 4930: Best = 6.162649e+02 FES = 493100 μ_F = 0.768 μ_{CR} = 0.208
 Diversity = 9.2137e-03 k_t = [1, 1, 1, 1]
 Generation 4940: Best = 6.162604e+02 FES = 494100 μ_F = 0.769 μ_{CR} = 0.211
 Diversity = 9.3416e-03 k_t = [4, 3, 3, 3]
 Generation 4950: Best = 6.162564e+02 FES = 495100 μ_F = 0.767 μ_{CR} = 0.220
 Diversity = 1.0091e-02 k_t = [1, 1, 1, 1]
 Generation 4960: Best = 6.162503e+02 FES = 496100 μ_F = 0.763 μ_{CR} = 0.209
 Diversity = 1.0306e-02 k_t = [4, 3, 2, 2]
 Generation 4970: Best = 6.162449e+02 FES = 497100 μ_F = 0.762 μ_{CR} = 0.212
 Diversity = 1.0932e-02 k_t = [1, 1, 1, 1]
 Generation 4980: Best = 6.162393e+02 FES = 498100 μ_F = 0.772 μ_{CR} = 0.223
 Diversity = 1.0603e-02 k_t = [3, 4, 4, 4]
 Generation 4990: Best = 6.162361e+02 FES = 499100 μ_F = 0.771 μ_{CR} = 0.227
 Diversity = 1.1239e-02 k_t = [1, 1, 1, 1]
 Generation 5000: Best = 6.162305e+02 FES = 500100 μ_F = 0.781 μ_{CR} = 0.224
 Diversity = 1.1760e-02 k_t = [4, 2, 3, 2]
 Generation 5010: Best = 6.162284e+02 FES = 501100 μ_F = 0.772 μ_{CR} = 0.229
 Diversity = 1.2173e-02 k_t = [1, 1, 1, 1]
 Generation 5020: Best = 6.162226e+02 FES = 502100 μ_F = 0.752 μ_{CR} = 0.227
 Diversity = 1.2721e-02 k_t = [2, 4, 2, 2]
 Generation 5030: Best = 6.162204e+02 FES = 503100 μ_F = 0.755 μ_{CR} = 0.218
 Diversity = 1.2107e-02 k_t = [1, 1, 1, 1]
 Generation 5040: Best = 6.162170e+02 FES = 504100 μ_F = 0.754 μ_{CR} = 0.210
 Diversity = 1.1330e-02 k_t = [2, 2, 3, 4]
 Generation 5050: Best = 6.162125e+02 FES = 505100 μ_F = 0.747 μ_{CR} = 0.203
 Diversity = 1.1665e-02 k_t = [1, 1, 1, 1]
 Generation 5060: Best = 6.162095e+02 FES = 506100 μ_F = 0.749 μ_{CR} = 0.202
 Diversity = 1.0856e-02 k_t = [3, 2, 4, 4]
 Generation 5070: Best = 6.162060e+02 FES = 507100 μ_F = 0.746 μ_{CR} = 0.199
 Diversity = 1.1380e-02 k_t = [1, 1, 1, 1]
 Generation 5080: Best = 6.162031e+02 FES = 508100 μ_F = 0.734 μ_{CR} = 0.202
 Diversity = 1.1079e-02 k_t = [2, 4, 3, 2]

Generation 5090: Best = 6.161987e+02 FES = 509100 μ_F = 0.733 μ_{CR} = 0.196
 Diversity = 1.0746e-02 k_t = [1, 1, 1, 1]
 Generation 5100: Best = 6.161964e+02 FES = 510100 μ_F = 0.742 μ_{CR} = 0.202
 Diversity = 1.0129e-02 k_t = [2, 2, 2, 2]
 Generation 5110: Best = 6.161942e+02 FES = 511100 μ_F = 0.726 μ_{CR} = 0.197
 Diversity = 1.0035e-02 k_t = [1, 1, 1, 1]
 Generation 5120: Best = 6.161912e+02 FES = 512100 μ_F = 0.722 μ_{CR} = 0.192
 Diversity = 9.0162e-03 k_t = [2, 4, 3, 3]
 Generation 5130: Best = 6.161892e+02 FES = 513100 μ_F = 0.722 μ_{CR} = 0.201
 Diversity = 8.9931e-03 k_t = [1, 1, 1, 1]
 Generation 5140: Best = 6.161861e+02 FES = 514100 μ_F = 0.725 μ_{CR} = 0.211
 Diversity = 9.4677e-03 k_t = [2, 4, 2, 2]
 Generation 5150: Best = 6.161824e+02 FES = 515100 μ_F = 0.721 μ_{CR} = 0.218
 Diversity = 8.9065e-03 k_t = [1, 1, 1, 1]
 Generation 5160: Best = 6.161810e+02 FES = 516100 μ_F = 0.729 μ_{CR} = 0.219
 Diversity = 8.1309e-03 k_t = [4, 2, 4, 2]
 Generation 5170: Best = 6.161785e+02 FES = 517100 μ_F = 0.740 μ_{CR} = 0.217
 Diversity = 7.3970e-03 k_t = [1, 1, 1, 1]
 Generation 5180: Best = 6.161753e+02 FES = 518100 μ_F = 0.745 μ_{CR} = 0.225
 Diversity = 7.5080e-03 k_t = [4, 4, 4, 4]
 Generation 5190: Best = 6.161727e+02 FES = 519100 μ_F = 0.753 μ_{CR} = 0.226
 Diversity = 7.8213e-03 k_t = [1, 1, 1, 1]
 Generation 5200: Best = 6.161702e+02 FES = 520100 μ_F = 0.759 μ_{CR} = 0.227
 Diversity = 7.9464e-03 k_t = [2, 3, 4, 4]
 Generation 5210: Best = 6.161677e+02 FES = 521100 μ_F = 0.773 μ_{CR} = 0.226
 Diversity = 7.9969e-03 k_t = [1, 1, 1, 1]
 Generation 5220: Best = 6.161653e+02 FES = 522100 μ_F = 0.778 μ_{CR} = 0.217
 Diversity = 7.8404e-03 k_t = [4, 4, 4, 4]
 Generation 5230: Best = 6.161632e+02 FES = 523100 μ_F = 0.784 μ_{CR} = 0.213
 Diversity = 7.9303e-03 k_t = [1, 1, 1, 1]
 Generation 5240: Best = 6.161615e+02 FES = 524100 μ_F = 0.776 μ_{CR} = 0.209
 Diversity = 7.7042e-03 k_t = [4, 2, 3, 3]
 Generation 5250: Best = 6.161590e+02 FES = 525100 μ_F = 0.777 μ_{CR} = 0.221
 Diversity = 8.1365e-03 k_t = [1, 1, 1, 1]
 Generation 5260: Best = 6.161552e+02 FES = 526100 μ_F = 0.775 μ_{CR} = 0.218
 Diversity = 8.3437e-03 k_t = [2, 3, 2, 4]
 Generation 5270: Best = 6.161523e+02 FES = 527100 μ_F = 0.776 μ_{CR} = 0.208
 Diversity = 9.3818e-03 k_t = [1, 1, 1, 1]
 Generation 5280: Best = 6.161490e+02 FES = 528100 μ_F = 0.777 μ_{CR} = 0.210
 Diversity = 1.0048e-02 k_t = [3, 4, 4, 3]
 Generation 5290: Best = 6.161455e+02 FES = 529100 μ_F = 0.772 μ_{CR} = 0.209
 Diversity = 1.0155e-02 k_t = [1, 1, 1, 1]
 Generation 5300: Best = 6.161423e+02 FES = 530100 μ_F = 0.767 μ_{CR} = 0.203
 Diversity = 9.2151e-03 k_t = [2, 2, 4, 4]
 Generation 5310: Best = 6.161396e+02 FES = 531100 μ_F = 0.757 μ_{CR} = 0.206
 Diversity = 8.9364e-03 k_t = [1, 1, 1, 1]
 Generation 5320: Best = 6.161369e+02 FES = 532100 μ_F = 0.756 μ_{CR} = 0.212
 Diversity = 8.6063e-03 k_t = [2, 4, 3, 3]
 Generation 5330: Best = 6.161345e+02 FES = 533100 μ_F = 0.768 μ_{CR} = 0.206
 Diversity = 9.1455e-03 k_t = [1, 1, 1, 1]
 Generation 5340: Best = 6.161316e+02 FES = 534100 μ_F = 0.756 μ_{CR} = 0.209
 Diversity = 9.3267e-03 k_t = [4, 2, 3, 2]
 Generation 5350: Best = 6.161294e+02 FES = 535100 μ_F = 0.760 μ_{CR} = 0.201
 Diversity = 9.2841e-03 k_t = [1, 1, 1, 1]

Generation 5360: Best = 6.161270e+02 FES = 536100 μ_F = 0.758 μ_{CR} = 0.202
 Diversity = 8.5813e-03 k_t = [3, 3, 3, 3]
 Generation 5370: Best = 6.161246e+02 FES = 537100 μ_F = 0.757 μ_{CR} = 0.198
 Diversity = 8.3449e-03 k_t = [1, 1, 1, 1]
 Generation 5380: Best = 6.161216e+02 FES = 538100 μ_F = 0.749 μ_{CR} = 0.195
 Diversity = 8.5860e-03 k_t = [4, 4, 2, 2]
 Generation 5390: Best = 6.161198e+02 FES = 539100 μ_F = 0.764 μ_{CR} = 0.196
 Diversity = 8.5712e-03 k_t = [1, 1, 1, 1]
 Generation 5400: Best = 6.161174e+02 FES = 540100 μ_F = 0.768 μ_{CR} = 0.201
 Diversity = 8.1944e-03 k_t = [3, 3, 4, 3]
 Generation 5410: Best = 6.161148e+02 FES = 541100 μ_F = 0.768 μ_{CR} = 0.205
 Diversity = 8.4631e-03 k_t = [1, 1, 1, 1]
 Generation 5420: Best = 6.161113e+02 FES = 542100 μ_F = 0.772 μ_{CR} = 0.202
 Diversity = 9.1613e-03 k_t = [3, 3, 3, 3]
 Generation 5430: Best = 6.161082e+02 FES = 543100 μ_F = 0.761 μ_{CR} = 0.204
 Diversity = 9.9550e-03 k_t = [1, 1, 1, 1]
 Generation 5440: Best = 6.161056e+02 FES = 544100 μ_F = 0.752 μ_{CR} = 0.209
 Diversity = 9.2792e-03 k_t = [2, 4, 4, 4]
 Generation 5450: Best = 6.161030e+02 FES = 545100 μ_F = 0.754 μ_{CR} = 0.217
 Diversity = 9.0191e-03 k_t = [1, 1, 1, 1]
 Generation 5460: Best = 6.161000e+02 FES = 546100 μ_F = 0.756 μ_{CR} = 0.211
 Diversity = 8.2920e-03 k_t = [2, 2, 3, 3]
 Generation 5470: Best = 6.160957e+02 FES = 547100 μ_F = 0.764 μ_{CR} = 0.213
 Diversity = 9.3613e-03 k_t = [1, 1, 1, 1]
 Generation 5480: Best = 6.160921e+02 FES = 548100 μ_F = 0.765 μ_{CR} = 0.217
 Diversity = 9.5333e-03 k_t = [4, 2, 2, 2]
 Generation 5490: Best = 6.160887e+02 FES = 549100 μ_F = 0.758 μ_{CR} = 0.219
 Diversity = 9.9712e-03 k_t = [1, 1, 1, 1]
 Generation 5500: Best = 6.160855e+02 FES = 550100 μ_F = 0.757 μ_{CR} = 0.223
 Diversity = 9.6172e-03 k_t = [2, 2, 4, 3]
 Generation 5510: Best = 6.160833e+02 FES = 551100 μ_F = 0.757 μ_{CR} = 0.214
 Diversity = 9.8762e-03 k_t = [1, 1, 1, 1]
 Generation 5520: Best = 6.160801e+02 FES = 552100 μ_F = 0.760 μ_{CR} = 0.221
 Diversity = 9.4462e-03 k_t = [2, 3, 4, 2]
 Generation 5530: Best = 6.160764e+02 FES = 553100 μ_F = 0.759 μ_{CR} = 0.220
 Diversity = 9.8895e-03 k_t = [1, 1, 1, 1]
 Generation 5540: Best = 6.160741e+02 FES = 554100 μ_F = 0.758 μ_{CR} = 0.212
 Diversity = 1.0017e-02 k_t = [3, 2, 3, 4]
 Generation 5550: Best = 6.160706e+02 FES = 555100 μ_F = 0.755 μ_{CR} = 0.209
 Diversity = 1.0005e-02 k_t = [1, 1, 1, 1]
 Generation 5560: Best = 6.160680e+02 FES = 556100 μ_F = 0.760 μ_{CR} = 0.210
 Diversity = 9.6689e-03 k_t = [4, 2, 2, 2]
 Generation 5570: Best = 6.160656e+02 FES = 557100 μ_F = 0.754 μ_{CR} = 0.210
 Diversity = 9.5602e-03 k_t = [1, 1, 1, 1]
 Generation 5580: Best = 6.160632e+02 FES = 558100 μ_F = 0.754 μ_{CR} = 0.200
 Diversity = 8.7295e-03 k_t = [3, 3, 4, 4]
 Generation 5590: Best = 6.160609e+02 FES = 559100 μ_F = 0.750 μ_{CR} = 0.194
 Diversity = 8.3563e-03 k_t = [1, 1, 1, 1]
 Generation 5600: Best = 6.160582e+02 FES = 560100 μ_F = 0.751 μ_{CR} = 0.198
 Diversity = 7.7121e-03 k_t = [2, 3, 2, 4]
 Generation 5610: Best = 6.160561e+02 FES = 561100 μ_F = 0.754 μ_{CR} = 0.200
 Diversity = 7.9828e-03 k_t = [1, 1, 1, 1]
 Generation 5620: Best = 6.160538e+02 FES = 562100 μ_F = 0.756 μ_{CR} = 0.191
 Diversity = 7.4006e-03 k_t = [2, 4, 4, 3]

Generation 5630: Best = 6.160511e+02 FES = 563100 μ_F = 0.762 μ_{CR} = 0.194
 Diversity = 8.0677e-03 k_t = [1, 1, 1, 1]
 Generation 5640: Best = 6.160486e+02 FES = 564100 μ_F = 0.773 μ_{CR} = 0.206
 Diversity = 8.2157e-03 k_t = [2, 4, 3, 2]
 Generation 5650: Best = 6.160463e+02 FES = 565100 μ_F = 0.760 μ_{CR} = 0.213
 Diversity = 8.5197e-03 k_t = [1, 1, 1, 1]
 Generation 5660: Best = 6.160424e+02 FES = 566100 μ_F = 0.758 μ_{CR} = 0.220
 Diversity = 8.2432e-03 k_t = [4, 4, 4, 3]
 Generation 5670: Best = 6.160407e+02 FES = 567100 μ_F = 0.758 μ_{CR} = 0.211
 Diversity = 8.3459e-03 k_t = [1, 1, 1, 1]
 Generation 5680: Best = 6.160380e+02 FES = 568100 μ_F = 0.755 μ_{CR} = 0.215
 Diversity = 8.4195e-03 k_t = [2, 4, 3, 4]
 Generation 5690: Best = 6.160352e+02 FES = 569100 μ_F = 0.759 μ_{CR} = 0.216
 Diversity = 8.2222e-03 k_t = [1, 1, 1, 1]
 Generation 5700: Best = 6.160322e+02 FES = 570100 μ_F = 0.768 μ_{CR} = 0.219
 Diversity = 7.8724e-03 k_t = [3, 2, 2, 4]
 Generation 5710: Best = 6.160282e+02 FES = 571100 μ_F = 0.773 μ_{CR} = 0.220
 Diversity = 8.1219e-03 k_t = [1, 1, 1, 1]
 Generation 5720: Best = 6.160260e+02 FES = 572100 μ_F = 0.770 μ_{CR} = 0.224
 Diversity = 8.0777e-03 k_t = [4, 2, 3, 4]
 Generation 5730: Best = 6.160239e+02 FES = 573100 μ_F = 0.762 μ_{CR} = 0.222
 Diversity = 8.0653e-03 k_t = [1, 1, 1, 1]
 Generation 5740: Best = 6.160205e+02 FES = 574100 μ_F = 0.771 μ_{CR} = 0.224
 Diversity = 8.3045e-03 k_t = [2, 2, 2, 4]
 Generation 5750: Best = 6.160163e+02 FES = 575100 μ_F = 0.782 μ_{CR} = 0.229
 Diversity = 9.4540e-03 k_t = [1, 1, 1, 1]
 Generation 5760: Best = 6.160137e+02 FES = 576100 μ_F = 0.778 μ_{CR} = 0.227
 Diversity = 9.3164e-03 k_t = [4, 2, 2, 4]
 Generation 5770: Best = 6.160094e+02 FES = 577100 μ_F = 0.771 μ_{CR} = 0.226
 Diversity = 9.4182e-03 k_t = [1, 1, 1, 1]
 Generation 5780: Best = 6.160077e+02 FES = 578100 μ_F = 0.766 μ_{CR} = 0.234
 Diversity = 9.1863e-03 k_t = [2, 2, 3, 2]
 Generation 5790: Best = 6.160052e+02 FES = 579100 μ_F = 0.739 μ_{CR} = 0.227
 Diversity = 9.4928e-03 k_t = [1, 1, 1, 1]
 Generation 5800: Best = 6.160027e+02 FES = 580100 μ_F = 0.738 μ_{CR} = 0.232
 Diversity = 8.7111e-03 k_t = [4, 3, 2, 3]
 Generation 5810: Best = 6.159998e+02 FES = 581100 μ_F = 0.741 μ_{CR} = 0.228
 Diversity = 8.6911e-03 k_t = [1, 1, 1, 1]
 Generation 5820: Best = 6.159972e+02 FES = 582100 μ_F = 0.742 μ_{CR} = 0.225
 Diversity = 8.3819e-03 k_t = [3, 3, 4, 4]
 Generation 5830: Best = 6.159941e+02 FES = 583100 μ_F = 0.752 μ_{CR} = 0.230
 Diversity = 8.1984e-03 k_t = [1, 1, 1, 1]
 Generation 5840: Best = 6.159919e+02 FES = 584100 μ_F = 0.753 μ_{CR} = 0.228
 Diversity = 8.6058e-03 k_t = [2, 2, 4, 3]
 Generation 5850: Best = 6.159899e+02 FES = 585100 μ_F = 0.753 μ_{CR} = 0.229
 Diversity = 8.4945e-03 k_t = [1, 1, 1, 1]
 Generation 5860: Best = 6.159871e+02 FES = 586100 μ_F = 0.758 μ_{CR} = 0.239
 Diversity = 8.0480e-03 k_t = [4, 4, 4, 3]
 Generation 5870: Best = 6.159842e+02 FES = 587100 μ_F = 0.761 μ_{CR} = 0.241
 Diversity = 8.4412e-03 k_t = [1, 1, 1, 1]
 Generation 5880: Best = 6.159792e+02 FES = 588100 μ_F = 0.763 μ_{CR} = 0.244
 Diversity = 9.7320e-03 k_t = [3, 4, 2, 2]
 Generation 5890: Best = 6.159750e+02 FES = 589100 μ_F = 0.761 μ_{CR} = 0.242
 Diversity = 1.0435e-02 k_t = [1, 1, 1, 1]

Generation 5900: Best = 6.159717e+02 FES = 590100 μ_F = 0.753 μ_{CR} = 0.240
 Diversity = 1.0085e-02 k_t = [3, 2, 3, 4]
 Generation 5910: Best = 6.159685e+02 FES = 591100 μ_F = 0.749 μ_{CR} = 0.227
 Diversity = 9.6460e-03 k_t = [1, 1, 1, 1]
 Generation 5920: Best = 6.159657e+02 FES = 592100 μ_F = 0.756 μ_{CR} = 0.234
 Diversity = 9.8572e-03 k_t = [3, 2, 4, 2]
 Generation 5930: Best = 6.159634e+02 FES = 593100 μ_F = 0.747 μ_{CR} = 0.236
 Diversity = 9.9279e-03 k_t = [1, 1, 1, 1]
 Generation 5940: Best = 6.159587e+02 FES = 594100 μ_F = 0.763 μ_{CR} = 0.232
 Diversity = 9.5758e-03 k_t = [3, 3, 2, 4]
 Generation 5950: Best = 6.159557e+02 FES = 595100 μ_F = 0.771 μ_{CR} = 0.240
 Diversity = 9.4615e-03 k_t = [1, 1, 1, 1]
 Generation 5960: Best = 6.159516e+02 FES = 596100 μ_F = 0.773 μ_{CR} = 0.244
 Diversity = 1.0196e-02 k_t = [2, 3, 4, 2]
 Generation 5970: Best = 6.159471e+02 FES = 597100 μ_F = 0.772 μ_{CR} = 0.238
 Diversity = 1.0349e-02 k_t = [1, 1, 1, 1]
 Generation 5980: Best = 6.159448e+02 FES = 598100 μ_F = 0.761 μ_{CR} = 0.233
 Diversity = 1.0296e-02 k_t = [2, 2, 3, 4]
 Generation 5990: Best = 6.159410e+02 FES = 599100 μ_F = 0.762 μ_{CR} = 0.231
 Diversity = 1.0461e-02 k_t = [1, 1, 1, 1]
 Generation 6000: Best = 6.159383e+02 FES = 600100 μ_F = 0.768 μ_{CR} = 0.225
 Diversity = 1.0460e-02 k_t = [4, 3, 3, 2]
 Generation 6010: Best = 6.159340e+02 FES = 601100 μ_F = 0.758 μ_{CR} = 0.227
 Diversity = 1.0551e-02 k_t = [1, 1, 1, 1]
 Generation 6020: Best = 6.159301e+02 FES = 602100 μ_F = 0.756 μ_{CR} = 0.228
 Diversity = 1.0472e-02 k_t = [2, 3, 4, 4]
 Generation 6030: Best = 6.159263e+02 FES = 603100 μ_F = 0.758 μ_{CR} = 0.232
 Diversity = 1.0263e-02 k_t = [1, 1, 1, 1]
 Generation 6040: Best = 6.159229e+02 FES = 604100 μ_F = 0.762 μ_{CR} = 0.229
 Diversity = 1.0411e-02 k_t = [2, 2, 3, 2]
 Generation 6050: Best = 6.159195e+02 FES = 605100 μ_F = 0.757 μ_{CR} = 0.223
 Diversity = 1.0601e-02 k_t = [1, 1, 1, 1]
 Generation 6060: Best = 6.159166e+02 FES = 606100 μ_F = 0.754 μ_{CR} = 0.217
 Diversity = 9.8746e-03 k_t = [3, 4, 4, 4]
 Generation 6070: Best = 6.159116e+02 FES = 607100 μ_F = 0.760 μ_{CR} = 0.225
 Diversity = 1.0338e-02 k_t = [1, 1, 1, 1]
 Generation 6080: Best = 6.159086e+02 FES = 608100 μ_F = 0.753 μ_{CR} = 0.222
 Diversity = 1.0451e-02 k_t = [3, 4, 4, 2]
 Generation 6090: Best = 6.159045e+02 FES = 609100 μ_F = 0.752 μ_{CR} = 0.217
 Diversity = 1.0261e-02 k_t = [1, 1, 1, 1]
 Generation 6100: Best = 6.159001e+02 FES = 610100 μ_F = 0.753 μ_{CR} = 0.208
 Diversity = 1.0268e-02 k_t = [2, 3, 2, 3]
 Generation 6110: Best = 6.158970e+02 FES = 611100 μ_F = 0.746 μ_{CR} = 0.205
 Diversity = 1.1191e-02 k_t = [1, 1, 1, 1]
 Generation 6120: Best = 6.158918e+02 FES = 612100 μ_F = 0.757 μ_{CR} = 0.204
 Diversity = 1.1226e-02 k_t = [4, 3, 2, 2]
 Generation 6130: Best = 6.158886e+02 FES = 613100 μ_F = 0.766 μ_{CR} = 0.208
 Diversity = 1.1207e-02 k_t = [1, 1, 1, 1]
 Generation 6140: Best = 6.158829e+02 FES = 614100 μ_F = 0.758 μ_{CR} = 0.204
 Diversity = 1.0906e-02 k_t = [3, 3, 2, 4]
 Generation 6150: Best = 6.158799e+02 FES = 615100 μ_F = 0.759 μ_{CR} = 0.198
 Diversity = 1.0397e-02 k_t = [1, 1, 1, 1]
 Generation 6160: Best = 6.158760e+02 FES = 616100 μ_F = 0.758 μ_{CR} = 0.201
 Diversity = 1.0230e-02 k_t = [2, 4, 3, 4]

Generation 6170: Best = 6.158730e+02 FES = 617100 μ_F = 0.765 μ_{CR} = 0.202
 Diversity = 9.5467e-03 k_t = [1, 1, 1, 1]
 Generation 6180: Best = 6.158693e+02 FES = 618100 μ_F = 0.771 μ_{CR} = 0.208
 Diversity = 9.2741e-03 k_t = [3, 2, 2, 3]
 Generation 6190: Best = 6.158675e+02 FES = 619100 μ_F = 0.761 μ_{CR} = 0.204
 Diversity = 9.1471e-03 k_t = [1, 1, 1, 1]
 Generation 6200: Best = 6.158641e+02 FES = 620100 μ_F = 0.762 μ_{CR} = 0.209
 Diversity = 9.1463e-03 k_t = [2, 4, 2, 2]
 Generation 6210: Best = 6.158617e+02 FES = 621100 μ_F = 0.762 μ_{CR} = 0.212
 Diversity = 9.1915e-03 k_t = [1, 1, 1, 1]
 Generation 6220: Best = 6.158594e+02 FES = 622100 μ_F = 0.767 μ_{CR} = 0.208
 Diversity = 9.0529e-03 k_t = [3, 3, 2, 2]
 Generation 6230: Best = 6.158575e+02 FES = 623100 μ_F = 0.765 μ_{CR} = 0.205
 Diversity = 9.2727e-03 k_t = [1, 1, 1, 1]
 Generation 6240: Best = 6.158544e+02 FES = 624100 μ_F = 0.772 μ_{CR} = 0.212
 Diversity = 9.3248e-03 k_t = [2, 3, 3, 2]
 Generation 6250: Best = 6.158513e+02 FES = 625100 μ_F = 0.770 μ_{CR} = 0.211
 Diversity = 9.6805e-03 k_t = [1, 1, 1, 1]
 Generation 6260: Best = 6.158495e+02 FES = 626100 μ_F = 0.744 μ_{CR} = 0.209
 Diversity = 8.4744e-03 k_t = [3, 3, 2, 3]
 Generation 6270: Best = 6.158466e+02 FES = 627100 μ_F = 0.755 μ_{CR} = 0.213
 Diversity = 8.9531e-03 k_t = [1, 1, 1, 1]
 Generation 6280: Best = 6.158446e+02 FES = 628100 μ_F = 0.754 μ_{CR} = 0.207
 Diversity = 8.1101e-03 k_t = [4, 4, 2, 4]
 Generation 6290: Best = 6.158424e+02 FES = 629100 μ_F = 0.764 μ_{CR} = 0.201
 Diversity = 8.0845e-03 k_t = [1, 1, 1, 1]
 Generation 6300: Best = 6.158393e+02 FES = 630100 μ_F = 0.760 μ_{CR} = 0.203
 Diversity = 8.9664e-03 k_t = [3, 3, 2, 2]
 Generation 6310: Best = 6.158376e+02 FES = 631100 μ_F = 0.756 μ_{CR} = 0.205
 Diversity = 8.7619e-03 k_t = [1, 1, 1, 1]
 Generation 6320: Best = 6.158356e+02 FES = 632100 μ_F = 0.760 μ_{CR} = 0.201
 Diversity = 8.0694e-03 k_t = [4, 4, 3, 4]
 Generation 6330: Best = 6.158339e+02 FES = 633100 μ_F = 0.767 μ_{CR} = 0.197
 Diversity = 7.9092e-03 k_t = [1, 1, 1, 1]
 Generation 6340: Best = 6.158316e+02 FES = 634100 μ_F = 0.756 μ_{CR} = 0.197
 Diversity = 7.9437e-03 k_t = [2, 4, 3, 3]
 Generation 6350: Best = 6.158300e+02 FES = 635100 μ_F = 0.760 μ_{CR} = 0.198
 Diversity = 8.0455e-03 k_t = [1, 1, 1, 1]
 Generation 6360: Best = 6.158267e+02 FES = 636100 μ_F = 0.772 μ_{CR} = 0.205
 Diversity = 8.6002e-03 k_t = [2, 4, 3, 2]
 Generation 6370: Best = 6.158242e+02 FES = 637100 μ_F = 0.769 μ_{CR} = 0.203
 Diversity = 9.3912e-03 k_t = [1, 1, 1, 1]
 Generation 6380: Best = 6.158221e+02 FES = 638100 μ_F = 0.760 μ_{CR} = 0.194
 Diversity = 9.1731e-03 k_t = [2, 2, 4, 2]
 Generation 6390: Best = 6.158201e+02 FES = 639100 μ_F = 0.761 μ_{CR} = 0.185
 Diversity = 8.8576e-03 k_t = [1, 1, 1, 1]
 Generation 6400: Best = 6.158179e+02 FES = 640100 μ_F = 0.762 μ_{CR} = 0.194
 Diversity = 8.3002e-03 k_t = [2, 2, 2, 3]
 Generation 6410: Best = 6.158149e+02 FES = 641100 μ_F = 0.758 μ_{CR} = 0.189
 Diversity = 8.6975e-03 k_t = [1, 1, 1, 1]
 Generation 6420: Best = 6.158139e+02 FES = 642100 μ_F = 0.746 μ_{CR} = 0.186
 Diversity = 7.7505e-03 k_t = [3, 4, 3, 4]
 Generation 6430: Best = 6.158125e+02 FES = 643100 μ_F = 0.756 μ_{CR} = 0.187
 Diversity = 7.3497e-03 k_t = [1, 1, 1, 1]

Generation 6440: Best = 6.158105e+02 FES = 644100 μ_F = 0.768 μ_{CR} = 0.203
 Diversity = 7.0114e-03 k_t = [4, 2, 3, 4]
 Generation 6450: Best = 6.158082e+02 FES = 645100 μ_F = 0.773 μ_{CR} = 0.198
 Diversity = 7.1279e-03 k_t = [1, 1, 1, 1]
 Generation 6460: Best = 6.158068e+02 FES = 646100 μ_F = 0.767 μ_{CR} = 0.193
 Diversity = 6.5982e-03 k_t = [3, 3, 3, 4]
 Generation 6470: Best = 6.158044e+02 FES = 647100 μ_F = 0.765 μ_{CR} = 0.200
 Diversity = 6.7761e-03 k_t = [1, 1, 1, 1]
 Generation 6480: Best = 6.158018e+02 FES = 648100 μ_F = 0.775 μ_{CR} = 0.198
 Diversity = 7.2737e-03 k_t = [3, 2, 2, 3]
 Generation 6490: Best = 6.157985e+02 FES = 649100 μ_F = 0.777 μ_{CR} = 0.210
 Diversity = 8.5677e-03 k_t = [1, 1, 1, 1]
 Generation 6500: Best = 6.157939e+02 FES = 650100 μ_F = 0.766 μ_{CR} = 0.208
 Diversity = 8.6311e-03 k_t = [2, 2, 4, 4]
 Generation 6510: Best = 6.157907e+02 FES = 651100 μ_F = 0.770 μ_{CR} = 0.203
 Diversity = 9.3270e-03 k_t = [1, 1, 1, 1]
 Generation 6520: Best = 6.157866e+02 FES = 652100 μ_F = 0.766 μ_{CR} = 0.203
 Diversity = 9.2872e-03 k_t = [2, 2, 3, 2]
 Generation 6530: Best = 6.157830e+02 FES = 653100 μ_F = 0.767 μ_{CR} = 0.204
 Diversity = 9.5398e-03 k_t = [1, 1, 1, 1]
 Generation 6540: Best = 6.157793e+02 FES = 654100 μ_F = 0.767 μ_{CR} = 0.212
 Diversity = 1.0595e-02 k_t = [3, 2, 2, 2]
 Generation 6550: Best = 6.157778e+02 FES = 655100 μ_F = 0.758 μ_{CR} = 0.198
 Diversity = 1.0637e-02 k_t = [1, 1, 1, 1]
 Generation 6560: Best = 6.157759e+02 FES = 656100 μ_F = 0.759 μ_{CR} = 0.191
 Diversity = 9.9051e-03 k_t = [2, 3, 2, 3]
 Generation 6570: Best = 6.157723e+02 FES = 657100 μ_F = 0.763 μ_{CR} = 0.195
 Diversity = 1.0879e-02 k_t = [1, 1, 1, 1]
 Generation 6580: Best = 6.157694e+02 FES = 658100 μ_F = 0.753 μ_{CR} = 0.196
 Diversity = 1.0408e-02 k_t = [3, 3, 3, 4]
 Generation 6590: Best = 6.157671e+02 FES = 659100 μ_F = 0.754 μ_{CR} = 0.201
 Diversity = 9.8063e-03 k_t = [1, 1, 1, 1]
 Generation 6600: Best = 6.157639e+02 FES = 660100 μ_F = 0.758 μ_{CR} = 0.201
 Diversity = 8.9605e-03 k_t = [3, 4, 4, 4]
 Generation 6610: Best = 6.157617e+02 FES = 661100 μ_F = 0.758 μ_{CR} = 0.219
 Diversity = 9.2515e-03 k_t = [1, 1, 1, 1]
 Generation 6620: Best = 6.157588e+02 FES = 662100 μ_F = 0.762 μ_{CR} = 0.208
 Diversity = 9.1150e-03 k_t = [3, 4, 4, 4]
 Generation 6630: Best = 6.157546e+02 FES = 663100 μ_F = 0.752 μ_{CR} = 0.216
 Diversity = 9.2159e-03 k_t = [1, 1, 1, 1]
 Generation 6640: Best = 6.157536e+02 FES = 664100 μ_F = 0.741 μ_{CR} = 0.221
 Diversity = 8.1782e-03 k_t = [4, 2, 2, 4]
 Generation 6650: Best = 6.157503e+02 FES = 665100 μ_F = 0.742 μ_{CR} = 0.233
 Diversity = 7.8813e-03 k_t = [1, 1, 1, 1]
 Generation 6660: Best = 6.157474e+02 FES = 666100 μ_F = 0.745 μ_{CR} = 0.226
 Diversity = 7.5122e-03 k_t = [2, 2, 3, 4]
 Generation 6670: Best = 6.157461e+02 FES = 667100 μ_F = 0.739 μ_{CR} = 0.226
 Diversity = 7.0689e-03 k_t = [1, 1, 1, 1]
 Generation 6680: Best = 6.157438e+02 FES = 668100 μ_F = 0.751 μ_{CR} = 0.236
 Diversity = 6.9230e-03 k_t = [3, 3, 2, 2]
 Generation 6690: Best = 6.157416e+02 FES = 669100 μ_F = 0.766 μ_{CR} = 0.239
 Diversity = 6.9925e-03 k_t = [1, 1, 1, 1]
 Generation 6700: Best = 6.157398e+02 FES = 670100 μ_F = 0.763 μ_{CR} = 0.237
 Diversity = 7.1738e-03 k_t = [4, 4, 2, 2]

Generation 6710: Best = 6.157368e+02 FES = 671100 μ_F = 0.761 μ_{CR} = 0.237
 Diversity = 7.7694e-03 k_t = [1, 1, 1, 1]
 Generation 6720: Best = 6.157338e+02 FES = 672100 μ_F = 0.765 μ_{CR} = 0.231
 Diversity = 8.5648e-03 k_t = [4, 2, 3, 2]
 Generation 6730: Best = 6.157305e+02 FES = 673100 μ_F = 0.772 μ_{CR} = 0.229
 Diversity = 9.1374e-03 k_t = [1, 1, 1, 1]
 Generation 6740: Best = 6.157281e+02 FES = 674100 μ_F = 0.769 μ_{CR} = 0.236
 Diversity = 9.0682e-03 k_t = [2, 3, 4, 2]
 Generation 6750: Best = 6.157258e+02 FES = 675100 μ_F = 0.759 μ_{CR} = 0.232
 Diversity = 9.2567e-03 k_t = [1, 1, 1, 1]
 Generation 6760: Best = 6.157223e+02 FES = 676100 μ_F = 0.761 μ_{CR} = 0.234
 Diversity = 9.8659e-03 k_t = [2, 3, 2, 2]
 Generation 6770: Best = 6.157196e+02 FES = 677100 μ_F = 0.752 μ_{CR} = 0.232
 Diversity = 9.5822e-03 k_t = [1, 1, 1, 1]
 Generation 6780: Best = 6.157182e+02 FES = 678100 μ_F = 0.747 μ_{CR} = 0.235
 Diversity = 9.0969e-03 k_t = [4, 4, 4, 4]
 Generation 6790: Best = 6.157157e+02 FES = 679100 μ_F = 0.740 μ_{CR} = 0.232
 Diversity = 7.9564e-03 k_t = [1, 1, 1, 1]
 Generation 6800: Best = 6.157130e+02 FES = 680100 μ_F = 0.757 μ_{CR} = 0.228
 Diversity = 7.8206e-03 k_t = [4, 4, 2, 3]
 Generation 6810: Best = 6.157099e+02 FES = 681100 μ_F = 0.767 μ_{CR} = 0.235
 Diversity = 8.2379e-03 k_t = [1, 1, 1, 1]
 Generation 6820: Best = 6.157064e+02 FES = 682100 μ_F = 0.777 μ_{CR} = 0.242
 Diversity = 8.7627e-03 k_t = [4, 3, 3, 2]
 Generation 6830: Best = 6.157049e+02 FES = 683100 μ_F = 0.751 μ_{CR} = 0.239
 Diversity = 8.1868e-03 k_t = [1, 1, 1, 1]
 Generation 6840: Best = 6.157016e+02 FES = 684100 μ_F = 0.754 μ_{CR} = 0.234
 Diversity = 8.0556e-03 k_t = [3, 2, 4, 4]
 Generation 6850: Best = 6.156989e+02 FES = 685100 μ_F = 0.758 μ_{CR} = 0.234
 Diversity = 8.2944e-03 k_t = [1, 1, 1, 1]
 Generation 6860: Best = 6.156963e+02 FES = 686100 μ_F = 0.743 μ_{CR} = 0.230
 Diversity = 7.5063e-03 k_t = [3, 2, 3, 3]
 Generation 6870: Best = 6.156937e+02 FES = 687100 μ_F = 0.737 μ_{CR} = 0.220
 Diversity = 7.5112e-03 k_t = [1, 1, 1, 1]
 Generation 6880: Best = 6.156921e+02 FES = 688100 μ_F = 0.754 μ_{CR} = 0.227
 Diversity = 7.4955e-03 k_t = [3, 3, 3, 3]
 Generation 6890: Best = 6.156892e+02 FES = 689100 μ_F = 0.751 μ_{CR} = 0.222
 Diversity = 7.9302e-03 k_t = [1, 1, 1, 1]
 Generation 6900: Best = 6.156864e+02 FES = 690100 μ_F = 0.740 μ_{CR} = 0.226
 Diversity = 7.7512e-03 k_t = [3, 2, 2, 2]
 Generation 6910: Best = 6.156849e+02 FES = 691100 μ_F = 0.743 μ_{CR} = 0.225
 Diversity = 7.9239e-03 k_t = [1, 1, 1, 1]
 Generation 6920: Best = 6.156839e+02 FES = 692100 μ_F = 0.748 μ_{CR} = 0.223
 Diversity = 7.3196e-03 k_t = [4, 4, 2, 2]
 Generation 6930: Best = 6.156824e+02 FES = 693100 μ_F = 0.744 μ_{CR} = 0.221
 Diversity = 7.2098e-03 k_t = [1, 1, 1, 1]
 Generation 6940: Best = 6.156800e+02 FES = 694100 μ_F = 0.750 μ_{CR} = 0.220
 Diversity = 6.9264e-03 k_t = [2, 3, 4, 4]
 Generation 6950: Best = 6.156779e+02 FES = 695100 μ_F = 0.745 μ_{CR} = 0.225
 Diversity = 7.1327e-03 k_t = [1, 1, 1, 1]
 Generation 6960: Best = 6.156755e+02 FES = 696100 μ_F = 0.745 μ_{CR} = 0.229
 Diversity = 7.1145e-03 k_t = [3, 3, 4, 2]
 Generation 6970: Best = 6.156737e+02 FES = 697100 μ_F = 0.759 μ_{CR} = 0.235
 Diversity = 7.2856e-03 k_t = [1, 1, 1, 1]

Generation 6980: Best = 6.156716e+02 FES = 698100 μ_F = 0.764 μ_{CR} = 0.235
 Diversity = 6.9695e-03 k_t = [2, 2, 4, 4]
 Generation 6990: Best = 6.156701e+02 FES = 699100 μ_F = 0.766 μ_{CR} = 0.236
 Diversity = 7.1967e-03 k_t = [1, 1, 1, 1]
 Generation 7000: Best = 6.156675e+02 FES = 700100 μ_F = 0.769 μ_{CR} = 0.237
 Diversity = 7.4278e-03 k_t = [2, 2, 4, 3]
 Generation 7010: Best = 6.156653e+02 FES = 701100 μ_F = 0.771 μ_{CR} = 0.228
 Diversity = 7.9744e-03 k_t = [1, 1, 1, 1]
 Generation 7020: Best = 6.156610e+02 FES = 702100 μ_F = 0.774 μ_{CR} = 0.227
 Diversity = 8.9191e-03 k_t = [4, 3, 3, 3]
 Generation 7030: Best = 6.156586e+02 FES = 703100 μ_F = 0.769 μ_{CR} = 0.233
 Diversity = 9.1975e-03 k_t = [1, 1, 1, 1]
 Generation 7040: Best = 6.156559e+02 FES = 704100 μ_F = 0.767 μ_{CR} = 0.227
 Diversity = 8.3512e-03 k_t = [4, 3, 4, 4]
 Generation 7050: Best = 6.156547e+02 FES = 705100 μ_F = 0.765 μ_{CR} = 0.235
 Diversity = 7.9164e-03 k_t = [1, 1, 1, 1]
 Generation 7060: Best = 6.156521e+02 FES = 706100 μ_F = 0.766 μ_{CR} = 0.235
 Diversity = 8.0734e-03 k_t = [2, 4, 4, 4]
 Generation 7070: Best = 6.156481e+02 FES = 707100 μ_F = 0.759 μ_{CR} = 0.231
 Diversity = 8.5813e-03 k_t = [1, 1, 1, 1]
 Generation 7080: Best = 6.156459e+02 FES = 708100 μ_F = 0.763 μ_{CR} = 0.227
 Diversity = 8.3322e-03 k_t = [4, 2, 3, 4]
 Generation 7090: Best = 6.156439e+02 FES = 709100 μ_F = 0.767 μ_{CR} = 0.222
 Diversity = 8.0775e-03 k_t = [1, 1, 1, 1]
 Generation 7100: Best = 6.156406e+02 FES = 710100 μ_F = 0.771 μ_{CR} = 0.224
 Diversity = 8.3397e-03 k_t = [3, 3, 3, 3]
 Generation 7110: Best = 6.156384e+02 FES = 711100 μ_F = 0.762 μ_{CR} = 0.218
 Diversity = 8.3600e-03 k_t = [1, 1, 1, 1]
 Generation 7120: Best = 6.156356e+02 FES = 712100 μ_F = 0.763 μ_{CR} = 0.219
 Diversity = 9.0396e-03 k_t = [4, 3, 2, 2]
 Generation 7130: Best = 6.156339e+02 FES = 713100 μ_F = 0.767 μ_{CR} = 0.218
 Diversity = 9.1614e-03 k_t = [1, 1, 1, 1]
 Generation 7140: Best = 6.156322e+02 FES = 714100 μ_F = 0.758 μ_{CR} = 0.204
 Diversity = 8.5807e-03 k_t = [4, 4, 3, 2]
 Generation 7150: Best = 6.156301e+02 FES = 715100 μ_F = 0.752 μ_{CR} = 0.196
 Diversity = 8.0539e-03 k_t = [1, 1, 1, 1]
 Generation 7160: Best = 6.156277e+02 FES = 716100 μ_F = 0.754 μ_{CR} = 0.201
 Diversity = 7.7223e-03 k_t = [2, 3, 3, 3]
 Generation 7170: Best = 6.156261e+02 FES = 717100 μ_F = 0.757 μ_{CR} = 0.201
 Diversity = 7.9049e-03 k_t = [1, 1, 1, 1]
 Generation 7180: Best = 6.156244e+02 FES = 718100 μ_F = 0.754 μ_{CR} = 0.200
 Diversity = 8.1045e-03 k_t = [2, 4, 4, 3]
 Generation 7190: Best = 6.156216e+02 FES = 719100 μ_F = 0.757 μ_{CR} = 0.197
 Diversity = 8.2293e-03 k_t = [1, 1, 1, 1]
 Generation 7200: Best = 6.156198e+02 FES = 720100 μ_F = 0.752 μ_{CR} = 0.200
 Diversity = 7.8947e-03 k_t = [4, 3, 4, 2]
 Generation 7210: Best = 6.156157e+02 FES = 721100 μ_F = 0.744 μ_{CR} = 0.203
 Diversity = 8.0869e-03 k_t = [1, 1, 1, 1]
 Generation 7220: Best = 6.156153e+02 FES = 722100 μ_F = 0.757 μ_{CR} = 0.204
 Diversity = 8.1521e-03 k_t = [2, 3, 4, 2]
 Generation 7230: Best = 6.156135e+02 FES = 723100 μ_F = 0.756 μ_{CR} = 0.200
 Diversity = 8.0081e-03 k_t = [1, 1, 1, 1]
 Generation 7240: Best = 6.156114e+02 FES = 724100 μ_F = 0.764 μ_{CR} = 0.207
 Diversity = 7.7045e-03 k_t = [2, 4, 4, 4]

Generation 7250: Best = 6.156086e+02 FES = 725100 μ_F = 0.765 μ_{CR} = 0.210
 Diversity = 8.1641e-03 k_t = [1, 1, 1, 1]
 Generation 7260: Best = 6.156068e+02 FES = 726100 μ_F = 0.776 μ_{CR} = 0.210
 Diversity = 8.8607e-03 k_t = [4, 2, 2, 2]
 Generation 7270: Best = 6.156047e+02 FES = 727100 μ_F = 0.769 μ_{CR} = 0.195
 Diversity = 8.4096e-03 k_t = [1, 1, 1, 1]
 Generation 7280: Best = 6.156016e+02 FES = 728100 μ_F = 0.771 μ_{CR} = 0.189
 Diversity = 7.7281e-03 k_t = [3, 2, 2, 4]
 Generation 7290: Best = 6.156013e+02 FES = 729100 μ_F = 0.771 μ_{CR} = 0.181
 Diversity = 7.7490e-03 k_t = [1, 1, 1, 1]
 Generation 7300: Best = 6.155989e+02 FES = 730100 μ_F = 0.783 μ_{CR} = 0.180
 Diversity = 8.0764e-03 k_t = [3, 3, 2, 3]
 Generation 7310: Best = 6.155955e+02 FES = 731100 μ_F = 0.779 μ_{CR} = 0.182
 Diversity = 8.3308e-03 k_t = [1, 1, 1, 1]
 Generation 7320: Best = 6.155938e+02 FES = 732100 μ_F = 0.771 μ_{CR} = 0.197
 Diversity = 7.7334e-03 k_t = [3, 4, 4, 4]
 Generation 7330: Best = 6.155916e+02 FES = 733100 μ_F = 0.776 μ_{CR} = 0.194
 Diversity = 7.6961e-03 k_t = [1, 1, 1, 1]
 Generation 7340: Best = 6.155890e+02 FES = 734100 μ_F = 0.782 μ_{CR} = 0.190
 Diversity = 8.1178e-03 k_t = [2, 3, 4, 2]
 Generation 7350: Best = 6.155877e+02 FES = 735100 μ_F = 0.775 μ_{CR} = 0.191
 Diversity = 8.3917e-03 k_t = [1, 1, 1, 1]
 Generation 7360: Best = 6.155855e+02 FES = 736100 μ_F = 0.768 μ_{CR} = 0.191
 Diversity = 7.7792e-03 k_t = [2, 4, 4, 4]
 Generation 7370: Best = 6.155829e+02 FES = 737100 μ_F = 0.768 μ_{CR} = 0.190
 Diversity = 8.1564e-03 k_t = [1, 1, 1, 1]
 Generation 7380: Best = 6.155808e+02 FES = 738100 μ_F = 0.780 μ_{CR} = 0.195
 Diversity = 8.6855e-03 k_t = [3, 2, 2, 2]
 Generation 7390: Best = 6.155779e+02 FES = 739100 μ_F = 0.776 μ_{CR} = 0.195
 Diversity = 9.1106e-03 k_t = [1, 1, 1, 1]
 Generation 7400: Best = 6.155766e+02 FES = 740100 μ_F = 0.767 μ_{CR} = 0.195
 Diversity = 9.2397e-03 k_t = [2, 4, 4, 2]
 Generation 7410: Best = 6.155736e+02 FES = 741100 μ_F = 0.760 μ_{CR} = 0.204
 Diversity = 8.8838e-03 k_t = [1, 1, 1, 1]
 Generation 7420: Best = 6.155715e+02 FES = 742100 μ_F = 0.761 μ_{CR} = 0.206
 Diversity = 8.8922e-03 k_t = [4, 2, 3, 2]
 Generation 7430: Best = 6.155695e+02 FES = 743100 μ_F = 0.754 μ_{CR} = 0.196
 Diversity = 8.9126e-03 k_t = [1, 1, 1, 1]
 Generation 7440: Best = 6.155662e+02 FES = 744100 μ_F = 0.766 μ_{CR} = 0.203
 Diversity = 9.0875e-03 k_t = [4, 4, 4, 4]
 Generation 7450: Best = 6.155629e+02 FES = 745100 μ_F = 0.762 μ_{CR} = 0.210
 Diversity = 8.8728e-03 k_t = [1, 1, 1, 1]
 Generation 7460: Best = 6.155613e+02 FES = 746100 μ_F = 0.767 μ_{CR} = 0.208
 Diversity = 8.7507e-03 k_t = [3, 4, 3, 4]
 Generation 7470: Best = 6.155593e+02 FES = 747100 μ_F = 0.770 μ_{CR} = 0.216
 Diversity = 8.5412e-03 k_t = [1, 1, 1, 1]
 Generation 7480: Best = 6.155574e+02 FES = 748100 μ_F = 0.765 μ_{CR} = 0.218
 Diversity = 8.4497e-03 k_t = [2, 4, 2, 3]
 Generation 7490: Best = 6.155549e+02 FES = 749100 μ_F = 0.750 μ_{CR} = 0.212
 Diversity = 7.9903e-03 k_t = [1, 1, 1, 1]
 Generation 7500: Best = 6.155531e+02 FES = 750100 μ_F = 0.747 μ_{CR} = 0.220
 Diversity = 7.7283e-03 k_t = [4, 4, 3, 2]
 Generation 7510: Best = 6.155515e+02 FES = 751100 μ_F = 0.753 μ_{CR} = 0.226
 Diversity = 7.3329e-03 k_t = [1, 1, 1, 1]

Generation 7520: Best = 6.155491e+02 FES = 752100 μ_F = 0.760 μ_{CR} = 0.220
 Diversity = 6.9271e-03 k_t = [3, 3, 4, 3]
 Generation 7530: Best = 6.155471e+02 FES = 753100 μ_F = 0.763 μ_{CR} = 0.215
 Diversity = 7.2878e-03 k_t = [1, 1, 1, 1]
 Generation 7540: Best = 6.155453e+02 FES = 754100 μ_F = 0.772 μ_{CR} = 0.209
 Diversity = 7.2257e-03 k_t = [4, 2, 3, 4]
 Generation 7550: Best = 6.155440e+02 FES = 755100 μ_F = 0.767 μ_{CR} = 0.209
 Diversity = 7.3758e-03 k_t = [1, 1, 1, 1]
 Generation 7560: Best = 6.155426e+02 FES = 756100 μ_F = 0.753 μ_{CR} = 0.217
 Diversity = 6.7055e-03 k_t = [2, 2, 3, 4]
 Generation 7570: Best = 6.155410e+02 FES = 757100 μ_F = 0.750 μ_{CR} = 0.207
 Diversity = 6.6861e-03 k_t = [1, 1, 1, 1]
 Generation 7580: Best = 6.155398e+02 FES = 758100 μ_F = 0.757 μ_{CR} = 0.199
 Diversity = 6.5496e-03 k_t = [3, 4, 4, 3]
 Generation 7590: Best = 6.155383e+02 FES = 759100 μ_F = 0.762 μ_{CR} = 0.196
 Diversity = 6.7966e-03 k_t = [1, 1, 1, 1]
 Generation 7600: Best = 6.155370e+02 FES = 760100 μ_F = 0.770 μ_{CR} = 0.200
 Diversity = 6.8604e-03 k_t = [2, 4, 2, 3]
 Generation 7610: Best = 6.155355e+02 FES = 761100 μ_F = 0.751 μ_{CR} = 0.190
 Diversity = 6.5541e-03 k_t = [1, 1, 1, 1]
 Generation 7620: Best = 6.155340e+02 FES = 762100 μ_F = 0.766 μ_{CR} = 0.195
 Diversity = 6.5427e-03 k_t = [4, 3, 2, 2]
 Generation 7630: Best = 6.155326e+02 FES = 763100 μ_F = 0.757 μ_{CR} = 0.194
 Diversity = 6.9135e-03 k_t = [1, 1, 1, 1]
 Generation 7640: Best = 6.155312e+02 FES = 764100 μ_F = 0.760 μ_{CR} = 0.193
 Diversity = 6.6038e-03 k_t = [3, 3, 3, 3]
 Generation 7650: Best = 6.155297e+02 FES = 765100 μ_F = 0.762 μ_{CR} = 0.193
 Diversity = 6.7887e-03 k_t = [1, 1, 1, 1]
 Generation 7660: Best = 6.155280e+02 FES = 766100 μ_F = 0.766 μ_{CR} = 0.188
 Diversity = 6.6580e-03 k_t = [3, 2, 2, 4]
 Generation 7670: Best = 6.155273e+02 FES = 767100 μ_F = 0.758 μ_{CR} = 0.192
 Diversity = 6.8134e-03 k_t = [1, 1, 1, 1]
 Generation 7680: Best = 6.155261e+02 FES = 768100 μ_F = 0.747 μ_{CR} = 0.190
 Diversity = 6.7253e-03 k_t = [3, 2, 3, 3]
 Generation 7690: Best = 6.155249e+02 FES = 769100 μ_F = 0.750 μ_{CR} = 0.197
 Diversity = 6.5362e-03 k_t = [1, 1, 1, 1]
 Generation 7700: Best = 6.155229e+02 FES = 770100 μ_F = 0.746 μ_{CR} = 0.193
 Diversity = 5.9454e-03 k_t = [3, 3, 4, 3]
 Generation 7710: Best = 6.155223e+02 FES = 771100 μ_F = 0.756 μ_{CR} = 0.194
 Diversity = 5.8878e-03 k_t = [1, 1, 1, 1]
 Generation 7720: Best = 6.155209e+02 FES = 772100 μ_F = 0.764 μ_{CR} = 0.197
 Diversity = 5.4618e-03 k_t = [4, 2, 2, 4]
 Generation 7730: Best = 6.155198e+02 FES = 773100 μ_F = 0.765 μ_{CR} = 0.191
 Diversity = 5.4629e-03 k_t = [1, 1, 1, 1]
 Generation 7740: Best = 6.155188e+02 FES = 774100 μ_F = 0.765 μ_{CR} = 0.198
 Diversity = 5.1734e-03 k_t = [3, 3, 2, 4]
 Generation 7750: Best = 6.155172e+02 FES = 775100 μ_F = 0.780 μ_{CR} = 0.202
 Diversity = 5.2850e-03 k_t = [1, 1, 1, 1]
 Generation 7760: Best = 6.155151e+02 FES = 776100 μ_F = 0.782 μ_{CR} = 0.205
 Diversity = 5.9888e-03 k_t = [3, 2, 3, 3]
 Generation 7770: Best = 6.155144e+02 FES = 777100 μ_F = 0.781 μ_{CR} = 0.211
 Diversity = 6.8222e-03 k_t = [1, 1, 1, 1]
 Generation 7780: Best = 6.155120e+02 FES = 778100 μ_F = 0.781 μ_{CR} = 0.212
 Diversity = 6.9546e-03 k_t = [2, 3, 4, 2]

Generation 7790: Best = 6.155108e+02 FES = 779100 μ_F = 0.773 μ_{CR} = 0.205
 Diversity = 6.8775e-03 k_t = [1, 1, 1, 1]
 Generation 7800: Best = 6.155091e+02 FES = 780100 μ_F = 0.776 μ_{CR} = 0.201
 Diversity = 6.4916e-03 k_t = [4, 4, 4, 4]
 Generation 7810: Best = 6.155073e+02 FES = 781100 μ_F = 0.772 μ_{CR} = 0.212
 Diversity = 6.4500e-03 k_t = [1, 1, 1, 1]
 Generation 7820: Best = 6.155066e+02 FES = 782100 μ_F = 0.772 μ_{CR} = 0.213
 Diversity = 6.4317e-03 k_t = [4, 3, 2, 4]
 Generation 7830: Best = 6.155050e+02 FES = 783100 μ_F = 0.772 μ_{CR} = 0.211
 Diversity = 6.2626e-03 k_t = [1, 1, 1, 1]
 Generation 7840: Best = 6.155021e+02 FES = 784100 μ_F = 0.777 μ_{CR} = 0.212
 Diversity = 6.3960e-03 k_t = [4, 2, 2, 3]
 Generation 7850: Best = 6.155007e+02 FES = 785100 μ_F = 0.766 μ_{CR} = 0.219
 Diversity = 6.8096e-03 k_t = [1, 1, 1, 1]
 Generation 7860: Best = 6.154985e+02 FES = 786100 μ_F = 0.764 μ_{CR} = 0.216
 Diversity = 6.9521e-03 k_t = [4, 4, 3, 3]
 Generation 7870: Best = 6.154969e+02 FES = 787100 μ_F = 0.770 μ_{CR} = 0.212
 Diversity = 7.2653e-03 k_t = [1, 1, 1, 1]
 Generation 7880: Best = 6.154946e+02 FES = 788100 μ_F = 0.769 μ_{CR} = 0.216
 Diversity = 7.7671e-03 k_t = [3, 3, 4, 3]
 Generation 7890: Best = 6.154921e+02 FES = 789100 μ_F = 0.769 μ_{CR} = 0.217
 Diversity = 8.6592e-03 k_t = [1, 1, 1, 1]
 Generation 7900: Best = 6.154851e+02 FES = 790100 μ_F = 0.766 μ_{CR} = 0.222
 Diversity = 1.0723e-02 k_t = [4, 4, 4, 2]
 Generation 7910: Best = 6.154835e+02 FES = 791100 μ_F = 0.767 μ_{CR} = 0.226
 Diversity = 1.0762e-02 k_t = [1, 1, 1, 1]
 Generation 7920: Best = 6.154809e+02 FES = 792100 μ_F = 0.761 μ_{CR} = 0.216
 Diversity = 9.6005e-03 k_t = [3, 4, 3, 4]
 Generation 7930: Best = 6.154782e+02 FES = 793100 μ_F = 0.751 μ_{CR} = 0.226
 Diversity = 9.0193e-03 k_t = [1, 1, 1, 1]
 Generation 7940: Best = 6.154753e+02 FES = 794100 μ_F = 0.750 μ_{CR} = 0.226
 Diversity = 9.2212e-03 k_t = [3, 3, 4, 2]
 Generation 7950: Best = 6.154732e+02 FES = 795100 μ_F = 0.766 μ_{CR} = 0.222
 Diversity = 9.1647e-03 k_t = [1, 1, 1, 1]
 Generation 7960: Best = 6.154707e+02 FES = 796100 μ_F = 0.769 μ_{CR} = 0.215
 Diversity = 9.1449e-03 k_t = [2, 3, 4, 2]
 Generation 7970: Best = 6.154682e+02 FES = 797100 μ_F = 0.772 μ_{CR} = 0.220
 Diversity = 9.2883e-03 k_t = [1, 1, 1, 1]
 Generation 7980: Best = 6.154653e+02 FES = 798100 μ_F = 0.764 μ_{CR} = 0.210
 Diversity = 9.2468e-03 k_t = [4, 2, 4, 3]
 Generation 7990: Best = 6.154646e+02 FES = 799100 μ_F = 0.767 μ_{CR} = 0.210
 Diversity = 9.2512e-03 k_t = [1, 1, 1, 1]
 Generation 8000: Best = 6.154619e+02 FES = 800100 μ_F = 0.765 μ_{CR} = 0.207
 Diversity = 9.5008e-03 k_t = [2, 3, 2, 3]
 Generation 8010: Best = 6.154591e+02 FES = 801100 μ_F = 0.740 μ_{CR} = 0.208
 Diversity = 9.1689e-03 k_t = [1, 1, 1, 1]
 Generation 8020: Best = 6.154577e+02 FES = 802100 μ_F = 0.725 μ_{CR} = 0.215
 Diversity = 8.4273e-03 k_t = [4, 3, 4, 2]
 Generation 8030: Best = 6.154549e+02 FES = 803100 μ_F = 0.732 μ_{CR} = 0.223
 Diversity = 8.4599e-03 k_t = [1, 1, 1, 1]
 Generation 8040: Best = 6.154534e+02 FES = 804100 μ_F = 0.735 μ_{CR} = 0.218
 Diversity = 7.6517e-03 k_t = [3, 2, 4, 3]
 Generation 8050: Best = 6.154512e+02 FES = 805100 μ_F = 0.746 μ_{CR} = 0.210
 Diversity = 8.2989e-03 k_t = [1, 1, 1, 1]

Generation 8060: Best = 6.154478e+02 FES = 806100 μ_F = 0.749 μ_{CR} = 0.211
 Diversity = 8.2414e-03 k_t = [3, 2, 4, 2]
 Generation 8070: Best = 6.154463e+02 FES = 807100 μ_F = 0.756 μ_{CR} = 0.208
 Diversity = 8.1773e-03 k_t = [1, 1, 1, 1]
 Generation 8080: Best = 6.154445e+02 FES = 808100 μ_F = 0.761 μ_{CR} = 0.202
 Diversity = 7.8099e-03 k_t = [4, 3, 3, 3]
 Generation 8090: Best = 6.154427e+02 FES = 809100 μ_F = 0.763 μ_{CR} = 0.214
 Diversity = 7.5472e-03 k_t = [1, 1, 1, 1]
 Generation 8100: Best = 6.154402e+02 FES = 810100 μ_F = 0.776 μ_{CR} = 0.220
 Diversity = 7.4839e-03 k_t = [3, 3, 4, 3]
 Generation 8110: Best = 6.154390e+02 FES = 811100 μ_F = 0.767 μ_{CR} = 0.218
 Diversity = 7.1602e-03 k_t = [1, 1, 1, 1]
 Generation 8120: Best = 6.154365e+02 FES = 812100 μ_F = 0.765 μ_{CR} = 0.220
 Diversity = 6.7390e-03 k_t = [4, 4, 2, 3]
 Generation 8130: Best = 6.154350e+02 FES = 813100 μ_F = 0.771 μ_{CR} = 0.215
 Diversity = 6.9039e-03 k_t = [1, 1, 1, 1]
 Generation 8140: Best = 6.154338e+02 FES = 814100 μ_F = 0.783 μ_{CR} = 0.212
 Diversity = 7.1246e-03 k_t = [2, 4, 2, 2]
 Generation 8150: Best = 6.154319e+02 FES = 815100 μ_F = 0.788 μ_{CR} = 0.208
 Diversity = 7.4514e-03 k_t = [1, 1, 1, 1]
 Generation 8160: Best = 6.154300e+02 FES = 816100 μ_F = 0.785 μ_{CR} = 0.205
 Diversity = 6.9427e-03 k_t = [2, 4, 3, 4]
 Generation 8170: Best = 6.154278e+02 FES = 817100 μ_F = 0.785 μ_{CR} = 0.208
 Diversity = 7.4849e-03 k_t = [1, 1, 1, 1]
 Generation 8180: Best = 6.154253e+02 FES = 818100 μ_F = 0.786 μ_{CR} = 0.203
 Diversity = 7.4007e-03 k_t = [2, 3, 3, 4]
 Generation 8190: Best = 6.154232e+02 FES = 819100 μ_F = 0.786 μ_{CR} = 0.209
 Diversity = 8.2355e-03 k_t = [1, 1, 1, 1]
 Generation 8200: Best = 6.154197e+02 FES = 820100 μ_F = 0.773 μ_{CR} = 0.212
 Diversity = 8.0190e-03 k_t = [3, 4, 4, 3]
 Generation 8210: Best = 6.154174e+02 FES = 821100 μ_F = 0.777 μ_{CR} = 0.215
 Diversity = 8.7477e-03 k_t = [1, 1, 1, 1]
 Generation 8220: Best = 6.154156e+02 FES = 822100 μ_F = 0.767 μ_{CR} = 0.226
 Diversity = 7.8321e-03 k_t = [3, 4, 3, 4]
 Generation 8230: Best = 6.154138e+02 FES = 823100 μ_F = 0.757 μ_{CR} = 0.225
 Diversity = 7.1589e-03 k_t = [1, 1, 1, 1]
 Generation 8240: Best = 6.154122e+02 FES = 824100 μ_F = 0.755 μ_{CR} = 0.223
 Diversity = 6.7101e-03 k_t = [2, 4, 4, 4]
 Generation 8250: Best = 6.154105e+02 FES = 825100 μ_F = 0.751 μ_{CR} = 0.219
 Diversity = 6.0625e-03 k_t = [1, 1, 1, 1]
 Generation 8260: Best = 6.154090e+02 FES = 826100 μ_F = 0.759 μ_{CR} = 0.219
 Diversity = 6.4684e-03 k_t = [3, 2, 4, 2]
 Generation 8270: Best = 6.154076e+02 FES = 827100 μ_F = 0.760 μ_{CR} = 0.221
 Diversity = 6.6525e-03 k_t = [1, 1, 1, 1]
 Generation 8280: Best = 6.154063e+02 FES = 828100 μ_F = 0.753 μ_{CR} = 0.231
 Diversity = 6.4418e-03 k_t = [4, 3, 2, 2]
 Generation 8290: Best = 6.154045e+02 FES = 829100 μ_F = 0.756 μ_{CR} = 0.225
 Diversity = 6.7474e-03 k_t = [1, 1, 1, 1]
 Generation 8300: Best = 6.154027e+02 FES = 830100 μ_F = 0.760 μ_{CR} = 0.232
 Diversity = 7.4814e-03 k_t = [4, 3, 3, 2]
 Generation 8310: Best = 6.154007e+02 FES = 831100 μ_F = 0.757 μ_{CR} = 0.229
 Diversity = 7.2473e-03 k_t = [1, 1, 1, 1]
 Generation 8320: Best = 6.153986e+02 FES = 832100 μ_F = 0.752 μ_{CR} = 0.223
 Diversity = 6.8827e-03 k_t = [4, 2, 4, 4]

Generation 8330: Best = 6.153971e+02 FES = 833100 μ_F = 0.755 μ_{CR} = 0.224
 Diversity = 6.8356e-03 k_t = [1, 1, 1, 1]
 Generation 8340: Best = 6.153958e+02 FES = 834100 μ_F = 0.754 μ_{CR} = 0.218
 Diversity = 6.3516e-03 k_t = [3, 2, 4, 3]
 Generation 8350: Best = 6.153942e+02 FES = 835100 μ_F = 0.755 μ_{CR} = 0.218
 Diversity = 6.5797e-03 k_t = [1, 1, 1, 1]
 Generation 8360: Best = 6.153930e+02 FES = 836100 μ_F = 0.751 μ_{CR} = 0.221
 Diversity = 6.9485e-03 k_t = [2, 3, 4, 2]
 Generation 8370: Best = 6.153917e+02 FES = 837100 μ_F = 0.736 μ_{CR} = 0.218
 Diversity = 6.7002e-03 k_t = [1, 1, 1, 1]
 Generation 8380: Best = 6.153902e+02 FES = 838100 μ_F = 0.748 μ_{CR} = 0.211
 Diversity = 6.5075e-03 k_t = [3, 4, 2, 3]
 Generation 8390: Best = 6.153890e+02 FES = 839100 μ_F = 0.747 μ_{CR} = 0.200
 Diversity = 6.4794e-03 k_t = [1, 1, 1, 1]
 Generation 8400: Best = 6.153874e+02 FES = 840100 μ_F = 0.741 μ_{CR} = 0.206
 Diversity = 6.3327e-03 k_t = [3, 4, 2, 2]
 Generation 8410: Best = 6.153866e+02 FES = 841100 μ_F = 0.751 μ_{CR} = 0.208
 Diversity = 7.0502e-03 k_t = [1, 1, 1, 1]
 Generation 8420: Best = 6.153849e+02 FES = 842100 μ_F = 0.756 μ_{CR} = 0.206
 Diversity = 6.5944e-03 k_t = [4, 4, 2, 2]
 Generation 8430: Best = 6.153842e+02 FES = 843100 μ_F = 0.747 μ_{CR} = 0.206
 Diversity = 6.3692e-03 k_t = [1, 1, 1, 1]
 Generation 8440: Best = 6.153824e+02 FES = 844100 μ_F = 0.750 μ_{CR} = 0.210
 Diversity = 6.0727e-03 k_t = [2, 3, 4, 2]
 Generation 8450: Best = 6.153808e+02 FES = 845100 μ_F = 0.748 μ_{CR} = 0.218
 Diversity = 6.2656e-03 k_t = [1, 1, 1, 1]
 Generation 8460: Best = 6.153793e+02 FES = 846100 μ_F = 0.747 μ_{CR} = 0.219
 Diversity = 6.0030e-03 k_t = [2, 2, 4, 4]
 Generation 8470: Best = 6.153775e+02 FES = 847100 μ_F = 0.753 μ_{CR} = 0.223
 Diversity = 6.1151e-03 k_t = [1, 1, 1, 1]
 Generation 8480: Best = 6.153744e+02 FES = 848100 μ_F = 0.765 μ_{CR} = 0.230
 Diversity = 6.6727e-03 k_t = [3, 3, 4, 3]
 Generation 8490: Best = 6.153719e+02 FES = 849100 μ_F = 0.764 μ_{CR} = 0.224
 Diversity = 7.2888e-03 k_t = [1, 1, 1, 1]
 Generation 8500: Best = 6.153691e+02 FES = 850100 μ_F = 0.775 μ_{CR} = 0.230
 Diversity = 8.0046e-03 k_t = [3, 4, 4, 4]
 Generation 8510: Best = 6.153667e+02 FES = 851100 μ_F = 0.783 μ_{CR} = 0.232
 Diversity = 8.6471e-03 k_t = [1, 1, 1, 1]
 Generation 8520: Best = 6.153633e+02 FES = 852100 μ_F = 0.780 μ_{CR} = 0.230
 Diversity = 8.3526e-03 k_t = [4, 4, 3, 3]
 Generation 8530: Best = 6.153610e+02 FES = 853100 μ_F = 0.775 μ_{CR} = 0.230
 Diversity = 8.5778e-03 k_t = [1, 1, 1, 1]
 Generation 8540: Best = 6.153580e+02 FES = 854100 μ_F = 0.766 μ_{CR} = 0.223
 Diversity = 8.8614e-03 k_t = [2, 4, 2, 2]
 Generation 8550: Best = 6.153563e+02 FES = 855100 μ_F = 0.758 μ_{CR} = 0.227
 Diversity = 8.3269e-03 k_t = [1, 1, 1, 1]
 Generation 8560: Best = 6.153537e+02 FES = 856100 μ_F = 0.760 μ_{CR} = 0.224
 Diversity = 8.0437e-03 k_t = [4, 4, 3, 3]
 Generation 8570: Best = 6.153517e+02 FES = 857100 μ_F = 0.757 μ_{CR} = 0.240
 Diversity = 7.8075e-03 k_t = [1, 1, 1, 1]
 Generation 8580: Best = 6.153496e+02 FES = 858100 μ_F = 0.768 μ_{CR} = 0.235
 Diversity = 7.9603e-03 k_t = [2, 3, 4, 3]
 Generation 8590: Best = 6.153478e+02 FES = 859100 μ_F = 0.768 μ_{CR} = 0.232
 Diversity = 8.2856e-03 k_t = [1, 1, 1, 1]

Generation 8600: Best = 6.153463e+02 FES = 860100 μ_F = 0.760 μ_{CR} = 0.218
 Diversity = 8.2201e-03 k_t = [2, 2, 2, 3]
 Generation 8610: Best = 6.153435e+02 FES = 861100 μ_F = 0.761 μ_{CR} = 0.222
 Diversity = 8.5175e-03 k_t = [1, 1, 1, 1]
 Generation 8620: Best = 6.153403e+02 FES = 862100 μ_F = 0.771 μ_{CR} = 0.233
 Diversity = 8.6441e-03 k_t = [4, 2, 4, 4]
 Generation 8630: Best = 6.153355e+02 FES = 863100 μ_F = 0.767 μ_{CR} = 0.231
 Diversity = 8.7892e-03 k_t = [1, 1, 1, 1]
 Generation 8640: Best = 6.153343e+02 FES = 864100 μ_F = 0.773 μ_{CR} = 0.227
 Diversity = 8.7213e-03 k_t = [4, 2, 2, 4]
 Generation 8650: Best = 6.153319e+02 FES = 865100 μ_F = 0.774 μ_{CR} = 0.237
 Diversity = 9.1726e-03 k_t = [1, 1, 1, 1]
 Generation 8660: Best = 6.153288e+02 FES = 866100 μ_F = 0.759 μ_{CR} = 0.245
 Diversity = 9.0389e-03 k_t = [4, 2, 4, 4]
 Generation 8670: Best = 6.153263e+02 FES = 867100 μ_F = 0.750 μ_{CR} = 0.247
 Diversity = 8.4483e-03 k_t = [1, 1, 1, 1]
 Generation 8680: Best = 6.153246e+02 FES = 868100 μ_F = 0.738 μ_{CR} = 0.245
 Diversity = 7.5386e-03 k_t = [4, 3, 4, 4]
 Generation 8690: Best = 6.153225e+02 FES = 869100 μ_F = 0.750 μ_{CR} = 0.245
 Diversity = 7.0243e-03 k_t = [1, 1, 1, 1]
 Generation 8700: Best = 6.153208e+02 FES = 870100 μ_F = 0.742 μ_{CR} = 0.237
 Diversity = 6.5745e-03 k_t = [4, 4, 3, 4]
 Generation 8710: Best = 6.153190e+02 FES = 871100 μ_F = 0.740 μ_{CR} = 0.237
 Diversity = 6.3664e-03 k_t = [1, 1, 1, 1]
 Generation 8720: Best = 6.153173e+02 FES = 872100 μ_F = 0.750 μ_{CR} = 0.230
 Diversity = 6.1268e-03 k_t = [2, 2, 4, 4]
 Generation 8730: Best = 6.153148e+02 FES = 873100 μ_F = 0.762 μ_{CR} = 0.232
 Diversity = 6.6164e-03 k_t = [1, 1, 1, 1]
 Generation 8740: Best = 6.153121e+02 FES = 874100 μ_F = 0.769 μ_{CR} = 0.238
 Diversity = 7.0136e-03 k_t = [4, 4, 3, 2]
 Generation 8750: Best = 6.153104e+02 FES = 875100 μ_F = 0.762 μ_{CR} = 0.236
 Diversity = 7.5304e-03 k_t = [1, 1, 1, 1]
 Generation 8760: Best = 6.153068e+02 FES = 876100 μ_F = 0.757 μ_{CR} = 0.236
 Diversity = 7.6107e-03 k_t = [2, 2, 4, 3]
 Generation 8770: Best = 6.153054e+02 FES = 877100 μ_F = 0.756 μ_{CR} = 0.232
 Diversity = 8.1945e-03 k_t = [1, 1, 1, 1]
 Generation 8780: Best = 6.153035e+02 FES = 878100 μ_F = 0.754 μ_{CR} = 0.225
 Diversity = 7.5796e-03 k_t = [4, 4, 4, 3]
 Generation 8790: Best = 6.153008e+02 FES = 879100 μ_F = 0.760 μ_{CR} = 0.235
 Diversity = 7.5282e-03 k_t = [1, 1, 1, 1]
 Generation 8800: Best = 6.152976e+02 FES = 880100 μ_F = 0.767 μ_{CR} = 0.229
 Diversity = 7.2265e-03 k_t = [3, 2, 3, 4]
 Generation 8810: Best = 6.152963e+02 FES = 881100 μ_F = 0.775 μ_{CR} = 0.223
 Diversity = 7.7041e-03 k_t = [1, 1, 1, 1]
 Generation 8820: Best = 6.152947e+02 FES = 882100 μ_F = 0.768 μ_{CR} = 0.220
 Diversity = 7.3717e-03 k_t = [2, 2, 2, 3]
 Generation 8830: Best = 6.152929e+02 FES = 883100 μ_F = 0.773 μ_{CR} = 0.219
 Diversity = 7.7692e-03 k_t = [1, 1, 1, 1]
 Generation 8840: Best = 6.152912e+02 FES = 884100 μ_F = 0.762 μ_{CR} = 0.212
 Diversity = 7.4021e-03 k_t = [4, 2, 3, 2]
 Generation 8850: Best = 6.152896e+02 FES = 885100 μ_F = 0.765 μ_{CR} = 0.219
 Diversity = 7.5919e-03 k_t = [1, 1, 1, 1]
 Generation 8860: Best = 6.152884e+02 FES = 886100 μ_F = 0.757 μ_{CR} = 0.211
 Diversity = 7.0215e-03 k_t = [4, 2, 3, 3]

Generation 8870: Best = 6.152865e+02 FES = 887100 μ_F = 0.755 μ_{CR} = 0.212
 Diversity = 7.2807e-03 k_t = [1, 1, 1, 1]
 Generation 8880: Best = 6.152846e+02 FES = 888100 μ_F = 0.760 μ_{CR} = 0.218
 Diversity = 7.0194e-03 k_t = [4, 2, 4, 2]
 Generation 8890: Best = 6.152832e+02 FES = 889100 μ_F = 0.768 μ_{CR} = 0.217
 Diversity = 6.7112e-03 k_t = [1, 1, 1, 1]
 Generation 8900: Best = 6.152819e+02 FES = 890100 μ_F = 0.776 μ_{CR} = 0.211
 Diversity = 6.3663e-03 k_t = [2, 4, 2, 4]
 Generation 8910: Best = 6.152808e+02 FES = 891100 μ_F = 0.760 μ_{CR} = 0.210
 Diversity = 6.0815e-03 k_t = [1, 1, 1, 1]
 Generation 8920: Best = 6.152792e+02 FES = 892100 μ_F = 0.755 μ_{CR} = 0.216
 Diversity = 5.7632e-03 k_t = [2, 3, 4, 4]
 Generation 8930: Best = 6.152773e+02 FES = 893100 μ_F = 0.765 μ_{CR} = 0.217
 Diversity = 5.8445e-03 k_t = [1, 1, 1, 1]
 Generation 8940: Best = 6.152764e+02 FES = 894100 μ_F = 0.773 μ_{CR} = 0.219
 Diversity = 5.8628e-03 k_t = [3, 4, 3, 3]
 Generation 8950: Best = 6.152739e+02 FES = 895100 μ_F = 0.764 μ_{CR} = 0.214
 Diversity = 6.2779e-03 k_t = [1, 1, 1, 1]
 Generation 8960: Best = 6.152731e+02 FES = 896100 μ_F = 0.769 μ_{CR} = 0.211
 Diversity = 6.2958e-03 k_t = [3, 4, 3, 2]
 Generation 8970: Best = 6.152712e+02 FES = 897100 μ_F = 0.776 μ_{CR} = 0.218
 Diversity = 6.4864e-03 k_t = [1, 1, 1, 1]
 Generation 8980: Best = 6.152696e+02 FES = 898100 μ_F = 0.776 μ_{CR} = 0.217
 Diversity = 6.5610e-03 k_t = [3, 3, 3, 2]
 Generation 8990: Best = 6.152675e+02 FES = 899100 μ_F = 0.766 μ_{CR} = 0.224
 Diversity = 7.0030e-03 k_t = [1, 1, 1, 1]
 Generation 9000: Best = 6.152659e+02 FES = 900100 μ_F = 0.766 μ_{CR} = 0.238
 Diversity = 7.1098e-03 k_t = [4, 3, 3, 2]
 Generation 9010: Best = 6.152638e+02 FES = 901100 μ_F = 0.755 μ_{CR} = 0.230
 Diversity = 7.0269e-03 k_t = [1, 1, 1, 1]
 Generation 9020: Best = 6.152623e+02 FES = 902100 μ_F = 0.759 μ_{CR} = 0.237
 Diversity = 6.6246e-03 k_t = [2, 2, 3, 4]
 Generation 9030: Best = 6.152612e+02 FES = 903100 μ_F = 0.752 μ_{CR} = 0.238
 Diversity = 6.0357e-03 k_t = [1, 1, 1, 1]
 Generation 9040: Best = 6.152596e+02 FES = 904100 μ_F = 0.745 μ_{CR} = 0.237
 Diversity = 5.7470e-03 k_t = [3, 2, 3, 2]
 Generation 9050: Best = 6.152587e+02 FES = 905100 μ_F = 0.751 μ_{CR} = 0.249
 Diversity = 5.9330e-03 k_t = [1, 1, 1, 1]
 Generation 9060: Best = 6.152574e+02 FES = 906100 μ_F = 0.758 μ_{CR} = 0.249
 Diversity = 6.1139e-03 k_t = [3, 2, 4, 2]
 Generation 9070: Best = 6.152563e+02 FES = 907100 μ_F = 0.746 μ_{CR} = 0.243
 Diversity = 6.1457e-03 k_t = [1, 1, 1, 1]
 Generation 9080: Best = 6.152554e+02 FES = 908100 μ_F = 0.738 μ_{CR} = 0.242
 Diversity = 5.4485e-03 k_t = [3, 2, 3, 3]
 Generation 9090: Best = 6.152544e+02 FES = 909100 μ_F = 0.741 μ_{CR} = 0.239
 Diversity = 5.3842e-03 k_t = [1, 1, 1, 1]
 Generation 9100: Best = 6.152532e+02 FES = 910100 μ_F = 0.735 μ_{CR} = 0.239
 Diversity = 4.8200e-03 k_t = [3, 3, 3, 2]
 Generation 9110: Best = 6.152523e+02 FES = 911100 μ_F = 0.741 μ_{CR} = 0.241
 Diversity = 5.0664e-03 k_t = [1, 1, 1, 1]
 Generation 9120: Best = 6.152510e+02 FES = 912100 μ_F = 0.740 μ_{CR} = 0.247
 Diversity = 4.9467e-03 k_t = [4, 2, 4, 2]
 Generation 9130: Best = 6.152501e+02 FES = 913100 μ_F = 0.743 μ_{CR} = 0.245
 Diversity = 5.2712e-03 k_t = [1, 1, 1, 1]

Generation 9140: Best = 6.152487e+02 FES = 914100 μ_F = 0.737 μ_{CR} = 0.235
 Diversity = 5.1415e-03 k_t = [4, 3, 4, 2]
 Generation 9150: Best = 6.152477e+02 FES = 915100 μ_F = 0.762 μ_{CR} = 0.229
 Diversity = 5.3756e-03 k_t = [1, 1, 1, 1]
 Generation 9160: Best = 6.152462e+02 FES = 916100 μ_F = 0.772 μ_{CR} = 0.229
 Diversity = 6.0656e-03 k_t = [4, 4, 2, 2]
 Generation 9170: Best = 6.152444e+02 FES = 917100 μ_F = 0.773 μ_{CR} = 0.232
 Diversity = 6.1911e-03 k_t = [1, 1, 1, 1]
 Generation 9180: Best = 6.152433e+02 FES = 918100 μ_F = 0.771 μ_{CR} = 0.229
 Diversity = 6.4682e-03 k_t = [2, 2, 4, 3]
 Generation 9190: Best = 6.152417e+02 FES = 919100 μ_F = 0.766 μ_{CR} = 0.233
 Diversity = 6.6703e-03 k_t = [1, 1, 1, 1]
 Generation 9200: Best = 6.152402e+02 FES = 920100 μ_F = 0.770 μ_{CR} = 0.241
 Diversity = 7.0686e-03 k_t = [3, 4, 2, 2]
 Generation 9210: Best = 6.152384e+02 FES = 921100 μ_F = 0.764 μ_{CR} = 0.243
 Diversity = 7.4952e-03 k_t = [1, 1, 1, 1]
 Generation 9220: Best = 6.152366e+02 FES = 922100 μ_F = 0.769 μ_{CR} = 0.242
 Diversity = 7.3453e-03 k_t = [2, 3, 2, 3]
 Generation 9230: Best = 6.152343e+02 FES = 923100 μ_F = 0.773 μ_{CR} = 0.240
 Diversity = 7.9625e-03 k_t = [1, 1, 1, 1]
 Generation 9240: Best = 6.152318e+02 FES = 924100 μ_F = 0.771 μ_{CR} = 0.239
 Diversity = 7.6846e-03 k_t = [4, 4, 2, 4]
 Generation 9250: Best = 6.152301e+02 FES = 925100 μ_F = 0.768 μ_{CR} = 0.234
 Diversity = 7.4312e-03 k_t = [1, 1, 1, 1]
 Generation 9260: Best = 6.152281e+02 FES = 926100 μ_F = 0.764 μ_{CR} = 0.241
 Diversity = 7.0964e-03 k_t = [3, 2, 3, 4]
 Generation 9270: Best = 6.152262e+02 FES = 927100 μ_F = 0.769 μ_{CR} = 0.237
 Diversity = 7.2652e-03 k_t = [1, 1, 1, 1]
 Generation 9280: Best = 6.152247e+02 FES = 928100 μ_F = 0.759 μ_{CR} = 0.229
 Diversity = 7.1709e-03 k_t = [4, 3, 4, 2]
 Generation 9290: Best = 6.152231e+02 FES = 929100 μ_F = 0.753 μ_{CR} = 0.227
 Diversity = 7.2178e-03 k_t = [1, 1, 1, 1]
 Generation 9300: Best = 6.152216e+02 FES = 930100 μ_F = 0.763 μ_{CR} = 0.229
 Diversity = 7.4614e-03 k_t = [4, 3, 2, 3]
 Generation 9310: Best = 6.152193e+02 FES = 931100 μ_F = 0.762 μ_{CR} = 0.233
 Diversity = 7.6233e-03 k_t = [1, 1, 1, 1]
 Generation 9320: Best = 6.152171e+02 FES = 932100 μ_F = 0.764 μ_{CR} = 0.228
 Diversity = 7.9970e-03 k_t = [3, 3, 2, 3]
 Generation 9330: Best = 6.152149e+02 FES = 933100 μ_F = 0.765 μ_{CR} = 0.218
 Diversity = 8.2793e-03 k_t = [1, 1, 1, 1]
 Generation 9340: Best = 6.152118e+02 FES = 934100 μ_F = 0.767 μ_{CR} = 0.220
 Diversity = 8.1407e-03 k_t = [2, 4, 4, 4]
 Generation 9350: Best = 6.152101e+02 FES = 935100 μ_F = 0.759 μ_{CR} = 0.213
 Diversity = 7.7934e-03 k_t = [1, 1, 1, 1]
 Generation 9360: Best = 6.152082e+02 FES = 936100 μ_F = 0.756 μ_{CR} = 0.224
 Diversity = 7.7154e-03 k_t = [3, 2, 4, 3]
 Generation 9370: Best = 6.152068e+02 FES = 937100 μ_F = 0.767 μ_{CR} = 0.227
 Diversity = 7.8045e-03 k_t = [1, 1, 1, 1]
 Generation 9380: Best = 6.152053e+02 FES = 938100 μ_F = 0.754 μ_{CR} = 0.219
 Diversity = 6.9133e-03 k_t = [4, 4, 2, 4]
 Generation 9390: Best = 6.152040e+02 FES = 939100 μ_F = 0.746 μ_{CR} = 0.221
 Diversity = 6.3462e-03 k_t = [1, 1, 1, 1]
 Generation 9400: Best = 6.152027e+02 FES = 940100 μ_F = 0.754 μ_{CR} = 0.225
 Diversity = 6.0862e-03 k_t = [2, 2, 2, 4]

Generation 9410: Best = 6.152012e+02 FES = 941100 μ_F = 0.752 μ_{CR} = 0.219
 Diversity = 6.3180e-03 k_t = [1, 1, 1, 1]
 Generation 9420: Best = 6.152000e+02 FES = 942100 μ_F = 0.757 μ_{CR} = 0.214
 Diversity = 5.8209e-03 k_t = [4, 4, 2, 4]
 Generation 9430: Best = 6.151988e+02 FES = 943100 μ_F = 0.754 μ_{CR} = 0.215
 Diversity = 5.7622e-03 k_t = [1, 1, 1, 1]
 Generation 9440: Best = 6.151980e+02 FES = 944100 μ_F = 0.760 μ_{CR} = 0.202
 Diversity = 5.1131e-03 k_t = [3, 3, 3, 4]
 Generation 9450: Best = 6.151970e+02 FES = 945100 μ_F = 0.752 μ_{CR} = 0.202
 Diversity = 5.1702e-03 k_t = [1, 1, 1, 1]
 Generation 9460: Best = 6.151960e+02 FES = 946100 μ_F = 0.759 μ_{CR} = 0.200
 Diversity = 5.1090e-03 k_t = [2, 3, 4, 3]
 Generation 9470: Best = 6.151952e+02 FES = 947100 μ_F = 0.771 μ_{CR} = 0.191
 Diversity = 4.9467e-03 k_t = [1, 1, 1, 1]
 Generation 9480: Best = 6.151937e+02 FES = 948100 μ_F = 0.779 μ_{CR} = 0.193
 Diversity = 5.5454e-03 k_t = [3, 4, 2, 2]
 Generation 9490: Best = 6.151919e+02 FES = 949100 μ_F = 0.781 μ_{CR} = 0.200
 Diversity = 6.1466e-03 k_t = [1, 1, 1, 1]
 Generation 9500: Best = 6.151904e+02 FES = 950100 μ_F = 0.788 μ_{CR} = 0.193
 Diversity = 6.2415e-03 k_t = [4, 3, 4, 4]
 Generation 9510: Best = 6.151896e+02 FES = 951100 μ_F = 0.791 μ_{CR} = 0.195
 Diversity = 6.5225e-03 k_t = [1, 1, 1, 1]
 Generation 9520: Best = 6.151878e+02 FES = 952100 μ_F = 0.786 μ_{CR} = 0.192
 Diversity = 7.1003e-03 k_t = [4, 4, 3, 2]
 Generation 9530: Best = 6.151863e+02 FES = 953100 μ_F = 0.782 μ_{CR} = 0.187
 Diversity = 7.2774e-03 k_t = [1, 1, 1, 1]
 Generation 9540: Best = 6.151850e+02 FES = 954100 μ_F = 0.789 μ_{CR} = 0.192
 Diversity = 7.3127e-03 k_t = [2, 4, 4, 4]
 Generation 9550: Best = 6.151834e+02 FES = 955100 μ_F = 0.774 μ_{CR} = 0.198
 Diversity = 7.1283e-03 k_t = [1, 1, 1, 1]
 Generation 9560: Best = 6.151813e+02 FES = 956100 μ_F = 0.762 μ_{CR} = 0.208
 Diversity = 6.5807e-03 k_t = [3, 3, 3, 3]
 Generation 9570: Best = 6.151796e+02 FES = 957100 μ_F = 0.778 μ_{CR} = 0.207
 Diversity = 6.8060e-03 k_t = [1, 1, 1, 1]
 Generation 9580: Best = 6.151782e+02 FES = 958100 μ_F = 0.777 μ_{CR} = 0.210
 Diversity = 6.7681e-03 k_t = [3, 2, 4, 2]
 Generation 9590: Best = 6.151767e+02 FES = 959100 μ_F = 0.768 μ_{CR} = 0.201
 Diversity = 6.8766e-03 k_t = [1, 1, 1, 1]
 Generation 9600: Best = 6.151750e+02 FES = 960100 μ_F = 0.768 μ_{CR} = 0.201
 Diversity = 6.1866e-03 k_t = [3, 2, 4, 4]
 Generation 9610: Best = 6.151735e+02 FES = 961100 μ_F = 0.761 μ_{CR} = 0.214
 Diversity = 6.5270e-03 k_t = [1, 1, 1, 1]
 Generation 9620: Best = 6.151723e+02 FES = 962100 μ_F = 0.754 μ_{CR} = 0.205
 Diversity = 6.5475e-03 k_t = [2, 2, 4, 3]
 Generation 9630: Best = 6.151694e+02 FES = 963100 μ_F = 0.748 μ_{CR} = 0.212
 Diversity = 7.0513e-03 k_t = [1, 1, 1, 1]
 Generation 9640: Best = 6.151687e+02 FES = 964100 μ_F = 0.752 μ_{CR} = 0.217
 Diversity = 6.8242e-03 k_t = [2, 2, 2, 3]
 Generation 9650: Best = 6.151673e+02 FES = 965100 μ_F = 0.753 μ_{CR} = 0.219
 Diversity = 6.7795e-03 k_t = [1, 1, 1, 1]
 Generation 9660: Best = 6.151664e+02 FES = 966100 μ_F = 0.748 μ_{CR} = 0.222
 Diversity = 6.1616e-03 k_t = [2, 3, 4, 4]
 Generation 9670: Best = 6.151656e+02 FES = 967100 μ_F = 0.754 μ_{CR} = 0.214
 Diversity = 6.0088e-03 k_t = [1, 1, 1, 1]

Generation 9680: Best = 6.151640e+02 FES = 968100 μ_F = 0.757 μ_{CR} = 0.212
 Diversity = 6.1883e-03 k_t = [3, 2, 3, 2]
 Generation 9690: Best = 6.151624e+02 FES = 969100 μ_F = 0.754 μ_{CR} = 0.214
 Diversity = 6.3392e-03 k_t = [1, 1, 1, 1]
 Generation 9700: Best = 6.151612e+02 FES = 970100 μ_F = 0.760 μ_{CR} = 0.218
 Diversity = 6.0066e-03 k_t = [4, 3, 4, 2]
 Generation 9710: Best = 6.151598e+02 FES = 971100 μ_F = 0.763 μ_{CR} = 0.226
 Diversity = 6.1704e-03 k_t = [1, 1, 1, 1]
 Generation 9720: Best = 6.151580e+02 FES = 972100 μ_F = 0.762 μ_{CR} = 0.230
 Diversity = 6.0360e-03 k_t = [2, 3, 4, 4]
 Generation 9730: Best = 6.151562e+02 FES = 973100 μ_F = 0.767 μ_{CR} = 0.231
 Diversity = 6.7635e-03 k_t = [1, 1, 1, 1]
 Generation 9740: Best = 6.151537e+02 FES = 974100 μ_F = 0.764 μ_{CR} = 0.235
 Diversity = 6.7564e-03 k_t = [4, 2, 4, 3]
 Generation 9750: Best = 6.151514e+02 FES = 975100 μ_F = 0.773 μ_{CR} = 0.243
 Diversity = 7.1035e-03 k_t = [1, 1, 1, 1]
 Generation 9760: Best = 6.151488e+02 FES = 976100 μ_F = 0.782 μ_{CR} = 0.238
 Diversity = 7.6089e-03 k_t = [2, 3, 4, 2]
 Generation 9770: Best = 6.151473e+02 FES = 977100 μ_F = 0.787 μ_{CR} = 0.233
 Diversity = 8.1419e-03 k_t = [1, 1, 1, 1]
 Generation 9780: Best = 6.151449e+02 FES = 978100 μ_F = 0.784 μ_{CR} = 0.236
 Diversity = 8.0370e-03 k_t = [3, 2, 4, 3]
 Generation 9790: Best = 6.151433e+02 FES = 979100 μ_F = 0.777 μ_{CR} = 0.227
 Diversity = 8.3344e-03 k_t = [1, 1, 1, 1]
 Generation 9800: Best = 6.151409e+02 FES = 980100 μ_F = 0.775 μ_{CR} = 0.233
 Diversity = 7.9360e-03 k_t = [3, 4, 3, 4]
 Generation 9810: Best = 6.151395e+02 FES = 981100 μ_F = 0.767 μ_{CR} = 0.240
 Diversity = 7.8675e-03 k_t = [1, 1, 1, 1]
 Generation 9820: Best = 6.151369e+02 FES = 982100 μ_F = 0.757 μ_{CR} = 0.231
 Diversity = 7.4941e-03 k_t = [3, 4, 4, 2]
 Generation 9830: Best = 6.151340e+02 FES = 983100 μ_F = 0.749 μ_{CR} = 0.231
 Diversity = 7.8128e-03 k_t = [1, 1, 1, 1]
 Generation 9840: Best = 6.151321e+02 FES = 984100 μ_F = 0.758 μ_{CR} = 0.234
 Diversity = 7.6384e-03 k_t = [2, 2, 3, 4]
 Generation 9850: Best = 6.151300e+02 FES = 985100 μ_F = 0.749 μ_{CR} = 0.238
 Diversity = 7.1021e-03 k_t = [1, 1, 1, 1]
 Generation 9860: Best = 6.151287e+02 FES = 986100 μ_F = 0.738 μ_{CR} = 0.227
 Diversity = 6.4351e-03 k_t = [3, 3, 2, 3]
 Generation 9870: Best = 6.151264e+02 FES = 987100 μ_F = 0.743 μ_{CR} = 0.231
 Diversity = 6.5947e-03 k_t = [1, 1, 1, 1]
 Generation 9880: Best = 6.151244e+02 FES = 988100 μ_F = 0.760 μ_{CR} = 0.225
 Diversity = 6.4278e-03 k_t = [4, 3, 3, 4]
 Generation 9890: Best = 6.151222e+02 FES = 989100 μ_F = 0.761 μ_{CR} = 0.228
 Diversity = 6.6640e-03 k_t = [1, 1, 1, 1]
 Generation 9900: Best = 6.151199e+02 FES = 990100 μ_F = 0.757 μ_{CR} = 0.228
 Diversity = 6.3821e-03 k_t = [4, 4, 3, 2]
 Generation 9910: Best = 6.151183e+02 FES = 991100 μ_F = 0.757 μ_{CR} = 0.225
 Diversity = 6.7748e-03 k_t = [1, 1, 1, 1]
 Generation 9920: Best = 6.151167e+02 FES = 992100 μ_F = 0.758 μ_{CR} = 0.230
 Diversity = 6.7916e-03 k_t = [4, 3, 2, 2]
 Generation 9930: Best = 6.151150e+02 FES = 993100 μ_F = 0.748 μ_{CR} = 0.222
 Diversity = 6.7176e-03 k_t = [1, 1, 1, 1]
 Generation 9940: Best = 6.151127e+02 FES = 994100 μ_F = 0.759 μ_{CR} = 0.212
 Diversity = 6.5766e-03 k_t = [4, 3, 4, 2]

Generation 9950: Best = 6.151115e+02 FES = 995100 μ_F = 0.764 μ_{CR} = 0.224
 Diversity = 6.2887e-03 k_t = [1, 1, 1, 1]
 Generation 9960: Best = 6.151104e+02 FES = 996100 μ_F = 0.765 μ_{CR} = 0.225
 Diversity = 6.2996e-03 k_t = [4, 2, 3, 2]
 Generation 9970: Best = 6.151087e+02 FES = 997100 μ_F = 0.776 μ_{CR} = 0.221
 Diversity = 6.4966e-03 k_t = [1, 1, 1, 1]
 Generation 9980: Best = 6.151070e+02 FES = 998100 μ_F = 0.764 μ_{CR} = 0.224
 Diversity = 6.4712e-03 k_t = [3, 4, 2, 2]
 Generation 9990: Best = 6.151060e+02 FES = 999100 μ_F = 0.766 μ_{CR} = 0.231
 Diversity = 7.0965e-03 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 6.151044e+02

Total Function Evals = 1000000

Total Successful Updates = 445776

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Results:

Best Fitness : 6.151044e+02

Global Optimum : 4.000000e+02

Error (|best-optimum|) : 2.151044e+02

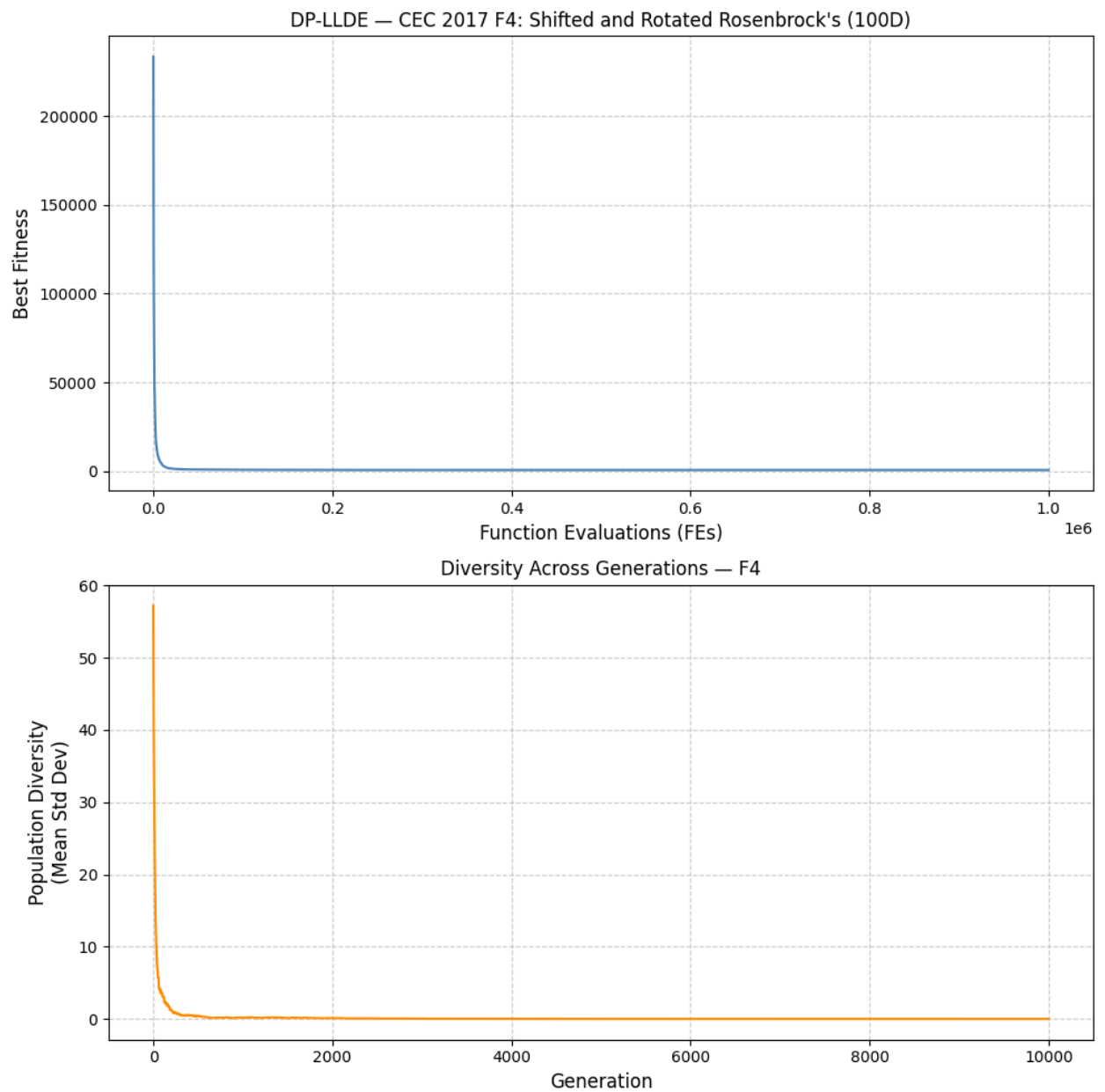
Best Solution (5 dim) : [39.8293261 37.1910649 -41.37437234 78.22520797
26.26696941]

$D^{(0)}$ (init. diversity): 5.7220e+01

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✓ Checkpoint saved to: /content/drive/My Drive/thesis_results/all_results_checkpoint.pkl



Graph saved as convergence_F4.png

✓ Plot saved to: /content/drive/My Drive/thesis_results/plots/convergence_F4.png

```
In [ ]: # =====
# F5 — Shifted and Rotated Rastrigin
# =====
run_and_plot('F5', make_wrapper(simple.f5), 500.0, "Shifted and Rotated Rastrigin")
```

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DP-LLDE on CEC 2017 F5 – Single Run

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Function : Shifted and Rotated Rastrigin's (100D)
Bounds : [-100, 100]
Global optimum : 500.0
Initial $D^*(0) = 5.7220e+01$ | Diversity threshold = 5.7220e+00
Generation 0: Best Fitness = 3.063799e+03

Generation 10:	Best = 2.201671e+03	FES = 1100	$\mu_F = 0.541$	$\mu_{CR} = 0.349$
Diversity = 3.0177e+01	$k_t = [1, 1, 1, 1]$			
Generation 20:	Best = 1.971778e+03	FES = 2100	$\mu_F = 0.587$	$\mu_{CR} = 0.344$
Diversity = 2.1426e+01	$k_t = [2, 3, 2, 3]$			
Generation 30:	Best = 1.846380e+03	FES = 3100	$\mu_F = 0.627$	$\mu_{CR} = 0.334$
Diversity = 1.7267e+01	$k_t = [1, 1, 1, 1]$			
Generation 40:	Best = 1.770489e+03	FES = 4100	$\mu_F = 0.624$	$\mu_{CR} = 0.320$
Diversity = 1.5387e+01	$k_t = [3, 2, 3, 3]$			
Generation 50:	Best = 1.705463e+03	FES = 5100	$\mu_F = 0.642$	$\mu_{CR} = 0.276$
Diversity = 1.4457e+01	$k_t = [1, 1, 1, 1]$			
Generation 60:	Best = 1.686024e+03	FES = 6100	$\mu_F = 0.675$	$\mu_{CR} = 0.232$
Diversity = 1.3916e+01	$k_t = [4, 2, 2, 2]$			
Generation 70:	Best = 1.672708e+03	FES = 7100	$\mu_F = 0.679$	$\mu_{CR} = 0.168$
Diversity = 1.3562e+01	$k_t = [1, 1, 1, 1]$			
Generation 80:	Best = 1.617591e+03	FES = 8100	$\mu_F = 0.708$	$\mu_{CR} = 0.127$
Diversity = 1.3051e+01	$k_t = [3, 3, 2, 4]$			
Generation 90:	Best = 1.617591e+03	FES = 9100	$\mu_F = 0.709$	$\mu_{CR} = 0.111$
Diversity = 1.2758e+01	$k_t = [1, 1, 1, 1]$			
Generation 100:	Best = 1.583332e+03	FES = 10100	$\mu_F = 0.705$	$\mu_{CR} = 0.093$
Diversity = 1.2798e+01	$k_t = [4, 3, 2, 2]$			
Generation 110:	Best = 1.576133e+03	FES = 11100	$\mu_F = 0.719$	$\mu_{CR} = 0.076$
Diversity = 1.2701e+01	$k_t = [1, 1, 1, 1]$			
Generation 120:	Best = 1.562904e+03	FES = 12100	$\mu_F = 0.736$	$\mu_{CR} = 0.063$
Diversity = 1.2507e+01	$k_t = [4, 3, 4, 4]$			
Generation 130:	Best = 1.553808e+03	FES = 13100	$\mu_F = 0.728$	$\mu_{CR} = 0.054$
Diversity = 1.2362e+01	$k_t = [1, 1, 1, 1]$			
Generation 140:	Best = 1.504341e+03	FES = 14100	$\mu_F = 0.666$	$\mu_{CR} = 0.056$
Diversity = 1.2301e+01	$k_t = [2, 4, 4, 2]$			
Generation 150:	Best = 1.504341e+03	FES = 15100	$\mu_F = 0.675$	$\mu_{CR} = 0.065$
Diversity = 1.2274e+01	$k_t = [1, 1, 1, 1]$			
Generation 160:	Best = 1.504341e+03	FES = 16100	$\mu_F = 0.653$	$\mu_{CR} = 0.052$
Diversity = 1.2117e+01	$k_t = [3, 3, 2, 3]$			
Generation 170:	Best = 1.504341e+03	FES = 17100	$\mu_F = 0.640$	$\mu_{CR} = 0.053$
Diversity = 1.2073e+01	$k_t = [1, 1, 1, 1]$			
Generation 180:	Best = 1.504341e+03	FES = 18100	$\mu_F = 0.732$	$\mu_{CR} = 0.044$
Diversity = 1.1918e+01	$k_t = [3, 4, 3, 4]$			
Generation 190:	Best = 1.496943e+03	FES = 19100	$\mu_F = 0.680$	$\mu_{CR} = 0.057$
Diversity = 1.1820e+01	$k_t = [1, 1, 1, 1]$			
Generation 200:	Best = 1.496943e+03	FES = 20100	$\mu_F = 0.693$	$\mu_{CR} = 0.053$
Diversity = 1.1809e+01	$k_t = [2, 3, 2, 3]$			
Generation 210:	Best = 1.485936e+03	FES = 21100	$\mu_F = 0.693$	$\mu_{CR} = 0.044$
Diversity = 1.1728e+01	$k_t = [1, 1, 1, 1]$			
Generation 220:	Best = 1.467755e+03	FES = 22100	$\mu_F = 0.701$	$\mu_{CR} = 0.043$
Diversity = 1.1713e+01	$k_t = [4, 4, 4, 3]$			

Generation 230: Best = 1.464898e+03	FES = 23100	$\mu_F = 0.657$	$\mu_{CR} = 0.036$
Diversity = 1.1671e+01	$k_t = [1, 1, 1, 1]$		
Generation 240: Best = 1.464898e+03	FES = 24100	$\mu_F = 0.568$	$\mu_{CR} = 0.035$
Diversity = 1.1611e+01	$k_t = [2, 4, 3, 3]$		
Generation 250: Best = 1.464898e+03	FES = 25100	$\mu_F = 0.663$	$\mu_{CR} = 0.037$
Diversity = 1.1589e+01	$k_t = [1, 1, 1, 1]$		
Generation 260: Best = 1.464898e+03	FES = 26100	$\mu_F = 0.635$	$\mu_{CR} = 0.038$
Diversity = 1.1564e+01	$k_t = [3, 2, 4, 2]$		
Generation 270: Best = 1.456949e+03	FES = 27100	$\mu_F = 0.579$	$\mu_{CR} = 0.042$
Diversity = 1.1506e+01	$k_t = [1, 1, 1, 1]$		
Generation 280: Best = 1.432337e+03	FES = 28100	$\mu_F = 0.552$	$\mu_{CR} = 0.037$
Diversity = 1.1519e+01	$k_t = [4, 3, 4, 3]$		
Generation 290: Best = 1.432337e+03	FES = 29100	$\mu_F = 0.508$	$\mu_{CR} = 0.035$
Diversity = 1.1470e+01	$k_t = [1, 1, 1, 1]$		
Generation 300: Best = 1.431552e+03	FES = 30100	$\mu_F = 0.517$	$\mu_{CR} = 0.037$
Diversity = 1.1446e+01	$k_t = [3, 3, 2, 2]$		
Generation 310: Best = 1.422896e+03	FES = 31100	$\mu_F = 0.503$	$\mu_{CR} = 0.034$
Diversity = 1.1414e+01	$k_t = [1, 1, 1, 1]$		
Generation 320: Best = 1.422896e+03	FES = 32100	$\mu_F = 0.513$	$\mu_{CR} = 0.037$
Diversity = 1.1386e+01	$k_t = [4, 3, 3, 4]$		
Generation 330: Best = 1.411484e+03	FES = 33100	$\mu_F = 0.519$	$\mu_{CR} = 0.039$
Diversity = 1.1355e+01	$k_t = [1, 1, 1, 1]$		
Generation 340: Best = 1.397638e+03	FES = 34100	$\mu_F = 0.568$	$\mu_{CR} = 0.041$
Diversity = 1.1320e+01	$k_t = [3, 4, 3, 4]$		
Generation 350: Best = 1.371265e+03	FES = 35100	$\mu_F = 0.532$	$\mu_{CR} = 0.031$
Diversity = 1.1289e+01	$k_t = [1, 1, 1, 1]$		
Generation 360: Best = 1.365642e+03	FES = 36100	$\mu_F = 0.527$	$\mu_{CR} = 0.035$
Diversity = 1.1241e+01	$k_t = [4, 3, 4, 4]$		
Generation 370: Best = 1.365642e+03	FES = 37100	$\mu_F = 0.478$	$\mu_{CR} = 0.038$
Diversity = 1.1198e+01	$k_t = [1, 1, 1, 1]$		
Generation 380: Best = 1.365642e+03	FES = 38100	$\mu_F = 0.456$	$\mu_{CR} = 0.029$
Diversity = 1.1202e+01	$k_t = [3, 4, 4, 3]$		
Generation 390: Best = 1.328699e+03	FES = 39100	$\mu_F = 0.426$	$\mu_{CR} = 0.034$
Diversity = 1.1178e+01	$k_t = [1, 1, 1, 1]$		
Generation 400: Best = 1.326289e+03	FES = 40100	$\mu_F = 0.436$	$\mu_{CR} = 0.046$
Diversity = 1.1133e+01	$k_t = [3, 3, 2, 4]$		
Generation 410: Best = 1.326289e+03	FES = 41100	$\mu_F = 0.469$	$\mu_{CR} = 0.044$
Diversity = 1.1103e+01	$k_t = [1, 1, 1, 1]$		
Generation 420: Best = 1.323640e+03	FES = 42100	$\mu_F = 0.422$	$\mu_{CR} = 0.047$
Diversity = 1.1095e+01	$k_t = [4, 2, 2, 3]$		
Generation 430: Best = 1.319290e+03	FES = 43100	$\mu_F = 0.420$	$\mu_{CR} = 0.037$
Diversity = 1.1083e+01	$k_t = [1, 1, 1, 1]$		
Generation 440: Best = 1.316324e+03	FES = 44100	$\mu_F = 0.434$	$\mu_{CR} = 0.038$
Diversity = 1.1043e+01	$k_t = [4, 3, 2, 2]$		
Generation 450: Best = 1.316324e+03	FES = 45100	$\mu_F = 0.494$	$\mu_{CR} = 0.040$
Diversity = 1.1077e+01	$k_t = [1, 1, 1, 1]$		
Generation 460: Best = 1.302597e+03	FES = 46100	$\mu_F = 0.526$	$\mu_{CR} = 0.039$
Diversity = 1.1053e+01	$k_t = [3, 4, 2, 2]$		
Generation 470: Best = 1.299818e+03	FES = 47100	$\mu_F = 0.494$	$\mu_{CR} = 0.040$
Diversity = 1.0999e+01	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 1.299818e+03	FES = 48100	$\mu_F = 0.507$	$\mu_{CR} = 0.048$
Diversity = 1.0986e+01	$k_t = [4, 3, 2, 3]$		
Generation 490: Best = 1.299818e+03	FES = 49100	$\mu_F = 0.439$	$\mu_{CR} = 0.043$
Diversity = 1.0960e+01	$k_t = [1, 1, 1, 1]$		

Generation 500: Best = 1.299818e+03	FES = 50100	$\mu_F = 0.449$	$\mu_{CR} = 0.049$
Diversity = 1.0947e+01	$k_t = [3, 3, 2, 3]$		
Generation 510: Best = 1.299818e+03	FES = 51100	$\mu_F = 0.415$	$\mu_{CR} = 0.043$
Diversity = 1.0936e+01	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 1.299818e+03	FES = 52100	$\mu_F = 0.408$	$\mu_{CR} = 0.046$
Diversity = 1.0925e+01	$k_t = [4, 2, 3, 2]$		
Generation 530: Best = 1.299818e+03	FES = 53100	$\mu_F = 0.410$	$\mu_{CR} = 0.055$
Diversity = 1.0920e+01	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 1.299818e+03	FES = 54100	$\mu_F = 0.383$	$\mu_{CR} = 0.045$
Diversity = 1.0900e+01	$k_t = [4, 3, 4, 3]$		
Generation 550: Best = 1.293175e+03	FES = 55100	$\mu_F = 0.389$	$\mu_{CR} = 0.045$
Diversity = 1.0901e+01	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 1.293175e+03	FES = 56100	$\mu_F = 0.403$	$\mu_{CR} = 0.042$
Diversity = 1.0887e+01	$k_t = [2, 3, 3, 3]$		
Generation 570: Best = 1.293175e+03	FES = 57100	$\mu_F = 0.432$	$\mu_{CR} = 0.043$
Diversity = 1.0877e+01	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 1.289948e+03	FES = 58100	$\mu_F = 0.430$	$\mu_{CR} = 0.045$
Diversity = 1.0856e+01	$k_t = [4, 2, 3, 2]$		
Generation 590: Best = 1.285694e+03	FES = 59100	$\mu_F = 0.412$	$\mu_{CR} = 0.041$
Diversity = 1.0828e+01	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 1.278318e+03	FES = 60100	$\mu_F = 0.365$	$\mu_{CR} = 0.052$
Diversity = 1.0813e+01	$k_t = [2, 3, 3, 3]$		
Generation 610: Best = 1.277364e+03	FES = 61100	$\mu_F = 0.311$	$\mu_{CR} = 0.044$
Diversity = 1.0787e+01	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 1.277364e+03	FES = 62100	$\mu_F = 0.325$	$\mu_{CR} = 0.044$
Diversity = 1.0772e+01	$k_t = [4, 3, 4, 2]$		
Generation 630: Best = 1.260467e+03	FES = 63100	$\mu_F = 0.300$	$\mu_{CR} = 0.048$
Diversity = 1.0748e+01	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 1.257856e+03	FES = 64100	$\mu_F = 0.339$	$\mu_{CR} = 0.048$
Diversity = 1.0740e+01	$k_t = [4, 2, 4, 4]$		
Generation 650: Best = 1.257591e+03	FES = 65100	$\mu_F = 0.393$	$\mu_{CR} = 0.043$
Diversity = 1.0754e+01	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 1.248357e+03	FES = 66100	$\mu_F = 0.338$	$\mu_{CR} = 0.045$
Diversity = 1.0741e+01	$k_t = [4, 4, 4, 3]$		
Generation 670: Best = 1.237239e+03	FES = 67100	$\mu_F = 0.323$	$\mu_{CR} = 0.051$
Diversity = 1.0724e+01	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 1.234336e+03	FES = 68100	$\mu_F = 0.431$	$\mu_{CR} = 0.054$
Diversity = 1.0751e+01	$k_t = [3, 2, 2, 4]$		
Generation 690: Best = 1.226283e+03	FES = 69100	$\mu_F = 0.407$	$\mu_{CR} = 0.043$
Diversity = 1.0743e+01	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 1.226283e+03	FES = 70100	$\mu_F = 0.434$	$\mu_{CR} = 0.043$
Diversity = 1.0733e+01	$k_t = [4, 2, 3, 3]$		
Generation 710: Best = 1.205346e+03	FES = 71100	$\mu_F = 0.406$	$\mu_{CR} = 0.045$
Diversity = 1.0721e+01	$k_t = [1, 1, 1, 1]$		
Generation 720: Best = 1.205346e+03	FES = 72100	$\mu_F = 0.458$	$\mu_{CR} = 0.045$
Diversity = 1.0709e+01	$k_t = [2, 3, 4, 3]$		
Generation 730: Best = 1.205346e+03	FES = 73100	$\mu_F = 0.364$	$\mu_{CR} = 0.038$
Diversity = 1.0707e+01	$k_t = [1, 1, 1, 1]$		
Generation 740: Best = 1.187335e+03	FES = 74100	$\mu_F = 0.432$	$\mu_{CR} = 0.036$
Diversity = 1.0681e+01	$k_t = [3, 4, 4, 4]$		
Generation 750: Best = 1.187335e+03	FES = 75100	$\mu_F = 0.385$	$\mu_{CR} = 0.029$
Diversity = 1.0672e+01	$k_t = [1, 1, 1, 1]$		
Generation 760: Best = 1.184653e+03	FES = 76100	$\mu_F = 0.352$	$\mu_{CR} = 0.039$
Diversity = 1.0653e+01	$k_t = [3, 4, 3, 2]$		

Generation 770:	Best = 1.180709e+03	FES = 77100	$\mu_F = 0.360$	$\mu_{CR} = 0.039$
Diversity = 1.0635e+01	$k_t = [1, 1, 1, 1]$			
Generation 780:	Best = 1.180709e+03	FES = 78100	$\mu_F = 0.310$	$\mu_{CR} = 0.043$
Diversity = 1.0622e+01	$k_t = [3, 3, 4, 3]$			
Generation 790:	Best = 1.180709e+03	FES = 79100	$\mu_F = 0.398$	$\mu_{CR} = 0.037$
Diversity = 1.0602e+01	$k_t = [1, 1, 1, 1]$			
Generation 800:	Best = 1.180709e+03	FES = 80100	$\mu_F = 0.459$	$\mu_{CR} = 0.040$
Diversity = 1.0620e+01	$k_t = [3, 3, 2, 4]$			
Generation 810:	Best = 1.163914e+03	FES = 81100	$\mu_F = 0.453$	$\mu_{CR} = 0.037$
Diversity = 1.0615e+01	$k_t = [1, 1, 1, 1]$			
Generation 820:	Best = 1.163914e+03	FES = 82100	$\mu_F = 0.421$	$\mu_{CR} = 0.036$
Diversity = 1.0615e+01	$k_t = [2, 4, 4, 4]$			
Generation 830:	Best = 1.163914e+03	FES = 83100	$\mu_F = 0.402$	$\mu_{CR} = 0.029$
Diversity = 1.0631e+01	$k_t = [1, 1, 1, 1]$			
Generation 840:	Best = 1.163914e+03	FES = 84100	$\mu_F = 0.553$	$\mu_{CR} = 0.033$
Diversity = 1.0619e+01	$k_t = [4, 2, 3, 4]$			
Generation 850:	Best = 1.157622e+03	FES = 85100	$\mu_F = 0.626$	$\mu_{CR} = 0.025$
Diversity = 1.0666e+01	$k_t = [1, 1, 1, 1]$			
Generation 860:	Best = 1.141573e+03	FES = 86100	$\mu_F = 0.628$	$\mu_{CR} = 0.029$
Diversity = 1.0655e+01	$k_t = [4, 2, 2, 3]$			
Generation 870:	Best = 1.141573e+03	FES = 87100	$\mu_F = 0.533$	$\mu_{CR} = 0.030$
Diversity = 1.0646e+01	$k_t = [1, 1, 1, 1]$			
Generation 880:	Best = 1.123826e+03	FES = 88100	$\mu_F = 0.456$	$\mu_{CR} = 0.048$
Diversity = 1.0645e+01	$k_t = [4, 4, 4, 3]$			
Generation 890:	Best = 1.123799e+03	FES = 89100	$\mu_F = 0.438$	$\mu_{CR} = 0.036$
Diversity = 1.0642e+01	$k_t = [1, 1, 1, 1]$			
Generation 900:	Best = 1.122834e+03	FES = 90100	$\mu_F = 0.441$	$\mu_{CR} = 0.042$
Diversity = 1.0642e+01	$k_t = [4, 3, 2, 2]$			
Generation 910:	Best = 1.122834e+03	FES = 91100	$\mu_F = 0.424$	$\mu_{CR} = 0.036$
Diversity = 1.0636e+01	$k_t = [1, 1, 1, 1]$			
Generation 920:	Best = 1.122834e+03	FES = 92100	$\mu_F = 0.429$	$\mu_{CR} = 0.034$
Diversity = 1.0636e+01	$k_t = [3, 3, 2, 3]$			
Generation 930:	Best = 1.122262e+03	FES = 93100	$\mu_F = 0.376$	$\mu_{CR} = 0.035$
Diversity = 1.0632e+01	$k_t = [1, 1, 1, 1]$			
Generation 940:	Best = 1.122103e+03	FES = 94100	$\mu_F = 0.399$	$\mu_{CR} = 0.036$
Diversity = 1.0615e+01	$k_t = [4, 4, 3, 2]$			
Generation 950:	Best = 1.120825e+03	FES = 95100	$\mu_F = 0.315$	$\mu_{CR} = 0.043$
Diversity = 1.0615e+01	$k_t = [1, 1, 1, 1]$			
Generation 960:	Best = 1.119427e+03	FES = 96100	$\mu_F = 0.313$	$\mu_{CR} = 0.045$
Diversity = 1.0637e+01	$k_t = [4, 4, 2, 2]$			
Generation 970:	Best = 1.101102e+03	FES = 97100	$\mu_F = 0.291$	$\mu_{CR} = 0.046$
Diversity = 1.0633e+01	$k_t = [1, 1, 1, 1]$			
Generation 980:	Best = 1.101102e+03	FES = 98100	$\mu_F = 0.321$	$\mu_{CR} = 0.053$
Diversity = 1.0616e+01	$k_t = [3, 4, 3, 2]$			
Generation 990:	Best = 1.087296e+03	FES = 99100	$\mu_F = 0.306$	$\mu_{CR} = 0.044$
Diversity = 1.0610e+01	$k_t = [1, 1, 1, 1]$			
Generation 1000:	Best = 1.087296e+03	FES = 100100	$\mu_F = 0.319$	$\mu_{CR} = 0.030$
Diversity = 1.0604e+01	$k_t = [3, 4, 3, 2]$			
Generation 1010:	Best = 1.087075e+03	FES = 101100	$\mu_F = 0.365$	$\mu_{CR} = 0.032$
Diversity = 1.0601e+01	$k_t = [1, 1, 1, 1]$			
Generation 1020:	Best = 1.087075e+03	FES = 102100	$\mu_F = 0.370$	$\mu_{CR} = 0.033$
Diversity = 1.0600e+01	$k_t = [2, 3, 4, 2]$			
Generation 1030:	Best = 1.087075e+03	FES = 103100	$\mu_F = 0.360$	$\mu_{CR} = 0.042$
Diversity = 1.0594e+01	$k_t = [1, 1, 1, 1]$			

Generation 1040: Best = 1.087075e+03 FES = 104100 μ_F = 0.386 μ_{CR} = 0.032
 Diversity = 1.0592e+01 k_t = [4, 2, 4, 3]
 Generation 1050: Best = 1.087075e+03 FES = 105100 μ_F = 0.307 μ_{CR} = 0.037
 Diversity = 1.0574e+01 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 1.087075e+03 FES = 106100 μ_F = 0.356 μ_{CR} = 0.036
 Diversity = 1.0523e+01 k_t = [2, 2, 3, 3]
 Generation 1070: Best = 1.087075e+03 FES = 107100 μ_F = 0.308 μ_{CR} = 0.036
 Diversity = 1.0520e+01 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 1.087075e+03 FES = 108100 μ_F = 0.348 μ_{CR} = 0.029
 Diversity = 1.0555e+01 k_t = [2, 2, 3, 2]
 Generation 1090: Best = 1.077218e+03 FES = 109100 μ_F = 0.255 μ_{CR} = 0.042
 Diversity = 1.0549e+01 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 1.077218e+03 FES = 110100 μ_F = 0.217 μ_{CR} = 0.044
 Diversity = 1.0537e+01 k_t = [4, 2, 2, 3]
 Generation 1110: Best = 1.071332e+03 FES = 111100 μ_F = 0.252 μ_{CR} = 0.055
 Diversity = 1.0535e+01 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 1.068552e+03 FES = 112100 μ_F = 0.347 μ_{CR} = 0.056
 Diversity = 1.0549e+01 k_t = [2, 3, 4, 4]
 Generation 1130: Best = 1.068401e+03 FES = 113100 μ_F = 0.345 μ_{CR} = 0.047
 Diversity = 1.0543e+01 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 1.067359e+03 FES = 114100 μ_F = 0.361 μ_{CR} = 0.037
 Diversity = 1.0541e+01 k_t = [4, 4, 4, 3]
 Generation 1150: Best = 1.062425e+03 FES = 115100 μ_F = 0.335 μ_{CR} = 0.037
 Diversity = 1.0540e+01 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 1.062425e+03 FES = 116100 μ_F = 0.445 μ_{CR} = 0.030
 Diversity = 1.0540e+01 k_t = [2, 2, 2, 3]
 Generation 1170: Best = 1.062425e+03 FES = 117100 μ_F = 0.446 μ_{CR} = 0.026
 Diversity = 1.0539e+01 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 1.058394e+03 FES = 118100 μ_F = 0.418 μ_{CR} = 0.044
 Diversity = 1.0514e+01 k_t = [4, 4, 3, 3]
 Generation 1190: Best = 1.049699e+03 FES = 119100 μ_F = 0.428 μ_{CR} = 0.036
 Diversity = 1.0511e+01 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 1.049699e+03 FES = 120100 μ_F = 0.439 μ_{CR} = 0.031
 Diversity = 1.0470e+01 k_t = [2, 3, 3, 4]
 Generation 1210: Best = 1.049595e+03 FES = 121100 μ_F = 0.411 μ_{CR} = 0.033
 Diversity = 1.0473e+01 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 1.048306e+03 FES = 122100 μ_F = 0.452 μ_{CR} = 0.035
 Diversity = 1.0444e+01 k_t = [2, 3, 4, 3]
 Generation 1230: Best = 1.042716e+03 FES = 123100 μ_F = 0.392 μ_{CR} = 0.033
 Diversity = 1.0437e+01 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 1.042716e+03 FES = 124100 μ_F = 0.429 μ_{CR} = 0.030
 Diversity = 1.0435e+01 k_t = [4, 3, 3, 2]
 Generation 1250: Best = 1.042716e+03 FES = 125100 μ_F = 0.451 μ_{CR} = 0.029
 Diversity = 1.0431e+01 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 1.041473e+03 FES = 126100 μ_F = 0.332 μ_{CR} = 0.030
 Diversity = 1.0429e+01 k_t = [2, 4, 4, 2]
 Generation 1270: Best = 1.040788e+03 FES = 127100 μ_F = 0.316 μ_{CR} = 0.030
 Diversity = 1.0428e+01 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 1.027882e+03 FES = 128100 μ_F = 0.283 μ_{CR} = 0.054
 Diversity = 1.0392e+01 k_t = [4, 2, 4, 2]
 Generation 1290: Best = 1.027882e+03 FES = 129100 μ_F = 0.216 μ_{CR} = 0.069
 Diversity = 1.0383e+01 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 1.027882e+03 FES = 130100 μ_F = 0.217 μ_{CR} = 0.070
 Diversity = 1.0351e+01 k_t = [3, 4, 4, 2]

Generation 1310: Best = 1.027730e+03 FES = 131100 μ_F = 0.169 μ_{CR} = 0.061
 Diversity = 1.0347e+01 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 1.027730e+03 FES = 132100 μ_F = 0.161 μ_{CR} = 0.083
 Diversity = 1.0338e+01 k_t = [2, 4, 3, 2]
 Generation 1330: Best = 1.025373e+03 FES = 133100 μ_F = 0.207 μ_{CR} = 0.064
 Diversity = 1.0333e+01 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 1.019109e+03 FES = 134100 μ_F = 0.188 μ_{CR} = 0.056
 Diversity = 1.0329e+01 k_t = [4, 4, 2, 3]
 Generation 1350: Best = 1.019109e+03 FES = 135100 μ_F = 0.224 μ_{CR} = 0.045
 Diversity = 1.0324e+01 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 1.018517e+03 FES = 136100 μ_F = 0.269 μ_{CR} = 0.037
 Diversity = 1.0253e+01 k_t = [4, 2, 4, 4]
 Generation 1370: Best = 1.018517e+03 FES = 137100 μ_F = 0.263 μ_{CR} = 0.029
 Diversity = 1.0247e+01 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 1.017003e+03 FES = 138100 μ_F = 0.268 μ_{CR} = 0.038
 Diversity = 1.0242e+01 k_t = [2, 4, 4, 2]
 Generation 1390: Best = 1.017003e+03 FES = 139100 μ_F = 0.377 μ_{CR} = 0.057
 Diversity = 1.0241e+01 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 1.015999e+03 FES = 140100 μ_F = 0.435 μ_{CR} = 0.050
 Diversity = 1.0184e+01 k_t = [3, 3, 4, 4]
 Generation 1410: Best = 1.015999e+03 FES = 141100 μ_F = 0.498 μ_{CR} = 0.041
 Diversity = 1.0178e+01 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 1.015510e+03 FES = 142100 μ_F = 0.497 μ_{CR} = 0.035
 Diversity = 1.0125e+01 k_t = [3, 2, 3, 3]
 Generation 1430: Best = 1.015510e+03 FES = 143100 μ_F = 0.442 μ_{CR} = 0.025
 Diversity = 1.0125e+01 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 1.006988e+03 FES = 144100 μ_F = 0.455 μ_{CR} = 0.020
 Diversity = 1.0098e+01 k_t = [3, 3, 4, 4]
 Generation 1450: Best = 1.006988e+03 FES = 145100 μ_F = 0.405 μ_{CR} = 0.019
 Diversity = 1.0097e+01 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 1.006988e+03 FES = 146100 μ_F = 0.432 μ_{CR} = 0.030
 Diversity = 1.0118e+01 k_t = [3, 4, 2, 2]
 Generation 1470: Best = 1.006988e+03 FES = 147100 μ_F = 0.390 μ_{CR} = 0.030
 Diversity = 1.0113e+01 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 1.006988e+03 FES = 148100 μ_F = 0.363 μ_{CR} = 0.045
 Diversity = 1.0105e+01 k_t = [2, 3, 2, 4]
 Generation 1490: Best = 1.006988e+03 FES = 149100 μ_F = 0.350 μ_{CR} = 0.034
 Diversity = 1.0104e+01 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 1.004690e+03 FES = 150100 μ_F = 0.383 μ_{CR} = 0.038
 Diversity = 1.0065e+01 k_t = [4, 3, 2, 4]
 Generation 1510: Best = 1.002564e+03 FES = 151100 μ_F = 0.387 μ_{CR} = 0.043
 Diversity = 1.0065e+01 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 1.002078e+03 FES = 152100 μ_F = 0.363 μ_{CR} = 0.036
 Diversity = 1.0063e+01 k_t = [3, 2, 2, 2]
 Generation 1530: Best = 1.002078e+03 FES = 153100 μ_F = 0.475 μ_{CR} = 0.034
 Diversity = 1.0063e+01 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 9.951038e+02 FES = 154100 μ_F = 0.562 μ_{CR} = 0.038
 Diversity = 1.0063e+01 k_t = [4, 2, 4, 2]
 Generation 1550: Best = 9.951038e+02 FES = 155100 μ_F = 0.450 μ_{CR} = 0.040
 Diversity = 1.0031e+01 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 9.879639e+02 FES = 156100 μ_F = 0.453 μ_{CR} = 0.045
 Diversity = 1.0032e+01 k_t = [3, 4, 4, 3]
 Generation 1570: Best = 9.879639e+02 FES = 157100 μ_F = 0.387 μ_{CR} = 0.032
 Diversity = 1.0024e+01 k_t = [1, 1, 1, 1]

Generation 1580: Best = 9.879639e+02 FES = 158100 μ_F = 0.425 μ_{CR} = 0.039
 Diversity = 1.0023e+01 k_t = [4, 4, 4, 3]
 Generation 1590: Best = 9.873735e+02 FES = 159100 μ_F = 0.354 μ_{CR} = 0.038
 Diversity = 1.0020e+01 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 9.863719e+02 FES = 160100 μ_F = 0.387 μ_{CR} = 0.049
 Diversity = 1.0022e+01 k_t = [4, 4, 2, 3]
 Generation 1610: Best = 9.860468e+02 FES = 161100 μ_F = 0.455 μ_{CR} = 0.034
 Diversity = 1.0022e+01 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 9.848691e+02 FES = 162100 μ_F = 0.331 μ_{CR} = 0.051
 Diversity = 1.0060e+01 k_t = [3, 4, 4, 4]
 Generation 1630: Best = 9.824827e+02 FES = 163100 μ_F = 0.340 μ_{CR} = 0.041
 Diversity = 1.0059e+01 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 9.823313e+02 FES = 164100 μ_F = 0.230 μ_{CR} = 0.030
 Diversity = 1.0058e+01 k_t = [3, 3, 2, 2]
 Generation 1650: Best = 9.823313e+02 FES = 165100 μ_F = 0.231 μ_{CR} = 0.042
 Diversity = 1.0054e+01 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 9.804448e+02 FES = 166100 μ_F = 0.242 μ_{CR} = 0.049
 Diversity = 1.0051e+01 k_t = [4, 2, 4, 2]
 Generation 1670: Best = 9.804448e+02 FES = 167100 μ_F = 0.327 μ_{CR} = 0.054
 Diversity = 1.0061e+01 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 9.789547e+02 FES = 168100 μ_F = 0.350 μ_{CR} = 0.053
 Diversity = 1.0063e+01 k_t = [3, 3, 4, 4]
 Generation 1690: Best = 9.786104e+02 FES = 169100 μ_F = 0.314 μ_{CR} = 0.050
 Diversity = 1.0061e+01 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 9.786104e+02 FES = 170100 μ_F = 0.404 μ_{CR} = 0.042
 Diversity = 9.9484e+00 k_t = [2, 3, 4, 4]
 Generation 1710: Best = 9.786104e+02 FES = 171100 μ_F = 0.406 μ_{CR} = 0.042
 Diversity = 9.9494e+00 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 9.786104e+02 FES = 172100 μ_F = 0.432 μ_{CR} = 0.038
 Diversity = 9.9158e+00 k_t = [2, 4, 2, 4]
 Generation 1730: Best = 9.786104e+02 FES = 173100 μ_F = 0.388 μ_{CR} = 0.037
 Diversity = 9.9144e+00 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 9.768979e+02 FES = 174100 μ_F = 0.388 μ_{CR} = 0.046
 Diversity = 9.9150e+00 k_t = [4, 4, 3, 3]
 Generation 1750: Best = 9.757636e+02 FES = 175100 μ_F = 0.400 μ_{CR} = 0.044
 Diversity = 9.9164e+00 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 9.751204e+02 FES = 176100 μ_F = 0.370 μ_{CR} = 0.037
 Diversity = 9.8442e+00 k_t = [2, 3, 2, 4]
 Generation 1770: Best = 9.743607e+02 FES = 177100 μ_F = 0.437 μ_{CR} = 0.038
 Diversity = 9.8110e+00 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 9.721549e+02 FES = 178100 μ_F = 0.420 μ_{CR} = 0.035
 Diversity = 9.8025e+00 k_t = [4, 3, 3, 2]
 Generation 1790: Best = 9.503089e+02 FES = 179100 μ_F = 0.425 μ_{CR} = 0.031
 Diversity = 9.7961e+00 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 9.503089e+02 FES = 180100 μ_F = 0.358 μ_{CR} = 0.032
 Diversity = 9.7955e+00 k_t = [3, 2, 2, 2]
 Generation 1810: Best = 9.484904e+02 FES = 181100 μ_F = 0.402 μ_{CR} = 0.034
 Diversity = 9.7916e+00 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 9.433424e+02 FES = 182100 μ_F = 0.461 μ_{CR} = 0.040
 Diversity = 9.7762e+00 k_t = [4, 2, 4, 4]
 Generation 1830: Best = 9.433424e+02 FES = 183100 μ_F = 0.420 μ_{CR} = 0.044
 Diversity = 9.7771e+00 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 9.380865e+02 FES = 184100 μ_F = 0.422 μ_{CR} = 0.052
 Diversity = 9.7715e+00 k_t = [4, 2, 4, 4]

Generation 1850: Best = 9.357928e+02 FES = 185100 μ_F = 0.374 μ_{CR} = 0.045
 Diversity = 9.7701e+00 k_t = [1, 1, 1, 1]
 Generation 1860: Best = 9.357928e+02 FES = 186100 μ_F = 0.478 μ_{CR} = 0.035
 Diversity = 9.7294e+00 k_t = [2, 3, 3, 2]
 Generation 1870: Best = 9.354847e+02 FES = 187100 μ_F = 0.399 μ_{CR} = 0.031
 Diversity = 9.7294e+00 k_t = [1, 1, 1, 1]
 Generation 1880: Best = 9.354847e+02 FES = 188100 μ_F = 0.399 μ_{CR} = 0.035
 Diversity = 9.7288e+00 k_t = [2, 3, 2, 2]
 Generation 1890: Best = 9.354847e+02 FES = 189100 μ_F = 0.418 μ_{CR} = 0.043
 Diversity = 9.7279e+00 k_t = [1, 1, 1, 1]
 Generation 1900: Best = 9.354847e+02 FES = 190100 μ_F = 0.417 μ_{CR} = 0.038
 Diversity = 9.7254e+00 k_t = [4, 2, 4, 2]
 Generation 1910: Best = 9.354847e+02 FES = 191100 μ_F = 0.419 μ_{CR} = 0.037
 Diversity = 9.7000e+00 k_t = [1, 1, 1, 1]
 Generation 1920: Best = 9.323225e+02 FES = 192100 μ_F = 0.370 μ_{CR} = 0.050
 Diversity = 9.6983e+00 k_t = [2, 4, 3, 2]
 Generation 1930: Best = 9.320953e+02 FES = 193100 μ_F = 0.316 μ_{CR} = 0.040
 Diversity = 9.6993e+00 k_t = [1, 1, 1, 1]
 Generation 1940: Best = 9.320953e+02 FES = 194100 μ_F = 0.329 μ_{CR} = 0.034
 Diversity = 9.7016e+00 k_t = [2, 2, 3, 3]
 Generation 1950: Best = 9.320953e+02 FES = 195100 μ_F = 0.405 μ_{CR} = 0.030
 Diversity = 9.7020e+00 k_t = [1, 1, 1, 1]
 Generation 1960: Best = 9.266529e+02 FES = 196100 μ_F = 0.381 μ_{CR} = 0.029
 Diversity = 9.7018e+00 k_t = [2, 3, 3, 2]
 Generation 1970: Best = 9.266529e+02 FES = 197100 μ_F = 0.334 μ_{CR} = 0.027
 Diversity = 9.7026e+00 k_t = [1, 1, 1, 1]
 Generation 1980: Best = 9.253267e+02 FES = 198100 μ_F = 0.382 μ_{CR} = 0.028
 Diversity = 9.6626e+00 k_t = [4, 2, 4, 4]
 Generation 1990: Best = 9.197357e+02 FES = 199100 μ_F = 0.357 μ_{CR} = 0.037
 Diversity = 9.6431e+00 k_t = [1, 1, 1, 1]
 Generation 2000: Best = 9.174797e+02 FES = 200100 μ_F = 0.302 μ_{CR} = 0.052
 Diversity = 9.6326e+00 k_t = [4, 3, 2, 4]
 Generation 2010: Best = 9.153236e+02 FES = 201100 μ_F = 0.232 μ_{CR} = 0.055
 Diversity = 9.6313e+00 k_t = [1, 1, 1, 1]
 Generation 2020: Best = 9.131121e+02 FES = 202100 μ_F = 0.256 μ_{CR} = 0.044
 Diversity = 9.6302e+00 k_t = [4, 4, 4, 3]
 Generation 2030: Best = 9.131121e+02 FES = 203100 μ_F = 0.301 μ_{CR} = 0.037
 Diversity = 9.6298e+00 k_t = [1, 1, 1, 1]
 Generation 2040: Best = 9.131121e+02 FES = 204100 μ_F = 0.353 μ_{CR} = 0.050
 Diversity = 9.6429e+00 k_t = [3, 4, 3, 3]
 Generation 2050: Best = 9.131121e+02 FES = 205100 μ_F = 0.296 μ_{CR} = 0.048
 Diversity = 9.6446e+00 k_t = [1, 1, 1, 1]
 Generation 2060: Best = 9.123381e+02 FES = 206100 μ_F = 0.412 μ_{CR} = 0.033
 Diversity = 9.6218e+00 k_t = [3, 2, 2, 3]
 Generation 2070: Best = 9.123381e+02 FES = 207100 μ_F = 0.375 μ_{CR} = 0.038
 Diversity = 9.6084e+00 k_t = [1, 1, 1, 1]
 Generation 2080: Best = 9.118918e+02 FES = 208100 μ_F = 0.423 μ_{CR} = 0.037
 Diversity = 9.6079e+00 k_t = [3, 2, 4, 3]
 Generation 2090: Best = 9.118918e+02 FES = 209100 μ_F = 0.443 μ_{CR} = 0.044
 Diversity = 9.6059e+00 k_t = [1, 1, 1, 1]
 Generation 2100: Best = 9.118918e+02 FES = 210100 μ_F = 0.379 μ_{CR} = 0.048
 Diversity = 9.6027e+00 k_t = [3, 4, 3, 2]
 Generation 2110: Best = 9.100204e+02 FES = 211100 μ_F = 0.433 μ_{CR} = 0.043
 Diversity = 9.6068e+00 k_t = [1, 1, 1, 1]

Generation 2120: Best = 9.089666e+02 FES = 212100 μ_F = 0.395 μ_{CR} = 0.039
 Diversity = 9.6051e+00 k_t = [4, 4, 4, 2]
 Generation 2130: Best = 9.070727e+02 FES = 213100 μ_F = 0.424 μ_{CR} = 0.031
 Diversity = 9.6068e+00 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 9.070727e+02 FES = 214100 μ_F = 0.493 μ_{CR} = 0.045
 Diversity = 9.6054e+00 k_t = [4, 3, 2, 2]
 Generation 2150: Best = 9.070727e+02 FES = 215100 μ_F = 0.444 μ_{CR} = 0.049
 Diversity = 9.6065e+00 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 9.070727e+02 FES = 216100 μ_F = 0.478 μ_{CR} = 0.049
 Diversity = 9.5489e+00 k_t = [3, 3, 4, 4]
 Generation 2170: Best = 9.070727e+02 FES = 217100 μ_F = 0.523 μ_{CR} = 0.051
 Diversity = 9.5441e+00 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 9.044470e+02 FES = 218100 μ_F = 0.586 μ_{CR} = 0.051
 Diversity = 9.5305e+00 k_t = [2, 4, 4, 3]
 Generation 2190: Best = 9.039943e+02 FES = 219100 μ_F = 0.556 μ_{CR} = 0.058
 Diversity = 9.5281e+00 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 9.032072e+02 FES = 220100 μ_F = 0.551 μ_{CR} = 0.051
 Diversity = 9.4700e+00 k_t = [3, 2, 4, 4]
 Generation 2210: Best = 8.948765e+02 FES = 221100 μ_F = 0.588 μ_{CR} = 0.060
 Diversity = 9.4236e+00 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 8.930666e+02 FES = 222100 μ_F = 0.582 μ_{CR} = 0.050
 Diversity = 9.4152e+00 k_t = [4, 3, 2, 2]
 Generation 2230: Best = 8.925384e+02 FES = 223100 μ_F = 0.611 μ_{CR} = 0.058
 Diversity = 9.4177e+00 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 8.874522e+02 FES = 224100 μ_F = 0.562 μ_{CR} = 0.067
 Diversity = 9.4118e+00 k_t = [4, 3, 3, 3]
 Generation 2250: Best = 8.874522e+02 FES = 225100 μ_F = 0.608 μ_{CR} = 0.058
 Diversity = 9.3809e+00 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 8.874522e+02 FES = 226100 μ_F = 0.617 μ_{CR} = 0.047
 Diversity = 9.3804e+00 k_t = [2, 2, 2, 3]
 Generation 2270: Best = 8.874522e+02 FES = 227100 μ_F = 0.633 μ_{CR} = 0.053
 Diversity = 9.3800e+00 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 8.872371e+02 FES = 228100 μ_F = 0.595 μ_{CR} = 0.066
 Diversity = 9.3831e+00 k_t = [3, 3, 4, 3]
 Generation 2290: Best = 8.832948e+02 FES = 229100 μ_F = 0.577 μ_{CR} = 0.079
 Diversity = 9.3532e+00 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 8.805172e+02 FES = 230100 μ_F = 0.607 μ_{CR} = 0.080
 Diversity = 9.3542e+00 k_t = [3, 3, 2, 2]
 Generation 2310: Best = 8.805172e+02 FES = 231100 μ_F = 0.643 μ_{CR} = 0.106
 Diversity = 9.3554e+00 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 8.790151e+02 FES = 232100 μ_F = 0.606 μ_{CR} = 0.095
 Diversity = 9.3549e+00 k_t = [2, 4, 4, 2]
 Generation 2330: Best = 8.779712e+02 FES = 233100 μ_F = 0.530 μ_{CR} = 0.082
 Diversity = 9.3545e+00 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 8.753491e+02 FES = 234100 μ_F = 0.456 μ_{CR} = 0.070
 Diversity = 9.3539e+00 k_t = [3, 3, 3, 2]
 Generation 2350: Best = 8.742746e+02 FES = 235100 μ_F = 0.485 μ_{CR} = 0.079
 Diversity = 9.3530e+00 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 8.742746e+02 FES = 236100 μ_F = 0.511 μ_{CR} = 0.072
 Diversity = 9.3471e+00 k_t = [4, 2, 2, 3]
 Generation 2370: Best = 8.737462e+02 FES = 237100 μ_F = 0.579 μ_{CR} = 0.092
 Diversity = 9.3256e+00 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 8.736804e+02 FES = 238100 μ_F = 0.519 μ_{CR} = 0.085
 Diversity = 9.3297e+00 k_t = [2, 2, 4, 4]

Generation 2390: Best = 8.728674e+02 FES = 239100 μ_F = 0.447 μ_{CR} = 0.086
 Diversity = 9.3296e+00 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 8.728674e+02 FES = 240100 μ_F = 0.447 μ_{CR} = 0.067
 Diversity = 9.2726e+00 k_t = [3, 3, 4, 4]
 Generation 2410: Best = 8.703081e+02 FES = 241100 μ_F = 0.425 μ_{CR} = 0.080
 Diversity = 9.2556e+00 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 8.703081e+02 FES = 242100 μ_F = 0.477 μ_{CR} = 0.068
 Diversity = 9.1689e+00 k_t = [2, 2, 2, 4]
 Generation 2430: Best = 8.698372e+02 FES = 243100 μ_F = 0.496 μ_{CR} = 0.101
 Diversity = 9.1503e+00 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 8.698372e+02 FES = 244100 μ_F = 0.423 μ_{CR} = 0.114
 Diversity = 9.1486e+00 k_t = [2, 3, 3, 3]
 Generation 2450: Best = 8.689959e+02 FES = 245100 μ_F = 0.317 μ_{CR} = 0.098
 Diversity = 9.1480e+00 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 8.681479e+02 FES = 246100 μ_F = 0.397 μ_{CR} = 0.089
 Diversity = 9.1140e+00 k_t = [2, 4, 4, 4]
 Generation 2470: Best = 8.681479e+02 FES = 247100 μ_F = 0.436 μ_{CR} = 0.079
 Diversity = 9.0770e+00 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 8.675680e+02 FES = 248100 μ_F = 0.439 μ_{CR} = 0.067
 Diversity = 9.0791e+00 k_t = [4, 2, 3, 3]
 Generation 2490: Best = 8.674662e+02 FES = 249100 μ_F = 0.512 μ_{CR} = 0.092
 Diversity = 9.0795e+00 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 8.663551e+02 FES = 250100 μ_F = 0.531 μ_{CR} = 0.084
 Diversity = 9.0773e+00 k_t = [4, 4, 3, 2]
 Generation 2510: Best = 8.663551e+02 FES = 251100 μ_F = 0.508 μ_{CR} = 0.079
 Diversity = 9.0495e+00 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 8.660991e+02 FES = 252100 μ_F = 0.493 μ_{CR} = 0.080
 Diversity = 8.9670e+00 k_t = [3, 2, 4, 4]
 Generation 2530: Best = 8.654440e+02 FES = 253100 μ_F = 0.541 μ_{CR} = 0.075
 Diversity = 8.9302e+00 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 8.654416e+02 FES = 254100 μ_F = 0.490 μ_{CR} = 0.069
 Diversity = 8.9313e+00 k_t = [3, 4, 2, 3]
 Generation 2550: Best = 8.654416e+02 FES = 255100 μ_F = 0.479 μ_{CR} = 0.055
 Diversity = 8.9321e+00 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 8.648731e+02 FES = 256100 μ_F = 0.435 μ_{CR} = 0.070
 Diversity = 8.9313e+00 k_t = [3, 2, 2, 3]
 Generation 2570: Best = 8.648731e+02 FES = 257100 μ_F = 0.332 μ_{CR} = 0.054
 Diversity = 8.9357e+00 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 8.648482e+02 FES = 258100 μ_F = 0.358 μ_{CR} = 0.053
 Diversity = 8.9336e+00 k_t = [2, 3, 2, 4]
 Generation 2590: Best = 8.648072e+02 FES = 259100 μ_F = 0.340 μ_{CR} = 0.080
 Diversity = 8.9302e+00 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 8.643440e+02 FES = 260100 μ_F = 0.288 μ_{CR} = 0.067
 Diversity = 8.9293e+00 k_t = [3, 2, 2, 3]
 Generation 2610: Best = 8.641480e+02 FES = 261100 μ_F = 0.275 μ_{CR} = 0.049
 Diversity = 8.9279e+00 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 8.639054e+02 FES = 262100 μ_F = 0.255 μ_{CR} = 0.057
 Diversity = 8.9260e+00 k_t = [2, 2, 4, 3]
 Generation 2630: Best = 8.629472e+02 FES = 263100 μ_F = 0.365 μ_{CR} = 0.053
 Diversity = 8.9036e+00 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 8.617737e+02 FES = 264100 μ_F = 0.388 μ_{CR} = 0.063
 Diversity = 8.9048e+00 k_t = [4, 3, 2, 2]
 Generation 2650: Best = 8.617737e+02 FES = 265100 μ_F = 0.369 μ_{CR} = 0.054
 Diversity = 8.9050e+00 k_t = [1, 1, 1, 1]

Generation 2660: Best = 8.608547e+02 FES = 266100 μ_F = 0.294 μ_{CR} = 0.065
 Diversity = 8.9039e+00 k_t = [4, 4, 4, 3]
 Generation 2670: Best = 8.605789e+02 FES = 267100 μ_F = 0.294 μ_{CR} = 0.068
 Diversity = 8.9068e+00 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 8.604537e+02 FES = 268100 μ_F = 0.385 μ_{CR} = 0.080
 Diversity = 8.8441e+00 k_t = [4, 4, 3, 3]
 Generation 2690: Best = 8.604537e+02 FES = 269100 μ_F = 0.326 μ_{CR} = 0.115
 Diversity = 8.8132e+00 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 8.604537e+02 FES = 270100 μ_F = 0.291 μ_{CR} = 0.086
 Diversity = 8.8113e+00 k_t = [2, 2, 3, 3]
 Generation 2710: Best = 8.599700e+02 FES = 271100 μ_F = 0.340 μ_{CR} = 0.072
 Diversity = 8.8107e+00 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 8.599545e+02 FES = 272100 μ_F = 0.304 μ_{CR} = 0.056
 Diversity = 8.8098e+00 k_t = [3, 2, 3, 4]
 Generation 2730: Best = 8.599545e+02 FES = 273100 μ_F = 0.385 μ_{CR} = 0.082
 Diversity = 8.8101e+00 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 8.599545e+02 FES = 274100 μ_F = 0.384 μ_{CR} = 0.066
 Diversity = 8.8102e+00 k_t = [2, 4, 4, 3]
 Generation 2750: Best = 8.595368e+02 FES = 275100 μ_F = 0.326 μ_{CR} = 0.070
 Diversity = 8.8106e+00 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 8.593965e+02 FES = 276100 μ_F = 0.369 μ_{CR} = 0.071
 Diversity = 8.7273e+00 k_t = [3, 2, 4, 2]
 Generation 2770: Best = 8.582528e+02 FES = 277100 μ_F = 0.342 μ_{CR} = 0.074
 Diversity = 8.7274e+00 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 8.580547e+02 FES = 278100 μ_F = 0.303 μ_{CR} = 0.064
 Diversity = 8.7266e+00 k_t = [2, 2, 2, 3]
 Generation 2790: Best = 8.580547e+02 FES = 279100 μ_F = 0.408 μ_{CR} = 0.086
 Diversity = 8.7183e+00 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 8.579672e+02 FES = 280100 μ_F = 0.408 μ_{CR} = 0.090
 Diversity = 8.7172e+00 k_t = [2, 4, 3, 2]
 Generation 2810: Best = 8.579672e+02 FES = 281100 μ_F = 0.317 μ_{CR} = 0.093
 Diversity = 8.7204e+00 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 8.579672e+02 FES = 282100 μ_F = 0.389 μ_{CR} = 0.096
 Diversity = 8.6938e+00 k_t = [4, 4, 4, 3]
 Generation 2830: Best = 8.577275e+02 FES = 283100 μ_F = 0.363 μ_{CR} = 0.100
 Diversity = 8.6927e+00 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 8.563924e+02 FES = 284100 μ_F = 0.466 μ_{CR} = 0.104
 Diversity = 8.6625e+00 k_t = [2, 3, 2, 4]
 Generation 2850: Best = 8.560526e+02 FES = 285100 μ_F = 0.511 μ_{CR} = 0.113
 Diversity = 8.6280e+00 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 8.546850e+02 FES = 286100 μ_F = 0.468 μ_{CR} = 0.080
 Diversity = 8.4699e+00 k_t = [4, 2, 2, 4]
 Generation 2870: Best = 8.538269e+02 FES = 287100 μ_F = 0.493 μ_{CR} = 0.093
 Diversity = 8.3683e+00 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 8.538269e+02 FES = 288100 μ_F = 0.531 μ_{CR} = 0.117
 Diversity = 8.3720e+00 k_t = [4, 2, 2, 4]
 Generation 2890: Best = 8.536757e+02 FES = 289100 μ_F = 0.619 μ_{CR} = 0.134
 Diversity = 8.2317e+00 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 8.521594e+02 FES = 290100 μ_F = 0.648 μ_{CR} = 0.162
 Diversity = 8.1991e+00 k_t = [4, 2, 2, 3]
 Generation 2910: Best = 8.521416e+02 FES = 291100 μ_F = 0.648 μ_{CR} = 0.149
 Diversity = 8.1204e+00 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 8.515662e+02 FES = 292100 μ_F = 0.670 μ_{CR} = 0.146
 Diversity = 8.0148e+00 k_t = [2, 3, 4, 3]

Generation 2930: Best = 8.500246e+02 FES = 293100 μ_F = 0.656 μ_{CR} = 0.170
 Diversity = 7.9368e+00 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 8.481173e+02 FES = 294100 μ_F = 0.670 μ_{CR} = 0.165
 Diversity = 7.7385e+00 k_t = [2, 2, 4, 4]
 Generation 2950: Best = 8.474933e+02 FES = 295100 μ_F = 0.697 μ_{CR} = 0.170
 Diversity = 7.6636e+00 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 8.459137e+02 FES = 296100 μ_F = 0.696 μ_{CR} = 0.176
 Diversity = 7.4142e+00 k_t = [3, 4, 3, 4]
 Generation 2970: Best = 8.456077e+02 FES = 297100 μ_F = 0.717 μ_{CR} = 0.179
 Diversity = 6.8910e+00 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 8.446421e+02 FES = 298100 μ_F = 0.754 μ_{CR} = 0.181
 Diversity = 5.8234e+00 k_t = [3, 4, 3, 2]
 Generation 2990: Best = 8.442071e+02 FES = 299100 μ_F = 0.769 μ_{CR} = 0.193
 Diversity = 3.8973e+00 k_t = [1, 1, 1, 1]
 Generation 3000: Best = 8.439364e+02 FES = 300100 μ_F = 0.775 μ_{CR} = 0.194
 Diversity = 1.5349e+00 k_t = [2, 4, 2, 4]
 Generation 3010: Best = 8.431447e+02 FES = 301100 μ_F = 0.762 μ_{CR} = 0.190
 Diversity = 1.8803e-01 k_t = [1, 1, 1, 1]
 Generation 3020: Best = 8.426592e+02 FES = 302100 μ_F = 0.735 μ_{CR} = 0.186
 Diversity = 1.2820e-01 k_t = [2, 4, 3, 3]
 Generation 3030: Best = 8.423529e+02 FES = 303100 μ_F = 0.703 μ_{CR} = 0.186
 Diversity = 1.0647e-01 k_t = [1, 1, 1, 1]
 Generation 3040: Best = 8.416851e+02 FES = 304100 μ_F = 0.680 μ_{CR} = 0.186
 Diversity = 9.1199e-02 k_t = [2, 3, 2, 2]
 Generation 3050: Best = 8.415010e+02 FES = 305100 μ_F = 0.676 μ_{CR} = 0.195
 Diversity = 8.1502e-02 k_t = [1, 1, 1, 1]
 Generation 3060: Best = 8.411788e+02 FES = 306100 μ_F = 0.679 μ_{CR} = 0.194
 Diversity = 7.0764e-02 k_t = [4, 4, 2, 2]
 Generation 3070: Best = 8.409723e+02 FES = 307100 μ_F = 0.665 μ_{CR} = 0.202
 Diversity = 5.7387e-02 k_t = [1, 1, 1, 1]
 Generation 3080: Best = 8.407061e+02 FES = 308100 μ_F = 0.660 μ_{CR} = 0.208
 Diversity = 4.7499e-02 k_t = [3, 2, 3, 4]
 Generation 3090: Best = 8.405823e+02 FES = 309100 μ_F = 0.674 μ_{CR} = 0.207
 Diversity = 4.1207e-02 k_t = [1, 1, 1, 1]
 Generation 3100: Best = 8.404157e+02 FES = 310100 μ_F = 0.684 μ_{CR} = 0.217
 Diversity = 3.6616e-02 k_t = [4, 4, 2, 2]
 Generation 3110: Best = 8.403261e+02 FES = 311100 μ_F = 0.681 μ_{CR} = 0.216
 Diversity = 3.4099e-02 k_t = [1, 1, 1, 1]
 Generation 3120: Best = 8.402410e+02 FES = 312100 μ_F = 0.689 μ_{CR} = 0.219
 Diversity = 2.9752e-02 k_t = [4, 3, 3, 3]
 Generation 3130: Best = 8.401721e+02 FES = 313100 μ_F = 0.694 μ_{CR} = 0.220
 Diversity = 2.7079e-02 k_t = [1, 1, 1, 1]
 Generation 3140: Best = 8.401084e+02 FES = 314100 μ_F = 0.683 μ_{CR} = 0.223
 Diversity = 2.3317e-02 k_t = [3, 3, 4, 4]
 Generation 3150: Best = 8.400455e+02 FES = 315100 μ_F = 0.691 μ_{CR} = 0.224
 Diversity = 2.1982e-02 k_t = [1, 1, 1, 1]
 Generation 3160: Best = 8.399867e+02 FES = 316100 μ_F = 0.702 μ_{CR} = 0.222
 Diversity = 2.0629e-02 k_t = [2, 3, 2, 3]
 Generation 3170: Best = 8.399260e+02 FES = 317100 μ_F = 0.709 μ_{CR} = 0.223
 Diversity = 2.1765e-02 k_t = [1, 1, 1, 1]
 Generation 3180: Best = 8.398952e+02 FES = 318100 μ_F = 0.696 μ_{CR} = 0.229
 Diversity = 1.9761e-02 k_t = [3, 2, 3, 3]
 Generation 3190: Best = 8.398442e+02 FES = 319100 μ_F = 0.697 μ_{CR} = 0.233
 Diversity = 2.0589e-02 k_t = [1, 1, 1, 1]

Generation 3200: Best = 8.398047e+02 FES = 320100 μ_F = 0.700 μ_{CR} = 0.229
 Diversity = 1.8476e-02 k_t = [3, 4, 3, 4]
 Generation 3210: Best = 8.397574e+02 FES = 321100 μ_F = 0.699 μ_{CR} = 0.238
 Diversity = 1.7517e-02 k_t = [1, 1, 1, 1]
 Generation 3220: Best = 8.397099e+02 FES = 322100 μ_F = 0.712 μ_{CR} = 0.243
 Diversity = 1.8231e-02 k_t = [4, 2, 4, 2]
 Generation 3230: Best = 8.396835e+02 FES = 323100 μ_F = 0.714 μ_{CR} = 0.234
 Diversity = 1.7590e-02 k_t = [1, 1, 1, 1]
 Generation 3240: Best = 8.396578e+02 FES = 324100 μ_F = 0.715 μ_{CR} = 0.220
 Diversity = 1.5955e-02 k_t = [2, 4, 4, 4]
 Generation 3250: Best = 8.396345e+02 FES = 325100 μ_F = 0.725 μ_{CR} = 0.216
 Diversity = 1.5530e-02 k_t = [1, 1, 1, 1]
 Generation 3260: Best = 8.396126e+02 FES = 326100 μ_F = 0.718 μ_{CR} = 0.215
 Diversity = 1.3533e-02 k_t = [4, 4, 4, 3]
 Generation 3270: Best = 8.395928e+02 FES = 327100 μ_F = 0.723 μ_{CR} = 0.214
 Diversity = 1.3265e-02 k_t = [1, 1, 1, 1]
 Generation 3280: Best = 8.395743e+02 FES = 328100 μ_F = 0.726 μ_{CR} = 0.210
 Diversity = 1.1880e-02 k_t = [3, 2, 3, 4]
 Generation 3290: Best = 8.395645e+02 FES = 329100 μ_F = 0.726 μ_{CR} = 0.206
 Diversity = 1.1174e-02 k_t = [1, 1, 1, 1]
 Generation 3300: Best = 8.395495e+02 FES = 330100 μ_F = 0.748 μ_{CR} = 0.222
 Diversity = 1.1090e-02 k_t = [3, 3, 4, 2]
 Generation 3310: Best = 8.395399e+02 FES = 331100 μ_F = 0.730 μ_{CR} = 0.216
 Diversity = 1.1044e-02 k_t = [1, 1, 1, 1]
 Generation 3320: Best = 8.395234e+02 FES = 332100 μ_F = 0.737 μ_{CR} = 0.214
 Diversity = 1.0496e-02 k_t = [2, 2, 2, 3]
 Generation 3330: Best = 8.395101e+02 FES = 333100 μ_F = 0.727 μ_{CR} = 0.217
 Diversity = 1.0623e-02 k_t = [1, 1, 1, 1]
 Generation 3340: Best = 8.394987e+02 FES = 334100 μ_F = 0.717 μ_{CR} = 0.221
 Diversity = 1.0333e-02 k_t = [2, 2, 3, 3]
 Generation 3350: Best = 8.394812e+02 FES = 335100 μ_F = 0.723 μ_{CR} = 0.226
 Diversity = 1.1135e-02 k_t = [1, 1, 1, 1]
 Generation 3360: Best = 8.394703e+02 FES = 336100 μ_F = 0.719 μ_{CR} = 0.231
 Diversity = 1.0855e-02 k_t = [3, 2, 3, 3]
 Generation 3370: Best = 8.394584e+02 FES = 337100 μ_F = 0.709 μ_{CR} = 0.230
 Diversity = 9.7997e-03 k_t = [1, 1, 1, 1]
 Generation 3380: Best = 8.394513e+02 FES = 338100 μ_F = 0.719 μ_{CR} = 0.232
 Diversity = 9.0579e-03 k_t = [2, 2, 2, 3]
 Generation 3390: Best = 8.394376e+02 FES = 339100 μ_F = 0.724 μ_{CR} = 0.227
 Diversity = 9.3985e-03 k_t = [1, 1, 1, 1]
 Generation 3400: Best = 8.394289e+02 FES = 340100 μ_F = 0.712 μ_{CR} = 0.229
 Diversity = 8.8342e-03 k_t = [2, 2, 4, 3]
 Generation 3410: Best = 8.394229e+02 FES = 341100 μ_F = 0.709 μ_{CR} = 0.228
 Diversity = 8.2710e-03 k_t = [1, 1, 1, 1]
 Generation 3420: Best = 8.394155e+02 FES = 342100 μ_F = 0.708 μ_{CR} = 0.228
 Diversity = 7.7428e-03 k_t = [4, 4, 3, 2]
 Generation 3430: Best = 8.394083e+02 FES = 343100 μ_F = 0.716 μ_{CR} = 0.230
 Diversity = 7.5766e-03 k_t = [1, 1, 1, 1]
 Generation 3440: Best = 8.393996e+02 FES = 344100 μ_F = 0.709 μ_{CR} = 0.235
 Diversity = 7.2318e-03 k_t = [2, 4, 3, 2]
 Generation 3450: Best = 8.393950e+02 FES = 345100 μ_F = 0.702 μ_{CR} = 0.217
 Diversity = 6.9819e-03 k_t = [1, 1, 1, 1]
 Generation 3460: Best = 8.393911e+02 FES = 346100 μ_F = 0.710 μ_{CR} = 0.215
 Diversity = 6.3407e-03 k_t = [3, 4, 4, 2]

Generation 3470: Best = 8.393848e+02 FES = 347100 μ_F = 0.712 μ_{CR} = 0.224
 Diversity = 6.3085e-03 k_t = [1, 1, 1, 1]
 Generation 3480: Best = 8.393789e+02 FES = 348100 μ_F = 0.714 μ_{CR} = 0.218
 Diversity = 5.9194e-03 k_t = [3, 3, 4, 2]
 Generation 3490: Best = 8.393745e+02 FES = 349100 μ_F = 0.728 μ_{CR} = 0.211
 Diversity = 6.1218e-03 k_t = [1, 1, 1, 1]
 Generation 3500: Best = 8.393700e+02 FES = 350100 μ_F = 0.723 μ_{CR} = 0.223
 Diversity = 5.8446e-03 k_t = [4, 2, 2, 2]
 Generation 3510: Best = 8.393658e+02 FES = 351100 μ_F = 0.716 μ_{CR} = 0.216
 Diversity = 5.9403e-03 k_t = [1, 1, 1, 1]
 Generation 3520: Best = 8.393588e+02 FES = 352100 μ_F = 0.724 μ_{CR} = 0.219
 Diversity = 6.1101e-03 k_t = [4, 2, 2, 3]
 Generation 3530: Best = 8.393553e+02 FES = 353100 μ_F = 0.727 μ_{CR} = 0.225
 Diversity = 6.0736e-03 k_t = [1, 1, 1, 1]
 Generation 3540: Best = 8.393529e+02 FES = 354100 μ_F = 0.733 μ_{CR} = 0.230
 Diversity = 5.6841e-03 k_t = [4, 3, 2, 3]
 Generation 3550: Best = 8.393484e+02 FES = 355100 μ_F = 0.731 μ_{CR} = 0.235
 Diversity = 5.4566e-03 k_t = [1, 1, 1, 1]
 Generation 3560: Best = 8.393450e+02 FES = 356100 μ_F = 0.729 μ_{CR} = 0.231
 Diversity = 5.4469e-03 k_t = [2, 3, 3, 2]
 Generation 3570: Best = 8.393410e+02 FES = 357100 μ_F = 0.731 μ_{CR} = 0.222
 Diversity = 5.4093e-03 k_t = [1, 1, 1, 1]
 Generation 3580: Best = 8.393385e+02 FES = 358100 μ_F = 0.720 μ_{CR} = 0.221
 Diversity = 5.2047e-03 k_t = [4, 2, 2, 4]
 Generation 3590: Best = 8.393366e+02 FES = 359100 μ_F = 0.728 μ_{CR} = 0.227
 Diversity = 4.9115e-03 k_t = [1, 1, 1, 1]
 Generation 3600: Best = 8.393331e+02 FES = 360100 μ_F = 0.728 μ_{CR} = 0.231
 Diversity = 4.8625e-03 k_t = [3, 2, 4, 3]
 Generation 3610: Best = 8.393295e+02 FES = 361100 μ_F = 0.721 μ_{CR} = 0.232
 Diversity = 4.9248e-03 k_t = [1, 1, 1, 1]
 Generation 3620: Best = 8.393266e+02 FES = 362100 μ_F = 0.734 μ_{CR} = 0.233
 Diversity = 4.4829e-03 k_t = [4, 2, 2, 4]
 Generation 3630: Best = 8.393232e+02 FES = 363100 μ_F = 0.731 μ_{CR} = 0.226
 Diversity = 4.5460e-03 k_t = [1, 1, 1, 1]
 Generation 3640: Best = 8.393212e+02 FES = 364100 μ_F = 0.727 μ_{CR} = 0.230
 Diversity = 4.0673e-03 k_t = [2, 4, 4, 4]
 Generation 3650: Best = 8.393192e+02 FES = 365100 μ_F = 0.728 μ_{CR} = 0.231
 Diversity = 4.0809e-03 k_t = [1, 1, 1, 1]
 Generation 3660: Best = 8.393176e+02 FES = 366100 μ_F = 0.721 μ_{CR} = 0.235
 Diversity = 3.6156e-03 k_t = [3, 2, 2, 4]
 Generation 3670: Best = 8.393163e+02 FES = 367100 μ_F = 0.715 μ_{CR} = 0.224
 Diversity = 3.5257e-03 k_t = [1, 1, 1, 1]
 Generation 3680: Best = 8.393145e+02 FES = 368100 μ_F = 0.707 μ_{CR} = 0.230
 Diversity = 3.2205e-03 k_t = [4, 3, 3, 3]
 Generation 3690: Best = 8.393135e+02 FES = 369100 μ_F = 0.698 μ_{CR} = 0.230
 Diversity = 2.9038e-03 k_t = [1, 1, 1, 1]
 Generation 3700: Best = 8.393125e+02 FES = 370100 μ_F = 0.702 μ_{CR} = 0.245
 Diversity = 2.8554e-03 k_t = [4, 2, 2, 2]
 Generation 3710: Best = 8.393116e+02 FES = 371100 μ_F = 0.689 μ_{CR} = 0.246
 Diversity = 2.7806e-03 k_t = [1, 1, 1, 1]
 Generation 3720: Best = 8.393098e+02 FES = 372100 μ_F = 0.690 μ_{CR} = 0.251
 Diversity = 2.5145e-03 k_t = [2, 4, 3, 2]
 Generation 3730: Best = 8.393088e+02 FES = 373100 μ_F = 0.693 μ_{CR} = 0.246
 Diversity = 2.7402e-03 k_t = [1, 1, 1, 1]

Generation 3740: Best = 8.393084e+02 FES = 374100 μ_F = 0.689 μ_{CR} = 0.243
 Diversity = 2.5829e-03 k_t = [2, 3, 2, 2]
 Generation 3750: Best = 8.393070e+02 FES = 375100 μ_F = 0.696 μ_{CR} = 0.234
 Diversity = 2.4706e-03 k_t = [1, 1, 1, 1]
 Generation 3760: Best = 8.393064e+02 FES = 376100 μ_F = 0.704 μ_{CR} = 0.232
 Diversity = 2.4099e-03 k_t = [2, 4, 2, 2]
 Generation 3770: Best = 8.393056e+02 FES = 377100 μ_F = 0.708 μ_{CR} = 0.241
 Diversity = 2.4164e-03 k_t = [1, 1, 1, 1]
 Generation 3780: Best = 8.393044e+02 FES = 378100 μ_F = 0.719 μ_{CR} = 0.244
 Diversity = 2.3533e-03 k_t = [3, 4, 3, 3]
 Generation 3790: Best = 8.393035e+02 FES = 379100 μ_F = 0.725 μ_{CR} = 0.243
 Diversity = 2.4716e-03 k_t = [1, 1, 1, 1]
 Generation 3800: Best = 8.393030e+02 FES = 380100 μ_F = 0.719 μ_{CR} = 0.239
 Diversity = 2.3820e-03 k_t = [4, 2, 3, 2]
 Generation 3810: Best = 8.393020e+02 FES = 381100 μ_F = 0.719 μ_{CR} = 0.250
 Diversity = 2.4358e-03 k_t = [1, 1, 1, 1]
 Generation 3820: Best = 8.393011e+02 FES = 382100 μ_F = 0.722 μ_{CR} = 0.238
 Diversity = 2.4356e-03 k_t = [3, 3, 3, 2]
 Generation 3830: Best = 8.393001e+02 FES = 383100 μ_F = 0.720 μ_{CR} = 0.240
 Diversity = 2.6398e-03 k_t = [1, 1, 1, 1]
 Generation 3840: Best = 8.392995e+02 FES = 384100 μ_F = 0.720 μ_{CR} = 0.238
 Diversity = 2.4221e-03 k_t = [3, 4, 2, 4]
 Generation 3850: Best = 8.392989e+02 FES = 385100 μ_F = 0.716 μ_{CR} = 0.235
 Diversity = 2.3102e-03 k_t = [1, 1, 1, 1]
 Generation 3860: Best = 8.392980e+02 FES = 386100 μ_F = 0.721 μ_{CR} = 0.235
 Diversity = 2.3150e-03 k_t = [4, 2, 4, 2]
 Generation 3870: Best = 8.392975e+02 FES = 387100 μ_F = 0.719 μ_{CR} = 0.229
 Diversity = 2.1997e-03 k_t = [1, 1, 1, 1]
 Generation 3880: Best = 8.392968e+02 FES = 388100 μ_F = 0.724 μ_{CR} = 0.228
 Diversity = 1.9763e-03 k_t = [2, 2, 4, 2]
 Generation 3890: Best = 8.392961e+02 FES = 389100 μ_F = 0.723 μ_{CR} = 0.220
 Diversity = 2.0063e-03 k_t = [1, 1, 1, 1]
 Generation 3900: Best = 8.392953e+02 FES = 390100 μ_F = 0.728 μ_{CR} = 0.209
 Diversity = 2.0573e-03 k_t = [3, 3, 2, 3]
 Generation 3910: Best = 8.392942e+02 FES = 391100 μ_F = 0.736 μ_{CR} = 0.212
 Diversity = 2.5124e-03 k_t = [1, 1, 1, 1]
 Generation 3920: Best = 8.392936e+02 FES = 392100 μ_F = 0.721 μ_{CR} = 0.211
 Diversity = 2.4616e-03 k_t = [4, 2, 2, 4]
 Generation 3930: Best = 8.392929e+02 FES = 393100 μ_F = 0.729 μ_{CR} = 0.210
 Diversity = 2.3305e-03 k_t = [1, 1, 1, 1]
 Generation 3940: Best = 8.392925e+02 FES = 394100 μ_F = 0.721 μ_{CR} = 0.211
 Diversity = 2.1906e-03 k_t = [4, 3, 2, 2]
 Generation 3950: Best = 8.392922e+02 FES = 395100 μ_F = 0.707 μ_{CR} = 0.211
 Diversity = 1.9753e-03 k_t = [1, 1, 1, 1]
 Generation 3960: Best = 8.392917e+02 FES = 396100 μ_F = 0.719 μ_{CR} = 0.212
 Diversity = 1.7630e-03 k_t = [2, 3, 3, 4]
 Generation 3970: Best = 8.392913e+02 FES = 397100 μ_F = 0.706 μ_{CR} = 0.212
 Diversity = 1.7034e-03 k_t = [1, 1, 1, 1]
 Generation 3980: Best = 8.392909e+02 FES = 398100 μ_F = 0.716 μ_{CR} = 0.214
 Diversity = 1.6295e-03 k_t = [3, 2, 3, 4]
 Generation 3990: Best = 8.392906e+02 FES = 399100 μ_F = 0.713 μ_{CR} = 0.200
 Diversity = 1.6048e-03 k_t = [1, 1, 1, 1]
 Generation 4000: Best = 8.392896e+02 FES = 400100 μ_F = 0.721 μ_{CR} = 0.214
 Diversity = 1.7763e-03 k_t = [4, 2, 3, 3]

Generation 4010: Best = 8.392889e+02 FES = 401100 μ_F = 0.734 μ_{CR} = 0.222
 Diversity = 2.1272e-03 k_t = [1, 1, 1, 1]
 Generation 4020: Best = 8.392886e+02 FES = 402100 μ_F = 0.732 μ_{CR} = 0.221
 Diversity = 2.1192e-03 k_t = [4, 4, 3, 2]
 Generation 4030: Best = 8.392882e+02 FES = 403100 μ_F = 0.730 μ_{CR} = 0.216
 Diversity = 2.0699e-03 k_t = [1, 1, 1, 1]
 Generation 4040: Best = 8.392877e+02 FES = 404100 μ_F = 0.732 μ_{CR} = 0.218
 Diversity = 1.7900e-03 k_t = [3, 4, 2, 4]
 Generation 4050: Best = 8.392874e+02 FES = 405100 μ_F = 0.734 μ_{CR} = 0.224
 Diversity = 1.7170e-03 k_t = [1, 1, 1, 1]
 Generation 4060: Best = 8.392872e+02 FES = 406100 μ_F = 0.726 μ_{CR} = 0.218
 Diversity = 1.5134e-03 k_t = [4, 2, 4, 4]
 Generation 4070: Best = 8.392870e+02 FES = 407100 μ_F = 0.728 μ_{CR} = 0.205
 Diversity = 1.3536e-03 k_t = [1, 1, 1, 1]
 Generation 4080: Best = 8.392868e+02 FES = 408100 μ_F = 0.730 μ_{CR} = 0.204
 Diversity = 1.3099e-03 k_t = [4, 4, 4, 2]
 Generation 4090: Best = 8.392867e+02 FES = 409100 μ_F = 0.742 μ_{CR} = 0.195
 Diversity = 1.2329e-03 k_t = [1, 1, 1, 1]
 Generation 4100: Best = 8.392863e+02 FES = 410100 μ_F = 0.735 μ_{CR} = 0.191
 Diversity = 1.1719e-03 k_t = [4, 3, 3, 3]
 Generation 4110: Best = 8.392862e+02 FES = 411100 μ_F = 0.734 μ_{CR} = 0.200
 Diversity = 1.1890e-03 k_t = [1, 1, 1, 1]
 Generation 4120: Best = 8.392861e+02 FES = 412100 μ_F = 0.739 μ_{CR} = 0.210
 Diversity = 1.0424e-03 k_t = [2, 3, 4, 4]
 Generation 4130: Best = 8.392860e+02 FES = 413100 μ_F = 0.735 μ_{CR} = 0.221
 Diversity = 1.0387e-03 k_t = [1, 1, 1, 1]
 Generation 4140: Best = 8.392858e+02 FES = 414100 μ_F = 0.728 μ_{CR} = 0.229
 Diversity = 1.0371e-03 k_t = [2, 2, 2, 4]
 Generation 4150: Best = 8.392855e+02 FES = 415100 μ_F = 0.735 μ_{CR} = 0.222
 Diversity = 1.1293e-03 k_t = [1, 1, 1, 1]
 Generation 4160: Best = 8.392853e+02 FES = 416100 μ_F = 0.746 μ_{CR} = 0.225
 Diversity = 1.1134e-03 k_t = [2, 3, 2, 3]
 Generation 4170: Best = 8.392851e+02 FES = 417100 μ_F = 0.733 μ_{CR} = 0.229
 Diversity = 1.1037e-03 k_t = [1, 1, 1, 1]
 Generation 4180: Best = 8.392849e+02 FES = 418100 μ_F = 0.734 μ_{CR} = 0.241
 Diversity = 1.1287e-03 k_t = [4, 4, 2, 2]
 Generation 4190: Best = 8.392848e+02 FES = 419100 μ_F = 0.729 μ_{CR} = 0.232
 Diversity = 1.0800e-03 k_t = [1, 1, 1, 1]
 Generation 4200: Best = 8.392847e+02 FES = 420100 μ_F = 0.730 μ_{CR} = 0.238
 Diversity = 1.0102e-03 k_t = [4, 3, 3, 3]
 Generation 4210: Best = 8.392846e+02 FES = 421100 μ_F = 0.723 μ_{CR} = 0.241
 Diversity = 9.7744e-04 k_t = [1, 1, 1, 1]
 Generation 4220: Best = 8.392845e+02 FES = 422100 μ_F = 0.721 μ_{CR} = 0.238
 Diversity = 8.3517e-04 k_t = [2, 3, 4, 3]
 Generation 4230: Best = 8.392844e+02 FES = 423100 μ_F = 0.709 μ_{CR} = 0.239
 Diversity = 8.2203e-04 k_t = [1, 1, 1, 1]
 Generation 4240: Best = 8.392843e+02 FES = 424100 μ_F = 0.711 μ_{CR} = 0.234
 Diversity = 7.8245e-04 k_t = [3, 2, 3, 2]
 Generation 4250: Best = 8.392841e+02 FES = 425100 μ_F = 0.717 μ_{CR} = 0.242
 Diversity = 9.1927e-04 k_t = [1, 1, 1, 1]
 Generation 4260: Best = 8.392840e+02 FES = 426100 μ_F = 0.712 μ_{CR} = 0.248
 Diversity = 8.6384e-04 k_t = [4, 2, 3, 4]
 Generation 4270: Best = 8.392840e+02 FES = 427100 μ_F = 0.722 μ_{CR} = 0.244
 Diversity = 7.4693e-04 k_t = [1, 1, 1, 1]

Generation 4280: Best = 8.392839e+02 FES = 428100 μ_F = 0.725 μ_{CR} = 0.237
 Diversity = 6.8521e-04 k_t = [2, 2, 4, 3]
 Generation 4290: Best = 8.392839e+02 FES = 429100 μ_F = 0.724 μ_{CR} = 0.235
 Diversity = 7.0654e-04 k_t = [1, 1, 1, 1]
 Generation 4300: Best = 8.392837e+02 FES = 430100 μ_F = 0.743 μ_{CR} = 0.233
 Diversity = 7.4574e-04 k_t = [2, 2, 4, 4]
 Generation 4310: Best = 8.392835e+02 FES = 431100 μ_F = 0.755 μ_{CR} = 0.232
 Diversity = 8.6255e-04 k_t = [1, 1, 1, 1]
 Generation 4320: Best = 8.392834e+02 FES = 432100 μ_F = 0.758 μ_{CR} = 0.222
 Diversity = 9.5006e-04 k_t = [2, 4, 4, 3]
 Generation 4330: Best = 8.392833e+02 FES = 433100 μ_F = 0.750 μ_{CR} = 0.218
 Diversity = 1.0286e-03 k_t = [1, 1, 1, 1]
 Generation 4340: Best = 8.392830e+02 FES = 434100 μ_F = 0.740 μ_{CR} = 0.223
 Diversity = 1.0307e-03 k_t = [2, 2, 2, 3]
 Generation 4350: Best = 8.392828e+02 FES = 435100 μ_F = 0.735 μ_{CR} = 0.229
 Diversity = 1.1670e-03 k_t = [1, 1, 1, 1]
 Generation 4360: Best = 8.392826e+02 FES = 436100 μ_F = 0.727 μ_{CR} = 0.229
 Diversity = 1.1957e-03 k_t = [4, 4, 3, 2]
 Generation 4370: Best = 8.392825e+02 FES = 437100 μ_F = 0.730 μ_{CR} = 0.229
 Diversity = 1.2143e-03 k_t = [1, 1, 1, 1]
 Generation 4380: Best = 8.392822e+02 FES = 438100 μ_F = 0.727 μ_{CR} = 0.235
 Diversity = 1.1500e-03 k_t = [4, 4, 4, 2]
 Generation 4390: Best = 8.392821e+02 FES = 439100 μ_F = 0.723 μ_{CR} = 0.235
 Diversity = 1.1195e-03 k_t = [1, 1, 1, 1]
 Generation 4400: Best = 8.392819e+02 FES = 440100 μ_F = 0.725 μ_{CR} = 0.235
 Diversity = 9.8522e-04 k_t = [3, 4, 3, 4]
 Generation 4410: Best = 8.392817e+02 FES = 441100 μ_F = 0.734 μ_{CR} = 0.233
 Diversity = 1.0051e-03 k_t = [1, 1, 1, 1]
 Generation 4420: Best = 8.392816e+02 FES = 442100 μ_F = 0.724 μ_{CR} = 0.238
 Diversity = 9.2792e-04 k_t = [3, 3, 4, 4]
 Generation 4430: Best = 8.392815e+02 FES = 443100 μ_F = 0.712 μ_{CR} = 0.237
 Diversity = 8.8054e-04 k_t = [1, 1, 1, 1]
 Generation 4440: Best = 8.392813e+02 FES = 444100 μ_F = 0.713 μ_{CR} = 0.245
 Diversity = 8.8247e-04 k_t = [3, 4, 3, 2]
 Generation 4450: Best = 8.392813e+02 FES = 445100 μ_F = 0.720 μ_{CR} = 0.238
 Diversity = 8.7863e-04 k_t = [1, 1, 1, 1]
 Generation 4460: Best = 8.392811e+02 FES = 446100 μ_F = 0.719 μ_{CR} = 0.246
 Diversity = 9.0844e-04 k_t = [3, 3, 4, 4]
 Generation 4470: Best = 8.392810e+02 FES = 447100 μ_F = 0.712 μ_{CR} = 0.245
 Diversity = 9.0617e-04 k_t = [1, 1, 1, 1]
 Generation 4480: Best = 8.392809e+02 FES = 448100 μ_F = 0.715 μ_{CR} = 0.240
 Diversity = 8.4661e-04 k_t = [2, 2, 3, 4]
 Generation 4490: Best = 8.392808e+02 FES = 449100 μ_F = 0.720 μ_{CR} = 0.240
 Diversity = 8.2736e-04 k_t = [1, 1, 1, 1]
 Generation 4500: Best = 8.392808e+02 FES = 450100 μ_F = 0.716 μ_{CR} = 0.240
 Diversity = 7.7226e-04 k_t = [4, 3, 2, 2]
 Generation 4510: Best = 8.392807e+02 FES = 451100 μ_F = 0.721 μ_{CR} = 0.244
 Diversity = 7.6798e-04 k_t = [1, 1, 1, 1]
 Generation 4520: Best = 8.392806e+02 FES = 452100 μ_F = 0.725 μ_{CR} = 0.240
 Diversity = 7.3581e-04 k_t = [4, 2, 4, 2]
 Generation 4530: Best = 8.392806e+02 FES = 453100 μ_F = 0.714 μ_{CR} = 0.246
 Diversity = 6.9033e-04 k_t = [1, 1, 1, 1]
 Generation 4540: Best = 8.392805e+02 FES = 454100 μ_F = 0.703 μ_{CR} = 0.246
 Diversity = 6.3284e-04 k_t = [2, 2, 2, 4]

Generation 4550: Best = 8.392805e+02 FES = 455100 μ_F = 0.714 μ_{CR} = 0.250
 Diversity = 6.1408e-04 k_t = [1, 1, 1, 1]
 Generation 4560: Best = 8.392804e+02 FES = 456100 μ_F = 0.711 μ_{CR} = 0.253
 Diversity = 6.1566e-04 k_t = [4, 4, 4, 2]
 Generation 4570: Best = 8.392804e+02 FES = 457100 μ_F = 0.709 μ_{CR} = 0.251
 Diversity = 5.5967e-04 k_t = [1, 1, 1, 1]
 Generation 4580: Best = 8.392804e+02 FES = 458100 μ_F = 0.704 μ_{CR} = 0.252
 Diversity = 5.0495e-04 k_t = [2, 2, 3, 2]
 Generation 4590: Best = 8.392803e+02 FES = 459100 μ_F = 0.689 μ_{CR} = 0.256
 Diversity = 4.9517e-04 k_t = [1, 1, 1, 1]
 Generation 4600: Best = 8.392802e+02 FES = 460100 μ_F = 0.702 μ_{CR} = 0.267
 Diversity = 5.3322e-04 k_t = [2, 3, 4, 2]
 Generation 4610: Best = 8.392801e+02 FES = 461100 μ_F = 0.715 μ_{CR} = 0.274
 Diversity = 6.9539e-04 k_t = [1, 1, 1, 1]
 Generation 4620: Best = 8.392799e+02 FES = 462100 μ_F = 0.718 μ_{CR} = 0.282
 Diversity = 7.7740e-04 k_t = [4, 4, 4, 4]
 Generation 4630: Best = 8.392799e+02 FES = 463100 μ_F = 0.719 μ_{CR} = 0.287
 Diversity = 7.4731e-04 k_t = [1, 1, 1, 1]
 Generation 4640: Best = 8.392798e+02 FES = 464100 μ_F = 0.701 μ_{CR} = 0.290
 Diversity = 6.1604e-04 k_t = [2, 4, 2, 3]
 Generation 4650: Best = 8.392798e+02 FES = 465100 μ_F = 0.693 μ_{CR} = 0.288
 Diversity = 5.7523e-04 k_t = [1, 1, 1, 1]
 Generation 4660: Best = 8.392798e+02 FES = 466100 μ_F = 0.688 μ_{CR} = 0.280
 Diversity = 4.4505e-04 k_t = [3, 3, 4, 3]
 Generation 4670: Best = 8.392798e+02 FES = 467100 μ_F = 0.688 μ_{CR} = 0.282
 Diversity = 3.7644e-04 k_t = [1, 1, 1, 1]
 Generation 4680: Best = 8.392798e+02 FES = 468100 μ_F = 0.684 μ_{CR} = 0.279
 Diversity = 2.8216e-04 k_t = [3, 4, 4, 4]
 Generation 4690: Best = 8.392797e+02 FES = 469100 μ_F = 0.695 μ_{CR} = 0.277
 Diversity = 2.5108e-04 k_t = [1, 1, 1, 1]
 Generation 4700: Best = 8.392797e+02 FES = 470100 μ_F = 0.685 μ_{CR} = 0.268
 Diversity = 2.3225e-04 k_t = [3, 2, 3, 2]
 Generation 4710: Best = 8.392797e+02 FES = 471100 μ_F = 0.685 μ_{CR} = 0.273
 Diversity = 2.3856e-04 k_t = [1, 1, 1, 1]
 Generation 4720: Best = 8.392797e+02 FES = 472100 μ_F = 0.686 μ_{CR} = 0.269
 Diversity = 2.1752e-04 k_t = [3, 2, 4, 2]
 Generation 4730: Best = 8.392797e+02 FES = 473100 μ_F = 0.696 μ_{CR} = 0.269
 Diversity = 2.1186e-04 k_t = [1, 1, 1, 1]
 Generation 4740: Best = 8.392797e+02 FES = 474100 μ_F = 0.707 μ_{CR} = 0.257
 Diversity = 1.8055e-04 k_t = [4, 4, 3, 4]
 Generation 4750: Best = 8.392797e+02 FES = 475100 μ_F = 0.712 μ_{CR} = 0.251
 Diversity = 1.6144e-04 k_t = [1, 1, 1, 1]
 Generation 4760: Best = 8.392797e+02 FES = 476100 μ_F = 0.726 μ_{CR} = 0.242
 Diversity = 1.5527e-04 k_t = [4, 3, 4, 3]
 Generation 4770: Best = 8.392797e+02 FES = 477100 μ_F = 0.729 μ_{CR} = 0.240
 Diversity = 1.5390e-04 k_t = [1, 1, 1, 1]
 Generation 4780: Best = 8.392797e+02 FES = 478100 μ_F = 0.727 μ_{CR} = 0.235
 Diversity = 1.4349e-04 k_t = [4, 4, 4, 3]
 Generation 4790: Best = 8.392797e+02 FES = 479100 μ_F = 0.726 μ_{CR} = 0.241
 Diversity = 1.3990e-04 k_t = [1, 1, 1, 1]
 Generation 4800: Best = 8.392797e+02 FES = 480100 μ_F = 0.729 μ_{CR} = 0.238
 Diversity = 1.3562e-04 k_t = [4, 4, 4, 2]
 Generation 4810: Best = 8.392797e+02 FES = 481100 μ_F = 0.718 μ_{CR} = 0.233
 Diversity = 1.3068e-04 k_t = [1, 1, 1, 1]

Generation 4820: Best = 8.392797e+02 FES = 482100 μ_F = 0.714 μ_{CR} = 0.231
 Diversity = 1.2366e-04 k_t = [4, 4, 4, 4]
 Generation 4830: Best = 8.392797e+02 FES = 483100 μ_F = 0.717 μ_{CR} = 0.223
 Diversity = 1.2019e-04 k_t = [1, 1, 1, 1]
 Generation 4840: Best = 8.392797e+02 FES = 484100 μ_F = 0.715 μ_{CR} = 0.228
 Diversity = 1.1789e-04 k_t = [3, 2, 2, 3]
 Generation 4850: Best = 8.392797e+02 FES = 485100 μ_F = 0.716 μ_{CR} = 0.247
 Diversity = 1.2501e-04 k_t = [1, 1, 1, 1]
 Generation 4860: Best = 8.392797e+02 FES = 486100 μ_F = 0.728 μ_{CR} = 0.252
 Diversity = 1.2925e-04 k_t = [3, 2, 4, 2]
 Generation 4870: Best = 8.392797e+02 FES = 487100 μ_F = 0.721 μ_{CR} = 0.255
 Diversity = 1.3499e-04 k_t = [1, 1, 1, 1]
 Generation 4880: Best = 8.392797e+02 FES = 488100 μ_F = 0.723 μ_{CR} = 0.262
 Diversity = 1.2747e-04 k_t = [3, 2, 4, 2]
 Generation 4890: Best = 8.392797e+02 FES = 489100 μ_F = 0.714 μ_{CR} = 0.260
 Diversity = 1.2899e-04 k_t = [1, 1, 1, 1]
 Generation 4900: Best = 8.392797e+02 FES = 490100 μ_F = 0.712 μ_{CR} = 0.255
 Diversity = 1.1857e-04 k_t = [4, 2, 4, 3]
 Generation 4910: Best = 8.392797e+02 FES = 491100 μ_F = 0.726 μ_{CR} = 0.255
 Diversity = 1.2301e-04 k_t = [1, 1, 1, 1]
 Generation 4920: Best = 8.392797e+02 FES = 492100 μ_F = 0.747 μ_{CR} = 0.254
 Diversity = 1.3813e-04 k_t = [4, 4, 2, 4]
 Generation 4930: Best = 8.392797e+02 FES = 493100 μ_F = 0.736 μ_{CR} = 0.241
 Diversity = 1.4360e-04 k_t = [1, 1, 1, 1]
 Generation 4940: Best = 8.392797e+02 FES = 494100 μ_F = 0.738 μ_{CR} = 0.243
 Diversity = 1.5191e-04 k_t = [4, 4, 2, 3]
 Generation 4950: Best = 8.392797e+02 FES = 495100 μ_F = 0.735 μ_{CR} = 0.252
 Diversity = 1.5842e-04 k_t = [1, 1, 1, 1]
 Generation 4960: Best = 8.392796e+02 FES = 496100 μ_F = 0.743 μ_{CR} = 0.247
 Diversity = 1.6009e-04 k_t = [2, 2, 4, 4]
 Generation 4970: Best = 8.392796e+02 FES = 497100 μ_F = 0.745 μ_{CR} = 0.239
 Diversity = 1.6410e-04 k_t = [1, 1, 1, 1]
 Generation 4980: Best = 8.392796e+02 FES = 498100 μ_F = 0.750 μ_{CR} = 0.235
 Diversity = 1.5658e-04 k_t = [4, 4, 4, 4]
 Generation 4990: Best = 8.392796e+02 FES = 499100 μ_F = 0.737 μ_{CR} = 0.232
 Diversity = 1.5288e-04 k_t = [1, 1, 1, 1]
 Generation 5000: Best = 8.392796e+02 FES = 500100 μ_F = 0.739 μ_{CR} = 0.227
 Diversity = 1.5365e-04 k_t = [3, 3, 2, 2]
 Generation 5010: Best = 8.392796e+02 FES = 501100 μ_F = 0.737 μ_{CR} = 0.220
 Diversity = 1.5090e-04 k_t = [1, 1, 1, 1]
 Generation 5020: Best = 8.392796e+02 FES = 502100 μ_F = 0.723 μ_{CR} = 0.216
 Diversity = 1.4027e-04 k_t = [3, 4, 4, 2]
 Generation 5030: Best = 8.392796e+02 FES = 503100 μ_F = 0.731 μ_{CR} = 0.215
 Diversity = 1.5020e-04 k_t = [1, 1, 1, 1]
 Generation 5040: Best = 8.392796e+02 FES = 504100 μ_F = 0.744 μ_{CR} = 0.221
 Diversity = 1.5197e-04 k_t = [4, 3, 3, 3]
 Generation 5050: Best = 8.392796e+02 FES = 505100 μ_F = 0.742 μ_{CR} = 0.231
 Diversity = 1.6692e-04 k_t = [1, 1, 1, 1]
 Generation 5060: Best = 8.392796e+02 FES = 506100 μ_F = 0.744 μ_{CR} = 0.231
 Diversity = 1.7177e-04 k_t = [3, 2, 2, 3]
 Generation 5070: Best = 8.392796e+02 FES = 507100 μ_F = 0.735 μ_{CR} = 0.228
 Diversity = 1.7635e-04 k_t = [1, 1, 1, 1]
 Generation 5080: Best = 8.392796e+02 FES = 508100 μ_F = 0.739 μ_{CR} = 0.234
 Diversity = 1.7559e-04 k_t = [2, 4, 3, 3]

Generation 5090: Best = 8.392796e+02 FES = 509100 μ_F = 0.739 μ_{CR} = 0.231
 Diversity = 1.9129e-04 k_t = [1, 1, 1, 1]
 Generation 5100: Best = 8.392796e+02 FES = 510100 μ_F = 0.740 μ_{CR} = 0.233
 Diversity = 2.0162e-04 k_t = [3, 4, 4, 2]
 Generation 5110: Best = 8.392796e+02 FES = 511100 μ_F = 0.740 μ_{CR} = 0.231
 Diversity = 1.9137e-04 k_t = [1, 1, 1, 1]
 Generation 5120: Best = 8.392796e+02 FES = 512100 μ_F = 0.742 μ_{CR} = 0.227
 Diversity = 1.7583e-04 k_t = [4, 3, 4, 4]
 Generation 5130: Best = 8.392796e+02 FES = 513100 μ_F = 0.730 μ_{CR} = 0.232
 Diversity = 1.6877e-04 k_t = [1, 1, 1, 1]
 Generation 5140: Best = 8.392796e+02 FES = 514100 μ_F = 0.725 μ_{CR} = 0.221
 Diversity = 1.5939e-04 k_t = [4, 3, 3, 3]
 Generation 5150: Best = 8.392796e+02 FES = 515100 μ_F = 0.724 μ_{CR} = 0.226
 Diversity = 1.6225e-04 k_t = [1, 1, 1, 1]
 Generation 5160: Best = 8.392796e+02 FES = 516100 μ_F = 0.719 μ_{CR} = 0.236
 Diversity = 1.5601e-04 k_t = [4, 3, 4, 2]
 Generation 5170: Best = 8.392796e+02 FES = 517100 μ_F = 0.709 μ_{CR} = 0.229
 Diversity = 1.5397e-04 k_t = [1, 1, 1, 1]
 Generation 5180: Best = 8.392796e+02 FES = 518100 μ_F = 0.727 μ_{CR} = 0.225
 Diversity = 1.6286e-04 k_t = [3, 3, 3, 2]
 Generation 5190: Best = 8.392796e+02 FES = 519100 μ_F = 0.720 μ_{CR} = 0.237
 Diversity = 1.6047e-04 k_t = [1, 1, 1, 1]
 Generation 5200: Best = 8.392796e+02 FES = 520100 μ_F = 0.712 μ_{CR} = 0.238
 Diversity = 1.4235e-04 k_t = [3, 2, 4, 4]
 Generation 5210: Best = 8.392796e+02 FES = 521100 μ_F = 0.714 μ_{CR} = 0.247
 Diversity = 1.4090e-04 k_t = [1, 1, 1, 1]
 Generation 5220: Best = 8.392795e+02 FES = 522100 μ_F = 0.715 μ_{CR} = 0.255
 Diversity = 1.3346e-04 k_t = [3, 2, 3, 4]
 Generation 5230: Best = 8.392795e+02 FES = 523100 μ_F = 0.708 μ_{CR} = 0.254
 Diversity = 1.2461e-04 k_t = [1, 1, 1, 1]
 Generation 5240: Best = 8.392795e+02 FES = 524100 μ_F = 0.730 μ_{CR} = 0.245
 Diversity = 1.2890e-04 k_t = [4, 2, 3, 2]
 Generation 5250: Best = 8.392795e+02 FES = 525100 μ_F = 0.725 μ_{CR} = 0.243
 Diversity = 1.3032e-04 k_t = [1, 1, 1, 1]
 Generation 5260: Best = 8.392795e+02 FES = 526100 μ_F = 0.723 μ_{CR} = 0.244
 Diversity = 1.2308e-04 k_t = [4, 2, 4, 4]
 Generation 5270: Best = 8.392795e+02 FES = 527100 μ_F = 0.726 μ_{CR} = 0.237
 Diversity = 1.1598e-04 k_t = [1, 1, 1, 1]
 Generation 5280: Best = 8.392795e+02 FES = 528100 μ_F = 0.727 μ_{CR} = 0.229
 Diversity = 1.1395e-04 k_t = [3, 4, 4, 4]
 Generation 5290: Best = 8.392795e+02 FES = 529100 μ_F = 0.733 μ_{CR} = 0.237
 Diversity = 1.2430e-04 k_t = [1, 1, 1, 1]
 Generation 5300: Best = 8.392795e+02 FES = 530100 μ_F = 0.731 μ_{CR} = 0.239
 Diversity = 1.3202e-04 k_t = [4, 3, 3, 2]
 Generation 5310: Best = 8.392795e+02 FES = 531100 μ_F = 0.727 μ_{CR} = 0.237
 Diversity = 1.2653e-04 k_t = [1, 1, 1, 1]
 Generation 5320: Best = 8.392795e+02 FES = 532100 μ_F = 0.740 μ_{CR} = 0.240
 Diversity = 1.2292e-04 k_t = [3, 3, 4, 2]
 Generation 5330: Best = 8.392795e+02 FES = 533100 μ_F = 0.731 μ_{CR} = 0.235
 Diversity = 1.2227e-04 k_t = [1, 1, 1, 1]
 Generation 5340: Best = 8.392795e+02 FES = 534100 μ_F = 0.733 μ_{CR} = 0.239
 Diversity = 1.2008e-04 k_t = [4, 2, 3, 4]
 Generation 5350: Best = 8.392795e+02 FES = 535100 μ_F = 0.734 μ_{CR} = 0.237
 Diversity = 1.2208e-04 k_t = [1, 1, 1, 1]

Generation 5360: Best = 8.392795e+02 FES = 536100 μ_F = 0.732 μ_{CR} = 0.233
 Diversity = 1.1011e-04 k_t = [4, 3, 4, 3]
 Generation 5370: Best = 8.392795e+02 FES = 537100 μ_F = 0.732 μ_{CR} = 0.227
 Diversity = 1.0799e-04 k_t = [1, 1, 1, 1]
 Generation 5380: Best = 8.392795e+02 FES = 538100 μ_F = 0.732 μ_{CR} = 0.226
 Diversity = 9.8184e-05 k_t = [4, 2, 4, 3]
 Generation 5390: Best = 8.392795e+02 FES = 539100 μ_F = 0.730 μ_{CR} = 0.229
 Diversity = 9.7673e-05 k_t = [1, 1, 1, 1]
 Generation 5400: Best = 8.392795e+02 FES = 540100 μ_F = 0.729 μ_{CR} = 0.233
 Diversity = 9.4896e-05 k_t = [4, 3, 4, 4]
 Generation 5410: Best = 8.392795e+02 FES = 541100 μ_F = 0.740 μ_{CR} = 0.243
 Diversity = 9.3536e-05 k_t = [1, 1, 1, 1]
 Generation 5420: Best = 8.392795e+02 FES = 542100 μ_F = 0.733 μ_{CR} = 0.237
 Diversity = 8.9568e-05 k_t = [3, 3, 2, 3]
 Generation 5430: Best = 8.392795e+02 FES = 543100 μ_F = 0.737 μ_{CR} = 0.239
 Diversity = 9.6545e-05 k_t = [1, 1, 1, 1]
 Generation 5440: Best = 8.392795e+02 FES = 544100 μ_F = 0.738 μ_{CR} = 0.241
 Diversity = 1.0472e-04 k_t = [2, 2, 3, 2]
 Generation 5450: Best = 8.392795e+02 FES = 545100 μ_F = 0.745 μ_{CR} = 0.247
 Diversity = 1.1890e-04 k_t = [1, 1, 1, 1]
 Generation 5460: Best = 8.392795e+02 FES = 546100 μ_F = 0.749 μ_{CR} = 0.248
 Diversity = 1.2227e-04 k_t = [4, 2, 3, 3]
 Generation 5470: Best = 8.392795e+02 FES = 547100 μ_F = 0.734 μ_{CR} = 0.243
 Diversity = 1.1848e-04 k_t = [1, 1, 1, 1]
 Generation 5480: Best = 8.392795e+02 FES = 548100 μ_F = 0.731 μ_{CR} = 0.243
 Diversity = 1.1696e-04 k_t = [2, 2, 3, 2]
 Generation 5490: Best = 8.392795e+02 FES = 549100 μ_F = 0.725 μ_{CR} = 0.239
 Diversity = 1.1517e-04 k_t = [1, 1, 1, 1]
 Generation 5500: Best = 8.392795e+02 FES = 550100 μ_F = 0.733 μ_{CR} = 0.238
 Diversity = 1.0611e-04 k_t = [4, 4, 2, 4]
 Generation 5510: Best = 8.392795e+02 FES = 551100 μ_F = 0.723 μ_{CR} = 0.239
 Diversity = 1.0307e-04 k_t = [1, 1, 1, 1]
 Generation 5520: Best = 8.392795e+02 FES = 552100 μ_F = 0.705 μ_{CR} = 0.219
 Diversity = 1.0793e-04 k_t = [3, 2, 3, 2]
 Generation 5530: Best = 8.392795e+02 FES = 553100 μ_F = 0.707 μ_{CR} = 0.219
 Diversity = 1.0814e-04 k_t = [1, 1, 1, 1]
 Generation 5540: Best = 8.392795e+02 FES = 554100 μ_F = 0.716 μ_{CR} = 0.220
 Diversity = 1.0088e-04 k_t = [2, 3, 3, 3]
 Generation 5550: Best = 8.392795e+02 FES = 555100 μ_F = 0.711 μ_{CR} = 0.225
 Diversity = 1.0551e-04 k_t = [1, 1, 1, 1]
 Generation 5560: Best = 8.392795e+02 FES = 556100 μ_F = 0.710 μ_{CR} = 0.231
 Diversity = 1.0870e-04 k_t = [4, 3, 3, 2]
 Generation 5570: Best = 8.392795e+02 FES = 557100 μ_F = 0.720 μ_{CR} = 0.232
 Diversity = 1.0526e-04 k_t = [1, 1, 1, 1]
 Generation 5580: Best = 8.392795e+02 FES = 558100 μ_F = 0.722 μ_{CR} = 0.233
 Diversity = 8.9488e-05 k_t = [3, 2, 2, 4]
 Generation 5590: Best = 8.392795e+02 FES = 559100 μ_F = 0.730 μ_{CR} = 0.225
 Diversity = 8.7873e-05 k_t = [1, 1, 1, 1]
 Generation 5600: Best = 8.392795e+02 FES = 560100 μ_F = 0.730 μ_{CR} = 0.227
 Diversity = 8.3027e-05 k_t = [4, 4, 2, 3]
 Generation 5610: Best = 8.392795e+02 FES = 561100 μ_F = 0.744 μ_{CR} = 0.229
 Diversity = 8.6461e-05 k_t = [1, 1, 1, 1]
 Generation 5620: Best = 8.392795e+02 FES = 562100 μ_F = 0.744 μ_{CR} = 0.218
 Diversity = 7.9574e-05 k_t = [4, 3, 2, 3]

Generation 5630: Best = 8.392795e+02 FES = 563100 μ_F = 0.747 μ_{CR} = 0.209
 Diversity = 7.6745e-05 k_t = [1, 1, 1, 1]
 Generation 5640: Best = 8.392795e+02 FES = 564100 μ_F = 0.738 μ_{CR} = 0.210
 Diversity = 7.2987e-05 k_t = [4, 3, 2, 2]
 Generation 5650: Best = 8.392795e+02 FES = 565100 μ_F = 0.727 μ_{CR} = 0.218
 Diversity = 7.2993e-05 k_t = [1, 1, 1, 1]
 Generation 5660: Best = 8.392795e+02 FES = 566100 μ_F = 0.734 μ_{CR} = 0.216
 Diversity = 6.7739e-05 k_t = [3, 2, 3, 2]
 Generation 5670: Best = 8.392795e+02 FES = 567100 μ_F = 0.740 μ_{CR} = 0.226
 Diversity = 7.2981e-05 k_t = [1, 1, 1, 1]
 Generation 5680: Best = 8.392795e+02 FES = 568100 μ_F = 0.730 μ_{CR} = 0.229
 Diversity = 6.5808e-05 k_t = [2, 2, 2, 3]
 Generation 5690: Best = 8.392795e+02 FES = 569100 μ_F = 0.716 μ_{CR} = 0.230
 Diversity = 6.8382e-05 k_t = [1, 1, 1, 1]
 Generation 5700: Best = 8.392795e+02 FES = 570100 μ_F = 0.728 μ_{CR} = 0.229
 Diversity = 6.3639e-05 k_t = [2, 3, 4, 4]
 Generation 5710: Best = 8.392795e+02 FES = 571100 μ_F = 0.725 μ_{CR} = 0.237
 Diversity = 6.3405e-05 k_t = [1, 1, 1, 1]
 Generation 5720: Best = 8.392795e+02 FES = 572100 μ_F = 0.740 μ_{CR} = 0.241
 Diversity = 6.2832e-05 k_t = [3, 2, 3, 3]
 Generation 5730: Best = 8.392795e+02 FES = 573100 μ_F = 0.748 μ_{CR} = 0.232
 Diversity = 6.6951e-05 k_t = [1, 1, 1, 1]
 Generation 5740: Best = 8.392795e+02 FES = 574100 μ_F = 0.747 μ_{CR} = 0.234
 Diversity = 7.0520e-05 k_t = [3, 4, 3, 3]
 Generation 5750: Best = 8.392795e+02 FES = 575100 μ_F = 0.755 μ_{CR} = 0.240
 Diversity = 7.6488e-05 k_t = [1, 1, 1, 1]
 Generation 5760: Best = 8.392795e+02 FES = 576100 μ_F = 0.743 μ_{CR} = 0.248
 Diversity = 7.6710e-05 k_t = [2, 4, 4, 3]
 Generation 5770: Best = 8.392795e+02 FES = 577100 μ_F = 0.729 μ_{CR} = 0.248
 Diversity = 7.1630e-05 k_t = [1, 1, 1, 1]
 Generation 5780: Best = 8.392795e+02 FES = 578100 μ_F = 0.739 μ_{CR} = 0.241
 Diversity = 6.8636e-05 k_t = [3, 3, 4, 2]
 Generation 5790: Best = 8.392795e+02 FES = 579100 μ_F = 0.739 μ_{CR} = 0.235
 Diversity = 7.3162e-05 k_t = [1, 1, 1, 1]
 Generation 5800: Best = 8.392795e+02 FES = 580100 μ_F = 0.731 μ_{CR} = 0.241
 Diversity = 7.1770e-05 k_t = [2, 2, 2, 2]
 Generation 5810: Best = 8.392795e+02 FES = 581100 μ_F = 0.716 μ_{CR} = 0.243
 Diversity = 7.5911e-05 k_t = [1, 1, 1, 1]
 Generation 5820: Best = 8.392795e+02 FES = 582100 μ_F = 0.703 μ_{CR} = 0.242
 Diversity = 7.2781e-05 k_t = [4, 4, 4, 4]
 Generation 5830: Best = 8.392795e+02 FES = 583100 μ_F = 0.713 μ_{CR} = 0.236
 Diversity = 7.6660e-05 k_t = [1, 1, 1, 1]
 Generation 5840: Best = 8.392795e+02 FES = 584100 μ_F = 0.715 μ_{CR} = 0.237
 Diversity = 8.1232e-05 k_t = [2, 3, 2, 3]
 Generation 5850: Best = 8.392795e+02 FES = 585100 μ_F = 0.713 μ_{CR} = 0.219
 Diversity = 7.6309e-05 k_t = [1, 1, 1, 1]
 Generation 5860: Best = 8.392795e+02 FES = 586100 μ_F = 0.715 μ_{CR} = 0.215
 Diversity = 6.7653e-05 k_t = [4, 4, 2, 3]
 Generation 5870: Best = 8.392795e+02 FES = 587100 μ_F = 0.712 μ_{CR} = 0.213
 Diversity = 6.3675e-05 k_t = [1, 1, 1, 1]
 Generation 5880: Best = 8.392795e+02 FES = 588100 μ_F = 0.720 μ_{CR} = 0.215
 Diversity = 5.4609e-05 k_t = [4, 2, 2, 4]
 Generation 5890: Best = 8.392795e+02 FES = 589100 μ_F = 0.716 μ_{CR} = 0.216
 Diversity = 5.0803e-05 k_t = [1, 1, 1, 1]

Generation 5900: Best = 8.392795e+02	FES = 590100	$\mu_F = 0.735$	$\mu_{CR} = 0.227$
Diversity = 5.1083e-05	k_t = [4, 4, 4, 4]		
Generation 5910: Best = 8.392795e+02	FES = 591100	$\mu_F = 0.747$	$\mu_{CR} = 0.224$
Diversity = 5.3649e-05	k_t = [1, 1, 1, 1]		
Generation 5920: Best = 8.392795e+02	FES = 592100	$\mu_F = 0.743$	$\mu_{CR} = 0.231$
Diversity = 5.0256e-05	k_t = [3, 3, 2, 4]		
Generation 5930: Best = 8.392795e+02	FES = 593100	$\mu_F = 0.740$	$\mu_{CR} = 0.223$
Diversity = 4.9898e-05	k_t = [1, 1, 1, 1]		
Generation 5940: Best = 8.392795e+02	FES = 594100	$\mu_F = 0.729$	$\mu_{CR} = 0.223$
Diversity = 4.7073e-05	k_t = [4, 3, 4, 3]		
Generation 5950: Best = 8.392795e+02	FES = 595100	$\mu_F = 0.724$	$\mu_{CR} = 0.223$
Diversity = 4.4178e-05	k_t = [1, 1, 1, 1]		
Generation 5960: Best = 8.392795e+02	FES = 596100	$\mu_F = 0.735$	$\mu_{CR} = 0.227$
Diversity = 4.2664e-05	k_t = [4, 3, 2, 3]		
Generation 5970: Best = 8.392795e+02	FES = 597100	$\mu_F = 0.721$	$\mu_{CR} = 0.224$
Diversity = 4.2671e-05	k_t = [1, 1, 1, 1]		
Generation 5980: Best = 8.392795e+02	FES = 598100	$\mu_F = 0.735$	$\mu_{CR} = 0.223$
Diversity = 4.2571e-05	k_t = [2, 3, 3, 4]		
Generation 5990: Best = 8.392795e+02	FES = 599100	$\mu_F = 0.744$	$\mu_{CR} = 0.228$
Diversity = 4.5326e-05	k_t = [1, 1, 1, 1]		
Generation 6000: Best = 8.392795e+02	FES = 600100	$\mu_F = 0.749$	$\mu_{CR} = 0.235$
Diversity = 4.5210e-05	k_t = [3, 4, 2, 4]		
Generation 6010: Best = 8.392795e+02	FES = 601100	$\mu_F = 0.735$	$\mu_{CR} = 0.234$
Diversity = 4.3822e-05	k_t = [1, 1, 1, 1]		
Generation 6020: Best = 8.392795e+02	FES = 602100	$\mu_F = 0.732$	$\mu_{CR} = 0.229$
Diversity = 3.9235e-05	k_t = [3, 4, 4, 3]		
Generation 6030: Best = 8.392795e+02	FES = 603100	$\mu_F = 0.735$	$\mu_{CR} = 0.231$
Diversity = 3.9218e-05	k_t = [1, 1, 1, 1]		
Generation 6040: Best = 8.392795e+02	FES = 604100	$\mu_F = 0.744$	$\mu_{CR} = 0.230$
Diversity = 3.6026e-05	k_t = [3, 4, 3, 4]		
Generation 6050: Best = 8.392795e+02	FES = 605100	$\mu_F = 0.745$	$\mu_{CR} = 0.229$
Diversity = 3.3917e-05	k_t = [1, 1, 1, 1]		
Generation 6060: Best = 8.392795e+02	FES = 606100	$\mu_F = 0.732$	$\mu_{CR} = 0.217$
Diversity = 3.0247e-05	k_t = [2, 2, 2, 4]		
Generation 6070: Best = 8.392795e+02	FES = 607100	$\mu_F = 0.723$	$\mu_{CR} = 0.214$
Diversity = 3.1180e-05	k_t = [1, 1, 1, 1]		
Generation 6080: Best = 8.392795e+02	FES = 608100	$\mu_F = 0.718$	$\mu_{CR} = 0.219$
Diversity = 3.0358e-05	k_t = [2, 2, 4, 3]		
Generation 6090: Best = 8.392795e+02	FES = 609100	$\mu_F = 0.724$	$\mu_{CR} = 0.221$
Diversity = 3.0112e-05	k_t = [1, 1, 1, 1]		
Generation 6100: Best = 8.392795e+02	FES = 610100	$\mu_F = 0.718$	$\mu_{CR} = 0.221$
Diversity = 2.6806e-05	k_t = [3, 3, 3, 4]		
Generation 6110: Best = 8.392795e+02	FES = 611100	$\mu_F = 0.728$	$\mu_{CR} = 0.214$
Diversity = 2.6201e-05	k_t = [1, 1, 1, 1]		
Generation 6120: Best = 8.392795e+02	FES = 612100	$\mu_F = 0.733$	$\mu_{CR} = 0.213$
Diversity = 2.6540e-05	k_t = [3, 4, 4, 2]		
Generation 6130: Best = 8.392795e+02	FES = 613100	$\mu_F = 0.717$	$\mu_{CR} = 0.220$
Diversity = 2.5563e-05	k_t = [1, 1, 1, 1]		
Generation 6140: Best = 8.392795e+02	FES = 614100	$\mu_F = 0.700$	$\mu_{CR} = 0.219$
Diversity = 2.1512e-05	k_t = [3, 4, 2, 4]		
Generation 6150: Best = 8.392795e+02	FES = 615100	$\mu_F = 0.708$	$\mu_{CR} = 0.225$
Diversity = 2.0799e-05	k_t = [1, 1, 1, 1]		
Generation 6160: Best = 8.392795e+02	FES = 616100	$\mu_F = 0.712$	$\mu_{CR} = 0.231$
Diversity = 1.9398e-05	k_t = [4, 2, 4, 4]		

Generation 6170: Best = 8.392795e+02 FES = 617100 μ_F = 0.719 μ_{CR} = 0.234
 Diversity = 1.8979e-05 k_t = [1, 1, 1, 1]
 Generation 6180: Best = 8.392795e+02 FES = 618100 μ_F = 0.715 μ_{CR} = 0.236
 Diversity = 1.6943e-05 k_t = [3, 3, 3, 4]
 Generation 6190: Best = 8.392795e+02 FES = 619100 μ_F = 0.710 μ_{CR} = 0.234
 Diversity = 1.6027e-05 k_t = [1, 1, 1, 1]
 Generation 6200: Best = 8.392795e+02 FES = 620100 μ_F = 0.717 μ_{CR} = 0.229
 Diversity = 1.4994e-05 k_t = [3, 2, 4, 4]
 Generation 6210: Best = 8.392795e+02 FES = 621100 μ_F = 0.724 μ_{CR} = 0.236
 Diversity = 1.6045e-05 k_t = [1, 1, 1, 1]
 Generation 6220: Best = 8.392795e+02 FES = 622100 μ_F = 0.719 μ_{CR} = 0.243
 Diversity = 1.5097e-05 k_t = [4, 2, 4, 2]
 Generation 6230: Best = 8.392795e+02 FES = 623100 μ_F = 0.724 μ_{CR} = 0.239
 Diversity = 1.4718e-05 k_t = [1, 1, 1, 1]
 Generation 6240: Best = 8.392795e+02 FES = 624100 μ_F = 0.711 μ_{CR} = 0.235
 Diversity = 1.3703e-05 k_t = [2, 2, 2, 4]
 Generation 6250: Best = 8.392795e+02 FES = 625100 μ_F = 0.714 μ_{CR} = 0.236
 Diversity = 1.3065e-05 k_t = [1, 1, 1, 1]
 Generation 6260: Best = 8.392795e+02 FES = 626100 μ_F = 0.713 μ_{CR} = 0.245
 Diversity = 1.1911e-05 k_t = [4, 2, 2, 2]
 Generation 6270: Best = 8.392795e+02 FES = 627100 μ_F = 0.712 μ_{CR} = 0.241
 Diversity = 1.1695e-05 k_t = [1, 1, 1, 1]
 Generation 6280: Best = 8.392795e+02 FES = 628100 μ_F = 0.706 μ_{CR} = 0.243
 Diversity = 1.0856e-05 k_t = [4, 2, 3, 3]
 Generation 6290: Best = 8.392795e+02 FES = 629100 μ_F = 0.720 μ_{CR} = 0.239
 Diversity = 1.1122e-05 k_t = [1, 1, 1, 1]
 Generation 6300: Best = 8.392795e+02 FES = 630100 μ_F = 0.733 μ_{CR} = 0.246
 Diversity = 1.1967e-05 k_t = [2, 4, 3, 2]
 Generation 6310: Best = 8.392795e+02 FES = 631100 μ_F = 0.731 μ_{CR} = 0.245
 Diversity = 1.2678e-05 k_t = [1, 1, 1, 1]
 Generation 6320: Best = 8.392795e+02 FES = 632100 μ_F = 0.718 μ_{CR} = 0.249
 Diversity = 1.1009e-05 k_t = [2, 4, 3, 4]
 Generation 6330: Best = 8.392795e+02 FES = 633100 μ_F = 0.711 μ_{CR} = 0.245
 Diversity = 1.0446e-05 k_t = [1, 1, 1, 1]
 Generation 6340: Best = 8.392795e+02 FES = 634100 μ_F = 0.714 μ_{CR} = 0.251
 Diversity = 9.6050e-06 k_t = [3, 3, 4, 4]
 Generation 6350: Best = 8.392795e+02 FES = 635100 μ_F = 0.718 μ_{CR} = 0.251
 Diversity = 9.7470e-06 k_t = [1, 1, 1, 1]
 Generation 6360: Best = 8.392795e+02 FES = 636100 μ_F = 0.715 μ_{CR} = 0.253
 Diversity = 9.1064e-06 k_t = [4, 4, 4, 3]
 Generation 6370: Best = 8.392795e+02 FES = 637100 μ_F = 0.728 μ_{CR} = 0.256
 Diversity = 9.3867e-06 k_t = [1, 1, 1, 1]
 Generation 6380: Best = 8.392795e+02 FES = 638100 μ_F = 0.721 μ_{CR} = 0.256
 Diversity = 8.5713e-06 k_t = [4, 4, 2, 2]
 Generation 6390: Best = 8.392795e+02 FES = 639100 μ_F = 0.723 μ_{CR} = 0.253
 Diversity = 8.1261e-06 k_t = [1, 1, 1, 1]
 Generation 6400: Best = 8.392795e+02 FES = 640100 μ_F = 0.729 μ_{CR} = 0.254
 Diversity = 7.8365e-06 k_t = [2, 4, 4, 3]
 Generation 6410: Best = 8.392795e+02 FES = 641100 μ_F = 0.713 μ_{CR} = 0.245
 Diversity = 8.1107e-06 k_t = [1, 1, 1, 1]
 Generation 6420: Best = 8.392795e+02 FES = 642100 μ_F = 0.717 μ_{CR} = 0.241
 Diversity = 7.7562e-06 k_t = [4, 3, 4, 3]
 Generation 6430: Best = 8.392795e+02 FES = 643100 μ_F = 0.717 μ_{CR} = 0.257
 Diversity = 8.1166e-06 k_t = [1, 1, 1, 1]

Generation 6440: Best = 8.392795e+02 FES = 644100 μ_F = 0.728 μ_{CR} = 0.261
 Diversity = 8.4190e-06 k_t = [2, 4, 3, 3]
 Generation 6450: Best = 8.392795e+02 FES = 645100 μ_F = 0.721 μ_{CR} = 0.260
 Diversity = 9.2222e-06 k_t = [1, 1, 1, 1]
 Generation 6460: Best = 8.392795e+02 FES = 646100 μ_F = 0.730 μ_{CR} = 0.258
 Diversity = 9.2626e-06 k_t = [2, 4, 3, 2]
 Generation 6470: Best = 8.392795e+02 FES = 647100 μ_F = 0.721 μ_{CR} = 0.260
 Diversity = 9.9254e-06 k_t = [1, 1, 1, 1]
 Generation 6480: Best = 8.392795e+02 FES = 648100 μ_F = 0.719 μ_{CR} = 0.263
 Diversity = 8.8793e-06 k_t = [2, 2, 4, 3]
 Generation 6490: Best = 8.392795e+02 FES = 649100 μ_F = 0.735 μ_{CR} = 0.267
 Diversity = 9.8603e-06 k_t = [1, 1, 1, 1]
 Generation 6500: Best = 8.392795e+02 FES = 650100 μ_F = 0.732 μ_{CR} = 0.268
 Diversity = 1.0070e-05 k_t = [4, 3, 4, 3]
 Generation 6510: Best = 8.392795e+02 FES = 651100 μ_F = 0.736 μ_{CR} = 0.272
 Diversity = 1.0423e-05 k_t = [1, 1, 1, 1]
 Generation 6520: Best = 8.392795e+02 FES = 652100 μ_F = 0.725 μ_{CR} = 0.254
 Diversity = 1.0755e-05 k_t = [3, 2, 4, 2]
 Generation 6530: Best = 8.392795e+02 FES = 653100 μ_F = 0.712 μ_{CR} = 0.253
 Diversity = 1.0769e-05 k_t = [1, 1, 1, 1]
 Generation 6540: Best = 8.392795e+02 FES = 654100 μ_F = 0.725 μ_{CR} = 0.260
 Diversity = 1.0897e-05 k_t = [2, 2, 4, 3]
 Generation 6550: Best = 8.392795e+02 FES = 655100 μ_F = 0.734 μ_{CR} = 0.255
 Diversity = 1.1380e-05 k_t = [1, 1, 1, 1]
 Generation 6560: Best = 8.392795e+02 FES = 656100 μ_F = 0.729 μ_{CR} = 0.250
 Diversity = 1.1337e-05 k_t = [2, 2, 4, 3]
 Generation 6570: Best = 8.392795e+02 FES = 657100 μ_F = 0.727 μ_{CR} = 0.248
 Diversity = 1.2127e-05 k_t = [1, 1, 1, 1]
 Generation 6580: Best = 8.392795e+02 FES = 658100 μ_F = 0.721 μ_{CR} = 0.251
 Diversity = 1.2059e-05 k_t = [2, 4, 4, 2]
 Generation 6590: Best = 8.392795e+02 FES = 659100 μ_F = 0.715 μ_{CR} = 0.247
 Diversity = 1.2583e-05 k_t = [1, 1, 1, 1]
 Generation 6600: Best = 8.392795e+02 FES = 660100 μ_F = 0.710 μ_{CR} = 0.245
 Diversity = 1.2055e-05 k_t = [4, 3, 3, 2]
 Generation 6610: Best = 8.392795e+02 FES = 661100 μ_F = 0.720 μ_{CR} = 0.238
 Diversity = 1.1933e-05 k_t = [1, 1, 1, 1]
 Generation 6620: Best = 8.392795e+02 FES = 662100 μ_F = 0.720 μ_{CR} = 0.236
 Diversity = 1.0961e-05 k_t = [4, 2, 3, 3]
 Generation 6630: Best = 8.392795e+02 FES = 663100 μ_F = 0.711 μ_{CR} = 0.233
 Diversity = 1.0991e-05 k_t = [1, 1, 1, 1]
 Generation 6640: Best = 8.392795e+02 FES = 664100 μ_F = 0.730 μ_{CR} = 0.234
 Diversity = 1.0578e-05 k_t = [3, 4, 2, 4]
 Generation 6650: Best = 8.392795e+02 FES = 665100 μ_F = 0.728 μ_{CR} = 0.225
 Diversity = 1.0430e-05 k_t = [1, 1, 1, 1]
 Generation 6660: Best = 8.392795e+02 FES = 666100 μ_F = 0.725 μ_{CR} = 0.221
 Diversity = 9.8043e-06 k_t = [4, 4, 3, 4]
 Generation 6670: Best = 8.392795e+02 FES = 667100 μ_F = 0.728 μ_{CR} = 0.227
 Diversity = 9.5076e-06 k_t = [1, 1, 1, 1]
 Generation 6680: Best = 8.392795e+02 FES = 668100 μ_F = 0.713 μ_{CR} = 0.228
 Diversity = 9.3478e-06 k_t = [4, 2, 4, 2]
 Generation 6690: Best = 8.392795e+02 FES = 669100 μ_F = 0.711 μ_{CR} = 0.222
 Diversity = 9.2714e-06 k_t = [1, 1, 1, 1]
 Generation 6700: Best = 8.392795e+02 FES = 670100 μ_F = 0.711 μ_{CR} = 0.221
 Diversity = 8.4230e-06 k_t = [3, 3, 3, 3]

Generation 6710: Best = 8.392795e+02 FES = 671100 μ_F = 0.694 μ_{CR} = 0.226
 Diversity = 8.2449e-06 k_t = [1, 1, 1, 1]
 Generation 6720: Best = 8.392795e+02 FES = 672100 μ_F = 0.690 μ_{CR} = 0.229
 Diversity = 7.5214e-06 k_t = [4, 3, 3, 2]
 Generation 6730: Best = 8.392795e+02 FES = 673100 μ_F = 0.699 μ_{CR} = 0.226
 Diversity = 7.1331e-06 k_t = [1, 1, 1, 1]
 Generation 6740: Best = 8.392794e+02 FES = 674100 μ_F = 0.717 μ_{CR} = 0.223
 Diversity = 6.8077e-06 k_t = [4, 4, 4, 2]
 Generation 6750: Best = 8.392794e+02 FES = 675100 μ_F = 0.713 μ_{CR} = 0.230
 Diversity = 7.1499e-06 k_t = [1, 1, 1, 1]
 Generation 6760: Best = 8.392794e+02 FES = 676100 μ_F = 0.719 μ_{CR} = 0.242
 Diversity = 7.5716e-06 k_t = [2, 2, 3, 2]
 Generation 6770: Best = 8.392794e+02 FES = 677100 μ_F = 0.713 μ_{CR} = 0.242
 Diversity = 7.7434e-06 k_t = [1, 1, 1, 1]
 Generation 6780: Best = 8.392794e+02 FES = 678100 μ_F = 0.704 μ_{CR} = 0.239
 Diversity = 7.3927e-06 k_t = [3, 2, 2, 2]
 Generation 6790: Best = 8.392794e+02 FES = 679100 μ_F = 0.714 μ_{CR} = 0.228
 Diversity = 7.4988e-06 k_t = [1, 1, 1, 1]
 Generation 6800: Best = 8.392794e+02 FES = 680100 μ_F = 0.712 μ_{CR} = 0.227
 Diversity = 6.8715e-06 k_t = [4, 3, 3, 4]
 Generation 6810: Best = 8.392794e+02 FES = 681100 μ_F = 0.718 μ_{CR} = 0.229
 Diversity = 7.2503e-06 k_t = [1, 1, 1, 1]
 Generation 6820: Best = 8.392794e+02 FES = 682100 μ_F = 0.727 μ_{CR} = 0.229
 Diversity = 6.5177e-06 k_t = [3, 2, 2, 4]
 Generation 6830: Best = 8.392794e+02 FES = 683100 μ_F = 0.733 μ_{CR} = 0.225
 Diversity = 6.4946e-06 k_t = [1, 1, 1, 1]
 Generation 6840: Best = 8.392794e+02 FES = 684100 μ_F = 0.727 μ_{CR} = 0.225
 Diversity = 6.9188e-06 k_t = [2, 4, 3, 2]
 Generation 6850: Best = 8.392794e+02 FES = 685100 μ_F = 0.727 μ_{CR} = 0.229
 Diversity = 6.9832e-06 k_t = [1, 1, 1, 1]
 Generation 6860: Best = 8.392794e+02 FES = 686100 μ_F = 0.726 μ_{CR} = 0.220
 Diversity = 6.7017e-06 k_t = [4, 4, 2, 3]
 Generation 6870: Best = 8.392794e+02 FES = 687100 μ_F = 0.735 μ_{CR} = 0.216
 Diversity = 6.4159e-06 k_t = [1, 1, 1, 1]
 Generation 6880: Best = 8.392794e+02 FES = 688100 μ_F = 0.733 μ_{CR} = 0.221
 Diversity = 6.3190e-06 k_t = [2, 2, 2, 4]
 Generation 6890: Best = 8.392794e+02 FES = 689100 μ_F = 0.740 μ_{CR} = 0.219
 Diversity = 6.8609e-06 k_t = [1, 1, 1, 1]
 Generation 6900: Best = 8.392794e+02 FES = 690100 μ_F = 0.751 μ_{CR} = 0.225
 Diversity = 7.0823e-06 k_t = [2, 3, 3, 3]
 Generation 6910: Best = 8.392794e+02 FES = 691100 μ_F = 0.738 μ_{CR} = 0.228
 Diversity = 6.9848e-06 k_t = [1, 1, 1, 1]
 Generation 6920: Best = 8.392794e+02 FES = 692100 μ_F = 0.735 μ_{CR} = 0.230
 Diversity = 6.7220e-06 k_t = [2, 4, 2, 4]
 Generation 6930: Best = 8.392794e+02 FES = 693100 μ_F = 0.725 μ_{CR} = 0.241
 Diversity = 6.3582e-06 k_t = [1, 1, 1, 1]
 Generation 6940: Best = 8.392794e+02 FES = 694100 μ_F = 0.728 μ_{CR} = 0.244
 Diversity = 6.2563e-06 k_t = [4, 2, 4, 3]
 Generation 6950: Best = 8.392794e+02 FES = 695100 μ_F = 0.729 μ_{CR} = 0.244
 Diversity = 6.3164e-06 k_t = [1, 1, 1, 1]
 Generation 6960: Best = 8.392794e+02 FES = 696100 μ_F = 0.731 μ_{CR} = 0.248
 Diversity = 6.0685e-06 k_t = [4, 4, 2, 3]
 Generation 6970: Best = 8.392794e+02 FES = 697100 μ_F = 0.718 μ_{CR} = 0.247
 Diversity = 5.6102e-06 k_t = [1, 1, 1, 1]

Generation 6980: Best = 8.392794e+02 FES = 698100 μ_F = 0.708 μ_{CR} = 0.249
 Diversity = 5.2759e-06 k_t = [3, 3, 3, 2]
 Generation 6990: Best = 8.392794e+02 FES = 699100 μ_F = 0.712 μ_{CR} = 0.255
 Diversity = 5.2156e-06 k_t = [1, 1, 1, 1]
 Generation 7000: Best = 8.392794e+02 FES = 700100 μ_F = 0.711 μ_{CR} = 0.248
 Diversity = 4.4022e-06 k_t = [2, 4, 4, 4]
 Generation 7010: Best = 8.392794e+02 FES = 701100 μ_F = 0.707 μ_{CR} = 0.237
 Diversity = 4.1009e-06 k_t = [1, 1, 1, 1]
 Generation 7020: Best = 8.392794e+02 FES = 702100 μ_F = 0.714 μ_{CR} = 0.232
 Diversity = 3.8577e-06 k_t = [3, 4, 2, 2]
 Generation 7030: Best = 8.392794e+02 FES = 703100 μ_F = 0.712 μ_{CR} = 0.230
 Diversity = 3.5493e-06 k_t = [1, 1, 1, 1]
 Generation 7040: Best = 8.392794e+02 FES = 704100 μ_F = 0.715 μ_{CR} = 0.236
 Diversity = 3.1559e-06 k_t = [2, 4, 4, 4]
 Generation 7050: Best = 8.392794e+02 FES = 705100 μ_F = 0.717 μ_{CR} = 0.236
 Diversity = 3.1773e-06 k_t = [1, 1, 1, 1]
 Generation 7060: Best = 8.392794e+02 FES = 706100 μ_F = 0.710 μ_{CR} = 0.237
 Diversity = 2.9073e-06 k_t = [4, 3, 3, 4]
 Generation 7070: Best = 8.392794e+02 FES = 707100 μ_F = 0.720 μ_{CR} = 0.235
 Diversity = 2.7441e-06 k_t = [1, 1, 1, 1]
 Generation 7080: Best = 8.392794e+02 FES = 708100 μ_F = 0.721 μ_{CR} = 0.239
 Diversity = 2.8037e-06 k_t = [2, 3, 4, 2]
 Generation 7090: Best = 8.392794e+02 FES = 709100 μ_F = 0.734 μ_{CR} = 0.234
 Diversity = 2.9758e-06 k_t = [1, 1, 1, 1]
 Generation 7100: Best = 8.392794e+02 FES = 710100 μ_F = 0.726 μ_{CR} = 0.228
 Diversity = 3.0059e-06 k_t = [2, 2, 2, 2]
 Generation 7110: Best = 8.392794e+02 FES = 711100 μ_F = 0.718 μ_{CR} = 0.217
 Diversity = 3.0449e-06 k_t = [1, 1, 1, 1]
 Generation 7120: Best = 8.392794e+02 FES = 712100 μ_F = 0.707 μ_{CR} = 0.223
 Diversity = 2.6896e-06 k_t = [4, 3, 3, 4]
 Generation 7130: Best = 8.392794e+02 FES = 713100 μ_F = 0.727 μ_{CR} = 0.220
 Diversity = 2.7358e-06 k_t = [1, 1, 1, 1]
 Generation 7140: Best = 8.392794e+02 FES = 714100 μ_F = 0.738 μ_{CR} = 0.231
 Diversity = 2.9370e-06 k_t = [2, 4, 4, 2]
 Generation 7150: Best = 8.392794e+02 FES = 715100 μ_F = 0.749 μ_{CR} = 0.229
 Diversity = 3.6039e-06 k_t = [1, 1, 1, 1]
 Generation 7160: Best = 8.392794e+02 FES = 716100 μ_F = 0.746 μ_{CR} = 0.228
 Diversity = 3.4598e-06 k_t = [2, 4, 2, 3]
 Generation 7170: Best = 8.392794e+02 FES = 717100 μ_F = 0.741 μ_{CR} = 0.232
 Diversity = 3.5389e-06 k_t = [1, 1, 1, 1]
 Generation 7180: Best = 8.392794e+02 FES = 718100 μ_F = 0.739 μ_{CR} = 0.222
 Diversity = 3.2982e-06 k_t = [2, 2, 3, 3]
 Generation 7190: Best = 8.392794e+02 FES = 719100 μ_F = 0.739 μ_{CR} = 0.218
 Diversity = 3.6353e-06 k_t = [1, 1, 1, 1]
 Generation 7200: Best = 8.392794e+02 FES = 720100 μ_F = 0.733 μ_{CR} = 0.207
 Diversity = 3.5258e-06 k_t = [3, 2, 2, 2]
 Generation 7210: Best = 8.392794e+02 FES = 721100 μ_F = 0.721 μ_{CR} = 0.209
 Diversity = 3.5212e-06 k_t = [1, 1, 1, 1]
 Generation 7220: Best = 8.392794e+02 FES = 722100 μ_F = 0.729 μ_{CR} = 0.206
 Diversity = 3.3268e-06 k_t = [4, 3, 3, 2]
 Generation 7230: Best = 8.392794e+02 FES = 723100 μ_F = 0.723 μ_{CR} = 0.201
 Diversity = 3.2356e-06 k_t = [1, 1, 1, 1]
 Generation 7240: Best = 8.392794e+02 FES = 724100 μ_F = 0.719 μ_{CR} = 0.199
 Diversity = 3.0229e-06 k_t = [2, 3, 3, 3]

Generation 7250: Best = 8.392794e+02 FES = 725100 μ_F = 0.726 μ_{CR} = 0.205
 Diversity = 3.1510e-06 k_t = [1, 1, 1, 1]
 Generation 7260: Best = 8.392794e+02 FES = 726100 μ_F = 0.721 μ_{CR} = 0.211
 Diversity = 2.9653e-06 k_t = [3, 4, 4, 3]
 Generation 7270: Best = 8.392794e+02 FES = 727100 μ_F = 0.729 μ_{CR} = 0.207
 Diversity = 2.8876e-06 k_t = [1, 1, 1, 1]
 Generation 7280: Best = 8.392794e+02 FES = 728100 μ_F = 0.722 μ_{CR} = 0.198
 Diversity = 2.8371e-06 k_t = [4, 4, 4, 2]
 Generation 7290: Best = 8.392794e+02 FES = 729100 μ_F = 0.719 μ_{CR} = 0.206
 Diversity = 2.8073e-06 k_t = [1, 1, 1, 1]
 Generation 7300: Best = 8.392794e+02 FES = 730100 μ_F = 0.716 μ_{CR} = 0.200
 Diversity = 2.4974e-06 k_t = [3, 2, 2, 4]
 Generation 7310: Best = 8.392794e+02 FES = 731100 μ_F = 0.733 μ_{CR} = 0.202
 Diversity = 2.5823e-06 k_t = [1, 1, 1, 1]
 Generation 7320: Best = 8.392794e+02 FES = 732100 μ_F = 0.740 μ_{CR} = 0.200
 Diversity = 2.5124e-06 k_t = [4, 4, 4, 4]
 Generation 7330: Best = 8.392794e+02 FES = 733100 μ_F = 0.742 μ_{CR} = 0.210
 Diversity = 2.6867e-06 k_t = [1, 1, 1, 1]
 Generation 7340: Best = 8.392794e+02 FES = 734100 μ_F = 0.737 μ_{CR} = 0.212
 Diversity = 2.6412e-06 k_t = [4, 2, 2, 4]
 Generation 7350: Best = 8.392794e+02 FES = 735100 μ_F = 0.739 μ_{CR} = 0.219
 Diversity = 2.8623e-06 k_t = [1, 1, 1, 1]
 Generation 7360: Best = 8.392794e+02 FES = 736100 μ_F = 0.740 μ_{CR} = 0.225
 Diversity = 2.6617e-06 k_t = [2, 2, 3, 4]
 Generation 7370: Best = 8.392794e+02 FES = 737100 μ_F = 0.738 μ_{CR} = 0.228
 Diversity = 2.8138e-06 k_t = [1, 1, 1, 1]
 Generation 7380: Best = 8.392794e+02 FES = 738100 μ_F = 0.736 μ_{CR} = 0.228
 Diversity = 2.9558e-06 k_t = [3, 4, 2, 2]
 Generation 7390: Best = 8.392794e+02 FES = 739100 μ_F = 0.746 μ_{CR} = 0.230
 Diversity = 3.0011e-06 k_t = [1, 1, 1, 1]
 Generation 7400: Best = 8.392794e+02 FES = 740100 μ_F = 0.730 μ_{CR} = 0.236
 Diversity = 2.7562e-06 k_t = [2, 3, 3, 3]
 Generation 7410: Best = 8.392794e+02 FES = 741100 μ_F = 0.728 μ_{CR} = 0.241
 Diversity = 2.9477e-06 k_t = [1, 1, 1, 1]
 Generation 7420: Best = 8.392794e+02 FES = 742100 μ_F = 0.729 μ_{CR} = 0.241
 Diversity = 2.6875e-06 k_t = [2, 4, 3, 4]
 Generation 7430: Best = 8.392794e+02 FES = 743100 μ_F = 0.727 μ_{CR} = 0.236
 Diversity = 2.7620e-06 k_t = [1, 1, 1, 1]
 Generation 7440: Best = 8.392794e+02 FES = 744100 μ_F = 0.725 μ_{CR} = 0.229
 Diversity = 2.7920e-06 k_t = [2, 2, 3, 4]
 Generation 7450: Best = 8.392794e+02 FES = 745100 μ_F = 0.742 μ_{CR} = 0.225
 Diversity = 3.0657e-06 k_t = [1, 1, 1, 1]
 Generation 7460: Best = 8.392794e+02 FES = 746100 μ_F = 0.738 μ_{CR} = 0.221
 Diversity = 2.8323e-06 k_t = [2, 4, 4, 4]
 Generation 7470: Best = 8.392794e+02 FES = 747100 μ_F = 0.765 μ_{CR} = 0.229
 Diversity = 3.1600e-06 k_t = [1, 1, 1, 1]
 Generation 7480: Best = 8.392794e+02 FES = 748100 μ_F = 0.752 μ_{CR} = 0.223
 Diversity = 3.4543e-06 k_t = [4, 4, 4, 2]
 Generation 7490: Best = 8.392794e+02 FES = 749100 μ_F = 0.750 μ_{CR} = 0.217
 Diversity = 3.4202e-06 k_t = [1, 1, 1, 1]
 Generation 7500: Best = 8.392794e+02 FES = 750100 μ_F = 0.750 μ_{CR} = 0.220
 Diversity = 3.1340e-06 k_t = [4, 3, 4, 4]
 Generation 7510: Best = 8.392794e+02 FES = 751100 μ_F = 0.739 μ_{CR} = 0.222
 Diversity = 3.0978e-06 k_t = [1, 1, 1, 1]

Generation 7520: Best = 8.392794e+02 FES = 752100 μ_F = 0.728 μ_{CR} = 0.222
 Diversity = 2.9323e-06 k_t = [3, 2, 2, 4]
 Generation 7530: Best = 8.392794e+02 FES = 753100 μ_F = 0.735 μ_{CR} = 0.225
 Diversity = 3.1233e-06 k_t = [1, 1, 1, 1]
 Generation 7540: Best = 8.392794e+02 FES = 754100 μ_F = 0.739 μ_{CR} = 0.230
 Diversity = 3.4584e-06 k_t = [3, 2, 3, 2]
 Generation 7550: Best = 8.392794e+02 FES = 755100 μ_F = 0.725 μ_{CR} = 0.223
 Diversity = 3.6227e-06 k_t = [1, 1, 1, 1]
 Generation 7560: Best = 8.392794e+02 FES = 756100 μ_F = 0.721 μ_{CR} = 0.221
 Diversity = 3.7149e-06 k_t = [4, 3, 4, 3]
 Generation 7570: Best = 8.392794e+02 FES = 757100 μ_F = 0.714 μ_{CR} = 0.227
 Diversity = 3.7179e-06 k_t = [1, 1, 1, 1]
 Generation 7580: Best = 8.392794e+02 FES = 758100 μ_F = 0.719 μ_{CR} = 0.215
 Diversity = 3.4914e-06 k_t = [4, 2, 3, 4]
 Generation 7590: Best = 8.392794e+02 FES = 759100 μ_F = 0.704 μ_{CR} = 0.209
 Diversity = 3.4010e-06 k_t = [1, 1, 1, 1]
 Generation 7600: Best = 8.392794e+02 FES = 760100 μ_F = 0.712 μ_{CR} = 0.219
 Diversity = 3.1044e-06 k_t = [4, 2, 3, 2]
 Generation 7610: Best = 8.392794e+02 FES = 761100 μ_F = 0.703 μ_{CR} = 0.225
 Diversity = 2.8864e-06 k_t = [1, 1, 1, 1]
 Generation 7620: Best = 8.392794e+02 FES = 762100 μ_F = 0.682 μ_{CR} = 0.215
 Diversity = 2.5182e-06 k_t = [2, 3, 3, 2]
 Generation 7630: Best = 8.392794e+02 FES = 763100 μ_F = 0.675 μ_{CR} = 0.212
 Diversity = 2.4160e-06 k_t = [1, 1, 1, 1]
 Generation 7640: Best = 8.392794e+02 FES = 764100 μ_F = 0.678 μ_{CR} = 0.223
 Diversity = 2.0999e-06 k_t = [2, 3, 3, 4]
 Generation 7650: Best = 8.392794e+02 FES = 765100 μ_F = 0.691 μ_{CR} = 0.226
 Diversity = 2.1430e-06 k_t = [1, 1, 1, 1]
 Generation 7660: Best = 8.392794e+02 FES = 766100 μ_F = 0.699 μ_{CR} = 0.224
 Diversity = 2.0345e-06 k_t = [4, 3, 3, 4]
 Generation 7670: Best = 8.392794e+02 FES = 767100 μ_F = 0.703 μ_{CR} = 0.236
 Diversity = 1.9344e-06 k_t = [1, 1, 1, 1]
 Generation 7680: Best = 8.392794e+02 FES = 768100 μ_F = 0.693 μ_{CR} = 0.243
 Diversity = 1.6869e-06 k_t = [3, 2, 3, 4]
 Generation 7690: Best = 8.392794e+02 FES = 769100 μ_F = 0.683 μ_{CR} = 0.242
 Diversity = 1.5851e-06 k_t = [1, 1, 1, 1]
 Generation 7700: Best = 8.392794e+02 FES = 770100 μ_F = 0.683 μ_{CR} = 0.245
 Diversity = 1.3594e-06 k_t = [4, 2, 2, 4]
 Generation 7710: Best = 8.392794e+02 FES = 771100 μ_F = 0.689 μ_{CR} = 0.256
 Diversity = 1.3536e-06 k_t = [1, 1, 1, 1]
 Generation 7720: Best = 8.392794e+02 FES = 772100 μ_F = 0.698 μ_{CR} = 0.259
 Diversity = 1.3122e-06 k_t = [2, 2, 3, 4]
 Generation 7730: Best = 8.392794e+02 FES = 773100 μ_F = 0.701 μ_{CR} = 0.263
 Diversity = 1.3549e-06 k_t = [1, 1, 1, 1]
 Generation 7740: Best = 8.392794e+02 FES = 774100 μ_F = 0.697 μ_{CR} = 0.262
 Diversity = 1.3142e-06 k_t = [2, 3, 3, 4]
 Generation 7750: Best = 8.392794e+02 FES = 775100 μ_F = 0.703 μ_{CR} = 0.268
 Diversity = 1.3104e-06 k_t = [1, 1, 1, 1]
 Generation 7760: Best = 8.392794e+02 FES = 776100 μ_F = 0.716 μ_{CR} = 0.265
 Diversity = 1.3441e-06 k_t = [2, 4, 4, 4]
 Generation 7770: Best = 8.392794e+02 FES = 777100 μ_F = 0.732 μ_{CR} = 0.267
 Diversity = 1.4635e-06 k_t = [1, 1, 1, 1]
 Generation 7780: Best = 8.392794e+02 FES = 778100 μ_F = 0.720 μ_{CR} = 0.280
 Diversity = 1.4897e-06 k_t = [3, 3, 2, 2]

Generation 7790: Best = 8.392794e+02 FES = 779100 μ_F = 0.735 μ_{CR} = 0.286
 Diversity = 1.6390e-06 k_t = [1, 1, 1, 1]
 Generation 7800: Best = 8.392794e+02 FES = 780100 μ_F = 0.723 μ_{CR} = 0.281
 Diversity = 1.6372e-06 k_t = [4, 2, 2, 2]
 Generation 7810: Best = 8.392794e+02 FES = 781100 μ_F = 0.718 μ_{CR} = 0.280
 Diversity = 1.6665e-06 k_t = [1, 1, 1, 1]
 Generation 7820: Best = 8.392794e+02 FES = 782100 μ_F = 0.715 μ_{CR} = 0.271
 Diversity = 1.4852e-06 k_t = [3, 3, 2, 4]
 Generation 7830: Best = 8.392794e+02 FES = 783100 μ_F = 0.714 μ_{CR} = 0.278
 Diversity = 1.4159e-06 k_t = [1, 1, 1, 1]
 Generation 7840: Best = 8.392794e+02 FES = 784100 μ_F = 0.702 μ_{CR} = 0.259
 Diversity = 1.3156e-06 k_t = [3, 2, 2, 2]
 Generation 7850: Best = 8.392794e+02 FES = 785100 μ_F = 0.691 μ_{CR} = 0.253
 Diversity = 1.2962e-06 k_t = [1, 1, 1, 1]
 Generation 7860: Best = 8.392794e+02 FES = 786100 μ_F = 0.691 μ_{CR} = 0.251
 Diversity = 1.1262e-06 k_t = [2, 4, 4, 4]
 Generation 7870: Best = 8.392794e+02 FES = 787100 μ_F = 0.701 μ_{CR} = 0.252
 Diversity = 1.0954e-06 k_t = [1, 1, 1, 1]
 Generation 7880: Best = 8.392794e+02 FES = 788100 μ_F = 0.706 μ_{CR} = 0.255
 Diversity = 1.0249e-06 k_t = [4, 2, 3, 3]
 Generation 7890: Best = 8.392794e+02 FES = 789100 μ_F = 0.713 μ_{CR} = 0.257
 Diversity = 1.1026e-06 k_t = [1, 1, 1, 1]
 Generation 7900: Best = 8.392794e+02 FES = 790100 μ_F = 0.712 μ_{CR} = 0.260
 Diversity = 1.0966e-06 k_t = [3, 3, 3, 4]
 Generation 7910: Best = 8.392794e+02 FES = 791100 μ_F = 0.719 μ_{CR} = 0.257
 Diversity = 1.1187e-06 k_t = [1, 1, 1, 1]
 Generation 7920: Best = 8.392794e+02 FES = 792100 μ_F = 0.715 μ_{CR} = 0.262
 Diversity = 1.0555e-06 k_t = [2, 4, 3, 3]
 Generation 7930: Best = 8.392794e+02 FES = 793100 μ_F = 0.711 μ_{CR} = 0.260
 Diversity = 1.0648e-06 k_t = [1, 1, 1, 1]
 Generation 7940: Best = 8.392794e+02 FES = 794100 μ_F = 0.699 μ_{CR} = 0.259
 Diversity = 1.1080e-06 k_t = [2, 2, 4, 2]
 Generation 7950: Best = 8.392794e+02 FES = 795100 μ_F = 0.693 μ_{CR} = 0.267
 Diversity = 1.0826e-06 k_t = [1, 1, 1, 1]
 Generation 7960: Best = 8.392794e+02 FES = 796100 μ_F = 0.694 μ_{CR} = 0.268
 Diversity = 9.8827e-07 k_t = [4, 2, 3, 3]
 Generation 7970: Best = 8.392794e+02 FES = 797100 μ_F = 0.692 μ_{CR} = 0.269
 Diversity = 9.0615e-07 k_t = [1, 1, 1, 1]
 Generation 7980: Best = 8.392794e+02 FES = 798100 μ_F = 0.694 μ_{CR} = 0.262
 Diversity = 8.3905e-07 k_t = [3, 2, 3, 3]
 Generation 7990: Best = 8.392794e+02 FES = 799100 μ_F = 0.704 μ_{CR} = 0.259
 Diversity = 9.0139e-07 k_t = [1, 1, 1, 1]
 Generation 8000: Best = 8.392794e+02 FES = 800100 μ_F = 0.699 μ_{CR} = 0.256
 Diversity = 7.8829e-07 k_t = [3, 4, 3, 3]
 Generation 8010: Best = 8.392794e+02 FES = 801100 μ_F = 0.708 μ_{CR} = 0.256
 Diversity = 8.0155e-07 k_t = [1, 1, 1, 1]
 Generation 8020: Best = 8.392794e+02 FES = 802100 μ_F = 0.710 μ_{CR} = 0.270
 Diversity = 7.4921e-07 k_t = [4, 2, 4, 2]
 Generation 8030: Best = 8.392794e+02 FES = 803100 μ_F = 0.710 μ_{CR} = 0.260
 Diversity = 7.1157e-07 k_t = [1, 1, 1, 1]
 Generation 8040: Best = 8.392794e+02 FES = 804100 μ_F = 0.709 μ_{CR} = 0.260
 Diversity = 6.4508e-07 k_t = [2, 4, 4, 3]
 Generation 8050: Best = 8.392794e+02 FES = 805100 μ_F = 0.714 μ_{CR} = 0.263
 Diversity = 6.7430e-07 k_t = [1, 1, 1, 1]

Generation 8060: Best = 8.392794e+02 FES = 806100 μ_F = 0.721 μ_{CR} = 0.259
 Diversity = 6.7210e-07 k_t = [2, 4, 3, 3]
 Generation 8070: Best = 8.392794e+02 FES = 807100 μ_F = 0.719 μ_{CR} = 0.260
 Diversity = 6.6683e-07 k_t = [1, 1, 1, 1]
 Generation 8080: Best = 8.392794e+02 FES = 808100 μ_F = 0.710 μ_{CR} = 0.264
 Diversity = 6.2267e-07 k_t = [3, 2, 4, 3]
 Generation 8090: Best = 8.392794e+02 FES = 809100 μ_F = 0.701 μ_{CR} = 0.271
 Diversity = 6.0323e-07 k_t = [1, 1, 1, 1]
 Generation 8100: Best = 8.392794e+02 FES = 810100 μ_F = 0.707 μ_{CR} = 0.270
 Diversity = 6.1852e-07 k_t = [2, 2, 3, 2]
 Generation 8110: Best = 8.392794e+02 FES = 811100 μ_F = 0.718 μ_{CR} = 0.264
 Diversity = 6.6707e-07 k_t = [1, 1, 1, 1]
 Generation 8120: Best = 8.392794e+02 FES = 812100 μ_F = 0.695 μ_{CR} = 0.265
 Diversity = 6.1560e-07 k_t = [3, 3, 4, 2]
 Generation 8130: Best = 8.392794e+02 FES = 813100 μ_F = 0.690 μ_{CR} = 0.256
 Diversity = 5.7557e-07 k_t = [1, 1, 1, 1]
 Generation 8140: Best = 8.392794e+02 FES = 814100 μ_F = 0.706 μ_{CR} = 0.255
 Diversity = 5.4508e-07 k_t = [4, 4, 4, 4]
 Generation 8150: Best = 8.392794e+02 FES = 815100 μ_F = 0.725 μ_{CR} = 0.254
 Diversity = 5.4818e-07 k_t = [1, 1, 1, 1]
 Generation 8160: Best = 8.392794e+02 FES = 816100 μ_F = 0.729 μ_{CR} = 0.256
 Diversity = 5.1746e-07 k_t = [4, 4, 3, 3]
 Generation 8170: Best = 8.392794e+02 FES = 817100 μ_F = 0.725 μ_{CR} = 0.256
 Diversity = 5.4001e-07 k_t = [1, 1, 1, 1]
 Generation 8180: Best = 8.392794e+02 FES = 818100 μ_F = 0.733 μ_{CR} = 0.263
 Diversity = 6.3576e-07 k_t = [3, 4, 4, 2]
 Generation 8190: Best = 8.392794e+02 FES = 819100 μ_F = 0.739 μ_{CR} = 0.270
 Diversity = 7.9046e-07 k_t = [1, 1, 1, 1]
 Generation 8200: Best = 8.392794e+02 FES = 820100 μ_F = 0.744 μ_{CR} = 0.273
 Diversity = 8.8211e-07 k_t = [3, 2, 2, 4]
 Generation 8210: Best = 8.392794e+02 FES = 821100 μ_F = 0.746 μ_{CR} = 0.267
 Diversity = 9.9004e-07 k_t = [1, 1, 1, 1]
 Generation 8220: Best = 8.392794e+02 FES = 822100 μ_F = 0.740 μ_{CR} = 0.265
 Diversity = 9.6243e-07 k_t = [2, 3, 2, 4]
 Generation 8230: Best = 8.392794e+02 FES = 823100 μ_F = 0.729 μ_{CR} = 0.263
 Diversity = 9.8430e-07 k_t = [1, 1, 1, 1]
 Generation 8240: Best = 8.392794e+02 FES = 824100 μ_F = 0.715 μ_{CR} = 0.260
 Diversity = 8.3976e-07 k_t = [2, 2, 4, 3]
 Generation 8250: Best = 8.392794e+02 FES = 825100 μ_F = 0.720 μ_{CR} = 0.262
 Diversity = 8.8285e-07 k_t = [1, 1, 1, 1]
 Generation 8260: Best = 8.392794e+02 FES = 826100 μ_F = 0.717 μ_{CR} = 0.260
 Diversity = 8.0391e-07 k_t = [2, 3, 2, 2]
 Generation 8270: Best = 8.392794e+02 FES = 827100 μ_F = 0.714 μ_{CR} = 0.252
 Diversity = 7.7545e-07 k_t = [1, 1, 1, 1]
 Generation 8280: Best = 8.392794e+02 FES = 828100 μ_F = 0.692 μ_{CR} = 0.255
 Diversity = 7.1480e-07 k_t = [4, 4, 3, 2]
 Generation 8290: Best = 8.392794e+02 FES = 829100 μ_F = 0.697 μ_{CR} = 0.263
 Diversity = 7.0334e-07 k_t = [1, 1, 1, 1]
 Generation 8300: Best = 8.392794e+02 FES = 830100 μ_F = 0.690 μ_{CR} = 0.262
 Diversity = 6.0524e-07 k_t = [3, 2, 3, 2]
 Generation 8310: Best = 8.392794e+02 FES = 831100 μ_F = 0.690 μ_{CR} = 0.249
 Diversity = 6.3169e-07 k_t = [1, 1, 1, 1]
 Generation 8320: Best = 8.392794e+02 FES = 832100 μ_F = 0.682 μ_{CR} = 0.246
 Diversity = 5.9715e-07 k_t = [2, 2, 4, 2]

Generation 8330: Best = 8.392794e+02 FES = 833100 μ_F = 0.690 μ_{CR} = 0.248
 Diversity = 5.8715e-07 k_t = [1, 1, 1, 1]
 Generation 8340: Best = 8.392794e+02 FES = 834100 μ_F = 0.682 μ_{CR} = 0.253
 Diversity = 5.6400e-07 k_t = [3, 3, 2, 2]
 Generation 8350: Best = 8.392794e+02 FES = 835100 μ_F = 0.697 μ_{CR} = 0.252
 Diversity = 6.0074e-07 k_t = [1, 1, 1, 1]
 Generation 8360: Best = 8.392794e+02 FES = 836100 μ_F = 0.725 μ_{CR} = 0.264
 Diversity = 7.1097e-07 k_t = [4, 2, 3, 2]
 Generation 8370: Best = 8.392794e+02 FES = 837100 μ_F = 0.743 μ_{CR} = 0.259
 Diversity = 7.8921e-07 k_t = [1, 1, 1, 1]
 Generation 8380: Best = 8.392794e+02 FES = 838100 μ_F = 0.750 μ_{CR} = 0.258
 Diversity = 8.5241e-07 k_t = [4, 3, 4, 3]
 Generation 8390: Best = 8.392794e+02 FES = 839100 μ_F = 0.756 μ_{CR} = 0.262
 Diversity = 1.0100e-06 k_t = [1, 1, 1, 1]
 Generation 8400: Best = 8.392794e+02 FES = 840100 μ_F = 0.750 μ_{CR} = 0.257
 Diversity = 1.0201e-06 k_t = [3, 3, 3, 4]
 Generation 8410: Best = 8.392794e+02 FES = 841100 μ_F = 0.752 μ_{CR} = 0.261
 Diversity = 1.1028e-06 k_t = [1, 1, 1, 1]
 Generation 8420: Best = 8.392794e+02 FES = 842100 μ_F = 0.763 μ_{CR} = 0.255
 Diversity = 1.0581e-06 k_t = [2, 4, 4, 4]
 Generation 8430: Best = 8.392794e+02 FES = 843100 μ_F = 0.752 μ_{CR} = 0.255
 Diversity = 1.0587e-06 k_t = [1, 1, 1, 1]
 Generation 8440: Best = 8.392794e+02 FES = 844100 μ_F = 0.739 μ_{CR} = 0.256
 Diversity = 9.9577e-07 k_t = [2, 4, 3, 4]
 Generation 8450: Best = 8.392794e+02 FES = 845100 μ_F = 0.732 μ_{CR} = 0.258
 Diversity = 9.9613e-07 k_t = [1, 1, 1, 1]
 Generation 8460: Best = 8.392794e+02 FES = 846100 μ_F = 0.727 μ_{CR} = 0.255
 Diversity = 9.3526e-07 k_t = [4, 3, 4, 3]
 Generation 8470: Best = 8.392794e+02 FES = 847100 μ_F = 0.726 μ_{CR} = 0.251
 Diversity = 9.3488e-07 k_t = [1, 1, 1, 1]
 Generation 8480: Best = 8.392794e+02 FES = 848100 μ_F = 0.727 μ_{CR} = 0.257
 Diversity = 8.6541e-07 k_t = [3, 2, 4, 2]
 Generation 8490: Best = 8.392794e+02 FES = 849100 μ_F = 0.718 μ_{CR} = 0.249
 Diversity = 8.0618e-07 k_t = [1, 1, 1, 1]
 Generation 8500: Best = 8.392794e+02 FES = 850100 μ_F = 0.707 μ_{CR} = 0.256
 Diversity = 7.5676e-07 k_t = [4, 2, 4, 2]
 Generation 8510: Best = 8.392794e+02 FES = 851100 μ_F = 0.711 μ_{CR} = 0.259
 Diversity = 7.2392e-07 k_t = [1, 1, 1, 1]
 Generation 8520: Best = 8.392794e+02 FES = 852100 μ_F = 0.708 μ_{CR} = 0.268
 Diversity = 6.0589e-07 k_t = [4, 2, 3, 4]
 Generation 8530: Best = 8.392794e+02 FES = 853100 μ_F = 0.713 μ_{CR} = 0.275
 Diversity = 5.3548e-07 k_t = [1, 1, 1, 1]
 Generation 8540: Best = 8.392794e+02 FES = 854100 μ_F = 0.706 μ_{CR} = 0.280
 Diversity = 4.7479e-07 k_t = [3, 3, 2, 2]
 Generation 8550: Best = 8.392794e+02 FES = 855100 μ_F = 0.709 μ_{CR} = 0.284
 Diversity = 4.7966e-07 k_t = [1, 1, 1, 1]
 Generation 8560: Best = 8.392794e+02 FES = 856100 μ_F = 0.724 μ_{CR} = 0.291
 Diversity = 5.2657e-07 k_t = [3, 2, 2, 3]
 Generation 8570: Best = 8.392794e+02 FES = 857100 μ_F = 0.741 μ_{CR} = 0.293
 Diversity = 6.4869e-07 k_t = [1, 1, 1, 1]
 Generation 8580: Best = 8.392794e+02 FES = 858100 μ_F = 0.734 μ_{CR} = 0.286
 Diversity = 6.1756e-07 k_t = [3, 4, 4, 2]
 Generation 8590: Best = 8.392794e+02 FES = 859100 μ_F = 0.728 μ_{CR} = 0.284
 Diversity = 6.7187e-07 k_t = [1, 1, 1, 1]

Generation 8600: Best = 8.392794e+02 FES = 860100 μ_F = 0.723 μ_{CR} = 0.282
 Diversity = 6.9162e-07 k_t = [2, 3, 4, 2]
 Generation 8610: Best = 8.392794e+02 FES = 861100 μ_F = 0.708 μ_{CR} = 0.277
 Diversity = 7.6242e-07 k_t = [1, 1, 1, 1]
 Generation 8620: Best = 8.392794e+02 FES = 862100 μ_F = 0.704 μ_{CR} = 0.277
 Diversity = 6.7939e-07 k_t = [3, 4, 3, 3]
 Generation 8630: Best = 8.392794e+02 FES = 863100 μ_F = 0.707 μ_{CR} = 0.280
 Diversity = 6.7055e-07 k_t = [1, 1, 1, 1]
 Generation 8640: Best = 8.392794e+02 FES = 864100 μ_F = 0.702 μ_{CR} = 0.277
 Diversity = 5.7507e-07 k_t = [2, 2, 3, 4]
 Generation 8650: Best = 8.392794e+02 FES = 865100 μ_F = 0.694 μ_{CR} = 0.283
 Diversity = 5.6522e-07 k_t = [1, 1, 1, 1]
 Generation 8660: Best = 8.392794e+02 FES = 866100 μ_F = 0.703 μ_{CR} = 0.287
 Diversity = 5.4411e-07 k_t = [3, 3, 3, 3]
 Generation 8670: Best = 8.392794e+02 FES = 867100 μ_F = 0.701 μ_{CR} = 0.298
 Diversity = 5.2551e-07 k_t = [1, 1, 1, 1]
 Generation 8680: Best = 8.392794e+02 FES = 868100 μ_F = 0.702 μ_{CR} = 0.300
 Diversity = 5.2400e-07 k_t = [4, 3, 3, 3]
 Generation 8690: Best = 8.392794e+02 FES = 869100 μ_F = 0.702 μ_{CR} = 0.299
 Diversity = 5.0280e-07 k_t = [1, 1, 1, 1]
 Generation 8700: Best = 8.392794e+02 FES = 870100 μ_F = 0.697 μ_{CR} = 0.281
 Diversity = 4.5526e-07 k_t = [2, 2, 3, 3]
 Generation 8710: Best = 8.392794e+02 FES = 871100 μ_F = 0.684 μ_{CR} = 0.286
 Diversity = 4.2594e-07 k_t = [1, 1, 1, 1]
 Generation 8720: Best = 8.392794e+02 FES = 872100 μ_F = 0.699 μ_{CR} = 0.284
 Diversity = 4.1812e-07 k_t = [4, 3, 4, 3]
 Generation 8730: Best = 8.392794e+02 FES = 873100 μ_F = 0.708 μ_{CR} = 0.288
 Diversity = 4.0333e-07 k_t = [1, 1, 1, 1]
 Generation 8740: Best = 8.392794e+02 FES = 874100 μ_F = 0.714 μ_{CR} = 0.295
 Diversity = 4.1548e-07 k_t = [2, 2, 3, 3]
 Generation 8750: Best = 8.392794e+02 FES = 875100 μ_F = 0.727 μ_{CR} = 0.295
 Diversity = 4.6587e-07 k_t = [1, 1, 1, 1]
 Generation 8760: Best = 8.392794e+02 FES = 876100 μ_F = 0.727 μ_{CR} = 0.297
 Diversity = 4.6323e-07 k_t = [3, 3, 4, 3]
 Generation 8770: Best = 8.392794e+02 FES = 877100 μ_F = 0.716 μ_{CR} = 0.310
 Diversity = 4.7688e-07 k_t = [1, 1, 1, 1]
 Generation 8780: Best = 8.392794e+02 FES = 878100 μ_F = 0.711 μ_{CR} = 0.301
 Diversity = 4.6134e-07 k_t = [4, 4, 2, 4]
 Generation 8790: Best = 8.392794e+02 FES = 879100 μ_F = 0.712 μ_{CR} = 0.308
 Diversity = 5.2093e-07 k_t = [1, 1, 1, 1]
 Generation 8800: Best = 8.392794e+02 FES = 880100 μ_F = 0.720 μ_{CR} = 0.314
 Diversity = 5.1129e-07 k_t = [2, 4, 4, 4]
 Generation 8810: Best = 8.392794e+02 FES = 881100 μ_F = 0.718 μ_{CR} = 0.318
 Diversity = 5.5060e-07 k_t = [1, 1, 1, 1]
 Generation 8820: Best = 8.392794e+02 FES = 882100 μ_F = 0.712 μ_{CR} = 0.323
 Diversity = 5.0081e-07 k_t = [2, 2, 3, 4]
 Generation 8830: Best = 8.392794e+02 FES = 883100 μ_F = 0.687 μ_{CR} = 0.331
 Diversity = 4.7968e-07 k_t = [1, 1, 1, 1]
 Generation 8840: Best = 8.392794e+02 FES = 884100 μ_F = 0.694 μ_{CR} = 0.332
 Diversity = 4.5286e-07 k_t = [3, 2, 3, 2]
 Generation 8850: Best = 8.392794e+02 FES = 885100 μ_F = 0.696 μ_{CR} = 0.336
 Diversity = 4.6361e-07 k_t = [1, 1, 1, 1]
 Generation 8860: Best = 8.392794e+02 FES = 886100 μ_F = 0.692 μ_{CR} = 0.340
 Diversity = 4.5514e-07 k_t = [4, 3, 2, 2]

Generation 8870: Best = 8.392794e+02	FES = 887100	$\mu_F = 0.703$	$\mu_{CR} = 0.338$
Diversity = 4.5438e-07	$k_t = [1, 1, 1, 1]$		
Generation 8880: Best = 8.392794e+02	FES = 888100	$\mu_F = 0.692$	$\mu_{CR} = 0.341$
Diversity = 4.0503e-07	$k_t = [2, 2, 2, 2]$		
Generation 8890: Best = 8.392794e+02	FES = 889100	$\mu_F = 0.683$	$\mu_{CR} = 0.346$
Diversity = 3.9584e-07	$k_t = [1, 1, 1, 1]$		
Generation 8900: Best = 8.392794e+02	FES = 890100	$\mu_F = 0.687$	$\mu_{CR} = 0.351$
Diversity = 3.5630e-07	$k_t = [4, 4, 3, 3]$		
Generation 8910: Best = 8.392794e+02	FES = 891100	$\mu_F = 0.703$	$\mu_{CR} = 0.347$
Diversity = 3.4044e-07	$k_t = [1, 1, 1, 1]$		
Generation 8920: Best = 8.392794e+02	FES = 892100	$\mu_F = 0.695$	$\mu_{CR} = 0.349$
Diversity = 3.2881e-07	$k_t = [3, 4, 4, 3]$		
Generation 8930: Best = 8.392794e+02	FES = 893100	$\mu_F = 0.690$	$\mu_{CR} = 0.339$
Diversity = 3.1034e-07	$k_t = [1, 1, 1, 1]$		
Generation 8940: Best = 8.392794e+02	FES = 894100	$\mu_F = 0.688$	$\mu_{CR} = 0.336$
Diversity = 2.6324e-07	$k_t = [2, 2, 2, 4]$		
Generation 8950: Best = 8.392794e+02	FES = 895100	$\mu_F = 0.682$	$\mu_{CR} = 0.330$
Diversity = 2.6234e-07	$k_t = [1, 1, 1, 1]$		
Generation 8960: Best = 8.392794e+02	FES = 896100	$\mu_F = 0.685$	$\mu_{CR} = 0.326$
Diversity = 2.1701e-07	$k_t = [3, 4, 3, 4]$		
Generation 8970: Best = 8.392794e+02	FES = 897100	$\mu_F = 0.687$	$\mu_{CR} = 0.323$
Diversity = 2.1856e-07	$k_t = [1, 1, 1, 1]$		
Generation 8980: Best = 8.392794e+02	FES = 898100	$\mu_F = 0.702$	$\mu_{CR} = 0.336$
Diversity = 2.3261e-07	$k_t = [2, 4, 4, 3]$		
Generation 8990: Best = 8.392794e+02	FES = 899100	$\mu_F = 0.687$	$\mu_{CR} = 0.328$
Diversity = 2.2013e-07	$k_t = [1, 1, 1, 1]$		
Generation 9000: Best = 8.392794e+02	FES = 900100	$\mu_F = 0.681$	$\mu_{CR} = 0.324$
Diversity = 1.9326e-07	$k_t = [4, 3, 2, 3]$		
Generation 9010: Best = 8.392794e+02	FES = 901100	$\mu_F = 0.684$	$\mu_{CR} = 0.330$
Diversity = 1.7612e-07	$k_t = [1, 1, 1, 1]$		
Generation 9020: Best = 8.392794e+02	FES = 902100	$\mu_F = 0.693$	$\mu_{CR} = 0.333$
Diversity = 1.6356e-07	$k_t = [3, 2, 3, 4]$		
Generation 9030: Best = 8.392794e+02	FES = 903100	$\mu_F = 0.685$	$\mu_{CR} = 0.337$
Diversity = 1.5944e-07	$k_t = [1, 1, 1, 1]$		
Generation 9040: Best = 8.392794e+02	FES = 904100	$\mu_F = 0.690$	$\mu_{CR} = 0.336$
Diversity = 1.4481e-07	$k_t = [3, 3, 3, 3]$		
Generation 9050: Best = 8.392794e+02	FES = 905100	$\mu_F = 0.695$	$\mu_{CR} = 0.324$
Diversity = 1.4536e-07	$k_t = [1, 1, 1, 1]$		
Generation 9060: Best = 8.392794e+02	FES = 906100	$\mu_F = 0.709$	$\mu_{CR} = 0.321$
Diversity = 1.4040e-07	$k_t = [2, 2, 4, 4]$		
Generation 9070: Best = 8.392794e+02	FES = 907100	$\mu_F = 0.706$	$\mu_{CR} = 0.325$
Diversity = 1.4721e-07	$k_t = [1, 1, 1, 1]$		
Generation 9080: Best = 8.392794e+02	FES = 908100	$\mu_F = 0.714$	$\mu_{CR} = 0.322$
Diversity = 1.6042e-07	$k_t = [3, 4, 4, 2]$		
Generation 9090: Best = 8.392794e+02	FES = 909100	$\mu_F = 0.723$	$\mu_{CR} = 0.314$
Diversity = 1.6799e-07	$k_t = [1, 1, 1, 1]$		
Generation 9100: Best = 8.392794e+02	FES = 910100	$\mu_F = 0.724$	$\mu_{CR} = 0.317$
Diversity = 1.7423e-07	$k_t = [3, 3, 4, 2]$		
Generation 9110: Best = 8.392794e+02	FES = 911100	$\mu_F = 0.735$	$\mu_{CR} = 0.329$
Diversity = 1.9579e-07	$k_t = [1, 1, 1, 1]$		
Generation 9120: Best = 8.392794e+02	FES = 912100	$\mu_F = 0.722$	$\mu_{CR} = 0.328$
Diversity = 1.9038e-07	$k_t = [4, 2, 3, 3]$		
Generation 9130: Best = 8.392794e+02	FES = 913100	$\mu_F = 0.726$	$\mu_{CR} = 0.327$
Diversity = 1.8242e-07	$k_t = [1, 1, 1, 1]$		

Generation 9140: Best = 8.392794e+02 FES = 914100 μ_F = 0.732 μ_{CR} = 0.316
 Diversity = 1.9088e-07 k_t = [4, 4, 4, 2]
 Generation 9150: Best = 8.392794e+02 FES = 915100 μ_F = 0.741 μ_{CR} = 0.307
 Diversity = 2.1793e-07 k_t = [1, 1, 1, 1]
 Generation 9160: Best = 8.392794e+02 FES = 916100 μ_F = 0.737 μ_{CR} = 0.305
 Diversity = 2.5436e-07 k_t = [3, 4, 3, 4]
 Generation 9170: Best = 8.392794e+02 FES = 917100 μ_F = 0.742 μ_{CR} = 0.301
 Diversity = 2.8192e-07 k_t = [1, 1, 1, 1]
 Generation 9180: Best = 8.392794e+02 FES = 918100 μ_F = 0.738 μ_{CR} = 0.301
 Diversity = 2.8381e-07 k_t = [2, 3, 4, 3]
 Generation 9190: Best = 8.392794e+02 FES = 919100 μ_F = 0.732 μ_{CR} = 0.305
 Diversity = 2.9599e-07 k_t = [1, 1, 1, 1]
 Generation 9200: Best = 8.392794e+02 FES = 920100 μ_F = 0.743 μ_{CR} = 0.303
 Diversity = 2.7333e-07 k_t = [2, 3, 3, 4]
 Generation 9210: Best = 8.392794e+02 FES = 921100 μ_F = 0.743 μ_{CR} = 0.295
 Diversity = 2.7576e-07 k_t = [1, 1, 1, 1]
 Generation 9220: Best = 8.392794e+02 FES = 922100 μ_F = 0.731 μ_{CR} = 0.282
 Diversity = 2.5966e-07 k_t = [4, 2, 3, 4]
 Generation 9230: Best = 8.392794e+02 FES = 923100 μ_F = 0.737 μ_{CR} = 0.289
 Diversity = 2.5122e-07 k_t = [1, 1, 1, 1]
 Generation 9240: Best = 8.392794e+02 FES = 924100 μ_F = 0.741 μ_{CR} = 0.287
 Diversity = 2.4742e-07 k_t = [3, 3, 2, 2]
 Generation 9250: Best = 8.392794e+02 FES = 925100 μ_F = 0.731 μ_{CR} = 0.288
 Diversity = 2.5814e-07 k_t = [1, 1, 1, 1]
 Generation 9260: Best = 8.392794e+02 FES = 926100 μ_F = 0.730 μ_{CR} = 0.283
 Diversity = 2.3755e-07 k_t = [4, 4, 2, 4]
 Generation 9270: Best = 8.392794e+02 FES = 927100 μ_F = 0.726 μ_{CR} = 0.291
 Diversity = 2.1689e-07 k_t = [1, 1, 1, 1]
 Generation 9280: Best = 8.392794e+02 FES = 928100 μ_F = 0.736 μ_{CR} = 0.297
 Diversity = 2.4151e-07 k_t = [2, 4, 4, 3]
 Generation 9290: Best = 8.392794e+02 FES = 929100 μ_F = 0.735 μ_{CR} = 0.296
 Diversity = 2.3842e-07 k_t = [1, 1, 1, 1]
 Generation 9300: Best = 8.392794e+02 FES = 930100 μ_F = 0.728 μ_{CR} = 0.293
 Diversity = 2.2273e-07 k_t = [3, 4, 3, 3]
 Generation 9310: Best = 8.392794e+02 FES = 931100 μ_F = 0.729 μ_{CR} = 0.291
 Diversity = 2.1681e-07 k_t = [1, 1, 1, 1]
 Generation 9320: Best = 8.392794e+02 FES = 932100 μ_F = 0.727 μ_{CR} = 0.294
 Diversity = 2.0259e-07 k_t = [4, 3, 4, 3]
 Generation 9330: Best = 8.392794e+02 FES = 933100 μ_F = 0.726 μ_{CR} = 0.293
 Diversity = 2.0066e-07 k_t = [1, 1, 1, 1]
 Generation 9340: Best = 8.392794e+02 FES = 934100 μ_F = 0.711 μ_{CR} = 0.281
 Diversity = 1.9066e-07 k_t = [3, 2, 3, 3]
 Generation 9350: Best = 8.392794e+02 FES = 935100 μ_F = 0.714 μ_{CR} = 0.284
 Diversity = 1.9832e-07 k_t = [1, 1, 1, 1]
 Generation 9360: Best = 8.392794e+02 FES = 936100 μ_F = 0.723 μ_{CR} = 0.290
 Diversity = 2.5149e-07 k_t = [3, 2, 3, 2]
 Generation 9370: Best = 8.392794e+02 FES = 937100 μ_F = 0.719 μ_{CR} = 0.288
 Diversity = 2.5123e-07 k_t = [1, 1, 1, 1]
 Generation 9380: Best = 8.392794e+02 FES = 938100 μ_F = 0.717 μ_{CR} = 0.295
 Diversity = 2.5052e-07 k_t = [2, 4, 4, 2]
 Generation 9390: Best = 8.392794e+02 FES = 939100 μ_F = 0.710 μ_{CR} = 0.293
 Diversity = 2.2774e-07 k_t = [1, 1, 1, 1]
 Generation 9400: Best = 8.392794e+02 FES = 940100 μ_F = 0.704 μ_{CR} = 0.291
 Diversity = 1.9998e-07 k_t = [2, 3, 3, 2]

Generation 9410: Best = 8.392794e+02 FES = 941100 μ_F = 0.689 μ_{CR} = 0.281
 Diversity = 1.9235e-07 k_t = [1, 1, 1, 1]
 Generation 9420: Best = 8.392794e+02 FES = 942100 μ_F = 0.689 μ_{CR} = 0.265
 Diversity = 1.7378e-07 k_t = [4, 4, 4, 3]
 Generation 9430: Best = 8.392794e+02 FES = 943100 μ_F = 0.706 μ_{CR} = 0.262
 Diversity = 1.5967e-07 k_t = [1, 1, 1, 1]
 Generation 9440: Best = 8.392794e+02 FES = 944100 μ_F = 0.704 μ_{CR} = 0.253
 Diversity = 1.4286e-07 k_t = [2, 3, 3, 3]
 Generation 9450: Best = 8.392794e+02 FES = 945100 μ_F = 0.684 μ_{CR} = 0.241
 Diversity = 1.2683e-07 k_t = [1, 1, 1, 1]
 Generation 9460: Best = 8.392794e+02 FES = 946100 μ_F = 0.693 μ_{CR} = 0.240
 Diversity = 1.1638e-07 k_t = [4, 3, 2, 2]
 Generation 9470: Best = 8.392794e+02 FES = 947100 μ_F = 0.699 μ_{CR} = 0.239
 Diversity = 1.0940e-07 k_t = [1, 1, 1, 1]
 Generation 9480: Best = 8.392794e+02 FES = 948100 μ_F = 0.704 μ_{CR} = 0.232
 Diversity = 9.9457e-08 k_t = [2, 3, 2, 2]
 Generation 9490: Best = 8.392794e+02 FES = 949100 μ_F = 0.710 μ_{CR} = 0.224
 Diversity = 1.0450e-07 k_t = [1, 1, 1, 1]
 Generation 9500: Best = 8.392794e+02 FES = 950100 μ_F = 0.706 μ_{CR} = 0.221
 Diversity = 9.0781e-08 k_t = [2, 4, 4, 3]
 Generation 9510: Best = 8.392794e+02 FES = 951100 μ_F = 0.702 μ_{CR} = 0.205
 Diversity = 8.7624e-08 k_t = [1, 1, 1, 1]
 Generation 9520: Best = 8.392794e+02 FES = 952100 μ_F = 0.717 μ_{CR} = 0.203
 Diversity = 8.3289e-08 k_t = [2, 2, 2, 3]
 Generation 9530: Best = 8.392794e+02 FES = 953100 μ_F = 0.721 μ_{CR} = 0.201
 Diversity = 8.8539e-08 k_t = [1, 1, 1, 1]
 Generation 9540: Best = 8.392794e+02 FES = 954100 μ_F = 0.735 μ_{CR} = 0.199
 Diversity = 8.9576e-08 k_t = [3, 3, 3, 3]
 Generation 9550: Best = 8.392794e+02 FES = 955100 μ_F = 0.734 μ_{CR} = 0.191
 Diversity = 1.0027e-07 k_t = [1, 1, 1, 1]
 Generation 9560: Best = 8.392794e+02 FES = 956100 μ_F = 0.739 μ_{CR} = 0.194
 Diversity = 1.0729e-07 k_t = [4, 4, 2, 2]
 Generation 9570: Best = 8.392794e+02 FES = 957100 μ_F = 0.742 μ_{CR} = 0.200
 Diversity = 1.1369e-07 k_t = [1, 1, 1, 1]
 Generation 9580: Best = 8.392794e+02 FES = 958100 μ_F = 0.749 μ_{CR} = 0.194
 Diversity = 1.1774e-07 k_t = [4, 4, 3, 4]
 Generation 9590: Best = 8.392794e+02 FES = 959100 μ_F = 0.741 μ_{CR} = 0.202
 Diversity = 1.1225e-07 k_t = [1, 1, 1, 1]
 Generation 9600: Best = 8.392794e+02 FES = 960100 μ_F = 0.745 μ_{CR} = 0.197
 Diversity = 1.0471e-07 k_t = [3, 2, 3, 4]
 Generation 9610: Best = 8.392794e+02 FES = 961100 μ_F = 0.728 μ_{CR} = 0.193
 Diversity = 9.8765e-08 k_t = [1, 1, 1, 1]
 Generation 9620: Best = 8.392794e+02 FES = 962100 μ_F = 0.727 μ_{CR} = 0.194
 Diversity = 9.3040e-08 k_t = [3, 2, 2, 3]
 Generation 9630: Best = 8.392794e+02 FES = 963100 μ_F = 0.730 μ_{CR} = 0.196
 Diversity = 9.9536e-08 k_t = [1, 1, 1, 1]
 Generation 9640: Best = 8.392794e+02 FES = 964100 μ_F = 0.724 μ_{CR} = 0.186
 Diversity = 9.6204e-08 k_t = [2, 2, 3, 3]
 Generation 9650: Best = 8.392794e+02 FES = 965100 μ_F = 0.728 μ_{CR} = 0.184
 Diversity = 1.0081e-07 k_t = [1, 1, 1, 1]
 Generation 9660: Best = 8.392794e+02 FES = 966100 μ_F = 0.736 μ_{CR} = 0.186
 Diversity = 1.0077e-07 k_t = [3, 2, 4, 2]
 Generation 9670: Best = 8.392794e+02 FES = 967100 μ_F = 0.731 μ_{CR} = 0.183
 Diversity = 1.1040e-07 k_t = [1, 1, 1, 1]

Generation 9680: Best = 8.392794e+02 FES = 968100 μ_F = 0.728 μ_{CR} = 0.179
 Diversity = 1.0585e-07 k_t = [4, 2, 3, 4]
 Generation 9690: Best = 8.392794e+02 FES = 969100 μ_F = 0.729 μ_{CR} = 0.178
 Diversity = 1.0182e-07 k_t = [1, 1, 1, 1]
 Generation 9700: Best = 8.392794e+02 FES = 970100 μ_F = 0.727 μ_{CR} = 0.176
 Diversity = 1.0675e-07 k_t = [2, 4, 3, 2]
 Generation 9710: Best = 8.392794e+02 FES = 971100 μ_F = 0.736 μ_{CR} = 0.176
 Diversity = 1.1614e-07 k_t = [1, 1, 1, 1]
 Generation 9720: Best = 8.392794e+02 FES = 972100 μ_F = 0.746 μ_{CR} = 0.175
 Diversity = 1.1944e-07 k_t = [2, 3, 3, 4]
 Generation 9730: Best = 8.392794e+02 FES = 973100 μ_F = 0.754 μ_{CR} = 0.159
 Diversity = 1.2109e-07 k_t = [1, 1, 1, 1]
 Generation 9740: Best = 8.392794e+02 FES = 974100 μ_F = 0.750 μ_{CR} = 0.156
 Diversity = 1.1576e-07 k_t = [4, 2, 3, 4]
 Generation 9750: Best = 8.392794e+02 FES = 975100 μ_F = 0.769 μ_{CR} = 0.161
 Diversity = 1.2461e-07 k_t = [1, 1, 1, 1]
 Generation 9760: Best = 8.392794e+02 FES = 976100 μ_F = 0.772 μ_{CR} = 0.167
 Diversity = 1.2327e-07 k_t = [2, 4, 3, 4]
 Generation 9770: Best = 8.392794e+02 FES = 977100 μ_F = 0.767 μ_{CR} = 0.173
 Diversity = 1.3726e-07 k_t = [1, 1, 1, 1]
 Generation 9780: Best = 8.392794e+02 FES = 978100 μ_F = 0.763 μ_{CR} = 0.178
 Diversity = 1.5344e-07 k_t = [2, 2, 2, 2]
 Generation 9790: Best = 8.392794e+02 FES = 979100 μ_F = 0.765 μ_{CR} = 0.176
 Diversity = 1.8611e-07 k_t = [1, 1, 1, 1]
 Generation 9800: Best = 8.392794e+02 FES = 980100 μ_F = 0.765 μ_{CR} = 0.174
 Diversity = 1.7567e-07 k_t = [4, 4, 4, 4]
 Generation 9810: Best = 8.392794e+02 FES = 981100 μ_F = 0.768 μ_{CR} = 0.175
 Diversity = 1.8359e-07 k_t = [1, 1, 1, 1]
 Generation 9820: Best = 8.392794e+02 FES = 982100 μ_F = 0.745 μ_{CR} = 0.183
 Diversity = 1.7225e-07 k_t = [2, 3, 3, 2]
 Generation 9830: Best = 8.392794e+02 FES = 983100 μ_F = 0.749 μ_{CR} = 0.186
 Diversity = 1.9157e-07 k_t = [1, 1, 1, 1]
 Generation 9840: Best = 8.392794e+02 FES = 984100 μ_F = 0.752 μ_{CR} = 0.191
 Diversity = 2.0805e-07 k_t = [2, 2, 2, 2]
 Generation 9850: Best = 8.392794e+02 FES = 985100 μ_F = 0.748 μ_{CR} = 0.180
 Diversity = 2.3235e-07 k_t = [1, 1, 1, 1]
 Generation 9860: Best = 8.392794e+02 FES = 986100 μ_F = 0.751 μ_{CR} = 0.180
 Diversity = 2.4034e-07 k_t = [3, 3, 4, 4]
 Generation 9870: Best = 8.392794e+02 FES = 987100 μ_F = 0.744 μ_{CR} = 0.180
 Diversity = 2.6057e-07 k_t = [1, 1, 1, 1]
 Generation 9880: Best = 8.392794e+02 FES = 988100 μ_F = 0.738 μ_{CR} = 0.193
 Diversity = 2.6455e-07 k_t = [3, 2, 3, 3]
 Generation 9890: Best = 8.392794e+02 FES = 989100 μ_F = 0.738 μ_{CR} = 0.188
 Diversity = 2.9285e-07 k_t = [1, 1, 1, 1]
 Generation 9900: Best = 8.392794e+02 FES = 990100 μ_F = 0.739 μ_{CR} = 0.194
 Diversity = 2.6407e-07 k_t = [4, 4, 4, 4]
 Generation 9910: Best = 8.392794e+02 FES = 991100 μ_F = 0.755 μ_{CR} = 0.197
 Diversity = 2.6575e-07 k_t = [1, 1, 1, 1]
 Generation 9920: Best = 8.392794e+02 FES = 992100 μ_F = 0.749 μ_{CR} = 0.199
 Diversity = 2.9775e-07 k_t = [3, 3, 2, 2]
 Generation 9930: Best = 8.392794e+02 FES = 993100 μ_F = 0.737 μ_{CR} = 0.188
 Diversity = 3.0422e-07 k_t = [1, 1, 1, 1]
 Generation 9940: Best = 8.392794e+02 FES = 994100 μ_F = 0.732 μ_{CR} = 0.180
 Diversity = 2.8620e-07 k_t = [2, 3, 4, 4]

Generation 9950: Best = 8.392794e+02 FES = 995100 μ_F = 0.732 μ_{CR} = 0.179
 Diversity = 3.0665e-07 k_t = [1, 1, 1, 1]
 Generation 9960: Best = 8.392794e+02 FES = 996100 μ_F = 0.738 μ_{CR} = 0.184
 Diversity = 3.3462e-07 k_t = [4, 4, 2, 4]
 Generation 9970: Best = 8.392794e+02 FES = 997100 μ_F = 0.731 μ_{CR} = 0.189
 Diversity = 3.3912e-07 k_t = [1, 1, 1, 1]
 Generation 9980: Best = 8.392794e+02 FES = 998100 μ_F = 0.737 μ_{CR} = 0.183
 Diversity = 3.4020e-07 k_t = [3, 2, 3, 2]
 Generation 9990: Best = 8.392794e+02 FES = 999100 μ_F = 0.730 μ_{CR} = 0.183
 Diversity = 3.2860e-07 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 8.392794e+02

Total Function Evals = 1000000

Total Successful Updates = 322936

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Results:

Best Fitness : 8.392794e+02

Global Optimum : 5.000000e+02

Error (|best-optimum|) : 3.392794e+02

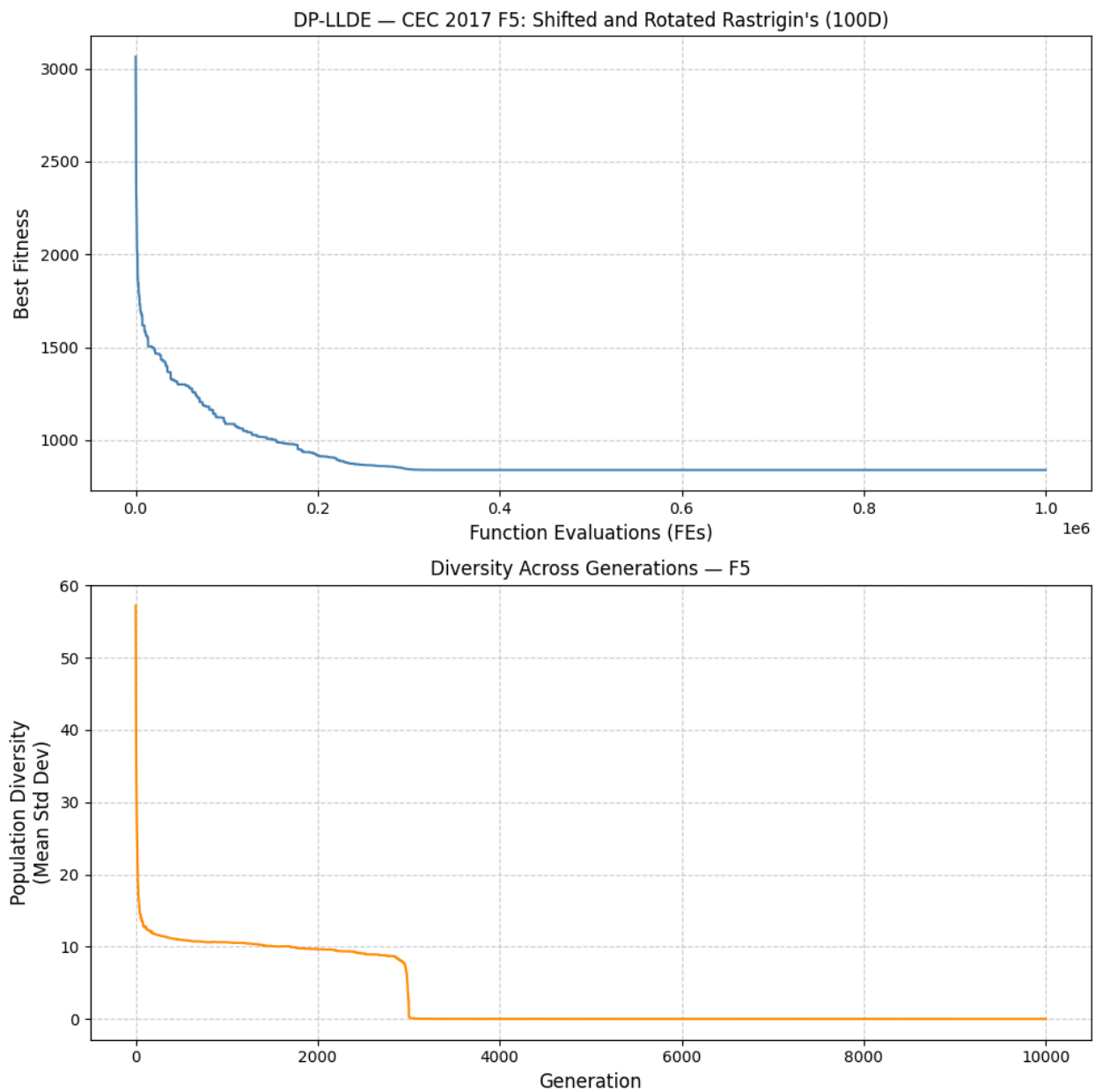
Best Solution (5 dim) : [-1.89842625 13.00419071 -25.89411105 -16.27126354
20.38846412]

D^(0) (init. diversity): 5.7220e+01

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✓ Checkpoint saved to: /content/drive/My Drive/thesis_results/all_results_checkpoint.pkl



Graph saved as convergence_F5.png

✓ Plot saved to: /content/drive/My Drive/thesis_results/plots/convergence_F5.png

```
In [ ]: # =====
# F6 – Shifted and Rotated Schaffer's F7
# =====
run_and_plot('F6', make_wrapper(simple.f6), 600.0, "Shifted and Rotated Schaff
```

```

=====
=
DP-LLDE on CEC 2017 F6 – Single Run
=====
=
Function      : Shifted and Rotated Schaffer's F7 (100D)
Bounds       : [-100, 100]
Global optimum : 600.0
Initial D^(0) = 5.7220e+01 | Diversity threshold = 5.7220e+00
Generation 0: Best Fitness = 7.936410e+02
Generation 10: Best = 7.299454e+02 FES = 1100  $\mu_F$  = 0.545  $\mu_{CR}$  = 0.353
Diversity = 3.3706e+01 k_t = [1, 1, 1, 1]
Generation 20: Best = 7.041565e+02 FES = 2100  $\mu_F$  = 0.593  $\mu_{CR}$  = 0.363
Diversity = 2.3770e+01 k_t = [2, 3, 2, 3]
Generation 30: Best = 6.978937e+02 FES = 3100  $\mu_F$  = 0.609  $\mu_{CR}$  = 0.359
Diversity = 1.8535e+01 k_t = [1, 1, 1, 1]
Generation 40: Best = 6.861867e+02 FES = 4100  $\mu_F$  = 0.618  $\mu_{CR}$  = 0.369
Diversity = 1.4980e+01 k_t = [3, 2, 3, 4]
Generation 50: Best = 6.829378e+02 FES = 5100  $\mu_F$  = 0.651  $\mu_{CR}$  = 0.370
Diversity = 1.2792e+01 k_t = [1, 1, 1, 1]
Generation 60: Best = 6.808393e+02 FES = 6100  $\mu_F$  = 0.684  $\mu_{CR}$  = 0.332
Diversity = 1.1805e+01 k_t = [4, 2, 2, 2]
Generation 70: Best = 6.784671e+02 FES = 7100  $\mu_F$  = 0.679  $\mu_{CR}$  = 0.320
Diversity = 1.1121e+01 k_t = [1, 1, 1, 1]
Generation 80: Best = 6.713181e+02 FES = 8100  $\mu_F$  = 0.686  $\mu_{CR}$  = 0.301
Diversity = 1.0110e+01 k_t = [4, 2, 2, 4]
Generation 90: Best = 6.689070e+02 FES = 9100  $\mu_F$  = 0.686  $\mu_{CR}$  = 0.288
Diversity = 1.0020e+01 k_t = [1, 1, 1, 1]
Generation 100: Best = 6.618232e+02 FES = 10100  $\mu_F$  = 0.701  $\mu_{CR}$  = 0.274
Diversity = 9.5619e+00 k_t = [2, 4, 4, 2]
Generation 110: Best = 6.562829e+02 FES = 11100  $\mu_F$  = 0.742  $\mu_{CR}$  = 0.263
Diversity = 9.1909e+00 k_t = [1, 1, 1, 1]
Generation 120: Best = 6.562829e+02 FES = 12100  $\mu_F$  = 0.752  $\mu_{CR}$  = 0.265
Diversity = 8.9640e+00 k_t = [3, 3, 3, 2]
Generation 130: Best = 6.526221e+02 FES = 13100  $\mu_F$  = 0.730  $\mu_{CR}$  = 0.244
Diversity = 8.6390e+00 k_t = [1, 1, 1, 1]
Generation 140: Best = 6.483663e+02 FES = 14100  $\mu_F$  = 0.716  $\mu_{CR}$  = 0.224
Diversity = 8.2555e+00 k_t = [2, 3, 4, 2]
Generation 150: Best = 6.483663e+02 FES = 15100  $\mu_F$  = 0.707  $\mu_{CR}$  = 0.198
Diversity = 7.7886e+00 k_t = [1, 1, 1, 1]
Generation 160: Best = 6.478516e+02 FES = 16100  $\mu_F$  = 0.675  $\mu_{CR}$  = 0.177
Diversity = 7.1821e+00 k_t = [3, 3, 4, 3]
Generation 170: Best = 6.462964e+02 FES = 17100  $\mu_F$  = 0.686  $\mu_{CR}$  = 0.142
Diversity = 6.7894e+00 k_t = [1, 1, 1, 1]
Generation 180: Best = 6.436623e+02 FES = 18100  $\mu_F$  = 0.703  $\mu_{CR}$  = 0.148
Diversity = 6.6628e+00 k_t = [3, 4, 3, 4]
Generation 190: Best = 6.436623e+02 FES = 19100  $\mu_F$  = 0.714  $\mu_{CR}$  = 0.120
Diversity = 6.3279e+00 k_t = [1, 1, 1, 1]
Generation 200: Best = 6.425124e+02 FES = 20100  $\mu_F$  = 0.743  $\mu_{CR}$  = 0.110
Diversity = 6.1112e+00 k_t = [4, 2, 3, 4]
Generation 210: Best = 6.418904e+02 FES = 21100  $\mu_F$  = 0.796  $\mu_{CR}$  = 0.087
Diversity = 5.9556e+00 k_t = [1, 1, 1, 1]
Generation 220: Best = 6.416186e+02 FES = 22100  $\mu_F$  = 0.789  $\mu_{CR}$  = 0.074
Diversity = 5.7268e+00 k_t = [3, 2, 3, 4]

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Generation 230: Best = 6.405329e+02	FES = 23100	$\mu_F = 0.760$	$\mu_{CR} = 0.059$
Diversity = 5.5893e+00	$k_t = [1, 1, 1, 1]$		
Generation 240: Best = 6.402847e+02	FES = 24100	$\mu_F = 0.786$	$\mu_{CR} = 0.058$
Diversity = 5.5335e+00	$k_t = [3, 3, 2, 2]$		
Generation 250: Best = 6.398497e+02	FES = 25100	$\mu_F = 0.770$	$\mu_{CR} = 0.051$
Diversity = 5.5068e+00	$k_t = [1, 1, 1, 1]$		
Generation 260: Best = 6.396646e+02	FES = 26100	$\mu_F = 0.765$	$\mu_{CR} = 0.053$
Diversity = 5.4747e+00	$k_t = [2, 4, 3, 2]$		
Generation 270: Best = 6.394513e+02	FES = 27100	$\mu_F = 0.735$	$\mu_{CR} = 0.048$
Diversity = 5.4919e+00	$k_t = [1, 1, 1, 1]$		
Generation 280: Best = 6.393498e+02	FES = 28100	$\mu_F = 0.685$	$\mu_{CR} = 0.053$
Diversity = 5.4304e+00	$k_t = [2, 3, 3, 3]$		
Generation 290: Best = 6.377280e+02	FES = 29100	$\mu_F = 0.695$	$\mu_{CR} = 0.050$
Diversity = 5.3547e+00	$k_t = [1, 1, 1, 1]$		
Generation 300: Best = 6.367522e+02	FES = 30100	$\mu_F = 0.688$	$\mu_{CR} = 0.051$
Diversity = 5.2632e+00	$k_t = [4, 2, 2, 2]$		
Generation 310: Best = 6.358844e+02	FES = 31100	$\mu_F = 0.735$	$\mu_{CR} = 0.059$
Diversity = 5.2266e+00	$k_t = [1, 1, 1, 1]$		
Generation 320: Best = 6.358844e+02	FES = 32100	$\mu_F = 0.776$	$\mu_{CR} = 0.038$
Diversity = 5.1378e+00	$k_t = [3, 3, 3, 4]$		
Generation 330: Best = 6.358625e+02	FES = 33100	$\mu_F = 0.788$	$\mu_{CR} = 0.047$
Diversity = 5.1288e+00	$k_t = [1, 1, 1, 1]$		
Generation 340: Best = 6.348762e+02	FES = 34100	$\mu_F = 0.753$	$\mu_{CR} = 0.045$
Diversity = 5.0447e+00	$k_t = [3, 4, 2, 3]$		
Generation 350: Best = 6.348655e+02	FES = 35100	$\mu_F = 0.764$	$\mu_{CR} = 0.040$
Diversity = 5.0604e+00	$k_t = [1, 1, 1, 1]$		
Generation 360: Best = 6.337866e+02	FES = 36100	$\mu_F = 0.756$	$\mu_{CR} = 0.032$
Diversity = 5.0033e+00	$k_t = [3, 4, 2, 3]$		
Generation 370: Best = 6.337866e+02	FES = 37100	$\mu_F = 0.804$	$\mu_{CR} = 0.035$
Diversity = 5.0058e+00	$k_t = [1, 1, 1, 1]$		
Generation 380: Best = 6.334844e+02	FES = 38100	$\mu_F = 0.808$	$\mu_{CR} = 0.033$
Diversity = 4.9855e+00	$k_t = [4, 4, 4, 3]$		
Generation 390: Best = 6.334844e+02	FES = 39100	$\mu_F = 0.788$	$\mu_{CR} = 0.033$
Diversity = 4.9330e+00	$k_t = [1, 1, 1, 1]$		
Generation 400: Best = 6.330891e+02	FES = 40100	$\mu_F = 0.761$	$\mu_{CR} = 0.043$
Diversity = 4.9095e+00	$k_t = [2, 2, 2, 3]$		
Generation 410: Best = 6.330891e+02	FES = 41100	$\mu_F = 0.712$	$\mu_{CR} = 0.040$
Diversity = 4.8965e+00	$k_t = [1, 1, 1, 1]$		
Generation 420: Best = 6.330891e+02	FES = 42100	$\mu_F = 0.716$	$\mu_{CR} = 0.048$
Diversity = 4.8764e+00	$k_t = [2, 3, 3, 4]$		
Generation 430: Best = 6.329333e+02	FES = 43100	$\mu_F = 0.764$	$\mu_{CR} = 0.039$
Diversity = 4.7969e+00	$k_t = [1, 1, 1, 1]$		
Generation 440: Best = 6.329333e+02	FES = 44100	$\mu_F = 0.645$	$\mu_{CR} = 0.035$
Diversity = 4.7662e+00	$k_t = [3, 2, 3, 2]$		
Generation 450: Best = 6.320072e+02	FES = 45100	$\mu_F = 0.665$	$\mu_{CR} = 0.037$
Diversity = 4.7258e+00	$k_t = [1, 1, 1, 1]$		
Generation 460: Best = 6.320072e+02	FES = 46100	$\mu_F = 0.714$	$\mu_{CR} = 0.041$
Diversity = 4.6795e+00	$k_t = [3, 2, 3, 4]$		
Generation 470: Best = 6.320072e+02	FES = 47100	$\mu_F = 0.698$	$\mu_{CR} = 0.039$
Diversity = 4.6569e+00	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 6.318164e+02	FES = 48100	$\mu_F = 0.652$	$\mu_{CR} = 0.041$
Diversity = 4.6688e+00	$k_t = [2, 4, 4, 3]$		
Generation 490: Best = 6.318164e+02	FES = 49100	$\mu_F = 0.622$	$\mu_{CR} = 0.042$
Diversity = 4.6668e+00	$k_t = [1, 1, 1, 1]$		

Generation 500: Best = 6.318164e+02	FES = 50100	$\mu_F = 0.644$	$\mu_{CR} = 0.040$
Diversity = 4.6775e+00	$k_t = [2, 4, 3, 4]$		
Generation 510: Best = 6.318164e+02	FES = 51100	$\mu_F = 0.733$	$\mu_{CR} = 0.039$
Diversity = 4.6656e+00	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 6.318164e+02	FES = 52100	$\mu_F = 0.650$	$\mu_{CR} = 0.041$
Diversity = 4.6464e+00	$k_t = [2, 2, 3, 3]$		
Generation 530: Best = 6.314713e+02	FES = 53100	$\mu_F = 0.678$	$\mu_{CR} = 0.039$
Diversity = 4.6480e+00	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 6.314713e+02	FES = 54100	$\mu_F = 0.680$	$\mu_{CR} = 0.033$
Diversity = 4.6443e+00	$k_t = [4, 4, 2, 3]$		
Generation 550: Best = 6.314713e+02	FES = 55100	$\mu_F = 0.662$	$\mu_{CR} = 0.039$
Diversity = 4.6164e+00	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 6.314713e+02	FES = 56100	$\mu_F = 0.626$	$\mu_{CR} = 0.039$
Diversity = 4.5220e+00	$k_t = [4, 2, 2, 4]$		
Generation 570: Best = 6.310094e+02	FES = 57100	$\mu_F = 0.583$	$\mu_{CR} = 0.036$
Diversity = 4.5168e+00	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 6.310094e+02	FES = 58100	$\mu_F = 0.554$	$\mu_{CR} = 0.038$
Diversity = 4.4958e+00	$k_t = [2, 3, 2, 4]$		
Generation 590: Best = 6.308107e+02	FES = 59100	$\mu_F = 0.616$	$\mu_{CR} = 0.028$
Diversity = 4.4763e+00	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 6.308107e+02	FES = 60100	$\mu_F = 0.576$	$\mu_{CR} = 0.036$
Diversity = 4.4653e+00	$k_t = [2, 2, 2, 4]$		
Generation 610: Best = 6.308107e+02	FES = 61100	$\mu_F = 0.649$	$\mu_{CR} = 0.026$
Diversity = 4.4540e+00	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 6.308107e+02	FES = 62100	$\mu_F = 0.637$	$\mu_{CR} = 0.039$
Diversity = 4.4404e+00	$k_t = [4, 4, 3, 3]$		
Generation 630: Best = 6.308107e+02	FES = 63100	$\mu_F = 0.722$	$\mu_{CR} = 0.033$
Diversity = 4.4422e+00	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 6.305280e+02	FES = 64100	$\mu_F = 0.661$	$\mu_{CR} = 0.024$
Diversity = 4.4223e+00	$k_t = [2, 3, 3, 4]$		
Generation 650: Best = 6.304600e+02	FES = 65100	$\mu_F = 0.617$	$\mu_{CR} = 0.022$
Diversity = 4.3408e+00	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 6.304403e+02	FES = 66100	$\mu_F = 0.604$	$\mu_{CR} = 0.035$
Diversity = 4.3401e+00	$k_t = [4, 3, 3, 2]$		
Generation 670: Best = 6.304403e+02	FES = 67100	$\mu_F = 0.617$	$\mu_{CR} = 0.035$
Diversity = 4.3596e+00	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 6.304403e+02	FES = 68100	$\mu_F = 0.542$	$\mu_{CR} = 0.033$
Diversity = 4.3455e+00	$k_t = [2, 3, 4, 2]$		
Generation 690: Best = 6.304403e+02	FES = 69100	$\mu_F = 0.585$	$\mu_{CR} = 0.033$
Diversity = 4.3396e+00	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 6.304403e+02	FES = 70100	$\mu_F = 0.597$	$\mu_{CR} = 0.038$
Diversity = 4.3380e+00	$k_t = [4, 3, 4, 3]$		
Generation 710: Best = 6.304403e+02	FES = 71100	$\mu_F = 0.487$	$\mu_{CR} = 0.039$
Diversity = 4.3183e+00	$k_t = [1, 1, 1, 1]$		
Generation 720: Best = 6.304403e+02	FES = 72100	$\mu_F = 0.500$	$\mu_{CR} = 0.042$
Diversity = 4.3084e+00	$k_t = [4, 2, 2, 3]$		
Generation 730: Best = 6.295541e+02	FES = 73100	$\mu_F = 0.511$	$\mu_{CR} = 0.036$
Diversity = 4.3095e+00	$k_t = [1, 1, 1, 1]$		
Generation 740: Best = 6.295541e+02	FES = 74100	$\mu_F = 0.495$	$\mu_{CR} = 0.037$
Diversity = 4.2942e+00	$k_t = [3, 4, 3, 4]$		
Generation 750: Best = 6.295541e+02	FES = 75100	$\mu_F = 0.518$	$\mu_{CR} = 0.041$
Diversity = 4.2957e+00	$k_t = [1, 1, 1, 1]$		
Generation 760: Best = 6.295541e+02	FES = 76100	$\mu_F = 0.526$	$\mu_{CR} = 0.032$
Diversity = 4.2502e+00	$k_t = [4, 4, 3, 4]$		

Generation 770: Best = 6.295346e+02 FES = 77100 μ_F = 0.502 μ_{CR} = 0.032
 Diversity = 4.2483e+00 k_t = [1, 1, 1, 1]
 Generation 780: Best = 6.293901e+02 FES = 78100 μ_F = 0.561 μ_{CR} = 0.036
 Diversity = 4.2404e+00 k_t = [2, 2, 2, 4]
 Generation 790: Best = 6.293901e+02 FES = 79100 μ_F = 0.582 μ_{CR} = 0.034
 Diversity = 4.2416e+00 k_t = [1, 1, 1, 1]
 Generation 800: Best = 6.293901e+02 FES = 80100 μ_F = 0.567 μ_{CR} = 0.028
 Diversity = 4.2266e+00 k_t = [3, 3, 4, 4]
 Generation 810: Best = 6.288229e+02 FES = 81100 μ_F = 0.597 μ_{CR} = 0.028
 Diversity = 4.1975e+00 k_t = [1, 1, 1, 1]
 Generation 820: Best = 6.288229e+02 FES = 82100 μ_F = 0.562 μ_{CR} = 0.035
 Diversity = 4.1113e+00 k_t = [3, 3, 4, 4]
 Generation 830: Best = 6.288070e+02 FES = 83100 μ_F = 0.548 μ_{CR} = 0.029
 Diversity = 4.1095e+00 k_t = [1, 1, 1, 1]
 Generation 840: Best = 6.287308e+02 FES = 84100 μ_F = 0.521 μ_{CR} = 0.023
 Diversity = 4.1170e+00 k_t = [3, 3, 2, 2]
 Generation 850: Best = 6.287308e+02 FES = 85100 μ_F = 0.533 μ_{CR} = 0.030
 Diversity = 4.1129e+00 k_t = [1, 1, 1, 1]
 Generation 860: Best = 6.286846e+02 FES = 86100 μ_F = 0.525 μ_{CR} = 0.031
 Diversity = 4.1126e+00 k_t = [3, 4, 3, 2]
 Generation 870: Best = 6.286846e+02 FES = 87100 μ_F = 0.528 μ_{CR} = 0.030
 Diversity = 4.1344e+00 k_t = [1, 1, 1, 1]
 Generation 880: Best = 6.284891e+02 FES = 88100 μ_F = 0.376 μ_{CR} = 0.038
 Diversity = 4.1299e+00 k_t = [2, 4, 2, 2]
 Generation 890: Best = 6.283626e+02 FES = 89100 μ_F = 0.350 μ_{CR} = 0.042
 Diversity = 4.1252e+00 k_t = [1, 1, 1, 1]
 Generation 900: Best = 6.283626e+02 FES = 90100 μ_F = 0.412 μ_{CR} = 0.035
 Diversity = 4.1016e+00 k_t = [3, 4, 4, 4]
 Generation 910: Best = 6.283626e+02 FES = 91100 μ_F = 0.351 μ_{CR} = 0.032
 Diversity = 4.0984e+00 k_t = [1, 1, 1, 1]
 Generation 920: Best = 6.283626e+02 FES = 92100 μ_F = 0.360 μ_{CR} = 0.031
 Diversity = 4.0950e+00 k_t = [3, 2, 4, 3]
 Generation 930: Best = 6.283626e+02 FES = 93100 μ_F = 0.382 μ_{CR} = 0.042
 Diversity = 4.0903e+00 k_t = [1, 1, 1, 1]
 Generation 940: Best = 6.283626e+02 FES = 94100 μ_F = 0.376 μ_{CR} = 0.040
 Diversity = 4.0882e+00 k_t = [2, 2, 2, 3]
 Generation 950: Best = 6.283626e+02 FES = 95100 μ_F = 0.433 μ_{CR} = 0.035
 Diversity = 4.0811e+00 k_t = [1, 1, 1, 1]
 Generation 960: Best = 6.282848e+02 FES = 96100 μ_F = 0.386 μ_{CR} = 0.029
 Diversity = 4.1052e+00 k_t = [4, 4, 3, 4]
 Generation 970: Best = 6.282848e+02 FES = 97100 μ_F = 0.334 μ_{CR} = 0.025
 Diversity = 4.1059e+00 k_t = [1, 1, 1, 1]
 Generation 980: Best = 6.282848e+02 FES = 98100 μ_F = 0.364 μ_{CR} = 0.039
 Diversity = 4.1038e+00 k_t = [4, 3, 2, 2]
 Generation 990: Best = 6.276936e+02 FES = 99100 μ_F = 0.343 μ_{CR} = 0.028
 Diversity = 4.1047e+00 k_t = [1, 1, 1, 1]
 Generation 1000: Best = 6.270230e+02 FES = 100100 μ_F = 0.337 μ_{CR} = 0.029
 Diversity = 4.1037e+00 k_t = [3, 4, 4, 2]
 Generation 1010: Best = 6.270230e+02 FES = 101100 μ_F = 0.426 μ_{CR} = 0.022
 Diversity = 4.0851e+00 k_t = [1, 1, 1, 1]
 Generation 1020: Best = 6.270230e+02 FES = 102100 μ_F = 0.360 μ_{CR} = 0.027
 Diversity = 4.0839e+00 k_t = [3, 3, 4, 2]
 Generation 1030: Best = 6.270230e+02 FES = 103100 μ_F = 0.379 μ_{CR} = 0.017
 Diversity = 4.0843e+00 k_t = [1, 1, 1, 1]

Generation 1040: Best = 6.270230e+02 FES = 104100 μ_F = 0.336 μ_{CR} = 0.030
 Diversity = 4.0810e+00 k_t = [3, 2, 3, 3]
 Generation 1050: Best = 6.270230e+02 FES = 105100 μ_F = 0.305 μ_{CR} = 0.031
 Diversity = 4.0787e+00 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 6.268489e+02 FES = 106100 μ_F = 0.282 μ_{CR} = 0.033
 Diversity = 4.0747e+00 k_t = [3, 2, 2, 2]
 Generation 1070: Best = 6.268489e+02 FES = 107100 μ_F = 0.246 μ_{CR} = 0.050
 Diversity = 4.0726e+00 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 6.268489e+02 FES = 108100 μ_F = 0.262 μ_{CR} = 0.047
 Diversity = 4.0452e+00 k_t = [4, 2, 2, 4]
 Generation 1090: Best = 6.268282e+02 FES = 109100 μ_F = 0.212 μ_{CR} = 0.044
 Diversity = 4.0419e+00 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 6.264289e+02 FES = 110100 μ_F = 0.272 μ_{CR} = 0.044
 Diversity = 4.0386e+00 k_t = [4, 4, 2, 4]
 Generation 1110: Best = 6.264289e+02 FES = 111100 μ_F = 0.267 μ_{CR} = 0.038
 Diversity = 4.0372e+00 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 6.264289e+02 FES = 112100 μ_F = 0.301 μ_{CR} = 0.034
 Diversity = 4.0242e+00 k_t = [3, 3, 3, 4]
 Generation 1130: Best = 6.264289e+02 FES = 113100 μ_F = 0.457 μ_{CR} = 0.027
 Diversity = 4.0252e+00 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 6.264289e+02 FES = 114100 μ_F = 0.434 μ_{CR} = 0.023
 Diversity = 4.0244e+00 k_t = [3, 3, 3, 3]
 Generation 1150: Best = 6.264289e+02 FES = 115100 μ_F = 0.407 μ_{CR} = 0.027
 Diversity = 4.0239e+00 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 6.264289e+02 FES = 116100 μ_F = 0.422 μ_{CR} = 0.024
 Diversity = 4.0251e+00 k_t = [2, 3, 3, 2]
 Generation 1170: Best = 6.263589e+02 FES = 117100 μ_F = 0.305 μ_{CR} = 0.028
 Diversity = 4.0251e+00 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 6.262651e+02 FES = 118100 μ_F = 0.305 μ_{CR} = 0.043
 Diversity = 4.0225e+00 k_t = [2, 3, 3, 2]
 Generation 1190: Best = 6.262651e+02 FES = 119100 μ_F = 0.328 μ_{CR} = 0.040
 Diversity = 4.0473e+00 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 6.262651e+02 FES = 120100 μ_F = 0.400 μ_{CR} = 0.031
 Diversity = 4.0322e+00 k_t = [3, 4, 2, 4]
 Generation 1210: Best = 6.262651e+02 FES = 121100 μ_F = 0.402 μ_{CR} = 0.035
 Diversity = 4.0358e+00 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 6.261578e+02 FES = 122100 μ_F = 0.366 μ_{CR} = 0.028
 Diversity = 4.0349e+00 k_t = [2, 4, 2, 3]
 Generation 1230: Best = 6.261213e+02 FES = 123100 μ_F = 0.394 μ_{CR} = 0.029
 Diversity = 4.0306e+00 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 6.261213e+02 FES = 124100 μ_F = 0.451 μ_{CR} = 0.028
 Diversity = 4.0305e+00 k_t = [4, 2, 3, 2]
 Generation 1250: Best = 6.261213e+02 FES = 125100 μ_F = 0.384 μ_{CR} = 0.028
 Diversity = 4.0302e+00 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 6.259072e+02 FES = 126100 μ_F = 0.394 μ_{CR} = 0.035
 Diversity = 4.0119e+00 k_t = [3, 4, 3, 4]
 Generation 1270: Best = 6.259072e+02 FES = 127100 μ_F = 0.434 μ_{CR} = 0.037
 Diversity = 4.0113e+00 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 6.259072e+02 FES = 128100 μ_F = 0.412 μ_{CR} = 0.034
 Diversity = 4.0113e+00 k_t = [2, 3, 3, 3]
 Generation 1290: Best = 6.259072e+02 FES = 129100 μ_F = 0.334 μ_{CR} = 0.043
 Diversity = 4.0110e+00 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 6.258986e+02 FES = 130100 μ_F = 0.319 μ_{CR} = 0.032
 Diversity = 4.0077e+00 k_t = [4, 4, 3, 3]

Generation 1310: Best = 6.258986e+02 FES = 131100 μ_F = 0.296 μ_{CR} = 0.048
 Diversity = 4.0075e+00 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 6.258986e+02 FES = 132100 μ_F = 0.242 μ_{CR} = 0.060
 Diversity = 4.0065e+00 k_t = [2, 3, 4, 2]
 Generation 1330: Best = 6.258009e+02 FES = 133100 μ_F = 0.202 μ_{CR} = 0.047
 Diversity = 4.0032e+00 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 6.255309e+02 FES = 134100 μ_F = 0.253 μ_{CR} = 0.037
 Diversity = 4.0000e+00 k_t = [2, 4, 4, 4]
 Generation 1350: Best = 6.255309e+02 FES = 135100 μ_F = 0.209 μ_{CR} = 0.040
 Diversity = 3.9972e+00 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 6.255144e+02 FES = 136100 μ_F = 0.270 μ_{CR} = 0.043
 Diversity = 3.9697e+00 k_t = [4, 4, 2, 4]
 Generation 1370: Best = 6.254683e+02 FES = 137100 μ_F = 0.254 μ_{CR} = 0.041
 Diversity = 3.9695e+00 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 6.254683e+02 FES = 138100 μ_F = 0.214 μ_{CR} = 0.038
 Diversity = 3.9670e+00 k_t = [2, 3, 4, 4]
 Generation 1390: Best = 6.253867e+02 FES = 139100 μ_F = 0.207 μ_{CR} = 0.032
 Diversity = 3.9660e+00 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 6.250756e+02 FES = 140100 μ_F = 0.202 μ_{CR} = 0.056
 Diversity = 3.9658e+00 k_t = [3, 3, 4, 2]
 Generation 1410: Best = 6.250585e+02 FES = 141100 μ_F = 0.196 μ_{CR} = 0.044
 Diversity = 3.9654e+00 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 6.250585e+02 FES = 142100 μ_F = 0.288 μ_{CR} = 0.042
 Diversity = 3.9638e+00 k_t = [4, 2, 3, 2]
 Generation 1430: Best = 6.249860e+02 FES = 143100 μ_F = 0.230 μ_{CR} = 0.036
 Diversity = 3.9607e+00 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 6.248622e+02 FES = 144100 μ_F = 0.184 μ_{CR} = 0.048
 Diversity = 3.9597e+00 k_t = [4, 4, 4, 3]
 Generation 1450: Best = 6.248622e+02 FES = 145100 μ_F = 0.173 μ_{CR} = 0.045
 Diversity = 3.9580e+00 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 6.248622e+02 FES = 146100 μ_F = 0.216 μ_{CR} = 0.049
 Diversity = 3.9553e+00 k_t = [4, 2, 2, 3]
 Generation 1470: Best = 6.248622e+02 FES = 147100 μ_F = 0.186 μ_{CR} = 0.050
 Diversity = 3.9534e+00 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 6.248622e+02 FES = 148100 μ_F = 0.170 μ_{CR} = 0.042
 Diversity = 3.9526e+00 k_t = [2, 2, 4, 3]
 Generation 1490: Best = 6.248622e+02 FES = 149100 μ_F = 0.124 μ_{CR} = 0.054
 Diversity = 3.9516e+00 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 6.248620e+02 FES = 150100 μ_F = 0.194 μ_{CR} = 0.051
 Diversity = 3.9483e+00 k_t = [3, 3, 3, 2]
 Generation 1510: Best = 6.248620e+02 FES = 151100 μ_F = 0.197 μ_{CR} = 0.037
 Diversity = 3.9487e+00 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 6.247523e+02 FES = 152100 μ_F = 0.219 μ_{CR} = 0.038
 Diversity = 3.9496e+00 k_t = [4, 2, 4, 4]
 Generation 1530: Best = 6.245752e+02 FES = 153100 μ_F = 0.230 μ_{CR} = 0.031
 Diversity = 3.9470e+00 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 6.245752e+02 FES = 154100 μ_F = 0.235 μ_{CR} = 0.031
 Diversity = 3.9470e+00 k_t = [2, 2, 2, 2]
 Generation 1550: Best = 6.245752e+02 FES = 155100 μ_F = 0.249 μ_{CR} = 0.035
 Diversity = 3.9468e+00 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 6.245752e+02 FES = 156100 μ_F = 0.317 μ_{CR} = 0.035
 Diversity = 3.9195e+00 k_t = [2, 4, 2, 4]
 Generation 1570: Best = 6.245612e+02 FES = 157100 μ_F = 0.242 μ_{CR} = 0.036
 Diversity = 3.9184e+00 k_t = [1, 1, 1, 1]

Generation 1580: Best = 6.245432e+02 FES = 158100 μ_F = 0.246 μ_{CR} = 0.022
 Diversity = 3.9168e+00 k_t = [4, 4, 3, 4]
 Generation 1590: Best = 6.245432e+02 FES = 159100 μ_F = 0.150 μ_{CR} = 0.049
 Diversity = 3.9158e+00 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 6.242277e+02 FES = 160100 μ_F = 0.206 μ_{CR} = 0.061
 Diversity = 3.9152e+00 k_t = [4, 3, 4, 2]
 Generation 1610: Best = 6.242277e+02 FES = 161100 μ_F = 0.163 μ_{CR} = 0.059
 Diversity = 3.9149e+00 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 6.239594e+02 FES = 162100 μ_F = 0.114 μ_{CR} = 0.073
 Diversity = 3.9138e+00 k_t = [2, 2, 4, 3]
 Generation 1630: Best = 6.236878e+02 FES = 163100 μ_F = 0.108 μ_{CR} = 0.061
 Diversity = 3.9127e+00 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 6.236878e+02 FES = 164100 μ_F = 0.187 μ_{CR} = 0.065
 Diversity = 3.8806e+00 k_t = [2, 3, 4, 4]
 Generation 1650: Best = 6.236878e+02 FES = 165100 μ_F = 0.172 μ_{CR} = 0.056
 Diversity = 3.8800e+00 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 6.236859e+02 FES = 166100 μ_F = 0.204 μ_{CR} = 0.041
 Diversity = 3.8803e+00 k_t = [4, 4, 3, 4]
 Generation 1670: Best = 6.236610e+02 FES = 167100 μ_F = 0.207 μ_{CR} = 0.030
 Diversity = 3.8806e+00 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 6.235093e+02 FES = 168100 μ_F = 0.242 μ_{CR} = 0.044
 Diversity = 3.8793e+00 k_t = [2, 2, 3, 2]
 Generation 1690: Best = 6.234116e+02 FES = 169100 μ_F = 0.187 μ_{CR} = 0.031
 Diversity = 3.8779e+00 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 6.234116e+02 FES = 170100 μ_F = 0.167 μ_{CR} = 0.035
 Diversity = 3.8773e+00 k_t = [3, 4, 3, 3]
 Generation 1710: Best = 6.233997e+02 FES = 171100 μ_F = 0.216 μ_{CR} = 0.035
 Diversity = 3.8760e+00 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 6.233997e+02 FES = 172100 μ_F = 0.206 μ_{CR} = 0.034
 Diversity = 3.8755e+00 k_t = [2, 3, 3, 3]
 Generation 1730: Best = 6.233672e+02 FES = 173100 μ_F = 0.187 μ_{CR} = 0.027
 Diversity = 3.8748e+00 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 6.232140e+02 FES = 174100 μ_F = 0.191 μ_{CR} = 0.040
 Diversity = 3.8732e+00 k_t = [4, 3, 3, 3]
 Generation 1750: Best = 6.232140e+02 FES = 175100 μ_F = 0.186 μ_{CR} = 0.037
 Diversity = 3.8725e+00 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 6.232140e+02 FES = 176100 μ_F = 0.134 μ_{CR} = 0.059
 Diversity = 3.8714e+00 k_t = [2, 3, 4, 2]
 Generation 1770: Best = 6.231489e+02 FES = 177100 μ_F = 0.107 μ_{CR} = 0.058
 Diversity = 3.8704e+00 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 6.231489e+02 FES = 178100 μ_F = 0.093 μ_{CR} = 0.073
 Diversity = 3.8689e+00 k_t = [3, 4, 4, 3]
 Generation 1790: Best = 6.231363e+02 FES = 179100 μ_F = 0.071 μ_{CR} = 0.069
 Diversity = 3.8670e+00 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 6.230646e+02 FES = 180100 μ_F = 0.174 μ_{CR} = 0.058
 Diversity = 3.8479e+00 k_t = [4, 3, 3, 4]
 Generation 1810: Best = 6.230646e+02 FES = 181100 μ_F = 0.151 μ_{CR} = 0.047
 Diversity = 3.8477e+00 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 6.230646e+02 FES = 182100 μ_F = 0.187 μ_{CR} = 0.044
 Diversity = 3.8467e+00 k_t = [2, 3, 4, 2]
 Generation 1830: Best = 6.230199e+02 FES = 183100 μ_F = 0.156 μ_{CR} = 0.038
 Diversity = 3.8459e+00 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 6.230199e+02 FES = 184100 μ_F = 0.243 μ_{CR} = 0.039
 Diversity = 3.8497e+00 k_t = [2, 4, 2, 4]

Generation 1850: Best = 6.230199e+02	FES = 185100	$\mu_F = 0.184$	$\mu_{CR} = 0.048$
Diversity = 3.8484e+00	$k_t = [1, 1, 1, 1]$		
Generation 1860: Best = 6.229655e+02	FES = 186100	$\mu_F = 0.271$	$\mu_{CR} = 0.040$
Diversity = 3.7545e+00	$k_t = [3, 2, 4, 4]$		
Generation 1870: Best = 6.229655e+02	FES = 187100	$\mu_F = 0.220$	$\mu_{CR} = 0.030$
Diversity = 3.7540e+00	$k_t = [1, 1, 1, 1]$		
Generation 1880: Best = 6.229634e+02	FES = 188100	$\mu_F = 0.181$	$\mu_{CR} = 0.029$
Diversity = 3.7532e+00	$k_t = [3, 3, 3, 3]$		
Generation 1890: Best = 6.229634e+02	FES = 189100	$\mu_F = 0.173$	$\mu_{CR} = 0.028$
Diversity = 3.7526e+00	$k_t = [1, 1, 1, 1]$		
Generation 1900: Best = 6.229252e+02	FES = 190100	$\mu_F = 0.297$	$\mu_{CR} = 0.043$
Diversity = 3.7438e+00	$k_t = [4, 4, 2, 4]$		
Generation 1910: Best = 6.228796e+02	FES = 191100	$\mu_F = 0.387$	$\mu_{CR} = 0.032$
Diversity = 3.7440e+00	$k_t = [1, 1, 1, 1]$		
Generation 1920: Best = 6.228796e+02	FES = 192100	$\mu_F = 0.334$	$\mu_{CR} = 0.032$
Diversity = 3.7436e+00	$k_t = [4, 3, 4, 2]$		
Generation 1930: Best = 6.228796e+02	FES = 193100	$\mu_F = 0.278$	$\mu_{CR} = 0.027$
Diversity = 3.7430e+00	$k_t = [1, 1, 1, 1]$		
Generation 1940: Best = 6.228796e+02	FES = 194100	$\mu_F = 0.253$	$\mu_{CR} = 0.029$
Diversity = 3.7423e+00	$k_t = [4, 2, 2, 3]$		
Generation 1950: Best = 6.228721e+02	FES = 195100	$\mu_F = 0.236$	$\mu_{CR} = 0.037$
Diversity = 3.7418e+00	$k_t = [1, 1, 1, 1]$		
Generation 1960: Best = 6.228368e+02	FES = 196100	$\mu_F = 0.188$	$\mu_{CR} = 0.041$
Diversity = 3.7415e+00	$k_t = [2, 4, 4, 3]$		
Generation 1970: Best = 6.228113e+02	FES = 197100	$\mu_F = 0.166$	$\mu_{CR} = 0.040$
Diversity = 3.7409e+00	$k_t = [1, 1, 1, 1]$		
Generation 1980: Best = 6.228113e+02	FES = 198100	$\mu_F = 0.135$	$\mu_{CR} = 0.032$
Diversity = 3.7403e+00	$k_t = [2, 3, 2, 3]$		
Generation 1990: Best = 6.228113e+02	FES = 199100	$\mu_F = 0.212$	$\mu_{CR} = 0.030$
Diversity = 3.7219e+00	$k_t = [1, 1, 1, 1]$		
Generation 2000: Best = 6.227629e+02	FES = 200100	$\mu_F = 0.221$	$\mu_{CR} = 0.026$
Diversity = 3.7218e+00	$k_t = [2, 3, 3, 2]$		
Generation 2010: Best = 6.227629e+02	FES = 201100	$\mu_F = 0.221$	$\mu_{CR} = 0.024$
Diversity = 3.7217e+00	$k_t = [1, 1, 1, 1]$		
Generation 2020: Best = 6.227629e+02	FES = 202100	$\mu_F = 0.301$	$\mu_{CR} = 0.024$
Diversity = 3.7214e+00	$k_t = [2, 3, 3, 4]$		
Generation 2030: Best = 6.227629e+02	FES = 203100	$\mu_F = 0.288$	$\mu_{CR} = 0.019$
Diversity = 3.7218e+00	$k_t = [1, 1, 1, 1]$		
Generation 2040: Best = 6.227531e+02	FES = 204100	$\mu_F = 0.258$	$\mu_{CR} = 0.026$
Diversity = 3.7210e+00	$k_t = [2, 2, 3, 3]$		
Generation 2050: Best = 6.227531e+02	FES = 205100	$\mu_F = 0.215$	$\mu_{CR} = 0.025$
Diversity = 3.7204e+00	$k_t = [1, 1, 1, 1]$		
Generation 2060: Best = 6.227531e+02	FES = 206100	$\mu_F = 0.131$	$\mu_{CR} = 0.036$
Diversity = 3.7200e+00	$k_t = [3, 4, 2, 3]$		
Generation 2070: Best = 6.227531e+02	FES = 207100	$\mu_F = 0.154$	$\mu_{CR} = 0.042$
Diversity = 3.7196e+00	$k_t = [1, 1, 1, 1]$		
Generation 2080: Best = 6.227303e+02	FES = 208100	$\mu_F = 0.248$	$\mu_{CR} = 0.051$
Diversity = 3.7184e+00	$k_t = [4, 4, 3, 2]$		
Generation 2090: Best = 6.227021e+02	FES = 209100	$\mu_F = 0.182$	$\mu_{CR} = 0.037$
Diversity = 3.7179e+00	$k_t = [1, 1, 1, 1]$		
Generation 2100: Best = 6.226970e+02	FES = 210100	$\mu_F = 0.211$	$\mu_{CR} = 0.043$
Diversity = 3.7225e+00	$k_t = [4, 4, 4, 4]$		
Generation 2110: Best = 6.226970e+02	FES = 211100	$\mu_F = 0.162$	$\mu_{CR} = 0.046$
Diversity = 3.7224e+00	$k_t = [1, 1, 1, 1]$		

Generation 2120: Best = 6.226970e+02 FES = 212100 μ_F = 0.203 μ_{CR} = 0.041
 Diversity = 3.7326e+00 k_t = [4, 2, 3, 4]
 Generation 2130: Best = 6.226970e+02 FES = 213100 μ_F = 0.207 μ_{CR} = 0.044
 Diversity = 3.7320e+00 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 6.226970e+02 FES = 214100 μ_F = 0.185 μ_{CR} = 0.032
 Diversity = 3.7318e+00 k_t = [2, 4, 4, 2]
 Generation 2150: Best = 6.226970e+02 FES = 215100 μ_F = 0.154 μ_{CR} = 0.029
 Diversity = 3.7313e+00 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 6.226861e+02 FES = 216100 μ_F = 0.136 μ_{CR} = 0.043
 Diversity = 3.7310e+00 k_t = [3, 4, 4, 2]
 Generation 2170: Best = 6.226750e+02 FES = 217100 μ_F = 0.110 μ_{CR} = 0.048
 Diversity = 3.7307e+00 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 6.226739e+02 FES = 218100 μ_F = 0.218 μ_{CR} = 0.044
 Diversity = 3.6960e+00 k_t = [2, 3, 3, 4]
 Generation 2190: Best = 6.226739e+02 FES = 219100 μ_F = 0.289 μ_{CR} = 0.031
 Diversity = 3.6954e+00 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 6.226739e+02 FES = 220100 μ_F = 0.325 μ_{CR} = 0.030
 Diversity = 3.6755e+00 k_t = [4, 4, 3, 4]
 Generation 2210: Best = 6.226739e+02 FES = 221100 μ_F = 0.296 μ_{CR} = 0.038
 Diversity = 3.6754e+00 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 6.226700e+02 FES = 222100 μ_F = 0.247 μ_{CR} = 0.039
 Diversity = 3.6748e+00 k_t = [4, 2, 2, 2]
 Generation 2230: Best = 6.226700e+02 FES = 223100 μ_F = 0.201 μ_{CR} = 0.041
 Diversity = 3.6746e+00 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 6.226700e+02 FES = 224100 μ_F = 0.192 μ_{CR} = 0.035
 Diversity = 3.6744e+00 k_t = [3, 2, 3, 2]
 Generation 2250: Best = 6.226700e+02 FES = 225100 μ_F = 0.137 μ_{CR} = 0.056
 Diversity = 3.6736e+00 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 6.226560e+02 FES = 226100 μ_F = 0.123 μ_{CR} = 0.045
 Diversity = 3.6726e+00 k_t = [3, 4, 3, 3]
 Generation 2270: Best = 6.226560e+02 FES = 227100 μ_F = 0.197 μ_{CR} = 0.040
 Diversity = 3.6586e+00 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 6.226560e+02 FES = 228100 μ_F = 0.198 μ_{CR} = 0.042
 Diversity = 3.6587e+00 k_t = [4, 3, 3, 2]
 Generation 2290: Best = 6.226560e+02 FES = 229100 μ_F = 0.177 μ_{CR} = 0.038
 Diversity = 3.6586e+00 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 6.226560e+02 FES = 230100 μ_F = 0.142 μ_{CR} = 0.041
 Diversity = 3.6580e+00 k_t = [3, 3, 4, 3]
 Generation 2310: Best = 6.226560e+02 FES = 231100 μ_F = 0.149 μ_{CR} = 0.034
 Diversity = 3.6576e+00 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 6.226253e+02 FES = 232100 μ_F = 0.190 μ_{CR} = 0.050
 Diversity = 3.6568e+00 k_t = [4, 4, 4, 3]
 Generation 2330: Best = 6.226104e+02 FES = 233100 μ_F = 0.148 μ_{CR} = 0.043
 Diversity = 3.6567e+00 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 6.226104e+02 FES = 234100 μ_F = 0.193 μ_{CR} = 0.044
 Diversity = 3.6327e+00 k_t = [3, 4, 3, 4]
 Generation 2350: Best = 6.225874e+02 FES = 235100 μ_F = 0.142 μ_{CR} = 0.054
 Diversity = 3.6312e+00 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 6.225871e+02 FES = 236100 μ_F = 0.224 μ_{CR} = 0.041
 Diversity = 3.6124e+00 k_t = [4, 3, 3, 4]
 Generation 2370: Best = 6.225853e+02 FES = 237100 μ_F = 0.274 μ_{CR} = 0.041
 Diversity = 3.6117e+00 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 6.225853e+02 FES = 238100 μ_F = 0.231 μ_{CR} = 0.025
 Diversity = 3.6108e+00 k_t = [3, 2, 3, 3]

Generation 2390: Best = 6.225853e+02 FES = 239100 μ_F = 0.249 μ_{CR} = 0.022
 Diversity = 3.6105e+00 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 6.225853e+02 FES = 240100 μ_F = 0.195 μ_{CR} = 0.028
 Diversity = 3.6100e+00 k_t = [3, 4, 3, 4]
 Generation 2410: Best = 6.225120e+02 FES = 241100 μ_F = 0.168 μ_{CR} = 0.052
 Diversity = 3.6100e+00 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 6.225120e+02 FES = 242100 μ_F = 0.159 μ_{CR} = 0.051
 Diversity = 3.6093e+00 k_t = [3, 2, 3, 3]
 Generation 2430: Best = 6.225120e+02 FES = 243100 μ_F = 0.197 μ_{CR} = 0.039
 Diversity = 3.6094e+00 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 6.225071e+02 FES = 244100 μ_F = 0.218 μ_{CR} = 0.040
 Diversity = 3.6094e+00 k_t = [2, 3, 4, 2]
 Generation 2450: Best = 6.224604e+02 FES = 245100 μ_F = 0.243 μ_{CR} = 0.027
 Diversity = 3.6094e+00 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 6.224604e+02 FES = 246100 μ_F = 0.230 μ_{CR} = 0.032
 Diversity = 3.6093e+00 k_t = [3, 3, 2, 4]
 Generation 2470: Best = 6.224604e+02 FES = 247100 μ_F = 0.294 μ_{CR} = 0.033
 Diversity = 3.6093e+00 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 6.224604e+02 FES = 248100 μ_F = 0.222 μ_{CR} = 0.038
 Diversity = 3.6090e+00 k_t = [2, 4, 2, 2]
 Generation 2490: Best = 6.224604e+02 FES = 249100 μ_F = 0.233 μ_{CR} = 0.029
 Diversity = 3.6089e+00 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 6.224604e+02 FES = 250100 μ_F = 0.205 μ_{CR} = 0.028
 Diversity = 3.6086e+00 k_t = [3, 3, 2, 2]
 Generation 2510: Best = 6.224466e+02 FES = 251100 μ_F = 0.246 μ_{CR} = 0.030
 Diversity = 3.6085e+00 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 6.224458e+02 FES = 252100 μ_F = 0.152 μ_{CR} = 0.029
 Diversity = 3.6081e+00 k_t = [2, 4, 3, 3]
 Generation 2530: Best = 6.224458e+02 FES = 253100 μ_F = 0.170 μ_{CR} = 0.036
 Diversity = 3.6078e+00 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 6.224458e+02 FES = 254100 μ_F = 0.212 μ_{CR} = 0.035
 Diversity = 3.6076e+00 k_t = [3, 4, 4, 2]
 Generation 2550: Best = 6.224458e+02 FES = 255100 μ_F = 0.204 μ_{CR} = 0.028
 Diversity = 3.6073e+00 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 6.224458e+02 FES = 256100 μ_F = 0.191 μ_{CR} = 0.032
 Diversity = 3.6070e+00 k_t = [2, 3, 3, 2]
 Generation 2570: Best = 6.224458e+02 FES = 257100 μ_F = 0.128 μ_{CR} = 0.040
 Diversity = 3.6070e+00 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 6.224458e+02 FES = 258100 μ_F = 0.197 μ_{CR} = 0.042
 Diversity = 3.6066e+00 k_t = [4, 2, 3, 2]
 Generation 2590: Best = 6.224458e+02 FES = 259100 μ_F = 0.172 μ_{CR} = 0.030
 Diversity = 3.6056e+00 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 6.224458e+02 FES = 260100 μ_F = 0.140 μ_{CR} = 0.054
 Diversity = 3.6053e+00 k_t = [4, 4, 3, 2]
 Generation 2610: Best = 6.223869e+02 FES = 261100 μ_F = 0.166 μ_{CR} = 0.038
 Diversity = 3.6016e+00 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 6.223523e+02 FES = 262100 μ_F = 0.146 μ_{CR} = 0.038
 Diversity = 3.6015e+00 k_t = [2, 2, 4, 3]
 Generation 2630: Best = 6.222964e+02 FES = 263100 μ_F = 0.221 μ_{CR} = 0.036
 Diversity = 3.6013e+00 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 6.222964e+02 FES = 264100 μ_F = 0.148 μ_{CR} = 0.052
 Diversity = 3.6007e+00 k_t = [4, 2, 4, 2]
 Generation 2650: Best = 6.222964e+02 FES = 265100 μ_F = 0.169 μ_{CR} = 0.041
 Diversity = 3.6004e+00 k_t = [1, 1, 1, 1]

Generation 2660: Best = 6.222769e+02 FES = 266100 μ_F = 0.226 μ_{CR} = 0.034
 Diversity = 3.5968e+00 k_t = [4, 2, 3, 4]
 Generation 2670: Best = 6.222033e+02 FES = 267100 μ_F = 0.207 μ_{CR} = 0.040
 Diversity = 3.5964e+00 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 6.222033e+02 FES = 268100 μ_F = 0.195 μ_{CR} = 0.041
 Diversity = 3.5962e+00 k_t = [3, 3, 2, 3]
 Generation 2690: Best = 6.222033e+02 FES = 269100 μ_F = 0.145 μ_{CR} = 0.037
 Diversity = 3.5957e+00 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 6.221905e+02 FES = 270100 μ_F = 0.134 μ_{CR} = 0.036
 Diversity = 3.5955e+00 k_t = [4, 4, 2, 2]
 Generation 2710: Best = 6.221670e+02 FES = 271100 μ_F = 0.093 μ_{CR} = 0.050
 Diversity = 3.5947e+00 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 6.221216e+02 FES = 272100 μ_F = 0.082 μ_{CR} = 0.041
 Diversity = 3.5928e+00 k_t = [3, 4, 4, 2]
 Generation 2730: Best = 6.220980e+02 FES = 273100 μ_F = 0.089 μ_{CR} = 0.043
 Diversity = 3.5919e+00 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 6.220980e+02 FES = 274100 μ_F = 0.161 μ_{CR} = 0.043
 Diversity = 3.5920e+00 k_t = [2, 2, 3, 3]
 Generation 2750: Best = 6.220642e+02 FES = 275100 μ_F = 0.146 μ_{CR} = 0.031
 Diversity = 3.5916e+00 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 6.220375e+02 FES = 276100 μ_F = 0.137 μ_{CR} = 0.047
 Diversity = 3.5899e+00 k_t = [4, 4, 2, 2]
 Generation 2770: Best = 6.218878e+02 FES = 277100 μ_F = 0.136 μ_{CR} = 0.047
 Diversity = 3.5896e+00 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 6.218878e+02 FES = 278100 μ_F = 0.207 μ_{CR} = 0.039
 Diversity = 3.5894e+00 k_t = [4, 4, 2, 3]
 Generation 2790: Best = 6.218878e+02 FES = 279100 μ_F = 0.142 μ_{CR} = 0.035
 Diversity = 3.5891e+00 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 6.218878e+02 FES = 280100 μ_F = 0.190 μ_{CR} = 0.032
 Diversity = 3.5885e+00 k_t = [4, 3, 3, 3]
 Generation 2810: Best = 6.218878e+02 FES = 281100 μ_F = 0.156 μ_{CR} = 0.033
 Diversity = 3.5883e+00 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 6.218878e+02 FES = 282100 μ_F = 0.263 μ_{CR} = 0.025
 Diversity = 3.5884e+00 k_t = [3, 2, 2, 3]
 Generation 2830: Best = 6.218344e+02 FES = 283100 μ_F = 0.203 μ_{CR} = 0.022
 Diversity = 3.5881e+00 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 6.218294e+02 FES = 284100 μ_F = 0.173 μ_{CR} = 0.030
 Diversity = 3.5882e+00 k_t = [2, 4, 2, 4]
 Generation 2850: Best = 6.218294e+02 FES = 285100 μ_F = 0.205 μ_{CR} = 0.032
 Diversity = 3.6030e+00 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 6.218294e+02 FES = 286100 μ_F = 0.137 μ_{CR} = 0.056
 Diversity = 3.6025e+00 k_t = [4, 3, 2, 3]
 Generation 2870: Best = 6.218294e+02 FES = 287100 μ_F = 0.204 μ_{CR} = 0.046
 Diversity = 3.6023e+00 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 6.218294e+02 FES = 288100 μ_F = 0.279 μ_{CR} = 0.035
 Diversity = 3.6023e+00 k_t = [3, 2, 4, 4]
 Generation 2890: Best = 6.218294e+02 FES = 289100 μ_F = 0.203 μ_{CR} = 0.044
 Diversity = 3.6023e+00 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 6.218294e+02 FES = 290100 μ_F = 0.231 μ_{CR} = 0.029
 Diversity = 3.6073e+00 k_t = [3, 2, 3, 3]
 Generation 2910: Best = 6.218294e+02 FES = 291100 μ_F = 0.251 μ_{CR} = 0.019
 Diversity = 3.5794e+00 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 6.218294e+02 FES = 292100 μ_F = 0.313 μ_{CR} = 0.022
 Diversity = 3.5793e+00 k_t = [4, 4, 4, 4]

Generation 2930: Best = 6.218294e+02 FES = 293100 μ_F = 0.322 μ_{CR} = 0.020
 Diversity = 3.5860e+00 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 6.218294e+02 FES = 294100 μ_F = 0.307 μ_{CR} = 0.023
 Diversity = 3.5860e+00 k_t = [2, 2, 4, 3]
 Generation 2950: Best = 6.218294e+02 FES = 295100 μ_F = 0.229 μ_{CR} = 0.016
 Diversity = 3.5859e+00 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 6.218294e+02 FES = 296100 μ_F = 0.264 μ_{CR} = 0.024
 Diversity = 3.5780e+00 k_t = [2, 4, 2, 4]
 Generation 2970: Best = 6.218294e+02 FES = 297100 μ_F = 0.266 μ_{CR} = 0.016
 Diversity = 3.5780e+00 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 6.217795e+02 FES = 298100 μ_F = 0.275 μ_{CR} = 0.018
 Diversity = 3.5717e+00 k_t = [2, 3, 4, 4]
 Generation 2990: Best = 6.217795e+02 FES = 299100 μ_F = 0.304 μ_{CR} = 0.020
 Diversity = 3.5717e+00 k_t = [1, 1, 1, 1]
 Generation 3000: Best = 6.217795e+02 FES = 300100 μ_F = 0.268 μ_{CR} = 0.021
 Diversity = 3.5718e+00 k_t = [4, 3, 2, 2]
 Generation 3010: Best = 6.217795e+02 FES = 301100 μ_F = 0.188 μ_{CR} = 0.034
 Diversity = 3.5718e+00 k_t = [1, 1, 1, 1]
 Generation 3020: Best = 6.217478e+02 FES = 302100 μ_F = 0.188 μ_{CR} = 0.041
 Diversity = 3.5423e+00 k_t = [3, 2, 4, 4]
 Generation 3030: Best = 6.217478e+02 FES = 303100 μ_F = 0.161 μ_{CR} = 0.035
 Diversity = 3.5389e+00 k_t = [1, 1, 1, 1]
 Generation 3040: Best = 6.217478e+02 FES = 304100 μ_F = 0.148 μ_{CR} = 0.035
 Diversity = 3.5388e+00 k_t = [3, 3, 3, 2]
 Generation 3050: Best = 6.217459e+02 FES = 305100 μ_F = 0.116 μ_{CR} = 0.041
 Diversity = 3.5387e+00 k_t = [1, 1, 1, 1]
 Generation 3060: Best = 6.217459e+02 FES = 306100 μ_F = 0.136 μ_{CR} = 0.032
 Diversity = 3.5386e+00 k_t = [3, 2, 2, 3]
 Generation 3070: Best = 6.217446e+02 FES = 307100 μ_F = 0.109 μ_{CR} = 0.041
 Diversity = 3.5381e+00 k_t = [1, 1, 1, 1]
 Generation 3080: Best = 6.217439e+02 FES = 308100 μ_F = 0.115 μ_{CR} = 0.036
 Diversity = 3.5381e+00 k_t = [4, 4, 3, 2]
 Generation 3090: Best = 6.217439e+02 FES = 309100 μ_F = 0.094 μ_{CR} = 0.034
 Diversity = 3.5381e+00 k_t = [1, 1, 1, 1]
 Generation 3100: Best = 6.217439e+02 FES = 310100 μ_F = 0.112 μ_{CR} = 0.040
 Diversity = 3.5376e+00 k_t = [4, 2, 3, 3]
 Generation 3110: Best = 6.217439e+02 FES = 311100 μ_F = 0.149 μ_{CR} = 0.038
 Diversity = 3.5374e+00 k_t = [1, 1, 1, 1]
 Generation 3120: Best = 6.217439e+02 FES = 312100 μ_F = 0.229 μ_{CR} = 0.031
 Diversity = 3.5374e+00 k_t = [2, 3, 3, 2]
 Generation 3130: Best = 6.217439e+02 FES = 313100 μ_F = 0.185 μ_{CR} = 0.032
 Diversity = 3.5327e+00 k_t = [1, 1, 1, 1]
 Generation 3140: Best = 6.217415e+02 FES = 314100 μ_F = 0.257 μ_{CR} = 0.029
 Diversity = 3.5173e+00 k_t = [2, 3, 3, 4]
 Generation 3150: Best = 6.217252e+02 FES = 315100 μ_F = 0.221 μ_{CR} = 0.024
 Diversity = 3.5173e+00 k_t = [1, 1, 1, 1]
 Generation 3160: Best = 6.217252e+02 FES = 316100 μ_F = 0.219 μ_{CR} = 0.032
 Diversity = 3.5037e+00 k_t = [2, 2, 2, 4]
 Generation 3170: Best = 6.217252e+02 FES = 317100 μ_F = 0.176 μ_{CR} = 0.035
 Diversity = 3.5038e+00 k_t = [1, 1, 1, 1]
 Generation 3180: Best = 6.217252e+02 FES = 318100 μ_F = 0.210 μ_{CR} = 0.028
 Diversity = 3.5034e+00 k_t = [2, 2, 3, 2]
 Generation 3190: Best = 6.217252e+02 FES = 319100 μ_F = 0.183 μ_{CR} = 0.035
 Diversity = 3.5031e+00 k_t = [1, 1, 1, 1]

Generation 3200: Best = 6.217252e+02 FES = 320100 μ_F = 0.159 μ_{CR} = 0.039
 Diversity = 3.5028e+00 k_t = [3, 3, 4, 4]
 Generation 3210: Best = 6.217252e+02 FES = 321100 μ_F = 0.142 μ_{CR} = 0.045
 Diversity = 3.5027e+00 k_t = [1, 1, 1, 1]
 Generation 3220: Best = 6.217234e+02 FES = 322100 μ_F = 0.112 μ_{CR} = 0.064
 Diversity = 3.5022e+00 k_t = [2, 3, 4, 4]
 Generation 3230: Best = 6.217234e+02 FES = 323100 μ_F = 0.095 μ_{CR} = 0.040
 Diversity = 3.5014e+00 k_t = [1, 1, 1, 1]
 Generation 3240: Best = 6.217234e+02 FES = 324100 μ_F = 0.112 μ_{CR} = 0.044
 Diversity = 3.5014e+00 k_t = [3, 4, 4, 3]
 Generation 3250: Best = 6.217063e+02 FES = 325100 μ_F = 0.104 μ_{CR} = 0.036
 Diversity = 3.5009e+00 k_t = [1, 1, 1, 1]
 Generation 3260: Best = 6.217063e+02 FES = 326100 μ_F = 0.109 μ_{CR} = 0.039
 Diversity = 3.5008e+00 k_t = [2, 3, 2, 3]
 Generation 3270: Best = 6.217063e+02 FES = 327100 μ_F = 0.111 μ_{CR} = 0.032
 Diversity = 3.5008e+00 k_t = [1, 1, 1, 1]
 Generation 3280: Best = 6.217009e+02 FES = 328100 μ_F = 0.082 μ_{CR} = 0.047
 Diversity = 3.5000e+00 k_t = [4, 2, 3, 3]
 Generation 3290: Best = 6.216597e+02 FES = 329100 μ_F = 0.134 μ_{CR} = 0.049
 Diversity = 3.4998e+00 k_t = [1, 1, 1, 1]
 Generation 3300: Best = 6.216597e+02 FES = 330100 μ_F = 0.193 μ_{CR} = 0.056
 Diversity = 3.5093e+00 k_t = [4, 3, 4, 4]
 Generation 3310: Best = 6.216597e+02 FES = 331100 μ_F = 0.183 μ_{CR} = 0.039
 Diversity = 3.5093e+00 k_t = [1, 1, 1, 1]
 Generation 3320: Best = 6.216597e+02 FES = 332100 μ_F = 0.214 μ_{CR} = 0.037
 Diversity = 3.5087e+00 k_t = [3, 4, 4, 2]
 Generation 3330: Best = 6.216597e+02 FES = 333100 μ_F = 0.329 μ_{CR} = 0.027
 Diversity = 3.5087e+00 k_t = [1, 1, 1, 1]
 Generation 3340: Best = 6.216597e+02 FES = 334100 μ_F = 0.354 μ_{CR} = 0.030
 Diversity = 3.5084e+00 k_t = [4, 3, 2, 3]
 Generation 3350: Best = 6.216597e+02 FES = 335100 μ_F = 0.304 μ_{CR} = 0.038
 Diversity = 3.5083e+00 k_t = [1, 1, 1, 1]
 Generation 3360: Best = 6.216597e+02 FES = 336100 μ_F = 0.281 μ_{CR} = 0.043
 Diversity = 3.4755e+00 k_t = [3, 2, 4, 4]
 Generation 3370: Best = 6.216597e+02 FES = 337100 μ_F = 0.214 μ_{CR} = 0.047
 Diversity = 3.4750e+00 k_t = [1, 1, 1, 1]
 Generation 3380: Best = 6.216597e+02 FES = 338100 μ_F = 0.272 μ_{CR} = 0.043
 Diversity = 3.4848e+00 k_t = [2, 2, 2, 4]
 Generation 3390: Best = 6.216597e+02 FES = 339100 μ_F = 0.215 μ_{CR} = 0.027
 Diversity = 3.4847e+00 k_t = [1, 1, 1, 1]
 Generation 3400: Best = 6.216597e+02 FES = 340100 μ_F = 0.261 μ_{CR} = 0.039
 Diversity = 3.4751e+00 k_t = [2, 3, 4, 4]
 Generation 3410: Best = 6.216597e+02 FES = 341100 μ_F = 0.334 μ_{CR} = 0.026
 Diversity = 3.4751e+00 k_t = [1, 1, 1, 1]
 Generation 3420: Best = 6.216597e+02 FES = 342100 μ_F = 0.350 μ_{CR} = 0.026
 Diversity = 3.4751e+00 k_t = [4, 4, 2, 3]
 Generation 3430: Best = 6.216597e+02 FES = 343100 μ_F = 0.375 μ_{CR} = 0.026
 Diversity = 3.4760e+00 k_t = [1, 1, 1, 1]
 Generation 3440: Best = 6.216597e+02 FES = 344100 μ_F = 0.327 μ_{CR} = 0.022
 Diversity = 3.4760e+00 k_t = [3, 3, 2, 3]
 Generation 3450: Best = 6.216597e+02 FES = 345100 μ_F = 0.268 μ_{CR} = 0.025
 Diversity = 3.4760e+00 k_t = [1, 1, 1, 1]
 Generation 3460: Best = 6.216597e+02 FES = 346100 μ_F = 0.342 μ_{CR} = 0.021
 Diversity = 3.4663e+00 k_t = [4, 3, 2, 4]

Generation 3470: Best = 6.216597e+02 FES = 347100 μ_F = 0.310 μ_{CR} = 0.023
 Diversity = 3.4664e+00 k_t = [1, 1, 1, 1]
 Generation 3480: Best = 6.216597e+02 FES = 348100 μ_F = 0.276 μ_{CR} = 0.030
 Diversity = 3.4501e+00 k_t = [2, 2, 4, 4]
 Generation 3490: Best = 6.216597e+02 FES = 349100 μ_F = 0.238 μ_{CR} = 0.033
 Diversity = 3.4502e+00 k_t = [1, 1, 1, 1]
 Generation 3500: Best = 6.216597e+02 FES = 350100 μ_F = 0.175 μ_{CR} = 0.041
 Diversity = 3.4501e+00 k_t = [2, 3, 4, 2]
 Generation 3510: Best = 6.216597e+02 FES = 351100 μ_F = 0.306 μ_{CR} = 0.034
 Diversity = 3.3868e+00 k_t = [1, 1, 1, 1]
 Generation 3520: Best = 6.216597e+02 FES = 352100 μ_F = 0.322 μ_{CR} = 0.045
 Diversity = 3.3557e+00 k_t = [2, 4, 2, 4]
 Generation 3530: Best = 6.216597e+02 FES = 353100 μ_F = 0.287 μ_{CR} = 0.041
 Diversity = 3.3557e+00 k_t = [1, 1, 1, 1]
 Generation 3540: Best = 6.216597e+02 FES = 354100 μ_F = 0.215 μ_{CR} = 0.049
 Diversity = 3.3555e+00 k_t = [2, 2, 2, 3]
 Generation 3550: Best = 6.216263e+02 FES = 355100 μ_F = 0.234 μ_{CR} = 0.039
 Diversity = 3.3454e+00 k_t = [1, 1, 1, 1]
 Generation 3560: Best = 6.216263e+02 FES = 356100 μ_F = 0.205 μ_{CR} = 0.049
 Diversity = 3.3449e+00 k_t = [2, 3, 2, 3]
 Generation 3570: Best = 6.216206e+02 FES = 357100 μ_F = 0.170 μ_{CR} = 0.042
 Diversity = 3.3449e+00 k_t = [1, 1, 1, 1]
 Generation 3580: Best = 6.216190e+02 FES = 358100 μ_F = 0.166 μ_{CR} = 0.035
 Diversity = 3.3449e+00 k_t = [4, 3, 2, 4]
 Generation 3590: Best = 6.215894e+02 FES = 359100 μ_F = 0.145 μ_{CR} = 0.036
 Diversity = 3.3436e+00 k_t = [1, 1, 1, 1]
 Generation 3600: Best = 6.215859e+02 FES = 360100 μ_F = 0.172 μ_{CR} = 0.051
 Diversity = 3.3176e+00 k_t = [3, 4, 2, 4]
 Generation 3610: Best = 6.215859e+02 FES = 361100 μ_F = 0.162 μ_{CR} = 0.046
 Diversity = 3.3167e+00 k_t = [1, 1, 1, 1]
 Generation 3620: Best = 6.215859e+02 FES = 362100 μ_F = 0.203 μ_{CR} = 0.045
 Diversity = 3.3216e+00 k_t = [4, 3, 4, 4]
 Generation 3630: Best = 6.215859e+02 FES = 363100 μ_F = 0.121 μ_{CR} = 0.043
 Diversity = 3.3215e+00 k_t = [1, 1, 1, 1]
 Generation 3640: Best = 6.215685e+02 FES = 364100 μ_F = 0.118 μ_{CR} = 0.044
 Diversity = 3.3213e+00 k_t = [3, 4, 3, 3]
 Generation 3650: Best = 6.215685e+02 FES = 365100 μ_F = 0.126 μ_{CR} = 0.044
 Diversity = 3.3208e+00 k_t = [1, 1, 1, 1]
 Generation 3660: Best = 6.215685e+02 FES = 366100 μ_F = 0.076 μ_{CR} = 0.050
 Diversity = 3.3200e+00 k_t = [4, 2, 4, 2]
 Generation 3670: Best = 6.215681e+02 FES = 367100 μ_F = 0.127 μ_{CR} = 0.041
 Diversity = 3.3196e+00 k_t = [1, 1, 1, 1]
 Generation 3680: Best = 6.215384e+02 FES = 368100 μ_F = 0.101 μ_{CR} = 0.057
 Diversity = 3.3197e+00 k_t = [2, 3, 2, 2]
 Generation 3690: Best = 6.215188e+02 FES = 369100 μ_F = 0.103 μ_{CR} = 0.045
 Diversity = 3.3197e+00 k_t = [1, 1, 1, 1]
 Generation 3700: Best = 6.215188e+02 FES = 370100 μ_F = 0.074 μ_{CR} = 0.061
 Diversity = 3.3187e+00 k_t = [4, 2, 2, 2]
 Generation 3710: Best = 6.215188e+02 FES = 371100 μ_F = 0.114 μ_{CR} = 0.059
 Diversity = 3.3186e+00 k_t = [1, 1, 1, 1]
 Generation 3720: Best = 6.215188e+02 FES = 372100 μ_F = 0.063 μ_{CR} = 0.051
 Diversity = 3.3186e+00 k_t = [2, 3, 3, 2]
 Generation 3730: Best = 6.215188e+02 FES = 373100 μ_F = 0.098 μ_{CR} = 0.048
 Diversity = 3.3176e+00 k_t = [1, 1, 1, 1]

Generation 3740: Best = 6.215122e+02 FES = 374100 μ_F = 0.194 μ_{CR} = 0.050
 Diversity = 3.3016e+00 k_t = [3, 2, 3, 4]
 Generation 3750: Best = 6.215122e+02 FES = 375100 μ_F = 0.158 μ_{CR} = 0.053
 Diversity = 3.3015e+00 k_t = [1, 1, 1, 1]
 Generation 3760: Best = 6.215122e+02 FES = 376100 μ_F = 0.180 μ_{CR} = 0.044
 Diversity = 3.2843e+00 k_t = [4, 2, 4, 4]
 Generation 3770: Best = 6.215032e+02 FES = 377100 μ_F = 0.128 μ_{CR} = 0.039
 Diversity = 3.2843e+00 k_t = [1, 1, 1, 1]
 Generation 3780: Best = 6.215032e+02 FES = 378100 μ_F = 0.111 μ_{CR} = 0.046
 Diversity = 3.2845e+00 k_t = [2, 4, 2, 3]
 Generation 3790: Best = 6.215032e+02 FES = 379100 μ_F = 0.168 μ_{CR} = 0.036
 Diversity = 3.2842e+00 k_t = [1, 1, 1, 1]
 Generation 3800: Best = 6.215010e+02 FES = 380100 μ_F = 0.218 μ_{CR} = 0.024
 Diversity = 3.2843e+00 k_t = [4, 3, 3, 2]
 Generation 3810: Best = 6.214672e+02 FES = 381100 μ_F = 0.160 μ_{CR} = 0.029
 Diversity = 3.2843e+00 k_t = [1, 1, 1, 1]
 Generation 3820: Best = 6.214672e+02 FES = 382100 μ_F = 0.137 μ_{CR} = 0.042
 Diversity = 3.2843e+00 k_t = [2, 4, 2, 3]
 Generation 3830: Best = 6.214672e+02 FES = 383100 μ_F = 0.100 μ_{CR} = 0.045
 Diversity = 3.2842e+00 k_t = [1, 1, 1, 1]
 Generation 3840: Best = 6.214672e+02 FES = 384100 μ_F = 0.080 μ_{CR} = 0.043
 Diversity = 3.2838e+00 k_t = [2, 4, 4, 2]
 Generation 3850: Best = 6.214672e+02 FES = 385100 μ_F = 0.061 μ_{CR} = 0.052
 Diversity = 3.2838e+00 k_t = [1, 1, 1, 1]
 Generation 3860: Best = 6.214672e+02 FES = 386100 μ_F = 0.079 μ_{CR} = 0.061
 Diversity = 3.2827e+00 k_t = [2, 4, 3, 3]
 Generation 3870: Best = 6.214672e+02 FES = 387100 μ_F = 0.075 μ_{CR} = 0.067
 Diversity = 3.2826e+00 k_t = [1, 1, 1, 1]
 Generation 3880: Best = 6.214672e+02 FES = 388100 μ_F = 0.164 μ_{CR} = 0.085
 Diversity = 3.2784e+00 k_t = [3, 4, 3, 4]
 Generation 3890: Best = 6.214643e+02 FES = 389100 μ_F = 0.130 μ_{CR} = 0.068
 Diversity = 3.2784e+00 k_t = [1, 1, 1, 1]
 Generation 3900: Best = 6.214643e+02 FES = 390100 μ_F = 0.100 μ_{CR} = 0.056
 Diversity = 3.2782e+00 k_t = [2, 3, 2, 2]
 Generation 3910: Best = 6.214581e+02 FES = 391100 μ_F = 0.075 μ_{CR} = 0.049
 Diversity = 3.2778e+00 k_t = [1, 1, 1, 1]
 Generation 3920: Best = 6.214217e+02 FES = 392100 μ_F = 0.085 μ_{CR} = 0.044
 Diversity = 3.2777e+00 k_t = [4, 3, 3, 4]
 Generation 3930: Best = 6.214217e+02 FES = 393100 μ_F = 0.181 μ_{CR} = 0.052
 Diversity = 3.2682e+00 k_t = [1, 1, 1, 1]
 Generation 3940: Best = 6.214217e+02 FES = 394100 μ_F = 0.189 μ_{CR} = 0.051
 Diversity = 3.2679e+00 k_t = [4, 3, 2, 3]
 Generation 3950: Best = 6.214130e+02 FES = 395100 μ_F = 0.166 μ_{CR} = 0.045
 Diversity = 3.2680e+00 k_t = [1, 1, 1, 1]
 Generation 3960: Best = 6.214020e+02 FES = 396100 μ_F = 0.155 μ_{CR} = 0.045
 Diversity = 3.2677e+00 k_t = [3, 2, 4, 2]
 Generation 3970: Best = 6.214020e+02 FES = 397100 μ_F = 0.134 μ_{CR} = 0.042
 Diversity = 3.2677e+00 k_t = [1, 1, 1, 1]
 Generation 3980: Best = 6.214020e+02 FES = 398100 μ_F = 0.113 μ_{CR} = 0.041
 Diversity = 3.2676e+00 k_t = [3, 4, 4, 3]
 Generation 3990: Best = 6.214016e+02 FES = 399100 μ_F = 0.110 μ_{CR} = 0.026
 Diversity = 3.2674e+00 k_t = [1, 1, 1, 1]
 Generation 4000: Best = 6.213967e+02 FES = 400100 μ_F = 0.101 μ_{CR} = 0.029
 Diversity = 3.2674e+00 k_t = [2, 2, 3, 2]

Generation 4010: Best = 6.213967e+02 FES = 401100 μ_F = 0.146 μ_{CR} = 0.036
 Diversity = 3.2676e+00 k_t = [1, 1, 1, 1]
 Generation 4020: Best = 6.213928e+02 FES = 402100 μ_F = 0.144 μ_{CR} = 0.043
 Diversity = 3.2674e+00 k_t = [4, 4, 3, 3]
 Generation 4030: Best = 6.213899e+02 FES = 403100 μ_F = 0.155 μ_{CR} = 0.040
 Diversity = 3.2670e+00 k_t = [1, 1, 1, 1]
 Generation 4040: Best = 6.213864e+02 FES = 404100 μ_F = 0.095 μ_{CR} = 0.059
 Diversity = 3.2670e+00 k_t = [2, 2, 3, 2]
 Generation 4050: Best = 6.213836e+02 FES = 405100 μ_F = 0.085 μ_{CR} = 0.040
 Diversity = 3.2668e+00 k_t = [1, 1, 1, 1]
 Generation 4060: Best = 6.213836e+02 FES = 406100 μ_F = 0.075 μ_{CR} = 0.050
 Diversity = 3.2663e+00 k_t = [4, 2, 4, 3]
 Generation 4070: Best = 6.213717e+02 FES = 407100 μ_F = 0.093 μ_{CR} = 0.041
 Diversity = 3.2661e+00 k_t = [1, 1, 1, 1]
 Generation 4080: Best = 6.213473e+02 FES = 408100 μ_F = 0.101 μ_{CR} = 0.040
 Diversity = 3.2654e+00 k_t = [4, 2, 2, 4]
 Generation 4090: Best = 6.213473e+02 FES = 409100 μ_F = 0.096 μ_{CR} = 0.049
 Diversity = 3.2653e+00 k_t = [1, 1, 1, 1]
 Generation 4100: Best = 6.213277e+02 FES = 410100 μ_F = 0.114 μ_{CR} = 0.051
 Diversity = 3.2652e+00 k_t = [4, 3, 4, 3]
 Generation 4110: Best = 6.213277e+02 FES = 411100 μ_F = 0.097 μ_{CR} = 0.049
 Diversity = 3.2649e+00 k_t = [1, 1, 1, 1]
 Generation 4120: Best = 6.212783e+02 FES = 412100 μ_F = 0.128 μ_{CR} = 0.072
 Diversity = 3.2546e+00 k_t = [2, 4, 3, 4]
 Generation 4130: Best = 6.212783e+02 FES = 413100 μ_F = 0.116 μ_{CR} = 0.056
 Diversity = 3.2545e+00 k_t = [1, 1, 1, 1]
 Generation 4140: Best = 6.212601e+02 FES = 414100 μ_F = 0.119 μ_{CR} = 0.058
 Diversity = 3.2545e+00 k_t = [3, 2, 3, 3]
 Generation 4150: Best = 6.212601e+02 FES = 415100 μ_F = 0.102 μ_{CR} = 0.050
 Diversity = 3.2546e+00 k_t = [1, 1, 1, 1]
 Generation 4160: Best = 6.212541e+02 FES = 416100 μ_F = 0.094 μ_{CR} = 0.039
 Diversity = 3.2541e+00 k_t = [3, 4, 4, 2]
 Generation 4170: Best = 6.212541e+02 FES = 417100 μ_F = 0.062 μ_{CR} = 0.048
 Diversity = 3.2538e+00 k_t = [1, 1, 1, 1]
 Generation 4180: Best = 6.212533e+02 FES = 418100 μ_F = 0.146 μ_{CR} = 0.044
 Diversity = 3.2490e+00 k_t = [3, 4, 3, 4]
 Generation 4190: Best = 6.212522e+02 FES = 419100 μ_F = 0.114 μ_{CR} = 0.050
 Diversity = 3.2490e+00 k_t = [1, 1, 1, 1]
 Generation 4200: Best = 6.212522e+02 FES = 420100 μ_F = 0.088 μ_{CR} = 0.047
 Diversity = 3.2504e+00 k_t = [3, 2, 3, 2]
 Generation 4210: Best = 6.212467e+02 FES = 421100 μ_F = 0.082 μ_{CR} = 0.061
 Diversity = 3.2502e+00 k_t = [1, 1, 1, 1]
 Generation 4220: Best = 6.212467e+02 FES = 422100 μ_F = 0.082 μ_{CR} = 0.057
 Diversity = 3.2500e+00 k_t = [4, 4, 2, 2]
 Generation 4230: Best = 6.212467e+02 FES = 423100 μ_F = 0.076 μ_{CR} = 0.065
 Diversity = 3.2497e+00 k_t = [1, 1, 1, 1]
 Generation 4240: Best = 6.212467e+02 FES = 424100 μ_F = 0.086 μ_{CR} = 0.055
 Diversity = 3.2491e+00 k_t = [2, 3, 2, 3]
 Generation 4250: Best = 6.212467e+02 FES = 425100 μ_F = 0.069 μ_{CR} = 0.049
 Diversity = 3.2491e+00 k_t = [1, 1, 1, 1]
 Generation 4260: Best = 6.212401e+02 FES = 426100 μ_F = 0.165 μ_{CR} = 0.068
 Diversity = 3.2444e+00 k_t = [2, 2, 4, 4]
 Generation 4270: Best = 6.212401e+02 FES = 427100 μ_F = 0.242 μ_{CR} = 0.057
 Diversity = 3.2370e+00 k_t = [1, 1, 1, 1]

Generation 4280: Best = 6.212401e+02 FES = 428100 μ_F = 0.196 μ_{CR} = 0.059
 Diversity = 3.2370e+00 k_t = [4, 4, 3, 2]
 Generation 4290: Best = 6.212401e+02 FES = 429100 μ_F = 0.156 μ_{CR} = 0.043
 Diversity = 3.2370e+00 k_t = [1, 1, 1, 1]
 Generation 4300: Best = 6.212378e+02 FES = 430100 μ_F = 0.132 μ_{CR} = 0.053
 Diversity = 3.2370e+00 k_t = [3, 2, 2, 2]
 Generation 4310: Best = 6.212340e+02 FES = 431100 μ_F = 0.170 μ_{CR} = 0.032
 Diversity = 3.2370e+00 k_t = [1, 1, 1, 1]
 Generation 4320: Best = 6.212340e+02 FES = 432100 μ_F = 0.169 μ_{CR} = 0.030
 Diversity = 3.2370e+00 k_t = [3, 3, 4, 2]
 Generation 4330: Best = 6.212340e+02 FES = 433100 μ_F = 0.181 μ_{CR} = 0.030
 Diversity = 3.2369e+00 k_t = [1, 1, 1, 1]
 Generation 4340: Best = 6.212340e+02 FES = 434100 μ_F = 0.175 μ_{CR} = 0.038
 Diversity = 3.2367e+00 k_t = [4, 3, 2, 2]
 Generation 4350: Best = 6.212340e+02 FES = 435100 μ_F = 0.205 μ_{CR} = 0.031
 Diversity = 3.2365e+00 k_t = [1, 1, 1, 1]
 Generation 4360: Best = 6.212246e+02 FES = 436100 μ_F = 0.237 μ_{CR} = 0.050
 Diversity = 3.2050e+00 k_t = [4, 4, 2, 4]
 Generation 4370: Best = 6.212240e+02 FES = 437100 μ_F = 0.188 μ_{CR} = 0.049
 Diversity = 3.2049e+00 k_t = [1, 1, 1, 1]
 Generation 4380: Best = 6.212240e+02 FES = 438100 μ_F = 0.129 μ_{CR} = 0.037
 Diversity = 3.2047e+00 k_t = [3, 4, 4, 4]
 Generation 4390: Best = 6.212240e+02 FES = 439100 μ_F = 0.236 μ_{CR} = 0.029
 Diversity = 3.2084e+00 k_t = [1, 1, 1, 1]
 Generation 4400: Best = 6.212227e+02 FES = 440100 μ_F = 0.268 μ_{CR} = 0.028
 Diversity = 3.2059e+00 k_t = [3, 3, 4, 4]
 Generation 4410: Best = 6.212227e+02 FES = 441100 μ_F = 0.227 μ_{CR} = 0.030
 Diversity = 3.2060e+00 k_t = [1, 1, 1, 1]
 Generation 4420: Best = 6.212227e+02 FES = 442100 μ_F = 0.239 μ_{CR} = 0.050
 Diversity = 3.1982e+00 k_t = [4, 3, 2, 4]
 Generation 4430: Best = 6.212227e+02 FES = 443100 μ_F = 0.196 μ_{CR} = 0.038
 Diversity = 3.1982e+00 k_t = [1, 1, 1, 1]
 Generation 4440: Best = 6.211983e+02 FES = 444100 μ_F = 0.161 μ_{CR} = 0.046
 Diversity = 3.1982e+00 k_t = [4, 4, 2, 3]
 Generation 4450: Best = 6.211983e+02 FES = 445100 μ_F = 0.087 μ_{CR} = 0.051
 Diversity = 3.1980e+00 k_t = [1, 1, 1, 1]
 Generation 4460: Best = 6.211983e+02 FES = 446100 μ_F = 0.079 μ_{CR} = 0.045
 Diversity = 3.1978e+00 k_t = [2, 2, 2, 3]
 Generation 4470: Best = 6.211964e+02 FES = 447100 μ_F = 0.164 μ_{CR} = 0.037
 Diversity = 3.1976e+00 k_t = [1, 1, 1, 1]
 Generation 4480: Best = 6.211935e+02 FES = 448100 μ_F = 0.229 μ_{CR} = 0.033
 Diversity = 3.1338e+00 k_t = [2, 4, 3, 4]
 Generation 4490: Best = 6.211770e+02 FES = 449100 μ_F = 0.248 μ_{CR} = 0.026
 Diversity = 3.1338e+00 k_t = [1, 1, 1, 1]
 Generation 4500: Best = 6.211770e+02 FES = 450100 μ_F = 0.302 μ_{CR} = 0.021
 Diversity = 3.0946e+00 k_t = [3, 3, 3, 4]
 Generation 4510: Best = 6.211770e+02 FES = 451100 μ_F = 0.183 μ_{CR} = 0.022
 Diversity = 3.0947e+00 k_t = [1, 1, 1, 1]
 Generation 4520: Best = 6.211770e+02 FES = 452100 μ_F = 0.191 μ_{CR} = 0.036
 Diversity = 3.0771e+00 k_t = [3, 2, 4, 4]
 Generation 4530: Best = 6.211334e+02 FES = 453100 μ_F = 0.201 μ_{CR} = 0.027
 Diversity = 3.0773e+00 k_t = [1, 1, 1, 1]
 Generation 4540: Best = 6.211334e+02 FES = 454100 μ_F = 0.197 μ_{CR} = 0.026
 Diversity = 3.0774e+00 k_t = [2, 2, 4, 3]

Generation 4550: Best = 6.211267e+02 FES = 455100 μ_F = 0.241 μ_{CR} = 0.023
 Diversity = 3.0775e+00 k_t = [1, 1, 1, 1]
 Generation 4560: Best = 6.211267e+02 FES = 456100 μ_F = 0.256 μ_{CR} = 0.023
 Diversity = 3.0622e+00 k_t = [3, 2, 3, 4]
 Generation 4570: Best = 6.211249e+02 FES = 457100 μ_F = 0.222 μ_{CR} = 0.020
 Diversity = 3.0622e+00 k_t = [1, 1, 1, 1]
 Generation 4580: Best = 6.211249e+02 FES = 458100 μ_F = 0.186 μ_{CR} = 0.034
 Diversity = 3.0624e+00 k_t = [2, 4, 4, 2]
 Generation 4590: Best = 6.211249e+02 FES = 459100 μ_F = 0.147 μ_{CR} = 0.044
 Diversity = 3.0623e+00 k_t = [1, 1, 1, 1]
 Generation 4600: Best = 6.211249e+02 FES = 460100 μ_F = 0.179 μ_{CR} = 0.040
 Diversity = 3.0577e+00 k_t = [3, 2, 3, 4]
 Generation 4610: Best = 6.211249e+02 FES = 461100 μ_F = 0.117 μ_{CR} = 0.039
 Diversity = 3.0577e+00 k_t = [1, 1, 1, 1]
 Generation 4620: Best = 6.211232e+02 FES = 462100 μ_F = 0.192 μ_{CR} = 0.034
 Diversity = 3.0431e+00 k_t = [3, 4, 2, 4]
 Generation 4630: Best = 6.211232e+02 FES = 463100 μ_F = 0.172 μ_{CR} = 0.041
 Diversity = 3.0428e+00 k_t = [1, 1, 1, 1]
 Generation 4640: Best = 6.211232e+02 FES = 464100 μ_F = 0.241 μ_{CR} = 0.032
 Diversity = 3.0103e+00 k_t = [2, 3, 4, 4]
 Generation 4650: Best = 6.211232e+02 FES = 465100 μ_F = 0.203 μ_{CR} = 0.033
 Diversity = 3.0103e+00 k_t = [1, 1, 1, 1]
 Generation 4660: Best = 6.211232e+02 FES = 466100 μ_F = 0.207 μ_{CR} = 0.028
 Diversity = 3.0102e+00 k_t = [3, 3, 3, 2]
 Generation 4670: Best = 6.211232e+02 FES = 467100 μ_F = 0.200 μ_{CR} = 0.027
 Diversity = 3.0103e+00 k_t = [1, 1, 1, 1]
 Generation 4680: Best = 6.211232e+02 FES = 468100 μ_F = 0.209 μ_{CR} = 0.043
 Diversity = 2.9728e+00 k_t = [4, 3, 4, 4]
 Generation 4690: Best = 6.211217e+02 FES = 469100 μ_F = 0.142 μ_{CR} = 0.049
 Diversity = 2.9728e+00 k_t = [1, 1, 1, 1]
 Generation 4700: Best = 6.211198e+02 FES = 470100 μ_F = 0.184 μ_{CR} = 0.043
 Diversity = 2.9352e+00 k_t = [2, 2, 2, 4]
 Generation 4710: Best = 6.211198e+02 FES = 471100 μ_F = 0.133 μ_{CR} = 0.033
 Diversity = 2.9352e+00 k_t = [1, 1, 1, 1]
 Generation 4720: Best = 6.211179e+02 FES = 472100 μ_F = 0.134 μ_{CR} = 0.042
 Diversity = 2.9349e+00 k_t = [4, 4, 4, 3]
 Generation 4730: Best = 6.211179e+02 FES = 473100 μ_F = 0.140 μ_{CR} = 0.043
 Diversity = 2.9349e+00 k_t = [1, 1, 1, 1]
 Generation 4740: Best = 6.211116e+02 FES = 474100 μ_F = 0.110 μ_{CR} = 0.050
 Diversity = 2.9349e+00 k_t = [4, 3, 4, 3]
 Generation 4750: Best = 6.211116e+02 FES = 475100 μ_F = 0.098 μ_{CR} = 0.047
 Diversity = 2.9349e+00 k_t = [1, 1, 1, 1]
 Generation 4760: Best = 6.211116e+02 FES = 476100 μ_F = 0.091 μ_{CR} = 0.041
 Diversity = 2.9347e+00 k_t = [4, 4, 2, 2]
 Generation 4770: Best = 6.211109e+02 FES = 477100 μ_F = 0.115 μ_{CR} = 0.035
 Diversity = 2.9348e+00 k_t = [1, 1, 1, 1]
 Generation 4780: Best = 6.211109e+02 FES = 478100 μ_F = 0.087 μ_{CR} = 0.048
 Diversity = 2.9347e+00 k_t = [2, 2, 2, 2]
 Generation 4790: Best = 6.211103e+02 FES = 479100 μ_F = 0.161 μ_{CR} = 0.065
 Diversity = 2.9347e+00 k_t = [1, 1, 1, 1]
 Generation 4800: Best = 6.210958e+02 FES = 480100 μ_F = 0.187 μ_{CR} = 0.058
 Diversity = 2.9294e+00 k_t = [4, 3, 2, 4]
 Generation 4810: Best = 6.210958e+02 FES = 481100 μ_F = 0.156 μ_{CR} = 0.044
 Diversity = 2.9294e+00 k_t = [1, 1, 1, 1]

Generation 4820: Best = 6.210958e+02 FES = 482100 μ_F = 0.173 μ_{CR} = 0.040
 Diversity = 2.9298e+00 k_t = [2, 4, 2, 3]
 Generation 4830: Best = 6.210958e+02 FES = 483100 μ_F = 0.193 μ_{CR} = 0.034
 Diversity = 2.9300e+00 k_t = [1, 1, 1, 1]
 Generation 4840: Best = 6.210958e+02 FES = 484100 μ_F = 0.235 μ_{CR} = 0.034
 Diversity = 2.9299e+00 k_t = [3, 4, 2, 2]
 Generation 4850: Best = 6.210958e+02 FES = 485100 μ_F = 0.143 μ_{CR} = 0.035
 Diversity = 2.9299e+00 k_t = [1, 1, 1, 1]
 Generation 4860: Best = 6.210958e+02 FES = 486100 μ_F = 0.195 μ_{CR} = 0.032
 Diversity = 2.9299e+00 k_t = [4, 3, 3, 2]
 Generation 4870: Best = 6.210890e+02 FES = 487100 μ_F = 0.216 μ_{CR} = 0.020
 Diversity = 2.9300e+00 k_t = [1, 1, 1, 1]
 Generation 4880: Best = 6.210756e+02 FES = 488100 μ_F = 0.271 μ_{CR} = 0.021
 Diversity = 2.9246e+00 k_t = [4, 3, 3, 4]
 Generation 4890: Best = 6.210756e+02 FES = 489100 μ_F = 0.252 μ_{CR} = 0.017
 Diversity = 2.9246e+00 k_t = [1, 1, 1, 1]
 Generation 4900: Best = 6.210756e+02 FES = 490100 μ_F = 0.289 μ_{CR} = 0.028
 Diversity = 2.9168e+00 k_t = [4, 4, 3, 4]
 Generation 4910: Best = 6.210756e+02 FES = 491100 μ_F = 0.292 μ_{CR} = 0.027
 Diversity = 2.9167e+00 k_t = [1, 1, 1, 1]
 Generation 4920: Best = 6.210726e+02 FES = 492100 μ_F = 0.226 μ_{CR} = 0.034
 Diversity = 2.9166e+00 k_t = [4, 4, 4, 2]
 Generation 4930: Best = 6.210726e+02 FES = 493100 μ_F = 0.163 μ_{CR} = 0.043
 Diversity = 2.9166e+00 k_t = [1, 1, 1, 1]
 Generation 4940: Best = 6.210688e+02 FES = 494100 μ_F = 0.126 μ_{CR} = 0.049
 Diversity = 2.9166e+00 k_t = [2, 4, 4, 2]
 Generation 4950: Best = 6.210671e+02 FES = 495100 μ_F = 0.121 μ_{CR} = 0.033
 Diversity = 2.9165e+00 k_t = [1, 1, 1, 1]
 Generation 4960: Best = 6.210671e+02 FES = 496100 μ_F = 0.231 μ_{CR} = 0.041
 Diversity = 2.9165e+00 k_t = [4, 4, 4, 3]
 Generation 4970: Best = 6.210671e+02 FES = 497100 μ_F = 0.165 μ_{CR} = 0.032
 Diversity = 2.9164e+00 k_t = [1, 1, 1, 1]
 Generation 4980: Best = 6.210671e+02 FES = 498100 μ_F = 0.128 μ_{CR} = 0.025
 Diversity = 2.9164e+00 k_t = [3, 4, 3, 2]
 Generation 4990: Best = 6.210671e+02 FES = 499100 μ_F = 0.181 μ_{CR} = 0.017
 Diversity = 2.9167e+00 k_t = [1, 1, 1, 1]
 Generation 5000: Best = 6.210671e+02 FES = 500100 μ_F = 0.328 μ_{CR} = 0.018
 Diversity = 2.8797e+00 k_t = [4, 3, 3, 4]
 Generation 5010: Best = 6.210671e+02 FES = 501100 μ_F = 0.322 μ_{CR} = 0.017
 Diversity = 2.8796e+00 k_t = [1, 1, 1, 1]
 Generation 5020: Best = 6.210671e+02 FES = 502100 μ_F = 0.335 μ_{CR} = 0.023
 Diversity = 2.8652e+00 k_t = [2, 3, 3, 4]
 Generation 5030: Best = 6.210671e+02 FES = 503100 μ_F = 0.319 μ_{CR} = 0.018
 Diversity = 2.8653e+00 k_t = [1, 1, 1, 1]
 Generation 5040: Best = 6.210482e+02 FES = 504100 μ_F = 0.250 μ_{CR} = 0.019
 Diversity = 2.8653e+00 k_t = [2, 3, 3, 4]
 Generation 5050: Best = 6.210482e+02 FES = 505100 μ_F = 0.232 μ_{CR} = 0.036
 Diversity = 2.8652e+00 k_t = [1, 1, 1, 1]
 Generation 5060: Best = 6.210482e+02 FES = 506100 μ_F = 0.185 μ_{CR} = 0.043
 Diversity = 2.8652e+00 k_t = [2, 2, 3, 3]
 Generation 5070: Best = 6.210482e+02 FES = 507100 μ_F = 0.141 μ_{CR} = 0.043
 Diversity = 2.8651e+00 k_t = [1, 1, 1, 1]
 Generation 5080: Best = 6.210441e+02 FES = 508100 μ_F = 0.180 μ_{CR} = 0.047
 Diversity = 2.8353e+00 k_t = [4, 4, 3, 4]

Generation 5090: Best = 6.210441e+02 FES = 509100 μ_F = 0.144 μ_{CR} = 0.043
 Diversity = 2.8353e+00 k_t = [1, 1, 1, 1]
 Generation 5100: Best = 6.210260e+02 FES = 510100 μ_F = 0.084 μ_{CR} = 0.033
 Diversity = 2.8352e+00 k_t = [2, 3, 4, 4]
 Generation 5110: Best = 6.210260e+02 FES = 511100 μ_F = 0.082 μ_{CR} = 0.039
 Diversity = 2.8351e+00 k_t = [1, 1, 1, 1]
 Generation 5120: Best = 6.210187e+02 FES = 512100 μ_F = 0.070 μ_{CR} = 0.042
 Diversity = 2.8352e+00 k_t = [4, 2, 3, 3]
 Generation 5130: Best = 6.210187e+02 FES = 513100 μ_F = 0.149 μ_{CR} = 0.063
 Diversity = 2.8362e+00 k_t = [1, 1, 1, 1]
 Generation 5140: Best = 6.210187e+02 FES = 514100 μ_F = 0.172 μ_{CR} = 0.061
 Diversity = 2.8362e+00 k_t = [4, 2, 3, 3]
 Generation 5150: Best = 6.210187e+02 FES = 515100 μ_F = 0.142 μ_{CR} = 0.055
 Diversity = 2.8362e+00 k_t = [1, 1, 1, 1]
 Generation 5160: Best = 6.210187e+02 FES = 516100 μ_F = 0.187 μ_{CR} = 0.076
 Diversity = 2.8139e+00 k_t = [3, 2, 2, 4]
 Generation 5170: Best = 6.210187e+02 FES = 517100 μ_F = 0.145 μ_{CR} = 0.062
 Diversity = 2.8139e+00 k_t = [1, 1, 1, 1]
 Generation 5180: Best = 6.210181e+02 FES = 518100 μ_F = 0.198 μ_{CR} = 0.048
 Diversity = 2.8190e+00 k_t = [4, 2, 3, 4]
 Generation 5190: Best = 6.210181e+02 FES = 519100 μ_F = 0.261 μ_{CR} = 0.039
 Diversity = 2.8190e+00 k_t = [1, 1, 1, 1]
 Generation 5200: Best = 6.210103e+02 FES = 520100 μ_F = 0.200 μ_{CR} = 0.033
 Diversity = 2.8189e+00 k_t = [3, 3, 2, 4]
 Generation 5210: Best = 6.210103e+02 FES = 521100 μ_F = 0.222 μ_{CR} = 0.031
 Diversity = 2.8190e+00 k_t = [1, 1, 1, 1]
 Generation 5220: Best = 6.210103e+02 FES = 522100 μ_F = 0.277 μ_{CR} = 0.026
 Diversity = 2.8153e+00 k_t = [4, 3, 4, 4]
 Generation 5230: Best = 6.210103e+02 FES = 523100 μ_F = 0.257 μ_{CR} = 0.021
 Diversity = 2.8153e+00 k_t = [1, 1, 1, 1]
 Generation 5240: Best = 6.210103e+02 FES = 524100 μ_F = 0.226 μ_{CR} = 0.022
 Diversity = 2.8153e+00 k_t = [3, 2, 3, 2]
 Generation 5250: Best = 6.210103e+02 FES = 525100 μ_F = 0.135 μ_{CR} = 0.028
 Diversity = 2.8154e+00 k_t = [1, 1, 1, 1]
 Generation 5260: Best = 6.210103e+02 FES = 526100 μ_F = 0.142 μ_{CR} = 0.038
 Diversity = 2.8153e+00 k_t = [4, 4, 4, 3]
 Generation 5270: Best = 6.210099e+02 FES = 527100 μ_F = 0.194 μ_{CR} = 0.040
 Diversity = 2.8154e+00 k_t = [1, 1, 1, 1]
 Generation 5280: Best = 6.210099e+02 FES = 528100 μ_F = 0.168 μ_{CR} = 0.034
 Diversity = 2.8154e+00 k_t = [3, 3, 2, 2]
 Generation 5290: Best = 6.210099e+02 FES = 529100 μ_F = 0.121 μ_{CR} = 0.032
 Diversity = 2.8153e+00 k_t = [1, 1, 1, 1]
 Generation 5300: Best = 6.209889e+02 FES = 530100 μ_F = 0.130 μ_{CR} = 0.037
 Diversity = 2.8152e+00 k_t = [2, 2, 4, 3]
 Generation 5310: Best = 6.209889e+02 FES = 531100 μ_F = 0.114 μ_{CR} = 0.033
 Diversity = 2.8152e+00 k_t = [1, 1, 1, 1]
 Generation 5320: Best = 6.209889e+02 FES = 532100 μ_F = 0.072 μ_{CR} = 0.044
 Diversity = 2.8151e+00 k_t = [4, 2, 3, 3]
 Generation 5330: Best = 6.209835e+02 FES = 533100 μ_F = 0.054 μ_{CR} = 0.054
 Diversity = 2.8151e+00 k_t = [1, 1, 1, 1]
 Generation 5340: Best = 6.209800e+02 FES = 534100 μ_F = 0.060 μ_{CR} = 0.059
 Diversity = 2.8149e+00 k_t = [2, 2, 2, 2]
 Generation 5350: Best = 6.209800e+02 FES = 535100 μ_F = 0.043 μ_{CR} = 0.047
 Diversity = 2.8149e+00 k_t = [1, 1, 1, 1]

Generation 5360: Best = 6.209758e+02 FES = 536100 μ_F = 0.080 μ_{CR} = 0.060
 Diversity = 2.8149e+00 k_t = [4, 2, 2, 2]
 Generation 5370: Best = 6.209758e+02 FES = 537100 μ_F = 0.156 μ_{CR} = 0.050
 Diversity = 2.8148e+00 k_t = [1, 1, 1, 1]
 Generation 5380: Best = 6.209758e+02 FES = 538100 μ_F = 0.103 μ_{CR} = 0.037
 Diversity = 2.8147e+00 k_t = [2, 4, 2, 3]
 Generation 5390: Best = 6.209758e+02 FES = 539100 μ_F = 0.081 μ_{CR} = 0.047
 Diversity = 2.8149e+00 k_t = [1, 1, 1, 1]
 Generation 5400: Best = 6.209743e+02 FES = 540100 μ_F = 0.167 μ_{CR} = 0.053
 Diversity = 2.8015e+00 k_t = [2, 4, 4, 4]
 Generation 5410: Best = 6.209743e+02 FES = 541100 μ_F = 0.215 μ_{CR} = 0.038
 Diversity = 2.8015e+00 k_t = [1, 1, 1, 1]
 Generation 5420: Best = 6.209728e+02 FES = 542100 μ_F = 0.195 μ_{CR} = 0.037
 Diversity = 2.8015e+00 k_t = [4, 3, 3, 2]
 Generation 5430: Best = 6.209728e+02 FES = 543100 μ_F = 0.253 μ_{CR} = 0.025
 Diversity = 2.8017e+00 k_t = [1, 1, 1, 1]
 Generation 5440: Best = 6.209728e+02 FES = 544100 μ_F = 0.330 μ_{CR} = 0.027
 Diversity = 2.8016e+00 k_t = [3, 3, 2, 2]
 Generation 5450: Best = 6.209728e+02 FES = 545100 μ_F = 0.295 μ_{CR} = 0.021
 Diversity = 2.8017e+00 k_t = [1, 1, 1, 1]
 Generation 5460: Best = 6.209728e+02 FES = 546100 μ_F = 0.245 μ_{CR} = 0.030
 Diversity = 2.8017e+00 k_t = [2, 3, 2, 2]
 Generation 5470: Best = 6.209728e+02 FES = 547100 μ_F = 0.234 μ_{CR} = 0.022
 Diversity = 2.8017e+00 k_t = [1, 1, 1, 1]
 Generation 5480: Best = 6.209727e+02 FES = 548100 μ_F = 0.166 μ_{CR} = 0.033
 Diversity = 2.8017e+00 k_t = [2, 4, 4, 3]
 Generation 5490: Best = 6.209727e+02 FES = 549100 μ_F = 0.152 μ_{CR} = 0.038
 Diversity = 2.8017e+00 k_t = [1, 1, 1, 1]
 Generation 5500: Best = 6.209727e+02 FES = 550100 μ_F = 0.161 μ_{CR} = 0.033
 Diversity = 2.8017e+00 k_t = [3, 2, 4, 2]
 Generation 5510: Best = 6.209727e+02 FES = 551100 μ_F = 0.152 μ_{CR} = 0.034
 Diversity = 2.8018e+00 k_t = [1, 1, 1, 1]
 Generation 5520: Best = 6.209727e+02 FES = 552100 μ_F = 0.183 μ_{CR} = 0.044
 Diversity = 2.7859e+00 k_t = [2, 4, 2, 4]
 Generation 5530: Best = 6.209727e+02 FES = 553100 μ_F = 0.174 μ_{CR} = 0.036
 Diversity = 2.7859e+00 k_t = [1, 1, 1, 1]
 Generation 5540: Best = 6.209727e+02 FES = 554100 μ_F = 0.142 μ_{CR} = 0.041
 Diversity = 2.7858e+00 k_t = [2, 4, 2, 2]
 Generation 5550: Best = 6.209727e+02 FES = 555100 μ_F = 0.173 μ_{CR} = 0.038
 Diversity = 2.7858e+00 k_t = [1, 1, 1, 1]
 Generation 5560: Best = 6.209727e+02 FES = 556100 μ_F = 0.216 μ_{CR} = 0.037
 Diversity = 2.7882e+00 k_t = [2, 2, 3, 4]
 Generation 5570: Best = 6.209727e+02 FES = 557100 μ_F = 0.197 μ_{CR} = 0.027
 Diversity = 2.7882e+00 k_t = [1, 1, 1, 1]
 Generation 5580: Best = 6.209727e+02 FES = 558100 μ_F = 0.181 μ_{CR} = 0.032
 Diversity = 2.7882e+00 k_t = [3, 4, 4, 2]
 Generation 5590: Best = 6.209727e+02 FES = 559100 μ_F = 0.135 μ_{CR} = 0.022
 Diversity = 2.7881e+00 k_t = [1, 1, 1, 1]
 Generation 5600: Best = 6.209727e+02 FES = 560100 μ_F = 0.159 μ_{CR} = 0.039
 Diversity = 2.7881e+00 k_t = [2, 4, 2, 4]
 Generation 5610: Best = 6.209727e+02 FES = 561100 μ_F = 0.138 μ_{CR} = 0.032
 Diversity = 2.7880e+00 k_t = [1, 1, 1, 1]
 Generation 5620: Best = 6.209727e+02 FES = 562100 μ_F = 0.190 μ_{CR} = 0.040
 Diversity = 2.7880e+00 k_t = [2, 4, 3, 4]

Generation 5630: Best = 6.209727e+02 FES = 563100 μ_F = 0.177 μ_{CR} = 0.038
 Diversity = 2.7880e+00 k_t = [1, 1, 1, 1]
 Generation 5640: Best = 6.209727e+02 FES = 564100 μ_F = 0.114 μ_{CR} = 0.047
 Diversity = 2.7882e+00 k_t = [2, 2, 3, 2]
 Generation 5650: Best = 6.209727e+02 FES = 565100 μ_F = 0.071 μ_{CR} = 0.043
 Diversity = 2.7881e+00 k_t = [1, 1, 1, 1]
 Generation 5660: Best = 6.209435e+02 FES = 566100 μ_F = 0.110 μ_{CR} = 0.057
 Diversity = 2.7880e+00 k_t = [4, 2, 2, 2]
 Generation 5670: Best = 6.209424e+02 FES = 567100 μ_F = 0.099 μ_{CR} = 0.052
 Diversity = 2.7880e+00 k_t = [1, 1, 1, 1]
 Generation 5680: Best = 6.209416e+02 FES = 568100 μ_F = 0.095 μ_{CR} = 0.050
 Diversity = 2.7880e+00 k_t = [4, 4, 2, 3]
 Generation 5690: Best = 6.209368e+02 FES = 569100 μ_F = 0.093 μ_{CR} = 0.033
 Diversity = 2.7881e+00 k_t = [1, 1, 1, 1]
 Generation 5700: Best = 6.209368e+02 FES = 570100 μ_F = 0.108 μ_{CR} = 0.030
 Diversity = 2.7882e+00 k_t = [3, 3, 2, 2]
 Generation 5710: Best = 6.209368e+02 FES = 571100 μ_F = 0.134 μ_{CR} = 0.032
 Diversity = 2.7882e+00 k_t = [1, 1, 1, 1]
 Generation 5720: Best = 6.209287e+02 FES = 572100 μ_F = 0.105 μ_{CR} = 0.035
 Diversity = 2.7882e+00 k_t = [3, 3, 2, 2]
 Generation 5730: Best = 6.209235e+02 FES = 573100 μ_F = 0.094 μ_{CR} = 0.038
 Diversity = 2.7882e+00 k_t = [1, 1, 1, 1]
 Generation 5740: Best = 6.209235e+02 FES = 574100 μ_F = 0.086 μ_{CR} = 0.046
 Diversity = 2.7881e+00 k_t = [3, 4, 4, 3]
 Generation 5750: Best = 6.209235e+02 FES = 575100 μ_F = 0.058 μ_{CR} = 0.043
 Diversity = 2.7884e+00 k_t = [1, 1, 1, 1]
 Generation 5760: Best = 6.209166e+02 FES = 576100 μ_F = 0.079 μ_{CR} = 0.048
 Diversity = 2.7884e+00 k_t = [4, 4, 3, 2]
 Generation 5770: Best = 6.209137e+02 FES = 577100 μ_F = 0.094 μ_{CR} = 0.039
 Diversity = 2.7883e+00 k_t = [1, 1, 1, 1]
 Generation 5780: Best = 6.209137e+02 FES = 578100 μ_F = 0.183 μ_{CR} = 0.040
 Diversity = 2.8282e+00 k_t = [4, 3, 4, 4]
 Generation 5790: Best = 6.209137e+02 FES = 579100 μ_F = 0.137 μ_{CR} = 0.041
 Diversity = 2.8282e+00 k_t = [1, 1, 1, 1]
 Generation 5800: Best = 6.209137e+02 FES = 580100 μ_F = 0.112 μ_{CR} = 0.048
 Diversity = 2.8282e+00 k_t = [2, 3, 2, 3]
 Generation 5810: Best = 6.209137e+02 FES = 581100 μ_F = 0.092 μ_{CR} = 0.037
 Diversity = 2.8281e+00 k_t = [1, 1, 1, 1]
 Generation 5820: Best = 6.209137e+02 FES = 582100 μ_F = 0.105 μ_{CR} = 0.032
 Diversity = 2.8282e+00 k_t = [2, 4, 2, 2]
 Generation 5830: Best = 6.209137e+02 FES = 583100 μ_F = 0.106 μ_{CR} = 0.028
 Diversity = 2.8282e+00 k_t = [1, 1, 1, 1]
 Generation 5840: Best = 6.209047e+02 FES = 584100 μ_F = 0.140 μ_{CR} = 0.035
 Diversity = 2.8281e+00 k_t = [4, 3, 2, 3]
 Generation 5850: Best = 6.208975e+02 FES = 585100 μ_F = 0.107 μ_{CR} = 0.048
 Diversity = 2.8280e+00 k_t = [1, 1, 1, 1]
 Generation 5860: Best = 6.208847e+02 FES = 586100 μ_F = 0.201 μ_{CR} = 0.044
 Diversity = 2.8200e+00 k_t = [3, 3, 3, 2]
 Generation 5870: Best = 6.208847e+02 FES = 587100 μ_F = 0.189 μ_{CR} = 0.037
 Diversity = 2.8200e+00 k_t = [1, 1, 1, 1]
 Generation 5880: Best = 6.208847e+02 FES = 588100 μ_F = 0.190 μ_{CR} = 0.041
 Diversity = 2.8200e+00 k_t = [3, 3, 3, 3]
 Generation 5890: Best = 6.208840e+02 FES = 589100 μ_F = 0.165 μ_{CR} = 0.040
 Diversity = 2.8200e+00 k_t = [1, 1, 1, 1]

Generation 5900: Best = 6.208840e+02 FES = 590100 μ_F = 0.200 μ_{CR} = 0.063
 Diversity = 2.7781e+00 k_t = [4, 4, 4, 4]
 Generation 5910: Best = 6.208822e+02 FES = 591100 μ_F = 0.156 μ_{CR} = 0.060
 Diversity = 2.7780e+00 k_t = [1, 1, 1, 1]
 Generation 5920: Best = 6.208822e+02 FES = 592100 μ_F = 0.169 μ_{CR} = 0.064
 Diversity = 2.7738e+00 k_t = [3, 4, 2, 4]
 Generation 5930: Best = 6.208822e+02 FES = 593100 μ_F = 0.170 μ_{CR} = 0.064
 Diversity = 2.7740e+00 k_t = [1, 1, 1, 1]
 Generation 5940: Best = 6.208786e+02 FES = 594100 μ_F = 0.187 μ_{CR} = 0.045
 Diversity = 2.7628e+00 k_t = [4, 3, 4, 4]
 Generation 5950: Best = 6.208785e+02 FES = 595100 μ_F = 0.120 μ_{CR} = 0.049
 Diversity = 2.7628e+00 k_t = [1, 1, 1, 1]
 Generation 5960: Best = 6.208586e+02 FES = 596100 μ_F = 0.105 μ_{CR} = 0.036
 Diversity = 2.7628e+00 k_t = [2, 2, 2, 2]
 Generation 5970: Best = 6.208538e+02 FES = 597100 μ_F = 0.090 μ_{CR} = 0.035
 Diversity = 2.7628e+00 k_t = [1, 1, 1, 1]
 Generation 5980: Best = 6.208538e+02 FES = 598100 μ_F = 0.089 μ_{CR} = 0.037
 Diversity = 2.7627e+00 k_t = [2, 4, 3, 3]
 Generation 5990: Best = 6.208538e+02 FES = 599100 μ_F = 0.153 μ_{CR} = 0.049
 Diversity = 2.7628e+00 k_t = [1, 1, 1, 1]
 Generation 6000: Best = 6.208538e+02 FES = 600100 μ_F = 0.107 μ_{CR} = 0.059
 Diversity = 2.7629e+00 k_t = [3, 4, 4, 3]
 Generation 6010: Best = 6.208538e+02 FES = 601100 μ_F = 0.090 μ_{CR} = 0.048
 Diversity = 2.7630e+00 k_t = [1, 1, 1, 1]
 Generation 6020: Best = 6.208538e+02 FES = 602100 μ_F = 0.083 μ_{CR} = 0.056
 Diversity = 2.7633e+00 k_t = [2, 3, 4, 3]
 Generation 6030: Best = 6.208494e+02 FES = 603100 μ_F = 0.085 μ_{CR} = 0.037
 Diversity = 2.7632e+00 k_t = [1, 1, 1, 1]
 Generation 6040: Best = 6.208360e+02 FES = 604100 μ_F = 0.107 μ_{CR} = 0.030
 Diversity = 2.7632e+00 k_t = [4, 4, 3, 2]
 Generation 6050: Best = 6.208360e+02 FES = 605100 μ_F = 0.198 μ_{CR} = 0.028
 Diversity = 2.7632e+00 k_t = [1, 1, 1, 1]
 Generation 6060: Best = 6.208360e+02 FES = 606100 μ_F = 0.248 μ_{CR} = 0.024
 Diversity = 2.7447e+00 k_t = [4, 4, 2, 4]
 Generation 6070: Best = 6.208348e+02 FES = 607100 μ_F = 0.258 μ_{CR} = 0.024
 Diversity = 2.7447e+00 k_t = [1, 1, 1, 1]
 Generation 6080: Best = 6.208067e+02 FES = 608100 μ_F = 0.340 μ_{CR} = 0.025
 Diversity = 2.7277e+00 k_t = [2, 3, 2, 4]
 Generation 6090: Best = 6.208067e+02 FES = 609100 μ_F = 0.313 μ_{CR} = 0.031
 Diversity = 2.7277e+00 k_t = [1, 1, 1, 1]
 Generation 6100: Best = 6.208067e+02 FES = 610100 μ_F = 0.301 μ_{CR} = 0.023
 Diversity = 2.7277e+00 k_t = [3, 3, 4, 2]
 Generation 6110: Best = 6.208067e+02 FES = 611100 μ_F = 0.280 μ_{CR} = 0.028
 Diversity = 2.7277e+00 k_t = [1, 1, 1, 1]
 Generation 6120: Best = 6.207937e+02 FES = 612100 μ_F = 0.215 μ_{CR} = 0.036
 Diversity = 2.7277e+00 k_t = [2, 2, 4, 3]
 Generation 6130: Best = 6.207937e+02 FES = 613100 μ_F = 0.181 μ_{CR} = 0.039
 Diversity = 2.7276e+00 k_t = [1, 1, 1, 1]
 Generation 6140: Best = 6.207870e+02 FES = 614100 μ_F = 0.122 μ_{CR} = 0.060
 Diversity = 2.7276e+00 k_t = [2, 4, 3, 2]
 Generation 6150: Best = 6.207808e+02 FES = 615100 μ_F = 0.127 μ_{CR} = 0.048
 Diversity = 2.7279e+00 k_t = [1, 1, 1, 1]
 Generation 6160: Best = 6.207808e+02 FES = 616100 μ_F = 0.192 μ_{CR} = 0.049
 Diversity = 2.7279e+00 k_t = [2, 3, 3, 3]

Generation 6170: Best = 6.207808e+02 FES = 617100 μ_F = 0.138 μ_{CR} = 0.045
 Diversity = 2.7276e+00 k_t = [1, 1, 1, 1]
 Generation 6180: Best = 6.207752e+02 FES = 618100 μ_F = 0.146 μ_{CR} = 0.057
 Diversity = 2.7277e+00 k_t = [3, 3, 4, 3]
 Generation 6190: Best = 6.207752e+02 FES = 619100 μ_F = 0.146 μ_{CR} = 0.059
 Diversity = 2.7276e+00 k_t = [1, 1, 1, 1]
 Generation 6200: Best = 6.207589e+02 FES = 620100 μ_F = 0.194 μ_{CR} = 0.068
 Diversity = 2.7268e+00 k_t = [4, 4, 3, 4]
 Generation 6210: Best = 6.207589e+02 FES = 621100 μ_F = 0.158 μ_{CR} = 0.043
 Diversity = 2.7268e+00 k_t = [1, 1, 1, 1]
 Generation 6220: Best = 6.207589e+02 FES = 622100 μ_F = 0.123 μ_{CR} = 0.033
 Diversity = 2.7267e+00 k_t = [3, 3, 4, 2]
 Generation 6230: Best = 6.207456e+02 FES = 623100 μ_F = 0.125 μ_{CR} = 0.042
 Diversity = 2.7265e+00 k_t = [1, 1, 1, 1]
 Generation 6240: Best = 6.207329e+02 FES = 624100 μ_F = 0.154 μ_{CR} = 0.030
 Diversity = 2.7265e+00 k_t = [3, 4, 4, 4]
 Generation 6250: Best = 6.207329e+02 FES = 625100 μ_F = 0.110 μ_{CR} = 0.058
 Diversity = 2.7265e+00 k_t = [1, 1, 1, 1]
 Generation 6260: Best = 6.207295e+02 FES = 626100 μ_F = 0.150 μ_{CR} = 0.055
 Diversity = 2.6885e+00 k_t = [4, 2, 3, 4]
 Generation 6270: Best = 6.207281e+02 FES = 627100 μ_F = 0.119 μ_{CR} = 0.035
 Diversity = 2.6886e+00 k_t = [1, 1, 1, 1]
 Generation 6280: Best = 6.207197e+02 FES = 628100 μ_F = 0.187 μ_{CR} = 0.033
 Diversity = 2.6683e+00 k_t = [4, 4, 2, 4]
 Generation 6290: Best = 6.207197e+02 FES = 629100 μ_F = 0.121 μ_{CR} = 0.045
 Diversity = 2.6681e+00 k_t = [1, 1, 1, 1]
 Generation 6300: Best = 6.207197e+02 FES = 630100 μ_F = 0.158 μ_{CR} = 0.048
 Diversity = 2.6652e+00 k_t = [2, 3, 4, 4]
 Generation 6310: Best = 6.207197e+02 FES = 631100 μ_F = 0.117 μ_{CR} = 0.068
 Diversity = 2.6653e+00 k_t = [1, 1, 1, 1]
 Generation 6320: Best = 6.207197e+02 FES = 632100 μ_F = 0.171 μ_{CR} = 0.048
 Diversity = 2.5989e+00 k_t = [2, 3, 4, 4]
 Generation 6330: Best = 6.207181e+02 FES = 633100 μ_F = 0.158 μ_{CR} = 0.036
 Diversity = 2.5989e+00 k_t = [1, 1, 1, 1]
 Generation 6340: Best = 6.207181e+02 FES = 634100 μ_F = 0.106 μ_{CR} = 0.051
 Diversity = 2.5988e+00 k_t = [2, 2, 3, 2]
 Generation 6350: Best = 6.207181e+02 FES = 635100 μ_F = 0.091 μ_{CR} = 0.037
 Diversity = 2.5988e+00 k_t = [1, 1, 1, 1]
 Generation 6360: Best = 6.206912e+02 FES = 636100 μ_F = 0.127 μ_{CR} = 0.056
 Diversity = 2.5987e+00 k_t = [2, 3, 2, 2]
 Generation 6370: Best = 6.206888e+02 FES = 637100 μ_F = 0.088 μ_{CR} = 0.057
 Diversity = 2.5987e+00 k_t = [1, 1, 1, 1]
 Generation 6380: Best = 6.206888e+02 FES = 638100 μ_F = 0.078 μ_{CR} = 0.049
 Diversity = 2.5987e+00 k_t = [2, 3, 4, 2]
 Generation 6390: Best = 6.206888e+02 FES = 639100 μ_F = 0.060 μ_{CR} = 0.055
 Diversity = 2.5986e+00 k_t = [1, 1, 1, 1]
 Generation 6400: Best = 6.206888e+02 FES = 640100 μ_F = 0.152 μ_{CR} = 0.057
 Diversity = 2.5994e+00 k_t = [2, 3, 3, 4]
 Generation 6410: Best = 6.206888e+02 FES = 641100 μ_F = 0.153 μ_{CR} = 0.045
 Diversity = 2.5994e+00 k_t = [1, 1, 1, 1]
 Generation 6420: Best = 6.206888e+02 FES = 642100 μ_F = 0.105 μ_{CR} = 0.050
 Diversity = 2.5994e+00 k_t = [2, 2, 3, 2]
 Generation 6430: Best = 6.206888e+02 FES = 643100 μ_F = 0.121 μ_{CR} = 0.035
 Diversity = 2.5995e+00 k_t = [1, 1, 1, 1]

Generation 6440: Best = 6.206888e+02 FES = 644100 μ_F = 0.105 μ_{CR} = 0.028
 Diversity = 2.5994e+00 k_t = [2, 3, 2, 2]
 Generation 6450: Best = 6.206888e+02 FES = 645100 μ_F = 0.087 μ_{CR} = 0.038
 Diversity = 2.5993e+00 k_t = [1, 1, 1, 1]
 Generation 6460: Best = 6.206888e+02 FES = 646100 μ_F = 0.076 μ_{CR} = 0.061
 Diversity = 2.5993e+00 k_t = [2, 2, 3, 2]
 Generation 6470: Best = 6.206888e+02 FES = 647100 μ_F = 0.082 μ_{CR} = 0.045
 Diversity = 2.5990e+00 k_t = [1, 1, 1, 1]
 Generation 6480: Best = 6.206574e+02 FES = 648100 μ_F = 0.121 μ_{CR} = 0.042
 Diversity = 2.5991e+00 k_t = [4, 2, 2, 3]
 Generation 6490: Best = 6.206551e+02 FES = 649100 μ_F = 0.102 μ_{CR} = 0.048
 Diversity = 2.5991e+00 k_t = [1, 1, 1, 1]
 Generation 6500: Best = 6.206551e+02 FES = 650100 μ_F = 0.215 μ_{CR} = 0.041
 Diversity = 2.5991e+00 k_t = [4, 2, 3, 3]
 Generation 6510: Best = 6.206551e+02 FES = 651100 μ_F = 0.236 μ_{CR} = 0.034
 Diversity = 2.5991e+00 k_t = [1, 1, 1, 1]
 Generation 6520: Best = 6.206551e+02 FES = 652100 μ_F = 0.325 μ_{CR} = 0.031
 Diversity = 2.5908e+00 k_t = [3, 4, 2, 4]
 Generation 6530: Best = 6.206551e+02 FES = 653100 μ_F = 0.342 μ_{CR} = 0.024
 Diversity = 2.5908e+00 k_t = [1, 1, 1, 1]
 Generation 6540: Best = 6.206551e+02 FES = 654100 μ_F = 0.285 μ_{CR} = 0.018
 Diversity = 2.5907e+00 k_t = [3, 4, 3, 3]
 Generation 6550: Best = 6.206551e+02 FES = 655100 μ_F = 0.271 μ_{CR} = 0.016
 Diversity = 2.5907e+00 k_t = [1, 1, 1, 1]
 Generation 6560: Best = 6.206493e+02 FES = 656100 μ_F = 0.259 μ_{CR} = 0.026
 Diversity = 2.5845e+00 k_t = [3, 3, 4, 4]
 Generation 6570: Best = 6.206493e+02 FES = 657100 μ_F = 0.231 μ_{CR} = 0.028
 Diversity = 2.5845e+00 k_t = [1, 1, 1, 1]
 Generation 6580: Best = 6.206493e+02 FES = 658100 μ_F = 0.169 μ_{CR} = 0.025
 Diversity = 2.5845e+00 k_t = [2, 2, 4, 2]
 Generation 6590: Best = 6.206493e+02 FES = 659100 μ_F = 0.162 μ_{CR} = 0.038
 Diversity = 2.5845e+00 k_t = [1, 1, 1, 1]
 Generation 6600: Best = 6.206493e+02 FES = 660100 μ_F = 0.210 μ_{CR} = 0.036
 Diversity = 2.5844e+00 k_t = [4, 2, 3, 3]
 Generation 6610: Best = 6.206394e+02 FES = 661100 μ_F = 0.199 μ_{CR} = 0.025
 Diversity = 2.5844e+00 k_t = [1, 1, 1, 1]
 Generation 6620: Best = 6.206394e+02 FES = 662100 μ_F = 0.235 μ_{CR} = 0.039
 Diversity = 2.5843e+00 k_t = [4, 4, 3, 4]
 Generation 6630: Best = 6.206394e+02 FES = 663100 μ_F = 0.196 μ_{CR} = 0.034
 Diversity = 2.5843e+00 k_t = [1, 1, 1, 1]
 Generation 6640: Best = 6.206394e+02 FES = 664100 μ_F = 0.270 μ_{CR} = 0.025
 Diversity = 2.5930e+00 k_t = [2, 4, 3, 4]
 Generation 6650: Best = 6.206394e+02 FES = 665100 μ_F = 0.214 μ_{CR} = 0.024
 Diversity = 2.5930e+00 k_t = [1, 1, 1, 1]
 Generation 6660: Best = 6.206394e+02 FES = 666100 μ_F = 0.244 μ_{CR} = 0.047
 Diversity = 2.5955e+00 k_t = [2, 4, 4, 4]
 Generation 6670: Best = 6.206394e+02 FES = 667100 μ_F = 0.225 μ_{CR} = 0.047
 Diversity = 2.5955e+00 k_t = [1, 1, 1, 1]
 Generation 6680: Best = 6.206394e+02 FES = 668100 μ_F = 0.194 μ_{CR} = 0.042
 Diversity = 2.5954e+00 k_t = [4, 3, 4, 3]
 Generation 6690: Best = 6.206367e+02 FES = 669100 μ_F = 0.114 μ_{CR} = 0.079
 Diversity = 2.5953e+00 k_t = [1, 1, 1, 1]
 Generation 6700: Best = 6.206367e+02 FES = 670100 μ_F = 0.163 μ_{CR} = 0.051
 Diversity = 2.5954e+00 k_t = [4, 3, 2, 3]

Generation 6710: Best = 6.206367e+02 FES = 671100 μ_F = 0.172 μ_{CR} = 0.040
 Diversity = 2.5953e+00 k_t = [1, 1, 1, 1]
 Generation 6720: Best = 6.206367e+02 FES = 672100 μ_F = 0.171 μ_{CR} = 0.041
 Diversity = 2.5955e+00 k_t = [4, 2, 3, 3]
 Generation 6730: Best = 6.206248e+02 FES = 673100 μ_F = 0.171 μ_{CR} = 0.043
 Diversity = 2.5956e+00 k_t = [1, 1, 1, 1]
 Generation 6740: Best = 6.206248e+02 FES = 674100 μ_F = 0.219 μ_{CR} = 0.058
 Diversity = 2.5827e+00 k_t = [4, 4, 2, 4]
 Generation 6750: Best = 6.206248e+02 FES = 675100 μ_F = 0.228 μ_{CR} = 0.057
 Diversity = 2.5827e+00 k_t = [1, 1, 1, 1]
 Generation 6760: Best = 6.206248e+02 FES = 676100 μ_F = 0.185 μ_{CR} = 0.050
 Diversity = 2.5828e+00 k_t = [4, 3, 4, 3]
 Generation 6770: Best = 6.206248e+02 FES = 677100 μ_F = 0.205 μ_{CR} = 0.051
 Diversity = 2.5828e+00 k_t = [1, 1, 1, 1]
 Generation 6780: Best = 6.206238e+02 FES = 678100 μ_F = 0.139 μ_{CR} = 0.038
 Diversity = 2.5828e+00 k_t = [3, 2, 3, 3]
 Generation 6790: Best = 6.206238e+02 FES = 679100 μ_F = 0.129 μ_{CR} = 0.037
 Diversity = 2.5827e+00 k_t = [1, 1, 1, 1]
 Generation 6800: Best = 6.206238e+02 FES = 680100 μ_F = 0.170 μ_{CR} = 0.038
 Diversity = 2.5716e+00 k_t = [3, 4, 3, 4]
 Generation 6810: Best = 6.206238e+02 FES = 681100 μ_F = 0.156 μ_{CR} = 0.036
 Diversity = 2.5716e+00 k_t = [1, 1, 1, 1]
 Generation 6820: Best = 6.206238e+02 FES = 682100 μ_F = 0.194 μ_{CR} = 0.036
 Diversity = 2.5678e+00 k_t = [2, 2, 2, 4]
 Generation 6830: Best = 6.206238e+02 FES = 683100 μ_F = 0.137 μ_{CR} = 0.029
 Diversity = 2.5678e+00 k_t = [1, 1, 1, 1]
 Generation 6840: Best = 6.206238e+02 FES = 684100 μ_F = 0.156 μ_{CR} = 0.021
 Diversity = 2.5678e+00 k_t = [3, 2, 4, 2]
 Generation 6850: Best = 6.206238e+02 FES = 685100 μ_F = 0.142 μ_{CR} = 0.030
 Diversity = 2.5679e+00 k_t = [1, 1, 1, 1]
 Generation 6860: Best = 6.206203e+02 FES = 686100 μ_F = 0.172 μ_{CR} = 0.035
 Diversity = 2.5679e+00 k_t = [2, 3, 2, 2]
 Generation 6870: Best = 6.206203e+02 FES = 687100 μ_F = 0.158 μ_{CR} = 0.044
 Diversity = 2.5679e+00 k_t = [1, 1, 1, 1]
 Generation 6880: Best = 6.206203e+02 FES = 688100 μ_F = 0.180 μ_{CR} = 0.045
 Diversity = 2.5679e+00 k_t = [3, 4, 4, 2]
 Generation 6890: Best = 6.206203e+02 FES = 689100 μ_F = 0.137 μ_{CR} = 0.032
 Diversity = 2.5679e+00 k_t = [1, 1, 1, 1]
 Generation 6900: Best = 6.206203e+02 FES = 690100 μ_F = 0.176 μ_{CR} = 0.042
 Diversity = 2.5642e+00 k_t = [2, 2, 4, 4]
 Generation 6910: Best = 6.206057e+02 FES = 691100 μ_F = 0.135 μ_{CR} = 0.038
 Diversity = 2.5643e+00 k_t = [1, 1, 1, 1]
 Generation 6920: Best = 6.206057e+02 FES = 692100 μ_F = 0.118 μ_{CR} = 0.040
 Diversity = 2.5645e+00 k_t = [2, 3, 3, 2]
 Generation 6930: Best = 6.206057e+02 FES = 693100 μ_F = 0.099 μ_{CR} = 0.041
 Diversity = 2.5645e+00 k_t = [1, 1, 1, 1]
 Generation 6940: Best = 6.206057e+02 FES = 694100 μ_F = 0.060 μ_{CR} = 0.034
 Diversity = 2.5645e+00 k_t = [2, 4, 2, 2]
 Generation 6950: Best = 6.206057e+02 FES = 695100 μ_F = 0.124 μ_{CR} = 0.032
 Diversity = 2.5643e+00 k_t = [1, 1, 1, 1]
 Generation 6960: Best = 6.206057e+02 FES = 696100 μ_F = 0.218 μ_{CR} = 0.028
 Diversity = 2.5646e+00 k_t = [3, 4, 4, 3]
 Generation 6970: Best = 6.206057e+02 FES = 697100 μ_F = 0.174 μ_{CR} = 0.027
 Diversity = 2.5646e+00 k_t = [1, 1, 1, 1]

Generation 6980: Best = 6.205833e+02 FES = 698100 μ_F = 0.246 μ_{CR} = 0.031
 Diversity = 2.5647e+00 k_t = [4, 3, 3, 2]
 Generation 6990: Best = 6.205821e+02 FES = 699100 μ_F = 0.225 μ_{CR} = 0.035
 Diversity = 2.5646e+00 k_t = [1, 1, 1, 1]
 Generation 7000: Best = 6.205779e+02 FES = 700100 μ_F = 0.157 μ_{CR} = 0.035
 Diversity = 2.5543e+00 k_t = [2, 3, 3, 3]
 Generation 7010: Best = 6.205779e+02 FES = 701100 μ_F = 0.186 μ_{CR} = 0.037
 Diversity = 2.5544e+00 k_t = [1, 1, 1, 1]
 Generation 7020: Best = 6.205779e+02 FES = 702100 μ_F = 0.182 μ_{CR} = 0.033
 Diversity = 2.5543e+00 k_t = [3, 3, 2, 4]
 Generation 7030: Best = 6.205310e+02 FES = 703100 μ_F = 0.219 μ_{CR} = 0.026
 Diversity = 2.5543e+00 k_t = [1, 1, 1, 1]
 Generation 7040: Best = 6.205310e+02 FES = 704100 μ_F = 0.168 μ_{CR} = 0.032
 Diversity = 2.5543e+00 k_t = [3, 2, 3, 2]
 Generation 7050: Best = 6.205310e+02 FES = 705100 μ_F = 0.155 μ_{CR} = 0.025
 Diversity = 2.5546e+00 k_t = [1, 1, 1, 1]
 Generation 7060: Best = 6.205310e+02 FES = 706100 μ_F = 0.237 μ_{CR} = 0.025
 Diversity = 2.5358e+00 k_t = [2, 4, 3, 4]
 Generation 7070: Best = 6.205310e+02 FES = 707100 μ_F = 0.206 μ_{CR} = 0.024
 Diversity = 2.5357e+00 k_t = [1, 1, 1, 1]
 Generation 7080: Best = 6.205310e+02 FES = 708100 μ_F = 0.267 μ_{CR} = 0.028
 Diversity = 2.5419e+00 k_t = [2, 2, 3, 4]
 Generation 7090: Best = 6.205310e+02 FES = 709100 μ_F = 0.269 μ_{CR} = 0.027
 Diversity = 2.5418e+00 k_t = [1, 1, 1, 1]
 Generation 7100: Best = 6.205310e+02 FES = 710100 μ_F = 0.311 μ_{CR} = 0.030
 Diversity = 2.5356e+00 k_t = [3, 4, 3, 3]
 Generation 7110: Best = 6.205147e+02 FES = 711100 μ_F = 0.269 μ_{CR} = 0.026
 Diversity = 2.5356e+00 k_t = [1, 1, 1, 1]
 Generation 7120: Best = 6.205079e+02 FES = 712100 μ_F = 0.233 μ_{CR} = 0.028
 Diversity = 2.5356e+00 k_t = [4, 4, 3, 3]
 Generation 7130: Best = 6.205034e+02 FES = 713100 μ_F = 0.204 μ_{CR} = 0.024
 Diversity = 2.5356e+00 k_t = [1, 1, 1, 1]
 Generation 7140: Best = 6.204936e+02 FES = 714100 μ_F = 0.192 μ_{CR} = 0.019
 Diversity = 2.5356e+00 k_t = [2, 2, 4, 2]
 Generation 7150: Best = 6.204936e+02 FES = 715100 μ_F = 0.147 μ_{CR} = 0.039
 Diversity = 2.5355e+00 k_t = [1, 1, 1, 1]
 Generation 7160: Best = 6.204911e+02 FES = 716100 μ_F = 0.112 μ_{CR} = 0.050
 Diversity = 2.5350e+00 k_t = [2, 3, 4, 4]
 Generation 7170: Best = 6.204911e+02 FES = 717100 μ_F = 0.087 μ_{CR} = 0.059
 Diversity = 2.5349e+00 k_t = [1, 1, 1, 1]
 Generation 7180: Best = 6.204886e+02 FES = 718100 μ_F = 0.173 μ_{CR} = 0.053
 Diversity = 2.5350e+00 k_t = [4, 3, 4, 2]
 Generation 7190: Best = 6.204777e+02 FES = 719100 μ_F = 0.096 μ_{CR} = 0.039
 Diversity = 2.5349e+00 k_t = [1, 1, 1, 1]
 Generation 7200: Best = 6.204690e+02 FES = 720100 μ_F = 0.168 μ_{CR} = 0.048
 Diversity = 2.5288e+00 k_t = [3, 4, 2, 4]
 Generation 7210: Best = 6.204690e+02 FES = 721100 μ_F = 0.200 μ_{CR} = 0.041
 Diversity = 2.5193e+00 k_t = [1, 1, 1, 1]
 Generation 7220: Best = 6.204414e+02 FES = 722100 μ_F = 0.163 μ_{CR} = 0.040
 Diversity = 2.5193e+00 k_t = [3, 2, 3, 3]
 Generation 7230: Best = 6.204414e+02 FES = 723100 μ_F = 0.166 μ_{CR} = 0.042
 Diversity = 2.5192e+00 k_t = [1, 1, 1, 1]
 Generation 7240: Best = 6.204414e+02 FES = 724100 μ_F = 0.166 μ_{CR} = 0.044
 Diversity = 2.5192e+00 k_t = [4, 4, 2, 4]

Generation 7250: Best = 6.204414e+02 FES = 725100 μ_F = 0.177 μ_{CR} = 0.038
 Diversity = 2.5191e+00 k_t = [1, 1, 1, 1]
 Generation 7260: Best = 6.204414e+02 FES = 726100 μ_F = 0.226 μ_{CR} = 0.044
 Diversity = 2.4817e+00 k_t = [4, 3, 2, 4]
 Generation 7270: Best = 6.204414e+02 FES = 727100 μ_F = 0.187 μ_{CR} = 0.050
 Diversity = 2.4817e+00 k_t = [1, 1, 1, 1]
 Generation 7280: Best = 6.204414e+02 FES = 728100 μ_F = 0.210 μ_{CR} = 0.039
 Diversity = 2.4639e+00 k_t = [4, 3, 2, 4]
 Generation 7290: Best = 6.204353e+02 FES = 729100 μ_F = 0.236 μ_{CR} = 0.058
 Diversity = 2.4638e+00 k_t = [1, 1, 1, 1]
 Generation 7300: Best = 6.204353e+02 FES = 730100 μ_F = 0.285 μ_{CR} = 0.065
 Diversity = 2.4638e+00 k_t = [3, 3, 3, 2]
 Generation 7310: Best = 6.204324e+02 FES = 731100 μ_F = 0.228 μ_{CR} = 0.056
 Diversity = 2.4638e+00 k_t = [1, 1, 1, 1]
 Generation 7320: Best = 6.204324e+02 FES = 732100 μ_F = 0.243 μ_{CR} = 0.061
 Diversity = 2.4638e+00 k_t = [3, 2, 2, 3]
 Generation 7330: Best = 6.204259e+02 FES = 733100 μ_F = 0.153 μ_{CR} = 0.059
 Diversity = 2.4638e+00 k_t = [1, 1, 1, 1]
 Generation 7340: Best = 6.204257e+02 FES = 734100 μ_F = 0.219 μ_{CR} = 0.051
 Diversity = 2.4638e+00 k_t = [4, 4, 2, 3]
 Generation 7350: Best = 6.204257e+02 FES = 735100 μ_F = 0.270 μ_{CR} = 0.036
 Diversity = 2.4639e+00 k_t = [1, 1, 1, 1]
 Generation 7360: Best = 6.204087e+02 FES = 736100 μ_F = 0.319 μ_{CR} = 0.035
 Diversity = 2.4124e+00 k_t = [4, 3, 2, 4]
 Generation 7370: Best = 6.204087e+02 FES = 737100 μ_F = 0.376 μ_{CR} = 0.031
 Diversity = 2.4124e+00 k_t = [1, 1, 1, 1]
 Generation 7380: Best = 6.204087e+02 FES = 738100 μ_F = 0.394 μ_{CR} = 0.025
 Diversity = 2.4124e+00 k_t = [3, 3, 2, 2]
 Generation 7390: Best = 6.204087e+02 FES = 739100 μ_F = 0.337 μ_{CR} = 0.030
 Diversity = 2.4124e+00 k_t = [1, 1, 1, 1]
 Generation 7400: Best = 6.204087e+02 FES = 740100 μ_F = 0.328 μ_{CR} = 0.040
 Diversity = 2.4124e+00 k_t = [3, 2, 2, 2]
 Generation 7410: Best = 6.204087e+02 FES = 741100 μ_F = 0.305 μ_{CR} = 0.028
 Diversity = 2.4124e+00 k_t = [1, 1, 1, 1]
 Generation 7420: Best = 6.204087e+02 FES = 742100 μ_F = 0.225 μ_{CR} = 0.032
 Diversity = 2.4125e+00 k_t = [2, 3, 4, 4]
 Generation 7430: Best = 6.204078e+02 FES = 743100 μ_F = 0.188 μ_{CR} = 0.045
 Diversity = 2.4126e+00 k_t = [1, 1, 1, 1]
 Generation 7440: Best = 6.203763e+02 FES = 744100 μ_F = 0.190 μ_{CR} = 0.040
 Diversity = 2.4126e+00 k_t = [4, 4, 3, 4]
 Generation 7450: Best = 6.203691e+02 FES = 745100 μ_F = 0.253 μ_{CR} = 0.032
 Diversity = 2.4031e+00 k_t = [1, 1, 1, 1]
 Generation 7460: Best = 6.203691e+02 FES = 746100 μ_F = 0.366 μ_{CR} = 0.031
 Diversity = 2.3916e+00 k_t = [2, 3, 2, 4]
 Generation 7470: Best = 6.203691e+02 FES = 747100 μ_F = 0.410 μ_{CR} = 0.032
 Diversity = 2.3917e+00 k_t = [1, 1, 1, 1]
 Generation 7480: Best = 6.203691e+02 FES = 748100 μ_F = 0.428 μ_{CR} = 0.045
 Diversity = 2.3876e+00 k_t = [3, 2, 3, 4]
 Generation 7490: Best = 6.203683e+02 FES = 749100 μ_F = 0.409 μ_{CR} = 0.034
 Diversity = 2.3876e+00 k_t = [1, 1, 1, 1]
 Generation 7500: Best = 6.203683e+02 FES = 750100 μ_F = 0.383 μ_{CR} = 0.031
 Diversity = 2.3876e+00 k_t = [3, 3, 2, 4]
 Generation 7510: Best = 6.203668e+02 FES = 751100 μ_F = 0.334 μ_{CR} = 0.027
 Diversity = 2.3876e+00 k_t = [1, 1, 1, 1]

Generation 7520: Best = 6.203668e+02 FES = 752100 μ_F = 0.294 μ_{CR} = 0.037
 Diversity = 2.3698e+00 k_t = [2, 2, 4, 4]
 Generation 7530: Best = 6.203668e+02 FES = 753100 μ_F = 0.296 μ_{CR} = 0.035
 Diversity = 2.3513e+00 k_t = [1, 1, 1, 1]
 Generation 7540: Best = 6.203632e+02 FES = 754100 μ_F = 0.287 μ_{CR} = 0.040
 Diversity = 2.3513e+00 k_t = [2, 3, 4, 3]
 Generation 7550: Best = 6.203632e+02 FES = 755100 μ_F = 0.273 μ_{CR} = 0.036
 Diversity = 2.3512e+00 k_t = [1, 1, 1, 1]
 Generation 7560: Best = 6.203632e+02 FES = 756100 μ_F = 0.222 μ_{CR} = 0.043
 Diversity = 2.3514e+00 k_t = [3, 2, 4, 2]
 Generation 7570: Best = 6.203632e+02 FES = 757100 μ_F = 0.212 μ_{CR} = 0.060
 Diversity = 2.3514e+00 k_t = [1, 1, 1, 1]
 Generation 7580: Best = 6.203632e+02 FES = 758100 μ_F = 0.230 μ_{CR} = 0.057
 Diversity = 2.3514e+00 k_t = [2, 4, 3, 2]
 Generation 7590: Best = 6.203632e+02 FES = 759100 μ_F = 0.171 μ_{CR} = 0.070
 Diversity = 2.3514e+00 k_t = [1, 1, 1, 1]
 Generation 7600: Best = 6.203632e+02 FES = 760100 μ_F = 0.223 μ_{CR} = 0.053
 Diversity = 2.3515e+00 k_t = [3, 2, 2, 3]
 Generation 7610: Best = 6.203627e+02 FES = 761100 μ_F = 0.214 μ_{CR} = 0.047
 Diversity = 2.3514e+00 k_t = [1, 1, 1, 1]
 Generation 7620: Best = 6.203627e+02 FES = 762100 μ_F = 0.275 μ_{CR} = 0.045
 Diversity = 2.3515e+00 k_t = [2, 4, 3, 2]
 Generation 7630: Best = 6.203522e+02 FES = 763100 μ_F = 0.264 μ_{CR} = 0.041
 Diversity = 2.3518e+00 k_t = [1, 1, 1, 1]
 Generation 7640: Best = 6.203522e+02 FES = 764100 μ_F = 0.288 μ_{CR} = 0.040
 Diversity = 2.3536e+00 k_t = [3, 4, 2, 4]
 Generation 7650: Best = 6.203437e+02 FES = 765100 μ_F = 0.226 μ_{CR} = 0.050
 Diversity = 2.3538e+00 k_t = [1, 1, 1, 1]
 Generation 7660: Best = 6.203437e+02 FES = 766100 μ_F = 0.318 μ_{CR} = 0.054
 Diversity = 2.3417e+00 k_t = [2, 3, 4, 4]
 Generation 7670: Best = 6.203437e+02 FES = 767100 μ_F = 0.266 μ_{CR} = 0.039
 Diversity = 2.3418e+00 k_t = [1, 1, 1, 1]
 Generation 7680: Best = 6.203122e+02 FES = 768100 μ_F = 0.250 μ_{CR} = 0.036
 Diversity = 2.3418e+00 k_t = [4, 2, 2, 4]
 Generation 7690: Best = 6.203122e+02 FES = 769100 μ_F = 0.226 μ_{CR} = 0.034
 Diversity = 2.3418e+00 k_t = [1, 1, 1, 1]
 Generation 7700: Best = 6.203122e+02 FES = 770100 μ_F = 0.273 μ_{CR} = 0.030
 Diversity = 2.3148e+00 k_t = [3, 4, 3, 4]
 Generation 7710: Best = 6.203122e+02 FES = 771100 μ_F = 0.212 μ_{CR} = 0.032
 Diversity = 2.3148e+00 k_t = [1, 1, 1, 1]
 Generation 7720: Best = 6.203122e+02 FES = 772100 μ_F = 0.239 μ_{CR} = 0.041
 Diversity = 2.3149e+00 k_t = [2, 3, 2, 2]
 Generation 7730: Best = 6.203030e+02 FES = 773100 μ_F = 0.234 μ_{CR} = 0.042
 Diversity = 2.3150e+00 k_t = [1, 1, 1, 1]
 Generation 7740: Best = 6.203030e+02 FES = 774100 μ_F = 0.292 μ_{CR} = 0.038
 Diversity = 2.3150e+00 k_t = [4, 3, 4, 3]
 Generation 7750: Best = 6.203030e+02 FES = 775100 μ_F = 0.241 μ_{CR} = 0.035
 Diversity = 2.3149e+00 k_t = [1, 1, 1, 1]
 Generation 7760: Best = 6.202725e+02 FES = 776100 μ_F = 0.288 μ_{CR} = 0.056
 Diversity = 2.3007e+00 k_t = [4, 2, 3, 4]
 Generation 7770: Best = 6.202725e+02 FES = 777100 μ_F = 0.218 μ_{CR} = 0.053
 Diversity = 2.3008e+00 k_t = [1, 1, 1, 1]
 Generation 7780: Best = 6.202725e+02 FES = 778100 μ_F = 0.214 μ_{CR} = 0.059
 Diversity = 2.3007e+00 k_t = [2, 3, 4, 3]

Generation 7790: Best = 6.202710e+02 FES = 779100 μ_F = 0.293 μ_{CR} = 0.038
 Diversity = 2.2880e+00 k_t = [1, 1, 1, 1]
 Generation 7800: Best = 6.202682e+02 FES = 780100 μ_F = 0.353 μ_{CR} = 0.031
 Diversity = 2.2881e+00 k_t = [3, 3, 3, 3]
 Generation 7810: Best = 6.202682e+02 FES = 781100 μ_F = 0.316 μ_{CR} = 0.032
 Diversity = 2.2881e+00 k_t = [1, 1, 1, 1]
 Generation 7820: Best = 6.202682e+02 FES = 782100 μ_F = 0.271 μ_{CR} = 0.040
 Diversity = 2.2881e+00 k_t = [3, 2, 3, 2]
 Generation 7830: Best = 6.202682e+02 FES = 783100 μ_F = 0.359 μ_{CR} = 0.045
 Diversity = 2.2881e+00 k_t = [1, 1, 1, 1]
 Generation 7840: Best = 6.202682e+02 FES = 784100 μ_F = 0.369 μ_{CR} = 0.035
 Diversity = 2.2372e+00 k_t = [2, 3, 3, 4]
 Generation 7850: Best = 6.202682e+02 FES = 785100 μ_F = 0.300 μ_{CR} = 0.023
 Diversity = 2.2373e+00 k_t = [1, 1, 1, 1]
 Generation 7860: Best = 6.202682e+02 FES = 786100 μ_F = 0.249 μ_{CR} = 0.035
 Diversity = 2.2374e+00 k_t = [4, 3, 2, 2]
 Generation 7870: Best = 6.202682e+02 FES = 787100 μ_F = 0.348 μ_{CR} = 0.031
 Diversity = 2.2374e+00 k_t = [1, 1, 1, 1]
 Generation 7880: Best = 6.202682e+02 FES = 788100 μ_F = 0.306 μ_{CR} = 0.038
 Diversity = 2.2374e+00 k_t = [3, 4, 2, 3]
 Generation 7890: Best = 6.202679e+02 FES = 789100 μ_F = 0.282 μ_{CR} = 0.040
 Diversity = 2.2375e+00 k_t = [1, 1, 1, 1]
 Generation 7900: Best = 6.202679e+02 FES = 790100 μ_F = 0.395 μ_{CR} = 0.037
 Diversity = 2.2068e+00 k_t = [3, 4, 3, 4]
 Generation 7910: Best = 6.202679e+02 FES = 791100 μ_F = 0.321 μ_{CR} = 0.033
 Diversity = 2.2068e+00 k_t = [1, 1, 1, 1]
 Generation 7920: Best = 6.202679e+02 FES = 792100 μ_F = 0.388 μ_{CR} = 0.029
 Diversity = 2.1713e+00 k_t = [4, 2, 3, 4]
 Generation 7930: Best = 6.202679e+02 FES = 793100 μ_F = 0.332 μ_{CR} = 0.032
 Diversity = 2.1714e+00 k_t = [1, 1, 1, 1]
 Generation 7940: Best = 6.202598e+02 FES = 794100 μ_F = 0.278 μ_{CR} = 0.035
 Diversity = 2.1714e+00 k_t = [2, 4, 4, 3]
 Generation 7950: Best = 6.202598e+02 FES = 795100 μ_F = 0.292 μ_{CR} = 0.049
 Diversity = 2.1714e+00 k_t = [1, 1, 1, 1]
 Generation 7960: Best = 6.202598e+02 FES = 796100 μ_F = 0.275 μ_{CR} = 0.057
 Diversity = 2.1715e+00 k_t = [2, 4, 4, 3]
 Generation 7970: Best = 6.202598e+02 FES = 797100 μ_F = 0.252 μ_{CR} = 0.054
 Diversity = 2.1717e+00 k_t = [1, 1, 1, 1]
 Generation 7980: Best = 6.202536e+02 FES = 798100 μ_F = 0.214 μ_{CR} = 0.046
 Diversity = 2.1717e+00 k_t = [2, 2, 3, 3]
 Generation 7990: Best = 6.202202e+02 FES = 799100 μ_F = 0.228 μ_{CR} = 0.040
 Diversity = 2.1718e+00 k_t = [1, 1, 1, 1]
 Generation 8000: Best = 6.202202e+02 FES = 800100 μ_F = 0.244 μ_{CR} = 0.055
 Diversity = 2.1720e+00 k_t = [2, 2, 2, 3]
 Generation 8010: Best = 6.202202e+02 FES = 801100 μ_F = 0.276 μ_{CR} = 0.042
 Diversity = 2.1721e+00 k_t = [1, 1, 1, 1]
 Generation 8020: Best = 6.202202e+02 FES = 802100 μ_F = 0.317 μ_{CR} = 0.037
 Diversity = 2.1509e+00 k_t = [4, 2, 3, 4]
 Generation 8030: Best = 6.202202e+02 FES = 803100 μ_F = 0.270 μ_{CR} = 0.031
 Diversity = 2.1509e+00 k_t = [1, 1, 1, 1]
 Generation 8040: Best = 6.202171e+02 FES = 804100 μ_F = 0.239 μ_{CR} = 0.037
 Diversity = 2.1510e+00 k_t = [2, 2, 2, 4]
 Generation 8050: Best = 6.202171e+02 FES = 805100 μ_F = 0.183 μ_{CR} = 0.037
 Diversity = 2.1511e+00 k_t = [1, 1, 1, 1]

Generation 8060: Best = 6.202171e+02 FES = 806100 μ_F = 0.181 μ_{CR} = 0.046
 Diversity = 2.1514e+00 k_t = [4, 3, 4, 3]
 Generation 8070: Best = 6.202171e+02 FES = 807100 μ_F = 0.220 μ_{CR} = 0.039
 Diversity = 2.1514e+00 k_t = [1, 1, 1, 1]
 Generation 8080: Best = 6.202122e+02 FES = 808100 μ_F = 0.279 μ_{CR} = 0.039
 Diversity = 2.1359e+00 k_t = [2, 3, 4, 3]
 Generation 8090: Best = 6.202122e+02 FES = 809100 μ_F = 0.326 μ_{CR} = 0.038
 Diversity = 2.1155e+00 k_t = [1, 1, 1, 1]
 Generation 8100: Best = 6.202122e+02 FES = 810100 μ_F = 0.296 μ_{CR} = 0.041
 Diversity = 2.1158e+00 k_t = [3, 3, 4, 2]
 Generation 8110: Best = 6.201847e+02 FES = 811100 μ_F = 0.253 μ_{CR} = 0.054
 Diversity = 2.1159e+00 k_t = [1, 1, 1, 1]
 Generation 8120: Best = 6.201798e+02 FES = 812100 μ_F = 0.206 μ_{CR} = 0.047
 Diversity = 2.1158e+00 k_t = [3, 3, 3, 2]
 Generation 8130: Best = 6.201751e+02 FES = 813100 μ_F = 0.186 μ_{CR} = 0.042
 Diversity = 2.1159e+00 k_t = [1, 1, 1, 1]
 Generation 8140: Best = 6.201751e+02 FES = 814100 μ_F = 0.197 μ_{CR} = 0.050
 Diversity = 2.1159e+00 k_t = [4, 2, 4, 2]
 Generation 8150: Best = 6.201751e+02 FES = 815100 μ_F = 0.194 μ_{CR} = 0.044
 Diversity = 2.1160e+00 k_t = [1, 1, 1, 1]
 Generation 8160: Best = 6.201751e+02 FES = 816100 μ_F = 0.232 μ_{CR} = 0.033
 Diversity = 2.1019e+00 k_t = [4, 3, 3, 4]
 Generation 8170: Best = 6.201731e+02 FES = 817100 μ_F = 0.252 μ_{CR} = 0.036
 Diversity = 2.1021e+00 k_t = [1, 1, 1, 1]
 Generation 8180: Best = 6.201731e+02 FES = 818100 μ_F = 0.228 μ_{CR} = 0.039
 Diversity = 2.1020e+00 k_t = [3, 3, 2, 4]
 Generation 8190: Best = 6.201674e+02 FES = 819100 μ_F = 0.203 μ_{CR} = 0.044
 Diversity = 2.1020e+00 k_t = [1, 1, 1, 1]
 Generation 8200: Best = 6.201674e+02 FES = 820100 μ_F = 0.212 μ_{CR} = 0.057
 Diversity = 2.1020e+00 k_t = [4, 2, 2, 2]
 Generation 8210: Best = 6.201674e+02 FES = 821100 μ_F = 0.171 μ_{CR} = 0.065
 Diversity = 2.1020e+00 k_t = [1, 1, 1, 1]
 Generation 8220: Best = 6.201674e+02 FES = 822100 μ_F = 0.260 μ_{CR} = 0.054
 Diversity = 2.0975e+00 k_t = [4, 2, 2, 4]
 Generation 8230: Best = 6.201611e+02 FES = 823100 μ_F = 0.226 μ_{CR} = 0.054
 Diversity = 2.0976e+00 k_t = [1, 1, 1, 1]
 Generation 8240: Best = 6.201590e+02 FES = 824100 μ_F = 0.237 μ_{CR} = 0.049
 Diversity = 2.0976e+00 k_t = [4, 2, 4, 2]
 Generation 8250: Best = 6.201541e+02 FES = 825100 μ_F = 0.166 μ_{CR} = 0.055
 Diversity = 2.0976e+00 k_t = [1, 1, 1, 1]
 Generation 8260: Best = 6.201541e+02 FES = 826100 μ_F = 0.180 μ_{CR} = 0.048
 Diversity = 2.0976e+00 k_t = [2, 2, 2, 2]
 Generation 8270: Best = 6.201364e+02 FES = 827100 μ_F = 0.205 μ_{CR} = 0.054
 Diversity = 2.0976e+00 k_t = [1, 1, 1, 1]
 Generation 8280: Best = 6.201364e+02 FES = 828100 μ_F = 0.309 μ_{CR} = 0.059
 Diversity = 2.0977e+00 k_t = [2, 3, 4, 4]
 Generation 8290: Best = 6.201364e+02 FES = 829100 μ_F = 0.291 μ_{CR} = 0.052
 Diversity = 2.0977e+00 k_t = [1, 1, 1, 1]
 Generation 8300: Best = 6.201364e+02 FES = 830100 μ_F = 0.343 μ_{CR} = 0.045
 Diversity = 2.0718e+00 k_t = [2, 2, 3, 4]
 Generation 8310: Best = 6.201364e+02 FES = 831100 μ_F = 0.299 μ_{CR} = 0.041
 Diversity = 2.0719e+00 k_t = [1, 1, 1, 1]
 Generation 8320: Best = 6.201356e+02 FES = 832100 μ_F = 0.333 μ_{CR} = 0.045
 Diversity = 2.0719e+00 k_t = [2, 4, 4, 3]

Generation 8330: Best = 6.201356e+02 FES = 833100 μ_F = 0.308 μ_{CR} = 0.030
 Diversity = 2.0719e+00 k_t = [1, 1, 1, 1]
 Generation 8340: Best = 6.201356e+02 FES = 834100 μ_F = 0.349 μ_{CR} = 0.046
 Diversity = 2.0301e+00 k_t = [2, 4, 3, 4]
 Generation 8350: Best = 6.201356e+02 FES = 835100 μ_F = 0.278 μ_{CR} = 0.054
 Diversity = 2.0300e+00 k_t = [1, 1, 1, 1]
 Generation 8360: Best = 6.201356e+02 FES = 836100 μ_F = 0.232 μ_{CR} = 0.051
 Diversity = 2.0300e+00 k_t = [2, 3, 3, 2]
 Generation 8370: Best = 6.201356e+02 FES = 837100 μ_F = 0.188 μ_{CR} = 0.047
 Diversity = 2.0300e+00 k_t = [1, 1, 1, 1]
 Generation 8380: Best = 6.201314e+02 FES = 838100 μ_F = 0.159 μ_{CR} = 0.060
 Diversity = 2.0301e+00 k_t = [3, 2, 2, 3]
 Generation 8390: Best = 6.201309e+02 FES = 839100 μ_F = 0.179 μ_{CR} = 0.052
 Diversity = 2.0302e+00 k_t = [1, 1, 1, 1]
 Generation 8400: Best = 6.201309e+02 FES = 840100 μ_F = 0.149 μ_{CR} = 0.058
 Diversity = 2.0302e+00 k_t = [3, 2, 4, 3]
 Generation 8410: Best = 6.201292e+02 FES = 841100 μ_F = 0.141 μ_{CR} = 0.050
 Diversity = 2.0302e+00 k_t = [1, 1, 1, 1]
 Generation 8420: Best = 6.201175e+02 FES = 842100 μ_F = 0.190 μ_{CR} = 0.056
 Diversity = 2.0302e+00 k_t = [3, 4, 4, 2]
 Generation 8430: Best = 6.201175e+02 FES = 843100 μ_F = 0.155 μ_{CR} = 0.050
 Diversity = 2.0302e+00 k_t = [1, 1, 1, 1]
 Generation 8440: Best = 6.201175e+02 FES = 844100 μ_F = 0.162 μ_{CR} = 0.048
 Diversity = 2.0302e+00 k_t = [2, 4, 4, 2]
 Generation 8450: Best = 6.201139e+02 FES = 845100 μ_F = 0.229 μ_{CR} = 0.048
 Diversity = 2.0302e+00 k_t = [1, 1, 1, 1]
 Generation 8460: Best = 6.201139e+02 FES = 846100 μ_F = 0.292 μ_{CR} = 0.046
 Diversity = 2.0145e+00 k_t = [3, 2, 3, 4]
 Generation 8470: Best = 6.201139e+02 FES = 847100 μ_F = 0.287 μ_{CR} = 0.043
 Diversity = 2.0145e+00 k_t = [1, 1, 1, 1]
 Generation 8480: Best = 6.201117e+02 FES = 848100 μ_F = 0.323 μ_{CR} = 0.063
 Diversity = 1.9926e+00 k_t = [3, 2, 3, 4]
 Generation 8490: Best = 6.201117e+02 FES = 849100 μ_F = 0.431 μ_{CR} = 0.043
 Diversity = 1.9733e+00 k_t = [1, 1, 1, 1]
 Generation 8500: Best = 6.201102e+02 FES = 850100 μ_F = 0.369 μ_{CR} = 0.044
 Diversity = 1.9733e+00 k_t = [3, 3, 2, 3]
 Generation 8510: Best = 6.201102e+02 FES = 851100 μ_F = 0.352 μ_{CR} = 0.054
 Diversity = 1.9733e+00 k_t = [1, 1, 1, 1]
 Generation 8520: Best = 6.201102e+02 FES = 852100 μ_F = 0.336 μ_{CR} = 0.054
 Diversity = 1.9734e+00 k_t = [3, 2, 2, 3]
 Generation 8530: Best = 6.201102e+02 FES = 853100 μ_F = 0.271 μ_{CR} = 0.055
 Diversity = 1.9412e+00 k_t = [1, 1, 1, 1]
 Generation 8540: Best = 6.201090e+02 FES = 854100 μ_F = 0.250 μ_{CR} = 0.061
 Diversity = 1.9412e+00 k_t = [2, 4, 2, 2]
 Generation 8550: Best = 6.201073e+02 FES = 855100 μ_F = 0.198 μ_{CR} = 0.047
 Diversity = 1.9412e+00 k_t = [1, 1, 1, 1]
 Generation 8560: Best = 6.201067e+02 FES = 856100 μ_F = 0.267 μ_{CR} = 0.043
 Diversity = 1.9412e+00 k_t = [2, 3, 2, 3]
 Generation 8570: Best = 6.201067e+02 FES = 857100 μ_F = 0.274 μ_{CR} = 0.031
 Diversity = 1.9411e+00 k_t = [1, 1, 1, 1]
 Generation 8580: Best = 6.201067e+02 FES = 858100 μ_F = 0.304 μ_{CR} = 0.038
 Diversity = 1.9214e+00 k_t = [2, 3, 2, 4]
 Generation 8590: Best = 6.201067e+02 FES = 859100 μ_F = 0.264 μ_{CR} = 0.045
 Diversity = 1.9215e+00 k_t = [1, 1, 1, 1]

Generation 8600: Best = 6.201067e+02 FES = 860100 μ_F = 0.276 μ_{CR} = 0.046
 Diversity = 1.8990e+00 k_t = [2, 4, 3, 2]
 Generation 8610: Best = 6.201011e+02 FES = 861100 μ_F = 0.326 μ_{CR} = 0.045
 Diversity = 1.8989e+00 k_t = [1, 1, 1, 1]
 Generation 8620: Best = 6.201006e+02 FES = 862100 μ_F = 0.298 μ_{CR} = 0.044
 Diversity = 1.8911e+00 k_t = [4, 2, 4, 4]
 Generation 8630: Best = 6.201006e+02 FES = 863100 μ_F = 0.300 μ_{CR} = 0.038
 Diversity = 1.8912e+00 k_t = [1, 1, 1, 1]
 Generation 8640: Best = 6.201006e+02 FES = 864100 μ_F = 0.286 μ_{CR} = 0.026
 Diversity = 1.8916e+00 k_t = [2, 3, 2, 3]
 Generation 8650: Best = 6.200902e+02 FES = 865100 μ_F = 0.273 μ_{CR} = 0.032
 Diversity = 1.8775e+00 k_t = [1, 1, 1, 1]
 Generation 8660: Best = 6.200902e+02 FES = 866100 μ_F = 0.295 μ_{CR} = 0.031
 Diversity = 1.8775e+00 k_t = [4, 2, 2, 3]
 Generation 8670: Best = 6.200902e+02 FES = 867100 μ_F = 0.283 μ_{CR} = 0.029
 Diversity = 1.8776e+00 k_t = [1, 1, 1, 1]
 Generation 8680: Best = 6.200887e+02 FES = 868100 μ_F = 0.235 μ_{CR} = 0.039
 Diversity = 1.8777e+00 k_t = [3, 2, 2, 4]
 Generation 8690: Best = 6.200839e+02 FES = 869100 μ_F = 0.196 μ_{CR} = 0.052
 Diversity = 1.8777e+00 k_t = [1, 1, 1, 1]
 Generation 8700: Best = 6.200832e+02 FES = 870100 μ_F = 0.146 μ_{CR} = 0.067
 Diversity = 1.8778e+00 k_t = [2, 2, 3, 2]
 Generation 8710: Best = 6.200832e+02 FES = 871100 μ_F = 0.203 μ_{CR} = 0.047
 Diversity = 1.8778e+00 k_t = [1, 1, 1, 1]
 Generation 8720: Best = 6.200817e+02 FES = 872100 μ_F = 0.206 μ_{CR} = 0.047
 Diversity = 1.8631e+00 k_t = [4, 4, 3, 2]
 Generation 8730: Best = 6.200817e+02 FES = 873100 μ_F = 0.156 μ_{CR} = 0.074
 Diversity = 1.8632e+00 k_t = [1, 1, 1, 1]
 Generation 8740: Best = 6.200817e+02 FES = 874100 μ_F = 0.218 μ_{CR} = 0.104
 Diversity = 1.8471e+00 k_t = [4, 2, 3, 4]
 Generation 8750: Best = 6.200817e+02 FES = 875100 μ_F = 0.210 μ_{CR} = 0.091
 Diversity = 1.8470e+00 k_t = [1, 1, 1, 1]
 Generation 8760: Best = 6.200591e+02 FES = 876100 μ_F = 0.244 μ_{CR} = 0.077
 Diversity = 1.8273e+00 k_t = [2, 4, 4, 2]
 Generation 8770: Best = 6.200591e+02 FES = 877100 μ_F = 0.228 μ_{CR} = 0.060
 Diversity = 1.8273e+00 k_t = [1, 1, 1, 1]
 Generation 8780: Best = 6.200591e+02 FES = 878100 μ_F = 0.246 μ_{CR} = 0.076
 Diversity = 1.8272e+00 k_t = [4, 2, 3, 2]
 Generation 8790: Best = 6.200530e+02 FES = 879100 μ_F = 0.234 μ_{CR} = 0.063
 Diversity = 1.8274e+00 k_t = [1, 1, 1, 1]
 Generation 8800: Best = 6.200530e+02 FES = 880100 μ_F = 0.287 μ_{CR} = 0.065
 Diversity = 1.8100e+00 k_t = [4, 4, 4, 4]
 Generation 8810: Best = 6.200530e+02 FES = 881100 μ_F = 0.312 μ_{CR} = 0.045
 Diversity = 1.7848e+00 k_t = [1, 1, 1, 1]
 Generation 8820: Best = 6.200530e+02 FES = 882100 μ_F = 0.449 μ_{CR} = 0.029
 Diversity = 1.7530e+00 k_t = [3, 4, 3, 4]
 Generation 8830: Best = 6.200530e+02 FES = 883100 μ_F = 0.358 μ_{CR} = 0.027
 Diversity = 1.7453e+00 k_t = [1, 1, 1, 1]
 Generation 8840: Best = 6.200530e+02 FES = 884100 μ_F = 0.359 μ_{CR} = 0.034
 Diversity = 1.7068e+00 k_t = [4, 3, 2, 4]
 Generation 8850: Best = 6.200471e+02 FES = 885100 μ_F = 0.301 μ_{CR} = 0.035
 Diversity = 1.7070e+00 k_t = [1, 1, 1, 1]
 Generation 8860: Best = 6.200431e+02 FES = 886100 μ_F = 0.205 μ_{CR} = 0.054
 Diversity = 1.7072e+00 k_t = [4, 2, 3, 2]

Generation 8870: Best = 6.200431e+02	FES = 887100	$\mu_F = 0.231$	$\mu_{CR} = 0.041$
Diversity = 1.7072e+00	$k_t = [1, 1, 1, 1]$		
Generation 8880: Best = 6.200431e+02	FES = 888100	$\mu_F = 0.246$	$\mu_{CR} = 0.043$
Diversity = 1.7074e+00	$k_t = [4, 4, 2, 2]$		
Generation 8890: Best = 6.200428e+02	FES = 889100	$\mu_F = 0.272$	$\mu_{CR} = 0.054$
Diversity = 1.6976e+00	$k_t = [1, 1, 1, 1]$		
Generation 8900: Best = 6.200427e+02	FES = 890100	$\mu_F = 0.308$	$\mu_{CR} = 0.041$
Diversity = 1.6977e+00	$k_t = [4, 4, 3, 2]$		
Generation 8910: Best = 6.200427e+02	FES = 891100	$\mu_F = 0.317$	$\mu_{CR} = 0.054$
Diversity = 1.6976e+00	$k_t = [1, 1, 1, 1]$		
Generation 8920: Best = 6.200427e+02	FES = 892100	$\mu_F = 0.306$	$\mu_{CR} = 0.050$
Diversity = 1.6976e+00	$k_t = [3, 2, 3, 4]$		
Generation 8930: Best = 6.200427e+02	FES = 893100	$\mu_F = 0.362$	$\mu_{CR} = 0.051$
Diversity = 1.6976e+00	$k_t = [1, 1, 1, 1]$		
Generation 8940: Best = 6.200427e+02	FES = 894100	$\mu_F = 0.322$	$\mu_{CR} = 0.046$
Diversity = 1.6976e+00	$k_t = [2, 3, 2, 3]$		
Generation 8950: Best = 6.200427e+02	FES = 895100	$\mu_F = 0.314$	$\mu_{CR} = 0.042$
Diversity = 1.6977e+00	$k_t = [1, 1, 1, 1]$		
Generation 8960: Best = 6.200252e+02	FES = 896100	$\mu_F = 0.257$	$\mu_{CR} = 0.040$
Diversity = 1.6976e+00	$k_t = [4, 2, 2, 3]$		
Generation 8970: Best = 6.200252e+02	FES = 897100	$\mu_F = 0.233$	$\mu_{CR} = 0.040$
Diversity = 1.6977e+00	$k_t = [1, 1, 1, 1]$		
Generation 8980: Best = 6.200252e+02	FES = 898100	$\mu_F = 0.228$	$\mu_{CR} = 0.046$
Diversity = 1.6977e+00	$k_t = [3, 2, 2, 2]$		
Generation 8990: Best = 6.200252e+02	FES = 899100	$\mu_F = 0.196$	$\mu_{CR} = 0.057$
Diversity = 1.6977e+00	$k_t = [1, 1, 1, 1]$		
Generation 9000: Best = 6.200252e+02	FES = 900100	$\mu_F = 0.213$	$\mu_{CR} = 0.055$
Diversity = 1.6977e+00	$k_t = [3, 4, 3, 3]$		
Generation 9010: Best = 6.200190e+02	FES = 901100	$\mu_F = 0.227$	$\mu_{CR} = 0.060$
Diversity = 1.6977e+00	$k_t = [1, 1, 1, 1]$		
Generation 9020: Best = 6.200190e+02	FES = 902100	$\mu_F = 0.285$	$\mu_{CR} = 0.046$
Diversity = 1.6977e+00	$k_t = [2, 4, 2, 2]$		
Generation 9030: Best = 6.200139e+02	FES = 903100	$\mu_F = 0.212$	$\mu_{CR} = 0.045$
Diversity = 1.6978e+00	$k_t = [1, 1, 1, 1]$		
Generation 9040: Best = 6.200127e+02	FES = 904100	$\mu_F = 0.245$	$\mu_{CR} = 0.038$
Diversity = 1.6978e+00	$k_t = [4, 4, 2, 3]$		
Generation 9050: Best = 6.200127e+02	FES = 905100	$\mu_F = 0.219$	$\mu_{CR} = 0.049$
Diversity = 1.6979e+00	$k_t = [1, 1, 1, 1]$		
Generation 9060: Best = 6.200127e+02	FES = 906100	$\mu_F = 0.199$	$\mu_{CR} = 0.055$
Diversity = 1.6979e+00	$k_t = [4, 4, 3, 3]$		
Generation 9070: Best = 6.200127e+02	FES = 907100	$\mu_F = 0.265$	$\mu_{CR} = 0.049$
Diversity = 1.6981e+00	$k_t = [1, 1, 1, 1]$		
Generation 9080: Best = 6.200127e+02	FES = 908100	$\mu_F = 0.281$	$\mu_{CR} = 0.035$
Diversity = 1.6980e+00	$k_t = [2, 4, 4, 3]$		
Generation 9090: Best = 6.200127e+02	FES = 909100	$\mu_F = 0.292$	$\mu_{CR} = 0.039$
Diversity = 1.6982e+00	$k_t = [1, 1, 1, 1]$		
Generation 9100: Best = 6.200078e+02	FES = 910100	$\mu_F = 0.262$	$\mu_{CR} = 0.043$
Diversity = 1.6982e+00	$k_t = [4, 2, 2, 2]$		
Generation 9110: Best = 6.200078e+02	FES = 911100	$\mu_F = 0.255$	$\mu_{CR} = 0.048$
Diversity = 1.6875e+00	$k_t = [1, 1, 1, 1]$		
Generation 9120: Best = 6.200057e+02	FES = 912100	$\mu_F = 0.308$	$\mu_{CR} = 0.042$
Diversity = 1.6876e+00	$k_t = [4, 4, 2, 3]$		
Generation 9130: Best = 6.200057e+02	FES = 913100	$\mu_F = 0.331$	$\mu_{CR} = 0.044$
Diversity = 1.6876e+00	$k_t = [1, 1, 1, 1]$		

Generation 9140: Best = 6.200057e+02 FES = 914100 μ_F = 0.330 μ_{CR} = 0.041
 Diversity = 1.6648e+00 k_t = [3, 3, 2, 4]
 Generation 9150: Best = 6.200057e+02 FES = 915100 μ_F = 0.381 μ_{CR} = 0.032
 Diversity = 1.6648e+00 k_t = [1, 1, 1, 1]
 Generation 9160: Best = 6.200057e+02 FES = 916100 μ_F = 0.341 μ_{CR} = 0.027
 Diversity = 1.6648e+00 k_t = [2, 2, 4, 2]
 Generation 9170: Best = 6.200048e+02 FES = 917100 μ_F = 0.281 μ_{CR} = 0.033
 Diversity = 1.6648e+00 k_t = [1, 1, 1, 1]
 Generation 9180: Best = 6.200013e+02 FES = 918100 μ_F = 0.329 μ_{CR} = 0.036
 Diversity = 1.6314e+00 k_t = [2, 2, 2, 2]
 Generation 9190: Best = 6.200013e+02 FES = 919100 μ_F = 0.271 μ_{CR} = 0.041
 Diversity = 1.6314e+00 k_t = [1, 1, 1, 1]
 Generation 9200: Best = 6.199985e+02 FES = 920100 μ_F = 0.314 μ_{CR} = 0.032
 Diversity = 1.6199e+00 k_t = [3, 4, 4, 4]
 Generation 9210: Best = 6.199982e+02 FES = 921100 μ_F = 0.372 μ_{CR} = 0.044
 Diversity = 1.6199e+00 k_t = [1, 1, 1, 1]
 Generation 9220: Best = 6.199978e+02 FES = 922100 μ_F = 0.342 μ_{CR} = 0.043
 Diversity = 1.6199e+00 k_t = [2, 2, 4, 2]
 Generation 9230: Best = 6.199970e+02 FES = 923100 μ_F = 0.368 μ_{CR} = 0.057
 Diversity = 1.6199e+00 k_t = [1, 1, 1, 1]
 Generation 9240: Best = 6.199968e+02 FES = 924100 μ_F = 0.413 μ_{CR} = 0.054
 Diversity = 1.6200e+00 k_t = [4, 2, 2, 2]
 Generation 9250: Best = 6.199968e+02 FES = 925100 μ_F = 0.375 μ_{CR} = 0.052
 Diversity = 1.6199e+00 k_t = [1, 1, 1, 1]
 Generation 9260: Best = 6.199953e+02 FES = 926100 μ_F = 0.329 μ_{CR} = 0.059
 Diversity = 1.6074e+00 k_t = [2, 4, 2, 2]
 Generation 9270: Best = 6.199953e+02 FES = 927100 μ_F = 0.343 μ_{CR} = 0.048
 Diversity = 1.6074e+00 k_t = [1, 1, 1, 1]
 Generation 9280: Best = 6.199953e+02 FES = 928100 μ_F = 0.425 μ_{CR} = 0.057
 Diversity = 1.5452e+00 k_t = [2, 2, 3, 4]
 Generation 9290: Best = 6.199953e+02 FES = 929100 μ_F = 0.339 μ_{CR} = 0.054
 Diversity = 1.5453e+00 k_t = [1, 1, 1, 1]
 Generation 9300: Best = 6.199953e+02 FES = 930100 μ_F = 0.461 μ_{CR} = 0.046
 Diversity = 1.4875e+00 k_t = [4, 2, 4, 4]
 Generation 9310: Best = 6.199936e+02 FES = 931100 μ_F = 0.385 μ_{CR} = 0.045
 Diversity = 1.4453e+00 k_t = [1, 1, 1, 1]
 Generation 9320: Best = 6.199936e+02 FES = 932100 μ_F = 0.396 μ_{CR} = 0.050
 Diversity = 1.4281e+00 k_t = [3, 2, 4, 2]
 Generation 9330: Best = 6.199813e+02 FES = 933100 μ_F = 0.384 μ_{CR} = 0.049
 Diversity = 1.4281e+00 k_t = [1, 1, 1, 1]
 Generation 9340: Best = 6.199748e+02 FES = 934100 μ_F = 0.448 μ_{CR} = 0.043
 Diversity = 1.4006e+00 k_t = [3, 2, 3, 3]
 Generation 9350: Best = 6.199707e+02 FES = 935100 μ_F = 0.462 μ_{CR} = 0.036
 Diversity = 1.3629e+00 k_t = [1, 1, 1, 1]
 Generation 9360: Best = 6.199689e+02 FES = 936100 μ_F = 0.386 μ_{CR} = 0.064
 Diversity = 1.3632e+00 k_t = [3, 3, 2, 3]
 Generation 9370: Best = 6.199659e+02 FES = 937100 μ_F = 0.422 μ_{CR} = 0.038
 Diversity = 1.3633e+00 k_t = [1, 1, 1, 1]
 Generation 9380: Best = 6.199639e+02 FES = 938100 μ_F = 0.428 μ_{CR} = 0.058
 Diversity = 1.3143e+00 k_t = [3, 3, 3, 3]
 Generation 9390: Best = 6.199639e+02 FES = 939100 μ_F = 0.514 μ_{CR} = 0.048
 Diversity = 1.2154e+00 k_t = [1, 1, 1, 1]
 Generation 9400: Best = 6.199639e+02 FES = 940100 μ_F = 0.524 μ_{CR} = 0.043
 Diversity = 1.2166e+00 k_t = [3, 2, 3, 3]

Generation 9410: Best = 6.199639e+02 FES = 941100 μ_F = 0.443 μ_{CR} = 0.043
 Diversity = 1.1820e+00 k_t = [1, 1, 1, 1]
 Generation 9420: Best = 6.199625e+02 FES = 942100 μ_F = 0.361 μ_{CR} = 0.055
 Diversity = 1.1821e+00 k_t = [2, 3, 3, 3]
 Generation 9430: Best = 6.199606e+02 FES = 943100 μ_F = 0.367 μ_{CR} = 0.067
 Diversity = 1.1455e+00 k_t = [1, 1, 1, 1]
 Generation 9440: Best = 6.199604e+02 FES = 944100 μ_F = 0.332 μ_{CR} = 0.078
 Diversity = 1.1456e+00 k_t = [2, 3, 2, 2]
 Generation 9450: Best = 6.199604e+02 FES = 945100 μ_F = 0.322 μ_{CR} = 0.072
 Diversity = 1.1455e+00 k_t = [1, 1, 1, 1]
 Generation 9460: Best = 6.199569e+02 FES = 946100 μ_F = 0.309 μ_{CR} = 0.065
 Diversity = 1.1455e+00 k_t = [2, 4, 3, 3]
 Generation 9470: Best = 6.199569e+02 FES = 947100 μ_F = 0.292 μ_{CR} = 0.060
 Diversity = 1.1455e+00 k_t = [1, 1, 1, 1]
 Generation 9480: Best = 6.199518e+02 FES = 948100 μ_F = 0.307 μ_{CR} = 0.067
 Diversity = 1.1458e+00 k_t = [4, 2, 2, 2]
 Generation 9490: Best = 6.199457e+02 FES = 949100 μ_F = 0.365 μ_{CR} = 0.058
 Diversity = 1.1457e+00 k_t = [1, 1, 1, 1]
 Generation 9500: Best = 6.199455e+02 FES = 950100 μ_F = 0.336 μ_{CR} = 0.047
 Diversity = 1.1035e+00 k_t = [2, 4, 2, 2]
 Generation 9510: Best = 6.199455e+02 FES = 951100 μ_F = 0.324 μ_{CR} = 0.052
 Diversity = 1.1038e+00 k_t = [1, 1, 1, 1]
 Generation 9520: Best = 6.199435e+02 FES = 952100 μ_F = 0.337 μ_{CR} = 0.057
 Diversity = 1.1038e+00 k_t = [3, 3, 4, 4]
 Generation 9530: Best = 6.199407e+02 FES = 953100 μ_F = 0.383 μ_{CR} = 0.059
 Diversity = 1.0551e+00 k_t = [1, 1, 1, 1]
 Generation 9540: Best = 6.199398e+02 FES = 954100 μ_F = 0.442 μ_{CR} = 0.046
 Diversity = 1.0548e+00 k_t = [3, 4, 4, 4]
 Generation 9550: Best = 6.199398e+02 FES = 955100 μ_F = 0.434 μ_{CR} = 0.042
 Diversity = 1.0542e+00 k_t = [1, 1, 1, 1]
 Generation 9560: Best = 6.199398e+02 FES = 956100 μ_F = 0.529 μ_{CR} = 0.049
 Diversity = 1.0222e+00 k_t = [2, 3, 3, 4]
 Generation 9570: Best = 6.199388e+02 FES = 957100 μ_F = 0.525 μ_{CR} = 0.039
 Diversity = 9.7606e-01 k_t = [1, 1, 1, 1]
 Generation 9580: Best = 6.199388e+02 FES = 958100 μ_F = 0.442 μ_{CR} = 0.061
 Diversity = 9.7565e-01 k_t = [4, 2, 2, 3]
 Generation 9590: Best = 6.199388e+02 FES = 959100 μ_F = 0.405 μ_{CR} = 0.053
 Diversity = 9.7299e-01 k_t = [1, 1, 1, 1]
 Generation 9600: Best = 6.199180e+02 FES = 960100 μ_F = 0.398 μ_{CR} = 0.056
 Diversity = 9.7183e-01 k_t = [2, 4, 3, 4]
 Generation 9610: Best = 6.199180e+02 FES = 961100 μ_F = 0.445 μ_{CR} = 0.080
 Diversity = 8.6741e-01 k_t = [1, 1, 1, 1]
 Generation 9620: Best = 6.199180e+02 FES = 962100 μ_F = 0.491 μ_{CR} = 0.089
 Diversity = 6.7297e-01 k_t = [2, 4, 2, 2]
 Generation 9630: Best = 6.199144e+02 FES = 963100 μ_F = 0.483 μ_{CR} = 0.086
 Diversity = 6.0135e-01 k_t = [1, 1, 1, 1]
 Generation 9640: Best = 6.199144e+02 FES = 964100 μ_F = 0.543 μ_{CR} = 0.063
 Diversity = 3.8256e-01 k_t = [2, 2, 3, 4]
 Generation 9650: Best = 6.199086e+02 FES = 965100 μ_F = 0.554 μ_{CR} = 0.072
 Diversity = 1.6358e-01 k_t = [1, 1, 1, 1]
 Generation 9660: Best = 6.199002e+02 FES = 966100 μ_F = 0.587 μ_{CR} = 0.077
 Diversity = 1.0651e-01 k_t = [3, 4, 4, 4]
 Generation 9670: Best = 6.198987e+02 FES = 967100 μ_F = 0.602 μ_{CR} = 0.071
 Diversity = 8.2645e-02 k_t = [1, 1, 1, 1]

Generation 9680: Best = 6.198955e+02 FES = 968100 μ_F = 0.577 μ_{CR} = 0.066
 Diversity = 6.4593e-02 k_t = [4, 4, 2, 4]
 Generation 9690: Best = 6.198889e+02 FES = 969100 μ_F = 0.577 μ_{CR} = 0.072
 Diversity = 5.1810e-02 k_t = [1, 1, 1, 1]
 Generation 9700: Best = 6.198889e+02 FES = 970100 μ_F = 0.599 μ_{CR} = 0.077
 Diversity = 4.7592e-02 k_t = [2, 4, 2, 3]
 Generation 9710: Best = 6.198713e+02 FES = 971100 μ_F = 0.619 μ_{CR} = 0.081
 Diversity = 4.4418e-02 k_t = [1, 1, 1, 1]
 Generation 9720: Best = 6.198543e+02 FES = 972100 μ_F = 0.626 μ_{CR} = 0.088
 Diversity = 3.9433e-02 k_t = [4, 3, 4, 4]
 Generation 9730: Best = 6.198220e+02 FES = 973100 μ_F = 0.622 μ_{CR} = 0.089
 Diversity = 3.8163e-02 k_t = [1, 1, 1, 1]
 Generation 9740: Best = 6.198104e+02 FES = 974100 μ_F = 0.656 μ_{CR} = 0.091
 Diversity = 3.7218e-02 k_t = [4, 3, 4, 3]
 Generation 9750: Best = 6.198081e+02 FES = 975100 μ_F = 0.636 μ_{CR} = 0.089
 Diversity = 3.6483e-02 k_t = [1, 1, 1, 1]
 Generation 9760: Best = 6.197695e+02 FES = 976100 μ_F = 0.637 μ_{CR} = 0.098
 Diversity = 3.6217e-02 k_t = [3, 3, 3, 2]
 Generation 9770: Best = 6.197634e+02 FES = 977100 μ_F = 0.649 μ_{CR} = 0.097
 Diversity = 3.6000e-02 k_t = [1, 1, 1, 1]
 Generation 9780: Best = 6.197608e+02 FES = 978100 μ_F = 0.648 μ_{CR} = 0.096
 Diversity = 2.9848e-02 k_t = [3, 3, 3, 4]
 Generation 9790: Best = 6.197496e+02 FES = 979100 μ_F = 0.644 μ_{CR} = 0.097
 Diversity = 2.7266e-02 k_t = [1, 1, 1, 1]
 Generation 9800: Best = 6.197318e+02 FES = 980100 μ_F = 0.628 μ_{CR} = 0.104
 Diversity = 2.3168e-02 k_t = [2, 2, 3, 2]
 Generation 9810: Best = 6.197260e+02 FES = 981100 μ_F = 0.627 μ_{CR} = 0.107
 Diversity = 2.0322e-02 k_t = [1, 1, 1, 1]
 Generation 9820: Best = 6.197134e+02 FES = 982100 μ_F = 0.619 μ_{CR} = 0.114
 Diversity = 1.6505e-02 k_t = [4, 2, 4, 3]
 Generation 9830: Best = 6.197050e+02 FES = 983100 μ_F = 0.625 μ_{CR} = 0.130
 Diversity = 1.4894e-02 k_t = [1, 1, 1, 1]
 Generation 9840: Best = 6.196890e+02 FES = 984100 μ_F = 0.633 μ_{CR} = 0.128
 Diversity = 1.4547e-02 k_t = [3, 3, 4, 2]
 Generation 9850: Best = 6.196802e+02 FES = 985100 μ_F = 0.628 μ_{CR} = 0.124
 Diversity = 1.2612e-02 k_t = [1, 1, 1, 1]
 Generation 9860: Best = 6.196765e+02 FES = 986100 μ_F = 0.634 μ_{CR} = 0.118
 Diversity = 1.0692e-02 k_t = [4, 3, 3, 4]
 Generation 9870: Best = 6.196699e+02 FES = 987100 μ_F = 0.647 μ_{CR} = 0.116
 Diversity = 8.8270e-03 k_t = [1, 1, 1, 1]
 Generation 9880: Best = 6.196649e+02 FES = 988100 μ_F = 0.667 μ_{CR} = 0.115
 Diversity = 8.8809e-03 k_t = [3, 4, 4, 2]
 Generation 9890: Best = 6.196608e+02 FES = 989100 μ_F = 0.658 μ_{CR} = 0.109
 Diversity = 8.7699e-03 k_t = [1, 1, 1, 1]
 Generation 9900: Best = 6.196576e+02 FES = 990100 μ_F = 0.658 μ_{CR} = 0.099
 Diversity = 7.2617e-03 k_t = [4, 4, 2, 3]
 Generation 9910: Best = 6.196534e+02 FES = 991100 μ_F = 0.660 μ_{CR} = 0.093
 Diversity = 6.9134e-03 k_t = [1, 1, 1, 1]
 Generation 9920: Best = 6.196474e+02 FES = 992100 μ_F = 0.645 μ_{CR} = 0.098
 Diversity = 6.4109e-03 k_t = [4, 3, 2, 4]
 Generation 9930: Best = 6.196454e+02 FES = 993100 μ_F = 0.669 μ_{CR} = 0.100
 Diversity = 6.8079e-03 k_t = [1, 1, 1, 1]
 Generation 9940: Best = 6.196449e+02 FES = 994100 μ_F = 0.703 μ_{CR} = 0.092
 Diversity = 6.2658e-03 k_t = [3, 3, 2, 4]

Generation 9950: Best = 6.196394e+02 FES = 995100 μ_F = 0.712 μ_{CR} = 0.099
 Diversity = 5.9486e-03 k_t = [1, 1, 1, 1]
 Generation 9960: Best = 6.196381e+02 FES = 996100 μ_F = 0.706 μ_{CR} = 0.096
 Diversity = 5.5063e-03 k_t = [4, 4, 3, 4]
 Generation 9970: Best = 6.196363e+02 FES = 997100 μ_F = 0.717 μ_{CR} = 0.087
 Diversity = 5.3583e-03 k_t = [1, 1, 1, 1]
 Generation 9980: Best = 6.196337e+02 FES = 998100 μ_F = 0.734 μ_{CR} = 0.099
 Diversity = 5.2764e-03 k_t = [2, 3, 4, 3]
 Generation 9990: Best = 6.196299e+02 FES = 999100 μ_F = 0.743 μ_{CR} = 0.094
 Diversity = 5.7353e-03 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 6.196271e+02

Total Function Evals = 1000000

Total Successful Updates = 28573

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Results:

Best Fitness : 6.196271e+02

Global Optimum : 6.000000e+02

Error (|best-optimum|) : 1.962708e+01

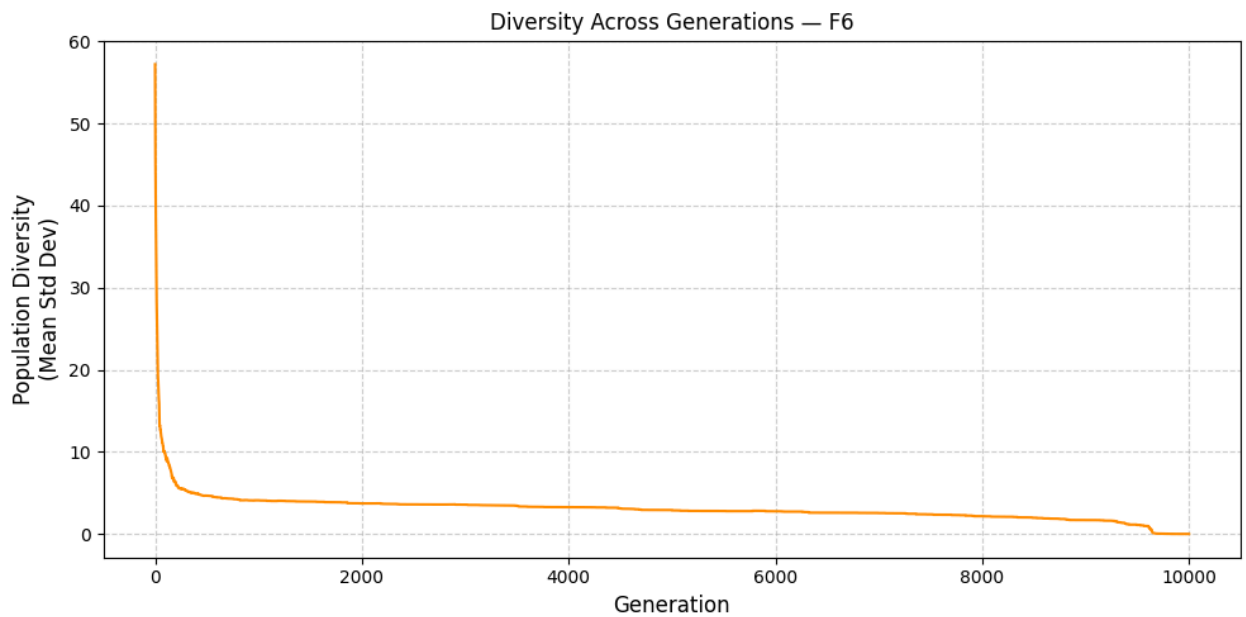
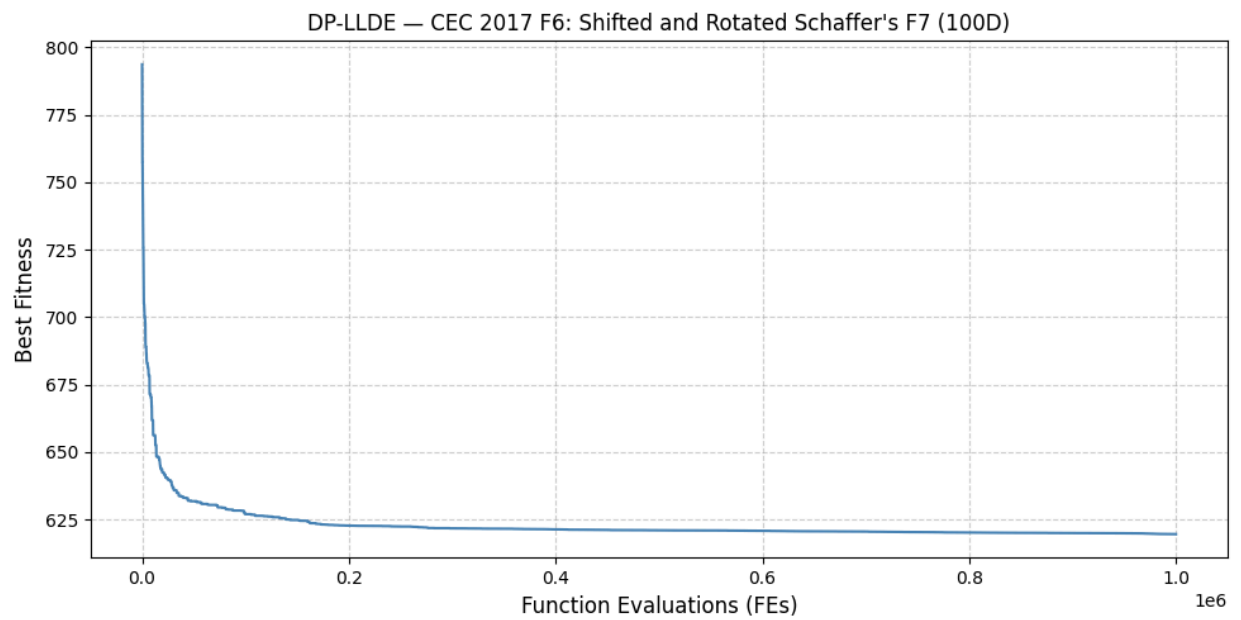
Best Solution (5 dim) : [69.27707114 -14.93703328 -16.58955674 -36.59993502
70.02339729]

$D^{(0)}$ (init. diversity): 5.7220e+01

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✓ Checkpoint saved to: /content/drive/My Drive/thesis_results/all_results_checkpoint.pkl



Graph saved as convergence_F6.png

✓ Plot saved to: /content/drive/My Drive/thesis_results/plots/convergence_F6.png

```
In [ ]: # =====
# F7 — Shifted and Rotated Lunacek Bi-Rastrigin
# =====
run_and_plot('F7', make_wrapper(simple.f7), 700.0, "Shifted and Rotated Lunacek Bi-Rastrigin")
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DP-LLDE on CEC 2017 F7 – Single Run

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Function : Shifted and Rotated Lunacek Bi-Rastrigin's (100D)
Bounds : [-100, 100]
Global optimum : 700.0
Initial $D^*(0) = 5.7220e+01$ | Diversity threshold = 5.7220e+00
Generation 0: Best Fitness = 1.400178e+04

Generation 10:	Best = 4.861021e+03	FES = 1100	$\mu_F = 0.537$	$\mu_{CR} = 0.353$
Diversity = 2.4329e+01	$k_t = [1, 1, 1, 1]$			
Generation 20:	Best = 3.226150e+03	FES = 2100	$\mu_F = 0.573$	$\mu_{CR} = 0.355$
Diversity = 1.2750e+01	$k_t = [2, 3, 2, 3]$			
Generation 30:	Best = 2.802508e+03	FES = 3100	$\mu_F = 0.590$	$\mu_{CR} = 0.361$
Diversity = 8.3282e+00	$k_t = [1, 1, 1, 1]$			
Generation 40:	Best = 2.463874e+03	FES = 4100	$\mu_F = 0.623$	$\mu_{CR} = 0.370$
Diversity = 5.8003e+00	$k_t = [3, 2, 3, 4]$			
Generation 50:	Best = 2.417695e+03	FES = 5100	$\mu_F = 0.663$	$\mu_{CR} = 0.365$
Diversity = 4.9934e+00	$k_t = [1, 1, 1, 1]$			
Generation 60:	Best = 2.371526e+03	FES = 6100	$\mu_F = 0.695$	$\mu_{CR} = 0.347$
Diversity = 4.6015e+00	$k_t = [4, 2, 2, 2]$			
Generation 70:	Best = 2.298990e+03	FES = 7100	$\mu_F = 0.702$	$\mu_{CR} = 0.316$
Diversity = 4.5007e+00	$k_t = [1, 1, 1, 1]$			
Generation 80:	Best = 2.206332e+03	FES = 8100	$\mu_F = 0.703$	$\mu_{CR} = 0.295$
Diversity = 4.2664e+00	$k_t = [4, 3, 3, 3]$			
Generation 90:	Best = 2.152500e+03	FES = 9100	$\mu_F = 0.729$	$\mu_{CR} = 0.255$
Diversity = 4.2031e+00	$k_t = [1, 1, 1, 1]$			
Generation 100:	Best = 2.125981e+03	FES = 10100	$\mu_F = 0.746$	$\mu_{CR} = 0.217$
Diversity = 4.2252e+00	$k_t = [2, 4, 4, 4]$			
Generation 110:	Best = 2.079890e+03	FES = 11100	$\mu_F = 0.726$	$\mu_{CR} = 0.165$
Diversity = 4.1526e+00	$k_t = [1, 1, 1, 1]$			
Generation 120:	Best = 2.057518e+03	FES = 12100	$\mu_F = 0.699$	$\mu_{CR} = 0.131$
Diversity = 4.0212e+00	$k_t = [2, 3, 3, 2]$			
Generation 130:	Best = 2.022304e+03	FES = 13100	$\mu_F = 0.691$	$\mu_{CR} = 0.110$
Diversity = 4.0545e+00	$k_t = [1, 1, 1, 1]$			
Generation 140:	Best = 1.996713e+03	FES = 14100	$\mu_F = 0.699$	$\mu_{CR} = 0.078$
Diversity = 4.0638e+00	$k_t = [3, 4, 4, 3]$			
Generation 150:	Best = 1.996713e+03	FES = 15100	$\mu_F = 0.690$	$\mu_{CR} = 0.068$
Diversity = 4.0209e+00	$k_t = [1, 1, 1, 1]$			
Generation 160:	Best = 1.996713e+03	FES = 16100	$\mu_F = 0.717$	$\mu_{CR} = 0.062$
Diversity = 3.9709e+00	$k_t = [3, 4, 2, 3]$			
Generation 170:	Best = 1.996278e+03	FES = 17100	$\mu_F = 0.741$	$\mu_{CR} = 0.052$
Diversity = 3.9219e+00	$k_t = [1, 1, 1, 1]$			
Generation 180:	Best = 1.950301e+03	FES = 18100	$\mu_F = 0.766$	$\mu_{CR} = 0.059$
Diversity = 3.8563e+00	$k_t = [2, 2, 3, 4]$			
Generation 190:	Best = 1.948319e+03	FES = 19100	$\mu_F = 0.762$	$\mu_{CR} = 0.051$
Diversity = 3.8543e+00	$k_t = [1, 1, 1, 1]$			
Generation 200:	Best = 1.948319e+03	FES = 20100	$\mu_F = 0.769$	$\mu_{CR} = 0.050$
Diversity = 3.8387e+00	$k_t = [3, 2, 3, 2]$			
Generation 210:	Best = 1.948319e+03	FES = 21100	$\mu_F = 0.792$	$\mu_{CR} = 0.046$
Diversity = 3.8561e+00	$k_t = [1, 1, 1, 1]$			
Generation 220:	Best = 1.927327e+03	FES = 22100	$\mu_F = 0.821$	$\mu_{CR} = 0.039$
Diversity = 3.7903e+00	$k_t = [2, 3, 4, 3]$			

Generation 230: Best = 1.927327e+03	FES = 23100	$\mu_F = 0.830$	$\mu_{CR} = 0.036$
Diversity = 3.8369e+00	$k_t = [1, 1, 1, 1]$		
Generation 240: Best = 1.925939e+03	FES = 24100	$\mu_F = 0.774$	$\mu_{CR} = 0.033$
Diversity = 3.8493e+00	$k_t = [3, 4, 4, 2]$		
Generation 250: Best = 1.892274e+03	FES = 25100	$\mu_F = 0.717$	$\mu_{CR} = 0.038$
Diversity = 3.8628e+00	$k_t = [1, 1, 1, 1]$		
Generation 260: Best = 1.889043e+03	FES = 26100	$\mu_F = 0.763$	$\mu_{CR} = 0.034$
Diversity = 3.8646e+00	$k_t = [3, 2, 3, 3]$		
Generation 270: Best = 1.889043e+03	FES = 27100	$\mu_F = 0.773$	$\mu_{CR} = 0.034$
Diversity = 3.8533e+00	$k_t = [1, 1, 1, 1]$		
Generation 280: Best = 1.889043e+03	FES = 28100	$\mu_F = 0.809$	$\mu_{CR} = 0.033$
Diversity = 3.7661e+00	$k_t = [3, 4, 3, 4]$		
Generation 290: Best = 1.876360e+03	FES = 29100	$\mu_F = 0.823$	$\mu_{CR} = 0.034$
Diversity = 3.7324e+00	$k_t = [1, 1, 1, 1]$		
Generation 300: Best = 1.871825e+03	FES = 30100	$\mu_F = 0.795$	$\mu_{CR} = 0.035$
Diversity = 3.7385e+00	$k_t = [4, 3, 3, 4]$		
Generation 310: Best = 1.861846e+03	FES = 31100	$\mu_F = 0.795$	$\mu_{CR} = 0.033$
Diversity = 3.7365e+00	$k_t = [1, 1, 1, 1]$		
Generation 320: Best = 1.861846e+03	FES = 32100	$\mu_F = 0.815$	$\mu_{CR} = 0.033$
Diversity = 3.7437e+00	$k_t = [2, 4, 3, 4]$		
Generation 330: Best = 1.861846e+03	FES = 33100	$\mu_F = 0.705$	$\mu_{CR} = 0.031$
Diversity = 3.7369e+00	$k_t = [1, 1, 1, 1]$		
Generation 340: Best = 1.836268e+03	FES = 34100	$\mu_F = 0.748$	$\mu_{CR} = 0.027$
Diversity = 3.7511e+00	$k_t = [2, 2, 4, 4]$		
Generation 350: Best = 1.831829e+03	FES = 35100	$\mu_F = 0.773$	$\mu_{CR} = 0.031$
Diversity = 3.7469e+00	$k_t = [1, 1, 1, 1]$		
Generation 360: Best = 1.822367e+03	FES = 36100	$\mu_F = 0.694$	$\mu_{CR} = 0.034$
Diversity = 3.7493e+00	$k_t = [2, 2, 2, 4]$		
Generation 370: Best = 1.821759e+03	FES = 37100	$\mu_F = 0.715$	$\mu_{CR} = 0.033$
Diversity = 3.7540e+00	$k_t = [1, 1, 1, 1]$		
Generation 380: Best = 1.821759e+03	FES = 38100	$\mu_F = 0.698$	$\mu_{CR} = 0.031$
Diversity = 3.7281e+00	$k_t = [2, 2, 3, 4]$		
Generation 390: Best = 1.821759e+03	FES = 39100	$\mu_F = 0.712$	$\mu_{CR} = 0.024$
Diversity = 3.7273e+00	$k_t = [1, 1, 1, 1]$		
Generation 400: Best = 1.821759e+03	FES = 40100	$\mu_F = 0.713$	$\mu_{CR} = 0.027$
Diversity = 3.7186e+00	$k_t = [2, 2, 4, 2]$		
Generation 410: Best = 1.780054e+03	FES = 41100	$\mu_F = 0.729$	$\mu_{CR} = 0.032$
Diversity = 3.7249e+00	$k_t = [1, 1, 1, 1]$		
Generation 420: Best = 1.780054e+03	FES = 42100	$\mu_F = 0.687$	$\mu_{CR} = 0.033$
Diversity = 3.7344e+00	$k_t = [3, 2, 3, 4]$		
Generation 430: Best = 1.780054e+03	FES = 43100	$\mu_F = 0.720$	$\mu_{CR} = 0.028$
Diversity = 3.7290e+00	$k_t = [1, 1, 1, 1]$		
Generation 440: Best = 1.780054e+03	FES = 44100	$\mu_F = 0.654$	$\mu_{CR} = 0.029$
Diversity = 3.7299e+00	$k_t = [2, 2, 4, 2]$		
Generation 450: Best = 1.776084e+03	FES = 45100	$\mu_F = 0.601$	$\mu_{CR} = 0.042$
Diversity = 3.7264e+00	$k_t = [1, 1, 1, 1]$		
Generation 460: Best = 1.769629e+03	FES = 46100	$\mu_F = 0.667$	$\mu_{CR} = 0.035$
Diversity = 3.7244e+00	$k_t = [3, 4, 2, 2]$		
Generation 470: Best = 1.746521e+03	FES = 47100	$\mu_F = 0.686$	$\mu_{CR} = 0.025$
Diversity = 3.7253e+00	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 1.746521e+03	FES = 48100	$\mu_F = 0.654$	$\mu_{CR} = 0.030$
Diversity = 3.7244e+00	$k_t = [4, 3, 4, 3]$		
Generation 490: Best = 1.746521e+03	FES = 49100	$\mu_F = 0.682$	$\mu_{CR} = 0.037$
Diversity = 3.7133e+00	$k_t = [1, 1, 1, 1]$		

Generation 500:	Best = 1.746521e+03	FES = 50100	$\mu_F = 0.677$	$\mu_{CR} = 0.042$
Diversity = 3.7142e+00	$k_t = [4, 2, 2, 3]$			
Generation 510:	Best = 1.744834e+03	FES = 51100	$\mu_F = 0.619$	$\mu_{CR} = 0.032$
Diversity = 3.7150e+00	$k_t = [1, 1, 1, 1]$			
Generation 520:	Best = 1.732410e+03	FES = 52100	$\mu_F = 0.555$	$\mu_{CR} = 0.040$
Diversity = 3.7108e+00	$k_t = [4, 2, 3, 2]$			
Generation 530:	Best = 1.732410e+03	FES = 53100	$\mu_F = 0.561$	$\mu_{CR} = 0.044$
Diversity = 3.7089e+00	$k_t = [1, 1, 1, 1]$			
Generation 540:	Best = 1.732410e+03	FES = 54100	$\mu_F = 0.602$	$\mu_{CR} = 0.038$
Diversity = 3.7061e+00	$k_t = [3, 2, 3, 2]$			
Generation 550:	Best = 1.728103e+03	FES = 55100	$\mu_F = 0.528$	$\mu_{CR} = 0.038$
Diversity = 3.7033e+00	$k_t = [1, 1, 1, 1]$			
Generation 560:	Best = 1.726673e+03	FES = 56100	$\mu_F = 0.569$	$\mu_{CR} = 0.035$
Diversity = 3.7232e+00	$k_t = [4, 2, 2, 4]$			
Generation 570:	Best = 1.723549e+03	FES = 57100	$\mu_F = 0.509$	$\mu_{CR} = 0.032$
Diversity = 3.7215e+00	$k_t = [1, 1, 1, 1]$			
Generation 580:	Best = 1.723549e+03	FES = 58100	$\mu_F = 0.367$	$\mu_{CR} = 0.039$
Diversity = 3.7185e+00	$k_t = [3, 3, 4, 3]$			
Generation 590:	Best = 1.716508e+03	FES = 59100	$\mu_F = 0.363$	$\mu_{CR} = 0.038$
Diversity = 3.7132e+00	$k_t = [1, 1, 1, 1]$			
Generation 600:	Best = 1.708458e+03	FES = 60100	$\mu_F = 0.322$	$\mu_{CR} = 0.044$
Diversity = 3.7090e+00	$k_t = [3, 3, 4, 2]$			
Generation 610:	Best = 1.708458e+03	FES = 61100	$\mu_F = 0.287$	$\mu_{CR} = 0.049$
Diversity = 3.6971e+00	$k_t = [1, 1, 1, 1]$			
Generation 620:	Best = 1.708458e+03	FES = 62100	$\mu_F = 0.280$	$\mu_{CR} = 0.044$
Diversity = 3.6882e+00	$k_t = [4, 4, 2, 2]$			
Generation 630:	Best = 1.699022e+03	FES = 63100	$\mu_F = 0.313$	$\mu_{CR} = 0.049$
Diversity = 3.6804e+00	$k_t = [1, 1, 1, 1]$			
Generation 640:	Best = 1.698409e+03	FES = 64100	$\mu_F = 0.257$	$\mu_{CR} = 0.059$
Diversity = 3.6741e+00	$k_t = [3, 3, 2, 2]$			
Generation 650:	Best = 1.696078e+03	FES = 65100	$\mu_F = 0.240$	$\mu_{CR} = 0.063$
Diversity = 3.6691e+00	$k_t = [1, 1, 1, 1]$			
Generation 660:	Best = 1.695134e+03	FES = 66100	$\mu_F = 0.334$	$\mu_{CR} = 0.067$
Diversity = 3.6527e+00	$k_t = [3, 3, 4, 4]$			
Generation 670:	Best = 1.694706e+03	FES = 67100	$\mu_F = 0.369$	$\mu_{CR} = 0.050$
Diversity = 3.6513e+00	$k_t = [1, 1, 1, 1]$			
Generation 680:	Best = 1.686946e+03	FES = 68100	$\mu_F = 0.354$	$\mu_{CR} = 0.044$
Diversity = 3.6482e+00	$k_t = [2, 3, 3, 4]$			
Generation 690:	Best = 1.686946e+03	FES = 69100	$\mu_F = 0.342$	$\mu_{CR} = 0.044$
Diversity = 3.6455e+00	$k_t = [1, 1, 1, 1]$			
Generation 700:	Best = 1.684633e+03	FES = 70100	$\mu_F = 0.395$	$\mu_{CR} = 0.037$
Diversity = 3.6464e+00	$k_t = [2, 2, 4, 2]$			
Generation 710:	Best = 1.684633e+03	FES = 71100	$\mu_F = 0.329$	$\mu_{CR} = 0.044$
Diversity = 3.6440e+00	$k_t = [1, 1, 1, 1]$			
Generation 720:	Best = 1.684633e+03	FES = 72100	$\mu_F = 0.377$	$\mu_{CR} = 0.046$
Diversity = 3.6395e+00	$k_t = [2, 4, 3, 4]$			
Generation 730:	Best = 1.678433e+03	FES = 73100	$\mu_F = 0.322$	$\mu_{CR} = 0.040$
Diversity = 3.6385e+00	$k_t = [1, 1, 1, 1]$			
Generation 740:	Best = 1.674648e+03	FES = 74100	$\mu_F = 0.299$	$\mu_{CR} = 0.049$
Diversity = 3.6233e+00	$k_t = [4, 4, 2, 4]$			
Generation 750:	Best = 1.669120e+03	FES = 75100	$\mu_F = 0.304$	$\mu_{CR} = 0.039$
Diversity = 3.6189e+00	$k_t = [1, 1, 1, 1]$			
Generation 760:	Best = 1.669120e+03	FES = 76100	$\mu_F = 0.281$	$\mu_{CR} = 0.045$
Diversity = 3.6170e+00	$k_t = [4, 4, 2, 2]$			

Generation 770: Best = 1.660182e+03	FES = 77100	$\mu_F = 0.252$	$\mu_{CR} = 0.049$
Diversity = 3.6131e+00	$k_t = [1, 1, 1, 1]$		
Generation 780: Best = 1.660182e+03	FES = 78100	$\mu_F = 0.290$	$\mu_{CR} = 0.043$
Diversity = 3.6084e+00	$k_t = [3, 4, 2, 2]$		
Generation 790: Best = 1.660182e+03	FES = 79100	$\mu_F = 0.393$	$\mu_{CR} = 0.044$
Diversity = 3.6046e+00	$k_t = [1, 1, 1, 1]$		
Generation 800: Best = 1.657734e+03	FES = 80100	$\mu_F = 0.327$	$\mu_{CR} = 0.031$
Diversity = 3.6021e+00	$k_t = [3, 3, 2, 4]$		
Generation 810: Best = 1.654106e+03	FES = 81100	$\mu_F = 0.300$	$\mu_{CR} = 0.042$
Diversity = 3.5989e+00	$k_t = [1, 1, 1, 1]$		
Generation 820: Best = 1.644732e+03	FES = 82100	$\mu_F = 0.322$	$\mu_{CR} = 0.039$
Diversity = 3.5954e+00	$k_t = [2, 4, 4, 3]$		
Generation 830: Best = 1.640855e+03	FES = 83100	$\mu_F = 0.333$	$\mu_{CR} = 0.040$
Diversity = 3.5927e+00	$k_t = [1, 1, 1, 1]$		
Generation 840: Best = 1.598058e+03	FES = 84100	$\mu_F = 0.343$	$\mu_{CR} = 0.037$
Diversity = 3.5813e+00	$k_t = [2, 3, 3, 4]$		
Generation 850: Best = 1.598058e+03	FES = 85100	$\mu_F = 0.360$	$\mu_{CR} = 0.028$
Diversity = 3.5794e+00	$k_t = [1, 1, 1, 1]$		
Generation 860: Best = 1.598058e+03	FES = 86100	$\mu_F = 0.334$	$\mu_{CR} = 0.044$
Diversity = 3.5796e+00	$k_t = [2, 4, 3, 3]$		
Generation 870: Best = 1.598058e+03	FES = 87100	$\mu_F = 0.273$	$\mu_{CR} = 0.038$
Diversity = 3.5786e+00	$k_t = [1, 1, 1, 1]$		
Generation 880: Best = 1.594968e+03	FES = 88100	$\mu_F = 0.302$	$\mu_{CR} = 0.048$
Diversity = 3.5604e+00	$k_t = [2, 2, 2, 4]$		
Generation 890: Best = 1.594244e+03	FES = 89100	$\mu_F = 0.307$	$\mu_{CR} = 0.035$
Diversity = 3.5597e+00	$k_t = [1, 1, 1, 1]$		
Generation 900: Best = 1.594244e+03	FES = 90100	$\mu_F = 0.308$	$\mu_{CR} = 0.046$
Diversity = 3.5566e+00	$k_t = [4, 3, 2, 2]$		
Generation 910: Best = 1.564652e+03	FES = 91100	$\mu_F = 0.395$	$\mu_{CR} = 0.033$
Diversity = 3.5551e+00	$k_t = [1, 1, 1, 1]$		
Generation 920: Best = 1.563582e+03	FES = 92100	$\mu_F = 0.400$	$\mu_{CR} = 0.039$
Diversity = 3.5544e+00	$k_t = [2, 4, 3, 4]$		
Generation 930: Best = 1.556685e+03	FES = 93100	$\mu_F = 0.343$	$\mu_{CR} = 0.044$
Diversity = 3.5535e+00	$k_t = [1, 1, 1, 1]$		
Generation 940: Best = 1.555477e+03	FES = 94100	$\mu_F = 0.344$	$\mu_{CR} = 0.043$
Diversity = 3.5380e+00	$k_t = [4, 3, 4, 4]$		
Generation 950: Best = 1.555477e+03	FES = 95100	$\mu_F = 0.336$	$\mu_{CR} = 0.043$
Diversity = 3.5342e+00	$k_t = [1, 1, 1, 1]$		
Generation 960: Best = 1.553701e+03	FES = 96100	$\mu_F = 0.368$	$\mu_{CR} = 0.032$
Diversity = 3.5280e+00	$k_t = [2, 3, 3, 4]$		
Generation 970: Best = 1.553620e+03	FES = 97100	$\mu_F = 0.359$	$\mu_{CR} = 0.031$
Diversity = 3.5273e+00	$k_t = [1, 1, 1, 1]$		
Generation 980: Best = 1.553620e+03	FES = 98100	$\mu_F = 0.318$	$\mu_{CR} = 0.031$
Diversity = 3.5265e+00	$k_t = [2, 2, 4, 3]$		
Generation 990: Best = 1.553620e+03	FES = 99100	$\mu_F = 0.340$	$\mu_{CR} = 0.034$
Diversity = 3.5250e+00	$k_t = [1, 1, 1, 1]$		
Generation 1000: Best = 1.553620e+03	FES = 100100	$\mu_F = 0.425$	$\mu_{CR} = 0.026$
Diversity = 3.5104e+00	$k_t = [2, 2, 4, 4]$		
Generation 1010: Best = 1.553620e+03	FES = 101100	$\mu_F = 0.520$	$\mu_{CR} = 0.027$
Diversity = 3.5029e+00	$k_t = [1, 1, 1, 1]$		
Generation 1020: Best = 1.530603e+03	FES = 102100	$\mu_F = 0.395$	$\mu_{CR} = 0.043$
Diversity = 3.5023e+00	$k_t = [2, 3, 2, 3]$		
Generation 1030: Best = 1.530603e+03	FES = 103100	$\mu_F = 0.354$	$\mu_{CR} = 0.039$
Diversity = 3.5009e+00	$k_t = [1, 1, 1, 1]$		

Generation 1040: Best = 1.530603e+03 FES = 104100 μ_F = 0.315 μ_{CR} = 0.033
 Diversity = 3.4994e+00 k_t = [2, 2, 3, 2]
 Generation 1050: Best = 1.530603e+03 FES = 105100 μ_F = 0.331 μ_{CR} = 0.031
 Diversity = 3.4972e+00 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 1.524196e+03 FES = 106100 μ_F = 0.343 μ_{CR} = 0.025
 Diversity = 3.4938e+00 k_t = [3, 2, 2, 3]
 Generation 1070: Best = 1.524196e+03 FES = 107100 μ_F = 0.283 μ_{CR} = 0.031
 Diversity = 3.4929e+00 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 1.524196e+03 FES = 108100 μ_F = 0.242 μ_{CR} = 0.038
 Diversity = 3.4911e+00 k_t = [2, 3, 2, 2]
 Generation 1090: Best = 1.521320e+03 FES = 109100 μ_F = 0.233 μ_{CR} = 0.046
 Diversity = 3.4898e+00 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 1.521320e+03 FES = 110100 μ_F = 0.214 μ_{CR} = 0.039
 Diversity = 3.4876e+00 k_t = [2, 3, 3, 2]
 Generation 1110: Best = 1.520992e+03 FES = 111100 μ_F = 0.210 μ_{CR} = 0.045
 Diversity = 3.4861e+00 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 1.520992e+03 FES = 112100 μ_F = 0.172 μ_{CR} = 0.041
 Diversity = 3.4847e+00 k_t = [4, 2, 3, 2]
 Generation 1130: Best = 1.519265e+03 FES = 113100 μ_F = 0.185 μ_{CR} = 0.047
 Diversity = 3.4816e+00 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 1.517235e+03 FES = 114100 μ_F = 0.170 μ_{CR} = 0.059
 Diversity = 3.4795e+00 k_t = [2, 2, 3, 3]
 Generation 1150: Best = 1.516233e+03 FES = 115100 μ_F = 0.174 μ_{CR} = 0.046
 Diversity = 3.4782e+00 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 1.516233e+03 FES = 116100 μ_F = 0.171 μ_{CR} = 0.045
 Diversity = 3.4731e+00 k_t = [4, 4, 4, 4]
 Generation 1170: Best = 1.515196e+03 FES = 117100 μ_F = 0.203 μ_{CR} = 0.056
 Diversity = 3.4696e+00 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 1.515196e+03 FES = 118100 μ_F = 0.156 μ_{CR} = 0.045
 Diversity = 3.4669e+00 k_t = [3, 3, 2, 2]
 Generation 1190: Best = 1.515196e+03 FES = 119100 μ_F = 0.226 μ_{CR} = 0.046
 Diversity = 3.4649e+00 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 1.510087e+03 FES = 120100 μ_F = 0.302 μ_{CR} = 0.042
 Diversity = 3.4641e+00 k_t = [3, 3, 4, 2]
 Generation 1210: Best = 1.508024e+03 FES = 121100 μ_F = 0.269 μ_{CR} = 0.060
 Diversity = 3.4614e+00 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 1.497628e+03 FES = 122100 μ_F = 0.338 μ_{CR} = 0.053
 Diversity = 3.4568e+00 k_t = [3, 3, 2, 4]
 Generation 1230: Best = 1.497628e+03 FES = 123100 μ_F = 0.359 μ_{CR} = 0.039
 Diversity = 3.4555e+00 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 1.497628e+03 FES = 124100 μ_F = 0.332 μ_{CR} = 0.038
 Diversity = 3.4547e+00 k_t = [3, 3, 4, 3]
 Generation 1250: Best = 1.497628e+03 FES = 125100 μ_F = 0.348 μ_{CR} = 0.029
 Diversity = 3.4538e+00 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 1.497628e+03 FES = 126100 μ_F = 0.312 μ_{CR} = 0.024
 Diversity = 3.4543e+00 k_t = [3, 3, 4, 2]
 Generation 1270: Best = 1.497628e+03 FES = 127100 μ_F = 0.293 μ_{CR} = 0.023
 Diversity = 3.4542e+00 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 1.497628e+03 FES = 128100 μ_F = 0.289 μ_{CR} = 0.026
 Diversity = 3.4532e+00 k_t = [3, 3, 3, 3]
 Generation 1290: Best = 1.497628e+03 FES = 129100 μ_F = 0.357 μ_{CR} = 0.032
 Diversity = 3.4372e+00 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 1.497628e+03 FES = 130100 μ_F = 0.293 μ_{CR} = 0.036
 Diversity = 3.4373e+00 k_t = [4, 3, 4, 2]

Generation 1310: Best = 1.497628e+03 FES = 131100 μ_F = 0.332 μ_{CR} = 0.028
 Diversity = 3.4369e+00 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 1.497628e+03 FES = 132100 μ_F = 0.439 μ_{CR} = 0.030
 Diversity = 3.4274e+00 k_t = [4, 4, 3, 4]
 Generation 1330: Best = 1.497628e+03 FES = 133100 μ_F = 0.323 μ_{CR} = 0.032
 Diversity = 3.4271e+00 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 1.497628e+03 FES = 134100 μ_F = 0.285 μ_{CR} = 0.035
 Diversity = 3.4382e+00 k_t = [4, 2, 3, 2]
 Generation 1350: Best = 1.486677e+03 FES = 135100 μ_F = 0.286 μ_{CR} = 0.028
 Diversity = 3.4311e+00 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 1.486677e+03 FES = 136100 μ_F = 0.227 μ_{CR} = 0.048
 Diversity = 3.4283e+00 k_t = [3, 3, 4, 2]
 Generation 1370: Best = 1.486677e+03 FES = 137100 μ_F = 0.250 μ_{CR} = 0.046
 Diversity = 3.4270e+00 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 1.486677e+03 FES = 138100 μ_F = 0.257 μ_{CR} = 0.038
 Diversity = 3.4249e+00 k_t = [2, 3, 3, 3]
 Generation 1390: Best = 1.486677e+03 FES = 139100 μ_F = 0.320 μ_{CR} = 0.032
 Diversity = 3.4247e+00 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 1.486677e+03 FES = 140100 μ_F = 0.351 μ_{CR} = 0.041
 Diversity = 3.4202e+00 k_t = [3, 2, 3, 4]
 Generation 1410: Best = 1.484124e+03 FES = 141100 μ_F = 0.401 μ_{CR} = 0.040
 Diversity = 3.4191e+00 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 1.484124e+03 FES = 142100 μ_F = 0.369 μ_{CR} = 0.035
 Diversity = 3.4248e+00 k_t = [2, 3, 3, 4]
 Generation 1430: Best = 1.483298e+03 FES = 143100 μ_F = 0.362 μ_{CR} = 0.031
 Diversity = 3.4233e+00 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 1.474751e+03 FES = 144100 μ_F = 0.294 μ_{CR} = 0.041
 Diversity = 3.4235e+00 k_t = [4, 3, 3, 2]
 Generation 1450: Best = 1.474424e+03 FES = 145100 μ_F = 0.248 μ_{CR} = 0.056
 Diversity = 3.4219e+00 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 1.474424e+03 FES = 146100 μ_F = 0.360 μ_{CR} = 0.043
 Diversity = 3.4211e+00 k_t = [3, 2, 3, 3]
 Generation 1470: Best = 1.474424e+03 FES = 147100 μ_F = 0.299 μ_{CR} = 0.042
 Diversity = 3.4202e+00 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 1.468620e+03 FES = 148100 μ_F = 0.297 μ_{CR} = 0.038
 Diversity = 3.4193e+00 k_t = [2, 4, 4, 4]
 Generation 1490: Best = 1.467766e+03 FES = 149100 μ_F = 0.296 μ_{CR} = 0.033
 Diversity = 3.4154e+00 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 1.465289e+03 FES = 150100 μ_F = 0.314 μ_{CR} = 0.039
 Diversity = 3.3972e+00 k_t = [3, 2, 3, 4]
 Generation 1510: Best = 1.457766e+03 FES = 151100 μ_F = 0.241 μ_{CR} = 0.036
 Diversity = 3.3969e+00 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 1.456944e+03 FES = 152100 μ_F = 0.300 μ_{CR} = 0.036
 Diversity = 3.3926e+00 k_t = [3, 3, 2, 4]
 Generation 1530: Best = 1.456083e+03 FES = 153100 μ_F = 0.233 μ_{CR} = 0.031
 Diversity = 3.3919e+00 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 1.453560e+03 FES = 154100 μ_F = 0.321 μ_{CR} = 0.035
 Diversity = 3.3970e+00 k_t = [2, 3, 4, 4]
 Generation 1550: Best = 1.453542e+03 FES = 155100 μ_F = 0.347 μ_{CR} = 0.027
 Diversity = 3.3960e+00 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 1.447208e+03 FES = 156100 μ_F = 0.382 μ_{CR} = 0.051
 Diversity = 3.4035e+00 k_t = [3, 2, 4, 4]
 Generation 1570: Best = 1.447208e+03 FES = 157100 μ_F = 0.314 μ_{CR} = 0.055
 Diversity = 3.4034e+00 k_t = [1, 1, 1, 1]

Generation 1580: Best = 1.447208e+03 FES = 158100 μ_F = 0.215 μ_{CR} = 0.056
 Diversity = 3.4024e+00 k_t = [4, 4, 4, 3]
 Generation 1590: Best = 1.437078e+03 FES = 159100 μ_F = 0.244 μ_{CR} = 0.053
 Diversity = 3.4010e+00 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 1.437078e+03 FES = 160100 μ_F = 0.230 μ_{CR} = 0.045
 Diversity = 3.3991e+00 k_t = [2, 4, 2, 2]
 Generation 1610: Best = 1.432327e+03 FES = 161100 μ_F = 0.229 μ_{CR} = 0.037
 Diversity = 3.3984e+00 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 1.432327e+03 FES = 162100 μ_F = 0.315 μ_{CR} = 0.034
 Diversity = 3.3982e+00 k_t = [3, 2, 2, 2]
 Generation 1630: Best = 1.431430e+03 FES = 163100 μ_F = 0.312 μ_{CR} = 0.029
 Diversity = 3.3989e+00 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 1.429261e+03 FES = 164100 μ_F = 0.338 μ_{CR} = 0.039
 Diversity = 3.3854e+00 k_t = [4, 4, 4, 4]
 Generation 1650: Best = 1.425072e+03 FES = 165100 μ_F = 0.295 μ_{CR} = 0.030
 Diversity = 3.3850e+00 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 1.425072e+03 FES = 166100 μ_F = 0.321 μ_{CR} = 0.035
 Diversity = 3.3856e+00 k_t = [3, 3, 4, 4]
 Generation 1670: Best = 1.419129e+03 FES = 167100 μ_F = 0.294 μ_{CR} = 0.036
 Diversity = 3.3865e+00 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 1.419119e+03 FES = 168100 μ_F = 0.274 μ_{CR} = 0.034
 Diversity = 3.3846e+00 k_t = [2, 3, 4, 2]
 Generation 1690: Best = 1.418944e+03 FES = 169100 μ_F = 0.336 μ_{CR} = 0.042
 Diversity = 3.3846e+00 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 1.418944e+03 FES = 170100 μ_F = 0.401 μ_{CR} = 0.027
 Diversity = 3.3852e+00 k_t = [2, 4, 4, 3]
 Generation 1710: Best = 1.418944e+03 FES = 171100 μ_F = 0.447 μ_{CR} = 0.030
 Diversity = 3.3830e+00 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 1.408615e+03 FES = 172100 μ_F = 0.382 μ_{CR} = 0.041
 Diversity = 3.3836e+00 k_t = [4, 2, 4, 3]
 Generation 1730: Best = 1.408615e+03 FES = 173100 μ_F = 0.313 μ_{CR} = 0.029
 Diversity = 3.3774e+00 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 1.408612e+03 FES = 174100 μ_F = 0.282 μ_{CR} = 0.032
 Diversity = 3.3776e+00 k_t = [2, 3, 4, 3]
 Generation 1750: Best = 1.408512e+03 FES = 175100 μ_F = 0.267 μ_{CR} = 0.036
 Diversity = 3.3759e+00 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 1.401188e+03 FES = 176100 μ_F = 0.254 μ_{CR} = 0.035
 Diversity = 3.3758e+00 k_t = [2, 4, 2, 3]
 Generation 1770: Best = 1.401188e+03 FES = 177100 μ_F = 0.293 μ_{CR} = 0.027
 Diversity = 3.3736e+00 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 1.400469e+03 FES = 178100 μ_F = 0.284 μ_{CR} = 0.021
 Diversity = 3.3728e+00 k_t = [3, 2, 3, 2]
 Generation 1790: Best = 1.400469e+03 FES = 179100 μ_F = 0.262 μ_{CR} = 0.054
 Diversity = 3.3722e+00 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 1.400469e+03 FES = 180100 μ_F = 0.342 μ_{CR} = 0.033
 Diversity = 3.3697e+00 k_t = [2, 2, 3, 4]
 Generation 1810: Best = 1.400469e+03 FES = 181100 μ_F = 0.313 μ_{CR} = 0.036
 Diversity = 3.3684e+00 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 1.400469e+03 FES = 182100 μ_F = 0.289 μ_{CR} = 0.045
 Diversity = 3.3684e+00 k_t = [4, 4, 4, 2]
 Generation 1830: Best = 1.398636e+03 FES = 183100 μ_F = 0.413 μ_{CR} = 0.037
 Diversity = 3.3672e+00 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 1.396078e+03 FES = 184100 μ_F = 0.402 μ_{CR} = 0.029
 Diversity = 3.3599e+00 k_t = [3, 3, 4, 3]

Generation 1850: Best = 1.395800e+03	FES = 185100	$\mu_F = 0.400$	$\mu_{CR} = 0.040$
Diversity = 3.3602e+00	$k_t = [1, 1, 1, 1]$		
Generation 1860: Best = 1.395767e+03	FES = 186100	$\mu_F = 0.358$	$\mu_{CR} = 0.053$
Diversity = 3.3605e+00	$k_t = [2, 4, 3, 4]$		
Generation 1870: Best = 1.395767e+03	FES = 187100	$\mu_F = 0.299$	$\mu_{CR} = 0.047$
Diversity = 3.3603e+00	$k_t = [1, 1, 1, 1]$		
Generation 1880: Best = 1.395275e+03	FES = 188100	$\mu_F = 0.314$	$\mu_{CR} = 0.036$
Diversity = 3.3588e+00	$k_t = [3, 3, 3, 2]$		
Generation 1890: Best = 1.386181e+03	FES = 189100	$\mu_F = 0.287$	$\mu_{CR} = 0.040$
Diversity = 3.3601e+00	$k_t = [1, 1, 1, 1]$		
Generation 1900: Best = 1.382973e+03	FES = 190100	$\mu_F = 0.311$	$\mu_{CR} = 0.046$
Diversity = 3.3350e+00	$k_t = [2, 2, 4, 4]$		
Generation 1910: Best = 1.382973e+03	FES = 191100	$\mu_F = 0.316$	$\mu_{CR} = 0.041$
Diversity = 3.3352e+00	$k_t = [1, 1, 1, 1]$		
Generation 1920: Best = 1.366128e+03	FES = 192100	$\mu_F = 0.312$	$\mu_{CR} = 0.037$
Diversity = 3.3350e+00	$k_t = [4, 4, 2, 3]$		
Generation 1930: Best = 1.361043e+03	FES = 193100	$\mu_F = 0.265$	$\mu_{CR} = 0.042$
Diversity = 3.3349e+00	$k_t = [1, 1, 1, 1]$		
Generation 1940: Best = 1.361043e+03	FES = 194100	$\mu_F = 0.220$	$\mu_{CR} = 0.046$
Diversity = 3.3341e+00	$k_t = [3, 4, 4, 2]$		
Generation 1950: Best = 1.359535e+03	FES = 195100	$\mu_F = 0.283$	$\mu_{CR} = 0.047$
Diversity = 3.3299e+00	$k_t = [1, 1, 1, 1]$		
Generation 1960: Best = 1.359289e+03	FES = 196100	$\mu_F = 0.256$	$\mu_{CR} = 0.058$
Diversity = 3.3296e+00	$k_t = [4, 3, 3, 2]$		
Generation 1970: Best = 1.358766e+03	FES = 197100	$\mu_F = 0.318$	$\mu_{CR} = 0.044$
Diversity = 3.3290e+00	$k_t = [1, 1, 1, 1]$		
Generation 1980: Best = 1.353798e+03	FES = 198100	$\mu_F = 0.361$	$\mu_{CR} = 0.036$
Diversity = 3.3286e+00	$k_t = [4, 4, 4, 4]$		
Generation 1990: Best = 1.353287e+03	FES = 199100	$\mu_F = 0.307$	$\mu_{CR} = 0.045$
Diversity = 3.3287e+00	$k_t = [1, 1, 1, 1]$		
Generation 2000: Best = 1.352681e+03	FES = 200100	$\mu_F = 0.317$	$\mu_{CR} = 0.040$
Diversity = 3.3282e+00	$k_t = [2, 2, 3, 2]$		
Generation 2010: Best = 1.352681e+03	FES = 201100	$\mu_F = 0.336$	$\mu_{CR} = 0.037$
Diversity = 3.3278e+00	$k_t = [1, 1, 1, 1]$		
Generation 2020: Best = 1.352681e+03	FES = 202100	$\mu_F = 0.347$	$\mu_{CR} = 0.039$
Diversity = 3.3236e+00	$k_t = [3, 3, 3, 4]$		
Generation 2030: Best = 1.352681e+03	FES = 203100	$\mu_F = 0.289$	$\mu_{CR} = 0.025$
Diversity = 3.3235e+00	$k_t = [1, 1, 1, 1]$		
Generation 2040: Best = 1.352681e+03	FES = 204100	$\mu_F = 0.306$	$\mu_{CR} = 0.023$
Diversity = 3.3239e+00	$k_t = [2, 3, 4, 4]$		
Generation 2050: Best = 1.350290e+03	FES = 205100	$\mu_F = 0.288$	$\mu_{CR} = 0.028$
Diversity = 3.3219e+00	$k_t = [1, 1, 1, 1]$		
Generation 2060: Best = 1.344607e+03	FES = 206100	$\mu_F = 0.228$	$\mu_{CR} = 0.030$
Diversity = 3.3207e+00	$k_t = [2, 3, 3, 2]$		
Generation 2070: Best = 1.340682e+03	FES = 207100	$\mu_F = 0.221$	$\mu_{CR} = 0.037$
Diversity = 3.3201e+00	$k_t = [1, 1, 1, 1]$		
Generation 2080: Best = 1.340682e+03	FES = 208100	$\mu_F = 0.168$	$\mu_{CR} = 0.048$
Diversity = 3.3187e+00	$k_t = [2, 2, 2, 3]$		
Generation 2090: Best = 1.340682e+03	FES = 209100	$\mu_F = 0.241$	$\mu_{CR} = 0.039$
Diversity = 3.3165e+00	$k_t = [1, 1, 1, 1]$		
Generation 2100: Best = 1.340001e+03	FES = 210100	$\mu_F = 0.282$	$\mu_{CR} = 0.034$
Diversity = 3.3109e+00	$k_t = [2, 2, 2, 2]$		
Generation 2110: Best = 1.340001e+03	FES = 211100	$\mu_F = 0.308$	$\mu_{CR} = 0.038$
Diversity = 3.3113e+00	$k_t = [1, 1, 1, 1]$		

Generation 2120: Best = 1.335021e+03 FES = 212100 μ_F = 0.358 μ_{CR} = 0.034
 Diversity = 3.3087e+00 k_t = [4, 2, 3, 4]
 Generation 2130: Best = 1.333519e+03 FES = 213100 μ_F = 0.320 μ_{CR} = 0.043
 Diversity = 3.3091e+00 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 1.333519e+03 FES = 214100 μ_F = 0.314 μ_{CR} = 0.031
 Diversity = 3.3089e+00 k_t = [2, 4, 4, 2]
 Generation 2150: Best = 1.333519e+03 FES = 215100 μ_F = 0.305 μ_{CR} = 0.033
 Diversity = 3.3093e+00 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 1.333519e+03 FES = 216100 μ_F = 0.267 μ_{CR} = 0.026
 Diversity = 3.3092e+00 k_t = [2, 2, 2, 2]
 Generation 2170: Best = 1.326483e+03 FES = 217100 μ_F = 0.251 μ_{CR} = 0.041
 Diversity = 3.3062e+00 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 1.325464e+03 FES = 218100 μ_F = 0.239 μ_{CR} = 0.046
 Diversity = 3.3062e+00 k_t = [2, 3, 4, 3]
 Generation 2190: Best = 1.325464e+03 FES = 219100 μ_F = 0.218 μ_{CR} = 0.046
 Diversity = 3.3063e+00 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 1.325464e+03 FES = 220100 μ_F = 0.199 μ_{CR} = 0.052
 Diversity = 3.3067e+00 k_t = [2, 3, 4, 2]
 Generation 2210: Best = 1.320080e+03 FES = 221100 μ_F = 0.253 μ_{CR} = 0.066
 Diversity = 3.3069e+00 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 1.319862e+03 FES = 222100 μ_F = 0.323 μ_{CR} = 0.050
 Diversity = 3.3064e+00 k_t = [4, 2, 4, 2]
 Generation 2230: Best = 1.318860e+03 FES = 223100 μ_F = 0.266 μ_{CR} = 0.038
 Diversity = 3.3064e+00 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 1.317324e+03 FES = 224100 μ_F = 0.260 μ_{CR} = 0.046
 Diversity = 3.3048e+00 k_t = [3, 3, 4, 3]
 Generation 2250: Best = 1.314343e+03 FES = 225100 μ_F = 0.226 μ_{CR} = 0.040
 Diversity = 3.3047e+00 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 1.304880e+03 FES = 226100 μ_F = 0.248 μ_{CR} = 0.044
 Diversity = 3.3049e+00 k_t = [4, 2, 3, 3]
 Generation 2270: Best = 1.304721e+03 FES = 227100 μ_F = 0.241 μ_{CR} = 0.046
 Diversity = 3.3053e+00 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 1.303403e+03 FES = 228100 μ_F = 0.304 μ_{CR} = 0.038
 Diversity = 3.2887e+00 k_t = [4, 4, 3, 4]
 Generation 2290: Best = 1.303403e+03 FES = 229100 μ_F = 0.326 μ_{CR} = 0.048
 Diversity = 3.2898e+00 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 1.303403e+03 FES = 230100 μ_F = 0.406 μ_{CR} = 0.043
 Diversity = 3.2804e+00 k_t = [4, 2, 4, 4]
 Generation 2310: Best = 1.300443e+03 FES = 231100 μ_F = 0.427 μ_{CR} = 0.036
 Diversity = 3.2807e+00 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 1.300395e+03 FES = 232100 μ_F = 0.478 μ_{CR} = 0.044
 Diversity = 3.2810e+00 k_t = [3, 4, 2, 2]
 Generation 2330: Best = 1.299108e+03 FES = 233100 μ_F = 0.460 μ_{CR} = 0.045
 Diversity = 3.2808e+00 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 1.299108e+03 FES = 234100 μ_F = 0.471 μ_{CR} = 0.035
 Diversity = 3.2599e+00 k_t = [4, 4, 3, 4]
 Generation 2350: Best = 1.294645e+03 FES = 235100 μ_F = 0.380 μ_{CR} = 0.056
 Diversity = 3.2602e+00 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 1.294645e+03 FES = 236100 μ_F = 0.408 μ_{CR} = 0.040
 Diversity = 3.2604e+00 k_t = [3, 2, 2, 3]
 Generation 2370: Best = 1.294645e+03 FES = 237100 μ_F = 0.465 μ_{CR} = 0.038
 Diversity = 3.2611e+00 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 1.294645e+03 FES = 238100 μ_F = 0.404 μ_{CR} = 0.047
 Diversity = 3.2617e+00 k_t = [2, 2, 3, 2]

Generation 2390: Best = 1.294645e+03 FES = 239100 μ_F = 0.329 μ_{CR} = 0.034
 Diversity = 3.2624e+00 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 1.294645e+03 FES = 240100 μ_F = 0.387 μ_{CR} = 0.052
 Diversity = 3.2623e+00 k_t = [3, 4, 3, 2]
 Generation 2410: Best = 1.292864e+03 FES = 241100 μ_F = 0.387 μ_{CR} = 0.034
 Diversity = 3.2622e+00 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 1.291892e+03 FES = 242100 μ_F = 0.374 μ_{CR} = 0.028
 Diversity = 3.2624e+00 k_t = [3, 3, 4, 4]
 Generation 2430: Best = 1.291892e+03 FES = 243100 μ_F = 0.332 μ_{CR} = 0.033
 Diversity = 3.2625e+00 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 1.291892e+03 FES = 244100 μ_F = 0.333 μ_{CR} = 0.038
 Diversity = 3.2626e+00 k_t = [3, 4, 3, 3]
 Generation 2450: Best = 1.290143e+03 FES = 245100 μ_F = 0.276 μ_{CR} = 0.044
 Diversity = 3.2634e+00 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 1.290143e+03 FES = 246100 μ_F = 0.360 μ_{CR} = 0.028
 Diversity = 3.2567e+00 k_t = [2, 2, 4, 4]
 Generation 2470: Best = 1.286984e+03 FES = 247100 μ_F = 0.342 μ_{CR} = 0.034
 Diversity = 3.2570e+00 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 1.282040e+03 FES = 248100 μ_F = 0.417 μ_{CR} = 0.048
 Diversity = 3.2572e+00 k_t = [4, 4, 3, 3]
 Generation 2490: Best = 1.282040e+03 FES = 249100 μ_F = 0.357 μ_{CR} = 0.039
 Diversity = 3.2581e+00 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 1.282040e+03 FES = 250100 μ_F = 0.396 μ_{CR} = 0.040
 Diversity = 3.2524e+00 k_t = [2, 4, 3, 4]
 Generation 2510: Best = 1.282040e+03 FES = 251100 μ_F = 0.380 μ_{CR} = 0.039
 Diversity = 3.2468e+00 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 1.275694e+03 FES = 252100 μ_F = 0.435 μ_{CR} = 0.034
 Diversity = 3.2241e+00 k_t = [3, 2, 2, 4]
 Generation 2530: Best = 1.272640e+03 FES = 253100 μ_F = 0.427 μ_{CR} = 0.034
 Diversity = 3.2244e+00 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 1.272640e+03 FES = 254100 μ_F = 0.323 μ_{CR} = 0.041
 Diversity = 3.2147e+00 k_t = [2, 3, 3, 4]
 Generation 2550: Best = 1.272640e+03 FES = 255100 μ_F = 0.273 μ_{CR} = 0.040
 Diversity = 3.2143e+00 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 1.269997e+03 FES = 256100 μ_F = 0.226 μ_{CR} = 0.054
 Diversity = 3.2142e+00 k_t = [3, 4, 2, 2]
 Generation 2570: Best = 1.267756e+03 FES = 257100 μ_F = 0.256 μ_{CR} = 0.052
 Diversity = 3.2141e+00 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 1.265532e+03 FES = 258100 μ_F = 0.236 μ_{CR} = 0.056
 Diversity = 3.2142e+00 k_t = [4, 2, 3, 2]
 Generation 2590: Best = 1.265532e+03 FES = 259100 μ_F = 0.250 μ_{CR} = 0.054
 Diversity = 3.2132e+00 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 1.265365e+03 FES = 260100 μ_F = 0.329 μ_{CR} = 0.041
 Diversity = 3.2113e+00 k_t = [2, 3, 4, 3]
 Generation 2610: Best = 1.265365e+03 FES = 261100 μ_F = 0.284 μ_{CR} = 0.044
 Diversity = 3.2114e+00 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 1.264179e+03 FES = 262100 μ_F = 0.225 μ_{CR} = 0.040
 Diversity = 3.2105e+00 k_t = [3, 3, 4, 2]
 Generation 2630: Best = 1.263259e+03 FES = 263100 μ_F = 0.172 μ_{CR} = 0.039
 Diversity = 3.2103e+00 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 1.260506e+03 FES = 264100 μ_F = 0.146 μ_{CR} = 0.057
 Diversity = 3.2096e+00 k_t = [2, 3, 4, 2]
 Generation 2650: Best = 1.260506e+03 FES = 265100 μ_F = 0.146 μ_{CR} = 0.071
 Diversity = 3.2089e+00 k_t = [1, 1, 1, 1]

Generation 2660: Best = 1.257850e+03 FES = 266100 μ_F = 0.134 μ_{CR} = 0.061
 Diversity = 3.2087e+00 k_t = [2, 4, 4, 3]
 Generation 2670: Best = 1.257664e+03 FES = 267100 μ_F = 0.167 μ_{CR} = 0.058
 Diversity = 3.2080e+00 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 1.257172e+03 FES = 268100 μ_F = 0.210 μ_{CR} = 0.049
 Diversity = 3.1971e+00 k_t = [3, 2, 3, 4]
 Generation 2690: Best = 1.257172e+03 FES = 269100 μ_F = 0.234 μ_{CR} = 0.045
 Diversity = 3.1977e+00 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 1.252343e+03 FES = 270100 μ_F = 0.255 μ_{CR} = 0.044
 Diversity = 3.1980e+00 k_t = [4, 2, 3, 3]
 Generation 2710: Best = 1.252343e+03 FES = 271100 μ_F = 0.310 μ_{CR} = 0.049
 Diversity = 3.1982e+00 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 1.252343e+03 FES = 272100 μ_F = 0.222 μ_{CR} = 0.062
 Diversity = 3.1978e+00 k_t = [2, 4, 4, 2]
 Generation 2730: Best = 1.247677e+03 FES = 273100 μ_F = 0.173 μ_{CR} = 0.056
 Diversity = 3.1976e+00 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 1.247677e+03 FES = 274100 μ_F = 0.217 μ_{CR} = 0.050
 Diversity = 3.1940e+00 k_t = [2, 4, 4, 4]
 Generation 2750: Best = 1.247677e+03 FES = 275100 μ_F = 0.234 μ_{CR} = 0.052
 Diversity = 3.1773e+00 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 1.247677e+03 FES = 276100 μ_F = 0.234 μ_{CR} = 0.053
 Diversity = 3.1767e+00 k_t = [2, 4, 4, 3]
 Generation 2770: Best = 1.247677e+03 FES = 277100 μ_F = 0.193 μ_{CR} = 0.054
 Diversity = 3.1767e+00 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 1.245170e+03 FES = 278100 μ_F = 0.223 μ_{CR} = 0.063
 Diversity = 3.1634e+00 k_t = [4, 2, 4, 4]
 Generation 2790: Best = 1.243226e+03 FES = 279100 μ_F = 0.203 μ_{CR} = 0.066
 Diversity = 3.1639e+00 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 1.242777e+03 FES = 280100 μ_F = 0.211 μ_{CR} = 0.041
 Diversity = 3.1635e+00 k_t = [3, 2, 2, 2]
 Generation 2810: Best = 1.242132e+03 FES = 281100 μ_F = 0.210 μ_{CR} = 0.052
 Diversity = 3.1632e+00 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 1.242132e+03 FES = 282100 μ_F = 0.258 μ_{CR} = 0.060
 Diversity = 3.1635e+00 k_t = [4, 4, 4, 3]
 Generation 2830: Best = 1.240330e+03 FES = 283100 μ_F = 0.228 μ_{CR} = 0.036
 Diversity = 3.1637e+00 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 1.240121e+03 FES = 284100 μ_F = 0.291 μ_{CR} = 0.068
 Diversity = 3.1360e+00 k_t = [4, 3, 2, 4]
 Generation 2850: Best = 1.239665e+03 FES = 285100 μ_F = 0.292 μ_{CR} = 0.059
 Diversity = 3.1359e+00 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 1.239665e+03 FES = 286100 μ_F = 0.265 μ_{CR} = 0.048
 Diversity = 3.1098e+00 k_t = [3, 2, 3, 4]
 Generation 2870: Best = 1.239665e+03 FES = 287100 μ_F = 0.285 μ_{CR} = 0.033
 Diversity = 3.1132e+00 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 1.239665e+03 FES = 288100 μ_F = 0.298 μ_{CR} = 0.034
 Diversity = 3.1136e+00 k_t = [3, 2, 4, 3]
 Generation 2890: Best = 1.239629e+03 FES = 289100 μ_F = 0.313 μ_{CR} = 0.032
 Diversity = 3.1138e+00 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 1.234737e+03 FES = 290100 μ_F = 0.338 μ_{CR} = 0.036
 Diversity = 3.1136e+00 k_t = [4, 4, 4, 2]
 Generation 2910: Best = 1.234706e+03 FES = 291100 μ_F = 0.331 μ_{CR} = 0.036
 Diversity = 3.1138e+00 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 1.232384e+03 FES = 292100 μ_F = 0.375 μ_{CR} = 0.033
 Diversity = 3.1137e+00 k_t = [2, 2, 2, 3]

Generation 2930: Best = 1.232384e+03 FES = 293100 μ_F = 0.383 μ_{CR} = 0.040
 Diversity = 3.1140e+00 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 1.232384e+03 FES = 294100 μ_F = 0.352 μ_{CR} = 0.053
 Diversity = 3.1134e+00 k_t = [3, 3, 4, 2]
 Generation 2950: Best = 1.231896e+03 FES = 295100 μ_F = 0.265 μ_{CR} = 0.049
 Diversity = 3.1130e+00 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 1.231896e+03 FES = 296100 μ_F = 0.274 μ_{CR} = 0.056
 Diversity = 3.1129e+00 k_t = [3, 4, 4, 3]
 Generation 2970: Best = 1.231865e+03 FES = 297100 μ_F = 0.258 μ_{CR} = 0.060
 Diversity = 3.1126e+00 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 1.228015e+03 FES = 298100 μ_F = 0.316 μ_{CR} = 0.061
 Diversity = 3.1090e+00 k_t = [2, 4, 3, 4]
 Generation 2990: Best = 1.227298e+03 FES = 299100 μ_F = 0.290 μ_{CR} = 0.063
 Diversity = 3.1105e+00 k_t = [1, 1, 1, 1]
 Generation 3000: Best = 1.227298e+03 FES = 300100 μ_F = 0.290 μ_{CR} = 0.063
 Diversity = 3.1072e+00 k_t = [3, 4, 2, 4]
 Generation 3010: Best = 1.225250e+03 FES = 301100 μ_F = 0.282 μ_{CR} = 0.066
 Diversity = 3.1073e+00 k_t = [1, 1, 1, 1]
 Generation 3020: Best = 1.224411e+03 FES = 302100 μ_F = 0.264 μ_{CR} = 0.053
 Diversity = 3.1071e+00 k_t = [3, 4, 4, 2]
 Generation 3030: Best = 1.224411e+03 FES = 303100 μ_F = 0.153 μ_{CR} = 0.058
 Diversity = 3.1070e+00 k_t = [1, 1, 1, 1]
 Generation 3040: Best = 1.224365e+03 FES = 304100 μ_F = 0.207 μ_{CR} = 0.065
 Diversity = 3.0925e+00 k_t = [2, 2, 2, 4]
 Generation 3050: Best = 1.224365e+03 FES = 305100 μ_F = 0.166 μ_{CR} = 0.067
 Diversity = 3.0921e+00 k_t = [1, 1, 1, 1]
 Generation 3060: Best = 1.224261e+03 FES = 306100 μ_F = 0.148 μ_{CR} = 0.079
 Diversity = 3.0914e+00 k_t = [2, 2, 3, 3]
 Generation 3070: Best = 1.224261e+03 FES = 307100 μ_F = 0.149 μ_{CR} = 0.063
 Diversity = 3.0911e+00 k_t = [1, 1, 1, 1]
 Generation 3080: Best = 1.223228e+03 FES = 308100 μ_F = 0.228 μ_{CR} = 0.055
 Diversity = 3.0904e+00 k_t = [4, 4, 3, 4]
 Generation 3090: Best = 1.222403e+03 FES = 309100 μ_F = 0.204 μ_{CR} = 0.048
 Diversity = 3.0908e+00 k_t = [1, 1, 1, 1]
 Generation 3100: Best = 1.222170e+03 FES = 310100 μ_F = 0.193 μ_{CR} = 0.048
 Diversity = 3.0907e+00 k_t = [4, 2, 3, 3]
 Generation 3110: Best = 1.218924e+03 FES = 311100 μ_F = 0.216 μ_{CR} = 0.053
 Diversity = 3.0909e+00 k_t = [1, 1, 1, 1]
 Generation 3120: Best = 1.218924e+03 FES = 312100 μ_F = 0.307 μ_{CR} = 0.051
 Diversity = 3.0909e+00 k_t = [3, 3, 2, 2]
 Generation 3130: Best = 1.218924e+03 FES = 313100 μ_F = 0.206 μ_{CR} = 0.049
 Diversity = 3.0911e+00 k_t = [1, 1, 1, 1]
 Generation 3140: Best = 1.218924e+03 FES = 314100 μ_F = 0.191 μ_{CR} = 0.047
 Diversity = 3.0911e+00 k_t = [2, 3, 4, 3]
 Generation 3150: Best = 1.218924e+03 FES = 315100 μ_F = 0.178 μ_{CR} = 0.040
 Diversity = 3.0907e+00 k_t = [1, 1, 1, 1]
 Generation 3160: Best = 1.217952e+03 FES = 316100 μ_F = 0.215 μ_{CR} = 0.047
 Diversity = 3.0904e+00 k_t = [4, 3, 3, 3]
 Generation 3170: Best = 1.217952e+03 FES = 317100 μ_F = 0.208 μ_{CR} = 0.040
 Diversity = 3.0902e+00 k_t = [1, 1, 1, 1]
 Generation 3180: Best = 1.216966e+03 FES = 318100 μ_F = 0.258 μ_{CR} = 0.040
 Diversity = 3.0905e+00 k_t = [3, 2, 2, 2]
 Generation 3190: Best = 1.216327e+03 FES = 319100 μ_F = 0.302 μ_{CR} = 0.033
 Diversity = 3.0907e+00 k_t = [1, 1, 1, 1]

Generation 3200: Best = 1.215968e+03 FES = 320100 μ_F = 0.254 μ_{CR} = 0.047
 Diversity = 3.0895e+00 k_t = [4, 2, 4, 4]
 Generation 3210: Best = 1.215968e+03 FES = 321100 μ_F = 0.226 μ_{CR} = 0.034
 Diversity = 3.0895e+00 k_t = [1, 1, 1, 1]
 Generation 3220: Best = 1.215686e+03 FES = 322100 μ_F = 0.223 μ_{CR} = 0.031
 Diversity = 3.0891e+00 k_t = [3, 3, 3, 3]
 Generation 3230: Best = 1.215686e+03 FES = 323100 μ_F = 0.244 μ_{CR} = 0.030
 Diversity = 3.0782e+00 k_t = [1, 1, 1, 1]
 Generation 3240: Best = 1.215686e+03 FES = 324100 μ_F = 0.221 μ_{CR} = 0.043
 Diversity = 3.0780e+00 k_t = [4, 4, 3, 2]
 Generation 3250: Best = 1.215686e+03 FES = 325100 μ_F = 0.195 μ_{CR} = 0.043
 Diversity = 3.0780e+00 k_t = [1, 1, 1, 1]
 Generation 3260: Best = 1.215657e+03 FES = 326100 μ_F = 0.176 μ_{CR} = 0.040
 Diversity = 3.0778e+00 k_t = [2, 4, 2, 3]
 Generation 3270: Best = 1.214745e+03 FES = 327100 μ_F = 0.181 μ_{CR} = 0.037
 Diversity = 3.0776e+00 k_t = [1, 1, 1, 1]
 Generation 3280: Best = 1.213592e+03 FES = 328100 μ_F = 0.187 μ_{CR} = 0.052
 Diversity = 3.0770e+00 k_t = [3, 3, 4, 2]
 Generation 3290: Best = 1.213592e+03 FES = 329100 μ_F = 0.165 μ_{CR} = 0.039
 Diversity = 3.0770e+00 k_t = [1, 1, 1, 1]
 Generation 3300: Best = 1.213592e+03 FES = 330100 μ_F = 0.185 μ_{CR} = 0.035
 Diversity = 3.0768e+00 k_t = [2, 4, 4, 3]
 Generation 3310: Best = 1.213086e+03 FES = 331100 μ_F = 0.218 μ_{CR} = 0.029
 Diversity = 3.0768e+00 k_t = [1, 1, 1, 1]
 Generation 3320: Best = 1.212258e+03 FES = 332100 μ_F = 0.197 μ_{CR} = 0.045
 Diversity = 3.0761e+00 k_t = [2, 2, 2, 4]
 Generation 3330: Best = 1.211938e+03 FES = 333100 μ_F = 0.205 μ_{CR} = 0.050
 Diversity = 3.0758e+00 k_t = [1, 1, 1, 1]
 Generation 3340: Best = 1.211600e+03 FES = 334100 μ_F = 0.257 μ_{CR} = 0.050
 Diversity = 3.0757e+00 k_t = [2, 3, 3, 3]
 Generation 3350: Best = 1.211388e+03 FES = 335100 μ_F = 0.310 μ_{CR} = 0.039
 Diversity = 3.0756e+00 k_t = [1, 1, 1, 1]
 Generation 3360: Best = 1.211340e+03 FES = 336100 μ_F = 0.406 μ_{CR} = 0.037
 Diversity = 3.0754e+00 k_t = [4, 3, 4, 2]
 Generation 3370: Best = 1.211340e+03 FES = 337100 μ_F = 0.410 μ_{CR} = 0.031
 Diversity = 3.0756e+00 k_t = [1, 1, 1, 1]
 Generation 3380: Best = 1.211340e+03 FES = 338100 μ_F = 0.398 μ_{CR} = 0.044
 Diversity = 3.0757e+00 k_t = [2, 4, 4, 4]
 Generation 3390: Best = 1.211340e+03 FES = 339100 μ_F = 0.394 μ_{CR} = 0.030
 Diversity = 3.0759e+00 k_t = [1, 1, 1, 1]
 Generation 3400: Best = 1.210723e+03 FES = 340100 μ_F = 0.434 μ_{CR} = 0.029
 Diversity = 3.0705e+00 k_t = [4, 3, 2, 4]
 Generation 3410: Best = 1.210723e+03 FES = 341100 μ_F = 0.414 μ_{CR} = 0.029
 Diversity = 3.0706e+00 k_t = [1, 1, 1, 1]
 Generation 3420: Best = 1.210723e+03 FES = 342100 μ_F = 0.506 μ_{CR} = 0.024
 Diversity = 3.0707e+00 k_t = [3, 2, 2, 3]
 Generation 3430: Best = 1.209609e+03 FES = 343100 μ_F = 0.365 μ_{CR} = 0.040
 Diversity = 3.0709e+00 k_t = [1, 1, 1, 1]
 Generation 3440: Best = 1.207479e+03 FES = 344100 μ_F = 0.429 μ_{CR} = 0.035
 Diversity = 3.0579e+00 k_t = [4, 4, 4, 4]
 Generation 3450: Best = 1.206966e+03 FES = 345100 μ_F = 0.511 μ_{CR} = 0.033
 Diversity = 3.0580e+00 k_t = [1, 1, 1, 1]
 Generation 3460: Best = 1.206966e+03 FES = 346100 μ_F = 0.574 μ_{CR} = 0.028
 Diversity = 3.0558e+00 k_t = [2, 4, 2, 4]

Generation 3470: Best = 1.206966e+03 FES = 347100 μ_F = 0.510 μ_{CR} = 0.034
 Diversity = 3.0558e+00 k_t = [1, 1, 1, 1]
 Generation 3480: Best = 1.206842e+03 FES = 348100 μ_F = 0.457 μ_{CR} = 0.022
 Diversity = 3.0557e+00 k_t = [3, 3, 2, 2]
 Generation 3490: Best = 1.206467e+03 FES = 349100 μ_F = 0.495 μ_{CR} = 0.023
 Diversity = 3.0561e+00 k_t = [1, 1, 1, 1]
 Generation 3500: Best = 1.206467e+03 FES = 350100 μ_F = 0.479 μ_{CR} = 0.022
 Diversity = 3.0564e+00 k_t = [3, 3, 4, 3]
 Generation 3510: Best = 1.206026e+03 FES = 351100 μ_F = 0.416 μ_{CR} = 0.026
 Diversity = 3.0566e+00 k_t = [1, 1, 1, 1]
 Generation 3520: Best = 1.205967e+03 FES = 352100 μ_F = 0.349 μ_{CR} = 0.031
 Diversity = 3.0565e+00 k_t = [2, 2, 4, 2]
 Generation 3530: Best = 1.205967e+03 FES = 353100 μ_F = 0.332 μ_{CR} = 0.029
 Diversity = 3.0566e+00 k_t = [1, 1, 1, 1]
 Generation 3540: Best = 1.205967e+03 FES = 354100 μ_F = 0.423 μ_{CR} = 0.023
 Diversity = 3.0565e+00 k_t = [3, 4, 4, 3]
 Generation 3550: Best = 1.205967e+03 FES = 355100 μ_F = 0.460 μ_{CR} = 0.020
 Diversity = 3.0566e+00 k_t = [1, 1, 1, 1]
 Generation 3560: Best = 1.205967e+03 FES = 356100 μ_F = 0.435 μ_{CR} = 0.024
 Diversity = 3.0567e+00 k_t = [3, 3, 2, 2]
 Generation 3570: Best = 1.205967e+03 FES = 357100 μ_F = 0.454 μ_{CR} = 0.025
 Diversity = 3.0577e+00 k_t = [1, 1, 1, 1]
 Generation 3580: Best = 1.205967e+03 FES = 358100 μ_F = 0.379 μ_{CR} = 0.053
 Diversity = 3.0589e+00 k_t = [2, 3, 2, 4]
 Generation 3590: Best = 1.205967e+03 FES = 359100 μ_F = 0.304 μ_{CR} = 0.046
 Diversity = 3.0589e+00 k_t = [1, 1, 1, 1]
 Generation 3600: Best = 1.205520e+03 FES = 360100 μ_F = 0.291 μ_{CR} = 0.046
 Diversity = 3.0489e+00 k_t = [4, 2, 4, 2]
 Generation 3610: Best = 1.205517e+03 FES = 361100 μ_F = 0.301 μ_{CR} = 0.053
 Diversity = 3.0490e+00 k_t = [1, 1, 1, 1]
 Generation 3620: Best = 1.205517e+03 FES = 362100 μ_F = 0.244 μ_{CR} = 0.042
 Diversity = 3.0484e+00 k_t = [3, 3, 4, 4]
 Generation 3630: Best = 1.205517e+03 FES = 363100 μ_F = 0.263 μ_{CR} = 0.032
 Diversity = 3.0479e+00 k_t = [1, 1, 1, 1]
 Generation 3640: Best = 1.204521e+03 FES = 364100 μ_F = 0.263 μ_{CR} = 0.034
 Diversity = 3.0480e+00 k_t = [4, 4, 2, 3]
 Generation 3650: Best = 1.204521e+03 FES = 365100 μ_F = 0.245 μ_{CR} = 0.031
 Diversity = 3.0479e+00 k_t = [1, 1, 1, 1]
 Generation 3660: Best = 1.204402e+03 FES = 366100 μ_F = 0.235 μ_{CR} = 0.034
 Diversity = 3.0478e+00 k_t = [4, 3, 3, 3]
 Generation 3670: Best = 1.204402e+03 FES = 367100 μ_F = 0.229 μ_{CR} = 0.031
 Diversity = 3.0481e+00 k_t = [1, 1, 1, 1]
 Generation 3680: Best = 1.202374e+03 FES = 368100 μ_F = 0.228 μ_{CR} = 0.077
 Diversity = 3.0332e+00 k_t = [4, 4, 2, 4]
 Generation 3690: Best = 1.202374e+03 FES = 369100 μ_F = 0.171 μ_{CR} = 0.066
 Diversity = 3.0331e+00 k_t = [1, 1, 1, 1]
 Generation 3700: Best = 1.200791e+03 FES = 370100 μ_F = 0.253 μ_{CR} = 0.059
 Diversity = 3.0333e+00 k_t = [4, 3, 3, 3]
 Generation 3710: Best = 1.200749e+03 FES = 371100 μ_F = 0.235 μ_{CR} = 0.053
 Diversity = 3.0331e+00 k_t = [1, 1, 1, 1]
 Generation 3720: Best = 1.200749e+03 FES = 372100 μ_F = 0.219 μ_{CR} = 0.049
 Diversity = 3.0329e+00 k_t = [2, 2, 2, 3]
 Generation 3730: Best = 1.200683e+03 FES = 373100 μ_F = 0.188 μ_{CR} = 0.038
 Diversity = 3.0252e+00 k_t = [1, 1, 1, 1]

Generation 3740: Best = 1.200479e+03 FES = 374100 μ_F = 0.273 μ_{CR} = 0.043
 Diversity = 3.0254e+00 k_t = [3, 2, 2, 2]
 Generation 3750: Best = 1.200354e+03 FES = 375100 μ_F = 0.317 μ_{CR} = 0.041
 Diversity = 3.0135e+00 k_t = [1, 1, 1, 1]
 Generation 3760: Best = 1.200354e+03 FES = 376100 μ_F = 0.305 μ_{CR} = 0.057
 Diversity = 3.0136e+00 k_t = [2, 3, 4, 3]
 Generation 3770: Best = 1.200354e+03 FES = 377100 μ_F = 0.260 μ_{CR} = 0.069
 Diversity = 3.0134e+00 k_t = [1, 1, 1, 1]
 Generation 3780: Best = 1.199466e+03 FES = 378100 μ_F = 0.259 μ_{CR} = 0.069
 Diversity = 3.0135e+00 k_t = [3, 3, 4, 3]
 Generation 3790: Best = 1.199466e+03 FES = 379100 μ_F = 0.244 μ_{CR} = 0.051
 Diversity = 3.0090e+00 k_t = [1, 1, 1, 1]
 Generation 3800: Best = 1.199197e+03 FES = 380100 μ_F = 0.195 μ_{CR} = 0.045
 Diversity = 3.0091e+00 k_t = [3, 2, 2, 3]
 Generation 3810: Best = 1.198865e+03 FES = 381100 μ_F = 0.156 μ_{CR} = 0.045
 Diversity = 3.0094e+00 k_t = [1, 1, 1, 1]
 Generation 3820: Best = 1.198567e+03 FES = 382100 μ_F = 0.263 μ_{CR} = 0.056
 Diversity = 3.0002e+00 k_t = [2, 4, 2, 4]
 Generation 3830: Best = 1.198567e+03 FES = 383100 μ_F = 0.214 μ_{CR} = 0.038
 Diversity = 3.0002e+00 k_t = [1, 1, 1, 1]
 Generation 3840: Best = 1.198560e+03 FES = 384100 μ_F = 0.241 μ_{CR} = 0.047
 Diversity = 2.9905e+00 k_t = [2, 4, 3, 4]
 Generation 3850: Best = 1.198560e+03 FES = 385100 μ_F = 0.269 μ_{CR} = 0.038
 Diversity = 2.9926e+00 k_t = [1, 1, 1, 1]
 Generation 3860: Best = 1.198560e+03 FES = 386100 μ_F = 0.437 μ_{CR} = 0.051
 Diversity = 2.9931e+00 k_t = [2, 2, 3, 4]
 Generation 3870: Best = 1.198560e+03 FES = 387100 μ_F = 0.384 μ_{CR} = 0.057
 Diversity = 2.9928e+00 k_t = [1, 1, 1, 1]
 Generation 3880: Best = 1.198560e+03 FES = 388100 μ_F = 0.381 μ_{CR} = 0.071
 Diversity = 2.9923e+00 k_t = [4, 2, 4, 3]
 Generation 3890: Best = 1.198554e+03 FES = 389100 μ_F = 0.346 μ_{CR} = 0.064
 Diversity = 2.9923e+00 k_t = [1, 1, 1, 1]
 Generation 3900: Best = 1.198554e+03 FES = 390100 μ_F = 0.316 μ_{CR} = 0.055
 Diversity = 2.9924e+00 k_t = [3, 2, 2, 3]
 Generation 3910: Best = 1.198554e+03 FES = 391100 μ_F = 0.305 μ_{CR} = 0.041
 Diversity = 2.9919e+00 k_t = [1, 1, 1, 1]
 Generation 3920: Best = 1.198485e+03 FES = 392100 μ_F = 0.290 μ_{CR} = 0.039
 Diversity = 2.9918e+00 k_t = [2, 2, 4, 2]
 Generation 3930: Best = 1.198485e+03 FES = 393100 μ_F = 0.262 μ_{CR} = 0.055
 Diversity = 2.9922e+00 k_t = [1, 1, 1, 1]
 Generation 3940: Best = 1.198485e+03 FES = 394100 μ_F = 0.264 μ_{CR} = 0.056
 Diversity = 2.9903e+00 k_t = [2, 4, 3, 2]
 Generation 3950: Best = 1.198485e+03 FES = 395100 μ_F = 0.282 μ_{CR} = 0.043
 Diversity = 2.9813e+00 k_t = [1, 1, 1, 1]
 Generation 3960: Best = 1.198440e+03 FES = 396100 μ_F = 0.292 μ_{CR} = 0.057
 Diversity = 2.9810e+00 k_t = [4, 4, 3, 3]
 Generation 3970: Best = 1.198383e+03 FES = 397100 μ_F = 0.364 μ_{CR} = 0.043
 Diversity = 2.9808e+00 k_t = [1, 1, 1, 1]
 Generation 3980: Best = 1.197702e+03 FES = 398100 μ_F = 0.319 μ_{CR} = 0.047
 Diversity = 2.9807e+00 k_t = [4, 3, 3, 3]
 Generation 3990: Best = 1.197408e+03 FES = 399100 μ_F = 0.377 μ_{CR} = 0.037
 Diversity = 2.9808e+00 k_t = [1, 1, 1, 1]
 Generation 4000: Best = 1.196974e+03 FES = 400100 μ_F = 0.429 μ_{CR} = 0.029
 Diversity = 2.9507e+00 k_t = [3, 3, 2, 4]

Generation 4010: Best = 1.196974e+03 FES = 401100 μ_F = 0.351 μ_{CR} = 0.033
 Diversity = 2.9508e+00 k_t = [1, 1, 1, 1]
 Generation 4020: Best = 1.196974e+03 FES = 402100 μ_F = 0.360 μ_{CR} = 0.037
 Diversity = 2.9509e+00 k_t = [3, 3, 3, 2]
 Generation 4030: Best = 1.196974e+03 FES = 403100 μ_F = 0.311 μ_{CR} = 0.035
 Diversity = 2.9509e+00 k_t = [1, 1, 1, 1]
 Generation 4040: Best = 1.196974e+03 FES = 404100 μ_F = 0.271 μ_{CR} = 0.047
 Diversity = 2.9514e+00 k_t = [2, 4, 4, 3]
 Generation 4050: Best = 1.196974e+03 FES = 405100 μ_F = 0.291 μ_{CR} = 0.038
 Diversity = 2.9514e+00 k_t = [1, 1, 1, 1]
 Generation 4060: Best = 1.196974e+03 FES = 406100 μ_F = 0.247 μ_{CR} = 0.042
 Diversity = 2.9519e+00 k_t = [2, 4, 4, 4]
 Generation 4070: Best = 1.196934e+03 FES = 407100 μ_F = 0.223 μ_{CR} = 0.044
 Diversity = 2.9520e+00 k_t = [1, 1, 1, 1]
 Generation 4080: Best = 1.196456e+03 FES = 408100 μ_F = 0.251 μ_{CR} = 0.049
 Diversity = 2.9541e+00 k_t = [3, 2, 4, 2]
 Generation 4090: Best = 1.196456e+03 FES = 409100 μ_F = 0.220 μ_{CR} = 0.048
 Diversity = 2.9538e+00 k_t = [1, 1, 1, 1]
 Generation 4100: Best = 1.195784e+03 FES = 410100 μ_F = 0.291 μ_{CR} = 0.049
 Diversity = 2.9524e+00 k_t = [2, 2, 2, 4]
 Generation 4110: Best = 1.195784e+03 FES = 411100 μ_F = 0.188 μ_{CR} = 0.051
 Diversity = 2.9518e+00 k_t = [1, 1, 1, 1]
 Generation 4120: Best = 1.195784e+03 FES = 412100 μ_F = 0.206 μ_{CR} = 0.044
 Diversity = 2.9515e+00 k_t = [3, 4, 3, 4]
 Generation 4130: Best = 1.195661e+03 FES = 413100 μ_F = 0.190 μ_{CR} = 0.045
 Diversity = 2.9515e+00 k_t = [1, 1, 1, 1]
 Generation 4140: Best = 1.195141e+03 FES = 414100 μ_F = 0.198 μ_{CR} = 0.050
 Diversity = 2.9514e+00 k_t = [4, 4, 3, 3]
 Generation 4150: Best = 1.194724e+03 FES = 415100 μ_F = 0.151 μ_{CR} = 0.054
 Diversity = 2.9513e+00 k_t = [1, 1, 1, 1]
 Generation 4160: Best = 1.194626e+03 FES = 416100 μ_F = 0.178 μ_{CR} = 0.050
 Diversity = 2.9510e+00 k_t = [4, 4, 2, 2]
 Generation 4170: Best = 1.194163e+03 FES = 417100 μ_F = 0.149 μ_{CR} = 0.055
 Diversity = 2.9506e+00 k_t = [1, 1, 1, 1]
 Generation 4180: Best = 1.193829e+03 FES = 418100 μ_F = 0.146 μ_{CR} = 0.066
 Diversity = 2.9503e+00 k_t = [4, 2, 4, 3]
 Generation 4190: Best = 1.193829e+03 FES = 419100 μ_F = 0.186 μ_{CR} = 0.060
 Diversity = 2.9505e+00 k_t = [1, 1, 1, 1]
 Generation 4200: Best = 1.193829e+03 FES = 420100 μ_F = 0.212 μ_{CR} = 0.050
 Diversity = 2.9504e+00 k_t = [3, 3, 2, 2]
 Generation 4210: Best = 1.193786e+03 FES = 421100 μ_F = 0.172 μ_{CR} = 0.058
 Diversity = 2.9501e+00 k_t = [1, 1, 1, 1]
 Generation 4220: Best = 1.193622e+03 FES = 422100 μ_F = 0.254 μ_{CR} = 0.066
 Diversity = 2.9505e+00 k_t = [3, 3, 2, 4]
 Generation 4230: Best = 1.193622e+03 FES = 423100 μ_F = 0.283 μ_{CR} = 0.061
 Diversity = 2.9507e+00 k_t = [1, 1, 1, 1]
 Generation 4240: Best = 1.193622e+03 FES = 424100 μ_F = 0.261 μ_{CR} = 0.071
 Diversity = 2.9507e+00 k_t = [2, 2, 3, 2]
 Generation 4250: Best = 1.193546e+03 FES = 425100 μ_F = 0.252 μ_{CR} = 0.064
 Diversity = 2.9508e+00 k_t = [1, 1, 1, 1]
 Generation 4260: Best = 1.192836e+03 FES = 426100 μ_F = 0.313 μ_{CR} = 0.063
 Diversity = 2.9510e+00 k_t = [3, 4, 3, 2]
 Generation 4270: Best = 1.192545e+03 FES = 427100 μ_F = 0.236 μ_{CR} = 0.060
 Diversity = 2.9511e+00 k_t = [1, 1, 1, 1]

Generation 4280: Best = 1.191699e+03 FES = 428100 μ_F = 0.189 μ_{CR} = 0.058
 Diversity = 2.9511e+00 k_t = [2, 3, 4, 2]
 Generation 4290: Best = 1.191699e+03 FES = 429100 μ_F = 0.129 μ_{CR} = 0.060
 Diversity = 2.9508e+00 k_t = [1, 1, 1, 1]
 Generation 4300: Best = 1.191699e+03 FES = 430100 μ_F = 0.130 μ_{CR} = 0.068
 Diversity = 2.9510e+00 k_t = [3, 4, 4, 3]
 Generation 4310: Best = 1.191676e+03 FES = 431100 μ_F = 0.205 μ_{CR} = 0.058
 Diversity = 2.9510e+00 k_t = [1, 1, 1, 1]
 Generation 4320: Best = 1.191676e+03 FES = 432100 μ_F = 0.271 μ_{CR} = 0.071
 Diversity = 2.9397e+00 k_t = [2, 3, 3, 4]
 Generation 4330: Best = 1.191676e+03 FES = 433100 μ_F = 0.281 μ_{CR} = 0.068
 Diversity = 2.9399e+00 k_t = [1, 1, 1, 1]
 Generation 4340: Best = 1.191226e+03 FES = 434100 μ_F = 0.296 μ_{CR} = 0.057
 Diversity = 2.9168e+00 k_t = [4, 2, 2, 4]
 Generation 4350: Best = 1.191226e+03 FES = 435100 μ_F = 0.286 μ_{CR} = 0.047
 Diversity = 2.9170e+00 k_t = [1, 1, 1, 1]
 Generation 4360: Best = 1.190114e+03 FES = 436100 μ_F = 0.246 μ_{CR} = 0.054
 Diversity = 2.9151e+00 k_t = [4, 3, 2, 4]
 Generation 4370: Best = 1.190114e+03 FES = 437100 μ_F = 0.207 μ_{CR} = 0.041
 Diversity = 2.9151e+00 k_t = [1, 1, 1, 1]
 Generation 4380: Best = 1.190114e+03 FES = 438100 μ_F = 0.182 μ_{CR} = 0.042
 Diversity = 2.9151e+00 k_t = [3, 3, 3, 2]
 Generation 4390: Best = 1.190114e+03 FES = 439100 μ_F = 0.202 μ_{CR} = 0.058
 Diversity = 2.9149e+00 k_t = [1, 1, 1, 1]
 Generation 4400: Best = 1.190114e+03 FES = 440100 μ_F = 0.218 μ_{CR} = 0.079
 Diversity = 2.9155e+00 k_t = [3, 3, 4, 3]
 Generation 4410: Best = 1.190039e+03 FES = 441100 μ_F = 0.153 μ_{CR} = 0.060
 Diversity = 2.9157e+00 k_t = [1, 1, 1, 1]
 Generation 4420: Best = 1.189551e+03 FES = 442100 μ_F = 0.117 μ_{CR} = 0.079
 Diversity = 2.9153e+00 k_t = [3, 3, 4, 3]
 Generation 4430: Best = 1.189365e+03 FES = 443100 μ_F = 0.192 μ_{CR} = 0.054
 Diversity = 2.9153e+00 k_t = [1, 1, 1, 1]
 Generation 4440: Best = 1.189124e+03 FES = 444100 μ_F = 0.198 μ_{CR} = 0.067
 Diversity = 2.9181e+00 k_t = [4, 3, 2, 3]
 Generation 4450: Best = 1.188818e+03 FES = 445100 μ_F = 0.204 μ_{CR} = 0.052
 Diversity = 2.9182e+00 k_t = [1, 1, 1, 1]
 Generation 4460: Best = 1.188743e+03 FES = 446100 μ_F = 0.195 μ_{CR} = 0.069
 Diversity = 2.9113e+00 k_t = [4, 3, 3, 4]
 Generation 4470: Best = 1.188714e+03 FES = 447100 μ_F = 0.177 μ_{CR} = 0.069
 Diversity = 2.9110e+00 k_t = [1, 1, 1, 1]
 Generation 4480: Best = 1.188714e+03 FES = 448100 μ_F = 0.144 μ_{CR} = 0.061
 Diversity = 2.9120e+00 k_t = [2, 3, 3, 2]
 Generation 4490: Best = 1.188714e+03 FES = 449100 μ_F = 0.148 μ_{CR} = 0.046
 Diversity = 2.9114e+00 k_t = [1, 1, 1, 1]
 Generation 4500: Best = 1.188438e+03 FES = 450100 μ_F = 0.123 μ_{CR} = 0.060
 Diversity = 2.9142e+00 k_t = [3, 4, 2, 4]
 Generation 4510: Best = 1.188000e+03 FES = 451100 μ_F = 0.205 μ_{CR} = 0.063
 Diversity = 2.9141e+00 k_t = [1, 1, 1, 1]
 Generation 4520: Best = 1.188000e+03 FES = 452100 μ_F = 0.257 μ_{CR} = 0.049
 Diversity = 2.9139e+00 k_t = [4, 3, 4, 2]
 Generation 4530: Best = 1.187999e+03 FES = 453100 μ_F = 0.295 μ_{CR} = 0.041
 Diversity = 2.9139e+00 k_t = [1, 1, 1, 1]
 Generation 4540: Best = 1.187682e+03 FES = 454100 μ_F = 0.297 μ_{CR} = 0.030
 Diversity = 2.9137e+00 k_t = [2, 4, 2, 3]

Generation 4550: Best = 1.187682e+03 FES = 455100 μ_F = 0.268 μ_{CR} = 0.045
 Diversity = 2.9127e+00 k_t = [1, 1, 1, 1]
 Generation 4560: Best = 1.187682e+03 FES = 456100 μ_F = 0.322 μ_{CR} = 0.077
 Diversity = 2.9111e+00 k_t = [2, 3, 3, 3]
 Generation 4570: Best = 1.187682e+03 FES = 457100 μ_F = 0.286 μ_{CR} = 0.061
 Diversity = 2.9115e+00 k_t = [1, 1, 1, 1]
 Generation 4580: Best = 1.187682e+03 FES = 458100 μ_F = 0.282 μ_{CR} = 0.081
 Diversity = 2.9054e+00 k_t = [2, 3, 2, 2]
 Generation 4590: Best = 1.187682e+03 FES = 459100 μ_F = 0.196 μ_{CR} = 0.070
 Diversity = 2.9053e+00 k_t = [1, 1, 1, 1]
 Generation 4600: Best = 1.187509e+03 FES = 460100 μ_F = 0.257 μ_{CR} = 0.074
 Diversity = 2.9053e+00 k_t = [3, 2, 3, 4]
 Generation 4610: Best = 1.187509e+03 FES = 461100 μ_F = 0.267 μ_{CR} = 0.072
 Diversity = 2.9057e+00 k_t = [1, 1, 1, 1]
 Generation 4620: Best = 1.187265e+03 FES = 462100 μ_F = 0.302 μ_{CR} = 0.072
 Diversity = 2.8935e+00 k_t = [3, 2, 2, 4]
 Generation 4630: Best = 1.186936e+03 FES = 463100 μ_F = 0.262 μ_{CR} = 0.050
 Diversity = 2.8938e+00 k_t = [1, 1, 1, 1]
 Generation 4640: Best = 1.186417e+03 FES = 464100 μ_F = 0.284 μ_{CR} = 0.052
 Diversity = 2.8939e+00 k_t = [4, 3, 2, 3]
 Generation 4650: Best = 1.186417e+03 FES = 465100 μ_F = 0.310 μ_{CR} = 0.067
 Diversity = 2.8941e+00 k_t = [1, 1, 1, 1]
 Generation 4660: Best = 1.186403e+03 FES = 466100 μ_F = 0.249 μ_{CR} = 0.067
 Diversity = 2.8943e+00 k_t = [2, 4, 2, 2]
 Generation 4670: Best = 1.186403e+03 FES = 467100 μ_F = 0.287 μ_{CR} = 0.074
 Diversity = 2.8943e+00 k_t = [1, 1, 1, 1]
 Generation 4680: Best = 1.186403e+03 FES = 468100 μ_F = 0.320 μ_{CR} = 0.071
 Diversity = 2.8915e+00 k_t = [3, 4, 2, 4]
 Generation 4690: Best = 1.186403e+03 FES = 469100 μ_F = 0.280 μ_{CR} = 0.075
 Diversity = 2.8916e+00 k_t = [1, 1, 1, 1]
 Generation 4700: Best = 1.185872e+03 FES = 470100 μ_F = 0.313 μ_{CR} = 0.059
 Diversity = 2.8966e+00 k_t = [4, 3, 2, 4]
 Generation 4710: Best = 1.185695e+03 FES = 471100 μ_F = 0.263 μ_{CR} = 0.048
 Diversity = 2.8965e+00 k_t = [1, 1, 1, 1]
 Generation 4720: Best = 1.185403e+03 FES = 472100 μ_F = 0.277 μ_{CR} = 0.047
 Diversity = 2.8964e+00 k_t = [4, 4, 4, 2]
 Generation 4730: Best = 1.185403e+03 FES = 473100 μ_F = 0.274 μ_{CR} = 0.056
 Diversity = 2.8961e+00 k_t = [1, 1, 1, 1]
 Generation 4740: Best = 1.185348e+03 FES = 474100 μ_F = 0.245 μ_{CR} = 0.053
 Diversity = 2.8958e+00 k_t = [2, 4, 2, 2]
 Generation 4750: Best = 1.184960e+03 FES = 475100 μ_F = 0.210 μ_{CR} = 0.048
 Diversity = 2.8957e+00 k_t = [1, 1, 1, 1]
 Generation 4760: Best = 1.183973e+03 FES = 476100 μ_F = 0.257 μ_{CR} = 0.068
 Diversity = 2.8957e+00 k_t = [4, 3, 2, 3]
 Generation 4770: Best = 1.183973e+03 FES = 477100 μ_F = 0.269 μ_{CR} = 0.047
 Diversity = 2.8958e+00 k_t = [1, 1, 1, 1]
 Generation 4780: Best = 1.183973e+03 FES = 478100 μ_F = 0.329 μ_{CR} = 0.037
 Diversity = 2.8874e+00 k_t = [2, 2, 2, 4]
 Generation 4790: Best = 1.183871e+03 FES = 479100 μ_F = 0.354 μ_{CR} = 0.060
 Diversity = 2.8876e+00 k_t = [1, 1, 1, 1]
 Generation 4800: Best = 1.183641e+03 FES = 480100 μ_F = 0.395 μ_{CR} = 0.055
 Diversity = 2.8874e+00 k_t = [3, 3, 4, 3]
 Generation 4810: Best = 1.183626e+03 FES = 481100 μ_F = 0.341 μ_{CR} = 0.055
 Diversity = 2.8875e+00 k_t = [1, 1, 1, 1]

Generation 4820: Best = 1.183285e+03 FES = 482100 μ_F = 0.314 μ_{CR} = 0.051
 Diversity = 2.8875e+00 k_t = [3, 4, 3, 3]
 Generation 4830: Best = 1.183285e+03 FES = 483100 μ_F = 0.381 μ_{CR} = 0.040
 Diversity = 2.8873e+00 k_t = [1, 1, 1, 1]
 Generation 4840: Best = 1.183137e+03 FES = 484100 μ_F = 0.423 μ_{CR} = 0.046
 Diversity = 2.8867e+00 k_t = [4, 3, 3, 2]
 Generation 4850: Best = 1.183137e+03 FES = 485100 μ_F = 0.393 μ_{CR} = 0.067
 Diversity = 2.8669e+00 k_t = [1, 1, 1, 1]
 Generation 4860: Best = 1.182998e+03 FES = 486100 μ_F = 0.402 μ_{CR} = 0.060
 Diversity = 2.8463e+00 k_t = [3, 3, 3, 4]
 Generation 4870: Best = 1.182998e+03 FES = 487100 μ_F = 0.434 μ_{CR} = 0.065
 Diversity = 2.8338e+00 k_t = [1, 1, 1, 1]
 Generation 4880: Best = 1.182994e+03 FES = 488100 μ_F = 0.474 μ_{CR} = 0.079
 Diversity = 2.8331e+00 k_t = [2, 3, 4, 2]
 Generation 4890: Best = 1.182877e+03 FES = 489100 μ_F = 0.557 μ_{CR} = 0.096
 Diversity = 2.8323e+00 k_t = [1, 1, 1, 1]
 Generation 4900: Best = 1.182730e+03 FES = 490100 μ_F = 0.597 μ_{CR} = 0.105
 Diversity = 2.8320e+00 k_t = [4, 3, 4, 4]
 Generation 4910: Best = 1.181621e+03 FES = 491100 μ_F = 0.632 μ_{CR} = 0.097
 Diversity = 2.8318e+00 k_t = [1, 1, 1, 1]
 Generation 4920: Best = 1.181456e+03 FES = 492100 μ_F = 0.658 μ_{CR} = 0.093
 Diversity = 2.8197e+00 k_t = [2, 2, 4, 4]
 Generation 4930: Best = 1.181340e+03 FES = 493100 μ_F = 0.712 μ_{CR} = 0.084
 Diversity = 2.8026e+00 k_t = [1, 1, 1, 1]
 Generation 4940: Best = 1.181006e+03 FES = 494100 μ_F = 0.742 μ_{CR} = 0.076
 Diversity = 2.7696e+00 k_t = [2, 3, 2, 3]
 Generation 4950: Best = 1.180859e+03 FES = 495100 μ_F = 0.761 μ_{CR} = 0.080
 Diversity = 2.7696e+00 k_t = [1, 1, 1, 1]
 Generation 4960: Best = 1.180778e+03 FES = 496100 μ_F = 0.722 μ_{CR} = 0.089
 Diversity = 2.7547e+00 k_t = [2, 2, 2, 2]
 Generation 4970: Best = 1.180755e+03 FES = 497100 μ_F = 0.739 μ_{CR} = 0.105
 Diversity = 2.7542e+00 k_t = [1, 1, 1, 1]
 Generation 4980: Best = 1.180551e+03 FES = 498100 μ_F = 0.705 μ_{CR} = 0.096
 Diversity = 2.7043e+00 k_t = [2, 4, 3, 4]
 Generation 4990: Best = 1.180397e+03 FES = 499100 μ_F = 0.675 μ_{CR} = 0.086
 Diversity = 2.7043e+00 k_t = [1, 1, 1, 1]
 Generation 5000: Best = 1.180389e+03 FES = 500100 μ_F = 0.640 μ_{CR} = 0.086
 Diversity = 2.6593e+00 k_t = [3, 2, 3, 2]
 Generation 5010: Best = 1.180040e+03 FES = 501100 μ_F = 0.715 μ_{CR} = 0.085
 Diversity = 2.6066e+00 k_t = [1, 1, 1, 1]
 Generation 5020: Best = 1.180014e+03 FES = 502100 μ_F = 0.728 μ_{CR} = 0.087
 Diversity = 2.5392e+00 k_t = [4, 4, 3, 3]
 Generation 5030: Best = 1.179888e+03 FES = 503100 μ_F = 0.763 μ_{CR} = 0.092
 Diversity = 2.5060e+00 k_t = [1, 1, 1, 1]
 Generation 5040: Best = 1.179542e+03 FES = 504100 μ_F = 0.771 μ_{CR} = 0.088
 Diversity = 2.4974e+00 k_t = [4, 2, 4, 2]
 Generation 5050: Best = 1.179503e+03 FES = 505100 μ_F = 0.775 μ_{CR} = 0.076
 Diversity = 2.4156e+00 k_t = [1, 1, 1, 1]
 Generation 5060: Best = 1.179367e+03 FES = 506100 μ_F = 0.797 μ_{CR} = 0.080
 Diversity = 2.2132e+00 k_t = [2, 4, 3, 2]
 Generation 5070: Best = 1.179274e+03 FES = 507100 μ_F = 0.801 μ_{CR} = 0.088
 Diversity = 1.9745e+00 k_t = [1, 1, 1, 1]
 Generation 5080: Best = 1.179124e+03 FES = 508100 μ_F = 0.807 μ_{CR} = 0.109
 Diversity = 1.8950e+00 k_t = [3, 2, 4, 2]

Generation 5090: Best = 1.178968e+03 FES = 509100 μ_F = 0.820 μ_{CR} = 0.112
 Diversity = 1.7687e+00 k_t = [1, 1, 1, 1]
 Generation 5100: Best = 1.178652e+03 FES = 510100 μ_F = 0.833 μ_{CR} = 0.115
 Diversity = 1.5980e+00 k_t = [4, 4, 4, 2]
 Generation 5110: Best = 1.178404e+03 FES = 511100 μ_F = 0.832 μ_{CR} = 0.131
 Diversity = 1.2033e+00 k_t = [1, 1, 1, 1]
 Generation 5120: Best = 1.178251e+03 FES = 512100 μ_F = 0.833 μ_{CR} = 0.134
 Diversity = 6.4020e-01 k_t = [4, 3, 3, 4]
 Generation 5130: Best = 1.177840e+03 FES = 513100 μ_F = 0.829 μ_{CR} = 0.137
 Diversity = 6.3725e-01 k_t = [1, 1, 1, 1]
 Generation 5140: Best = 1.177661e+03 FES = 514100 μ_F = 0.822 μ_{CR} = 0.139
 Diversity = 5.0451e-01 k_t = [2, 3, 2, 4]
 Generation 5150: Best = 1.177527e+03 FES = 515100 μ_F = 0.814 μ_{CR} = 0.140
 Diversity = 6.9804e-02 k_t = [1, 1, 1, 1]
 Generation 5160: Best = 1.177364e+03 FES = 516100 μ_F = 0.787 μ_{CR} = 0.155
 Diversity = 2.9271e-02 k_t = [2, 3, 4, 3]
 Generation 5170: Best = 1.177269e+03 FES = 517100 μ_F = 0.759 μ_{CR} = 0.159
 Diversity = 2.2620e-02 k_t = [1, 1, 1, 1]
 Generation 5180: Best = 1.177055e+03 FES = 518100 μ_F = 0.716 μ_{CR} = 0.157
 Diversity = 1.9424e-02 k_t = [2, 4, 3, 2]
 Generation 5190: Best = 1.176878e+03 FES = 519100 μ_F = 0.683 μ_{CR} = 0.163
 Diversity = 1.6536e-02 k_t = [1, 1, 1, 1]
 Generation 5200: Best = 1.176752e+03 FES = 520100 μ_F = 0.668 μ_{CR} = 0.168
 Diversity = 1.2690e-02 k_t = [4, 2, 2, 2]
 Generation 5210: Best = 1.176614e+03 FES = 521100 μ_F = 0.668 μ_{CR} = 0.166
 Diversity = 1.0750e-02 k_t = [1, 1, 1, 1]
 Generation 5220: Best = 1.176545e+03 FES = 522100 μ_F = 0.656 μ_{CR} = 0.178
 Diversity = 8.8438e-03 k_t = [3, 3, 4, 2]
 Generation 5230: Best = 1.176510e+03 FES = 523100 μ_F = 0.646 μ_{CR} = 0.186
 Diversity = 7.5681e-03 k_t = [1, 1, 1, 1]
 Generation 5240: Best = 1.176453e+03 FES = 524100 μ_F = 0.645 μ_{CR} = 0.187
 Diversity = 6.3724e-03 k_t = [2, 2, 4, 2]
 Generation 5250: Best = 1.176415e+03 FES = 525100 μ_F = 0.649 μ_{CR} = 0.196
 Diversity = 5.5120e-03 k_t = [1, 1, 1, 1]
 Generation 5260: Best = 1.176393e+03 FES = 526100 μ_F = 0.660 μ_{CR} = 0.203
 Diversity = 4.7120e-03 k_t = [3, 4, 4, 2]
 Generation 5270: Best = 1.176363e+03 FES = 527100 μ_F = 0.663 μ_{CR} = 0.214
 Diversity = 4.0847e-03 k_t = [1, 1, 1, 1]
 Generation 5280: Best = 1.176346e+03 FES = 528100 μ_F = 0.666 μ_{CR} = 0.218
 Diversity = 3.5197e-03 k_t = [2, 4, 4, 3]
 Generation 5290: Best = 1.176330e+03 FES = 529100 μ_F = 0.676 μ_{CR} = 0.217
 Diversity = 3.3678e-03 k_t = [1, 1, 1, 1]
 Generation 5300: Best = 1.176315e+03 FES = 530100 μ_F = 0.675 μ_{CR} = 0.220
 Diversity = 3.0387e-03 k_t = [3, 3, 4, 2]
 Generation 5310: Best = 1.176303e+03 FES = 531100 μ_F = 0.685 μ_{CR} = 0.221
 Diversity = 2.7936e-03 k_t = [1, 1, 1, 1]
 Generation 5320: Best = 1.176289e+03 FES = 532100 μ_F = 0.689 μ_{CR} = 0.227
 Diversity = 2.6724e-03 k_t = [4, 4, 4, 2]
 Generation 5330: Best = 1.176278e+03 FES = 533100 μ_F = 0.695 μ_{CR} = 0.233
 Diversity = 2.5751e-03 k_t = [1, 1, 1, 1]
 Generation 5340: Best = 1.176268e+03 FES = 534100 μ_F = 0.689 μ_{CR} = 0.233
 Diversity = 2.4582e-03 k_t = [2, 2, 2, 2]
 Generation 5350: Best = 1.176262e+03 FES = 535100 μ_F = 0.679 μ_{CR} = 0.232
 Diversity = 2.2103e-03 k_t = [1, 1, 1, 1]

Generation 5360: Best = 1.176256e+03 FES = 536100 μ_F = 0.681 μ_{CR} = 0.224
 Diversity = 2.0007e-03 k_t = [2, 4, 3, 3]
 Generation 5370: Best = 1.176248e+03 FES = 537100 μ_F = 0.690 μ_{CR} = 0.229
 Diversity = 1.8930e-03 k_t = [1, 1, 1, 1]
 Generation 5380: Best = 1.176241e+03 FES = 538100 μ_F = 0.713 μ_{CR} = 0.226
 Diversity = 1.8131e-03 k_t = [4, 3, 4, 4]
 Generation 5390: Best = 1.176235e+03 FES = 539100 μ_F = 0.731 μ_{CR} = 0.237
 Diversity = 1.8558e-03 k_t = [1, 1, 1, 1]
 Generation 5400: Best = 1.176229e+03 FES = 540100 μ_F = 0.730 μ_{CR} = 0.237
 Diversity = 1.7746e-03 k_t = [2, 2, 3, 4]
 Generation 5410: Best = 1.176224e+03 FES = 541100 μ_F = 0.722 μ_{CR} = 0.237
 Diversity = 1.7062e-03 k_t = [1, 1, 1, 1]
 Generation 5420: Best = 1.176218e+03 FES = 542100 μ_F = 0.719 μ_{CR} = 0.245
 Diversity = 1.6273e-03 k_t = [3, 4, 3, 2]
 Generation 5430: Best = 1.176214e+03 FES = 543100 μ_F = 0.717 μ_{CR} = 0.236
 Diversity = 1.6384e-03 k_t = [1, 1, 1, 1]
 Generation 5440: Best = 1.176212e+03 FES = 544100 μ_F = 0.706 μ_{CR} = 0.221
 Diversity = 1.5318e-03 k_t = [2, 3, 3, 3]
 Generation 5450: Best = 1.176209e+03 FES = 545100 μ_F = 0.701 μ_{CR} = 0.222
 Diversity = 1.4135e-03 k_t = [1, 1, 1, 1]
 Generation 5460: Best = 1.176206e+03 FES = 546100 μ_F = 0.708 μ_{CR} = 0.219
 Diversity = 1.3033e-03 k_t = [2, 3, 4, 2]
 Generation 5470: Best = 1.176203e+03 FES = 547100 μ_F = 0.705 μ_{CR} = 0.218
 Diversity = 1.1994e-03 k_t = [1, 1, 1, 1]
 Generation 5480: Best = 1.176201e+03 FES = 548100 μ_F = 0.703 μ_{CR} = 0.221
 Diversity = 1.0248e-03 k_t = [2, 4, 2, 4]
 Generation 5490: Best = 1.176200e+03 FES = 549100 μ_F = 0.705 μ_{CR} = 0.211
 Diversity = 1.0056e-03 k_t = [1, 1, 1, 1]
 Generation 5500: Best = 1.176198e+03 FES = 550100 μ_F = 0.715 μ_{CR} = 0.209
 Diversity = 9.8074e-04 k_t = [2, 3, 2, 4]
 Generation 5510: Best = 1.176196e+03 FES = 551100 μ_F = 0.710 μ_{CR} = 0.217
 Diversity = 9.8491e-04 k_t = [1, 1, 1, 1]
 Generation 5520: Best = 1.176195e+03 FES = 552100 μ_F = 0.705 μ_{CR} = 0.213
 Diversity = 9.2883e-04 k_t = [4, 3, 3, 4]
 Generation 5530: Best = 1.176194e+03 FES = 553100 μ_F = 0.704 μ_{CR} = 0.208
 Diversity = 8.1473e-04 k_t = [1, 1, 1, 1]
 Generation 5540: Best = 1.176192e+03 FES = 554100 μ_F = 0.720 μ_{CR} = 0.208
 Diversity = 7.8685e-04 k_t = [4, 3, 2, 3]
 Generation 5550: Best = 1.176191e+03 FES = 555100 μ_F = 0.730 μ_{CR} = 0.218
 Diversity = 8.2986e-04 k_t = [1, 1, 1, 1]
 Generation 5560: Best = 1.176187e+03 FES = 556100 μ_F = 0.738 μ_{CR} = 0.219
 Diversity = 9.1880e-04 k_t = [3, 3, 2, 2]
 Generation 5570: Best = 1.176186e+03 FES = 557100 μ_F = 0.732 μ_{CR} = 0.217
 Diversity = 9.8019e-04 k_t = [1, 1, 1, 1]
 Generation 5580: Best = 1.176185e+03 FES = 558100 μ_F = 0.719 μ_{CR} = 0.215
 Diversity = 9.1979e-04 k_t = [3, 4, 4, 2]
 Generation 5590: Best = 1.176182e+03 FES = 559100 μ_F = 0.727 μ_{CR} = 0.225
 Diversity = 9.7660e-04 k_t = [1, 1, 1, 1]
 Generation 5600: Best = 1.176180e+03 FES = 560100 μ_F = 0.722 μ_{CR} = 0.230
 Diversity = 9.8585e-04 k_t = [2, 2, 4, 2]
 Generation 5610: Best = 1.176178e+03 FES = 561100 μ_F = 0.723 μ_{CR} = 0.228
 Diversity = 9.7440e-04 k_t = [1, 1, 1, 1]
 Generation 5620: Best = 1.176177e+03 FES = 562100 μ_F = 0.724 μ_{CR} = 0.231
 Diversity = 9.2261e-04 k_t = [4, 4, 3, 2]

Generation 5630: Best = 1.176175e+03 FES = 563100 μ_F = 0.723 μ_{CR} = 0.231
 Diversity = 9.3189e-04 k_t = [1, 1, 1, 1]
 Generation 5640: Best = 1.176174e+03 FES = 564100 μ_F = 0.714 μ_{CR} = 0.236
 Diversity = 8.7391e-04 k_t = [4, 4, 3, 3]
 Generation 5650: Best = 1.176173e+03 FES = 565100 μ_F = 0.710 μ_{CR} = 0.230
 Diversity = 8.2914e-04 k_t = [1, 1, 1, 1]
 Generation 5660: Best = 1.176172e+03 FES = 566100 μ_F = 0.714 μ_{CR} = 0.231
 Diversity = 7.2541e-04 k_t = [2, 2, 4, 3]
 Generation 5670: Best = 1.176170e+03 FES = 567100 μ_F = 0.719 μ_{CR} = 0.223
 Diversity = 7.6749e-04 k_t = [1, 1, 1, 1]
 Generation 5680: Best = 1.176167e+03 FES = 568100 μ_F = 0.719 μ_{CR} = 0.224
 Diversity = 8.0915e-04 k_t = [2, 2, 3, 4]
 Generation 5690: Best = 1.176165e+03 FES = 569100 μ_F = 0.722 μ_{CR} = 0.223
 Diversity = 8.6034e-04 k_t = [1, 1, 1, 1]
 Generation 5700: Best = 1.176164e+03 FES = 570100 μ_F = 0.734 μ_{CR} = 0.220
 Diversity = 8.5825e-04 k_t = [2, 4, 2, 4]
 Generation 5710: Best = 1.176163e+03 FES = 571100 μ_F = 0.728 μ_{CR} = 0.220
 Diversity = 8.3830e-04 k_t = [1, 1, 1, 1]
 Generation 5720: Best = 1.176162e+03 FES = 572100 μ_F = 0.712 μ_{CR} = 0.218
 Diversity = 7.3909e-04 k_t = [4, 4, 2, 3]
 Generation 5730: Best = 1.176161e+03 FES = 573100 μ_F = 0.717 μ_{CR} = 0.212
 Diversity = 6.8793e-04 k_t = [1, 1, 1, 1]
 Generation 5740: Best = 1.176161e+03 FES = 574100 μ_F = 0.702 μ_{CR} = 0.213
 Diversity = 6.1197e-04 k_t = [4, 4, 3, 2]
 Generation 5750: Best = 1.176160e+03 FES = 575100 μ_F = 0.701 μ_{CR} = 0.212
 Diversity = 5.5675e-04 k_t = [1, 1, 1, 1]
 Generation 5760: Best = 1.176160e+03 FES = 576100 μ_F = 0.694 μ_{CR} = 0.213
 Diversity = 4.6845e-04 k_t = [2, 4, 4, 3]
 Generation 5770: Best = 1.176159e+03 FES = 577100 μ_F = 0.696 μ_{CR} = 0.212
 Diversity = 4.1445e-04 k_t = [1, 1, 1, 1]
 Generation 5780: Best = 1.176159e+03 FES = 578100 μ_F = 0.708 μ_{CR} = 0.220
 Diversity = 3.8998e-04 k_t = [3, 3, 4, 3]
 Generation 5790: Best = 1.176159e+03 FES = 579100 μ_F = 0.720 μ_{CR} = 0.221
 Diversity = 4.0972e-04 k_t = [1, 1, 1, 1]
 Generation 5800: Best = 1.176158e+03 FES = 580100 μ_F = 0.714 μ_{CR} = 0.218
 Diversity = 3.9634e-04 k_t = [2, 3, 2, 4]
 Generation 5810: Best = 1.176158e+03 FES = 581100 μ_F = 0.720 μ_{CR} = 0.219
 Diversity = 4.2196e-04 k_t = [1, 1, 1, 1]
 Generation 5820: Best = 1.176158e+03 FES = 582100 μ_F = 0.728 μ_{CR} = 0.213
 Diversity = 4.1226e-04 k_t = [2, 4, 3, 3]
 Generation 5830: Best = 1.176157e+03 FES = 583100 μ_F = 0.733 μ_{CR} = 0.215
 Diversity = 3.9690e-04 k_t = [1, 1, 1, 1]
 Generation 5840: Best = 1.176157e+03 FES = 584100 μ_F = 0.726 μ_{CR} = 0.217
 Diversity = 3.9493e-04 k_t = [4, 2, 4, 2]
 Generation 5850: Best = 1.176157e+03 FES = 585100 μ_F = 0.719 μ_{CR} = 0.214
 Diversity = 3.6743e-04 k_t = [1, 1, 1, 1]
 Generation 5860: Best = 1.176156e+03 FES = 586100 μ_F = 0.725 μ_{CR} = 0.214
 Diversity = 3.6168e-04 k_t = [4, 2, 2, 3]
 Generation 5870: Best = 1.176155e+03 FES = 587100 μ_F = 0.737 μ_{CR} = 0.223
 Diversity = 4.1407e-04 k_t = [1, 1, 1, 1]
 Generation 5880: Best = 1.176155e+03 FES = 588100 μ_F = 0.737 μ_{CR} = 0.223
 Diversity = 4.4102e-04 k_t = [4, 3, 4, 3]
 Generation 5890: Best = 1.176155e+03 FES = 589100 μ_F = 0.748 μ_{CR} = 0.224
 Diversity = 4.2970e-04 k_t = [1, 1, 1, 1]

Generation 5900: Best = 1.176154e+03 FES = 590100 μ_F = 0.737 μ_{CR} = 0.227
 Diversity = 3.8905e-04 k_t = [4, 2, 2, 4]
 Generation 5910: Best = 1.176154e+03 FES = 591100 μ_F = 0.724 μ_{CR} = 0.225
 Diversity = 3.4749e-04 k_t = [1, 1, 1, 1]
 Generation 5920: Best = 1.176154e+03 FES = 592100 μ_F = 0.727 μ_{CR} = 0.216
 Diversity = 3.3229e-04 k_t = [3, 3, 3, 4]
 Generation 5930: Best = 1.176154e+03 FES = 593100 μ_F = 0.702 μ_{CR} = 0.221
 Diversity = 3.1252e-04 k_t = [1, 1, 1, 1]
 Generation 5940: Best = 1.176153e+03 FES = 594100 μ_F = 0.702 μ_{CR} = 0.229
 Diversity = 3.1169e-04 k_t = [3, 4, 4, 3]
 Generation 5950: Best = 1.176153e+03 FES = 595100 μ_F = 0.698 μ_{CR} = 0.228
 Diversity = 3.0252e-04 k_t = [1, 1, 1, 1]
 Generation 5960: Best = 1.176153e+03 FES = 596100 μ_F = 0.704 μ_{CR} = 0.227
 Diversity = 3.1080e-04 k_t = [2, 4, 2, 2]
 Generation 5970: Best = 1.176153e+03 FES = 597100 μ_F = 0.714 μ_{CR} = 0.226
 Diversity = 3.1794e-04 k_t = [1, 1, 1, 1]
 Generation 5980: Best = 1.176153e+03 FES = 598100 μ_F = 0.694 μ_{CR} = 0.224
 Diversity = 2.6812e-04 k_t = [2, 2, 3, 4]
 Generation 5990: Best = 1.176153e+03 FES = 599100 μ_F = 0.710 μ_{CR} = 0.217
 Diversity = 2.4881e-04 k_t = [1, 1, 1, 1]
 Generation 6000: Best = 1.176152e+03 FES = 600100 μ_F = 0.714 μ_{CR} = 0.221
 Diversity = 2.4006e-04 k_t = [4, 3, 3, 2]
 Generation 6010: Best = 1.176152e+03 FES = 601100 μ_F = 0.716 μ_{CR} = 0.225
 Diversity = 2.5233e-04 k_t = [1, 1, 1, 1]
 Generation 6020: Best = 1.176152e+03 FES = 602100 μ_F = 0.721 μ_{CR} = 0.234
 Diversity = 2.5775e-04 k_t = [3, 2, 4, 2]
 Generation 6030: Best = 1.176152e+03 FES = 603100 μ_F = 0.742 μ_{CR} = 0.221
 Diversity = 2.7344e-04 k_t = [1, 1, 1, 1]
 Generation 6040: Best = 1.176152e+03 FES = 604100 μ_F = 0.736 μ_{CR} = 0.222
 Diversity = 2.7206e-04 k_t = [4, 3, 2, 3]
 Generation 6050: Best = 1.176151e+03 FES = 605100 μ_F = 0.741 μ_{CR} = 0.228
 Diversity = 2.7962e-04 k_t = [1, 1, 1, 1]
 Generation 6060: Best = 1.176151e+03 FES = 606100 μ_F = 0.746 μ_{CR} = 0.231
 Diversity = 2.7387e-04 k_t = [4, 3, 4, 4]
 Generation 6070: Best = 1.176151e+03 FES = 607100 μ_F = 0.741 μ_{CR} = 0.224
 Diversity = 2.7451e-04 k_t = [1, 1, 1, 1]
 Generation 6080: Best = 1.176151e+03 FES = 608100 μ_F = 0.741 μ_{CR} = 0.227
 Diversity = 2.6069e-04 k_t = [3, 4, 2, 4]
 Generation 6090: Best = 1.176151e+03 FES = 609100 μ_F = 0.749 μ_{CR} = 0.228
 Diversity = 2.5316e-04 k_t = [1, 1, 1, 1]
 Generation 6100: Best = 1.176151e+03 FES = 610100 μ_F = 0.744 μ_{CR} = 0.233
 Diversity = 2.4987e-04 k_t = [3, 2, 4, 2]
 Generation 6110: Best = 1.176151e+03 FES = 611100 μ_F = 0.738 μ_{CR} = 0.239
 Diversity = 2.5249e-04 k_t = [1, 1, 1, 1]
 Generation 6120: Best = 1.176150e+03 FES = 612100 μ_F = 0.737 μ_{CR} = 0.246
 Diversity = 2.4503e-04 k_t = [4, 3, 2, 3]
 Generation 6130: Best = 1.176150e+03 FES = 613100 μ_F = 0.753 μ_{CR} = 0.247
 Diversity = 2.6080e-04 k_t = [1, 1, 1, 1]
 Generation 6140: Best = 1.176150e+03 FES = 614100 μ_F = 0.753 μ_{CR} = 0.258
 Diversity = 3.0642e-04 k_t = [3, 3, 3, 2]
 Generation 6150: Best = 1.176149e+03 FES = 615100 μ_F = 0.747 μ_{CR} = 0.259
 Diversity = 3.4112e-04 k_t = [1, 1, 1, 1]
 Generation 6160: Best = 1.176149e+03 FES = 616100 μ_F = 0.749 μ_{CR} = 0.266
 Diversity = 3.6675e-04 k_t = [2, 3, 4, 2]

Generation 6170: Best = 1.176148e+03 FES = 617100 μ_F = 0.745 μ_{CR} = 0.266
 Diversity = 4.3715e-04 k_t = [1, 1, 1, 1]
 Generation 6180: Best = 1.176148e+03 FES = 618100 μ_F = 0.735 μ_{CR} = 0.264
 Diversity = 4.0249e-04 k_t = [3, 4, 2, 3]
 Generation 6190: Best = 1.176147e+03 FES = 619100 μ_F = 0.735 μ_{CR} = 0.256
 Diversity = 4.0028e-04 k_t = [1, 1, 1, 1]
 Generation 6200: Best = 1.176147e+03 FES = 620100 μ_F = 0.737 μ_{CR} = 0.251
 Diversity = 3.8334e-04 k_t = [3, 3, 2, 2]
 Generation 6210: Best = 1.176147e+03 FES = 621100 μ_F = 0.745 μ_{CR} = 0.257
 Diversity = 3.6879e-04 k_t = [1, 1, 1, 1]
 Generation 6220: Best = 1.176147e+03 FES = 622100 μ_F = 0.724 μ_{CR} = 0.256
 Diversity = 3.4693e-04 k_t = [4, 2, 2, 2]
 Generation 6230: Best = 1.176146e+03 FES = 623100 μ_F = 0.711 μ_{CR} = 0.253
 Diversity = 3.2295e-04 k_t = [1, 1, 1, 1]
 Generation 6240: Best = 1.176146e+03 FES = 624100 μ_F = 0.697 μ_{CR} = 0.254
 Diversity = 2.6692e-04 k_t = [4, 4, 4, 2]
 Generation 6250: Best = 1.176146e+03 FES = 625100 μ_F = 0.689 μ_{CR} = 0.255
 Diversity = 2.3591e-04 k_t = [1, 1, 1, 1]
 Generation 6260: Best = 1.176146e+03 FES = 626100 μ_F = 0.695 μ_{CR} = 0.251
 Diversity = 2.0692e-04 k_t = [3, 3, 2, 4]
 Generation 6270: Best = 1.176146e+03 FES = 627100 μ_F = 0.699 μ_{CR} = 0.249
 Diversity = 1.8887e-04 k_t = [1, 1, 1, 1]
 Generation 6280: Best = 1.176146e+03 FES = 628100 μ_F = 0.713 μ_{CR} = 0.247
 Diversity = 1.7809e-04 k_t = [2, 4, 4, 3]
 Generation 6290: Best = 1.176146e+03 FES = 629100 μ_F = 0.721 μ_{CR} = 0.247
 Diversity = 1.8695e-04 k_t = [1, 1, 1, 1]
 Generation 6300: Best = 1.176146e+03 FES = 630100 μ_F = 0.714 μ_{CR} = 0.246
 Diversity = 1.6926e-04 k_t = [2, 2, 4, 4]
 Generation 6310: Best = 1.176146e+03 FES = 631100 μ_F = 0.707 μ_{CR} = 0.256
 Diversity = 1.6072e-04 k_t = [1, 1, 1, 1]
 Generation 6320: Best = 1.176146e+03 FES = 632100 μ_F = 0.722 μ_{CR} = 0.260
 Diversity = 1.5250e-04 k_t = [2, 3, 3, 4]
 Generation 6330: Best = 1.176146e+03 FES = 633100 μ_F = 0.725 μ_{CR} = 0.256
 Diversity = 1.5877e-04 k_t = [1, 1, 1, 1]
 Generation 6340: Best = 1.176146e+03 FES = 634100 μ_F = 0.736 μ_{CR} = 0.251
 Diversity = 1.6090e-04 k_t = [2, 3, 2, 4]
 Generation 6350: Best = 1.176146e+03 FES = 635100 μ_F = 0.740 μ_{CR} = 0.248
 Diversity = 1.6142e-04 k_t = [1, 1, 1, 1]
 Generation 6360: Best = 1.176145e+03 FES = 636100 μ_F = 0.751 μ_{CR} = 0.245
 Diversity = 1.7567e-04 k_t = [2, 3, 4, 2]
 Generation 6370: Best = 1.176145e+03 FES = 637100 μ_F = 0.737 μ_{CR} = 0.242
 Diversity = 1.8148e-04 k_t = [1, 1, 1, 1]
 Generation 6380: Best = 1.176145e+03 FES = 638100 μ_F = 0.716 μ_{CR} = 0.247
 Diversity = 1.6767e-04 k_t = [2, 3, 2, 2]
 Generation 6390: Best = 1.176145e+03 FES = 639100 μ_F = 0.708 μ_{CR} = 0.237
 Diversity = 1.5234e-04 k_t = [1, 1, 1, 1]
 Generation 6400: Best = 1.176145e+03 FES = 640100 μ_F = 0.706 μ_{CR} = 0.233
 Diversity = 1.3596e-04 k_t = [3, 3, 2, 4]
 Generation 6410: Best = 1.176145e+03 FES = 641100 μ_F = 0.697 μ_{CR} = 0.225
 Diversity = 1.1885e-04 k_t = [1, 1, 1, 1]
 Generation 6420: Best = 1.176145e+03 FES = 642100 μ_F = 0.698 μ_{CR} = 0.238
 Diversity = 1.0753e-04 k_t = [3, 3, 2, 4]
 Generation 6430: Best = 1.176145e+03 FES = 643100 μ_F = 0.694 μ_{CR} = 0.242
 Diversity = 1.0084e-04 k_t = [1, 1, 1, 1]

Generation 6440: Best = 1.176145e+03 FES = 644100 μ_F = 0.695 μ_{CR} = 0.235
 Diversity = 8.7336e-05 k_t = [4, 3, 2, 4]
 Generation 6450: Best = 1.176145e+03 FES = 645100 μ_F = 0.705 μ_{CR} = 0.247
 Diversity = 9.0951e-05 k_t = [1, 1, 1, 1]
 Generation 6460: Best = 1.176145e+03 FES = 646100 μ_F = 0.714 μ_{CR} = 0.238
 Diversity = 7.7236e-05 k_t = [2, 4, 4, 4]
 Generation 6470: Best = 1.176145e+03 FES = 647100 μ_F = 0.720 μ_{CR} = 0.246
 Diversity = 7.9672e-05 k_t = [1, 1, 1, 1]
 Generation 6480: Best = 1.176145e+03 FES = 648100 μ_F = 0.735 μ_{CR} = 0.246
 Diversity = 8.2685e-05 k_t = [4, 3, 4, 2]
 Generation 6490: Best = 1.176145e+03 FES = 649100 μ_F = 0.746 μ_{CR} = 0.252
 Diversity = 8.6202e-05 k_t = [1, 1, 1, 1]
 Generation 6500: Best = 1.176145e+03 FES = 650100 μ_F = 0.734 μ_{CR} = 0.247
 Diversity = 8.2656e-05 k_t = [3, 4, 2, 4]
 Generation 6510: Best = 1.176145e+03 FES = 651100 μ_F = 0.734 μ_{CR} = 0.248
 Diversity = 9.1391e-05 k_t = [1, 1, 1, 1]
 Generation 6520: Best = 1.176145e+03 FES = 652100 μ_F = 0.706 μ_{CR} = 0.239
 Diversity = 8.8171e-05 k_t = [4, 4, 2, 2]
 Generation 6530: Best = 1.176145e+03 FES = 653100 μ_F = 0.711 μ_{CR} = 0.236
 Diversity = 8.9469e-05 k_t = [1, 1, 1, 1]
 Generation 6540: Best = 1.176145e+03 FES = 654100 μ_F = 0.712 μ_{CR} = 0.237
 Diversity = 8.6503e-05 k_t = [2, 4, 2, 3]
 Generation 6550: Best = 1.176145e+03 FES = 655100 μ_F = 0.714 μ_{CR} = 0.238
 Diversity = 8.4960e-05 k_t = [1, 1, 1, 1]
 Generation 6560: Best = 1.176145e+03 FES = 656100 μ_F = 0.718 μ_{CR} = 0.238
 Diversity = 7.6687e-05 k_t = [3, 3, 3, 2]
 Generation 6570: Best = 1.176145e+03 FES = 657100 μ_F = 0.722 μ_{CR} = 0.232
 Diversity = 7.3981e-05 k_t = [1, 1, 1, 1]
 Generation 6580: Best = 1.176145e+03 FES = 658100 μ_F = 0.712 μ_{CR} = 0.238
 Diversity = 6.8997e-05 k_t = [3, 2, 3, 4]
 Generation 6590: Best = 1.176145e+03 FES = 659100 μ_F = 0.709 μ_{CR} = 0.244
 Diversity = 6.9971e-05 k_t = [1, 1, 1, 1]
 Generation 6600: Best = 1.176145e+03 FES = 660100 μ_F = 0.704 μ_{CR} = 0.240
 Diversity = 6.5249e-05 k_t = [2, 4, 2, 2]
 Generation 6610: Best = 1.176145e+03 FES = 661100 μ_F = 0.705 μ_{CR} = 0.240
 Diversity = 6.2635e-05 k_t = [1, 1, 1, 1]
 Generation 6620: Best = 1.176145e+03 FES = 662100 μ_F = 0.703 μ_{CR} = 0.241
 Diversity = 5.6748e-05 k_t = [4, 4, 4, 4]
 Generation 6630: Best = 1.176145e+03 FES = 663100 μ_F = 0.721 μ_{CR} = 0.244
 Diversity = 5.7936e-05 k_t = [1, 1, 1, 1]
 Generation 6640: Best = 1.176145e+03 FES = 664100 μ_F = 0.729 μ_{CR} = 0.251
 Diversity = 6.0032e-05 k_t = [2, 4, 3, 3]
 Generation 6650: Best = 1.176145e+03 FES = 665100 μ_F = 0.725 μ_{CR} = 0.265
 Diversity = 6.3833e-05 k_t = [1, 1, 1, 1]
 Generation 6660: Best = 1.176145e+03 FES = 666100 μ_F = 0.733 μ_{CR} = 0.270
 Diversity = 6.4122e-05 k_t = [4, 3, 4, 2]
 Generation 6670: Best = 1.176145e+03 FES = 667100 μ_F = 0.732 μ_{CR} = 0.267
 Diversity = 6.6906e-05 k_t = [1, 1, 1, 1]
 Generation 6680: Best = 1.176145e+03 FES = 668100 μ_F = 0.728 μ_{CR} = 0.263
 Diversity = 6.4118e-05 k_t = [4, 2, 3, 3]
 Generation 6690: Best = 1.176145e+03 FES = 669100 μ_F = 0.736 μ_{CR} = 0.263
 Diversity = 6.4904e-05 k_t = [1, 1, 1, 1]
 Generation 6700: Best = 1.176145e+03 FES = 670100 μ_F = 0.729 μ_{CR} = 0.273
 Diversity = 6.0116e-05 k_t = [2, 4, 2, 3]

Generation 6710: Best = 1.176145e+03 FES = 671100 μ_F = 0.729 μ_{CR} = 0.270
 Diversity = 6.1995e-05 k_t = [1, 1, 1, 1]
 Generation 6720: Best = 1.176145e+03 FES = 672100 μ_F = 0.731 μ_{CR} = 0.276
 Diversity = 6.2031e-05 k_t = [3, 4, 3, 2]
 Generation 6730: Best = 1.176145e+03 FES = 673100 μ_F = 0.727 μ_{CR} = 0.274
 Diversity = 6.7619e-05 k_t = [1, 1, 1, 1]
 Generation 6740: Best = 1.176145e+03 FES = 674100 μ_F = 0.718 μ_{CR} = 0.281
 Diversity = 6.1389e-05 k_t = [4, 3, 3, 4]
 Generation 6750: Best = 1.176145e+03 FES = 675100 μ_F = 0.718 μ_{CR} = 0.280
 Diversity = 6.1513e-05 k_t = [1, 1, 1, 1]
 Generation 6760: Best = 1.176145e+03 FES = 676100 μ_F = 0.705 μ_{CR} = 0.278
 Diversity = 5.6067e-05 k_t = [3, 3, 2, 3]
 Generation 6770: Best = 1.176145e+03 FES = 677100 μ_F = 0.698 μ_{CR} = 0.279
 Diversity = 5.0904e-05 k_t = [1, 1, 1, 1]
 Generation 6780: Best = 1.176145e+03 FES = 678100 μ_F = 0.689 μ_{CR} = 0.277
 Diversity = 4.4603e-05 k_t = [4, 2, 3, 2]
 Generation 6790: Best = 1.176145e+03 FES = 679100 μ_F = 0.684 μ_{CR} = 0.264
 Diversity = 4.1117e-05 k_t = [1, 1, 1, 1]
 Generation 6800: Best = 1.176145e+03 FES = 680100 μ_F = 0.683 μ_{CR} = 0.252
 Diversity = 3.4653e-05 k_t = [2, 4, 3, 3]
 Generation 6810: Best = 1.176145e+03 FES = 681100 μ_F = 0.696 μ_{CR} = 0.251
 Diversity = 3.2859e-05 k_t = [1, 1, 1, 1]
 Generation 6820: Best = 1.176145e+03 FES = 682100 μ_F = 0.699 μ_{CR} = 0.251
 Diversity = 2.9761e-05 k_t = [4, 4, 2, 4]
 Generation 6830: Best = 1.176145e+03 FES = 683100 μ_F = 0.723 μ_{CR} = 0.255
 Diversity = 2.9118e-05 k_t = [1, 1, 1, 1]
 Generation 6840: Best = 1.176145e+03 FES = 684100 μ_F = 0.723 μ_{CR} = 0.243
 Diversity = 3.2278e-05 k_t = [3, 4, 2, 2]
 Generation 6850: Best = 1.176145e+03 FES = 685100 μ_F = 0.728 μ_{CR} = 0.246
 Diversity = 4.1147e-05 k_t = [1, 1, 1, 1]
 Generation 6860: Best = 1.176145e+03 FES = 686100 μ_F = 0.717 μ_{CR} = 0.243
 Diversity = 4.0140e-05 k_t = [4, 3, 4, 2]
 Generation 6870: Best = 1.176145e+03 FES = 687100 μ_F = 0.723 μ_{CR} = 0.245
 Diversity = 3.8791e-05 k_t = [1, 1, 1, 1]
 Generation 6880: Best = 1.176145e+03 FES = 688100 μ_F = 0.701 μ_{CR} = 0.239
 Diversity = 3.5074e-05 k_t = [2, 3, 2, 2]
 Generation 6890: Best = 1.176145e+03 FES = 689100 μ_F = 0.685 μ_{CR} = 0.234
 Diversity = 3.2992e-05 k_t = [1, 1, 1, 1]
 Generation 6900: Best = 1.176145e+03 FES = 690100 μ_F = 0.694 μ_{CR} = 0.237
 Diversity = 2.9511e-05 k_t = [2, 2, 3, 3]
 Generation 6910: Best = 1.176145e+03 FES = 691100 μ_F = 0.704 μ_{CR} = 0.237
 Diversity = 2.9250e-05 k_t = [1, 1, 1, 1]
 Generation 6920: Best = 1.176145e+03 FES = 692100 μ_F = 0.712 μ_{CR} = 0.240
 Diversity = 2.8917e-05 k_t = [4, 2, 3, 2]
 Generation 6930: Best = 1.176145e+03 FES = 693100 μ_F = 0.709 μ_{CR} = 0.240
 Diversity = 2.8316e-05 k_t = [1, 1, 1, 1]
 Generation 6940: Best = 1.176145e+03 FES = 694100 μ_F = 0.717 μ_{CR} = 0.242
 Diversity = 2.5117e-05 k_t = [3, 4, 4, 4]
 Generation 6950: Best = 1.176145e+03 FES = 695100 μ_F = 0.724 μ_{CR} = 0.240
 Diversity = 2.5631e-05 k_t = [1, 1, 1, 1]
 Generation 6960: Best = 1.176145e+03 FES = 696100 μ_F = 0.732 μ_{CR} = 0.242
 Diversity = 2.8292e-05 k_t = [2, 2, 4, 2]
 Generation 6970: Best = 1.176145e+03 FES = 697100 μ_F = 0.724 μ_{CR} = 0.226
 Diversity = 2.9021e-05 k_t = [1, 1, 1, 1]

Generation 6980: Best = 1.176145e+03 FES = 698100 μ_F = 0.711 μ_{CR} = 0.229
 Diversity = 2.8492e-05 k_t = [2, 4, 2, 2]
 Generation 6990: Best = 1.176145e+03 FES = 699100 μ_F = 0.710 μ_{CR} = 0.233
 Diversity = 2.9215e-05 k_t = [1, 1, 1, 1]
 Generation 7000: Best = 1.176145e+03 FES = 700100 μ_F = 0.714 μ_{CR} = 0.239
 Diversity = 2.8514e-05 k_t = [3, 3, 2, 2]
 Generation 7010: Best = 1.176145e+03 FES = 701100 μ_F = 0.703 μ_{CR} = 0.241
 Diversity = 2.9115e-05 k_t = [1, 1, 1, 1]
 Generation 7020: Best = 1.176145e+03 FES = 702100 μ_F = 0.716 μ_{CR} = 0.243
 Diversity = 2.7803e-05 k_t = [2, 4, 2, 2]
 Generation 7030: Best = 1.176145e+03 FES = 703100 μ_F = 0.737 μ_{CR} = 0.243
 Diversity = 3.0853e-05 k_t = [1, 1, 1, 1]
 Generation 7040: Best = 1.176145e+03 FES = 704100 μ_F = 0.732 μ_{CR} = 0.240
 Diversity = 2.9452e-05 k_t = [3, 4, 4, 3]
 Generation 7050: Best = 1.176145e+03 FES = 705100 μ_F = 0.739 μ_{CR} = 0.244
 Diversity = 3.3223e-05 k_t = [1, 1, 1, 1]
 Generation 7060: Best = 1.176145e+03 FES = 706100 μ_F = 0.742 μ_{CR} = 0.257
 Diversity = 3.8395e-05 k_t = [3, 4, 2, 3]
 Generation 7070: Best = 1.176145e+03 FES = 707100 μ_F = 0.735 μ_{CR} = 0.251
 Diversity = 3.8915e-05 k_t = [1, 1, 1, 1]
 Generation 7080: Best = 1.176145e+03 FES = 708100 μ_F = 0.729 μ_{CR} = 0.236
 Diversity = 3.7368e-05 k_t = [4, 4, 2, 2]
 Generation 7090: Best = 1.176145e+03 FES = 709100 μ_F = 0.741 μ_{CR} = 0.237
 Diversity = 3.8092e-05 k_t = [1, 1, 1, 1]
 Generation 7100: Best = 1.176145e+03 FES = 710100 μ_F = 0.727 μ_{CR} = 0.225
 Diversity = 3.7012e-05 k_t = [4, 2, 2, 4]
 Generation 7110: Best = 1.176145e+03 FES = 711100 μ_F = 0.735 μ_{CR} = 0.233
 Diversity = 3.7566e-05 k_t = [1, 1, 1, 1]
 Generation 7120: Best = 1.176145e+03 FES = 712100 μ_F = 0.730 μ_{CR} = 0.229
 Diversity = 3.9085e-05 k_t = [3, 4, 4, 3]
 Generation 7130: Best = 1.176145e+03 FES = 713100 μ_F = 0.741 μ_{CR} = 0.234
 Diversity = 4.0952e-05 k_t = [1, 1, 1, 1]
 Generation 7140: Best = 1.176145e+03 FES = 714100 μ_F = 0.749 μ_{CR} = 0.235
 Diversity = 4.2674e-05 k_t = [3, 2, 4, 3]
 Generation 7150: Best = 1.176145e+03 FES = 715100 μ_F = 0.742 μ_{CR} = 0.233
 Diversity = 4.6242e-05 k_t = [1, 1, 1, 1]
 Generation 7160: Best = 1.176145e+03 FES = 716100 μ_F = 0.737 μ_{CR} = 0.239
 Diversity = 4.6656e-05 k_t = [2, 3, 3, 2]
 Generation 7170: Best = 1.176145e+03 FES = 717100 μ_F = 0.741 μ_{CR} = 0.238
 Diversity = 5.2196e-05 k_t = [1, 1, 1, 1]
 Generation 7180: Best = 1.176145e+03 FES = 718100 μ_F = 0.731 μ_{CR} = 0.244
 Diversity = 5.3161e-05 k_t = [2, 3, 4, 4]
 Generation 7190: Best = 1.176145e+03 FES = 719100 μ_F = 0.736 μ_{CR} = 0.251
 Diversity = 5.5173e-05 k_t = [1, 1, 1, 1]
 Generation 7200: Best = 1.176145e+03 FES = 720100 μ_F = 0.744 μ_{CR} = 0.248
 Diversity = 5.5551e-05 k_t = [4, 4, 3, 2]
 Generation 7210: Best = 1.176145e+03 FES = 721100 μ_F = 0.744 μ_{CR} = 0.246
 Diversity = 5.9384e-05 k_t = [1, 1, 1, 1]
 Generation 7220: Best = 1.176145e+03 FES = 722100 μ_F = 0.748 μ_{CR} = 0.251
 Diversity = 5.7808e-05 k_t = [2, 2, 2, 3]
 Generation 7230: Best = 1.176145e+03 FES = 723100 μ_F = 0.743 μ_{CR} = 0.247
 Diversity = 5.6737e-05 k_t = [1, 1, 1, 1]
 Generation 7240: Best = 1.176145e+03 FES = 724100 μ_F = 0.737 μ_{CR} = 0.253
 Diversity = 5.3882e-05 k_t = [4, 4, 4, 3]

Generation 7250: Best = 1.176145e+03 FES = 725100 μ_F = 0.719 μ_{CR} = 0.251
 Diversity = 5.0269e-05 k_t = [1, 1, 1, 1]
 Generation 7260: Best = 1.176145e+03 FES = 726100 μ_F = 0.707 μ_{CR} = 0.248
 Diversity = 4.2623e-05 k_t = [2, 2, 3, 4]
 Generation 7270: Best = 1.176145e+03 FES = 727100 μ_F = 0.702 μ_{CR} = 0.244
 Diversity = 3.7700e-05 k_t = [1, 1, 1, 1]
 Generation 7280: Best = 1.176145e+03 FES = 728100 μ_F = 0.699 μ_{CR} = 0.248
 Diversity = 3.4820e-05 k_t = [4, 3, 2, 2]
 Generation 7290: Best = 1.176145e+03 FES = 729100 μ_F = 0.701 μ_{CR} = 0.241
 Diversity = 3.2799e-05 k_t = [1, 1, 1, 1]
 Generation 7300: Best = 1.176145e+03 FES = 730100 μ_F = 0.698 μ_{CR} = 0.249
 Diversity = 2.8442e-05 k_t = [3, 3, 2, 4]
 Generation 7310: Best = 1.176145e+03 FES = 731100 μ_F = 0.705 μ_{CR} = 0.250
 Diversity = 2.7705e-05 k_t = [1, 1, 1, 1]
 Generation 7320: Best = 1.176145e+03 FES = 732100 μ_F = 0.715 μ_{CR} = 0.238
 Diversity = 2.4998e-05 k_t = [4, 2, 3, 4]
 Generation 7330: Best = 1.176145e+03 FES = 733100 μ_F = 0.731 μ_{CR} = 0.261
 Diversity = 2.7469e-05 k_t = [1, 1, 1, 1]
 Generation 7340: Best = 1.176145e+03 FES = 734100 μ_F = 0.718 μ_{CR} = 0.268
 Diversity = 2.5450e-05 k_t = [3, 4, 4, 3]
 Generation 7350: Best = 1.176145e+03 FES = 735100 μ_F = 0.721 μ_{CR} = 0.272
 Diversity = 2.6608e-05 k_t = [1, 1, 1, 1]
 Generation 7360: Best = 1.176145e+03 FES = 736100 μ_F = 0.716 μ_{CR} = 0.279
 Diversity = 2.4214e-05 k_t = [4, 3, 3, 4]
 Generation 7370: Best = 1.176145e+03 FES = 737100 μ_F = 0.710 μ_{CR} = 0.275
 Diversity = 2.3076e-05 k_t = [1, 1, 1, 1]
 Generation 7380: Best = 1.176145e+03 FES = 738100 μ_F = 0.702 μ_{CR} = 0.271
 Diversity = 2.3311e-05 k_t = [2, 3, 2, 2]
 Generation 7390: Best = 1.176145e+03 FES = 739100 μ_F = 0.696 μ_{CR} = 0.267
 Diversity = 2.3046e-05 k_t = [1, 1, 1, 1]
 Generation 7400: Best = 1.176145e+03 FES = 740100 μ_F = 0.700 μ_{CR} = 0.265
 Diversity = 2.1878e-05 k_t = [2, 3, 3, 4]
 Generation 7410: Best = 1.176145e+03 FES = 741100 μ_F = 0.706 μ_{CR} = 0.264
 Diversity = 2.1531e-05 k_t = [1, 1, 1, 1]
 Generation 7420: Best = 1.176145e+03 FES = 742100 μ_F = 0.708 μ_{CR} = 0.265
 Diversity = 2.2430e-05 k_t = [4, 2, 4, 4]
 Generation 7430: Best = 1.176145e+03 FES = 743100 μ_F = 0.715 μ_{CR} = 0.262
 Diversity = 2.2954e-05 k_t = [1, 1, 1, 1]
 Generation 7440: Best = 1.176145e+03 FES = 744100 μ_F = 0.731 μ_{CR} = 0.266
 Diversity = 2.6858e-05 k_t = [4, 2, 3, 2]
 Generation 7450: Best = 1.176145e+03 FES = 745100 μ_F = 0.727 μ_{CR} = 0.269
 Diversity = 2.7925e-05 k_t = [1, 1, 1, 1]
 Generation 7460: Best = 1.176145e+03 FES = 746100 μ_F = 0.739 μ_{CR} = 0.266
 Diversity = 2.7272e-05 k_t = [2, 4, 2, 4]
 Generation 7470: Best = 1.176145e+03 FES = 747100 μ_F = 0.726 μ_{CR} = 0.261
 Diversity = 3.0001e-05 k_t = [1, 1, 1, 1]
 Generation 7480: Best = 1.176145e+03 FES = 748100 μ_F = 0.729 μ_{CR} = 0.265
 Diversity = 2.8840e-05 k_t = [3, 3, 2, 3]
 Generation 7490: Best = 1.176145e+03 FES = 749100 μ_F = 0.726 μ_{CR} = 0.260
 Diversity = 3.0404e-05 k_t = [1, 1, 1, 1]
 Generation 7500: Best = 1.176145e+03 FES = 750100 μ_F = 0.729 μ_{CR} = 0.269
 Diversity = 3.1926e-05 k_t = [3, 3, 4, 2]
 Generation 7510: Best = 1.176145e+03 FES = 751100 μ_F = 0.726 μ_{CR} = 0.262
 Diversity = 3.2558e-05 k_t = [1, 1, 1, 1]

Generation 7520: Best = 1.176145e+03 FES = 752100 μ_F = 0.728 μ_{CR} = 0.262
 Diversity = 3.0631e-05 k_t = [4, 4, 4, 3]
 Generation 7530: Best = 1.176145e+03 FES = 753100 μ_F = 0.730 μ_{CR} = 0.267
 Diversity = 3.3419e-05 k_t = [1, 1, 1, 1]
 Generation 7540: Best = 1.176145e+03 FES = 754100 μ_F = 0.736 μ_{CR} = 0.270
 Diversity = 3.3999e-05 k_t = [4, 2, 4, 2]
 Generation 7550: Best = 1.176145e+03 FES = 755100 μ_F = 0.729 μ_{CR} = 0.267
 Diversity = 3.4810e-05 k_t = [1, 1, 1, 1]
 Generation 7560: Best = 1.176145e+03 FES = 756100 μ_F = 0.732 μ_{CR} = 0.267
 Diversity = 3.1129e-05 k_t = [2, 4, 4, 4]
 Generation 7570: Best = 1.176145e+03 FES = 757100 μ_F = 0.730 μ_{CR} = 0.258
 Diversity = 2.9323e-05 k_t = [1, 1, 1, 1]
 Generation 7580: Best = 1.176145e+03 FES = 758100 μ_F = 0.730 μ_{CR} = 0.260
 Diversity = 2.9048e-05 k_t = [4, 4, 2, 2]
 Generation 7590: Best = 1.176145e+03 FES = 759100 μ_F = 0.721 μ_{CR} = 0.264
 Diversity = 2.8338e-05 k_t = [1, 1, 1, 1]
 Generation 7600: Best = 1.176145e+03 FES = 760100 μ_F = 0.730 μ_{CR} = 0.267
 Diversity = 2.7700e-05 k_t = [2, 2, 2, 2]
 Generation 7610: Best = 1.176145e+03 FES = 761100 μ_F = 0.720 μ_{CR} = 0.268
 Diversity = 3.0243e-05 k_t = [1, 1, 1, 1]
 Generation 7620: Best = 1.176145e+03 FES = 762100 μ_F = 0.724 μ_{CR} = 0.276
 Diversity = 3.0321e-05 k_t = [4, 2, 3, 2]
 Generation 7630: Best = 1.176145e+03 FES = 763100 μ_F = 0.730 μ_{CR} = 0.279
 Diversity = 3.2310e-05 k_t = [1, 1, 1, 1]
 Generation 7640: Best = 1.176145e+03 FES = 764100 μ_F = 0.721 μ_{CR} = 0.273
 Diversity = 2.7245e-05 k_t = [2, 4, 4, 4]
 Generation 7650: Best = 1.176145e+03 FES = 765100 μ_F = 0.740 μ_{CR} = 0.272
 Diversity = 2.6752e-05 k_t = [1, 1, 1, 1]
 Generation 7660: Best = 1.176145e+03 FES = 766100 μ_F = 0.730 μ_{CR} = 0.270
 Diversity = 2.9051e-05 k_t = [2, 2, 3, 2]
 Generation 7670: Best = 1.176145e+03 FES = 767100 μ_F = 0.729 μ_{CR} = 0.286
 Diversity = 3.2159e-05 k_t = [1, 1, 1, 1]
 Generation 7680: Best = 1.176145e+03 FES = 768100 μ_F = 0.725 μ_{CR} = 0.285
 Diversity = 3.5032e-05 k_t = [3, 3, 2, 2]
 Generation 7690: Best = 1.176145e+03 FES = 769100 μ_F = 0.720 μ_{CR} = 0.278
 Diversity = 3.7572e-05 k_t = [1, 1, 1, 1]
 Generation 7700: Best = 1.176145e+03 FES = 770100 μ_F = 0.726 μ_{CR} = 0.281
 Diversity = 3.4113e-05 k_t = [4, 3, 3, 4]
 Generation 7710: Best = 1.176145e+03 FES = 771100 μ_F = 0.728 μ_{CR} = 0.274
 Diversity = 3.3646e-05 k_t = [1, 1, 1, 1]
 Generation 7720: Best = 1.176145e+03 FES = 772100 μ_F = 0.734 μ_{CR} = 0.264
 Diversity = 3.6270e-05 k_t = [4, 4, 2, 2]
 Generation 7730: Best = 1.176144e+03 FES = 773100 μ_F = 0.731 μ_{CR} = 0.259
 Diversity = 3.7544e-05 k_t = [1, 1, 1, 1]
 Generation 7740: Best = 1.176144e+03 FES = 774100 μ_F = 0.725 μ_{CR} = 0.256
 Diversity = 3.2324e-05 k_t = [3, 3, 3, 4]
 Generation 7750: Best = 1.176144e+03 FES = 775100 μ_F = 0.722 μ_{CR} = 0.251
 Diversity = 3.3044e-05 k_t = [1, 1, 1, 1]
 Generation 7760: Best = 1.176144e+03 FES = 776100 μ_F = 0.713 μ_{CR} = 0.251
 Diversity = 2.9422e-05 k_t = [2, 3, 3, 3]
 Generation 7770: Best = 1.176144e+03 FES = 777100 μ_F = 0.720 μ_{CR} = 0.248
 Diversity = 2.9194e-05 k_t = [1, 1, 1, 1]
 Generation 7780: Best = 1.176144e+03 FES = 778100 μ_F = 0.714 μ_{CR} = 0.258
 Diversity = 2.6555e-05 k_t = [4, 4, 3, 3]

Generation 7790: Best = 1.176144e+03 FES = 779100 μ_F = 0.713 μ_{CR} = 0.250
 Diversity = 2.5488e-05 k_t = [1, 1, 1, 1]
 Generation 7800: Best = 1.176144e+03 FES = 780100 μ_F = 0.701 μ_{CR} = 0.254
 Diversity = 2.1861e-05 k_t = [4, 2, 2, 3]
 Generation 7810: Best = 1.176144e+03 FES = 781100 μ_F = 0.705 μ_{CR} = 0.243
 Diversity = 2.0450e-05 k_t = [1, 1, 1, 1]
 Generation 7820: Best = 1.176144e+03 FES = 782100 μ_F = 0.700 μ_{CR} = 0.239
 Diversity = 2.0035e-05 k_t = [3, 4, 4, 2]
 Generation 7830: Best = 1.176144e+03 FES = 783100 μ_F = 0.698 μ_{CR} = 0.240
 Diversity = 1.7195e-05 k_t = [1, 1, 1, 1]
 Generation 7840: Best = 1.176144e+03 FES = 784100 μ_F = 0.698 μ_{CR} = 0.236
 Diversity = 1.6406e-05 k_t = [4, 2, 3, 2]
 Generation 7850: Best = 1.176144e+03 FES = 785100 μ_F = 0.707 μ_{CR} = 0.244
 Diversity = 1.8831e-05 k_t = [1, 1, 1, 1]
 Generation 7860: Best = 1.176144e+03 FES = 786100 μ_F = 0.723 μ_{CR} = 0.241
 Diversity = 2.0876e-05 k_t = [4, 4, 3, 3]
 Generation 7870: Best = 1.176144e+03 FES = 787100 μ_F = 0.727 μ_{CR} = 0.244
 Diversity = 2.1929e-05 k_t = [1, 1, 1, 1]
 Generation 7880: Best = 1.176144e+03 FES = 788100 μ_F = 0.715 μ_{CR} = 0.245
 Diversity = 2.0116e-05 k_t = [2, 2, 4, 3]
 Generation 7890: Best = 1.176144e+03 FES = 789100 μ_F = 0.703 μ_{CR} = 0.251
 Diversity = 2.0349e-05 k_t = [1, 1, 1, 1]
 Generation 7900: Best = 1.176144e+03 FES = 790100 μ_F = 0.711 μ_{CR} = 0.247
 Diversity = 1.7230e-05 k_t = [3, 4, 3, 4]
 Generation 7910: Best = 1.176144e+03 FES = 791100 μ_F = 0.719 μ_{CR} = 0.255
 Diversity = 1.5792e-05 k_t = [1, 1, 1, 1]
 Generation 7920: Best = 1.176144e+03 FES = 792100 μ_F = 0.732 μ_{CR} = 0.252
 Diversity = 1.6414e-05 k_t = [4, 4, 4, 3]
 Generation 7930: Best = 1.176144e+03 FES = 793100 μ_F = 0.739 μ_{CR} = 0.253
 Diversity = 1.6272e-05 k_t = [1, 1, 1, 1]
 Generation 7940: Best = 1.176144e+03 FES = 794100 μ_F = 0.742 μ_{CR} = 0.251
 Diversity = 1.4989e-05 k_t = [3, 2, 3, 3]
 Generation 7950: Best = 1.176144e+03 FES = 795100 μ_F = 0.736 μ_{CR} = 0.250
 Diversity = 1.5646e-05 k_t = [1, 1, 1, 1]
 Generation 7960: Best = 1.176144e+03 FES = 796100 μ_F = 0.732 μ_{CR} = 0.250
 Diversity = 1.3915e-05 k_t = [4, 3, 4, 4]
 Generation 7970: Best = 1.176144e+03 FES = 797100 μ_F = 0.732 μ_{CR} = 0.244
 Diversity = 1.3268e-05 k_t = [1, 1, 1, 1]
 Generation 7980: Best = 1.176144e+03 FES = 798100 μ_F = 0.733 μ_{CR} = 0.240
 Diversity = 1.2929e-05 k_t = [2, 4, 2, 3]
 Generation 7990: Best = 1.176144e+03 FES = 799100 μ_F = 0.732 μ_{CR} = 0.242
 Diversity = 1.2264e-05 k_t = [1, 1, 1, 1]
 Generation 8000: Best = 1.176144e+03 FES = 800100 μ_F = 0.745 μ_{CR} = 0.244
 Diversity = 1.1694e-05 k_t = [4, 4, 2, 3]
 Generation 8010: Best = 1.176144e+03 FES = 801100 μ_F = 0.765 μ_{CR} = 0.235
 Diversity = 1.2910e-05 k_t = [1, 1, 1, 1]
 Generation 8020: Best = 1.176144e+03 FES = 802100 μ_F = 0.776 μ_{CR} = 0.243
 Diversity = 1.4050e-05 k_t = [4, 3, 4, 4]
 Generation 8030: Best = 1.176144e+03 FES = 803100 μ_F = 0.783 μ_{CR} = 0.246
 Diversity = 1.6588e-05 k_t = [1, 1, 1, 1]
 Generation 8040: Best = 1.176144e+03 FES = 804100 μ_F = 0.778 μ_{CR} = 0.251
 Diversity = 1.7925e-05 k_t = [4, 3, 2, 2]
 Generation 8050: Best = 1.176144e+03 FES = 805100 μ_F = 0.764 μ_{CR} = 0.250
 Diversity = 1.9354e-05 k_t = [1, 1, 1, 1]

Generation 8060: Best = 1.176144e+03 FES = 806100 μ_F = 0.749 μ_{CR} = 0.242
 Diversity = 1.7937e-05 k_t = [4, 2, 4, 4]
 Generation 8070: Best = 1.176144e+03 FES = 807100 μ_F = 0.749 μ_{CR} = 0.240
 Diversity = 1.7924e-05 k_t = [1, 1, 1, 1]
 Generation 8080: Best = 1.176144e+03 FES = 808100 μ_F = 0.751 μ_{CR} = 0.247
 Diversity = 1.8531e-05 k_t = [4, 3, 2, 3]
 Generation 8090: Best = 1.176144e+03 FES = 809100 μ_F = 0.761 μ_{CR} = 0.256
 Diversity = 2.0176e-05 k_t = [1, 1, 1, 1]
 Generation 8100: Best = 1.176144e+03 FES = 810100 μ_F = 0.768 μ_{CR} = 0.256
 Diversity = 2.3038e-05 k_t = [2, 2, 4, 3]
 Generation 8110: Best = 1.176144e+03 FES = 811100 μ_F = 0.754 μ_{CR} = 0.246
 Diversity = 2.3942e-05 k_t = [1, 1, 1, 1]
 Generation 8120: Best = 1.176144e+03 FES = 812100 μ_F = 0.744 μ_{CR} = 0.244
 Diversity = 2.1300e-05 k_t = [4, 3, 4, 3]
 Generation 8130: Best = 1.176144e+03 FES = 813100 μ_F = 0.734 μ_{CR} = 0.245
 Diversity = 2.2433e-05 k_t = [1, 1, 1, 1]
 Generation 8140: Best = 1.176144e+03 FES = 814100 μ_F = 0.726 μ_{CR} = 0.240
 Diversity = 2.1335e-05 k_t = [2, 4, 4, 4]
 Generation 8150: Best = 1.176144e+03 FES = 815100 μ_F = 0.732 μ_{CR} = 0.238
 Diversity = 2.3936e-05 k_t = [1, 1, 1, 1]
 Generation 8160: Best = 1.176144e+03 FES = 816100 μ_F = 0.714 μ_{CR} = 0.246
 Diversity = 2.3312e-05 k_t = [4, 2, 2, 3]
 Generation 8170: Best = 1.176144e+03 FES = 817100 μ_F = 0.711 μ_{CR} = 0.236
 Diversity = 2.1967e-05 k_t = [1, 1, 1, 1]
 Generation 8180: Best = 1.176144e+03 FES = 818100 μ_F = 0.712 μ_{CR} = 0.234
 Diversity = 1.9690e-05 k_t = [2, 3, 4, 3]
 Generation 8190: Best = 1.176144e+03 FES = 819100 μ_F = 0.714 μ_{CR} = 0.233
 Diversity = 1.8353e-05 k_t = [1, 1, 1, 1]
 Generation 8200: Best = 1.176144e+03 FES = 820100 μ_F = 0.706 μ_{CR} = 0.236
 Diversity = 1.5427e-05 k_t = [2, 4, 4, 4]
 Generation 8210: Best = 1.176144e+03 FES = 821100 μ_F = 0.703 μ_{CR} = 0.236
 Diversity = 1.4825e-05 k_t = [1, 1, 1, 1]
 Generation 8220: Best = 1.176144e+03 FES = 822100 μ_F = 0.698 μ_{CR} = 0.232
 Diversity = 1.2619e-05 k_t = [4, 2, 4, 4]
 Generation 8230: Best = 1.176144e+03 FES = 823100 μ_F = 0.703 μ_{CR} = 0.235
 Diversity = 1.1192e-05 k_t = [1, 1, 1, 1]
 Generation 8240: Best = 1.176144e+03 FES = 824100 μ_F = 0.696 μ_{CR} = 0.236
 Diversity = 9.8219e-06 k_t = [3, 4, 2, 2]
 Generation 8250: Best = 1.176144e+03 FES = 825100 μ_F = 0.694 μ_{CR} = 0.236
 Diversity = 9.3197e-06 k_t = [1, 1, 1, 1]
 Generation 8260: Best = 1.176144e+03 FES = 826100 μ_F = 0.698 μ_{CR} = 0.234
 Diversity = 8.7013e-06 k_t = [3, 2, 2, 3]
 Generation 8270: Best = 1.176144e+03 FES = 827100 μ_F = 0.703 μ_{CR} = 0.232
 Diversity = 8.7546e-06 k_t = [1, 1, 1, 1]
 Generation 8280: Best = 1.176144e+03 FES = 828100 μ_F = 0.707 μ_{CR} = 0.232
 Diversity = 7.6399e-06 k_t = [3, 4, 2, 4]
 Generation 8290: Best = 1.176144e+03 FES = 829100 μ_F = 0.724 μ_{CR} = 0.228
 Diversity = 7.3708e-06 k_t = [1, 1, 1, 1]
 Generation 8300: Best = 1.176144e+03 FES = 830100 μ_F = 0.721 μ_{CR} = 0.231
 Diversity = 7.0228e-06 k_t = [2, 2, 3, 2]
 Generation 8310: Best = 1.176144e+03 FES = 831100 μ_F = 0.703 μ_{CR} = 0.227
 Diversity = 6.8709e-06 k_t = [1, 1, 1, 1]
 Generation 8320: Best = 1.176144e+03 FES = 832100 μ_F = 0.691 μ_{CR} = 0.222
 Diversity = 5.6491e-06 k_t = [4, 3, 3, 4]

Generation 8330: Best = 1.176144e+03 FES = 833100 μ_F = 0.701 μ_{CR} = 0.226
 Diversity = 5.4082e-06 k_t = [1, 1, 1, 1]
 Generation 8340: Best = 1.176144e+03 FES = 834100 μ_F = 0.699 μ_{CR} = 0.221
 Diversity = 5.1526e-06 k_t = [4, 4, 2, 3]
 Generation 8350: Best = 1.176144e+03 FES = 835100 μ_F = 0.698 μ_{CR} = 0.214
 Diversity = 5.0610e-06 k_t = [1, 1, 1, 1]
 Generation 8360: Best = 1.176144e+03 FES = 836100 μ_F = 0.698 μ_{CR} = 0.218
 Diversity = 4.7117e-06 k_t = [2, 2, 2, 3]
 Generation 8370: Best = 1.176144e+03 FES = 837100 μ_F = 0.708 μ_{CR} = 0.230
 Diversity = 5.2715e-06 k_t = [1, 1, 1, 1]
 Generation 8380: Best = 1.176144e+03 FES = 838100 μ_F = 0.714 μ_{CR} = 0.222
 Diversity = 5.0672e-06 k_t = [2, 4, 4, 4]
 Generation 8390: Best = 1.176144e+03 FES = 839100 μ_F = 0.725 μ_{CR} = 0.228
 Diversity = 5.2417e-06 k_t = [1, 1, 1, 1]
 Generation 8400: Best = 1.176144e+03 FES = 840100 μ_F = 0.722 μ_{CR} = 0.225
 Diversity = 5.0956e-06 k_t = [3, 3, 4, 2]
 Generation 8410: Best = 1.176144e+03 FES = 841100 μ_F = 0.709 μ_{CR} = 0.231
 Diversity = 4.9413e-06 k_t = [1, 1, 1, 1]
 Generation 8420: Best = 1.176144e+03 FES = 842100 μ_F = 0.713 μ_{CR} = 0.227
 Diversity = 4.4892e-06 k_t = [3, 4, 3, 4]
 Generation 8430: Best = 1.176144e+03 FES = 843100 μ_F = 0.729 μ_{CR} = 0.229
 Diversity = 4.9745e-06 k_t = [1, 1, 1, 1]
 Generation 8440: Best = 1.176144e+03 FES = 844100 μ_F = 0.719 μ_{CR} = 0.231
 Diversity = 4.4932e-06 k_t = [4, 4, 4, 4]
 Generation 8450: Best = 1.176144e+03 FES = 845100 μ_F = 0.734 μ_{CR} = 0.236
 Diversity = 4.4868e-06 k_t = [1, 1, 1, 1]
 Generation 8460: Best = 1.176144e+03 FES = 846100 μ_F = 0.739 μ_{CR} = 0.242
 Diversity = 4.7992e-06 k_t = [2, 3, 3, 4]
 Generation 8470: Best = 1.176144e+03 FES = 847100 μ_F = 0.718 μ_{CR} = 0.234
 Diversity = 4.6874e-06 k_t = [1, 1, 1, 1]
 Generation 8480: Best = 1.176144e+03 FES = 848100 μ_F = 0.710 μ_{CR} = 0.238
 Diversity = 4.3708e-06 k_t = [3, 2, 4, 3]
 Generation 8490: Best = 1.176144e+03 FES = 849100 μ_F = 0.712 μ_{CR} = 0.243
 Diversity = 4.2773e-06 k_t = [1, 1, 1, 1]
 Generation 8500: Best = 1.176144e+03 FES = 850100 μ_F = 0.702 μ_{CR} = 0.241
 Diversity = 4.0501e-06 k_t = [4, 2, 2, 2]
 Generation 8510: Best = 1.176144e+03 FES = 851100 μ_F = 0.693 μ_{CR} = 0.251
 Diversity = 3.8058e-06 k_t = [1, 1, 1, 1]
 Generation 8520: Best = 1.176144e+03 FES = 852100 μ_F = 0.687 μ_{CR} = 0.255
 Diversity = 3.1929e-06 k_t = [4, 3, 3, 3]
 Generation 8530: Best = 1.176144e+03 FES = 853100 μ_F = 0.691 μ_{CR} = 0.247
 Diversity = 2.8835e-06 k_t = [1, 1, 1, 1]
 Generation 8540: Best = 1.176144e+03 FES = 854100 μ_F = 0.700 μ_{CR} = 0.257
 Diversity = 2.5502e-06 k_t = [3, 4, 2, 3]
 Generation 8550: Best = 1.176144e+03 FES = 855100 μ_F = 0.709 μ_{CR} = 0.250
 Diversity = 2.6100e-06 k_t = [1, 1, 1, 1]
 Generation 8560: Best = 1.176144e+03 FES = 856100 μ_F = 0.707 μ_{CR} = 0.253
 Diversity = 2.4414e-06 k_t = [4, 2, 4, 3]
 Generation 8570: Best = 1.176144e+03 FES = 857100 μ_F = 0.720 μ_{CR} = 0.254
 Diversity = 2.4785e-06 k_t = [1, 1, 1, 1]
 Generation 8580: Best = 1.176144e+03 FES = 858100 μ_F = 0.724 μ_{CR} = 0.255
 Diversity = 2.6236e-06 k_t = [2, 4, 3, 2]
 Generation 8590: Best = 1.176144e+03 FES = 859100 μ_F = 0.743 μ_{CR} = 0.258
 Diversity = 2.9813e-06 k_t = [1, 1, 1, 1]

Generation 8600: Best = 1.176144e+03 FES = 860100 μ_F = 0.752 μ_{CR} = 0.260
 Diversity = 3.3408e-06 k_t = [3, 4, 2, 2]
 Generation 8610: Best = 1.176144e+03 FES = 861100 μ_F = 0.735 μ_{CR} = 0.261
 Diversity = 3.3128e-06 k_t = [1, 1, 1, 1]
 Generation 8620: Best = 1.176144e+03 FES = 862100 μ_F = 0.733 μ_{CR} = 0.260
 Diversity = 2.9647e-06 k_t = [4, 3, 3, 4]
 Generation 8630: Best = 1.176144e+03 FES = 863100 μ_F = 0.734 μ_{CR} = 0.263
 Diversity = 2.7760e-06 k_t = [1, 1, 1, 1]
 Generation 8640: Best = 1.176144e+03 FES = 864100 μ_F = 0.724 μ_{CR} = 0.263
 Diversity = 2.6260e-06 k_t = [4, 4, 4, 3]
 Generation 8650: Best = 1.176144e+03 FES = 865100 μ_F = 0.723 μ_{CR} = 0.254
 Diversity = 2.5131e-06 k_t = [1, 1, 1, 1]
 Generation 8660: Best = 1.176144e+03 FES = 866100 μ_F = 0.715 μ_{CR} = 0.259
 Diversity = 2.5079e-06 k_t = [2, 2, 2, 4]
 Generation 8670: Best = 1.176144e+03 FES = 867100 μ_F = 0.712 μ_{CR} = 0.262
 Diversity = 2.5060e-06 k_t = [1, 1, 1, 1]
 Generation 8680: Best = 1.176144e+03 FES = 868100 μ_F = 0.712 μ_{CR} = 0.264
 Diversity = 2.4839e-06 k_t = [2, 2, 4, 2]
 Generation 8690: Best = 1.176144e+03 FES = 869100 μ_F = 0.710 μ_{CR} = 0.261
 Diversity = 2.5684e-06 k_t = [1, 1, 1, 1]
 Generation 8700: Best = 1.176144e+03 FES = 870100 μ_F = 0.721 μ_{CR} = 0.265
 Diversity = 2.6856e-06 k_t = [2, 2, 4, 3]
 Generation 8710: Best = 1.176144e+03 FES = 871100 μ_F = 0.706 μ_{CR} = 0.254
 Diversity = 2.5489e-06 k_t = [1, 1, 1, 1]
 Generation 8720: Best = 1.176144e+03 FES = 872100 μ_F = 0.699 μ_{CR} = 0.265
 Diversity = 2.0647e-06 k_t = [3, 4, 4, 4]
 Generation 8730: Best = 1.176144e+03 FES = 873100 μ_F = 0.709 μ_{CR} = 0.257
 Diversity = 1.9373e-06 k_t = [1, 1, 1, 1]
 Generation 8740: Best = 1.176144e+03 FES = 874100 μ_F = 0.711 μ_{CR} = 0.266
 Diversity = 2.0186e-06 k_t = [4, 2, 2, 2]
 Generation 8750: Best = 1.176144e+03 FES = 875100 μ_F = 0.716 μ_{CR} = 0.268
 Diversity = 1.9774e-06 k_t = [1, 1, 1, 1]
 Generation 8760: Best = 1.176144e+03 FES = 876100 μ_F = 0.715 μ_{CR} = 0.268
 Diversity = 1.7702e-06 k_t = [4, 2, 3, 4]
 Generation 8770: Best = 1.176144e+03 FES = 877100 μ_F = 0.718 μ_{CR} = 0.276
 Diversity = 1.7205e-06 k_t = [1, 1, 1, 1]
 Generation 8780: Best = 1.176144e+03 FES = 878100 μ_F = 0.722 μ_{CR} = 0.280
 Diversity = 1.8019e-06 k_t = [2, 4, 2, 3]
 Generation 8790: Best = 1.176144e+03 FES = 879100 μ_F = 0.724 μ_{CR} = 0.286
 Diversity = 1.9644e-06 k_t = [1, 1, 1, 1]
 Generation 8800: Best = 1.176144e+03 FES = 880100 μ_F = 0.728 μ_{CR} = 0.281
 Diversity = 1.8064e-06 k_t = [2, 4, 3, 4]
 Generation 8810: Best = 1.176144e+03 FES = 881100 μ_F = 0.727 μ_{CR} = 0.277
 Diversity = 1.8867e-06 k_t = [1, 1, 1, 1]
 Generation 8820: Best = 1.176144e+03 FES = 882100 μ_F = 0.728 μ_{CR} = 0.283
 Diversity = 1.8199e-06 k_t = [4, 2, 2, 4]
 Generation 8830: Best = 1.176144e+03 FES = 883100 μ_F = 0.730 μ_{CR} = 0.286
 Diversity = 1.8051e-06 k_t = [1, 1, 1, 1]
 Generation 8840: Best = 1.176144e+03 FES = 884100 μ_F = 0.740 μ_{CR} = 0.283
 Diversity = 1.8222e-06 k_t = [2, 4, 4, 2]
 Generation 8850: Best = 1.176144e+03 FES = 885100 μ_F = 0.742 μ_{CR} = 0.273
 Diversity = 1.9115e-06 k_t = [1, 1, 1, 1]
 Generation 8860: Best = 1.176144e+03 FES = 886100 μ_F = 0.734 μ_{CR} = 0.259
 Diversity = 1.7617e-06 k_t = [3, 3, 4, 3]

Generation 8870: Best = 1.176144e+03 FES = 887100 μ_F = 0.734 μ_{CR} = 0.245
 Diversity = 1.7575e-06 k_t = [1, 1, 1, 1]
 Generation 8880: Best = 1.176144e+03 FES = 888100 μ_F = 0.737 μ_{CR} = 0.253
 Diversity = 1.7181e-06 k_t = [2, 2, 3, 3]
 Generation 8890: Best = 1.176144e+03 FES = 889100 μ_F = 0.736 μ_{CR} = 0.248
 Diversity = 1.8350e-06 k_t = [1, 1, 1, 1]
 Generation 8900: Best = 1.176144e+03 FES = 890100 μ_F = 0.726 μ_{CR} = 0.246
 Diversity = 1.7449e-06 k_t = [2, 3, 3, 3]
 Generation 8910: Best = 1.176144e+03 FES = 891100 μ_F = 0.730 μ_{CR} = 0.237
 Diversity = 1.8676e-06 k_t = [1, 1, 1, 1]
 Generation 8920: Best = 1.176144e+03 FES = 892100 μ_F = 0.725 μ_{CR} = 0.237
 Diversity = 1.8231e-06 k_t = [4, 3, 2, 4]
 Generation 8930: Best = 1.176144e+03 FES = 893100 μ_F = 0.724 μ_{CR} = 0.244
 Diversity = 1.8055e-06 k_t = [1, 1, 1, 1]
 Generation 8940: Best = 1.176144e+03 FES = 894100 μ_F = 0.720 μ_{CR} = 0.253
 Diversity = 1.7111e-06 k_t = [4, 2, 2, 2]
 Generation 8950: Best = 1.176144e+03 FES = 895100 μ_F = 0.720 μ_{CR} = 0.257
 Diversity = 1.6668e-06 k_t = [1, 1, 1, 1]
 Generation 8960: Best = 1.176144e+03 FES = 896100 μ_F = 0.719 μ_{CR} = 0.262
 Diversity = 1.6896e-06 k_t = [2, 2, 3, 3]
 Generation 8970: Best = 1.176144e+03 FES = 897100 μ_F = 0.707 μ_{CR} = 0.261
 Diversity = 1.7086e-06 k_t = [1, 1, 1, 1]
 Generation 8980: Best = 1.176144e+03 FES = 898100 μ_F = 0.705 μ_{CR} = 0.269
 Diversity = 1.6469e-06 k_t = [4, 3, 3, 2]
 Generation 8990: Best = 1.176144e+03 FES = 899100 μ_F = 0.698 μ_{CR} = 0.262
 Diversity = 1.4731e-06 k_t = [1, 1, 1, 1]
 Generation 9000: Best = 1.176144e+03 FES = 900100 μ_F = 0.697 μ_{CR} = 0.262
 Diversity = 1.2765e-06 k_t = [3, 4, 3, 2]
 Generation 9010: Best = 1.176144e+03 FES = 901100 μ_F = 0.706 μ_{CR} = 0.264
 Diversity = 1.2689e-06 k_t = [1, 1, 1, 1]
 Generation 9020: Best = 1.176144e+03 FES = 902100 μ_F = 0.699 μ_{CR} = 0.260
 Diversity = 1.1412e-06 k_t = [4, 2, 3, 4]
 Generation 9030: Best = 1.176144e+03 FES = 903100 μ_F = 0.705 μ_{CR} = 0.259
 Diversity = 1.0303e-06 k_t = [1, 1, 1, 1]
 Generation 9040: Best = 1.176144e+03 FES = 904100 μ_F = 0.712 μ_{CR} = 0.255
 Diversity = 9.4948e-07 k_t = [4, 2, 2, 3]
 Generation 9050: Best = 1.176144e+03 FES = 905100 μ_F = 0.717 μ_{CR} = 0.256
 Diversity = 1.0162e-06 k_t = [1, 1, 1, 1]
 Generation 9060: Best = 1.176144e+03 FES = 906100 μ_F = 0.718 μ_{CR} = 0.256
 Diversity = 9.2800e-07 k_t = [3, 3, 4, 4]
 Generation 9070: Best = 1.176144e+03 FES = 907100 μ_F = 0.715 μ_{CR} = 0.257
 Diversity = 9.5365e-07 k_t = [1, 1, 1, 1]
 Generation 9080: Best = 1.176144e+03 FES = 908100 μ_F = 0.710 μ_{CR} = 0.252
 Diversity = 8.5562e-07 k_t = [2, 4, 2, 4]
 Generation 9090: Best = 1.176144e+03 FES = 909100 μ_F = 0.701 μ_{CR} = 0.255
 Diversity = 8.3490e-07 k_t = [1, 1, 1, 1]
 Generation 9100: Best = 1.176144e+03 FES = 910100 μ_F = 0.710 μ_{CR} = 0.253
 Diversity = 8.3364e-07 k_t = [2, 2, 3, 3]
 Generation 9110: Best = 1.176144e+03 FES = 911100 μ_F = 0.715 μ_{CR} = 0.248
 Diversity = 9.0409e-07 k_t = [1, 1, 1, 1]
 Generation 9120: Best = 1.176144e+03 FES = 912100 μ_F = 0.714 μ_{CR} = 0.249
 Diversity = 8.5997e-07 k_t = [2, 4, 2, 4]
 Generation 9130: Best = 1.176144e+03 FES = 913100 μ_F = 0.707 μ_{CR} = 0.246
 Diversity = 8.0909e-07 k_t = [1, 1, 1, 1]

Generation 9140: Best = 1.176144e+03 FES = 914100 μ_F = 0.713 μ_{CR} = 0.244
 Diversity = 7.6215e-07 k_t = [4, 3, 4, 2]
 Generation 9150: Best = 1.176144e+03 FES = 915100 μ_F = 0.716 μ_{CR} = 0.239
 Diversity = 7.8554e-07 k_t = [1, 1, 1, 1]
 Generation 9160: Best = 1.176144e+03 FES = 916100 μ_F = 0.717 μ_{CR} = 0.238
 Diversity = 7.7585e-07 k_t = [4, 4, 3, 4]
 Generation 9170: Best = 1.176144e+03 FES = 917100 μ_F = 0.735 μ_{CR} = 0.228
 Diversity = 8.6897e-07 k_t = [1, 1, 1, 1]
 Generation 9180: Best = 1.176144e+03 FES = 918100 μ_F = 0.729 μ_{CR} = 0.232
 Diversity = 8.1427e-07 k_t = [3, 4, 4, 4]
 Generation 9190: Best = 1.176144e+03 FES = 919100 μ_F = 0.728 μ_{CR} = 0.234
 Diversity = 7.6187e-07 k_t = [1, 1, 1, 1]
 Generation 9200: Best = 1.176144e+03 FES = 920100 μ_F = 0.723 μ_{CR} = 0.230
 Diversity = 7.0362e-07 k_t = [2, 4, 4, 3]
 Generation 9210: Best = 1.176144e+03 FES = 921100 μ_F = 0.725 μ_{CR} = 0.229
 Diversity = 7.4872e-07 k_t = [1, 1, 1, 1]
 Generation 9220: Best = 1.176144e+03 FES = 922100 μ_F = 0.712 μ_{CR} = 0.228
 Diversity = 6.6162e-07 k_t = [4, 4, 4, 4]
 Generation 9230: Best = 1.176144e+03 FES = 923100 μ_F = 0.718 μ_{CR} = 0.227
 Diversity = 6.0944e-07 k_t = [1, 1, 1, 1]
 Generation 9240: Best = 1.176144e+03 FES = 924100 μ_F = 0.714 μ_{CR} = 0.223
 Diversity = 5.3754e-07 k_t = [2, 4, 4, 3]
 Generation 9250: Best = 1.176144e+03 FES = 925100 μ_F = 0.730 μ_{CR} = 0.217
 Diversity = 5.6134e-07 k_t = [1, 1, 1, 1]
 Generation 9260: Best = 1.176144e+03 FES = 926100 μ_F = 0.726 μ_{CR} = 0.224
 Diversity = 5.1715e-07 k_t = [2, 2, 4, 4]
 Generation 9270: Best = 1.176144e+03 FES = 927100 μ_F = 0.729 μ_{CR} = 0.227
 Diversity = 5.6796e-07 k_t = [1, 1, 1, 1]
 Generation 9280: Best = 1.176144e+03 FES = 928100 μ_F = 0.752 μ_{CR} = 0.232
 Diversity = 6.0727e-07 k_t = [2, 4, 4, 4]
 Generation 9290: Best = 1.176144e+03 FES = 929100 μ_F = 0.743 μ_{CR} = 0.227
 Diversity = 6.0398e-07 k_t = [1, 1, 1, 1]
 Generation 9300: Best = 1.176144e+03 FES = 930100 μ_F = 0.735 μ_{CR} = 0.227
 Diversity = 5.9576e-07 k_t = [3, 2, 2, 3]
 Generation 9310: Best = 1.176144e+03 FES = 931100 μ_F = 0.733 μ_{CR} = 0.226
 Diversity = 6.2869e-07 k_t = [1, 1, 1, 1]
 Generation 9320: Best = 1.176144e+03 FES = 932100 μ_F = 0.728 μ_{CR} = 0.229
 Diversity = 6.0899e-07 k_t = [2, 3, 4, 3]
 Generation 9330: Best = 1.176144e+03 FES = 933100 μ_F = 0.721 μ_{CR} = 0.234
 Diversity = 6.6812e-07 k_t = [1, 1, 1, 1]
 Generation 9340: Best = 1.176144e+03 FES = 934100 μ_F = 0.722 μ_{CR} = 0.237
 Diversity = 6.7569e-07 k_t = [3, 4, 4, 2]
 Generation 9350: Best = 1.176144e+03 FES = 935100 μ_F = 0.730 μ_{CR} = 0.239
 Diversity = 6.7738e-07 k_t = [1, 1, 1, 1]
 Generation 9360: Best = 1.176144e+03 FES = 936100 μ_F = 0.733 μ_{CR} = 0.237
 Diversity = 6.5968e-07 k_t = [4, 4, 2, 3]
 Generation 9370: Best = 1.176144e+03 FES = 937100 μ_F = 0.731 μ_{CR} = 0.239
 Diversity = 6.2143e-07 k_t = [1, 1, 1, 1]
 Generation 9380: Best = 1.176144e+03 FES = 938100 μ_F = 0.724 μ_{CR} = 0.234
 Diversity = 5.4561e-07 k_t = [2, 3, 2, 4]
 Generation 9390: Best = 1.176144e+03 FES = 939100 μ_F = 0.720 μ_{CR} = 0.234
 Diversity = 5.2570e-07 k_t = [1, 1, 1, 1]
 Generation 9400: Best = 1.176144e+03 FES = 940100 μ_F = 0.709 μ_{CR} = 0.231
 Diversity = 4.8939e-07 k_t = [4, 4, 3, 2]

Generation 9410: Best = 1.176144e+03 FES = 941100 μ_F = 0.716 μ_{CR} = 0.219
 Diversity = 4.8405e-07 k_t = [1, 1, 1, 1]
 Generation 9420: Best = 1.176144e+03 FES = 942100 μ_F = 0.717 μ_{CR} = 0.216
 Diversity = 4.8052e-07 k_t = [3, 3, 4, 2]
 Generation 9430: Best = 1.176144e+03 FES = 943100 μ_F = 0.701 μ_{CR} = 0.224
 Diversity = 4.6290e-07 k_t = [1, 1, 1, 1]
 Generation 9440: Best = 1.176144e+03 FES = 944100 μ_F = 0.706 μ_{CR} = 0.229
 Diversity = 4.0571e-07 k_t = [4, 2, 2, 4]
 Generation 9450: Best = 1.176144e+03 FES = 945100 μ_F = 0.703 μ_{CR} = 0.230
 Diversity = 3.9202e-07 k_t = [1, 1, 1, 1]
 Generation 9460: Best = 1.176144e+03 FES = 946100 μ_F = 0.693 μ_{CR} = 0.239
 Diversity = 3.3846e-07 k_t = [4, 2, 2, 3]
 Generation 9470: Best = 1.176144e+03 FES = 947100 μ_F = 0.704 μ_{CR} = 0.237
 Diversity = 3.4247e-07 k_t = [1, 1, 1, 1]
 Generation 9480: Best = 1.176144e+03 FES = 948100 μ_F = 0.721 μ_{CR} = 0.235
 Diversity = 3.4176e-07 k_t = [2, 3, 4, 3]
 Generation 9490: Best = 1.176144e+03 FES = 949100 μ_F = 0.731 μ_{CR} = 0.227
 Diversity = 3.6548e-07 k_t = [1, 1, 1, 1]
 Generation 9500: Best = 1.176144e+03 FES = 950100 μ_F = 0.744 μ_{CR} = 0.231
 Diversity = 3.7434e-07 k_t = [4, 3, 3, 3]
 Generation 9510: Best = 1.176144e+03 FES = 951100 μ_F = 0.741 μ_{CR} = 0.226
 Diversity = 3.7719e-07 k_t = [1, 1, 1, 1]
 Generation 9520: Best = 1.176144e+03 FES = 952100 μ_F = 0.755 μ_{CR} = 0.230
 Diversity = 3.8439e-07 k_t = [4, 3, 4, 3]
 Generation 9530: Best = 1.176144e+03 FES = 953100 μ_F = 0.754 μ_{CR} = 0.239
 Diversity = 4.1023e-07 k_t = [1, 1, 1, 1]
 Generation 9540: Best = 1.176144e+03 FES = 954100 μ_F = 0.745 μ_{CR} = 0.244
 Diversity = 4.3037e-07 k_t = [2, 4, 2, 3]
 Generation 9550: Best = 1.176144e+03 FES = 955100 μ_F = 0.738 μ_{CR} = 0.251
 Diversity = 4.3624e-07 k_t = [1, 1, 1, 1]
 Generation 9560: Best = 1.176144e+03 FES = 956100 μ_F = 0.732 μ_{CR} = 0.254
 Diversity = 4.3481e-07 k_t = [2, 2, 4, 2]
 Generation 9570: Best = 1.176144e+03 FES = 957100 μ_F = 0.718 μ_{CR} = 0.245
 Diversity = 4.4663e-07 k_t = [1, 1, 1, 1]
 Generation 9580: Best = 1.176144e+03 FES = 958100 μ_F = 0.703 μ_{CR} = 0.245
 Diversity = 4.2073e-07 k_t = [4, 3, 2, 2]
 Generation 9590: Best = 1.176144e+03 FES = 959100 μ_F = 0.695 μ_{CR} = 0.249
 Diversity = 3.8957e-07 k_t = [1, 1, 1, 1]
 Generation 9600: Best = 1.176144e+03 FES = 960100 μ_F = 0.696 μ_{CR} = 0.243
 Diversity = 3.3759e-07 k_t = [3, 4, 3, 4]
 Generation 9610: Best = 1.176144e+03 FES = 961100 μ_F = 0.690 μ_{CR} = 0.246
 Diversity = 3.1744e-07 k_t = [1, 1, 1, 1]
 Generation 9620: Best = 1.176144e+03 FES = 962100 μ_F = 0.710 μ_{CR} = 0.251
 Diversity = 3.5955e-07 k_t = [2, 3, 3, 3]
 Generation 9630: Best = 1.176144e+03 FES = 963100 μ_F = 0.711 μ_{CR} = 0.260
 Diversity = 3.8258e-07 k_t = [1, 1, 1, 1]
 Generation 9640: Best = 1.176144e+03 FES = 964100 μ_F = 0.713 μ_{CR} = 0.266
 Diversity = 3.5934e-07 k_t = [2, 4, 2, 4]
 Generation 9650: Best = 1.176144e+03 FES = 965100 μ_F = 0.729 μ_{CR} = 0.259
 Diversity = 3.5637e-07 k_t = [1, 1, 1, 1]
 Generation 9660: Best = 1.176144e+03 FES = 966100 μ_F = 0.742 μ_{CR} = 0.261
 Diversity = 3.5001e-07 k_t = [3, 3, 3, 3]
 Generation 9670: Best = 1.176144e+03 FES = 967100 μ_F = 0.740 μ_{CR} = 0.264
 Diversity = 4.0403e-07 k_t = [1, 1, 1, 1]

Generation 9680: Best = 1.176144e+03 FES = 968100 μ_F = 0.743 μ_{CR} = 0.267
 Diversity = 3.7982e-07 k_t = [4, 4, 3, 3]
 Generation 9690: Best = 1.176144e+03 FES = 969100 μ_F = 0.745 μ_{CR} = 0.275
 Diversity = 4.3773e-07 k_t = [1, 1, 1, 1]
 Generation 9700: Best = 1.176144e+03 FES = 970100 μ_F = 0.737 μ_{CR} = 0.269
 Diversity = 4.6033e-07 k_t = [2, 2, 4, 2]
 Generation 9710: Best = 1.176144e+03 FES = 971100 μ_F = 0.723 μ_{CR} = 0.257
 Diversity = 4.5632e-07 k_t = [1, 1, 1, 1]
 Generation 9720: Best = 1.176144e+03 FES = 972100 μ_F = 0.725 μ_{CR} = 0.254
 Diversity = 3.5905e-07 k_t = [3, 3, 3, 4]
 Generation 9730: Best = 1.176144e+03 FES = 973100 μ_F = 0.715 μ_{CR} = 0.249
 Diversity = 3.0871e-07 k_t = [1, 1, 1, 1]
 Generation 9740: Best = 1.176144e+03 FES = 974100 μ_F = 0.706 μ_{CR} = 0.255
 Diversity = 2.4793e-07 k_t = [3, 3, 3, 4]
 Generation 9750: Best = 1.176144e+03 FES = 975100 μ_F = 0.709 μ_{CR} = 0.253
 Diversity = 2.2615e-07 k_t = [1, 1, 1, 1]
 Generation 9760: Best = 1.176144e+03 FES = 976100 μ_F = 0.725 μ_{CR} = 0.256
 Diversity = 2.1676e-07 k_t = [3, 2, 4, 4]
 Generation 9770: Best = 1.176144e+03 FES = 977100 μ_F = 0.723 μ_{CR} = 0.258
 Diversity = 2.3322e-07 k_t = [1, 1, 1, 1]
 Generation 9780: Best = 1.176144e+03 FES = 978100 μ_F = 0.729 μ_{CR} = 0.265
 Diversity = 2.1512e-07 k_t = [3, 4, 3, 3]
 Generation 9790: Best = 1.176144e+03 FES = 979100 μ_F = 0.727 μ_{CR} = 0.261
 Diversity = 2.1953e-07 k_t = [1, 1, 1, 1]
 Generation 9800: Best = 1.176144e+03 FES = 980100 μ_F = 0.716 μ_{CR} = 0.268
 Diversity = 1.9771e-07 k_t = [4, 3, 2, 3]
 Generation 9810: Best = 1.176144e+03 FES = 981100 μ_F = 0.710 μ_{CR} = 0.268
 Diversity = 1.9382e-07 k_t = [1, 1, 1, 1]
 Generation 9820: Best = 1.176144e+03 FES = 982100 μ_F = 0.703 μ_{CR} = 0.268
 Diversity = 1.7598e-07 k_t = [4, 4, 4, 2]
 Generation 9830: Best = 1.176144e+03 FES = 983100 μ_F = 0.711 μ_{CR} = 0.264
 Diversity = 1.6628e-07 k_t = [1, 1, 1, 1]
 Generation 9840: Best = 1.176144e+03 FES = 984100 μ_F = 0.712 μ_{CR} = 0.262
 Diversity = 1.5832e-07 k_t = [4, 3, 4, 2]
 Generation 9850: Best = 1.176144e+03 FES = 985100 μ_F = 0.710 μ_{CR} = 0.266
 Diversity = 1.5395e-07 k_t = [1, 1, 1, 1]
 Generation 9860: Best = 1.176144e+03 FES = 986100 μ_F = 0.710 μ_{CR} = 0.261
 Diversity = 1.4772e-07 k_t = [4, 4, 2, 4]
 Generation 9870: Best = 1.176144e+03 FES = 987100 μ_F = 0.700 μ_{CR} = 0.275
 Diversity = 1.5403e-07 k_t = [1, 1, 1, 1]
 Generation 9880: Best = 1.176144e+03 FES = 988100 μ_F = 0.706 μ_{CR} = 0.280
 Diversity = 1.3995e-07 k_t = [4, 3, 2, 3]
 Generation 9890: Best = 1.176144e+03 FES = 989100 μ_F = 0.713 μ_{CR} = 0.286
 Diversity = 1.4854e-07 k_t = [1, 1, 1, 1]
 Generation 9900: Best = 1.176144e+03 FES = 990100 μ_F = 0.721 μ_{CR} = 0.287
 Diversity = 1.6072e-07 k_t = [2, 3, 3, 4]
 Generation 9910: Best = 1.176144e+03 FES = 991100 μ_F = 0.730 μ_{CR} = 0.285
 Diversity = 1.8823e-07 k_t = [1, 1, 1, 1]
 Generation 9920: Best = 1.176144e+03 FES = 992100 μ_F = 0.739 μ_{CR} = 0.280
 Diversity = 2.2615e-07 k_t = [4, 2, 3, 4]
 Generation 9930: Best = 1.176144e+03 FES = 993100 μ_F = 0.748 μ_{CR} = 0.282
 Diversity = 2.5062e-07 k_t = [1, 1, 1, 1]
 Generation 9940: Best = 1.176144e+03 FES = 994100 μ_F = 0.736 μ_{CR} = 0.279
 Diversity = 2.3828e-07 k_t = [3, 4, 2, 3]

Generation 9950: Best = 1.176144e+03 FES = 995100 μ_F = 0.728 μ_{CR} = 0.283
 Diversity = 2.2172e-07 k_t = [1, 1, 1, 1]
 Generation 9960: Best = 1.176144e+03 FES = 996100 μ_F = 0.727 μ_{CR} = 0.276
 Diversity = 1.9003e-07 k_t = [3, 3, 4, 4]
 Generation 9970: Best = 1.176144e+03 FES = 997100 μ_F = 0.716 μ_{CR} = 0.282
 Diversity = 1.7063e-07 k_t = [1, 1, 1, 1]
 Generation 9980: Best = 1.176144e+03 FES = 998100 μ_F = 0.726 μ_{CR} = 0.277
 Diversity = 1.5718e-07 k_t = [4, 2, 4, 3]
 Generation 9990: Best = 1.176144e+03 FES = 999100 μ_F = 0.720 μ_{CR} = 0.265
 Diversity = 1.5379e-07 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 1.176144e+03
 Total Function Evals = 1000000
 Total Successful Updates = 233263

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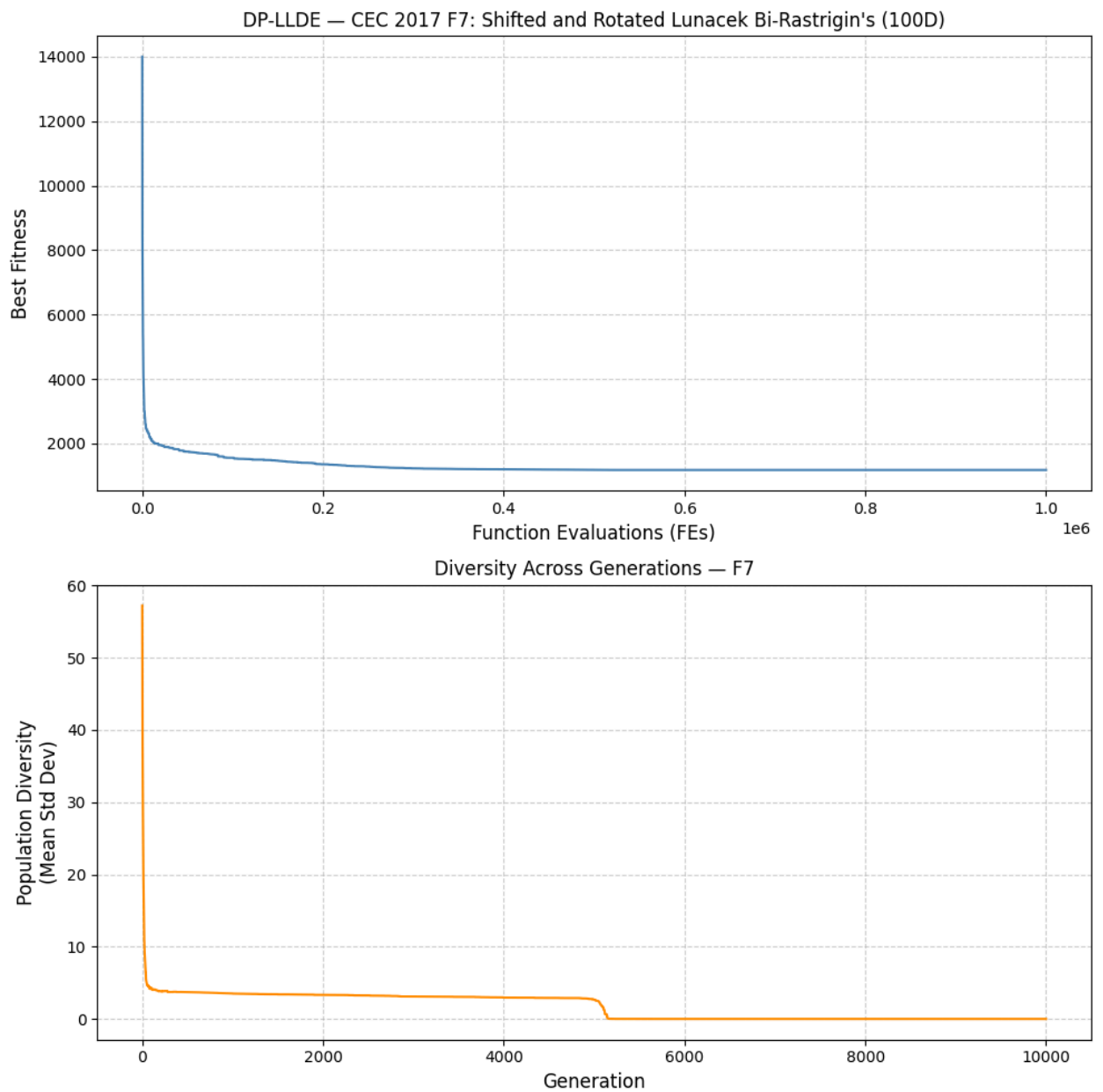
Results:

Best Fitness : 1.176144e+03
 Global Optimum : 7.000000e+02
 Error (|best-optimum|) : 4.761444e+02
 Best Solution (5 dim) : [-13.21925438 16.73708936 -16.05057161 -5.55999237
 15.45208261]
 $D^{(0)}$ (init. diversity): 5.7220e+01

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✓ Checkpoint saved to: /content/drive/My Drive/thesis_results/all_results_checkpoint.pkl



Graph saved as convergence_F7.png

✓ Plot saved to: /content/drive/My Drive/thesis_results/plots/convergence_F7.png

```
In [ ]: # =====
# F8 — Shifted and Rotated Non-Continuous Rastrigin
# =====
run_and_plot('F8', make_wrapper(simple.f8), 800.0, "Shifted and Rotated Non-Co
```

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DP-LLDE on CEC 2017 F8 – Single Run

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Function : Shifted and Rotated Non-Continuous Rastrigin's (100D)
Bounds : [-100, 100]
Global optimum : 800.0
Initial $D^*(0) = 5.7220e+01$ | Diversity threshold = 5.7220e+00
Generation 0: Best Fitness = 3.562373e+03

Generation 10:	Best = 2.559948e+03	FES = 1100	$\mu_F = 0.541$	$\mu_{CR} = 0.352$
Diversity = 2.9438e+01	$k_t = [1, 1, 1, 1]$			
Generation 20:	Best = 2.287780e+03	FES = 2100	$\mu_F = 0.579$	$\mu_{CR} = 0.354$
Diversity = 1.9714e+01	$k_t = [2, 3, 2, 3]$			
Generation 30:	Best = 2.131139e+03	FES = 3100	$\mu_F = 0.610$	$\mu_{CR} = 0.379$
Diversity = 1.6365e+01	$k_t = [1, 1, 1, 1]$			
Generation 40:	Best = 2.073456e+03	FES = 4100	$\mu_F = 0.618$	$\mu_{CR} = 0.362$
Diversity = 1.3759e+01	$k_t = [3, 2, 3, 4]$			
Generation 50:	Best = 2.073456e+03	FES = 5100	$\mu_F = 0.623$	$\mu_{CR} = 0.323$
Diversity = 1.2733e+01	$k_t = [1, 1, 1, 1]$			
Generation 60:	Best = 2.052885e+03	FES = 6100	$\mu_F = 0.655$	$\mu_{CR} = 0.272$
Diversity = 1.2057e+01	$k_t = [3, 4, 4, 3]$			
Generation 70:	Best = 1.978515e+03	FES = 7100	$\mu_F = 0.650$	$\mu_{CR} = 0.255$
Diversity = 1.1345e+01	$k_t = [1, 1, 1, 1]$			
Generation 80:	Best = 1.939225e+03	FES = 8100	$\mu_F = 0.659$	$\mu_{CR} = 0.237$
Diversity = 1.0827e+01	$k_t = [4, 3, 3, 3]$			
Generation 90:	Best = 1.927494e+03	FES = 9100	$\mu_F = 0.645$	$\mu_{CR} = 0.201$
Diversity = 1.0513e+01	$k_t = [1, 1, 1, 1]$			
Generation 100:	Best = 1.827016e+03	FES = 10100	$\mu_F = 0.657$	$\mu_{CR} = 0.147$
Diversity = 1.0361e+01	$k_t = [3, 2, 2, 2]$			
Generation 110:	Best = 1.814512e+03	FES = 11100	$\mu_F = 0.641$	$\mu_{CR} = 0.102$
Diversity = 1.0290e+01	$k_t = [1, 1, 1, 1]$			
Generation 120:	Best = 1.814512e+03	FES = 12100	$\mu_F = 0.673$	$\mu_{CR} = 0.086$
Diversity = 9.9927e+00	$k_t = [4, 4, 2, 4]$			
Generation 130:	Best = 1.814512e+03	FES = 13100	$\mu_F = 0.725$	$\mu_{CR} = 0.077$
Diversity = 9.9700e+00	$k_t = [1, 1, 1, 1]$			
Generation 140:	Best = 1.814512e+03	FES = 14100	$\mu_F = 0.738$	$\mu_{CR} = 0.063$
Diversity = 9.9951e+00	$k_t = [3, 3, 4, 2]$			
Generation 150:	Best = 1.814512e+03	FES = 15100	$\mu_F = 0.748$	$\mu_{CR} = 0.063$
Diversity = 1.0109e+01	$k_t = [1, 1, 1, 1]$			
Generation 160:	Best = 1.814512e+03	FES = 16100	$\mu_F = 0.759$	$\mu_{CR} = 0.052$
Diversity = 1.0023e+01	$k_t = [4, 3, 4, 2]$			
Generation 170:	Best = 1.783867e+03	FES = 17100	$\mu_F = 0.735$	$\mu_{CR} = 0.047$
Diversity = 1.0065e+01	$k_t = [1, 1, 1, 1]$			
Generation 180:	Best = 1.730398e+03	FES = 18100	$\mu_F = 0.711$	$\mu_{CR} = 0.047$
Diversity = 1.0014e+01	$k_t = [4, 3, 4, 3]$			
Generation 190:	Best = 1.730398e+03	FES = 19100	$\mu_F = 0.718$	$\mu_{CR} = 0.037$
Diversity = 1.0000e+01	$k_t = [1, 1, 1, 1]$			
Generation 200:	Best = 1.730398e+03	FES = 20100	$\mu_F = 0.744$	$\mu_{CR} = 0.036$
Diversity = 9.9474e+00	$k_t = [2, 4, 4, 2]$			
Generation 210:	Best = 1.730398e+03	FES = 21100	$\mu_F = 0.737$	$\mu_{CR} = 0.044$
Diversity = 9.9211e+00	$k_t = [1, 1, 1, 1]$			
Generation 220:	Best = 1.730398e+03	FES = 22100	$\mu_F = 0.743$	$\mu_{CR} = 0.043$
Diversity = 9.9593e+00	$k_t = [4, 3, 4, 2]$			

Generation	230:	Best = 1.730398e+03	FES = 23100	μ_F = 0.735	μ_{CR} = 0.045
Diversity	= 9.9869e+00	k_t = [1, 1, 1, 1]			
Generation	240:	Best = 1.730143e+03	FES = 24100	μ_F = 0.713	μ_{CR} = 0.046
Diversity	= 1.0009e+01	k_t = [2, 2, 3, 4]			
Generation	250:	Best = 1.724791e+03	FES = 25100	μ_F = 0.726	μ_{CR} = 0.035
Diversity	= 1.0005e+01	k_t = [1, 1, 1, 1]			
Generation	260:	Best = 1.723340e+03	FES = 26100	μ_F = 0.759	μ_{CR} = 0.038
Diversity	= 9.7481e+00	k_t = [3, 4, 4, 4]			
Generation	270:	Best = 1.723340e+03	FES = 27100	μ_F = 0.669	μ_{CR} = 0.039
Diversity	= 9.7371e+00	k_t = [1, 1, 1, 1]			
Generation	280:	Best = 1.719181e+03	FES = 28100	μ_F = 0.675	μ_{CR} = 0.038
Diversity	= 9.7147e+00	k_t = [4, 4, 4, 3]			
Generation	290:	Best = 1.716576e+03	FES = 29100	μ_F = 0.658	μ_{CR} = 0.034
Diversity	= 9.7054e+00	k_t = [1, 1, 1, 1]			
Generation	300:	Best = 1.716576e+03	FES = 30100	μ_F = 0.675	μ_{CR} = 0.038
Diversity	= 9.6781e+00	k_t = [2, 4, 2, 2]			
Generation	310:	Best = 1.708641e+03	FES = 31100	μ_F = 0.581	μ_{CR} = 0.051
Diversity	= 9.6859e+00	k_t = [1, 1, 1, 1]			
Generation	320:	Best = 1.698460e+03	FES = 32100	μ_F = 0.602	μ_{CR} = 0.043
Diversity	= 9.6616e+00	k_t = [4, 3, 2, 2]			
Generation	330:	Best = 1.698460e+03	FES = 33100	μ_F = 0.555	μ_{CR} = 0.048
Diversity	= 9.6229e+00	k_t = [1, 1, 1, 1]			
Generation	340:	Best = 1.698460e+03	FES = 34100	μ_F = 0.607	μ_{CR} = 0.040
Diversity	= 9.4255e+00	k_t = [4, 4, 4, 4]			
Generation	350:	Best = 1.688370e+03	FES = 35100	μ_F = 0.623	μ_{CR} = 0.041
Diversity	= 9.4174e+00	k_t = [1, 1, 1, 1]			
Generation	360:	Best = 1.688370e+03	FES = 36100	μ_F = 0.595	μ_{CR} = 0.046
Diversity	= 9.3678e+00	k_t = [2, 2, 2, 4]			
Generation	370:	Best = 1.685155e+03	FES = 37100	μ_F = 0.594	μ_{CR} = 0.054
Diversity	= 9.3678e+00	k_t = [1, 1, 1, 1]			
Generation	380:	Best = 1.685155e+03	FES = 38100	μ_F = 0.668	μ_{CR} = 0.039
Diversity	= 9.3291e+00	k_t = [4, 4, 3, 4]			
Generation	390:	Best = 1.685155e+03	FES = 39100	μ_F = 0.709	μ_{CR} = 0.034
Diversity	= 9.2857e+00	k_t = [1, 1, 1, 1]			
Generation	400:	Best = 1.684641e+03	FES = 40100	μ_F = 0.732	μ_{CR} = 0.032
Diversity	= 9.2406e+00	k_t = [4, 2, 4, 4]			
Generation	410:	Best = 1.674785e+03	FES = 41100	μ_F = 0.715	μ_{CR} = 0.028
Diversity	= 9.2213e+00	k_t = [1, 1, 1, 1]			
Generation	420:	Best = 1.674785e+03	FES = 42100	μ_F = 0.677	μ_{CR} = 0.030
Diversity	= 9.2127e+00	k_t = [2, 4, 2, 3]			
Generation	430:	Best = 1.671891e+03	FES = 43100	μ_F = 0.695	μ_{CR} = 0.032
Diversity	= 9.2598e+00	k_t = [1, 1, 1, 1]			
Generation	440:	Best = 1.671891e+03	FES = 44100	μ_F = 0.727	μ_{CR} = 0.027
Diversity	= 9.2917e+00	k_t = [3, 3, 3, 2]			
Generation	450:	Best = 1.649249e+03	FES = 45100	μ_F = 0.755	μ_{CR} = 0.025
Diversity	= 9.2891e+00	k_t = [1, 1, 1, 1]			
Generation	460:	Best = 1.649249e+03	FES = 46100	μ_F = 0.748	μ_{CR} = 0.035
Diversity	= 9.1800e+00	k_t = [4, 4, 4, 4]			
Generation	470:	Best = 1.649249e+03	FES = 47100	μ_F = 0.692	μ_{CR} = 0.032
Diversity	= 9.1598e+00	k_t = [1, 1, 1, 1]			
Generation	480:	Best = 1.648144e+03	FES = 48100	μ_F = 0.650	μ_{CR} = 0.028
Diversity	= 9.0671e+00	k_t = [2, 4, 3, 4]			
Generation	490:	Best = 1.645807e+03	FES = 49100	μ_F = 0.690	μ_{CR} = 0.025
Diversity	= 9.0630e+00	k_t = [1, 1, 1, 1]			

Generation 500: Best = 1.645807e+03	FES = 50100	$\mu_F = 0.701$	$\mu_{CR} = 0.030$
Diversity = 9.0531e+00	$k_t = [2, 4, 3, 4]$		
Generation 510: Best = 1.645807e+03	FES = 51100	$\mu_F = 0.591$	$\mu_{CR} = 0.030$
Diversity = 9.0412e+00	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 1.633886e+03	FES = 52100	$\mu_F = 0.593$	$\mu_{CR} = 0.036$
Diversity = 9.0453e+00	$k_t = [2, 3, 2, 4]$		
Generation 530: Best = 1.633886e+03	FES = 53100	$\mu_F = 0.540$	$\mu_{CR} = 0.040$
Diversity = 9.0400e+00	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 1.627401e+03	FES = 54100	$\mu_F = 0.534$	$\mu_{CR} = 0.036$
Diversity = 9.0440e+00	$k_t = [4, 2, 3, 4]$		
Generation 550: Best = 1.612241e+03	FES = 55100	$\mu_F = 0.482$	$\mu_{CR} = 0.037$
Diversity = 9.0181e+00	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 1.588536e+03	FES = 56100	$\mu_F = 0.542$	$\mu_{CR} = 0.033$
Diversity = 9.0158e+00	$k_t = [3, 3, 3, 3]$		
Generation 570: Best = 1.588536e+03	FES = 57100	$\mu_F = 0.579$	$\mu_{CR} = 0.027$
Diversity = 9.0006e+00	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 1.588536e+03	FES = 58100	$\mu_F = 0.636$	$\mu_{CR} = 0.035$
Diversity = 8.9929e+00	$k_t = [4, 4, 4, 2]$		
Generation 590: Best = 1.588536e+03	FES = 59100	$\mu_F = 0.570$	$\mu_{CR} = 0.035$
Diversity = 8.9830e+00	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 1.579189e+03	FES = 60100	$\mu_F = 0.565$	$\mu_{CR} = 0.037$
Diversity = 8.9800e+00	$k_t = [2, 2, 2, 2]$		
Generation 610: Best = 1.576775e+03	FES = 61100	$\mu_F = 0.492$	$\mu_{CR} = 0.051$
Diversity = 8.9670e+00	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 1.576775e+03	FES = 62100	$\mu_F = 0.491$	$\mu_{CR} = 0.063$
Diversity = 8.9539e+00	$k_t = [4, 3, 3, 2]$		
Generation 630: Best = 1.574998e+03	FES = 63100	$\mu_F = 0.421$	$\mu_{CR} = 0.074$
Diversity = 8.9420e+00	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 1.574998e+03	FES = 64100	$\mu_F = 0.459$	$\mu_{CR} = 0.058$
Diversity = 8.8954e+00	$k_t = [3, 2, 4, 4]$		
Generation 650: Best = 1.574998e+03	FES = 65100	$\mu_F = 0.407$	$\mu_{CR} = 0.050$
Diversity = 8.8869e+00	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 1.574998e+03	FES = 66100	$\mu_F = 0.351$	$\mu_{CR} = 0.053$
Diversity = 8.8800e+00	$k_t = [2, 4, 3, 2]$		
Generation 670: Best = 1.560539e+03	FES = 67100	$\mu_F = 0.338$	$\mu_{CR} = 0.053$
Diversity = 8.8605e+00	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 1.556000e+03	FES = 68100	$\mu_F = 0.350$	$\mu_{CR} = 0.055$
Diversity = 8.8346e+00	$k_t = [3, 4, 3, 2]$		
Generation 690: Best = 1.553810e+03	FES = 69100	$\mu_F = 0.317$	$\mu_{CR} = 0.051$
Diversity = 8.8211e+00	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 1.525763e+03	FES = 70100	$\mu_F = 0.287$	$\mu_{CR} = 0.052$
Diversity = 8.8093e+00	$k_t = [3, 4, 3, 2]$		
Generation 710: Best = 1.512974e+03	FES = 71100	$\mu_F = 0.318$	$\mu_{CR} = 0.066$
Diversity = 8.7917e+00	$k_t = [1, 1, 1, 1]$		
Generation 720: Best = 1.512974e+03	FES = 72100	$\mu_F = 0.404$	$\mu_{CR} = 0.069$
Diversity = 8.7578e+00	$k_t = [2, 4, 4, 3]$		
Generation 730: Best = 1.512677e+03	FES = 73100	$\mu_F = 0.404$	$\mu_{CR} = 0.059$
Diversity = 8.7575e+00	$k_t = [1, 1, 1, 1]$		
Generation 740: Best = 1.512677e+03	FES = 74100	$\mu_F = 0.411$	$\mu_{CR} = 0.044$
Diversity = 8.7759e+00	$k_t = [2, 4, 3, 4]$		
Generation 750: Best = 1.512677e+03	FES = 75100	$\mu_F = 0.489$	$\mu_{CR} = 0.042$
Diversity = 8.7264e+00	$k_t = [1, 1, 1, 1]$		
Generation 760: Best = 1.512677e+03	FES = 76100	$\mu_F = 0.515$	$\mu_{CR} = 0.039$
Diversity = 8.6355e+00	$k_t = [4, 4, 2, 4]$		

Generation 770: Best = 1.512086e+03	FES = 77100	$\mu_F = 0.379$	$\mu_{CR} = 0.050$
Diversity = 8.6252e+00	$k_t = [1, 1, 1, 1]$		
Generation 780: Best = 1.512086e+03	FES = 78100	$\mu_F = 0.332$	$\mu_{CR} = 0.054$
Diversity = 8.6230e+00	$k_t = [4, 4, 4, 3]$		
Generation 790: Best = 1.512086e+03	FES = 79100	$\mu_F = 0.284$	$\mu_{CR} = 0.051$
Diversity = 8.6160e+00	$k_t = [1, 1, 1, 1]$		
Generation 800: Best = 1.500904e+03	FES = 80100	$\mu_F = 0.331$	$\mu_{CR} = 0.053$
Diversity = 8.5721e+00	$k_t = [4, 2, 2, 4]$		
Generation 810: Best = 1.497558e+03	FES = 81100	$\mu_F = 0.370$	$\mu_{CR} = 0.047$
Diversity = 8.5604e+00	$k_t = [1, 1, 1, 1]$		
Generation 820: Best = 1.493841e+03	FES = 82100	$\mu_F = 0.433$	$\mu_{CR} = 0.054$
Diversity = 8.4811e+00	$k_t = [2, 4, 3, 4]$		
Generation 830: Best = 1.491413e+03	FES = 83100	$\mu_F = 0.505$	$\mu_{CR} = 0.047$
Diversity = 8.4738e+00	$k_t = [1, 1, 1, 1]$		
Generation 840: Best = 1.491413e+03	FES = 84100	$\mu_F = 0.455$	$\mu_{CR} = 0.041$
Diversity = 8.4728e+00	$k_t = [4, 2, 3, 2]$		
Generation 850: Best = 1.491413e+03	FES = 85100	$\mu_F = 0.452$	$\mu_{CR} = 0.037$
Diversity = 8.4721e+00	$k_t = [1, 1, 1, 1]$		
Generation 860: Best = 1.491413e+03	FES = 86100	$\mu_F = 0.402$	$\mu_{CR} = 0.045$
Diversity = 8.4647e+00	$k_t = [2, 4, 3, 3]$		
Generation 870: Best = 1.490542e+03	FES = 87100	$\mu_F = 0.428$	$\mu_{CR} = 0.044$
Diversity = 8.4590e+00	$k_t = [1, 1, 1, 1]$		
Generation 880: Best = 1.490542e+03	FES = 88100	$\mu_F = 0.418$	$\mu_{CR} = 0.046$
Diversity = 8.4244e+00	$k_t = [4, 2, 3, 4]$		
Generation 890: Best = 1.483470e+03	FES = 89100	$\mu_F = 0.412$	$\mu_{CR} = 0.038$
Diversity = 8.4233e+00	$k_t = [1, 1, 1, 1]$		
Generation 900: Best = 1.481766e+03	FES = 90100	$\mu_F = 0.375$	$\mu_{CR} = 0.032$
Diversity = 8.4190e+00	$k_t = [3, 4, 3, 2]$		
Generation 910: Best = 1.475615e+03	FES = 91100	$\mu_F = 0.374$	$\mu_{CR} = 0.038$
Diversity = 8.4044e+00	$k_t = [1, 1, 1, 1]$		
Generation 920: Best = 1.475615e+03	FES = 92100	$\mu_F = 0.344$	$\mu_{CR} = 0.046$
Diversity = 8.3991e+00	$k_t = [3, 4, 3, 2]$		
Generation 930: Best = 1.471949e+03	FES = 93100	$\mu_F = 0.431$	$\mu_{CR} = 0.053$
Diversity = 8.3945e+00	$k_t = [1, 1, 1, 1]$		
Generation 940: Best = 1.471949e+03	FES = 94100	$\mu_F = 0.440$	$\mu_{CR} = 0.065$
Diversity = 8.3572e+00	$k_t = [4, 2, 4, 4]$		
Generation 950: Best = 1.471949e+03	FES = 95100	$\mu_F = 0.449$	$\mu_{CR} = 0.053$
Diversity = 8.3367e+00	$k_t = [1, 1, 1, 1]$		
Generation 960: Best = 1.471949e+03	FES = 96100	$\mu_F = 0.509$	$\mu_{CR} = 0.041$
Diversity = 8.3178e+00	$k_t = [2, 4, 3, 3]$		
Generation 970: Best = 1.462343e+03	FES = 97100	$\mu_F = 0.494$	$\mu_{CR} = 0.038$
Diversity = 8.3153e+00	$k_t = [1, 1, 1, 1]$		
Generation 980: Best = 1.462343e+03	FES = 98100	$\mu_F = 0.563$	$\mu_{CR} = 0.039$
Diversity = 8.3084e+00	$k_t = [3, 4, 2, 3]$		
Generation 990: Best = 1.462343e+03	FES = 99100	$\mu_F = 0.526$	$\mu_{CR} = 0.046$
Diversity = 8.3097e+00	$k_t = [1, 1, 1, 1]$		
Generation 1000: Best = 1.457430e+03	FES = 100100	$\mu_F = 0.555$	$\mu_{CR} = 0.041$
Diversity = 8.3013e+00	$k_t = [4, 2, 2, 4]$		
Generation 1010: Best = 1.450668e+03	FES = 101100	$\mu_F = 0.549$	$\mu_{CR} = 0.044$
Diversity = 8.2557e+00	$k_t = [1, 1, 1, 1]$		
Generation 1020: Best = 1.449650e+03	FES = 102100	$\mu_F = 0.486$	$\mu_{CR} = 0.042$
Diversity = 8.2536e+00	$k_t = [3, 3, 4, 2]$		
Generation 1030: Best = 1.449650e+03	FES = 103100	$\mu_F = 0.523$	$\mu_{CR} = 0.045$
Diversity = 8.2450e+00	$k_t = [1, 1, 1, 1]$		

Generation 1040: Best = 1.442754e+03 FES = 104100 μ_F = 0.570 μ_{CR} = 0.038
 Diversity = 8.2442e+00 k_t = [3, 2, 3, 2]
 Generation 1050: Best = 1.442754e+03 FES = 105100 μ_F = 0.552 μ_{CR} = 0.034
 Diversity = 8.2433e+00 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 1.442754e+03 FES = 106100 μ_F = 0.505 μ_{CR} = 0.034
 Diversity = 8.2245e+00 k_t = [2, 3, 3, 3]
 Generation 1070: Best = 1.440514e+03 FES = 107100 μ_F = 0.554 μ_{CR} = 0.030
 Diversity = 8.2227e+00 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 1.431936e+03 FES = 108100 μ_F = 0.624 μ_{CR} = 0.030
 Diversity = 8.2250e+00 k_t = [4, 4, 3, 2]
 Generation 1090: Best = 1.431936e+03 FES = 109100 μ_F = 0.502 μ_{CR} = 0.041
 Diversity = 8.2242e+00 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 1.426133e+03 FES = 110100 μ_F = 0.488 μ_{CR} = 0.040
 Diversity = 8.2213e+00 k_t = [4, 2, 4, 2]
 Generation 1110: Best = 1.426133e+03 FES = 111100 μ_F = 0.472 μ_{CR} = 0.052
 Diversity = 8.2089e+00 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 1.419957e+03 FES = 112100 μ_F = 0.448 μ_{CR} = 0.039
 Diversity = 8.2010e+00 k_t = [3, 4, 3, 2]
 Generation 1130: Best = 1.419957e+03 FES = 113100 μ_F = 0.452 μ_{CR} = 0.040
 Diversity = 8.2012e+00 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 1.419957e+03 FES = 114100 μ_F = 0.458 μ_{CR} = 0.033
 Diversity = 8.2008e+00 k_t = [4, 3, 2, 3]
 Generation 1150: Best = 1.418605e+03 FES = 115100 μ_F = 0.418 μ_{CR} = 0.033
 Diversity = 8.1973e+00 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 1.417095e+03 FES = 116100 μ_F = 0.442 μ_{CR} = 0.036
 Diversity = 8.2056e+00 k_t = [2, 4, 3, 4]
 Generation 1170: Best = 1.417095e+03 FES = 117100 μ_F = 0.423 μ_{CR} = 0.038
 Diversity = 8.2035e+00 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 1.417095e+03 FES = 118100 μ_F = 0.416 μ_{CR} = 0.040
 Diversity = 8.1151e+00 k_t = [2, 4, 2, 4]
 Generation 1190: Best = 1.416485e+03 FES = 119100 μ_F = 0.423 μ_{CR} = 0.048
 Diversity = 8.1088e+00 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 1.416406e+03 FES = 120100 μ_F = 0.414 μ_{CR} = 0.034
 Diversity = 8.0818e+00 k_t = [2, 2, 3, 2]
 Generation 1210: Best = 1.416106e+03 FES = 121100 μ_F = 0.417 μ_{CR} = 0.033
 Diversity = 8.0797e+00 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 1.415783e+03 FES = 122100 μ_F = 0.455 μ_{CR} = 0.035
 Diversity = 8.0786e+00 k_t = [2, 2, 4, 2]
 Generation 1230: Best = 1.415511e+03 FES = 123100 μ_F = 0.411 μ_{CR} = 0.040
 Diversity = 8.0787e+00 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 1.415287e+03 FES = 124100 μ_F = 0.402 μ_{CR} = 0.046
 Diversity = 8.0713e+00 k_t = [3, 3, 2, 2]
 Generation 1250: Best = 1.407716e+03 FES = 125100 μ_F = 0.411 μ_{CR} = 0.051
 Diversity = 8.0794e+00 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 1.407716e+03 FES = 126100 μ_F = 0.405 μ_{CR} = 0.053
 Diversity = 8.0705e+00 k_t = [2, 3, 4, 3]
 Generation 1270: Best = 1.406309e+03 FES = 127100 μ_F = 0.458 μ_{CR} = 0.060
 Diversity = 8.0599e+00 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 1.405861e+03 FES = 128100 μ_F = 0.485 μ_{CR} = 0.049
 Diversity = 8.0525e+00 k_t = [2, 4, 3, 2]
 Generation 1290: Best = 1.405861e+03 FES = 129100 μ_F = 0.406 μ_{CR} = 0.048
 Diversity = 8.0508e+00 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 1.405861e+03 FES = 130100 μ_F = 0.359 μ_{CR} = 0.056
 Diversity = 8.0173e+00 k_t = [2, 2, 3, 2]

Generation 1310: Best = 1.398076e+03 FES = 131100 μ_F = 0.309 μ_{CR} = 0.067
 Diversity = 8.0097e+00 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 1.394557e+03 FES = 132100 μ_F = 0.288 μ_{CR} = 0.063
 Diversity = 8.0033e+00 k_t = [2, 3, 3, 3]
 Generation 1330: Best = 1.392168e+03 FES = 133100 μ_F = 0.286 μ_{CR} = 0.047
 Diversity = 7.9988e+00 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 1.392168e+03 FES = 134100 μ_F = 0.322 μ_{CR} = 0.051
 Diversity = 7.9913e+00 k_t = [3, 4, 3, 2]
 Generation 1350: Best = 1.389499e+03 FES = 135100 μ_F = 0.287 μ_{CR} = 0.047
 Diversity = 7.9850e+00 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 1.388108e+03 FES = 136100 μ_F = 0.274 μ_{CR} = 0.042
 Diversity = 7.9776e+00 k_t = [2, 4, 2, 2]
 Generation 1370: Best = 1.384200e+03 FES = 137100 μ_F = 0.332 μ_{CR} = 0.044
 Diversity = 7.9725e+00 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 1.379291e+03 FES = 138100 μ_F = 0.268 μ_{CR} = 0.062
 Diversity = 7.9665e+00 k_t = [3, 4, 4, 3]
 Generation 1390: Best = 1.375043e+03 FES = 139100 μ_F = 0.293 μ_{CR} = 0.057
 Diversity = 7.9628e+00 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 1.372757e+03 FES = 140100 μ_F = 0.328 μ_{CR} = 0.042
 Diversity = 7.9612e+00 k_t = [4, 4, 3, 3]
 Generation 1410: Best = 1.372371e+03 FES = 141100 μ_F = 0.342 μ_{CR} = 0.036
 Diversity = 7.9568e+00 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 1.369902e+03 FES = 142100 μ_F = 0.423 μ_{CR} = 0.036
 Diversity = 7.9589e+00 k_t = [2, 3, 2, 2]
 Generation 1430: Best = 1.369902e+03 FES = 143100 μ_F = 0.426 μ_{CR} = 0.038
 Diversity = 7.9610e+00 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 1.368553e+03 FES = 144100 μ_F = 0.365 μ_{CR} = 0.053
 Diversity = 7.9616e+00 k_t = [3, 2, 4, 2]
 Generation 1450: Best = 1.367382e+03 FES = 145100 μ_F = 0.422 μ_{CR} = 0.053
 Diversity = 7.9085e+00 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 1.363092e+03 FES = 146100 μ_F = 0.461 μ_{CR} = 0.053
 Diversity = 7.9042e+00 k_t = [2, 2, 2, 4]
 Generation 1470: Best = 1.363092e+03 FES = 147100 μ_F = 0.491 μ_{CR} = 0.041
 Diversity = 7.8965e+00 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 1.363092e+03 FES = 148100 μ_F = 0.519 μ_{CR} = 0.029
 Diversity = 7.8458e+00 k_t = [2, 2, 3, 3]
 Generation 1490: Best = 1.363092e+03 FES = 149100 μ_F = 0.461 μ_{CR} = 0.032
 Diversity = 7.8449e+00 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 1.363092e+03 FES = 150100 μ_F = 0.455 μ_{CR} = 0.030
 Diversity = 7.8426e+00 k_t = [3, 3, 2, 4]
 Generation 1510: Best = 1.358949e+03 FES = 151100 μ_F = 0.476 μ_{CR} = 0.028
 Diversity = 7.8472e+00 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 1.349452e+03 FES = 152100 μ_F = 0.493 μ_{CR} = 0.037
 Diversity = 7.8488e+00 k_t = [2, 3, 3, 3]
 Generation 1530: Best = 1.348127e+03 FES = 153100 μ_F = 0.426 μ_{CR} = 0.034
 Diversity = 7.7912e+00 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 1.348127e+03 FES = 154100 μ_F = 0.364 μ_{CR} = 0.035
 Diversity = 7.7898e+00 k_t = [4, 2, 2, 2]
 Generation 1550: Best = 1.348127e+03 FES = 155100 μ_F = 0.404 μ_{CR} = 0.040
 Diversity = 7.7587e+00 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 1.339945e+03 FES = 156100 μ_F = 0.471 μ_{CR} = 0.039
 Diversity = 7.7238e+00 k_t = [3, 2, 3, 3]
 Generation 1570: Best = 1.339945e+03 FES = 157100 μ_F = 0.395 μ_{CR} = 0.033
 Diversity = 7.7242e+00 k_t = [1, 1, 1, 1]

Generation 1580: Best = 1.339945e+03 FES = 158100 μ_F = 0.440 μ_{CR} = 0.041
 Diversity = 7.7225e+00 k_t = [3, 4, 4, 2]
 Generation 1590: Best = 1.339945e+03 FES = 159100 μ_F = 0.363 μ_{CR} = 0.039
 Diversity = 7.7185e+00 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 1.331336e+03 FES = 160100 μ_F = 0.427 μ_{CR} = 0.046
 Diversity = 7.7197e+00 k_t = [2, 3, 4, 3]
 Generation 1610: Best = 1.331336e+03 FES = 161100 μ_F = 0.405 μ_{CR} = 0.047
 Diversity = 7.7207e+00 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 1.329850e+03 FES = 162100 μ_F = 0.455 μ_{CR} = 0.045
 Diversity = 7.7233e+00 k_t = [4, 2, 4, 3]
 Generation 1630: Best = 1.325753e+03 FES = 163100 μ_F = 0.435 μ_{CR} = 0.045
 Diversity = 7.7200e+00 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 1.325753e+03 FES = 164100 μ_F = 0.479 μ_{CR} = 0.033
 Diversity = 7.7300e+00 k_t = [2, 4, 2, 3]
 Generation 1650: Best = 1.324921e+03 FES = 165100 μ_F = 0.565 μ_{CR} = 0.028
 Diversity = 7.7263e+00 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 1.321758e+03 FES = 166100 μ_F = 0.512 μ_{CR} = 0.036
 Diversity = 7.6941e+00 k_t = [4, 3, 2, 4]
 Generation 1670: Best = 1.321723e+03 FES = 167100 μ_F = 0.445 μ_{CR} = 0.028
 Diversity = 7.6293e+00 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 1.319686e+03 FES = 168100 μ_F = 0.452 μ_{CR} = 0.040
 Diversity = 7.6270e+00 k_t = [4, 2, 2, 3]
 Generation 1690: Best = 1.319575e+03 FES = 169100 μ_F = 0.473 μ_{CR} = 0.049
 Diversity = 7.6261e+00 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 1.319575e+03 FES = 170100 μ_F = 0.456 μ_{CR} = 0.048
 Diversity = 7.6209e+00 k_t = [3, 4, 4, 2]
 Generation 1710: Best = 1.319575e+03 FES = 171100 μ_F = 0.514 μ_{CR} = 0.039
 Diversity = 7.6051e+00 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 1.319575e+03 FES = 172100 μ_F = 0.546 μ_{CR} = 0.051
 Diversity = 7.5439e+00 k_t = [2, 4, 3, 4]
 Generation 1730: Best = 1.318156e+03 FES = 173100 μ_F = 0.529 μ_{CR} = 0.047
 Diversity = 7.5118e+00 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 1.310332e+03 FES = 174100 μ_F = 0.505 μ_{CR} = 0.042
 Diversity = 7.5137e+00 k_t = [4, 2, 3, 3]
 Generation 1750: Best = 1.308295e+03 FES = 175100 μ_F = 0.577 μ_{CR} = 0.038
 Diversity = 7.4589e+00 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 1.308295e+03 FES = 176100 μ_F = 0.551 μ_{CR} = 0.052
 Diversity = 7.4064e+00 k_t = [3, 4, 3, 4]
 Generation 1770: Best = 1.306776e+03 FES = 177100 μ_F = 0.533 μ_{CR} = 0.044
 Diversity = 7.3818e+00 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 1.303758e+03 FES = 178100 μ_F = 0.522 μ_{CR} = 0.046
 Diversity = 7.3868e+00 k_t = [3, 2, 2, 2]
 Generation 1790: Best = 1.300525e+03 FES = 179100 μ_F = 0.483 μ_{CR} = 0.060
 Diversity = 7.3928e+00 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 1.299179e+03 FES = 180100 μ_F = 0.513 μ_{CR} = 0.055
 Diversity = 7.4003e+00 k_t = [3, 2, 3, 4]
 Generation 1810: Best = 1.298776e+03 FES = 181100 μ_F = 0.480 μ_{CR} = 0.052
 Diversity = 7.4006e+00 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 1.296528e+03 FES = 182100 μ_F = 0.536 μ_{CR} = 0.057
 Diversity = 7.4029e+00 k_t = [4, 2, 4, 3]
 Generation 1830: Best = 1.294128e+03 FES = 183100 μ_F = 0.592 μ_{CR} = 0.067
 Diversity = 7.4063e+00 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 1.294128e+03 FES = 184100 μ_F = 0.601 μ_{CR} = 0.051
 Diversity = 7.4056e+00 k_t = [2, 3, 3, 2]

Generation 1850: Best = 1.292270e+03 FES = 185100 μ_F = 0.601 μ_{CR} = 0.053
 Diversity = 7.3630e+00 k_t = [1, 1, 1, 1]
 Generation 1860: Best = 1.286106e+03 FES = 186100 μ_F = 0.677 μ_{CR} = 0.045
 Diversity = 7.2408e+00 k_t = [2, 4, 3, 4]
 Generation 1870: Best = 1.280632e+03 FES = 187100 μ_F = 0.750 μ_{CR} = 0.048
 Diversity = 7.2325e+00 k_t = [1, 1, 1, 1]
 Generation 1880: Best = 1.277545e+03 FES = 188100 μ_F = 0.694 μ_{CR} = 0.041
 Diversity = 7.2371e+00 k_t = [2, 2, 2, 2]
 Generation 1890: Best = 1.277545e+03 FES = 189100 μ_F = 0.741 μ_{CR} = 0.062
 Diversity = 7.2319e+00 k_t = [1, 1, 1, 1]
 Generation 1900: Best = 1.272290e+03 FES = 190100 μ_F = 0.768 μ_{CR} = 0.053
 Diversity = 7.1774e+00 k_t = [4, 3, 2, 3]
 Generation 1910: Best = 1.268338e+03 FES = 191100 μ_F = 0.702 μ_{CR} = 0.090
 Diversity = 7.1737e+00 k_t = [1, 1, 1, 1]
 Generation 1920: Best = 1.268187e+03 FES = 192100 μ_F = 0.740 μ_{CR} = 0.090
 Diversity = 7.1510e+00 k_t = [2, 4, 2, 3]
 Generation 1930: Best = 1.266662e+03 FES = 193100 μ_F = 0.727 μ_{CR} = 0.082
 Diversity = 7.1350e+00 k_t = [1, 1, 1, 1]
 Generation 1940: Best = 1.261056e+03 FES = 194100 μ_F = 0.640 μ_{CR} = 0.101
 Diversity = 7.1359e+00 k_t = [4, 2, 3, 3]
 Generation 1950: Best = 1.255574e+03 FES = 195100 μ_F = 0.665 μ_{CR} = 0.115
 Diversity = 7.1123e+00 k_t = [1, 1, 1, 1]
 Generation 1960: Best = 1.246548e+03 FES = 196100 μ_F = 0.687 μ_{CR} = 0.131
 Diversity = 7.0864e+00 k_t = [4, 2, 3, 3]
 Generation 1970: Best = 1.243428e+03 FES = 197100 μ_F = 0.710 μ_{CR} = 0.138
 Diversity = 7.0486e+00 k_t = [1, 1, 1, 1]
 Generation 1980: Best = 1.235689e+03 FES = 198100 μ_F = 0.707 μ_{CR} = 0.135
 Diversity = 7.0299e+00 k_t = [3, 2, 4, 3]
 Generation 1990: Best = 1.225784e+03 FES = 199100 μ_F = 0.722 μ_{CR} = 0.154
 Diversity = 6.8393e+00 k_t = [1, 1, 1, 1]
 Generation 2000: Best = 1.222100e+03 FES = 200100 μ_F = 0.710 μ_{CR} = 0.168
 Diversity = 6.6675e+00 k_t = [3, 3, 3, 4]
 Generation 2010: Best = 1.212527e+03 FES = 201100 μ_F = 0.738 μ_{CR} = 0.171
 Diversity = 6.5173e+00 k_t = [1, 1, 1, 1]
 Generation 2020: Best = 1.206178e+03 FES = 202100 μ_F = 0.752 μ_{CR} = 0.170
 Diversity = 6.1524e+00 k_t = [3, 4, 2, 3]
 Generation 2030: Best = 1.199233e+03 FES = 203100 μ_F = 0.765 μ_{CR} = 0.179
 Diversity = 5.6410e+00 k_t = [1, 1, 1, 1]
 Generation 2040: Best = 1.192714e+03 FES = 204100 μ_F = 0.780 μ_{CR} = 0.186
 Diversity = 3.3697e+00 k_t = [2, 2, 3, 3]
 Generation 2050: Best = 1.191577e+03 FES = 205100 μ_F = 0.776 μ_{CR} = 0.191
 Diversity = 4.4691e-01 k_t = [1, 1, 1, 1]
 Generation 2060: Best = 1.185673e+03 FES = 206100 μ_F = 0.740 μ_{CR} = 0.191
 Diversity = 3.4382e-01 k_t = [2, 3, 2, 2]
 Generation 2070: Best = 1.179265e+03 FES = 207100 μ_F = 0.723 μ_{CR} = 0.187
 Diversity = 3.1453e-01 k_t = [1, 1, 1, 1]
 Generation 2080: Best = 1.174595e+03 FES = 208100 μ_F = 0.711 μ_{CR} = 0.193
 Diversity = 2.7996e-01 k_t = [4, 2, 2, 2]
 Generation 2090: Best = 1.173042e+03 FES = 209100 μ_F = 0.698 μ_{CR} = 0.192
 Diversity = 2.4626e-01 k_t = [1, 1, 1, 1]
 Generation 2100: Best = 1.169993e+03 FES = 210100 μ_F = 0.678 μ_{CR} = 0.188
 Diversity = 1.9591e-01 k_t = [3, 4, 2, 4]
 Generation 2110: Best = 1.167952e+03 FES = 211100 μ_F = 0.681 μ_{CR} = 0.184
 Diversity = 1.6492e-01 k_t = [1, 1, 1, 1]

Generation 2120: Best = 1.166666e+03 FES = 212100 μ_F = 0.673 μ_{CR} = 0.192
 Diversity = 1.4452e-01 k_t = [2, 3, 3, 2]
 Generation 2130: Best = 1.165257e+03 FES = 213100 μ_F = 0.667 μ_{CR} = 0.191
 Diversity = 1.3152e-01 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 1.164307e+03 FES = 214100 μ_F = 0.672 μ_{CR} = 0.195
 Diversity = 1.1587e-01 k_t = [3, 4, 2, 2]
 Generation 2150: Best = 1.163463e+03 FES = 215100 μ_F = 0.682 μ_{CR} = 0.196
 Diversity = 1.0606e-01 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 1.162872e+03 FES = 216100 μ_F = 0.684 μ_{CR} = 0.208
 Diversity = 9.3639e-02 k_t = [2, 4, 2, 2]
 Generation 2170: Best = 1.162018e+03 FES = 217100 μ_F = 0.687 μ_{CR} = 0.210
 Diversity = 8.5102e-02 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 1.161840e+03 FES = 218100 μ_F = 0.679 μ_{CR} = 0.216
 Diversity = 7.5559e-02 k_t = [3, 4, 3, 2]
 Generation 2190: Best = 1.161511e+03 FES = 219100 μ_F = 0.671 μ_{CR} = 0.214
 Diversity = 6.7332e-02 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 1.161345e+03 FES = 220100 μ_F = 0.677 μ_{CR} = 0.215
 Diversity = 5.8030e-02 k_t = [2, 3, 4, 3]
 Generation 2210: Best = 1.160983e+03 FES = 221100 μ_F = 0.676 μ_{CR} = 0.211
 Diversity = 5.2989e-02 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 1.160831e+03 FES = 222100 μ_F = 0.686 μ_{CR} = 0.211
 Diversity = 5.0559e-02 k_t = [4, 2, 2, 2]
 Generation 2230: Best = 1.160542e+03 FES = 223100 μ_F = 0.691 μ_{CR} = 0.208
 Diversity = 4.6438e-02 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 1.160411e+03 FES = 224100 μ_F = 0.687 μ_{CR} = 0.210
 Diversity = 4.0508e-02 k_t = [3, 2, 3, 3]
 Generation 2250: Best = 1.160263e+03 FES = 225100 μ_F = 0.679 μ_{CR} = 0.225
 Diversity = 3.7365e-02 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 1.159942e+03 FES = 226100 μ_F = 0.694 μ_{CR} = 0.221
 Diversity = 3.5504e-02 k_t = [3, 4, 4, 4]
 Generation 2270: Best = 1.159897e+03 FES = 227100 μ_F = 0.705 μ_{CR} = 0.223
 Diversity = 3.2459e-02 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 1.159779e+03 FES = 228100 μ_F = 0.720 μ_{CR} = 0.222
 Diversity = 2.9674e-02 k_t = [4, 2, 3, 4]
 Generation 2290: Best = 1.159644e+03 FES = 229100 μ_F = 0.731 μ_{CR} = 0.219
 Diversity = 3.1459e-02 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 1.159512e+03 FES = 230100 μ_F = 0.727 μ_{CR} = 0.221
 Diversity = 3.1198e-02 k_t = [2, 4, 2, 3]
 Generation 2310: Best = 1.159445e+03 FES = 231100 μ_F = 0.714 μ_{CR} = 0.221
 Diversity = 3.2338e-02 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 1.159293e+03 FES = 232100 μ_F = 0.706 μ_{CR} = 0.224
 Diversity = 2.9997e-02 k_t = [3, 3, 4, 2]
 Generation 2330: Best = 1.159212e+03 FES = 233100 μ_F = 0.713 μ_{CR} = 0.224
 Diversity = 2.9780e-02 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 1.159122e+03 FES = 234100 μ_F = 0.693 μ_{CR} = 0.229
 Diversity = 2.7267e-02 k_t = [3, 4, 3, 2]
 Generation 2350: Best = 1.159054e+03 FES = 235100 μ_F = 0.702 μ_{CR} = 0.221
 Diversity = 2.4626e-02 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 1.159008e+03 FES = 236100 μ_F = 0.689 μ_{CR} = 0.219
 Diversity = 2.2245e-02 k_t = [2, 3, 2, 3]
 Generation 2370: Best = 1.158949e+03 FES = 237100 μ_F = 0.700 μ_{CR} = 0.224
 Diversity = 2.2440e-02 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 1.158916e+03 FES = 238100 μ_F = 0.693 μ_{CR} = 0.219
 Diversity = 2.0804e-02 k_t = [2, 2, 2, 3]

Generation 2390: Best = 1.158892e+03 FES = 239100 μ_F = 0.692 μ_{CR} = 0.219
 Diversity = 1.9764e-02 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 1.158843e+03 FES = 240100 μ_F = 0.688 μ_{CR} = 0.220
 Diversity = 1.6334e-02 k_t = [4, 2, 2, 4]
 Generation 2410: Best = 1.158818e+03 FES = 241100 μ_F = 0.699 μ_{CR} = 0.241
 Diversity = 1.4383e-02 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 1.158798e+03 FES = 242100 μ_F = 0.715 μ_{CR} = 0.243
 Diversity = 1.3266e-02 k_t = [4, 3, 2, 3]
 Generation 2430: Best = 1.158779e+03 FES = 243100 μ_F = 0.714 μ_{CR} = 0.248
 Diversity = 1.3248e-02 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 1.158759e+03 FES = 244100 μ_F = 0.717 μ_{CR} = 0.253
 Diversity = 1.2881e-02 k_t = [3, 4, 4, 2]
 Generation 2450: Best = 1.158741e+03 FES = 245100 μ_F = 0.720 μ_{CR} = 0.247
 Diversity = 1.2367e-02 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 1.158721e+03 FES = 246100 μ_F = 0.725 μ_{CR} = 0.231
 Diversity = 1.2286e-02 k_t = [3, 4, 3, 3]
 Generation 2470: Best = 1.158692e+03 FES = 247100 μ_F = 0.726 μ_{CR} = 0.231
 Diversity = 1.2147e-02 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 1.158673e+03 FES = 248100 μ_F = 0.714 μ_{CR} = 0.238
 Diversity = 1.2936e-02 k_t = [4, 4, 4, 2]
 Generation 2490: Best = 1.158659e+03 FES = 249100 μ_F = 0.712 μ_{CR} = 0.228
 Diversity = 1.2818e-02 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 1.158642e+03 FES = 250100 μ_F = 0.727 μ_{CR} = 0.227
 Diversity = 1.3690e-02 k_t = [2, 2, 3, 4]
 Generation 2510: Best = 1.158619e+03 FES = 251100 μ_F = 0.715 μ_{CR} = 0.222
 Diversity = 1.4389e-02 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 1.158588e+03 FES = 252100 μ_F = 0.720 μ_{CR} = 0.229
 Diversity = 1.2805e-02 k_t = [2, 2, 2, 3]
 Generation 2530: Best = 1.158575e+03 FES = 253100 μ_F = 0.711 μ_{CR} = 0.223
 Diversity = 1.2682e-02 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 1.158551e+03 FES = 254100 μ_F = 0.703 μ_{CR} = 0.229
 Diversity = 1.2229e-02 k_t = [4, 2, 4, 2]
 Generation 2550: Best = 1.158524e+03 FES = 255100 μ_F = 0.711 μ_{CR} = 0.234
 Diversity = 1.2817e-02 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 1.158503e+03 FES = 256100 μ_F = 0.706 μ_{CR} = 0.235
 Diversity = 1.2217e-02 k_t = [4, 3, 4, 4]
 Generation 2570: Best = 1.158476e+03 FES = 257100 μ_F = 0.717 μ_{CR} = 0.236
 Diversity = 1.2115e-02 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 1.158454e+03 FES = 258100 μ_F = 0.729 μ_{CR} = 0.243
 Diversity = 1.1842e-02 k_t = [2, 2, 2, 4]
 Generation 2590: Best = 1.158425e+03 FES = 259100 μ_F = 0.729 μ_{CR} = 0.236
 Diversity = 1.1698e-02 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 1.158419e+03 FES = 260100 μ_F = 0.737 μ_{CR} = 0.230
 Diversity = 1.1228e-02 k_t = [3, 4, 3, 3]
 Generation 2610: Best = 1.158396e+03 FES = 261100 μ_F = 0.742 μ_{CR} = 0.230
 Diversity = 1.1217e-02 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 1.158388e+03 FES = 262100 μ_F = 0.730 μ_{CR} = 0.229
 Diversity = 1.0527e-02 k_t = [2, 3, 4, 4]
 Generation 2630: Best = 1.158369e+03 FES = 263100 μ_F = 0.736 μ_{CR} = 0.234
 Diversity = 1.0932e-02 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 1.158349e+03 FES = 264100 μ_F = 0.731 μ_{CR} = 0.228
 Diversity = 1.0466e-02 k_t = [3, 3, 3, 4]
 Generation 2650: Best = 1.158335e+03 FES = 265100 μ_F = 0.728 μ_{CR} = 0.228
 Diversity = 9.7742e-03 k_t = [1, 1, 1, 1]

Generation 2660: Best = 1.158324e+03 FES = 266100 μ_F = 0.719 μ_{CR} = 0.231
 Diversity = 9.0502e-03 k_t = [3, 4, 4, 2]
 Generation 2670: Best = 1.158316e+03 FES = 267100 μ_F = 0.709 μ_{CR} = 0.235
 Diversity = 8.5859e-03 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 1.158302e+03 FES = 268100 μ_F = 0.719 μ_{CR} = 0.236
 Diversity = 8.4534e-03 k_t = [4, 4, 2, 3]
 Generation 2690: Best = 1.158294e+03 FES = 269100 μ_F = 0.715 μ_{CR} = 0.237
 Diversity = 8.5836e-03 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 1.158286e+03 FES = 270100 μ_F = 0.717 μ_{CR} = 0.240
 Diversity = 8.1291e-03 k_t = [3, 2, 4, 3]
 Generation 2710: Best = 1.158278e+03 FES = 271100 μ_F = 0.721 μ_{CR} = 0.238
 Diversity = 8.3883e-03 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 1.158269e+03 FES = 272100 μ_F = 0.718 μ_{CR} = 0.228
 Diversity = 7.4927e-03 k_t = [3, 4, 3, 3]
 Generation 2730: Best = 1.158263e+03 FES = 273100 μ_F = 0.718 μ_{CR} = 0.224
 Diversity = 6.7448e-03 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 1.158260e+03 FES = 274100 μ_F = 0.715 μ_{CR} = 0.220
 Diversity = 5.7836e-03 k_t = [4, 4, 4, 3]
 Generation 2750: Best = 1.158256e+03 FES = 275100 μ_F = 0.708 μ_{CR} = 0.223
 Diversity = 4.9543e-03 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 1.158253e+03 FES = 276100 μ_F = 0.721 μ_{CR} = 0.224
 Diversity = 4.5071e-03 k_t = [2, 3, 2, 4]
 Generation 2770: Best = 1.158251e+03 FES = 277100 μ_F = 0.716 μ_{CR} = 0.209
 Diversity = 4.4489e-03 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 1.158248e+03 FES = 278100 μ_F = 0.716 μ_{CR} = 0.217
 Diversity = 4.2044e-03 k_t = [4, 4, 4, 4]
 Generation 2790: Best = 1.158245e+03 FES = 279100 μ_F = 0.723 μ_{CR} = 0.217
 Diversity = 4.1352e-03 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 1.158243e+03 FES = 280100 μ_F = 0.735 μ_{CR} = 0.220
 Diversity = 3.8799e-03 k_t = [3, 3, 4, 4]
 Generation 2810: Best = 1.158240e+03 FES = 281100 μ_F = 0.736 μ_{CR} = 0.210
 Diversity = 4.2346e-03 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 1.158237e+03 FES = 282100 μ_F = 0.751 μ_{CR} = 0.199
 Diversity = 4.1607e-03 k_t = [3, 3, 2, 3]
 Generation 2830: Best = 1.158233e+03 FES = 283100 μ_F = 0.743 μ_{CR} = 0.198
 Diversity = 4.3652e-03 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 1.158231e+03 FES = 284100 μ_F = 0.736 μ_{CR} = 0.203
 Diversity = 4.1403e-03 k_t = [3, 2, 3, 3]
 Generation 2850: Best = 1.158229e+03 FES = 285100 μ_F = 0.734 μ_{CR} = 0.202
 Diversity = 4.2744e-03 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 1.158227e+03 FES = 286100 μ_F = 0.737 μ_{CR} = 0.199
 Diversity = 3.8242e-03 k_t = [3, 3, 3, 4]
 Generation 2870: Best = 1.158225e+03 FES = 287100 μ_F = 0.747 μ_{CR} = 0.202
 Diversity = 3.7683e-03 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 1.158224e+03 FES = 288100 μ_F = 0.750 μ_{CR} = 0.209
 Diversity = 3.7586e-03 k_t = [3, 3, 4, 2]
 Generation 2890: Best = 1.158222e+03 FES = 289100 μ_F = 0.738 μ_{CR} = 0.205
 Diversity = 3.6390e-03 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 1.158221e+03 FES = 290100 μ_F = 0.738 μ_{CR} = 0.201
 Diversity = 3.4030e-03 k_t = [4, 4, 4, 3]
 Generation 2910: Best = 1.158219e+03 FES = 291100 μ_F = 0.738 μ_{CR} = 0.207
 Diversity = 3.5077e-03 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 1.158217e+03 FES = 292100 μ_F = 0.739 μ_{CR} = 0.215
 Diversity = 3.3762e-03 k_t = [3, 2, 4, 3]

Generation 2930: Best = 1.158216e+03 FES = 293100 μ_F = 0.735 μ_{CR} = 0.223
 Diversity = 3.4606e-03 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 1.158214e+03 FES = 294100 μ_F = 0.735 μ_{CR} = 0.225
 Diversity = 3.5157e-03 k_t = [4, 4, 3, 2]
 Generation 2950: Best = 1.158212e+03 FES = 295100 μ_F = 0.711 μ_{CR} = 0.223
 Diversity = 3.3412e-03 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 1.158210e+03 FES = 296100 μ_F = 0.717 μ_{CR} = 0.229
 Diversity = 3.3735e-03 k_t = [4, 3, 3, 3]
 Generation 2970: Best = 1.158210e+03 FES = 297100 μ_F = 0.708 μ_{CR} = 0.221
 Diversity = 3.3364e-03 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 1.158208e+03 FES = 298100 μ_F = 0.713 μ_{CR} = 0.224
 Diversity = 2.9422e-03 k_t = [3, 3, 3, 4]
 Generation 2990: Best = 1.158207e+03 FES = 299100 μ_F = 0.723 μ_{CR} = 0.222
 Diversity = 2.9307e-03 k_t = [1, 1, 1, 1]
 Generation 3000: Best = 1.158206e+03 FES = 300100 μ_F = 0.729 μ_{CR} = 0.224
 Diversity = 3.0769e-03 k_t = [3, 3, 3, 2]
 Generation 3010: Best = 1.158205e+03 FES = 301100 μ_F = 0.729 μ_{CR} = 0.222
 Diversity = 2.8961e-03 k_t = [1, 1, 1, 1]
 Generation 3020: Best = 1.158204e+03 FES = 302100 μ_F = 0.720 μ_{CR} = 0.218
 Diversity = 2.5635e-03 k_t = [4, 3, 3, 3]
 Generation 3030: Best = 1.158203e+03 FES = 303100 μ_F = 0.722 μ_{CR} = 0.220
 Diversity = 2.4917e-03 k_t = [1, 1, 1, 1]
 Generation 3040: Best = 1.158203e+03 FES = 304100 μ_F = 0.707 μ_{CR} = 0.213
 Diversity = 2.3054e-03 k_t = [2, 2, 3, 4]
 Generation 3050: Best = 1.158202e+03 FES = 305100 μ_F = 0.713 μ_{CR} = 0.217
 Diversity = 2.1964e-03 k_t = [1, 1, 1, 1]
 Generation 3060: Best = 1.158202e+03 FES = 306100 μ_F = 0.713 μ_{CR} = 0.221
 Diversity = 2.0176e-03 k_t = [4, 3, 4, 2]
 Generation 3070: Best = 1.158202e+03 FES = 307100 μ_F = 0.726 μ_{CR} = 0.210
 Diversity = 1.9839e-03 k_t = [1, 1, 1, 1]
 Generation 3080: Best = 1.158201e+03 FES = 308100 μ_F = 0.723 μ_{CR} = 0.208
 Diversity = 1.8144e-03 k_t = [4, 3, 2, 4]
 Generation 3090: Best = 1.158201e+03 FES = 309100 μ_F = 0.711 μ_{CR} = 0.211
 Diversity = 1.8803e-03 k_t = [1, 1, 1, 1]
 Generation 3100: Best = 1.158200e+03 FES = 310100 μ_F = 0.707 μ_{CR} = 0.212
 Diversity = 1.7232e-03 k_t = [4, 2, 2, 3]
 Generation 3110: Best = 1.158200e+03 FES = 311100 μ_F = 0.715 μ_{CR} = 0.204
 Diversity = 1.7039e-03 k_t = [1, 1, 1, 1]
 Generation 3120: Best = 1.158200e+03 FES = 312100 μ_F = 0.720 μ_{CR} = 0.209
 Diversity = 1.6143e-03 k_t = [2, 2, 4, 2]
 Generation 3130: Best = 1.158199e+03 FES = 313100 μ_F = 0.724 μ_{CR} = 0.203
 Diversity = 1.5865e-03 k_t = [1, 1, 1, 1]
 Generation 3140: Best = 1.158199e+03 FES = 314100 μ_F = 0.722 μ_{CR} = 0.213
 Diversity = 1.6014e-03 k_t = [3, 4, 2, 3]
 Generation 3150: Best = 1.158199e+03 FES = 315100 μ_F = 0.705 μ_{CR} = 0.221
 Diversity = 1.5152e-03 k_t = [1, 1, 1, 1]
 Generation 3160: Best = 1.158198e+03 FES = 316100 μ_F = 0.705 μ_{CR} = 0.226
 Diversity = 1.3675e-03 k_t = [4, 2, 4, 3]
 Generation 3170: Best = 1.158198e+03 FES = 317100 μ_F = 0.710 μ_{CR} = 0.229
 Diversity = 1.3331e-03 k_t = [1, 1, 1, 1]
 Generation 3180: Best = 1.158198e+03 FES = 318100 μ_F = 0.712 μ_{CR} = 0.225
 Diversity = 1.3041e-03 k_t = [3, 4, 4, 3]
 Generation 3190: Best = 1.158198e+03 FES = 319100 μ_F = 0.717 μ_{CR} = 0.227
 Diversity = 1.2414e-03 k_t = [1, 1, 1, 1]

Generation 3200: Best = 1.158198e+03 FES = 320100 μ_F = 0.720 μ_{CR} = 0.231
 Diversity = 1.1908e-03 k_t = [3, 3, 2, 3]
 Generation 3210: Best = 1.158197e+03 FES = 321100 μ_F = 0.719 μ_{CR} = 0.225
 Diversity = 1.2162e-03 k_t = [1, 1, 1, 1]
 Generation 3220: Best = 1.158197e+03 FES = 322100 μ_F = 0.710 μ_{CR} = 0.228
 Diversity = 1.2294e-03 k_t = [2, 4, 3, 3]
 Generation 3230: Best = 1.158197e+03 FES = 323100 μ_F = 0.708 μ_{CR} = 0.233
 Diversity = 1.2382e-03 k_t = [1, 1, 1, 1]
 Generation 3240: Best = 1.158197e+03 FES = 324100 μ_F = 0.703 μ_{CR} = 0.221
 Diversity = 1.1319e-03 k_t = [3, 2, 3, 2]
 Generation 3250: Best = 1.158197e+03 FES = 325100 μ_F = 0.705 μ_{CR} = 0.220
 Diversity = 1.2072e-03 k_t = [1, 1, 1, 1]
 Generation 3260: Best = 1.158196e+03 FES = 326100 μ_F = 0.691 μ_{CR} = 0.216
 Diversity = 1.1619e-03 k_t = [3, 4, 4, 2]
 Generation 3270: Best = 1.158196e+03 FES = 327100 μ_F = 0.687 μ_{CR} = 0.226
 Diversity = 1.1001e-03 k_t = [1, 1, 1, 1]
 Generation 3280: Best = 1.158196e+03 FES = 328100 μ_F = 0.696 μ_{CR} = 0.230
 Diversity = 9.3619e-04 k_t = [4, 2, 3, 3]
 Generation 3290: Best = 1.158196e+03 FES = 329100 μ_F = 0.696 μ_{CR} = 0.228
 Diversity = 8.8152e-04 k_t = [1, 1, 1, 1]
 Generation 3300: Best = 1.158196e+03 FES = 330100 μ_F = 0.703 μ_{CR} = 0.228
 Diversity = 8.0122e-04 k_t = [2, 3, 4, 2]
 Generation 3310: Best = 1.158196e+03 FES = 331100 μ_F = 0.706 μ_{CR} = 0.239
 Diversity = 7.9261e-04 k_t = [1, 1, 1, 1]
 Generation 3320: Best = 1.158196e+03 FES = 332100 μ_F = 0.702 μ_{CR} = 0.241
 Diversity = 7.6748e-04 k_t = [3, 4, 2, 4]
 Generation 3330: Best = 1.158196e+03 FES = 333100 μ_F = 0.706 μ_{CR} = 0.243
 Diversity = 7.7644e-04 k_t = [1, 1, 1, 1]
 Generation 3340: Best = 1.158196e+03 FES = 334100 μ_F = 0.712 μ_{CR} = 0.246
 Diversity = 7.6057e-04 k_t = [3, 4, 3, 4]
 Generation 3350: Best = 1.158195e+03 FES = 335100 μ_F = 0.722 μ_{CR} = 0.248
 Diversity = 7.6117e-04 k_t = [1, 1, 1, 1]
 Generation 3360: Best = 1.158195e+03 FES = 336100 μ_F = 0.721 μ_{CR} = 0.248
 Diversity = 6.8002e-04 k_t = [2, 4, 4, 4]
 Generation 3370: Best = 1.158195e+03 FES = 337100 μ_F = 0.728 μ_{CR} = 0.249
 Diversity = 7.3250e-04 k_t = [1, 1, 1, 1]
 Generation 3380: Best = 1.158195e+03 FES = 338100 μ_F = 0.734 μ_{CR} = 0.239
 Diversity = 6.9158e-04 k_t = [2, 3, 3, 3]
 Generation 3390: Best = 1.158195e+03 FES = 339100 μ_F = 0.726 μ_{CR} = 0.240
 Diversity = 7.2159e-04 k_t = [1, 1, 1, 1]
 Generation 3400: Best = 1.158195e+03 FES = 340100 μ_F = 0.741 μ_{CR} = 0.234
 Diversity = 7.5749e-04 k_t = [3, 4, 3, 3]
 Generation 3410: Best = 1.158195e+03 FES = 341100 μ_F = 0.752 μ_{CR} = 0.238
 Diversity = 8.9051e-04 k_t = [1, 1, 1, 1]
 Generation 3420: Best = 1.158195e+03 FES = 342100 μ_F = 0.745 μ_{CR} = 0.237
 Diversity = 9.9813e-04 k_t = [3, 4, 4, 2]
 Generation 3430: Best = 1.158194e+03 FES = 343100 μ_F = 0.737 μ_{CR} = 0.235
 Diversity = 1.0142e-03 k_t = [1, 1, 1, 1]
 Generation 3440: Best = 1.158194e+03 FES = 344100 μ_F = 0.735 μ_{CR} = 0.241
 Diversity = 1.0021e-03 k_t = [4, 2, 4, 3]
 Generation 3450: Best = 1.158194e+03 FES = 345100 μ_F = 0.723 μ_{CR} = 0.248
 Diversity = 1.0236e-03 k_t = [1, 1, 1, 1]
 Generation 3460: Best = 1.158194e+03 FES = 346100 μ_F = 0.722 μ_{CR} = 0.256
 Diversity = 9.1624e-04 k_t = [2, 3, 2, 4]

Generation 3470: Best = 1.158194e+03 FES = 347100 μ_F = 0.714 μ_{CR} = 0.257
 Diversity = 8.9741e-04 k_t = [1, 1, 1, 1]
 Generation 3480: Best = 1.158194e+03 FES = 348100 μ_F = 0.725 μ_{CR} = 0.258
 Diversity = 8.5010e-04 k_t = [3, 2, 3, 2]
 Generation 3490: Best = 1.158194e+03 FES = 349100 μ_F = 0.721 μ_{CR} = 0.258
 Diversity = 9.3952e-04 k_t = [1, 1, 1, 1]
 Generation 3500: Best = 1.158194e+03 FES = 350100 μ_F = 0.722 μ_{CR} = 0.256
 Diversity = 8.8855e-04 k_t = [2, 4, 3, 3]
 Generation 3510: Best = 1.158193e+03 FES = 351100 μ_F = 0.711 μ_{CR} = 0.252
 Diversity = 8.6527e-04 k_t = [1, 1, 1, 1]
 Generation 3520: Best = 1.158193e+03 FES = 352100 μ_F = 0.711 μ_{CR} = 0.253
 Diversity = 8.1246e-04 k_t = [4, 2, 2, 3]
 Generation 3530: Best = 1.158193e+03 FES = 353100 μ_F = 0.709 μ_{CR} = 0.247
 Diversity = 8.2051e-04 k_t = [1, 1, 1, 1]
 Generation 3540: Best = 1.158193e+03 FES = 354100 μ_F = 0.722 μ_{CR} = 0.238
 Diversity = 7.8529e-04 k_t = [4, 4, 3, 4]
 Generation 3550: Best = 1.158193e+03 FES = 355100 μ_F = 0.725 μ_{CR} = 0.240
 Diversity = 7.3976e-04 k_t = [1, 1, 1, 1]
 Generation 3560: Best = 1.158193e+03 FES = 356100 μ_F = 0.723 μ_{CR} = 0.249
 Diversity = 7.0283e-04 k_t = [4, 4, 4, 2]
 Generation 3570: Best = 1.158193e+03 FES = 357100 μ_F = 0.711 μ_{CR} = 0.237
 Diversity = 6.7580e-04 k_t = [1, 1, 1, 1]
 Generation 3580: Best = 1.158193e+03 FES = 358100 μ_F = 0.717 μ_{CR} = 0.236
 Diversity = 5.8596e-04 k_t = [4, 4, 3, 4]
 Generation 3590: Best = 1.158193e+03 FES = 359100 μ_F = 0.727 μ_{CR} = 0.240
 Diversity = 6.4030e-04 k_t = [1, 1, 1, 1]
 Generation 3600: Best = 1.158193e+03 FES = 360100 μ_F = 0.734 μ_{CR} = 0.251
 Diversity = 6.5371e-04 k_t = [3, 3, 3, 4]
 Generation 3610: Best = 1.158193e+03 FES = 361100 μ_F = 0.739 μ_{CR} = 0.242
 Diversity = 7.3402e-04 k_t = [1, 1, 1, 1]
 Generation 3620: Best = 1.158193e+03 FES = 362100 μ_F = 0.742 μ_{CR} = 0.244
 Diversity = 7.2482e-04 k_t = [3, 3, 2, 2]
 Generation 3630: Best = 1.158192e+03 FES = 363100 μ_F = 0.740 μ_{CR} = 0.229
 Diversity = 7.2973e-04 k_t = [1, 1, 1, 1]
 Generation 3640: Best = 1.158192e+03 FES = 364100 μ_F = 0.718 μ_{CR} = 0.229
 Diversity = 6.7380e-04 k_t = [2, 2, 2, 3]
 Generation 3650: Best = 1.158192e+03 FES = 365100 μ_F = 0.714 μ_{CR} = 0.224
 Diversity = 7.0431e-04 k_t = [1, 1, 1, 1]
 Generation 3660: Best = 1.158192e+03 FES = 366100 μ_F = 0.731 μ_{CR} = 0.238
 Diversity = 7.5094e-04 k_t = [4, 3, 3, 4]
 Generation 3670: Best = 1.158192e+03 FES = 367100 μ_F = 0.734 μ_{CR} = 0.252
 Diversity = 7.6981e-04 k_t = [1, 1, 1, 1]
 Generation 3680: Best = 1.158192e+03 FES = 368100 μ_F = 0.731 μ_{CR} = 0.258
 Diversity = 7.6255e-04 k_t = [4, 3, 4, 2]
 Generation 3690: Best = 1.158192e+03 FES = 369100 μ_F = 0.724 μ_{CR} = 0.259
 Diversity = 8.3747e-04 k_t = [1, 1, 1, 1]
 Generation 3700: Best = 1.158192e+03 FES = 370100 μ_F = 0.730 μ_{CR} = 0.250
 Diversity = 8.0656e-04 k_t = [2, 2, 4, 4]
 Generation 3710: Best = 1.158192e+03 FES = 371100 μ_F = 0.726 μ_{CR} = 0.253
 Diversity = 8.8599e-04 k_t = [1, 1, 1, 1]
 Generation 3720: Best = 1.158191e+03 FES = 372100 μ_F = 0.732 μ_{CR} = 0.261
 Diversity = 9.4143e-04 k_t = [4, 3, 3, 3]
 Generation 3730: Best = 1.158191e+03 FES = 373100 μ_F = 0.741 μ_{CR} = 0.265
 Diversity = 1.1831e-03 k_t = [1, 1, 1, 1]

Generation 3740: Best = 1.158190e+03 FES = 374100 μ_F = 0.733 μ_{CR} = 0.261
 Diversity = 1.1929e-03 k_t = [3, 4, 2, 2]
 Generation 3750: Best = 1.158190e+03 FES = 375100 μ_F = 0.736 μ_{CR} = 0.266
 Diversity = 1.1864e-03 k_t = [1, 1, 1, 1]
 Generation 3760: Best = 1.158190e+03 FES = 376100 μ_F = 0.732 μ_{CR} = 0.274
 Diversity = 1.0323e-03 k_t = [4, 3, 2, 4]
 Generation 3770: Best = 1.158190e+03 FES = 377100 μ_F = 0.730 μ_{CR} = 0.268
 Diversity = 9.9264e-04 k_t = [1, 1, 1, 1]
 Generation 3780: Best = 1.158190e+03 FES = 378100 μ_F = 0.720 μ_{CR} = 0.265
 Diversity = 1.0237e-03 k_t = [4, 2, 3, 3]
 Generation 3790: Best = 1.158190e+03 FES = 379100 μ_F = 0.710 μ_{CR} = 0.266
 Diversity = 1.0078e-03 k_t = [1, 1, 1, 1]
 Generation 3800: Best = 1.158189e+03 FES = 380100 μ_F = 0.709 μ_{CR} = 0.272
 Diversity = 8.9228e-04 k_t = [3, 2, 3, 3]
 Generation 3810: Best = 1.158189e+03 FES = 381100 μ_F = 0.691 μ_{CR} = 0.264
 Diversity = 8.3737e-04 k_t = [1, 1, 1, 1]
 Generation 3820: Best = 1.158189e+03 FES = 382100 μ_F = 0.681 μ_{CR} = 0.258
 Diversity = 6.8416e-04 k_t = [4, 3, 2, 4]
 Generation 3830: Best = 1.158189e+03 FES = 383100 μ_F = 0.684 μ_{CR} = 0.243
 Diversity = 5.5796e-04 k_t = [1, 1, 1, 1]
 Generation 3840: Best = 1.158189e+03 FES = 384100 μ_F = 0.692 μ_{CR} = 0.234
 Diversity = 4.6546e-04 k_t = [3, 3, 4, 3]
 Generation 3850: Best = 1.158189e+03 FES = 385100 μ_F = 0.674 μ_{CR} = 0.230
 Diversity = 4.4062e-04 k_t = [1, 1, 1, 1]
 Generation 3860: Best = 1.158189e+03 FES = 386100 μ_F = 0.701 μ_{CR} = 0.218
 Diversity = 4.3788e-04 k_t = [3, 2, 4, 3]
 Generation 3870: Best = 1.158189e+03 FES = 387100 μ_F = 0.715 μ_{CR} = 0.216
 Diversity = 4.5080e-04 k_t = [1, 1, 1, 1]
 Generation 3880: Best = 1.158189e+03 FES = 388100 μ_F = 0.710 μ_{CR} = 0.216
 Diversity = 4.2050e-04 k_t = [2, 2, 4, 4]
 Generation 3890: Best = 1.158189e+03 FES = 389100 μ_F = 0.705 μ_{CR} = 0.222
 Diversity = 4.2833e-04 k_t = [1, 1, 1, 1]
 Generation 3900: Best = 1.158189e+03 FES = 390100 μ_F = 0.716 μ_{CR} = 0.224
 Diversity = 4.0189e-04 k_t = [2, 3, 3, 3]
 Generation 3910: Best = 1.158189e+03 FES = 391100 μ_F = 0.709 μ_{CR} = 0.227
 Diversity = 3.8788e-04 k_t = [1, 1, 1, 1]
 Generation 3920: Best = 1.158189e+03 FES = 392100 μ_F = 0.712 μ_{CR} = 0.239
 Diversity = 3.7857e-04 k_t = [4, 4, 3, 3]
 Generation 3930: Best = 1.158189e+03 FES = 393100 μ_F = 0.718 μ_{CR} = 0.247
 Diversity = 3.5387e-04 k_t = [1, 1, 1, 1]
 Generation 3940: Best = 1.158189e+03 FES = 394100 μ_F = 0.718 μ_{CR} = 0.246
 Diversity = 3.5044e-04 k_t = [2, 4, 3, 2]
 Generation 3950: Best = 1.158189e+03 FES = 395100 μ_F = 0.729 μ_{CR} = 0.239
 Diversity = 3.8292e-04 k_t = [1, 1, 1, 1]
 Generation 3960: Best = 1.158189e+03 FES = 396100 μ_F = 0.739 μ_{CR} = 0.237
 Diversity = 4.0536e-04 k_t = [2, 4, 3, 2]
 Generation 3970: Best = 1.158189e+03 FES = 397100 μ_F = 0.727 μ_{CR} = 0.249
 Diversity = 4.2779e-04 k_t = [1, 1, 1, 1]
 Generation 3980: Best = 1.158189e+03 FES = 398100 μ_F = 0.713 μ_{CR} = 0.248
 Diversity = 3.8627e-04 k_t = [2, 3, 2, 4]
 Generation 3990: Best = 1.158189e+03 FES = 399100 μ_F = 0.702 μ_{CR} = 0.241
 Diversity = 3.5980e-04 k_t = [1, 1, 1, 1]
 Generation 4000: Best = 1.158189e+03 FES = 400100 μ_F = 0.715 μ_{CR} = 0.239
 Diversity = 3.5015e-04 k_t = [3, 2, 3, 2]

Generation 4010: Best = 1.158189e+03 FES = 401100 μ_F = 0.720 μ_{CR} = 0.230
 Diversity = 3.4815e-04 k_t = [1, 1, 1, 1]
 Generation 4020: Best = 1.158189e+03 FES = 402100 μ_F = 0.707 μ_{CR} = 0.238
 Diversity = 3.2235e-04 k_t = [3, 4, 2, 3]
 Generation 4030: Best = 1.158189e+03 FES = 403100 μ_F = 0.712 μ_{CR} = 0.235
 Diversity = 3.0101e-04 k_t = [1, 1, 1, 1]
 Generation 4040: Best = 1.158189e+03 FES = 404100 μ_F = 0.711 μ_{CR} = 0.237
 Diversity = 2.8451e-04 k_t = [3, 4, 4, 2]
 Generation 4050: Best = 1.158189e+03 FES = 405100 μ_F = 0.721 μ_{CR} = 0.240
 Diversity = 2.9195e-04 k_t = [1, 1, 1, 1]
 Generation 4060: Best = 1.158189e+03 FES = 406100 μ_F = 0.714 μ_{CR} = 0.232
 Diversity = 2.6741e-04 k_t = [4, 4, 4, 3]
 Generation 4070: Best = 1.158189e+03 FES = 407100 μ_F = 0.714 μ_{CR} = 0.234
 Diversity = 2.5861e-04 k_t = [1, 1, 1, 1]
 Generation 4080: Best = 1.158189e+03 FES = 408100 μ_F = 0.711 μ_{CR} = 0.240
 Diversity = 2.3396e-04 k_t = [4, 4, 2, 4]
 Generation 4090: Best = 1.158189e+03 FES = 409100 μ_F = 0.710 μ_{CR} = 0.246
 Diversity = 2.3154e-04 k_t = [1, 1, 1, 1]
 Generation 4100: Best = 1.158189e+03 FES = 410100 μ_F = 0.712 μ_{CR} = 0.250
 Diversity = 2.1442e-04 k_t = [3, 2, 3, 3]
 Generation 4110: Best = 1.158189e+03 FES = 411100 μ_F = 0.720 μ_{CR} = 0.253
 Diversity = 2.3496e-04 k_t = [1, 1, 1, 1]
 Generation 4120: Best = 1.158189e+03 FES = 412100 μ_F = 0.725 μ_{CR} = 0.248
 Diversity = 2.3928e-04 k_t = [4, 3, 4, 4]
 Generation 4130: Best = 1.158189e+03 FES = 413100 μ_F = 0.728 μ_{CR} = 0.248
 Diversity = 2.6665e-04 k_t = [1, 1, 1, 1]
 Generation 4140: Best = 1.158189e+03 FES = 414100 μ_F = 0.725 μ_{CR} = 0.248
 Diversity = 2.5849e-04 k_t = [4, 3, 3, 4]
 Generation 4150: Best = 1.158189e+03 FES = 415100 μ_F = 0.732 μ_{CR} = 0.247
 Diversity = 2.7265e-04 k_t = [1, 1, 1, 1]
 Generation 4160: Best = 1.158189e+03 FES = 416100 μ_F = 0.743 μ_{CR} = 0.248
 Diversity = 2.7441e-04 k_t = [4, 4, 3, 3]
 Generation 4170: Best = 1.158189e+03 FES = 417100 μ_F = 0.735 μ_{CR} = 0.251
 Diversity = 2.8651e-04 k_t = [1, 1, 1, 1]
 Generation 4180: Best = 1.158189e+03 FES = 418100 μ_F = 0.737 μ_{CR} = 0.248
 Diversity = 2.6016e-04 k_t = [3, 2, 3, 3]
 Generation 4190: Best = 1.158189e+03 FES = 419100 μ_F = 0.741 μ_{CR} = 0.245
 Diversity = 2.8237e-04 k_t = [1, 1, 1, 1]
 Generation 4200: Best = 1.158189e+03 FES = 420100 μ_F = 0.750 μ_{CR} = 0.242
 Diversity = 2.8889e-04 k_t = [4, 2, 4, 4]
 Generation 4210: Best = 1.158189e+03 FES = 421100 μ_F = 0.755 μ_{CR} = 0.239
 Diversity = 2.7504e-04 k_t = [1, 1, 1, 1]
 Generation 4220: Best = 1.158189e+03 FES = 422100 μ_F = 0.754 μ_{CR} = 0.239
 Diversity = 2.7188e-04 k_t = [3, 2, 3, 4]
 Generation 4230: Best = 1.158189e+03 FES = 423100 μ_F = 0.765 μ_{CR} = 0.246
 Diversity = 3.0832e-04 k_t = [1, 1, 1, 1]
 Generation 4240: Best = 1.158189e+03 FES = 424100 μ_F = 0.758 μ_{CR} = 0.237
 Diversity = 2.9493e-04 k_t = [4, 4, 2, 3]
 Generation 4250: Best = 1.158189e+03 FES = 425100 μ_F = 0.767 μ_{CR} = 0.233
 Diversity = 3.5016e-04 k_t = [1, 1, 1, 1]
 Generation 4260: Best = 1.158189e+03 FES = 426100 μ_F = 0.756 μ_{CR} = 0.228
 Diversity = 3.4059e-04 k_t = [3, 4, 2, 2]
 Generation 4270: Best = 1.158189e+03 FES = 427100 μ_F = 0.744 μ_{CR} = 0.221
 Diversity = 3.8293e-04 k_t = [1, 1, 1, 1]

Generation 4280: Best = 1.158189e+03 FES = 428100 μ_F = 0.736 μ_{CR} = 0.227
 Diversity = 3.8228e-04 k_t = [3, 4, 3, 4]
 Generation 4290: Best = 1.158188e+03 FES = 429100 μ_F = 0.741 μ_{CR} = 0.232
 Diversity = 4.0072e-04 k_t = [1, 1, 1, 1]
 Generation 4300: Best = 1.158188e+03 FES = 430100 μ_F = 0.747 μ_{CR} = 0.239
 Diversity = 4.0619e-04 k_t = [4, 4, 4, 4]
 Generation 4310: Best = 1.158188e+03 FES = 431100 μ_F = 0.720 μ_{CR} = 0.242
 Diversity = 3.8653e-04 k_t = [1, 1, 1, 1]
 Generation 4320: Best = 1.158188e+03 FES = 432100 μ_F = 0.710 μ_{CR} = 0.244
 Diversity = 3.5041e-04 k_t = [2, 3, 4, 3]
 Generation 4330: Best = 1.158188e+03 FES = 433100 μ_F = 0.711 μ_{CR} = 0.238
 Diversity = 3.6144e-04 k_t = [1, 1, 1, 1]
 Generation 4340: Best = 1.158188e+03 FES = 434100 μ_F = 0.712 μ_{CR} = 0.228
 Diversity = 3.3572e-04 k_t = [2, 2, 3, 4]
 Generation 4350: Best = 1.158188e+03 FES = 435100 μ_F = 0.715 μ_{CR} = 0.217
 Diversity = 3.5642e-04 k_t = [1, 1, 1, 1]
 Generation 4360: Best = 1.158188e+03 FES = 436100 μ_F = 0.710 μ_{CR} = 0.221
 Diversity = 3.2791e-04 k_t = [3, 2, 3, 3]
 Generation 4370: Best = 1.158188e+03 FES = 437100 μ_F = 0.708 μ_{CR} = 0.224
 Diversity = 3.0053e-04 k_t = [1, 1, 1, 1]
 Generation 4380: Best = 1.158188e+03 FES = 438100 μ_F = 0.692 μ_{CR} = 0.221
 Diversity = 2.7339e-04 k_t = [2, 3, 4, 2]
 Generation 4390: Best = 1.158188e+03 FES = 439100 μ_F = 0.684 μ_{CR} = 0.223
 Diversity = 2.7069e-04 k_t = [1, 1, 1, 1]
 Generation 4400: Best = 1.158188e+03 FES = 440100 μ_F = 0.687 μ_{CR} = 0.227
 Diversity = 2.5479e-04 k_t = [3, 4, 3, 2]
 Generation 4410: Best = 1.158188e+03 FES = 441100 μ_F = 0.681 μ_{CR} = 0.228
 Diversity = 2.4525e-04 k_t = [1, 1, 1, 1]
 Generation 4420: Best = 1.158188e+03 FES = 442100 μ_F = 0.700 μ_{CR} = 0.231
 Diversity = 2.4222e-04 k_t = [3, 2, 4, 3]
 Generation 4430: Best = 1.158188e+03 FES = 443100 μ_F = 0.702 μ_{CR} = 0.230
 Diversity = 2.5025e-04 k_t = [1, 1, 1, 1]
 Generation 4440: Best = 1.158188e+03 FES = 444100 μ_F = 0.710 μ_{CR} = 0.238
 Diversity = 2.4860e-04 k_t = [3, 4, 2, 2]
 Generation 4450: Best = 1.158188e+03 FES = 445100 μ_F = 0.718 μ_{CR} = 0.232
 Diversity = 2.4585e-04 k_t = [1, 1, 1, 1]
 Generation 4460: Best = 1.158188e+03 FES = 446100 μ_F = 0.720 μ_{CR} = 0.222
 Diversity = 2.3019e-04 k_t = [4, 4, 4, 4]
 Generation 4470: Best = 1.158188e+03 FES = 447100 μ_F = 0.742 μ_{CR} = 0.222
 Diversity = 2.3531e-04 k_t = [1, 1, 1, 1]
 Generation 4480: Best = 1.158188e+03 FES = 448100 μ_F = 0.748 μ_{CR} = 0.215
 Diversity = 2.4621e-04 k_t = [3, 3, 2, 2]
 Generation 4490: Best = 1.158188e+03 FES = 449100 μ_F = 0.741 μ_{CR} = 0.210
 Diversity = 2.5088e-04 k_t = [1, 1, 1, 1]
 Generation 4500: Best = 1.158188e+03 FES = 450100 μ_F = 0.746 μ_{CR} = 0.216
 Diversity = 2.7512e-04 k_t = [3, 4, 4, 2]
 Generation 4510: Best = 1.158188e+03 FES = 451100 μ_F = 0.751 μ_{CR} = 0.213
 Diversity = 3.0214e-04 k_t = [1, 1, 1, 1]
 Generation 4520: Best = 1.158188e+03 FES = 452100 μ_F = 0.754 μ_{CR} = 0.219
 Diversity = 3.1317e-04 k_t = [4, 3, 2, 2]
 Generation 4530: Best = 1.158188e+03 FES = 453100 μ_F = 0.752 μ_{CR} = 0.215
 Diversity = 3.2691e-04 k_t = [1, 1, 1, 1]
 Generation 4540: Best = 1.158188e+03 FES = 454100 μ_F = 0.744 μ_{CR} = 0.212
 Diversity = 3.4099e-04 k_t = [2, 2, 2, 4]

Generation 4550: Best = 1.158188e+03 FES = 455100 μ_F = 0.750 μ_{CR} = 0.205
 Diversity = 3.9150e-04 k_t = [1, 1, 1, 1]
 Generation 4560: Best = 1.158188e+03 FES = 456100 μ_F = 0.748 μ_{CR} = 0.202
 Diversity = 4.2614e-04 k_t = [4, 4, 3, 2]
 Generation 4570: Best = 1.158188e+03 FES = 457100 μ_F = 0.747 μ_{CR} = 0.211
 Diversity = 4.1934e-04 k_t = [1, 1, 1, 1]
 Generation 4580: Best = 1.158188e+03 FES = 458100 μ_F = 0.747 μ_{CR} = 0.213
 Diversity = 4.0243e-04 k_t = [4, 4, 4, 3]
 Generation 4590: Best = 1.158188e+03 FES = 459100 μ_F = 0.746 μ_{CR} = 0.221
 Diversity = 4.1764e-04 k_t = [1, 1, 1, 1]
 Generation 4600: Best = 1.158188e+03 FES = 460100 μ_F = 0.743 μ_{CR} = 0.229
 Diversity = 3.8834e-04 k_t = [4, 3, 3, 3]
 Generation 4610: Best = 1.158188e+03 FES = 461100 μ_F = 0.734 μ_{CR} = 0.237
 Diversity = 3.7858e-04 k_t = [1, 1, 1, 1]
 Generation 4620: Best = 1.158188e+03 FES = 462100 μ_F = 0.713 μ_{CR} = 0.240
 Diversity = 3.7359e-04 k_t = [4, 3, 2, 2]
 Generation 4630: Best = 1.158188e+03 FES = 463100 μ_F = 0.717 μ_{CR} = 0.242
 Diversity = 3.3958e-04 k_t = [1, 1, 1, 1]
 Generation 4640: Best = 1.158188e+03 FES = 464100 μ_F = 0.708 μ_{CR} = 0.248
 Diversity = 3.1997e-04 k_t = [4, 2, 3, 2]
 Generation 4650: Best = 1.158188e+03 FES = 465100 μ_F = 0.691 μ_{CR} = 0.243
 Diversity = 2.9576e-04 k_t = [1, 1, 1, 1]
 Generation 4660: Best = 1.158188e+03 FES = 466100 μ_F = 0.692 μ_{CR} = 0.242
 Diversity = 2.5719e-04 k_t = [2, 2, 2, 2]
 Generation 4670: Best = 1.158188e+03 FES = 467100 μ_F = 0.685 μ_{CR} = 0.244
 Diversity = 2.5079e-04 k_t = [1, 1, 1, 1]
 Generation 4680: Best = 1.158188e+03 FES = 468100 μ_F = 0.683 μ_{CR} = 0.245
 Diversity = 2.2648e-04 k_t = [3, 4, 3, 4]
 Generation 4690: Best = 1.158188e+03 FES = 469100 μ_F = 0.689 μ_{CR} = 0.245
 Diversity = 2.1067e-04 k_t = [1, 1, 1, 1]
 Generation 4700: Best = 1.158188e+03 FES = 470100 μ_F = 0.704 μ_{CR} = 0.248
 Diversity = 2.1603e-04 k_t = [3, 3, 2, 2]
 Generation 4710: Best = 1.158188e+03 FES = 471100 μ_F = 0.699 μ_{CR} = 0.252
 Diversity = 2.1343e-04 k_t = [1, 1, 1, 1]
 Generation 4720: Best = 1.158188e+03 FES = 472100 μ_F = 0.700 μ_{CR} = 0.261
 Diversity = 1.8109e-04 k_t = [3, 2, 4, 4]
 Generation 4730: Best = 1.158188e+03 FES = 473100 μ_F = 0.687 μ_{CR} = 0.264
 Diversity = 1.6251e-04 k_t = [1, 1, 1, 1]
 Generation 4740: Best = 1.158188e+03 FES = 474100 μ_F = 0.697 μ_{CR} = 0.261
 Diversity = 1.4684e-04 k_t = [3, 4, 4, 4]
 Generation 4750: Best = 1.158188e+03 FES = 475100 μ_F = 0.712 μ_{CR} = 0.257
 Diversity = 1.4345e-04 k_t = [1, 1, 1, 1]
 Generation 4760: Best = 1.158188e+03 FES = 476100 μ_F = 0.729 μ_{CR} = 0.261
 Diversity = 1.5171e-04 k_t = [3, 4, 4, 2]
 Generation 4770: Best = 1.158188e+03 FES = 477100 μ_F = 0.735 μ_{CR} = 0.271
 Diversity = 1.5015e-04 k_t = [1, 1, 1, 1]
 Generation 4780: Best = 1.158188e+03 FES = 478100 μ_F = 0.729 μ_{CR} = 0.264
 Diversity = 1.4297e-04 k_t = [4, 3, 4, 3]
 Generation 4790: Best = 1.158188e+03 FES = 479100 μ_F = 0.725 μ_{CR} = 0.260
 Diversity = 1.4663e-04 k_t = [1, 1, 1, 1]
 Generation 4800: Best = 1.158188e+03 FES = 480100 μ_F = 0.736 μ_{CR} = 0.263
 Diversity = 1.2799e-04 k_t = [2, 4, 3, 4]
 Generation 4810: Best = 1.158188e+03 FES = 481100 μ_F = 0.745 μ_{CR} = 0.262
 Diversity = 1.3734e-04 k_t = [1, 1, 1, 1]

Generation 4820: Best = 1.158188e+03 FES = 482100 μ_F = 0.744 μ_{CR} = 0.263
 Diversity = 1.5192e-04 k_t = [4, 3, 4, 4]
 Generation 4830: Best = 1.158188e+03 FES = 483100 μ_F = 0.739 μ_{CR} = 0.272
 Diversity = 1.7173e-04 k_t = [1, 1, 1, 1]
 Generation 4840: Best = 1.158188e+03 FES = 484100 μ_F = 0.738 μ_{CR} = 0.279
 Diversity = 1.7074e-04 k_t = [4, 3, 2, 4]
 Generation 4850: Best = 1.158188e+03 FES = 485100 μ_F = 0.730 μ_{CR} = 0.286
 Diversity = 1.7111e-04 k_t = [1, 1, 1, 1]
 Generation 4860: Best = 1.158188e+03 FES = 486100 μ_F = 0.732 μ_{CR} = 0.280
 Diversity = 1.8335e-04 k_t = [3, 3, 3, 2]
 Generation 4870: Best = 1.158188e+03 FES = 487100 μ_F = 0.723 μ_{CR} = 0.279
 Diversity = 1.7531e-04 k_t = [1, 1, 1, 1]
 Generation 4880: Best = 1.158188e+03 FES = 488100 μ_F = 0.718 μ_{CR} = 0.275
 Diversity = 1.7072e-04 k_t = [2, 4, 3, 3]
 Generation 4890: Best = 1.158188e+03 FES = 489100 μ_F = 0.711 μ_{CR} = 0.279
 Diversity = 1.7998e-04 k_t = [1, 1, 1, 1]
 Generation 4900: Best = 1.158188e+03 FES = 490100 μ_F = 0.715 μ_{CR} = 0.276
 Diversity = 1.7862e-04 k_t = [2, 2, 2, 4]
 Generation 4910: Best = 1.158188e+03 FES = 491100 μ_F = 0.712 μ_{CR} = 0.286
 Diversity = 1.7898e-04 k_t = [1, 1, 1, 1]
 Generation 4920: Best = 1.158188e+03 FES = 492100 μ_F = 0.716 μ_{CR} = 0.282
 Diversity = 1.7593e-04 k_t = [3, 3, 4, 3]
 Generation 4930: Best = 1.158188e+03 FES = 493100 μ_F = 0.715 μ_{CR} = 0.276
 Diversity = 1.7062e-04 k_t = [1, 1, 1, 1]
 Generation 4940: Best = 1.158188e+03 FES = 494100 μ_F = 0.710 μ_{CR} = 0.270
 Diversity = 1.6189e-04 k_t = [3, 2, 3, 2]
 Generation 4950: Best = 1.158188e+03 FES = 495100 μ_F = 0.705 μ_{CR} = 0.263
 Diversity = 1.5110e-04 k_t = [1, 1, 1, 1]
 Generation 4960: Best = 1.158188e+03 FES = 496100 μ_F = 0.712 μ_{CR} = 0.271
 Diversity = 1.3661e-04 k_t = [4, 3, 3, 4]
 Generation 4970: Best = 1.158188e+03 FES = 497100 μ_F = 0.704 μ_{CR} = 0.279
 Diversity = 1.3028e-04 k_t = [1, 1, 1, 1]
 Generation 4980: Best = 1.158188e+03 FES = 498100 μ_F = 0.707 μ_{CR} = 0.279
 Diversity = 1.2436e-04 k_t = [3, 3, 2, 2]
 Generation 4990: Best = 1.158188e+03 FES = 499100 μ_F = 0.707 μ_{CR} = 0.271
 Diversity = 1.2553e-04 k_t = [1, 1, 1, 1]
 Generation 5000: Best = 1.158188e+03 FES = 500100 μ_F = 0.709 μ_{CR} = 0.276
 Diversity = 1.3029e-04 k_t = [4, 4, 2, 3]
 Generation 5010: Best = 1.158188e+03 FES = 501100 μ_F = 0.713 μ_{CR} = 0.288
 Diversity = 1.3553e-04 k_t = [1, 1, 1, 1]
 Generation 5020: Best = 1.158188e+03 FES = 502100 μ_F = 0.725 μ_{CR} = 0.287
 Diversity = 1.3975e-04 k_t = [3, 4, 2, 3]
 Generation 5030: Best = 1.158188e+03 FES = 503100 μ_F = 0.720 μ_{CR} = 0.283
 Diversity = 1.5105e-04 k_t = [1, 1, 1, 1]
 Generation 5040: Best = 1.158188e+03 FES = 504100 μ_F = 0.693 μ_{CR} = 0.280
 Diversity = 1.3399e-04 k_t = [2, 3, 3, 2]
 Generation 5050: Best = 1.158188e+03 FES = 505100 μ_F = 0.689 μ_{CR} = 0.271
 Diversity = 1.2394e-04 k_t = [1, 1, 1, 1]
 Generation 5060: Best = 1.158188e+03 FES = 506100 μ_F = 0.687 μ_{CR} = 0.272
 Diversity = 9.8066e-05 k_t = [4, 3, 4, 4]
 Generation 5070: Best = 1.158188e+03 FES = 507100 μ_F = 0.682 μ_{CR} = 0.278
 Diversity = 8.1053e-05 k_t = [1, 1, 1, 1]
 Generation 5080: Best = 1.158188e+03 FES = 508100 μ_F = 0.684 μ_{CR} = 0.275
 Diversity = 7.5450e-05 k_t = [2, 4, 3, 4]

Generation 5090: Best = 1.158188e+03 FES = 509100 μ_F = 0.689 μ_{CR} = 0.280
 Diversity = 7.0653e-05 k_t = [1, 1, 1, 1]
 Generation 5100: Best = 1.158188e+03 FES = 510100 μ_F = 0.673 μ_{CR} = 0.265
 Diversity = 6.2696e-05 k_t = [2, 3, 4, 3]
 Generation 5110: Best = 1.158188e+03 FES = 511100 μ_F = 0.659 μ_{CR} = 0.268
 Diversity = 6.0032e-05 k_t = [1, 1, 1, 1]
 Generation 5120: Best = 1.158188e+03 FES = 512100 μ_F = 0.675 μ_{CR} = 0.275
 Diversity = 5.8924e-05 k_t = [2, 2, 3, 3]
 Generation 5130: Best = 1.158188e+03 FES = 513100 μ_F = 0.692 μ_{CR} = 0.270
 Diversity = 6.4562e-05 k_t = [1, 1, 1, 1]
 Generation 5140: Best = 1.158188e+03 FES = 514100 μ_F = 0.699 μ_{CR} = 0.271
 Diversity = 6.1858e-05 k_t = [3, 2, 2, 3]
 Generation 5150: Best = 1.158188e+03 FES = 515100 μ_F = 0.703 μ_{CR} = 0.268
 Diversity = 6.6019e-05 k_t = [1, 1, 1, 1]
 Generation 5160: Best = 1.158188e+03 FES = 516100 μ_F = 0.707 μ_{CR} = 0.271
 Diversity = 6.5930e-05 k_t = [2, 4, 3, 2]
 Generation 5170: Best = 1.158188e+03 FES = 517100 μ_F = 0.703 μ_{CR} = 0.277
 Diversity = 6.6091e-05 k_t = [1, 1, 1, 1]
 Generation 5180: Best = 1.158188e+03 FES = 518100 μ_F = 0.699 μ_{CR} = 0.277
 Diversity = 6.7671e-05 k_t = [4, 4, 4, 2]
 Generation 5190: Best = 1.158188e+03 FES = 519100 μ_F = 0.702 μ_{CR} = 0.278
 Diversity = 6.6160e-05 k_t = [1, 1, 1, 1]
 Generation 5200: Best = 1.158188e+03 FES = 520100 μ_F = 0.707 μ_{CR} = 0.276
 Diversity = 6.2688e-05 k_t = [4, 2, 2, 4]
 Generation 5210: Best = 1.158188e+03 FES = 521100 μ_F = 0.732 μ_{CR} = 0.278
 Diversity = 7.6003e-05 k_t = [1, 1, 1, 1]
 Generation 5220: Best = 1.158188e+03 FES = 522100 μ_F = 0.716 μ_{CR} = 0.268
 Diversity = 7.3318e-05 k_t = [4, 3, 2, 3]
 Generation 5230: Best = 1.158188e+03 FES = 523100 μ_F = 0.716 μ_{CR} = 0.261
 Diversity = 7.2971e-05 k_t = [1, 1, 1, 1]
 Generation 5240: Best = 1.158188e+03 FES = 524100 μ_F = 0.714 μ_{CR} = 0.258
 Diversity = 7.9811e-05 k_t = [4, 2, 3, 2]
 Generation 5250: Best = 1.158188e+03 FES = 525100 μ_F = 0.717 μ_{CR} = 0.253
 Diversity = 7.5199e-05 k_t = [1, 1, 1, 1]
 Generation 5260: Best = 1.158188e+03 FES = 526100 μ_F = 0.729 μ_{CR} = 0.260
 Diversity = 7.9789e-05 k_t = [4, 2, 4, 2]
 Generation 5270: Best = 1.158188e+03 FES = 527100 μ_F = 0.729 μ_{CR} = 0.260
 Diversity = 9.0540e-05 k_t = [1, 1, 1, 1]
 Generation 5280: Best = 1.158188e+03 FES = 528100 μ_F = 0.751 μ_{CR} = 0.262
 Diversity = 9.2891e-05 k_t = [2, 4, 4, 4]
 Generation 5290: Best = 1.158188e+03 FES = 529100 μ_F = 0.747 μ_{CR} = 0.252
 Diversity = 9.2524e-05 k_t = [1, 1, 1, 1]
 Generation 5300: Best = 1.158188e+03 FES = 530100 μ_F = 0.769 μ_{CR} = 0.242
 Diversity = 1.0010e-04 k_t = [4, 4, 2, 4]
 Generation 5310: Best = 1.158188e+03 FES = 531100 μ_F = 0.770 μ_{CR} = 0.241
 Diversity = 1.0918e-04 k_t = [1, 1, 1, 1]
 Generation 5320: Best = 1.158188e+03 FES = 532100 μ_F = 0.764 μ_{CR} = 0.244
 Diversity = 1.2890e-04 k_t = [3, 4, 3, 3]
 Generation 5330: Best = 1.158188e+03 FES = 533100 μ_F = 0.761 μ_{CR} = 0.243
 Diversity = 1.3361e-04 k_t = [1, 1, 1, 1]
 Generation 5340: Best = 1.158188e+03 FES = 534100 μ_F = 0.765 μ_{CR} = 0.243
 Diversity = 1.5494e-04 k_t = [3, 2, 2, 4]
 Generation 5350: Best = 1.158188e+03 FES = 535100 μ_F = 0.767 μ_{CR} = 0.242
 Diversity = 1.9547e-04 k_t = [1, 1, 1, 1]

Generation 5360: Best = 1.158188e+03 FES = 536100 μ_F = 0.773 μ_{CR} = 0.255
 Diversity = 2.3610e-04 k_t = [3, 2, 2, 2]
 Generation 5370: Best = 1.158188e+03 FES = 537100 μ_F = 0.767 μ_{CR} = 0.248
 Diversity = 2.6704e-04 k_t = [1, 1, 1, 1]
 Generation 5380: Best = 1.158188e+03 FES = 538100 μ_F = 0.778 μ_{CR} = 0.260
 Diversity = 3.0980e-04 k_t = [4, 4, 2, 4]
 Generation 5390: Best = 1.158188e+03 FES = 539100 μ_F = 0.774 μ_{CR} = 0.254
 Diversity = 3.4573e-04 k_t = [1, 1, 1, 1]
 Generation 5400: Best = 1.158188e+03 FES = 540100 μ_F = 0.774 μ_{CR} = 0.248
 Diversity = 3.6412e-04 k_t = [3, 3, 3, 2]
 Generation 5410: Best = 1.158187e+03 FES = 541100 μ_F = 0.759 μ_{CR} = 0.245
 Diversity = 3.7163e-04 k_t = [1, 1, 1, 1]
 Generation 5420: Best = 1.158187e+03 FES = 542100 μ_F = 0.757 μ_{CR} = 0.250
 Diversity = 3.7459e-04 k_t = [2, 4, 2, 3]
 Generation 5430: Best = 1.158187e+03 FES = 543100 μ_F = 0.754 μ_{CR} = 0.257
 Diversity = 4.1271e-04 k_t = [1, 1, 1, 1]
 Generation 5440: Best = 1.158187e+03 FES = 544100 μ_F = 0.748 μ_{CR} = 0.258
 Diversity = 4.3429e-04 k_t = [2, 2, 2, 4]
 Generation 5450: Best = 1.158187e+03 FES = 545100 μ_F = 0.736 μ_{CR} = 0.248
 Diversity = 4.2096e-04 k_t = [1, 1, 1, 1]
 Generation 5460: Best = 1.158187e+03 FES = 546100 μ_F = 0.738 μ_{CR} = 0.256
 Diversity = 4.1400e-04 k_t = [3, 4, 3, 2]
 Generation 5470: Best = 1.158187e+03 FES = 547100 μ_F = 0.739 μ_{CR} = 0.248
 Diversity = 4.1141e-04 k_t = [1, 1, 1, 1]
 Generation 5480: Best = 1.158187e+03 FES = 548100 μ_F = 0.744 μ_{CR} = 0.251
 Diversity = 3.8528e-04 k_t = [3, 4, 3, 3]
 Generation 5490: Best = 1.158187e+03 FES = 549100 μ_F = 0.743 μ_{CR} = 0.254
 Diversity = 3.7158e-04 k_t = [1, 1, 1, 1]
 Generation 5500: Best = 1.158187e+03 FES = 550100 μ_F = 0.728 μ_{CR} = 0.262
 Diversity = 3.3770e-04 k_t = [3, 3, 4, 4]
 Generation 5510: Best = 1.158187e+03 FES = 551100 μ_F = 0.738 μ_{CR} = 0.265
 Diversity = 3.8275e-04 k_t = [1, 1, 1, 1]
 Generation 5520: Best = 1.158187e+03 FES = 552100 μ_F = 0.743 μ_{CR} = 0.273
 Diversity = 4.7328e-04 k_t = [3, 4, 4, 2]
 Generation 5530: Best = 1.158187e+03 FES = 553100 μ_F = 0.737 μ_{CR} = 0.278
 Diversity = 5.0281e-04 k_t = [1, 1, 1, 1]
 Generation 5540: Best = 1.158187e+03 FES = 554100 μ_F = 0.736 μ_{CR} = 0.280
 Diversity = 5.0791e-04 k_t = [2, 4, 3, 4]
 Generation 5550: Best = 1.158187e+03 FES = 555100 μ_F = 0.732 μ_{CR} = 0.274
 Diversity = 4.8443e-04 k_t = [1, 1, 1, 1]
 Generation 5560: Best = 1.158187e+03 FES = 556100 μ_F = 0.711 μ_{CR} = 0.278
 Diversity = 4.6759e-04 k_t = [4, 3, 2, 2]
 Generation 5570: Best = 1.158187e+03 FES = 557100 μ_F = 0.717 μ_{CR} = 0.285
 Diversity = 5.0534e-04 k_t = [1, 1, 1, 1]
 Generation 5580: Best = 1.158187e+03 FES = 558100 μ_F = 0.714 μ_{CR} = 0.283
 Diversity = 4.6483e-04 k_t = [2, 3, 2, 2]
 Generation 5590: Best = 1.158187e+03 FES = 559100 μ_F = 0.705 μ_{CR} = 0.288
 Diversity = 4.4428e-04 k_t = [1, 1, 1, 1]
 Generation 5600: Best = 1.158187e+03 FES = 560100 μ_F = 0.705 μ_{CR} = 0.287
 Diversity = 4.0166e-04 k_t = [2, 4, 3, 2]
 Generation 5610: Best = 1.158187e+03 FES = 561100 μ_F = 0.695 μ_{CR} = 0.292
 Diversity = 3.6339e-04 k_t = [1, 1, 1, 1]
 Generation 5620: Best = 1.158187e+03 FES = 562100 μ_F = 0.685 μ_{CR} = 0.287
 Diversity = 3.1969e-04 k_t = [3, 3, 2, 2]

Generation 5630: Best = 1.158187e+03 FES = 563100 μ_F = 0.684 μ_{CR} = 0.280
 Diversity = 3.0381e-04 k_t = [1, 1, 1, 1]
 Generation 5640: Best = 1.158187e+03 FES = 564100 μ_F = 0.685 μ_{CR} = 0.296
 Diversity = 2.7922e-04 k_t = [2, 4, 3, 4]
 Generation 5650: Best = 1.158187e+03 FES = 565100 μ_F = 0.696 μ_{CR} = 0.295
 Diversity = 2.7530e-04 k_t = [1, 1, 1, 1]
 Generation 5660: Best = 1.158187e+03 FES = 566100 μ_F = 0.693 μ_{CR} = 0.287
 Diversity = 2.4428e-04 k_t = [4, 2, 4, 3]
 Generation 5670: Best = 1.158187e+03 FES = 567100 μ_F = 0.693 μ_{CR} = 0.283
 Diversity = 2.2359e-04 k_t = [1, 1, 1, 1]
 Generation 5680: Best = 1.158187e+03 FES = 568100 μ_F = 0.698 μ_{CR} = 0.283
 Diversity = 1.8938e-04 k_t = [3, 4, 2, 4]
 Generation 5690: Best = 1.158187e+03 FES = 569100 μ_F = 0.696 μ_{CR} = 0.275
 Diversity = 1.6361e-04 k_t = [1, 1, 1, 1]
 Generation 5700: Best = 1.158187e+03 FES = 570100 μ_F = 0.706 μ_{CR} = 0.280
 Diversity = 1.6712e-04 k_t = [2, 2, 4, 2]
 Generation 5710: Best = 1.158187e+03 FES = 571100 μ_F = 0.708 μ_{CR} = 0.280
 Diversity = 1.6892e-04 k_t = [1, 1, 1, 1]
 Generation 5720: Best = 1.158187e+03 FES = 572100 μ_F = 0.700 μ_{CR} = 0.275
 Diversity = 1.5939e-04 k_t = [2, 4, 2, 3]
 Generation 5730: Best = 1.158187e+03 FES = 573100 μ_F = 0.692 μ_{CR} = 0.266
 Diversity = 1.6228e-04 k_t = [1, 1, 1, 1]
 Generation 5740: Best = 1.158187e+03 FES = 574100 μ_F = 0.687 μ_{CR} = 0.273
 Diversity = 1.4745e-04 k_t = [3, 2, 4, 2]
 Generation 5750: Best = 1.158187e+03 FES = 575100 μ_F = 0.703 μ_{CR} = 0.277
 Diversity = 1.4951e-04 k_t = [1, 1, 1, 1]
 Generation 5760: Best = 1.158187e+03 FES = 576100 μ_F = 0.702 μ_{CR} = 0.267
 Diversity = 1.3802e-04 k_t = [4, 3, 3, 3]
 Generation 5770: Best = 1.158187e+03 FES = 577100 μ_F = 0.689 μ_{CR} = 0.272
 Diversity = 1.3017e-04 k_t = [1, 1, 1, 1]
 Generation 5780: Best = 1.158187e+03 FES = 578100 μ_F = 0.691 μ_{CR} = 0.262
 Diversity = 1.1247e-04 k_t = [3, 4, 2, 3]
 Generation 5790: Best = 1.158187e+03 FES = 579100 μ_F = 0.689 μ_{CR} = 0.276
 Diversity = 1.0721e-04 k_t = [1, 1, 1, 1]
 Generation 5800: Best = 1.158187e+03 FES = 580100 μ_F = 0.690 μ_{CR} = 0.273
 Diversity = 9.5700e-05 k_t = [3, 2, 4, 3]
 Generation 5810: Best = 1.158187e+03 FES = 581100 μ_F = 0.697 μ_{CR} = 0.270
 Diversity = 9.1933e-05 k_t = [1, 1, 1, 1]
 Generation 5820: Best = 1.158187e+03 FES = 582100 μ_F = 0.691 μ_{CR} = 0.273
 Diversity = 8.4296e-05 k_t = [2, 2, 2, 3]
 Generation 5830: Best = 1.158187e+03 FES = 583100 μ_F = 0.695 μ_{CR} = 0.278
 Diversity = 8.2556e-05 k_t = [1, 1, 1, 1]
 Generation 5840: Best = 1.158187e+03 FES = 584100 μ_F = 0.705 μ_{CR} = 0.277
 Diversity = 7.3467e-05 k_t = [4, 2, 4, 3]
 Generation 5850: Best = 1.158187e+03 FES = 585100 μ_F = 0.711 μ_{CR} = 0.270
 Diversity = 7.0882e-05 k_t = [1, 1, 1, 1]
 Generation 5860: Best = 1.158187e+03 FES = 586100 μ_F = 0.714 μ_{CR} = 0.270
 Diversity = 7.4517e-05 k_t = [2, 2, 2, 2]
 Generation 5870: Best = 1.158187e+03 FES = 587100 μ_F = 0.718 μ_{CR} = 0.260
 Diversity = 8.0629e-05 k_t = [1, 1, 1, 1]
 Generation 5880: Best = 1.158187e+03 FES = 588100 μ_F = 0.734 μ_{CR} = 0.262
 Diversity = 8.0166e-05 k_t = [3, 2, 3, 4]
 Generation 5890: Best = 1.158187e+03 FES = 589100 μ_F = 0.739 μ_{CR} = 0.264
 Diversity = 8.9521e-05 k_t = [1, 1, 1, 1]

Generation 5900: Best = 1.158187e+03 FES = 590100 μ_F = 0.741 μ_{CR} = 0.264
 Diversity = 9.2704e-05 k_t = [2, 3, 4, 3]
 Generation 5910: Best = 1.158187e+03 FES = 591100 μ_F = 0.741 μ_{CR} = 0.264
 Diversity = 1.0731e-04 k_t = [1, 1, 1, 1]
 Generation 5920: Best = 1.158187e+03 FES = 592100 μ_F = 0.735 μ_{CR} = 0.270
 Diversity = 1.1133e-04 k_t = [4, 2, 4, 2]
 Generation 5930: Best = 1.158187e+03 FES = 593100 μ_F = 0.720 μ_{CR} = 0.268
 Diversity = 1.0969e-04 k_t = [1, 1, 1, 1]
 Generation 5940: Best = 1.158187e+03 FES = 594100 μ_F = 0.719 μ_{CR} = 0.259
 Diversity = 1.0297e-04 k_t = [2, 2, 4, 2]
 Generation 5950: Best = 1.158187e+03 FES = 595100 μ_F = 0.712 μ_{CR} = 0.258
 Diversity = 1.0461e-04 k_t = [1, 1, 1, 1]
 Generation 5960: Best = 1.158187e+03 FES = 596100 μ_F = 0.712 μ_{CR} = 0.256
 Diversity = 9.8852e-05 k_t = [3, 3, 4, 2]
 Generation 5970: Best = 1.158187e+03 FES = 597100 μ_F = 0.714 μ_{CR} = 0.253
 Diversity = 9.7735e-05 k_t = [1, 1, 1, 1]
 Generation 5980: Best = 1.158187e+03 FES = 598100 μ_F = 0.720 μ_{CR} = 0.257
 Diversity = 9.3024e-05 k_t = [3, 4, 3, 3]
 Generation 5990: Best = 1.158187e+03 FES = 599100 μ_F = 0.721 μ_{CR} = 0.247
 Diversity = 9.8138e-05 k_t = [1, 1, 1, 1]
 Generation 6000: Best = 1.158187e+03 FES = 600100 μ_F = 0.718 μ_{CR} = 0.243
 Diversity = 9.6971e-05 k_t = [2, 2, 4, 4]
 Generation 6010: Best = 1.158187e+03 FES = 601100 μ_F = 0.716 μ_{CR} = 0.235
 Diversity = 1.1396e-04 k_t = [1, 1, 1, 1]
 Generation 6020: Best = 1.158187e+03 FES = 602100 μ_F = 0.722 μ_{CR} = 0.234
 Diversity = 1.0564e-04 k_t = [4, 4, 3, 4]
 Generation 6030: Best = 1.158187e+03 FES = 603100 μ_F = 0.725 μ_{CR} = 0.238
 Diversity = 9.6875e-05 k_t = [1, 1, 1, 1]
 Generation 6040: Best = 1.158187e+03 FES = 604100 μ_F = 0.706 μ_{CR} = 0.238
 Diversity = 8.8837e-05 k_t = [4, 2, 2, 4]
 Generation 6050: Best = 1.158187e+03 FES = 605100 μ_F = 0.707 μ_{CR} = 0.239
 Diversity = 7.7436e-05 k_t = [1, 1, 1, 1]
 Generation 6060: Best = 1.158187e+03 FES = 606100 μ_F = 0.697 μ_{CR} = 0.236
 Diversity = 6.2962e-05 k_t = [4, 4, 4, 4]
 Generation 6070: Best = 1.158187e+03 FES = 607100 μ_F = 0.697 μ_{CR} = 0.241
 Diversity = 5.3338e-05 k_t = [1, 1, 1, 1]
 Generation 6080: Best = 1.158187e+03 FES = 608100 μ_F = 0.689 μ_{CR} = 0.233
 Diversity = 4.7802e-05 k_t = [4, 3, 3, 3]
 Generation 6090: Best = 1.158187e+03 FES = 609100 μ_F = 0.693 μ_{CR} = 0.229
 Diversity = 4.5281e-05 k_t = [1, 1, 1, 1]
 Generation 6100: Best = 1.158187e+03 FES = 610100 μ_F = 0.702 μ_{CR} = 0.228
 Diversity = 4.4382e-05 k_t = [2, 4, 4, 4]
 Generation 6110: Best = 1.158187e+03 FES = 611100 μ_F = 0.698 μ_{CR} = 0.241
 Diversity = 4.4440e-05 k_t = [1, 1, 1, 1]
 Generation 6120: Best = 1.158187e+03 FES = 612100 μ_F = 0.706 μ_{CR} = 0.241
 Diversity = 4.1818e-05 k_t = [4, 2, 2, 4]
 Generation 6130: Best = 1.158187e+03 FES = 613100 μ_F = 0.702 μ_{CR} = 0.243
 Diversity = 4.1954e-05 k_t = [1, 1, 1, 1]
 Generation 6140: Best = 1.158187e+03 FES = 614100 μ_F = 0.698 μ_{CR} = 0.244
 Diversity = 3.6910e-05 k_t = [4, 4, 4, 3]
 Generation 6150: Best = 1.158187e+03 FES = 615100 μ_F = 0.684 μ_{CR} = 0.249
 Diversity = 3.3953e-05 k_t = [1, 1, 1, 1]
 Generation 6160: Best = 1.158187e+03 FES = 616100 μ_F = 0.682 μ_{CR} = 0.258
 Diversity = 2.9997e-05 k_t = [3, 4, 3, 3]

Generation 6170: Best = 1.158187e+03 FES = 617100 μ_F = 0.668 μ_{CR} = 0.264
 Diversity = 2.7568e-05 k_t = [1, 1, 1, 1]
 Generation 6180: Best = 1.158187e+03 FES = 618100 μ_F = 0.686 μ_{CR} = 0.264
 Diversity = 2.8142e-05 k_t = [2, 4, 4, 2]
 Generation 6190: Best = 1.158187e+03 FES = 619100 μ_F = 0.706 μ_{CR} = 0.276
 Diversity = 3.2807e-05 k_t = [1, 1, 1, 1]
 Generation 6200: Best = 1.158187e+03 FES = 620100 μ_F = 0.729 μ_{CR} = 0.277
 Diversity = 3.6660e-05 k_t = [3, 3, 3, 2]
 Generation 6210: Best = 1.158187e+03 FES = 621100 μ_F = 0.731 μ_{CR} = 0.278
 Diversity = 4.0925e-05 k_t = [1, 1, 1, 1]
 Generation 6220: Best = 1.158187e+03 FES = 622100 μ_F = 0.723 μ_{CR} = 0.278
 Diversity = 3.7954e-05 k_t = [4, 2, 3, 3]
 Generation 6230: Best = 1.158187e+03 FES = 623100 μ_F = 0.715 μ_{CR} = 0.281
 Diversity = 3.5692e-05 k_t = [1, 1, 1, 1]
 Generation 6240: Best = 1.158187e+03 FES = 624100 μ_F = 0.713 μ_{CR} = 0.285
 Diversity = 3.3163e-05 k_t = [3, 3, 2, 2]
 Generation 6250: Best = 1.158187e+03 FES = 625100 μ_F = 0.701 μ_{CR} = 0.276
 Diversity = 3.0048e-05 k_t = [1, 1, 1, 1]
 Generation 6260: Best = 1.158187e+03 FES = 626100 μ_F = 0.701 μ_{CR} = 0.277
 Diversity = 2.5153e-05 k_t = [2, 3, 2, 4]
 Generation 6270: Best = 1.158187e+03 FES = 627100 μ_F = 0.705 μ_{CR} = 0.272
 Diversity = 2.3994e-05 k_t = [1, 1, 1, 1]
 Generation 6280: Best = 1.158187e+03 FES = 628100 μ_F = 0.709 μ_{CR} = 0.275
 Diversity = 2.4477e-05 k_t = [2, 2, 2, 2]
 Generation 6290: Best = 1.158187e+03 FES = 629100 μ_F = 0.709 μ_{CR} = 0.275
 Diversity = 2.5895e-05 k_t = [1, 1, 1, 1]
 Generation 6300: Best = 1.158187e+03 FES = 630100 μ_F = 0.709 μ_{CR} = 0.258
 Diversity = 2.3379e-05 k_t = [2, 4, 2, 4]
 Generation 6310: Best = 1.158187e+03 FES = 631100 μ_F = 0.726 μ_{CR} = 0.248
 Diversity = 2.3946e-05 k_t = [1, 1, 1, 1]
 Generation 6320: Best = 1.158187e+03 FES = 632100 μ_F = 0.733 μ_{CR} = 0.250
 Diversity = 2.3068e-05 k_t = [3, 4, 3, 4]
 Generation 6330: Best = 1.158187e+03 FES = 633100 μ_F = 0.735 μ_{CR} = 0.249
 Diversity = 2.2139e-05 k_t = [1, 1, 1, 1]
 Generation 6340: Best = 1.158187e+03 FES = 634100 μ_F = 0.732 μ_{CR} = 0.247
 Diversity = 2.0548e-05 k_t = [3, 2, 4, 4]
 Generation 6350: Best = 1.158187e+03 FES = 635100 μ_F = 0.722 μ_{CR} = 0.258
 Diversity = 2.0186e-05 k_t = [1, 1, 1, 1]
 Generation 6360: Best = 1.158187e+03 FES = 636100 μ_F = 0.721 μ_{CR} = 0.254
 Diversity = 1.8539e-05 k_t = [2, 4, 4, 4]
 Generation 6370: Best = 1.158187e+03 FES = 637100 μ_F = 0.732 μ_{CR} = 0.252
 Diversity = 1.8721e-05 k_t = [1, 1, 1, 1]
 Generation 6380: Best = 1.158187e+03 FES = 638100 μ_F = 0.725 μ_{CR} = 0.258
 Diversity = 2.0199e-05 k_t = [3, 4, 2, 2]
 Generation 6390: Best = 1.158187e+03 FES = 639100 μ_F = 0.726 μ_{CR} = 0.266
 Diversity = 2.2359e-05 k_t = [1, 1, 1, 1]
 Generation 6400: Best = 1.158187e+03 FES = 640100 μ_F = 0.734 μ_{CR} = 0.265
 Diversity = 2.3671e-05 k_t = [3, 2, 4, 3]
 Generation 6410: Best = 1.158187e+03 FES = 641100 μ_F = 0.726 μ_{CR} = 0.253
 Diversity = 2.4469e-05 k_t = [1, 1, 1, 1]
 Generation 6420: Best = 1.158187e+03 FES = 642100 μ_F = 0.734 μ_{CR} = 0.250
 Diversity = 2.3644e-05 k_t = [2, 3, 2, 3]
 Generation 6430: Best = 1.158187e+03 FES = 643100 μ_F = 0.724 μ_{CR} = 0.245
 Diversity = 2.3605e-05 k_t = [1, 1, 1, 1]

Generation 6440: Best = 1.158187e+03 FES = 644100 μ_F = 0.716 μ_{CR} = 0.242
 Diversity = 2.2747e-05 k_t = [3, 4, 3, 2]
 Generation 6450: Best = 1.158187e+03 FES = 645100 μ_F = 0.715 μ_{CR} = 0.242
 Diversity = 2.0410e-05 k_t = [1, 1, 1, 1]
 Generation 6460: Best = 1.158187e+03 FES = 646100 μ_F = 0.710 μ_{CR} = 0.246
 Diversity = 1.7798e-05 k_t = [3, 3, 3, 3]
 Generation 6470: Best = 1.158187e+03 FES = 647100 μ_F = 0.705 μ_{CR} = 0.240
 Diversity = 1.6896e-05 k_t = [1, 1, 1, 1]
 Generation 6480: Best = 1.158187e+03 FES = 648100 μ_F = 0.706 μ_{CR} = 0.241
 Diversity = 1.6115e-05 k_t = [4, 4, 3, 3]
 Generation 6490: Best = 1.158187e+03 FES = 649100 μ_F = 0.711 μ_{CR} = 0.226
 Diversity = 1.5408e-05 k_t = [1, 1, 1, 1]
 Generation 6500: Best = 1.158187e+03 FES = 650100 μ_F = 0.711 μ_{CR} = 0.224
 Diversity = 1.3732e-05 k_t = [4, 4, 4, 4]
 Generation 6510: Best = 1.158187e+03 FES = 651100 μ_F = 0.708 μ_{CR} = 0.224
 Diversity = 1.2292e-05 k_t = [1, 1, 1, 1]
 Generation 6520: Best = 1.158187e+03 FES = 652100 μ_F = 0.711 μ_{CR} = 0.219
 Diversity = 1.2217e-05 k_t = [4, 2, 3, 2]
 Generation 6530: Best = 1.158187e+03 FES = 653100 μ_F = 0.712 μ_{CR} = 0.213
 Diversity = 1.2713e-05 k_t = [1, 1, 1, 1]
 Generation 6540: Best = 1.158187e+03 FES = 654100 μ_F = 0.719 μ_{CR} = 0.217
 Diversity = 1.2447e-05 k_t = [4, 4, 2, 3]
 Generation 6550: Best = 1.158187e+03 FES = 655100 μ_F = 0.719 μ_{CR} = 0.222
 Diversity = 1.2386e-05 k_t = [1, 1, 1, 1]
 Generation 6560: Best = 1.158187e+03 FES = 656100 μ_F = 0.715 μ_{CR} = 0.227
 Diversity = 1.1177e-05 k_t = [2, 3, 4, 4]
 Generation 6570: Best = 1.158187e+03 FES = 657100 μ_F = 0.724 μ_{CR} = 0.226
 Diversity = 1.1036e-05 k_t = [1, 1, 1, 1]
 Generation 6580: Best = 1.158187e+03 FES = 658100 μ_F = 0.729 μ_{CR} = 0.236
 Diversity = 1.0594e-05 k_t = [3, 4, 3, 4]
 Generation 6590: Best = 1.158187e+03 FES = 659100 μ_F = 0.736 μ_{CR} = 0.222
 Diversity = 9.5034e-06 k_t = [1, 1, 1, 1]
 Generation 6600: Best = 1.158187e+03 FES = 660100 μ_F = 0.751 μ_{CR} = 0.226
 Diversity = 9.9013e-06 k_t = [2, 3, 4, 3]
 Generation 6610: Best = 1.158187e+03 FES = 661100 μ_F = 0.758 μ_{CR} = 0.225
 Diversity = 1.1574e-05 k_t = [1, 1, 1, 1]
 Generation 6620: Best = 1.158187e+03 FES = 662100 μ_F = 0.756 μ_{CR} = 0.230
 Diversity = 1.1947e-05 k_t = [3, 2, 3, 4]
 Generation 6630: Best = 1.158187e+03 FES = 663100 μ_F = 0.751 μ_{CR} = 0.236
 Diversity = 1.2751e-05 k_t = [1, 1, 1, 1]
 Generation 6640: Best = 1.158187e+03 FES = 664100 μ_F = 0.754 μ_{CR} = 0.234
 Diversity = 1.4826e-05 k_t = [3, 3, 2, 2]
 Generation 6650: Best = 1.158187e+03 FES = 665100 μ_F = 0.762 μ_{CR} = 0.229
 Diversity = 1.6881e-05 k_t = [1, 1, 1, 1]
 Generation 6660: Best = 1.158187e+03 FES = 666100 μ_F = 0.743 μ_{CR} = 0.231
 Diversity = 1.8853e-05 k_t = [3, 3, 2, 2]
 Generation 6670: Best = 1.158187e+03 FES = 667100 μ_F = 0.744 μ_{CR} = 0.235
 Diversity = 2.1918e-05 k_t = [1, 1, 1, 1]
 Generation 6680: Best = 1.158187e+03 FES = 668100 μ_F = 0.744 μ_{CR} = 0.240
 Diversity = 2.4571e-05 k_t = [2, 4, 2, 3]
 Generation 6690: Best = 1.158187e+03 FES = 669100 μ_F = 0.744 μ_{CR} = 0.236
 Diversity = 2.6885e-05 k_t = [1, 1, 1, 1]
 Generation 6700: Best = 1.158187e+03 FES = 670100 μ_F = 0.739 μ_{CR} = 0.234
 Diversity = 2.8541e-05 k_t = [3, 3, 2, 3]

Generation 6710: Best = 1.158187e+03 FES = 671100 μ_F = 0.736 μ_{CR} = 0.243
 Diversity = 2.8595e-05 k_t = [1, 1, 1, 1]
 Generation 6720: Best = 1.158187e+03 FES = 672100 μ_F = 0.729 μ_{CR} = 0.245
 Diversity = 2.7226e-05 k_t = [4, 4, 3, 3]
 Generation 6730: Best = 1.158187e+03 FES = 673100 μ_F = 0.728 μ_{CR} = 0.249
 Diversity = 2.8434e-05 k_t = [1, 1, 1, 1]
 Generation 6740: Best = 1.158187e+03 FES = 674100 μ_F = 0.734 μ_{CR} = 0.255
 Diversity = 2.8665e-05 k_t = [3, 4, 4, 4]
 Generation 6750: Best = 1.158187e+03 FES = 675100 μ_F = 0.726 μ_{CR} = 0.259
 Diversity = 2.8981e-05 k_t = [1, 1, 1, 1]
 Generation 6760: Best = 1.158187e+03 FES = 676100 μ_F = 0.718 μ_{CR} = 0.266
 Diversity = 2.8905e-05 k_t = [2, 2, 3, 2]
 Generation 6770: Best = 1.158187e+03 FES = 677100 μ_F = 0.709 μ_{CR} = 0.265
 Diversity = 2.7792e-05 k_t = [1, 1, 1, 1]
 Generation 6780: Best = 1.158187e+03 FES = 678100 μ_F = 0.699 μ_{CR} = 0.264
 Diversity = 2.3112e-05 k_t = [3, 4, 3, 4]
 Generation 6790: Best = 1.158187e+03 FES = 679100 μ_F = 0.702 μ_{CR} = 0.250
 Diversity = 2.0679e-05 k_t = [1, 1, 1, 1]
 Generation 6800: Best = 1.158187e+03 FES = 680100 μ_F = 0.687 μ_{CR} = 0.248
 Diversity = 1.9836e-05 k_t = [3, 2, 2, 2]
 Generation 6810: Best = 1.158187e+03 FES = 681100 μ_F = 0.695 μ_{CR} = 0.251
 Diversity = 2.0281e-05 k_t = [1, 1, 1, 1]
 Generation 6820: Best = 1.158187e+03 FES = 682100 μ_F = 0.697 μ_{CR} = 0.248
 Diversity = 1.7465e-05 k_t = [3, 2, 4, 3]
 Generation 6830: Best = 1.158187e+03 FES = 683100 μ_F = 0.700 μ_{CR} = 0.246
 Diversity = 1.7440e-05 k_t = [1, 1, 1, 1]
 Generation 6840: Best = 1.158187e+03 FES = 684100 μ_F = 0.692 μ_{CR} = 0.248
 Diversity = 1.5098e-05 k_t = [4, 2, 3, 4]
 Generation 6850: Best = 1.158187e+03 FES = 685100 μ_F = 0.690 μ_{CR} = 0.255
 Diversity = 1.2616e-05 k_t = [1, 1, 1, 1]
 Generation 6860: Best = 1.158187e+03 FES = 686100 μ_F = 0.691 μ_{CR} = 0.259
 Diversity = 1.0996e-05 k_t = [4, 4, 3, 3]
 Generation 6870: Best = 1.158187e+03 FES = 687100 μ_F = 0.693 μ_{CR} = 0.259
 Diversity = 1.0321e-05 k_t = [1, 1, 1, 1]
 Generation 6880: Best = 1.158187e+03 FES = 688100 μ_F = 0.710 μ_{CR} = 0.260
 Diversity = 1.0680e-05 k_t = [2, 3, 3, 4]
 Generation 6890: Best = 1.158187e+03 FES = 689100 μ_F = 0.712 μ_{CR} = 0.269
 Diversity = 1.0225e-05 k_t = [1, 1, 1, 1]
 Generation 6900: Best = 1.158187e+03 FES = 690100 μ_F = 0.705 μ_{CR} = 0.276
 Diversity = 8.6780e-06 k_t = [4, 4, 2, 4]
 Generation 6910: Best = 1.158187e+03 FES = 691100 μ_F = 0.710 μ_{CR} = 0.272
 Diversity = 7.5592e-06 k_t = [1, 1, 1, 1]
 Generation 6920: Best = 1.158187e+03 FES = 692100 μ_F = 0.714 μ_{CR} = 0.261
 Diversity = 6.8315e-06 k_t = [2, 4, 4, 4]
 Generation 6930: Best = 1.158187e+03 FES = 693100 μ_F = 0.715 μ_{CR} = 0.251
 Diversity = 6.2514e-06 k_t = [1, 1, 1, 1]
 Generation 6940: Best = 1.158187e+03 FES = 694100 μ_F = 0.731 μ_{CR} = 0.247
 Diversity = 6.0244e-06 k_t = [2, 4, 3, 4]
 Generation 6950: Best = 1.158187e+03 FES = 695100 μ_F = 0.732 μ_{CR} = 0.243
 Diversity = 6.2863e-06 k_t = [1, 1, 1, 1]
 Generation 6960: Best = 1.158187e+03 FES = 696100 μ_F = 0.708 μ_{CR} = 0.252
 Diversity = 5.6986e-06 k_t = [2, 4, 3, 4]
 Generation 6970: Best = 1.158187e+03 FES = 697100 μ_F = 0.707 μ_{CR} = 0.255
 Diversity = 5.2871e-06 k_t = [1, 1, 1, 1]

Generation 6980: Best = 1.158187e+03 FES = 698100 μ_F = 0.710 μ_{CR} = 0.256
 Diversity = 4.5417e-06 k_t = [4, 4, 4, 4]
 Generation 6990: Best = 1.158187e+03 FES = 699100 μ_F = 0.718 μ_{CR} = 0.253
 Diversity = 4.1412e-06 k_t = [1, 1, 1, 1]
 Generation 7000: Best = 1.158187e+03 FES = 700100 μ_F = 0.722 μ_{CR} = 0.246
 Diversity = 3.9647e-06 k_t = [4, 3, 3, 3]
 Generation 7010: Best = 1.158187e+03 FES = 701100 μ_F = 0.729 μ_{CR} = 0.243
 Diversity = 4.0926e-06 k_t = [1, 1, 1, 1]
 Generation 7020: Best = 1.158187e+03 FES = 702100 μ_F = 0.735 μ_{CR} = 0.249
 Diversity = 4.5173e-06 k_t = [3, 2, 4, 3]
 Generation 7030: Best = 1.158187e+03 FES = 703100 μ_F = 0.724 μ_{CR} = 0.246
 Diversity = 4.9382e-06 k_t = [1, 1, 1, 1]
 Generation 7040: Best = 1.158187e+03 FES = 704100 μ_F = 0.725 μ_{CR} = 0.238
 Diversity = 5.1286e-06 k_t = [3, 3, 4, 2]
 Generation 7050: Best = 1.158187e+03 FES = 705100 μ_F = 0.703 μ_{CR} = 0.231
 Diversity = 5.0694e-06 k_t = [1, 1, 1, 1]
 Generation 7060: Best = 1.158187e+03 FES = 706100 μ_F = 0.706 μ_{CR} = 0.235
 Diversity = 4.6066e-06 k_t = [2, 2, 2, 4]
 Generation 7070: Best = 1.158187e+03 FES = 707100 μ_F = 0.714 μ_{CR} = 0.227
 Diversity = 4.7072e-06 k_t = [1, 1, 1, 1]
 Generation 7080: Best = 1.158187e+03 FES = 708100 μ_F = 0.723 μ_{CR} = 0.225
 Diversity = 4.4993e-06 k_t = [2, 4, 3, 4]
 Generation 7090: Best = 1.158187e+03 FES = 709100 μ_F = 0.705 μ_{CR} = 0.223
 Diversity = 4.2260e-06 k_t = [1, 1, 1, 1]
 Generation 7100: Best = 1.158187e+03 FES = 710100 μ_F = 0.718 μ_{CR} = 0.227
 Diversity = 4.1375e-06 k_t = [4, 2, 4, 4]
 Generation 7110: Best = 1.158187e+03 FES = 711100 μ_F = 0.717 μ_{CR} = 0.225
 Diversity = 4.1554e-06 k_t = [1, 1, 1, 1]
 Generation 7120: Best = 1.158187e+03 FES = 712100 μ_F = 0.718 μ_{CR} = 0.230
 Diversity = 3.8686e-06 k_t = [2, 3, 2, 4]
 Generation 7130: Best = 1.158187e+03 FES = 713100 μ_F = 0.710 μ_{CR} = 0.225
 Diversity = 3.4796e-06 k_t = [1, 1, 1, 1]
 Generation 7140: Best = 1.158187e+03 FES = 714100 μ_F = 0.718 μ_{CR} = 0.226
 Diversity = 3.2204e-06 k_t = [3, 3, 4, 4]
 Generation 7150: Best = 1.158187e+03 FES = 715100 μ_F = 0.728 μ_{CR} = 0.223
 Diversity = 3.1179e-06 k_t = [1, 1, 1, 1]
 Generation 7160: Best = 1.158187e+03 FES = 716100 μ_F = 0.724 μ_{CR} = 0.227
 Diversity = 3.0537e-06 k_t = [2, 4, 2, 3]
 Generation 7170: Best = 1.158187e+03 FES = 717100 μ_F = 0.724 μ_{CR} = 0.223
 Diversity = 3.0541e-06 k_t = [1, 1, 1, 1]
 Generation 7180: Best = 1.158187e+03 FES = 718100 μ_F = 0.728 μ_{CR} = 0.218
 Diversity = 2.9452e-06 k_t = [3, 2, 3, 2]
 Generation 7190: Best = 1.158187e+03 FES = 719100 μ_F = 0.728 μ_{CR} = 0.215
 Diversity = 3.1552e-06 k_t = [1, 1, 1, 1]
 Generation 7200: Best = 1.158187e+03 FES = 720100 μ_F = 0.727 μ_{CR} = 0.217
 Diversity = 2.9874e-06 k_t = [3, 4, 2, 4]
 Generation 7210: Best = 1.158187e+03 FES = 721100 μ_F = 0.739 μ_{CR} = 0.211
 Diversity = 3.0702e-06 k_t = [1, 1, 1, 1]
 Generation 7220: Best = 1.158187e+03 FES = 722100 μ_F = 0.743 μ_{CR} = 0.211
 Diversity = 2.8980e-06 k_t = [3, 4, 2, 3]
 Generation 7230: Best = 1.158187e+03 FES = 723100 μ_F = 0.747 μ_{CR} = 0.218
 Diversity = 2.9631e-06 k_t = [1, 1, 1, 1]
 Generation 7240: Best = 1.158187e+03 FES = 724100 μ_F = 0.752 μ_{CR} = 0.223
 Diversity = 3.2115e-06 k_t = [4, 4, 2, 3]

Generation 7250: Best = 1.158187e+03 FES = 725100 μ_F = 0.765 μ_{CR} = 0.224
 Diversity = 3.5156e-06 k_t = [1, 1, 1, 1]
 Generation 7260: Best = 1.158187e+03 FES = 726100 μ_F = 0.770 μ_{CR} = 0.231
 Diversity = 3.8625e-06 k_t = [3, 3, 3, 3]
 Generation 7270: Best = 1.158187e+03 FES = 727100 μ_F = 0.771 μ_{CR} = 0.227
 Diversity = 4.1529e-06 k_t = [1, 1, 1, 1]
 Generation 7280: Best = 1.158187e+03 FES = 728100 μ_F = 0.781 μ_{CR} = 0.233
 Diversity = 3.8457e-06 k_t = [3, 3, 3, 4]
 Generation 7290: Best = 1.158187e+03 FES = 729100 μ_F = 0.763 μ_{CR} = 0.231
 Diversity = 3.5973e-06 k_t = [1, 1, 1, 1]
 Generation 7300: Best = 1.158187e+03 FES = 730100 μ_F = 0.735 μ_{CR} = 0.231
 Diversity = 3.3777e-06 k_t = [4, 2, 4, 3]
 Generation 7310: Best = 1.158187e+03 FES = 731100 μ_F = 0.732 μ_{CR} = 0.233
 Diversity = 3.6592e-06 k_t = [1, 1, 1, 1]
 Generation 7320: Best = 1.158187e+03 FES = 732100 μ_F = 0.724 μ_{CR} = 0.239
 Diversity = 3.9134e-06 k_t = [3, 2, 2, 4]
 Generation 7330: Best = 1.158187e+03 FES = 733100 μ_F = 0.716 μ_{CR} = 0.238
 Diversity = 3.8663e-06 k_t = [1, 1, 1, 1]
 Generation 7340: Best = 1.158187e+03 FES = 734100 μ_F = 0.696 μ_{CR} = 0.235
 Diversity = 3.3350e-06 k_t = [2, 4, 3, 3]
 Generation 7350: Best = 1.158187e+03 FES = 735100 μ_F = 0.702 μ_{CR} = 0.226
 Diversity = 3.3573e-06 k_t = [1, 1, 1, 1]
 Generation 7360: Best = 1.158187e+03 FES = 736100 μ_F = 0.703 μ_{CR} = 0.230
 Diversity = 3.0989e-06 k_t = [4, 3, 4, 3]
 Generation 7370: Best = 1.158187e+03 FES = 737100 μ_F = 0.711 μ_{CR} = 0.235
 Diversity = 3.1587e-06 k_t = [1, 1, 1, 1]
 Generation 7380: Best = 1.158187e+03 FES = 738100 μ_F = 0.708 μ_{CR} = 0.236
 Diversity = 3.0374e-06 k_t = [2, 2, 2, 2]
 Generation 7390: Best = 1.158187e+03 FES = 739100 μ_F = 0.703 μ_{CR} = 0.238
 Diversity = 2.8947e-06 k_t = [1, 1, 1, 1]
 Generation 7400: Best = 1.158187e+03 FES = 740100 μ_F = 0.715 μ_{CR} = 0.239
 Diversity = 2.9371e-06 k_t = [2, 2, 4, 2]
 Generation 7410: Best = 1.158187e+03 FES = 741100 μ_F = 0.721 μ_{CR} = 0.237
 Diversity = 3.0313e-06 k_t = [1, 1, 1, 1]
 Generation 7420: Best = 1.158187e+03 FES = 742100 μ_F = 0.720 μ_{CR} = 0.242
 Diversity = 3.0004e-06 k_t = [4, 2, 4, 4]
 Generation 7430: Best = 1.158187e+03 FES = 743100 μ_F = 0.723 μ_{CR} = 0.246
 Diversity = 2.7790e-06 k_t = [1, 1, 1, 1]
 Generation 7440: Best = 1.158187e+03 FES = 744100 μ_F = 0.714 μ_{CR} = 0.251
 Diversity = 2.5968e-06 k_t = [2, 3, 2, 3]
 Generation 7450: Best = 1.158187e+03 FES = 745100 μ_F = 0.719 μ_{CR} = 0.258
 Diversity = 2.8324e-06 k_t = [1, 1, 1, 1]
 Generation 7460: Best = 1.158187e+03 FES = 746100 μ_F = 0.737 μ_{CR} = 0.266
 Diversity = 3.0513e-06 k_t = [2, 3, 4, 4]
 Generation 7470: Best = 1.158187e+03 FES = 747100 μ_F = 0.748 μ_{CR} = 0.263
 Diversity = 3.4799e-06 k_t = [1, 1, 1, 1]
 Generation 7480: Best = 1.158187e+03 FES = 748100 μ_F = 0.740 μ_{CR} = 0.258
 Diversity = 3.2950e-06 k_t = [4, 2, 4, 3]
 Generation 7490: Best = 1.158187e+03 FES = 749100 μ_F = 0.732 μ_{CR} = 0.258
 Diversity = 3.1864e-06 k_t = [1, 1, 1, 1]
 Generation 7500: Best = 1.158187e+03 FES = 750100 μ_F = 0.729 μ_{CR} = 0.251
 Diversity = 2.7387e-06 k_t = [2, 2, 4, 4]
 Generation 7510: Best = 1.158187e+03 FES = 751100 μ_F = 0.728 μ_{CR} = 0.259
 Diversity = 2.6800e-06 k_t = [1, 1, 1, 1]

Generation 7520: Best = 1.158187e+03 FES = 752100 μ_F = 0.722 μ_{CR} = 0.263
 Diversity = 2.6435e-06 k_t = [3, 4, 4, 2]
 Generation 7530: Best = 1.158187e+03 FES = 753100 μ_F = 0.708 μ_{CR} = 0.269
 Diversity = 2.3942e-06 k_t = [1, 1, 1, 1]
 Generation 7540: Best = 1.158187e+03 FES = 754100 μ_F = 0.710 μ_{CR} = 0.269
 Diversity = 2.1535e-06 k_t = [2, 2, 3, 4]
 Generation 7550: Best = 1.158187e+03 FES = 755100 μ_F = 0.723 μ_{CR} = 0.269
 Diversity = 2.3699e-06 k_t = [1, 1, 1, 1]
 Generation 7560: Best = 1.158187e+03 FES = 756100 μ_F = 0.730 μ_{CR} = 0.266
 Diversity = 2.6878e-06 k_t = [3, 4, 3, 2]
 Generation 7570: Best = 1.158187e+03 FES = 757100 μ_F = 0.731 μ_{CR} = 0.274
 Diversity = 2.8135e-06 k_t = [1, 1, 1, 1]
 Generation 7580: Best = 1.158187e+03 FES = 758100 μ_F = 0.737 μ_{CR} = 0.278
 Diversity = 3.0050e-06 k_t = [4, 4, 4, 3]
 Generation 7590: Best = 1.158187e+03 FES = 759100 μ_F = 0.742 μ_{CR} = 0.277
 Diversity = 3.4106e-06 k_t = [1, 1, 1, 1]
 Generation 7600: Best = 1.158187e+03 FES = 760100 μ_F = 0.737 μ_{CR} = 0.275
 Diversity = 3.7053e-06 k_t = [2, 2, 3, 2]
 Generation 7610: Best = 1.158187e+03 FES = 761100 μ_F = 0.733 μ_{CR} = 0.277
 Diversity = 4.0798e-06 k_t = [1, 1, 1, 1]
 Generation 7620: Best = 1.158187e+03 FES = 762100 μ_F = 0.734 μ_{CR} = 0.287
 Diversity = 3.8998e-06 k_t = [4, 3, 3, 4]
 Generation 7630: Best = 1.158187e+03 FES = 763100 μ_F = 0.750 μ_{CR} = 0.293
 Diversity = 4.8799e-06 k_t = [1, 1, 1, 1]
 Generation 7640: Best = 1.158187e+03 FES = 764100 μ_F = 0.747 μ_{CR} = 0.298
 Diversity = 5.6900e-06 k_t = [4, 4, 2, 4]
 Generation 7650: Best = 1.158187e+03 FES = 765100 μ_F = 0.746 μ_{CR} = 0.293
 Diversity = 5.3099e-06 k_t = [1, 1, 1, 1]
 Generation 7660: Best = 1.158187e+03 FES = 766100 μ_F = 0.720 μ_{CR} = 0.290
 Diversity = 4.7404e-06 k_t = [2, 4, 2, 4]
 Generation 7670: Best = 1.158187e+03 FES = 767100 μ_F = 0.716 μ_{CR} = 0.288
 Diversity = 4.9651e-06 k_t = [1, 1, 1, 1]
 Generation 7680: Best = 1.158187e+03 FES = 768100 μ_F = 0.721 μ_{CR} = 0.294
 Diversity = 4.9533e-06 k_t = [2, 4, 3, 4]
 Generation 7690: Best = 1.158187e+03 FES = 769100 μ_F = 0.735 μ_{CR} = 0.295
 Diversity = 5.2855e-06 k_t = [1, 1, 1, 1]
 Generation 7700: Best = 1.158187e+03 FES = 770100 μ_F = 0.739 μ_{CR} = 0.297
 Diversity = 5.6292e-06 k_t = [3, 2, 2, 4]
 Generation 7710: Best = 1.158187e+03 FES = 771100 μ_F = 0.728 μ_{CR} = 0.289
 Diversity = 5.6620e-06 k_t = [1, 1, 1, 1]
 Generation 7720: Best = 1.158187e+03 FES = 772100 μ_F = 0.720 μ_{CR} = 0.281
 Diversity = 4.7580e-06 k_t = [2, 3, 3, 4]
 Generation 7730: Best = 1.158187e+03 FES = 773100 μ_F = 0.713 μ_{CR} = 0.281
 Diversity = 4.4228e-06 k_t = [1, 1, 1, 1]
 Generation 7740: Best = 1.158187e+03 FES = 774100 μ_F = 0.709 μ_{CR} = 0.281
 Diversity = 3.9660e-06 k_t = [3, 2, 2, 4]
 Generation 7750: Best = 1.158187e+03 FES = 775100 μ_F = 0.694 μ_{CR} = 0.275
 Diversity = 3.7723e-06 k_t = [1, 1, 1, 1]
 Generation 7760: Best = 1.158187e+03 FES = 776100 μ_F = 0.697 μ_{CR} = 0.265
 Diversity = 3.4717e-06 k_t = [3, 2, 3, 2]
 Generation 7770: Best = 1.158187e+03 FES = 777100 μ_F = 0.687 μ_{CR} = 0.279
 Diversity = 3.0799e-06 k_t = [1, 1, 1, 1]
 Generation 7780: Best = 1.158187e+03 FES = 778100 μ_F = 0.689 μ_{CR} = 0.276
 Diversity = 2.7448e-06 k_t = [2, 4, 2, 4]

Generation 7790: Best = 1.158187e+03 FES = 779100 μ_F = 0.698 μ_{CR} = 0.268
 Diversity = 2.8411e-06 k_t = [1, 1, 1, 1]
 Generation 7800: Best = 1.158187e+03 FES = 780100 μ_F = 0.716 μ_{CR} = 0.260
 Diversity = 2.6780e-06 k_t = [2, 4, 2, 3]
 Generation 7810: Best = 1.158187e+03 FES = 781100 μ_F = 0.718 μ_{CR} = 0.250
 Diversity = 2.5928e-06 k_t = [1, 1, 1, 1]
 Generation 7820: Best = 1.158187e+03 FES = 782100 μ_F = 0.715 μ_{CR} = 0.254
 Diversity = 2.6014e-06 k_t = [4, 4, 2, 3]
 Generation 7830: Best = 1.158187e+03 FES = 783100 μ_F = 0.718 μ_{CR} = 0.256
 Diversity = 2.7352e-06 k_t = [1, 1, 1, 1]
 Generation 7840: Best = 1.158187e+03 FES = 784100 μ_F = 0.724 μ_{CR} = 0.259
 Diversity = 2.5816e-06 k_t = [3, 4, 2, 3]
 Generation 7850: Best = 1.158187e+03 FES = 785100 μ_F = 0.715 μ_{CR} = 0.252
 Diversity = 2.7124e-06 k_t = [1, 1, 1, 1]
 Generation 7860: Best = 1.158187e+03 FES = 786100 μ_F = 0.712 μ_{CR} = 0.249
 Diversity = 2.4931e-06 k_t = [2, 2, 2, 4]
 Generation 7870: Best = 1.158187e+03 FES = 787100 μ_F = 0.704 μ_{CR} = 0.249
 Diversity = 2.3771e-06 k_t = [1, 1, 1, 1]
 Generation 7880: Best = 1.158187e+03 FES = 788100 μ_F = 0.698 μ_{CR} = 0.247
 Diversity = 2.3058e-06 k_t = [2, 4, 2, 4]
 Generation 7890: Best = 1.158187e+03 FES = 789100 μ_F = 0.693 μ_{CR} = 0.245
 Diversity = 2.1969e-06 k_t = [1, 1, 1, 1]
 Generation 7900: Best = 1.158187e+03 FES = 790100 μ_F = 0.694 μ_{CR} = 0.249
 Diversity = 1.9087e-06 k_t = [2, 2, 2, 3]
 Generation 7910: Best = 1.158187e+03 FES = 791100 μ_F = 0.684 μ_{CR} = 0.257
 Diversity = 1.8527e-06 k_t = [1, 1, 1, 1]
 Generation 7920: Best = 1.158187e+03 FES = 792100 μ_F = 0.699 μ_{CR} = 0.259
 Diversity = 1.7056e-06 k_t = [2, 3, 3, 3]
 Generation 7930: Best = 1.158187e+03 FES = 793100 μ_F = 0.680 μ_{CR} = 0.264
 Diversity = 1.6660e-06 k_t = [1, 1, 1, 1]
 Generation 7940: Best = 1.158187e+03 FES = 794100 μ_F = 0.702 μ_{CR} = 0.259
 Diversity = 1.7143e-06 k_t = [3, 3, 3, 2]
 Generation 7950: Best = 1.158187e+03 FES = 795100 μ_F = 0.701 μ_{CR} = 0.269
 Diversity = 1.8109e-06 k_t = [1, 1, 1, 1]
 Generation 7960: Best = 1.158187e+03 FES = 796100 μ_F = 0.710 μ_{CR} = 0.273
 Diversity = 1.6435e-06 k_t = [2, 2, 4, 4]
 Generation 7970: Best = 1.158187e+03 FES = 797100 μ_F = 0.707 μ_{CR} = 0.284
 Diversity = 1.6646e-06 k_t = [1, 1, 1, 1]
 Generation 7980: Best = 1.158187e+03 FES = 798100 μ_F = 0.704 μ_{CR} = 0.283
 Diversity = 1.5130e-06 k_t = [4, 4, 3, 4]
 Generation 7990: Best = 1.158187e+03 FES = 799100 μ_F = 0.709 μ_{CR} = 0.281
 Diversity = 1.5114e-06 k_t = [1, 1, 1, 1]
 Generation 8000: Best = 1.158187e+03 FES = 800100 μ_F = 0.708 μ_{CR} = 0.276
 Diversity = 1.4412e-06 k_t = [3, 2, 4, 4]
 Generation 8010: Best = 1.158187e+03 FES = 801100 μ_F = 0.732 μ_{CR} = 0.272
 Diversity = 1.5341e-06 k_t = [1, 1, 1, 1]
 Generation 8020: Best = 1.158187e+03 FES = 802100 μ_F = 0.723 μ_{CR} = 0.276
 Diversity = 1.4888e-06 k_t = [4, 4, 4, 4]
 Generation 8030: Best = 1.158187e+03 FES = 803100 μ_F = 0.744 μ_{CR} = 0.289
 Diversity = 1.5619e-06 k_t = [1, 1, 1, 1]
 Generation 8040: Best = 1.158187e+03 FES = 804100 μ_F = 0.735 μ_{CR} = 0.288
 Diversity = 1.5838e-06 k_t = [2, 3, 3, 2]
 Generation 8050: Best = 1.158187e+03 FES = 805100 μ_F = 0.725 μ_{CR} = 0.281
 Diversity = 1.5886e-06 k_t = [1, 1, 1, 1]

Generation 8060: Best = 1.158187e+03 FES = 806100 μ_F = 0.727 μ_{CR} = 0.276
 Diversity = 1.5256e-06 k_t = [4, 4, 4, 2]
 Generation 8070: Best = 1.158187e+03 FES = 807100 μ_F = 0.719 μ_{CR} = 0.270
 Diversity = 1.4919e-06 k_t = [1, 1, 1, 1]
 Generation 8080: Best = 1.158187e+03 FES = 808100 μ_F = 0.716 μ_{CR} = 0.267
 Diversity = 1.4738e-06 k_t = [2, 4, 4, 3]
 Generation 8090: Best = 1.158187e+03 FES = 809100 μ_F = 0.718 μ_{CR} = 0.263
 Diversity = 1.5910e-06 k_t = [1, 1, 1, 1]
 Generation 8100: Best = 1.158187e+03 FES = 810100 μ_F = 0.714 μ_{CR} = 0.256
 Diversity = 1.4819e-06 k_t = [3, 4, 2, 3]
 Generation 8110: Best = 1.158187e+03 FES = 811100 μ_F = 0.712 μ_{CR} = 0.256
 Diversity = 1.4001e-06 k_t = [1, 1, 1, 1]
 Generation 8120: Best = 1.158187e+03 FES = 812100 μ_F = 0.725 μ_{CR} = 0.246
 Diversity = 1.3224e-06 k_t = [4, 3, 2, 4]
 Generation 8130: Best = 1.158187e+03 FES = 813100 μ_F = 0.729 μ_{CR} = 0.248
 Diversity = 1.3310e-06 k_t = [1, 1, 1, 1]
 Generation 8140: Best = 1.158187e+03 FES = 814100 μ_F = 0.711 μ_{CR} = 0.241
 Diversity = 1.3362e-06 k_t = [3, 4, 3, 2]
 Generation 8150: Best = 1.158187e+03 FES = 815100 μ_F = 0.716 μ_{CR} = 0.243
 Diversity = 1.3424e-06 k_t = [1, 1, 1, 1]
 Generation 8160: Best = 1.158187e+03 FES = 816100 μ_F = 0.712 μ_{CR} = 0.249
 Diversity = 1.2015e-06 k_t = [2, 2, 2, 3]
 Generation 8170: Best = 1.158187e+03 FES = 817100 μ_F = 0.709 μ_{CR} = 0.251
 Diversity = 1.1777e-06 k_t = [1, 1, 1, 1]
 Generation 8180: Best = 1.158187e+03 FES = 818100 μ_F = 0.716 μ_{CR} = 0.249
 Diversity = 1.1066e-06 k_t = [4, 3, 4, 3]
 Generation 8190: Best = 1.158187e+03 FES = 819100 μ_F = 0.717 μ_{CR} = 0.253
 Diversity = 1.1310e-06 k_t = [1, 1, 1, 1]
 Generation 8200: Best = 1.158187e+03 FES = 820100 μ_F = 0.708 μ_{CR} = 0.251
 Diversity = 1.0671e-06 k_t = [3, 4, 2, 2]
 Generation 8210: Best = 1.158187e+03 FES = 821100 μ_F = 0.705 μ_{CR} = 0.251
 Diversity = 1.0816e-06 k_t = [1, 1, 1, 1]
 Generation 8220: Best = 1.158187e+03 FES = 822100 μ_F = 0.693 μ_{CR} = 0.246
 Diversity = 9.5774e-07 k_t = [2, 3, 2, 2]
 Generation 8230: Best = 1.158187e+03 FES = 823100 μ_F = 0.695 μ_{CR} = 0.243
 Diversity = 9.2920e-07 k_t = [1, 1, 1, 1]
 Generation 8240: Best = 1.158187e+03 FES = 824100 μ_F = 0.706 μ_{CR} = 0.249
 Diversity = 8.6813e-07 k_t = [4, 3, 3, 3]
 Generation 8250: Best = 1.158187e+03 FES = 825100 μ_F = 0.712 μ_{CR} = 0.250
 Diversity = 9.4524e-07 k_t = [1, 1, 1, 1]
 Generation 8260: Best = 1.158187e+03 FES = 826100 μ_F = 0.718 μ_{CR} = 0.250
 Diversity = 9.7396e-07 k_t = [3, 2, 3, 2]
 Generation 8270: Best = 1.158187e+03 FES = 827100 μ_F = 0.717 μ_{CR} = 0.245
 Diversity = 1.0666e-06 k_t = [1, 1, 1, 1]
 Generation 8280: Best = 1.158187e+03 FES = 828100 μ_F = 0.727 μ_{CR} = 0.244
 Diversity = 1.1253e-06 k_t = [4, 4, 4, 2]
 Generation 8290: Best = 1.158187e+03 FES = 829100 μ_F = 0.731 μ_{CR} = 0.246
 Diversity = 1.1551e-06 k_t = [1, 1, 1, 1]
 Generation 8300: Best = 1.158187e+03 FES = 830100 μ_F = 0.738 μ_{CR} = 0.234
 Diversity = 1.2034e-06 k_t = [2, 3, 3, 2]
 Generation 8310: Best = 1.158187e+03 FES = 831100 μ_F = 0.752 μ_{CR} = 0.239
 Diversity = 1.4678e-06 k_t = [1, 1, 1, 1]
 Generation 8320: Best = 1.158187e+03 FES = 832100 μ_F = 0.744 μ_{CR} = 0.240
 Diversity = 1.5287e-06 k_t = [2, 3, 4, 4]

Generation 8330: Best = 1.158187e+03 FES = 833100 μ_F = 0.741 μ_{CR} = 0.242
 Diversity = 1.6428e-06 k_t = [1, 1, 1, 1]
 Generation 8340: Best = 1.158187e+03 FES = 834100 μ_F = 0.747 μ_{CR} = 0.252
 Diversity = 1.6365e-06 k_t = [2, 4, 2, 3]
 Generation 8350: Best = 1.158187e+03 FES = 835100 μ_F = 0.733 μ_{CR} = 0.247
 Diversity = 1.6282e-06 k_t = [1, 1, 1, 1]
 Generation 8360: Best = 1.158187e+03 FES = 836100 μ_F = 0.718 μ_{CR} = 0.240
 Diversity = 1.4378e-06 k_t = [3, 3, 4, 2]
 Generation 8370: Best = 1.158187e+03 FES = 837100 μ_F = 0.697 μ_{CR} = 0.239
 Diversity = 1.3476e-06 k_t = [1, 1, 1, 1]
 Generation 8380: Best = 1.158187e+03 FES = 838100 μ_F = 0.704 μ_{CR} = 0.229
 Diversity = 1.1992e-06 k_t = [2, 4, 3, 4]
 Generation 8390: Best = 1.158187e+03 FES = 839100 μ_F = 0.697 μ_{CR} = 0.231
 Diversity = 1.1574e-06 k_t = [1, 1, 1, 1]
 Generation 8400: Best = 1.158187e+03 FES = 840100 μ_F = 0.687 μ_{CR} = 0.223
 Diversity = 9.9175e-07 k_t = [4, 4, 3, 3]
 Generation 8410: Best = 1.158187e+03 FES = 841100 μ_F = 0.690 μ_{CR} = 0.228
 Diversity = 9.3704e-07 k_t = [1, 1, 1, 1]
 Generation 8420: Best = 1.158187e+03 FES = 842100 μ_F = 0.701 μ_{CR} = 0.229
 Diversity = 8.4358e-07 k_t = [2, 3, 4, 3]
 Generation 8430: Best = 1.158187e+03 FES = 843100 μ_F = 0.710 μ_{CR} = 0.239
 Diversity = 9.0250e-07 k_t = [1, 1, 1, 1]
 Generation 8440: Best = 1.158187e+03 FES = 844100 μ_F = 0.723 μ_{CR} = 0.234
 Diversity = 9.2459e-07 k_t = [4, 2, 3, 3]
 Generation 8450: Best = 1.158187e+03 FES = 845100 μ_F = 0.722 μ_{CR} = 0.248
 Diversity = 9.3112e-07 k_t = [1, 1, 1, 1]
 Generation 8460: Best = 1.158187e+03 FES = 846100 μ_F = 0.728 μ_{CR} = 0.240
 Diversity = 8.8848e-07 k_t = [4, 4, 3, 4]
 Generation 8470: Best = 1.158187e+03 FES = 847100 μ_F = 0.743 μ_{CR} = 0.239
 Diversity = 9.0355e-07 k_t = [1, 1, 1, 1]
 Generation 8480: Best = 1.158187e+03 FES = 848100 μ_F = 0.750 μ_{CR} = 0.240
 Diversity = 9.0316e-07 k_t = [3, 3, 2, 3]
 Generation 8490: Best = 1.158187e+03 FES = 849100 μ_F = 0.746 μ_{CR} = 0.239
 Diversity = 1.1069e-06 k_t = [1, 1, 1, 1]
 Generation 8500: Best = 1.158187e+03 FES = 850100 μ_F = 0.747 μ_{CR} = 0.238
 Diversity = 1.0495e-06 k_t = [3, 4, 3, 3]
 Generation 8510: Best = 1.158187e+03 FES = 851100 μ_F = 0.753 μ_{CR} = 0.235
 Diversity = 1.1516e-06 k_t = [1, 1, 1, 1]
 Generation 8520: Best = 1.158187e+03 FES = 852100 μ_F = 0.752 μ_{CR} = 0.240
 Diversity = 1.1300e-06 k_t = [3, 3, 4, 4]
 Generation 8530: Best = 1.158187e+03 FES = 853100 μ_F = 0.730 μ_{CR} = 0.241
 Diversity = 1.0749e-06 k_t = [1, 1, 1, 1]
 Generation 8540: Best = 1.158187e+03 FES = 854100 μ_F = 0.723 μ_{CR} = 0.247
 Diversity = 1.0995e-06 k_t = [2, 3, 2, 2]
 Generation 8550: Best = 1.158187e+03 FES = 855100 μ_F = 0.718 μ_{CR} = 0.255
 Diversity = 1.2156e-06 k_t = [1, 1, 1, 1]
 Generation 8560: Best = 1.158187e+03 FES = 856100 μ_F = 0.717 μ_{CR} = 0.261
 Diversity = 1.0909e-06 k_t = [3, 3, 3, 3]
 Generation 8570: Best = 1.158187e+03 FES = 857100 μ_F = 0.702 μ_{CR} = 0.265
 Diversity = 1.0602e-06 k_t = [1, 1, 1, 1]
 Generation 8580: Best = 1.158187e+03 FES = 858100 μ_F = 0.703 μ_{CR} = 0.272
 Diversity = 9.4523e-07 k_t = [3, 2, 4, 3]
 Generation 8590: Best = 1.158187e+03 FES = 859100 μ_F = 0.695 μ_{CR} = 0.273
 Diversity = 9.4955e-07 k_t = [1, 1, 1, 1]

Generation 8600: Best = 1.158187e+03 FES = 860100 μ_F = 0.699 μ_{CR} = 0.278
 Diversity = 8.4001e-07 k_t = [3, 4, 3, 4]
 Generation 8610: Best = 1.158187e+03 FES = 861100 μ_F = 0.698 μ_{CR} = 0.275
 Diversity = 7.7203e-07 k_t = [1, 1, 1, 1]
 Generation 8620: Best = 1.158187e+03 FES = 862100 μ_F = 0.687 μ_{CR} = 0.270
 Diversity = 6.5211e-07 k_t = [4, 3, 2, 4]
 Generation 8630: Best = 1.158187e+03 FES = 863100 μ_F = 0.691 μ_{CR} = 0.281
 Diversity = 6.2421e-07 k_t = [1, 1, 1, 1]
 Generation 8640: Best = 1.158187e+03 FES = 864100 μ_F = 0.700 μ_{CR} = 0.281
 Diversity = 6.4071e-07 k_t = [3, 4, 2, 3]
 Generation 8650: Best = 1.158187e+03 FES = 865100 μ_F = 0.705 μ_{CR} = 0.290
 Diversity = 6.5868e-07 k_t = [1, 1, 1, 1]
 Generation 8660: Best = 1.158187e+03 FES = 866100 μ_F = 0.700 μ_{CR} = 0.289
 Diversity = 5.6605e-07 k_t = [4, 2, 4, 4]
 Generation 8670: Best = 1.158187e+03 FES = 867100 μ_F = 0.712 μ_{CR} = 0.296
 Diversity = 5.7120e-07 k_t = [1, 1, 1, 1]
 Generation 8680: Best = 1.158187e+03 FES = 868100 μ_F = 0.721 μ_{CR} = 0.291
 Diversity = 5.6251e-07 k_t = [3, 3, 4, 3]
 Generation 8690: Best = 1.158187e+03 FES = 869100 μ_F = 0.726 μ_{CR} = 0.293
 Diversity = 5.6981e-07 k_t = [1, 1, 1, 1]
 Generation 8700: Best = 1.158187e+03 FES = 870100 μ_F = 0.708 μ_{CR} = 0.302
 Diversity = 5.2667e-07 k_t = [2, 2, 3, 3]
 Generation 8710: Best = 1.158187e+03 FES = 871100 μ_F = 0.702 μ_{CR} = 0.292
 Diversity = 5.5761e-07 k_t = [1, 1, 1, 1]
 Generation 8720: Best = 1.158187e+03 FES = 872100 μ_F = 0.697 μ_{CR} = 0.296
 Diversity = 5.0209e-07 k_t = [2, 3, 2, 2]
 Generation 8730: Best = 1.158187e+03 FES = 873100 μ_F = 0.693 μ_{CR} = 0.288
 Diversity = 4.8042e-07 k_t = [1, 1, 1, 1]
 Generation 8740: Best = 1.158187e+03 FES = 874100 μ_F = 0.700 μ_{CR} = 0.292
 Diversity = 4.1605e-07 k_t = [3, 3, 2, 4]
 Generation 8750: Best = 1.158187e+03 FES = 875100 μ_F = 0.694 μ_{CR} = 0.285
 Diversity = 3.6700e-07 k_t = [1, 1, 1, 1]
 Generation 8760: Best = 1.158187e+03 FES = 876100 μ_F = 0.696 μ_{CR} = 0.296
 Diversity = 3.5558e-07 k_t = [4, 3, 2, 3]
 Generation 8770: Best = 1.158187e+03 FES = 877100 μ_F = 0.709 μ_{CR} = 0.283
 Diversity = 3.6579e-07 k_t = [1, 1, 1, 1]
 Generation 8780: Best = 1.158187e+03 FES = 878100 μ_F = 0.714 μ_{CR} = 0.272
 Diversity = 3.7845e-07 k_t = [3, 3, 4, 2]
 Generation 8790: Best = 1.158187e+03 FES = 879100 μ_F = 0.715 μ_{CR} = 0.269
 Diversity = 4.1258e-07 k_t = [1, 1, 1, 1]
 Generation 8800: Best = 1.158187e+03 FES = 880100 μ_F = 0.715 μ_{CR} = 0.265
 Diversity = 3.8139e-07 k_t = [2, 4, 4, 3]
 Generation 8810: Best = 1.158187e+03 FES = 881100 μ_F = 0.710 μ_{CR} = 0.259
 Diversity = 3.9635e-07 k_t = [1, 1, 1, 1]
 Generation 8820: Best = 1.158187e+03 FES = 882100 μ_F = 0.717 μ_{CR} = 0.254
 Diversity = 3.7456e-07 k_t = [4, 4, 3, 2]
 Generation 8830: Best = 1.158187e+03 FES = 883100 μ_F = 0.711 μ_{CR} = 0.256
 Diversity = 3.6030e-07 k_t = [1, 1, 1, 1]
 Generation 8840: Best = 1.158187e+03 FES = 884100 μ_F = 0.716 μ_{CR} = 0.257
 Diversity = 3.0579e-07 k_t = [2, 2, 4, 4]
 Generation 8850: Best = 1.158187e+03 FES = 885100 μ_F = 0.717 μ_{CR} = 0.247
 Diversity = 2.9628e-07 k_t = [1, 1, 1, 1]
 Generation 8860: Best = 1.158187e+03 FES = 886100 μ_F = 0.727 μ_{CR} = 0.235
 Diversity = 2.6270e-07 k_t = [3, 3, 3, 4]

Generation 8870: Best = 1.158187e+03 FES = 887100 μ_F = 0.732 μ_{CR} = 0.223
 Diversity = 2.4679e-07 k_t = [1, 1, 1, 1]
 Generation 8880: Best = 1.158187e+03 FES = 888100 μ_F = 0.716 μ_{CR} = 0.227
 Diversity = 2.1918e-07 k_t = [4, 3, 3, 4]
 Generation 8890: Best = 1.158187e+03 FES = 889100 μ_F = 0.721 μ_{CR} = 0.225
 Diversity = 2.1411e-07 k_t = [1, 1, 1, 1]
 Generation 8900: Best = 1.158187e+03 FES = 890100 μ_F = 0.742 μ_{CR} = 0.224
 Diversity = 2.1374e-07 k_t = [3, 4, 3, 4]
 Generation 8910: Best = 1.158187e+03 FES = 891100 μ_F = 0.746 μ_{CR} = 0.222
 Diversity = 2.1923e-07 k_t = [1, 1, 1, 1]
 Generation 8920: Best = 1.158187e+03 FES = 892100 μ_F = 0.738 μ_{CR} = 0.215
 Diversity = 2.2213e-07 k_t = [3, 3, 3, 2]
 Generation 8930: Best = 1.158187e+03 FES = 893100 μ_F = 0.733 μ_{CR} = 0.212
 Diversity = 2.1929e-07 k_t = [1, 1, 1, 1]
 Generation 8940: Best = 1.158187e+03 FES = 894100 μ_F = 0.736 μ_{CR} = 0.210
 Diversity = 2.0246e-07 k_t = [4, 2, 4, 3]
 Generation 8950: Best = 1.158187e+03 FES = 895100 μ_F = 0.730 μ_{CR} = 0.201
 Diversity = 2.0156e-07 k_t = [1, 1, 1, 1]
 Generation 8960: Best = 1.158187e+03 FES = 896100 μ_F = 0.726 μ_{CR} = 0.202
 Diversity = 2.0544e-07 k_t = [2, 4, 2, 2]
 Generation 8970: Best = 1.158187e+03 FES = 897100 μ_F = 0.739 μ_{CR} = 0.200
 Diversity = 2.1382e-07 k_t = [1, 1, 1, 1]
 Generation 8980: Best = 1.158187e+03 FES = 898100 μ_F = 0.730 μ_{CR} = 0.207
 Diversity = 2.0167e-07 k_t = [4, 2, 3, 2]
 Generation 8990: Best = 1.158187e+03 FES = 899100 μ_F = 0.719 μ_{CR} = 0.209
 Diversity = 1.8800e-07 k_t = [1, 1, 1, 1]
 Generation 9000: Best = 1.158187e+03 FES = 900100 μ_F = 0.711 μ_{CR} = 0.210
 Diversity = 1.7146e-07 k_t = [3, 2, 4, 2]
 Generation 9010: Best = 1.158187e+03 FES = 901100 μ_F = 0.704 μ_{CR} = 0.203
 Diversity = 1.6577e-07 k_t = [1, 1, 1, 1]
 Generation 9020: Best = 1.158187e+03 FES = 902100 μ_F = 0.715 μ_{CR} = 0.204
 Diversity = 1.5418e-07 k_t = [4, 4, 4, 3]
 Generation 9030: Best = 1.158187e+03 FES = 903100 μ_F = 0.712 μ_{CR} = 0.202
 Diversity = 1.5051e-07 k_t = [1, 1, 1, 1]
 Generation 9040: Best = 1.158187e+03 FES = 904100 μ_F = 0.695 μ_{CR} = 0.197
 Diversity = 1.4072e-07 k_t = [3, 2, 4, 3]
 Generation 9050: Best = 1.158187e+03 FES = 905100 μ_F = 0.703 μ_{CR} = 0.195
 Diversity = 1.3487e-07 k_t = [1, 1, 1, 1]
 Generation 9060: Best = 1.158187e+03 FES = 906100 μ_F = 0.711 μ_{CR} = 0.204
 Diversity = 1.1714e-07 k_t = [3, 3, 3, 4]
 Generation 9070: Best = 1.158187e+03 FES = 907100 μ_F = 0.726 μ_{CR} = 0.203
 Diversity = 1.1962e-07 k_t = [1, 1, 1, 1]
 Generation 9080: Best = 1.158187e+03 FES = 908100 μ_F = 0.731 μ_{CR} = 0.197
 Diversity = 1.2020e-07 k_t = [2, 2, 3, 3]
 Generation 9090: Best = 1.158187e+03 FES = 909100 μ_F = 0.727 μ_{CR} = 0.196
 Diversity = 1.3235e-07 k_t = [1, 1, 1, 1]
 Generation 9100: Best = 1.158187e+03 FES = 910100 μ_F = 0.726 μ_{CR} = 0.185
 Diversity = 1.2921e-07 k_t = [2, 4, 4, 2]
 Generation 9110: Best = 1.158187e+03 FES = 911100 μ_F = 0.714 μ_{CR} = 0.184
 Diversity = 1.3441e-07 k_t = [1, 1, 1, 1]
 Generation 9120: Best = 1.158187e+03 FES = 912100 μ_F = 0.723 μ_{CR} = 0.190
 Diversity = 1.2088e-07 k_t = [4, 2, 3, 4]
 Generation 9130: Best = 1.158187e+03 FES = 913100 μ_F = 0.720 μ_{CR} = 0.180
 Diversity = 1.1319e-07 k_t = [1, 1, 1, 1]

Generation 9140: Best = 1.158187e+03 FES = 914100 μ_F = 0.734 μ_{CR} = 0.176
 Diversity = 1.0259e-07 k_t = [2, 2, 4, 4]
 Generation 9150: Best = 1.158187e+03 FES = 915100 μ_F = 0.742 μ_{CR} = 0.164
 Diversity = 1.0298e-07 k_t = [1, 1, 1, 1]
 Generation 9160: Best = 1.158187e+03 FES = 916100 μ_F = 0.754 μ_{CR} = 0.165
 Diversity = 1.0510e-07 k_t = [3, 4, 2, 3]
 Generation 9170: Best = 1.158187e+03 FES = 917100 μ_F = 0.761 μ_{CR} = 0.162
 Diversity = 1.1783e-07 k_t = [1, 1, 1, 1]
 Generation 9180: Best = 1.158187e+03 FES = 918100 μ_F = 0.760 μ_{CR} = 0.153
 Diversity = 1.2171e-07 k_t = [2, 2, 3, 2]
 Generation 9190: Best = 1.158187e+03 FES = 919100 μ_F = 0.754 μ_{CR} = 0.150
 Diversity = 1.2646e-07 k_t = [1, 1, 1, 1]
 Generation 9200: Best = 1.158187e+03 FES = 920100 μ_F = 0.752 μ_{CR} = 0.151
 Diversity = 1.2647e-07 k_t = [3, 4, 3, 2]
 Generation 9210: Best = 1.158187e+03 FES = 921100 μ_F = 0.740 μ_{CR} = 0.151
 Diversity = 1.3165e-07 k_t = [1, 1, 1, 1]
 Generation 9220: Best = 1.158187e+03 FES = 922100 μ_F = 0.745 μ_{CR} = 0.153
 Diversity = 1.2121e-07 k_t = [3, 3, 3, 3]
 Generation 9230: Best = 1.158187e+03 FES = 923100 μ_F = 0.741 μ_{CR} = 0.161
 Diversity = 1.2876e-07 k_t = [1, 1, 1, 1]
 Generation 9240: Best = 1.158187e+03 FES = 924100 μ_F = 0.744 μ_{CR} = 0.160
 Diversity = 1.2317e-07 k_t = [2, 3, 2, 4]
 Generation 9250: Best = 1.158187e+03 FES = 925100 μ_F = 0.744 μ_{CR} = 0.163
 Diversity = 1.2691e-07 k_t = [1, 1, 1, 1]
 Generation 9260: Best = 1.158187e+03 FES = 926100 μ_F = 0.760 μ_{CR} = 0.162
 Diversity = 1.3763e-07 k_t = [4, 3, 4, 2]
 Generation 9270: Best = 1.158187e+03 FES = 927100 μ_F = 0.755 μ_{CR} = 0.162
 Diversity = 1.3371e-07 k_t = [1, 1, 1, 1]
 Generation 9280: Best = 1.158187e+03 FES = 928100 μ_F = 0.747 μ_{CR} = 0.161
 Diversity = 1.2908e-07 k_t = [4, 3, 4, 3]
 Generation 9290: Best = 1.158187e+03 FES = 929100 μ_F = 0.761 μ_{CR} = 0.157
 Diversity = 1.2975e-07 k_t = [1, 1, 1, 1]
 Generation 9300: Best = 1.158187e+03 FES = 930100 μ_F = 0.761 μ_{CR} = 0.149
 Diversity = 1.1735e-07 k_t = [2, 3, 3, 3]
 Generation 9310: Best = 1.158187e+03 FES = 931100 μ_F = 0.755 μ_{CR} = 0.158
 Diversity = 1.1550e-07 k_t = [1, 1, 1, 1]
 Generation 9320: Best = 1.158187e+03 FES = 932100 μ_F = 0.753 μ_{CR} = 0.157
 Diversity = 1.0998e-07 k_t = [4, 3, 3, 2]
 Generation 9330: Best = 1.158187e+03 FES = 933100 μ_F = 0.737 μ_{CR} = 0.169
 Diversity = 1.0809e-07 k_t = [1, 1, 1, 1]
 Generation 9340: Best = 1.158187e+03 FES = 934100 μ_F = 0.735 μ_{CR} = 0.160
 Diversity = 9.6987e-08 k_t = [2, 4, 3, 3]
 Generation 9350: Best = 1.158187e+03 FES = 935100 μ_F = 0.741 μ_{CR} = 0.161
 Diversity = 9.7026e-08 k_t = [1, 1, 1, 1]
 Generation 9360: Best = 1.158187e+03 FES = 936100 μ_F = 0.735 μ_{CR} = 0.171
 Diversity = 9.8094e-08 k_t = [3, 4, 4, 2]
 Generation 9370: Best = 1.158187e+03 FES = 937100 μ_F = 0.751 μ_{CR} = 0.166
 Diversity = 9.9458e-08 k_t = [1, 1, 1, 1]
 Generation 9380: Best = 1.158187e+03 FES = 938100 μ_F = 0.749 μ_{CR} = 0.155
 Diversity = 9.4471e-08 k_t = [2, 4, 4, 3]
 Generation 9390: Best = 1.158187e+03 FES = 939100 μ_F = 0.750 μ_{CR} = 0.160
 Diversity = 1.0063e-07 k_t = [1, 1, 1, 1]
 Generation 9400: Best = 1.158187e+03 FES = 940100 μ_F = 0.746 μ_{CR} = 0.154
 Diversity = 9.9937e-08 k_t = [3, 4, 3, 2]

Generation 9410: Best = 1.158187e+03 FES = 941100 μ_F = 0.755 μ_{CR} = 0.157
 Diversity = 1.0063e-07 k_t = [1, 1, 1, 1]
 Generation 9420: Best = 1.158187e+03 FES = 942100 μ_F = 0.761 μ_{CR} = 0.158
 Diversity = 9.7351e-08 k_t = [3, 4, 3, 4]
 Generation 9430: Best = 1.158187e+03 FES = 943100 μ_F = 0.757 μ_{CR} = 0.159
 Diversity = 9.2475e-08 k_t = [1, 1, 1, 1]
 Generation 9440: Best = 1.158187e+03 FES = 944100 μ_F = 0.752 μ_{CR} = 0.164
 Diversity = 8.8248e-08 k_t = [2, 4, 2, 2]
 Generation 9450: Best = 1.158187e+03 FES = 945100 μ_F = 0.758 μ_{CR} = 0.157
 Diversity = 9.1119e-08 k_t = [1, 1, 1, 1]
 Generation 9460: Best = 1.158187e+03 FES = 946100 μ_F = 0.762 μ_{CR} = 0.152
 Diversity = 9.3629e-08 k_t = [3, 4, 3, 3]
 Generation 9470: Best = 1.158187e+03 FES = 947100 μ_F = 0.763 μ_{CR} = 0.156
 Diversity = 9.8242e-08 k_t = [1, 1, 1, 1]
 Generation 9480: Best = 1.158187e+03 FES = 948100 μ_F = 0.776 μ_{CR} = 0.158
 Diversity = 1.0861e-07 k_t = [2, 2, 3, 2]
 Generation 9490: Best = 1.158187e+03 FES = 949100 μ_F = 0.777 μ_{CR} = 0.152
 Diversity = 1.1610e-07 k_t = [1, 1, 1, 1]
 Generation 9500: Best = 1.158187e+03 FES = 950100 μ_F = 0.777 μ_{CR} = 0.150
 Diversity = 1.1066e-07 k_t = [2, 2, 2, 3]
 Generation 9510: Best = 1.158187e+03 FES = 951100 μ_F = 0.768 μ_{CR} = 0.150
 Diversity = 1.1655e-07 k_t = [1, 1, 1, 1]
 Generation 9520: Best = 1.158187e+03 FES = 952100 μ_F = 0.772 μ_{CR} = 0.151
 Diversity = 1.0875e-07 k_t = [2, 3, 4, 4]
 Generation 9530: Best = 1.158187e+03 FES = 953100 μ_F = 0.766 μ_{CR} = 0.150
 Diversity = 1.0857e-07 k_t = [1, 1, 1, 1]
 Generation 9540: Best = 1.158187e+03 FES = 954100 μ_F = 0.770 μ_{CR} = 0.149
 Diversity = 1.0512e-07 k_t = [2, 3, 4, 4]
 Generation 9550: Best = 1.158187e+03 FES = 955100 μ_F = 0.771 μ_{CR} = 0.154
 Diversity = 1.1709e-07 k_t = [1, 1, 1, 1]
 Generation 9560: Best = 1.158187e+03 FES = 956100 μ_F = 0.769 μ_{CR} = 0.149
 Diversity = 1.3756e-07 k_t = [4, 2, 2, 2]
 Generation 9570: Best = 1.158187e+03 FES = 957100 μ_F = 0.769 μ_{CR} = 0.151
 Diversity = 1.4772e-07 k_t = [1, 1, 1, 1]
 Generation 9580: Best = 1.158187e+03 FES = 958100 μ_F = 0.771 μ_{CR} = 0.149
 Diversity = 1.5084e-07 k_t = [2, 2, 4, 2]
 Generation 9590: Best = 1.158187e+03 FES = 959100 μ_F = 0.770 μ_{CR} = 0.157
 Diversity = 1.6695e-07 k_t = [1, 1, 1, 1]
 Generation 9600: Best = 1.158187e+03 FES = 960100 μ_F = 0.772 μ_{CR} = 0.158
 Diversity = 1.7824e-07 k_t = [2, 4, 4, 2]
 Generation 9610: Best = 1.158187e+03 FES = 961100 μ_F = 0.763 μ_{CR} = 0.149
 Diversity = 1.8787e-07 k_t = [1, 1, 1, 1]
 Generation 9620: Best = 1.158187e+03 FES = 962100 μ_F = 0.747 μ_{CR} = 0.148
 Diversity = 1.7439e-07 k_t = [2, 3, 3, 3]
 Generation 9630: Best = 1.158187e+03 FES = 963100 μ_F = 0.735 μ_{CR} = 0.143
 Diversity = 1.7353e-07 k_t = [1, 1, 1, 1]
 Generation 9640: Best = 1.158187e+03 FES = 964100 μ_F = 0.737 μ_{CR} = 0.157
 Diversity = 1.5866e-07 k_t = [3, 3, 2, 4]
 Generation 9650: Best = 1.158187e+03 FES = 965100 μ_F = 0.729 μ_{CR} = 0.152
 Diversity = 1.5461e-07 k_t = [1, 1, 1, 1]
 Generation 9660: Best = 1.158187e+03 FES = 966100 μ_F = 0.738 μ_{CR} = 0.156
 Diversity = 1.5744e-07 k_t = [2, 2, 2, 2]
 Generation 9670: Best = 1.158187e+03 FES = 967100 μ_F = 0.738 μ_{CR} = 0.141
 Diversity = 1.5675e-07 k_t = [1, 1, 1, 1]

Generation 9680: Best = 1.158187e+03 FES = 968100 μ_F = 0.738 μ_{CR} = 0.142
 Diversity = 1.4533e-07 k_t = [2, 3, 4, 3]
 Generation 9690: Best = 1.158187e+03 FES = 969100 μ_F = 0.737 μ_{CR} = 0.150
 Diversity = 1.4637e-07 k_t = [1, 1, 1, 1]
 Generation 9700: Best = 1.158187e+03 FES = 970100 μ_F = 0.737 μ_{CR} = 0.143
 Diversity = 1.3314e-07 k_t = [3, 3, 2, 3]
 Generation 9710: Best = 1.158187e+03 FES = 971100 μ_F = 0.742 μ_{CR} = 0.142
 Diversity = 1.3214e-07 k_t = [1, 1, 1, 1]
 Generation 9720: Best = 1.158187e+03 FES = 972100 μ_F = 0.752 μ_{CR} = 0.144
 Diversity = 1.2223e-07 k_t = [4, 3, 4, 4]
 Generation 9730: Best = 1.158187e+03 FES = 973100 μ_F = 0.757 μ_{CR} = 0.154
 Diversity = 1.2002e-07 k_t = [1, 1, 1, 1]
 Generation 9740: Best = 1.158187e+03 FES = 974100 μ_F = 0.754 μ_{CR} = 0.151
 Diversity = 1.1479e-07 k_t = [3, 3, 2, 3]
 Generation 9750: Best = 1.158187e+03 FES = 975100 μ_F = 0.755 μ_{CR} = 0.154
 Diversity = 1.1893e-07 k_t = [1, 1, 1, 1]
 Generation 9760: Best = 1.158187e+03 FES = 976100 μ_F = 0.754 μ_{CR} = 0.157
 Diversity = 1.2007e-07 k_t = [2, 4, 4, 4]
 Generation 9770: Best = 1.158187e+03 FES = 977100 μ_F = 0.750 μ_{CR} = 0.164
 Diversity = 1.3131e-07 k_t = [1, 1, 1, 1]
 Generation 9780: Best = 1.158187e+03 FES = 978100 μ_F = 0.754 μ_{CR} = 0.167
 Diversity = 1.2305e-07 k_t = [3, 4, 4, 4]
 Generation 9790: Best = 1.158187e+03 FES = 979100 μ_F = 0.752 μ_{CR} = 0.162
 Diversity = 1.2746e-07 k_t = [1, 1, 1, 1]
 Generation 9800: Best = 1.158187e+03 FES = 980100 μ_F = 0.767 μ_{CR} = 0.169
 Diversity = 1.3885e-07 k_t = [4, 2, 3, 3]
 Generation 9810: Best = 1.158187e+03 FES = 981100 μ_F = 0.757 μ_{CR} = 0.171
 Diversity = 1.4693e-07 k_t = [1, 1, 1, 1]
 Generation 9820: Best = 1.158187e+03 FES = 982100 μ_F = 0.755 μ_{CR} = 0.171
 Diversity = 1.4165e-07 k_t = [4, 2, 4, 3]
 Generation 9830: Best = 1.158187e+03 FES = 983100 μ_F = 0.751 μ_{CR} = 0.162
 Diversity = 1.5092e-07 k_t = [1, 1, 1, 1]
 Generation 9840: Best = 1.158187e+03 FES = 984100 μ_F = 0.756 μ_{CR} = 0.172
 Diversity = 1.5208e-07 k_t = [2, 3, 2, 3]
 Generation 9850: Best = 1.158187e+03 FES = 985100 μ_F = 0.753 μ_{CR} = 0.174
 Diversity = 1.5037e-07 k_t = [1, 1, 1, 1]
 Generation 9860: Best = 1.158187e+03 FES = 986100 μ_F = 0.745 μ_{CR} = 0.169
 Diversity = 1.4020e-07 k_t = [3, 3, 3, 3]
 Generation 9870: Best = 1.158187e+03 FES = 987100 μ_F = 0.744 μ_{CR} = 0.166
 Diversity = 1.3831e-07 k_t = [1, 1, 1, 1]
 Generation 9880: Best = 1.158187e+03 FES = 988100 μ_F = 0.735 μ_{CR} = 0.171
 Diversity = 1.1762e-07 k_t = [2, 3, 3, 3]
 Generation 9890: Best = 1.158187e+03 FES = 989100 μ_F = 0.726 μ_{CR} = 0.162
 Diversity = 1.1389e-07 k_t = [1, 1, 1, 1]
 Generation 9900: Best = 1.158187e+03 FES = 990100 μ_F = 0.732 μ_{CR} = 0.164
 Diversity = 9.8214e-08 k_t = [3, 4, 3, 4]
 Generation 9910: Best = 1.158187e+03 FES = 991100 μ_F = 0.740 μ_{CR} = 0.158
 Diversity = 9.5683e-08 k_t = [1, 1, 1, 1]
 Generation 9920: Best = 1.158187e+03 FES = 992100 μ_F = 0.735 μ_{CR} = 0.160
 Diversity = 8.9077e-08 k_t = [2, 3, 2, 3]
 Generation 9930: Best = 1.158187e+03 FES = 993100 μ_F = 0.734 μ_{CR} = 0.163
 Diversity = 9.0975e-08 k_t = [1, 1, 1, 1]
 Generation 9940: Best = 1.158187e+03 FES = 994100 μ_F = 0.745 μ_{CR} = 0.166
 Diversity = 8.2790e-08 k_t = [3, 3, 4, 3]

Generation 9950: Best = 1.158187e+03 FES = 995100 μ_F = 0.751 μ_{CR} = 0.168
 Diversity = 9.6390e-08 k_t = [1, 1, 1, 1]
 Generation 9960: Best = 1.158187e+03 FES = 996100 μ_F = 0.751 μ_{CR} = 0.174
 Diversity = 1.0435e-07 k_t = [4, 4, 3, 4]
 Generation 9970: Best = 1.158187e+03 FES = 997100 μ_F = 0.761 μ_{CR} = 0.173
 Diversity = 1.0813e-07 k_t = [1, 1, 1, 1]
 Generation 9980: Best = 1.158187e+03 FES = 998100 μ_F = 0.763 μ_{CR} = 0.163
 Diversity = 1.3316e-07 k_t = [4, 3, 4, 2]
 Generation 9990: Best = 1.158187e+03 FES = 999100 μ_F = 0.753 μ_{CR} = 0.161
 Diversity = 1.4510e-07 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 1.158187e+03
 Total Function Evals = 1000000
 Total Successful Updates = 366767

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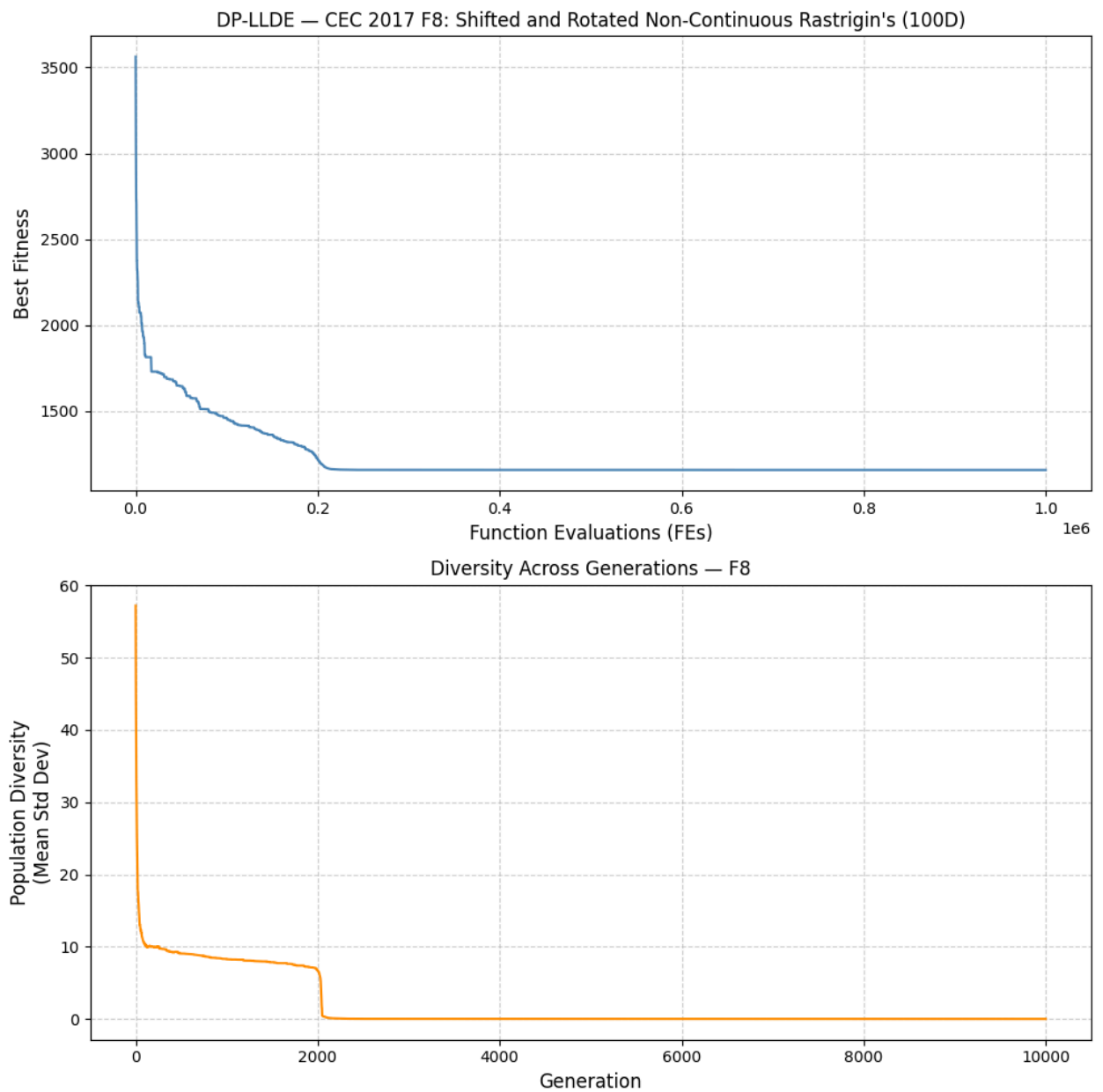
Results:

Best Fitness : 1.158187e+03
 Global Optimum : 8.000000e+02
 Error (|best-optimum|) : 3.581867e+02
 Best Solution (5 dim) : [-3.6926127 -45.42958733 16.70490378 -29.84325561
 -31.28761667]
 D^(0) (init. diversity): 5.7220e+01

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✓ Checkpoint saved to: /content/drive/My Drive/thesis_results/all_results_checkpoint.pkl



Graph saved as convergence_F8.png

✓ Plot saved to: /content/drive/My Drive/thesis_results/plots/convergence_F8.png

```
In [ ]: # =====
# F9 – Shifted and Rotated Levy
# =====
run_and_plot('F9', make_wrapper(simple.f9), 900.0, 'Shifted and Rotated Levy')
```

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DP-LLDE on CEC 2017 F9 – Single Run

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Function : Shifted and Rotated Levy (100D)
Bounds : [-100, 100]
Global optimum : 900.0
Initial $D^*(0) = 5.7220e+01$ | Diversity threshold = 5.7220e+00
Generation 0: Best Fitness = 2.313486e+05

Generation 10:	Best = 1.143948e+05	FES = 1100	$\mu_F = 0.546$	$\mu_{CR} = 0.357$
Diversity = 3.4392e+01	$k_t = [1, 1, 1, 1]$			
Generation 20:	Best = 8.451099e+04	FES = 2100	$\mu_F = 0.589$	$\mu_{CR} = 0.378$
Diversity = 2.4638e+01	$k_t = [2, 3, 2, 3]$			
Generation 30:	Best = 6.428864e+04	FES = 3100	$\mu_F = 0.608$	$\mu_{CR} = 0.380$
Diversity = 1.8877e+01	$k_t = [1, 1, 1, 1]$			
Generation 40:	Best = 5.104895e+04	FES = 4100	$\mu_F = 0.635$	$\mu_{CR} = 0.397$
Diversity = 1.4989e+01	$k_t = [3, 2, 3, 4]$			
Generation 50:	Best = 4.567143e+04	FES = 5100	$\mu_F = 0.669$	$\mu_{CR} = 0.402$
Diversity = 1.3589e+01	$k_t = [1, 1, 1, 1]$			
Generation 60:	Best = 3.633245e+04	FES = 6100	$\mu_F = 0.673$	$\mu_{CR} = 0.396$
Diversity = 1.2349e+01	$k_t = [4, 2, 2, 2]$			
Generation 70:	Best = 3.633245e+04	FES = 7100	$\mu_F = 0.695$	$\mu_{CR} = 0.398$
Diversity = 1.1417e+01	$k_t = [1, 1, 1, 1]$			
Generation 80:	Best = 3.121626e+04	FES = 8100	$\mu_F = 0.724$	$\mu_{CR} = 0.425$
Diversity = 1.1390e+01	$k_t = [4, 3, 3, 3]$			
Generation 90:	Best = 2.631786e+04	FES = 9100	$\mu_F = 0.731$	$\mu_{CR} = 0.436$
Diversity = 1.1169e+01	$k_t = [1, 1, 1, 1]$			
Generation 100:	Best = 2.381856e+04	FES = 10100	$\mu_F = 0.749$	$\mu_{CR} = 0.439$
Diversity = 1.0941e+01	$k_t = [2, 4, 4, 2]$			
Generation 110:	Best = 2.381856e+04	FES = 11100	$\mu_F = 0.728$	$\mu_{CR} = 0.454$
Diversity = 1.0628e+01	$k_t = [1, 1, 1, 1]$			
Generation 120:	Best = 2.002288e+04	FES = 12100	$\mu_F = 0.701$	$\mu_{CR} = 0.445$
Diversity = 9.9815e+00	$k_t = [3, 3, 2, 4]$			
Generation 130:	Best = 1.787862e+04	FES = 13100	$\mu_F = 0.742$	$\mu_{CR} = 0.445$
Diversity = 9.3324e+00	$k_t = [1, 1, 1, 1]$			
Generation 140:	Best = 1.737462e+04	FES = 14100	$\mu_F = 0.723$	$\mu_{CR} = 0.452$
Diversity = 8.5820e+00	$k_t = [2, 4, 2, 3]$			
Generation 150:	Best = 1.635322e+04	FES = 15100	$\mu_F = 0.727$	$\mu_{CR} = 0.454$
Diversity = 8.1764e+00	$k_t = [1, 1, 1, 1]$			
Generation 160:	Best = 1.595533e+04	FES = 16100	$\mu_F = 0.713$	$\mu_{CR} = 0.454$
Diversity = 7.9226e+00	$k_t = [4, 3, 2, 2]$			
Generation 170:	Best = 1.499947e+04	FES = 17100	$\mu_F = 0.698$	$\mu_{CR} = 0.468$
Diversity = 7.4975e+00	$k_t = [1, 1, 1, 1]$			
Generation 180:	Best = 1.296689e+04	FES = 18100	$\mu_F = 0.669$	$\mu_{CR} = 0.485$
Diversity = 7.1495e+00	$k_t = [3, 4, 2, 3]$			
Generation 190:	Best = 1.248810e+04	FES = 19100	$\mu_F = 0.658$	$\mu_{CR} = 0.495$
Diversity = 6.4612e+00	$k_t = [1, 1, 1, 1]$			
Generation 200:	Best = 1.192328e+04	FES = 20100	$\mu_F = 0.673$	$\mu_{CR} = 0.484$
Diversity = 6.0701e+00	$k_t = [2, 3, 3, 3]$			
Generation 210:	Best = 1.171097e+04	FES = 21100	$\mu_F = 0.685$	$\mu_{CR} = 0.524$
Diversity = 5.6953e+00	$k_t = [1, 1, 1, 1]$			
Generation 220:	Best = 1.102901e+04	FES = 22100	$\mu_F = 0.712$	$\mu_{CR} = 0.510$
Diversity = 5.4925e+00	$k_t = [4, 2, 2, 3]$			

Generation 230: Best = 1.053212e+04	FES = 23100	$\mu_F = 0.708$	$\mu_{CR} = 0.518$
Diversity = 5.4945e+00	k_t = [1, 1, 1, 1]		
Generation 240: Best = 8.994444e+03	FES = 24100	$\mu_F = 0.732$	$\mu_{CR} = 0.487$
Diversity = 5.4170e+00	k_t = [4, 2, 4, 2]		
Generation 250: Best = 8.994444e+03	FES = 25100	$\mu_F = 0.731$	$\mu_{CR} = 0.464$
Diversity = 5.3482e+00	k_t = [1, 1, 1, 1]		
Generation 260: Best = 8.789302e+03	FES = 26100	$\mu_F = 0.737$	$\mu_{CR} = 0.433$
Diversity = 5.3827e+00	k_t = [2, 4, 4, 3]		
Generation 270: Best = 8.789302e+03	FES = 27100	$\mu_F = 0.741$	$\mu_{CR} = 0.417$
Diversity = 5.2452e+00	k_t = [1, 1, 1, 1]		
Generation 280: Best = 7.904893e+03	FES = 28100	$\mu_F = 0.778$	$\mu_{CR} = 0.409$
Diversity = 5.2848e+00	k_t = [4, 2, 2, 4]		
Generation 290: Best = 7.904893e+03	FES = 29100	$\mu_F = 0.743$	$\mu_{CR} = 0.422$
Diversity = 5.1136e+00	k_t = [1, 1, 1, 1]		
Generation 300: Best = 7.471476e+03	FES = 30100	$\mu_F = 0.700$	$\mu_{CR} = 0.401$
Diversity = 4.9745e+00	k_t = [3, 3, 2, 4]		
Generation 310: Best = 7.471476e+03	FES = 31100	$\mu_F = 0.732$	$\mu_{CR} = 0.371$
Diversity = 4.8363e+00	k_t = [1, 1, 1, 1]		
Generation 320: Best = 7.403199e+03	FES = 32100	$\mu_F = 0.663$	$\mu_{CR} = 0.354$
Diversity = 4.7173e+00	k_t = [4, 2, 4, 3]		
Generation 330: Best = 6.929785e+03	FES = 33100	$\mu_F = 0.652$	$\mu_{CR} = 0.332$
Diversity = 4.5931e+00	k_t = [1, 1, 1, 1]		
Generation 340: Best = 6.557447e+03	FES = 34100	$\mu_F = 0.616$	$\mu_{CR} = 0.306$
Diversity = 4.4583e+00	k_t = [3, 3, 4, 3]		
Generation 350: Best = 6.557447e+03	FES = 35100	$\mu_F = 0.655$	$\mu_{CR} = 0.226$
Diversity = 4.3993e+00	k_t = [1, 1, 1, 1]		
Generation 360: Best = 6.557447e+03	FES = 36100	$\mu_F = 0.628$	$\mu_{CR} = 0.172$
Diversity = 4.3251e+00	k_t = [4, 4, 2, 3]		
Generation 370: Best = 6.222837e+03	FES = 37100	$\mu_F = 0.646$	$\mu_{CR} = 0.104$
Diversity = 4.2500e+00	k_t = [1, 1, 1, 1]		
Generation 380: Best = 6.222837e+03	FES = 38100	$\mu_F = 0.653$	$\mu_{CR} = 0.089$
Diversity = 4.1891e+00	k_t = [4, 3, 3, 4]		
Generation 390: Best = 5.798385e+03	FES = 39100	$\mu_F = 0.643$	$\mu_{CR} = 0.078$
Diversity = 4.1097e+00	k_t = [1, 1, 1, 1]		
Generation 400: Best = 5.364273e+03	FES = 40100	$\mu_F = 0.660$	$\mu_{CR} = 0.061$
Diversity = 4.0699e+00	k_t = [3, 4, 4, 3]		
Generation 410: Best = 5.364273e+03	FES = 41100	$\mu_F = 0.743$	$\mu_{CR} = 0.049$
Diversity = 4.0635e+00	k_t = [1, 1, 1, 1]		
Generation 420: Best = 5.364273e+03	FES = 42100	$\mu_F = 0.789$	$\mu_{CR} = 0.042$
Diversity = 4.0158e+00	k_t = [3, 2, 2, 4]		
Generation 430: Best = 4.853860e+03	FES = 43100	$\mu_F = 0.805$	$\mu_{CR} = 0.037$
Diversity = 3.9863e+00	k_t = [1, 1, 1, 1]		
Generation 440: Best = 4.830526e+03	FES = 44100	$\mu_F = 0.829$	$\mu_{CR} = 0.037$
Diversity = 3.9932e+00	k_t = [3, 2, 2, 4]		
Generation 450: Best = 4.813649e+03	FES = 45100	$\mu_F = 0.823$	$\mu_{CR} = 0.032$
Diversity = 3.9718e+00	k_t = [1, 1, 1, 1]		
Generation 460: Best = 4.813649e+03	FES = 46100	$\mu_F = 0.802$	$\mu_{CR} = 0.032$
Diversity = 3.9784e+00	k_t = [3, 3, 2, 2]		
Generation 470: Best = 4.813649e+03	FES = 47100	$\mu_F = 0.798$	$\mu_{CR} = 0.031$
Diversity = 3.9652e+00	k_t = [1, 1, 1, 1]		
Generation 480: Best = 4.813649e+03	FES = 48100	$\mu_F = 0.804$	$\mu_{CR} = 0.022$
Diversity = 3.9754e+00	k_t = [2, 4, 2, 3]		
Generation 490: Best = 4.708214e+03	FES = 49100	$\mu_F = 0.731$	$\mu_{CR} = 0.024$
Diversity = 3.9616e+00	k_t = [1, 1, 1, 1]		

Generation 500: Best = 4.707006e+03	FES = 50100	$\mu_F = 0.772$	$\mu_{CR} = 0.028$
Diversity = 3.9319e+00	$k_t = [3, 4, 3, 2]$		
Generation 510: Best = 4.707006e+03	FES = 51100	$\mu_F = 0.751$	$\mu_{CR} = 0.025$
Diversity = 3.9145e+00	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 4.707006e+03	FES = 52100	$\mu_F = 0.776$	$\mu_{CR} = 0.039$
Diversity = 3.8990e+00	$k_t = [2, 2, 2, 4]$		
Generation 530: Best = 4.670614e+03	FES = 53100	$\mu_F = 0.746$	$\mu_{CR} = 0.032$
Diversity = 3.8743e+00	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 4.603554e+03	FES = 54100	$\mu_F = 0.774$	$\mu_{CR} = 0.031$
Diversity = 3.8215e+00	$k_t = [3, 4, 3, 4]$		
Generation 550: Best = 4.603554e+03	FES = 55100	$\mu_F = 0.747$	$\mu_{CR} = 0.033$
Diversity = 3.8220e+00	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 4.603554e+03	FES = 56100	$\mu_F = 0.776$	$\mu_{CR} = 0.031$
Diversity = 3.8181e+00	$k_t = [4, 3, 3, 3]$		
Generation 570: Best = 4.603554e+03	FES = 57100	$\mu_F = 0.782$	$\mu_{CR} = 0.029$
Diversity = 3.8202e+00	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 4.603554e+03	FES = 58100	$\mu_F = 0.783$	$\mu_{CR} = 0.036$
Diversity = 3.8003e+00	$k_t = [3, 4, 3, 2]$		
Generation 590: Best = 4.603554e+03	FES = 59100	$\mu_F = 0.720$	$\mu_{CR} = 0.033$
Diversity = 3.8070e+00	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 4.575833e+03	FES = 60100	$\mu_F = 0.712$	$\mu_{CR} = 0.043$
Diversity = 3.8135e+00	$k_t = [4, 2, 3, 2]$		
Generation 610: Best = 4.398993e+03	FES = 61100	$\mu_F = 0.686$	$\mu_{CR} = 0.046$
Diversity = 3.7949e+00	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 4.358205e+03	FES = 62100	$\mu_F = 0.728$	$\mu_{CR} = 0.043$
Diversity = 3.7986e+00	$k_t = [2, 4, 4, 3]$		
Generation 630: Best = 4.358205e+03	FES = 63100	$\mu_F = 0.640$	$\mu_{CR} = 0.043$
Diversity = 3.7832e+00	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 4.358205e+03	FES = 64100	$\mu_F = 0.655$	$\mu_{CR} = 0.036$
Diversity = 3.7760e+00	$k_t = [4, 2, 2, 2]$		
Generation 650: Best = 4.358205e+03	FES = 65100	$\mu_F = 0.617$	$\mu_{CR} = 0.061$
Diversity = 3.7844e+00	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 4.267593e+03	FES = 66100	$\mu_F = 0.597$	$\mu_{CR} = 0.040$
Diversity = 3.7718e+00	$k_t = [2, 2, 2, 2]$		
Generation 670: Best = 4.242051e+03	FES = 67100	$\mu_F = 0.628$	$\mu_{CR} = 0.032$
Diversity = 3.7710e+00	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 4.227254e+03	FES = 68100	$\mu_F = 0.588$	$\mu_{CR} = 0.028$
Diversity = 3.7553e+00	$k_t = [3, 3, 2, 3]$		
Generation 690: Best = 4.227254e+03	FES = 69100	$\mu_F = 0.519$	$\mu_{CR} = 0.032$
Diversity = 3.7502e+00	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 4.149127e+03	FES = 70100	$\mu_F = 0.567$	$\mu_{CR} = 0.037$
Diversity = 3.7074e+00	$k_t = [4, 2, 3, 4]$		
Generation 710: Best = 4.149127e+03	FES = 71100	$\mu_F = 0.544$	$\mu_{CR} = 0.032$
Diversity = 3.7051e+00	$k_t = [1, 1, 1, 1]$		
Generation 720: Best = 4.149127e+03	FES = 72100	$\mu_F = 0.490$	$\mu_{CR} = 0.033$
Diversity = 3.6974e+00	$k_t = [4, 2, 4, 4]$		
Generation 730: Best = 4.149127e+03	FES = 73100	$\mu_F = 0.458$	$\mu_{CR} = 0.040$
Diversity = 3.6846e+00	$k_t = [1, 1, 1, 1]$		
Generation 740: Best = 3.980135e+03	FES = 74100	$\mu_F = 0.451$	$\mu_{CR} = 0.047$
Diversity = 3.6920e+00	$k_t = [4, 2, 4, 4]$		
Generation 750: Best = 3.972739e+03	FES = 75100	$\mu_F = 0.457$	$\mu_{CR} = 0.032$
Diversity = 3.6900e+00	$k_t = [1, 1, 1, 1]$		
Generation 760: Best = 3.893489e+03	FES = 76100	$\mu_F = 0.536$	$\mu_{CR} = 0.034$
Diversity = 3.7024e+00	$k_t = [3, 4, 4, 4]$		

Generation 770: Best = 3.893489e+03	FES = 77100	$\mu_F = 0.508$	$\mu_{CR} = 0.031$
Diversity = 3.6974e+00	$k_t = [1, 1, 1, 1]$		
Generation 780: Best = 3.853935e+03	FES = 78100	$\mu_F = 0.426$	$\mu_{CR} = 0.038$
Diversity = 3.6914e+00	$k_t = [4, 2, 2, 2]$		
Generation 790: Best = 3.808553e+03	FES = 79100	$\mu_F = 0.425$	$\mu_{CR} = 0.042$
Diversity = 3.6831e+00	$k_t = [1, 1, 1, 1]$		
Generation 800: Best = 3.808553e+03	FES = 80100	$\mu_F = 0.404$	$\mu_{CR} = 0.040$
Diversity = 3.6783e+00	$k_t = [3, 4, 4, 3]$		
Generation 810: Best = 3.808553e+03	FES = 81100	$\mu_F = 0.394$	$\mu_{CR} = 0.038$
Diversity = 3.6660e+00	$k_t = [1, 1, 1, 1]$		
Generation 820: Best = 3.808553e+03	FES = 82100	$\mu_F = 0.388$	$\mu_{CR} = 0.033$
Diversity = 3.6590e+00	$k_t = [3, 4, 4, 4]$		
Generation 830: Best = 3.808553e+03	FES = 83100	$\mu_F = 0.296$	$\mu_{CR} = 0.028$
Diversity = 3.6583e+00	$k_t = [1, 1, 1, 1]$		
Generation 840: Best = 3.664393e+03	FES = 84100	$\mu_F = 0.394$	$\mu_{CR} = 0.030$
Diversity = 3.6545e+00	$k_t = [2, 2, 3, 2]$		
Generation 850: Best = 3.660976e+03	FES = 85100	$\mu_F = 0.309$	$\mu_{CR} = 0.050$
Diversity = 3.6498e+00	$k_t = [1, 1, 1, 1]$		
Generation 860: Best = 3.660976e+03	FES = 86100	$\mu_F = 0.266$	$\mu_{CR} = 0.050$
Diversity = 3.6459e+00	$k_t = [4, 2, 2, 3]$		
Generation 870: Best = 3.660976e+03	FES = 87100	$\mu_F = 0.220$	$\mu_{CR} = 0.055$
Diversity = 3.6409e+00	$k_t = [1, 1, 1, 1]$		
Generation 880: Best = 3.621724e+03	FES = 88100	$\mu_F = 0.252$	$\mu_{CR} = 0.049$
Diversity = 3.6336e+00	$k_t = [3, 3, 2, 4]$		
Generation 890: Best = 3.621724e+03	FES = 89100	$\mu_F = 0.300$	$\mu_{CR} = 0.042$
Diversity = 3.6310e+00	$k_t = [1, 1, 1, 1]$		
Generation 900: Best = 3.621724e+03	FES = 90100	$\mu_F = 0.279$	$\mu_{CR} = 0.065$
Diversity = 3.6209e+00	$k_t = [2, 4, 4, 2]$		
Generation 910: Best = 3.621724e+03	FES = 91100	$\mu_F = 0.330$	$\mu_{CR} = 0.046$
Diversity = 3.6179e+00	$k_t = [1, 1, 1, 1]$		
Generation 920: Best = 3.621724e+03	FES = 92100	$\mu_F = 0.250$	$\mu_{CR} = 0.059$
Diversity = 3.6149e+00	$k_t = [2, 3, 4, 3]$		
Generation 930: Best = 3.621724e+03	FES = 93100	$\mu_F = 0.222$	$\mu_{CR} = 0.063$
Diversity = 3.6105e+00	$k_t = [1, 1, 1, 1]$		
Generation 940: Best = 3.611612e+03	FES = 94100	$\mu_F = 0.265$	$\mu_{CR} = 0.052$
Diversity = 3.6063e+00	$k_t = [4, 2, 2, 2]$		
Generation 950: Best = 3.606810e+03	FES = 95100	$\mu_F = 0.248$	$\mu_{CR} = 0.043$
Diversity = 3.5998e+00	$k_t = [1, 1, 1, 1]$		
Generation 960: Best = 3.606810e+03	FES = 96100	$\mu_F = 0.292$	$\mu_{CR} = 0.052$
Diversity = 3.5950e+00	$k_t = [3, 4, 2, 4]$		
Generation 970: Best = 3.606810e+03	FES = 97100	$\mu_F = 0.257$	$\mu_{CR} = 0.038$
Diversity = 3.5952e+00	$k_t = [1, 1, 1, 1]$		
Generation 980: Best = 3.606810e+03	FES = 98100	$\mu_F = 0.264$	$\mu_{CR} = 0.041$
Diversity = 3.5912e+00	$k_t = [3, 3, 2, 2]$		
Generation 990: Best = 3.589403e+03	FES = 99100	$\mu_F = 0.236$	$\mu_{CR} = 0.039$
Diversity = 3.5900e+00	$k_t = [1, 1, 1, 1]$		
Generation 1000: Best = 3.583533e+03	FES = 100100	$\mu_F = 0.311$	$\mu_{CR} = 0.039$
Diversity = 3.5859e+00	$k_t = [3, 3, 3, 2]$		
Generation 1010: Best = 3.583533e+03	FES = 101100	$\mu_F = 0.356$	$\mu_{CR} = 0.037$
Diversity = 3.5843e+00	$k_t = [1, 1, 1, 1]$		
Generation 1020: Best = 3.564831e+03	FES = 102100	$\mu_F = 0.253$	$\mu_{CR} = 0.034$
Diversity = 3.5825e+00	$k_t = [2, 3, 3, 4]$		
Generation 1030: Best = 3.564831e+03	FES = 103100	$\mu_F = 0.289$	$\mu_{CR} = 0.044$
Diversity = 3.5784e+00	$k_t = [1, 1, 1, 1]$		

Generation 1040: Best = 3.564831e+03 FES = 104100 μ_F = 0.388 μ_{CR} = 0.037
 Diversity = 3.5702e+00 k_t = [3, 3, 4, 4]
 Generation 1050: Best = 3.510989e+03 FES = 105100 μ_F = 0.365 μ_{CR} = 0.029
 Diversity = 3.5663e+00 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 3.482804e+03 FES = 106100 μ_F = 0.284 μ_{CR} = 0.050
 Diversity = 3.5644e+00 k_t = [3, 3, 4, 3]
 Generation 1070: Best = 3.476721e+03 FES = 107100 μ_F = 0.287 μ_{CR} = 0.048
 Diversity = 3.5635e+00 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 3.476721e+03 FES = 108100 μ_F = 0.344 μ_{CR} = 0.044
 Diversity = 3.5481e+00 k_t = [2, 3, 4, 4]
 Generation 1090: Best = 3.457913e+03 FES = 109100 μ_F = 0.309 μ_{CR} = 0.036
 Diversity = 3.5451e+00 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 3.432769e+03 FES = 110100 μ_F = 0.307 μ_{CR} = 0.036
 Diversity = 3.5432e+00 k_t = [2, 2, 2, 3]
 Generation 1110: Best = 3.432769e+03 FES = 111100 μ_F = 0.378 μ_{CR} = 0.040
 Diversity = 3.5421e+00 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 3.417246e+03 FES = 112100 μ_F = 0.451 μ_{CR} = 0.034
 Diversity = 3.5219e+00 k_t = [3, 3, 3, 4]
 Generation 1130: Best = 3.417246e+03 FES = 113100 μ_F = 0.427 μ_{CR} = 0.024
 Diversity = 3.5213e+00 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 3.391419e+03 FES = 114100 μ_F = 0.489 μ_{CR} = 0.024
 Diversity = 3.5144e+00 k_t = [4, 2, 2, 3]
 Generation 1150: Best = 3.391419e+03 FES = 115100 μ_F = 0.521 μ_{CR} = 0.029
 Diversity = 3.5127e+00 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 3.347078e+03 FES = 116100 μ_F = 0.578 μ_{CR} = 0.030
 Diversity = 3.4821e+00 k_t = [3, 3, 4, 4]
 Generation 1170: Best = 3.347078e+03 FES = 117100 μ_F = 0.450 μ_{CR} = 0.029
 Diversity = 3.4827e+00 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 3.347078e+03 FES = 118100 μ_F = 0.485 μ_{CR} = 0.042
 Diversity = 3.4400e+00 k_t = [4, 4, 3, 4]
 Generation 1190: Best = 3.310309e+03 FES = 119100 μ_F = 0.570 μ_{CR} = 0.031
 Diversity = 3.4449e+00 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 3.309282e+03 FES = 120100 μ_F = 0.605 μ_{CR} = 0.027
 Diversity = 3.4479e+00 k_t = [4, 3, 3, 3]
 Generation 1210: Best = 3.265885e+03 FES = 121100 μ_F = 0.588 μ_{CR} = 0.029
 Diversity = 3.4487e+00 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 3.251594e+03 FES = 122100 μ_F = 0.433 μ_{CR} = 0.042
 Diversity = 3.4470e+00 k_t = [3, 2, 4, 3]
 Generation 1230: Best = 3.251594e+03 FES = 123100 μ_F = 0.398 μ_{CR} = 0.034
 Diversity = 3.4457e+00 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 3.251594e+03 FES = 124100 μ_F = 0.395 μ_{CR} = 0.037
 Diversity = 3.4450e+00 k_t = [2, 4, 2, 2]
 Generation 1250: Best = 3.251594e+03 FES = 125100 μ_F = 0.422 μ_{CR} = 0.030
 Diversity = 3.4439e+00 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 3.197726e+03 FES = 126100 μ_F = 0.411 μ_{CR} = 0.028
 Diversity = 3.4394e+00 k_t = [2, 2, 3, 3]
 Generation 1270: Best = 3.197726e+03 FES = 127100 μ_F = 0.342 μ_{CR} = 0.044
 Diversity = 3.4382e+00 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 3.196221e+03 FES = 128100 μ_F = 0.337 μ_{CR} = 0.044
 Diversity = 3.4272e+00 k_t = [4, 4, 3, 4]
 Generation 1290: Best = 3.186578e+03 FES = 129100 μ_F = 0.404 μ_{CR} = 0.033
 Diversity = 3.4287e+00 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 3.186578e+03 FES = 130100 μ_F = 0.452 μ_{CR} = 0.040
 Diversity = 3.4137e+00 k_t = [4, 4, 4, 4]

Generation 1310: Best = 3.186578e+03 FES = 131100 μ_F = 0.456 μ_{CR} = 0.036
 Diversity = 3.4143e+00 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 3.186578e+03 FES = 132100 μ_F = 0.567 μ_{CR} = 0.030
 Diversity = 3.4007e+00 k_t = [4, 3, 2, 2]
 Generation 1330: Best = 3.186578e+03 FES = 133100 μ_F = 0.493 μ_{CR} = 0.030
 Diversity = 3.4015e+00 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 3.173854e+03 FES = 134100 μ_F = 0.531 μ_{CR} = 0.032
 Diversity = 3.4022e+00 k_t = [4, 4, 4, 3]
 Generation 1350: Best = 3.173854e+03 FES = 135100 μ_F = 0.548 μ_{CR} = 0.030
 Diversity = 3.4029e+00 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 3.170777e+03 FES = 136100 μ_F = 0.571 μ_{CR} = 0.040
 Diversity = 3.3891e+00 k_t = [3, 4, 2, 3]
 Generation 1370: Best = 3.162825e+03 FES = 137100 μ_F = 0.444 μ_{CR} = 0.037
 Diversity = 3.3882e+00 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 3.162825e+03 FES = 138100 μ_F = 0.542 μ_{CR} = 0.025
 Diversity = 3.3900e+00 k_t = [2, 4, 3, 4]
 Generation 1390: Best = 3.162825e+03 FES = 139100 μ_F = 0.487 μ_{CR} = 0.027
 Diversity = 3.3908e+00 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 3.162825e+03 FES = 140100 μ_F = 0.485 μ_{CR} = 0.026
 Diversity = 3.3892e+00 k_t = [2, 3, 2, 2]
 Generation 1410: Best = 3.162825e+03 FES = 141100 μ_F = 0.374 μ_{CR} = 0.047
 Diversity = 3.3886e+00 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 3.143987e+03 FES = 142100 μ_F = 0.350 μ_{CR} = 0.049
 Diversity = 3.3763e+00 k_t = [2, 4, 4, 4]
 Generation 1430: Best = 3.143987e+03 FES = 143100 μ_F = 0.299 μ_{CR} = 0.040
 Diversity = 3.3752e+00 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 3.143987e+03 FES = 144100 μ_F = 0.274 μ_{CR} = 0.034
 Diversity = 3.3748e+00 k_t = [4, 3, 4, 3]
 Generation 1450: Best = 3.133019e+03 FES = 145100 μ_F = 0.299 μ_{CR} = 0.040
 Diversity = 3.3693e+00 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 3.105612e+03 FES = 146100 μ_F = 0.335 μ_{CR} = 0.040
 Diversity = 3.3477e+00 k_t = [3, 4, 4, 4]
 Generation 1470: Best = 3.105612e+03 FES = 147100 μ_F = 0.304 μ_{CR} = 0.053
 Diversity = 3.3453e+00 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 3.105612e+03 FES = 148100 μ_F = 0.229 μ_{CR} = 0.066
 Diversity = 3.3433e+00 k_t = [2, 3, 4, 2]
 Generation 1490: Best = 3.083042e+03 FES = 149100 μ_F = 0.227 μ_{CR} = 0.056
 Diversity = 3.3387e+00 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 3.083042e+03 FES = 150100 μ_F = 0.225 μ_{CR} = 0.049
 Diversity = 3.3370e+00 k_t = [3, 2, 2, 2]
 Generation 1510: Best = 3.081389e+03 FES = 151100 μ_F = 0.239 μ_{CR} = 0.046
 Diversity = 3.3365e+00 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 3.064790e+03 FES = 152100 μ_F = 0.211 μ_{CR} = 0.043
 Diversity = 3.3332e+00 k_t = [4, 2, 2, 3]
 Generation 1530: Best = 3.061275e+03 FES = 153100 μ_F = 0.187 μ_{CR} = 0.052
 Diversity = 3.3325e+00 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 3.061275e+03 FES = 154100 μ_F = 0.279 μ_{CR} = 0.062
 Diversity = 3.3190e+00 k_t = [4, 2, 2, 4]
 Generation 1550: Best = 3.060330e+03 FES = 155100 μ_F = 0.271 μ_{CR} = 0.039
 Diversity = 3.3180e+00 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 3.056114e+03 FES = 156100 μ_F = 0.294 μ_{CR} = 0.041
 Diversity = 3.3134e+00 k_t = [4, 2, 4, 2]
 Generation 1570: Best = 3.056114e+03 FES = 157100 μ_F = 0.257 μ_{CR} = 0.029
 Diversity = 3.3130e+00 k_t = [1, 1, 1, 1]

Generation 1580: Best = 3.054003e+03 FES = 158100 μ_F = 0.231 μ_{CR} = 0.038
 Diversity = 3.3118e+00 k_t = [3, 4, 3, 2]
 Generation 1590: Best = 3.012937e+03 FES = 159100 μ_F = 0.311 μ_{CR} = 0.029
 Diversity = 3.3089e+00 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 2.926465e+03 FES = 160100 μ_F = 0.306 μ_{CR} = 0.045
 Diversity = 3.2895e+00 k_t = [4, 4, 3, 4]
 Generation 1610: Best = 2.926465e+03 FES = 161100 μ_F = 0.312 μ_{CR} = 0.035
 Diversity = 3.2884e+00 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 2.926077e+03 FES = 162100 μ_F = 0.304 μ_{CR} = 0.032
 Diversity = 3.2881e+00 k_t = [2, 3, 3, 3]
 Generation 1630: Best = 2.926077e+03 FES = 163100 μ_F = 0.300 μ_{CR} = 0.024
 Diversity = 3.2876e+00 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 2.905706e+03 FES = 164100 μ_F = 0.308 μ_{CR} = 0.025
 Diversity = 3.2913e+00 k_t = [3, 2, 3, 4]
 Generation 1650: Best = 2.900418e+03 FES = 165100 μ_F = 0.251 μ_{CR} = 0.045
 Diversity = 3.2909e+00 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 2.896484e+03 FES = 166100 μ_F = 0.352 μ_{CR} = 0.035
 Diversity = 3.2896e+00 k_t = [3, 4, 3, 4]
 Generation 1670: Best = 2.862605e+03 FES = 167100 μ_F = 0.324 μ_{CR} = 0.037
 Diversity = 3.2891e+00 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 2.850301e+03 FES = 168100 μ_F = 0.250 μ_{CR} = 0.038
 Diversity = 3.2882e+00 k_t = [4, 4, 2, 3]
 Generation 1690: Best = 2.808146e+03 FES = 169100 μ_F = 0.308 μ_{CR} = 0.042
 Diversity = 3.2870e+00 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 2.808146e+03 FES = 170100 μ_F = 0.380 μ_{CR} = 0.041
 Diversity = 3.2660e+00 k_t = [4, 2, 3, 2]
 Generation 1710: Best = 2.796769e+03 FES = 171100 μ_F = 0.370 μ_{CR} = 0.028
 Diversity = 3.2661e+00 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 2.796769e+03 FES = 172100 μ_F = 0.452 μ_{CR} = 0.025
 Diversity = 3.2663e+00 k_t = [4, 2, 3, 2]
 Generation 1730: Best = 2.789984e+03 FES = 173100 μ_F = 0.502 μ_{CR} = 0.034
 Diversity = 3.2663e+00 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 2.789984e+03 FES = 174100 μ_F = 0.532 μ_{CR} = 0.029
 Diversity = 3.2660e+00 k_t = [3, 2, 2, 3]
 Generation 1750: Best = 2.789098e+03 FES = 175100 μ_F = 0.527 μ_{CR} = 0.023
 Diversity = 3.2521e+00 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 2.785335e+03 FES = 176100 μ_F = 0.658 μ_{CR} = 0.020
 Diversity = 3.2509e+00 k_t = [2, 3, 2, 4]
 Generation 1770: Best = 2.785335e+03 FES = 177100 μ_F = 0.627 μ_{CR} = 0.022
 Diversity = 3.2508e+00 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 2.785335e+03 FES = 178100 μ_F = 0.613 μ_{CR} = 0.022
 Diversity = 3.2508e+00 k_t = [2, 4, 2, 4]
 Generation 1790: Best = 2.785335e+03 FES = 179100 μ_F = 0.562 μ_{CR} = 0.020
 Diversity = 3.2505e+00 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 2.785335e+03 FES = 180100 μ_F = 0.627 μ_{CR} = 0.019
 Diversity = 3.2498e+00 k_t = [2, 2, 2, 4]
 Generation 1810: Best = 2.771900e+03 FES = 181100 μ_F = 0.617 μ_{CR} = 0.019
 Diversity = 3.2499e+00 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 2.771900e+03 FES = 182100 μ_F = 0.653 μ_{CR} = 0.020
 Diversity = 3.2448e+00 k_t = [4, 4, 2, 4]
 Generation 1830: Best = 2.771900e+03 FES = 183100 μ_F = 0.703 μ_{CR} = 0.016
 Diversity = 3.2363e+00 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 2.771900e+03 FES = 184100 μ_F = 0.693 μ_{CR} = 0.016
 Diversity = 3.2279e+00 k_t = [4, 4, 4, 3]

Generation 1850: Best = 2.771900e+03 FES = 185100 μ_F = 0.707 μ_{CR} = 0.015
 Diversity = 3.2279e+00 k_t = [1, 1, 1, 1]
 Generation 1860: Best = 2.771900e+03 FES = 186100 μ_F = 0.698 μ_{CR} = 0.024
 Diversity = 3.2036e+00 k_t = [3, 3, 3, 2]
 Generation 1870: Best = 2.771900e+03 FES = 187100 μ_F = 0.727 μ_{CR} = 0.017
 Diversity = 3.2036e+00 k_t = [1, 1, 1, 1]
 Generation 1880: Best = 2.771900e+03 FES = 188100 μ_F = 0.694 μ_{CR} = 0.025
 Diversity = 3.2031e+00 k_t = [4, 2, 4, 3]
 Generation 1890: Best = 2.771900e+03 FES = 189100 μ_F = 0.669 μ_{CR} = 0.019
 Diversity = 3.2006e+00 k_t = [1, 1, 1, 1]
 Generation 1900: Best = 2.771900e+03 FES = 190100 μ_F = 0.561 μ_{CR} = 0.034
 Diversity = 3.2012e+00 k_t = [4, 4, 3, 2]
 Generation 1910: Best = 2.771900e+03 FES = 191100 μ_F = 0.573 μ_{CR} = 0.032
 Diversity = 3.2013e+00 k_t = [1, 1, 1, 1]
 Generation 1920: Best = 2.771900e+03 FES = 192100 μ_F = 0.609 μ_{CR} = 0.037
 Diversity = 3.2113e+00 k_t = [2, 2, 2, 4]
 Generation 1930: Best = 2.771900e+03 FES = 193100 μ_F = 0.611 μ_{CR} = 0.026
 Diversity = 3.2110e+00 k_t = [1, 1, 1, 1]
 Generation 1940: Best = 2.771900e+03 FES = 194100 μ_F = 0.610 μ_{CR} = 0.029
 Diversity = 3.2246e+00 k_t = [4, 4, 3, 4]
 Generation 1950: Best = 2.771900e+03 FES = 195100 μ_F = 0.545 μ_{CR} = 0.045
 Diversity = 3.2253e+00 k_t = [1, 1, 1, 1]
 Generation 1960: Best = 2.771900e+03 FES = 196100 μ_F = 0.582 μ_{CR} = 0.032
 Diversity = 3.2249e+00 k_t = [4, 2, 2, 2]
 Generation 1970: Best = 2.771900e+03 FES = 197100 μ_F = 0.579 μ_{CR} = 0.036
 Diversity = 3.2250e+00 k_t = [1, 1, 1, 1]
 Generation 1980: Best = 2.771900e+03 FES = 198100 μ_F = 0.617 μ_{CR} = 0.032
 Diversity = 3.2155e+00 k_t = [4, 3, 2, 3]
 Generation 1990: Best = 2.771900e+03 FES = 199100 μ_F = 0.536 μ_{CR} = 0.026
 Diversity = 3.2153e+00 k_t = [1, 1, 1, 1]
 Generation 2000: Best = 2.771900e+03 FES = 200100 μ_F = 0.411 μ_{CR} = 0.053
 Diversity = 3.2154e+00 k_t = [2, 4, 3, 2]
 Generation 2010: Best = 2.771900e+03 FES = 201100 μ_F = 0.339 μ_{CR} = 0.050
 Diversity = 3.2158e+00 k_t = [1, 1, 1, 1]
 Generation 2020: Best = 2.771900e+03 FES = 202100 μ_F = 0.250 μ_{CR} = 0.046
 Diversity = 3.2150e+00 k_t = [2, 4, 3, 3]
 Generation 2030: Best = 2.771900e+03 FES = 203100 μ_F = 0.303 μ_{CR} = 0.039
 Diversity = 3.2144e+00 k_t = [1, 1, 1, 1]
 Generation 2040: Best = 2.771899e+03 FES = 204100 μ_F = 0.280 μ_{CR} = 0.028
 Diversity = 3.2128e+00 k_t = [4, 4, 3, 4]
 Generation 2050: Best = 2.771899e+03 FES = 205100 μ_F = 0.253 μ_{CR} = 0.037
 Diversity = 3.2119e+00 k_t = [1, 1, 1, 1]
 Generation 2060: Best = 2.771720e+03 FES = 206100 μ_F = 0.297 μ_{CR} = 0.038
 Diversity = 3.2117e+00 k_t = [4, 3, 2, 3]
 Generation 2070: Best = 2.758129e+03 FES = 207100 μ_F = 0.320 μ_{CR} = 0.035
 Diversity = 3.2110e+00 k_t = [1, 1, 1, 1]
 Generation 2080: Best = 2.746174e+03 FES = 208100 μ_F = 0.325 μ_{CR} = 0.035
 Diversity = 3.2106e+00 k_t = [4, 3, 3, 2]
 Generation 2090: Best = 2.746174e+03 FES = 209100 μ_F = 0.297 μ_{CR} = 0.036
 Diversity = 3.2105e+00 k_t = [1, 1, 1, 1]
 Generation 2100: Best = 2.738580e+03 FES = 210100 μ_F = 0.350 μ_{CR} = 0.030
 Diversity = 3.2103e+00 k_t = [2, 4, 2, 2]
 Generation 2110: Best = 2.737920e+03 FES = 211100 μ_F = 0.309 μ_{CR} = 0.023
 Diversity = 3.2099e+00 k_t = [1, 1, 1, 1]

Generation 2120: Best = 2.736739e+03 FES = 212100 μ_F = 0.351 μ_{CR} = 0.027
 Diversity = 3.2098e+00 k_t = [3, 4, 2, 3]
 Generation 2130: Best = 2.736739e+03 FES = 213100 μ_F = 0.330 μ_{CR} = 0.030
 Diversity = 3.2091e+00 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 2.736548e+03 FES = 214100 μ_F = 0.354 μ_{CR} = 0.034
 Diversity = 3.2094e+00 k_t = [2, 4, 2, 4]
 Generation 2150: Best = 2.736548e+03 FES = 215100 μ_F = 0.293 μ_{CR} = 0.050
 Diversity = 3.2083e+00 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 2.736548e+03 FES = 216100 μ_F = 0.319 μ_{CR} = 0.045
 Diversity = 3.2082e+00 k_t = [3, 3, 4, 2]
 Generation 2170: Best = 2.723900e+03 FES = 217100 μ_F = 0.268 μ_{CR} = 0.046
 Diversity = 3.2080e+00 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 2.723900e+03 FES = 218100 μ_F = 0.293 μ_{CR} = 0.045
 Diversity = 3.1782e+00 k_t = [3, 2, 2, 4]
 Generation 2190: Best = 2.723900e+03 FES = 219100 μ_F = 0.258 μ_{CR} = 0.040
 Diversity = 3.1779e+00 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 2.723900e+03 FES = 220100 μ_F = 0.274 μ_{CR} = 0.046
 Diversity = 3.1763e+00 k_t = [2, 3, 3, 2]
 Generation 2210: Best = 2.717715e+03 FES = 221100 μ_F = 0.362 μ_{CR} = 0.028
 Diversity = 3.1796e+00 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 2.717715e+03 FES = 222100 μ_F = 0.379 μ_{CR} = 0.030
 Diversity = 3.1676e+00 k_t = [4, 2, 3, 4]
 Generation 2230: Best = 2.717715e+03 FES = 223100 μ_F = 0.291 μ_{CR} = 0.033
 Diversity = 3.1664e+00 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 2.717095e+03 FES = 224100 μ_F = 0.278 μ_{CR} = 0.021
 Diversity = 3.1658e+00 k_t = [2, 3, 3, 2]
 Generation 2250: Best = 2.717095e+03 FES = 225100 μ_F = 0.211 μ_{CR} = 0.039
 Diversity = 3.1651e+00 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 2.715796e+03 FES = 226100 μ_F = 0.229 μ_{CR} = 0.035
 Diversity = 3.1645e+00 k_t = [3, 2, 2, 3]
 Generation 2270: Best = 2.715796e+03 FES = 227100 μ_F = 0.296 μ_{CR} = 0.057
 Diversity = 3.1645e+00 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 2.715530e+03 FES = 228100 μ_F = 0.256 μ_{CR} = 0.049
 Diversity = 3.1636e+00 k_t = [2, 4, 2, 2]
 Generation 2290: Best = 2.715384e+03 FES = 229100 μ_F = 0.233 μ_{CR} = 0.043
 Diversity = 3.1632e+00 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 2.715384e+03 FES = 230100 μ_F = 0.219 μ_{CR} = 0.042
 Diversity = 3.1624e+00 k_t = [4, 2, 4, 4]
 Generation 2310: Best = 2.715384e+03 FES = 231100 μ_F = 0.187 μ_{CR} = 0.050
 Diversity = 3.1623e+00 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 2.709121e+03 FES = 232100 μ_F = 0.227 μ_{CR} = 0.046
 Diversity = 3.1581e+00 k_t = [4, 2, 3, 3]
 Generation 2330: Best = 2.709121e+03 FES = 233100 μ_F = 0.266 μ_{CR} = 0.047
 Diversity = 3.1572e+00 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 2.709083e+03 FES = 234100 μ_F = 0.234 μ_{CR} = 0.040
 Diversity = 3.1571e+00 k_t = [4, 3, 3, 3]
 Generation 2350: Best = 2.703780e+03 FES = 235100 μ_F = 0.285 μ_{CR} = 0.054
 Diversity = 3.1562e+00 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 2.703780e+03 FES = 236100 μ_F = 0.242 μ_{CR} = 0.037
 Diversity = 3.1562e+00 k_t = [4, 3, 2, 3]
 Generation 2370: Best = 2.702909e+03 FES = 237100 μ_F = 0.308 μ_{CR} = 0.028
 Diversity = 3.1544e+00 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 2.702909e+03 FES = 238100 μ_F = 0.333 μ_{CR} = 0.030
 Diversity = 3.1328e+00 k_t = [4, 3, 3, 4]

Generation 2390: Best = 2.693192e+03 FES = 239100 μ_F = 0.254 μ_{CR} = 0.033
 Diversity = 3.1323e+00 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 2.693192e+03 FES = 240100 μ_F = 0.264 μ_{CR} = 0.038
 Diversity = 3.1322e+00 k_t = [3, 3, 3, 2]
 Generation 2410: Best = 2.693192e+03 FES = 241100 μ_F = 0.300 μ_{CR} = 0.054
 Diversity = 3.1318e+00 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 2.693192e+03 FES = 242100 μ_F = 0.355 μ_{CR} = 0.038
 Diversity = 3.1176e+00 k_t = [2, 2, 4, 2]
 Generation 2430: Best = 2.693192e+03 FES = 243100 μ_F = 0.353 μ_{CR} = 0.040
 Diversity = 3.1175e+00 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 2.682016e+03 FES = 244100 μ_F = 0.397 μ_{CR} = 0.031
 Diversity = 3.1167e+00 k_t = [2, 2, 2, 2]
 Generation 2450: Best = 2.682016e+03 FES = 245100 μ_F = 0.387 μ_{CR} = 0.028
 Diversity = 3.1120e+00 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 2.682016e+03 FES = 246100 μ_F = 0.363 μ_{CR} = 0.027
 Diversity = 3.1035e+00 k_t = [4, 4, 4, 4]
 Generation 2470: Best = 2.682016e+03 FES = 247100 μ_F = 0.374 μ_{CR} = 0.021
 Diversity = 3.0918e+00 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 2.677011e+03 FES = 248100 μ_F = 0.406 μ_{CR} = 0.021
 Diversity = 3.0924e+00 k_t = [3, 2, 3, 3]
 Generation 2490: Best = 2.677011e+03 FES = 249100 μ_F = 0.356 μ_{CR} = 0.026
 Diversity = 3.0918e+00 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 2.677011e+03 FES = 250100 μ_F = 0.375 μ_{CR} = 0.038
 Diversity = 3.0902e+00 k_t = [2, 2, 3, 4]
 Generation 2510: Best = 2.677011e+03 FES = 251100 μ_F = 0.326 μ_{CR} = 0.039
 Diversity = 3.0892e+00 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 2.670454e+03 FES = 252100 μ_F = 0.371 μ_{CR} = 0.035
 Diversity = 3.0888e+00 k_t = [4, 4, 4, 2]
 Generation 2530: Best = 2.670454e+03 FES = 253100 μ_F = 0.325 μ_{CR} = 0.037
 Diversity = 3.0828e+00 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 2.666381e+03 FES = 254100 μ_F = 0.319 μ_{CR} = 0.041
 Diversity = 3.0810e+00 k_t = [3, 4, 4, 4]
 Generation 2550: Best = 2.663680e+03 FES = 255100 μ_F = 0.438 μ_{CR} = 0.035
 Diversity = 3.0808e+00 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 2.663680e+03 FES = 256100 μ_F = 0.419 μ_{CR} = 0.031
 Diversity = 3.0810e+00 k_t = [2, 3, 2, 2]
 Generation 2570: Best = 2.663680e+03 FES = 257100 μ_F = 0.481 μ_{CR} = 0.023
 Diversity = 3.0830e+00 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 2.663680e+03 FES = 258100 μ_F = 0.423 μ_{CR} = 0.038
 Diversity = 3.0692e+00 k_t = [2, 2, 3, 4]
 Generation 2590: Best = 2.663680e+03 FES = 259100 μ_F = 0.346 μ_{CR} = 0.033
 Diversity = 3.0688e+00 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 2.658378e+03 FES = 260100 μ_F = 0.396 μ_{CR} = 0.028
 Diversity = 3.0451e+00 k_t = [3, 3, 2, 4]
 Generation 2610: Best = 2.658378e+03 FES = 261100 μ_F = 0.416 μ_{CR} = 0.028
 Diversity = 3.0442e+00 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 2.657421e+03 FES = 262100 μ_F = 0.530 μ_{CR} = 0.022
 Diversity = 3.0317e+00 k_t = [2, 2, 3, 4]
 Generation 2630: Best = 2.657421e+03 FES = 263100 μ_F = 0.469 μ_{CR} = 0.024
 Diversity = 3.0320e+00 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 2.651505e+03 FES = 264100 μ_F = 0.477 μ_{CR} = 0.026
 Diversity = 3.0233e+00 k_t = [2, 2, 2, 2]
 Generation 2650: Best = 2.648509e+03 FES = 265100 μ_F = 0.435 μ_{CR} = 0.027
 Diversity = 3.0236e+00 k_t = [1, 1, 1, 1]

Generation 2660: Best = 2.648509e+03 FES = 266100 μ_F = 0.396 μ_{CR} = 0.035
 Diversity = 3.0226e+00 k_t = [4, 4, 4, 4]
 Generation 2670: Best = 2.647886e+03 FES = 267100 μ_F = 0.430 μ_{CR} = 0.047
 Diversity = 3.0214e+00 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 2.647886e+03 FES = 268100 μ_F = 0.418 μ_{CR} = 0.038
 Diversity = 3.0216e+00 k_t = [3, 4, 2, 2]
 Generation 2690: Best = 2.639138e+03 FES = 269100 μ_F = 0.459 μ_{CR} = 0.023
 Diversity = 3.0215e+00 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 2.639138e+03 FES = 270100 μ_F = 0.395 μ_{CR} = 0.035
 Diversity = 3.0208e+00 k_t = [3, 3, 2, 2]
 Generation 2710: Best = 2.639138e+03 FES = 271100 μ_F = 0.437 μ_{CR} = 0.030
 Diversity = 3.0211e+00 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 2.639138e+03 FES = 272100 μ_F = 0.436 μ_{CR} = 0.033
 Diversity = 3.0197e+00 k_t = [2, 2, 4, 2]
 Generation 2730: Best = 2.639138e+03 FES = 273100 μ_F = 0.457 μ_{CR} = 0.038
 Diversity = 3.0023e+00 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 2.639138e+03 FES = 274100 μ_F = 0.539 μ_{CR} = 0.033
 Diversity = 2.9828e+00 k_t = [3, 2, 2, 4]
 Generation 2750: Best = 2.639138e+03 FES = 275100 μ_F = 0.571 μ_{CR} = 0.023
 Diversity = 2.9825e+00 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 2.639138e+03 FES = 276100 μ_F = 0.594 μ_{CR} = 0.029
 Diversity = 2.9562e+00 k_t = [2, 4, 4, 2]
 Generation 2770: Best = 2.639138e+03 FES = 277100 μ_F = 0.596 μ_{CR} = 0.028
 Diversity = 2.9463e+00 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 2.639138e+03 FES = 278100 μ_F = 0.583 μ_{CR} = 0.020
 Diversity = 2.9466e+00 k_t = [3, 3, 2, 3]
 Generation 2790: Best = 2.639138e+03 FES = 279100 μ_F = 0.624 μ_{CR} = 0.018
 Diversity = 2.9373e+00 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 2.639138e+03 FES = 280100 μ_F = 0.629 μ_{CR} = 0.016
 Diversity = 2.9149e+00 k_t = [2, 4, 3, 4]
 Generation 2810: Best = 2.639138e+03 FES = 281100 μ_F = 0.571 μ_{CR} = 0.023
 Diversity = 2.8843e+00 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 2.639138e+03 FES = 282100 μ_F = 0.446 μ_{CR} = 0.050
 Diversity = 2.8840e+00 k_t = [4, 2, 4, 3]
 Generation 2830: Best = 2.639138e+03 FES = 283100 μ_F = 0.406 μ_{CR} = 0.046
 Diversity = 2.8839e+00 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 2.636090e+03 FES = 284100 μ_F = 0.486 μ_{CR} = 0.047
 Diversity = 2.8722e+00 k_t = [4, 4, 2, 3]
 Generation 2850: Best = 2.635919e+03 FES = 285100 μ_F = 0.392 μ_{CR} = 0.044
 Diversity = 2.8617e+00 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 2.619836e+03 FES = 286100 μ_F = 0.399 μ_{CR} = 0.034
 Diversity = 2.8611e+00 k_t = [4, 2, 4, 3]
 Generation 2870: Best = 2.619836e+03 FES = 287100 μ_F = 0.447 μ_{CR} = 0.039
 Diversity = 2.8217e+00 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 2.619836e+03 FES = 288100 μ_F = 0.524 μ_{CR} = 0.037
 Diversity = 2.8185e+00 k_t = [4, 2, 3, 4]
 Generation 2890: Best = 2.618183e+03 FES = 289100 μ_F = 0.455 μ_{CR} = 0.037
 Diversity = 2.8063e+00 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 2.609428e+03 FES = 290100 μ_F = 0.429 μ_{CR} = 0.039
 Diversity = 2.7833e+00 k_t = [2, 4, 3, 4]
 Generation 2910: Best = 2.609428e+03 FES = 291100 μ_F = 0.422 μ_{CR} = 0.035
 Diversity = 2.7776e+00 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 2.609428e+03 FES = 292100 μ_F = 0.473 μ_{CR} = 0.035
 Diversity = 2.7723e+00 k_t = [2, 2, 3, 4]

Generation 2930: Best = 2.609428e+03 FES = 293100 μ_F = 0.463 μ_{CR} = 0.023
 Diversity = 2.7228e+00 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 2.608516e+03 FES = 294100 μ_F = 0.517 μ_{CR} = 0.029
 Diversity = 2.6385e+00 k_t = [3, 4, 4, 4]
 Generation 2950: Best = 2.608516e+03 FES = 295100 μ_F = 0.553 μ_{CR} = 0.038
 Diversity = 2.6298e+00 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 2.608516e+03 FES = 296100 μ_F = 0.454 μ_{CR} = 0.048
 Diversity = 2.6200e+00 k_t = [4, 3, 3, 3]
 Generation 2970: Best = 2.608516e+03 FES = 297100 μ_F = 0.400 μ_{CR} = 0.066
 Diversity = 2.6200e+00 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 2.608516e+03 FES = 298100 μ_F = 0.358 μ_{CR} = 0.048
 Diversity = 2.6198e+00 k_t = [3, 4, 2, 2]
 Generation 2990: Best = 2.600549e+03 FES = 299100 μ_F = 0.335 μ_{CR} = 0.048
 Diversity = 2.6196e+00 k_t = [1, 1, 1, 1]
 Generation 3000: Best = 2.600526e+03 FES = 300100 μ_F = 0.381 μ_{CR} = 0.043
 Diversity = 2.6193e+00 k_t = [4, 3, 3, 2]
 Generation 3010: Best = 2.600526e+03 FES = 301100 μ_F = 0.435 μ_{CR} = 0.040
 Diversity = 2.5948e+00 k_t = [1, 1, 1, 1]
 Generation 3020: Best = 2.600526e+03 FES = 302100 μ_F = 0.406 μ_{CR} = 0.058
 Diversity = 2.5609e+00 k_t = [4, 3, 4, 4]
 Generation 3030: Best = 2.597214e+03 FES = 303100 μ_F = 0.401 μ_{CR} = 0.043
 Diversity = 2.5550e+00 k_t = [1, 1, 1, 1]
 Generation 3040: Best = 2.596087e+03 FES = 304100 μ_F = 0.442 μ_{CR} = 0.045
 Diversity = 2.5366e+00 k_t = [4, 3, 4, 4]
 Generation 3050: Best = 2.593837e+03 FES = 305100 μ_F = 0.437 μ_{CR} = 0.048
 Diversity = 2.5069e+00 k_t = [1, 1, 1, 1]
 Generation 3060: Best = 2.593837e+03 FES = 306100 μ_F = 0.472 μ_{CR} = 0.049
 Diversity = 2.4493e+00 k_t = [2, 4, 3, 3]
 Generation 3070: Best = 2.593837e+03 FES = 307100 μ_F = 0.566 μ_{CR} = 0.038
 Diversity = 2.4493e+00 k_t = [1, 1, 1, 1]
 Generation 3080: Best = 2.590535e+03 FES = 308100 μ_F = 0.474 μ_{CR} = 0.048
 Diversity = 2.4503e+00 k_t = [2, 3, 2, 3]
 Generation 3090: Best = 2.585358e+03 FES = 309100 μ_F = 0.375 μ_{CR} = 0.051
 Diversity = 2.4495e+00 k_t = [1, 1, 1, 1]
 Generation 3100: Best = 2.580101e+03 FES = 310100 μ_F = 0.395 μ_{CR} = 0.072
 Diversity = 2.3922e+00 k_t = [4, 3, 3, 4]
 Generation 3110: Best = 2.570557e+03 FES = 311100 μ_F = 0.415 μ_{CR} = 0.056
 Diversity = 2.3921e+00 k_t = [1, 1, 1, 1]
 Generation 3120: Best = 2.557766e+03 FES = 312100 μ_F = 0.460 μ_{CR} = 0.063
 Diversity = 2.3549e+00 k_t = [3, 4, 3, 4]
 Generation 3130: Best = 2.550779e+03 FES = 313100 μ_F = 0.423 μ_{CR} = 0.054
 Diversity = 2.3546e+00 k_t = [1, 1, 1, 1]
 Generation 3140: Best = 2.550779e+03 FES = 314100 μ_F = 0.415 μ_{CR} = 0.052
 Diversity = 2.3483e+00 k_t = [2, 2, 2, 3]
 Generation 3150: Best = 2.545574e+03 FES = 315100 μ_F = 0.418 μ_{CR} = 0.041
 Diversity = 2.3491e+00 k_t = [1, 1, 1, 1]
 Generation 3160: Best = 2.545574e+03 FES = 316100 μ_F = 0.430 μ_{CR} = 0.040
 Diversity = 2.2778e+00 k_t = [2, 3, 3, 4]
 Generation 3170: Best = 2.545574e+03 FES = 317100 μ_F = 0.418 μ_{CR} = 0.031
 Diversity = 2.2753e+00 k_t = [1, 1, 1, 1]
 Generation 3180: Best = 2.544030e+03 FES = 318100 μ_F = 0.446 μ_{CR} = 0.035
 Diversity = 2.2696e+00 k_t = [4, 3, 4, 4]
 Generation 3190: Best = 2.544030e+03 FES = 319100 μ_F = 0.412 μ_{CR} = 0.036
 Diversity = 2.2684e+00 k_t = [1, 1, 1, 1]

Generation 3200: Best = 2.544030e+03 FES = 320100 μ_F = 0.423 μ_{CR} = 0.036
 Diversity = 2.2681e+00 k_t = [2, 3, 3, 2]
 Generation 3210: Best = 2.541798e+03 FES = 321100 μ_F = 0.487 μ_{CR} = 0.063
 Diversity = 2.2174e+00 k_t = [1, 1, 1, 1]
 Generation 3220: Best = 2.541798e+03 FES = 322100 μ_F = 0.433 μ_{CR} = 0.053
 Diversity = 2.2014e+00 k_t = [4, 3, 3, 2]
 Generation 3230: Best = 2.541798e+03 FES = 323100 μ_F = 0.408 μ_{CR} = 0.053
 Diversity = 2.2011e+00 k_t = [1, 1, 1, 1]
 Generation 3240: Best = 2.541798e+03 FES = 324100 μ_F = 0.483 μ_{CR} = 0.044
 Diversity = 2.2023e+00 k_t = [4, 2, 2, 3]
 Generation 3250: Best = 2.541798e+03 FES = 325100 μ_F = 0.428 μ_{CR} = 0.061
 Diversity = 2.1795e+00 k_t = [1, 1, 1, 1]
 Generation 3260: Best = 2.541798e+03 FES = 326100 μ_F = 0.553 μ_{CR} = 0.040
 Diversity = 2.1621e+00 k_t = [3, 4, 4, 3]
 Generation 3270: Best = 2.541798e+03 FES = 327100 μ_F = 0.514 μ_{CR} = 0.053
 Diversity = 2.1618e+00 k_t = [1, 1, 1, 1]
 Generation 3280: Best = 2.541798e+03 FES = 328100 μ_F = 0.483 μ_{CR} = 0.041
 Diversity = 2.1454e+00 k_t = [4, 3, 2, 4]
 Generation 3290: Best = 2.541798e+03 FES = 329100 μ_F = 0.472 μ_{CR} = 0.033
 Diversity = 2.1374e+00 k_t = [1, 1, 1, 1]
 Generation 3300: Best = 2.541798e+03 FES = 330100 μ_F = 0.549 μ_{CR} = 0.038
 Diversity = 2.1241e+00 k_t = [4, 2, 3, 3]
 Generation 3310: Best = 2.533180e+03 FES = 331100 μ_F = 0.593 μ_{CR} = 0.028
 Diversity = 2.1114e+00 k_t = [1, 1, 1, 1]
 Generation 3320: Best = 2.533180e+03 FES = 332100 μ_F = 0.568 μ_{CR} = 0.035
 Diversity = 2.1089e+00 k_t = [3, 3, 4, 3]
 Generation 3330: Best = 2.533180e+03 FES = 333100 μ_F = 0.543 μ_{CR} = 0.028
 Diversity = 2.1074e+00 k_t = [1, 1, 1, 1]
 Generation 3340: Best = 2.525963e+03 FES = 334100 μ_F = 0.469 μ_{CR} = 0.042
 Diversity = 2.0859e+00 k_t = [4, 2, 3, 4]
 Generation 3350: Best = 2.525963e+03 FES = 335100 μ_F = 0.455 μ_{CR} = 0.041
 Diversity = 2.0637e+00 k_t = [1, 1, 1, 1]
 Generation 3360: Best = 2.525963e+03 FES = 336100 μ_F = 0.418 μ_{CR} = 0.039
 Diversity = 2.0666e+00 k_t = [3, 3, 4, 3]
 Generation 3370: Best = 2.525963e+03 FES = 337100 μ_F = 0.440 μ_{CR} = 0.032
 Diversity = 2.0661e+00 k_t = [1, 1, 1, 1]
 Generation 3380: Best = 2.525963e+03 FES = 338100 μ_F = 0.481 μ_{CR} = 0.034
 Diversity = 2.0181e+00 k_t = [4, 2, 3, 4]
 Generation 3390: Best = 2.525963e+03 FES = 339100 μ_F = 0.423 μ_{CR} = 0.031
 Diversity = 1.9997e+00 k_t = [1, 1, 1, 1]
 Generation 3400: Best = 2.525963e+03 FES = 340100 μ_F = 0.433 μ_{CR} = 0.039
 Diversity = 1.9753e+00 k_t = [4, 4, 2, 3]
 Generation 3410: Best = 2.522857e+03 FES = 341100 μ_F = 0.435 μ_{CR} = 0.031
 Diversity = 1.9754e+00 k_t = [1, 1, 1, 1]
 Generation 3420: Best = 2.522843e+03 FES = 342100 μ_F = 0.382 μ_{CR} = 0.032
 Diversity = 1.9754e+00 k_t = [2, 3, 2, 4]
 Generation 3430: Best = 2.519926e+03 FES = 343100 μ_F = 0.337 μ_{CR} = 0.046
 Diversity = 1.9753e+00 k_t = [1, 1, 1, 1]
 Generation 3440: Best = 2.514461e+03 FES = 344100 μ_F = 0.307 μ_{CR} = 0.060
 Diversity = 1.9741e+00 k_t = [3, 4, 3, 3]
 Generation 3450: Best = 2.514461e+03 FES = 345100 μ_F = 0.265 μ_{CR} = 0.046
 Diversity = 1.9741e+00 k_t = [1, 1, 1, 1]
 Generation 3460: Best = 2.514458e+03 FES = 346100 μ_F = 0.294 μ_{CR} = 0.060
 Diversity = 1.9730e+00 k_t = [3, 3, 4, 3]

Generation 3470: Best = 2.514458e+03 FES = 347100 μ_F = 0.241 μ_{CR} = 0.083
 Diversity = 1.9707e+00 k_t = [1, 1, 1, 1]
 Generation 3480: Best = 2.514458e+03 FES = 348100 μ_F = 0.281 μ_{CR} = 0.091
 Diversity = 1.9701e+00 k_t = [2, 2, 4, 2]
 Generation 3490: Best = 2.512621e+03 FES = 349100 μ_F = 0.376 μ_{CR} = 0.068
 Diversity = 1.9516e+00 k_t = [1, 1, 1, 1]
 Generation 3500: Best = 2.512621e+03 FES = 350100 μ_F = 0.347 μ_{CR} = 0.058
 Diversity = 1.9505e+00 k_t = [3, 4, 4, 3]
 Generation 3510: Best = 2.512621e+03 FES = 351100 μ_F = 0.346 μ_{CR} = 0.050
 Diversity = 1.9508e+00 k_t = [1, 1, 1, 1]
 Generation 3520: Best = 2.511882e+03 FES = 352100 μ_F = 0.385 μ_{CR} = 0.040
 Diversity = 1.9504e+00 k_t = [2, 4, 3, 2]
 Generation 3530: Best = 2.511882e+03 FES = 353100 μ_F = 0.401 μ_{CR} = 0.047
 Diversity = 1.9222e+00 k_t = [1, 1, 1, 1]
 Generation 3540: Best = 2.511882e+03 FES = 354100 μ_F = 0.446 μ_{CR} = 0.048
 Diversity = 1.9181e+00 k_t = [2, 3, 4, 2]
 Generation 3550: Best = 2.511616e+03 FES = 355100 μ_F = 0.526 μ_{CR} = 0.043
 Diversity = 1.8896e+00 k_t = [1, 1, 1, 1]
 Generation 3560: Best = 2.510654e+03 FES = 356100 μ_F = 0.567 μ_{CR} = 0.031
 Diversity = 1.8888e+00 k_t = [4, 2, 3, 3]
 Generation 3570: Best = 2.510654e+03 FES = 357100 μ_F = 0.519 μ_{CR} = 0.034
 Diversity = 1.8874e+00 k_t = [1, 1, 1, 1]
 Generation 3580: Best = 2.510654e+03 FES = 358100 μ_F = 0.512 μ_{CR} = 0.049
 Diversity = 1.8539e+00 k_t = [3, 4, 2, 4]
 Generation 3590: Best = 2.510491e+03 FES = 359100 μ_F = 0.377 μ_{CR} = 0.064
 Diversity = 1.8525e+00 k_t = [1, 1, 1, 1]
 Generation 3600: Best = 2.510491e+03 FES = 360100 μ_F = 0.311 μ_{CR} = 0.057
 Diversity = 1.8522e+00 k_t = [3, 3, 2, 2]
 Generation 3610: Best = 2.510491e+03 FES = 361100 μ_F = 0.370 μ_{CR} = 0.050
 Diversity = 1.8516e+00 k_t = [1, 1, 1, 1]
 Generation 3620: Best = 2.502795e+03 FES = 362100 μ_F = 0.363 μ_{CR} = 0.040
 Diversity = 1.8523e+00 k_t = [3, 4, 3, 2]
 Generation 3630: Best = 2.502795e+03 FES = 363100 μ_F = 0.334 μ_{CR} = 0.038
 Diversity = 1.8520e+00 k_t = [1, 1, 1, 1]
 Generation 3640: Best = 2.502763e+03 FES = 364100 μ_F = 0.398 μ_{CR} = 0.063
 Diversity = 1.8276e+00 k_t = [4, 3, 2, 3]
 Generation 3650: Best = 2.502142e+03 FES = 365100 μ_F = 0.388 μ_{CR} = 0.049
 Diversity = 1.7972e+00 k_t = [1, 1, 1, 1]
 Generation 3660: Best = 2.499545e+03 FES = 366100 μ_F = 0.426 μ_{CR} = 0.036
 Diversity = 1.7429e+00 k_t = [3, 3, 3, 4]
 Generation 3670: Best = 2.499545e+03 FES = 367100 μ_F = 0.417 μ_{CR} = 0.039
 Diversity = 1.7420e+00 k_t = [1, 1, 1, 1]
 Generation 3680: Best = 2.499545e+03 FES = 368100 μ_F = 0.473 μ_{CR} = 0.038
 Diversity = 1.7439e+00 k_t = [2, 3, 4, 3]
 Generation 3690: Best = 2.499545e+03 FES = 369100 μ_F = 0.453 μ_{CR} = 0.030
 Diversity = 1.6899e+00 k_t = [1, 1, 1, 1]
 Generation 3700: Best = 2.499545e+03 FES = 370100 μ_F = 0.446 μ_{CR} = 0.029
 Diversity = 1.6905e+00 k_t = [2, 4, 2, 2]
 Generation 3710: Best = 2.495471e+03 FES = 371100 μ_F = 0.537 μ_{CR} = 0.030
 Diversity = 1.6622e+00 k_t = [1, 1, 1, 1]
 Generation 3720: Best = 2.494972e+03 FES = 372100 μ_F = 0.542 μ_{CR} = 0.037
 Diversity = 1.6238e+00 k_t = [4, 3, 4, 2]
 Generation 3730: Best = 2.492720e+03 FES = 373100 μ_F = 0.502 μ_{CR} = 0.026
 Diversity = 1.6244e+00 k_t = [1, 1, 1, 1]

Generation 3740: Best = 2.492720e+03 FES = 374100 μ_F = 0.470 μ_{CR} = 0.024
 Diversity = 1.6248e+00 k_t = [2, 3, 3, 4]
 Generation 3750: Best = 2.492720e+03 FES = 375100 μ_F = 0.474 μ_{CR} = 0.033
 Diversity = 1.6008e+00 k_t = [1, 1, 1, 1]
 Generation 3760: Best = 2.492720e+03 FES = 376100 μ_F = 0.471 μ_{CR} = 0.037
 Diversity = 1.6029e+00 k_t = [2, 4, 4, 4]
 Generation 3770: Best = 2.492720e+03 FES = 377100 μ_F = 0.559 μ_{CR} = 0.023
 Diversity = 1.6028e+00 k_t = [1, 1, 1, 1]
 Generation 3780: Best = 2.492720e+03 FES = 378100 μ_F = 0.564 μ_{CR} = 0.033
 Diversity = 1.6035e+00 k_t = [4, 3, 4, 2]
 Generation 3790: Best = 2.492720e+03 FES = 379100 μ_F = 0.605 μ_{CR} = 0.041
 Diversity = 1.5671e+00 k_t = [1, 1, 1, 1]
 Generation 3800: Best = 2.492720e+03 FES = 380100 μ_F = 0.560 μ_{CR} = 0.041
 Diversity = 1.5631e+00 k_t = [4, 4, 2, 4]
 Generation 3810: Best = 2.492720e+03 FES = 381100 μ_F = 0.507 μ_{CR} = 0.038
 Diversity = 1.5623e+00 k_t = [1, 1, 1, 1]
 Generation 3820: Best = 2.474902e+03 FES = 382100 μ_F = 0.487 μ_{CR} = 0.050
 Diversity = 1.5627e+00 k_t = [2, 2, 3, 4]
 Generation 3830: Best = 2.474902e+03 FES = 383100 μ_F = 0.429 μ_{CR} = 0.046
 Diversity = 1.5590e+00 k_t = [1, 1, 1, 1]
 Generation 3840: Best = 2.474902e+03 FES = 384100 μ_F = 0.408 μ_{CR} = 0.061
 Diversity = 1.5059e+00 k_t = [2, 4, 4, 2]
 Generation 3850: Best = 2.474902e+03 FES = 385100 μ_F = 0.496 μ_{CR} = 0.065
 Diversity = 1.5061e+00 k_t = [1, 1, 1, 1]
 Generation 3860: Best = 2.474902e+03 FES = 386100 μ_F = 0.515 μ_{CR} = 0.044
 Diversity = 1.5138e+00 k_t = [2, 4, 3, 2]
 Generation 3870: Best = 2.474902e+03 FES = 387100 μ_F = 0.437 μ_{CR} = 0.039
 Diversity = 1.5131e+00 k_t = [1, 1, 1, 1]
 Generation 3880: Best = 2.474902e+03 FES = 388100 μ_F = 0.356 μ_{CR} = 0.039
 Diversity = 1.5131e+00 k_t = [4, 4, 4, 3]
 Generation 3890: Best = 2.474523e+03 FES = 389100 μ_F = 0.342 μ_{CR} = 0.037
 Diversity = 1.5127e+00 k_t = [1, 1, 1, 1]
 Generation 3900: Best = 2.474523e+03 FES = 390100 μ_F = 0.292 μ_{CR} = 0.060
 Diversity = 1.5118e+00 k_t = [2, 3, 2, 4]
 Generation 3910: Best = 2.474523e+03 FES = 391100 μ_F = 0.323 μ_{CR} = 0.057
 Diversity = 1.5038e+00 k_t = [1, 1, 1, 1]
 Generation 3920: Best = 2.474523e+03 FES = 392100 μ_F = 0.302 μ_{CR} = 0.057
 Diversity = 1.5040e+00 k_t = [4, 3, 2, 4]
 Generation 3930: Best = 2.474523e+03 FES = 393100 μ_F = 0.326 μ_{CR} = 0.042
 Diversity = 1.5012e+00 k_t = [1, 1, 1, 1]
 Generation 3940: Best = 2.474523e+03 FES = 394100 μ_F = 0.324 μ_{CR} = 0.042
 Diversity = 1.5008e+00 k_t = [2, 2, 3, 4]
 Generation 3950: Best = 2.474523e+03 FES = 395100 μ_F = 0.291 μ_{CR} = 0.062
 Diversity = 1.4991e+00 k_t = [1, 1, 1, 1]
 Generation 3960: Best = 2.474523e+03 FES = 396100 μ_F = 0.323 μ_{CR} = 0.054
 Diversity = 1.4987e+00 k_t = [3, 2, 3, 2]
 Generation 3970: Best = 2.468173e+03 FES = 397100 μ_F = 0.308 μ_{CR} = 0.045
 Diversity = 1.4986e+00 k_t = [1, 1, 1, 1]
 Generation 3980: Best = 2.468173e+03 FES = 398100 μ_F = 0.273 μ_{CR} = 0.071
 Diversity = 1.4972e+00 k_t = [2, 4, 4, 3]
 Generation 3990: Best = 2.467828e+03 FES = 399100 μ_F = 0.303 μ_{CR} = 0.060
 Diversity = 1.4972e+00 k_t = [1, 1, 1, 1]
 Generation 4000: Best = 2.467828e+03 FES = 400100 μ_F = 0.301 μ_{CR} = 0.059
 Diversity = 1.4971e+00 k_t = [3, 4, 4, 4]

Generation 4010: Best = 2.467828e+03 FES = 401100 μ_F = 0.252 μ_{CR} = 0.063
 Diversity = 1.4960e+00 k_t = [1, 1, 1, 1]
 Generation 4020: Best = 2.467828e+03 FES = 402100 μ_F = 0.326 μ_{CR} = 0.055
 Diversity = 1.4933e+00 k_t = [2, 2, 3, 4]
 Generation 4030: Best = 2.467828e+03 FES = 403100 μ_F = 0.337 μ_{CR} = 0.042
 Diversity = 1.4935e+00 k_t = [1, 1, 1, 1]
 Generation 4040: Best = 2.467701e+03 FES = 404100 μ_F = 0.344 μ_{CR} = 0.045
 Diversity = 1.4933e+00 k_t = [4, 4, 4, 3]
 Generation 4050: Best = 2.467701e+03 FES = 405100 μ_F = 0.355 μ_{CR} = 0.039
 Diversity = 1.4932e+00 k_t = [1, 1, 1, 1]
 Generation 4060: Best = 2.467701e+03 FES = 406100 μ_F = 0.332 μ_{CR} = 0.040
 Diversity = 1.5004e+00 k_t = [2, 4, 2, 2]
 Generation 4070: Best = 2.467701e+03 FES = 407100 μ_F = 0.306 μ_{CR} = 0.042
 Diversity = 1.4996e+00 k_t = [1, 1, 1, 1]
 Generation 4080: Best = 2.466773e+03 FES = 408100 μ_F = 0.327 μ_{CR} = 0.057
 Diversity = 1.4994e+00 k_t = [4, 4, 4, 2]
 Generation 4090: Best = 2.465372e+03 FES = 409100 μ_F = 0.277 μ_{CR} = 0.057
 Diversity = 1.4994e+00 k_t = [1, 1, 1, 1]
 Generation 4100: Best = 2.461886e+03 FES = 410100 μ_F = 0.340 μ_{CR} = 0.052
 Diversity = 1.5102e+00 k_t = [3, 4, 3, 3]
 Generation 4110: Best = 2.461886e+03 FES = 411100 μ_F = 0.328 μ_{CR} = 0.040
 Diversity = 1.5098e+00 k_t = [1, 1, 1, 1]
 Generation 4120: Best = 2.461886e+03 FES = 412100 μ_F = 0.336 μ_{CR} = 0.051
 Diversity = 1.5094e+00 k_t = [2, 2, 2, 3]
 Generation 4130: Best = 2.461886e+03 FES = 413100 μ_F = 0.369 μ_{CR} = 0.040
 Diversity = 1.5062e+00 k_t = [1, 1, 1, 1]
 Generation 4140: Best = 2.461886e+03 FES = 414100 μ_F = 0.388 μ_{CR} = 0.054
 Diversity = 1.5061e+00 k_t = [3, 4, 2, 3]
 Generation 4150: Best = 2.461886e+03 FES = 415100 μ_F = 0.347 μ_{CR} = 0.042
 Diversity = 1.5062e+00 k_t = [1, 1, 1, 1]
 Generation 4160: Best = 2.455483e+03 FES = 416100 μ_F = 0.362 μ_{CR} = 0.046
 Diversity = 1.5141e+00 k_t = [3, 3, 4, 3]
 Generation 4170: Best = 2.455483e+03 FES = 417100 μ_F = 0.406 μ_{CR} = 0.051
 Diversity = 1.5132e+00 k_t = [1, 1, 1, 1]
 Generation 4180: Best = 2.455483e+03 FES = 418100 μ_F = 0.448 μ_{CR} = 0.053
 Diversity = 1.5136e+00 k_t = [2, 3, 2, 3]
 Generation 4190: Best = 2.455483e+03 FES = 419100 μ_F = 0.442 μ_{CR} = 0.045
 Diversity = 1.5140e+00 k_t = [1, 1, 1, 1]
 Generation 4200: Best = 2.455483e+03 FES = 420100 μ_F = 0.394 μ_{CR} = 0.051
 Diversity = 1.4641e+00 k_t = [4, 2, 2, 4]
 Generation 4210: Best = 2.455483e+03 FES = 421100 μ_F = 0.359 μ_{CR} = 0.059
 Diversity = 1.4639e+00 k_t = [1, 1, 1, 1]
 Generation 4220: Best = 2.455483e+03 FES = 422100 μ_F = 0.338 μ_{CR} = 0.047
 Diversity = 1.4690e+00 k_t = [4, 4, 2, 4]
 Generation 4230: Best = 2.455483e+03 FES = 423100 μ_F = 0.286 μ_{CR} = 0.042
 Diversity = 1.4688e+00 k_t = [1, 1, 1, 1]
 Generation 4240: Best = 2.455483e+03 FES = 424100 μ_F = 0.293 μ_{CR} = 0.042
 Diversity = 1.4680e+00 k_t = [2, 2, 2, 4]
 Generation 4250: Best = 2.455483e+03 FES = 425100 μ_F = 0.269 μ_{CR} = 0.039
 Diversity = 1.4672e+00 k_t = [1, 1, 1, 1]
 Generation 4260: Best = 2.454973e+03 FES = 426100 μ_F = 0.283 μ_{CR} = 0.039
 Diversity = 1.4521e+00 k_t = [3, 3, 4, 2]
 Generation 4270: Best = 2.453315e+03 FES = 427100 μ_F = 0.251 μ_{CR} = 0.040
 Diversity = 1.4515e+00 k_t = [1, 1, 1, 1]

Generation 4280: Best = 2.450204e+03 FES = 428100 μ_F = 0.239 μ_{CR} = 0.037
 Diversity = 1.4514e+00 k_t = [3, 3, 4, 2]
 Generation 4290: Best = 2.449315e+03 FES = 429100 μ_F = 0.309 μ_{CR} = 0.040
 Diversity = 1.4513e+00 k_t = [1, 1, 1, 1]
 Generation 4300: Best = 2.448037e+03 FES = 430100 μ_F = 0.303 μ_{CR} = 0.039
 Diversity = 1.4514e+00 k_t = [4, 3, 4, 2]
 Generation 4310: Best = 2.448037e+03 FES = 431100 μ_F = 0.369 μ_{CR} = 0.049
 Diversity = 1.4508e+00 k_t = [1, 1, 1, 1]
 Generation 4320: Best = 2.448037e+03 FES = 432100 μ_F = 0.283 μ_{CR} = 0.043
 Diversity = 1.4505e+00 k_t = [3, 2, 4, 2]
 Generation 4330: Best = 2.448037e+03 FES = 433100 μ_F = 0.277 μ_{CR} = 0.039
 Diversity = 1.4500e+00 k_t = [1, 1, 1, 1]
 Generation 4340: Best = 2.448037e+03 FES = 434100 μ_F = 0.305 μ_{CR} = 0.037
 Diversity = 1.4499e+00 k_t = [3, 2, 4, 3]
 Generation 4350: Best = 2.448037e+03 FES = 435100 μ_F = 0.307 μ_{CR} = 0.039
 Diversity = 1.4497e+00 k_t = [1, 1, 1, 1]
 Generation 4360: Best = 2.448037e+03 FES = 436100 μ_F = 0.311 μ_{CR} = 0.050
 Diversity = 1.4504e+00 k_t = [4, 2, 3, 3]
 Generation 4370: Best = 2.447777e+03 FES = 437100 μ_F = 0.360 μ_{CR} = 0.048
 Diversity = 1.4494e+00 k_t = [1, 1, 1, 1]
 Generation 4380: Best = 2.445745e+03 FES = 438100 μ_F = 0.379 μ_{CR} = 0.047
 Diversity = 1.4326e+00 k_t = [2, 4, 3, 4]
 Generation 4390: Best = 2.445745e+03 FES = 439100 μ_F = 0.405 μ_{CR} = 0.036
 Diversity = 1.4331e+00 k_t = [1, 1, 1, 1]
 Generation 4400: Best = 2.445745e+03 FES = 440100 μ_F = 0.353 μ_{CR} = 0.050
 Diversity = 1.4299e+00 k_t = [4, 2, 3, 2]
 Generation 4410: Best = 2.445469e+03 FES = 441100 μ_F = 0.286 μ_{CR} = 0.049
 Diversity = 1.4300e+00 k_t = [1, 1, 1, 1]
 Generation 4420: Best = 2.445469e+03 FES = 442100 μ_F = 0.323 μ_{CR} = 0.061
 Diversity = 1.4299e+00 k_t = [4, 4, 2, 2]
 Generation 4430: Best = 2.445469e+03 FES = 443100 μ_F = 0.297 μ_{CR} = 0.057
 Diversity = 1.4299e+00 k_t = [1, 1, 1, 1]
 Generation 4440: Best = 2.445469e+03 FES = 444100 μ_F = 0.277 μ_{CR} = 0.047
 Diversity = 1.4305e+00 k_t = [2, 2, 3, 2]
 Generation 4450: Best = 2.444844e+03 FES = 445100 μ_F = 0.310 μ_{CR} = 0.045
 Diversity = 1.4295e+00 k_t = [1, 1, 1, 1]
 Generation 4460: Best = 2.444180e+03 FES = 446100 μ_F = 0.271 μ_{CR} = 0.044
 Diversity = 1.4292e+00 k_t = [2, 4, 4, 2]
 Generation 4470: Best = 2.443796e+03 FES = 447100 μ_F = 0.269 μ_{CR} = 0.036
 Diversity = 1.4286e+00 k_t = [1, 1, 1, 1]
 Generation 4480: Best = 2.443796e+03 FES = 448100 μ_F = 0.352 μ_{CR} = 0.056
 Diversity = 1.4284e+00 k_t = [2, 4, 2, 2]
 Generation 4490: Best = 2.443796e+03 FES = 449100 μ_F = 0.347 μ_{CR} = 0.042
 Diversity = 1.4283e+00 k_t = [1, 1, 1, 1]
 Generation 4500: Best = 2.443585e+03 FES = 450100 μ_F = 0.298 μ_{CR} = 0.054
 Diversity = 1.4285e+00 k_t = [4, 4, 3, 4]
 Generation 4510: Best = 2.440845e+03 FES = 451100 μ_F = 0.276 μ_{CR} = 0.064
 Diversity = 1.4281e+00 k_t = [1, 1, 1, 1]
 Generation 4520: Best = 2.440845e+03 FES = 452100 μ_F = 0.259 μ_{CR} = 0.072
 Diversity = 1.4281e+00 k_t = [4, 3, 3, 3]
 Generation 4530: Best = 2.439550e+03 FES = 453100 μ_F = 0.236 μ_{CR} = 0.055
 Diversity = 1.4278e+00 k_t = [1, 1, 1, 1]
 Generation 4540: Best = 2.439550e+03 FES = 454100 μ_F = 0.193 μ_{CR} = 0.059
 Diversity = 1.4265e+00 k_t = [2, 2, 4, 4]

Generation 4550: Best = 2.439550e+03 FES = 455100 μ_F = 0.200 μ_{CR} = 0.074
 Diversity = 1.4259e+00 k_t = [1, 1, 1, 1]
 Generation 4560: Best = 2.439550e+03 FES = 456100 μ_F = 0.200 μ_{CR} = 0.074
 Diversity = 1.4254e+00 k_t = [2, 4, 2, 3]
 Generation 4570: Best = 2.437395e+03 FES = 457100 μ_F = 0.191 μ_{CR} = 0.062
 Diversity = 1.4254e+00 k_t = [1, 1, 1, 1]
 Generation 4580: Best = 2.437395e+03 FES = 458100 μ_F = 0.239 μ_{CR} = 0.051
 Diversity = 1.4219e+00 k_t = [2, 2, 3, 4]
 Generation 4590: Best = 2.433004e+03 FES = 459100 μ_F = 0.238 μ_{CR} = 0.053
 Diversity = 1.4215e+00 k_t = [1, 1, 1, 1]
 Generation 4600: Best = 2.433004e+03 FES = 460100 μ_F = 0.290 μ_{CR} = 0.066
 Diversity = 1.4205e+00 k_t = [4, 2, 4, 4]
 Generation 4610: Best = 2.432944e+03 FES = 461100 μ_F = 0.344 μ_{CR} = 0.058
 Diversity = 1.4204e+00 k_t = [1, 1, 1, 1]
 Generation 4620: Best = 2.432184e+03 FES = 462100 μ_F = 0.359 μ_{CR} = 0.043
 Diversity = 1.4205e+00 k_t = [2, 2, 4, 2]
 Generation 4630: Best = 2.432184e+03 FES = 463100 μ_F = 0.294 μ_{CR} = 0.041
 Diversity = 1.4205e+00 k_t = [1, 1, 1, 1]
 Generation 4640: Best = 2.432184e+03 FES = 464100 μ_F = 0.253 μ_{CR} = 0.028
 Diversity = 1.4203e+00 k_t = [2, 2, 2, 3]
 Generation 4650: Best = 2.432184e+03 FES = 465100 μ_F = 0.244 μ_{CR} = 0.037
 Diversity = 1.4201e+00 k_t = [1, 1, 1, 1]
 Generation 4660: Best = 2.430950e+03 FES = 466100 μ_F = 0.240 μ_{CR} = 0.029
 Diversity = 1.4199e+00 k_t = [2, 3, 4, 4]
 Generation 4670: Best = 2.430950e+03 FES = 467100 μ_F = 0.212 μ_{CR} = 0.037
 Diversity = 1.4198e+00 k_t = [1, 1, 1, 1]
 Generation 4680: Best = 2.428759e+03 FES = 468100 μ_F = 0.265 μ_{CR} = 0.050
 Diversity = 1.4200e+00 k_t = [2, 2, 3, 3]
 Generation 4690: Best = 2.428601e+03 FES = 469100 μ_F = 0.262 μ_{CR} = 0.059
 Diversity = 1.4171e+00 k_t = [1, 1, 1, 1]
 Generation 4700: Best = 2.428028e+03 FES = 470100 μ_F = 0.232 μ_{CR} = 0.057
 Diversity = 1.4170e+00 k_t = [4, 4, 4, 2]
 Generation 4710: Best = 2.428028e+03 FES = 471100 μ_F = 0.206 μ_{CR} = 0.040
 Diversity = 1.4172e+00 k_t = [1, 1, 1, 1]
 Generation 4720: Best = 2.428028e+03 FES = 472100 μ_F = 0.288 μ_{CR} = 0.038
 Diversity = 1.4171e+00 k_t = [2, 3, 4, 3]
 Generation 4730: Best = 2.428028e+03 FES = 473100 μ_F = 0.261 μ_{CR} = 0.054
 Diversity = 1.4177e+00 k_t = [1, 1, 1, 1]
 Generation 4740: Best = 2.425757e+03 FES = 474100 μ_F = 0.244 μ_{CR} = 0.059
 Diversity = 1.4175e+00 k_t = [2, 4, 4, 3]
 Generation 4750: Best = 2.424463e+03 FES = 475100 μ_F = 0.278 μ_{CR} = 0.058
 Diversity = 1.4247e+00 k_t = [1, 1, 1, 1]
 Generation 4760: Best = 2.418327e+03 FES = 476100 μ_F = 0.419 μ_{CR} = 0.067
 Diversity = 1.4239e+00 k_t = [4, 4, 2, 4]
 Generation 4770: Best = 2.418327e+03 FES = 477100 μ_F = 0.331 μ_{CR} = 0.070
 Diversity = 1.4242e+00 k_t = [1, 1, 1, 1]
 Generation 4780: Best = 2.418327e+03 FES = 478100 μ_F = 0.262 μ_{CR} = 0.055
 Diversity = 1.4237e+00 k_t = [2, 3, 4, 2]
 Generation 4790: Best = 2.418327e+03 FES = 479100 μ_F = 0.249 μ_{CR} = 0.046
 Diversity = 1.4235e+00 k_t = [1, 1, 1, 1]
 Generation 4800: Best = 2.418327e+03 FES = 480100 μ_F = 0.223 μ_{CR} = 0.057
 Diversity = 1.4232e+00 k_t = [4, 3, 3, 3]
 Generation 4810: Best = 2.418327e+03 FES = 481100 μ_F = 0.218 μ_{CR} = 0.056
 Diversity = 1.4230e+00 k_t = [1, 1, 1, 1]

Generation 4820: Best = 2.417860e+03 FES = 482100 μ_F = 0.264 μ_{CR} = 0.065
 Diversity = 1.4263e+00 k_t = [2, 3, 2, 4]
 Generation 4830: Best = 2.415633e+03 FES = 483100 μ_F = 0.288 μ_{CR} = 0.055
 Diversity = 1.4264e+00 k_t = [1, 1, 1, 1]
 Generation 4840: Best = 2.411801e+03 FES = 484100 μ_F = 0.260 μ_{CR} = 0.068
 Diversity = 1.4262e+00 k_t = [4, 3, 4, 2]
 Generation 4850: Best = 2.411349e+03 FES = 485100 μ_F = 0.228 μ_{CR} = 0.071
 Diversity = 1.4265e+00 k_t = [1, 1, 1, 1]
 Generation 4860: Best = 2.411349e+03 FES = 486100 μ_F = 0.269 μ_{CR} = 0.074
 Diversity = 1.4303e+00 k_t = [3, 3, 4, 3]
 Generation 4870: Best = 2.411349e+03 FES = 487100 μ_F = 0.276 μ_{CR} = 0.075
 Diversity = 1.4302e+00 k_t = [1, 1, 1, 1]
 Generation 4880: Best = 2.411349e+03 FES = 488100 μ_F = 0.223 μ_{CR} = 0.066
 Diversity = 1.4303e+00 k_t = [4, 2, 2, 3]
 Generation 4890: Best = 2.411228e+03 FES = 489100 μ_F = 0.191 μ_{CR} = 0.054
 Diversity = 1.4301e+00 k_t = [1, 1, 1, 1]
 Generation 4900: Best = 2.410846e+03 FES = 490100 μ_F = 0.266 μ_{CR} = 0.054
 Diversity = 1.4302e+00 k_t = [3, 3, 4, 2]
 Generation 4910: Best = 2.408937e+03 FES = 491100 μ_F = 0.256 μ_{CR} = 0.044
 Diversity = 1.4299e+00 k_t = [1, 1, 1, 1]
 Generation 4920: Best = 2.408937e+03 FES = 492100 μ_F = 0.228 μ_{CR} = 0.050
 Diversity = 1.4298e+00 k_t = [3, 3, 2, 2]
 Generation 4930: Best = 2.408937e+03 FES = 493100 μ_F = 0.310 μ_{CR} = 0.038
 Diversity = 1.4301e+00 k_t = [1, 1, 1, 1]
 Generation 4940: Best = 2.408937e+03 FES = 494100 μ_F = 0.295 μ_{CR} = 0.035
 Diversity = 1.4299e+00 k_t = [2, 4, 2, 2]
 Generation 4950: Best = 2.408937e+03 FES = 495100 μ_F = 0.253 μ_{CR} = 0.052
 Diversity = 1.4300e+00 k_t = [1, 1, 1, 1]
 Generation 4960: Best = 2.403630e+03 FES = 496100 μ_F = 0.301 μ_{CR} = 0.080
 Diversity = 1.4331e+00 k_t = [4, 2, 4, 4]
 Generation 4970: Best = 2.401482e+03 FES = 497100 μ_F = 0.213 μ_{CR} = 0.080
 Diversity = 1.4328e+00 k_t = [1, 1, 1, 1]
 Generation 4980: Best = 2.401482e+03 FES = 498100 μ_F = 0.282 μ_{CR} = 0.070
 Diversity = 1.4327e+00 k_t = [4, 4, 3, 3]
 Generation 4990: Best = 2.399205e+03 FES = 499100 μ_F = 0.334 μ_{CR} = 0.056
 Diversity = 1.4326e+00 k_t = [1, 1, 1, 1]
 Generation 5000: Best = 2.398989e+03 FES = 500100 μ_F = 0.360 μ_{CR} = 0.044
 Diversity = 1.4289e+00 k_t = [2, 4, 2, 4]
 Generation 5010: Best = 2.398989e+03 FES = 501100 μ_F = 0.411 μ_{CR} = 0.040
 Diversity = 1.4292e+00 k_t = [1, 1, 1, 1]
 Generation 5020: Best = 2.398989e+03 FES = 502100 μ_F = 0.402 μ_{CR} = 0.046
 Diversity = 1.4297e+00 k_t = [3, 2, 4, 2]
 Generation 5030: Best = 2.398989e+03 FES = 503100 μ_F = 0.348 μ_{CR} = 0.038
 Diversity = 1.4300e+00 k_t = [1, 1, 1, 1]
 Generation 5040: Best = 2.398368e+03 FES = 504100 μ_F = 0.348 μ_{CR} = 0.053
 Diversity = 1.4080e+00 k_t = [3, 2, 4, 4]
 Generation 5050: Best = 2.398368e+03 FES = 505100 μ_F = 0.380 μ_{CR} = 0.045
 Diversity = 1.4082e+00 k_t = [1, 1, 1, 1]
 Generation 5060: Best = 2.398368e+03 FES = 506100 μ_F = 0.376 μ_{CR} = 0.041
 Diversity = 1.4039e+00 k_t = [2, 3, 2, 4]
 Generation 5070: Best = 2.398257e+03 FES = 507100 μ_F = 0.393 μ_{CR} = 0.030
 Diversity = 1.4037e+00 k_t = [1, 1, 1, 1]
 Generation 5080: Best = 2.397905e+03 FES = 508100 μ_F = 0.442 μ_{CR} = 0.036
 Diversity = 1.4037e+00 k_t = [3, 4, 3, 3]

Generation 5090: Best = 2.397854e+03 FES = 509100 μ_F = 0.374 μ_{CR} = 0.036
 Diversity = 1.4037e+00 k_t = [1, 1, 1, 1]
 Generation 5100: Best = 2.397854e+03 FES = 510100 μ_F = 0.418 μ_{CR} = 0.040
 Diversity = 1.3972e+00 k_t = [3, 3, 4, 4]
 Generation 5110: Best = 2.397854e+03 FES = 511100 μ_F = 0.462 μ_{CR} = 0.027
 Diversity = 1.3972e+00 k_t = [1, 1, 1, 1]
 Generation 5120: Best = 2.397854e+03 FES = 512100 μ_F = 0.424 μ_{CR} = 0.032
 Diversity = 1.3971e+00 k_t = [2, 2, 4, 2]
 Generation 5130: Best = 2.397854e+03 FES = 513100 μ_F = 0.451 μ_{CR} = 0.032
 Diversity = 1.3908e+00 k_t = [1, 1, 1, 1]
 Generation 5140: Best = 2.397854e+03 FES = 514100 μ_F = 0.366 μ_{CR} = 0.028
 Diversity = 1.3907e+00 k_t = [3, 2, 2, 4]
 Generation 5150: Best = 2.397854e+03 FES = 515100 μ_F = 0.394 μ_{CR} = 0.026
 Diversity = 1.3907e+00 k_t = [1, 1, 1, 1]
 Generation 5160: Best = 2.397854e+03 FES = 516100 μ_F = 0.325 μ_{CR} = 0.042
 Diversity = 1.3901e+00 k_t = [4, 4, 4, 4]
 Generation 5170: Best = 2.397666e+03 FES = 517100 μ_F = 0.316 μ_{CR} = 0.035
 Diversity = 1.3901e+00 k_t = [1, 1, 1, 1]
 Generation 5180: Best = 2.397666e+03 FES = 518100 μ_F = 0.279 μ_{CR} = 0.062
 Diversity = 1.3902e+00 k_t = [2, 2, 3, 3]
 Generation 5190: Best = 2.397666e+03 FES = 519100 μ_F = 0.247 μ_{CR} = 0.062
 Diversity = 1.3904e+00 k_t = [1, 1, 1, 1]
 Generation 5200: Best = 2.397666e+03 FES = 520100 μ_F = 0.268 μ_{CR} = 0.071
 Diversity = 1.3899e+00 k_t = [4, 3, 2, 3]
 Generation 5210: Best = 2.397449e+03 FES = 521100 μ_F = 0.191 μ_{CR} = 0.070
 Diversity = 1.3898e+00 k_t = [1, 1, 1, 1]
 Generation 5220: Best = 2.396323e+03 FES = 522100 μ_F = 0.173 μ_{CR} = 0.064
 Diversity = 1.3890e+00 k_t = [4, 3, 3, 3]
 Generation 5230: Best = 2.394743e+03 FES = 523100 μ_F = 0.141 μ_{CR} = 0.066
 Diversity = 1.3890e+00 k_t = [1, 1, 1, 1]
 Generation 5240: Best = 2.394743e+03 FES = 524100 μ_F = 0.240 μ_{CR} = 0.068
 Diversity = 1.3881e+00 k_t = [3, 2, 2, 3]
 Generation 5250: Best = 2.393690e+03 FES = 525100 μ_F = 0.289 μ_{CR} = 0.068
 Diversity = 1.3881e+00 k_t = [1, 1, 1, 1]
 Generation 5260: Best = 2.390431e+03 FES = 526100 μ_F = 0.284 μ_{CR} = 0.071
 Diversity = 1.3868e+00 k_t = [4, 3, 4, 4]
 Generation 5270: Best = 2.390431e+03 FES = 527100 μ_F = 0.286 μ_{CR} = 0.058
 Diversity = 1.3866e+00 k_t = [1, 1, 1, 1]
 Generation 5280: Best = 2.390431e+03 FES = 528100 μ_F = 0.367 μ_{CR} = 0.056
 Diversity = 1.3865e+00 k_t = [2, 4, 4, 2]
 Generation 5290: Best = 2.390431e+03 FES = 529100 μ_F = 0.325 μ_{CR} = 0.049
 Diversity = 1.3864e+00 k_t = [1, 1, 1, 1]
 Generation 5300: Best = 2.390011e+03 FES = 530100 μ_F = 0.374 μ_{CR} = 0.055
 Diversity = 1.3862e+00 k_t = [3, 4, 4, 4]
 Generation 5310: Best = 2.390011e+03 FES = 531100 μ_F = 0.428 μ_{CR} = 0.045
 Diversity = 1.3861e+00 k_t = [1, 1, 1, 1]
 Generation 5320: Best = 2.390011e+03 FES = 532100 μ_F = 0.438 μ_{CR} = 0.044
 Diversity = 1.3916e+00 k_t = [3, 3, 4, 4]
 Generation 5330: Best = 2.390011e+03 FES = 533100 μ_F = 0.470 μ_{CR} = 0.040
 Diversity = 1.3916e+00 k_t = [1, 1, 1, 1]
 Generation 5340: Best = 2.390011e+03 FES = 534100 μ_F = 0.492 μ_{CR} = 0.046
 Diversity = 1.3855e+00 k_t = [3, 4, 2, 2]
 Generation 5350: Best = 2.390011e+03 FES = 535100 μ_F = 0.455 μ_{CR} = 0.039
 Diversity = 1.3858e+00 k_t = [1, 1, 1, 1]

Generation 5360: Best = 2.390011e+03 FES = 536100 μ_F = 0.453 μ_{CR} = 0.038
 Diversity = 1.3872e+00 k_t = [4, 2, 3, 2]
 Generation 5370: Best = 2.388988e+03 FES = 537100 μ_F = 0.342 μ_{CR} = 0.057
 Diversity = 1.3873e+00 k_t = [1, 1, 1, 1]
 Generation 5380: Best = 2.388988e+03 FES = 538100 μ_F = 0.289 μ_{CR} = 0.047
 Diversity = 1.3867e+00 k_t = [2, 4, 4, 3]
 Generation 5390: Best = 2.388988e+03 FES = 539100 μ_F = 0.215 μ_{CR} = 0.064
 Diversity = 1.3867e+00 k_t = [1, 1, 1, 1]
 Generation 5400: Best = 2.388738e+03 FES = 540100 μ_F = 0.226 μ_{CR} = 0.069
 Diversity = 1.3831e+00 k_t = [2, 3, 4, 4]
 Generation 5410: Best = 2.388738e+03 FES = 541100 μ_F = 0.212 μ_{CR} = 0.062
 Diversity = 1.3831e+00 k_t = [1, 1, 1, 1]
 Generation 5420: Best = 2.386490e+03 FES = 542100 μ_F = 0.217 μ_{CR} = 0.068
 Diversity = 1.3831e+00 k_t = [3, 4, 2, 4]
 Generation 5430: Best = 2.386490e+03 FES = 543100 μ_F = 0.228 μ_{CR} = 0.053
 Diversity = 1.3833e+00 k_t = [1, 1, 1, 1]
 Generation 5440: Best = 2.386490e+03 FES = 544100 μ_F = 0.211 μ_{CR} = 0.064
 Diversity = 1.3834e+00 k_t = [4, 4, 3, 3]
 Generation 5450: Best = 2.386490e+03 FES = 545100 μ_F = 0.222 μ_{CR} = 0.048
 Diversity = 1.3836e+00 k_t = [1, 1, 1, 1]
 Generation 5460: Best = 2.384660e+03 FES = 546100 μ_F = 0.214 μ_{CR} = 0.055
 Diversity = 1.3836e+00 k_t = [4, 3, 3, 4]
 Generation 5470: Best = 2.384660e+03 FES = 547100 μ_F = 0.188 μ_{CR} = 0.071
 Diversity = 1.3842e+00 k_t = [1, 1, 1, 1]
 Generation 5480: Best = 2.384660e+03 FES = 548100 μ_F = 0.237 μ_{CR} = 0.063
 Diversity = 1.3759e+00 k_t = [4, 3, 2, 4]
 Generation 5490: Best = 2.383910e+03 FES = 549100 μ_F = 0.349 μ_{CR} = 0.054
 Diversity = 1.3759e+00 k_t = [1, 1, 1, 1]
 Generation 5500: Best = 2.383910e+03 FES = 550100 μ_F = 0.489 μ_{CR} = 0.034
 Diversity = 1.3604e+00 k_t = [2, 2, 3, 4]
 Generation 5510: Best = 2.383910e+03 FES = 551100 μ_F = 0.522 μ_{CR} = 0.039
 Diversity = 1.3526e+00 k_t = [1, 1, 1, 1]
 Generation 5520: Best = 2.383184e+03 FES = 552100 μ_F = 0.512 μ_{CR} = 0.033
 Diversity = 1.3530e+00 k_t = [3, 2, 3, 3]
 Generation 5530: Best = 2.383184e+03 FES = 553100 μ_F = 0.412 μ_{CR} = 0.037
 Diversity = 1.3546e+00 k_t = [1, 1, 1, 1]
 Generation 5540: Best = 2.383184e+03 FES = 554100 μ_F = 0.428 μ_{CR} = 0.033
 Diversity = 1.3481e+00 k_t = [2, 2, 3, 3]
 Generation 5550: Best = 2.383092e+03 FES = 555100 μ_F = 0.367 μ_{CR} = 0.038
 Diversity = 1.3485e+00 k_t = [1, 1, 1, 1]
 Generation 5560: Best = 2.383092e+03 FES = 556100 μ_F = 0.379 μ_{CR} = 0.040
 Diversity = 1.3505e+00 k_t = [4, 2, 4, 4]
 Generation 5570: Best = 2.382772e+03 FES = 557100 μ_F = 0.337 μ_{CR} = 0.048
 Diversity = 1.3503e+00 k_t = [1, 1, 1, 1]
 Generation 5580: Best = 2.382772e+03 FES = 558100 μ_F = 0.320 μ_{CR} = 0.069
 Diversity = 1.3336e+00 k_t = [2, 3, 3, 4]
 Generation 5590: Best = 2.382772e+03 FES = 559100 μ_F = 0.244 μ_{CR} = 0.052
 Diversity = 1.3339e+00 k_t = [1, 1, 1, 1]
 Generation 5600: Best = 2.382395e+03 FES = 560100 μ_F = 0.258 μ_{CR} = 0.040
 Diversity = 1.3338e+00 k_t = [4, 2, 2, 3]
 Generation 5610: Best = 2.382395e+03 FES = 561100 μ_F = 0.188 μ_{CR} = 0.053
 Diversity = 1.3339e+00 k_t = [1, 1, 1, 1]
 Generation 5620: Best = 2.382395e+03 FES = 562100 μ_F = 0.174 μ_{CR} = 0.047
 Diversity = 1.3337e+00 k_t = [3, 3, 4, 3]

Generation 5630: Best = 2.382395e+03 FES = 563100 μ_F = 0.173 μ_{CR} = 0.049
 Diversity = 1.3340e+00 k_t = [1, 1, 1, 1]
 Generation 5640: Best = 2.382395e+03 FES = 564100 μ_F = 0.199 μ_{CR} = 0.041
 Diversity = 1.3336e+00 k_t = [4, 4, 4, 2]
 Generation 5650: Best = 2.382395e+03 FES = 565100 μ_F = 0.216 μ_{CR} = 0.031
 Diversity = 1.3339e+00 k_t = [1, 1, 1, 1]
 Generation 5660: Best = 2.381065e+03 FES = 566100 μ_F = 0.188 μ_{CR} = 0.043
 Diversity = 1.3338e+00 k_t = [4, 4, 3, 2]
 Generation 5670: Best = 2.381065e+03 FES = 567100 μ_F = 0.227 μ_{CR} = 0.056
 Diversity = 1.3336e+00 k_t = [1, 1, 1, 1]
 Generation 5680: Best = 2.380526e+03 FES = 568100 μ_F = 0.238 μ_{CR} = 0.054
 Diversity = 1.3338e+00 k_t = [3, 4, 2, 3]
 Generation 5690: Best = 2.380526e+03 FES = 569100 μ_F = 0.232 μ_{CR} = 0.039
 Diversity = 1.3339e+00 k_t = [1, 1, 1, 1]
 Generation 5700: Best = 2.380526e+03 FES = 570100 μ_F = 0.323 μ_{CR} = 0.036
 Diversity = 1.3300e+00 k_t = [2, 4, 2, 4]
 Generation 5710: Best = 2.379575e+03 FES = 571100 μ_F = 0.262 μ_{CR} = 0.025
 Diversity = 1.3301e+00 k_t = [1, 1, 1, 1]
 Generation 5720: Best = 2.379575e+03 FES = 572100 μ_F = 0.263 μ_{CR} = 0.051
 Diversity = 1.3271e+00 k_t = [3, 4, 3, 4]
 Generation 5730: Best = 2.379575e+03 FES = 573100 μ_F = 0.208 μ_{CR} = 0.069
 Diversity = 1.3270e+00 k_t = [1, 1, 1, 1]
 Generation 5740: Best = 2.379575e+03 FES = 574100 μ_F = 0.246 μ_{CR} = 0.057
 Diversity = 1.3319e+00 k_t = [2, 2, 2, 4]
 Generation 5750: Best = 2.379575e+03 FES = 575100 μ_F = 0.213 μ_{CR} = 0.047
 Diversity = 1.3322e+00 k_t = [1, 1, 1, 1]
 Generation 5760: Best = 2.379575e+03 FES = 576100 μ_F = 0.189 μ_{CR} = 0.050
 Diversity = 1.3321e+00 k_t = [4, 4, 4, 3]
 Generation 5770: Best = 2.379575e+03 FES = 577100 μ_F = 0.203 μ_{CR} = 0.035
 Diversity = 1.3322e+00 k_t = [1, 1, 1, 1]
 Generation 5780: Best = 2.379575e+03 FES = 578100 μ_F = 0.204 μ_{CR} = 0.051
 Diversity = 1.3388e+00 k_t = [4, 4, 4, 4]
 Generation 5790: Best = 2.379575e+03 FES = 579100 μ_F = 0.138 μ_{CR} = 0.049
 Diversity = 1.3389e+00 k_t = [1, 1, 1, 1]
 Generation 5800: Best = 2.379575e+03 FES = 580100 μ_F = 0.106 μ_{CR} = 0.061
 Diversity = 1.3390e+00 k_t = [3, 3, 2, 3]
 Generation 5810: Best = 2.379575e+03 FES = 581100 μ_F = 0.090 μ_{CR} = 0.062
 Diversity = 1.3390e+00 k_t = [1, 1, 1, 1]
 Generation 5820: Best = 2.379525e+03 FES = 582100 μ_F = 0.094 μ_{CR} = 0.072
 Diversity = 1.3390e+00 k_t = [4, 4, 2, 3]
 Generation 5830: Best = 2.379525e+03 FES = 583100 μ_F = 0.223 μ_{CR} = 0.048
 Diversity = 1.3393e+00 k_t = [1, 1, 1, 1]
 Generation 5840: Best = 2.377748e+03 FES = 584100 μ_F = 0.313 μ_{CR} = 0.046
 Diversity = 1.3374e+00 k_t = [4, 4, 3, 2]
 Generation 5850: Best = 2.377748e+03 FES = 585100 μ_F = 0.268 μ_{CR} = 0.037
 Diversity = 1.3376e+00 k_t = [1, 1, 1, 1]
 Generation 5860: Best = 2.377342e+03 FES = 586100 μ_F = 0.274 μ_{CR} = 0.054
 Diversity = 1.3374e+00 k_t = [3, 3, 4, 2]
 Generation 5870: Best = 2.377342e+03 FES = 587100 μ_F = 0.203 μ_{CR} = 0.056
 Diversity = 1.3374e+00 k_t = [1, 1, 1, 1]
 Generation 5880: Best = 2.377002e+03 FES = 588100 μ_F = 0.171 μ_{CR} = 0.054
 Diversity = 1.3376e+00 k_t = [4, 3, 4, 3]
 Generation 5890: Best = 2.377002e+03 FES = 589100 μ_F = 0.198 μ_{CR} = 0.041
 Diversity = 1.3378e+00 k_t = [1, 1, 1, 1]

Generation 5900: Best = 2.377002e+03 FES = 590100 μ_F = 0.171 μ_{CR} = 0.044
 Diversity = 1.3380e+00 k_t = [3, 4, 2, 3]
 Generation 5910: Best = 2.377002e+03 FES = 591100 μ_F = 0.191 μ_{CR} = 0.046
 Diversity = 1.3379e+00 k_t = [1, 1, 1, 1]
 Generation 5920: Best = 2.375921e+03 FES = 592100 μ_F = 0.230 μ_{CR} = 0.050
 Diversity = 1.3371e+00 k_t = [4, 2, 2, 4]
 Generation 5930: Best = 2.374574e+03 FES = 593100 μ_F = 0.170 μ_{CR} = 0.040
 Diversity = 1.3370e+00 k_t = [1, 1, 1, 1]
 Generation 5940: Best = 2.374574e+03 FES = 594100 μ_F = 0.201 μ_{CR} = 0.047
 Diversity = 1.3303e+00 k_t = [3, 3, 4, 4]
 Generation 5950: Best = 2.374574e+03 FES = 595100 μ_F = 0.158 μ_{CR} = 0.044
 Diversity = 1.3305e+00 k_t = [1, 1, 1, 1]
 Generation 5960: Best = 2.373307e+03 FES = 596100 μ_F = 0.228 μ_{CR} = 0.059
 Diversity = 1.3144e+00 k_t = [4, 4, 4, 4]
 Generation 5970: Best = 2.373170e+03 FES = 597100 μ_F = 0.203 μ_{CR} = 0.051
 Diversity = 1.3153e+00 k_t = [1, 1, 1, 1]
 Generation 5980: Best = 2.373170e+03 FES = 598100 μ_F = 0.230 μ_{CR} = 0.037
 Diversity = 1.3118e+00 k_t = [3, 4, 3, 2]
 Generation 5990: Best = 2.373170e+03 FES = 599100 μ_F = 0.220 μ_{CR} = 0.033
 Diversity = 1.3105e+00 k_t = [1, 1, 1, 1]
 Generation 6000: Best = 2.373170e+03 FES = 600100 μ_F = 0.257 μ_{CR} = 0.031
 Diversity = 1.3105e+00 k_t = [3, 3, 4, 2]
 Generation 6010: Best = 2.373170e+03 FES = 601100 μ_F = 0.263 μ_{CR} = 0.031
 Diversity = 1.3105e+00 k_t = [1, 1, 1, 1]
 Generation 6020: Best = 2.373170e+03 FES = 602100 μ_F = 0.305 μ_{CR} = 0.048
 Diversity = 1.2929e+00 k_t = [2, 2, 4, 4]
 Generation 6030: Best = 2.373170e+03 FES = 603100 μ_F = 0.285 μ_{CR} = 0.029
 Diversity = 1.2929e+00 k_t = [1, 1, 1, 1]
 Generation 6040: Best = 2.373170e+03 FES = 604100 μ_F = 0.340 μ_{CR} = 0.041
 Diversity = 1.2921e+00 k_t = [2, 3, 4, 4]
 Generation 6050: Best = 2.373170e+03 FES = 605100 μ_F = 0.308 μ_{CR} = 0.044
 Diversity = 1.2922e+00 k_t = [1, 1, 1, 1]
 Generation 6060: Best = 2.373170e+03 FES = 606100 μ_F = 0.279 μ_{CR} = 0.038
 Diversity = 1.2860e+00 k_t = [4, 2, 4, 4]
 Generation 6070: Best = 2.373170e+03 FES = 607100 μ_F = 0.221 μ_{CR} = 0.034
 Diversity = 1.2861e+00 k_t = [1, 1, 1, 1]
 Generation 6080: Best = 2.373170e+03 FES = 608100 μ_F = 0.198 μ_{CR} = 0.053
 Diversity = 1.2862e+00 k_t = [4, 2, 4, 2]
 Generation 6090: Best = 2.372839e+03 FES = 609100 μ_F = 0.176 μ_{CR} = 0.067
 Diversity = 1.2862e+00 k_t = [1, 1, 1, 1]
 Generation 6100: Best = 2.372839e+03 FES = 610100 μ_F = 0.193 μ_{CR} = 0.055
 Diversity = 1.2861e+00 k_t = [4, 3, 4, 2]
 Generation 6110: Best = 2.371897e+03 FES = 611100 μ_F = 0.142 μ_{CR} = 0.063
 Diversity = 1.2861e+00 k_t = [1, 1, 1, 1]
 Generation 6120: Best = 2.371897e+03 FES = 612100 μ_F = 0.148 μ_{CR} = 0.075
 Diversity = 1.2865e+00 k_t = [3, 4, 3, 2]
 Generation 6130: Best = 2.370422e+03 FES = 613100 μ_F = 0.154 μ_{CR} = 0.072
 Diversity = 1.2867e+00 k_t = [1, 1, 1, 1]
 Generation 6140: Best = 2.370422e+03 FES = 614100 μ_F = 0.173 μ_{CR} = 0.067
 Diversity = 1.2866e+00 k_t = [4, 3, 2, 4]
 Generation 6150: Best = 2.370422e+03 FES = 615100 μ_F = 0.142 μ_{CR} = 0.050
 Diversity = 1.2866e+00 k_t = [1, 1, 1, 1]
 Generation 6160: Best = 2.370422e+03 FES = 616100 μ_F = 0.190 μ_{CR} = 0.051
 Diversity = 1.2869e+00 k_t = [3, 4, 4, 3]

Generation 6170: Best = 2.370422e+03 FES = 617100 μ_F = 0.161 μ_{CR} = 0.058
 Diversity = 1.2872e+00 k_t = [1, 1, 1, 1]
 Generation 6180: Best = 2.370298e+03 FES = 618100 μ_F = 0.104 μ_{CR} = 0.041
 Diversity = 1.2872e+00 k_t = [3, 4, 3, 3]
 Generation 6190: Best = 2.370298e+03 FES = 619100 μ_F = 0.125 μ_{CR} = 0.054
 Diversity = 1.2873e+00 k_t = [1, 1, 1, 1]
 Generation 6200: Best = 2.370298e+03 FES = 620100 μ_F = 0.134 μ_{CR} = 0.055
 Diversity = 1.2874e+00 k_t = [3, 3, 2, 2]
 Generation 6210: Best = 2.370298e+03 FES = 621100 μ_F = 0.207 μ_{CR} = 0.041
 Diversity = 1.2875e+00 k_t = [1, 1, 1, 1]
 Generation 6220: Best = 2.369884e+03 FES = 622100 μ_F = 0.214 μ_{CR} = 0.039
 Diversity = 1.2876e+00 k_t = [4, 3, 4, 2]
 Generation 6230: Best = 2.369762e+03 FES = 623100 μ_F = 0.210 μ_{CR} = 0.038
 Diversity = 1.2876e+00 k_t = [1, 1, 1, 1]
 Generation 6240: Best = 2.369762e+03 FES = 624100 μ_F = 0.161 μ_{CR} = 0.051
 Diversity = 1.2876e+00 k_t = [2, 4, 3, 2]
 Generation 6250: Best = 2.369762e+03 FES = 625100 μ_F = 0.185 μ_{CR} = 0.049
 Diversity = 1.2876e+00 k_t = [1, 1, 1, 1]
 Generation 6260: Best = 2.369762e+03 FES = 626100 μ_F = 0.126 μ_{CR} = 0.048
 Diversity = 1.2882e+00 k_t = [2, 3, 3, 2]
 Generation 6270: Best = 2.369762e+03 FES = 627100 μ_F = 0.150 μ_{CR} = 0.041
 Diversity = 1.2882e+00 k_t = [1, 1, 1, 1]
 Generation 6280: Best = 2.368245e+03 FES = 628100 μ_F = 0.197 μ_{CR} = 0.052
 Diversity = 1.2943e+00 k_t = [3, 2, 4, 4]
 Generation 6290: Best = 2.364111e+03 FES = 629100 μ_F = 0.248 μ_{CR} = 0.097
 Diversity = 1.2943e+00 k_t = [1, 1, 1, 1]
 Generation 6300: Best = 2.362252e+03 FES = 630100 μ_F = 0.287 μ_{CR} = 0.117
 Diversity = 1.2942e+00 k_t = [3, 3, 4, 2]
 Generation 6310: Best = 2.356896e+03 FES = 631100 μ_F = 0.299 μ_{CR} = 0.139
 Diversity = 1.2945e+00 k_t = [1, 1, 1, 1]
 Generation 6320: Best = 2.356328e+03 FES = 632100 μ_F = 0.357 μ_{CR} = 0.153
 Diversity = 1.2946e+00 k_t = [2, 4, 2, 3]
 Generation 6330: Best = 2.353282e+03 FES = 633100 μ_F = 0.354 μ_{CR} = 0.169
 Diversity = 1.2942e+00 k_t = [1, 1, 1, 1]
 Generation 6340: Best = 2.352072e+03 FES = 634100 μ_F = 0.406 μ_{CR} = 0.175
 Diversity = 1.2929e+00 k_t = [2, 4, 3, 2]
 Generation 6350: Best = 2.348609e+03 FES = 635100 μ_F = 0.438 μ_{CR} = 0.181
 Diversity = 1.2931e+00 k_t = [1, 1, 1, 1]
 Generation 6360: Best = 2.347223e+03 FES = 636100 μ_F = 0.456 μ_{CR} = 0.179
 Diversity = 1.2930e+00 k_t = [4, 2, 3, 2]
 Generation 6370: Best = 2.346087e+03 FES = 637100 μ_F = 0.519 μ_{CR} = 0.163
 Diversity = 1.2803e+00 k_t = [1, 1, 1, 1]
 Generation 6380: Best = 2.344110e+03 FES = 638100 μ_F = 0.576 μ_{CR} = 0.166
 Diversity = 1.2801e+00 k_t = [3, 3, 2, 4]
 Generation 6390: Best = 2.342212e+03 FES = 639100 μ_F = 0.594 μ_{CR} = 0.173
 Diversity = 1.2923e+00 k_t = [1, 1, 1, 1]
 Generation 6400: Best = 2.341435e+03 FES = 640100 μ_F = 0.629 μ_{CR} = 0.167
 Diversity = 1.2977e+00 k_t = [4, 2, 3, 3]
 Generation 6410: Best = 2.339876e+03 FES = 641100 μ_F = 0.621 μ_{CR} = 0.167
 Diversity = 1.2950e+00 k_t = [1, 1, 1, 1]
 Generation 6420: Best = 2.337925e+03 FES = 642100 μ_F = 0.641 μ_{CR} = 0.178
 Diversity = 1.2954e+00 k_t = [4, 4, 3, 4]
 Generation 6430: Best = 2.336910e+03 FES = 643100 μ_F = 0.693 μ_{CR} = 0.189
 Diversity = 1.2956e+00 k_t = [1, 1, 1, 1]

Generation 6440: Best = 2.335591e+03 FES = 644100 μ_F = 0.722 μ_{CR} = 0.183
 Diversity = 1.2954e+00 k_t = [2, 2, 2, 2]
 Generation 6450: Best = 2.334938e+03 FES = 645100 μ_F = 0.721 μ_{CR} = 0.181
 Diversity = 1.2992e+00 k_t = [1, 1, 1, 1]
 Generation 6460: Best = 2.333924e+03 FES = 646100 μ_F = 0.727 μ_{CR} = 0.188
 Diversity = 1.2929e+00 k_t = [4, 4, 3, 4]
 Generation 6470: Best = 2.333236e+03 FES = 647100 μ_F = 0.711 μ_{CR} = 0.169
 Diversity = 1.2991e+00 k_t = [1, 1, 1, 1]
 Generation 6480: Best = 2.332387e+03 FES = 648100 μ_F = 0.704 μ_{CR} = 0.179
 Diversity = 1.3005e+00 k_t = [4, 3, 2, 4]
 Generation 6490: Best = 2.331781e+03 FES = 649100 μ_F = 0.745 μ_{CR} = 0.184
 Diversity = 1.3039e+00 k_t = [1, 1, 1, 1]
 Generation 6500: Best = 2.331237e+03 FES = 650100 μ_F = 0.743 μ_{CR} = 0.157
 Diversity = 1.2983e+00 k_t = [2, 3, 4, 4]
 Generation 6510: Best = 2.330465e+03 FES = 651100 μ_F = 0.783 μ_{CR} = 0.164
 Diversity = 1.2985e+00 k_t = [1, 1, 1, 1]
 Generation 6520: Best = 2.329934e+03 FES = 652100 μ_F = 0.752 μ_{CR} = 0.175
 Diversity = 1.2980e+00 k_t = [4, 2, 2, 2]
 Generation 6530: Best = 2.329464e+03 FES = 653100 μ_F = 0.773 μ_{CR} = 0.166
 Diversity = 1.2981e+00 k_t = [1, 1, 1, 1]
 Generation 6540: Best = 2.328451e+03 FES = 654100 μ_F = 0.786 μ_{CR} = 0.171
 Diversity = 1.2982e+00 k_t = [3, 2, 2, 3]
 Generation 6550: Best = 2.328154e+03 FES = 655100 μ_F = 0.808 μ_{CR} = 0.164
 Diversity = 1.2981e+00 k_t = [1, 1, 1, 1]
 Generation 6560: Best = 2.327350e+03 FES = 656100 μ_F = 0.811 μ_{CR} = 0.178
 Diversity = 1.2912e+00 k_t = [4, 4, 4, 3]
 Generation 6570: Best = 2.326619e+03 FES = 657100 μ_F = 0.802 μ_{CR} = 0.173
 Diversity = 1.2825e+00 k_t = [1, 1, 1, 1]
 Generation 6580: Best = 2.326353e+03 FES = 658100 μ_F = 0.804 μ_{CR} = 0.164
 Diversity = 1.2817e+00 k_t = [3, 4, 4, 2]
 Generation 6590: Best = 2.326013e+03 FES = 659100 μ_F = 0.807 μ_{CR} = 0.160
 Diversity = 1.2817e+00 k_t = [1, 1, 1, 1]
 Generation 6600: Best = 2.325389e+03 FES = 660100 μ_F = 0.808 μ_{CR} = 0.177
 Diversity = 1.2826e+00 k_t = [4, 4, 3, 3]
 Generation 6610: Best = 2.325053e+03 FES = 661100 μ_F = 0.824 μ_{CR} = 0.189
 Diversity = 1.2802e+00 k_t = [1, 1, 1, 1]
 Generation 6620: Best = 2.324615e+03 FES = 662100 μ_F = 0.823 μ_{CR} = 0.181
 Diversity = 1.2805e+00 k_t = [2, 4, 4, 2]
 Generation 6630: Best = 2.324200e+03 FES = 663100 μ_F = 0.828 μ_{CR} = 0.183
 Diversity = 1.2805e+00 k_t = [1, 1, 1, 1]
 Generation 6640: Best = 2.323635e+03 FES = 664100 μ_F = 0.824 μ_{CR} = 0.191
 Diversity = 1.2803e+00 k_t = [4, 2, 4, 2]
 Generation 6650: Best = 2.323016e+03 FES = 665100 μ_F = 0.816 μ_{CR} = 0.209
 Diversity = 1.2804e+00 k_t = [1, 1, 1, 1]
 Generation 6660: Best = 2.322615e+03 FES = 666100 μ_F = 0.814 μ_{CR} = 0.192
 Diversity = 1.2671e+00 k_t = [2, 2, 3, 3]
 Generation 6670: Best = 2.322144e+03 FES = 667100 μ_F = 0.801 μ_{CR} = 0.192
 Diversity = 1.2671e+00 k_t = [1, 1, 1, 1]
 Generation 6680: Best = 2.321560e+03 FES = 668100 μ_F = 0.819 μ_{CR} = 0.184
 Diversity = 1.2672e+00 k_t = [3, 3, 2, 2]
 Generation 6690: Best = 2.321452e+03 FES = 669100 μ_F = 0.845 μ_{CR} = 0.172
 Diversity = 1.2672e+00 k_t = [1, 1, 1, 1]
 Generation 6700: Best = 2.321008e+03 FES = 670100 μ_F = 0.829 μ_{CR} = 0.174
 Diversity = 1.2671e+00 k_t = [2, 3, 2, 2]

Generation 6710: Best = 2.320901e+03 FES = 671100 μ_F = 0.815 μ_{CR} = 0.170
 Diversity = 1.2672e+00 k_t = [1, 1, 1, 1]
 Generation 6720: Best = 2.320559e+03 FES = 672100 μ_F = 0.795 μ_{CR} = 0.164
 Diversity = 1.2512e+00 k_t = [2, 4, 4, 3]
 Generation 6730: Best = 2.320053e+03 FES = 673100 μ_F = 0.805 μ_{CR} = 0.167
 Diversity = 1.2519e+00 k_t = [1, 1, 1, 1]
 Generation 6740: Best = 2.319874e+03 FES = 674100 μ_F = 0.827 μ_{CR} = 0.175
 Diversity = 1.2526e+00 k_t = [3, 2, 2, 2]
 Generation 6750: Best = 2.319323e+03 FES = 675100 μ_F = 0.835 μ_{CR} = 0.176
 Diversity = 1.2463e+00 k_t = [1, 1, 1, 1]
 Generation 6760: Best = 2.318636e+03 FES = 676100 μ_F = 0.813 μ_{CR} = 0.180
 Diversity = 1.2462e+00 k_t = [3, 2, 3, 3]
 Generation 6770: Best = 2.317693e+03 FES = 677100 μ_F = 0.820 μ_{CR} = 0.158
 Diversity = 1.2464e+00 k_t = [1, 1, 1, 1]
 Generation 6780: Best = 2.317538e+03 FES = 678100 μ_F = 0.802 μ_{CR} = 0.155
 Diversity = 1.2395e+00 k_t = [4, 3, 3, 2]
 Generation 6790: Best = 2.317097e+03 FES = 679100 μ_F = 0.797 μ_{CR} = 0.159
 Diversity = 1.2395e+00 k_t = [1, 1, 1, 1]
 Generation 6800: Best = 2.316506e+03 FES = 680100 μ_F = 0.786 μ_{CR} = 0.138
 Diversity = 1.2143e+00 k_t = [2, 3, 2, 3]
 Generation 6810: Best = 2.315766e+03 FES = 681100 μ_F = 0.761 μ_{CR} = 0.140
 Diversity = 1.2143e+00 k_t = [1, 1, 1, 1]
 Generation 6820: Best = 2.315455e+03 FES = 682100 μ_F = 0.766 μ_{CR} = 0.137
 Diversity = 1.2061e+00 k_t = [3, 3, 3, 2]
 Generation 6830: Best = 2.314978e+03 FES = 683100 μ_F = 0.782 μ_{CR} = 0.138
 Diversity = 1.1965e+00 k_t = [1, 1, 1, 1]
 Generation 6840: Best = 2.314571e+03 FES = 684100 μ_F = 0.771 μ_{CR} = 0.149
 Diversity = 1.1761e+00 k_t = [2, 4, 3, 3]
 Generation 6850: Best = 2.313925e+03 FES = 685100 μ_F = 0.765 μ_{CR} = 0.147
 Diversity = 1.1512e+00 k_t = [1, 1, 1, 1]
 Generation 6860: Best = 2.313588e+03 FES = 686100 μ_F = 0.752 μ_{CR} = 0.148
 Diversity = 1.0871e+00 k_t = [4, 3, 3, 4]
 Generation 6870: Best = 2.312620e+03 FES = 687100 μ_F = 0.765 μ_{CR} = 0.160
 Diversity = 1.0200e+00 k_t = [1, 1, 1, 1]
 Generation 6880: Best = 2.312298e+03 FES = 688100 μ_F = 0.787 μ_{CR} = 0.159
 Diversity = 9.8038e-01 k_t = [2, 2, 2, 3]
 Generation 6890: Best = 2.311862e+03 FES = 689100 μ_F = 0.787 μ_{CR} = 0.170
 Diversity = 9.5903e-01 k_t = [1, 1, 1, 1]
 Generation 6900: Best = 2.311319e+03 FES = 690100 μ_F = 0.784 μ_{CR} = 0.170
 Diversity = 9.0449e-01 k_t = [3, 3, 3, 4]
 Generation 6910: Best = 2.310917e+03 FES = 691100 μ_F = 0.788 μ_{CR} = 0.172
 Diversity = 8.7983e-01 k_t = [1, 1, 1, 1]
 Generation 6920: Best = 2.310543e+03 FES = 692100 μ_F = 0.784 μ_{CR} = 0.173
 Diversity = 7.9407e-01 k_t = [3, 4, 3, 3]
 Generation 6930: Best = 2.310126e+03 FES = 693100 μ_F = 0.771 μ_{CR} = 0.182
 Diversity = 7.6113e-01 k_t = [1, 1, 1, 1]
 Generation 6940: Best = 2.309766e+03 FES = 694100 μ_F = 0.777 μ_{CR} = 0.185
 Diversity = 6.4691e-01 k_t = [3, 4, 3, 3]
 Generation 6950: Best = 2.309383e+03 FES = 695100 μ_F = 0.783 μ_{CR} = 0.176
 Diversity = 6.0247e-01 k_t = [1, 1, 1, 1]
 Generation 6960: Best = 2.308848e+03 FES = 696100 μ_F = 0.760 μ_{CR} = 0.184
 Diversity = 6.0246e-01 k_t = [2, 2, 4, 3]
 Generation 6970: Best = 2.308505e+03 FES = 697100 μ_F = 0.767 μ_{CR} = 0.180
 Diversity = 5.5458e-01 k_t = [1, 1, 1, 1]

Generation 6980: Best = 2.307965e+03 FES = 698100 μ_F = 0.767 μ_{CR} = 0.183
 Diversity = 5.5464e-01 k_t = [3, 3, 3, 2]
 Generation 6990: Best = 2.307589e+03 FES = 699100 μ_F = 0.767 μ_{CR} = 0.176
 Diversity = 4.9894e-01 k_t = [1, 1, 1, 1]
 Generation 7000: Best = 2.307095e+03 FES = 700100 μ_F = 0.763 μ_{CR} = 0.177
 Diversity = 4.3606e-01 k_t = [3, 2, 3, 3]
 Generation 7010: Best = 2.306742e+03 FES = 701100 μ_F = 0.775 μ_{CR} = 0.172
 Diversity = 3.6144e-01 k_t = [1, 1, 1, 1]
 Generation 7020: Best = 2.306409e+03 FES = 702100 μ_F = 0.757 μ_{CR} = 0.168
 Diversity = 4.3520e-02 k_t = [4, 3, 3, 2]
 Generation 7030: Best = 2.302619e+03 FES = 703100 μ_F = 0.752 μ_{CR} = 0.167
 Diversity = 3.2068e-02 k_t = [1, 1, 1, 1]
 Generation 7040: Best = 2.300767e+03 FES = 704100 μ_F = 0.735 μ_{CR} = 0.152
 Diversity = 2.6992e-02 k_t = [4, 4, 4, 4]
 Generation 7050: Best = 2.300334e+03 FES = 705100 μ_F = 0.734 μ_{CR} = 0.150
 Diversity = 2.1874e-02 k_t = [1, 1, 1, 1]
 Generation 7060: Best = 2.299506e+03 FES = 706100 μ_F = 0.739 μ_{CR} = 0.164
 Diversity = 1.8136e-02 k_t = [2, 4, 4, 3]
 Generation 7070: Best = 2.298545e+03 FES = 707100 μ_F = 0.733 μ_{CR} = 0.171
 Diversity = 1.7147e-02 k_t = [1, 1, 1, 1]
 Generation 7080: Best = 2.297762e+03 FES = 708100 μ_F = 0.720 μ_{CR} = 0.172
 Diversity = 1.4542e-02 k_t = [2, 4, 3, 4]
 Generation 7090: Best = 2.297219e+03 FES = 709100 μ_F = 0.714 μ_{CR} = 0.175
 Diversity = 1.3994e-02 k_t = [1, 1, 1, 1]
 Generation 7100: Best = 2.296966e+03 FES = 710100 μ_F = 0.722 μ_{CR} = 0.166
 Diversity = 1.2962e-02 k_t = [4, 2, 3, 2]
 Generation 7110: Best = 2.296428e+03 FES = 711100 μ_F = 0.717 μ_{CR} = 0.168
 Diversity = 1.1972e-02 k_t = [1, 1, 1, 1]
 Generation 7120: Best = 2.296047e+03 FES = 712100 μ_F = 0.716 μ_{CR} = 0.162
 Diversity = 1.0746e-02 k_t = [3, 2, 2, 4]
 Generation 7130: Best = 2.295559e+03 FES = 713100 μ_F = 0.710 μ_{CR} = 0.162
 Diversity = 1.0886e-02 k_t = [1, 1, 1, 1]
 Generation 7140: Best = 2.295209e+03 FES = 714100 μ_F = 0.731 μ_{CR} = 0.155
 Diversity = 1.0794e-02 k_t = [3, 3, 2, 3]
 Generation 7150: Best = 2.294944e+03 FES = 715100 μ_F = 0.746 μ_{CR} = 0.158
 Diversity = 1.1314e-02 k_t = [1, 1, 1, 1]
 Generation 7160: Best = 2.294457e+03 FES = 716100 μ_F = 0.738 μ_{CR} = 0.164
 Diversity = 9.9604e-03 k_t = [3, 3, 4, 4]
 Generation 7170: Best = 2.294161e+03 FES = 717100 μ_F = 0.747 μ_{CR} = 0.161
 Diversity = 9.7008e-03 k_t = [1, 1, 1, 1]
 Generation 7180: Best = 2.293806e+03 FES = 718100 μ_F = 0.748 μ_{CR} = 0.166
 Diversity = 9.1478e-03 k_t = [2, 2, 3, 3]
 Generation 7190: Best = 2.293307e+03 FES = 719100 μ_F = 0.759 μ_{CR} = 0.167
 Diversity = 1.0205e-02 k_t = [1, 1, 1, 1]
 Generation 7200: Best = 2.293075e+03 FES = 720100 μ_F = 0.763 μ_{CR} = 0.167
 Diversity = 1.0187e-02 k_t = [3, 4, 2, 2]
 Generation 7210: Best = 2.292628e+03 FES = 721100 μ_F = 0.761 μ_{CR} = 0.164
 Diversity = 1.1056e-02 k_t = [1, 1, 1, 1]
 Generation 7220: Best = 2.292318e+03 FES = 722100 μ_F = 0.758 μ_{CR} = 0.163
 Diversity = 1.2106e-02 k_t = [4, 3, 3, 2]
 Generation 7230: Best = 2.291892e+03 FES = 723100 μ_F = 0.759 μ_{CR} = 0.154
 Diversity = 1.2466e-02 k_t = [1, 1, 1, 1]
 Generation 7240: Best = 2.291490e+03 FES = 724100 μ_F = 0.761 μ_{CR} = 0.150
 Diversity = 1.2012e-02 k_t = [2, 3, 2, 2]

Generation 7250: Best = 2.291167e+03 FES = 725100 μ_F = 0.765 μ_{CR} = 0.139
 Diversity = 1.3302e-02 k_t = [1, 1, 1, 1]
 Generation 7260: Best = 2.290921e+03 FES = 726100 μ_F = 0.762 μ_{CR} = 0.146
 Diversity = 1.2903e-02 k_t = [4, 4, 3, 4]
 Generation 7270: Best = 2.290695e+03 FES = 727100 μ_F = 0.761 μ_{CR} = 0.152
 Diversity = 1.2390e-02 k_t = [1, 1, 1, 1]
 Generation 7280: Best = 2.290368e+03 FES = 728100 μ_F = 0.770 μ_{CR} = 0.149
 Diversity = 1.1744e-02 k_t = [4, 2, 2, 4]
 Generation 7290: Best = 2.290202e+03 FES = 729100 μ_F = 0.774 μ_{CR} = 0.139
 Diversity = 1.1190e-02 k_t = [1, 1, 1, 1]
 Generation 7300: Best = 2.289887e+03 FES = 730100 μ_F = 0.764 μ_{CR} = 0.140
 Diversity = 1.1179e-02 k_t = [2, 3, 4, 3]
 Generation 7310: Best = 2.289683e+03 FES = 731100 μ_F = 0.751 μ_{CR} = 0.135
 Diversity = 1.2397e-02 k_t = [1, 1, 1, 1]
 Generation 7320: Best = 2.289516e+03 FES = 732100 μ_F = 0.751 μ_{CR} = 0.140
 Diversity = 1.1249e-02 k_t = [4, 4, 2, 3]
 Generation 7330: Best = 2.289201e+03 FES = 733100 μ_F = 0.756 μ_{CR} = 0.140
 Diversity = 1.1773e-02 k_t = [1, 1, 1, 1]
 Generation 7340: Best = 2.289028e+03 FES = 734100 μ_F = 0.772 μ_{CR} = 0.132
 Diversity = 1.2005e-02 k_t = [2, 3, 3, 2]
 Generation 7350: Best = 2.288781e+03 FES = 735100 μ_F = 0.771 μ_{CR} = 0.137
 Diversity = 1.4035e-02 k_t = [1, 1, 1, 1]
 Generation 7360: Best = 2.288563e+03 FES = 736100 μ_F = 0.777 μ_{CR} = 0.138
 Diversity = 1.3554e-02 k_t = [4, 2, 4, 3]
 Generation 7370: Best = 2.288354e+03 FES = 737100 μ_F = 0.770 μ_{CR} = 0.141
 Diversity = 1.3610e-02 k_t = [1, 1, 1, 1]
 Generation 7380: Best = 2.288138e+03 FES = 738100 μ_F = 0.766 μ_{CR} = 0.142
 Diversity = 1.2125e-02 k_t = [4, 2, 3, 4]
 Generation 7390: Best = 2.287964e+03 FES = 739100 μ_F = 0.771 μ_{CR} = 0.142
 Diversity = 1.2033e-02 k_t = [1, 1, 1, 1]
 Generation 7400: Best = 2.287647e+03 FES = 740100 μ_F = 0.758 μ_{CR} = 0.137
 Diversity = 1.2685e-02 k_t = [4, 4, 2, 2]
 Generation 7410: Best = 2.287484e+03 FES = 741100 μ_F = 0.762 μ_{CR} = 0.142
 Diversity = 1.2908e-02 k_t = [1, 1, 1, 1]
 Generation 7420: Best = 2.287233e+03 FES = 742100 μ_F = 0.774 μ_{CR} = 0.141
 Diversity = 1.2429e-02 k_t = [3, 3, 3, 4]
 Generation 7430: Best = 2.287156e+03 FES = 743100 μ_F = 0.790 μ_{CR} = 0.129
 Diversity = 1.1671e-02 k_t = [1, 1, 1, 1]
 Generation 7440: Best = 2.286829e+03 FES = 744100 μ_F = 0.786 μ_{CR} = 0.123
 Diversity = 1.1547e-02 k_t = [3, 2, 4, 2]
 Generation 7450: Best = 2.286690e+03 FES = 745100 μ_F = 0.777 μ_{CR} = 0.109
 Diversity = 1.2170e-02 k_t = [1, 1, 1, 1]
 Generation 7460: Best = 2.286529e+03 FES = 746100 μ_F = 0.772 μ_{CR} = 0.116
 Diversity = 1.3275e-02 k_t = [4, 4, 4, 3]
 Generation 7470: Best = 2.286372e+03 FES = 747100 μ_F = 0.774 μ_{CR} = 0.116
 Diversity = 1.4604e-02 k_t = [1, 1, 1, 1]
 Generation 7480: Best = 2.286165e+03 FES = 748100 μ_F = 0.777 μ_{CR} = 0.117
 Diversity = 1.4034e-02 k_t = [4, 4, 3, 2]
 Generation 7490: Best = 2.285922e+03 FES = 749100 μ_F = 0.782 μ_{CR} = 0.112
 Diversity = 1.4122e-02 k_t = [1, 1, 1, 1]
 Generation 7500: Best = 2.285827e+03 FES = 750100 μ_F = 0.774 μ_{CR} = 0.110
 Diversity = 1.3542e-02 k_t = [4, 4, 3, 3]
 Generation 7510: Best = 2.285725e+03 FES = 751100 μ_F = 0.762 μ_{CR} = 0.112
 Diversity = 1.3018e-02 k_t = [1, 1, 1, 1]

Generation 7520: Best = 2.285565e+03 FES = 752100 μ_F = 0.761 μ_{CR} = 0.113
 Diversity = 1.3853e-02 k_t = [3, 2, 2, 2]
 Generation 7530: Best = 2.285310e+03 FES = 753100 μ_F = 0.773 μ_{CR} = 0.114
 Diversity = 1.3620e-02 k_t = [1, 1, 1, 1]
 Generation 7540: Best = 2.285146e+03 FES = 754100 μ_F = 0.775 μ_{CR} = 0.106
 Diversity = 1.3186e-02 k_t = [3, 2, 2, 2]
 Generation 7550: Best = 2.285106e+03 FES = 755100 μ_F = 0.786 μ_{CR} = 0.107
 Diversity = 1.3299e-02 k_t = [1, 1, 1, 1]
 Generation 7560: Best = 2.284969e+03 FES = 756100 μ_F = 0.800 μ_{CR} = 0.102
 Diversity = 1.2298e-02 k_t = [2, 4, 4, 4]
 Generation 7570: Best = 2.284592e+03 FES = 757100 μ_F = 0.793 μ_{CR} = 0.107
 Diversity = 1.0948e-02 k_t = [1, 1, 1, 1]
 Generation 7580: Best = 2.284376e+03 FES = 758100 μ_F = 0.806 μ_{CR} = 0.121
 Diversity = 1.0940e-02 k_t = [2, 2, 4, 3]
 Generation 7590: Best = 2.284264e+03 FES = 759100 μ_F = 0.814 μ_{CR} = 0.120
 Diversity = 1.3145e-02 k_t = [1, 1, 1, 1]
 Generation 7600: Best = 2.284162e+03 FES = 760100 μ_F = 0.804 μ_{CR} = 0.123
 Diversity = 1.2664e-02 k_t = [4, 3, 3, 4]
 Generation 7610: Best = 2.283977e+03 FES = 761100 μ_F = 0.801 μ_{CR} = 0.116
 Diversity = 1.1895e-02 k_t = [1, 1, 1, 1]
 Generation 7620: Best = 2.283859e+03 FES = 762100 μ_F = 0.776 μ_{CR} = 0.124
 Diversity = 1.0736e-02 k_t = [3, 2, 4, 3]
 Generation 7630: Best = 2.283617e+03 FES = 763100 μ_F = 0.779 μ_{CR} = 0.132
 Diversity = 1.2807e-02 k_t = [1, 1, 1, 1]
 Generation 7640: Best = 2.283335e+03 FES = 764100 μ_F = 0.781 μ_{CR} = 0.128
 Diversity = 1.3365e-02 k_t = [4, 4, 2, 3]
 Generation 7650: Best = 2.283227e+03 FES = 765100 μ_F = 0.770 μ_{CR} = 0.123
 Diversity = 1.2492e-02 k_t = [1, 1, 1, 1]
 Generation 7660: Best = 2.283115e+03 FES = 766100 μ_F = 0.764 μ_{CR} = 0.128
 Diversity = 1.1706e-02 k_t = [4, 4, 2, 2]
 Generation 7670: Best = 2.282886e+03 FES = 767100 μ_F = 0.761 μ_{CR} = 0.123
 Diversity = 1.1951e-02 k_t = [1, 1, 1, 1]
 Generation 7680: Best = 2.282713e+03 FES = 768100 μ_F = 0.761 μ_{CR} = 0.123
 Diversity = 1.1006e-02 k_t = [3, 2, 3, 3]
 Generation 7690: Best = 2.282451e+03 FES = 769100 μ_F = 0.766 μ_{CR} = 0.121
 Diversity = 1.2018e-02 k_t = [1, 1, 1, 1]
 Generation 7700: Best = 2.282359e+03 FES = 770100 μ_F = 0.759 μ_{CR} = 0.122
 Diversity = 1.0972e-02 k_t = [3, 4, 2, 3]
 Generation 7710: Best = 2.282258e+03 FES = 771100 μ_F = 0.756 μ_{CR} = 0.127
 Diversity = 1.1277e-02 k_t = [1, 1, 1, 1]
 Generation 7720: Best = 2.281903e+03 FES = 772100 μ_F = 0.772 μ_{CR} = 0.126
 Diversity = 1.1178e-02 k_t = [2, 2, 4, 2]
 Generation 7730: Best = 2.281808e+03 FES = 773100 μ_F = 0.781 μ_{CR} = 0.133
 Diversity = 1.2654e-02 k_t = [1, 1, 1, 1]
 Generation 7740: Best = 2.281467e+03 FES = 774100 μ_F = 0.774 μ_{CR} = 0.130
 Diversity = 1.1058e-02 k_t = [3, 4, 2, 3]
 Generation 7750: Best = 2.281267e+03 FES = 775100 μ_F = 0.776 μ_{CR} = 0.132
 Diversity = 1.2076e-02 k_t = [1, 1, 1, 1]
 Generation 7760: Best = 2.281150e+03 FES = 776100 μ_F = 0.762 μ_{CR} = 0.130
 Diversity = 1.1479e-02 k_t = [2, 3, 4, 3]
 Generation 7770: Best = 2.281004e+03 FES = 777100 μ_F = 0.758 μ_{CR} = 0.118
 Diversity = 1.1888e-02 k_t = [1, 1, 1, 1]
 Generation 7780: Best = 2.280746e+03 FES = 778100 μ_F = 0.763 μ_{CR} = 0.121
 Diversity = 1.1113e-02 k_t = [4, 4, 2, 2]

Generation 7790: Best = 2.280630e+03 FES = 779100 μ_F = 0.760 μ_{CR} = 0.116
 Diversity = 1.2230e-02 k_t = [1, 1, 1, 1]
 Generation 7800: Best = 2.280388e+03 FES = 780100 μ_F = 0.771 μ_{CR} = 0.115
 Diversity = 1.2219e-02 k_t = [4, 4, 3, 2]
 Generation 7810: Best = 2.280183e+03 FES = 781100 μ_F = 0.757 μ_{CR} = 0.113
 Diversity = 1.1654e-02 k_t = [1, 1, 1, 1]
 Generation 7820: Best = 2.280153e+03 FES = 782100 μ_F = 0.754 μ_{CR} = 0.114
 Diversity = 1.2158e-02 k_t = [2, 4, 3, 3]
 Generation 7830: Best = 2.279945e+03 FES = 783100 μ_F = 0.755 μ_{CR} = 0.110
 Diversity = 1.1317e-02 k_t = [1, 1, 1, 1]
 Generation 7840: Best = 2.279781e+03 FES = 784100 μ_F = 0.759 μ_{CR} = 0.117
 Diversity = 1.1341e-02 k_t = [2, 3, 3, 3]
 Generation 7850: Best = 2.279586e+03 FES = 785100 μ_F = 0.776 μ_{CR} = 0.115
 Diversity = 1.1196e-02 k_t = [1, 1, 1, 1]
 Generation 7860: Best = 2.279441e+03 FES = 786100 μ_F = 0.766 μ_{CR} = 0.125
 Diversity = 1.0773e-02 k_t = [4, 4, 3, 3]
 Generation 7870: Best = 2.279325e+03 FES = 787100 μ_F = 0.777 μ_{CR} = 0.126
 Diversity = 1.0285e-02 k_t = [1, 1, 1, 1]
 Generation 7880: Best = 2.279113e+03 FES = 788100 μ_F = 0.757 μ_{CR} = 0.135
 Diversity = 1.0703e-02 k_t = [2, 4, 2, 2]
 Generation 7890: Best = 2.278994e+03 FES = 789100 μ_F = 0.763 μ_{CR} = 0.131
 Diversity = 1.2442e-02 k_t = [1, 1, 1, 1]
 Generation 7900: Best = 2.278812e+03 FES = 790100 μ_F = 0.769 μ_{CR} = 0.132
 Diversity = 1.1776e-02 k_t = [4, 3, 3, 3]
 Generation 7910: Best = 2.278658e+03 FES = 791100 μ_F = 0.763 μ_{CR} = 0.130
 Diversity = 1.1047e-02 k_t = [1, 1, 1, 1]
 Generation 7920: Best = 2.278502e+03 FES = 792100 μ_F = 0.777 μ_{CR} = 0.131
 Diversity = 9.4552e-03 k_t = [2, 3, 3, 4]
 Generation 7930: Best = 2.278392e+03 FES = 793100 μ_F = 0.776 μ_{CR} = 0.130
 Diversity = 7.7922e-03 k_t = [1, 1, 1, 1]
 Generation 7940: Best = 2.278338e+03 FES = 794100 μ_F = 0.774 μ_{CR} = 0.129
 Diversity = 7.4302e-03 k_t = [4, 2, 4, 4]
 Generation 7950: Best = 2.278209e+03 FES = 795100 μ_F = 0.780 μ_{CR} = 0.133
 Diversity = 8.2847e-03 k_t = [1, 1, 1, 1]
 Generation 7960: Best = 2.278091e+03 FES = 796100 μ_F = 0.779 μ_{CR} = 0.127
 Diversity = 7.5884e-03 k_t = [2, 2, 4, 4]
 Generation 7970: Best = 2.277982e+03 FES = 797100 μ_F = 0.786 μ_{CR} = 0.120
 Diversity = 7.9164e-03 k_t = [1, 1, 1, 1]
 Generation 7980: Best = 2.277818e+03 FES = 798100 μ_F = 0.786 μ_{CR} = 0.127
 Diversity = 7.8971e-03 k_t = [4, 2, 2, 3]
 Generation 7990: Best = 2.277712e+03 FES = 799100 μ_F = 0.795 μ_{CR} = 0.133
 Diversity = 7.6799e-03 k_t = [1, 1, 1, 1]
 Generation 8000: Best = 2.277545e+03 FES = 800100 μ_F = 0.802 μ_{CR} = 0.132
 Diversity = 9.0786e-03 k_t = [4, 3, 2, 3]
 Generation 8010: Best = 2.277222e+03 FES = 801100 μ_F = 0.800 μ_{CR} = 0.137
 Diversity = 9.6043e-03 k_t = [1, 1, 1, 1]
 Generation 8020: Best = 2.277169e+03 FES = 802100 μ_F = 0.803 μ_{CR} = 0.133
 Diversity = 9.5642e-03 k_t = [2, 2, 4, 3]
 Generation 8030: Best = 2.276967e+03 FES = 803100 μ_F = 0.799 μ_{CR} = 0.135
 Diversity = 1.2348e-02 k_t = [1, 1, 1, 1]
 Generation 8040: Best = 2.276877e+03 FES = 804100 μ_F = 0.792 μ_{CR} = 0.121
 Diversity = 1.1318e-02 k_t = [4, 3, 2, 4]
 Generation 8050: Best = 2.276696e+03 FES = 805100 μ_F = 0.802 μ_{CR} = 0.121
 Diversity = 1.1998e-02 k_t = [1, 1, 1, 1]

Generation 8060: Best = 2.276482e+03 FES = 806100 μ_F = 0.785 μ_{CR} = 0.112
 Diversity = 1.1107e-02 k_t = [2, 4, 3, 2]
 Generation 8070: Best = 2.276392e+03 FES = 807100 μ_F = 0.771 μ_{CR} = 0.108
 Diversity = 1.1252e-02 k_t = [1, 1, 1, 1]
 Generation 8080: Best = 2.276214e+03 FES = 808100 μ_F = 0.772 μ_{CR} = 0.105
 Diversity = 1.2260e-02 k_t = [4, 3, 2, 2]
 Generation 8090: Best = 2.276126e+03 FES = 809100 μ_F = 0.772 μ_{CR} = 0.113
 Diversity = 1.2796e-02 k_t = [1, 1, 1, 1]
 Generation 8100: Best = 2.275979e+03 FES = 810100 μ_F = 0.779 μ_{CR} = 0.108
 Diversity = 1.1098e-02 k_t = [3, 2, 3, 3]
 Generation 8110: Best = 2.275894e+03 FES = 811100 μ_F = 0.771 μ_{CR} = 0.108
 Diversity = 9.6665e-03 k_t = [1, 1, 1, 1]
 Generation 8120: Best = 2.275689e+03 FES = 812100 μ_F = 0.777 μ_{CR} = 0.115
 Diversity = 9.7312e-03 k_t = [2, 2, 3, 3]
 Generation 8130: Best = 2.275561e+03 FES = 813100 μ_F = 0.775 μ_{CR} = 0.109
 Diversity = 9.9426e-03 k_t = [1, 1, 1, 1]
 Generation 8140: Best = 2.275445e+03 FES = 814100 μ_F = 0.763 μ_{CR} = 0.112
 Diversity = 9.7412e-03 k_t = [4, 3, 3, 4]
 Generation 8150: Best = 2.275302e+03 FES = 815100 μ_F = 0.767 μ_{CR} = 0.121
 Diversity = 8.8370e-03 k_t = [1, 1, 1, 1]
 Generation 8160: Best = 2.275151e+03 FES = 816100 μ_F = 0.772 μ_{CR} = 0.126
 Diversity = 8.3897e-03 k_t = [4, 4, 3, 2]
 Generation 8170: Best = 2.275045e+03 FES = 817100 μ_F = 0.773 μ_{CR} = 0.132
 Diversity = 8.7374e-03 k_t = [1, 1, 1, 1]
 Generation 8180: Best = 2.274924e+03 FES = 818100 μ_F = 0.755 μ_{CR} = 0.139
 Diversity = 7.2941e-03 k_t = [4, 4, 4, 4]
 Generation 8190: Best = 2.274756e+03 FES = 819100 μ_F = 0.759 μ_{CR} = 0.152
 Diversity = 7.1640e-03 k_t = [1, 1, 1, 1]
 Generation 8200: Best = 2.274609e+03 FES = 820100 μ_F = 0.763 μ_{CR} = 0.143
 Diversity = 6.3150e-03 k_t = [2, 4, 4, 4]
 Generation 8210: Best = 2.274557e+03 FES = 821100 μ_F = 0.752 μ_{CR} = 0.143
 Diversity = 6.1610e-03 k_t = [1, 1, 1, 1]
 Generation 8220: Best = 2.274439e+03 FES = 822100 μ_F = 0.766 μ_{CR} = 0.149
 Diversity = 6.2541e-03 k_t = [3, 4, 3, 2]
 Generation 8230: Best = 2.274321e+03 FES = 823100 μ_F = 0.767 μ_{CR} = 0.153
 Diversity = 6.3279e-03 k_t = [1, 1, 1, 1]
 Generation 8240: Best = 2.274256e+03 FES = 824100 μ_F = 0.752 μ_{CR} = 0.149
 Diversity = 6.3089e-03 k_t = [2, 4, 4, 2]
 Generation 8250: Best = 2.274108e+03 FES = 825100 μ_F = 0.751 μ_{CR} = 0.149
 Diversity = 6.1850e-03 k_t = [1, 1, 1, 1]
 Generation 8260: Best = 2.274005e+03 FES = 826100 μ_F = 0.747 μ_{CR} = 0.159
 Diversity = 5.7091e-03 k_t = [2, 4, 4, 4]
 Generation 8270: Best = 2.273918e+03 FES = 827100 μ_F = 0.751 μ_{CR} = 0.154
 Diversity = 5.9379e-03 k_t = [1, 1, 1, 1]
 Generation 8280: Best = 2.273799e+03 FES = 828100 μ_F = 0.757 μ_{CR} = 0.154
 Diversity = 6.4614e-03 k_t = [4, 3, 2, 2]
 Generation 8290: Best = 2.273662e+03 FES = 829100 μ_F = 0.765 μ_{CR} = 0.157
 Diversity = 6.7621e-03 k_t = [1, 1, 1, 1]
 Generation 8300: Best = 2.273515e+03 FES = 830100 μ_F = 0.765 μ_{CR} = 0.159
 Diversity = 6.8317e-03 k_t = [2, 4, 2, 2]
 Generation 8310: Best = 2.273348e+03 FES = 831100 μ_F = 0.771 μ_{CR} = 0.163
 Diversity = 7.4260e-03 k_t = [1, 1, 1, 1]
 Generation 8320: Best = 2.273247e+03 FES = 832100 μ_F = 0.764 μ_{CR} = 0.156
 Diversity = 7.1320e-03 k_t = [2, 2, 4, 4]

Generation 8330: Best = 2.273077e+03 FES = 833100 μ_F = 0.771 μ_{CR} = 0.146
 Diversity = 7.1545e-03 k_t = [1, 1, 1, 1]
 Generation 8340: Best = 2.272967e+03 FES = 834100 μ_F = 0.791 μ_{CR} = 0.150
 Diversity = 7.8368e-03 k_t = [3, 3, 4, 2]
 Generation 8350: Best = 2.272824e+03 FES = 835100 μ_F = 0.776 μ_{CR} = 0.148
 Diversity = 8.1066e-03 k_t = [1, 1, 1, 1]
 Generation 8360: Best = 2.272660e+03 FES = 836100 μ_F = 0.785 μ_{CR} = 0.158
 Diversity = 7.5445e-03 k_t = [2, 4, 4, 4]
 Generation 8370: Best = 2.272449e+03 FES = 837100 μ_F = 0.788 μ_{CR} = 0.161
 Diversity = 8.2178e-03 k_t = [1, 1, 1, 1]
 Generation 8380: Best = 2.272308e+03 FES = 838100 μ_F = 0.781 μ_{CR} = 0.154
 Diversity = 8.4341e-03 k_t = [4, 4, 3, 4]
 Generation 8390: Best = 2.272130e+03 FES = 839100 μ_F = 0.779 μ_{CR} = 0.152
 Diversity = 7.8542e-03 k_t = [1, 1, 1, 1]
 Generation 8400: Best = 2.271954e+03 FES = 840100 μ_F = 0.777 μ_{CR} = 0.155
 Diversity = 7.5771e-03 k_t = [2, 2, 2, 4]
 Generation 8410: Best = 2.271843e+03 FES = 841100 μ_F = 0.763 μ_{CR} = 0.160
 Diversity = 7.5494e-03 k_t = [1, 1, 1, 1]
 Generation 8420: Best = 2.271776e+03 FES = 842100 μ_F = 0.761 μ_{CR} = 0.151
 Diversity = 7.5256e-03 k_t = [3, 2, 3, 2]
 Generation 8430: Best = 2.271637e+03 FES = 843100 μ_F = 0.757 μ_{CR} = 0.146
 Diversity = 7.6533e-03 k_t = [1, 1, 1, 1]
 Generation 8440: Best = 2.271531e+03 FES = 844100 μ_F = 0.761 μ_{CR} = 0.145
 Diversity = 7.2778e-03 k_t = [3, 3, 4, 4]
 Generation 8450: Best = 2.271425e+03 FES = 845100 μ_F = 0.770 μ_{CR} = 0.141
 Diversity = 7.7442e-03 k_t = [1, 1, 1, 1]
 Generation 8460: Best = 2.271335e+03 FES = 846100 μ_F = 0.775 μ_{CR} = 0.143
 Diversity = 7.0713e-03 k_t = [4, 2, 2, 2]
 Generation 8470: Best = 2.271250e+03 FES = 847100 μ_F = 0.775 μ_{CR} = 0.138
 Diversity = 7.2178e-03 k_t = [1, 1, 1, 1]
 Generation 8480: Best = 2.271078e+03 FES = 848100 μ_F = 0.779 μ_{CR} = 0.139
 Diversity = 7.0184e-03 k_t = [3, 4, 4, 4]
 Generation 8490: Best = 2.271007e+03 FES = 849100 μ_F = 0.777 μ_{CR} = 0.141
 Diversity = 6.9381e-03 k_t = [1, 1, 1, 1]
 Generation 8500: Best = 2.270771e+03 FES = 850100 μ_F = 0.773 μ_{CR} = 0.143
 Diversity = 7.1045e-03 k_t = [2, 4, 2, 2]
 Generation 8510: Best = 2.270655e+03 FES = 851100 μ_F = 0.772 μ_{CR} = 0.143
 Diversity = 7.8260e-03 k_t = [1, 1, 1, 1]
 Generation 8520: Best = 2.270492e+03 FES = 852100 μ_F = 0.769 μ_{CR} = 0.135
 Diversity = 7.0690e-03 k_t = [4, 4, 2, 3]
 Generation 8530: Best = 2.270359e+03 FES = 853100 μ_F = 0.759 μ_{CR} = 0.135
 Diversity = 6.7577e-03 k_t = [1, 1, 1, 1]
 Generation 8540: Best = 2.270162e+03 FES = 854100 μ_F = 0.752 μ_{CR} = 0.136
 Diversity = 7.6021e-03 k_t = [3, 2, 4, 2]
 Generation 8550: Best = 2.270070e+03 FES = 855100 μ_F = 0.766 μ_{CR} = 0.135
 Diversity = 7.4516e-03 k_t = [1, 1, 1, 1]
 Generation 8560: Best = 2.269911e+03 FES = 856100 μ_F = 0.769 μ_{CR} = 0.130
 Diversity = 6.9716e-03 k_t = [3, 3, 4, 2]
 Generation 8570: Best = 2.269744e+03 FES = 857100 μ_F = 0.775 μ_{CR} = 0.127
 Diversity = 7.3893e-03 k_t = [1, 1, 1, 1]
 Generation 8580: Best = 2.269589e+03 FES = 858100 μ_F = 0.776 μ_{CR} = 0.141
 Diversity = 8.8397e-03 k_t = [3, 4, 2, 2]
 Generation 8590: Best = 2.269489e+03 FES = 859100 μ_F = 0.761 μ_{CR} = 0.135
 Diversity = 8.8375e-03 k_t = [1, 1, 1, 1]

Generation 8600: Best = 2.269223e+03 FES = 860100 μ_F = 0.765 μ_{CR} = 0.149
 Diversity = 8.7485e-03 k_t = [4, 3, 4, 4]
 Generation 8610: Best = 2.268943e+03 FES = 861100 μ_F = 0.760 μ_{CR} = 0.154
 Diversity = 8.2649e-03 k_t = [1, 1, 1, 1]
 Generation 8620: Best = 2.268825e+03 FES = 862100 μ_F = 0.760 μ_{CR} = 0.151
 Diversity = 7.8178e-03 k_t = [3, 2, 3, 4]
 Generation 8630: Best = 2.268676e+03 FES = 863100 μ_F = 0.750 μ_{CR} = 0.155
 Diversity = 7.4649e-03 k_t = [1, 1, 1, 1]
 Generation 8640: Best = 2.268582e+03 FES = 864100 μ_F = 0.755 μ_{CR} = 0.157
 Diversity = 7.1602e-03 k_t = [3, 2, 3, 3]
 Generation 8650: Best = 2.268340e+03 FES = 865100 μ_F = 0.754 μ_{CR} = 0.155
 Diversity = 6.9423e-03 k_t = [1, 1, 1, 1]
 Generation 8660: Best = 2.268292e+03 FES = 866100 μ_F = 0.758 μ_{CR} = 0.145
 Diversity = 7.5152e-03 k_t = [4, 3, 3, 2]
 Generation 8670: Best = 2.268150e+03 FES = 867100 μ_F = 0.759 μ_{CR} = 0.142
 Diversity = 7.0593e-03 k_t = [1, 1, 1, 1]
 Generation 8680: Best = 2.268030e+03 FES = 868100 μ_F = 0.767 μ_{CR} = 0.146
 Diversity = 6.7792e-03 k_t = [3, 4, 2, 2]
 Generation 8690: Best = 2.267832e+03 FES = 869100 μ_F = 0.767 μ_{CR} = 0.153
 Diversity = 7.0341e-03 k_t = [1, 1, 1, 1]
 Generation 8700: Best = 2.267655e+03 FES = 870100 μ_F = 0.757 μ_{CR} = 0.165
 Diversity = 7.0381e-03 k_t = [4, 3, 4, 2]
 Generation 8710: Best = 2.267534e+03 FES = 871100 μ_F = 0.755 μ_{CR} = 0.171
 Diversity = 6.5651e-03 k_t = [1, 1, 1, 1]
 Generation 8720: Best = 2.267408e+03 FES = 872100 μ_F = 0.748 μ_{CR} = 0.167
 Diversity = 6.3035e-03 k_t = [3, 3, 4, 3]
 Generation 8730: Best = 2.267290e+03 FES = 873100 μ_F = 0.760 μ_{CR} = 0.164
 Diversity = 6.3266e-03 k_t = [1, 1, 1, 1]
 Generation 8740: Best = 2.267080e+03 FES = 874100 μ_F = 0.765 μ_{CR} = 0.165
 Diversity = 7.0076e-03 k_t = [3, 2, 3, 2]
 Generation 8750: Best = 2.266960e+03 FES = 875100 μ_F = 0.753 μ_{CR} = 0.159
 Diversity = 7.0418e-03 k_t = [1, 1, 1, 1]
 Generation 8760: Best = 2.266791e+03 FES = 876100 μ_F = 0.761 μ_{CR} = 0.163
 Diversity = 6.9358e-03 k_t = [3, 3, 2, 3]
 Generation 8770: Best = 2.266661e+03 FES = 877100 μ_F = 0.761 μ_{CR} = 0.163
 Diversity = 6.8024e-03 k_t = [1, 1, 1, 1]
 Generation 8780: Best = 2.266498e+03 FES = 878100 μ_F = 0.766 μ_{CR} = 0.157
 Diversity = 7.0903e-03 k_t = [2, 3, 3, 3]
 Generation 8790: Best = 2.266299e+03 FES = 879100 μ_F = 0.755 μ_{CR} = 0.147
 Diversity = 7.1619e-03 k_t = [1, 1, 1, 1]
 Generation 8800: Best = 2.266144e+03 FES = 880100 μ_F = 0.749 μ_{CR} = 0.152
 Diversity = 6.6972e-03 k_t = [4, 2, 3, 4]
 Generation 8810: Best = 2.266006e+03 FES = 881100 μ_F = 0.758 μ_{CR} = 0.147
 Diversity = 6.4407e-03 k_t = [1, 1, 1, 1]
 Generation 8820: Best = 2.265847e+03 FES = 882100 μ_F = 0.763 μ_{CR} = 0.146
 Diversity = 5.9021e-03 k_t = [4, 4, 2, 4]
 Generation 8830: Best = 2.265752e+03 FES = 883100 μ_F = 0.772 μ_{CR} = 0.142
 Diversity = 5.5660e-03 k_t = [1, 1, 1, 1]
 Generation 8840: Best = 2.265605e+03 FES = 884100 μ_F = 0.769 μ_{CR} = 0.152
 Diversity = 5.9255e-03 k_t = [3, 3, 4, 3]
 Generation 8850: Best = 2.265524e+03 FES = 885100 μ_F = 0.765 μ_{CR} = 0.146
 Diversity = 6.7404e-03 k_t = [1, 1, 1, 1]
 Generation 8860: Best = 2.265395e+03 FES = 886100 μ_F = 0.760 μ_{CR} = 0.152
 Diversity = 6.3421e-03 k_t = [4, 2, 3, 2]

Generation 8870: Best = 2.265278e+03 FES = 887100 μ_F = 0.761 μ_{CR} = 0.166
 Diversity = 6.7232e-03 k_t = [1, 1, 1, 1]
 Generation 8880: Best = 2.265169e+03 FES = 888100 μ_F = 0.747 μ_{CR} = 0.156
 Diversity = 6.1580e-03 k_t = [4, 4, 2, 4]
 Generation 8890: Best = 2.265063e+03 FES = 889100 μ_F = 0.741 μ_{CR} = 0.152
 Diversity = 6.4127e-03 k_t = [1, 1, 1, 1]
 Generation 8900: Best = 2.265016e+03 FES = 890100 μ_F = 0.753 μ_{CR} = 0.150
 Diversity = 6.2998e-03 k_t = [2, 2, 4, 3]
 Generation 8910: Best = 2.264911e+03 FES = 891100 μ_F = 0.757 μ_{CR} = 0.138
 Diversity = 6.0902e-03 k_t = [1, 1, 1, 1]
 Generation 8920: Best = 2.264852e+03 FES = 892100 μ_F = 0.764 μ_{CR} = 0.131
 Diversity = 5.9804e-03 k_t = [2, 3, 4, 3]
 Generation 8930: Best = 2.264758e+03 FES = 893100 μ_F = 0.755 μ_{CR} = 0.123
 Diversity = 6.2433e-03 k_t = [1, 1, 1, 1]
 Generation 8940: Best = 2.264686e+03 FES = 894100 μ_F = 0.748 μ_{CR} = 0.129
 Diversity = 5.7198e-03 k_t = [4, 3, 4, 3]
 Generation 8950: Best = 2.264665e+03 FES = 895100 μ_F = 0.745 μ_{CR} = 0.131
 Diversity = 5.0681e-03 k_t = [1, 1, 1, 1]
 Generation 8960: Best = 2.264618e+03 FES = 896100 μ_F = 0.764 μ_{CR} = 0.124
 Diversity = 5.2271e-03 k_t = [2, 3, 4, 2]
 Generation 8970: Best = 2.264482e+03 FES = 897100 μ_F = 0.771 μ_{CR} = 0.124
 Diversity = 5.4513e-03 k_t = [1, 1, 1, 1]
 Generation 8980: Best = 2.264450e+03 FES = 898100 μ_F = 0.775 μ_{CR} = 0.137
 Diversity = 5.2291e-03 k_t = [2, 4, 3, 3]
 Generation 8990: Best = 2.264360e+03 FES = 899100 μ_F = 0.786 μ_{CR} = 0.142
 Diversity = 5.5526e-03 k_t = [1, 1, 1, 1]
 Generation 9000: Best = 2.264260e+03 FES = 900100 μ_F = 0.785 μ_{CR} = 0.149
 Diversity = 5.8770e-03 k_t = [4, 2, 4, 3]
 Generation 9010: Best = 2.264220e+03 FES = 901100 μ_F = 0.778 μ_{CR} = 0.143
 Diversity = 6.0191e-03 k_t = [1, 1, 1, 1]
 Generation 9020: Best = 2.264139e+03 FES = 902100 μ_F = 0.774 μ_{CR} = 0.145
 Diversity = 5.6020e-03 k_t = [4, 2, 2, 3]
 Generation 9030: Best = 2.264053e+03 FES = 903100 μ_F = 0.761 μ_{CR} = 0.143
 Diversity = 5.8618e-03 k_t = [1, 1, 1, 1]
 Generation 9040: Best = 2.263992e+03 FES = 904100 μ_F = 0.774 μ_{CR} = 0.136
 Diversity = 5.5377e-03 k_t = [4, 4, 4, 4]
 Generation 9050: Best = 2.263900e+03 FES = 905100 μ_F = 0.763 μ_{CR} = 0.138
 Diversity = 5.3308e-03 k_t = [1, 1, 1, 1]
 Generation 9060: Best = 2.263802e+03 FES = 906100 μ_F = 0.769 μ_{CR} = 0.139
 Diversity = 5.6249e-03 k_t = [2, 2, 2, 3]
 Generation 9070: Best = 2.263783e+03 FES = 907100 μ_F = 0.772 μ_{CR} = 0.126
 Diversity = 6.0480e-03 k_t = [1, 1, 1, 1]
 Generation 9080: Best = 2.263696e+03 FES = 908100 μ_F = 0.766 μ_{CR} = 0.128
 Diversity = 5.8638e-03 k_t = [2, 4, 4, 3]
 Generation 9090: Best = 2.263652e+03 FES = 909100 μ_F = 0.766 μ_{CR} = 0.130
 Diversity = 5.4392e-03 k_t = [1, 1, 1, 1]
 Generation 9100: Best = 2.263568e+03 FES = 910100 μ_F = 0.777 μ_{CR} = 0.129
 Diversity = 5.4101e-03 k_t = [2, 2, 4, 2]
 Generation 9110: Best = 2.263523e+03 FES = 911100 μ_F = 0.787 μ_{CR} = 0.129
 Diversity = 5.7005e-03 k_t = [1, 1, 1, 1]
 Generation 9120: Best = 2.263442e+03 FES = 912100 μ_F = 0.782 μ_{CR} = 0.146
 Diversity = 5.1237e-03 k_t = [3, 4, 4, 4]
 Generation 9130: Best = 2.263397e+03 FES = 913100 μ_F = 0.765 μ_{CR} = 0.142
 Diversity = 4.8799e-03 k_t = [1, 1, 1, 1]

Generation 9140: Best = 2.263340e+03 FES = 914100 μ_F = 0.767 μ_{CR} = 0.144
 Diversity = 4.8755e-03 k_t = [3, 2, 3, 2]
 Generation 9150: Best = 2.263292e+03 FES = 915100 μ_F = 0.763 μ_{CR} = 0.152
 Diversity = 4.9192e-03 k_t = [1, 1, 1, 1]
 Generation 9160: Best = 2.263237e+03 FES = 916100 μ_F = 0.758 μ_{CR} = 0.154
 Diversity = 4.6321e-03 k_t = [3, 4, 2, 3]
 Generation 9170: Best = 2.263189e+03 FES = 917100 μ_F = 0.744 μ_{CR} = 0.146
 Diversity = 4.5885e-03 k_t = [1, 1, 1, 1]
 Generation 9180: Best = 2.263149e+03 FES = 918100 μ_F = 0.754 μ_{CR} = 0.146
 Diversity = 4.1515e-03 k_t = [2, 2, 3, 4]
 Generation 9190: Best = 2.263095e+03 FES = 919100 μ_F = 0.758 μ_{CR} = 0.141
 Diversity = 4.4629e-03 k_t = [1, 1, 1, 1]
 Generation 9200: Best = 2.263039e+03 FES = 920100 μ_F = 0.759 μ_{CR} = 0.139
 Diversity = 4.8526e-03 k_t = [3, 4, 4, 3]
 Generation 9210: Best = 2.262989e+03 FES = 921100 μ_F = 0.760 μ_{CR} = 0.149
 Diversity = 4.8182e-03 k_t = [1, 1, 1, 1]
 Generation 9220: Best = 2.262946e+03 FES = 922100 μ_F = 0.765 μ_{CR} = 0.154
 Diversity = 4.7242e-03 k_t = [4, 3, 3, 3]
 Generation 9230: Best = 2.262890e+03 FES = 923100 μ_F = 0.770 μ_{CR} = 0.140
 Diversity = 4.8402e-03 k_t = [1, 1, 1, 1]
 Generation 9240: Best = 2.262829e+03 FES = 924100 μ_F = 0.771 μ_{CR} = 0.141
 Diversity = 4.8821e-03 k_t = [2, 2, 2, 3]
 Generation 9250: Best = 2.262731e+03 FES = 925100 μ_F = 0.771 μ_{CR} = 0.142
 Diversity = 5.0622e-03 k_t = [1, 1, 1, 1]
 Generation 9260: Best = 2.262631e+03 FES = 926100 μ_F = 0.777 μ_{CR} = 0.153
 Diversity = 5.0132e-03 k_t = [3, 4, 2, 3]
 Generation 9270: Best = 2.262559e+03 FES = 927100 μ_F = 0.778 μ_{CR} = 0.147
 Diversity = 5.4627e-03 k_t = [1, 1, 1, 1]
 Generation 9280: Best = 2.262497e+03 FES = 928100 μ_F = 0.782 μ_{CR} = 0.141
 Diversity = 5.9093e-03 k_t = [2, 2, 4, 2]
 Generation 9290: Best = 2.262367e+03 FES = 929100 μ_F = 0.781 μ_{CR} = 0.139
 Diversity = 5.9613e-03 k_t = [1, 1, 1, 1]
 Generation 9300: Best = 2.262276e+03 FES = 930100 μ_F = 0.782 μ_{CR} = 0.143
 Diversity = 5.5915e-03 k_t = [3, 2, 4, 3]
 Generation 9310: Best = 2.262226e+03 FES = 931100 μ_F = 0.767 μ_{CR} = 0.146
 Diversity = 5.8931e-03 k_t = [1, 1, 1, 1]
 Generation 9320: Best = 2.262158e+03 FES = 932100 μ_F = 0.769 μ_{CR} = 0.142
 Diversity = 5.8208e-03 k_t = [2, 4, 4, 2]
 Generation 9330: Best = 2.262110e+03 FES = 933100 μ_F = 0.772 μ_{CR} = 0.143
 Diversity = 5.5207e-03 k_t = [1, 1, 1, 1]
 Generation 9340: Best = 2.262030e+03 FES = 934100 μ_F = 0.773 μ_{CR} = 0.145
 Diversity = 5.2012e-03 k_t = [2, 4, 2, 4]
 Generation 9350: Best = 2.261992e+03 FES = 935100 μ_F = 0.768 μ_{CR} = 0.142
 Diversity = 5.4837e-03 k_t = [1, 1, 1, 1]
 Generation 9360: Best = 2.261937e+03 FES = 936100 μ_F = 0.744 μ_{CR} = 0.150
 Diversity = 5.5446e-03 k_t = [3, 2, 3, 2]
 Generation 9370: Best = 2.261852e+03 FES = 937100 μ_F = 0.742 μ_{CR} = 0.144
 Diversity = 5.0802e-03 k_t = [1, 1, 1, 1]
 Generation 9380: Best = 2.261804e+03 FES = 938100 μ_F = 0.755 μ_{CR} = 0.141
 Diversity = 4.9238e-03 k_t = [2, 2, 2, 3]
 Generation 9390: Best = 2.261741e+03 FES = 939100 μ_F = 0.776 μ_{CR} = 0.141
 Diversity = 5.5168e-03 k_t = [1, 1, 1, 1]
 Generation 9400: Best = 2.261670e+03 FES = 940100 μ_F = 0.766 μ_{CR} = 0.139
 Diversity = 5.2403e-03 k_t = [3, 2, 4, 4]

Generation 9410: Best = 2.261611e+03 FES = 941100 μ_F = 0.769 μ_{CR} = 0.142
 Diversity = 4.8611e-03 k_t = [1, 1, 1, 1]
 Generation 9420: Best = 2.261530e+03 FES = 942100 μ_F = 0.771 μ_{CR} = 0.143
 Diversity = 4.7351e-03 k_t = [3, 2, 2, 4]
 Generation 9430: Best = 2.261477e+03 FES = 943100 μ_F = 0.772 μ_{CR} = 0.143
 Diversity = 4.8199e-03 k_t = [1, 1, 1, 1]
 Generation 9440: Best = 2.261442e+03 FES = 944100 μ_F = 0.758 μ_{CR} = 0.145
 Diversity = 4.6551e-03 k_t = [4, 3, 4, 4]
 Generation 9450: Best = 2.261359e+03 FES = 945100 μ_F = 0.778 μ_{CR} = 0.146
 Diversity = 4.5049e-03 k_t = [1, 1, 1, 1]
 Generation 9460: Best = 2.261310e+03 FES = 946100 μ_F = 0.789 μ_{CR} = 0.147
 Diversity = 5.0272e-03 k_t = [3, 4, 2, 3]
 Generation 9470: Best = 2.261252e+03 FES = 947100 μ_F = 0.787 μ_{CR} = 0.145
 Diversity = 5.2660e-03 k_t = [1, 1, 1, 1]
 Generation 9480: Best = 2.261151e+03 FES = 948100 μ_F = 0.778 μ_{CR} = 0.149
 Diversity = 5.1119e-03 k_t = [2, 3, 3, 3]
 Generation 9490: Best = 2.261112e+03 FES = 949100 μ_F = 0.760 μ_{CR} = 0.153
 Diversity = 5.7553e-03 k_t = [1, 1, 1, 1]
 Generation 9500: Best = 2.260992e+03 FES = 950100 μ_F = 0.769 μ_{CR} = 0.155
 Diversity = 5.5099e-03 k_t = [2, 4, 4, 4]
 Generation 9510: Best = 2.260911e+03 FES = 951100 μ_F = 0.768 μ_{CR} = 0.150
 Diversity = 5.3532e-03 k_t = [1, 1, 1, 1]
 Generation 9520: Best = 2.260842e+03 FES = 952100 μ_F = 0.774 μ_{CR} = 0.153
 Diversity = 4.9045e-03 k_t = [4, 3, 2, 4]
 Generation 9530: Best = 2.260748e+03 FES = 953100 μ_F = 0.770 μ_{CR} = 0.143
 Diversity = 4.4603e-03 k_t = [1, 1, 1, 1]
 Generation 9540: Best = 2.260736e+03 FES = 954100 μ_F = 0.774 μ_{CR} = 0.146
 Diversity = 4.3950e-03 k_t = [2, 3, 2, 3]
 Generation 9550: Best = 2.260691e+03 FES = 955100 μ_F = 0.756 μ_{CR} = 0.148
 Diversity = 4.6286e-03 k_t = [1, 1, 1, 1]
 Generation 9560: Best = 2.260623e+03 FES = 956100 μ_F = 0.750 μ_{CR} = 0.147
 Diversity = 4.4034e-03 k_t = [2, 2, 3, 2]
 Generation 9570: Best = 2.260582e+03 FES = 957100 μ_F = 0.744 μ_{CR} = 0.145
 Diversity = 4.6875e-03 k_t = [1, 1, 1, 1]
 Generation 9580: Best = 2.260536e+03 FES = 958100 μ_F = 0.744 μ_{CR} = 0.145
 Diversity = 4.1459e-03 k_t = [2, 2, 3, 4]
 Generation 9590: Best = 2.260468e+03 FES = 959100 μ_F = 0.745 μ_{CR} = 0.154
 Diversity = 3.9850e-03 k_t = [1, 1, 1, 1]
 Generation 9600: Best = 2.260441e+03 FES = 960100 μ_F = 0.743 μ_{CR} = 0.155
 Diversity = 3.6868e-03 k_t = [4, 4, 4, 2]
 Generation 9610: Best = 2.260413e+03 FES = 961100 μ_F = 0.767 μ_{CR} = 0.146
 Diversity = 3.9351e-03 k_t = [1, 1, 1, 1]
 Generation 9620: Best = 2.260358e+03 FES = 962100 μ_F = 0.761 μ_{CR} = 0.146
 Diversity = 3.5978e-03 k_t = [3, 2, 2, 4]
 Generation 9630: Best = 2.260331e+03 FES = 963100 μ_F = 0.754 μ_{CR} = 0.141
 Diversity = 3.3699e-03 k_t = [1, 1, 1, 1]
 Generation 9640: Best = 2.260291e+03 FES = 964100 μ_F = 0.757 μ_{CR} = 0.138
 Diversity = 3.3473e-03 k_t = [4, 4, 4, 3]
 Generation 9650: Best = 2.260243e+03 FES = 965100 μ_F = 0.758 μ_{CR} = 0.143
 Diversity = 3.5367e-03 k_t = [1, 1, 1, 1]
 Generation 9660: Best = 2.260186e+03 FES = 966100 μ_F = 0.759 μ_{CR} = 0.141
 Diversity = 3.5974e-03 k_t = [4, 3, 2, 2]
 Generation 9670: Best = 2.260150e+03 FES = 967100 μ_F = 0.769 μ_{CR} = 0.144
 Diversity = 4.1797e-03 k_t = [1, 1, 1, 1]

Generation 9680: Best = 2.260105e+03 FES = 968100 μ_F = 0.761 μ_{CR} = 0.154
 Diversity = 4.1530e-03 k_t = [2, 2, 3, 3]
 Generation 9690: Best = 2.260059e+03 FES = 969100 μ_F = 0.746 μ_{CR} = 0.154
 Diversity = 4.0823e-03 k_t = [1, 1, 1, 1]
 Generation 9700: Best = 2.260004e+03 FES = 970100 μ_F = 0.750 μ_{CR} = 0.158
 Diversity = 4.1920e-03 k_t = [4, 2, 4, 2]
 Generation 9710: Best = 2.259966e+03 FES = 971100 μ_F = 0.743 μ_{CR} = 0.153
 Diversity = 4.4831e-03 k_t = [1, 1, 1, 1]
 Generation 9720: Best = 2.259913e+03 FES = 972100 μ_F = 0.746 μ_{CR} = 0.158
 Diversity = 4.0266e-03 k_t = [3, 3, 2, 2]
 Generation 9730: Best = 2.259890e+03 FES = 973100 μ_F = 0.729 μ_{CR} = 0.156
 Diversity = 3.9238e-03 k_t = [1, 1, 1, 1]
 Generation 9740: Best = 2.259858e+03 FES = 974100 μ_F = 0.735 μ_{CR} = 0.151
 Diversity = 3.4549e-03 k_t = [4, 2, 4, 3]
 Generation 9750: Best = 2.259818e+03 FES = 975100 μ_F = 0.747 μ_{CR} = 0.150
 Diversity = 3.3841e-03 k_t = [1, 1, 1, 1]
 Generation 9760: Best = 2.259758e+03 FES = 976100 μ_F = 0.769 μ_{CR} = 0.150
 Diversity = 3.2006e-03 k_t = [2, 2, 4, 4]
 Generation 9770: Best = 2.259673e+03 FES = 977100 μ_F = 0.771 μ_{CR} = 0.149
 Diversity = 3.3185e-03 k_t = [1, 1, 1, 1]
 Generation 9780: Best = 2.259662e+03 FES = 978100 μ_F = 0.778 μ_{CR} = 0.145
 Diversity = 3.7705e-03 k_t = [3, 4, 4, 3]
 Generation 9790: Best = 2.259644e+03 FES = 979100 μ_F = 0.778 μ_{CR} = 0.147
 Diversity = 3.7793e-03 k_t = [1, 1, 1, 1]
 Generation 9800: Best = 2.259624e+03 FES = 980100 μ_F = 0.782 μ_{CR} = 0.141
 Diversity = 3.5987e-03 k_t = [4, 3, 3, 2]
 Generation 9810: Best = 2.259584e+03 FES = 981100 μ_F = 0.774 μ_{CR} = 0.150
 Diversity = 3.6351e-03 k_t = [1, 1, 1, 1]
 Generation 9820: Best = 2.259528e+03 FES = 982100 μ_F = 0.770 μ_{CR} = 0.146
 Diversity = 3.5691e-03 k_t = [2, 3, 4, 3]
 Generation 9830: Best = 2.259478e+03 FES = 983100 μ_F = 0.787 μ_{CR} = 0.152
 Diversity = 3.7506e-03 k_t = [1, 1, 1, 1]
 Generation 9840: Best = 2.259430e+03 FES = 984100 μ_F = 0.793 μ_{CR} = 0.143
 Diversity = 4.2010e-03 k_t = [2, 2, 4, 4]
 Generation 9850: Best = 2.259393e+03 FES = 985100 μ_F = 0.790 μ_{CR} = 0.142
 Diversity = 3.6107e-03 k_t = [1, 1, 1, 1]
 Generation 9860: Best = 2.259351e+03 FES = 986100 μ_F = 0.804 μ_{CR} = 0.144
 Diversity = 3.6397e-03 k_t = [4, 4, 3, 4]
 Generation 9870: Best = 2.259325e+03 FES = 987100 μ_F = 0.809 μ_{CR} = 0.137
 Diversity = 3.8568e-03 k_t = [1, 1, 1, 1]
 Generation 9880: Best = 2.259270e+03 FES = 988100 μ_F = 0.801 μ_{CR} = 0.143
 Diversity = 3.6832e-03 k_t = [2, 2, 2, 4]
 Generation 9890: Best = 2.259209e+03 FES = 989100 μ_F = 0.802 μ_{CR} = 0.142
 Diversity = 4.0210e-03 k_t = [1, 1, 1, 1]
 Generation 9900: Best = 2.259180e+03 FES = 990100 μ_F = 0.797 μ_{CR} = 0.139
 Diversity = 4.1094e-03 k_t = [4, 4, 4, 2]
 Generation 9910: Best = 2.259097e+03 FES = 991100 μ_F = 0.785 μ_{CR} = 0.139
 Diversity = 4.3331e-03 k_t = [1, 1, 1, 1]
 Generation 9920: Best = 2.259089e+03 FES = 992100 μ_F = 0.785 μ_{CR} = 0.147
 Diversity = 4.2982e-03 k_t = [3, 2, 4, 3]
 Generation 9930: Best = 2.259044e+03 FES = 993100 μ_F = 0.776 μ_{CR} = 0.146
 Diversity = 4.3470e-03 k_t = [1, 1, 1, 1]
 Generation 9940: Best = 2.259005e+03 FES = 994100 μ_F = 0.774 μ_{CR} = 0.146
 Diversity = 3.8481e-03 k_t = [3, 4, 4, 4]

Generation 9950: Best = 2.258964e+03 FES = 995100 μ_F = 0.772 μ_{CR} = 0.137
 Diversity = 3.8135e-03 k_t = [1, 1, 1, 1]
 Generation 9960: Best = 2.258933e+03 FES = 996100 μ_F = 0.770 μ_{CR} = 0.137
 Diversity = 3.7940e-03 k_t = [3, 3, 3, 2]
 Generation 9970: Best = 2.258917e+03 FES = 997100 μ_F = 0.773 μ_{CR} = 0.136
 Diversity = 3.7514e-03 k_t = [1, 1, 1, 1]
 Generation 9980: Best = 2.258875e+03 FES = 998100 μ_F = 0.789 μ_{CR} = 0.141
 Diversity = 3.6779e-03 k_t = [3, 3, 4, 3]
 Generation 9990: Best = 2.258828e+03 FES = 999100 μ_F = 0.795 μ_{CR} = 0.136
 Diversity = 3.9775e-03 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 2.258795e+03

Total Function Evals = 1000000

Total Successful Updates = 129875

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Results:

Best Fitness : 2.258795e+03

Global Optimum : 9.000000e+02

Error (|best-optimum|) : 1.358795e+03

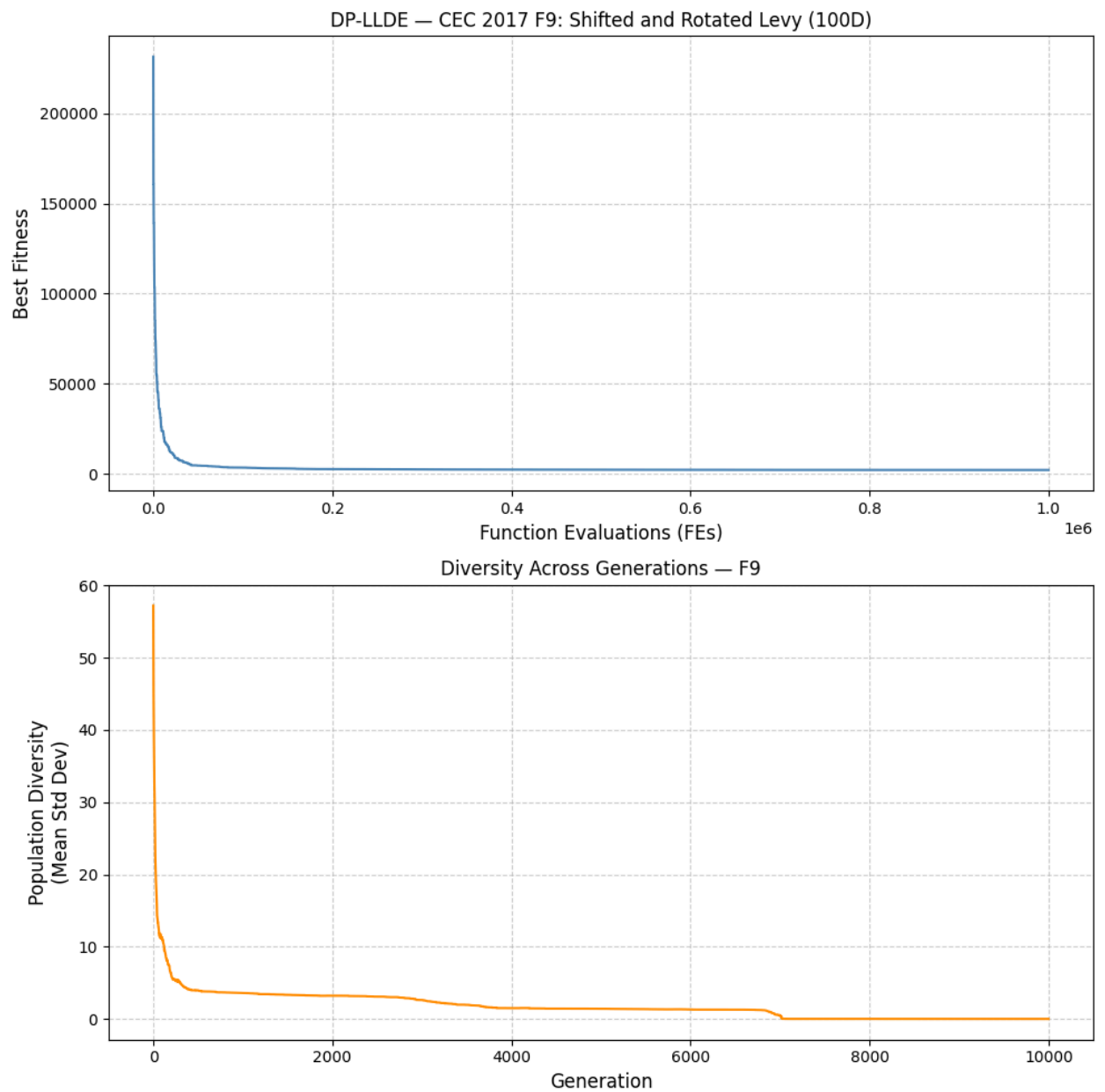
Best Solution (5 dim) : [-14.76452635 -4.85389551 23.33453221 47.45803234
-2.24922442]

D^(0) (init. diversity): 5.7220e+01

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✓ Checkpoint saved to: /content/drive/My Drive/thesis_results/all_results_checkpoint.pkl



Graph saved as convergence_F9.png

✓ Plot saved to: /content/drive/My Drive/thesis_results/plots/convergence_F9.png

```
In [ ]: # =====
# F10 — Shifted and Rotated Schwefel
# =====
run_and_plot('F10', make_wrapper(simple.f10), 1000.0, "Shifted and Rotated Sch
```

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DP-LLDE on CEC 2017 F10 – Single Run

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Function : Shifted and Rotated Schwefel's (100D)
Bounds : [-100, 100]
Global optimum : 1000.0
Initial $D^*(0) = 5.7220e+01$ | Diversity threshold = 5.7220e+00
Generation 0: Best Fitness = 3.685326e+04
Generation 10: Best = 3.425313e+04 FES = 1100 $\mu_F = 0.592$ $\mu_{CR} = 0.336$
Diversity = 5.0203e+01 $k_t = [1, 1, 1, 1]$
Generation 20: Best = 3.315520e+04 FES = 2100 $\mu_F = 0.655$ $\mu_{CR} = 0.308$
Diversity = 4.8224e+01 $k_t = [2, 3, 2, 3]$
Generation 30: Best = 3.315520e+04 FES = 3100 $\mu_F = 0.688$ $\mu_{CR} = 0.240$
Diversity = 4.8062e+01 $k_t = [1, 1, 1, 1]$
Generation 40: Best = 3.223527e+04 FES = 4100 $\mu_F = 0.712$ $\mu_{CR} = 0.213$
Diversity = 4.7822e+01 $k_t = [3, 3, 2, 2]$
Generation 50: Best = 3.223527e+04 FES = 5100 $\mu_F = 0.746$ $\mu_{CR} = 0.154$
Diversity = 4.7318e+01 $k_t = [1, 1, 1, 1]$
Generation 60: Best = 3.223527e+04 FES = 6100 $\mu_F = 0.752$ $\mu_{CR} = 0.089$
Diversity = 4.7092e+01 $k_t = [3, 3, 2, 3]$
Generation 70: Best = 3.112246e+04 FES = 7100 $\mu_F = 0.732$ $\mu_{CR} = 0.071$
Diversity = 4.7036e+01 $k_t = [1, 1, 1, 1]$
Generation 80: Best = 3.112246e+04 FES = 8100 $\mu_F = 0.723$ $\mu_{CR} = 0.050$
Diversity = 4.7312e+01 $k_t = [4, 3, 4, 2]$
Generation 90: Best = 3.026316e+04 FES = 9100 $\mu_F = 0.736$ $\mu_{CR} = 0.046$
Diversity = 4.6645e+01 $k_t = [1, 1, 1, 1]$
Generation 100: Best = 3.026316e+04 FES = 10100 $\mu_F = 0.735$ $\mu_{CR} = 0.037$
Diversity = 4.6266e+01 $k_t = [2, 3, 2, 4]$
Generation 110: Best = 3.026316e+04 FES = 11100 $\mu_F = 0.729$ $\mu_{CR} = 0.035$
Diversity = 4.5864e+01 $k_t = [1, 1, 1, 1]$
Generation 120: Best = 3.026316e+04 FES = 12100 $\mu_F = 0.769$ $\mu_{CR} = 0.024$
Diversity = 4.5736e+01 $k_t = [3, 2, 2, 4]$
Generation 130: Best = 3.014850e+04 FES = 13100 $\mu_F = 0.797$ $\mu_{CR} = 0.025$
Diversity = 4.5730e+01 $k_t = [1, 1, 1, 1]$
Generation 140: Best = 2.967095e+04 FES = 14100 $\mu_F = 0.720$ $\mu_{CR} = 0.042$
Diversity = 4.5794e+01 $k_t = [3, 3, 2, 3]$
Generation 150: Best = 2.967095e+04 FES = 15100 $\mu_F = 0.692$ $\mu_{CR} = 0.038$
Diversity = 4.5773e+01 $k_t = [1, 1, 1, 1]$
Generation 160: Best = 2.967095e+04 FES = 16100 $\mu_F = 0.763$ $\mu_{CR} = 0.035$
Diversity = 4.5724e+01 $k_t = [3, 4, 3, 2]$
Generation 170: Best = 2.962927e+04 FES = 17100 $\mu_F = 0.754$ $\mu_{CR} = 0.038$
Diversity = 4.5689e+01 $k_t = [1, 1, 1, 1]$
Generation 180: Best = 2.962927e+04 FES = 18100 $\mu_F = 0.757$ $\mu_{CR} = 0.038$
Diversity = 4.5703e+01 $k_t = [3, 4, 4, 2]$
Generation 190: Best = 2.959093e+04 FES = 19100 $\mu_F = 0.769$ $\mu_{CR} = 0.031$
Diversity = 4.5650e+01 $k_t = [1, 1, 1, 1]$
Generation 200: Best = 2.959093e+04 FES = 20100 $\mu_F = 0.841$ $\mu_{CR} = 0.029$
Diversity = 4.5360e+01 $k_t = [3, 3, 4, 4]$
Generation 210: Best = 2.949357e+04 FES = 21100 $\mu_F = 0.841$ $\mu_{CR} = 0.048$
Diversity = 4.5246e+01 $k_t = [1, 1, 1, 1]$
Generation 220: Best = 2.944622e+04 FES = 22100 $\mu_F = 0.833$ $\mu_{CR} = 0.034$
Diversity = 4.5272e+01 $k_t = [4, 3, 4, 3]$

Generation 230: Best = 2.944622e+04	FES = 23100	$\mu_F = 0.850$	$\mu_{CR} = 0.026$
Diversity = 4.5361e+01	$k_t = [1, 1, 1, 1]$		
Generation 240: Best = 2.930562e+04	FES = 24100	$\mu_F = 0.793$	$\mu_{CR} = 0.022$
Diversity = 4.5514e+01	$k_t = [2, 4, 3, 4]$		
Generation 250: Best = 2.871297e+04	FES = 25100	$\mu_F = 0.742$	$\mu_{CR} = 0.024$
Diversity = 4.5509e+01	$k_t = [1, 1, 1, 1]$		
Generation 260: Best = 2.815024e+04	FES = 26100	$\mu_F = 0.733$	$\mu_{CR} = 0.023$
Diversity = 4.5748e+01	$k_t = [3, 3, 3, 3]$		
Generation 270: Best = 2.815024e+04	FES = 27100	$\mu_F = 0.711$	$\mu_{CR} = 0.027$
Diversity = 4.5815e+01	$k_t = [1, 1, 1, 1]$		
Generation 280: Best = 2.815024e+04	FES = 28100	$\mu_F = 0.761$	$\mu_{CR} = 0.020$
Diversity = 4.5840e+01	$k_t = [2, 2, 3, 3]$		
Generation 290: Best = 2.712335e+04	FES = 29100	$\mu_F = 0.717$	$\mu_{CR} = 0.031$
Diversity = 4.5831e+01	$k_t = [1, 1, 1, 1]$		
Generation 300: Best = 2.712335e+04	FES = 30100	$\mu_F = 0.678$	$\mu_{CR} = 0.024$
Diversity = 4.5810e+01	$k_t = [3, 2, 3, 3]$		
Generation 310: Best = 2.712335e+04	FES = 31100	$\mu_F = 0.664$	$\mu_{CR} = 0.019$
Diversity = 4.5821e+01	$k_t = [1, 1, 1, 1]$		
Generation 320: Best = 2.712335e+04	FES = 32100	$\mu_F = 0.681$	$\mu_{CR} = 0.031$
Diversity = 4.5794e+01	$k_t = [2, 3, 3, 2]$		
Generation 330: Best = 2.712335e+04	FES = 33100	$\mu_F = 0.608$	$\mu_{CR} = 0.029$
Diversity = 4.5767e+01	$k_t = [1, 1, 1, 1]$		
Generation 340: Best = 2.712335e+04	FES = 34100	$\mu_F = 0.525$	$\mu_{CR} = 0.029$
Diversity = 4.5696e+01	$k_t = [4, 4, 2, 3]$		
Generation 350: Best = 2.704238e+04	FES = 35100	$\mu_F = 0.571$	$\mu_{CR} = 0.026$
Diversity = 4.5721e+01	$k_t = [1, 1, 1, 1]$		
Generation 360: Best = 2.704238e+04	FES = 36100	$\mu_F = 0.570$	$\mu_{CR} = 0.025$
Diversity = 4.5545e+01	$k_t = [2, 4, 3, 2]$		
Generation 370: Best = 2.704238e+04	FES = 37100	$\mu_F = 0.522$	$\mu_{CR} = 0.032$
Diversity = 4.5541e+01	$k_t = [1, 1, 1, 1]$		
Generation 380: Best = 2.590735e+04	FES = 38100	$\mu_F = 0.502$	$\mu_{CR} = 0.025$
Diversity = 4.5370e+01	$k_t = [3, 4, 4, 4]$		
Generation 390: Best = 2.590735e+04	FES = 39100	$\mu_F = 0.500$	$\mu_{CR} = 0.023$
Diversity = 4.5316e+01	$k_t = [1, 1, 1, 1]$		
Generation 400: Best = 2.585424e+04	FES = 40100	$\mu_F = 0.500$	$\mu_{CR} = 0.043$
Diversity = 4.5100e+01	$k_t = [3, 4, 4, 2]$		
Generation 410: Best = 2.585424e+04	FES = 41100	$\mu_F = 0.501$	$\mu_{CR} = 0.040$
Diversity = 4.5086e+01	$k_t = [1, 1, 1, 1]$		
Generation 420: Best = 2.585424e+04	FES = 42100	$\mu_F = 0.488$	$\mu_{CR} = 0.031$
Diversity = 4.4663e+01	$k_t = [3, 3, 4, 4]$		
Generation 430: Best = 2.585424e+04	FES = 43100	$\mu_F = 0.483$	$\mu_{CR} = 0.038$
Diversity = 4.4634e+01	$k_t = [1, 1, 1, 1]$		
Generation 440: Best = 2.582989e+04	FES = 44100	$\mu_F = 0.485$	$\mu_{CR} = 0.037$
Diversity = 4.4639e+01	$k_t = [4, 2, 2, 3]$		
Generation 450: Best = 2.582989e+04	FES = 45100	$\mu_F = 0.450$	$\mu_{CR} = 0.022$
Diversity = 4.4579e+01	$k_t = [1, 1, 1, 1]$		
Generation 460: Best = 2.582989e+04	FES = 46100	$\mu_F = 0.401$	$\mu_{CR} = 0.024$
Diversity = 4.4566e+01	$k_t = [3, 2, 2, 2]$		
Generation 470: Best = 2.582989e+04	FES = 47100	$\mu_F = 0.363$	$\mu_{CR} = 0.033$
Diversity = 4.4544e+01	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 2.582989e+04	FES = 48100	$\mu_F = 0.311$	$\mu_{CR} = 0.033$
Diversity = 4.4476e+01	$k_t = [2, 3, 2, 3]$		
Generation 490: Best = 2.582989e+04	FES = 49100	$\mu_F = 0.315$	$\mu_{CR} = 0.027$
Diversity = 4.4297e+01	$k_t = [1, 1, 1, 1]$		

Generation 500: Best = 2.582989e+04	FES = 50100	$\mu_F = 0.468$	$\mu_{CR} = 0.031$
Diversity = 4.4307e+01	k_t = [4, 4, 3, 3]		
Generation 510: Best = 2.582989e+04	FES = 51100	$\mu_F = 0.390$	$\mu_{CR} = 0.037$
Diversity = 4.4286e+01	k_t = [1, 1, 1, 1]		
Generation 520: Best = 2.582989e+04	FES = 52100	$\mu_F = 0.356$	$\mu_{CR} = 0.029$
Diversity = 4.4370e+01	k_t = [3, 3, 4, 3]		
Generation 530: Best = 2.499947e+04	FES = 53100	$\mu_F = 0.403$	$\mu_{CR} = 0.035$
Diversity = 4.4344e+01	k_t = [1, 1, 1, 1]		
Generation 540: Best = 2.499947e+04	FES = 54100	$\mu_F = 0.411$	$\mu_{CR} = 0.032$
Diversity = 4.4309e+01	k_t = [4, 2, 3, 3]		
Generation 550: Best = 2.499947e+04	FES = 55100	$\mu_F = 0.355$	$\mu_{CR} = 0.050$
Diversity = 4.4127e+01	k_t = [1, 1, 1, 1]		
Generation 560: Best = 2.499947e+04	FES = 56100	$\mu_F = 0.426$	$\mu_{CR} = 0.038$
Diversity = 4.4111e+01	k_t = [4, 2, 4, 4]		
Generation 570: Best = 2.439214e+04	FES = 57100	$\mu_F = 0.452$	$\mu_{CR} = 0.039$
Diversity = 4.4095e+01	k_t = [1, 1, 1, 1]		
Generation 580: Best = 2.439214e+04	FES = 58100	$\mu_F = 0.439$	$\mu_{CR} = 0.033$
Diversity = 4.4092e+01	k_t = [3, 2, 4, 3]		
Generation 590: Best = 2.439214e+04	FES = 59100	$\mu_F = 0.446$	$\mu_{CR} = 0.030$
Diversity = 4.4087e+01	k_t = [1, 1, 1, 1]		
Generation 600: Best = 2.439214e+04	FES = 60100	$\mu_F = 0.366$	$\mu_{CR} = 0.021$
Diversity = 4.4090e+01	k_t = [4, 4, 4, 3]		
Generation 610: Best = 2.430948e+04	FES = 61100	$\mu_F = 0.322$	$\mu_{CR} = 0.031$
Diversity = 4.4069e+01	k_t = [1, 1, 1, 1]		
Generation 620: Best = 2.430948e+04	FES = 62100	$\mu_F = 0.357$	$\mu_{CR} = 0.033$
Diversity = 4.4036e+01	k_t = [2, 3, 2, 2]		
Generation 630: Best = 2.430948e+04	FES = 63100	$\mu_F = 0.322$	$\mu_{CR} = 0.024$
Diversity = 4.4029e+01	k_t = [1, 1, 1, 1]		
Generation 640: Best = 2.430948e+04	FES = 64100	$\mu_F = 0.298$	$\mu_{CR} = 0.019$
Diversity = 4.4007e+01	k_t = [3, 4, 3, 3]		
Generation 650: Best = 2.430948e+04	FES = 65100	$\mu_F = 0.326$	$\mu_{CR} = 0.022$
Diversity = 4.3983e+01	k_t = [1, 1, 1, 1]		
Generation 660: Best = 2.430948e+04	FES = 66100	$\mu_F = 0.352$	$\mu_{CR} = 0.021$
Diversity = 4.3972e+01	k_t = [3, 3, 2, 2]		
Generation 670: Best = 2.430948e+04	FES = 67100	$\mu_F = 0.334$	$\mu_{CR} = 0.021$
Diversity = 4.3960e+01	k_t = [1, 1, 1, 1]		
Generation 680: Best = 2.430948e+04	FES = 68100	$\mu_F = 0.373$	$\mu_{CR} = 0.026$
Diversity = 4.3891e+01	k_t = [4, 2, 3, 2]		
Generation 690: Best = 2.430948e+04	FES = 69100	$\mu_F = 0.302$	$\mu_{CR} = 0.026$
Diversity = 4.3852e+01	k_t = [1, 1, 1, 1]		
Generation 700: Best = 2.430948e+04	FES = 70100	$\mu_F = 0.206$	$\mu_{CR} = 0.039$
Diversity = 4.3849e+01	k_t = [4, 4, 2, 2]		
Generation 710: Best = 2.430948e+04	FES = 71100	$\mu_F = 0.222$	$\mu_{CR} = 0.036$
Diversity = 4.3773e+01	k_t = [1, 1, 1, 1]		
Generation 720: Best = 2.430948e+04	FES = 72100	$\mu_F = 0.236$	$\mu_{CR} = 0.046$
Diversity = 4.3608e+01	k_t = [4, 4, 4, 2]		
Generation 730: Best = 2.424220e+04	FES = 73100	$\mu_F = 0.204$	$\mu_{CR} = 0.037$
Diversity = 4.3576e+01	k_t = [1, 1, 1, 1]		
Generation 740: Best = 2.416223e+04	FES = 74100	$\mu_F = 0.219$	$\mu_{CR} = 0.033$
Diversity = 4.3626e+01	k_t = [3, 2, 4, 4]		
Generation 750: Best = 2.416223e+04	FES = 75100	$\mu_F = 0.225$	$\mu_{CR} = 0.031$
Diversity = 4.3614e+01	k_t = [1, 1, 1, 1]		
Generation 760: Best = 2.403438e+04	FES = 76100	$\mu_F = 0.271$	$\mu_{CR} = 0.032$
Diversity = 4.3439e+01	k_t = [2, 4, 2, 4]		

Generation 770:	Best = 2.403438e+04	FES = 77100	$\mu_F = 0.253$	$\mu_{CR} = 0.027$
Diversity = 4.3426e+01	$k_t = [1, 1, 1, 1]$			
Generation 780:	Best = 2.403438e+04	FES = 78100	$\mu_F = 0.245$	$\mu_{CR} = 0.040$
Diversity = 4.3366e+01	$k_t = [3, 2, 4, 2]$			
Generation 790:	Best = 2.403438e+04	FES = 79100	$\mu_F = 0.266$	$\mu_{CR} = 0.033$
Diversity = 4.3368e+01	$k_t = [1, 1, 1, 1]$			
Generation 800:	Best = 2.403438e+04	FES = 80100	$\mu_F = 0.269$	$\mu_{CR} = 0.031$
Diversity = 4.3366e+01	$k_t = [4, 2, 3, 3]$			
Generation 810:	Best = 2.402805e+04	FES = 81100	$\mu_F = 0.262$	$\mu_{CR} = 0.026$
Diversity = 4.3351e+01	$k_t = [1, 1, 1, 1]$			
Generation 820:	Best = 2.402805e+04	FES = 82100	$\mu_F = 0.266$	$\mu_{CR} = 0.029$
Diversity = 4.3332e+01	$k_t = [4, 2, 4, 2]$			
Generation 830:	Best = 2.402805e+04	FES = 83100	$\mu_F = 0.210$	$\mu_{CR} = 0.044$
Diversity = 4.3301e+01	$k_t = [1, 1, 1, 1]$			
Generation 840:	Best = 2.402805e+04	FES = 84100	$\mu_F = 0.236$	$\mu_{CR} = 0.037$
Diversity = 4.3264e+01	$k_t = [3, 3, 3, 3]$			
Generation 850:	Best = 2.340698e+04	FES = 85100	$\mu_F = 0.159$	$\mu_{CR} = 0.038$
Diversity = 4.3261e+01	$k_t = [1, 1, 1, 1]$			
Generation 860:	Best = 2.340698e+04	FES = 86100	$\mu_F = 0.168$	$\mu_{CR} = 0.044$
Diversity = 4.3262e+01	$k_t = [3, 3, 4, 3]$			
Generation 870:	Best = 2.340698e+04	FES = 87100	$\mu_F = 0.140$	$\mu_{CR} = 0.037$
Diversity = 4.3207e+01	$k_t = [1, 1, 1, 1]$			
Generation 880:	Best = 2.338240e+04	FES = 88100	$\mu_F = 0.146$	$\mu_{CR} = 0.046$
Diversity = 4.3178e+01	$k_t = [3, 4, 3, 3]$			
Generation 890:	Best = 2.338240e+04	FES = 89100	$\mu_F = 0.195$	$\mu_{CR} = 0.051$
Diversity = 4.3170e+01	$k_t = [1, 1, 1, 1]$			
Generation 900:	Best = 2.338240e+04	FES = 90100	$\mu_F = 0.139$	$\mu_{CR} = 0.070$
Diversity = 4.3155e+01	$k_t = [4, 3, 3, 3]$			
Generation 910:	Best = 2.338240e+04	FES = 91100	$\mu_F = 0.181$	$\mu_{CR} = 0.058$
Diversity = 4.3131e+01	$k_t = [1, 1, 1, 1]$			
Generation 920:	Best = 2.338240e+04	FES = 92100	$\mu_F = 0.325$	$\mu_{CR} = 0.044$
Diversity = 4.3107e+01	$k_t = [3, 3, 4, 4]$			
Generation 930:	Best = 2.338240e+04	FES = 93100	$\mu_F = 0.288$	$\mu_{CR} = 0.039$
Diversity = 4.3102e+01	$k_t = [1, 1, 1, 1]$			
Generation 940:	Best = 2.338240e+04	FES = 94100	$\mu_F = 0.369$	$\mu_{CR} = 0.028$
Diversity = 4.3108e+01	$k_t = [3, 3, 2, 2]$			
Generation 950:	Best = 2.338240e+04	FES = 95100	$\mu_F = 0.392$	$\mu_{CR} = 0.027$
Diversity = 4.3104e+01	$k_t = [1, 1, 1, 1]$			
Generation 960:	Best = 2.338240e+04	FES = 96100	$\mu_F = 0.425$	$\mu_{CR} = 0.025$
Diversity = 4.3153e+01	$k_t = [2, 2, 2, 4]$			
Generation 970:	Best = 2.338240e+04	FES = 97100	$\mu_F = 0.380$	$\mu_{CR} = 0.019$
Diversity = 4.3157e+01	$k_t = [1, 1, 1, 1]$			
Generation 980:	Best = 2.338240e+04	FES = 98100	$\mu_F = 0.296$	$\mu_{CR} = 0.021$
Diversity = 4.3146e+01	$k_t = [2, 3, 3, 3]$			
Generation 990:	Best = 2.338240e+04	FES = 99100	$\mu_F = 0.251$	$\mu_{CR} = 0.023$
Diversity = 4.3139e+01	$k_t = [1, 1, 1, 1]$			
Generation 1000:	Best = 2.338240e+04	FES = 100100	$\mu_F = 0.283$	$\mu_{CR} = 0.021$
Diversity = 4.3128e+01	$k_t = [4, 3, 2, 2]$			
Generation 1010:	Best = 2.338240e+04	FES = 101100	$\mu_F = 0.286$	$\mu_{CR} = 0.020$
Diversity = 4.3125e+01	$k_t = [1, 1, 1, 1]$			
Generation 1020:	Best = 2.338240e+04	FES = 102100	$\mu_F = 0.298$	$\mu_{CR} = 0.038$
Diversity = 4.3116e+01	$k_t = [4, 3, 4, 3]$			
Generation 1030:	Best = 2.338240e+04	FES = 103100	$\mu_F = 0.291$	$\mu_{CR} = 0.030$
Diversity = 4.3105e+01	$k_t = [1, 1, 1, 1]$			

Generation 1040: Best = 2.338240e+04 FES = 104100 μ_F = 0.280 μ_{CR} = 0.035
 Diversity = 4.3285e+01 k_t = [2, 4, 3, 4]
 Generation 1050: Best = 2.338240e+04 FES = 105100 μ_F = 0.305 μ_{CR} = 0.038
 Diversity = 4.3267e+01 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 2.327453e+04 FES = 106100 μ_F = 0.358 μ_{CR} = 0.036
 Diversity = 4.3257e+01 k_t = [3, 3, 3, 4]
 Generation 1070: Best = 2.327453e+04 FES = 107100 μ_F = 0.272 μ_{CR} = 0.034
 Diversity = 4.3244e+01 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 2.327453e+04 FES = 108100 μ_F = 0.294 μ_{CR} = 0.044
 Diversity = 4.3177e+01 k_t = [4, 2, 4, 4]
 Generation 1090: Best = 2.327453e+04 FES = 109100 μ_F = 0.263 μ_{CR} = 0.031
 Diversity = 4.3174e+01 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 2.273496e+04 FES = 110100 μ_F = 0.266 μ_{CR} = 0.033
 Diversity = 4.3172e+01 k_t = [4, 2, 4, 4]
 Generation 1110: Best = 2.273496e+04 FES = 111100 μ_F = 0.283 μ_{CR} = 0.032
 Diversity = 4.3167e+01 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 2.273496e+04 FES = 112100 μ_F = 0.273 μ_{CR} = 0.038
 Diversity = 4.3161e+01 k_t = [3, 3, 4, 2]
 Generation 1130: Best = 2.273496e+04 FES = 113100 μ_F = 0.192 μ_{CR} = 0.032
 Diversity = 4.3140e+01 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 2.270862e+04 FES = 114100 μ_F = 0.145 μ_{CR} = 0.044
 Diversity = 4.3132e+01 k_t = [3, 2, 2, 2]
 Generation 1150: Best = 2.270862e+04 FES = 115100 μ_F = 0.124 μ_{CR} = 0.047
 Diversity = 4.3100e+01 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 2.262828e+04 FES = 116100 μ_F = 0.166 μ_{CR} = 0.069
 Diversity = 4.2857e+01 k_t = [3, 2, 4, 4]
 Generation 1170: Best = 2.262828e+04 FES = 117100 μ_F = 0.097 μ_{CR} = 0.081
 Diversity = 4.2840e+01 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 2.262828e+04 FES = 118100 μ_F = 0.157 μ_{CR} = 0.072
 Diversity = 4.2820e+01 k_t = [3, 4, 3, 3]
 Generation 1190: Best = 2.262828e+04 FES = 119100 μ_F = 0.098 μ_{CR} = 0.053
 Diversity = 4.2813e+01 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 2.262828e+04 FES = 120100 μ_F = 0.168 μ_{CR} = 0.066
 Diversity = 4.2853e+01 k_t = [3, 3, 3, 4]
 Generation 1210: Best = 2.262828e+04 FES = 121100 μ_F = 0.151 μ_{CR} = 0.054
 Diversity = 4.2842e+01 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 2.241701e+04 FES = 122100 μ_F = 0.156 μ_{CR} = 0.030
 Diversity = 4.2834e+01 k_t = [4, 3, 2, 2]
 Generation 1230: Best = 2.241701e+04 FES = 123100 μ_F = 0.113 μ_{CR} = 0.048
 Diversity = 4.2825e+01 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 2.241701e+04 FES = 124100 μ_F = 0.099 μ_{CR} = 0.049
 Diversity = 4.2796e+01 k_t = [3, 2, 2, 3]
 Generation 1250: Best = 2.237441e+04 FES = 125100 μ_F = 0.107 μ_{CR} = 0.055
 Diversity = 4.2791e+01 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 2.210406e+04 FES = 126100 μ_F = 0.090 μ_{CR} = 0.071
 Diversity = 4.2772e+01 k_t = [3, 2, 3, 3]
 Generation 1270: Best = 2.210397e+04 FES = 127100 μ_F = 0.091 μ_{CR} = 0.095
 Diversity = 4.2769e+01 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 2.210397e+04 FES = 128100 μ_F = 0.076 μ_{CR} = 0.082
 Diversity = 4.2752e+01 k_t = [3, 3, 3, 3]
 Generation 1290: Best = 2.190752e+04 FES = 129100 μ_F = 0.060 μ_{CR} = 0.075
 Diversity = 4.2739e+01 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 2.190752e+04 FES = 130100 μ_F = 0.138 μ_{CR} = 0.078
 Diversity = 4.2833e+01 k_t = [4, 4, 3, 4]

Generation 1310: Best = 2.190752e+04 FES = 131100 μ_F = 0.101 μ_{CR} = 0.074
 Diversity = 4.2823e+01 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 2.190752e+04 FES = 132100 μ_F = 0.187 μ_{CR} = 0.044
 Diversity = 4.2819e+01 k_t = [4, 4, 4, 3]
 Generation 1330: Best = 2.190752e+04 FES = 133100 μ_F = 0.150 μ_{CR} = 0.046
 Diversity = 4.2813e+01 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 2.190752e+04 FES = 134100 μ_F = 0.261 μ_{CR} = 0.035
 Diversity = 4.2747e+01 k_t = [4, 4, 2, 4]
 Generation 1350: Best = 2.190752e+04 FES = 135100 μ_F = 0.354 μ_{CR} = 0.023
 Diversity = 4.2731e+01 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 2.190752e+04 FES = 136100 μ_F = 0.288 μ_{CR} = 0.020
 Diversity = 4.2730e+01 k_t = [3, 3, 4, 4]
 Generation 1370: Best = 2.190752e+04 FES = 137100 μ_F = 0.249 μ_{CR} = 0.022
 Diversity = 4.2726e+01 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 2.173950e+04 FES = 138100 μ_F = 0.214 μ_{CR} = 0.032
 Diversity = 4.2717e+01 k_t = [4, 3, 4, 3]
 Generation 1390: Best = 2.173950e+04 FES = 139100 μ_F = 0.172 μ_{CR} = 0.033
 Diversity = 4.2712e+01 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 2.173950e+04 FES = 140100 μ_F = 0.145 μ_{CR} = 0.056
 Diversity = 4.2704e+01 k_t = [3, 4, 2, 2]
 Generation 1410: Best = 2.173950e+04 FES = 141100 μ_F = 0.107 μ_{CR} = 0.055
 Diversity = 4.2701e+01 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 2.173950e+04 FES = 142100 μ_F = 0.115 μ_{CR} = 0.055
 Diversity = 4.2698e+01 k_t = [2, 3, 3, 2]
 Generation 1430: Best = 2.173950e+04 FES = 143100 μ_F = 0.087 μ_{CR} = 0.060
 Diversity = 4.2677e+01 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 2.173950e+04 FES = 144100 μ_F = 0.100 μ_{CR} = 0.069
 Diversity = 4.2661e+01 k_t = [4, 4, 3, 3]
 Generation 1450: Best = 2.170621e+04 FES = 145100 μ_F = 0.146 μ_{CR} = 0.044
 Diversity = 4.2654e+01 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 2.154023e+04 FES = 146100 μ_F = 0.222 μ_{CR} = 0.040
 Diversity = 4.2517e+01 k_t = [4, 4, 3, 4]
 Generation 1470: Best = 2.150840e+04 FES = 147100 μ_F = 0.219 μ_{CR} = 0.034
 Diversity = 4.2510e+01 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 2.150840e+04 FES = 148100 μ_F = 0.173 μ_{CR} = 0.031
 Diversity = 4.2505e+01 k_t = [4, 3, 2, 3]
 Generation 1490: Best = 2.150840e+04 FES = 149100 μ_F = 0.103 μ_{CR} = 0.032
 Diversity = 4.2505e+01 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 2.150840e+04 FES = 150100 μ_F = 0.130 μ_{CR} = 0.030
 Diversity = 4.2498e+01 k_t = [2, 3, 4, 3]
 Generation 1510: Best = 2.127360e+04 FES = 151100 μ_F = 0.102 μ_{CR} = 0.046
 Diversity = 4.2492e+01 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 2.125315e+04 FES = 152100 μ_F = 0.083 μ_{CR} = 0.066
 Diversity = 4.2483e+01 k_t = [2, 4, 4, 3]
 Generation 1530: Best = 2.125315e+04 FES = 153100 μ_F = 0.082 μ_{CR} = 0.065
 Diversity = 4.2464e+01 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 2.109804e+04 FES = 154100 μ_F = 0.102 μ_{CR} = 0.064
 Diversity = 4.2453e+01 k_t = [4, 4, 3, 3]
 Generation 1550: Best = 2.107486e+04 FES = 155100 μ_F = 0.086 μ_{CR} = 0.078
 Diversity = 4.2443e+01 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 2.107486e+04 FES = 156100 μ_F = 0.073 μ_{CR} = 0.064
 Diversity = 4.2434e+01 k_t = [2, 2, 2, 3]
 Generation 1570: Best = 2.107486e+04 FES = 157100 μ_F = 0.089 μ_{CR} = 0.067
 Diversity = 4.2418e+01 k_t = [1, 1, 1, 1]

Generation 1580: Best = 2.079826e+04 FES = 158100 μ_F = 0.158 μ_{CR} = 0.078
 Diversity = 4.2321e+01 k_t = [4, 3, 3, 4]
 Generation 1590: Best = 2.079826e+04 FES = 159100 μ_F = 0.109 μ_{CR} = 0.065
 Diversity = 4.2318e+01 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 2.079826e+04 FES = 160100 μ_F = 0.253 μ_{CR} = 0.060
 Diversity = 4.2467e+01 k_t = [2, 4, 4, 4]
 Generation 1610: Best = 2.079826e+04 FES = 161100 μ_F = 0.287 μ_{CR} = 0.052
 Diversity = 4.2467e+01 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 2.079826e+04 FES = 162100 μ_F = 0.312 μ_{CR} = 0.060
 Diversity = 4.2333e+01 k_t = [2, 3, 4, 4]
 Generation 1630: Best = 2.079826e+04 FES = 163100 μ_F = 0.382 μ_{CR} = 0.051
 Diversity = 4.2407e+01 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 2.079826e+04 FES = 164100 μ_F = 0.322 μ_{CR} = 0.053
 Diversity = 4.2407e+01 k_t = [3, 4, 3, 3]
 Generation 1650: Best = 2.079826e+04 FES = 165100 μ_F = 0.254 μ_{CR} = 0.051
 Diversity = 4.2406e+01 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 2.079826e+04 FES = 166100 μ_F = 0.280 μ_{CR} = 0.035
 Diversity = 4.2280e+01 k_t = [2, 4, 2, 4]
 Generation 1670: Best = 2.079826e+04 FES = 167100 μ_F = 0.263 μ_{CR} = 0.030
 Diversity = 4.2279e+01 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 2.079826e+04 FES = 168100 μ_F = 0.177 μ_{CR} = 0.048
 Diversity = 4.2273e+01 k_t = [2, 3, 2, 3]
 Generation 1690: Best = 2.079826e+04 FES = 169100 μ_F = 0.237 μ_{CR} = 0.035
 Diversity = 4.2273e+01 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 2.079826e+04 FES = 170100 μ_F = 0.346 μ_{CR} = 0.023
 Diversity = 4.2272e+01 k_t = [2, 3, 4, 3]
 Generation 1710: Best = 2.079826e+04 FES = 171100 μ_F = 0.356 μ_{CR} = 0.021
 Diversity = 4.2273e+01 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 2.034838e+04 FES = 172100 μ_F = 0.324 μ_{CR} = 0.023
 Diversity = 4.2276e+01 k_t = [4, 4, 4, 3]
 Generation 1730: Best = 2.034838e+04 FES = 173100 μ_F = 0.247 μ_{CR} = 0.033
 Diversity = 4.2275e+01 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 2.034838e+04 FES = 174100 μ_F = 0.196 μ_{CR} = 0.028
 Diversity = 4.2273e+01 k_t = [4, 2, 3, 3]
 Generation 1750: Best = 2.034328e+04 FES = 175100 μ_F = 0.129 μ_{CR} = 0.033
 Diversity = 4.2267e+01 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 2.032479e+04 FES = 176100 μ_F = 0.147 μ_{CR} = 0.030
 Diversity = 4.2257e+01 k_t = [2, 2, 3, 2]
 Generation 1770: Best = 2.032479e+04 FES = 177100 μ_F = 0.100 μ_{CR} = 0.050
 Diversity = 4.2252e+01 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 2.032479e+04 FES = 178100 μ_F = 0.145 μ_{CR} = 0.050
 Diversity = 4.2240e+01 k_t = [3, 4, 2, 3]
 Generation 1790: Best = 2.030090e+04 FES = 179100 μ_F = 0.121 μ_{CR} = 0.048
 Diversity = 4.2236e+01 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 2.027373e+04 FES = 180100 μ_F = 0.122 μ_{CR} = 0.094
 Diversity = 4.2229e+01 k_t = [2, 3, 2, 3]
 Generation 1810: Best = 2.026575e+04 FES = 181100 μ_F = 0.097 μ_{CR} = 0.061
 Diversity = 4.2220e+01 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 2.026575e+04 FES = 182100 μ_F = 0.122 μ_{CR} = 0.038
 Diversity = 4.2216e+01 k_t = [3, 4, 2, 3]
 Generation 1830: Best = 2.026575e+04 FES = 183100 μ_F = 0.162 μ_{CR} = 0.038
 Diversity = 4.2215e+01 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 2.026575e+04 FES = 184100 μ_F = 0.143 μ_{CR} = 0.041
 Diversity = 4.2213e+01 k_t = [2, 3, 2, 3]

Generation 1850: Best = 2.026575e+04	FES = 185100	$\mu_F = 0.093$	$\mu_{CR} = 0.055$
Diversity = 4.2208e+01	k_t = [1, 1, 1, 1]		
Generation 1860: Best = 1.999253e+04	FES = 186100	$\mu_F = 0.142$	$\mu_{CR} = 0.043$
Diversity = 4.2200e+01	k_t = [4, 3, 2, 2]		
Generation 1870: Best = 1.999253e+04	FES = 187100	$\mu_F = 0.121$	$\mu_{CR} = 0.044$
Diversity = 4.2190e+01	k_t = [1, 1, 1, 1]		
Generation 1880: Best = 1.999253e+04	FES = 188100	$\mu_F = 0.200$	$\mu_{CR} = 0.040$
Diversity = 4.2261e+01	k_t = [3, 4, 3, 4]		
Generation 1890: Best = 1.999253e+04	FES = 189100	$\mu_F = 0.167$	$\mu_{CR} = 0.027$
Diversity = 4.2258e+01	k_t = [1, 1, 1, 1]		
Generation 1900: Best = 1.971483e+04	FES = 190100	$\mu_F = 0.151$	$\mu_{CR} = 0.032$
Diversity = 4.2256e+01	k_t = [3, 4, 4, 4]		
Generation 1910: Best = 1.971483e+04	FES = 191100	$\mu_F = 0.116$	$\mu_{CR} = 0.041$
Diversity = 4.2249e+01	k_t = [1, 1, 1, 1]		
Generation 1920: Best = 1.971483e+04	FES = 192100	$\mu_F = 0.087$	$\mu_{CR} = 0.054$
Diversity = 4.2240e+01	k_t = [3, 2, 2, 3]		
Generation 1930: Best = 1.971483e+04	FES = 193100	$\mu_F = 0.048$	$\mu_{CR} = 0.060$
Diversity = 4.2232e+01	k_t = [1, 1, 1, 1]		
Generation 1940: Best = 1.956810e+04	FES = 194100	$\mu_F = 0.045$	$\mu_{CR} = 0.070$
Diversity = 4.2219e+01	k_t = [4, 3, 3, 2]		
Generation 1950: Best = 1.956810e+04	FES = 195100	$\mu_F = 0.074$	$\mu_{CR} = 0.058$
Diversity = 4.2213e+01	k_t = [1, 1, 1, 1]		
Generation 1960: Best = 1.956810e+04	FES = 196100	$\mu_F = 0.058$	$\mu_{CR} = 0.064$
Diversity = 4.2206e+01	k_t = [2, 3, 3, 3]		
Generation 1970: Best = 1.956810e+04	FES = 197100	$\mu_F = 0.072$	$\mu_{CR} = 0.052$
Diversity = 4.2196e+01	k_t = [1, 1, 1, 1]		
Generation 1980: Best = 1.956810e+04	FES = 198100	$\mu_F = 0.157$	$\mu_{CR} = 0.046$
Diversity = 4.2228e+01	k_t = [2, 2, 3, 4]		
Generation 1990: Best = 1.956810e+04	FES = 199100	$\mu_F = 0.231$	$\mu_{CR} = 0.039$
Diversity = 4.2173e+01	k_t = [1, 1, 1, 1]		
Generation 2000: Best = 1.956810e+04	FES = 200100	$\mu_F = 0.187$	$\mu_{CR} = 0.040$
Diversity = 4.2171e+01	k_t = [4, 2, 2, 2]		
Generation 2010: Best = 1.956810e+04	FES = 201100	$\mu_F = 0.193$	$\mu_{CR} = 0.033$
Diversity = 4.2167e+01	k_t = [1, 1, 1, 1]		
Generation 2020: Best = 1.956810e+04	FES = 202100	$\mu_F = 0.168$	$\mu_{CR} = 0.043$
Diversity = 4.2166e+01	k_t = [2, 4, 3, 2]		
Generation 2030: Best = 1.956810e+04	FES = 203100	$\mu_F = 0.162$	$\mu_{CR} = 0.037$
Diversity = 4.2160e+01	k_t = [1, 1, 1, 1]		
Generation 2040: Best = 1.956810e+04	FES = 204100	$\mu_F = 0.192$	$\mu_{CR} = 0.045$
Diversity = 4.2064e+01	k_t = [2, 4, 4, 4]		
Generation 2050: Best = 1.956810e+04	FES = 205100	$\mu_F = 0.157$	$\mu_{CR} = 0.043$
Diversity = 4.2063e+01	k_t = [1, 1, 1, 1]		
Generation 2060: Best = 1.956754e+04	FES = 206100	$\mu_F = 0.119$	$\mu_{CR} = 0.033$
Diversity = 4.2061e+01	k_t = [3, 3, 4, 4]		
Generation 2070: Best = 1.956640e+04	FES = 207100	$\mu_F = 0.132$	$\mu_{CR} = 0.025$
Diversity = 4.2062e+01	k_t = [1, 1, 1, 1]		
Generation 2080: Best = 1.956640e+04	FES = 208100	$\mu_F = 0.124$	$\mu_{CR} = 0.032$
Diversity = 4.2057e+01	k_t = [2, 4, 2, 2]		
Generation 2090: Best = 1.954390e+04	FES = 209100	$\mu_F = 0.112$	$\mu_{CR} = 0.035$
Diversity = 4.2053e+01	k_t = [1, 1, 1, 1]		
Generation 2100: Best = 1.954390e+04	FES = 210100	$\mu_F = 0.185$	$\mu_{CR} = 0.045$
Diversity = 4.2046e+01	k_t = [4, 2, 3, 4]		
Generation 2110: Best = 1.954390e+04	FES = 211100	$\mu_F = 0.171$	$\mu_{CR} = 0.038$
Diversity = 4.2047e+01	k_t = [1, 1, 1, 1]		

Generation 2120: Best = 1.954390e+04 FES = 212100 μ_F = 0.165 μ_{CR} = 0.030
 Diversity = 4.2046e+01 k_t = [2, 3, 3, 3]
 Generation 2130: Best = 1.954390e+04 FES = 213100 μ_F = 0.102 μ_{CR} = 0.056
 Diversity = 4.2045e+01 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 1.954390e+04 FES = 214100 μ_F = 0.178 μ_{CR} = 0.054
 Diversity = 4.2010e+01 k_t = [2, 4, 2, 4]
 Generation 2150: Best = 1.954390e+04 FES = 215100 μ_F = 0.118 μ_{CR} = 0.047
 Diversity = 4.2009e+01 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 1.940514e+04 FES = 216100 μ_F = 0.184 μ_{CR} = 0.047
 Diversity = 4.1887e+01 k_t = [2, 4, 4, 4]
 Generation 2170: Best = 1.940514e+04 FES = 217100 μ_F = 0.147 μ_{CR} = 0.038
 Diversity = 4.1885e+01 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 1.939534e+04 FES = 218100 μ_F = 0.213 μ_{CR} = 0.040
 Diversity = 4.1932e+01 k_t = [2, 3, 4, 4]
 Generation 2190: Best = 1.939534e+04 FES = 219100 μ_F = 0.161 μ_{CR} = 0.045
 Diversity = 4.1931e+01 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 1.936287e+04 FES = 220100 μ_F = 0.153 μ_{CR} = 0.044
 Diversity = 4.1929e+01 k_t = [4, 4, 2, 3]
 Generation 2210: Best = 1.936287e+04 FES = 221100 μ_F = 0.133 μ_{CR} = 0.045
 Diversity = 4.1925e+01 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 1.935883e+04 FES = 222100 μ_F = 0.091 μ_{CR} = 0.068
 Diversity = 4.1922e+01 k_t = [4, 4, 2, 2]
 Generation 2230: Best = 1.935883e+04 FES = 223100 μ_F = 0.077 μ_{CR} = 0.071
 Diversity = 4.1920e+01 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 1.935883e+04 FES = 224100 μ_F = 0.126 μ_{CR} = 0.051
 Diversity = 4.1918e+01 k_t = [4, 2, 3, 4]
 Generation 2250: Best = 1.934010e+04 FES = 225100 μ_F = 0.098 μ_{CR} = 0.051
 Diversity = 4.1914e+01 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 1.934010e+04 FES = 226100 μ_F = 0.155 μ_{CR} = 0.035
 Diversity = 4.1827e+01 k_t = [2, 4, 2, 4]
 Generation 2270: Best = 1.934010e+04 FES = 227100 μ_F = 0.186 μ_{CR} = 0.055
 Diversity = 4.1825e+01 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 1.932433e+04 FES = 228100 μ_F = 0.223 μ_{CR} = 0.058
 Diversity = 4.1679e+01 k_t = [3, 2, 3, 4]
 Generation 2290: Best = 1.932433e+04 FES = 229100 μ_F = 0.336 μ_{CR} = 0.046
 Diversity = 4.1600e+01 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 1.932433e+04 FES = 230100 μ_F = 0.280 μ_{CR} = 0.067
 Diversity = 4.1527e+01 k_t = [3, 3, 4, 4]
 Generation 2310: Best = 1.932433e+04 FES = 231100 μ_F = 0.201 μ_{CR} = 0.048
 Diversity = 4.1526e+01 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 1.932433e+04 FES = 232100 μ_F = 0.255 μ_{CR} = 0.040
 Diversity = 4.1526e+01 k_t = [4, 4, 2, 2]
 Generation 2330: Best = 1.932433e+04 FES = 233100 μ_F = 0.276 μ_{CR} = 0.047
 Diversity = 4.1517e+01 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 1.932433e+04 FES = 234100 μ_F = 0.223 μ_{CR} = 0.031
 Diversity = 4.1516e+01 k_t = [2, 2, 4, 4]
 Generation 2350: Best = 1.932433e+04 FES = 235100 μ_F = 0.207 μ_{CR} = 0.032
 Diversity = 4.1517e+01 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 1.932433e+04 FES = 236100 μ_F = 0.192 μ_{CR} = 0.024
 Diversity = 4.1512e+01 k_t = [3, 4, 2, 2]
 Generation 2370: Best = 1.903277e+04 FES = 237100 μ_F = 0.111 μ_{CR} = 0.038
 Diversity = 4.1510e+01 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 1.903277e+04 FES = 238100 μ_F = 0.135 μ_{CR} = 0.044
 Diversity = 4.1493e+01 k_t = [4, 3, 3, 2]

Generation 2390: Best = 1.903277e+04 FES = 239100 μ_F = 0.122 μ_{CR} = 0.048
 Diversity = 4.1486e+01 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 1.897144e+04 FES = 240100 μ_F = 0.175 μ_{CR} = 0.045
 Diversity = 4.1431e+01 k_t = [4, 2, 4, 4]
 Generation 2410: Best = 1.896675e+04 FES = 241100 μ_F = 0.147 μ_{CR} = 0.039
 Diversity = 4.1429e+01 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 1.896675e+04 FES = 242100 μ_F = 0.259 μ_{CR} = 0.040
 Diversity = 4.1111e+01 k_t = [4, 2, 2, 4]
 Generation 2430: Best = 1.896664e+04 FES = 243100 μ_F = 0.250 μ_{CR} = 0.038
 Diversity = 4.1110e+01 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 1.894682e+04 FES = 244100 μ_F = 0.136 μ_{CR} = 0.053
 Diversity = 4.1108e+01 k_t = [3, 3, 4, 2]
 Generation 2450: Best = 1.888429e+04 FES = 245100 μ_F = 0.180 μ_{CR} = 0.053
 Diversity = 4.1107e+01 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 1.888429e+04 FES = 246100 μ_F = 0.117 μ_{CR} = 0.047
 Diversity = 4.1105e+01 k_t = [4, 3, 4, 3]
 Generation 2470: Best = 1.888429e+04 FES = 247100 μ_F = 0.079 μ_{CR} = 0.041
 Diversity = 4.1105e+01 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 1.886311e+04 FES = 248100 μ_F = 0.162 μ_{CR} = 0.035
 Diversity = 4.1104e+01 k_t = [4, 2, 4, 2]
 Generation 2490: Best = 1.886311e+04 FES = 249100 μ_F = 0.176 μ_{CR} = 0.046
 Diversity = 4.1103e+01 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 1.872117e+04 FES = 250100 μ_F = 0.207 μ_{CR} = 0.046
 Diversity = 4.1101e+01 k_t = [3, 4, 2, 2]
 Generation 2510: Best = 1.872117e+04 FES = 251100 μ_F = 0.227 μ_{CR} = 0.031
 Diversity = 4.1102e+01 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 1.872117e+04 FES = 252100 μ_F = 0.135 μ_{CR} = 0.033
 Diversity = 4.1098e+01 k_t = [2, 2, 4, 3]
 Generation 2530: Best = 1.872117e+04 FES = 253100 μ_F = 0.178 μ_{CR} = 0.023
 Diversity = 4.1093e+01 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 1.872117e+04 FES = 254100 μ_F = 0.164 μ_{CR} = 0.023
 Diversity = 4.1092e+01 k_t = [2, 4, 2, 4]
 Generation 2550: Best = 1.872117e+04 FES = 255100 μ_F = 0.199 μ_{CR} = 0.021
 Diversity = 4.1088e+01 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 1.838297e+04 FES = 256100 μ_F = 0.259 μ_{CR} = 0.028
 Diversity = 4.1007e+01 k_t = [3, 2, 4, 4]
 Generation 2570: Best = 1.838297e+04 FES = 257100 μ_F = 0.363 μ_{CR} = 0.033
 Diversity = 4.0887e+01 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 1.838297e+04 FES = 258100 μ_F = 0.296 μ_{CR} = 0.035
 Diversity = 4.0886e+01 k_t = [2, 3, 3, 3]
 Generation 2590: Best = 1.838297e+04 FES = 259100 μ_F = 0.306 μ_{CR} = 0.031
 Diversity = 4.0885e+01 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 1.838297e+04 FES = 260100 μ_F = 0.275 μ_{CR} = 0.038
 Diversity = 4.0884e+01 k_t = [4, 3, 2, 2]
 Generation 2610: Best = 1.838297e+04 FES = 261100 μ_F = 0.272 μ_{CR} = 0.034
 Diversity = 4.0886e+01 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 1.833712e+04 FES = 262100 μ_F = 0.218 μ_{CR} = 0.040
 Diversity = 4.0886e+01 k_t = [3, 3, 3, 3]
 Generation 2630: Best = 1.824637e+04 FES = 263100 μ_F = 0.161 μ_{CR} = 0.042
 Diversity = 4.0885e+01 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 1.820828e+04 FES = 264100 μ_F = 0.147 μ_{CR} = 0.044
 Diversity = 4.0884e+01 k_t = [4, 4, 2, 3]
 Generation 2650: Best = 1.817614e+04 FES = 265100 μ_F = 0.198 μ_{CR} = 0.040
 Diversity = 4.0911e+01 k_t = [1, 1, 1, 1]

Generation 2660: Best = 1.817614e+04 FES = 266100 μ_F = 0.154 μ_{CR} = 0.035
 Diversity = 4.0911e+01 k_t = [2, 3, 4, 2]
 Generation 2670: Best = 1.817614e+04 FES = 267100 μ_F = 0.187 μ_{CR} = 0.024
 Diversity = 4.0910e+01 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 1.817405e+04 FES = 268100 μ_F = 0.280 μ_{CR} = 0.023
 Diversity = 4.0742e+01 k_t = [4, 4, 3, 4]
 Generation 2690: Best = 1.817405e+04 FES = 269100 μ_F = 0.290 μ_{CR} = 0.022
 Diversity = 4.0741e+01 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 1.817405e+04 FES = 270100 μ_F = 0.245 μ_{CR} = 0.023
 Diversity = 4.0743e+01 k_t = [2, 4, 4, 2]
 Generation 2710: Best = 1.817405e+04 FES = 271100 μ_F = 0.274 μ_{CR} = 0.023
 Diversity = 4.0862e+01 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 1.809585e+04 FES = 272100 μ_F = 0.255 μ_{CR} = 0.029
 Diversity = 4.0777e+01 k_t = [4, 3, 3, 4]
 Generation 2730: Best = 1.809585e+04 FES = 273100 μ_F = 0.168 μ_{CR} = 0.051
 Diversity = 4.0775e+01 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 1.804774e+04 FES = 274100 μ_F = 0.191 μ_{CR} = 0.054
 Diversity = 4.0865e+01 k_t = [4, 4, 4, 4]
 Generation 2750: Best = 1.804774e+04 FES = 275100 μ_F = 0.145 μ_{CR} = 0.046
 Diversity = 4.0863e+01 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 1.804774e+04 FES = 276100 μ_F = 0.118 μ_{CR} = 0.042
 Diversity = 4.0862e+01 k_t = [2, 2, 2, 3]
 Generation 2770: Best = 1.804774e+04 FES = 277100 μ_F = 0.116 μ_{CR} = 0.049
 Diversity = 4.0858e+01 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 1.799506e+04 FES = 278100 μ_F = 0.126 μ_{CR} = 0.049
 Diversity = 4.0857e+01 k_t = [4, 4, 2, 3]
 Generation 2790: Best = 1.799506e+04 FES = 279100 μ_F = 0.145 μ_{CR} = 0.056
 Diversity = 4.0856e+01 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 1.799506e+04 FES = 280100 μ_F = 0.093 μ_{CR} = 0.068
 Diversity = 4.0852e+01 k_t = [3, 4, 2, 2]
 Generation 2810: Best = 1.799506e+04 FES = 281100 μ_F = 0.079 μ_{CR} = 0.066
 Diversity = 4.0845e+01 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 1.786893e+04 FES = 282100 μ_F = 0.127 μ_{CR} = 0.065
 Diversity = 4.0765e+01 k_t = [3, 4, 4, 4]
 Generation 2830: Best = 1.786893e+04 FES = 283100 μ_F = 0.107 μ_{CR} = 0.044
 Diversity = 4.0762e+01 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 1.786893e+04 FES = 284100 μ_F = 0.190 μ_{CR} = 0.056
 Diversity = 4.0759e+01 k_t = [4, 3, 3, 2]
 Generation 2850: Best = 1.774570e+04 FES = 285100 μ_F = 0.147 μ_{CR} = 0.038
 Diversity = 4.0761e+01 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 1.774354e+04 FES = 286100 μ_F = 0.154 μ_{CR} = 0.034
 Diversity = 4.0759e+01 k_t = [3, 4, 4, 2]
 Generation 2870: Best = 1.773918e+04 FES = 287100 μ_F = 0.114 μ_{CR} = 0.044
 Diversity = 4.0759e+01 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 1.773918e+04 FES = 288100 μ_F = 0.128 μ_{CR} = 0.038
 Diversity = 4.0756e+01 k_t = [3, 4, 4, 2]
 Generation 2890: Best = 1.773918e+04 FES = 289100 μ_F = 0.223 μ_{CR} = 0.037
 Diversity = 4.0754e+01 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 1.772506e+04 FES = 290100 μ_F = 0.269 μ_{CR} = 0.034
 Diversity = 4.0755e+01 k_t = [3, 4, 4, 2]
 Generation 2910: Best = 1.772506e+04 FES = 291100 μ_F = 0.160 μ_{CR} = 0.036
 Diversity = 4.0755e+01 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 1.772506e+04 FES = 292100 μ_F = 0.194 μ_{CR} = 0.051
 Diversity = 4.0634e+01 k_t = [3, 2, 4, 4]

Generation 2930: Best = 1.772506e+04 FES = 293100 μ_F = 0.138 μ_{CR} = 0.050
 Diversity = 4.0634e+01 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 1.772506e+04 FES = 294100 μ_F = 0.176 μ_{CR} = 0.056
 Diversity = 4.0659e+01 k_t = [4, 4, 2, 4]
 Generation 2950: Best = 1.772506e+04 FES = 295100 μ_F = 0.169 μ_{CR} = 0.052
 Diversity = 4.0659e+01 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 1.772506e+04 FES = 296100 μ_F = 0.274 μ_{CR} = 0.052
 Diversity = 4.0465e+01 k_t = [3, 4, 4, 2]
 Generation 2970: Best = 1.772506e+04 FES = 297100 μ_F = 0.252 μ_{CR} = 0.039
 Diversity = 4.0466e+01 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 1.772506e+04 FES = 298100 μ_F = 0.218 μ_{CR} = 0.042
 Diversity = 4.0464e+01 k_t = [4, 4, 2, 3]
 Generation 2990: Best = 1.772506e+04 FES = 299100 μ_F = 0.162 μ_{CR} = 0.042
 Diversity = 4.0466e+01 k_t = [1, 1, 1, 1]
 Generation 3000: Best = 1.772506e+04 FES = 300100 μ_F = 0.205 μ_{CR} = 0.036
 Diversity = 4.0599e+01 k_t = [4, 2, 3, 4]
 Generation 3010: Best = 1.772506e+04 FES = 301100 μ_F = 0.278 μ_{CR} = 0.033
 Diversity = 4.0694e+01 k_t = [1, 1, 1, 1]
 Generation 3020: Best = 1.772506e+04 FES = 302100 μ_F = 0.255 μ_{CR} = 0.033
 Diversity = 4.0691e+01 k_t = [4, 4, 2, 3]
 Generation 3030: Best = 1.772506e+04 FES = 303100 μ_F = 0.263 μ_{CR} = 0.035
 Diversity = 4.0691e+01 k_t = [1, 1, 1, 1]
 Generation 3040: Best = 1.772506e+04 FES = 304100 μ_F = 0.281 μ_{CR} = 0.034
 Diversity = 4.0464e+01 k_t = [4, 4, 4, 4]
 Generation 3050: Best = 1.772506e+04 FES = 305100 μ_F = 0.291 μ_{CR} = 0.031
 Diversity = 4.0463e+01 k_t = [1, 1, 1, 1]
 Generation 3060: Best = 1.772506e+04 FES = 306100 μ_F = 0.283 μ_{CR} = 0.038
 Diversity = 4.0464e+01 k_t = [4, 3, 4, 4]
 Generation 3070: Best = 1.772506e+04 FES = 307100 μ_F = 0.232 μ_{CR} = 0.032
 Diversity = 4.0463e+01 k_t = [1, 1, 1, 1]
 Generation 3080: Best = 1.772506e+04 FES = 308100 μ_F = 0.157 μ_{CR} = 0.026
 Diversity = 4.0463e+01 k_t = [2, 4, 3, 3]
 Generation 3090: Best = 1.772506e+04 FES = 309100 μ_F = 0.136 μ_{CR} = 0.026
 Diversity = 4.0465e+01 k_t = [1, 1, 1, 1]
 Generation 3100: Best = 1.772506e+04 FES = 310100 μ_F = 0.219 μ_{CR} = 0.033
 Diversity = 4.0441e+01 k_t = [2, 4, 3, 2]
 Generation 3110: Best = 1.772506e+04 FES = 311100 μ_F = 0.218 μ_{CR} = 0.026
 Diversity = 4.0440e+01 k_t = [1, 1, 1, 1]
 Generation 3120: Best = 1.772506e+04 FES = 312100 μ_F = 0.248 μ_{CR} = 0.037
 Diversity = 4.0443e+01 k_t = [2, 4, 2, 2]
 Generation 3130: Best = 1.772506e+04 FES = 313100 μ_F = 0.200 μ_{CR} = 0.045
 Diversity = 4.0441e+01 k_t = [1, 1, 1, 1]
 Generation 3140: Best = 1.771893e+04 FES = 314100 μ_F = 0.251 μ_{CR} = 0.041
 Diversity = 4.0441e+01 k_t = [4, 3, 4, 3]
 Generation 3150: Best = 1.771893e+04 FES = 315100 μ_F = 0.241 μ_{CR} = 0.030
 Diversity = 4.0441e+01 k_t = [1, 1, 1, 1]
 Generation 3160: Best = 1.771893e+04 FES = 316100 μ_F = 0.287 μ_{CR} = 0.034
 Diversity = 4.0441e+01 k_t = [3, 4, 2, 3]
 Generation 3170: Best = 1.771870e+04 FES = 317100 μ_F = 0.281 μ_{CR} = 0.035
 Diversity = 4.0444e+01 k_t = [1, 1, 1, 1]
 Generation 3180: Best = 1.771870e+04 FES = 318100 μ_F = 0.195 μ_{CR} = 0.050
 Diversity = 4.0441e+01 k_t = [2, 3, 3, 2]
 Generation 3190: Best = 1.756320e+04 FES = 319100 μ_F = 0.174 μ_{CR} = 0.042
 Diversity = 4.0441e+01 k_t = [1, 1, 1, 1]

Generation 3200: Best = 1.756320e+04 FES = 320100 μ_F = 0.243 μ_{CR} = 0.028
 Diversity = 4.0461e+01 k_t = [3, 3, 4, 4]
 Generation 3210: Best = 1.754768e+04 FES = 321100 μ_F = 0.164 μ_{CR} = 0.043
 Diversity = 4.0460e+01 k_t = [1, 1, 1, 1]
 Generation 3220: Best = 1.746508e+04 FES = 322100 μ_F = 0.192 μ_{CR} = 0.051
 Diversity = 4.0461e+01 k_t = [2, 3, 4, 2]
 Generation 3230: Best = 1.746508e+04 FES = 323100 μ_F = 0.161 μ_{CR} = 0.051
 Diversity = 4.0461e+01 k_t = [1, 1, 1, 1]
 Generation 3240: Best = 1.746508e+04 FES = 324100 μ_F = 0.194 μ_{CR} = 0.054
 Diversity = 4.0459e+01 k_t = [4, 2, 2, 2]
 Generation 3250: Best = 1.746508e+04 FES = 325100 μ_F = 0.195 μ_{CR} = 0.074
 Diversity = 4.0458e+01 k_t = [1, 1, 1, 1]
 Generation 3260: Best = 1.746243e+04 FES = 326100 μ_F = 0.145 μ_{CR} = 0.060
 Diversity = 4.0457e+01 k_t = [3, 3, 4, 2]
 Generation 3270: Best = 1.746243e+04 FES = 327100 μ_F = 0.155 μ_{CR} = 0.057
 Diversity = 4.0451e+01 k_t = [1, 1, 1, 1]
 Generation 3280: Best = 1.746243e+04 FES = 328100 μ_F = 0.103 μ_{CR} = 0.051
 Diversity = 4.0449e+01 k_t = [3, 4, 2, 3]
 Generation 3290: Best = 1.746243e+04 FES = 329100 μ_F = 0.145 μ_{CR} = 0.054
 Diversity = 4.0445e+01 k_t = [1, 1, 1, 1]
 Generation 3300: Best = 1.746243e+04 FES = 330100 μ_F = 0.131 μ_{CR} = 0.043
 Diversity = 4.0446e+01 k_t = [2, 4, 4, 3]
 Generation 3310: Best = 1.746243e+04 FES = 331100 μ_F = 0.094 μ_{CR} = 0.053
 Diversity = 4.0443e+01 k_t = [1, 1, 1, 1]
 Generation 3320: Best = 1.746243e+04 FES = 332100 μ_F = 0.092 μ_{CR} = 0.053
 Diversity = 4.0439e+01 k_t = [2, 2, 2, 3]
 Generation 3330: Best = 1.736476e+04 FES = 333100 μ_F = 0.135 μ_{CR} = 0.050
 Diversity = 4.0263e+01 k_t = [1, 1, 1, 1]
 Generation 3340: Best = 1.736476e+04 FES = 334100 μ_F = 0.124 μ_{CR} = 0.040
 Diversity = 4.0256e+01 k_t = [2, 3, 3, 2]
 Generation 3350: Best = 1.736476e+04 FES = 335100 μ_F = 0.074 μ_{CR} = 0.052
 Diversity = 4.0252e+01 k_t = [1, 1, 1, 1]
 Generation 3360: Best = 1.727521e+04 FES = 336100 μ_F = 0.061 μ_{CR} = 0.066
 Diversity = 4.0245e+01 k_t = [3, 3, 2, 3]
 Generation 3370: Best = 1.727521e+04 FES = 337100 μ_F = 0.069 μ_{CR} = 0.053
 Diversity = 4.0242e+01 k_t = [1, 1, 1, 1]
 Generation 3380: Best = 1.725859e+04 FES = 338100 μ_F = 0.107 μ_{CR} = 0.047
 Diversity = 4.0239e+01 k_t = [3, 3, 2, 2]
 Generation 3390: Best = 1.725859e+04 FES = 339100 μ_F = 0.068 μ_{CR} = 0.045
 Diversity = 4.0238e+01 k_t = [1, 1, 1, 1]
 Generation 3400: Best = 1.725859e+04 FES = 340100 μ_F = 0.092 μ_{CR} = 0.034
 Diversity = 4.0234e+01 k_t = [3, 3, 4, 3]
 Generation 3410: Best = 1.725859e+04 FES = 341100 μ_F = 0.131 μ_{CR} = 0.048
 Diversity = 4.0272e+01 k_t = [1, 1, 1, 1]
 Generation 3420: Best = 1.725859e+04 FES = 342100 μ_F = 0.135 μ_{CR} = 0.039
 Diversity = 4.0270e+01 k_t = [2, 4, 3, 3]
 Generation 3430: Best = 1.725859e+04 FES = 343100 μ_F = 0.102 μ_{CR} = 0.041
 Diversity = 4.0267e+01 k_t = [1, 1, 1, 1]
 Generation 3440: Best = 1.719045e+04 FES = 344100 μ_F = 0.089 μ_{CR} = 0.047
 Diversity = 4.0265e+01 k_t = [2, 3, 3, 2]
 Generation 3450: Best = 1.719045e+04 FES = 345100 μ_F = 0.087 μ_{CR} = 0.037
 Diversity = 4.0263e+01 k_t = [1, 1, 1, 1]
 Generation 3460: Best = 1.719045e+04 FES = 346100 μ_F = 0.170 μ_{CR} = 0.040
 Diversity = 4.0238e+01 k_t = [3, 4, 4, 3]

Generation 3470: Best = 1.719045e+04 FES = 347100 μ_F = 0.197 μ_{CR} = 0.039
 Diversity = 4.0243e+01 k_t = [1, 1, 1, 1]
 Generation 3480: Best = 1.719045e+04 FES = 348100 μ_F = 0.275 μ_{CR} = 0.031
 Diversity = 4.0276e+01 k_t = [2, 4, 4, 4]
 Generation 3490: Best = 1.719045e+04 FES = 349100 μ_F = 0.279 μ_{CR} = 0.027
 Diversity = 4.0276e+01 k_t = [1, 1, 1, 1]
 Generation 3500: Best = 1.712993e+04 FES = 350100 μ_F = 0.261 μ_{CR} = 0.027
 Diversity = 4.0276e+01 k_t = [4, 4, 4, 2]
 Generation 3510: Best = 1.712993e+04 FES = 351100 μ_F = 0.298 μ_{CR} = 0.025
 Diversity = 4.0276e+01 k_t = [1, 1, 1, 1]
 Generation 3520: Best = 1.712993e+04 FES = 352100 μ_F = 0.276 μ_{CR} = 0.041
 Diversity = 4.0276e+01 k_t = [4, 4, 4, 2]
 Generation 3530: Best = 1.712993e+04 FES = 353100 μ_F = 0.306 μ_{CR} = 0.044
 Diversity = 4.0230e+01 k_t = [1, 1, 1, 1]
 Generation 3540: Best = 1.712993e+04 FES = 354100 μ_F = 0.264 μ_{CR} = 0.044
 Diversity = 4.0230e+01 k_t = [3, 3, 2, 3]
 Generation 3550: Best = 1.712993e+04 FES = 355100 μ_F = 0.214 μ_{CR} = 0.038
 Diversity = 4.0222e+01 k_t = [1, 1, 1, 1]
 Generation 3560: Best = 1.712993e+04 FES = 356100 μ_F = 0.151 μ_{CR} = 0.028
 Diversity = 4.0219e+01 k_t = [2, 3, 3, 3]
 Generation 3570: Best = 1.712993e+04 FES = 357100 μ_F = 0.164 μ_{CR} = 0.044
 Diversity = 4.0218e+01 k_t = [1, 1, 1, 1]
 Generation 3580: Best = 1.712993e+04 FES = 358100 μ_F = 0.126 μ_{CR} = 0.028
 Diversity = 4.0218e+01 k_t = [2, 3, 3, 2]
 Generation 3590: Best = 1.712993e+04 FES = 359100 μ_F = 0.106 μ_{CR} = 0.032
 Diversity = 4.0217e+01 k_t = [1, 1, 1, 1]
 Generation 3600: Best = 1.712993e+04 FES = 360100 μ_F = 0.082 μ_{CR} = 0.045
 Diversity = 4.0218e+01 k_t = [2, 4, 2, 2]
 Generation 3610: Best = 1.712993e+04 FES = 361100 μ_F = 0.052 μ_{CR} = 0.043
 Diversity = 4.0215e+01 k_t = [1, 1, 1, 1]
 Generation 3620: Best = 1.712993e+04 FES = 362100 μ_F = 0.077 μ_{CR} = 0.045
 Diversity = 4.0204e+01 k_t = [3, 3, 4, 3]
 Generation 3630: Best = 1.712993e+04 FES = 363100 μ_F = 0.049 μ_{CR} = 0.057
 Diversity = 4.0199e+01 k_t = [1, 1, 1, 1]
 Generation 3640: Best = 1.712993e+04 FES = 364100 μ_F = 0.073 μ_{CR} = 0.067
 Diversity = 4.0196e+01 k_t = [2, 2, 3, 3]
 Generation 3650: Best = 1.712993e+04 FES = 365100 μ_F = 0.099 μ_{CR} = 0.074
 Diversity = 4.0195e+01 k_t = [1, 1, 1, 1]
 Generation 3660: Best = 1.712993e+04 FES = 366100 μ_F = 0.174 μ_{CR} = 0.058
 Diversity = 4.0321e+01 k_t = [4, 2, 2, 4]
 Generation 3670: Best = 1.712993e+04 FES = 367100 μ_F = 0.109 μ_{CR} = 0.061
 Diversity = 4.0318e+01 k_t = [1, 1, 1, 1]
 Generation 3680: Best = 1.711271e+04 FES = 368100 μ_F = 0.074 μ_{CR} = 0.049
 Diversity = 4.0320e+01 k_t = [3, 4, 3, 4]
 Generation 3690: Best = 1.705080e+04 FES = 369100 μ_F = 0.089 μ_{CR} = 0.043
 Diversity = 4.0317e+01 k_t = [1, 1, 1, 1]
 Generation 3700: Best = 1.705080e+04 FES = 370100 μ_F = 0.086 μ_{CR} = 0.051
 Diversity = 4.0311e+01 k_t = [2, 2, 4, 2]
 Generation 3710: Best = 1.703560e+04 FES = 371100 μ_F = 0.078 μ_{CR} = 0.042
 Diversity = 4.0309e+01 k_t = [1, 1, 1, 1]
 Generation 3720: Best = 1.703560e+04 FES = 372100 μ_F = 0.090 μ_{CR} = 0.036
 Diversity = 4.0306e+01 k_t = [3, 3, 4, 4]
 Generation 3730: Best = 1.703560e+04 FES = 373100 μ_F = 0.076 μ_{CR} = 0.057
 Diversity = 4.0304e+01 k_t = [1, 1, 1, 1]

Generation 3740: Best = 1.703560e+04 FES = 374100 μ_F = 0.095 μ_{CR} = 0.050
 Diversity = 4.0302e+01 k_t = [4, 4, 3, 2]
 Generation 3750: Best = 1.703560e+04 FES = 375100 μ_F = 0.128 μ_{CR} = 0.053
 Diversity = 4.0300e+01 k_t = [1, 1, 1, 1]
 Generation 3760: Best = 1.702899e+04 FES = 376100 μ_F = 0.078 μ_{CR} = 0.057
 Diversity = 4.0296e+01 k_t = [4, 2, 3, 2]
 Generation 3770: Best = 1.702899e+04 FES = 377100 μ_F = 0.146 μ_{CR} = 0.064
 Diversity = 4.0295e+01 k_t = [1, 1, 1, 1]
 Generation 3780: Best = 1.702899e+04 FES = 378100 μ_F = 0.110 μ_{CR} = 0.063
 Diversity = 4.0296e+01 k_t = [4, 4, 3, 2]
 Generation 3790: Best = 1.702899e+04 FES = 379100 μ_F = 0.087 μ_{CR} = 0.068
 Diversity = 4.0293e+01 k_t = [1, 1, 1, 1]
 Generation 3800: Best = 1.695321e+04 FES = 380100 μ_F = 0.094 μ_{CR} = 0.057
 Diversity = 4.0291e+01 k_t = [3, 2, 2, 2]
 Generation 3810: Best = 1.695321e+04 FES = 381100 μ_F = 0.097 μ_{CR} = 0.045
 Diversity = 4.0289e+01 k_t = [1, 1, 1, 1]
 Generation 3820: Best = 1.695321e+04 FES = 382100 μ_F = 0.090 μ_{CR} = 0.041
 Diversity = 4.0287e+01 k_t = [3, 2, 3, 3]
 Generation 3830: Best = 1.695321e+04 FES = 383100 μ_F = 0.106 μ_{CR} = 0.040
 Diversity = 4.0288e+01 k_t = [1, 1, 1, 1]
 Generation 3840: Best = 1.694975e+04 FES = 384100 μ_F = 0.077 μ_{CR} = 0.042
 Diversity = 4.0286e+01 k_t = [2, 2, 2, 2]
 Generation 3850: Best = 1.694975e+04 FES = 385100 μ_F = 0.101 μ_{CR} = 0.035
 Diversity = 4.0287e+01 k_t = [1, 1, 1, 1]
 Generation 3860: Best = 1.694975e+04 FES = 386100 μ_F = 0.092 μ_{CR} = 0.058
 Diversity = 4.0285e+01 k_t = [2, 3, 4, 2]
 Generation 3870: Best = 1.694975e+04 FES = 387100 μ_F = 0.080 μ_{CR} = 0.058
 Diversity = 4.0282e+01 k_t = [1, 1, 1, 1]
 Generation 3880: Best = 1.694975e+04 FES = 388100 μ_F = 0.071 μ_{CR} = 0.047
 Diversity = 4.0279e+01 k_t = [3, 2, 4, 3]
 Generation 3890: Best = 1.694975e+04 FES = 389100 μ_F = 0.095 μ_{CR} = 0.060
 Diversity = 4.0283e+01 k_t = [1, 1, 1, 1]
 Generation 3900: Best = 1.694975e+04 FES = 390100 μ_F = 0.234 μ_{CR} = 0.047
 Diversity = 3.9992e+01 k_t = [4, 4, 4, 4]
 Generation 3910: Best = 1.694975e+04 FES = 391100 μ_F = 0.258 μ_{CR} = 0.050
 Diversity = 3.9992e+01 k_t = [1, 1, 1, 1]
 Generation 3920: Best = 1.694975e+04 FES = 392100 μ_F = 0.358 μ_{CR} = 0.046
 Diversity = 3.9993e+01 k_t = [2, 3, 2, 2]
 Generation 3930: Best = 1.694975e+04 FES = 393100 μ_F = 0.313 μ_{CR} = 0.041
 Diversity = 3.9991e+01 k_t = [1, 1, 1, 1]
 Generation 3940: Best = 1.694975e+04 FES = 394100 μ_F = 0.260 μ_{CR} = 0.042
 Diversity = 3.9991e+01 k_t = [4, 2, 3, 4]
 Generation 3950: Best = 1.694975e+04 FES = 395100 μ_F = 0.232 μ_{CR} = 0.046
 Diversity = 3.9990e+01 k_t = [1, 1, 1, 1]
 Generation 3960: Best = 1.694975e+04 FES = 396100 μ_F = 0.248 μ_{CR} = 0.039
 Diversity = 3.9990e+01 k_t = [2, 3, 4, 2]
 Generation 3970: Best = 1.694975e+04 FES = 397100 μ_F = 0.236 μ_{CR} = 0.031
 Diversity = 3.9991e+01 k_t = [1, 1, 1, 1]
 Generation 3980: Best = 1.694975e+04 FES = 398100 μ_F = 0.249 μ_{CR} = 0.029
 Diversity = 3.9990e+01 k_t = [3, 4, 3, 2]
 Generation 3990: Best = 1.694975e+04 FES = 399100 μ_F = 0.191 μ_{CR} = 0.025
 Diversity = 3.9991e+01 k_t = [1, 1, 1, 1]
 Generation 4000: Best = 1.694975e+04 FES = 400100 μ_F = 0.268 μ_{CR} = 0.023
 Diversity = 3.9992e+01 k_t = [2, 3, 4, 4]

Generation 4010: Best = 1.694975e+04 FES = 401100 μ_F = 0.277 μ_{CR} = 0.028
 Diversity = 3.9863e+01 k_t = [1, 1, 1, 1]
 Generation 4020: Best = 1.694975e+04 FES = 402100 μ_F = 0.300 μ_{CR} = 0.028
 Diversity = 3.9864e+01 k_t = [2, 3, 4, 4]
 Generation 4030: Best = 1.694975e+04 FES = 403100 μ_F = 0.245 μ_{CR} = 0.024
 Diversity = 3.9865e+01 k_t = [1, 1, 1, 1]
 Generation 4040: Best = 1.694975e+04 FES = 404100 μ_F = 0.209 μ_{CR} = 0.028
 Diversity = 3.9865e+01 k_t = [4, 2, 2, 2]
 Generation 4050: Best = 1.694975e+04 FES = 405100 μ_F = 0.255 μ_{CR} = 0.034
 Diversity = 3.9864e+01 k_t = [1, 1, 1, 1]
 Generation 4060: Best = 1.694975e+04 FES = 406100 μ_F = 0.251 μ_{CR} = 0.039
 Diversity = 3.9776e+01 k_t = [3, 3, 3, 2]
 Generation 4070: Best = 1.694975e+04 FES = 407100 μ_F = 0.187 μ_{CR} = 0.043
 Diversity = 3.9776e+01 k_t = [1, 1, 1, 1]
 Generation 4080: Best = 1.687064e+04 FES = 408100 μ_F = 0.191 μ_{CR} = 0.038
 Diversity = 3.9775e+01 k_t = [2, 4, 2, 2]
 Generation 4090: Best = 1.687064e+04 FES = 409100 μ_F = 0.121 μ_{CR} = 0.072
 Diversity = 3.9774e+01 k_t = [1, 1, 1, 1]
 Generation 4100: Best = 1.687064e+04 FES = 410100 μ_F = 0.085 μ_{CR} = 0.070
 Diversity = 3.9768e+01 k_t = [3, 3, 2, 3]
 Generation 4110: Best = 1.687064e+04 FES = 411100 μ_F = 0.066 μ_{CR} = 0.093
 Diversity = 3.9766e+01 k_t = [1, 1, 1, 1]
 Generation 4120: Best = 1.682415e+04 FES = 412100 μ_F = 0.123 μ_{CR} = 0.071
 Diversity = 3.9630e+01 k_t = [4, 2, 4, 4]
 Generation 4130: Best = 1.682415e+04 FES = 413100 μ_F = 0.083 μ_{CR} = 0.064
 Diversity = 3.9629e+01 k_t = [1, 1, 1, 1]
 Generation 4140: Best = 1.682415e+04 FES = 414100 μ_F = 0.109 μ_{CR} = 0.058
 Diversity = 3.9628e+01 k_t = [2, 2, 2, 2]
 Generation 4150: Best = 1.682415e+04 FES = 415100 μ_F = 0.077 μ_{CR} = 0.043
 Diversity = 3.9628e+01 k_t = [1, 1, 1, 1]
 Generation 4160: Best = 1.663460e+04 FES = 416100 μ_F = 0.126 μ_{CR} = 0.058
 Diversity = 3.9627e+01 k_t = [4, 3, 4, 3]
 Generation 4170: Best = 1.663460e+04 FES = 417100 μ_F = 0.113 μ_{CR} = 0.053
 Diversity = 3.9626e+01 k_t = [1, 1, 1, 1]
 Generation 4180: Best = 1.662843e+04 FES = 418100 μ_F = 0.190 μ_{CR} = 0.058
 Diversity = 3.9586e+01 k_t = [3, 3, 2, 2]
 Generation 4190: Best = 1.662843e+04 FES = 419100 μ_F = 0.152 μ_{CR} = 0.053
 Diversity = 3.9587e+01 k_t = [1, 1, 1, 1]
 Generation 4200: Best = 1.662843e+04 FES = 420100 μ_F = 0.201 μ_{CR} = 0.049
 Diversity = 3.9586e+01 k_t = [4, 4, 4, 3]
 Generation 4210: Best = 1.662625e+04 FES = 421100 μ_F = 0.195 μ_{CR} = 0.034
 Diversity = 3.9587e+01 k_t = [1, 1, 1, 1]
 Generation 4220: Best = 1.662625e+04 FES = 422100 μ_F = 0.217 μ_{CR} = 0.049
 Diversity = 3.9590e+01 k_t = [3, 3, 3, 4]
 Generation 4230: Best = 1.662625e+04 FES = 423100 μ_F = 0.254 μ_{CR} = 0.039
 Diversity = 3.9589e+01 k_t = [1, 1, 1, 1]
 Generation 4240: Best = 1.650367e+04 FES = 424100 μ_F = 0.238 μ_{CR} = 0.035
 Diversity = 3.9590e+01 k_t = [2, 4, 4, 3]
 Generation 4250: Best = 1.643035e+04 FES = 425100 μ_F = 0.267 μ_{CR} = 0.039
 Diversity = 3.9590e+01 k_t = [1, 1, 1, 1]
 Generation 4260: Best = 1.631084e+04 FES = 426100 μ_F = 0.221 μ_{CR} = 0.043
 Diversity = 3.9590e+01 k_t = [4, 3, 4, 2]
 Generation 4270: Best = 1.629346e+04 FES = 427100 μ_F = 0.169 μ_{CR} = 0.031
 Diversity = 3.9589e+01 k_t = [1, 1, 1, 1]

Generation 4280: Best = 1.629346e+04 FES = 428100 μ_F = 0.136 μ_{CR} = 0.029
 Diversity = 3.9589e+01 k_t = [4, 4, 3, 2]
 Generation 4290: Best = 1.629346e+04 FES = 429100 μ_F = 0.107 μ_{CR} = 0.023
 Diversity = 3.9589e+01 k_t = [1, 1, 1, 1]
 Generation 4300: Best = 1.628245e+04 FES = 430100 μ_F = 0.143 μ_{CR} = 0.025
 Diversity = 3.9587e+01 k_t = [2, 3, 2, 4]
 Generation 4310: Best = 1.608060e+04 FES = 431100 μ_F = 0.216 μ_{CR} = 0.038
 Diversity = 3.9476e+01 k_t = [1, 1, 1, 1]
 Generation 4320: Best = 1.608060e+04 FES = 432100 μ_F = 0.266 μ_{CR} = 0.036
 Diversity = 3.9290e+01 k_t = [2, 4, 3, 4]
 Generation 4330: Best = 1.608060e+04 FES = 433100 μ_F = 0.295 μ_{CR} = 0.023
 Diversity = 3.9289e+01 k_t = [1, 1, 1, 1]
 Generation 4340: Best = 1.608060e+04 FES = 434100 μ_F = 0.274 μ_{CR} = 0.020
 Diversity = 3.9289e+01 k_t = [3, 3, 2, 3]
 Generation 4350: Best = 1.608060e+04 FES = 435100 μ_F = 0.258 μ_{CR} = 0.023
 Diversity = 3.9289e+01 k_t = [1, 1, 1, 1]
 Generation 4360: Best = 1.608060e+04 FES = 436100 μ_F = 0.319 μ_{CR} = 0.032
 Diversity = 3.9372e+01 k_t = [2, 2, 3, 4]
 Generation 4370: Best = 1.608060e+04 FES = 437100 μ_F = 0.255 μ_{CR} = 0.029
 Diversity = 3.9372e+01 k_t = [1, 1, 1, 1]
 Generation 4380: Best = 1.608060e+04 FES = 438100 μ_F = 0.179 μ_{CR} = 0.039
 Diversity = 3.9372e+01 k_t = [2, 3, 4, 2]
 Generation 4390: Best = 1.608060e+04 FES = 439100 μ_F = 0.134 μ_{CR} = 0.043
 Diversity = 3.9372e+01 k_t = [1, 1, 1, 1]
 Generation 4400: Best = 1.608060e+04 FES = 440100 μ_F = 0.099 μ_{CR} = 0.042
 Diversity = 3.9371e+01 k_t = [2, 3, 3, 4]
 Generation 4410: Best = 1.608060e+04 FES = 441100 μ_F = 0.056 μ_{CR} = 0.065
 Diversity = 3.9371e+01 k_t = [1, 1, 1, 1]
 Generation 4420: Best = 1.608060e+04 FES = 442100 μ_F = 0.124 μ_{CR} = 0.085
 Diversity = 3.9350e+01 k_t = [2, 4, 4, 4]
 Generation 4430: Best = 1.608060e+04 FES = 443100 μ_F = 0.173 μ_{CR} = 0.073
 Diversity = 3.9349e+01 k_t = [1, 1, 1, 1]
 Generation 4440: Best = 1.568582e+04 FES = 444100 μ_F = 0.295 μ_{CR} = 0.059
 Diversity = 3.9240e+01 k_t = [4, 4, 3, 4]
 Generation 4450: Best = 1.561437e+04 FES = 445100 μ_F = 0.347 μ_{CR} = 0.052
 Diversity = 3.9240e+01 k_t = [1, 1, 1, 1]
 Generation 4460: Best = 1.561437e+04 FES = 446100 μ_F = 0.369 μ_{CR} = 0.053
 Diversity = 3.9239e+01 k_t = [4, 3, 4, 4]
 Generation 4470: Best = 1.555602e+04 FES = 447100 μ_F = 0.316 μ_{CR} = 0.045
 Diversity = 3.9238e+01 k_t = [1, 1, 1, 1]
 Generation 4480: Best = 1.555602e+04 FES = 448100 μ_F = 0.271 μ_{CR} = 0.041
 Diversity = 3.9238e+01 k_t = [3, 2, 2, 3]
 Generation 4490: Best = 1.546320e+04 FES = 449100 μ_F = 0.223 μ_{CR} = 0.047
 Diversity = 3.9239e+01 k_t = [1, 1, 1, 1]
 Generation 4500: Best = 1.539990e+04 FES = 450100 μ_F = 0.193 μ_{CR} = 0.040
 Diversity = 3.9239e+01 k_t = [3, 4, 3, 3]
 Generation 4510: Best = 1.538209e+04 FES = 451100 μ_F = 0.157 μ_{CR} = 0.062
 Diversity = 3.9239e+01 k_t = [1, 1, 1, 1]
 Generation 4520: Best = 1.538132e+04 FES = 452100 μ_F = 0.100 μ_{CR} = 0.054
 Diversity = 3.9235e+01 k_t = [2, 3, 3, 2]
 Generation 4530: Best = 1.537898e+04 FES = 453100 μ_F = 0.113 μ_{CR} = 0.070
 Diversity = 3.9235e+01 k_t = [1, 1, 1, 1]
 Generation 4540: Best = 1.537898e+04 FES = 454100 μ_F = 0.131 μ_{CR} = 0.066
 Diversity = 3.9235e+01 k_t = [2, 4, 3, 2]

Generation 4550: Best = 1.537898e+04 FES = 455100 μ_F = 0.110 μ_{CR} = 0.061
 Diversity = 3.9236e+01 k_t = [1, 1, 1, 1]
 Generation 4560: Best = 1.537898e+04 FES = 456100 μ_F = 0.165 μ_{CR} = 0.072
 Diversity = 3.9121e+01 k_t = [3, 2, 4, 4]
 Generation 4570: Best = 1.537898e+04 FES = 457100 μ_F = 0.175 μ_{CR} = 0.069
 Diversity = 3.9121e+01 k_t = [1, 1, 1, 1]
 Generation 4580: Best = 1.530656e+04 FES = 458100 μ_F = 0.198 μ_{CR} = 0.042
 Diversity = 3.9120e+01 k_t = [4, 4, 2, 2]
 Generation 4590: Best = 1.530656e+04 FES = 459100 μ_F = 0.182 μ_{CR} = 0.036
 Diversity = 3.9119e+01 k_t = [1, 1, 1, 1]
 Generation 4600: Best = 1.530656e+04 FES = 460100 μ_F = 0.126 μ_{CR} = 0.043
 Diversity = 3.9112e+01 k_t = [2, 3, 4, 2]
 Generation 4610: Best = 1.527394e+04 FES = 461100 μ_F = 0.091 μ_{CR} = 0.050
 Diversity = 3.9109e+01 k_t = [1, 1, 1, 1]
 Generation 4620: Best = 1.527394e+04 FES = 462100 μ_F = 0.198 μ_{CR} = 0.042
 Diversity = 3.8916e+01 k_t = [4, 3, 4, 4]
 Generation 4630: Best = 1.527394e+04 FES = 463100 μ_F = 0.182 μ_{CR} = 0.032
 Diversity = 3.8916e+01 k_t = [1, 1, 1, 1]
 Generation 4640: Best = 1.527394e+04 FES = 464100 μ_F = 0.147 μ_{CR} = 0.031
 Diversity = 3.8915e+01 k_t = [2, 3, 3, 3]
 Generation 4650: Best = 1.527394e+04 FES = 465100 μ_F = 0.104 μ_{CR} = 0.047
 Diversity = 3.8912e+01 k_t = [1, 1, 1, 1]
 Generation 4660: Best = 1.527394e+04 FES = 466100 μ_F = 0.183 μ_{CR} = 0.056
 Diversity = 3.8834e+01 k_t = [3, 3, 4, 4]
 Generation 4670: Best = 1.527394e+04 FES = 467100 μ_F = 0.192 μ_{CR} = 0.051
 Diversity = 3.8834e+01 k_t = [1, 1, 1, 1]
 Generation 4680: Best = 1.525975e+04 FES = 468100 μ_F = 0.218 μ_{CR} = 0.050
 Diversity = 3.8725e+01 k_t = [4, 3, 2, 4]
 Generation 4690: Best = 1.511426e+04 FES = 469100 μ_F = 0.185 μ_{CR} = 0.051
 Diversity = 3.8723e+01 k_t = [1, 1, 1, 1]
 Generation 4700: Best = 1.511426e+04 FES = 470100 μ_F = 0.258 μ_{CR} = 0.049
 Diversity = 3.8723e+01 k_t = [2, 4, 4, 3]
 Generation 4710: Best = 1.511049e+04 FES = 471100 μ_F = 0.210 μ_{CR} = 0.043
 Diversity = 3.8722e+01 k_t = [1, 1, 1, 1]
 Generation 4720: Best = 1.511049e+04 FES = 472100 μ_F = 0.277 μ_{CR} = 0.041
 Diversity = 3.8696e+01 k_t = [4, 3, 3, 2]
 Generation 4730: Best = 1.511049e+04 FES = 473100 μ_F = 0.203 μ_{CR} = 0.037
 Diversity = 3.8695e+01 k_t = [1, 1, 1, 1]
 Generation 4740: Best = 1.504803e+04 FES = 474100 μ_F = 0.229 μ_{CR} = 0.045
 Diversity = 3.8492e+01 k_t = [3, 4, 4, 4]
 Generation 4750: Best = 1.504803e+04 FES = 475100 μ_F = 0.244 μ_{CR} = 0.031
 Diversity = 3.8491e+01 k_t = [1, 1, 1, 1]
 Generation 4760: Best = 1.504803e+04 FES = 476100 μ_F = 0.299 μ_{CR} = 0.024
 Diversity = 3.8350e+01 k_t = [2, 4, 4, 4]
 Generation 4770: Best = 1.504803e+04 FES = 477100 μ_F = 0.366 μ_{CR} = 0.018
 Diversity = 3.8350e+01 k_t = [1, 1, 1, 1]
 Generation 4780: Best = 1.504355e+04 FES = 478100 μ_F = 0.362 μ_{CR} = 0.019
 Diversity = 3.8351e+01 k_t = [4, 2, 3, 4]
 Generation 4790: Best = 1.504355e+04 FES = 479100 μ_F = 0.301 μ_{CR} = 0.046
 Diversity = 3.8352e+01 k_t = [1, 1, 1, 1]
 Generation 4800: Best = 1.501484e+04 FES = 480100 μ_F = 0.223 μ_{CR} = 0.035
 Diversity = 3.8353e+01 k_t = [4, 2, 2, 2]
 Generation 4810: Best = 1.501484e+04 FES = 481100 μ_F = 0.190 μ_{CR} = 0.034
 Diversity = 3.8352e+01 k_t = [1, 1, 1, 1]

Generation 4820: Best = 1.482369e+04 FES = 482100 μ_F = 0.298 μ_{CR} = 0.039
 Diversity = 3.8346e+01 k_t = [2, 4, 2, 4]
 Generation 4830: Best = 1.475816e+04 FES = 483100 μ_F = 0.353 μ_{CR} = 0.047
 Diversity = 3.8345e+01 k_t = [1, 1, 1, 1]
 Generation 4840: Best = 1.467675e+04 FES = 484100 μ_F = 0.397 μ_{CR} = 0.050
 Diversity = 3.8195e+01 k_t = [4, 4, 2, 2]
 Generation 4850: Best = 1.459231e+04 FES = 485100 μ_F = 0.373 μ_{CR} = 0.063
 Diversity = 3.8195e+01 k_t = [1, 1, 1, 1]
 Generation 4860: Best = 1.459231e+04 FES = 486100 μ_F = 0.400 μ_{CR} = 0.053
 Diversity = 3.8158e+01 k_t = [4, 3, 3, 4]
 Generation 4870: Best = 1.459231e+04 FES = 487100 μ_F = 0.386 μ_{CR} = 0.046
 Diversity = 3.8158e+01 k_t = [1, 1, 1, 1]
 Generation 4880: Best = 1.459231e+04 FES = 488100 μ_F = 0.448 μ_{CR} = 0.037
 Diversity = 3.8090e+01 k_t = [2, 3, 2, 4]
 Generation 4890: Best = 1.459231e+04 FES = 489100 μ_F = 0.421 μ_{CR} = 0.049
 Diversity = 3.8090e+01 k_t = [1, 1, 1, 1]
 Generation 4900: Best = 1.440486e+04 FES = 490100 μ_F = 0.439 μ_{CR} = 0.058
 Diversity = 3.8091e+01 k_t = [4, 4, 3, 3]
 Generation 4910: Best = 1.440486e+04 FES = 491100 μ_F = 0.338 μ_{CR} = 0.048
 Diversity = 3.8090e+01 k_t = [1, 1, 1, 1]
 Generation 4920: Best = 1.440486e+04 FES = 492100 μ_F = 0.393 μ_{CR} = 0.039
 Diversity = 3.7950e+01 k_t = [3, 2, 4, 4]
 Generation 4930: Best = 1.440486e+04 FES = 493100 μ_F = 0.381 μ_{CR} = 0.031
 Diversity = 3.7949e+01 k_t = [1, 1, 1, 1]
 Generation 4940: Best = 1.435360e+04 FES = 494100 μ_F = 0.382 μ_{CR} = 0.028
 Diversity = 3.7972e+01 k_t = [4, 3, 3, 4]
 Generation 4950: Best = 1.435360e+04 FES = 495100 μ_F = 0.343 μ_{CR} = 0.026
 Diversity = 3.7959e+01 k_t = [1, 1, 1, 1]
 Generation 4960: Best = 1.428951e+04 FES = 496100 μ_F = 0.332 μ_{CR} = 0.048
 Diversity = 3.7958e+01 k_t = [2, 2, 2, 3]
 Generation 4970: Best = 1.428951e+04 FES = 497100 μ_F = 0.275 μ_{CR} = 0.042
 Diversity = 3.7958e+01 k_t = [1, 1, 1, 1]
 Generation 4980: Best = 1.428723e+04 FES = 498100 μ_F = 0.263 μ_{CR} = 0.038
 Diversity = 3.7957e+01 k_t = [3, 3, 3, 2]
 Generation 4990: Best = 1.428723e+04 FES = 499100 μ_F = 0.229 μ_{CR} = 0.037
 Diversity = 3.7958e+01 k_t = [1, 1, 1, 1]
 Generation 5000: Best = 1.427116e+04 FES = 500100 μ_F = 0.361 μ_{CR} = 0.040
 Diversity = 3.7739e+01 k_t = [4, 4, 3, 4]
 Generation 5010: Best = 1.427116e+04 FES = 501100 μ_F = 0.329 μ_{CR} = 0.060
 Diversity = 3.7669e+01 k_t = [1, 1, 1, 1]
 Generation 5020: Best = 1.427116e+04 FES = 502100 μ_F = 0.311 μ_{CR} = 0.049
 Diversity = 3.7669e+01 k_t = [3, 2, 2, 3]
 Generation 5030: Best = 1.427116e+04 FES = 503100 μ_F = 0.292 μ_{CR} = 0.058
 Diversity = 3.7669e+01 k_t = [1, 1, 1, 1]
 Generation 5040: Best = 1.427116e+04 FES = 504100 μ_F = 0.351 μ_{CR} = 0.060
 Diversity = 3.7690e+01 k_t = [2, 3, 2, 3]
 Generation 5050: Best = 1.427116e+04 FES = 505100 μ_F = 0.397 μ_{CR} = 0.053
 Diversity = 3.7691e+01 k_t = [1, 1, 1, 1]
 Generation 5060: Best = 1.427116e+04 FES = 506100 μ_F = 0.416 μ_{CR} = 0.062
 Diversity = 3.7496e+01 k_t = [3, 3, 4, 4]
 Generation 5070: Best = 1.425940e+04 FES = 507100 μ_F = 0.404 μ_{CR} = 0.057
 Diversity = 3.7492e+01 k_t = [1, 1, 1, 1]
 Generation 5080: Best = 1.423574e+04 FES = 508100 μ_F = 0.396 μ_{CR} = 0.046
 Diversity = 3.7493e+01 k_t = [3, 3, 4, 2]

Generation 5090: Best = 1.423574e+04 FES = 509100 μ_F = 0.332 μ_{CR} = 0.045
 Diversity = 3.7493e+01 k_t = [1, 1, 1, 1]
 Generation 5100: Best = 1.423574e+04 FES = 510100 μ_F = 0.273 μ_{CR} = 0.039
 Diversity = 3.7494e+01 k_t = [3, 3, 2, 2]
 Generation 5110: Best = 1.423574e+04 FES = 511100 μ_F = 0.239 μ_{CR} = 0.038
 Diversity = 3.7495e+01 k_t = [1, 1, 1, 1]
 Generation 5120: Best = 1.423405e+04 FES = 512100 μ_F = 0.217 μ_{CR} = 0.050
 Diversity = 3.7494e+01 k_t = [3, 3, 3, 3]
 Generation 5130: Best = 1.420682e+04 FES = 513100 μ_F = 0.174 μ_{CR} = 0.070
 Diversity = 3.7494e+01 k_t = [1, 1, 1, 1]
 Generation 5140: Best = 1.420682e+04 FES = 514100 μ_F = 0.214 μ_{CR} = 0.061
 Diversity = 3.7494e+01 k_t = [3, 3, 2, 2]
 Generation 5150: Best = 1.419901e+04 FES = 515100 μ_F = 0.145 μ_{CR} = 0.078
 Diversity = 3.7494e+01 k_t = [1, 1, 1, 1]
 Generation 5160: Best = 1.419901e+04 FES = 516100 μ_F = 0.162 μ_{CR} = 0.087
 Diversity = 3.7493e+01 k_t = [2, 4, 2, 3]
 Generation 5170: Best = 1.419901e+04 FES = 517100 μ_F = 0.168 μ_{CR} = 0.088
 Diversity = 3.7492e+01 k_t = [1, 1, 1, 1]
 Generation 5180: Best = 1.414621e+04 FES = 518100 μ_F = 0.331 μ_{CR} = 0.079
 Diversity = 3.7490e+01 k_t = [4, 4, 2, 2]
 Generation 5190: Best = 1.414621e+04 FES = 519100 μ_F = 0.390 μ_{CR} = 0.070
 Diversity = 3.7491e+01 k_t = [1, 1, 1, 1]
 Generation 5200: Best = 1.414621e+04 FES = 520100 μ_F = 0.436 μ_{CR} = 0.081
 Diversity = 3.7121e+01 k_t = [2, 4, 4, 4]
 Generation 5210: Best = 1.410363e+04 FES = 521100 μ_F = 0.499 μ_{CR} = 0.072
 Diversity = 3.7007e+01 k_t = [1, 1, 1, 1]
 Generation 5220: Best = 1.410363e+04 FES = 522100 μ_F = 0.618 μ_{CR} = 0.065
 Diversity = 3.6458e+01 k_t = [2, 2, 2, 4]
 Generation 5230: Best = 1.410363e+04 FES = 523100 μ_F = 0.543 μ_{CR} = 0.069
 Diversity = 3.6458e+01 k_t = [1, 1, 1, 1]
 Generation 5240: Best = 1.410363e+04 FES = 524100 μ_F = 0.527 μ_{CR} = 0.069
 Diversity = 3.6504e+01 k_t = [3, 2, 4, 4]
 Generation 5250: Best = 1.410363e+04 FES = 525100 μ_F = 0.600 μ_{CR} = 0.056
 Diversity = 3.6413e+01 k_t = [1, 1, 1, 1]
 Generation 5260: Best = 1.410363e+04 FES = 526100 μ_F = 0.620 μ_{CR} = 0.062
 Diversity = 3.6414e+01 k_t = [4, 4, 3, 2]
 Generation 5270: Best = 1.410363e+04 FES = 527100 μ_F = 0.620 μ_{CR} = 0.058
 Diversity = 3.6411e+01 k_t = [1, 1, 1, 1]
 Generation 5280: Best = 1.410363e+04 FES = 528100 μ_F = 0.563 μ_{CR} = 0.049
 Diversity = 3.6412e+01 k_t = [2, 4, 2, 3]
 Generation 5290: Best = 1.410185e+04 FES = 529100 μ_F = 0.589 μ_{CR} = 0.054
 Diversity = 3.6412e+01 k_t = [1, 1, 1, 1]
 Generation 5300: Best = 1.410185e+04 FES = 530100 μ_F = 0.510 μ_{CR} = 0.044
 Diversity = 3.6413e+01 k_t = [2, 3, 3, 3]
 Generation 5310: Best = 1.410185e+04 FES = 531100 μ_F = 0.535 μ_{CR} = 0.040
 Diversity = 3.6413e+01 k_t = [1, 1, 1, 1]
 Generation 5320: Best = 1.410185e+04 FES = 532100 μ_F = 0.556 μ_{CR} = 0.070
 Diversity = 3.6413e+01 k_t = [4, 3, 4, 2]
 Generation 5330: Best = 1.410185e+04 FES = 533100 μ_F = 0.497 μ_{CR} = 0.086
 Diversity = 3.6412e+01 k_t = [1, 1, 1, 1]
 Generation 5340: Best = 1.408163e+04 FES = 534100 μ_F = 0.556 μ_{CR} = 0.077
 Diversity = 3.6412e+01 k_t = [4, 4, 3, 2]
 Generation 5350: Best = 1.408163e+04 FES = 535100 μ_F = 0.455 μ_{CR} = 0.064
 Diversity = 3.6412e+01 k_t = [1, 1, 1, 1]

Generation 5360: Best = 1.408163e+04 FES = 536100 μ_F = 0.502 μ_{CR} = 0.048
 Diversity = 3.6343e+01 k_t = [4, 3, 3, 4]
 Generation 5370: Best = 1.408163e+04 FES = 537100 μ_F = 0.553 μ_{CR} = 0.048
 Diversity = 3.6343e+01 k_t = [1, 1, 1, 1]
 Generation 5380: Best = 1.408134e+04 FES = 538100 μ_F = 0.561 μ_{CR} = 0.041
 Diversity = 3.6214e+01 k_t = [3, 4, 4, 2]
 Generation 5390: Best = 1.408134e+04 FES = 539100 μ_F = 0.529 μ_{CR} = 0.047
 Diversity = 3.6215e+01 k_t = [1, 1, 1, 1]
 Generation 5400: Best = 1.406134e+04 FES = 540100 μ_F = 0.539 μ_{CR} = 0.079
 Diversity = 3.6195e+01 k_t = [3, 2, 3, 4]
 Generation 5410: Best = 1.406134e+04 FES = 541100 μ_F = 0.522 μ_{CR} = 0.077
 Diversity = 3.6106e+01 k_t = [1, 1, 1, 1]
 Generation 5420: Best = 1.406134e+04 FES = 542100 μ_F = 0.504 μ_{CR} = 0.083
 Diversity = 3.6106e+01 k_t = [3, 4, 2, 3]
 Generation 5430: Best = 1.400392e+04 FES = 543100 μ_F = 0.439 μ_{CR} = 0.066
 Diversity = 3.6107e+01 k_t = [1, 1, 1, 1]
 Generation 5440: Best = 1.400384e+04 FES = 544100 μ_F = 0.476 μ_{CR} = 0.070
 Diversity = 3.6217e+01 k_t = [3, 4, 4, 4]
 Generation 5450: Best = 1.397446e+04 FES = 545100 μ_F = 0.548 μ_{CR} = 0.051
 Diversity = 3.6218e+01 k_t = [1, 1, 1, 1]
 Generation 5460: Best = 1.397446e+04 FES = 546100 μ_F = 0.551 μ_{CR} = 0.044
 Diversity = 3.6071e+01 k_t = [2, 3, 2, 3]
 Generation 5470: Best = 1.397446e+04 FES = 547100 μ_F = 0.575 μ_{CR} = 0.036
 Diversity = 3.5826e+01 k_t = [1, 1, 1, 1]
 Generation 5480: Best = 1.397446e+04 FES = 548100 μ_F = 0.555 μ_{CR} = 0.033
 Diversity = 3.5827e+01 k_t = [4, 4, 3, 3]
 Generation 5490: Best = 1.394446e+04 FES = 549100 μ_F = 0.481 μ_{CR} = 0.042
 Diversity = 3.5595e+01 k_t = [1, 1, 1, 1]
 Generation 5500: Best = 1.394446e+04 FES = 550100 μ_F = 0.493 μ_{CR} = 0.045
 Diversity = 3.5595e+01 k_t = [3, 2, 3, 3]
 Generation 5510: Best = 1.394446e+04 FES = 551100 μ_F = 0.492 μ_{CR} = 0.056
 Diversity = 3.5595e+01 k_t = [1, 1, 1, 1]
 Generation 5520: Best = 1.392756e+04 FES = 552100 μ_F = 0.616 μ_{CR} = 0.050
 Diversity = 3.5604e+01 k_t = [3, 2, 2, 4]
 Generation 5530: Best = 1.387413e+04 FES = 553100 μ_F = 0.662 μ_{CR} = 0.080
 Diversity = 3.5551e+01 k_t = [1, 1, 1, 1]
 Generation 5540: Best = 1.387392e+04 FES = 554100 μ_F = 0.627 μ_{CR} = 0.116
 Diversity = 3.5553e+01 k_t = [4, 4, 3, 3]
 Generation 5550: Best = 1.387392e+04 FES = 555100 μ_F = 0.577 μ_{CR} = 0.118
 Diversity = 3.5441e+01 k_t = [1, 1, 1, 1]
 Generation 5560: Best = 1.387392e+04 FES = 556100 μ_F = 0.524 μ_{CR} = 0.124
 Diversity = 3.5443e+01 k_t = [2, 4, 4, 2]
 Generation 5570: Best = 1.387392e+04 FES = 557100 μ_F = 0.530 μ_{CR} = 0.103
 Diversity = 3.5281e+01 k_t = [1, 1, 1, 1]
 Generation 5580: Best = 1.383354e+04 FES = 558100 μ_F = 0.534 μ_{CR} = 0.103
 Diversity = 3.5128e+01 k_t = [2, 2, 2, 4]
 Generation 5590: Best = 1.383110e+04 FES = 559100 μ_F = 0.600 μ_{CR} = 0.107
 Diversity = 3.4804e+01 k_t = [1, 1, 1, 1]
 Generation 5600: Best = 1.372837e+04 FES = 560100 μ_F = 0.704 μ_{CR} = 0.097
 Diversity = 3.4804e+01 k_t = [4, 2, 4, 2]
 Generation 5610: Best = 1.372837e+04 FES = 561100 μ_F = 0.636 μ_{CR} = 0.085
 Diversity = 3.4796e+01 k_t = [1, 1, 1, 1]
 Generation 5620: Best = 1.359262e+04 FES = 562100 μ_F = 0.670 μ_{CR} = 0.083
 Diversity = 3.4645e+01 k_t = [3, 3, 4, 2]

Generation 5630: Best = 1.357417e+04 FES = 563100 μ_F = 0.690 μ_{CR} = 0.105
 Diversity = 3.4649e+01 k_t = [1, 1, 1, 1]
 Generation 5640: Best = 1.357417e+04 FES = 564100 μ_F = 0.675 μ_{CR} = 0.095
 Diversity = 3.4621e+01 k_t = [4, 3, 4, 4]
 Generation 5650: Best = 1.357417e+04 FES = 565100 μ_F = 0.709 μ_{CR} = 0.111
 Diversity = 3.4490e+01 k_t = [1, 1, 1, 1]
 Generation 5660: Best = 1.355335e+04 FES = 566100 μ_F = 0.719 μ_{CR} = 0.093
 Diversity = 3.4491e+01 k_t = [3, 2, 2, 3]
 Generation 5670: Best = 1.350984e+04 FES = 567100 μ_F = 0.695 μ_{CR} = 0.094
 Diversity = 3.4191e+01 k_t = [1, 1, 1, 1]
 Generation 5680: Best = 1.331221e+04 FES = 568100 μ_F = 0.745 μ_{CR} = 0.095
 Diversity = 3.4018e+01 k_t = [2, 2, 3, 2]
 Generation 5690: Best = 1.331221e+04 FES = 569100 μ_F = 0.696 μ_{CR} = 0.094
 Diversity = 3.4031e+01 k_t = [1, 1, 1, 1]
 Generation 5700: Best = 1.331221e+04 FES = 570100 μ_F = 0.705 μ_{CR} = 0.090
 Diversity = 3.3758e+01 k_t = [2, 3, 4, 4]
 Generation 5710: Best = 1.331144e+04 FES = 571100 μ_F = 0.719 μ_{CR} = 0.094
 Diversity = 3.3017e+01 k_t = [1, 1, 1, 1]
 Generation 5720: Best = 1.330637e+04 FES = 572100 μ_F = 0.774 μ_{CR} = 0.097
 Diversity = 3.1979e+01 k_t = [3, 2, 2, 2]
 Generation 5730: Best = 1.321024e+04 FES = 573100 μ_F = 0.790 μ_{CR} = 0.110
 Diversity = 3.1417e+01 k_t = [1, 1, 1, 1]
 Generation 5740: Best = 1.289871e+04 FES = 574100 μ_F = 0.793 μ_{CR} = 0.109
 Diversity = 3.1572e+01 k_t = [2, 4, 2, 2]
 Generation 5750: Best = 1.289871e+04 FES = 575100 μ_F = 0.815 μ_{CR} = 0.114
 Diversity = 3.0761e+01 k_t = [1, 1, 1, 1]
 Generation 5760: Best = 1.281772e+04 FES = 576100 μ_F = 0.809 μ_{CR} = 0.109
 Diversity = 2.9772e+01 k_t = [3, 2, 3, 2]
 Generation 5770: Best = 1.280752e+04 FES = 577100 μ_F = 0.808 μ_{CR} = 0.105
 Diversity = 2.8646e+01 k_t = [1, 1, 1, 1]
 Generation 5780: Best = 1.268311e+04 FES = 578100 μ_F = 0.838 μ_{CR} = 0.108
 Diversity = 2.5462e+01 k_t = [4, 3, 2, 4]
 Generation 5790: Best = 1.268216e+04 FES = 579100 μ_F = 0.841 μ_{CR} = 0.104
 Diversity = 1.1556e+01 k_t = [1, 1, 1, 1]
 Generation 5800: Best = 1.263143e+04 FES = 580100 μ_F = 0.819 μ_{CR} = 0.104
 Diversity = 6.7738e+00 k_t = [4, 3, 4, 4]
 Generation 5810: Best = 1.253379e+04 FES = 581100 μ_F = 0.814 μ_{CR} = 0.102
 Diversity = 5.1375e+00 k_t = [1, 1, 1, 1]
 Generation 5820: Best = 1.249965e+04 FES = 582100 μ_F = 0.809 μ_{CR} = 0.102
 Diversity = 1.1804e+00 k_t = [4, 4, 4, 3]
 Generation 5830: Best = 1.249658e+04 FES = 583100 μ_F = 0.803 μ_{CR} = 0.105
 Diversity = 1.1566e+00 k_t = [1, 1, 1, 1]
 Generation 5840: Best = 1.248131e+04 FES = 584100 μ_F = 0.795 μ_{CR} = 0.102
 Diversity = 1.1319e+00 k_t = [4, 3, 3, 3]
 Generation 5850: Best = 1.236717e+04 FES = 585100 μ_F = 0.783 μ_{CR} = 0.099
 Diversity = 1.0794e+00 k_t = [1, 1, 1, 1]
 Generation 5860: Best = 1.231384e+04 FES = 586100 μ_F = 0.758 μ_{CR} = 0.094
 Diversity = 1.0423e+00 k_t = [3, 4, 2, 3]
 Generation 5870: Best = 1.230095e+04 FES = 587100 μ_F = 0.760 μ_{CR} = 0.100
 Diversity = 1.0317e+00 k_t = [1, 1, 1, 1]
 Generation 5880: Best = 1.223272e+04 FES = 588100 μ_F = 0.765 μ_{CR} = 0.109
 Diversity = 9.9171e-01 k_t = [2, 4, 3, 4]
 Generation 5890: Best = 1.223238e+04 FES = 589100 μ_F = 0.759 μ_{CR} = 0.105
 Diversity = 9.6269e-01 k_t = [1, 1, 1, 1]

Generation 5900: Best = 1.219665e+04 FES = 590100 μ_F = 0.754 μ_{CR} = 0.099
 Diversity = 9.3299e-01 k_t = [4, 3, 3, 4]
 Generation 5910: Best = 1.215589e+04 FES = 591100 μ_F = 0.758 μ_{CR} = 0.100
 Diversity = 9.1493e-01 k_t = [1, 1, 1, 1]
 Generation 5920: Best = 1.214213e+04 FES = 592100 μ_F = 0.768 μ_{CR} = 0.095
 Diversity = 8.0994e-01 k_t = [3, 2, 2, 3]
 Generation 5930: Best = 1.212963e+04 FES = 593100 μ_F = 0.777 μ_{CR} = 0.096
 Diversity = 7.8649e-01 k_t = [1, 1, 1, 1]
 Generation 5940: Best = 1.211397e+04 FES = 594100 μ_F = 0.795 μ_{CR} = 0.097
 Diversity = 7.3620e-01 k_t = [3, 2, 3, 2]
 Generation 5950: Best = 1.209655e+04 FES = 595100 μ_F = 0.768 μ_{CR} = 0.094
 Diversity = 6.9227e-01 k_t = [1, 1, 1, 1]
 Generation 5960: Best = 1.208743e+04 FES = 596100 μ_F = 0.773 μ_{CR} = 0.087
 Diversity = 6.0961e-01 k_t = [4, 3, 3, 4]
 Generation 5970: Best = 1.206701e+04 FES = 597100 μ_F = 0.761 μ_{CR} = 0.084
 Diversity = 5.5355e-01 k_t = [1, 1, 1, 1]
 Generation 5980: Best = 1.206380e+04 FES = 598100 μ_F = 0.776 μ_{CR} = 0.074
 Diversity = 4.9103e-01 k_t = [3, 2, 2, 2]
 Generation 5990: Best = 1.205146e+04 FES = 599100 μ_F = 0.771 μ_{CR} = 0.072
 Diversity = 4.8682e-01 k_t = [1, 1, 1, 1]
 Generation 6000: Best = 1.203343e+04 FES = 600100 μ_F = 0.773 μ_{CR} = 0.074
 Diversity = 4.5658e-01 k_t = [3, 3, 3, 2]
 Generation 6010: Best = 1.203343e+04 FES = 601100 μ_F = 0.744 μ_{CR} = 0.073
 Diversity = 4.4494e-01 k_t = [1, 1, 1, 1]
 Generation 6020: Best = 1.203321e+04 FES = 602100 μ_F = 0.737 μ_{CR} = 0.070
 Diversity = 4.0197e-01 k_t = [2, 3, 2, 2]
 Generation 6030: Best = 1.202714e+04 FES = 603100 μ_F = 0.724 μ_{CR} = 0.069
 Diversity = 3.8455e-01 k_t = [1, 1, 1, 1]
 Generation 6040: Best = 1.199711e+04 FES = 604100 μ_F = 0.725 μ_{CR} = 0.073
 Diversity = 3.4380e-01 k_t = [2, 2, 4, 4]
 Generation 6050: Best = 1.199711e+04 FES = 605100 μ_F = 0.746 μ_{CR} = 0.074
 Diversity = 2.5604e-01 k_t = [1, 1, 1, 1]
 Generation 6060: Best = 1.198983e+04 FES = 606100 μ_F = 0.732 μ_{CR} = 0.082
 Diversity = 2.5067e-01 k_t = [3, 2, 4, 3]
 Generation 6070: Best = 1.197866e+04 FES = 607100 μ_F = 0.735 μ_{CR} = 0.095
 Diversity = 2.4732e-01 k_t = [1, 1, 1, 1]
 Generation 6080: Best = 1.194826e+04 FES = 608100 μ_F = 0.736 μ_{CR} = 0.101
 Diversity = 2.4261e-01 k_t = [3, 2, 4, 3]
 Generation 6090: Best = 1.194292e+04 FES = 609100 μ_F = 0.739 μ_{CR} = 0.099
 Diversity = 2.4485e-01 k_t = [1, 1, 1, 1]
 Generation 6100: Best = 1.193571e+04 FES = 610100 μ_F = 0.737 μ_{CR} = 0.104
 Diversity = 2.2271e-01 k_t = [2, 4, 2, 4]
 Generation 6110: Best = 1.192769e+04 FES = 611100 μ_F = 0.736 μ_{CR} = 0.114
 Diversity = 2.1997e-01 k_t = [1, 1, 1, 1]
 Generation 6120: Best = 1.191984e+04 FES = 612100 μ_F = 0.737 μ_{CR} = 0.116
 Diversity = 1.9486e-01 k_t = [3, 3, 3, 4]
 Generation 6130: Best = 1.191021e+04 FES = 613100 μ_F = 0.735 μ_{CR} = 0.128
 Diversity = 1.9132e-01 k_t = [1, 1, 1, 1]
 Generation 6140: Best = 1.189634e+04 FES = 614100 μ_F = 0.724 μ_{CR} = 0.138
 Diversity = 1.3716e-01 k_t = [4, 4, 3, 4]
 Generation 6150: Best = 1.189100e+04 FES = 615100 μ_F = 0.693 μ_{CR} = 0.135
 Diversity = 1.0775e-01 k_t = [1, 1, 1, 1]
 Generation 6160: Best = 1.188596e+04 FES = 616100 μ_F = 0.668 μ_{CR} = 0.127
 Diversity = 9.1862e-02 k_t = [3, 4, 3, 4]

Generation 6170: Best = 1.187647e+04 FES = 617100 μ_F = 0.644 μ_{CR} = 0.125
 Diversity = 7.8200e-02 k_t = [1, 1, 1, 1]
 Generation 6180: Best = 1.187036e+04 FES = 618100 μ_F = 0.639 μ_{CR} = 0.123
 Diversity = 6.9129e-02 k_t = [4, 3, 3, 2]
 Generation 6190: Best = 1.186692e+04 FES = 619100 μ_F = 0.621 μ_{CR} = 0.126
 Diversity = 5.8202e-02 k_t = [1, 1, 1, 1]
 Generation 6200: Best = 1.186006e+04 FES = 620100 μ_F = 0.623 μ_{CR} = 0.125
 Diversity = 4.8389e-02 k_t = [2, 3, 2, 4]
 Generation 6210: Best = 1.185705e+04 FES = 621100 μ_F = 0.634 μ_{CR} = 0.134
 Diversity = 4.1871e-02 k_t = [1, 1, 1, 1]
 Generation 6220: Best = 1.185504e+04 FES = 622100 μ_F = 0.632 μ_{CR} = 0.130
 Diversity = 3.6339e-02 k_t = [4, 4, 2, 4]
 Generation 6230: Best = 1.185278e+04 FES = 623100 μ_F = 0.629 μ_{CR} = 0.120
 Diversity = 3.0087e-02 k_t = [1, 1, 1, 1]
 Generation 6240: Best = 1.185132e+04 FES = 624100 μ_F = 0.636 μ_{CR} = 0.129
 Diversity = 2.6877e-02 k_t = [4, 4, 4, 4]
 Generation 6250: Best = 1.184794e+04 FES = 625100 μ_F = 0.649 μ_{CR} = 0.125
 Diversity = 2.6090e-02 k_t = [1, 1, 1, 1]
 Generation 6260: Best = 1.184522e+04 FES = 626100 μ_F = 0.681 μ_{CR} = 0.133
 Diversity = 2.7782e-02 k_t = [3, 4, 4, 4]
 Generation 6270: Best = 1.184445e+04 FES = 627100 μ_F = 0.700 μ_{CR} = 0.128
 Diversity = 2.8040e-02 k_t = [1, 1, 1, 1]
 Generation 6280: Best = 1.184146e+04 FES = 628100 μ_F = 0.679 μ_{CR} = 0.122
 Diversity = 2.8082e-02 k_t = [2, 4, 4, 2]
 Generation 6290: Best = 1.184122e+04 FES = 629100 μ_F = 0.692 μ_{CR} = 0.128
 Diversity = 2.8387e-02 k_t = [1, 1, 1, 1]
 Generation 6300: Best = 1.184043e+04 FES = 630100 μ_F = 0.697 μ_{CR} = 0.128
 Diversity = 2.6963e-02 k_t = [4, 2, 3, 4]
 Generation 6310: Best = 1.183757e+04 FES = 631100 μ_F = 0.697 μ_{CR} = 0.143
 Diversity = 2.3372e-02 k_t = [1, 1, 1, 1]
 Generation 6320: Best = 1.183757e+04 FES = 632100 μ_F = 0.679 μ_{CR} = 0.136
 Diversity = 2.3929e-02 k_t = [3, 3, 4, 2]
 Generation 6330: Best = 1.183466e+04 FES = 633100 μ_F = 0.680 μ_{CR} = 0.133
 Diversity = 2.4166e-02 k_t = [1, 1, 1, 1]
 Generation 6340: Best = 1.183328e+04 FES = 634100 μ_F = 0.688 μ_{CR} = 0.144
 Diversity = 2.4237e-02 k_t = [4, 4, 2, 3]
 Generation 6350: Best = 1.183304e+04 FES = 635100 μ_F = 0.674 μ_{CR} = 0.144
 Diversity = 2.4029e-02 k_t = [1, 1, 1, 1]
 Generation 6360: Best = 1.183195e+04 FES = 636100 μ_F = 0.692 μ_{CR} = 0.133
 Diversity = 2.1244e-02 k_t = [2, 2, 3, 4]
 Generation 6370: Best = 1.183116e+04 FES = 637100 μ_F = 0.689 μ_{CR} = 0.119
 Diversity = 1.9433e-02 k_t = [1, 1, 1, 1]
 Generation 6380: Best = 1.183082e+04 FES = 638100 μ_F = 0.688 μ_{CR} = 0.121
 Diversity = 1.9367e-02 k_t = [4, 4, 2, 2]
 Generation 6390: Best = 1.183007e+04 FES = 639100 μ_F = 0.686 μ_{CR} = 0.114
 Diversity = 1.8932e-02 k_t = [1, 1, 1, 1]
 Generation 6400: Best = 1.182929e+04 FES = 640100 μ_F = 0.692 μ_{CR} = 0.116
 Diversity = 1.6431e-02 k_t = [2, 4, 3, 3]
 Generation 6410: Best = 1.182883e+04 FES = 641100 μ_F = 0.698 μ_{CR} = 0.117
 Diversity = 1.6059e-02 k_t = [1, 1, 1, 1]
 Generation 6420: Best = 1.182817e+04 FES = 642100 μ_F = 0.705 μ_{CR} = 0.111
 Diversity = 1.5033e-02 k_t = [4, 3, 4, 3]
 Generation 6430: Best = 1.182774e+04 FES = 643100 μ_F = 0.701 μ_{CR} = 0.106
 Diversity = 1.4421e-02 k_t = [1, 1, 1, 1]

Generation 6440: Best = 1.182757e+04 FES = 644100 μ_F = 0.695 μ_{CR} = 0.104
 Diversity = 1.3415e-02 k_t = [3, 4, 3, 4]
 Generation 6450: Best = 1.182703e+04 FES = 645100 μ_F = 0.700 μ_{CR} = 0.108
 Diversity = 1.2616e-02 k_t = [1, 1, 1, 1]
 Generation 6460: Best = 1.182660e+04 FES = 646100 μ_F = 0.701 μ_{CR} = 0.118
 Diversity = 1.2330e-02 k_t = [2, 2, 3, 2]
 Generation 6470: Best = 1.182573e+04 FES = 647100 μ_F = 0.707 μ_{CR} = 0.118
 Diversity = 1.2307e-02 k_t = [1, 1, 1, 1]
 Generation 6480: Best = 1.182544e+04 FES = 648100 μ_F = 0.727 μ_{CR} = 0.111
 Diversity = 1.2391e-02 k_t = [3, 3, 2, 4]
 Generation 6490: Best = 1.182538e+04 FES = 649100 μ_F = 0.716 μ_{CR} = 0.101
 Diversity = 1.1219e-02 k_t = [1, 1, 1, 1]
 Generation 6500: Best = 1.182530e+04 FES = 650100 μ_F = 0.728 μ_{CR} = 0.090
 Diversity = 1.1683e-02 k_t = [2, 3, 3, 2]
 Generation 6510: Best = 1.182507e+04 FES = 651100 μ_F = 0.716 μ_{CR} = 0.090
 Diversity = 1.1840e-02 k_t = [1, 1, 1, 1]
 Generation 6520: Best = 1.182468e+04 FES = 652100 μ_F = 0.737 μ_{CR} = 0.092
 Diversity = 1.1233e-02 k_t = [3, 3, 3, 4]
 Generation 6530: Best = 1.182401e+04 FES = 653100 μ_F = 0.722 μ_{CR} = 0.101
 Diversity = 1.0861e-02 k_t = [1, 1, 1, 1]
 Generation 6540: Best = 1.182375e+04 FES = 654100 μ_F = 0.746 μ_{CR} = 0.107
 Diversity = 1.0949e-02 k_t = [3, 4, 2, 4]
 Generation 6550: Best = 1.182345e+04 FES = 655100 μ_F = 0.740 μ_{CR} = 0.114
 Diversity = 1.1021e-02 k_t = [1, 1, 1, 1]
 Generation 6560: Best = 1.182326e+04 FES = 656100 μ_F = 0.734 μ_{CR} = 0.122
 Diversity = 1.0706e-02 k_t = [2, 4, 4, 3]
 Generation 6570: Best = 1.182291e+04 FES = 657100 μ_F = 0.725 μ_{CR} = 0.106
 Diversity = 1.0739e-02 k_t = [1, 1, 1, 1]
 Generation 6580: Best = 1.182230e+04 FES = 658100 μ_F = 0.745 μ_{CR} = 0.107
 Diversity = 1.0825e-02 k_t = [2, 3, 2, 2]
 Generation 6590: Best = 1.182159e+04 FES = 659100 μ_F = 0.754 μ_{CR} = 0.111
 Diversity = 1.2119e-02 k_t = [1, 1, 1, 1]
 Generation 6600: Best = 1.182148e+04 FES = 660100 μ_F = 0.754 μ_{CR} = 0.107
 Diversity = 1.2550e-02 k_t = [4, 4, 3, 4]
 Generation 6610: Best = 1.182100e+04 FES = 661100 μ_F = 0.743 μ_{CR} = 0.114
 Diversity = 1.2838e-02 k_t = [1, 1, 1, 1]
 Generation 6620: Best = 1.182058e+04 FES = 662100 μ_F = 0.740 μ_{CR} = 0.107
 Diversity = 1.3223e-02 k_t = [3, 2, 4, 3]
 Generation 6630: Best = 1.182038e+04 FES = 663100 μ_F = 0.725 μ_{CR} = 0.108
 Diversity = 1.3476e-02 k_t = [1, 1, 1, 1]
 Generation 6640: Best = 1.182000e+04 FES = 664100 μ_F = 0.715 μ_{CR} = 0.097
 Diversity = 1.3538e-02 k_t = [2, 2, 2, 2]
 Generation 6650: Best = 1.181965e+04 FES = 665100 μ_F = 0.698 μ_{CR} = 0.096
 Diversity = 1.2801e-02 k_t = [1, 1, 1, 1]
 Generation 6660: Best = 1.181913e+04 FES = 666100 μ_F = 0.683 μ_{CR} = 0.108
 Diversity = 1.1312e-02 k_t = [3, 4, 2, 4]
 Generation 6670: Best = 1.181913e+04 FES = 667100 μ_F = 0.672 μ_{CR} = 0.120
 Diversity = 1.0135e-02 k_t = [1, 1, 1, 1]
 Generation 6680: Best = 1.181872e+04 FES = 668100 μ_F = 0.684 μ_{CR} = 0.111
 Diversity = 1.0186e-02 k_t = [2, 4, 2, 2]
 Generation 6690: Best = 1.181847e+04 FES = 669100 μ_F = 0.686 μ_{CR} = 0.108
 Diversity = 1.0253e-02 k_t = [1, 1, 1, 1]
 Generation 6700: Best = 1.181825e+04 FES = 670100 μ_F = 0.684 μ_{CR} = 0.108
 Diversity = 9.5793e-03 k_t = [4, 3, 2, 4]

Generation 6710: Best = 1.181817e+04 FES = 671100 μ_F = 0.691 μ_{CR} = 0.119
 Diversity = 9.1006e-03 k_t = [1, 1, 1, 1]
 Generation 6720: Best = 1.181799e+04 FES = 672100 μ_F = 0.703 μ_{CR} = 0.109
 Diversity = 8.8490e-03 k_t = [3, 2, 3, 3]
 Generation 6730: Best = 1.181796e+04 FES = 673100 μ_F = 0.684 μ_{CR} = 0.096
 Diversity = 8.8135e-03 k_t = [1, 1, 1, 1]
 Generation 6740: Best = 1.181769e+04 FES = 674100 μ_F = 0.679 μ_{CR} = 0.097
 Diversity = 8.2369e-03 k_t = [4, 3, 3, 4]
 Generation 6750: Best = 1.181761e+04 FES = 675100 μ_F = 0.681 μ_{CR} = 0.097
 Diversity = 7.0779e-03 k_t = [1, 1, 1, 1]
 Generation 6760: Best = 1.181743e+04 FES = 676100 μ_F = 0.701 μ_{CR} = 0.097
 Diversity = 6.8654e-03 k_t = [3, 2, 2, 3]
 Generation 6770: Best = 1.181725e+04 FES = 677100 μ_F = 0.696 μ_{CR} = 0.098
 Diversity = 7.2423e-03 k_t = [1, 1, 1, 1]
 Generation 6780: Best = 1.181703e+04 FES = 678100 μ_F = 0.687 μ_{CR} = 0.101
 Diversity = 6.9618e-03 k_t = [3, 3, 3, 4]
 Generation 6790: Best = 1.181678e+04 FES = 679100 μ_F = 0.718 μ_{CR} = 0.094
 Diversity = 7.2629e-03 k_t = [1, 1, 1, 1]
 Generation 6800: Best = 1.181665e+04 FES = 680100 μ_F = 0.722 μ_{CR} = 0.085
 Diversity = 7.6682e-03 k_t = [4, 3, 3, 3]
 Generation 6810: Best = 1.181659e+04 FES = 681100 μ_F = 0.732 μ_{CR} = 0.084
 Diversity = 8.2151e-03 k_t = [1, 1, 1, 1]
 Generation 6820: Best = 1.181639e+04 FES = 682100 μ_F = 0.735 μ_{CR} = 0.087
 Diversity = 7.9589e-03 k_t = [2, 2, 3, 3]
 Generation 6830: Best = 1.181625e+04 FES = 683100 μ_F = 0.720 μ_{CR} = 0.091
 Diversity = 7.8670e-03 k_t = [1, 1, 1, 1]
 Generation 6840: Best = 1.181610e+04 FES = 684100 μ_F = 0.732 μ_{CR} = 0.093
 Diversity = 7.9602e-03 k_t = [3, 4, 2, 3]
 Generation 6850: Best = 1.181589e+04 FES = 685100 μ_F = 0.739 μ_{CR} = 0.093
 Diversity = 8.2714e-03 k_t = [1, 1, 1, 1]
 Generation 6860: Best = 1.181577e+04 FES = 686100 μ_F = 0.736 μ_{CR} = 0.095
 Diversity = 7.8759e-03 k_t = [2, 4, 3, 3]
 Generation 6870: Best = 1.181558e+04 FES = 687100 μ_F = 0.736 μ_{CR} = 0.096
 Diversity = 7.8244e-03 k_t = [1, 1, 1, 1]
 Generation 6880: Best = 1.181530e+04 FES = 688100 μ_F = 0.742 μ_{CR} = 0.097
 Diversity = 7.5711e-03 k_t = [2, 2, 4, 4]
 Generation 6890: Best = 1.181517e+04 FES = 689100 μ_F = 0.752 μ_{CR} = 0.101
 Diversity = 7.5967e-03 k_t = [1, 1, 1, 1]
 Generation 6900: Best = 1.181507e+04 FES = 690100 μ_F = 0.758 μ_{CR} = 0.095
 Diversity = 7.7685e-03 k_t = [2, 2, 3, 3]
 Generation 6910: Best = 1.181491e+04 FES = 691100 μ_F = 0.774 μ_{CR} = 0.105
 Diversity = 7.8267e-03 k_t = [1, 1, 1, 1]
 Generation 6920: Best = 1.181471e+04 FES = 692100 μ_F = 0.765 μ_{CR} = 0.104
 Diversity = 7.4838e-03 k_t = [4, 4, 2, 4]
 Generation 6930: Best = 1.181459e+04 FES = 693100 μ_F = 0.770 μ_{CR} = 0.108
 Diversity = 7.0928e-03 k_t = [1, 1, 1, 1]
 Generation 6940: Best = 1.181441e+04 FES = 694100 μ_F = 0.772 μ_{CR} = 0.108
 Diversity = 7.1381e-03 k_t = [3, 4, 4, 3]
 Generation 6950: Best = 1.181421e+04 FES = 695100 μ_F = 0.762 μ_{CR} = 0.114
 Diversity = 7.4379e-03 k_t = [1, 1, 1, 1]
 Generation 6960: Best = 1.181405e+04 FES = 696100 μ_F = 0.756 μ_{CR} = 0.112
 Diversity = 7.7066e-03 k_t = [3, 3, 3, 2]
 Generation 6970: Best = 1.181398e+04 FES = 697100 μ_F = 0.739 μ_{CR} = 0.102
 Diversity = 7.9584e-03 k_t = [1, 1, 1, 1]

Generation 6980: Best = 1.181381e+04 FES = 698100 μ_F = 0.734 μ_{CR} = 0.091
 Diversity = 7.4079e-03 k_t = [2, 3, 3, 4]
 Generation 6990: Best = 1.181368e+04 FES = 699100 μ_F = 0.744 μ_{CR} = 0.085
 Diversity = 7.1785e-03 k_t = [1, 1, 1, 1]
 Generation 7000: Best = 1.181361e+04 FES = 700100 μ_F = 0.723 μ_{CR} = 0.085
 Diversity = 6.6937e-03 k_t = [3, 3, 4, 4]
 Generation 7010: Best = 1.181359e+04 FES = 701100 μ_F = 0.715 μ_{CR} = 0.080
 Diversity = 6.3669e-03 k_t = [1, 1, 1, 1]
 Generation 7020: Best = 1.181349e+04 FES = 702100 μ_F = 0.733 μ_{CR} = 0.083
 Diversity = 6.0916e-03 k_t = [3, 2, 4, 3]
 Generation 7030: Best = 1.181341e+04 FES = 703100 μ_F = 0.727 μ_{CR} = 0.087
 Diversity = 6.0733e-03 k_t = [1, 1, 1, 1]
 Generation 7040: Best = 1.181329e+04 FES = 704100 μ_F = 0.710 μ_{CR} = 0.088
 Diversity = 5.9269e-03 k_t = [4, 3, 4, 2]
 Generation 7050: Best = 1.181324e+04 FES = 705100 μ_F = 0.704 μ_{CR} = 0.086
 Diversity = 6.0930e-03 k_t = [1, 1, 1, 1]
 Generation 7060: Best = 1.181317e+04 FES = 706100 μ_F = 0.719 μ_{CR} = 0.085
 Diversity = 6.0087e-03 k_t = [2, 4, 3, 2]
 Generation 7070: Best = 1.181301e+04 FES = 707100 μ_F = 0.719 μ_{CR} = 0.080
 Diversity = 6.4963e-03 k_t = [1, 1, 1, 1]
 Generation 7080: Best = 1.181297e+04 FES = 708100 μ_F = 0.723 μ_{CR} = 0.087
 Diversity = 5.9146e-03 k_t = [4, 3, 2, 4]
 Generation 7090: Best = 1.181289e+04 FES = 709100 μ_F = 0.730 μ_{CR} = 0.093
 Diversity = 5.5934e-03 k_t = [1, 1, 1, 1]
 Generation 7100: Best = 1.181284e+04 FES = 710100 μ_F = 0.719 μ_{CR} = 0.103
 Diversity = 5.5830e-03 k_t = [2, 3, 3, 2]
 Generation 7110: Best = 1.181278e+04 FES = 711100 μ_F = 0.716 μ_{CR} = 0.105
 Diversity = 5.4061e-03 k_t = [1, 1, 1, 1]
 Generation 7120: Best = 1.181273e+04 FES = 712100 μ_F = 0.724 μ_{CR} = 0.097
 Diversity = 5.1872e-03 k_t = [2, 2, 2, 4]
 Generation 7130: Best = 1.181266e+04 FES = 713100 μ_F = 0.725 μ_{CR} = 0.091
 Diversity = 5.3821e-03 k_t = [1, 1, 1, 1]
 Generation 7140: Best = 1.181261e+04 FES = 714100 μ_F = 0.736 μ_{CR} = 0.093
 Diversity = 5.3344e-03 k_t = [2, 3, 4, 4]
 Generation 7150: Best = 1.181251e+04 FES = 715100 μ_F = 0.721 μ_{CR} = 0.095
 Diversity = 5.0317e-03 k_t = [1, 1, 1, 1]
 Generation 7160: Best = 1.181244e+04 FES = 716100 μ_F = 0.724 μ_{CR} = 0.100
 Diversity = 4.9631e-03 k_t = [2, 2, 3, 4]
 Generation 7170: Best = 1.181227e+04 FES = 717100 μ_F = 0.734 μ_{CR} = 0.103
 Diversity = 5.2068e-03 k_t = [1, 1, 1, 1]
 Generation 7180: Best = 1.181209e+04 FES = 718100 μ_F = 0.754 μ_{CR} = 0.112
 Diversity = 5.6738e-03 k_t = [3, 4, 3, 2]
 Generation 7190: Best = 1.181208e+04 FES = 719100 μ_F = 0.762 μ_{CR} = 0.107
 Diversity = 6.1696e-03 k_t = [1, 1, 1, 1]
 Generation 7200: Best = 1.181201e+04 FES = 720100 μ_F = 0.758 μ_{CR} = 0.103
 Diversity = 6.3168e-03 k_t = [3, 4, 3, 2]
 Generation 7210: Best = 1.181189e+04 FES = 721100 μ_F = 0.750 μ_{CR} = 0.103
 Diversity = 6.2824e-03 k_t = [1, 1, 1, 1]
 Generation 7220: Best = 1.181176e+04 FES = 722100 μ_F = 0.749 μ_{CR} = 0.106
 Diversity = 6.2228e-03 k_t = [4, 3, 2, 3]
 Generation 7230: Best = 1.181168e+04 FES = 723100 μ_F = 0.748 μ_{CR} = 0.100
 Diversity = 6.3502e-03 k_t = [1, 1, 1, 1]
 Generation 7240: Best = 1.181162e+04 FES = 724100 μ_F = 0.744 μ_{CR} = 0.107
 Diversity = 6.2659e-03 k_t = [3, 2, 2, 4]

Generation 7250: Best = 1.181154e+04 FES = 725100 μ_F = 0.757 μ_{CR} = 0.102
 Diversity = 5.5299e-03 k_t = [1, 1, 1, 1]
 Generation 7260: Best = 1.181147e+04 FES = 726100 μ_F = 0.764 μ_{CR} = 0.109
 Diversity = 5.3969e-03 k_t = [4, 2, 2, 3]
 Generation 7270: Best = 1.181141e+04 FES = 727100 μ_F = 0.757 μ_{CR} = 0.104
 Diversity = 5.3356e-03 k_t = [1, 1, 1, 1]
 Generation 7280: Best = 1.181136e+04 FES = 728100 μ_F = 0.752 μ_{CR} = 0.097
 Diversity = 5.2831e-03 k_t = [3, 4, 4, 2]
 Generation 7290: Best = 1.181127e+04 FES = 729100 μ_F = 0.753 μ_{CR} = 0.101
 Diversity = 5.4003e-03 k_t = [1, 1, 1, 1]
 Generation 7300: Best = 1.181127e+04 FES = 730100 μ_F = 0.746 μ_{CR} = 0.106
 Diversity = 5.3028e-03 k_t = [2, 2, 3, 3]
 Generation 7310: Best = 1.181114e+04 FES = 731100 μ_F = 0.733 μ_{CR} = 0.096
 Diversity = 5.4121e-03 k_t = [1, 1, 1, 1]
 Generation 7320: Best = 1.181113e+04 FES = 732100 μ_F = 0.740 μ_{CR} = 0.102
 Diversity = 5.3345e-03 k_t = [2, 4, 4, 2]
 Generation 7330: Best = 1.181110e+04 FES = 733100 μ_F = 0.718 μ_{CR} = 0.105
 Diversity = 5.0734e-03 k_t = [1, 1, 1, 1]
 Generation 7340: Best = 1.181105e+04 FES = 734100 μ_F = 0.725 μ_{CR} = 0.097
 Diversity = 4.7113e-03 k_t = [3, 4, 4, 4]
 Generation 7350: Best = 1.181099e+04 FES = 735100 μ_F = 0.735 μ_{CR} = 0.098
 Diversity = 4.3869e-03 k_t = [1, 1, 1, 1]
 Generation 7360: Best = 1.181096e+04 FES = 736100 μ_F = 0.761 μ_{CR} = 0.095
 Diversity = 4.3041e-03 k_t = [4, 4, 2, 3]
 Generation 7370: Best = 1.181091e+04 FES = 737100 μ_F = 0.776 μ_{CR} = 0.086
 Diversity = 4.2944e-03 k_t = [1, 1, 1, 1]
 Generation 7380: Best = 1.181089e+04 FES = 738100 μ_F = 0.763 μ_{CR} = 0.090
 Diversity = 4.2453e-03 k_t = [4, 4, 4, 4]
 Generation 7390: Best = 1.181085e+04 FES = 739100 μ_F = 0.756 μ_{CR} = 0.099
 Diversity = 4.0299e-03 k_t = [1, 1, 1, 1]
 Generation 7400: Best = 1.181081e+04 FES = 740100 μ_F = 0.758 μ_{CR} = 0.095
 Diversity = 4.0070e-03 k_t = [3, 3, 4, 4]
 Generation 7410: Best = 1.181078e+04 FES = 741100 μ_F = 0.746 μ_{CR} = 0.097
 Diversity = 3.9697e-03 k_t = [1, 1, 1, 1]
 Generation 7420: Best = 1.181073e+04 FES = 742100 μ_F = 0.740 μ_{CR} = 0.103
 Diversity = 4.0132e-03 k_t = [3, 4, 2, 2]
 Generation 7430: Best = 1.181071e+04 FES = 743100 μ_F = 0.724 μ_{CR} = 0.108
 Diversity = 4.0548e-03 k_t = [1, 1, 1, 1]
 Generation 7440: Best = 1.181068e+04 FES = 744100 μ_F = 0.740 μ_{CR} = 0.096
 Diversity = 4.0328e-03 k_t = [3, 3, 4, 4]
 Generation 7450: Best = 1.181063e+04 FES = 745100 μ_F = 0.745 μ_{CR} = 0.098
 Diversity = 3.7670e-03 k_t = [1, 1, 1, 1]
 Generation 7460: Best = 1.181061e+04 FES = 746100 μ_F = 0.748 μ_{CR} = 0.098
 Diversity = 3.6994e-03 k_t = [3, 4, 4, 3]
 Generation 7470: Best = 1.181057e+04 FES = 747100 μ_F = 0.726 μ_{CR} = 0.099
 Diversity = 3.6804e-03 k_t = [1, 1, 1, 1]
 Generation 7480: Best = 1.181056e+04 FES = 748100 μ_F = 0.707 μ_{CR} = 0.104
 Diversity = 3.4823e-03 k_t = [3, 4, 2, 2]
 Generation 7490: Best = 1.181054e+04 FES = 749100 μ_F = 0.706 μ_{CR} = 0.103
 Diversity = 3.6383e-03 k_t = [1, 1, 1, 1]
 Generation 7500: Best = 1.181050e+04 FES = 750100 μ_F = 0.690 μ_{CR} = 0.103
 Diversity = 3.3919e-03 k_t = [2, 3, 3, 4]
 Generation 7510: Best = 1.181046e+04 FES = 751100 μ_F = 0.690 μ_{CR} = 0.106
 Diversity = 3.3340e-03 k_t = [1, 1, 1, 1]

Generation 7520: Best = 1.181044e+04 FES = 752100 μ_F = 0.707 μ_{CR} = 0.115
 Diversity = 3.2452e-03 k_t = [3, 2, 3, 3]
 Generation 7530: Best = 1.181042e+04 FES = 753100 μ_F = 0.693 μ_{CR} = 0.100
 Diversity = 3.1932e-03 k_t = [1, 1, 1, 1]
 Generation 7540: Best = 1.181038e+04 FES = 754100 μ_F = 0.702 μ_{CR} = 0.098
 Diversity = 3.0670e-03 k_t = [4, 4, 3, 3]
 Generation 7550: Best = 1.181034e+04 FES = 755100 μ_F = 0.724 μ_{CR} = 0.099
 Diversity = 3.2886e-03 k_t = [1, 1, 1, 1]
 Generation 7560: Best = 1.181033e+04 FES = 756100 μ_F = 0.710 μ_{CR} = 0.105
 Diversity = 3.3346e-03 k_t = [2, 3, 3, 2]
 Generation 7570: Best = 1.181030e+04 FES = 757100 μ_F = 0.709 μ_{CR} = 0.099
 Diversity = 3.3624e-03 k_t = [1, 1, 1, 1]
 Generation 7580: Best = 1.181027e+04 FES = 758100 μ_F = 0.728 μ_{CR} = 0.100
 Diversity = 3.2772e-03 k_t = [2, 3, 2, 2]
 Generation 7590: Best = 1.181026e+04 FES = 759100 μ_F = 0.710 μ_{CR} = 0.093
 Diversity = 3.2147e-03 k_t = [1, 1, 1, 1]
 Generation 7600: Best = 1.181024e+04 FES = 760100 μ_F = 0.702 μ_{CR} = 0.099
 Diversity = 2.9196e-03 k_t = [4, 4, 3, 3]
 Generation 7610: Best = 1.181019e+04 FES = 761100 μ_F = 0.683 μ_{CR} = 0.093
 Diversity = 2.8201e-03 k_t = [1, 1, 1, 1]
 Generation 7620: Best = 1.181019e+04 FES = 762100 μ_F = 0.684 μ_{CR} = 0.094
 Diversity = 2.6517e-03 k_t = [2, 4, 4, 4]
 Generation 7630: Best = 1.181017e+04 FES = 763100 μ_F = 0.674 μ_{CR} = 0.098
 Diversity = 2.4886e-03 k_t = [1, 1, 1, 1]
 Generation 7640: Best = 1.181016e+04 FES = 764100 μ_F = 0.673 μ_{CR} = 0.104
 Diversity = 2.2678e-03 k_t = [2, 4, 4, 4]
 Generation 7650: Best = 1.181015e+04 FES = 765100 μ_F = 0.693 μ_{CR} = 0.087
 Diversity = 2.0289e-03 k_t = [1, 1, 1, 1]
 Generation 7660: Best = 1.181013e+04 FES = 766100 μ_F = 0.718 μ_{CR} = 0.091
 Diversity = 2.0299e-03 k_t = [2, 3, 2, 3]
 Generation 7670: Best = 1.181012e+04 FES = 767100 μ_F = 0.732 μ_{CR} = 0.090
 Diversity = 2.1426e-03 k_t = [1, 1, 1, 1]
 Generation 7680: Best = 1.181011e+04 FES = 768100 μ_F = 0.719 μ_{CR} = 0.087
 Diversity = 2.0688e-03 k_t = [2, 2, 3, 4]
 Generation 7690: Best = 1.181009e+04 FES = 769100 μ_F = 0.704 μ_{CR} = 0.094
 Diversity = 1.9034e-03 k_t = [1, 1, 1, 1]
 Generation 7700: Best = 1.181009e+04 FES = 770100 μ_F = 0.703 μ_{CR} = 0.091
 Diversity = 1.7577e-03 k_t = [2, 2, 4, 4]
 Generation 7710: Best = 1.181007e+04 FES = 771100 μ_F = 0.712 μ_{CR} = 0.086
 Diversity = 1.8139e-03 k_t = [1, 1, 1, 1]
 Generation 7720: Best = 1.181006e+04 FES = 772100 μ_F = 0.719 μ_{CR} = 0.084
 Diversity = 1.7632e-03 k_t = [2, 2, 4, 4]
 Generation 7730: Best = 1.181005e+04 FES = 773100 μ_F = 0.721 μ_{CR} = 0.088
 Diversity = 1.7957e-03 k_t = [1, 1, 1, 1]
 Generation 7740: Best = 1.181003e+04 FES = 774100 μ_F = 0.725 μ_{CR} = 0.095
 Diversity = 1.8623e-03 k_t = [2, 2, 3, 2]
 Generation 7750: Best = 1.181000e+04 FES = 775100 μ_F = 0.731 μ_{CR} = 0.089
 Diversity = 2.0038e-03 k_t = [1, 1, 1, 1]
 Generation 7760: Best = 1.180998e+04 FES = 776100 μ_F = 0.737 μ_{CR} = 0.095
 Diversity = 2.0956e-03 k_t = [4, 4, 4, 2]
 Generation 7770: Best = 1.180997e+04 FES = 777100 μ_F = 0.753 μ_{CR} = 0.106
 Diversity = 2.2435e-03 k_t = [1, 1, 1, 1]
 Generation 7780: Best = 1.180997e+04 FES = 778100 μ_F = 0.756 μ_{CR} = 0.107
 Diversity = 2.3008e-03 k_t = [3, 3, 2, 2]

Generation 7790: Best = 1.180996e+04 FES = 779100 μ_F = 0.756 μ_{CR} = 0.104
 Diversity = 2.3790e-03 k_t = [1, 1, 1, 1]
 Generation 7800: Best = 1.180995e+04 FES = 780100 μ_F = 0.731 μ_{CR} = 0.111
 Diversity = 2.3576e-03 k_t = [2, 2, 4, 2]
 Generation 7810: Best = 1.180993e+04 FES = 781100 μ_F = 0.734 μ_{CR} = 0.107
 Diversity = 2.2938e-03 k_t = [1, 1, 1, 1]
 Generation 7820: Best = 1.180991e+04 FES = 782100 μ_F = 0.752 μ_{CR} = 0.109
 Diversity = 2.2652e-03 k_t = [3, 4, 3, 3]
 Generation 7830: Best = 1.180989e+04 FES = 783100 μ_F = 0.758 μ_{CR} = 0.110
 Diversity = 2.3280e-03 k_t = [1, 1, 1, 1]
 Generation 7840: Best = 1.180985e+04 FES = 784100 μ_F = 0.757 μ_{CR} = 0.105
 Diversity = 2.3985e-03 k_t = [4, 4, 2, 3]
 Generation 7850: Best = 1.180981e+04 FES = 785100 μ_F = 0.750 μ_{CR} = 0.108
 Diversity = 2.5380e-03 k_t = [1, 1, 1, 1]
 Generation 7860: Best = 1.180981e+04 FES = 786100 μ_F = 0.756 μ_{CR} = 0.110
 Diversity = 2.5242e-03 k_t = [2, 4, 4, 4]
 Generation 7870: Best = 1.180978e+04 FES = 787100 μ_F = 0.756 μ_{CR} = 0.104
 Diversity = 2.5105e-03 k_t = [1, 1, 1, 1]
 Generation 7880: Best = 1.180976e+04 FES = 788100 μ_F = 0.752 μ_{CR} = 0.099
 Diversity = 2.5521e-03 k_t = [3, 3, 4, 3]
 Generation 7890: Best = 1.180970e+04 FES = 789100 μ_F = 0.752 μ_{CR} = 0.103
 Diversity = 2.6981e-03 k_t = [1, 1, 1, 1]
 Generation 7900: Best = 1.180967e+04 FES = 790100 μ_F = 0.754 μ_{CR} = 0.112
 Diversity = 2.8017e-03 k_t = [4, 4, 2, 2]
 Generation 7910: Best = 1.180963e+04 FES = 791100 μ_F = 0.752 μ_{CR} = 0.110
 Diversity = 3.0494e-03 k_t = [1, 1, 1, 1]
 Generation 7920: Best = 1.180958e+04 FES = 792100 μ_F = 0.758 μ_{CR} = 0.121
 Diversity = 2.9393e-03 k_t = [3, 4, 4, 4]
 Generation 7930: Best = 1.180955e+04 FES = 793100 μ_F = 0.749 μ_{CR} = 0.121
 Diversity = 2.8839e-03 k_t = [1, 1, 1, 1]
 Generation 7940: Best = 1.180952e+04 FES = 794100 μ_F = 0.741 μ_{CR} = 0.125
 Diversity = 2.8434e-03 k_t = [3, 2, 3, 4]
 Generation 7950: Best = 1.180948e+04 FES = 795100 μ_F = 0.754 μ_{CR} = 0.116
 Diversity = 2.8694e-03 k_t = [1, 1, 1, 1]
 Generation 7960: Best = 1.180945e+04 FES = 796100 μ_F = 0.741 μ_{CR} = 0.120
 Diversity = 2.8885e-03 k_t = [4, 4, 3, 2]
 Generation 7970: Best = 1.180941e+04 FES = 797100 μ_F = 0.747 μ_{CR} = 0.119
 Diversity = 2.9949e-03 k_t = [1, 1, 1, 1]
 Generation 7980: Best = 1.180939e+04 FES = 798100 μ_F = 0.755 μ_{CR} = 0.119
 Diversity = 2.8708e-03 k_t = [4, 3, 4, 4]
 Generation 7990: Best = 1.180934e+04 FES = 799100 μ_F = 0.749 μ_{CR} = 0.123
 Diversity = 2.8677e-03 k_t = [1, 1, 1, 1]
 Generation 8000: Best = 1.180930e+04 FES = 800100 μ_F = 0.738 μ_{CR} = 0.127
 Diversity = 3.0102e-03 k_t = [4, 3, 4, 2]
 Generation 8010: Best = 1.180922e+04 FES = 801100 μ_F = 0.741 μ_{CR} = 0.137
 Diversity = 3.2480e-03 k_t = [1, 1, 1, 1]
 Generation 8020: Best = 1.180912e+04 FES = 802100 μ_F = 0.758 μ_{CR} = 0.149
 Diversity = 3.7683e-03 k_t = [3, 4, 2, 3]
 Generation 8030: Best = 1.180899e+04 FES = 803100 μ_F = 0.749 μ_{CR} = 0.152
 Diversity = 4.0981e-03 k_t = [1, 1, 1, 1]
 Generation 8040: Best = 1.180892e+04 FES = 804100 μ_F = 0.755 μ_{CR} = 0.157
 Diversity = 4.0570e-03 k_t = [4, 4, 3, 4]
 Generation 8050: Best = 1.180885e+04 FES = 805100 μ_F = 0.761 μ_{CR} = 0.159
 Diversity = 3.6975e-03 k_t = [1, 1, 1, 1]

Generation 8060: Best = 1.180878e+04 FES = 806100 μ_F = 0.750 μ_{CR} = 0.162
 Diversity = 3.5820e-03 k_t = [2, 4, 3, 3]
 Generation 8070: Best = 1.180871e+04 FES = 807100 μ_F = 0.734 μ_{CR} = 0.164
 Diversity = 3.5547e-03 k_t = [1, 1, 1, 1]
 Generation 8080: Best = 1.180867e+04 FES = 808100 μ_F = 0.742 μ_{CR} = 0.164
 Diversity = 3.2799e-03 k_t = [3, 3, 2, 4]
 Generation 8090: Best = 1.180864e+04 FES = 809100 μ_F = 0.749 μ_{CR} = 0.152
 Diversity = 3.2065e-03 k_t = [1, 1, 1, 1]
 Generation 8100: Best = 1.180859e+04 FES = 810100 μ_F = 0.748 μ_{CR} = 0.141
 Diversity = 3.2974e-03 k_t = [4, 3, 2, 2]
 Generation 8110: Best = 1.180856e+04 FES = 811100 μ_F = 0.737 μ_{CR} = 0.131
 Diversity = 3.3276e-03 k_t = [1, 1, 1, 1]
 Generation 8120: Best = 1.180851e+04 FES = 812100 μ_F = 0.732 μ_{CR} = 0.132
 Diversity = 3.3556e-03 k_t = [4, 3, 3, 3]
 Generation 8130: Best = 1.180845e+04 FES = 813100 μ_F = 0.747 μ_{CR} = 0.131
 Diversity = 3.4221e-03 k_t = [1, 1, 1, 1]
 Generation 8140: Best = 1.180843e+04 FES = 814100 μ_F = 0.736 μ_{CR} = 0.135
 Diversity = 3.2110e-03 k_t = [3, 2, 2, 4]
 Generation 8150: Best = 1.180842e+04 FES = 815100 μ_F = 0.732 μ_{CR} = 0.139
 Diversity = 3.0080e-03 k_t = [1, 1, 1, 1]
 Generation 8160: Best = 1.180839e+04 FES = 816100 μ_F = 0.718 μ_{CR} = 0.140
 Diversity = 2.8324e-03 k_t = [4, 2, 2, 4]
 Generation 8170: Best = 1.180838e+04 FES = 817100 μ_F = 0.707 μ_{CR} = 0.133
 Diversity = 2.6689e-03 k_t = [1, 1, 1, 1]
 Generation 8180: Best = 1.180836e+04 FES = 818100 μ_F = 0.710 μ_{CR} = 0.128
 Diversity = 2.5911e-03 k_t = [3, 4, 4, 3]
 Generation 8190: Best = 1.180834e+04 FES = 819100 μ_F = 0.703 μ_{CR} = 0.116
 Diversity = 2.4689e-03 k_t = [1, 1, 1, 1]
 Generation 8200: Best = 1.180833e+04 FES = 820100 μ_F = 0.694 μ_{CR} = 0.115
 Diversity = 2.2668e-03 k_t = [2, 3, 2, 4]
 Generation 8210: Best = 1.180831e+04 FES = 821100 μ_F = 0.694 μ_{CR} = 0.115
 Diversity = 2.0181e-03 k_t = [1, 1, 1, 1]
 Generation 8220: Best = 1.180831e+04 FES = 822100 μ_F = 0.686 μ_{CR} = 0.120
 Diversity = 1.9415e-03 k_t = [3, 3, 3, 3]
 Generation 8230: Best = 1.180831e+04 FES = 823100 μ_F = 0.686 μ_{CR} = 0.113
 Diversity = 1.9538e-03 k_t = [1, 1, 1, 1]
 Generation 8240: Best = 1.180830e+04 FES = 824100 μ_F = 0.714 μ_{CR} = 0.099
 Diversity = 1.8034e-03 k_t = [4, 4, 3, 3]
 Generation 8250: Best = 1.180829e+04 FES = 825100 μ_F = 0.716 μ_{CR} = 0.102
 Diversity = 1.7172e-03 k_t = [1, 1, 1, 1]
 Generation 8260: Best = 1.180828e+04 FES = 826100 μ_F = 0.721 μ_{CR} = 0.105
 Diversity = 1.6305e-03 k_t = [4, 3, 4, 4]
 Generation 8270: Best = 1.180826e+04 FES = 827100 μ_F = 0.741 μ_{CR} = 0.103
 Diversity = 1.5615e-03 k_t = [1, 1, 1, 1]
 Generation 8280: Best = 1.180824e+04 FES = 828100 μ_F = 0.753 μ_{CR} = 0.105
 Diversity = 1.7209e-03 k_t = [2, 3, 4, 3]
 Generation 8290: Best = 1.180823e+04 FES = 829100 μ_F = 0.772 μ_{CR} = 0.116
 Diversity = 2.0173e-03 k_t = [1, 1, 1, 1]
 Generation 8300: Best = 1.180822e+04 FES = 830100 μ_F = 0.766 μ_{CR} = 0.119
 Diversity = 2.1097e-03 k_t = [2, 2, 2, 2]
 Generation 8310: Best = 1.180820e+04 FES = 831100 μ_F = 0.737 μ_{CR} = 0.130
 Diversity = 2.0848e-03 k_t = [1, 1, 1, 1]
 Generation 8320: Best = 1.180818e+04 FES = 832100 μ_F = 0.719 μ_{CR} = 0.127
 Diversity = 2.0232e-03 k_t = [4, 3, 2, 2]

Generation 8330: Best = 1.180817e+04 FES = 833100 μ_F = 0.725 μ_{CR} = 0.132
 Diversity = 2.0689e-03 k_t = [1, 1, 1, 1]
 Generation 8340: Best = 1.180816e+04 FES = 834100 μ_F = 0.731 μ_{CR} = 0.126
 Diversity = 1.9590e-03 k_t = [4, 2, 4, 4]
 Generation 8350: Best = 1.180814e+04 FES = 835100 μ_F = 0.738 μ_{CR} = 0.126
 Diversity = 1.9631e-03 k_t = [1, 1, 1, 1]
 Generation 8360: Best = 1.180814e+04 FES = 836100 μ_F = 0.743 μ_{CR} = 0.126
 Diversity = 1.9055e-03 k_t = [2, 3, 3, 4]
 Generation 8370: Best = 1.180811e+04 FES = 837100 μ_F = 0.740 μ_{CR} = 0.120
 Diversity = 1.8934e-03 k_t = [1, 1, 1, 1]
 Generation 8380: Best = 1.180810e+04 FES = 838100 μ_F = 0.736 μ_{CR} = 0.123
 Diversity = 1.8871e-03 k_t = [4, 4, 4, 3]
 Generation 8390: Best = 1.180809e+04 FES = 839100 μ_F = 0.725 μ_{CR} = 0.126
 Diversity = 1.9175e-03 k_t = [1, 1, 1, 1]
 Generation 8400: Best = 1.180809e+04 FES = 840100 μ_F = 0.719 μ_{CR} = 0.115
 Diversity = 1.8182e-03 k_t = [4, 2, 4, 3]
 Generation 8410: Best = 1.180808e+04 FES = 841100 μ_F = 0.702 μ_{CR} = 0.105
 Diversity = 1.6797e-03 k_t = [1, 1, 1, 1]
 Generation 8420: Best = 1.180807e+04 FES = 842100 μ_F = 0.687 μ_{CR} = 0.104
 Diversity = 1.5428e-03 k_t = [2, 3, 2, 3]
 Generation 8430: Best = 1.180807e+04 FES = 843100 μ_F = 0.684 μ_{CR} = 0.094
 Diversity = 1.4458e-03 k_t = [1, 1, 1, 1]
 Generation 8440: Best = 1.180807e+04 FES = 844100 μ_F = 0.670 μ_{CR} = 0.088
 Diversity = 1.3298e-03 k_t = [2, 4, 4, 3]
 Generation 8450: Best = 1.180806e+04 FES = 845100 μ_F = 0.700 μ_{CR} = 0.073
 Diversity = 1.2852e-03 k_t = [1, 1, 1, 1]
 Generation 8460: Best = 1.180806e+04 FES = 846100 μ_F = 0.721 μ_{CR} = 0.085
 Diversity = 1.2584e-03 k_t = [3, 3, 4, 4]
 Generation 8470: Best = 1.180805e+04 FES = 847100 μ_F = 0.731 μ_{CR} = 0.087
 Diversity = 1.2671e-03 k_t = [1, 1, 1, 1]
 Generation 8480: Best = 1.180805e+04 FES = 848100 μ_F = 0.744 μ_{CR} = 0.083
 Diversity = 1.3165e-03 k_t = [4, 2, 2, 3]
 Generation 8490: Best = 1.180805e+04 FES = 849100 μ_F = 0.743 μ_{CR} = 0.079
 Diversity = 1.3397e-03 k_t = [1, 1, 1, 1]
 Generation 8500: Best = 1.180804e+04 FES = 850100 μ_F = 0.741 μ_{CR} = 0.087
 Diversity = 1.3038e-03 k_t = [4, 3, 2, 2]
 Generation 8510: Best = 1.180804e+04 FES = 851100 μ_F = 0.737 μ_{CR} = 0.091
 Diversity = 1.3060e-03 k_t = [1, 1, 1, 1]
 Generation 8520: Best = 1.180803e+04 FES = 852100 μ_F = 0.737 μ_{CR} = 0.096
 Diversity = 1.3237e-03 k_t = [4, 4, 2, 3]
 Generation 8530: Best = 1.180802e+04 FES = 853100 μ_F = 0.720 μ_{CR} = 0.099
 Diversity = 1.3047e-03 k_t = [1, 1, 1, 1]
 Generation 8540: Best = 1.180801e+04 FES = 854100 μ_F = 0.716 μ_{CR} = 0.100
 Diversity = 1.3105e-03 k_t = [3, 2, 4, 2]
 Generation 8550: Best = 1.180801e+04 FES = 855100 μ_F = 0.717 μ_{CR} = 0.095
 Diversity = 1.3637e-03 k_t = [1, 1, 1, 1]
 Generation 8560: Best = 1.180800e+04 FES = 856100 μ_F = 0.708 μ_{CR} = 0.092
 Diversity = 1.2584e-03 k_t = [2, 2, 3, 4]
 Generation 8570: Best = 1.180800e+04 FES = 857100 μ_F = 0.697 μ_{CR} = 0.087
 Diversity = 1.2825e-03 k_t = [1, 1, 1, 1]
 Generation 8580: Best = 1.180799e+04 FES = 858100 μ_F = 0.720 μ_{CR} = 0.094
 Diversity = 1.3106e-03 k_t = [2, 4, 4, 2]
 Generation 8590: Best = 1.180798e+04 FES = 859100 μ_F = 0.737 μ_{CR} = 0.097
 Diversity = 1.3842e-03 k_t = [1, 1, 1, 1]

Generation 8600: Best = 1.180798e+04 FES = 860100 μ_F = 0.742 μ_{CR} = 0.101
 Diversity = 1.4418e-03 k_t = [4, 4, 3, 3]
 Generation 8610: Best = 1.180797e+04 FES = 861100 μ_F = 0.724 μ_{CR} = 0.103
 Diversity = 1.4267e-03 k_t = [1, 1, 1, 1]
 Generation 8620: Best = 1.180797e+04 FES = 862100 μ_F = 0.732 μ_{CR} = 0.096
 Diversity = 1.3493e-03 k_t = [4, 4, 2, 4]
 Generation 8630: Best = 1.180796e+04 FES = 863100 μ_F = 0.729 μ_{CR} = 0.102
 Diversity = 1.3158e-03 k_t = [1, 1, 1, 1]
 Generation 8640: Best = 1.180796e+04 FES = 864100 μ_F = 0.741 μ_{CR} = 0.095
 Diversity = 1.3503e-03 k_t = [2, 2, 2, 3]
 Generation 8650: Best = 1.180795e+04 FES = 865100 μ_F = 0.746 μ_{CR} = 0.090
 Diversity = 1.4344e-03 k_t = [1, 1, 1, 1]
 Generation 8660: Best = 1.180794e+04 FES = 866100 μ_F = 0.764 μ_{CR} = 0.096
 Diversity = 1.3918e-03 k_t = [4, 2, 3, 3]
 Generation 8670: Best = 1.180794e+04 FES = 867100 μ_F = 0.744 μ_{CR} = 0.097
 Diversity = 1.4351e-03 k_t = [1, 1, 1, 1]
 Generation 8680: Best = 1.180793e+04 FES = 868100 μ_F = 0.743 μ_{CR} = 0.099
 Diversity = 1.3872e-03 k_t = [3, 4, 4, 4]
 Generation 8690: Best = 1.180792e+04 FES = 869100 μ_F = 0.739 μ_{CR} = 0.095
 Diversity = 1.3721e-03 k_t = [1, 1, 1, 1]
 Generation 8700: Best = 1.180792e+04 FES = 870100 μ_F = 0.727 μ_{CR} = 0.092
 Diversity = 1.4694e-03 k_t = [4, 4, 2, 2]
 Generation 8710: Best = 1.180791e+04 FES = 871100 μ_F = 0.729 μ_{CR} = 0.089
 Diversity = 1.4477e-03 k_t = [1, 1, 1, 1]
 Generation 8720: Best = 1.180791e+04 FES = 872100 μ_F = 0.714 μ_{CR} = 0.086
 Diversity = 1.3185e-03 k_t = [4, 4, 3, 3]
 Generation 8730: Best = 1.180790e+04 FES = 873100 μ_F = 0.709 μ_{CR} = 0.089
 Diversity = 1.2611e-03 k_t = [1, 1, 1, 1]
 Generation 8740: Best = 1.180790e+04 FES = 874100 μ_F = 0.725 μ_{CR} = 0.083
 Diversity = 1.1814e-03 k_t = [2, 2, 2, 3]
 Generation 8750: Best = 1.180790e+04 FES = 875100 μ_F = 0.751 μ_{CR} = 0.090
 Diversity = 1.2068e-03 k_t = [1, 1, 1, 1]
 Generation 8760: Best = 1.180789e+04 FES = 876100 μ_F = 0.770 μ_{CR} = 0.096
 Diversity = 1.2562e-03 k_t = [4, 3, 3, 4]
 Generation 8770: Best = 1.180788e+04 FES = 877100 μ_F = 0.756 μ_{CR} = 0.095
 Diversity = 1.2854e-03 k_t = [1, 1, 1, 1]
 Generation 8780: Best = 1.180788e+04 FES = 878100 μ_F = 0.751 μ_{CR} = 0.086
 Diversity = 1.2821e-03 k_t = [4, 4, 2, 2]
 Generation 8790: Best = 1.180787e+04 FES = 879100 μ_F = 0.752 μ_{CR} = 0.080
 Diversity = 1.3683e-03 k_t = [1, 1, 1, 1]
 Generation 8800: Best = 1.180787e+04 FES = 880100 μ_F = 0.753 μ_{CR} = 0.080
 Diversity = 1.3644e-03 k_t = [3, 2, 3, 3]
 Generation 8810: Best = 1.180786e+04 FES = 881100 μ_F = 0.749 μ_{CR} = 0.086
 Diversity = 1.3141e-03 k_t = [1, 1, 1, 1]
 Generation 8820: Best = 1.180786e+04 FES = 882100 μ_F = 0.743 μ_{CR} = 0.085
 Diversity = 1.3759e-03 k_t = [2, 4, 3, 2]
 Generation 8830: Best = 1.180786e+04 FES = 883100 μ_F = 0.759 μ_{CR} = 0.085
 Diversity = 1.4385e-03 k_t = [1, 1, 1, 1]
 Generation 8840: Best = 1.180785e+04 FES = 884100 μ_F = 0.755 μ_{CR} = 0.088
 Diversity = 1.3492e-03 k_t = [4, 4, 3, 3]
 Generation 8850: Best = 1.180784e+04 FES = 885100 μ_F = 0.754 μ_{CR} = 0.088
 Diversity = 1.3377e-03 k_t = [1, 1, 1, 1]
 Generation 8860: Best = 1.180784e+04 FES = 886100 μ_F = 0.748 μ_{CR} = 0.088
 Diversity = 1.3065e-03 k_t = [4, 2, 3, 4]

Generation 8870: Best = 1.180784e+04 FES = 887100 μ_F = 0.748 μ_{CR} = 0.084
 Diversity = 1.2986e-03 k_t = [1, 1, 1, 1]
 Generation 8880: Best = 1.180783e+04 FES = 888100 μ_F = 0.754 μ_{CR} = 0.084
 Diversity = 1.2956e-03 k_t = [4, 2, 2, 3]
 Generation 8890: Best = 1.180783e+04 FES = 889100 μ_F = 0.758 μ_{CR} = 0.088
 Diversity = 1.2749e-03 k_t = [1, 1, 1, 1]
 Generation 8900: Best = 1.180783e+04 FES = 890100 μ_F = 0.746 μ_{CR} = 0.093
 Diversity = 1.1627e-03 k_t = [4, 2, 2, 4]
 Generation 8910: Best = 1.180782e+04 FES = 891100 μ_F = 0.754 μ_{CR} = 0.087
 Diversity = 1.1600e-03 k_t = [1, 1, 1, 1]
 Generation 8920: Best = 1.180782e+04 FES = 892100 μ_F = 0.744 μ_{CR} = 0.085
 Diversity = 1.1423e-03 k_t = [4, 3, 2, 2]
 Generation 8930: Best = 1.180781e+04 FES = 893100 μ_F = 0.739 μ_{CR} = 0.083
 Diversity = 1.1337e-03 k_t = [1, 1, 1, 1]
 Generation 8940: Best = 1.180781e+04 FES = 894100 μ_F = 0.739 μ_{CR} = 0.091
 Diversity = 1.1637e-03 k_t = [3, 4, 3, 2]
 Generation 8950: Best = 1.180780e+04 FES = 895100 μ_F = 0.723 μ_{CR} = 0.095
 Diversity = 1.1500e-03 k_t = [1, 1, 1, 1]
 Generation 8960: Best = 1.180780e+04 FES = 896100 μ_F = 0.708 μ_{CR} = 0.094
 Diversity = 1.0815e-03 k_t = [4, 3, 2, 4]
 Generation 8970: Best = 1.180779e+04 FES = 897100 μ_F = 0.727 μ_{CR} = 0.085
 Diversity = 1.0089e-03 k_t = [1, 1, 1, 1]
 Generation 8980: Best = 1.180778e+04 FES = 898100 μ_F = 0.728 μ_{CR} = 0.093
 Diversity = 9.8803e-04 k_t = [2, 3, 2, 4]
 Generation 8990: Best = 1.180778e+04 FES = 899100 μ_F = 0.740 μ_{CR} = 0.100
 Diversity = 9.9054e-04 k_t = [1, 1, 1, 1]
 Generation 9000: Best = 1.180778e+04 FES = 900100 μ_F = 0.744 μ_{CR} = 0.101
 Diversity = 1.1089e-03 k_t = [2, 4, 3, 4]
 Generation 9010: Best = 1.180777e+04 FES = 901100 μ_F = 0.736 μ_{CR} = 0.117
 Diversity = 1.2148e-03 k_t = [1, 1, 1, 1]
 Generation 9020: Best = 1.180775e+04 FES = 902100 μ_F = 0.723 μ_{CR} = 0.117
 Diversity = 1.2504e-03 k_t = [3, 3, 4, 2]
 Generation 9030: Best = 1.180775e+04 FES = 903100 μ_F = 0.721 μ_{CR} = 0.113
 Diversity = 1.3650e-03 k_t = [1, 1, 1, 1]
 Generation 9040: Best = 1.180774e+04 FES = 904100 μ_F = 0.732 μ_{CR} = 0.108
 Diversity = 1.3436e-03 k_t = [3, 2, 4, 2]
 Generation 9050: Best = 1.180774e+04 FES = 905100 μ_F = 0.722 μ_{CR} = 0.101
 Diversity = 1.2654e-03 k_t = [1, 1, 1, 1]
 Generation 9060: Best = 1.180773e+04 FES = 906100 μ_F = 0.732 μ_{CR} = 0.094
 Diversity = 1.1629e-03 k_t = [3, 3, 3, 4]
 Generation 9070: Best = 1.180773e+04 FES = 907100 μ_F = 0.744 μ_{CR} = 0.102
 Diversity = 1.1000e-03 k_t = [1, 1, 1, 1]
 Generation 9080: Best = 1.180770e+04 FES = 908100 μ_F = 0.746 μ_{CR} = 0.124
 Diversity = 1.1835e-03 k_t = [4, 3, 2, 4]
 Generation 9090: Best = 1.180769e+04 FES = 909100 μ_F = 0.746 μ_{CR} = 0.129
 Diversity = 1.3437e-03 k_t = [1, 1, 1, 1]
 Generation 9100: Best = 1.180768e+04 FES = 910100 μ_F = 0.738 μ_{CR} = 0.134
 Diversity = 1.3547e-03 k_t = [3, 4, 3, 2]
 Generation 9110: Best = 1.180768e+04 FES = 911100 μ_F = 0.733 μ_{CR} = 0.128
 Diversity = 1.3577e-03 k_t = [1, 1, 1, 1]
 Generation 9120: Best = 1.180766e+04 FES = 912100 μ_F = 0.732 μ_{CR} = 0.128
 Diversity = 1.3109e-03 k_t = [4, 4, 2, 4]
 Generation 9130: Best = 1.180766e+04 FES = 913100 μ_F = 0.734 μ_{CR} = 0.125
 Diversity = 1.2501e-03 k_t = [1, 1, 1, 1]

Generation 9140: Best = 1.180765e+04 FES = 914100 μ_F = 0.730 μ_{CR} = 0.130
 Diversity = 1.3349e-03 k_t = [2, 4, 3, 2]
 Generation 9150: Best = 1.180764e+04 FES = 915100 μ_F = 0.752 μ_{CR} = 0.119
 Diversity = 1.4147e-03 k_t = [1, 1, 1, 1]
 Generation 9160: Best = 1.180763e+04 FES = 916100 μ_F = 0.758 μ_{CR} = 0.125
 Diversity = 1.4139e-03 k_t = [3, 2, 3, 2]
 Generation 9170: Best = 1.180760e+04 FES = 917100 μ_F = 0.764 μ_{CR} = 0.133
 Diversity = 1.5605e-03 k_t = [1, 1, 1, 1]
 Generation 9180: Best = 1.180758e+04 FES = 918100 μ_F = 0.761 μ_{CR} = 0.145
 Diversity = 1.6629e-03 k_t = [3, 2, 3, 3]
 Generation 9190: Best = 1.180757e+04 FES = 919100 μ_F = 0.747 μ_{CR} = 0.139
 Diversity = 1.6479e-03 k_t = [1, 1, 1, 1]
 Generation 9200: Best = 1.180756e+04 FES = 920100 μ_F = 0.743 μ_{CR} = 0.128
 Diversity = 1.5517e-03 k_t = [4, 4, 2, 4]
 Generation 9210: Best = 1.180756e+04 FES = 921100 μ_F = 0.736 μ_{CR} = 0.124
 Diversity = 1.4791e-03 k_t = [1, 1, 1, 1]
 Generation 9220: Best = 1.180755e+04 FES = 922100 μ_F = 0.736 μ_{CR} = 0.112
 Diversity = 1.3803e-03 k_t = [3, 3, 4, 3]
 Generation 9230: Best = 1.180755e+04 FES = 923100 μ_F = 0.721 μ_{CR} = 0.122
 Diversity = 1.3691e-03 k_t = [1, 1, 1, 1]
 Generation 9240: Best = 1.180754e+04 FES = 924100 μ_F = 0.730 μ_{CR} = 0.119
 Diversity = 1.3573e-03 k_t = [4, 2, 4, 2]
 Generation 9250: Best = 1.180753e+04 FES = 925100 μ_F = 0.752 μ_{CR} = 0.115
 Diversity = 1.4124e-03 k_t = [1, 1, 1, 1]
 Generation 9260: Best = 1.180753e+04 FES = 926100 μ_F = 0.755 μ_{CR} = 0.116
 Diversity = 1.5055e-03 k_t = [2, 3, 3, 3]
 Generation 9270: Best = 1.180752e+04 FES = 927100 μ_F = 0.766 μ_{CR} = 0.116
 Diversity = 1.5692e-03 k_t = [1, 1, 1, 1]
 Generation 9280: Best = 1.180751e+04 FES = 928100 μ_F = 0.761 μ_{CR} = 0.109
 Diversity = 1.5229e-03 k_t = [3, 2, 2, 3]
 Generation 9290: Best = 1.180751e+04 FES = 929100 μ_F = 0.774 μ_{CR} = 0.102
 Diversity = 1.5816e-03 k_t = [1, 1, 1, 1]
 Generation 9300: Best = 1.180750e+04 FES = 930100 μ_F = 0.760 μ_{CR} = 0.100
 Diversity = 1.5648e-03 k_t = [3, 4, 4, 4]
 Generation 9310: Best = 1.180749e+04 FES = 931100 μ_F = 0.760 μ_{CR} = 0.098
 Diversity = 1.4349e-03 k_t = [1, 1, 1, 1]
 Generation 9320: Best = 1.180749e+04 FES = 932100 μ_F = 0.754 μ_{CR} = 0.092
 Diversity = 1.3941e-03 k_t = [4, 3, 4, 3]
 Generation 9330: Best = 1.180749e+04 FES = 933100 μ_F = 0.739 μ_{CR} = 0.102
 Diversity = 1.4598e-03 k_t = [1, 1, 1, 1]
 Generation 9340: Best = 1.180748e+04 FES = 934100 μ_F = 0.722 μ_{CR} = 0.100
 Diversity = 1.4884e-03 k_t = [3, 2, 3, 2]
 Generation 9350: Best = 1.180748e+04 FES = 935100 μ_F = 0.724 μ_{CR} = 0.111
 Diversity = 1.5990e-03 k_t = [1, 1, 1, 1]
 Generation 9360: Best = 1.180747e+04 FES = 936100 μ_F = 0.726 μ_{CR} = 0.110
 Diversity = 1.4881e-03 k_t = [2, 2, 3, 4]
 Generation 9370: Best = 1.180747e+04 FES = 937100 μ_F = 0.720 μ_{CR} = 0.108
 Diversity = 1.3655e-03 k_t = [1, 1, 1, 1]
 Generation 9380: Best = 1.180746e+04 FES = 938100 μ_F = 0.719 μ_{CR} = 0.104
 Diversity = 1.3448e-03 k_t = [4, 2, 4, 2]
 Generation 9390: Best = 1.180746e+04 FES = 939100 μ_F = 0.710 μ_{CR} = 0.092
 Diversity = 1.3616e-03 k_t = [1, 1, 1, 1]
 Generation 9400: Best = 1.180745e+04 FES = 940100 μ_F = 0.711 μ_{CR} = 0.095
 Diversity = 1.3020e-03 k_t = [4, 3, 3, 2]

Generation 9410: Best = 1.180745e+04 FES = 941100 μ_F = 0.715 μ_{CR} = 0.096
 Diversity = 1.2977e-03 k_t = [1, 1, 1, 1]
 Generation 9420: Best = 1.180744e+04 FES = 942100 μ_F = 0.709 μ_{CR} = 0.094
 Diversity = 1.2327e-03 k_t = [2, 4, 4, 4]
 Generation 9430: Best = 1.180744e+04 FES = 943100 μ_F = 0.716 μ_{CR} = 0.098
 Diversity = 1.1466e-03 k_t = [1, 1, 1, 1]
 Generation 9440: Best = 1.180743e+04 FES = 944100 μ_F = 0.738 μ_{CR} = 0.101
 Diversity = 1.1813e-03 k_t = [4, 4, 4, 2]
 Generation 9450: Best = 1.180743e+04 FES = 945100 μ_F = 0.745 μ_{CR} = 0.102
 Diversity = 1.2254e-03 k_t = [1, 1, 1, 1]
 Generation 9460: Best = 1.180742e+04 FES = 946100 μ_F = 0.748 μ_{CR} = 0.100
 Diversity = 1.2174e-03 k_t = [2, 2, 3, 3]
 Generation 9470: Best = 1.180742e+04 FES = 947100 μ_F = 0.762 μ_{CR} = 0.100
 Diversity = 1.1723e-03 k_t = [1, 1, 1, 1]
 Generation 9480: Best = 1.180741e+04 FES = 948100 μ_F = 0.755 μ_{CR} = 0.100
 Diversity = 1.0979e-03 k_t = [3, 3, 2, 4]
 Generation 9490: Best = 1.180741e+04 FES = 949100 μ_F = 0.736 μ_{CR} = 0.097
 Diversity = 9.9232e-04 k_t = [1, 1, 1, 1]
 Generation 9500: Best = 1.180741e+04 FES = 950100 μ_F = 0.740 μ_{CR} = 0.088
 Diversity = 9.4469e-04 k_t = [2, 4, 4, 3]
 Generation 9510: Best = 1.180741e+04 FES = 951100 μ_F = 0.738 μ_{CR} = 0.091
 Diversity = 9.4421e-04 k_t = [1, 1, 1, 1]
 Generation 9520: Best = 1.180741e+04 FES = 952100 μ_F = 0.725 μ_{CR} = 0.085
 Diversity = 9.2576e-04 k_t = [4, 2, 2, 4]
 Generation 9530: Best = 1.180740e+04 FES = 953100 μ_F = 0.762 μ_{CR} = 0.083
 Diversity = 8.9962e-04 k_t = [1, 1, 1, 1]
 Generation 9540: Best = 1.180740e+04 FES = 954100 μ_F = 0.764 μ_{CR} = 0.080
 Diversity = 8.6735e-04 k_t = [3, 2, 3, 3]
 Generation 9550: Best = 1.180740e+04 FES = 955100 μ_F = 0.750 μ_{CR} = 0.085
 Diversity = 8.5781e-04 k_t = [1, 1, 1, 1]
 Generation 9560: Best = 1.180740e+04 FES = 956100 μ_F = 0.754 μ_{CR} = 0.087
 Diversity = 8.9796e-04 k_t = [2, 2, 3, 2]
 Generation 9570: Best = 1.180740e+04 FES = 957100 μ_F = 0.746 μ_{CR} = 0.080
 Diversity = 9.1847e-04 k_t = [1, 1, 1, 1]
 Generation 9580: Best = 1.180739e+04 FES = 958100 μ_F = 0.731 μ_{CR} = 0.079
 Diversity = 9.3790e-04 k_t = [4, 4, 2, 4]
 Generation 9590: Best = 1.180739e+04 FES = 959100 μ_F = 0.727 μ_{CR} = 0.080
 Diversity = 9.6146e-04 k_t = [1, 1, 1, 1]
 Generation 9600: Best = 1.180739e+04 FES = 960100 μ_F = 0.730 μ_{CR} = 0.086
 Diversity = 9.9859e-04 k_t = [2, 4, 4, 4]
 Generation 9610: Best = 1.180738e+04 FES = 961100 μ_F = 0.736 μ_{CR} = 0.090
 Diversity = 9.4915e-04 k_t = [1, 1, 1, 1]
 Generation 9620: Best = 1.180738e+04 FES = 962100 μ_F = 0.745 μ_{CR} = 0.087
 Diversity = 9.5846e-04 k_t = [3, 3, 2, 2]
 Generation 9630: Best = 1.180737e+04 FES = 963100 μ_F = 0.763 μ_{CR} = 0.082
 Diversity = 1.0358e-03 k_t = [1, 1, 1, 1]
 Generation 9640: Best = 1.180737e+04 FES = 964100 μ_F = 0.768 μ_{CR} = 0.086
 Diversity = 1.1568e-03 k_t = [4, 2, 4, 2]
 Generation 9650: Best = 1.180736e+04 FES = 965100 μ_F = 0.767 μ_{CR} = 0.090
 Diversity = 1.2668e-03 k_t = [1, 1, 1, 1]
 Generation 9660: Best = 1.180735e+04 FES = 966100 μ_F = 0.780 μ_{CR} = 0.086
 Diversity = 1.1764e-03 k_t = [3, 4, 2, 4]
 Generation 9670: Best = 1.180735e+04 FES = 967100 μ_F = 0.764 μ_{CR} = 0.089
 Diversity = 1.2060e-03 k_t = [1, 1, 1, 1]

Generation 9680: Best = 1.180735e+04 FES = 968100 μ_F = 0.766 μ_{CR} = 0.089
 Diversity = 1.1546e-03 k_t = [2, 2, 4, 4]
 Generation 9690: Best = 1.180735e+04 FES = 969100 μ_F = 0.762 μ_{CR} = 0.083
 Diversity = 1.1054e-03 k_t = [1, 1, 1, 1]
 Generation 9700: Best = 1.180735e+04 FES = 970100 μ_F = 0.748 μ_{CR} = 0.083
 Diversity = 1.0975e-03 k_t = [2, 3, 2, 3]
 Generation 9710: Best = 1.180734e+04 FES = 971100 μ_F = 0.739 μ_{CR} = 0.083
 Diversity = 1.1720e-03 k_t = [1, 1, 1, 1]
 Generation 9720: Best = 1.180734e+04 FES = 972100 μ_F = 0.721 μ_{CR} = 0.078
 Diversity = 1.1559e-03 k_t = [2, 3, 3, 3]
 Generation 9730: Best = 1.180733e+04 FES = 973100 μ_F = 0.744 μ_{CR} = 0.075
 Diversity = 1.2240e-03 k_t = [1, 1, 1, 1]
 Generation 9740: Best = 1.180733e+04 FES = 974100 μ_F = 0.749 μ_{CR} = 0.080
 Diversity = 1.2880e-03 k_t = [2, 2, 2, 4]
 Generation 9750: Best = 1.180732e+04 FES = 975100 μ_F = 0.759 μ_{CR} = 0.083
 Diversity = 1.3647e-03 k_t = [1, 1, 1, 1]
 Generation 9760: Best = 1.180731e+04 FES = 976100 μ_F = 0.753 μ_{CR} = 0.085
 Diversity = 1.2841e-03 k_t = [4, 3, 3, 4]
 Generation 9770: Best = 1.180730e+04 FES = 977100 μ_F = 0.746 μ_{CR} = 0.093
 Diversity = 1.2503e-03 k_t = [1, 1, 1, 1]
 Generation 9780: Best = 1.180730e+04 FES = 978100 μ_F = 0.727 μ_{CR} = 0.089
 Diversity = 1.1391e-03 k_t = [3, 2, 2, 4]
 Generation 9790: Best = 1.180730e+04 FES = 979100 μ_F = 0.736 μ_{CR} = 0.087
 Diversity = 1.1034e-03 k_t = [1, 1, 1, 1]
 Generation 9800: Best = 1.180730e+04 FES = 980100 μ_F = 0.729 μ_{CR} = 0.087
 Diversity = 1.1117e-03 k_t = [3, 4, 3, 4]
 Generation 9810: Best = 1.180728e+04 FES = 981100 μ_F = 0.742 μ_{CR} = 0.089
 Diversity = 1.1170e-03 k_t = [1, 1, 1, 1]
 Generation 9820: Best = 1.180727e+04 FES = 982100 μ_F = 0.742 μ_{CR} = 0.095
 Diversity = 1.2293e-03 k_t = [2, 3, 3, 2]
 Generation 9830: Best = 1.180725e+04 FES = 983100 μ_F = 0.752 μ_{CR} = 0.099
 Diversity = 1.4572e-03 k_t = [1, 1, 1, 1]
 Generation 9840: Best = 1.180724e+04 FES = 984100 μ_F = 0.743 μ_{CR} = 0.103
 Diversity = 1.6142e-03 k_t = [4, 2, 3, 2]
 Generation 9850: Best = 1.180724e+04 FES = 985100 μ_F = 0.737 μ_{CR} = 0.102
 Diversity = 1.7410e-03 k_t = [1, 1, 1, 1]
 Generation 9860: Best = 1.180722e+04 FES = 986100 μ_F = 0.732 μ_{CR} = 0.106
 Diversity = 1.8152e-03 k_t = [3, 3, 4, 2]
 Generation 9870: Best = 1.180720e+04 FES = 987100 μ_F = 0.718 μ_{CR} = 0.114
 Diversity = 1.9120e-03 k_t = [1, 1, 1, 1]
 Generation 9880: Best = 1.180720e+04 FES = 988100 μ_F = 0.738 μ_{CR} = 0.114
 Diversity = 1.8675e-03 k_t = [2, 4, 2, 4]
 Generation 9890: Best = 1.180719e+04 FES = 989100 μ_F = 0.726 μ_{CR} = 0.117
 Diversity = 1.8447e-03 k_t = [1, 1, 1, 1]
 Generation 9900: Best = 1.180718e+04 FES = 990100 μ_F = 0.742 μ_{CR} = 0.106
 Diversity = 1.7350e-03 k_t = [4, 3, 4, 4]
 Generation 9910: Best = 1.180717e+04 FES = 991100 μ_F = 0.742 μ_{CR} = 0.109
 Diversity = 1.6504e-03 k_t = [1, 1, 1, 1]
 Generation 9920: Best = 1.180716e+04 FES = 992100 μ_F = 0.725 μ_{CR} = 0.112
 Diversity = 1.6231e-03 k_t = [4, 4, 2, 3]
 Generation 9930: Best = 1.180715e+04 FES = 993100 μ_F = 0.726 μ_{CR} = 0.115
 Diversity = 1.6445e-03 k_t = [1, 1, 1, 1]
 Generation 9940: Best = 1.180715e+04 FES = 994100 μ_F = 0.701 μ_{CR} = 0.108
 Diversity = 1.4659e-03 k_t = [4, 3, 3, 4]

Generation 9950: Best = 1.180714e+04 FES = 995100 μ_F = 0.715 μ_{CR} = 0.099
 Diversity = 1.2851e-03 k_t = [1, 1, 1, 1]
 Generation 9960: Best = 1.180714e+04 FES = 996100 μ_F = 0.718 μ_{CR} = 0.095
 Diversity = 1.2425e-03 k_t = [4, 2, 4, 2]
 Generation 9970: Best = 1.180714e+04 FES = 997100 μ_F = 0.719 μ_{CR} = 0.092
 Diversity = 1.2407e-03 k_t = [1, 1, 1, 1]
 Generation 9980: Best = 1.180712e+04 FES = 998100 μ_F = 0.737 μ_{CR} = 0.104
 Diversity = 1.2731e-03 k_t = [4, 2, 2, 4]
 Generation 9990: Best = 1.180712e+04 FES = 999100 μ_F = 0.749 μ_{CR} = 0.095
 Diversity = 1.3266e-03 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 1.180711e+04
 Total Function Evals = 1000000
 Total Successful Updates = 111872

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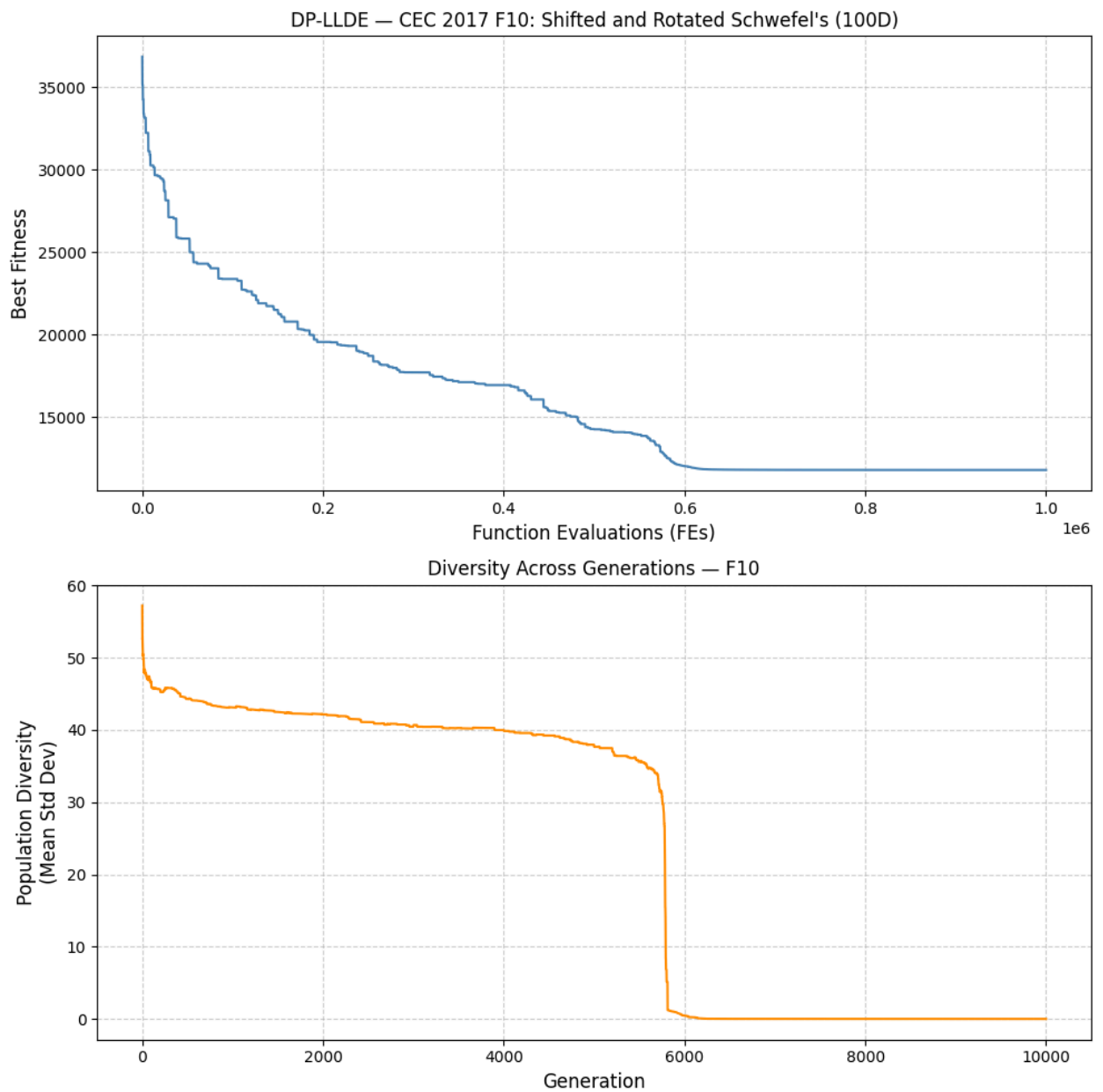
Results:

Best Fitness : 1.180711e+04
 Global Optimum : 1.000000e+03
 Error (|best-optimum|) : 1.080711e+04
 Best Solution (5 dim) : [-43.82491411 -6.3491841 21.45575997 -85.26571693
 -17.66699895]
 D^(0) (init. diversity): 5.7220e+01

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✓ Checkpoint saved to: /content/drive/My Drive/thesis_results/all_results_checkpoint.pkl



Graph saved as convergence_F10.png

✓ Plot saved to: /content/drive/My Drive/thesis_results/plots/convergence_F10.png

6.3 Hybrid Functions (F11-F20)

```
In [ ]: # =====
# F11 – Hybrid Function 1 (Zakharov; Rosenbrock; Rastrigin)
# =====
run_and_plot('F11', make_wrapper(hybrid.f11), 1100.0, 'Hybrid: Zakharov; Rosenbrock; Rastrigin')
```

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DP-LLDE on CEC 2017 F11 – Single Run

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Function : Hybrid: Zakharov; Rosenbrock; Rastrigin (100D)

Bounds : [-100, 100]

Global optimum : 1100.0

Initial $D^*(0) = 5.7220e+01$ | Diversity threshold = 5.7220e+00

Generation 0: Best Fitness = 7.262506e+05

Generation 10: Best = 3.327204e+05 FES = 1100 $\mu_F = 0.576$ $\mu_{CR} = 0.336$

Diversity = 4.7442e+01 $k_t = [1, 1, 1, 1]$

Generation 20: Best = 1.947581e+05 FES = 2100 $\mu_F = 0.616$ $\mu_{CR} = 0.282$

Diversity = 3.9181e+01 $k_t = [2, 3, 2, 3]$

Generation 30: Best = 1.418370e+05 FES = 3100 $\mu_F = 0.631$ $\mu_{CR} = 0.204$

Diversity = 2.9444e+01 $k_t = [1, 1, 1, 1]$

Generation 40: Best = 1.032780e+05 FES = 4100 $\mu_F = 0.641$ $\mu_{CR} = 0.134$

Diversity = 2.0180e+01 $k_t = [3, 2, 2, 2]$

Generation 50: Best = 9.626732e+04 FES = 5100 $\mu_F = 0.655$ $\mu_{CR} = 0.087$

Diversity = 1.5544e+01 $k_t = [1, 1, 1, 1]$

Generation 60: Best = 9.014468e+04 FES = 6100 $\mu_F = 0.677$ $\mu_{CR} = 0.065$

Diversity = 1.2602e+01 $k_t = [3, 2, 4, 4]$

Generation 70: Best = 8.238324e+04 FES = 7100 $\mu_F = 0.711$ $\mu_{CR} = 0.063$

Diversity = 1.0728e+01 $k_t = [1, 1, 1, 1]$

Generation 80: Best = 7.577205e+04 FES = 8100 $\mu_F = 0.735$ $\mu_{CR} = 0.057$

Diversity = 9.5472e+00 $k_t = [4, 4, 2, 4]$

Generation 90: Best = 7.164213e+04 FES = 9100 $\mu_F = 0.767$ $\mu_{CR} = 0.055$

Diversity = 7.8067e+00 $k_t = [1, 1, 1, 1]$

Generation 100: Best = 6.752657e+04 FES = 10100 $\mu_F = 0.776$ $\mu_{CR} = 0.055$

Diversity = 7.5707e+00 $k_t = [2, 3, 4, 4]$

Generation 110: Best = 6.434790e+04 FES = 11100 $\mu_F = 0.806$ $\mu_{CR} = 0.057$

Diversity = 7.8702e+00 $k_t = [1, 1, 1, 1]$

Generation 120: Best = 6.038858e+04 FES = 12100 $\mu_F = 0.808$ $\mu_{CR} = 0.056$

Diversity = 8.2253e+00 $k_t = [3, 4, 2, 4]$

Generation 130: Best = 5.396304e+04 FES = 13100 $\mu_F = 0.808$ $\mu_{CR} = 0.060$

Diversity = 8.1799e+00 $k_t = [1, 1, 1, 1]$

Generation 140: Best = 5.129990e+04 FES = 14100 $\mu_F = 0.810$ $\mu_{CR} = 0.061$

Diversity = 8.9142e+00 $k_t = [2, 2, 2, 2]$

Generation 150: Best = 4.754524e+04 FES = 15100 $\mu_F = 0.816$ $\mu_{CR} = 0.056$

Diversity = 9.7390e+00 $k_t = [1, 1, 1, 1]$

Generation 160: Best = 4.531268e+04 FES = 16100 $\mu_F = 0.798$ $\mu_{CR} = 0.073$

Diversity = 8.8436e+00 $k_t = [3, 2, 3, 2]$

Generation 170: Best = 4.249470e+04 FES = 17100 $\mu_F = 0.782$ $\mu_{CR} = 0.069$

Diversity = 9.0428e+00 $k_t = [1, 1, 1, 1]$

Generation 180: Best = 4.023060e+04 FES = 18100 $\mu_F = 0.775$ $\mu_{CR} = 0.073$

Diversity = 8.3809e+00 $k_t = [3, 3, 3, 2]$

Generation 190: Best = 3.809035e+04 FES = 19100 $\mu_F = 0.770$ $\mu_{CR} = 0.068$

Diversity = 8.1036e+00 $k_t = [1, 1, 1, 1]$

Generation 200: Best = 3.651398e+04 FES = 20100 $\mu_F = 0.773$ $\mu_{CR} = 0.063$

Diversity = 8.2070e+00 $k_t = [2, 3, 2, 2]$

Generation 210: Best = 3.512450e+04 FES = 21100 $\mu_F = 0.764$ $\mu_{CR} = 0.064$

Diversity = 7.9499e+00 $k_t = [1, 1, 1, 1]$

Generation 220: Best = 3.355724e+04 FES = 22100 $\mu_F = 0.759$ $\mu_{CR} = 0.065$

Diversity = 7.5405e+00 $k_t = [4, 3, 4, 4]$

Generation 230: Best = 3.273020e+04	FES = 23100	$\mu_F = 0.766$	$\mu_{CR} = 0.075$
Diversity = 6.3220e+00	$k_t = [1, 1, 1, 1]$		
Generation 240: Best = 3.115299e+04	FES = 24100	$\mu_F = 0.782$	$\mu_{CR} = 0.082$
Diversity = 6.3848e+00	$k_t = [4, 4, 4, 3]$		
Generation 250: Best = 3.041712e+04	FES = 25100	$\mu_F = 0.787$	$\mu_{CR} = 0.074$
Diversity = 6.2128e+00	$k_t = [1, 1, 1, 1]$		
Generation 260: Best = 2.962695e+04	FES = 26100	$\mu_F = 0.776$	$\mu_{CR} = 0.076$
Diversity = 6.3533e+00	$k_t = [2, 4, 3, 2]$		
Generation 270: Best = 2.858063e+04	FES = 27100	$\mu_F = 0.777$	$\mu_{CR} = 0.072$
Diversity = 6.6505e+00	$k_t = [1, 1, 1, 1]$		
Generation 280: Best = 2.783691e+04	FES = 28100	$\mu_F = 0.775$	$\mu_{CR} = 0.072$
Diversity = 6.5894e+00	$k_t = [3, 2, 4, 3]$		
Generation 290: Best = 2.714190e+04	FES = 29100	$\mu_F = 0.781$	$\mu_{CR} = 0.076$
Diversity = 6.1426e+00	$k_t = [1, 1, 1, 1]$		
Generation 300: Best = 2.642878e+04	FES = 30100	$\mu_F = 0.792$	$\mu_{CR} = 0.075$
Diversity = 6.1735e+00	$k_t = [4, 4, 4, 4]$		
Generation 310: Best = 2.590836e+04	FES = 31100	$\mu_F = 0.801$	$\mu_{CR} = 0.067$
Diversity = 6.0477e+00	$k_t = [1, 1, 1, 1]$		
Generation 320: Best = 2.522866e+04	FES = 32100	$\mu_F = 0.798$	$\mu_{CR} = 0.074$
Diversity = 6.3280e+00	$k_t = [3, 4, 2, 3]$		
Generation 330: Best = 2.442154e+04	FES = 33100	$\mu_F = 0.790$	$\mu_{CR} = 0.077$
Diversity = 6.4594e+00	$k_t = [1, 1, 1, 1]$		
Generation 340: Best = 2.373580e+04	FES = 34100	$\mu_F = 0.802$	$\mu_{CR} = 0.079$
Diversity = 6.7936e+00	$k_t = [4, 3, 3, 2]$		
Generation 350: Best = 2.304485e+04	FES = 35100	$\mu_F = 0.809$	$\mu_{CR} = 0.083$
Diversity = 7.1615e+00	$k_t = [1, 1, 1, 1]$		
Generation 360: Best = 2.252290e+04	FES = 36100	$\mu_F = 0.820$	$\mu_{CR} = 0.082$
Diversity = 6.9859e+00	$k_t = [2, 2, 4, 2]$		
Generation 370: Best = 2.175896e+04	FES = 37100	$\mu_F = 0.801$	$\mu_{CR} = 0.086$
Diversity = 6.7478e+00	$k_t = [1, 1, 1, 1]$		
Generation 380: Best = 2.097002e+04	FES = 38100	$\mu_F = 0.803$	$\mu_{CR} = 0.088$
Diversity = 7.5218e+00	$k_t = [4, 4, 2, 3]$		
Generation 390: Best = 1.986533e+04	FES = 39100	$\mu_F = 0.801$	$\mu_{CR} = 0.084$
Diversity = 7.4651e+00	$k_t = [1, 1, 1, 1]$		
Generation 400: Best = 1.903639e+04	FES = 40100	$\mu_F = 0.802$	$\mu_{CR} = 0.083$
Diversity = 7.5789e+00	$k_t = [2, 3, 2, 2]$		
Generation 410: Best = 1.848093e+04	FES = 41100	$\mu_F = 0.815$	$\mu_{CR} = 0.086$
Diversity = 8.0335e+00	$k_t = [1, 1, 1, 1]$		
Generation 420: Best = 1.786577e+04	FES = 42100	$\mu_F = 0.818$	$\mu_{CR} = 0.084$
Diversity = 8.3748e+00	$k_t = [4, 4, 3, 2]$		
Generation 430: Best = 1.726112e+04	FES = 43100	$\mu_F = 0.801$	$\mu_{CR} = 0.070$
Diversity = 9.0639e+00	$k_t = [1, 1, 1, 1]$		
Generation 440: Best = 1.694319e+04	FES = 44100	$\mu_F = 0.781$	$\mu_{CR} = 0.069$
Diversity = 8.7095e+00	$k_t = [4, 2, 2, 2]$		
Generation 450: Best = 1.657944e+04	FES = 45100	$\mu_F = 0.762$	$\mu_{CR} = 0.072$
Diversity = 7.9311e+00	$k_t = [1, 1, 1, 1]$		
Generation 460: Best = 1.608382e+04	FES = 46100	$\mu_F = 0.766$	$\mu_{CR} = 0.084$
Diversity = 7.8698e+00	$k_t = [3, 2, 3, 3]$		
Generation 470: Best = 1.572775e+04	FES = 47100	$\mu_F = 0.754$	$\mu_{CR} = 0.092$
Diversity = 7.6829e+00	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 1.524520e+04	FES = 48100	$\mu_F = 0.752$	$\mu_{CR} = 0.099$
Diversity = 7.3253e+00	$k_t = [4, 3, 3, 2]$		
Generation 490: Best = 1.493333e+04	FES = 49100	$\mu_F = 0.757$	$\mu_{CR} = 0.087$
Diversity = 6.9214e+00	$k_t = [1, 1, 1, 1]$		

Generation 500: Best = 1.425250e+04	FES = 50100	$\mu_F = 0.763$	$\mu_{CR} = 0.082$
Diversity = 6.3625e+00	$k_t = [3, 4, 2, 3]$		
Generation 510: Best = 1.410401e+04	FES = 51100	$\mu_F = 0.777$	$\mu_{CR} = 0.093$
Diversity = 6.9499e+00	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 1.371928e+04	FES = 52100	$\mu_F = 0.788$	$\mu_{CR} = 0.094$
Diversity = 6.9758e+00	$k_t = [4, 2, 3, 2]$		
Generation 530: Best = 1.342237e+04	FES = 53100	$\mu_F = 0.778$	$\mu_{CR} = 0.103$
Diversity = 6.7204e+00	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 1.307511e+04	FES = 54100	$\mu_F = 0.775$	$\mu_{CR} = 0.100$
Diversity = 6.7342e+00	$k_t = [4, 4, 3, 3]$		
Generation 550: Best = 1.271172e+04	FES = 55100	$\mu_F = 0.771$	$\mu_{CR} = 0.096$
Diversity = 6.1198e+00	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 1.248203e+04	FES = 56100	$\mu_F = 0.784$	$\mu_{CR} = 0.082$
Diversity = 5.9578e+00	$k_t = [2, 4, 3, 3]$		
Generation 570: Best = 1.207696e+04	FES = 57100	$\mu_F = 0.779$	$\mu_{CR} = 0.075$
Diversity = 6.0673e+00	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 1.203841e+04	FES = 58100	$\mu_F = 0.772$	$\mu_{CR} = 0.079$
Diversity = 5.9265e+00	$k_t = [3, 4, 4, 2]$		
Generation 590: Best = 1.180711e+04	FES = 59100	$\mu_F = 0.769$	$\mu_{CR} = 0.093$
Diversity = 5.7499e+00	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 1.158711e+04	FES = 60100	$\mu_F = 0.765$	$\mu_{CR} = 0.092$
Diversity = 5.4692e+00	$k_t = [2, 3, 4, 4]$		
Generation 610: Best = 1.150166e+04	FES = 61100	$\mu_F = 0.762$	$\mu_{CR} = 0.079$
Diversity = 5.3715e+00	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 1.103462e+04	FES = 62100	$\mu_F = 0.776$	$\mu_{CR} = 0.093$
Diversity = 5.5459e+00	$k_t = [4, 4, 4, 2]$		
Generation 630: Best = 1.088928e+04	FES = 63100	$\mu_F = 0.791$	$\mu_{CR} = 0.092$
Diversity = 5.8879e+00	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 1.054281e+04	FES = 64100	$\mu_F = 0.789$	$\mu_{CR} = 0.084$
Diversity = 5.8256e+00	$k_t = [3, 3, 3, 3]$		
Generation 650: Best = 1.037421e+04	FES = 65100	$\mu_F = 0.791$	$\mu_{CR} = 0.090$
Diversity = 5.6912e+00	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 1.011693e+04	FES = 66100	$\mu_F = 0.785$	$\mu_{CR} = 0.101$
Diversity = 5.5843e+00	$k_t = [4, 4, 3, 3]$		
Generation 670: Best = 9.948562e+03	FES = 67100	$\mu_F = 0.775$	$\mu_{CR} = 0.106$
Diversity = 5.4894e+00	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 9.698223e+03	FES = 68100	$\mu_F = 0.789$	$\mu_{CR} = 0.110$
Diversity = 5.8054e+00	$k_t = [3, 2, 4, 3]$		
Generation 690: Best = 9.576364e+03	FES = 69100	$\mu_F = 0.795$	$\mu_{CR} = 0.115$
Diversity = 5.9095e+00	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 9.314128e+03	FES = 70100	$\mu_F = 0.791$	$\mu_{CR} = 0.104$
Diversity = 6.0427e+00	$k_t = [2, 4, 2, 4]$		
Generation 710: Best = 9.111026e+03	FES = 71100	$\mu_F = 0.792$	$\mu_{CR} = 0.096$
Diversity = 5.8166e+00	$k_t = [1, 1, 1, 1]$		
Generation 720: Best = 8.992090e+03	FES = 72100	$\mu_F = 0.787$	$\mu_{CR} = 0.089$
Diversity = 5.6709e+00	$k_t = [4, 3, 3, 2]$		
Generation 730: Best = 8.811273e+03	FES = 73100	$\mu_F = 0.791$	$\mu_{CR} = 0.095$
Diversity = 5.8616e+00	$k_t = [1, 1, 1, 1]$		
Generation 740: Best = 8.701309e+03	FES = 74100	$\mu_F = 0.793$	$\mu_{CR} = 0.095$
Diversity = 5.6607e+00	$k_t = [3, 4, 4, 2]$		
Generation 750: Best = 8.547365e+03	FES = 75100	$\mu_F = 0.790$	$\mu_{CR} = 0.089$
Diversity = 5.8873e+00	$k_t = [1, 1, 1, 1]$		
Generation 760: Best = 8.376244e+03	FES = 76100	$\mu_F = 0.772$	$\mu_{CR} = 0.092$
Diversity = 5.7691e+00	$k_t = [3, 4, 2, 4]$		

Generation 770: Best = 8.271246e+03 FES = 77100 μ_F = 0.789 μ_{CR} = 0.091
 Diversity = 5.4141e+00 k_t = [1, 1, 1, 1]
 Generation 780: Best = 8.129868e+03 FES = 78100 μ_F = 0.791 μ_{CR} = 0.090
 Diversity = 5.9879e+00 k_t = [3, 4, 4, 4]
 Generation 790: Best = 7.958821e+03 FES = 79100 μ_F = 0.789 μ_{CR} = 0.094
 Diversity = 5.6533e+00 k_t = [1, 1, 1, 1]
 Generation 800: Best = 7.894712e+03 FES = 80100 μ_F = 0.800 μ_{CR} = 0.089
 Diversity = 5.6125e+00 k_t = [3, 4, 2, 2]
 Generation 810: Best = 7.780257e+03 FES = 81100 μ_F = 0.795 μ_{CR} = 0.096
 Diversity = 5.9819e+00 k_t = [1, 1, 1, 1]
 Generation 820: Best = 7.705835e+03 FES = 82100 μ_F = 0.800 μ_{CR} = 0.109
 Diversity = 6.0547e+00 k_t = [4, 4, 3, 2]
 Generation 830: Best = 7.480457e+03 FES = 83100 μ_F = 0.774 μ_{CR} = 0.115
 Diversity = 6.2234e+00 k_t = [1, 1, 1, 1]
 Generation 840: Best = 7.263018e+03 FES = 84100 μ_F = 0.778 μ_{CR} = 0.126
 Diversity = 7.2584e+00 k_t = [4, 2, 3, 2]
 Generation 850: Best = 6.999338e+03 FES = 85100 μ_F = 0.780 μ_{CR} = 0.109
 Diversity = 7.3015e+00 k_t = [1, 1, 1, 1]
 Generation 860: Best = 6.980210e+03 FES = 86100 μ_F = 0.778 μ_{CR} = 0.095
 Diversity = 6.9002e+00 k_t = [4, 4, 3, 2]
 Generation 870: Best = 6.777244e+03 FES = 87100 μ_F = 0.793 μ_{CR} = 0.095
 Diversity = 7.2062e+00 k_t = [1, 1, 1, 1]
 Generation 880: Best = 6.601238e+03 FES = 88100 μ_F = 0.796 μ_{CR} = 0.100
 Diversity = 7.2447e+00 k_t = [2, 2, 3, 4]
 Generation 890: Best = 6.499015e+03 FES = 89100 μ_F = 0.781 μ_{CR} = 0.090
 Diversity = 6.4633e+00 k_t = [1, 1, 1, 1]
 Generation 900: Best = 6.387186e+03 FES = 90100 μ_F = 0.768 μ_{CR} = 0.083
 Diversity = 6.1625e+00 k_t = [3, 4, 2, 2]
 Generation 910: Best = 6.191894e+03 FES = 91100 μ_F = 0.776 μ_{CR} = 0.090
 Diversity = 6.0256e+00 k_t = [1, 1, 1, 1]
 Generation 920: Best = 6.172087e+03 FES = 92100 μ_F = 0.773 μ_{CR} = 0.079
 Diversity = 5.6902e+00 k_t = [2, 2, 4, 4]
 Generation 930: Best = 6.050556e+03 FES = 93100 μ_F = 0.764 μ_{CR} = 0.095
 Diversity = 5.5926e+00 k_t = [1, 1, 1, 1]
 Generation 940: Best = 5.938015e+03 FES = 94100 μ_F = 0.763 μ_{CR} = 0.080
 Diversity = 5.4163e+00 k_t = [2, 2, 4, 3]
 Generation 950: Best = 5.859584e+03 FES = 95100 μ_F = 0.773 μ_{CR} = 0.075
 Diversity = 5.3397e+00 k_t = [1, 1, 1, 1]
 Generation 960: Best = 5.816396e+03 FES = 96100 μ_F = 0.777 μ_{CR} = 0.081
 Diversity = 5.4236e+00 k_t = [4, 4, 2, 3]
 Generation 970: Best = 5.700827e+03 FES = 97100 μ_F = 0.766 μ_{CR} = 0.085
 Diversity = 5.5382e+00 k_t = [1, 1, 1, 1]
 Generation 980: Best = 5.646523e+03 FES = 98100 μ_F = 0.782 μ_{CR} = 0.091
 Diversity = 5.6099e+00 k_t = [4, 4, 4, 3]
 Generation 990: Best = 5.537454e+03 FES = 99100 μ_F = 0.788 μ_{CR} = 0.081
 Diversity = 5.5282e+00 k_t = [1, 1, 1, 1]
 Generation 1000: Best = 5.434931e+03 FES = 100100 μ_F = 0.795 μ_{CR} = 0.083
 Diversity = 5.3492e+00 k_t = [4, 2, 2, 4]
 Generation 1010: Best = 5.312961e+03 FES = 101100 μ_F = 0.794 μ_{CR} = 0.092
 Diversity = 5.0069e+00 k_t = [1, 1, 1, 1]
 Generation 1020: Best = 5.287145e+03 FES = 102100 μ_F = 0.794 μ_{CR} = 0.084
 Diversity = 5.1694e+00 k_t = [2, 2, 4, 3]
 Generation 1030: Best = 5.096325e+03 FES = 103100 μ_F = 0.788 μ_{CR} = 0.084
 Diversity = 5.4147e+00 k_t = [1, 1, 1, 1]

Generation 1040: Best = 5.096325e+03 FES = 104100 μ_F = 0.791 μ_{CR} = 0.079
 Diversity = 5.1035e+00 k_t = [2, 2, 3, 3]
 Generation 1050: Best = 4.921327e+03 FES = 105100 μ_F = 0.795 μ_{CR} = 0.078
 Diversity = 5.5594e+00 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 4.882230e+03 FES = 106100 μ_F = 0.762 μ_{CR} = 0.092
 Diversity = 5.7144e+00 k_t = [2, 3, 2, 2]
 Generation 1070: Best = 4.780852e+03 FES = 107100 μ_F = 0.759 μ_{CR} = 0.077
 Diversity = 5.5969e+00 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 4.768420e+03 FES = 108100 μ_F = 0.765 μ_{CR} = 0.075
 Diversity = 4.9227e+00 k_t = [4, 4, 3, 3]
 Generation 1090: Best = 4.697016e+03 FES = 109100 μ_F = 0.762 μ_{CR} = 0.083
 Diversity = 4.5608e+00 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 4.648077e+03 FES = 110100 μ_F = 0.751 μ_{CR} = 0.080
 Diversity = 4.5886e+00 k_t = [2, 4, 2, 2]
 Generation 1110: Best = 4.566301e+03 FES = 111100 μ_F = 0.761 μ_{CR} = 0.091
 Diversity = 4.6370e+00 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 4.517175e+03 FES = 112100 μ_F = 0.773 μ_{CR} = 0.082
 Diversity = 4.6323e+00 k_t = [3, 3, 3, 3]
 Generation 1130: Best = 4.446076e+03 FES = 113100 μ_F = 0.760 μ_{CR} = 0.071
 Diversity = 4.5325e+00 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 4.370544e+03 FES = 114100 μ_F = 0.750 μ_{CR} = 0.075
 Diversity = 4.3508e+00 k_t = [3, 4, 4, 3]
 Generation 1150: Best = 4.347982e+03 FES = 115100 μ_F = 0.760 μ_{CR} = 0.066
 Diversity = 4.3481e+00 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 4.315002e+03 FES = 116100 μ_F = 0.756 μ_{CR} = 0.064
 Diversity = 4.1754e+00 k_t = [4, 3, 4, 4]
 Generation 1170: Best = 4.221068e+03 FES = 117100 μ_F = 0.750 μ_{CR} = 0.075
 Diversity = 4.0075e+00 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 4.186181e+03 FES = 118100 μ_F = 0.774 μ_{CR} = 0.072
 Diversity = 4.2730e+00 k_t = [3, 4, 3, 2]
 Generation 1190: Best = 4.115329e+03 FES = 119100 μ_F = 0.771 μ_{CR} = 0.081
 Diversity = 4.4186e+00 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 4.068183e+03 FES = 120100 μ_F = 0.778 μ_{CR} = 0.087
 Diversity = 4.3240e+00 k_t = [4, 2, 4, 4]
 Generation 1210: Best = 4.026672e+03 FES = 121100 μ_F = 0.784 μ_{CR} = 0.092
 Diversity = 4.3265e+00 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 3.960687e+03 FES = 122100 μ_F = 0.772 μ_{CR} = 0.082
 Diversity = 4.3574e+00 k_t = [3, 3, 2, 3]
 Generation 1230: Best = 3.929333e+03 FES = 123100 μ_F = 0.773 μ_{CR} = 0.073
 Diversity = 4.3359e+00 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 3.879857e+03 FES = 124100 μ_F = 0.774 μ_{CR} = 0.079
 Diversity = 4.4345e+00 k_t = [3, 4, 4, 3]
 Generation 1250: Best = 3.838364e+03 FES = 125100 μ_F = 0.767 μ_{CR} = 0.074
 Diversity = 4.4817e+00 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 3.792223e+03 FES = 126100 μ_F = 0.774 μ_{CR} = 0.080
 Diversity = 4.4964e+00 k_t = [4, 3, 4, 4]
 Generation 1270: Best = 3.777565e+03 FES = 127100 μ_F = 0.775 μ_{CR} = 0.082
 Diversity = 4.4788e+00 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 3.701450e+03 FES = 128100 μ_F = 0.761 μ_{CR} = 0.087
 Diversity = 4.2971e+00 k_t = [2, 3, 4, 2]
 Generation 1290: Best = 3.675873e+03 FES = 129100 μ_F = 0.763 μ_{CR} = 0.074
 Diversity = 4.2321e+00 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 3.657038e+03 FES = 130100 μ_F = 0.773 μ_{CR} = 0.069
 Diversity = 4.2725e+00 k_t = [4, 4, 3, 2]

Generation 1310: Best = 3.640234e+03 FES = 131100 μ_F = 0.779 μ_{CR} = 0.060
 Diversity = 4.4925e+00 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 3.579495e+03 FES = 132100 μ_F = 0.777 μ_{CR} = 0.066
 Diversity = 4.4809e+00 k_t = [4, 3, 2, 2]
 Generation 1330: Best = 3.549480e+03 FES = 133100 μ_F = 0.769 μ_{CR} = 0.064
 Diversity = 4.4670e+00 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 3.441228e+03 FES = 134100 μ_F = 0.765 μ_{CR} = 0.067
 Diversity = 4.5240e+00 k_t = [2, 2, 4, 3]
 Generation 1350: Best = 3.400936e+03 FES = 135100 μ_F = 0.745 μ_{CR} = 0.064
 Diversity = 4.5237e+00 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 3.340696e+03 FES = 136100 μ_F = 0.767 μ_{CR} = 0.079
 Diversity = 4.1704e+00 k_t = [4, 2, 2, 2]
 Generation 1370: Best = 3.301488e+03 FES = 137100 μ_F = 0.770 μ_{CR} = 0.071
 Diversity = 4.2331e+00 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 3.260930e+03 FES = 138100 μ_F = 0.767 μ_{CR} = 0.069
 Diversity = 4.2279e+00 k_t = [4, 4, 3, 4]
 Generation 1390: Best = 3.208429e+03 FES = 139100 μ_F = 0.762 μ_{CR} = 0.073
 Diversity = 4.0059e+00 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 3.123655e+03 FES = 140100 μ_F = 0.747 μ_{CR} = 0.073
 Diversity = 4.1659e+00 k_t = [3, 3, 3, 2]
 Generation 1410: Best = 3.087630e+03 FES = 141100 μ_F = 0.752 μ_{CR} = 0.069
 Diversity = 4.1128e+00 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 3.084692e+03 FES = 142100 μ_F = 0.768 μ_{CR} = 0.073
 Diversity = 4.0834e+00 k_t = [3, 3, 2, 4]
 Generation 1430: Best = 3.036964e+03 FES = 143100 μ_F = 0.776 μ_{CR} = 0.069
 Diversity = 4.0701e+00 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 3.021349e+03 FES = 144100 μ_F = 0.787 μ_{CR} = 0.086
 Diversity = 4.3324e+00 k_t = [2, 2, 4, 2]
 Generation 1450: Best = 2.951648e+03 FES = 145100 μ_F = 0.756 μ_{CR} = 0.074
 Diversity = 4.4704e+00 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 2.935515e+03 FES = 146100 μ_F = 0.771 μ_{CR} = 0.065
 Diversity = 4.2159e+00 k_t = [3, 2, 2, 3]
 Generation 1470: Best = 2.882119e+03 FES = 147100 μ_F = 0.768 μ_{CR} = 0.066
 Diversity = 3.9094e+00 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 2.860034e+03 FES = 148100 μ_F = 0.769 μ_{CR} = 0.062
 Diversity = 3.6988e+00 k_t = [3, 4, 2, 3]
 Generation 1490: Best = 2.830578e+03 FES = 149100 μ_F = 0.759 μ_{CR} = 0.070
 Diversity = 3.6150e+00 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 2.781701e+03 FES = 150100 μ_F = 0.770 μ_{CR} = 0.072
 Diversity = 3.4894e+00 k_t = [2, 2, 4, 2]
 Generation 1510: Best = 2.749918e+03 FES = 151100 μ_F = 0.783 μ_{CR} = 0.073
 Diversity = 3.6452e+00 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 2.739046e+03 FES = 152100 μ_F = 0.792 μ_{CR} = 0.066
 Diversity = 3.6737e+00 k_t = [2, 4, 4, 3]
 Generation 1530: Best = 2.717510e+03 FES = 153100 μ_F = 0.792 μ_{CR} = 0.068
 Diversity = 3.7182e+00 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 2.613803e+03 FES = 154100 μ_F = 0.787 μ_{CR} = 0.063
 Diversity = 3.7619e+00 k_t = [4, 3, 4, 2]
 Generation 1550: Best = 2.613803e+03 FES = 155100 μ_F = 0.771 μ_{CR} = 0.060
 Diversity = 3.8661e+00 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 2.604125e+03 FES = 156100 μ_F = 0.756 μ_{CR} = 0.060
 Diversity = 3.8544e+00 k_t = [3, 2, 3, 3]
 Generation 1570: Best = 2.591327e+03 FES = 157100 μ_F = 0.746 μ_{CR} = 0.064
 Diversity = 3.7840e+00 k_t = [1, 1, 1, 1]

Generation 1580: Best = 2.571647e+03 FES = 158100 μ_F = 0.733 μ_{CR} = 0.068
 Diversity = 3.7325e+00 k_t = [4, 4, 2, 3]
 Generation 1590: Best = 2.567809e+03 FES = 159100 μ_F = 0.734 μ_{CR} = 0.072
 Diversity = 3.6352e+00 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 2.542310e+03 FES = 160100 μ_F = 0.729 μ_{CR} = 0.060
 Diversity = 3.6192e+00 k_t = [4, 2, 4, 3]
 Generation 1610: Best = 2.502540e+03 FES = 161100 μ_F = 0.762 μ_{CR} = 0.053
 Diversity = 3.5134e+00 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 2.467646e+03 FES = 162100 μ_F = 0.745 μ_{CR} = 0.057
 Diversity = 3.4296e+00 k_t = [4, 4, 4, 3]
 Generation 1630: Best = 2.450602e+03 FES = 163100 μ_F = 0.737 μ_{CR} = 0.060
 Diversity = 3.3273e+00 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 2.450602e+03 FES = 164100 μ_F = 0.728 μ_{CR} = 0.066
 Diversity = 3.1133e+00 k_t = [3, 2, 4, 4]
 Generation 1650: Best = 2.450224e+03 FES = 165100 μ_F = 0.713 μ_{CR} = 0.063
 Diversity = 3.0899e+00 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 2.448669e+03 FES = 166100 μ_F = 0.723 μ_{CR} = 0.068
 Diversity = 3.0090e+00 k_t = [2, 4, 2, 2]
 Generation 1670: Best = 2.429937e+03 FES = 167100 μ_F = 0.723 μ_{CR} = 0.059
 Diversity = 2.9332e+00 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 2.429937e+03 FES = 168100 μ_F = 0.731 μ_{CR} = 0.057
 Diversity = 2.8673e+00 k_t = [4, 2, 4, 3]
 Generation 1690: Best = 2.414490e+03 FES = 169100 μ_F = 0.718 μ_{CR} = 0.055
 Diversity = 2.8497e+00 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 2.398214e+03 FES = 170100 μ_F = 0.725 μ_{CR} = 0.052
 Diversity = 2.8198e+00 k_t = [2, 4, 4, 3]
 Generation 1710: Best = 2.398214e+03 FES = 171100 μ_F = 0.710 μ_{CR} = 0.055
 Diversity = 2.7441e+00 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 2.398214e+03 FES = 172100 μ_F = 0.754 μ_{CR} = 0.052
 Diversity = 2.7290e+00 k_t = [4, 3, 2, 3]
 Generation 1730: Best = 2.385491e+03 FES = 173100 μ_F = 0.765 μ_{CR} = 0.057
 Diversity = 2.7069e+00 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 2.376832e+03 FES = 174100 μ_F = 0.760 μ_{CR} = 0.059
 Diversity = 2.6541e+00 k_t = [2, 2, 2, 3]
 Generation 1750: Best = 2.372913e+03 FES = 175100 μ_F = 0.763 μ_{CR} = 0.046
 Diversity = 2.6608e+00 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 2.371843e+03 FES = 176100 μ_F = 0.738 μ_{CR} = 0.047
 Diversity = 2.6109e+00 k_t = [2, 3, 3, 3]
 Generation 1770: Best = 2.371843e+03 FES = 177100 μ_F = 0.698 μ_{CR} = 0.059
 Diversity = 2.6136e+00 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 2.355015e+03 FES = 178100 μ_F = 0.677 μ_{CR} = 0.061
 Diversity = 2.5592e+00 k_t = [2, 4, 2, 3]
 Generation 1790: Best = 2.345567e+03 FES = 179100 μ_F = 0.726 μ_{CR} = 0.058
 Diversity = 2.5461e+00 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 2.345567e+03 FES = 180100 μ_F = 0.702 μ_{CR} = 0.054
 Diversity = 2.5182e+00 k_t = [3, 2, 2, 3]
 Generation 1810: Best = 2.331563e+03 FES = 181100 μ_F = 0.715 μ_{CR} = 0.060
 Diversity = 2.4816e+00 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 2.330969e+03 FES = 182100 μ_F = 0.713 μ_{CR} = 0.058
 Diversity = 2.4444e+00 k_t = [3, 3, 3, 3]
 Generation 1830: Best = 2.322801e+03 FES = 183100 μ_F = 0.717 μ_{CR} = 0.046
 Diversity = 2.4293e+00 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 2.322722e+03 FES = 184100 μ_F = 0.744 μ_{CR} = 0.044
 Diversity = 2.3906e+00 k_t = [2, 4, 4, 3]

Generation 1850: Best = 2.322722e+03 FES = 185100 μ_F = 0.757 μ_{CR} = 0.048
 Diversity = 2.3787e+00 k_t = [1, 1, 1, 1]
 Generation 1860: Best = 2.299632e+03 FES = 186100 μ_F = 0.754 μ_{CR} = 0.042
 Diversity = 2.3513e+00 k_t = [2, 4, 4, 4]
 Generation 1870: Best = 2.298715e+03 FES = 187100 μ_F = 0.734 μ_{CR} = 0.046
 Diversity = 2.3443e+00 k_t = [1, 1, 1, 1]
 Generation 1880: Best = 2.298715e+03 FES = 188100 μ_F = 0.754 μ_{CR} = 0.040
 Diversity = 2.2746e+00 k_t = [2, 4, 4, 4]
 Generation 1890: Best = 2.298715e+03 FES = 189100 μ_F = 0.770 μ_{CR} = 0.044
 Diversity = 2.2539e+00 k_t = [1, 1, 1, 1]
 Generation 1900: Best = 2.293973e+03 FES = 190100 μ_F = 0.795 μ_{CR} = 0.046
 Diversity = 2.2348e+00 k_t = [2, 3, 3, 4]
 Generation 1910: Best = 2.293973e+03 FES = 191100 μ_F = 0.810 μ_{CR} = 0.046
 Diversity = 2.2347e+00 k_t = [1, 1, 1, 1]
 Generation 1920: Best = 2.293973e+03 FES = 192100 μ_F = 0.758 μ_{CR} = 0.044
 Diversity = 2.1724e+00 k_t = [3, 4, 2, 4]
 Generation 1930: Best = 2.288358e+03 FES = 193100 μ_F = 0.756 μ_{CR} = 0.043
 Diversity = 2.1554e+00 k_t = [1, 1, 1, 1]
 Generation 1940: Best = 2.288358e+03 FES = 194100 μ_F = 0.748 μ_{CR} = 0.056
 Diversity = 2.1206e+00 k_t = [2, 4, 3, 4]
 Generation 1950: Best = 2.284377e+03 FES = 195100 μ_F = 0.731 μ_{CR} = 0.052
 Diversity = 2.0872e+00 k_t = [1, 1, 1, 1]
 Generation 1960: Best = 2.278111e+03 FES = 196100 μ_F = 0.737 μ_{CR} = 0.052
 Diversity = 2.0788e+00 k_t = [2, 4, 3, 2]
 Generation 1970: Best = 2.277815e+03 FES = 197100 μ_F = 0.728 μ_{CR} = 0.055
 Diversity = 2.1024e+00 k_t = [1, 1, 1, 1]
 Generation 1980: Best = 2.262154e+03 FES = 198100 μ_F = 0.730 μ_{CR} = 0.048
 Diversity = 2.0741e+00 k_t = [2, 2, 3, 3]
 Generation 1990: Best = 2.262154e+03 FES = 199100 μ_F = 0.782 μ_{CR} = 0.048
 Diversity = 2.0410e+00 k_t = [1, 1, 1, 1]
 Generation 2000: Best = 2.244535e+03 FES = 200100 μ_F = 0.762 μ_{CR} = 0.062
 Diversity = 1.9209e+00 k_t = [4, 3, 3, 4]
 Generation 2010: Best = 2.244535e+03 FES = 201100 μ_F = 0.772 μ_{CR} = 0.055
 Diversity = 1.8776e+00 k_t = [1, 1, 1, 1]
 Generation 2020: Best = 2.241417e+03 FES = 202100 μ_F = 0.757 μ_{CR} = 0.057
 Diversity = 1.8384e+00 k_t = [4, 4, 2, 3]
 Generation 2030: Best = 2.239114e+03 FES = 203100 μ_F = 0.750 μ_{CR} = 0.062
 Diversity = 1.7989e+00 k_t = [1, 1, 1, 1]
 Generation 2040: Best = 2.227846e+03 FES = 204100 μ_F = 0.744 μ_{CR} = 0.067
 Diversity = 1.7595e+00 k_t = [4, 3, 4, 2]
 Generation 2050: Best = 2.218198e+03 FES = 205100 μ_F = 0.746 μ_{CR} = 0.066
 Diversity = 1.7333e+00 k_t = [1, 1, 1, 1]
 Generation 2060: Best = 2.189683e+03 FES = 206100 μ_F = 0.759 μ_{CR} = 0.067
 Diversity = 1.5698e+00 k_t = [4, 2, 4, 3]
 Generation 2070: Best = 2.165659e+03 FES = 207100 μ_F = 0.754 μ_{CR} = 0.078
 Diversity = 1.3948e+00 k_t = [1, 1, 1, 1]
 Generation 2080: Best = 2.165659e+03 FES = 208100 μ_F = 0.735 μ_{CR} = 0.102
 Diversity = 8.7697e-01 k_t = [2, 2, 4, 2]
 Generation 2090: Best = 2.134234e+03 FES = 209100 μ_F = 0.725 μ_{CR} = 0.115
 Diversity = 5.6384e-01 k_t = [1, 1, 1, 1]
 Generation 2100: Best = 2.115466e+03 FES = 210100 μ_F = 0.730 μ_{CR} = 0.119
 Diversity = 4.5049e-01 k_t = [3, 4, 4, 4]
 Generation 2110: Best = 2.101637e+03 FES = 211100 μ_F = 0.730 μ_{CR} = 0.141
 Diversity = 3.7396e-01 k_t = [1, 1, 1, 1]

Generation 2120: Best = 2.079811e+03 FES = 212100 μ_F = 0.721 μ_{CR} = 0.139
 Diversity = 3.2511e-01 k_t = [3, 4, 3, 4]
 Generation 2130: Best = 2.068815e+03 FES = 213100 μ_F = 0.719 μ_{CR} = 0.149
 Diversity = 3.0473e-01 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 2.060665e+03 FES = 214100 μ_F = 0.712 μ_{CR} = 0.155
 Diversity = 2.9367e-01 k_t = [4, 3, 3, 3]
 Generation 2150: Best = 2.056963e+03 FES = 215100 μ_F = 0.703 μ_{CR} = 0.152
 Diversity = 2.9035e-01 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 2.048825e+03 FES = 216100 μ_F = 0.717 μ_{CR} = 0.147
 Diversity = 2.5959e-01 k_t = [2, 4, 2, 4]
 Generation 2170: Best = 2.043960e+03 FES = 217100 μ_F = 0.736 μ_{CR} = 0.146
 Diversity = 2.5401e-01 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 2.036769e+03 FES = 218100 μ_F = 0.725 μ_{CR} = 0.150
 Diversity = 2.4909e-01 k_t = [3, 3, 3, 2]
 Generation 2190: Best = 2.031916e+03 FES = 219100 μ_F = 0.729 μ_{CR} = 0.143
 Diversity = 2.5841e-01 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 2.026107e+03 FES = 220100 μ_F = 0.722 μ_{CR} = 0.141
 Diversity = 2.5050e-01 k_t = [3, 3, 2, 3]
 Generation 2210: Best = 2.020370e+03 FES = 221100 μ_F = 0.738 μ_{CR} = 0.132
 Diversity = 2.6143e-01 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 2.012465e+03 FES = 222100 μ_F = 0.743 μ_{CR} = 0.137
 Diversity = 2.5983e-01 k_t = [2, 4, 4, 3]
 Generation 2230: Best = 2.005266e+03 FES = 223100 μ_F = 0.743 μ_{CR} = 0.137
 Diversity = 2.6559e-01 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 1.993418e+03 FES = 224100 μ_F = 0.742 μ_{CR} = 0.127
 Diversity = 2.8494e-01 k_t = [3, 2, 4, 2]
 Generation 2250: Best = 1.984673e+03 FES = 225100 μ_F = 0.746 μ_{CR} = 0.133
 Diversity = 2.9184e-01 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 1.981464e+03 FES = 226100 μ_F = 0.746 μ_{CR} = 0.137
 Diversity = 2.8282e-01 k_t = [4, 4, 4, 2]
 Generation 2270: Best = 1.970001e+03 FES = 227100 μ_F = 0.741 μ_{CR} = 0.139
 Diversity = 2.8406e-01 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 1.963224e+03 FES = 228100 μ_F = 0.766 μ_{CR} = 0.141
 Diversity = 2.9620e-01 k_t = [2, 3, 3, 2]
 Generation 2290: Best = 1.956572e+03 FES = 229100 μ_F = 0.763 μ_{CR} = 0.139
 Diversity = 3.2481e-01 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 1.939591e+03 FES = 230100 μ_F = 0.763 μ_{CR} = 0.136
 Diversity = 3.1742e-01 k_t = [2, 3, 4, 4]
 Generation 2310: Best = 1.932294e+03 FES = 231100 μ_F = 0.783 μ_{CR} = 0.132
 Diversity = 3.6857e-01 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 1.920711e+03 FES = 232100 μ_F = 0.775 μ_{CR} = 0.135
 Diversity = 3.6405e-01 k_t = [4, 4, 3, 3]
 Generation 2330: Best = 1.913410e+03 FES = 233100 μ_F = 0.771 μ_{CR} = 0.132
 Diversity = 3.5408e-01 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 1.902820e+03 FES = 234100 μ_F = 0.763 μ_{CR} = 0.133
 Diversity = 3.5632e-01 k_t = [2, 4, 2, 2]
 Generation 2350: Best = 1.892998e+03 FES = 235100 μ_F = 0.747 μ_{CR} = 0.142
 Diversity = 3.6232e-01 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 1.884169e+03 FES = 236100 μ_F = 0.744 μ_{CR} = 0.143
 Diversity = 3.4763e-01 k_t = [4, 2, 2, 2]
 Generation 2370: Best = 1.880188e+03 FES = 237100 μ_F = 0.734 μ_{CR} = 0.141
 Diversity = 3.2838e-01 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 1.871713e+03 FES = 238100 μ_F = 0.731 μ_{CR} = 0.134
 Diversity = 3.1489e-01 k_t = [4, 2, 2, 3]

Generation 2390: Best = 1.866711e+03 FES = 239100 μ_F = 0.723 μ_{CR} = 0.132
 Diversity = 3.0795e-01 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 1.857219e+03 FES = 240100 μ_F = 0.737 μ_{CR} = 0.134
 Diversity = 3.1450e-01 k_t = [3, 2, 2, 2]
 Generation 2410: Best = 1.855489e+03 FES = 241100 μ_F = 0.721 μ_{CR} = 0.127
 Diversity = 3.1194e-01 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 1.845447e+03 FES = 242100 μ_F = 0.727 μ_{CR} = 0.129
 Diversity = 2.9745e-01 k_t = [3, 4, 4, 3]
 Generation 2430: Best = 1.842595e+03 FES = 243100 μ_F = 0.731 μ_{CR} = 0.134
 Diversity = 2.8489e-01 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 1.833353e+03 FES = 244100 μ_F = 0.732 μ_{CR} = 0.129
 Diversity = 2.6360e-01 k_t = [3, 4, 3, 4]
 Generation 2450: Best = 1.830717e+03 FES = 245100 μ_F = 0.733 μ_{CR} = 0.129
 Diversity = 2.6419e-01 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 1.826579e+03 FES = 246100 μ_F = 0.725 μ_{CR} = 0.125
 Diversity = 2.5371e-01 k_t = [3, 3, 3, 4]
 Generation 2470: Best = 1.822591e+03 FES = 247100 μ_F = 0.721 μ_{CR} = 0.133
 Diversity = 2.6353e-01 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 1.820011e+03 FES = 248100 μ_F = 0.716 μ_{CR} = 0.130
 Diversity = 2.6492e-01 k_t = [2, 3, 3, 2]
 Generation 2490: Best = 1.815613e+03 FES = 249100 μ_F = 0.724 μ_{CR} = 0.134
 Diversity = 2.8304e-01 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 1.811266e+03 FES = 250100 μ_F = 0.732 μ_{CR} = 0.133
 Diversity = 2.5721e-01 k_t = [2, 4, 4, 3]
 Generation 2510: Best = 1.808413e+03 FES = 251100 μ_F = 0.743 μ_{CR} = 0.132
 Diversity = 2.5459e-01 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 1.806964e+03 FES = 252100 μ_F = 0.740 μ_{CR} = 0.139
 Diversity = 2.4951e-01 k_t = [2, 3, 4, 3]
 Generation 2530: Best = 1.803315e+03 FES = 253100 μ_F = 0.739 μ_{CR} = 0.146
 Diversity = 2.2700e-01 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 1.801241e+03 FES = 254100 μ_F = 0.738 μ_{CR} = 0.138
 Diversity = 2.2845e-01 k_t = [4, 2, 3, 2]
 Generation 2550: Best = 1.799055e+03 FES = 255100 μ_F = 0.709 μ_{CR} = 0.148
 Diversity = 2.3313e-01 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 1.797227e+03 FES = 256100 μ_F = 0.697 μ_{CR} = 0.149
 Diversity = 1.8556e-01 k_t = [3, 3, 4, 4]
 Generation 2570: Best = 1.794628e+03 FES = 257100 μ_F = 0.700 μ_{CR} = 0.158
 Diversity = 1.7405e-01 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 1.791972e+03 FES = 258100 μ_F = 0.708 μ_{CR} = 0.161
 Diversity = 1.9106e-01 k_t = [4, 2, 2, 4]
 Generation 2590: Best = 1.787178e+03 FES = 259100 μ_F = 0.729 μ_{CR} = 0.157
 Diversity = 1.9354e-01 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 1.785909e+03 FES = 260100 μ_F = 0.725 μ_{CR} = 0.159
 Diversity = 2.0702e-01 k_t = [4, 4, 3, 4]
 Generation 2610: Best = 1.782995e+03 FES = 261100 μ_F = 0.722 μ_{CR} = 0.163
 Diversity = 1.9062e-01 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 1.780126e+03 FES = 262100 μ_F = 0.717 μ_{CR} = 0.166
 Diversity = 1.9682e-01 k_t = [2, 4, 4, 2]
 Generation 2630: Best = 1.777971e+03 FES = 263100 μ_F = 0.717 μ_{CR} = 0.159
 Diversity = 2.0887e-01 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 1.775122e+03 FES = 264100 μ_F = 0.714 μ_{CR} = 0.150
 Diversity = 2.0174e-01 k_t = [4, 2, 4, 3]
 Generation 2650: Best = 1.773171e+03 FES = 265100 μ_F = 0.721 μ_{CR} = 0.144
 Diversity = 2.1564e-01 k_t = [1, 1, 1, 1]

Generation 2660: Best = 1.770160e+03 FES = 266100 μ_F = 0.722 μ_{CR} = 0.147
 Diversity = 2.0334e-01 k_t = [4, 3, 2, 4]
 Generation 2670: Best = 1.767689e+03 FES = 267100 μ_F = 0.728 μ_{CR} = 0.142
 Diversity = 2.0472e-01 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 1.765697e+03 FES = 268100 μ_F = 0.735 μ_{CR} = 0.143
 Diversity = 1.9716e-01 k_t = [4, 3, 3, 4]
 Generation 2690: Best = 1.764194e+03 FES = 269100 μ_F = 0.734 μ_{CR} = 0.159
 Diversity = 2.1117e-01 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 1.762519e+03 FES = 270100 μ_F = 0.705 μ_{CR} = 0.158
 Diversity = 1.9802e-01 k_t = [2, 3, 2, 4]
 Generation 2710: Best = 1.760393e+03 FES = 271100 μ_F = 0.719 μ_{CR} = 0.161
 Diversity = 1.8527e-01 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 1.760068e+03 FES = 272100 μ_F = 0.721 μ_{CR} = 0.159
 Diversity = 1.8890e-01 k_t = [3, 4, 3, 3]
 Generation 2730: Best = 1.758333e+03 FES = 273100 μ_F = 0.710 μ_{CR} = 0.138
 Diversity = 1.9553e-01 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 1.757952e+03 FES = 274100 μ_F = 0.707 μ_{CR} = 0.133
 Diversity = 1.9465e-01 k_t = [2, 4, 4, 2]
 Generation 2750: Best = 1.756886e+03 FES = 275100 μ_F = 0.716 μ_{CR} = 0.123
 Diversity = 1.9357e-01 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 1.755540e+03 FES = 276100 μ_F = 0.721 μ_{CR} = 0.132
 Diversity = 1.9051e-01 k_t = [4, 4, 3, 3]
 Generation 2770: Best = 1.754602e+03 FES = 277100 μ_F = 0.715 μ_{CR} = 0.132
 Diversity = 1.9160e-01 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 1.752575e+03 FES = 278100 μ_F = 0.719 μ_{CR} = 0.130
 Diversity = 1.8784e-01 k_t = [2, 4, 2, 3]
 Generation 2790: Best = 1.752107e+03 FES = 279100 μ_F = 0.704 μ_{CR} = 0.136
 Diversity = 1.7837e-01 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 1.749940e+03 FES = 280100 μ_F = 0.711 μ_{CR} = 0.136
 Diversity = 1.8403e-01 k_t = [2, 2, 4, 3]
 Generation 2810: Best = 1.749214e+03 FES = 281100 μ_F = 0.714 μ_{CR} = 0.143
 Diversity = 1.7977e-01 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 1.745815e+03 FES = 282100 μ_F = 0.722 μ_{CR} = 0.143
 Diversity = 1.7153e-01 k_t = [4, 4, 4, 2]
 Generation 2830: Best = 1.744676e+03 FES = 283100 μ_F = 0.735 μ_{CR} = 0.143
 Diversity = 2.0710e-01 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 1.743295e+03 FES = 284100 μ_F = 0.741 μ_{CR} = 0.147
 Diversity = 1.7384e-01 k_t = [3, 2, 4, 4]
 Generation 2850: Best = 1.741670e+03 FES = 285100 μ_F = 0.739 μ_{CR} = 0.139
 Diversity = 1.8008e-01 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 1.739444e+03 FES = 286100 μ_F = 0.748 μ_{CR} = 0.139
 Diversity = 1.7030e-01 k_t = [4, 3, 3, 2]
 Generation 2870: Best = 1.737615e+03 FES = 287100 μ_F = 0.744 μ_{CR} = 0.140
 Diversity = 2.1846e-01 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 1.736023e+03 FES = 288100 μ_F = 0.731 μ_{CR} = 0.141
 Diversity = 1.8186e-01 k_t = [2, 3, 3, 3]
 Generation 2890: Best = 1.734187e+03 FES = 289100 μ_F = 0.731 μ_{CR} = 0.134
 Diversity = 1.8147e-01 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 1.731119e+03 FES = 290100 μ_F = 0.733 μ_{CR} = 0.140
 Diversity = 1.8794e-01 k_t = [2, 2, 3, 3]
 Generation 2910: Best = 1.730255e+03 FES = 291100 μ_F = 0.738 μ_{CR} = 0.132
 Diversity = 2.0341e-01 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 1.727769e+03 FES = 292100 μ_F = 0.743 μ_{CR} = 0.133
 Diversity = 1.8544e-01 k_t = [3, 4, 2, 4]

Generation 2930: Best = 1.726107e+03 FES = 293100 μ_F = 0.743 μ_{CR} = 0.128
 Diversity = 1.7642e-01 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 1.725246e+03 FES = 294100 μ_F = 0.742 μ_{CR} = 0.124
 Diversity = 1.7762e-01 k_t = [3, 2, 3, 3]
 Generation 2950: Best = 1.723720e+03 FES = 295100 μ_F = 0.731 μ_{CR} = 0.116
 Diversity = 1.7625e-01 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 1.722578e+03 FES = 296100 μ_F = 0.741 μ_{CR} = 0.129
 Diversity = 1.6772e-01 k_t = [4, 4, 2, 3]
 Generation 2970: Best = 1.721285e+03 FES = 297100 μ_F = 0.728 μ_{CR} = 0.128
 Diversity = 1.8371e-01 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 1.720478e+03 FES = 298100 μ_F = 0.735 μ_{CR} = 0.127
 Diversity = 1.7110e-01 k_t = [3, 3, 2, 4]
 Generation 2990: Best = 1.719373e+03 FES = 299100 μ_F = 0.726 μ_{CR} = 0.113
 Diversity = 1.7074e-01 k_t = [1, 1, 1, 1]
 Generation 3000: Best = 1.718733e+03 FES = 300100 μ_F = 0.729 μ_{CR} = 0.118
 Diversity = 1.7818e-01 k_t = [3, 3, 3, 2]
 Generation 3010: Best = 1.717328e+03 FES = 301100 μ_F = 0.730 μ_{CR} = 0.118
 Diversity = 1.8362e-01 k_t = [1, 1, 1, 1]
 Generation 3020: Best = 1.716027e+03 FES = 302100 μ_F = 0.723 μ_{CR} = 0.123
 Diversity = 1.6687e-01 k_t = [3, 3, 4, 3]
 Generation 3030: Best = 1.714709e+03 FES = 303100 μ_F = 0.737 μ_{CR} = 0.129
 Diversity = 1.7724e-01 k_t = [1, 1, 1, 1]
 Generation 3040: Best = 1.713686e+03 FES = 304100 μ_F = 0.729 μ_{CR} = 0.119
 Diversity = 1.6573e-01 k_t = [4, 3, 3, 3]
 Generation 3050: Best = 1.712743e+03 FES = 305100 μ_F = 0.747 μ_{CR} = 0.118
 Diversity = 1.6250e-01 k_t = [1, 1, 1, 1]
 Generation 3060: Best = 1.711259e+03 FES = 306100 μ_F = 0.747 μ_{CR} = 0.123
 Diversity = 1.6246e-01 k_t = [2, 3, 3, 2]
 Generation 3070: Best = 1.710575e+03 FES = 307100 μ_F = 0.744 μ_{CR} = 0.126
 Diversity = 1.7077e-01 k_t = [1, 1, 1, 1]
 Generation 3080: Best = 1.708898e+03 FES = 308100 μ_F = 0.752 μ_{CR} = 0.129
 Diversity = 1.8653e-01 k_t = [3, 2, 4, 3]
 Generation 3090: Best = 1.707272e+03 FES = 309100 μ_F = 0.745 μ_{CR} = 0.117
 Diversity = 2.1304e-01 k_t = [1, 1, 1, 1]
 Generation 3100: Best = 1.706207e+03 FES = 310100 μ_F = 0.760 μ_{CR} = 0.112
 Diversity = 2.1964e-01 k_t = [2, 4, 3, 2]
 Generation 3110: Best = 1.705205e+03 FES = 311100 μ_F = 0.766 μ_{CR} = 0.113
 Diversity = 2.1423e-01 k_t = [1, 1, 1, 1]
 Generation 3120: Best = 1.703256e+03 FES = 312100 μ_F = 0.751 μ_{CR} = 0.116
 Diversity = 1.9327e-01 k_t = [4, 4, 2, 4]
 Generation 3130: Best = 1.702175e+03 FES = 313100 μ_F = 0.748 μ_{CR} = 0.126
 Diversity = 1.7372e-01 k_t = [1, 1, 1, 1]
 Generation 3140: Best = 1.701055e+03 FES = 314100 μ_F = 0.755 μ_{CR} = 0.130
 Diversity = 1.5752e-01 k_t = [4, 3, 2, 4]
 Generation 3150: Best = 1.700628e+03 FES = 315100 μ_F = 0.763 μ_{CR} = 0.129
 Diversity = 1.6227e-01 k_t = [1, 1, 1, 1]
 Generation 3160: Best = 1.699315e+03 FES = 316100 μ_F = 0.760 μ_{CR} = 0.135
 Diversity = 1.6460e-01 k_t = [2, 4, 3, 2]
 Generation 3170: Best = 1.698265e+03 FES = 317100 μ_F = 0.760 μ_{CR} = 0.136
 Diversity = 1.6832e-01 k_t = [1, 1, 1, 1]
 Generation 3180: Best = 1.697092e+03 FES = 318100 μ_F = 0.750 μ_{CR} = 0.138
 Diversity = 1.5601e-01 k_t = [3, 4, 4, 4]
 Generation 3190: Best = 1.694581e+03 FES = 319100 μ_F = 0.744 μ_{CR} = 0.142
 Diversity = 1.7901e-01 k_t = [1, 1, 1, 1]

Generation 3200: Best = 1.690750e+03 FES = 320100 μ_F = 0.743 μ_{CR} = 0.150
 Diversity = 2.0980e-01 k_t = [4, 4, 4, 2]
 Generation 3210: Best = 1.690268e+03 FES = 321100 μ_F = 0.751 μ_{CR} = 0.145
 Diversity = 2.2401e-01 k_t = [1, 1, 1, 1]
 Generation 3220: Best = 1.687133e+03 FES = 322100 μ_F = 0.764 μ_{CR} = 0.148
 Diversity = 2.2674e-01 k_t = [2, 2, 3, 2]
 Generation 3230: Best = 1.686553e+03 FES = 323100 μ_F = 0.758 μ_{CR} = 0.139
 Diversity = 2.2653e-01 k_t = [1, 1, 1, 1]
 Generation 3240: Best = 1.682818e+03 FES = 324100 μ_F = 0.750 μ_{CR} = 0.138
 Diversity = 2.2481e-01 k_t = [3, 2, 2, 2]
 Generation 3250: Best = 1.681497e+03 FES = 325100 μ_F = 0.752 μ_{CR} = 0.143
 Diversity = 2.4190e-01 k_t = [1, 1, 1, 1]
 Generation 3260: Best = 1.678829e+03 FES = 326100 μ_F = 0.745 μ_{CR} = 0.138
 Diversity = 2.3581e-01 k_t = [2, 4, 4, 3]
 Generation 3270: Best = 1.675023e+03 FES = 327100 μ_F = 0.744 μ_{CR} = 0.136
 Diversity = 2.7274e-01 k_t = [1, 1, 1, 1]
 Generation 3280: Best = 1.673262e+03 FES = 328100 μ_F = 0.746 μ_{CR} = 0.134
 Diversity = 2.7123e-01 k_t = [4, 4, 2, 3]
 Generation 3290: Best = 1.670259e+03 FES = 329100 μ_F = 0.739 μ_{CR} = 0.132
 Diversity = 2.3137e-01 k_t = [1, 1, 1, 1]
 Generation 3300: Best = 1.667316e+03 FES = 330100 μ_F = 0.746 μ_{CR} = 0.144
 Diversity = 2.4116e-01 k_t = [4, 2, 4, 2]
 Generation 3310: Best = 1.663729e+03 FES = 331100 μ_F = 0.752 μ_{CR} = 0.152
 Diversity = 2.5647e-01 k_t = [1, 1, 1, 1]
 Generation 3320: Best = 1.661002e+03 FES = 332100 μ_F = 0.746 μ_{CR} = 0.151
 Diversity = 2.5482e-01 k_t = [2, 4, 4, 2]
 Generation 3330: Best = 1.659732e+03 FES = 333100 μ_F = 0.738 μ_{CR} = 0.148
 Diversity = 2.2728e-01 k_t = [1, 1, 1, 1]
 Generation 3340: Best = 1.657041e+03 FES = 334100 μ_F = 0.744 μ_{CR} = 0.152
 Diversity = 2.2060e-01 k_t = [3, 3, 3, 4]
 Generation 3350: Best = 1.652372e+03 FES = 335100 μ_F = 0.738 μ_{CR} = 0.150
 Diversity = 2.2223e-01 k_t = [1, 1, 1, 1]
 Generation 3360: Best = 1.651635e+03 FES = 336100 μ_F = 0.735 μ_{CR} = 0.156
 Diversity = 2.3166e-01 k_t = [4, 2, 3, 2]
 Generation 3370: Best = 1.649212e+03 FES = 337100 μ_F = 0.728 μ_{CR} = 0.150
 Diversity = 2.5086e-01 k_t = [1, 1, 1, 1]
 Generation 3380: Best = 1.646736e+03 FES = 338100 μ_F = 0.737 μ_{CR} = 0.156
 Diversity = 2.4859e-01 k_t = [2, 2, 2, 2]
 Generation 3390: Best = 1.645983e+03 FES = 339100 μ_F = 0.724 μ_{CR} = 0.153
 Diversity = 2.4951e-01 k_t = [1, 1, 1, 1]
 Generation 3400: Best = 1.644113e+03 FES = 340100 μ_F = 0.718 μ_{CR} = 0.140
 Diversity = 2.2416e-01 k_t = [4, 3, 2, 4]
 Generation 3410: Best = 1.641683e+03 FES = 341100 μ_F = 0.705 μ_{CR} = 0.140
 Diversity = 1.9169e-01 k_t = [1, 1, 1, 1]
 Generation 3420: Best = 1.639671e+03 FES = 342100 μ_F = 0.703 μ_{CR} = 0.148
 Diversity = 1.9906e-01 k_t = [4, 3, 2, 2]
 Generation 3430: Best = 1.639007e+03 FES = 343100 μ_F = 0.700 μ_{CR} = 0.144
 Diversity = 2.0424e-01 k_t = [1, 1, 1, 1]
 Generation 3440: Best = 1.637833e+03 FES = 344100 μ_F = 0.704 μ_{CR} = 0.145
 Diversity = 2.1053e-01 k_t = [2, 3, 4, 2]
 Generation 3450: Best = 1.635086e+03 FES = 345100 μ_F = 0.710 μ_{CR} = 0.142
 Diversity = 2.0448e-01 k_t = [1, 1, 1, 1]
 Generation 3460: Best = 1.633999e+03 FES = 346100 μ_F = 0.720 μ_{CR} = 0.143
 Diversity = 2.1390e-01 k_t = [2, 3, 2, 2]

Generation 3470: Best = 1.633096e+03 FES = 347100 μ_F = 0.721 μ_{CR} = 0.135
 Diversity = 2.1552e-01 k_t = [1, 1, 1, 1]
 Generation 3480: Best = 1.632382e+03 FES = 348100 μ_F = 0.709 μ_{CR} = 0.133
 Diversity = 2.0081e-01 k_t = [2, 4, 2, 3]
 Generation 3490: Best = 1.631051e+03 FES = 349100 μ_F = 0.700 μ_{CR} = 0.134
 Diversity = 1.8555e-01 k_t = [1, 1, 1, 1]
 Generation 3500: Best = 1.629467e+03 FES = 350100 μ_F = 0.702 μ_{CR} = 0.130
 Diversity = 1.7998e-01 k_t = [3, 4, 4, 2]
 Generation 3510: Best = 1.628727e+03 FES = 351100 μ_F = 0.719 μ_{CR} = 0.126
 Diversity = 1.8675e-01 k_t = [1, 1, 1, 1]
 Generation 3520: Best = 1.626817e+03 FES = 352100 μ_F = 0.741 μ_{CR} = 0.116
 Diversity = 1.7293e-01 k_t = [3, 2, 3, 4]
 Generation 3530: Best = 1.626125e+03 FES = 353100 μ_F = 0.736 μ_{CR} = 0.115
 Diversity = 1.7688e-01 k_t = [1, 1, 1, 1]
 Generation 3540: Best = 1.624655e+03 FES = 354100 μ_F = 0.764 μ_{CR} = 0.119
 Diversity = 1.7472e-01 k_t = [2, 2, 4, 4]
 Generation 3550: Best = 1.624042e+03 FES = 355100 μ_F = 0.763 μ_{CR} = 0.123
 Diversity = 1.8986e-01 k_t = [1, 1, 1, 1]
 Generation 3560: Best = 1.622688e+03 FES = 356100 μ_F = 0.741 μ_{CR} = 0.109
 Diversity = 1.9567e-01 k_t = [3, 2, 2, 3]
 Generation 3570: Best = 1.621347e+03 FES = 357100 μ_F = 0.744 μ_{CR} = 0.110
 Diversity = 1.9842e-01 k_t = [1, 1, 1, 1]
 Generation 3580: Best = 1.620424e+03 FES = 358100 μ_F = 0.753 μ_{CR} = 0.114
 Diversity = 1.8735e-01 k_t = [4, 2, 3, 3]
 Generation 3590: Best = 1.618571e+03 FES = 359100 μ_F = 0.757 μ_{CR} = 0.119
 Diversity = 1.6478e-01 k_t = [1, 1, 1, 1]
 Generation 3600: Best = 1.617485e+03 FES = 360100 μ_F = 0.763 μ_{CR} = 0.124
 Diversity = 1.8058e-01 k_t = [3, 3, 4, 3]
 Generation 3610: Best = 1.615573e+03 FES = 361100 μ_F = 0.747 μ_{CR} = 0.126
 Diversity = 1.9313e-01 k_t = [1, 1, 1, 1]
 Generation 3620: Best = 1.613620e+03 FES = 362100 μ_F = 0.753 μ_{CR} = 0.130
 Diversity = 2.0390e-01 k_t = [3, 4, 2, 4]
 Generation 3630: Best = 1.612066e+03 FES = 363100 μ_F = 0.744 μ_{CR} = 0.133
 Diversity = 1.9974e-01 k_t = [1, 1, 1, 1]
 Generation 3640: Best = 1.610018e+03 FES = 364100 μ_F = 0.745 μ_{CR} = 0.132
 Diversity = 1.8878e-01 k_t = [4, 2, 2, 4]
 Generation 3650: Best = 1.609101e+03 FES = 365100 μ_F = 0.732 μ_{CR} = 0.132
 Diversity = 1.9348e-01 k_t = [1, 1, 1, 1]
 Generation 3660: Best = 1.607938e+03 FES = 366100 μ_F = 0.739 μ_{CR} = 0.127
 Diversity = 1.9078e-01 k_t = [4, 4, 4, 3]
 Generation 3670: Best = 1.605989e+03 FES = 367100 μ_F = 0.748 μ_{CR} = 0.125
 Diversity = 1.9529e-01 k_t = [1, 1, 1, 1]
 Generation 3680: Best = 1.604988e+03 FES = 368100 μ_F = 0.744 μ_{CR} = 0.121
 Diversity = 2.0330e-01 k_t = [2, 3, 3, 2]
 Generation 3690: Best = 1.603198e+03 FES = 369100 μ_F = 0.746 μ_{CR} = 0.130
 Diversity = 2.1166e-01 k_t = [1, 1, 1, 1]
 Generation 3700: Best = 1.601633e+03 FES = 370100 μ_F = 0.744 μ_{CR} = 0.136
 Diversity = 2.0483e-01 k_t = [4, 3, 2, 3]
 Generation 3710: Best = 1.599922e+03 FES = 371100 μ_F = 0.730 μ_{CR} = 0.138
 Diversity = 2.1323e-01 k_t = [1, 1, 1, 1]
 Generation 3720: Best = 1.599182e+03 FES = 372100 μ_F = 0.724 μ_{CR} = 0.145
 Diversity = 1.9117e-01 k_t = [2, 3, 2, 4]
 Generation 3730: Best = 1.598087e+03 FES = 373100 μ_F = 0.726 μ_{CR} = 0.138
 Diversity = 1.7152e-01 k_t = [1, 1, 1, 1]

Generation 3740: Best = 1.596985e+03 FES = 374100 μ_F = 0.721 μ_{CR} = 0.146
 Diversity = 1.7216e-01 k_t = [4, 2, 4, 4]
 Generation 3750: Best = 1.595656e+03 FES = 375100 μ_F = 0.729 μ_{CR} = 0.132
 Diversity = 1.7921e-01 k_t = [1, 1, 1, 1]
 Generation 3760: Best = 1.594936e+03 FES = 376100 μ_F = 0.723 μ_{CR} = 0.132
 Diversity = 1.6862e-01 k_t = [4, 2, 2, 3]
 Generation 3770: Best = 1.593634e+03 FES = 377100 μ_F = 0.736 μ_{CR} = 0.141
 Diversity = 1.7794e-01 k_t = [1, 1, 1, 1]
 Generation 3780: Best = 1.593287e+03 FES = 378100 μ_F = 0.730 μ_{CR} = 0.137
 Diversity = 1.5805e-01 k_t = [2, 4, 2, 3]
 Generation 3790: Best = 1.592721e+03 FES = 379100 μ_F = 0.720 μ_{CR} = 0.126
 Diversity = 1.4890e-01 k_t = [1, 1, 1, 1]
 Generation 3800: Best = 1.591876e+03 FES = 380100 μ_F = 0.717 μ_{CR} = 0.122
 Diversity = 1.4735e-01 k_t = [4, 3, 2, 2]
 Generation 3810: Best = 1.591064e+03 FES = 381100 μ_F = 0.725 μ_{CR} = 0.134
 Diversity = 1.5159e-01 k_t = [1, 1, 1, 1]
 Generation 3820: Best = 1.590567e+03 FES = 382100 μ_F = 0.714 μ_{CR} = 0.141
 Diversity = 1.4578e-01 k_t = [2, 4, 4, 2]
 Generation 3830: Best = 1.589980e+03 FES = 383100 μ_F = 0.715 μ_{CR} = 0.149
 Diversity = 1.4451e-01 k_t = [1, 1, 1, 1]
 Generation 3840: Best = 1.588638e+03 FES = 384100 μ_F = 0.721 μ_{CR} = 0.152
 Diversity = 1.3463e-01 k_t = [2, 4, 2, 4]
 Generation 3850: Best = 1.586914e+03 FES = 385100 μ_F = 0.743 μ_{CR} = 0.145
 Diversity = 1.5312e-01 k_t = [1, 1, 1, 1]
 Generation 3860: Best = 1.586476e+03 FES = 386100 μ_F = 0.740 μ_{CR} = 0.153
 Diversity = 1.3228e-01 k_t = [4, 2, 3, 3]
 Generation 3870: Best = 1.585814e+03 FES = 387100 μ_F = 0.740 μ_{CR} = 0.152
 Diversity = 1.4332e-01 k_t = [1, 1, 1, 1]
 Generation 3880: Best = 1.585061e+03 FES = 388100 μ_F = 0.733 μ_{CR} = 0.147
 Diversity = 1.3681e-01 k_t = [2, 3, 4, 4]
 Generation 3890: Best = 1.583771e+03 FES = 389100 μ_F = 0.746 μ_{CR} = 0.149
 Diversity = 1.4609e-01 k_t = [1, 1, 1, 1]
 Generation 3900: Best = 1.582679e+03 FES = 390100 μ_F = 0.739 μ_{CR} = 0.145
 Diversity = 1.4233e-01 k_t = [3, 2, 3, 2]
 Generation 3910: Best = 1.582064e+03 FES = 391100 μ_F = 0.740 μ_{CR} = 0.139
 Diversity = 1.7753e-01 k_t = [1, 1, 1, 1]
 Generation 3920: Best = 1.580695e+03 FES = 392100 μ_F = 0.743 μ_{CR} = 0.128
 Diversity = 1.6136e-01 k_t = [3, 2, 2, 3]
 Generation 3930: Best = 1.579000e+03 FES = 393100 μ_F = 0.744 μ_{CR} = 0.129
 Diversity = 1.7914e-01 k_t = [1, 1, 1, 1]
 Generation 3940: Best = 1.576794e+03 FES = 394100 μ_F = 0.737 μ_{CR} = 0.130
 Diversity = 1.8024e-01 k_t = [2, 4, 3, 2]
 Generation 3950: Best = 1.576375e+03 FES = 395100 μ_F = 0.735 μ_{CR} = 0.130
 Diversity = 2.0550e-01 k_t = [1, 1, 1, 1]
 Generation 3960: Best = 1.575345e+03 FES = 396100 μ_F = 0.758 μ_{CR} = 0.129
 Diversity = 1.9164e-01 k_t = [3, 2, 2, 4]
 Generation 3970: Best = 1.574496e+03 FES = 397100 μ_F = 0.759 μ_{CR} = 0.126
 Diversity = 1.9307e-01 k_t = [1, 1, 1, 1]
 Generation 3980: Best = 1.573824e+03 FES = 398100 μ_F = 0.743 μ_{CR} = 0.122
 Diversity = 1.8529e-01 k_t = [4, 4, 3, 3]
 Generation 3990: Best = 1.573066e+03 FES = 399100 μ_F = 0.738 μ_{CR} = 0.131
 Diversity = 1.7491e-01 k_t = [1, 1, 1, 1]
 Generation 4000: Best = 1.571812e+03 FES = 400100 μ_F = 0.724 μ_{CR} = 0.133
 Diversity = 1.8762e-01 k_t = [4, 4, 3, 2]

Generation 4010: Best = 1.571470e+03 FES = 401100 μ_F = 0.729 μ_{CR} = 0.136
 Diversity = 2.0156e-01 k_t = [1, 1, 1, 1]
 Generation 4020: Best = 1.570828e+03 FES = 402100 μ_F = 0.732 μ_{CR} = 0.125
 Diversity = 2.2225e-01 k_t = [2, 2, 3, 2]
 Generation 4030: Best = 1.569962e+03 FES = 403100 μ_F = 0.727 μ_{CR} = 0.127
 Diversity = 2.1509e-01 k_t = [1, 1, 1, 1]
 Generation 4040: Best = 1.569250e+03 FES = 404100 μ_F = 0.737 μ_{CR} = 0.121
 Diversity = 2.2962e-01 k_t = [4, 2, 3, 2]
 Generation 4050: Best = 1.568542e+03 FES = 405100 μ_F = 0.733 μ_{CR} = 0.115
 Diversity = 2.4934e-01 k_t = [1, 1, 1, 1]
 Generation 4060: Best = 1.568016e+03 FES = 406100 μ_F = 0.724 μ_{CR} = 0.116
 Diversity = 2.6020e-01 k_t = [4, 4, 4, 4]
 Generation 4070: Best = 1.567217e+03 FES = 407100 μ_F = 0.734 μ_{CR} = 0.122
 Diversity = 2.7228e-01 k_t = [1, 1, 1, 1]
 Generation 4080: Best = 1.566103e+03 FES = 408100 μ_F = 0.734 μ_{CR} = 0.107
 Diversity = 2.4054e-01 k_t = [3, 4, 2, 4]
 Generation 4090: Best = 1.564415e+03 FES = 409100 μ_F = 0.737 μ_{CR} = 0.111
 Diversity = 2.0741e-01 k_t = [1, 1, 1, 1]
 Generation 4100: Best = 1.562839e+03 FES = 410100 μ_F = 0.749 μ_{CR} = 0.124
 Diversity = 2.1783e-01 k_t = [2, 4, 2, 2]
 Generation 4110: Best = 1.561328e+03 FES = 411100 μ_F = 0.739 μ_{CR} = 0.126
 Diversity = 2.3953e-01 k_t = [1, 1, 1, 1]
 Generation 4120: Best = 1.559668e+03 FES = 412100 μ_F = 0.729 μ_{CR} = 0.139
 Diversity = 2.0360e-01 k_t = [3, 3, 3, 4]
 Generation 4130: Best = 1.558051e+03 FES = 413100 μ_F = 0.744 μ_{CR} = 0.141
 Diversity = 1.8577e-01 k_t = [1, 1, 1, 1]
 Generation 4140: Best = 1.556513e+03 FES = 414100 μ_F = 0.731 μ_{CR} = 0.139
 Diversity = 1.7946e-01 k_t = [3, 2, 4, 2]
 Generation 4150: Best = 1.554705e+03 FES = 415100 μ_F = 0.721 μ_{CR} = 0.139
 Diversity = 1.7862e-01 k_t = [1, 1, 1, 1]
 Generation 4160: Best = 1.553423e+03 FES = 416100 μ_F = 0.722 μ_{CR} = 0.141
 Diversity = 1.6220e-01 k_t = [4, 3, 3, 2]
 Generation 4170: Best = 1.551423e+03 FES = 417100 μ_F = 0.720 μ_{CR} = 0.145
 Diversity = 1.6989e-01 k_t = [1, 1, 1, 1]
 Generation 4180: Best = 1.550835e+03 FES = 418100 μ_F = 0.724 μ_{CR} = 0.144
 Diversity = 1.6466e-01 k_t = [4, 3, 3, 2]
 Generation 4190: Best = 1.549010e+03 FES = 419100 μ_F = 0.709 μ_{CR} = 0.152
 Diversity = 1.6902e-01 k_t = [1, 1, 1, 1]
 Generation 4200: Best = 1.548500e+03 FES = 420100 μ_F = 0.711 μ_{CR} = 0.139
 Diversity = 1.4495e-01 k_t = [4, 2, 2, 4]
 Generation 4210: Best = 1.547819e+03 FES = 421100 μ_F = 0.716 μ_{CR} = 0.144
 Diversity = 1.3124e-01 k_t = [1, 1, 1, 1]
 Generation 4220: Best = 1.546684e+03 FES = 422100 μ_F = 0.729 μ_{CR} = 0.143
 Diversity = 1.3152e-01 k_t = [4, 2, 4, 2]
 Generation 4230: Best = 1.546054e+03 FES = 423100 μ_F = 0.739 μ_{CR} = 0.140
 Diversity = 1.3182e-01 k_t = [1, 1, 1, 1]
 Generation 4240: Best = 1.545109e+03 FES = 424100 μ_F = 0.748 μ_{CR} = 0.145
 Diversity = 1.1466e-01 k_t = [2, 2, 4, 4]
 Generation 4250: Best = 1.544181e+03 FES = 425100 μ_F = 0.730 μ_{CR} = 0.140
 Diversity = 1.1201e-01 k_t = [1, 1, 1, 1]
 Generation 4260: Best = 1.543796e+03 FES = 426100 μ_F = 0.728 μ_{CR} = 0.139
 Diversity = 1.1668e-01 k_t = [2, 3, 2, 2]
 Generation 4270: Best = 1.542545e+03 FES = 427100 μ_F = 0.739 μ_{CR} = 0.144
 Diversity = 1.2957e-01 k_t = [1, 1, 1, 1]

Generation 4280: Best = 1.541157e+03 FES = 428100 μ_F = 0.741 μ_{CR} = 0.147
 Diversity = 1.2682e-01 k_t = [3, 4, 3, 3]
 Generation 4290: Best = 1.540349e+03 FES = 429100 μ_F = 0.720 μ_{CR} = 0.149
 Diversity = 1.3110e-01 k_t = [1, 1, 1, 1]
 Generation 4300: Best = 1.539410e+03 FES = 430100 μ_F = 0.726 μ_{CR} = 0.153
 Diversity = 1.3775e-01 k_t = [4, 4, 4, 2]
 Generation 4310: Best = 1.538858e+03 FES = 431100 μ_F = 0.724 μ_{CR} = 0.142
 Diversity = 1.3450e-01 k_t = [1, 1, 1, 1]
 Generation 4320: Best = 1.537339e+03 FES = 432100 μ_F = 0.740 μ_{CR} = 0.157
 Diversity = 1.3022e-01 k_t = [2, 3, 3, 4]
 Generation 4330: Best = 1.535522e+03 FES = 433100 μ_F = 0.752 μ_{CR} = 0.155
 Diversity = 1.4274e-01 k_t = [1, 1, 1, 1]
 Generation 4340: Best = 1.535178e+03 FES = 434100 μ_F = 0.746 μ_{CR} = 0.155
 Diversity = 1.3669e-01 k_t = [3, 2, 4, 4]
 Generation 4350: Best = 1.532397e+03 FES = 435100 μ_F = 0.754 μ_{CR} = 0.163
 Diversity = 1.4491e-01 k_t = [1, 1, 1, 1]
 Generation 4360: Best = 1.531513e+03 FES = 436100 μ_F = 0.759 μ_{CR} = 0.175
 Diversity = 1.4159e-01 k_t = [2, 3, 2, 4]
 Generation 4370: Best = 1.530579e+03 FES = 437100 μ_F = 0.746 μ_{CR} = 0.184
 Diversity = 1.2904e-01 k_t = [1, 1, 1, 1]
 Generation 4380: Best = 1.529543e+03 FES = 438100 μ_F = 0.737 μ_{CR} = 0.187
 Diversity = 1.1999e-01 k_t = [3, 3, 2, 4]
 Generation 4390: Best = 1.528753e+03 FES = 439100 μ_F = 0.743 μ_{CR} = 0.187
 Diversity = 1.1700e-01 k_t = [1, 1, 1, 1]
 Generation 4400: Best = 1.527705e+03 FES = 440100 μ_F = 0.751 μ_{CR} = 0.174
 Diversity = 1.0960e-01 k_t = [2, 2, 4, 3]
 Generation 4410: Best = 1.526589e+03 FES = 441100 μ_F = 0.746 μ_{CR} = 0.161
 Diversity = 1.1495e-01 k_t = [1, 1, 1, 1]
 Generation 4420: Best = 1.525491e+03 FES = 442100 μ_F = 0.751 μ_{CR} = 0.166
 Diversity = 1.2056e-01 k_t = [3, 3, 2, 3]
 Generation 4430: Best = 1.524603e+03 FES = 443100 μ_F = 0.746 μ_{CR} = 0.159
 Diversity = 1.2877e-01 k_t = [1, 1, 1, 1]
 Generation 4440: Best = 1.523338e+03 FES = 444100 μ_F = 0.751 μ_{CR} = 0.158
 Diversity = 1.4144e-01 k_t = [3, 2, 2, 2]
 Generation 4450: Best = 1.522933e+03 FES = 445100 μ_F = 0.737 μ_{CR} = 0.157
 Diversity = 1.4471e-01 k_t = [1, 1, 1, 1]
 Generation 4460: Best = 1.522193e+03 FES = 446100 μ_F = 0.732 μ_{CR} = 0.157
 Diversity = 1.3790e-01 k_t = [3, 3, 3, 3]
 Generation 4470: Best = 1.520703e+03 FES = 447100 μ_F = 0.732 μ_{CR} = 0.156
 Diversity = 1.3766e-01 k_t = [1, 1, 1, 1]
 Generation 4480: Best = 1.519864e+03 FES = 448100 μ_F = 0.736 μ_{CR} = 0.161
 Diversity = 1.3539e-01 k_t = [4, 3, 3, 2]
 Generation 4490: Best = 1.518910e+03 FES = 449100 μ_F = 0.741 μ_{CR} = 0.150
 Diversity = 1.4081e-01 k_t = [1, 1, 1, 1]
 Generation 4500: Best = 1.518064e+03 FES = 450100 μ_F = 0.743 μ_{CR} = 0.148
 Diversity = 1.3558e-01 k_t = [3, 4, 4, 2]
 Generation 4510: Best = 1.516735e+03 FES = 451100 μ_F = 0.739 μ_{CR} = 0.147
 Diversity = 1.4152e-01 k_t = [1, 1, 1, 1]
 Generation 4520: Best = 1.516252e+03 FES = 452100 μ_F = 0.734 μ_{CR} = 0.152
 Diversity = 1.3590e-01 k_t = [2, 4, 3, 3]
 Generation 4530: Best = 1.515112e+03 FES = 453100 μ_F = 0.722 μ_{CR} = 0.164
 Diversity = 1.3589e-01 k_t = [1, 1, 1, 1]
 Generation 4540: Best = 1.513836e+03 FES = 454100 μ_F = 0.716 μ_{CR} = 0.166
 Diversity = 1.3176e-01 k_t = [4, 4, 3, 2]

Generation 4550: Best = 1.513036e+03 FES = 455100 μ_F = 0.723 μ_{CR} = 0.179
 Diversity = 1.2703e-01 k_t = [1, 1, 1, 1]
 Generation 4560: Best = 1.512049e+03 FES = 456100 μ_F = 0.730 μ_{CR} = 0.180
 Diversity = 1.1494e-01 k_t = [4, 3, 3, 4]
 Generation 4570: Best = 1.511578e+03 FES = 457100 μ_F = 0.724 μ_{CR} = 0.168
 Diversity = 1.1069e-01 k_t = [1, 1, 1, 1]
 Generation 4580: Best = 1.510947e+03 FES = 458100 μ_F = 0.736 μ_{CR} = 0.168
 Diversity = 1.0371e-01 k_t = [4, 2, 2, 3]
 Generation 4590: Best = 1.510095e+03 FES = 459100 μ_F = 0.733 μ_{CR} = 0.171
 Diversity = 1.0125e-01 k_t = [1, 1, 1, 1]
 Generation 4600: Best = 1.509521e+03 FES = 460100 μ_F = 0.731 μ_{CR} = 0.166
 Diversity = 1.0035e-01 k_t = [3, 3, 2, 2]
 Generation 4610: Best = 1.509086e+03 FES = 461100 μ_F = 0.719 μ_{CR} = 0.163
 Diversity = 1.0540e-01 k_t = [1, 1, 1, 1]
 Generation 4620: Best = 1.508531e+03 FES = 462100 μ_F = 0.728 μ_{CR} = 0.159
 Diversity = 9.6106e-02 k_t = [2, 4, 4, 4]
 Generation 4630: Best = 1.507917e+03 FES = 463100 μ_F = 0.723 μ_{CR} = 0.160
 Diversity = 9.4898e-02 k_t = [1, 1, 1, 1]
 Generation 4640: Best = 1.507539e+03 FES = 464100 μ_F = 0.740 μ_{CR} = 0.161
 Diversity = 1.0040e-01 k_t = [4, 3, 3, 2]
 Generation 4650: Best = 1.506891e+03 FES = 465100 μ_F = 0.739 μ_{CR} = 0.161
 Diversity = 1.0187e-01 k_t = [1, 1, 1, 1]
 Generation 4660: Best = 1.506205e+03 FES = 466100 μ_F = 0.739 μ_{CR} = 0.170
 Diversity = 9.1880e-02 k_t = [4, 3, 2, 4]
 Generation 4670: Best = 1.505937e+03 FES = 467100 μ_F = 0.744 μ_{CR} = 0.167
 Diversity = 8.3497e-02 k_t = [1, 1, 1, 1]
 Generation 4680: Best = 1.505453e+03 FES = 468100 μ_F = 0.745 μ_{CR} = 0.169
 Diversity = 8.0046e-02 k_t = [3, 4, 3, 2]
 Generation 4690: Best = 1.505173e+03 FES = 469100 μ_F = 0.743 μ_{CR} = 0.171
 Diversity = 7.9325e-02 k_t = [1, 1, 1, 1]
 Generation 4700: Best = 1.504576e+03 FES = 470100 μ_F = 0.742 μ_{CR} = 0.176
 Diversity = 7.6808e-02 k_t = [4, 3, 4, 2]
 Generation 4710: Best = 1.504367e+03 FES = 471100 μ_F = 0.756 μ_{CR} = 0.182
 Diversity = 7.8122e-02 k_t = [1, 1, 1, 1]
 Generation 4720: Best = 1.503890e+03 FES = 472100 μ_F = 0.737 μ_{CR} = 0.179
 Diversity = 7.6297e-02 k_t = [4, 3, 3, 2]
 Generation 4730: Best = 1.503381e+03 FES = 473100 μ_F = 0.734 μ_{CR} = 0.182
 Diversity = 7.8027e-02 k_t = [1, 1, 1, 1]
 Generation 4740: Best = 1.502905e+03 FES = 474100 μ_F = 0.751 μ_{CR} = 0.180
 Diversity = 8.1949e-02 k_t = [4, 3, 2, 2]
 Generation 4750: Best = 1.502491e+03 FES = 475100 μ_F = 0.741 μ_{CR} = 0.176
 Diversity = 8.1348e-02 k_t = [1, 1, 1, 1]
 Generation 4760: Best = 1.502160e+03 FES = 476100 μ_F = 0.737 μ_{CR} = 0.173
 Diversity = 8.1877e-02 k_t = [4, 4, 3, 2]
 Generation 4770: Best = 1.501822e+03 FES = 477100 μ_F = 0.730 μ_{CR} = 0.182
 Diversity = 8.1208e-02 k_t = [1, 1, 1, 1]
 Generation 4780: Best = 1.501273e+03 FES = 478100 μ_F = 0.726 μ_{CR} = 0.184
 Diversity = 7.1762e-02 k_t = [4, 2, 2, 4]
 Generation 4790: Best = 1.500957e+03 FES = 479100 μ_F = 0.737 μ_{CR} = 0.187
 Diversity = 7.2062e-02 k_t = [1, 1, 1, 1]
 Generation 4800: Best = 1.500497e+03 FES = 480100 μ_F = 0.732 μ_{CR} = 0.187
 Diversity = 6.8710e-02 k_t = [4, 2, 4, 4]
 Generation 4810: Best = 1.500142e+03 FES = 481100 μ_F = 0.742 μ_{CR} = 0.188
 Diversity = 6.6795e-02 k_t = [1, 1, 1, 1]

Generation 4820: Best = 1.499978e+03 FES = 482100 μ_F = 0.735 μ_{CR} = 0.184
 Diversity = 6.2538e-02 k_t = [3, 4, 4, 4]
 Generation 4830: Best = 1.499641e+03 FES = 483100 μ_F = 0.741 μ_{CR} = 0.190
 Diversity = 6.1353e-02 k_t = [1, 1, 1, 1]
 Generation 4840: Best = 1.499384e+03 FES = 484100 μ_F = 0.748 μ_{CR} = 0.190
 Diversity = 6.3495e-02 k_t = [3, 4, 2, 2]
 Generation 4850: Best = 1.499114e+03 FES = 485100 μ_F = 0.750 μ_{CR} = 0.198
 Diversity = 6.8133e-02 k_t = [1, 1, 1, 1]
 Generation 4860: Best = 1.498766e+03 FES = 486100 μ_F = 0.752 μ_{CR} = 0.204
 Diversity = 6.2652e-02 k_t = [2, 4, 2, 4]
 Generation 4870: Best = 1.498440e+03 FES = 487100 μ_F = 0.753 μ_{CR} = 0.195
 Diversity = 6.3864e-02 k_t = [1, 1, 1, 1]
 Generation 4880: Best = 1.498160e+03 FES = 488100 μ_F = 0.753 μ_{CR} = 0.187
 Diversity = 6.3571e-02 k_t = [3, 3, 3, 2]
 Generation 4890: Best = 1.497820e+03 FES = 489100 μ_F = 0.748 μ_{CR} = 0.187
 Diversity = 6.5808e-02 k_t = [1, 1, 1, 1]
 Generation 4900: Best = 1.497675e+03 FES = 490100 μ_F = 0.725 μ_{CR} = 0.192
 Diversity = 6.2552e-02 k_t = [2, 3, 2, 3]
 Generation 4910: Best = 1.497480e+03 FES = 491100 μ_F = 0.711 μ_{CR} = 0.190
 Diversity = 6.0677e-02 k_t = [1, 1, 1, 1]
 Generation 4920: Best = 1.497230e+03 FES = 492100 μ_F = 0.712 μ_{CR} = 0.186
 Diversity = 5.9372e-02 k_t = [2, 3, 4, 3]
 Generation 4930: Best = 1.496931e+03 FES = 493100 μ_F = 0.695 μ_{CR} = 0.188
 Diversity = 5.8450e-02 k_t = [1, 1, 1, 1]
 Generation 4940: Best = 1.496672e+03 FES = 494100 μ_F = 0.708 μ_{CR} = 0.188
 Diversity = 5.2143e-02 k_t = [3, 2, 3, 4]
 Generation 4950: Best = 1.496436e+03 FES = 495100 μ_F = 0.712 μ_{CR} = 0.189
 Diversity = 5.0882e-02 k_t = [1, 1, 1, 1]
 Generation 4960: Best = 1.495977e+03 FES = 496100 μ_F = 0.723 μ_{CR} = 0.184
 Diversity = 5.2491e-02 k_t = [2, 2, 3, 4]
 Generation 4970: Best = 1.495768e+03 FES = 497100 μ_F = 0.736 μ_{CR} = 0.186
 Diversity = 5.6590e-02 k_t = [1, 1, 1, 1]
 Generation 4980: Best = 1.495502e+03 FES = 498100 μ_F = 0.730 μ_{CR} = 0.171
 Diversity = 5.7527e-02 k_t = [2, 4, 4, 2]
 Generation 4990: Best = 1.495385e+03 FES = 499100 μ_F = 0.717 μ_{CR} = 0.172
 Diversity = 6.3897e-02 k_t = [1, 1, 1, 1]
 Generation 5000: Best = 1.494961e+03 FES = 500100 μ_F = 0.711 μ_{CR} = 0.168
 Diversity = 5.9260e-02 k_t = [4, 3, 4, 3]
 Generation 5010: Best = 1.494745e+03 FES = 501100 μ_F = 0.734 μ_{CR} = 0.172
 Diversity = 6.1977e-02 k_t = [1, 1, 1, 1]
 Generation 5020: Best = 1.494562e+03 FES = 502100 μ_F = 0.744 μ_{CR} = 0.171
 Diversity = 6.0686e-02 k_t = [3, 3, 2, 4]
 Generation 5030: Best = 1.494435e+03 FES = 503100 μ_F = 0.736 μ_{CR} = 0.154
 Diversity = 5.6066e-02 k_t = [1, 1, 1, 1]
 Generation 5040: Best = 1.494288e+03 FES = 504100 μ_F = 0.736 μ_{CR} = 0.158
 Diversity = 5.0490e-02 k_t = [3, 2, 4, 4]
 Generation 5050: Best = 1.494156e+03 FES = 505100 μ_F = 0.725 μ_{CR} = 0.153
 Diversity = 4.9774e-02 k_t = [1, 1, 1, 1]
 Generation 5060: Best = 1.493929e+03 FES = 506100 μ_F = 0.714 μ_{CR} = 0.157
 Diversity = 4.7875e-02 k_t = [2, 2, 2, 2]
 Generation 5070: Best = 1.493842e+03 FES = 507100 μ_F = 0.722 μ_{CR} = 0.161
 Diversity = 5.0271e-02 k_t = [1, 1, 1, 1]
 Generation 5080: Best = 1.493604e+03 FES = 508100 μ_F = 0.724 μ_{CR} = 0.167
 Diversity = 4.8824e-02 k_t = [3, 2, 3, 3]

Generation 5090: Best = 1.493416e+03 FES = 509100 μ_F = 0.715 μ_{CR} = 0.164
 Diversity = 4.7979e-02 k_t = [1, 1, 1, 1]
 Generation 5100: Best = 1.493332e+03 FES = 510100 μ_F = 0.715 μ_{CR} = 0.160
 Diversity = 4.8401e-02 k_t = [3, 4, 4, 3]
 Generation 5110: Best = 1.493096e+03 FES = 511100 μ_F = 0.721 μ_{CR} = 0.161
 Diversity = 4.8209e-02 k_t = [1, 1, 1, 1]
 Generation 5120: Best = 1.492978e+03 FES = 512100 μ_F = 0.731 μ_{CR} = 0.164
 Diversity = 4.7837e-02 k_t = [2, 4, 4, 3]
 Generation 5130: Best = 1.492845e+03 FES = 513100 μ_F = 0.734 μ_{CR} = 0.163
 Diversity = 4.4934e-02 k_t = [1, 1, 1, 1]
 Generation 5140: Best = 1.492720e+03 FES = 514100 μ_F = 0.730 μ_{CR} = 0.165
 Diversity = 4.3533e-02 k_t = [3, 4, 2, 3]
 Generation 5150: Best = 1.492553e+03 FES = 515100 μ_F = 0.728 μ_{CR} = 0.172
 Diversity = 4.1112e-02 k_t = [1, 1, 1, 1]
 Generation 5160: Best = 1.492351e+03 FES = 516100 μ_F = 0.726 μ_{CR} = 0.182
 Diversity = 4.0523e-02 k_t = [3, 2, 2, 3]
 Generation 5170: Best = 1.492179e+03 FES = 517100 μ_F = 0.740 μ_{CR} = 0.187
 Diversity = 4.3452e-02 k_t = [1, 1, 1, 1]
 Generation 5180: Best = 1.492033e+03 FES = 518100 μ_F = 0.729 μ_{CR} = 0.188
 Diversity = 4.2558e-02 k_t = [4, 3, 3, 3]
 Generation 5190: Best = 1.491890e+03 FES = 519100 μ_F = 0.743 μ_{CR} = 0.192
 Diversity = 4.2656e-02 k_t = [1, 1, 1, 1]
 Generation 5200: Best = 1.491689e+03 FES = 520100 μ_F = 0.746 μ_{CR} = 0.193
 Diversity = 4.2252e-02 k_t = [3, 4, 4, 4]
 Generation 5210: Best = 1.491590e+03 FES = 521100 μ_F = 0.738 μ_{CR} = 0.199
 Diversity = 4.3141e-02 k_t = [1, 1, 1, 1]
 Generation 5220: Best = 1.491295e+03 FES = 522100 μ_F = 0.744 μ_{CR} = 0.195
 Diversity = 4.3856e-02 k_t = [3, 3, 2, 4]
 Generation 5230: Best = 1.490936e+03 FES = 523100 μ_F = 0.756 μ_{CR} = 0.196
 Diversity = 4.6439e-02 k_t = [1, 1, 1, 1]
 Generation 5240: Best = 1.490690e+03 FES = 524100 μ_F = 0.758 μ_{CR} = 0.197
 Diversity = 5.3346e-02 k_t = [3, 3, 3, 2]
 Generation 5250: Best = 1.490522e+03 FES = 525100 μ_F = 0.752 μ_{CR} = 0.200
 Diversity = 5.4920e-02 k_t = [1, 1, 1, 1]
 Generation 5260: Best = 1.490248e+03 FES = 526100 μ_F = 0.747 μ_{CR} = 0.195
 Diversity = 5.4505e-02 k_t = [2, 2, 4, 2]
 Generation 5270: Best = 1.489830e+03 FES = 527100 μ_F = 0.736 μ_{CR} = 0.200
 Diversity = 5.5412e-02 k_t = [1, 1, 1, 1]
 Generation 5280: Best = 1.489592e+03 FES = 528100 μ_F = 0.741 μ_{CR} = 0.201
 Diversity = 5.8864e-02 k_t = [2, 4, 2, 2]
 Generation 5290: Best = 1.489151e+03 FES = 529100 μ_F = 0.752 μ_{CR} = 0.213
 Diversity = 6.2287e-02 k_t = [1, 1, 1, 1]
 Generation 5300: Best = 1.488792e+03 FES = 530100 μ_F = 0.750 μ_{CR} = 0.214
 Diversity = 6.9034e-02 k_t = [3, 3, 2, 2]
 Generation 5310: Best = 1.488093e+03 FES = 531100 μ_F = 0.746 μ_{CR} = 0.211
 Diversity = 7.5584e-02 k_t = [1, 1, 1, 1]
 Generation 5320: Best = 1.487623e+03 FES = 532100 μ_F = 0.748 μ_{CR} = 0.212
 Diversity = 7.9458e-02 k_t = [2, 3, 3, 2]
 Generation 5330: Best = 1.486850e+03 FES = 533100 μ_F = 0.744 μ_{CR} = 0.208
 Diversity = 8.9274e-02 k_t = [1, 1, 1, 1]
 Generation 5340: Best = 1.486397e+03 FES = 534100 μ_F = 0.740 μ_{CR} = 0.205
 Diversity = 8.4920e-02 k_t = [3, 2, 4, 3]
 Generation 5350: Best = 1.485424e+03 FES = 535100 μ_F = 0.737 μ_{CR} = 0.211
 Diversity = 8.6754e-02 k_t = [1, 1, 1, 1]

Generation 5360: Best = 1.484593e+03 FES = 536100 μ_F = 0.733 μ_{CR} = 0.210
 Diversity = 8.7666e-02 k_t = [4, 3, 3, 2]
 Generation 5370: Best = 1.483398e+03 FES = 537100 μ_F = 0.726 μ_{CR} = 0.208
 Diversity = 9.5947e-02 k_t = [1, 1, 1, 1]
 Generation 5380: Best = 1.483049e+03 FES = 538100 μ_F = 0.717 μ_{CR} = 0.192
 Diversity = 8.7878e-02 k_t = [2, 3, 4, 4]
 Generation 5390: Best = 1.482732e+03 FES = 539100 μ_F = 0.703 μ_{CR} = 0.192
 Diversity = 7.9139e-02 k_t = [1, 1, 1, 1]
 Generation 5400: Best = 1.481909e+03 FES = 540100 μ_F = 0.711 μ_{CR} = 0.191
 Diversity = 7.2688e-02 k_t = [3, 2, 3, 2]
 Generation 5410: Best = 1.481263e+03 FES = 541100 μ_F = 0.725 μ_{CR} = 0.189
 Diversity = 7.6870e-02 k_t = [1, 1, 1, 1]
 Generation 5420: Best = 1.480765e+03 FES = 542100 μ_F = 0.723 μ_{CR} = 0.189
 Diversity = 7.5553e-02 k_t = [4, 3, 3, 3]
 Generation 5430: Best = 1.480346e+03 FES = 543100 μ_F = 0.730 μ_{CR} = 0.192
 Diversity = 7.4961e-02 k_t = [1, 1, 1, 1]
 Generation 5440: Best = 1.479994e+03 FES = 544100 μ_F = 0.736 μ_{CR} = 0.175
 Diversity = 7.1510e-02 k_t = [2, 3, 4, 2]
 Generation 5450: Best = 1.479601e+03 FES = 545100 μ_F = 0.722 μ_{CR} = 0.183
 Diversity = 7.0295e-02 k_t = [1, 1, 1, 1]
 Generation 5460: Best = 1.478823e+03 FES = 546100 μ_F = 0.730 μ_{CR} = 0.186
 Diversity = 7.1880e-02 k_t = [3, 2, 2, 2]
 Generation 5470: Best = 1.478668e+03 FES = 547100 μ_F = 0.736 μ_{CR} = 0.184
 Diversity = 7.0343e-02 k_t = [1, 1, 1, 1]
 Generation 5480: Best = 1.478048e+03 FES = 548100 μ_F = 0.738 μ_{CR} = 0.188
 Diversity = 6.8639e-02 k_t = [3, 4, 2, 2]
 Generation 5490: Best = 1.477058e+03 FES = 549100 μ_F = 0.728 μ_{CR} = 0.190
 Diversity = 7.4857e-02 k_t = [1, 1, 1, 1]
 Generation 5500: Best = 1.476801e+03 FES = 550100 μ_F = 0.714 μ_{CR} = 0.185
 Diversity = 7.6716e-02 k_t = [2, 4, 4, 3]
 Generation 5510: Best = 1.476153e+03 FES = 551100 μ_F = 0.720 μ_{CR} = 0.164
 Diversity = 7.7546e-02 k_t = [1, 1, 1, 1]
 Generation 5520: Best = 1.475497e+03 FES = 552100 μ_F = 0.731 μ_{CR} = 0.162
 Diversity = 7.8243e-02 k_t = [2, 3, 2, 4]
 Generation 5530: Best = 1.474358e+03 FES = 553100 μ_F = 0.742 μ_{CR} = 0.160
 Diversity = 8.1699e-02 k_t = [1, 1, 1, 1]
 Generation 5540: Best = 1.473720e+03 FES = 554100 μ_F = 0.750 μ_{CR} = 0.159
 Diversity = 8.5105e-02 k_t = [2, 2, 2, 4]
 Generation 5550: Best = 1.473081e+03 FES = 555100 μ_F = 0.750 μ_{CR} = 0.158
 Diversity = 7.7001e-02 k_t = [1, 1, 1, 1]
 Generation 5560: Best = 1.472421e+03 FES = 556100 μ_F = 0.747 μ_{CR} = 0.164
 Diversity = 8.1530e-02 k_t = [2, 4, 3, 4]
 Generation 5570: Best = 1.471714e+03 FES = 557100 μ_F = 0.744 μ_{CR} = 0.158
 Diversity = 8.1546e-02 k_t = [1, 1, 1, 1]
 Generation 5580: Best = 1.471055e+03 FES = 558100 μ_F = 0.747 μ_{CR} = 0.160
 Diversity = 8.1033e-02 k_t = [4, 4, 4, 3]
 Generation 5590: Best = 1.470286e+03 FES = 559100 μ_F = 0.743 μ_{CR} = 0.152
 Diversity = 8.1650e-02 k_t = [1, 1, 1, 1]
 Generation 5600: Best = 1.469328e+03 FES = 560100 μ_F = 0.755 μ_{CR} = 0.157
 Diversity = 8.1097e-02 k_t = [4, 4, 2, 3]
 Generation 5610: Best = 1.469161e+03 FES = 561100 μ_F = 0.749 μ_{CR} = 0.157
 Diversity = 8.2363e-02 k_t = [1, 1, 1, 1]
 Generation 5620: Best = 1.468654e+03 FES = 562100 μ_F = 0.743 μ_{CR} = 0.147
 Diversity = 7.7508e-02 k_t = [4, 4, 4, 3]

Generation 5630: Best = 1.468385e+03 FES = 563100 μ_F = 0.740 μ_{CR} = 0.154
 Diversity = 7.7440e-02 k_t = [1, 1, 1, 1]
 Generation 5640: Best = 1.467948e+03 FES = 564100 μ_F = 0.727 μ_{CR} = 0.159
 Diversity = 7.6487e-02 k_t = [4, 4, 4, 2]
 Generation 5650: Best = 1.467177e+03 FES = 565100 μ_F = 0.726 μ_{CR} = 0.168
 Diversity = 8.0990e-02 k_t = [1, 1, 1, 1]
 Generation 5660: Best = 1.466519e+03 FES = 566100 μ_F = 0.728 μ_{CR} = 0.167
 Diversity = 7.1784e-02 k_t = [2, 4, 4, 3]
 Generation 5670: Best = 1.465805e+03 FES = 567100 μ_F = 0.737 μ_{CR} = 0.169
 Diversity = 8.0152e-02 k_t = [1, 1, 1, 1]
 Generation 5680: Best = 1.464922e+03 FES = 568100 μ_F = 0.727 μ_{CR} = 0.161
 Diversity = 8.4635e-02 k_t = [3, 4, 2, 2]
 Generation 5690: Best = 1.464294e+03 FES = 569100 μ_F = 0.737 μ_{CR} = 0.165
 Diversity = 8.8894e-02 k_t = [1, 1, 1, 1]
 Generation 5700: Best = 1.463276e+03 FES = 570100 μ_F = 0.739 μ_{CR} = 0.170
 Diversity = 9.1549e-02 k_t = [2, 3, 4, 3]
 Generation 5710: Best = 1.462660e+03 FES = 571100 μ_F = 0.744 μ_{CR} = 0.173
 Diversity = 9.7759e-02 k_t = [1, 1, 1, 1]
 Generation 5720: Best = 1.462010e+03 FES = 572100 μ_F = 0.743 μ_{CR} = 0.172
 Diversity = 9.3285e-02 k_t = [2, 4, 3, 2]
 Generation 5730: Best = 1.461362e+03 FES = 573100 μ_F = 0.728 μ_{CR} = 0.165
 Diversity = 9.4730e-02 k_t = [1, 1, 1, 1]
 Generation 5740: Best = 1.460878e+03 FES = 574100 μ_F = 0.724 μ_{CR} = 0.170
 Diversity = 8.1306e-02 k_t = [2, 4, 4, 3]
 Generation 5750: Best = 1.460505e+03 FES = 575100 μ_F = 0.723 μ_{CR} = 0.178
 Diversity = 8.4268e-02 k_t = [1, 1, 1, 1]
 Generation 5760: Best = 1.459914e+03 FES = 576100 μ_F = 0.724 μ_{CR} = 0.182
 Diversity = 8.3759e-02 k_t = [3, 4, 2, 2]
 Generation 5770: Best = 1.459452e+03 FES = 577100 μ_F = 0.720 μ_{CR} = 0.177
 Diversity = 8.0551e-02 k_t = [1, 1, 1, 1]
 Generation 5780: Best = 1.459117e+03 FES = 578100 μ_F = 0.703 μ_{CR} = 0.172
 Diversity = 7.1520e-02 k_t = [2, 3, 4, 4]
 Generation 5790: Best = 1.458856e+03 FES = 579100 μ_F = 0.701 μ_{CR} = 0.170
 Diversity = 6.8482e-02 k_t = [1, 1, 1, 1]
 Generation 5800: Best = 1.458274e+03 FES = 580100 μ_F = 0.703 μ_{CR} = 0.172
 Diversity = 6.6446e-02 k_t = [3, 3, 4, 3]
 Generation 5810: Best = 1.457938e+03 FES = 581100 μ_F = 0.718 μ_{CR} = 0.169
 Diversity = 6.9855e-02 k_t = [1, 1, 1, 1]
 Generation 5820: Best = 1.457694e+03 FES = 582100 μ_F = 0.701 μ_{CR} = 0.165
 Diversity = 6.0732e-02 k_t = [3, 2, 2, 3]
 Generation 5830: Best = 1.457440e+03 FES = 583100 μ_F = 0.711 μ_{CR} = 0.158
 Diversity = 6.1658e-02 k_t = [1, 1, 1, 1]
 Generation 5840: Best = 1.457126e+03 FES = 584100 μ_F = 0.716 μ_{CR} = 0.147
 Diversity = 6.3971e-02 k_t = [4, 4, 2, 2]
 Generation 5850: Best = 1.456670e+03 FES = 585100 μ_F = 0.723 μ_{CR} = 0.136
 Diversity = 6.1020e-02 k_t = [1, 1, 1, 1]
 Generation 5860: Best = 1.456334e+03 FES = 586100 μ_F = 0.718 μ_{CR} = 0.134
 Diversity = 6.0324e-02 k_t = [2, 3, 3, 3]
 Generation 5870: Best = 1.456036e+03 FES = 587100 μ_F = 0.719 μ_{CR} = 0.131
 Diversity = 6.6243e-02 k_t = [1, 1, 1, 1]
 Generation 5880: Best = 1.455740e+03 FES = 588100 μ_F = 0.716 μ_{CR} = 0.144
 Diversity = 6.7835e-02 k_t = [2, 2, 3, 4]
 Generation 5890: Best = 1.455507e+03 FES = 589100 μ_F = 0.716 μ_{CR} = 0.147
 Diversity = 6.8906e-02 k_t = [1, 1, 1, 1]

Generation 5900: Best = 1.455374e+03 FES = 590100 μ_F = 0.733 μ_{CR} = 0.146
 Diversity = 6.6925e-02 k_t = [3, 3, 3, 4]
 Generation 5910: Best = 1.455146e+03 FES = 591100 μ_F = 0.736 μ_{CR} = 0.146
 Diversity = 5.9159e-02 k_t = [1, 1, 1, 1]
 Generation 5920: Best = 1.454982e+03 FES = 592100 μ_F = 0.738 μ_{CR} = 0.151
 Diversity = 6.0077e-02 k_t = [3, 4, 4, 4]
 Generation 5930: Best = 1.454772e+03 FES = 593100 μ_F = 0.738 μ_{CR} = 0.155
 Diversity = 5.7509e-02 k_t = [1, 1, 1, 1]
 Generation 5940: Best = 1.454596e+03 FES = 594100 μ_F = 0.719 μ_{CR} = 0.148
 Diversity = 5.4404e-02 k_t = [2, 2, 3, 3]
 Generation 5950: Best = 1.454366e+03 FES = 595100 μ_F = 0.712 μ_{CR} = 0.143
 Diversity = 5.4655e-02 k_t = [1, 1, 1, 1]
 Generation 5960: Best = 1.454029e+03 FES = 596100 μ_F = 0.722 μ_{CR} = 0.160
 Diversity = 5.1749e-02 k_t = [4, 4, 4, 3]
 Generation 5970: Best = 1.453789e+03 FES = 597100 μ_F = 0.739 μ_{CR} = 0.169
 Diversity = 5.5137e-02 k_t = [1, 1, 1, 1]
 Generation 5980: Best = 1.453673e+03 FES = 598100 μ_F = 0.734 μ_{CR} = 0.166
 Diversity = 4.9816e-02 k_t = [4, 2, 4, 3]
 Generation 5990: Best = 1.453485e+03 FES = 599100 μ_F = 0.745 μ_{CR} = 0.171
 Diversity = 5.2460e-02 k_t = [1, 1, 1, 1]
 Generation 6000: Best = 1.453243e+03 FES = 600100 μ_F = 0.748 μ_{CR} = 0.180
 Diversity = 5.1301e-02 k_t = [4, 4, 3, 2]
 Generation 6010: Best = 1.452733e+03 FES = 601100 μ_F = 0.742 μ_{CR} = 0.175
 Diversity = 5.6935e-02 k_t = [1, 1, 1, 1]
 Generation 6020: Best = 1.452601e+03 FES = 602100 μ_F = 0.741 μ_{CR} = 0.172
 Diversity = 5.8051e-02 k_t = [3, 4, 4, 4]
 Generation 6030: Best = 1.452008e+03 FES = 603100 μ_F = 0.742 μ_{CR} = 0.170
 Diversity = 5.6512e-02 k_t = [1, 1, 1, 1]
 Generation 6040: Best = 1.451785e+03 FES = 604100 μ_F = 0.754 μ_{CR} = 0.169
 Diversity = 6.0297e-02 k_t = [2, 2, 2, 4]
 Generation 6050: Best = 1.451258e+03 FES = 605100 μ_F = 0.750 μ_{CR} = 0.173
 Diversity = 6.0298e-02 k_t = [1, 1, 1, 1]
 Generation 6060: Best = 1.450865e+03 FES = 606100 μ_F = 0.743 μ_{CR} = 0.181
 Diversity = 5.7644e-02 k_t = [2, 4, 2, 2]
 Generation 6070: Best = 1.450426e+03 FES = 607100 μ_F = 0.757 μ_{CR} = 0.182
 Diversity = 6.3951e-02 k_t = [1, 1, 1, 1]
 Generation 6080: Best = 1.449709e+03 FES = 608100 μ_F = 0.768 μ_{CR} = 0.184
 Diversity = 7.0110e-02 k_t = [4, 2, 3, 2]
 Generation 6090: Best = 1.449224e+03 FES = 609100 μ_F = 0.765 μ_{CR} = 0.192
 Diversity = 7.8352e-02 k_t = [1, 1, 1, 1]
 Generation 6100: Best = 1.448500e+03 FES = 610100 μ_F = 0.763 μ_{CR} = 0.195
 Diversity = 7.9817e-02 k_t = [2, 2, 2, 3]
 Generation 6110: Best = 1.447866e+03 FES = 611100 μ_F = 0.747 μ_{CR} = 0.204
 Diversity = 8.9759e-02 k_t = [1, 1, 1, 1]
 Generation 6120: Best = 1.447166e+03 FES = 612100 μ_F = 0.732 μ_{CR} = 0.204
 Diversity = 8.8074e-02 k_t = [3, 2, 3, 2]
 Generation 6130: Best = 1.446838e+03 FES = 613100 μ_F = 0.720 μ_{CR} = 0.192
 Diversity = 8.5318e-02 k_t = [1, 1, 1, 1]
 Generation 6140: Best = 1.446244e+03 FES = 614100 μ_F = 0.718 μ_{CR} = 0.196
 Diversity = 7.5626e-02 k_t = [3, 4, 4, 3]
 Generation 6150: Best = 1.445335e+03 FES = 615100 μ_F = 0.720 μ_{CR} = 0.206
 Diversity = 8.7552e-02 k_t = [1, 1, 1, 1]
 Generation 6160: Best = 1.444521e+03 FES = 616100 μ_F = 0.738 μ_{CR} = 0.198
 Diversity = 9.0028e-02 k_t = [2, 3, 2, 4]

Generation 6170: Best = 1.443906e+03 FES = 617100 μ_F = 0.741 μ_{CR} = 0.186
 Diversity = 8.6106e-02 k_t = [1, 1, 1, 1]
 Generation 6180: Best = 1.443548e+03 FES = 618100 μ_F = 0.741 μ_{CR} = 0.184
 Diversity = 8.2785e-02 k_t = [2, 2, 2, 4]
 Generation 6190: Best = 1.443016e+03 FES = 619100 μ_F = 0.742 μ_{CR} = 0.195
 Diversity = 7.4065e-02 k_t = [1, 1, 1, 1]
 Generation 6200: Best = 1.442836e+03 FES = 620100 μ_F = 0.737 μ_{CR} = 0.181
 Diversity = 6.8411e-02 k_t = [4, 4, 2, 3]
 Generation 6210: Best = 1.442630e+03 FES = 621100 μ_F = 0.722 μ_{CR} = 0.181
 Diversity = 6.8476e-02 k_t = [1, 1, 1, 1]
 Generation 6220: Best = 1.442374e+03 FES = 622100 μ_F = 0.712 μ_{CR} = 0.175
 Diversity = 5.9112e-02 k_t = [2, 4, 4, 4]
 Generation 6230: Best = 1.442114e+03 FES = 623100 μ_F = 0.704 μ_{CR} = 0.179
 Diversity = 5.4355e-02 k_t = [1, 1, 1, 1]
 Generation 6240: Best = 1.441838e+03 FES = 624100 μ_F = 0.712 μ_{CR} = 0.175
 Diversity = 5.4993e-02 k_t = [4, 2, 3, 4]
 Generation 6250: Best = 1.441617e+03 FES = 625100 μ_F = 0.717 μ_{CR} = 0.177
 Diversity = 4.9309e-02 k_t = [1, 1, 1, 1]
 Generation 6260: Best = 1.441420e+03 FES = 626100 μ_F = 0.717 μ_{CR} = 0.172
 Diversity = 5.4909e-02 k_t = [3, 2, 3, 2]
 Generation 6270: Best = 1.441270e+03 FES = 627100 μ_F = 0.711 μ_{CR} = 0.174
 Diversity = 5.0388e-02 k_t = [1, 1, 1, 1]
 Generation 6280: Best = 1.441131e+03 FES = 628100 μ_F = 0.715 μ_{CR} = 0.170
 Diversity = 4.7049e-02 k_t = [2, 2, 3, 3]
 Generation 6290: Best = 1.440936e+03 FES = 629100 μ_F = 0.725 μ_{CR} = 0.162
 Diversity = 4.4546e-02 k_t = [1, 1, 1, 1]
 Generation 6300: Best = 1.440855e+03 FES = 630100 μ_F = 0.725 μ_{CR} = 0.161
 Diversity = 4.0323e-02 k_t = [2, 2, 2, 4]
 Generation 6310: Best = 1.440721e+03 FES = 631100 μ_F = 0.732 μ_{CR} = 0.150
 Diversity = 3.7217e-02 k_t = [1, 1, 1, 1]
 Generation 6320: Best = 1.440490e+03 FES = 632100 μ_F = 0.747 μ_{CR} = 0.155
 Diversity = 3.7015e-02 k_t = [2, 4, 2, 4]
 Generation 6330: Best = 1.440328e+03 FES = 633100 μ_F = 0.744 μ_{CR} = 0.166
 Diversity = 3.6697e-02 k_t = [1, 1, 1, 1]
 Generation 6340: Best = 1.440134e+03 FES = 634100 μ_F = 0.736 μ_{CR} = 0.177
 Diversity = 3.9608e-02 k_t = [3, 3, 4, 2]
 Generation 6350: Best = 1.439867e+03 FES = 635100 μ_F = 0.749 μ_{CR} = 0.178
 Diversity = 4.5759e-02 k_t = [1, 1, 1, 1]
 Generation 6360: Best = 1.439602e+03 FES = 636100 μ_F = 0.764 μ_{CR} = 0.186
 Diversity = 4.7988e-02 k_t = [3, 2, 4, 4]
 Generation 6370: Best = 1.439491e+03 FES = 637100 μ_F = 0.771 μ_{CR} = 0.179
 Diversity = 4.7212e-02 k_t = [1, 1, 1, 1]
 Generation 6380: Best = 1.439253e+03 FES = 638100 μ_F = 0.758 μ_{CR} = 0.179
 Diversity = 4.7558e-02 k_t = [2, 4, 3, 4]
 Generation 6390: Best = 1.438934e+03 FES = 639100 μ_F = 0.748 μ_{CR} = 0.184
 Diversity = 5.2311e-02 k_t = [1, 1, 1, 1]
 Generation 6400: Best = 1.438558e+03 FES = 640100 μ_F = 0.764 μ_{CR} = 0.177
 Diversity = 5.6205e-02 k_t = [2, 3, 2, 4]
 Generation 6410: Best = 1.438076e+03 FES = 641100 μ_F = 0.764 μ_{CR} = 0.177
 Diversity = 5.8605e-02 k_t = [1, 1, 1, 1]
 Generation 6420: Best = 1.437710e+03 FES = 642100 μ_F = 0.761 μ_{CR} = 0.182
 Diversity = 5.9550e-02 k_t = [2, 3, 4, 4]
 Generation 6430: Best = 1.437366e+03 FES = 643100 μ_F = 0.749 μ_{CR} = 0.180
 Diversity = 6.0428e-02 k_t = [1, 1, 1, 1]

Generation 6440: Best = 1.437176e+03 FES = 644100 μ_F = 0.747 μ_{CR} = 0.186
 Diversity = 5.5801e-02 k_t = [2, 3, 3, 4]
 Generation 6450: Best = 1.436928e+03 FES = 645100 μ_F = 0.740 μ_{CR} = 0.182
 Diversity = 5.6544e-02 k_t = [1, 1, 1, 1]
 Generation 6460: Best = 1.436553e+03 FES = 646100 μ_F = 0.750 μ_{CR} = 0.191
 Diversity = 5.1141e-02 k_t = [2, 2, 4, 3]
 Generation 6470: Best = 1.436333e+03 FES = 647100 μ_F = 0.750 μ_{CR} = 0.197
 Diversity = 6.0329e-02 k_t = [1, 1, 1, 1]
 Generation 6480: Best = 1.436061e+03 FES = 648100 μ_F = 0.749 μ_{CR} = 0.196
 Diversity = 5.3183e-02 k_t = [3, 4, 4, 3]
 Generation 6490: Best = 1.435869e+03 FES = 649100 μ_F = 0.732 μ_{CR} = 0.189
 Diversity = 5.4648e-02 k_t = [1, 1, 1, 1]
 Generation 6500: Best = 1.435679e+03 FES = 650100 μ_F = 0.728 μ_{CR} = 0.189
 Diversity = 5.1469e-02 k_t = [2, 4, 4, 4]
 Generation 6510: Best = 1.435535e+03 FES = 651100 μ_F = 0.711 μ_{CR} = 0.180
 Diversity = 4.6349e-02 k_t = [1, 1, 1, 1]
 Generation 6520: Best = 1.435369e+03 FES = 652100 μ_F = 0.697 μ_{CR} = 0.178
 Diversity = 4.0797e-02 k_t = [3, 4, 3, 4]
 Generation 6530: Best = 1.435261e+03 FES = 653100 μ_F = 0.709 μ_{CR} = 0.180
 Diversity = 4.0129e-02 k_t = [1, 1, 1, 1]
 Generation 6540: Best = 1.435200e+03 FES = 654100 μ_F = 0.710 μ_{CR} = 0.178
 Diversity = 3.6334e-02 k_t = [3, 2, 2, 3]
 Generation 6550: Best = 1.435062e+03 FES = 655100 μ_F = 0.708 μ_{CR} = 0.177
 Diversity = 3.8837e-02 k_t = [1, 1, 1, 1]
 Generation 6560: Best = 1.434843e+03 FES = 656100 μ_F = 0.726 μ_{CR} = 0.181
 Diversity = 3.6710e-02 k_t = [2, 4, 3, 2]
 Generation 6570: Best = 1.434665e+03 FES = 657100 μ_F = 0.740 μ_{CR} = 0.181
 Diversity = 4.5355e-02 k_t = [1, 1, 1, 1]
 Generation 6580: Best = 1.434593e+03 FES = 658100 μ_F = 0.742 μ_{CR} = 0.167
 Diversity = 4.0376e-02 k_t = [4, 4, 3, 4]
 Generation 6590: Best = 1.434511e+03 FES = 659100 μ_F = 0.746 μ_{CR} = 0.162
 Diversity = 3.8460e-02 k_t = [1, 1, 1, 1]
 Generation 6600: Best = 1.434457e+03 FES = 660100 μ_F = 0.745 μ_{CR} = 0.154
 Diversity = 3.7025e-02 k_t = [2, 3, 2, 2]
 Generation 6610: Best = 1.434348e+03 FES = 661100 μ_F = 0.730 μ_{CR} = 0.155
 Diversity = 3.9173e-02 k_t = [1, 1, 1, 1]
 Generation 6620: Best = 1.434228e+03 FES = 662100 μ_F = 0.729 μ_{CR} = 0.151
 Diversity = 3.8809e-02 k_t = [4, 3, 3, 2]
 Generation 6630: Best = 1.434046e+03 FES = 663100 μ_F = 0.731 μ_{CR} = 0.158
 Diversity = 4.1519e-02 k_t = [1, 1, 1, 1]
 Generation 6640: Best = 1.434005e+03 FES = 664100 μ_F = 0.715 μ_{CR} = 0.157
 Diversity = 3.8193e-02 k_t = [4, 3, 3, 4]
 Generation 6650: Best = 1.433801e+03 FES = 665100 μ_F = 0.728 μ_{CR} = 0.170
 Diversity = 3.6757e-02 k_t = [1, 1, 1, 1]
 Generation 6660: Best = 1.433581e+03 FES = 666100 μ_F = 0.733 μ_{CR} = 0.172
 Diversity = 3.8409e-02 k_t = [3, 2, 2, 3]
 Generation 6670: Best = 1.433136e+03 FES = 667100 μ_F = 0.740 μ_{CR} = 0.172
 Diversity = 4.6524e-02 k_t = [1, 1, 1, 1]
 Generation 6680: Best = 1.432474e+03 FES = 668100 μ_F = 0.758 μ_{CR} = 0.186
 Diversity = 5.4184e-02 k_t = [2, 3, 3, 4]
 Generation 6690: Best = 1.432333e+03 FES = 669100 μ_F = 0.751 μ_{CR} = 0.182
 Diversity = 5.7212e-02 k_t = [1, 1, 1, 1]
 Generation 6700: Best = 1.431788e+03 FES = 670100 μ_F = 0.751 μ_{CR} = 0.179
 Diversity = 5.7450e-02 k_t = [4, 3, 4, 4]

Generation 6710: Best = 1.431573e+03 FES = 671100 μ_F = 0.743 μ_{CR} = 0.169
 Diversity = 5.2800e-02 k_t = [1, 1, 1, 1]
 Generation 6720: Best = 1.431007e+03 FES = 672100 μ_F = 0.753 μ_{CR} = 0.171
 Diversity = 5.4798e-02 k_t = [2, 2, 3, 2]
 Generation 6730: Best = 1.430179e+03 FES = 673100 μ_F = 0.758 μ_{CR} = 0.181
 Diversity = 6.7921e-02 k_t = [1, 1, 1, 1]
 Generation 6740: Best = 1.429941e+03 FES = 674100 μ_F = 0.766 μ_{CR} = 0.174
 Diversity = 6.4389e-02 k_t = [2, 2, 3, 4]
 Generation 6750: Best = 1.429332e+03 FES = 675100 μ_F = 0.767 μ_{CR} = 0.183
 Diversity = 6.9427e-02 k_t = [1, 1, 1, 1]
 Generation 6760: Best = 1.428958e+03 FES = 676100 μ_F = 0.763 μ_{CR} = 0.172
 Diversity = 6.5487e-02 k_t = [2, 4, 3, 4]
 Generation 6770: Best = 1.428543e+03 FES = 677100 μ_F = 0.764 μ_{CR} = 0.182
 Diversity = 6.5451e-02 k_t = [1, 1, 1, 1]
 Generation 6780: Best = 1.428182e+03 FES = 678100 μ_F = 0.750 μ_{CR} = 0.187
 Diversity = 6.4379e-02 k_t = [4, 2, 2, 3]
 Generation 6790: Best = 1.427784e+03 FES = 679100 μ_F = 0.736 μ_{CR} = 0.186
 Diversity = 6.5144e-02 k_t = [1, 1, 1, 1]
 Generation 6800: Best = 1.427161e+03 FES = 680100 μ_F = 0.737 μ_{CR} = 0.184
 Diversity = 5.6238e-02 k_t = [2, 2, 2, 4]
 Generation 6810: Best = 1.427061e+03 FES = 681100 μ_F = 0.736 μ_{CR} = 0.183
 Diversity = 5.7233e-02 k_t = [1, 1, 1, 1]
 Generation 6820: Best = 1.426723e+03 FES = 682100 μ_F = 0.738 μ_{CR} = 0.185
 Diversity = 5.5155e-02 k_t = [2, 4, 2, 3]
 Generation 6830: Best = 1.426320e+03 FES = 683100 μ_F = 0.738 μ_{CR} = 0.190
 Diversity = 6.2238e-02 k_t = [1, 1, 1, 1]
 Generation 6840: Best = 1.425861e+03 FES = 684100 μ_F = 0.747 μ_{CR} = 0.192
 Diversity = 5.8827e-02 k_t = [4, 4, 2, 3]
 Generation 6850: Best = 1.425320e+03 FES = 685100 μ_F = 0.747 μ_{CR} = 0.194
 Diversity = 6.0934e-02 k_t = [1, 1, 1, 1]
 Generation 6860: Best = 1.424959e+03 FES = 686100 μ_F = 0.744 μ_{CR} = 0.186
 Diversity = 6.1544e-02 k_t = [2, 3, 4, 3]
 Generation 6870: Best = 1.424698e+03 FES = 687100 μ_F = 0.729 μ_{CR} = 0.178
 Diversity = 6.2817e-02 k_t = [1, 1, 1, 1]
 Generation 6880: Best = 1.424602e+03 FES = 688100 μ_F = 0.728 μ_{CR} = 0.178
 Diversity = 6.1967e-02 k_t = [2, 2, 3, 3]
 Generation 6890: Best = 1.424215e+03 FES = 689100 μ_F = 0.717 μ_{CR} = 0.180
 Diversity = 6.5248e-02 k_t = [1, 1, 1, 1]
 Generation 6900: Best = 1.423766e+03 FES = 690100 μ_F = 0.728 μ_{CR} = 0.176
 Diversity = 6.5191e-02 k_t = [3, 3, 3, 2]
 Generation 6910: Best = 1.423474e+03 FES = 691100 μ_F = 0.735 μ_{CR} = 0.179
 Diversity = 6.5098e-02 k_t = [1, 1, 1, 1]
 Generation 6920: Best = 1.422889e+03 FES = 692100 μ_F = 0.740 μ_{CR} = 0.168
 Diversity = 6.0135e-02 k_t = [2, 3, 4, 3]
 Generation 6930: Best = 1.422525e+03 FES = 693100 μ_F = 0.742 μ_{CR} = 0.171
 Diversity = 6.0615e-02 k_t = [1, 1, 1, 1]
 Generation 6940: Best = 1.422181e+03 FES = 694100 μ_F = 0.761 μ_{CR} = 0.167
 Diversity = 7.2384e-02 k_t = [2, 3, 3, 2]
 Generation 6950: Best = 1.421098e+03 FES = 695100 μ_F = 0.763 μ_{CR} = 0.162
 Diversity = 8.3149e-02 k_t = [1, 1, 1, 1]
 Generation 6960: Best = 1.420993e+03 FES = 696100 μ_F = 0.741 μ_{CR} = 0.159
 Diversity = 8.5292e-02 k_t = [3, 4, 3, 3]
 Generation 6970: Best = 1.420568e+03 FES = 697100 μ_F = 0.742 μ_{CR} = 0.155
 Diversity = 7.7222e-02 k_t = [1, 1, 1, 1]

Generation 6980: Best = 1.420161e+03 FES = 698100 μ_F = 0.736 μ_{CR} = 0.166
 Diversity = 7.2098e-02 k_t = [2, 3, 2, 4]
 Generation 6990: Best = 1.419854e+03 FES = 699100 μ_F = 0.733 μ_{CR} = 0.158
 Diversity = 7.2502e-02 k_t = [1, 1, 1, 1]
 Generation 7000: Best = 1.419667e+03 FES = 700100 μ_F = 0.721 μ_{CR} = 0.155
 Diversity = 6.9649e-02 k_t = [2, 2, 4, 2]
 Generation 7010: Best = 1.419402e+03 FES = 701100 μ_F = 0.714 μ_{CR} = 0.154
 Diversity = 7.1080e-02 k_t = [1, 1, 1, 1]
 Generation 7020: Best = 1.419030e+03 FES = 702100 μ_F = 0.712 μ_{CR} = 0.151
 Diversity = 6.2768e-02 k_t = [3, 3, 3, 2]
 Generation 7030: Best = 1.418905e+03 FES = 703100 μ_F = 0.714 μ_{CR} = 0.148
 Diversity = 6.2182e-02 k_t = [1, 1, 1, 1]
 Generation 7040: Best = 1.418684e+03 FES = 704100 μ_F = 0.720 μ_{CR} = 0.143
 Diversity = 5.6519e-02 k_t = [2, 2, 3, 4]
 Generation 7050: Best = 1.418481e+03 FES = 705100 μ_F = 0.730 μ_{CR} = 0.143
 Diversity = 5.3418e-02 k_t = [1, 1, 1, 1]
 Generation 7060: Best = 1.418225e+03 FES = 706100 μ_F = 0.729 μ_{CR} = 0.150
 Diversity = 5.1966e-02 k_t = [2, 3, 4, 3]
 Generation 7070: Best = 1.418195e+03 FES = 707100 μ_F = 0.729 μ_{CR} = 0.148
 Diversity = 5.0764e-02 k_t = [1, 1, 1, 1]
 Generation 7080: Best = 1.418021e+03 FES = 708100 μ_F = 0.728 μ_{CR} = 0.148
 Diversity = 4.8246e-02 k_t = [3, 3, 4, 3]
 Generation 7090: Best = 1.417831e+03 FES = 709100 μ_F = 0.718 μ_{CR} = 0.144
 Diversity = 4.7129e-02 k_t = [1, 1, 1, 1]
 Generation 7100: Best = 1.417699e+03 FES = 710100 μ_F = 0.730 μ_{CR} = 0.153
 Diversity = 4.6783e-02 k_t = [4, 3, 4, 2]
 Generation 7110: Best = 1.417555e+03 FES = 711100 μ_F = 0.736 μ_{CR} = 0.156
 Diversity = 4.5904e-02 k_t = [1, 1, 1, 1]
 Generation 7120: Best = 1.417441e+03 FES = 712100 μ_F = 0.748 μ_{CR} = 0.159
 Diversity = 4.2570e-02 k_t = [2, 3, 3, 4]
 Generation 7130: Best = 1.417175e+03 FES = 713100 μ_F = 0.735 μ_{CR} = 0.165
 Diversity = 4.2848e-02 k_t = [1, 1, 1, 1]
 Generation 7140: Best = 1.417021e+03 FES = 714100 μ_F = 0.746 μ_{CR} = 0.165
 Diversity = 4.4893e-02 k_t = [3, 4, 3, 2]
 Generation 7150: Best = 1.416906e+03 FES = 715100 μ_F = 0.761 μ_{CR} = 0.162
 Diversity = 4.9779e-02 k_t = [1, 1, 1, 1]
 Generation 7160: Best = 1.416781e+03 FES = 716100 μ_F = 0.756 μ_{CR} = 0.156
 Diversity = 5.1383e-02 k_t = [3, 2, 4, 2]
 Generation 7170: Best = 1.416610e+03 FES = 717100 μ_F = 0.754 μ_{CR} = 0.149
 Diversity = 5.5374e-02 k_t = [1, 1, 1, 1]
 Generation 7180: Best = 1.416444e+03 FES = 718100 μ_F = 0.749 μ_{CR} = 0.149
 Diversity = 5.2100e-02 k_t = [3, 4, 3, 3]
 Generation 7190: Best = 1.416240e+03 FES = 719100 μ_F = 0.749 μ_{CR} = 0.150
 Diversity = 5.4804e-02 k_t = [1, 1, 1, 1]
 Generation 7200: Best = 1.415969e+03 FES = 720100 μ_F = 0.742 μ_{CR} = 0.152
 Diversity = 5.8918e-02 k_t = [2, 4, 2, 2]
 Generation 7210: Best = 1.415741e+03 FES = 721100 μ_F = 0.728 μ_{CR} = 0.154
 Diversity = 6.1727e-02 k_t = [1, 1, 1, 1]
 Generation 7220: Best = 1.415578e+03 FES = 722100 μ_F = 0.734 μ_{CR} = 0.153
 Diversity = 6.2159e-02 k_t = [4, 4, 4, 2]
 Generation 7230: Best = 1.415426e+03 FES = 723100 μ_F = 0.725 μ_{CR} = 0.155
 Diversity = 5.7850e-02 k_t = [1, 1, 1, 1]
 Generation 7240: Best = 1.415239e+03 FES = 724100 μ_F = 0.710 μ_{CR} = 0.152
 Diversity = 4.9336e-02 k_t = [4, 2, 2, 4]

Generation 7250: Best = 1.415030e+03 FES = 725100 μ_F = 0.711 μ_{CR} = 0.166
 Diversity = 4.4644e-02 k_t = [1, 1, 1, 1]
 Generation 7260: Best = 1.414846e+03 FES = 726100 μ_F = 0.720 μ_{CR} = 0.164
 Diversity = 4.4646e-02 k_t = [2, 2, 4, 4]
 Generation 7270: Best = 1.414545e+03 FES = 727100 μ_F = 0.733 μ_{CR} = 0.166
 Diversity = 5.0905e-02 k_t = [1, 1, 1, 1]
 Generation 7280: Best = 1.414392e+03 FES = 728100 μ_F = 0.721 μ_{CR} = 0.168
 Diversity = 4.9157e-02 k_t = [3, 2, 3, 3]
 Generation 7290: Best = 1.414123e+03 FES = 729100 μ_F = 0.720 μ_{CR} = 0.161
 Diversity = 5.0425e-02 k_t = [1, 1, 1, 1]
 Generation 7300: Best = 1.413981e+03 FES = 730100 μ_F = 0.725 μ_{CR} = 0.161
 Diversity = 5.0678e-02 k_t = [4, 2, 2, 4]
 Generation 7310: Best = 1.413872e+03 FES = 731100 μ_F = 0.746 μ_{CR} = 0.157
 Diversity = 5.3426e-02 k_t = [1, 1, 1, 1]
 Generation 7320: Best = 1.413736e+03 FES = 732100 μ_F = 0.740 μ_{CR} = 0.159
 Diversity = 4.7597e-02 k_t = [2, 3, 4, 2]
 Generation 7330: Best = 1.413432e+03 FES = 733100 μ_F = 0.741 μ_{CR} = 0.161
 Diversity = 5.1714e-02 k_t = [1, 1, 1, 1]
 Generation 7340: Best = 1.413241e+03 FES = 734100 μ_F = 0.740 μ_{CR} = 0.165
 Diversity = 5.4304e-02 k_t = [4, 3, 2, 3]
 Generation 7350: Best = 1.412899e+03 FES = 735100 μ_F = 0.734 μ_{CR} = 0.157
 Diversity = 5.7717e-02 k_t = [1, 1, 1, 1]
 Generation 7360: Best = 1.412813e+03 FES = 736100 μ_F = 0.744 μ_{CR} = 0.162
 Diversity = 5.9227e-02 k_t = [4, 3, 3, 2]
 Generation 7370: Best = 1.412610e+03 FES = 737100 μ_F = 0.727 μ_{CR} = 0.150
 Diversity = 5.8884e-02 k_t = [1, 1, 1, 1]
 Generation 7380: Best = 1.412474e+03 FES = 738100 μ_F = 0.737 μ_{CR} = 0.151
 Diversity = 6.1441e-02 k_t = [4, 4, 4, 2]
 Generation 7390: Best = 1.412296e+03 FES = 739100 μ_F = 0.718 μ_{CR} = 0.156
 Diversity = 5.6748e-02 k_t = [1, 1, 1, 1]
 Generation 7400: Best = 1.412077e+03 FES = 740100 μ_F = 0.721 μ_{CR} = 0.153
 Diversity = 5.0925e-02 k_t = [2, 3, 3, 4]
 Generation 7410: Best = 1.411866e+03 FES = 741100 μ_F = 0.724 μ_{CR} = 0.141
 Diversity = 5.5200e-02 k_t = [1, 1, 1, 1]
 Generation 7420: Best = 1.411577e+03 FES = 742100 μ_F = 0.723 μ_{CR} = 0.147
 Diversity = 5.1591e-02 k_t = [3, 4, 2, 4]
 Generation 7430: Best = 1.411345e+03 FES = 743100 μ_F = 0.736 μ_{CR} = 0.150
 Diversity = 4.9606e-02 k_t = [1, 1, 1, 1]
 Generation 7440: Best = 1.411166e+03 FES = 744100 μ_F = 0.731 μ_{CR} = 0.152
 Diversity = 5.3840e-02 k_t = [2, 2, 4, 3]
 Generation 7450: Best = 1.410956e+03 FES = 745100 μ_F = 0.713 μ_{CR} = 0.153
 Diversity = 5.7439e-02 k_t = [1, 1, 1, 1]
 Generation 7460: Best = 1.410710e+03 FES = 746100 μ_F = 0.725 μ_{CR} = 0.162
 Diversity = 5.8704e-02 k_t = [2, 3, 3, 2]
 Generation 7470: Best = 1.410489e+03 FES = 747100 μ_F = 0.707 μ_{CR} = 0.162
 Diversity = 5.9973e-02 k_t = [1, 1, 1, 1]
 Generation 7480: Best = 1.410185e+03 FES = 748100 μ_F = 0.717 μ_{CR} = 0.156
 Diversity = 5.6360e-02 k_t = [4, 3, 3, 3]
 Generation 7490: Best = 1.409824e+03 FES = 749100 μ_F = 0.711 μ_{CR} = 0.150
 Diversity = 5.8169e-02 k_t = [1, 1, 1, 1]
 Generation 7500: Best = 1.409467e+03 FES = 750100 μ_F = 0.713 μ_{CR} = 0.151
 Diversity = 5.8961e-02 k_t = [3, 2, 3, 2]
 Generation 7510: Best = 1.409327e+03 FES = 751100 μ_F = 0.725 μ_{CR} = 0.143
 Diversity = 6.4150e-02 k_t = [1, 1, 1, 1]

Generation 7520: Best = 1.409224e+03 FES = 752100 μ_F = 0.737 μ_{CR} = 0.135
 Diversity = 6.3337e-02 k_t = [4, 2, 2, 2]
 Generation 7530: Best = 1.408837e+03 FES = 753100 μ_F = 0.729 μ_{CR} = 0.138
 Diversity = 6.1848e-02 k_t = [1, 1, 1, 1]
 Generation 7540: Best = 1.408538e+03 FES = 754100 μ_F = 0.732 μ_{CR} = 0.135
 Diversity = 5.9897e-02 k_t = [2, 3, 3, 3]
 Generation 7550: Best = 1.408279e+03 FES = 755100 μ_F = 0.749 μ_{CR} = 0.137
 Diversity = 6.5670e-02 k_t = [1, 1, 1, 1]
 Generation 7560: Best = 1.408148e+03 FES = 756100 μ_F = 0.741 μ_{CR} = 0.145
 Diversity = 5.8130e-02 k_t = [4, 4, 3, 4]
 Generation 7570: Best = 1.407941e+03 FES = 757100 μ_F = 0.746 μ_{CR} = 0.141
 Diversity = 5.5708e-02 k_t = [1, 1, 1, 1]
 Generation 7580: Best = 1.407821e+03 FES = 758100 μ_F = 0.739 μ_{CR} = 0.126
 Diversity = 5.9484e-02 k_t = [3, 2, 4, 2]
 Generation 7590: Best = 1.407560e+03 FES = 759100 μ_F = 0.734 μ_{CR} = 0.125
 Diversity = 6.1238e-02 k_t = [1, 1, 1, 1]
 Generation 7600: Best = 1.407342e+03 FES = 760100 μ_F = 0.731 μ_{CR} = 0.120
 Diversity = 5.9251e-02 k_t = [2, 4, 3, 4]
 Generation 7610: Best = 1.407238e+03 FES = 761100 μ_F = 0.723 μ_{CR} = 0.116
 Diversity = 5.3419e-02 k_t = [1, 1, 1, 1]
 Generation 7620: Best = 1.407078e+03 FES = 762100 μ_F = 0.716 μ_{CR} = 0.119
 Diversity = 5.3360e-02 k_t = [4, 2, 2, 2]
 Generation 7630: Best = 1.406948e+03 FES = 763100 μ_F = 0.727 μ_{CR} = 0.124
 Diversity = 5.4895e-02 k_t = [1, 1, 1, 1]
 Generation 7640: Best = 1.406892e+03 FES = 764100 μ_F = 0.733 μ_{CR} = 0.125
 Diversity = 5.1588e-02 k_t = [2, 4, 2, 3]
 Generation 7650: Best = 1.406724e+03 FES = 765100 μ_F = 0.727 μ_{CR} = 0.128
 Diversity = 4.9923e-02 k_t = [1, 1, 1, 1]
 Generation 7660: Best = 1.406539e+03 FES = 766100 μ_F = 0.733 μ_{CR} = 0.136
 Diversity = 5.0324e-02 k_t = [4, 3, 3, 2]
 Generation 7670: Best = 1.406365e+03 FES = 767100 μ_F = 0.739 μ_{CR} = 0.138
 Diversity = 5.0014e-02 k_t = [1, 1, 1, 1]
 Generation 7680: Best = 1.406364e+03 FES = 768100 μ_F = 0.730 μ_{CR} = 0.140
 Diversity = 4.5932e-02 k_t = [3, 3, 4, 2]
 Generation 7690: Best = 1.406186e+03 FES = 769100 μ_F = 0.735 μ_{CR} = 0.138
 Diversity = 4.5544e-02 k_t = [1, 1, 1, 1]
 Generation 7700: Best = 1.406121e+03 FES = 770100 μ_F = 0.738 μ_{CR} = 0.149
 Diversity = 4.2591e-02 k_t = [3, 3, 3, 2]
 Generation 7710: Best = 1.405925e+03 FES = 771100 μ_F = 0.729 μ_{CR} = 0.150
 Diversity = 4.3468e-02 k_t = [1, 1, 1, 1]
 Generation 7720: Best = 1.405737e+03 FES = 772100 μ_F = 0.738 μ_{CR} = 0.160
 Diversity = 4.7538e-02 k_t = [2, 3, 4, 4]
 Generation 7730: Best = 1.405702e+03 FES = 773100 μ_F = 0.731 μ_{CR} = 0.154
 Diversity = 4.5757e-02 k_t = [1, 1, 1, 1]
 Generation 7740: Best = 1.405590e+03 FES = 774100 μ_F = 0.720 μ_{CR} = 0.144
 Diversity = 4.2422e-02 k_t = [2, 2, 3, 2]
 Generation 7750: Best = 1.405467e+03 FES = 775100 μ_F = 0.719 μ_{CR} = 0.144
 Diversity = 4.2260e-02 k_t = [1, 1, 1, 1]
 Generation 7760: Best = 1.405305e+03 FES = 776100 μ_F = 0.726 μ_{CR} = 0.145
 Diversity = 3.9260e-02 k_t = [3, 3, 4, 3]
 Generation 7770: Best = 1.405156e+03 FES = 777100 μ_F = 0.743 μ_{CR} = 0.145
 Diversity = 4.3932e-02 k_t = [1, 1, 1, 1]
 Generation 7780: Best = 1.405027e+03 FES = 778100 μ_F = 0.739 μ_{CR} = 0.152
 Diversity = 4.0390e-02 k_t = [2, 2, 2, 4]

Generation 7790: Best = 1.404955e+03 FES = 779100 μ_F = 0.734 μ_{CR} = 0.166
 Diversity = 3.8084e-02 k_t = [1, 1, 1, 1]
 Generation 7800: Best = 1.404614e+03 FES = 780100 μ_F = 0.741 μ_{CR} = 0.169
 Diversity = 4.0010e-02 k_t = [2, 2, 2, 3]
 Generation 7810: Best = 1.404309e+03 FES = 781100 μ_F = 0.750 μ_{CR} = 0.176
 Diversity = 5.5311e-02 k_t = [1, 1, 1, 1]
 Generation 7820: Best = 1.403596e+03 FES = 782100 μ_F = 0.750 μ_{CR} = 0.179
 Diversity = 7.5651e-02 k_t = [2, 3, 4, 2]
 Generation 7830: Best = 1.402763e+03 FES = 783100 μ_F = 0.764 μ_{CR} = 0.158
 Diversity = 8.5394e-02 k_t = [1, 1, 1, 1]
 Generation 7840: Best = 1.402088e+03 FES = 784100 μ_F = 0.756 μ_{CR} = 0.157
 Diversity = 8.9857e-02 k_t = [2, 2, 3, 4]
 Generation 7850: Best = 1.401393e+03 FES = 785100 μ_F = 0.755 μ_{CR} = 0.154
 Diversity = 1.0141e-01 k_t = [1, 1, 1, 1]
 Generation 7860: Best = 1.400976e+03 FES = 786100 μ_F = 0.733 μ_{CR} = 0.158
 Diversity = 1.0366e-01 k_t = [4, 4, 2, 3]
 Generation 7870: Best = 1.400565e+03 FES = 787100 μ_F = 0.737 μ_{CR} = 0.149
 Diversity = 1.0720e-01 k_t = [1, 1, 1, 1]
 Generation 7880: Best = 1.400237e+03 FES = 788100 μ_F = 0.719 μ_{CR} = 0.149
 Diversity = 9.1344e-02 k_t = [4, 3, 3, 3]
 Generation 7890: Best = 1.399687e+03 FES = 789100 μ_F = 0.721 μ_{CR} = 0.140
 Diversity = 8.6510e-02 k_t = [1, 1, 1, 1]
 Generation 7900: Best = 1.399239e+03 FES = 790100 μ_F = 0.707 μ_{CR} = 0.146
 Diversity = 8.5952e-02 k_t = [4, 2, 4, 4]
 Generation 7910: Best = 1.398856e+03 FES = 791100 μ_F = 0.714 μ_{CR} = 0.142
 Diversity = 8.3944e-02 k_t = [1, 1, 1, 1]
 Generation 7920: Best = 1.398721e+03 FES = 792100 μ_F = 0.714 μ_{CR} = 0.132
 Diversity = 8.0844e-02 k_t = [4, 4, 2, 3]
 Generation 7930: Best = 1.398532e+03 FES = 793100 μ_F = 0.713 μ_{CR} = 0.134
 Diversity = 7.2752e-02 k_t = [1, 1, 1, 1]
 Generation 7940: Best = 1.398192e+03 FES = 794100 μ_F = 0.706 μ_{CR} = 0.131
 Diversity = 6.6353e-02 k_t = [3, 2, 3, 4]
 Generation 7950: Best = 1.398028e+03 FES = 795100 μ_F = 0.702 μ_{CR} = 0.139
 Diversity = 6.5017e-02 k_t = [1, 1, 1, 1]
 Generation 7960: Best = 1.397796e+03 FES = 796100 μ_F = 0.707 μ_{CR} = 0.142
 Diversity = 6.3048e-02 k_t = [2, 3, 4, 3]
 Generation 7970: Best = 1.397611e+03 FES = 797100 μ_F = 0.708 μ_{CR} = 0.136
 Diversity = 6.7034e-02 k_t = [1, 1, 1, 1]
 Generation 7980: Best = 1.397441e+03 FES = 798100 μ_F = 0.707 μ_{CR} = 0.130
 Diversity = 6.1748e-02 k_t = [3, 4, 4, 4]
 Generation 7990: Best = 1.397222e+03 FES = 799100 μ_F = 0.714 μ_{CR} = 0.128
 Diversity = 5.8747e-02 k_t = [1, 1, 1, 1]
 Generation 8000: Best = 1.397196e+03 FES = 800100 μ_F = 0.705 μ_{CR} = 0.127
 Diversity = 5.2778e-02 k_t = [3, 2, 3, 4]
 Generation 8010: Best = 1.397117e+03 FES = 801100 μ_F = 0.710 μ_{CR} = 0.122
 Diversity = 5.1092e-02 k_t = [1, 1, 1, 1]
 Generation 8020: Best = 1.396998e+03 FES = 802100 μ_F = 0.717 μ_{CR} = 0.121
 Diversity = 4.3941e-02 k_t = [4, 3, 2, 3]
 Generation 8030: Best = 1.396921e+03 FES = 803100 μ_F = 0.703 μ_{CR} = 0.124
 Diversity = 4.4980e-02 k_t = [1, 1, 1, 1]
 Generation 8040: Best = 1.396843e+03 FES = 804100 μ_F = 0.714 μ_{CR} = 0.126
 Diversity = 3.9443e-02 k_t = [4, 3, 4, 4]
 Generation 8050: Best = 1.396713e+03 FES = 805100 μ_F = 0.714 μ_{CR} = 0.126
 Diversity = 3.9164e-02 k_t = [1, 1, 1, 1]

Generation 8060: Best = 1.396700e+03 FES = 806100 μ_F = 0.708 μ_{CR} = 0.126
 Diversity = 3.6527e-02 k_t = [3, 2, 4, 3]
 Generation 8070: Best = 1.396678e+03 FES = 807100 μ_F = 0.715 μ_{CR} = 0.125
 Diversity = 3.3120e-02 k_t = [1, 1, 1, 1]
 Generation 8080: Best = 1.396598e+03 FES = 808100 μ_F = 0.706 μ_{CR} = 0.122
 Diversity = 3.4364e-02 k_t = [2, 2, 4, 2]
 Generation 8090: Best = 1.396577e+03 FES = 809100 μ_F = 0.714 μ_{CR} = 0.125
 Diversity = 3.4944e-02 k_t = [1, 1, 1, 1]
 Generation 8100: Best = 1.396531e+03 FES = 810100 μ_F = 0.730 μ_{CR} = 0.127
 Diversity = 3.2864e-02 k_t = [3, 3, 3, 3]
 Generation 8110: Best = 1.396491e+03 FES = 811100 μ_F = 0.732 μ_{CR} = 0.129
 Diversity = 3.2220e-02 k_t = [1, 1, 1, 1]
 Generation 8120: Best = 1.396413e+03 FES = 812100 μ_F = 0.726 μ_{CR} = 0.137
 Diversity = 2.9124e-02 k_t = [4, 2, 4, 2]
 Generation 8130: Best = 1.396336e+03 FES = 813100 μ_F = 0.732 μ_{CR} = 0.143
 Diversity = 2.9895e-02 k_t = [1, 1, 1, 1]
 Generation 8140: Best = 1.396305e+03 FES = 814100 μ_F = 0.738 μ_{CR} = 0.138
 Diversity = 3.1287e-02 k_t = [3, 4, 2, 4]
 Generation 8150: Best = 1.396243e+03 FES = 815100 μ_F = 0.727 μ_{CR} = 0.129
 Diversity = 3.1561e-02 k_t = [1, 1, 1, 1]
 Generation 8160: Best = 1.396210e+03 FES = 816100 μ_F = 0.726 μ_{CR} = 0.135
 Diversity = 3.1027e-02 k_t = [4, 2, 3, 3]
 Generation 8170: Best = 1.396105e+03 FES = 817100 μ_F = 0.730 μ_{CR} = 0.129
 Diversity = 3.1828e-02 k_t = [1, 1, 1, 1]
 Generation 8180: Best = 1.396098e+03 FES = 818100 μ_F = 0.722 μ_{CR} = 0.131
 Diversity = 3.0960e-02 k_t = [3, 3, 3, 2]
 Generation 8190: Best = 1.395997e+03 FES = 819100 μ_F = 0.712 μ_{CR} = 0.126
 Diversity = 2.9658e-02 k_t = [1, 1, 1, 1]
 Generation 8200: Best = 1.395942e+03 FES = 820100 μ_F = 0.728 μ_{CR} = 0.126
 Diversity = 2.8613e-02 k_t = [2, 4, 2, 4]
 Generation 8210: Best = 1.395851e+03 FES = 821100 μ_F = 0.737 μ_{CR} = 0.136
 Diversity = 3.1510e-02 k_t = [1, 1, 1, 1]
 Generation 8220: Best = 1.395755e+03 FES = 822100 μ_F = 0.741 μ_{CR} = 0.140
 Diversity = 3.2233e-02 k_t = [4, 4, 2, 4]
 Generation 8230: Best = 1.395614e+03 FES = 823100 μ_F = 0.750 μ_{CR} = 0.138
 Diversity = 3.4627e-02 k_t = [1, 1, 1, 1]
 Generation 8240: Best = 1.395499e+03 FES = 824100 μ_F = 0.746 μ_{CR} = 0.146
 Diversity = 3.7204e-02 k_t = [2, 4, 4, 4]
 Generation 8250: Best = 1.395422e+03 FES = 825100 μ_F = 0.747 μ_{CR} = 0.139
 Diversity = 3.6355e-02 k_t = [1, 1, 1, 1]
 Generation 8260: Best = 1.395369e+03 FES = 826100 μ_F = 0.746 μ_{CR} = 0.146
 Diversity = 3.6068e-02 k_t = [4, 2, 4, 3]
 Generation 8270: Best = 1.395299e+03 FES = 827100 μ_F = 0.738 μ_{CR} = 0.138
 Diversity = 3.4647e-02 k_t = [1, 1, 1, 1]
 Generation 8280: Best = 1.395240e+03 FES = 828100 μ_F = 0.745 μ_{CR} = 0.134
 Diversity = 3.2131e-02 k_t = [3, 3, 4, 3]
 Generation 8290: Best = 1.395178e+03 FES = 829100 μ_F = 0.752 μ_{CR} = 0.143
 Diversity = 3.0946e-02 k_t = [1, 1, 1, 1]
 Generation 8300: Best = 1.395133e+03 FES = 830100 μ_F = 0.742 μ_{CR} = 0.140
 Diversity = 3.1246e-02 k_t = [2, 3, 4, 2]
 Generation 8310: Best = 1.395092e+03 FES = 831100 μ_F = 0.738 μ_{CR} = 0.136
 Diversity = 3.1320e-02 k_t = [1, 1, 1, 1]
 Generation 8320: Best = 1.395054e+03 FES = 832100 μ_F = 0.734 μ_{CR} = 0.146
 Diversity = 2.9203e-02 k_t = [4, 4, 2, 2]

Generation 8330: Best = 1.394997e+03 FES = 833100 μ_F = 0.739 μ_{CR} = 0.146
 Diversity = 3.0403e-02 k_t = [1, 1, 1, 1]
 Generation 8340: Best = 1.394955e+03 FES = 834100 μ_F = 0.743 μ_{CR} = 0.145
 Diversity = 2.8720e-02 k_t = [4, 2, 2, 2]
 Generation 8350: Best = 1.394919e+03 FES = 835100 μ_F = 0.749 μ_{CR} = 0.139
 Diversity = 2.9608e-02 k_t = [1, 1, 1, 1]
 Generation 8360: Best = 1.394851e+03 FES = 836100 μ_F = 0.755 μ_{CR} = 0.139
 Diversity = 2.9757e-02 k_t = [2, 4, 2, 3]
 Generation 8370: Best = 1.394817e+03 FES = 837100 μ_F = 0.746 μ_{CR} = 0.141
 Diversity = 3.3116e-02 k_t = [1, 1, 1, 1]
 Generation 8380: Best = 1.394772e+03 FES = 838100 μ_F = 0.746 μ_{CR} = 0.138
 Diversity = 3.3977e-02 k_t = [2, 2, 4, 2]
 Generation 8390: Best = 1.394737e+03 FES = 839100 μ_F = 0.738 μ_{CR} = 0.117
 Diversity = 3.1335e-02 k_t = [1, 1, 1, 1]
 Generation 8400: Best = 1.394606e+03 FES = 840100 μ_F = 0.760 μ_{CR} = 0.127
 Diversity = 3.3526e-02 k_t = [3, 3, 3, 4]
 Generation 8410: Best = 1.394437e+03 FES = 841100 μ_F = 0.768 μ_{CR} = 0.140
 Diversity = 4.1073e-02 k_t = [1, 1, 1, 1]
 Generation 8420: Best = 1.394210e+03 FES = 842100 μ_F = 0.760 μ_{CR} = 0.150
 Diversity = 4.9863e-02 k_t = [3, 3, 3, 2]
 Generation 8430: Best = 1.394098e+03 FES = 843100 μ_F = 0.766 μ_{CR} = 0.146
 Diversity = 5.5068e-02 k_t = [1, 1, 1, 1]
 Generation 8440: Best = 1.393935e+03 FES = 844100 μ_F = 0.752 μ_{CR} = 0.140
 Diversity = 5.0627e-02 k_t = [3, 4, 3, 2]
 Generation 8450: Best = 1.393862e+03 FES = 845100 μ_F = 0.739 μ_{CR} = 0.127
 Diversity = 5.8049e-02 k_t = [1, 1, 1, 1]
 Generation 8460: Best = 1.393675e+03 FES = 846100 μ_F = 0.727 μ_{CR} = 0.126
 Diversity = 5.5040e-02 k_t = [2, 3, 2, 4]
 Generation 8470: Best = 1.393513e+03 FES = 847100 μ_F = 0.727 μ_{CR} = 0.122
 Diversity = 5.3380e-02 k_t = [1, 1, 1, 1]
 Generation 8480: Best = 1.393344e+03 FES = 848100 μ_F = 0.737 μ_{CR} = 0.125
 Diversity = 5.8939e-02 k_t = [4, 4, 3, 3]
 Generation 8490: Best = 1.393251e+03 FES = 849100 μ_F = 0.731 μ_{CR} = 0.133
 Diversity = 5.9860e-02 k_t = [1, 1, 1, 1]
 Generation 8500: Best = 1.393098e+03 FES = 850100 μ_F = 0.726 μ_{CR} = 0.138
 Diversity = 5.3824e-02 k_t = [3, 4, 4, 2]
 Generation 8510: Best = 1.392988e+03 FES = 851100 μ_F = 0.719 μ_{CR} = 0.146
 Diversity = 5.7046e-02 k_t = [1, 1, 1, 1]
 Generation 8520: Best = 1.392890e+03 FES = 852100 μ_F = 0.693 μ_{CR} = 0.145
 Diversity = 4.8158e-02 k_t = [4, 2, 4, 4]
 Generation 8530: Best = 1.392815e+03 FES = 853100 μ_F = 0.702 μ_{CR} = 0.147
 Diversity = 4.2953e-02 k_t = [1, 1, 1, 1]
 Generation 8540: Best = 1.392707e+03 FES = 854100 μ_F = 0.716 μ_{CR} = 0.152
 Diversity = 3.9820e-02 k_t = [4, 3, 3, 4]
 Generation 8550: Best = 1.392701e+03 FES = 855100 μ_F = 0.706 μ_{CR} = 0.151
 Diversity = 3.3553e-02 k_t = [1, 1, 1, 1]
 Generation 8560: Best = 1.392672e+03 FES = 856100 μ_F = 0.692 μ_{CR} = 0.159
 Diversity = 3.0100e-02 k_t = [4, 2, 2, 3]
 Generation 8570: Best = 1.392634e+03 FES = 857100 μ_F = 0.681 μ_{CR} = 0.162
 Diversity = 2.6001e-02 k_t = [1, 1, 1, 1]
 Generation 8580: Best = 1.392598e+03 FES = 858100 μ_F = 0.684 μ_{CR} = 0.162
 Diversity = 2.3669e-02 k_t = [2, 4, 3, 3]
 Generation 8590: Best = 1.392562e+03 FES = 859100 μ_F = 0.694 μ_{CR} = 0.168
 Diversity = 2.3415e-02 k_t = [1, 1, 1, 1]

Generation 8600: Best = 1.392541e+03 FES = 860100 μ_F = 0.692 μ_{CR} = 0.164
 Diversity = 2.2133e-02 k_t = [3, 3, 3, 3]
 Generation 8610: Best = 1.392527e+03 FES = 861100 μ_F = 0.691 μ_{CR} = 0.161
 Diversity = 1.9666e-02 k_t = [1, 1, 1, 1]
 Generation 8620: Best = 1.392512e+03 FES = 862100 μ_F = 0.690 μ_{CR} = 0.154
 Diversity = 1.7810e-02 k_t = [4, 3, 4, 3]
 Generation 8630: Best = 1.392497e+03 FES = 863100 μ_F = 0.707 μ_{CR} = 0.157
 Diversity = 1.6904e-02 k_t = [1, 1, 1, 1]
 Generation 8640: Best = 1.392487e+03 FES = 864100 μ_F = 0.706 μ_{CR} = 0.157
 Diversity = 1.5258e-02 k_t = [4, 2, 4, 4]
 Generation 8650: Best = 1.392461e+03 FES = 865100 μ_F = 0.716 μ_{CR} = 0.156
 Diversity = 1.4827e-02 k_t = [1, 1, 1, 1]
 Generation 8660: Best = 1.392447e+03 FES = 866100 μ_F = 0.724 μ_{CR} = 0.156
 Diversity = 1.4767e-02 k_t = [3, 4, 3, 3]
 Generation 8670: Best = 1.392422e+03 FES = 867100 μ_F = 0.729 μ_{CR} = 0.159
 Diversity = 1.5338e-02 k_t = [1, 1, 1, 1]
 Generation 8680: Best = 1.392417e+03 FES = 868100 μ_F = 0.729 μ_{CR} = 0.148
 Diversity = 1.5323e-02 k_t = [2, 4, 3, 4]
 Generation 8690: Best = 1.392403e+03 FES = 869100 μ_F = 0.744 μ_{CR} = 0.148
 Diversity = 1.4008e-02 k_t = [1, 1, 1, 1]
 Generation 8700: Best = 1.392390e+03 FES = 870100 μ_F = 0.755 μ_{CR} = 0.145
 Diversity = 1.2925e-02 k_t = [3, 4, 3, 4]
 Generation 8710: Best = 1.392373e+03 FES = 871100 μ_F = 0.772 μ_{CR} = 0.147
 Diversity = 1.3562e-02 k_t = [1, 1, 1, 1]
 Generation 8720: Best = 1.392363e+03 FES = 872100 μ_F = 0.775 μ_{CR} = 0.139
 Diversity = 1.4379e-02 k_t = [3, 2, 3, 3]
 Generation 8730: Best = 1.392353e+03 FES = 873100 μ_F = 0.774 μ_{CR} = 0.136
 Diversity = 1.3740e-02 k_t = [1, 1, 1, 1]
 Generation 8740: Best = 1.392334e+03 FES = 874100 μ_F = 0.765 μ_{CR} = 0.143
 Diversity = 1.2880e-02 k_t = [4, 3, 3, 4]
 Generation 8750: Best = 1.392317e+03 FES = 875100 μ_F = 0.770 μ_{CR} = 0.146
 Diversity = 1.4643e-02 k_t = [1, 1, 1, 1]
 Generation 8760: Best = 1.392308e+03 FES = 876100 μ_F = 0.753 μ_{CR} = 0.154
 Diversity = 1.6721e-02 k_t = [2, 3, 4, 3]
 Generation 8770: Best = 1.392295e+03 FES = 877100 μ_F = 0.748 μ_{CR} = 0.151
 Diversity = 1.6921e-02 k_t = [1, 1, 1, 1]
 Generation 8780: Best = 1.392286e+03 FES = 878100 μ_F = 0.756 μ_{CR} = 0.140
 Diversity = 1.6302e-02 k_t = [4, 3, 3, 4]
 Generation 8790: Best = 1.392271e+03 FES = 879100 μ_F = 0.752 μ_{CR} = 0.135
 Diversity = 1.4737e-02 k_t = [1, 1, 1, 1]
 Generation 8800: Best = 1.392252e+03 FES = 880100 μ_F = 0.757 μ_{CR} = 0.138
 Diversity = 1.5007e-02 k_t = [4, 3, 3, 2]
 Generation 8810: Best = 1.392236e+03 FES = 881100 μ_F = 0.759 μ_{CR} = 0.145
 Diversity = 1.6630e-02 k_t = [1, 1, 1, 1]
 Generation 8820: Best = 1.392212e+03 FES = 882100 μ_F = 0.767 μ_{CR} = 0.148
 Diversity = 1.8073e-02 k_t = [4, 2, 4, 2]
 Generation 8830: Best = 1.392145e+03 FES = 883100 μ_F = 0.770 μ_{CR} = 0.148
 Diversity = 2.1029e-02 k_t = [1, 1, 1, 1]
 Generation 8840: Best = 1.392118e+03 FES = 884100 μ_F = 0.765 μ_{CR} = 0.137
 Diversity = 2.0879e-02 k_t = [2, 3, 4, 3]
 Generation 8850: Best = 1.392072e+03 FES = 885100 μ_F = 0.754 μ_{CR} = 0.142
 Diversity = 2.3356e-02 k_t = [1, 1, 1, 1]
 Generation 8860: Best = 1.392038e+03 FES = 886100 μ_F = 0.748 μ_{CR} = 0.154
 Diversity = 2.4168e-02 k_t = [3, 3, 3, 4]

Generation 8870: Best = 1.391995e+03 FES = 887100 μ_F = 0.752 μ_{CR} = 0.155
 Diversity = 2.3864e-02 k_t = [1, 1, 1, 1]
 Generation 8880: Best = 1.391982e+03 FES = 888100 μ_F = 0.746 μ_{CR} = 0.149
 Diversity = 2.2785e-02 k_t = [4, 3, 4, 3]
 Generation 8890: Best = 1.391946e+03 FES = 889100 μ_F = 0.735 μ_{CR} = 0.150
 Diversity = 2.2138e-02 k_t = [1, 1, 1, 1]
 Generation 8900: Best = 1.391888e+03 FES = 890100 μ_F = 0.741 μ_{CR} = 0.150
 Diversity = 2.4087e-02 k_t = [4, 4, 2, 2]
 Generation 8910: Best = 1.391860e+03 FES = 891100 μ_F = 0.741 μ_{CR} = 0.150
 Diversity = 2.5839e-02 k_t = [1, 1, 1, 1]
 Generation 8920: Best = 1.391829e+03 FES = 892100 μ_F = 0.744 μ_{CR} = 0.148
 Diversity = 2.3129e-02 k_t = [3, 2, 4, 4]
 Generation 8930: Best = 1.391792e+03 FES = 893100 μ_F = 0.739 μ_{CR} = 0.151
 Diversity = 2.2707e-02 k_t = [1, 1, 1, 1]
 Generation 8940: Best = 1.391779e+03 FES = 894100 μ_F = 0.722 μ_{CR} = 0.158
 Diversity = 2.0929e-02 k_t = [2, 2, 2, 2]
 Generation 8950: Best = 1.391757e+03 FES = 895100 μ_F = 0.719 μ_{CR} = 0.153
 Diversity = 2.1147e-02 k_t = [1, 1, 1, 1]
 Generation 8960: Best = 1.391736e+03 FES = 896100 μ_F = 0.712 μ_{CR} = 0.154
 Diversity = 1.9542e-02 k_t = [3, 3, 3, 2]
 Generation 8970: Best = 1.391715e+03 FES = 897100 μ_F = 0.700 μ_{CR} = 0.158
 Diversity = 1.8610e-02 k_t = [1, 1, 1, 1]
 Generation 8980: Best = 1.391703e+03 FES = 898100 μ_F = 0.709 μ_{CR} = 0.152
 Diversity = 1.6809e-02 k_t = [3, 3, 4, 3]
 Generation 8990: Best = 1.391680e+03 FES = 899100 μ_F = 0.710 μ_{CR} = 0.155
 Diversity = 1.6315e-02 k_t = [1, 1, 1, 1]
 Generation 9000: Best = 1.391666e+03 FES = 900100 μ_F = 0.727 μ_{CR} = 0.147
 Diversity = 1.7032e-02 k_t = [2, 2, 4, 3]
 Generation 9010: Best = 1.391650e+03 FES = 901100 μ_F = 0.729 μ_{CR} = 0.162
 Diversity = 1.5573e-02 k_t = [1, 1, 1, 1]
 Generation 9020: Best = 1.391637e+03 FES = 902100 μ_F = 0.728 μ_{CR} = 0.162
 Diversity = 1.6445e-02 k_t = [2, 4, 2, 3]
 Generation 9030: Best = 1.391631e+03 FES = 903100 μ_F = 0.726 μ_{CR} = 0.162
 Diversity = 1.6072e-02 k_t = [1, 1, 1, 1]
 Generation 9040: Best = 1.391608e+03 FES = 904100 μ_F = 0.740 μ_{CR} = 0.144
 Diversity = 1.4853e-02 k_t = [2, 2, 3, 4]
 Generation 9050: Best = 1.391591e+03 FES = 905100 μ_F = 0.734 μ_{CR} = 0.141
 Diversity = 1.5409e-02 k_t = [1, 1, 1, 1]
 Generation 9060: Best = 1.391585e+03 FES = 906100 μ_F = 0.744 μ_{CR} = 0.135
 Diversity = 1.4174e-02 k_t = [3, 4, 4, 4]
 Generation 9070: Best = 1.391559e+03 FES = 907100 μ_F = 0.740 μ_{CR} = 0.143
 Diversity = 1.4869e-02 k_t = [1, 1, 1, 1]
 Generation 9080: Best = 1.391541e+03 FES = 908100 μ_F = 0.751 μ_{CR} = 0.150
 Diversity = 1.5758e-02 k_t = [2, 4, 3, 3]
 Generation 9090: Best = 1.391529e+03 FES = 909100 μ_F = 0.745 μ_{CR} = 0.149
 Diversity = 1.6788e-02 k_t = [1, 1, 1, 1]
 Generation 9100: Best = 1.391504e+03 FES = 910100 μ_F = 0.757 μ_{CR} = 0.156
 Diversity = 1.7922e-02 k_t = [4, 4, 4, 4]
 Generation 9110: Best = 1.391473e+03 FES = 911100 μ_F = 0.771 μ_{CR} = 0.143
 Diversity = 1.7564e-02 k_t = [1, 1, 1, 1]
 Generation 9120: Best = 1.391464e+03 FES = 912100 μ_F = 0.765 μ_{CR} = 0.141
 Diversity = 1.8298e-02 k_t = [4, 2, 2, 4]
 Generation 9130: Best = 1.391425e+03 FES = 913100 μ_F = 0.768 μ_{CR} = 0.150
 Diversity = 1.8318e-02 k_t = [1, 1, 1, 1]

Generation 9140: Best = 1.391390e+03 FES = 914100 μ_F = 0.754 μ_{CR} = 0.158
 Diversity = 2.0924e-02 k_t = [3, 3, 3, 2]
 Generation 9150: Best = 1.391337e+03 FES = 915100 μ_F = 0.755 μ_{CR} = 0.162
 Diversity = 2.3539e-02 k_t = [1, 1, 1, 1]
 Generation 9160: Best = 1.391311e+03 FES = 916100 μ_F = 0.748 μ_{CR} = 0.156
 Diversity = 2.4938e-02 k_t = [2, 2, 3, 2]
 Generation 9170: Best = 1.391309e+03 FES = 917100 μ_F = 0.739 μ_{CR} = 0.154
 Diversity = 2.2876e-02 k_t = [1, 1, 1, 1]
 Generation 9180: Best = 1.391293e+03 FES = 918100 μ_F = 0.728 μ_{CR} = 0.149
 Diversity = 2.1584e-02 k_t = [3, 2, 2, 4]
 Generation 9190: Best = 1.391252e+03 FES = 919100 μ_F = 0.737 μ_{CR} = 0.140
 Diversity = 1.9477e-02 k_t = [1, 1, 1, 1]
 Generation 9200: Best = 1.391245e+03 FES = 920100 μ_F = 0.754 μ_{CR} = 0.142
 Diversity = 1.8928e-02 k_t = [4, 3, 3, 2]
 Generation 9210: Best = 1.391218e+03 FES = 921100 μ_F = 0.741 μ_{CR} = 0.150
 Diversity = 1.8749e-02 k_t = [1, 1, 1, 1]
 Generation 9220: Best = 1.391206e+03 FES = 922100 μ_F = 0.733 μ_{CR} = 0.136
 Diversity = 1.6464e-02 k_t = [3, 4, 3, 2]
 Generation 9230: Best = 1.391200e+03 FES = 923100 μ_F = 0.730 μ_{CR} = 0.133
 Diversity = 1.6896e-02 k_t = [1, 1, 1, 1]
 Generation 9240: Best = 1.391183e+03 FES = 924100 μ_F = 0.719 μ_{CR} = 0.141
 Diversity = 1.4972e-02 k_t = [2, 4, 3, 4]
 Generation 9250: Best = 1.391167e+03 FES = 925100 μ_F = 0.723 μ_{CR} = 0.142
 Diversity = 1.5539e-02 k_t = [1, 1, 1, 1]
 Generation 9260: Best = 1.391154e+03 FES = 926100 μ_F = 0.734 μ_{CR} = 0.135
 Diversity = 1.5854e-02 k_t = [4, 4, 4, 3]
 Generation 9270: Best = 1.391150e+03 FES = 927100 μ_F = 0.748 μ_{CR} = 0.141
 Diversity = 1.5228e-02 k_t = [1, 1, 1, 1]
 Generation 9280: Best = 1.391135e+03 FES = 928100 μ_F = 0.734 μ_{CR} = 0.146
 Diversity = 1.4678e-02 k_t = [2, 3, 2, 2]
 Generation 9290: Best = 1.391109e+03 FES = 929100 μ_F = 0.733 μ_{CR} = 0.140
 Diversity = 1.5485e-02 k_t = [1, 1, 1, 1]
 Generation 9300: Best = 1.391083e+03 FES = 930100 μ_F = 0.730 μ_{CR} = 0.151
 Diversity = 1.5995e-02 k_t = [3, 2, 2, 4]
 Generation 9310: Best = 1.391070e+03 FES = 931100 μ_F = 0.736 μ_{CR} = 0.148
 Diversity = 1.7489e-02 k_t = [1, 1, 1, 1]
 Generation 9320: Best = 1.391056e+03 FES = 932100 μ_F = 0.726 μ_{CR} = 0.150
 Diversity = 1.6997e-02 k_t = [4, 3, 3, 4]
 Generation 9330: Best = 1.391038e+03 FES = 933100 μ_F = 0.728 μ_{CR} = 0.151
 Diversity = 1.5164e-02 k_t = [1, 1, 1, 1]
 Generation 9340: Best = 1.391027e+03 FES = 934100 μ_F = 0.717 μ_{CR} = 0.158
 Diversity = 1.4168e-02 k_t = [4, 3, 3, 3]
 Generation 9350: Best = 1.391009e+03 FES = 935100 μ_F = 0.715 μ_{CR} = 0.158
 Diversity = 1.3175e-02 k_t = [1, 1, 1, 1]
 Generation 9360: Best = 1.390995e+03 FES = 936100 μ_F = 0.735 μ_{CR} = 0.167
 Diversity = 1.3162e-02 k_t = [2, 4, 4, 3]
 Generation 9370: Best = 1.390985e+03 FES = 937100 μ_F = 0.755 μ_{CR} = 0.161
 Diversity = 1.3664e-02 k_t = [1, 1, 1, 1]
 Generation 9380: Best = 1.390967e+03 FES = 938100 μ_F = 0.754 μ_{CR} = 0.154
 Diversity = 1.3447e-02 k_t = [2, 3, 4, 2]
 Generation 9390: Best = 1.390957e+03 FES = 939100 μ_F = 0.741 μ_{CR} = 0.164
 Diversity = 1.3705e-02 k_t = [1, 1, 1, 1]
 Generation 9400: Best = 1.390942e+03 FES = 940100 μ_F = 0.755 μ_{CR} = 0.165
 Diversity = 1.3041e-02 k_t = [3, 4, 2, 4]

Generation 9410: Best = 1.390932e+03 FES = 941100 μ_F = 0.759 μ_{CR} = 0.162
 Diversity = 1.3536e-02 k_t = [1, 1, 1, 1]
 Generation 9420: Best = 1.390912e+03 FES = 942100 μ_F = 0.756 μ_{CR} = 0.156
 Diversity = 1.2655e-02 k_t = [3, 2, 3, 4]
 Generation 9430: Best = 1.390896e+03 FES = 943100 μ_F = 0.747 μ_{CR} = 0.153
 Diversity = 1.4366e-02 k_t = [1, 1, 1, 1]
 Generation 9440: Best = 1.390880e+03 FES = 944100 μ_F = 0.752 μ_{CR} = 0.160
 Diversity = 1.4589e-02 k_t = [3, 4, 3, 3]
 Generation 9450: Best = 1.390872e+03 FES = 945100 μ_F = 0.746 μ_{CR} = 0.155
 Diversity = 1.5007e-02 k_t = [1, 1, 1, 1]
 Generation 9460: Best = 1.390858e+03 FES = 946100 μ_F = 0.753 μ_{CR} = 0.154
 Diversity = 1.4658e-02 k_t = [4, 2, 2, 3]
 Generation 9470: Best = 1.390847e+03 FES = 947100 μ_F = 0.742 μ_{CR} = 0.152
 Diversity = 1.4341e-02 k_t = [1, 1, 1, 1]
 Generation 9480: Best = 1.390836e+03 FES = 948100 μ_F = 0.751 μ_{CR} = 0.154
 Diversity = 1.3451e-02 k_t = [2, 4, 4, 4]
 Generation 9490: Best = 1.390827e+03 FES = 949100 μ_F = 0.750 μ_{CR} = 0.147
 Diversity = 1.3677e-02 k_t = [1, 1, 1, 1]
 Generation 9500: Best = 1.390823e+03 FES = 950100 μ_F = 0.750 μ_{CR} = 0.139
 Diversity = 1.3482e-02 k_t = [2, 3, 2, 2]
 Generation 9510: Best = 1.390809e+03 FES = 951100 μ_F = 0.749 μ_{CR} = 0.140
 Diversity = 1.2791e-02 k_t = [1, 1, 1, 1]
 Generation 9520: Best = 1.390801e+03 FES = 952100 μ_F = 0.745 μ_{CR} = 0.139
 Diversity = 1.2727e-02 k_t = [2, 4, 2, 3]
 Generation 9530: Best = 1.390791e+03 FES = 953100 μ_F = 0.753 μ_{CR} = 0.134
 Diversity = 1.2771e-02 k_t = [1, 1, 1, 1]
 Generation 9540: Best = 1.390775e+03 FES = 954100 μ_F = 0.747 μ_{CR} = 0.130
 Diversity = 1.3258e-02 k_t = [2, 3, 2, 4]
 Generation 9550: Best = 1.390763e+03 FES = 955100 μ_F = 0.750 μ_{CR} = 0.134
 Diversity = 1.3834e-02 k_t = [1, 1, 1, 1]
 Generation 9560: Best = 1.390753e+03 FES = 956100 μ_F = 0.757 μ_{CR} = 0.145
 Diversity = 1.4196e-02 k_t = [4, 4, 4, 4]
 Generation 9570: Best = 1.390734e+03 FES = 957100 μ_F = 0.755 μ_{CR} = 0.147
 Diversity = 1.3725e-02 k_t = [1, 1, 1, 1]
 Generation 9580: Best = 1.390718e+03 FES = 958100 μ_F = 0.765 μ_{CR} = 0.148
 Diversity = 1.4845e-02 k_t = [3, 4, 4, 3]
 Generation 9590: Best = 1.390707e+03 FES = 959100 μ_F = 0.748 μ_{CR} = 0.148
 Diversity = 1.6137e-02 k_t = [1, 1, 1, 1]
 Generation 9600: Best = 1.390691e+03 FES = 960100 μ_F = 0.749 μ_{CR} = 0.139
 Diversity = 1.6073e-02 k_t = [4, 4, 4, 3]
 Generation 9610: Best = 1.390677e+03 FES = 961100 μ_F = 0.751 μ_{CR} = 0.126
 Diversity = 1.6780e-02 k_t = [1, 1, 1, 1]
 Generation 9620: Best = 1.390662e+03 FES = 962100 μ_F = 0.750 μ_{CR} = 0.136
 Diversity = 1.6375e-02 k_t = [3, 2, 2, 4]
 Generation 9630: Best = 1.390657e+03 FES = 963100 μ_F = 0.735 μ_{CR} = 0.138
 Diversity = 1.5687e-02 k_t = [1, 1, 1, 1]
 Generation 9640: Best = 1.390639e+03 FES = 964100 μ_F = 0.728 μ_{CR} = 0.136
 Diversity = 1.4745e-02 k_t = [2, 2, 2, 2]
 Generation 9650: Best = 1.390633e+03 FES = 965100 μ_F = 0.741 μ_{CR} = 0.137
 Diversity = 1.4142e-02 k_t = [1, 1, 1, 1]
 Generation 9660: Best = 1.390623e+03 FES = 966100 μ_F = 0.726 μ_{CR} = 0.127
 Diversity = 1.2548e-02 k_t = [3, 2, 3, 2]
 Generation 9670: Best = 1.390611e+03 FES = 967100 μ_F = 0.717 μ_{CR} = 0.128
 Diversity = 1.3074e-02 k_t = [1, 1, 1, 1]

Generation 9680: Best = 1.390602e+03 FES = 968100 μ_F = 0.721 μ_{CR} = 0.130
 Diversity = 1.1388e-02 k_t = [4, 4, 2, 4]
 Generation 9690: Best = 1.390582e+03 FES = 969100 μ_F = 0.713 μ_{CR} = 0.140
 Diversity = 1.1975e-02 k_t = [1, 1, 1, 1]
 Generation 9700: Best = 1.390572e+03 FES = 970100 μ_F = 0.711 μ_{CR} = 0.150
 Diversity = 1.2577e-02 k_t = [4, 3, 3, 3]
 Generation 9710: Best = 1.390559e+03 FES = 971100 μ_F = 0.724 μ_{CR} = 0.153
 Diversity = 1.2652e-02 k_t = [1, 1, 1, 1]
 Generation 9720: Best = 1.390522e+03 FES = 972100 μ_F = 0.728 μ_{CR} = 0.161
 Diversity = 1.4170e-02 k_t = [2, 2, 3, 2]
 Generation 9730: Best = 1.390499e+03 FES = 973100 μ_F = 0.748 μ_{CR} = 0.157
 Diversity = 2.0014e-02 k_t = [1, 1, 1, 1]
 Generation 9740: Best = 1.390483e+03 FES = 974100 μ_F = 0.751 μ_{CR} = 0.150
 Diversity = 1.8216e-02 k_t = [2, 4, 2, 4]
 Generation 9750: Best = 1.390458e+03 FES = 975100 μ_F = 0.750 μ_{CR} = 0.154
 Diversity = 1.6522e-02 k_t = [1, 1, 1, 1]
 Generation 9760: Best = 1.390440e+03 FES = 976100 μ_F = 0.753 μ_{CR} = 0.150
 Diversity = 1.7530e-02 k_t = [2, 4, 2, 3]
 Generation 9770: Best = 1.390436e+03 FES = 977100 μ_F = 0.741 μ_{CR} = 0.151
 Diversity = 1.7320e-02 k_t = [1, 1, 1, 1]
 Generation 9780: Best = 1.390429e+03 FES = 978100 μ_F = 0.732 μ_{CR} = 0.143
 Diversity = 1.5660e-02 k_t = [2, 4, 3, 2]
 Generation 9790: Best = 1.390418e+03 FES = 979100 μ_F = 0.720 μ_{CR} = 0.135
 Diversity = 1.4784e-02 k_t = [1, 1, 1, 1]
 Generation 9800: Best = 1.390414e+03 FES = 980100 μ_F = 0.716 μ_{CR} = 0.137
 Diversity = 1.3898e-02 k_t = [4, 2, 3, 3]
 Generation 9810: Best = 1.390401e+03 FES = 981100 μ_F = 0.712 μ_{CR} = 0.145
 Diversity = 1.2684e-02 k_t = [1, 1, 1, 1]
 Generation 9820: Best = 1.390387e+03 FES = 982100 μ_F = 0.706 μ_{CR} = 0.153
 Diversity = 1.2549e-02 k_t = [2, 2, 3, 2]
 Generation 9830: Best = 1.390368e+03 FES = 983100 μ_F = 0.722 μ_{CR} = 0.152
 Diversity = 1.4626e-02 k_t = [1, 1, 1, 1]
 Generation 9840: Best = 1.390357e+03 FES = 984100 μ_F = 0.739 μ_{CR} = 0.153
 Diversity = 1.6292e-02 k_t = [3, 4, 2, 2]
 Generation 9850: Best = 1.390347e+03 FES = 985100 μ_F = 0.747 μ_{CR} = 0.139
 Diversity = 1.6189e-02 k_t = [1, 1, 1, 1]
 Generation 9860: Best = 1.390342e+03 FES = 986100 μ_F = 0.750 μ_{CR} = 0.132
 Diversity = 1.5031e-02 k_t = [2, 4, 4, 4]
 Generation 9870: Best = 1.390335e+03 FES = 987100 μ_F = 0.750 μ_{CR} = 0.129
 Diversity = 1.4245e-02 k_t = [1, 1, 1, 1]
 Generation 9880: Best = 1.390326e+03 FES = 988100 μ_F = 0.740 μ_{CR} = 0.127
 Diversity = 1.3087e-02 k_t = [2, 3, 3, 3]
 Generation 9890: Best = 1.390322e+03 FES = 989100 μ_F = 0.739 μ_{CR} = 0.128
 Diversity = 1.3707e-02 k_t = [1, 1, 1, 1]
 Generation 9900: Best = 1.390318e+03 FES = 990100 μ_F = 0.728 μ_{CR} = 0.140
 Diversity = 1.2897e-02 k_t = [4, 4, 2, 3]
 Generation 9910: Best = 1.390313e+03 FES = 991100 μ_F = 0.718 μ_{CR} = 0.144
 Diversity = 1.0859e-02 k_t = [1, 1, 1, 1]
 Generation 9920: Best = 1.390306e+03 FES = 992100 μ_F = 0.719 μ_{CR} = 0.139
 Diversity = 1.0098e-02 k_t = [4, 4, 2, 3]
 Generation 9930: Best = 1.390299e+03 FES = 993100 μ_F = 0.727 μ_{CR} = 0.149
 Diversity = 9.8642e-03 k_t = [1, 1, 1, 1]
 Generation 9940: Best = 1.390292e+03 FES = 994100 μ_F = 0.728 μ_{CR} = 0.147
 Diversity = 9.9104e-03 k_t = [4, 3, 4, 4]

Generation 9950: Best = 1.390290e+03 FES = 995100 μ_F = 0.730 μ_{CR} = 0.146
Diversity = 9.9059e-03 k_t = [1, 1, 1, 1]
Generation 9960: Best = 1.390283e+03 FES = 996100 μ_F = 0.732 μ_{CR} = 0.145
Diversity = 9.4230e-03 k_t = [4, 2, 4, 3]
Generation 9970: Best = 1.390276e+03 FES = 997100 μ_F = 0.735 μ_{CR} = 0.159
Diversity = 9.5924e-03 k_t = [1, 1, 1, 1]
Generation 9980: Best = 1.390270e+03 FES = 998100 μ_F = 0.726 μ_{CR} = 0.163
Diversity = 9.1457e-03 k_t = [3, 3, 2, 3]
Generation 9990: Best = 1.390265e+03 FES = 999100 μ_F = 0.714 μ_{CR} = 0.162
Diversity = 8.1049e-03 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 1.390262e+03

Total Function Evals = 1000000

Total Successful Updates = 303736

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Results:

Best Fitness : 1.390262e+03

Global Optimum : 1.100000e+03

Error (|best-optimum|) : 2.902615e+02

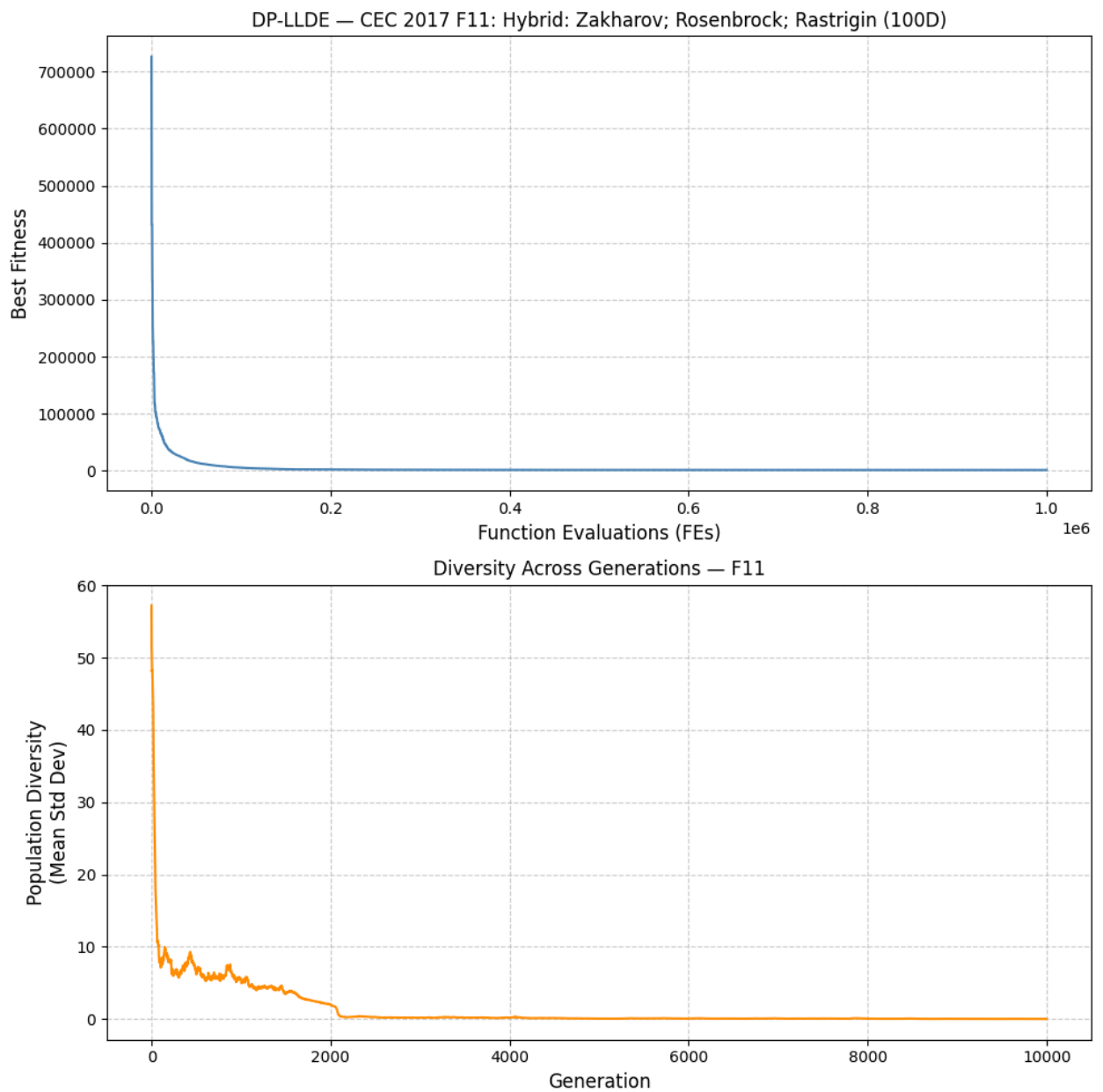
Best Solution (5 dim) : [-20.06233441 -51.51829695 -55.07714748 8.77208384
19.96135998]

D^(0) (init. diversity): 5.7220e+01

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✓ Checkpoint saved to: /content/drive/My Drive/thesis_results/all_results_checkpoint.pkl



Graph saved as convergence_F11.png

✓ Plot saved to: /content/drive/My Drive/thesis_results/plots/convergence_F11.png

```
In [ ]: # =====
# F12 — Hybrid Function 2 (Elliptic; Schwefel; Bent Cigar)
# =====
run_and_plot('F12', make_wrapper(hybrid.f12), 1200.0, 'Hybrid: Elliptic; Schwe
```


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DP-LLDE on CEC 2017 F12 – Single Run

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Function : Hybrid: Elliptic; Schwefel; Bent Cigar (100D)

Bounds : [-100, 100]

Global optimum : 1200.0

Initial $D^*(0) = 5.7220e+01$ | Diversity threshold = 5.7220e+00

Generation 0: Best Fitness = 3.383138e+12

Generation 10: Best = 1.123553e+12 FES = 1100 $\mu_F = 0.546$ $\mu_{CR} = 0.353$

Diversity = 3.1708e+01 $k_t = [1, 1, 1, 1]$

Generation 20: Best = 4.762008e+11 FES = 2100 $\mu_F = 0.592$ $\mu_{CR} = 0.354$

Diversity = 2.0068e+01 $k_t = [2, 3, 2, 3]$

Generation 30: Best = 2.206001e+11 FES = 3100 $\mu_F = 0.628$ $\mu_{CR} = 0.360$

Diversity = 1.4803e+01 $k_t = [1, 1, 1, 1]$

Generation 40: Best = 1.478131e+11 FES = 4100 $\mu_F = 0.655$ $\mu_{CR} = 0.363$

Diversity = 1.0342e+01 $k_t = [3, 2, 3, 4]$

Generation 50: Best = 9.706953e+10 FES = 5100 $\mu_F = 0.697$ $\mu_{CR} = 0.365$

Diversity = 1.0157e+01 $k_t = [1, 1, 1, 1]$

Generation 60: Best = 6.565081e+10 FES = 6100 $\mu_F = 0.716$ $\mu_{CR} = 0.361$

Diversity = 1.0003e+01 $k_t = [4, 2, 2, 2]$

Generation 70: Best = 4.675340e+10 FES = 7100 $\mu_F = 0.730$ $\mu_{CR} = 0.358$

Diversity = 9.3807e+00 $k_t = [1, 1, 1, 1]$

Generation 80: Best = 3.848243e+10 FES = 8100 $\mu_F = 0.747$ $\mu_{CR} = 0.357$

Diversity = 9.0455e+00 $k_t = [3, 2, 3, 4]$

Generation 90: Best = 3.118978e+10 FES = 9100 $\mu_F = 0.748$ $\mu_{CR} = 0.356$

Diversity = 9.0084e+00 $k_t = [1, 1, 1, 1]$

Generation 100: Best = 2.491860e+10 FES = 10100 $\mu_F = 0.748$ $\mu_{CR} = 0.361$

Diversity = 8.2922e+00 $k_t = [2, 3, 2, 4]$

Generation 110: Best = 1.978156e+10 FES = 11100 $\mu_F = 0.753$ $\mu_{CR} = 0.364$

Diversity = 8.8684e+00 $k_t = [1, 1, 1, 1]$

Generation 120: Best = 1.535439e+10 FES = 12100 $\mu_F = 0.758$ $\mu_{CR} = 0.378$

Diversity = 9.2922e+00 $k_t = [3, 2, 2, 3]$

Generation 130: Best = 1.201482e+10 FES = 13100 $\mu_F = 0.749$ $\mu_{CR} = 0.381$

Diversity = 9.3071e+00 $k_t = [1, 1, 1, 1]$

Generation 140: Best = 8.003732e+09 FES = 14100 $\mu_F = 0.759$ $\mu_{CR} = 0.383$

Diversity = 9.8011e+00 $k_t = [2, 4, 2, 4]$

Generation 150: Best = 4.421687e+09 FES = 15100 $\mu_F = 0.766$ $\mu_{CR} = 0.394$

Diversity = 1.0424e+01 $k_t = [1, 1, 1, 1]$

Generation 160: Best = 2.993415e+09 FES = 16100 $\mu_F = 0.758$ $\mu_{CR} = 0.408$

Diversity = 1.0133e+01 $k_t = [4, 4, 2, 2]$

Generation 170: Best = 2.003496e+09 FES = 17100 $\mu_F = 0.745$ $\mu_{CR} = 0.410$

Diversity = 8.6401e+00 $k_t = [1, 1, 1, 1]$

Generation 180: Best = 1.392312e+09 FES = 18100 $\mu_F = 0.741$ $\mu_{CR} = 0.417$

Diversity = 7.4787e+00 $k_t = [3, 2, 3, 3]$

Generation 190: Best = 1.079326e+09 FES = 19100 $\mu_F = 0.734$ $\mu_{CR} = 0.416$

Diversity = 6.9658e+00 $k_t = [1, 1, 1, 1]$

Generation 200: Best = 8.252257e+08 FES = 20100 $\mu_F = 0.727$ $\mu_{CR} = 0.421$

Diversity = 5.4915e+00 $k_t = [4, 4, 3, 4]$

Generation 210: Best = 6.644401e+08 FES = 21100 $\mu_F = 0.736$ $\mu_{CR} = 0.426$

Diversity = 4.5711e+00 $k_t = [1, 1, 1, 1]$

Generation 220: Best = 5.547998e+08 FES = 22100 $\mu_F = 0.740$ $\mu_{CR} = 0.427$

Diversity = 4.0152e+00 $k_t = [2, 4, 3, 4]$

Generation 230: Best = 4.595230e+08	FES = 23100	$\mu_F = 0.747$	$\mu_{CR} = 0.428$
Diversity = 4.3602e+00	$k_t = [1, 1, 1, 1]$		
Generation 240: Best = 3.831166e+08	FES = 24100	$\mu_F = 0.745$	$\mu_{CR} = 0.439$
Diversity = 4.1728e+00	$k_t = [3, 4, 3, 3]$		
Generation 250: Best = 3.079256e+08	FES = 25100	$\mu_F = 0.754$	$\mu_{CR} = 0.439$
Diversity = 4.5362e+00	$k_t = [1, 1, 1, 1]$		
Generation 260: Best = 2.666606e+08	FES = 26100	$\mu_F = 0.767$	$\mu_{CR} = 0.436$
Diversity = 5.1622e+00	$k_t = [2, 3, 4, 2]$		
Generation 270: Best = 2.249422e+08	FES = 27100	$\mu_F = 0.762$	$\mu_{CR} = 0.449$
Diversity = 5.3500e+00	$k_t = [1, 1, 1, 1]$		
Generation 280: Best = 1.829354e+08	FES = 28100	$\mu_F = 0.767$	$\mu_{CR} = 0.449$
Diversity = 5.7139e+00	$k_t = [4, 4, 2, 4]$		
Generation 290: Best = 1.619510e+08	FES = 29100	$\mu_F = 0.761$	$\mu_{CR} = 0.456$
Diversity = 5.7173e+00	$k_t = [1, 1, 1, 1]$		
Generation 300: Best = 1.344744e+08	FES = 30100	$\mu_F = 0.766$	$\mu_{CR} = 0.457$
Diversity = 5.9601e+00	$k_t = [4, 2, 2, 3]$		
Generation 310: Best = 1.119480e+08	FES = 31100	$\mu_F = 0.767$	$\mu_{CR} = 0.463$
Diversity = 5.9578e+00	$k_t = [1, 1, 1, 1]$		
Generation 320: Best = 9.605813e+07	FES = 32100	$\mu_F = 0.768$	$\mu_{CR} = 0.467$
Diversity = 5.3722e+00	$k_t = [2, 3, 3, 2]$		
Generation 330: Best = 8.407747e+07	FES = 33100	$\mu_F = 0.759$	$\mu_{CR} = 0.484$
Diversity = 5.9272e+00	$k_t = [1, 1, 1, 1]$		
Generation 340: Best = 7.250458e+07	FES = 34100	$\mu_F = 0.752$	$\mu_{CR} = 0.479$
Diversity = 5.2110e+00	$k_t = [4, 4, 2, 2]$		
Generation 350: Best = 6.426541e+07	FES = 35100	$\mu_F = 0.752$	$\mu_{CR} = 0.483$
Diversity = 5.1404e+00	$k_t = [1, 1, 1, 1]$		
Generation 360: Best = 5.915408e+07	FES = 36100	$\mu_F = 0.753$	$\mu_{CR} = 0.489$
Diversity = 4.7622e+00	$k_t = [3, 3, 2, 4]$		
Generation 370: Best = 5.381738e+07	FES = 37100	$\mu_F = 0.767$	$\mu_{CR} = 0.496$
Diversity = 5.1064e+00	$k_t = [1, 1, 1, 1]$		
Generation 380: Best = 5.075052e+07	FES = 38100	$\mu_F = 0.767$	$\mu_{CR} = 0.505$
Diversity = 5.0467e+00	$k_t = [3, 3, 4, 2]$		
Generation 390: Best = 4.777916e+07	FES = 39100	$\mu_F = 0.757$	$\mu_{CR} = 0.507$
Diversity = 5.4246e+00	$k_t = [1, 1, 1, 1]$		
Generation 400: Best = 4.416590e+07	FES = 40100	$\mu_F = 0.750$	$\mu_{CR} = 0.507$
Diversity = 4.7874e+00	$k_t = [2, 4, 2, 2]$		
Generation 410: Best = 4.162499e+07	FES = 41100	$\mu_F = 0.745$	$\mu_{CR} = 0.517$
Diversity = 5.2403e+00	$k_t = [1, 1, 1, 1]$		
Generation 420: Best = 3.920473e+07	FES = 42100	$\mu_F = 0.740$	$\mu_{CR} = 0.520$
Diversity = 4.5034e+00	$k_t = [3, 3, 3, 3]$		
Generation 430: Best = 3.684843e+07	FES = 43100	$\mu_F = 0.743$	$\mu_{CR} = 0.522$
Diversity = 4.5443e+00	$k_t = [1, 1, 1, 1]$		
Generation 440: Best = 3.512479e+07	FES = 44100	$\mu_F = 0.754$	$\mu_{CR} = 0.523$
Diversity = 4.7355e+00	$k_t = [4, 2, 2, 3]$		
Generation 450: Best = 3.347243e+07	FES = 45100	$\mu_F = 0.756$	$\mu_{CR} = 0.527$
Diversity = 4.5857e+00	$k_t = [1, 1, 1, 1]$		
Generation 460: Best = 3.193517e+07	FES = 46100	$\mu_F = 0.763$	$\mu_{CR} = 0.528$
Diversity = 4.4942e+00	$k_t = [2, 2, 3, 2]$		
Generation 470: Best = 3.077676e+07	FES = 47100	$\mu_F = 0.763$	$\mu_{CR} = 0.532$
Diversity = 4.8308e+00	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 2.946404e+07	FES = 48100	$\mu_F = 0.758$	$\mu_{CR} = 0.547$
Diversity = 5.2918e+00	$k_t = [2, 2, 3, 4]$		
Generation 490: Best = 2.818086e+07	FES = 49100	$\mu_F = 0.763$	$\mu_{CR} = 0.560$
Diversity = 5.5869e+00	$k_t = [1, 1, 1, 1]$		

Generation 500: Best = 2.689694e+07	FES = 50100	$\mu_F = 0.765$	$\mu_{CR} = 0.558$
Diversity = 5.6315e+00	$k_t = [2, 3, 3, 2]$		
Generation 510: Best = 2.583234e+07	FES = 51100	$\mu_F = 0.755$	$\mu_{CR} = 0.568$
Diversity = 5.0803e+00	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 2.475429e+07	FES = 52100	$\mu_F = 0.740$	$\mu_{CR} = 0.575$
Diversity = 4.4118e+00	$k_t = [3, 2, 2, 2]$		
Generation 530: Best = 2.369777e+07	FES = 53100	$\mu_F = 0.734$	$\mu_{CR} = 0.575$
Diversity = 4.1687e+00	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 2.278757e+07	FES = 54100	$\mu_F = 0.737$	$\mu_{CR} = 0.574$
Diversity = 3.9300e+00	$k_t = [3, 3, 3, 2]$		
Generation 550: Best = 2.228230e+07	FES = 55100	$\mu_F = 0.735$	$\mu_{CR} = 0.586$
Diversity = 3.4936e+00	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 2.180531e+07	FES = 56100	$\mu_F = 0.745$	$\mu_{CR} = 0.592$
Diversity = 3.7858e+00	$k_t = [2, 3, 2, 2]$		
Generation 570: Best = 2.131825e+07	FES = 57100	$\mu_F = 0.744$	$\mu_{CR} = 0.605$
Diversity = 3.6558e+00	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 2.097949e+07	FES = 58100	$\mu_F = 0.738$	$\mu_{CR} = 0.600$
Diversity = 3.0162e+00	$k_t = [2, 3, 3, 3]$		
Generation 590: Best = 2.064918e+07	FES = 59100	$\mu_F = 0.741$	$\mu_{CR} = 0.606$
Diversity = 3.3150e+00	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 2.033106e+07	FES = 60100	$\mu_F = 0.754$	$\mu_{CR} = 0.605$
Diversity = 3.5116e+00	$k_t = [2, 3, 3, 2]$		
Generation 610: Best = 2.006381e+07	FES = 61100	$\mu_F = 0.756$	$\mu_{CR} = 0.612$
Diversity = 4.2615e+00	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 1.971479e+07	FES = 62100	$\mu_F = 0.755$	$\mu_{CR} = 0.615$
Diversity = 3.9862e+00	$k_t = [3, 2, 4, 4]$		
Generation 630: Best = 1.944426e+07	FES = 63100	$\mu_F = 0.762$	$\mu_{CR} = 0.620$
Diversity = 3.8287e+00	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 1.918481e+07	FES = 64100	$\mu_F = 0.764$	$\mu_{CR} = 0.629$
Diversity = 3.5156e+00	$k_t = [3, 3, 4, 3]$		
Generation 650: Best = 1.891549e+07	FES = 65100	$\mu_F = 0.771$	$\mu_{CR} = 0.640$
Diversity = 4.0311e+00	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 1.865913e+07	FES = 66100	$\mu_F = 0.771$	$\mu_{CR} = 0.653$
Diversity = 3.5333e+00	$k_t = [2, 4, 4, 2]$		
Generation 670: Best = 1.840476e+07	FES = 67100	$\mu_F = 0.766$	$\mu_{CR} = 0.661$
Diversity = 3.7865e+00	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 1.803566e+07	FES = 68100	$\mu_F = 0.767$	$\mu_{CR} = 0.673$
Diversity = 3.9915e+00	$k_t = [4, 2, 4, 4]$		
Generation 690: Best = 1.787085e+07	FES = 69100	$\mu_F = 0.776$	$\mu_{CR} = 0.678$
Diversity = 3.7131e+00	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 1.764745e+07	FES = 70100	$\mu_F = 0.778$	$\mu_{CR} = 0.685$
Diversity = 3.5105e+00	$k_t = [4, 4, 2, 3]$		
Generation 710: Best = 1.748465e+07	FES = 71100	$\mu_F = 0.781$	$\mu_{CR} = 0.695$
Diversity = 3.1470e+00	$k_t = [1, 1, 1, 1]$		
Generation 720: Best = 1.731956e+07	FES = 72100	$\mu_F = 0.785$	$\mu_{CR} = 0.692$
Diversity = 3.6257e+00	$k_t = [2, 2, 2, 4]$		
Generation 730: Best = 1.713140e+07	FES = 73100	$\mu_F = 0.789$	$\mu_{CR} = 0.696$
Diversity = 4.1964e+00	$k_t = [1, 1, 1, 1]$		
Generation 740: Best = 1.690989e+07	FES = 74100	$\mu_F = 0.780$	$\mu_{CR} = 0.711$
Diversity = 3.6890e+00	$k_t = [3, 2, 2, 4]$		
Generation 750: Best = 1.677965e+07	FES = 75100	$\mu_F = 0.793$	$\mu_{CR} = 0.717$
Diversity = 3.9100e+00	$k_t = [1, 1, 1, 1]$		
Generation 760: Best = 1.658324e+07	FES = 76100	$\mu_F = 0.788$	$\mu_{CR} = 0.722$
Diversity = 4.1921e+00	$k_t = [2, 3, 2, 4]$		

Generation 770:	Best = 1.638640e+07	FES = 77100	$\mu_F = 0.792$	$\mu_{CR} = 0.733$
Diversity = 4.8311e+00	$k_t = [1, 1, 1, 1]$			
Generation 780:	Best = 1.622856e+07	FES = 78100	$\mu_F = 0.777$	$\mu_{CR} = 0.732$
Diversity = 4.3702e+00	$k_t = [4, 4, 3, 2]$			
Generation 790:	Best = 1.610612e+07	FES = 79100	$\mu_F = 0.772$	$\mu_{CR} = 0.727$
Diversity = 3.5569e+00	$k_t = [1, 1, 1, 1]$			
Generation 800:	Best = 1.597003e+07	FES = 80100	$\mu_F = 0.774$	$\mu_{CR} = 0.732$
Diversity = 3.0123e+00	$k_t = [2, 4, 2, 2]$			
Generation 810:	Best = 1.584644e+07	FES = 81100	$\mu_F = 0.781$	$\mu_{CR} = 0.741$
Diversity = 2.8192e+00	$k_t = [1, 1, 1, 1]$			
Generation 820:	Best = 1.569725e+07	FES = 82100	$\mu_F = 0.778$	$\mu_{CR} = 0.744$
Diversity = 2.9568e+00	$k_t = [3, 3, 4, 2]$			
Generation 830:	Best = 1.556275e+07	FES = 83100	$\mu_F = 0.771$	$\mu_{CR} = 0.764$
Diversity = 3.5414e+00	$k_t = [1, 1, 1, 1]$			
Generation 840:	Best = 1.538820e+07	FES = 84100	$\mu_F = 0.777$	$\mu_{CR} = 0.767$
Diversity = 3.3751e+00	$k_t = [4, 2, 2, 3]$			
Generation 850:	Best = 1.522574e+07	FES = 85100	$\mu_F = 0.790$	$\mu_{CR} = 0.762$
Diversity = 3.3448e+00	$k_t = [1, 1, 1, 1]$			
Generation 860:	Best = 1.504100e+07	FES = 86100	$\mu_F = 0.793$	$\mu_{CR} = 0.764$
Diversity = 2.9815e+00	$k_t = [2, 3, 4, 2]$			
Generation 870:	Best = 1.485854e+07	FES = 87100	$\mu_F = 0.793$	$\mu_{CR} = 0.764$
Diversity = 3.5613e+00	$k_t = [1, 1, 1, 1]$			
Generation 880:	Best = 1.469578e+07	FES = 88100	$\mu_F = 0.794$	$\mu_{CR} = 0.774$
Diversity = 3.3865e+00	$k_t = [4, 4, 3, 3]$			
Generation 890:	Best = 1.458637e+07	FES = 89100	$\mu_F = 0.787$	$\mu_{CR} = 0.780$
Diversity = 3.3160e+00	$k_t = [1, 1, 1, 1]$			
Generation 900:	Best = 1.446545e+07	FES = 90100	$\mu_F = 0.790$	$\mu_{CR} = 0.783$
Diversity = 2.8481e+00	$k_t = [4, 3, 3, 2]$			
Generation 910:	Best = 1.439337e+07	FES = 91100	$\mu_F = 0.790$	$\mu_{CR} = 0.779$
Diversity = 2.6437e+00	$k_t = [1, 1, 1, 1]$			
Generation 920:	Best = 1.428539e+07	FES = 92100	$\mu_F = 0.794$	$\mu_{CR} = 0.797$
Diversity = 3.0545e+00	$k_t = [4, 2, 3, 2]$			
Generation 930:	Best = 1.421133e+07	FES = 93100	$\mu_F = 0.794$	$\mu_{CR} = 0.799$
Diversity = 3.5217e+00	$k_t = [1, 1, 1, 1]$			
Generation 940:	Best = 1.412898e+07	FES = 94100	$\mu_F = 0.792$	$\mu_{CR} = 0.807$
Diversity = 3.2156e+00	$k_t = [3, 2, 2, 3]$			
Generation 950:	Best = 1.405680e+07	FES = 95100	$\mu_F = 0.791$	$\mu_{CR} = 0.805$
Diversity = 2.9273e+00	$k_t = [1, 1, 1, 1]$			
Generation 960:	Best = 1.397962e+07	FES = 96100	$\mu_F = 0.791$	$\mu_{CR} = 0.809$
Diversity = 3.2766e+00	$k_t = [2, 4, 2, 4]$			
Generation 970:	Best = 1.391638e+07	FES = 97100	$\mu_F = 0.801$	$\mu_{CR} = 0.816$
Diversity = 3.7450e+00	$k_t = [1, 1, 1, 1]$			
Generation 980:	Best = 1.384074e+07	FES = 98100	$\mu_F = 0.795$	$\mu_{CR} = 0.816$
Diversity = 3.4787e+00	$k_t = [2, 4, 2, 4]$			
Generation 990:	Best = 1.376921e+07	FES = 99100	$\mu_F = 0.796$	$\mu_{CR} = 0.813$
Diversity = 4.5149e+00	$k_t = [1, 1, 1, 1]$			
Generation 1000:	Best = 1.368992e+07	FES = 100100	$\mu_F = 0.797$	$\mu_{CR} = 0.817$
Diversity = 3.5406e+00	$k_t = [4, 3, 3, 3]$			
Generation 1010:	Best = 1.363550e+07	FES = 101100	$\mu_F = 0.797$	$\mu_{CR} = 0.826$
Diversity = 3.0417e+00	$k_t = [1, 1, 1, 1]$			
Generation 1020:	Best = 1.358029e+07	FES = 102100	$\mu_F = 0.802$	$\mu_{CR} = 0.829$
Diversity = 2.7803e+00	$k_t = [4, 2, 4, 2]$			
Generation 1030:	Best = 1.354116e+07	FES = 103100	$\mu_F = 0.797$	$\mu_{CR} = 0.836$
Diversity = 2.6716e+00	$k_t = [1, 1, 1, 1]$			

Generation 1040: Best = 1.350026e+07 FES = 104100 μ_F = 0.787 μ_{CR} = 0.839
 Diversity = 2.8443e+00 k_t = [2, 4, 3, 4]
 Generation 1050: Best = 1.346321e+07 FES = 105100 μ_F = 0.799 μ_{CR} = 0.848
 Diversity = 2.9061e+00 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 1.340550e+07 FES = 106100 μ_F = 0.800 μ_{CR} = 0.857
 Diversity = 2.4571e+00 k_t = [3, 2, 3, 3]
 Generation 1070: Best = 1.335575e+07 FES = 107100 μ_F = 0.794 μ_{CR} = 0.859
 Diversity = 2.9124e+00 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 1.328528e+07 FES = 108100 μ_F = 0.805 μ_{CR} = 0.867
 Diversity = 3.1986e+00 k_t = [4, 3, 4, 2]
 Generation 1090: Best = 1.323969e+07 FES = 109100 μ_F = 0.804 μ_{CR} = 0.870
 Diversity = 3.4034e+00 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 1.318105e+07 FES = 110100 μ_F = 0.808 μ_{CR} = 0.870
 Diversity = 3.2308e+00 k_t = [3, 4, 3, 3]
 Generation 1110: Best = 1.311214e+07 FES = 111100 μ_F = 0.817 μ_{CR} = 0.877
 Diversity = 3.6988e+00 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 1.303642e+07 FES = 112100 μ_F = 0.810 μ_{CR} = 0.881
 Diversity = 2.8409e+00 k_t = [2, 4, 3, 4]
 Generation 1130: Best = 1.297399e+07 FES = 113100 μ_F = 0.813 μ_{CR} = 0.886
 Diversity = 2.9230e+00 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 1.290648e+07 FES = 114100 μ_F = 0.822 μ_{CR} = 0.887
 Diversity = 2.8856e+00 k_t = [4, 3, 4, 3]
 Generation 1150: Best = 1.284712e+07 FES = 115100 μ_F = 0.826 μ_{CR} = 0.883
 Diversity = 3.0036e+00 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 1.277511e+07 FES = 116100 μ_F = 0.830 μ_{CR} = 0.887
 Diversity = 3.0432e+00 k_t = [2, 3, 4, 4]
 Generation 1170: Best = 1.271153e+07 FES = 117100 μ_F = 0.827 μ_{CR} = 0.894
 Diversity = 2.9109e+00 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 1.264470e+07 FES = 118100 μ_F = 0.829 μ_{CR} = 0.893
 Diversity = 3.0906e+00 k_t = [4, 3, 3, 2]
 Generation 1190: Best = 1.256082e+07 FES = 119100 μ_F = 0.825 μ_{CR} = 0.887
 Diversity = 2.9107e+00 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 1.248195e+07 FES = 120100 μ_F = 0.826 μ_{CR} = 0.888
 Diversity = 2.5600e+00 k_t = [2, 2, 3, 2]
 Generation 1210: Best = 1.240991e+07 FES = 121100 μ_F = 0.813 μ_{CR} = 0.888
 Diversity = 2.4517e+00 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 1.234871e+07 FES = 122100 μ_F = 0.811 μ_{CR} = 0.891
 Diversity = 1.7966e+00 k_t = [4, 2, 3, 2]
 Generation 1230: Best = 1.229721e+07 FES = 123100 μ_F = 0.818 μ_{CR} = 0.888
 Diversity = 1.2545e+00 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 1.224769e+07 FES = 124100 μ_F = 0.821 μ_{CR} = 0.890
 Diversity = 1.3526e+00 k_t = [3, 4, 2, 3]
 Generation 1250: Best = 1.217721e+07 FES = 125100 μ_F = 0.820 μ_{CR} = 0.884
 Diversity = 1.6886e+00 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 1.210527e+07 FES = 126100 μ_F = 0.819 μ_{CR} = 0.891
 Diversity = 1.1909e+00 k_t = [2, 2, 4, 4]
 Generation 1270: Best = 1.204510e+07 FES = 127100 μ_F = 0.816 μ_{CR} = 0.890
 Diversity = 1.4088e+00 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 1.196595e+07 FES = 128100 μ_F = 0.807 μ_{CR} = 0.899
 Diversity = 1.6115e+00 k_t = [3, 4, 4, 2]
 Generation 1290: Best = 1.189731e+07 FES = 129100 μ_F = 0.808 μ_{CR} = 0.896
 Diversity = 1.5336e+00 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 1.182866e+07 FES = 130100 μ_F = 0.810 μ_{CR} = 0.894
 Diversity = 1.4849e+00 k_t = [2, 2, 4, 2]

Generation 1310: Best = 1.176917e+07 FES = 131100 μ_F = 0.811 μ_{CR} = 0.896
 Diversity = 1.4197e+00 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 1.168464e+07 FES = 132100 μ_F = 0.809 μ_{CR} = 0.894
 Diversity = 1.7476e+00 k_t = [2, 2, 3, 2]
 Generation 1330: Best = 1.158445e+07 FES = 133100 μ_F = 0.809 μ_{CR} = 0.893
 Diversity = 1.9019e+00 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 1.147723e+07 FES = 134100 μ_F = 0.809 μ_{CR} = 0.894
 Diversity = 1.4446e+00 k_t = [2, 4, 3, 3]
 Generation 1350: Best = 1.141784e+07 FES = 135100 μ_F = 0.806 μ_{CR} = 0.896
 Diversity = 1.4061e+00 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 1.134346e+07 FES = 136100 μ_F = 0.809 μ_{CR} = 0.897
 Diversity = 1.3886e+00 k_t = [4, 3, 4, 2]
 Generation 1370: Best = 1.129435e+07 FES = 137100 μ_F = 0.818 μ_{CR} = 0.899
 Diversity = 1.4819e+00 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 1.123422e+07 FES = 138100 μ_F = 0.811 μ_{CR} = 0.893
 Diversity = 1.2910e+00 k_t = [3, 3, 4, 2]
 Generation 1390: Best = 1.118528e+07 FES = 139100 μ_F = 0.813 μ_{CR} = 0.896
 Diversity = 1.7591e+00 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 1.113328e+07 FES = 140100 μ_F = 0.813 μ_{CR} = 0.901
 Diversity = 2.0368e+00 k_t = [3, 2, 4, 3]
 Generation 1410: Best = 1.109414e+07 FES = 141100 μ_F = 0.818 μ_{CR} = 0.906
 Diversity = 1.4162e+00 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 1.105343e+07 FES = 142100 μ_F = 0.822 μ_{CR} = 0.905
 Diversity = 1.3876e+00 k_t = [4, 3, 2, 2]
 Generation 1430: Best = 1.101412e+07 FES = 143100 μ_F = 0.821 μ_{CR} = 0.903
 Diversity = 1.0441e+00 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 1.097402e+07 FES = 144100 μ_F = 0.819 μ_{CR} = 0.904
 Diversity = 1.2315e+00 k_t = [3, 2, 3, 4]
 Generation 1450: Best = 1.095199e+07 FES = 145100 μ_F = 0.815 μ_{CR} = 0.899
 Diversity = 1.1030e+00 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 1.092688e+07 FES = 146100 μ_F = 0.810 μ_{CR} = 0.901
 Diversity = 1.0281e+00 k_t = [4, 2, 4, 4]
 Generation 1470: Best = 1.090945e+07 FES = 147100 μ_F = 0.825 μ_{CR} = 0.900
 Diversity = 9.8262e-01 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 1.088262e+07 FES = 148100 μ_F = 0.825 μ_{CR} = 0.896
 Diversity = 1.2661e+00 k_t = [2, 3, 2, 3]
 Generation 1490: Best = 1.086551e+07 FES = 149100 μ_F = 0.825 μ_{CR} = 0.899
 Diversity = 1.7673e+00 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 1.084590e+07 FES = 150100 μ_F = 0.818 μ_{CR} = 0.901
 Diversity = 1.3641e+00 k_t = [2, 3, 2, 3]
 Generation 1510: Best = 1.082646e+07 FES = 151100 μ_F = 0.816 μ_{CR} = 0.891
 Diversity = 1.7057e+00 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 1.080215e+07 FES = 152100 μ_F = 0.812 μ_{CR} = 0.896
 Diversity = 1.2678e+00 k_t = [3, 2, 3, 4]
 Generation 1530: Best = 1.078245e+07 FES = 153100 μ_F = 0.804 μ_{CR} = 0.886
 Diversity = 1.0026e+00 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 1.075791e+07 FES = 154100 μ_F = 0.816 μ_{CR} = 0.880
 Diversity = 1.5251e+00 k_t = [4, 2, 3, 4]
 Generation 1550: Best = 1.073727e+07 FES = 155100 μ_F = 0.823 μ_{CR} = 0.883
 Diversity = 1.5322e+00 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 1.071313e+07 FES = 156100 μ_F = 0.828 μ_{CR} = 0.883
 Diversity = 1.4864e+00 k_t = [4, 4, 3, 3]
 Generation 1570: Best = 1.068699e+07 FES = 157100 μ_F = 0.829 μ_{CR} = 0.882
 Diversity = 1.9063e+00 k_t = [1, 1, 1, 1]

Generation 1580: Best = 1.064918e+07 FES = 158100 μ_F = 0.828 μ_{CR} = 0.878
 Diversity = 1.8830e+00 k_t = [3, 2, 2, 3]
 Generation 1590: Best = 1.060787e+07 FES = 159100 μ_F = 0.821 μ_{CR} = 0.872
 Diversity = 1.4681e+00 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 1.057328e+07 FES = 160100 μ_F = 0.816 μ_{CR} = 0.876
 Diversity = 1.2609e+00 k_t = [3, 2, 3, 3]
 Generation 1610: Best = 1.053035e+07 FES = 161100 μ_F = 0.811 μ_{CR} = 0.880
 Diversity = 1.4767e+00 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 1.050153e+07 FES = 162100 μ_F = 0.799 μ_{CR} = 0.885
 Diversity = 1.2969e+00 k_t = [3, 2, 3, 4]
 Generation 1630: Best = 1.047389e+07 FES = 163100 μ_F = 0.786 μ_{CR} = 0.888
 Diversity = 1.3017e+00 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 1.044681e+07 FES = 164100 μ_F = 0.795 μ_{CR} = 0.893
 Diversity = 1.4181e+00 k_t = [3, 3, 2, 2]
 Generation 1650: Best = 1.042654e+07 FES = 165100 μ_F = 0.801 μ_{CR} = 0.894
 Diversity = 1.3988e+00 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 1.040285e+07 FES = 166100 μ_F = 0.804 μ_{CR} = 0.892
 Diversity = 1.4257e+00 k_t = [3, 2, 2, 3]
 Generation 1670: Best = 1.038062e+07 FES = 167100 μ_F = 0.804 μ_{CR} = 0.889
 Diversity = 1.3258e+00 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 1.034570e+07 FES = 168100 μ_F = 0.807 μ_{CR} = 0.891
 Diversity = 7.6995e-01 k_t = [3, 2, 3, 3]
 Generation 1690: Best = 1.031506e+07 FES = 169100 μ_F = 0.806 μ_{CR} = 0.890
 Diversity = 7.8954e-01 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 1.029134e+07 FES = 170100 μ_F = 0.800 μ_{CR} = 0.893
 Diversity = 7.3206e-01 k_t = [2, 3, 4, 4]
 Generation 1710: Best = 1.026920e+07 FES = 171100 μ_F = 0.807 μ_{CR} = 0.888
 Diversity = 6.5256e-01 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 1.024581e+07 FES = 172100 μ_F = 0.810 μ_{CR} = 0.881
 Diversity = 6.9177e-01 k_t = [3, 2, 4, 2]
 Generation 1730: Best = 1.021466e+07 FES = 173100 μ_F = 0.816 μ_{CR} = 0.884
 Diversity = 8.4313e-01 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 1.018460e+07 FES = 174100 μ_F = 0.821 μ_{CR} = 0.886
 Diversity = 9.8481e-01 k_t = [2, 4, 2, 3]
 Generation 1750: Best = 1.015864e+07 FES = 175100 μ_F = 0.821 μ_{CR} = 0.878
 Diversity = 9.8784e-01 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 1.011231e+07 FES = 176100 μ_F = 0.825 μ_{CR} = 0.879
 Diversity = 9.6836e-01 k_t = [4, 3, 3, 2]
 Generation 1770: Best = 1.008514e+07 FES = 177100 μ_F = 0.813 μ_{CR} = 0.881
 Diversity = 1.0319e+00 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 1.003382e+07 FES = 178100 μ_F = 0.811 μ_{CR} = 0.881
 Diversity = 1.0182e+00 k_t = [4, 2, 3, 3]
 Generation 1790: Best = 9.954777e+06 FES = 179100 μ_F = 0.812 μ_{CR} = 0.872
 Diversity = 1.2339e+00 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 9.880603e+06 FES = 180100 μ_F = 0.808 μ_{CR} = 0.874
 Diversity = 9.2805e-01 k_t = [4, 2, 2, 2]
 Generation 1810: Best = 9.828952e+06 FES = 181100 μ_F = 0.811 μ_{CR} = 0.875
 Diversity = 8.6376e-01 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 9.768781e+06 FES = 182100 μ_F = 0.815 μ_{CR} = 0.880
 Diversity = 6.6423e-01 k_t = [4, 4, 3, 2]
 Generation 1830: Best = 9.721076e+06 FES = 183100 μ_F = 0.822 μ_{CR} = 0.889
 Diversity = 5.7879e-01 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 9.679577e+06 FES = 184100 μ_F = 0.812 μ_{CR} = 0.889
 Diversity = 6.1565e-01 k_t = [4, 4, 2, 4]

Generation 1850: Best = 9.658709e+06 FES = 185100 μ_F = 0.816 μ_{CR} = 0.888
 Diversity = 7.4144e-01 k_t = [1, 1, 1, 1]
 Generation 1860: Best = 9.619209e+06 FES = 186100 μ_F = 0.821 μ_{CR} = 0.884
 Diversity = 1.0099e+00 k_t = [3, 2, 4, 4]
 Generation 1870: Best = 9.594970e+06 FES = 187100 μ_F = 0.818 μ_{CR} = 0.887
 Diversity = 1.1767e+00 k_t = [1, 1, 1, 1]
 Generation 1880: Best = 9.557441e+06 FES = 188100 μ_F = 0.820 μ_{CR} = 0.886
 Diversity = 1.4434e+00 k_t = [3, 2, 2, 3]
 Generation 1890: Best = 9.524575e+06 FES = 189100 μ_F = 0.820 μ_{CR} = 0.892
 Diversity = 1.7293e+00 k_t = [1, 1, 1, 1]
 Generation 1900: Best = 9.473794e+06 FES = 190100 μ_F = 0.828 μ_{CR} = 0.888
 Diversity = 1.5301e+00 k_t = [2, 3, 4, 3]
 Generation 1910: Best = 9.394094e+06 FES = 191100 μ_F = 0.820 μ_{CR} = 0.890
 Diversity = 1.7918e+00 k_t = [1, 1, 1, 1]
 Generation 1920: Best = 9.325451e+06 FES = 192100 μ_F = 0.819 μ_{CR} = 0.886
 Diversity = 1.4207e+00 k_t = [3, 3, 2, 4]
 Generation 1930: Best = 9.273464e+06 FES = 193100 μ_F = 0.818 μ_{CR} = 0.878
 Diversity = 1.6600e+00 k_t = [1, 1, 1, 1]
 Generation 1940: Best = 9.233253e+06 FES = 194100 μ_F = 0.807 μ_{CR} = 0.876
 Diversity = 1.1527e+00 k_t = [2, 4, 3, 2]
 Generation 1950: Best = 9.192413e+06 FES = 195100 μ_F = 0.812 μ_{CR} = 0.885
 Diversity = 1.1181e+00 k_t = [1, 1, 1, 1]
 Generation 1960: Best = 9.155191e+06 FES = 196100 μ_F = 0.821 μ_{CR} = 0.890
 Diversity = 1.0744e+00 k_t = [2, 4, 2, 3]
 Generation 1970: Best = 9.114266e+06 FES = 197100 μ_F = 0.822 μ_{CR} = 0.893
 Diversity = 1.1710e+00 k_t = [1, 1, 1, 1]
 Generation 1980: Best = 9.065591e+06 FES = 198100 μ_F = 0.815 μ_{CR} = 0.899
 Diversity = 1.2090e+00 k_t = [4, 2, 2, 4]
 Generation 1990: Best = 9.008380e+06 FES = 199100 μ_F = 0.819 μ_{CR} = 0.899
 Diversity = 1.0311e+00 k_t = [1, 1, 1, 1]
 Generation 2000: Best = 8.965874e+06 FES = 200100 μ_F = 0.826 μ_{CR} = 0.902
 Diversity = 9.5407e-01 k_t = [3, 3, 4, 4]
 Generation 2010: Best = 8.926218e+06 FES = 201100 μ_F = 0.822 μ_{CR} = 0.903
 Diversity = 9.1325e-01 k_t = [1, 1, 1, 1]
 Generation 2020: Best = 8.895413e+06 FES = 202100 μ_F = 0.819 μ_{CR} = 0.902
 Diversity = 6.9597e-01 k_t = [4, 3, 4, 2]
 Generation 2030: Best = 8.864733e+06 FES = 203100 μ_F = 0.817 μ_{CR} = 0.899
 Diversity = 7.4647e-01 k_t = [1, 1, 1, 1]
 Generation 2040: Best = 8.839295e+06 FES = 204100 μ_F = 0.816 μ_{CR} = 0.896
 Diversity = 8.8872e-01 k_t = [4, 3, 4, 4]
 Generation 2050: Best = 8.811732e+06 FES = 205100 μ_F = 0.821 μ_{CR} = 0.899
 Diversity = 8.2871e-01 k_t = [1, 1, 1, 1]
 Generation 2060: Best = 8.777376e+06 FES = 206100 μ_F = 0.832 μ_{CR} = 0.892
 Diversity = 8.1812e-01 k_t = [3, 2, 2, 3]
 Generation 2070: Best = 8.745351e+06 FES = 207100 μ_F = 0.826 μ_{CR} = 0.889
 Diversity = 7.2639e-01 k_t = [1, 1, 1, 1]
 Generation 2080: Best = 8.709054e+06 FES = 208100 μ_F = 0.815 μ_{CR} = 0.894
 Diversity = 7.1229e-01 k_t = [3, 4, 3, 2]
 Generation 2090: Best = 8.677353e+06 FES = 209100 μ_F = 0.812 μ_{CR} = 0.888
 Diversity = 7.7136e-01 k_t = [1, 1, 1, 1]
 Generation 2100: Best = 8.645987e+06 FES = 210100 μ_F = 0.814 μ_{CR} = 0.891
 Diversity = 5.9853e-01 k_t = [3, 4, 2, 4]
 Generation 2110: Best = 8.614426e+06 FES = 211100 μ_F = 0.817 μ_{CR} = 0.892
 Diversity = 8.0023e-01 k_t = [1, 1, 1, 1]

Generation 2120: Best = 8.579130e+06 FES = 212100 μ_F = 0.814 μ_{CR} = 0.898
 Diversity = 7.3747e-01 k_t = [3, 4, 4, 2]
 Generation 2130: Best = 8.547812e+06 FES = 213100 μ_F = 0.812 μ_{CR} = 0.905
 Diversity = 7.4544e-01 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 8.507831e+06 FES = 214100 μ_F = 0.820 μ_{CR} = 0.899
 Diversity = 8.6681e-01 k_t = [2, 2, 4, 4]
 Generation 2150: Best = 8.454240e+06 FES = 215100 μ_F = 0.825 μ_{CR} = 0.898
 Diversity = 6.8351e-01 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 8.391584e+06 FES = 216100 μ_F = 0.826 μ_{CR} = 0.897
 Diversity = 7.1817e-01 k_t = [3, 4, 2, 2]
 Generation 2170: Best = 8.346648e+06 FES = 217100 μ_F = 0.818 μ_{CR} = 0.898
 Diversity = 6.9038e-01 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 8.298548e+06 FES = 218100 μ_F = 0.816 μ_{CR} = 0.894
 Diversity = 7.8596e-01 k_t = [4, 4, 4, 2]
 Generation 2190: Best = 8.256936e+06 FES = 219100 μ_F = 0.819 μ_{CR} = 0.896
 Diversity = 1.0113e+00 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 8.203326e+06 FES = 220100 μ_F = 0.814 μ_{CR} = 0.902
 Diversity = 1.0800e+00 k_t = [2, 3, 2, 4]
 Generation 2210: Best = 8.161179e+06 FES = 221100 μ_F = 0.813 μ_{CR} = 0.901
 Diversity = 8.9699e-01 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 8.112984e+06 FES = 222100 μ_F = 0.811 μ_{CR} = 0.893
 Diversity = 7.2921e-01 k_t = [4, 3, 3, 3]
 Generation 2230: Best = 8.087098e+06 FES = 223100 μ_F = 0.814 μ_{CR} = 0.897
 Diversity = 6.2240e-01 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 8.051024e+06 FES = 224100 μ_F = 0.806 μ_{CR} = 0.897
 Diversity = 5.6061e-01 k_t = [4, 2, 2, 3]
 Generation 2250: Best = 8.026515e+06 FES = 225100 μ_F = 0.810 μ_{CR} = 0.886
 Diversity = 4.1787e-01 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 7.993040e+06 FES = 226100 μ_F = 0.818 μ_{CR} = 0.880
 Diversity = 4.6725e-01 k_t = [3, 4, 3, 3]
 Generation 2270: Best = 7.958282e+06 FES = 227100 μ_F = 0.825 μ_{CR} = 0.877
 Diversity = 4.4822e-01 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 7.904159e+06 FES = 228100 μ_F = 0.826 μ_{CR} = 0.876
 Diversity = 6.4804e-01 k_t = [3, 4, 2, 4]
 Generation 2290: Best = 7.863143e+06 FES = 229100 μ_F = 0.826 μ_{CR} = 0.879
 Diversity = 8.8328e-01 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 7.812271e+06 FES = 230100 μ_F = 0.824 μ_{CR} = 0.879
 Diversity = 1.2349e+00 k_t = [2, 3, 3, 3]
 Generation 2310: Best = 7.764143e+06 FES = 231100 μ_F = 0.817 μ_{CR} = 0.867
 Diversity = 1.4528e+00 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 7.726940e+06 FES = 232100 μ_F = 0.820 μ_{CR} = 0.874
 Diversity = 1.0248e+00 k_t = [2, 3, 2, 2]
 Generation 2330: Best = 7.681803e+06 FES = 233100 μ_F = 0.812 μ_{CR} = 0.876
 Diversity = 1.3462e+00 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 7.639681e+06 FES = 234100 μ_F = 0.799 μ_{CR} = 0.882
 Diversity = 1.2749e+00 k_t = [3, 2, 3, 2]
 Generation 2350: Best = 7.590960e+06 FES = 235100 μ_F = 0.797 μ_{CR} = 0.887
 Diversity = 1.3281e+00 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 7.528657e+06 FES = 236100 μ_F = 0.803 μ_{CR} = 0.886
 Diversity = 1.1356e+00 k_t = [4, 3, 2, 2]
 Generation 2370: Best = 7.491604e+06 FES = 237100 μ_F = 0.817 μ_{CR} = 0.889
 Diversity = 1.0326e+00 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 7.450114e+06 FES = 238100 μ_F = 0.812 μ_{CR} = 0.892
 Diversity = 9.2560e-01 k_t = [3, 4, 3, 2]

Generation 2390: Best = 7.422958e+06 FES = 239100 μ_F = 0.813 μ_{CR} = 0.894
 Diversity = 6.6115e-01 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 7.394051e+06 FES = 240100 μ_F = 0.810 μ_{CR} = 0.896
 Diversity = 4.8706e-01 k_t = [3, 4, 3, 3]
 Generation 2410: Best = 7.358867e+06 FES = 241100 μ_F = 0.816 μ_{CR} = 0.892
 Diversity = 5.8144e-01 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 7.333106e+06 FES = 242100 μ_F = 0.820 μ_{CR} = 0.895
 Diversity = 7.1965e-01 k_t = [3, 2, 4, 3]
 Generation 2430: Best = 7.305997e+06 FES = 243100 μ_F = 0.822 μ_{CR} = 0.897
 Diversity = 8.5289e-01 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 7.274871e+06 FES = 244100 μ_F = 0.821 μ_{CR} = 0.892
 Diversity = 8.6694e-01 k_t = [4, 3, 4, 4]
 Generation 2450: Best = 7.251144e+06 FES = 245100 μ_F = 0.821 μ_{CR} = 0.892
 Diversity = 1.0707e+00 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 7.204648e+06 FES = 246100 μ_F = 0.821 μ_{CR} = 0.890
 Diversity = 8.8296e-01 k_t = [4, 3, 4, 4]
 Generation 2470: Best = 7.183250e+06 FES = 247100 μ_F = 0.820 μ_{CR} = 0.890
 Diversity = 6.9232e-01 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 7.166505e+06 FES = 248100 μ_F = 0.824 μ_{CR} = 0.885
 Diversity = 6.1257e-01 k_t = [4, 3, 4, 4]
 Generation 2490: Best = 7.150044e+06 FES = 249100 μ_F = 0.823 μ_{CR} = 0.877
 Diversity = 4.5239e-01 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 7.134049e+06 FES = 250100 μ_F = 0.818 μ_{CR} = 0.874
 Diversity = 3.5117e-01 k_t = [2, 2, 2, 2]
 Generation 2510: Best = 7.114095e+06 FES = 251100 μ_F = 0.817 μ_{CR} = 0.878
 Diversity = 5.5535e-01 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 7.087752e+06 FES = 252100 μ_F = 0.815 μ_{CR} = 0.878
 Diversity = 6.3237e-01 k_t = [4, 2, 3, 4]
 Generation 2530: Best = 7.070467e+06 FES = 253100 μ_F = 0.817 μ_{CR} = 0.879
 Diversity = 3.9348e-01 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 7.055753e+06 FES = 254100 μ_F = 0.818 μ_{CR} = 0.878
 Diversity = 3.7853e-01 k_t = [3, 2, 3, 2]
 Generation 2550: Best = 7.041711e+06 FES = 255100 μ_F = 0.811 μ_{CR} = 0.872
 Diversity = 3.0145e-01 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 7.031137e+06 FES = 256100 μ_F = 0.811 μ_{CR} = 0.876
 Diversity = 2.5165e-01 k_t = [4, 4, 4, 4]
 Generation 2570: Best = 7.021307e+06 FES = 257100 μ_F = 0.818 μ_{CR} = 0.872
 Diversity = 2.3809e-01 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 7.003746e+06 FES = 258100 μ_F = 0.818 μ_{CR} = 0.871
 Diversity = 3.3932e-01 k_t = [3, 4, 4, 4]
 Generation 2590: Best = 6.984629e+06 FES = 259100 μ_F = 0.816 μ_{CR} = 0.874
 Diversity = 3.7366e-01 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 6.952384e+06 FES = 260100 μ_F = 0.813 μ_{CR} = 0.869
 Diversity = 4.3621e-01 k_t = [3, 4, 3, 2]
 Generation 2610: Best = 6.927756e+06 FES = 261100 μ_F = 0.810 μ_{CR} = 0.874
 Diversity = 4.9343e-01 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 6.902725e+06 FES = 262100 μ_F = 0.812 μ_{CR} = 0.879
 Diversity = 5.6293e-01 k_t = [2, 2, 3, 2]
 Generation 2630: Best = 6.867962e+06 FES = 263100 μ_F = 0.807 μ_{CR} = 0.879
 Diversity = 7.1168e-01 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 6.844104e+06 FES = 264100 μ_F = 0.799 μ_{CR} = 0.878
 Diversity = 5.9478e-01 k_t = [4, 4, 3, 4]
 Generation 2650: Best = 6.820110e+06 FES = 265100 μ_F = 0.808 μ_{CR} = 0.876
 Diversity = 6.8483e-01 k_t = [1, 1, 1, 1]

Generation 2660: Best = 6.792582e+06 FES = 266100 μ_F = 0.816 μ_{CR} = 0.871
 Diversity = 8.0426e-01 k_t = [3, 4, 2, 3]
 Generation 2670: Best = 6.775590e+06 FES = 267100 μ_F = 0.808 μ_{CR} = 0.868
 Diversity = 6.5259e-01 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 6.759104e+06 FES = 268100 μ_F = 0.817 μ_{CR} = 0.864
 Diversity = 6.2632e-01 k_t = [3, 3, 2, 4]
 Generation 2690: Best = 6.736002e+06 FES = 269100 μ_F = 0.825 μ_{CR} = 0.864
 Diversity = 1.1880e+00 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 6.711527e+06 FES = 270100 μ_F = 0.818 μ_{CR} = 0.869
 Diversity = 1.2098e+00 k_t = [3, 3, 3, 2]
 Generation 2710: Best = 6.687741e+06 FES = 271100 μ_F = 0.812 μ_{CR} = 0.872
 Diversity = 1.1109e+00 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 6.664556e+06 FES = 272100 μ_F = 0.810 μ_{CR} = 0.869
 Diversity = 1.3184e+00 k_t = [3, 3, 2, 4]
 Generation 2730: Best = 6.646159e+06 FES = 273100 μ_F = 0.805 μ_{CR} = 0.873
 Diversity = 1.4927e+00 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 6.614881e+06 FES = 274100 μ_F = 0.809 μ_{CR} = 0.880
 Diversity = 1.3692e+00 k_t = [4, 2, 2, 3]
 Generation 2750: Best = 6.593232e+06 FES = 275100 μ_F = 0.803 μ_{CR} = 0.883
 Diversity = 1.2265e+00 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 6.569593e+06 FES = 276100 μ_F = 0.806 μ_{CR} = 0.879
 Diversity = 1.1028e+00 k_t = [2, 4, 2, 3]
 Generation 2770: Best = 6.540743e+06 FES = 277100 μ_F = 0.810 μ_{CR} = 0.882
 Diversity = 1.1459e+00 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 6.510702e+06 FES = 278100 μ_F = 0.816 μ_{CR} = 0.887
 Diversity = 1.0842e+00 k_t = [2, 2, 3, 4]
 Generation 2790: Best = 6.491271e+06 FES = 279100 μ_F = 0.816 μ_{CR} = 0.883
 Diversity = 1.0347e+00 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 6.464858e+06 FES = 280100 μ_F = 0.820 μ_{CR} = 0.888
 Diversity = 1.0331e+00 k_t = [2, 3, 4, 4]
 Generation 2810: Best = 6.446736e+06 FES = 281100 μ_F = 0.814 μ_{CR} = 0.885
 Diversity = 8.5017e-01 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 6.418954e+06 FES = 282100 μ_F = 0.807 μ_{CR} = 0.884
 Diversity = 4.1035e-01 k_t = [3, 3, 4, 3]
 Generation 2830: Best = 6.403000e+06 FES = 283100 μ_F = 0.805 μ_{CR} = 0.881
 Diversity = 4.8819e-01 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 6.382859e+06 FES = 284100 μ_F = 0.799 μ_{CR} = 0.885
 Diversity = 6.1806e-01 k_t = [3, 4, 3, 2]
 Generation 2850: Best = 6.362487e+06 FES = 285100 μ_F = 0.813 μ_{CR} = 0.885
 Diversity = 4.5668e-01 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 6.339254e+06 FES = 286100 μ_F = 0.808 μ_{CR} = 0.884
 Diversity = 5.9756e-01 k_t = [3, 4, 3, 2]
 Generation 2870: Best = 6.319162e+06 FES = 287100 μ_F = 0.813 μ_{CR} = 0.887
 Diversity = 5.5046e-01 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 6.300270e+06 FES = 288100 μ_F = 0.806 μ_{CR} = 0.884
 Diversity = 5.4079e-01 k_t = [2, 2, 4, 2]
 Generation 2890: Best = 6.287230e+06 FES = 289100 μ_F = 0.800 μ_{CR} = 0.876
 Diversity = 5.6158e-01 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 6.272824e+06 FES = 290100 μ_F = 0.805 μ_{CR} = 0.876
 Diversity = 4.1858e-01 k_t = [4, 2, 2, 3]
 Generation 2910: Best = 6.262432e+06 FES = 291100 μ_F = 0.794 μ_{CR} = 0.872
 Diversity = 3.9602e-01 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 6.249926e+06 FES = 292100 μ_F = 0.794 μ_{CR} = 0.880
 Diversity = 3.6593e-01 k_t = [4, 2, 3, 4]

Generation 2930: Best = 6.238485e+06	FES = 293100	$\mu_F = 0.795$	$\mu_{CR} = 0.876$
Diversity = 4.4656e-01	k_t = [1, 1, 1, 1]		
Generation 2940: Best = 6.223679e+06	FES = 294100	$\mu_F = 0.803$	$\mu_{CR} = 0.875$
Diversity = 3.5355e-01	k_t = [2, 2, 4, 4]		
Generation 2950: Best = 6.209134e+06	FES = 295100	$\mu_F = 0.803$	$\mu_{CR} = 0.869$
Diversity = 3.4985e-01	k_t = [1, 1, 1, 1]		
Generation 2960: Best = 6.186832e+06	FES = 296100	$\mu_F = 0.797$	$\mu_{CR} = 0.868$
Diversity = 4.7618e-01	k_t = [4, 4, 3, 2]		
Generation 2970: Best = 6.170470e+06	FES = 297100	$\mu_F = 0.794$	$\mu_{CR} = 0.871$
Diversity = 3.8569e-01	k_t = [1, 1, 1, 1]		
Generation 2980: Best = 6.150386e+06	FES = 298100	$\mu_F = 0.806$	$\mu_{CR} = 0.875$
Diversity = 3.9536e-01	k_t = [3, 4, 4, 4]		
Generation 2990: Best = 6.136570e+06	FES = 299100	$\mu_F = 0.806$	$\mu_{CR} = 0.880$
Diversity = 4.4678e-01	k_t = [1, 1, 1, 1]		
Generation 3000: Best = 6.119018e+06	FES = 300100	$\mu_F = 0.801$	$\mu_{CR} = 0.885$
Diversity = 3.6309e-01	k_t = [2, 2, 4, 2]		
Generation 3010: Best = 6.107960e+06	FES = 301100	$\mu_F = 0.794$	$\mu_{CR} = 0.881$
Diversity = 3.0330e-01	k_t = [1, 1, 1, 1]		
Generation 3020: Best = 6.093909e+06	FES = 302100	$\mu_F = 0.806$	$\mu_{CR} = 0.882$
Diversity = 3.8122e-01	k_t = [4, 4, 3, 2]		
Generation 3030: Best = 6.083915e+06	FES = 303100	$\mu_F = 0.810$	$\mu_{CR} = 0.879$
Diversity = 2.1319e-01	k_t = [1, 1, 1, 1]		
Generation 3040: Best = 6.073204e+06	FES = 304100	$\mu_F = 0.808$	$\mu_{CR} = 0.882$
Diversity = 1.9291e-01	k_t = [2, 3, 3, 3]		
Generation 3050: Best = 6.058012e+06	FES = 305100	$\mu_F = 0.809$	$\mu_{CR} = 0.889$
Diversity = 2.4348e-01	k_t = [1, 1, 1, 1]		
Generation 3060: Best = 6.044666e+06	FES = 306100	$\mu_F = 0.810$	$\mu_{CR} = 0.885$
Diversity = 2.2066e-01	k_t = [3, 3, 2, 2]		
Generation 3070: Best = 6.032327e+06	FES = 307100	$\mu_F = 0.803$	$\mu_{CR} = 0.874$
Diversity = 1.8342e-01	k_t = [1, 1, 1, 1]		
Generation 3080: Best = 6.020590e+06	FES = 308100	$\mu_F = 0.803$	$\mu_{CR} = 0.872$
Diversity = 2.2765e-01	k_t = [3, 3, 2, 3]		
Generation 3090: Best = 6.013944e+06	FES = 309100	$\mu_F = 0.799$	$\mu_{CR} = 0.870$
Diversity = 2.3007e-01	k_t = [1, 1, 1, 1]		
Generation 3100: Best = 6.004437e+06	FES = 310100	$\mu_F = 0.791$	$\mu_{CR} = 0.866$
Diversity = 1.5618e-01	k_t = [2, 3, 3, 4]		
Generation 3110: Best = 5.994360e+06	FES = 311100	$\mu_F = 0.792$	$\mu_{CR} = 0.879$
Diversity = 1.9232e-01	k_t = [1, 1, 1, 1]		
Generation 3120: Best = 5.984428e+06	FES = 312100	$\mu_F = 0.793$	$\mu_{CR} = 0.879$
Diversity = 2.0673e-01	k_t = [3, 3, 2, 3]		
Generation 3130: Best = 5.975236e+06	FES = 313100	$\mu_F = 0.797$	$\mu_{CR} = 0.880$
Diversity = 2.3199e-01	k_t = [1, 1, 1, 1]		
Generation 3140: Best = 5.962195e+06	FES = 314100	$\mu_F = 0.799$	$\mu_{CR} = 0.879$
Diversity = 2.0093e-01	k_t = [4, 3, 4, 2]		
Generation 3150: Best = 5.950901e+06	FES = 315100	$\mu_F = 0.812$	$\mu_{CR} = 0.877$
Diversity = 2.0609e-01	k_t = [1, 1, 1, 1]		
Generation 3160: Best = 5.939948e+06	FES = 316100	$\mu_F = 0.806$	$\mu_{CR} = 0.872$
Diversity = 1.9070e-01	k_t = [4, 4, 2, 2]		
Generation 3170: Best = 5.929480e+06	FES = 317100	$\mu_F = 0.806$	$\mu_{CR} = 0.859$
Diversity = 2.1673e-01	k_t = [1, 1, 1, 1]		
Generation 3180: Best = 5.921971e+06	FES = 318100	$\mu_F = 0.798$	$\mu_{CR} = 0.858$
Diversity = 1.6180e-01	k_t = [3, 4, 2, 4]		
Generation 3190: Best = 5.914545e+06	FES = 319100	$\mu_F = 0.799$	$\mu_{CR} = 0.865$
Diversity = 1.8979e-01	k_t = [1, 1, 1, 1]		

Generation 3200: Best = 5.905408e+06 FES = 320100 μ_F = 0.800 μ_{CR} = 0.857
 Diversity = 1.8278e-01 k_t = [2, 2, 2, 4]
 Generation 3210: Best = 5.896478e+06 FES = 321100 μ_F = 0.786 μ_{CR} = 0.861
 Diversity = 2.0534e-01 k_t = [1, 1, 1, 1]
 Generation 3220: Best = 5.878595e+06 FES = 322100 μ_F = 0.793 μ_{CR} = 0.866
 Diversity = 1.7802e-01 k_t = [4, 4, 3, 2]
 Generation 3230: Best = 5.865838e+06 FES = 323100 μ_F = 0.799 μ_{CR} = 0.859
 Diversity = 2.1309e-01 k_t = [1, 1, 1, 1]
 Generation 3240: Best = 5.849167e+06 FES = 324100 μ_F = 0.792 μ_{CR} = 0.863
 Diversity = 2.2907e-01 k_t = [4, 4, 3, 4]
 Generation 3250: Best = 5.837300e+06 FES = 325100 μ_F = 0.799 μ_{CR} = 0.860
 Diversity = 2.3413e-01 k_t = [1, 1, 1, 1]
 Generation 3260: Best = 5.825933e+06 FES = 326100 μ_F = 0.782 μ_{CR} = 0.859
 Diversity = 1.9534e-01 k_t = [2, 2, 3, 4]
 Generation 3270: Best = 5.815193e+06 FES = 327100 μ_F = 0.783 μ_{CR} = 0.863
 Diversity = 2.2437e-01 k_t = [1, 1, 1, 1]
 Generation 3280: Best = 5.803345e+06 FES = 328100 μ_F = 0.781 μ_{CR} = 0.859
 Diversity = 1.8008e-01 k_t = [3, 3, 2, 2]
 Generation 3290: Best = 5.791379e+06 FES = 329100 μ_F = 0.783 μ_{CR} = 0.868
 Diversity = 2.1914e-01 k_t = [1, 1, 1, 1]
 Generation 3300: Best = 5.775420e+06 FES = 330100 μ_F = 0.791 μ_{CR} = 0.869
 Diversity = 2.3553e-01 k_t = [4, 2, 3, 4]
 Generation 3310: Best = 5.759821e+06 FES = 331100 μ_F = 0.800 μ_{CR} = 0.865
 Diversity = 2.6205e-01 k_t = [1, 1, 1, 1]
 Generation 3320: Best = 5.745982e+06 FES = 332100 μ_F = 0.801 μ_{CR} = 0.867
 Diversity = 2.0856e-01 k_t = [4, 4, 3, 3]
 Generation 3330: Best = 5.736540e+06 FES = 333100 μ_F = 0.807 μ_{CR} = 0.874
 Diversity = 2.3686e-01 k_t = [1, 1, 1, 1]
 Generation 3340: Best = 5.724241e+06 FES = 334100 μ_F = 0.801 μ_{CR} = 0.879
 Diversity = 2.1527e-01 k_t = [4, 2, 3, 3]
 Generation 3350: Best = 5.716155e+06 FES = 335100 μ_F = 0.809 μ_{CR} = 0.873
 Diversity = 2.7735e-01 k_t = [1, 1, 1, 1]
 Generation 3360: Best = 5.705258e+06 FES = 336100 μ_F = 0.811 μ_{CR} = 0.877
 Diversity = 3.3747e-01 k_t = [2, 2, 2, 2]
 Generation 3370: Best = 5.692989e+06 FES = 337100 μ_F = 0.818 μ_{CR} = 0.878
 Diversity = 3.2560e-01 k_t = [1, 1, 1, 1]
 Generation 3380: Best = 5.675099e+06 FES = 338100 μ_F = 0.812 μ_{CR} = 0.871
 Diversity = 3.0460e-01 k_t = [4, 4, 3, 4]
 Generation 3390: Best = 5.661179e+06 FES = 339100 μ_F = 0.815 μ_{CR} = 0.865
 Diversity = 3.5491e-01 k_t = [1, 1, 1, 1]
 Generation 3400: Best = 5.643293e+06 FES = 340100 μ_F = 0.812 μ_{CR} = 0.864
 Diversity = 3.0677e-01 k_t = [4, 3, 2, 4]
 Generation 3410: Best = 5.626422e+06 FES = 341100 μ_F = 0.812 μ_{CR} = 0.874
 Diversity = 3.5418e-01 k_t = [1, 1, 1, 1]
 Generation 3420: Best = 5.613496e+06 FES = 342100 μ_F = 0.795 μ_{CR} = 0.879
 Diversity = 3.0704e-01 k_t = [3, 2, 3, 4]
 Generation 3430: Best = 5.599995e+06 FES = 343100 μ_F = 0.796 μ_{CR} = 0.879
 Diversity = 3.6266e-01 k_t = [1, 1, 1, 1]
 Generation 3440: Best = 5.587529e+06 FES = 344100 μ_F = 0.793 μ_{CR} = 0.877
 Diversity = 3.2462e-01 k_t = [3, 4, 3, 2]
 Generation 3450: Best = 5.576406e+06 FES = 345100 μ_F = 0.798 μ_{CR} = 0.877
 Diversity = 3.6090e-01 k_t = [1, 1, 1, 1]
 Generation 3460: Best = 5.565021e+06 FES = 346100 μ_F = 0.794 μ_{CR} = 0.877
 Diversity = 3.0377e-01 k_t = [4, 2, 2, 2]

Generation 3470: Best = 5.552384e+06 FES = 347100 μ_F = 0.793 μ_{CR} = 0.873
 Diversity = 4.4377e-01 k_t = [1, 1, 1, 1]
 Generation 3480: Best = 5.536294e+06 FES = 348100 μ_F = 0.802 μ_{CR} = 0.867
 Diversity = 7.4199e-01 k_t = [4, 4, 4, 3]
 Generation 3490: Best = 5.520824e+06 FES = 349100 μ_F = 0.811 μ_{CR} = 0.870
 Diversity = 9.3630e-01 k_t = [1, 1, 1, 1]
 Generation 3500: Best = 5.498036e+06 FES = 350100 μ_F = 0.810 μ_{CR} = 0.873
 Diversity = 6.3253e-01 k_t = [2, 2, 2, 3]
 Generation 3510: Best = 5.474736e+06 FES = 351100 μ_F = 0.804 μ_{CR} = 0.878
 Diversity = 6.6797e-01 k_t = [1, 1, 1, 1]
 Generation 3520: Best = 5.460620e+06 FES = 352100 μ_F = 0.802 μ_{CR} = 0.878
 Diversity = 5.6759e-01 k_t = [2, 2, 2, 4]
 Generation 3530: Best = 5.440366e+06 FES = 353100 μ_F = 0.802 μ_{CR} = 0.885
 Diversity = 4.0298e-01 k_t = [1, 1, 1, 1]
 Generation 3540: Best = 5.426664e+06 FES = 354100 μ_F = 0.800 μ_{CR} = 0.881
 Diversity = 4.4327e-01 k_t = [2, 3, 2, 2]
 Generation 3550: Best = 5.416166e+06 FES = 355100 μ_F = 0.800 μ_{CR} = 0.887
 Diversity = 4.5171e-01 k_t = [1, 1, 1, 1]
 Generation 3560: Best = 5.404613e+06 FES = 356100 μ_F = 0.787 μ_{CR} = 0.883
 Diversity = 3.7110e-01 k_t = [4, 4, 4, 2]
 Generation 3570: Best = 5.394810e+06 FES = 357100 μ_F = 0.797 μ_{CR} = 0.881
 Diversity = 3.4989e-01 k_t = [1, 1, 1, 1]
 Generation 3580: Best = 5.383615e+06 FES = 358100 μ_F = 0.804 μ_{CR} = 0.879
 Diversity = 2.7789e-01 k_t = [2, 3, 4, 3]
 Generation 3590: Best = 5.369641e+06 FES = 359100 μ_F = 0.808 μ_{CR} = 0.883
 Diversity = 3.3891e-01 k_t = [1, 1, 1, 1]
 Generation 3600: Best = 5.355365e+06 FES = 360100 μ_F = 0.804 μ_{CR} = 0.883
 Diversity = 2.8260e-01 k_t = [2, 2, 3, 2]
 Generation 3610: Best = 5.342609e+06 FES = 361100 μ_F = 0.803 μ_{CR} = 0.883
 Diversity = 3.2104e-01 k_t = [1, 1, 1, 1]
 Generation 3620: Best = 5.329832e+06 FES = 362100 μ_F = 0.805 μ_{CR} = 0.879
 Diversity = 2.2381e-01 k_t = [3, 3, 4, 4]
 Generation 3630: Best = 5.318728e+06 FES = 363100 μ_F = 0.806 μ_{CR} = 0.876
 Diversity = 2.3140e-01 k_t = [1, 1, 1, 1]
 Generation 3640: Best = 5.308294e+06 FES = 364100 μ_F = 0.807 μ_{CR} = 0.875
 Diversity = 2.5851e-01 k_t = [2, 3, 3, 4]
 Generation 3650: Best = 5.297241e+06 FES = 365100 μ_F = 0.813 μ_{CR} = 0.875
 Diversity = 3.0729e-01 k_t = [1, 1, 1, 1]
 Generation 3660: Best = 5.280491e+06 FES = 366100 μ_F = 0.813 μ_{CR} = 0.875
 Diversity = 3.0655e-01 k_t = [2, 3, 2, 4]
 Generation 3670: Best = 5.263061e+06 FES = 367100 μ_F = 0.814 μ_{CR} = 0.876
 Diversity = 3.7687e-01 k_t = [1, 1, 1, 1]
 Generation 3680: Best = 5.248889e+06 FES = 368100 μ_F = 0.809 μ_{CR} = 0.874
 Diversity = 3.9697e-01 k_t = [2, 3, 3, 2]
 Generation 3690: Best = 5.236374e+06 FES = 369100 μ_F = 0.811 μ_{CR} = 0.876
 Diversity = 3.9539e-01 k_t = [1, 1, 1, 1]
 Generation 3700: Best = 5.226519e+06 FES = 370100 μ_F = 0.805 μ_{CR} = 0.877
 Diversity = 2.9720e-01 k_t = [2, 2, 3, 2]
 Generation 3710: Best = 5.212748e+06 FES = 371100 μ_F = 0.810 μ_{CR} = 0.875
 Diversity = 3.6242e-01 k_t = [1, 1, 1, 1]
 Generation 3720: Best = 5.202648e+06 FES = 372100 μ_F = 0.803 μ_{CR} = 0.880
 Diversity = 4.5574e-01 k_t = [2, 4, 2, 4]
 Generation 3730: Best = 5.185623e+06 FES = 373100 μ_F = 0.804 μ_{CR} = 0.884
 Diversity = 6.7427e-01 k_t = [1, 1, 1, 1]

Generation 3740: Best = 5.161100e+06 FES = 374100 μ_F = 0.809 μ_{CR} = 0.880
 Diversity = 6.1628e-01 k_t = [4, 4, 2, 3]
 Generation 3750: Best = 5.144156e+06 FES = 375100 μ_F = 0.814 μ_{CR} = 0.877
 Diversity = 4.8407e-01 k_t = [1, 1, 1, 1]
 Generation 3760: Best = 5.124259e+06 FES = 376100 μ_F = 0.808 μ_{CR} = 0.877
 Diversity = 5.9041e-01 k_t = [2, 3, 3, 4]
 Generation 3770: Best = 5.103702e+06 FES = 377100 μ_F = 0.811 μ_{CR} = 0.872
 Diversity = 8.3776e-01 k_t = [1, 1, 1, 1]
 Generation 3780: Best = 5.087061e+06 FES = 378100 μ_F = 0.809 μ_{CR} = 0.878
 Diversity = 6.7023e-01 k_t = [3, 3, 2, 3]
 Generation 3790: Best = 5.067431e+06 FES = 379100 μ_F = 0.809 μ_{CR} = 0.881
 Diversity = 6.1776e-01 k_t = [1, 1, 1, 1]
 Generation 3800: Best = 5.043547e+06 FES = 380100 μ_F = 0.812 μ_{CR} = 0.884
 Diversity = 8.3459e-01 k_t = [2, 3, 4, 3]
 Generation 3810: Best = 5.025386e+06 FES = 381100 μ_F = 0.815 μ_{CR} = 0.888
 Diversity = 7.8795e-01 k_t = [1, 1, 1, 1]
 Generation 3820: Best = 5.011684e+06 FES = 382100 μ_F = 0.813 μ_{CR} = 0.886
 Diversity = 5.1193e-01 k_t = [2, 4, 4, 3]
 Generation 3830: Best = 4.993851e+06 FES = 383100 μ_F = 0.812 μ_{CR} = 0.884
 Diversity = 4.1835e-01 k_t = [1, 1, 1, 1]
 Generation 3840: Best = 4.976478e+06 FES = 384100 μ_F = 0.809 μ_{CR} = 0.880
 Diversity = 3.1463e-01 k_t = [3, 2, 3, 3]
 Generation 3850: Best = 4.962345e+06 FES = 385100 μ_F = 0.808 μ_{CR} = 0.875
 Diversity = 3.1295e-01 k_t = [1, 1, 1, 1]
 Generation 3860: Best = 4.949534e+06 FES = 386100 μ_F = 0.806 μ_{CR} = 0.879
 Diversity = 2.5915e-01 k_t = [3, 2, 4, 4]
 Generation 3870: Best = 4.939400e+06 FES = 387100 μ_F = 0.806 μ_{CR} = 0.883
 Diversity = 2.4161e-01 k_t = [1, 1, 1, 1]
 Generation 3880: Best = 4.925195e+06 FES = 388100 μ_F = 0.801 μ_{CR} = 0.880
 Diversity = 2.4911e-01 k_t = [3, 3, 2, 4]
 Generation 3890: Best = 4.912926e+06 FES = 389100 μ_F = 0.806 μ_{CR} = 0.871
 Diversity = 2.8791e-01 k_t = [1, 1, 1, 1]
 Generation 3900: Best = 4.898175e+06 FES = 390100 μ_F = 0.795 μ_{CR} = 0.871
 Diversity = 2.5945e-01 k_t = [4, 4, 4, 2]
 Generation 3910: Best = 4.891869e+06 FES = 391100 μ_F = 0.795 μ_{CR} = 0.868
 Diversity = 2.0346e-01 k_t = [1, 1, 1, 1]
 Generation 3920: Best = 4.883110e+06 FES = 392100 μ_F = 0.783 μ_{CR} = 0.861
 Diversity = 1.9107e-01 k_t = [4, 3, 3, 2]
 Generation 3930: Best = 4.875396e+06 FES = 393100 μ_F = 0.789 μ_{CR} = 0.855
 Diversity = 2.0769e-01 k_t = [1, 1, 1, 1]
 Generation 3940: Best = 4.865705e+06 FES = 394100 μ_F = 0.794 μ_{CR} = 0.858
 Diversity = 1.5371e-01 k_t = [4, 4, 4, 3]
 Generation 3950: Best = 4.859272e+06 FES = 395100 μ_F = 0.799 μ_{CR} = 0.854
 Diversity = 1.9994e-01 k_t = [1, 1, 1, 1]
 Generation 3960: Best = 4.851025e+06 FES = 396100 μ_F = 0.801 μ_{CR} = 0.857
 Diversity = 1.9821e-01 k_t = [3, 2, 2, 4]
 Generation 3970: Best = 4.844704e+06 FES = 397100 μ_F = 0.805 μ_{CR} = 0.867
 Diversity = 2.6593e-01 k_t = [1, 1, 1, 1]
 Generation 3980: Best = 4.837901e+06 FES = 398100 μ_F = 0.813 μ_{CR} = 0.859
 Diversity = 1.8459e-01 k_t = [4, 3, 4, 2]
 Generation 3990: Best = 4.831090e+06 FES = 399100 μ_F = 0.812 μ_{CR} = 0.867
 Diversity = 2.1599e-01 k_t = [1, 1, 1, 1]
 Generation 4000: Best = 4.823638e+06 FES = 400100 μ_F = 0.807 μ_{CR} = 0.863
 Diversity = 1.9772e-01 k_t = [2, 3, 3, 2]

Generation 4010: Best = 4.816444e+06 FES = 401100 μ_F = 0.802 μ_{CR} = 0.861
 Diversity = 1.9485e-01 k_t = [1, 1, 1, 1]
 Generation 4020: Best = 4.804919e+06 FES = 402100 μ_F = 0.796 μ_{CR} = 0.864
 Diversity = 1.6717e-01 k_t = [3, 3, 4, 2]
 Generation 4030: Best = 4.792605e+06 FES = 403100 μ_F = 0.796 μ_{CR} = 0.862
 Diversity = 1.9393e-01 k_t = [1, 1, 1, 1]
 Generation 4040: Best = 4.780707e+06 FES = 404100 μ_F = 0.790 μ_{CR} = 0.859
 Diversity = 2.1948e-01 k_t = [4, 2, 2, 2]
 Generation 4050: Best = 4.767753e+06 FES = 405100 μ_F = 0.801 μ_{CR} = 0.853
 Diversity = 3.4692e-01 k_t = [1, 1, 1, 1]
 Generation 4060: Best = 4.759695e+06 FES = 406100 μ_F = 0.797 μ_{CR} = 0.870
 Diversity = 1.6938e-01 k_t = [2, 4, 3, 3]
 Generation 4070: Best = 4.749336e+06 FES = 407100 μ_F = 0.795 μ_{CR} = 0.866
 Diversity = 2.1373e-01 k_t = [1, 1, 1, 1]
 Generation 4080: Best = 4.740414e+06 FES = 408100 μ_F = 0.802 μ_{CR} = 0.870
 Diversity = 2.1563e-01 k_t = [3, 4, 3, 3]
 Generation 4090: Best = 4.731996e+06 FES = 409100 μ_F = 0.802 μ_{CR} = 0.869
 Diversity = 1.9359e-01 k_t = [1, 1, 1, 1]
 Generation 4100: Best = 4.724656e+06 FES = 410100 μ_F = 0.792 μ_{CR} = 0.870
 Diversity = 1.4726e-01 k_t = [3, 3, 3, 3]
 Generation 4110: Best = 4.715716e+06 FES = 411100 μ_F = 0.792 μ_{CR} = 0.867
 Diversity = 2.0499e-01 k_t = [1, 1, 1, 1]
 Generation 4120: Best = 4.706264e+06 FES = 412100 μ_F = 0.792 μ_{CR} = 0.866
 Diversity = 2.3109e-01 k_t = [4, 2, 2, 2]
 Generation 4130: Best = 4.690785e+06 FES = 413100 μ_F = 0.790 μ_{CR} = 0.867
 Diversity = 2.6263e-01 k_t = [1, 1, 1, 1]
 Generation 4140: Best = 4.672908e+06 FES = 414100 μ_F = 0.801 μ_{CR} = 0.874
 Diversity = 3.7686e-01 k_t = [4, 2, 4, 3]
 Generation 4150: Best = 4.660573e+06 FES = 415100 μ_F = 0.807 μ_{CR} = 0.879
 Diversity = 3.9871e-01 k_t = [1, 1, 1, 1]
 Generation 4160: Best = 4.649178e+06 FES = 416100 μ_F = 0.807 μ_{CR} = 0.885
 Diversity = 3.5587e-01 k_t = [4, 3, 4, 4]
 Generation 4170: Best = 4.637758e+06 FES = 417100 μ_F = 0.812 μ_{CR} = 0.879
 Diversity = 4.8290e-01 k_t = [1, 1, 1, 1]
 Generation 4180: Best = 4.618746e+06 FES = 418100 μ_F = 0.817 μ_{CR} = 0.877
 Diversity = 5.1962e-01 k_t = [2, 4, 3, 2]
 Generation 4190: Best = 4.600971e+06 FES = 419100 μ_F = 0.818 μ_{CR} = 0.889
 Diversity = 7.5717e-01 k_t = [1, 1, 1, 1]
 Generation 4200: Best = 4.585024e+06 FES = 420100 μ_F = 0.816 μ_{CR} = 0.894
 Diversity = 8.0186e-01 k_t = [4, 4, 3, 2]
 Generation 4210: Best = 4.568241e+06 FES = 421100 μ_F = 0.822 μ_{CR} = 0.892
 Diversity = 1.1911e+00 k_t = [1, 1, 1, 1]
 Generation 4220: Best = 4.542238e+06 FES = 422100 μ_F = 0.816 μ_{CR} = 0.893
 Diversity = 8.0283e-01 k_t = [4, 3, 3, 2]
 Generation 4230: Best = 4.529581e+06 FES = 423100 μ_F = 0.815 μ_{CR} = 0.892
 Diversity = 5.3518e-01 k_t = [1, 1, 1, 1]
 Generation 4240: Best = 4.513828e+06 FES = 424100 μ_F = 0.813 μ_{CR} = 0.891
 Diversity = 4.0564e-01 k_t = [3, 3, 3, 3]
 Generation 4250: Best = 4.498329e+06 FES = 425100 μ_F = 0.822 μ_{CR} = 0.887
 Diversity = 3.8118e-01 k_t = [1, 1, 1, 1]
 Generation 4260: Best = 4.486657e+06 FES = 426100 μ_F = 0.817 μ_{CR} = 0.887
 Diversity = 2.7018e-01 k_t = [4, 4, 4, 4]
 Generation 4270: Best = 4.477688e+06 FES = 427100 μ_F = 0.814 μ_{CR} = 0.884
 Diversity = 2.1293e-01 k_t = [1, 1, 1, 1]

Generation 4280: Best = 4.467603e+06 FES = 428100 μ_F = 0.818 μ_{CR} = 0.889
 Diversity = 2.3636e-01 k_t = [3, 4, 3, 4]
 Generation 4290: Best = 4.457157e+06 FES = 429100 μ_F = 0.819 μ_{CR} = 0.889
 Diversity = 2.3931e-01 k_t = [1, 1, 1, 1]
 Generation 4300: Best = 4.439857e+06 FES = 430100 μ_F = 0.820 μ_{CR} = 0.891
 Diversity = 2.2449e-01 k_t = [2, 4, 4, 2]
 Generation 4310: Best = 4.424240e+06 FES = 431100 μ_F = 0.823 μ_{CR} = 0.893
 Diversity = 4.3505e-01 k_t = [1, 1, 1, 1]
 Generation 4320: Best = 4.407731e+06 FES = 432100 μ_F = 0.825 μ_{CR} = 0.892
 Diversity = 4.7173e-01 k_t = [2, 2, 4, 3]
 Generation 4330: Best = 4.389918e+06 FES = 433100 μ_F = 0.829 μ_{CR} = 0.891
 Diversity = 4.2858e-01 k_t = [1, 1, 1, 1]
 Generation 4340: Best = 4.372373e+06 FES = 434100 μ_F = 0.811 μ_{CR} = 0.891
 Diversity = 2.9424e-01 k_t = [2, 3, 2, 2]
 Generation 4350: Best = 4.357143e+06 FES = 435100 μ_F = 0.809 μ_{CR} = 0.888
 Diversity = 2.8149e-01 k_t = [1, 1, 1, 1]
 Generation 4360: Best = 4.339329e+06 FES = 436100 μ_F = 0.806 μ_{CR} = 0.896
 Diversity = 2.4210e-01 k_t = [4, 2, 3, 2]
 Generation 4370: Best = 4.325734e+06 FES = 437100 μ_F = 0.808 μ_{CR} = 0.893
 Diversity = 2.4342e-01 k_t = [1, 1, 1, 1]
 Generation 4380: Best = 4.312205e+06 FES = 438100 μ_F = 0.805 μ_{CR} = 0.893
 Diversity = 2.3414e-01 k_t = [4, 4, 4, 3]
 Generation 4390: Best = 4.303531e+06 FES = 439100 μ_F = 0.809 μ_{CR} = 0.898
 Diversity = 1.9688e-01 k_t = [1, 1, 1, 1]
 Generation 4400: Best = 4.291956e+06 FES = 440100 μ_F = 0.814 μ_{CR} = 0.899
 Diversity = 3.0708e-01 k_t = [2, 3, 2, 3]
 Generation 4410: Best = 4.281094e+06 FES = 441100 μ_F = 0.808 μ_{CR} = 0.902
 Diversity = 2.5372e-01 k_t = [1, 1, 1, 1]
 Generation 4420: Best = 4.268557e+06 FES = 442100 μ_F = 0.804 μ_{CR} = 0.899
 Diversity = 2.0747e-01 k_t = [2, 4, 4, 3]
 Generation 4430: Best = 4.258154e+06 FES = 443100 μ_F = 0.803 μ_{CR} = 0.893
 Diversity = 2.9058e-01 k_t = [1, 1, 1, 1]
 Generation 4440: Best = 4.249935e+06 FES = 444100 μ_F = 0.799 μ_{CR} = 0.893
 Diversity = 2.5571e-01 k_t = [3, 3, 3, 4]
 Generation 4450: Best = 4.243147e+06 FES = 445100 μ_F = 0.798 μ_{CR} = 0.896
 Diversity = 3.1456e-01 k_t = [1, 1, 1, 1]
 Generation 4460: Best = 4.230526e+06 FES = 446100 μ_F = 0.811 μ_{CR} = 0.889
 Diversity = 3.2806e-01 k_t = [2, 2, 4, 3]
 Generation 4470: Best = 4.221010e+06 FES = 447100 μ_F = 0.810 μ_{CR} = 0.890
 Diversity = 2.4160e-01 k_t = [1, 1, 1, 1]
 Generation 4480: Best = 4.210802e+06 FES = 448100 μ_F = 0.818 μ_{CR} = 0.882
 Diversity = 2.7802e-01 k_t = [4, 3, 3, 3]
 Generation 4490: Best = 4.198087e+06 FES = 449100 μ_F = 0.822 μ_{CR} = 0.883
 Diversity = 3.1779e-01 k_t = [1, 1, 1, 1]
 Generation 4500: Best = 4.181191e+06 FES = 450100 μ_F = 0.806 μ_{CR} = 0.881
 Diversity = 3.3109e-01 k_t = [3, 2, 4, 3]
 Generation 4510: Best = 4.165428e+06 FES = 451100 μ_F = 0.794 μ_{CR} = 0.879
 Diversity = 2.6901e-01 k_t = [1, 1, 1, 1]
 Generation 4520: Best = 4.155486e+06 FES = 452100 μ_F = 0.784 μ_{CR} = 0.886
 Diversity = 1.6306e-01 k_t = [3, 2, 3, 3]
 Generation 4530: Best = 4.146260e+06 FES = 453100 μ_F = 0.792 μ_{CR} = 0.882
 Diversity = 1.9607e-01 k_t = [1, 1, 1, 1]
 Generation 4540: Best = 4.131545e+06 FES = 454100 μ_F = 0.800 μ_{CR} = 0.883
 Diversity = 1.9237e-01 k_t = [3, 3, 4, 3]

Generation 4550: Best = 4.121179e+06 FES = 455100 μ_F = 0.804 μ_{CR} = 0.876
 Diversity = 2.0833e-01 k_t = [1, 1, 1, 1]
 Generation 4560: Best = 4.108298e+06 FES = 456100 μ_F = 0.801 μ_{CR} = 0.873
 Diversity = 2.1661e-01 k_t = [2, 2, 2, 3]
 Generation 4570: Best = 4.097818e+06 FES = 457100 μ_F = 0.795 μ_{CR} = 0.868
 Diversity = 1.9754e-01 k_t = [1, 1, 1, 1]
 Generation 4580: Best = 4.088927e+06 FES = 458100 μ_F = 0.792 μ_{CR} = 0.867
 Diversity = 1.5912e-01 k_t = [2, 3, 4, 3]
 Generation 4590: Best = 4.079704e+06 FES = 459100 μ_F = 0.797 μ_{CR} = 0.864
 Diversity = 2.1807e-01 k_t = [1, 1, 1, 1]
 Generation 4600: Best = 4.073198e+06 FES = 460100 μ_F = 0.806 μ_{CR} = 0.868
 Diversity = 1.9757e-01 k_t = [2, 3, 4, 4]
 Generation 4610: Best = 4.064034e+06 FES = 461100 μ_F = 0.811 μ_{CR} = 0.869
 Diversity = 2.6754e-01 k_t = [1, 1, 1, 1]
 Generation 4620: Best = 4.053387e+06 FES = 462100 μ_F = 0.808 μ_{CR} = 0.867
 Diversity = 2.2291e-01 k_t = [3, 4, 2, 4]
 Generation 4630: Best = 4.041414e+06 FES = 463100 μ_F = 0.812 μ_{CR} = 0.865
 Diversity = 2.2602e-01 k_t = [1, 1, 1, 1]
 Generation 4640: Best = 4.023939e+06 FES = 464100 μ_F = 0.817 μ_{CR} = 0.865
 Diversity = 2.8748e-01 k_t = [4, 4, 4, 3]
 Generation 4650: Best = 4.012424e+06 FES = 465100 μ_F = 0.813 μ_{CR} = 0.866
 Diversity = 2.7654e-01 k_t = [1, 1, 1, 1]
 Generation 4660: Best = 3.994915e+06 FES = 466100 μ_F = 0.815 μ_{CR} = 0.863
 Diversity = 3.5531e-01 k_t = [3, 2, 3, 4]
 Generation 4670: Best = 3.975350e+06 FES = 467100 μ_F = 0.812 μ_{CR} = 0.861
 Diversity = 3.7806e-01 k_t = [1, 1, 1, 1]
 Generation 4680: Best = 3.959186e+06 FES = 468100 μ_F = 0.806 μ_{CR} = 0.863
 Diversity = 3.5834e-01 k_t = [2, 4, 3, 3]
 Generation 4690: Best = 3.943047e+06 FES = 469100 μ_F = 0.803 μ_{CR} = 0.871
 Diversity = 3.8859e-01 k_t = [1, 1, 1, 1]
 Generation 4700: Best = 3.926155e+06 FES = 470100 μ_F = 0.803 μ_{CR} = 0.872
 Diversity = 3.0911e-01 k_t = [3, 4, 4, 3]
 Generation 4710: Best = 3.911290e+06 FES = 471100 μ_F = 0.804 μ_{CR} = 0.873
 Diversity = 2.8644e-01 k_t = [1, 1, 1, 1]
 Generation 4720: Best = 3.898056e+06 FES = 472100 μ_F = 0.799 μ_{CR} = 0.870
 Diversity = 3.0265e-01 k_t = [3, 2, 3, 4]
 Generation 4730: Best = 3.882282e+06 FES = 473100 μ_F = 0.801 μ_{CR} = 0.865
 Diversity = 3.1010e-01 k_t = [1, 1, 1, 1]
 Generation 4740: Best = 3.870743e+06 FES = 474100 μ_F = 0.803 μ_{CR} = 0.872
 Diversity = 2.4245e-01 k_t = [4, 3, 2, 2]
 Generation 4750: Best = 3.860118e+06 FES = 475100 μ_F = 0.799 μ_{CR} = 0.881
 Diversity = 2.8741e-01 k_t = [1, 1, 1, 1]
 Generation 4760: Best = 3.846007e+06 FES = 476100 μ_F = 0.800 μ_{CR} = 0.885
 Diversity = 2.4228e-01 k_t = [2, 2, 2, 3]
 Generation 4770: Best = 3.834928e+06 FES = 477100 μ_F = 0.805 μ_{CR} = 0.888
 Diversity = 2.5959e-01 k_t = [1, 1, 1, 1]
 Generation 4780: Best = 3.825629e+06 FES = 478100 μ_F = 0.807 μ_{CR} = 0.888
 Diversity = 2.2375e-01 k_t = [2, 2, 4, 4]
 Generation 4790: Best = 3.816010e+06 FES = 479100 μ_F = 0.800 μ_{CR} = 0.887
 Diversity = 2.1978e-01 k_t = [1, 1, 1, 1]
 Generation 4800: Best = 3.807869e+06 FES = 480100 μ_F = 0.804 μ_{CR} = 0.894
 Diversity = 2.6797e-01 k_t = [4, 2, 2, 3]
 Generation 4810: Best = 3.799663e+06 FES = 481100 μ_F = 0.801 μ_{CR} = 0.891
 Diversity = 3.3564e-01 k_t = [1, 1, 1, 1]

Generation 4820: Best = 3.792952e+06 FES = 482100 μ_F = 0.792 μ_{CR} = 0.889
 Diversity = 2.2239e-01 k_t = [3, 4, 3, 2]
 Generation 4830: Best = 3.786678e+06 FES = 483100 μ_F = 0.792 μ_{CR} = 0.893
 Diversity = 2.3600e-01 k_t = [1, 1, 1, 1]
 Generation 4840: Best = 3.778242e+06 FES = 484100 μ_F = 0.798 μ_{CR} = 0.895
 Diversity = 2.1336e-01 k_t = [2, 4, 3, 4]
 Generation 4850: Best = 3.772086e+06 FES = 485100 μ_F = 0.797 μ_{CR} = 0.884
 Diversity = 2.8033e-01 k_t = [1, 1, 1, 1]
 Generation 4860: Best = 3.765239e+06 FES = 486100 μ_F = 0.807 μ_{CR} = 0.884
 Diversity = 2.3080e-01 k_t = [3, 2, 2, 3]
 Generation 4870: Best = 3.760705e+06 FES = 487100 μ_F = 0.795 μ_{CR} = 0.874
 Diversity = 2.5284e-01 k_t = [1, 1, 1, 1]
 Generation 4880: Best = 3.756075e+06 FES = 488100 μ_F = 0.790 μ_{CR} = 0.872
 Diversity = 1.6700e-01 k_t = [3, 2, 2, 2]
 Generation 4890: Best = 3.752471e+06 FES = 489100 μ_F = 0.779 μ_{CR} = 0.874
 Diversity = 1.7306e-01 k_t = [1, 1, 1, 1]
 Generation 4900: Best = 3.747763e+06 FES = 490100 μ_F = 0.771 μ_{CR} = 0.878
 Diversity = 1.4612e-01 k_t = [4, 4, 3, 2]
 Generation 4910: Best = 3.745271e+06 FES = 491100 μ_F = 0.778 μ_{CR} = 0.874
 Diversity = 1.5858e-01 k_t = [1, 1, 1, 1]
 Generation 4920: Best = 3.739200e+06 FES = 492100 μ_F = 0.768 μ_{CR} = 0.874
 Diversity = 1.4645e-01 k_t = [4, 4, 4, 3]
 Generation 4930: Best = 3.736506e+06 FES = 493100 μ_F = 0.769 μ_{CR} = 0.869
 Diversity = 1.7322e-01 k_t = [1, 1, 1, 1]
 Generation 4940: Best = 3.732794e+06 FES = 494100 μ_F = 0.769 μ_{CR} = 0.868
 Diversity = 1.3895e-01 k_t = [3, 4, 4, 3]
 Generation 4950: Best = 3.730088e+06 FES = 495100 μ_F = 0.768 μ_{CR} = 0.861
 Diversity = 1.3159e-01 k_t = [1, 1, 1, 1]
 Generation 4960: Best = 3.727557e+06 FES = 496100 μ_F = 0.757 μ_{CR} = 0.863
 Diversity = 1.4401e-01 k_t = [3, 2, 2, 2]
 Generation 4970: Best = 3.724907e+06 FES = 497100 μ_F = 0.759 μ_{CR} = 0.857
 Diversity = 1.6453e-01 k_t = [1, 1, 1, 1]
 Generation 4980: Best = 3.722599e+06 FES = 498100 μ_F = 0.766 μ_{CR} = 0.859
 Diversity = 1.3548e-01 k_t = [2, 2, 3, 4]
 Generation 4990: Best = 3.720108e+06 FES = 499100 μ_F = 0.758 μ_{CR} = 0.860
 Diversity = 1.4695e-01 k_t = [1, 1, 1, 1]
 Generation 5000: Best = 3.716457e+06 FES = 500100 μ_F = 0.768 μ_{CR} = 0.855
 Diversity = 1.4068e-01 k_t = [4, 4, 2, 4]
 Generation 5010: Best = 3.713396e+06 FES = 501100 μ_F = 0.776 μ_{CR} = 0.856
 Diversity = 1.8057e-01 k_t = [1, 1, 1, 1]
 Generation 5020: Best = 3.709536e+06 FES = 502100 μ_F = 0.780 μ_{CR} = 0.861
 Diversity = 1.3542e-01 k_t = [3, 3, 2, 3]
 Generation 5030: Best = 3.706551e+06 FES = 503100 μ_F = 0.775 μ_{CR} = 0.861
 Diversity = 1.5728e-01 k_t = [1, 1, 1, 1]
 Generation 5040: Best = 3.703218e+06 FES = 504100 μ_F = 0.779 μ_{CR} = 0.866
 Diversity = 1.2053e-01 k_t = [4, 3, 4, 4]
 Generation 5050: Best = 3.700127e+06 FES = 505100 μ_F = 0.791 μ_{CR} = 0.865
 Diversity = 1.3138e-01 k_t = [1, 1, 1, 1]
 Generation 5060: Best = 3.693729e+06 FES = 506100 μ_F = 0.793 μ_{CR} = 0.868
 Diversity = 1.4691e-01 k_t = [3, 4, 2, 3]
 Generation 5070: Best = 3.686534e+06 FES = 507100 μ_F = 0.798 μ_{CR} = 0.866
 Diversity = 1.8192e-01 k_t = [1, 1, 1, 1]
 Generation 5080: Best = 3.676969e+06 FES = 508100 μ_F = 0.799 μ_{CR} = 0.866
 Diversity = 1.6538e-01 k_t = [4, 4, 2, 3]

Generation 5090: Best = 3.669178e+06 FES = 509100 μ_F = 0.796 μ_{CR} = 0.861
 Diversity = 1.9508e-01 k_t = [1, 1, 1, 1]
 Generation 5100: Best = 3.665550e+06 FES = 510100 μ_F = 0.790 μ_{CR} = 0.856
 Diversity = 1.7254e-01 k_t = [3, 3, 3, 2]
 Generation 5110: Best = 3.661432e+06 FES = 511100 μ_F = 0.787 μ_{CR} = 0.858
 Diversity = 1.6329e-01 k_t = [1, 1, 1, 1]
 Generation 5120: Best = 3.655348e+06 FES = 512100 μ_F = 0.788 μ_{CR} = 0.861
 Diversity = 1.6817e-01 k_t = [4, 4, 3, 3]
 Generation 5130: Best = 3.650384e+06 FES = 513100 μ_F = 0.775 μ_{CR} = 0.858
 Diversity = 1.7892e-01 k_t = [1, 1, 1, 1]
 Generation 5140: Best = 3.644667e+06 FES = 514100 μ_F = 0.782 μ_{CR} = 0.862
 Diversity = 1.2933e-01 k_t = [4, 4, 3, 2]
 Generation 5150: Best = 3.641864e+06 FES = 515100 μ_F = 0.769 μ_{CR} = 0.859
 Diversity = 1.1711e-01 k_t = [1, 1, 1, 1]
 Generation 5160: Best = 3.638864e+06 FES = 516100 μ_F = 0.765 μ_{CR} = 0.861
 Diversity = 1.2060e-01 k_t = [4, 2, 2, 3]
 Generation 5170: Best = 3.636755e+06 FES = 517100 μ_F = 0.775 μ_{CR} = 0.860
 Diversity = 1.4642e-01 k_t = [1, 1, 1, 1]
 Generation 5180: Best = 3.633829e+06 FES = 518100 μ_F = 0.773 μ_{CR} = 0.855
 Diversity = 1.2450e-01 k_t = [4, 3, 4, 2]
 Generation 5190: Best = 3.631089e+06 FES = 519100 μ_F = 0.772 μ_{CR} = 0.847
 Diversity = 1.4326e-01 k_t = [1, 1, 1, 1]
 Generation 5200: Best = 3.628298e+06 FES = 520100 μ_F = 0.775 μ_{CR} = 0.837
 Diversity = 1.0372e-01 k_t = [2, 4, 2, 3]
 Generation 5210: Best = 3.624822e+06 FES = 521100 μ_F = 0.781 μ_{CR} = 0.835
 Diversity = 2.0399e-01 k_t = [1, 1, 1, 1]
 Generation 5220: Best = 3.618947e+06 FES = 522100 μ_F = 0.781 μ_{CR} = 0.835
 Diversity = 1.9238e-01 k_t = [3, 4, 4, 4]
 Generation 5230: Best = 3.613553e+06 FES = 523100 μ_F = 0.785 μ_{CR} = 0.841
 Diversity = 1.7397e-01 k_t = [1, 1, 1, 1]
 Generation 5240: Best = 3.608125e+06 FES = 524100 μ_F = 0.783 μ_{CR} = 0.842
 Diversity = 1.5408e-01 k_t = [3, 3, 2, 4]
 Generation 5250: Best = 3.604143e+06 FES = 525100 μ_F = 0.776 μ_{CR} = 0.839
 Diversity = 1.3412e-01 k_t = [1, 1, 1, 1]
 Generation 5260: Best = 3.600037e+06 FES = 526100 μ_F = 0.773 μ_{CR} = 0.835
 Diversity = 1.1095e-01 k_t = [2, 3, 3, 2]
 Generation 5270: Best = 3.596003e+06 FES = 527100 μ_F = 0.773 μ_{CR} = 0.841
 Diversity = 1.4803e-01 k_t = [1, 1, 1, 1]
 Generation 5280: Best = 3.589053e+06 FES = 528100 μ_F = 0.776 μ_{CR} = 0.852
 Diversity = 1.5957e-01 k_t = [3, 3, 2, 2]
 Generation 5290: Best = 3.581746e+06 FES = 529100 μ_F = 0.774 μ_{CR} = 0.858
 Diversity = 1.6857e-01 k_t = [1, 1, 1, 1]
 Generation 5300: Best = 3.576486e+06 FES = 530100 μ_F = 0.779 μ_{CR} = 0.855
 Diversity = 1.6429e-01 k_t = [2, 4, 3, 2]
 Generation 5310: Best = 3.570933e+06 FES = 531100 μ_F = 0.776 μ_{CR} = 0.857
 Diversity = 1.9584e-01 k_t = [1, 1, 1, 1]
 Generation 5320: Best = 3.563595e+06 FES = 532100 μ_F = 0.780 μ_{CR} = 0.854
 Diversity = 1.7610e-01 k_t = [4, 4, 4, 2]
 Generation 5330: Best = 3.557918e+06 FES = 533100 μ_F = 0.787 μ_{CR} = 0.854
 Diversity = 1.8138e-01 k_t = [1, 1, 1, 1]
 Generation 5340: Best = 3.552514e+06 FES = 534100 μ_F = 0.786 μ_{CR} = 0.850
 Diversity = 1.6599e-01 k_t = [3, 2, 4, 4]
 Generation 5350: Best = 3.546839e+06 FES = 535100 μ_F = 0.784 μ_{CR} = 0.852
 Diversity = 1.8192e-01 k_t = [1, 1, 1, 1]

Generation 5360: Best = 3.540715e+06 FES = 536100 μ_F = 0.782 μ_{CR} = 0.860
 Diversity = 1.4315e-01 k_t = [4, 4, 2, 3]
 Generation 5370: Best = 3.536150e+06 FES = 537100 μ_F = 0.785 μ_{CR} = 0.856
 Diversity = 2.1369e-01 k_t = [1, 1, 1, 1]
 Generation 5380: Best = 3.530815e+06 FES = 538100 μ_F = 0.788 μ_{CR} = 0.865
 Diversity = 1.6201e-01 k_t = [2, 2, 4, 4]
 Generation 5390: Best = 3.527442e+06 FES = 539100 μ_F = 0.776 μ_{CR} = 0.875
 Diversity = 1.5867e-01 k_t = [1, 1, 1, 1]
 Generation 5400: Best = 3.523195e+06 FES = 540100 μ_F = 0.773 μ_{CR} = 0.871
 Diversity = 1.6052e-01 k_t = [3, 2, 4, 3]
 Generation 5410: Best = 3.518369e+06 FES = 541100 μ_F = 0.776 μ_{CR} = 0.870
 Diversity = 1.9654e-01 k_t = [1, 1, 1, 1]
 Generation 5420: Best = 3.512547e+06 FES = 542100 μ_F = 0.779 μ_{CR} = 0.870
 Diversity = 1.6082e-01 k_t = [3, 4, 2, 3]
 Generation 5430: Best = 3.508996e+06 FES = 543100 μ_F = 0.789 μ_{CR} = 0.877
 Diversity = 1.7500e-01 k_t = [1, 1, 1, 1]
 Generation 5440: Best = 3.505729e+06 FES = 544100 μ_F = 0.777 μ_{CR} = 0.876
 Diversity = 1.2785e-01 k_t = [3, 3, 2, 2]
 Generation 5450: Best = 3.502016e+06 FES = 545100 μ_F = 0.786 μ_{CR} = 0.874
 Diversity = 1.4934e-01 k_t = [1, 1, 1, 1]
 Generation 5460: Best = 3.498572e+06 FES = 546100 μ_F = 0.788 μ_{CR} = 0.869
 Diversity = 1.1978e-01 k_t = [2, 3, 2, 4]
 Generation 5470: Best = 3.495617e+06 FES = 547100 μ_F = 0.784 μ_{CR} = 0.868
 Diversity = 1.2889e-01 k_t = [1, 1, 1, 1]
 Generation 5480: Best = 3.492541e+06 FES = 548100 μ_F = 0.787 μ_{CR} = 0.871
 Diversity = 1.1965e-01 k_t = [4, 3, 4, 3]
 Generation 5490: Best = 3.489717e+06 FES = 549100 μ_F = 0.784 μ_{CR} = 0.862
 Diversity = 1.5643e-01 k_t = [1, 1, 1, 1]
 Generation 5500: Best = 3.486486e+06 FES = 550100 μ_F = 0.792 μ_{CR} = 0.862
 Diversity = 1.4303e-01 k_t = [4, 3, 3, 4]
 Generation 5510: Best = 3.484490e+06 FES = 551100 μ_F = 0.789 μ_{CR} = 0.868
 Diversity = 1.4302e-01 k_t = [1, 1, 1, 1]
 Generation 5520: Best = 3.482914e+06 FES = 552100 μ_F = 0.782 μ_{CR} = 0.870
 Diversity = 1.1943e-01 k_t = [2, 2, 2, 4]
 Generation 5530: Best = 3.480819e+06 FES = 553100 μ_F = 0.782 μ_{CR} = 0.874
 Diversity = 1.5955e-01 k_t = [1, 1, 1, 1]
 Generation 5540: Best = 3.477973e+06 FES = 554100 μ_F = 0.782 μ_{CR} = 0.877
 Diversity = 1.6181e-01 k_t = [3, 2, 4, 3]
 Generation 5550: Best = 3.476234e+06 FES = 555100 μ_F = 0.790 μ_{CR} = 0.871
 Diversity = 1.5838e-01 k_t = [1, 1, 1, 1]
 Generation 5560: Best = 3.474522e+06 FES = 556100 μ_F = 0.790 μ_{CR} = 0.874
 Diversity = 1.1676e-01 k_t = [4, 3, 4, 4]
 Generation 5570: Best = 3.473036e+06 FES = 557100 μ_F = 0.790 μ_{CR} = 0.879
 Diversity = 1.1822e-01 k_t = [1, 1, 1, 1]
 Generation 5580: Best = 3.471554e+06 FES = 558100 μ_F = 0.784 μ_{CR} = 0.864
 Diversity = 1.0803e-01 k_t = [2, 2, 2, 3]
 Generation 5590: Best = 3.467297e+06 FES = 559100 μ_F = 0.793 μ_{CR} = 0.860
 Diversity = 1.9993e-01 k_t = [1, 1, 1, 1]
 Generation 5600: Best = 3.461756e+06 FES = 560100 μ_F = 0.793 μ_{CR} = 0.856
 Diversity = 1.8587e-01 k_t = [2, 3, 4, 4]
 Generation 5610: Best = 3.457830e+06 FES = 561100 μ_F = 0.789 μ_{CR} = 0.861
 Diversity = 2.2050e-01 k_t = [1, 1, 1, 1]
 Generation 5620: Best = 3.453248e+06 FES = 562100 μ_F = 0.789 μ_{CR} = 0.866
 Diversity = 1.9454e-01 k_t = [4, 3, 3, 4]

Generation 5630: Best = 3.449854e+06 FES = 563100 μ_F = 0.785 μ_{CR} = 0.862
 Diversity = 1.8140e-01 k_t = [1, 1, 1, 1]
 Generation 5640: Best = 3.444377e+06 FES = 564100 μ_F = 0.787 μ_{CR} = 0.864
 Diversity = 1.3749e-01 k_t = [3, 4, 3, 4]
 Generation 5650: Best = 3.440838e+06 FES = 565100 μ_F = 0.778 μ_{CR} = 0.861
 Diversity = 1.4249e-01 k_t = [1, 1, 1, 1]
 Generation 5660: Best = 3.436446e+06 FES = 566100 μ_F = 0.765 μ_{CR} = 0.859
 Diversity = 1.4086e-01 k_t = [3, 2, 2, 4]
 Generation 5670: Best = 3.433784e+06 FES = 567100 μ_F = 0.757 μ_{CR} = 0.858
 Diversity = 1.4226e-01 k_t = [1, 1, 1, 1]
 Generation 5680: Best = 3.431544e+06 FES = 568100 μ_F = 0.749 μ_{CR} = 0.863
 Diversity = 1.0271e-01 k_t = [4, 3, 2, 2]
 Generation 5690: Best = 3.430082e+06 FES = 569100 μ_F = 0.754 μ_{CR} = 0.862
 Diversity = 1.2039e-01 k_t = [1, 1, 1, 1]
 Generation 5700: Best = 3.428220e+06 FES = 570100 μ_F = 0.752 μ_{CR} = 0.857
 Diversity = 1.4276e-01 k_t = [3, 2, 2, 2]
 Generation 5710: Best = 3.426633e+06 FES = 571100 μ_F = 0.755 μ_{CR} = 0.854
 Diversity = 1.5568e-01 k_t = [1, 1, 1, 1]
 Generation 5720: Best = 3.425589e+06 FES = 572100 μ_F = 0.749 μ_{CR} = 0.860
 Diversity = 1.4423e-01 k_t = [4, 2, 2, 2]
 Generation 5730: Best = 3.424394e+06 FES = 573100 μ_F = 0.764 μ_{CR} = 0.866
 Diversity = 1.7482e-01 k_t = [1, 1, 1, 1]
 Generation 5740: Best = 3.423086e+06 FES = 574100 μ_F = 0.757 μ_{CR} = 0.876
 Diversity = 1.3466e-01 k_t = [2, 3, 3, 4]
 Generation 5750: Best = 3.422431e+06 FES = 575100 μ_F = 0.759 μ_{CR} = 0.877
 Diversity = 1.4616e-01 k_t = [1, 1, 1, 1]
 Generation 5760: Best = 3.421767e+06 FES = 576100 μ_F = 0.753 μ_{CR} = 0.893
 Diversity = 1.1948e-01 k_t = [2, 3, 4, 3]
 Generation 5770: Best = 3.421089e+06 FES = 577100 μ_F = 0.758 μ_{CR} = 0.896
 Diversity = 1.4174e-01 k_t = [1, 1, 1, 1]
 Generation 5780: Best = 3.420343e+06 FES = 578100 μ_F = 0.760 μ_{CR} = 0.908
 Diversity = 1.6606e-01 k_t = [2, 4, 3, 2]
 Generation 5790: Best = 3.419629e+06 FES = 579100 μ_F = 0.751 μ_{CR} = 0.897
 Diversity = 1.0961e-01 k_t = [1, 1, 1, 1]
 Generation 5800: Best = 3.418874e+06 FES = 580100 μ_F = 0.752 μ_{CR} = 0.891
 Diversity = 8.4557e-02 k_t = [4, 2, 3, 2]
 Generation 5810: Best = 3.418325e+06 FES = 581100 μ_F = 0.753 μ_{CR} = 0.885
 Diversity = 7.0967e-02 k_t = [1, 1, 1, 1]
 Generation 5820: Best = 3.418082e+06 FES = 582100 μ_F = 0.744 μ_{CR} = 0.879
 Diversity = 5.8378e-02 k_t = [3, 4, 4, 2]
 Generation 5830: Best = 3.417907e+06 FES = 583100 μ_F = 0.737 μ_{CR} = 0.881
 Diversity = 4.6597e-02 k_t = [1, 1, 1, 1]
 Generation 5840: Best = 3.417628e+06 FES = 584100 μ_F = 0.736 μ_{CR} = 0.884
 Diversity = 4.9154e-02 k_t = [4, 2, 4, 4]
 Generation 5850: Best = 3.417512e+06 FES = 585100 μ_F = 0.735 μ_{CR} = 0.882
 Diversity = 3.2406e-02 k_t = [1, 1, 1, 1]
 Generation 5860: Best = 3.417420e+06 FES = 586100 μ_F = 0.741 μ_{CR} = 0.869
 Diversity = 3.4134e-02 k_t = [4, 2, 3, 2]
 Generation 5870: Best = 3.417312e+06 FES = 587100 μ_F = 0.751 μ_{CR} = 0.864
 Diversity = 3.7478e-02 k_t = [1, 1, 1, 1]
 Generation 5880: Best = 3.417183e+06 FES = 588100 μ_F = 0.749 μ_{CR} = 0.858
 Diversity = 3.2194e-02 k_t = [3, 4, 2, 3]
 Generation 5890: Best = 3.417077e+06 FES = 589100 μ_F = 0.753 μ_{CR} = 0.850
 Diversity = 3.6032e-02 k_t = [1, 1, 1, 1]

Generation 5900: Best = 3.416828e+06 FES = 590100 μ_F = 0.768 μ_{CR} = 0.855
 Diversity = 3.7670e-02 k_t = [4, 4, 2, 2]
 Generation 5910: Best = 3.416586e+06 FES = 591100 μ_F = 0.773 μ_{CR} = 0.856
 Diversity = 5.1287e-02 k_t = [1, 1, 1, 1]
 Generation 5920: Best = 3.416354e+06 FES = 592100 μ_F = 0.783 μ_{CR} = 0.847
 Diversity = 4.3119e-02 k_t = [4, 2, 3, 2]
 Generation 5930: Best = 3.416206e+06 FES = 593100 μ_F = 0.776 μ_{CR} = 0.840
 Diversity = 3.7034e-02 k_t = [1, 1, 1, 1]
 Generation 5940: Best = 3.416008e+06 FES = 594100 μ_F = 0.758 μ_{CR} = 0.828
 Diversity = 3.1242e-02 k_t = [2, 3, 3, 3]
 Generation 5950: Best = 3.415785e+06 FES = 595100 μ_F = 0.763 μ_{CR} = 0.824
 Diversity = 3.6036e-02 k_t = [1, 1, 1, 1]
 Generation 5960: Best = 3.415647e+06 FES = 596100 μ_F = 0.752 μ_{CR} = 0.812
 Diversity = 3.2284e-02 k_t = [2, 3, 4, 3]
 Generation 5970: Best = 3.415443e+06 FES = 597100 μ_F = 0.761 μ_{CR} = 0.811
 Diversity = 3.6823e-02 k_t = [1, 1, 1, 1]
 Generation 5980: Best = 3.415248e+06 FES = 598100 μ_F = 0.753 μ_{CR} = 0.801
 Diversity = 3.5002e-02 k_t = [2, 4, 3, 2]
 Generation 5990: Best = 3.415019e+06 FES = 599100 μ_F = 0.750 μ_{CR} = 0.803
 Diversity = 4.4848e-02 k_t = [1, 1, 1, 1]
 Generation 6000: Best = 3.414734e+06 FES = 600100 μ_F = 0.738 μ_{CR} = 0.805
 Diversity = 3.9328e-02 k_t = [2, 3, 3, 2]
 Generation 6010: Best = 3.414610e+06 FES = 601100 μ_F = 0.729 μ_{CR} = 0.802
 Diversity = 4.2050e-02 k_t = [1, 1, 1, 1]
 Generation 6020: Best = 3.414364e+06 FES = 602100 μ_F = 0.728 μ_{CR} = 0.797
 Diversity = 3.4989e-02 k_t = [2, 2, 4, 4]
 Generation 6030: Best = 3.414299e+06 FES = 603100 μ_F = 0.722 μ_{CR} = 0.789
 Diversity = 3.3190e-02 k_t = [1, 1, 1, 1]
 Generation 6040: Best = 3.414148e+06 FES = 604100 μ_F = 0.736 μ_{CR} = 0.786
 Diversity = 3.0347e-02 k_t = [3, 3, 2, 4]
 Generation 6050: Best = 3.413948e+06 FES = 605100 μ_F = 0.735 μ_{CR} = 0.798
 Diversity = 3.9406e-02 k_t = [1, 1, 1, 1]
 Generation 6060: Best = 3.413778e+06 FES = 606100 μ_F = 0.731 μ_{CR} = 0.793
 Diversity = 3.3448e-02 k_t = [4, 4, 4, 3]
 Generation 6070: Best = 3.413690e+06 FES = 607100 μ_F = 0.736 μ_{CR} = 0.802
 Diversity = 2.8813e-02 k_t = [1, 1, 1, 1]
 Generation 6080: Best = 3.413639e+06 FES = 608100 μ_F = 0.736 μ_{CR} = 0.796
 Diversity = 2.4522e-02 k_t = [2, 4, 3, 2]
 Generation 6090: Best = 3.413575e+06 FES = 609100 μ_F = 0.738 μ_{CR} = 0.800
 Diversity = 2.2261e-02 k_t = [1, 1, 1, 1]
 Generation 6100: Best = 3.413515e+06 FES = 610100 μ_F = 0.745 μ_{CR} = 0.801
 Diversity = 2.1413e-02 k_t = [3, 4, 2, 3]
 Generation 6110: Best = 3.413426e+06 FES = 611100 μ_F = 0.752 μ_{CR} = 0.803
 Diversity = 2.1601e-02 k_t = [1, 1, 1, 1]
 Generation 6120: Best = 3.413363e+06 FES = 612100 μ_F = 0.747 μ_{CR} = 0.792
 Diversity = 2.0175e-02 k_t = [3, 2, 3, 4]
 Generation 6130: Best = 3.413260e+06 FES = 613100 μ_F = 0.761 μ_{CR} = 0.791
 Diversity = 2.4605e-02 k_t = [1, 1, 1, 1]
 Generation 6140: Best = 3.413135e+06 FES = 614100 μ_F = 0.769 μ_{CR} = 0.789
 Diversity = 2.1505e-02 k_t = [4, 4, 4, 4]
 Generation 6150: Best = 3.413095e+06 FES = 615100 μ_F = 0.765 μ_{CR} = 0.781
 Diversity = 2.1599e-02 k_t = [1, 1, 1, 1]
 Generation 6160: Best = 3.413037e+06 FES = 616100 μ_F = 0.750 μ_{CR} = 0.783
 Diversity = 1.9025e-02 k_t = [3, 2, 3, 4]

Generation 6170: Best = 3.412989e+06 FES = 617100 μ_F = 0.727 μ_{CR} = 0.787
 Diversity = 2.2938e-02 k_t = [1, 1, 1, 1]
 Generation 6180: Best = 3.412920e+06 FES = 618100 μ_F = 0.737 μ_{CR} = 0.792
 Diversity = 2.2680e-02 k_t = [2, 3, 2, 3]
 Generation 6190: Best = 3.412840e+06 FES = 619100 μ_F = 0.745 μ_{CR} = 0.794
 Diversity = 2.7463e-02 k_t = [1, 1, 1, 1]
 Generation 6200: Best = 3.412661e+06 FES = 620100 μ_F = 0.737 μ_{CR} = 0.787
 Diversity = 2.7624e-02 k_t = [2, 2, 3, 4]
 Generation 6210: Best = 3.412526e+06 FES = 621100 μ_F = 0.741 μ_{CR} = 0.786
 Diversity = 3.8950e-02 k_t = [1, 1, 1, 1]
 Generation 6220: Best = 3.412401e+06 FES = 622100 μ_F = 0.727 μ_{CR} = 0.793
 Diversity = 3.3196e-02 k_t = [4, 3, 2, 3]
 Generation 6230: Best = 3.412228e+06 FES = 623100 μ_F = 0.732 μ_{CR} = 0.795
 Diversity = 3.8882e-02 k_t = [1, 1, 1, 1]
 Generation 6240: Best = 3.412058e+06 FES = 624100 μ_F = 0.742 μ_{CR} = 0.792
 Diversity = 2.6596e-02 k_t = [4, 2, 4, 3]
 Generation 6250: Best = 3.411996e+06 FES = 625100 μ_F = 0.738 μ_{CR} = 0.789
 Diversity = 2.3568e-02 k_t = [1, 1, 1, 1]
 Generation 6260: Best = 3.411900e+06 FES = 626100 μ_F = 0.749 μ_{CR} = 0.782
 Diversity = 2.3279e-02 k_t = [2, 2, 2, 3]
 Generation 6270: Best = 3.411815e+06 FES = 627100 μ_F = 0.751 μ_{CR} = 0.780
 Diversity = 3.0797e-02 k_t = [1, 1, 1, 1]
 Generation 6280: Best = 3.411674e+06 FES = 628100 μ_F = 0.756 μ_{CR} = 0.783
 Diversity = 2.6816e-02 k_t = [2, 3, 3, 4]
 Generation 6290: Best = 3.411552e+06 FES = 629100 μ_F = 0.746 μ_{CR} = 0.782
 Diversity = 3.1193e-02 k_t = [1, 1, 1, 1]
 Generation 6300: Best = 3.411477e+06 FES = 630100 μ_F = 0.734 μ_{CR} = 0.780
 Diversity = 2.5263e-02 k_t = [2, 4, 3, 4]
 Generation 6310: Best = 3.411413e+06 FES = 631100 μ_F = 0.735 μ_{CR} = 0.774
 Diversity = 2.3751e-02 k_t = [1, 1, 1, 1]
 Generation 6320: Best = 3.411252e+06 FES = 632100 μ_F = 0.743 μ_{CR} = 0.785
 Diversity = 2.7075e-02 k_t = [3, 3, 2, 3]
 Generation 6330: Best = 3.411182e+06 FES = 633100 μ_F = 0.738 μ_{CR} = 0.794
 Diversity = 2.4686e-02 k_t = [1, 1, 1, 1]
 Generation 6340: Best = 3.411081e+06 FES = 634100 μ_F = 0.754 μ_{CR} = 0.795
 Diversity = 2.3235e-02 k_t = [4, 4, 2, 2]
 Generation 6350: Best = 3.410997e+06 FES = 635100 μ_F = 0.769 μ_{CR} = 0.797
 Diversity = 2.4429e-02 k_t = [1, 1, 1, 1]
 Generation 6360: Best = 3.410881e+06 FES = 636100 μ_F = 0.754 μ_{CR} = 0.795
 Diversity = 2.3804e-02 k_t = [4, 2, 3, 4]
 Generation 6370: Best = 3.410765e+06 FES = 637100 μ_F = 0.759 μ_{CR} = 0.800
 Diversity = 2.8060e-02 k_t = [1, 1, 1, 1]
 Generation 6380: Best = 3.410681e+06 FES = 638100 μ_F = 0.748 μ_{CR} = 0.803
 Diversity = 2.1445e-02 k_t = [4, 4, 3, 4]
 Generation 6390: Best = 3.410600e+06 FES = 639100 μ_F = 0.747 μ_{CR} = 0.809
 Diversity = 2.4042e-02 k_t = [1, 1, 1, 1]
 Generation 6400: Best = 3.410527e+06 FES = 640100 μ_F = 0.744 μ_{CR} = 0.817
 Diversity = 2.0007e-02 k_t = [3, 3, 3, 3]
 Generation 6410: Best = 3.410459e+06 FES = 641100 μ_F = 0.744 μ_{CR} = 0.823
 Diversity = 2.5838e-02 k_t = [1, 1, 1, 1]
 Generation 6420: Best = 3.410377e+06 FES = 642100 μ_F = 0.744 μ_{CR} = 0.822
 Diversity = 2.9627e-02 k_t = [2, 2, 2, 2]
 Generation 6430: Best = 3.410321e+06 FES = 643100 μ_F = 0.735 μ_{CR} = 0.828
 Diversity = 2.7245e-02 k_t = [1, 1, 1, 1]

Generation 6440: Best = 3.410243e+06 FES = 644100 μ_F = 0.746 μ_{CR} = 0.827
 Diversity = 3.0587e-02 k_t = [3, 2, 3, 3]
 Generation 6450: Best = 3.410181e+06 FES = 645100 μ_F = 0.742 μ_{CR} = 0.833
 Diversity = 3.1380e-02 k_t = [1, 1, 1, 1]
 Generation 6460: Best = 3.410150e+06 FES = 646100 μ_F = 0.740 μ_{CR} = 0.841
 Diversity = 2.1040e-02 k_t = [3, 4, 3, 3]
 Generation 6470: Best = 3.410092e+06 FES = 647100 μ_F = 0.750 μ_{CR} = 0.851
 Diversity = 2.5092e-02 k_t = [1, 1, 1, 1]
 Generation 6480: Best = 3.410038e+06 FES = 648100 μ_F = 0.764 μ_{CR} = 0.847
 Diversity = 2.3275e-02 k_t = [4, 3, 4, 4]
 Generation 6490: Best = 3.409990e+06 FES = 649100 μ_F = 0.758 μ_{CR} = 0.836
 Diversity = 2.2841e-02 k_t = [1, 1, 1, 1]
 Generation 6500: Best = 3.409909e+06 FES = 650100 μ_F = 0.760 μ_{CR} = 0.833
 Diversity = 2.4168e-02 k_t = [4, 3, 3, 4]
 Generation 6510: Best = 3.409837e+06 FES = 651100 μ_F = 0.756 μ_{CR} = 0.832
 Diversity = 2.7644e-02 k_t = [1, 1, 1, 1]
 Generation 6520: Best = 3.409776e+06 FES = 652100 μ_F = 0.752 μ_{CR} = 0.842
 Diversity = 2.1956e-02 k_t = [2, 3, 4, 4]
 Generation 6530: Best = 3.409721e+06 FES = 653100 μ_F = 0.753 μ_{CR} = 0.827
 Diversity = 2.1765e-02 k_t = [1, 1, 1, 1]
 Generation 6540: Best = 3.409668e+06 FES = 654100 μ_F = 0.759 μ_{CR} = 0.826
 Diversity = 2.4766e-02 k_t = [3, 4, 2, 3]
 Generation 6550: Best = 3.409628e+06 FES = 655100 μ_F = 0.745 μ_{CR} = 0.839
 Diversity = 2.5460e-02 k_t = [1, 1, 1, 1]
 Generation 6560: Best = 3.409567e+06 FES = 656100 μ_F = 0.757 μ_{CR} = 0.842
 Diversity = 2.2899e-02 k_t = [4, 2, 4, 4]
 Generation 6570: Best = 3.409494e+06 FES = 657100 μ_F = 0.759 μ_{CR} = 0.854
 Diversity = 3.2887e-02 k_t = [1, 1, 1, 1]
 Generation 6580: Best = 3.409463e+06 FES = 658100 μ_F = 0.755 μ_{CR} = 0.848
 Diversity = 2.8991e-02 k_t = [2, 4, 2, 2]
 Generation 6590: Best = 3.409422e+06 FES = 659100 μ_F = 0.752 μ_{CR} = 0.852
 Diversity = 2.0950e-02 k_t = [1, 1, 1, 1]
 Generation 6600: Best = 3.409350e+06 FES = 660100 μ_F = 0.750 μ_{CR} = 0.847
 Diversity = 2.3240e-02 k_t = [2, 2, 3, 3]
 Generation 6610: Best = 3.409259e+06 FES = 661100 μ_F = 0.766 μ_{CR} = 0.849
 Diversity = 3.2050e-02 k_t = [1, 1, 1, 1]
 Generation 6620: Best = 3.409194e+06 FES = 662100 μ_F = 0.770 μ_{CR} = 0.848
 Diversity = 2.2753e-02 k_t = [4, 2, 3, 4]
 Generation 6630: Best = 3.409102e+06 FES = 663100 μ_F = 0.774 μ_{CR} = 0.844
 Diversity = 2.4561e-02 k_t = [1, 1, 1, 1]
 Generation 6640: Best = 3.409052e+06 FES = 664100 μ_F = 0.762 μ_{CR} = 0.839
 Diversity = 2.0689e-02 k_t = [2, 3, 2, 4]
 Generation 6650: Best = 3.408990e+06 FES = 665100 μ_F = 0.741 μ_{CR} = 0.836
 Diversity = 2.3886e-02 k_t = [1, 1, 1, 1]
 Generation 6660: Best = 3.408946e+06 FES = 666100 μ_F = 0.745 μ_{CR} = 0.842
 Diversity = 1.8224e-02 k_t = [3, 3, 4, 4]
 Generation 6670: Best = 3.408899e+06 FES = 667100 μ_F = 0.738 μ_{CR} = 0.833
 Diversity = 2.2416e-02 k_t = [1, 1, 1, 1]
 Generation 6680: Best = 3.408856e+06 FES = 668100 μ_F = 0.741 μ_{CR} = 0.823
 Diversity = 1.8384e-02 k_t = [4, 2, 2, 3]
 Generation 6690: Best = 3.408811e+06 FES = 669100 μ_F = 0.744 μ_{CR} = 0.826
 Diversity = 2.2153e-02 k_t = [1, 1, 1, 1]
 Generation 6700: Best = 3.408759e+06 FES = 670100 μ_F = 0.743 μ_{CR} = 0.825
 Diversity = 1.8746e-02 k_t = [4, 2, 3, 2]

Generation 6710: Best = 3.408741e+06 FES = 671100 μ_F = 0.760 μ_{CR} = 0.839
 Diversity = 2.0884e-02 k_t = [1, 1, 1, 1]
 Generation 6720: Best = 3.408692e+06 FES = 672100 μ_F = 0.757 μ_{CR} = 0.827
 Diversity = 1.7912e-02 k_t = [3, 3, 4, 3]
 Generation 6730: Best = 3.408648e+06 FES = 673100 μ_F = 0.761 μ_{CR} = 0.808
 Diversity = 2.3394e-02 k_t = [1, 1, 1, 1]
 Generation 6740: Best = 3.408592e+06 FES = 674100 μ_F = 0.751 μ_{CR} = 0.804
 Diversity = 1.8698e-02 k_t = [4, 3, 3, 3]
 Generation 6750: Best = 3.408556e+06 FES = 675100 μ_F = 0.745 μ_{CR} = 0.806
 Diversity = 1.8793e-02 k_t = [1, 1, 1, 1]
 Generation 6760: Best = 3.408489e+06 FES = 676100 μ_F = 0.756 μ_{CR} = 0.811
 Diversity = 2.4381e-02 k_t = [2, 2, 3, 2]
 Generation 6770: Best = 3.408435e+06 FES = 677100 μ_F = 0.758 μ_{CR} = 0.806
 Diversity = 2.8309e-02 k_t = [1, 1, 1, 1]
 Generation 6780: Best = 3.408372e+06 FES = 678100 μ_F = 0.757 μ_{CR} = 0.802
 Diversity = 2.5549e-02 k_t = [2, 2, 4, 3]
 Generation 6790: Best = 3.408276e+06 FES = 679100 μ_F = 0.752 μ_{CR} = 0.802
 Diversity = 3.1677e-02 k_t = [1, 1, 1, 1]
 Generation 6800: Best = 3.408205e+06 FES = 680100 μ_F = 0.739 μ_{CR} = 0.802
 Diversity = 2.4278e-02 k_t = [2, 4, 3, 3]
 Generation 6810: Best = 3.408130e+06 FES = 681100 μ_F = 0.737 μ_{CR} = 0.806
 Diversity = 2.3808e-02 k_t = [1, 1, 1, 1]
 Generation 6820: Best = 3.408091e+06 FES = 682100 μ_F = 0.736 μ_{CR} = 0.803
 Diversity = 2.1383e-02 k_t = [4, 3, 2, 3]
 Generation 6830: Best = 3.408020e+06 FES = 683100 μ_F = 0.744 μ_{CR} = 0.800
 Diversity = 2.2398e-02 k_t = [1, 1, 1, 1]
 Generation 6840: Best = 3.407961e+06 FES = 684100 μ_F = 0.748 μ_{CR} = 0.802
 Diversity = 2.3607e-02 k_t = [4, 4, 2, 4]
 Generation 6850: Best = 3.407869e+06 FES = 685100 μ_F = 0.745 μ_{CR} = 0.801
 Diversity = 2.7982e-02 k_t = [1, 1, 1, 1]
 Generation 6860: Best = 3.407725e+06 FES = 686100 μ_F = 0.750 μ_{CR} = 0.808
 Diversity = 3.5603e-02 k_t = [4, 3, 2, 2]
 Generation 6870: Best = 3.407632e+06 FES = 687100 μ_F = 0.749 μ_{CR} = 0.796
 Diversity = 3.1536e-02 k_t = [1, 1, 1, 1]
 Generation 6880: Best = 3.407566e+06 FES = 688100 μ_F = 0.744 μ_{CR} = 0.796
 Diversity = 2.2063e-02 k_t = [4, 3, 3, 4]
 Generation 6890: Best = 3.407512e+06 FES = 689100 μ_F = 0.757 μ_{CR} = 0.799
 Diversity = 2.2305e-02 k_t = [1, 1, 1, 1]
 Generation 6900: Best = 3.407461e+06 FES = 690100 μ_F = 0.754 μ_{CR} = 0.799
 Diversity = 2.1218e-02 k_t = [2, 2, 2, 3]
 Generation 6910: Best = 3.407412e+06 FES = 691100 μ_F = 0.734 μ_{CR} = 0.795
 Diversity = 2.0974e-02 k_t = [1, 1, 1, 1]
 Generation 6920: Best = 3.407348e+06 FES = 692100 μ_F = 0.734 μ_{CR} = 0.786
 Diversity = 2.2291e-02 k_t = [3, 2, 3, 2]
 Generation 6930: Best = 3.407289e+06 FES = 693100 μ_F = 0.748 μ_{CR} = 0.784
 Diversity = 2.2710e-02 k_t = [1, 1, 1, 1]
 Generation 6940: Best = 3.407181e+06 FES = 694100 μ_F = 0.740 μ_{CR} = 0.780
 Diversity = 2.3403e-02 k_t = [2, 4, 2, 3]
 Generation 6950: Best = 3.407050e+06 FES = 695100 μ_F = 0.750 μ_{CR} = 0.778
 Diversity = 2.8962e-02 k_t = [1, 1, 1, 1]
 Generation 6960: Best = 3.406935e+06 FES = 696100 μ_F = 0.754 μ_{CR} = 0.780
 Diversity = 2.7435e-02 k_t = [2, 2, 3, 3]
 Generation 6970: Best = 3.406839e+06 FES = 697100 μ_F = 0.749 μ_{CR} = 0.785
 Diversity = 2.7817e-02 k_t = [1, 1, 1, 1]

Generation 6980: Best = 3.406693e+06 FES = 698100 μ_F = 0.752 μ_{CR} = 0.778
 Diversity = 2.5487e-02 k_t = [3, 4, 3, 4]
 Generation 6990: Best = 3.406588e+06 FES = 699100 μ_F = 0.756 μ_{CR} = 0.778
 Diversity = 3.0768e-02 k_t = [1, 1, 1, 1]
 Generation 7000: Best = 3.406489e+06 FES = 700100 μ_F = 0.747 μ_{CR} = 0.776
 Diversity = 2.7310e-02 k_t = [2, 4, 3, 4]
 Generation 7010: Best = 3.406347e+06 FES = 701100 μ_F = 0.752 μ_{CR} = 0.775
 Diversity = 3.4733e-02 k_t = [1, 1, 1, 1]
 Generation 7020: Best = 3.406197e+06 FES = 702100 μ_F = 0.749 μ_{CR} = 0.777
 Diversity = 3.7976e-02 k_t = [4, 4, 2, 3]
 Generation 7030: Best = 3.406124e+06 FES = 703100 μ_F = 0.751 μ_{CR} = 0.782
 Diversity = 3.2298e-02 k_t = [1, 1, 1, 1]
 Generation 7040: Best = 3.406050e+06 FES = 704100 μ_F = 0.744 μ_{CR} = 0.771
 Diversity = 2.6568e-02 k_t = [2, 3, 3, 4]
 Generation 7050: Best = 3.405940e+06 FES = 705100 μ_F = 0.736 μ_{CR} = 0.766
 Diversity = 2.9791e-02 k_t = [1, 1, 1, 1]
 Generation 7060: Best = 3.405850e+06 FES = 706100 μ_F = 0.733 μ_{CR} = 0.756
 Diversity = 2.7203e-02 k_t = [2, 4, 3, 4]
 Generation 7070: Best = 3.405786e+06 FES = 707100 μ_F = 0.733 μ_{CR} = 0.754
 Diversity = 2.6929e-02 k_t = [1, 1, 1, 1]
 Generation 7080: Best = 3.405722e+06 FES = 708100 μ_F = 0.737 μ_{CR} = 0.749
 Diversity = 2.6505e-02 k_t = [3, 4, 3, 3]
 Generation 7090: Best = 3.405627e+06 FES = 709100 μ_F = 0.748 μ_{CR} = 0.749
 Diversity = 3.0446e-02 k_t = [1, 1, 1, 1]
 Generation 7100: Best = 3.405521e+06 FES = 710100 μ_F = 0.749 μ_{CR} = 0.742
 Diversity = 2.6602e-02 k_t = [4, 2, 4, 4]
 Generation 7110: Best = 3.405472e+06 FES = 711100 μ_F = 0.750 μ_{CR} = 0.740
 Diversity = 3.0967e-02 k_t = [1, 1, 1, 1]
 Generation 7120: Best = 3.405408e+06 FES = 712100 μ_F = 0.746 μ_{CR} = 0.744
 Diversity = 2.3357e-02 k_t = [2, 2, 4, 3]
 Generation 7130: Best = 3.405379e+06 FES = 713100 μ_F = 0.751 μ_{CR} = 0.739
 Diversity = 2.6242e-02 k_t = [1, 1, 1, 1]
 Generation 7140: Best = 3.405346e+06 FES = 714100 μ_F = 0.732 μ_{CR} = 0.736
 Diversity = 1.9132e-02 k_t = [4, 4, 2, 4]
 Generation 7150: Best = 3.405314e+06 FES = 715100 μ_F = 0.727 μ_{CR} = 0.735
 Diversity = 2.0809e-02 k_t = [1, 1, 1, 1]
 Generation 7160: Best = 3.405281e+06 FES = 716100 μ_F = 0.734 μ_{CR} = 0.725
 Diversity = 1.5827e-02 k_t = [2, 4, 3, 3]
 Generation 7170: Best = 3.405245e+06 FES = 717100 μ_F = 0.733 μ_{CR} = 0.726
 Diversity = 1.8789e-02 k_t = [1, 1, 1, 1]
 Generation 7180: Best = 3.405199e+06 FES = 718100 μ_F = 0.734 μ_{CR} = 0.730
 Diversity = 1.7639e-02 k_t = [4, 2, 3, 4]
 Generation 7190: Best = 3.405172e+06 FES = 719100 μ_F = 0.721 μ_{CR} = 0.724
 Diversity = 1.7273e-02 k_t = [1, 1, 1, 1]
 Generation 7200: Best = 3.405129e+06 FES = 720100 μ_F = 0.719 μ_{CR} = 0.724
 Diversity = 1.5584e-02 k_t = [3, 3, 3, 2]
 Generation 7210: Best = 3.405070e+06 FES = 721100 μ_F = 0.732 μ_{CR} = 0.721
 Diversity = 2.0186e-02 k_t = [1, 1, 1, 1]
 Generation 7220: Best = 3.405027e+06 FES = 722100 μ_F = 0.735 μ_{CR} = 0.727
 Diversity = 2.0811e-02 k_t = [4, 3, 2, 2]
 Generation 7230: Best = 3.404987e+06 FES = 723100 μ_F = 0.746 μ_{CR} = 0.731
 Diversity = 1.8823e-02 k_t = [1, 1, 1, 1]
 Generation 7240: Best = 3.404933e+06 FES = 724100 μ_F = 0.745 μ_{CR} = 0.733
 Diversity = 1.6300e-02 k_t = [3, 2, 3, 4]

Generation 7250: Best = 3.404881e+06 FES = 725100 μ_F = 0.756 μ_{CR} = 0.733
 Diversity = 2.0343e-02 k_t = [1, 1, 1, 1]
 Generation 7260: Best = 3.404723e+06 FES = 726100 μ_F = 0.763 μ_{CR} = 0.730
 Diversity = 2.1306e-02 k_t = [4, 3, 2, 3]
 Generation 7270: Best = 3.404665e+06 FES = 727100 μ_F = 0.786 μ_{CR} = 0.727
 Diversity = 2.7465e-02 k_t = [1, 1, 1, 1]
 Generation 7280: Best = 3.404559e+06 FES = 728100 μ_F = 0.775 μ_{CR} = 0.718
 Diversity = 2.8180e-02 k_t = [2, 3, 2, 2]
 Generation 7290: Best = 3.404453e+06 FES = 729100 μ_F = 0.768 μ_{CR} = 0.716
 Diversity = 3.3533e-02 k_t = [1, 1, 1, 1]
 Generation 7300: Best = 3.404378e+06 FES = 730100 μ_F = 0.759 μ_{CR} = 0.714
 Diversity = 2.9359e-02 k_t = [3, 4, 3, 3]
 Generation 7310: Best = 3.404306e+06 FES = 731100 μ_F = 0.755 μ_{CR} = 0.708
 Diversity = 2.5700e-02 k_t = [1, 1, 1, 1]
 Generation 7320: Best = 3.404239e+06 FES = 732100 μ_F = 0.756 μ_{CR} = 0.693
 Diversity = 2.3395e-02 k_t = [3, 4, 3, 2]
 Generation 7330: Best = 3.404195e+06 FES = 733100 μ_F = 0.750 μ_{CR} = 0.697
 Diversity = 2.2172e-02 k_t = [1, 1, 1, 1]
 Generation 7340: Best = 3.404133e+06 FES = 734100 μ_F = 0.762 μ_{CR} = 0.697
 Diversity = 2.1917e-02 k_t = [3, 4, 2, 4]
 Generation 7350: Best = 3.404021e+06 FES = 735100 μ_F = 0.772 μ_{CR} = 0.694
 Diversity = 3.0460e-02 k_t = [1, 1, 1, 1]
 Generation 7360: Best = 3.403946e+06 FES = 736100 μ_F = 0.757 μ_{CR} = 0.696
 Diversity = 3.1174e-02 k_t = [2, 2, 3, 2]
 Generation 7370: Best = 3.403850e+06 FES = 737100 μ_F = 0.754 μ_{CR} = 0.687
 Diversity = 3.2793e-02 k_t = [1, 1, 1, 1]
 Generation 7380: Best = 3.403759e+06 FES = 738100 μ_F = 0.756 μ_{CR} = 0.692
 Diversity = 2.6791e-02 k_t = [3, 3, 3, 3]
 Generation 7390: Best = 3.403681e+06 FES = 739100 μ_F = 0.746 μ_{CR} = 0.685
 Diversity = 2.9490e-02 k_t = [1, 1, 1, 1]
 Generation 7400: Best = 3.403613e+06 FES = 740100 μ_F = 0.745 μ_{CR} = 0.682
 Diversity = 2.3415e-02 k_t = [3, 3, 2, 4]
 Generation 7410: Best = 3.403569e+06 FES = 741100 μ_F = 0.754 μ_{CR} = 0.674
 Diversity = 2.9753e-02 k_t = [1, 1, 1, 1]
 Generation 7420: Best = 3.403484e+06 FES = 742100 μ_F = 0.749 μ_{CR} = 0.687
 Diversity = 2.3529e-02 k_t = [2, 3, 4, 3]
 Generation 7430: Best = 3.403430e+06 FES = 743100 μ_F = 0.737 μ_{CR} = 0.687
 Diversity = 2.7762e-02 k_t = [1, 1, 1, 1]
 Generation 7440: Best = 3.403354e+06 FES = 744100 μ_F = 0.729 μ_{CR} = 0.662
 Diversity = 2.4707e-02 k_t = [4, 4, 3, 4]
 Generation 7450: Best = 3.403304e+06 FES = 745100 μ_F = 0.723 μ_{CR} = 0.669
 Diversity = 2.1987e-02 k_t = [1, 1, 1, 1]
 Generation 7460: Best = 3.403264e+06 FES = 746100 μ_F = 0.728 μ_{CR} = 0.674
 Diversity = 1.9914e-02 k_t = [4, 2, 4, 3]
 Generation 7470: Best = 3.403227e+06 FES = 747100 μ_F = 0.721 μ_{CR} = 0.674
 Diversity = 1.9549e-02 k_t = [1, 1, 1, 1]
 Generation 7480: Best = 3.403182e+06 FES = 748100 μ_F = 0.722 μ_{CR} = 0.681
 Diversity = 1.6837e-02 k_t = [3, 2, 3, 3]
 Generation 7490: Best = 3.403137e+06 FES = 749100 μ_F = 0.725 μ_{CR} = 0.683
 Diversity = 2.3339e-02 k_t = [1, 1, 1, 1]
 Generation 7500: Best = 3.403100e+06 FES = 750100 μ_F = 0.730 μ_{CR} = 0.683
 Diversity = 1.8636e-02 k_t = [3, 3, 4, 3]
 Generation 7510: Best = 3.403074e+06 FES = 751100 μ_F = 0.722 μ_{CR} = 0.680
 Diversity = 1.8630e-02 k_t = [1, 1, 1, 1]

Generation 7520: Best = 3.403020e+06 FES = 752100 μ_F = 0.735 μ_{CR} = 0.692
 Diversity = 1.6122e-02 k_t = [4, 4, 2, 4]
 Generation 7530: Best = 3.402964e+06 FES = 753100 μ_F = 0.752 μ_{CR} = 0.686
 Diversity = 2.1978e-02 k_t = [1, 1, 1, 1]
 Generation 7540: Best = 3.402906e+06 FES = 754100 μ_F = 0.760 μ_{CR} = 0.694
 Diversity = 2.6813e-02 k_t = [3, 2, 3, 4]
 Generation 7550: Best = 3.402854e+06 FES = 755100 μ_F = 0.769 μ_{CR} = 0.698
 Diversity = 3.1881e-02 k_t = [1, 1, 1, 1]
 Generation 7560: Best = 3.402762e+06 FES = 756100 μ_F = 0.745 μ_{CR} = 0.709
 Diversity = 2.9596e-02 k_t = [2, 4, 3, 3]
 Generation 7570: Best = 3.402670e+06 FES = 757100 μ_F = 0.743 μ_{CR} = 0.722
 Diversity = 3.4397e-02 k_t = [1, 1, 1, 1]
 Generation 7580: Best = 3.402604e+06 FES = 758100 μ_F = 0.734 μ_{CR} = 0.721
 Diversity = 2.6944e-02 k_t = [2, 3, 3, 4]
 Generation 7590: Best = 3.402510e+06 FES = 759100 μ_F = 0.747 μ_{CR} = 0.718
 Diversity = 3.4105e-02 k_t = [1, 1, 1, 1]
 Generation 7600: Best = 3.402477e+06 FES = 760100 μ_F = 0.745 μ_{CR} = 0.721
 Diversity = 3.3572e-02 k_t = [4, 2, 2, 2]
 Generation 7610: Best = 3.402421e+06 FES = 761100 μ_F = 0.744 μ_{CR} = 0.721
 Diversity = 3.2493e-02 k_t = [1, 1, 1, 1]
 Generation 7620: Best = 3.402375e+06 FES = 762100 μ_F = 0.737 μ_{CR} = 0.717
 Diversity = 2.9465e-02 k_t = [2, 4, 4, 3]
 Generation 7630: Best = 3.402294e+06 FES = 763100 μ_F = 0.727 μ_{CR} = 0.722
 Diversity = 3.9295e-02 k_t = [1, 1, 1, 1]
 Generation 7640: Best = 3.402243e+06 FES = 764100 μ_F = 0.705 μ_{CR} = 0.726
 Diversity = 3.6491e-02 k_t = [4, 2, 4, 2]
 Generation 7650: Best = 3.402177e+06 FES = 765100 μ_F = 0.709 μ_{CR} = 0.729
 Diversity = 3.5429e-02 k_t = [1, 1, 1, 1]
 Generation 7660: Best = 3.402169e+06 FES = 766100 μ_F = 0.693 μ_{CR} = 0.728
 Diversity = 2.6625e-02 k_t = [3, 3, 4, 4]
 Generation 7670: Best = 3.402135e+06 FES = 767100 μ_F = 0.698 μ_{CR} = 0.739
 Diversity = 2.5117e-02 k_t = [1, 1, 1, 1]
 Generation 7680: Best = 3.402091e+06 FES = 768100 μ_F = 0.718 μ_{CR} = 0.748
 Diversity = 2.9181e-02 k_t = [3, 2, 2, 2]
 Generation 7690: Best = 3.402068e+06 FES = 769100 μ_F = 0.706 μ_{CR} = 0.755
 Diversity = 2.7246e-02 k_t = [1, 1, 1, 1]
 Generation 7700: Best = 3.402045e+06 FES = 770100 μ_F = 0.718 μ_{CR} = 0.760
 Diversity = 2.6826e-02 k_t = [3, 3, 4, 3]
 Generation 7710: Best = 3.402035e+06 FES = 771100 μ_F = 0.704 μ_{CR} = 0.766
 Diversity = 2.2288e-02 k_t = [1, 1, 1, 1]
 Generation 7720: Best = 3.402022e+06 FES = 772100 μ_F = 0.699 μ_{CR} = 0.778
 Diversity = 1.7125e-02 k_t = [2, 4, 3, 2]
 Generation 7730: Best = 3.402010e+06 FES = 773100 μ_F = 0.696 μ_{CR} = 0.785
 Diversity = 1.6128e-02 k_t = [1, 1, 1, 1]
 Generation 7740: Best = 3.402001e+06 FES = 774100 μ_F = 0.710 μ_{CR} = 0.794
 Diversity = 1.2120e-02 k_t = [4, 3, 4, 2]
 Generation 7750: Best = 3.401998e+06 FES = 775100 μ_F = 0.722 μ_{CR} = 0.815
 Diversity = 9.3233e-03 k_t = [1, 1, 1, 1]
 Generation 7760: Best = 3.401994e+06 FES = 776100 μ_F = 0.729 μ_{CR} = 0.826
 Diversity = 8.4208e-03 k_t = [3, 2, 3, 3]
 Generation 7770: Best = 3.401988e+06 FES = 777100 μ_F = 0.728 μ_{CR} = 0.830
 Diversity = 1.0019e-02 k_t = [1, 1, 1, 1]
 Generation 7780: Best = 3.401980e+06 FES = 778100 μ_F = 0.725 μ_{CR} = 0.829
 Diversity = 1.2274e-02 k_t = [2, 4, 2, 2]

Generation 7790: Best = 3.401972e+06 FES = 779100 μ_F = 0.721 μ_{CR} = 0.839
 Diversity = 1.3540e-02 k_t = [1, 1, 1, 1]
 Generation 7800: Best = 3.401969e+06 FES = 780100 μ_F = 0.715 μ_{CR} = 0.844
 Diversity = 1.0582e-02 k_t = [3, 3, 4, 3]
 Generation 7810: Best = 3.401966e+06 FES = 781100 μ_F = 0.713 μ_{CR} = 0.848
 Diversity = 9.6176e-03 k_t = [1, 1, 1, 1]
 Generation 7820: Best = 3.401961e+06 FES = 782100 μ_F = 0.719 μ_{CR} = 0.843
 Diversity = 6.9138e-03 k_t = [4, 4, 4, 3]
 Generation 7830: Best = 3.401958e+06 FES = 783100 μ_F = 0.737 μ_{CR} = 0.844
 Diversity = 7.1847e-03 k_t = [1, 1, 1, 1]
 Generation 7840: Best = 3.401948e+06 FES = 784100 μ_F = 0.742 μ_{CR} = 0.844
 Diversity = 8.5835e-03 k_t = [4, 4, 4, 4]
 Generation 7850: Best = 3.401943e+06 FES = 785100 μ_F = 0.755 μ_{CR} = 0.854
 Diversity = 9.3852e-03 k_t = [1, 1, 1, 1]
 Generation 7860: Best = 3.401938e+06 FES = 786100 μ_F = 0.758 μ_{CR} = 0.857
 Diversity = 7.8035e-03 k_t = [2, 4, 4, 2]
 Generation 7870: Best = 3.401935e+06 FES = 787100 μ_F = 0.767 μ_{CR} = 0.857
 Diversity = 8.3413e-03 k_t = [1, 1, 1, 1]
 Generation 7880: Best = 3.401931e+06 FES = 788100 μ_F = 0.759 μ_{CR} = 0.865
 Diversity = 6.4111e-03 k_t = [3, 4, 4, 2]
 Generation 7890: Best = 3.401929e+06 FES = 789100 μ_F = 0.752 μ_{CR} = 0.870
 Diversity = 6.5359e-03 k_t = [1, 1, 1, 1]
 Generation 7900: Best = 3.401927e+06 FES = 790100 μ_F = 0.748 μ_{CR} = 0.852
 Diversity = 6.0401e-03 k_t = [2, 3, 3, 3]
 Generation 7910: Best = 3.401924e+06 FES = 791100 μ_F = 0.742 μ_{CR} = 0.859
 Diversity = 7.2819e-03 k_t = [1, 1, 1, 1]
 Generation 7920: Best = 3.401923e+06 FES = 792100 μ_F = 0.727 μ_{CR} = 0.863
 Diversity = 4.7937e-03 k_t = [2, 2, 4, 4]
 Generation 7930: Best = 3.401922e+06 FES = 793100 μ_F = 0.741 μ_{CR} = 0.859
 Diversity = 4.9514e-03 k_t = [1, 1, 1, 1]
 Generation 7940: Best = 3.401920e+06 FES = 794100 μ_F = 0.753 μ_{CR} = 0.863
 Diversity = 3.7752e-03 k_t = [3, 2, 4, 4]
 Generation 7950: Best = 3.401918e+06 FES = 795100 μ_F = 0.765 μ_{CR} = 0.858
 Diversity = 4.8812e-03 k_t = [1, 1, 1, 1]
 Generation 7960: Best = 3.401916e+06 FES = 796100 μ_F = 0.773 μ_{CR} = 0.863
 Diversity = 6.0473e-03 k_t = [3, 3, 2, 3]
 Generation 7970: Best = 3.401912e+06 FES = 797100 μ_F = 0.774 μ_{CR} = 0.868
 Diversity = 7.8527e-03 k_t = [1, 1, 1, 1]
 Generation 7980: Best = 3.401908e+06 FES = 798100 μ_F = 0.767 μ_{CR} = 0.865
 Diversity = 9.0944e-03 k_t = [2, 2, 4, 3]
 Generation 7990: Best = 3.401903e+06 FES = 799100 μ_F = 0.756 μ_{CR} = 0.851
 Diversity = 1.0874e-02 k_t = [1, 1, 1, 1]
 Generation 8000: Best = 3.401898e+06 FES = 800100 μ_F = 0.751 μ_{CR} = 0.853
 Diversity = 9.4677e-03 k_t = [2, 2, 4, 4]
 Generation 8010: Best = 3.401894e+06 FES = 801100 μ_F = 0.740 μ_{CR} = 0.864
 Diversity = 8.3702e-03 k_t = [1, 1, 1, 1]
 Generation 8020: Best = 3.401886e+06 FES = 802100 μ_F = 0.749 μ_{CR} = 0.873
 Diversity = 8.9252e-03 k_t = [4, 4, 2, 2]
 Generation 8030: Best = 3.401881e+06 FES = 803100 μ_F = 0.751 μ_{CR} = 0.877
 Diversity = 9.2101e-03 k_t = [1, 1, 1, 1]
 Generation 8040: Best = 3.401876e+06 FES = 804100 μ_F = 0.745 μ_{CR} = 0.881
 Diversity = 8.0892e-03 k_t = [2, 4, 4, 2]
 Generation 8050: Best = 3.401869e+06 FES = 805100 μ_F = 0.749 μ_{CR} = 0.877
 Diversity = 8.5632e-03 k_t = [1, 1, 1, 1]

Generation 8060: Best = 3.401863e+06 FES = 806100 μ_F = 0.743 μ_{CR} = 0.880
 Diversity = 7.5728e-03 k_t = [2, 2, 2, 2]
 Generation 8070: Best = 3.401861e+06 FES = 807100 μ_F = 0.732 μ_{CR} = 0.870
 Diversity = 9.1094e-03 k_t = [1, 1, 1, 1]
 Generation 8080: Best = 3.401857e+06 FES = 808100 μ_F = 0.739 μ_{CR} = 0.867
 Diversity = 5.6497e-03 k_t = [4, 4, 4, 4]
 Generation 8090: Best = 3.401852e+06 FES = 809100 μ_F = 0.747 μ_{CR} = 0.872
 Diversity = 7.5209e-03 k_t = [1, 1, 1, 1]
 Generation 8100: Best = 3.401850e+06 FES = 810100 μ_F = 0.735 μ_{CR} = 0.870
 Diversity = 5.9149e-03 k_t = [3, 2, 4, 2]
 Generation 8110: Best = 3.401847e+06 FES = 811100 μ_F = 0.740 μ_{CR} = 0.867
 Diversity = 6.3273e-03 k_t = [1, 1, 1, 1]
 Generation 8120: Best = 3.401843e+06 FES = 812100 μ_F = 0.726 μ_{CR} = 0.869
 Diversity = 5.8921e-03 k_t = [3, 3, 2, 3]
 Generation 8130: Best = 3.401841e+06 FES = 813100 μ_F = 0.730 μ_{CR} = 0.875
 Diversity = 7.2756e-03 k_t = [1, 1, 1, 1]
 Generation 8140: Best = 3.401839e+06 FES = 814100 μ_F = 0.744 μ_{CR} = 0.879
 Diversity = 5.2877e-03 k_t = [4, 3, 4, 2]
 Generation 8150: Best = 3.401835e+06 FES = 815100 μ_F = 0.729 μ_{CR} = 0.887
 Diversity = 7.8484e-03 k_t = [1, 1, 1, 1]
 Generation 8160: Best = 3.401833e+06 FES = 816100 μ_F = 0.725 μ_{CR} = 0.888
 Diversity = 6.5396e-03 k_t = [3, 4, 2, 2]
 Generation 8170: Best = 3.401832e+06 FES = 817100 μ_F = 0.723 μ_{CR} = 0.891
 Diversity = 5.4533e-03 k_t = [1, 1, 1, 1]
 Generation 8180: Best = 3.401830e+06 FES = 818100 μ_F = 0.731 μ_{CR} = 0.892
 Diversity = 4.3087e-03 k_t = [2, 4, 3, 2]
 Generation 8190: Best = 3.401829e+06 FES = 819100 μ_F = 0.730 μ_{CR} = 0.890
 Diversity = 3.9965e-03 k_t = [1, 1, 1, 1]
 Generation 8200: Best = 3.401828e+06 FES = 820100 μ_F = 0.733 μ_{CR} = 0.899
 Diversity = 3.9043e-03 k_t = [2, 4, 2, 3]
 Generation 8210: Best = 3.401827e+06 FES = 821100 μ_F = 0.739 μ_{CR} = 0.888
 Diversity = 5.3395e-03 k_t = [1, 1, 1, 1]
 Generation 8220: Best = 3.401826e+06 FES = 822100 μ_F = 0.746 μ_{CR} = 0.890
 Diversity = 4.0830e-03 k_t = [4, 2, 2, 3]
 Generation 8230: Best = 3.401825e+06 FES = 823100 μ_F = 0.746 μ_{CR} = 0.888
 Diversity = 4.3317e-03 k_t = [1, 1, 1, 1]
 Generation 8240: Best = 3.401823e+06 FES = 824100 μ_F = 0.759 μ_{CR} = 0.892
 Diversity = 3.5370e-03 k_t = [4, 4, 4, 3]
 Generation 8250: Best = 3.401822e+06 FES = 825100 μ_F = 0.754 μ_{CR} = 0.895
 Diversity = 3.7139e-03 k_t = [1, 1, 1, 1]
 Generation 8260: Best = 3.401821e+06 FES = 826100 μ_F = 0.760 μ_{CR} = 0.902
 Diversity = 2.9419e-03 k_t = [2, 4, 4, 3]
 Generation 8270: Best = 3.401821e+06 FES = 827100 μ_F = 0.754 μ_{CR} = 0.894
 Diversity = 3.7620e-03 k_t = [1, 1, 1, 1]
 Generation 8280: Best = 3.401820e+06 FES = 828100 μ_F = 0.745 μ_{CR} = 0.899
 Diversity = 2.7321e-03 k_t = [3, 2, 2, 3]
 Generation 8290: Best = 3.401820e+06 FES = 829100 μ_F = 0.741 μ_{CR} = 0.898
 Diversity = 2.5176e-03 k_t = [1, 1, 1, 1]
 Generation 8300: Best = 3.401820e+06 FES = 830100 μ_F = 0.722 μ_{CR} = 0.894
 Diversity = 2.0935e-03 k_t = [2, 3, 3, 3]
 Generation 8310: Best = 3.401819e+06 FES = 831100 μ_F = 0.735 μ_{CR} = 0.887
 Diversity = 2.3147e-03 k_t = [1, 1, 1, 1]
 Generation 8320: Best = 3.401819e+06 FES = 832100 μ_F = 0.735 μ_{CR} = 0.888
 Diversity = 2.3357e-03 k_t = [3, 3, 2, 4]

Generation 8330: Best = 3.401819e+06 FES = 833100 μ_F = 0.741 μ_{CR} = 0.891
 Diversity = 2.1479e-03 k_t = [1, 1, 1, 1]
 Generation 8340: Best = 3.401819e+06 FES = 834100 μ_F = 0.743 μ_{CR} = 0.888
 Diversity = 1.5288e-03 k_t = [2, 2, 4, 4]
 Generation 8350: Best = 3.401819e+06 FES = 835100 μ_F = 0.743 μ_{CR} = 0.893
 Diversity = 1.7287e-03 k_t = [1, 1, 1, 1]
 Generation 8360: Best = 3.401818e+06 FES = 836100 μ_F = 0.745 μ_{CR} = 0.895
 Diversity = 1.2854e-03 k_t = [4, 3, 4, 4]
 Generation 8370: Best = 3.401818e+06 FES = 837100 μ_F = 0.751 μ_{CR} = 0.889
 Diversity = 1.4728e-03 k_t = [1, 1, 1, 1]
 Generation 8380: Best = 3.401818e+06 FES = 838100 μ_F = 0.754 μ_{CR} = 0.881
 Diversity = 1.4554e-03 k_t = [4, 4, 3, 4]
 Generation 8390: Best = 3.401818e+06 FES = 839100 μ_F = 0.759 μ_{CR} = 0.870
 Diversity = 1.5414e-03 k_t = [1, 1, 1, 1]
 Generation 8400: Best = 3.401818e+06 FES = 840100 μ_F = 0.772 μ_{CR} = 0.863
 Diversity = 1.5224e-03 k_t = [4, 4, 3, 2]
 Generation 8410: Best = 3.401817e+06 FES = 841100 μ_F = 0.767 μ_{CR} = 0.868
 Diversity = 1.8962e-03 k_t = [1, 1, 1, 1]
 Generation 8420: Best = 3.401817e+06 FES = 842100 μ_F = 0.762 μ_{CR} = 0.877
 Diversity = 2.1599e-03 k_t = [2, 3, 2, 2]
 Generation 8430: Best = 3.401817e+06 FES = 843100 μ_F = 0.753 μ_{CR} = 0.878
 Diversity = 2.0058e-03 k_t = [1, 1, 1, 1]
 Generation 8440: Best = 3.401817e+06 FES = 844100 μ_F = 0.755 μ_{CR} = 0.878
 Diversity = 1.1559e-03 k_t = [3, 3, 2, 3]
 Generation 8450: Best = 3.401817e+06 FES = 845100 μ_F = 0.759 μ_{CR} = 0.880
 Diversity = 1.8185e-03 k_t = [1, 1, 1, 1]
 Generation 8460: Best = 3.401816e+06 FES = 846100 μ_F = 0.759 μ_{CR} = 0.883
 Diversity = 1.8124e-03 k_t = [3, 3, 4, 2]
 Generation 8470: Best = 3.401816e+06 FES = 847100 μ_F = 0.757 μ_{CR} = 0.886
 Diversity = 2.1862e-03 k_t = [1, 1, 1, 1]
 Generation 8480: Best = 3.401816e+06 FES = 848100 μ_F = 0.745 μ_{CR} = 0.891
 Diversity = 1.7285e-03 k_t = [2, 3, 2, 2]
 Generation 8490: Best = 3.401816e+06 FES = 849100 μ_F = 0.747 μ_{CR} = 0.885
 Diversity = 2.1300e-03 k_t = [1, 1, 1, 1]
 Generation 8500: Best = 3.401816e+06 FES = 850100 μ_F = 0.738 μ_{CR} = 0.888
 Diversity = 1.7306e-03 k_t = [2, 3, 2, 3]
 Generation 8510: Best = 3.401816e+06 FES = 851100 μ_F = 0.748 μ_{CR} = 0.883
 Diversity = 1.8293e-03 k_t = [1, 1, 1, 1]
 Generation 8520: Best = 3.401815e+06 FES = 852100 μ_F = 0.749 μ_{CR} = 0.896
 Diversity = 1.5219e-03 k_t = [3, 4, 4, 2]
 Generation 8530: Best = 3.401815e+06 FES = 853100 μ_F = 0.737 μ_{CR} = 0.885
 Diversity = 1.4626e-03 k_t = [1, 1, 1, 1]
 Generation 8540: Best = 3.401815e+06 FES = 854100 μ_F = 0.739 μ_{CR} = 0.887
 Diversity = 1.2683e-03 k_t = [4, 2, 2, 4]
 Generation 8550: Best = 3.401815e+06 FES = 855100 μ_F = 0.738 μ_{CR} = 0.899
 Diversity = 1.3590e-03 k_t = [1, 1, 1, 1]
 Generation 8560: Best = 3.401815e+06 FES = 856100 μ_F = 0.744 μ_{CR} = 0.899
 Diversity = 1.1593e-03 k_t = [2, 2, 4, 4]
 Generation 8570: Best = 3.401815e+06 FES = 857100 μ_F = 0.735 μ_{CR} = 0.903
 Diversity = 9.0945e-04 k_t = [1, 1, 1, 1]
 Generation 8580: Best = 3.401815e+06 FES = 858100 μ_F = 0.744 μ_{CR} = 0.907
 Diversity = 9.1011e-04 k_t = [2, 2, 2, 4]
 Generation 8590: Best = 3.401815e+06 FES = 859100 μ_F = 0.750 μ_{CR} = 0.908
 Diversity = 1.2037e-03 k_t = [1, 1, 1, 1]

Generation 8600: Best = 3.401815e+06 FES = 860100 μ_F = 0.742 μ_{CR} = 0.914
 Diversity = 6.5032e-04 k_t = [4, 2, 4, 3]
 Generation 8610: Best = 3.401815e+06 FES = 861100 μ_F = 0.737 μ_{CR} = 0.899
 Diversity = 5.6672e-04 k_t = [1, 1, 1, 1]
 Generation 8620: Best = 3.401815e+06 FES = 862100 μ_F = 0.734 μ_{CR} = 0.908
 Diversity = 4.8585e-04 k_t = [2, 2, 4, 4]
 Generation 8630: Best = 3.401815e+06 FES = 863100 μ_F = 0.730 μ_{CR} = 0.895
 Diversity = 5.1875e-04 k_t = [1, 1, 1, 1]
 Generation 8640: Best = 3.401815e+06 FES = 864100 μ_F = 0.726 μ_{CR} = 0.901
 Diversity = 4.5572e-04 k_t = [2, 3, 4, 3]
 Generation 8650: Best = 3.401815e+06 FES = 865100 μ_F = 0.717 μ_{CR} = 0.904
 Diversity = 4.1752e-04 k_t = [1, 1, 1, 1]
 Generation 8660: Best = 3.401815e+06 FES = 866100 μ_F = 0.714 μ_{CR} = 0.901
 Diversity = 3.3484e-04 k_t = [2, 3, 2, 3]
 Generation 8670: Best = 3.401815e+06 FES = 867100 μ_F = 0.727 μ_{CR} = 0.908
 Diversity = 3.6196e-04 k_t = [1, 1, 1, 1]
 Generation 8680: Best = 3.401815e+06 FES = 868100 μ_F = 0.734 μ_{CR} = 0.896
 Diversity = 3.7183e-04 k_t = [3, 3, 3, 3]
 Generation 8690: Best = 3.401815e+06 FES = 869100 μ_F = 0.727 μ_{CR} = 0.893
 Diversity = 4.0542e-04 k_t = [1, 1, 1, 1]
 Generation 8700: Best = 3.401815e+06 FES = 870100 μ_F = 0.729 μ_{CR} = 0.885
 Diversity = 4.5943e-04 k_t = [3, 2, 3, 4]
 Generation 8710: Best = 3.401815e+06 FES = 871100 μ_F = 0.722 μ_{CR} = 0.895
 Diversity = 4.4582e-04 k_t = [1, 1, 1, 1]
 Generation 8720: Best = 3.401815e+06 FES = 872100 μ_F = 0.727 μ_{CR} = 0.897
 Diversity = 3.8243e-04 k_t = [4, 2, 3, 3]
 Generation 8730: Best = 3.401815e+06 FES = 873100 μ_F = 0.734 μ_{CR} = 0.897
 Diversity = 4.0846e-04 k_t = [1, 1, 1, 1]
 Generation 8740: Best = 3.401815e+06 FES = 874100 μ_F = 0.743 μ_{CR} = 0.900
 Diversity = 3.2582e-04 k_t = [4, 2, 3, 3]
 Generation 8750: Best = 3.401815e+06 FES = 875100 μ_F = 0.756 μ_{CR} = 0.906
 Diversity = 2.9751e-04 k_t = [1, 1, 1, 1]
 Generation 8760: Best = 3.401815e+06 FES = 876100 μ_F = 0.754 μ_{CR} = 0.893
 Diversity = 2.7091e-04 k_t = [2, 4, 2, 2]
 Generation 8770: Best = 3.401815e+06 FES = 877100 μ_F = 0.731 μ_{CR} = 0.892
 Diversity = 2.4897e-04 k_t = [1, 1, 1, 1]
 Generation 8780: Best = 3.401815e+06 FES = 878100 μ_F = 0.734 μ_{CR} = 0.889
 Diversity = 1.8565e-04 k_t = [4, 2, 4, 3]
 Generation 8790: Best = 3.401815e+06 FES = 879100 μ_F = 0.742 μ_{CR} = 0.894
 Diversity = 1.8165e-04 k_t = [1, 1, 1, 1]
 Generation 8800: Best = 3.401815e+06 FES = 880100 μ_F = 0.762 μ_{CR} = 0.887
 Diversity = 1.9053e-04 k_t = [3, 4, 3, 4]
 Generation 8810: Best = 3.401815e+06 FES = 881100 μ_F = 0.763 μ_{CR} = 0.892
 Diversity = 1.9814e-04 k_t = [1, 1, 1, 1]
 Generation 8820: Best = 3.401815e+06 FES = 882100 μ_F = 0.760 μ_{CR} = 0.894
 Diversity = 1.7720e-04 k_t = [3, 3, 2, 4]
 Generation 8830: Best = 3.401815e+06 FES = 883100 μ_F = 0.759 μ_{CR} = 0.883
 Diversity = 1.9222e-04 k_t = [1, 1, 1, 1]
 Generation 8840: Best = 3.401815e+06 FES = 884100 μ_F = 0.761 μ_{CR} = 0.879
 Diversity = 2.1500e-04 k_t = [4, 3, 3, 2]
 Generation 8850: Best = 3.401815e+06 FES = 885100 μ_F = 0.759 μ_{CR} = 0.874
 Diversity = 2.3496e-04 k_t = [1, 1, 1, 1]
 Generation 8860: Best = 3.401815e+06 FES = 886100 μ_F = 0.758 μ_{CR} = 0.871
 Diversity = 2.8672e-04 k_t = [3, 2, 3, 4]

Generation 8870: Best = 3.401815e+06 FES = 887100 μ_F = 0.758 μ_{CR} = 0.870
 Diversity = 2.9150e-04 k_t = [1, 1, 1, 1]
 Generation 8880: Best = 3.401815e+06 FES = 888100 μ_F = 0.757 μ_{CR} = 0.878
 Diversity = 3.1238e-04 k_t = [3, 2, 3, 3]
 Generation 8890: Best = 3.401815e+06 FES = 889100 μ_F = 0.751 μ_{CR} = 0.881
 Diversity = 3.4295e-04 k_t = [1, 1, 1, 1]
 Generation 8900: Best = 3.401815e+06 FES = 890100 μ_F = 0.761 μ_{CR} = 0.884
 Diversity = 2.8757e-04 k_t = [3, 2, 4, 4]
 Generation 8910: Best = 3.401815e+06 FES = 891100 μ_F = 0.763 μ_{CR} = 0.885
 Diversity = 3.5128e-04 k_t = [1, 1, 1, 1]
 Generation 8920: Best = 3.401815e+06 FES = 892100 μ_F = 0.745 μ_{CR} = 0.896
 Diversity = 3.5170e-04 k_t = [4, 4, 4, 2]
 Generation 8930: Best = 3.401815e+06 FES = 893100 μ_F = 0.747 μ_{CR} = 0.896
 Diversity = 3.3672e-04 k_t = [1, 1, 1, 1]
 Generation 8940: Best = 3.401815e+06 FES = 894100 μ_F = 0.750 μ_{CR} = 0.901
 Diversity = 3.3934e-04 k_t = [2, 3, 2, 4]
 Generation 8950: Best = 3.401815e+06 FES = 895100 μ_F = 0.746 μ_{CR} = 0.896
 Diversity = 3.1302e-04 k_t = [1, 1, 1, 1]
 Generation 8960: Best = 3.401815e+06 FES = 896100 μ_F = 0.734 μ_{CR} = 0.895
 Diversity = 2.6541e-04 k_t = [4, 2, 3, 3]
 Generation 8970: Best = 3.401815e+06 FES = 897100 μ_F = 0.728 μ_{CR} = 0.897
 Diversity = 2.0898e-04 k_t = [1, 1, 1, 1]
 Generation 8980: Best = 3.401815e+06 FES = 898100 μ_F = 0.727 μ_{CR} = 0.904
 Diversity = 1.8253e-04 k_t = [3, 4, 2, 3]
 Generation 8990: Best = 3.401815e+06 FES = 899100 μ_F = 0.732 μ_{CR} = 0.908
 Diversity = 1.6254e-04 k_t = [1, 1, 1, 1]
 Generation 9000: Best = 3.401815e+06 FES = 900100 μ_F = 0.740 μ_{CR} = 0.903
 Diversity = 1.6151e-04 k_t = [3, 4, 4, 3]
 Generation 9010: Best = 3.401815e+06 FES = 901100 μ_F = 0.752 μ_{CR} = 0.906
 Diversity = 1.8848e-04 k_t = [1, 1, 1, 1]
 Generation 9020: Best = 3.401815e+06 FES = 902100 μ_F = 0.750 μ_{CR} = 0.901
 Diversity = 1.4818e-04 k_t = [2, 3, 4, 2]
 Generation 9030: Best = 3.401815e+06 FES = 903100 μ_F = 0.751 μ_{CR} = 0.915
 Diversity = 1.6657e-04 k_t = [1, 1, 1, 1]
 Generation 9040: Best = 3.401815e+06 FES = 904100 μ_F = 0.754 μ_{CR} = 0.915
 Diversity = 1.7337e-04 k_t = [3, 4, 2, 2]
 Generation 9050: Best = 3.401815e+06 FES = 905100 μ_F = 0.750 μ_{CR} = 0.909
 Diversity = 1.7551e-04 k_t = [1, 1, 1, 1]
 Generation 9060: Best = 3.401815e+06 FES = 906100 μ_F = 0.743 μ_{CR} = 0.914
 Diversity = 1.3949e-04 k_t = [4, 2, 2, 3]
 Generation 9070: Best = 3.401815e+06 FES = 907100 μ_F = 0.760 μ_{CR} = 0.908
 Diversity = 1.4518e-04 k_t = [1, 1, 1, 1]
 Generation 9080: Best = 3.401815e+06 FES = 908100 μ_F = 0.742 μ_{CR} = 0.912
 Diversity = 1.1420e-04 k_t = [2, 3, 3, 3]
 Generation 9090: Best = 3.401815e+06 FES = 909100 μ_F = 0.746 μ_{CR} = 0.908
 Diversity = 1.5223e-04 k_t = [1, 1, 1, 1]
 Generation 9100: Best = 3.401815e+06 FES = 910100 μ_F = 0.739 μ_{CR} = 0.916
 Diversity = 1.5719e-04 k_t = [2, 2, 4, 4]
 Generation 9110: Best = 3.401815e+06 FES = 911100 μ_F = 0.719 μ_{CR} = 0.917
 Diversity = 1.3662e-04 k_t = [1, 1, 1, 1]
 Generation 9120: Best = 3.401815e+06 FES = 912100 μ_F = 0.732 μ_{CR} = 0.916
 Diversity = 1.2502e-04 k_t = [2, 4, 3, 2]
 Generation 9130: Best = 3.401815e+06 FES = 913100 μ_F = 0.725 μ_{CR} = 0.913
 Diversity = 1.1990e-04 k_t = [1, 1, 1, 1]

Generation 9140: Best = 3.401815e+06 FES = 914100 μ_F = 0.735 μ_{CR} = 0.917
 Diversity = 1.1742e-04 k_t = [3, 2, 2, 3]
 Generation 9150: Best = 3.401815e+06 FES = 915100 μ_F = 0.741 μ_{CR} = 0.917
 Diversity = 1.1292e-04 k_t = [1, 1, 1, 1]
 Generation 9160: Best = 3.401815e+06 FES = 916100 μ_F = 0.737 μ_{CR} = 0.902
 Diversity = 1.0438e-04 k_t = [2, 3, 4, 2]
 Generation 9170: Best = 3.401815e+06 FES = 917100 μ_F = 0.730 μ_{CR} = 0.900
 Diversity = 1.1694e-04 k_t = [1, 1, 1, 1]
 Generation 9180: Best = 3.401815e+06 FES = 918100 μ_F = 0.736 μ_{CR} = 0.901
 Diversity = 1.3659e-04 k_t = [4, 2, 2, 2]
 Generation 9190: Best = 3.401815e+06 FES = 919100 μ_F = 0.731 μ_{CR} = 0.896
 Diversity = 1.3672e-04 k_t = [1, 1, 1, 1]
 Generation 9200: Best = 3.401815e+06 FES = 920100 μ_F = 0.728 μ_{CR} = 0.895
 Diversity = 1.3251e-04 k_t = [4, 2, 2, 3]
 Generation 9210: Best = 3.401815e+06 FES = 921100 μ_F = 0.740 μ_{CR} = 0.897
 Diversity = 1.2813e-04 k_t = [1, 1, 1, 1]
 Generation 9220: Best = 3.401815e+06 FES = 922100 μ_F = 0.740 μ_{CR} = 0.903
 Diversity = 9.2579e-05 k_t = [3, 4, 3, 3]
 Generation 9230: Best = 3.401815e+06 FES = 923100 μ_F = 0.753 μ_{CR} = 0.901
 Diversity = 1.1420e-04 k_t = [1, 1, 1, 1]
 Generation 9240: Best = 3.401815e+06 FES = 924100 μ_F = 0.766 μ_{CR} = 0.904
 Diversity = 1.3190e-04 k_t = [2, 3, 4, 3]
 Generation 9250: Best = 3.401815e+06 FES = 925100 μ_F = 0.763 μ_{CR} = 0.908
 Diversity = 1.8921e-04 k_t = [1, 1, 1, 1]
 Generation 9260: Best = 3.401815e+06 FES = 926100 μ_F = 0.750 μ_{CR} = 0.904
 Diversity = 1.5270e-04 k_t = [3, 4, 4, 2]
 Generation 9270: Best = 3.401815e+06 FES = 927100 μ_F = 0.760 μ_{CR} = 0.897
 Diversity = 1.7764e-04 k_t = [1, 1, 1, 1]
 Generation 9280: Best = 3.401815e+06 FES = 928100 μ_F = 0.765 μ_{CR} = 0.890
 Diversity = 2.3250e-04 k_t = [2, 2, 4, 3]
 Generation 9290: Best = 3.401815e+06 FES = 929100 μ_F = 0.761 μ_{CR} = 0.887
 Diversity = 2.2353e-04 k_t = [1, 1, 1, 1]
 Generation 9300: Best = 3.401815e+06 FES = 930100 μ_F = 0.750 μ_{CR} = 0.882
 Diversity = 2.0833e-04 k_t = [3, 3, 3, 2]
 Generation 9310: Best = 3.401815e+06 FES = 931100 μ_F = 0.739 μ_{CR} = 0.886
 Diversity = 2.0729e-04 k_t = [1, 1, 1, 1]
 Generation 9320: Best = 3.401815e+06 FES = 932100 μ_F = 0.744 μ_{CR} = 0.883
 Diversity = 2.0952e-04 k_t = [2, 2, 4, 3]
 Generation 9330: Best = 3.401815e+06 FES = 933100 μ_F = 0.746 μ_{CR} = 0.875
 Diversity = 3.2407e-04 k_t = [1, 1, 1, 1]
 Generation 9340: Best = 3.401815e+06 FES = 934100 μ_F = 0.747 μ_{CR} = 0.887
 Diversity = 2.5763e-04 k_t = [4, 4, 3, 2]
 Generation 9350: Best = 3.401815e+06 FES = 935100 μ_F = 0.756 μ_{CR} = 0.883
 Diversity = 2.2017e-04 k_t = [1, 1, 1, 1]
 Generation 9360: Best = 3.401815e+06 FES = 936100 μ_F = 0.760 μ_{CR} = 0.892
 Diversity = 2.2918e-04 k_t = [3, 4, 3, 3]
 Generation 9370: Best = 3.401815e+06 FES = 937100 μ_F = 0.761 μ_{CR} = 0.889
 Diversity = 3.0008e-04 k_t = [1, 1, 1, 1]
 Generation 9380: Best = 3.401815e+06 FES = 938100 μ_F = 0.766 μ_{CR} = 0.891
 Diversity = 2.6233e-04 k_t = [2, 2, 4, 4]
 Generation 9390: Best = 3.401815e+06 FES = 939100 μ_F = 0.768 μ_{CR} = 0.882
 Diversity = 2.9348e-04 k_t = [1, 1, 1, 1]
 Generation 9400: Best = 3.401815e+06 FES = 940100 μ_F = 0.762 μ_{CR} = 0.882
 Diversity = 3.1085e-04 k_t = [3, 3, 4, 2]

Generation 9410: Best = 3.401815e+06 FES = 941100 μ_F = 0.764 μ_{CR} = 0.890
 Diversity = 2.9137e-04 k_t = [1, 1, 1, 1]
 Generation 9420: Best = 3.401815e+06 FES = 942100 μ_F = 0.776 μ_{CR} = 0.892
 Diversity = 2.9123e-04 k_t = [3, 4, 2, 3]
 Generation 9430: Best = 3.401815e+06 FES = 943100 μ_F = 0.773 μ_{CR} = 0.894
 Diversity = 3.2596e-04 k_t = [1, 1, 1, 1]
 Generation 9440: Best = 3.401815e+06 FES = 944100 μ_F = 0.755 μ_{CR} = 0.902
 Diversity = 2.6254e-04 k_t = [2, 3, 3, 3]
 Generation 9450: Best = 3.401815e+06 FES = 945100 μ_F = 0.750 μ_{CR} = 0.894
 Diversity = 4.2923e-04 k_t = [1, 1, 1, 1]
 Generation 9460: Best = 3.401815e+06 FES = 946100 μ_F = 0.750 μ_{CR} = 0.890
 Diversity = 2.8806e-04 k_t = [2, 4, 4, 4]
 Generation 9470: Best = 3.401815e+06 FES = 947100 μ_F = 0.741 μ_{CR} = 0.896
 Diversity = 2.7366e-04 k_t = [1, 1, 1, 1]
 Generation 9480: Best = 3.401814e+06 FES = 948100 μ_F = 0.732 μ_{CR} = 0.896
 Diversity = 2.9549e-04 k_t = [3, 4, 2, 3]
 Generation 9490: Best = 3.401814e+06 FES = 949100 μ_F = 0.746 μ_{CR} = 0.898
 Diversity = 3.1339e-04 k_t = [1, 1, 1, 1]
 Generation 9500: Best = 3.401814e+06 FES = 950100 μ_F = 0.739 μ_{CR} = 0.898
 Diversity = 2.7576e-04 k_t = [2, 4, 3, 4]
 Generation 9510: Best = 3.401814e+06 FES = 951100 μ_F = 0.744 μ_{CR} = 0.895
 Diversity = 3.0607e-04 k_t = [1, 1, 1, 1]
 Generation 9520: Best = 3.401814e+06 FES = 952100 μ_F = 0.740 μ_{CR} = 0.901
 Diversity = 3.2178e-04 k_t = [3, 4, 4, 3]
 Generation 9530: Best = 3.401814e+06 FES = 953100 μ_F = 0.734 μ_{CR} = 0.896
 Diversity = 3.3739e-04 k_t = [1, 1, 1, 1]
 Generation 9540: Best = 3.401814e+06 FES = 954100 μ_F = 0.737 μ_{CR} = 0.895
 Diversity = 3.2745e-04 k_t = [4, 3, 3, 4]
 Generation 9550: Best = 3.401814e+06 FES = 955100 μ_F = 0.749 μ_{CR} = 0.900
 Diversity = 3.6578e-04 k_t = [1, 1, 1, 1]
 Generation 9560: Best = 3.401814e+06 FES = 956100 μ_F = 0.732 μ_{CR} = 0.898
 Diversity = 3.6170e-04 k_t = [2, 2, 4, 2]
 Generation 9570: Best = 3.401814e+06 FES = 957100 μ_F = 0.734 μ_{CR} = 0.890
 Diversity = 5.4036e-04 k_t = [1, 1, 1, 1]
 Generation 9580: Best = 3.401814e+06 FES = 958100 μ_F = 0.729 μ_{CR} = 0.890
 Diversity = 4.1771e-04 k_t = [2, 3, 3, 2]
 Generation 9590: Best = 3.401814e+06 FES = 959100 μ_F = 0.739 μ_{CR} = 0.888
 Diversity = 4.3032e-04 k_t = [1, 1, 1, 1]
 Generation 9600: Best = 3.401814e+06 FES = 960100 μ_F = 0.747 μ_{CR} = 0.883
 Diversity = 2.8197e-04 k_t = [4, 3, 3, 4]
 Generation 9610: Best = 3.401814e+06 FES = 961100 μ_F = 0.754 μ_{CR} = 0.875
 Diversity = 3.9634e-04 k_t = [1, 1, 1, 1]
 Generation 9620: Best = 3.401814e+06 FES = 962100 μ_F = 0.764 μ_{CR} = 0.880
 Diversity = 2.9421e-04 k_t = [3, 3, 3, 4]
 Generation 9630: Best = 3.401814e+06 FES = 963100 μ_F = 0.774 μ_{CR} = 0.883
 Diversity = 4.2297e-04 k_t = [1, 1, 1, 1]
 Generation 9640: Best = 3.401814e+06 FES = 964100 μ_F = 0.764 μ_{CR} = 0.878
 Diversity = 3.1765e-04 k_t = [2, 2, 3, 4]
 Generation 9650: Best = 3.401814e+06 FES = 965100 μ_F = 0.749 μ_{CR} = 0.871
 Diversity = 3.6861e-04 k_t = [1, 1, 1, 1]
 Generation 9660: Best = 3.401814e+06 FES = 966100 μ_F = 0.738 μ_{CR} = 0.876
 Diversity = 3.1070e-04 k_t = [4, 4, 2, 2]
 Generation 9670: Best = 3.401814e+06 FES = 967100 μ_F = 0.751 μ_{CR} = 0.872
 Diversity = 3.3285e-04 k_t = [1, 1, 1, 1]

Generation 9680: Best = 3.401814e+06 FES = 968100 μ_F = 0.754 μ_{CR} = 0.870
 Diversity = 2.5130e-04 k_t = [4, 3, 4, 3]
 Generation 9690: Best = 3.401814e+06 FES = 969100 μ_F = 0.753 μ_{CR} = 0.874
 Diversity = 3.2511e-04 k_t = [1, 1, 1, 1]
 Generation 9700: Best = 3.401814e+06 FES = 970100 μ_F = 0.757 μ_{CR} = 0.880
 Diversity = 3.1731e-04 k_t = [3, 3, 2, 2]
 Generation 9710: Best = 3.401814e+06 FES = 971100 μ_F = 0.750 μ_{CR} = 0.874
 Diversity = 4.2050e-04 k_t = [1, 1, 1, 1]
 Generation 9720: Best = 3.401814e+06 FES = 972100 μ_F = 0.742 μ_{CR} = 0.873
 Diversity = 3.5220e-04 k_t = [4, 3, 4, 3]
 Generation 9730: Best = 3.401814e+06 FES = 973100 μ_F = 0.757 μ_{CR} = 0.878
 Diversity = 5.2399e-04 k_t = [1, 1, 1, 1]
 Generation 9740: Best = 3.401814e+06 FES = 974100 μ_F = 0.747 μ_{CR} = 0.876
 Diversity = 5.0460e-04 k_t = [3, 4, 3, 4]
 Generation 9750: Best = 3.401814e+06 FES = 975100 μ_F = 0.734 μ_{CR} = 0.875
 Diversity = 4.7740e-04 k_t = [1, 1, 1, 1]
 Generation 9760: Best = 3.401814e+06 FES = 976100 μ_F = 0.736 μ_{CR} = 0.877
 Diversity = 4.2898e-04 k_t = [4, 4, 2, 4]
 Generation 9770: Best = 3.401814e+06 FES = 977100 μ_F = 0.743 μ_{CR} = 0.872
 Diversity = 4.7756e-04 k_t = [1, 1, 1, 1]
 Generation 9780: Best = 3.401814e+06 FES = 978100 μ_F = 0.721 μ_{CR} = 0.861
 Diversity = 3.9508e-04 k_t = [2, 2, 2, 3]
 Generation 9790: Best = 3.401814e+06 FES = 979100 μ_F = 0.733 μ_{CR} = 0.859
 Diversity = 4.7451e-04 k_t = [1, 1, 1, 1]
 Generation 9800: Best = 3.401814e+06 FES = 980100 μ_F = 0.735 μ_{CR} = 0.866
 Diversity = 5.3382e-04 k_t = [2, 4, 4, 3]
 Generation 9810: Best = 3.401814e+06 FES = 981100 μ_F = 0.723 μ_{CR} = 0.863
 Diversity = 5.1402e-04 k_t = [1, 1, 1, 1]
 Generation 9820: Best = 3.401814e+06 FES = 982100 μ_F = 0.716 μ_{CR} = 0.861
 Diversity = 5.1341e-04 k_t = [2, 4, 2, 4]
 Generation 9830: Best = 3.401814e+06 FES = 983100 μ_F = 0.711 μ_{CR} = 0.869
 Diversity = 4.3364e-04 k_t = [1, 1, 1, 1]
 Generation 9840: Best = 3.401814e+06 FES = 984100 μ_F = 0.732 μ_{CR} = 0.873
 Diversity = 3.3738e-04 k_t = [4, 4, 4, 2]
 Generation 9850: Best = 3.401814e+06 FES = 985100 μ_F = 0.745 μ_{CR} = 0.882
 Diversity = 3.5093e-04 k_t = [1, 1, 1, 1]
 Generation 9860: Best = 3.401814e+06 FES = 986100 μ_F = 0.753 μ_{CR} = 0.880
 Diversity = 3.8854e-04 k_t = [3, 2, 4, 2]
 Generation 9870: Best = 3.401814e+06 FES = 987100 μ_F = 0.764 μ_{CR} = 0.890
 Diversity = 6.1141e-04 k_t = [1, 1, 1, 1]
 Generation 9880: Best = 3.401814e+06 FES = 988100 μ_F = 0.770 μ_{CR} = 0.886
 Diversity = 5.1119e-04 k_t = [2, 3, 4, 3]
 Generation 9890: Best = 3.401814e+06 FES = 989100 μ_F = 0.766 μ_{CR} = 0.889
 Diversity = 6.8604e-04 k_t = [1, 1, 1, 1]
 Generation 9900: Best = 3.401814e+06 FES = 990100 μ_F = 0.758 μ_{CR} = 0.878
 Diversity = 6.7315e-04 k_t = [3, 3, 4, 3]
 Generation 9910: Best = 3.401814e+06 FES = 991100 μ_F = 0.754 μ_{CR} = 0.879
 Diversity = 7.0025e-04 k_t = [1, 1, 1, 1]
 Generation 9920: Best = 3.401814e+06 FES = 992100 μ_F = 0.758 μ_{CR} = 0.880
 Diversity = 6.2178e-04 k_t = [3, 3, 3, 2]
 Generation 9930: Best = 3.401814e+06 FES = 993100 μ_F = 0.746 μ_{CR} = 0.882
 Diversity = 7.7050e-04 k_t = [1, 1, 1, 1]
 Generation 9940: Best = 3.401814e+06 FES = 994100 μ_F = 0.747 μ_{CR} = 0.883
 Diversity = 5.1747e-04 k_t = [4, 4, 4, 3]

Generation 9950: Best = 3.401814e+06 FES = 995100 μ_F = 0.754 μ_{CR} = 0.881
 Diversity = 5.0533e-04 k_t = [1, 1, 1, 1]
 Generation 9960: Best = 3.401814e+06 FES = 996100 μ_F = 0.757 μ_{CR} = 0.887
 Diversity = 5.5111e-04 k_t = [4, 3, 3, 2]
 Generation 9970: Best = 3.401814e+06 FES = 997100 μ_F = 0.764 μ_{CR} = 0.888
 Diversity = 6.9975e-04 k_t = [1, 1, 1, 1]
 Generation 9980: Best = 3.401814e+06 FES = 998100 μ_F = 0.762 μ_{CR} = 0.898
 Diversity = 5.7237e-04 k_t = [3, 2, 4, 2]
 Generation 9990: Best = 3.401814e+06 FES = 999100 μ_F = 0.767 μ_{CR} = 0.895
 Diversity = 5.9089e-04 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 3.401814e+06

Total Function Evals = 1000000

Total Successful Updates = 531369

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Results:

Best Fitness : 3.401814e+06

Global Optimum : 1.200000e+03

Error (|best-optimum|) : 3.400614e+06

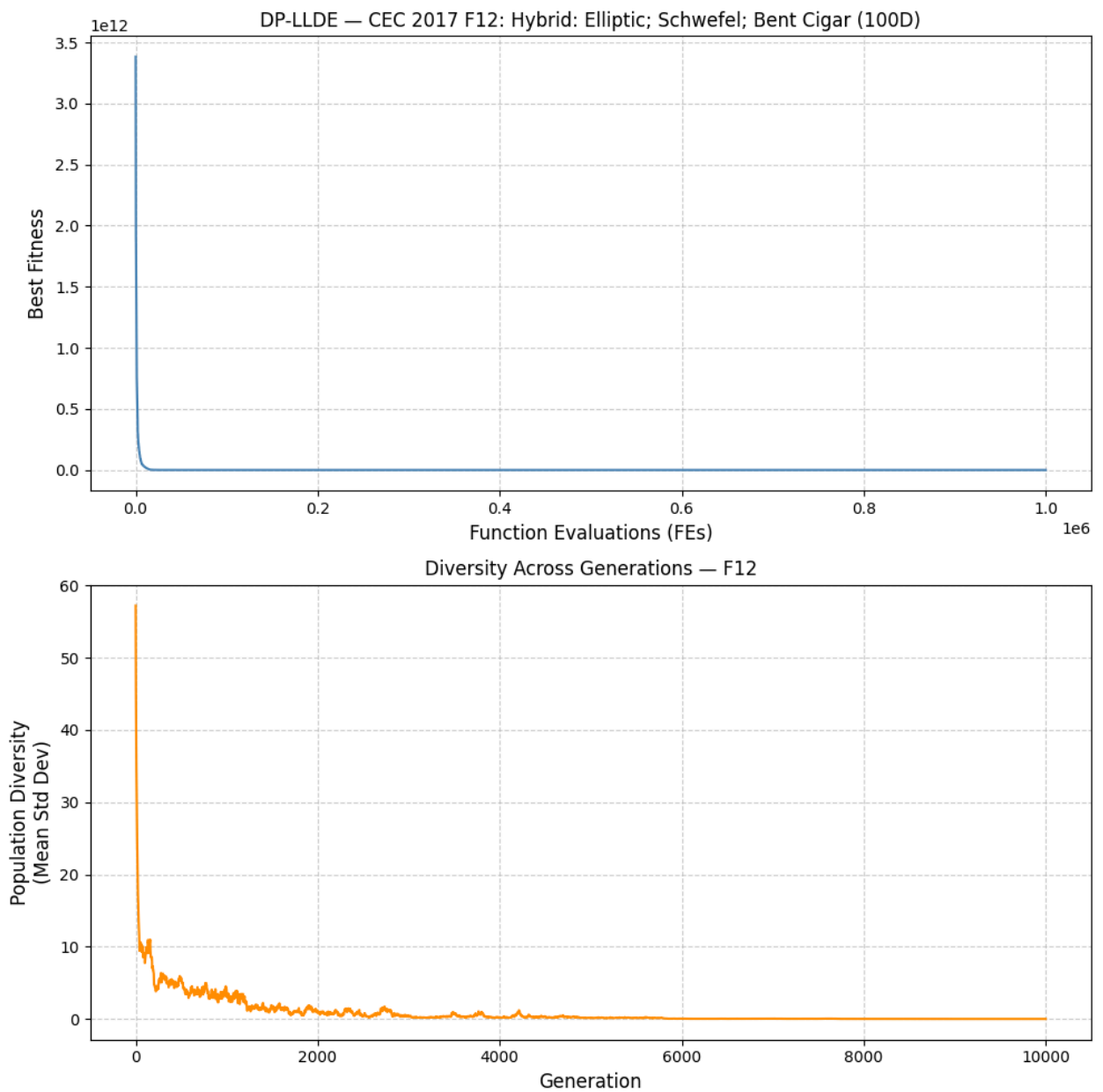
Best Solution (5 dim) : [54.12846105 -99.99066096 -85.75051012 28.27401148
-30.93984586]

$D^{(0)}$ (init. diversity): 5.7220e+01

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✓ Checkpoint saved to: /content/drive/My Drive/thesis_results/all_results_checkpoint.pkl



Graph saved as convergence_F12.png

✓ Plot saved to: /content/drive/My Drive/thesis_results/plots/convergence_F12.png

```
In [ ]: # =====
# F13 — Hybrid Function 3 (Bent Cigar; Rosenbrock; Bi-Rastrigin)
# =====
run_and_plot('F13', make_wrapper(hybrid.f13), 1300.0, 'Hybrid: Bent Cigar; Ros
```

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DP-LLDE on CEC 2017 F13 – Single Run

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Function : Hybrid: Bent Cigar; Rosenbrock; Bi-Rastrigin (100D)
Bounds : [-100, 100]
Global optimum : 1300.0
Initial $D^*(0) = 5.7220e+01$ | Diversity threshold = 5.7220e+00
Generation 0: Best Fitness = 9.446380e+11

Generation 10:	Best = 2.587278e+11	FES = 1100	$\mu_F = 0.560$	$\mu_{CR} = 0.353$
Diversity = 3.3077e+01	$k_t = [1, 1, 1, 1]$			
Generation 20:	Best = 8.171367e+10	FES = 2100	$\mu_F = 0.614$	$\mu_{CR} = 0.358$
Diversity = 2.2792e+01	$k_t = [2, 3, 2, 3]$			
Generation 30:	Best = 2.769933e+10	FES = 3100	$\mu_F = 0.646$	$\mu_{CR} = 0.361$
Diversity = 1.8462e+01	$k_t = [1, 1, 1, 1]$			
Generation 40:	Best = 9.141575e+09	FES = 4100	$\mu_F = 0.676$	$\mu_{CR} = 0.363$
Diversity = 1.3782e+01	$k_t = [3, 2, 3, 4]$			
Generation 50:	Best = 2.887347e+09	FES = 5100	$\mu_F = 0.700$	$\mu_{CR} = 0.369$
Diversity = 1.2853e+01	$k_t = [1, 1, 1, 1]$			
Generation 60:	Best = 1.434958e+09	FES = 6100	$\mu_F = 0.699$	$\mu_{CR} = 0.360$
Diversity = 1.1749e+01	$k_t = [2, 2, 3, 3]$			
Generation 70:	Best = 6.698339e+08	FES = 7100	$\mu_F = 0.709$	$\mu_{CR} = 0.359$
Diversity = 1.1109e+01	$k_t = [1, 1, 1, 1]$			
Generation 80:	Best = 2.887501e+08	FES = 8100	$\mu_F = 0.731$	$\mu_{CR} = 0.362$
Diversity = 1.1041e+01	$k_t = [4, 3, 4, 2]$			
Generation 90:	Best = 1.285517e+08	FES = 9100	$\mu_F = 0.741$	$\mu_{CR} = 0.363$
Diversity = 1.0843e+01	$k_t = [1, 1, 1, 1]$			
Generation 100:	Best = 3.551438e+07	FES = 10100	$\mu_F = 0.747$	$\mu_{CR} = 0.368$
Diversity = 1.1029e+01	$k_t = [2, 2, 4, 3]$			
Generation 110:	Best = 1.252172e+07	FES = 11100	$\mu_F = 0.739$	$\mu_{CR} = 0.378$
Diversity = 1.0875e+01	$k_t = [1, 1, 1, 1]$			
Generation 120:	Best = 4.434795e+06	FES = 12100	$\mu_F = 0.737$	$\mu_{CR} = 0.380$
Diversity = 1.0404e+01	$k_t = [2, 4, 2, 2]$			
Generation 130:	Best = 1.174551e+06	FES = 13100	$\mu_F = 0.735$	$\mu_{CR} = 0.377$
Diversity = 9.7248e+00	$k_t = [1, 1, 1, 1]$			
Generation 140:	Best = 3.882286e+05	FES = 14100	$\mu_F = 0.724$	$\mu_{CR} = 0.387$
Diversity = 8.0414e+00	$k_t = [2, 3, 4, 2]$			
Generation 150:	Best = 1.313816e+05	FES = 15100	$\mu_F = 0.716$	$\mu_{CR} = 0.390$
Diversity = 7.2579e+00	$k_t = [1, 1, 1, 1]$			
Generation 160:	Best = 4.993540e+04	FES = 16100	$\mu_F = 0.711$	$\mu_{CR} = 0.396$
Diversity = 6.4857e+00	$k_t = [4, 2, 3, 2]$			
Generation 170:	Best = 2.984895e+04	FES = 17100	$\mu_F = 0.702$	$\mu_{CR} = 0.396$
Diversity = 5.3173e+00	$k_t = [1, 1, 1, 1]$			
Generation 180:	Best = 2.256238e+04	FES = 18100	$\mu_F = 0.691$	$\mu_{CR} = 0.400$
Diversity = 4.0004e+00	$k_t = [3, 2, 3, 3]$			
Generation 190:	Best = 1.900215e+04	FES = 19100	$\mu_F = 0.708$	$\mu_{CR} = 0.405$
Diversity = 3.4601e+00	$k_t = [1, 1, 1, 1]$			
Generation 200:	Best = 1.647952e+04	FES = 20100	$\mu_F = 0.711$	$\mu_{CR} = 0.411$
Diversity = 3.1616e+00	$k_t = [4, 4, 2, 4]$			
Generation 210:	Best = 1.526056e+04	FES = 21100	$\mu_F = 0.736$	$\mu_{CR} = 0.412$
Diversity = 2.8863e+00	$k_t = [1, 1, 1, 1]$			
Generation 220:	Best = 1.368740e+04	FES = 22100	$\mu_F = 0.754$	$\mu_{CR} = 0.412$
Diversity = 2.8878e+00	$k_t = [4, 4, 2, 3]$			

Generation 230: Best = 1.290139e+04	FES = 23100	$\mu_F = 0.756$	$\mu_{CR} = 0.414$
Diversity = 2.5082e+00	$k_t = [1, 1, 1, 1]$		
Generation 240: Best = 1.220561e+04	FES = 24100	$\mu_F = 0.760$	$\mu_{CR} = 0.415$
Diversity = 2.6498e+00	$k_t = [4, 2, 2, 4]$		
Generation 250: Best = 1.131367e+04	FES = 25100	$\mu_F = 0.778$	$\mu_{CR} = 0.425$
Diversity = 2.8536e+00	$k_t = [1, 1, 1, 1]$		
Generation 260: Best = 1.066031e+04	FES = 26100	$\mu_F = 0.777$	$\mu_{CR} = 0.428$
Diversity = 3.2262e+00	$k_t = [4, 4, 4, 2]$		
Generation 270: Best = 1.011901e+04	FES = 27100	$\mu_F = 0.778$	$\mu_{CR} = 0.421$
Diversity = 3.2242e+00	$k_t = [1, 1, 1, 1]$		
Generation 280: Best = 9.458496e+03	FES = 28100	$\mu_F = 0.780$	$\mu_{CR} = 0.418$
Diversity = 3.5843e+00	$k_t = [3, 2, 3, 3]$		
Generation 290: Best = 9.004650e+03	FES = 29100	$\mu_F = 0.778$	$\mu_{CR} = 0.427$
Diversity = 3.7061e+00	$k_t = [1, 1, 1, 1]$		
Generation 300: Best = 8.481805e+03	FES = 30100	$\mu_F = 0.779$	$\mu_{CR} = 0.429$
Diversity = 3.7351e+00	$k_t = [3, 4, 2, 2]$		
Generation 310: Best = 7.970717e+03	FES = 31100	$\mu_F = 0.774$	$\mu_{CR} = 0.437$
Diversity = 3.9492e+00	$k_t = [1, 1, 1, 1]$		
Generation 320: Best = 7.383285e+03	FES = 32100	$\mu_F = 0.773$	$\mu_{CR} = 0.435$
Diversity = 3.6290e+00	$k_t = [2, 3, 3, 4]$		
Generation 330: Best = 7.136410e+03	FES = 33100	$\mu_F = 0.766$	$\mu_{CR} = 0.443$
Diversity = 3.3976e+00	$k_t = [1, 1, 1, 1]$		
Generation 340: Best = 6.692826e+03	FES = 34100	$\mu_F = 0.765$	$\mu_{CR} = 0.435$
Diversity = 3.3372e+00	$k_t = [2, 2, 2, 4]$		
Generation 350: Best = 6.504313e+03	FES = 35100	$\mu_F = 0.762$	$\mu_{CR} = 0.445$
Diversity = 3.3080e+00	$k_t = [1, 1, 1, 1]$		
Generation 360: Best = 6.288234e+03	FES = 36100	$\mu_F = 0.762$	$\mu_{CR} = 0.446$
Diversity = 2.8333e+00	$k_t = [3, 3, 2, 4]$		
Generation 370: Best = 6.169757e+03	FES = 37100	$\mu_F = 0.742$	$\mu_{CR} = 0.448$
Diversity = 2.3857e+00	$k_t = [1, 1, 1, 1]$		
Generation 380: Best = 6.114261e+03	FES = 38100	$\mu_F = 0.738$	$\mu_{CR} = 0.436$
Diversity = 2.1491e+00	$k_t = [3, 2, 4, 4]$		
Generation 390: Best = 5.985881e+03	FES = 39100	$\mu_F = 0.734$	$\mu_{CR} = 0.434$
Diversity = 2.1325e+00	$k_t = [1, 1, 1, 1]$		
Generation 400: Best = 5.936003e+03	FES = 40100	$\mu_F = 0.724$	$\mu_{CR} = 0.429$
Diversity = 1.8470e+00	$k_t = [2, 2, 4, 4]$		
Generation 410: Best = 5.936003e+03	FES = 41100	$\mu_F = 0.703$	$\mu_{CR} = 0.434$
Diversity = 1.7493e+00	$k_t = [1, 1, 1, 1]$		
Generation 420: Best = 5.924672e+03	FES = 42100	$\mu_F = 0.708$	$\mu_{CR} = 0.400$
Diversity = 1.6487e+00	$k_t = [4, 4, 2, 4]$		
Generation 430: Best = 5.899899e+03	FES = 43100	$\mu_F = 0.679$	$\mu_{CR} = 0.408$
Diversity = 1.5646e+00	$k_t = [1, 1, 1, 1]$		
Generation 440: Best = 5.880691e+03	FES = 44100	$\mu_F = 0.660$	$\mu_{CR} = 0.401$
Diversity = 1.4581e+00	$k_t = [3, 3, 4, 4]$		
Generation 450: Best = 5.848617e+03	FES = 45100	$\mu_F = 0.682$	$\mu_{CR} = 0.350$
Diversity = 1.3419e+00	$k_t = [1, 1, 1, 1]$		
Generation 460: Best = 5.848617e+03	FES = 46100	$\mu_F = 0.669$	$\mu_{CR} = 0.331$
Diversity = 1.3021e+00	$k_t = [2, 2, 3, 2]$		
Generation 470: Best = 5.838190e+03	FES = 47100	$\mu_F = 0.677$	$\mu_{CR} = 0.252$
Diversity = 1.2822e+00	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 5.808583e+03	FES = 48100	$\mu_F = 0.681$	$\mu_{CR} = 0.232$
Diversity = 1.2325e+00	$k_t = [2, 3, 3, 4]$		
Generation 490: Best = 5.801604e+03	FES = 49100	$\mu_F = 0.666$	$\mu_{CR} = 0.180$
Diversity = 1.2055e+00	$k_t = [1, 1, 1, 1]$		

Generation 500: Best = 5.789415e+03	FES = 50100	$\mu_F = 0.675$	$\mu_{CR} = 0.144$
Diversity = 1.1698e+00	$k_t = [3, 2, 2, 3]$		
Generation 510: Best = 5.786907e+03	FES = 51100	$\mu_F = 0.698$	$\mu_{CR} = 0.113$
Diversity = 1.1669e+00	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 5.776975e+03	FES = 52100	$\mu_F = 0.734$	$\mu_{CR} = 0.091$
Diversity = 1.1485e+00	$k_t = [2, 2, 2, 4]$		
Generation 530: Best = 5.775134e+03	FES = 53100	$\mu_F = 0.729$	$\mu_{CR} = 0.072$
Diversity = 1.1395e+00	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 5.749559e+03	FES = 54100	$\mu_F = 0.725$	$\mu_{CR} = 0.070$
Diversity = 1.1417e+00	$k_t = [4, 3, 2, 3]$		
Generation 550: Best = 5.736132e+03	FES = 55100	$\mu_F = 0.715$	$\mu_{CR} = 0.075$
Diversity = 1.1483e+00	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 5.736132e+03	FES = 56100	$\mu_F = 0.699$	$\mu_{CR} = 0.067$
Diversity = 1.1242e+00	$k_t = [4, 2, 4, 4]$		
Generation 570: Best = 5.727413e+03	FES = 57100	$\mu_F = 0.717$	$\mu_{CR} = 0.063$
Diversity = 1.1183e+00	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 5.726674e+03	FES = 58100	$\mu_F = 0.721$	$\mu_{CR} = 0.062$
Diversity = 1.1328e+00	$k_t = [3, 3, 2, 2]$		
Generation 590: Best = 5.726028e+03	FES = 59100	$\mu_F = 0.726$	$\mu_{CR} = 0.062$
Diversity = 1.1186e+00	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 5.690782e+03	FES = 60100	$\mu_F = 0.735$	$\mu_{CR} = 0.062$
Diversity = 1.0932e+00	$k_t = [3, 2, 4, 3]$		
Generation 610: Best = 5.690195e+03	FES = 61100	$\mu_F = 0.740$	$\mu_{CR} = 0.073$
Diversity = 1.0828e+00	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 5.682117e+03	FES = 62100	$\mu_F = 0.743$	$\mu_{CR} = 0.074$
Diversity = 1.0750e+00	$k_t = [3, 3, 3, 2]$		
Generation 630: Best = 5.662646e+03	FES = 63100	$\mu_F = 0.754$	$\mu_{CR} = 0.078$
Diversity = 1.0673e+00	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 5.662601e+03	FES = 64100	$\mu_F = 0.770$	$\mu_{CR} = 0.076$
Diversity = 1.0060e+00	$k_t = [3, 4, 3, 4]$		
Generation 650: Best = 5.641161e+03	FES = 65100	$\mu_F = 0.764$	$\mu_{CR} = 0.076$
Diversity = 9.8374e-01	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 5.627781e+03	FES = 66100	$\mu_F = 0.765$	$\mu_{CR} = 0.077$
Diversity = 9.5000e-01	$k_t = [4, 3, 3, 3]$		
Generation 670: Best = 5.627686e+03	FES = 67100	$\mu_F = 0.755$	$\mu_{CR} = 0.094$
Diversity = 8.8313e-01	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 5.611999e+03	FES = 68100	$\mu_F = 0.753$	$\mu_{CR} = 0.107$
Diversity = 7.4368e-01	$k_t = [3, 2, 3, 3]$		
Generation 690: Best = 5.592168e+03	FES = 69100	$\mu_F = 0.752$	$\mu_{CR} = 0.122$
Diversity = 6.3283e-01	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 5.561645e+03	FES = 70100	$\mu_F = 0.750$	$\mu_{CR} = 0.135$
Diversity = 4.6658e-01	$k_t = [3, 2, 4, 2]$		
Generation 710: Best = 5.522985e+03	FES = 71100	$\mu_F = 0.752$	$\mu_{CR} = 0.152$
Diversity = 4.1830e-01	$k_t = [1, 1, 1, 1]$		
Generation 720: Best = 5.486051e+03	FES = 72100	$\mu_F = 0.764$	$\mu_{CR} = 0.164$
Diversity = 4.0591e-01	$k_t = [3, 2, 3, 4]$		
Generation 730: Best = 5.442670e+03	FES = 73100	$\mu_F = 0.769$	$\mu_{CR} = 0.177$
Diversity = 4.0527e-01	$k_t = [1, 1, 1, 1]$		
Generation 740: Best = 5.399057e+03	FES = 74100	$\mu_F = 0.773$	$\mu_{CR} = 0.186$
Diversity = 3.9756e-01	$k_t = [4, 4, 3, 4]$		
Generation 750: Best = 5.355290e+03	FES = 75100	$\mu_F = 0.767$	$\mu_{CR} = 0.196$
Diversity = 3.9393e-01	$k_t = [1, 1, 1, 1]$		
Generation 760: Best = 5.327541e+03	FES = 76100	$\mu_F = 0.773$	$\mu_{CR} = 0.201$
Diversity = 3.5062e-01	$k_t = [3, 2, 2, 4]$		

Generation 770: Best = 5.302352e+03	FES = 77100	$\mu_F = 0.775$	$\mu_{CR} = 0.208$
Diversity = 3.4322e-01	$k_t = [1, 1, 1, 1]$		
Generation 780: Best = 5.278833e+03	FES = 78100	$\mu_F = 0.780$	$\mu_{CR} = 0.210$
Diversity = 3.2977e-01	$k_t = [3, 2, 2, 4]$		
Generation 790: Best = 5.262368e+03	FES = 79100	$\mu_F = 0.774$	$\mu_{CR} = 0.217$
Diversity = 3.1486e-01	$k_t = [1, 1, 1, 1]$		
Generation 800: Best = 5.248104e+03	FES = 80100	$\mu_F = 0.771$	$\mu_{CR} = 0.219$
Diversity = 2.8132e-01	$k_t = [3, 4, 4, 3]$		
Generation 810: Best = 5.233178e+03	FES = 81100	$\mu_F = 0.765$	$\mu_{CR} = 0.210$
Diversity = 2.8632e-01	$k_t = [1, 1, 1, 1]$		
Generation 820: Best = 5.221435e+03	FES = 82100	$\mu_F = 0.756$	$\mu_{CR} = 0.208$
Diversity = 2.7859e-01	$k_t = [4, 3, 4, 4]$		
Generation 830: Best = 5.211929e+03	FES = 83100	$\mu_F = 0.752$	$\mu_{CR} = 0.219$
Diversity = 2.7193e-01	$k_t = [1, 1, 1, 1]$		
Generation 840: Best = 5.202237e+03	FES = 84100	$\mu_F = 0.752$	$\mu_{CR} = 0.218$
Diversity = 2.8015e-01	$k_t = [3, 3, 4, 2]$		
Generation 850: Best = 5.191369e+03	FES = 85100	$\mu_F = 0.752$	$\mu_{CR} = 0.220$
Diversity = 2.8621e-01	$k_t = [1, 1, 1, 1]$		
Generation 860: Best = 5.185840e+03	FES = 86100	$\mu_F = 0.743$	$\mu_{CR} = 0.222$
Diversity = 2.4936e-01	$k_t = [4, 3, 4, 2]$		
Generation 870: Best = 5.177125e+03	FES = 87100	$\mu_F = 0.737$	$\mu_{CR} = 0.226$
Diversity = 2.6018e-01	$k_t = [1, 1, 1, 1]$		
Generation 880: Best = 5.170743e+03	FES = 88100	$\mu_F = 0.740$	$\mu_{CR} = 0.225$
Diversity = 2.3821e-01	$k_t = [4, 4, 2, 3]$		
Generation 890: Best = 5.165807e+03	FES = 89100	$\mu_F = 0.749$	$\mu_{CR} = 0.230$
Diversity = 2.4036e-01	$k_t = [1, 1, 1, 1]$		
Generation 900: Best = 5.156927e+03	FES = 90100	$\mu_F = 0.748$	$\mu_{CR} = 0.226$
Diversity = 2.1833e-01	$k_t = [3, 3, 2, 2]$		
Generation 910: Best = 5.153412e+03	FES = 91100	$\mu_F = 0.747$	$\mu_{CR} = 0.228$
Diversity = 2.4825e-01	$k_t = [1, 1, 1, 1]$		
Generation 920: Best = 5.145933e+03	FES = 92100	$\mu_F = 0.749$	$\mu_{CR} = 0.221$
Diversity = 2.3855e-01	$k_t = [3, 2, 4, 3]$		
Generation 930: Best = 5.139744e+03	FES = 93100	$\mu_F = 0.752$	$\mu_{CR} = 0.224$
Diversity = 2.6368e-01	$k_t = [1, 1, 1, 1]$		
Generation 940: Best = 5.132028e+03	FES = 94100	$\mu_F = 0.767$	$\mu_{CR} = 0.226$
Diversity = 2.4634e-01	$k_t = [4, 4, 4, 4]$		
Generation 950: Best = 5.128854e+03	FES = 95100	$\mu_F = 0.760$	$\mu_{CR} = 0.212$
Diversity = 2.3761e-01	$k_t = [1, 1, 1, 1]$		
Generation 960: Best = 5.123115e+03	FES = 96100	$\mu_F = 0.753$	$\mu_{CR} = 0.219$
Diversity = 2.3776e-01	$k_t = [3, 2, 2, 2]$		
Generation 970: Best = 5.118240e+03	FES = 97100	$\mu_F = 0.746$	$\mu_{CR} = 0.224$
Diversity = 2.4304e-01	$k_t = [1, 1, 1, 1]$		
Generation 980: Best = 5.112954e+03	FES = 98100	$\mu_F = 0.748$	$\mu_{CR} = 0.221$
Diversity = 2.2332e-01	$k_t = [3, 4, 3, 3]$		
Generation 990: Best = 5.107971e+03	FES = 99100	$\mu_F = 0.735$	$\mu_{CR} = 0.223$
Diversity = 2.1774e-01	$k_t = [1, 1, 1, 1]$		
Generation 1000: Best = 5.104768e+03	FES = 100100	$\mu_F = 0.732$	$\mu_{CR} = 0.226$
Diversity = 2.2078e-01	$k_t = [4, 3, 4, 2]$		
Generation 1010: Best = 5.100890e+03	FES = 101100	$\mu_F = 0.734$	$\mu_{CR} = 0.228$
Diversity = 2.1154e-01	$k_t = [1, 1, 1, 1]$		
Generation 1020: Best = 5.097420e+03	FES = 102100	$\mu_F = 0.731$	$\mu_{CR} = 0.232$
Diversity = 2.0784e-01	$k_t = [2, 3, 4, 4]$		
Generation 1030: Best = 5.094082e+03	FES = 103100	$\mu_F = 0.728$	$\mu_{CR} = 0.230$
Diversity = 2.1001e-01	$k_t = [1, 1, 1, 1]$		

Generation 1040: Best = 5.091565e+03 FES = 104100 μ_F = 0.723 μ_{CR} = 0.220
 Diversity = 1.8294e-01 k_t = [2, 3, 4, 3]
 Generation 1050: Best = 5.088768e+03 FES = 105100 μ_F = 0.722 μ_{CR} = 0.214
 Diversity = 1.8380e-01 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 5.086637e+03 FES = 106100 μ_F = 0.736 μ_{CR} = 0.210
 Diversity = 1.8450e-01 k_t = [3, 4, 2, 3]
 Generation 1070: Best = 5.083491e+03 FES = 107100 μ_F = 0.742 μ_{CR} = 0.206
 Diversity = 1.8490e-01 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 5.081500e+03 FES = 108100 μ_F = 0.740 μ_{CR} = 0.199
 Diversity = 1.8086e-01 k_t = [2, 4, 3, 3]
 Generation 1090: Best = 5.080295e+03 FES = 109100 μ_F = 0.723 μ_{CR} = 0.206
 Diversity = 1.6546e-01 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 5.078820e+03 FES = 110100 μ_F = 0.709 μ_{CR} = 0.196
 Diversity = 1.5570e-01 k_t = [2, 4, 4, 4]
 Generation 1110: Best = 5.077724e+03 FES = 111100 μ_F = 0.706 μ_{CR} = 0.195
 Diversity = 1.2972e-01 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 5.077317e+03 FES = 112100 μ_F = 0.712 μ_{CR} = 0.193
 Diversity = 1.1681e-01 k_t = [4, 4, 3, 3]
 Generation 1130: Best = 5.076131e+03 FES = 113100 μ_F = 0.714 μ_{CR} = 0.192
 Diversity = 1.1172e-01 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 5.075398e+03 FES = 114100 μ_F = 0.724 μ_{CR} = 0.187
 Diversity = 1.0037e-01 k_t = [3, 3, 2, 4]
 Generation 1150: Best = 5.075135e+03 FES = 115100 μ_F = 0.741 μ_{CR} = 0.191
 Diversity = 9.7181e-02 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 5.074275e+03 FES = 116100 μ_F = 0.743 μ_{CR} = 0.200
 Diversity = 1.0077e-01 k_t = [3, 3, 2, 2]
 Generation 1170: Best = 5.073463e+03 FES = 117100 μ_F = 0.758 μ_{CR} = 0.203
 Diversity = 1.2027e-01 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 5.072538e+03 FES = 118100 μ_F = 0.764 μ_{CR} = 0.210
 Diversity = 1.1332e-01 k_t = [4, 2, 3, 4]
 Generation 1190: Best = 5.071876e+03 FES = 119100 μ_F = 0.767 μ_{CR} = 0.206
 Diversity = 1.1509e-01 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 5.071554e+03 FES = 120100 μ_F = 0.763 μ_{CR} = 0.202
 Diversity = 1.0364e-01 k_t = [3, 4, 4, 4]
 Generation 1210: Best = 5.070891e+03 FES = 121100 μ_F = 0.753 μ_{CR} = 0.200
 Diversity = 9.9834e-02 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 5.070167e+03 FES = 122100 μ_F = 0.751 μ_{CR} = 0.195
 Diversity = 1.0141e-01 k_t = [4, 3, 4, 2]
 Generation 1230: Best = 5.069542e+03 FES = 123100 μ_F = 0.745 μ_{CR} = 0.202
 Diversity = 9.9626e-02 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 5.068927e+03 FES = 124100 μ_F = 0.734 μ_{CR} = 0.192
 Diversity = 9.3335e-02 k_t = [2, 3, 2, 2]
 Generation 1250: Best = 5.067966e+03 FES = 125100 μ_F = 0.744 μ_{CR} = 0.201
 Diversity = 9.9927e-02 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 5.067586e+03 FES = 126100 μ_F = 0.735 μ_{CR} = 0.195
 Diversity = 9.5980e-02 k_t = [2, 3, 2, 3]
 Generation 1270: Best = 5.067450e+03 FES = 127100 μ_F = 0.728 μ_{CR} = 0.189
 Diversity = 9.1790e-02 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 5.067016e+03 FES = 128100 μ_F = 0.730 μ_{CR} = 0.187
 Diversity = 8.4821e-02 k_t = [4, 4, 3, 3]
 Generation 1290: Best = 5.066645e+03 FES = 129100 μ_F = 0.734 μ_{CR} = 0.187
 Diversity = 8.1417e-02 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 5.066273e+03 FES = 130100 μ_F = 0.734 μ_{CR} = 0.188
 Diversity = 8.3313e-02 k_t = [2, 4, 3, 2]

Generation 1310: Best = 5.065850e+03 FES = 131100 μ_F = 0.732 μ_{CR} = 0.185
 Diversity = 8.7735e-02 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 5.064878e+03 FES = 132100 μ_F = 0.752 μ_{CR} = 0.193
 Diversity = 9.8020e-02 k_t = [3, 4, 2, 3]
 Generation 1330: Best = 5.063955e+03 FES = 133100 μ_F = 0.744 μ_{CR} = 0.189
 Diversity = 1.1575e-01 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 5.063156e+03 FES = 134100 μ_F = 0.746 μ_{CR} = 0.192
 Diversity = 1.1994e-01 k_t = [2, 4, 3, 3]
 Generation 1350: Best = 5.062328e+03 FES = 135100 μ_F = 0.758 μ_{CR} = 0.182
 Diversity = 1.3193e-01 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 5.061297e+03 FES = 136100 μ_F = 0.746 μ_{CR} = 0.176
 Diversity = 1.3974e-01 k_t = [2, 4, 4, 3]
 Generation 1370: Best = 5.060443e+03 FES = 137100 μ_F = 0.751 μ_{CR} = 0.177
 Diversity = 1.4497e-01 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 5.058520e+03 FES = 138100 μ_F = 0.750 μ_{CR} = 0.176
 Diversity = 1.4266e-01 k_t = [3, 3, 2, 3]
 Generation 1390: Best = 5.057770e+03 FES = 139100 μ_F = 0.756 μ_{CR} = 0.181
 Diversity = 1.5675e-01 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 5.056559e+03 FES = 140100 μ_F = 0.749 μ_{CR} = 0.177
 Diversity = 1.5397e-01 k_t = [4, 4, 3, 2]
 Generation 1410: Best = 5.056173e+03 FES = 141100 μ_F = 0.739 μ_{CR} = 0.167
 Diversity = 1.3531e-01 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 5.054821e+03 FES = 142100 μ_F = 0.753 μ_{CR} = 0.166
 Diversity = 1.3608e-01 k_t = [4, 4, 2, 4]
 Generation 1430: Best = 5.054176e+03 FES = 143100 μ_F = 0.759 μ_{CR} = 0.172
 Diversity = 1.3956e-01 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 5.053473e+03 FES = 144100 μ_F = 0.750 μ_{CR} = 0.170
 Diversity = 1.4300e-01 k_t = [4, 2, 4, 2]
 Generation 1450: Best = 5.052884e+03 FES = 145100 μ_F = 0.742 μ_{CR} = 0.159
 Diversity = 1.5021e-01 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 5.051579e+03 FES = 146100 μ_F = 0.728 μ_{CR} = 0.154
 Diversity = 1.5628e-01 k_t = [4, 3, 2, 2]
 Generation 1470: Best = 5.050705e+03 FES = 147100 μ_F = 0.736 μ_{CR} = 0.159
 Diversity = 1.5958e-01 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 5.049602e+03 FES = 148100 μ_F = 0.737 μ_{CR} = 0.163
 Diversity = 1.4511e-01 k_t = [3, 3, 4, 4]
 Generation 1490: Best = 5.048582e+03 FES = 149100 μ_F = 0.728 μ_{CR} = 0.164
 Diversity = 1.4288e-01 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 5.047887e+03 FES = 150100 μ_F = 0.726 μ_{CR} = 0.163
 Diversity = 1.3594e-01 k_t = [2, 3, 3, 3]
 Generation 1510: Best = 5.047381e+03 FES = 151100 μ_F = 0.732 μ_{CR} = 0.150
 Diversity = 1.4479e-01 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 5.046676e+03 FES = 152100 μ_F = 0.732 μ_{CR} = 0.152
 Diversity = 1.2644e-01 k_t = [4, 4, 4, 3]
 Generation 1530: Best = 5.046104e+03 FES = 153100 μ_F = 0.735 μ_{CR} = 0.154
 Diversity = 1.1836e-01 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 5.045494e+03 FES = 154100 μ_F = 0.738 μ_{CR} = 0.152
 Diversity = 1.0496e-01 k_t = [2, 4, 4, 3]
 Generation 1550: Best = 5.045192e+03 FES = 155100 μ_F = 0.745 μ_{CR} = 0.164
 Diversity = 1.0660e-01 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 5.044591e+03 FES = 156100 μ_F = 0.734 μ_{CR} = 0.168
 Diversity = 1.1220e-01 k_t = [3, 4, 2, 2]
 Generation 1570: Best = 5.043833e+03 FES = 157100 μ_F = 0.726 μ_{CR} = 0.167
 Diversity = 1.1207e-01 k_t = [1, 1, 1, 1]

Generation 1580: Best = 5.042755e+03 FES = 158100 μ_F = 0.730 μ_{CR} = 0.175
 Diversity = 1.2134e-01 k_t = [4, 2, 3, 3]
 Generation 1590: Best = 5.042019e+03 FES = 159100 μ_F = 0.719 μ_{CR} = 0.177
 Diversity = 1.1875e-01 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 5.041742e+03 FES = 160100 μ_F = 0.717 μ_{CR} = 0.175
 Diversity = 1.0354e-01 k_t = [3, 2, 3, 3]
 Generation 1610: Best = 5.041485e+03 FES = 161100 μ_F = 0.713 μ_{CR} = 0.163
 Diversity = 9.8510e-02 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 5.040863e+03 FES = 162100 μ_F = 0.724 μ_{CR} = 0.166
 Diversity = 1.0466e-01 k_t = [4, 3, 4, 3]
 Generation 1630: Best = 5.039973e+03 FES = 163100 μ_F = 0.729 μ_{CR} = 0.163
 Diversity = 1.1041e-01 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 5.039727e+03 FES = 164100 μ_F = 0.741 μ_{CR} = 0.163
 Diversity = 1.1571e-01 k_t = [3, 4, 2, 2]
 Generation 1650: Best = 5.039111e+03 FES = 165100 μ_F = 0.746 μ_{CR} = 0.175
 Diversity = 1.1109e-01 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 5.038705e+03 FES = 166100 μ_F = 0.747 μ_{CR} = 0.174
 Diversity = 1.0048e-01 k_t = [2, 2, 4, 2]
 Generation 1670: Best = 5.037551e+03 FES = 167100 μ_F = 0.751 μ_{CR} = 0.173
 Diversity = 1.1062e-01 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 5.036505e+03 FES = 168100 μ_F = 0.752 μ_{CR} = 0.166
 Diversity = 1.3689e-01 k_t = [2, 2, 3, 2]
 Generation 1690: Best = 5.035549e+03 FES = 169100 μ_F = 0.754 μ_{CR} = 0.169
 Diversity = 1.4759e-01 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 5.034971e+03 FES = 170100 μ_F = 0.753 μ_{CR} = 0.151
 Diversity = 1.4245e-01 k_t = [3, 4, 4, 2]
 Generation 1710: Best = 5.033854e+03 FES = 171100 μ_F = 0.755 μ_{CR} = 0.153
 Diversity = 1.4524e-01 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 5.032875e+03 FES = 172100 μ_F = 0.762 μ_{CR} = 0.161
 Diversity = 1.4093e-01 k_t = [3, 3, 3, 3]
 Generation 1730: Best = 5.032325e+03 FES = 173100 μ_F = 0.769 μ_{CR} = 0.165
 Diversity = 1.4270e-01 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 5.031549e+03 FES = 174100 μ_F = 0.758 μ_{CR} = 0.165
 Diversity = 1.2528e-01 k_t = [2, 4, 4, 4]
 Generation 1750: Best = 5.031020e+03 FES = 175100 μ_F = 0.758 μ_{CR} = 0.160
 Diversity = 1.3029e-01 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 5.030447e+03 FES = 176100 μ_F = 0.754 μ_{CR} = 0.154
 Diversity = 1.2700e-01 k_t = [3, 2, 3, 2]
 Generation 1770: Best = 5.029387e+03 FES = 177100 μ_F = 0.753 μ_{CR} = 0.149
 Diversity = 1.3168e-01 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 5.029014e+03 FES = 178100 μ_F = 0.756 μ_{CR} = 0.154
 Diversity = 1.2482e-01 k_t = [3, 4, 4, 3]
 Generation 1790: Best = 5.028416e+03 FES = 179100 μ_F = 0.759 μ_{CR} = 0.156
 Diversity = 1.2357e-01 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 5.027775e+03 FES = 180100 μ_F = 0.759 μ_{CR} = 0.144
 Diversity = 1.2183e-01 k_t = [2, 2, 4, 2]
 Generation 1810: Best = 5.027097e+03 FES = 181100 μ_F = 0.759 μ_{CR} = 0.146
 Diversity = 1.4084e-01 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 5.026205e+03 FES = 182100 μ_F = 0.738 μ_{CR} = 0.159
 Diversity = 1.3645e-01 k_t = [4, 4, 2, 3]
 Generation 1830: Best = 5.025655e+03 FES = 183100 μ_F = 0.731 μ_{CR} = 0.157
 Diversity = 1.3453e-01 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 5.024499e+03 FES = 184100 μ_F = 0.734 μ_{CR} = 0.159
 Diversity = 1.3577e-01 k_t = [2, 3, 2, 2]

Generation 1850: Best = 5.023817e+03	FES = 185100	$\mu_F = 0.738$	$\mu_{CR} = 0.164$
Diversity = 1.4974e-01	k_t = [1, 1, 1, 1]		
Generation 1860: Best = 5.022723e+03	FES = 186100	$\mu_F = 0.741$	$\mu_{CR} = 0.160$
Diversity = 1.4702e-01	k_t = [2, 4, 3, 2]		
Generation 1870: Best = 5.021895e+03	FES = 187100	$\mu_F = 0.747$	$\mu_{CR} = 0.162$
Diversity = 1.5137e-01	k_t = [1, 1, 1, 1]		
Generation 1880: Best = 5.021048e+03	FES = 188100	$\mu_F = 0.735$	$\mu_{CR} = 0.161$
Diversity = 1.3365e-01	k_t = [3, 4, 4, 4]		
Generation 1890: Best = 5.019960e+03	FES = 189100	$\mu_F = 0.737$	$\mu_{CR} = 0.169$
Diversity = 1.2744e-01	k_t = [1, 1, 1, 1]		
Generation 1900: Best = 5.019164e+03	FES = 190100	$\mu_F = 0.741$	$\mu_{CR} = 0.167$
Diversity = 1.2220e-01	k_t = [2, 2, 4, 4]		
Generation 1910: Best = 5.018721e+03	FES = 191100	$\mu_F = 0.749$	$\mu_{CR} = 0.173$
Diversity = 1.2211e-01	k_t = [1, 1, 1, 1]		
Generation 1920: Best = 5.017880e+03	FES = 192100	$\mu_F = 0.745$	$\mu_{CR} = 0.178$
Diversity = 1.1276e-01	k_t = [4, 3, 2, 4]		
Generation 1930: Best = 5.016785e+03	FES = 193100	$\mu_F = 0.744$	$\mu_{CR} = 0.179$
Diversity = 1.2113e-01	k_t = [1, 1, 1, 1]		
Generation 1940: Best = 5.016344e+03	FES = 194100	$\mu_F = 0.758$	$\mu_{CR} = 0.182$
Diversity = 1.2286e-01	k_t = [3, 3, 2, 4]		
Generation 1950: Best = 5.015783e+03	FES = 195100	$\mu_F = 0.757$	$\mu_{CR} = 0.178$
Diversity = 1.3020e-01	k_t = [1, 1, 1, 1]		
Generation 1960: Best = 5.014841e+03	FES = 196100	$\mu_F = 0.750$	$\mu_{CR} = 0.176$
Diversity = 1.2300e-01	k_t = [2, 2, 4, 3]		
Generation 1970: Best = 5.014029e+03	FES = 197100	$\mu_F = 0.745$	$\mu_{CR} = 0.181$
Diversity = 1.2557e-01	k_t = [1, 1, 1, 1]		
Generation 1980: Best = 5.013608e+03	FES = 198100	$\mu_F = 0.727$	$\mu_{CR} = 0.174$
Diversity = 1.1971e-01	k_t = [2, 4, 4, 2]		
Generation 1990: Best = 5.012966e+03	FES = 199100	$\mu_F = 0.734$	$\mu_{CR} = 0.171$
Diversity = 1.2872e-01	k_t = [1, 1, 1, 1]		
Generation 2000: Best = 5.012306e+03	FES = 200100	$\mu_F = 0.741$	$\mu_{CR} = 0.168$
Diversity = 1.1458e-01	k_t = [3, 3, 3, 3]		
Generation 2010: Best = 5.011566e+03	FES = 201100	$\mu_F = 0.735$	$\mu_{CR} = 0.182$
Diversity = 1.2154e-01	k_t = [1, 1, 1, 1]		
Generation 2020: Best = 5.010414e+03	FES = 202100	$\mu_F = 0.740$	$\mu_{CR} = 0.176$
Diversity = 1.3003e-01	k_t = [4, 2, 4, 3]		
Generation 2030: Best = 5.009813e+03	FES = 203100	$\mu_F = 0.746$	$\mu_{CR} = 0.180$
Diversity = 1.3684e-01	k_t = [1, 1, 1, 1]		
Generation 2040: Best = 5.008704e+03	FES = 204100	$\mu_F = 0.746$	$\mu_{CR} = 0.177$
Diversity = 1.3984e-01	k_t = [3, 3, 3, 3]		
Generation 2050: Best = 5.007882e+03	FES = 205100	$\mu_F = 0.740$	$\mu_{CR} = 0.174$
Diversity = 1.3966e-01	k_t = [1, 1, 1, 1]		
Generation 2060: Best = 5.007059e+03	FES = 206100	$\mu_F = 0.740$	$\mu_{CR} = 0.176$
Diversity = 1.3477e-01	k_t = [4, 2, 3, 4]		
Generation 2070: Best = 5.005996e+03	FES = 207100	$\mu_F = 0.736$	$\mu_{CR} = 0.175$
Diversity = 1.3946e-01	k_t = [1, 1, 1, 1]		
Generation 2080: Best = 5.005855e+03	FES = 208100	$\mu_F = 0.741$	$\mu_{CR} = 0.166$
Diversity = 1.3532e-01	k_t = [4, 2, 4, 4]		
Generation 2090: Best = 5.004910e+03	FES = 209100	$\mu_F = 0.735$	$\mu_{CR} = 0.167$
Diversity = 1.2659e-01	k_t = [1, 1, 1, 1]		
Generation 2100: Best = 5.004555e+03	FES = 210100	$\mu_F = 0.745$	$\mu_{CR} = 0.165$
Diversity = 1.2099e-01	k_t = [3, 2, 2, 3]		
Generation 2110: Best = 5.004097e+03	FES = 211100	$\mu_F = 0.751$	$\mu_{CR} = 0.163$
Diversity = 1.3274e-01	k_t = [1, 1, 1, 1]		

Generation 2120: Best = 5.003421e+03 FES = 212100 μ_F = 0.743 μ_{CR} = 0.167
 Diversity = 1.2470e-01 k_t = [2, 3, 3, 2]
 Generation 2130: Best = 5.003108e+03 FES = 213100 μ_F = 0.729 μ_{CR} = 0.168
 Diversity = 1.1670e-01 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 5.002587e+03 FES = 214100 μ_F = 0.721 μ_{CR} = 0.162
 Diversity = 1.0060e-01 k_t = [2, 4, 2, 3]
 Generation 2150: Best = 5.002067e+03 FES = 215100 μ_F = 0.732 μ_{CR} = 0.169
 Diversity = 1.0399e-01 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 5.001794e+03 FES = 216100 μ_F = 0.722 μ_{CR} = 0.168
 Diversity = 8.8246e-02 k_t = [3, 3, 2, 4]
 Generation 2170: Best = 5.001291e+03 FES = 217100 μ_F = 0.728 μ_{CR} = 0.173
 Diversity = 9.2019e-02 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 5.001029e+03 FES = 218100 μ_F = 0.734 μ_{CR} = 0.174
 Diversity = 8.8955e-02 k_t = [3, 2, 3, 3]
 Generation 2190: Best = 5.000293e+03 FES = 219100 μ_F = 0.730 μ_{CR} = 0.176
 Diversity = 8.5406e-02 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 4.999863e+03 FES = 220100 μ_F = 0.744 μ_{CR} = 0.177
 Diversity = 9.1549e-02 k_t = [4, 2, 4, 2]
 Generation 2210: Best = 4.999255e+03 FES = 221100 μ_F = 0.743 μ_{CR} = 0.180
 Diversity = 9.4929e-02 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 4.999203e+03 FES = 222100 μ_F = 0.745 μ_{CR} = 0.180
 Diversity = 8.8090e-02 k_t = [4, 4, 4, 4]
 Generation 2230: Best = 4.998716e+03 FES = 223100 μ_F = 0.751 μ_{CR} = 0.181
 Diversity = 8.3220e-02 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 4.998425e+03 FES = 224100 μ_F = 0.759 μ_{CR} = 0.186
 Diversity = 7.7408e-02 k_t = [3, 4, 4, 4]
 Generation 2250: Best = 4.998091e+03 FES = 225100 μ_F = 0.732 μ_{CR} = 0.198
 Diversity = 7.7385e-02 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 4.997826e+03 FES = 226100 μ_F = 0.736 μ_{CR} = 0.197
 Diversity = 7.5193e-02 k_t = [3, 3, 2, 4]
 Generation 2270: Best = 4.997383e+03 FES = 227100 μ_F = 0.738 μ_{CR} = 0.189
 Diversity = 7.4199e-02 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 4.997291e+03 FES = 228100 μ_F = 0.731 μ_{CR} = 0.186
 Diversity = 6.9504e-02 k_t = [4, 2, 3, 3]
 Generation 2290: Best = 4.997058e+03 FES = 229100 μ_F = 0.732 μ_{CR} = 0.205
 Diversity = 6.6057e-02 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 4.996920e+03 FES = 230100 μ_F = 0.721 μ_{CR} = 0.205
 Diversity = 6.1114e-02 k_t = [4, 2, 4, 2]
 Generation 2310: Best = 4.996791e+03 FES = 231100 μ_F = 0.720 μ_{CR} = 0.211
 Diversity = 5.6975e-02 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 4.996505e+03 FES = 232100 μ_F = 0.702 μ_{CR} = 0.211
 Diversity = 5.2853e-02 k_t = [3, 3, 2, 4]
 Generation 2330: Best = 4.996432e+03 FES = 233100 μ_F = 0.698 μ_{CR} = 0.212
 Diversity = 4.7340e-02 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 4.996347e+03 FES = 234100 μ_F = 0.711 μ_{CR} = 0.215
 Diversity = 4.6162e-02 k_t = [2, 2, 3, 2]
 Generation 2350: Best = 4.996202e+03 FES = 235100 μ_F = 0.714 μ_{CR} = 0.210
 Diversity = 4.5660e-02 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 4.996140e+03 FES = 236100 μ_F = 0.708 μ_{CR} = 0.213
 Diversity = 3.8642e-02 k_t = [2, 2, 3, 2]
 Generation 2370: Best = 4.996038e+03 FES = 237100 μ_F = 0.708 μ_{CR} = 0.212
 Diversity = 3.6223e-02 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 4.995961e+03 FES = 238100 μ_F = 0.720 μ_{CR} = 0.214
 Diversity = 3.4010e-02 k_t = [2, 2, 2, 4]

Generation 2390: Best = 4.995883e+03 FES = 239100 μ_F = 0.717 μ_{CR} = 0.212
 Diversity = 3.3535e-02 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 4.995816e+03 FES = 240100 μ_F = 0.726 μ_{CR} = 0.210
 Diversity = 3.1976e-02 k_t = [2, 2, 4, 3]
 Generation 2410: Best = 4.995772e+03 FES = 241100 μ_F = 0.727 μ_{CR} = 0.211
 Diversity = 3.1779e-02 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 4.995685e+03 FES = 242100 μ_F = 0.740 μ_{CR} = 0.211
 Diversity = 3.2013e-02 k_t = [4, 4, 4, 3]
 Generation 2430: Best = 4.995602e+03 FES = 243100 μ_F = 0.747 μ_{CR} = 0.223
 Diversity = 3.4543e-02 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 4.995490e+03 FES = 244100 μ_F = 0.749 μ_{CR} = 0.230
 Diversity = 3.7483e-02 k_t = [2, 2, 3, 3]
 Generation 2450: Best = 4.995246e+03 FES = 245100 μ_F = 0.763 μ_{CR} = 0.235
 Diversity = 4.7391e-02 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 4.995018e+03 FES = 246100 μ_F = 0.758 μ_{CR} = 0.228
 Diversity = 5.2470e-02 k_t = [2, 3, 4, 2]
 Generation 2470: Best = 4.994730e+03 FES = 247100 μ_F = 0.759 μ_{CR} = 0.229
 Diversity = 6.2595e-02 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 4.994413e+03 FES = 248100 μ_F = 0.769 μ_{CR} = 0.247
 Diversity = 7.3120e-02 k_t = [4, 3, 2, 2]
 Generation 2490: Best = 4.993994e+03 FES = 249100 μ_F = 0.762 μ_{CR} = 0.232
 Diversity = 7.6813e-02 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 4.993676e+03 FES = 250100 μ_F = 0.766 μ_{CR} = 0.230
 Diversity = 7.1777e-02 k_t = [3, 3, 2, 4]
 Generation 2510: Best = 4.993331e+03 FES = 251100 μ_F = 0.764 μ_{CR} = 0.230
 Diversity = 7.7698e-02 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 4.993043e+03 FES = 252100 μ_F = 0.755 μ_{CR} = 0.233
 Diversity = 6.6757e-02 k_t = [4, 4, 2, 4]
 Generation 2530: Best = 4.992779e+03 FES = 253100 μ_F = 0.753 μ_{CR} = 0.229
 Diversity = 6.2690e-02 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 4.992486e+03 FES = 254100 μ_F = 0.737 μ_{CR} = 0.229
 Diversity = 6.1109e-02 k_t = [2, 3, 4, 2]
 Generation 2550: Best = 4.992212e+03 FES = 255100 μ_F = 0.744 μ_{CR} = 0.225
 Diversity = 5.9143e-02 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 4.991364e+03 FES = 256100 μ_F = 0.753 μ_{CR} = 0.231
 Diversity = 6.6415e-02 k_t = [2, 4, 2, 3]
 Generation 2570: Best = 4.990742e+03 FES = 257100 μ_F = 0.770 μ_{CR} = 0.235
 Diversity = 8.0224e-02 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 4.990488e+03 FES = 258100 μ_F = 0.774 μ_{CR} = 0.239
 Diversity = 7.0694e-02 k_t = [4, 3, 4, 3]
 Generation 2590: Best = 4.989978e+03 FES = 259100 μ_F = 0.777 μ_{CR} = 0.235
 Diversity = 8.0499e-02 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 4.989691e+03 FES = 260100 μ_F = 0.768 μ_{CR} = 0.243
 Diversity = 7.3041e-02 k_t = [2, 4, 3, 3]
 Generation 2610: Best = 4.989309e+03 FES = 261100 μ_F = 0.747 μ_{CR} = 0.241
 Diversity = 7.6491e-02 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 4.988988e+03 FES = 262100 μ_F = 0.740 μ_{CR} = 0.245
 Diversity = 6.7958e-02 k_t = [2, 3, 4, 4]
 Generation 2630: Best = 4.988723e+03 FES = 263100 μ_F = 0.739 μ_{CR} = 0.234
 Diversity = 6.3772e-02 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 4.988585e+03 FES = 264100 μ_F = 0.724 μ_{CR} = 0.227
 Diversity = 5.6511e-02 k_t = [4, 3, 4, 2]
 Generation 2650: Best = 4.988399e+03 FES = 265100 μ_F = 0.723 μ_{CR} = 0.230
 Diversity = 5.4531e-02 k_t = [1, 1, 1, 1]

Generation 2660: Best = 4.988253e+03 FES = 266100 μ_F = 0.723 μ_{CR} = 0.216
 Diversity = 4.6352e-02 k_t = [4, 4, 3, 3]
 Generation 2670: Best = 4.988161e+03 FES = 267100 μ_F = 0.730 μ_{CR} = 0.209
 Diversity = 4.3172e-02 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 4.988032e+03 FES = 268100 μ_F = 0.744 μ_{CR} = 0.203
 Diversity = 4.0800e-02 k_t = [4, 2, 2, 3]
 Generation 2690: Best = 4.987906e+03 FES = 269100 μ_F = 0.751 μ_{CR} = 0.204
 Diversity = 4.0140e-02 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 4.987728e+03 FES = 270100 μ_F = 0.768 μ_{CR} = 0.201
 Diversity = 4.1176e-02 k_t = [2, 3, 2, 2]
 Generation 2710: Best = 4.987584e+03 FES = 271100 μ_F = 0.770 μ_{CR} = 0.208
 Diversity = 4.2490e-02 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 4.987416e+03 FES = 272100 μ_F = 0.764 μ_{CR} = 0.201
 Diversity = 4.5403e-02 k_t = [3, 3, 2, 4]
 Generation 2730: Best = 4.987278e+03 FES = 273100 μ_F = 0.753 μ_{CR} = 0.212
 Diversity = 4.5567e-02 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 4.987129e+03 FES = 274100 μ_F = 0.762 μ_{CR} = 0.208
 Diversity = 4.7348e-02 k_t = [4, 3, 4, 3]
 Generation 2750: Best = 4.986901e+03 FES = 275100 μ_F = 0.753 μ_{CR} = 0.218
 Diversity = 4.6283e-02 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 4.986842e+03 FES = 276100 μ_F = 0.745 μ_{CR} = 0.212
 Diversity = 4.1849e-02 k_t = [2, 4, 4, 2]
 Generation 2770: Best = 4.986691e+03 FES = 277100 μ_F = 0.747 μ_{CR} = 0.212
 Diversity = 4.2508e-02 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 4.986532e+03 FES = 278100 μ_F = 0.746 μ_{CR} = 0.215
 Diversity = 4.3248e-02 k_t = [4, 2, 4, 3]
 Generation 2790: Best = 4.986432e+03 FES = 279100 μ_F = 0.752 μ_{CR} = 0.215
 Diversity = 4.7607e-02 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 4.986267e+03 FES = 280100 μ_F = 0.745 μ_{CR} = 0.219
 Diversity = 4.4117e-02 k_t = [4, 4, 3, 2]
 Generation 2810: Best = 4.986177e+03 FES = 281100 μ_F = 0.745 μ_{CR} = 0.214
 Diversity = 4.3486e-02 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 4.986048e+03 FES = 282100 μ_F = 0.742 μ_{CR} = 0.213
 Diversity = 4.0025e-02 k_t = [3, 3, 2, 2]
 Generation 2830: Best = 4.985965e+03 FES = 283100 μ_F = 0.741 μ_{CR} = 0.221
 Diversity = 3.9172e-02 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 4.985793e+03 FES = 284100 μ_F = 0.755 μ_{CR} = 0.218
 Diversity = 3.9875e-02 k_t = [4, 3, 4, 4]
 Generation 2850: Best = 4.985693e+03 FES = 285100 μ_F = 0.755 μ_{CR} = 0.221
 Diversity = 3.9650e-02 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 4.985569e+03 FES = 286100 μ_F = 0.738 μ_{CR} = 0.232
 Diversity = 4.1993e-02 k_t = [4, 3, 2, 2]
 Generation 2870: Best = 4.985449e+03 FES = 287100 μ_F = 0.733 μ_{CR} = 0.229
 Diversity = 4.1249e-02 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 4.985347e+03 FES = 288100 μ_F = 0.729 μ_{CR} = 0.232
 Diversity = 3.8520e-02 k_t = [4, 3, 2, 2]
 Generation 2890: Best = 4.985215e+03 FES = 289100 μ_F = 0.734 μ_{CR} = 0.237
 Diversity = 3.9136e-02 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 4.985108e+03 FES = 290100 μ_F = 0.748 μ_{CR} = 0.231
 Diversity = 3.7242e-02 k_t = [3, 2, 4, 4]
 Generation 2910: Best = 4.985028e+03 FES = 291100 μ_F = 0.744 μ_{CR} = 0.229
 Diversity = 3.4469e-02 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 4.984897e+03 FES = 292100 μ_F = 0.758 μ_{CR} = 0.223
 Diversity = 3.3219e-02 k_t = [4, 4, 2, 3]

Generation 2930: Best = 4.984792e+03 FES = 293100 μ_F = 0.754 μ_{CR} = 0.224
 Diversity = 3.2747e-02 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 4.984705e+03 FES = 294100 μ_F = 0.759 μ_{CR} = 0.222
 Diversity = 3.3649e-02 k_t = [3, 4, 3, 2]
 Generation 2950: Best = 4.984615e+03 FES = 295100 μ_F = 0.760 μ_{CR} = 0.219
 Diversity = 3.4741e-02 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 4.984500e+03 FES = 296100 μ_F = 0.763 μ_{CR} = 0.231
 Diversity = 3.6226e-02 k_t = [4, 3, 3, 3]
 Generation 2970: Best = 4.984417e+03 FES = 297100 μ_F = 0.743 μ_{CR} = 0.232
 Diversity = 3.8675e-02 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 4.984290e+03 FES = 298100 μ_F = 0.743 μ_{CR} = 0.236
 Diversity = 3.6895e-02 k_t = [2, 2, 3, 2]
 Generation 2990: Best = 4.984220e+03 FES = 299100 μ_F = 0.735 μ_{CR} = 0.239
 Diversity = 3.7202e-02 k_t = [1, 1, 1, 1]
 Generation 3000: Best = 4.984146e+03 FES = 300100 μ_F = 0.736 μ_{CR} = 0.238
 Diversity = 3.2596e-02 k_t = [3, 4, 4, 3]
 Generation 3010: Best = 4.984066e+03 FES = 301100 μ_F = 0.726 μ_{CR} = 0.234
 Diversity = 2.9057e-02 k_t = [1, 1, 1, 1]
 Generation 3020: Best = 4.983997e+03 FES = 302100 μ_F = 0.740 μ_{CR} = 0.226
 Diversity = 2.8418e-02 k_t = [3, 3, 4, 4]
 Generation 3030: Best = 4.983863e+03 FES = 303100 μ_F = 0.745 μ_{CR} = 0.216
 Diversity = 3.0160e-02 k_t = [1, 1, 1, 1]
 Generation 3040: Best = 4.983790e+03 FES = 304100 μ_F = 0.745 μ_{CR} = 0.210
 Diversity = 3.1447e-02 k_t = [3, 3, 4, 4]
 Generation 3050: Best = 4.983628e+03 FES = 305100 μ_F = 0.755 μ_{CR} = 0.208
 Diversity = 3.2126e-02 k_t = [1, 1, 1, 1]
 Generation 3060: Best = 4.983535e+03 FES = 306100 μ_F = 0.765 μ_{CR} = 0.221
 Diversity = 3.2772e-02 k_t = [2, 4, 4, 4]
 Generation 3070: Best = 4.983390e+03 FES = 307100 μ_F = 0.770 μ_{CR} = 0.220
 Diversity = 3.3833e-02 k_t = [1, 1, 1, 1]
 Generation 3080: Best = 4.983179e+03 FES = 308100 μ_F = 0.769 μ_{CR} = 0.222
 Diversity = 3.8063e-02 k_t = [4, 2, 2, 2]
 Generation 3090: Best = 4.982901e+03 FES = 309100 μ_F = 0.769 μ_{CR} = 0.209
 Diversity = 4.1387e-02 k_t = [1, 1, 1, 1]
 Generation 3100: Best = 4.982614e+03 FES = 310100 μ_F = 0.779 μ_{CR} = 0.214
 Diversity = 4.6141e-02 k_t = [3, 4, 2, 3]
 Generation 3110: Best = 4.982383e+03 FES = 311100 μ_F = 0.778 μ_{CR} = 0.221
 Diversity = 5.3085e-02 k_t = [1, 1, 1, 1]
 Generation 3120: Best = 4.982131e+03 FES = 312100 μ_F = 0.777 μ_{CR} = 0.219
 Diversity = 5.9761e-02 k_t = [2, 4, 4, 3]
 Generation 3130: Best = 4.982046e+03 FES = 313100 μ_F = 0.755 μ_{CR} = 0.204
 Diversity = 5.7233e-02 k_t = [1, 1, 1, 1]
 Generation 3140: Best = 4.981835e+03 FES = 314100 μ_F = 0.753 μ_{CR} = 0.187
 Diversity = 5.1480e-02 k_t = [3, 3, 2, 3]
 Generation 3150: Best = 4.981697e+03 FES = 315100 μ_F = 0.746 μ_{CR} = 0.190
 Diversity = 5.0600e-02 k_t = [1, 1, 1, 1]
 Generation 3160: Best = 4.981562e+03 FES = 316100 μ_F = 0.745 μ_{CR} = 0.187
 Diversity = 4.6971e-02 k_t = [4, 4, 3, 2]
 Generation 3170: Best = 4.981391e+03 FES = 317100 μ_F = 0.740 μ_{CR} = 0.191
 Diversity = 5.0344e-02 k_t = [1, 1, 1, 1]
 Generation 3180: Best = 4.981081e+03 FES = 318100 μ_F = 0.738 μ_{CR} = 0.197
 Diversity = 5.8032e-02 k_t = [3, 2, 2, 2]
 Generation 3190: Best = 4.980828e+03 FES = 319100 μ_F = 0.741 μ_{CR} = 0.193
 Diversity = 6.7854e-02 k_t = [1, 1, 1, 1]

Generation 3200: Best = 4.980544e+03 FES = 320100 μ_F = 0.747 μ_{CR} = 0.208
 Diversity = 6.4334e-02 k_t = [4, 2, 4, 3]
 Generation 3210: Best = 4.980272e+03 FES = 321100 μ_F = 0.745 μ_{CR} = 0.214
 Diversity = 6.8537e-02 k_t = [1, 1, 1, 1]
 Generation 3220: Best = 4.980173e+03 FES = 322100 μ_F = 0.746 μ_{CR} = 0.219
 Diversity = 6.0717e-02 k_t = [3, 3, 2, 3]
 Generation 3230: Best = 4.980056e+03 FES = 323100 μ_F = 0.739 μ_{CR} = 0.206
 Diversity = 5.9430e-02 k_t = [1, 1, 1, 1]
 Generation 3240: Best = 4.979900e+03 FES = 324100 μ_F = 0.726 μ_{CR} = 0.194
 Diversity = 5.0891e-02 k_t = [4, 4, 2, 4]
 Generation 3250: Best = 4.979791e+03 FES = 325100 μ_F = 0.728 μ_{CR} = 0.193
 Diversity = 4.6957e-02 k_t = [1, 1, 1, 1]
 Generation 3260: Best = 4.979714e+03 FES = 326100 μ_F = 0.734 μ_{CR} = 0.196
 Diversity = 4.5871e-02 k_t = [3, 3, 2, 2]
 Generation 3270: Best = 4.979634e+03 FES = 327100 μ_F = 0.737 μ_{CR} = 0.190
 Diversity = 4.1284e-02 k_t = [1, 1, 1, 1]
 Generation 3280: Best = 4.979545e+03 FES = 328100 μ_F = 0.750 μ_{CR} = 0.185
 Diversity = 4.2676e-02 k_t = [3, 3, 2, 2]
 Generation 3290: Best = 4.979442e+03 FES = 329100 μ_F = 0.748 μ_{CR} = 0.187
 Diversity = 4.4910e-02 k_t = [1, 1, 1, 1]
 Generation 3300: Best = 4.979310e+03 FES = 330100 μ_F = 0.743 μ_{CR} = 0.192
 Diversity = 4.0573e-02 k_t = [2, 4, 3, 3]
 Generation 3310: Best = 4.979246e+03 FES = 331100 μ_F = 0.745 μ_{CR} = 0.189
 Diversity = 4.0367e-02 k_t = [1, 1, 1, 1]
 Generation 3320: Best = 4.979129e+03 FES = 332100 μ_F = 0.738 μ_{CR} = 0.185
 Diversity = 3.8751e-02 k_t = [3, 3, 4, 3]
 Generation 3330: Best = 4.979008e+03 FES = 333100 μ_F = 0.740 μ_{CR} = 0.174
 Diversity = 4.1473e-02 k_t = [1, 1, 1, 1]
 Generation 3340: Best = 4.978902e+03 FES = 334100 μ_F = 0.748 μ_{CR} = 0.175
 Diversity = 4.0924e-02 k_t = [2, 2, 4, 3]
 Generation 3350: Best = 4.978800e+03 FES = 335100 μ_F = 0.752 μ_{CR} = 0.179
 Diversity = 4.4097e-02 k_t = [1, 1, 1, 1]
 Generation 3360: Best = 4.978705e+03 FES = 336100 μ_F = 0.753 μ_{CR} = 0.185
 Diversity = 4.2454e-02 k_t = [3, 4, 4, 2]
 Generation 3370: Best = 4.978620e+03 FES = 337100 μ_F = 0.753 μ_{CR} = 0.178
 Diversity = 4.0711e-02 k_t = [1, 1, 1, 1]
 Generation 3380: Best = 4.978506e+03 FES = 338100 μ_F = 0.746 μ_{CR} = 0.180
 Diversity = 3.4303e-02 k_t = [3, 3, 3, 3]
 Generation 3390: Best = 4.978426e+03 FES = 339100 μ_F = 0.739 μ_{CR} = 0.171
 Diversity = 3.9948e-02 k_t = [1, 1, 1, 1]
 Generation 3400: Best = 4.978406e+03 FES = 340100 μ_F = 0.731 μ_{CR} = 0.167
 Diversity = 3.5986e-02 k_t = [4, 4, 3, 2]
 Generation 3410: Best = 4.978356e+03 FES = 341100 μ_F = 0.734 μ_{CR} = 0.171
 Diversity = 3.4605e-02 k_t = [1, 1, 1, 1]
 Generation 3420: Best = 4.978309e+03 FES = 342100 μ_F = 0.734 μ_{CR} = 0.184
 Diversity = 3.1261e-02 k_t = [2, 2, 2, 2]
 Generation 3430: Best = 4.978259e+03 FES = 343100 μ_F = 0.744 μ_{CR} = 0.175
 Diversity = 3.1932e-02 k_t = [1, 1, 1, 1]
 Generation 3440: Best = 4.978183e+03 FES = 344100 μ_F = 0.744 μ_{CR} = 0.183
 Diversity = 2.8923e-02 k_t = [2, 4, 2, 4]
 Generation 3450: Best = 4.978123e+03 FES = 345100 μ_F = 0.748 μ_{CR} = 0.184
 Diversity = 2.8242e-02 k_t = [1, 1, 1, 1]
 Generation 3460: Best = 4.978081e+03 FES = 346100 μ_F = 0.748 μ_{CR} = 0.181
 Diversity = 2.7541e-02 k_t = [3, 4, 4, 4]

Generation 3470: Best = 4.978032e+03 FES = 347100 μ_F = 0.762 μ_{CR} = 0.180
 Diversity = 2.5890e-02 k_t = [1, 1, 1, 1]
 Generation 3480: Best = 4.977983e+03 FES = 348100 μ_F = 0.767 μ_{CR} = 0.178
 Diversity = 2.6854e-02 k_t = [3, 3, 2, 2]
 Generation 3490: Best = 4.977896e+03 FES = 349100 μ_F = 0.750 μ_{CR} = 0.177
 Diversity = 2.7489e-02 k_t = [1, 1, 1, 1]
 Generation 3500: Best = 4.977833e+03 FES = 350100 μ_F = 0.742 μ_{CR} = 0.173
 Diversity = 2.8792e-02 k_t = [4, 2, 2, 3]
 Generation 3510: Best = 4.977790e+03 FES = 351100 μ_F = 0.742 μ_{CR} = 0.179
 Diversity = 3.0064e-02 k_t = [1, 1, 1, 1]
 Generation 3520: Best = 4.977737e+03 FES = 352100 μ_F = 0.739 μ_{CR} = 0.184
 Diversity = 2.7854e-02 k_t = [2, 3, 3, 4]
 Generation 3530: Best = 4.977656e+03 FES = 353100 μ_F = 0.728 μ_{CR} = 0.190
 Diversity = 2.7550e-02 k_t = [1, 1, 1, 1]
 Generation 3540: Best = 4.977572e+03 FES = 354100 μ_F = 0.736 μ_{CR} = 0.191
 Diversity = 3.0167e-02 k_t = [2, 2, 2, 2]
 Generation 3550: Best = 4.977525e+03 FES = 355100 μ_F = 0.732 μ_{CR} = 0.194
 Diversity = 3.3075e-02 k_t = [1, 1, 1, 1]
 Generation 3560: Best = 4.977456e+03 FES = 356100 μ_F = 0.723 μ_{CR} = 0.198
 Diversity = 2.9982e-02 k_t = [4, 4, 3, 3]
 Generation 3570: Best = 4.977415e+03 FES = 357100 μ_F = 0.722 μ_{CR} = 0.199
 Diversity = 2.7328e-02 k_t = [1, 1, 1, 1]
 Generation 3580: Best = 4.977384e+03 FES = 358100 μ_F = 0.721 μ_{CR} = 0.189
 Diversity = 2.5150e-02 k_t = [3, 2, 3, 4]
 Generation 3590: Best = 4.977347e+03 FES = 359100 μ_F = 0.738 μ_{CR} = 0.202
 Diversity = 2.5031e-02 k_t = [1, 1, 1, 1]
 Generation 3600: Best = 4.977278e+03 FES = 360100 μ_F = 0.745 μ_{CR} = 0.195
 Diversity = 2.5158e-02 k_t = [4, 3, 4, 2]
 Generation 3610: Best = 4.977195e+03 FES = 361100 μ_F = 0.758 μ_{CR} = 0.199
 Diversity = 2.8283e-02 k_t = [1, 1, 1, 1]
 Generation 3620: Best = 4.977091e+03 FES = 362100 μ_F = 0.748 μ_{CR} = 0.201
 Diversity = 2.6665e-02 k_t = [3, 3, 3, 3]
 Generation 3630: Best = 4.977021e+03 FES = 363100 μ_F = 0.766 μ_{CR} = 0.200
 Diversity = 3.1296e-02 k_t = [1, 1, 1, 1]
 Generation 3640: Best = 4.976954e+03 FES = 364100 μ_F = 0.775 μ_{CR} = 0.190
 Diversity = 3.0371e-02 k_t = [2, 2, 4, 4]
 Generation 3650: Best = 4.976904e+03 FES = 365100 μ_F = 0.773 μ_{CR} = 0.186
 Diversity = 3.3133e-02 k_t = [1, 1, 1, 1]
 Generation 3660: Best = 4.976846e+03 FES = 366100 μ_F = 0.770 μ_{CR} = 0.189
 Diversity = 3.0856e-02 k_t = [2, 3, 2, 4]
 Generation 3670: Best = 4.976807e+03 FES = 367100 μ_F = 0.760 μ_{CR} = 0.182
 Diversity = 3.0380e-02 k_t = [1, 1, 1, 1]
 Generation 3680: Best = 4.976760e+03 FES = 368100 μ_F = 0.758 μ_{CR} = 0.181
 Diversity = 2.6944e-02 k_t = [3, 4, 4, 3]
 Generation 3690: Best = 4.976699e+03 FES = 369100 μ_F = 0.756 μ_{CR} = 0.182
 Diversity = 2.7301e-02 k_t = [1, 1, 1, 1]
 Generation 3700: Best = 4.976548e+03 FES = 370100 μ_F = 0.750 μ_{CR} = 0.192
 Diversity = 2.9537e-02 k_t = [2, 3, 2, 4]
 Generation 3710: Best = 4.976442e+03 FES = 371100 μ_F = 0.753 μ_{CR} = 0.192
 Diversity = 3.3254e-02 k_t = [1, 1, 1, 1]
 Generation 3720: Best = 4.976311e+03 FES = 372100 μ_F = 0.756 μ_{CR} = 0.199
 Diversity = 4.0546e-02 k_t = [3, 2, 2, 3]
 Generation 3730: Best = 4.976191e+03 FES = 373100 μ_F = 0.752 μ_{CR} = 0.201
 Diversity = 3.8631e-02 k_t = [1, 1, 1, 1]

Generation 3740: Best = 4.976119e+03 FES = 374100 μ_F = 0.755 μ_{CR} = 0.193
 Diversity = 3.7598e-02 k_t = [3, 2, 4, 3]
 Generation 3750: Best = 4.976008e+03 FES = 375100 μ_F = 0.752 μ_{CR} = 0.191
 Diversity = 3.8363e-02 k_t = [1, 1, 1, 1]
 Generation 3760: Best = 4.975921e+03 FES = 376100 μ_F = 0.747 μ_{CR} = 0.189
 Diversity = 3.7935e-02 k_t = [4, 4, 2, 2]
 Generation 3770: Best = 4.975854e+03 FES = 377100 μ_F = 0.735 μ_{CR} = 0.197
 Diversity = 3.8444e-02 k_t = [1, 1, 1, 1]
 Generation 3780: Best = 4.975790e+03 FES = 378100 μ_F = 0.732 μ_{CR} = 0.196
 Diversity = 3.9255e-02 k_t = [4, 2, 2, 2]
 Generation 3790: Best = 4.975706e+03 FES = 379100 μ_F = 0.740 μ_{CR} = 0.192
 Diversity = 4.2064e-02 k_t = [1, 1, 1, 1]
 Generation 3800: Best = 4.975609e+03 FES = 380100 μ_F = 0.740 μ_{CR} = 0.176
 Diversity = 4.1951e-02 k_t = [2, 2, 3, 2]
 Generation 3810: Best = 4.975535e+03 FES = 381100 μ_F = 0.737 μ_{CR} = 0.177
 Diversity = 4.1786e-02 k_t = [1, 1, 1, 1]
 Generation 3820: Best = 4.975410e+03 FES = 382100 μ_F = 0.727 μ_{CR} = 0.182
 Diversity = 3.6032e-02 k_t = [3, 4, 4, 3]
 Generation 3830: Best = 4.975277e+03 FES = 383100 μ_F = 0.739 μ_{CR} = 0.189
 Diversity = 4.1582e-02 k_t = [1, 1, 1, 1]
 Generation 3840: Best = 4.975207e+03 FES = 384100 μ_F = 0.739 μ_{CR} = 0.182
 Diversity = 4.1046e-02 k_t = [4, 2, 4, 2]
 Generation 3850: Best = 4.975007e+03 FES = 385100 μ_F = 0.727 μ_{CR} = 0.188
 Diversity = 4.2664e-02 k_t = [1, 1, 1, 1]
 Generation 3860: Best = 4.974864e+03 FES = 386100 μ_F = 0.723 μ_{CR} = 0.183
 Diversity = 3.7988e-02 k_t = [2, 2, 4, 4]
 Generation 3870: Best = 4.974684e+03 FES = 387100 μ_F = 0.740 μ_{CR} = 0.199
 Diversity = 4.6267e-02 k_t = [1, 1, 1, 1]
 Generation 3880: Best = 4.974578e+03 FES = 388100 μ_F = 0.737 μ_{CR} = 0.201
 Diversity = 4.3830e-02 k_t = [3, 2, 4, 4]
 Generation 3890: Best = 4.974465e+03 FES = 389100 μ_F = 0.747 μ_{CR} = 0.201
 Diversity = 4.0612e-02 k_t = [1, 1, 1, 1]
 Generation 3900: Best = 4.974356e+03 FES = 390100 μ_F = 0.753 μ_{CR} = 0.189
 Diversity = 3.8323e-02 k_t = [4, 2, 4, 2]
 Generation 3910: Best = 4.974240e+03 FES = 391100 μ_F = 0.746 μ_{CR} = 0.201
 Diversity = 4.0093e-02 k_t = [1, 1, 1, 1]
 Generation 3920: Best = 4.974117e+03 FES = 392100 μ_F = 0.746 μ_{CR} = 0.206
 Diversity = 3.8866e-02 k_t = [2, 4, 2, 2]
 Generation 3930: Best = 4.973990e+03 FES = 393100 μ_F = 0.746 μ_{CR} = 0.193
 Diversity = 4.1679e-02 k_t = [1, 1, 1, 1]
 Generation 3940: Best = 4.973910e+03 FES = 394100 μ_F = 0.746 μ_{CR} = 0.194
 Diversity = 4.0475e-02 k_t = [3, 3, 2, 4]
 Generation 3950: Best = 4.973864e+03 FES = 395100 μ_F = 0.740 μ_{CR} = 0.180
 Diversity = 3.5242e-02 k_t = [1, 1, 1, 1]
 Generation 3960: Best = 4.973792e+03 FES = 396100 μ_F = 0.720 μ_{CR} = 0.174
 Diversity = 3.5277e-02 k_t = [3, 2, 2, 2]
 Generation 3970: Best = 4.973762e+03 FES = 397100 μ_F = 0.734 μ_{CR} = 0.174
 Diversity = 3.4413e-02 k_t = [1, 1, 1, 1]
 Generation 3980: Best = 4.973727e+03 FES = 398100 μ_F = 0.737 μ_{CR} = 0.161
 Diversity = 2.9213e-02 k_t = [3, 2, 3, 4]
 Generation 3990: Best = 4.973686e+03 FES = 399100 μ_F = 0.741 μ_{CR} = 0.163
 Diversity = 2.7053e-02 k_t = [1, 1, 1, 1]
 Generation 4000: Best = 4.973647e+03 FES = 400100 μ_F = 0.740 μ_{CR} = 0.164
 Diversity = 2.6346e-02 k_t = [4, 2, 4, 3]

Generation 4010: Best = 4.973596e+03 FES = 401100 μ_F = 0.744 μ_{CR} = 0.180
 Diversity = 2.5590e-02 k_t = [1, 1, 1, 1]
 Generation 4020: Best = 4.973571e+03 FES = 402100 μ_F = 0.741 μ_{CR} = 0.172
 Diversity = 2.2576e-02 k_t = [4, 4, 4, 3]
 Generation 4030: Best = 4.973546e+03 FES = 403100 μ_F = 0.742 μ_{CR} = 0.172
 Diversity = 1.9355e-02 k_t = [1, 1, 1, 1]
 Generation 4040: Best = 4.973515e+03 FES = 404100 μ_F = 0.735 μ_{CR} = 0.177
 Diversity = 1.7788e-02 k_t = [4, 4, 2, 4]
 Generation 4050: Best = 4.973474e+03 FES = 405100 μ_F = 0.744 μ_{CR} = 0.180
 Diversity = 1.8952e-02 k_t = [1, 1, 1, 1]
 Generation 4060: Best = 4.973442e+03 FES = 406100 μ_F = 0.741 μ_{CR} = 0.195
 Diversity = 1.8643e-02 k_t = [2, 2, 4, 4]
 Generation 4070: Best = 4.973419e+03 FES = 407100 μ_F = 0.735 μ_{CR} = 0.194
 Diversity = 1.8656e-02 k_t = [1, 1, 1, 1]
 Generation 4080: Best = 4.973382e+03 FES = 408100 μ_F = 0.737 μ_{CR} = 0.202
 Diversity = 1.8389e-02 k_t = [4, 4, 3, 4]
 Generation 4090: Best = 4.973344e+03 FES = 409100 μ_F = 0.743 μ_{CR} = 0.202
 Diversity = 2.0128e-02 k_t = [1, 1, 1, 1]
 Generation 4100: Best = 4.973320e+03 FES = 410100 μ_F = 0.743 μ_{CR} = 0.203
 Diversity = 1.9546e-02 k_t = [3, 4, 2, 2]
 Generation 4110: Best = 4.973291e+03 FES = 411100 μ_F = 0.728 μ_{CR} = 0.198
 Diversity = 1.7350e-02 k_t = [1, 1, 1, 1]
 Generation 4120: Best = 4.973262e+03 FES = 412100 μ_F = 0.740 μ_{CR} = 0.203
 Diversity = 1.6728e-02 k_t = [2, 4, 4, 3]
 Generation 4130: Best = 4.973253e+03 FES = 413100 μ_F = 0.741 μ_{CR} = 0.202
 Diversity = 1.6621e-02 k_t = [1, 1, 1, 1]
 Generation 4140: Best = 4.973217e+03 FES = 414100 μ_F = 0.746 μ_{CR} = 0.199
 Diversity = 1.7018e-02 k_t = [4, 4, 4, 2]
 Generation 4150: Best = 4.973195e+03 FES = 415100 μ_F = 0.748 μ_{CR} = 0.194
 Diversity = 1.7919e-02 k_t = [1, 1, 1, 1]
 Generation 4160: Best = 4.973169e+03 FES = 416100 μ_F = 0.763 μ_{CR} = 0.194
 Diversity = 1.8108e-02 k_t = [2, 4, 3, 2]
 Generation 4170: Best = 4.973136e+03 FES = 417100 μ_F = 0.762 μ_{CR} = 0.198
 Diversity = 1.7822e-02 k_t = [1, 1, 1, 1]
 Generation 4180: Best = 4.973120e+03 FES = 418100 μ_F = 0.767 μ_{CR} = 0.215
 Diversity = 1.8303e-02 k_t = [3, 2, 4, 2]
 Generation 4190: Best = 4.973093e+03 FES = 419100 μ_F = 0.748 μ_{CR} = 0.218
 Diversity = 1.7484e-02 k_t = [1, 1, 1, 1]
 Generation 4200: Best = 4.973075e+03 FES = 420100 μ_F = 0.735 μ_{CR} = 0.226
 Diversity = 1.5862e-02 k_t = [3, 4, 2, 3]
 Generation 4210: Best = 4.973057e+03 FES = 421100 μ_F = 0.735 μ_{CR} = 0.219
 Diversity = 1.5895e-02 k_t = [1, 1, 1, 1]
 Generation 4220: Best = 4.973040e+03 FES = 422100 μ_F = 0.738 μ_{CR} = 0.231
 Diversity = 1.3647e-02 k_t = [3, 4, 2, 3]
 Generation 4230: Best = 4.973027e+03 FES = 423100 μ_F = 0.731 μ_{CR} = 0.228
 Diversity = 1.2766e-02 k_t = [1, 1, 1, 1]
 Generation 4240: Best = 4.973011e+03 FES = 424100 μ_F = 0.732 μ_{CR} = 0.224
 Diversity = 1.1753e-02 k_t = [3, 3, 4, 3]
 Generation 4250: Best = 4.972995e+03 FES = 425100 μ_F = 0.739 μ_{CR} = 0.226
 Diversity = 1.4252e-02 k_t = [1, 1, 1, 1]
 Generation 4260: Best = 4.972969e+03 FES = 426100 μ_F = 0.750 μ_{CR} = 0.227
 Diversity = 1.5707e-02 k_t = [3, 3, 3, 2]
 Generation 4270: Best = 4.972941e+03 FES = 427100 μ_F = 0.759 μ_{CR} = 0.226
 Diversity = 1.6956e-02 k_t = [1, 1, 1, 1]

Generation 4280: Best = 4.972917e+03 FES = 428100 μ_F = 0.766 μ_{CR} = 0.222
 Diversity = 1.6284e-02 k_t = [2, 2, 3, 3]
 Generation 4290: Best = 4.972888e+03 FES = 429100 μ_F = 0.773 μ_{CR} = 0.219
 Diversity = 1.7149e-02 k_t = [1, 1, 1, 1]
 Generation 4300: Best = 4.972877e+03 FES = 430100 μ_F = 0.768 μ_{CR} = 0.204
 Diversity = 1.6845e-02 k_t = [4, 4, 2, 4]
 Generation 4310: Best = 4.972866e+03 FES = 431100 μ_F = 0.759 μ_{CR} = 0.197
 Diversity = 1.4626e-02 k_t = [1, 1, 1, 1]
 Generation 4320: Best = 4.972854e+03 FES = 432100 μ_F = 0.750 μ_{CR} = 0.204
 Diversity = 1.3932e-02 k_t = [2, 3, 4, 3]
 Generation 4330: Best = 4.972841e+03 FES = 433100 μ_F = 0.754 μ_{CR} = 0.200
 Diversity = 1.3959e-02 k_t = [1, 1, 1, 1]
 Generation 4340: Best = 4.972827e+03 FES = 434100 μ_F = 0.764 μ_{CR} = 0.207
 Diversity = 1.4505e-02 k_t = [2, 4, 2, 2]
 Generation 4350: Best = 4.972807e+03 FES = 435100 μ_F = 0.759 μ_{CR} = 0.219
 Diversity = 1.6160e-02 k_t = [1, 1, 1, 1]
 Generation 4360: Best = 4.972786e+03 FES = 436100 μ_F = 0.767 μ_{CR} = 0.219
 Diversity = 1.5936e-02 k_t = [4, 2, 3, 3]
 Generation 4370: Best = 4.972769e+03 FES = 437100 μ_F = 0.758 μ_{CR} = 0.218
 Diversity = 1.5988e-02 k_t = [1, 1, 1, 1]
 Generation 4380: Best = 4.972749e+03 FES = 438100 μ_F = 0.753 μ_{CR} = 0.212
 Diversity = 1.3519e-02 k_t = [3, 4, 4, 4]
 Generation 4390: Best = 4.972736e+03 FES = 439100 μ_F = 0.750 μ_{CR} = 0.216
 Diversity = 1.4591e-02 k_t = [1, 1, 1, 1]
 Generation 4400: Best = 4.972730e+03 FES = 440100 μ_F = 0.745 μ_{CR} = 0.216
 Diversity = 1.3277e-02 k_t = [4, 3, 4, 4]
 Generation 4410: Best = 4.972711e+03 FES = 441100 μ_F = 0.747 μ_{CR} = 0.217
 Diversity = 1.3415e-02 k_t = [1, 1, 1, 1]
 Generation 4420: Best = 4.972693e+03 FES = 442100 μ_F = 0.746 μ_{CR} = 0.213
 Diversity = 1.4414e-02 k_t = [3, 2, 3, 2]
 Generation 4430: Best = 4.972672e+03 FES = 443100 μ_F = 0.753 μ_{CR} = 0.213
 Diversity = 1.6062e-02 k_t = [1, 1, 1, 1]
 Generation 4440: Best = 4.972643e+03 FES = 444100 μ_F = 0.758 μ_{CR} = 0.220
 Diversity = 1.6156e-02 k_t = [4, 4, 2, 4]
 Generation 4450: Best = 4.972623e+03 FES = 445100 μ_F = 0.761 μ_{CR} = 0.218
 Diversity = 1.6315e-02 k_t = [1, 1, 1, 1]
 Generation 4460: Best = 4.972601e+03 FES = 446100 μ_F = 0.758 μ_{CR} = 0.231
 Diversity = 1.5808e-02 k_t = [4, 2, 2, 4]
 Generation 4470: Best = 4.972587e+03 FES = 447100 μ_F = 0.759 μ_{CR} = 0.215
 Diversity = 1.5137e-02 k_t = [1, 1, 1, 1]
 Generation 4480: Best = 4.972578e+03 FES = 448100 μ_F = 0.757 μ_{CR} = 0.223
 Diversity = 1.2891e-02 k_t = [2, 2, 4, 3]
 Generation 4490: Best = 4.972562e+03 FES = 449100 μ_F = 0.759 μ_{CR} = 0.225
 Diversity = 1.3517e-02 k_t = [1, 1, 1, 1]
 Generation 4500: Best = 4.972546e+03 FES = 450100 μ_F = 0.757 μ_{CR} = 0.215
 Diversity = 1.3927e-02 k_t = [3, 4, 4, 4]
 Generation 4510: Best = 4.972536e+03 FES = 451100 μ_F = 0.758 μ_{CR} = 0.210
 Diversity = 1.3408e-02 k_t = [1, 1, 1, 1]
 Generation 4520: Best = 4.972523e+03 FES = 452100 μ_F = 0.737 μ_{CR} = 0.213
 Diversity = 1.3126e-02 k_t = [4, 2, 4, 3]
 Generation 4530: Best = 4.972510e+03 FES = 453100 μ_F = 0.731 μ_{CR} = 0.217
 Diversity = 1.2809e-02 k_t = [1, 1, 1, 1]
 Generation 4540: Best = 4.972502e+03 FES = 454100 μ_F = 0.720 μ_{CR} = 0.211
 Diversity = 1.1275e-02 k_t = [4, 4, 2, 4]

Generation 4550: Best = 4.972495e+03 FES = 455100 μ_F = 0.708 μ_{CR} = 0.208
 Diversity = 9.3542e-03 k_t = [1, 1, 1, 1]
 Generation 4560: Best = 4.972484e+03 FES = 456100 μ_F = 0.714 μ_{CR} = 0.202
 Diversity = 8.9301e-03 k_t = [2, 3, 4, 4]
 Generation 4570: Best = 4.972471e+03 FES = 457100 μ_F = 0.738 μ_{CR} = 0.208
 Diversity = 1.0208e-02 k_t = [1, 1, 1, 1]
 Generation 4580: Best = 4.972462e+03 FES = 458100 μ_F = 0.756 μ_{CR} = 0.204
 Diversity = 1.1225e-02 k_t = [3, 4, 4, 2]
 Generation 4590: Best = 4.972456e+03 FES = 459100 μ_F = 0.774 μ_{CR} = 0.208
 Diversity = 1.1015e-02 k_t = [1, 1, 1, 1]
 Generation 4600: Best = 4.972445e+03 FES = 460100 μ_F = 0.777 μ_{CR} = 0.213
 Diversity = 1.0269e-02 k_t = [3, 2, 2, 3]
 Generation 4610: Best = 4.972432e+03 FES = 461100 μ_F = 0.778 μ_{CR} = 0.211
 Diversity = 1.0290e-02 k_t = [1, 1, 1, 1]
 Generation 4620: Best = 4.972422e+03 FES = 462100 μ_F = 0.779 μ_{CR} = 0.212
 Diversity = 1.0380e-02 k_t = [3, 2, 4, 4]
 Generation 4630: Best = 4.972413e+03 FES = 463100 μ_F = 0.774 μ_{CR} = 0.210
 Diversity = 1.0897e-02 k_t = [1, 1, 1, 1]
 Generation 4640: Best = 4.972402e+03 FES = 464100 μ_F = 0.772 μ_{CR} = 0.201
 Diversity = 1.1237e-02 k_t = [2, 2, 2, 3]
 Generation 4650: Best = 4.972389e+03 FES = 465100 μ_F = 0.762 μ_{CR} = 0.208
 Diversity = 1.1071e-02 k_t = [1, 1, 1, 1]
 Generation 4660: Best = 4.972380e+03 FES = 466100 μ_F = 0.753 μ_{CR} = 0.204
 Diversity = 1.0757e-02 k_t = [2, 2, 4, 4]
 Generation 4670: Best = 4.972370e+03 FES = 467100 μ_F = 0.756 μ_{CR} = 0.201
 Diversity = 1.0882e-02 k_t = [1, 1, 1, 1]
 Generation 4680: Best = 4.972360e+03 FES = 468100 μ_F = 0.756 μ_{CR} = 0.190
 Diversity = 1.0641e-02 k_t = [2, 3, 4, 2]
 Generation 4690: Best = 4.972346e+03 FES = 469100 μ_F = 0.760 μ_{CR} = 0.177
 Diversity = 1.2248e-02 k_t = [1, 1, 1, 1]
 Generation 4700: Best = 4.972334e+03 FES = 470100 μ_F = 0.769 μ_{CR} = 0.183
 Diversity = 1.2557e-02 k_t = [3, 4, 3, 3]
 Generation 4710: Best = 4.972314e+03 FES = 471100 μ_F = 0.769 μ_{CR} = 0.189
 Diversity = 1.5007e-02 k_t = [1, 1, 1, 1]
 Generation 4720: Best = 4.972302e+03 FES = 472100 μ_F = 0.774 μ_{CR} = 0.190
 Diversity = 1.5987e-02 k_t = [3, 2, 4, 2]
 Generation 4730: Best = 4.972292e+03 FES = 473100 μ_F = 0.765 μ_{CR} = 0.194
 Diversity = 1.7969e-02 k_t = [1, 1, 1, 1]
 Generation 4740: Best = 4.972274e+03 FES = 474100 μ_F = 0.761 μ_{CR} = 0.196
 Diversity = 1.7257e-02 k_t = [2, 3, 4, 2]
 Generation 4750: Best = 4.972253e+03 FES = 475100 μ_F = 0.758 μ_{CR} = 0.201
 Diversity = 1.7603e-02 k_t = [1, 1, 1, 1]
 Generation 4760: Best = 4.972230e+03 FES = 476100 μ_F = 0.758 μ_{CR} = 0.200
 Diversity = 1.7061e-02 k_t = [4, 4, 3, 3]
 Generation 4770: Best = 4.972220e+03 FES = 477100 μ_F = 0.762 μ_{CR} = 0.204
 Diversity = 1.7871e-02 k_t = [1, 1, 1, 1]
 Generation 4780: Best = 4.972199e+03 FES = 478100 μ_F = 0.756 μ_{CR} = 0.200
 Diversity = 1.6643e-02 k_t = [2, 2, 4, 2]
 Generation 4790: Best = 4.972189e+03 FES = 479100 μ_F = 0.747 μ_{CR} = 0.195
 Diversity = 1.6267e-02 k_t = [1, 1, 1, 1]
 Generation 4800: Best = 4.972178e+03 FES = 480100 μ_F = 0.740 μ_{CR} = 0.187
 Diversity = 1.4323e-02 k_t = [2, 2, 2, 3]
 Generation 4810: Best = 4.972170e+03 FES = 481100 μ_F = 0.737 μ_{CR} = 0.178
 Diversity = 1.3348e-02 k_t = [1, 1, 1, 1]

Generation 4820: Best = 4.972164e+03 FES = 482100 μ_F = 0.728 μ_{CR} = 0.179
 Diversity = 1.1877e-02 k_t = [4, 3, 4, 4]
 Generation 4830: Best = 4.972154e+03 FES = 483100 μ_F = 0.734 μ_{CR} = 0.180
 Diversity = 1.1010e-02 k_t = [1, 1, 1, 1]
 Generation 4840: Best = 4.972146e+03 FES = 484100 μ_F = 0.746 μ_{CR} = 0.185
 Diversity = 9.7040e-03 k_t = [4, 2, 2, 4]
 Generation 4850: Best = 4.972143e+03 FES = 485100 μ_F = 0.752 μ_{CR} = 0.185
 Diversity = 9.5105e-03 k_t = [1, 1, 1, 1]
 Generation 4860: Best = 4.972130e+03 FES = 486100 μ_F = 0.756 μ_{CR} = 0.187
 Diversity = 9.9125e-03 k_t = [4, 4, 4, 4]
 Generation 4870: Best = 4.972125e+03 FES = 487100 μ_F = 0.739 μ_{CR} = 0.191
 Diversity = 1.0204e-02 k_t = [1, 1, 1, 1]
 Generation 4880: Best = 4.972115e+03 FES = 488100 μ_F = 0.736 μ_{CR} = 0.193
 Diversity = 1.0829e-02 k_t = [3, 2, 4, 4]
 Generation 4890: Best = 4.972105e+03 FES = 489100 μ_F = 0.748 μ_{CR} = 0.194
 Diversity = 1.1960e-02 k_t = [1, 1, 1, 1]
 Generation 4900: Best = 4.972097e+03 FES = 490100 μ_F = 0.772 μ_{CR} = 0.196
 Diversity = 1.2540e-02 k_t = [3, 2, 2, 3]
 Generation 4910: Best = 4.972090e+03 FES = 491100 μ_F = 0.770 μ_{CR} = 0.188
 Diversity = 1.2440e-02 k_t = [1, 1, 1, 1]
 Generation 4920: Best = 4.972085e+03 FES = 492100 μ_F = 0.755 μ_{CR} = 0.192
 Diversity = 1.1203e-02 k_t = [3, 3, 4, 2]
 Generation 4930: Best = 4.972075e+03 FES = 493100 μ_F = 0.752 μ_{CR} = 0.198
 Diversity = 1.1419e-02 k_t = [1, 1, 1, 1]
 Generation 4940: Best = 4.972066e+03 FES = 494100 μ_F = 0.743 μ_{CR} = 0.192
 Diversity = 1.1973e-02 k_t = [3, 4, 4, 2]
 Generation 4950: Best = 4.972059e+03 FES = 495100 μ_F = 0.746 μ_{CR} = 0.201
 Diversity = 1.2392e-02 k_t = [1, 1, 1, 1]
 Generation 4960: Best = 4.972048e+03 FES = 496100 μ_F = 0.750 μ_{CR} = 0.197
 Diversity = 1.1599e-02 k_t = [3, 4, 4, 4]
 Generation 4970: Best = 4.972042e+03 FES = 497100 μ_F = 0.752 μ_{CR} = 0.195
 Diversity = 1.1242e-02 k_t = [1, 1, 1, 1]
 Generation 4980: Best = 4.972033e+03 FES = 498100 μ_F = 0.743 μ_{CR} = 0.194
 Diversity = 1.0714e-02 k_t = [4, 2, 3, 3]
 Generation 4990: Best = 4.972026e+03 FES = 499100 μ_F = 0.739 μ_{CR} = 0.178
 Diversity = 1.0992e-02 k_t = [1, 1, 1, 1]
 Generation 5000: Best = 4.972018e+03 FES = 500100 μ_F = 0.750 μ_{CR} = 0.182
 Diversity = 1.0872e-02 k_t = [3, 4, 4, 2]
 Generation 5010: Best = 4.972006e+03 FES = 501100 μ_F = 0.749 μ_{CR} = 0.180
 Diversity = 1.1223e-02 k_t = [1, 1, 1, 1]
 Generation 5020: Best = 4.972001e+03 FES = 502100 μ_F = 0.749 μ_{CR} = 0.178
 Diversity = 1.1275e-02 k_t = [2, 4, 2, 2]
 Generation 5030: Best = 4.971987e+03 FES = 503100 μ_F = 0.743 μ_{CR} = 0.183
 Diversity = 1.2099e-02 k_t = [1, 1, 1, 1]
 Generation 5040: Best = 4.971983e+03 FES = 504100 μ_F = 0.737 μ_{CR} = 0.190
 Diversity = 1.1531e-02 k_t = [3, 3, 2, 4]
 Generation 5050: Best = 4.971976e+03 FES = 505100 μ_F = 0.741 μ_{CR} = 0.193
 Diversity = 9.7295e-03 k_t = [1, 1, 1, 1]
 Generation 5060: Best = 4.971970e+03 FES = 506100 μ_F = 0.747 μ_{CR} = 0.200
 Diversity = 8.4509e-03 k_t = [3, 3, 2, 3]
 Generation 5070: Best = 4.971963e+03 FES = 507100 μ_F = 0.759 μ_{CR} = 0.209
 Diversity = 8.9333e-03 k_t = [1, 1, 1, 1]
 Generation 5080: Best = 4.971955e+03 FES = 508100 μ_F = 0.745 μ_{CR} = 0.196
 Diversity = 9.0738e-03 k_t = [2, 2, 3, 2]

Generation 5090: Best = 4.971950e+03 FES = 509100 μ_F = 0.743 μ_{CR} = 0.198
 Diversity = 8.9314e-03 k_t = [1, 1, 1, 1]
 Generation 5100: Best = 4.971945e+03 FES = 510100 μ_F = 0.754 μ_{CR} = 0.195
 Diversity = 8.1020e-03 k_t = [4, 2, 4, 2]
 Generation 5110: Best = 4.971939e+03 FES = 511100 μ_F = 0.752 μ_{CR} = 0.198
 Diversity = 8.6877e-03 k_t = [1, 1, 1, 1]
 Generation 5120: Best = 4.971935e+03 FES = 512100 μ_F = 0.742 μ_{CR} = 0.203
 Diversity = 7.8476e-03 k_t = [3, 4, 2, 3]
 Generation 5130: Best = 4.971929e+03 FES = 513100 μ_F = 0.740 μ_{CR} = 0.207
 Diversity = 7.8774e-03 k_t = [1, 1, 1, 1]
 Generation 5140: Best = 4.971923e+03 FES = 514100 μ_F = 0.738 μ_{CR} = 0.205
 Diversity = 6.9070e-03 k_t = [2, 2, 3, 4]
 Generation 5150: Best = 4.971916e+03 FES = 515100 μ_F = 0.750 μ_{CR} = 0.214
 Diversity = 7.0614e-03 k_t = [1, 1, 1, 1]
 Generation 5160: Best = 4.971914e+03 FES = 516100 μ_F = 0.765 μ_{CR} = 0.218
 Diversity = 7.1343e-03 k_t = [3, 2, 2, 4]
 Generation 5170: Best = 4.971907e+03 FES = 517100 μ_F = 0.787 μ_{CR} = 0.223
 Diversity = 7.4149e-03 k_t = [1, 1, 1, 1]
 Generation 5180: Best = 4.971905e+03 FES = 518100 μ_F = 0.776 μ_{CR} = 0.216
 Diversity = 7.2145e-03 k_t = [2, 2, 3, 2]
 Generation 5190: Best = 4.971902e+03 FES = 519100 μ_F = 0.763 μ_{CR} = 0.205
 Diversity = 7.4571e-03 k_t = [1, 1, 1, 1]
 Generation 5200: Best = 4.971896e+03 FES = 520100 μ_F = 0.762 μ_{CR} = 0.201
 Diversity = 7.8662e-03 k_t = [4, 2, 3, 3]
 Generation 5210: Best = 4.971888e+03 FES = 521100 μ_F = 0.753 μ_{CR} = 0.194
 Diversity = 8.9203e-03 k_t = [1, 1, 1, 1]
 Generation 5220: Best = 4.971883e+03 FES = 522100 μ_F = 0.748 μ_{CR} = 0.198
 Diversity = 8.2932e-03 k_t = [3, 2, 2, 4]
 Generation 5230: Best = 4.971873e+03 FES = 523100 μ_F = 0.763 μ_{CR} = 0.198
 Diversity = 8.4945e-03 k_t = [1, 1, 1, 1]
 Generation 5240: Best = 4.971868e+03 FES = 524100 μ_F = 0.756 μ_{CR} = 0.200
 Diversity = 9.2391e-03 k_t = [2, 4, 4, 2]
 Generation 5250: Best = 4.971859e+03 FES = 525100 μ_F = 0.758 μ_{CR} = 0.198
 Diversity = 9.5012e-03 k_t = [1, 1, 1, 1]
 Generation 5260: Best = 4.971856e+03 FES = 526100 μ_F = 0.765 μ_{CR} = 0.203
 Diversity = 9.0445e-03 k_t = [3, 3, 3, 4]
 Generation 5270: Best = 4.971849e+03 FES = 527100 μ_F = 0.776 μ_{CR} = 0.195
 Diversity = 8.8248e-03 k_t = [1, 1, 1, 1]
 Generation 5280: Best = 4.971845e+03 FES = 528100 μ_F = 0.781 μ_{CR} = 0.200
 Diversity = 8.3986e-03 k_t = [4, 4, 2, 4]
 Generation 5290: Best = 4.971841e+03 FES = 529100 μ_F = 0.785 μ_{CR} = 0.196
 Diversity = 7.9071e-03 k_t = [1, 1, 1, 1]
 Generation 5300: Best = 4.971836e+03 FES = 530100 μ_F = 0.777 μ_{CR} = 0.189
 Diversity = 7.8812e-03 k_t = [2, 2, 2, 3]
 Generation 5310: Best = 4.971833e+03 FES = 531100 μ_F = 0.774 μ_{CR} = 0.191
 Diversity = 8.0673e-03 k_t = [1, 1, 1, 1]
 Generation 5320: Best = 4.971829e+03 FES = 532100 μ_F = 0.780 μ_{CR} = 0.196
 Diversity = 7.2166e-03 k_t = [3, 3, 3, 4]
 Generation 5330: Best = 4.971824e+03 FES = 533100 μ_F = 0.772 μ_{CR} = 0.200
 Diversity = 8.0548e-03 k_t = [1, 1, 1, 1]
 Generation 5340: Best = 4.971823e+03 FES = 534100 μ_F = 0.759 μ_{CR} = 0.201
 Diversity = 7.9047e-03 k_t = [3, 4, 2, 2]
 Generation 5350: Best = 4.971818e+03 FES = 535100 μ_F = 0.756 μ_{CR} = 0.200
 Diversity = 8.0343e-03 k_t = [1, 1, 1, 1]

Generation 5360: Best = 4.971816e+03 FES = 536100 μ_F = 0.763 μ_{CR} = 0.189
 Diversity = 7.2615e-03 k_t = [3, 2, 4, 4]
 Generation 5370: Best = 4.971812e+03 FES = 537100 μ_F = 0.759 μ_{CR} = 0.197
 Diversity = 6.9227e-03 k_t = [1, 1, 1, 1]
 Generation 5380: Best = 4.971809e+03 FES = 538100 μ_F = 0.761 μ_{CR} = 0.196
 Diversity = 6.8433e-03 k_t = [4, 3, 2, 4]
 Generation 5390: Best = 4.971806e+03 FES = 539100 μ_F = 0.766 μ_{CR} = 0.198
 Diversity = 6.9019e-03 k_t = [1, 1, 1, 1]
 Generation 5400: Best = 4.971802e+03 FES = 540100 μ_F = 0.762 μ_{CR} = 0.204
 Diversity = 6.7729e-03 k_t = [2, 2, 3, 2]
 Generation 5410: Best = 4.971798e+03 FES = 541100 μ_F = 0.757 μ_{CR} = 0.208
 Diversity = 7.2367e-03 k_t = [1, 1, 1, 1]
 Generation 5420: Best = 4.971793e+03 FES = 542100 μ_F = 0.774 μ_{CR} = 0.210
 Diversity = 7.2014e-03 k_t = [3, 2, 4, 3]
 Generation 5430: Best = 4.971791e+03 FES = 543100 μ_F = 0.761 μ_{CR} = 0.207
 Diversity = 7.0659e-03 k_t = [1, 1, 1, 1]
 Generation 5440: Best = 4.971786e+03 FES = 544100 μ_F = 0.768 μ_{CR} = 0.208
 Diversity = 6.7886e-03 k_t = [3, 4, 4, 3]
 Generation 5450: Best = 4.971782e+03 FES = 545100 μ_F = 0.784 μ_{CR} = 0.208
 Diversity = 6.9149e-03 k_t = [1, 1, 1, 1]
 Generation 5460: Best = 4.971779e+03 FES = 546100 μ_F = 0.782 μ_{CR} = 0.218
 Diversity = 6.8776e-03 k_t = [2, 4, 2, 3]
 Generation 5470: Best = 4.971772e+03 FES = 547100 μ_F = 0.775 μ_{CR} = 0.213
 Diversity = 7.6081e-03 k_t = [1, 1, 1, 1]
 Generation 5480: Best = 4.971768e+03 FES = 548100 μ_F = 0.776 μ_{CR} = 0.218
 Diversity = 7.6439e-03 k_t = [4, 4, 3, 3]
 Generation 5490: Best = 4.971762e+03 FES = 549100 μ_F = 0.764 μ_{CR} = 0.226
 Diversity = 7.5969e-03 k_t = [1, 1, 1, 1]
 Generation 5500: Best = 4.971756e+03 FES = 550100 μ_F = 0.768 μ_{CR} = 0.232
 Diversity = 7.7197e-03 k_t = [2, 4, 2, 4]
 Generation 5510: Best = 4.971750e+03 FES = 551100 μ_F = 0.753 μ_{CR} = 0.236
 Diversity = 8.4694e-03 k_t = [1, 1, 1, 1]
 Generation 5520: Best = 4.971744e+03 FES = 552100 μ_F = 0.764 μ_{CR} = 0.232
 Diversity = 8.2944e-03 k_t = [3, 2, 4, 3]
 Generation 5530: Best = 4.971738e+03 FES = 553100 μ_F = 0.752 μ_{CR} = 0.234
 Diversity = 8.1354e-03 k_t = [1, 1, 1, 1]
 Generation 5540: Best = 4.971735e+03 FES = 554100 μ_F = 0.757 μ_{CR} = 0.239
 Diversity = 7.3209e-03 k_t = [2, 4, 3, 3]
 Generation 5550: Best = 4.971731e+03 FES = 555100 μ_F = 0.762 μ_{CR} = 0.230
 Diversity = 8.2543e-03 k_t = [1, 1, 1, 1]
 Generation 5560: Best = 4.971724e+03 FES = 556100 μ_F = 0.756 μ_{CR} = 0.228
 Diversity = 7.6471e-03 k_t = [4, 3, 4, 3]
 Generation 5570: Best = 4.971720e+03 FES = 557100 μ_F = 0.760 μ_{CR} = 0.224
 Diversity = 8.0713e-03 k_t = [1, 1, 1, 1]
 Generation 5580: Best = 4.971713e+03 FES = 558100 μ_F = 0.754 μ_{CR} = 0.232
 Diversity = 8.8021e-03 k_t = [2, 2, 4, 4]
 Generation 5590: Best = 4.971708e+03 FES = 559100 μ_F = 0.749 μ_{CR} = 0.228
 Diversity = 9.2276e-03 k_t = [1, 1, 1, 1]
 Generation 5600: Best = 4.971706e+03 FES = 560100 μ_F = 0.746 μ_{CR} = 0.219
 Diversity = 8.0451e-03 k_t = [3, 4, 3, 4]
 Generation 5610: Best = 4.971702e+03 FES = 561100 μ_F = 0.740 μ_{CR} = 0.211
 Diversity = 7.7749e-03 k_t = [1, 1, 1, 1]
 Generation 5620: Best = 4.971699e+03 FES = 562100 μ_F = 0.732 μ_{CR} = 0.192
 Diversity = 7.1284e-03 k_t = [3, 2, 4, 3]

Generation 5630: Best = 4.971698e+03	FES = 563100	$\mu_F = 0.742$	$\mu_{CR} = 0.191$
Diversity = 6.1951e-03	$k_t = [1, 1, 1, 1]$		
Generation 5640: Best = 4.971694e+03	FES = 564100	$\mu_F = 0.744$	$\mu_{CR} = 0.192$
Diversity = 5.5066e-03	$k_t = [4, 3, 3, 4]$		
Generation 5650: Best = 4.971692e+03	FES = 565100	$\mu_F = 0.743$	$\mu_{CR} = 0.188$
Diversity = 5.5356e-03	$k_t = [1, 1, 1, 1]$		
Generation 5660: Best = 4.971688e+03	FES = 566100	$\mu_F = 0.747$	$\mu_{CR} = 0.195$
Diversity = 6.0245e-03	$k_t = [4, 4, 3, 3]$		
Generation 5670: Best = 4.971686e+03	FES = 567100	$\mu_F = 0.750$	$\mu_{CR} = 0.198$
Diversity = 6.3512e-03	$k_t = [1, 1, 1, 1]$		
Generation 5680: Best = 4.971682e+03	FES = 568100	$\mu_F = 0.748$	$\mu_{CR} = 0.194$
Diversity = 5.9515e-03	$k_t = [3, 3, 4, 2]$		
Generation 5690: Best = 4.971680e+03	FES = 569100	$\mu_F = 0.748$	$\mu_{CR} = 0.192$
Diversity = 5.6869e-03	$k_t = [1, 1, 1, 1]$		
Generation 5700: Best = 4.971678e+03	FES = 570100	$\mu_F = 0.741$	$\mu_{CR} = 0.191$
Diversity = 5.5403e-03	$k_t = [2, 2, 3, 3]$		
Generation 5710: Best = 4.971673e+03	FES = 571100	$\mu_F = 0.751$	$\mu_{CR} = 0.185$
Diversity = 6.0102e-03	$k_t = [1, 1, 1, 1]$		
Generation 5720: Best = 4.971670e+03	FES = 572100	$\mu_F = 0.750$	$\mu_{CR} = 0.190$
Diversity = 6.1569e-03	$k_t = [4, 4, 3, 3]$		
Generation 5730: Best = 4.971665e+03	FES = 573100	$\mu_F = 0.756$	$\mu_{CR} = 0.191$
Diversity = 7.0855e-03	$k_t = [1, 1, 1, 1]$		
Generation 5740: Best = 4.971661e+03	FES = 574100	$\mu_F = 0.754$	$\mu_{CR} = 0.198$
Diversity = 7.3929e-03	$k_t = [4, 2, 3, 3]$		
Generation 5750: Best = 4.971656e+03	FES = 575100	$\mu_F = 0.759$	$\mu_{CR} = 0.199$
Diversity = 7.7424e-03	$k_t = [1, 1, 1, 1]$		
Generation 5760: Best = 4.971649e+03	FES = 576100	$\mu_F = 0.763$	$\mu_{CR} = 0.197$
Diversity = 7.3103e-03	$k_t = [3, 2, 4, 3]$		
Generation 5770: Best = 4.971646e+03	FES = 577100	$\mu_F = 0.762$	$\mu_{CR} = 0.198$
Diversity = 8.0280e-03	$k_t = [1, 1, 1, 1]$		
Generation 5780: Best = 4.971638e+03	FES = 578100	$\mu_F = 0.760$	$\mu_{CR} = 0.202$
Diversity = 8.3685e-03	$k_t = [4, 3, 3, 3]$		
Generation 5790: Best = 4.971632e+03	FES = 579100	$\mu_F = 0.762$	$\mu_{CR} = 0.207$
Diversity = 8.6906e-03	$k_t = [1, 1, 1, 1]$		
Generation 5800: Best = 4.971627e+03	FES = 580100	$\mu_F = 0.760$	$\mu_{CR} = 0.206$
Diversity = 8.5713e-03	$k_t = [4, 3, 4, 2]$		
Generation 5810: Best = 4.971623e+03	FES = 581100	$\mu_F = 0.760$	$\mu_{CR} = 0.208$
Diversity = 8.9197e-03	$k_t = [1, 1, 1, 1]$		
Generation 5820: Best = 4.971619e+03	FES = 582100	$\mu_F = 0.753$	$\mu_{CR} = 0.205$
Diversity = 8.1643e-03	$k_t = [2, 3, 4, 4]$		
Generation 5830: Best = 4.971615e+03	FES = 583100	$\mu_F = 0.759$	$\mu_{CR} = 0.194$
Diversity = 7.5468e-03	$k_t = [1, 1, 1, 1]$		
Generation 5840: Best = 4.971612e+03	FES = 584100	$\mu_F = 0.764$	$\mu_{CR} = 0.193$
Diversity = 7.4396e-03	$k_t = [3, 3, 3, 2]$		
Generation 5850: Best = 4.971610e+03	FES = 585100	$\mu_F = 0.753$	$\mu_{CR} = 0.204$
Diversity = 8.2334e-03	$k_t = [1, 1, 1, 1]$		
Generation 5860: Best = 4.971607e+03	FES = 586100	$\mu_F = 0.735$	$\mu_{CR} = 0.203$
Diversity = 6.4823e-03	$k_t = [4, 2, 4, 4]$		
Generation 5870: Best = 4.971604e+03	FES = 587100	$\mu_F = 0.738$	$\mu_{CR} = 0.207$
Diversity = 6.2289e-03	$k_t = [1, 1, 1, 1]$		
Generation 5880: Best = 4.971601e+03	FES = 588100	$\mu_F = 0.737$	$\mu_{CR} = 0.216$
Diversity = 6.1346e-03	$k_t = [3, 2, 3, 3]$		
Generation 5890: Best = 4.971598e+03	FES = 589100	$\mu_F = 0.746$	$\mu_{CR} = 0.214$
Diversity = 6.0135e-03	$k_t = [1, 1, 1, 1]$		

Generation 5900: Best = 4.971597e+03 FES = 590100 μ_F = 0.745 μ_{CR} = 0.202
 Diversity = 5.3678e-03 k_t = [2, 4, 3, 2]
 Generation 5910: Best = 4.971594e+03 FES = 591100 μ_F = 0.744 μ_{CR} = 0.203
 Diversity = 6.0504e-03 k_t = [1, 1, 1, 1]
 Generation 5920: Best = 4.971591e+03 FES = 592100 μ_F = 0.744 μ_{CR} = 0.206
 Diversity = 5.8323e-03 k_t = [4, 4, 2, 2]
 Generation 5930: Best = 4.971590e+03 FES = 593100 μ_F = 0.747 μ_{CR} = 0.213
 Diversity = 5.9394e-03 k_t = [1, 1, 1, 1]
 Generation 5940: Best = 4.971586e+03 FES = 594100 μ_F = 0.753 μ_{CR} = 0.213
 Diversity = 6.3893e-03 k_t = [3, 3, 4, 2]
 Generation 5950: Best = 4.971584e+03 FES = 595100 μ_F = 0.748 μ_{CR} = 0.211
 Diversity = 6.4406e-03 k_t = [1, 1, 1, 1]
 Generation 5960: Best = 4.971582e+03 FES = 596100 μ_F = 0.754 μ_{CR} = 0.210
 Diversity = 5.9812e-03 k_t = [3, 3, 3, 4]
 Generation 5970: Best = 4.971580e+03 FES = 597100 μ_F = 0.764 μ_{CR} = 0.208
 Diversity = 6.1078e-03 k_t = [1, 1, 1, 1]
 Generation 5980: Best = 4.971579e+03 FES = 598100 μ_F = 0.762 μ_{CR} = 0.201
 Diversity = 6.1816e-03 k_t = [4, 2, 4, 2]
 Generation 5990: Best = 4.971575e+03 FES = 599100 μ_F = 0.762 μ_{CR} = 0.200
 Diversity = 6.2996e-03 k_t = [1, 1, 1, 1]
 Generation 6000: Best = 4.971572e+03 FES = 600100 μ_F = 0.761 μ_{CR} = 0.197
 Diversity = 6.1530e-03 k_t = [2, 3, 4, 4]
 Generation 6010: Best = 4.971570e+03 FES = 601100 μ_F = 0.768 μ_{CR} = 0.193
 Diversity = 6.0044e-03 k_t = [1, 1, 1, 1]
 Generation 6020: Best = 4.971567e+03 FES = 602100 μ_F = 0.754 μ_{CR} = 0.196
 Diversity = 6.0127e-03 k_t = [2, 3, 2, 4]
 Generation 6030: Best = 4.971565e+03 FES = 603100 μ_F = 0.745 μ_{CR} = 0.197
 Diversity = 6.2361e-03 k_t = [1, 1, 1, 1]
 Generation 6040: Best = 4.971563e+03 FES = 604100 μ_F = 0.738 μ_{CR} = 0.194
 Diversity = 5.3865e-03 k_t = [3, 4, 2, 4]
 Generation 6050: Best = 4.971561e+03 FES = 605100 μ_F = 0.744 μ_{CR} = 0.191
 Diversity = 5.1820e-03 k_t = [1, 1, 1, 1]
 Generation 6060: Best = 4.971560e+03 FES = 606100 μ_F = 0.743 μ_{CR} = 0.195
 Diversity = 5.3546e-03 k_t = [3, 4, 3, 2]
 Generation 6070: Best = 4.971558e+03 FES = 607100 μ_F = 0.746 μ_{CR} = 0.199
 Diversity = 5.1861e-03 k_t = [1, 1, 1, 1]
 Generation 6080: Best = 4.971555e+03 FES = 608100 μ_F = 0.756 μ_{CR} = 0.200
 Diversity = 5.6200e-03 k_t = [4, 3, 3, 3]
 Generation 6090: Best = 4.971552e+03 FES = 609100 μ_F = 0.777 μ_{CR} = 0.199
 Diversity = 6.7703e-03 k_t = [1, 1, 1, 1]
 Generation 6100: Best = 4.971549e+03 FES = 610100 μ_F = 0.779 μ_{CR} = 0.186
 Diversity = 6.4304e-03 k_t = [3, 4, 2, 3]
 Generation 6110: Best = 4.971545e+03 FES = 611100 μ_F = 0.774 μ_{CR} = 0.176
 Diversity = 6.5595e-03 k_t = [1, 1, 1, 1]
 Generation 6120: Best = 4.971543e+03 FES = 612100 μ_F = 0.767 μ_{CR} = 0.176
 Diversity = 6.5616e-03 k_t = [3, 4, 2, 4]
 Generation 6130: Best = 4.971541e+03 FES = 613100 μ_F = 0.764 μ_{CR} = 0.174
 Diversity = 6.6058e-03 k_t = [1, 1, 1, 1]
 Generation 6140: Best = 4.971538e+03 FES = 614100 μ_F = 0.770 μ_{CR} = 0.171
 Diversity = 6.7130e-03 k_t = [3, 2, 4, 4]
 Generation 6150: Best = 4.971536e+03 FES = 615100 μ_F = 0.751 μ_{CR} = 0.170
 Diversity = 6.3918e-03 k_t = [1, 1, 1, 1]
 Generation 6160: Best = 4.971533e+03 FES = 616100 μ_F = 0.736 μ_{CR} = 0.177
 Diversity = 5.9856e-03 k_t = [3, 4, 3, 3]

Generation 6170: Best = 4.971531e+03 FES = 617100 μ_F = 0.728 μ_{CR} = 0.181
 Diversity = 5.9359e-03 k_t = [1, 1, 1, 1]
 Generation 6180: Best = 4.971528e+03 FES = 618100 μ_F = 0.734 μ_{CR} = 0.175
 Diversity = 5.8079e-03 k_t = [4, 2, 2, 2]
 Generation 6190: Best = 4.971525e+03 FES = 619100 μ_F = 0.738 μ_{CR} = 0.184
 Diversity = 5.8235e-03 k_t = [1, 1, 1, 1]
 Generation 6200: Best = 4.971523e+03 FES = 620100 μ_F = 0.747 μ_{CR} = 0.185
 Diversity = 5.3173e-03 k_t = [2, 2, 4, 4]
 Generation 6210: Best = 4.971522e+03 FES = 621100 μ_F = 0.749 μ_{CR} = 0.183
 Diversity = 5.2739e-03 k_t = [1, 1, 1, 1]
 Generation 6220: Best = 4.971519e+03 FES = 622100 μ_F = 0.762 μ_{CR} = 0.182
 Diversity = 5.4592e-03 k_t = [3, 4, 3, 3]
 Generation 6230: Best = 4.971517e+03 FES = 623100 μ_F = 0.760 μ_{CR} = 0.179
 Diversity = 5.4840e-03 k_t = [1, 1, 1, 1]
 Generation 6240: Best = 4.971515e+03 FES = 624100 μ_F = 0.771 μ_{CR} = 0.183
 Diversity = 5.1535e-03 k_t = [3, 3, 2, 4]
 Generation 6250: Best = 4.971511e+03 FES = 625100 μ_F = 0.773 μ_{CR} = 0.187
 Diversity = 5.6234e-03 k_t = [1, 1, 1, 1]
 Generation 6260: Best = 4.971508e+03 FES = 626100 μ_F = 0.763 μ_{CR} = 0.194
 Diversity = 5.8798e-03 k_t = [3, 4, 2, 4]
 Generation 6270: Best = 4.971502e+03 FES = 627100 μ_F = 0.760 μ_{CR} = 0.202
 Diversity = 6.4698e-03 k_t = [1, 1, 1, 1]
 Generation 6280: Best = 4.971496e+03 FES = 628100 μ_F = 0.760 μ_{CR} = 0.212
 Diversity = 6.4935e-03 k_t = [3, 3, 2, 2]
 Generation 6290: Best = 4.971490e+03 FES = 629100 μ_F = 0.755 μ_{CR} = 0.213
 Diversity = 7.2869e-03 k_t = [1, 1, 1, 1]
 Generation 6300: Best = 4.971487e+03 FES = 630100 μ_F = 0.756 μ_{CR} = 0.220
 Diversity = 6.7053e-03 k_t = [2, 4, 3, 2]
 Generation 6310: Best = 4.971480e+03 FES = 631100 μ_F = 0.757 μ_{CR} = 0.211
 Diversity = 7.9303e-03 k_t = [1, 1, 1, 1]
 Generation 6320: Best = 4.971471e+03 FES = 632100 μ_F = 0.774 μ_{CR} = 0.203
 Diversity = 8.1757e-03 k_t = [3, 3, 2, 3]
 Generation 6330: Best = 4.971465e+03 FES = 633100 μ_F = 0.775 μ_{CR} = 0.207
 Diversity = 9.0763e-03 k_t = [1, 1, 1, 1]
 Generation 6340: Best = 4.971457e+03 FES = 634100 μ_F = 0.781 μ_{CR} = 0.208
 Diversity = 1.0181e-02 k_t = [4, 3, 3, 2]
 Generation 6350: Best = 4.971451e+03 FES = 635100 μ_F = 0.777 μ_{CR} = 0.209
 Diversity = 1.0406e-02 k_t = [1, 1, 1, 1]
 Generation 6360: Best = 4.971444e+03 FES = 636100 μ_F = 0.766 μ_{CR} = 0.210
 Diversity = 9.8009e-03 k_t = [4, 2, 2, 3]
 Generation 6370: Best = 4.971439e+03 FES = 637100 μ_F = 0.760 μ_{CR} = 0.210
 Diversity = 1.0342e-02 k_t = [1, 1, 1, 1]
 Generation 6380: Best = 4.971434e+03 FES = 638100 μ_F = 0.759 μ_{CR} = 0.213
 Diversity = 9.5058e-03 k_t = [3, 2, 2, 4]
 Generation 6390: Best = 4.971431e+03 FES = 639100 μ_F = 0.749 μ_{CR} = 0.203
 Diversity = 8.7802e-03 k_t = [1, 1, 1, 1]
 Generation 6400: Best = 4.971427e+03 FES = 640100 μ_F = 0.742 μ_{CR} = 0.206
 Diversity = 7.6659e-03 k_t = [2, 3, 3, 4]
 Generation 6410: Best = 4.971424e+03 FES = 641100 μ_F = 0.735 μ_{CR} = 0.193
 Diversity = 7.2227e-03 k_t = [1, 1, 1, 1]
 Generation 6420: Best = 4.971421e+03 FES = 642100 μ_F = 0.724 μ_{CR} = 0.191
 Diversity = 6.3672e-03 k_t = [3, 2, 2, 3]
 Generation 6430: Best = 4.971417e+03 FES = 643100 μ_F = 0.734 μ_{CR} = 0.196
 Diversity = 6.5962e-03 k_t = [1, 1, 1, 1]

Generation 6440: Best = 4.971412e+03 FES = 644100 μ_F = 0.742 μ_{CR} = 0.197
 Diversity = 6.7954e-03 k_t = [2, 3, 3, 2]
 Generation 6450: Best = 4.971407e+03 FES = 645100 μ_F = 0.745 μ_{CR} = 0.195
 Diversity = 7.2760e-03 k_t = [1, 1, 1, 1]
 Generation 6460: Best = 4.971404e+03 FES = 646100 μ_F = 0.750 μ_{CR} = 0.200
 Diversity = 6.9822e-03 k_t = [3, 3, 4, 2]
 Generation 6470: Best = 4.971402e+03 FES = 647100 μ_F = 0.754 μ_{CR} = 0.195
 Diversity = 7.0667e-03 k_t = [1, 1, 1, 1]
 Generation 6480: Best = 4.971395e+03 FES = 648100 μ_F = 0.761 μ_{CR} = 0.197
 Diversity = 7.4167e-03 k_t = [2, 4, 2, 3]
 Generation 6490: Best = 4.971391e+03 FES = 649100 μ_F = 0.757 μ_{CR} = 0.195
 Diversity = 8.5571e-03 k_t = [1, 1, 1, 1]
 Generation 6500: Best = 4.971383e+03 FES = 650100 μ_F = 0.771 μ_{CR} = 0.200
 Diversity = 9.1411e-03 k_t = [4, 2, 4, 3]
 Generation 6510: Best = 4.971380e+03 FES = 651100 μ_F = 0.758 μ_{CR} = 0.205
 Diversity = 8.9187e-03 k_t = [1, 1, 1, 1]
 Generation 6520: Best = 4.971375e+03 FES = 652100 μ_F = 0.751 μ_{CR} = 0.210
 Diversity = 8.3424e-03 k_t = [3, 3, 3, 3]
 Generation 6530: Best = 4.971369e+03 FES = 653100 μ_F = 0.758 μ_{CR} = 0.209
 Diversity = 9.3472e-03 k_t = [1, 1, 1, 1]
 Generation 6540: Best = 4.971365e+03 FES = 654100 μ_F = 0.767 μ_{CR} = 0.205
 Diversity = 8.8167e-03 k_t = [2, 2, 2, 4]
 Generation 6550: Best = 4.971360e+03 FES = 655100 μ_F = 0.759 μ_{CR} = 0.206
 Diversity = 8.2342e-03 k_t = [1, 1, 1, 1]
 Generation 6560: Best = 4.971357e+03 FES = 656100 μ_F = 0.744 μ_{CR} = 0.207
 Diversity = 7.6918e-03 k_t = [2, 4, 3, 4]
 Generation 6570: Best = 4.971355e+03 FES = 657100 μ_F = 0.743 μ_{CR} = 0.212
 Diversity = 6.9745e-03 k_t = [1, 1, 1, 1]
 Generation 6580: Best = 4.971351e+03 FES = 658100 μ_F = 0.738 μ_{CR} = 0.221
 Diversity = 6.9369e-03 k_t = [4, 4, 2, 2]
 Generation 6590: Best = 4.971348e+03 FES = 659100 μ_F = 0.746 μ_{CR} = 0.222
 Diversity = 7.4201e-03 k_t = [1, 1, 1, 1]
 Generation 6600: Best = 4.971344e+03 FES = 660100 μ_F = 0.747 μ_{CR} = 0.220
 Diversity = 7.3236e-03 k_t = [2, 4, 3, 2]
 Generation 6610: Best = 4.971340e+03 FES = 661100 μ_F = 0.749 μ_{CR} = 0.214
 Diversity = 7.8195e-03 k_t = [1, 1, 1, 1]
 Generation 6620: Best = 4.971336e+03 FES = 662100 μ_F = 0.751 μ_{CR} = 0.211
 Diversity = 7.4074e-03 k_t = [4, 4, 4, 4]
 Generation 6630: Best = 4.971333e+03 FES = 663100 μ_F = 0.756 μ_{CR} = 0.214
 Diversity = 7.5781e-03 k_t = [1, 1, 1, 1]
 Generation 6640: Best = 4.971327e+03 FES = 664100 μ_F = 0.765 μ_{CR} = 0.222
 Diversity = 7.4730e-03 k_t = [2, 4, 3, 4]
 Generation 6650: Best = 4.971324e+03 FES = 665100 μ_F = 0.758 μ_{CR} = 0.217
 Diversity = 7.4354e-03 k_t = [1, 1, 1, 1]
 Generation 6660: Best = 4.971321e+03 FES = 666100 μ_F = 0.755 μ_{CR} = 0.218
 Diversity = 7.6094e-03 k_t = [3, 3, 3, 3]
 Generation 6670: Best = 4.971317e+03 FES = 667100 μ_F = 0.747 μ_{CR} = 0.214
 Diversity = 7.4389e-03 k_t = [1, 1, 1, 1]
 Generation 6680: Best = 4.971314e+03 FES = 668100 μ_F = 0.750 μ_{CR} = 0.210
 Diversity = 7.9527e-03 k_t = [3, 3, 4, 3]
 Generation 6690: Best = 4.971311e+03 FES = 669100 μ_F = 0.742 μ_{CR} = 0.210
 Diversity = 7.6438e-03 k_t = [1, 1, 1, 1]
 Generation 6700: Best = 4.971308e+03 FES = 670100 μ_F = 0.723 μ_{CR} = 0.215
 Diversity = 6.7970e-03 k_t = [2, 3, 3, 4]

Generation 6710: Best = 4.971306e+03 FES = 671100 μ_F = 0.719 μ_{CR} = 0.209
 Diversity = 6.3994e-03 k_t = [1, 1, 1, 1]
 Generation 6720: Best = 4.971302e+03 FES = 672100 μ_F = 0.723 μ_{CR} = 0.218
 Diversity = 6.0580e-03 k_t = [4, 3, 4, 2]
 Generation 6730: Best = 4.971301e+03 FES = 673100 μ_F = 0.744 μ_{CR} = 0.216
 Diversity = 6.0821e-03 k_t = [1, 1, 1, 1]
 Generation 6740: Best = 4.971296e+03 FES = 674100 μ_F = 0.755 μ_{CR} = 0.219
 Diversity = 5.6232e-03 k_t = [2, 4, 3, 4]
 Generation 6750: Best = 4.971293e+03 FES = 675100 μ_F = 0.761 μ_{CR} = 0.211
 Diversity = 5.4644e-03 k_t = [1, 1, 1, 1]
 Generation 6760: Best = 4.971291e+03 FES = 676100 μ_F = 0.761 μ_{CR} = 0.193
 Diversity = 5.7940e-03 k_t = [2, 2, 4, 4]
 Generation 6770: Best = 4.971286e+03 FES = 677100 μ_F = 0.762 μ_{CR} = 0.188
 Diversity = 6.0419e-03 k_t = [1, 1, 1, 1]
 Generation 6780: Best = 4.971281e+03 FES = 678100 μ_F = 0.767 μ_{CR} = 0.192
 Diversity = 7.0667e-03 k_t = [4, 3, 3, 4]
 Generation 6790: Best = 4.971275e+03 FES = 679100 μ_F = 0.768 μ_{CR} = 0.204
 Diversity = 7.8451e-03 k_t = [1, 1, 1, 1]
 Generation 6800: Best = 4.971269e+03 FES = 680100 μ_F = 0.772 μ_{CR} = 0.198
 Diversity = 8.0691e-03 k_t = [2, 2, 4, 4]
 Generation 6810: Best = 4.971263e+03 FES = 681100 μ_F = 0.776 μ_{CR} = 0.199
 Diversity = 9.2313e-03 k_t = [1, 1, 1, 1]
 Generation 6820: Best = 4.971255e+03 FES = 682100 μ_F = 0.767 μ_{CR} = 0.198
 Diversity = 9.1722e-03 k_t = [3, 2, 3, 3]
 Generation 6830: Best = 4.971250e+03 FES = 683100 μ_F = 0.772 μ_{CR} = 0.198
 Diversity = 9.2282e-03 k_t = [1, 1, 1, 1]
 Generation 6840: Best = 4.971246e+03 FES = 684100 μ_F = 0.782 μ_{CR} = 0.196
 Diversity = 9.6464e-03 k_t = [2, 4, 2, 3]
 Generation 6850: Best = 4.971239e+03 FES = 685100 μ_F = 0.778 μ_{CR} = 0.189
 Diversity = 1.0434e-02 k_t = [1, 1, 1, 1]
 Generation 6860: Best = 4.971233e+03 FES = 686100 μ_F = 0.788 μ_{CR} = 0.192
 Diversity = 1.0788e-02 k_t = [3, 3, 3, 2]
 Generation 6870: Best = 4.971227e+03 FES = 687100 μ_F = 0.767 μ_{CR} = 0.190
 Diversity = 1.0952e-02 k_t = [1, 1, 1, 1]
 Generation 6880: Best = 4.971222e+03 FES = 688100 μ_F = 0.747 μ_{CR} = 0.194
 Diversity = 1.0053e-02 k_t = [2, 3, 4, 2]
 Generation 6890: Best = 4.971216e+03 FES = 689100 μ_F = 0.750 μ_{CR} = 0.203
 Diversity = 1.0818e-02 k_t = [1, 1, 1, 1]
 Generation 6900: Best = 4.971209e+03 FES = 690100 μ_F = 0.751 μ_{CR} = 0.193
 Diversity = 9.9124e-03 k_t = [4, 4, 3, 4]
 Generation 6910: Best = 4.971202e+03 FES = 691100 μ_F = 0.756 μ_{CR} = 0.187
 Diversity = 9.9593e-03 k_t = [1, 1, 1, 1]
 Generation 6920: Best = 4.971197e+03 FES = 692100 μ_F = 0.770 μ_{CR} = 0.184
 Diversity = 1.0588e-02 k_t = [2, 2, 3, 2]
 Generation 6930: Best = 4.971193e+03 FES = 693100 μ_F = 0.758 μ_{CR} = 0.179
 Diversity = 1.0833e-02 k_t = [1, 1, 1, 1]
 Generation 6940: Best = 4.971188e+03 FES = 694100 μ_F = 0.743 μ_{CR} = 0.178
 Diversity = 9.9432e-03 k_t = [2, 4, 3, 2]
 Generation 6950: Best = 4.971184e+03 FES = 695100 μ_F = 0.743 μ_{CR} = 0.176
 Diversity = 9.4223e-03 k_t = [1, 1, 1, 1]
 Generation 6960: Best = 4.971179e+03 FES = 696100 μ_F = 0.731 μ_{CR} = 0.178
 Diversity = 8.9460e-03 k_t = [3, 4, 2, 4]
 Generation 6970: Best = 4.971175e+03 FES = 697100 μ_F = 0.730 μ_{CR} = 0.179
 Diversity = 8.5618e-03 k_t = [1, 1, 1, 1]

Generation 6980: Best = 4.971172e+03 FES = 698100 μ_F = 0.726 μ_{CR} = 0.182
 Diversity = 8.1033e-03 k_t = [2, 2, 4, 3]
 Generation 6990: Best = 4.971167e+03 FES = 699100 μ_F = 0.722 μ_{CR} = 0.190
 Diversity = 7.6870e-03 k_t = [1, 1, 1, 1]
 Generation 7000: Best = 4.971164e+03 FES = 700100 μ_F = 0.736 μ_{CR} = 0.187
 Diversity = 7.1606e-03 k_t = [3, 2, 2, 4]
 Generation 7010: Best = 4.971160e+03 FES = 701100 μ_F = 0.743 μ_{CR} = 0.182
 Diversity = 6.8113e-03 k_t = [1, 1, 1, 1]
 Generation 7020: Best = 4.971156e+03 FES = 702100 μ_F = 0.737 μ_{CR} = 0.189
 Diversity = 7.4484e-03 k_t = [3, 2, 4, 2]
 Generation 7030: Best = 4.971152e+03 FES = 703100 μ_F = 0.739 μ_{CR} = 0.183
 Diversity = 9.1541e-03 k_t = [1, 1, 1, 1]
 Generation 7040: Best = 4.971147e+03 FES = 704100 μ_F = 0.741 μ_{CR} = 0.175
 Diversity = 8.9690e-03 k_t = [2, 2, 4, 4]
 Generation 7050: Best = 4.971146e+03 FES = 705100 μ_F = 0.736 μ_{CR} = 0.164
 Diversity = 8.8764e-03 k_t = [1, 1, 1, 1]
 Generation 7060: Best = 4.971140e+03 FES = 706100 μ_F = 0.741 μ_{CR} = 0.164
 Diversity = 8.6101e-03 k_t = [3, 3, 2, 3]
 Generation 7070: Best = 4.971138e+03 FES = 707100 μ_F = 0.751 μ_{CR} = 0.156
 Diversity = 9.5377e-03 k_t = [1, 1, 1, 1]
 Generation 7080: Best = 4.971131e+03 FES = 708100 μ_F = 0.765 μ_{CR} = 0.159
 Diversity = 9.8271e-03 k_t = [2, 2, 2, 2]
 Generation 7090: Best = 4.971128e+03 FES = 709100 μ_F = 0.756 μ_{CR} = 0.157
 Diversity = 1.0405e-02 k_t = [1, 1, 1, 1]
 Generation 7100: Best = 4.971125e+03 FES = 710100 μ_F = 0.744 μ_{CR} = 0.164
 Diversity = 1.0154e-02 k_t = [2, 4, 4, 2]
 Generation 7110: Best = 4.971120e+03 FES = 711100 μ_F = 0.733 μ_{CR} = 0.164
 Diversity = 1.1193e-02 k_t = [1, 1, 1, 1]
 Generation 7120: Best = 4.971116e+03 FES = 712100 μ_F = 0.732 μ_{CR} = 0.164
 Diversity = 1.0705e-02 k_t = [3, 3, 4, 2]
 Generation 7130: Best = 4.971111e+03 FES = 713100 μ_F = 0.736 μ_{CR} = 0.165
 Diversity = 1.1392e-02 k_t = [1, 1, 1, 1]
 Generation 7140: Best = 4.971108e+03 FES = 714100 μ_F = 0.737 μ_{CR} = 0.161
 Diversity = 1.0759e-02 k_t = [4, 2, 3, 3]
 Generation 7150: Best = 4.971108e+03 FES = 715100 μ_F = 0.712 μ_{CR} = 0.141
 Diversity = 1.0456e-02 k_t = [1, 1, 1, 1]
 Generation 7160: Best = 4.971105e+03 FES = 716100 μ_F = 0.714 μ_{CR} = 0.128
 Diversity = 1.0302e-02 k_t = [4, 3, 4, 2]
 Generation 7170: Best = 4.971102e+03 FES = 717100 μ_F = 0.725 μ_{CR} = 0.130
 Diversity = 1.0256e-02 k_t = [1, 1, 1, 1]
 Generation 7180: Best = 4.971098e+03 FES = 718100 μ_F = 0.728 μ_{CR} = 0.138
 Diversity = 9.6757e-03 k_t = [3, 4, 3, 3]
 Generation 7190: Best = 4.971095e+03 FES = 719100 μ_F = 0.736 μ_{CR} = 0.141
 Diversity = 8.6277e-03 k_t = [1, 1, 1, 1]
 Generation 7200: Best = 4.971092e+03 FES = 720100 μ_F = 0.759 μ_{CR} = 0.147
 Diversity = 8.7257e-03 k_t = [3, 2, 2, 3]
 Generation 7210: Best = 4.971087e+03 FES = 721100 μ_F = 0.765 μ_{CR} = 0.154
 Diversity = 9.9373e-03 k_t = [1, 1, 1, 1]
 Generation 7220: Best = 4.971084e+03 FES = 722100 μ_F = 0.768 μ_{CR} = 0.154
 Diversity = 1.0154e-02 k_t = [2, 3, 2, 3]
 Generation 7230: Best = 4.971082e+03 FES = 723100 μ_F = 0.756 μ_{CR} = 0.144
 Diversity = 9.6218e-03 k_t = [1, 1, 1, 1]
 Generation 7240: Best = 4.971079e+03 FES = 724100 μ_F = 0.756 μ_{CR} = 0.152
 Diversity = 9.3098e-03 k_t = [3, 4, 2, 3]

Generation 7250: Best = 4.971074e+03 FES = 725100 μ_F = 0.769 μ_{CR} = 0.157
 Diversity = 9.5485e-03 k_t = [1, 1, 1, 1]
 Generation 7260: Best = 4.971071e+03 FES = 726100 μ_F = 0.781 μ_{CR} = 0.156
 Diversity = 9.1818e-03 k_t = [4, 3, 2, 4]
 Generation 7270: Best = 4.971068e+03 FES = 727100 μ_F = 0.784 μ_{CR} = 0.157
 Diversity = 9.1212e-03 k_t = [1, 1, 1, 1]
 Generation 7280: Best = 4.971065e+03 FES = 728100 μ_F = 0.780 μ_{CR} = 0.160
 Diversity = 1.0147e-02 k_t = [2, 2, 2, 3]
 Generation 7290: Best = 4.971062e+03 FES = 729100 μ_F = 0.769 μ_{CR} = 0.150
 Diversity = 1.0289e-02 k_t = [1, 1, 1, 1]
 Generation 7300: Best = 4.971060e+03 FES = 730100 μ_F = 0.754 μ_{CR} = 0.153
 Diversity = 9.2903e-03 k_t = [2, 3, 2, 4]
 Generation 7310: Best = 4.971059e+03 FES = 731100 μ_F = 0.754 μ_{CR} = 0.150
 Diversity = 8.0250e-03 k_t = [1, 1, 1, 1]
 Generation 7320: Best = 4.971056e+03 FES = 732100 μ_F = 0.754 μ_{CR} = 0.139
 Diversity = 7.7025e-03 k_t = [4, 4, 2, 2]
 Generation 7330: Best = 4.971054e+03 FES = 733100 μ_F = 0.755 μ_{CR} = 0.144
 Diversity = 7.6504e-03 k_t = [1, 1, 1, 1]
 Generation 7340: Best = 4.971052e+03 FES = 734100 μ_F = 0.756 μ_{CR} = 0.153
 Diversity = 8.3608e-03 k_t = [4, 4, 2, 2]
 Generation 7350: Best = 4.971050e+03 FES = 735100 μ_F = 0.745 μ_{CR} = 0.152
 Diversity = 8.5134e-03 k_t = [1, 1, 1, 1]
 Generation 7360: Best = 4.971048e+03 FES = 736100 μ_F = 0.748 μ_{CR} = 0.155
 Diversity = 7.4814e-03 k_t = [4, 4, 3, 3]
 Generation 7370: Best = 4.971044e+03 FES = 737100 μ_F = 0.734 μ_{CR} = 0.153
 Diversity = 7.4195e-03 k_t = [1, 1, 1, 1]
 Generation 7380: Best = 4.971043e+03 FES = 738100 μ_F = 0.749 μ_{CR} = 0.154
 Diversity = 6.3671e-03 k_t = [2, 3, 4, 4]
 Generation 7390: Best = 4.971042e+03 FES = 739100 μ_F = 0.751 μ_{CR} = 0.163
 Diversity = 5.7317e-03 k_t = [1, 1, 1, 1]
 Generation 7400: Best = 4.971040e+03 FES = 740100 μ_F = 0.754 μ_{CR} = 0.173
 Diversity = 5.5914e-03 k_t = [3, 4, 2, 2]
 Generation 7410: Best = 4.971037e+03 FES = 741100 μ_F = 0.747 μ_{CR} = 0.173
 Diversity = 6.1688e-03 k_t = [1, 1, 1, 1]
 Generation 7420: Best = 4.971034e+03 FES = 742100 μ_F = 0.759 μ_{CR} = 0.174
 Diversity = 7.2272e-03 k_t = [3, 2, 2, 3]
 Generation 7430: Best = 4.971032e+03 FES = 743100 μ_F = 0.754 μ_{CR} = 0.168
 Diversity = 6.7678e-03 k_t = [1, 1, 1, 1]
 Generation 7440: Best = 4.971031e+03 FES = 744100 μ_F = 0.750 μ_{CR} = 0.169
 Diversity = 5.7866e-03 k_t = [3, 4, 4, 3]
 Generation 7450: Best = 4.971028e+03 FES = 745100 μ_F = 0.758 μ_{CR} = 0.166
 Diversity = 6.2592e-03 k_t = [1, 1, 1, 1]
 Generation 7460: Best = 4.971027e+03 FES = 746100 μ_F = 0.742 μ_{CR} = 0.162
 Diversity = 6.1854e-03 k_t = [4, 4, 2, 3]
 Generation 7470: Best = 4.971026e+03 FES = 747100 μ_F = 0.733 μ_{CR} = 0.170
 Diversity = 6.3216e-03 k_t = [1, 1, 1, 1]
 Generation 7480: Best = 4.971023e+03 FES = 748100 μ_F = 0.736 μ_{CR} = 0.168
 Diversity = 6.4879e-03 k_t = [2, 3, 2, 4]
 Generation 7490: Best = 4.971023e+03 FES = 749100 μ_F = 0.735 μ_{CR} = 0.167
 Diversity = 5.9104e-03 k_t = [1, 1, 1, 1]
 Generation 7500: Best = 4.971020e+03 FES = 750100 μ_F = 0.746 μ_{CR} = 0.163
 Diversity = 5.8594e-03 k_t = [2, 2, 2, 4]
 Generation 7510: Best = 4.971016e+03 FES = 751100 μ_F = 0.761 μ_{CR} = 0.167
 Diversity = 6.0892e-03 k_t = [1, 1, 1, 1]

Generation 7520: Best = 4.971012e+03 FES = 752100 μ_F = 0.774 μ_{CR} = 0.166
 Diversity = 6.9647e-03 k_t = [3, 3, 2, 4]
 Generation 7530: Best = 4.971009e+03 FES = 753100 μ_F = 0.764 μ_{CR} = 0.174
 Diversity = 7.6046e-03 k_t = [1, 1, 1, 1]
 Generation 7540: Best = 4.971008e+03 FES = 754100 μ_F = 0.769 μ_{CR} = 0.170
 Diversity = 6.7690e-03 k_t = [2, 4, 4, 3]
 Generation 7550: Best = 4.971006e+03 FES = 755100 μ_F = 0.777 μ_{CR} = 0.167
 Diversity = 7.0124e-03 k_t = [1, 1, 1, 1]
 Generation 7560: Best = 4.971003e+03 FES = 756100 μ_F = 0.772 μ_{CR} = 0.170
 Diversity = 7.1275e-03 k_t = [3, 3, 4, 2]
 Generation 7570: Best = 4.971001e+03 FES = 757100 μ_F = 0.742 μ_{CR} = 0.173
 Diversity = 6.9191e-03 k_t = [1, 1, 1, 1]
 Generation 7580: Best = 4.970998e+03 FES = 758100 μ_F = 0.737 μ_{CR} = 0.174
 Diversity = 6.6943e-03 k_t = [3, 2, 2, 4]
 Generation 7590: Best = 4.970995e+03 FES = 759100 μ_F = 0.733 μ_{CR} = 0.179
 Diversity = 7.4466e-03 k_t = [1, 1, 1, 1]
 Generation 7600: Best = 4.970992e+03 FES = 760100 μ_F = 0.735 μ_{CR} = 0.184
 Diversity = 7.0350e-03 k_t = [4, 4, 3, 2]
 Generation 7610: Best = 4.970989e+03 FES = 761100 μ_F = 0.752 μ_{CR} = 0.189
 Diversity = 7.7368e-03 k_t = [1, 1, 1, 1]
 Generation 7620: Best = 4.970987e+03 FES = 762100 μ_F = 0.744 μ_{CR} = 0.179
 Diversity = 6.8931e-03 k_t = [4, 2, 4, 2]
 Generation 7630: Best = 4.970984e+03 FES = 763100 μ_F = 0.746 μ_{CR} = 0.177
 Diversity = 6.8474e-03 k_t = [1, 1, 1, 1]
 Generation 7640: Best = 4.970982e+03 FES = 764100 μ_F = 0.736 μ_{CR} = 0.175
 Diversity = 6.0336e-03 k_t = [4, 4, 3, 2]
 Generation 7650: Best = 4.970980e+03 FES = 765100 μ_F = 0.743 μ_{CR} = 0.175
 Diversity = 6.1566e-03 k_t = [1, 1, 1, 1]
 Generation 7660: Best = 4.970976e+03 FES = 766100 μ_F = 0.741 μ_{CR} = 0.186
 Diversity = 6.3602e-03 k_t = [3, 3, 3, 2]
 Generation 7670: Best = 4.970975e+03 FES = 767100 μ_F = 0.738 μ_{CR} = 0.180
 Diversity = 6.8803e-03 k_t = [1, 1, 1, 1]
 Generation 7680: Best = 4.970972e+03 FES = 768100 μ_F = 0.739 μ_{CR} = 0.175
 Diversity = 6.8181e-03 k_t = [2, 3, 4, 3]
 Generation 7690: Best = 4.970970e+03 FES = 769100 μ_F = 0.739 μ_{CR} = 0.178
 Diversity = 6.2624e-03 k_t = [1, 1, 1, 1]
 Generation 7700: Best = 4.970967e+03 FES = 770100 μ_F = 0.749 μ_{CR} = 0.180
 Diversity = 6.1254e-03 k_t = [3, 4, 3, 3]
 Generation 7710: Best = 4.970965e+03 FES = 771100 μ_F = 0.753 μ_{CR} = 0.182
 Diversity = 6.5399e-03 k_t = [1, 1, 1, 1]
 Generation 7720: Best = 4.970963e+03 FES = 772100 μ_F = 0.751 μ_{CR} = 0.183
 Diversity = 5.9124e-03 k_t = [4, 2, 4, 4]
 Generation 7730: Best = 4.970961e+03 FES = 773100 μ_F = 0.756 μ_{CR} = 0.186
 Diversity = 5.5157e-03 k_t = [1, 1, 1, 1]
 Generation 7740: Best = 4.970959e+03 FES = 774100 μ_F = 0.747 μ_{CR} = 0.194
 Diversity = 5.0882e-03 k_t = [3, 4, 2, 3]
 Generation 7750: Best = 4.970956e+03 FES = 775100 μ_F = 0.749 μ_{CR} = 0.200
 Diversity = 5.3270e-03 k_t = [1, 1, 1, 1]
 Generation 7760: Best = 4.970955e+03 FES = 776100 μ_F = 0.758 μ_{CR} = 0.202
 Diversity = 5.4245e-03 k_t = [3, 2, 2, 3]
 Generation 7770: Best = 4.970954e+03 FES = 777100 μ_F = 0.752 μ_{CR} = 0.198
 Diversity = 5.6166e-03 k_t = [1, 1, 1, 1]
 Generation 7780: Best = 4.970951e+03 FES = 778100 μ_F = 0.750 μ_{CR} = 0.196
 Diversity = 6.4647e-03 k_t = [4, 2, 3, 2]

Generation 7790: Best = 4.970948e+03 FES = 779100 μ_F = 0.736 μ_{CR} = 0.193
 Diversity = 6.3263e-03 k_t = [1, 1, 1, 1]
 Generation 7800: Best = 4.970946e+03 FES = 780100 μ_F = 0.744 μ_{CR} = 0.187
 Diversity = 5.7859e-03 k_t = [4, 3, 2, 4]
 Generation 7810: Best = 4.970944e+03 FES = 781100 μ_F = 0.751 μ_{CR} = 0.196
 Diversity = 5.6835e-03 k_t = [1, 1, 1, 1]
 Generation 7820: Best = 4.970942e+03 FES = 782100 μ_F = 0.742 μ_{CR} = 0.195
 Diversity = 5.6865e-03 k_t = [4, 4, 3, 2]
 Generation 7830: Best = 4.970940e+03 FES = 783100 μ_F = 0.750 μ_{CR} = 0.189
 Diversity = 5.7563e-03 k_t = [1, 1, 1, 1]
 Generation 7840: Best = 4.970937e+03 FES = 784100 μ_F = 0.741 μ_{CR} = 0.190
 Diversity = 5.1963e-03 k_t = [4, 3, 4, 4]
 Generation 7850: Best = 4.970935e+03 FES = 785100 μ_F = 0.750 μ_{CR} = 0.194
 Diversity = 4.8958e-03 k_t = [1, 1, 1, 1]
 Generation 7860: Best = 4.970933e+03 FES = 786100 μ_F = 0.746 μ_{CR} = 0.196
 Diversity = 5.2021e-03 k_t = [3, 3, 2, 3]
 Generation 7870: Best = 4.970931e+03 FES = 787100 μ_F = 0.758 μ_{CR} = 0.196
 Diversity = 5.5759e-03 k_t = [1, 1, 1, 1]
 Generation 7880: Best = 4.970929e+03 FES = 788100 μ_F = 0.758 μ_{CR} = 0.189
 Diversity = 5.4712e-03 k_t = [4, 3, 4, 3]
 Generation 7890: Best = 4.970927e+03 FES = 789100 μ_F = 0.751 μ_{CR} = 0.203
 Diversity = 5.1934e-03 k_t = [1, 1, 1, 1]
 Generation 7900: Best = 4.970926e+03 FES = 790100 μ_F = 0.744 μ_{CR} = 0.197
 Diversity = 4.6269e-03 k_t = [2, 4, 2, 3]
 Generation 7910: Best = 4.970924e+03 FES = 791100 μ_F = 0.751 μ_{CR} = 0.193
 Diversity = 4.7894e-03 k_t = [1, 1, 1, 1]
 Generation 7920: Best = 4.970922e+03 FES = 792100 μ_F = 0.752 μ_{CR} = 0.191
 Diversity = 4.4362e-03 k_t = [3, 2, 3, 4]
 Generation 7930: Best = 4.970919e+03 FES = 793100 μ_F = 0.757 μ_{CR} = 0.194
 Diversity = 4.7511e-03 k_t = [1, 1, 1, 1]
 Generation 7940: Best = 4.970918e+03 FES = 794100 μ_F = 0.751 μ_{CR} = 0.187
 Diversity = 5.0006e-03 k_t = [3, 2, 4, 2]
 Generation 7950: Best = 4.970917e+03 FES = 795100 μ_F = 0.747 μ_{CR} = 0.194
 Diversity = 4.6788e-03 k_t = [1, 1, 1, 1]
 Generation 7960: Best = 4.970916e+03 FES = 796100 μ_F = 0.747 μ_{CR} = 0.179
 Diversity = 4.2355e-03 k_t = [2, 3, 4, 4]
 Generation 7970: Best = 4.970914e+03 FES = 797100 μ_F = 0.755 μ_{CR} = 0.178
 Diversity = 3.8857e-03 k_t = [1, 1, 1, 1]
 Generation 7980: Best = 4.970912e+03 FES = 798100 μ_F = 0.762 μ_{CR} = 0.179
 Diversity = 3.8732e-03 k_t = [2, 4, 3, 3]
 Generation 7990: Best = 4.970911e+03 FES = 799100 μ_F = 0.771 μ_{CR} = 0.183
 Diversity = 4.4278e-03 k_t = [1, 1, 1, 1]
 Generation 8000: Best = 4.970910e+03 FES = 800100 μ_F = 0.775 μ_{CR} = 0.174
 Diversity = 4.4534e-03 k_t = [3, 4, 4, 4]
 Generation 8010: Best = 4.970908e+03 FES = 801100 μ_F = 0.770 μ_{CR} = 0.177
 Diversity = 4.3386e-03 k_t = [1, 1, 1, 1]
 Generation 8020: Best = 4.970906e+03 FES = 802100 μ_F = 0.773 μ_{CR} = 0.173
 Diversity = 4.5660e-03 k_t = [3, 2, 4, 4]
 Generation 8030: Best = 4.970905e+03 FES = 803100 μ_F = 0.767 μ_{CR} = 0.178
 Diversity = 4.6243e-03 k_t = [1, 1, 1, 1]
 Generation 8040: Best = 4.970903e+03 FES = 804100 μ_F = 0.767 μ_{CR} = 0.178
 Diversity = 4.7622e-03 k_t = [3, 4, 2, 2]
 Generation 8050: Best = 4.970902e+03 FES = 805100 μ_F = 0.771 μ_{CR} = 0.169
 Diversity = 4.8839e-03 k_t = [1, 1, 1, 1]

Generation 8060: Best = 4.970901e+03 FES = 806100 μ_F = 0.765 μ_{CR} = 0.170
 Diversity = 4.6610e-03 k_t = [3, 4, 4, 3]
 Generation 8070: Best = 4.970898e+03 FES = 807100 μ_F = 0.761 μ_{CR} = 0.169
 Diversity = 5.0199e-03 k_t = [1, 1, 1, 1]
 Generation 8080: Best = 4.970897e+03 FES = 808100 μ_F = 0.774 μ_{CR} = 0.167
 Diversity = 5.1637e-03 k_t = [4, 3, 3, 3]
 Generation 8090: Best = 4.970896e+03 FES = 809100 μ_F = 0.768 μ_{CR} = 0.168
 Diversity = 5.4324e-03 k_t = [1, 1, 1, 1]
 Generation 8100: Best = 4.970894e+03 FES = 810100 μ_F = 0.772 μ_{CR} = 0.165
 Diversity = 5.1910e-03 k_t = [3, 4, 2, 4]
 Generation 8110: Best = 4.970893e+03 FES = 811100 μ_F = 0.769 μ_{CR} = 0.168
 Diversity = 5.0528e-03 k_t = [1, 1, 1, 1]
 Generation 8120: Best = 4.970891e+03 FES = 812100 μ_F = 0.767 μ_{CR} = 0.161
 Diversity = 5.4617e-03 k_t = [4, 4, 2, 3]
 Generation 8130: Best = 4.970888e+03 FES = 813100 μ_F = 0.772 μ_{CR} = 0.163
 Diversity = 6.0028e-03 k_t = [1, 1, 1, 1]
 Generation 8140: Best = 4.970887e+03 FES = 814100 μ_F = 0.752 μ_{CR} = 0.169
 Diversity = 5.2746e-03 k_t = [4, 3, 2, 3]
 Generation 8150: Best = 4.970886e+03 FES = 815100 μ_F = 0.760 μ_{CR} = 0.172
 Diversity = 4.9036e-03 k_t = [1, 1, 1, 1]
 Generation 8160: Best = 4.970885e+03 FES = 816100 μ_F = 0.774 μ_{CR} = 0.163
 Diversity = 4.6922e-03 k_t = [3, 3, 2, 3]
 Generation 8170: Best = 4.970883e+03 FES = 817100 μ_F = 0.771 μ_{CR} = 0.157
 Diversity = 5.1788e-03 k_t = [1, 1, 1, 1]
 Generation 8180: Best = 4.970882e+03 FES = 818100 μ_F = 0.776 μ_{CR} = 0.159
 Diversity = 5.3853e-03 k_t = [2, 4, 4, 3]
 Generation 8190: Best = 4.970881e+03 FES = 819100 μ_F = 0.772 μ_{CR} = 0.169
 Diversity = 5.2775e-03 k_t = [1, 1, 1, 1]
 Generation 8200: Best = 4.970879e+03 FES = 820100 μ_F = 0.784 μ_{CR} = 0.167
 Diversity = 5.3760e-03 k_t = [3, 4, 2, 3]
 Generation 8210: Best = 4.970877e+03 FES = 821100 μ_F = 0.788 μ_{CR} = 0.168
 Diversity = 5.4687e-03 k_t = [1, 1, 1, 1]
 Generation 8220: Best = 4.970876e+03 FES = 822100 μ_F = 0.793 μ_{CR} = 0.172
 Diversity = 5.4510e-03 k_t = [4, 4, 2, 4]
 Generation 8230: Best = 4.970874e+03 FES = 823100 μ_F = 0.786 μ_{CR} = 0.173
 Diversity = 5.1626e-03 k_t = [1, 1, 1, 1]
 Generation 8240: Best = 4.970873e+03 FES = 824100 μ_F = 0.767 μ_{CR} = 0.159
 Diversity = 4.5323e-03 k_t = [2, 4, 4, 3]
 Generation 8250: Best = 4.970872e+03 FES = 825100 μ_F = 0.758 μ_{CR} = 0.169
 Diversity = 4.5172e-03 k_t = [1, 1, 1, 1]
 Generation 8260: Best = 4.970871e+03 FES = 826100 μ_F = 0.744 μ_{CR} = 0.173
 Diversity = 4.2522e-03 k_t = [2, 3, 3, 3]
 Generation 8270: Best = 4.970870e+03 FES = 827100 μ_F = 0.744 μ_{CR} = 0.178
 Diversity = 4.3181e-03 k_t = [1, 1, 1, 1]
 Generation 8280: Best = 4.970869e+03 FES = 828100 μ_F = 0.740 μ_{CR} = 0.176
 Diversity = 4.1968e-03 k_t = [4, 3, 2, 3]
 Generation 8290: Best = 4.970868e+03 FES = 829100 μ_F = 0.749 μ_{CR} = 0.175
 Diversity = 3.8782e-03 k_t = [1, 1, 1, 1]
 Generation 8300: Best = 4.970867e+03 FES = 830100 μ_F = 0.753 μ_{CR} = 0.182
 Diversity = 3.6860e-03 k_t = [2, 4, 3, 2]
 Generation 8310: Best = 4.970866e+03 FES = 831100 μ_F = 0.757 μ_{CR} = 0.181
 Diversity = 4.1038e-03 k_t = [1, 1, 1, 1]
 Generation 8320: Best = 4.970865e+03 FES = 832100 μ_F = 0.756 μ_{CR} = 0.182
 Diversity = 4.0842e-03 k_t = [3, 3, 4, 2]

Generation 8330: Best = 4.970864e+03 FES = 833100 μ_F = 0.751 μ_{CR} = 0.180
 Diversity = 3.8274e-03 k_t = [1, 1, 1, 1]
 Generation 8340: Best = 4.970863e+03 FES = 834100 μ_F = 0.753 μ_{CR} = 0.176
 Diversity = 3.9723e-03 k_t = [2, 4, 4, 3]
 Generation 8350: Best = 4.970862e+03 FES = 835100 μ_F = 0.758 μ_{CR} = 0.176
 Diversity = 4.3996e-03 k_t = [1, 1, 1, 1]
 Generation 8360: Best = 4.970861e+03 FES = 836100 μ_F = 0.741 μ_{CR} = 0.164
 Diversity = 4.1453e-03 k_t = [2, 2, 2, 4]
 Generation 8370: Best = 4.970860e+03 FES = 837100 μ_F = 0.746 μ_{CR} = 0.157
 Diversity = 3.9655e-03 k_t = [1, 1, 1, 1]
 Generation 8380: Best = 4.970859e+03 FES = 838100 μ_F = 0.748 μ_{CR} = 0.160
 Diversity = 4.7669e-03 k_t = [3, 3, 3, 2]
 Generation 8390: Best = 4.970857e+03 FES = 839100 μ_F = 0.743 μ_{CR} = 0.175
 Diversity = 5.1050e-03 k_t = [1, 1, 1, 1]
 Generation 8400: Best = 4.970856e+03 FES = 840100 μ_F = 0.751 μ_{CR} = 0.171
 Diversity = 4.7693e-03 k_t = [4, 4, 2, 3]
 Generation 8410: Best = 4.970855e+03 FES = 841100 μ_F = 0.757 μ_{CR} = 0.165
 Diversity = 5.1972e-03 k_t = [1, 1, 1, 1]
 Generation 8420: Best = 4.970854e+03 FES = 842100 μ_F = 0.756 μ_{CR} = 0.158
 Diversity = 4.5100e-03 k_t = [4, 3, 4, 4]
 Generation 8430: Best = 4.970853e+03 FES = 843100 μ_F = 0.771 μ_{CR} = 0.157
 Diversity = 4.9455e-03 k_t = [1, 1, 1, 1]
 Generation 8440: Best = 4.970851e+03 FES = 844100 μ_F = 0.749 μ_{CR} = 0.164
 Diversity = 4.8347e-03 k_t = [4, 2, 4, 2]
 Generation 8450: Best = 4.970850e+03 FES = 845100 μ_F = 0.754 μ_{CR} = 0.165
 Diversity = 4.6295e-03 k_t = [1, 1, 1, 1]
 Generation 8460: Best = 4.970849e+03 FES = 846100 μ_F = 0.760 μ_{CR} = 0.148
 Diversity = 4.7055e-03 k_t = [3, 3, 3, 4]
 Generation 8470: Best = 4.970848e+03 FES = 847100 μ_F = 0.765 μ_{CR} = 0.142
 Diversity = 4.7200e-03 k_t = [1, 1, 1, 1]
 Generation 8480: Best = 4.970848e+03 FES = 848100 μ_F = 0.754 μ_{CR} = 0.150
 Diversity = 4.7805e-03 k_t = [4, 3, 2, 3]
 Generation 8490: Best = 4.970847e+03 FES = 849100 μ_F = 0.756 μ_{CR} = 0.154
 Diversity = 4.8827e-03 k_t = [1, 1, 1, 1]
 Generation 8500: Best = 4.970845e+03 FES = 850100 μ_F = 0.752 μ_{CR} = 0.160
 Diversity = 5.1161e-03 k_t = [2, 3, 2, 2]
 Generation 8510: Best = 4.970844e+03 FES = 851100 μ_F = 0.753 μ_{CR} = 0.153
 Diversity = 5.9299e-03 k_t = [1, 1, 1, 1]
 Generation 8520: Best = 4.970843e+03 FES = 852100 μ_F = 0.754 μ_{CR} = 0.140
 Diversity = 5.6426e-03 k_t = [3, 2, 3, 3]
 Generation 8530: Best = 4.970842e+03 FES = 853100 μ_F = 0.750 μ_{CR} = 0.138
 Diversity = 5.4651e-03 k_t = [1, 1, 1, 1]
 Generation 8540: Best = 4.970841e+03 FES = 854100 μ_F = 0.746 μ_{CR} = 0.144
 Diversity = 5.1468e-03 k_t = [2, 4, 4, 3]
 Generation 8550: Best = 4.970840e+03 FES = 855100 μ_F = 0.749 μ_{CR} = 0.149
 Diversity = 5.6235e-03 k_t = [1, 1, 1, 1]
 Generation 8560: Best = 4.970839e+03 FES = 856100 μ_F = 0.746 μ_{CR} = 0.145
 Diversity = 5.3580e-03 k_t = [3, 2, 3, 2]
 Generation 8570: Best = 4.970838e+03 FES = 857100 μ_F = 0.748 μ_{CR} = 0.153
 Diversity = 5.0362e-03 k_t = [1, 1, 1, 1]
 Generation 8580: Best = 4.970837e+03 FES = 858100 μ_F = 0.752 μ_{CR} = 0.154
 Diversity = 4.4803e-03 k_t = [3, 2, 2, 4]
 Generation 8590: Best = 4.970836e+03 FES = 859100 μ_F = 0.748 μ_{CR} = 0.148
 Diversity = 4.6512e-03 k_t = [1, 1, 1, 1]

Generation 8600: Best = 4.970835e+03 FES = 860100 μ_F = 0.746 μ_{CR} = 0.143
 Diversity = 4.9644e-03 k_t = [4, 3, 2, 3]
 Generation 8610: Best = 4.970835e+03 FES = 861100 μ_F = 0.745 μ_{CR} = 0.142
 Diversity = 4.7065e-03 k_t = [1, 1, 1, 1]
 Generation 8620: Best = 4.970834e+03 FES = 862100 μ_F = 0.754 μ_{CR} = 0.137
 Diversity = 4.2215e-03 k_t = [4, 2, 4, 4]
 Generation 8630: Best = 4.970834e+03 FES = 863100 μ_F = 0.740 μ_{CR} = 0.134
 Diversity = 3.9411e-03 k_t = [1, 1, 1, 1]
 Generation 8640: Best = 4.970834e+03 FES = 864100 μ_F = 0.744 μ_{CR} = 0.132
 Diversity = 3.2735e-03 k_t = [4, 4, 2, 4]
 Generation 8650: Best = 4.970833e+03 FES = 865100 μ_F = 0.747 μ_{CR} = 0.135
 Diversity = 3.3561e-03 k_t = [1, 1, 1, 1]
 Generation 8660: Best = 4.970832e+03 FES = 866100 μ_F = 0.746 μ_{CR} = 0.139
 Diversity = 3.5538e-03 k_t = [4, 4, 2, 3]
 Generation 8670: Best = 4.970831e+03 FES = 867100 μ_F = 0.751 μ_{CR} = 0.146
 Diversity = 4.4042e-03 k_t = [1, 1, 1, 1]
 Generation 8680: Best = 4.970830e+03 FES = 868100 μ_F = 0.752 μ_{CR} = 0.158
 Diversity = 3.9681e-03 k_t = [2, 2, 3, 3]
 Generation 8690: Best = 4.970829e+03 FES = 869100 μ_F = 0.756 μ_{CR} = 0.172
 Diversity = 4.9310e-03 k_t = [1, 1, 1, 1]
 Generation 8700: Best = 4.970827e+03 FES = 870100 μ_F = 0.753 μ_{CR} = 0.159
 Diversity = 4.6398e-03 k_t = [2, 3, 3, 4]
 Generation 8710: Best = 4.970827e+03 FES = 871100 μ_F = 0.758 μ_{CR} = 0.154
 Diversity = 4.5537e-03 k_t = [1, 1, 1, 1]
 Generation 8720: Best = 4.970826e+03 FES = 872100 μ_F = 0.767 μ_{CR} = 0.153
 Diversity = 4.2489e-03 k_t = [4, 2, 2, 4]
 Generation 8730: Best = 4.970825e+03 FES = 873100 μ_F = 0.763 μ_{CR} = 0.153
 Diversity = 3.9990e-03 k_t = [1, 1, 1, 1]
 Generation 8740: Best = 4.970824e+03 FES = 874100 μ_F = 0.764 μ_{CR} = 0.149
 Diversity = 3.8002e-03 k_t = [3, 2, 2, 3]
 Generation 8750: Best = 4.970824e+03 FES = 875100 μ_F = 0.753 μ_{CR} = 0.152
 Diversity = 4.0023e-03 k_t = [1, 1, 1, 1]
 Generation 8760: Best = 4.970824e+03 FES = 876100 μ_F = 0.757 μ_{CR} = 0.146
 Diversity = 4.3083e-03 k_t = [4, 2, 3, 2]
 Generation 8770: Best = 4.970823e+03 FES = 877100 μ_F = 0.753 μ_{CR} = 0.147
 Diversity = 4.5361e-03 k_t = [1, 1, 1, 1]
 Generation 8780: Best = 4.970822e+03 FES = 878100 μ_F = 0.744 μ_{CR} = 0.145
 Diversity = 4.2979e-03 k_t = [4, 2, 3, 2]
 Generation 8790: Best = 4.970821e+03 FES = 879100 μ_F = 0.730 μ_{CR} = 0.152
 Diversity = 4.2281e-03 k_t = [1, 1, 1, 1]
 Generation 8800: Best = 4.970820e+03 FES = 880100 μ_F = 0.744 μ_{CR} = 0.159
 Diversity = 4.3808e-03 k_t = [4, 2, 2, 2]
 Generation 8810: Best = 4.970820e+03 FES = 881100 μ_F = 0.726 μ_{CR} = 0.156
 Diversity = 4.5597e-03 k_t = [1, 1, 1, 1]
 Generation 8820: Best = 4.970819e+03 FES = 882100 μ_F = 0.718 μ_{CR} = 0.141
 Diversity = 4.0647e-03 k_t = [4, 2, 2, 2]
 Generation 8830: Best = 4.970819e+03 FES = 883100 μ_F = 0.728 μ_{CR} = 0.146
 Diversity = 4.0398e-03 k_t = [1, 1, 1, 1]
 Generation 8840: Best = 4.970818e+03 FES = 884100 μ_F = 0.737 μ_{CR} = 0.145
 Diversity = 3.6298e-03 k_t = [4, 4, 2, 2]
 Generation 8850: Best = 4.970818e+03 FES = 885100 μ_F = 0.739 μ_{CR} = 0.149
 Diversity = 3.9760e-03 k_t = [1, 1, 1, 1]
 Generation 8860: Best = 4.970817e+03 FES = 886100 μ_F = 0.740 μ_{CR} = 0.150
 Diversity = 3.6064e-03 k_t = [2, 3, 3, 2]

Generation 8870: Best = 4.970817e+03 FES = 887100 μ_F = 0.748 μ_{CR} = 0.148
 Diversity = 3.3799e-03 k_t = [1, 1, 1, 1]
 Generation 8880: Best = 4.970816e+03 FES = 888100 μ_F = 0.752 μ_{CR} = 0.152
 Diversity = 3.5434e-03 k_t = [4, 4, 3, 2]
 Generation 8890: Best = 4.970815e+03 FES = 889100 μ_F = 0.755 μ_{CR} = 0.152
 Diversity = 3.5497e-03 k_t = [1, 1, 1, 1]
 Generation 8900: Best = 4.970815e+03 FES = 890100 μ_F = 0.755 μ_{CR} = 0.154
 Diversity = 3.2422e-03 k_t = [2, 3, 4, 3]
 Generation 8910: Best = 4.970815e+03 FES = 891100 μ_F = 0.728 μ_{CR} = 0.159
 Diversity = 3.3582e-03 k_t = [1, 1, 1, 1]
 Generation 8920: Best = 4.970814e+03 FES = 892100 μ_F = 0.744 μ_{CR} = 0.167
 Diversity = 3.0798e-03 k_t = [2, 3, 2, 4]
 Generation 8930: Best = 4.970813e+03 FES = 893100 μ_F = 0.739 μ_{CR} = 0.165
 Diversity = 3.0211e-03 k_t = [1, 1, 1, 1]
 Generation 8940: Best = 4.970813e+03 FES = 894100 μ_F = 0.738 μ_{CR} = 0.169
 Diversity = 3.1722e-03 k_t = [3, 2, 3, 2]
 Generation 8950: Best = 4.970812e+03 FES = 895100 μ_F = 0.733 μ_{CR} = 0.168
 Diversity = 3.1334e-03 k_t = [1, 1, 1, 1]
 Generation 8960: Best = 4.970812e+03 FES = 896100 μ_F = 0.738 μ_{CR} = 0.169
 Diversity = 2.7685e-03 k_t = [4, 3, 4, 4]
 Generation 8970: Best = 4.970812e+03 FES = 897100 μ_F = 0.741 μ_{CR} = 0.160
 Diversity = 2.5268e-03 k_t = [1, 1, 1, 1]
 Generation 8980: Best = 4.970812e+03 FES = 898100 μ_F = 0.749 μ_{CR} = 0.153
 Diversity = 2.3176e-03 k_t = [4, 4, 2, 4]
 Generation 8990: Best = 4.970811e+03 FES = 899100 μ_F = 0.749 μ_{CR} = 0.140
 Diversity = 2.4139e-03 k_t = [1, 1, 1, 1]
 Generation 9000: Best = 4.970811e+03 FES = 900100 μ_F = 0.750 μ_{CR} = 0.142
 Diversity = 2.2767e-03 k_t = [3, 2, 3, 4]
 Generation 9010: Best = 4.970811e+03 FES = 901100 μ_F = 0.747 μ_{CR} = 0.148
 Diversity = 2.1573e-03 k_t = [1, 1, 1, 1]
 Generation 9020: Best = 4.970811e+03 FES = 902100 μ_F = 0.740 μ_{CR} = 0.157
 Diversity = 2.1661e-03 k_t = [4, 2, 2, 3]
 Generation 9030: Best = 4.970810e+03 FES = 903100 μ_F = 0.755 μ_{CR} = 0.169
 Diversity = 2.2032e-03 k_t = [1, 1, 1, 1]
 Generation 9040: Best = 4.970810e+03 FES = 904100 μ_F = 0.756 μ_{CR} = 0.158
 Diversity = 2.2522e-03 k_t = [3, 2, 4, 3]
 Generation 9050: Best = 4.970809e+03 FES = 905100 μ_F = 0.768 μ_{CR} = 0.164
 Diversity = 2.4496e-03 k_t = [1, 1, 1, 1]
 Generation 9060: Best = 4.970808e+03 FES = 906100 μ_F = 0.771 μ_{CR} = 0.174
 Diversity = 2.5041e-03 k_t = [4, 4, 2, 4]
 Generation 9070: Best = 4.970808e+03 FES = 907100 μ_F = 0.769 μ_{CR} = 0.170
 Diversity = 2.5003e-03 k_t = [1, 1, 1, 1]
 Generation 9080: Best = 4.970807e+03 FES = 908100 μ_F = 0.778 μ_{CR} = 0.161
 Diversity = 2.4654e-03 k_t = [4, 4, 2, 4]
 Generation 9090: Best = 4.970807e+03 FES = 909100 μ_F = 0.775 μ_{CR} = 0.168
 Diversity = 2.7744e-03 k_t = [1, 1, 1, 1]
 Generation 9100: Best = 4.970806e+03 FES = 910100 μ_F = 0.778 μ_{CR} = 0.171
 Diversity = 2.7977e-03 k_t = [3, 4, 3, 3]
 Generation 9110: Best = 4.970805e+03 FES = 911100 μ_F = 0.781 μ_{CR} = 0.167
 Diversity = 3.0094e-03 k_t = [1, 1, 1, 1]
 Generation 9120: Best = 4.970804e+03 FES = 912100 μ_F = 0.785 μ_{CR} = 0.170
 Diversity = 3.0365e-03 k_t = [4, 2, 2, 3]
 Generation 9130: Best = 4.970804e+03 FES = 913100 μ_F = 0.783 μ_{CR} = 0.169
 Diversity = 3.3599e-03 k_t = [1, 1, 1, 1]

Generation 9140: Best = 4.970803e+03 FES = 914100 μ_F = 0.781 μ_{CR} = 0.173
 Diversity = 3.6089e-03 k_t = [2, 4, 3, 2]
 Generation 9150: Best = 4.970801e+03 FES = 915100 μ_F = 0.779 μ_{CR} = 0.179
 Diversity = 4.0862e-03 k_t = [1, 1, 1, 1]
 Generation 9160: Best = 4.970801e+03 FES = 916100 μ_F = 0.763 μ_{CR} = 0.178
 Diversity = 3.5298e-03 k_t = [2, 4, 2, 4]
 Generation 9170: Best = 4.970800e+03 FES = 917100 μ_F = 0.773 μ_{CR} = 0.185
 Diversity = 3.4055e-03 k_t = [1, 1, 1, 1]
 Generation 9180: Best = 4.970800e+03 FES = 918100 μ_F = 0.775 μ_{CR} = 0.174
 Diversity = 3.2216e-03 k_t = [4, 2, 2, 3]
 Generation 9190: Best = 4.970799e+03 FES = 919100 μ_F = 0.761 μ_{CR} = 0.176
 Diversity = 3.2956e-03 k_t = [1, 1, 1, 1]
 Generation 9200: Best = 4.970798e+03 FES = 920100 μ_F = 0.770 μ_{CR} = 0.185
 Diversity = 3.4744e-03 k_t = [3, 4, 3, 2]
 Generation 9210: Best = 4.970797e+03 FES = 921100 μ_F = 0.783 μ_{CR} = 0.187
 Diversity = 4.0249e-03 k_t = [1, 1, 1, 1]
 Generation 9220: Best = 4.970796e+03 FES = 922100 μ_F = 0.781 μ_{CR} = 0.182
 Diversity = 3.8811e-03 k_t = [4, 2, 4, 2]
 Generation 9230: Best = 4.970795e+03 FES = 923100 μ_F = 0.789 μ_{CR} = 0.182
 Diversity = 4.5560e-03 k_t = [1, 1, 1, 1]
 Generation 9240: Best = 4.970794e+03 FES = 924100 μ_F = 0.784 μ_{CR} = 0.177
 Diversity = 4.2987e-03 k_t = [4, 2, 3, 2]
 Generation 9250: Best = 4.970793e+03 FES = 925100 μ_F = 0.780 μ_{CR} = 0.182
 Diversity = 4.1144e-03 k_t = [1, 1, 1, 1]
 Generation 9260: Best = 4.970792e+03 FES = 926100 μ_F = 0.773 μ_{CR} = 0.182
 Diversity = 3.9199e-03 k_t = [3, 3, 2, 4]
 Generation 9270: Best = 4.970791e+03 FES = 927100 μ_F = 0.776 μ_{CR} = 0.178
 Diversity = 4.0614e-03 k_t = [1, 1, 1, 1]
 Generation 9280: Best = 4.970790e+03 FES = 928100 μ_F = 0.768 μ_{CR} = 0.180
 Diversity = 4.3344e-03 k_t = [2, 3, 2, 2]
 Generation 9290: Best = 4.970788e+03 FES = 929100 μ_F = 0.770 μ_{CR} = 0.177
 Diversity = 4.7416e-03 k_t = [1, 1, 1, 1]
 Generation 9300: Best = 4.970786e+03 FES = 930100 μ_F = 0.759 μ_{CR} = 0.186
 Diversity = 4.2937e-03 k_t = [4, 3, 2, 4]
 Generation 9310: Best = 4.970784e+03 FES = 931100 μ_F = 0.763 μ_{CR} = 0.193
 Diversity = 4.7896e-03 k_t = [1, 1, 1, 1]
 Generation 9320: Best = 4.970783e+03 FES = 932100 μ_F = 0.760 μ_{CR} = 0.204
 Diversity = 4.4533e-03 k_t = [2, 2, 4, 4]
 Generation 9330: Best = 4.970782e+03 FES = 933100 μ_F = 0.753 μ_{CR} = 0.201
 Diversity = 4.2927e-03 k_t = [1, 1, 1, 1]
 Generation 9340: Best = 4.970781e+03 FES = 934100 μ_F = 0.743 μ_{CR} = 0.211
 Diversity = 3.9922e-03 k_t = [3, 4, 3, 3]
 Generation 9350: Best = 4.970780e+03 FES = 935100 μ_F = 0.735 μ_{CR} = 0.210
 Diversity = 4.0480e-03 k_t = [1, 1, 1, 1]
 Generation 9360: Best = 4.970779e+03 FES = 936100 μ_F = 0.739 μ_{CR} = 0.215
 Diversity = 3.6108e-03 k_t = [4, 2, 4, 4]
 Generation 9370: Best = 4.970778e+03 FES = 937100 μ_F = 0.726 μ_{CR} = 0.213
 Diversity = 3.2306e-03 k_t = [1, 1, 1, 1]
 Generation 9380: Best = 4.970778e+03 FES = 938100 μ_F = 0.725 μ_{CR} = 0.208
 Diversity = 2.7714e-03 k_t = [2, 2, 4, 3]
 Generation 9390: Best = 4.970777e+03 FES = 939100 μ_F = 0.718 μ_{CR} = 0.205
 Diversity = 2.7681e-03 k_t = [1, 1, 1, 1]
 Generation 9400: Best = 4.970776e+03 FES = 940100 μ_F = 0.734 μ_{CR} = 0.208
 Diversity = 2.8353e-03 k_t = [2, 3, 2, 2]

Generation 9410: Best = 4.970775e+03 FES = 941100 μ_F = 0.733 μ_{CR} = 0.207
 Diversity = 3.0951e-03 k_t = [1, 1, 1, 1]
 Generation 9420: Best = 4.970775e+03 FES = 942100 μ_F = 0.728 μ_{CR} = 0.212
 Diversity = 3.0568e-03 k_t = [4, 2, 4, 3]
 Generation 9430: Best = 4.970775e+03 FES = 943100 μ_F = 0.735 μ_{CR} = 0.211
 Diversity = 2.9422e-03 k_t = [1, 1, 1, 1]
 Generation 9440: Best = 4.970774e+03 FES = 944100 μ_F = 0.724 μ_{CR} = 0.205
 Diversity = 2.6211e-03 k_t = [2, 3, 4, 2]
 Generation 9450: Best = 4.970773e+03 FES = 945100 μ_F = 0.726 μ_{CR} = 0.203
 Diversity = 2.8262e-03 k_t = [1, 1, 1, 1]
 Generation 9460: Best = 4.970773e+03 FES = 946100 μ_F = 0.727 μ_{CR} = 0.207
 Diversity = 2.8303e-03 k_t = [4, 4, 4, 4]
 Generation 9470: Best = 4.970771e+03 FES = 947100 μ_F = 0.746 μ_{CR} = 0.207
 Diversity = 3.1678e-03 k_t = [1, 1, 1, 1]
 Generation 9480: Best = 4.970771e+03 FES = 948100 μ_F = 0.744 μ_{CR} = 0.210
 Diversity = 3.5088e-03 k_t = [4, 2, 4, 2]
 Generation 9490: Best = 4.970770e+03 FES = 949100 μ_F = 0.735 μ_{CR} = 0.198
 Diversity = 3.5083e-03 k_t = [1, 1, 1, 1]
 Generation 9500: Best = 4.970770e+03 FES = 950100 μ_F = 0.746 μ_{CR} = 0.190
 Diversity = 3.3364e-03 k_t = [2, 2, 2, 2]
 Generation 9510: Best = 4.970769e+03 FES = 951100 μ_F = 0.748 μ_{CR} = 0.182
 Diversity = 3.3714e-03 k_t = [1, 1, 1, 1]
 Generation 9520: Best = 4.970768e+03 FES = 952100 μ_F = 0.754 μ_{CR} = 0.182
 Diversity = 3.1510e-03 k_t = [3, 2, 3, 3]
 Generation 9530: Best = 4.970768e+03 FES = 953100 μ_F = 0.759 μ_{CR} = 0.192
 Diversity = 3.4507e-03 k_t = [1, 1, 1, 1]
 Generation 9540: Best = 4.970767e+03 FES = 954100 μ_F = 0.753 μ_{CR} = 0.213
 Diversity = 3.3710e-03 k_t = [3, 2, 4, 3]
 Generation 9550: Best = 4.970766e+03 FES = 955100 μ_F = 0.752 μ_{CR} = 0.196
 Diversity = 3.7820e-03 k_t = [1, 1, 1, 1]
 Generation 9560: Best = 4.970765e+03 FES = 956100 μ_F = 0.737 μ_{CR} = 0.192
 Diversity = 3.7547e-03 k_t = [2, 4, 3, 2]
 Generation 9570: Best = 4.970764e+03 FES = 957100 μ_F = 0.732 μ_{CR} = 0.194
 Diversity = 3.4494e-03 k_t = [1, 1, 1, 1]
 Generation 9580: Best = 4.970763e+03 FES = 958100 μ_F = 0.741 μ_{CR} = 0.186
 Diversity = 3.4179e-03 k_t = [3, 4, 3, 4]
 Generation 9590: Best = 4.970762e+03 FES = 959100 μ_F = 0.733 μ_{CR} = 0.179
 Diversity = 3.4937e-03 k_t = [1, 1, 1, 1]
 Generation 9600: Best = 4.970762e+03 FES = 960100 μ_F = 0.726 μ_{CR} = 0.178
 Diversity = 3.6093e-03 k_t = [3, 3, 4, 2]
 Generation 9610: Best = 4.970761e+03 FES = 961100 μ_F = 0.745 μ_{CR} = 0.177
 Diversity = 3.7112e-03 k_t = [1, 1, 1, 1]
 Generation 9620: Best = 4.970760e+03 FES = 962100 μ_F = 0.749 μ_{CR} = 0.170
 Diversity = 3.6224e-03 k_t = [2, 3, 3, 3]
 Generation 9630: Best = 4.970759e+03 FES = 963100 μ_F = 0.739 μ_{CR} = 0.173
 Diversity = 3.4215e-03 k_t = [1, 1, 1, 1]
 Generation 9640: Best = 4.970759e+03 FES = 964100 μ_F = 0.745 μ_{CR} = 0.174
 Diversity = 3.3006e-03 k_t = [4, 2, 3, 2]
 Generation 9650: Best = 4.970757e+03 FES = 965100 μ_F = 0.744 μ_{CR} = 0.177
 Diversity = 3.9860e-03 k_t = [1, 1, 1, 1]
 Generation 9660: Best = 4.970757e+03 FES = 966100 μ_F = 0.741 μ_{CR} = 0.193
 Diversity = 3.5309e-03 k_t = [2, 4, 4, 4]
 Generation 9670: Best = 4.970756e+03 FES = 967100 μ_F = 0.741 μ_{CR} = 0.184
 Diversity = 3.2720e-03 k_t = [1, 1, 1, 1]

Generation 9680: Best = 4.970754e+03 FES = 968100 μ_F = 0.743 μ_{CR} = 0.186
 Diversity = 3.6500e-03 k_t = [2, 3, 4, 2]
 Generation 9690: Best = 4.970753e+03 FES = 969100 μ_F = 0.748 μ_{CR} = 0.184
 Diversity = 4.1156e-03 k_t = [1, 1, 1, 1]
 Generation 9700: Best = 4.970752e+03 FES = 970100 μ_F = 0.760 μ_{CR} = 0.182
 Diversity = 4.1242e-03 k_t = [4, 4, 3, 2]
 Generation 9710: Best = 4.970751e+03 FES = 971100 μ_F = 0.753 μ_{CR} = 0.175
 Diversity = 5.2512e-03 k_t = [1, 1, 1, 1]
 Generation 9720: Best = 4.970749e+03 FES = 972100 μ_F = 0.738 μ_{CR} = 0.174
 Diversity = 5.2966e-03 k_t = [3, 2, 2, 2]
 Generation 9730: Best = 4.970749e+03 FES = 973100 μ_F = 0.731 μ_{CR} = 0.170
 Diversity = 5.4427e-03 k_t = [1, 1, 1, 1]
 Generation 9740: Best = 4.970748e+03 FES = 974100 μ_F = 0.730 μ_{CR} = 0.170
 Diversity = 4.9514e-03 k_t = [4, 4, 2, 2]
 Generation 9750: Best = 4.970747e+03 FES = 975100 μ_F = 0.748 μ_{CR} = 0.170
 Diversity = 4.6169e-03 k_t = [1, 1, 1, 1]
 Generation 9760: Best = 4.970746e+03 FES = 976100 μ_F = 0.741 μ_{CR} = 0.171
 Diversity = 4.3631e-03 k_t = [3, 4, 3, 3]
 Generation 9770: Best = 4.970745e+03 FES = 977100 μ_F = 0.750 μ_{CR} = 0.174
 Diversity = 5.1568e-03 k_t = [1, 1, 1, 1]
 Generation 9780: Best = 4.970744e+03 FES = 978100 μ_F = 0.753 μ_{CR} = 0.167
 Diversity = 5.0741e-03 k_t = [4, 3, 3, 2]
 Generation 9790: Best = 4.970743e+03 FES = 979100 μ_F = 0.750 μ_{CR} = 0.164
 Diversity = 4.8131e-03 k_t = [1, 1, 1, 1]
 Generation 9800: Best = 4.970742e+03 FES = 980100 μ_F = 0.756 μ_{CR} = 0.161
 Diversity = 4.5154e-03 k_t = [2, 2, 3, 4]
 Generation 9810: Best = 4.970741e+03 FES = 981100 μ_F = 0.754 μ_{CR} = 0.159
 Diversity = 4.5797e-03 k_t = [1, 1, 1, 1]
 Generation 9820: Best = 4.970740e+03 FES = 982100 μ_F = 0.751 μ_{CR} = 0.151
 Diversity = 4.4145e-03 k_t = [2, 2, 2, 4]
 Generation 9830: Best = 4.970739e+03 FES = 983100 μ_F = 0.743 μ_{CR} = 0.161
 Diversity = 4.1308e-03 k_t = [1, 1, 1, 1]
 Generation 9840: Best = 4.970738e+03 FES = 984100 μ_F = 0.747 μ_{CR} = 0.160
 Diversity = 3.8591e-03 k_t = [3, 3, 4, 4]
 Generation 9850: Best = 4.970738e+03 FES = 985100 μ_F = 0.747 μ_{CR} = 0.153
 Diversity = 3.8111e-03 k_t = [1, 1, 1, 1]
 Generation 9860: Best = 4.970737e+03 FES = 986100 μ_F = 0.747 μ_{CR} = 0.149
 Diversity = 3.6446e-03 k_t = [3, 3, 2, 3]
 Generation 9870: Best = 4.970736e+03 FES = 987100 μ_F = 0.750 μ_{CR} = 0.150
 Diversity = 3.8430e-03 k_t = [1, 1, 1, 1]
 Generation 9880: Best = 4.970736e+03 FES = 988100 μ_F = 0.758 μ_{CR} = 0.154
 Diversity = 3.5189e-03 k_t = [2, 2, 3, 4]
 Generation 9890: Best = 4.970735e+03 FES = 989100 μ_F = 0.763 μ_{CR} = 0.157
 Diversity = 3.5897e-03 k_t = [1, 1, 1, 1]
 Generation 9900: Best = 4.970735e+03 FES = 990100 μ_F = 0.771 μ_{CR} = 0.153
 Diversity = 3.8568e-03 k_t = [3, 4, 4, 2]
 Generation 9910: Best = 4.970734e+03 FES = 991100 μ_F = 0.781 μ_{CR} = 0.154
 Diversity = 4.0109e-03 k_t = [1, 1, 1, 1]
 Generation 9920: Best = 4.970734e+03 FES = 992100 μ_F = 0.769 μ_{CR} = 0.152
 Diversity = 3.4204e-03 k_t = [2, 4, 4, 3]
 Generation 9930: Best = 4.970733e+03 FES = 993100 μ_F = 0.758 μ_{CR} = 0.156
 Diversity = 3.1859e-03 k_t = [1, 1, 1, 1]
 Generation 9940: Best = 4.970731e+03 FES = 994100 μ_F = 0.756 μ_{CR} = 0.173
 Diversity = 3.6548e-03 k_t = [2, 2, 2, 4]

Generation 9950: Best = 4.970729e+03 FES = 995100 μ_F = 0.760 μ_{CR} = 0.170
 Diversity = 4.3213e-03 k_t = [1, 1, 1, 1]
 Generation 9960: Best = 4.970726e+03 FES = 996100 μ_F = 0.772 μ_{CR} = 0.175
 Diversity = 4.8542e-03 k_t = [3, 2, 4, 4]
 Generation 9970: Best = 4.970720e+03 FES = 997100 μ_F = 0.780 μ_{CR} = 0.174
 Diversity = 5.8502e-03 k_t = [1, 1, 1, 1]
 Generation 9980: Best = 4.970710e+03 FES = 998100 μ_F = 0.784 μ_{CR} = 0.189
 Diversity = 7.9849e-03 k_t = [4, 3, 2, 3]
 Generation 9990: Best = 4.970707e+03 FES = 999100 μ_F = 0.775 μ_{CR} = 0.202
 Diversity = 8.2632e-03 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 4.970705e+03

Total Function Evals = 1000000

Total Successful Updates = 392451

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Results:

Best Fitness : 4.970705e+03

Global Optimum : 1.300000e+03

Error (|best-optimum|) : 3.670705e+03

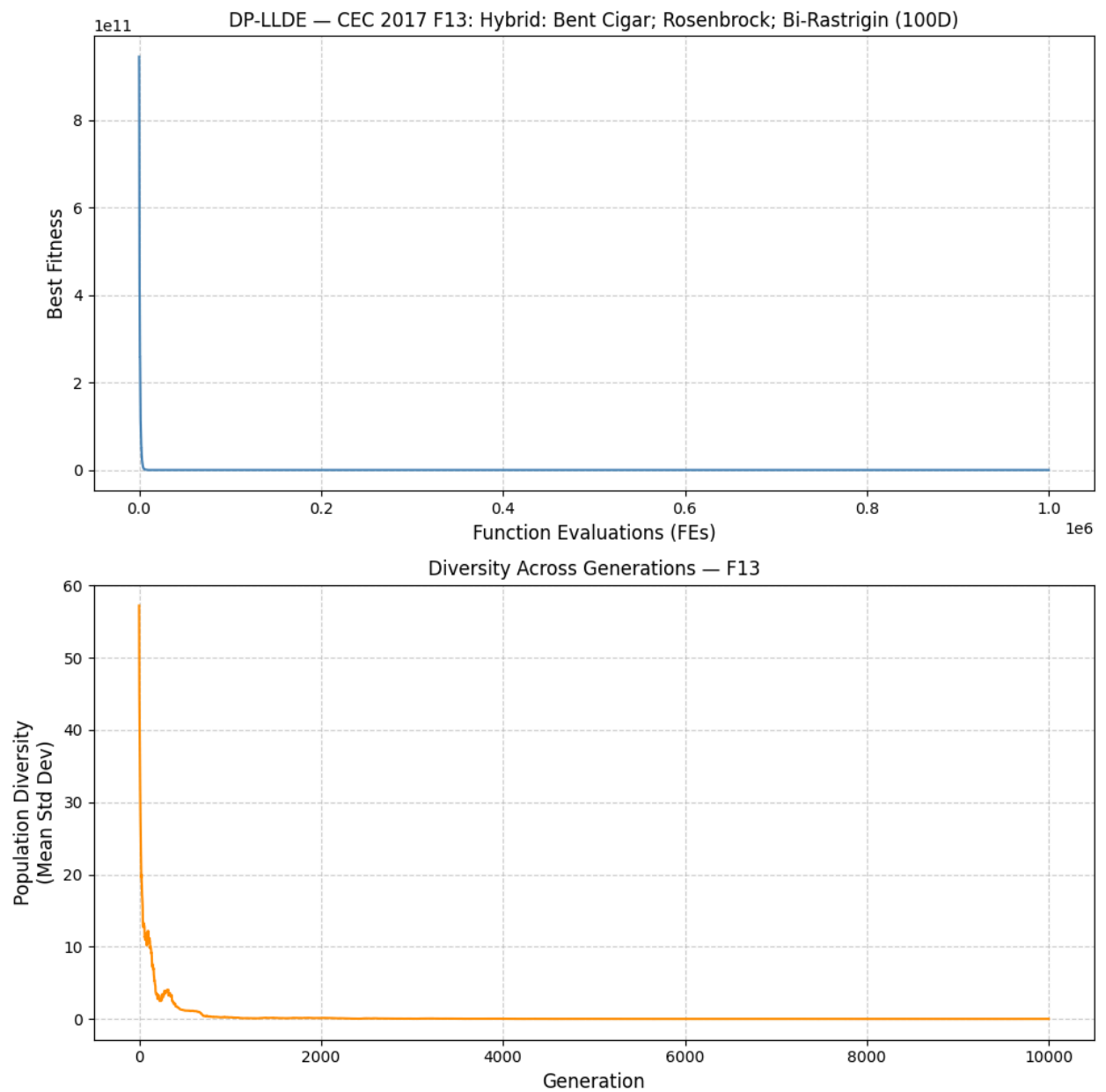
Best Solution (5 dim) : [-20.07291905 -69.62858139 -20.82606005 -12.48738395
17.42780205]

D^(0) (init. diversity): 5.7220e+01

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✓ Checkpoint saved to: /content/drive/My Drive/thesis_results/all_results_checkpoint.pkl



Graph saved as convergence_F13.png

✓ Plot saved to: /content/drive/My Drive/thesis_results/plots/convergence_F13.png

```
In [ ]: # =====
# F14 — Hybrid Function 4 (Elliptic; Ackley; Schaffer; Rastrigin)
# =====
run_and_plot('F14', make_wrapper(hybrid.f14), 1400.0, 'Hybrid: Elliptic; Ackle
```

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DP-LLDE on CEC 2017 F14 – Single Run

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Function : Hybrid: Elliptic; Ackley; Schaffer; Rastrigin (100D)

Bounds : [-100, 100]

Global optimum : 1400.0

Initial $D^*(0) = 5.7220e+01$ | Diversity threshold = 5.7220e+00

Generation 0: Best Fitness = 7.040180e+08

Generation 10:	Best = 5.710935e+07	FES = 1100	$\mu_F = 0.566$	$\mu_{CR} = 0.348$
Diversity = 3.9321e+01	$k_t = [1, 1, 1, 1]$			
Generation 20:	Best = 2.160261e+07	FES = 2100	$\mu_F = 0.611$	$\mu_{CR} = 0.322$
Diversity = 2.8972e+01	$k_t = [2, 3, 2, 3]$			
Generation 30:	Best = 1.663751e+07	FES = 3100	$\mu_F = 0.626$	$\mu_{CR} = 0.310$
Diversity = 2.2719e+01	$k_t = [1, 1, 1, 1]$			
Generation 40:	Best = 9.209763e+06	FES = 4100	$\mu_F = 0.655$	$\mu_{CR} = 0.225$
Diversity = 1.9384e+01	$k_t = [3, 3, 4, 2]$			
Generation 50:	Best = 6.254699e+06	FES = 5100	$\mu_F = 0.678$	$\mu_{CR} = 0.135$
Diversity = 1.6499e+01	$k_t = [1, 1, 1, 1]$			
Generation 60:	Best = 5.715815e+06	FES = 6100	$\mu_F = 0.696$	$\mu_{CR} = 0.089$
Diversity = 1.4774e+01	$k_t = [2, 2, 4, 4]$			
Generation 70:	Best = 4.582327e+06	FES = 7100	$\mu_F = 0.728$	$\mu_{CR} = 0.067$
Diversity = 1.2162e+01	$k_t = [1, 1, 1, 1]$			
Generation 80:	Best = 4.222586e+06	FES = 8100	$\mu_F = 0.748$	$\mu_{CR} = 0.057$
Diversity = 1.1862e+01	$k_t = [2, 4, 4, 2]$			
Generation 90:	Best = 3.610253e+06	FES = 9100	$\mu_F = 0.752$	$\mu_{CR} = 0.050$
Diversity = 1.1396e+01	$k_t = [1, 1, 1, 1]$			
Generation 100:	Best = 3.568825e+06	FES = 10100	$\mu_F = 0.766$	$\mu_{CR} = 0.050$
Diversity = 1.1702e+01	$k_t = [4, 4, 3, 2]$			
Generation 110:	Best = 3.336036e+06	FES = 11100	$\mu_F = 0.773$	$\mu_{CR} = 0.045$
Diversity = 1.0967e+01	$k_t = [1, 1, 1, 1]$			
Generation 120:	Best = 3.204007e+06	FES = 12100	$\mu_F = 0.780$	$\mu_{CR} = 0.047$
Diversity = 1.0199e+01	$k_t = [3, 4, 2, 4]$			
Generation 130:	Best = 3.094642e+06	FES = 13100	$\mu_F = 0.794$	$\mu_{CR} = 0.045$
Diversity = 1.0789e+01	$k_t = [1, 1, 1, 1]$			
Generation 140:	Best = 2.924934e+06	FES = 14100	$\mu_F = 0.794$	$\mu_{CR} = 0.048$
Diversity = 1.1719e+01	$k_t = [4, 2, 4, 4]$			
Generation 150:	Best = 2.823433e+06	FES = 15100	$\mu_F = 0.795$	$\mu_{CR} = 0.043$
Diversity = 1.0089e+01	$k_t = [1, 1, 1, 1]$			
Generation 160:	Best = 2.786699e+06	FES = 16100	$\mu_F = 0.800$	$\mu_{CR} = 0.056$
Diversity = 9.7889e+00	$k_t = [2, 4, 2, 3]$			
Generation 170:	Best = 2.668411e+06	FES = 17100	$\mu_F = 0.810$	$\mu_{CR} = 0.046$
Diversity = 1.0789e+01	$k_t = [1, 1, 1, 1]$			
Generation 180:	Best = 2.560197e+06	FES = 18100	$\mu_F = 0.808$	$\mu_{CR} = 0.047$
Diversity = 1.0979e+01	$k_t = [2, 4, 2, 3]$			
Generation 190:	Best = 2.510507e+06	FES = 19100	$\mu_F = 0.821$	$\mu_{CR} = 0.049$
Diversity = 1.1704e+01	$k_t = [1, 1, 1, 1]$			
Generation 200:	Best = 2.450323e+06	FES = 20100	$\mu_F = 0.817$	$\mu_{CR} = 0.044$
Diversity = 1.2021e+01	$k_t = [2, 4, 2, 2]$			
Generation 210:	Best = 2.379533e+06	FES = 21100	$\mu_F = 0.819$	$\mu_{CR} = 0.052$
Diversity = 1.2719e+01	$k_t = [1, 1, 1, 1]$			
Generation 220:	Best = 2.351942e+06	FES = 22100	$\mu_F = 0.814$	$\mu_{CR} = 0.046$
Diversity = 1.4707e+01	$k_t = [4, 4, 4, 2]$			

Generation 230: Best = 2.242014e+06	FES = 23100	$\mu_F = 0.819$	$\mu_{CR} = 0.048$
Diversity = 1.5188e+01	$k_t = [1, 1, 1, 1]$		
Generation 240: Best = 2.153859e+06	FES = 24100	$\mu_F = 0.829$	$\mu_{CR} = 0.049$
Diversity = 1.7338e+01	$k_t = [2, 3, 2, 3]$		
Generation 250: Best = 2.032841e+06	FES = 25100	$\mu_F = 0.830$	$\mu_{CR} = 0.047$
Diversity = 2.0839e+01	$k_t = [1, 1, 1, 1]$		
Generation 260: Best = 1.890699e+06	FES = 26100	$\mu_F = 0.830$	$\mu_{CR} = 0.046$
Diversity = 2.2884e+01	$k_t = [3, 2, 2, 2]$		
Generation 270: Best = 1.799970e+06	FES = 27100	$\mu_F = 0.833$	$\mu_{CR} = 0.044$
Diversity = 2.2311e+01	$k_t = [1, 1, 1, 1]$		
Generation 280: Best = 1.743340e+06	FES = 28100	$\mu_F = 0.828$	$\mu_{CR} = 0.044$
Diversity = 2.0512e+01	$k_t = [3, 3, 2, 4]$		
Generation 290: Best = 1.727581e+06	FES = 29100	$\mu_F = 0.825$	$\mu_{CR} = 0.049$
Diversity = 1.9973e+01	$k_t = [1, 1, 1, 1]$		
Generation 300: Best = 1.620905e+06	FES = 30100	$\mu_F = 0.831$	$\mu_{CR} = 0.052$
Diversity = 2.0649e+01	$k_t = [3, 4, 4, 2]$		
Generation 310: Best = 1.530633e+06	FES = 31100	$\mu_F = 0.821$	$\mu_{CR} = 0.059$
Diversity = 2.1844e+01	$k_t = [1, 1, 1, 1]$		
Generation 320: Best = 1.506749e+06	FES = 32100	$\mu_F = 0.814$	$\mu_{CR} = 0.051$
Diversity = 2.1311e+01	$k_t = [3, 3, 3, 2]$		
Generation 330: Best = 1.487684e+06	FES = 33100	$\mu_F = 0.816$	$\mu_{CR} = 0.053$
Diversity = 2.1914e+01	$k_t = [1, 1, 1, 1]$		
Generation 340: Best = 1.395527e+06	FES = 34100	$\mu_F = 0.813$	$\mu_{CR} = 0.046$
Diversity = 2.2852e+01	$k_t = [3, 4, 4, 4]$		
Generation 350: Best = 1.257817e+06	FES = 35100	$\mu_F = 0.824$	$\mu_{CR} = 0.052$
Diversity = 2.1824e+01	$k_t = [1, 1, 1, 1]$		
Generation 360: Best = 1.167380e+06	FES = 36100	$\mu_F = 0.821$	$\mu_{CR} = 0.053$
Diversity = 2.0868e+01	$k_t = [3, 4, 3, 4]$		
Generation 370: Best = 1.123911e+06	FES = 37100	$\mu_F = 0.823$	$\mu_{CR} = 0.052$
Diversity = 2.1504e+01	$k_t = [1, 1, 1, 1]$		
Generation 380: Best = 1.089681e+06	FES = 38100	$\mu_F = 0.821$	$\mu_{CR} = 0.053$
Diversity = 2.1852e+01	$k_t = [4, 2, 4, 4]$		
Generation 390: Best = 1.035429e+06	FES = 39100	$\mu_F = 0.831$	$\mu_{CR} = 0.056$
Diversity = 2.1865e+01	$k_t = [1, 1, 1, 1]$		
Generation 400: Best = 9.872019e+05	FES = 40100	$\mu_F = 0.818$	$\mu_{CR} = 0.052$
Diversity = 2.1742e+01	$k_t = [4, 4, 2, 2]$		
Generation 410: Best = 9.660168e+05	FES = 41100	$\mu_F = 0.818$	$\mu_{CR} = 0.050$
Diversity = 1.9160e+01	$k_t = [1, 1, 1, 1]$		
Generation 420: Best = 9.488995e+05	FES = 42100	$\mu_F = 0.808$	$\mu_{CR} = 0.049$
Diversity = 1.6334e+01	$k_t = [3, 3, 2, 3]$		
Generation 430: Best = 9.011110e+05	FES = 43100	$\mu_F = 0.792$	$\mu_{CR} = 0.050$
Diversity = 1.5736e+01	$k_t = [1, 1, 1, 1]$		
Generation 440: Best = 8.610972e+05	FES = 44100	$\mu_F = 0.793$	$\mu_{CR} = 0.048$
Diversity = 1.4911e+01	$k_t = [2, 2, 3, 3]$		
Generation 450: Best = 8.387656e+05	FES = 45100	$\mu_F = 0.789$	$\mu_{CR} = 0.052$
Diversity = 1.3460e+01	$k_t = [1, 1, 1, 1]$		
Generation 460: Best = 8.313677e+05	FES = 46100	$\mu_F = 0.799$	$\mu_{CR} = 0.046$
Diversity = 1.3917e+01	$k_t = [2, 2, 3, 2]$		
Generation 470: Best = 8.242241e+05	FES = 47100	$\mu_F = 0.799$	$\mu_{CR} = 0.045$
Diversity = 1.2990e+01	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 8.101224e+05	FES = 48100	$\mu_F = 0.813$	$\mu_{CR} = 0.053$
Diversity = 1.2632e+01	$k_t = [4, 3, 2, 4]$		
Generation 490: Best = 7.814307e+05	FES = 49100	$\mu_F = 0.817$	$\mu_{CR} = 0.054$
Diversity = 1.1732e+01	$k_t = [1, 1, 1, 1]$		

Generation 500: Best = 7.775769e+05	FES = 50100	$\mu_F = 0.824$	$\mu_{CR} = 0.047$
Diversity = 1.2073e+01	k_t = [4, 3, 3, 4]		
Generation 510: Best = 7.709497e+05	FES = 51100	$\mu_F = 0.834$	$\mu_{CR} = 0.047$
Diversity = 1.1913e+01	k_t = [1, 1, 1, 1]		
Generation 520: Best = 7.630894e+05	FES = 52100	$\mu_F = 0.829$	$\mu_{CR} = 0.046$
Diversity = 1.2504e+01	k_t = [2, 4, 4, 3]		
Generation 530: Best = 7.513296e+05	FES = 53100	$\mu_F = 0.843$	$\mu_{CR} = 0.041$
Diversity = 1.2283e+01	k_t = [1, 1, 1, 1]		
Generation 540: Best = 7.470572e+05	FES = 54100	$\mu_F = 0.843$	$\mu_{CR} = 0.047$
Diversity = 1.2702e+01	k_t = [2, 3, 4, 4]		
Generation 550: Best = 7.432457e+05	FES = 55100	$\mu_F = 0.840$	$\mu_{CR} = 0.051$
Diversity = 1.2110e+01	k_t = [1, 1, 1, 1]		
Generation 560: Best = 7.371103e+05	FES = 56100	$\mu_F = 0.820$	$\mu_{CR} = 0.056$
Diversity = 1.2232e+01	k_t = [2, 2, 2, 2]		
Generation 570: Best = 7.306423e+05	FES = 57100	$\mu_F = 0.822$	$\mu_{CR} = 0.053$
Diversity = 1.2867e+01	k_t = [1, 1, 1, 1]		
Generation 580: Best = 7.170076e+05	FES = 58100	$\mu_F = 0.839$	$\mu_{CR} = 0.047$
Diversity = 1.3649e+01	k_t = [2, 2, 2, 4]		
Generation 590: Best = 7.075009e+05	FES = 59100	$\mu_F = 0.839$	$\mu_{CR} = 0.042$
Diversity = 1.2825e+01	k_t = [1, 1, 1, 1]		
Generation 600: Best = 7.013343e+05	FES = 60100	$\mu_F = 0.821$	$\mu_{CR} = 0.046$
Diversity = 1.3142e+01	k_t = [3, 3, 3, 2]		
Generation 610: Best = 6.952712e+05	FES = 61100	$\mu_F = 0.819$	$\mu_{CR} = 0.043$
Diversity = 1.3771e+01	k_t = [1, 1, 1, 1]		
Generation 620: Best = 6.930604e+05	FES = 62100	$\mu_F = 0.822$	$\mu_{CR} = 0.046$
Diversity = 1.2732e+01	k_t = [4, 3, 4, 4]		
Generation 630: Best = 6.610982e+05	FES = 63100	$\mu_F = 0.814$	$\mu_{CR} = 0.048$
Diversity = 1.1943e+01	k_t = [1, 1, 1, 1]		
Generation 640: Best = 6.487261e+05	FES = 64100	$\mu_F = 0.814$	$\mu_{CR} = 0.050$
Diversity = 1.2943e+01	k_t = [2, 4, 3, 3]		
Generation 650: Best = 6.248362e+05	FES = 65100	$\mu_F = 0.827$	$\mu_{CR} = 0.051$
Diversity = 1.3264e+01	k_t = [1, 1, 1, 1]		
Generation 660: Best = 6.146284e+05	FES = 66100	$\mu_F = 0.830$	$\mu_{CR} = 0.051$
Diversity = 1.3889e+01	k_t = [3, 3, 2, 2]		
Generation 670: Best = 5.968978e+05	FES = 67100	$\mu_F = 0.826$	$\mu_{CR} = 0.055$
Diversity = 1.5424e+01	k_t = [1, 1, 1, 1]		
Generation 680: Best = 5.944251e+05	FES = 68100	$\mu_F = 0.828$	$\mu_{CR} = 0.053$
Diversity = 1.3868e+01	k_t = [3, 4, 4, 3]		
Generation 690: Best = 5.855281e+05	FES = 69100	$\mu_F = 0.824$	$\mu_{CR} = 0.051$
Diversity = 1.4541e+01	k_t = [1, 1, 1, 1]		
Generation 700: Best = 5.774975e+05	FES = 70100	$\mu_F = 0.820$	$\mu_{CR} = 0.056$
Diversity = 1.4456e+01	k_t = [4, 4, 4, 4]		
Generation 710: Best = 5.654889e+05	FES = 71100	$\mu_F = 0.818$	$\mu_{CR} = 0.052$
Diversity = 1.4103e+01	k_t = [1, 1, 1, 1]		
Generation 720: Best = 5.608630e+05	FES = 72100	$\mu_F = 0.820$	$\mu_{CR} = 0.047$
Diversity = 1.3495e+01	k_t = [2, 4, 3, 3]		
Generation 730: Best = 5.554989e+05	FES = 73100	$\mu_F = 0.825$	$\mu_{CR} = 0.042$
Diversity = 1.3524e+01	k_t = [1, 1, 1, 1]		
Generation 740: Best = 5.454573e+05	FES = 74100	$\mu_F = 0.823$	$\mu_{CR} = 0.049$
Diversity = 1.1739e+01	k_t = [3, 4, 4, 3]		
Generation 750: Best = 5.427441e+05	FES = 75100	$\mu_F = 0.823$	$\mu_{CR} = 0.050$
Diversity = 1.1762e+01	k_t = [1, 1, 1, 1]		
Generation 760: Best = 5.331586e+05	FES = 76100	$\mu_F = 0.821$	$\mu_{CR} = 0.052$
Diversity = 1.3255e+01	k_t = [2, 4, 2, 3]		

Generation 770:	Best = 5.302118e+05	FES = 77100	$\mu_F = 0.824$	$\mu_{CR} = 0.053$
Diversity = 1.3666e+01	$k_t = [1, 1, 1, 1]$			
Generation 780:	Best = 5.292506e+05	FES = 78100	$\mu_F = 0.828$	$\mu_{CR} = 0.052$
Diversity = 1.2973e+01	$k_t = [3, 4, 2, 4]$			
Generation 790:	Best = 5.271148e+05	FES = 79100	$\mu_F = 0.813$	$\mu_{CR} = 0.054$
Diversity = 1.1081e+01	$k_t = [1, 1, 1, 1]$			
Generation 800:	Best = 5.238879e+05	FES = 80100	$\mu_F = 0.814$	$\mu_{CR} = 0.055$
Diversity = 1.0735e+01	$k_t = [4, 4, 3, 3]$			
Generation 810:	Best = 5.174419e+05	FES = 81100	$\mu_F = 0.805$	$\mu_{CR} = 0.048$
Diversity = 1.1561e+01	$k_t = [1, 1, 1, 1]$			
Generation 820:	Best = 5.153267e+05	FES = 82100	$\mu_F = 0.820$	$\mu_{CR} = 0.050$
Diversity = 1.1265e+01	$k_t = [2, 3, 3, 4]$			
Generation 830:	Best = 5.093657e+05	FES = 83100	$\mu_F = 0.824$	$\mu_{CR} = 0.056$
Diversity = 1.1665e+01	$k_t = [1, 1, 1, 1]$			
Generation 840:	Best = 5.014773e+05	FES = 84100	$\mu_F = 0.822$	$\mu_{CR} = 0.064$
Diversity = 1.2822e+01	$k_t = [3, 4, 3, 4]$			
Generation 850:	Best = 4.914436e+05	FES = 85100	$\mu_F = 0.840$	$\mu_{CR} = 0.060$
Diversity = 1.2722e+01	$k_t = [1, 1, 1, 1]$			
Generation 860:	Best = 4.871722e+05	FES = 86100	$\mu_F = 0.842$	$\mu_{CR} = 0.051$
Diversity = 1.4098e+01	$k_t = [4, 2, 3, 2]$			
Generation 870:	Best = 4.840595e+05	FES = 87100	$\mu_F = 0.831$	$\mu_{CR} = 0.047$
Diversity = 1.4308e+01	$k_t = [1, 1, 1, 1]$			
Generation 880:	Best = 4.798372e+05	FES = 88100	$\mu_F = 0.828$	$\mu_{CR} = 0.054$
Diversity = 1.3274e+01	$k_t = [3, 4, 4, 3]$			
Generation 890:	Best = 4.775409e+05	FES = 89100	$\mu_F = 0.827$	$\mu_{CR} = 0.056$
Diversity = 1.3713e+01	$k_t = [1, 1, 1, 1]$			
Generation 900:	Best = 4.746616e+05	FES = 90100	$\mu_F = 0.820$	$\mu_{CR} = 0.052$
Diversity = 1.3035e+01	$k_t = [3, 4, 4, 2]$			
Generation 910:	Best = 4.728515e+05	FES = 91100	$\mu_F = 0.808$	$\mu_{CR} = 0.050$
Diversity = 1.3097e+01	$k_t = [1, 1, 1, 1]$			
Generation 920:	Best = 4.691366e+05	FES = 92100	$\mu_F = 0.811$	$\mu_{CR} = 0.044$
Diversity = 1.2141e+01	$k_t = [2, 2, 4, 3]$			
Generation 930:	Best = 4.659440e+05	FES = 93100	$\mu_F = 0.814$	$\mu_{CR} = 0.047$
Diversity = 1.2471e+01	$k_t = [1, 1, 1, 1]$			
Generation 940:	Best = 4.587274e+05	FES = 94100	$\mu_F = 0.823$	$\mu_{CR} = 0.043$
Diversity = 1.3561e+01	$k_t = [2, 3, 3, 4]$			
Generation 950:	Best = 4.567658e+05	FES = 95100	$\mu_F = 0.834$	$\mu_{CR} = 0.052$
Diversity = 1.1081e+01	$k_t = [1, 1, 1, 1]$			
Generation 960:	Best = 4.539577e+05	FES = 96100	$\mu_F = 0.837$	$\mu_{CR} = 0.054$
Diversity = 1.1564e+01	$k_t = [2, 2, 2, 4]$			
Generation 970:	Best = 4.518993e+05	FES = 97100	$\mu_F = 0.811$	$\mu_{CR} = 0.057$
Diversity = 1.1318e+01	$k_t = [1, 1, 1, 1]$			
Generation 980:	Best = 4.477706e+05	FES = 98100	$\mu_F = 0.811$	$\mu_{CR} = 0.061$
Diversity = 1.1179e+01	$k_t = [3, 3, 3, 4]$			
Generation 990:	Best = 4.428590e+05	FES = 99100	$\mu_F = 0.823$	$\mu_{CR} = 0.060$
Diversity = 1.1259e+01	$k_t = [1, 1, 1, 1]$			
Generation 1000:	Best = 4.402904e+05	FES = 100100	$\mu_F = 0.822$	$\mu_{CR} = 0.061$
Diversity = 1.1607e+01	$k_t = [3, 2, 3, 3]$			
Generation 1010:	Best = 4.376645e+05	FES = 101100	$\mu_F = 0.831$	$\mu_{CR} = 0.058$
Diversity = 1.1861e+01	$k_t = [1, 1, 1, 1]$			
Generation 1020:	Best = 4.358780e+05	FES = 102100	$\mu_F = 0.818$	$\mu_{CR} = 0.054$
Diversity = 1.1687e+01	$k_t = [4, 2, 2, 2]$			
Generation 1030:	Best = 4.305144e+05	FES = 103100	$\mu_F = 0.820$	$\mu_{CR} = 0.056$
Diversity = 1.1673e+01	$k_t = [1, 1, 1, 1]$			

Generation 1040: Best = 4.283511e+05 FES = 104100 μ_F = 0.819 μ_{CR} = 0.063
 Diversity = 1.1205e+01 k_t = [2, 4, 3, 3]
 Generation 1050: Best = 4.243300e+05 FES = 105100 μ_F = 0.814 μ_{CR} = 0.066
 Diversity = 1.1559e+01 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 4.224242e+05 FES = 106100 μ_F = 0.827 μ_{CR} = 0.056
 Diversity = 1.1151e+01 k_t = [3, 4, 3, 3]
 Generation 1070: Best = 4.192451e+05 FES = 107100 μ_F = 0.822 μ_{CR} = 0.052
 Diversity = 1.1444e+01 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 4.174181e+05 FES = 108100 μ_F = 0.819 μ_{CR} = 0.045
 Diversity = 1.1507e+01 k_t = [2, 4, 2, 4]
 Generation 1090: Best = 4.138731e+05 FES = 109100 μ_F = 0.814 μ_{CR} = 0.049
 Diversity = 1.1627e+01 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 4.120506e+05 FES = 110100 μ_F = 0.817 μ_{CR} = 0.050
 Diversity = 1.0926e+01 k_t = [3, 2, 3, 3]
 Generation 1110: Best = 4.085186e+05 FES = 111100 μ_F = 0.827 μ_{CR} = 0.042
 Diversity = 1.1411e+01 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 4.058918e+05 FES = 112100 μ_F = 0.831 μ_{CR} = 0.046
 Diversity = 1.1126e+01 k_t = [3, 4, 4, 2]
 Generation 1130: Best = 4.037429e+05 FES = 113100 μ_F = 0.830 μ_{CR} = 0.051
 Diversity = 1.0762e+01 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 4.019273e+05 FES = 114100 μ_F = 0.829 μ_{CR} = 0.047
 Diversity = 1.0992e+01 k_t = [2, 2, 3, 3]
 Generation 1150: Best = 3.997328e+05 FES = 115100 μ_F = 0.829 μ_{CR} = 0.048
 Diversity = 1.0427e+01 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 3.980266e+05 FES = 116100 μ_F = 0.829 μ_{CR} = 0.044
 Diversity = 1.1338e+01 k_t = [3, 2, 3, 2]
 Generation 1170: Best = 3.962537e+05 FES = 117100 μ_F = 0.837 μ_{CR} = 0.049
 Diversity = 1.1622e+01 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 3.933506e+05 FES = 118100 μ_F = 0.829 μ_{CR} = 0.049
 Diversity = 1.1729e+01 k_t = [3, 2, 4, 4]
 Generation 1190: Best = 3.896215e+05 FES = 119100 μ_F = 0.824 μ_{CR} = 0.049
 Diversity = 1.1641e+01 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 3.859900e+05 FES = 120100 μ_F = 0.826 μ_{CR} = 0.047
 Diversity = 1.3490e+01 k_t = [4, 4, 3, 3]
 Generation 1210: Best = 3.851692e+05 FES = 121100 μ_F = 0.831 μ_{CR} = 0.052
 Diversity = 1.2258e+01 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 3.847959e+05 FES = 122100 μ_F = 0.822 μ_{CR} = 0.053
 Diversity = 1.1548e+01 k_t = [4, 4, 4, 2]
 Generation 1230: Best = 3.817397e+05 FES = 123100 μ_F = 0.820 μ_{CR} = 0.055
 Diversity = 1.2132e+01 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 3.806045e+05 FES = 124100 μ_F = 0.827 μ_{CR} = 0.047
 Diversity = 1.1629e+01 k_t = [4, 3, 3, 4]
 Generation 1250: Best = 3.788004e+05 FES = 125100 μ_F = 0.819 μ_{CR} = 0.046
 Diversity = 1.0851e+01 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 3.752040e+05 FES = 126100 μ_F = 0.817 μ_{CR} = 0.044
 Diversity = 1.1092e+01 k_t = [3, 4, 4, 4]
 Generation 1270: Best = 3.748413e+05 FES = 127100 μ_F = 0.826 μ_{CR} = 0.050
 Diversity = 1.0069e+01 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 3.729423e+05 FES = 128100 μ_F = 0.821 μ_{CR} = 0.052
 Diversity = 1.0079e+01 k_t = [4, 4, 3, 3]
 Generation 1290: Best = 3.714757e+05 FES = 129100 μ_F = 0.828 μ_{CR} = 0.056
 Diversity = 1.0513e+01 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 3.688999e+05 FES = 130100 μ_F = 0.823 μ_{CR} = 0.058
 Diversity = 1.0618e+01 k_t = [2, 3, 2, 2]

Generation 1310: Best = 3.668069e+05 FES = 131100 μ_F = 0.836 μ_{CR} = 0.050
 Diversity = 1.1249e+01 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 3.646108e+05 FES = 132100 μ_F = 0.822 μ_{CR} = 0.059
 Diversity = 9.5831e+00 k_t = [4, 4, 3, 4]
 Generation 1330: Best = 3.632275e+05 FES = 133100 μ_F = 0.821 μ_{CR} = 0.052
 Diversity = 9.3068e+00 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 3.617151e+05 FES = 134100 μ_F = 0.819 μ_{CR} = 0.048
 Diversity = 9.4835e+00 k_t = [4, 4, 2, 3]
 Generation 1350: Best = 3.593951e+05 FES = 135100 μ_F = 0.815 μ_{CR} = 0.048
 Diversity = 9.4505e+00 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 3.585032e+05 FES = 136100 μ_F = 0.804 μ_{CR} = 0.045
 Diversity = 1.0018e+01 k_t = [4, 2, 4, 2]
 Generation 1370: Best = 3.566807e+05 FES = 137100 μ_F = 0.794 μ_{CR} = 0.055
 Diversity = 9.6725e+00 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 3.560963e+05 FES = 138100 μ_F = 0.803 μ_{CR} = 0.049
 Diversity = 1.0476e+01 k_t = [3, 3, 3, 4]
 Generation 1390: Best = 3.558361e+05 FES = 139100 μ_F = 0.795 μ_{CR} = 0.040
 Diversity = 8.8941e+00 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 3.548479e+05 FES = 140100 μ_F = 0.805 μ_{CR} = 0.041
 Diversity = 8.5679e+00 k_t = [2, 3, 3, 4]
 Generation 1410: Best = 3.537238e+05 FES = 141100 μ_F = 0.817 μ_{CR} = 0.044
 Diversity = 8.2593e+00 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 3.521976e+05 FES = 142100 μ_F = 0.832 μ_{CR} = 0.045
 Diversity = 8.1275e+00 k_t = [4, 4, 3, 3]
 Generation 1430: Best = 3.501608e+05 FES = 143100 μ_F = 0.842 μ_{CR} = 0.051
 Diversity = 8.8634e+00 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 3.471727e+05 FES = 144100 μ_F = 0.843 μ_{CR} = 0.051
 Diversity = 8.9882e+00 k_t = [4, 4, 2, 4]
 Generation 1450: Best = 3.452114e+05 FES = 145100 μ_F = 0.837 μ_{CR} = 0.056
 Diversity = 8.7054e+00 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 3.448863e+05 FES = 146100 μ_F = 0.842 μ_{CR} = 0.053
 Diversity = 9.1156e+00 k_t = [4, 2, 2, 4]
 Generation 1470: Best = 3.430490e+05 FES = 147100 μ_F = 0.843 μ_{CR} = 0.045
 Diversity = 9.3977e+00 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 3.420969e+05 FES = 148100 μ_F = 0.827 μ_{CR} = 0.041
 Diversity = 9.0926e+00 k_t = [3, 2, 4, 3]
 Generation 1490: Best = 3.417312e+05 FES = 149100 μ_F = 0.822 μ_{CR} = 0.047
 Diversity = 9.2090e+00 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 3.378610e+05 FES = 150100 μ_F = 0.827 μ_{CR} = 0.051
 Diversity = 9.7414e+00 k_t = [3, 3, 2, 4]
 Generation 1510: Best = 3.375696e+05 FES = 151100 μ_F = 0.837 μ_{CR} = 0.057
 Diversity = 9.9897e+00 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 3.365304e+05 FES = 152100 μ_F = 0.841 μ_{CR} = 0.052
 Diversity = 8.6620e+00 k_t = [2, 3, 3, 3]
 Generation 1530: Best = 3.356637e+05 FES = 153100 μ_F = 0.821 μ_{CR} = 0.053
 Diversity = 9.0648e+00 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 3.331032e+05 FES = 154100 μ_F = 0.829 μ_{CR} = 0.052
 Diversity = 8.7280e+00 k_t = [3, 2, 2, 4]
 Generation 1550: Best = 3.325302e+05 FES = 155100 μ_F = 0.833 μ_{CR} = 0.057
 Diversity = 8.4238e+00 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 3.308761e+05 FES = 156100 μ_F = 0.833 μ_{CR} = 0.051
 Diversity = 8.4310e+00 k_t = [2, 3, 4, 3]
 Generation 1570: Best = 3.300959e+05 FES = 157100 μ_F = 0.821 μ_{CR} = 0.045
 Diversity = 8.6626e+00 k_t = [1, 1, 1, 1]

Generation 1580: Best = 3.284703e+05 FES = 158100 μ_F = 0.820 μ_{CR} = 0.049
 Diversity = 8.5206e+00 k_t = [3, 3, 2, 3]
 Generation 1590: Best = 3.262068e+05 FES = 159100 μ_F = 0.823 μ_{CR} = 0.051
 Diversity = 1.0355e+01 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 3.246521e+05 FES = 160100 μ_F = 0.820 μ_{CR} = 0.050
 Diversity = 9.8821e+00 k_t = [4, 2, 4, 2]
 Generation 1610: Best = 3.244648e+05 FES = 161100 μ_F = 0.811 μ_{CR} = 0.055
 Diversity = 9.9714e+00 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 3.237066e+05 FES = 162100 μ_F = 0.811 μ_{CR} = 0.048
 Diversity = 9.7797e+00 k_t = [2, 3, 4, 4]
 Generation 1630: Best = 3.231938e+05 FES = 163100 μ_F = 0.831 μ_{CR} = 0.054
 Diversity = 9.0268e+00 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 3.228944e+05 FES = 164100 μ_F = 0.807 μ_{CR} = 0.051
 Diversity = 8.6541e+00 k_t = [4, 2, 4, 4]
 Generation 1650: Best = 3.224365e+05 FES = 165100 μ_F = 0.819 μ_{CR} = 0.051
 Diversity = 8.8434e+00 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 3.215109e+05 FES = 166100 μ_F = 0.817 μ_{CR} = 0.047
 Diversity = 8.8572e+00 k_t = [2, 3, 2, 3]
 Generation 1670: Best = 3.213524e+05 FES = 167100 μ_F = 0.815 μ_{CR} = 0.050
 Diversity = 9.1862e+00 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 3.204530e+05 FES = 168100 μ_F = 0.816 μ_{CR} = 0.051
 Diversity = 9.4787e+00 k_t = [2, 2, 2, 2]
 Generation 1690: Best = 3.195799e+05 FES = 169100 μ_F = 0.823 μ_{CR} = 0.049
 Diversity = 1.0741e+01 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 3.193420e+05 FES = 170100 μ_F = 0.808 μ_{CR} = 0.056
 Diversity = 1.0503e+01 k_t = [2, 4, 3, 3]
 Generation 1710: Best = 3.185221e+05 FES = 171100 μ_F = 0.814 μ_{CR} = 0.053
 Diversity = 9.5371e+00 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 3.180251e+05 FES = 172100 μ_F = 0.818 μ_{CR} = 0.051
 Diversity = 9.8092e+00 k_t = [4, 4, 2, 3]
 Generation 1730: Best = 3.172928e+05 FES = 173100 μ_F = 0.816 μ_{CR} = 0.060
 Diversity = 1.0171e+01 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 3.169539e+05 FES = 174100 μ_F = 0.817 μ_{CR} = 0.054
 Diversity = 1.1529e+01 k_t = [2, 3, 3, 2]
 Generation 1750: Best = 3.163687e+05 FES = 175100 μ_F = 0.811 μ_{CR} = 0.048
 Diversity = 1.0882e+01 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 3.156149e+05 FES = 176100 μ_F = 0.808 μ_{CR} = 0.059
 Diversity = 1.1420e+01 k_t = [3, 2, 4, 2]
 Generation 1770: Best = 3.153244e+05 FES = 177100 μ_F = 0.803 μ_{CR} = 0.054
 Diversity = 1.1231e+01 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 3.145183e+05 FES = 178100 μ_F = 0.812 μ_{CR} = 0.058
 Diversity = 1.0744e+01 k_t = [2, 2, 2, 3]
 Generation 1790: Best = 3.143278e+05 FES = 179100 μ_F = 0.814 μ_{CR} = 0.053
 Diversity = 1.1502e+01 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 3.139294e+05 FES = 180100 μ_F = 0.818 μ_{CR} = 0.049
 Diversity = 1.2176e+01 k_t = [4, 3, 4, 2]
 Generation 1810: Best = 3.134205e+05 FES = 181100 μ_F = 0.820 μ_{CR} = 0.049
 Diversity = 1.1941e+01 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 3.128621e+05 FES = 182100 μ_F = 0.830 μ_{CR} = 0.052
 Diversity = 1.1915e+01 k_t = [2, 4, 3, 4]
 Generation 1830: Best = 3.124842e+05 FES = 183100 μ_F = 0.824 μ_{CR} = 0.045
 Diversity = 1.1814e+01 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 3.120040e+05 FES = 184100 μ_F = 0.820 μ_{CR} = 0.046
 Diversity = 1.1130e+01 k_t = [4, 2, 2, 3]

Generation 1850: Best = 3.116271e+05	FES = 185100	$\mu_F = 0.812$	$\mu_{CR} = 0.055$
Diversity = 9.6581e+00	k_t = [1, 1, 1, 1]		
Generation 1860: Best = 3.110845e+05	FES = 186100	$\mu_F = 0.810$	$\mu_{CR} = 0.045$
Diversity = 1.0399e+01	k_t = [2, 4, 2, 4]		
Generation 1870: Best = 3.108454e+05	FES = 187100	$\mu_F = 0.797$	$\mu_{CR} = 0.045$
Diversity = 1.0681e+01	k_t = [1, 1, 1, 1]		
Generation 1880: Best = 3.103899e+05	FES = 188100	$\mu_F = 0.809$	$\mu_{CR} = 0.050$
Diversity = 1.0385e+01	k_t = [3, 4, 2, 2]		
Generation 1890: Best = 3.097035e+05	FES = 189100	$\mu_F = 0.807$	$\mu_{CR} = 0.054$
Diversity = 1.0658e+01	k_t = [1, 1, 1, 1]		
Generation 1900: Best = 3.095387e+05	FES = 190100	$\mu_F = 0.795$	$\mu_{CR} = 0.047$
Diversity = 9.3919e+00	k_t = [2, 4, 2, 3]		
Generation 1910: Best = 3.089162e+05	FES = 191100	$\mu_F = 0.808$	$\mu_{CR} = 0.050$
Diversity = 9.8436e+00	k_t = [1, 1, 1, 1]		
Generation 1920: Best = 3.084888e+05	FES = 192100	$\mu_F = 0.813$	$\mu_{CR} = 0.047$
Diversity = 1.1024e+01	k_t = [3, 2, 4, 2]		
Generation 1930: Best = 3.082243e+05	FES = 193100	$\mu_F = 0.807$	$\mu_{CR} = 0.043$
Diversity = 1.0986e+01	k_t = [1, 1, 1, 1]		
Generation 1940: Best = 3.079421e+05	FES = 194100	$\mu_F = 0.790$	$\mu_{CR} = 0.044$
Diversity = 1.0426e+01	k_t = [2, 2, 4, 2]		
Generation 1950: Best = 3.078488e+05	FES = 195100	$\mu_F = 0.792$	$\mu_{CR} = 0.044$
Diversity = 1.0234e+01	k_t = [1, 1, 1, 1]		
Generation 1960: Best = 3.075564e+05	FES = 196100	$\mu_F = 0.808$	$\mu_{CR} = 0.037$
Diversity = 9.9947e+00	k_t = [4, 4, 2, 3]		
Generation 1970: Best = 3.073946e+05	FES = 197100	$\mu_F = 0.805$	$\mu_{CR} = 0.042$
Diversity = 9.9081e+00	k_t = [1, 1, 1, 1]		
Generation 1980: Best = 3.070744e+05	FES = 198100	$\mu_F = 0.814$	$\mu_{CR} = 0.045$
Diversity = 9.6544e+00	k_t = [3, 4, 3, 4]		
Generation 1990: Best = 3.064995e+05	FES = 199100	$\mu_F = 0.818$	$\mu_{CR} = 0.043$
Diversity = 9.3217e+00	k_t = [1, 1, 1, 1]		
Generation 2000: Best = 3.060381e+05	FES = 200100	$\mu_F = 0.807$	$\mu_{CR} = 0.042$
Diversity = 8.7655e+00	k_t = [4, 4, 3, 4]		
Generation 2010: Best = 3.056427e+05	FES = 201100	$\mu_F = 0.815$	$\mu_{CR} = 0.047$
Diversity = 8.5946e+00	k_t = [1, 1, 1, 1]		
Generation 2020: Best = 3.049403e+05	FES = 202100	$\mu_F = 0.827$	$\mu_{CR} = 0.050$
Diversity = 8.4431e+00	k_t = [4, 4, 2, 4]		
Generation 2030: Best = 3.036568e+05	FES = 203100	$\mu_F = 0.850$	$\mu_{CR} = 0.049$
Diversity = 8.6881e+00	k_t = [1, 1, 1, 1]		
Generation 2040: Best = 3.031603e+05	FES = 204100	$\mu_F = 0.852$	$\mu_{CR} = 0.051$
Diversity = 8.4493e+00	k_t = [3, 2, 3, 3]		
Generation 2050: Best = 3.024895e+05	FES = 205100	$\mu_F = 0.851$	$\mu_{CR} = 0.048$
Diversity = 9.3679e+00	k_t = [1, 1, 1, 1]		
Generation 2060: Best = 3.018281e+05	FES = 206100	$\mu_F = 0.838$	$\mu_{CR} = 0.051$
Diversity = 9.5136e+00	k_t = [3, 2, 3, 2]		
Generation 2070: Best = 3.016310e+05	FES = 207100	$\mu_F = 0.833$	$\mu_{CR} = 0.050$
Diversity = 1.0354e+01	k_t = [1, 1, 1, 1]		
Generation 2080: Best = 3.008971e+05	FES = 208100	$\mu_F = 0.824$	$\mu_{CR} = 0.047$
Diversity = 9.7975e+00	k_t = [3, 4, 4, 2]		
Generation 2090: Best = 3.005057e+05	FES = 209100	$\mu_F = 0.826$	$\mu_{CR} = 0.049$
Diversity = 9.5512e+00	k_t = [1, 1, 1, 1]		
Generation 2100: Best = 3.000924e+05	FES = 210100	$\mu_F = 0.827$	$\mu_{CR} = 0.063$
Diversity = 9.6325e+00	k_t = [3, 3, 4, 2]		
Generation 2110: Best = 2.995324e+05	FES = 211100	$\mu_F = 0.818$	$\mu_{CR} = 0.063$
Diversity = 9.2425e+00	k_t = [1, 1, 1, 1]		

Generation 2120: Best = 2.989941e+05 FES = 212100 μ_F = 0.825 μ_{CR} = 0.061
 Diversity = 9.9627e+00 k_t = [4, 4, 2, 4]
 Generation 2130: Best = 2.986048e+05 FES = 213100 μ_F = 0.832 μ_{CR} = 0.056
 Diversity = 9.1793e+00 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 2.982676e+05 FES = 214100 μ_F = 0.832 μ_{CR} = 0.054
 Diversity = 9.3373e+00 k_t = [2, 3, 3, 4]
 Generation 2150: Best = 2.977472e+05 FES = 215100 μ_F = 0.834 μ_{CR} = 0.049
 Diversity = 9.4920e+00 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 2.976008e+05 FES = 216100 μ_F = 0.829 μ_{CR} = 0.044
 Diversity = 8.9732e+00 k_t = [2, 4, 4, 3]
 Generation 2170: Best = 2.971371e+05 FES = 217100 μ_F = 0.827 μ_{CR} = 0.037
 Diversity = 8.3097e+00 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 2.968231e+05 FES = 218100 μ_F = 0.821 μ_{CR} = 0.043
 Diversity = 7.5787e+00 k_t = [3, 4, 2, 2]
 Generation 2190: Best = 2.958341e+05 FES = 219100 μ_F = 0.808 μ_{CR} = 0.047
 Diversity = 8.0771e+00 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 2.955406e+05 FES = 220100 μ_F = 0.812 μ_{CR} = 0.049
 Diversity = 8.4090e+00 k_t = [2, 2, 4, 2]
 Generation 2210: Best = 2.948171e+05 FES = 221100 μ_F = 0.811 μ_{CR} = 0.049
 Diversity = 8.5949e+00 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 2.940157e+05 FES = 222100 μ_F = 0.815 μ_{CR} = 0.046
 Diversity = 7.8917e+00 k_t = [2, 4, 4, 3]
 Generation 2230: Best = 2.937927e+05 FES = 223100 μ_F = 0.797 μ_{CR} = 0.051
 Diversity = 7.4945e+00 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 2.931032e+05 FES = 224100 μ_F = 0.812 μ_{CR} = 0.054
 Diversity = 7.0675e+00 k_t = [2, 4, 4, 4]
 Generation 2250: Best = 2.919053e+05 FES = 225100 μ_F = 0.823 μ_{CR} = 0.052
 Diversity = 6.8135e+00 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 2.919010e+05 FES = 226100 μ_F = 0.815 μ_{CR} = 0.047
 Diversity = 7.5393e+00 k_t = [3, 4, 4, 3]
 Generation 2270: Best = 2.912361e+05 FES = 227100 μ_F = 0.810 μ_{CR} = 0.041
 Diversity = 7.5353e+00 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 2.909086e+05 FES = 228100 μ_F = 0.814 μ_{CR} = 0.045
 Diversity = 7.1125e+00 k_t = [2, 3, 3, 3]
 Generation 2290: Best = 2.905556e+05 FES = 229100 μ_F = 0.801 μ_{CR} = 0.041
 Diversity = 6.9413e+00 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 2.901934e+05 FES = 230100 μ_F = 0.802 μ_{CR} = 0.042
 Diversity = 6.8901e+00 k_t = [2, 4, 4, 2]
 Generation 2310: Best = 2.900209e+05 FES = 231100 μ_F = 0.812 μ_{CR} = 0.048
 Diversity = 7.4616e+00 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 2.896113e+05 FES = 232100 μ_F = 0.825 μ_{CR} = 0.053
 Diversity = 7.8782e+00 k_t = [3, 4, 3, 4]
 Generation 2330: Best = 2.891023e+05 FES = 233100 μ_F = 0.826 μ_{CR} = 0.049
 Diversity = 8.0077e+00 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 2.886779e+05 FES = 234100 μ_F = 0.814 μ_{CR} = 0.050
 Diversity = 7.3133e+00 k_t = [3, 4, 4, 4]
 Generation 2350: Best = 2.883364e+05 FES = 235100 μ_F = 0.813 μ_{CR} = 0.050
 Diversity = 7.2892e+00 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 2.874868e+05 FES = 236100 μ_F = 0.805 μ_{CR} = 0.052
 Diversity = 6.7851e+00 k_t = [3, 2, 4, 4]
 Generation 2370: Best = 2.871460e+05 FES = 237100 μ_F = 0.814 μ_{CR} = 0.044
 Diversity = 6.8693e+00 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 2.868590e+05 FES = 238100 μ_F = 0.809 μ_{CR} = 0.044
 Diversity = 6.2479e+00 k_t = [2, 2, 4, 4]

Generation 2390: Best = 2.866072e+05 FES = 239100 μ_F = 0.815 μ_{CR} = 0.052
 Diversity = 6.8413e+00 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 2.861625e+05 FES = 240100 μ_F = 0.807 μ_{CR} = 0.054
 Diversity = 7.3155e+00 k_t = [3, 2, 2, 4]
 Generation 2410: Best = 2.856734e+05 FES = 241100 μ_F = 0.802 μ_{CR} = 0.052
 Diversity = 6.3035e+00 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 2.853302e+05 FES = 242100 μ_F = 0.822 μ_{CR} = 0.051
 Diversity = 6.7421e+00 k_t = [3, 4, 2, 4]
 Generation 2430: Best = 2.850799e+05 FES = 243100 μ_F = 0.810 μ_{CR} = 0.050
 Diversity = 6.6398e+00 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 2.847490e+05 FES = 244100 μ_F = 0.798 μ_{CR} = 0.053
 Diversity = 6.3164e+00 k_t = [2, 4, 4, 4]
 Generation 2450: Best = 2.845204e+05 FES = 245100 μ_F = 0.795 μ_{CR} = 0.050
 Diversity = 5.9653e+00 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 2.836646e+05 FES = 246100 μ_F = 0.788 μ_{CR} = 0.053
 Diversity = 6.0906e+00 k_t = [3, 4, 2, 3]
 Generation 2470: Best = 2.830261e+05 FES = 247100 μ_F = 0.797 μ_{CR} = 0.058
 Diversity = 6.4883e+00 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 2.826921e+05 FES = 248100 μ_F = 0.787 μ_{CR} = 0.057
 Diversity = 6.1434e+00 k_t = [4, 2, 4, 3]
 Generation 2490: Best = 2.823075e+05 FES = 249100 μ_F = 0.795 μ_{CR} = 0.052
 Diversity = 6.8746e+00 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 2.818628e+05 FES = 250100 μ_F = 0.793 μ_{CR} = 0.057
 Diversity = 7.1916e+00 k_t = [2, 2, 3, 4]
 Generation 2510: Best = 2.816632e+05 FES = 251100 μ_F = 0.794 μ_{CR} = 0.054
 Diversity = 6.1716e+00 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 2.808949e+05 FES = 252100 μ_F = 0.796 μ_{CR} = 0.063
 Diversity = 6.1000e+00 k_t = [2, 3, 4, 4]
 Generation 2530: Best = 2.796871e+05 FES = 253100 μ_F = 0.818 μ_{CR} = 0.058
 Diversity = 6.3078e+00 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 2.793199e+05 FES = 254100 μ_F = 0.817 μ_{CR} = 0.064
 Diversity = 6.6206e+00 k_t = [2, 3, 4, 2]
 Generation 2550: Best = 2.786091e+05 FES = 255100 μ_F = 0.814 μ_{CR} = 0.065
 Diversity = 6.6703e+00 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 2.779075e+05 FES = 256100 μ_F = 0.815 μ_{CR} = 0.067
 Diversity = 6.2330e+00 k_t = [4, 3, 3, 4]
 Generation 2570: Best = 2.770331e+05 FES = 257100 μ_F = 0.832 μ_{CR} = 0.067
 Diversity = 6.8075e+00 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 2.761453e+05 FES = 258100 μ_F = 0.825 μ_{CR} = 0.062
 Diversity = 7.3452e+00 k_t = [3, 2, 2, 4]
 Generation 2590: Best = 2.756138e+05 FES = 259100 μ_F = 0.822 μ_{CR} = 0.055
 Diversity = 7.8802e+00 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 2.753842e+05 FES = 260100 μ_F = 0.838 μ_{CR} = 0.062
 Diversity = 8.2463e+00 k_t = [4, 4, 4, 3]
 Generation 2610: Best = 2.748459e+05 FES = 261100 μ_F = 0.826 μ_{CR} = 0.053
 Diversity = 8.4862e+00 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 2.745768e+05 FES = 262100 μ_F = 0.826 μ_{CR} = 0.055
 Diversity = 7.9278e+00 k_t = [3, 3, 4, 2]
 Generation 2630: Best = 2.738185e+05 FES = 263100 μ_F = 0.825 μ_{CR} = 0.052
 Diversity = 8.5902e+00 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 2.737990e+05 FES = 264100 μ_F = 0.821 μ_{CR} = 0.056
 Diversity = 8.4371e+00 k_t = [4, 4, 2, 3]
 Generation 2650: Best = 2.724643e+05 FES = 265100 μ_F = 0.828 μ_{CR} = 0.050
 Diversity = 9.1933e+00 k_t = [1, 1, 1, 1]

Generation 2660: Best = 2.718428e+05 FES = 266100 μ_F = 0.831 μ_{CR} = 0.045
 Diversity = 9.4126e+00 k_t = [3, 4, 3, 3]
 Generation 2670: Best = 2.714393e+05 FES = 267100 μ_F = 0.835 μ_{CR} = 0.053
 Diversity = 9.4939e+00 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 2.707853e+05 FES = 268100 μ_F = 0.836 μ_{CR} = 0.052
 Diversity = 9.4598e+00 k_t = [4, 4, 4, 4]
 Generation 2690: Best = 2.701393e+05 FES = 269100 μ_F = 0.839 μ_{CR} = 0.052
 Diversity = 9.5149e+00 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 2.694672e+05 FES = 270100 μ_F = 0.823 μ_{CR} = 0.053
 Diversity = 9.0687e+00 k_t = [3, 3, 4, 4]
 Generation 2710: Best = 2.691362e+05 FES = 271100 μ_F = 0.829 μ_{CR} = 0.052
 Diversity = 8.4122e+00 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 2.684628e+05 FES = 272100 μ_F = 0.838 μ_{CR} = 0.047
 Diversity = 8.5071e+00 k_t = [2, 3, 3, 2]
 Generation 2730: Best = 2.675982e+05 FES = 273100 μ_F = 0.837 μ_{CR} = 0.048
 Diversity = 8.9889e+00 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 2.672041e+05 FES = 274100 μ_F = 0.826 μ_{CR} = 0.050
 Diversity = 9.3553e+00 k_t = [2, 4, 2, 2]
 Generation 2750: Best = 2.666027e+05 FES = 275100 μ_F = 0.827 μ_{CR} = 0.049
 Diversity = 1.0608e+01 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 2.662746e+05 FES = 276100 μ_F = 0.827 μ_{CR} = 0.050
 Diversity = 1.0736e+01 k_t = [2, 2, 3, 3]
 Generation 2770: Best = 2.660785e+05 FES = 277100 μ_F = 0.829 μ_{CR} = 0.048
 Diversity = 1.0349e+01 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 2.655807e+05 FES = 278100 μ_F = 0.826 μ_{CR} = 0.043
 Diversity = 1.0523e+01 k_t = [3, 4, 4, 2]
 Generation 2790: Best = 2.650274e+05 FES = 279100 μ_F = 0.840 μ_{CR} = 0.049
 Diversity = 1.0624e+01 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 2.642744e+05 FES = 280100 μ_F = 0.834 μ_{CR} = 0.048
 Diversity = 1.0764e+01 k_t = [3, 2, 4, 4]
 Generation 2810: Best = 2.632827e+05 FES = 281100 μ_F = 0.832 μ_{CR} = 0.044
 Diversity = 1.1439e+01 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 2.629037e+05 FES = 282100 μ_F = 0.829 μ_{CR} = 0.048
 Diversity = 1.0521e+01 k_t = [2, 3, 3, 4]
 Generation 2830: Best = 2.627581e+05 FES = 283100 μ_F = 0.828 μ_{CR} = 0.042
 Diversity = 1.0675e+01 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 2.624879e+05 FES = 284100 μ_F = 0.830 μ_{CR} = 0.043
 Diversity = 1.0721e+01 k_t = [4, 3, 4, 2]
 Generation 2850: Best = 2.620645e+05 FES = 285100 μ_F = 0.813 μ_{CR} = 0.050
 Diversity = 1.0456e+01 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 2.618180e+05 FES = 286100 μ_F = 0.796 μ_{CR} = 0.047
 Diversity = 9.6877e+00 k_t = [4, 3, 3, 4]
 Generation 2870: Best = 2.608849e+05 FES = 287100 μ_F = 0.797 μ_{CR} = 0.043
 Diversity = 9.1384e+00 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 2.605246e+05 FES = 288100 μ_F = 0.805 μ_{CR} = 0.046
 Diversity = 9.3135e+00 k_t = [3, 4, 2, 2]
 Generation 2890: Best = 2.604011e+05 FES = 289100 μ_F = 0.819 μ_{CR} = 0.046
 Diversity = 9.2026e+00 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 2.599672e+05 FES = 290100 μ_F = 0.818 μ_{CR} = 0.046
 Diversity = 9.0406e+00 k_t = [4, 2, 2, 3]
 Generation 2910: Best = 2.594951e+05 FES = 291100 μ_F = 0.807 μ_{CR} = 0.044
 Diversity = 8.6308e+00 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 2.590714e+05 FES = 292100 μ_F = 0.819 μ_{CR} = 0.054
 Diversity = 8.5267e+00 k_t = [3, 3, 3, 4]

Generation 2930: Best = 2.586097e+05 FES = 293100 μ_F = 0.826 μ_{CR} = 0.053
 Diversity = 8.4148e+00 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 2.580550e+05 FES = 294100 μ_F = 0.826 μ_{CR} = 0.061
 Diversity = 9.0625e+00 k_t = [4, 3, 4, 3]
 Generation 2950: Best = 2.572947e+05 FES = 295100 μ_F = 0.826 μ_{CR} = 0.057
 Diversity = 9.5316e+00 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 2.569246e+05 FES = 296100 μ_F = 0.823 μ_{CR} = 0.045
 Diversity = 9.4417e+00 k_t = [2, 3, 4, 3]
 Generation 2970: Best = 2.565374e+05 FES = 297100 μ_F = 0.819 μ_{CR} = 0.045
 Diversity = 9.2659e+00 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 2.561842e+05 FES = 298100 μ_F = 0.835 μ_{CR} = 0.047
 Diversity = 1.0177e+01 k_t = [4, 4, 2, 3]
 Generation 2990: Best = 2.557452e+05 FES = 299100 μ_F = 0.828 μ_{CR} = 0.039
 Diversity = 9.5504e+00 k_t = [1, 1, 1, 1]
 Generation 3000: Best = 2.553699e+05 FES = 300100 μ_F = 0.820 μ_{CR} = 0.044
 Diversity = 8.7206e+00 k_t = [3, 3, 2, 3]
 Generation 3010: Best = 2.552889e+05 FES = 301100 μ_F = 0.819 μ_{CR} = 0.051
 Diversity = 1.0362e+01 k_t = [1, 1, 1, 1]
 Generation 3020: Best = 2.549112e+05 FES = 302100 μ_F = 0.829 μ_{CR} = 0.045
 Diversity = 1.0743e+01 k_t = [2, 4, 2, 3]
 Generation 3030: Best = 2.546183e+05 FES = 303100 μ_F = 0.822 μ_{CR} = 0.046
 Diversity = 1.0951e+01 k_t = [1, 1, 1, 1]
 Generation 3040: Best = 2.543970e+05 FES = 304100 μ_F = 0.817 μ_{CR} = 0.041
 Diversity = 1.1721e+01 k_t = [2, 3, 2, 2]
 Generation 3050: Best = 2.538422e+05 FES = 305100 μ_F = 0.813 μ_{CR} = 0.040
 Diversity = 1.1507e+01 k_t = [1, 1, 1, 1]
 Generation 3060: Best = 2.537923e+05 FES = 306100 μ_F = 0.809 μ_{CR} = 0.047
 Diversity = 1.0475e+01 k_t = [4, 4, 4, 4]
 Generation 3070: Best = 2.533435e+05 FES = 307100 μ_F = 0.820 μ_{CR} = 0.046
 Diversity = 9.8972e+00 k_t = [1, 1, 1, 1]
 Generation 3080: Best = 2.530505e+05 FES = 308100 μ_F = 0.821 μ_{CR} = 0.051
 Diversity = 1.0469e+01 k_t = [2, 3, 4, 4]
 Generation 3090: Best = 2.526505e+05 FES = 309100 μ_F = 0.806 μ_{CR} = 0.054
 Diversity = 9.4629e+00 k_t = [1, 1, 1, 1]
 Generation 3100: Best = 2.521354e+05 FES = 310100 μ_F = 0.816 μ_{CR} = 0.060
 Diversity = 1.0181e+01 k_t = [4, 4, 3, 2]
 Generation 3110: Best = 2.516998e+05 FES = 311100 μ_F = 0.816 μ_{CR} = 0.056
 Diversity = 1.0735e+01 k_t = [1, 1, 1, 1]
 Generation 3120: Best = 2.509396e+05 FES = 312100 μ_F = 0.810 μ_{CR} = 0.052
 Diversity = 1.0813e+01 k_t = [2, 4, 2, 2]
 Generation 3130: Best = 2.503263e+05 FES = 313100 μ_F = 0.802 μ_{CR} = 0.051
 Diversity = 1.2419e+01 k_t = [1, 1, 1, 1]
 Generation 3140: Best = 2.499115e+05 FES = 314100 μ_F = 0.802 μ_{CR} = 0.046
 Diversity = 1.0760e+01 k_t = [4, 2, 3, 4]
 Generation 3150: Best = 2.490533e+05 FES = 315100 μ_F = 0.820 μ_{CR} = 0.044
 Diversity = 1.0539e+01 k_t = [1, 1, 1, 1]
 Generation 3160: Best = 2.485258e+05 FES = 316100 μ_F = 0.821 μ_{CR} = 0.045
 Diversity = 1.0835e+01 k_t = [4, 3, 3, 2]
 Generation 3170: Best = 2.466295e+05 FES = 317100 μ_F = 0.837 μ_{CR} = 0.051
 Diversity = 1.1496e+01 k_t = [1, 1, 1, 1]
 Generation 3180: Best = 2.460116e+05 FES = 318100 μ_F = 0.828 μ_{CR} = 0.048
 Diversity = 1.2309e+01 k_t = [4, 2, 4, 3]
 Generation 3190: Best = 2.453649e+05 FES = 319100 μ_F = 0.826 μ_{CR} = 0.046
 Diversity = 1.2587e+01 k_t = [1, 1, 1, 1]

Generation 3200: Best = 2.450444e+05 FES = 320100 μ_F = 0.822 μ_{CR} = 0.042
 Diversity = 1.2660e+01 k_t = [3, 2, 4, 4]
 Generation 3210: Best = 2.444240e+05 FES = 321100 μ_F = 0.833 μ_{CR} = 0.042
 Diversity = 1.0697e+01 k_t = [1, 1, 1, 1]
 Generation 3220: Best = 2.441293e+05 FES = 322100 μ_F = 0.836 μ_{CR} = 0.047
 Diversity = 1.1491e+01 k_t = [4, 2, 3, 2]
 Generation 3230: Best = 2.436538e+05 FES = 323100 μ_F = 0.828 μ_{CR} = 0.050
 Diversity = 1.2629e+01 k_t = [1, 1, 1, 1]
 Generation 3240: Best = 2.431186e+05 FES = 324100 μ_F = 0.820 μ_{CR} = 0.044
 Diversity = 1.2552e+01 k_t = [4, 4, 2, 3]
 Generation 3250: Best = 2.423693e+05 FES = 325100 μ_F = 0.827 μ_{CR} = 0.043
 Diversity = 1.3664e+01 k_t = [1, 1, 1, 1]
 Generation 3260: Best = 2.419008e+05 FES = 326100 μ_F = 0.831 μ_{CR} = 0.049
 Diversity = 1.2910e+01 k_t = [2, 2, 2, 2]
 Generation 3270: Best = 2.416214e+05 FES = 327100 μ_F = 0.825 μ_{CR} = 0.042
 Diversity = 1.3383e+01 k_t = [1, 1, 1, 1]
 Generation 3280: Best = 2.410589e+05 FES = 328100 μ_F = 0.826 μ_{CR} = 0.039
 Diversity = 1.2057e+01 k_t = [2, 3, 2, 3]
 Generation 3290: Best = 2.405689e+05 FES = 329100 μ_F = 0.829 μ_{CR} = 0.041
 Diversity = 1.2028e+01 k_t = [1, 1, 1, 1]
 Generation 3300: Best = 2.401918e+05 FES = 330100 μ_F = 0.826 μ_{CR} = 0.042
 Diversity = 1.2457e+01 k_t = [4, 4, 4, 4]
 Generation 3310: Best = 2.398588e+05 FES = 331100 μ_F = 0.825 μ_{CR} = 0.043
 Diversity = 1.1832e+01 k_t = [1, 1, 1, 1]
 Generation 3320: Best = 2.395546e+05 FES = 332100 μ_F = 0.818 μ_{CR} = 0.046
 Diversity = 1.2445e+01 k_t = [2, 3, 3, 2]
 Generation 3330: Best = 2.394657e+05 FES = 333100 μ_F = 0.804 μ_{CR} = 0.039
 Diversity = 1.1909e+01 k_t = [1, 1, 1, 1]
 Generation 3340: Best = 2.390332e+05 FES = 334100 μ_F = 0.797 μ_{CR} = 0.043
 Diversity = 1.1769e+01 k_t = [2, 3, 2, 3]
 Generation 3350: Best = 2.388611e+05 FES = 335100 μ_F = 0.803 μ_{CR} = 0.049
 Diversity = 1.1619e+01 k_t = [1, 1, 1, 1]
 Generation 3360: Best = 2.386198e+05 FES = 336100 μ_F = 0.811 μ_{CR} = 0.038
 Diversity = 1.1319e+01 k_t = [2, 2, 3, 3]
 Generation 3370: Best = 2.385006e+05 FES = 337100 μ_F = 0.810 μ_{CR} = 0.045
 Diversity = 1.0727e+01 k_t = [1, 1, 1, 1]
 Generation 3380: Best = 2.382239e+05 FES = 338100 μ_F = 0.825 μ_{CR} = 0.047
 Diversity = 1.1102e+01 k_t = [4, 3, 4, 3]
 Generation 3390: Best = 2.377760e+05 FES = 339100 μ_F = 0.818 μ_{CR} = 0.044
 Diversity = 1.1328e+01 k_t = [1, 1, 1, 1]
 Generation 3400: Best = 2.373674e+05 FES = 340100 μ_F = 0.838 μ_{CR} = 0.047
 Diversity = 1.1410e+01 k_t = [4, 2, 4, 2]
 Generation 3410: Best = 2.371241e+05 FES = 341100 μ_F = 0.833 μ_{CR} = 0.047
 Diversity = 1.1908e+01 k_t = [1, 1, 1, 1]
 Generation 3420: Best = 2.369535e+05 FES = 342100 μ_F = 0.838 μ_{CR} = 0.043
 Diversity = 1.1246e+01 k_t = [2, 4, 4, 3]
 Generation 3430: Best = 2.366595e+05 FES = 343100 μ_F = 0.828 μ_{CR} = 0.050
 Diversity = 1.0405e+01 k_t = [1, 1, 1, 1]
 Generation 3440: Best = 2.362490e+05 FES = 344100 μ_F = 0.824 μ_{CR} = 0.045
 Diversity = 1.0679e+01 k_t = [2, 3, 2, 4]
 Generation 3450: Best = 2.361380e+05 FES = 345100 μ_F = 0.824 μ_{CR} = 0.045
 Diversity = 1.1445e+01 k_t = [1, 1, 1, 1]
 Generation 3460: Best = 2.358242e+05 FES = 346100 μ_F = 0.819 μ_{CR} = 0.045
 Diversity = 1.1559e+01 k_t = [2, 3, 2, 3]

Generation 3470: Best = 2.352890e+05 FES = 347100 μ_F = 0.823 μ_{CR} = 0.044
 Diversity = 1.1943e+01 k_t = [1, 1, 1, 1]
 Generation 3480: Best = 2.350974e+05 FES = 348100 μ_F = 0.827 μ_{CR} = 0.043
 Diversity = 1.1652e+01 k_t = [2, 3, 4, 3]
 Generation 3490: Best = 2.349328e+05 FES = 349100 μ_F = 0.829 μ_{CR} = 0.044
 Diversity = 1.2376e+01 k_t = [1, 1, 1, 1]
 Generation 3500: Best = 2.346885e+05 FES = 350100 μ_F = 0.824 μ_{CR} = 0.043
 Diversity = 1.1932e+01 k_t = [2, 2, 2, 2]
 Generation 3510: Best = 2.343922e+05 FES = 351100 μ_F = 0.831 μ_{CR} = 0.045
 Diversity = 1.2235e+01 k_t = [1, 1, 1, 1]
 Generation 3520: Best = 2.342533e+05 FES = 352100 μ_F = 0.831 μ_{CR} = 0.046
 Diversity = 1.1937e+01 k_t = [3, 4, 3, 3]
 Generation 3530: Best = 2.339357e+05 FES = 353100 μ_F = 0.821 μ_{CR} = 0.041
 Diversity = 1.2043e+01 k_t = [1, 1, 1, 1]
 Generation 3540: Best = 2.338692e+05 FES = 354100 μ_F = 0.825 μ_{CR} = 0.043
 Diversity = 1.2053e+01 k_t = [4, 4, 2, 4]
 Generation 3550: Best = 2.337097e+05 FES = 355100 μ_F = 0.825 μ_{CR} = 0.039
 Diversity = 1.1259e+01 k_t = [1, 1, 1, 1]
 Generation 3560: Best = 2.335519e+05 FES = 356100 μ_F = 0.834 μ_{CR} = 0.046
 Diversity = 1.2693e+01 k_t = [4, 2, 2, 3]
 Generation 3570: Best = 2.331625e+05 FES = 357100 μ_F = 0.834 μ_{CR} = 0.045
 Diversity = 1.2994e+01 k_t = [1, 1, 1, 1]
 Generation 3580: Best = 2.331625e+05 FES = 358100 μ_F = 0.826 μ_{CR} = 0.043
 Diversity = 1.1927e+01 k_t = [2, 2, 2, 4]
 Generation 3590: Best = 2.328282e+05 FES = 359100 μ_F = 0.806 μ_{CR} = 0.046
 Diversity = 1.1707e+01 k_t = [1, 1, 1, 1]
 Generation 3600: Best = 2.323232e+05 FES = 360100 μ_F = 0.808 μ_{CR} = 0.047
 Diversity = 1.2515e+01 k_t = [4, 4, 2, 3]
 Generation 3610: Best = 2.316332e+05 FES = 361100 μ_F = 0.818 μ_{CR} = 0.042
 Diversity = 1.3447e+01 k_t = [1, 1, 1, 1]
 Generation 3620: Best = 2.312726e+05 FES = 362100 μ_F = 0.806 μ_{CR} = 0.050
 Diversity = 1.1618e+01 k_t = [3, 2, 4, 4]
 Generation 3630: Best = 2.309182e+05 FES = 363100 μ_F = 0.815 μ_{CR} = 0.051
 Diversity = 1.0794e+01 k_t = [1, 1, 1, 1]
 Generation 3640: Best = 2.306915e+05 FES = 364100 μ_F = 0.826 μ_{CR} = 0.050
 Diversity = 1.0594e+01 k_t = [2, 3, 3, 4]
 Generation 3650: Best = 2.303760e+05 FES = 365100 μ_F = 0.827 μ_{CR} = 0.049
 Diversity = 1.0218e+01 k_t = [1, 1, 1, 1]
 Generation 3660: Best = 2.300621e+05 FES = 366100 μ_F = 0.837 μ_{CR} = 0.046
 Diversity = 1.0488e+01 k_t = [4, 4, 4, 4]
 Generation 3670: Best = 2.298072e+05 FES = 367100 μ_F = 0.839 μ_{CR} = 0.048
 Diversity = 1.0972e+01 k_t = [1, 1, 1, 1]
 Generation 3680: Best = 2.296642e+05 FES = 368100 μ_F = 0.837 μ_{CR} = 0.038
 Diversity = 1.3477e+01 k_t = [3, 3, 2, 2]
 Generation 3690: Best = 2.294128e+05 FES = 369100 μ_F = 0.827 μ_{CR} = 0.044
 Diversity = 1.3448e+01 k_t = [1, 1, 1, 1]
 Generation 3700: Best = 2.285180e+05 FES = 370100 μ_F = 0.838 μ_{CR} = 0.039
 Diversity = 1.5231e+01 k_t = [4, 4, 3, 4]
 Generation 3710: Best = 2.284452e+05 FES = 371100 μ_F = 0.845 μ_{CR} = 0.043
 Diversity = 1.4857e+01 k_t = [1, 1, 1, 1]
 Generation 3720: Best = 2.282286e+05 FES = 372100 μ_F = 0.845 μ_{CR} = 0.045
 Diversity = 1.2712e+01 k_t = [3, 3, 4, 3]
 Generation 3730: Best = 2.281925e+05 FES = 373100 μ_F = 0.832 μ_{CR} = 0.041
 Diversity = 1.2294e+01 k_t = [1, 1, 1, 1]

Generation 3740: Best = 2.280566e+05 FES = 374100 μ_F = 0.828 μ_{CR} = 0.040
 Diversity = 1.1796e+01 k_t = [2, 2, 4, 3]
 Generation 3750: Best = 2.279248e+05 FES = 375100 μ_F = 0.837 μ_{CR} = 0.036
 Diversity = 1.1407e+01 k_t = [1, 1, 1, 1]
 Generation 3760: Best = 2.277082e+05 FES = 376100 μ_F = 0.842 μ_{CR} = 0.042
 Diversity = 1.1895e+01 k_t = [2, 2, 2, 4]
 Generation 3770: Best = 2.273792e+05 FES = 377100 μ_F = 0.851 μ_{CR} = 0.045
 Diversity = 1.2143e+01 k_t = [1, 1, 1, 1]
 Generation 3780: Best = 2.272523e+05 FES = 378100 μ_F = 0.844 μ_{CR} = 0.045
 Diversity = 1.3568e+01 k_t = [2, 3, 4, 2]
 Generation 3790: Best = 2.269169e+05 FES = 379100 μ_F = 0.840 μ_{CR} = 0.046
 Diversity = 1.3619e+01 k_t = [1, 1, 1, 1]
 Generation 3800: Best = 2.266722e+05 FES = 380100 μ_F = 0.854 μ_{CR} = 0.044
 Diversity = 1.3877e+01 k_t = [2, 4, 3, 3]
 Generation 3810: Best = 2.261752e+05 FES = 381100 μ_F = 0.860 μ_{CR} = 0.041
 Diversity = 1.3573e+01 k_t = [1, 1, 1, 1]
 Generation 3820: Best = 2.260400e+05 FES = 382100 μ_F = 0.862 μ_{CR} = 0.046
 Diversity = 1.3605e+01 k_t = [2, 4, 4, 3]
 Generation 3830: Best = 2.257808e+05 FES = 383100 μ_F = 0.847 μ_{CR} = 0.041
 Diversity = 1.4140e+01 k_t = [1, 1, 1, 1]
 Generation 3840: Best = 2.256359e+05 FES = 384100 μ_F = 0.844 μ_{CR} = 0.041
 Diversity = 1.3279e+01 k_t = [3, 4, 3, 4]
 Generation 3850: Best = 2.254111e+05 FES = 385100 μ_F = 0.837 μ_{CR} = 0.041
 Diversity = 1.3052e+01 k_t = [1, 1, 1, 1]
 Generation 3860: Best = 2.253809e+05 FES = 386100 μ_F = 0.832 μ_{CR} = 0.044
 Diversity = 1.2438e+01 k_t = [4, 2, 2, 4]
 Generation 3870: Best = 2.251308e+05 FES = 387100 μ_F = 0.832 μ_{CR} = 0.048
 Diversity = 1.1405e+01 k_t = [1, 1, 1, 1]
 Generation 3880: Best = 2.250744e+05 FES = 388100 μ_F = 0.828 μ_{CR} = 0.050
 Diversity = 1.1065e+01 k_t = [2, 2, 3, 4]
 Generation 3890: Best = 2.248336e+05 FES = 389100 μ_F = 0.814 μ_{CR} = 0.049
 Diversity = 9.8869e+00 k_t = [1, 1, 1, 1]
 Generation 3900: Best = 2.247870e+05 FES = 390100 μ_F = 0.805 μ_{CR} = 0.050
 Diversity = 1.0114e+01 k_t = [3, 4, 2, 3]
 Generation 3910: Best = 2.245821e+05 FES = 391100 μ_F = 0.810 μ_{CR} = 0.051
 Diversity = 1.1614e+01 k_t = [1, 1, 1, 1]
 Generation 3920: Best = 2.243216e+05 FES = 392100 μ_F = 0.825 μ_{CR} = 0.048
 Diversity = 1.2951e+01 k_t = [2, 4, 3, 3]
 Generation 3930: Best = 2.241091e+05 FES = 393100 μ_F = 0.826 μ_{CR} = 0.044
 Diversity = 1.3464e+01 k_t = [1, 1, 1, 1]
 Generation 3940: Best = 2.239183e+05 FES = 394100 μ_F = 0.840 μ_{CR} = 0.047
 Diversity = 1.2362e+01 k_t = [2, 3, 4, 4]
 Generation 3950: Best = 2.237482e+05 FES = 395100 μ_F = 0.833 μ_{CR} = 0.046
 Diversity = 1.2169e+01 k_t = [1, 1, 1, 1]
 Generation 3960: Best = 2.235232e+05 FES = 396100 μ_F = 0.845 μ_{CR} = 0.048
 Diversity = 1.3256e+01 k_t = [3, 2, 4, 2]
 Generation 3970: Best = 2.231678e+05 FES = 397100 μ_F = 0.838 μ_{CR} = 0.053
 Diversity = 1.4050e+01 k_t = [1, 1, 1, 1]
 Generation 3980: Best = 2.229273e+05 FES = 398100 μ_F = 0.832 μ_{CR} = 0.040
 Diversity = 1.4607e+01 k_t = [3, 2, 2, 3]
 Generation 3990: Best = 2.228815e+05 FES = 399100 μ_F = 0.833 μ_{CR} = 0.038
 Diversity = 1.3729e+01 k_t = [1, 1, 1, 1]
 Generation 4000: Best = 2.223985e+05 FES = 400100 μ_F = 0.826 μ_{CR} = 0.039
 Diversity = 1.3252e+01 k_t = [3, 3, 3, 3]

Generation 4010: Best = 2.221816e+05	FES = 401100	$\mu_F = 0.810$	$\mu_{CR} = 0.046$
Diversity = 1.2918e+01	$k_t = [1, 1, 1, 1]$		
Generation 4020: Best = 2.219035e+05	FES = 402100	$\mu_F = 0.826$	$\mu_{CR} = 0.044$
Diversity = 1.2600e+01	$k_t = [2, 3, 2, 3]$		
Generation 4030: Best = 2.216612e+05	FES = 403100	$\mu_F = 0.834$	$\mu_{CR} = 0.047$
Diversity = 1.2602e+01	$k_t = [1, 1, 1, 1]$		
Generation 4040: Best = 2.214534e+05	FES = 404100	$\mu_F = 0.830$	$\mu_{CR} = 0.044$
Diversity = 1.2046e+01	$k_t = [3, 2, 3, 4]$		
Generation 4050: Best = 2.212006e+05	FES = 405100	$\mu_F = 0.815$	$\mu_{CR} = 0.053$
Diversity = 1.1374e+01	$k_t = [1, 1, 1, 1]$		
Generation 4060: Best = 2.209853e+05	FES = 406100	$\mu_F = 0.795$	$\mu_{CR} = 0.050$
Diversity = 1.0423e+01	$k_t = [3, 4, 3, 4]$		
Generation 4070: Best = 2.209401e+05	FES = 407100	$\mu_F = 0.806$	$\mu_{CR} = 0.051$
Diversity = 9.9239e+00	$k_t = [1, 1, 1, 1]$		
Generation 4080: Best = 2.207824e+05	FES = 408100	$\mu_F = 0.802$	$\mu_{CR} = 0.049$
Diversity = 1.0212e+01	$k_t = [2, 3, 4, 2]$		
Generation 4090: Best = 2.206360e+05	FES = 409100	$\mu_F = 0.813$	$\mu_{CR} = 0.044$
Diversity = 1.0230e+01	$k_t = [1, 1, 1, 1]$		
Generation 4100: Best = 2.204064e+05	FES = 410100	$\mu_F = 0.817$	$\mu_{CR} = 0.045$
Diversity = 9.7897e+00	$k_t = [2, 2, 4, 4]$		
Generation 4110: Best = 2.202561e+05	FES = 411100	$\mu_F = 0.829$	$\mu_{CR} = 0.041$
Diversity = 9.0105e+00	$k_t = [1, 1, 1, 1]$		
Generation 4120: Best = 2.201745e+05	FES = 412100	$\mu_F = 0.829$	$\mu_{CR} = 0.041$
Diversity = 8.5638e+00	$k_t = [4, 3, 4, 4]$		
Generation 4130: Best = 2.198979e+05	FES = 413100	$\mu_F = 0.842$	$\mu_{CR} = 0.043$
Diversity = 8.6552e+00	$k_t = [1, 1, 1, 1]$		
Generation 4140: Best = 2.196258e+05	FES = 414100	$\mu_F = 0.831$	$\mu_{CR} = 0.048$
Diversity = 9.5448e+00	$k_t = [4, 2, 2, 2]$		
Generation 4150: Best = 2.193404e+05	FES = 415100	$\mu_F = 0.830$	$\mu_{CR} = 0.046$
Diversity = 1.0799e+01	$k_t = [1, 1, 1, 1]$		
Generation 4160: Best = 2.186418e+05	FES = 416100	$\mu_F = 0.837$	$\mu_{CR} = 0.047$
Diversity = 1.1480e+01	$k_t = [3, 2, 2, 3]$		
Generation 4170: Best = 2.184421e+05	FES = 417100	$\mu_F = 0.830$	$\mu_{CR} = 0.050$
Diversity = 1.2324e+01	$k_t = [1, 1, 1, 1]$		
Generation 4180: Best = 2.181311e+05	FES = 418100	$\mu_F = 0.826$	$\mu_{CR} = 0.054$
Diversity = 1.1808e+01	$k_t = [2, 2, 4, 4]$		
Generation 4190: Best = 2.178543e+05	FES = 419100	$\mu_F = 0.829$	$\mu_{CR} = 0.062$
Diversity = 1.1009e+01	$k_t = [1, 1, 1, 1]$		
Generation 4200: Best = 2.177004e+05	FES = 420100	$\mu_F = 0.830$	$\mu_{CR} = 0.058$
Diversity = 1.1153e+01	$k_t = [4, 3, 2, 3]$		
Generation 4210: Best = 2.172136e+05	FES = 421100	$\mu_F = 0.825$	$\mu_{CR} = 0.058$
Diversity = 1.1032e+01	$k_t = [1, 1, 1, 1]$		
Generation 4220: Best = 2.170117e+05	FES = 422100	$\mu_F = 0.838$	$\mu_{CR} = 0.051$
Diversity = 1.1210e+01	$k_t = [3, 2, 4, 3]$		
Generation 4230: Best = 2.168506e+05	FES = 423100	$\mu_F = 0.826$	$\mu_{CR} = 0.050$
Diversity = 1.0687e+01	$k_t = [1, 1, 1, 1]$		
Generation 4240: Best = 2.165991e+05	FES = 424100	$\mu_F = 0.829$	$\mu_{CR} = 0.048$
Diversity = 1.0815e+01	$k_t = [3, 4, 3, 4]$		
Generation 4250: Best = 2.163874e+05	FES = 425100	$\mu_F = 0.830$	$\mu_{CR} = 0.057$
Diversity = 1.0141e+01	$k_t = [1, 1, 1, 1]$		
Generation 4260: Best = 2.162823e+05	FES = 426100	$\mu_F = 0.843$	$\mu_{CR} = 0.042$
Diversity = 1.0130e+01	$k_t = [2, 4, 2, 4]$		
Generation 4270: Best = 2.160559e+05	FES = 427100	$\mu_F = 0.841$	$\mu_{CR} = 0.044$
Diversity = 1.1170e+01	$k_t = [1, 1, 1, 1]$		

Generation 4280: Best = 2.158223e+05 FES = 428100 μ_F = 0.833 μ_{CR} = 0.042
 Diversity = 1.0764e+01 k_t = [3, 3, 4, 4]
 Generation 4290: Best = 2.156301e+05 FES = 429100 μ_F = 0.836 μ_{CR} = 0.042
 Diversity = 1.0255e+01 k_t = [1, 1, 1, 1]
 Generation 4300: Best = 2.154478e+05 FES = 430100 μ_F = 0.824 μ_{CR} = 0.058
 Diversity = 1.0227e+01 k_t = [4, 2, 4, 3]
 Generation 4310: Best = 2.152303e+05 FES = 431100 μ_F = 0.801 μ_{CR} = 0.073
 Diversity = 1.0421e+01 k_t = [1, 1, 1, 1]
 Generation 4320: Best = 2.151392e+05 FES = 432100 μ_F = 0.798 μ_{CR} = 0.056
 Diversity = 9.8309e+00 k_t = [2, 3, 4, 2]
 Generation 4330: Best = 2.149674e+05 FES = 433100 μ_F = 0.785 μ_{CR} = 0.047
 Diversity = 1.0629e+01 k_t = [1, 1, 1, 1]
 Generation 4340: Best = 2.147426e+05 FES = 434100 μ_F = 0.789 μ_{CR} = 0.045
 Diversity = 9.9350e+00 k_t = [4, 3, 4, 3]
 Generation 4350: Best = 2.146648e+05 FES = 435100 μ_F = 0.781 μ_{CR} = 0.051
 Diversity = 8.8678e+00 k_t = [1, 1, 1, 1]
 Generation 4360: Best = 2.142620e+05 FES = 436100 μ_F = 0.785 μ_{CR} = 0.053
 Diversity = 8.4834e+00 k_t = [4, 4, 2, 3]
 Generation 4370: Best = 2.142582e+05 FES = 437100 μ_F = 0.795 μ_{CR} = 0.051
 Diversity = 8.5214e+00 k_t = [1, 1, 1, 1]
 Generation 4380: Best = 2.141433e+05 FES = 438100 μ_F = 0.804 μ_{CR} = 0.052
 Diversity = 8.5359e+00 k_t = [2, 4, 2, 3]
 Generation 4390: Best = 2.139894e+05 FES = 439100 μ_F = 0.812 μ_{CR} = 0.054
 Diversity = 9.3929e+00 k_t = [1, 1, 1, 1]
 Generation 4400: Best = 2.138240e+05 FES = 440100 μ_F = 0.806 μ_{CR} = 0.057
 Diversity = 9.5873e+00 k_t = [2, 4, 3, 3]
 Generation 4410: Best = 2.137132e+05 FES = 441100 μ_F = 0.794 μ_{CR} = 0.055
 Diversity = 1.0166e+01 k_t = [1, 1, 1, 1]
 Generation 4420: Best = 2.132452e+05 FES = 442100 μ_F = 0.809 μ_{CR} = 0.050
 Diversity = 1.0084e+01 k_t = [4, 3, 3, 2]
 Generation 4430: Best = 2.130209e+05 FES = 443100 μ_F = 0.802 μ_{CR} = 0.050
 Diversity = 1.1211e+01 k_t = [1, 1, 1, 1]
 Generation 4440: Best = 2.129346e+05 FES = 444100 μ_F = 0.797 μ_{CR} = 0.042
 Diversity = 1.0439e+01 k_t = [4, 2, 2, 4]
 Generation 4450: Best = 2.128000e+05 FES = 445100 μ_F = 0.822 μ_{CR} = 0.043
 Diversity = 1.0364e+01 k_t = [1, 1, 1, 1]
 Generation 4460: Best = 2.125449e+05 FES = 446100 μ_F = 0.827 μ_{CR} = 0.051
 Diversity = 1.0512e+01 k_t = [4, 4, 4, 3]
 Generation 4470: Best = 2.123302e+05 FES = 447100 μ_F = 0.836 μ_{CR} = 0.045
 Diversity = 1.1213e+01 k_t = [1, 1, 1, 1]
 Generation 4480: Best = 2.121337e+05 FES = 448100 μ_F = 0.825 μ_{CR} = 0.048
 Diversity = 1.1119e+01 k_t = [2, 2, 2, 4]
 Generation 4490: Best = 2.118903e+05 FES = 449100 μ_F = 0.827 μ_{CR} = 0.060
 Diversity = 1.1063e+01 k_t = [1, 1, 1, 1]
 Generation 4500: Best = 2.115761e+05 FES = 450100 μ_F = 0.820 μ_{CR} = 0.049
 Diversity = 1.0423e+01 k_t = [2, 3, 3, 4]
 Generation 4510: Best = 2.113496e+05 FES = 451100 μ_F = 0.824 μ_{CR} = 0.043
 Diversity = 1.0775e+01 k_t = [1, 1, 1, 1]
 Generation 4520: Best = 2.111219e+05 FES = 452100 μ_F = 0.818 μ_{CR} = 0.049
 Diversity = 9.9856e+00 k_t = [2, 4, 2, 4]
 Generation 4530: Best = 2.109620e+05 FES = 453100 μ_F = 0.825 μ_{CR} = 0.043
 Diversity = 1.0982e+01 k_t = [1, 1, 1, 1]
 Generation 4540: Best = 2.106213e+05 FES = 454100 μ_F = 0.816 μ_{CR} = 0.050
 Diversity = 1.0406e+01 k_t = [2, 4, 3, 2]

Generation 4550: Best = 2.103999e+05 FES = 455100 μ_F = 0.834 μ_{CR} = 0.048
 Diversity = 1.0256e+01 k_t = [1, 1, 1, 1]
 Generation 4560: Best = 2.101731e+05 FES = 456100 μ_F = 0.840 μ_{CR} = 0.048
 Diversity = 9.9042e+00 k_t = [2, 4, 2, 2]
 Generation 4570: Best = 2.100145e+05 FES = 457100 μ_F = 0.829 μ_{CR} = 0.052
 Diversity = 1.0488e+01 k_t = [1, 1, 1, 1]
 Generation 4580: Best = 2.097270e+05 FES = 458100 μ_F = 0.830 μ_{CR} = 0.051
 Diversity = 1.2370e+01 k_t = [2, 2, 2, 2]
 Generation 4590: Best = 2.095099e+05 FES = 459100 μ_F = 0.823 μ_{CR} = 0.046
 Diversity = 1.2104e+01 k_t = [1, 1, 1, 1]
 Generation 4600: Best = 2.091452e+05 FES = 460100 μ_F = 0.825 μ_{CR} = 0.047
 Diversity = 1.2176e+01 k_t = [2, 2, 2, 4]
 Generation 4610: Best = 2.091026e+05 FES = 461100 μ_F = 0.822 μ_{CR} = 0.043
 Diversity = 1.2260e+01 k_t = [1, 1, 1, 1]
 Generation 4620: Best = 2.089355e+05 FES = 462100 μ_F = 0.836 μ_{CR} = 0.045
 Diversity = 1.2428e+01 k_t = [4, 3, 4, 2]
 Generation 4630: Best = 2.087803e+05 FES = 463100 μ_F = 0.830 μ_{CR} = 0.052
 Diversity = 1.2377e+01 k_t = [1, 1, 1, 1]
 Generation 4640: Best = 2.082699e+05 FES = 464100 μ_F = 0.838 μ_{CR} = 0.050
 Diversity = 1.2892e+01 k_t = [4, 4, 4, 3]
 Generation 4650: Best = 2.079402e+05 FES = 465100 μ_F = 0.834 μ_{CR} = 0.049
 Diversity = 1.4941e+01 k_t = [1, 1, 1, 1]
 Generation 4660: Best = 2.076927e+05 FES = 466100 μ_F = 0.829 μ_{CR} = 0.047
 Diversity = 1.4754e+01 k_t = [4, 4, 4, 2]
 Generation 4670: Best = 2.071184e+05 FES = 467100 μ_F = 0.835 μ_{CR} = 0.048
 Diversity = 1.6654e+01 k_t = [1, 1, 1, 1]
 Generation 4680: Best = 2.068785e+05 FES = 468100 μ_F = 0.819 μ_{CR} = 0.048
 Diversity = 1.6176e+01 k_t = [2, 2, 4, 4]
 Generation 4690: Best = 2.066816e+05 FES = 469100 μ_F = 0.810 μ_{CR} = 0.042
 Diversity = 1.4372e+01 k_t = [1, 1, 1, 1]
 Generation 4700: Best = 2.065363e+05 FES = 470100 μ_F = 0.810 μ_{CR} = 0.045
 Diversity = 1.3504e+01 k_t = [4, 4, 2, 2]
 Generation 4710: Best = 2.064002e+05 FES = 471100 μ_F = 0.815 μ_{CR} = 0.042
 Diversity = 1.3045e+01 k_t = [1, 1, 1, 1]
 Generation 4720: Best = 2.063175e+05 FES = 472100 μ_F = 0.832 μ_{CR} = 0.050
 Diversity = 1.4155e+01 k_t = [4, 4, 4, 3]
 Generation 4730: Best = 2.061111e+05 FES = 473100 μ_F = 0.814 μ_{CR} = 0.053
 Diversity = 1.4142e+01 k_t = [1, 1, 1, 1]
 Generation 4740: Best = 2.056947e+05 FES = 474100 μ_F = 0.828 μ_{CR} = 0.053
 Diversity = 1.3212e+01 k_t = [4, 3, 4, 4]
 Generation 4750: Best = 2.055555e+05 FES = 475100 μ_F = 0.829 μ_{CR} = 0.054
 Diversity = 1.3752e+01 k_t = [1, 1, 1, 1]
 Generation 4760: Best = 2.053860e+05 FES = 476100 μ_F = 0.818 μ_{CR} = 0.055
 Diversity = 1.3609e+01 k_t = [2, 3, 4, 3]
 Generation 4770: Best = 2.051748e+05 FES = 477100 μ_F = 0.806 μ_{CR} = 0.052
 Diversity = 1.4002e+01 k_t = [1, 1, 1, 1]
 Generation 4780: Best = 2.049263e+05 FES = 478100 μ_F = 0.820 μ_{CR} = 0.050
 Diversity = 1.4922e+01 k_t = [2, 4, 2, 4]
 Generation 4790: Best = 2.047141e+05 FES = 479100 μ_F = 0.809 μ_{CR} = 0.057
 Diversity = 1.4420e+01 k_t = [1, 1, 1, 1]
 Generation 4800: Best = 2.045367e+05 FES = 480100 μ_F = 0.814 μ_{CR} = 0.056
 Diversity = 1.4405e+01 k_t = [3, 2, 4, 2]
 Generation 4810: Best = 2.041032e+05 FES = 481100 μ_F = 0.815 μ_{CR} = 0.055
 Diversity = 1.5358e+01 k_t = [1, 1, 1, 1]

Generation 4820: Best = 2.040355e+05 FES = 482100 μ_F = 0.814 μ_{CR} = 0.047
 Diversity = 1.4238e+01 k_t = [2, 4, 3, 4]
 Generation 4830: Best = 2.036811e+05 FES = 483100 μ_F = 0.799 μ_{CR} = 0.049
 Diversity = 1.3287e+01 k_t = [1, 1, 1, 1]
 Generation 4840: Best = 2.035495e+05 FES = 484100 μ_F = 0.812 μ_{CR} = 0.052
 Diversity = 1.4760e+01 k_t = [4, 3, 4, 2]
 Generation 4850: Best = 2.033094e+05 FES = 485100 μ_F = 0.800 μ_{CR} = 0.051
 Diversity = 1.4257e+01 k_t = [1, 1, 1, 1]
 Generation 4860: Best = 2.030448e+05 FES = 486100 μ_F = 0.791 μ_{CR} = 0.054
 Diversity = 1.3264e+01 k_t = [2, 2, 2, 4]
 Generation 4870: Best = 2.028665e+05 FES = 487100 μ_F = 0.800 μ_{CR} = 0.048
 Diversity = 1.3068e+01 k_t = [1, 1, 1, 1]
 Generation 4880: Best = 2.025800e+05 FES = 488100 μ_F = 0.794 μ_{CR} = 0.049
 Diversity = 1.3270e+01 k_t = [3, 3, 2, 3]
 Generation 4890: Best = 2.019435e+05 FES = 489100 μ_F = 0.818 μ_{CR} = 0.051
 Diversity = 1.3887e+01 k_t = [1, 1, 1, 1]
 Generation 4900: Best = 2.015144e+05 FES = 490100 μ_F = 0.827 μ_{CR} = 0.050
 Diversity = 1.3268e+01 k_t = [4, 2, 2, 3]
 Generation 4910: Best = 2.008354e+05 FES = 491100 μ_F = 0.820 μ_{CR} = 0.071
 Diversity = 1.2180e+01 k_t = [1, 1, 1, 1]
 Generation 4920: Best = 2.005350e+05 FES = 492100 μ_F = 0.817 μ_{CR} = 0.064
 Diversity = 1.2153e+01 k_t = [3, 2, 3, 3]
 Generation 4930: Best = 2.002647e+05 FES = 493100 μ_F = 0.804 μ_{CR} = 0.058
 Diversity = 1.3061e+01 k_t = [1, 1, 1, 1]
 Generation 4940: Best = 1.997528e+05 FES = 494100 μ_F = 0.807 μ_{CR} = 0.059
 Diversity = 1.2903e+01 k_t = [4, 4, 2, 4]
 Generation 4950: Best = 1.993130e+05 FES = 495100 μ_F = 0.815 μ_{CR} = 0.053
 Diversity = 1.3357e+01 k_t = [1, 1, 1, 1]
 Generation 4960: Best = 1.990328e+05 FES = 496100 μ_F = 0.808 μ_{CR} = 0.056
 Diversity = 1.3243e+01 k_t = [3, 2, 4, 4]
 Generation 4970: Best = 1.988138e+05 FES = 497100 μ_F = 0.800 μ_{CR} = 0.066
 Diversity = 1.2186e+01 k_t = [1, 1, 1, 1]
 Generation 4980: Best = 1.984634e+05 FES = 498100 μ_F = 0.818 μ_{CR} = 0.058
 Diversity = 1.2814e+01 k_t = [4, 4, 2, 4]
 Generation 4990: Best = 1.980270e+05 FES = 499100 μ_F = 0.825 μ_{CR} = 0.056
 Diversity = 1.3200e+01 k_t = [1, 1, 1, 1]
 Generation 5000: Best = 1.977632e+05 FES = 500100 μ_F = 0.809 μ_{CR} = 0.061
 Diversity = 1.3124e+01 k_t = [2, 4, 3, 2]
 Generation 5010: Best = 1.976070e+05 FES = 501100 μ_F = 0.800 μ_{CR} = 0.059
 Diversity = 1.3127e+01 k_t = [1, 1, 1, 1]
 Generation 5020: Best = 1.974677e+05 FES = 502100 μ_F = 0.788 μ_{CR} = 0.056
 Diversity = 1.1800e+01 k_t = [4, 2, 3, 3]
 Generation 5030: Best = 1.973226e+05 FES = 503100 μ_F = 0.784 μ_{CR} = 0.059
 Diversity = 1.1666e+01 k_t = [1, 1, 1, 1]
 Generation 5040: Best = 1.971727e+05 FES = 504100 μ_F = 0.798 μ_{CR} = 0.054
 Diversity = 1.2500e+01 k_t = [4, 3, 4, 2]
 Generation 5050: Best = 1.969193e+05 FES = 505100 μ_F = 0.798 μ_{CR} = 0.060
 Diversity = 1.2634e+01 k_t = [1, 1, 1, 1]
 Generation 5060: Best = 1.966964e+05 FES = 506100 μ_F = 0.800 μ_{CR} = 0.059
 Diversity = 1.3309e+01 k_t = [4, 2, 4, 2]
 Generation 5070: Best = 1.965672e+05 FES = 507100 μ_F = 0.792 μ_{CR} = 0.061
 Diversity = 1.3257e+01 k_t = [1, 1, 1, 1]
 Generation 5080: Best = 1.963688e+05 FES = 508100 μ_F = 0.796 μ_{CR} = 0.056
 Diversity = 1.1923e+01 k_t = [2, 4, 4, 4]

Generation 5090: Best = 1.961303e+05 FES = 509100 μ_F = 0.811 μ_{CR} = 0.045
 Diversity = 1.3033e+01 k_t = [1, 1, 1, 1]
 Generation 5100: Best = 1.959475e+05 FES = 510100 μ_F = 0.824 μ_{CR} = 0.047
 Diversity = 1.4020e+01 k_t = [3, 3, 4, 2]
 Generation 5110: Best = 1.958889e+05 FES = 511100 μ_F = 0.818 μ_{CR} = 0.050
 Diversity = 1.4199e+01 k_t = [1, 1, 1, 1]
 Generation 5120: Best = 1.957719e+05 FES = 512100 μ_F = 0.808 μ_{CR} = 0.048
 Diversity = 1.3407e+01 k_t = [4, 3, 3, 2]
 Generation 5130: Best = 1.956049e+05 FES = 513100 μ_F = 0.810 μ_{CR} = 0.053
 Diversity = 1.3951e+01 k_t = [1, 1, 1, 1]
 Generation 5140: Best = 1.951095e+05 FES = 514100 μ_F = 0.815 μ_{CR} = 0.049
 Diversity = 1.4601e+01 k_t = [2, 4, 2, 4]
 Generation 5150: Best = 1.950203e+05 FES = 515100 μ_F = 0.820 μ_{CR} = 0.058
 Diversity = 1.3839e+01 k_t = [1, 1, 1, 1]
 Generation 5160: Best = 1.947141e+05 FES = 516100 μ_F = 0.819 μ_{CR} = 0.064
 Diversity = 1.3711e+01 k_t = [3, 3, 4, 3]
 Generation 5170: Best = 1.944627e+05 FES = 517100 μ_F = 0.804 μ_{CR} = 0.056
 Diversity = 1.5159e+01 k_t = [1, 1, 1, 1]
 Generation 5180: Best = 1.940753e+05 FES = 518100 μ_F = 0.813 μ_{CR} = 0.054
 Diversity = 1.5720e+01 k_t = [4, 4, 2, 3]
 Generation 5190: Best = 1.938340e+05 FES = 519100 μ_F = 0.809 μ_{CR} = 0.054
 Diversity = 1.5989e+01 k_t = [1, 1, 1, 1]
 Generation 5200: Best = 1.936331e+05 FES = 520100 μ_F = 0.802 μ_{CR} = 0.055
 Diversity = 1.4178e+01 k_t = [2, 4, 2, 4]
 Generation 5210: Best = 1.933005e+05 FES = 521100 μ_F = 0.797 μ_{CR} = 0.051
 Diversity = 1.3122e+01 k_t = [1, 1, 1, 1]
 Generation 5220: Best = 1.932827e+05 FES = 522100 μ_F = 0.801 μ_{CR} = 0.057
 Diversity = 1.2875e+01 k_t = [4, 4, 2, 2]
 Generation 5230: Best = 1.930299e+05 FES = 523100 μ_F = 0.799 μ_{CR} = 0.052
 Diversity = 1.2089e+01 k_t = [1, 1, 1, 1]
 Generation 5240: Best = 1.929754e+05 FES = 524100 μ_F = 0.778 μ_{CR} = 0.057
 Diversity = 1.0933e+01 k_t = [4, 3, 3, 2]
 Generation 5250: Best = 1.926083e+05 FES = 525100 μ_F = 0.779 μ_{CR} = 0.053
 Diversity = 1.0591e+01 k_t = [1, 1, 1, 1]
 Generation 5260: Best = 1.925115e+05 FES = 526100 μ_F = 0.771 μ_{CR} = 0.055
 Diversity = 1.0918e+01 k_t = [4, 2, 4, 3]
 Generation 5270: Best = 1.922417e+05 FES = 527100 μ_F = 0.787 μ_{CR} = 0.057
 Diversity = 1.0718e+01 k_t = [1, 1, 1, 1]
 Generation 5280: Best = 1.921152e+05 FES = 528100 μ_F = 0.789 μ_{CR} = 0.056
 Diversity = 1.1634e+01 k_t = [4, 4, 2, 3]
 Generation 5290: Best = 1.918848e+05 FES = 529100 μ_F = 0.803 μ_{CR} = 0.058
 Diversity = 1.2385e+01 k_t = [1, 1, 1, 1]
 Generation 5300: Best = 1.916661e+05 FES = 530100 μ_F = 0.801 μ_{CR} = 0.061
 Diversity = 1.1537e+01 k_t = [4, 3, 2, 3]
 Generation 5310: Best = 1.914447e+05 FES = 531100 μ_F = 0.786 μ_{CR} = 0.061
 Diversity = 1.1762e+01 k_t = [1, 1, 1, 1]
 Generation 5320: Best = 1.913312e+05 FES = 532100 μ_F = 0.799 μ_{CR} = 0.060
 Diversity = 1.1128e+01 k_t = [2, 4, 4, 4]
 Generation 5330: Best = 1.912194e+05 FES = 533100 μ_F = 0.804 μ_{CR} = 0.058
 Diversity = 1.1202e+01 k_t = [1, 1, 1, 1]
 Generation 5340: Best = 1.910024e+05 FES = 534100 μ_F = 0.802 μ_{CR} = 0.064
 Diversity = 1.1290e+01 k_t = [2, 4, 4, 4]
 Generation 5350: Best = 1.908759e+05 FES = 535100 μ_F = 0.802 μ_{CR} = 0.059
 Diversity = 1.0046e+01 k_t = [1, 1, 1, 1]

Generation 5360: Best = 1.906623e+05 FES = 536100 μ_F = 0.814 μ_{CR} = 0.058
 Diversity = 1.1307e+01 k_t = [4, 3, 3, 3]
 Generation 5370: Best = 1.904966e+05 FES = 537100 μ_F = 0.799 μ_{CR} = 0.056
 Diversity = 1.0706e+01 k_t = [1, 1, 1, 1]
 Generation 5380: Best = 1.902968e+05 FES = 538100 μ_F = 0.799 μ_{CR} = 0.054
 Diversity = 1.0985e+01 k_t = [3, 4, 4, 2]
 Generation 5390: Best = 1.900910e+05 FES = 539100 μ_F = 0.783 μ_{CR} = 0.062
 Diversity = 1.1436e+01 k_t = [1, 1, 1, 1]
 Generation 5400: Best = 1.898395e+05 FES = 540100 μ_F = 0.786 μ_{CR} = 0.072
 Diversity = 1.0529e+01 k_t = [2, 3, 4, 4]
 Generation 5410: Best = 1.895981e+05 FES = 541100 μ_F = 0.798 μ_{CR} = 0.071
 Diversity = 9.8579e+00 k_t = [1, 1, 1, 1]
 Generation 5420: Best = 1.891039e+05 FES = 542100 μ_F = 0.814 μ_{CR} = 0.067
 Diversity = 1.0518e+01 k_t = [4, 4, 3, 3]
 Generation 5430: Best = 1.889341e+05 FES = 543100 μ_F = 0.810 μ_{CR} = 0.061
 Diversity = 1.1509e+01 k_t = [1, 1, 1, 1]
 Generation 5440: Best = 1.887582e+05 FES = 544100 μ_F = 0.804 μ_{CR} = 0.064
 Diversity = 1.1564e+01 k_t = [4, 3, 4, 4]
 Generation 5450: Best = 1.884889e+05 FES = 545100 μ_F = 0.805 μ_{CR} = 0.063
 Diversity = 1.0663e+01 k_t = [1, 1, 1, 1]
 Generation 5460: Best = 1.882600e+05 FES = 546100 μ_F = 0.799 μ_{CR} = 0.063
 Diversity = 1.0525e+01 k_t = [2, 3, 2, 3]
 Generation 5470: Best = 1.879160e+05 FES = 547100 μ_F = 0.790 μ_{CR} = 0.061
 Diversity = 1.1259e+01 k_t = [1, 1, 1, 1]
 Generation 5480: Best = 1.874896e+05 FES = 548100 μ_F = 0.782 μ_{CR} = 0.056
 Diversity = 1.1047e+01 k_t = [3, 2, 2, 3]
 Generation 5490: Best = 1.873859e+05 FES = 549100 μ_F = 0.798 μ_{CR} = 0.060
 Diversity = 1.0764e+01 k_t = [1, 1, 1, 1]
 Generation 5500: Best = 1.871622e+05 FES = 550100 μ_F = 0.802 μ_{CR} = 0.063
 Diversity = 1.0492e+01 k_t = [3, 4, 2, 2]
 Generation 5510: Best = 1.869067e+05 FES = 551100 μ_F = 0.807 μ_{CR} = 0.060
 Diversity = 1.1431e+01 k_t = [1, 1, 1, 1]
 Generation 5520: Best = 1.866897e+05 FES = 552100 μ_F = 0.815 μ_{CR} = 0.050
 Diversity = 1.1714e+01 k_t = [3, 4, 3, 3]
 Generation 5530: Best = 1.864605e+05 FES = 553100 μ_F = 0.826 μ_{CR} = 0.056
 Diversity = 1.2788e+01 k_t = [1, 1, 1, 1]
 Generation 5540: Best = 1.860727e+05 FES = 554100 μ_F = 0.827 μ_{CR} = 0.060
 Diversity = 1.2927e+01 k_t = [4, 4, 4, 4]
 Generation 5550: Best = 1.859812e+05 FES = 555100 μ_F = 0.821 μ_{CR} = 0.061
 Diversity = 1.2029e+01 k_t = [1, 1, 1, 1]
 Generation 5560: Best = 1.856483e+05 FES = 556100 μ_F = 0.814 μ_{CR} = 0.058
 Diversity = 1.1981e+01 k_t = [3, 4, 2, 2]
 Generation 5570: Best = 1.852029e+05 FES = 557100 μ_F = 0.814 μ_{CR} = 0.062
 Diversity = 1.4129e+01 k_t = [1, 1, 1, 1]
 Generation 5580: Best = 1.850588e+05 FES = 558100 μ_F = 0.813 μ_{CR} = 0.060
 Diversity = 1.3456e+01 k_t = [2, 2, 3, 4]
 Generation 5590: Best = 1.847366e+05 FES = 559100 μ_F = 0.813 μ_{CR} = 0.060
 Diversity = 1.1917e+01 k_t = [1, 1, 1, 1]
 Generation 5600: Best = 1.845856e+05 FES = 560100 μ_F = 0.812 μ_{CR} = 0.056
 Diversity = 1.1303e+01 k_t = [2, 3, 2, 3]
 Generation 5610: Best = 1.843694e+05 FES = 561100 μ_F = 0.803 μ_{CR} = 0.058
 Diversity = 1.0621e+01 k_t = [1, 1, 1, 1]
 Generation 5620: Best = 1.841478e+05 FES = 562100 μ_F = 0.808 μ_{CR} = 0.051
 Diversity = 1.0906e+01 k_t = [3, 2, 2, 2]

Generation 5630: Best = 1.840643e+05 FES = 563100 μ_F = 0.803 μ_{CR} = 0.058
 Diversity = 1.1675e+01 k_t = [1, 1, 1, 1]
 Generation 5640: Best = 1.838942e+05 FES = 564100 μ_F = 0.814 μ_{CR} = 0.059
 Diversity = 1.1917e+01 k_t = [4, 2, 2, 4]
 Generation 5650: Best = 1.836833e+05 FES = 565100 μ_F = 0.816 μ_{CR} = 0.063
 Diversity = 1.2374e+01 k_t = [1, 1, 1, 1]
 Generation 5660: Best = 1.833509e+05 FES = 566100 μ_F = 0.813 μ_{CR} = 0.061
 Diversity = 1.2726e+01 k_t = [2, 4, 2, 3]
 Generation 5670: Best = 1.829426e+05 FES = 567100 μ_F = 0.821 μ_{CR} = 0.064
 Diversity = 1.2968e+01 k_t = [1, 1, 1, 1]
 Generation 5680: Best = 1.826249e+05 FES = 568100 μ_F = 0.805 μ_{CR} = 0.064
 Diversity = 1.2361e+01 k_t = [4, 4, 4, 4]
 Generation 5690: Best = 1.822960e+05 FES = 569100 μ_F = 0.809 μ_{CR} = 0.067
 Diversity = 1.1841e+01 k_t = [1, 1, 1, 1]
 Generation 5700: Best = 1.821081e+05 FES = 570100 μ_F = 0.818 μ_{CR} = 0.063
 Diversity = 1.1755e+01 k_t = [3, 2, 3, 3]
 Generation 5710: Best = 1.818750e+05 FES = 571100 μ_F = 0.813 μ_{CR} = 0.068
 Diversity = 1.2235e+01 k_t = [1, 1, 1, 1]
 Generation 5720: Best = 1.815291e+05 FES = 572100 μ_F = 0.805 μ_{CR} = 0.070
 Diversity = 1.1695e+01 k_t = [3, 2, 3, 4]
 Generation 5730: Best = 1.812904e+05 FES = 573100 μ_F = 0.816 μ_{CR} = 0.063
 Diversity = 1.0673e+01 k_t = [1, 1, 1, 1]
 Generation 5740: Best = 1.807630e+05 FES = 574100 μ_F = 0.829 μ_{CR} = 0.058
 Diversity = 1.0742e+01 k_t = [3, 3, 4, 2]
 Generation 5750: Best = 1.805788e+05 FES = 575100 μ_F = 0.814 μ_{CR} = 0.055
 Diversity = 1.1117e+01 k_t = [1, 1, 1, 1]
 Generation 5760: Best = 1.802318e+05 FES = 576100 μ_F = 0.814 μ_{CR} = 0.048
 Diversity = 1.1614e+01 k_t = [3, 3, 3, 4]
 Generation 5770: Best = 1.797824e+05 FES = 577100 μ_F = 0.808 μ_{CR} = 0.049
 Diversity = 1.1197e+01 k_t = [1, 1, 1, 1]
 Generation 5780: Best = 1.795303e+05 FES = 578100 μ_F = 0.796 μ_{CR} = 0.052
 Diversity = 1.0103e+01 k_t = [2, 3, 2, 4]
 Generation 5790: Best = 1.794857e+05 FES = 579100 μ_F = 0.797 μ_{CR} = 0.054
 Diversity = 9.2216e+00 k_t = [1, 1, 1, 1]
 Generation 5800: Best = 1.794189e+05 FES = 580100 μ_F = 0.809 μ_{CR} = 0.053
 Diversity = 9.1048e+00 k_t = [3, 3, 3, 3]
 Generation 5810: Best = 1.793074e+05 FES = 581100 μ_F = 0.796 μ_{CR} = 0.048
 Diversity = 9.1002e+00 k_t = [1, 1, 1, 1]
 Generation 5820: Best = 1.791112e+05 FES = 582100 μ_F = 0.796 μ_{CR} = 0.046
 Diversity = 9.4759e+00 k_t = [4, 4, 4, 2]
 Generation 5830: Best = 1.790650e+05 FES = 583100 μ_F = 0.806 μ_{CR} = 0.043
 Diversity = 9.9785e+00 k_t = [1, 1, 1, 1]
 Generation 5840: Best = 1.788444e+05 FES = 584100 μ_F = 0.814 μ_{CR} = 0.050
 Diversity = 9.4931e+00 k_t = [4, 3, 3, 4]
 Generation 5850: Best = 1.786340e+05 FES = 585100 μ_F = 0.822 μ_{CR} = 0.051
 Diversity = 1.0023e+01 k_t = [1, 1, 1, 1]
 Generation 5860: Best = 1.783464e+05 FES = 586100 μ_F = 0.819 μ_{CR} = 0.058
 Diversity = 1.1661e+01 k_t = [4, 3, 4, 2]
 Generation 5870: Best = 1.781322e+05 FES = 587100 μ_F = 0.826 μ_{CR} = 0.054
 Diversity = 1.3641e+01 k_t = [1, 1, 1, 1]
 Generation 5880: Best = 1.778167e+05 FES = 588100 μ_F = 0.824 μ_{CR} = 0.055
 Diversity = 1.3069e+01 k_t = [4, 4, 3, 4]
 Generation 5890: Best = 1.774108e+05 FES = 589100 μ_F = 0.820 μ_{CR} = 0.059
 Diversity = 1.2773e+01 k_t = [1, 1, 1, 1]

Generation 5900: Best = 1.772339e+05 FES = 590100 μ_F = 0.818 μ_{CR} = 0.055
 Diversity = 1.1421e+01 k_t = [4, 2, 3, 3]
 Generation 5910: Best = 1.770638e+05 FES = 591100 μ_F = 0.814 μ_{CR} = 0.054
 Diversity = 1.1817e+01 k_t = [1, 1, 1, 1]
 Generation 5920: Best = 1.767910e+05 FES = 592100 μ_F = 0.814 μ_{CR} = 0.052
 Diversity = 1.1440e+01 k_t = [4, 2, 4, 2]
 Generation 5930: Best = 1.765115e+05 FES = 593100 μ_F = 0.805 μ_{CR} = 0.053
 Diversity = 1.1219e+01 k_t = [1, 1, 1, 1]
 Generation 5940: Best = 1.763904e+05 FES = 594100 μ_F = 0.811 μ_{CR} = 0.051
 Diversity = 1.1511e+01 k_t = [3, 2, 2, 4]
 Generation 5950: Best = 1.760034e+05 FES = 595100 μ_F = 0.809 μ_{CR} = 0.055
 Diversity = 1.0838e+01 k_t = [1, 1, 1, 1]
 Generation 5960: Best = 1.758139e+05 FES = 596100 μ_F = 0.810 μ_{CR} = 0.058
 Diversity = 1.0983e+01 k_t = [3, 4, 4, 4]
 Generation 5970: Best = 1.756304e+05 FES = 597100 μ_F = 0.812 μ_{CR} = 0.057
 Diversity = 1.0701e+01 k_t = [1, 1, 1, 1]
 Generation 5980: Best = 1.753817e+05 FES = 598100 μ_F = 0.821 μ_{CR} = 0.060
 Diversity = 1.1767e+01 k_t = [3, 2, 3, 3]
 Generation 5990: Best = 1.751759e+05 FES = 599100 μ_F = 0.815 μ_{CR} = 0.064
 Diversity = 1.1874e+01 k_t = [1, 1, 1, 1]
 Generation 6000: Best = 1.749880e+05 FES = 600100 μ_F = 0.826 μ_{CR} = 0.058
 Diversity = 1.1019e+01 k_t = [3, 4, 4, 3]
 Generation 6010: Best = 1.746019e+05 FES = 601100 μ_F = 0.821 μ_{CR} = 0.059
 Diversity = 1.0867e+01 k_t = [1, 1, 1, 1]
 Generation 6020: Best = 1.743826e+05 FES = 602100 μ_F = 0.821 μ_{CR} = 0.061
 Diversity = 1.1573e+01 k_t = [2, 4, 3, 2]
 Generation 6030: Best = 1.742638e+05 FES = 603100 μ_F = 0.804 μ_{CR} = 0.052
 Diversity = 1.1121e+01 k_t = [1, 1, 1, 1]
 Generation 6040: Best = 1.739949e+05 FES = 604100 μ_F = 0.821 μ_{CR} = 0.049
 Diversity = 1.1219e+01 k_t = [3, 2, 3, 4]
 Generation 6050: Best = 1.739185e+05 FES = 605100 μ_F = 0.825 μ_{CR} = 0.053
 Diversity = 1.1209e+01 k_t = [1, 1, 1, 1]
 Generation 6060: Best = 1.737083e+05 FES = 606100 μ_F = 0.824 μ_{CR} = 0.055
 Diversity = 1.1439e+01 k_t = [4, 4, 2, 3]
 Generation 6070: Best = 1.735195e+05 FES = 607100 μ_F = 0.819 μ_{CR} = 0.058
 Diversity = 1.1736e+01 k_t = [1, 1, 1, 1]
 Generation 6080: Best = 1.733520e+05 FES = 608100 μ_F = 0.804 μ_{CR} = 0.052
 Diversity = 1.0806e+01 k_t = [4, 4, 3, 4]
 Generation 6090: Best = 1.731309e+05 FES = 609100 μ_F = 0.806 μ_{CR} = 0.052
 Diversity = 1.0715e+01 k_t = [1, 1, 1, 1]
 Generation 6100: Best = 1.729676e+05 FES = 610100 μ_F = 0.809 μ_{CR} = 0.063
 Diversity = 9.8344e+00 k_t = [4, 4, 3, 4]
 Generation 6110: Best = 1.728212e+05 FES = 611100 μ_F = 0.817 μ_{CR} = 0.063
 Diversity = 9.1999e+00 k_t = [1, 1, 1, 1]
 Generation 6120: Best = 1.726672e+05 FES = 612100 μ_F = 0.815 μ_{CR} = 0.061
 Diversity = 9.2573e+00 k_t = [4, 3, 3, 3]
 Generation 6130: Best = 1.723845e+05 FES = 613100 μ_F = 0.831 μ_{CR} = 0.057
 Diversity = 1.0619e+01 k_t = [1, 1, 1, 1]
 Generation 6140: Best = 1.722231e+05 FES = 614100 μ_F = 0.826 μ_{CR} = 0.061
 Diversity = 1.0772e+01 k_t = [2, 3, 2, 2]
 Generation 6150: Best = 1.721651e+05 FES = 615100 μ_F = 0.838 μ_{CR} = 0.060
 Diversity = 1.1532e+01 k_t = [1, 1, 1, 1]
 Generation 6160: Best = 1.716351e+05 FES = 616100 μ_F = 0.840 μ_{CR} = 0.062
 Diversity = 1.2734e+01 k_t = [2, 4, 4, 4]

Generation 6170: Best = 1.715502e+05 FES = 617100 μ_F = 0.847 μ_{CR} = 0.054
 Diversity = 1.2450e+01 k_t = [1, 1, 1, 1]
 Generation 6180: Best = 1.713299e+05 FES = 618100 μ_F = 0.845 μ_{CR} = 0.054
 Diversity = 1.2757e+01 k_t = [4, 4, 4, 3]
 Generation 6190: Best = 1.711051e+05 FES = 619100 μ_F = 0.828 μ_{CR} = 0.056
 Diversity = 1.3148e+01 k_t = [1, 1, 1, 1]
 Generation 6200: Best = 1.708246e+05 FES = 620100 μ_F = 0.821 μ_{CR} = 0.049
 Diversity = 1.2562e+01 k_t = [3, 2, 3, 3]
 Generation 6210: Best = 1.705242e+05 FES = 621100 μ_F = 0.815 μ_{CR} = 0.051
 Diversity = 1.2093e+01 k_t = [1, 1, 1, 1]
 Generation 6220: Best = 1.703472e+05 FES = 622100 μ_F = 0.811 μ_{CR} = 0.058
 Diversity = 1.1646e+01 k_t = [4, 2, 2, 3]
 Generation 6230: Best = 1.702373e+05 FES = 623100 μ_F = 0.819 μ_{CR} = 0.054
 Diversity = 1.1849e+01 k_t = [1, 1, 1, 1]
 Generation 6240: Best = 1.700269e+05 FES = 624100 μ_F = 0.815 μ_{CR} = 0.059
 Diversity = 1.2078e+01 k_t = [4, 3, 3, 2]
 Generation 6250: Best = 1.697678e+05 FES = 625100 μ_F = 0.816 μ_{CR} = 0.061
 Diversity = 1.3038e+01 k_t = [1, 1, 1, 1]
 Generation 6260: Best = 1.693977e+05 FES = 626100 μ_F = 0.819 μ_{CR} = 0.057
 Diversity = 1.3381e+01 k_t = [3, 3, 4, 2]
 Generation 6270: Best = 1.691599e+05 FES = 627100 μ_F = 0.827 μ_{CR} = 0.054
 Diversity = 1.3256e+01 k_t = [1, 1, 1, 1]
 Generation 6280: Best = 1.690574e+05 FES = 628100 μ_F = 0.806 μ_{CR} = 0.049
 Diversity = 1.2915e+01 k_t = [3, 4, 2, 2]
 Generation 6290: Best = 1.689501e+05 FES = 629100 μ_F = 0.801 μ_{CR} = 0.054
 Diversity = 1.3176e+01 k_t = [1, 1, 1, 1]
 Generation 6300: Best = 1.687541e+05 FES = 630100 μ_F = 0.810 μ_{CR} = 0.053
 Diversity = 1.3139e+01 k_t = [3, 2, 2, 3]
 Generation 6310: Best = 1.685164e+05 FES = 631100 μ_F = 0.815 μ_{CR} = 0.049
 Diversity = 1.4337e+01 k_t = [1, 1, 1, 1]
 Generation 6320: Best = 1.682251e+05 FES = 632100 μ_F = 0.817 μ_{CR} = 0.056
 Diversity = 1.3681e+01 k_t = [3, 3, 3, 4]
 Generation 6330: Best = 1.680424e+05 FES = 633100 μ_F = 0.830 μ_{CR} = 0.056
 Diversity = 1.2989e+01 k_t = [1, 1, 1, 1]
 Generation 6340: Best = 1.679350e+05 FES = 634100 μ_F = 0.827 μ_{CR} = 0.050
 Diversity = 1.3558e+01 k_t = [2, 2, 2, 4]
 Generation 6350: Best = 1.677476e+05 FES = 635100 μ_F = 0.824 μ_{CR} = 0.048
 Diversity = 1.2742e+01 k_t = [1, 1, 1, 1]
 Generation 6360: Best = 1.676471e+05 FES = 636100 μ_F = 0.825 μ_{CR} = 0.046
 Diversity = 1.3697e+01 k_t = [4, 2, 2, 2]
 Generation 6370: Best = 1.674727e+05 FES = 637100 μ_F = 0.808 μ_{CR} = 0.052
 Diversity = 1.4056e+01 k_t = [1, 1, 1, 1]
 Generation 6380: Best = 1.673744e+05 FES = 638100 μ_F = 0.796 μ_{CR} = 0.056
 Diversity = 1.3489e+01 k_t = [3, 4, 2, 4]
 Generation 6390: Best = 1.671752e+05 FES = 639100 μ_F = 0.803 μ_{CR} = 0.051
 Diversity = 1.3138e+01 k_t = [1, 1, 1, 1]
 Generation 6400: Best = 1.671590e+05 FES = 640100 μ_F = 0.810 μ_{CR} = 0.051
 Diversity = 1.2811e+01 k_t = [3, 4, 2, 3]
 Generation 6410: Best = 1.669337e+05 FES = 641100 μ_F = 0.814 μ_{CR} = 0.051
 Diversity = 1.3541e+01 k_t = [1, 1, 1, 1]
 Generation 6420: Best = 1.667874e+05 FES = 642100 μ_F = 0.814 μ_{CR} = 0.052
 Diversity = 1.2344e+01 k_t = [2, 2, 2, 3]
 Generation 6430: Best = 1.667470e+05 FES = 643100 μ_F = 0.821 μ_{CR} = 0.043
 Diversity = 1.2892e+01 k_t = [1, 1, 1, 1]

Generation 6440: Best = 1.666031e+05 FES = 644100 μ_F = 0.832 μ_{CR} = 0.049
 Diversity = 1.2526e+01 k_t = [4, 3, 2, 4]
 Generation 6450: Best = 1.665612e+05 FES = 645100 μ_F = 0.830 μ_{CR} = 0.052
 Diversity = 1.2619e+01 k_t = [1, 1, 1, 1]
 Generation 6460: Best = 1.665001e+05 FES = 646100 μ_F = 0.826 μ_{CR} = 0.054
 Diversity = 1.2258e+01 k_t = [3, 2, 2, 4]
 Generation 6470: Best = 1.663174e+05 FES = 647100 μ_F = 0.826 μ_{CR} = 0.049
 Diversity = 1.2100e+01 k_t = [1, 1, 1, 1]
 Generation 6480: Best = 1.662310e+05 FES = 648100 μ_F = 0.838 μ_{CR} = 0.057
 Diversity = 1.2921e+01 k_t = [2, 2, 2, 2]
 Generation 6490: Best = 1.660892e+05 FES = 649100 μ_F = 0.828 μ_{CR} = 0.051
 Diversity = 1.2649e+01 k_t = [1, 1, 1, 1]
 Generation 6500: Best = 1.659287e+05 FES = 650100 μ_F = 0.829 μ_{CR} = 0.053
 Diversity = 1.2660e+01 k_t = [4, 2, 4, 4]
 Generation 6510: Best = 1.658129e+05 FES = 651100 μ_F = 0.821 μ_{CR} = 0.050
 Diversity = 1.2315e+01 k_t = [1, 1, 1, 1]
 Generation 6520: Best = 1.656890e+05 FES = 652100 μ_F = 0.813 μ_{CR} = 0.051
 Diversity = 1.1597e+01 k_t = [3, 3, 2, 2]
 Generation 6530: Best = 1.654894e+05 FES = 653100 μ_F = 0.793 μ_{CR} = 0.047
 Diversity = 1.2366e+01 k_t = [1, 1, 1, 1]
 Generation 6540: Best = 1.652691e+05 FES = 654100 μ_F = 0.814 μ_{CR} = 0.055
 Diversity = 1.2402e+01 k_t = [3, 4, 3, 3]
 Generation 6550: Best = 1.650898e+05 FES = 655100 μ_F = 0.811 μ_{CR} = 0.061
 Diversity = 1.3018e+01 k_t = [1, 1, 1, 1]
 Generation 6560: Best = 1.647883e+05 FES = 656100 μ_F = 0.788 μ_{CR} = 0.064
 Diversity = 1.2993e+01 k_t = [4, 2, 4, 2]
 Generation 6570: Best = 1.645685e+05 FES = 657100 μ_F = 0.802 μ_{CR} = 0.060
 Diversity = 1.4459e+01 k_t = [1, 1, 1, 1]
 Generation 6580: Best = 1.642072e+05 FES = 658100 μ_F = 0.810 μ_{CR} = 0.057
 Diversity = 1.5226e+01 k_t = [4, 3, 2, 2]
 Generation 6590: Best = 1.638362e+05 FES = 659100 μ_F = 0.801 μ_{CR} = 0.057
 Diversity = 1.5241e+01 k_t = [1, 1, 1, 1]
 Generation 6600: Best = 1.635045e+05 FES = 660100 μ_F = 0.800 μ_{CR} = 0.059
 Diversity = 1.4488e+01 k_t = [3, 2, 3, 4]
 Generation 6610: Best = 1.634623e+05 FES = 661100 μ_F = 0.796 μ_{CR} = 0.054
 Diversity = 1.3526e+01 k_t = [1, 1, 1, 1]
 Generation 6620: Best = 1.631746e+05 FES = 662100 μ_F = 0.803 μ_{CR} = 0.061
 Diversity = 1.2777e+01 k_t = [2, 4, 2, 3]
 Generation 6630: Best = 1.631632e+05 FES = 663100 μ_F = 0.823 μ_{CR} = 0.056
 Diversity = 1.2883e+01 k_t = [1, 1, 1, 1]
 Generation 6640: Best = 1.630398e+05 FES = 664100 μ_F = 0.817 μ_{CR} = 0.051
 Diversity = 1.2303e+01 k_t = [4, 2, 4, 3]
 Generation 6650: Best = 1.629192e+05 FES = 665100 μ_F = 0.809 μ_{CR} = 0.047
 Diversity = 1.2075e+01 k_t = [1, 1, 1, 1]
 Generation 6660: Best = 1.628269e+05 FES = 666100 μ_F = 0.795 μ_{CR} = 0.050
 Diversity = 1.1335e+01 k_t = [2, 4, 2, 4]
 Generation 6670: Best = 1.626150e+05 FES = 667100 μ_F = 0.803 μ_{CR} = 0.061
 Diversity = 1.0973e+01 k_t = [1, 1, 1, 1]
 Generation 6680: Best = 1.624620e+05 FES = 668100 μ_F = 0.803 μ_{CR} = 0.059
 Diversity = 1.0864e+01 k_t = [3, 4, 2, 4]
 Generation 6690: Best = 1.623268e+05 FES = 669100 μ_F = 0.808 μ_{CR} = 0.054
 Diversity = 1.1530e+01 k_t = [1, 1, 1, 1]
 Generation 6700: Best = 1.622270e+05 FES = 670100 μ_F = 0.825 μ_{CR} = 0.052
 Diversity = 1.1980e+01 k_t = [2, 3, 3, 3]

Generation 6710: Best = 1.619662e+05 FES = 671100 μ_F = 0.829 μ_{CR} = 0.061
 Diversity = 1.2906e+01 k_t = [1, 1, 1, 1]
 Generation 6720: Best = 1.615336e+05 FES = 672100 μ_F = 0.831 μ_{CR} = 0.066
 Diversity = 1.3882e+01 k_t = [2, 2, 4, 2]
 Generation 6730: Best = 1.614233e+05 FES = 673100 μ_F = 0.830 μ_{CR} = 0.057
 Diversity = 1.4092e+01 k_t = [1, 1, 1, 1]
 Generation 6740: Best = 1.610846e+05 FES = 674100 μ_F = 0.842 μ_{CR} = 0.052
 Diversity = 1.4426e+01 k_t = [3, 2, 3, 3]
 Generation 6750: Best = 1.609520e+05 FES = 675100 μ_F = 0.833 μ_{CR} = 0.057
 Diversity = 1.3295e+01 k_t = [1, 1, 1, 1]
 Generation 6760: Best = 1.607287e+05 FES = 676100 μ_F = 0.809 μ_{CR} = 0.054
 Diversity = 1.3383e+01 k_t = [3, 3, 2, 2]
 Generation 6770: Best = 1.605026e+05 FES = 677100 μ_F = 0.821 μ_{CR} = 0.057
 Diversity = 1.4464e+01 k_t = [1, 1, 1, 1]
 Generation 6780: Best = 1.603166e+05 FES = 678100 μ_F = 0.814 μ_{CR} = 0.065
 Diversity = 1.3756e+01 k_t = [4, 3, 4, 2]
 Generation 6790: Best = 1.601228e+05 FES = 679100 μ_F = 0.805 μ_{CR} = 0.070
 Diversity = 1.3926e+01 k_t = [1, 1, 1, 1]
 Generation 6800: Best = 1.598862e+05 FES = 680100 μ_F = 0.799 μ_{CR} = 0.061
 Diversity = 1.3062e+01 k_t = [2, 4, 2, 3]
 Generation 6810: Best = 1.595860e+05 FES = 681100 μ_F = 0.811 μ_{CR} = 0.067
 Diversity = 1.3159e+01 k_t = [1, 1, 1, 1]
 Generation 6820: Best = 1.592993e+05 FES = 682100 μ_F = 0.810 μ_{CR} = 0.066
 Diversity = 1.4441e+01 k_t = [4, 3, 3, 2]
 Generation 6830: Best = 1.590533e+05 FES = 683100 μ_F = 0.804 μ_{CR} = 0.064
 Diversity = 1.5064e+01 k_t = [1, 1, 1, 1]
 Generation 6840: Best = 1.588544e+05 FES = 684100 μ_F = 0.817 μ_{CR} = 0.065
 Diversity = 1.4740e+01 k_t = [3, 4, 3, 2]
 Generation 6850: Best = 1.587127e+05 FES = 685100 μ_F = 0.814 μ_{CR} = 0.056
 Diversity = 1.4800e+01 k_t = [1, 1, 1, 1]
 Generation 6860: Best = 1.585963e+05 FES = 686100 μ_F = 0.810 μ_{CR} = 0.057
 Diversity = 1.4023e+01 k_t = [3, 4, 4, 2]
 Generation 6870: Best = 1.583416e+05 FES = 687100 μ_F = 0.804 μ_{CR} = 0.062
 Diversity = 1.4300e+01 k_t = [1, 1, 1, 1]
 Generation 6880: Best = 1.582281e+05 FES = 688100 μ_F = 0.809 μ_{CR} = 0.059
 Diversity = 1.3086e+01 k_t = [3, 2, 2, 4]
 Generation 6890: Best = 1.580717e+05 FES = 689100 μ_F = 0.808 μ_{CR} = 0.056
 Diversity = 1.2745e+01 k_t = [1, 1, 1, 1]
 Generation 6900: Best = 1.578689e+05 FES = 690100 μ_F = 0.804 μ_{CR} = 0.058
 Diversity = 1.2509e+01 k_t = [3, 4, 2, 3]
 Generation 6910: Best = 1.575928e+05 FES = 691100 μ_F = 0.797 μ_{CR} = 0.061
 Diversity = 1.2785e+01 k_t = [1, 1, 1, 1]
 Generation 6920: Best = 1.574510e+05 FES = 692100 μ_F = 0.812 μ_{CR} = 0.059
 Diversity = 1.2967e+01 k_t = [4, 4, 2, 2]
 Generation 6930: Best = 1.570696e+05 FES = 693100 μ_F = 0.812 μ_{CR} = 0.058
 Diversity = 1.3295e+01 k_t = [1, 1, 1, 1]
 Generation 6940: Best = 1.569989e+05 FES = 694100 μ_F = 0.815 μ_{CR} = 0.054
 Diversity = 1.3142e+01 k_t = [4, 4, 2, 3]
 Generation 6950: Best = 1.566863e+05 FES = 695100 μ_F = 0.812 μ_{CR} = 0.053
 Diversity = 1.3554e+01 k_t = [1, 1, 1, 1]
 Generation 6960: Best = 1.565188e+05 FES = 696100 μ_F = 0.821 μ_{CR} = 0.049
 Diversity = 1.3978e+01 k_t = [4, 2, 3, 2]
 Generation 6970: Best = 1.560532e+05 FES = 697100 μ_F = 0.831 μ_{CR} = 0.058
 Diversity = 1.4415e+01 k_t = [1, 1, 1, 1]

Generation 6980: Best = 1.557540e+05 FES = 698100 μ_F = 0.835 μ_{CR} = 0.062
 Diversity = 1.5108e+01 k_t = [4, 4, 2, 3]
 Generation 6990: Best = 1.553671e+05 FES = 699100 μ_F = 0.825 μ_{CR} = 0.069
 Diversity = 1.4406e+01 k_t = [1, 1, 1, 1]
 Generation 7000: Best = 1.549782e+05 FES = 700100 μ_F = 0.821 μ_{CR} = 0.061
 Diversity = 1.3433e+01 k_t = [2, 4, 4, 4]
 Generation 7010: Best = 1.548496e+05 FES = 701100 μ_F = 0.827 μ_{CR} = 0.054
 Diversity = 1.2602e+01 k_t = [1, 1, 1, 1]
 Generation 7020: Best = 1.544125e+05 FES = 702100 μ_F = 0.832 μ_{CR} = 0.060
 Diversity = 1.3415e+01 k_t = [4, 3, 2, 2]
 Generation 7030: Best = 1.538047e+05 FES = 703100 μ_F = 0.822 μ_{CR} = 0.056
 Diversity = 1.4538e+01 k_t = [1, 1, 1, 1]
 Generation 7040: Best = 1.536099e+05 FES = 704100 μ_F = 0.820 μ_{CR} = 0.056
 Diversity = 1.3196e+01 k_t = [4, 3, 2, 2]
 Generation 7050: Best = 1.530448e+05 FES = 705100 μ_F = 0.824 μ_{CR} = 0.053
 Diversity = 1.3089e+01 k_t = [1, 1, 1, 1]
 Generation 7060: Best = 1.527370e+05 FES = 706100 μ_F = 0.822 μ_{CR} = 0.057
 Diversity = 1.2194e+01 k_t = [2, 3, 2, 4]
 Generation 7070: Best = 1.525964e+05 FES = 707100 μ_F = 0.827 μ_{CR} = 0.051
 Diversity = 1.3047e+01 k_t = [1, 1, 1, 1]
 Generation 7080: Best = 1.521634e+05 FES = 708100 μ_F = 0.821 μ_{CR} = 0.057
 Diversity = 1.2952e+01 k_t = [3, 3, 2, 4]
 Generation 7090: Best = 1.519350e+05 FES = 709100 μ_F = 0.807 μ_{CR} = 0.051
 Diversity = 1.2830e+01 k_t = [1, 1, 1, 1]
 Generation 7100: Best = 1.517573e+05 FES = 710100 μ_F = 0.811 μ_{CR} = 0.055
 Diversity = 1.2155e+01 k_t = [3, 4, 4, 4]
 Generation 7110: Best = 1.516059e+05 FES = 711100 μ_F = 0.821 μ_{CR} = 0.058
 Diversity = 1.1922e+01 k_t = [1, 1, 1, 1]
 Generation 7120: Best = 1.512654e+05 FES = 712100 μ_F = 0.824 μ_{CR} = 0.052
 Diversity = 1.1792e+01 k_t = [4, 4, 4, 3]
 Generation 7130: Best = 1.510001e+05 FES = 713100 μ_F = 0.815 μ_{CR} = 0.062
 Diversity = 1.1914e+01 k_t = [1, 1, 1, 1]
 Generation 7140: Best = 1.506777e+05 FES = 714100 μ_F = 0.806 μ_{CR} = 0.062
 Diversity = 1.2262e+01 k_t = [4, 2, 2, 2]
 Generation 7150: Best = 1.505596e+05 FES = 715100 μ_F = 0.808 μ_{CR} = 0.052
 Diversity = 1.2923e+01 k_t = [1, 1, 1, 1]
 Generation 7160: Best = 1.502725e+05 FES = 716100 μ_F = 0.801 μ_{CR} = 0.054
 Diversity = 1.2421e+01 k_t = [4, 4, 3, 2]
 Generation 7170: Best = 1.501696e+05 FES = 717100 μ_F = 0.807 μ_{CR} = 0.059
 Diversity = 1.1534e+01 k_t = [1, 1, 1, 1]
 Generation 7180: Best = 1.499445e+05 FES = 718100 μ_F = 0.812 μ_{CR} = 0.051
 Diversity = 1.1838e+01 k_t = [4, 3, 2, 2]
 Generation 7190: Best = 1.497492e+05 FES = 719100 μ_F = 0.827 μ_{CR} = 0.050
 Diversity = 1.2864e+01 k_t = [1, 1, 1, 1]
 Generation 7200: Best = 1.494211e+05 FES = 720100 μ_F = 0.825 μ_{CR} = 0.054
 Diversity = 1.2262e+01 k_t = [4, 4, 4, 3]
 Generation 7210: Best = 1.492929e+05 FES = 721100 μ_F = 0.828 μ_{CR} = 0.053
 Diversity = 1.2043e+01 k_t = [1, 1, 1, 1]
 Generation 7220: Best = 1.491513e+05 FES = 722100 μ_F = 0.816 μ_{CR} = 0.049
 Diversity = 1.1741e+01 k_t = [3, 3, 4, 2]
 Generation 7230: Best = 1.488044e+05 FES = 723100 μ_F = 0.805 μ_{CR} = 0.049
 Diversity = 1.2506e+01 k_t = [1, 1, 1, 1]
 Generation 7240: Best = 1.487674e+05 FES = 724100 μ_F = 0.800 μ_{CR} = 0.051
 Diversity = 1.1427e+01 k_t = [3, 4, 3, 3]

Generation 7250: Best = 1.486422e+05 FES = 725100 μ_F = 0.788 μ_{CR} = 0.048
 Diversity = 1.0597e+01 k_t = [1, 1, 1, 1]
 Generation 7260: Best = 1.484496e+05 FES = 726100 μ_F = 0.801 μ_{CR} = 0.060
 Diversity = 1.1326e+01 k_t = [3, 3, 2, 2]
 Generation 7270: Best = 1.484202e+05 FES = 727100 μ_F = 0.816 μ_{CR} = 0.048
 Diversity = 1.2142e+01 k_t = [1, 1, 1, 1]
 Generation 7280: Best = 1.482518e+05 FES = 728100 μ_F = 0.813 μ_{CR} = 0.045
 Diversity = 1.1921e+01 k_t = [2, 2, 3, 2]
 Generation 7290: Best = 1.480739e+05 FES = 729100 μ_F = 0.814 μ_{CR} = 0.051
 Diversity = 1.2484e+01 k_t = [1, 1, 1, 1]
 Generation 7300: Best = 1.479034e+05 FES = 730100 μ_F = 0.820 μ_{CR} = 0.054
 Diversity = 1.1628e+01 k_t = [2, 2, 3, 3]
 Generation 7310: Best = 1.475759e+05 FES = 731100 μ_F = 0.834 μ_{CR} = 0.047
 Diversity = 1.1431e+01 k_t = [1, 1, 1, 1]
 Generation 7320: Best = 1.474672e+05 FES = 732100 μ_F = 0.818 μ_{CR} = 0.051
 Diversity = 1.0855e+01 k_t = [4, 4, 2, 4]
 Generation 7330: Best = 1.473007e+05 FES = 733100 μ_F = 0.813 μ_{CR} = 0.049
 Diversity = 1.0627e+01 k_t = [1, 1, 1, 1]
 Generation 7340: Best = 1.470900e+05 FES = 734100 μ_F = 0.799 μ_{CR} = 0.052
 Diversity = 1.0449e+01 k_t = [3, 3, 3, 2]
 Generation 7350: Best = 1.469790e+05 FES = 735100 μ_F = 0.805 μ_{CR} = 0.051
 Diversity = 1.0759e+01 k_t = [1, 1, 1, 1]
 Generation 7360: Best = 1.467574e+05 FES = 736100 μ_F = 0.800 μ_{CR} = 0.057
 Diversity = 1.0392e+01 k_t = [2, 4, 4, 3]
 Generation 7370: Best = 1.466517e+05 FES = 737100 μ_F = 0.805 μ_{CR} = 0.045
 Diversity = 1.1075e+01 k_t = [1, 1, 1, 1]
 Generation 7380: Best = 1.465141e+05 FES = 738100 μ_F = 0.814 μ_{CR} = 0.052
 Diversity = 1.0826e+01 k_t = [3, 2, 4, 3]
 Generation 7390: Best = 1.462582e+05 FES = 739100 μ_F = 0.825 μ_{CR} = 0.049
 Diversity = 1.1352e+01 k_t = [1, 1, 1, 1]
 Generation 7400: Best = 1.458813e+05 FES = 740100 μ_F = 0.831 μ_{CR} = 0.057
 Diversity = 1.2488e+01 k_t = [3, 2, 4, 3]
 Generation 7410: Best = 1.456215e+05 FES = 741100 μ_F = 0.828 μ_{CR} = 0.050
 Diversity = 1.2361e+01 k_t = [1, 1, 1, 1]
 Generation 7420: Best = 1.451342e+05 FES = 742100 μ_F = 0.833 μ_{CR} = 0.051
 Diversity = 1.3408e+01 k_t = [2, 3, 2, 4]
 Generation 7430: Best = 1.449312e+05 FES = 743100 μ_F = 0.818 μ_{CR} = 0.046
 Diversity = 1.2057e+01 k_t = [1, 1, 1, 1]
 Generation 7440: Best = 1.448617e+05 FES = 744100 μ_F = 0.827 μ_{CR} = 0.039
 Diversity = 1.1992e+01 k_t = [4, 2, 4, 2]
 Generation 7450: Best = 1.446981e+05 FES = 745100 μ_F = 0.818 μ_{CR} = 0.041
 Diversity = 1.1430e+01 k_t = [1, 1, 1, 1]
 Generation 7460: Best = 1.444587e+05 FES = 746100 μ_F = 0.809 μ_{CR} = 0.053
 Diversity = 1.1019e+01 k_t = [2, 2, 3, 2]
 Generation 7470: Best = 1.443074e+05 FES = 747100 μ_F = 0.811 μ_{CR} = 0.042
 Diversity = 1.2637e+01 k_t = [1, 1, 1, 1]
 Generation 7480: Best = 1.440920e+05 FES = 748100 μ_F = 0.808 μ_{CR} = 0.043
 Diversity = 1.2956e+01 k_t = [3, 3, 3, 2]
 Generation 7490: Best = 1.438535e+05 FES = 749100 μ_F = 0.808 μ_{CR} = 0.049
 Diversity = 1.2792e+01 k_t = [1, 1, 1, 1]
 Generation 7500: Best = 1.436714e+05 FES = 750100 μ_F = 0.803 μ_{CR} = 0.043
 Diversity = 1.1624e+01 k_t = [2, 4, 4, 3]
 Generation 7510: Best = 1.433720e+05 FES = 751100 μ_F = 0.811 μ_{CR} = 0.041
 Diversity = 1.1484e+01 k_t = [1, 1, 1, 1]

Generation 7520: Best = 1.428950e+05 FES = 752100 μ_F = 0.828 μ_{CR} = 0.039
 Diversity = 1.1858e+01 k_t = [2, 3, 3, 2]
 Generation 7530: Best = 1.428008e+05 FES = 753100 μ_F = 0.840 μ_{CR} = 0.046
 Diversity = 1.0495e+01 k_t = [1, 1, 1, 1]
 Generation 7540: Best = 1.421683e+05 FES = 754100 μ_F = 0.836 μ_{CR} = 0.048
 Diversity = 1.0611e+01 k_t = [4, 3, 2, 2]
 Generation 7550: Best = 1.420755e+05 FES = 755100 μ_F = 0.837 μ_{CR} = 0.052
 Diversity = 1.1535e+01 k_t = [1, 1, 1, 1]
 Generation 7560: Best = 1.419290e+05 FES = 756100 μ_F = 0.831 μ_{CR} = 0.046
 Diversity = 1.1897e+01 k_t = [3, 2, 4, 2]
 Generation 7570: Best = 1.417853e+05 FES = 757100 μ_F = 0.828 μ_{CR} = 0.047
 Diversity = 1.2215e+01 k_t = [1, 1, 1, 1]
 Generation 7580: Best = 1.413838e+05 FES = 758100 μ_F = 0.813 μ_{CR} = 0.049
 Diversity = 1.1443e+01 k_t = [4, 4, 4, 4]
 Generation 7590: Best = 1.409220e+05 FES = 759100 μ_F = 0.813 μ_{CR} = 0.050
 Diversity = 1.0633e+01 k_t = [1, 1, 1, 1]
 Generation 7600: Best = 1.409220e+05 FES = 760100 μ_F = 0.823 μ_{CR} = 0.056
 Diversity = 1.1283e+01 k_t = [2, 3, 3, 3]
 Generation 7610: Best = 1.406967e+05 FES = 761100 μ_F = 0.832 μ_{CR} = 0.057
 Diversity = 1.1283e+01 k_t = [1, 1, 1, 1]
 Generation 7620: Best = 1.402108e+05 FES = 762100 μ_F = 0.823 μ_{CR} = 0.052
 Diversity = 1.0880e+01 k_t = [3, 3, 4, 3]
 Generation 7630: Best = 1.400141e+05 FES = 763100 μ_F = 0.834 μ_{CR} = 0.050
 Diversity = 1.0918e+01 k_t = [1, 1, 1, 1]
 Generation 7640: Best = 1.395223e+05 FES = 764100 μ_F = 0.838 μ_{CR} = 0.047
 Diversity = 1.2860e+01 k_t = [2, 3, 3, 2]
 Generation 7650: Best = 1.392350e+05 FES = 765100 μ_F = 0.832 μ_{CR} = 0.046
 Diversity = 1.3139e+01 k_t = [1, 1, 1, 1]
 Generation 7660: Best = 1.389763e+05 FES = 766100 μ_F = 0.832 μ_{CR} = 0.046
 Diversity = 1.3371e+01 k_t = [4, 4, 2, 2]
 Generation 7670: Best = 1.387082e+05 FES = 767100 μ_F = 0.802 μ_{CR} = 0.049
 Diversity = 1.3625e+01 k_t = [1, 1, 1, 1]
 Generation 7680: Best = 1.384892e+05 FES = 768100 μ_F = 0.803 μ_{CR} = 0.045
 Diversity = 1.3129e+01 k_t = [2, 4, 2, 4]
 Generation 7690: Best = 1.382352e+05 FES = 769100 μ_F = 0.807 μ_{CR} = 0.045
 Diversity = 1.2332e+01 k_t = [1, 1, 1, 1]
 Generation 7700: Best = 1.382090e+05 FES = 770100 μ_F = 0.808 μ_{CR} = 0.041
 Diversity = 1.1193e+01 k_t = [2, 4, 4, 4]
 Generation 7710: Best = 1.379577e+05 FES = 771100 μ_F = 0.803 μ_{CR} = 0.046
 Diversity = 1.0866e+01 k_t = [1, 1, 1, 1]
 Generation 7720: Best = 1.378227e+05 FES = 772100 μ_F = 0.807 μ_{CR} = 0.051
 Diversity = 1.2169e+01 k_t = [3, 2, 2, 2]
 Generation 7730: Best = 1.376721e+05 FES = 773100 μ_F = 0.800 μ_{CR} = 0.052
 Diversity = 1.2047e+01 k_t = [1, 1, 1, 1]
 Generation 7740: Best = 1.374034e+05 FES = 774100 μ_F = 0.812 μ_{CR} = 0.053
 Diversity = 1.2566e+01 k_t = [2, 2, 2, 4]
 Generation 7750: Best = 1.373295e+05 FES = 775100 μ_F = 0.814 μ_{CR} = 0.048
 Diversity = 1.1624e+01 k_t = [1, 1, 1, 1]
 Generation 7760: Best = 1.371448e+05 FES = 776100 μ_F = 0.811 μ_{CR} = 0.050
 Diversity = 1.0816e+01 k_t = [2, 4, 2, 4]
 Generation 7770: Best = 1.370403e+05 FES = 777100 μ_F = 0.810 μ_{CR} = 0.051
 Diversity = 9.8080e+00 k_t = [1, 1, 1, 1]
 Generation 7780: Best = 1.369726e+05 FES = 778100 μ_F = 0.815 μ_{CR} = 0.051
 Diversity = 9.2317e+00 k_t = [4, 2, 3, 4]

Generation 7790: Best = 1.368211e+05 FES = 779100 μ_F = 0.801 μ_{CR} = 0.050
 Diversity = 9.6924e+00 k_t = [1, 1, 1, 1]
 Generation 7800: Best = 1.365931e+05 FES = 780100 μ_F = 0.815 μ_{CR} = 0.043
 Diversity = 1.0479e+01 k_t = [4, 2, 4, 2]
 Generation 7810: Best = 1.363381e+05 FES = 781100 μ_F = 0.813 μ_{CR} = 0.048
 Diversity = 1.1241e+01 k_t = [1, 1, 1, 1]
 Generation 7820: Best = 1.362114e+05 FES = 782100 μ_F = 0.823 μ_{CR} = 0.048
 Diversity = 1.1749e+01 k_t = [4, 2, 3, 4]
 Generation 7830: Best = 1.359619e+05 FES = 783100 μ_F = 0.825 μ_{CR} = 0.047
 Diversity = 1.2569e+01 k_t = [1, 1, 1, 1]
 Generation 7840: Best = 1.356835e+05 FES = 784100 μ_F = 0.824 μ_{CR} = 0.048
 Diversity = 1.2911e+01 k_t = [4, 3, 3, 3]
 Generation 7850: Best = 1.355252e+05 FES = 785100 μ_F = 0.830 μ_{CR} = 0.043
 Diversity = 1.2835e+01 k_t = [1, 1, 1, 1]
 Generation 7860: Best = 1.351465e+05 FES = 786100 μ_F = 0.814 μ_{CR} = 0.051
 Diversity = 1.1345e+01 k_t = [4, 2, 4, 2]
 Generation 7870: Best = 1.350054e+05 FES = 787100 μ_F = 0.819 μ_{CR} = 0.055
 Diversity = 1.1745e+01 k_t = [1, 1, 1, 1]
 Generation 7880: Best = 1.348935e+05 FES = 788100 μ_F = 0.830 μ_{CR} = 0.042
 Diversity = 1.1646e+01 k_t = [2, 2, 4, 4]
 Generation 7890: Best = 1.347339e+05 FES = 789100 μ_F = 0.810 μ_{CR} = 0.044
 Diversity = 1.1064e+01 k_t = [1, 1, 1, 1]
 Generation 7900: Best = 1.344338e+05 FES = 790100 μ_F = 0.817 μ_{CR} = 0.044
 Diversity = 1.1603e+01 k_t = [4, 2, 4, 2]
 Generation 7910: Best = 1.342361e+05 FES = 791100 μ_F = 0.819 μ_{CR} = 0.044
 Diversity = 1.2468e+01 k_t = [1, 1, 1, 1]
 Generation 7920: Best = 1.340002e+05 FES = 792100 μ_F = 0.817 μ_{CR} = 0.043
 Diversity = 1.1258e+01 k_t = [2, 4, 3, 3]
 Generation 7930: Best = 1.338617e+05 FES = 793100 μ_F = 0.823 μ_{CR} = 0.039
 Diversity = 1.0139e+01 k_t = [1, 1, 1, 1]
 Generation 7940: Best = 1.337075e+05 FES = 794100 μ_F = 0.815 μ_{CR} = 0.044
 Diversity = 9.4639e+00 k_t = [3, 4, 3, 3]
 Generation 7950: Best = 1.336099e+05 FES = 795100 μ_F = 0.805 μ_{CR} = 0.046
 Diversity = 8.9822e+00 k_t = [1, 1, 1, 1]
 Generation 7960: Best = 1.334850e+05 FES = 796100 μ_F = 0.813 μ_{CR} = 0.049
 Diversity = 8.8331e+00 k_t = [4, 4, 3, 2]
 Generation 7970: Best = 1.332349e+05 FES = 797100 μ_F = 0.834 μ_{CR} = 0.049
 Diversity = 1.0737e+01 k_t = [1, 1, 1, 1]
 Generation 7980: Best = 1.328440e+05 FES = 798100 μ_F = 0.833 μ_{CR} = 0.050
 Diversity = 1.0993e+01 k_t = [2, 4, 4, 3]
 Generation 7990: Best = 1.325992e+05 FES = 799100 μ_F = 0.819 μ_{CR} = 0.050
 Diversity = 1.0972e+01 k_t = [1, 1, 1, 1]
 Generation 8000: Best = 1.324756e+05 FES = 800100 μ_F = 0.804 μ_{CR} = 0.056
 Diversity = 1.0876e+01 k_t = [4, 3, 2, 2]
 Generation 8010: Best = 1.323281e+05 FES = 801100 μ_F = 0.808 μ_{CR} = 0.049
 Diversity = 9.8190e+00 k_t = [1, 1, 1, 1]
 Generation 8020: Best = 1.322273e+05 FES = 802100 μ_F = 0.802 μ_{CR} = 0.050
 Diversity = 9.0098e+00 k_t = [3, 4, 2, 4]
 Generation 8030: Best = 1.321443e+05 FES = 803100 μ_F = 0.804 μ_{CR} = 0.050
 Diversity = 8.8765e+00 k_t = [1, 1, 1, 1]
 Generation 8040: Best = 1.320545e+05 FES = 804100 μ_F = 0.793 μ_{CR} = 0.049
 Diversity = 8.6167e+00 k_t = [2, 4, 4, 2]
 Generation 8050: Best = 1.318827e+05 FES = 805100 μ_F = 0.799 μ_{CR} = 0.045
 Diversity = 8.5368e+00 k_t = [1, 1, 1, 1]

Generation 8060: Best = 1.318099e+05 FES = 806100 μ_F = 0.811 μ_{CR} = 0.048
 Diversity = 9.2902e+00 k_t = [2, 2, 3, 2]
 Generation 8070: Best = 1.317299e+05 FES = 807100 μ_F = 0.805 μ_{CR} = 0.053
 Diversity = 8.9439e+00 k_t = [1, 1, 1, 1]
 Generation 8080: Best = 1.316111e+05 FES = 808100 μ_F = 0.812 μ_{CR} = 0.050
 Diversity = 8.8044e+00 k_t = [2, 3, 3, 2]
 Generation 8090: Best = 1.315150e+05 FES = 809100 μ_F = 0.822 μ_{CR} = 0.051
 Diversity = 9.5133e+00 k_t = [1, 1, 1, 1]
 Generation 8100: Best = 1.313993e+05 FES = 810100 μ_F = 0.814 μ_{CR} = 0.049
 Diversity = 9.2697e+00 k_t = [2, 4, 3, 2]
 Generation 8110: Best = 1.313128e+05 FES = 811100 μ_F = 0.824 μ_{CR} = 0.046
 Diversity = 9.3732e+00 k_t = [1, 1, 1, 1]
 Generation 8120: Best = 1.311793e+05 FES = 812100 μ_F = 0.823 μ_{CR} = 0.048
 Diversity = 9.7799e+00 k_t = [4, 4, 3, 3]
 Generation 8130: Best = 1.311117e+05 FES = 813100 μ_F = 0.811 μ_{CR} = 0.052
 Diversity = 9.5280e+00 k_t = [1, 1, 1, 1]
 Generation 8140: Best = 1.308694e+05 FES = 814100 μ_F = 0.816 μ_{CR} = 0.056
 Diversity = 1.0215e+01 k_t = [4, 2, 2, 3]
 Generation 8150: Best = 1.308649e+05 FES = 815100 μ_F = 0.806 μ_{CR} = 0.049
 Diversity = 1.1225e+01 k_t = [1, 1, 1, 1]
 Generation 8160: Best = 1.308342e+05 FES = 816100 μ_F = 0.816 μ_{CR} = 0.050
 Diversity = 1.1156e+01 k_t = [3, 2, 4, 4]
 Generation 8170: Best = 1.307298e+05 FES = 817100 μ_F = 0.827 μ_{CR} = 0.046
 Diversity = 1.0202e+01 k_t = [1, 1, 1, 1]
 Generation 8180: Best = 1.305165e+05 FES = 818100 μ_F = 0.817 μ_{CR} = 0.053
 Diversity = 1.0970e+01 k_t = [4, 2, 2, 4]
 Generation 8190: Best = 1.304861e+05 FES = 819100 μ_F = 0.816 μ_{CR} = 0.049
 Diversity = 9.6322e+00 k_t = [1, 1, 1, 1]
 Generation 8200: Best = 1.302757e+05 FES = 820100 μ_F = 0.829 μ_{CR} = 0.045
 Diversity = 1.0249e+01 k_t = [2, 3, 3, 4]
 Generation 8210: Best = 1.301843e+05 FES = 821100 μ_F = 0.834 μ_{CR} = 0.051
 Diversity = 1.0178e+01 k_t = [1, 1, 1, 1]
 Generation 8220: Best = 1.300004e+05 FES = 822100 μ_F = 0.846 μ_{CR} = 0.044
 Diversity = 1.0028e+01 k_t = [2, 3, 3, 4]
 Generation 8230: Best = 1.299642e+05 FES = 823100 μ_F = 0.834 μ_{CR} = 0.040
 Diversity = 1.0220e+01 k_t = [1, 1, 1, 1]
 Generation 8240: Best = 1.298106e+05 FES = 824100 μ_F = 0.827 μ_{CR} = 0.040
 Diversity = 1.0840e+01 k_t = [3, 3, 3, 3]
 Generation 8250: Best = 1.296979e+05 FES = 825100 μ_F = 0.833 μ_{CR} = 0.049
 Diversity = 1.1402e+01 k_t = [1, 1, 1, 1]
 Generation 8260: Best = 1.296030e+05 FES = 826100 μ_F = 0.834 μ_{CR} = 0.043
 Diversity = 1.1724e+01 k_t = [4, 4, 3, 2]
 Generation 8270: Best = 1.294289e+05 FES = 827100 μ_F = 0.824 μ_{CR} = 0.043
 Diversity = 1.1981e+01 k_t = [1, 1, 1, 1]
 Generation 8280: Best = 1.292822e+05 FES = 828100 μ_F = 0.824 μ_{CR} = 0.038
 Diversity = 1.2152e+01 k_t = [2, 3, 4, 2]
 Generation 8290: Best = 1.292451e+05 FES = 829100 μ_F = 0.824 μ_{CR} = 0.045
 Diversity = 1.2674e+01 k_t = [1, 1, 1, 1]
 Generation 8300: Best = 1.291334e+05 FES = 830100 μ_F = 0.830 μ_{CR} = 0.055
 Diversity = 1.1695e+01 k_t = [4, 4, 3, 2]
 Generation 8310: Best = 1.289876e+05 FES = 831100 μ_F = 0.808 μ_{CR} = 0.054
 Diversity = 1.1469e+01 k_t = [1, 1, 1, 1]
 Generation 8320: Best = 1.289634e+05 FES = 832100 μ_F = 0.814 μ_{CR} = 0.051
 Diversity = 1.0681e+01 k_t = [3, 3, 4, 2]

Generation 8330: Best = 1.288292e+05 FES = 833100 μ_F = 0.803 μ_{CR} = 0.049
 Diversity = 1.0926e+01 k_t = [1, 1, 1, 1]
 Generation 8340: Best = 1.288036e+05 FES = 834100 μ_F = 0.803 μ_{CR} = 0.048
 Diversity = 1.1127e+01 k_t = [2, 2, 4, 2]
 Generation 8350: Best = 1.286591e+05 FES = 835100 μ_F = 0.815 μ_{CR} = 0.049
 Diversity = 1.1683e+01 k_t = [1, 1, 1, 1]
 Generation 8360: Best = 1.284664e+05 FES = 836100 μ_F = 0.829 μ_{CR} = 0.049
 Diversity = 1.1548e+01 k_t = [4, 3, 4, 4]
 Generation 8370: Best = 1.283653e+05 FES = 837100 μ_F = 0.844 μ_{CR} = 0.050
 Diversity = 1.2623e+01 k_t = [1, 1, 1, 1]
 Generation 8380: Best = 1.281953e+05 FES = 838100 μ_F = 0.848 μ_{CR} = 0.051
 Diversity = 1.3844e+01 k_t = [2, 2, 4, 3]
 Generation 8390: Best = 1.280369e+05 FES = 839100 μ_F = 0.840 μ_{CR} = 0.050
 Diversity = 1.3532e+01 k_t = [1, 1, 1, 1]
 Generation 8400: Best = 1.277522e+05 FES = 840100 μ_F = 0.835 μ_{CR} = 0.047
 Diversity = 1.3978e+01 k_t = [4, 2, 3, 2]
 Generation 8410: Best = 1.273385e+05 FES = 841100 μ_F = 0.833 μ_{CR} = 0.055
 Diversity = 1.5316e+01 k_t = [1, 1, 1, 1]
 Generation 8420: Best = 1.269915e+05 FES = 842100 μ_F = 0.832 μ_{CR} = 0.058
 Diversity = 1.5660e+01 k_t = [2, 4, 4, 3]
 Generation 8430: Best = 1.265724e+05 FES = 843100 μ_F = 0.835 μ_{CR} = 0.058
 Diversity = 1.5989e+01 k_t = [1, 1, 1, 1]
 Generation 8440: Best = 1.264997e+05 FES = 844100 μ_F = 0.835 μ_{CR} = 0.047
 Diversity = 1.7025e+01 k_t = [4, 2, 2, 3]
 Generation 8450: Best = 1.260800e+05 FES = 845100 μ_F = 0.838 μ_{CR} = 0.051
 Diversity = 1.7316e+01 k_t = [1, 1, 1, 1]
 Generation 8460: Best = 1.258655e+05 FES = 846100 μ_F = 0.821 μ_{CR} = 0.046
 Diversity = 1.7200e+01 k_t = [4, 2, 2, 2]
 Generation 8470: Best = 1.255868e+05 FES = 847100 μ_F = 0.833 μ_{CR} = 0.053
 Diversity = 1.7527e+01 k_t = [1, 1, 1, 1]
 Generation 8480: Best = 1.252370e+05 FES = 848100 μ_F = 0.827 μ_{CR} = 0.051
 Diversity = 1.6198e+01 k_t = [3, 2, 4, 2]
 Generation 8490: Best = 1.250755e+05 FES = 849100 μ_F = 0.830 μ_{CR} = 0.051
 Diversity = 1.7556e+01 k_t = [1, 1, 1, 1]
 Generation 8500: Best = 1.247851e+05 FES = 850100 μ_F = 0.816 μ_{CR} = 0.060
 Diversity = 1.7362e+01 k_t = [2, 2, 2, 4]
 Generation 8510: Best = 1.246017e+05 FES = 851100 μ_F = 0.822 μ_{CR} = 0.051
 Diversity = 1.5736e+01 k_t = [1, 1, 1, 1]
 Generation 8520: Best = 1.244634e+05 FES = 852100 μ_F = 0.822 μ_{CR} = 0.047
 Diversity = 1.4974e+01 k_t = [2, 3, 3, 3]
 Generation 8530: Best = 1.241596e+05 FES = 853100 μ_F = 0.812 μ_{CR} = 0.050
 Diversity = 1.4857e+01 k_t = [1, 1, 1, 1]
 Generation 8540: Best = 1.238587e+05 FES = 854100 μ_F = 0.803 μ_{CR} = 0.047
 Diversity = 1.4409e+01 k_t = [2, 3, 2, 3]
 Generation 8550: Best = 1.237273e+05 FES = 855100 μ_F = 0.798 μ_{CR} = 0.042
 Diversity = 1.3914e+01 k_t = [1, 1, 1, 1]
 Generation 8560: Best = 1.233627e+05 FES = 856100 μ_F = 0.813 μ_{CR} = 0.047
 Diversity = 1.3913e+01 k_t = [4, 3, 4, 4]
 Generation 8570: Best = 1.232416e+05 FES = 857100 μ_F = 0.794 μ_{CR} = 0.047
 Diversity = 1.2483e+01 k_t = [1, 1, 1, 1]
 Generation 8580: Best = 1.231505e+05 FES = 858100 μ_F = 0.795 μ_{CR} = 0.052
 Diversity = 1.1481e+01 k_t = [3, 3, 3, 3]
 Generation 8590: Best = 1.230146e+05 FES = 859100 μ_F = 0.787 μ_{CR} = 0.051
 Diversity = 1.0878e+01 k_t = [1, 1, 1, 1]

Generation 8600: Best = 1.229131e+05 FES = 860100 μ_F = 0.788 μ_{CR} = 0.049
 Diversity = 1.0569e+01 k_t = [2, 4, 4, 2]
 Generation 8610: Best = 1.227331e+05 FES = 861100 μ_F = 0.791 μ_{CR} = 0.061
 Diversity = 1.1292e+01 k_t = [1, 1, 1, 1]
 Generation 8620: Best = 1.224474e+05 FES = 862100 μ_F = 0.796 μ_{CR} = 0.046
 Diversity = 1.1420e+01 k_t = [2, 4, 3, 3]
 Generation 8630: Best = 1.222486e+05 FES = 863100 μ_F = 0.803 μ_{CR} = 0.050
 Diversity = 1.1871e+01 k_t = [1, 1, 1, 1]
 Generation 8640: Best = 1.220142e+05 FES = 864100 μ_F = 0.811 μ_{CR} = 0.050
 Diversity = 1.1597e+01 k_t = [2, 2, 4, 3]
 Generation 8650: Best = 1.217295e+05 FES = 865100 μ_F = 0.801 μ_{CR} = 0.047
 Diversity = 1.1012e+01 k_t = [1, 1, 1, 1]
 Generation 8660: Best = 1.216237e+05 FES = 866100 μ_F = 0.807 μ_{CR} = 0.048
 Diversity = 1.1441e+01 k_t = [2, 4, 3, 2]
 Generation 8670: Best = 1.215572e+05 FES = 867100 μ_F = 0.804 μ_{CR} = 0.049
 Diversity = 1.1764e+01 k_t = [1, 1, 1, 1]
 Generation 8680: Best = 1.213847e+05 FES = 868100 μ_F = 0.823 μ_{CR} = 0.059
 Diversity = 1.1728e+01 k_t = [2, 2, 3, 3]
 Generation 8690: Best = 1.213491e+05 FES = 869100 μ_F = 0.823 μ_{CR} = 0.050
 Diversity = 1.2050e+01 k_t = [1, 1, 1, 1]
 Generation 8700: Best = 1.212108e+05 FES = 870100 μ_F = 0.829 μ_{CR} = 0.042
 Diversity = 1.2041e+01 k_t = [3, 2, 2, 4]
 Generation 8710: Best = 1.210740e+05 FES = 871100 μ_F = 0.826 μ_{CR} = 0.048
 Diversity = 1.2167e+01 k_t = [1, 1, 1, 1]
 Generation 8720: Best = 1.208846e+05 FES = 872100 μ_F = 0.828 μ_{CR} = 0.048
 Diversity = 1.2636e+01 k_t = [4, 4, 2, 3]
 Generation 8730: Best = 1.207782e+05 FES = 873100 μ_F = 0.823 μ_{CR} = 0.054
 Diversity = 1.1364e+01 k_t = [1, 1, 1, 1]
 Generation 8740: Best = 1.205770e+05 FES = 874100 μ_F = 0.837 μ_{CR} = 0.049
 Diversity = 1.1366e+01 k_t = [4, 3, 3, 3]
 Generation 8750: Best = 1.203148e+05 FES = 875100 μ_F = 0.834 μ_{CR} = 0.045
 Diversity = 1.1610e+01 k_t = [1, 1, 1, 1]
 Generation 8760: Best = 1.202236e+05 FES = 876100 μ_F = 0.839 μ_{CR} = 0.045
 Diversity = 1.1959e+01 k_t = [4, 3, 4, 3]
 Generation 8770: Best = 1.200050e+05 FES = 877100 μ_F = 0.832 μ_{CR} = 0.045
 Diversity = 1.1567e+01 k_t = [1, 1, 1, 1]
 Generation 8780: Best = 1.197920e+05 FES = 878100 μ_F = 0.827 μ_{CR} = 0.049
 Diversity = 1.1555e+01 k_t = [2, 3, 2, 4]
 Generation 8790: Best = 1.196932e+05 FES = 879100 μ_F = 0.818 μ_{CR} = 0.049
 Diversity = 1.1683e+01 k_t = [1, 1, 1, 1]
 Generation 8800: Best = 1.195817e+05 FES = 880100 μ_F = 0.802 μ_{CR} = 0.048
 Diversity = 1.2495e+01 k_t = [2, 2, 3, 2]
 Generation 8810: Best = 1.194880e+05 FES = 881100 μ_F = 0.811 μ_{CR} = 0.049
 Diversity = 1.2709e+01 k_t = [1, 1, 1, 1]
 Generation 8820: Best = 1.192856e+05 FES = 882100 μ_F = 0.805 μ_{CR} = 0.051
 Diversity = 1.1706e+01 k_t = [3, 3, 3, 3]
 Generation 8830: Best = 1.191897e+05 FES = 883100 μ_F = 0.818 μ_{CR} = 0.046
 Diversity = 1.2695e+01 k_t = [1, 1, 1, 1]
 Generation 8840: Best = 1.189933e+05 FES = 884100 μ_F = 0.815 μ_{CR} = 0.052
 Diversity = 1.2959e+01 k_t = [3, 3, 3, 3]
 Generation 8850: Best = 1.188825e+05 FES = 885100 μ_F = 0.825 μ_{CR} = 0.048
 Diversity = 1.2937e+01 k_t = [1, 1, 1, 1]
 Generation 8860: Best = 1.186613e+05 FES = 886100 μ_F = 0.820 μ_{CR} = 0.041
 Diversity = 1.1950e+01 k_t = [2, 2, 4, 2]

Generation 8870: Best = 1.186378e+05 FES = 887100 μ_F = 0.809 μ_{CR} = 0.041
 Diversity = 1.2448e+01 k_t = [1, 1, 1, 1]
 Generation 8880: Best = 1.185402e+05 FES = 888100 μ_F = 0.813 μ_{CR} = 0.038
 Diversity = 1.2134e+01 k_t = [4, 2, 4, 3]
 Generation 8890: Best = 1.184399e+05 FES = 889100 μ_F = 0.794 μ_{CR} = 0.041
 Diversity = 1.1847e+01 k_t = [1, 1, 1, 1]
 Generation 8900: Best = 1.183449e+05 FES = 890100 μ_F = 0.805 μ_{CR} = 0.043
 Diversity = 1.1515e+01 k_t = [2, 4, 2, 3]
 Generation 8910: Best = 1.182372e+05 FES = 891100 μ_F = 0.799 μ_{CR} = 0.051
 Diversity = 1.1230e+01 k_t = [1, 1, 1, 1]
 Generation 8920: Best = 1.181408e+05 FES = 892100 μ_F = 0.810 μ_{CR} = 0.059
 Diversity = 1.1728e+01 k_t = [3, 4, 4, 4]
 Generation 8930: Best = 1.180198e+05 FES = 893100 μ_F = 0.817 μ_{CR} = 0.056
 Diversity = 1.1654e+01 k_t = [1, 1, 1, 1]
 Generation 8940: Best = 1.177711e+05 FES = 894100 μ_F = 0.821 μ_{CR} = 0.059
 Diversity = 1.1328e+01 k_t = [2, 2, 4, 3]
 Generation 8950: Best = 1.176413e+05 FES = 895100 μ_F = 0.822 μ_{CR} = 0.059
 Diversity = 1.2200e+01 k_t = [1, 1, 1, 1]
 Generation 8960: Best = 1.173771e+05 FES = 896100 μ_F = 0.829 μ_{CR} = 0.054
 Diversity = 1.2498e+01 k_t = [4, 2, 4, 3]
 Generation 8970: Best = 1.172520e+05 FES = 897100 μ_F = 0.822 μ_{CR} = 0.061
 Diversity = 1.2632e+01 k_t = [1, 1, 1, 1]
 Generation 8980: Best = 1.170198e+05 FES = 898100 μ_F = 0.814 μ_{CR} = 0.058
 Diversity = 1.3197e+01 k_t = [3, 4, 3, 4]
 Generation 8990: Best = 1.168931e+05 FES = 899100 μ_F = 0.809 μ_{CR} = 0.055
 Diversity = 1.2407e+01 k_t = [1, 1, 1, 1]
 Generation 9000: Best = 1.168397e+05 FES = 900100 μ_F = 0.803 μ_{CR} = 0.049
 Diversity = 1.2940e+01 k_t = [2, 3, 2, 2]
 Generation 9010: Best = 1.166614e+05 FES = 901100 μ_F = 0.816 μ_{CR} = 0.054
 Diversity = 1.2648e+01 k_t = [1, 1, 1, 1]
 Generation 9020: Best = 1.165138e+05 FES = 902100 μ_F = 0.825 μ_{CR} = 0.055
 Diversity = 1.2660e+01 k_t = [2, 3, 4, 3]
 Generation 9030: Best = 1.163696e+05 FES = 903100 μ_F = 0.813 μ_{CR} = 0.054
 Diversity = 1.2464e+01 k_t = [1, 1, 1, 1]
 Generation 9040: Best = 1.163246e+05 FES = 904100 μ_F = 0.815 μ_{CR} = 0.054
 Diversity = 1.2296e+01 k_t = [4, 3, 3, 4]
 Generation 9050: Best = 1.162145e+05 FES = 905100 μ_F = 0.818 μ_{CR} = 0.055
 Diversity = 1.2248e+01 k_t = [1, 1, 1, 1]
 Generation 9060: Best = 1.161009e+05 FES = 906100 μ_F = 0.809 μ_{CR} = 0.054
 Diversity = 1.2292e+01 k_t = [4, 3, 2, 2]
 Generation 9070: Best = 1.160118e+05 FES = 907100 μ_F = 0.825 μ_{CR} = 0.054
 Diversity = 1.2645e+01 k_t = [1, 1, 1, 1]
 Generation 9080: Best = 1.159177e+05 FES = 908100 μ_F = 0.814 μ_{CR} = 0.052
 Diversity = 1.2399e+01 k_t = [2, 3, 3, 3]
 Generation 9090: Best = 1.158187e+05 FES = 909100 μ_F = 0.790 μ_{CR} = 0.046
 Diversity = 1.0874e+01 k_t = [1, 1, 1, 1]
 Generation 9100: Best = 1.156740e+05 FES = 910100 μ_F = 0.809 μ_{CR} = 0.049
 Diversity = 1.1040e+01 k_t = [4, 3, 4, 2]
 Generation 9110: Best = 1.156739e+05 FES = 911100 μ_F = 0.803 μ_{CR} = 0.043
 Diversity = 1.0290e+01 k_t = [1, 1, 1, 1]
 Generation 9120: Best = 1.155742e+05 FES = 912100 μ_F = 0.810 μ_{CR} = 0.046
 Diversity = 1.0482e+01 k_t = [4, 3, 2, 4]
 Generation 9130: Best = 1.154816e+05 FES = 913100 μ_F = 0.806 μ_{CR} = 0.049
 Diversity = 1.0399e+01 k_t = [1, 1, 1, 1]

Generation 9140: Best = 1.153295e+05 FES = 914100 μ_F = 0.803 μ_{CR} = 0.053
 Diversity = 9.8679e+00 k_t = [2, 2, 3, 4]
 Generation 9150: Best = 1.151746e+05 FES = 915100 μ_F = 0.802 μ_{CR} = 0.051
 Diversity = 1.0043e+01 k_t = [1, 1, 1, 1]
 Generation 9160: Best = 1.151276e+05 FES = 916100 μ_F = 0.807 μ_{CR} = 0.049
 Diversity = 9.9749e+00 k_t = [3, 4, 2, 4]
 Generation 9170: Best = 1.149992e+05 FES = 917100 μ_F = 0.810 μ_{CR} = 0.059
 Diversity = 1.0230e+01 k_t = [1, 1, 1, 1]
 Generation 9180: Best = 1.148049e+05 FES = 918100 μ_F = 0.813 μ_{CR} = 0.059
 Diversity = 1.0696e+01 k_t = [3, 4, 2, 2]
 Generation 9190: Best = 1.145874e+05 FES = 919100 μ_F = 0.796 μ_{CR} = 0.063
 Diversity = 1.1209e+01 k_t = [1, 1, 1, 1]
 Generation 9200: Best = 1.144084e+05 FES = 920100 μ_F = 0.816 μ_{CR} = 0.055
 Diversity = 1.3010e+01 k_t = [3, 4, 2, 3]
 Generation 9210: Best = 1.142570e+05 FES = 921100 μ_F = 0.819 μ_{CR} = 0.057
 Diversity = 1.2782e+01 k_t = [1, 1, 1, 1]
 Generation 9220: Best = 1.141256e+05 FES = 922100 μ_F = 0.817 μ_{CR} = 0.061
 Diversity = 1.1979e+01 k_t = [2, 3, 4, 4]
 Generation 9230: Best = 1.140408e+05 FES = 923100 μ_F = 0.812 μ_{CR} = 0.057
 Diversity = 1.1821e+01 k_t = [1, 1, 1, 1]
 Generation 9240: Best = 1.137887e+05 FES = 924100 μ_F = 0.809 μ_{CR} = 0.061
 Diversity = 1.1340e+01 k_t = [2, 4, 2, 4]
 Generation 9250: Best = 1.136872e+05 FES = 925100 μ_F = 0.812 μ_{CR} = 0.060
 Diversity = 1.0884e+01 k_t = [1, 1, 1, 1]
 Generation 9260: Best = 1.135666e+05 FES = 926100 μ_F = 0.802 μ_{CR} = 0.057
 Diversity = 1.0926e+01 k_t = [2, 4, 2, 4]
 Generation 9270: Best = 1.134176e+05 FES = 927100 μ_F = 0.793 μ_{CR} = 0.061
 Diversity = 1.0966e+01 k_t = [1, 1, 1, 1]
 Generation 9280: Best = 1.132492e+05 FES = 928100 μ_F = 0.800 μ_{CR} = 0.065
 Diversity = 1.1417e+01 k_t = [3, 3, 2, 4]
 Generation 9290: Best = 1.130827e+05 FES = 929100 μ_F = 0.815 μ_{CR} = 0.065
 Diversity = 1.1306e+01 k_t = [1, 1, 1, 1]
 Generation 9300: Best = 1.130154e+05 FES = 930100 μ_F = 0.816 μ_{CR} = 0.059
 Diversity = 1.1745e+01 k_t = [3, 2, 2, 3]
 Generation 9310: Best = 1.129003e+05 FES = 931100 μ_F = 0.818 μ_{CR} = 0.057
 Diversity = 1.2574e+01 k_t = [1, 1, 1, 1]
 Generation 9320: Best = 1.125964e+05 FES = 932100 μ_F = 0.798 μ_{CR} = 0.056
 Diversity = 1.1887e+01 k_t = [3, 3, 4, 2]
 Generation 9330: Best = 1.124196e+05 FES = 933100 μ_F = 0.815 μ_{CR} = 0.063
 Diversity = 1.2566e+01 k_t = [1, 1, 1, 1]
 Generation 9340: Best = 1.122348e+05 FES = 934100 μ_F = 0.823 μ_{CR} = 0.056
 Diversity = 1.2551e+01 k_t = [3, 2, 4, 3]
 Generation 9350: Best = 1.121003e+05 FES = 935100 μ_F = 0.814 μ_{CR} = 0.051
 Diversity = 1.2648e+01 k_t = [1, 1, 1, 1]
 Generation 9360: Best = 1.119142e+05 FES = 936100 μ_F = 0.814 μ_{CR} = 0.052
 Diversity = 1.1254e+01 k_t = [4, 2, 4, 4]
 Generation 9370: Best = 1.118646e+05 FES = 937100 μ_F = 0.806 μ_{CR} = 0.046
 Diversity = 1.0488e+01 k_t = [1, 1, 1, 1]
 Generation 9380: Best = 1.116017e+05 FES = 938100 μ_F = 0.795 μ_{CR} = 0.049
 Diversity = 1.0458e+01 k_t = [4, 2, 4, 2]
 Generation 9390: Best = 1.116017e+05 FES = 939100 μ_F = 0.788 μ_{CR} = 0.054
 Diversity = 1.0525e+01 k_t = [1, 1, 1, 1]
 Generation 9400: Best = 1.115670e+05 FES = 940100 μ_F = 0.803 μ_{CR} = 0.056
 Diversity = 1.0010e+01 k_t = [2, 3, 2, 4]

Generation 9410: Best = 1.114483e+05 FES = 941100 μ_F = 0.793 μ_{CR} = 0.052
 Diversity = 9.3906e+00 k_t = [1, 1, 1, 1]
 Generation 9420: Best = 1.114483e+05 FES = 942100 μ_F = 0.798 μ_{CR} = 0.044
 Diversity = 8.9978e+00 k_t = [3, 2, 4, 2]
 Generation 9430: Best = 1.113594e+05 FES = 943100 μ_F = 0.789 μ_{CR} = 0.044
 Diversity = 9.2118e+00 k_t = [1, 1, 1, 1]
 Generation 9440: Best = 1.112688e+05 FES = 944100 μ_F = 0.791 μ_{CR} = 0.044
 Diversity = 9.6393e+00 k_t = [4, 2, 4, 4]
 Generation 9450: Best = 1.111018e+05 FES = 945100 μ_F = 0.795 μ_{CR} = 0.039
 Diversity = 9.6753e+00 k_t = [1, 1, 1, 1]
 Generation 9460: Best = 1.109909e+05 FES = 946100 μ_F = 0.798 μ_{CR} = 0.051
 Diversity = 1.0048e+01 k_t = [2, 3, 2, 3]
 Generation 9470: Best = 1.108356e+05 FES = 947100 μ_F = 0.819 μ_{CR} = 0.052
 Diversity = 1.1439e+01 k_t = [1, 1, 1, 1]
 Generation 9480: Best = 1.106409e+05 FES = 948100 μ_F = 0.823 μ_{CR} = 0.050
 Diversity = 1.1390e+01 k_t = [3, 2, 4, 4]
 Generation 9490: Best = 1.103893e+05 FES = 949100 μ_F = 0.835 μ_{CR} = 0.050
 Diversity = 1.2384e+01 k_t = [1, 1, 1, 1]
 Generation 9500: Best = 1.102583e+05 FES = 950100 μ_F = 0.836 μ_{CR} = 0.048
 Diversity = 1.2947e+01 k_t = [2, 4, 3, 3]
 Generation 9510: Best = 1.100626e+05 FES = 951100 μ_F = 0.821 μ_{CR} = 0.050
 Diversity = 1.2613e+01 k_t = [1, 1, 1, 1]
 Generation 9520: Best = 1.098696e+05 FES = 952100 μ_F = 0.824 μ_{CR} = 0.050
 Diversity = 1.3272e+01 k_t = [2, 4, 4, 3]
 Generation 9530: Best = 1.097617e+05 FES = 953100 μ_F = 0.817 μ_{CR} = 0.043
 Diversity = 1.3486e+01 k_t = [1, 1, 1, 1]
 Generation 9540: Best = 1.096330e+05 FES = 954100 μ_F = 0.820 μ_{CR} = 0.055
 Diversity = 1.2537e+01 k_t = [3, 2, 2, 4]
 Generation 9550: Best = 1.094141e+05 FES = 955100 μ_F = 0.826 μ_{CR} = 0.057
 Diversity = 1.1958e+01 k_t = [1, 1, 1, 1]
 Generation 9560: Best = 1.092670e+05 FES = 956100 μ_F = 0.822 μ_{CR} = 0.056
 Diversity = 1.1168e+01 k_t = [4, 3, 2, 2]
 Generation 9570: Best = 1.091861e+05 FES = 957100 μ_F = 0.797 μ_{CR} = 0.055
 Diversity = 1.0934e+01 k_t = [1, 1, 1, 1]
 Generation 9580: Best = 1.090536e+05 FES = 958100 μ_F = 0.802 μ_{CR} = 0.052
 Diversity = 1.1094e+01 k_t = [3, 2, 2, 2]
 Generation 9590: Best = 1.090157e+05 FES = 959100 μ_F = 0.797 μ_{CR} = 0.053
 Diversity = 1.1778e+01 k_t = [1, 1, 1, 1]
 Generation 9600: Best = 1.088218e+05 FES = 960100 μ_F = 0.809 μ_{CR} = 0.053
 Diversity = 1.1745e+01 k_t = [4, 2, 4, 4]
 Generation 9610: Best = 1.087712e+05 FES = 961100 μ_F = 0.816 μ_{CR} = 0.067
 Diversity = 1.1396e+01 k_t = [1, 1, 1, 1]
 Generation 9620: Best = 1.087103e+05 FES = 962100 μ_F = 0.829 μ_{CR} = 0.051
 Diversity = 1.0804e+01 k_t = [4, 4, 4, 4]
 Generation 9630: Best = 1.085485e+05 FES = 963100 μ_F = 0.830 μ_{CR} = 0.051
 Diversity = 1.0669e+01 k_t = [1, 1, 1, 1]
 Generation 9640: Best = 1.084420e+05 FES = 964100 μ_F = 0.817 μ_{CR} = 0.047
 Diversity = 1.0907e+01 k_t = [2, 4, 3, 2]
 Generation 9650: Best = 1.083415e+05 FES = 965100 μ_F = 0.820 μ_{CR} = 0.053
 Diversity = 1.0495e+01 k_t = [1, 1, 1, 1]
 Generation 9660: Best = 1.082750e+05 FES = 966100 μ_F = 0.834 μ_{CR} = 0.053
 Diversity = 1.0249e+01 k_t = [4, 4, 2, 2]
 Generation 9670: Best = 1.080330e+05 FES = 967100 μ_F = 0.825 μ_{CR} = 0.054
 Diversity = 1.1608e+01 k_t = [1, 1, 1, 1]

Generation 9680: Best = 1.079445e+05 FES = 968100 μ_F = 0.826 μ_{CR} = 0.061
 Diversity = 1.1060e+01 k_t = [3, 4, 3, 4]
 Generation 9690: Best = 1.078015e+05 FES = 969100 μ_F = 0.825 μ_{CR} = 0.052
 Diversity = 1.0938e+01 k_t = [1, 1, 1, 1]
 Generation 9700: Best = 1.076603e+05 FES = 970100 μ_F = 0.841 μ_{CR} = 0.048
 Diversity = 1.1372e+01 k_t = [4, 3, 4, 3]
 Generation 9710: Best = 1.075326e+05 FES = 971100 μ_F = 0.832 μ_{CR} = 0.049
 Diversity = 1.2434e+01 k_t = [1, 1, 1, 1]
 Generation 9720: Best = 1.074589e+05 FES = 972100 μ_F = 0.827 μ_{CR} = 0.052
 Diversity = 1.2007e+01 k_t = [2, 3, 2, 2]
 Generation 9730: Best = 1.071260e+05 FES = 973100 μ_F = 0.816 μ_{CR} = 0.050
 Diversity = 1.2343e+01 k_t = [1, 1, 1, 1]
 Generation 9740: Best = 1.070853e+05 FES = 974100 μ_F = 0.812 μ_{CR} = 0.046
 Diversity = 1.2989e+01 k_t = [2, 3, 3, 2]
 Generation 9750: Best = 1.069784e+05 FES = 975100 μ_F = 0.808 μ_{CR} = 0.046
 Diversity = 1.3006e+01 k_t = [1, 1, 1, 1]
 Generation 9760: Best = 1.067359e+05 FES = 976100 μ_F = 0.815 μ_{CR} = 0.055
 Diversity = 1.3619e+01 k_t = [2, 4, 3, 2]
 Generation 9770: Best = 1.066235e+05 FES = 977100 μ_F = 0.827 μ_{CR} = 0.054
 Diversity = 1.3293e+01 k_t = [1, 1, 1, 1]
 Generation 9780: Best = 1.064723e+05 FES = 978100 μ_F = 0.818 μ_{CR} = 0.049
 Diversity = 1.2748e+01 k_t = [2, 3, 4, 4]
 Generation 9790: Best = 1.063934e+05 FES = 979100 μ_F = 0.811 μ_{CR} = 0.057
 Diversity = 1.2506e+01 k_t = [1, 1, 1, 1]
 Generation 9800: Best = 1.062386e+05 FES = 980100 μ_F = 0.829 μ_{CR} = 0.053
 Diversity = 1.2380e+01 k_t = [2, 4, 2, 4]
 Generation 9810: Best = 1.061275e+05 FES = 981100 μ_F = 0.813 μ_{CR} = 0.053
 Diversity = 1.1373e+01 k_t = [1, 1, 1, 1]
 Generation 9820: Best = 1.058018e+05 FES = 982100 μ_F = 0.813 μ_{CR} = 0.053
 Diversity = 1.1836e+01 k_t = [3, 3, 4, 2]
 Generation 9830: Best = 1.057366e+05 FES = 983100 μ_F = 0.808 μ_{CR} = 0.056
 Diversity = 1.1597e+01 k_t = [1, 1, 1, 1]
 Generation 9840: Best = 1.054792e+05 FES = 984100 μ_F = 0.809 μ_{CR} = 0.056
 Diversity = 1.1489e+01 k_t = [4, 3, 3, 4]
 Generation 9850: Best = 1.054568e+05 FES = 985100 μ_F = 0.819 μ_{CR} = 0.046
 Diversity = 1.1939e+01 k_t = [1, 1, 1, 1]
 Generation 9860: Best = 1.052620e+05 FES = 986100 μ_F = 0.823 μ_{CR} = 0.053
 Diversity = 1.1898e+01 k_t = [2, 4, 2, 4]
 Generation 9870: Best = 1.051054e+05 FES = 987100 μ_F = 0.836 μ_{CR} = 0.057
 Diversity = 1.1939e+01 k_t = [1, 1, 1, 1]
 Generation 9880: Best = 1.048651e+05 FES = 988100 μ_F = 0.841 μ_{CR} = 0.048
 Diversity = 1.2009e+01 k_t = [4, 4, 4, 3]
 Generation 9890: Best = 1.047629e+05 FES = 989100 μ_F = 0.835 μ_{CR} = 0.051
 Diversity = 1.2571e+01 k_t = [1, 1, 1, 1]
 Generation 9900: Best = 1.046879e+05 FES = 990100 μ_F = 0.835 μ_{CR} = 0.052
 Diversity = 1.1912e+01 k_t = [2, 2, 4, 4]
 Generation 9910: Best = 1.045989e+05 FES = 991100 μ_F = 0.824 μ_{CR} = 0.048
 Diversity = 1.1948e+01 k_t = [1, 1, 1, 1]
 Generation 9920: Best = 1.044452e+05 FES = 992100 μ_F = 0.797 μ_{CR} = 0.062
 Diversity = 1.1331e+01 k_t = [4, 2, 4, 2]
 Generation 9930: Best = 1.043828e+05 FES = 993100 μ_F = 0.799 μ_{CR} = 0.063
 Diversity = 1.1105e+01 k_t = [1, 1, 1, 1]
 Generation 9940: Best = 1.042391e+05 FES = 994100 μ_F = 0.802 μ_{CR} = 0.060
 Diversity = 1.1721e+01 k_t = [4, 3, 3, 2]

Generation 9950: Best = 1.040838e+05 FES = 995100 μ_F = 0.799 μ_{CR} = 0.050
 Diversity = 1.2159e+01 k_t = [1, 1, 1, 1]
 Generation 9960: Best = 1.037548e+05 FES = 996100 μ_F = 0.798 μ_{CR} = 0.063
 Diversity = 1.2264e+01 k_t = [3, 2, 4, 2]
 Generation 9970: Best = 1.036049e+05 FES = 997100 μ_F = 0.818 μ_{CR} = 0.064
 Diversity = 1.2463e+01 k_t = [1, 1, 1, 1]
 Generation 9980: Best = 1.035926e+05 FES = 998100 μ_F = 0.815 μ_{CR} = 0.065
 Diversity = 1.2248e+01 k_t = [4, 2, 4, 3]
 Generation 9990: Best = 1.034949e+05 FES = 999100 μ_F = 0.811 μ_{CR} = 0.059
 Diversity = 1.2531e+01 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 1.034320e+05

Total Function Evals = 1000000

Total Successful Updates = 229860

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Results:

Best Fitness : 1.034320e+05

Global Optimum : 1.400000e+03

Error (|best-optimum|) : 1.020320e+05

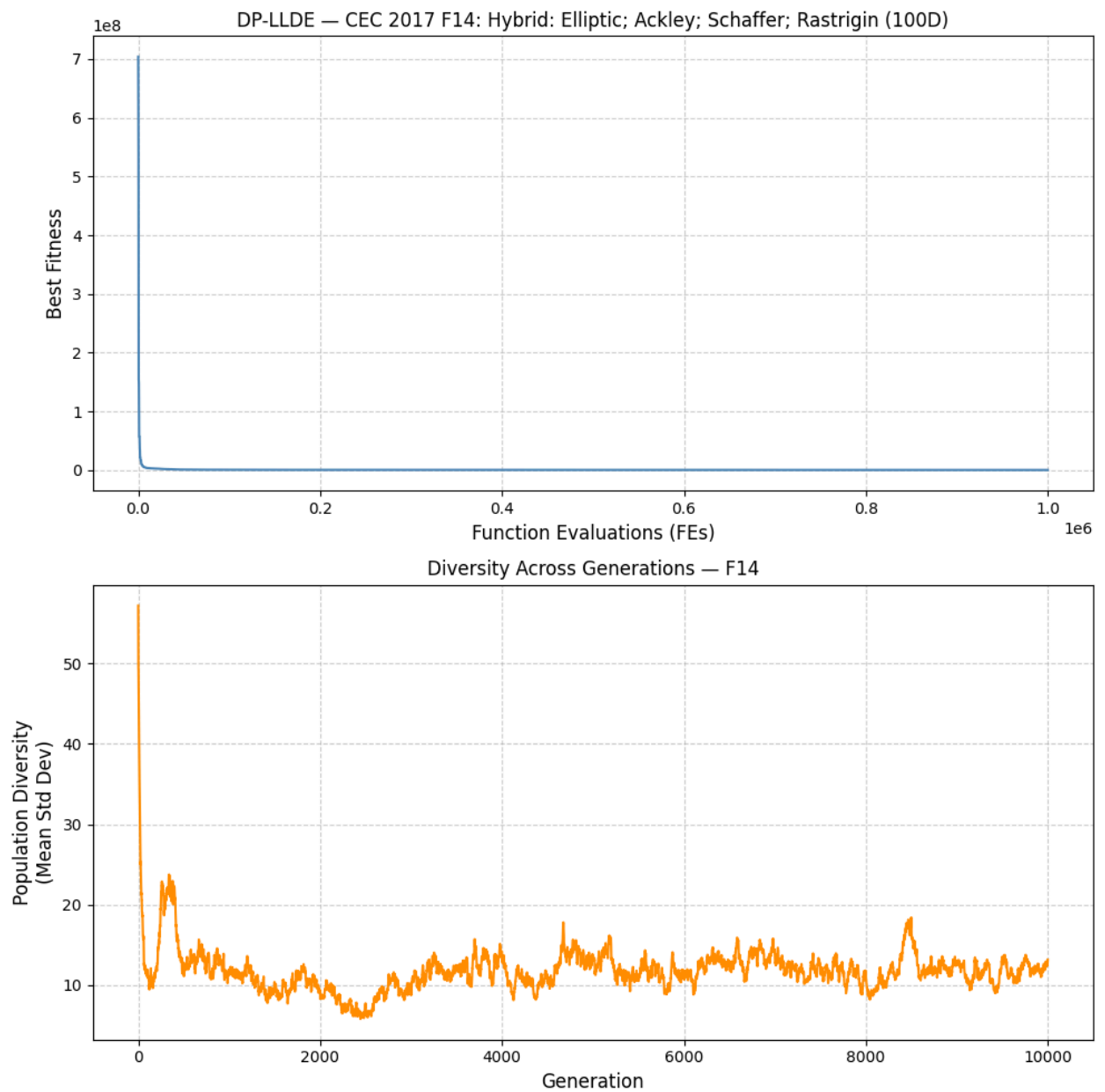
Best Solution (5 dim) : [90.69807899 -37.66928778 -81.03470468 -30.84329056
-48.93186709]

D^(0) (init. diversity): 5.7220e+01

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✓ Checkpoint saved to: /content/drive/My Drive/thesis_results/all_results_checkpoint.pkl



Graph saved as convergence_F14.png

✓ Plot saved to: /content/drive/My Drive/thesis_results/plots/convergence_F14.png

```
In [ ]: # =====
# F15 — Hybrid Function 5 (Bent Cigar; HGBat; Rastrigin; Rosenbrock)
# =====
run_and_plot('F15', make_wrapper(hybrid.f15), 1500.0, 'Hybrid: Bent Cigar; HGE
```

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DP-LLDE on CEC 2017 F15 – Single Run

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Function : Hybrid: Bent Cigar; HGBat; Rastrigin; Rosenbrock (100D)

Bounds : [-100, 100]

Global optimum : 1500.0

Initial $D^{(0)} = 5.7220e+01$ | Diversity threshold = 5.7220e+00

Generation 0: Best Fitness = 3.781522e+11

Generation 10: Best = 9.140550e+10 FES = 1100 $\mu_F = 0.565$ $\mu_{CR} = 0.356$

Diversity = 3.2997e+01 $k_t = [1, 1, 1, 1]$

Generation 20: Best = 1.813583e+10 FES = 2100 $\mu_F = 0.620$ $\mu_{CR} = 0.355$

Diversity = 2.2583e+01 $k_t = [2, 3, 2, 3]$

Generation 30: Best = 2.421706e+09 FES = 3100 $\mu_F = 0.648$ $\mu_{CR} = 0.358$

Diversity = 1.7148e+01 $k_t = [1, 1, 1, 1]$

Generation 40: Best = 8.152482e+08 FES = 4100 $\mu_F = 0.672$ $\mu_{CR} = 0.366$

Diversity = 1.2900e+01 $k_t = [3, 2, 3, 4]$

Generation 50: Best = 1.267728e+08 FES = 5100 $\mu_F = 0.689$ $\mu_{CR} = 0.373$

Diversity = 1.1455e+01 $k_t = [1, 1, 1, 1]$

Generation 60: Best = 3.071815e+07 FES = 6100 $\mu_F = 0.703$ $\mu_{CR} = 0.365$

Diversity = 1.0749e+01 $k_t = [2, 2, 3, 3]$

Generation 70: Best = 6.065255e+06 FES = 7100 $\mu_F = 0.709$ $\mu_{CR} = 0.366$

Diversity = 9.8930e+00 $k_t = [1, 1, 1, 1]$

Generation 80: Best = 1.235220e+06 FES = 8100 $\mu_F = 0.712$ $\mu_{CR} = 0.374$

Diversity = 8.1335e+00 $k_t = [4, 3, 4, 2]$

Generation 90: Best = 2.705395e+05 FES = 9100 $\mu_F = 0.721$ $\mu_{CR} = 0.382$

Diversity = 7.9049e+00 $k_t = [1, 1, 1, 1]$

Generation 100: Best = 8.176965e+04 FES = 10100 $\mu_F = 0.725$ $\mu_{CR} = 0.385$

Diversity = 6.8197e+00 $k_t = [2, 2, 4, 3]$

Generation 110: Best = 4.094242e+04 FES = 11100 $\mu_F = 0.725$ $\mu_{CR} = 0.395$

Diversity = 6.3724e+00 $k_t = [1, 1, 1, 1]$

Generation 120: Best = 2.329544e+04 FES = 12100 $\mu_F = 0.737$ $\mu_{CR} = 0.396$

Diversity = 5.7535e+00 $k_t = [2, 4, 2, 2]$

Generation 130: Best = 1.488176e+04 FES = 13100 $\mu_F = 0.742$ $\mu_{CR} = 0.403$

Diversity = 6.6452e+00 $k_t = [1, 1, 1, 1]$

Generation 140: Best = 1.100518e+04 FES = 14100 $\mu_F = 0.743$ $\mu_{CR} = 0.407$

Diversity = 6.6129e+00 $k_t = [2, 3, 4, 2]$

Generation 150: Best = 8.204829e+03 FES = 15100 $\mu_F = 0.748$ $\mu_{CR} = 0.409$

Diversity = 7.4625e+00 $k_t = [1, 1, 1, 1]$

Generation 160: Best = 6.304167e+03 FES = 16100 $\mu_F = 0.737$ $\mu_{CR} = 0.414$

Diversity = 7.8770e+00 $k_t = [4, 2, 3, 2]$

Generation 170: Best = 5.275166e+03 FES = 17100 $\mu_F = 0.733$ $\mu_{CR} = 0.408$

Diversity = 7.8952e+00 $k_t = [1, 1, 1, 1]$

Generation 180: Best = 4.539938e+03 FES = 18100 $\mu_F = 0.724$ $\mu_{CR} = 0.404$

Diversity = 6.9146e+00 $k_t = [3, 2, 3, 3]$

Generation 190: Best = 4.214716e+03 FES = 19100 $\mu_F = 0.743$ $\mu_{CR} = 0.410$

Diversity = 7.2379e+00 $k_t = [1, 1, 1, 1]$

Generation 200: Best = 3.811365e+03 FES = 20100 $\mu_F = 0.746$ $\mu_{CR} = 0.416$

Diversity = 6.8734e+00 $k_t = [4, 4, 2, 4]$

Generation 210: Best = 3.558872e+03 FES = 21100 $\mu_F = 0.757$ $\mu_{CR} = 0.413$

Diversity = 6.5610e+00 $k_t = [1, 1, 1, 1]$

Generation 220: Best = 3.171044e+03 FES = 22100 $\mu_F = 0.764$ $\mu_{CR} = 0.414$

Diversity = 7.3838e+00 $k_t = [4, 4, 2, 3]$

Generation	230: Best = 3.048418e+03	FES = 23100	$\mu_F = 0.763$	$\mu_{CR} = 0.417$
Diversity =	6.8395e+00	$k_t = [1, 1, 1, 1]$		
Generation	240: Best = 2.885777e+03	FES = 24100	$\mu_F = 0.766$	$\mu_{CR} = 0.404$
Diversity =	7.0123e+00	$k_t = [4, 2, 2, 4]$		
Generation	250: Best = 2.830255e+03	FES = 25100	$\mu_F = 0.773$	$\mu_{CR} = 0.397$
Diversity =	6.6307e+00	$k_t = [1, 1, 1, 1]$		
Generation	260: Best = 2.759103e+03	FES = 26100	$\mu_F = 0.749$	$\mu_{CR} = 0.391$
Diversity =	6.1859e+00	$k_t = [3, 3, 4, 3]$		
Generation	270: Best = 2.664457e+03	FES = 27100	$\mu_F = 0.738$	$\mu_{CR} = 0.394$
Diversity =	5.6143e+00	$k_t = [1, 1, 1, 1]$		
Generation	280: Best = 2.637698e+03	FES = 28100	$\mu_F = 0.725$	$\mu_{CR} = 0.376$
Diversity =	5.3130e+00	$k_t = [3, 2, 3, 3]$		
Generation	290: Best = 2.631323e+03	FES = 29100	$\mu_F = 0.736$	$\mu_{CR} = 0.362$
Diversity =	5.2120e+00	$k_t = [1, 1, 1, 1]$		
Generation	300: Best = 2.574181e+03	FES = 30100	$\mu_F = 0.743$	$\mu_{CR} = 0.324$
Diversity =	5.0390e+00	$k_t = [2, 3, 4, 4]$		
Generation	310: Best = 2.574181e+03	FES = 31100	$\mu_F = 0.728$	$\mu_{CR} = 0.288$
Diversity =	4.7946e+00	$k_t = [1, 1, 1, 1]$		
Generation	320: Best = 2.570774e+03	FES = 32100	$\mu_F = 0.729$	$\mu_{CR} = 0.256$
Diversity =	4.5362e+00	$k_t = [2, 3, 3, 4]$		
Generation	330: Best = 2.512457e+03	FES = 33100	$\mu_F = 0.696$	$\mu_{CR} = 0.217$
Diversity =	4.1822e+00	$k_t = [1, 1, 1, 1]$		
Generation	340: Best = 2.502113e+03	FES = 34100	$\mu_F = 0.700$	$\mu_{CR} = 0.185$
Diversity =	3.9137e+00	$k_t = [4, 2, 4, 3]$		
Generation	350: Best = 2.481888e+03	FES = 35100	$\mu_F = 0.695$	$\mu_{CR} = 0.159$
Diversity =	3.6695e+00	$k_t = [1, 1, 1, 1]$		
Generation	360: Best = 2.476293e+03	FES = 36100	$\mu_F = 0.716$	$\mu_{CR} = 0.127$
Diversity =	3.4815e+00	$k_t = [3, 3, 4, 2]$		
Generation	370: Best = 2.441458e+03	FES = 37100	$\mu_F = 0.742$	$\mu_{CR} = 0.108$
Diversity =	3.3966e+00	$k_t = [1, 1, 1, 1]$		
Generation	380: Best = 2.441458e+03	FES = 38100	$\mu_F = 0.752$	$\mu_{CR} = 0.099$
Diversity =	3.3836e+00	$k_t = [4, 3, 2, 2]$		
Generation	390: Best = 2.428445e+03	FES = 39100	$\mu_F = 0.737$	$\mu_{CR} = 0.085$
Diversity =	3.3503e+00	$k_t = [1, 1, 1, 1]$		
Generation	400: Best = 2.427898e+03	FES = 40100	$\mu_F = 0.747$	$\mu_{CR} = 0.082$
Diversity =	3.1511e+00	$k_t = [3, 4, 3, 2]$		
Generation	410: Best = 2.427898e+03	FES = 41100	$\mu_F = 0.721$	$\mu_{CR} = 0.082$
Diversity =	3.0489e+00	$k_t = [1, 1, 1, 1]$		
Generation	420: Best = 2.423145e+03	FES = 42100	$\mu_F = 0.719$	$\mu_{CR} = 0.082$
Diversity =	2.8533e+00	$k_t = [2, 4, 3, 4]$		
Generation	430: Best = 2.404913e+03	FES = 43100	$\mu_F = 0.723$	$\mu_{CR} = 0.079$
Diversity =	2.6702e+00	$k_t = [1, 1, 1, 1]$		
Generation	440: Best = 2.384909e+03	FES = 44100	$\mu_F = 0.724$	$\mu_{CR} = 0.083$
Diversity =	2.6261e+00	$k_t = [3, 2, 2, 3]$		
Generation	450: Best = 2.377631e+03	FES = 45100	$\mu_F = 0.720$	$\mu_{CR} = 0.083$
Diversity =	2.5704e+00	$k_t = [1, 1, 1, 1]$		
Generation	460: Best = 2.351661e+03	FES = 46100	$\mu_F = 0.723$	$\mu_{CR} = 0.100$
Diversity =	2.3567e+00	$k_t = [4, 4, 3, 4]$		
Generation	470: Best = 2.332001e+03	FES = 47100	$\mu_F = 0.730$	$\mu_{CR} = 0.114$
Diversity =	1.9500e+00	$k_t = [1, 1, 1, 1]$		
Generation	480: Best = 2.310577e+03	FES = 48100	$\mu_F = 0.745$	$\mu_{CR} = 0.138$
Diversity =	1.4017e+00	$k_t = [2, 3, 4, 2]$		
Generation	490: Best = 2.297767e+03	FES = 49100	$\mu_F = 0.754$	$\mu_{CR} = 0.156$
Diversity =	1.2601e+00	$k_t = [1, 1, 1, 1]$		

Generation 500:	Best = 2.274546e+03	FES = 50100	$\mu_F = 0.757$	$\mu_{CR} = 0.170$
Diversity = 1.1837e+00	$k_t = [3, 2, 2, 2]$			
Generation 510:	Best = 2.260659e+03	FES = 51100	$\mu_F = 0.762$	$\mu_{CR} = 0.169$
Diversity = 1.1645e+00	$k_t = [1, 1, 1, 1]$			
Generation 520:	Best = 2.247339e+03	FES = 52100	$\mu_F = 0.769$	$\mu_{CR} = 0.180$
Diversity = 1.1056e+00	$k_t = [2, 2, 2, 3]$			
Generation 530:	Best = 2.228602e+03	FES = 53100	$\mu_F = 0.775$	$\mu_{CR} = 0.186$
Diversity = 1.1544e+00	$k_t = [1, 1, 1, 1]$			
Generation 540:	Best = 2.199079e+03	FES = 54100	$\mu_F = 0.787$	$\mu_{CR} = 0.201$
Diversity = 1.2780e+00	$k_t = [4, 3, 3, 4]$			
Generation 550:	Best = 2.176890e+03	FES = 55100	$\mu_F = 0.796$	$\mu_{CR} = 0.212$
Diversity = 1.3337e+00	$k_t = [1, 1, 1, 1]$			
Generation 560:	Best = 2.157691e+03	FES = 56100	$\mu_F = 0.787$	$\mu_{CR} = 0.230$
Diversity = 1.3929e+00	$k_t = [4, 3, 2, 3]$			
Generation 570:	Best = 2.141528e+03	FES = 57100	$\mu_F = 0.779$	$\mu_{CR} = 0.226$
Diversity = 1.3105e+00	$k_t = [1, 1, 1, 1]$			
Generation 580:	Best = 2.129565e+03	FES = 58100	$\mu_F = 0.784$	$\mu_{CR} = 0.231$
Diversity = 1.2621e+00	$k_t = [2, 4, 3, 3]$			
Generation 590:	Best = 2.113638e+03	FES = 59100	$\mu_F = 0.792$	$\mu_{CR} = 0.237$
Diversity = 1.3921e+00	$k_t = [1, 1, 1, 1]$			
Generation 600:	Best = 2.101748e+03	FES = 60100	$\mu_F = 0.774$	$\mu_{CR} = 0.243$
Diversity = 1.3621e+00	$k_t = [4, 4, 3, 2]$			
Generation 610:	Best = 2.091650e+03	FES = 61100	$\mu_F = 0.769$	$\mu_{CR} = 0.250$
Diversity = 1.1823e+00	$k_t = [1, 1, 1, 1]$			
Generation 620:	Best = 2.081736e+03	FES = 62100	$\mu_F = 0.761$	$\mu_{CR} = 0.253$
Diversity = 1.0435e+00	$k_t = [2, 3, 2, 4]$			
Generation 630:	Best = 2.074149e+03	FES = 63100	$\mu_F = 0.760$	$\mu_{CR} = 0.259$
Diversity = 9.3459e-01	$k_t = [1, 1, 1, 1]$			
Generation 640:	Best = 2.066402e+03	FES = 64100	$\mu_F = 0.760$	$\mu_{CR} = 0.261$
Diversity = 9.3790e-01	$k_t = [4, 4, 2, 3]$			
Generation 650:	Best = 2.056643e+03	FES = 65100	$\mu_F = 0.762$	$\mu_{CR} = 0.272$
Diversity = 9.3414e-01	$k_t = [1, 1, 1, 1]$			
Generation 660:	Best = 2.050332e+03	FES = 66100	$\mu_F = 0.768$	$\mu_{CR} = 0.269$
Diversity = 9.4557e-01	$k_t = [2, 4, 2, 3]$			
Generation 670:	Best = 2.038581e+03	FES = 67100	$\mu_F = 0.769$	$\mu_{CR} = 0.273$
Diversity = 1.0140e+00	$k_t = [1, 1, 1, 1]$			
Generation 680:	Best = 2.030074e+03	FES = 68100	$\mu_F = 0.768$	$\mu_{CR} = 0.279$
Diversity = 1.0756e+00	$k_t = [2, 4, 3, 2]$			
Generation 690:	Best = 2.020296e+03	FES = 69100	$\mu_F = 0.772$	$\mu_{CR} = 0.277$
Diversity = 1.0651e+00	$k_t = [1, 1, 1, 1]$			
Generation 700:	Best = 2.014355e+03	FES = 70100	$\mu_F = 0.756$	$\mu_{CR} = 0.269$
Diversity = 9.3483e-01	$k_t = [3, 2, 2, 3]$			
Generation 710:	Best = 2.013617e+03	FES = 71100	$\mu_F = 0.733$	$\mu_{CR} = 0.263$
Diversity = 8.1549e-01	$k_t = [1, 1, 1, 1]$			
Generation 720:	Best = 2.011895e+03	FES = 72100	$\mu_F = 0.715$	$\mu_{CR} = 0.248$
Diversity = 7.8139e-01	$k_t = [2, 3, 2, 2]$			
Generation 730:	Best = 2.010636e+03	FES = 73100	$\mu_F = 0.720$	$\mu_{CR} = 0.222$
Diversity = 7.5799e-01	$k_t = [1, 1, 1, 1]$			
Generation 740:	Best = 2.010081e+03	FES = 74100	$\mu_F = 0.716$	$\mu_{CR} = 0.196$
Diversity = 7.3180e-01	$k_t = [3, 2, 3, 2]$			
Generation 750:	Best = 2.009018e+03	FES = 75100	$\mu_F = 0.732$	$\mu_{CR} = 0.175$
Diversity = 7.0740e-01	$k_t = [1, 1, 1, 1]$			
Generation 760:	Best = 2.008204e+03	FES = 76100	$\mu_F = 0.728$	$\mu_{CR} = 0.150$
Diversity = 6.8538e-01	$k_t = [2, 3, 4, 3]$			

Generation 770: Best = 2.007359e+03	FES = 77100	$\mu_F = 0.715$	$\mu_{CR} = 0.140$
Diversity = 6.5552e-01	$k_t = [1, 1, 1, 1]$		
Generation 780: Best = 2.006453e+03	FES = 78100	$\mu_F = 0.723$	$\mu_{CR} = 0.131$
Diversity = 6.1924e-01	$k_t = [2, 4, 4, 4]$		
Generation 790: Best = 2.006453e+03	FES = 79100	$\mu_F = 0.739$	$\mu_{CR} = 0.114$
Diversity = 6.3053e-01	$k_t = [1, 1, 1, 1]$		
Generation 800: Best = 2.006365e+03	FES = 80100	$\mu_F = 0.736$	$\mu_{CR} = 0.098$
Diversity = 6.2404e-01	$k_t = [3, 4, 4, 2]$		
Generation 810: Best = 2.006027e+03	FES = 81100	$\mu_F = 0.733$	$\mu_{CR} = 0.087$
Diversity = 6.1458e-01	$k_t = [1, 1, 1, 1]$		
Generation 820: Best = 2.005959e+03	FES = 82100	$\mu_F = 0.750$	$\mu_{CR} = 0.077$
Diversity = 6.1643e-01	$k_t = [4, 3, 4, 2]$		
Generation 830: Best = 2.005600e+03	FES = 83100	$\mu_F = 0.735$	$\mu_{CR} = 0.076$
Diversity = 6.0464e-01	$k_t = [1, 1, 1, 1]$		
Generation 840: Best = 2.005220e+03	FES = 84100	$\mu_F = 0.739$	$\mu_{CR} = 0.072$
Diversity = 5.9825e-01	$k_t = [2, 4, 4, 2]$		
Generation 850: Best = 2.005195e+03	FES = 85100	$\mu_F = 0.727$	$\mu_{CR} = 0.066$
Diversity = 5.9793e-01	$k_t = [1, 1, 1, 1]$		
Generation 860: Best = 2.005046e+03	FES = 86100	$\mu_F = 0.750$	$\mu_{CR} = 0.065$
Diversity = 6.0378e-01	$k_t = [2, 2, 2, 3]$		
Generation 870: Best = 2.004835e+03	FES = 87100	$\mu_F = 0.753$	$\mu_{CR} = 0.064$
Diversity = 6.0957e-01	$k_t = [1, 1, 1, 1]$		
Generation 880: Best = 2.004523e+03	FES = 88100	$\mu_F = 0.774$	$\mu_{CR} = 0.064$
Diversity = 6.0869e-01	$k_t = [4, 3, 2, 3]$		
Generation 890: Best = 2.004314e+03	FES = 89100	$\mu_F = 0.774$	$\mu_{CR} = 0.070$
Diversity = 6.1555e-01	$k_t = [1, 1, 1, 1]$		
Generation 900: Best = 2.003728e+03	FES = 90100	$\mu_F = 0.776$	$\mu_{CR} = 0.066$
Diversity = 6.1882e-01	$k_t = [4, 4, 2, 4]$		
Generation 910: Best = 2.003698e+03	FES = 91100	$\mu_F = 0.770$	$\mu_{CR} = 0.067$
Diversity = 6.1946e-01	$k_t = [1, 1, 1, 1]$		
Generation 920: Best = 2.003517e+03	FES = 92100	$\mu_F = 0.758$	$\mu_{CR} = 0.066$
Diversity = 6.1722e-01	$k_t = [3, 2, 3, 2]$		
Generation 930: Best = 2.003276e+03	FES = 93100	$\mu_F = 0.752$	$\mu_{CR} = 0.078$
Diversity = 6.2876e-01	$k_t = [1, 1, 1, 1]$		
Generation 940: Best = 2.002746e+03	FES = 94100	$\mu_F = 0.779$	$\mu_{CR} = 0.074$
Diversity = 6.1821e-01	$k_t = [3, 3, 2, 3]$		
Generation 950: Best = 2.002581e+03	FES = 95100	$\mu_F = 0.792$	$\mu_{CR} = 0.060$
Diversity = 6.0315e-01	$k_t = [1, 1, 1, 1]$		
Generation 960: Best = 2.002383e+03	FES = 96100	$\mu_F = 0.802$	$\mu_{CR} = 0.069$
Diversity = 6.0406e-01	$k_t = [2, 4, 4, 4]$		
Generation 970: Best = 2.002348e+03	FES = 97100	$\mu_F = 0.814$	$\mu_{CR} = 0.071$
Diversity = 5.8766e-01	$k_t = [1, 1, 1, 1]$		
Generation 980: Best = 2.002234e+03	FES = 98100	$\mu_F = 0.811$	$\mu_{CR} = 0.060$
Diversity = 5.9087e-01	$k_t = [3, 2, 4, 4]$		
Generation 990: Best = 2.001922e+03	FES = 99100	$\mu_F = 0.813$	$\mu_{CR} = 0.063$
Diversity = 5.8988e-01	$k_t = [1, 1, 1, 1]$		
Generation 1000: Best = 2.001756e+03	FES = 100100	$\mu_F = 0.809$	$\mu_{CR} = 0.059$
Diversity = 5.8682e-01	$k_t = [3, 4, 4, 4]$		
Generation 1010: Best = 2.001640e+03	FES = 101100	$\mu_F = 0.806$	$\mu_{CR} = 0.053$
Diversity = 5.8386e-01	$k_t = [1, 1, 1, 1]$		
Generation 1020: Best = 2.001405e+03	FES = 102100	$\mu_F = 0.805$	$\mu_{CR} = 0.066$
Diversity = 5.7676e-01	$k_t = [2, 4, 2, 4]$		
Generation 1030: Best = 2.001312e+03	FES = 103100	$\mu_F = 0.830$	$\mu_{CR} = 0.065$
Diversity = 5.6600e-01	$k_t = [1, 1, 1, 1]$		

Generation 1040: Best = 2.001119e+03 FES = 104100 μ_F = 0.805 μ_{CR} = 0.064
 Diversity = 5.6120e-01 k_t = [4, 4, 2, 2]
 Generation 1050: Best = 2.000975e+03 FES = 105100 μ_F = 0.821 μ_{CR} = 0.060
 Diversity = 5.5626e-01 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 2.000871e+03 FES = 106100 μ_F = 0.826 μ_{CR} = 0.056
 Diversity = 5.4184e-01 k_t = [3, 2, 4, 4]
 Generation 1070: Best = 2.000504e+03 FES = 107100 μ_F = 0.807 μ_{CR} = 0.065
 Diversity = 5.3213e-01 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 2.000397e+03 FES = 108100 μ_F = 0.834 μ_{CR} = 0.060
 Diversity = 5.2465e-01 k_t = [3, 2, 3, 4]
 Generation 1090: Best = 1.999960e+03 FES = 109100 μ_F = 0.823 μ_{CR} = 0.062
 Diversity = 5.2845e-01 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 1.999921e+03 FES = 110100 μ_F = 0.830 μ_{CR} = 0.074
 Diversity = 5.2404e-01 k_t = [2, 4, 3, 3]
 Generation 1110: Best = 1.999502e+03 FES = 111100 μ_F = 0.811 μ_{CR} = 0.073
 Diversity = 5.1897e-01 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 1.999085e+03 FES = 112100 μ_F = 0.811 μ_{CR} = 0.064
 Diversity = 5.3635e-01 k_t = [4, 4, 2, 2]
 Generation 1130: Best = 1.999085e+03 FES = 113100 μ_F = 0.809 μ_{CR} = 0.075
 Diversity = 5.4135e-01 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 1.998960e+03 FES = 114100 μ_F = 0.800 μ_{CR} = 0.071
 Diversity = 5.3399e-01 k_t = [2, 3, 4, 4]
 Generation 1150: Best = 1.998781e+03 FES = 115100 μ_F = 0.809 μ_{CR} = 0.066
 Diversity = 5.3110e-01 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 1.998604e+03 FES = 116100 μ_F = 0.810 μ_{CR} = 0.061
 Diversity = 5.2618e-01 k_t = [2, 3, 3, 3]
 Generation 1170: Best = 1.998047e+03 FES = 117100 μ_F = 0.803 μ_{CR} = 0.066
 Diversity = 5.1048e-01 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 1.997564e+03 FES = 118100 μ_F = 0.791 μ_{CR} = 0.070
 Diversity = 4.9576e-01 k_t = [3, 2, 3, 4]
 Generation 1190: Best = 1.997346e+03 FES = 119100 μ_F = 0.793 μ_{CR} = 0.075
 Diversity = 4.9030e-01 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 1.997290e+03 FES = 120100 μ_F = 0.789 μ_{CR} = 0.072
 Diversity = 4.6511e-01 k_t = [4, 4, 3, 3]
 Generation 1210: Best = 1.997084e+03 FES = 121100 μ_F = 0.791 μ_{CR} = 0.079
 Diversity = 4.6088e-01 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 1.996347e+03 FES = 122100 μ_F = 0.789 μ_{CR} = 0.082
 Diversity = 4.7390e-01 k_t = [3, 4, 3, 2]
 Generation 1230: Best = 1.996234e+03 FES = 123100 μ_F = 0.793 μ_{CR} = 0.082
 Diversity = 4.7573e-01 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 1.996167e+03 FES = 124100 μ_F = 0.782 μ_{CR} = 0.073
 Diversity = 4.6325e-01 k_t = [3, 4, 4, 3]
 Generation 1250: Best = 1.995766e+03 FES = 125100 μ_F = 0.773 μ_{CR} = 0.068
 Diversity = 4.6875e-01 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 1.995574e+03 FES = 126100 μ_F = 0.759 μ_{CR} = 0.070
 Diversity = 4.5756e-01 k_t = [3, 3, 2, 2]
 Generation 1270: Best = 1.995435e+03 FES = 127100 μ_F = 0.770 μ_{CR} = 0.064
 Diversity = 4.4549e-01 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 1.995160e+03 FES = 128100 μ_F = 0.777 μ_{CR} = 0.079
 Diversity = 4.3254e-01 k_t = [4, 3, 2, 3]
 Generation 1290: Best = 1.994799e+03 FES = 129100 μ_F = 0.783 μ_{CR} = 0.068
 Diversity = 4.3307e-01 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 1.994692e+03 FES = 130100 μ_F = 0.778 μ_{CR} = 0.068
 Diversity = 4.3242e-01 k_t = [3, 3, 2, 2]

Generation 1310: Best = 1.994379e+03 FES = 131100 μ_F = 0.772 μ_{CR} = 0.073
 Diversity = 4.2124e-01 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 1.994127e+03 FES = 132100 μ_F = 0.768 μ_{CR} = 0.068
 Diversity = 4.1436e-01 k_t = [3, 2, 2, 2]
 Generation 1330: Best = 1.994009e+03 FES = 133100 μ_F = 0.761 μ_{CR} = 0.069
 Diversity = 4.0428e-01 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 1.993657e+03 FES = 134100 μ_F = 0.762 μ_{CR} = 0.075
 Diversity = 3.9252e-01 k_t = [4, 3, 3, 2]
 Generation 1350: Best = 1.993386e+03 FES = 135100 μ_F = 0.759 μ_{CR} = 0.071
 Diversity = 3.9171e-01 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 1.993124e+03 FES = 136100 μ_F = 0.763 μ_{CR} = 0.077
 Diversity = 3.7970e-01 k_t = [3, 3, 4, 3]
 Generation 1370: Best = 1.992723e+03 FES = 137100 μ_F = 0.765 μ_{CR} = 0.084
 Diversity = 3.6918e-01 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 1.992571e+03 FES = 138100 μ_F = 0.770 μ_{CR} = 0.082
 Diversity = 3.5941e-01 k_t = [2, 3, 3, 4]
 Generation 1390: Best = 1.992415e+03 FES = 139100 μ_F = 0.768 μ_{CR} = 0.076
 Diversity = 3.6283e-01 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 1.992283e+03 FES = 140100 μ_F = 0.776 μ_{CR} = 0.076
 Diversity = 3.5235e-01 k_t = [2, 3, 4, 2]
 Generation 1410: Best = 1.992031e+03 FES = 141100 μ_F = 0.779 μ_{CR} = 0.082
 Diversity = 3.5219e-01 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 1.991680e+03 FES = 142100 μ_F = 0.788 μ_{CR} = 0.090
 Diversity = 3.4066e-01 k_t = [2, 4, 3, 2]
 Generation 1430: Best = 1.991571e+03 FES = 143100 μ_F = 0.781 μ_{CR} = 0.085
 Diversity = 3.4204e-01 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 1.991300e+03 FES = 144100 μ_F = 0.797 μ_{CR} = 0.088
 Diversity = 3.4562e-01 k_t = [2, 2, 4, 2]
 Generation 1450: Best = 1.991143e+03 FES = 145100 μ_F = 0.801 μ_{CR} = 0.078
 Diversity = 3.5584e-01 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 1.990921e+03 FES = 146100 μ_F = 0.815 μ_{CR} = 0.074
 Diversity = 3.4404e-01 k_t = [4, 4, 3, 4]
 Generation 1470: Best = 1.990519e+03 FES = 147100 μ_F = 0.801 μ_{CR} = 0.069
 Diversity = 3.4658e-01 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 1.990519e+03 FES = 148100 μ_F = 0.783 μ_{CR} = 0.064
 Diversity = 3.5267e-01 k_t = [4, 2, 3, 4]
 Generation 1490: Best = 1.990276e+03 FES = 149100 μ_F = 0.788 μ_{CR} = 0.067
 Diversity = 3.5684e-01 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 1.990148e+03 FES = 150100 μ_F = 0.778 μ_{CR} = 0.072
 Diversity = 3.5384e-01 k_t = [3, 3, 3, 2]
 Generation 1510: Best = 1.989980e+03 FES = 151100 μ_F = 0.782 μ_{CR} = 0.070
 Diversity = 3.4726e-01 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 1.989730e+03 FES = 152100 μ_F = 0.772 μ_{CR} = 0.073
 Diversity = 3.4401e-01 k_t = [3, 2, 3, 3]
 Generation 1530: Best = 1.989517e+03 FES = 153100 μ_F = 0.771 μ_{CR} = 0.063
 Diversity = 3.3954e-01 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 1.989223e+03 FES = 154100 μ_F = 0.771 μ_{CR} = 0.070
 Diversity = 3.3369e-01 k_t = [2, 2, 3, 2]
 Generation 1550: Best = 1.989223e+03 FES = 155100 μ_F = 0.767 μ_{CR} = 0.069
 Diversity = 3.2575e-01 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 1.988948e+03 FES = 156100 μ_F = 0.778 μ_{CR} = 0.068
 Diversity = 3.3002e-01 k_t = [3, 2, 2, 2]
 Generation 1570: Best = 1.988842e+03 FES = 157100 μ_F = 0.733 μ_{CR} = 0.075
 Diversity = 3.2321e-01 k_t = [1, 1, 1, 1]

Generation 1580: Best = 1.988746e+03 FES = 158100 μ_F = 0.746 μ_{CR} = 0.069
 Diversity = 3.1453e-01 k_t = [3, 4, 3, 2]
 Generation 1590: Best = 1.988519e+03 FES = 159100 μ_F = 0.737 μ_{CR} = 0.067
 Diversity = 3.0882e-01 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 1.988359e+03 FES = 160100 μ_F = 0.751 μ_{CR} = 0.064
 Diversity = 3.0081e-01 k_t = [3, 2, 3, 4]
 Generation 1610: Best = 1.988320e+03 FES = 161100 μ_F = 0.762 μ_{CR} = 0.067
 Diversity = 2.9630e-01 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 1.988140e+03 FES = 162100 μ_F = 0.783 μ_{CR} = 0.059
 Diversity = 2.9533e-01 k_t = [3, 3, 3, 2]
 Generation 1630: Best = 1.987959e+03 FES = 163100 μ_F = 0.782 μ_{CR} = 0.061
 Diversity = 2.8813e-01 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 1.987948e+03 FES = 164100 μ_F = 0.782 μ_{CR} = 0.061
 Diversity = 2.8380e-01 k_t = [3, 4, 4, 2]
 Generation 1650: Best = 1.987864e+03 FES = 165100 μ_F = 0.744 μ_{CR} = 0.068
 Diversity = 2.7854e-01 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 1.987536e+03 FES = 166100 μ_F = 0.755 μ_{CR} = 0.073
 Diversity = 2.7374e-01 k_t = [2, 2, 4, 4]
 Generation 1670: Best = 1.987496e+03 FES = 167100 μ_F = 0.746 μ_{CR} = 0.058
 Diversity = 2.6194e-01 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 1.987496e+03 FES = 168100 μ_F = 0.762 μ_{CR} = 0.058
 Diversity = 2.5438e-01 k_t = [3, 4, 3, 3]
 Generation 1690: Best = 1.987325e+03 FES = 169100 μ_F = 0.795 μ_{CR} = 0.060
 Diversity = 2.5330e-01 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 1.987295e+03 FES = 170100 μ_F = 0.787 μ_{CR} = 0.057
 Diversity = 2.4939e-01 k_t = [3, 2, 2, 4]
 Generation 1710: Best = 1.987161e+03 FES = 171100 μ_F = 0.795 μ_{CR} = 0.054
 Diversity = 2.4035e-01 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 1.987044e+03 FES = 172100 μ_F = 0.785 μ_{CR} = 0.053
 Diversity = 2.3360e-01 k_t = [4, 3, 3, 4]
 Generation 1730: Best = 1.987027e+03 FES = 173100 μ_F = 0.772 μ_{CR} = 0.053
 Diversity = 2.2772e-01 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 1.987004e+03 FES = 174100 μ_F = 0.767 μ_{CR} = 0.052
 Diversity = 2.2533e-01 k_t = [2, 4, 3, 3]
 Generation 1750: Best = 1.986965e+03 FES = 175100 μ_F = 0.770 μ_{CR} = 0.047
 Diversity = 2.2047e-01 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 1.986895e+03 FES = 176100 μ_F = 0.760 μ_{CR} = 0.056
 Diversity = 2.1567e-01 k_t = [4, 2, 3, 2]
 Generation 1770: Best = 1.986829e+03 FES = 177100 μ_F = 0.769 μ_{CR} = 0.058
 Diversity = 2.1765e-01 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 1.986700e+03 FES = 178100 μ_F = 0.760 μ_{CR} = 0.056
 Diversity = 2.1360e-01 k_t = [2, 4, 4, 4]
 Generation 1790: Best = 1.986699e+03 FES = 179100 μ_F = 0.752 μ_{CR} = 0.067
 Diversity = 2.0799e-01 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 1.986537e+03 FES = 180100 μ_F = 0.765 μ_{CR} = 0.066
 Diversity = 2.0882e-01 k_t = [4, 2, 2, 3]
 Generation 1810: Best = 1.986461e+03 FES = 181100 μ_F = 0.782 μ_{CR} = 0.064
 Diversity = 2.1898e-01 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 1.986461e+03 FES = 182100 μ_F = 0.799 μ_{CR} = 0.056
 Diversity = 2.1588e-01 k_t = [2, 2, 4, 4]
 Generation 1830: Best = 1.986370e+03 FES = 183100 μ_F = 0.793 μ_{CR} = 0.063
 Diversity = 2.1649e-01 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 1.986318e+03 FES = 184100 μ_F = 0.799 μ_{CR} = 0.054
 Diversity = 2.1228e-01 k_t = [3, 3, 2, 4]

Generation 1850: Best = 1.986172e+03	FES = 185100	$\mu_F = 0.813$	$\mu_{CR} = 0.054$
Diversity = 2.0982e-01	$k_t = [1, 1, 1, 1]$		
Generation 1860: Best = 1.986156e+03	FES = 186100	$\mu_F = 0.810$	$\mu_{CR} = 0.059$
Diversity = 2.1071e-01	$k_t = [3, 2, 3, 3]$		
Generation 1870: Best = 1.986130e+03	FES = 187100	$\mu_F = 0.807$	$\mu_{CR} = 0.058$
Diversity = 2.1742e-01	$k_t = [1, 1, 1, 1]$		
Generation 1880: Best = 1.986022e+03	FES = 188100	$\mu_F = 0.805$	$\mu_{CR} = 0.055$
Diversity = 2.2103e-01	$k_t = [3, 3, 3, 3]$		
Generation 1890: Best = 1.985926e+03	FES = 189100	$\mu_F = 0.799$	$\mu_{CR} = 0.064$
Diversity = 2.2912e-01	$k_t = [1, 1, 1, 1]$		
Generation 1900: Best = 1.985902e+03	FES = 190100	$\mu_F = 0.801$	$\mu_{CR} = 0.067$
Diversity = 2.3431e-01	$k_t = [3, 3, 3, 3]$		
Generation 1910: Best = 1.985783e+03	FES = 191100	$\mu_F = 0.808$	$\mu_{CR} = 0.061$
Diversity = 2.4587e-01	$k_t = [1, 1, 1, 1]$		
Generation 1920: Best = 1.985692e+03	FES = 192100	$\mu_F = 0.806$	$\mu_{CR} = 0.057$
Diversity = 2.4854e-01	$k_t = [2, 2, 2, 2]$		
Generation 1930: Best = 1.985622e+03	FES = 193100	$\mu_F = 0.814$	$\mu_{CR} = 0.063$
Diversity = 2.5653e-01	$k_t = [1, 1, 1, 1]$		
Generation 1940: Best = 1.985535e+03	FES = 194100	$\mu_F = 0.817$	$\mu_{CR} = 0.063$
Diversity = 2.5808e-01	$k_t = [2, 2, 2, 4]$		
Generation 1950: Best = 1.985370e+03	FES = 195100	$\mu_F = 0.821$	$\mu_{CR} = 0.066$
Diversity = 2.6345e-01	$k_t = [1, 1, 1, 1]$		
Generation 1960: Best = 1.985150e+03	FES = 196100	$\mu_F = 0.821$	$\mu_{CR} = 0.073$
Diversity = 2.6990e-01	$k_t = [4, 3, 2, 3]$		
Generation 1970: Best = 1.985071e+03	FES = 197100	$\mu_F = 0.844$	$\mu_{CR} = 0.068$
Diversity = 2.6660e-01	$k_t = [1, 1, 1, 1]$		
Generation 1980: Best = 1.984944e+03	FES = 198100	$\mu_F = 0.850$	$\mu_{CR} = 0.063$
Diversity = 2.7371e-01	$k_t = [2, 2, 4, 2]$		
Generation 1990: Best = 1.984852e+03	FES = 199100	$\mu_F = 0.832$	$\mu_{CR} = 0.070$
Diversity = 2.8278e-01	$k_t = [1, 1, 1, 1]$		
Generation 2000: Best = 1.984722e+03	FES = 200100	$\mu_F = 0.837$	$\mu_{CR} = 0.070$
Diversity = 2.8259e-01	$k_t = [4, 3, 4, 3]$		
Generation 2010: Best = 1.984543e+03	FES = 201100	$\mu_F = 0.804$	$\mu_{CR} = 0.074$
Diversity = 2.8223e-01	$k_t = [1, 1, 1, 1]$		
Generation 2020: Best = 1.984327e+03	FES = 202100	$\mu_F = 0.813$	$\mu_{CR} = 0.072$
Diversity = 2.8364e-01	$k_t = [4, 2, 4, 2]$		
Generation 2030: Best = 1.984217e+03	FES = 203100	$\mu_F = 0.814$	$\mu_{CR} = 0.076$
Diversity = 2.9063e-01	$k_t = [1, 1, 1, 1]$		
Generation 2040: Best = 1.983421e+03	FES = 204100	$\mu_F = 0.801$	$\mu_{CR} = 0.076$
Diversity = 2.9393e-01	$k_t = [2, 2, 4, 2]$		
Generation 2050: Best = 1.983391e+03	FES = 205100	$\mu_F = 0.786$	$\mu_{CR} = 0.077$
Diversity = 2.8699e-01	$k_t = [1, 1, 1, 1]$		
Generation 2060: Best = 1.983118e+03	FES = 206100	$\mu_F = 0.781$	$\mu_{CR} = 0.074$
Diversity = 2.7720e-01	$k_t = [3, 3, 2, 2]$		
Generation 2070: Best = 1.982991e+03	FES = 207100	$\mu_F = 0.776$	$\mu_{CR} = 0.078$
Diversity = 2.7532e-01	$k_t = [1, 1, 1, 1]$		
Generation 2080: Best = 1.982769e+03	FES = 208100	$\mu_F = 0.785$	$\mu_{CR} = 0.078$
Diversity = 2.6440e-01	$k_t = [2, 4, 4, 4]$		
Generation 2090: Best = 1.982639e+03	FES = 209100	$\mu_F = 0.781$	$\mu_{CR} = 0.078$
Diversity = 2.6200e-01	$k_t = [1, 1, 1, 1]$		
Generation 2100: Best = 1.982334e+03	FES = 210100	$\mu_F = 0.788$	$\mu_{CR} = 0.076$
Diversity = 2.5166e-01	$k_t = [2, 4, 4, 3]$		
Generation 2110: Best = 1.982299e+03	FES = 211100	$\mu_F = 0.761$	$\mu_{CR} = 0.076$
Diversity = 2.5251e-01	$k_t = [1, 1, 1, 1]$		

Generation 2120: Best = 1.982067e+03 FES = 212100 μ_F = 0.759 μ_{CR} = 0.075
 Diversity = 2.3715e-01 k_t = [3, 3, 2, 3]
 Generation 2130: Best = 1.981981e+03 FES = 213100 μ_F = 0.753 μ_{CR} = 0.068
 Diversity = 2.3141e-01 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 1.981885e+03 FES = 214100 μ_F = 0.776 μ_{CR} = 0.071
 Diversity = 2.2708e-01 k_t = [3, 3, 4, 3]
 Generation 2150: Best = 1.981669e+03 FES = 215100 μ_F = 0.777 μ_{CR} = 0.063
 Diversity = 2.3832e-01 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 1.981586e+03 FES = 216100 μ_F = 0.776 μ_{CR} = 0.066
 Diversity = 2.4261e-01 k_t = [3, 4, 2, 4]
 Generation 2170: Best = 1.981262e+03 FES = 217100 μ_F = 0.769 μ_{CR} = 0.070
 Diversity = 2.4198e-01 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 1.981125e+03 FES = 218100 μ_F = 0.772 μ_{CR} = 0.061
 Diversity = 2.3570e-01 k_t = [2, 2, 4, 3]
 Generation 2190: Best = 1.981120e+03 FES = 219100 μ_F = 0.756 μ_{CR} = 0.063
 Diversity = 2.3534e-01 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 1.980933e+03 FES = 220100 μ_F = 0.755 μ_{CR} = 0.061
 Diversity = 2.2459e-01 k_t = [4, 2, 3, 4]
 Generation 2210: Best = 1.980783e+03 FES = 221100 μ_F = 0.735 μ_{CR} = 0.060
 Diversity = 2.1469e-01 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 1.980651e+03 FES = 222100 μ_F = 0.755 μ_{CR} = 0.061
 Diversity = 2.1188e-01 k_t = [3, 3, 2, 3]
 Generation 2230: Best = 1.980436e+03 FES = 223100 μ_F = 0.764 μ_{CR} = 0.065
 Diversity = 2.1046e-01 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 1.980335e+03 FES = 224100 μ_F = 0.761 μ_{CR} = 0.081
 Diversity = 2.0853e-01 k_t = [4, 4, 2, 2]
 Generation 2250: Best = 1.980077e+03 FES = 225100 μ_F = 0.776 μ_{CR} = 0.072
 Diversity = 2.0778e-01 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 1.980070e+03 FES = 226100 μ_F = 0.765 μ_{CR} = 0.065
 Diversity = 2.0570e-01 k_t = [2, 4, 4, 2]
 Generation 2270: Best = 1.980070e+03 FES = 227100 μ_F = 0.735 μ_{CR} = 0.064
 Diversity = 2.0806e-01 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 1.979848e+03 FES = 228100 μ_F = 0.765 μ_{CR} = 0.073
 Diversity = 2.0749e-01 k_t = [4, 2, 4, 4]
 Generation 2290: Best = 1.979791e+03 FES = 229100 μ_F = 0.767 μ_{CR} = 0.073
 Diversity = 2.0645e-01 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 1.979682e+03 FES = 230100 μ_F = 0.780 μ_{CR} = 0.074
 Diversity = 2.2167e-01 k_t = [4, 3, 3, 2]
 Generation 2310: Best = 1.979603e+03 FES = 231100 μ_F = 0.785 μ_{CR} = 0.063
 Diversity = 2.2158e-01 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 1.979553e+03 FES = 232100 μ_F = 0.766 μ_{CR} = 0.054
 Diversity = 2.1983e-01 k_t = [3, 3, 4, 3]
 Generation 2330: Best = 1.979462e+03 FES = 233100 μ_F = 0.744 μ_{CR} = 0.055
 Diversity = 2.2287e-01 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 1.979272e+03 FES = 234100 μ_F = 0.752 μ_{CR} = 0.063
 Diversity = 2.2513e-01 k_t = [3, 3, 3, 2]
 Generation 2350: Best = 1.979183e+03 FES = 235100 μ_F = 0.756 μ_{CR} = 0.062
 Diversity = 2.2463e-01 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 1.979083e+03 FES = 236100 μ_F = 0.754 μ_{CR} = 0.067
 Diversity = 2.2908e-01 k_t = [4, 2, 3, 2]
 Generation 2370: Best = 1.978930e+03 FES = 237100 μ_F = 0.766 μ_{CR} = 0.068
 Diversity = 2.3382e-01 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 1.978796e+03 FES = 238100 μ_F = 0.770 μ_{CR} = 0.060
 Diversity = 2.3345e-01 k_t = [2, 3, 2, 3]

Generation 2390: Best = 1.978459e+03 FES = 239100 μ_F = 0.764 μ_{CR} = 0.062
 Diversity = 2.3789e-01 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 1.978459e+03 FES = 240100 μ_F = 0.769 μ_{CR} = 0.058
 Diversity = 2.3524e-01 k_t = [4, 4, 2, 2]
 Generation 2410: Best = 1.978457e+03 FES = 241100 μ_F = 0.778 μ_{CR} = 0.060
 Diversity = 2.3912e-01 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 1.978325e+03 FES = 242100 μ_F = 0.787 μ_{CR} = 0.070
 Diversity = 2.4049e-01 k_t = [2, 3, 3, 2]
 Generation 2430: Best = 1.978200e+03 FES = 243100 μ_F = 0.764 μ_{CR} = 0.073
 Diversity = 2.4079e-01 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 1.978175e+03 FES = 244100 μ_F = 0.766 μ_{CR} = 0.073
 Diversity = 2.3811e-01 k_t = [4, 3, 4, 3]
 Generation 2450: Best = 1.978031e+03 FES = 245100 μ_F = 0.768 μ_{CR} = 0.072
 Diversity = 2.3589e-01 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 1.977836e+03 FES = 246100 μ_F = 0.765 μ_{CR} = 0.072
 Diversity = 2.3021e-01 k_t = [3, 2, 4, 3]
 Generation 2470: Best = 1.977748e+03 FES = 247100 μ_F = 0.744 μ_{CR} = 0.068
 Diversity = 2.2425e-01 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 1.977600e+03 FES = 248100 μ_F = 0.748 μ_{CR} = 0.067
 Diversity = 2.1460e-01 k_t = [3, 2, 4, 4]
 Generation 2490: Best = 1.977404e+03 FES = 249100 μ_F = 0.786 μ_{CR} = 0.066
 Diversity = 2.1477e-01 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 1.977379e+03 FES = 250100 μ_F = 0.785 μ_{CR} = 0.064
 Diversity = 2.1171e-01 k_t = [2, 2, 4, 3]
 Generation 2510: Best = 1.977318e+03 FES = 251100 μ_F = 0.804 μ_{CR} = 0.058
 Diversity = 2.1481e-01 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 1.977252e+03 FES = 252100 μ_F = 0.799 μ_{CR} = 0.064
 Diversity = 2.1212e-01 k_t = [3, 2, 2, 3]
 Generation 2530: Best = 1.977052e+03 FES = 253100 μ_F = 0.784 μ_{CR} = 0.067
 Diversity = 2.1065e-01 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 1.976967e+03 FES = 254100 μ_F = 0.785 μ_{CR} = 0.061
 Diversity = 2.0755e-01 k_t = [2, 4, 4, 4]
 Generation 2550: Best = 1.976962e+03 FES = 255100 μ_F = 0.772 μ_{CR} = 0.065
 Diversity = 2.0626e-01 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 1.976742e+03 FES = 256100 μ_F = 0.778 μ_{CR} = 0.072
 Diversity = 2.0284e-01 k_t = [3, 3, 3, 4]
 Generation 2570: Best = 1.976742e+03 FES = 257100 μ_F = 0.752 μ_{CR} = 0.075
 Diversity = 1.9858e-01 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 1.976710e+03 FES = 258100 μ_F = 0.752 μ_{CR} = 0.075
 Diversity = 1.9498e-01 k_t = [2, 2, 3, 3]
 Generation 2590: Best = 1.976470e+03 FES = 259100 μ_F = 0.721 μ_{CR} = 0.072
 Diversity = 1.9698e-01 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 1.976470e+03 FES = 260100 μ_F = 0.705 μ_{CR} = 0.075
 Diversity = 1.9911e-01 k_t = [2, 3, 2, 2]
 Generation 2610: Best = 1.976383e+03 FES = 261100 μ_F = 0.741 μ_{CR} = 0.068
 Diversity = 1.9708e-01 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 1.976236e+03 FES = 262100 μ_F = 0.740 μ_{CR} = 0.065
 Diversity = 1.9450e-01 k_t = [3, 3, 2, 4]
 Generation 2630: Best = 1.976192e+03 FES = 263100 μ_F = 0.748 μ_{CR} = 0.080
 Diversity = 1.9650e-01 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 1.975972e+03 FES = 264100 μ_F = 0.765 μ_{CR} = 0.068
 Diversity = 2.0157e-01 k_t = [3, 3, 2, 4]
 Generation 2650: Best = 1.975965e+03 FES = 265100 μ_F = 0.753 μ_{CR} = 0.069
 Diversity = 1.9790e-01 k_t = [1, 1, 1, 1]

Generation 2660: Best = 1.975858e+03 FES = 266100 μ_F = 0.745 μ_{CR} = 0.070
 Diversity = 1.9363e-01 k_t = [2, 2, 2, 2]
 Generation 2670: Best = 1.975832e+03 FES = 267100 μ_F = 0.752 μ_{CR} = 0.063
 Diversity = 1.9426e-01 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 1.975778e+03 FES = 268100 μ_F = 0.744 μ_{CR} = 0.058
 Diversity = 1.8856e-01 k_t = [3, 2, 4, 3]
 Generation 2690: Best = 1.975762e+03 FES = 269100 μ_F = 0.757 μ_{CR} = 0.051
 Diversity = 1.8639e-01 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 1.975648e+03 FES = 270100 μ_F = 0.734 μ_{CR} = 0.062
 Diversity = 1.7867e-01 k_t = [4, 3, 2, 3]
 Generation 2710: Best = 1.975499e+03 FES = 271100 μ_F = 0.707 μ_{CR} = 0.064
 Diversity = 1.7417e-01 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 1.975484e+03 FES = 272100 μ_F = 0.717 μ_{CR} = 0.063
 Diversity = 1.7005e-01 k_t = [2, 2, 4, 4]
 Generation 2730: Best = 1.975482e+03 FES = 273100 μ_F = 0.729 μ_{CR} = 0.057
 Diversity = 1.6588e-01 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 1.975362e+03 FES = 274100 μ_F = 0.739 μ_{CR} = 0.054
 Diversity = 1.7023e-01 k_t = [3, 3, 3, 2]
 Generation 2750: Best = 1.975362e+03 FES = 275100 μ_F = 0.751 μ_{CR} = 0.059
 Diversity = 1.7004e-01 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 1.975362e+03 FES = 276100 μ_F = 0.765 μ_{CR} = 0.063
 Diversity = 1.6898e-01 k_t = [4, 4, 3, 2]
 Generation 2770: Best = 1.975303e+03 FES = 277100 μ_F = 0.794 μ_{CR} = 0.059
 Diversity = 1.7446e-01 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 1.975109e+03 FES = 278100 μ_F = 0.798 μ_{CR} = 0.056
 Diversity = 1.7490e-01 k_t = [3, 4, 2, 2]
 Generation 2790: Best = 1.975109e+03 FES = 279100 μ_F = 0.771 μ_{CR} = 0.050
 Diversity = 1.7495e-01 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 1.975052e+03 FES = 280100 μ_F = 0.758 μ_{CR} = 0.060
 Diversity = 1.7650e-01 k_t = [2, 4, 3, 3]
 Generation 2810: Best = 1.975051e+03 FES = 281100 μ_F = 0.777 μ_{CR} = 0.056
 Diversity = 1.7609e-01 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 1.974948e+03 FES = 282100 μ_F = 0.794 μ_{CR} = 0.051
 Diversity = 1.7572e-01 k_t = [3, 4, 3, 3]
 Generation 2830: Best = 1.974922e+03 FES = 283100 μ_F = 0.810 μ_{CR} = 0.046
 Diversity = 1.8109e-01 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 1.974892e+03 FES = 284100 μ_F = 0.818 μ_{CR} = 0.054
 Diversity = 1.8658e-01 k_t = [4, 3, 4, 2]
 Generation 2850: Best = 1.974735e+03 FES = 285100 μ_F = 0.837 μ_{CR} = 0.048
 Diversity = 1.9368e-01 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 1.974693e+03 FES = 286100 μ_F = 0.825 μ_{CR} = 0.050
 Diversity = 1.9946e-01 k_t = [2, 3, 2, 2]
 Generation 2870: Best = 1.974544e+03 FES = 287100 μ_F = 0.824 μ_{CR} = 0.057
 Diversity = 2.0488e-01 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 1.974535e+03 FES = 288100 μ_F = 0.837 μ_{CR} = 0.058
 Diversity = 2.0811e-01 k_t = [3, 2, 4, 3]
 Generation 2890: Best = 1.974468e+03 FES = 289100 μ_F = 0.846 μ_{CR} = 0.058
 Diversity = 2.2131e-01 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 1.974276e+03 FES = 290100 μ_F = 0.841 μ_{CR} = 0.058
 Diversity = 2.2778e-01 k_t = [4, 4, 2, 2]
 Generation 2910: Best = 1.974241e+03 FES = 291100 μ_F = 0.849 μ_{CR} = 0.064
 Diversity = 2.4087e-01 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 1.974143e+03 FES = 292100 μ_F = 0.819 μ_{CR} = 0.064
 Diversity = 2.4983e-01 k_t = [2, 4, 2, 2]

Generation 2930: Best = 1.973975e+03 FES = 293100 μ_F = 0.826 μ_{CR} = 0.067
 Diversity = 2.6245e-01 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 1.973837e+03 FES = 294100 μ_F = 0.832 μ_{CR} = 0.068
 Diversity = 2.7270e-01 k_t = [3, 2, 3, 2]
 Generation 2950: Best = 1.973477e+03 FES = 295100 μ_F = 0.813 μ_{CR} = 0.063
 Diversity = 2.8860e-01 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 1.973326e+03 FES = 296100 μ_F = 0.807 μ_{CR} = 0.078
 Diversity = 2.9907e-01 k_t = [4, 4, 2, 2]
 Generation 2970: Best = 1.973030e+03 FES = 297100 μ_F = 0.807 μ_{CR} = 0.083
 Diversity = 3.0749e-01 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 1.973021e+03 FES = 298100 μ_F = 0.807 μ_{CR} = 0.087
 Diversity = 3.0022e-01 k_t = [2, 3, 3, 4]
 Generation 2990: Best = 1.972668e+03 FES = 299100 μ_F = 0.815 μ_{CR} = 0.084
 Diversity = 2.8761e-01 k_t = [1, 1, 1, 1]
 Generation 3000: Best = 1.972566e+03 FES = 300100 μ_F = 0.789 μ_{CR} = 0.074
 Diversity = 2.9110e-01 k_t = [3, 2, 4, 4]
 Generation 3010: Best = 1.972524e+03 FES = 301100 μ_F = 0.795 μ_{CR} = 0.074
 Diversity = 2.9220e-01 k_t = [1, 1, 1, 1]
 Generation 3020: Best = 1.972424e+03 FES = 302100 μ_F = 0.776 μ_{CR} = 0.069
 Diversity = 2.8834e-01 k_t = [4, 3, 2, 4]
 Generation 3030: Best = 1.972301e+03 FES = 303100 μ_F = 0.787 μ_{CR} = 0.067
 Diversity = 2.9084e-01 k_t = [1, 1, 1, 1]
 Generation 3040: Best = 1.972137e+03 FES = 304100 μ_F = 0.781 μ_{CR} = 0.072
 Diversity = 2.8672e-01 k_t = [2, 4, 4, 4]
 Generation 3050: Best = 1.972019e+03 FES = 305100 μ_F = 0.799 μ_{CR} = 0.067
 Diversity = 2.8490e-01 k_t = [1, 1, 1, 1]
 Generation 3060: Best = 1.971788e+03 FES = 306100 μ_F = 0.787 μ_{CR} = 0.065
 Diversity = 2.7344e-01 k_t = [2, 4, 3, 3]
 Generation 3070: Best = 1.971744e+03 FES = 307100 μ_F = 0.781 μ_{CR} = 0.063
 Diversity = 2.5930e-01 k_t = [1, 1, 1, 1]
 Generation 3080: Best = 1.971468e+03 FES = 308100 μ_F = 0.770 μ_{CR} = 0.063
 Diversity = 2.4605e-01 k_t = [3, 4, 4, 2]
 Generation 3090: Best = 1.971432e+03 FES = 309100 μ_F = 0.772 μ_{CR} = 0.062
 Diversity = 2.4813e-01 k_t = [1, 1, 1, 1]
 Generation 3100: Best = 1.971386e+03 FES = 310100 μ_F = 0.764 μ_{CR} = 0.063
 Diversity = 2.4607e-01 k_t = [3, 2, 2, 2]
 Generation 3110: Best = 1.971250e+03 FES = 311100 μ_F = 0.782 μ_{CR} = 0.061
 Diversity = 2.5028e-01 k_t = [1, 1, 1, 1]
 Generation 3120: Best = 1.971159e+03 FES = 312100 μ_F = 0.783 μ_{CR} = 0.063
 Diversity = 2.4325e-01 k_t = [2, 2, 2, 3]
 Generation 3130: Best = 1.971044e+03 FES = 313100 μ_F = 0.764 μ_{CR} = 0.061
 Diversity = 2.4029e-01 k_t = [1, 1, 1, 1]
 Generation 3140: Best = 1.970907e+03 FES = 314100 μ_F = 0.763 μ_{CR} = 0.065
 Diversity = 2.4058e-01 k_t = [4, 4, 2, 3]
 Generation 3150: Best = 1.970821e+03 FES = 315100 μ_F = 0.765 μ_{CR} = 0.068
 Diversity = 2.3883e-01 k_t = [1, 1, 1, 1]
 Generation 3160: Best = 1.970654e+03 FES = 316100 μ_F = 0.782 μ_{CR} = 0.056
 Diversity = 2.4016e-01 k_t = [2, 3, 3, 2]
 Generation 3170: Best = 1.970572e+03 FES = 317100 μ_F = 0.786 μ_{CR} = 0.059
 Diversity = 2.4269e-01 k_t = [1, 1, 1, 1]
 Generation 3180: Best = 1.970563e+03 FES = 318100 μ_F = 0.779 μ_{CR} = 0.069
 Diversity = 2.3623e-01 k_t = [4, 3, 2, 4]
 Generation 3190: Best = 1.970548e+03 FES = 319100 μ_F = 0.783 μ_{CR} = 0.062
 Diversity = 2.3671e-01 k_t = [1, 1, 1, 1]

Generation 3200: Best = 1.970472e+03 FES = 320100 μ_F = 0.781 μ_{CR} = 0.065
 Diversity = 2.3961e-01 k_t = [2, 4, 4, 2]
 Generation 3210: Best = 1.970327e+03 FES = 321100 μ_F = 0.776 μ_{CR} = 0.064
 Diversity = 2.4443e-01 k_t = [1, 1, 1, 1]
 Generation 3220: Best = 1.970203e+03 FES = 322100 μ_F = 0.774 μ_{CR} = 0.064
 Diversity = 2.4599e-01 k_t = [4, 2, 3, 3]
 Generation 3230: Best = 1.970148e+03 FES = 323100 μ_F = 0.770 μ_{CR} = 0.063
 Diversity = 2.4481e-01 k_t = [1, 1, 1, 1]
 Generation 3240: Best = 1.970100e+03 FES = 324100 μ_F = 0.758 μ_{CR} = 0.060
 Diversity = 2.3899e-01 k_t = [4, 4, 4, 3]
 Generation 3250: Best = 1.969962e+03 FES = 325100 μ_F = 0.749 μ_{CR} = 0.065
 Diversity = 2.3562e-01 k_t = [1, 1, 1, 1]
 Generation 3260: Best = 1.969862e+03 FES = 326100 μ_F = 0.768 μ_{CR} = 0.060
 Diversity = 2.3871e-01 k_t = [2, 4, 2, 3]
 Generation 3270: Best = 1.969599e+03 FES = 327100 μ_F = 0.782 μ_{CR} = 0.060
 Diversity = 2.4432e-01 k_t = [1, 1, 1, 1]
 Generation 3280: Best = 1.969599e+03 FES = 328100 μ_F = 0.782 μ_{CR} = 0.065
 Diversity = 2.5029e-01 k_t = [4, 4, 2, 4]
 Generation 3290: Best = 1.969469e+03 FES = 329100 μ_F = 0.783 μ_{CR} = 0.061
 Diversity = 2.4910e-01 k_t = [1, 1, 1, 1]
 Generation 3300: Best = 1.969394e+03 FES = 330100 μ_F = 0.794 μ_{CR} = 0.059
 Diversity = 2.5142e-01 k_t = [4, 3, 3, 2]
 Generation 3310: Best = 1.969288e+03 FES = 331100 μ_F = 0.763 μ_{CR} = 0.062
 Diversity = 2.5192e-01 k_t = [1, 1, 1, 1]
 Generation 3320: Best = 1.969231e+03 FES = 332100 μ_F = 0.777 μ_{CR} = 0.060
 Diversity = 2.5439e-01 k_t = [2, 3, 4, 4]
 Generation 3330: Best = 1.969122e+03 FES = 333100 μ_F = 0.781 μ_{CR} = 0.062
 Diversity = 2.4863e-01 k_t = [1, 1, 1, 1]
 Generation 3340: Best = 1.969068e+03 FES = 334100 μ_F = 0.759 μ_{CR} = 0.064
 Diversity = 2.4255e-01 k_t = [4, 4, 2, 3]
 Generation 3350: Best = 1.968828e+03 FES = 335100 μ_F = 0.761 μ_{CR} = 0.055
 Diversity = 2.3778e-01 k_t = [1, 1, 1, 1]
 Generation 3360: Best = 1.968812e+03 FES = 336100 μ_F = 0.745 μ_{CR} = 0.070
 Diversity = 2.3428e-01 k_t = [3, 4, 2, 2]
 Generation 3370: Best = 1.968812e+03 FES = 337100 μ_F = 0.729 μ_{CR} = 0.067
 Diversity = 2.3067e-01 k_t = [1, 1, 1, 1]
 Generation 3380: Best = 1.968702e+03 FES = 338100 μ_F = 0.729 μ_{CR} = 0.065
 Diversity = 2.2694e-01 k_t = [4, 2, 4, 2]
 Generation 3390: Best = 1.968638e+03 FES = 339100 μ_F = 0.717 μ_{CR} = 0.063
 Diversity = 2.2828e-01 k_t = [1, 1, 1, 1]
 Generation 3400: Best = 1.968507e+03 FES = 340100 μ_F = 0.740 μ_{CR} = 0.056
 Diversity = 2.2135e-01 k_t = [4, 3, 2, 2]
 Generation 3410: Best = 1.968507e+03 FES = 341100 μ_F = 0.753 μ_{CR} = 0.055
 Diversity = 2.2301e-01 k_t = [1, 1, 1, 1]
 Generation 3420: Best = 1.968400e+03 FES = 342100 μ_F = 0.753 μ_{CR} = 0.063
 Diversity = 2.2390e-01 k_t = [2, 2, 4, 3]
 Generation 3430: Best = 1.968400e+03 FES = 343100 μ_F = 0.747 μ_{CR} = 0.060
 Diversity = 2.2311e-01 k_t = [1, 1, 1, 1]
 Generation 3440: Best = 1.968279e+03 FES = 344100 μ_F = 0.768 μ_{CR} = 0.064
 Diversity = 2.2184e-01 k_t = [3, 4, 3, 2]
 Generation 3450: Best = 1.968276e+03 FES = 345100 μ_F = 0.762 μ_{CR} = 0.055
 Diversity = 2.2309e-01 k_t = [1, 1, 1, 1]
 Generation 3460: Best = 1.968206e+03 FES = 346100 μ_F = 0.744 μ_{CR} = 0.057
 Diversity = 2.2058e-01 k_t = [2, 4, 4, 2]

Generation 3470: Best = 1.968187e+03 FES = 347100 μ_F = 0.743 μ_{CR} = 0.058
 Diversity = 2.1802e-01 k_t = [1, 1, 1, 1]
 Generation 3480: Best = 1.968183e+03 FES = 348100 μ_F = 0.770 μ_{CR} = 0.053
 Diversity = 2.1889e-01 k_t = [4, 4, 3, 4]
 Generation 3490: Best = 1.968042e+03 FES = 349100 μ_F = 0.790 μ_{CR} = 0.053
 Diversity = 2.1380e-01 k_t = [1, 1, 1, 1]
 Generation 3500: Best = 1.968036e+03 FES = 350100 μ_F = 0.806 μ_{CR} = 0.047
 Diversity = 2.1295e-01 k_t = [3, 3, 4, 2]
 Generation 3510: Best = 1.967895e+03 FES = 351100 μ_F = 0.818 μ_{CR} = 0.047
 Diversity = 2.1375e-01 k_t = [1, 1, 1, 1]
 Generation 3520: Best = 1.967895e+03 FES = 352100 μ_F = 0.790 μ_{CR} = 0.047
 Diversity = 2.1520e-01 k_t = [4, 3, 4, 3]
 Generation 3530: Best = 1.967891e+03 FES = 353100 μ_F = 0.803 μ_{CR} = 0.054
 Diversity = 2.1732e-01 k_t = [1, 1, 1, 1]
 Generation 3540: Best = 1.967702e+03 FES = 354100 μ_F = 0.794 μ_{CR} = 0.050
 Diversity = 2.1898e-01 k_t = [3, 2, 4, 2]
 Generation 3550: Best = 1.967651e+03 FES = 355100 μ_F = 0.766 μ_{CR} = 0.063
 Diversity = 2.1872e-01 k_t = [1, 1, 1, 1]
 Generation 3560: Best = 1.967589e+03 FES = 356100 μ_F = 0.790 μ_{CR} = 0.059
 Diversity = 2.1916e-01 k_t = [4, 2, 3, 4]
 Generation 3570: Best = 1.967579e+03 FES = 357100 μ_F = 0.763 μ_{CR} = 0.052
 Diversity = 2.1802e-01 k_t = [1, 1, 1, 1]
 Generation 3580: Best = 1.967546e+03 FES = 358100 μ_F = 0.773 μ_{CR} = 0.050
 Diversity = 2.1506e-01 k_t = [3, 3, 2, 3]
 Generation 3590: Best = 1.967441e+03 FES = 359100 μ_F = 0.750 μ_{CR} = 0.051
 Diversity = 2.1233e-01 k_t = [1, 1, 1, 1]
 Generation 3600: Best = 1.967441e+03 FES = 360100 μ_F = 0.750 μ_{CR} = 0.058
 Diversity = 2.1145e-01 k_t = [4, 2, 2, 2]
 Generation 3610: Best = 1.967441e+03 FES = 361100 μ_F = 0.766 μ_{CR} = 0.056
 Diversity = 2.0977e-01 k_t = [1, 1, 1, 1]
 Generation 3620: Best = 1.967281e+03 FES = 362100 μ_F = 0.773 μ_{CR} = 0.053
 Diversity = 2.1312e-01 k_t = [3, 3, 4, 2]
 Generation 3630: Best = 1.967216e+03 FES = 363100 μ_F = 0.765 μ_{CR} = 0.061
 Diversity = 2.1648e-01 k_t = [1, 1, 1, 1]
 Generation 3640: Best = 1.967126e+03 FES = 364100 μ_F = 0.747 μ_{CR} = 0.073
 Diversity = 2.2262e-01 k_t = [3, 2, 4, 3]
 Generation 3650: Best = 1.967068e+03 FES = 365100 μ_F = 0.755 μ_{CR} = 0.061
 Diversity = 2.2233e-01 k_t = [1, 1, 1, 1]
 Generation 3660: Best = 1.967032e+03 FES = 366100 μ_F = 0.774 μ_{CR} = 0.062
 Diversity = 2.2404e-01 k_t = [2, 2, 2, 3]
 Generation 3670: Best = 1.966977e+03 FES = 367100 μ_F = 0.763 μ_{CR} = 0.050
 Diversity = 2.2325e-01 k_t = [1, 1, 1, 1]
 Generation 3680: Best = 1.966977e+03 FES = 368100 μ_F = 0.762 μ_{CR} = 0.054
 Diversity = 2.2127e-01 k_t = [4, 3, 4, 4]
 Generation 3690: Best = 1.966977e+03 FES = 369100 μ_F = 0.774 μ_{CR} = 0.054
 Diversity = 2.2146e-01 k_t = [1, 1, 1, 1]
 Generation 3700: Best = 1.966830e+03 FES = 370100 μ_F = 0.773 μ_{CR} = 0.055
 Diversity = 2.2150e-01 k_t = [4, 4, 3, 3]
 Generation 3710: Best = 1.966625e+03 FES = 371100 μ_F = 0.758 μ_{CR} = 0.057
 Diversity = 2.2115e-01 k_t = [1, 1, 1, 1]
 Generation 3720: Best = 1.966625e+03 FES = 372100 μ_F = 0.758 μ_{CR} = 0.060
 Diversity = 2.1972e-01 k_t = [2, 3, 2, 3]
 Generation 3730: Best = 1.966571e+03 FES = 373100 μ_F = 0.754 μ_{CR} = 0.054
 Diversity = 2.1437e-01 k_t = [1, 1, 1, 1]

Generation 3740: Best = 1.966453e+03 FES = 374100 μ_F = 0.764 μ_{CR} = 0.052
 Diversity = 2.1059e-01 k_t = [4, 3, 4, 2]
 Generation 3750: Best = 1.966430e+03 FES = 375100 μ_F = 0.782 μ_{CR} = 0.063
 Diversity = 2.0964e-01 k_t = [1, 1, 1, 1]
 Generation 3760: Best = 1.966376e+03 FES = 376100 μ_F = 0.765 μ_{CR} = 0.066
 Diversity = 2.0415e-01 k_t = [4, 3, 2, 4]
 Generation 3770: Best = 1.966376e+03 FES = 377100 μ_F = 0.768 μ_{CR} = 0.052
 Diversity = 2.0158e-01 k_t = [1, 1, 1, 1]
 Generation 3780: Best = 1.966263e+03 FES = 378100 μ_F = 0.786 μ_{CR} = 0.056
 Diversity = 1.9864e-01 k_t = [2, 3, 3, 3]
 Generation 3790: Best = 1.966261e+03 FES = 379100 μ_F = 0.777 μ_{CR} = 0.058
 Diversity = 1.9868e-01 k_t = [1, 1, 1, 1]
 Generation 3800: Best = 1.966224e+03 FES = 380100 μ_F = 0.764 μ_{CR} = 0.058
 Diversity = 1.9426e-01 k_t = [4, 2, 2, 4]
 Generation 3810: Best = 1.966131e+03 FES = 381100 μ_F = 0.772 μ_{CR} = 0.060
 Diversity = 1.9778e-01 k_t = [1, 1, 1, 1]
 Generation 3820: Best = 1.966131e+03 FES = 382100 μ_F = 0.780 μ_{CR} = 0.054
 Diversity = 2.0029e-01 k_t = [3, 3, 4, 2]
 Generation 3830: Best = 1.965981e+03 FES = 383100 μ_F = 0.771 μ_{CR} = 0.059
 Diversity = 2.0158e-01 k_t = [1, 1, 1, 1]
 Generation 3840: Best = 1.965963e+03 FES = 384100 μ_F = 0.768 μ_{CR} = 0.064
 Diversity = 2.0095e-01 k_t = [3, 2, 4, 2]
 Generation 3850: Best = 1.965919e+03 FES = 385100 μ_F = 0.793 μ_{CR} = 0.058
 Diversity = 2.0422e-01 k_t = [1, 1, 1, 1]
 Generation 3860: Best = 1.965860e+03 FES = 386100 μ_F = 0.778 μ_{CR} = 0.061
 Diversity = 2.0525e-01 k_t = [2, 4, 3, 3]
 Generation 3870: Best = 1.965854e+03 FES = 387100 μ_F = 0.763 μ_{CR} = 0.062
 Diversity = 2.0199e-01 k_t = [1, 1, 1, 1]
 Generation 3880: Best = 1.965788e+03 FES = 388100 μ_F = 0.779 μ_{CR} = 0.056
 Diversity = 2.0077e-01 k_t = [4, 3, 4, 4]
 Generation 3890: Best = 1.965734e+03 FES = 389100 μ_F = 0.804 μ_{CR} = 0.053
 Diversity = 2.0369e-01 k_t = [1, 1, 1, 1]
 Generation 3900: Best = 1.965734e+03 FES = 390100 μ_F = 0.813 μ_{CR} = 0.056
 Diversity = 2.0502e-01 k_t = [4, 4, 3, 4]
 Generation 3910: Best = 1.965703e+03 FES = 391100 μ_F = 0.823 μ_{CR} = 0.055
 Diversity = 2.0119e-01 k_t = [1, 1, 1, 1]
 Generation 3920: Best = 1.965665e+03 FES = 392100 μ_F = 0.799 μ_{CR} = 0.057
 Diversity = 2.0059e-01 k_t = [4, 3, 2, 2]
 Generation 3930: Best = 1.965660e+03 FES = 393100 μ_F = 0.783 μ_{CR} = 0.060
 Diversity = 2.0027e-01 k_t = [1, 1, 1, 1]
 Generation 3940: Best = 1.965591e+03 FES = 394100 μ_F = 0.804 μ_{CR} = 0.054
 Diversity = 1.9769e-01 k_t = [3, 4, 3, 3]
 Generation 3950: Best = 1.965483e+03 FES = 395100 μ_F = 0.802 μ_{CR} = 0.056
 Diversity = 1.9879e-01 k_t = [1, 1, 1, 1]
 Generation 3960: Best = 1.965483e+03 FES = 396100 μ_F = 0.809 μ_{CR} = 0.056
 Diversity = 1.9975e-01 k_t = [3, 4, 2, 2]
 Generation 3970: Best = 1.965483e+03 FES = 397100 μ_F = 0.819 μ_{CR} = 0.051
 Diversity = 1.9988e-01 k_t = [1, 1, 1, 1]
 Generation 3980: Best = 1.965464e+03 FES = 398100 μ_F = 0.836 μ_{CR} = 0.046
 Diversity = 1.9961e-01 k_t = [3, 4, 3, 4]
 Generation 3990: Best = 1.965347e+03 FES = 399100 μ_F = 0.826 μ_{CR} = 0.042
 Diversity = 1.9814e-01 k_t = [1, 1, 1, 1]
 Generation 4000: Best = 1.965347e+03 FES = 400100 μ_F = 0.837 μ_{CR} = 0.045
 Diversity = 1.9756e-01 k_t = [3, 2, 2, 2]

Generation 4010: Best = 1.965343e+03 FES = 401100 μ_F = 0.803 μ_{CR} = 0.040
 Diversity = 1.9847e-01 k_t = [1, 1, 1, 1]
 Generation 4020: Best = 1.965324e+03 FES = 402100 μ_F = 0.807 μ_{CR} = 0.048
 Diversity = 1.9877e-01 k_t = [4, 4, 4, 2]
 Generation 4030: Best = 1.965271e+03 FES = 403100 μ_F = 0.789 μ_{CR} = 0.053
 Diversity = 2.0162e-01 k_t = [1, 1, 1, 1]
 Generation 4040: Best = 1.965166e+03 FES = 404100 μ_F = 0.805 μ_{CR} = 0.056
 Diversity = 1.9990e-01 k_t = [2, 2, 2, 4]
 Generation 4050: Best = 1.965163e+03 FES = 405100 μ_F = 0.808 μ_{CR} = 0.049
 Diversity = 1.9986e-01 k_t = [1, 1, 1, 1]
 Generation 4060: Best = 1.965053e+03 FES = 406100 μ_F = 0.800 μ_{CR} = 0.052
 Diversity = 1.9769e-01 k_t = [4, 2, 4, 3]
 Generation 4070: Best = 1.965026e+03 FES = 407100 μ_F = 0.802 μ_{CR} = 0.060
 Diversity = 2.0144e-01 k_t = [1, 1, 1, 1]
 Generation 4080: Best = 1.964801e+03 FES = 408100 μ_F = 0.814 μ_{CR} = 0.052
 Diversity = 2.0476e-01 k_t = [4, 2, 2, 2]
 Generation 4090: Best = 1.964801e+03 FES = 409100 μ_F = 0.815 μ_{CR} = 0.053
 Diversity = 2.0425e-01 k_t = [1, 1, 1, 1]
 Generation 4100: Best = 1.964800e+03 FES = 410100 μ_F = 0.815 μ_{CR} = 0.056
 Diversity = 2.0914e-01 k_t = [2, 4, 4, 2]
 Generation 4110: Best = 1.964800e+03 FES = 411100 μ_F = 0.812 μ_{CR} = 0.059
 Diversity = 2.1133e-01 k_t = [1, 1, 1, 1]
 Generation 4120: Best = 1.964792e+03 FES = 412100 μ_F = 0.820 μ_{CR} = 0.058
 Diversity = 2.1125e-01 k_t = [4, 4, 3, 4]
 Generation 4130: Best = 1.964792e+03 FES = 413100 μ_F = 0.842 μ_{CR} = 0.053
 Diversity = 2.1097e-01 k_t = [1, 1, 1, 1]
 Generation 4140: Best = 1.964561e+03 FES = 414100 μ_F = 0.835 μ_{CR} = 0.051
 Diversity = 2.1180e-01 k_t = [2, 4, 4, 2]
 Generation 4150: Best = 1.964561e+03 FES = 415100 μ_F = 0.822 μ_{CR} = 0.046
 Diversity = 2.1699e-01 k_t = [1, 1, 1, 1]
 Generation 4160: Best = 1.964465e+03 FES = 416100 μ_F = 0.815 μ_{CR} = 0.051
 Diversity = 2.2532e-01 k_t = [4, 3, 2, 4]
 Generation 4170: Best = 1.964394e+03 FES = 417100 μ_F = 0.818 μ_{CR} = 0.048
 Diversity = 2.3030e-01 k_t = [1, 1, 1, 1]
 Generation 4180: Best = 1.964394e+03 FES = 418100 μ_F = 0.828 μ_{CR} = 0.049
 Diversity = 2.3220e-01 k_t = [4, 4, 2, 4]
 Generation 4190: Best = 1.964361e+03 FES = 419100 μ_F = 0.845 μ_{CR} = 0.056
 Diversity = 2.3215e-01 k_t = [1, 1, 1, 1]
 Generation 4200: Best = 1.964349e+03 FES = 420100 μ_F = 0.834 μ_{CR} = 0.049
 Diversity = 2.3234e-01 k_t = [3, 4, 3, 4]
 Generation 4210: Best = 1.964313e+03 FES = 421100 μ_F = 0.826 μ_{CR} = 0.053
 Diversity = 2.3498e-01 k_t = [1, 1, 1, 1]
 Generation 4220: Best = 1.964157e+03 FES = 422100 μ_F = 0.835 μ_{CR} = 0.049
 Diversity = 2.3239e-01 k_t = [3, 2, 2, 4]
 Generation 4230: Best = 1.964042e+03 FES = 423100 μ_F = 0.824 μ_{CR} = 0.051
 Diversity = 2.3042e-01 k_t = [1, 1, 1, 1]
 Generation 4240: Best = 1.964038e+03 FES = 424100 μ_F = 0.819 μ_{CR} = 0.056
 Diversity = 2.2886e-01 k_t = [3, 2, 4, 2]
 Generation 4250: Best = 1.963978e+03 FES = 425100 μ_F = 0.824 μ_{CR} = 0.053
 Diversity = 2.2817e-01 k_t = [1, 1, 1, 1]
 Generation 4260: Best = 1.963941e+03 FES = 426100 μ_F = 0.822 μ_{CR} = 0.058
 Diversity = 2.3049e-01 k_t = [2, 4, 2, 3]
 Generation 4270: Best = 1.963866e+03 FES = 427100 μ_F = 0.794 μ_{CR} = 0.055
 Diversity = 2.2864e-01 k_t = [1, 1, 1, 1]

Generation 4280: Best = 1.963866e+03 FES = 428100 μ_F = 0.807 μ_{CR} = 0.056
 Diversity = 2.2542e-01 k_t = [3, 3, 3, 4]
 Generation 4290: Best = 1.963792e+03 FES = 429100 μ_F = 0.813 μ_{CR} = 0.054
 Diversity = 2.2370e-01 k_t = [1, 1, 1, 1]
 Generation 4300: Best = 1.963671e+03 FES = 430100 μ_F = 0.805 μ_{CR} = 0.058
 Diversity = 2.2185e-01 k_t = [3, 2, 2, 4]
 Generation 4310: Best = 1.963622e+03 FES = 431100 μ_F = 0.797 μ_{CR} = 0.051
 Diversity = 2.1831e-01 k_t = [1, 1, 1, 1]
 Generation 4320: Best = 1.963618e+03 FES = 432100 μ_F = 0.804 μ_{CR} = 0.052
 Diversity = 2.1554e-01 k_t = [4, 3, 3, 4]
 Generation 4330: Best = 1.963615e+03 FES = 433100 μ_F = 0.789 μ_{CR} = 0.052
 Diversity = 2.1087e-01 k_t = [1, 1, 1, 1]
 Generation 4340: Best = 1.963513e+03 FES = 434100 μ_F = 0.797 μ_{CR} = 0.053
 Diversity = 2.0492e-01 k_t = [3, 3, 2, 4]
 Generation 4350: Best = 1.963487e+03 FES = 435100 μ_F = 0.760 μ_{CR} = 0.051
 Diversity = 2.0251e-01 k_t = [1, 1, 1, 1]
 Generation 4360: Best = 1.963431e+03 FES = 436100 μ_F = 0.765 μ_{CR} = 0.047
 Diversity = 2.0462e-01 k_t = [4, 2, 2, 4]
 Generation 4370: Best = 1.963395e+03 FES = 437100 μ_F = 0.774 μ_{CR} = 0.053
 Diversity = 2.0382e-01 k_t = [1, 1, 1, 1]
 Generation 4380: Best = 1.963353e+03 FES = 438100 μ_F = 0.774 μ_{CR} = 0.047
 Diversity = 2.0343e-01 k_t = [2, 4, 3, 3]
 Generation 4390: Best = 1.963321e+03 FES = 439100 μ_F = 0.787 μ_{CR} = 0.049
 Diversity = 2.0583e-01 k_t = [1, 1, 1, 1]
 Generation 4400: Best = 1.963207e+03 FES = 440100 μ_F = 0.788 μ_{CR} = 0.054
 Diversity = 2.0280e-01 k_t = [3, 2, 3, 3]
 Generation 4410: Best = 1.963183e+03 FES = 441100 μ_F = 0.774 μ_{CR} = 0.054
 Diversity = 2.0276e-01 k_t = [1, 1, 1, 1]
 Generation 4420: Best = 1.963158e+03 FES = 442100 μ_F = 0.773 μ_{CR} = 0.053
 Diversity = 2.0134e-01 k_t = [4, 4, 4, 2]
 Generation 4430: Best = 1.963119e+03 FES = 443100 μ_F = 0.774 μ_{CR} = 0.054
 Diversity = 2.0291e-01 k_t = [1, 1, 1, 1]
 Generation 4440: Best = 1.963117e+03 FES = 444100 μ_F = 0.770 μ_{CR} = 0.051
 Diversity = 1.9901e-01 k_t = [4, 2, 4, 3]
 Generation 4450: Best = 1.963113e+03 FES = 445100 μ_F = 0.766 μ_{CR} = 0.049
 Diversity = 2.0077e-01 k_t = [1, 1, 1, 1]
 Generation 4460: Best = 1.963083e+03 FES = 446100 μ_F = 0.768 μ_{CR} = 0.054
 Diversity = 1.9650e-01 k_t = [4, 2, 3, 2]
 Generation 4470: Best = 1.962959e+03 FES = 447100 μ_F = 0.755 μ_{CR} = 0.055
 Diversity = 1.9457e-01 k_t = [1, 1, 1, 1]
 Generation 4480: Best = 1.962879e+03 FES = 448100 μ_F = 0.772 μ_{CR} = 0.053
 Diversity = 1.9139e-01 k_t = [2, 4, 2, 4]
 Generation 4490: Best = 1.962879e+03 FES = 449100 μ_F = 0.781 μ_{CR} = 0.045
 Diversity = 1.8890e-01 k_t = [1, 1, 1, 1]
 Generation 4500: Best = 1.962879e+03 FES = 450100 μ_F = 0.764 μ_{CR} = 0.046
 Diversity = 1.8610e-01 k_t = [3, 3, 2, 2]
 Generation 4510: Best = 1.962838e+03 FES = 451100 μ_F = 0.760 μ_{CR} = 0.049
 Diversity = 1.8461e-01 k_t = [1, 1, 1, 1]
 Generation 4520: Best = 1.962825e+03 FES = 452100 μ_F = 0.762 μ_{CR} = 0.050
 Diversity = 1.8239e-01 k_t = [4, 4, 2, 2]
 Generation 4530: Best = 1.962823e+03 FES = 453100 μ_F = 0.777 μ_{CR} = 0.045
 Diversity = 1.8590e-01 k_t = [1, 1, 1, 1]
 Generation 4540: Best = 1.962808e+03 FES = 454100 μ_F = 0.788 μ_{CR} = 0.044
 Diversity = 1.8522e-01 k_t = [3, 2, 2, 2]

Generation 4550: Best = 1.962747e+03 FES = 455100 μ_F = 0.798 μ_{CR} = 0.051
 Diversity = 1.8522e-01 k_t = [1, 1, 1, 1]
 Generation 4560: Best = 1.962729e+03 FES = 456100 μ_F = 0.799 μ_{CR} = 0.046
 Diversity = 1.8525e-01 k_t = [2, 3, 4, 3]
 Generation 4570: Best = 1.962687e+03 FES = 457100 μ_F = 0.782 μ_{CR} = 0.040
 Diversity = 1.8254e-01 k_t = [1, 1, 1, 1]
 Generation 4580: Best = 1.962687e+03 FES = 458100 μ_F = 0.777 μ_{CR} = 0.039
 Diversity = 1.8316e-01 k_t = [2, 2, 2, 3]
 Generation 4590: Best = 1.962627e+03 FES = 459100 μ_F = 0.763 μ_{CR} = 0.054
 Diversity = 1.8406e-01 k_t = [1, 1, 1, 1]
 Generation 4600: Best = 1.962583e+03 FES = 460100 μ_F = 0.762 μ_{CR} = 0.052
 Diversity = 1.8338e-01 k_t = [2, 3, 3, 2]
 Generation 4610: Best = 1.962566e+03 FES = 461100 μ_F = 0.789 μ_{CR} = 0.052
 Diversity = 1.8471e-01 k_t = [1, 1, 1, 1]
 Generation 4620: Best = 1.962503e+03 FES = 462100 μ_F = 0.804 μ_{CR} = 0.054
 Diversity = 1.8375e-01 k_t = [2, 3, 3, 4]
 Generation 4630: Best = 1.962483e+03 FES = 463100 μ_F = 0.816 μ_{CR} = 0.048
 Diversity = 1.8221e-01 k_t = [1, 1, 1, 1]
 Generation 4640: Best = 1.962465e+03 FES = 464100 μ_F = 0.830 μ_{CR} = 0.046
 Diversity = 1.8554e-01 k_t = [3, 3, 3, 3]
 Generation 4650: Best = 1.962406e+03 FES = 465100 μ_F = 0.823 μ_{CR} = 0.046
 Diversity = 1.8587e-01 k_t = [1, 1, 1, 1]
 Generation 4660: Best = 1.962398e+03 FES = 466100 μ_F = 0.807 μ_{CR} = 0.046
 Diversity = 1.8753e-01 k_t = [4, 2, 3, 3]
 Generation 4670: Best = 1.962378e+03 FES = 467100 μ_F = 0.813 μ_{CR} = 0.043
 Diversity = 1.9035e-01 k_t = [1, 1, 1, 1]
 Generation 4680: Best = 1.962358e+03 FES = 468100 μ_F = 0.804 μ_{CR} = 0.047
 Diversity = 1.8880e-01 k_t = [2, 2, 4, 4]
 Generation 4690: Best = 1.962324e+03 FES = 469100 μ_F = 0.804 μ_{CR} = 0.048
 Diversity = 1.8962e-01 k_t = [1, 1, 1, 1]
 Generation 4700: Best = 1.962324e+03 FES = 470100 μ_F = 0.805 μ_{CR} = 0.052
 Diversity = 1.8722e-01 k_t = [2, 2, 2, 3]
 Generation 4710: Best = 1.962244e+03 FES = 471100 μ_F = 0.798 μ_{CR} = 0.054
 Diversity = 1.8477e-01 k_t = [1, 1, 1, 1]
 Generation 4720: Best = 1.962222e+03 FES = 472100 μ_F = 0.803 μ_{CR} = 0.050
 Diversity = 1.9053e-01 k_t = [4, 3, 3, 3]
 Generation 4730: Best = 1.962197e+03 FES = 473100 μ_F = 0.827 μ_{CR} = 0.045
 Diversity = 1.9314e-01 k_t = [1, 1, 1, 1]
 Generation 4740: Best = 1.962174e+03 FES = 474100 μ_F = 0.809 μ_{CR} = 0.046
 Diversity = 1.9631e-01 k_t = [2, 3, 3, 3]
 Generation 4750: Best = 1.962136e+03 FES = 475100 μ_F = 0.776 μ_{CR} = 0.049
 Diversity = 1.9700e-01 k_t = [1, 1, 1, 1]
 Generation 4760: Best = 1.962104e+03 FES = 476100 μ_F = 0.795 μ_{CR} = 0.049
 Diversity = 2.0001e-01 k_t = [3, 2, 3, 2]
 Generation 4770: Best = 1.962021e+03 FES = 477100 μ_F = 0.805 μ_{CR} = 0.057
 Diversity = 2.0212e-01 k_t = [1, 1, 1, 1]
 Generation 4780: Best = 1.961995e+03 FES = 478100 μ_F = 0.796 μ_{CR} = 0.054
 Diversity = 2.0305e-01 k_t = [4, 2, 2, 2]
 Generation 4790: Best = 1.961991e+03 FES = 479100 μ_F = 0.808 μ_{CR} = 0.054
 Diversity = 2.0470e-01 k_t = [1, 1, 1, 1]
 Generation 4800: Best = 1.961960e+03 FES = 480100 μ_F = 0.818 μ_{CR} = 0.057
 Diversity = 2.0691e-01 k_t = [4, 4, 2, 4]
 Generation 4810: Best = 1.961817e+03 FES = 481100 μ_F = 0.801 μ_{CR} = 0.058
 Diversity = 2.0719e-01 k_t = [1, 1, 1, 1]

Generation 4820: Best = 1.961811e+03 FES = 482100 μ_F = 0.816 μ_{CR} = 0.054
 Diversity = 2.0527e-01 k_t = [3, 3, 3, 4]
 Generation 4830: Best = 1.961791e+03 FES = 483100 μ_F = 0.785 μ_{CR} = 0.045
 Diversity = 2.0557e-01 k_t = [1, 1, 1, 1]
 Generation 4840: Best = 1.961713e+03 FES = 484100 μ_F = 0.795 μ_{CR} = 0.056
 Diversity = 2.0925e-01 k_t = [3, 2, 3, 2]
 Generation 4850: Best = 1.961685e+03 FES = 485100 μ_F = 0.781 μ_{CR} = 0.055
 Diversity = 2.0868e-01 k_t = [1, 1, 1, 1]
 Generation 4860: Best = 1.961685e+03 FES = 486100 μ_F = 0.783 μ_{CR} = 0.061
 Diversity = 2.0790e-01 k_t = [2, 2, 3, 2]
 Generation 4870: Best = 1.961619e+03 FES = 487100 μ_F = 0.818 μ_{CR} = 0.049
 Diversity = 2.0987e-01 k_t = [1, 1, 1, 1]
 Generation 4880: Best = 1.961523e+03 FES = 488100 μ_F = 0.834 μ_{CR} = 0.058
 Diversity = 2.1010e-01 k_t = [4, 2, 2, 3]
 Generation 4890: Best = 1.961501e+03 FES = 489100 μ_F = 0.820 μ_{CR} = 0.054
 Diversity = 2.1148e-01 k_t = [1, 1, 1, 1]
 Generation 4900: Best = 1.961501e+03 FES = 490100 μ_F = 0.839 μ_{CR} = 0.058
 Diversity = 2.1292e-01 k_t = [3, 2, 3, 4]
 Generation 4910: Best = 1.961495e+03 FES = 491100 μ_F = 0.795 μ_{CR} = 0.055
 Diversity = 2.1350e-01 k_t = [1, 1, 1, 1]
 Generation 4920: Best = 1.961494e+03 FES = 492100 μ_F = 0.802 μ_{CR} = 0.057
 Diversity = 2.1596e-01 k_t = [2, 2, 3, 4]
 Generation 4930: Best = 1.961493e+03 FES = 493100 μ_F = 0.790 μ_{CR} = 0.058
 Diversity = 2.1671e-01 k_t = [1, 1, 1, 1]
 Generation 4940: Best = 1.961427e+03 FES = 494100 μ_F = 0.803 μ_{CR} = 0.062
 Diversity = 2.1668e-01 k_t = [3, 2, 2, 3]
 Generation 4950: Best = 1.961312e+03 FES = 495100 μ_F = 0.796 μ_{CR} = 0.055
 Diversity = 2.1945e-01 k_t = [1, 1, 1, 1]
 Generation 4960: Best = 1.961312e+03 FES = 496100 μ_F = 0.817 μ_{CR} = 0.055
 Diversity = 2.1826e-01 k_t = [4, 4, 4, 4]
 Generation 4970: Best = 1.961308e+03 FES = 497100 μ_F = 0.838 μ_{CR} = 0.052
 Diversity = 2.2101e-01 k_t = [1, 1, 1, 1]
 Generation 4980: Best = 1.961274e+03 FES = 498100 μ_F = 0.830 μ_{CR} = 0.051
 Diversity = 2.2032e-01 k_t = [2, 3, 4, 2]
 Generation 4990: Best = 1.961219e+03 FES = 499100 μ_F = 0.856 μ_{CR} = 0.053
 Diversity = 2.2379e-01 k_t = [1, 1, 1, 1]
 Generation 5000: Best = 1.961149e+03 FES = 500100 μ_F = 0.855 μ_{CR} = 0.051
 Diversity = 2.2347e-01 k_t = [3, 4, 2, 3]
 Generation 5010: Best = 1.961149e+03 FES = 501100 μ_F = 0.845 μ_{CR} = 0.049
 Diversity = 2.2503e-01 k_t = [1, 1, 1, 1]
 Generation 5020: Best = 1.961117e+03 FES = 502100 μ_F = 0.818 μ_{CR} = 0.041
 Diversity = 2.2456e-01 k_t = [3, 3, 3, 3]
 Generation 5030: Best = 1.961116e+03 FES = 503100 μ_F = 0.820 μ_{CR} = 0.040
 Diversity = 2.2824e-01 k_t = [1, 1, 1, 1]
 Generation 5040: Best = 1.961024e+03 FES = 504100 μ_F = 0.812 μ_{CR} = 0.043
 Diversity = 2.3063e-01 k_t = [4, 2, 4, 3]
 Generation 5050: Best = 1.961010e+03 FES = 505100 μ_F = 0.805 μ_{CR} = 0.045
 Diversity = 2.3145e-01 k_t = [1, 1, 1, 1]
 Generation 5060: Best = 1.960990e+03 FES = 506100 μ_F = 0.817 μ_{CR} = 0.053
 Diversity = 2.2902e-01 k_t = [3, 4, 2, 4]
 Generation 5070: Best = 1.960979e+03 FES = 507100 μ_F = 0.812 μ_{CR} = 0.056
 Diversity = 2.2937e-01 k_t = [1, 1, 1, 1]
 Generation 5080: Best = 1.960935e+03 FES = 508100 μ_F = 0.820 μ_{CR} = 0.053
 Diversity = 2.2643e-01 k_t = [2, 3, 4, 4]

Generation 5090: Best = 1.960926e+03 FES = 509100 μ_F = 0.826 μ_{CR} = 0.056
 Diversity = 2.2967e-01 k_t = [1, 1, 1, 1]
 Generation 5100: Best = 1.960817e+03 FES = 510100 μ_F = 0.810 μ_{CR} = 0.055
 Diversity = 2.3162e-01 k_t = [2, 4, 2, 3]
 Generation 5110: Best = 1.960809e+03 FES = 511100 μ_F = 0.811 μ_{CR} = 0.048
 Diversity = 2.3503e-01 k_t = [1, 1, 1, 1]
 Generation 5120: Best = 1.960805e+03 FES = 512100 μ_F = 0.786 μ_{CR} = 0.048
 Diversity = 2.3306e-01 k_t = [4, 2, 3, 3]
 Generation 5130: Best = 1.960805e+03 FES = 513100 μ_F = 0.803 μ_{CR} = 0.049
 Diversity = 2.3216e-01 k_t = [1, 1, 1, 1]
 Generation 5140: Best = 1.960758e+03 FES = 514100 μ_F = 0.809 μ_{CR} = 0.057
 Diversity = 2.2901e-01 k_t = [2, 3, 3, 4]
 Generation 5150: Best = 1.960709e+03 FES = 515100 μ_F = 0.834 μ_{CR} = 0.051
 Diversity = 2.2980e-01 k_t = [1, 1, 1, 1]
 Generation 5160: Best = 1.960703e+03 FES = 516100 μ_F = 0.844 μ_{CR} = 0.054
 Diversity = 2.3474e-01 k_t = [2, 3, 2, 2]
 Generation 5170: Best = 1.960696e+03 FES = 517100 μ_F = 0.824 μ_{CR} = 0.051
 Diversity = 2.3276e-01 k_t = [1, 1, 1, 1]
 Generation 5180: Best = 1.960667e+03 FES = 518100 μ_F = 0.807 μ_{CR} = 0.051
 Diversity = 2.2765e-01 k_t = [3, 3, 2, 4]
 Generation 5190: Best = 1.960653e+03 FES = 519100 μ_F = 0.788 μ_{CR} = 0.050
 Diversity = 2.2351e-01 k_t = [1, 1, 1, 1]
 Generation 5200: Best = 1.960583e+03 FES = 520100 μ_F = 0.817 μ_{CR} = 0.049
 Diversity = 2.2712e-01 k_t = [3, 3, 2, 2]
 Generation 5210: Best = 1.960547e+03 FES = 521100 μ_F = 0.786 μ_{CR} = 0.051
 Diversity = 2.2602e-01 k_t = [1, 1, 1, 1]
 Generation 5220: Best = 1.960543e+03 FES = 522100 μ_F = 0.789 μ_{CR} = 0.049
 Diversity = 2.2383e-01 k_t = [4, 3, 2, 3]
 Generation 5230: Best = 1.960539e+03 FES = 523100 μ_F = 0.770 μ_{CR} = 0.051
 Diversity = 2.2569e-01 k_t = [1, 1, 1, 1]
 Generation 5240: Best = 1.960461e+03 FES = 524100 μ_F = 0.769 μ_{CR} = 0.047
 Diversity = 2.2658e-01 k_t = [3, 2, 3, 2]
 Generation 5250: Best = 1.960458e+03 FES = 525100 μ_F = 0.787 μ_{CR} = 0.045
 Diversity = 2.2532e-01 k_t = [1, 1, 1, 1]
 Generation 5260: Best = 1.960413e+03 FES = 526100 μ_F = 0.793 μ_{CR} = 0.050
 Diversity = 2.2323e-01 k_t = [4, 3, 2, 3]
 Generation 5270: Best = 1.960394e+03 FES = 527100 μ_F = 0.799 μ_{CR} = 0.052
 Diversity = 2.2538e-01 k_t = [1, 1, 1, 1]
 Generation 5280: Best = 1.960373e+03 FES = 528100 μ_F = 0.795 μ_{CR} = 0.053
 Diversity = 2.2219e-01 k_t = [2, 4, 2, 4]
 Generation 5290: Best = 1.960334e+03 FES = 529100 μ_F = 0.773 μ_{CR} = 0.045
 Diversity = 2.2073e-01 k_t = [1, 1, 1, 1]
 Generation 5300: Best = 1.960334e+03 FES = 530100 μ_F = 0.770 μ_{CR} = 0.052
 Diversity = 2.1482e-01 k_t = [4, 2, 2, 4]
 Generation 5310: Best = 1.960266e+03 FES = 531100 μ_F = 0.796 μ_{CR} = 0.047
 Diversity = 2.1473e-01 k_t = [1, 1, 1, 1]
 Generation 5320: Best = 1.960261e+03 FES = 532100 μ_F = 0.805 μ_{CR} = 0.044
 Diversity = 2.1389e-01 k_t = [4, 4, 4, 3]
 Generation 5330: Best = 1.960261e+03 FES = 533100 μ_F = 0.810 μ_{CR} = 0.045
 Diversity = 2.1372e-01 k_t = [1, 1, 1, 1]
 Generation 5340: Best = 1.960256e+03 FES = 534100 μ_F = 0.812 μ_{CR} = 0.047
 Diversity = 2.1310e-01 k_t = [4, 2, 4, 4]
 Generation 5350: Best = 1.960222e+03 FES = 535100 μ_F = 0.828 μ_{CR} = 0.043
 Diversity = 2.1484e-01 k_t = [1, 1, 1, 1]

Generation 5360: Best = 1.960194e+03 FES = 536100 μ_F = 0.859 μ_{CR} = 0.048
 Diversity = 2.1002e-01 k_t = [3, 4, 2, 4]
 Generation 5370: Best = 1.960166e+03 FES = 537100 μ_F = 0.858 μ_{CR} = 0.047
 Diversity = 2.1147e-01 k_t = [1, 1, 1, 1]
 Generation 5380: Best = 1.960131e+03 FES = 538100 μ_F = 0.836 μ_{CR} = 0.041
 Diversity = 2.1129e-01 k_t = [2, 3, 4, 3]
 Generation 5390: Best = 1.960131e+03 FES = 539100 μ_F = 0.829 μ_{CR} = 0.046
 Diversity = 2.1193e-01 k_t = [1, 1, 1, 1]
 Generation 5400: Best = 1.960118e+03 FES = 540100 μ_F = 0.832 μ_{CR} = 0.045
 Diversity = 2.1208e-01 k_t = [4, 2, 4, 4]
 Generation 5410: Best = 1.960029e+03 FES = 541100 μ_F = 0.828 μ_{CR} = 0.046
 Diversity = 2.1145e-01 k_t = [1, 1, 1, 1]
 Generation 5420: Best = 1.960029e+03 FES = 542100 μ_F = 0.832 μ_{CR} = 0.047
 Diversity = 2.1107e-01 k_t = [4, 3, 3, 3]
 Generation 5430: Best = 1.960029e+03 FES = 543100 μ_F = 0.820 μ_{CR} = 0.045
 Diversity = 2.1252e-01 k_t = [1, 1, 1, 1]
 Generation 5440: Best = 1.960005e+03 FES = 544100 μ_F = 0.794 μ_{CR} = 0.047
 Diversity = 2.1119e-01 k_t = [2, 3, 2, 3]
 Generation 5450: Best = 1.960005e+03 FES = 545100 μ_F = 0.806 μ_{CR} = 0.041
 Diversity = 2.1021e-01 k_t = [1, 1, 1, 1]
 Generation 5460: Best = 1.959995e+03 FES = 546100 μ_F = 0.827 μ_{CR} = 0.048
 Diversity = 2.1231e-01 k_t = [2, 2, 4, 2]
 Generation 5470: Best = 1.959964e+03 FES = 547100 μ_F = 0.828 μ_{CR} = 0.048
 Diversity = 2.1045e-01 k_t = [1, 1, 1, 1]
 Generation 5480: Best = 1.959959e+03 FES = 548100 μ_F = 0.818 μ_{CR} = 0.047
 Diversity = 2.0950e-01 k_t = [3, 2, 2, 2]
 Generation 5490: Best = 1.959959e+03 FES = 549100 μ_F = 0.835 μ_{CR} = 0.043
 Diversity = 2.1040e-01 k_t = [1, 1, 1, 1]
 Generation 5500: Best = 1.959955e+03 FES = 550100 μ_F = 0.812 μ_{CR} = 0.046
 Diversity = 2.1046e-01 k_t = [2, 2, 2, 4]
 Generation 5510: Best = 1.959914e+03 FES = 551100 μ_F = 0.808 μ_{CR} = 0.045
 Diversity = 2.1162e-01 k_t = [1, 1, 1, 1]
 Generation 5520: Best = 1.959913e+03 FES = 552100 μ_F = 0.836 μ_{CR} = 0.045
 Diversity = 2.1422e-01 k_t = [3, 4, 4, 3]
 Generation 5530: Best = 1.959908e+03 FES = 553100 μ_F = 0.832 μ_{CR} = 0.043
 Diversity = 2.1520e-01 k_t = [1, 1, 1, 1]
 Generation 5540: Best = 1.959903e+03 FES = 554100 μ_F = 0.839 μ_{CR} = 0.048
 Diversity = 2.1636e-01 k_t = [2, 3, 2, 4]
 Generation 5550: Best = 1.959881e+03 FES = 555100 μ_F = 0.845 μ_{CR} = 0.046
 Diversity = 2.1611e-01 k_t = [1, 1, 1, 1]
 Generation 5560: Best = 1.959869e+03 FES = 556100 μ_F = 0.837 μ_{CR} = 0.046
 Diversity = 2.1922e-01 k_t = [3, 3, 3, 4]
 Generation 5570: Best = 1.959869e+03 FES = 557100 μ_F = 0.833 μ_{CR} = 0.047
 Diversity = 2.2141e-01 k_t = [1, 1, 1, 1]
 Generation 5580: Best = 1.959847e+03 FES = 558100 μ_F = 0.836 μ_{CR} = 0.041
 Diversity = 2.2257e-01 k_t = [2, 2, 3, 3]
 Generation 5590: Best = 1.959763e+03 FES = 559100 μ_F = 0.814 μ_{CR} = 0.047
 Diversity = 2.2222e-01 k_t = [1, 1, 1, 1]
 Generation 5600: Best = 1.959746e+03 FES = 560100 μ_F = 0.816 μ_{CR} = 0.053
 Diversity = 2.2256e-01 k_t = [4, 2, 3, 3]
 Generation 5610: Best = 1.959746e+03 FES = 561100 μ_F = 0.836 μ_{CR} = 0.043
 Diversity = 2.2464e-01 k_t = [1, 1, 1, 1]
 Generation 5620: Best = 1.959746e+03 FES = 562100 μ_F = 0.831 μ_{CR} = 0.040
 Diversity = 2.2481e-01 k_t = [3, 3, 2, 2]

Generation 5630: Best = 1.959740e+03 FES = 563100 μ_F = 0.838 μ_{CR} = 0.040
 Diversity = 2.2577e-01 k_t = [1, 1, 1, 1]
 Generation 5640: Best = 1.959732e+03 FES = 564100 μ_F = 0.828 μ_{CR} = 0.043
 Diversity = 2.2414e-01 k_t = [3, 3, 3, 2]
 Generation 5650: Best = 1.959730e+03 FES = 565100 μ_F = 0.811 μ_{CR} = 0.043
 Diversity = 2.2372e-01 k_t = [1, 1, 1, 1]
 Generation 5660: Best = 1.959679e+03 FES = 566100 μ_F = 0.784 μ_{CR} = 0.046
 Diversity = 2.2324e-01 k_t = [3, 2, 4, 2]
 Generation 5670: Best = 1.959672e+03 FES = 567100 μ_F = 0.786 μ_{CR} = 0.045
 Diversity = 2.2386e-01 k_t = [1, 1, 1, 1]
 Generation 5680: Best = 1.959653e+03 FES = 568100 μ_F = 0.806 μ_{CR} = 0.047
 Diversity = 2.2380e-01 k_t = [2, 4, 2, 2]
 Generation 5690: Best = 1.959650e+03 FES = 569100 μ_F = 0.836 μ_{CR} = 0.042
 Diversity = 2.2755e-01 k_t = [1, 1, 1, 1]
 Generation 5700: Best = 1.959635e+03 FES = 570100 μ_F = 0.850 μ_{CR} = 0.040
 Diversity = 2.2611e-01 k_t = [4, 2, 3, 3]
 Generation 5710: Best = 1.959635e+03 FES = 571100 μ_F = 0.824 μ_{CR} = 0.039
 Diversity = 2.2622e-01 k_t = [1, 1, 1, 1]
 Generation 5720: Best = 1.959614e+03 FES = 572100 μ_F = 0.830 μ_{CR} = 0.042
 Diversity = 2.2693e-01 k_t = [4, 3, 3, 4]
 Generation 5730: Best = 1.959591e+03 FES = 573100 μ_F = 0.840 μ_{CR} = 0.044
 Diversity = 2.2560e-01 k_t = [1, 1, 1, 1]
 Generation 5740: Best = 1.959581e+03 FES = 574100 μ_F = 0.857 μ_{CR} = 0.041
 Diversity = 2.2357e-01 k_t = [4, 4, 3, 4]
 Generation 5750: Best = 1.959577e+03 FES = 575100 μ_F = 0.831 μ_{CR} = 0.039
 Diversity = 2.2496e-01 k_t = [1, 1, 1, 1]
 Generation 5760: Best = 1.959549e+03 FES = 576100 μ_F = 0.831 μ_{CR} = 0.039
 Diversity = 2.2509e-01 k_t = [2, 2, 2, 4]
 Generation 5770: Best = 1.959519e+03 FES = 577100 μ_F = 0.816 μ_{CR} = 0.041
 Diversity = 2.2672e-01 k_t = [1, 1, 1, 1]
 Generation 5780: Best = 1.959519e+03 FES = 578100 μ_F = 0.811 μ_{CR} = 0.040
 Diversity = 2.2721e-01 k_t = [2, 3, 3, 3]
 Generation 5790: Best = 1.959516e+03 FES = 579100 μ_F = 0.805 μ_{CR} = 0.040
 Diversity = 2.2579e-01 k_t = [1, 1, 1, 1]
 Generation 5800: Best = 1.959512e+03 FES = 580100 μ_F = 0.816 μ_{CR} = 0.047
 Diversity = 2.2306e-01 k_t = [3, 3, 3, 4]
 Generation 5810: Best = 1.959493e+03 FES = 581100 μ_F = 0.827 μ_{CR} = 0.039
 Diversity = 2.2181e-01 k_t = [1, 1, 1, 1]
 Generation 5820: Best = 1.959475e+03 FES = 582100 μ_F = 0.844 μ_{CR} = 0.037
 Diversity = 2.2126e-01 k_t = [3, 2, 4, 2]
 Generation 5830: Best = 1.959428e+03 FES = 583100 μ_F = 0.835 μ_{CR} = 0.039
 Diversity = 2.2121e-01 k_t = [1, 1, 1, 1]
 Generation 5840: Best = 1.959410e+03 FES = 584100 μ_F = 0.821 μ_{CR} = 0.041
 Diversity = 2.1949e-01 k_t = [2, 4, 4, 2]
 Generation 5850: Best = 1.959410e+03 FES = 585100 μ_F = 0.814 μ_{CR} = 0.044
 Diversity = 2.1957e-01 k_t = [1, 1, 1, 1]
 Generation 5860: Best = 1.959410e+03 FES = 586100 μ_F = 0.843 μ_{CR} = 0.046
 Diversity = 2.1981e-01 k_t = [2, 4, 2, 2]
 Generation 5870: Best = 1.959403e+03 FES = 587100 μ_F = 0.856 μ_{CR} = 0.044
 Diversity = 2.1985e-01 k_t = [1, 1, 1, 1]
 Generation 5880: Best = 1.959394e+03 FES = 588100 μ_F = 0.855 μ_{CR} = 0.044
 Diversity = 2.2104e-01 k_t = [3, 2, 2, 4]
 Generation 5890: Best = 1.959390e+03 FES = 589100 μ_F = 0.864 μ_{CR} = 0.042
 Diversity = 2.2127e-01 k_t = [1, 1, 1, 1]

Generation 5900: Best = 1.959390e+03 FES = 590100 μ_F = 0.862 μ_{CR} = 0.040
 Diversity = 2.2172e-01 k_t = [4, 3, 3, 3]
 Generation 5910: Best = 1.959389e+03 FES = 591100 μ_F = 0.849 μ_{CR} = 0.043
 Diversity = 2.2178e-01 k_t = [1, 1, 1, 1]
 Generation 5920: Best = 1.959348e+03 FES = 592100 μ_F = 0.835 μ_{CR} = 0.038
 Diversity = 2.2044e-01 k_t = [4, 2, 4, 3]
 Generation 5930: Best = 1.959323e+03 FES = 593100 μ_F = 0.836 μ_{CR} = 0.039
 Diversity = 2.2162e-01 k_t = [1, 1, 1, 1]
 Generation 5940: Best = 1.959273e+03 FES = 594100 μ_F = 0.816 μ_{CR} = 0.048
 Diversity = 2.2073e-01 k_t = [4, 2, 3, 2]
 Generation 5950: Best = 1.959273e+03 FES = 595100 μ_F = 0.825 μ_{CR} = 0.044
 Diversity = 2.2273e-01 k_t = [1, 1, 1, 1]
 Generation 5960: Best = 1.959268e+03 FES = 596100 μ_F = 0.829 μ_{CR} = 0.047
 Diversity = 2.2651e-01 k_t = [3, 4, 4, 2]
 Generation 5970: Best = 1.959253e+03 FES = 597100 μ_F = 0.832 μ_{CR} = 0.043
 Diversity = 2.2900e-01 k_t = [1, 1, 1, 1]
 Generation 5980: Best = 1.959206e+03 FES = 598100 μ_F = 0.812 μ_{CR} = 0.043
 Diversity = 2.2860e-01 k_t = [2, 2, 3, 2]
 Generation 5990: Best = 1.959191e+03 FES = 599100 μ_F = 0.816 μ_{CR} = 0.042
 Diversity = 2.2893e-01 k_t = [1, 1, 1, 1]
 Generation 6000: Best = 1.959191e+03 FES = 600100 μ_F = 0.832 μ_{CR} = 0.046
 Diversity = 2.2854e-01 k_t = [4, 3, 2, 4]
 Generation 6010: Best = 1.959191e+03 FES = 601100 μ_F = 0.832 μ_{CR} = 0.038
 Diversity = 2.2822e-01 k_t = [1, 1, 1, 1]
 Generation 6020: Best = 1.959189e+03 FES = 602100 μ_F = 0.841 μ_{CR} = 0.040
 Diversity = 2.2770e-01 k_t = [3, 3, 3, 4]
 Generation 6030: Best = 1.959171e+03 FES = 603100 μ_F = 0.855 μ_{CR} = 0.045
 Diversity = 2.2768e-01 k_t = [1, 1, 1, 1]
 Generation 6040: Best = 1.959159e+03 FES = 604100 μ_F = 0.850 μ_{CR} = 0.042
 Diversity = 2.2964e-01 k_t = [4, 4, 3, 3]
 Generation 6050: Best = 1.959139e+03 FES = 605100 μ_F = 0.842 μ_{CR} = 0.046
 Diversity = 2.3077e-01 k_t = [1, 1, 1, 1]
 Generation 6060: Best = 1.959114e+03 FES = 606100 μ_F = 0.831 μ_{CR} = 0.043
 Diversity = 2.3064e-01 k_t = [4, 2, 3, 2]
 Generation 6070: Best = 1.959114e+03 FES = 607100 μ_F = 0.811 μ_{CR} = 0.049
 Diversity = 2.2971e-01 k_t = [1, 1, 1, 1]
 Generation 6080: Best = 1.959110e+03 FES = 608100 μ_F = 0.839 μ_{CR} = 0.053
 Diversity = 2.3052e-01 k_t = [3, 4, 4, 3]
 Generation 6090: Best = 1.959084e+03 FES = 609100 μ_F = 0.847 μ_{CR} = 0.050
 Diversity = 2.3230e-01 k_t = [1, 1, 1, 1]
 Generation 6100: Best = 1.959041e+03 FES = 610100 μ_F = 0.814 μ_{CR} = 0.048
 Diversity = 2.3111e-01 k_t = [3, 4, 4, 4]
 Generation 6110: Best = 1.959004e+03 FES = 611100 μ_F = 0.781 μ_{CR} = 0.044
 Diversity = 2.2785e-01 k_t = [1, 1, 1, 1]
 Generation 6120: Best = 1.959002e+03 FES = 612100 μ_F = 0.804 μ_{CR} = 0.046
 Diversity = 2.3093e-01 k_t = [4, 4, 3, 4]
 Generation 6130: Best = 1.958911e+03 FES = 613100 μ_F = 0.816 μ_{CR} = 0.039
 Diversity = 2.3136e-01 k_t = [1, 1, 1, 1]
 Generation 6140: Best = 1.958909e+03 FES = 614100 μ_F = 0.826 μ_{CR} = 0.047
 Diversity = 2.3231e-01 k_t = [4, 3, 2, 2]
 Generation 6150: Best = 1.958906e+03 FES = 615100 μ_F = 0.803 μ_{CR} = 0.039
 Diversity = 2.3264e-01 k_t = [1, 1, 1, 1]
 Generation 6160: Best = 1.958906e+03 FES = 616100 μ_F = 0.825 μ_{CR} = 0.042
 Diversity = 2.3231e-01 k_t = [3, 3, 4, 2]

Generation 6170: Best = 1.958889e+03 FES = 617100 μ_F = 0.831 μ_{CR} = 0.045
 Diversity = 2.3387e-01 k_t = [1, 1, 1, 1]
 Generation 6180: Best = 1.958863e+03 FES = 618100 μ_F = 0.819 μ_{CR} = 0.054
 Diversity = 2.3569e-01 k_t = [2, 3, 2, 4]
 Generation 6190: Best = 1.958861e+03 FES = 619100 μ_F = 0.840 μ_{CR} = 0.050
 Diversity = 2.3719e-01 k_t = [1, 1, 1, 1]
 Generation 6200: Best = 1.958861e+03 FES = 620100 μ_F = 0.859 μ_{CR} = 0.049
 Diversity = 2.3630e-01 k_t = [2, 2, 4, 4]
 Generation 6210: Best = 1.958859e+03 FES = 621100 μ_F = 0.825 μ_{CR} = 0.049
 Diversity = 2.3633e-01 k_t = [1, 1, 1, 1]
 Generation 6220: Best = 1.958859e+03 FES = 622100 μ_F = 0.806 μ_{CR} = 0.045
 Diversity = 2.3488e-01 k_t = [4, 4, 4, 3]
 Generation 6230: Best = 1.958859e+03 FES = 623100 μ_F = 0.824 μ_{CR} = 0.050
 Diversity = 2.3347e-01 k_t = [1, 1, 1, 1]
 Generation 6240: Best = 1.958832e+03 FES = 624100 μ_F = 0.831 μ_{CR} = 0.047
 Diversity = 2.3480e-01 k_t = [2, 2, 4, 2]
 Generation 6250: Best = 1.958832e+03 FES = 625100 μ_F = 0.827 μ_{CR} = 0.041
 Diversity = 2.3504e-01 k_t = [1, 1, 1, 1]
 Generation 6260: Best = 1.958825e+03 FES = 626100 μ_F = 0.830 μ_{CR} = 0.041
 Diversity = 2.3427e-01 k_t = [4, 2, 4, 4]
 Generation 6270: Best = 1.958806e+03 FES = 627100 μ_F = 0.844 μ_{CR} = 0.042
 Diversity = 2.3502e-01 k_t = [1, 1, 1, 1]
 Generation 6280: Best = 1.958791e+03 FES = 628100 μ_F = 0.854 μ_{CR} = 0.043
 Diversity = 2.3399e-01 k_t = [4, 2, 2, 3]
 Generation 6290: Best = 1.958734e+03 FES = 629100 μ_F = 0.874 μ_{CR} = 0.042
 Diversity = 2.3542e-01 k_t = [1, 1, 1, 1]
 Generation 6300: Best = 1.958731e+03 FES = 630100 μ_F = 0.850 μ_{CR} = 0.038
 Diversity = 2.3379e-01 k_t = [3, 4, 4, 4]
 Generation 6310: Best = 1.958726e+03 FES = 631100 μ_F = 0.842 μ_{CR} = 0.039
 Diversity = 2.3273e-01 k_t = [1, 1, 1, 1]
 Generation 6320: Best = 1.958696e+03 FES = 632100 μ_F = 0.848 μ_{CR} = 0.044
 Diversity = 2.3437e-01 k_t = [3, 3, 4, 4]
 Generation 6330: Best = 1.958696e+03 FES = 633100 μ_F = 0.834 μ_{CR} = 0.041
 Diversity = 2.3129e-01 k_t = [1, 1, 1, 1]
 Generation 6340: Best = 1.958677e+03 FES = 634100 μ_F = 0.814 μ_{CR} = 0.044
 Diversity = 2.3036e-01 k_t = [3, 4, 2, 3]
 Generation 6350: Best = 1.958647e+03 FES = 635100 μ_F = 0.820 μ_{CR} = 0.048
 Diversity = 2.2948e-01 k_t = [1, 1, 1, 1]
 Generation 6360: Best = 1.958630e+03 FES = 636100 μ_F = 0.830 μ_{CR} = 0.039
 Diversity = 2.2718e-01 k_t = [2, 4, 4, 3]
 Generation 6370: Best = 1.958627e+03 FES = 637100 μ_F = 0.805 μ_{CR} = 0.039
 Diversity = 2.2525e-01 k_t = [1, 1, 1, 1]
 Generation 6380: Best = 1.958621e+03 FES = 638100 μ_F = 0.797 μ_{CR} = 0.046
 Diversity = 2.2681e-01 k_t = [2, 3, 2, 3]
 Generation 6390: Best = 1.958620e+03 FES = 639100 μ_F = 0.822 μ_{CR} = 0.049
 Diversity = 2.2800e-01 k_t = [1, 1, 1, 1]
 Generation 6400: Best = 1.958607e+03 FES = 640100 μ_F = 0.837 μ_{CR} = 0.046
 Diversity = 2.2853e-01 k_t = [2, 4, 4, 4]
 Generation 6410: Best = 1.958596e+03 FES = 641100 μ_F = 0.834 μ_{CR} = 0.042
 Diversity = 2.2807e-01 k_t = [1, 1, 1, 1]
 Generation 6420: Best = 1.958582e+03 FES = 642100 μ_F = 0.833 μ_{CR} = 0.051
 Diversity = 2.2518e-01 k_t = [4, 3, 2, 3]
 Generation 6430: Best = 1.958581e+03 FES = 643100 μ_F = 0.837 μ_{CR} = 0.044
 Diversity = 2.2699e-01 k_t = [1, 1, 1, 1]

Generation 6440: Best = 1.958561e+03 FES = 644100 μ_F = 0.837 μ_{CR} = 0.042
 Diversity = 2.2536e-01 k_t = [4, 3, 2, 2]
 Generation 6450: Best = 1.958543e+03 FES = 645100 μ_F = 0.830 μ_{CR} = 0.042
 Diversity = 2.2523e-01 k_t = [1, 1, 1, 1]
 Generation 6460: Best = 1.958542e+03 FES = 646100 μ_F = 0.829 μ_{CR} = 0.040
 Diversity = 2.2734e-01 k_t = [3, 4, 2, 2]
 Generation 6470: Best = 1.958511e+03 FES = 647100 μ_F = 0.805 μ_{CR} = 0.039
 Diversity = 2.2653e-01 k_t = [1, 1, 1, 1]
 Generation 6480: Best = 1.958511e+03 FES = 648100 μ_F = 0.829 μ_{CR} = 0.042
 Diversity = 2.2771e-01 k_t = [4, 2, 2, 3]
 Generation 6490: Best = 1.958511e+03 FES = 649100 μ_F = 0.814 μ_{CR} = 0.038
 Diversity = 2.2860e-01 k_t = [1, 1, 1, 1]
 Generation 6500: Best = 1.958420e+03 FES = 650100 μ_F = 0.820 μ_{CR} = 0.040
 Diversity = 2.2665e-01 k_t = [2, 3, 4, 2]
 Generation 6510: Best = 1.958412e+03 FES = 651100 μ_F = 0.800 μ_{CR} = 0.043
 Diversity = 2.2704e-01 k_t = [1, 1, 1, 1]
 Generation 6520: Best = 1.958412e+03 FES = 652100 μ_F = 0.811 μ_{CR} = 0.045
 Diversity = 2.2744e-01 k_t = [3, 3, 4, 4]
 Generation 6530: Best = 1.958412e+03 FES = 653100 μ_F = 0.817 μ_{CR} = 0.047
 Diversity = 2.2586e-01 k_t = [1, 1, 1, 1]
 Generation 6540: Best = 1.958412e+03 FES = 654100 μ_F = 0.822 μ_{CR} = 0.045
 Diversity = 2.2793e-01 k_t = [4, 2, 4, 3]
 Generation 6550: Best = 1.958412e+03 FES = 655100 μ_F = 0.853 μ_{CR} = 0.043
 Diversity = 2.2800e-01 k_t = [1, 1, 1, 1]
 Generation 6560: Best = 1.958407e+03 FES = 656100 μ_F = 0.864 μ_{CR} = 0.043
 Diversity = 2.2754e-01 k_t = [3, 4, 4, 4]
 Generation 6570: Best = 1.958402e+03 FES = 657100 μ_F = 0.874 μ_{CR} = 0.045
 Diversity = 2.2550e-01 k_t = [1, 1, 1, 1]
 Generation 6580: Best = 1.958401e+03 FES = 658100 μ_F = 0.881 μ_{CR} = 0.044
 Diversity = 2.2605e-01 k_t = [2, 2, 3, 2]
 Generation 6590: Best = 1.958400e+03 FES = 659100 μ_F = 0.845 μ_{CR} = 0.041
 Diversity = 2.2600e-01 k_t = [1, 1, 1, 1]
 Generation 6600: Best = 1.958378e+03 FES = 660100 μ_F = 0.831 μ_{CR} = 0.051
 Diversity = 2.2393e-01 k_t = [4, 4, 3, 3]
 Generation 6610: Best = 1.958337e+03 FES = 661100 μ_F = 0.803 μ_{CR} = 0.046
 Diversity = 2.2418e-01 k_t = [1, 1, 1, 1]
 Generation 6620: Best = 1.958333e+03 FES = 662100 μ_F = 0.803 μ_{CR} = 0.041
 Diversity = 2.2452e-01 k_t = [3, 4, 2, 2]
 Generation 6630: Best = 1.958283e+03 FES = 663100 μ_F = 0.824 μ_{CR} = 0.039
 Diversity = 2.2524e-01 k_t = [1, 1, 1, 1]
 Generation 6640: Best = 1.958283e+03 FES = 664100 μ_F = 0.828 μ_{CR} = 0.042
 Diversity = 2.2510e-01 k_t = [2, 4, 2, 3]
 Generation 6650: Best = 1.958283e+03 FES = 665100 μ_F = 0.841 μ_{CR} = 0.038
 Diversity = 2.2505e-01 k_t = [1, 1, 1, 1]
 Generation 6660: Best = 1.958283e+03 FES = 666100 μ_F = 0.829 μ_{CR} = 0.035
 Diversity = 2.2759e-01 k_t = [2, 2, 3, 4]
 Generation 6670: Best = 1.958226e+03 FES = 667100 μ_F = 0.831 μ_{CR} = 0.043
 Diversity = 2.2943e-01 k_t = [1, 1, 1, 1]
 Generation 6680: Best = 1.958217e+03 FES = 668100 μ_F = 0.811 μ_{CR} = 0.042
 Diversity = 2.2999e-01 k_t = [2, 2, 4, 2]
 Generation 6690: Best = 1.958194e+03 FES = 669100 μ_F = 0.832 μ_{CR} = 0.040
 Diversity = 2.3061e-01 k_t = [1, 1, 1, 1]
 Generation 6700: Best = 1.958187e+03 FES = 670100 μ_F = 0.808 μ_{CR} = 0.038
 Diversity = 2.2932e-01 k_t = [2, 3, 4, 4]

Generation 6710: Best = 1.958185e+03 FES = 671100 μ_F = 0.831 μ_{CR} = 0.038
 Diversity = 2.2768e-01 k_t = [1, 1, 1, 1]
 Generation 6720: Best = 1.958184e+03 FES = 672100 μ_F = 0.839 μ_{CR} = 0.039
 Diversity = 2.2623e-01 k_t = [3, 4, 4, 3]
 Generation 6730: Best = 1.958180e+03 FES = 673100 μ_F = 0.832 μ_{CR} = 0.036
 Diversity = 2.2990e-01 k_t = [1, 1, 1, 1]
 Generation 6740: Best = 1.958178e+03 FES = 674100 μ_F = 0.859 μ_{CR} = 0.040
 Diversity = 2.3104e-01 k_t = [4, 3, 2, 3]
 Generation 6750: Best = 1.958143e+03 FES = 675100 μ_F = 0.870 μ_{CR} = 0.038
 Diversity = 2.3128e-01 k_t = [1, 1, 1, 1]
 Generation 6760: Best = 1.958117e+03 FES = 676100 μ_F = 0.853 μ_{CR} = 0.049
 Diversity = 2.3013e-01 k_t = [3, 4, 2, 2]
 Generation 6770: Best = 1.958117e+03 FES = 677100 μ_F = 0.853 μ_{CR} = 0.045
 Diversity = 2.3045e-01 k_t = [1, 1, 1, 1]
 Generation 6780: Best = 1.958086e+03 FES = 678100 μ_F = 0.842 μ_{CR} = 0.046
 Diversity = 2.2798e-01 k_t = [2, 3, 4, 3]
 Generation 6790: Best = 1.958086e+03 FES = 679100 μ_F = 0.861 μ_{CR} = 0.043
 Diversity = 2.2787e-01 k_t = [1, 1, 1, 1]
 Generation 6800: Best = 1.958086e+03 FES = 680100 μ_F = 0.827 μ_{CR} = 0.055
 Diversity = 2.2877e-01 k_t = [3, 4, 3, 2]
 Generation 6810: Best = 1.958086e+03 FES = 681100 μ_F = 0.815 μ_{CR} = 0.044
 Diversity = 2.2825e-01 k_t = [1, 1, 1, 1]
 Generation 6820: Best = 1.958085e+03 FES = 682100 μ_F = 0.788 μ_{CR} = 0.041
 Diversity = 2.2907e-01 k_t = [2, 2, 4, 2]
 Generation 6830: Best = 1.958078e+03 FES = 683100 μ_F = 0.802 μ_{CR} = 0.057
 Diversity = 2.2788e-01 k_t = [1, 1, 1, 1]
 Generation 6840: Best = 1.958074e+03 FES = 684100 μ_F = 0.822 μ_{CR} = 0.055
 Diversity = 2.2716e-01 k_t = [2, 4, 2, 2]
 Generation 6850: Best = 1.958074e+03 FES = 685100 μ_F = 0.802 μ_{CR} = 0.049
 Diversity = 2.2630e-01 k_t = [1, 1, 1, 1]
 Generation 6860: Best = 1.958072e+03 FES = 686100 μ_F = 0.809 μ_{CR} = 0.044
 Diversity = 2.2556e-01 k_t = [4, 3, 4, 3]
 Generation 6870: Best = 1.958068e+03 FES = 687100 μ_F = 0.815 μ_{CR} = 0.046
 Diversity = 2.2625e-01 k_t = [1, 1, 1, 1]
 Generation 6880: Best = 1.958000e+03 FES = 688100 μ_F = 0.793 μ_{CR} = 0.040
 Diversity = 2.2780e-01 k_t = [3, 4, 2, 2]
 Generation 6890: Best = 1.957997e+03 FES = 689100 μ_F = 0.761 μ_{CR} = 0.045
 Diversity = 2.2609e-01 k_t = [1, 1, 1, 1]
 Generation 6900: Best = 1.957991e+03 FES = 690100 μ_F = 0.780 μ_{CR} = 0.044
 Diversity = 2.2447e-01 k_t = [3, 4, 2, 4]
 Generation 6910: Best = 1.957967e+03 FES = 691100 μ_F = 0.790 μ_{CR} = 0.041
 Diversity = 2.2069e-01 k_t = [1, 1, 1, 1]
 Generation 6920: Best = 1.957967e+03 FES = 692100 μ_F = 0.806 μ_{CR} = 0.041
 Diversity = 2.1895e-01 k_t = [3, 4, 3, 2]
 Generation 6930: Best = 1.957962e+03 FES = 693100 μ_F = 0.816 μ_{CR} = 0.041
 Diversity = 2.1804e-01 k_t = [1, 1, 1, 1]
 Generation 6940: Best = 1.957960e+03 FES = 694100 μ_F = 0.821 μ_{CR} = 0.045
 Diversity = 2.1669e-01 k_t = [3, 3, 4, 3]
 Generation 6950: Best = 1.957960e+03 FES = 695100 μ_F = 0.812 μ_{CR} = 0.044
 Diversity = 2.1563e-01 k_t = [1, 1, 1, 1]
 Generation 6960: Best = 1.957960e+03 FES = 696100 μ_F = 0.825 μ_{CR} = 0.046
 Diversity = 2.1437e-01 k_t = [3, 2, 4, 3]
 Generation 6970: Best = 1.957958e+03 FES = 697100 μ_F = 0.861 μ_{CR} = 0.041
 Diversity = 2.1491e-01 k_t = [1, 1, 1, 1]

Generation 6980: Best = 1.957953e+03 FES = 698100 μ_F = 0.855 μ_{CR} = 0.035
 Diversity = 2.1543e-01 k_t = [4, 3, 4, 2]
 Generation 6990: Best = 1.957951e+03 FES = 699100 μ_F = 0.829 μ_{CR} = 0.039
 Diversity = 2.1556e-01 k_t = [1, 1, 1, 1]
 Generation 7000: Best = 1.957940e+03 FES = 700100 μ_F = 0.812 μ_{CR} = 0.045
 Diversity = 2.1457e-01 k_t = [4, 2, 3, 2]
 Generation 7010: Best = 1.957932e+03 FES = 701100 μ_F = 0.803 μ_{CR} = 0.046
 Diversity = 2.1543e-01 k_t = [1, 1, 1, 1]
 Generation 7020: Best = 1.957922e+03 FES = 702100 μ_F = 0.789 μ_{CR} = 0.051
 Diversity = 2.1477e-01 k_t = [4, 4, 2, 4]
 Generation 7030: Best = 1.957912e+03 FES = 703100 μ_F = 0.790 μ_{CR} = 0.045
 Diversity = 2.1432e-01 k_t = [1, 1, 1, 1]
 Generation 7040: Best = 1.957908e+03 FES = 704100 μ_F = 0.825 μ_{CR} = 0.038
 Diversity = 2.1578e-01 k_t = [3, 2, 3, 3]
 Generation 7050: Best = 1.957888e+03 FES = 705100 μ_F = 0.826 μ_{CR} = 0.039
 Diversity = 2.1425e-01 k_t = [1, 1, 1, 1]
 Generation 7060: Best = 1.957867e+03 FES = 706100 μ_F = 0.835 μ_{CR} = 0.039
 Diversity = 2.1324e-01 k_t = [2, 4, 3, 4]
 Generation 7070: Best = 1.957849e+03 FES = 707100 μ_F = 0.845 μ_{CR} = 0.042
 Diversity = 2.1366e-01 k_t = [1, 1, 1, 1]
 Generation 7080: Best = 1.957849e+03 FES = 708100 μ_F = 0.847 μ_{CR} = 0.049
 Diversity = 2.1363e-01 k_t = [4, 2, 3, 4]
 Generation 7090: Best = 1.957847e+03 FES = 709100 μ_F = 0.845 μ_{CR} = 0.046
 Diversity = 2.1456e-01 k_t = [1, 1, 1, 1]
 Generation 7100: Best = 1.957847e+03 FES = 710100 μ_F = 0.851 μ_{CR} = 0.046
 Diversity = 2.1451e-01 k_t = [4, 4, 4, 2]
 Generation 7110: Best = 1.957840e+03 FES = 711100 μ_F = 0.866 μ_{CR} = 0.043
 Diversity = 2.1539e-01 k_t = [1, 1, 1, 1]
 Generation 7120: Best = 1.957836e+03 FES = 712100 μ_F = 0.833 μ_{CR} = 0.041
 Diversity = 2.1517e-01 k_t = [3, 3, 4, 2]
 Generation 7130: Best = 1.957823e+03 FES = 713100 μ_F = 0.831 μ_{CR} = 0.040
 Diversity = 2.1493e-01 k_t = [1, 1, 1, 1]
 Generation 7140: Best = 1.957796e+03 FES = 714100 μ_F = 0.849 μ_{CR} = 0.039
 Diversity = 2.1491e-01 k_t = [2, 2, 2, 4]
 Generation 7150: Best = 1.957780e+03 FES = 715100 μ_F = 0.817 μ_{CR} = 0.041
 Diversity = 2.1558e-01 k_t = [1, 1, 1, 1]
 Generation 7160: Best = 1.957774e+03 FES = 716100 μ_F = 0.803 μ_{CR} = 0.037
 Diversity = 2.1574e-01 k_t = [4, 3, 3, 3]
 Generation 7170: Best = 1.957767e+03 FES = 717100 μ_F = 0.816 μ_{CR} = 0.046
 Diversity = 2.1546e-01 k_t = [1, 1, 1, 1]
 Generation 7180: Best = 1.957767e+03 FES = 718100 μ_F = 0.827 μ_{CR} = 0.043
 Diversity = 2.1569e-01 k_t = [4, 2, 2, 3]
 Generation 7190: Best = 1.957744e+03 FES = 719100 μ_F = 0.843 μ_{CR} = 0.042
 Diversity = 2.1544e-01 k_t = [1, 1, 1, 1]
 Generation 7200: Best = 1.957744e+03 FES = 720100 μ_F = 0.863 μ_{CR} = 0.040
 Diversity = 2.1453e-01 k_t = [3, 2, 3, 2]
 Generation 7210: Best = 1.957744e+03 FES = 721100 μ_F = 0.874 μ_{CR} = 0.042
 Diversity = 2.1531e-01 k_t = [1, 1, 1, 1]
 Generation 7220: Best = 1.957744e+03 FES = 722100 μ_F = 0.862 μ_{CR} = 0.043
 Diversity = 2.1532e-01 k_t = [2, 3, 4, 3]
 Generation 7230: Best = 1.957737e+03 FES = 723100 μ_F = 0.858 μ_{CR} = 0.044
 Diversity = 2.1604e-01 k_t = [1, 1, 1, 1]
 Generation 7240: Best = 1.957714e+03 FES = 724100 μ_F = 0.869 μ_{CR} = 0.044
 Diversity = 2.1618e-01 k_t = [3, 2, 2, 4]

Generation 7250: Best = 1.957711e+03 FES = 725100 μ_F = 0.867 μ_{CR} = 0.039
 Diversity = 2.1676e-01 k_t = [1, 1, 1, 1]
 Generation 7260: Best = 1.957696e+03 FES = 726100 μ_F = 0.865 μ_{CR} = 0.040
 Diversity = 2.1456e-01 k_t = [4, 4, 3, 4]
 Generation 7270: Best = 1.957689e+03 FES = 727100 μ_F = 0.869 μ_{CR} = 0.042
 Diversity = 2.1332e-01 k_t = [1, 1, 1, 1]
 Generation 7280: Best = 1.957687e+03 FES = 728100 μ_F = 0.841 μ_{CR} = 0.043
 Diversity = 2.1354e-01 k_t = [2, 3, 4, 2]
 Generation 7290: Best = 1.957682e+03 FES = 729100 μ_F = 0.830 μ_{CR} = 0.036
 Diversity = 2.1371e-01 k_t = [1, 1, 1, 1]
 Generation 7300: Best = 1.957675e+03 FES = 730100 μ_F = 0.822 μ_{CR} = 0.037
 Diversity = 2.1186e-01 k_t = [3, 3, 4, 4]
 Generation 7310: Best = 1.957665e+03 FES = 731100 μ_F = 0.806 μ_{CR} = 0.038
 Diversity = 2.1199e-01 k_t = [1, 1, 1, 1]
 Generation 7320: Best = 1.957665e+03 FES = 732100 μ_F = 0.795 μ_{CR} = 0.044
 Diversity = 2.1135e-01 k_t = [3, 2, 4, 4]
 Generation 7330: Best = 1.957643e+03 FES = 733100 μ_F = 0.801 μ_{CR} = 0.043
 Diversity = 2.1090e-01 k_t = [1, 1, 1, 1]
 Generation 7340: Best = 1.957643e+03 FES = 734100 μ_F = 0.821 μ_{CR} = 0.042
 Diversity = 2.1044e-01 k_t = [4, 3, 4, 2]
 Generation 7350: Best = 1.957643e+03 FES = 735100 μ_F = 0.810 μ_{CR} = 0.042
 Diversity = 2.0946e-01 k_t = [1, 1, 1, 1]
 Generation 7360: Best = 1.957639e+03 FES = 736100 μ_F = 0.811 μ_{CR} = 0.047
 Diversity = 2.0899e-01 k_t = [2, 2, 4, 2]
 Generation 7370: Best = 1.957636e+03 FES = 737100 μ_F = 0.828 μ_{CR} = 0.051
 Diversity = 2.0911e-01 k_t = [1, 1, 1, 1]
 Generation 7380: Best = 1.957636e+03 FES = 738100 μ_F = 0.821 μ_{CR} = 0.042
 Diversity = 2.0831e-01 k_t = [4, 4, 4, 4]
 Generation 7390: Best = 1.957625e+03 FES = 739100 μ_F = 0.808 μ_{CR} = 0.039
 Diversity = 2.0763e-01 k_t = [1, 1, 1, 1]
 Generation 7400: Best = 1.957621e+03 FES = 740100 μ_F = 0.809 μ_{CR} = 0.039
 Diversity = 2.0795e-01 k_t = [2, 3, 2, 2]
 Generation 7410: Best = 1.957610e+03 FES = 741100 μ_F = 0.828 μ_{CR} = 0.038
 Diversity = 2.0842e-01 k_t = [1, 1, 1, 1]
 Generation 7420: Best = 1.957610e+03 FES = 742100 μ_F = 0.804 μ_{CR} = 0.038
 Diversity = 2.0827e-01 k_t = [2, 2, 4, 2]
 Generation 7430: Best = 1.957610e+03 FES = 743100 μ_F = 0.813 μ_{CR} = 0.040
 Diversity = 2.0914e-01 k_t = [1, 1, 1, 1]
 Generation 7440: Best = 1.957601e+03 FES = 744100 μ_F = 0.814 μ_{CR} = 0.048
 Diversity = 2.0905e-01 k_t = [4, 3, 3, 4]
 Generation 7450: Best = 1.957598e+03 FES = 745100 μ_F = 0.847 μ_{CR} = 0.043
 Diversity = 2.0851e-01 k_t = [1, 1, 1, 1]
 Generation 7460: Best = 1.957583e+03 FES = 746100 μ_F = 0.863 μ_{CR} = 0.038
 Diversity = 2.0889e-01 k_t = [4, 4, 2, 2]
 Generation 7470: Best = 1.957583e+03 FES = 747100 μ_F = 0.846 μ_{CR} = 0.040
 Diversity = 2.1006e-01 k_t = [1, 1, 1, 1]
 Generation 7480: Best = 1.957583e+03 FES = 748100 μ_F = 0.832 μ_{CR} = 0.044
 Diversity = 2.1043e-01 k_t = [2, 3, 4, 2]
 Generation 7490: Best = 1.957580e+03 FES = 749100 μ_F = 0.849 μ_{CR} = 0.040
 Diversity = 2.1055e-01 k_t = [1, 1, 1, 1]
 Generation 7500: Best = 1.957572e+03 FES = 750100 μ_F = 0.831 μ_{CR} = 0.036
 Diversity = 2.1019e-01 k_t = [3, 4, 2, 3]
 Generation 7510: Best = 1.957572e+03 FES = 751100 μ_F = 0.860 μ_{CR} = 0.035
 Diversity = 2.1089e-01 k_t = [1, 1, 1, 1]

Generation 7520: Best = 1.957566e+03 FES = 752100 μ_F = 0.861 μ_{CR} = 0.038
 Diversity = 2.1285e-01 k_t = [4, 2, 3, 2]
 Generation 7530: Best = 1.957563e+03 FES = 753100 μ_F = 0.814 μ_{CR} = 0.042
 Diversity = 2.1337e-01 k_t = [1, 1, 1, 1]
 Generation 7540: Best = 1.957537e+03 FES = 754100 μ_F = 0.826 μ_{CR} = 0.040
 Diversity = 2.1590e-01 k_t = [3, 3, 4, 3]
 Generation 7550: Best = 1.957535e+03 FES = 755100 μ_F = 0.822 μ_{CR} = 0.038
 Diversity = 2.1624e-01 k_t = [1, 1, 1, 1]
 Generation 7560: Best = 1.957518e+03 FES = 756100 μ_F = 0.813 μ_{CR} = 0.040
 Diversity = 2.1512e-01 k_t = [4, 3, 2, 2]
 Generation 7570: Best = 1.957517e+03 FES = 757100 μ_F = 0.800 μ_{CR} = 0.040
 Diversity = 2.1648e-01 k_t = [1, 1, 1, 1]
 Generation 7580: Best = 1.957516e+03 FES = 758100 μ_F = 0.792 μ_{CR} = 0.044
 Diversity = 2.1565e-01 k_t = [3, 4, 4, 4]
 Generation 7590: Best = 1.957513e+03 FES = 759100 μ_F = 0.807 μ_{CR} = 0.043
 Diversity = 2.1554e-01 k_t = [1, 1, 1, 1]
 Generation 7600: Best = 1.957505e+03 FES = 760100 μ_F = 0.841 μ_{CR} = 0.048
 Diversity = 2.1506e-01 k_t = [4, 2, 3, 4]
 Generation 7610: Best = 1.957497e+03 FES = 761100 μ_F = 0.858 μ_{CR} = 0.046
 Diversity = 2.1427e-01 k_t = [1, 1, 1, 1]
 Generation 7620: Best = 1.957496e+03 FES = 762100 μ_F = 0.833 μ_{CR} = 0.043
 Diversity = 2.1477e-01 k_t = [2, 2, 3, 3]
 Generation 7630: Best = 1.957496e+03 FES = 763100 μ_F = 0.827 μ_{CR} = 0.044
 Diversity = 2.1408e-01 k_t = [1, 1, 1, 1]
 Generation 7640: Best = 1.957493e+03 FES = 764100 μ_F = 0.834 μ_{CR} = 0.045
 Diversity = 2.1365e-01 k_t = [4, 3, 3, 2]
 Generation 7650: Best = 1.957490e+03 FES = 765100 μ_F = 0.818 μ_{CR} = 0.039
 Diversity = 2.1424e-01 k_t = [1, 1, 1, 1]
 Generation 7660: Best = 1.957483e+03 FES = 766100 μ_F = 0.824 μ_{CR} = 0.042
 Diversity = 2.1348e-01 k_t = [3, 2, 3, 2]
 Generation 7670: Best = 1.957480e+03 FES = 767100 μ_F = 0.805 μ_{CR} = 0.037
 Diversity = 2.1418e-01 k_t = [1, 1, 1, 1]
 Generation 7680: Best = 1.957468e+03 FES = 768100 μ_F = 0.802 μ_{CR} = 0.039
 Diversity = 2.1381e-01 k_t = [3, 4, 4, 4]
 Generation 7690: Best = 1.957457e+03 FES = 769100 μ_F = 0.795 μ_{CR} = 0.041
 Diversity = 2.1148e-01 k_t = [1, 1, 1, 1]
 Generation 7700: Best = 1.957456e+03 FES = 770100 μ_F = 0.782 μ_{CR} = 0.041
 Diversity = 2.1072e-01 k_t = [4, 2, 2, 2]
 Generation 7710: Best = 1.957444e+03 FES = 771100 μ_F = 0.754 μ_{CR} = 0.047
 Diversity = 2.1157e-01 k_t = [1, 1, 1, 1]
 Generation 7720: Best = 1.957431e+03 FES = 772100 μ_F = 0.758 μ_{CR} = 0.042
 Diversity = 2.1080e-01 k_t = [4, 4, 2, 4]
 Generation 7730: Best = 1.957426e+03 FES = 773100 μ_F = 0.768 μ_{CR} = 0.041
 Diversity = 2.1280e-01 k_t = [1, 1, 1, 1]
 Generation 7740: Best = 1.957419e+03 FES = 774100 μ_F = 0.805 μ_{CR} = 0.040
 Diversity = 2.1437e-01 k_t = [3, 2, 2, 4]
 Generation 7750: Best = 1.957412e+03 FES = 775100 μ_F = 0.781 μ_{CR} = 0.037
 Diversity = 2.1490e-01 k_t = [1, 1, 1, 1]
 Generation 7760: Best = 1.957384e+03 FES = 776100 μ_F = 0.790 μ_{CR} = 0.038
 Diversity = 2.1528e-01 k_t = [3, 3, 3, 3]
 Generation 7770: Best = 1.957383e+03 FES = 777100 μ_F = 0.808 μ_{CR} = 0.035
 Diversity = 2.1601e-01 k_t = [1, 1, 1, 1]
 Generation 7780: Best = 1.957383e+03 FES = 778100 μ_F = 0.797 μ_{CR} = 0.040
 Diversity = 2.1581e-01 k_t = [2, 3, 2, 2]

Generation 7790: Best = 1.957377e+03 FES = 779100 μ_F = 0.801 μ_{CR} = 0.038
 Diversity = 2.1517e-01 k_t = [1, 1, 1, 1]
 Generation 7800: Best = 1.957377e+03 FES = 780100 μ_F = 0.787 μ_{CR} = 0.048
 Diversity = 2.1442e-01 k_t = [3, 2, 3, 3]
 Generation 7810: Best = 1.957361e+03 FES = 781100 μ_F = 0.799 μ_{CR} = 0.045
 Diversity = 2.1361e-01 k_t = [1, 1, 1, 1]
 Generation 7820: Best = 1.957361e+03 FES = 782100 μ_F = 0.772 μ_{CR} = 0.041
 Diversity = 2.1360e-01 k_t = [2, 4, 2, 3]
 Generation 7830: Best = 1.957359e+03 FES = 783100 μ_F = 0.779 μ_{CR} = 0.037
 Diversity = 2.1259e-01 k_t = [1, 1, 1, 1]
 Generation 7840: Best = 1.957356e+03 FES = 784100 μ_F = 0.775 μ_{CR} = 0.039
 Diversity = 2.1336e-01 k_t = [4, 4, 2, 3]
 Generation 7850: Best = 1.957347e+03 FES = 785100 μ_F = 0.756 μ_{CR} = 0.040
 Diversity = 2.1291e-01 k_t = [1, 1, 1, 1]
 Generation 7860: Best = 1.957346e+03 FES = 786100 μ_F = 0.762 μ_{CR} = 0.044
 Diversity = 2.1291e-01 k_t = [3, 4, 4, 3]
 Generation 7870: Best = 1.957344e+03 FES = 787100 μ_F = 0.785 μ_{CR} = 0.040
 Diversity = 2.1238e-01 k_t = [1, 1, 1, 1]
 Generation 7880: Best = 1.957344e+03 FES = 788100 μ_F = 0.812 μ_{CR} = 0.043
 Diversity = 2.1127e-01 k_t = [4, 2, 3, 4]
 Generation 7890: Best = 1.957341e+03 FES = 789100 μ_F = 0.812 μ_{CR} = 0.044
 Diversity = 2.1097e-01 k_t = [1, 1, 1, 1]
 Generation 7900: Best = 1.957334e+03 FES = 790100 μ_F = 0.825 μ_{CR} = 0.038
 Diversity = 2.1156e-01 k_t = [3, 4, 2, 2]
 Generation 7910: Best = 1.957334e+03 FES = 791100 μ_F = 0.826 μ_{CR} = 0.051
 Diversity = 2.1164e-01 k_t = [1, 1, 1, 1]
 Generation 7920: Best = 1.957334e+03 FES = 792100 μ_F = 0.819 μ_{CR} = 0.038
 Diversity = 2.0990e-01 k_t = [2, 4, 2, 2]
 Generation 7930: Best = 1.957333e+03 FES = 793100 μ_F = 0.803 μ_{CR} = 0.041
 Diversity = 2.1037e-01 k_t = [1, 1, 1, 1]
 Generation 7940: Best = 1.957331e+03 FES = 794100 μ_F = 0.817 μ_{CR} = 0.039
 Diversity = 2.1040e-01 k_t = [3, 3, 4, 3]
 Generation 7950: Best = 1.957331e+03 FES = 795100 μ_F = 0.807 μ_{CR} = 0.042
 Diversity = 2.1094e-01 k_t = [1, 1, 1, 1]
 Generation 7960: Best = 1.957330e+03 FES = 796100 μ_F = 0.786 μ_{CR} = 0.039
 Diversity = 2.1070e-01 k_t = [2, 4, 2, 3]
 Generation 7970: Best = 1.957308e+03 FES = 797100 μ_F = 0.806 μ_{CR} = 0.038
 Diversity = 2.1101e-01 k_t = [1, 1, 1, 1]
 Generation 7980: Best = 1.957308e+03 FES = 798100 μ_F = 0.806 μ_{CR} = 0.035
 Diversity = 2.1130e-01 k_t = [2, 4, 4, 2]
 Generation 7990: Best = 1.957301e+03 FES = 799100 μ_F = 0.829 μ_{CR} = 0.045
 Diversity = 2.1065e-01 k_t = [1, 1, 1, 1]
 Generation 8000: Best = 1.957301e+03 FES = 800100 μ_F = 0.855 μ_{CR} = 0.038
 Diversity = 2.1138e-01 k_t = [2, 2, 2, 4]
 Generation 8010: Best = 1.957264e+03 FES = 801100 μ_F = 0.840 μ_{CR} = 0.049
 Diversity = 2.1087e-01 k_t = [1, 1, 1, 1]
 Generation 8020: Best = 1.957263e+03 FES = 802100 μ_F = 0.839 μ_{CR} = 0.045
 Diversity = 2.1152e-01 k_t = [3, 4, 4, 3]
 Generation 8030: Best = 1.957262e+03 FES = 803100 μ_F = 0.829 μ_{CR} = 0.042
 Diversity = 2.1077e-01 k_t = [1, 1, 1, 1]
 Generation 8040: Best = 1.957262e+03 FES = 804100 μ_F = 0.828 μ_{CR} = 0.046
 Diversity = 2.1130e-01 k_t = [2, 3, 3, 4]
 Generation 8050: Best = 1.957262e+03 FES = 805100 μ_F = 0.820 μ_{CR} = 0.041
 Diversity = 2.1175e-01 k_t = [1, 1, 1, 1]

Generation 8060: Best = 1.957252e+03 FES = 806100 μ_F = 0.825 μ_{CR} = 0.044
 Diversity = 2.1161e-01 k_t = [2, 4, 3, 3]
 Generation 8070: Best = 1.957246e+03 FES = 807100 μ_F = 0.840 μ_{CR} = 0.046
 Diversity = 2.1208e-01 k_t = [1, 1, 1, 1]
 Generation 8080: Best = 1.957245e+03 FES = 808100 μ_F = 0.826 μ_{CR} = 0.044
 Diversity = 2.1077e-01 k_t = [4, 2, 2, 3]
 Generation 8090: Best = 1.957241e+03 FES = 809100 μ_F = 0.847 μ_{CR} = 0.036
 Diversity = 2.1010e-01 k_t = [1, 1, 1, 1]
 Generation 8100: Best = 1.957241e+03 FES = 810100 μ_F = 0.827 μ_{CR} = 0.037
 Diversity = 2.0910e-01 k_t = [3, 4, 2, 3]
 Generation 8110: Best = 1.957238e+03 FES = 811100 μ_F = 0.801 μ_{CR} = 0.037
 Diversity = 2.1043e-01 k_t = [1, 1, 1, 1]
 Generation 8120: Best = 1.957237e+03 FES = 812100 μ_F = 0.803 μ_{CR} = 0.036
 Diversity = 2.1088e-01 k_t = [2, 2, 3, 3]
 Generation 8130: Best = 1.957231e+03 FES = 813100 μ_F = 0.821 μ_{CR} = 0.037
 Diversity = 2.1116e-01 k_t = [1, 1, 1, 1]
 Generation 8140: Best = 1.957223e+03 FES = 814100 μ_F = 0.827 μ_{CR} = 0.040
 Diversity = 2.1150e-01 k_t = [2, 2, 4, 4]
 Generation 8150: Best = 1.957223e+03 FES = 815100 μ_F = 0.818 μ_{CR} = 0.041
 Diversity = 2.1180e-01 k_t = [1, 1, 1, 1]
 Generation 8160: Best = 1.957210e+03 FES = 816100 μ_F = 0.810 μ_{CR} = 0.044
 Diversity = 2.1239e-01 k_t = [4, 2, 3, 3]
 Generation 8170: Best = 1.957200e+03 FES = 817100 μ_F = 0.814 μ_{CR} = 0.042
 Diversity = 2.1247e-01 k_t = [1, 1, 1, 1]
 Generation 8180: Best = 1.957183e+03 FES = 818100 μ_F = 0.812 μ_{CR} = 0.044
 Diversity = 2.1295e-01 k_t = [2, 2, 4, 3]
 Generation 8190: Best = 1.957180e+03 FES = 819100 μ_F = 0.804 μ_{CR} = 0.042
 Diversity = 2.1134e-01 k_t = [1, 1, 1, 1]
 Generation 8200: Best = 1.957160e+03 FES = 820100 μ_F = 0.817 μ_{CR} = 0.038
 Diversity = 2.1125e-01 k_t = [3, 4, 4, 2]
 Generation 8210: Best = 1.957156e+03 FES = 821100 μ_F = 0.834 μ_{CR} = 0.038
 Diversity = 2.1143e-01 k_t = [1, 1, 1, 1]
 Generation 8220: Best = 1.957153e+03 FES = 822100 μ_F = 0.858 μ_{CR} = 0.038
 Diversity = 2.0956e-01 k_t = [2, 3, 4, 4]
 Generation 8230: Best = 1.957151e+03 FES = 823100 μ_F = 0.845 μ_{CR} = 0.043
 Diversity = 2.1040e-01 k_t = [1, 1, 1, 1]
 Generation 8240: Best = 1.957151e+03 FES = 824100 μ_F = 0.825 μ_{CR} = 0.043
 Diversity = 2.1078e-01 k_t = [2, 3, 2, 2]
 Generation 8250: Best = 1.957150e+03 FES = 825100 μ_F = 0.838 μ_{CR} = 0.040
 Diversity = 2.1030e-01 k_t = [1, 1, 1, 1]
 Generation 8260: Best = 1.957148e+03 FES = 826100 μ_F = 0.826 μ_{CR} = 0.042
 Diversity = 2.1144e-01 k_t = [2, 4, 4, 3]
 Generation 8270: Best = 1.957139e+03 FES = 827100 μ_F = 0.808 μ_{CR} = 0.039
 Diversity = 2.1228e-01 k_t = [1, 1, 1, 1]
 Generation 8280: Best = 1.957139e+03 FES = 828100 μ_F = 0.830 μ_{CR} = 0.047
 Diversity = 2.1294e-01 k_t = [3, 3, 4, 4]
 Generation 8290: Best = 1.957128e+03 FES = 829100 μ_F = 0.831 μ_{CR} = 0.042
 Diversity = 2.1361e-01 k_t = [1, 1, 1, 1]
 Generation 8300: Best = 1.957117e+03 FES = 830100 μ_F = 0.849 μ_{CR} = 0.040
 Diversity = 2.1384e-01 k_t = [2, 4, 4, 3]
 Generation 8310: Best = 1.957115e+03 FES = 831100 μ_F = 0.821 μ_{CR} = 0.043
 Diversity = 2.1408e-01 k_t = [1, 1, 1, 1]
 Generation 8320: Best = 1.957107e+03 FES = 832100 μ_F = 0.793 μ_{CR} = 0.044
 Diversity = 2.1415e-01 k_t = [2, 2, 3, 3]

Generation 8330: Best = 1.957098e+03 FES = 833100 μ_F = 0.781 μ_{CR} = 0.043
 Diversity = 2.1462e-01 k_t = [1, 1, 1, 1]
 Generation 8340: Best = 1.957098e+03 FES = 834100 μ_F = 0.789 μ_{CR} = 0.044
 Diversity = 2.1428e-01 k_t = [4, 4, 2, 2]
 Generation 8350: Best = 1.957098e+03 FES = 835100 μ_F = 0.805 μ_{CR} = 0.038
 Diversity = 2.1444e-01 k_t = [1, 1, 1, 1]
 Generation 8360: Best = 1.957098e+03 FES = 836100 μ_F = 0.830 μ_{CR} = 0.038
 Diversity = 2.1351e-01 k_t = [2, 2, 4, 2]
 Generation 8370: Best = 1.957091e+03 FES = 837100 μ_F = 0.816 μ_{CR} = 0.039
 Diversity = 2.1392e-01 k_t = [1, 1, 1, 1]
 Generation 8380: Best = 1.957077e+03 FES = 838100 μ_F = 0.826 μ_{CR} = 0.038
 Diversity = 2.1551e-01 k_t = [3, 4, 3, 2]
 Generation 8390: Best = 1.957077e+03 FES = 839100 μ_F = 0.815 μ_{CR} = 0.040
 Diversity = 2.1416e-01 k_t = [1, 1, 1, 1]
 Generation 8400: Best = 1.957077e+03 FES = 840100 μ_F = 0.781 μ_{CR} = 0.042
 Diversity = 2.1551e-01 k_t = [3, 3, 2, 3]
 Generation 8410: Best = 1.957073e+03 FES = 841100 μ_F = 0.790 μ_{CR} = 0.047
 Diversity = 2.1507e-01 k_t = [1, 1, 1, 1]
 Generation 8420: Best = 1.957068e+03 FES = 842100 μ_F = 0.797 μ_{CR} = 0.042
 Diversity = 2.1606e-01 k_t = [3, 2, 2, 3]
 Generation 8430: Best = 1.957048e+03 FES = 843100 μ_F = 0.808 μ_{CR} = 0.041
 Diversity = 2.1598e-01 k_t = [1, 1, 1, 1]
 Generation 8440: Best = 1.957046e+03 FES = 844100 μ_F = 0.839 μ_{CR} = 0.036
 Diversity = 2.1533e-01 k_t = [4, 2, 3, 4]
 Generation 8450: Best = 1.957046e+03 FES = 845100 μ_F = 0.838 μ_{CR} = 0.046
 Diversity = 2.1426e-01 k_t = [1, 1, 1, 1]
 Generation 8460: Best = 1.957046e+03 FES = 846100 μ_F = 0.842 μ_{CR} = 0.045
 Diversity = 2.1334e-01 k_t = [2, 3, 3, 4]
 Generation 8470: Best = 1.957038e+03 FES = 847100 μ_F = 0.831 μ_{CR} = 0.049
 Diversity = 2.1278e-01 k_t = [1, 1, 1, 1]
 Generation 8480: Best = 1.957038e+03 FES = 848100 μ_F = 0.826 μ_{CR} = 0.052
 Diversity = 2.1279e-01 k_t = [3, 4, 4, 4]
 Generation 8490: Best = 1.957031e+03 FES = 849100 μ_F = 0.824 μ_{CR} = 0.045
 Diversity = 2.1293e-01 k_t = [1, 1, 1, 1]
 Generation 8500: Best = 1.957029e+03 FES = 850100 μ_F = 0.785 μ_{CR} = 0.042
 Diversity = 2.1306e-01 k_t = [2, 3, 4, 3]
 Generation 8510: Best = 1.957027e+03 FES = 851100 μ_F = 0.788 μ_{CR} = 0.037
 Diversity = 2.1297e-01 k_t = [1, 1, 1, 1]
 Generation 8520: Best = 1.957013e+03 FES = 852100 μ_F = 0.799 μ_{CR} = 0.040
 Diversity = 2.1347e-01 k_t = [2, 2, 4, 2]
 Generation 8530: Best = 1.957013e+03 FES = 853100 μ_F = 0.818 μ_{CR} = 0.043
 Diversity = 2.1355e-01 k_t = [1, 1, 1, 1]
 Generation 8540: Best = 1.957012e+03 FES = 854100 μ_F = 0.830 μ_{CR} = 0.045
 Diversity = 2.1213e-01 k_t = [3, 4, 2, 4]
 Generation 8550: Best = 1.957011e+03 FES = 855100 μ_F = 0.826 μ_{CR} = 0.043
 Diversity = 2.1204e-01 k_t = [1, 1, 1, 1]
 Generation 8560: Best = 1.957006e+03 FES = 856100 μ_F = 0.836 μ_{CR} = 0.042
 Diversity = 2.1032e-01 k_t = [3, 3, 3, 4]
 Generation 8570: Best = 1.956987e+03 FES = 857100 μ_F = 0.814 μ_{CR} = 0.037
 Diversity = 2.1055e-01 k_t = [1, 1, 1, 1]
 Generation 8580: Best = 1.956987e+03 FES = 858100 μ_F = 0.789 μ_{CR} = 0.045
 Diversity = 2.0977e-01 k_t = [3, 2, 4, 3]
 Generation 8590: Best = 1.956987e+03 FES = 859100 μ_F = 0.790 μ_{CR} = 0.039
 Diversity = 2.0965e-01 k_t = [1, 1, 1, 1]

Generation 8600: Best = 1.956987e+03 FES = 860100 μ_F = 0.806 μ_{CR} = 0.041
 Diversity = 2.0960e-01 k_t = [2, 2, 3, 4]
 Generation 8610: Best = 1.956987e+03 FES = 861100 μ_F = 0.807 μ_{CR} = 0.040
 Diversity = 2.0946e-01 k_t = [1, 1, 1, 1]
 Generation 8620: Best = 1.956968e+03 FES = 862100 μ_F = 0.823 μ_{CR} = 0.042
 Diversity = 2.0942e-01 k_t = [3, 3, 2, 3]
 Generation 8630: Best = 1.956968e+03 FES = 863100 μ_F = 0.829 μ_{CR} = 0.041
 Diversity = 2.0944e-01 k_t = [1, 1, 1, 1]
 Generation 8640: Best = 1.956968e+03 FES = 864100 μ_F = 0.834 μ_{CR} = 0.037
 Diversity = 2.0667e-01 k_t = [4, 2, 2, 4]
 Generation 8650: Best = 1.956965e+03 FES = 865100 μ_F = 0.810 μ_{CR} = 0.043
 Diversity = 2.0616e-01 k_t = [1, 1, 1, 1]
 Generation 8660: Best = 1.956965e+03 FES = 866100 μ_F = 0.819 μ_{CR} = 0.046
 Diversity = 2.0601e-01 k_t = [3, 2, 2, 4]
 Generation 8670: Best = 1.956959e+03 FES = 867100 μ_F = 0.821 μ_{CR} = 0.042
 Diversity = 2.0498e-01 k_t = [1, 1, 1, 1]
 Generation 8680: Best = 1.956946e+03 FES = 868100 μ_F = 0.850 μ_{CR} = 0.041
 Diversity = 2.0134e-01 k_t = [2, 4, 3, 4]
 Generation 8690: Best = 1.956945e+03 FES = 869100 μ_F = 0.859 μ_{CR} = 0.036
 Diversity = 2.0140e-01 k_t = [1, 1, 1, 1]
 Generation 8700: Best = 1.956944e+03 FES = 870100 μ_F = 0.840 μ_{CR} = 0.041
 Diversity = 2.0164e-01 k_t = [4, 2, 4, 2]
 Generation 8710: Best = 1.956944e+03 FES = 871100 μ_F = 0.848 μ_{CR} = 0.039
 Diversity = 2.0151e-01 k_t = [1, 1, 1, 1]
 Generation 8720: Best = 1.956938e+03 FES = 872100 μ_F = 0.848 μ_{CR} = 0.043
 Diversity = 2.0091e-01 k_t = [3, 2, 3, 4]
 Generation 8730: Best = 1.956938e+03 FES = 873100 μ_F = 0.805 μ_{CR} = 0.041
 Diversity = 2.0044e-01 k_t = [1, 1, 1, 1]
 Generation 8740: Best = 1.956936e+03 FES = 874100 μ_F = 0.822 μ_{CR} = 0.042
 Diversity = 2.0119e-01 k_t = [3, 2, 4, 2]
 Generation 8750: Best = 1.956934e+03 FES = 875100 μ_F = 0.820 μ_{CR} = 0.037
 Diversity = 2.0180e-01 k_t = [1, 1, 1, 1]
 Generation 8760: Best = 1.956932e+03 FES = 876100 μ_F = 0.827 μ_{CR} = 0.041
 Diversity = 2.0177e-01 k_t = [2, 4, 4, 3]
 Generation 8770: Best = 1.956931e+03 FES = 877100 μ_F = 0.847 μ_{CR} = 0.042
 Diversity = 2.0172e-01 k_t = [1, 1, 1, 1]
 Generation 8780: Best = 1.956920e+03 FES = 878100 μ_F = 0.822 μ_{CR} = 0.040
 Diversity = 2.0310e-01 k_t = [2, 3, 2, 2]
 Generation 8790: Best = 1.956920e+03 FES = 879100 μ_F = 0.837 μ_{CR} = 0.041
 Diversity = 2.0326e-01 k_t = [1, 1, 1, 1]
 Generation 8800: Best = 1.956919e+03 FES = 880100 μ_F = 0.834 μ_{CR} = 0.039
 Diversity = 2.0287e-01 k_t = [2, 3, 2, 2]
 Generation 8810: Best = 1.956919e+03 FES = 881100 μ_F = 0.845 μ_{CR} = 0.043
 Diversity = 2.0319e-01 k_t = [1, 1, 1, 1]
 Generation 8820: Best = 1.956910e+03 FES = 882100 μ_F = 0.845 μ_{CR} = 0.045
 Diversity = 2.0403e-01 k_t = [3, 4, 4, 3]
 Generation 8830: Best = 1.956903e+03 FES = 883100 μ_F = 0.838 μ_{CR} = 0.043
 Diversity = 2.0409e-01 k_t = [1, 1, 1, 1]
 Generation 8840: Best = 1.956878e+03 FES = 884100 μ_F = 0.841 μ_{CR} = 0.040
 Diversity = 2.0450e-01 k_t = [4, 4, 3, 3]
 Generation 8850: Best = 1.956877e+03 FES = 885100 μ_F = 0.845 μ_{CR} = 0.037
 Diversity = 2.0546e-01 k_t = [1, 1, 1, 1]
 Generation 8860: Best = 1.956876e+03 FES = 886100 μ_F = 0.836 μ_{CR} = 0.040
 Diversity = 2.0492e-01 k_t = [3, 3, 4, 3]

Generation 8870: Best = 1.956876e+03	FES = 887100	$\mu_F = 0.825$	$\mu_{CR} = 0.045$
Diversity = 2.0435e-01	$k_t = [1, 1, 1, 1]$		
Generation 8880: Best = 1.956876e+03	FES = 888100	$\mu_F = 0.831$	$\mu_{CR} = 0.049$
Diversity = 2.0500e-01	$k_t = [4, 3, 4, 2]$		
Generation 8890: Best = 1.956872e+03	FES = 889100	$\mu_F = 0.827$	$\mu_{CR} = 0.042$
Diversity = 2.0478e-01	$k_t = [1, 1, 1, 1]$		
Generation 8900: Best = 1.956872e+03	FES = 890100	$\mu_F = 0.854$	$\mu_{CR} = 0.044$
Diversity = 2.0483e-01	$k_t = [2, 3, 4, 4]$		
Generation 8910: Best = 1.956870e+03	FES = 891100	$\mu_F = 0.874$	$\mu_{CR} = 0.039$
Diversity = 2.0545e-01	$k_t = [1, 1, 1, 1]$		
Generation 8920: Best = 1.956868e+03	FES = 892100	$\mu_F = 0.873$	$\mu_{CR} = 0.041$
Diversity = 2.0660e-01	$k_t = [4, 4, 4, 4]$		
Generation 8930: Best = 1.956863e+03	FES = 893100	$\mu_F = 0.840$	$\mu_{CR} = 0.039$
Diversity = 2.0571e-01	$k_t = [1, 1, 1, 1]$		
Generation 8940: Best = 1.956861e+03	FES = 894100	$\mu_F = 0.833$	$\mu_{CR} = 0.044$
Diversity = 2.0550e-01	$k_t = [3, 3, 4, 3]$		
Generation 8950: Best = 1.956860e+03	FES = 895100	$\mu_F = 0.850$	$\mu_{CR} = 0.041$
Diversity = 2.0528e-01	$k_t = [1, 1, 1, 1]$		
Generation 8960: Best = 1.956855e+03	FES = 896100	$\mu_F = 0.869$	$\mu_{CR} = 0.045$
Diversity = 2.0591e-01	$k_t = [2, 2, 2, 4]$		
Generation 8970: Best = 1.956850e+03	FES = 897100	$\mu_F = 0.849$	$\mu_{CR} = 0.040$
Diversity = 2.0777e-01	$k_t = [1, 1, 1, 1]$		
Generation 8980: Best = 1.956850e+03	FES = 898100	$\mu_F = 0.848$	$\mu_{CR} = 0.034$
Diversity = 2.0442e-01	$k_t = [4, 2, 2, 4]$		
Generation 8990: Best = 1.956843e+03	FES = 899100	$\mu_F = 0.858$	$\mu_{CR} = 0.037$
Diversity = 2.0518e-01	$k_t = [1, 1, 1, 1]$		
Generation 9000: Best = 1.956838e+03	FES = 900100	$\mu_F = 0.859$	$\mu_{CR} = 0.041$
Diversity = 2.0524e-01	$k_t = [4, 2, 3, 3]$		
Generation 9010: Best = 1.956838e+03	FES = 901100	$\mu_F = 0.844$	$\mu_{CR} = 0.046$
Diversity = 2.0568e-01	$k_t = [1, 1, 1, 1]$		
Generation 9020: Best = 1.956825e+03	FES = 902100	$\mu_F = 0.856$	$\mu_{CR} = 0.039$
Diversity = 2.0323e-01	$k_t = [2, 2, 3, 4]$		
Generation 9030: Best = 1.956825e+03	FES = 903100	$\mu_F = 0.829$	$\mu_{CR} = 0.041$
Diversity = 2.0312e-01	$k_t = [1, 1, 1, 1]$		
Generation 9040: Best = 1.956823e+03	FES = 904100	$\mu_F = 0.808$	$\mu_{CR} = 0.038$
Diversity = 2.0306e-01	$k_t = [2, 4, 4, 3]$		
Generation 9050: Best = 1.956769e+03	FES = 905100	$\mu_F = 0.802$	$\mu_{CR} = 0.039$
Diversity = 2.0324e-01	$k_t = [1, 1, 1, 1]$		
Generation 9060: Best = 1.956764e+03	FES = 906100	$\mu_F = 0.803$	$\mu_{CR} = 0.042$
Diversity = 2.0404e-01	$k_t = [3, 4, 2, 3]$		
Generation 9070: Best = 1.956763e+03	FES = 907100	$\mu_F = 0.789$	$\mu_{CR} = 0.039$
Diversity = 2.0403e-01	$k_t = [1, 1, 1, 1]$		
Generation 9080: Best = 1.956756e+03	FES = 908100	$\mu_F = 0.817$	$\mu_{CR} = 0.043$
Diversity = 2.0422e-01	$k_t = [2, 2, 2, 4]$		
Generation 9090: Best = 1.956755e+03	FES = 909100	$\mu_F = 0.827$	$\mu_{CR} = 0.038$
Diversity = 2.0454e-01	$k_t = [1, 1, 1, 1]$		
Generation 9100: Best = 1.956755e+03	FES = 910100	$\mu_F = 0.830$	$\mu_{CR} = 0.036$
Diversity = 2.0494e-01	$k_t = [2, 4, 4, 2]$		
Generation 9110: Best = 1.956755e+03	FES = 911100	$\mu_F = 0.839$	$\mu_{CR} = 0.036$
Diversity = 2.0504e-01	$k_t = [1, 1, 1, 1]$		
Generation 9120: Best = 1.956755e+03	FES = 912100	$\mu_F = 0.855$	$\mu_{CR} = 0.042$
Diversity = 2.0285e-01	$k_t = [4, 4, 2, 4]$		
Generation 9130: Best = 1.956754e+03	FES = 913100	$\mu_F = 0.840$	$\mu_{CR} = 0.047$
Diversity = 2.0310e-01	$k_t = [1, 1, 1, 1]$		

Generation 9140: Best = 1.956754e+03 FES = 914100 μ_F = 0.827 μ_{CR} = 0.048
 Diversity = 2.0320e-01 k_t = [3, 4, 4, 3]
 Generation 9150: Best = 1.956754e+03 FES = 915100 μ_F = 0.809 μ_{CR} = 0.048
 Diversity = 2.0326e-01 k_t = [1, 1, 1, 1]
 Generation 9160: Best = 1.956754e+03 FES = 916100 μ_F = 0.803 μ_{CR} = 0.052
 Diversity = 2.0313e-01 k_t = [4, 2, 2, 2]
 Generation 9170: Best = 1.956751e+03 FES = 917100 μ_F = 0.800 μ_{CR} = 0.040
 Diversity = 2.0240e-01 k_t = [1, 1, 1, 1]
 Generation 9180: Best = 1.956751e+03 FES = 918100 μ_F = 0.801 μ_{CR} = 0.049
 Diversity = 2.0264e-01 k_t = [2, 3, 3, 2]
 Generation 9190: Best = 1.956749e+03 FES = 919100 μ_F = 0.821 μ_{CR} = 0.043
 Diversity = 2.0293e-01 k_t = [1, 1, 1, 1]
 Generation 9200: Best = 1.956749e+03 FES = 920100 μ_F = 0.820 μ_{CR} = 0.036
 Diversity = 2.0334e-01 k_t = [2, 4, 4, 3]
 Generation 9210: Best = 1.956747e+03 FES = 921100 μ_F = 0.790 μ_{CR} = 0.034
 Diversity = 2.0184e-01 k_t = [1, 1, 1, 1]
 Generation 9220: Best = 1.956747e+03 FES = 922100 μ_F = 0.799 μ_{CR} = 0.036
 Diversity = 2.0141e-01 k_t = [4, 2, 3, 3]
 Generation 9230: Best = 1.956746e+03 FES = 923100 μ_F = 0.816 μ_{CR} = 0.037
 Diversity = 2.0147e-01 k_t = [1, 1, 1, 1]
 Generation 9240: Best = 1.956745e+03 FES = 924100 μ_F = 0.828 μ_{CR} = 0.045
 Diversity = 2.0190e-01 k_t = [3, 4, 2, 3]
 Generation 9250: Best = 1.956745e+03 FES = 925100 μ_F = 0.803 μ_{CR} = 0.044
 Diversity = 2.0198e-01 k_t = [1, 1, 1, 1]
 Generation 9260: Best = 1.956742e+03 FES = 926100 μ_F = 0.788 μ_{CR} = 0.044
 Diversity = 2.0170e-01 k_t = [3, 3, 3, 4]
 Generation 9270: Best = 1.956737e+03 FES = 927100 μ_F = 0.779 μ_{CR} = 0.041
 Diversity = 2.0174e-01 k_t = [1, 1, 1, 1]
 Generation 9280: Best = 1.956733e+03 FES = 928100 μ_F = 0.771 μ_{CR} = 0.044
 Diversity = 2.0105e-01 k_t = [2, 4, 4, 3]
 Generation 9290: Best = 1.956732e+03 FES = 929100 μ_F = 0.774 μ_{CR} = 0.039
 Diversity = 2.0046e-01 k_t = [1, 1, 1, 1]
 Generation 9300: Best = 1.956732e+03 FES = 930100 μ_F = 0.791 μ_{CR} = 0.036
 Diversity = 2.0052e-01 k_t = [2, 4, 3, 3]
 Generation 9310: Best = 1.956732e+03 FES = 931100 μ_F = 0.822 μ_{CR} = 0.045
 Diversity = 2.0049e-01 k_t = [1, 1, 1, 1]
 Generation 9320: Best = 1.956724e+03 FES = 932100 μ_F = 0.824 μ_{CR} = 0.046
 Diversity = 1.9970e-01 k_t = [4, 4, 4, 4]
 Generation 9330: Best = 1.956723e+03 FES = 933100 μ_F = 0.806 μ_{CR} = 0.046
 Diversity = 1.9970e-01 k_t = [1, 1, 1, 1]
 Generation 9340: Best = 1.956723e+03 FES = 934100 μ_F = 0.824 μ_{CR} = 0.040
 Diversity = 1.9832e-01 k_t = [2, 2, 4, 3]
 Generation 9350: Best = 1.956712e+03 FES = 935100 μ_F = 0.814 μ_{CR} = 0.043
 Diversity = 1.9857e-01 k_t = [1, 1, 1, 1]
 Generation 9360: Best = 1.956712e+03 FES = 936100 μ_F = 0.846 μ_{CR} = 0.045
 Diversity = 1.9852e-01 k_t = [4, 3, 2, 2]
 Generation 9370: Best = 1.956712e+03 FES = 937100 μ_F = 0.823 μ_{CR} = 0.041
 Diversity = 1.9857e-01 k_t = [1, 1, 1, 1]
 Generation 9380: Best = 1.956708e+03 FES = 938100 μ_F = 0.838 μ_{CR} = 0.044
 Diversity = 1.9614e-01 k_t = [4, 4, 2, 4]
 Generation 9390: Best = 1.956703e+03 FES = 939100 μ_F = 0.827 μ_{CR} = 0.044
 Diversity = 1.9574e-01 k_t = [1, 1, 1, 1]
 Generation 9400: Best = 1.956703e+03 FES = 940100 μ_F = 0.824 μ_{CR} = 0.047
 Diversity = 1.9779e-01 k_t = [4, 2, 4, 3]

Generation 9410: Best = 1.956703e+03 FES = 941100 μ_F = 0.856 μ_{CR} = 0.037
 Diversity = 1.9642e-01 k_t = [1, 1, 1, 1]
 Generation 9420: Best = 1.956703e+03 FES = 942100 μ_F = 0.849 μ_{CR} = 0.039
 Diversity = 1.9654e-01 k_t = [2, 3, 2, 3]
 Generation 9430: Best = 1.956699e+03 FES = 943100 μ_F = 0.861 μ_{CR} = 0.036
 Diversity = 1.9603e-01 k_t = [1, 1, 1, 1]
 Generation 9440: Best = 1.956685e+03 FES = 944100 μ_F = 0.859 μ_{CR} = 0.040
 Diversity = 1.9653e-01 k_t = [4, 3, 4, 2]
 Generation 9450: Best = 1.956685e+03 FES = 945100 μ_F = 0.861 μ_{CR} = 0.043
 Diversity = 1.9673e-01 k_t = [1, 1, 1, 1]
 Generation 9460: Best = 1.956685e+03 FES = 946100 μ_F = 0.829 μ_{CR} = 0.045
 Diversity = 1.9684e-01 k_t = [3, 4, 2, 2]
 Generation 9470: Best = 1.956681e+03 FES = 947100 μ_F = 0.842 μ_{CR} = 0.043
 Diversity = 1.9604e-01 k_t = [1, 1, 1, 1]
 Generation 9480: Best = 1.956673e+03 FES = 948100 μ_F = 0.848 μ_{CR} = 0.040
 Diversity = 1.9633e-01 k_t = [2, 3, 4, 3]
 Generation 9490: Best = 1.956673e+03 FES = 949100 μ_F = 0.853 μ_{CR} = 0.040
 Diversity = 1.9649e-01 k_t = [1, 1, 1, 1]
 Generation 9500: Best = 1.956670e+03 FES = 950100 μ_F = 0.841 μ_{CR} = 0.041
 Diversity = 1.9686e-01 k_t = [2, 4, 4, 3]
 Generation 9510: Best = 1.956662e+03 FES = 951100 μ_F = 0.847 μ_{CR} = 0.043
 Diversity = 1.9682e-01 k_t = [1, 1, 1, 1]
 Generation 9520: Best = 1.956661e+03 FES = 952100 μ_F = 0.847 μ_{CR} = 0.039
 Diversity = 1.9575e-01 k_t = [2, 4, 2, 2]
 Generation 9530: Best = 1.956661e+03 FES = 953100 μ_F = 0.824 μ_{CR} = 0.043
 Diversity = 1.9712e-01 k_t = [1, 1, 1, 1]
 Generation 9540: Best = 1.956655e+03 FES = 954100 μ_F = 0.807 μ_{CR} = 0.041
 Diversity = 1.9750e-01 k_t = [3, 3, 3, 2]
 Generation 9550: Best = 1.956651e+03 FES = 955100 μ_F = 0.816 μ_{CR} = 0.046
 Diversity = 1.9758e-01 k_t = [1, 1, 1, 1]
 Generation 9560: Best = 1.956650e+03 FES = 956100 μ_F = 0.831 μ_{CR} = 0.045
 Diversity = 1.9767e-01 k_t = [3, 2, 2, 2]
 Generation 9570: Best = 1.956650e+03 FES = 957100 μ_F = 0.834 μ_{CR} = 0.044
 Diversity = 1.9664e-01 k_t = [1, 1, 1, 1]
 Generation 9580: Best = 1.956650e+03 FES = 958100 μ_F = 0.852 μ_{CR} = 0.044
 Diversity = 1.9610e-01 k_t = [3, 2, 2, 4]
 Generation 9590: Best = 1.956650e+03 FES = 959100 μ_F = 0.854 μ_{CR} = 0.037
 Diversity = 1.9609e-01 k_t = [1, 1, 1, 1]
 Generation 9600: Best = 1.956650e+03 FES = 960100 μ_F = 0.857 μ_{CR} = 0.040
 Diversity = 1.9562e-01 k_t = [4, 4, 2, 3]
 Generation 9610: Best = 1.956649e+03 FES = 961100 μ_F = 0.838 μ_{CR} = 0.042
 Diversity = 1.9581e-01 k_t = [1, 1, 1, 1]
 Generation 9620: Best = 1.956645e+03 FES = 962100 μ_F = 0.820 μ_{CR} = 0.045
 Diversity = 1.9485e-01 k_t = [2, 4, 2, 2]
 Generation 9630: Best = 1.956643e+03 FES = 963100 μ_F = 0.807 μ_{CR} = 0.039
 Diversity = 1.9516e-01 k_t = [1, 1, 1, 1]
 Generation 9640: Best = 1.956643e+03 FES = 964100 μ_F = 0.826 μ_{CR} = 0.039
 Diversity = 1.9523e-01 k_t = [4, 2, 4, 3]
 Generation 9650: Best = 1.956633e+03 FES = 965100 μ_F = 0.820 μ_{CR} = 0.041
 Diversity = 1.9549e-01 k_t = [1, 1, 1, 1]
 Generation 9660: Best = 1.956633e+03 FES = 966100 μ_F = 0.811 μ_{CR} = 0.039
 Diversity = 1.9547e-01 k_t = [2, 2, 4, 2]
 Generation 9670: Best = 1.956633e+03 FES = 967100 μ_F = 0.822 μ_{CR} = 0.037
 Diversity = 1.9404e-01 k_t = [1, 1, 1, 1]

Generation 9680: Best = 1.956633e+03 FES = 968100 μ_F = 0.807 μ_{CR} = 0.039
 Diversity = 1.9430e-01 k_t = [3, 2, 2, 3]
 Generation 9690: Best = 1.956627e+03 FES = 969100 μ_F = 0.848 μ_{CR} = 0.040
 Diversity = 1.9457e-01 k_t = [1, 1, 1, 1]
 Generation 9700: Best = 1.956627e+03 FES = 970100 μ_F = 0.820 μ_{CR} = 0.041
 Diversity = 1.9360e-01 k_t = [3, 4, 4, 4]
 Generation 9710: Best = 1.956627e+03 FES = 971100 μ_F = 0.825 μ_{CR} = 0.041
 Diversity = 1.9290e-01 k_t = [1, 1, 1, 1]
 Generation 9720: Best = 1.956620e+03 FES = 972100 μ_F = 0.835 μ_{CR} = 0.040
 Diversity = 1.9254e-01 k_t = [3, 3, 2, 3]
 Generation 9730: Best = 1.956618e+03 FES = 973100 μ_F = 0.867 μ_{CR} = 0.044
 Diversity = 1.9266e-01 k_t = [1, 1, 1, 1]
 Generation 9740: Best = 1.956615e+03 FES = 974100 μ_F = 0.867 μ_{CR} = 0.043
 Diversity = 1.9215e-01 k_t = [3, 3, 4, 2]
 Generation 9750: Best = 1.956615e+03 FES = 975100 μ_F = 0.870 μ_{CR} = 0.040
 Diversity = 1.9301e-01 k_t = [1, 1, 1, 1]
 Generation 9760: Best = 1.956611e+03 FES = 976100 μ_F = 0.867 μ_{CR} = 0.040
 Diversity = 1.9285e-01 k_t = [3, 3, 3, 3]
 Generation 9770: Best = 1.956611e+03 FES = 977100 μ_F = 0.838 μ_{CR} = 0.045
 Diversity = 1.9267e-01 k_t = [1, 1, 1, 1]
 Generation 9780: Best = 1.956607e+03 FES = 978100 μ_F = 0.878 μ_{CR} = 0.042
 Diversity = 1.9162e-01 k_t = [2, 2, 4, 2]
 Generation 9790: Best = 1.956604e+03 FES = 979100 μ_F = 0.872 μ_{CR} = 0.044
 Diversity = 1.9243e-01 k_t = [1, 1, 1, 1]
 Generation 9800: Best = 1.956604e+03 FES = 980100 μ_F = 0.834 μ_{CR} = 0.044
 Diversity = 1.9169e-01 k_t = [3, 2, 3, 4]
 Generation 9810: Best = 1.956601e+03 FES = 981100 μ_F = 0.840 μ_{CR} = 0.050
 Diversity = 1.9189e-01 k_t = [1, 1, 1, 1]
 Generation 9820: Best = 1.956598e+03 FES = 982100 μ_F = 0.823 μ_{CR} = 0.048
 Diversity = 1.9192e-01 k_t = [3, 2, 4, 2]
 Generation 9830: Best = 1.956598e+03 FES = 983100 μ_F = 0.850 μ_{CR} = 0.037
 Diversity = 1.9184e-01 k_t = [1, 1, 1, 1]
 Generation 9840: Best = 1.956596e+03 FES = 984100 μ_F = 0.856 μ_{CR} = 0.040
 Diversity = 1.9124e-01 k_t = [4, 4, 3, 3]
 Generation 9850: Best = 1.956591e+03 FES = 985100 μ_F = 0.823 μ_{CR} = 0.041
 Diversity = 1.9200e-01 k_t = [1, 1, 1, 1]
 Generation 9860: Best = 1.956577e+03 FES = 986100 μ_F = 0.824 μ_{CR} = 0.042
 Diversity = 1.9183e-01 k_t = [2, 3, 3, 3]
 Generation 9870: Best = 1.956576e+03 FES = 987100 μ_F = 0.829 μ_{CR} = 0.040
 Diversity = 1.9100e-01 k_t = [1, 1, 1, 1]
 Generation 9880: Best = 1.956576e+03 FES = 988100 μ_F = 0.791 μ_{CR} = 0.043
 Diversity = 1.9146e-01 k_t = [2, 2, 2, 2]
 Generation 9890: Best = 1.956575e+03 FES = 989100 μ_F = 0.804 μ_{CR} = 0.044
 Diversity = 1.9207e-01 k_t = [1, 1, 1, 1]
 Generation 9900: Best = 1.956575e+03 FES = 990100 μ_F = 0.848 μ_{CR} = 0.042
 Diversity = 1.9048e-01 k_t = [2, 2, 2, 4]
 Generation 9910: Best = 1.956575e+03 FES = 991100 μ_F = 0.860 μ_{CR} = 0.036
 Diversity = 1.9053e-01 k_t = [1, 1, 1, 1]
 Generation 9920: Best = 1.956575e+03 FES = 992100 μ_F = 0.808 μ_{CR} = 0.046
 Diversity = 1.9046e-01 k_t = [4, 2, 4, 3]
 Generation 9930: Best = 1.956572e+03 FES = 993100 μ_F = 0.830 μ_{CR} = 0.039
 Diversity = 1.8858e-01 k_t = [1, 1, 1, 1]
 Generation 9940: Best = 1.956572e+03 FES = 994100 μ_F = 0.847 μ_{CR} = 0.039
 Diversity = 1.8866e-01 k_t = [3, 2, 3, 4]

Generation 9950: Best = 1.956571e+03 FES = 995100 μ_F = 0.817 μ_{CR} = 0.040
 Diversity = 1.8811e-01 k_t = [1, 1, 1, 1]
 Generation 9960: Best = 1.956569e+03 FES = 996100 μ_F = 0.840 μ_{CR} = 0.041
 Diversity = 1.8838e-01 k_t = [2, 2, 3, 3]
 Generation 9970: Best = 1.956568e+03 FES = 997100 μ_F = 0.852 μ_{CR} = 0.042
 Diversity = 1.8887e-01 k_t = [1, 1, 1, 1]
 Generation 9980: Best = 1.956565e+03 FES = 998100 μ_F = 0.843 μ_{CR} = 0.051
 Diversity = 1.8963e-01 k_t = [2, 2, 4, 3]
 Generation 9990: Best = 1.956554e+03 FES = 999100 μ_F = 0.850 μ_{CR} = 0.046
 Diversity = 1.8964e-01 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 1.956554e+03

Total Function Evals = 1000000

Total Successful Updates = 125306

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Results:

Best Fitness : 1.956554e+03

Global Optimum : 1.500000e+03

Error (|best-optimum|) : 4.565540e+02

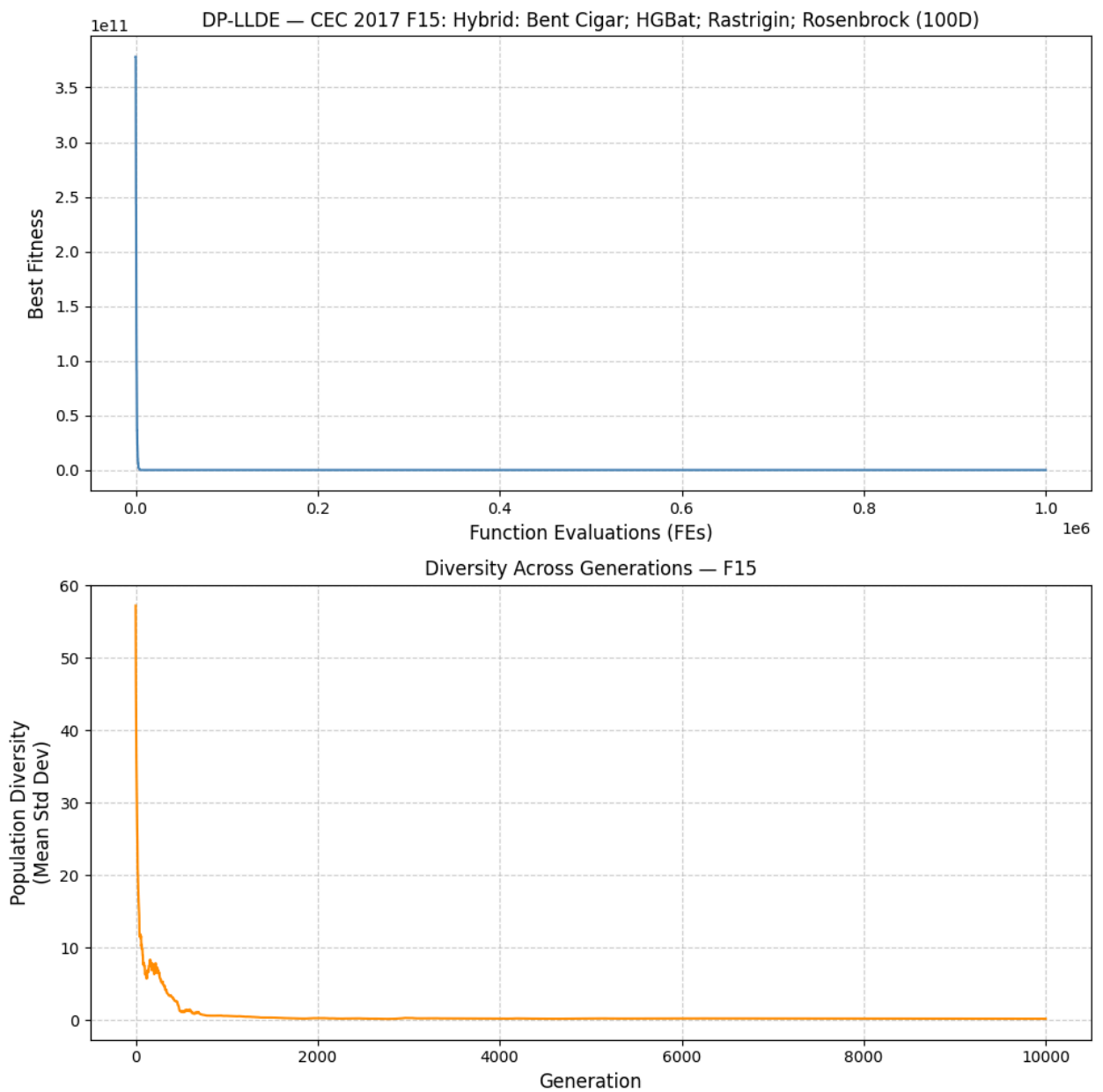
Best Solution (5 dim) : [-10.42854555 10.05583918 46.11967532 11.56771397
64.74910987]

$D^{(0)}$ (init. diversity): 5.7220e+01

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✓ Checkpoint saved to: /content/drive/My Drive/thesis_results/all_results_checkpoint.pkl



Graph saved as convergence_F15.png

✓ Plot saved to: /content/drive/My Drive/thesis_results/plots/convergence_F15.png

```
In [ ]: # =====
# F16 — Hybrid Function 6 (Schaffer; HGBat; Rosenbrock; Schwefel)
# =====
run_and_plot('F16', make_wrapper(hybrid.f16), 1600.0, 'Hybrid: Schaffer; HGBat
```

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DP-LLDE on CEC 2017 F16 – Single Run

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Function : Hybrid: Schaffer; HGBat; Rosenbrock; Schwefel (100D)
Bounds : [-100, 100]
Global optimum : 1600.0
Initial $D^*(0) = 5.7220e+01$ | Diversity threshold = 5.7220e+00
Generation 0: Best Fitness = 3.124977e+04
Generation 10: Best = 1.731862e+04 FES = 1100 $\mu_F = 0.558$ $\mu_{CR} = 0.349$
Diversity = 3.4935e+01 $k_t = [1, 1, 1, 1]$
Generation 20: Best = 1.282641e+04 FES = 2100 $\mu_F = 0.605$ $\mu_{CR} = 0.353$
Diversity = 2.7191e+01 $k_t = [2, 3, 2, 3]$
Generation 30: Best = 1.240204e+04 FES = 3100 $\mu_F = 0.625$ $\mu_{CR} = 0.362$
Diversity = 2.4292e+01 $k_t = [1, 1, 1, 1]$
Generation 40: Best = 1.213756e+04 FES = 4100 $\mu_F = 0.614$ $\mu_{CR} = 0.326$
Diversity = 2.1959e+01 $k_t = [3, 2, 3, 3]$
Generation 50: Best = 1.179390e+04 FES = 5100 $\mu_F = 0.602$ $\mu_{CR} = 0.277$
Diversity = 2.0693e+01 $k_t = [1, 1, 1, 1]$
Generation 60: Best = 1.129648e+04 FES = 6100 $\mu_F = 0.604$ $\mu_{CR} = 0.236$
Diversity = 2.0145e+01 $k_t = [4, 2, 2, 2]$
Generation 70: Best = 1.127944e+04 FES = 7100 $\mu_F = 0.594$ $\mu_{CR} = 0.179$
Diversity = 1.9931e+01 $k_t = [1, 1, 1, 1]$
Generation 80: Best = 1.105922e+04 FES = 8100 $\mu_F = 0.651$ $\mu_{CR} = 0.106$
Diversity = 1.9421e+01 $k_t = [3, 3, 2, 4]$
Generation 90: Best = 1.030399e+04 FES = 9100 $\mu_F = 0.674$ $\mu_{CR} = 0.086$
Diversity = 1.9287e+01 $k_t = [1, 1, 1, 1]$
Generation 100: Best = 9.963436e+03 FES = 10100 $\mu_F = 0.707$ $\mu_{CR} = 0.068$
Diversity = 1.9088e+01 $k_t = [3, 4, 4, 4]$
Generation 110: Best = 9.963436e+03 FES = 11100 $\mu_F = 0.723$ $\mu_{CR} = 0.068$
Diversity = 1.9255e+01 $k_t = [1, 1, 1, 1]$
Generation 120: Best = 9.601684e+03 FES = 12100 $\mu_F = 0.756$ $\mu_{CR} = 0.065$
Diversity = 1.9238e+01 $k_t = [2, 2, 4, 4]$
Generation 130: Best = 9.601684e+03 FES = 13100 $\mu_F = 0.757$ $\mu_{CR} = 0.051$
Diversity = 1.9181e+01 $k_t = [1, 1, 1, 1]$
Generation 140: Best = 9.543913e+03 FES = 14100 $\mu_F = 0.777$ $\mu_{CR} = 0.057$
Diversity = 1.9184e+01 $k_t = [3, 4, 4, 4]$
Generation 150: Best = 9.302038e+03 FES = 15100 $\mu_F = 0.764$ $\mu_{CR} = 0.052$
Diversity = 1.9026e+01 $k_t = [1, 1, 1, 1]$
Generation 160: Best = 9.004078e+03 FES = 16100 $\mu_F = 0.780$ $\mu_{CR} = 0.045$
Diversity = 1.9044e+01 $k_t = [3, 4, 2, 2]$
Generation 170: Best = 8.395821e+03 FES = 17100 $\mu_F = 0.754$ $\mu_{CR} = 0.047$
Diversity = 1.9291e+01 $k_t = [1, 1, 1, 1]$
Generation 180: Best = 8.245472e+03 FES = 18100 $\mu_F = 0.769$ $\mu_{CR} = 0.042$
Diversity = 1.9227e+01 $k_t = [4, 2, 2, 3]$
Generation 190: Best = 8.243618e+03 FES = 19100 $\mu_F = 0.743$ $\mu_{CR} = 0.044$
Diversity = 1.9182e+01 $k_t = [1, 1, 1, 1]$
Generation 200: Best = 8.212881e+03 FES = 20100 $\mu_F = 0.741$ $\mu_{CR} = 0.039$
Diversity = 1.8864e+01 $k_t = [3, 4, 2, 4]$
Generation 210: Best = 8.179632e+03 FES = 21100 $\mu_F = 0.744$ $\mu_{CR} = 0.044$
Diversity = 1.8745e+01 $k_t = [1, 1, 1, 1]$
Generation 220: Best = 8.179632e+03 FES = 22100 $\mu_F = 0.749$ $\mu_{CR} = 0.045$
Diversity = 1.8739e+01 $k_t = [3, 4, 4, 2]$

Generation	230:	Best = 8.179632e+03	FES = 23100	$\mu_F = 0.751$	$\mu_{CR} = 0.046$
Diversity =	1.8699e+01	k_t = [1, 1, 1, 1]			
Generation	240:	Best = 8.179342e+03	FES = 24100	$\mu_F = 0.745$	$\mu_{CR} = 0.041$
Diversity =	1.8620e+01	k_t = [4, 3, 3, 2]			
Generation	250:	Best = 8.179309e+03	FES = 25100	$\mu_F = 0.738$	$\mu_{CR} = 0.045$
Diversity =	1.8563e+01	k_t = [1, 1, 1, 1]			
Generation	260:	Best = 8.002069e+03	FES = 26100	$\mu_F = 0.730$	$\mu_{CR} = 0.040$
Diversity =	1.8386e+01	k_t = [4, 2, 3, 3]			
Generation	270:	Best = 8.002069e+03	FES = 27100	$\mu_F = 0.732$	$\mu_{CR} = 0.041$
Diversity =	1.8379e+01	k_t = [1, 1, 1, 1]			
Generation	280:	Best = 8.002069e+03	FES = 28100	$\mu_F = 0.768$	$\mu_{CR} = 0.043$
Diversity =	1.8374e+01	k_t = [2, 2, 4, 2]			
Generation	290:	Best = 7.725808e+03	FES = 29100	$\mu_F = 0.790$	$\mu_{CR} = 0.044$
Diversity =	1.8366e+01	k_t = [1, 1, 1, 1]			
Generation	300:	Best = 7.725511e+03	FES = 30100	$\mu_F = 0.797$	$\mu_{CR} = 0.041$
Diversity =	1.8361e+01	k_t = [3, 3, 2, 3]			
Generation	310:	Best = 7.718796e+03	FES = 31100	$\mu_F = 0.768$	$\mu_{CR} = 0.045$
Diversity =	1.8336e+01	k_t = [1, 1, 1, 1]			
Generation	320:	Best = 7.718796e+03	FES = 32100	$\mu_F = 0.753$	$\mu_{CR} = 0.047$
Diversity =	1.8424e+01	k_t = [4, 4, 3, 3]			
Generation	330:	Best = 7.325168e+03	FES = 33100	$\mu_F = 0.749$	$\mu_{CR} = 0.041$
Diversity =	1.8408e+01	k_t = [1, 1, 1, 1]			
Generation	340:	Best = 7.141557e+03	FES = 34100	$\mu_F = 0.775$	$\mu_{CR} = 0.039$
Diversity =	1.8381e+01	k_t = [4, 4, 4, 4]			
Generation	350:	Best = 7.141557e+03	FES = 35100	$\mu_F = 0.802$	$\mu_{CR} = 0.033$
Diversity =	1.8351e+01	k_t = [1, 1, 1, 1]			
Generation	360:	Best = 7.141557e+03	FES = 36100	$\mu_F = 0.820$	$\mu_{CR} = 0.038$
Diversity =	1.8428e+01	k_t = [4, 3, 3, 2]			
Generation	370:	Best = 7.141557e+03	FES = 37100	$\mu_F = 0.830$	$\mu_{CR} = 0.042$
Diversity =	1.8387e+01	k_t = [1, 1, 1, 1]			
Generation	380:	Best = 7.124036e+03	FES = 38100	$\mu_F = 0.853$	$\mu_{CR} = 0.042$
Diversity =	1.8417e+01	k_t = [4, 4, 4, 2]			
Generation	390:	Best = 7.124036e+03	FES = 39100	$\mu_F = 0.843$	$\mu_{CR} = 0.033$
Diversity =	1.8456e+01	k_t = [1, 1, 1, 1]			
Generation	400:	Best = 7.096427e+03	FES = 40100	$\mu_F = 0.819$	$\mu_{CR} = 0.034$
Diversity =	1.8318e+01	k_t = [3, 4, 4, 4]			
Generation	410:	Best = 7.082004e+03	FES = 41100	$\mu_F = 0.829$	$\mu_{CR} = 0.035$
Diversity =	1.8291e+01	k_t = [1, 1, 1, 1]			
Generation	420:	Best = 7.044769e+03	FES = 42100	$\mu_F = 0.821$	$\mu_{CR} = 0.034$
Diversity =	1.8271e+01	k_t = [4, 4, 4, 3]			
Generation	430:	Best = 7.044769e+03	FES = 43100	$\mu_F = 0.830$	$\mu_{CR} = 0.032$
Diversity =	1.8268e+01	k_t = [1, 1, 1, 1]			
Generation	440:	Best = 7.044769e+03	FES = 44100	$\mu_F = 0.857$	$\mu_{CR} = 0.038$
Diversity =	1.8316e+01	k_t = [3, 2, 2, 3]			
Generation	450:	Best = 7.044769e+03	FES = 45100	$\mu_F = 0.835$	$\mu_{CR} = 0.030$
Diversity =	1.8280e+01	k_t = [1, 1, 1, 1]			
Generation	460:	Best = 6.846449e+03	FES = 46100	$\mu_F = 0.829$	$\mu_{CR} = 0.032$
Diversity =	1.8258e+01	k_t = [3, 4, 2, 3]			
Generation	470:	Best = 6.808466e+03	FES = 47100	$\mu_F = 0.823$	$\mu_{CR} = 0.034$
Diversity =	1.8239e+01	k_t = [1, 1, 1, 1]			
Generation	480:	Best = 6.799820e+03	FES = 48100	$\mu_F = 0.793$	$\mu_{CR} = 0.036$
Diversity =	1.8178e+01	k_t = [2, 4, 3, 4]			
Generation	490:	Best = 6.790655e+03	FES = 49100	$\mu_F = 0.814$	$\mu_{CR} = 0.030$
Diversity =	1.8190e+01	k_t = [1, 1, 1, 1]			

Generation 500: Best = 6.755728e+03	FES = 50100	$\mu_F = 0.815$	$\mu_{CR} = 0.036$
Diversity = 1.8204e+01	k_t = [2, 2, 4, 3]		
Generation 510: Best = 6.741780e+03	FES = 51100	$\mu_F = 0.834$	$\mu_{CR} = 0.034$
Diversity = 1.8223e+01	k_t = [1, 1, 1, 1]		
Generation 520: Best = 6.740688e+03	FES = 52100	$\mu_F = 0.854$	$\mu_{CR} = 0.030$
Diversity = 1.8222e+01	k_t = [2, 4, 4, 3]		
Generation 530: Best = 6.740688e+03	FES = 53100	$\mu_F = 0.847$	$\mu_{CR} = 0.030$
Diversity = 1.8285e+01	k_t = [1, 1, 1, 1]		
Generation 540: Best = 6.740688e+03	FES = 54100	$\mu_F = 0.857$	$\mu_{CR} = 0.038$
Diversity = 1.8342e+01	k_t = [2, 4, 4, 3]		
Generation 550: Best = 6.689252e+03	FES = 55100	$\mu_F = 0.866$	$\mu_{CR} = 0.036$
Diversity = 1.8445e+01	k_t = [1, 1, 1, 1]		
Generation 560: Best = 6.680059e+03	FES = 56100	$\mu_F = 0.823$	$\mu_{CR} = 0.032$
Diversity = 1.8447e+01	k_t = [3, 4, 2, 2]		
Generation 570: Best = 6.680059e+03	FES = 57100	$\mu_F = 0.785$	$\mu_{CR} = 0.035$
Diversity = 1.8428e+01	k_t = [1, 1, 1, 1]		
Generation 580: Best = 6.680059e+03	FES = 58100	$\mu_F = 0.789$	$\mu_{CR} = 0.033$
Diversity = 1.8361e+01	k_t = [3, 2, 3, 2]		
Generation 590: Best = 6.680059e+03	FES = 59100	$\mu_F = 0.779$	$\mu_{CR} = 0.030$
Diversity = 1.8369e+01	k_t = [1, 1, 1, 1]		
Generation 600: Best = 6.679723e+03	FES = 60100	$\mu_F = 0.793$	$\mu_{CR} = 0.033$
Diversity = 1.8330e+01	k_t = [3, 4, 2, 3]		
Generation 610: Best = 6.582542e+03	FES = 61100	$\mu_F = 0.760$	$\mu_{CR} = 0.039$
Diversity = 1.8348e+01	k_t = [1, 1, 1, 1]		
Generation 620: Best = 6.559267e+03	FES = 62100	$\mu_F = 0.771$	$\mu_{CR} = 0.041$
Diversity = 1.8250e+01	k_t = [3, 3, 2, 3]		
Generation 630: Best = 6.559267e+03	FES = 63100	$\mu_F = 0.768$	$\mu_{CR} = 0.036$
Diversity = 1.8311e+01	k_t = [1, 1, 1, 1]		
Generation 640: Best = 6.493380e+03	FES = 64100	$\mu_F = 0.767$	$\mu_{CR} = 0.034$
Diversity = 1.8331e+01	k_t = [3, 2, 4, 2]		
Generation 650: Best = 6.319895e+03	FES = 65100	$\mu_F = 0.758$	$\mu_{CR} = 0.032$
Diversity = 1.8388e+01	k_t = [1, 1, 1, 1]		
Generation 660: Best = 6.314225e+03	FES = 66100	$\mu_F = 0.773$	$\mu_{CR} = 0.033$
Diversity = 1.8309e+01	k_t = [3, 4, 2, 4]		
Generation 670: Best = 6.314225e+03	FES = 67100	$\mu_F = 0.800$	$\mu_{CR} = 0.034$
Diversity = 1.8369e+01	k_t = [1, 1, 1, 1]		
Generation 680: Best = 6.314225e+03	FES = 68100	$\mu_F = 0.814$	$\mu_{CR} = 0.035$
Diversity = 1.8403e+01	k_t = [2, 2, 3, 2]		
Generation 690: Best = 6.305686e+03	FES = 69100	$\mu_F = 0.789$	$\mu_{CR} = 0.030$
Diversity = 1.8387e+01	k_t = [1, 1, 1, 1]		
Generation 700: Best = 6.285115e+03	FES = 70100	$\mu_F = 0.765$	$\mu_{CR} = 0.036$
Diversity = 1.8416e+01	k_t = [4, 2, 2, 3]		
Generation 710: Best = 6.212010e+03	FES = 71100	$\mu_F = 0.762$	$\mu_{CR} = 0.037$
Diversity = 1.8414e+01	k_t = [1, 1, 1, 1]		
Generation 720: Best = 6.204883e+03	FES = 72100	$\mu_F = 0.785$	$\mu_{CR} = 0.041$
Diversity = 1.8368e+01	k_t = [2, 2, 2, 3]		
Generation 730: Best = 6.179939e+03	FES = 73100	$\mu_F = 0.795$	$\mu_{CR} = 0.033$
Diversity = 1.8416e+01	k_t = [1, 1, 1, 1]		
Generation 740: Best = 6.170815e+03	FES = 74100	$\mu_F = 0.760$	$\mu_{CR} = 0.028$
Diversity = 1.8465e+01	k_t = [3, 4, 4, 2]		
Generation 750: Best = 6.145604e+03	FES = 75100	$\mu_F = 0.784$	$\mu_{CR} = 0.031$
Diversity = 1.8489e+01	k_t = [1, 1, 1, 1]		
Generation 760: Best = 6.145604e+03	FES = 76100	$\mu_F = 0.786$	$\mu_{CR} = 0.033$
Diversity = 1.8567e+01	k_t = [3, 2, 2, 2]		

Generation 770: Best = 6.135079e+03	FES = 77100	$\mu_F = 0.798$	$\mu_{CR} = 0.032$
Diversity = 1.8671e+01	k_t = [1, 1, 1, 1]		
Generation 780: Best = 6.061814e+03	FES = 78100	$\mu_F = 0.814$	$\mu_{CR} = 0.033$
Diversity = 1.8701e+01	k_t = [4, 2, 2, 3]		
Generation 790: Best = 6.061814e+03	FES = 79100	$\mu_F = 0.832$	$\mu_{CR} = 0.031$
Diversity = 1.8742e+01	k_t = [1, 1, 1, 1]		
Generation 800: Best = 6.061814e+03	FES = 80100	$\mu_F = 0.845$	$\mu_{CR} = 0.033$
Diversity = 1.8788e+01	k_t = [3, 2, 2, 3]		
Generation 810: Best = 6.061466e+03	FES = 81100	$\mu_F = 0.825$	$\mu_{CR} = 0.026$
Diversity = 1.8714e+01	k_t = [1, 1, 1, 1]		
Generation 820: Best = 6.051927e+03	FES = 82100	$\mu_F = 0.824$	$\mu_{CR} = 0.031$
Diversity = 1.8673e+01	k_t = [4, 3, 4, 2]		
Generation 830: Best = 6.020468e+03	FES = 83100	$\mu_F = 0.806$	$\mu_{CR} = 0.032$
Diversity = 1.8666e+01	k_t = [1, 1, 1, 1]		
Generation 840: Best = 6.020468e+03	FES = 84100	$\mu_F = 0.783$	$\mu_{CR} = 0.038$
Diversity = 1.8630e+01	k_t = [2, 2, 2, 2]		
Generation 850: Best = 6.013951e+03	FES = 85100	$\mu_F = 0.832$	$\mu_{CR} = 0.035$
Diversity = 1.8662e+01	k_t = [1, 1, 1, 1]		
Generation 860: Best = 6.013951e+03	FES = 86100	$\mu_F = 0.763$	$\mu_{CR} = 0.046$
Diversity = 1.8686e+01	k_t = [2, 4, 3, 3]		
Generation 870: Best = 6.011842e+03	FES = 87100	$\mu_F = 0.823$	$\mu_{CR} = 0.039$
Diversity = 1.8678e+01	k_t = [1, 1, 1, 1]		
Generation 880: Best = 6.010275e+03	FES = 88100	$\mu_F = 0.800$	$\mu_{CR} = 0.036$
Diversity = 1.8660e+01	k_t = [4, 4, 4, 2]		
Generation 890: Best = 6.010134e+03	FES = 89100	$\mu_F = 0.806$	$\mu_{CR} = 0.036$
Diversity = 1.8671e+01	k_t = [1, 1, 1, 1]		
Generation 900: Best = 6.010134e+03	FES = 90100	$\mu_F = 0.791$	$\mu_{CR} = 0.029$
Diversity = 1.8664e+01	k_t = [4, 2, 4, 2]		
Generation 910: Best = 5.991260e+03	FES = 91100	$\mu_F = 0.777$	$\mu_{CR} = 0.037$
Diversity = 1.8799e+01	k_t = [1, 1, 1, 1]		
Generation 920: Best = 5.991260e+03	FES = 92100	$\mu_F = 0.809$	$\mu_{CR} = 0.035$
Diversity = 1.8394e+01	k_t = [4, 2, 2, 4]		
Generation 930: Best = 5.979530e+03	FES = 93100	$\mu_F = 0.835$	$\mu_{CR} = 0.031$
Diversity = 1.8388e+01	k_t = [1, 1, 1, 1]		
Generation 940: Best = 5.979530e+03	FES = 94100	$\mu_F = 0.797$	$\mu_{CR} = 0.035$
Diversity = 1.7906e+01	k_t = [4, 4, 2, 4]		
Generation 950: Best = 5.970222e+03	FES = 95100	$\mu_F = 0.795$	$\mu_{CR} = 0.026$
Diversity = 1.7848e+01	k_t = [1, 1, 1, 1]		
Generation 960: Best = 5.968348e+03	FES = 96100	$\mu_F = 0.797$	$\mu_{CR} = 0.025$
Diversity = 1.7839e+01	k_t = [4, 2, 3, 2]		
Generation 970: Best = 5.967145e+03	FES = 97100	$\mu_F = 0.769$	$\mu_{CR} = 0.027$
Diversity = 1.7874e+01	k_t = [1, 1, 1, 1]		
Generation 980: Best = 5.967145e+03	FES = 98100	$\mu_F = 0.788$	$\mu_{CR} = 0.035$
Diversity = 1.7919e+01	k_t = [4, 4, 3, 3]		
Generation 990: Best = 5.967145e+03	FES = 99100	$\mu_F = 0.800$	$\mu_{CR} = 0.034$
Diversity = 1.7922e+01	k_t = [1, 1, 1, 1]		
Generation 1000: Best = 5.967145e+03	FES = 100100	$\mu_F = 0.822$	$\mu_{CR} = 0.035$
Diversity = 1.7765e+01	k_t = [2, 3, 4, 4]		
Generation 1010: Best = 5.967145e+03	FES = 101100	$\mu_F = 0.830$	$\mu_{CR} = 0.029$
Diversity = 1.7768e+01	k_t = [1, 1, 1, 1]		
Generation 1020: Best = 5.925052e+03	FES = 102100	$\mu_F = 0.821$	$\mu_{CR} = 0.039$
Diversity = 1.7727e+01	k_t = [4, 4, 2, 2]		
Generation 1030: Best = 5.897332e+03	FES = 103100	$\mu_F = 0.818$	$\mu_{CR} = 0.037$
Diversity = 1.7806e+01	k_t = [1, 1, 1, 1]		

Generation 1040: Best = 5.832229e+03 FES = 104100 μ_F = 0.822 μ_{CR} = 0.049
 Diversity = 1.7785e+01 k_t = [4, 4, 3, 3]
 Generation 1050: Best = 5.604532e+03 FES = 105100 μ_F = 0.805 μ_{CR} = 0.038
 Diversity = 1.7766e+01 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 5.378521e+03 FES = 106100 μ_F = 0.805 μ_{CR} = 0.035
 Diversity = 1.7708e+01 k_t = [2, 4, 4, 3]
 Generation 1070: Best = 5.367416e+03 FES = 107100 μ_F = 0.825 μ_{CR} = 0.037
 Diversity = 1.7766e+01 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 5.362344e+03 FES = 108100 μ_F = 0.834 μ_{CR} = 0.033
 Diversity = 1.7816e+01 k_t = [3, 3, 3, 3]
 Generation 1090: Best = 5.310622e+03 FES = 109100 μ_F = 0.764 μ_{CR} = 0.037
 Diversity = 1.7802e+01 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 5.310622e+03 FES = 110100 μ_F = 0.731 μ_{CR} = 0.040
 Diversity = 1.7714e+01 k_t = [4, 3, 4, 4]
 Generation 1110: Best = 5.310562e+03 FES = 111100 μ_F = 0.732 μ_{CR} = 0.040
 Diversity = 1.7593e+01 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 5.310562e+03 FES = 112100 μ_F = 0.728 μ_{CR} = 0.037
 Diversity = 1.7555e+01 k_t = [2, 2, 2, 2]
 Generation 1130: Best = 5.290631e+03 FES = 113100 μ_F = 0.750 μ_{CR} = 0.036
 Diversity = 1.7571e+01 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 5.290631e+03 FES = 114100 μ_F = 0.753 μ_{CR} = 0.045
 Diversity = 1.7503e+01 k_t = [4, 4, 2, 2]
 Generation 1150: Best = 5.290631e+03 FES = 115100 μ_F = 0.749 μ_{CR} = 0.036
 Diversity = 1.7440e+01 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 5.257680e+03 FES = 116100 μ_F = 0.732 μ_{CR} = 0.045
 Diversity = 1.7444e+01 k_t = [2, 3, 3, 2]
 Generation 1170: Best = 5.257680e+03 FES = 117100 μ_F = 0.752 μ_{CR} = 0.039
 Diversity = 1.7431e+01 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 5.123283e+03 FES = 118100 μ_F = 0.734 μ_{CR} = 0.042
 Diversity = 1.7363e+01 k_t = [3, 2, 2, 2]
 Generation 1190: Best = 5.076401e+03 FES = 119100 μ_F = 0.774 μ_{CR} = 0.046
 Diversity = 1.7318e+01 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 5.074187e+03 FES = 120100 μ_F = 0.798 μ_{CR} = 0.049
 Diversity = 1.7291e+01 k_t = [2, 2, 4, 2]
 Generation 1210: Best = 5.053079e+03 FES = 121100 μ_F = 0.812 μ_{CR} = 0.043
 Diversity = 1.7319e+01 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 5.052899e+03 FES = 122100 μ_F = 0.738 μ_{CR} = 0.036
 Diversity = 1.7255e+01 k_t = [2, 3, 4, 3]
 Generation 1230: Best = 5.022169e+03 FES = 123100 μ_F = 0.773 μ_{CR} = 0.034
 Diversity = 1.7247e+01 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 5.008848e+03 FES = 124100 μ_F = 0.768 μ_{CR} = 0.042
 Diversity = 1.7132e+01 k_t = [3, 4, 2, 4]
 Generation 1250: Best = 4.668323e+03 FES = 125100 μ_F = 0.782 μ_{CR} = 0.045
 Diversity = 1.7145e+01 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 4.634778e+03 FES = 126100 μ_F = 0.811 μ_{CR} = 0.052
 Diversity = 1.7125e+01 k_t = [4, 2, 4, 4]
 Generation 1270: Best = 4.616815e+03 FES = 127100 μ_F = 0.817 μ_{CR} = 0.042
 Diversity = 1.7134e+01 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 4.609137e+03 FES = 128100 μ_F = 0.824 μ_{CR} = 0.042
 Diversity = 1.7149e+01 k_t = [3, 3, 2, 2]
 Generation 1290: Best = 4.607727e+03 FES = 129100 μ_F = 0.871 μ_{CR} = 0.045
 Diversity = 1.7261e+01 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 4.607021e+03 FES = 130100 μ_F = 0.853 μ_{CR} = 0.062
 Diversity = 1.7262e+01 k_t = [2, 4, 3, 2]

Generation 1310: Best = 4.607021e+03 FES = 131100 μ_F = 0.840 μ_{CR} = 0.046
 Diversity = 1.7274e+01 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 4.585398e+03 FES = 132100 μ_F = 0.811 μ_{CR} = 0.047
 Diversity = 1.6849e+01 k_t = [2, 3, 2, 4]
 Generation 1330: Best = 4.585398e+03 FES = 133100 μ_F = 0.830 μ_{CR} = 0.050
 Diversity = 1.6720e+01 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 4.560069e+03 FES = 134100 μ_F = 0.836 μ_{CR} = 0.055
 Diversity = 1.6731e+01 k_t = [3, 2, 3, 3]
 Generation 1350: Best = 4.560069e+03 FES = 135100 μ_F = 0.804 μ_{CR} = 0.057
 Diversity = 1.6682e+01 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 4.556640e+03 FES = 136100 μ_F = 0.808 μ_{CR} = 0.066
 Diversity = 1.6561e+01 k_t = [4, 3, 4, 4]
 Generation 1370: Best = 4.542691e+03 FES = 137100 μ_F = 0.778 μ_{CR} = 0.070
 Diversity = 1.6389e+01 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 4.494159e+03 FES = 138100 μ_F = 0.784 μ_{CR} = 0.080
 Diversity = 1.6255e+01 k_t = [4, 4, 2, 2]
 Generation 1390: Best = 4.473094e+03 FES = 139100 μ_F = 0.795 μ_{CR} = 0.094
 Diversity = 1.6211e+01 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 4.467039e+03 FES = 140100 μ_F = 0.801 μ_{CR} = 0.107
 Diversity = 1.6251e+01 k_t = [4, 2, 3, 2]
 Generation 1410: Best = 4.436188e+03 FES = 141100 μ_F = 0.815 μ_{CR} = 0.114
 Diversity = 1.6088e+01 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 4.397681e+03 FES = 142100 μ_F = 0.827 μ_{CR} = 0.109
 Diversity = 1.5666e+01 k_t = [2, 2, 4, 3]
 Generation 1430: Best = 4.389888e+03 FES = 143100 μ_F = 0.831 μ_{CR} = 0.122
 Diversity = 1.5001e+01 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 4.350174e+03 FES = 144100 μ_F = 0.837 μ_{CR} = 0.141
 Diversity = 1.4167e+01 k_t = [3, 2, 2, 2]
 Generation 1450: Best = 4.330083e+03 FES = 145100 μ_F = 0.843 μ_{CR} = 0.165
 Diversity = 8.6437e+00 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 4.297637e+03 FES = 146100 μ_F = 0.817 μ_{CR} = 0.182
 Diversity = 7.7970e+00 k_t = [2, 4, 4, 3]
 Generation 1470: Best = 4.267981e+03 FES = 147100 μ_F = 0.793 μ_{CR} = 0.179
 Diversity = 6.2432e+00 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 4.242341e+03 FES = 148100 μ_F = 0.781 μ_{CR} = 0.179
 Diversity = 5.6085e+00 k_t = [2, 3, 4, 2]
 Generation 1490: Best = 4.225931e+03 FES = 149100 μ_F = 0.774 μ_{CR} = 0.187
 Diversity = 5.5452e+00 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 4.202925e+03 FES = 150100 μ_F = 0.778 μ_{CR} = 0.193
 Diversity = 5.0833e+00 k_t = [2, 4, 2, 3]
 Generation 1510: Best = 4.190725e+03 FES = 151100 μ_F = 0.766 μ_{CR} = 0.204
 Diversity = 4.6633e+00 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 4.172783e+03 FES = 152100 μ_F = 0.769 μ_{CR} = 0.206
 Diversity = 4.5293e+00 k_t = [4, 3, 3, 2]
 Generation 1530: Best = 4.163638e+03 FES = 153100 μ_F = 0.770 μ_{CR} = 0.206
 Diversity = 4.5729e+00 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 4.154439e+03 FES = 154100 μ_F = 0.763 μ_{CR} = 0.207
 Diversity = 4.4100e+00 k_t = [2, 3, 3, 2]
 Generation 1550: Best = 4.144564e+03 FES = 155100 μ_F = 0.764 μ_{CR} = 0.206
 Diversity = 4.5498e+00 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 4.133948e+03 FES = 156100 μ_F = 0.769 μ_{CR} = 0.212
 Diversity = 4.7898e+00 k_t = [4, 4, 3, 3]
 Generation 1570: Best = 4.128086e+03 FES = 157100 μ_F = 0.767 μ_{CR} = 0.220
 Diversity = 4.3732e+00 k_t = [1, 1, 1, 1]

Generation 1580: Best = 4.120024e+03 FES = 158100 μ_F = 0.767 μ_{CR} = 0.214
 Diversity = 4.2781e+00 k_t = [4, 4, 4, 3]
 Generation 1590: Best = 4.113512e+03 FES = 159100 μ_F = 0.758 μ_{CR} = 0.219
 Diversity = 3.9399e+00 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 4.107584e+03 FES = 160100 μ_F = 0.754 μ_{CR} = 0.221
 Diversity = 4.0932e+00 k_t = [2, 3, 3, 2]
 Generation 1610: Best = 4.103461e+03 FES = 161100 μ_F = 0.762 μ_{CR} = 0.228
 Diversity = 4.0986e+00 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 4.097734e+03 FES = 162100 μ_F = 0.764 μ_{CR} = 0.225
 Diversity = 3.7784e+00 k_t = [4, 2, 2, 4]
 Generation 1630: Best = 4.096839e+03 FES = 163100 μ_F = 0.756 μ_{CR} = 0.218
 Diversity = 3.6783e+00 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 4.093978e+03 FES = 164100 μ_F = 0.740 μ_{CR} = 0.219
 Diversity = 3.5739e+00 k_t = [2, 4, 3, 3]
 Generation 1650: Best = 4.091762e+03 FES = 165100 μ_F = 0.753 μ_{CR} = 0.215
 Diversity = 3.7221e+00 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 4.089922e+03 FES = 166100 μ_F = 0.754 μ_{CR} = 0.222
 Diversity = 3.6084e+00 k_t = [2, 4, 4, 3]
 Generation 1670: Best = 4.088019e+03 FES = 167100 μ_F = 0.751 μ_{CR} = 0.225
 Diversity = 3.6035e+00 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 4.086535e+03 FES = 168100 μ_F = 0.763 μ_{CR} = 0.222
 Diversity = 3.6226e+00 k_t = [4, 3, 2, 3]
 Generation 1690: Best = 4.086089e+03 FES = 169100 μ_F = 0.758 μ_{CR} = 0.202
 Diversity = 3.7911e+00 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 4.085143e+03 FES = 170100 μ_F = 0.753 μ_{CR} = 0.195
 Diversity = 3.7499e+00 k_t = [4, 2, 2, 4]
 Generation 1710: Best = 4.084100e+03 FES = 171100 μ_F = 0.754 μ_{CR} = 0.188
 Diversity = 3.7089e+00 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 4.082959e+03 FES = 172100 μ_F = 0.763 μ_{CR} = 0.181
 Diversity = 3.6471e+00 k_t = [2, 3, 4, 3]
 Generation 1730: Best = 4.082225e+03 FES = 173100 μ_F = 0.758 μ_{CR} = 0.160
 Diversity = 3.7615e+00 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 4.080971e+03 FES = 174100 μ_F = 0.759 μ_{CR} = 0.142
 Diversity = 3.7512e+00 k_t = [3, 3, 3, 2]
 Generation 1750: Best = 4.080724e+03 FES = 175100 μ_F = 0.767 μ_{CR} = 0.136
 Diversity = 3.8188e+00 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 4.079454e+03 FES = 176100 μ_F = 0.763 μ_{CR} = 0.133
 Diversity = 3.6441e+00 k_t = [4, 2, 2, 3]
 Generation 1770: Best = 4.079424e+03 FES = 177100 μ_F = 0.756 μ_{CR} = 0.122
 Diversity = 3.5281e+00 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 4.078820e+03 FES = 178100 μ_F = 0.752 μ_{CR} = 0.111
 Diversity = 3.2420e+00 k_t = [2, 2, 2, 3]
 Generation 1790: Best = 4.078689e+03 FES = 179100 μ_F = 0.760 μ_{CR} = 0.089
 Diversity = 3.1799e+00 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 4.078379e+03 FES = 180100 μ_F = 0.770 μ_{CR} = 0.092
 Diversity = 3.1660e+00 k_t = [3, 3, 3, 4]
 Generation 1810: Best = 4.077893e+03 FES = 181100 μ_F = 0.790 μ_{CR} = 0.080
 Diversity = 3.1481e+00 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 4.077849e+03 FES = 182100 μ_F = 0.823 μ_{CR} = 0.071
 Diversity = 3.1041e+00 k_t = [4, 4, 4, 4]
 Generation 1830: Best = 4.077423e+03 FES = 183100 μ_F = 0.815 μ_{CR} = 0.070
 Diversity = 3.0145e+00 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 4.077220e+03 FES = 184100 μ_F = 0.816 μ_{CR} = 0.065
 Diversity = 2.9738e+00 k_t = [2, 3, 3, 2]

Generation 1850: Best = 4.077102e+03	FES = 185100	$\mu_F = 0.800$	$\mu_{CR} = 0.059$
Diversity = 2.9572e+00	$k_t = [1, 1, 1, 1]$		
Generation 1860: Best = 4.076610e+03	FES = 186100	$\mu_F = 0.790$	$\mu_{CR} = 0.068$
Diversity = 2.9914e+00	$k_t = [4, 4, 4, 3]$		
Generation 1870: Best = 4.076173e+03	FES = 187100	$\mu_F = 0.791$	$\mu_{CR} = 0.079$
Diversity = 3.0668e+00	$k_t = [1, 1, 1, 1]$		
Generation 1880: Best = 4.075569e+03	FES = 188100	$\mu_F = 0.786$	$\mu_{CR} = 0.080$
Diversity = 2.9876e+00	$k_t = [3, 2, 4, 2]$		
Generation 1890: Best = 4.074915e+03	FES = 189100	$\mu_F = 0.793$	$\mu_{CR} = 0.082$
Diversity = 3.0279e+00	$k_t = [1, 1, 1, 1]$		
Generation 1900: Best = 4.074835e+03	FES = 190100	$\mu_F = 0.802$	$\mu_{CR} = 0.081$
Diversity = 2.9030e+00	$k_t = [2, 4, 4, 2]$		
Generation 1910: Best = 4.073933e+03	FES = 191100	$\mu_F = 0.788$	$\mu_{CR} = 0.095$
Diversity = 3.1284e+00	$k_t = [1, 1, 1, 1]$		
Generation 1920: Best = 4.073107e+03	FES = 192100	$\mu_F = 0.788$	$\mu_{CR} = 0.092$
Diversity = 3.1919e+00	$k_t = [3, 4, 3, 2]$		
Generation 1930: Best = 4.073107e+03	FES = 193100	$\mu_F = 0.786$	$\mu_{CR} = 0.092$
Diversity = 3.1639e+00	$k_t = [1, 1, 1, 1]$		
Generation 1940: Best = 4.073107e+03	FES = 194100	$\mu_F = 0.778$	$\mu_{CR} = 0.102$
Diversity = 3.1442e+00	$k_t = [2, 3, 4, 3]$		
Generation 1950: Best = 4.072727e+03	FES = 195100	$\mu_F = 0.783$	$\mu_{CR} = 0.093$
Diversity = 3.2406e+00	$k_t = [1, 1, 1, 1]$		
Generation 1960: Best = 4.072162e+03	FES = 196100	$\mu_F = 0.779$	$\mu_{CR} = 0.084$
Diversity = 3.2043e+00	$k_t = [3, 2, 2, 2]$		
Generation 1970: Best = 4.072014e+03	FES = 197100	$\mu_F = 0.785$	$\mu_{CR} = 0.076$
Diversity = 3.1172e+00	$k_t = [1, 1, 1, 1]$		
Generation 1980: Best = 4.071968e+03	FES = 198100	$\mu_F = 0.788$	$\mu_{CR} = 0.072$
Diversity = 2.9972e+00	$k_t = [4, 4, 4, 3]$		
Generation 1990: Best = 4.071610e+03	FES = 199100	$\mu_F = 0.771$	$\mu_{CR} = 0.073$
Diversity = 2.8881e+00	$k_t = [1, 1, 1, 1]$		
Generation 2000: Best = 4.071022e+03	FES = 200100	$\mu_F = 0.768$	$\mu_{CR} = 0.075$
Diversity = 2.8753e+00	$k_t = [4, 4, 4, 4]$		
Generation 2010: Best = 4.070863e+03	FES = 201100	$\mu_F = 0.780$	$\mu_{CR} = 0.083$
Diversity = 2.8077e+00	$k_t = [1, 1, 1, 1]$		
Generation 2020: Best = 4.070495e+03	FES = 202100	$\mu_F = 0.804$	$\mu_{CR} = 0.078$
Diversity = 2.8270e+00	$k_t = [2, 4, 3, 3]$		
Generation 2030: Best = 4.070040e+03	FES = 203100	$\mu_F = 0.797$	$\mu_{CR} = 0.079$
Diversity = 2.8392e+00	$k_t = [1, 1, 1, 1]$		
Generation 2040: Best = 4.069860e+03	FES = 204100	$\mu_F = 0.796$	$\mu_{CR} = 0.089$
Diversity = 2.8844e+00	$k_t = [3, 3, 2, 2]$		
Generation 2050: Best = 4.069404e+03	FES = 205100	$\mu_F = 0.800$	$\mu_{CR} = 0.093$
Diversity = 2.9594e+00	$k_t = [1, 1, 1, 1]$		
Generation 2060: Best = 4.068977e+03	FES = 206100	$\mu_F = 0.777$	$\mu_{CR} = 0.095$
Diversity = 3.0393e+00	$k_t = [4, 3, 3, 3]$		
Generation 2070: Best = 4.068409e+03	FES = 207100	$\mu_F = 0.772$	$\mu_{CR} = 0.100$
Diversity = 3.2024e+00	$k_t = [1, 1, 1, 1]$		
Generation 2080: Best = 4.068119e+03	FES = 208100	$\mu_F = 0.766$	$\mu_{CR} = 0.097$
Diversity = 3.1750e+00	$k_t = [3, 2, 3, 4]$		
Generation 2090: Best = 4.067967e+03	FES = 209100	$\mu_F = 0.778$	$\mu_{CR} = 0.095$
Diversity = 3.2901e+00	$k_t = [1, 1, 1, 1]$		
Generation 2100: Best = 4.067681e+03	FES = 210100	$\mu_F = 0.776$	$\mu_{CR} = 0.094$
Diversity = 3.3929e+00	$k_t = [3, 3, 4, 2]$		
Generation 2110: Best = 4.067240e+03	FES = 211100	$\mu_F = 0.771$	$\mu_{CR} = 0.095$
Diversity = 3.3817e+00	$k_t = [1, 1, 1, 1]$		

Generation 2120: Best = 4.066736e+03 FES = 212100 μ_F = 0.774 μ_{CR} = 0.098
 Diversity = 3.2990e+00 k_t = [2, 2, 2, 4]
 Generation 2130: Best = 4.066412e+03 FES = 213100 μ_F = 0.764 μ_{CR} = 0.090
 Diversity = 3.4076e+00 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 4.065785e+03 FES = 214100 μ_F = 0.773 μ_{CR} = 0.092
 Diversity = 3.3215e+00 k_t = [3, 4, 3, 4]
 Generation 2150: Best = 4.065054e+03 FES = 215100 μ_F = 0.783 μ_{CR} = 0.085
 Diversity = 3.3900e+00 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 4.065054e+03 FES = 216100 μ_F = 0.780 μ_{CR} = 0.086
 Diversity = 3.3348e+00 k_t = [2, 3, 3, 4]
 Generation 2170: Best = 4.064620e+03 FES = 217100 μ_F = 0.779 μ_{CR} = 0.084
 Diversity = 3.1408e+00 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 4.064053e+03 FES = 218100 μ_F = 0.775 μ_{CR} = 0.085
 Diversity = 3.1047e+00 k_t = [3, 2, 3, 2]
 Generation 2190: Best = 4.063790e+03 FES = 219100 μ_F = 0.763 μ_{CR} = 0.079
 Diversity = 3.0645e+00 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 4.063556e+03 FES = 220100 μ_F = 0.766 μ_{CR} = 0.078
 Diversity = 2.9341e+00 k_t = [2, 2, 2, 4]
 Generation 2210: Best = 4.063117e+03 FES = 221100 μ_F = 0.763 μ_{CR} = 0.079
 Diversity = 2.8848e+00 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 4.063117e+03 FES = 222100 μ_F = 0.767 μ_{CR} = 0.078
 Diversity = 2.7991e+00 k_t = [3, 3, 2, 3]
 Generation 2230: Best = 4.062638e+03 FES = 223100 μ_F = 0.766 μ_{CR} = 0.080
 Diversity = 2.7397e+00 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 4.062566e+03 FES = 224100 μ_F = 0.765 μ_{CR} = 0.077
 Diversity = 2.6404e+00 k_t = [2, 3, 2, 4]
 Generation 2250: Best = 4.062325e+03 FES = 225100 μ_F = 0.788 μ_{CR} = 0.075
 Diversity = 2.7026e+00 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 4.062195e+03 FES = 226100 μ_F = 0.797 μ_{CR} = 0.075
 Diversity = 2.8262e+00 k_t = [2, 4, 3, 3]
 Generation 2270: Best = 4.061998e+03 FES = 227100 μ_F = 0.787 μ_{CR} = 0.086
 Diversity = 2.8718e+00 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 4.061922e+03 FES = 228100 μ_F = 0.767 μ_{CR} = 0.088
 Diversity = 2.7775e+00 k_t = [3, 3, 2, 3]
 Generation 2290: Best = 4.061700e+03 FES = 229100 μ_F = 0.762 μ_{CR} = 0.081
 Diversity = 2.8258e+00 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 4.061585e+03 FES = 230100 μ_F = 0.762 μ_{CR} = 0.078
 Diversity = 2.9847e+00 k_t = [2, 2, 3, 2]
 Generation 2310: Best = 4.061480e+03 FES = 231100 μ_F = 0.777 μ_{CR} = 0.072
 Diversity = 3.0958e+00 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 4.061177e+03 FES = 232100 μ_F = 0.774 μ_{CR} = 0.069
 Diversity = 3.1680e+00 k_t = [4, 4, 3, 2]
 Generation 2330: Best = 4.061038e+03 FES = 233100 μ_F = 0.767 μ_{CR} = 0.080
 Diversity = 3.1841e+00 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 4.060701e+03 FES = 234100 μ_F = 0.765 μ_{CR} = 0.085
 Diversity = 3.1439e+00 k_t = [2, 3, 4, 2]
 Generation 2350: Best = 4.060463e+03 FES = 235100 μ_F = 0.778 μ_{CR} = 0.082
 Diversity = 3.1873e+00 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 4.060275e+03 FES = 236100 μ_F = 0.780 μ_{CR} = 0.088
 Diversity = 3.1746e+00 k_t = [2, 2, 4, 2]
 Generation 2370: Best = 4.059942e+03 FES = 237100 μ_F = 0.772 μ_{CR} = 0.089
 Diversity = 3.2269e+00 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 4.059686e+03 FES = 238100 μ_F = 0.783 μ_{CR} = 0.091
 Diversity = 3.1868e+00 k_t = [2, 2, 4, 3]

Generation 2390: Best = 4.059422e+03 FES = 239100 μ_F = 0.771 μ_{CR} = 0.082
 Diversity = 3.2658e+00 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 4.059124e+03 FES = 240100 μ_F = 0.757 μ_{CR} = 0.083
 Diversity = 3.2452e+00 k_t = [3, 2, 3, 3]
 Generation 2410: Best = 4.058965e+03 FES = 241100 μ_F = 0.761 μ_{CR} = 0.083
 Diversity = 3.2964e+00 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 4.058721e+03 FES = 242100 μ_F = 0.767 μ_{CR} = 0.087
 Diversity = 3.2812e+00 k_t = [2, 2, 3, 2]
 Generation 2430: Best = 4.058290e+03 FES = 243100 μ_F = 0.770 μ_{CR} = 0.088
 Diversity = 3.2047e+00 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 4.058039e+03 FES = 244100 μ_F = 0.749 μ_{CR} = 0.085
 Diversity = 3.3318e+00 k_t = [4, 2, 4, 4]
 Generation 2450: Best = 4.057739e+03 FES = 245100 μ_F = 0.762 μ_{CR} = 0.086
 Diversity = 3.4634e+00 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 4.057436e+03 FES = 246100 μ_F = 0.773 μ_{CR} = 0.077
 Diversity = 3.4385e+00 k_t = [2, 2, 3, 3]
 Generation 2470: Best = 4.057201e+03 FES = 247100 μ_F = 0.754 μ_{CR} = 0.075
 Diversity = 3.4540e+00 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 4.057201e+03 FES = 248100 μ_F = 0.757 μ_{CR} = 0.088
 Diversity = 3.4106e+00 k_t = [3, 3, 2, 4]
 Generation 2490: Best = 4.056600e+03 FES = 249100 μ_F = 0.762 μ_{CR} = 0.091
 Diversity = 3.4606e+00 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 4.056586e+03 FES = 250100 μ_F = 0.760 μ_{CR} = 0.078
 Diversity = 3.3690e+00 k_t = [3, 4, 3, 2]
 Generation 2510: Best = 4.056315e+03 FES = 251100 μ_F = 0.763 μ_{CR} = 0.068
 Diversity = 3.4061e+00 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 4.056315e+03 FES = 252100 μ_F = 0.768 μ_{CR} = 0.068
 Diversity = 3.3933e+00 k_t = [4, 2, 3, 4]
 Generation 2530: Best = 4.056281e+03 FES = 253100 μ_F = 0.786 μ_{CR} = 0.061
 Diversity = 3.3954e+00 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 4.055958e+03 FES = 254100 μ_F = 0.814 μ_{CR} = 0.064
 Diversity = 3.4337e+00 k_t = [4, 2, 4, 4]
 Generation 2550: Best = 4.055309e+03 FES = 255100 μ_F = 0.829 μ_{CR} = 0.052
 Diversity = 3.5691e+00 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 4.055227e+03 FES = 256100 μ_F = 0.828 μ_{CR} = 0.062
 Diversity = 3.7350e+00 k_t = [3, 4, 2, 4]
 Generation 2570: Best = 4.055186e+03 FES = 257100 μ_F = 0.831 μ_{CR} = 0.063
 Diversity = 3.6263e+00 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 4.055181e+03 FES = 258100 μ_F = 0.814 μ_{CR} = 0.067
 Diversity = 3.5731e+00 k_t = [2, 4, 2, 3]
 Generation 2590: Best = 4.055025e+03 FES = 259100 μ_F = 0.823 μ_{CR} = 0.052
 Diversity = 3.6228e+00 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 4.054857e+03 FES = 260100 μ_F = 0.818 μ_{CR} = 0.057
 Diversity = 3.4318e+00 k_t = [4, 3, 4, 3]
 Generation 2610: Best = 4.054836e+03 FES = 261100 μ_F = 0.820 μ_{CR} = 0.058
 Diversity = 3.4618e+00 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 4.053978e+03 FES = 262100 μ_F = 0.816 μ_{CR} = 0.064
 Diversity = 3.4703e+00 k_t = [3, 2, 4, 3]
 Generation 2630: Best = 4.053065e+03 FES = 263100 μ_F = 0.807 μ_{CR} = 0.075
 Diversity = 3.5600e+00 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 4.052995e+03 FES = 264100 μ_F = 0.800 μ_{CR} = 0.092
 Diversity = 3.4632e+00 k_t = [2, 2, 4, 3]
 Generation 2650: Best = 4.052913e+03 FES = 265100 μ_F = 0.793 μ_{CR} = 0.088
 Diversity = 3.5255e+00 k_t = [1, 1, 1, 1]

Generation 2660: Best = 4.052160e+03 FES = 266100 μ_F = 0.787 μ_{CR} = 0.090
 Diversity = 3.6549e+00 k_t = [4, 2, 4, 2]
 Generation 2670: Best = 4.051607e+03 FES = 267100 μ_F = 0.781 μ_{CR} = 0.103
 Diversity = 3.7759e+00 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 4.051074e+03 FES = 268100 μ_F = 0.802 μ_{CR} = 0.105
 Diversity = 3.6458e+00 k_t = [4, 2, 3, 4]
 Generation 2690: Best = 4.050479e+03 FES = 269100 μ_F = 0.787 μ_{CR} = 0.102
 Diversity = 3.5257e+00 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 4.050479e+03 FES = 270100 μ_F = 0.793 μ_{CR} = 0.079
 Diversity = 3.4815e+00 k_t = [3, 3, 4, 2]
 Generation 2710: Best = 4.050215e+03 FES = 271100 μ_F = 0.787 μ_{CR} = 0.083
 Diversity = 3.5132e+00 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 4.049488e+03 FES = 272100 μ_F = 0.783 μ_{CR} = 0.084
 Diversity = 3.6648e+00 k_t = [3, 2, 4, 4]
 Generation 2730: Best = 4.049289e+03 FES = 273100 μ_F = 0.784 μ_{CR} = 0.086
 Diversity = 3.5151e+00 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 4.048844e+03 FES = 274100 μ_F = 0.778 μ_{CR} = 0.084
 Diversity = 3.5784e+00 k_t = [2, 4, 4, 3]
 Generation 2750: Best = 4.048506e+03 FES = 275100 μ_F = 0.781 μ_{CR} = 0.079
 Diversity = 3.7190e+00 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 4.048254e+03 FES = 276100 μ_F = 0.781 μ_{CR} = 0.081
 Diversity = 3.5451e+00 k_t = [3, 4, 4, 3]
 Generation 2770: Best = 4.047972e+03 FES = 277100 μ_F = 0.776 μ_{CR} = 0.069
 Diversity = 3.6085e+00 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 4.047567e+03 FES = 278100 μ_F = 0.776 μ_{CR} = 0.072
 Diversity = 3.5023e+00 k_t = [2, 2, 2, 4]
 Generation 2790: Best = 4.047412e+03 FES = 279100 μ_F = 0.776 μ_{CR} = 0.066
 Diversity = 3.5291e+00 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 4.047296e+03 FES = 280100 μ_F = 0.763 μ_{CR} = 0.074
 Diversity = 3.5907e+00 k_t = [4, 4, 2, 2]
 Generation 2810: Best = 4.047071e+03 FES = 281100 μ_F = 0.783 μ_{CR} = 0.079
 Diversity = 3.8061e+00 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 4.046680e+03 FES = 282100 μ_F = 0.785 μ_{CR} = 0.076
 Diversity = 3.6865e+00 k_t = [3, 3, 3, 3]
 Generation 2830: Best = 4.046334e+03 FES = 283100 μ_F = 0.764 μ_{CR} = 0.090
 Diversity = 3.7754e+00 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 4.045781e+03 FES = 284100 μ_F = 0.773 μ_{CR} = 0.089
 Diversity = 3.9356e+00 k_t = [4, 2, 3, 2]
 Generation 2850: Best = 4.045391e+03 FES = 285100 μ_F = 0.777 μ_{CR} = 0.097
 Diversity = 4.3378e+00 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 4.045030e+03 FES = 286100 μ_F = 0.778 μ_{CR} = 0.105
 Diversity = 4.3311e+00 k_t = [3, 2, 3, 4]
 Generation 2870: Best = 4.044543e+03 FES = 287100 μ_F = 0.786 μ_{CR} = 0.097
 Diversity = 4.1204e+00 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 4.044008e+03 FES = 288100 μ_F = 0.780 μ_{CR} = 0.094
 Diversity = 3.8693e+00 k_t = [4, 2, 4, 3]
 Generation 2890: Best = 4.043660e+03 FES = 289100 μ_F = 0.781 μ_{CR} = 0.115
 Diversity = 3.9501e+00 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 4.043470e+03 FES = 290100 μ_F = 0.792 μ_{CR} = 0.100
 Diversity = 3.9763e+00 k_t = [3, 4, 3, 4]
 Generation 2910: Best = 4.043141e+03 FES = 291100 μ_F = 0.799 μ_{CR} = 0.094
 Diversity = 4.0086e+00 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 4.043037e+03 FES = 292100 μ_F = 0.799 μ_{CR} = 0.088
 Diversity = 3.9026e+00 k_t = [4, 4, 2, 4]

Generation 2930: Best = 4.042693e+03 FES = 293100 μ_F = 0.785 μ_{CR} = 0.071
 Diversity = 3.9524e+00 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 4.042405e+03 FES = 294100 μ_F = 0.767 μ_{CR} = 0.074
 Diversity = 4.0217e+00 k_t = [2, 2, 2, 3]
 Generation 2950: Best = 4.042219e+03 FES = 295100 μ_F = 0.782 μ_{CR} = 0.078
 Diversity = 4.1258e+00 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 4.041953e+03 FES = 296100 μ_F = 0.782 μ_{CR} = 0.061
 Diversity = 3.9157e+00 k_t = [2, 3, 3, 3]
 Generation 2970: Best = 4.041475e+03 FES = 297100 μ_F = 0.775 μ_{CR} = 0.070
 Diversity = 3.7348e+00 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 4.041405e+03 FES = 298100 μ_F = 0.791 μ_{CR} = 0.079
 Diversity = 3.7011e+00 k_t = [4, 3, 2, 3]
 Generation 2990: Best = 4.041405e+03 FES = 299100 μ_F = 0.781 μ_{CR} = 0.071
 Diversity = 3.6508e+00 k_t = [1, 1, 1, 1]
 Generation 3000: Best = 4.041077e+03 FES = 300100 μ_F = 0.787 μ_{CR} = 0.076
 Diversity = 3.6848e+00 k_t = [2, 3, 2, 2]
 Generation 3010: Best = 4.040500e+03 FES = 301100 μ_F = 0.780 μ_{CR} = 0.092
 Diversity = 3.9627e+00 k_t = [1, 1, 1, 1]
 Generation 3020: Best = 4.040180e+03 FES = 302100 μ_F = 0.800 μ_{CR} = 0.095
 Diversity = 4.1342e+00 k_t = [2, 4, 2, 3]
 Generation 3030: Best = 4.039660e+03 FES = 303100 μ_F = 0.799 μ_{CR} = 0.090
 Diversity = 4.1529e+00 k_t = [1, 1, 1, 1]
 Generation 3040: Best = 4.039053e+03 FES = 304100 μ_F = 0.794 μ_{CR} = 0.089
 Diversity = 4.5697e+00 k_t = [4, 2, 3, 4]
 Generation 3050: Best = 4.038353e+03 FES = 305100 μ_F = 0.804 μ_{CR} = 0.095
 Diversity = 4.5763e+00 k_t = [1, 1, 1, 1]
 Generation 3060: Best = 4.038082e+03 FES = 306100 μ_F = 0.798 μ_{CR} = 0.087
 Diversity = 4.4252e+00 k_t = [3, 4, 4, 3]
 Generation 3070: Best = 4.037558e+03 FES = 307100 μ_F = 0.781 μ_{CR} = 0.089
 Diversity = 4.3441e+00 k_t = [1, 1, 1, 1]
 Generation 3080: Best = 4.037138e+03 FES = 308100 μ_F = 0.780 μ_{CR} = 0.091
 Diversity = 4.0940e+00 k_t = [4, 4, 4, 2]
 Generation 3090: Best = 4.036663e+03 FES = 309100 μ_F = 0.791 μ_{CR} = 0.100
 Diversity = 4.1619e+00 k_t = [1, 1, 1, 1]
 Generation 3100: Best = 4.036536e+03 FES = 310100 μ_F = 0.797 μ_{CR} = 0.098
 Diversity = 4.2452e+00 k_t = [4, 2, 2, 2]
 Generation 3110: Best = 4.035954e+03 FES = 311100 μ_F = 0.802 μ_{CR} = 0.103
 Diversity = 4.4315e+00 k_t = [1, 1, 1, 1]
 Generation 3120: Best = 4.035544e+03 FES = 312100 μ_F = 0.805 μ_{CR} = 0.090
 Diversity = 4.1518e+00 k_t = [2, 2, 4, 4]
 Generation 3130: Best = 4.035047e+03 FES = 313100 μ_F = 0.812 μ_{CR} = 0.094
 Diversity = 4.1903e+00 k_t = [1, 1, 1, 1]
 Generation 3140: Best = 4.034925e+03 FES = 314100 μ_F = 0.811 μ_{CR} = 0.087
 Diversity = 4.2775e+00 k_t = [3, 4, 2, 2]
 Generation 3150: Best = 4.034732e+03 FES = 315100 μ_F = 0.808 μ_{CR} = 0.062
 Diversity = 4.4142e+00 k_t = [1, 1, 1, 1]
 Generation 3160: Best = 4.034698e+03 FES = 316100 μ_F = 0.790 μ_{CR} = 0.071
 Diversity = 4.2060e+00 k_t = [3, 2, 3, 4]
 Generation 3170: Best = 4.034602e+03 FES = 317100 μ_F = 0.781 μ_{CR} = 0.061
 Diversity = 4.0084e+00 k_t = [1, 1, 1, 1]
 Generation 3180: Best = 4.034602e+03 FES = 318100 μ_F = 0.758 μ_{CR} = 0.055
 Diversity = 3.8638e+00 k_t = [3, 3, 3, 2]
 Generation 3190: Best = 4.034396e+03 FES = 319100 μ_F = 0.765 μ_{CR} = 0.055
 Diversity = 3.9305e+00 k_t = [1, 1, 1, 1]

Generation 3200: Best = 4.034244e+03 FES = 320100 μ_F = 0.780 μ_{CR} = 0.054
 Diversity = 3.8055e+00 k_t = [2, 2, 2, 2]
 Generation 3210: Best = 4.034123e+03 FES = 321100 μ_F = 0.793 μ_{CR} = 0.062
 Diversity = 3.7382e+00 k_t = [1, 1, 1, 1]
 Generation 3220: Best = 4.033887e+03 FES = 322100 μ_F = 0.810 μ_{CR} = 0.067
 Diversity = 3.8851e+00 k_t = [3, 3, 4, 3]
 Generation 3230: Best = 4.033693e+03 FES = 323100 μ_F = 0.821 μ_{CR} = 0.053
 Diversity = 4.0133e+00 k_t = [1, 1, 1, 1]
 Generation 3240: Best = 4.033358e+03 FES = 324100 μ_F = 0.814 μ_{CR} = 0.065
 Diversity = 3.9484e+00 k_t = [4, 2, 4, 4]
 Generation 3250: Best = 4.033094e+03 FES = 325100 μ_F = 0.814 μ_{CR} = 0.072
 Diversity = 4.1998e+00 k_t = [1, 1, 1, 1]
 Generation 3260: Best = 4.032393e+03 FES = 326100 μ_F = 0.809 μ_{CR} = 0.080
 Diversity = 3.8220e+00 k_t = [3, 4, 2, 3]
 Generation 3270: Best = 4.032153e+03 FES = 327100 μ_F = 0.803 μ_{CR} = 0.095
 Diversity = 3.7723e+00 k_t = [1, 1, 1, 1]
 Generation 3280: Best = 4.031589e+03 FES = 328100 μ_F = 0.797 μ_{CR} = 0.095
 Diversity = 3.5750e+00 k_t = [3, 2, 3, 3]
 Generation 3290: Best = 4.031357e+03 FES = 329100 μ_F = 0.796 μ_{CR} = 0.089
 Diversity = 3.5554e+00 k_t = [1, 1, 1, 1]
 Generation 3300: Best = 4.030840e+03 FES = 330100 μ_F = 0.792 μ_{CR} = 0.094
 Diversity = 3.5469e+00 k_t = [2, 4, 3, 4]
 Generation 3310: Best = 4.030333e+03 FES = 331100 μ_F = 0.804 μ_{CR} = 0.107
 Diversity = 3.7896e+00 k_t = [1, 1, 1, 1]
 Generation 3320: Best = 4.030031e+03 FES = 332100 μ_F = 0.797 μ_{CR} = 0.095
 Diversity = 3.7973e+00 k_t = [4, 3, 3, 3]
 Generation 3330: Best = 4.029556e+03 FES = 333100 μ_F = 0.803 μ_{CR} = 0.094
 Diversity = 4.0368e+00 k_t = [1, 1, 1, 1]
 Generation 3340: Best = 4.029334e+03 FES = 334100 μ_F = 0.790 μ_{CR} = 0.101
 Diversity = 4.0254e+00 k_t = [3, 2, 4, 2]
 Generation 3350: Best = 4.028685e+03 FES = 335100 μ_F = 0.784 μ_{CR} = 0.097
 Diversity = 4.0965e+00 k_t = [1, 1, 1, 1]
 Generation 3360: Best = 4.028338e+03 FES = 336100 μ_F = 0.788 μ_{CR} = 0.092
 Diversity = 3.9933e+00 k_t = [3, 3, 3, 4]
 Generation 3370: Best = 4.027976e+03 FES = 337100 μ_F = 0.798 μ_{CR} = 0.093
 Diversity = 3.8897e+00 k_t = [1, 1, 1, 1]
 Generation 3380: Best = 4.027679e+03 FES = 338100 μ_F = 0.790 μ_{CR} = 0.097
 Diversity = 3.8761e+00 k_t = [4, 4, 3, 3]
 Generation 3390: Best = 4.027145e+03 FES = 339100 μ_F = 0.795 μ_{CR} = 0.103
 Diversity = 4.1375e+00 k_t = [1, 1, 1, 1]
 Generation 3400: Best = 4.026631e+03 FES = 340100 μ_F = 0.792 μ_{CR} = 0.106
 Diversity = 4.1906e+00 k_t = [3, 2, 3, 3]
 Generation 3410: Best = 4.026373e+03 FES = 341100 μ_F = 0.780 μ_{CR} = 0.104
 Diversity = 4.1246e+00 k_t = [1, 1, 1, 1]
 Generation 3420: Best = 4.025962e+03 FES = 342100 μ_F = 0.780 μ_{CR} = 0.091
 Diversity = 3.8794e+00 k_t = [4, 4, 2, 3]
 Generation 3430: Best = 4.025399e+03 FES = 343100 μ_F = 0.780 μ_{CR} = 0.116
 Diversity = 3.8543e+00 k_t = [1, 1, 1, 1]
 Generation 3440: Best = 4.024832e+03 FES = 344100 μ_F = 0.781 μ_{CR} = 0.126
 Diversity = 3.7220e+00 k_t = [4, 3, 3, 3]
 Generation 3450: Best = 4.024517e+03 FES = 345100 μ_F = 0.772 μ_{CR} = 0.118
 Diversity = 3.6505e+00 k_t = [1, 1, 1, 1]
 Generation 3460: Best = 4.023985e+03 FES = 346100 μ_F = 0.771 μ_{CR} = 0.116
 Diversity = 3.6866e+00 k_t = [4, 3, 2, 4]

Generation 3470: Best = 4.023237e+03 FES = 347100 μ_F = 0.775 μ_{CR} = 0.107
 Diversity = 3.7087e+00 k_t = [1, 1, 1, 1]
 Generation 3480: Best = 4.022958e+03 FES = 348100 μ_F = 0.775 μ_{CR} = 0.114
 Diversity = 3.5285e+00 k_t = [2, 4, 3, 3]
 Generation 3490: Best = 4.022614e+03 FES = 349100 μ_F = 0.774 μ_{CR} = 0.116
 Diversity = 3.5754e+00 k_t = [1, 1, 1, 1]
 Generation 3500: Best = 4.022346e+03 FES = 350100 μ_F = 0.772 μ_{CR} = 0.093
 Diversity = 3.4726e+00 k_t = [3, 3, 3, 3]
 Generation 3510: Best = 4.022091e+03 FES = 351100 μ_F = 0.763 μ_{CR} = 0.079
 Diversity = 3.5658e+00 k_t = [1, 1, 1, 1]
 Generation 3520: Best = 4.021951e+03 FES = 352100 μ_F = 0.777 μ_{CR} = 0.071
 Diversity = 3.4936e+00 k_t = [4, 3, 2, 4]
 Generation 3530: Best = 4.021877e+03 FES = 353100 μ_F = 0.765 μ_{CR} = 0.075
 Diversity = 3.4631e+00 k_t = [1, 1, 1, 1]
 Generation 3540: Best = 4.021665e+03 FES = 354100 μ_F = 0.764 μ_{CR} = 0.086
 Diversity = 3.4743e+00 k_t = [3, 4, 2, 2]
 Generation 3550: Best = 4.021639e+03 FES = 355100 μ_F = 0.778 μ_{CR} = 0.073
 Diversity = 3.5521e+00 k_t = [1, 1, 1, 1]
 Generation 3560: Best = 4.021420e+03 FES = 356100 μ_F = 0.766 μ_{CR} = 0.071
 Diversity = 3.5050e+00 k_t = [2, 2, 2, 3]
 Generation 3570: Best = 4.021154e+03 FES = 357100 μ_F = 0.745 μ_{CR} = 0.072
 Diversity = 3.5253e+00 k_t = [1, 1, 1, 1]
 Generation 3580: Best = 4.021009e+03 FES = 358100 μ_F = 0.758 μ_{CR} = 0.073
 Diversity = 3.5555e+00 k_t = [4, 2, 4, 2]
 Generation 3590: Best = 4.020787e+03 FES = 359100 μ_F = 0.761 μ_{CR} = 0.069
 Diversity = 3.6910e+00 k_t = [1, 1, 1, 1]
 Generation 3600: Best = 4.020581e+03 FES = 360100 μ_F = 0.782 μ_{CR} = 0.070
 Diversity = 3.7523e+00 k_t = [4, 4, 3, 4]
 Generation 3610: Best = 4.020128e+03 FES = 361100 μ_F = 0.807 μ_{CR} = 0.067
 Diversity = 3.7845e+00 k_t = [1, 1, 1, 1]
 Generation 3620: Best = 4.019816e+03 FES = 362100 μ_F = 0.792 μ_{CR} = 0.073
 Diversity = 3.7979e+00 k_t = [4, 3, 4, 4]
 Generation 3630: Best = 4.019816e+03 FES = 363100 μ_F = 0.787 μ_{CR} = 0.077
 Diversity = 3.5862e+00 k_t = [1, 1, 1, 1]
 Generation 3640: Best = 4.019407e+03 FES = 364100 μ_F = 0.778 μ_{CR} = 0.079
 Diversity = 3.4220e+00 k_t = [3, 4, 3, 4]
 Generation 3650: Best = 4.019196e+03 FES = 365100 μ_F = 0.784 μ_{CR} = 0.069
 Diversity = 3.2866e+00 k_t = [1, 1, 1, 1]
 Generation 3660: Best = 4.019196e+03 FES = 366100 μ_F = 0.798 μ_{CR} = 0.061
 Diversity = 3.2687e+00 k_t = [3, 4, 2, 2]
 Generation 3670: Best = 4.018765e+03 FES = 367100 μ_F = 0.811 μ_{CR} = 0.059
 Diversity = 3.4104e+00 k_t = [1, 1, 1, 1]
 Generation 3680: Best = 4.018532e+03 FES = 368100 μ_F = 0.805 μ_{CR} = 0.059
 Diversity = 3.3477e+00 k_t = [2, 3, 4, 4]
 Generation 3690: Best = 4.018259e+03 FES = 369100 μ_F = 0.823 μ_{CR} = 0.078
 Diversity = 3.5308e+00 k_t = [1, 1, 1, 1]
 Generation 3700: Best = 4.018056e+03 FES = 370100 μ_F = 0.801 μ_{CR} = 0.088
 Diversity = 3.4198e+00 k_t = [3, 4, 2, 3]
 Generation 3710: Best = 4.017187e+03 FES = 371100 μ_F = 0.788 μ_{CR} = 0.089
 Diversity = 3.2268e+00 k_t = [1, 1, 1, 1]
 Generation 3720: Best = 4.016479e+03 FES = 372100 μ_F = 0.786 μ_{CR} = 0.097
 Diversity = 3.1880e+00 k_t = [4, 3, 2, 3]
 Generation 3730: Best = 4.015959e+03 FES = 373100 μ_F = 0.789 μ_{CR} = 0.111
 Diversity = 3.4348e+00 k_t = [1, 1, 1, 1]

Generation 3740: Best = 4.015387e+03 FES = 374100 μ_F = 0.799 μ_{CR} = 0.111
 Diversity = 3.5033e+00 k_t = [4, 3, 2, 3]
 Generation 3750: Best = 4.014953e+03 FES = 375100 μ_F = 0.803 μ_{CR} = 0.105
 Diversity = 3.3622e+00 k_t = [1, 1, 1, 1]
 Generation 3760: Best = 4.014603e+03 FES = 376100 μ_F = 0.792 μ_{CR} = 0.107
 Diversity = 3.3350e+00 k_t = [4, 2, 3, 2]
 Generation 3770: Best = 4.014020e+03 FES = 377100 μ_F = 0.804 μ_{CR} = 0.116
 Diversity = 3.3064e+00 k_t = [1, 1, 1, 1]
 Generation 3780: Best = 4.013441e+03 FES = 378100 μ_F = 0.790 μ_{CR} = 0.119
 Diversity = 3.3177e+00 k_t = [4, 3, 2, 2]
 Generation 3790: Best = 4.012741e+03 FES = 379100 μ_F = 0.794 μ_{CR} = 0.109
 Diversity = 3.2915e+00 k_t = [1, 1, 1, 1]
 Generation 3800: Best = 4.012409e+03 FES = 380100 μ_F = 0.792 μ_{CR} = 0.118
 Diversity = 3.5708e+00 k_t = [4, 4, 3, 2]
 Generation 3810: Best = 4.011923e+03 FES = 381100 μ_F = 0.787 μ_{CR} = 0.105
 Diversity = 3.5812e+00 k_t = [1, 1, 1, 1]
 Generation 3820: Best = 4.011564e+03 FES = 382100 μ_F = 0.764 μ_{CR} = 0.108
 Diversity = 3.5497e+00 k_t = [2, 3, 3, 2]
 Generation 3830: Best = 4.011060e+03 FES = 383100 μ_F = 0.735 μ_{CR} = 0.106
 Diversity = 3.3914e+00 k_t = [1, 1, 1, 1]
 Generation 3840: Best = 4.010804e+03 FES = 384100 μ_F = 0.741 μ_{CR} = 0.096
 Diversity = 3.2306e+00 k_t = [3, 4, 4, 3]
 Generation 3850: Best = 4.010511e+03 FES = 385100 μ_F = 0.730 μ_{CR} = 0.108
 Diversity = 3.2566e+00 k_t = [1, 1, 1, 1]
 Generation 3860: Best = 4.010221e+03 FES = 386100 μ_F = 0.739 μ_{CR} = 0.106
 Diversity = 3.2200e+00 k_t = [4, 2, 3, 3]
 Generation 3870: Best = 4.009917e+03 FES = 387100 μ_F = 0.748 μ_{CR} = 0.086
 Diversity = 3.3262e+00 k_t = [1, 1, 1, 1]
 Generation 3880: Best = 4.009492e+03 FES = 388100 μ_F = 0.749 μ_{CR} = 0.086
 Diversity = 3.3061e+00 k_t = [2, 4, 2, 3]
 Generation 3890: Best = 4.009478e+03 FES = 389100 μ_F = 0.769 μ_{CR} = 0.084
 Diversity = 3.4315e+00 k_t = [1, 1, 1, 1]
 Generation 3900: Best = 4.009263e+03 FES = 390100 μ_F = 0.786 μ_{CR} = 0.082
 Diversity = 3.3191e+00 k_t = [4, 2, 2, 3]
 Generation 3910: Best = 4.008818e+03 FES = 391100 μ_F = 0.784 μ_{CR} = 0.085
 Diversity = 3.3789e+00 k_t = [1, 1, 1, 1]
 Generation 3920: Best = 4.008616e+03 FES = 392100 μ_F = 0.781 μ_{CR} = 0.073
 Diversity = 3.1636e+00 k_t = [2, 2, 4, 2]
 Generation 3930: Best = 4.008489e+03 FES = 393100 μ_F = 0.761 μ_{CR} = 0.081
 Diversity = 3.2691e+00 k_t = [1, 1, 1, 1]
 Generation 3940: Best = 4.008129e+03 FES = 394100 μ_F = 0.763 μ_{CR} = 0.068
 Diversity = 3.2237e+00 k_t = [2, 3, 4, 2]
 Generation 3950: Best = 4.008129e+03 FES = 395100 μ_F = 0.768 μ_{CR} = 0.072
 Diversity = 3.3018e+00 k_t = [1, 1, 1, 1]
 Generation 3960: Best = 4.008032e+03 FES = 396100 μ_F = 0.771 μ_{CR} = 0.065
 Diversity = 3.2383e+00 k_t = [4, 3, 2, 2]
 Generation 3970: Best = 4.008032e+03 FES = 397100 μ_F = 0.740 μ_{CR} = 0.062
 Diversity = 3.1991e+00 k_t = [1, 1, 1, 1]
 Generation 3980: Best = 4.008027e+03 FES = 398100 μ_F = 0.747 μ_{CR} = 0.061
 Diversity = 3.2168e+00 k_t = [3, 3, 2, 2]
 Generation 3990: Best = 4.007794e+03 FES = 399100 μ_F = 0.772 μ_{CR} = 0.060
 Diversity = 3.2465e+00 k_t = [1, 1, 1, 1]
 Generation 4000: Best = 4.007355e+03 FES = 400100 μ_F = 0.792 μ_{CR} = 0.064
 Diversity = 3.3739e+00 k_t = [3, 2, 3, 4]

Generation 4010: Best = 4.007345e+03 FES = 401100 μ_F = 0.787 μ_{CR} = 0.059
 Diversity = 3.5598e+00 k_t = [1, 1, 1, 1]
 Generation 4020: Best = 4.007345e+03 FES = 402100 μ_F = 0.784 μ_{CR} = 0.060
 Diversity = 3.5685e+00 k_t = [4, 3, 3, 3]
 Generation 4030: Best = 4.007300e+03 FES = 403100 μ_F = 0.795 μ_{CR} = 0.061
 Diversity = 3.7956e+00 k_t = [1, 1, 1, 1]
 Generation 4040: Best = 4.006908e+03 FES = 404100 μ_F = 0.766 μ_{CR} = 0.066
 Diversity = 3.9788e+00 k_t = [3, 4, 3, 2]
 Generation 4050: Best = 4.006669e+03 FES = 405100 μ_F = 0.760 μ_{CR} = 0.080
 Diversity = 3.9256e+00 k_t = [1, 1, 1, 1]
 Generation 4060: Best = 4.006191e+03 FES = 406100 μ_F = 0.759 μ_{CR} = 0.079
 Diversity = 3.6816e+00 k_t = [2, 3, 4, 2]
 Generation 4070: Best = 4.005769e+03 FES = 407100 μ_F = 0.761 μ_{CR} = 0.095
 Diversity = 3.4909e+00 k_t = [1, 1, 1, 1]
 Generation 4080: Best = 4.005739e+03 FES = 408100 μ_F = 0.768 μ_{CR} = 0.088
 Diversity = 3.3874e+00 k_t = [2, 3, 2, 3]
 Generation 4090: Best = 4.005421e+03 FES = 409100 μ_F = 0.749 μ_{CR} = 0.092
 Diversity = 3.3150e+00 k_t = [1, 1, 1, 1]
 Generation 4100: Best = 4.005101e+03 FES = 410100 μ_F = 0.762 μ_{CR} = 0.085
 Diversity = 3.3795e+00 k_t = [3, 4, 2, 2]
 Generation 4110: Best = 4.004824e+03 FES = 411100 μ_F = 0.774 μ_{CR} = 0.085
 Diversity = 3.3687e+00 k_t = [1, 1, 1, 1]
 Generation 4120: Best = 4.004626e+03 FES = 412100 μ_F = 0.776 μ_{CR} = 0.089
 Diversity = 3.7320e+00 k_t = [4, 4, 3, 2]
 Generation 4130: Best = 4.004469e+03 FES = 413100 μ_F = 0.773 μ_{CR} = 0.089
 Diversity = 3.8045e+00 k_t = [1, 1, 1, 1]
 Generation 4140: Best = 4.004328e+03 FES = 414100 μ_F = 0.768 μ_{CR} = 0.072
 Diversity = 3.6554e+00 k_t = [3, 3, 4, 4]
 Generation 4150: Best = 4.004155e+03 FES = 415100 μ_F = 0.767 μ_{CR} = 0.074
 Diversity = 3.6299e+00 k_t = [1, 1, 1, 1]
 Generation 4160: Best = 4.004038e+03 FES = 416100 μ_F = 0.764 μ_{CR} = 0.069
 Diversity = 3.6179e+00 k_t = [2, 3, 4, 4]
 Generation 4170: Best = 4.004038e+03 FES = 417100 μ_F = 0.768 μ_{CR} = 0.065
 Diversity = 3.4375e+00 k_t = [1, 1, 1, 1]
 Generation 4180: Best = 4.003634e+03 FES = 418100 μ_F = 0.767 μ_{CR} = 0.074
 Diversity = 3.5471e+00 k_t = [3, 2, 4, 2]
 Generation 4190: Best = 4.003508e+03 FES = 419100 μ_F = 0.766 μ_{CR} = 0.059
 Diversity = 3.5860e+00 k_t = [1, 1, 1, 1]
 Generation 4200: Best = 4.003304e+03 FES = 420100 μ_F = 0.768 μ_{CR} = 0.066
 Diversity = 3.7355e+00 k_t = [4, 2, 4, 3]
 Generation 4210: Best = 4.003177e+03 FES = 421100 μ_F = 0.747 μ_{CR} = 0.069
 Diversity = 3.8385e+00 k_t = [1, 1, 1, 1]
 Generation 4220: Best = 4.003123e+03 FES = 422100 μ_F = 0.757 μ_{CR} = 0.068
 Diversity = 3.7364e+00 k_t = [4, 4, 2, 4]
 Generation 4230: Best = 4.002668e+03 FES = 423100 μ_F = 0.776 μ_{CR} = 0.070
 Diversity = 3.6033e+00 k_t = [1, 1, 1, 1]
 Generation 4240: Best = 4.001950e+03 FES = 424100 μ_F = 0.774 μ_{CR} = 0.068
 Diversity = 3.5752e+00 k_t = [2, 4, 2, 4]
 Generation 4250: Best = 4.001950e+03 FES = 425100 μ_F = 0.774 μ_{CR} = 0.062
 Diversity = 3.4075e+00 k_t = [1, 1, 1, 1]
 Generation 4260: Best = 4.001781e+03 FES = 426100 μ_F = 0.792 μ_{CR} = 0.060
 Diversity = 3.3788e+00 k_t = [4, 4, 2, 2]
 Generation 4270: Best = 4.001781e+03 FES = 427100 μ_F = 0.781 μ_{CR} = 0.058
 Diversity = 3.1932e+00 k_t = [1, 1, 1, 1]

Generation 4280: Best = 4.001578e+03 FES = 428100 μ_F = 0.797 μ_{CR} = 0.069
 Diversity = 2.9274e+00 k_t = [2, 4, 3, 4]
 Generation 4290: Best = 4.001297e+03 FES = 429100 μ_F = 0.794 μ_{CR} = 0.067
 Diversity = 2.7576e+00 k_t = [1, 1, 1, 1]
 Generation 4300: Best = 4.001297e+03 FES = 430100 μ_F = 0.801 μ_{CR} = 0.076
 Diversity = 2.6813e+00 k_t = [4, 2, 4, 3]
 Generation 4310: Best = 4.001204e+03 FES = 431100 μ_F = 0.797 μ_{CR} = 0.066
 Diversity = 2.7352e+00 k_t = [1, 1, 1, 1]
 Generation 4320: Best = 4.001137e+03 FES = 432100 μ_F = 0.798 μ_{CR} = 0.058
 Diversity = 2.6577e+00 k_t = [3, 2, 3, 4]
 Generation 4330: Best = 4.001021e+03 FES = 433100 μ_F = 0.798 μ_{CR} = 0.055
 Diversity = 2.6611e+00 k_t = [1, 1, 1, 1]
 Generation 4340: Best = 4.001021e+03 FES = 434100 μ_F = 0.813 μ_{CR} = 0.052
 Diversity = 2.6128e+00 k_t = [2, 4, 4, 3]
 Generation 4350: Best = 4.000744e+03 FES = 435100 μ_F = 0.819 μ_{CR} = 0.056
 Diversity = 2.6594e+00 k_t = [1, 1, 1, 1]
 Generation 4360: Best = 4.000662e+03 FES = 436100 μ_F = 0.809 μ_{CR} = 0.061
 Diversity = 2.6576e+00 k_t = [4, 2, 2, 3]
 Generation 4370: Best = 4.000497e+03 FES = 437100 μ_F = 0.813 μ_{CR} = 0.056
 Diversity = 2.5827e+00 k_t = [1, 1, 1, 1]
 Generation 4380: Best = 4.000420e+03 FES = 438100 μ_F = 0.822 μ_{CR} = 0.045
 Diversity = 2.4537e+00 k_t = [4, 3, 4, 4]
 Generation 4390: Best = 4.000246e+03 FES = 439100 μ_F = 0.831 μ_{CR} = 0.048
 Diversity = 2.5354e+00 k_t = [1, 1, 1, 1]
 Generation 4400: Best = 4.000016e+03 FES = 440100 μ_F = 0.808 μ_{CR} = 0.058
 Diversity = 2.6232e+00 k_t = [4, 2, 4, 3]
 Generation 4410: Best = 3.999818e+03 FES = 441100 μ_F = 0.785 μ_{CR} = 0.060
 Diversity = 2.6271e+00 k_t = [1, 1, 1, 1]
 Generation 4420: Best = 3.999355e+03 FES = 442100 μ_F = 0.776 μ_{CR} = 0.064
 Diversity = 2.5963e+00 k_t = [2, 2, 3, 2]
 Generation 4430: Best = 3.999303e+03 FES = 443100 μ_F = 0.780 μ_{CR} = 0.064
 Diversity = 2.5814e+00 k_t = [1, 1, 1, 1]
 Generation 4440: Best = 3.999303e+03 FES = 444100 μ_F = 0.797 μ_{CR} = 0.061
 Diversity = 2.6695e+00 k_t = [2, 3, 3, 4]
 Generation 4450: Best = 3.999119e+03 FES = 445100 μ_F = 0.799 μ_{CR} = 0.054
 Diversity = 2.6444e+00 k_t = [1, 1, 1, 1]
 Generation 4460: Best = 3.999040e+03 FES = 446100 μ_F = 0.810 μ_{CR} = 0.053
 Diversity = 2.5244e+00 k_t = [3, 2, 2, 3]
 Generation 4470: Best = 3.998792e+03 FES = 447100 μ_F = 0.793 μ_{CR} = 0.060
 Diversity = 2.5896e+00 k_t = [1, 1, 1, 1]
 Generation 4480: Best = 3.998687e+03 FES = 448100 μ_F = 0.789 μ_{CR} = 0.059
 Diversity = 2.7035e+00 k_t = [4, 4, 4, 4]
 Generation 4490: Best = 3.998231e+03 FES = 449100 μ_F = 0.804 μ_{CR} = 0.057
 Diversity = 2.6841e+00 k_t = [1, 1, 1, 1]
 Generation 4500: Best = 3.997831e+03 FES = 450100 μ_F = 0.794 μ_{CR} = 0.064
 Diversity = 2.6507e+00 k_t = [4, 4, 4, 3]
 Generation 4510: Best = 3.997318e+03 FES = 451100 μ_F = 0.778 μ_{CR} = 0.066
 Diversity = 2.4653e+00 k_t = [1, 1, 1, 1]
 Generation 4520: Best = 3.997149e+03 FES = 452100 μ_F = 0.770 μ_{CR} = 0.071
 Diversity = 2.3699e+00 k_t = [3, 2, 2, 3]
 Generation 4530: Best = 3.996960e+03 FES = 453100 μ_F = 0.786 μ_{CR} = 0.077
 Diversity = 2.2785e+00 k_t = [1, 1, 1, 1]
 Generation 4540: Best = 3.996947e+03 FES = 454100 μ_F = 0.792 μ_{CR} = 0.067
 Diversity = 2.2633e+00 k_t = [2, 4, 2, 3]

Generation 4550: Best = 3.996906e+03 FES = 455100 μ_F = 0.783 μ_{CR} = 0.064
 Diversity = 2.1938e+00 k_t = [1, 1, 1, 1]
 Generation 4560: Best = 3.996866e+03 FES = 456100 μ_F = 0.785 μ_{CR} = 0.058
 Diversity = 2.1015e+00 k_t = [2, 3, 2, 3]
 Generation 4570: Best = 3.996832e+03 FES = 457100 μ_F = 0.765 μ_{CR} = 0.055
 Diversity = 2.1169e+00 k_t = [1, 1, 1, 1]
 Generation 4580: Best = 3.996650e+03 FES = 458100 μ_F = 0.770 μ_{CR} = 0.050
 Diversity = 2.0907e+00 k_t = [4, 4, 4, 2]
 Generation 4590: Best = 3.996618e+03 FES = 459100 μ_F = 0.757 μ_{CR} = 0.057
 Diversity = 2.0610e+00 k_t = [1, 1, 1, 1]
 Generation 4600: Best = 3.996560e+03 FES = 460100 μ_F = 0.765 μ_{CR} = 0.045
 Diversity = 2.0678e+00 k_t = [2, 2, 2, 2]
 Generation 4610: Best = 3.996548e+03 FES = 461100 μ_F = 0.769 μ_{CR} = 0.045
 Diversity = 2.0645e+00 k_t = [1, 1, 1, 1]
 Generation 4620: Best = 3.996538e+03 FES = 462100 μ_F = 0.783 μ_{CR} = 0.042
 Diversity = 1.9971e+00 k_t = [2, 2, 3, 2]
 Generation 4630: Best = 3.996519e+03 FES = 463100 μ_F = 0.774 μ_{CR} = 0.041
 Diversity = 2.0330e+00 k_t = [1, 1, 1, 1]
 Generation 4640: Best = 3.996343e+03 FES = 464100 μ_F = 0.767 μ_{CR} = 0.042
 Diversity = 2.0325e+00 k_t = [3, 4, 3, 4]
 Generation 4650: Best = 3.996343e+03 FES = 465100 μ_F = 0.743 μ_{CR} = 0.041
 Diversity = 2.0107e+00 k_t = [1, 1, 1, 1]
 Generation 4660: Best = 3.996307e+03 FES = 466100 μ_F = 0.736 μ_{CR} = 0.042
 Diversity = 1.9365e+00 k_t = [3, 2, 3, 4]
 Generation 4670: Best = 3.996298e+03 FES = 467100 μ_F = 0.726 μ_{CR} = 0.037
 Diversity = 1.9387e+00 k_t = [1, 1, 1, 1]
 Generation 4680: Best = 3.996215e+03 FES = 468100 μ_F = 0.740 μ_{CR} = 0.040
 Diversity = 1.9351e+00 k_t = [4, 3, 2, 3]
 Generation 4690: Best = 3.996203e+03 FES = 469100 μ_F = 0.760 μ_{CR} = 0.037
 Diversity = 1.9004e+00 k_t = [1, 1, 1, 1]
 Generation 4700: Best = 3.996203e+03 FES = 470100 μ_F = 0.747 μ_{CR} = 0.041
 Diversity = 1.8860e+00 k_t = [3, 4, 2, 3]
 Generation 4710: Best = 3.996140e+03 FES = 471100 μ_F = 0.734 μ_{CR} = 0.034
 Diversity = 1.9009e+00 k_t = [1, 1, 1, 1]
 Generation 4720: Best = 3.996034e+03 FES = 472100 μ_F = 0.759 μ_{CR} = 0.033
 Diversity = 1.8933e+00 k_t = [4, 3, 2, 2]
 Generation 4730: Best = 3.996019e+03 FES = 473100 μ_F = 0.756 μ_{CR} = 0.033
 Diversity = 1.9070e+00 k_t = [1, 1, 1, 1]
 Generation 4740: Best = 3.995977e+03 FES = 474100 μ_F = 0.762 μ_{CR} = 0.035
 Diversity = 1.8979e+00 k_t = [4, 2, 2, 3]
 Generation 4750: Best = 3.995924e+03 FES = 475100 μ_F = 0.785 μ_{CR} = 0.037
 Diversity = 1.8659e+00 k_t = [1, 1, 1, 1]
 Generation 4760: Best = 3.995924e+03 FES = 476100 μ_F = 0.793 μ_{CR} = 0.044
 Diversity = 1.8716e+00 k_t = [2, 4, 3, 3]
 Generation 4770: Best = 3.995897e+03 FES = 477100 μ_F = 0.813 μ_{CR} = 0.037
 Diversity = 1.8855e+00 k_t = [1, 1, 1, 1]
 Generation 4780: Best = 3.995747e+03 FES = 478100 μ_F = 0.819 μ_{CR} = 0.034
 Diversity = 1.8760e+00 k_t = [2, 3, 4, 4]
 Generation 4790: Best = 3.995712e+03 FES = 479100 μ_F = 0.836 μ_{CR} = 0.035
 Diversity = 1.8784e+00 k_t = [1, 1, 1, 1]
 Generation 4800: Best = 3.995712e+03 FES = 480100 μ_F = 0.845 μ_{CR} = 0.034
 Diversity = 1.8956e+00 k_t = [3, 2, 4, 3]
 Generation 4810: Best = 3.995712e+03 FES = 481100 μ_F = 0.831 μ_{CR} = 0.035
 Diversity = 1.9067e+00 k_t = [1, 1, 1, 1]

Generation 4820: Best = 3.995707e+03 FES = 482100 μ_F = 0.827 μ_{CR} = 0.039
 Diversity = 1.8716e+00 k_t = [3, 3, 4, 4]
 Generation 4830: Best = 3.995697e+03 FES = 483100 μ_F = 0.822 μ_{CR} = 0.036
 Diversity = 1.8610e+00 k_t = [1, 1, 1, 1]
 Generation 4840: Best = 3.995697e+03 FES = 484100 μ_F = 0.828 μ_{CR} = 0.033
 Diversity = 1.8310e+00 k_t = [2, 2, 3, 4]
 Generation 4850: Best = 3.995662e+03 FES = 485100 μ_F = 0.834 μ_{CR} = 0.030
 Diversity = 1.8301e+00 k_t = [1, 1, 1, 1]
 Generation 4860: Best = 3.995646e+03 FES = 486100 μ_F = 0.835 μ_{CR} = 0.031
 Diversity = 1.8347e+00 k_t = [2, 2, 2, 4]
 Generation 4870: Best = 3.995646e+03 FES = 487100 μ_F = 0.861 μ_{CR} = 0.030
 Diversity = 1.8397e+00 k_t = [1, 1, 1, 1]
 Generation 4880: Best = 3.995571e+03 FES = 488100 μ_F = 0.853 μ_{CR} = 0.031
 Diversity = 1.8525e+00 k_t = [3, 3, 2, 2]
 Generation 4890: Best = 3.995571e+03 FES = 489100 μ_F = 0.832 μ_{CR} = 0.033
 Diversity = 1.8552e+00 k_t = [1, 1, 1, 1]
 Generation 4900: Best = 3.995566e+03 FES = 490100 μ_F = 0.832 μ_{CR} = 0.039
 Diversity = 1.8436e+00 k_t = [4, 4, 2, 3]
 Generation 4910: Best = 3.995566e+03 FES = 491100 μ_F = 0.823 μ_{CR} = 0.044
 Diversity = 1.8363e+00 k_t = [1, 1, 1, 1]
 Generation 4920: Best = 3.995566e+03 FES = 492100 μ_F = 0.824 μ_{CR} = 0.033
 Diversity = 1.8253e+00 k_t = [2, 4, 4, 4]
 Generation 4930: Best = 3.995564e+03 FES = 493100 μ_F = 0.804 μ_{CR} = 0.030
 Diversity = 1.8143e+00 k_t = [1, 1, 1, 1]
 Generation 4940: Best = 3.995518e+03 FES = 494100 μ_F = 0.823 μ_{CR} = 0.033
 Diversity = 1.7860e+00 k_t = [3, 2, 4, 4]
 Generation 4950: Best = 3.995518e+03 FES = 495100 μ_F = 0.793 μ_{CR} = 0.034
 Diversity = 1.7638e+00 k_t = [1, 1, 1, 1]
 Generation 4960: Best = 3.995480e+03 FES = 496100 μ_F = 0.858 μ_{CR} = 0.031
 Diversity = 1.7827e+00 k_t = [3, 3, 4, 4]
 Generation 4970: Best = 3.995451e+03 FES = 497100 μ_F = 0.852 μ_{CR} = 0.035
 Diversity = 1.7837e+00 k_t = [1, 1, 1, 1]
 Generation 4980: Best = 3.995451e+03 FES = 498100 μ_F = 0.844 μ_{CR} = 0.031
 Diversity = 1.7702e+00 k_t = [2, 3, 4, 2]
 Generation 4990: Best = 3.995380e+03 FES = 499100 μ_F = 0.793 μ_{CR} = 0.028
 Diversity = 1.7705e+00 k_t = [1, 1, 1, 1]
 Generation 5000: Best = 3.995305e+03 FES = 500100 μ_F = 0.755 μ_{CR} = 0.030
 Diversity = 1.7712e+00 k_t = [3, 3, 3, 3]
 Generation 5010: Best = 3.995280e+03 FES = 501100 μ_F = 0.763 μ_{CR} = 0.027
 Diversity = 1.7636e+00 k_t = [1, 1, 1, 1]
 Generation 5020: Best = 3.995280e+03 FES = 502100 μ_F = 0.803 μ_{CR} = 0.034
 Diversity = 1.7424e+00 k_t = [3, 3, 2, 3]
 Generation 5030: Best = 3.995277e+03 FES = 503100 μ_F = 0.784 μ_{CR} = 0.032
 Diversity = 1.7399e+00 k_t = [1, 1, 1, 1]
 Generation 5040: Best = 3.995246e+03 FES = 504100 μ_F = 0.775 μ_{CR} = 0.029
 Diversity = 1.7389e+00 k_t = [4, 3, 4, 2]
 Generation 5050: Best = 3.995228e+03 FES = 505100 μ_F = 0.784 μ_{CR} = 0.026
 Diversity = 1.7579e+00 k_t = [1, 1, 1, 1]
 Generation 5060: Best = 3.995226e+03 FES = 506100 μ_F = 0.786 μ_{CR} = 0.033
 Diversity = 1.7638e+00 k_t = [2, 3, 4, 3]
 Generation 5070: Best = 3.995179e+03 FES = 507100 μ_F = 0.794 μ_{CR} = 0.034
 Diversity = 1.7598e+00 k_t = [1, 1, 1, 1]
 Generation 5080: Best = 3.995108e+03 FES = 508100 μ_F = 0.801 μ_{CR} = 0.033
 Diversity = 1.7597e+00 k_t = [4, 2, 3, 2]

Generation 5090: Best = 3.995108e+03 FES = 509100 μ_F = 0.840 μ_{CR} = 0.026
 Diversity = 1.7684e+00 k_t = [1, 1, 1, 1]
 Generation 5100: Best = 3.995108e+03 FES = 510100 μ_F = 0.828 μ_{CR} = 0.025
 Diversity = 1.7882e+00 k_t = [4, 4, 3, 4]
 Generation 5110: Best = 3.995107e+03 FES = 511100 μ_F = 0.821 μ_{CR} = 0.025
 Diversity = 1.7725e+00 k_t = [1, 1, 1, 1]
 Generation 5120: Best = 3.995107e+03 FES = 512100 μ_F = 0.761 μ_{CR} = 0.036
 Diversity = 1.7763e+00 k_t = [4, 2, 3, 3]
 Generation 5130: Best = 3.995075e+03 FES = 513100 μ_F = 0.687 μ_{CR} = 0.029
 Diversity = 1.7784e+00 k_t = [1, 1, 1, 1]
 Generation 5140: Best = 3.995036e+03 FES = 514100 μ_F = 0.703 μ_{CR} = 0.034
 Diversity = 1.7673e+00 k_t = [2, 3, 3, 4]
 Generation 5150: Best = 3.995012e+03 FES = 515100 μ_F = 0.733 μ_{CR} = 0.030
 Diversity = 1.7957e+00 k_t = [1, 1, 1, 1]
 Generation 5160: Best = 3.995003e+03 FES = 516100 μ_F = 0.732 μ_{CR} = 0.028
 Diversity = 1.8081e+00 k_t = [4, 2, 3, 3]
 Generation 5170: Best = 3.994962e+03 FES = 517100 μ_F = 0.744 μ_{CR} = 0.030
 Diversity = 1.7962e+00 k_t = [1, 1, 1, 1]
 Generation 5180: Best = 3.994962e+03 FES = 518100 μ_F = 0.758 μ_{CR} = 0.033
 Diversity = 1.8008e+00 k_t = [3, 3, 4, 3]
 Generation 5190: Best = 3.994949e+03 FES = 519100 μ_F = 0.723 μ_{CR} = 0.041
 Diversity = 1.7809e+00 k_t = [1, 1, 1, 1]
 Generation 5200: Best = 3.994946e+03 FES = 520100 μ_F = 0.721 μ_{CR} = 0.036
 Diversity = 1.7744e+00 k_t = [3, 4, 4, 2]
 Generation 5210: Best = 3.994944e+03 FES = 521100 μ_F = 0.706 μ_{CR} = 0.035
 Diversity = 1.7764e+00 k_t = [1, 1, 1, 1]
 Generation 5220: Best = 3.994944e+03 FES = 522100 μ_F = 0.767 μ_{CR} = 0.031
 Diversity = 1.7648e+00 k_t = [2, 4, 4, 4]
 Generation 5230: Best = 3.994908e+03 FES = 523100 μ_F = 0.767 μ_{CR} = 0.031
 Diversity = 1.7848e+00 k_t = [1, 1, 1, 1]
 Generation 5240: Best = 3.994908e+03 FES = 524100 μ_F = 0.758 μ_{CR} = 0.028
 Diversity = 1.8011e+00 k_t = [2, 3, 3, 2]
 Generation 5250: Best = 3.994900e+03 FES = 525100 μ_F = 0.786 μ_{CR} = 0.030
 Diversity = 1.8009e+00 k_t = [1, 1, 1, 1]
 Generation 5260: Best = 3.994900e+03 FES = 526100 μ_F = 0.820 μ_{CR} = 0.030
 Diversity = 1.7811e+00 k_t = [4, 2, 2, 3]
 Generation 5270: Best = 3.994900e+03 FES = 527100 μ_F = 0.811 μ_{CR} = 0.034
 Diversity = 1.7931e+00 k_t = [1, 1, 1, 1]
 Generation 5280: Best = 3.994864e+03 FES = 528100 μ_F = 0.805 μ_{CR} = 0.037
 Diversity = 1.7548e+00 k_t = [3, 2, 3, 3]
 Generation 5290: Best = 3.994864e+03 FES = 529100 μ_F = 0.797 μ_{CR} = 0.036
 Diversity = 1.7539e+00 k_t = [1, 1, 1, 1]
 Generation 5300: Best = 3.994864e+03 FES = 530100 μ_F = 0.751 μ_{CR} = 0.033
 Diversity = 1.7272e+00 k_t = [2, 2, 4, 4]
 Generation 5310: Best = 3.994831e+03 FES = 531100 μ_F = 0.759 μ_{CR} = 0.029
 Diversity = 1.7336e+00 k_t = [1, 1, 1, 1]
 Generation 5320: Best = 3.994831e+03 FES = 532100 μ_F = 0.763 μ_{CR} = 0.030
 Diversity = 1.6975e+00 k_t = [3, 3, 3, 4]
 Generation 5330: Best = 3.994831e+03 FES = 533100 μ_F = 0.755 μ_{CR} = 0.031
 Diversity = 1.6971e+00 k_t = [1, 1, 1, 1]
 Generation 5340: Best = 3.994831e+03 FES = 534100 μ_F = 0.783 μ_{CR} = 0.026
 Diversity = 1.6969e+00 k_t = [2, 2, 2, 2]
 Generation 5350: Best = 3.994823e+03 FES = 535100 μ_F = 0.813 μ_{CR} = 0.027
 Diversity = 1.7027e+00 k_t = [1, 1, 1, 1]

Generation 5360: Best = 3.994800e+03 FES = 536100 μ_F = 0.808 μ_{CR} = 0.029
 Diversity = 1.6921e+00 k_t = [2, 4, 3, 3]
 Generation 5370: Best = 3.994799e+03 FES = 537100 μ_F = 0.824 μ_{CR} = 0.032
 Diversity = 1.6937e+00 k_t = [1, 1, 1, 1]
 Generation 5380: Best = 3.994799e+03 FES = 538100 μ_F = 0.808 μ_{CR} = 0.029
 Diversity = 1.7063e+00 k_t = [2, 2, 2, 3]
 Generation 5390: Best = 3.994759e+03 FES = 539100 μ_F = 0.799 μ_{CR} = 0.029
 Diversity = 1.7062e+00 k_t = [1, 1, 1, 1]
 Generation 5400: Best = 3.994759e+03 FES = 540100 μ_F = 0.818 μ_{CR} = 0.026
 Diversity = 1.7078e+00 k_t = [4, 3, 4, 2]
 Generation 5410: Best = 3.994759e+03 FES = 541100 μ_F = 0.803 μ_{CR} = 0.024
 Diversity = 1.6943e+00 k_t = [1, 1, 1, 1]
 Generation 5420: Best = 3.994755e+03 FES = 542100 μ_F = 0.825 μ_{CR} = 0.024
 Diversity = 1.6951e+00 k_t = [4, 2, 3, 4]
 Generation 5430: Best = 3.994755e+03 FES = 543100 μ_F = 0.825 μ_{CR} = 0.025
 Diversity = 1.6937e+00 k_t = [1, 1, 1, 1]
 Generation 5440: Best = 3.994750e+03 FES = 544100 μ_F = 0.841 μ_{CR} = 0.025
 Diversity = 1.6871e+00 k_t = [3, 4, 3, 4]
 Generation 5450: Best = 3.994726e+03 FES = 545100 μ_F = 0.842 μ_{CR} = 0.031
 Diversity = 1.6705e+00 k_t = [1, 1, 1, 1]
 Generation 5460: Best = 3.994726e+03 FES = 546100 μ_F = 0.852 μ_{CR} = 0.029
 Diversity = 1.5962e+00 k_t = [4, 3, 3, 4]
 Generation 5470: Best = 3.994715e+03 FES = 547100 μ_F = 0.827 μ_{CR} = 0.031
 Diversity = 1.5775e+00 k_t = [1, 1, 1, 1]
 Generation 5480: Best = 3.994692e+03 FES = 548100 μ_F = 0.856 μ_{CR} = 0.024
 Diversity = 1.6129e+00 k_t = [4, 2, 2, 4]
 Generation 5490: Best = 3.994692e+03 FES = 549100 μ_F = 0.869 μ_{CR} = 0.023
 Diversity = 1.5997e+00 k_t = [1, 1, 1, 1]
 Generation 5500: Best = 3.994689e+03 FES = 550100 μ_F = 0.869 μ_{CR} = 0.023
 Diversity = 1.5181e+00 k_t = [3, 2, 2, 4]
 Generation 5510: Best = 3.994625e+03 FES = 551100 μ_F = 0.866 μ_{CR} = 0.026
 Diversity = 1.5181e+00 k_t = [1, 1, 1, 1]
 Generation 5520: Best = 3.994594e+03 FES = 552100 μ_F = 0.869 μ_{CR} = 0.025
 Diversity = 1.5247e+00 k_t = [2, 4, 4, 2]
 Generation 5530: Best = 3.994594e+03 FES = 553100 μ_F = 0.868 μ_{CR} = 0.029
 Diversity = 1.5246e+00 k_t = [1, 1, 1, 1]
 Generation 5540: Best = 3.994577e+03 FES = 554100 μ_F = 0.864 μ_{CR} = 0.026
 Diversity = 1.5036e+00 k_t = [2, 2, 4, 4]
 Generation 5550: Best = 3.994577e+03 FES = 555100 μ_F = 0.855 μ_{CR} = 0.030
 Diversity = 1.4972e+00 k_t = [1, 1, 1, 1]
 Generation 5560: Best = 3.994577e+03 FES = 556100 μ_F = 0.849 μ_{CR} = 0.027
 Diversity = 1.4807e+00 k_t = [3, 4, 3, 2]
 Generation 5570: Best = 3.994577e+03 FES = 557100 μ_F = 0.841 μ_{CR} = 0.028
 Diversity = 1.4805e+00 k_t = [1, 1, 1, 1]
 Generation 5580: Best = 3.994577e+03 FES = 558100 μ_F = 0.826 μ_{CR} = 0.032
 Diversity = 1.4691e+00 k_t = [2, 3, 4, 4]
 Generation 5590: Best = 3.994577e+03 FES = 559100 μ_F = 0.866 μ_{CR} = 0.031
 Diversity = 1.4657e+00 k_t = [1, 1, 1, 1]
 Generation 5600: Best = 3.994571e+03 FES = 560100 μ_F = 0.868 μ_{CR} = 0.039
 Diversity = 1.4662e+00 k_t = [4, 4, 2, 3]
 Generation 5610: Best = 3.994570e+03 FES = 561100 μ_F = 0.850 μ_{CR} = 0.032
 Diversity = 1.4265e+00 k_t = [1, 1, 1, 1]
 Generation 5620: Best = 3.994570e+03 FES = 562100 μ_F = 0.856 μ_{CR} = 0.031
 Diversity = 1.3419e+00 k_t = [2, 2, 2, 4]

Generation 5630: Best = 3.994556e+03 FES = 563100 μ_F = 0.860 μ_{CR} = 0.033
 Diversity = 1.3388e+00 k_t = [1, 1, 1, 1]
 Generation 5640: Best = 3.994522e+03 FES = 564100 μ_F = 0.845 μ_{CR} = 0.029
 Diversity = 1.2889e+00 k_t = [2, 4, 2, 4]
 Generation 5650: Best = 3.994522e+03 FES = 565100 μ_F = 0.856 μ_{CR} = 0.027
 Diversity = 1.2900e+00 k_t = [1, 1, 1, 1]
 Generation 5660: Best = 3.994522e+03 FES = 566100 μ_F = 0.835 μ_{CR} = 0.039
 Diversity = 1.2946e+00 k_t = [4, 4, 4, 3]
 Generation 5670: Best = 3.994522e+03 FES = 567100 μ_F = 0.852 μ_{CR} = 0.034
 Diversity = 1.2925e+00 k_t = [1, 1, 1, 1]
 Generation 5680: Best = 3.994522e+03 FES = 568100 μ_F = 0.863 μ_{CR} = 0.034
 Diversity = 1.2926e+00 k_t = [2, 4, 4, 3]
 Generation 5690: Best = 3.994501e+03 FES = 569100 μ_F = 0.860 μ_{CR} = 0.037
 Diversity = 1.2195e+00 k_t = [1, 1, 1, 1]
 Generation 5700: Best = 3.994501e+03 FES = 570100 μ_F = 0.857 μ_{CR} = 0.028
 Diversity = 1.2196e+00 k_t = [2, 4, 2, 2]
 Generation 5710: Best = 3.994501e+03 FES = 571100 μ_F = 0.805 μ_{CR} = 0.030
 Diversity = 1.2171e+00 k_t = [1, 1, 1, 1]
 Generation 5720: Best = 3.994496e+03 FES = 572100 μ_F = 0.819 μ_{CR} = 0.026
 Diversity = 1.1227e+00 k_t = [3, 2, 4, 4]
 Generation 5730: Best = 3.994493e+03 FES = 573100 μ_F = 0.810 μ_{CR} = 0.027
 Diversity = 1.0942e+00 k_t = [1, 1, 1, 1]
 Generation 5740: Best = 3.994488e+03 FES = 574100 μ_F = 0.811 μ_{CR} = 0.030
 Diversity = 1.0983e+00 k_t = [4, 2, 2, 2]
 Generation 5750: Best = 3.994461e+03 FES = 575100 μ_F = 0.836 μ_{CR} = 0.029
 Diversity = 9.9204e-01 k_t = [1, 1, 1, 1]
 Generation 5760: Best = 3.994459e+03 FES = 576100 μ_F = 0.813 μ_{CR} = 0.026
 Diversity = 9.8482e-01 k_t = [2, 4, 3, 3]
 Generation 5770: Best = 3.994429e+03 FES = 577100 μ_F = 0.836 μ_{CR} = 0.027
 Diversity = 9.7829e-01 k_t = [1, 1, 1, 1]
 Generation 5780: Best = 3.994427e+03 FES = 578100 μ_F = 0.808 μ_{CR} = 0.031
 Diversity = 9.7176e-01 k_t = [2, 3, 4, 2]
 Generation 5790: Best = 3.994421e+03 FES = 579100 μ_F = 0.805 μ_{CR} = 0.030
 Diversity = 9.8170e-01 k_t = [1, 1, 1, 1]
 Generation 5800: Best = 3.994416e+03 FES = 580100 μ_F = 0.759 μ_{CR} = 0.028
 Diversity = 9.8187e-01 k_t = [4, 4, 2, 2]
 Generation 5810: Best = 3.994416e+03 FES = 581100 μ_F = 0.731 μ_{CR} = 0.029
 Diversity = 9.8645e-01 k_t = [1, 1, 1, 1]
 Generation 5820: Best = 3.994402e+03 FES = 582100 μ_F = 0.725 μ_{CR} = 0.029
 Diversity = 9.8357e-01 k_t = [2, 2, 3, 3]
 Generation 5830: Best = 3.994385e+03 FES = 583100 μ_F = 0.720 μ_{CR} = 0.030
 Diversity = 9.9279e-01 k_t = [1, 1, 1, 1]
 Generation 5840: Best = 3.994379e+03 FES = 584100 μ_F = 0.738 μ_{CR} = 0.025
 Diversity = 9.9030e-01 k_t = [2, 3, 2, 2]
 Generation 5850: Best = 3.994379e+03 FES = 585100 μ_F = 0.756 μ_{CR} = 0.021
 Diversity = 9.8438e-01 k_t = [1, 1, 1, 1]
 Generation 5860: Best = 3.994332e+03 FES = 586100 μ_F = 0.748 μ_{CR} = 0.032
 Diversity = 9.9214e-01 k_t = [4, 4, 4, 4]
 Generation 5870: Best = 3.994332e+03 FES = 587100 μ_F = 0.748 μ_{CR} = 0.028
 Diversity = 9.9195e-01 k_t = [1, 1, 1, 1]
 Generation 5880: Best = 3.994315e+03 FES = 588100 μ_F = 0.741 μ_{CR} = 0.036
 Diversity = 9.8806e-01 k_t = [4, 4, 2, 2]
 Generation 5890: Best = 3.994308e+03 FES = 589100 μ_F = 0.750 μ_{CR} = 0.033
 Diversity = 9.8481e-01 k_t = [1, 1, 1, 1]

Generation 5900: Best = 3.994299e+03 FES = 590100 μ_F = 0.788 μ_{CR} = 0.030
 Diversity = 9.8870e-01 k_t = [2, 2, 2, 4]
 Generation 5910: Best = 3.994261e+03 FES = 591100 μ_F = 0.819 μ_{CR} = 0.029
 Diversity = 9.9241e-01 k_t = [1, 1, 1, 1]
 Generation 5920: Best = 3.994261e+03 FES = 592100 μ_F = 0.828 μ_{CR} = 0.031
 Diversity = 9.9441e-01 k_t = [3, 2, 2, 2]
 Generation 5930: Best = 3.994261e+03 FES = 593100 μ_F = 0.840 μ_{CR} = 0.029
 Diversity = 9.8136e-01 k_t = [1, 1, 1, 1]
 Generation 5940: Best = 3.994261e+03 FES = 594100 μ_F = 0.826 μ_{CR} = 0.023
 Diversity = 9.7893e-01 k_t = [4, 3, 4, 3]
 Generation 5950: Best = 3.994259e+03 FES = 595100 μ_F = 0.782 μ_{CR} = 0.023
 Diversity = 9.7951e-01 k_t = [1, 1, 1, 1]
 Generation 5960: Best = 3.994259e+03 FES = 596100 μ_F = 0.813 μ_{CR} = 0.025
 Diversity = 9.8861e-01 k_t = [3, 2, 3, 4]
 Generation 5970: Best = 3.994255e+03 FES = 597100 μ_F = 0.847 μ_{CR} = 0.027
 Diversity = 9.7216e-01 k_t = [1, 1, 1, 1]
 Generation 5980: Best = 3.994255e+03 FES = 598100 μ_F = 0.779 μ_{CR} = 0.022
 Diversity = 9.7010e-01 k_t = [3, 3, 4, 3]
 Generation 5990: Best = 3.994255e+03 FES = 599100 μ_F = 0.790 μ_{CR} = 0.027
 Diversity = 9.6969e-01 k_t = [1, 1, 1, 1]
 Generation 6000: Best = 3.994255e+03 FES = 600100 μ_F = 0.794 μ_{CR} = 0.039
 Diversity = 9.6633e-01 k_t = [4, 3, 4, 3]
 Generation 6010: Best = 3.994254e+03 FES = 601100 μ_F = 0.749 μ_{CR} = 0.032
 Diversity = 9.5516e-01 k_t = [1, 1, 1, 1]
 Generation 6020: Best = 3.994254e+03 FES = 602100 μ_F = 0.749 μ_{CR} = 0.033
 Diversity = 9.5451e-01 k_t = [4, 4, 2, 4]
 Generation 6030: Best = 3.994253e+03 FES = 603100 μ_F = 0.737 μ_{CR} = 0.032
 Diversity = 9.5425e-01 k_t = [1, 1, 1, 1]
 Generation 6040: Best = 3.994253e+03 FES = 604100 μ_F = 0.749 μ_{CR} = 0.030
 Diversity = 9.5384e-01 k_t = [4, 3, 2, 2]
 Generation 6050: Best = 3.994253e+03 FES = 605100 μ_F = 0.782 μ_{CR} = 0.027
 Diversity = 9.5069e-01 k_t = [1, 1, 1, 1]
 Generation 6060: Best = 3.994201e+03 FES = 606100 μ_F = 0.780 μ_{CR} = 0.028
 Diversity = 9.5065e-01 k_t = [2, 3, 3, 2]
 Generation 6070: Best = 3.994201e+03 FES = 607100 μ_F = 0.790 μ_{CR} = 0.026
 Diversity = 9.5051e-01 k_t = [1, 1, 1, 1]
 Generation 6080: Best = 3.994197e+03 FES = 608100 μ_F = 0.813 μ_{CR} = 0.027
 Diversity = 9.4929e-01 k_t = [4, 2, 2, 2]
 Generation 6090: Best = 3.994197e+03 FES = 609100 μ_F = 0.808 μ_{CR} = 0.028
 Diversity = 9.4758e-01 k_t = [1, 1, 1, 1]
 Generation 6100: Best = 3.994197e+03 FES = 610100 μ_F = 0.824 μ_{CR} = 0.033
 Diversity = 9.4859e-01 k_t = [4, 2, 4, 3]
 Generation 6110: Best = 3.994197e+03 FES = 611100 μ_F = 0.826 μ_{CR} = 0.027
 Diversity = 9.3741e-01 k_t = [1, 1, 1, 1]
 Generation 6120: Best = 3.994197e+03 FES = 612100 μ_F = 0.780 μ_{CR} = 0.031
 Diversity = 9.3390e-01 k_t = [4, 2, 2, 4]
 Generation 6130: Best = 3.994193e+03 FES = 613100 μ_F = 0.773 μ_{CR} = 0.038
 Diversity = 9.2939e-01 k_t = [1, 1, 1, 1]
 Generation 6140: Best = 3.994145e+03 FES = 614100 μ_F = 0.698 μ_{CR} = 0.038
 Diversity = 9.2311e-01 k_t = [4, 3, 3, 3]
 Generation 6150: Best = 3.994145e+03 FES = 615100 μ_F = 0.689 μ_{CR} = 0.027
 Diversity = 9.2441e-01 k_t = [1, 1, 1, 1]
 Generation 6160: Best = 3.994138e+03 FES = 616100 μ_F = 0.742 μ_{CR} = 0.031
 Diversity = 9.3151e-01 k_t = [4, 3, 2, 4]

Generation 6170: Best = 3.994135e+03 FES = 617100 μ_F = 0.796 μ_{CR} = 0.028
 Diversity = 9.3365e-01 k_t = [1, 1, 1, 1]
 Generation 6180: Best = 3.994134e+03 FES = 618100 μ_F = 0.791 μ_{CR} = 0.023
 Diversity = 9.3352e-01 k_t = [2, 2, 3, 2]
 Generation 6190: Best = 3.994134e+03 FES = 619100 μ_F = 0.768 μ_{CR} = 0.028
 Diversity = 9.2922e-01 k_t = [1, 1, 1, 1]
 Generation 6200: Best = 3.994134e+03 FES = 620100 μ_F = 0.772 μ_{CR} = 0.024
 Diversity = 9.3428e-01 k_t = [2, 3, 3, 3]
 Generation 6210: Best = 3.994133e+03 FES = 621100 μ_F = 0.786 μ_{CR} = 0.027
 Diversity = 9.3434e-01 k_t = [1, 1, 1, 1]
 Generation 6220: Best = 3.994133e+03 FES = 622100 μ_F = 0.812 μ_{CR} = 0.035
 Diversity = 9.3185e-01 k_t = [4, 2, 4, 4]
 Generation 6230: Best = 3.994127e+03 FES = 623100 μ_F = 0.808 μ_{CR} = 0.024
 Diversity = 9.3276e-01 k_t = [1, 1, 1, 1]
 Generation 6240: Best = 3.994127e+03 FES = 624100 μ_F = 0.791 μ_{CR} = 0.029
 Diversity = 9.3035e-01 k_t = [2, 2, 3, 3]
 Generation 6250: Best = 3.994123e+03 FES = 625100 μ_F = 0.834 μ_{CR} = 0.030
 Diversity = 9.3154e-01 k_t = [1, 1, 1, 1]
 Generation 6260: Best = 3.994123e+03 FES = 626100 μ_F = 0.841 μ_{CR} = 0.029
 Diversity = 9.0843e-01 k_t = [3, 3, 4, 4]
 Generation 6270: Best = 3.994122e+03 FES = 627100 μ_F = 0.835 μ_{CR} = 0.024
 Diversity = 9.0826e-01 k_t = [1, 1, 1, 1]
 Generation 6280: Best = 3.994122e+03 FES = 628100 μ_F = 0.848 μ_{CR} = 0.026
 Diversity = 9.0838e-01 k_t = [2, 2, 3, 2]
 Generation 6290: Best = 3.994122e+03 FES = 629100 μ_F = 0.835 μ_{CR} = 0.030
 Diversity = 9.0824e-01 k_t = [1, 1, 1, 1]
 Generation 6300: Best = 3.994120e+03 FES = 630100 μ_F = 0.826 μ_{CR} = 0.024
 Diversity = 9.0381e-01 k_t = [3, 3, 2, 3]
 Generation 6310: Best = 3.994110e+03 FES = 631100 μ_F = 0.798 μ_{CR} = 0.027
 Diversity = 9.0346e-01 k_t = [1, 1, 1, 1]
 Generation 6320: Best = 3.994110e+03 FES = 632100 μ_F = 0.808 μ_{CR} = 0.030
 Diversity = 9.2074e-01 k_t = [2, 3, 4, 4]
 Generation 6330: Best = 3.994103e+03 FES = 633100 μ_F = 0.770 μ_{CR} = 0.029
 Diversity = 9.2068e-01 k_t = [1, 1, 1, 1]
 Generation 6340: Best = 3.994102e+03 FES = 634100 μ_F = 0.788 μ_{CR} = 0.026
 Diversity = 9.1645e-01 k_t = [4, 2, 2, 4]
 Generation 6350: Best = 3.994102e+03 FES = 635100 μ_F = 0.720 μ_{CR} = 0.032
 Diversity = 9.1656e-01 k_t = [1, 1, 1, 1]
 Generation 6360: Best = 3.994102e+03 FES = 636100 μ_F = 0.664 μ_{CR} = 0.028
 Diversity = 9.1467e-01 k_t = [3, 2, 2, 4]
 Generation 6370: Best = 3.994092e+03 FES = 637100 μ_F = 0.709 μ_{CR} = 0.027
 Diversity = 9.1333e-01 k_t = [1, 1, 1, 1]
 Generation 6380: Best = 3.994092e+03 FES = 638100 μ_F = 0.724 μ_{CR} = 0.031
 Diversity = 9.1257e-01 k_t = [2, 4, 2, 3]
 Generation 6390: Best = 3.994077e+03 FES = 639100 μ_F = 0.694 μ_{CR} = 0.032
 Diversity = 9.1228e-01 k_t = [1, 1, 1, 1]
 Generation 6400: Best = 3.994066e+03 FES = 640100 μ_F = 0.725 μ_{CR} = 0.032
 Diversity = 9.1082e-01 k_t = [4, 4, 3, 2]
 Generation 6410: Best = 3.994054e+03 FES = 641100 μ_F = 0.727 μ_{CR} = 0.027
 Diversity = 9.1241e-01 k_t = [1, 1, 1, 1]
 Generation 6420: Best = 3.994051e+03 FES = 642100 μ_F = 0.755 μ_{CR} = 0.031
 Diversity = 9.1156e-01 k_t = [4, 3, 2, 4]
 Generation 6430: Best = 3.994051e+03 FES = 643100 μ_F = 0.745 μ_{CR} = 0.032
 Diversity = 9.1154e-01 k_t = [1, 1, 1, 1]

Generation 6440: Best = 3.994051e+03 FES = 644100 μ_F = 0.728 μ_{CR} = 0.034
 Diversity = 9.1154e-01 k_t = [4, 4, 4, 2]
 Generation 6450: Best = 3.994042e+03 FES = 645100 μ_F = 0.747 μ_{CR} = 0.027
 Diversity = 9.1146e-01 k_t = [1, 1, 1, 1]
 Generation 6460: Best = 3.994042e+03 FES = 646100 μ_F = 0.776 μ_{CR} = 0.028
 Diversity = 9.0954e-01 k_t = [3, 3, 3, 3]
 Generation 6470: Best = 3.994042e+03 FES = 647100 μ_F = 0.792 μ_{CR} = 0.024
 Diversity = 9.0929e-01 k_t = [1, 1, 1, 1]
 Generation 6480: Best = 3.994042e+03 FES = 648100 μ_F = 0.754 μ_{CR} = 0.026
 Diversity = 9.0881e-01 k_t = [3, 4, 2, 3]
 Generation 6490: Best = 3.994042e+03 FES = 649100 μ_F = 0.770 μ_{CR} = 0.028
 Diversity = 9.1310e-01 k_t = [1, 1, 1, 1]
 Generation 6500: Best = 3.994042e+03 FES = 650100 μ_F = 0.786 μ_{CR} = 0.026
 Diversity = 9.1112e-01 k_t = [4, 4, 2, 3]
 Generation 6510: Best = 3.994042e+03 FES = 651100 μ_F = 0.783 μ_{CR} = 0.023
 Diversity = 9.1071e-01 k_t = [1, 1, 1, 1]
 Generation 6520: Best = 3.994042e+03 FES = 652100 μ_F = 0.818 μ_{CR} = 0.022
 Diversity = 9.0628e-01 k_t = [2, 3, 4, 4]
 Generation 6530: Best = 3.994042e+03 FES = 653100 μ_F = 0.846 μ_{CR} = 0.023
 Diversity = 9.0633e-01 k_t = [1, 1, 1, 1]
 Generation 6540: Best = 3.994042e+03 FES = 654100 μ_F = 0.838 μ_{CR} = 0.025
 Diversity = 9.0417e-01 k_t = [2, 4, 2, 2]
 Generation 6550: Best = 3.994036e+03 FES = 655100 μ_F = 0.855 μ_{CR} = 0.029
 Diversity = 9.0419e-01 k_t = [1, 1, 1, 1]
 Generation 6560: Best = 3.994036e+03 FES = 656100 μ_F = 0.860 μ_{CR} = 0.027
 Diversity = 9.0369e-01 k_t = [4, 4, 3, 2]
 Generation 6570: Best = 3.994036e+03 FES = 657100 μ_F = 0.852 μ_{CR} = 0.023
 Diversity = 9.0369e-01 k_t = [1, 1, 1, 1]
 Generation 6580: Best = 3.994036e+03 FES = 658100 μ_F = 0.847 μ_{CR} = 0.021
 Diversity = 8.9778e-01 k_t = [3, 3, 2, 2]
 Generation 6590: Best = 3.994036e+03 FES = 659100 μ_F = 0.805 μ_{CR} = 0.022
 Diversity = 8.9045e-01 k_t = [1, 1, 1, 1]
 Generation 6600: Best = 3.994036e+03 FES = 660100 μ_F = 0.767 μ_{CR} = 0.024
 Diversity = 8.9140e-01 k_t = [2, 3, 3, 2]
 Generation 6610: Best = 3.994034e+03 FES = 661100 μ_F = 0.756 μ_{CR} = 0.036
 Diversity = 8.9143e-01 k_t = [1, 1, 1, 1]
 Generation 6620: Best = 3.994034e+03 FES = 662100 μ_F = 0.685 μ_{CR} = 0.035
 Diversity = 8.9021e-01 k_t = [2, 2, 4, 4]
 Generation 6630: Best = 3.994034e+03 FES = 663100 μ_F = 0.680 μ_{CR} = 0.024
 Diversity = 8.9124e-01 k_t = [1, 1, 1, 1]
 Generation 6640: Best = 3.994025e+03 FES = 664100 μ_F = 0.720 μ_{CR} = 0.024
 Diversity = 8.9029e-01 k_t = [2, 2, 4, 3]
 Generation 6650: Best = 3.994012e+03 FES = 665100 μ_F = 0.772 μ_{CR} = 0.025
 Diversity = 8.8782e-01 k_t = [1, 1, 1, 1]
 Generation 6660: Best = 3.994012e+03 FES = 666100 μ_F = 0.793 μ_{CR} = 0.030
 Diversity = 8.9177e-01 k_t = [3, 4, 2, 4]
 Generation 6670: Best = 3.994012e+03 FES = 667100 μ_F = 0.825 μ_{CR} = 0.030
 Diversity = 8.8839e-01 k_t = [1, 1, 1, 1]
 Generation 6680: Best = 3.994008e+03 FES = 668100 μ_F = 0.789 μ_{CR} = 0.024
 Diversity = 8.8836e-01 k_t = [3, 3, 3, 2]
 Generation 6690: Best = 3.994008e+03 FES = 669100 μ_F = 0.819 μ_{CR} = 0.025
 Diversity = 8.8854e-01 k_t = [1, 1, 1, 1]
 Generation 6700: Best = 3.994008e+03 FES = 670100 μ_F = 0.824 μ_{CR} = 0.025
 Diversity = 8.8894e-01 k_t = [3, 4, 4, 3]

Generation 6710: Best = 3.994008e+03 FES = 671100 μ_F = 0.800 μ_{CR} = 0.026
 Diversity = 8.9113e-01 k_t = [1, 1, 1, 1]
 Generation 6720: Best = 3.994004e+03 FES = 672100 μ_F = 0.826 μ_{CR} = 0.028
 Diversity = 8.8961e-01 k_t = [3, 4, 2, 2]
 Generation 6730: Best = 3.994004e+03 FES = 673100 μ_F = 0.851 μ_{CR} = 0.025
 Diversity = 8.8947e-01 k_t = [1, 1, 1, 1]
 Generation 6740: Best = 3.994004e+03 FES = 674100 μ_F = 0.848 μ_{CR} = 0.028
 Diversity = 8.8868e-01 k_t = [2, 2, 3, 3]
 Generation 6750: Best = 3.994004e+03 FES = 675100 μ_F = 0.785 μ_{CR} = 0.020
 Diversity = 8.8856e-01 k_t = [1, 1, 1, 1]
 Generation 6760: Best = 3.994001e+03 FES = 676100 μ_F = 0.801 μ_{CR} = 0.018
 Diversity = 8.8400e-01 k_t = [3, 4, 3, 2]
 Generation 6770: Best = 3.994001e+03 FES = 677100 μ_F = 0.836 μ_{CR} = 0.020
 Diversity = 8.8376e-01 k_t = [1, 1, 1, 1]
 Generation 6780: Best = 3.994001e+03 FES = 678100 μ_F = 0.788 μ_{CR} = 0.026
 Diversity = 8.8079e-01 k_t = [2, 3, 4, 3]
 Generation 6790: Best = 3.994001e+03 FES = 679100 μ_F = 0.760 μ_{CR} = 0.030
 Diversity = 8.8338e-01 k_t = [1, 1, 1, 1]
 Generation 6800: Best = 3.993984e+03 FES = 680100 μ_F = 0.766 μ_{CR} = 0.027
 Diversity = 8.7142e-01 k_t = [3, 2, 3, 4]
 Generation 6810: Best = 3.993984e+03 FES = 681100 μ_F = 0.750 μ_{CR} = 0.031
 Diversity = 8.7080e-01 k_t = [1, 1, 1, 1]
 Generation 6820: Best = 3.993984e+03 FES = 682100 μ_F = 0.753 μ_{CR} = 0.027
 Diversity = 8.5362e-01 k_t = [4, 2, 3, 4]
 Generation 6830: Best = 3.993983e+03 FES = 683100 μ_F = 0.726 μ_{CR} = 0.026
 Diversity = 8.5937e-01 k_t = [1, 1, 1, 1]
 Generation 6840: Best = 3.993976e+03 FES = 684100 μ_F = 0.754 μ_{CR} = 0.026
 Diversity = 8.5849e-01 k_t = [4, 3, 2, 3]
 Generation 6850: Best = 3.993976e+03 FES = 685100 μ_F = 0.787 μ_{CR} = 0.022
 Diversity = 8.5917e-01 k_t = [1, 1, 1, 1]
 Generation 6860: Best = 3.993952e+03 FES = 686100 μ_F = 0.771 μ_{CR} = 0.025
 Diversity = 8.5910e-01 k_t = [3, 2, 3, 2]
 Generation 6870: Best = 3.993952e+03 FES = 687100 μ_F = 0.753 μ_{CR} = 0.029
 Diversity = 8.5906e-01 k_t = [1, 1, 1, 1]
 Generation 6880: Best = 3.993952e+03 FES = 688100 μ_F = 0.766 μ_{CR} = 0.029
 Diversity = 8.5924e-01 k_t = [3, 2, 4, 2]
 Generation 6890: Best = 3.993941e+03 FES = 689100 μ_F = 0.797 μ_{CR} = 0.027
 Diversity = 8.5910e-01 k_t = [1, 1, 1, 1]
 Generation 6900: Best = 3.993940e+03 FES = 690100 μ_F = 0.799 μ_{CR} = 0.030
 Diversity = 8.4908e-01 k_t = [3, 3, 2, 4]
 Generation 6910: Best = 3.993940e+03 FES = 691100 μ_F = 0.747 μ_{CR} = 0.028
 Diversity = 8.4905e-01 k_t = [1, 1, 1, 1]
 Generation 6920: Best = 3.993940e+03 FES = 692100 μ_F = 0.725 μ_{CR} = 0.034
 Diversity = 8.4936e-01 k_t = [2, 3, 2, 3]
 Generation 6930: Best = 3.993934e+03 FES = 693100 μ_F = 0.764 μ_{CR} = 0.027
 Diversity = 8.4905e-01 k_t = [1, 1, 1, 1]
 Generation 6940: Best = 3.993934e+03 FES = 694100 μ_F = 0.776 μ_{CR} = 0.024
 Diversity = 8.6104e-01 k_t = [3, 3, 3, 4]
 Generation 6950: Best = 3.993934e+03 FES = 695100 μ_F = 0.807 μ_{CR} = 0.027
 Diversity = 8.6117e-01 k_t = [1, 1, 1, 1]
 Generation 6960: Best = 3.993934e+03 FES = 696100 μ_F = 0.812 μ_{CR} = 0.031
 Diversity = 8.6462e-01 k_t = [3, 4, 4, 4]
 Generation 6970: Best = 3.993934e+03 FES = 697100 μ_F = 0.803 μ_{CR} = 0.026
 Diversity = 8.6457e-01 k_t = [1, 1, 1, 1]

Generation 6980: Best = 3.993922e+03 FES = 698100 μ_F = 0.828 μ_{CR} = 0.027
 Diversity = 8.7168e-01 k_t = [2, 3, 2, 4]
 Generation 6990: Best = 3.993922e+03 FES = 699100 μ_F = 0.847 μ_{CR} = 0.025
 Diversity = 8.7172e-01 k_t = [1, 1, 1, 1]
 Generation 7000: Best = 3.993921e+03 FES = 700100 μ_F = 0.818 μ_{CR} = 0.027
 Diversity = 8.7550e-01 k_t = [3, 4, 3, 3]
 Generation 7010: Best = 3.993916e+03 FES = 701100 μ_F = 0.768 μ_{CR} = 0.026
 Diversity = 8.7541e-01 k_t = [1, 1, 1, 1]
 Generation 7020: Best = 3.993916e+03 FES = 702100 μ_F = 0.761 μ_{CR} = 0.034
 Diversity = 8.7530e-01 k_t = [2, 4, 4, 2]
 Generation 7030: Best = 3.993916e+03 FES = 703100 μ_F = 0.713 μ_{CR} = 0.024
 Diversity = 8.7526e-01 k_t = [1, 1, 1, 1]
 Generation 7040: Best = 3.993916e+03 FES = 704100 μ_F = 0.665 μ_{CR} = 0.030
 Diversity = 8.6983e-01 k_t = [3, 2, 2, 4]
 Generation 7050: Best = 3.993916e+03 FES = 705100 μ_F = 0.635 μ_{CR} = 0.030
 Diversity = 8.6993e-01 k_t = [1, 1, 1, 1]
 Generation 7060: Best = 3.993915e+03 FES = 706100 μ_F = 0.647 μ_{CR} = 0.022
 Diversity = 8.6992e-01 k_t = [3, 4, 4, 3]
 Generation 7070: Best = 3.993915e+03 FES = 707100 μ_F = 0.612 μ_{CR} = 0.023
 Diversity = 8.6976e-01 k_t = [1, 1, 1, 1]
 Generation 7080: Best = 3.993915e+03 FES = 708100 μ_F = 0.680 μ_{CR} = 0.026
 Diversity = 8.6585e-01 k_t = [3, 2, 3, 4]
 Generation 7090: Best = 3.993915e+03 FES = 709100 μ_F = 0.610 μ_{CR} = 0.027
 Diversity = 8.6570e-01 k_t = [1, 1, 1, 1]
 Generation 7100: Best = 3.993905e+03 FES = 710100 μ_F = 0.681 μ_{CR} = 0.025
 Diversity = 8.6535e-01 k_t = [2, 3, 2, 3]
 Generation 7110: Best = 3.993905e+03 FES = 711100 μ_F = 0.684 μ_{CR} = 0.024
 Diversity = 8.6522e-01 k_t = [1, 1, 1, 1]
 Generation 7120: Best = 3.993904e+03 FES = 712100 μ_F = 0.692 μ_{CR} = 0.027
 Diversity = 8.4513e-01 k_t = [2, 3, 4, 2]
 Generation 7130: Best = 3.993904e+03 FES = 713100 μ_F = 0.729 μ_{CR} = 0.031
 Diversity = 8.4514e-01 k_t = [1, 1, 1, 1]
 Generation 7140: Best = 3.993904e+03 FES = 714100 μ_F = 0.752 μ_{CR} = 0.023
 Diversity = 8.4500e-01 k_t = [3, 2, 3, 3]
 Generation 7150: Best = 3.993891e+03 FES = 715100 μ_F = 0.733 μ_{CR} = 0.026
 Diversity = 8.4764e-01 k_t = [1, 1, 1, 1]
 Generation 7160: Best = 3.993889e+03 FES = 716100 μ_F = 0.702 μ_{CR} = 0.024
 Diversity = 8.4369e-01 k_t = [4, 3, 3, 4]
 Generation 7170: Best = 3.993889e+03 FES = 717100 μ_F = 0.744 μ_{CR} = 0.026
 Diversity = 8.4362e-01 k_t = [1, 1, 1, 1]
 Generation 7180: Best = 3.993889e+03 FES = 718100 μ_F = 0.749 μ_{CR} = 0.022
 Diversity = 8.4361e-01 k_t = [4, 4, 4, 3]
 Generation 7190: Best = 3.993884e+03 FES = 719100 μ_F = 0.703 μ_{CR} = 0.021
 Diversity = 8.4358e-01 k_t = [1, 1, 1, 1]
 Generation 7200: Best = 3.993883e+03 FES = 720100 μ_F = 0.693 μ_{CR} = 0.026
 Diversity = 8.4339e-01 k_t = [3, 4, 4, 3]
 Generation 7210: Best = 3.993883e+03 FES = 721100 μ_F = 0.636 μ_{CR} = 0.025
 Diversity = 8.4327e-01 k_t = [1, 1, 1, 1]
 Generation 7220: Best = 3.993870e+03 FES = 722100 μ_F = 0.689 μ_{CR} = 0.030
 Diversity = 8.5807e-01 k_t = [4, 2, 4, 4]
 Generation 7230: Best = 3.993868e+03 FES = 723100 μ_F = 0.733 μ_{CR} = 0.031
 Diversity = 8.5811e-01 k_t = [1, 1, 1, 1]
 Generation 7240: Best = 3.993856e+03 FES = 724100 μ_F = 0.799 μ_{CR} = 0.033
 Diversity = 8.5814e-01 k_t = [4, 2, 3, 2]

Generation 7250: Best = 3.993856e+03 FES = 725100 μ_F = 0.792 μ_{CR} = 0.033
 Diversity = 8.5833e-01 k_t = [1, 1, 1, 1]
 Generation 7260: Best = 3.993842e+03 FES = 726100 μ_F = 0.727 μ_{CR} = 0.029
 Diversity = 8.5790e-01 k_t = [3, 3, 2, 3]
 Generation 7270: Best = 3.993839e+03 FES = 727100 μ_F = 0.756 μ_{CR} = 0.025
 Diversity = 8.5795e-01 k_t = [1, 1, 1, 1]
 Generation 7280: Best = 3.993839e+03 FES = 728100 μ_F = 0.741 μ_{CR} = 0.026
 Diversity = 8.5799e-01 k_t = [2, 3, 3, 2]
 Generation 7290: Best = 3.993822e+03 FES = 729100 μ_F = 0.749 μ_{CR} = 0.025
 Diversity = 8.5800e-01 k_t = [1, 1, 1, 1]
 Generation 7300: Best = 3.993822e+03 FES = 730100 μ_F = 0.725 μ_{CR} = 0.028
 Diversity = 8.4754e-01 k_t = [4, 3, 4, 4]
 Generation 7310: Best = 3.993820e+03 FES = 731100 μ_F = 0.743 μ_{CR} = 0.030
 Diversity = 8.4713e-01 k_t = [1, 1, 1, 1]
 Generation 7320: Best = 3.993820e+03 FES = 732100 μ_F = 0.768 μ_{CR} = 0.025
 Diversity = 8.4707e-01 k_t = [4, 4, 3, 2]
 Generation 7330: Best = 3.993820e+03 FES = 733100 μ_F = 0.801 μ_{CR} = 0.026
 Diversity = 8.4710e-01 k_t = [1, 1, 1, 1]
 Generation 7340: Best = 3.993820e+03 FES = 734100 μ_F = 0.810 μ_{CR} = 0.024
 Diversity = 8.4962e-01 k_t = [4, 2, 2, 3]
 Generation 7350: Best = 3.993817e+03 FES = 735100 μ_F = 0.772 μ_{CR} = 0.027
 Diversity = 8.4952e-01 k_t = [1, 1, 1, 1]
 Generation 7360: Best = 3.993817e+03 FES = 736100 μ_F = 0.796 μ_{CR} = 0.025
 Diversity = 8.5222e-01 k_t = [2, 3, 3, 3]
 Generation 7370: Best = 3.993812e+03 FES = 737100 μ_F = 0.831 μ_{CR} = 0.024
 Diversity = 8.5228e-01 k_t = [1, 1, 1, 1]
 Generation 7380: Best = 3.993812e+03 FES = 738100 μ_F = 0.752 μ_{CR} = 0.032
 Diversity = 8.5221e-01 k_t = [3, 4, 3, 2]
 Generation 7390: Best = 3.993811e+03 FES = 739100 μ_F = 0.802 μ_{CR} = 0.031
 Diversity = 8.5194e-01 k_t = [1, 1, 1, 1]
 Generation 7400: Best = 3.993811e+03 FES = 740100 μ_F = 0.837 μ_{CR} = 0.026
 Diversity = 8.5199e-01 k_t = [2, 2, 2, 2]
 Generation 7410: Best = 3.993811e+03 FES = 741100 μ_F = 0.740 μ_{CR} = 0.019
 Diversity = 8.5202e-01 k_t = [1, 1, 1, 1]
 Generation 7420: Best = 3.993811e+03 FES = 742100 μ_F = 0.776 μ_{CR} = 0.024
 Diversity = 8.5058e-01 k_t = [3, 4, 4, 4]
 Generation 7430: Best = 3.993797e+03 FES = 743100 μ_F = 0.807 μ_{CR} = 0.027
 Diversity = 8.5047e-01 k_t = [1, 1, 1, 1]
 Generation 7440: Best = 3.993797e+03 FES = 744100 μ_F = 0.831 μ_{CR} = 0.028
 Diversity = 8.5055e-01 k_t = [2, 3, 3, 2]
 Generation 7450: Best = 3.993797e+03 FES = 745100 μ_F = 0.840 μ_{CR} = 0.026
 Diversity = 8.5075e-01 k_t = [1, 1, 1, 1]
 Generation 7460: Best = 3.993797e+03 FES = 746100 μ_F = 0.865 μ_{CR} = 0.024
 Diversity = 8.5083e-01 k_t = [2, 3, 3, 3]
 Generation 7470: Best = 3.993797e+03 FES = 747100 μ_F = 0.856 μ_{CR} = 0.026
 Diversity = 8.5080e-01 k_t = [1, 1, 1, 1]
 Generation 7480: Best = 3.993797e+03 FES = 748100 μ_F = 0.834 μ_{CR} = 0.026
 Diversity = 8.4147e-01 k_t = [2, 2, 2, 4]
 Generation 7490: Best = 3.993797e+03 FES = 749100 μ_F = 0.854 μ_{CR} = 0.028
 Diversity = 8.3819e-01 k_t = [1, 1, 1, 1]
 Generation 7500: Best = 3.993795e+03 FES = 750100 μ_F = 0.859 μ_{CR} = 0.029
 Diversity = 8.3545e-01 k_t = [2, 4, 2, 4]
 Generation 7510: Best = 3.993795e+03 FES = 751100 μ_F = 0.861 μ_{CR} = 0.022
 Diversity = 8.3553e-01 k_t = [1, 1, 1, 1]

Generation 7520: Best = 3.993795e+03 FES = 752100 μ_F = 0.827 μ_{CR} = 0.029
 Diversity = 8.3558e-01 k_t = [4, 3, 2, 3]
 Generation 7530: Best = 3.993795e+03 FES = 753100 μ_F = 0.806 μ_{CR} = 0.029
 Diversity = 8.3563e-01 k_t = [1, 1, 1, 1]
 Generation 7540: Best = 3.993795e+03 FES = 754100 μ_F = 0.805 μ_{CR} = 0.025
 Diversity = 8.3403e-01 k_t = [2, 2, 4, 4]
 Generation 7550: Best = 3.993783e+03 FES = 755100 μ_F = 0.814 μ_{CR} = 0.022
 Diversity = 8.3238e-01 k_t = [1, 1, 1, 1]
 Generation 7560: Best = 3.993783e+03 FES = 756100 μ_F = 0.818 μ_{CR} = 0.025
 Diversity = 8.3248e-01 k_t = [3, 3, 2, 2]
 Generation 7570: Best = 3.993783e+03 FES = 757100 μ_F = 0.830 μ_{CR} = 0.019
 Diversity = 8.3250e-01 k_t = [1, 1, 1, 1]
 Generation 7580: Best = 3.993783e+03 FES = 758100 μ_F = 0.836 μ_{CR} = 0.028
 Diversity = 8.3472e-01 k_t = [2, 4, 2, 3]
 Generation 7590: Best = 3.993783e+03 FES = 759100 μ_F = 0.834 μ_{CR} = 0.025
 Diversity = 8.3682e-01 k_t = [1, 1, 1, 1]
 Generation 7600: Best = 3.993783e+03 FES = 760100 μ_F = 0.842 μ_{CR} = 0.027
 Diversity = 8.3533e-01 k_t = [3, 3, 2, 2]
 Generation 7610: Best = 3.993774e+03 FES = 761100 μ_F = 0.792 μ_{CR} = 0.026
 Diversity = 8.3533e-01 k_t = [1, 1, 1, 1]
 Generation 7620: Best = 3.993755e+03 FES = 762100 μ_F = 0.811 μ_{CR} = 0.035
 Diversity = 8.3524e-01 k_t = [3, 3, 2, 2]
 Generation 7630: Best = 3.993744e+03 FES = 763100 μ_F = 0.802 μ_{CR} = 0.026
 Diversity = 8.3520e-01 k_t = [1, 1, 1, 1]
 Generation 7640: Best = 3.993744e+03 FES = 764100 μ_F = 0.771 μ_{CR} = 0.025
 Diversity = 8.3520e-01 k_t = [3, 4, 2, 3]
 Generation 7650: Best = 3.993744e+03 FES = 765100 μ_F = 0.801 μ_{CR} = 0.026
 Diversity = 8.3521e-01 k_t = [1, 1, 1, 1]
 Generation 7660: Best = 3.993744e+03 FES = 766100 μ_F = 0.806 μ_{CR} = 0.023
 Diversity = 8.3518e-01 k_t = [2, 2, 4, 3]
 Generation 7670: Best = 3.993744e+03 FES = 767100 μ_F = 0.752 μ_{CR} = 0.027
 Diversity = 8.3436e-01 k_t = [1, 1, 1, 1]
 Generation 7680: Best = 3.993744e+03 FES = 768100 μ_F = 0.760 μ_{CR} = 0.029
 Diversity = 8.3381e-01 k_t = [4, 2, 4, 3]
 Generation 7690: Best = 3.993744e+03 FES = 769100 μ_F = 0.750 μ_{CR} = 0.024
 Diversity = 8.3385e-01 k_t = [1, 1, 1, 1]
 Generation 7700: Best = 3.993744e+03 FES = 770100 μ_F = 0.761 μ_{CR} = 0.028
 Diversity = 8.3381e-01 k_t = [2, 2, 3, 2]
 Generation 7710: Best = 3.993741e+03 FES = 771100 μ_F = 0.818 μ_{CR} = 0.029
 Diversity = 8.3006e-01 k_t = [1, 1, 1, 1]
 Generation 7720: Best = 3.993719e+03 FES = 772100 μ_F = 0.828 μ_{CR} = 0.032
 Diversity = 8.3013e-01 k_t = [4, 4, 3, 3]
 Generation 7730: Best = 3.993718e+03 FES = 773100 μ_F = 0.821 μ_{CR} = 0.029
 Diversity = 8.3041e-01 k_t = [1, 1, 1, 1]
 Generation 7740: Best = 3.993714e+03 FES = 774100 μ_F = 0.804 μ_{CR} = 0.037
 Diversity = 8.3066e-01 k_t = [3, 3, 3, 2]
 Generation 7750: Best = 3.993713e+03 FES = 775100 μ_F = 0.791 μ_{CR} = 0.031
 Diversity = 8.3199e-01 k_t = [1, 1, 1, 1]
 Generation 7760: Best = 3.993713e+03 FES = 776100 μ_F = 0.826 μ_{CR} = 0.024
 Diversity = 8.1637e-01 k_t = [4, 2, 4, 4]
 Generation 7770: Best = 3.993712e+03 FES = 777100 μ_F = 0.834 μ_{CR} = 0.026
 Diversity = 8.1622e-01 k_t = [1, 1, 1, 1]
 Generation 7780: Best = 3.993712e+03 FES = 778100 μ_F = 0.825 μ_{CR} = 0.028
 Diversity = 8.1623e-01 k_t = [2, 3, 4, 3]

Generation 7790: Best = 3.993706e+03 FES = 779100 μ_F = 0.823 μ_{CR} = 0.025
 Diversity = 8.1631e-01 k_t = [1, 1, 1, 1]
 Generation 7800: Best = 3.993706e+03 FES = 780100 μ_F = 0.840 μ_{CR} = 0.023
 Diversity = 8.1624e-01 k_t = [3, 3, 3, 2]
 Generation 7810: Best = 3.993706e+03 FES = 781100 μ_F = 0.819 μ_{CR} = 0.028
 Diversity = 8.1632e-01 k_t = [1, 1, 1, 1]
 Generation 7820: Best = 3.993706e+03 FES = 782100 μ_F = 0.844 μ_{CR} = 0.026
 Diversity = 8.1202e-01 k_t = [4, 2, 3, 3]
 Generation 7830: Best = 3.993704e+03 FES = 783100 μ_F = 0.848 μ_{CR} = 0.023
 Diversity = 8.1210e-01 k_t = [1, 1, 1, 1]
 Generation 7840: Best = 3.993704e+03 FES = 784100 μ_F = 0.805 μ_{CR} = 0.031
 Diversity = 8.1215e-01 k_t = [3, 3, 3, 2]
 Generation 7850: Best = 3.993704e+03 FES = 785100 μ_F = 0.793 μ_{CR} = 0.020
 Diversity = 8.1227e-01 k_t = [1, 1, 1, 1]
 Generation 7860: Best = 3.993704e+03 FES = 786100 μ_F = 0.816 μ_{CR} = 0.021
 Diversity = 8.0506e-01 k_t = [2, 3, 4, 4]
 Generation 7870: Best = 3.993699e+03 FES = 787100 μ_F = 0.832 μ_{CR} = 0.024
 Diversity = 7.9825e-01 k_t = [1, 1, 1, 1]
 Generation 7880: Best = 3.993699e+03 FES = 788100 μ_F = 0.826 μ_{CR} = 0.027
 Diversity = 7.9826e-01 k_t = [4, 4, 2, 2]
 Generation 7890: Best = 3.993690e+03 FES = 789100 μ_F = 0.834 μ_{CR} = 0.032
 Diversity = 7.9827e-01 k_t = [1, 1, 1, 1]
 Generation 7900: Best = 3.993690e+03 FES = 790100 μ_F = 0.840 μ_{CR} = 0.028
 Diversity = 7.9823e-01 k_t = [3, 3, 2, 3]
 Generation 7910: Best = 3.993686e+03 FES = 791100 μ_F = 0.848 μ_{CR} = 0.023
 Diversity = 7.9841e-01 k_t = [1, 1, 1, 1]
 Generation 7920: Best = 3.993686e+03 FES = 792100 μ_F = 0.850 μ_{CR} = 0.022
 Diversity = 7.9843e-01 k_t = [2, 4, 4, 3]
 Generation 7930: Best = 3.993686e+03 FES = 793100 μ_F = 0.830 μ_{CR} = 0.022
 Diversity = 7.9841e-01 k_t = [1, 1, 1, 1]
 Generation 7940: Best = 3.993686e+03 FES = 794100 μ_F = 0.794 μ_{CR} = 0.030
 Diversity = 7.9838e-01 k_t = [3, 4, 3, 2]
 Generation 7950: Best = 3.993686e+03 FES = 795100 μ_F = 0.806 μ_{CR} = 0.026
 Diversity = 7.9851e-01 k_t = [1, 1, 1, 1]
 Generation 7960: Best = 3.993686e+03 FES = 796100 μ_F = 0.793 μ_{CR} = 0.023
 Diversity = 8.0003e-01 k_t = [4, 4, 4, 3]
 Generation 7970: Best = 3.993684e+03 FES = 797100 μ_F = 0.748 μ_{CR} = 0.027
 Diversity = 8.0009e-01 k_t = [1, 1, 1, 1]
 Generation 7980: Best = 3.993684e+03 FES = 798100 μ_F = 0.760 μ_{CR} = 0.027
 Diversity = 8.0011e-01 k_t = [2, 3, 3, 2]
 Generation 7990: Best = 3.993684e+03 FES = 799100 μ_F = 0.815 μ_{CR} = 0.030
 Diversity = 8.0022e-01 k_t = [1, 1, 1, 1]
 Generation 8000: Best = 3.993652e+03 FES = 800100 μ_F = 0.856 μ_{CR} = 0.027
 Diversity = 8.0016e-01 k_t = [2, 3, 4, 2]
 Generation 8010: Best = 3.993639e+03 FES = 801100 μ_F = 0.866 μ_{CR} = 0.028
 Diversity = 8.0021e-01 k_t = [1, 1, 1, 1]
 Generation 8020: Best = 3.993639e+03 FES = 802100 μ_F = 0.844 μ_{CR} = 0.035
 Diversity = 8.0083e-01 k_t = [4, 4, 2, 2]
 Generation 8030: Best = 3.993639e+03 FES = 803100 μ_F = 0.865 μ_{CR} = 0.032
 Diversity = 8.0085e-01 k_t = [1, 1, 1, 1]
 Generation 8040: Best = 3.993639e+03 FES = 804100 μ_F = 0.857 μ_{CR} = 0.035
 Diversity = 8.0094e-01 k_t = [3, 4, 4, 2]
 Generation 8050: Best = 3.993639e+03 FES = 805100 μ_F = 0.800 μ_{CR} = 0.027
 Diversity = 8.0095e-01 k_t = [1, 1, 1, 1]

Generation 8060: Best = 3.993639e+03 FES = 806100 μ_F = 0.779 μ_{CR} = 0.022
 Diversity = 7.9347e-01 k_t = [3, 4, 4, 4]
 Generation 8070: Best = 3.993638e+03 FES = 807100 μ_F = 0.801 μ_{CR} = 0.033
 Diversity = 7.9345e-01 k_t = [1, 1, 1, 1]
 Generation 8080: Best = 3.993638e+03 FES = 808100 μ_F = 0.745 μ_{CR} = 0.028
 Diversity = 7.9338e-01 k_t = [4, 3, 3, 2]
 Generation 8090: Best = 3.993638e+03 FES = 809100 μ_F = 0.773 μ_{CR} = 0.024
 Diversity = 7.8553e-01 k_t = [1, 1, 1, 1]
 Generation 8100: Best = 3.993638e+03 FES = 810100 μ_F = 0.758 μ_{CR} = 0.023
 Diversity = 7.8557e-01 k_t = [2, 3, 2, 2]
 Generation 8110: Best = 3.993632e+03 FES = 811100 μ_F = 0.758 μ_{CR} = 0.026
 Diversity = 7.8557e-01 k_t = [1, 1, 1, 1]
 Generation 8120: Best = 3.993630e+03 FES = 812100 μ_F = 0.790 μ_{CR} = 0.030
 Diversity = 7.8361e-01 k_t = [3, 3, 4, 4]
 Generation 8130: Best = 3.993630e+03 FES = 813100 μ_F = 0.811 μ_{CR} = 0.027
 Diversity = 7.8363e-01 k_t = [1, 1, 1, 1]
 Generation 8140: Best = 3.993630e+03 FES = 814100 μ_F = 0.844 μ_{CR} = 0.026
 Diversity = 7.9345e-01 k_t = [4, 3, 4, 4]
 Generation 8150: Best = 3.993630e+03 FES = 815100 μ_F = 0.815 μ_{CR} = 0.028
 Diversity = 7.9337e-01 k_t = [1, 1, 1, 1]
 Generation 8160: Best = 3.993629e+03 FES = 816100 μ_F = 0.839 μ_{CR} = 0.026
 Diversity = 7.9345e-01 k_t = [2, 3, 2, 2]
 Generation 8170: Best = 3.993624e+03 FES = 817100 μ_F = 0.808 μ_{CR} = 0.030
 Diversity = 7.9346e-01 k_t = [1, 1, 1, 1]
 Generation 8180: Best = 3.993618e+03 FES = 818100 μ_F = 0.807 μ_{CR} = 0.033
 Diversity = 7.9346e-01 k_t = [4, 4, 2, 3]
 Generation 8190: Best = 3.993616e+03 FES = 819100 μ_F = 0.792 μ_{CR} = 0.030
 Diversity = 7.9317e-01 k_t = [1, 1, 1, 1]
 Generation 8200: Best = 3.993616e+03 FES = 820100 μ_F = 0.796 μ_{CR} = 0.029
 Diversity = 7.8515e-01 k_t = [3, 2, 4, 3]
 Generation 8210: Best = 3.993608e+03 FES = 821100 μ_F = 0.802 μ_{CR} = 0.026
 Diversity = 7.8521e-01 k_t = [1, 1, 1, 1]
 Generation 8220: Best = 3.993607e+03 FES = 822100 μ_F = 0.777 μ_{CR} = 0.027
 Diversity = 7.8145e-01 k_t = [4, 4, 3, 4]
 Generation 8230: Best = 3.993595e+03 FES = 823100 μ_F = 0.786 μ_{CR} = 0.029
 Diversity = 7.8131e-01 k_t = [1, 1, 1, 1]
 Generation 8240: Best = 3.993595e+03 FES = 824100 μ_F = 0.787 μ_{CR} = 0.029
 Diversity = 7.8122e-01 k_t = [3, 3, 3, 3]
 Generation 8250: Best = 3.993595e+03 FES = 825100 μ_F = 0.791 μ_{CR} = 0.029
 Diversity = 7.8124e-01 k_t = [1, 1, 1, 1]
 Generation 8260: Best = 3.993595e+03 FES = 826100 μ_F = 0.819 μ_{CR} = 0.031
 Diversity = 7.8068e-01 k_t = [4, 3, 2, 4]
 Generation 8270: Best = 3.993592e+03 FES = 827100 μ_F = 0.876 μ_{CR} = 0.032
 Diversity = 7.8064e-01 k_t = [1, 1, 1, 1]
 Generation 8280: Best = 3.993582e+03 FES = 828100 μ_F = 0.861 μ_{CR} = 0.028
 Diversity = 7.8073e-01 k_t = [2, 2, 2, 2]
 Generation 8290: Best = 3.993580e+03 FES = 829100 μ_F = 0.869 μ_{CR} = 0.030
 Diversity = 7.8081e-01 k_t = [1, 1, 1, 1]
 Generation 8300: Best = 3.993576e+03 FES = 830100 μ_F = 0.852 μ_{CR} = 0.033
 Diversity = 7.8087e-01 k_t = [4, 2, 2, 3]
 Generation 8310: Best = 3.993576e+03 FES = 831100 μ_F = 0.856 μ_{CR} = 0.027
 Diversity = 7.8089e-01 k_t = [1, 1, 1, 1]
 Generation 8320: Best = 3.993576e+03 FES = 832100 μ_F = 0.817 μ_{CR} = 0.024
 Diversity = 7.8314e-01 k_t = [3, 3, 4, 4]

Generation 8330: Best = 3.993576e+03 FES = 833100 μ_F = 0.833 μ_{CR} = 0.021
 Diversity = 7.8317e-01 k_t = [1, 1, 1, 1]
 Generation 8340: Best = 3.993576e+03 FES = 834100 μ_F = 0.833 μ_{CR} = 0.026
 Diversity = 7.8153e-01 k_t = [4, 4, 4, 3]
 Generation 8350: Best = 3.993574e+03 FES = 835100 μ_F = 0.838 μ_{CR} = 0.023
 Diversity = 7.8153e-01 k_t = [1, 1, 1, 1]
 Generation 8360: Best = 3.993574e+03 FES = 836100 μ_F = 0.814 μ_{CR} = 0.022
 Diversity = 7.8146e-01 k_t = [2, 3, 4, 3]
 Generation 8370: Best = 3.993572e+03 FES = 837100 μ_F = 0.784 μ_{CR} = 0.024
 Diversity = 7.8145e-01 k_t = [1, 1, 1, 1]
 Generation 8380: Best = 3.993572e+03 FES = 838100 μ_F = 0.798 μ_{CR} = 0.029
 Diversity = 7.5283e-01 k_t = [3, 3, 4, 4]
 Generation 8390: Best = 3.993572e+03 FES = 839100 μ_F = 0.777 μ_{CR} = 0.023
 Diversity = 7.5283e-01 k_t = [1, 1, 1, 1]
 Generation 8400: Best = 3.993572e+03 FES = 840100 μ_F = 0.725 μ_{CR} = 0.025
 Diversity = 7.5286e-01 k_t = [3, 3, 2, 2]
 Generation 8410: Best = 3.993572e+03 FES = 841100 μ_F = 0.759 μ_{CR} = 0.026
 Diversity = 7.5236e-01 k_t = [1, 1, 1, 1]
 Generation 8420: Best = 3.993572e+03 FES = 842100 μ_F = 0.746 μ_{CR} = 0.034
 Diversity = 7.5236e-01 k_t = [2, 2, 3, 3]
 Generation 8430: Best = 3.993563e+03 FES = 843100 μ_F = 0.747 μ_{CR} = 0.029
 Diversity = 7.5030e-01 k_t = [1, 1, 1, 1]
 Generation 8440: Best = 3.993560e+03 FES = 844100 μ_F = 0.770 μ_{CR} = 0.029
 Diversity = 7.5030e-01 k_t = [2, 3, 2, 2]
 Generation 8450: Best = 3.993560e+03 FES = 845100 μ_F = 0.780 μ_{CR} = 0.035
 Diversity = 7.5022e-01 k_t = [1, 1, 1, 1]
 Generation 8460: Best = 3.993560e+03 FES = 846100 μ_F = 0.757 μ_{CR} = 0.037
 Diversity = 7.2769e-01 k_t = [4, 2, 4, 4]
 Generation 8470: Best = 3.993560e+03 FES = 847100 μ_F = 0.713 μ_{CR} = 0.031
 Diversity = 7.2779e-01 k_t = [1, 1, 1, 1]
 Generation 8480: Best = 3.993556e+03 FES = 848100 μ_F = 0.734 μ_{CR} = 0.033
 Diversity = 7.1170e-01 k_t = [4, 4, 3, 4]
 Generation 8490: Best = 3.993553e+03 FES = 849100 μ_F = 0.718 μ_{CR} = 0.031
 Diversity = 7.1172e-01 k_t = [1, 1, 1, 1]
 Generation 8500: Best = 3.993553e+03 FES = 850100 μ_F = 0.696 μ_{CR} = 0.031
 Diversity = 7.1165e-01 k_t = [2, 4, 2, 2]
 Generation 8510: Best = 3.993553e+03 FES = 851100 μ_F = 0.748 μ_{CR} = 0.022
 Diversity = 7.1154e-01 k_t = [1, 1, 1, 1]
 Generation 8520: Best = 3.993553e+03 FES = 852100 μ_F = 0.727 μ_{CR} = 0.023
 Diversity = 7.1149e-01 k_t = [3, 3, 3, 2]
 Generation 8530: Best = 3.993553e+03 FES = 853100 μ_F = 0.733 μ_{CR} = 0.021
 Diversity = 7.1148e-01 k_t = [1, 1, 1, 1]
 Generation 8540: Best = 3.993553e+03 FES = 854100 μ_F = 0.678 μ_{CR} = 0.024
 Diversity = 7.0778e-01 k_t = [4, 2, 3, 3]
 Generation 8550: Best = 3.993552e+03 FES = 855100 μ_F = 0.720 μ_{CR} = 0.027
 Diversity = 7.0706e-01 k_t = [1, 1, 1, 1]
 Generation 8560: Best = 3.993552e+03 FES = 856100 μ_F = 0.740 μ_{CR} = 0.031
 Diversity = 7.0621e-01 k_t = [4, 3, 3, 2]
 Generation 8570: Best = 3.993552e+03 FES = 857100 μ_F = 0.772 μ_{CR} = 0.033
 Diversity = 7.0636e-01 k_t = [1, 1, 1, 1]
 Generation 8580: Best = 3.993544e+03 FES = 858100 μ_F = 0.832 μ_{CR} = 0.032
 Diversity = 7.0648e-01 k_t = [4, 3, 2, 3]
 Generation 8590: Best = 3.993544e+03 FES = 859100 μ_F = 0.863 μ_{CR} = 0.029
 Diversity = 7.0645e-01 k_t = [1, 1, 1, 1]

Generation 8600: Best = 3.993541e+03 FES = 860100 μ_F = 0.833 μ_{CR} = 0.025
 Diversity = 7.0643e-01 k_t = [2, 4, 3, 2]
 Generation 8610: Best = 3.993540e+03 FES = 861100 μ_F = 0.850 μ_{CR} = 0.025
 Diversity = 6.9878e-01 k_t = [1, 1, 1, 1]
 Generation 8620: Best = 3.993540e+03 FES = 862100 μ_F = 0.841 μ_{CR} = 0.026
 Diversity = 6.8618e-01 k_t = [2, 2, 3, 4]
 Generation 8630: Best = 3.993532e+03 FES = 863100 μ_F = 0.877 μ_{CR} = 0.022
 Diversity = 6.8487e-01 k_t = [1, 1, 1, 1]
 Generation 8640: Best = 3.993532e+03 FES = 864100 μ_F = 0.827 μ_{CR} = 0.024
 Diversity = 6.8484e-01 k_t = [2, 4, 4, 2]
 Generation 8650: Best = 3.993532e+03 FES = 865100 μ_F = 0.837 μ_{CR} = 0.022
 Diversity = 6.8484e-01 k_t = [1, 1, 1, 1]
 Generation 8660: Best = 3.993532e+03 FES = 866100 μ_F = 0.820 μ_{CR} = 0.026
 Diversity = 6.6809e-01 k_t = [3, 3, 4, 4]
 Generation 8670: Best = 3.993531e+03 FES = 867100 μ_F = 0.786 μ_{CR} = 0.027
 Diversity = 6.6803e-01 k_t = [1, 1, 1, 1]
 Generation 8680: Best = 3.993531e+03 FES = 868100 μ_F = 0.789 μ_{CR} = 0.028
 Diversity = 6.4754e-01 k_t = [4, 2, 4, 4]
 Generation 8690: Best = 3.993531e+03 FES = 869100 μ_F = 0.779 μ_{CR} = 0.024
 Diversity = 6.4755e-01 k_t = [1, 1, 1, 1]
 Generation 8700: Best = 3.993531e+03 FES = 870100 μ_F = 0.826 μ_{CR} = 0.024
 Diversity = 6.0570e-01 k_t = [4, 3, 3, 4]
 Generation 8710: Best = 3.993528e+03 FES = 871100 μ_F = 0.799 μ_{CR} = 0.024
 Diversity = 5.7824e-01 k_t = [1, 1, 1, 1]
 Generation 8720: Best = 3.993525e+03 FES = 872100 μ_F = 0.831 μ_{CR} = 0.031
 Diversity = 5.8549e-01 k_t = [4, 3, 3, 3]
 Generation 8730: Best = 3.993525e+03 FES = 873100 μ_F = 0.856 μ_{CR} = 0.026
 Diversity = 5.8548e-01 k_t = [1, 1, 1, 1]
 Generation 8740: Best = 3.993525e+03 FES = 874100 μ_F = 0.849 μ_{CR} = 0.024
 Diversity = 5.8545e-01 k_t = [2, 2, 2, 3]
 Generation 8750: Best = 3.993525e+03 FES = 875100 μ_F = 0.795 μ_{CR} = 0.028
 Diversity = 5.8543e-01 k_t = [1, 1, 1, 1]
 Generation 8760: Best = 3.993525e+03 FES = 876100 μ_F = 0.811 μ_{CR} = 0.028
 Diversity = 5.7837e-01 k_t = [3, 2, 4, 4]
 Generation 8770: Best = 3.993525e+03 FES = 877100 μ_F = 0.829 μ_{CR} = 0.025
 Diversity = 5.8301e-01 k_t = [1, 1, 1, 1]
 Generation 8780: Best = 3.993524e+03 FES = 878100 μ_F = 0.780 μ_{CR} = 0.029
 Diversity = 5.5162e-01 k_t = [4, 4, 2, 4]
 Generation 8790: Best = 3.993524e+03 FES = 879100 μ_F = 0.824 μ_{CR} = 0.024
 Diversity = 5.5769e-01 k_t = [1, 1, 1, 1]
 Generation 8800: Best = 3.993524e+03 FES = 880100 μ_F = 0.841 μ_{CR} = 0.029
 Diversity = 5.5765e-01 k_t = [2, 3, 2, 3]
 Generation 8810: Best = 3.993521e+03 FES = 881100 μ_F = 0.783 μ_{CR} = 0.029
 Diversity = 5.5765e-01 k_t = [1, 1, 1, 1]
 Generation 8820: Best = 3.993521e+03 FES = 882100 μ_F = 0.792 μ_{CR} = 0.025
 Diversity = 5.4476e-01 k_t = [3, 2, 2, 4]
 Generation 8830: Best = 3.993520e+03 FES = 883100 μ_F = 0.780 μ_{CR} = 0.021
 Diversity = 5.4470e-01 k_t = [1, 1, 1, 1]
 Generation 8840: Best = 3.993520e+03 FES = 884100 μ_F = 0.774 μ_{CR} = 0.025
 Diversity = 5.4474e-01 k_t = [2, 3, 3, 2]
 Generation 8850: Best = 3.993520e+03 FES = 885100 μ_F = 0.780 μ_{CR} = 0.020
 Diversity = 5.4156e-01 k_t = [1, 1, 1, 1]
 Generation 8860: Best = 3.993520e+03 FES = 886100 μ_F = 0.798 μ_{CR} = 0.023
 Diversity = 5.4013e-01 k_t = [3, 4, 2, 4]

Generation 8870: Best = 3.993520e+03 FES = 887100 μ_F = 0.776 μ_{CR} = 0.026
 Diversity = 5.3679e-01 k_t = [1, 1, 1, 1]
 Generation 8880: Best = 3.993518e+03 FES = 888100 μ_F = 0.778 μ_{CR} = 0.025
 Diversity = 5.3697e-01 k_t = [4, 4, 4, 2]
 Generation 8890: Best = 3.993518e+03 FES = 889100 μ_F = 0.740 μ_{CR} = 0.030
 Diversity = 5.3698e-01 k_t = [1, 1, 1, 1]
 Generation 8900: Best = 3.993515e+03 FES = 890100 μ_F = 0.741 μ_{CR} = 0.031
 Diversity = 5.3692e-01 k_t = [4, 2, 4, 3]
 Generation 8910: Best = 3.993515e+03 FES = 891100 μ_F = 0.672 μ_{CR} = 0.034
 Diversity = 5.2573e-01 k_t = [1, 1, 1, 1]
 Generation 8920: Best = 3.993515e+03 FES = 892100 μ_F = 0.743 μ_{CR} = 0.034
 Diversity = 5.2571e-01 k_t = [2, 3, 3, 4]
 Generation 8930: Best = 3.993515e+03 FES = 893100 μ_F = 0.682 μ_{CR} = 0.031
 Diversity = 5.2570e-01 k_t = [1, 1, 1, 1]
 Generation 8940: Best = 3.993514e+03 FES = 894100 μ_F = 0.679 μ_{CR} = 0.044
 Diversity = 5.3301e-01 k_t = [4, 3, 4, 2]
 Generation 8950: Best = 3.993514e+03 FES = 895100 μ_F = 0.710 μ_{CR} = 0.030
 Diversity = 5.3297e-01 k_t = [1, 1, 1, 1]
 Generation 8960: Best = 3.993513e+03 FES = 896100 μ_F = 0.724 μ_{CR} = 0.024
 Diversity = 5.3304e-01 k_t = [3, 4, 2, 2]
 Generation 8970: Best = 3.993513e+03 FES = 897100 μ_F = 0.704 μ_{CR} = 0.025
 Diversity = 5.3296e-01 k_t = [1, 1, 1, 1]
 Generation 8980: Best = 3.993513e+03 FES = 898100 μ_F = 0.758 μ_{CR} = 0.026
 Diversity = 5.3300e-01 k_t = [3, 2, 2, 3]
 Generation 8990: Best = 3.993512e+03 FES = 899100 μ_F = 0.760 μ_{CR} = 0.026
 Diversity = 5.3560e-01 k_t = [1, 1, 1, 1]
 Generation 9000: Best = 3.993510e+03 FES = 900100 μ_F = 0.726 μ_{CR} = 0.035
 Diversity = 5.2384e-01 k_t = [2, 3, 2, 2]
 Generation 9010: Best = 3.993510e+03 FES = 901100 μ_F = 0.742 μ_{CR} = 0.032
 Diversity = 5.2376e-01 k_t = [1, 1, 1, 1]
 Generation 9020: Best = 3.993510e+03 FES = 902100 μ_F = 0.749 μ_{CR} = 0.028
 Diversity = 5.2374e-01 k_t = [3, 2, 2, 2]
 Generation 9030: Best = 3.993510e+03 FES = 903100 μ_F = 0.769 μ_{CR} = 0.028
 Diversity = 5.2371e-01 k_t = [1, 1, 1, 1]
 Generation 9040: Best = 3.993506e+03 FES = 904100 μ_F = 0.830 μ_{CR} = 0.032
 Diversity = 5.2362e-01 k_t = [4, 4, 3, 3]
 Generation 9050: Best = 3.993506e+03 FES = 905100 μ_F = 0.844 μ_{CR} = 0.027
 Diversity = 5.2357e-01 k_t = [1, 1, 1, 1]
 Generation 9060: Best = 3.993506e+03 FES = 906100 μ_F = 0.811 μ_{CR} = 0.026
 Diversity = 5.2350e-01 k_t = [4, 4, 4, 2]
 Generation 9070: Best = 3.993506e+03 FES = 907100 μ_F = 0.773 μ_{CR} = 0.023
 Diversity = 5.2351e-01 k_t = [1, 1, 1, 1]
 Generation 9080: Best = 3.993505e+03 FES = 908100 μ_F = 0.746 μ_{CR} = 0.026
 Diversity = 5.1931e-01 k_t = [4, 4, 2, 3]
 Generation 9090: Best = 3.993504e+03 FES = 909100 μ_F = 0.772 μ_{CR} = 0.028
 Diversity = 5.1937e-01 k_t = [1, 1, 1, 1]
 Generation 9100: Best = 3.993497e+03 FES = 910100 μ_F = 0.771 μ_{CR} = 0.027
 Diversity = 5.1938e-01 k_t = [2, 2, 2, 3]
 Generation 9110: Best = 3.993496e+03 FES = 911100 μ_F = 0.792 μ_{CR} = 0.031
 Diversity = 5.2597e-01 k_t = [1, 1, 1, 1]
 Generation 9120: Best = 3.993496e+03 FES = 912100 μ_F = 0.788 μ_{CR} = 0.029
 Diversity = 5.2597e-01 k_t = [3, 2, 2, 2]
 Generation 9130: Best = 3.993496e+03 FES = 913100 μ_F = 0.804 μ_{CR} = 0.026
 Diversity = 5.2257e-01 k_t = [1, 1, 1, 1]

Generation 9140: Best = 3.993496e+03 FES = 914100 μ_F = 0.831 μ_{CR} = 0.026
 Diversity = 5.1541e-01 k_t = [3, 4, 2, 4]
 Generation 9150: Best = 3.993496e+03 FES = 915100 μ_F = 0.843 μ_{CR} = 0.025
 Diversity = 5.1544e-01 k_t = [1, 1, 1, 1]
 Generation 9160: Best = 3.993496e+03 FES = 916100 μ_F = 0.776 μ_{CR} = 0.027
 Diversity = 5.1534e-01 k_t = [4, 4, 3, 2]
 Generation 9170: Best = 3.993496e+03 FES = 917100 μ_F = 0.748 μ_{CR} = 0.021
 Diversity = 5.1544e-01 k_t = [1, 1, 1, 1]
 Generation 9180: Best = 3.993492e+03 FES = 918100 μ_F = 0.760 μ_{CR} = 0.027
 Diversity = 5.1543e-01 k_t = [4, 2, 3, 3]
 Generation 9190: Best = 3.993492e+03 FES = 919100 μ_F = 0.782 μ_{CR} = 0.022
 Diversity = 5.2606e-01 k_t = [1, 1, 1, 1]
 Generation 9200: Best = 3.993492e+03 FES = 920100 μ_F = 0.758 μ_{CR} = 0.022
 Diversity = 5.2603e-01 k_t = [2, 2, 4, 2]
 Generation 9210: Best = 3.993491e+03 FES = 921100 μ_F = 0.773 μ_{CR} = 0.023
 Diversity = 5.2607e-01 k_t = [1, 1, 1, 1]
 Generation 9220: Best = 3.993491e+03 FES = 922100 μ_F = 0.778 μ_{CR} = 0.025
 Diversity = 5.2602e-01 k_t = [2, 2, 4, 3]
 Generation 9230: Best = 3.993491e+03 FES = 923100 μ_F = 0.768 μ_{CR} = 0.025
 Diversity = 5.2594e-01 k_t = [1, 1, 1, 1]
 Generation 9240: Best = 3.993489e+03 FES = 924100 μ_F = 0.790 μ_{CR} = 0.022
 Diversity = 5.0663e-01 k_t = [4, 3, 4, 4]
 Generation 9250: Best = 3.993489e+03 FES = 925100 μ_F = 0.860 μ_{CR} = 0.021
 Diversity = 5.0664e-01 k_t = [1, 1, 1, 1]
 Generation 9260: Best = 3.993489e+03 FES = 926100 μ_F = 0.852 μ_{CR} = 0.023
 Diversity = 5.0671e-01 k_t = [3, 4, 4, 2]
 Generation 9270: Best = 3.993488e+03 FES = 927100 μ_F = 0.832 μ_{CR} = 0.023
 Diversity = 5.0678e-01 k_t = [1, 1, 1, 1]
 Generation 9280: Best = 3.993488e+03 FES = 928100 μ_F = 0.811 μ_{CR} = 0.029
 Diversity = 4.7810e-01 k_t = [4, 4, 2, 3]
 Generation 9290: Best = 3.993488e+03 FES = 929100 μ_F = 0.845 μ_{CR} = 0.025
 Diversity = 4.7808e-01 k_t = [1, 1, 1, 1]
 Generation 9300: Best = 3.993486e+03 FES = 930100 μ_F = 0.778 μ_{CR} = 0.024
 Diversity = 4.7815e-01 k_t = [4, 2, 3, 2]
 Generation 9310: Best = 3.993486e+03 FES = 931100 μ_F = 0.761 μ_{CR} = 0.026
 Diversity = 4.8342e-01 k_t = [1, 1, 1, 1]
 Generation 9320: Best = 3.993486e+03 FES = 932100 μ_F = 0.710 μ_{CR} = 0.023
 Diversity = 4.8348e-01 k_t = [2, 3, 2, 2]
 Generation 9330: Best = 3.993486e+03 FES = 933100 μ_F = 0.731 μ_{CR} = 0.022
 Diversity = 4.8343e-01 k_t = [1, 1, 1, 1]
 Generation 9340: Best = 3.993486e+03 FES = 934100 μ_F = 0.752 μ_{CR} = 0.026
 Diversity = 4.6421e-01 k_t = [2, 3, 3, 4]
 Generation 9350: Best = 3.993483e+03 FES = 935100 μ_F = 0.742 μ_{CR} = 0.022
 Diversity = 4.6816e-01 k_t = [1, 1, 1, 1]
 Generation 9360: Best = 3.993483e+03 FES = 936100 μ_F = 0.753 μ_{CR} = 0.028
 Diversity = 4.3713e-01 k_t = [3, 4, 3, 3]
 Generation 9370: Best = 3.993483e+03 FES = 937100 μ_F = 0.747 μ_{CR} = 0.029
 Diversity = 4.4232e-01 k_t = [1, 1, 1, 1]
 Generation 9380: Best = 3.993483e+03 FES = 938100 μ_F = 0.704 μ_{CR} = 0.037
 Diversity = 4.5094e-01 k_t = [3, 4, 4, 2]
 Generation 9390: Best = 3.993480e+03 FES = 939100 μ_F = 0.712 μ_{CR} = 0.031
 Diversity = 4.5099e-01 k_t = [1, 1, 1, 1]
 Generation 9400: Best = 3.993479e+03 FES = 940100 μ_F = 0.730 μ_{CR} = 0.028
 Diversity = 4.5107e-01 k_t = [2, 2, 4, 2]

Generation 9410: Best = 3.993479e+03 FES = 941100 μ_F = 0.712 μ_{CR} = 0.028
 Diversity = 4.5792e-01 k_t = [1, 1, 1, 1]
 Generation 9420: Best = 3.993478e+03 FES = 942100 μ_F = 0.739 μ_{CR} = 0.026
 Diversity = 4.6752e-01 k_t = [2, 2, 3, 3]
 Generation 9430: Best = 3.993475e+03 FES = 943100 μ_F = 0.763 μ_{CR} = 0.027
 Diversity = 4.6792e-01 k_t = [1, 1, 1, 1]
 Generation 9440: Best = 3.993473e+03 FES = 944100 μ_F = 0.780 μ_{CR} = 0.028
 Diversity = 4.6818e-01 k_t = [3, 4, 2, 2]
 Generation 9450: Best = 3.993473e+03 FES = 945100 μ_F = 0.777 μ_{CR} = 0.023
 Diversity = 4.6815e-01 k_t = [1, 1, 1, 1]
 Generation 9460: Best = 3.993471e+03 FES = 946100 μ_F = 0.756 μ_{CR} = 0.037
 Diversity = 4.6826e-01 k_t = [4, 2, 2, 3]
 Generation 9470: Best = 3.993468e+03 FES = 947100 μ_F = 0.771 μ_{CR} = 0.035
 Diversity = 4.7358e-01 k_t = [1, 1, 1, 1]
 Generation 9480: Best = 3.993467e+03 FES = 948100 μ_F = 0.802 μ_{CR} = 0.029
 Diversity = 4.7956e-01 k_t = [4, 2, 2, 4]
 Generation 9490: Best = 3.993467e+03 FES = 949100 μ_F = 0.795 μ_{CR} = 0.032
 Diversity = 4.8432e-01 k_t = [1, 1, 1, 1]
 Generation 9500: Best = 3.993466e+03 FES = 950100 μ_F = 0.832 μ_{CR} = 0.031
 Diversity = 4.7059e-01 k_t = [3, 2, 3, 4]
 Generation 9510: Best = 3.993462e+03 FES = 951100 μ_F = 0.812 μ_{CR} = 0.030
 Diversity = 4.7127e-01 k_t = [1, 1, 1, 1]
 Generation 9520: Best = 3.993461e+03 FES = 952100 μ_F = 0.818 μ_{CR} = 0.040
 Diversity = 4.7021e-01 k_t = [2, 3, 2, 3]
 Generation 9530: Best = 3.993459e+03 FES = 953100 μ_F = 0.826 μ_{CR} = 0.041
 Diversity = 4.6980e-01 k_t = [1, 1, 1, 1]
 Generation 9540: Best = 3.993458e+03 FES = 954100 μ_F = 0.828 μ_{CR} = 0.048
 Diversity = 4.6636e-01 k_t = [2, 4, 2, 4]
 Generation 9550: Best = 3.993457e+03 FES = 955100 μ_F = 0.818 μ_{CR} = 0.041
 Diversity = 4.5723e-01 k_t = [1, 1, 1, 1]
 Generation 9560: Best = 3.993457e+03 FES = 956100 μ_F = 0.801 μ_{CR} = 0.038
 Diversity = 4.5746e-01 k_t = [2, 3, 2, 2]
 Generation 9570: Best = 3.993457e+03 FES = 957100 μ_F = 0.787 μ_{CR} = 0.035
 Diversity = 4.5255e-01 k_t = [1, 1, 1, 1]
 Generation 9580: Best = 3.993449e+03 FES = 958100 μ_F = 0.799 μ_{CR} = 0.037
 Diversity = 4.4315e-01 k_t = [3, 2, 3, 2]
 Generation 9590: Best = 3.993449e+03 FES = 959100 μ_F = 0.792 μ_{CR} = 0.035
 Diversity = 4.3182e-01 k_t = [1, 1, 1, 1]
 Generation 9600: Best = 3.993449e+03 FES = 960100 μ_F = 0.784 μ_{CR} = 0.041
 Diversity = 4.2325e-01 k_t = [2, 4, 2, 2]
 Generation 9610: Best = 3.993445e+03 FES = 961100 μ_F = 0.755 μ_{CR} = 0.040
 Diversity = 3.9670e-01 k_t = [1, 1, 1, 1]
 Generation 9620: Best = 3.993445e+03 FES = 962100 μ_F = 0.744 μ_{CR} = 0.034
 Diversity = 3.6953e-01 k_t = [2, 2, 2, 2]
 Generation 9630: Best = 3.993445e+03 FES = 963100 μ_F = 0.723 μ_{CR} = 0.045
 Diversity = 3.1580e-01 k_t = [1, 1, 1, 1]
 Generation 9640: Best = 3.993445e+03 FES = 964100 μ_F = 0.747 μ_{CR} = 0.049
 Diversity = 2.9244e-01 k_t = [3, 3, 4, 4]
 Generation 9650: Best = 3.993441e+03 FES = 965100 μ_F = 0.761 μ_{CR} = 0.056
 Diversity = 2.6658e-01 k_t = [1, 1, 1, 1]
 Generation 9660: Best = 3.993441e+03 FES = 966100 μ_F = 0.803 μ_{CR} = 0.048
 Diversity = 2.5191e-01 k_t = [2, 4, 3, 2]
 Generation 9670: Best = 3.993441e+03 FES = 967100 μ_F = 0.813 μ_{CR} = 0.052
 Diversity = 2.3615e-01 k_t = [1, 1, 1, 1]

Generation 9680: Best = 3.993441e+03 FES = 968100 μ_F = 0.778 μ_{CR} = 0.050
 Diversity = 2.1769e-01 k_t = [3, 2, 2, 4]
 Generation 9690: Best = 3.993441e+03 FES = 969100 μ_F = 0.813 μ_{CR} = 0.044
 Diversity = 1.4926e-01 k_t = [1, 1, 1, 1]
 Generation 9700: Best = 3.993441e+03 FES = 970100 μ_F = 0.850 μ_{CR} = 0.035
 Diversity = 1.1433e-01 k_t = [4, 2, 3, 2]
 Generation 9710: Best = 3.993440e+03 FES = 971100 μ_F = 0.856 μ_{CR} = 0.031
 Diversity = 1.1442e-01 k_t = [1, 1, 1, 1]
 Generation 9720: Best = 3.993440e+03 FES = 972100 μ_F = 0.842 μ_{CR} = 0.035
 Diversity = 1.1444e-01 k_t = [3, 3, 3, 2]
 Generation 9730: Best = 3.993440e+03 FES = 973100 μ_F = 0.800 μ_{CR} = 0.041
 Diversity = 5.4774e-02 k_t = [1, 1, 1, 1]
 Generation 9740: Best = 3.993436e+03 FES = 974100 μ_F = 0.749 μ_{CR} = 0.035
 Diversity = 4.8860e-02 k_t = [2, 3, 3, 3]
 Generation 9750: Best = 3.993433e+03 FES = 975100 μ_F = 0.701 μ_{CR} = 0.040
 Diversity = 4.4474e-02 k_t = [1, 1, 1, 1]
 Generation 9760: Best = 3.993432e+03 FES = 976100 μ_F = 0.709 μ_{CR} = 0.035
 Diversity = 4.4604e-02 k_t = [2, 3, 4, 2]
 Generation 9770: Best = 3.993431e+03 FES = 977100 μ_F = 0.690 μ_{CR} = 0.039
 Diversity = 4.2852e-02 k_t = [1, 1, 1, 1]
 Generation 9780: Best = 3.993427e+03 FES = 978100 μ_F = 0.656 μ_{CR} = 0.039
 Diversity = 4.1128e-02 k_t = [3, 3, 3, 4]
 Generation 9790: Best = 3.993422e+03 FES = 979100 μ_F = 0.667 μ_{CR} = 0.046
 Diversity = 3.6935e-02 k_t = [1, 1, 1, 1]
 Generation 9800: Best = 3.993422e+03 FES = 980100 μ_F = 0.669 μ_{CR} = 0.032
 Diversity = 3.2817e-02 k_t = [3, 4, 3, 2]
 Generation 9810: Best = 3.993418e+03 FES = 981100 μ_F = 0.686 μ_{CR} = 0.026
 Diversity = 3.4720e-02 k_t = [1, 1, 1, 1]
 Generation 9820: Best = 3.993418e+03 FES = 982100 μ_F = 0.681 μ_{CR} = 0.034
 Diversity = 3.1663e-02 k_t = [2, 3, 2, 2]
 Generation 9830: Best = 3.993414e+03 FES = 983100 μ_F = 0.700 μ_{CR} = 0.038
 Diversity = 3.5394e-02 k_t = [1, 1, 1, 1]
 Generation 9840: Best = 3.993413e+03 FES = 984100 μ_F = 0.712 μ_{CR} = 0.034
 Diversity = 3.4436e-02 k_t = [3, 3, 4, 4]
 Generation 9850: Best = 3.993410e+03 FES = 985100 μ_F = 0.714 μ_{CR} = 0.028
 Diversity = 3.3602e-02 k_t = [1, 1, 1, 1]
 Generation 9860: Best = 3.993410e+03 FES = 986100 μ_F = 0.743 μ_{CR} = 0.040
 Diversity = 3.1388e-02 k_t = [2, 3, 4, 4]
 Generation 9870: Best = 3.993408e+03 FES = 987100 μ_F = 0.742 μ_{CR} = 0.034
 Diversity = 3.1303e-02 k_t = [1, 1, 1, 1]
 Generation 9880: Best = 3.993407e+03 FES = 988100 μ_F = 0.737 μ_{CR} = 0.032
 Diversity = 3.0402e-02 k_t = [2, 2, 3, 2]
 Generation 9890: Best = 3.993404e+03 FES = 989100 μ_F = 0.737 μ_{CR} = 0.034
 Diversity = 3.4060e-02 k_t = [1, 1, 1, 1]
 Generation 9900: Best = 3.993403e+03 FES = 990100 μ_F = 0.732 μ_{CR} = 0.039
 Diversity = 3.3049e-02 k_t = [4, 3, 2, 3]
 Generation 9910: Best = 3.993403e+03 FES = 991100 μ_F = 0.714 μ_{CR} = 0.036
 Diversity = 2.9604e-02 k_t = [1, 1, 1, 1]
 Generation 9920: Best = 3.993399e+03 FES = 992100 μ_F = 0.702 μ_{CR} = 0.033
 Diversity = 2.6874e-02 k_t = [2, 3, 3, 3]
 Generation 9930: Best = 3.993399e+03 FES = 993100 μ_F = 0.687 μ_{CR} = 0.026
 Diversity = 2.4852e-02 k_t = [1, 1, 1, 1]
 Generation 9940: Best = 3.993399e+03 FES = 994100 μ_F = 0.669 μ_{CR} = 0.026
 Diversity = 2.3600e-02 k_t = [2, 2, 4, 2]

Generation 9950: Best = 3.993399e+03 FES = 995100 μ_F = 0.665 μ_{CR} = 0.029
 Diversity = 2.0746e-02 k_t = [1, 1, 1, 1]
 Generation 9960: Best = 3.993398e+03 FES = 996100 μ_F = 0.682 μ_{CR} = 0.030
 Diversity = 2.1303e-02 k_t = [2, 4, 2, 2]
 Generation 9970: Best = 3.993397e+03 FES = 997100 μ_F = 0.630 μ_{CR} = 0.037
 Diversity = 1.8949e-02 k_t = [1, 1, 1, 1]
 Generation 9980: Best = 3.993397e+03 FES = 998100 μ_F = 0.655 μ_{CR} = 0.034
 Diversity = 1.8451e-02 k_t = [3, 3, 3, 2]
 Generation 9990: Best = 3.993395e+03 FES = 999100 μ_F = 0.666 μ_{CR} = 0.037
 Diversity = 2.0251e-02 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 3.993394e+03

Total Function Evals = 1000000

Total Successful Updates = 94985

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Results:

Best Fitness : 3.993394e+03

Global Optimum : 1.600000e+03

Error (|best-optimum|) : 2.393394e+03

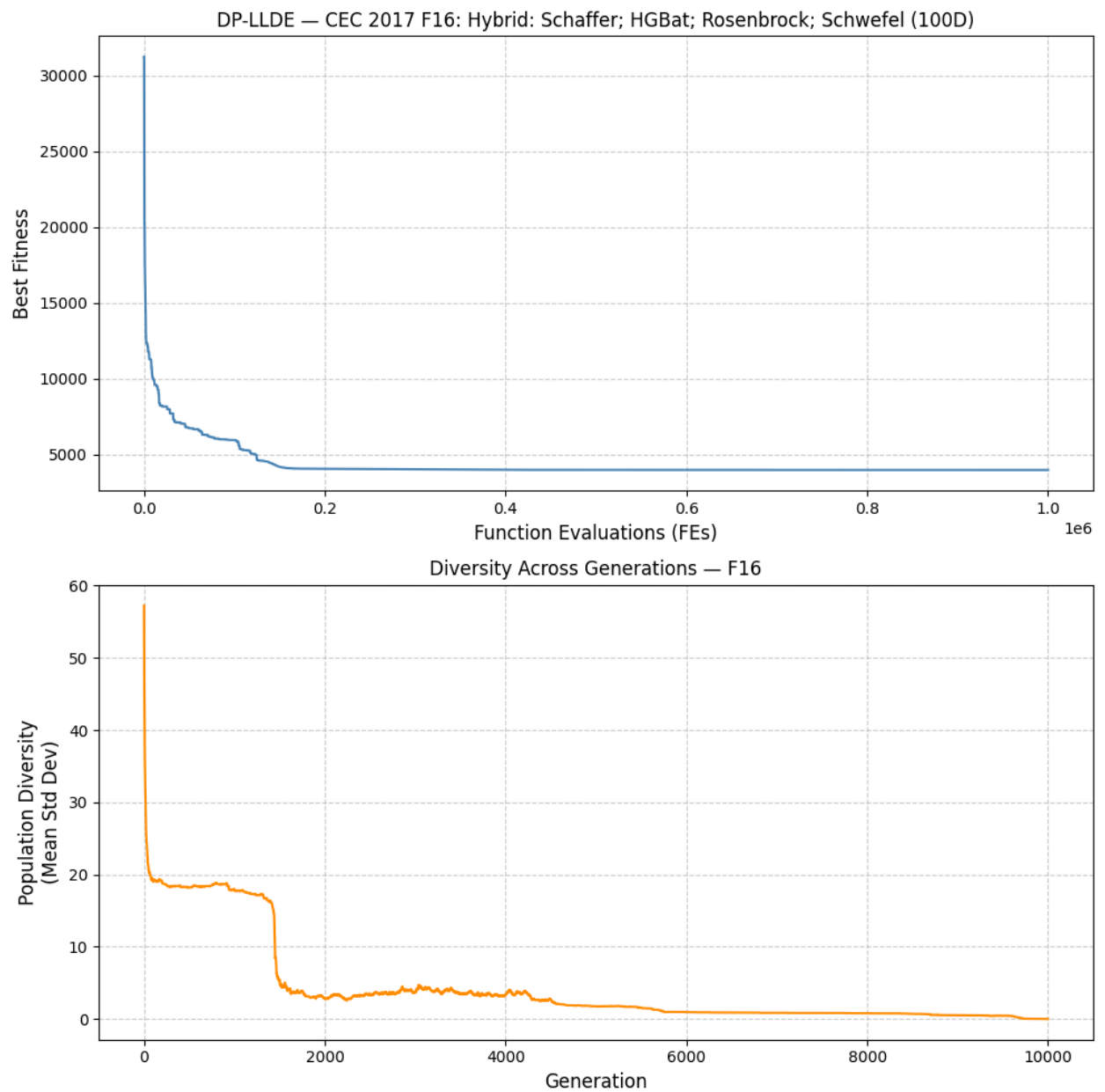
Best Solution (5 dim) : [17.08199352 73.83984366 -14.6605705 -49.96920289
84.92609794]

D^(0) (init. diversity): 5.7220e+01

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✓ Checkpoint saved to: /content/drive/My Drive/thesis_results/all_results_checkpoint.pkl



Graph saved as convergence_F16.png

✓ Plot saved to: /content/drive/My Drive/thesis_results/plots/convergence_F16.png

```
In [ ]: # =====
# F17 — Hybrid Function 7 (Katsuura; Ackley; Griewank+Rosenbrock; Schwefel; Ra
# =====
run_and_plot('F17', make_wrapper(hybrid.f17), 1700.0, 'Hybrid: Katsuura; Ackle
```

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DP-LLDE on CEC 2017 F17 – Single Run

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Function : Hybrid: Katsuura; Ackley; Griewank+Rosenbrock; Schwefel; Rastrigin (100D)

Bounds : [-100, 100]

Global optimum : 1700.0

Initial $D^*(0) = 5.7220e+01$ | Diversity threshold = 5.7220e+00

Generation 0: Best Fitness = 3.179716e+07

Generation 10: Best = 3.232794e+05	FES = 1100	$\mu_F = 0.568$	$\mu_{CR} = 0.345$
Diversity = 3.7851e+01	$k_t = [1, 1, 1, 1]$		
Generation 20: Best = 1.907743e+04	FES = 2100	$\mu_F = 0.610$	$\mu_{CR} = 0.339$
Diversity = 2.4951e+01	$k_t = [2, 3, 2, 3]$		
Generation 30: Best = 9.760879e+03	FES = 3100	$\mu_F = 0.641$	$\mu_{CR} = 0.332$
Diversity = 2.1131e+01	$k_t = [1, 1, 1, 1]$		
Generation 40: Best = 8.242354e+03	FES = 4100	$\mu_F = 0.662$	$\mu_{CR} = 0.303$
Diversity = 1.9293e+01	$k_t = [3, 2, 3, 3]$		
Generation 50: Best = 8.217751e+03	FES = 5100	$\mu_F = 0.679$	$\mu_{CR} = 0.267$
Diversity = 1.9002e+01	$k_t = [1, 1, 1, 1]$		
Generation 60: Best = 7.864680e+03	FES = 6100	$\mu_F = 0.646$	$\mu_{CR} = 0.220$
Diversity = 1.8445e+01	$k_t = [2, 3, 3, 2]$		
Generation 70: Best = 7.836707e+03	FES = 7100	$\mu_F = 0.664$	$\mu_{CR} = 0.171$
Diversity = 1.8143e+01	$k_t = [1, 1, 1, 1]$		
Generation 80: Best = 7.249157e+03	FES = 8100	$\mu_F = 0.685$	$\mu_{CR} = 0.132$
Diversity = 1.7688e+01	$k_t = [3, 4, 3, 3]$		
Generation 90: Best = 7.206076e+03	FES = 9100	$\mu_F = 0.707$	$\mu_{CR} = 0.093$
Diversity = 1.7840e+01	$k_t = [1, 1, 1, 1]$		
Generation 100: Best = 7.160374e+03	FES = 10100	$\mu_F = 0.749$	$\mu_{CR} = 0.072$
Diversity = 1.7828e+01	$k_t = [2, 3, 4, 2]$		
Generation 110: Best = 7.031274e+03	FES = 11100	$\mu_F = 0.743$	$\mu_{CR} = 0.066$
Diversity = 1.7796e+01	$k_t = [1, 1, 1, 1]$		
Generation 120: Best = 7.031274e+03	FES = 12100	$\mu_F = 0.784$	$\mu_{CR} = 0.057$
Diversity = 1.8063e+01	$k_t = [2, 4, 2, 2]$		
Generation 130: Best = 7.026904e+03	FES = 13100	$\mu_F = 0.728$	$\mu_{CR} = 0.050$
Diversity = 1.8084e+01	$k_t = [1, 1, 1, 1]$		
Generation 140: Best = 6.544950e+03	FES = 14100	$\mu_F = 0.749$	$\mu_{CR} = 0.058$
Diversity = 1.7947e+01	$k_t = [4, 4, 2, 4]$		
Generation 150: Best = 6.522048e+03	FES = 15100	$\mu_F = 0.768$	$\mu_{CR} = 0.058$
Diversity = 1.7908e+01	$k_t = [1, 1, 1, 1]$		
Generation 160: Best = 6.492096e+03	FES = 16100	$\mu_F = 0.776$	$\mu_{CR} = 0.060$
Diversity = 1.7816e+01	$k_t = [4, 4, 2, 4]$		
Generation 170: Best = 6.446644e+03	FES = 17100	$\mu_F = 0.752$	$\mu_{CR} = 0.061$
Diversity = 1.7780e+01	$k_t = [1, 1, 1, 1]$		
Generation 180: Best = 6.446644e+03	FES = 18100	$\mu_F = 0.776$	$\mu_{CR} = 0.054$
Diversity = 1.7770e+01	$k_t = [4, 3, 3, 2]$		
Generation 190: Best = 6.446644e+03	FES = 19100	$\mu_F = 0.765$	$\mu_{CR} = 0.041$
Diversity = 1.7676e+01	$k_t = [1, 1, 1, 1]$		
Generation 200: Best = 6.423839e+03	FES = 20100	$\mu_F = 0.798$	$\mu_{CR} = 0.046$
Diversity = 1.7588e+01	$k_t = [2, 3, 2, 3]$		
Generation 210: Best = 6.423839e+03	FES = 21100	$\mu_F = 0.733$	$\mu_{CR} = 0.046$
Diversity = 1.7488e+01	$k_t = [1, 1, 1, 1]$		
Generation 220: Best = 6.294276e+03	FES = 22100	$\mu_F = 0.701$	$\mu_{CR} = 0.045$

Diversity = 1.7426e+01	k_t = [4, 2, 2, 3]				
Generation 230: Best = 6.294276e+03	FES = 23100	$\mu_F = 0.703$	$\mu_{CR} = 0.047$		
Diversity = 1.7362e+01	k_t = [1, 1, 1, 1]				
Generation 240: Best = 6.294276e+03	FES = 24100	$\mu_F = 0.698$	$\mu_{CR} = 0.039$		
Diversity = 1.7448e+01	k_t = [4, 4, 4, 3]				
Generation 250: Best = 6.282099e+03	FES = 25100	$\mu_F = 0.706$	$\mu_{CR} = 0.041$		
Diversity = 1.7609e+01	k_t = [1, 1, 1, 1]				
Generation 260: Best = 6.231967e+03	FES = 26100	$\mu_F = 0.721$	$\mu_{CR} = 0.040$		
Diversity = 1.7589e+01	k_t = [2, 3, 4, 2]				
Generation 270: Best = 6.208637e+03	FES = 27100	$\mu_F = 0.744$	$\mu_{CR} = 0.046$		
Diversity = 1.7705e+01	k_t = [1, 1, 1, 1]				
Generation 280: Best = 6.208637e+03	FES = 28100	$\mu_F = 0.767$	$\mu_{CR} = 0.046$		
Diversity = 1.7584e+01	k_t = [4, 2, 4, 4]				
Generation 290: Best = 6.118605e+03	FES = 29100	$\mu_F = 0.749$	$\mu_{CR} = 0.047$		
Diversity = 1.7673e+01	k_t = [1, 1, 1, 1]				
Generation 300: Best = 5.817971e+03	FES = 30100	$\mu_F = 0.783$	$\mu_{CR} = 0.053$		
Diversity = 1.7481e+01	k_t = [2, 3, 3, 4]				
Generation 310: Best = 5.817971e+03	FES = 31100	$\mu_F = 0.792$	$\mu_{CR} = 0.040$		
Diversity = 1.7494e+01	k_t = [1, 1, 1, 1]				
Generation 320: Best = 5.708423e+03	FES = 32100	$\mu_F = 0.802$	$\mu_{CR} = 0.039$		
Diversity = 1.7442e+01	k_t = [3, 4, 3, 2]				
Generation 330: Best = 5.708423e+03	FES = 33100	$\mu_F = 0.790$	$\mu_{CR} = 0.041$		
Diversity = 1.7435e+01	k_t = [1, 1, 1, 1]				
Generation 340: Best = 5.681092e+03	FES = 34100	$\mu_F = 0.798$	$\mu_{CR} = 0.039$		
Diversity = 1.7368e+01	k_t = [3, 3, 2, 4]				
Generation 350: Best = 5.663236e+03	FES = 35100	$\mu_F = 0.728$	$\mu_{CR} = 0.038$		
Diversity = 1.7369e+01	k_t = [1, 1, 1, 1]				
Generation 360: Best = 5.574233e+03	FES = 36100	$\mu_F = 0.740$	$\mu_{CR} = 0.043$		
Diversity = 1.7279e+01	k_t = [4, 2, 4, 3]				
Generation 370: Best = 5.574233e+03	FES = 37100	$\mu_F = 0.731$	$\mu_{CR} = 0.040$		
Diversity = 1.7202e+01	k_t = [1, 1, 1, 1]				
Generation 380: Best = 5.531409e+03	FES = 38100	$\mu_F = 0.736$	$\mu_{CR} = 0.037$		
Diversity = 1.7147e+01	k_t = [4, 4, 2, 3]				
Generation 390: Best = 5.531409e+03	FES = 39100	$\mu_F = 0.740$	$\mu_{CR} = 0.042$		
Diversity = 1.7160e+01	k_t = [1, 1, 1, 1]				
Generation 400: Best = 5.527888e+03	FES = 40100	$\mu_F = 0.672$	$\mu_{CR} = 0.041$		
Diversity = 1.7134e+01	k_t = [4, 4, 3, 4]				
Generation 410: Best = 5.527888e+03	FES = 41100	$\mu_F = 0.675$	$\mu_{CR} = 0.035$		
Diversity = 1.7070e+01	k_t = [1, 1, 1, 1]				
Generation 420: Best = 5.527888e+03	FES = 42100	$\mu_F = 0.628$	$\mu_{CR} = 0.035$		
Diversity = 1.7013e+01	k_t = [2, 3, 2, 3]				
Generation 430: Best = 5.527888e+03	FES = 43100	$\mu_F = 0.665$	$\mu_{CR} = 0.037$		
Diversity = 1.6974e+01	k_t = [1, 1, 1, 1]				
Generation 440: Best = 5.527888e+03	FES = 44100	$\mu_F = 0.712$	$\mu_{CR} = 0.035$		
Diversity = 1.6900e+01	k_t = [3, 3, 3, 3]				
Generation 450: Best = 5.527888e+03	FES = 45100	$\mu_F = 0.731$	$\mu_{CR} = 0.040$		
Diversity = 1.6902e+01	k_t = [1, 1, 1, 1]				
Generation 460: Best = 5.527888e+03	FES = 46100	$\mu_F = 0.715$	$\mu_{CR} = 0.038$		
Diversity = 1.6964e+01	k_t = [3, 4, 3, 2]				
Generation 470: Best = 5.464269e+03	FES = 47100	$\mu_F = 0.717$	$\mu_{CR} = 0.040$		
Diversity = 1.6961e+01	k_t = [1, 1, 1, 1]				
Generation 480: Best = 5.378250e+03	FES = 48100	$\mu_F = 0.683$	$\mu_{CR} = 0.043$		
Diversity = 1.6887e+01	k_t = [3, 2, 2, 4]				
Generation 490: Best = 5.355632e+03	FES = 49100	$\mu_F = 0.718$	$\mu_{CR} = 0.041$		

Diversity = 1.6851e+01	k_t = [1, 1, 1, 1]					
Generation 500: Best = 5.355632e+03	FES = 50100	$\mu_F = 0.688$	$\mu_{CR} = 0.039$			
Diversity = 1.6818e+01	k_t = [4, 2, 2, 3]					
Generation 510: Best = 5.355632e+03	FES = 51100	$\mu_F = 0.643$	$\mu_{CR} = 0.050$			
Diversity = 1.6790e+01	k_t = [1, 1, 1, 1]					
Generation 520: Best = 5.309138e+03	FES = 52100	$\mu_F = 0.700$	$\mu_{CR} = 0.039$			
Diversity = 1.6758e+01	k_t = [2, 4, 4, 3]					
Generation 530: Best = 5.088205e+03	FES = 53100	$\mu_F = 0.680$	$\mu_{CR} = 0.030$			
Diversity = 1.6753e+01	k_t = [1, 1, 1, 1]					
Generation 540: Best = 5.088205e+03	FES = 54100	$\mu_F = 0.708$	$\mu_{CR} = 0.036$			
Diversity = 1.6741e+01	k_t = [2, 4, 4, 3]					
Generation 550: Best = 5.069560e+03	FES = 55100	$\mu_F = 0.722$	$\mu_{CR} = 0.036$			
Diversity = 1.6712e+01	k_t = [1, 1, 1, 1]					
Generation 560: Best = 5.057728e+03	FES = 56100	$\mu_F = 0.756$	$\mu_{CR} = 0.045$			
Diversity = 1.6572e+01	k_t = [2, 4, 3, 4]					
Generation 570: Best = 5.057728e+03	FES = 57100	$\mu_F = 0.800$	$\mu_{CR} = 0.033$			
Diversity = 1.6620e+01	k_t = [1, 1, 1, 1]					
Generation 580: Best = 5.054270e+03	FES = 58100	$\mu_F = 0.788$	$\mu_{CR} = 0.038$			
Diversity = 1.6636e+01	k_t = [2, 3, 2, 2]					
Generation 590: Best = 5.054270e+03	FES = 59100	$\mu_F = 0.736$	$\mu_{CR} = 0.039$			
Diversity = 1.6586e+01	k_t = [1, 1, 1, 1]					
Generation 600: Best = 5.054270e+03	FES = 60100	$\mu_F = 0.715$	$\mu_{CR} = 0.042$			
Diversity = 1.6564e+01	k_t = [3, 4, 2, 3]					
Generation 610: Best = 5.019430e+03	FES = 61100	$\mu_F = 0.706$	$\mu_{CR} = 0.029$			
Diversity = 1.6619e+01	k_t = [1, 1, 1, 1]					
Generation 620: Best = 5.019430e+03	FES = 62100	$\mu_F = 0.738$	$\mu_{CR} = 0.031$			
Diversity = 1.6630e+01	k_t = [2, 4, 2, 2]					
Generation 630: Best = 5.019430e+03	FES = 63100	$\mu_F = 0.768$	$\mu_{CR} = 0.029$			
Diversity = 1.6643e+01	k_t = [1, 1, 1, 1]					
Generation 640: Best = 5.019430e+03	FES = 64100	$\mu_F = 0.691$	$\mu_{CR} = 0.034$			
Diversity = 1.6614e+01	k_t = [4, 2, 4, 2]					
Generation 650: Best = 5.019430e+03	FES = 65100	$\mu_F = 0.678$	$\mu_{CR} = 0.033$			
Diversity = 1.6617e+01	k_t = [1, 1, 1, 1]					
Generation 660: Best = 5.019430e+03	FES = 66100	$\mu_F = 0.738$	$\mu_{CR} = 0.045$			
Diversity = 1.6584e+01	k_t = [3, 4, 3, 4]					
Generation 670: Best = 5.019430e+03	FES = 67100	$\mu_F = 0.700$	$\mu_{CR} = 0.044$			
Diversity = 1.6598e+01	k_t = [1, 1, 1, 1]					
Generation 680: Best = 5.019430e+03	FES = 68100	$\mu_F = 0.748$	$\mu_{CR} = 0.035$			
Diversity = 1.6570e+01	k_t = [3, 3, 3, 3]					
Generation 690: Best = 5.007182e+03	FES = 69100	$\mu_F = 0.778$	$\mu_{CR} = 0.033$			
Diversity = 1.6556e+01	k_t = [1, 1, 1, 1]					
Generation 700: Best = 4.972306e+03	FES = 70100	$\mu_F = 0.735$	$\mu_{CR} = 0.039$			
Diversity = 1.6519e+01	k_t = [4, 2, 4, 3]					
Generation 710: Best = 4.957307e+03	FES = 71100	$\mu_F = 0.739$	$\mu_{CR} = 0.031$			
Diversity = 1.6565e+01	k_t = [1, 1, 1, 1]					
Generation 720: Best = 4.915186e+03	FES = 72100	$\mu_F = 0.739$	$\mu_{CR} = 0.033$			
Diversity = 1.6493e+01	k_t = [3, 3, 4, 2]					
Generation 730: Best = 4.909590e+03	FES = 73100	$\mu_F = 0.754$	$\mu_{CR} = 0.040$			
Diversity = 1.6562e+01	k_t = [1, 1, 1, 1]					
Generation 740: Best = 4.909590e+03	FES = 74100	$\mu_F = 0.755$	$\mu_{CR} = 0.038$			
Diversity = 1.6574e+01	k_t = [4, 3, 4, 3]					
Generation 750: Best = 4.909590e+03	FES = 75100	$\mu_F = 0.757$	$\mu_{CR} = 0.035$			
Diversity = 1.6581e+01	k_t = [1, 1, 1, 1]					
Generation 760: Best = 4.909590e+03	FES = 76100	$\mu_F = 0.797$	$\mu_{CR} = 0.025$			

Diversity = 1.6527e+01	k_t = [2, 4, 3, 4]				
Generation 770: Best = 4.886552e+03	FES = 77100	$\mu_F = 0.748$	$\mu_{CR} = 0.034$		
Diversity = 1.6515e+01	k_t = [1, 1, 1, 1]				
Generation 780: Best = 4.886552e+03	FES = 78100	$\mu_F = 0.757$	$\mu_{CR} = 0.036$		
Diversity = 1.6473e+01	k_t = [2, 4, 3, 2]				
Generation 790: Best = 4.886552e+03	FES = 79100	$\mu_F = 0.791$	$\mu_{CR} = 0.031$		
Diversity = 1.6500e+01	k_t = [1, 1, 1, 1]				
Generation 800: Best = 4.886552e+03	FES = 80100	$\mu_F = 0.723$	$\mu_{CR} = 0.041$		
Diversity = 1.6359e+01	k_t = [2, 2, 4, 4]				
Generation 810: Best = 4.886494e+03	FES = 81100	$\mu_F = 0.724$	$\mu_{CR} = 0.058$		
Diversity = 1.6324e+01	k_t = [1, 1, 1, 1]				
Generation 820: Best = 4.886494e+03	FES = 82100	$\mu_F = 0.753$	$\mu_{CR} = 0.040$		
Diversity = 1.6325e+01	k_t = [4, 4, 3, 3]				
Generation 830: Best = 4.886494e+03	FES = 83100	$\mu_F = 0.780$	$\mu_{CR} = 0.040$		
Diversity = 1.6362e+01	k_t = [1, 1, 1, 1]				
Generation 840: Best = 4.886494e+03	FES = 84100	$\mu_F = 0.753$	$\mu_{CR} = 0.036$		
Diversity = 1.6357e+01	k_t = [2, 3, 3, 2]				
Generation 850: Best = 4.886494e+03	FES = 85100	$\mu_F = 0.747$	$\mu_{CR} = 0.027$		
Diversity = 1.6368e+01	k_t = [1, 1, 1, 1]				
Generation 860: Best = 4.886494e+03	FES = 86100	$\mu_F = 0.770$	$\mu_{CR} = 0.029$		
Diversity = 1.6376e+01	k_t = [2, 4, 3, 3]				
Generation 870: Best = 4.886494e+03	FES = 87100	$\mu_F = 0.751$	$\mu_{CR} = 0.036$		
Diversity = 1.6392e+01	k_t = [1, 1, 1, 1]				
Generation 880: Best = 4.886494e+03	FES = 88100	$\mu_F = 0.714$	$\mu_{CR} = 0.033$		
Diversity = 1.6373e+01	k_t = [3, 3, 3, 4]				
Generation 890: Best = 4.846294e+03	FES = 89100	$\mu_F = 0.700$	$\mu_{CR} = 0.044$		
Diversity = 1.6374e+01	k_t = [1, 1, 1, 1]				
Generation 900: Best = 4.820042e+03	FES = 90100	$\mu_F = 0.691$	$\mu_{CR} = 0.041$		
Diversity = 1.6459e+01	k_t = [4, 2, 4, 2]				
Generation 910: Best = 4.785817e+03	FES = 91100	$\mu_F = 0.768$	$\mu_{CR} = 0.031$		
Diversity = 1.6483e+01	k_t = [1, 1, 1, 1]				
Generation 920: Best = 4.785817e+03	FES = 92100	$\mu_F = 0.712$	$\mu_{CR} = 0.025$		
Diversity = 1.6449e+01	k_t = [2, 3, 3, 2]				
Generation 930: Best = 4.785817e+03	FES = 93100	$\mu_F = 0.696$	$\mu_{CR} = 0.031$		
Diversity = 1.6349e+01	k_t = [1, 1, 1, 1]				
Generation 940: Best = 4.782489e+03	FES = 94100	$\mu_F = 0.730$	$\mu_{CR} = 0.044$		
Diversity = 1.6413e+01	k_t = [4, 4, 4, 2]				
Generation 950: Best = 4.782489e+03	FES = 95100	$\mu_F = 0.743$	$\mu_{CR} = 0.046$		
Diversity = 1.6412e+01	k_t = [1, 1, 1, 1]				
Generation 960: Best = 4.782489e+03	FES = 96100	$\mu_F = 0.800$	$\mu_{CR} = 0.046$		
Diversity = 1.6301e+01	k_t = [4, 3, 3, 4]				
Generation 970: Best = 4.782489e+03	FES = 97100	$\mu_F = 0.754$	$\mu_{CR} = 0.041$		
Diversity = 1.6303e+01	k_t = [1, 1, 1, 1]				
Generation 980: Best = 4.758402e+03	FES = 98100	$\mu_F = 0.734$	$\mu_{CR} = 0.050$		
Diversity = 1.6235e+01	k_t = [3, 4, 4, 4]				
Generation 990: Best = 4.758402e+03	FES = 99100	$\mu_F = 0.771$	$\mu_{CR} = 0.041$		
Diversity = 1.6247e+01	k_t = [1, 1, 1, 1]				
Generation 1000: Best = 4.758402e+03	FES = 100100	$\mu_F = 0.798$	$\mu_{CR} = 0.045$		
Diversity = 1.6169e+01	k_t = [4, 3, 2, 4]				
Generation 1010: Best = 4.749425e+03	FES = 101100	$\mu_F = 0.802$	$\mu_{CR} = 0.045$		
Diversity = 1.6214e+01	k_t = [1, 1, 1, 1]				
Generation 1020: Best = 4.749425e+03	FES = 102100	$\mu_F = 0.784$	$\mu_{CR} = 0.045$		
Diversity = 1.6239e+01	k_t = [2, 3, 3, 3]				
Generation 1030: Best = 4.749425e+03	FES = 103100	$\mu_F = 0.736$	$\mu_{CR} = 0.048$		

Diversity = 1.6276e+01	k_t = [1, 1, 1, 1]			
Generation 1040: Best = 4.749425e+03	FES = 104100	$\mu_F = 0.685$	$\mu_{CR} = 0.063$	
Diversity = 1.6216e+01	k_t = [4, 2, 3, 4]			
Generation 1050: Best = 4.741011e+03	FES = 105100	$\mu_F = 0.671$	$\mu_{CR} = 0.046$	
Diversity = 1.6179e+01	k_t = [1, 1, 1, 1]			
Generation 1060: Best = 4.708419e+03	FES = 106100	$\mu_F = 0.669$	$\mu_{CR} = 0.035$	
Diversity = 1.6244e+01	k_t = [3, 4, 2, 2]			
Generation 1070: Best = 4.708419e+03	FES = 107100	$\mu_F = 0.686$	$\mu_{CR} = 0.036$	
Diversity = 1.6226e+01	k_t = [1, 1, 1, 1]			
Generation 1080: Best = 4.696463e+03	FES = 108100	$\mu_F = 0.691$	$\mu_{CR} = 0.043$	
Diversity = 1.6190e+01	k_t = [4, 2, 4, 4]			
Generation 1090: Best = 4.577930e+03	FES = 109100	$\mu_F = 0.688$	$\mu_{CR} = 0.051$	
Diversity = 1.6232e+01	k_t = [1, 1, 1, 1]			
Generation 1100: Best = 4.577837e+03	FES = 110100	$\mu_F = 0.653$	$\mu_{CR} = 0.045$	
Diversity = 1.6224e+01	k_t = [3, 3, 3, 4]			
Generation 1110: Best = 4.577837e+03	FES = 111100	$\mu_F = 0.609$	$\mu_{CR} = 0.040$	
Diversity = 1.6212e+01	k_t = [1, 1, 1, 1]			
Generation 1120: Best = 4.577512e+03	FES = 112100	$\mu_F = 0.585$	$\mu_{CR} = 0.039$	
Diversity = 1.6210e+01	k_t = [2, 3, 3, 4]			
Generation 1130: Best = 4.577512e+03	FES = 113100	$\mu_F = 0.583$	$\mu_{CR} = 0.042$	
Diversity = 1.6247e+01	k_t = [1, 1, 1, 1]			
Generation 1140: Best = 4.577512e+03	FES = 114100	$\mu_F = 0.618$	$\mu_{CR} = 0.039$	
Diversity = 1.6206e+01	k_t = [3, 2, 4, 4]			
Generation 1150: Best = 4.534051e+03	FES = 115100	$\mu_F = 0.621$	$\mu_{CR} = 0.040$	
Diversity = 1.6216e+01	k_t = [1, 1, 1, 1]			
Generation 1160: Best = 4.534051e+03	FES = 116100	$\mu_F = 0.618$	$\mu_{CR} = 0.046$	
Diversity = 1.6175e+01	k_t = [3, 3, 4, 4]			
Generation 1170: Best = 4.534051e+03	FES = 117100	$\mu_F = 0.674$	$\mu_{CR} = 0.046$	
Diversity = 1.6189e+01	k_t = [1, 1, 1, 1]			
Generation 1180: Best = 4.534051e+03	FES = 118100	$\mu_F = 0.700$	$\mu_{CR} = 0.041$	
Diversity = 1.6276e+01	k_t = [3, 4, 4, 4]			
Generation 1190: Best = 4.534051e+03	FES = 119100	$\mu_F = 0.705$	$\mu_{CR} = 0.050$	
Diversity = 1.6340e+01	k_t = [1, 1, 1, 1]			
Generation 1200: Best = 4.534051e+03	FES = 120100	$\mu_F = 0.737$	$\mu_{CR} = 0.040$	
Diversity = 1.6304e+01	k_t = [4, 4, 2, 4]			
Generation 1210: Best = 4.534051e+03	FES = 121100	$\mu_F = 0.743$	$\mu_{CR} = 0.052$	
Diversity = 1.6357e+01	k_t = [1, 1, 1, 1]			
Generation 1220: Best = 4.529622e+03	FES = 122100	$\mu_F = 0.705$	$\mu_{CR} = 0.056$	
Diversity = 1.6348e+01	k_t = [4, 2, 3, 2]			
Generation 1230: Best = 4.517058e+03	FES = 123100	$\mu_F = 0.715$	$\mu_{CR} = 0.060$	
Diversity = 1.6340e+01	k_t = [1, 1, 1, 1]			
Generation 1240: Best = 4.466019e+03	FES = 124100	$\mu_F = 0.680$	$\mu_{CR} = 0.047$	
Diversity = 1.6495e+01	k_t = [2, 4, 3, 3]			
Generation 1250: Best = 4.431741e+03	FES = 125100	$\mu_F = 0.693$	$\mu_{CR} = 0.043$	
Diversity = 1.6539e+01	k_t = [1, 1, 1, 1]			
Generation 1260: Best = 4.431586e+03	FES = 126100	$\mu_F = 0.750$	$\mu_{CR} = 0.040$	
Diversity = 1.6486e+01	k_t = [2, 4, 2, 2]			
Generation 1270: Best = 4.431586e+03	FES = 127100	$\mu_F = 0.725$	$\mu_{CR} = 0.048$	
Diversity = 1.6450e+01	k_t = [1, 1, 1, 1]			
Generation 1280: Best = 4.364752e+03	FES = 128100	$\mu_F = 0.717$	$\mu_{CR} = 0.044$	
Diversity = 1.6423e+01	k_t = [3, 3, 2, 3]			
Generation 1290: Best = 4.348350e+03	FES = 129100	$\mu_F = 0.677$	$\mu_{CR} = 0.048$	
Diversity = 1.6373e+01	k_t = [1, 1, 1, 1]			
Generation 1300: Best = 4.348350e+03	FES = 130100	$\mu_F = 0.683$	$\mu_{CR} = 0.059$	

Diversity = 1.6316e+01	k_t = [2, 4, 2, 3]			
Generation 1310: Best = 4.348350e+03	FES = 131100	$\mu_F = 0.721$	$\mu_{CR} = 0.059$	
Diversity = 1.6288e+01	k_t = [1, 1, 1, 1]			
Generation 1320: Best = 4.328441e+03	FES = 132100	$\mu_F = 0.668$	$\mu_{CR} = 0.070$	
Diversity = 1.6334e+01	k_t = [4, 3, 2, 2]			
Generation 1330: Best = 4.328441e+03	FES = 133100	$\mu_F = 0.656$	$\mu_{CR} = 0.058$	
Diversity = 1.6311e+01	k_t = [1, 1, 1, 1]			
Generation 1340: Best = 4.327446e+03	FES = 134100	$\mu_F = 0.700$	$\mu_{CR} = 0.057$	
Diversity = 1.6420e+01	k_t = [4, 4, 3, 3]			
Generation 1350: Best = 4.325721e+03	FES = 135100	$\mu_F = 0.737$	$\mu_{CR} = 0.060$	
Diversity = 1.6394e+01	k_t = [1, 1, 1, 1]			
Generation 1360: Best = 4.325721e+03	FES = 136100	$\mu_F = 0.772$	$\mu_{CR} = 0.056$	
Diversity = 1.6082e+01	k_t = [3, 4, 2, 4]			
Generation 1370: Best = 4.320278e+03	FES = 137100	$\mu_F = 0.819$	$\mu_{CR} = 0.054$	
Diversity = 1.6013e+01	k_t = [1, 1, 1, 1]			
Generation 1380: Best = 4.280337e+03	FES = 138100	$\mu_F = 0.822$	$\mu_{CR} = 0.066$	
Diversity = 1.5617e+01	k_t = [3, 3, 4, 4]			
Generation 1390: Best = 4.279732e+03	FES = 139100	$\mu_F = 0.826$	$\mu_{CR} = 0.079$	
Diversity = 1.5484e+01	k_t = [1, 1, 1, 1]			
Generation 1400: Best = 4.240753e+03	FES = 140100	$\mu_F = 0.841$	$\mu_{CR} = 0.089$	
Diversity = 1.5096e+01	k_t = [3, 2, 2, 2]			
Generation 1410: Best = 4.240753e+03	FES = 141100	$\mu_F = 0.846$	$\mu_{CR} = 0.093$	
Diversity = 1.4899e+01	k_t = [1, 1, 1, 1]			
Generation 1420: Best = 4.240753e+03	FES = 142100	$\mu_F = 0.847$	$\mu_{CR} = 0.099$	
Diversity = 1.4429e+01	k_t = [4, 4, 4, 2]			
Generation 1430: Best = 4.240753e+03	FES = 143100	$\mu_F = 0.874$	$\mu_{CR} = 0.104$	
Diversity = 1.2349e+01	k_t = [1, 1, 1, 1]			
Generation 1440: Best = 4.240753e+03	FES = 144100	$\mu_F = 0.833$	$\mu_{CR} = 0.097$	
Diversity = 9.8179e+00	k_t = [3, 4, 2, 4]			
Generation 1450: Best = 4.239539e+03	FES = 145100	$\mu_F = 0.810$	$\mu_{CR} = 0.097$	
Diversity = 8.7379e+00	k_t = [1, 1, 1, 1]			
Generation 1460: Best = 4.177814e+03	FES = 146100	$\mu_F = 0.778$	$\mu_{CR} = 0.088$	
Diversity = 8.4944e+00	k_t = [4, 3, 4, 3]			
Generation 1470: Best = 4.175930e+03	FES = 147100	$\mu_F = 0.750$	$\mu_{CR} = 0.079$	
Diversity = 8.2511e+00	k_t = [1, 1, 1, 1]			
Generation 1480: Best = 4.171334e+03	FES = 148100	$\mu_F = 0.765$	$\mu_{CR} = 0.080$	
Diversity = 7.8039e+00	k_t = [3, 3, 2, 4]			
Generation 1490: Best = 4.151815e+03	FES = 149100	$\mu_F = 0.754$	$\mu_{CR} = 0.072$	
Diversity = 7.6094e+00	k_t = [1, 1, 1, 1]			
Generation 1500: Best = 4.137319e+03	FES = 150100	$\mu_F = 0.748$	$\mu_{CR} = 0.066$	
Diversity = 7.4619e+00	k_t = [4, 3, 4, 4]			
Generation 1510: Best = 4.124456e+03	FES = 151100	$\mu_F = 0.727$	$\mu_{CR} = 0.062$	
Diversity = 7.1949e+00	k_t = [1, 1, 1, 1]			
Generation 1520: Best = 4.108777e+03	FES = 152100	$\mu_F = 0.704$	$\mu_{CR} = 0.071$	
Diversity = 6.9064e+00	k_t = [3, 3, 4, 2]			
Generation 1530: Best = 4.107071e+03	FES = 153100	$\mu_F = 0.691$	$\mu_{CR} = 0.056$	
Diversity = 6.7746e+00	k_t = [1, 1, 1, 1]			
Generation 1540: Best = 4.095217e+03	FES = 154100	$\mu_F = 0.702$	$\mu_{CR} = 0.061$	
Diversity = 6.7551e+00	k_t = [2, 2, 3, 2]			
Generation 1550: Best = 4.094143e+03	FES = 155100	$\mu_F = 0.698$	$\mu_{CR} = 0.062$	
Diversity = 6.6910e+00	k_t = [1, 1, 1, 1]			
Generation 1560: Best = 4.090218e+03	FES = 156100	$\mu_F = 0.667$	$\mu_{CR} = 0.061$	
Diversity = 6.5468e+00	k_t = [3, 2, 4, 3]			
Generation 1570: Best = 4.090218e+03	FES = 157100	$\mu_F = 0.659$	$\mu_{CR} = 0.063$	

Diversity = 6.4266e+00	k_t = [1, 1, 1, 1]			
Generation 1580: Best = 4.062536e+03	FES = 158100	$\mu_F = 0.660$	$\mu_{CR} = 0.055$	
Diversity = 6.3982e+00	k_t = [2, 3, 3, 4]			
Generation 1590: Best = 4.057587e+03	FES = 159100	$\mu_F = 0.671$	$\mu_{CR} = 0.049$	
Diversity = 6.3680e+00	k_t = [1, 1, 1, 1]			
Generation 1600: Best = 4.056483e+03	FES = 160100	$\mu_F = 0.695$	$\mu_{CR} = 0.055$	
Diversity = 6.3269e+00	k_t = [4, 2, 2, 2]			
Generation 1610: Best = 4.050948e+03	FES = 161100	$\mu_F = 0.712$	$\mu_{CR} = 0.051$	
Diversity = 6.2799e+00	k_t = [1, 1, 1, 1]			
Generation 1620: Best = 4.039116e+03	FES = 162100	$\mu_F = 0.711$	$\mu_{CR} = 0.058$	
Diversity = 6.2751e+00	k_t = [4, 2, 4, 4]			
Generation 1630: Best = 4.019258e+03	FES = 163100	$\mu_F = 0.731$	$\mu_{CR} = 0.045$	
Diversity = 6.2311e+00	k_t = [1, 1, 1, 1]			
Generation 1640: Best = 4.015542e+03	FES = 164100	$\mu_F = 0.744$	$\mu_{CR} = 0.049$	
Diversity = 6.2334e+00	k_t = [3, 2, 2, 3]			
Generation 1650: Best = 4.015542e+03	FES = 165100	$\mu_F = 0.756$	$\mu_{CR} = 0.041$	
Diversity = 6.2482e+00	k_t = [1, 1, 1, 1]			
Generation 1660: Best = 4.015542e+03	FES = 166100	$\mu_F = 0.732$	$\mu_{CR} = 0.039$	
Diversity = 6.1177e+00	k_t = [3, 4, 4, 4]			
Generation 1670: Best = 4.013581e+03	FES = 167100	$\mu_F = 0.734$	$\mu_{CR} = 0.036$	
Diversity = 6.0604e+00	k_t = [1, 1, 1, 1]			
Generation 1680: Best = 4.013581e+03	FES = 168100	$\mu_F = 0.727$	$\mu_{CR} = 0.050$	
Diversity = 6.0552e+00	k_t = [2, 4, 2, 4]			
Generation 1690: Best = 4.013581e+03	FES = 169100	$\mu_F = 0.763$	$\mu_{CR} = 0.041$	
Diversity = 6.0580e+00	k_t = [1, 1, 1, 1]			
Generation 1700: Best = 4.013581e+03	FES = 170100	$\mu_F = 0.727$	$\mu_{CR} = 0.035$	
Diversity = 6.0229e+00	k_t = [2, 4, 4, 3]			
Generation 1710: Best = 4.013364e+03	FES = 171100	$\mu_F = 0.705$	$\mu_{CR} = 0.032$	
Diversity = 6.0255e+00	k_t = [1, 1, 1, 1]			
Generation 1720: Best = 4.010342e+03	FES = 172100	$\mu_F = 0.680$	$\mu_{CR} = 0.032$	
Diversity = 6.0022e+00	k_t = [2, 2, 2, 2]			
Generation 1730: Best = 4.010342e+03	FES = 173100	$\mu_F = 0.671$	$\mu_{CR} = 0.042$	
Diversity = 6.0242e+00	k_t = [1, 1, 1, 1]			
Generation 1740: Best = 4.010342e+03	FES = 174100	$\mu_F = 0.735$	$\mu_{CR} = 0.041$	
Diversity = 6.0485e+00	k_t = [4, 4, 2, 2]			
Generation 1750: Best = 4.008294e+03	FES = 175100	$\mu_F = 0.738$	$\mu_{CR} = 0.044$	
Diversity = 6.0332e+00	k_t = [1, 1, 1, 1]			
Generation 1760: Best = 4.003970e+03	FES = 176100	$\mu_F = 0.698$	$\mu_{CR} = 0.058$	
Diversity = 6.0716e+00	k_t = [3, 3, 2, 3]			
Generation 1770: Best = 4.003970e+03	FES = 177100	$\mu_F = 0.713$	$\mu_{CR} = 0.042$	
Diversity = 6.0474e+00	k_t = [1, 1, 1, 1]			
Generation 1780: Best = 3.998995e+03	FES = 178100	$\mu_F = 0.715$	$\mu_{CR} = 0.037$	
Diversity = 6.0212e+00	k_t = [3, 2, 4, 2]			
Generation 1790: Best = 3.998995e+03	FES = 179100	$\mu_F = 0.728$	$\mu_{CR} = 0.032$	
Diversity = 5.9694e+00	k_t = [1, 1, 1, 1]			
Generation 1800: Best = 3.998995e+03	FES = 180100	$\mu_F = 0.711$	$\mu_{CR} = 0.033$	
Diversity = 5.9939e+00	k_t = [4, 2, 3, 3]			
Generation 1810: Best = 3.998995e+03	FES = 181100	$\mu_F = 0.703$	$\mu_{CR} = 0.036$	
Diversity = 5.9978e+00	k_t = [1, 1, 1, 1]			
Generation 1820: Best = 3.996910e+03	FES = 182100	$\mu_F = 0.732$	$\mu_{CR} = 0.031$	
Diversity = 5.9487e+00	k_t = [4, 2, 2, 4]			
Generation 1830: Best = 3.996101e+03	FES = 183100	$\mu_F = 0.764$	$\mu_{CR} = 0.033$	
Diversity = 5.9528e+00	k_t = [1, 1, 1, 1]			
Generation 1840: Best = 3.991802e+03	FES = 184100	$\mu_F = 0.757$	$\mu_{CR} = 0.044$	

Diversity = 5.8740e+00	k_t = [3, 2, 4, 4]			
Generation 1850: Best = 3.991802e+03	FES = 185100	$\mu_F = 0.732$	$\mu_{CR} = 0.040$	
Diversity = 5.8712e+00	k_t = [1, 1, 1, 1]			
Generation 1860: Best = 3.991802e+03	FES = 186100	$\mu_F = 0.747$	$\mu_{CR} = 0.042$	
Diversity = 5.8701e+00	k_t = [2, 4, 4, 3]			
Generation 1870: Best = 3.990133e+03	FES = 187100	$\mu_F = 0.723$	$\mu_{CR} = 0.043$	
Diversity = 5.8619e+00	k_t = [1, 1, 1, 1]			
Generation 1880: Best = 3.990133e+03	FES = 188100	$\mu_F = 0.708$	$\mu_{CR} = 0.039$	
Diversity = 5.8430e+00	k_t = [4, 3, 4, 2]			
Generation 1890: Best = 3.988382e+03	FES = 189100	$\mu_F = 0.716$	$\mu_{CR} = 0.035$	
Diversity = 5.8387e+00	k_t = [1, 1, 1, 1]			
Generation 1900: Best = 3.973212e+03	FES = 190100	$\mu_F = 0.702$	$\mu_{CR} = 0.044$	
Diversity = 5.8352e+00	k_t = [2, 2, 4, 2]			
Generation 1910: Best = 3.973212e+03	FES = 191100	$\mu_F = 0.682$	$\mu_{CR} = 0.040$	
Diversity = 5.8411e+00	k_t = [1, 1, 1, 1]			
Generation 1920: Best = 3.972224e+03	FES = 192100	$\mu_F = 0.638$	$\mu_{CR} = 0.058$	
Diversity = 5.8271e+00	k_t = [4, 4, 2, 4]			
Generation 1930: Best = 3.971242e+03	FES = 193100	$\mu_F = 0.688$	$\mu_{CR} = 0.047$	
Diversity = 5.8411e+00	k_t = [1, 1, 1, 1]			
Generation 1940: Best = 3.971242e+03	FES = 194100	$\mu_F = 0.727$	$\mu_{CR} = 0.039$	
Diversity = 5.7274e+00	k_t = [4, 2, 2, 4]			
Generation 1950: Best = 3.971242e+03	FES = 195100	$\mu_F = 0.613$	$\mu_{CR} = 0.038$	
Diversity = 5.7277e+00	k_t = [1, 1, 1, 1]			
Generation 1960: Best = 3.971242e+03	FES = 196100	$\mu_F = 0.612$	$\mu_{CR} = 0.048$	
Diversity = 5.7451e+00	k_t = [4, 2, 2, 2]			
Generation 1970: Best = 3.970587e+03	FES = 197100	$\mu_F = 0.655$	$\mu_{CR} = 0.036$	
Diversity = 5.7471e+00	k_t = [1, 1, 1, 1]			
Generation 1980: Best = 3.964197e+03	FES = 198100	$\mu_F = 0.659$	$\mu_{CR} = 0.037$	
Diversity = 5.7024e+00	k_t = [2, 3, 3, 3]			
Generation 1990: Best = 3.963814e+03	FES = 199100	$\mu_F = 0.658$	$\mu_{CR} = 0.043$	
Diversity = 5.7284e+00	k_t = [1, 1, 1, 1]			
Generation 2000: Best = 3.962861e+03	FES = 200100	$\mu_F = 0.677$	$\mu_{CR} = 0.039$	
Diversity = 5.5939e+00	k_t = [3, 4, 4, 4]			
Generation 2010: Best = 3.958220e+03	FES = 201100	$\mu_F = 0.721$	$\mu_{CR} = 0.049$	
Diversity = 5.5398e+00	k_t = [1, 1, 1, 1]			
Generation 2020: Best = 3.957122e+03	FES = 202100	$\mu_F = 0.724$	$\mu_{CR} = 0.045$	
Diversity = 5.5312e+00	k_t = [2, 4, 2, 3]			
Generation 2030: Best = 3.948754e+03	FES = 203100	$\mu_F = 0.696$	$\mu_{CR} = 0.044$	
Diversity = 5.5333e+00	k_t = [1, 1, 1, 1]			
Generation 2040: Best = 3.948754e+03	FES = 204100	$\mu_F = 0.748$	$\mu_{CR} = 0.035$	
Diversity = 5.5316e+00	k_t = [4, 3, 2, 4]			
Generation 2050: Best = 3.948754e+03	FES = 205100	$\mu_F = 0.738$	$\mu_{CR} = 0.026$	
Diversity = 5.5035e+00	k_t = [1, 1, 1, 1]			
Generation 2060: Best = 3.948754e+03	FES = 206100	$\mu_F = 0.779$	$\mu_{CR} = 0.031$	
Diversity = 5.4620e+00	k_t = [4, 3, 3, 3]			
Generation 2070: Best = 3.945505e+03	FES = 207100	$\mu_F = 0.779$	$\mu_{CR} = 0.035$	
Diversity = 5.4520e+00	k_t = [1, 1, 1, 1]			
Generation 2080: Best = 3.943665e+03	FES = 208100	$\mu_F = 0.830$	$\mu_{CR} = 0.041$	
Diversity = 5.4480e+00	k_t = [4, 4, 4, 2]			
Generation 2090: Best = 3.943309e+03	FES = 209100	$\mu_F = 0.818$	$\mu_{CR} = 0.043$	
Diversity = 5.4155e+00	k_t = [1, 1, 1, 1]			
Generation 2100: Best = 3.943309e+03	FES = 210100	$\mu_F = 0.817$	$\mu_{CR} = 0.031$	
Diversity = 5.3868e+00	k_t = [2, 4, 4, 2]			
Generation 2110: Best = 3.943309e+03	FES = 211100	$\mu_F = 0.793$	$\mu_{CR} = 0.043$	

Diversity = 5.3806e+00	k_t = [1, 1, 1, 1]				
Generation 2120: Best = 3.943309e+03	FES = 212100	$\mu_F = 0.741$	$\mu_{CR} = 0.047$		
Diversity = 5.3745e+00	k_t = [3, 4, 2, 3]				
Generation 2130: Best = 3.939976e+03	FES = 213100	$\mu_F = 0.770$	$\mu_{CR} = 0.041$		
Diversity = 5.3647e+00	k_t = [1, 1, 1, 1]				
Generation 2140: Best = 3.939976e+03	FES = 214100	$\mu_F = 0.792$	$\mu_{CR} = 0.040$		
Diversity = 5.3302e+00	k_t = [3, 4, 2, 4]				
Generation 2150: Best = 3.939976e+03	FES = 215100	$\mu_F = 0.781$	$\mu_{CR} = 0.035$		
Diversity = 5.2854e+00	k_t = [1, 1, 1, 1]				
Generation 2160: Best = 3.939976e+03	FES = 216100	$\mu_F = 0.762$	$\mu_{CR} = 0.030$		
Diversity = 5.2167e+00	k_t = [4, 2, 2, 2]				
Generation 2170: Best = 3.936057e+03	FES = 217100	$\mu_F = 0.814$	$\mu_{CR} = 0.031$		
Diversity = 5.1946e+00	k_t = [1, 1, 1, 1]				
Generation 2180: Best = 3.935302e+03	FES = 218100	$\mu_F = 0.803$	$\mu_{CR} = 0.038$		
Diversity = 5.1953e+00	k_t = [2, 4, 2, 2]				
Generation 2190: Best = 3.935302e+03	FES = 219100	$\mu_F = 0.806$	$\mu_{CR} = 0.039$		
Diversity = 5.1478e+00	k_t = [1, 1, 1, 1]				
Generation 2200: Best = 3.934925e+03	FES = 220100	$\mu_F = 0.813$	$\mu_{CR} = 0.047$		
Diversity = 5.1508e+00	k_t = [4, 3, 3, 3]				
Generation 2210: Best = 3.934925e+03	FES = 221100	$\mu_F = 0.817$	$\mu_{CR} = 0.044$		
Diversity = 5.1473e+00	k_t = [1, 1, 1, 1]				
Generation 2220: Best = 3.934925e+03	FES = 222100	$\mu_F = 0.772$	$\mu_{CR} = 0.051$		
Diversity = 5.0080e+00	k_t = [4, 4, 3, 4]				
Generation 2230: Best = 3.934925e+03	FES = 223100	$\mu_F = 0.787$	$\mu_{CR} = 0.036$		
Diversity = 5.0032e+00	k_t = [1, 1, 1, 1]				
Generation 2240: Best = 3.934700e+03	FES = 224100	$\mu_F = 0.781$	$\mu_{CR} = 0.050$		
Diversity = 4.9878e+00	k_t = [4, 2, 3, 3]				
Generation 2250: Best = 3.932633e+03	FES = 225100	$\mu_F = 0.678$	$\mu_{CR} = 0.055$		
Diversity = 4.9855e+00	k_t = [1, 1, 1, 1]				
Generation 2260: Best = 3.932633e+03	FES = 226100	$\mu_F = 0.688$	$\mu_{CR} = 0.041$		
Diversity = 4.9725e+00	k_t = [2, 4, 2, 2]				
Generation 2270: Best = 3.932633e+03	FES = 227100	$\mu_F = 0.707$	$\mu_{CR} = 0.040$		
Diversity = 4.9571e+00	k_t = [1, 1, 1, 1]				
Generation 2280: Best = 3.926006e+03	FES = 228100	$\mu_F = 0.721$	$\mu_{CR} = 0.050$		
Diversity = 4.9078e+00	k_t = [4, 2, 3, 4]				
Generation 2290: Best = 3.923213e+03	FES = 229100	$\mu_F = 0.690$	$\mu_{CR} = 0.060$		
Diversity = 4.8854e+00	k_t = [1, 1, 1, 1]				
Generation 2300: Best = 3.915940e+03	FES = 230100	$\mu_F = 0.700$	$\mu_{CR} = 0.048$		
Diversity = 4.8219e+00	k_t = [3, 3, 4, 3]				
Generation 2310: Best = 3.915940e+03	FES = 231100	$\mu_F = 0.695$	$\mu_{CR} = 0.051$		
Diversity = 4.7222e+00	k_t = [1, 1, 1, 1]				
Generation 2320: Best = 3.915792e+03	FES = 232100	$\mu_F = 0.705$	$\mu_{CR} = 0.062$		
Diversity = 4.6052e+00	k_t = [4, 2, 3, 2]				
Generation 2330: Best = 3.913921e+03	FES = 233100	$\mu_F = 0.673$	$\mu_{CR} = 0.069$		
Diversity = 4.6016e+00	k_t = [1, 1, 1, 1]				
Generation 2340: Best = 3.913099e+03	FES = 234100	$\mu_F = 0.687$	$\mu_{CR} = 0.078$		
Diversity = 4.3368e+00	k_t = [4, 3, 4, 4]				
Generation 2350: Best = 3.912378e+03	FES = 235100	$\mu_F = 0.724$	$\mu_{CR} = 0.093$		
Diversity = 4.0872e+00	k_t = [1, 1, 1, 1]				
Generation 2360: Best = 3.910148e+03	FES = 236100	$\mu_F = 0.746$	$\mu_{CR} = 0.092$		
Diversity = 3.4114e+00	k_t = [4, 2, 3, 4]				
Generation 2370: Best = 3.903425e+03	FES = 237100	$\mu_F = 0.769$	$\mu_{CR} = 0.089$		
Diversity = 2.9427e+00	k_t = [1, 1, 1, 1]				
Generation 2380: Best = 3.897945e+03	FES = 238100	$\mu_F = 0.761$	$\mu_{CR} = 0.085$		

Diversity = 2.1473e+00	k_t = [4, 3, 3, 3]			
Generation 2390: Best = 3.895224e+03	FES = 239100	$\mu_F = 0.742$	$\mu_{CR} = 0.100$	
Diversity = 1.8163e+00	k_t = [1, 1, 1, 1]			
Generation 2400: Best = 3.893588e+03	FES = 240100	$\mu_F = 0.738$	$\mu_{CR} = 0.095$	
Diversity = 1.7660e+00	k_t = [3, 2, 3, 3]			
Generation 2410: Best = 3.891584e+03	FES = 241100	$\mu_F = 0.750$	$\mu_{CR} = 0.095$	
Diversity = 1.7446e+00	k_t = [1, 1, 1, 1]			
Generation 2420: Best = 3.883448e+03	FES = 242100	$\mu_F = 0.735$	$\mu_{CR} = 0.084$	
Diversity = 1.6558e+00	k_t = [3, 2, 3, 4]			
Generation 2430: Best = 3.883346e+03	FES = 243100	$\mu_F = 0.720$	$\mu_{CR} = 0.076$	
Diversity = 1.6134e+00	k_t = [1, 1, 1, 1]			
Generation 2440: Best = 3.880368e+03	FES = 244100	$\mu_F = 0.733$	$\mu_{CR} = 0.073$	
Diversity = 1.6018e+00	k_t = [4, 4, 3, 4]			
Generation 2450: Best = 3.879384e+03	FES = 245100	$\mu_F = 0.723$	$\mu_{CR} = 0.067$	
Diversity = 1.5553e+00	k_t = [1, 1, 1, 1]			
Generation 2460: Best = 3.879027e+03	FES = 246100	$\mu_F = 0.741$	$\mu_{CR} = 0.065$	
Diversity = 1.5371e+00	k_t = [2, 4, 2, 3]			
Generation 2470: Best = 3.878095e+03	FES = 247100	$\mu_F = 0.744$	$\mu_{CR} = 0.067$	
Diversity = 1.5458e+00	k_t = [1, 1, 1, 1]			
Generation 2480: Best = 3.877828e+03	FES = 248100	$\mu_F = 0.748$	$\mu_{CR} = 0.060$	
Diversity = 1.5287e+00	k_t = [3, 2, 2, 2]			
Generation 2490: Best = 3.876829e+03	FES = 249100	$\mu_F = 0.759$	$\mu_{CR} = 0.066$	
Diversity = 1.5296e+00	k_t = [1, 1, 1, 1]			
Generation 2500: Best = 3.875928e+03	FES = 250100	$\mu_F = 0.776$	$\mu_{CR} = 0.056$	
Diversity = 1.4954e+00	k_t = [3, 3, 4, 3]			
Generation 2510: Best = 3.874884e+03	FES = 251100	$\mu_F = 0.759$	$\mu_{CR} = 0.055$	
Diversity = 1.4964e+00	k_t = [1, 1, 1, 1]			
Generation 2520: Best = 3.874884e+03	FES = 252100	$\mu_F = 0.744$	$\mu_{CR} = 0.047$	
Diversity = 1.4819e+00	k_t = [4, 4, 2, 3]			
Generation 2530: Best = 3.874884e+03	FES = 253100	$\mu_F = 0.720$	$\mu_{CR} = 0.047$	
Diversity = 1.4764e+00	k_t = [1, 1, 1, 1]			
Generation 2540: Best = 3.874352e+03	FES = 254100	$\mu_F = 0.739$	$\mu_{CR} = 0.049$	
Diversity = 1.4714e+00	k_t = [4, 3, 4, 2]			
Generation 2550: Best = 3.874266e+03	FES = 255100	$\mu_F = 0.771$	$\mu_{CR} = 0.051$	
Diversity = 1.4768e+00	k_t = [1, 1, 1, 1]			
Generation 2560: Best = 3.874011e+03	FES = 256100	$\mu_F = 0.772$	$\mu_{CR} = 0.048$	
Diversity = 1.4854e+00	k_t = [4, 2, 2, 4]			
Generation 2570: Best = 3.872692e+03	FES = 257100	$\mu_F = 0.740$	$\mu_{CR} = 0.052$	
Diversity = 1.4867e+00	k_t = [1, 1, 1, 1]			
Generation 2580: Best = 3.871465e+03	FES = 258100	$\mu_F = 0.733$	$\mu_{CR} = 0.052$	
Diversity = 1.5148e+00	k_t = [4, 4, 2, 3]			
Generation 2590: Best = 3.870202e+03	FES = 259100	$\mu_F = 0.729$	$\mu_{CR} = 0.049$	
Diversity = 1.5152e+00	k_t = [1, 1, 1, 1]			
Generation 2600: Best = 3.870202e+03	FES = 260100	$\mu_F = 0.699$	$\mu_{CR} = 0.049$	
Diversity = 1.5071e+00	k_t = [2, 2, 4, 2]			
Generation 2610: Best = 3.870133e+03	FES = 261100	$\mu_F = 0.727$	$\mu_{CR} = 0.050$	
Diversity = 1.5037e+00	k_t = [1, 1, 1, 1]			
Generation 2620: Best = 3.869766e+03	FES = 262100	$\mu_F = 0.721$	$\mu_{CR} = 0.052$	
Diversity = 1.4787e+00	k_t = [2, 3, 3, 4]			
Generation 2630: Best = 3.869766e+03	FES = 263100	$\mu_F = 0.745$	$\mu_{CR} = 0.047$	
Diversity = 1.5036e+00	k_t = [1, 1, 1, 1]			
Generation 2640: Best = 3.869766e+03	FES = 264100	$\mu_F = 0.755$	$\mu_{CR} = 0.045$	
Diversity = 1.4908e+00	k_t = [4, 3, 3, 4]			
Generation 2650: Best = 3.869766e+03	FES = 265100	$\mu_F = 0.777$	$\mu_{CR} = 0.038$	

Diversity = 1.4804e+00	k_t = [1, 1, 1, 1]			
Generation 2660: Best = 3.869440e+03	FES = 266100	$\mu_F = 0.776$	$\mu_{CR} = 0.035$	
Diversity = 1.4674e+00	k_t = [2, 2, 3, 2]			
Generation 2670: Best = 3.869025e+03	FES = 267100	$\mu_F = 0.766$	$\mu_{CR} = 0.031$	
Diversity = 1.4728e+00	k_t = [1, 1, 1, 1]			
Generation 2680: Best = 3.868621e+03	FES = 268100	$\mu_F = 0.751$	$\mu_{CR} = 0.045$	
Diversity = 1.4764e+00	k_t = [2, 4, 3, 2]			
Generation 2690: Best = 3.868621e+03	FES = 269100	$\mu_F = 0.740$	$\mu_{CR} = 0.044$	
Diversity = 1.4795e+00	k_t = [1, 1, 1, 1]			
Generation 2700: Best = 3.867619e+03	FES = 270100	$\mu_F = 0.770$	$\mu_{CR} = 0.052$	
Diversity = 1.5113e+00	k_t = [2, 4, 2, 4]			
Generation 2710: Best = 3.867149e+03	FES = 271100	$\mu_F = 0.765$	$\mu_{CR} = 0.039$	
Diversity = 1.5118e+00	k_t = [1, 1, 1, 1]			
Generation 2720: Best = 3.867060e+03	FES = 272100	$\mu_F = 0.733$	$\mu_{CR} = 0.046$	
Diversity = 1.5108e+00	k_t = [4, 2, 3, 2]			
Generation 2730: Best = 3.867025e+03	FES = 273100	$\mu_F = 0.744$	$\mu_{CR} = 0.043$	
Diversity = 1.5109e+00	k_t = [1, 1, 1, 1]			
Generation 2740: Best = 3.866661e+03	FES = 274100	$\mu_F = 0.775$	$\mu_{CR} = 0.042$	
Diversity = 1.4978e+00	k_t = [4, 4, 3, 4]			
Generation 2750: Best = 3.866661e+03	FES = 275100	$\mu_F = 0.749$	$\mu_{CR} = 0.042$	
Diversity = 1.4903e+00	k_t = [1, 1, 1, 1]			
Generation 2760: Best = 3.866417e+03	FES = 276100	$\mu_F = 0.757$	$\mu_{CR} = 0.042$	
Diversity = 1.4713e+00	k_t = [4, 3, 4, 4]			
Generation 2770: Best = 3.866086e+03	FES = 277100	$\mu_F = 0.696$	$\mu_{CR} = 0.042$	
Diversity = 1.4610e+00	k_t = [1, 1, 1, 1]			
Generation 2780: Best = 3.866086e+03	FES = 278100	$\mu_F = 0.750$	$\mu_{CR} = 0.041$	
Diversity = 1.4622e+00	k_t = [2, 4, 4, 3]			
Generation 2790: Best = 3.865904e+03	FES = 279100	$\mu_F = 0.716$	$\mu_{CR} = 0.038$	
Diversity = 1.4608e+00	k_t = [1, 1, 1, 1]			
Generation 2800: Best = 3.865343e+03	FES = 280100	$\mu_F = 0.735$	$\mu_{CR} = 0.039$	
Diversity = 1.4493e+00	k_t = [2, 4, 4, 2]			
Generation 2810: Best = 3.865216e+03	FES = 281100	$\mu_F = 0.717$	$\mu_{CR} = 0.058$	
Diversity = 1.4561e+00	k_t = [1, 1, 1, 1]			
Generation 2820: Best = 3.865191e+03	FES = 282100	$\mu_F = 0.692$	$\mu_{CR} = 0.041$	
Diversity = 1.4461e+00	k_t = [3, 4, 3, 3]			
Generation 2830: Best = 3.864634e+03	FES = 283100	$\mu_F = 0.700$	$\mu_{CR} = 0.041$	
Diversity = 1.4419e+00	k_t = [1, 1, 1, 1]			
Generation 2840: Best = 3.864634e+03	FES = 284100	$\mu_F = 0.741$	$\mu_{CR} = 0.032$	
Diversity = 1.4381e+00	k_t = [2, 3, 2, 3]			
Generation 2850: Best = 3.864634e+03	FES = 285100	$\mu_F = 0.757$	$\mu_{CR} = 0.037$	
Diversity = 1.4317e+00	k_t = [1, 1, 1, 1]			
Generation 2860: Best = 3.864205e+03	FES = 286100	$\mu_F = 0.791$	$\mu_{CR} = 0.035$	
Diversity = 1.4347e+00	k_t = [2, 4, 3, 2]			
Generation 2870: Best = 3.864151e+03	FES = 287100	$\mu_F = 0.797$	$\mu_{CR} = 0.033$	
Diversity = 1.4312e+00	k_t = [1, 1, 1, 1]			
Generation 2880: Best = 3.864151e+03	FES = 288100	$\mu_F = 0.789$	$\mu_{CR} = 0.025$	
Diversity = 1.4328e+00	k_t = [2, 4, 2, 3]			
Generation 2890: Best = 3.864027e+03	FES = 289100	$\mu_F = 0.696$	$\mu_{CR} = 0.023$	
Diversity = 1.4274e+00	k_t = [1, 1, 1, 1]			
Generation 2900: Best = 3.864027e+03	FES = 290100	$\mu_F = 0.680$	$\mu_{CR} = 0.033$	
Diversity = 1.4242e+00	k_t = [3, 3, 2, 4]			
Generation 2910: Best = 3.864027e+03	FES = 291100	$\mu_F = 0.702$	$\mu_{CR} = 0.042$	
Diversity = 1.4218e+00	k_t = [1, 1, 1, 1]			
Generation 2920: Best = 3.863841e+03	FES = 292100	$\mu_F = 0.746$	$\mu_{CR} = 0.038$	

Diversity = 1.4253e+00	k_t = [2, 4, 3, 3]			
Generation 2930: Best = 3.863408e+03	FES = 293100	$\mu_F = 0.737$	$\mu_{CR} = 0.033$	
Diversity = 1.4240e+00	k_t = [1, 1, 1, 1]			
Generation 2940: Best = 3.863408e+03	FES = 294100	$\mu_F = 0.706$	$\mu_{CR} = 0.041$	
Diversity = 1.4248e+00	k_t = [4, 3, 3, 2]			
Generation 2950: Best = 3.863118e+03	FES = 295100	$\mu_F = 0.716$	$\mu_{CR} = 0.037$	
Diversity = 1.4258e+00	k_t = [1, 1, 1, 1]			
Generation 2960: Best = 3.863048e+03	FES = 296100	$\mu_F = 0.781$	$\mu_{CR} = 0.036$	
Diversity = 1.4164e+00	k_t = [4, 2, 2, 2]			
Generation 2970: Best = 3.863048e+03	FES = 297100	$\mu_F = 0.754$	$\mu_{CR} = 0.031$	
Diversity = 1.4190e+00	k_t = [1, 1, 1, 1]			
Generation 2980: Best = 3.863037e+03	FES = 298100	$\mu_F = 0.779$	$\mu_{CR} = 0.034$	
Diversity = 1.4269e+00	k_t = [3, 2, 4, 2]			
Generation 2990: Best = 3.862396e+03	FES = 299100	$\mu_F = 0.727$	$\mu_{CR} = 0.033$	
Diversity = 1.4292e+00	k_t = [1, 1, 1, 1]			
Generation 3000: Best = 3.862396e+03	FES = 300100	$\mu_F = 0.754$	$\mu_{CR} = 0.045$	
Diversity = 1.4411e+00	k_t = [3, 3, 3, 3]			
Generation 3010: Best = 3.862396e+03	FES = 301100	$\mu_F = 0.699$	$\mu_{CR} = 0.040$	
Diversity = 1.4445e+00	k_t = [1, 1, 1, 1]			
Generation 3020: Best = 3.862396e+03	FES = 302100	$\mu_F = 0.750$	$\mu_{CR} = 0.047$	
Diversity = 1.4358e+00	k_t = [3, 2, 2, 4]			
Generation 3030: Best = 3.862286e+03	FES = 303100	$\mu_F = 0.775$	$\mu_{CR} = 0.036$	
Diversity = 1.4419e+00	k_t = [1, 1, 1, 1]			
Generation 3040: Best = 3.862286e+03	FES = 304100	$\mu_F = 0.794$	$\mu_{CR} = 0.034$	
Diversity = 1.4410e+00	k_t = [2, 4, 2, 4]			
Generation 3050: Best = 3.862286e+03	FES = 305100	$\mu_F = 0.752$	$\mu_{CR} = 0.032$	
Diversity = 1.4417e+00	k_t = [1, 1, 1, 1]			
Generation 3060: Best = 3.862286e+03	FES = 306100	$\mu_F = 0.752$	$\mu_{CR} = 0.032$	
Diversity = 1.4570e+00	k_t = [2, 4, 4, 2]			
Generation 3070: Best = 3.862286e+03	FES = 307100	$\mu_F = 0.727$	$\mu_{CR} = 0.043$	
Diversity = 1.4554e+00	k_t = [1, 1, 1, 1]			
Generation 3080: Best = 3.862214e+03	FES = 308100	$\mu_F = 0.740$	$\mu_{CR} = 0.040$	
Diversity = 1.4492e+00	k_t = [2, 3, 2, 3]			
Generation 3090: Best = 3.862214e+03	FES = 309100	$\mu_F = 0.756$	$\mu_{CR} = 0.036$	
Diversity = 1.4404e+00	k_t = [1, 1, 1, 1]			
Generation 3100: Best = 3.861496e+03	FES = 310100	$\mu_F = 0.790$	$\mu_{CR} = 0.030$	
Diversity = 1.4374e+00	k_t = [2, 4, 4, 4]			
Generation 3110: Best = 3.861496e+03	FES = 311100	$\mu_F = 0.809$	$\mu_{CR} = 0.030$	
Diversity = 1.4375e+00	k_t = [1, 1, 1, 1]			
Generation 3120: Best = 3.861173e+03	FES = 312100	$\mu_F = 0.808$	$\mu_{CR} = 0.032$	
Diversity = 1.4395e+00	k_t = [4, 4, 4, 3]			
Generation 3130: Best = 3.861144e+03	FES = 313100	$\mu_F = 0.773$	$\mu_{CR} = 0.028$	
Diversity = 1.4432e+00	k_t = [1, 1, 1, 1]			
Generation 3140: Best = 3.861144e+03	FES = 314100	$\mu_F = 0.764$	$\mu_{CR} = 0.028$	
Diversity = 1.4549e+00	k_t = [3, 4, 4, 2]			
Generation 3150: Best = 3.861126e+03	FES = 315100	$\mu_F = 0.760$	$\mu_{CR} = 0.029$	
Diversity = 1.4569e+00	k_t = [1, 1, 1, 1]			
Generation 3160: Best = 3.861079e+03	FES = 316100	$\mu_F = 0.774$	$\mu_{CR} = 0.030$	
Diversity = 1.4429e+00	k_t = [3, 3, 3, 4]			
Generation 3170: Best = 3.860998e+03	FES = 317100	$\mu_F = 0.782$	$\mu_{CR} = 0.032$	
Diversity = 1.4301e+00	k_t = [1, 1, 1, 1]			
Generation 3180: Best = 3.860998e+03	FES = 318100	$\mu_F = 0.791$	$\mu_{CR} = 0.040$	
Diversity = 1.4389e+00	k_t = [4, 4, 3, 2]			
Generation 3190: Best = 3.860998e+03	FES = 319100	$\mu_F = 0.778$	$\mu_{CR} = 0.040$	

Diversity = 1.4355e+00	k_t = [1, 1, 1, 1]			
Generation 3200: Best = 3.860998e+03	FES = 320100	$\mu_F = 0.793$	$\mu_{CR} = 0.040$	
Diversity = 1.4064e+00	k_t = [3, 4, 4, 4]			
Generation 3210: Best = 3.860998e+03	FES = 321100	$\mu_F = 0.823$	$\mu_{CR} = 0.033$	
Diversity = 1.4168e+00	k_t = [1, 1, 1, 1]			
Generation 3220: Best = 3.860998e+03	FES = 322100	$\mu_F = 0.850$	$\mu_{CR} = 0.032$	
Diversity = 1.4188e+00	k_t = [4, 4, 4, 2]			
Generation 3230: Best = 3.860920e+03	FES = 323100	$\mu_F = 0.767$	$\mu_{CR} = 0.033$	
Diversity = 1.4204e+00	k_t = [1, 1, 1, 1]			
Generation 3240: Best = 3.860920e+03	FES = 324100	$\mu_F = 0.714$	$\mu_{CR} = 0.033$	
Diversity = 1.4266e+00	k_t = [2, 3, 2, 3]			
Generation 3250: Best = 3.860842e+03	FES = 325100	$\mu_F = 0.677$	$\mu_{CR} = 0.035$	
Diversity = 1.4213e+00	k_t = [1, 1, 1, 1]			
Generation 3260: Best = 3.860842e+03	FES = 326100	$\mu_F = 0.722$	$\mu_{CR} = 0.044$	
Diversity = 1.4222e+00	k_t = [4, 2, 4, 4]			
Generation 3270: Best = 3.860716e+03	FES = 327100	$\mu_F = 0.741$	$\mu_{CR} = 0.044$	
Diversity = 1.4219e+00	k_t = [1, 1, 1, 1]			
Generation 3280: Best = 3.860716e+03	FES = 328100	$\mu_F = 0.757$	$\mu_{CR} = 0.038$	
Diversity = 1.4244e+00	k_t = [3, 2, 4, 3]			
Generation 3290: Best = 3.860716e+03	FES = 329100	$\mu_F = 0.773$	$\mu_{CR} = 0.034$	
Diversity = 1.4378e+00	k_t = [1, 1, 1, 1]			
Generation 3300: Best = 3.860452e+03	FES = 330100	$\mu_F = 0.712$	$\mu_{CR} = 0.035$	
Diversity = 1.4331e+00	k_t = [2, 2, 2, 3]			
Generation 3310: Best = 3.860281e+03	FES = 331100	$\mu_F = 0.686$	$\mu_{CR} = 0.040$	
Diversity = 1.4405e+00	k_t = [1, 1, 1, 1]			
Generation 3320: Best = 3.860203e+03	FES = 332100	$\mu_F = 0.656$	$\mu_{CR} = 0.034$	
Diversity = 1.4429e+00	k_t = [3, 2, 4, 3]			
Generation 3330: Best = 3.860203e+03	FES = 333100	$\mu_F = 0.678$	$\mu_{CR} = 0.029$	
Diversity = 1.4429e+00	k_t = [1, 1, 1, 1]			
Generation 3340: Best = 3.860203e+03	FES = 334100	$\mu_F = 0.628$	$\mu_{CR} = 0.038$	
Diversity = 1.4430e+00	k_t = [4, 4, 3, 2]			
Generation 3350: Best = 3.860088e+03	FES = 335100	$\mu_F = 0.685$	$\mu_{CR} = 0.027$	
Diversity = 1.4433e+00	k_t = [1, 1, 1, 1]			
Generation 3360: Best = 3.860088e+03	FES = 336100	$\mu_F = 0.709$	$\mu_{CR} = 0.029$	
Diversity = 1.4403e+00	k_t = [3, 3, 3, 4]			
Generation 3370: Best = 3.860088e+03	FES = 337100	$\mu_F = 0.725$	$\mu_{CR} = 0.031$	
Diversity = 1.4385e+00	k_t = [1, 1, 1, 1]			
Generation 3380: Best = 3.860088e+03	FES = 338100	$\mu_F = 0.716$	$\mu_{CR} = 0.033$	
Diversity = 1.4374e+00	k_t = [2, 3, 4, 2]			
Generation 3390: Best = 3.859619e+03	FES = 339100	$\mu_F = 0.709$	$\mu_{CR} = 0.041$	
Diversity = 1.4371e+00	k_t = [1, 1, 1, 1]			
Generation 3400: Best = 3.859306e+03	FES = 340100	$\mu_F = 0.731$	$\mu_{CR} = 0.042$	
Diversity = 1.4157e+00	k_t = [3, 2, 3, 4]			
Generation 3410: Best = 3.859004e+03	FES = 341100	$\mu_F = 0.728$	$\mu_{CR} = 0.049$	
Diversity = 1.4246e+00	k_t = [1, 1, 1, 1]			
Generation 3420: Best = 3.859004e+03	FES = 342100	$\mu_F = 0.759$	$\mu_{CR} = 0.042$	
Diversity = 1.4111e+00	k_t = [3, 4, 3, 2]			
Generation 3430: Best = 3.859004e+03	FES = 343100	$\mu_F = 0.772$	$\mu_{CR} = 0.038$	
Diversity = 1.4278e+00	k_t = [1, 1, 1, 1]			
Generation 3440: Best = 3.859004e+03	FES = 344100	$\mu_F = 0.787$	$\mu_{CR} = 0.038$	
Diversity = 1.4308e+00	k_t = [4, 4, 3, 3]			
Generation 3450: Best = 3.859004e+03	FES = 345100	$\mu_F = 0.718$	$\mu_{CR} = 0.038$	
Diversity = 1.4302e+00	k_t = [1, 1, 1, 1]			
Generation 3460: Best = 3.858943e+03	FES = 346100	$\mu_F = 0.733$	$\mu_{CR} = 0.034$	

Diversity = 1.4293e+00	k_t = [3, 3, 3, 2]			
Generation 3470: Best = 3.858943e+03	FES = 347100	$\mu_F = 0.728$	$\mu_{CR} = 0.038$	
Diversity = 1.4230e+00	k_t = [1, 1, 1, 1]			
Generation 3480: Best = 3.858943e+03	FES = 348100	$\mu_F = 0.728$	$\mu_{CR} = 0.029$	
Diversity = 1.4222e+00	k_t = [4, 3, 4, 2]			
Generation 3490: Best = 3.858922e+03	FES = 349100	$\mu_F = 0.749$	$\mu_{CR} = 0.033$	
Diversity = 1.4234e+00	k_t = [1, 1, 1, 1]			
Generation 3500: Best = 3.858868e+03	FES = 350100	$\mu_F = 0.653$	$\mu_{CR} = 0.037$	
Diversity = 1.4200e+00	k_t = [3, 3, 2, 2]			
Generation 3510: Best = 3.858737e+03	FES = 351100	$\mu_F = 0.605$	$\mu_{CR} = 0.036$	
Diversity = 1.4173e+00	k_t = [1, 1, 1, 1]			
Generation 3520: Best = 3.858590e+03	FES = 352100	$\mu_F = 0.682$	$\mu_{CR} = 0.039$	
Diversity = 1.4170e+00	k_t = [4, 4, 4, 2]			
Generation 3530: Best = 3.858585e+03	FES = 353100	$\mu_F = 0.688$	$\mu_{CR} = 0.030$	
Diversity = 1.4176e+00	k_t = [1, 1, 1, 1]			
Generation 3540: Best = 3.858585e+03	FES = 354100	$\mu_F = 0.705$	$\mu_{CR} = 0.034$	
Diversity = 1.4119e+00	k_t = [3, 3, 3, 4]			
Generation 3550: Best = 3.858585e+03	FES = 355100	$\mu_F = 0.732$	$\mu_{CR} = 0.042$	
Diversity = 1.3932e+00	k_t = [1, 1, 1, 1]			
Generation 3560: Best = 3.858161e+03	FES = 356100	$\mu_F = 0.721$	$\mu_{CR} = 0.042$	
Diversity = 1.3840e+00	k_t = [2, 3, 2, 4]			
Generation 3570: Best = 3.858161e+03	FES = 357100	$\mu_F = 0.744$	$\mu_{CR} = 0.038$	
Diversity = 1.3708e+00	k_t = [1, 1, 1, 1]			
Generation 3580: Best = 3.858099e+03	FES = 358100	$\mu_F = 0.787$	$\mu_{CR} = 0.032$	
Diversity = 1.3802e+00	k_t = [3, 2, 4, 3]			
Generation 3590: Best = 3.858099e+03	FES = 359100	$\mu_F = 0.741$	$\mu_{CR} = 0.033$	
Diversity = 1.4063e+00	k_t = [1, 1, 1, 1]			
Generation 3600: Best = 3.858058e+03	FES = 360100	$\mu_F = 0.796$	$\mu_{CR} = 0.046$	
Diversity = 1.3914e+00	k_t = [4, 4, 2, 4]			
Generation 3610: Best = 3.858058e+03	FES = 361100	$\mu_F = 0.662$	$\mu_{CR} = 0.039$	
Diversity = 1.3931e+00	k_t = [1, 1, 1, 1]			
Generation 3620: Best = 3.858058e+03	FES = 362100	$\mu_F = 0.603$	$\mu_{CR} = 0.039$	
Diversity = 1.3916e+00	k_t = [4, 3, 2, 4]			
Generation 3630: Best = 3.858058e+03	FES = 363100	$\mu_F = 0.587$	$\mu_{CR} = 0.038$	
Diversity = 1.3921e+00	k_t = [1, 1, 1, 1]			
Generation 3640: Best = 3.857989e+03	FES = 364100	$\mu_F = 0.609$	$\mu_{CR} = 0.031$	
Diversity = 1.3908e+00	k_t = [2, 2, 2, 3]			
Generation 3650: Best = 3.857989e+03	FES = 365100	$\mu_F = 0.640$	$\mu_{CR} = 0.026$	
Diversity = 1.3822e+00	k_t = [1, 1, 1, 1]			
Generation 3660: Best = 3.857989e+03	FES = 366100	$\mu_F = 0.705$	$\mu_{CR} = 0.025$	
Diversity = 1.3779e+00	k_t = [2, 4, 3, 2]			
Generation 3670: Best = 3.857989e+03	FES = 367100	$\mu_F = 0.738$	$\mu_{CR} = 0.029$	
Diversity = 1.3835e+00	k_t = [1, 1, 1, 1]			
Generation 3680: Best = 3.857957e+03	FES = 368100	$\mu_F = 0.670$	$\mu_{CR} = 0.031$	
Diversity = 1.3741e+00	k_t = [4, 2, 3, 3]			
Generation 3690: Best = 3.857853e+03	FES = 369100	$\mu_F = 0.658$	$\mu_{CR} = 0.029$	
Diversity = 1.3737e+00	k_t = [1, 1, 1, 1]			
Generation 3700: Best = 3.857853e+03	FES = 370100	$\mu_F = 0.658$	$\mu_{CR} = 0.030$	
Diversity = 1.3740e+00	k_t = [4, 4, 2, 2]			
Generation 3710: Best = 3.857853e+03	FES = 371100	$\mu_F = 0.697$	$\mu_{CR} = 0.027$	
Diversity = 1.3740e+00	k_t = [1, 1, 1, 1]			
Generation 3720: Best = 3.857853e+03	FES = 372100	$\mu_F = 0.755$	$\mu_{CR} = 0.032$	
Diversity = 1.3685e+00	k_t = [4, 4, 2, 4]			
Generation 3730: Best = 3.857853e+03	FES = 373100	$\mu_F = 0.759$	$\mu_{CR} = 0.025$	

Diversity = 1.3763e+00	k_t = [1, 1, 1, 1]			
Generation 3740: Best = 3.857853e+03	FES = 374100	$\mu_F = 0.733$	$\mu_{CR} = 0.027$	
Diversity = 1.3762e+00	k_t = [2, 3, 3, 2]			
Generation 3750: Best = 3.857853e+03	FES = 375100	$\mu_F = 0.794$	$\mu_{CR} = 0.030$	
Diversity = 1.3751e+00	k_t = [1, 1, 1, 1]			
Generation 3760: Best = 3.857853e+03	FES = 376100	$\mu_F = 0.798$	$\mu_{CR} = 0.028$	
Diversity = 1.3722e+00	k_t = [4, 4, 2, 2]			
Generation 3770: Best = 3.857853e+03	FES = 377100	$\mu_F = 0.775$	$\mu_{CR} = 0.022$	
Diversity = 1.3715e+00	k_t = [1, 1, 1, 1]			
Generation 3780: Best = 3.857834e+03	FES = 378100	$\mu_F = 0.709$	$\mu_{CR} = 0.027$	
Diversity = 1.3727e+00	k_t = [3, 4, 2, 4]			
Generation 3790: Best = 3.857664e+03	FES = 379100	$\mu_F = 0.746$	$\mu_{CR} = 0.022$	
Diversity = 1.3810e+00	k_t = [1, 1, 1, 1]			
Generation 3800: Best = 3.857649e+03	FES = 380100	$\mu_F = 0.780$	$\mu_{CR} = 0.025$	
Diversity = 1.3655e+00	k_t = [3, 3, 2, 4]			
Generation 3810: Best = 3.857649e+03	FES = 381100	$\mu_F = 0.693$	$\mu_{CR} = 0.027$	
Diversity = 1.3626e+00	k_t = [1, 1, 1, 1]			
Generation 3820: Best = 3.857512e+03	FES = 382100	$\mu_F = 0.648$	$\mu_{CR} = 0.038$	
Diversity = 1.3704e+00	k_t = [4, 4, 4, 3]			
Generation 3830: Best = 3.857196e+03	FES = 383100	$\mu_F = 0.678$	$\mu_{CR} = 0.036$	
Diversity = 1.3594e+00	k_t = [1, 1, 1, 1]			
Generation 3840: Best = 3.857196e+03	FES = 384100	$\mu_F = 0.709$	$\mu_{CR} = 0.030$	
Diversity = 1.3395e+00	k_t = [3, 4, 2, 4]			
Generation 3850: Best = 3.857196e+03	FES = 385100	$\mu_F = 0.685$	$\mu_{CR} = 0.028$	
Diversity = 1.3405e+00	k_t = [1, 1, 1, 1]			
Generation 3860: Best = 3.857196e+03	FES = 386100	$\mu_F = 0.783$	$\mu_{CR} = 0.027$	
Diversity = 1.3566e+00	k_t = [2, 4, 3, 4]			
Generation 3870: Best = 3.857196e+03	FES = 387100	$\mu_F = 0.798$	$\mu_{CR} = 0.030$	
Diversity = 1.3648e+00	k_t = [1, 1, 1, 1]			
Generation 3880: Best = 3.856686e+03	FES = 388100	$\mu_F = 0.760$	$\mu_{CR} = 0.028$	
Diversity = 1.3682e+00	k_t = [3, 2, 2, 2]			
Generation 3890: Best = 3.856686e+03	FES = 389100	$\mu_F = 0.746$	$\mu_{CR} = 0.038$	
Diversity = 1.3718e+00	k_t = [1, 1, 1, 1]			
Generation 3900: Best = 3.856686e+03	FES = 390100	$\mu_F = 0.765$	$\mu_{CR} = 0.031$	
Diversity = 1.3643e+00	k_t = [4, 4, 2, 3]			
Generation 3910: Best = 3.856640e+03	FES = 391100	$\mu_F = 0.764$	$\mu_{CR} = 0.030$	
Diversity = 1.3644e+00	k_t = [1, 1, 1, 1]			
Generation 3920: Best = 3.856605e+03	FES = 392100	$\mu_F = 0.735$	$\mu_{CR} = 0.021$	
Diversity = 1.3605e+00	k_t = [3, 2, 2, 3]			
Generation 3930: Best = 3.856449e+03	FES = 393100	$\mu_F = 0.724$	$\mu_{CR} = 0.028$	
Diversity = 1.3607e+00	k_t = [1, 1, 1, 1]			
Generation 3940: Best = 3.856449e+03	FES = 394100	$\mu_F = 0.763$	$\mu_{CR} = 0.028$	
Diversity = 1.3599e+00	k_t = [3, 3, 3, 2]			
Generation 3950: Best = 3.856226e+03	FES = 395100	$\mu_F = 0.675$	$\mu_{CR} = 0.038$	
Diversity = 1.3602e+00	k_t = [1, 1, 1, 1]			
Generation 3960: Best = 3.856226e+03	FES = 396100	$\mu_F = 0.675$	$\mu_{CR} = 0.045$	
Diversity = 1.3560e+00	k_t = [2, 2, 2, 4]			
Generation 3970: Best = 3.856064e+03	FES = 397100	$\mu_F = 0.669$	$\mu_{CR} = 0.043$	
Diversity = 1.3534e+00	k_t = [1, 1, 1, 1]			
Generation 3980: Best = 3.856064e+03	FES = 398100	$\mu_F = 0.691$	$\mu_{CR} = 0.038$	
Diversity = 1.3340e+00	k_t = [3, 4, 2, 2]			
Generation 3990: Best = 3.855992e+03	FES = 399100	$\mu_F = 0.709$	$\mu_{CR} = 0.032$	
Diversity = 1.3412e+00	k_t = [1, 1, 1, 1]			
Generation 4000: Best = 3.855992e+03	FES = 400100	$\mu_F = 0.702$	$\mu_{CR} = 0.025$	

Diversity = 1.3319e+00	k_t = [4, 4, 4, 2]			
Generation 4010: Best = 3.855979e+03	FES = 401100	$\mu_F = 0.713$	$\mu_{CR} = 0.033$	
Diversity = 1.3331e+00	k_t = [1, 1, 1, 1]			
Generation 4020: Best = 3.855979e+03	FES = 402100	$\mu_F = 0.720$	$\mu_{CR} = 0.034$	
Diversity = 1.3391e+00	k_t = [3, 2, 2, 4]			
Generation 4030: Best = 3.855925e+03	FES = 403100	$\mu_F = 0.735$	$\mu_{CR} = 0.033$	
Diversity = 1.3262e+00	k_t = [1, 1, 1, 1]			
Generation 4040: Best = 3.855565e+03	FES = 404100	$\mu_F = 0.629$	$\mu_{CR} = 0.038$	
Diversity = 1.3203e+00	k_t = [2, 4, 2, 2]			
Generation 4050: Best = 3.855503e+03	FES = 405100	$\mu_F = 0.673$	$\mu_{CR} = 0.035$	
Diversity = 1.3092e+00	k_t = [1, 1, 1, 1]			
Generation 4060: Best = 3.855503e+03	FES = 406100	$\mu_F = 0.704$	$\mu_{CR} = 0.041$	
Diversity = 1.2673e+00	k_t = [3, 3, 3, 4]			
Generation 4070: Best = 3.855503e+03	FES = 407100	$\mu_F = 0.736$	$\mu_{CR} = 0.037$	
Diversity = 1.2659e+00	k_t = [1, 1, 1, 1]			
Generation 4080: Best = 3.855334e+03	FES = 408100	$\mu_F = 0.776$	$\mu_{CR} = 0.038$	
Diversity = 1.2143e+00	k_t = [3, 2, 3, 4]			
Generation 4090: Best = 3.855316e+03	FES = 409100	$\mu_F = 0.776$	$\mu_{CR} = 0.042$	
Diversity = 1.2217e+00	k_t = [1, 1, 1, 1]			
Generation 4100: Best = 3.855307e+03	FES = 410100	$\mu_F = 0.784$	$\mu_{CR} = 0.055$	
Diversity = 1.1934e+00	k_t = [4, 4, 4, 3]			
Generation 4110: Best = 3.855193e+03	FES = 411100	$\mu_F = 0.779$	$\mu_{CR} = 0.041$	
Diversity = 1.1675e+00	k_t = [1, 1, 1, 1]			
Generation 4120: Best = 3.855009e+03	FES = 412100	$\mu_F = 0.791$	$\mu_{CR} = 0.041$	
Diversity = 1.1388e+00	k_t = [3, 2, 2, 3]			
Generation 4130: Best = 3.855009e+03	FES = 413100	$\mu_F = 0.785$	$\mu_{CR} = 0.043$	
Diversity = 1.1005e+00	k_t = [1, 1, 1, 1]			
Generation 4140: Best = 3.854531e+03	FES = 414100	$\mu_F = 0.814$	$\mu_{CR} = 0.046$	
Diversity = 1.0304e+00	k_t = [3, 4, 2, 3]			
Generation 4150: Best = 3.854531e+03	FES = 415100	$\mu_F = 0.811$	$\mu_{CR} = 0.046$	
Diversity = 9.8745e-01	k_t = [1, 1, 1, 1]			
Generation 4160: Best = 3.854531e+03	FES = 416100	$\mu_F = 0.795$	$\mu_{CR} = 0.045$	
Diversity = 9.3882e-01	k_t = [3, 3, 4, 3]			
Generation 4170: Best = 3.853857e+03	FES = 417100	$\mu_F = 0.777$	$\mu_{CR} = 0.055$	
Diversity = 8.3188e-01	k_t = [1, 1, 1, 1]			
Generation 4180: Best = 3.853436e+03	FES = 418100	$\mu_F = 0.759$	$\mu_{CR} = 0.059$	
Diversity = 7.8320e-01	k_t = [3, 3, 4, 3]			
Generation 4190: Best = 3.853356e+03	FES = 419100	$\mu_F = 0.773$	$\mu_{CR} = 0.063$	
Diversity = 7.2001e-01	k_t = [1, 1, 1, 1]			
Generation 4200: Best = 3.853154e+03	FES = 420100	$\mu_F = 0.769$	$\mu_{CR} = 0.062$	
Diversity = 7.2699e-01	k_t = [4, 2, 4, 2]			
Generation 4210: Best = 3.852950e+03	FES = 421100	$\mu_F = 0.742$	$\mu_{CR} = 0.066$	
Diversity = 6.9536e-01	k_t = [1, 1, 1, 1]			
Generation 4220: Best = 3.852950e+03	FES = 422100	$\mu_F = 0.731$	$\mu_{CR} = 0.061$	
Diversity = 6.9369e-01	k_t = [2, 4, 3, 4]			
Generation 4230: Best = 3.852424e+03	FES = 423100	$\mu_F = 0.731$	$\mu_{CR} = 0.061$	
Diversity = 7.0303e-01	k_t = [1, 1, 1, 1]			
Generation 4240: Best = 3.852264e+03	FES = 424100	$\mu_F = 0.754$	$\mu_{CR} = 0.068$	
Diversity = 7.0523e-01	k_t = [3, 2, 2, 3]			
Generation 4250: Best = 3.851796e+03	FES = 425100	$\mu_F = 0.756$	$\mu_{CR} = 0.076$	
Diversity = 6.8211e-01	k_t = [1, 1, 1, 1]			
Generation 4260: Best = 3.851671e+03	FES = 426100	$\mu_F = 0.746$	$\mu_{CR} = 0.075$	
Diversity = 6.4519e-01	k_t = [2, 4, 3, 2]			
Generation 4270: Best = 3.851450e+03	FES = 427100	$\mu_F = 0.730$	$\mu_{CR} = 0.068$	

Diversity = 5.8552e-01	k_t = [1, 1, 1, 1]			
Generation 4280: Best = 3.851036e+03	FES = 428100	$\mu_F = 0.734$	$\mu_{CR} = 0.076$	
Diversity = 5.5829e-01	k_t = [2, 2, 2, 4]			
Generation 4290: Best = 3.850713e+03	FES = 429100	$\mu_F = 0.716$	$\mu_{CR} = 0.070$	
Diversity = 4.6194e-01	k_t = [1, 1, 1, 1]			
Generation 4300: Best = 3.850414e+03	FES = 430100	$\mu_F = 0.699$	$\mu_{CR} = 0.071$	
Diversity = 4.0607e-01	k_t = [3, 4, 4, 4]			
Generation 4310: Best = 3.850263e+03	FES = 431100	$\mu_F = 0.719$	$\mu_{CR} = 0.064$	
Diversity = 3.6415e-01	k_t = [1, 1, 1, 1]			
Generation 4320: Best = 3.850263e+03	FES = 432100	$\mu_F = 0.723$	$\mu_{CR} = 0.057$	
Diversity = 3.2994e-01	k_t = [3, 2, 3, 3]			
Generation 4330: Best = 3.850239e+03	FES = 433100	$\mu_F = 0.726$	$\mu_{CR} = 0.055$	
Diversity = 3.3520e-01	k_t = [1, 1, 1, 1]			
Generation 4340: Best = 3.850035e+03	FES = 434100	$\mu_F = 0.709$	$\mu_{CR} = 0.061$	
Diversity = 3.5686e-01	k_t = [2, 2, 2, 3]			
Generation 4350: Best = 3.849883e+03	FES = 435100	$\mu_F = 0.733$	$\mu_{CR} = 0.053$	
Diversity = 3.3925e-01	k_t = [1, 1, 1, 1]			
Generation 4360: Best = 3.849883e+03	FES = 436100	$\mu_F = 0.752$	$\mu_{CR} = 0.049$	
Diversity = 3.3261e-01	k_t = [3, 4, 2, 4]			
Generation 4370: Best = 3.849670e+03	FES = 437100	$\mu_F = 0.778$	$\mu_{CR} = 0.053$	
Diversity = 3.5934e-01	k_t = [1, 1, 1, 1]			
Generation 4380: Best = 3.849551e+03	FES = 438100	$\mu_F = 0.772$	$\mu_{CR} = 0.045$	
Diversity = 4.1948e-01	k_t = [3, 3, 2, 2]			
Generation 4390: Best = 3.849546e+03	FES = 439100	$\mu_F = 0.773$	$\mu_{CR} = 0.036$	
Diversity = 4.2073e-01	k_t = [1, 1, 1, 1]			
Generation 4400: Best = 3.849450e+03	FES = 440100	$\mu_F = 0.794$	$\mu_{CR} = 0.041$	
Diversity = 4.1277e-01	k_t = [4, 3, 4, 3]			
Generation 4410: Best = 3.849258e+03	FES = 441100	$\mu_F = 0.788$	$\mu_{CR} = 0.051$	
Diversity = 4.5554e-01	k_t = [1, 1, 1, 1]			
Generation 4420: Best = 3.849232e+03	FES = 442100	$\mu_F = 0.777$	$\mu_{CR} = 0.053$	
Diversity = 4.6100e-01	k_t = [3, 2, 2, 2]			
Generation 4430: Best = 3.848939e+03	FES = 443100	$\mu_F = 0.783$	$\mu_{CR} = 0.047$	
Diversity = 4.3829e-01	k_t = [1, 1, 1, 1]			
Generation 4440: Best = 3.848939e+03	FES = 444100	$\mu_F = 0.771$	$\mu_{CR} = 0.056$	
Diversity = 4.5541e-01	k_t = [4, 2, 4, 3]			
Generation 4450: Best = 3.848939e+03	FES = 445100	$\mu_F = 0.780$	$\mu_{CR} = 0.052$	
Diversity = 4.8193e-01	k_t = [1, 1, 1, 1]			
Generation 4460: Best = 3.848897e+03	FES = 446100	$\mu_F = 0.766$	$\mu_{CR} = 0.052$	
Diversity = 4.7691e-01	k_t = [3, 4, 4, 4]			
Generation 4470: Best = 3.848624e+03	FES = 447100	$\mu_F = 0.763$	$\mu_{CR} = 0.055$	
Diversity = 4.7805e-01	k_t = [1, 1, 1, 1]			
Generation 4480: Best = 3.848497e+03	FES = 448100	$\mu_F = 0.748$	$\mu_{CR} = 0.056$	
Diversity = 4.7957e-01	k_t = [3, 3, 2, 4]			
Generation 4490: Best = 3.848460e+03	FES = 449100	$\mu_F = 0.735$	$\mu_{CR} = 0.058$	
Diversity = 4.4636e-01	k_t = [1, 1, 1, 1]			
Generation 4500: Best = 3.848414e+03	FES = 450100	$\mu_F = 0.762$	$\mu_{CR} = 0.056$	
Diversity = 4.3784e-01	k_t = [2, 2, 4, 4]			
Generation 4510: Best = 3.848405e+03	FES = 451100	$\mu_F = 0.762$	$\mu_{CR} = 0.045$	
Diversity = 4.1518e-01	k_t = [1, 1, 1, 1]			
Generation 4520: Best = 3.848176e+03	FES = 452100	$\mu_F = 0.771$	$\mu_{CR} = 0.051$	
Diversity = 3.9448e-01	k_t = [2, 3, 4, 4]			
Generation 4530: Best = 3.848176e+03	FES = 453100	$\mu_F = 0.793$	$\mu_{CR} = 0.062$	
Diversity = 3.8727e-01	k_t = [1, 1, 1, 1]			
Generation 4540: Best = 3.848085e+03	FES = 454100	$\mu_F = 0.770$	$\mu_{CR} = 0.065$	

Diversity = 3.8685e-01 k_t = [4, 2, 2, 3]
 Generation 4550: Best = 3.848075e+03 FES = 455100 μ_F = 0.765 μ_{CR} = 0.065
 Diversity = 3.8933e-01 k_t = [1, 1, 1, 1]
 Generation 4560: Best = 3.847997e+03 FES = 456100 μ_F = 0.788 μ_{CR} = 0.057
 Diversity = 3.6881e-01 k_t = [3, 3, 3, 4]
 Generation 4570: Best = 3.847997e+03 FES = 457100 μ_F = 0.774 μ_{CR} = 0.049
 Diversity = 3.6725e-01 k_t = [1, 1, 1, 1]
 Generation 4580: Best = 3.847805e+03 FES = 458100 μ_F = 0.768 μ_{CR} = 0.046
 Diversity = 3.5587e-01 k_t = [2, 3, 4, 3]
 Generation 4590: Best = 3.847779e+03 FES = 459100 μ_F = 0.784 μ_{CR} = 0.045
 Diversity = 3.5615e-01 k_t = [1, 1, 1, 1]
 Generation 4600: Best = 3.847727e+03 FES = 460100 μ_F = 0.763 μ_{CR} = 0.048
 Diversity = 3.5115e-01 k_t = [2, 4, 4, 4]
 Generation 4610: Best = 3.847611e+03 FES = 461100 μ_F = 0.773 μ_{CR} = 0.043
 Diversity = 3.4199e-01 k_t = [1, 1, 1, 1]
 Generation 4620: Best = 3.847611e+03 FES = 462100 μ_F = 0.783 μ_{CR} = 0.045
 Diversity = 3.4419e-01 k_t = [4, 3, 4, 4]
 Generation 4630: Best = 3.847607e+03 FES = 463100 μ_F = 0.819 μ_{CR} = 0.038
 Diversity = 3.4415e-01 k_t = [1, 1, 1, 1]
 Generation 4640: Best = 3.847531e+03 FES = 464100 μ_F = 0.805 μ_{CR} = 0.037
 Diversity = 3.4611e-01 k_t = [4, 3, 4, 2]
 Generation 4650: Best = 3.847518e+03 FES = 465100 μ_F = 0.805 μ_{CR} = 0.047
 Diversity = 3.7630e-01 k_t = [1, 1, 1, 1]
 Generation 4660: Best = 3.847479e+03 FES = 466100 μ_F = 0.822 μ_{CR} = 0.037
 Diversity = 3.9222e-01 k_t = [4, 2, 2, 3]
 Generation 4670: Best = 3.847461e+03 FES = 467100 μ_F = 0.825 μ_{CR} = 0.044
 Diversity = 3.9404e-01 k_t = [1, 1, 1, 1]
 Generation 4680: Best = 3.847110e+03 FES = 468100 μ_F = 0.827 μ_{CR} = 0.055
 Diversity = 4.0863e-01 k_t = [3, 2, 2, 2]
 Generation 4690: Best = 3.847086e+03 FES = 469100 μ_F = 0.814 μ_{CR} = 0.053
 Diversity = 4.3649e-01 k_t = [1, 1, 1, 1]
 Generation 4700: Best = 3.847075e+03 FES = 470100 μ_F = 0.805 μ_{CR} = 0.057
 Diversity = 4.3918e-01 k_t = [4, 3, 4, 2]
 Generation 4710: Best = 3.846813e+03 FES = 471100 μ_F = 0.796 μ_{CR} = 0.064
 Diversity = 4.5230e-01 k_t = [1, 1, 1, 1]
 Generation 4720: Best = 3.846813e+03 FES = 472100 μ_F = 0.799 μ_{CR} = 0.058
 Diversity = 4.5150e-01 k_t = [3, 3, 4, 4]
 Generation 4730: Best = 3.846813e+03 FES = 473100 μ_F = 0.801 μ_{CR} = 0.050
 Diversity = 4.6623e-01 k_t = [1, 1, 1, 1]
 Generation 4740: Best = 3.846703e+03 FES = 474100 μ_F = 0.788 μ_{CR} = 0.046
 Diversity = 4.4257e-01 k_t = [3, 4, 3, 3]
 Generation 4750: Best = 3.846670e+03 FES = 475100 μ_F = 0.789 μ_{CR} = 0.047
 Diversity = 4.3728e-01 k_t = [1, 1, 1, 1]
 Generation 4760: Best = 3.846570e+03 FES = 476100 μ_F = 0.748 μ_{CR} = 0.050
 Diversity = 4.3823e-01 k_t = [3, 2, 4, 2]
 Generation 4770: Best = 3.846395e+03 FES = 477100 μ_F = 0.773 μ_{CR} = 0.050
 Diversity = 4.6228e-01 k_t = [1, 1, 1, 1]
 Generation 4780: Best = 3.846384e+03 FES = 478100 μ_F = 0.785 μ_{CR} = 0.043
 Diversity = 4.4579e-01 k_t = [3, 3, 4, 3]
 Generation 4790: Best = 3.846341e+03 FES = 479100 μ_F = 0.812 μ_{CR} = 0.052
 Diversity = 4.2899e-01 k_t = [1, 1, 1, 1]
 Generation 4800: Best = 3.846341e+03 FES = 480100 μ_F = 0.788 μ_{CR} = 0.052
 Diversity = 4.1767e-01 k_t = [2, 3, 3, 4]
 Generation 4810: Best = 3.846332e+03 FES = 481100 μ_F = 0.768 μ_{CR} = 0.048

Diversity = 4.2411e-01	k_t = [1, 1, 1, 1]			
Generation 4820: Best = 3.846205e+03	FES = 482100	$\mu_F = 0.768$	$\mu_{CR} = 0.051$	
Diversity = 4.2871e-01	k_t = [2, 3, 4, 3]			
Generation 4830: Best = 3.846189e+03	FES = 483100	$\mu_F = 0.762$	$\mu_{CR} = 0.049$	
Diversity = 4.4147e-01	k_t = [1, 1, 1, 1]			
Generation 4840: Best = 3.846045e+03	FES = 484100	$\mu_F = 0.748$	$\mu_{CR} = 0.053$	
Diversity = 4.4149e-01	k_t = [4, 2, 2, 2]			
Generation 4850: Best = 3.845966e+03	FES = 485100	$\mu_F = 0.757$	$\mu_{CR} = 0.050$	
Diversity = 4.5055e-01	k_t = [1, 1, 1, 1]			
Generation 4860: Best = 3.845966e+03	FES = 486100	$\mu_F = 0.759$	$\mu_{CR} = 0.049$	
Diversity = 4.3521e-01	k_t = [2, 3, 3, 3]			
Generation 4870: Best = 3.845896e+03	FES = 487100	$\mu_F = 0.740$	$\mu_{CR} = 0.048$	
Diversity = 4.4100e-01	k_t = [1, 1, 1, 1]			
Generation 4880: Best = 3.845780e+03	FES = 488100	$\mu_F = 0.724$	$\mu_{CR} = 0.049$	
Diversity = 4.2947e-01	k_t = [3, 2, 4, 2]			
Generation 4890: Best = 3.845780e+03	FES = 489100	$\mu_F = 0.759$	$\mu_{CR} = 0.044$	
Diversity = 4.4128e-01	k_t = [1, 1, 1, 1]			
Generation 4900: Best = 3.845689e+03	FES = 490100	$\mu_F = 0.765$	$\mu_{CR} = 0.048$	
Diversity = 4.3797e-01	k_t = [3, 2, 2, 2]			
Generation 4910: Best = 3.845689e+03	FES = 491100	$\mu_F = 0.772$	$\mu_{CR} = 0.039$	
Diversity = 4.2916e-01	k_t = [1, 1, 1, 1]			
Generation 4920: Best = 3.845689e+03	FES = 492100	$\mu_F = 0.749$	$\mu_{CR} = 0.040$	
Diversity = 4.3044e-01	k_t = [3, 3, 3, 3]			
Generation 4930: Best = 3.845606e+03	FES = 493100	$\mu_F = 0.746$	$\mu_{CR} = 0.038$	
Diversity = 4.1838e-01	k_t = [1, 1, 1, 1]			
Generation 4940: Best = 3.845543e+03	FES = 494100	$\mu_F = 0.743$	$\mu_{CR} = 0.042$	
Diversity = 4.1279e-01	k_t = [4, 4, 3, 4]			
Generation 4950: Best = 3.845404e+03	FES = 495100	$\mu_F = 0.727$	$\mu_{CR} = 0.053$	
Diversity = 4.1280e-01	k_t = [1, 1, 1, 1]			
Generation 4960: Best = 3.845404e+03	FES = 496100	$\mu_F = 0.733$	$\mu_{CR} = 0.046$	
Diversity = 4.1492e-01	k_t = [2, 3, 3, 4]			
Generation 4970: Best = 3.845317e+03	FES = 497100	$\mu_F = 0.730$	$\mu_{CR} = 0.052$	
Diversity = 4.0526e-01	k_t = [1, 1, 1, 1]			
Generation 4980: Best = 3.845314e+03	FES = 498100	$\mu_F = 0.694$	$\mu_{CR} = 0.051$	
Diversity = 4.0137e-01	k_t = [4, 2, 3, 2]			
Generation 4990: Best = 3.845269e+03	FES = 499100	$\mu_F = 0.686$	$\mu_{CR} = 0.047$	
Diversity = 4.1069e-01	k_t = [1, 1, 1, 1]			
Generation 5000: Best = 3.845261e+03	FES = 500100	$\mu_F = 0.733$	$\mu_{CR} = 0.050$	
Diversity = 4.0922e-01	k_t = [2, 2, 4, 4]			
Generation 5010: Best = 3.845203e+03	FES = 501100	$\mu_F = 0.736$	$\mu_{CR} = 0.046$	
Diversity = 4.0481e-01	k_t = [1, 1, 1, 1]			
Generation 5020: Best = 3.845138e+03	FES = 502100	$\mu_F = 0.744$	$\mu_{CR} = 0.046$	
Diversity = 4.0626e-01	k_t = [4, 4, 3, 2]			
Generation 5030: Best = 3.845121e+03	FES = 503100	$\mu_F = 0.743$	$\mu_{CR} = 0.042$	
Diversity = 4.0134e-01	k_t = [1, 1, 1, 1]			
Generation 5040: Best = 3.845121e+03	FES = 504100	$\mu_F = 0.765$	$\mu_{CR} = 0.041$	
Diversity = 4.1135e-01	k_t = [4, 2, 3, 4]			
Generation 5050: Best = 3.845115e+03	FES = 505100	$\mu_F = 0.727$	$\mu_{CR} = 0.051$	
Diversity = 4.0663e-01	k_t = [1, 1, 1, 1]			
Generation 5060: Best = 3.845105e+03	FES = 506100	$\mu_F = 0.758$	$\mu_{CR} = 0.039$	
Diversity = 3.9832e-01	k_t = [2, 2, 2, 4]			
Generation 5070: Best = 3.845085e+03	FES = 507100	$\mu_F = 0.765$	$\mu_{CR} = 0.036$	
Diversity = 3.9595e-01	k_t = [1, 1, 1, 1]			
Generation 5080: Best = 3.845046e+03	FES = 508100	$\mu_F = 0.757$	$\mu_{CR} = 0.038$	

Diversity = 4.0662e-01	k_t = [2, 4, 4, 3]			
Generation 5090: Best = 3.844953e+03	FES = 509100	$\mu_F = 0.759$	$\mu_{CR} = 0.040$	
Diversity = 4.0970e-01	k_t = [1, 1, 1, 1]			
Generation 5100: Best = 3.844953e+03	FES = 510100	$\mu_F = 0.738$	$\mu_{CR} = 0.037$	
Diversity = 4.1077e-01	k_t = [4, 4, 4, 2]			
Generation 5110: Best = 3.844953e+03	FES = 511100	$\mu_F = 0.710$	$\mu_{CR} = 0.037$	
Diversity = 4.1284e-01	k_t = [1, 1, 1, 1]			
Generation 5120: Best = 3.844939e+03	FES = 512100	$\mu_F = 0.732$	$\mu_{CR} = 0.035$	
Diversity = 4.1830e-01	k_t = [3, 2, 4, 2]			
Generation 5130: Best = 3.844937e+03	FES = 513100	$\mu_F = 0.728$	$\mu_{CR} = 0.035$	
Diversity = 4.1763e-01	k_t = [1, 1, 1, 1]			
Generation 5140: Best = 3.844921e+03	FES = 514100	$\mu_F = 0.742$	$\mu_{CR} = 0.035$	
Diversity = 3.9720e-01	k_t = [3, 4, 4, 4]			
Generation 5150: Best = 3.844921e+03	FES = 515100	$\mu_F = 0.762$	$\mu_{CR} = 0.031$	
Diversity = 3.9940e-01	k_t = [1, 1, 1, 1]			
Generation 5160: Best = 3.844888e+03	FES = 516100	$\mu_F = 0.725$	$\mu_{CR} = 0.038$	
Diversity = 3.9897e-01	k_t = [2, 3, 3, 2]			
Generation 5170: Best = 3.844888e+03	FES = 517100	$\mu_F = 0.786$	$\mu_{CR} = 0.038$	
Diversity = 3.9872e-01	k_t = [1, 1, 1, 1]			
Generation 5180: Best = 3.844731e+03	FES = 518100	$\mu_F = 0.780$	$\mu_{CR} = 0.034$	
Diversity = 3.9782e-01	k_t = [4, 2, 2, 3]			
Generation 5190: Best = 3.844731e+03	FES = 519100	$\mu_F = 0.770$	$\mu_{CR} = 0.032$	
Diversity = 3.9798e-01	k_t = [1, 1, 1, 1]			
Generation 5200: Best = 3.844730e+03	FES = 520100	$\mu_F = 0.806$	$\mu_{CR} = 0.034$	
Diversity = 3.9745e-01	k_t = [3, 3, 2, 3]			
Generation 5210: Best = 3.844727e+03	FES = 521100	$\mu_F = 0.813$	$\mu_{CR} = 0.035$	
Diversity = 4.0259e-01	k_t = [1, 1, 1, 1]			
Generation 5220: Best = 3.844727e+03	FES = 522100	$\mu_F = 0.811$	$\mu_{CR} = 0.028$	
Diversity = 4.0300e-01	k_t = [4, 3, 2, 3]			
Generation 5230: Best = 3.844664e+03	FES = 523100	$\mu_F = 0.811$	$\mu_{CR} = 0.027$	
Diversity = 4.0361e-01	k_t = [1, 1, 1, 1]			
Generation 5240: Best = 3.844633e+03	FES = 524100	$\mu_F = 0.808$	$\mu_{CR} = 0.027$	
Diversity = 4.1194e-01	k_t = [3, 2, 4, 2]			
Generation 5250: Best = 3.844633e+03	FES = 525100	$\mu_F = 0.763$	$\mu_{CR} = 0.030$	
Diversity = 4.1503e-01	k_t = [1, 1, 1, 1]			
Generation 5260: Best = 3.844633e+03	FES = 526100	$\mu_F = 0.732$	$\mu_{CR} = 0.033$	
Diversity = 4.1489e-01	k_t = [2, 4, 2, 3]			
Generation 5270: Best = 3.844633e+03	FES = 527100	$\mu_F = 0.778$	$\mu_{CR} = 0.033$	
Diversity = 4.1832e-01	k_t = [1, 1, 1, 1]			
Generation 5280: Best = 3.844623e+03	FES = 528100	$\mu_F = 0.780$	$\mu_{CR} = 0.028$	
Diversity = 4.1945e-01	k_t = [3, 4, 3, 4]			
Generation 5290: Best = 3.844623e+03	FES = 529100	$\mu_F = 0.728$	$\mu_{CR} = 0.033$	
Diversity = 4.1709e-01	k_t = [1, 1, 1, 1]			
Generation 5300: Best = 3.844622e+03	FES = 530100	$\mu_F = 0.757$	$\mu_{CR} = 0.032$	
Diversity = 4.2094e-01	k_t = [3, 4, 4, 3]			
Generation 5310: Best = 3.844525e+03	FES = 531100	$\mu_F = 0.751$	$\mu_{CR} = 0.030$	
Diversity = 4.2389e-01	k_t = [1, 1, 1, 1]			
Generation 5320: Best = 3.844525e+03	FES = 532100	$\mu_F = 0.725$	$\mu_{CR} = 0.036$	
Diversity = 4.2759e-01	k_t = [2, 4, 2, 3]			
Generation 5330: Best = 3.844457e+03	FES = 533100	$\mu_F = 0.691$	$\mu_{CR} = 0.031$	
Diversity = 4.2367e-01	k_t = [1, 1, 1, 1]			
Generation 5340: Best = 3.844457e+03	FES = 534100	$\mu_F = 0.673$	$\mu_{CR} = 0.033$	
Diversity = 4.2336e-01	k_t = [3, 2, 2, 3]			
Generation 5350: Best = 3.844457e+03	FES = 535100	$\mu_F = 0.664$	$\mu_{CR} = 0.035$	

Diversity = 4.2494e-01 k_t = [1, 1, 1, 1]
 Generation 5360: Best = 3.844457e+03 FES = 536100 μ_F = 0.697 μ_{CR} = 0.030
 Diversity = 4.3578e-01 k_t = [2, 3, 3, 4]
 Generation 5370: Best = 3.844457e+03 FES = 537100 μ_F = 0.733 μ_{CR} = 0.034
 Diversity = 4.3516e-01 k_t = [1, 1, 1, 1]
 Generation 5380: Best = 3.844441e+03 FES = 538100 μ_F = 0.751 μ_{CR} = 0.034
 Diversity = 4.3031e-01 k_t = [4, 2, 4, 2]
 Generation 5390: Best = 3.844435e+03 FES = 539100 μ_F = 0.767 μ_{CR} = 0.037
 Diversity = 4.3304e-01 k_t = [1, 1, 1, 1]
 Generation 5400: Best = 3.844435e+03 FES = 540100 μ_F = 0.770 μ_{CR} = 0.037
 Diversity = 4.3609e-01 k_t = [2, 3, 3, 4]
 Generation 5410: Best = 3.844435e+03 FES = 541100 μ_F = 0.795 μ_{CR} = 0.030
 Diversity = 4.4263e-01 k_t = [1, 1, 1, 1]
 Generation 5420: Best = 3.844435e+03 FES = 542100 μ_F = 0.811 μ_{CR} = 0.027
 Diversity = 4.6042e-01 k_t = [4, 2, 4, 2]
 Generation 5430: Best = 3.844435e+03 FES = 543100 μ_F = 0.816 μ_{CR} = 0.036
 Diversity = 4.5200e-01 k_t = [1, 1, 1, 1]
 Generation 5440: Best = 3.844435e+03 FES = 544100 μ_F = 0.790 μ_{CR} = 0.037
 Diversity = 4.3676e-01 k_t = [4, 4, 2, 4]
 Generation 5450: Best = 3.844434e+03 FES = 545100 μ_F = 0.763 μ_{CR} = 0.034
 Diversity = 4.4158e-01 k_t = [1, 1, 1, 1]
 Generation 5460: Best = 3.844434e+03 FES = 546100 μ_F = 0.779 μ_{CR} = 0.042
 Diversity = 4.3557e-01 k_t = [4, 4, 2, 4]
 Generation 5470: Best = 3.844434e+03 FES = 547100 μ_F = 0.787 μ_{CR} = 0.034
 Diversity = 4.3562e-01 k_t = [1, 1, 1, 1]
 Generation 5480: Best = 3.844432e+03 FES = 548100 μ_F = 0.661 μ_{CR} = 0.040
 Diversity = 4.3613e-01 k_t = [2, 2, 3, 3]
 Generation 5490: Best = 3.844432e+03 FES = 549100 μ_F = 0.668 μ_{CR} = 0.035
 Diversity = 4.2881e-01 k_t = [1, 1, 1, 1]
 Generation 5500: Best = 3.844432e+03 FES = 550100 μ_F = 0.730 μ_{CR} = 0.041
 Diversity = 4.2882e-01 k_t = [4, 4, 3, 2]
 Generation 5510: Best = 3.844432e+03 FES = 551100 μ_F = 0.710 μ_{CR} = 0.043
 Diversity = 4.2786e-01 k_t = [1, 1, 1, 1]
 Generation 5520: Best = 3.844432e+03 FES = 552100 μ_F = 0.737 μ_{CR} = 0.042
 Diversity = 4.2891e-01 k_t = [3, 3, 2, 4]
 Generation 5530: Best = 3.844432e+03 FES = 553100 μ_F = 0.733 μ_{CR} = 0.029
 Diversity = 4.2891e-01 k_t = [1, 1, 1, 1]
 Generation 5540: Best = 3.844432e+03 FES = 554100 μ_F = 0.644 μ_{CR} = 0.032
 Diversity = 4.2588e-01 k_t = [4, 4, 3, 4]
 Generation 5550: Best = 3.844429e+03 FES = 555100 μ_F = 0.672 μ_{CR} = 0.041
 Diversity = 4.2589e-01 k_t = [1, 1, 1, 1]
 Generation 5560: Best = 3.844423e+03 FES = 556100 μ_F = 0.659 μ_{CR} = 0.035
 Diversity = 4.2586e-01 k_t = [4, 3, 4, 3]
 Generation 5570: Best = 3.844423e+03 FES = 557100 μ_F = 0.664 μ_{CR} = 0.035
 Diversity = 4.2577e-01 k_t = [1, 1, 1, 1]
 Generation 5580: Best = 3.844421e+03 FES = 558100 μ_F = 0.672 μ_{CR} = 0.035
 Diversity = 4.2572e-01 k_t = [4, 2, 3, 2]
 Generation 5590: Best = 3.844417e+03 FES = 559100 μ_F = 0.701 μ_{CR} = 0.035
 Diversity = 4.2570e-01 k_t = [1, 1, 1, 1]
 Generation 5600: Best = 3.844403e+03 FES = 560100 μ_F = 0.681 μ_{CR} = 0.037
 Diversity = 4.2569e-01 k_t = [3, 2, 3, 3]
 Generation 5610: Best = 3.844403e+03 FES = 561100 μ_F = 0.680 μ_{CR} = 0.034
 Diversity = 4.2556e-01 k_t = [1, 1, 1, 1]
 Generation 5620: Best = 3.844403e+03 FES = 562100 μ_F = 0.717 μ_{CR} = 0.036

Diversity = 4.2281e-01 k_t = [2, 4, 2, 4]
 Generation 5630: Best = 3.844400e+03 FES = 563100 μ_F = 0.748 μ_{CR} = 0.036
 Diversity = 4.2280e-01 k_t = [1, 1, 1, 1]
 Generation 5640: Best = 3.844395e+03 FES = 564100 μ_F = 0.780 μ_{CR} = 0.034
 Diversity = 4.2250e-01 k_t = [3, 2, 3, 3]
 Generation 5650: Best = 3.844392e+03 FES = 565100 μ_F = 0.791 μ_{CR} = 0.027
 Diversity = 4.2260e-01 k_t = [1, 1, 1, 1]
 Generation 5660: Best = 3.844382e+03 FES = 566100 μ_F = 0.779 μ_{CR} = 0.033
 Diversity = 4.2252e-01 k_t = [2, 3, 4, 2]
 Generation 5670: Best = 3.844377e+03 FES = 567100 μ_F = 0.757 μ_{CR} = 0.029
 Diversity = 4.1935e-01 k_t = [1, 1, 1, 1]
 Generation 5680: Best = 3.844377e+03 FES = 568100 μ_F = 0.800 μ_{CR} = 0.032
 Diversity = 4.2447e-01 k_t = [4, 4, 2, 4]
 Generation 5690: Best = 3.844299e+03 FES = 569100 μ_F = 0.818 μ_{CR} = 0.030
 Diversity = 4.2447e-01 k_t = [1, 1, 1, 1]
 Generation 5700: Best = 3.844291e+03 FES = 570100 μ_F = 0.787 μ_{CR} = 0.028
 Diversity = 4.2536e-01 k_t = [4, 2, 3, 4]
 Generation 5710: Best = 3.844287e+03 FES = 571100 μ_F = 0.759 μ_{CR} = 0.020
 Diversity = 4.2407e-01 k_t = [1, 1, 1, 1]
 Generation 5720: Best = 3.844278e+03 FES = 572100 μ_F = 0.805 μ_{CR} = 0.028
 Diversity = 4.4469e-01 k_t = [3, 3, 3, 3]
 Generation 5730: Best = 3.844278e+03 FES = 573100 μ_F = 0.848 μ_{CR} = 0.027
 Diversity = 4.4524e-01 k_t = [1, 1, 1, 1]
 Generation 5740: Best = 3.844278e+03 FES = 574100 μ_F = 0.818 μ_{CR} = 0.030
 Diversity = 4.4842e-01 k_t = [2, 2, 2, 3]
 Generation 5750: Best = 3.844278e+03 FES = 575100 μ_F = 0.798 μ_{CR} = 0.029
 Diversity = 4.4841e-01 k_t = [1, 1, 1, 1]
 Generation 5760: Best = 3.844278e+03 FES = 576100 μ_F = 0.794 μ_{CR} = 0.028
 Diversity = 4.4701e-01 k_t = [2, 4, 4, 2]
 Generation 5770: Best = 3.844235e+03 FES = 577100 μ_F = 0.843 μ_{CR} = 0.027
 Diversity = 4.4888e-01 k_t = [1, 1, 1, 1]
 Generation 5780: Best = 3.844207e+03 FES = 578100 μ_F = 0.840 μ_{CR} = 0.028
 Diversity = 4.5435e-01 k_t = [2, 3, 4, 4]
 Generation 5790: Best = 3.844207e+03 FES = 579100 μ_F = 0.812 μ_{CR} = 0.025
 Diversity = 4.5503e-01 k_t = [1, 1, 1, 1]
 Generation 5800: Best = 3.844207e+03 FES = 580100 μ_F = 0.841 μ_{CR} = 0.030
 Diversity = 4.5506e-01 k_t = [2, 3, 3, 2]
 Generation 5810: Best = 3.844201e+03 FES = 581100 μ_F = 0.797 μ_{CR} = 0.033
 Diversity = 4.5319e-01 k_t = [1, 1, 1, 1]
 Generation 5820: Best = 3.844090e+03 FES = 582100 μ_F = 0.784 μ_{CR} = 0.033
 Diversity = 4.5313e-01 k_t = [4, 3, 3, 2]
 Generation 5830: Best = 3.844078e+03 FES = 583100 μ_F = 0.713 μ_{CR} = 0.033
 Diversity = 4.5278e-01 k_t = [1, 1, 1, 1]
 Generation 5840: Best = 3.844059e+03 FES = 584100 μ_F = 0.721 μ_{CR} = 0.035
 Diversity = 4.4417e-01 k_t = [4, 4, 4, 4]
 Generation 5850: Best = 3.844017e+03 FES = 585100 μ_F = 0.725 μ_{CR} = 0.030
 Diversity = 4.3923e-01 k_t = [1, 1, 1, 1]
 Generation 5860: Best = 3.844017e+03 FES = 586100 μ_F = 0.648 μ_{CR} = 0.031
 Diversity = 4.3794e-01 k_t = [3, 4, 3, 4]
 Generation 5870: Best = 3.844013e+03 FES = 587100 μ_F = 0.645 μ_{CR} = 0.031
 Diversity = 4.3789e-01 k_t = [1, 1, 1, 1]
 Generation 5880: Best = 3.844013e+03 FES = 588100 μ_F = 0.713 μ_{CR} = 0.034
 Diversity = 4.3596e-01 k_t = [4, 4, 3, 2]
 Generation 5890: Best = 3.844004e+03 FES = 589100 μ_F = 0.770 μ_{CR} = 0.032

Diversity = 4.3241e-01	k_t = [1, 1, 1, 1]			
Generation 5900: Best = 3.844004e+03	FES = 590100	$\mu_F = 0.804$	$\mu_{CR} = 0.030$	
Diversity = 4.3650e-01	k_t = [4, 2, 2, 3]			
Generation 5910: Best = 3.844004e+03	FES = 591100	$\mu_F = 0.845$	$\mu_{CR} = 0.028$	
Diversity = 4.3647e-01	k_t = [1, 1, 1, 1]			
Generation 5920: Best = 3.843958e+03	FES = 592100	$\mu_F = 0.784$	$\mu_{CR} = 0.036$	
Diversity = 4.2820e-01	k_t = [3, 4, 4, 2]			
Generation 5930: Best = 3.843958e+03	FES = 593100	$\mu_F = 0.784$	$\mu_{CR} = 0.031$	
Diversity = 4.2633e-01	k_t = [1, 1, 1, 1]			
Generation 5940: Best = 3.843947e+03	FES = 594100	$\mu_F = 0.779$	$\mu_{CR} = 0.031$	
Diversity = 4.2633e-01	k_t = [3, 3, 4, 3]			
Generation 5950: Best = 3.843947e+03	FES = 595100	$\mu_F = 0.786$	$\mu_{CR} = 0.030$	
Diversity = 4.2634e-01	k_t = [1, 1, 1, 1]			
Generation 5960: Best = 3.843943e+03	FES = 596100	$\mu_F = 0.795$	$\mu_{CR} = 0.026$	
Diversity = 4.2633e-01	k_t = [3, 3, 4, 2]			
Generation 5970: Best = 3.843935e+03	FES = 597100	$\mu_F = 0.736$	$\mu_{CR} = 0.022$	
Diversity = 4.2629e-01	k_t = [1, 1, 1, 1]			
Generation 5980: Best = 3.843935e+03	FES = 598100	$\mu_F = 0.701$	$\mu_{CR} = 0.026$	
Diversity = 4.2624e-01	k_t = [3, 2, 3, 3]			
Generation 5990: Best = 3.843890e+03	FES = 599100	$\mu_F = 0.699$	$\mu_{CR} = 0.034$	
Diversity = 4.2612e-01	k_t = [1, 1, 1, 1]			
Generation 6000: Best = 3.843890e+03	FES = 600100	$\mu_F = 0.763$	$\mu_{CR} = 0.025$	
Diversity = 4.2608e-01	k_t = [2, 2, 3, 3]			
Generation 6010: Best = 3.843840e+03	FES = 601100	$\mu_F = 0.816$	$\mu_{CR} = 0.026$	
Diversity = 4.2763e-01	k_t = [1, 1, 1, 1]			
Generation 6020: Best = 3.843793e+03	FES = 602100	$\mu_F = 0.766$	$\mu_{CR} = 0.029$	
Diversity = 4.3635e-01	k_t = [4, 3, 4, 2]			
Generation 6030: Best = 3.843790e+03	FES = 603100	$\mu_F = 0.798$	$\mu_{CR} = 0.028$	
Diversity = 4.3391e-01	k_t = [1, 1, 1, 1]			
Generation 6040: Best = 3.843790e+03	FES = 604100	$\mu_F = 0.789$	$\mu_{CR} = 0.032$	
Diversity = 4.2317e-01	k_t = [3, 3, 4, 2]			
Generation 6050: Best = 3.843781e+03	FES = 605100	$\mu_F = 0.789$	$\mu_{CR} = 0.035$	
Diversity = 4.2315e-01	k_t = [1, 1, 1, 1]			
Generation 6060: Best = 3.843781e+03	FES = 606100	$\mu_F = 0.799$	$\mu_{CR} = 0.035$	
Diversity = 4.2878e-01	k_t = [3, 4, 2, 4]			
Generation 6070: Best = 3.843770e+03	FES = 607100	$\mu_F = 0.757$	$\mu_{CR} = 0.029$	
Diversity = 4.2874e-01	k_t = [1, 1, 1, 1]			
Generation 6080: Best = 3.843770e+03	FES = 608100	$\mu_F = 0.761$	$\mu_{CR} = 0.033$	
Diversity = 4.2870e-01	k_t = [2, 3, 4, 2]			
Generation 6090: Best = 3.843718e+03	FES = 609100	$\mu_F = 0.728$	$\mu_{CR} = 0.023$	
Diversity = 4.2680e-01	k_t = [1, 1, 1, 1]			
Generation 6100: Best = 3.843718e+03	FES = 610100	$\mu_F = 0.738$	$\mu_{CR} = 0.028$	
Diversity = 4.2501e-01	k_t = [3, 2, 2, 4]			
Generation 6110: Best = 3.843686e+03	FES = 611100	$\mu_F = 0.743$	$\mu_{CR} = 0.025$	
Diversity = 4.2497e-01	k_t = [1, 1, 1, 1]			
Generation 6120: Best = 3.843686e+03	FES = 612100	$\mu_F = 0.738$	$\mu_{CR} = 0.029$	
Diversity = 4.2528e-01	k_t = [2, 4, 3, 4]			
Generation 6130: Best = 3.843676e+03	FES = 613100	$\mu_F = 0.759$	$\mu_{CR} = 0.029$	
Diversity = 4.2533e-01	k_t = [1, 1, 1, 1]			
Generation 6140: Best = 3.843676e+03	FES = 614100	$\mu_F = 0.617$	$\mu_{CR} = 0.030$	
Diversity = 4.2529e-01	k_t = [2, 4, 4, 3]			
Generation 6150: Best = 3.843649e+03	FES = 615100	$\mu_F = 0.609$	$\mu_{CR} = 0.028$	
Diversity = 4.2467e-01	k_t = [1, 1, 1, 1]			
Generation 6160: Best = 3.843642e+03	FES = 616100	$\mu_F = 0.588$	$\mu_{CR} = 0.028$	

Diversity = 4.2457e-01 k_t = [2, 3, 2, 3]
 Generation 6170: Best = 3.843623e+03 FES = 617100 μ_F = 0.616 μ_{CR} = 0.031
 Diversity = 4.2448e-01 k_t = [1, 1, 1, 1]
 Generation 6180: Best = 3.843623e+03 FES = 618100 μ_F = 0.639 μ_{CR} = 0.033
 Diversity = 4.2087e-01 k_t = [4, 2, 4, 4]
 Generation 6190: Best = 3.843623e+03 FES = 619100 μ_F = 0.717 μ_{CR} = 0.030
 Diversity = 4.2083e-01 k_t = [1, 1, 1, 1]
 Generation 6200: Best = 3.843623e+03 FES = 620100 μ_F = 0.709 μ_{CR} = 0.027
 Diversity = 4.2072e-01 k_t = [4, 3, 4, 2]
 Generation 6210: Best = 3.843623e+03 FES = 621100 μ_F = 0.758 μ_{CR} = 0.029
 Diversity = 4.2070e-01 k_t = [1, 1, 1, 1]
 Generation 6220: Best = 3.843610e+03 FES = 622100 μ_F = 0.744 μ_{CR} = 0.031
 Diversity = 4.2067e-01 k_t = [4, 4, 2, 2]
 Generation 6230: Best = 3.843589e+03 FES = 623100 μ_F = 0.756 μ_{CR} = 0.026
 Diversity = 4.2104e-01 k_t = [1, 1, 1, 1]
 Generation 6240: Best = 3.843589e+03 FES = 624100 μ_F = 0.747 μ_{CR} = 0.025
 Diversity = 4.2099e-01 k_t = [4, 4, 4, 2]
 Generation 6250: Best = 3.843589e+03 FES = 625100 μ_F = 0.764 μ_{CR} = 0.022
 Diversity = 4.2101e-01 k_t = [1, 1, 1, 1]
 Generation 6260: Best = 3.843589e+03 FES = 626100 μ_F = 0.777 μ_{CR} = 0.026
 Diversity = 4.2096e-01 k_t = [2, 2, 2, 3]
 Generation 6270: Best = 3.843589e+03 FES = 627100 μ_F = 0.692 μ_{CR} = 0.030
 Diversity = 4.2092e-01 k_t = [1, 1, 1, 1]
 Generation 6280: Best = 3.843588e+03 FES = 628100 μ_F = 0.645 μ_{CR} = 0.034
 Diversity = 4.1432e-01 k_t = [3, 4, 2, 4]
 Generation 6290: Best = 3.843588e+03 FES = 629100 μ_F = 0.631 μ_{CR} = 0.028
 Diversity = 4.1435e-01 k_t = [1, 1, 1, 1]
 Generation 6300: Best = 3.843588e+03 FES = 630100 μ_F = 0.682 μ_{CR} = 0.029
 Diversity = 4.1430e-01 k_t = [2, 2, 3, 2]
 Generation 6310: Best = 3.843588e+03 FES = 631100 μ_F = 0.667 μ_{CR} = 0.034
 Diversity = 4.1494e-01 k_t = [1, 1, 1, 1]
 Generation 6320: Best = 3.843585e+03 FES = 632100 μ_F = 0.698 μ_{CR} = 0.026
 Diversity = 4.1490e-01 k_t = [2, 2, 4, 2]
 Generation 6330: Best = 3.843585e+03 FES = 633100 μ_F = 0.673 μ_{CR} = 0.027
 Diversity = 4.1491e-01 k_t = [1, 1, 1, 1]
 Generation 6340: Best = 3.843585e+03 FES = 634100 μ_F = 0.638 μ_{CR} = 0.029
 Diversity = 4.1359e-01 k_t = [2, 2, 3, 4]
 Generation 6350: Best = 3.843585e+03 FES = 635100 μ_F = 0.726 μ_{CR} = 0.022
 Diversity = 4.1351e-01 k_t = [1, 1, 1, 1]
 Generation 6360: Best = 3.843585e+03 FES = 636100 μ_F = 0.757 μ_{CR} = 0.033
 Diversity = 4.1378e-01 k_t = [3, 2, 3, 2]
 Generation 6370: Best = 3.843585e+03 FES = 637100 μ_F = 0.689 μ_{CR} = 0.028
 Diversity = 4.1373e-01 k_t = [1, 1, 1, 1]
 Generation 6380: Best = 3.843585e+03 FES = 638100 μ_F = 0.652 μ_{CR} = 0.030
 Diversity = 4.1254e-01 k_t = [4, 3, 4, 2]
 Generation 6390: Best = 3.843585e+03 FES = 639100 μ_F = 0.710 μ_{CR} = 0.035
 Diversity = 4.1261e-01 k_t = [1, 1, 1, 1]
 Generation 6400: Best = 3.843584e+03 FES = 640100 μ_F = 0.746 μ_{CR} = 0.037
 Diversity = 4.1263e-01 k_t = [2, 3, 4, 2]
 Generation 6410: Best = 3.843584e+03 FES = 641100 μ_F = 0.770 μ_{CR} = 0.030
 Diversity = 4.1263e-01 k_t = [1, 1, 1, 1]
 Generation 6420: Best = 3.843583e+03 FES = 642100 μ_F = 0.739 μ_{CR} = 0.029
 Diversity = 4.1261e-01 k_t = [3, 4, 4, 2]
 Generation 6430: Best = 3.843578e+03 FES = 643100 μ_F = 0.785 μ_{CR} = 0.025

Diversity = 4.1256e-01	k_t = [1, 1, 1, 1]			
Generation 6440: Best = 3.843578e+03	FES = 644100	$\mu_F = 0.758$	$\mu_{CR} = 0.027$	
Diversity = 4.1290e-01	k_t = [3, 2, 4, 2]			
Generation 6450: Best = 3.843578e+03	FES = 645100	$\mu_F = 0.734$	$\mu_{CR} = 0.025$	
Diversity = 4.1289e-01	k_t = [1, 1, 1, 1]			
Generation 6460: Best = 3.843570e+03	FES = 646100	$\mu_F = 0.653$	$\mu_{CR} = 0.032$	
Diversity = 4.1279e-01	k_t = [4, 2, 2, 4]			
Generation 6470: Best = 3.843570e+03	FES = 647100	$\mu_F = 0.649$	$\mu_{CR} = 0.037$	
Diversity = 4.1185e-01	k_t = [1, 1, 1, 1]			
Generation 6480: Best = 3.843570e+03	FES = 648100	$\mu_F = 0.670$	$\mu_{CR} = 0.031$	
Diversity = 4.1027e-01	k_t = [3, 4, 2, 4]			
Generation 6490: Best = 3.843570e+03	FES = 649100	$\mu_F = 0.697$	$\mu_{CR} = 0.032$	
Diversity = 4.1031e-01	k_t = [1, 1, 1, 1]			
Generation 6500: Best = 3.843565e+03	FES = 650100	$\mu_F = 0.739$	$\mu_{CR} = 0.029$	
Diversity = 4.0477e-01	k_t = [4, 4, 4, 4]			
Generation 6510: Best = 3.843561e+03	FES = 651100	$\mu_F = 0.713$	$\mu_{CR} = 0.025$	
Diversity = 4.0477e-01	k_t = [1, 1, 1, 1]			
Generation 6520: Best = 3.843557e+03	FES = 652100	$\mu_F = 0.744$	$\mu_{CR} = 0.028$	
Diversity = 4.0477e-01	k_t = [3, 3, 3, 2]			
Generation 6530: Best = 3.843542e+03	FES = 653100	$\mu_F = 0.755$	$\mu_{CR} = 0.025$	
Diversity = 4.0466e-01	k_t = [1, 1, 1, 1]			
Generation 6540: Best = 3.843542e+03	FES = 654100	$\mu_F = 0.756$	$\mu_{CR} = 0.025$	
Diversity = 3.9744e-01	k_t = [2, 4, 4, 4]			
Generation 6550: Best = 3.843542e+03	FES = 655100	$\mu_F = 0.770$	$\mu_{CR} = 0.026$	
Diversity = 3.9742e-01	k_t = [1, 1, 1, 1]			
Generation 6560: Best = 3.843510e+03	FES = 656100	$\mu_F = 0.713$	$\mu_{CR} = 0.026$	
Diversity = 3.9743e-01	k_t = [3, 3, 2, 3]			
Generation 6570: Best = 3.843510e+03	FES = 657100	$\mu_F = 0.692$	$\mu_{CR} = 0.031$	
Diversity = 3.9749e-01	k_t = [1, 1, 1, 1]			
Generation 6580: Best = 3.843510e+03	FES = 658100	$\mu_F = 0.734$	$\mu_{CR} = 0.025$	
Diversity = 4.0661e-01	k_t = [3, 4, 3, 4]			
Generation 6590: Best = 3.843510e+03	FES = 659100	$\mu_F = 0.669$	$\mu_{CR} = 0.035$	
Diversity = 4.0657e-01	k_t = [1, 1, 1, 1]			
Generation 6600: Best = 3.843510e+03	FES = 660100	$\mu_F = 0.682$	$\mu_{CR} = 0.038$	
Diversity = 4.0609e-01	k_t = [2, 4, 2, 4]			
Generation 6610: Best = 3.843510e+03	FES = 661100	$\mu_F = 0.692$	$\mu_{CR} = 0.030$	
Diversity = 4.0610e-01	k_t = [1, 1, 1, 1]			
Generation 6620: Best = 3.843510e+03	FES = 662100	$\mu_F = 0.708$	$\mu_{CR} = 0.022$	
Diversity = 4.0550e-01	k_t = [2, 2, 4, 4]			
Generation 6630: Best = 3.843510e+03	FES = 663100	$\mu_F = 0.723$	$\mu_{CR} = 0.021$	
Diversity = 4.0551e-01	k_t = [1, 1, 1, 1]			
Generation 6640: Best = 3.843510e+03	FES = 664100	$\mu_F = 0.753$	$\mu_{CR} = 0.021$	
Diversity = 4.0463e-01	k_t = [3, 2, 4, 4]			
Generation 6650: Best = 3.843510e+03	FES = 665100	$\mu_F = 0.761$	$\mu_{CR} = 0.023$	
Diversity = 4.0462e-01	k_t = [1, 1, 1, 1]			
Generation 6660: Best = 3.843510e+03	FES = 666100	$\mu_F = 0.729$	$\mu_{CR} = 0.033$	
Diversity = 4.0461e-01	k_t = [3, 3, 2, 3]			
Generation 6670: Best = 3.843510e+03	FES = 667100	$\mu_F = 0.723$	$\mu_{CR} = 0.030$	
Diversity = 4.0460e-01	k_t = [1, 1, 1, 1]			
Generation 6680: Best = 3.843510e+03	FES = 668100	$\mu_F = 0.717$	$\mu_{CR} = 0.033$	
Diversity = 4.0458e-01	k_t = [2, 4, 4, 2]			
Generation 6690: Best = 3.843510e+03	FES = 669100	$\mu_F = 0.623$	$\mu_{CR} = 0.037$	
Diversity = 4.0727e-01	k_t = [1, 1, 1, 1]			
Generation 6700: Best = 3.843510e+03	FES = 670100	$\mu_F = 0.662$	$\mu_{CR} = 0.041$	

Diversity = 4.0721e-01	k_t = [2, 4, 4, 2]			
Generation 6710: Best = 3.843510e+03	FES = 671100	$\mu_F = 0.704$	$\mu_{CR} = 0.032$	
Diversity = 4.0721e-01	k_t = [1, 1, 1, 1]			
Generation 6720: Best = 3.843498e+03	FES = 672100	$\mu_F = 0.765$	$\mu_{CR} = 0.038$	
Diversity = 4.0719e-01	k_t = [4, 3, 4, 3]			
Generation 6730: Best = 3.843498e+03	FES = 673100	$\mu_F = 0.709$	$\mu_{CR} = 0.033$	
Diversity = 4.0720e-01	k_t = [1, 1, 1, 1]			
Generation 6740: Best = 3.843494e+03	FES = 674100	$\mu_F = 0.751$	$\mu_{CR} = 0.032$	
Diversity = 4.0719e-01	k_t = [2, 2, 3, 2]			
Generation 6750: Best = 3.843494e+03	FES = 675100	$\mu_F = 0.718$	$\mu_{CR} = 0.027$	
Diversity = 4.0717e-01	k_t = [1, 1, 1, 1]			
Generation 6760: Best = 3.843492e+03	FES = 676100	$\mu_F = 0.764$	$\mu_{CR} = 0.032$	
Diversity = 4.0707e-01	k_t = [4, 3, 2, 2]			
Generation 6770: Best = 3.843492e+03	FES = 677100	$\mu_F = 0.763$	$\mu_{CR} = 0.023$	
Diversity = 4.0740e-01	k_t = [1, 1, 1, 1]			
Generation 6780: Best = 3.843492e+03	FES = 678100	$\mu_F = 0.767$	$\mu_{CR} = 0.021$	
Diversity = 4.0739e-01	k_t = [4, 2, 4, 4]			
Generation 6790: Best = 3.843488e+03	FES = 679100	$\mu_F = 0.774$	$\mu_{CR} = 0.027$	
Diversity = 4.0736e-01	k_t = [1, 1, 1, 1]			
Generation 6800: Best = 3.843481e+03	FES = 680100	$\mu_F = 0.760$	$\mu_{CR} = 0.035$	
Diversity = 4.0739e-01	k_t = [3, 4, 2, 2]			
Generation 6810: Best = 3.843465e+03	FES = 681100	$\mu_F = 0.768$	$\mu_{CR} = 0.028$	
Diversity = 4.0737e-01	k_t = [1, 1, 1, 1]			
Generation 6820: Best = 3.843465e+03	FES = 682100	$\mu_F = 0.816$	$\mu_{CR} = 0.027$	
Diversity = 4.0738e-01	k_t = [3, 2, 4, 4]			
Generation 6830: Best = 3.843450e+03	FES = 683100	$\mu_F = 0.819$	$\mu_{CR} = 0.024$	
Diversity = 4.0738e-01	k_t = [1, 1, 1, 1]			
Generation 6840: Best = 3.843450e+03	FES = 684100	$\mu_F = 0.818$	$\mu_{CR} = 0.029$	
Diversity = 4.0729e-01	k_t = [4, 3, 4, 4]			
Generation 6850: Best = 3.843450e+03	FES = 685100	$\mu_F = 0.787$	$\mu_{CR} = 0.030$	
Diversity = 4.0726e-01	k_t = [1, 1, 1, 1]			
Generation 6860: Best = 3.843447e+03	FES = 686100	$\mu_F = 0.780$	$\mu_{CR} = 0.027$	
Diversity = 4.0728e-01	k_t = [2, 3, 3, 2]			
Generation 6870: Best = 3.843447e+03	FES = 687100	$\mu_F = 0.752$	$\mu_{CR} = 0.032$	
Diversity = 4.0728e-01	k_t = [1, 1, 1, 1]			
Generation 6880: Best = 3.843447e+03	FES = 688100	$\mu_F = 0.733$	$\mu_{CR} = 0.022$	
Diversity = 4.0729e-01	k_t = [3, 2, 2, 3]			
Generation 6890: Best = 3.843447e+03	FES = 689100	$\mu_F = 0.716$	$\mu_{CR} = 0.019$	
Diversity = 4.0729e-01	k_t = [1, 1, 1, 1]			
Generation 6900: Best = 3.843447e+03	FES = 690100	$\mu_F = 0.794$	$\mu_{CR} = 0.019$	
Diversity = 3.9885e-01	k_t = [3, 3, 2, 4]			
Generation 6910: Best = 3.843447e+03	FES = 691100	$\mu_F = 0.832$	$\mu_{CR} = 0.031$	
Diversity = 3.9883e-01	k_t = [1, 1, 1, 1]			
Generation 6920: Best = 3.843447e+03	FES = 692100	$\mu_F = 0.842$	$\mu_{CR} = 0.026$	
Diversity = 3.9885e-01	k_t = [4, 4, 4, 3]			
Generation 6930: Best = 3.843444e+03	FES = 693100	$\mu_F = 0.804$	$\mu_{CR} = 0.024$	
Diversity = 3.9887e-01	k_t = [1, 1, 1, 1]			
Generation 6940: Best = 3.843444e+03	FES = 694100	$\mu_F = 0.836$	$\mu_{CR} = 0.025$	
Diversity = 3.9884e-01	k_t = [4, 3, 4, 3]			
Generation 6950: Best = 3.843444e+03	FES = 695100	$\mu_F = 0.865$	$\mu_{CR} = 0.024$	
Diversity = 3.9884e-01	k_t = [1, 1, 1, 1]			
Generation 6960: Best = 3.843444e+03	FES = 696100	$\mu_F = 0.875$	$\mu_{CR} = 0.022$	
Diversity = 3.9881e-01	k_t = [2, 4, 3, 2]			
Generation 6970: Best = 3.843442e+03	FES = 697100	$\mu_F = 0.849$	$\mu_{CR} = 0.028$	

Diversity = 3.9885e-01	k_t = [1, 1, 1, 1]			
Generation 6980: Best = 3.843442e+03	FES = 698100	$\mu_F = 0.869$	$\mu_{CR} = 0.032$	
Diversity = 3.8841e-01	k_t = [2, 3, 4, 4]			
Generation 6990: Best = 3.843441e+03	FES = 699100	$\mu_F = 0.880$	$\mu_{CR} = 0.030$	
Diversity = 3.8841e-01	k_t = [1, 1, 1, 1]			
Generation 7000: Best = 3.843441e+03	FES = 700100	$\mu_F = 0.886$	$\mu_{CR} = 0.025$	
Diversity = 3.9086e-01	k_t = [3, 3, 2, 4]			
Generation 7010: Best = 3.843423e+03	FES = 701100	$\mu_F = 0.842$	$\mu_{CR} = 0.024$	
Diversity = 3.9084e-01	k_t = [1, 1, 1, 1]			
Generation 7020: Best = 3.843414e+03	FES = 702100	$\mu_F = 0.792$	$\mu_{CR} = 0.026$	
Diversity = 3.9086e-01	k_t = [3, 2, 4, 3]			
Generation 7030: Best = 3.843414e+03	FES = 703100	$\mu_F = 0.818$	$\mu_{CR} = 0.027$	
Diversity = 3.9083e-01	k_t = [1, 1, 1, 1]			
Generation 7040: Best = 3.843414e+03	FES = 704100	$\mu_F = 0.798$	$\mu_{CR} = 0.026$	
Diversity = 3.9081e-01	k_t = [3, 2, 4, 2]			
Generation 7050: Best = 3.843414e+03	FES = 705100	$\mu_F = 0.801$	$\mu_{CR} = 0.025$	
Diversity = 3.9079e-01	k_t = [1, 1, 1, 1]			
Generation 7060: Best = 3.843414e+03	FES = 706100	$\mu_F = 0.830$	$\mu_{CR} = 0.026$	
Diversity = 3.9583e-01	k_t = [2, 3, 2, 4]			
Generation 7070: Best = 3.843414e+03	FES = 707100	$\mu_F = 0.827$	$\mu_{CR} = 0.021$	
Diversity = 3.9581e-01	k_t = [1, 1, 1, 1]			
Generation 7080: Best = 3.843410e+03	FES = 708100	$\mu_F = 0.847$	$\mu_{CR} = 0.028$	
Diversity = 3.9581e-01	k_t = [4, 2, 2, 2]			
Generation 7090: Best = 3.843410e+03	FES = 709100	$\mu_F = 0.840$	$\mu_{CR} = 0.028$	
Diversity = 3.9583e-01	k_t = [1, 1, 1, 1]			
Generation 7100: Best = 3.843407e+03	FES = 710100	$\mu_F = 0.811$	$\mu_{CR} = 0.025$	
Diversity = 3.9581e-01	k_t = [4, 4, 3, 2]			
Generation 7110: Best = 3.843404e+03	FES = 711100	$\mu_F = 0.832$	$\mu_{CR} = 0.019$	
Diversity = 3.9590e-01	k_t = [1, 1, 1, 1]			
Generation 7120: Best = 3.843404e+03	FES = 712100	$\mu_F = 0.855$	$\mu_{CR} = 0.027$	
Diversity = 3.9899e-01	k_t = [4, 2, 4, 4]			
Generation 7130: Best = 3.843404e+03	FES = 713100	$\mu_F = 0.830$	$\mu_{CR} = 0.028$	
Diversity = 3.9901e-01	k_t = [1, 1, 1, 1]			
Generation 7140: Best = 3.843404e+03	FES = 714100	$\mu_F = 0.813$	$\mu_{CR} = 0.024$	
Diversity = 3.9158e-01	k_t = [4, 2, 2, 4]			
Generation 7150: Best = 3.843402e+03	FES = 715100	$\mu_F = 0.737$	$\mu_{CR} = 0.021$	
Diversity = 3.9154e-01	k_t = [1, 1, 1, 1]			
Generation 7160: Best = 3.843402e+03	FES = 716100	$\mu_F = 0.741$	$\mu_{CR} = 0.024$	
Diversity = 3.9638e-01	k_t = [4, 3, 3, 4]			
Generation 7170: Best = 3.843402e+03	FES = 717100	$\mu_F = 0.721$	$\mu_{CR} = 0.023$	
Diversity = 3.9634e-01	k_t = [1, 1, 1, 1]			
Generation 7180: Best = 3.843375e+03	FES = 718100	$\mu_F = 0.767$	$\mu_{CR} = 0.023$	
Diversity = 3.9883e-01	k_t = [4, 3, 2, 4]			
Generation 7190: Best = 3.843367e+03	FES = 719100	$\mu_F = 0.776$	$\mu_{CR} = 0.020$	
Diversity = 3.9882e-01	k_t = [1, 1, 1, 1]			
Generation 7200: Best = 3.843367e+03	FES = 720100	$\mu_F = 0.779$	$\mu_{CR} = 0.021$	
Diversity = 3.9853e-01	k_t = [4, 4, 4, 4]			
Generation 7210: Best = 3.843364e+03	FES = 721100	$\mu_F = 0.796$	$\mu_{CR} = 0.026$	
Diversity = 3.9856e-01	k_t = [1, 1, 1, 1]			
Generation 7220: Best = 3.843364e+03	FES = 722100	$\mu_F = 0.831$	$\mu_{CR} = 0.022$	
Diversity = 4.0028e-01	k_t = [2, 2, 3, 4]			
Generation 7230: Best = 3.843358e+03	FES = 723100	$\mu_F = 0.824$	$\mu_{CR} = 0.029$	
Diversity = 4.0021e-01	k_t = [1, 1, 1, 1]			
Generation 7240: Best = 3.843358e+03	FES = 724100	$\mu_F = 0.844$	$\mu_{CR} = 0.026$	

Diversity = 4.0019e-01	k_t = [3, 4, 4, 3]			
Generation 7250: Best = 3.843358e+03	FES = 725100	$\mu_F = 0.760$	$\mu_{CR} = 0.028$	
Diversity = 4.0025e-01	k_t = [1, 1, 1, 1]			
Generation 7260: Best = 3.843358e+03	FES = 726100	$\mu_F = 0.702$	$\mu_{CR} = 0.029$	
Diversity = 4.0022e-01	k_t = [3, 3, 2, 2]			
Generation 7270: Best = 3.843356e+03	FES = 727100	$\mu_F = 0.732$	$\mu_{CR} = 0.032$	
Diversity = 4.0020e-01	k_t = [1, 1, 1, 1]			
Generation 7280: Best = 3.843355e+03	FES = 728100	$\mu_F = 0.718$	$\mu_{CR} = 0.029$	
Diversity = 4.0017e-01	k_t = [2, 3, 2, 2]			
Generation 7290: Best = 3.843355e+03	FES = 729100	$\mu_F = 0.749$	$\mu_{CR} = 0.028$	
Diversity = 4.0015e-01	k_t = [1, 1, 1, 1]			
Generation 7300: Best = 3.843355e+03	FES = 730100	$\mu_F = 0.706$	$\mu_{CR} = 0.031$	
Diversity = 3.9826e-01	k_t = [3, 2, 4, 4]			
Generation 7310: Best = 3.843355e+03	FES = 731100	$\mu_F = 0.754$	$\mu_{CR} = 0.030$	
Diversity = 3.9826e-01	k_t = [1, 1, 1, 1]			
Generation 7320: Best = 3.843337e+03	FES = 732100	$\mu_F = 0.680$	$\mu_{CR} = 0.034$	
Diversity = 3.9641e-01	k_t = [4, 4, 4, 3]			
Generation 7330: Best = 3.843337e+03	FES = 733100	$\mu_F = 0.681$	$\mu_{CR} = 0.034$	
Diversity = 3.9612e-01	k_t = [1, 1, 1, 1]			
Generation 7340: Best = 3.843337e+03	FES = 734100	$\mu_F = 0.698$	$\mu_{CR} = 0.029$	
Diversity = 3.9611e-01	k_t = [4, 3, 3, 2]			
Generation 7350: Best = 3.843337e+03	FES = 735100	$\mu_F = 0.646$	$\mu_{CR} = 0.027$	
Diversity = 3.9610e-01	k_t = [1, 1, 1, 1]			
Generation 7360: Best = 3.843333e+03	FES = 736100	$\mu_F = 0.607$	$\mu_{CR} = 0.032$	
Diversity = 3.9667e-01	k_t = [2, 3, 2, 2]			
Generation 7370: Best = 3.843333e+03	FES = 737100	$\mu_F = 0.606$	$\mu_{CR} = 0.027$	
Diversity = 3.9666e-01	k_t = [1, 1, 1, 1]			
Generation 7380: Best = 3.843333e+03	FES = 738100	$\mu_F = 0.608$	$\mu_{CR} = 0.035$	
Diversity = 3.9665e-01	k_t = [4, 2, 2, 3]			
Generation 7390: Best = 3.843333e+03	FES = 739100	$\mu_F = 0.562$	$\mu_{CR} = 0.030$	
Diversity = 3.9662e-01	k_t = [1, 1, 1, 1]			
Generation 7400: Best = 3.843333e+03	FES = 740100	$\mu_F = 0.588$	$\mu_{CR} = 0.034$	
Diversity = 3.9869e-01	k_t = [2, 4, 3, 4]			
Generation 7410: Best = 3.843333e+03	FES = 741100	$\mu_F = 0.619$	$\mu_{CR} = 0.035$	
Diversity = 3.9875e-01	k_t = [1, 1, 1, 1]			
Generation 7420: Best = 3.843333e+03	FES = 742100	$\mu_F = 0.669$	$\mu_{CR} = 0.031$	
Diversity = 3.9873e-01	k_t = [3, 3, 3, 4]			
Generation 7430: Best = 3.843333e+03	FES = 743100	$\mu_F = 0.684$	$\mu_{CR} = 0.034$	
Diversity = 3.9867e-01	k_t = [1, 1, 1, 1]			
Generation 7440: Best = 3.843333e+03	FES = 744100	$\mu_F = 0.696$	$\mu_{CR} = 0.027$	
Diversity = 3.9870e-01	k_t = [3, 4, 2, 3]			
Generation 7450: Best = 3.843325e+03	FES = 745100	$\mu_F = 0.741$	$\mu_{CR} = 0.020$	
Diversity = 3.9868e-01	k_t = [1, 1, 1, 1]			
Generation 7460: Best = 3.843324e+03	FES = 746100	$\mu_F = 0.809$	$\mu_{CR} = 0.029$	
Diversity = 3.9738e-01	k_t = [4, 4, 2, 4]			
Generation 7470: Best = 3.843322e+03	FES = 747100	$\mu_F = 0.738$	$\mu_{CR} = 0.044$	
Diversity = 3.9737e-01	k_t = [1, 1, 1, 1]			
Generation 7480: Best = 3.843315e+03	FES = 748100	$\mu_F = 0.681$	$\mu_{CR} = 0.029$	
Diversity = 3.9816e-01	k_t = [2, 2, 3, 4]			
Generation 7490: Best = 3.843315e+03	FES = 749100	$\mu_F = 0.655$	$\mu_{CR} = 0.029$	
Diversity = 3.9815e-01	k_t = [1, 1, 1, 1]			
Generation 7500: Best = 3.843301e+03	FES = 750100	$\mu_F = 0.657$	$\mu_{CR} = 0.031$	
Diversity = 3.9816e-01	k_t = [4, 2, 4, 2]			
Generation 7510: Best = 3.843301e+03	FES = 751100	$\mu_F = 0.684$	$\mu_{CR} = 0.030$	

Diversity = 3.9815e-01	k_t = [1, 1, 1, 1]			
Generation 7520: Best = 3.843301e+03	FES = 752100	$\mu_F = 0.737$	$\mu_{CR} = 0.027$	
Diversity = 3.9816e-01	k_t = [2, 2, 3, 3]			
Generation 7530: Best = 3.843300e+03	FES = 753100	$\mu_F = 0.764$	$\mu_{CR} = 0.025$	
Diversity = 3.9814e-01	k_t = [1, 1, 1, 1]			
Generation 7540: Best = 3.843300e+03	FES = 754100	$\mu_F = 0.787$	$\mu_{CR} = 0.026$	
Diversity = 3.9812e-01	k_t = [2, 2, 4, 2]			
Generation 7550: Best = 3.843300e+03	FES = 755100	$\mu_F = 0.789$	$\mu_{CR} = 0.023$	
Diversity = 3.9810e-01	k_t = [1, 1, 1, 1]			
Generation 7560: Best = 3.843299e+03	FES = 756100	$\mu_F = 0.724$	$\mu_{CR} = 0.050$	
Diversity = 3.9807e-01	k_t = [4, 3, 4, 3]			
Generation 7570: Best = 3.843299e+03	FES = 757100	$\mu_F = 0.708$	$\mu_{CR} = 0.036$	
Diversity = 3.9806e-01	k_t = [1, 1, 1, 1]			
Generation 7580: Best = 3.843293e+03	FES = 758100	$\mu_F = 0.745$	$\mu_{CR} = 0.033$	
Diversity = 3.9806e-01	k_t = [3, 4, 2, 2]			
Generation 7590: Best = 3.843293e+03	FES = 759100	$\mu_F = 0.760$	$\mu_{CR} = 0.025$	
Diversity = 3.9804e-01	k_t = [1, 1, 1, 1]			
Generation 7600: Best = 3.843293e+03	FES = 760100	$\mu_F = 0.749$	$\mu_{CR} = 0.032$	
Diversity = 3.9803e-01	k_t = [4, 4, 3, 2]			
Generation 7610: Best = 3.843285e+03	FES = 761100	$\mu_F = 0.730$	$\mu_{CR} = 0.035$	
Diversity = 3.9799e-01	k_t = [1, 1, 1, 1]			
Generation 7620: Best = 3.843285e+03	FES = 762100	$\mu_F = 0.735$	$\mu_{CR} = 0.036$	
Diversity = 3.9799e-01	k_t = [4, 2, 3, 2]			
Generation 7630: Best = 3.843284e+03	FES = 763100	$\mu_F = 0.750$	$\mu_{CR} = 0.028$	
Diversity = 4.0006e-01	k_t = [1, 1, 1, 1]			
Generation 7640: Best = 3.843284e+03	FES = 764100	$\mu_F = 0.725$	$\mu_{CR} = 0.031$	
Diversity = 4.0008e-01	k_t = [3, 2, 4, 3]			
Generation 7650: Best = 3.843284e+03	FES = 765100	$\mu_F = 0.773$	$\mu_{CR} = 0.024$	
Diversity = 4.0009e-01	k_t = [1, 1, 1, 1]			
Generation 7660: Best = 3.843277e+03	FES = 766100	$\mu_F = 0.777$	$\mu_{CR} = 0.031$	
Diversity = 3.9835e-01	k_t = [2, 4, 4, 4]			
Generation 7670: Best = 3.843277e+03	FES = 767100	$\mu_F = 0.747$	$\mu_{CR} = 0.027$	
Diversity = 3.9833e-01	k_t = [1, 1, 1, 1]			
Generation 7680: Best = 3.843271e+03	FES = 768100	$\mu_F = 0.728$	$\mu_{CR} = 0.023$	
Diversity = 3.9834e-01	k_t = [3, 3, 2, 2]			
Generation 7690: Best = 3.843271e+03	FES = 769100	$\mu_F = 0.786$	$\mu_{CR} = 0.025$	
Diversity = 3.9832e-01	k_t = [1, 1, 1, 1]			
Generation 7700: Best = 3.843271e+03	FES = 770100	$\mu_F = 0.805$	$\mu_{CR} = 0.026$	
Diversity = 4.0296e-01	k_t = [4, 3, 2, 4]			
Generation 7710: Best = 3.843270e+03	FES = 771100	$\mu_F = 0.757$	$\mu_{CR} = 0.021$	
Diversity = 4.0341e-01	k_t = [1, 1, 1, 1]			
Generation 7720: Best = 3.843270e+03	FES = 772100	$\mu_F = 0.794$	$\mu_{CR} = 0.024$	
Diversity = 3.9716e-01	k_t = [4, 2, 2, 4]			
Generation 7730: Best = 3.843270e+03	FES = 773100	$\mu_F = 0.740$	$\mu_{CR} = 0.018$	
Diversity = 3.9713e-01	k_t = [1, 1, 1, 1]			
Generation 7740: Best = 3.843270e+03	FES = 774100	$\mu_F = 0.720$	$\mu_{CR} = 0.020$	
Diversity = 3.9750e-01	k_t = [3, 2, 2, 4]			
Generation 7750: Best = 3.843267e+03	FES = 775100	$\mu_F = 0.730$	$\mu_{CR} = 0.027$	
Diversity = 3.9750e-01	k_t = [1, 1, 1, 1]			
Generation 7760: Best = 3.843267e+03	FES = 776100	$\mu_F = 0.684$	$\mu_{CR} = 0.030$	
Diversity = 3.9746e-01	k_t = [2, 2, 3, 2]			
Generation 7770: Best = 3.843267e+03	FES = 777100	$\mu_F = 0.690$	$\mu_{CR} = 0.025$	
Diversity = 3.9748e-01	k_t = [1, 1, 1, 1]			
Generation 7780: Best = 3.843267e+03	FES = 778100	$\mu_F = 0.675$	$\mu_{CR} = 0.031$	

Diversity = 3.9746e-01	k_t = [4, 3, 2, 3]			
Generation 7790: Best = 3.843267e+03	FES = 779100	$\mu_F = 0.586$	$\mu_{CR} = 0.032$	
Diversity = 3.9748e-01	k_t = [1, 1, 1, 1]			
Generation 7800: Best = 3.843267e+03	FES = 780100	$\mu_F = 0.530$	$\mu_{CR} = 0.033$	
Diversity = 3.9747e-01	k_t = [4, 4, 2, 3]			
Generation 7810: Best = 3.843267e+03	FES = 781100	$\mu_F = 0.549$	$\mu_{CR} = 0.033$	
Diversity = 3.9746e-01	k_t = [1, 1, 1, 1]			
Generation 7820: Best = 3.843267e+03	FES = 782100	$\mu_F = 0.542$	$\mu_{CR} = 0.027$	
Diversity = 3.9741e-01	k_t = [2, 4, 4, 3]			
Generation 7830: Best = 3.843267e+03	FES = 783100	$\mu_F = 0.622$	$\mu_{CR} = 0.020$	
Diversity = 3.9738e-01	k_t = [1, 1, 1, 1]			
Generation 7840: Best = 3.843263e+03	FES = 784100	$\mu_F = 0.582$	$\mu_{CR} = 0.031$	
Diversity = 3.9732e-01	k_t = [3, 4, 4, 2]			
Generation 7850: Best = 3.843263e+03	FES = 785100	$\mu_F = 0.572$	$\mu_{CR} = 0.029$	
Diversity = 3.9732e-01	k_t = [1, 1, 1, 1]			
Generation 7860: Best = 3.843262e+03	FES = 786100	$\mu_F = 0.587$	$\mu_{CR} = 0.029$	
Diversity = 3.9734e-01	k_t = [2, 3, 2, 3]			
Generation 7870: Best = 3.843262e+03	FES = 787100	$\mu_F = 0.596$	$\mu_{CR} = 0.030$	
Diversity = 3.9730e-01	k_t = [1, 1, 1, 1]			
Generation 7880: Best = 3.843262e+03	FES = 788100	$\mu_F = 0.587$	$\mu_{CR} = 0.034$	
Diversity = 3.9438e-01	k_t = [4, 2, 3, 4]			
Generation 7890: Best = 3.843262e+03	FES = 789100	$\mu_F = 0.642$	$\mu_{CR} = 0.024$	
Diversity = 3.9437e-01	k_t = [1, 1, 1, 1]			
Generation 7900: Best = 3.843251e+03	FES = 790100	$\mu_F = 0.727$	$\mu_{CR} = 0.023$	
Diversity = 3.9432e-01	k_t = [2, 2, 3, 3]			
Generation 7910: Best = 3.843249e+03	FES = 791100	$\mu_F = 0.763$	$\mu_{CR} = 0.021$	
Diversity = 3.9431e-01	k_t = [1, 1, 1, 1]			
Generation 7920: Best = 3.843249e+03	FES = 792100	$\mu_F = 0.784$	$\mu_{CR} = 0.024$	
Diversity = 3.9699e-01	k_t = [2, 4, 2, 4]			
Generation 7930: Best = 3.843249e+03	FES = 793100	$\mu_F = 0.841$	$\mu_{CR} = 0.021$	
Diversity = 3.9699e-01	k_t = [1, 1, 1, 1]			
Generation 7940: Best = 3.843249e+03	FES = 794100	$\mu_F = 0.804$	$\mu_{CR} = 0.025$	
Diversity = 4.0266e-01	k_t = [3, 4, 2, 4]			
Generation 7950: Best = 3.843249e+03	FES = 795100	$\mu_F = 0.804$	$\mu_{CR} = 0.024$	
Diversity = 4.0266e-01	k_t = [1, 1, 1, 1]			
Generation 7960: Best = 3.843249e+03	FES = 796100	$\mu_F = 0.843$	$\mu_{CR} = 0.024$	
Diversity = 4.0266e-01	k_t = [4, 4, 3, 3]			
Generation 7970: Best = 3.843249e+03	FES = 797100	$\mu_F = 0.850$	$\mu_{CR} = 0.026$	
Diversity = 4.0266e-01	k_t = [1, 1, 1, 1]			
Generation 7980: Best = 3.843249e+03	FES = 798100	$\mu_F = 0.825$	$\mu_{CR} = 0.027$	
Diversity = 4.0516e-01	k_t = [4, 2, 3, 4]			
Generation 7990: Best = 3.843249e+03	FES = 799100	$\mu_F = 0.813$	$\mu_{CR} = 0.021$	
Diversity = 4.0514e-01	k_t = [1, 1, 1, 1]			
Generation 8000: Best = 3.843249e+03	FES = 800100	$\mu_F = 0.773$	$\mu_{CR} = 0.024$	
Diversity = 4.0507e-01	k_t = [2, 2, 4, 4]			
Generation 8010: Best = 3.843241e+03	FES = 801100	$\mu_F = 0.775$	$\mu_{CR} = 0.027$	
Diversity = 4.0508e-01	k_t = [1, 1, 1, 1]			
Generation 8020: Best = 3.843241e+03	FES = 802100	$\mu_F = 0.797$	$\mu_{CR} = 0.026$	
Diversity = 4.0645e-01	k_t = [2, 3, 2, 4]			
Generation 8030: Best = 3.843239e+03	FES = 803100	$\mu_F = 0.827$	$\mu_{CR} = 0.020$	
Diversity = 4.0647e-01	k_t = [1, 1, 1, 1]			
Generation 8040: Best = 3.843239e+03	FES = 804100	$\mu_F = 0.729$	$\mu_{CR} = 0.022$	
Diversity = 4.0640e-01	k_t = [3, 3, 2, 2]			
Generation 8050: Best = 3.843239e+03	FES = 805100	$\mu_F = 0.705$	$\mu_{CR} = 0.019$	

Diversity = 4.0640e-01	k_t = [1, 1, 1, 1]			
Generation 8060: Best = 3.843239e+03	FES = 806100	$\mu_F = 0.727$	$\mu_{CR} = 0.028$	
Diversity = 4.0748e-01	k_t = [3, 3, 2, 4]			
Generation 8070: Best = 3.843239e+03	FES = 807100	$\mu_F = 0.733$	$\mu_{CR} = 0.024$	
Diversity = 4.0746e-01	k_t = [1, 1, 1, 1]			
Generation 8080: Best = 3.843236e+03	FES = 808100	$\mu_F = 0.675$	$\mu_{CR} = 0.027$	
Diversity = 4.0744e-01	k_t = [4, 4, 4, 3]			
Generation 8090: Best = 3.843235e+03	FES = 809100	$\mu_F = 0.677$	$\mu_{CR} = 0.021$	
Diversity = 4.0743e-01	k_t = [1, 1, 1, 1]			
Generation 8100: Best = 3.843221e+03	FES = 810100	$\mu_F = 0.703$	$\mu_{CR} = 0.024$	
Diversity = 4.0743e-01	k_t = [4, 3, 2, 2]			
Generation 8110: Best = 3.843221e+03	FES = 811100	$\mu_F = 0.701$	$\mu_{CR} = 0.026$	
Diversity = 4.0742e-01	k_t = [1, 1, 1, 1]			
Generation 8120: Best = 3.843221e+03	FES = 812100	$\mu_F = 0.743$	$\mu_{CR} = 0.027$	
Diversity = 4.0752e-01	k_t = [2, 2, 4, 4]			
Generation 8130: Best = 3.843221e+03	FES = 813100	$\mu_F = 0.706$	$\mu_{CR} = 0.030$	
Diversity = 4.0752e-01	k_t = [1, 1, 1, 1]			
Generation 8140: Best = 3.843221e+03	FES = 814100	$\mu_F = 0.735$	$\mu_{CR} = 0.032$	
Diversity = 4.0751e-01	k_t = [4, 3, 4, 3]			
Generation 8150: Best = 3.843221e+03	FES = 815100	$\mu_F = 0.737$	$\mu_{CR} = 0.030$	
Diversity = 4.0754e-01	k_t = [1, 1, 1, 1]			
Generation 8160: Best = 3.843221e+03	FES = 816100	$\mu_F = 0.710$	$\mu_{CR} = 0.024$	
Diversity = 4.0757e-01	k_t = [2, 2, 3, 3]			
Generation 8170: Best = 3.843221e+03	FES = 817100	$\mu_F = 0.712$	$\mu_{CR} = 0.022$	
Diversity = 4.0755e-01	k_t = [1, 1, 1, 1]			
Generation 8180: Best = 3.843221e+03	FES = 818100	$\mu_F = 0.627$	$\mu_{CR} = 0.024$	
Diversity = 4.1221e-01	k_t = [2, 2, 2, 4]			
Generation 8190: Best = 3.843220e+03	FES = 819100	$\mu_F = 0.679$	$\mu_{CR} = 0.023$	
Diversity = 4.1221e-01	k_t = [1, 1, 1, 1]			
Generation 8200: Best = 3.843219e+03	FES = 820100	$\mu_F = 0.740$	$\mu_{CR} = 0.027$	
Diversity = 4.0520e-01	k_t = [2, 3, 4, 4]			
Generation 8210: Best = 3.843209e+03	FES = 821100	$\mu_F = 0.694$	$\mu_{CR} = 0.030$	
Diversity = 4.0520e-01	k_t = [1, 1, 1, 1]			
Generation 8220: Best = 3.843209e+03	FES = 822100	$\mu_F = 0.696$	$\mu_{CR} = 0.027$	
Diversity = 4.0082e-01	k_t = [2, 3, 3, 4]			
Generation 8230: Best = 3.843209e+03	FES = 823100	$\mu_F = 0.736$	$\mu_{CR} = 0.024$	
Diversity = 4.0081e-01	k_t = [1, 1, 1, 1]			
Generation 8240: Best = 3.843207e+03	FES = 824100	$\mu_F = 0.736$	$\mu_{CR} = 0.027$	
Diversity = 3.9815e-01	k_t = [2, 2, 3, 4]			
Generation 8250: Best = 3.843205e+03	FES = 825100	$\mu_F = 0.735$	$\mu_{CR} = 0.031$	
Diversity = 3.9812e-01	k_t = [1, 1, 1, 1]			
Generation 8260: Best = 3.843205e+03	FES = 826100	$\mu_F = 0.766$	$\mu_{CR} = 0.025$	
Diversity = 3.9806e-01	k_t = [4, 3, 3, 2]			
Generation 8270: Best = 3.843205e+03	FES = 827100	$\mu_F = 0.759$	$\mu_{CR} = 0.028$	
Diversity = 3.9805e-01	k_t = [1, 1, 1, 1]			
Generation 8280: Best = 3.843205e+03	FES = 828100	$\mu_F = 0.720$	$\mu_{CR} = 0.023$	
Diversity = 3.9308e-01	k_t = [2, 3, 2, 4]			
Generation 8290: Best = 3.843205e+03	FES = 829100	$\mu_F = 0.768$	$\mu_{CR} = 0.025$	
Diversity = 3.9307e-01	k_t = [1, 1, 1, 1]			
Generation 8300: Best = 3.843205e+03	FES = 830100	$\mu_F = 0.706$	$\mu_{CR} = 0.024$	
Diversity = 3.9306e-01	k_t = [3, 3, 2, 3]			
Generation 8310: Best = 3.843199e+03	FES = 831100	$\mu_F = 0.734$	$\mu_{CR} = 0.019$	
Diversity = 3.9305e-01	k_t = [1, 1, 1, 1]			
Generation 8320: Best = 3.843199e+03	FES = 832100	$\mu_F = 0.722$	$\mu_{CR} = 0.029$	

Diversity = 3.9304e-01	k_t = [3, 2, 3, 3]			
Generation 8330: Best = 3.843199e+03	FES = 833100	$\mu_F = 0.686$	$\mu_{CR} = 0.022$	
Diversity = 3.9289e-01	k_t = [1, 1, 1, 1]			
Generation 8340: Best = 3.843197e+03	FES = 834100	$\mu_F = 0.681$	$\mu_{CR} = 0.024$	
Diversity = 3.9286e-01	k_t = [4, 3, 3, 2]			
Generation 8350: Best = 3.843197e+03	FES = 835100	$\mu_F = 0.637$	$\mu_{CR} = 0.025$	
Diversity = 3.9295e-01	k_t = [1, 1, 1, 1]			
Generation 8360: Best = 3.843197e+03	FES = 836100	$\mu_F = 0.659$	$\mu_{CR} = 0.030$	
Diversity = 3.9294e-01	k_t = [3, 2, 3, 3]			
Generation 8370: Best = 3.843197e+03	FES = 837100	$\mu_F = 0.696$	$\mu_{CR} = 0.020$	
Diversity = 3.9287e-01	k_t = [1, 1, 1, 1]			
Generation 8380: Best = 3.843195e+03	FES = 838100	$\mu_F = 0.685$	$\mu_{CR} = 0.023$	
Diversity = 3.9285e-01	k_t = [3, 2, 2, 3]			
Generation 8390: Best = 3.843195e+03	FES = 839100	$\mu_F = 0.687$	$\mu_{CR} = 0.025$	
Diversity = 3.9283e-01	k_t = [1, 1, 1, 1]			
Generation 8400: Best = 3.843195e+03	FES = 840100	$\mu_F = 0.688$	$\mu_{CR} = 0.027$	
Diversity = 3.9722e-01	k_t = [4, 3, 2, 4]			
Generation 8410: Best = 3.843193e+03	FES = 841100	$\mu_F = 0.695$	$\mu_{CR} = 0.027$	
Diversity = 3.9723e-01	k_t = [1, 1, 1, 1]			
Generation 8420: Best = 3.843193e+03	FES = 842100	$\mu_F = 0.771$	$\mu_{CR} = 0.027$	
Diversity = 3.9342e-01	k_t = [4, 3, 2, 4]			
Generation 8430: Best = 3.843193e+03	FES = 843100	$\mu_F = 0.750$	$\mu_{CR} = 0.029$	
Diversity = 3.9341e-01	k_t = [1, 1, 1, 1]			
Generation 8440: Best = 3.843192e+03	FES = 844100	$\mu_F = 0.752$	$\mu_{CR} = 0.030$	
Diversity = 3.9340e-01	k_t = [3, 3, 3, 3]			
Generation 8450: Best = 3.843191e+03	FES = 845100	$\mu_F = 0.745$	$\mu_{CR} = 0.025$	
Diversity = 3.9341e-01	k_t = [1, 1, 1, 1]			
Generation 8460: Best = 3.843189e+03	FES = 846100	$\mu_F = 0.736$	$\mu_{CR} = 0.026$	
Diversity = 3.9287e-01	k_t = [2, 2, 4, 4]			
Generation 8470: Best = 3.843189e+03	FES = 847100	$\mu_F = 0.717$	$\mu_{CR} = 0.019$	
Diversity = 3.9287e-01	k_t = [1, 1, 1, 1]			
Generation 8480: Best = 3.843189e+03	FES = 848100	$\mu_F = 0.722$	$\mu_{CR} = 0.020$	
Diversity = 3.9284e-01	k_t = [2, 4, 4, 2]			
Generation 8490: Best = 3.843186e+03	FES = 849100	$\mu_F = 0.708$	$\mu_{CR} = 0.024$	
Diversity = 3.9285e-01	k_t = [1, 1, 1, 1]			
Generation 8500: Best = 3.843186e+03	FES = 850100	$\mu_F = 0.703$	$\mu_{CR} = 0.033$	
Diversity = 3.9277e-01	k_t = [4, 2, 2, 4]			
Generation 8510: Best = 3.843175e+03	FES = 851100	$\mu_F = 0.643$	$\mu_{CR} = 0.034$	
Diversity = 3.9277e-01	k_t = [1, 1, 1, 1]			
Generation 8520: Best = 3.843174e+03	FES = 852100	$\mu_F = 0.706$	$\mu_{CR} = 0.023$	
Diversity = 3.8794e-01	k_t = [2, 2, 3, 4]			
Generation 8530: Best = 3.843170e+03	FES = 853100	$\mu_F = 0.696$	$\mu_{CR} = 0.029$	
Diversity = 3.8793e-01	k_t = [1, 1, 1, 1]			
Generation 8540: Best = 3.843170e+03	FES = 854100	$\mu_F = 0.689$	$\mu_{CR} = 0.024$	
Diversity = 3.8809e-01	k_t = [2, 4, 4, 4]			
Generation 8550: Best = 3.843170e+03	FES = 855100	$\mu_F = 0.697$	$\mu_{CR} = 0.028$	
Diversity = 3.8807e-01	k_t = [1, 1, 1, 1]			
Generation 8560: Best = 3.843170e+03	FES = 856100	$\mu_F = 0.703$	$\mu_{CR} = 0.029$	
Diversity = 3.8805e-01	k_t = [4, 2, 3, 3]			
Generation 8570: Best = 3.843170e+03	FES = 857100	$\mu_F = 0.753$	$\mu_{CR} = 0.023$	
Diversity = 3.8805e-01	k_t = [1, 1, 1, 1]			
Generation 8580: Best = 3.843170e+03	FES = 858100	$\mu_F = 0.778$	$\mu_{CR} = 0.024$	
Diversity = 3.8804e-01	k_t = [2, 3, 2, 3]			
Generation 8590: Best = 3.843170e+03	FES = 859100	$\mu_F = 0.835$	$\mu_{CR} = 0.023$	

Diversity = 3.8804e-01	k_t = [1, 1, 1, 1]			
Generation 8600: Best = 3.843169e+03	FES = 860100	$\mu_F = 0.788$	$\mu_{CR} = 0.029$	
Diversity = 3.8811e-01	k_t = [4, 4, 2, 3]			
Generation 8610: Best = 3.843169e+03	FES = 861100	$\mu_F = 0.744$	$\mu_{CR} = 0.023$	
Diversity = 3.8811e-01	k_t = [1, 1, 1, 1]			
Generation 8620: Best = 3.843169e+03	FES = 862100	$\mu_F = 0.751$	$\mu_{CR} = 0.021$	
Diversity = 3.8810e-01	k_t = [3, 3, 4, 3]			
Generation 8630: Best = 3.843169e+03	FES = 863100	$\mu_F = 0.727$	$\mu_{CR} = 0.020$	
Diversity = 3.8812e-01	k_t = [1, 1, 1, 1]			
Generation 8640: Best = 3.843169e+03	FES = 864100	$\mu_F = 0.718$	$\mu_{CR} = 0.018$	
Diversity = 3.8812e-01	k_t = [3, 2, 3, 3]			
Generation 8650: Best = 3.843168e+03	FES = 865100	$\mu_F = 0.728$	$\mu_{CR} = 0.022$	
Diversity = 3.8813e-01	k_t = [1, 1, 1, 1]			
Generation 8660: Best = 3.843168e+03	FES = 866100	$\mu_F = 0.767$	$\mu_{CR} = 0.022$	
Diversity = 3.8813e-01	k_t = [3, 3, 2, 3]			
Generation 8670: Best = 3.843168e+03	FES = 867100	$\mu_F = 0.800$	$\mu_{CR} = 0.025$	
Diversity = 3.8814e-01	k_t = [1, 1, 1, 1]			
Generation 8680: Best = 3.843165e+03	FES = 868100	$\mu_F = 0.835$	$\mu_{CR} = 0.029$	
Diversity = 3.8812e-01	k_t = [2, 3, 4, 2]			
Generation 8690: Best = 3.843161e+03	FES = 869100	$\mu_F = 0.832$	$\mu_{CR} = 0.028$	
Diversity = 3.8810e-01	k_t = [1, 1, 1, 1]			
Generation 8700: Best = 3.843161e+03	FES = 870100	$\mu_F = 0.818$	$\mu_{CR} = 0.026$	
Diversity = 3.8810e-01	k_t = [4, 2, 4, 2]			
Generation 8710: Best = 3.843160e+03	FES = 871100	$\mu_F = 0.764$	$\mu_{CR} = 0.031$	
Diversity = 3.8811e-01	k_t = [1, 1, 1, 1]			
Generation 8720: Best = 3.843160e+03	FES = 872100	$\mu_F = 0.802$	$\mu_{CR} = 0.023$	
Diversity = 3.8811e-01	k_t = [2, 3, 2, 3]			
Generation 8730: Best = 3.843159e+03	FES = 873100	$\mu_F = 0.822$	$\mu_{CR} = 0.024$	
Diversity = 3.8809e-01	k_t = [1, 1, 1, 1]			
Generation 8740: Best = 3.843156e+03	FES = 874100	$\mu_F = 0.832$	$\mu_{CR} = 0.027$	
Diversity = 3.8809e-01	k_t = [3, 4, 3, 2]			
Generation 8750: Best = 3.843156e+03	FES = 875100	$\mu_F = 0.719$	$\mu_{CR} = 0.022$	
Diversity = 3.8781e-01	k_t = [1, 1, 1, 1]			
Generation 8760: Best = 3.843156e+03	FES = 876100	$\mu_F = 0.741$	$\mu_{CR} = 0.027$	
Diversity = 3.7149e-01	k_t = [2, 2, 4, 4]			
Generation 8770: Best = 3.843156e+03	FES = 877100	$\mu_F = 0.718$	$\mu_{CR} = 0.029$	
Diversity = 3.7150e-01	k_t = [1, 1, 1, 1]			
Generation 8780: Best = 3.843156e+03	FES = 878100	$\mu_F = 0.706$	$\mu_{CR} = 0.027$	
Diversity = 3.7148e-01	k_t = [2, 4, 4, 2]			
Generation 8790: Best = 3.843156e+03	FES = 879100	$\mu_F = 0.713$	$\mu_{CR} = 0.027$	
Diversity = 3.7144e-01	k_t = [1, 1, 1, 1]			
Generation 8800: Best = 3.843156e+03	FES = 880100	$\mu_F = 0.704$	$\mu_{CR} = 0.030$	
Diversity = 3.7141e-01	k_t = [4, 2, 2, 2]			
Generation 8810: Best = 3.843156e+03	FES = 881100	$\mu_F = 0.710$	$\mu_{CR} = 0.034$	
Diversity = 3.7140e-01	k_t = [1, 1, 1, 1]			
Generation 8820: Best = 3.843145e+03	FES = 882100	$\mu_F = 0.747$	$\mu_{CR} = 0.028$	
Diversity = 3.7607e-01	k_t = [4, 3, 2, 4]			
Generation 8830: Best = 3.843145e+03	FES = 883100	$\mu_F = 0.722$	$\mu_{CR} = 0.026$	
Diversity = 3.7607e-01	k_t = [1, 1, 1, 1]			
Generation 8840: Best = 3.843145e+03	FES = 884100	$\mu_F = 0.738$	$\mu_{CR} = 0.023$	
Diversity = 3.7606e-01	k_t = [4, 4, 3, 2]			
Generation 8850: Best = 3.843145e+03	FES = 885100	$\mu_F = 0.768$	$\mu_{CR} = 0.019$	
Diversity = 3.7606e-01	k_t = [1, 1, 1, 1]			
Generation 8860: Best = 3.843143e+03	FES = 886100	$\mu_F = 0.755$	$\mu_{CR} = 0.021$	

Diversity = 3.7176e-01	k_t = [2, 4, 2, 4]			
Generation 8870: Best = 3.843143e+03	FES = 887100	$\mu_F = 0.690$	$\mu_{CR} = 0.018$	
Diversity = 3.7174e-01	k_t = [1, 1, 1, 1]			
Generation 8880: Best = 3.843142e+03	FES = 888100	$\mu_F = 0.730$	$\mu_{CR} = 0.019$	
Diversity = 3.7174e-01	k_t = [4, 2, 3, 3]			
Generation 8890: Best = 3.843142e+03	FES = 889100	$\mu_F = 0.718$	$\mu_{CR} = 0.022$	
Diversity = 3.7173e-01	k_t = [1, 1, 1, 1]			
Generation 8900: Best = 3.843141e+03	FES = 890100	$\mu_F = 0.747$	$\mu_{CR} = 0.021$	
Diversity = 3.7191e-01	k_t = [2, 2, 4, 4]			
Generation 8910: Best = 3.843141e+03	FES = 891100	$\mu_F = 0.758$	$\mu_{CR} = 0.018$	
Diversity = 3.7190e-01	k_t = [1, 1, 1, 1]			
Generation 8920: Best = 3.843141e+03	FES = 892100	$\mu_F = 0.763$	$\mu_{CR} = 0.025$	
Diversity = 3.7200e-01	k_t = [2, 4, 4, 4]			
Generation 8930: Best = 3.843141e+03	FES = 893100	$\mu_F = 0.764$	$\mu_{CR} = 0.024$	
Diversity = 3.7184e-01	k_t = [1, 1, 1, 1]			
Generation 8940: Best = 3.843140e+03	FES = 894100	$\mu_F = 0.824$	$\mu_{CR} = 0.026$	
Diversity = 3.7185e-01	k_t = [3, 4, 2, 3]			
Generation 8950: Best = 3.843139e+03	FES = 895100	$\mu_F = 0.788$	$\mu_{CR} = 0.023$	
Diversity = 3.7185e-01	k_t = [1, 1, 1, 1]			
Generation 8960: Best = 3.843139e+03	FES = 896100	$\mu_F = 0.827$	$\mu_{CR} = 0.022$	
Diversity = 3.7186e-01	k_t = [4, 3, 3, 2]			
Generation 8970: Best = 3.843137e+03	FES = 897100	$\mu_F = 0.831$	$\mu_{CR} = 0.018$	
Diversity = 3.7188e-01	k_t = [1, 1, 1, 1]			
Generation 8980: Best = 3.843137e+03	FES = 898100	$\mu_F = 0.748$	$\mu_{CR} = 0.023$	
Diversity = 3.7187e-01	k_t = [3, 3, 3, 2]			
Generation 8990: Best = 3.843137e+03	FES = 899100	$\mu_F = 0.737$	$\mu_{CR} = 0.025$	
Diversity = 3.7190e-01	k_t = [1, 1, 1, 1]			
Generation 9000: Best = 3.843137e+03	FES = 900100	$\mu_F = 0.758$	$\mu_{CR} = 0.030$	
Diversity = 3.7190e-01	k_t = [4, 3, 4, 2]			
Generation 9010: Best = 3.843137e+03	FES = 901100	$\mu_F = 0.767$	$\mu_{CR} = 0.025$	
Diversity = 3.7189e-01	k_t = [1, 1, 1, 1]			
Generation 9020: Best = 3.843136e+03	FES = 902100	$\mu_F = 0.775$	$\mu_{CR} = 0.026$	
Diversity = 3.7189e-01	k_t = [4, 3, 4, 3]			
Generation 9030: Best = 3.843136e+03	FES = 903100	$\mu_F = 0.776$	$\mu_{CR} = 0.031$	
Diversity = 3.7191e-01	k_t = [1, 1, 1, 1]			
Generation 9040: Best = 3.843136e+03	FES = 904100	$\mu_F = 0.702$	$\mu_{CR} = 0.034$	
Diversity = 3.7192e-01	k_t = [3, 2, 2, 2]			
Generation 9050: Best = 3.843136e+03	FES = 905100	$\mu_F = 0.706$	$\mu_{CR} = 0.028$	
Diversity = 3.7191e-01	k_t = [1, 1, 1, 1]			
Generation 9060: Best = 3.843134e+03	FES = 906100	$\mu_F = 0.742$	$\mu_{CR} = 0.020$	
Diversity = 3.7175e-01	k_t = [3, 3, 3, 2]			
Generation 9070: Best = 3.843132e+03	FES = 907100	$\mu_F = 0.730$	$\mu_{CR} = 0.027$	
Diversity = 3.7178e-01	k_t = [1, 1, 1, 1]			
Generation 9080: Best = 3.843132e+03	FES = 908100	$\mu_F = 0.761$	$\mu_{CR} = 0.028$	
Diversity = 3.7647e-01	k_t = [3, 2, 2, 4]			
Generation 9090: Best = 3.843132e+03	FES = 909100	$\mu_F = 0.740$	$\mu_{CR} = 0.037$	
Diversity = 3.7647e-01	k_t = [1, 1, 1, 1]			
Generation 9100: Best = 3.843130e+03	FES = 910100	$\mu_F = 0.781$	$\mu_{CR} = 0.025$	
Diversity = 3.7647e-01	k_t = [4, 2, 2, 2]			
Generation 9110: Best = 3.843130e+03	FES = 911100	$\mu_F = 0.799$	$\mu_{CR} = 0.022$	
Diversity = 3.7646e-01	k_t = [1, 1, 1, 1]			
Generation 9120: Best = 3.843130e+03	FES = 912100	$\mu_F = 0.754$	$\mu_{CR} = 0.025$	
Diversity = 3.7648e-01	k_t = [4, 3, 2, 3]			
Generation 9130: Best = 3.843130e+03	FES = 913100	$\mu_F = 0.797$	$\mu_{CR} = 0.022$	

Diversity = 3.7647e-01	k_t = [1, 1, 1, 1]			
Generation 9140: Best = 3.843130e+03	FES = 914100	$\mu_F = 0.837$	$\mu_{CR} = 0.021$	
Diversity = 3.7666e-01	k_t = [3, 3, 2, 4]			
Generation 9150: Best = 3.843130e+03	FES = 915100	$\mu_F = 0.791$	$\mu_{CR} = 0.023$	
Diversity = 3.7664e-01	k_t = [1, 1, 1, 1]			
Generation 9160: Best = 3.843123e+03	FES = 916100	$\mu_F = 0.842$	$\mu_{CR} = 0.027$	
Diversity = 3.7128e-01	k_t = [4, 4, 4, 3]			
Generation 9170: Best = 3.843120e+03	FES = 917100	$\mu_F = 0.760$	$\mu_{CR} = 0.024$	
Diversity = 3.7127e-01	k_t = [1, 1, 1, 1]			
Generation 9180: Best = 3.843118e+03	FES = 918100	$\mu_F = 0.752$	$\mu_{CR} = 0.030$	
Diversity = 3.7088e-01	k_t = [4, 3, 2, 4]			
Generation 9190: Best = 3.843115e+03	FES = 919100	$\mu_F = 0.747$	$\mu_{CR} = 0.023$	
Diversity = 3.7089e-01	k_t = [1, 1, 1, 1]			
Generation 9200: Best = 3.843114e+03	FES = 920100	$\mu_F = 0.744$	$\mu_{CR} = 0.023$	
Diversity = 3.7087e-01	k_t = [4, 4, 4, 3]			
Generation 9210: Best = 3.843114e+03	FES = 921100	$\mu_F = 0.772$	$\mu_{CR} = 0.025$	
Diversity = 3.7085e-01	k_t = [1, 1, 1, 1]			
Generation 9220: Best = 3.843114e+03	FES = 922100	$\mu_F = 0.750$	$\mu_{CR} = 0.025$	
Diversity = 3.7086e-01	k_t = [3, 2, 3, 2]			
Generation 9230: Best = 3.843114e+03	FES = 923100	$\mu_F = 0.728$	$\mu_{CR} = 0.027$	
Diversity = 3.7086e-01	k_t = [1, 1, 1, 1]			
Generation 9240: Best = 3.843114e+03	FES = 924100	$\mu_F = 0.780$	$\mu_{CR} = 0.021$	
Diversity = 3.7085e-01	k_t = [3, 4, 3, 3]			
Generation 9250: Best = 3.843114e+03	FES = 925100	$\mu_F = 0.762$	$\mu_{CR} = 0.023$	
Diversity = 3.7085e-01	k_t = [1, 1, 1, 1]			
Generation 9260: Best = 3.843114e+03	FES = 926100	$\mu_F = 0.732$	$\mu_{CR} = 0.024$	
Diversity = 3.5935e-01	k_t = [2, 2, 2, 4]			
Generation 9270: Best = 3.843114e+03	FES = 927100	$\mu_F = 0.758$	$\mu_{CR} = 0.027$	
Diversity = 3.5935e-01	k_t = [1, 1, 1, 1]			
Generation 9280: Best = 3.843114e+03	FES = 928100	$\mu_F = 0.760$	$\mu_{CR} = 0.026$	
Diversity = 3.5953e-01	k_t = [3, 2, 3, 4]			
Generation 9290: Best = 3.843114e+03	FES = 929100	$\mu_F = 0.767$	$\mu_{CR} = 0.021$	
Diversity = 3.5953e-01	k_t = [1, 1, 1, 1]			
Generation 9300: Best = 3.843107e+03	FES = 930100	$\mu_F = 0.805$	$\mu_{CR} = 0.024$	
Diversity = 3.5954e-01	k_t = [4, 3, 3, 2]			
Generation 9310: Best = 3.843105e+03	FES = 931100	$\mu_F = 0.782$	$\mu_{CR} = 0.033$	
Diversity = 3.5954e-01	k_t = [1, 1, 1, 1]			
Generation 9320: Best = 3.843105e+03	FES = 932100	$\mu_F = 0.803$	$\mu_{CR} = 0.032$	
Diversity = 3.5954e-01	k_t = [3, 4, 2, 2]			
Generation 9330: Best = 3.843105e+03	FES = 933100	$\mu_F = 0.744$	$\mu_{CR} = 0.028$	
Diversity = 3.5955e-01	k_t = [1, 1, 1, 1]			
Generation 9340: Best = 3.843105e+03	FES = 934100	$\mu_F = 0.763$	$\mu_{CR} = 0.022$	
Diversity = 3.5673e-01	k_t = [2, 4, 2, 4]			
Generation 9350: Best = 3.843098e+03	FES = 935100	$\mu_F = 0.748$	$\mu_{CR} = 0.018$	
Diversity = 3.5673e-01	k_t = [1, 1, 1, 1]			
Generation 9360: Best = 3.843097e+03	FES = 936100	$\mu_F = 0.808$	$\mu_{CR} = 0.021$	
Diversity = 3.5674e-01	k_t = [3, 2, 4, 2]			
Generation 9370: Best = 3.843097e+03	FES = 937100	$\mu_F = 0.826$	$\mu_{CR} = 0.023$	
Diversity = 3.5673e-01	k_t = [1, 1, 1, 1]			
Generation 9380: Best = 3.843097e+03	FES = 938100	$\mu_F = 0.844$	$\mu_{CR} = 0.022$	
Diversity = 3.5672e-01	k_t = [3, 4, 3, 2]			
Generation 9390: Best = 3.843097e+03	FES = 939100	$\mu_F = 0.830$	$\mu_{CR} = 0.025$	
Diversity = 3.5671e-01	k_t = [1, 1, 1, 1]			
Generation 9400: Best = 3.843097e+03	FES = 940100	$\mu_F = 0.822$	$\mu_{CR} = 0.025$	

Diversity = 3.4801e-01	k_t = [4, 2, 4, 4]			
Generation 9410: Best = 3.843097e+03	FES = 941100	$\mu_F = 0.850$	$\mu_{CR} = 0.030$	
Diversity = 3.4801e-01	k_t = [1, 1, 1, 1]			
Generation 9420: Best = 3.843094e+03	FES = 942100	$\mu_F = 0.886$	$\mu_{CR} = 0.027$	
Diversity = 3.4801e-01	k_t = [2, 4, 4, 3]			
Generation 9430: Best = 3.843094e+03	FES = 943100	$\mu_F = 0.808$	$\mu_{CR} = 0.023$	
Diversity = 3.4800e-01	k_t = [1, 1, 1, 1]			
Generation 9440: Best = 3.843093e+03	FES = 944100	$\mu_F = 0.754$	$\mu_{CR} = 0.028$	
Diversity = 3.4568e-01	k_t = [4, 2, 2, 4]			
Generation 9450: Best = 3.843093e+03	FES = 945100	$\mu_F = 0.782$	$\mu_{CR} = 0.023$	
Diversity = 3.4568e-01	k_t = [1, 1, 1, 1]			
Generation 9460: Best = 3.843093e+03	FES = 946100	$\mu_F = 0.759$	$\mu_{CR} = 0.029$	
Diversity = 3.4567e-01	k_t = [2, 3, 4, 2]			
Generation 9470: Best = 3.843093e+03	FES = 947100	$\mu_F = 0.790$	$\mu_{CR} = 0.033$	
Diversity = 3.4565e-01	k_t = [1, 1, 1, 1]			
Generation 9480: Best = 3.843093e+03	FES = 948100	$\mu_F = 0.732$	$\mu_{CR} = 0.029$	
Diversity = 3.4565e-01	k_t = [3, 2, 3, 3]			
Generation 9490: Best = 3.843090e+03	FES = 949100	$\mu_F = 0.727$	$\mu_{CR} = 0.028$	
Diversity = 3.4566e-01	k_t = [1, 1, 1, 1]			
Generation 9500: Best = 3.843090e+03	FES = 950100	$\mu_F = 0.733$	$\mu_{CR} = 0.028$	
Diversity = 3.4566e-01	k_t = [4, 3, 2, 3]			
Generation 9510: Best = 3.843090e+03	FES = 951100	$\mu_F = 0.696$	$\mu_{CR} = 0.023$	
Diversity = 3.4567e-01	k_t = [1, 1, 1, 1]			
Generation 9520: Best = 3.843088e+03	FES = 952100	$\mu_F = 0.648$	$\mu_{CR} = 0.023$	
Diversity = 3.4569e-01	k_t = [3, 2, 3, 3]			
Generation 9530: Best = 3.843087e+03	FES = 953100	$\mu_F = 0.641$	$\mu_{CR} = 0.023$	
Diversity = 3.4567e-01	k_t = [1, 1, 1, 1]			
Generation 9540: Best = 3.843084e+03	FES = 954100	$\mu_F = 0.706$	$\mu_{CR} = 0.029$	
Diversity = 3.4567e-01	k_t = [4, 2, 2, 3]			
Generation 9550: Best = 3.843063e+03	FES = 955100	$\mu_F = 0.718$	$\mu_{CR} = 0.028$	
Diversity = 3.4566e-01	k_t = [1, 1, 1, 1]			
Generation 9560: Best = 3.843063e+03	FES = 956100	$\mu_F = 0.740$	$\mu_{CR} = 0.022$	
Diversity = 3.4566e-01	k_t = [2, 3, 3, 3]			
Generation 9570: Best = 3.843063e+03	FES = 957100	$\mu_F = 0.746$	$\mu_{CR} = 0.021$	
Diversity = 3.4566e-01	k_t = [1, 1, 1, 1]			
Generation 9580: Best = 3.843063e+03	FES = 958100	$\mu_F = 0.744$	$\mu_{CR} = 0.025$	
Diversity = 3.4566e-01	k_t = [4, 3, 3, 3]			
Generation 9590: Best = 3.843063e+03	FES = 959100	$\mu_F = 0.697$	$\mu_{CR} = 0.021$	
Diversity = 3.4568e-01	k_t = [1, 1, 1, 1]			
Generation 9600: Best = 3.843063e+03	FES = 960100	$\mu_F = 0.727$	$\mu_{CR} = 0.024$	
Diversity = 3.2964e-01	k_t = [3, 4, 2, 4]			
Generation 9610: Best = 3.843058e+03	FES = 961100	$\mu_F = 0.751$	$\mu_{CR} = 0.029$	
Diversity = 3.2965e-01	k_t = [1, 1, 1, 1]			
Generation 9620: Best = 3.843058e+03	FES = 962100	$\mu_F = 0.790$	$\mu_{CR} = 0.035$	
Diversity = 3.2965e-01	k_t = [4, 4, 2, 3]			
Generation 9630: Best = 3.843058e+03	FES = 963100	$\mu_F = 0.732$	$\mu_{CR} = 0.022$	
Diversity = 3.2966e-01	k_t = [1, 1, 1, 1]			
Generation 9640: Best = 3.843051e+03	FES = 964100	$\mu_F = 0.711$	$\mu_{CR} = 0.021$	
Diversity = 3.2965e-01	k_t = [4, 2, 3, 3]			
Generation 9650: Best = 3.843051e+03	FES = 965100	$\mu_F = 0.658$	$\mu_{CR} = 0.021$	
Diversity = 3.2964e-01	k_t = [1, 1, 1, 1]			
Generation 9660: Best = 3.843051e+03	FES = 966100	$\mu_F = 0.721$	$\mu_{CR} = 0.028$	
Diversity = 3.2964e-01	k_t = [4, 4, 2, 3]			
Generation 9670: Best = 3.843051e+03	FES = 967100	$\mu_F = 0.746$	$\mu_{CR} = 0.024$	

Diversity = 3.2965e-01	k_t = [1, 1, 1, 1]			
Generation 9680: Best = 3.843051e+03	FES = 968100	$\mu_F = 0.730$	$\mu_{CR} = 0.019$	
Diversity = 3.2965e-01	k_t = [4, 2, 2, 3]			
Generation 9690: Best = 3.843051e+03	FES = 969100	$\mu_F = 0.768$	$\mu_{CR} = 0.020$	
Diversity = 3.2966e-01	k_t = [1, 1, 1, 1]			
Generation 9700: Best = 3.843051e+03	FES = 970100	$\mu_F = 0.787$	$\mu_{CR} = 0.017$	
Diversity = 3.2755e-01	k_t = [3, 2, 3, 4]			
Generation 9710: Best = 3.843051e+03	FES = 971100	$\mu_F = 0.742$	$\mu_{CR} = 0.024$	
Diversity = 3.2756e-01	k_t = [1, 1, 1, 1]			
Generation 9720: Best = 3.843042e+03	FES = 972100	$\mu_F = 0.728$	$\mu_{CR} = 0.018$	
Diversity = 3.2669e-01	k_t = [3, 2, 4, 4]			
Generation 9730: Best = 3.843041e+03	FES = 973100	$\mu_F = 0.773$	$\mu_{CR} = 0.017$	
Diversity = 3.2669e-01	k_t = [1, 1, 1, 1]			
Generation 9740: Best = 3.843040e+03	FES = 974100	$\mu_F = 0.761$	$\mu_{CR} = 0.018$	
Diversity = 3.2668e-01	k_t = [2, 3, 2, 2]			
Generation 9750: Best = 3.843039e+03	FES = 975100	$\mu_F = 0.792$	$\mu_{CR} = 0.021$	
Diversity = 3.2822e-01	k_t = [1, 1, 1, 1]			
Generation 9760: Best = 3.843039e+03	FES = 976100	$\mu_F = 0.801$	$\mu_{CR} = 0.027$	
Diversity = 3.2821e-01	k_t = [2, 2, 2, 3]			
Generation 9770: Best = 3.843039e+03	FES = 977100	$\mu_F = 0.748$	$\mu_{CR} = 0.021$	
Diversity = 3.2822e-01	k_t = [1, 1, 1, 1]			
Generation 9780: Best = 3.843036e+03	FES = 978100	$\mu_F = 0.774$	$\mu_{CR} = 0.020$	
Diversity = 3.1560e-01	k_t = [3, 4, 3, 4]			
Generation 9790: Best = 3.843036e+03	FES = 979100	$\mu_F = 0.686$	$\mu_{CR} = 0.021$	
Diversity = 3.1561e-01	k_t = [1, 1, 1, 1]			
Generation 9800: Best = 3.843036e+03	FES = 980100	$\mu_F = 0.728$	$\mu_{CR} = 0.023$	
Diversity = 3.0787e-01	k_t = [4, 4, 4, 4]			
Generation 9810: Best = 3.843036e+03	FES = 981100	$\mu_F = 0.783$	$\mu_{CR} = 0.034$	
Diversity = 3.0787e-01	k_t = [1, 1, 1, 1]			
Generation 9820: Best = 3.843036e+03	FES = 982100	$\mu_F = 0.760$	$\mu_{CR} = 0.033$	
Diversity = 3.0787e-01	k_t = [4, 4, 4, 3]			
Generation 9830: Best = 3.843036e+03	FES = 983100	$\mu_F = 0.777$	$\mu_{CR} = 0.025$	
Diversity = 3.0787e-01	k_t = [1, 1, 1, 1]			
Generation 9840: Best = 3.843036e+03	FES = 984100	$\mu_F = 0.714$	$\mu_{CR} = 0.029$	
Diversity = 2.9810e-01	k_t = [2, 4, 4, 3]			
Generation 9850: Best = 3.843036e+03	FES = 985100	$\mu_F = 0.649$	$\mu_{CR} = 0.029$	
Diversity = 2.7607e-01	k_t = [1, 1, 1, 1]			
Generation 9860: Best = 3.843036e+03	FES = 986100	$\mu_F = 0.679$	$\mu_{CR} = 0.031$	
Diversity = 2.6509e-01	k_t = [4, 4, 2, 4]			
Generation 9870: Best = 3.843036e+03	FES = 987100	$\mu_F = 0.726$	$\mu_{CR} = 0.029$	
Diversity = 2.6509e-01	k_t = [1, 1, 1, 1]			
Generation 9880: Best = 3.843036e+03	FES = 988100	$\mu_F = 0.744$	$\mu_{CR} = 0.032$	
Diversity = 2.5415e-01	k_t = [4, 3, 2, 4]			
Generation 9890: Best = 3.843036e+03	FES = 989100	$\mu_F = 0.743$	$\mu_{CR} = 0.021$	
Diversity = 2.5261e-01	k_t = [1, 1, 1, 1]			
Generation 9900: Best = 3.843036e+03	FES = 990100	$\mu_F = 0.753$	$\mu_{CR} = 0.030$	
Diversity = 2.5261e-01	k_t = [4, 4, 2, 2]			
Generation 9910: Best = 3.843036e+03	FES = 991100	$\mu_F = 0.760$	$\mu_{CR} = 0.027$	
Diversity = 2.5260e-01	k_t = [1, 1, 1, 1]			
Generation 9920: Best = 3.843036e+03	FES = 992100	$\mu_F = 0.733$	$\mu_{CR} = 0.023$	
Diversity = 2.5090e-01	k_t = [4, 2, 2, 4]			
Generation 9930: Best = 3.843036e+03	FES = 993100	$\mu_F = 0.716$	$\mu_{CR} = 0.027$	
Diversity = 2.5089e-01	k_t = [1, 1, 1, 1]			
Generation 9940: Best = 3.843035e+03	FES = 994100	$\mu_F = 0.709$	$\mu_{CR} = 0.032$	


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Diversity = 2.5243e-01 k_t = [2, 2, 3, 4]
Generation 9950: Best = 3.843035e+03 FES = 995100  $\mu_F$  = 0.661  $\mu_{CR}$  = 0.032
Diversity = 2.5072e-01 k_t = [1, 1, 1, 1]
Generation 9960: Best = 3.843030e+03 FES = 996100  $\mu_F$  = 0.668  $\mu_{CR}$  = 0.028
Diversity = 2.5071e-01 k_t = [4, 4, 4, 2]
Generation 9970: Best = 3.843030e+03 FES = 997100  $\mu_F$  = 0.630  $\mu_{CR}$  = 0.035
Diversity = 2.5070e-01 k_t = [1, 1, 1, 1]
Generation 9980: Best = 3.843030e+03 FES = 998100  $\mu_F$  = 0.676  $\mu_{CR}$  = 0.030
Diversity = 2.3309e-01 k_t = [3, 2, 2, 4]
Generation 9990: Best = 3.843025e+03 FES = 999100  $\mu_F$  = 0.719  $\mu_{CR}$  = 0.024
Diversity = 2.3310e-01 k_t = [1, 1, 1, 1]

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Optimization complete.

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Best Fitness          = 3.843024e+03
Total Function Evals  = 1000000
Total Successful Updates = 48930

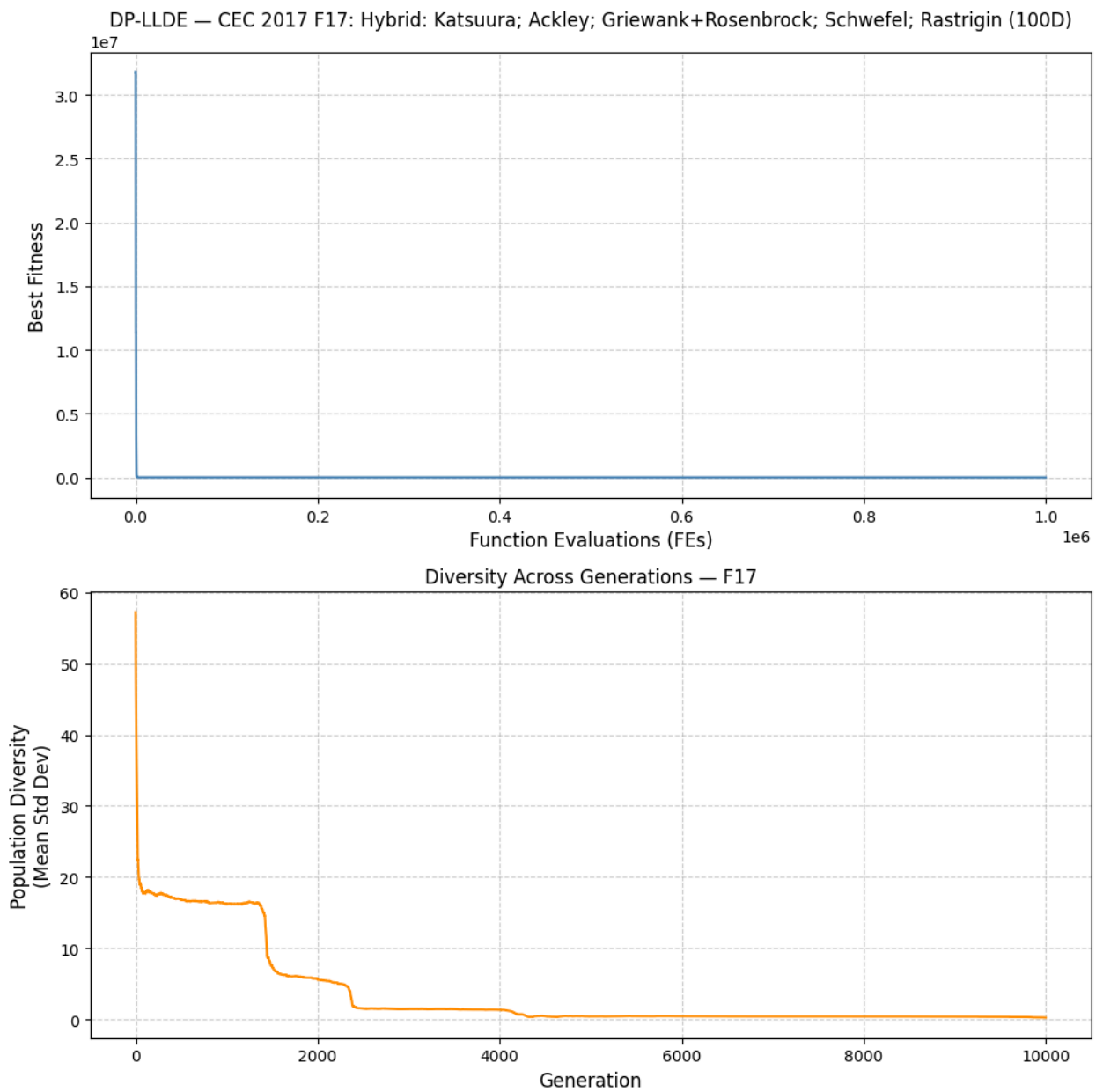
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Results:
  Best Fitness          : 3.843024e+03
  Global Optimum        : 1.700000e+03
  Error (|best-optimum|) : 2.143024e+03
  Best Solution (5 dim) : [-25.06409835   6.69180086 -33.32252734   5.53235039
-41.3972775 ]
  D^(0) (init. diversity): 5.7220e+01
=====
=

```

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results/all_results_checkpoint.pkl



Graph saved as convergence_F17.png

✓ Plot saved to: /content/drive/My Drive/thesis_results/plots/convergence_F17.png

```
In [ ]: # =====
# F18 — Hybrid Function 8 (Elliptic; Ackley; Rastrigin; HGBat; Discus)
# =====
run_and_plot('F18', make_wrapper(hybrid.f18), 1800.0, 'Hybrid: Elliptic; Ackle
```

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DP-LLDE on CEC 2017 F18 – Single Run

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Function : Hybrid: Elliptic; Ackley; Rastrigin; HGBat; Discus (100D)
Bounds : [-100, 100]
Global optimum : 1800.0
Initial $D^*(0) = 5.7220e+01$ | Diversity threshold = 5.7220e+00
Generation 0: Best Fitness = 1.395496e+09
Generation 10: Best = 6.130873e+07 FES = 1100 $\mu_F = 0.561$ $\mu_{CR} = 0.354$
Diversity = 3.8982e+01 $k_t = [1, 1, 1, 1]$
Generation 20: Best = 4.601408e+07 FES = 2100 $\mu_F = 0.607$ $\mu_{CR} = 0.332$
Diversity = 2.9858e+01 $k_t = [2, 3, 2, 3]$
Generation 30: Best = 1.787334e+07 FES = 3100 $\mu_F = 0.608$ $\mu_{CR} = 0.292$
Diversity = 2.3596e+01 $k_t = [1, 1, 1, 1]$
Generation 40: Best = 7.102130e+06 FES = 4100 $\mu_F = 0.633$ $\mu_{CR} = 0.254$
Diversity = 1.6583e+01 $k_t = [3, 3, 4, 2]$
Generation 50: Best = 6.785617e+06 FES = 5100 $\mu_F = 0.664$ $\mu_{CR} = 0.174$
Diversity = 1.3736e+01 $k_t = [1, 1, 1, 1]$
Generation 60: Best = 4.137281e+06 FES = 6100 $\mu_F = 0.683$ $\mu_{CR} = 0.146$
Diversity = 1.1549e+01 $k_t = [2, 3, 3, 3]$
Generation 70: Best = 3.304633e+06 FES = 7100 $\mu_F = 0.698$ $\mu_{CR} = 0.084$
Diversity = 9.4466e+00 $k_t = [1, 1, 1, 1]$
Generation 80: Best = 3.047326e+06 FES = 8100 $\mu_F = 0.726$ $\mu_{CR} = 0.054$
Diversity = 8.8816e+00 $k_t = [3, 3, 3, 3]$
Generation 90: Best = 2.893977e+06 FES = 9100 $\mu_F = 0.730$ $\mu_{CR} = 0.035$
Diversity = 7.7805e+00 $k_t = [1, 1, 1, 1]$
Generation 100: Best = 2.654738e+06 FES = 10100 $\mu_F = 0.734$ $\mu_{CR} = 0.043$
Diversity = 5.9692e+00 $k_t = [2, 2, 2, 2]$
Generation 110: Best = 2.444683e+06 FES = 11100 $\mu_F = 0.735$ $\mu_{CR} = 0.041$
Diversity = 5.5610e+00 $k_t = [1, 1, 1, 1]$
Generation 120: Best = 2.227622e+06 FES = 12100 $\mu_F = 0.758$ $\mu_{CR} = 0.037$
Diversity = 5.8240e+00 $k_t = [3, 2, 4, 3]$
Generation 130: Best = 2.016275e+06 FES = 13100 $\mu_F = 0.764$ $\mu_{CR} = 0.042$
Diversity = 5.2138e+00 $k_t = [1, 1, 1, 1]$
Generation 140: Best = 1.963372e+06 FES = 14100 $\mu_F = 0.786$ $\mu_{CR} = 0.041$
Diversity = 5.1700e+00 $k_t = [4, 4, 4, 3]$
Generation 150: Best = 1.894726e+06 FES = 15100 $\mu_F = 0.795$ $\mu_{CR} = 0.035$
Diversity = 4.9639e+00 $k_t = [1, 1, 1, 1]$
Generation 160: Best = 1.810438e+06 FES = 16100 $\mu_F = 0.792$ $\mu_{CR} = 0.042$
Diversity = 4.4079e+00 $k_t = [4, 3, 4, 2]$
Generation 170: Best = 1.763671e+06 FES = 17100 $\mu_F = 0.795$ $\mu_{CR} = 0.036$
Diversity = 4.3811e+00 $k_t = [1, 1, 1, 1]$
Generation 180: Best = 1.695178e+06 FES = 18100 $\mu_F = 0.803$ $\mu_{CR} = 0.036$
Diversity = 4.6546e+00 $k_t = [2, 2, 4, 4]$
Generation 190: Best = 1.577206e+06 FES = 19100 $\mu_F = 0.811$ $\mu_{CR} = 0.033$
Diversity = 4.8841e+00 $k_t = [1, 1, 1, 1]$
Generation 200: Best = 1.542067e+06 FES = 20100 $\mu_F = 0.818$ $\mu_{CR} = 0.038$
Diversity = 4.9628e+00 $k_t = [4, 2, 3, 4]$
Generation 210: Best = 1.491383e+06 FES = 21100 $\mu_F = 0.810$ $\mu_{CR} = 0.041$
Diversity = 4.7751e+00 $k_t = [1, 1, 1, 1]$
Generation 220: Best = 1.452861e+06 FES = 22100 $\mu_F = 0.805$ $\mu_{CR} = 0.035$
Diversity = 5.0545e+00 $k_t = [3, 2, 3, 3]$

Generation 230: Best = 1.422697e+06	FES = 23100	$\mu_F = 0.791$	$\mu_{CR} = 0.037$
Diversity = 4.8410e+00	$k_t = [1, 1, 1, 1]$		
Generation 240: Best = 1.394368e+06	FES = 24100	$\mu_F = 0.805$	$\mu_{CR} = 0.039$
Diversity = 5.1645e+00	$k_t = [3, 4, 2, 4]$		
Generation 250: Best = 1.365214e+06	FES = 25100	$\mu_F = 0.804$	$\mu_{CR} = 0.035$
Diversity = 5.3139e+00	$k_t = [1, 1, 1, 1]$		
Generation 260: Best = 1.323744e+06	FES = 26100	$\mu_F = 0.801$	$\mu_{CR} = 0.041$
Diversity = 5.0963e+00	$k_t = [3, 3, 3, 3]$		
Generation 270: Best = 1.269962e+06	FES = 27100	$\mu_F = 0.817$	$\mu_{CR} = 0.037$
Diversity = 6.0161e+00	$k_t = [1, 1, 1, 1]$		
Generation 280: Best = 1.249968e+06	FES = 28100	$\mu_F = 0.817$	$\mu_{CR} = 0.039$
Diversity = 7.1949e+00	$k_t = [4, 4, 3, 4]$		
Generation 290: Best = 1.190816e+06	FES = 29100	$\mu_F = 0.819$	$\mu_{CR} = 0.036$
Diversity = 7.2336e+00	$k_t = [1, 1, 1, 1]$		
Generation 300: Best = 1.160230e+06	FES = 30100	$\mu_F = 0.818$	$\mu_{CR} = 0.033$
Diversity = 6.0570e+00	$k_t = [3, 2, 4, 4]$		
Generation 310: Best = 1.144401e+06	FES = 31100	$\mu_F = 0.834$	$\mu_{CR} = 0.032$
Diversity = 6.4775e+00	$k_t = [1, 1, 1, 1]$		
Generation 320: Best = 1.111021e+06	FES = 32100	$\mu_F = 0.827$	$\mu_{CR} = 0.033$
Diversity = 5.3417e+00	$k_t = [4, 4, 3, 3]$		
Generation 330: Best = 1.075945e+06	FES = 33100	$\mu_F = 0.829$	$\mu_{CR} = 0.028$
Diversity = 6.4765e+00	$k_t = [1, 1, 1, 1]$		
Generation 340: Best = 1.071360e+06	FES = 34100	$\mu_F = 0.824$	$\mu_{CR} = 0.037$
Diversity = 6.5890e+00	$k_t = [4, 3, 2, 3]$		
Generation 350: Best = 1.045389e+06	FES = 35100	$\mu_F = 0.832$	$\mu_{CR} = 0.043$
Diversity = 6.3495e+00	$k_t = [1, 1, 1, 1]$		
Generation 360: Best = 1.032998e+06	FES = 36100	$\mu_F = 0.819$	$\mu_{CR} = 0.041$
Diversity = 5.9806e+00	$k_t = [2, 2, 3, 4]$		
Generation 370: Best = 1.020849e+06	FES = 37100	$\mu_F = 0.826$	$\mu_{CR} = 0.030$
Diversity = 5.6971e+00	$k_t = [1, 1, 1, 1]$		
Generation 380: Best = 1.003340e+06	FES = 38100	$\mu_F = 0.824$	$\mu_{CR} = 0.035$
Diversity = 4.7675e+00	$k_t = [4, 3, 4, 3]$		
Generation 390: Best = 9.852687e+05	FES = 39100	$\mu_F = 0.828$	$\mu_{CR} = 0.036$
Diversity = 5.1384e+00	$k_t = [1, 1, 1, 1]$		
Generation 400: Best = 9.797901e+05	FES = 40100	$\mu_F = 0.823$	$\mu_{CR} = 0.033$
Diversity = 5.5358e+00	$k_t = [3, 4, 2, 3]$		
Generation 410: Best = 9.702364e+05	FES = 41100	$\mu_F = 0.827$	$\mu_{CR} = 0.030$
Diversity = 6.3352e+00	$k_t = [1, 1, 1, 1]$		
Generation 420: Best = 9.545135e+05	FES = 42100	$\mu_F = 0.813$	$\mu_{CR} = 0.036$
Diversity = 6.1469e+00	$k_t = [4, 4, 3, 4]$		
Generation 430: Best = 9.457260e+05	FES = 43100	$\mu_F = 0.792$	$\mu_{CR} = 0.037$
Diversity = 6.3468e+00	$k_t = [1, 1, 1, 1]$		
Generation 440: Best = 9.333265e+05	FES = 44100	$\mu_F = 0.785$	$\mu_{CR} = 0.031$
Diversity = 6.6361e+00	$k_t = [3, 2, 2, 4]$		
Generation 450: Best = 9.288295e+05	FES = 45100	$\mu_F = 0.773$	$\mu_{CR} = 0.026$
Diversity = 5.2505e+00	$k_t = [1, 1, 1, 1]$		
Generation 460: Best = 9.191451e+05	FES = 46100	$\mu_F = 0.791$	$\mu_{CR} = 0.037$
Diversity = 5.0178e+00	$k_t = [2, 4, 2, 3]$		
Generation 470: Best = 9.191451e+05	FES = 47100	$\mu_F = 0.792$	$\mu_{CR} = 0.034$
Diversity = 4.9759e+00	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 9.085445e+05	FES = 48100	$\mu_F = 0.779$	$\mu_{CR} = 0.042$
Diversity = 5.4682e+00	$k_t = [4, 3, 2, 2]$		
Generation 490: Best = 8.950790e+05	FES = 49100	$\mu_F = 0.777$	$\mu_{CR} = 0.033$
Diversity = 5.9130e+00	$k_t = [1, 1, 1, 1]$		

Generation 500: Best = 8.903143e+05	FES = 50100	$\mu_F = 0.788$	$\mu_{CR} = 0.038$
Diversity = 5.9586e+00	$k_t = [2, 4, 2, 3]$		
Generation 510: Best = 8.828254e+05	FES = 51100	$\mu_F = 0.788$	$\mu_{CR} = 0.038$
Diversity = 5.9293e+00	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 8.759490e+05	FES = 52100	$\mu_F = 0.800$	$\mu_{CR} = 0.038$
Diversity = 5.8335e+00	$k_t = [2, 2, 3, 4]$		
Generation 530: Best = 8.710247e+05	FES = 53100	$\mu_F = 0.793$	$\mu_{CR} = 0.038$
Diversity = 5.5304e+00	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 8.642647e+05	FES = 54100	$\mu_F = 0.796$	$\mu_{CR} = 0.031$
Diversity = 5.6312e+00	$k_t = [3, 2, 2, 4]$		
Generation 550: Best = 8.545694e+05	FES = 55100	$\mu_F = 0.808$	$\mu_{CR} = 0.038$
Diversity = 6.4593e+00	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 8.397526e+05	FES = 56100	$\mu_F = 0.819$	$\mu_{CR} = 0.039$
Diversity = 7.4200e+00	$k_t = [3, 3, 4, 3]$		
Generation 570: Best = 8.191261e+05	FES = 57100	$\mu_F = 0.831$	$\mu_{CR} = 0.047$
Diversity = 7.9038e+00	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 8.146500e+05	FES = 58100	$\mu_F = 0.840$	$\mu_{CR} = 0.047$
Diversity = 8.2619e+00	$k_t = [4, 4, 2, 2]$		
Generation 590: Best = 8.020086e+05	FES = 59100	$\mu_F = 0.837$	$\mu_{CR} = 0.043$
Diversity = 8.8657e+00	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 7.927008e+05	FES = 60100	$\mu_F = 0.819$	$\mu_{CR} = 0.049$
Diversity = 8.6209e+00	$k_t = [4, 4, 2, 4]$		
Generation 610: Best = 7.757849e+05	FES = 61100	$\mu_F = 0.821$	$\mu_{CR} = 0.051$
Diversity = 8.6597e+00	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 7.640745e+05	FES = 62100	$\mu_F = 0.822$	$\mu_{CR} = 0.040$
Diversity = 8.3195e+00	$k_t = [4, 2, 3, 4]$		
Generation 630: Best = 7.536959e+05	FES = 63100	$\mu_F = 0.822$	$\mu_{CR} = 0.039$
Diversity = 8.4437e+00	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 7.442760e+05	FES = 64100	$\mu_F = 0.812$	$\mu_{CR} = 0.036$
Diversity = 7.3766e+00	$k_t = [2, 3, 4, 3]$		
Generation 650: Best = 7.302064e+05	FES = 65100	$\mu_F = 0.814$	$\mu_{CR} = 0.040$
Diversity = 7.6638e+00	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 7.212830e+05	FES = 66100	$\mu_F = 0.814$	$\mu_{CR} = 0.040$
Diversity = 7.2005e+00	$k_t = [4, 4, 3, 4]$		
Generation 670: Best = 7.166489e+05	FES = 67100	$\mu_F = 0.803$	$\mu_{CR} = 0.033$
Diversity = 7.3359e+00	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 7.114288e+05	FES = 68100	$\mu_F = 0.815$	$\mu_{CR} = 0.031$
Diversity = 6.6333e+00	$k_t = [3, 3, 3, 3]$		
Generation 690: Best = 7.086883e+05	FES = 69100	$\mu_F = 0.819$	$\mu_{CR} = 0.035$
Diversity = 6.1273e+00	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 7.047710e+05	FES = 70100	$\mu_F = 0.819$	$\mu_{CR} = 0.034$
Diversity = 6.2314e+00	$k_t = [2, 4, 2, 3]$		
Generation 710: Best = 7.021610e+05	FES = 71100	$\mu_F = 0.813$	$\mu_{CR} = 0.043$
Diversity = 6.1763e+00	$k_t = [1, 1, 1, 1]$		
Generation 720: Best = 6.980443e+05	FES = 72100	$\mu_F = 0.821$	$\mu_{CR} = 0.041$
Diversity = 6.4414e+00	$k_t = [4, 3, 4, 3]$		
Generation 730: Best = 6.955908e+05	FES = 73100	$\mu_F = 0.828$	$\mu_{CR} = 0.038$
Diversity = 6.8419e+00	$k_t = [1, 1, 1, 1]$		
Generation 740: Best = 6.921195e+05	FES = 74100	$\mu_F = 0.830$	$\mu_{CR} = 0.042$
Diversity = 7.4932e+00	$k_t = [2, 3, 4, 3]$		
Generation 750: Best = 6.904588e+05	FES = 75100	$\mu_F = 0.811$	$\mu_{CR} = 0.039$
Diversity = 7.3298e+00	$k_t = [1, 1, 1, 1]$		
Generation 760: Best = 6.883165e+05	FES = 76100	$\mu_F = 0.806$	$\mu_{CR} = 0.038$
Diversity = 6.8944e+00	$k_t = [4, 3, 2, 4]$		

Generation 770: Best = 6.854122e+05	FES = 77100	$\mu_F = 0.802$	$\mu_{CR} = 0.043$
Diversity = 6.8525e+00	k_t = [1, 1, 1, 1]		
Generation 780: Best = 6.831057e+05	FES = 78100	$\mu_F = 0.804$	$\mu_{CR} = 0.039$
Diversity = 6.7367e+00	k_t = [2, 2, 2, 2]		
Generation 790: Best = 6.802968e+05	FES = 79100	$\mu_F = 0.794$	$\mu_{CR} = 0.036$
Diversity = 7.2139e+00	k_t = [1, 1, 1, 1]		
Generation 800: Best = 6.770810e+05	FES = 80100	$\mu_F = 0.776$	$\mu_{CR} = 0.043$
Diversity = 6.3708e+00	k_t = [3, 4, 3, 3]		
Generation 810: Best = 6.743720e+05	FES = 81100	$\mu_F = 0.779$	$\mu_{CR} = 0.043$
Diversity = 7.5152e+00	k_t = [1, 1, 1, 1]		
Generation 820: Best = 6.720957e+05	FES = 82100	$\mu_F = 0.781$	$\mu_{CR} = 0.040$
Diversity = 7.0007e+00	k_t = [4, 3, 3, 3]		
Generation 830: Best = 6.677737e+05	FES = 83100	$\mu_F = 0.802$	$\mu_{CR} = 0.037$
Diversity = 7.6745e+00	k_t = [1, 1, 1, 1]		
Generation 840: Best = 6.653970e+05	FES = 84100	$\mu_F = 0.804$	$\mu_{CR} = 0.046$
Diversity = 7.1725e+00	k_t = [3, 2, 4, 2]		
Generation 850: Best = 6.633703e+05	FES = 85100	$\mu_F = 0.812$	$\mu_{CR} = 0.032$
Diversity = 8.5913e+00	k_t = [1, 1, 1, 1]		
Generation 860: Best = 6.601567e+05	FES = 86100	$\mu_F = 0.810$	$\mu_{CR} = 0.033$
Diversity = 8.3588e+00	k_t = [2, 3, 2, 4]		
Generation 870: Best = 6.564728e+05	FES = 87100	$\mu_F = 0.812$	$\mu_{CR} = 0.035$
Diversity = 8.9324e+00	k_t = [1, 1, 1, 1]		
Generation 880: Best = 6.512545e+05	FES = 88100	$\mu_F = 0.816$	$\mu_{CR} = 0.030$
Diversity = 8.5182e+00	k_t = [2, 4, 3, 4]		
Generation 890: Best = 6.489568e+05	FES = 89100	$\mu_F = 0.812$	$\mu_{CR} = 0.041$
Diversity = 9.7632e+00	k_t = [1, 1, 1, 1]		
Generation 900: Best = 6.439087e+05	FES = 90100	$\mu_F = 0.830$	$\mu_{CR} = 0.039$
Diversity = 9.7915e+00	k_t = [3, 2, 4, 4]		
Generation 910: Best = 6.409149e+05	FES = 91100	$\mu_F = 0.831$	$\mu_{CR} = 0.035$
Diversity = 1.0693e+01	k_t = [1, 1, 1, 1]		
Generation 920: Best = 6.398936e+05	FES = 92100	$\mu_F = 0.826$	$\mu_{CR} = 0.046$
Diversity = 9.7741e+00	k_t = [4, 2, 2, 3]		
Generation 930: Best = 6.361127e+05	FES = 93100	$\mu_F = 0.818$	$\mu_{CR} = 0.043$
Diversity = 9.7055e+00	k_t = [1, 1, 1, 1]		
Generation 940: Best = 6.325013e+05	FES = 94100	$\mu_F = 0.824$	$\mu_{CR} = 0.037$
Diversity = 1.1036e+01	k_t = [4, 3, 4, 3]		
Generation 950: Best = 6.314889e+05	FES = 95100	$\mu_F = 0.822$	$\mu_{CR} = 0.040$
Diversity = 1.1370e+01	k_t = [1, 1, 1, 1]		
Generation 960: Best = 6.272224e+05	FES = 96100	$\mu_F = 0.806$	$\mu_{CR} = 0.036$
Diversity = 1.1738e+01	k_t = [2, 3, 2, 2]		
Generation 970: Best = 6.247661e+05	FES = 97100	$\mu_F = 0.805$	$\mu_{CR} = 0.036$
Diversity = 1.1703e+01	k_t = [1, 1, 1, 1]		
Generation 980: Best = 6.197161e+05	FES = 98100	$\mu_F = 0.804$	$\mu_{CR} = 0.042$
Diversity = 1.1623e+01	k_t = [3, 2, 4, 2]		
Generation 990: Best = 6.161481e+05	FES = 99100	$\mu_F = 0.789$	$\mu_{CR} = 0.033$
Diversity = 1.2461e+01	k_t = [1, 1, 1, 1]		
Generation 1000: Best = 6.126405e+05	FES = 100100	$\mu_F = 0.809$	$\mu_{CR} = 0.035$
Diversity = 1.2501e+01	k_t = [4, 3, 3, 3]		
Generation 1010: Best = 6.085957e+05	FES = 101100	$\mu_F = 0.801$	$\mu_{CR} = 0.039$
Diversity = 1.1693e+01	k_t = [1, 1, 1, 1]		
Generation 1020: Best = 6.069101e+05	FES = 102100	$\mu_F = 0.813$	$\mu_{CR} = 0.032$
Diversity = 1.1500e+01	k_t = [2, 3, 3, 2]		
Generation 1030: Best = 6.005706e+05	FES = 103100	$\mu_F = 0.798$	$\mu_{CR} = 0.043$
Diversity = 1.1895e+01	k_t = [1, 1, 1, 1]		

Generation 1040: Best = 5.950457e+05 FES = 104100 μ_F = 0.808 μ_{CR} = 0.039
 Diversity = 1.2156e+01 k_t = [3, 4, 4, 3]
 Generation 1050: Best = 5.950457e+05 FES = 105100 μ_F = 0.815 μ_{CR} = 0.037
 Diversity = 1.2153e+01 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 5.896092e+05 FES = 106100 μ_F = 0.791 μ_{CR} = 0.037
 Diversity = 1.1341e+01 k_t = [2, 3, 4, 2]
 Generation 1070: Best = 5.880832e+05 FES = 107100 μ_F = 0.801 μ_{CR} = 0.038
 Diversity = 1.1649e+01 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 5.818615e+05 FES = 108100 μ_F = 0.810 μ_{CR} = 0.041
 Diversity = 1.2524e+01 k_t = [4, 4, 3, 2]
 Generation 1090: Best = 5.783183e+05 FES = 109100 μ_F = 0.795 μ_{CR} = 0.039
 Diversity = 1.1730e+01 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 5.778257e+05 FES = 110100 μ_F = 0.805 μ_{CR} = 0.039
 Diversity = 1.0978e+01 k_t = [2, 4, 4, 4]
 Generation 1110: Best = 5.756841e+05 FES = 111100 μ_F = 0.810 μ_{CR} = 0.040
 Diversity = 1.1260e+01 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 5.735335e+05 FES = 112100 μ_F = 0.800 μ_{CR} = 0.035
 Diversity = 1.0845e+01 k_t = [2, 2, 4, 2]
 Generation 1130: Best = 5.683737e+05 FES = 113100 μ_F = 0.786 μ_{CR} = 0.042
 Diversity = 1.1644e+01 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 5.635891e+05 FES = 114100 μ_F = 0.781 μ_{CR} = 0.044
 Diversity = 1.0786e+01 k_t = [3, 2, 3, 3]
 Generation 1150: Best = 5.593503e+05 FES = 115100 μ_F = 0.775 μ_{CR} = 0.050
 Diversity = 1.1211e+01 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 5.575085e+05 FES = 116100 μ_F = 0.774 μ_{CR} = 0.042
 Diversity = 1.0477e+01 k_t = [3, 3, 3, 2]
 Generation 1170: Best = 5.562605e+05 FES = 117100 μ_F = 0.778 μ_{CR} = 0.045
 Diversity = 9.7283e+00 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 5.510680e+05 FES = 118100 μ_F = 0.788 μ_{CR} = 0.048
 Diversity = 9.3846e+00 k_t = [4, 4, 2, 2]
 Generation 1190: Best = 5.506821e+05 FES = 119100 μ_F = 0.788 μ_{CR} = 0.044
 Diversity = 1.0486e+01 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 5.482866e+05 FES = 120100 μ_F = 0.787 μ_{CR} = 0.049
 Diversity = 1.0054e+01 k_t = [3, 2, 3, 4]
 Generation 1210: Best = 5.454647e+05 FES = 121100 μ_F = 0.795 μ_{CR} = 0.045
 Diversity = 8.9109e+00 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 5.427518e+05 FES = 122100 μ_F = 0.793 μ_{CR} = 0.045
 Diversity = 9.0205e+00 k_t = [3, 4, 4, 4]
 Generation 1230: Best = 5.405508e+05 FES = 123100 μ_F = 0.801 μ_{CR} = 0.053
 Diversity = 8.3768e+00 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 5.386859e+05 FES = 124100 μ_F = 0.808 μ_{CR} = 0.049
 Diversity = 8.8584e+00 k_t = [2, 4, 2, 2]
 Generation 1250: Best = 5.378087e+05 FES = 125100 μ_F = 0.799 μ_{CR} = 0.045
 Diversity = 9.3325e+00 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 5.362130e+05 FES = 126100 μ_F = 0.796 μ_{CR} = 0.042
 Diversity = 9.0626e+00 k_t = [4, 3, 4, 2]
 Generation 1270: Best = 5.324176e+05 FES = 127100 μ_F = 0.796 μ_{CR} = 0.042
 Diversity = 8.8130e+00 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 5.306636e+05 FES = 128100 μ_F = 0.793 μ_{CR} = 0.040
 Diversity = 8.6914e+00 k_t = [4, 3, 2, 4]
 Generation 1290: Best = 5.286926e+05 FES = 129100 μ_F = 0.807 μ_{CR} = 0.036
 Diversity = 8.9601e+00 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 5.263041e+05 FES = 130100 μ_F = 0.817 μ_{CR} = 0.038
 Diversity = 9.2315e+00 k_t = [2, 4, 4, 3]

Generation 1310: Best = 5.258762e+05 FES = 131100 μ_F = 0.822 μ_{CR} = 0.027
 Diversity = 9.8360e+00 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 5.234111e+05 FES = 132100 μ_F = 0.832 μ_{CR} = 0.033
 Diversity = 8.7973e+00 k_t = [2, 3, 3, 3]
 Generation 1330: Best = 5.207282e+05 FES = 133100 μ_F = 0.838 μ_{CR} = 0.035
 Diversity = 9.5110e+00 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 5.200189e+05 FES = 134100 μ_F = 0.820 μ_{CR} = 0.039
 Diversity = 9.4617e+00 k_t = [4, 2, 4, 2]
 Generation 1350: Best = 5.177473e+05 FES = 135100 μ_F = 0.821 μ_{CR} = 0.034
 Diversity = 1.0272e+01 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 5.169608e+05 FES = 136100 μ_F = 0.816 μ_{CR} = 0.032
 Diversity = 1.0885e+01 k_t = [3, 2, 4, 2]
 Generation 1370: Best = 5.139501e+05 FES = 137100 μ_F = 0.822 μ_{CR} = 0.036
 Diversity = 1.1843e+01 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 5.110003e+05 FES = 138100 μ_F = 0.808 μ_{CR} = 0.039
 Diversity = 1.0622e+01 k_t = [3, 4, 4, 4]
 Generation 1390: Best = 5.077443e+05 FES = 139100 μ_F = 0.795 μ_{CR} = 0.035
 Diversity = 9.3863e+00 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 5.067189e+05 FES = 140100 μ_F = 0.804 μ_{CR} = 0.038
 Diversity = 9.1553e+00 k_t = [2, 4, 4, 4]
 Generation 1410: Best = 5.045301e+05 FES = 141100 μ_F = 0.811 μ_{CR} = 0.041
 Diversity = 1.0079e+01 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 5.023176e+05 FES = 142100 μ_F = 0.827 μ_{CR} = 0.042
 Diversity = 1.1716e+01 k_t = [2, 3, 3, 2]
 Generation 1430: Best = 5.008652e+05 FES = 143100 μ_F = 0.804 μ_{CR} = 0.039
 Diversity = 1.2037e+01 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 4.994607e+05 FES = 144100 μ_F = 0.807 μ_{CR} = 0.037
 Diversity = 1.1530e+01 k_t = [2, 4, 2, 2]
 Generation 1450: Best = 4.978490e+05 FES = 145100 μ_F = 0.789 μ_{CR} = 0.046
 Diversity = 1.0447e+01 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 4.962798e+05 FES = 146100 μ_F = 0.793 μ_{CR} = 0.043
 Diversity = 1.0286e+01 k_t = [4, 2, 2, 4]
 Generation 1470: Best = 4.942177e+05 FES = 147100 μ_F = 0.792 μ_{CR} = 0.038
 Diversity = 9.3384e+00 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 4.940152e+05 FES = 148100 μ_F = 0.786 μ_{CR} = 0.039
 Diversity = 8.7928e+00 k_t = [2, 2, 3, 3]
 Generation 1490: Best = 4.929011e+05 FES = 149100 μ_F = 0.777 μ_{CR} = 0.043
 Diversity = 8.1297e+00 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 4.926811e+05 FES = 150100 μ_F = 0.788 μ_{CR} = 0.040
 Diversity = 8.8639e+00 k_t = [4, 4, 2, 4]
 Generation 1510: Best = 4.914997e+05 FES = 151100 μ_F = 0.805 μ_{CR} = 0.050
 Diversity = 8.6620e+00 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 4.900056e+05 FES = 152100 μ_F = 0.807 μ_{CR} = 0.043
 Diversity = 9.8825e+00 k_t = [2, 3, 3, 4]
 Generation 1530: Best = 4.893078e+05 FES = 153100 μ_F = 0.801 μ_{CR} = 0.045
 Diversity = 8.8012e+00 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 4.886518e+05 FES = 154100 μ_F = 0.806 μ_{CR} = 0.040
 Diversity = 8.2563e+00 k_t = [3, 4, 4, 4]
 Generation 1550: Best = 4.873483e+05 FES = 155100 μ_F = 0.813 μ_{CR} = 0.048
 Diversity = 8.2537e+00 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 4.866224e+05 FES = 156100 μ_F = 0.828 μ_{CR} = 0.039
 Diversity = 8.9530e+00 k_t = [3, 4, 3, 3]
 Generation 1570: Best = 4.847772e+05 FES = 157100 μ_F = 0.842 μ_{CR} = 0.040
 Diversity = 9.4859e+00 k_t = [1, 1, 1, 1]

Generation 1580: Best = 4.845663e+05 FES = 158100 μ_F = 0.823 μ_{CR} = 0.045
 Diversity = 9.3124e+00 k_t = [4, 2, 3, 2]
 Generation 1590: Best = 4.834314e+05 FES = 159100 μ_F = 0.807 μ_{CR} = 0.051
 Diversity = 8.8619e+00 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 4.821050e+05 FES = 160100 μ_F = 0.816 μ_{CR} = 0.048
 Diversity = 9.5620e+00 k_t = [3, 3, 4, 3]
 Generation 1610: Best = 4.813554e+05 FES = 161100 μ_F = 0.808 μ_{CR} = 0.051
 Diversity = 9.8248e+00 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 4.801524e+05 FES = 162100 μ_F = 0.817 μ_{CR} = 0.046
 Diversity = 9.9011e+00 k_t = [3, 2, 2, 2]
 Generation 1630: Best = 4.792344e+05 FES = 163100 μ_F = 0.800 μ_{CR} = 0.040
 Diversity = 1.0338e+01 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 4.785886e+05 FES = 164100 μ_F = 0.803 μ_{CR} = 0.047
 Diversity = 1.0769e+01 k_t = [2, 3, 2, 2]
 Generation 1650: Best = 4.769743e+05 FES = 165100 μ_F = 0.793 μ_{CR} = 0.042
 Diversity = 1.1715e+01 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 4.758867e+05 FES = 166100 μ_F = 0.800 μ_{CR} = 0.048
 Diversity = 1.2018e+01 k_t = [4, 3, 3, 2]
 Generation 1670: Best = 4.749520e+05 FES = 167100 μ_F = 0.787 μ_{CR} = 0.045
 Diversity = 1.1109e+01 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 4.739174e+05 FES = 168100 μ_F = 0.783 μ_{CR} = 0.041
 Diversity = 9.9540e+00 k_t = [2, 4, 2, 2]
 Generation 1690: Best = 4.730060e+05 FES = 169100 μ_F = 0.790 μ_{CR} = 0.037
 Diversity = 1.0495e+01 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 4.725045e+05 FES = 170100 μ_F = 0.782 μ_{CR} = 0.031
 Diversity = 9.7423e+00 k_t = [3, 4, 4, 3]
 Generation 1710: Best = 4.717677e+05 FES = 171100 μ_F = 0.799 μ_{CR} = 0.028
 Diversity = 9.6257e+00 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 4.705375e+05 FES = 172100 μ_F = 0.802 μ_{CR} = 0.034
 Diversity = 8.7522e+00 k_t = [2, 3, 4, 3]
 Generation 1730: Best = 4.681605e+05 FES = 173100 μ_F = 0.805 μ_{CR} = 0.038
 Diversity = 9.0667e+00 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 4.668969e+05 FES = 174100 μ_F = 0.817 μ_{CR} = 0.040
 Diversity = 9.4881e+00 k_t = [4, 3, 4, 3]
 Generation 1750: Best = 4.648846e+05 FES = 175100 μ_F = 0.823 μ_{CR} = 0.046
 Diversity = 9.8317e+00 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 4.637963e+05 FES = 176100 μ_F = 0.826 μ_{CR} = 0.044
 Diversity = 1.0368e+01 k_t = [3, 2, 3, 3]
 Generation 1770: Best = 4.626108e+05 FES = 177100 μ_F = 0.823 μ_{CR} = 0.041
 Diversity = 1.0016e+01 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 4.602643e+05 FES = 178100 μ_F = 0.836 μ_{CR} = 0.049
 Diversity = 9.4616e+00 k_t = [4, 2, 4, 2]
 Generation 1790: Best = 4.579864e+05 FES = 179100 μ_F = 0.812 μ_{CR} = 0.051
 Diversity = 9.5692e+00 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 4.565401e+05 FES = 180100 μ_F = 0.820 μ_{CR} = 0.049
 Diversity = 9.5633e+00 k_t = [3, 2, 2, 4]
 Generation 1810: Best = 4.556280e+05 FES = 181100 μ_F = 0.828 μ_{CR} = 0.042
 Diversity = 8.2103e+00 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 4.545101e+05 FES = 182100 μ_F = 0.814 μ_{CR} = 0.038
 Diversity = 7.5701e+00 k_t = [2, 4, 4, 3]
 Generation 1830: Best = 4.536387e+05 FES = 183100 μ_F = 0.814 μ_{CR} = 0.036
 Diversity = 7.6437e+00 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 4.527603e+05 FES = 184100 μ_F = 0.792 μ_{CR} = 0.040
 Diversity = 7.4854e+00 k_t = [4, 3, 2, 2]

Generation 1850: Best = 4.519463e+05 FES = 185100 μ_F = 0.809 μ_{CR} = 0.038
 Diversity = 7.6547e+00 k_t = [1, 1, 1, 1]
 Generation 1860: Best = 4.515529e+05 FES = 186100 μ_F = 0.822 μ_{CR} = 0.031
 Diversity = 7.0797e+00 k_t = [3, 2, 3, 4]
 Generation 1870: Best = 4.506268e+05 FES = 187100 μ_F = 0.814 μ_{CR} = 0.035
 Diversity = 7.3341e+00 k_t = [1, 1, 1, 1]
 Generation 1880: Best = 4.499795e+05 FES = 188100 μ_F = 0.819 μ_{CR} = 0.040
 Diversity = 8.1619e+00 k_t = [3, 4, 2, 3]
 Generation 1890: Best = 4.456947e+05 FES = 189100 μ_F = 0.816 μ_{CR} = 0.041
 Diversity = 7.9418e+00 k_t = [1, 1, 1, 1]
 Generation 1900: Best = 4.456312e+05 FES = 190100 μ_F = 0.813 μ_{CR} = 0.043
 Diversity = 8.4153e+00 k_t = [2, 3, 4, 3]
 Generation 1910: Best = 4.449586e+05 FES = 191100 μ_F = 0.804 μ_{CR} = 0.053
 Diversity = 7.6858e+00 k_t = [1, 1, 1, 1]
 Generation 1920: Best = 4.441666e+05 FES = 192100 μ_F = 0.813 μ_{CR} = 0.041
 Diversity = 7.3138e+00 k_t = [2, 4, 3, 4]
 Generation 1930: Best = 4.434910e+05 FES = 193100 μ_F = 0.804 μ_{CR} = 0.036
 Diversity = 7.2284e+00 k_t = [1, 1, 1, 1]
 Generation 1940: Best = 4.429119e+05 FES = 194100 μ_F = 0.790 μ_{CR} = 0.043
 Diversity = 6.5825e+00 k_t = [4, 2, 3, 2]
 Generation 1950: Best = 4.419140e+05 FES = 195100 μ_F = 0.798 μ_{CR} = 0.045
 Diversity = 7.4888e+00 k_t = [1, 1, 1, 1]
 Generation 1960: Best = 4.416209e+05 FES = 196100 μ_F = 0.814 μ_{CR} = 0.040
 Diversity = 7.0043e+00 k_t = [4, 4, 2, 4]
 Generation 1970: Best = 4.400038e+05 FES = 197100 μ_F = 0.823 μ_{CR} = 0.042
 Diversity = 7.5780e+00 k_t = [1, 1, 1, 1]
 Generation 1980: Best = 4.391296e+05 FES = 198100 μ_F = 0.819 μ_{CR} = 0.034
 Diversity = 7.6347e+00 k_t = [4, 2, 3, 2]
 Generation 1990: Best = 4.385782e+05 FES = 199100 μ_F = 0.822 μ_{CR} = 0.037
 Diversity = 7.8737e+00 k_t = [1, 1, 1, 1]
 Generation 2000: Best = 4.372658e+05 FES = 200100 μ_F = 0.824 μ_{CR} = 0.030
 Diversity = 7.6661e+00 k_t = [2, 4, 2, 4]
 Generation 2010: Best = 4.361380e+05 FES = 201100 μ_F = 0.816 μ_{CR} = 0.035
 Diversity = 8.0423e+00 k_t = [1, 1, 1, 1]
 Generation 2020: Best = 4.336107e+05 FES = 202100 μ_F = 0.813 μ_{CR} = 0.035
 Diversity = 7.7834e+00 k_t = [3, 4, 4, 3]
 Generation 2030: Best = 4.329053e+05 FES = 203100 μ_F = 0.822 μ_{CR} = 0.036
 Diversity = 7.9915e+00 k_t = [1, 1, 1, 1]
 Generation 2040: Best = 4.309529e+05 FES = 204100 μ_F = 0.830 μ_{CR} = 0.046
 Diversity = 8.0995e+00 k_t = [2, 3, 3, 4]
 Generation 2050: Best = 4.295140e+05 FES = 205100 μ_F = 0.840 μ_{CR} = 0.038
 Diversity = 7.9561e+00 k_t = [1, 1, 1, 1]
 Generation 2060: Best = 4.276664e+05 FES = 206100 μ_F = 0.827 μ_{CR} = 0.036
 Diversity = 8.0531e+00 k_t = [2, 3, 3, 2]
 Generation 2070: Best = 4.271145e+05 FES = 207100 μ_F = 0.804 μ_{CR} = 0.036
 Diversity = 7.5776e+00 k_t = [1, 1, 1, 1]
 Generation 2080: Best = 4.265689e+05 FES = 208100 μ_F = 0.813 μ_{CR} = 0.043
 Diversity = 7.5497e+00 k_t = [3, 4, 3, 3]
 Generation 2090: Best = 4.250441e+05 FES = 209100 μ_F = 0.816 μ_{CR} = 0.042
 Diversity = 7.2667e+00 k_t = [1, 1, 1, 1]
 Generation 2100: Best = 4.233480e+05 FES = 210100 μ_F = 0.824 μ_{CR} = 0.040
 Diversity = 8.1096e+00 k_t = [2, 4, 3, 4]
 Generation 2110: Best = 4.215701e+05 FES = 211100 μ_F = 0.807 μ_{CR} = 0.034
 Diversity = 7.8896e+00 k_t = [1, 1, 1, 1]

Generation 2120: Best = 4.196440e+05 FES = 212100 μ_F = 0.808 μ_{CR} = 0.036
 Diversity = 7.5129e+00 k_t = [3, 2, 2, 2]
 Generation 2130: Best = 4.193306e+05 FES = 213100 μ_F = 0.809 μ_{CR} = 0.040
 Diversity = 8.2129e+00 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 4.187511e+05 FES = 214100 μ_F = 0.808 μ_{CR} = 0.037
 Diversity = 7.3830e+00 k_t = [2, 4, 2, 4]
 Generation 2150: Best = 4.177973e+05 FES = 215100 μ_F = 0.808 μ_{CR} = 0.032
 Diversity = 6.9836e+00 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 4.176377e+05 FES = 216100 μ_F = 0.804 μ_{CR} = 0.035
 Diversity = 6.3197e+00 k_t = [3, 4, 2, 3]
 Generation 2170: Best = 4.161690e+05 FES = 217100 μ_F = 0.810 μ_{CR} = 0.043
 Diversity = 6.7129e+00 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 4.150344e+05 FES = 218100 μ_F = 0.807 μ_{CR} = 0.044
 Diversity = 7.0985e+00 k_t = [3, 3, 3, 2]
 Generation 2190: Best = 4.143287e+05 FES = 219100 μ_F = 0.811 μ_{CR} = 0.034
 Diversity = 8.0790e+00 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 4.132944e+05 FES = 220100 μ_F = 0.813 μ_{CR} = 0.029
 Diversity = 8.3650e+00 k_t = [4, 2, 3, 3]
 Generation 2210: Best = 4.116776e+05 FES = 221100 μ_F = 0.805 μ_{CR} = 0.035
 Diversity = 8.1071e+00 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 4.108367e+05 FES = 222100 μ_F = 0.809 μ_{CR} = 0.039
 Diversity = 8.7559e+00 k_t = [2, 2, 3, 3]
 Generation 2230: Best = 4.101076e+05 FES = 223100 μ_F = 0.809 μ_{CR} = 0.037
 Diversity = 9.3596e+00 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 4.088837e+05 FES = 224100 μ_F = 0.806 μ_{CR} = 0.035
 Diversity = 8.7900e+00 k_t = [2, 2, 2, 4]
 Generation 2250: Best = 4.079873e+05 FES = 225100 μ_F = 0.820 μ_{CR} = 0.032
 Diversity = 7.6173e+00 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 4.069895e+05 FES = 226100 μ_F = 0.836 μ_{CR} = 0.029
 Diversity = 8.2131e+00 k_t = [3, 2, 4, 2]
 Generation 2270: Best = 4.062218e+05 FES = 227100 μ_F = 0.827 μ_{CR} = 0.029
 Diversity = 8.0344e+00 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 4.057900e+05 FES = 228100 μ_F = 0.832 μ_{CR} = 0.025
 Diversity = 8.0658e+00 k_t = [3, 3, 4, 4]
 Generation 2290: Best = 4.048617e+05 FES = 229100 μ_F = 0.830 μ_{CR} = 0.026
 Diversity = 8.5290e+00 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 4.047035e+05 FES = 230100 μ_F = 0.810 μ_{CR} = 0.031
 Diversity = 8.0233e+00 k_t = [2, 4, 2, 4]
 Generation 2310: Best = 4.044105e+05 FES = 231100 μ_F = 0.819 μ_{CR} = 0.036
 Diversity = 7.1418e+00 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 4.036973e+05 FES = 232100 μ_F = 0.809 μ_{CR} = 0.038
 Diversity = 6.2714e+00 k_t = [4, 4, 3, 3]
 Generation 2330: Best = 4.033271e+05 FES = 233100 μ_F = 0.791 μ_{CR} = 0.037
 Diversity = 5.6985e+00 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 4.030410e+05 FES = 234100 μ_F = 0.806 μ_{CR} = 0.041
 Diversity = 5.6182e+00 k_t = [4, 3, 4, 2]
 Generation 2350: Best = 4.026922e+05 FES = 235100 μ_F = 0.794 μ_{CR} = 0.038
 Diversity = 5.6471e+00 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 4.024130e+05 FES = 236100 μ_F = 0.792 μ_{CR} = 0.038
 Diversity = 5.6945e+00 k_t = [2, 2, 2, 4]
 Generation 2370: Best = 4.020951e+05 FES = 237100 μ_F = 0.810 μ_{CR} = 0.032
 Diversity = 5.2233e+00 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 4.017825e+05 FES = 238100 μ_F = 0.815 μ_{CR} = 0.031
 Diversity = 5.0886e+00 k_t = [4, 3, 2, 4]

Generation 2390: Best = 4.014468e+05 FES = 239100 μ_F = 0.813 μ_{CR} = 0.032
 Diversity = 4.8217e+00 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 4.010890e+05 FES = 240100 μ_F = 0.808 μ_{CR} = 0.036
 Diversity = 5.1415e+00 k_t = [2, 4, 3, 4]
 Generation 2410: Best = 4.006232e+05 FES = 241100 μ_F = 0.808 μ_{CR} = 0.034
 Diversity = 5.9001e+00 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 4.003385e+05 FES = 242100 μ_F = 0.805 μ_{CR} = 0.037
 Diversity = 6.5499e+00 k_t = [4, 2, 4, 2]
 Generation 2430: Best = 3.996357e+05 FES = 243100 μ_F = 0.800 μ_{CR} = 0.041
 Diversity = 5.9385e+00 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 3.972227e+05 FES = 244100 μ_F = 0.815 μ_{CR} = 0.045
 Diversity = 7.3361e+00 k_t = [3, 4, 2, 4]
 Generation 2450: Best = 3.970835e+05 FES = 245100 μ_F = 0.810 μ_{CR} = 0.040
 Diversity = 7.0376e+00 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 3.965305e+05 FES = 246100 μ_F = 0.799 μ_{CR} = 0.044
 Diversity = 7.0031e+00 k_t = [2, 4, 4, 4]
 Generation 2470: Best = 3.961221e+05 FES = 247100 μ_F = 0.797 μ_{CR} = 0.045
 Diversity = 5.9741e+00 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 3.955318e+05 FES = 248100 μ_F = 0.819 μ_{CR} = 0.038
 Diversity = 5.7221e+00 k_t = [2, 4, 2, 4]
 Generation 2490: Best = 3.949421e+05 FES = 249100 μ_F = 0.820 μ_{CR} = 0.036
 Diversity = 5.6931e+00 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 3.947652e+05 FES = 250100 μ_F = 0.826 μ_{CR} = 0.038
 Diversity = 5.2808e+00 k_t = [4, 3, 3, 3]
 Generation 2510: Best = 3.940133e+05 FES = 251100 μ_F = 0.810 μ_{CR} = 0.050
 Diversity = 5.2014e+00 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 3.931103e+05 FES = 252100 μ_F = 0.816 μ_{CR} = 0.050
 Diversity = 5.4141e+00 k_t = [2, 2, 2, 4]
 Generation 2530: Best = 3.924064e+05 FES = 253100 μ_F = 0.827 μ_{CR} = 0.039
 Diversity = 5.9576e+00 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 3.917939e+05 FES = 254100 μ_F = 0.814 μ_{CR} = 0.041
 Diversity = 5.8324e+00 k_t = [3, 4, 4, 3]
 Generation 2550: Best = 3.914529e+05 FES = 255100 μ_F = 0.817 μ_{CR} = 0.033
 Diversity = 6.0718e+00 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 3.911540e+05 FES = 256100 μ_F = 0.800 μ_{CR} = 0.036
 Diversity = 5.4838e+00 k_t = [4, 4, 4, 4]
 Generation 2570: Best = 3.904021e+05 FES = 257100 μ_F = 0.799 μ_{CR} = 0.038
 Diversity = 4.8809e+00 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 3.900743e+05 FES = 258100 μ_F = 0.798 μ_{CR} = 0.038
 Diversity = 5.5569e+00 k_t = [2, 3, 2, 2]
 Generation 2590: Best = 3.893415e+05 FES = 259100 μ_F = 0.794 μ_{CR} = 0.047
 Diversity = 5.7189e+00 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 3.891615e+05 FES = 260100 μ_F = 0.797 μ_{CR} = 0.041
 Diversity = 5.6694e+00 k_t = [4, 2, 2, 3]
 Generation 2610: Best = 3.884756e+05 FES = 261100 μ_F = 0.798 μ_{CR} = 0.035
 Diversity = 6.1315e+00 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 3.883395e+05 FES = 262100 μ_F = 0.804 μ_{CR} = 0.036
 Diversity = 6.0717e+00 k_t = [4, 3, 3, 2]
 Generation 2630: Best = 3.878644e+05 FES = 263100 μ_F = 0.811 μ_{CR} = 0.036
 Diversity = 6.3431e+00 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 3.873213e+05 FES = 264100 μ_F = 0.805 μ_{CR} = 0.039
 Diversity = 5.9414e+00 k_t = [3, 2, 3, 4]
 Generation 2650: Best = 3.863328e+05 FES = 265100 μ_F = 0.795 μ_{CR} = 0.037
 Diversity = 5.5722e+00 k_t = [1, 1, 1, 1]

Generation 2660: Best = 3.858063e+05 FES = 266100 μ_F = 0.793 μ_{CR} = 0.036
 Diversity = 5.8574e+00 k_t = [2, 3, 2, 4]
 Generation 2670: Best = 3.854034e+05 FES = 267100 μ_F = 0.811 μ_{CR} = 0.033
 Diversity = 5.7344e+00 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 3.848849e+05 FES = 268100 μ_F = 0.799 μ_{CR} = 0.047
 Diversity = 5.5904e+00 k_t = [2, 2, 4, 3]
 Generation 2690: Best = 3.846523e+05 FES = 269100 μ_F = 0.802 μ_{CR} = 0.040
 Diversity = 5.4960e+00 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 3.844167e+05 FES = 270100 μ_F = 0.812 μ_{CR} = 0.041
 Diversity = 5.3117e+00 k_t = [3, 4, 4, 3]
 Generation 2710: Best = 3.841965e+05 FES = 271100 μ_F = 0.812 μ_{CR} = 0.045
 Diversity = 5.7539e+00 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 3.837364e+05 FES = 272100 μ_F = 0.811 μ_{CR} = 0.049
 Diversity = 5.6005e+00 k_t = [2, 4, 3, 2]
 Generation 2730: Best = 3.833202e+05 FES = 273100 μ_F = 0.808 μ_{CR} = 0.045
 Diversity = 6.0623e+00 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 3.832433e+05 FES = 274100 μ_F = 0.794 μ_{CR} = 0.043
 Diversity = 6.4565e+00 k_t = [4, 3, 2, 2]
 Generation 2750: Best = 3.827926e+05 FES = 275100 μ_F = 0.788 μ_{CR} = 0.050
 Diversity = 7.0927e+00 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 3.824526e+05 FES = 276100 μ_F = 0.788 μ_{CR} = 0.048
 Diversity = 6.5094e+00 k_t = [2, 2, 2, 2]
 Generation 2770: Best = 3.819574e+05 FES = 277100 μ_F = 0.779 μ_{CR} = 0.042
 Diversity = 6.1955e+00 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 3.817084e+05 FES = 278100 μ_F = 0.794 μ_{CR} = 0.038
 Diversity = 6.1080e+00 k_t = [3, 2, 4, 2]
 Generation 2790: Best = 3.814838e+05 FES = 279100 μ_F = 0.781 μ_{CR} = 0.032
 Diversity = 5.9390e+00 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 3.805450e+05 FES = 280100 μ_F = 0.779 μ_{CR} = 0.033
 Diversity = 4.6759e+00 k_t = [2, 3, 3, 4]
 Generation 2810: Best = 3.805450e+05 FES = 281100 μ_F = 0.799 μ_{CR} = 0.038
 Diversity = 4.3317e+00 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 3.801604e+05 FES = 282100 μ_F = 0.797 μ_{CR} = 0.038
 Diversity = 4.7056e+00 k_t = [3, 4, 4, 4]
 Generation 2830: Best = 3.798058e+05 FES = 283100 μ_F = 0.809 μ_{CR} = 0.043
 Diversity = 4.3083e+00 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 3.794743e+05 FES = 284100 μ_F = 0.803 μ_{CR} = 0.038
 Diversity = 4.4081e+00 k_t = [3, 4, 3, 4]
 Generation 2850: Best = 3.790137e+05 FES = 285100 μ_F = 0.796 μ_{CR} = 0.041
 Diversity = 4.0373e+00 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 3.787820e+05 FES = 286100 μ_F = 0.806 μ_{CR} = 0.041
 Diversity = 4.3907e+00 k_t = [2, 3, 4, 4]
 Generation 2870: Best = 3.784597e+05 FES = 287100 μ_F = 0.815 μ_{CR} = 0.046
 Diversity = 4.4455e+00 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 3.780835e+05 FES = 288100 μ_F = 0.804 μ_{CR} = 0.044
 Diversity = 4.5233e+00 k_t = [4, 3, 3, 4]
 Generation 2890: Best = 3.771134e+05 FES = 289100 μ_F = 0.809 μ_{CR} = 0.054
 Diversity = 4.6466e+00 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 3.771134e+05 FES = 290100 μ_F = 0.813 μ_{CR} = 0.048
 Diversity = 5.0087e+00 k_t = [4, 3, 4, 2]
 Generation 2910: Best = 3.762822e+05 FES = 291100 μ_F = 0.801 μ_{CR} = 0.050
 Diversity = 4.4156e+00 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 3.760465e+05 FES = 292100 μ_F = 0.806 μ_{CR} = 0.049
 Diversity = 4.6451e+00 k_t = [3, 3, 4, 3]

Generation 2930: Best = 3.757994e+05 FES = 293100 μ_F = 0.823 μ_{CR} = 0.037
 Diversity = 5.0558e+00 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 3.754370e+05 FES = 294100 μ_F = 0.814 μ_{CR} = 0.034
 Diversity = 4.6399e+00 k_t = [2, 3, 3, 3]
 Generation 2950: Best = 3.749896e+05 FES = 295100 μ_F = 0.808 μ_{CR} = 0.044
 Diversity = 4.4723e+00 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 3.746482e+05 FES = 296100 μ_F = 0.804 μ_{CR} = 0.037
 Diversity = 4.3340e+00 k_t = [2, 3, 3, 3]
 Generation 2970: Best = 3.741296e+05 FES = 297100 μ_F = 0.793 μ_{CR} = 0.049
 Diversity = 4.1757e+00 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 3.736018e+05 FES = 298100 μ_F = 0.810 μ_{CR} = 0.045
 Diversity = 4.4871e+00 k_t = [3, 2, 2, 2]
 Generation 2990: Best = 3.730281e+05 FES = 299100 μ_F = 0.796 μ_{CR} = 0.041
 Diversity = 5.2559e+00 k_t = [1, 1, 1, 1]
 Generation 3000: Best = 3.725486e+05 FES = 300100 μ_F = 0.794 μ_{CR} = 0.039
 Diversity = 5.4844e+00 k_t = [3, 2, 4, 3]
 Generation 3010: Best = 3.722291e+05 FES = 301100 μ_F = 0.802 μ_{CR} = 0.032
 Diversity = 5.3842e+00 k_t = [1, 1, 1, 1]
 Generation 3020: Best = 3.717816e+05 FES = 302100 μ_F = 0.796 μ_{CR} = 0.040
 Diversity = 5.3116e+00 k_t = [2, 3, 4, 2]
 Generation 3030: Best = 3.713056e+05 FES = 303100 μ_F = 0.795 μ_{CR} = 0.041
 Diversity = 4.9670e+00 k_t = [1, 1, 1, 1]
 Generation 3040: Best = 3.705141e+05 FES = 304100 μ_F = 0.794 μ_{CR} = 0.042
 Diversity = 4.8520e+00 k_t = [4, 4, 2, 4]
 Generation 3050: Best = 3.701561e+05 FES = 305100 μ_F = 0.809 μ_{CR} = 0.042
 Diversity = 4.2906e+00 k_t = [1, 1, 1, 1]
 Generation 3060: Best = 3.700110e+05 FES = 306100 μ_F = 0.802 μ_{CR} = 0.054
 Diversity = 4.3616e+00 k_t = [2, 2, 2, 4]
 Generation 3070: Best = 3.696839e+05 FES = 307100 μ_F = 0.805 μ_{CR} = 0.052
 Diversity = 4.1862e+00 k_t = [1, 1, 1, 1]
 Generation 3080: Best = 3.693846e+05 FES = 308100 μ_F = 0.804 μ_{CR} = 0.045
 Diversity = 4.5482e+00 k_t = [4, 2, 3, 2]
 Generation 3090: Best = 3.688831e+05 FES = 309100 μ_F = 0.803 μ_{CR} = 0.040
 Diversity = 4.9267e+00 k_t = [1, 1, 1, 1]
 Generation 3100: Best = 3.685481e+05 FES = 310100 μ_F = 0.804 μ_{CR} = 0.040
 Diversity = 4.6845e+00 k_t = [4, 4, 3, 2]
 Generation 3110: Best = 3.679429e+05 FES = 311100 μ_F = 0.780 μ_{CR} = 0.039
 Diversity = 4.3179e+00 k_t = [1, 1, 1, 1]
 Generation 3120: Best = 3.679429e+05 FES = 312100 μ_F = 0.787 μ_{CR} = 0.046
 Diversity = 4.4300e+00 k_t = [2, 2, 2, 3]
 Generation 3130: Best = 3.674066e+05 FES = 313100 μ_F = 0.790 μ_{CR} = 0.043
 Diversity = 4.8018e+00 k_t = [1, 1, 1, 1]
 Generation 3140: Best = 3.670483e+05 FES = 314100 μ_F = 0.806 μ_{CR} = 0.040
 Diversity = 4.6113e+00 k_t = [4, 4, 3, 3]
 Generation 3150: Best = 3.668194e+05 FES = 315100 μ_F = 0.814 μ_{CR} = 0.040
 Diversity = 4.4267e+00 k_t = [1, 1, 1, 1]
 Generation 3160: Best = 3.665254e+05 FES = 316100 μ_F = 0.811 μ_{CR} = 0.042
 Diversity = 4.8031e+00 k_t = [3, 2, 3, 2]
 Generation 3170: Best = 3.658319e+05 FES = 317100 μ_F = 0.806 μ_{CR} = 0.045
 Diversity = 5.1761e+00 k_t = [1, 1, 1, 1]
 Generation 3180: Best = 3.654335e+05 FES = 318100 μ_F = 0.808 μ_{CR} = 0.049
 Diversity = 5.0868e+00 k_t = [4, 4, 2, 4]
 Generation 3190: Best = 3.649532e+05 FES = 319100 μ_F = 0.808 μ_{CR} = 0.049
 Diversity = 5.4957e+00 k_t = [1, 1, 1, 1]

Generation 3200: Best = 3.647815e+05 FES = 320100 μ_F = 0.828 μ_{CR} = 0.051
 Diversity = 5.5170e+00 k_t = [2, 2, 4, 3]
 Generation 3210: Best = 3.642855e+05 FES = 321100 μ_F = 0.808 μ_{CR} = 0.048
 Diversity = 4.9262e+00 k_t = [1, 1, 1, 1]
 Generation 3220: Best = 3.642693e+05 FES = 322100 μ_F = 0.795 μ_{CR} = 0.046
 Diversity = 5.0171e+00 k_t = [4, 2, 3, 3]
 Generation 3230: Best = 3.637702e+05 FES = 323100 μ_F = 0.803 μ_{CR} = 0.046
 Diversity = 5.1375e+00 k_t = [1, 1, 1, 1]
 Generation 3240: Best = 3.635611e+05 FES = 324100 μ_F = 0.801 μ_{CR} = 0.042
 Diversity = 5.0108e+00 k_t = [4, 2, 4, 2]
 Generation 3250: Best = 3.629826e+05 FES = 325100 μ_F = 0.802 μ_{CR} = 0.040
 Diversity = 4.9553e+00 k_t = [1, 1, 1, 1]
 Generation 3260: Best = 3.627147e+05 FES = 326100 μ_F = 0.808 μ_{CR} = 0.042
 Diversity = 4.7578e+00 k_t = [4, 2, 4, 3]
 Generation 3270: Best = 3.623672e+05 FES = 327100 μ_F = 0.821 μ_{CR} = 0.045
 Diversity = 5.3510e+00 k_t = [1, 1, 1, 1]
 Generation 3280: Best = 3.621090e+05 FES = 328100 μ_F = 0.829 μ_{CR} = 0.046
 Diversity = 5.6030e+00 k_t = [4, 4, 3, 4]
 Generation 3290: Best = 3.615239e+05 FES = 329100 μ_F = 0.825 μ_{CR} = 0.040
 Diversity = 5.7137e+00 k_t = [1, 1, 1, 1]
 Generation 3300: Best = 3.612214e+05 FES = 330100 μ_F = 0.817 μ_{CR} = 0.039
 Diversity = 5.7093e+00 k_t = [2, 3, 4, 3]
 Generation 3310: Best = 3.610515e+05 FES = 331100 μ_F = 0.814 μ_{CR} = 0.040
 Diversity = 5.7079e+00 k_t = [1, 1, 1, 1]
 Generation 3320: Best = 3.606995e+05 FES = 332100 μ_F = 0.826 μ_{CR} = 0.043
 Diversity = 5.7308e+00 k_t = [4, 3, 2, 4]
 Generation 3330: Best = 3.603407e+05 FES = 333100 μ_F = 0.816 μ_{CR} = 0.039
 Diversity = 5.7218e+00 k_t = [1, 1, 1, 1]
 Generation 3340: Best = 3.601005e+05 FES = 334100 μ_F = 0.816 μ_{CR} = 0.041
 Diversity = 5.4065e+00 k_t = [4, 3, 4, 3]
 Generation 3350: Best = 3.597216e+05 FES = 335100 μ_F = 0.818 μ_{CR} = 0.034
 Diversity = 6.0083e+00 k_t = [1, 1, 1, 1]
 Generation 3360: Best = 3.594981e+05 FES = 336100 μ_F = 0.829 μ_{CR} = 0.027
 Diversity = 6.4898e+00 k_t = [4, 4, 2, 2]
 Generation 3370: Best = 3.593121e+05 FES = 337100 μ_F = 0.831 μ_{CR} = 0.035
 Diversity = 6.3471e+00 k_t = [1, 1, 1, 1]
 Generation 3380: Best = 3.588723e+05 FES = 338100 μ_F = 0.811 μ_{CR} = 0.038
 Diversity = 5.7861e+00 k_t = [2, 3, 4, 3]
 Generation 3390: Best = 3.582815e+05 FES = 339100 μ_F = 0.823 μ_{CR} = 0.042
 Diversity = 6.2777e+00 k_t = [1, 1, 1, 1]
 Generation 3400: Best = 3.580840e+05 FES = 340100 μ_F = 0.834 μ_{CR} = 0.039
 Diversity = 6.8013e+00 k_t = [2, 3, 3, 4]
 Generation 3410: Best = 3.575921e+05 FES = 341100 μ_F = 0.825 μ_{CR} = 0.041
 Diversity = 7.1070e+00 k_t = [1, 1, 1, 1]
 Generation 3420: Best = 3.573689e+05 FES = 342100 μ_F = 0.835 μ_{CR} = 0.044
 Diversity = 6.6491e+00 k_t = [4, 3, 2, 4]
 Generation 3430: Best = 3.569264e+05 FES = 343100 μ_F = 0.841 μ_{CR} = 0.040
 Diversity = 6.6925e+00 k_t = [1, 1, 1, 1]
 Generation 3440: Best = 3.564512e+05 FES = 344100 μ_F = 0.837 μ_{CR} = 0.040
 Diversity = 6.9473e+00 k_t = [2, 2, 3, 2]
 Generation 3450: Best = 3.560171e+05 FES = 345100 μ_F = 0.826 μ_{CR} = 0.040
 Diversity = 6.4428e+00 k_t = [1, 1, 1, 1]
 Generation 3460: Best = 3.555324e+05 FES = 346100 μ_F = 0.837 μ_{CR} = 0.036
 Diversity = 6.9431e+00 k_t = [3, 2, 4, 2]

Generation 3470: Best = 3.553986e+05 FES = 347100 μ_F = 0.813 μ_{CR} = 0.037
 Diversity = 6.9392e+00 k_t = [1, 1, 1, 1]
 Generation 3480: Best = 3.550978e+05 FES = 348100 μ_F = 0.813 μ_{CR} = 0.034
 Diversity = 6.9758e+00 k_t = [2, 4, 3, 3]
 Generation 3490: Best = 3.543831e+05 FES = 349100 μ_F = 0.801 μ_{CR} = 0.046
 Diversity = 7.0460e+00 k_t = [1, 1, 1, 1]
 Generation 3500: Best = 3.540221e+05 FES = 350100 μ_F = 0.806 μ_{CR} = 0.050
 Diversity = 6.7401e+00 k_t = [2, 4, 3, 3]
 Generation 3510: Best = 3.538283e+05 FES = 351100 μ_F = 0.807 μ_{CR} = 0.048
 Diversity = 6.5447e+00 k_t = [1, 1, 1, 1]
 Generation 3520: Best = 3.534845e+05 FES = 352100 μ_F = 0.812 μ_{CR} = 0.042
 Diversity = 6.4378e+00 k_t = [3, 4, 2, 4]
 Generation 3530: Best = 3.530291e+05 FES = 353100 μ_F = 0.821 μ_{CR} = 0.035
 Diversity = 6.2947e+00 k_t = [1, 1, 1, 1]
 Generation 3540: Best = 3.526044e+05 FES = 354100 μ_F = 0.815 μ_{CR} = 0.035
 Diversity = 5.9520e+00 k_t = [3, 4, 3, 3]
 Generation 3550: Best = 3.525221e+05 FES = 355100 μ_F = 0.810 μ_{CR} = 0.035
 Diversity = 5.4340e+00 k_t = [1, 1, 1, 1]
 Generation 3560: Best = 3.522290e+05 FES = 356100 μ_F = 0.813 μ_{CR} = 0.035
 Diversity = 5.3071e+00 k_t = [3, 2, 3, 4]
 Generation 3570: Best = 3.519769e+05 FES = 357100 μ_F = 0.808 μ_{CR} = 0.042
 Diversity = 5.0730e+00 k_t = [1, 1, 1, 1]
 Generation 3580: Best = 3.516251e+05 FES = 358100 μ_F = 0.811 μ_{CR} = 0.040
 Diversity = 5.3843e+00 k_t = [2, 4, 2, 3]
 Generation 3590: Best = 3.515081e+05 FES = 359100 μ_F = 0.811 μ_{CR} = 0.043
 Diversity = 5.6015e+00 k_t = [1, 1, 1, 1]
 Generation 3600: Best = 3.511980e+05 FES = 360100 μ_F = 0.801 μ_{CR} = 0.047
 Diversity = 5.4737e+00 k_t = [4, 2, 2, 3]
 Generation 3610: Best = 3.508612e+05 FES = 361100 μ_F = 0.788 μ_{CR} = 0.047
 Diversity = 5.2322e+00 k_t = [1, 1, 1, 1]
 Generation 3620: Best = 3.502908e+05 FES = 362100 μ_F = 0.797 μ_{CR} = 0.051
 Diversity = 5.5350e+00 k_t = [3, 2, 4, 2]
 Generation 3630: Best = 3.500655e+05 FES = 363100 μ_F = 0.800 μ_{CR} = 0.047
 Diversity = 6.1665e+00 k_t = [1, 1, 1, 1]
 Generation 3640: Best = 3.493699e+05 FES = 364100 μ_F = 0.807 μ_{CR} = 0.041
 Diversity = 6.8548e+00 k_t = [2, 4, 2, 2]
 Generation 3650: Best = 3.492841e+05 FES = 365100 μ_F = 0.818 μ_{CR} = 0.040
 Diversity = 7.0107e+00 k_t = [1, 1, 1, 1]
 Generation 3660: Best = 3.484179e+05 FES = 366100 μ_F = 0.824 μ_{CR} = 0.041
 Diversity = 6.4463e+00 k_t = [2, 3, 3, 3]
 Generation 3670: Best = 3.477238e+05 FES = 367100 μ_F = 0.834 μ_{CR} = 0.036
 Diversity = 6.9034e+00 k_t = [1, 1, 1, 1]
 Generation 3680: Best = 3.470341e+05 FES = 368100 μ_F = 0.838 μ_{CR} = 0.040
 Diversity = 7.2327e+00 k_t = [4, 3, 4, 3]
 Generation 3690: Best = 3.466197e+05 FES = 369100 μ_F = 0.835 μ_{CR} = 0.035
 Diversity = 7.0488e+00 k_t = [1, 1, 1, 1]
 Generation 3700: Best = 3.459246e+05 FES = 370100 μ_F = 0.825 μ_{CR} = 0.034
 Diversity = 7.3592e+00 k_t = [3, 3, 4, 4]
 Generation 3710: Best = 3.454179e+05 FES = 371100 μ_F = 0.804 μ_{CR} = 0.039
 Diversity = 7.1583e+00 k_t = [1, 1, 1, 1]
 Generation 3720: Best = 3.447513e+05 FES = 372100 μ_F = 0.809 μ_{CR} = 0.039
 Diversity = 7.4055e+00 k_t = [3, 4, 2, 2]
 Generation 3730: Best = 3.445105e+05 FES = 373100 μ_F = 0.801 μ_{CR} = 0.044
 Diversity = 7.3666e+00 k_t = [1, 1, 1, 1]

Generation 3740: Best = 3.441669e+05 FES = 374100 μ_F = 0.795 μ_{CR} = 0.039
 Diversity = 6.9297e+00 k_t = [3, 4, 3, 3]
 Generation 3750: Best = 3.439011e+05 FES = 375100 μ_F = 0.801 μ_{CR} = 0.052
 Diversity = 7.1846e+00 k_t = [1, 1, 1, 1]
 Generation 3760: Best = 3.434381e+05 FES = 376100 μ_F = 0.791 μ_{CR} = 0.042
 Diversity = 6.5139e+00 k_t = [2, 2, 3, 3]
 Generation 3770: Best = 3.431869e+05 FES = 377100 μ_F = 0.786 μ_{CR} = 0.041
 Diversity = 6.1600e+00 k_t = [1, 1, 1, 1]
 Generation 3780: Best = 3.423492e+05 FES = 378100 μ_F = 0.784 μ_{CR} = 0.035
 Diversity = 5.4916e+00 k_t = [4, 3, 2, 3]
 Generation 3790: Best = 3.420564e+05 FES = 379100 μ_F = 0.784 μ_{CR} = 0.036
 Diversity = 5.7252e+00 k_t = [1, 1, 1, 1]
 Generation 3800: Best = 3.412780e+05 FES = 380100 μ_F = 0.788 μ_{CR} = 0.042
 Diversity = 5.7358e+00 k_t = [2, 4, 3, 2]
 Generation 3810: Best = 3.409920e+05 FES = 381100 μ_F = 0.800 μ_{CR} = 0.040
 Diversity = 6.9875e+00 k_t = [1, 1, 1, 1]
 Generation 3820: Best = 3.405053e+05 FES = 382100 μ_F = 0.778 μ_{CR} = 0.034
 Diversity = 5.8123e+00 k_t = [2, 4, 2, 3]
 Generation 3830: Best = 3.400407e+05 FES = 383100 μ_F = 0.790 μ_{CR} = 0.042
 Diversity = 5.3144e+00 k_t = [1, 1, 1, 1]
 Generation 3840: Best = 3.393361e+05 FES = 384100 μ_F = 0.791 μ_{CR} = 0.039
 Diversity = 4.9421e+00 k_t = [3, 2, 2, 2]
 Generation 3850: Best = 3.392629e+05 FES = 385100 μ_F = 0.789 μ_{CR} = 0.043
 Diversity = 5.0661e+00 k_t = [1, 1, 1, 1]
 Generation 3860: Best = 3.388349e+05 FES = 386100 μ_F = 0.809 μ_{CR} = 0.051
 Diversity = 4.5720e+00 k_t = [3, 3, 2, 3]
 Generation 3870: Best = 3.387522e+05 FES = 387100 μ_F = 0.812 μ_{CR} = 0.045
 Diversity = 4.8973e+00 k_t = [1, 1, 1, 1]
 Generation 3880: Best = 3.382181e+05 FES = 388100 μ_F = 0.819 μ_{CR} = 0.041
 Diversity = 5.4881e+00 k_t = [4, 2, 3, 4]
 Generation 3890: Best = 3.377284e+05 FES = 389100 μ_F = 0.822 μ_{CR} = 0.045
 Diversity = 6.0582e+00 k_t = [1, 1, 1, 1]
 Generation 3900: Best = 3.376854e+05 FES = 390100 μ_F = 0.813 μ_{CR} = 0.044
 Diversity = 5.6995e+00 k_t = [4, 2, 2, 3]
 Generation 3910: Best = 3.375304e+05 FES = 391100 μ_F = 0.812 μ_{CR} = 0.040
 Diversity = 5.2167e+00 k_t = [1, 1, 1, 1]
 Generation 3920: Best = 3.370944e+05 FES = 392100 μ_F = 0.796 μ_{CR} = 0.040
 Diversity = 5.3135e+00 k_t = [3, 4, 4, 2]
 Generation 3930: Best = 3.364469e+05 FES = 393100 μ_F = 0.811 μ_{CR} = 0.043
 Diversity = 6.3315e+00 k_t = [1, 1, 1, 1]
 Generation 3940: Best = 3.362264e+05 FES = 394100 μ_F = 0.810 μ_{CR} = 0.038
 Diversity = 5.4832e+00 k_t = [2, 2, 2, 4]
 Generation 3950: Best = 3.356372e+05 FES = 395100 μ_F = 0.819 μ_{CR} = 0.041
 Diversity = 5.1839e+00 k_t = [1, 1, 1, 1]
 Generation 3960: Best = 3.352969e+05 FES = 396100 μ_F = 0.825 μ_{CR} = 0.031
 Diversity = 4.7646e+00 k_t = [3, 2, 3, 3]
 Generation 3970: Best = 3.349306e+05 FES = 397100 μ_F = 0.819 μ_{CR} = 0.045
 Diversity = 4.9101e+00 k_t = [1, 1, 1, 1]
 Generation 3980: Best = 3.347519e+05 FES = 398100 μ_F = 0.811 μ_{CR} = 0.045
 Diversity = 5.5022e+00 k_t = [3, 2, 2, 2]
 Generation 3990: Best = 3.342824e+05 FES = 399100 μ_F = 0.795 μ_{CR} = 0.035
 Diversity = 4.9393e+00 k_t = [1, 1, 1, 1]
 Generation 4000: Best = 3.341257e+05 FES = 400100 μ_F = 0.798 μ_{CR} = 0.033
 Diversity = 5.3250e+00 k_t = [4, 3, 2, 2]

Generation 4010: Best = 3.336278e+05 FES = 401100 μ_F = 0.795 μ_{CR} = 0.041
 Diversity = 5.2701e+00 k_t = [1, 1, 1, 1]
 Generation 4020: Best = 3.335532e+05 FES = 402100 μ_F = 0.806 μ_{CR} = 0.040
 Diversity = 5.7563e+00 k_t = [3, 3, 4, 2]
 Generation 4030: Best = 3.331852e+05 FES = 403100 μ_F = 0.805 μ_{CR} = 0.037
 Diversity = 5.7787e+00 k_t = [1, 1, 1, 1]
 Generation 4040: Best = 3.329495e+05 FES = 404100 μ_F = 0.802 μ_{CR} = 0.042
 Diversity = 5.8541e+00 k_t = [4, 2, 4, 2]
 Generation 4050: Best = 3.327981e+05 FES = 405100 μ_F = 0.816 μ_{CR} = 0.036
 Diversity = 5.4194e+00 k_t = [1, 1, 1, 1]
 Generation 4060: Best = 3.324429e+05 FES = 406100 μ_F = 0.820 μ_{CR} = 0.032
 Diversity = 5.5622e+00 k_t = [2, 3, 3, 4]
 Generation 4070: Best = 3.319092e+05 FES = 407100 μ_F = 0.815 μ_{CR} = 0.043
 Diversity = 5.1261e+00 k_t = [1, 1, 1, 1]
 Generation 4080: Best = 3.316052e+05 FES = 408100 μ_F = 0.814 μ_{CR} = 0.047
 Diversity = 5.4538e+00 k_t = [3, 3, 3, 4]
 Generation 4090: Best = 3.311372e+05 FES = 409100 μ_F = 0.804 μ_{CR} = 0.052
 Diversity = 5.2777e+00 k_t = [1, 1, 1, 1]
 Generation 4100: Best = 3.308820e+05 FES = 410100 μ_F = 0.793 μ_{CR} = 0.047
 Diversity = 5.8082e+00 k_t = [3, 3, 4, 2]
 Generation 4110: Best = 3.306941e+05 FES = 411100 μ_F = 0.783 μ_{CR} = 0.042
 Diversity = 6.2758e+00 k_t = [1, 1, 1, 1]
 Generation 4120: Best = 3.303145e+05 FES = 412100 μ_F = 0.780 μ_{CR} = 0.044
 Diversity = 5.5995e+00 k_t = [4, 4, 2, 2]
 Generation 4130: Best = 3.300887e+05 FES = 413100 μ_F = 0.799 μ_{CR} = 0.038
 Diversity = 5.5052e+00 k_t = [1, 1, 1, 1]
 Generation 4140: Best = 3.298117e+05 FES = 414100 μ_F = 0.816 μ_{CR} = 0.043
 Diversity = 5.3380e+00 k_t = [3, 2, 2, 4]
 Generation 4150: Best = 3.295024e+05 FES = 415100 μ_F = 0.832 μ_{CR} = 0.037
 Diversity = 5.6951e+00 k_t = [1, 1, 1, 1]
 Generation 4160: Best = 3.288470e+05 FES = 416100 μ_F = 0.829 μ_{CR} = 0.044
 Diversity = 6.8047e+00 k_t = [2, 3, 4, 3]
 Generation 4170: Best = 3.286781e+05 FES = 417100 μ_F = 0.822 μ_{CR} = 0.039
 Diversity = 6.5576e+00 k_t = [1, 1, 1, 1]
 Generation 4180: Best = 3.283197e+05 FES = 418100 μ_F = 0.806 μ_{CR} = 0.041
 Diversity = 6.4877e+00 k_t = [4, 4, 2, 2]
 Generation 4190: Best = 3.278180e+05 FES = 419100 μ_F = 0.791 μ_{CR} = 0.052
 Diversity = 6.2682e+00 k_t = [1, 1, 1, 1]
 Generation 4200: Best = 3.276320e+05 FES = 420100 μ_F = 0.783 μ_{CR} = 0.047
 Diversity = 5.9583e+00 k_t = [4, 3, 4, 3]
 Generation 4210: Best = 3.272837e+05 FES = 421100 μ_F = 0.794 μ_{CR} = 0.048
 Diversity = 5.7538e+00 k_t = [1, 1, 1, 1]
 Generation 4220: Best = 3.270014e+05 FES = 422100 μ_F = 0.787 μ_{CR} = 0.047
 Diversity = 5.6030e+00 k_t = [2, 4, 2, 4]
 Generation 4230: Best = 3.266121e+05 FES = 423100 μ_F = 0.806 μ_{CR} = 0.041
 Diversity = 5.3440e+00 k_t = [1, 1, 1, 1]
 Generation 4240: Best = 3.265690e+05 FES = 424100 μ_F = 0.803 μ_{CR} = 0.038
 Diversity = 4.7010e+00 k_t = [2, 4, 4, 3]
 Generation 4250: Best = 3.262850e+05 FES = 425100 μ_F = 0.798 μ_{CR} = 0.043
 Diversity = 4.3962e+00 k_t = [1, 1, 1, 1]
 Generation 4260: Best = 3.259607e+05 FES = 426100 μ_F = 0.811 μ_{CR} = 0.047
 Diversity = 4.7115e+00 k_t = [3, 4, 4, 2]
 Generation 4270: Best = 3.253533e+05 FES = 427100 μ_F = 0.822 μ_{CR} = 0.039
 Diversity = 5.5422e+00 k_t = [1, 1, 1, 1]

Generation 4280: Best = 3.250474e+05 FES = 428100 μ_F = 0.806 μ_{CR} = 0.043
 Diversity = 5.8087e+00 k_t = [4, 4, 2, 2]
 Generation 4290: Best = 3.248038e+05 FES = 429100 μ_F = 0.794 μ_{CR} = 0.040
 Diversity = 6.2416e+00 k_t = [1, 1, 1, 1]
 Generation 4300: Best = 3.241829e+05 FES = 430100 μ_F = 0.794 μ_{CR} = 0.040
 Diversity = 5.7294e+00 k_t = [2, 3, 2, 2]
 Generation 4310: Best = 3.235491e+05 FES = 431100 μ_F = 0.808 μ_{CR} = 0.044
 Diversity = 5.7204e+00 k_t = [1, 1, 1, 1]
 Generation 4320: Best = 3.233081e+05 FES = 432100 μ_F = 0.803 μ_{CR} = 0.046
 Diversity = 5.7951e+00 k_t = [3, 4, 3, 3]
 Generation 4330: Best = 3.230469e+05 FES = 433100 μ_F = 0.819 μ_{CR} = 0.042
 Diversity = 5.8208e+00 k_t = [1, 1, 1, 1]
 Generation 4340: Best = 3.227548e+05 FES = 434100 μ_F = 0.802 μ_{CR} = 0.037
 Diversity = 5.2822e+00 k_t = [3, 2, 2, 3]
 Generation 4350: Best = 3.223908e+05 FES = 435100 μ_F = 0.800 μ_{CR} = 0.039
 Diversity = 5.2190e+00 k_t = [1, 1, 1, 1]
 Generation 4360: Best = 3.220743e+05 FES = 436100 μ_F = 0.795 μ_{CR} = 0.042
 Diversity = 5.0276e+00 k_t = [3, 2, 3, 3]
 Generation 4370: Best = 3.217171e+05 FES = 437100 μ_F = 0.799 μ_{CR} = 0.039
 Diversity = 5.1691e+00 k_t = [1, 1, 1, 1]
 Generation 4380: Best = 3.215576e+05 FES = 438100 μ_F = 0.807 μ_{CR} = 0.034
 Diversity = 4.7923e+00 k_t = [3, 4, 4, 2]
 Generation 4390: Best = 3.213121e+05 FES = 439100 μ_F = 0.814 μ_{CR} = 0.032
 Diversity = 5.0355e+00 k_t = [1, 1, 1, 1]
 Generation 4400: Best = 3.208781e+05 FES = 440100 μ_F = 0.808 μ_{CR} = 0.037
 Diversity = 5.6881e+00 k_t = [3, 4, 3, 2]
 Generation 4410: Best = 3.207911e+05 FES = 441100 μ_F = 0.796 μ_{CR} = 0.039
 Diversity = 5.8395e+00 k_t = [1, 1, 1, 1]
 Generation 4420: Best = 3.205224e+05 FES = 442100 μ_F = 0.793 μ_{CR} = 0.043
 Diversity = 5.4859e+00 k_t = [3, 3, 2, 3]
 Generation 4430: Best = 3.202151e+05 FES = 443100 μ_F = 0.794 μ_{CR} = 0.044
 Diversity = 5.5959e+00 k_t = [1, 1, 1, 1]
 Generation 4440: Best = 3.199359e+05 FES = 444100 μ_F = 0.818 μ_{CR} = 0.051
 Diversity = 5.4667e+00 k_t = [3, 2, 2, 3]
 Generation 4450: Best = 3.196136e+05 FES = 445100 μ_F = 0.832 μ_{CR} = 0.049
 Diversity = 6.1771e+00 k_t = [1, 1, 1, 1]
 Generation 4460: Best = 3.191406e+05 FES = 446100 μ_F = 0.820 μ_{CR} = 0.044
 Diversity = 6.0554e+00 k_t = [4, 3, 4, 3]
 Generation 4470: Best = 3.184208e+05 FES = 447100 μ_F = 0.821 μ_{CR} = 0.040
 Diversity = 7.1658e+00 k_t = [1, 1, 1, 1]
 Generation 4480: Best = 3.181493e+05 FES = 448100 μ_F = 0.812 μ_{CR} = 0.040
 Diversity = 7.2823e+00 k_t = [3, 3, 4, 2]
 Generation 4490: Best = 3.176584e+05 FES = 449100 μ_F = 0.806 μ_{CR} = 0.035
 Diversity = 7.2837e+00 k_t = [1, 1, 1, 1]
 Generation 4500: Best = 3.172617e+05 FES = 450100 μ_F = 0.794 μ_{CR} = 0.038
 Diversity = 7.0479e+00 k_t = [2, 2, 4, 2]
 Generation 4510: Best = 3.169843e+05 FES = 451100 μ_F = 0.784 μ_{CR} = 0.039
 Diversity = 6.2943e+00 k_t = [1, 1, 1, 1]
 Generation 4520: Best = 3.166501e+05 FES = 452100 μ_F = 0.782 μ_{CR} = 0.037
 Diversity = 5.3972e+00 k_t = [2, 2, 3, 2]
 Generation 4530: Best = 3.164302e+05 FES = 453100 μ_F = 0.776 μ_{CR} = 0.037
 Diversity = 5.8118e+00 k_t = [1, 1, 1, 1]
 Generation 4540: Best = 3.161490e+05 FES = 454100 μ_F = 0.795 μ_{CR} = 0.037
 Diversity = 5.6664e+00 k_t = [4, 4, 3, 2]

Generation 4550: Best = 3.157574e+05 FES = 455100 μ_F = 0.798 μ_{CR} = 0.037
 Diversity = 5.7228e+00 k_t = [1, 1, 1, 1]
 Generation 4560: Best = 3.157061e+05 FES = 456100 μ_F = 0.800 μ_{CR} = 0.034
 Diversity = 5.6440e+00 k_t = [2, 4, 2, 3]
 Generation 4570: Best = 3.155232e+05 FES = 457100 μ_F = 0.778 μ_{CR} = 0.040
 Diversity = 5.0866e+00 k_t = [1, 1, 1, 1]
 Generation 4580: Best = 3.153328e+05 FES = 458100 μ_F = 0.798 μ_{CR} = 0.045
 Diversity = 5.0429e+00 k_t = [2, 2, 2, 2]
 Generation 4590: Best = 3.150718e+05 FES = 459100 μ_F = 0.791 μ_{CR} = 0.044
 Diversity = 5.3967e+00 k_t = [1, 1, 1, 1]
 Generation 4600: Best = 3.146359e+05 FES = 460100 μ_F = 0.795 μ_{CR} = 0.034
 Diversity = 4.8067e+00 k_t = [3, 2, 4, 3]
 Generation 4610: Best = 3.141959e+05 FES = 461100 μ_F = 0.808 μ_{CR} = 0.041
 Diversity = 5.1339e+00 k_t = [1, 1, 1, 1]
 Generation 4620: Best = 3.137718e+05 FES = 462100 μ_F = 0.804 μ_{CR} = 0.041
 Diversity = 4.8845e+00 k_t = [4, 2, 3, 4]
 Generation 4630: Best = 3.131347e+05 FES = 463100 μ_F = 0.799 μ_{CR} = 0.050
 Diversity = 4.4731e+00 k_t = [1, 1, 1, 1]
 Generation 4640: Best = 3.126099e+05 FES = 464100 μ_F = 0.791 μ_{CR} = 0.051
 Diversity = 5.0557e+00 k_t = [2, 2, 4, 2]
 Generation 4650: Best = 3.121743e+05 FES = 465100 μ_F = 0.786 μ_{CR} = 0.052
 Diversity = 5.1501e+00 k_t = [1, 1, 1, 1]
 Generation 4660: Best = 3.119490e+05 FES = 466100 μ_F = 0.794 μ_{CR} = 0.046
 Diversity = 5.0626e+00 k_t = [3, 3, 2, 2]
 Generation 4670: Best = 3.117346e+05 FES = 467100 μ_F = 0.803 μ_{CR} = 0.038
 Diversity = 5.1341e+00 k_t = [1, 1, 1, 1]
 Generation 4680: Best = 3.111408e+05 FES = 468100 μ_F = 0.803 μ_{CR} = 0.042
 Diversity = 4.8133e+00 k_t = [3, 3, 4, 4]
 Generation 4690: Best = 3.111407e+05 FES = 469100 μ_F = 0.816 μ_{CR} = 0.044
 Diversity = 4.4775e+00 k_t = [1, 1, 1, 1]
 Generation 4700: Best = 3.107577e+05 FES = 470100 μ_F = 0.806 μ_{CR} = 0.041
 Diversity = 5.2433e+00 k_t = [3, 4, 4, 4]
 Generation 4710: Best = 3.103664e+05 FES = 471100 μ_F = 0.811 μ_{CR} = 0.039
 Diversity = 4.9559e+00 k_t = [1, 1, 1, 1]
 Generation 4720: Best = 3.101000e+05 FES = 472100 μ_F = 0.813 μ_{CR} = 0.034
 Diversity = 5.2690e+00 k_t = [2, 4, 3, 3]
 Generation 4730: Best = 3.097085e+05 FES = 473100 μ_F = 0.802 μ_{CR} = 0.052
 Diversity = 4.9905e+00 k_t = [1, 1, 1, 1]
 Generation 4740: Best = 3.093323e+05 FES = 474100 μ_F = 0.805 μ_{CR} = 0.048
 Diversity = 5.2863e+00 k_t = [3, 2, 2, 3]
 Generation 4750: Best = 3.088851e+05 FES = 475100 μ_F = 0.805 μ_{CR} = 0.042
 Diversity = 5.6552e+00 k_t = [1, 1, 1, 1]
 Generation 4760: Best = 3.086743e+05 FES = 476100 μ_F = 0.817 μ_{CR} = 0.047
 Diversity = 5.5760e+00 k_t = [4, 4, 4, 3]
 Generation 4770: Best = 3.078670e+05 FES = 477100 μ_F = 0.816 μ_{CR} = 0.049
 Diversity = 5.7056e+00 k_t = [1, 1, 1, 1]
 Generation 4780: Best = 3.075079e+05 FES = 478100 μ_F = 0.802 μ_{CR} = 0.045
 Diversity = 5.2710e+00 k_t = [3, 2, 4, 4]
 Generation 4790: Best = 3.070800e+05 FES = 479100 μ_F = 0.808 μ_{CR} = 0.045
 Diversity = 4.9856e+00 k_t = [1, 1, 1, 1]
 Generation 4800: Best = 3.067473e+05 FES = 480100 μ_F = 0.804 μ_{CR} = 0.039
 Diversity = 4.9193e+00 k_t = [4, 2, 2, 4]
 Generation 4810: Best = 3.066289e+05 FES = 481100 μ_F = 0.804 μ_{CR} = 0.041
 Diversity = 4.3257e+00 k_t = [1, 1, 1, 1]

Generation 4820: Best = 3.063937e+05 FES = 482100 μ_F = 0.803 μ_{CR} = 0.043
 Diversity = 4.5312e+00 k_t = [3, 4, 3, 3]
 Generation 4830: Best = 3.062271e+05 FES = 483100 μ_F = 0.800 μ_{CR} = 0.042
 Diversity = 4.3898e+00 k_t = [1, 1, 1, 1]
 Generation 4840: Best = 3.060344e+05 FES = 484100 μ_F = 0.802 μ_{CR} = 0.046
 Diversity = 4.2099e+00 k_t = [3, 3, 2, 2]
 Generation 4850: Best = 3.057618e+05 FES = 485100 μ_F = 0.804 μ_{CR} = 0.042
 Diversity = 3.9214e+00 k_t = [1, 1, 1, 1]
 Generation 4860: Best = 3.056150e+05 FES = 486100 μ_F = 0.808 μ_{CR} = 0.032
 Diversity = 3.8033e+00 k_t = [2, 2, 3, 3]
 Generation 4870: Best = 3.054531e+05 FES = 487100 μ_F = 0.813 μ_{CR} = 0.040
 Diversity = 3.8378e+00 k_t = [1, 1, 1, 1]
 Generation 4880: Best = 3.052693e+05 FES = 488100 μ_F = 0.816 μ_{CR} = 0.042
 Diversity = 3.9408e+00 k_t = [4, 3, 3, 3]
 Generation 4890: Best = 3.051129e+05 FES = 489100 μ_F = 0.811 μ_{CR} = 0.050
 Diversity = 4.0965e+00 k_t = [1, 1, 1, 1]
 Generation 4900: Best = 3.048967e+05 FES = 490100 μ_F = 0.808 μ_{CR} = 0.043
 Diversity = 4.3660e+00 k_t = [2, 4, 4, 4]
 Generation 4910: Best = 3.046995e+05 FES = 491100 μ_F = 0.813 μ_{CR} = 0.044
 Diversity = 3.8120e+00 k_t = [1, 1, 1, 1]
 Generation 4920: Best = 3.044571e+05 FES = 492100 μ_F = 0.816 μ_{CR} = 0.041
 Diversity = 4.2212e+00 k_t = [3, 2, 2, 4]
 Generation 4930: Best = 3.042967e+05 FES = 493100 μ_F = 0.821 μ_{CR} = 0.042
 Diversity = 4.1186e+00 k_t = [1, 1, 1, 1]
 Generation 4940: Best = 3.039261e+05 FES = 494100 μ_F = 0.827 μ_{CR} = 0.049
 Diversity = 4.4624e+00 k_t = [3, 3, 2, 2]
 Generation 4950: Best = 3.036945e+05 FES = 495100 μ_F = 0.845 μ_{CR} = 0.038
 Diversity = 5.7046e+00 k_t = [1, 1, 1, 1]
 Generation 4960: Best = 3.036713e+05 FES = 496100 μ_F = 0.832 μ_{CR} = 0.044
 Diversity = 5.3891e+00 k_t = [3, 3, 2, 3]
 Generation 4970: Best = 3.033157e+05 FES = 497100 μ_F = 0.819 μ_{CR} = 0.041
 Diversity = 5.3810e+00 k_t = [1, 1, 1, 1]
 Generation 4980: Best = 3.031598e+05 FES = 498100 μ_F = 0.804 μ_{CR} = 0.048
 Diversity = 5.1748e+00 k_t = [3, 4, 4, 2]
 Generation 4990: Best = 3.030173e+05 FES = 499100 μ_F = 0.803 μ_{CR} = 0.042
 Diversity = 5.2603e+00 k_t = [1, 1, 1, 1]
 Generation 5000: Best = 3.027409e+05 FES = 500100 μ_F = 0.807 μ_{CR} = 0.041
 Diversity = 5.2650e+00 k_t = [3, 2, 4, 2]
 Generation 5010: Best = 3.024339e+05 FES = 501100 μ_F = 0.800 μ_{CR} = 0.037
 Diversity = 5.7000e+00 k_t = [1, 1, 1, 1]
 Generation 5020: Best = 3.022623e+05 FES = 502100 μ_F = 0.805 μ_{CR} = 0.045
 Diversity = 5.7254e+00 k_t = [3, 4, 2, 4]
 Generation 5030: Best = 3.018211e+05 FES = 503100 μ_F = 0.816 μ_{CR} = 0.042
 Diversity = 5.8285e+00 k_t = [1, 1, 1, 1]
 Generation 5040: Best = 3.016115e+05 FES = 504100 μ_F = 0.831 μ_{CR} = 0.045
 Diversity = 6.0982e+00 k_t = [4, 4, 3, 3]
 Generation 5050: Best = 3.013228e+05 FES = 505100 μ_F = 0.820 μ_{CR} = 0.038
 Diversity = 5.9726e+00 k_t = [1, 1, 1, 1]
 Generation 5060: Best = 3.008655e+05 FES = 506100 μ_F = 0.814 μ_{CR} = 0.039
 Diversity = 6.2787e+00 k_t = [4, 2, 3, 2]
 Generation 5070: Best = 3.006337e+05 FES = 507100 μ_F = 0.826 μ_{CR} = 0.032
 Diversity = 6.9894e+00 k_t = [1, 1, 1, 1]
 Generation 5080: Best = 3.002540e+05 FES = 508100 μ_F = 0.825 μ_{CR} = 0.035
 Diversity = 6.5962e+00 k_t = [3, 4, 4, 3]

Generation 5090: Best = 2.998810e+05 FES = 509100 μ_F = 0.820 μ_{CR} = 0.032
 Diversity = 5.5868e+00 k_t = [1, 1, 1, 1]
 Generation 5100: Best = 2.998776e+05 FES = 510100 μ_F = 0.827 μ_{CR} = 0.031
 Diversity = 5.4617e+00 k_t = [3, 2, 4, 3]
 Generation 5110: Best = 2.997217e+05 FES = 511100 μ_F = 0.817 μ_{CR} = 0.034
 Diversity = 4.9989e+00 k_t = [1, 1, 1, 1]
 Generation 5120: Best = 2.995511e+05 FES = 512100 μ_F = 0.810 μ_{CR} = 0.034
 Diversity = 4.9251e+00 k_t = [2, 2, 4, 2]
 Generation 5130: Best = 2.993307e+05 FES = 513100 μ_F = 0.813 μ_{CR} = 0.036
 Diversity = 4.9116e+00 k_t = [1, 1, 1, 1]
 Generation 5140: Best = 2.991480e+05 FES = 514100 μ_F = 0.822 μ_{CR} = 0.039
 Diversity = 4.9063e+00 k_t = [3, 4, 3, 4]
 Generation 5150: Best = 2.990174e+05 FES = 515100 μ_F = 0.825 μ_{CR} = 0.042
 Diversity = 4.6873e+00 k_t = [1, 1, 1, 1]
 Generation 5160: Best = 2.988360e+05 FES = 516100 μ_F = 0.815 μ_{CR} = 0.041
 Diversity = 5.0022e+00 k_t = [4, 2, 4, 2]
 Generation 5170: Best = 2.985983e+05 FES = 517100 μ_F = 0.823 μ_{CR} = 0.046
 Diversity = 5.4556e+00 k_t = [1, 1, 1, 1]
 Generation 5180: Best = 2.983726e+05 FES = 518100 μ_F = 0.816 μ_{CR} = 0.041
 Diversity = 5.6612e+00 k_t = [3, 4, 4, 4]
 Generation 5190: Best = 2.979942e+05 FES = 519100 μ_F = 0.815 μ_{CR} = 0.044
 Diversity = 5.9659e+00 k_t = [1, 1, 1, 1]
 Generation 5200: Best = 2.978361e+05 FES = 520100 μ_F = 0.819 μ_{CR} = 0.044
 Diversity = 5.4200e+00 k_t = [3, 4, 4, 3]
 Generation 5210: Best = 2.976021e+05 FES = 521100 μ_F = 0.819 μ_{CR} = 0.040
 Diversity = 6.1845e+00 k_t = [1, 1, 1, 1]
 Generation 5220: Best = 2.973640e+05 FES = 522100 μ_F = 0.826 μ_{CR} = 0.039
 Diversity = 5.8805e+00 k_t = [2, 2, 3, 3]
 Generation 5230: Best = 2.972607e+05 FES = 523100 μ_F = 0.813 μ_{CR} = 0.045
 Diversity = 5.2388e+00 k_t = [1, 1, 1, 1]
 Generation 5240: Best = 2.970728e+05 FES = 524100 μ_F = 0.813 μ_{CR} = 0.046
 Diversity = 5.2476e+00 k_t = [3, 2, 2, 3]
 Generation 5250: Best = 2.968911e+05 FES = 525100 μ_F = 0.805 μ_{CR} = 0.041
 Diversity = 5.2292e+00 k_t = [1, 1, 1, 1]
 Generation 5260: Best = 2.966281e+05 FES = 526100 μ_F = 0.813 μ_{CR} = 0.038
 Diversity = 5.7863e+00 k_t = [2, 2, 2, 2]
 Generation 5270: Best = 2.960284e+05 FES = 527100 μ_F = 0.815 μ_{CR} = 0.036
 Diversity = 6.4778e+00 k_t = [1, 1, 1, 1]
 Generation 5280: Best = 2.955946e+05 FES = 528100 μ_F = 0.810 μ_{CR} = 0.043
 Diversity = 7.0074e+00 k_t = [3, 3, 2, 3]
 Generation 5290: Best = 2.951071e+05 FES = 529100 μ_F = 0.802 μ_{CR} = 0.044
 Diversity = 6.7009e+00 k_t = [1, 1, 1, 1]
 Generation 5300: Best = 2.949134e+05 FES = 530100 μ_F = 0.798 μ_{CR} = 0.046
 Diversity = 6.3263e+00 k_t = [3, 3, 2, 2]
 Generation 5310: Best = 2.946798e+05 FES = 531100 μ_F = 0.804 μ_{CR} = 0.047
 Diversity = 5.8192e+00 k_t = [1, 1, 1, 1]
 Generation 5320: Best = 2.942819e+05 FES = 532100 μ_F = 0.803 μ_{CR} = 0.042
 Diversity = 5.6394e+00 k_t = [4, 3, 3, 2]
 Generation 5330: Best = 2.941592e+05 FES = 533100 μ_F = 0.800 μ_{CR} = 0.039
 Diversity = 5.9250e+00 k_t = [1, 1, 1, 1]
 Generation 5340: Best = 2.940119e+05 FES = 534100 μ_F = 0.798 μ_{CR} = 0.043
 Diversity = 5.6530e+00 k_t = [2, 2, 3, 4]
 Generation 5350: Best = 2.938292e+05 FES = 535100 μ_F = 0.806 μ_{CR} = 0.041
 Diversity = 5.0438e+00 k_t = [1, 1, 1, 1]

Generation 5360: Best = 2.937879e+05 FES = 536100 μ_F = 0.810 μ_{CR} = 0.037
 Diversity = 5.3982e+00 k_t = [3, 2, 3, 2]
 Generation 5370: Best = 2.936864e+05 FES = 537100 μ_F = 0.803 μ_{CR} = 0.044
 Diversity = 4.6376e+00 k_t = [1, 1, 1, 1]
 Generation 5380: Best = 2.935005e+05 FES = 538100 μ_F = 0.776 μ_{CR} = 0.040
 Diversity = 4.0088e+00 k_t = [2, 2, 2, 3]
 Generation 5390: Best = 2.932114e+05 FES = 539100 μ_F = 0.775 μ_{CR} = 0.046
 Diversity = 4.1348e+00 k_t = [1, 1, 1, 1]
 Generation 5400: Best = 2.930039e+05 FES = 540100 μ_F = 0.774 μ_{CR} = 0.045
 Diversity = 3.9310e+00 k_t = [4, 4, 4, 3]
 Generation 5410: Best = 2.929286e+05 FES = 541100 μ_F = 0.772 μ_{CR} = 0.041
 Diversity = 3.6838e+00 k_t = [1, 1, 1, 1]
 Generation 5420: Best = 2.925519e+05 FES = 542100 μ_F = 0.770 μ_{CR} = 0.053
 Diversity = 3.9282e+00 k_t = [4, 3, 2, 2]
 Generation 5430: Best = 2.924490e+05 FES = 543100 μ_F = 0.776 μ_{CR} = 0.056
 Diversity = 4.3903e+00 k_t = [1, 1, 1, 1]
 Generation 5440: Best = 2.921231e+05 FES = 544100 μ_F = 0.784 μ_{CR} = 0.058
 Diversity = 4.5018e+00 k_t = [3, 3, 3, 2]
 Generation 5450: Best = 2.918769e+05 FES = 545100 μ_F = 0.781 μ_{CR} = 0.051
 Diversity = 4.8843e+00 k_t = [1, 1, 1, 1]
 Generation 5460: Best = 2.917664e+05 FES = 546100 μ_F = 0.779 μ_{CR} = 0.055
 Diversity = 4.5982e+00 k_t = [2, 4, 2, 4]
 Generation 5470: Best = 2.915797e+05 FES = 547100 μ_F = 0.776 μ_{CR} = 0.050
 Diversity = 4.1242e+00 k_t = [1, 1, 1, 1]
 Generation 5480: Best = 2.914302e+05 FES = 548100 μ_F = 0.780 μ_{CR} = 0.053
 Diversity = 3.5795e+00 k_t = [4, 2, 4, 4]
 Generation 5490: Best = 2.913160e+05 FES = 549100 μ_F = 0.799 μ_{CR} = 0.048
 Diversity = 3.4434e+00 k_t = [1, 1, 1, 1]
 Generation 5500: Best = 2.912285e+05 FES = 550100 μ_F = 0.801 μ_{CR} = 0.055
 Diversity = 3.4654e+00 k_t = [4, 2, 4, 2]
 Generation 5510: Best = 2.909407e+05 FES = 551100 μ_F = 0.805 μ_{CR} = 0.056
 Diversity = 3.9583e+00 k_t = [1, 1, 1, 1]
 Generation 5520: Best = 2.907571e+05 FES = 552100 μ_F = 0.800 μ_{CR} = 0.046
 Diversity = 4.1603e+00 k_t = [3, 2, 2, 3]
 Generation 5530: Best = 2.905924e+05 FES = 553100 μ_F = 0.802 μ_{CR} = 0.044
 Diversity = 4.6316e+00 k_t = [1, 1, 1, 1]
 Generation 5540: Best = 2.905193e+05 FES = 554100 μ_F = 0.785 μ_{CR} = 0.040
 Diversity = 4.4787e+00 k_t = [3, 4, 2, 4]
 Generation 5550: Best = 2.904032e+05 FES = 555100 μ_F = 0.794 μ_{CR} = 0.042
 Diversity = 4.5328e+00 k_t = [1, 1, 1, 1]
 Generation 5560: Best = 2.902742e+05 FES = 556100 μ_F = 0.793 μ_{CR} = 0.044
 Diversity = 4.3608e+00 k_t = [4, 2, 2, 3]
 Generation 5570: Best = 2.901473e+05 FES = 557100 μ_F = 0.801 μ_{CR} = 0.038
 Diversity = 4.8791e+00 k_t = [1, 1, 1, 1]
 Generation 5580: Best = 2.898121e+05 FES = 558100 μ_F = 0.794 μ_{CR} = 0.042
 Diversity = 5.0754e+00 k_t = [4, 3, 2, 3]
 Generation 5590: Best = 2.897499e+05 FES = 559100 μ_F = 0.815 μ_{CR} = 0.043
 Diversity = 5.2212e+00 k_t = [1, 1, 1, 1]
 Generation 5600: Best = 2.896076e+05 FES = 560100 μ_F = 0.817 μ_{CR} = 0.043
 Diversity = 4.7766e+00 k_t = [3, 3, 3, 3]
 Generation 5610: Best = 2.894839e+05 FES = 561100 μ_F = 0.818 μ_{CR} = 0.042
 Diversity = 4.6476e+00 k_t = [1, 1, 1, 1]
 Generation 5620: Best = 2.893416e+05 FES = 562100 μ_F = 0.803 μ_{CR} = 0.039
 Diversity = 4.8899e+00 k_t = [2, 4, 3, 3]

Generation 5630: Best = 2.892647e+05 FES = 563100 μ_F = 0.803 μ_{CR} = 0.036
 Diversity = 4.5033e+00 k_t = [1, 1, 1, 1]
 Generation 5640: Best = 2.891330e+05 FES = 564100 μ_F = 0.791 μ_{CR} = 0.036
 Diversity = 4.2383e+00 k_t = [2, 2, 3, 3]
 Generation 5650: Best = 2.890728e+05 FES = 565100 μ_F = 0.797 μ_{CR} = 0.039
 Diversity = 3.9641e+00 k_t = [1, 1, 1, 1]
 Generation 5660: Best = 2.889171e+05 FES = 566100 μ_F = 0.799 μ_{CR} = 0.043
 Diversity = 3.7728e+00 k_t = [2, 3, 3, 4]
 Generation 5670: Best = 2.887797e+05 FES = 567100 μ_F = 0.815 μ_{CR} = 0.052
 Diversity = 3.7920e+00 k_t = [1, 1, 1, 1]
 Generation 5680: Best = 2.885637e+05 FES = 568100 μ_F = 0.806 μ_{CR} = 0.044
 Diversity = 3.8395e+00 k_t = [2, 4, 2, 3]
 Generation 5690: Best = 2.885262e+05 FES = 569100 μ_F = 0.797 μ_{CR} = 0.039
 Diversity = 4.1824e+00 k_t = [1, 1, 1, 1]
 Generation 5700: Best = 2.884123e+05 FES = 570100 μ_F = 0.813 μ_{CR} = 0.037
 Diversity = 4.5977e+00 k_t = [2, 2, 3, 4]
 Generation 5710: Best = 2.881432e+05 FES = 571100 μ_F = 0.794 μ_{CR} = 0.044
 Diversity = 4.3757e+00 k_t = [1, 1, 1, 1]
 Generation 5720: Best = 2.880235e+05 FES = 572100 μ_F = 0.793 μ_{CR} = 0.048
 Diversity = 4.1067e+00 k_t = [3, 3, 3, 4]
 Generation 5730: Best = 2.877770e+05 FES = 573100 μ_F = 0.800 μ_{CR} = 0.051
 Diversity = 4.0076e+00 k_t = [1, 1, 1, 1]
 Generation 5740: Best = 2.877335e+05 FES = 574100 μ_F = 0.809 μ_{CR} = 0.042
 Diversity = 4.1696e+00 k_t = [2, 2, 3, 3]
 Generation 5750: Best = 2.876017e+05 FES = 575100 μ_F = 0.800 μ_{CR} = 0.046
 Diversity = 4.4799e+00 k_t = [1, 1, 1, 1]
 Generation 5760: Best = 2.874466e+05 FES = 576100 μ_F = 0.790 μ_{CR} = 0.057
 Diversity = 4.2070e+00 k_t = [3, 4, 4, 3]
 Generation 5770: Best = 2.874109e+05 FES = 577100 μ_F = 0.789 μ_{CR} = 0.047
 Diversity = 4.0994e+00 k_t = [1, 1, 1, 1]
 Generation 5780: Best = 2.871631e+05 FES = 578100 μ_F = 0.793 μ_{CR} = 0.052
 Diversity = 4.2714e+00 k_t = [3, 4, 3, 2]
 Generation 5790: Best = 2.871284e+05 FES = 579100 μ_F = 0.807 μ_{CR} = 0.058
 Diversity = 4.4211e+00 k_t = [1, 1, 1, 1]
 Generation 5800: Best = 2.869865e+05 FES = 580100 μ_F = 0.815 μ_{CR} = 0.043
 Diversity = 4.6628e+00 k_t = [3, 4, 4, 4]
 Generation 5810: Best = 2.867969e+05 FES = 581100 μ_F = 0.800 μ_{CR} = 0.039
 Diversity = 4.7163e+00 k_t = [1, 1, 1, 1]
 Generation 5820: Best = 2.865523e+05 FES = 582100 μ_F = 0.798 μ_{CR} = 0.052
 Diversity = 4.7074e+00 k_t = [4, 2, 3, 2]
 Generation 5830: Best = 2.865421e+05 FES = 583100 μ_F = 0.797 μ_{CR} = 0.042
 Diversity = 5.3539e+00 k_t = [1, 1, 1, 1]
 Generation 5840: Best = 2.864023e+05 FES = 584100 μ_F = 0.802 μ_{CR} = 0.037
 Diversity = 5.2713e+00 k_t = [3, 2, 3, 3]
 Generation 5850: Best = 2.862590e+05 FES = 585100 μ_F = 0.805 μ_{CR} = 0.035
 Diversity = 5.4747e+00 k_t = [1, 1, 1, 1]
 Generation 5860: Best = 2.860743e+05 FES = 586100 μ_F = 0.814 μ_{CR} = 0.039
 Diversity = 5.4914e+00 k_t = [4, 4, 3, 4]
 Generation 5870: Best = 2.859864e+05 FES = 587100 μ_F = 0.815 μ_{CR} = 0.040
 Diversity = 5.1673e+00 k_t = [1, 1, 1, 1]
 Generation 5880: Best = 2.857849e+05 FES = 588100 μ_F = 0.814 μ_{CR} = 0.040
 Diversity = 5.2586e+00 k_t = [2, 2, 2, 2]
 Generation 5890: Best = 2.856621e+05 FES = 589100 μ_F = 0.803 μ_{CR} = 0.046
 Diversity = 5.0390e+00 k_t = [1, 1, 1, 1]

Generation 5900: Best = 2.855600e+05 FES = 590100 μ_F = 0.807 μ_{CR} = 0.049
 Diversity = 5.1707e+00 k_t = [2, 4, 3, 2]
 Generation 5910: Best = 2.853175e+05 FES = 591100 μ_F = 0.790 μ_{CR} = 0.050
 Diversity = 5.0271e+00 k_t = [1, 1, 1, 1]
 Generation 5920: Best = 2.851541e+05 FES = 592100 μ_F = 0.803 μ_{CR} = 0.046
 Diversity = 5.0244e+00 k_t = [4, 4, 3, 4]
 Generation 5930: Best = 2.849855e+05 FES = 593100 μ_F = 0.791 μ_{CR} = 0.045
 Diversity = 5.0936e+00 k_t = [1, 1, 1, 1]
 Generation 5940: Best = 2.848965e+05 FES = 594100 μ_F = 0.803 μ_{CR} = 0.042
 Diversity = 4.7316e+00 k_t = [3, 4, 2, 4]
 Generation 5950: Best = 2.847532e+05 FES = 595100 μ_F = 0.799 μ_{CR} = 0.051
 Diversity = 4.4931e+00 k_t = [1, 1, 1, 1]
 Generation 5960: Best = 2.846844e+05 FES = 596100 μ_F = 0.803 μ_{CR} = 0.043
 Diversity = 4.6058e+00 k_t = [4, 4, 3, 4]
 Generation 5970: Best = 2.845826e+05 FES = 597100 μ_F = 0.814 μ_{CR} = 0.044
 Diversity = 4.0903e+00 k_t = [1, 1, 1, 1]
 Generation 5980: Best = 2.844817e+05 FES = 598100 μ_F = 0.800 μ_{CR} = 0.046
 Diversity = 4.2020e+00 k_t = [3, 3, 3, 3]
 Generation 5990: Best = 2.843931e+05 FES = 599100 μ_F = 0.819 μ_{CR} = 0.037
 Diversity = 4.1422e+00 k_t = [1, 1, 1, 1]
 Generation 6000: Best = 2.843122e+05 FES = 600100 μ_F = 0.810 μ_{CR} = 0.040
 Diversity = 4.3584e+00 k_t = [3, 4, 4, 2]
 Generation 6010: Best = 2.841773e+05 FES = 601100 μ_F = 0.802 μ_{CR} = 0.043
 Diversity = 4.6905e+00 k_t = [1, 1, 1, 1]
 Generation 6020: Best = 2.840679e+05 FES = 602100 μ_F = 0.795 μ_{CR} = 0.041
 Diversity = 4.5611e+00 k_t = [3, 2, 3, 2]
 Generation 6030: Best = 2.839705e+05 FES = 603100 μ_F = 0.791 μ_{CR} = 0.041
 Diversity = 4.8732e+00 k_t = [1, 1, 1, 1]
 Generation 6040: Best = 2.838666e+05 FES = 604100 μ_F = 0.804 μ_{CR} = 0.044
 Diversity = 5.0412e+00 k_t = [4, 4, 2, 2]
 Generation 6050: Best = 2.837789e+05 FES = 605100 μ_F = 0.801 μ_{CR} = 0.052
 Diversity = 5.3774e+00 k_t = [1, 1, 1, 1]
 Generation 6060: Best = 2.836592e+05 FES = 606100 μ_F = 0.805 μ_{CR} = 0.039
 Diversity = 5.1694e+00 k_t = [3, 3, 2, 4]
 Generation 6070: Best = 2.834667e+05 FES = 607100 μ_F = 0.789 μ_{CR} = 0.044
 Diversity = 4.9502e+00 k_t = [1, 1, 1, 1]
 Generation 6080: Best = 2.833909e+05 FES = 608100 μ_F = 0.788 μ_{CR} = 0.036
 Diversity = 4.8762e+00 k_t = [2, 2, 2, 3]
 Generation 6090: Best = 2.832067e+05 FES = 609100 μ_F = 0.790 μ_{CR} = 0.043
 Diversity = 4.6030e+00 k_t = [1, 1, 1, 1]
 Generation 6100: Best = 2.831487e+05 FES = 610100 μ_F = 0.802 μ_{CR} = 0.050
 Diversity = 4.7347e+00 k_t = [4, 4, 4, 3]
 Generation 6110: Best = 2.829733e+05 FES = 611100 μ_F = 0.800 μ_{CR} = 0.045
 Diversity = 4.5940e+00 k_t = [1, 1, 1, 1]
 Generation 6120: Best = 2.826578e+05 FES = 612100 μ_F = 0.799 μ_{CR} = 0.053
 Diversity = 4.5467e+00 k_t = [2, 3, 3, 2]
 Generation 6130: Best = 2.823452e+05 FES = 613100 μ_F = 0.805 μ_{CR} = 0.060
 Diversity = 5.0256e+00 k_t = [1, 1, 1, 1]
 Generation 6140: Best = 2.820755e+05 FES = 614100 μ_F = 0.818 μ_{CR} = 0.058
 Diversity = 5.5990e+00 k_t = [2, 4, 4, 2]
 Generation 6150: Best = 2.816410e+05 FES = 615100 μ_F = 0.824 μ_{CR} = 0.046
 Diversity = 6.6019e+00 k_t = [1, 1, 1, 1]
 Generation 6160: Best = 2.815470e+05 FES = 616100 μ_F = 0.824 μ_{CR} = 0.052
 Diversity = 6.1650e+00 k_t = [4, 3, 4, 3]

Generation 6170: Best = 2.813460e+05 FES = 617100 μ_F = 0.817 μ_{CR} = 0.053
 Diversity = 6.1036e+00 k_t = [1, 1, 1, 1]
 Generation 6180: Best = 2.811037e+05 FES = 618100 μ_F = 0.814 μ_{CR} = 0.047
 Diversity = 5.9767e+00 k_t = [4, 2, 3, 4]
 Generation 6190: Best = 2.809298e+05 FES = 619100 μ_F = 0.808 μ_{CR} = 0.045
 Diversity = 5.8941e+00 k_t = [1, 1, 1, 1]
 Generation 6200: Best = 2.805645e+05 FES = 620100 μ_F = 0.802 μ_{CR} = 0.048
 Diversity = 5.3420e+00 k_t = [2, 3, 2, 4]
 Generation 6210: Best = 2.805504e+05 FES = 621100 μ_F = 0.804 μ_{CR} = 0.042
 Diversity = 5.3375e+00 k_t = [1, 1, 1, 1]
 Generation 6220: Best = 2.803873e+05 FES = 622100 μ_F = 0.795 μ_{CR} = 0.043
 Diversity = 5.2017e+00 k_t = [4, 3, 4, 3]
 Generation 6230: Best = 2.801316e+05 FES = 623100 μ_F = 0.787 μ_{CR} = 0.049
 Diversity = 5.0440e+00 k_t = [1, 1, 1, 1]
 Generation 6240: Best = 2.800313e+05 FES = 624100 μ_F = 0.787 μ_{CR} = 0.056
 Diversity = 5.0858e+00 k_t = [4, 2, 4, 3]
 Generation 6250: Best = 2.798908e+05 FES = 625100 μ_F = 0.792 μ_{CR} = 0.051
 Diversity = 5.1216e+00 k_t = [1, 1, 1, 1]
 Generation 6260: Best = 2.797762e+05 FES = 626100 μ_F = 0.797 μ_{CR} = 0.042
 Diversity = 5.3106e+00 k_t = [2, 3, 3, 2]
 Generation 6270: Best = 2.795417e+05 FES = 627100 μ_F = 0.803 μ_{CR} = 0.042
 Diversity = 5.7462e+00 k_t = [1, 1, 1, 1]
 Generation 6280: Best = 2.794643e+05 FES = 628100 μ_F = 0.800 μ_{CR} = 0.039
 Diversity = 6.2539e+00 k_t = [2, 4, 4, 4]
 Generation 6290: Best = 2.791307e+05 FES = 629100 μ_F = 0.798 μ_{CR} = 0.039
 Diversity = 5.3617e+00 k_t = [1, 1, 1, 1]
 Generation 6300: Best = 2.790213e+05 FES = 630100 μ_F = 0.792 μ_{CR} = 0.041
 Diversity = 5.4721e+00 k_t = [2, 4, 4, 2]
 Generation 6310: Best = 2.789685e+05 FES = 631100 μ_F = 0.804 μ_{CR} = 0.037
 Diversity = 5.1406e+00 k_t = [1, 1, 1, 1]
 Generation 6320: Best = 2.787763e+05 FES = 632100 μ_F = 0.804 μ_{CR} = 0.045
 Diversity = 5.1582e+00 k_t = [2, 3, 2, 2]
 Generation 6330: Best = 2.787053e+05 FES = 633100 μ_F = 0.811 μ_{CR} = 0.041
 Diversity = 5.4300e+00 k_t = [1, 1, 1, 1]
 Generation 6340: Best = 2.785088e+05 FES = 634100 μ_F = 0.812 μ_{CR} = 0.045
 Diversity = 5.5761e+00 k_t = [2, 3, 2, 3]
 Generation 6350: Best = 2.782737e+05 FES = 635100 μ_F = 0.806 μ_{CR} = 0.043
 Diversity = 6.1064e+00 k_t = [1, 1, 1, 1]
 Generation 6360: Best = 2.781229e+05 FES = 636100 μ_F = 0.805 μ_{CR} = 0.045
 Diversity = 6.2529e+00 k_t = [4, 4, 2, 4]
 Generation 6370: Best = 2.778881e+05 FES = 637100 μ_F = 0.806 μ_{CR} = 0.036
 Diversity = 5.9340e+00 k_t = [1, 1, 1, 1]
 Generation 6380: Best = 2.776619e+05 FES = 638100 μ_F = 0.812 μ_{CR} = 0.044
 Diversity = 6.7414e+00 k_t = [2, 4, 3, 2]
 Generation 6390: Best = 2.774185e+05 FES = 639100 μ_F = 0.802 μ_{CR} = 0.046
 Diversity = 6.3569e+00 k_t = [1, 1, 1, 1]
 Generation 6400: Best = 2.772624e+05 FES = 640100 μ_F = 0.805 μ_{CR} = 0.051
 Diversity = 6.0881e+00 k_t = [4, 3, 4, 4]
 Generation 6410: Best = 2.770543e+05 FES = 641100 μ_F = 0.809 μ_{CR} = 0.054
 Diversity = 5.7263e+00 k_t = [1, 1, 1, 1]
 Generation 6420: Best = 2.769010e+05 FES = 642100 μ_F = 0.804 μ_{CR} = 0.043
 Diversity = 5.4163e+00 k_t = [4, 4, 3, 3]
 Generation 6430: Best = 2.765595e+05 FES = 643100 μ_F = 0.797 μ_{CR} = 0.043
 Diversity = 5.3768e+00 k_t = [1, 1, 1, 1]

Generation 6440: Best = 2.762779e+05 FES = 644100 μ_F = 0.803 μ_{CR} = 0.040
 Diversity = 5.0423e+00 k_t = [4, 3, 3, 4]
 Generation 6450: Best = 2.759727e+05 FES = 645100 μ_F = 0.804 μ_{CR} = 0.044
 Diversity = 4.6105e+00 k_t = [1, 1, 1, 1]
 Generation 6460: Best = 2.758765e+05 FES = 646100 μ_F = 0.805 μ_{CR} = 0.049
 Diversity = 5.2882e+00 k_t = [3, 2, 3, 3]
 Generation 6470: Best = 2.755936e+05 FES = 647100 μ_F = 0.799 μ_{CR} = 0.042
 Diversity = 5.6904e+00 k_t = [1, 1, 1, 1]
 Generation 6480: Best = 2.754837e+05 FES = 648100 μ_F = 0.795 μ_{CR} = 0.045
 Diversity = 5.4351e+00 k_t = [4, 4, 2, 2]
 Generation 6490: Best = 2.753398e+05 FES = 649100 μ_F = 0.811 μ_{CR} = 0.041
 Diversity = 5.6060e+00 k_t = [1, 1, 1, 1]
 Generation 6500: Best = 2.751269e+05 FES = 650100 μ_F = 0.811 μ_{CR} = 0.047
 Diversity = 5.5363e+00 k_t = [3, 4, 3, 3]
 Generation 6510: Best = 2.748964e+05 FES = 651100 μ_F = 0.810 μ_{CR} = 0.049
 Diversity = 6.1231e+00 k_t = [1, 1, 1, 1]
 Generation 6520: Best = 2.746397e+05 FES = 652100 μ_F = 0.808 μ_{CR} = 0.045
 Diversity = 6.0069e+00 k_t = [4, 4, 4, 2]
 Generation 6530: Best = 2.745062e+05 FES = 653100 μ_F = 0.806 μ_{CR} = 0.052
 Diversity = 6.3749e+00 k_t = [1, 1, 1, 1]
 Generation 6540: Best = 2.742537e+05 FES = 654100 μ_F = 0.794 μ_{CR} = 0.046
 Diversity = 5.3364e+00 k_t = [3, 3, 2, 3]
 Generation 6550: Best = 2.740123e+05 FES = 655100 μ_F = 0.790 μ_{CR} = 0.040
 Diversity = 5.5036e+00 k_t = [1, 1, 1, 1]
 Generation 6560: Best = 2.737801e+05 FES = 656100 μ_F = 0.792 μ_{CR} = 0.044
 Diversity = 5.1522e+00 k_t = [4, 4, 3, 3]
 Generation 6570: Best = 2.736608e+05 FES = 657100 μ_F = 0.804 μ_{CR} = 0.039
 Diversity = 5.2715e+00 k_t = [1, 1, 1, 1]
 Generation 6580: Best = 2.734647e+05 FES = 658100 μ_F = 0.801 μ_{CR} = 0.043
 Diversity = 5.1025e+00 k_t = [4, 4, 2, 4]
 Generation 6590: Best = 2.732562e+05 FES = 659100 μ_F = 0.798 μ_{CR} = 0.044
 Diversity = 4.6084e+00 k_t = [1, 1, 1, 1]
 Generation 6600: Best = 2.731239e+05 FES = 660100 μ_F = 0.792 μ_{CR} = 0.043
 Diversity = 4.0760e+00 k_t = [4, 2, 4, 3]
 Generation 6610: Best = 2.730362e+05 FES = 661100 μ_F = 0.801 μ_{CR} = 0.042
 Diversity = 4.0535e+00 k_t = [1, 1, 1, 1]
 Generation 6620: Best = 2.728917e+05 FES = 662100 μ_F = 0.796 μ_{CR} = 0.034
 Diversity = 3.9278e+00 k_t = [2, 2, 2, 2]
 Generation 6630: Best = 2.728195e+05 FES = 663100 μ_F = 0.779 μ_{CR} = 0.036
 Diversity = 4.0407e+00 k_t = [1, 1, 1, 1]
 Generation 6640: Best = 2.726310e+05 FES = 664100 μ_F = 0.792 μ_{CR} = 0.044
 Diversity = 3.9761e+00 k_t = [4, 2, 3, 4]
 Generation 6650: Best = 2.725314e+05 FES = 665100 μ_F = 0.795 μ_{CR} = 0.047
 Diversity = 3.7740e+00 k_t = [1, 1, 1, 1]
 Generation 6660: Best = 2.723518e+05 FES = 666100 μ_F = 0.807 μ_{CR} = 0.045
 Diversity = 3.9179e+00 k_t = [4, 3, 2, 2]
 Generation 6670: Best = 2.722692e+05 FES = 667100 μ_F = 0.813 μ_{CR} = 0.045
 Diversity = 4.1215e+00 k_t = [1, 1, 1, 1]
 Generation 6680: Best = 2.721115e+05 FES = 668100 μ_F = 0.820 μ_{CR} = 0.046
 Diversity = 4.7970e+00 k_t = [3, 4, 4, 2]
 Generation 6690: Best = 2.719330e+05 FES = 669100 μ_F = 0.828 μ_{CR} = 0.040
 Diversity = 4.8405e+00 k_t = [1, 1, 1, 1]
 Generation 6700: Best = 2.718474e+05 FES = 670100 μ_F = 0.818 μ_{CR} = 0.043
 Diversity = 4.4011e+00 k_t = [3, 3, 4, 3]

Generation 6710: Best = 2.716602e+05 FES = 671100 μ_F = 0.810 μ_{CR} = 0.038
 Diversity = 4.5022e+00 k_t = [1, 1, 1, 1]
 Generation 6720: Best = 2.715000e+05 FES = 672100 μ_F = 0.818 μ_{CR} = 0.038
 Diversity = 4.5308e+00 k_t = [3, 4, 2, 4]
 Generation 6730: Best = 2.714151e+05 FES = 673100 μ_F = 0.816 μ_{CR} = 0.046
 Diversity = 4.0223e+00 k_t = [1, 1, 1, 1]
 Generation 6740: Best = 2.713171e+05 FES = 674100 μ_F = 0.811 μ_{CR} = 0.044
 Diversity = 3.8776e+00 k_t = [2, 3, 4, 4]
 Generation 6750: Best = 2.711572e+05 FES = 675100 μ_F = 0.817 μ_{CR} = 0.048
 Diversity = 3.5650e+00 k_t = [1, 1, 1, 1]
 Generation 6760: Best = 2.709335e+05 FES = 676100 μ_F = 0.822 μ_{CR} = 0.044
 Diversity = 3.4797e+00 k_t = [4, 4, 4, 3]
 Generation 6770: Best = 2.707608e+05 FES = 677100 μ_F = 0.826 μ_{CR} = 0.048
 Diversity = 3.4188e+00 k_t = [1, 1, 1, 1]
 Generation 6780: Best = 2.707348e+05 FES = 678100 μ_F = 0.807 μ_{CR} = 0.052
 Diversity = 3.4621e+00 k_t = [2, 3, 3, 4]
 Generation 6790: Best = 2.705555e+05 FES = 679100 μ_F = 0.798 μ_{CR} = 0.049
 Diversity = 3.2844e+00 k_t = [1, 1, 1, 1]
 Generation 6800: Best = 2.705069e+05 FES = 680100 μ_F = 0.805 μ_{CR} = 0.049
 Diversity = 3.2819e+00 k_t = [3, 3, 3, 3]
 Generation 6810: Best = 2.703300e+05 FES = 681100 μ_F = 0.804 μ_{CR} = 0.040
 Diversity = 3.4582e+00 k_t = [1, 1, 1, 1]
 Generation 6820: Best = 2.701802e+05 FES = 682100 μ_F = 0.816 μ_{CR} = 0.044
 Diversity = 3.4582e+00 k_t = [4, 4, 2, 3]
 Generation 6830: Best = 2.700420e+05 FES = 683100 μ_F = 0.828 μ_{CR} = 0.035
 Diversity = 3.9363e+00 k_t = [1, 1, 1, 1]
 Generation 6840: Best = 2.698548e+05 FES = 684100 μ_F = 0.820 μ_{CR} = 0.039
 Diversity = 4.3350e+00 k_t = [2, 2, 3, 2]
 Generation 6850: Best = 2.697614e+05 FES = 685100 μ_F = 0.824 μ_{CR} = 0.038
 Diversity = 4.5334e+00 k_t = [1, 1, 1, 1]
 Generation 6860: Best = 2.696496e+05 FES = 686100 μ_F = 0.822 μ_{CR} = 0.041
 Diversity = 4.3582e+00 k_t = [4, 3, 4, 3]
 Generation 6870: Best = 2.694496e+05 FES = 687100 μ_F = 0.801 μ_{CR} = 0.055
 Diversity = 4.4775e+00 k_t = [1, 1, 1, 1]
 Generation 6880: Best = 2.693306e+05 FES = 688100 μ_F = 0.797 μ_{CR} = 0.051
 Diversity = 4.6673e+00 k_t = [2, 3, 2, 3]
 Generation 6890: Best = 2.691662e+05 FES = 689100 μ_F = 0.813 μ_{CR} = 0.041
 Diversity = 4.8901e+00 k_t = [1, 1, 1, 1]
 Generation 6900: Best = 2.690551e+05 FES = 690100 μ_F = 0.809 μ_{CR} = 0.045
 Diversity = 4.6967e+00 k_t = [2, 4, 4, 2]
 Generation 6910: Best = 2.689796e+05 FES = 691100 μ_F = 0.807 μ_{CR} = 0.037
 Diversity = 4.6098e+00 k_t = [1, 1, 1, 1]
 Generation 6920: Best = 2.687958e+05 FES = 692100 μ_F = 0.800 μ_{CR} = 0.047
 Diversity = 4.7588e+00 k_t = [4, 2, 2, 4]
 Generation 6930: Best = 2.686655e+05 FES = 693100 μ_F = 0.802 μ_{CR} = 0.044
 Diversity = 4.8012e+00 k_t = [1, 1, 1, 1]
 Generation 6940: Best = 2.685173e+05 FES = 694100 μ_F = 0.822 μ_{CR} = 0.041
 Diversity = 4.8839e+00 k_t = [4, 4, 2, 4]
 Generation 6950: Best = 2.683939e+05 FES = 695100 μ_F = 0.805 μ_{CR} = 0.044
 Diversity = 4.5767e+00 k_t = [1, 1, 1, 1]
 Generation 6960: Best = 2.682274e+05 FES = 696100 μ_F = 0.806 μ_{CR} = 0.049
 Diversity = 4.3502e+00 k_t = [4, 2, 2, 3]
 Generation 6970: Best = 2.680059e+05 FES = 697100 μ_F = 0.805 μ_{CR} = 0.038
 Diversity = 4.4492e+00 k_t = [1, 1, 1, 1]

Generation 6980: Best = 2.679352e+05 FES = 698100 μ_F = 0.797 μ_{CR} = 0.038
 Diversity = 4.3766e+00 k_t = [3, 4, 2, 2]
 Generation 6990: Best = 2.678343e+05 FES = 699100 μ_F = 0.795 μ_{CR} = 0.039
 Diversity = 4.5782e+00 k_t = [1, 1, 1, 1]
 Generation 7000: Best = 2.676761e+05 FES = 700100 μ_F = 0.796 μ_{CR} = 0.039
 Diversity = 4.4528e+00 k_t = [3, 4, 3, 2]
 Generation 7010: Best = 2.674932e+05 FES = 701100 μ_F = 0.787 μ_{CR} = 0.050
 Diversity = 4.5348e+00 k_t = [1, 1, 1, 1]
 Generation 7020: Best = 2.674139e+05 FES = 702100 μ_F = 0.793 μ_{CR} = 0.039
 Diversity = 4.5754e+00 k_t = [2, 3, 3, 3]
 Generation 7030: Best = 2.673219e+05 FES = 703100 μ_F = 0.795 μ_{CR} = 0.045
 Diversity = 4.3336e+00 k_t = [1, 1, 1, 1]
 Generation 7040: Best = 2.671299e+05 FES = 704100 μ_F = 0.793 μ_{CR} = 0.037
 Diversity = 3.9719e+00 k_t = [3, 3, 3, 4]
 Generation 7050: Best = 2.669718e+05 FES = 705100 μ_F = 0.807 μ_{CR} = 0.041
 Diversity = 4.2492e+00 k_t = [1, 1, 1, 1]
 Generation 7060: Best = 2.668761e+05 FES = 706100 μ_F = 0.814 μ_{CR} = 0.040
 Diversity = 4.6495e+00 k_t = [3, 4, 4, 2]
 Generation 7070: Best = 2.665819e+05 FES = 707100 μ_F = 0.818 μ_{CR} = 0.042
 Diversity = 5.0187e+00 k_t = [1, 1, 1, 1]
 Generation 7080: Best = 2.662857e+05 FES = 708100 μ_F = 0.813 μ_{CR} = 0.050
 Diversity = 5.2348e+00 k_t = [2, 4, 4, 3]
 Generation 7090: Best = 2.661891e+05 FES = 709100 μ_F = 0.801 μ_{CR} = 0.047
 Diversity = 4.7854e+00 k_t = [1, 1, 1, 1]
 Generation 7100: Best = 2.660113e+05 FES = 710100 μ_F = 0.783 μ_{CR} = 0.036
 Diversity = 4.7024e+00 k_t = [4, 3, 2, 4]
 Generation 7110: Best = 2.656251e+05 FES = 711100 μ_F = 0.790 μ_{CR} = 0.048
 Diversity = 4.6630e+00 k_t = [1, 1, 1, 1]
 Generation 7120: Best = 2.653040e+05 FES = 712100 μ_F = 0.812 μ_{CR} = 0.050
 Diversity = 4.4197e+00 k_t = [3, 3, 2, 4]
 Generation 7130: Best = 2.651243e+05 FES = 713100 μ_F = 0.811 μ_{CR} = 0.041
 Diversity = 4.3880e+00 k_t = [1, 1, 1, 1]
 Generation 7140: Best = 2.651192e+05 FES = 714100 μ_F = 0.814 μ_{CR} = 0.037
 Diversity = 4.6834e+00 k_t = [2, 3, 4, 4]
 Generation 7150: Best = 2.649310e+05 FES = 715100 μ_F = 0.811 μ_{CR} = 0.038
 Diversity = 4.3740e+00 k_t = [1, 1, 1, 1]
 Generation 7160: Best = 2.648212e+05 FES = 716100 μ_F = 0.801 μ_{CR} = 0.043
 Diversity = 4.1359e+00 k_t = [4, 4, 2, 2]
 Generation 7170: Best = 2.647154e+05 FES = 717100 μ_F = 0.802 μ_{CR} = 0.051
 Diversity = 4.2904e+00 k_t = [1, 1, 1, 1]
 Generation 7180: Best = 2.644526e+05 FES = 718100 μ_F = 0.798 μ_{CR} = 0.048
 Diversity = 3.9289e+00 k_t = [2, 3, 2, 2]
 Generation 7190: Best = 2.643685e+05 FES = 719100 μ_F = 0.786 μ_{CR} = 0.044
 Diversity = 4.0903e+00 k_t = [1, 1, 1, 1]
 Generation 7200: Best = 2.642409e+05 FES = 720100 μ_F = 0.795 μ_{CR} = 0.045
 Diversity = 4.0030e+00 k_t = [2, 4, 3, 4]
 Generation 7210: Best = 2.640476e+05 FES = 721100 μ_F = 0.803 μ_{CR} = 0.058
 Diversity = 3.6820e+00 k_t = [1, 1, 1, 1]
 Generation 7220: Best = 2.638796e+05 FES = 722100 μ_F = 0.810 μ_{CR} = 0.059
 Diversity = 4.3380e+00 k_t = [3, 2, 4, 2]
 Generation 7230: Best = 2.637140e+05 FES = 723100 μ_F = 0.801 μ_{CR} = 0.053
 Diversity = 4.6866e+00 k_t = [1, 1, 1, 1]
 Generation 7240: Best = 2.636311e+05 FES = 724100 μ_F = 0.797 μ_{CR} = 0.055
 Diversity = 4.5733e+00 k_t = [4, 4, 4, 4]

Generation 7250: Best = 2.635533e+05 FES = 725100 μ_F = 0.800 μ_{CR} = 0.042
 Diversity = 4.4047e+00 k_t = [1, 1, 1, 1]
 Generation 7260: Best = 2.633567e+05 FES = 726100 μ_F = 0.788 μ_{CR} = 0.037
 Diversity = 3.9813e+00 k_t = [3, 2, 2, 4]
 Generation 7270: Best = 2.632731e+05 FES = 727100 μ_F = 0.793 μ_{CR} = 0.040
 Diversity = 3.7296e+00 k_t = [1, 1, 1, 1]
 Generation 7280: Best = 2.630261e+05 FES = 728100 μ_F = 0.800 μ_{CR} = 0.039
 Diversity = 3.7538e+00 k_t = [2, 2, 2, 3]
 Generation 7290: Best = 2.629117e+05 FES = 729100 μ_F = 0.788 μ_{CR} = 0.040
 Diversity = 3.7948e+00 k_t = [1, 1, 1, 1]
 Generation 7300: Best = 2.627586e+05 FES = 730100 μ_F = 0.800 μ_{CR} = 0.041
 Diversity = 4.1208e+00 k_t = [3, 3, 3, 2]
 Generation 7310: Best = 2.622865e+05 FES = 731100 μ_F = 0.801 μ_{CR} = 0.041
 Diversity = 4.7243e+00 k_t = [1, 1, 1, 1]
 Generation 7320: Best = 2.622110e+05 FES = 732100 μ_F = 0.807 μ_{CR} = 0.052
 Diversity = 4.9117e+00 k_t = [4, 3, 2, 3]
 Generation 7330: Best = 2.621045e+05 FES = 733100 μ_F = 0.817 μ_{CR} = 0.051
 Diversity = 4.9277e+00 k_t = [1, 1, 1, 1]
 Generation 7340: Best = 2.619072e+05 FES = 734100 μ_F = 0.807 μ_{CR} = 0.051
 Diversity = 4.7154e+00 k_t = [2, 4, 2, 3]
 Generation 7350: Best = 2.617811e+05 FES = 735100 μ_F = 0.808 μ_{CR} = 0.047
 Diversity = 4.8627e+00 k_t = [1, 1, 1, 1]
 Generation 7360: Best = 2.616516e+05 FES = 736100 μ_F = 0.811 μ_{CR} = 0.053
 Diversity = 4.5103e+00 k_t = [2, 3, 4, 4]
 Generation 7370: Best = 2.614027e+05 FES = 737100 μ_F = 0.820 μ_{CR} = 0.056
 Diversity = 4.2141e+00 k_t = [1, 1, 1, 1]
 Generation 7380: Best = 2.613249e+05 FES = 738100 μ_F = 0.815 μ_{CR} = 0.048
 Diversity = 4.7462e+00 k_t = [2, 2, 4, 3]
 Generation 7390: Best = 2.612776e+05 FES = 739100 μ_F = 0.815 μ_{CR} = 0.046
 Diversity = 4.8577e+00 k_t = [1, 1, 1, 1]
 Generation 7400: Best = 2.611452e+05 FES = 740100 μ_F = 0.809 μ_{CR} = 0.040
 Diversity = 5.5617e+00 k_t = [4, 2, 2, 2]
 Generation 7410: Best = 2.610374e+05 FES = 741100 μ_F = 0.805 μ_{CR} = 0.039
 Diversity = 5.5643e+00 k_t = [1, 1, 1, 1]
 Generation 7420: Best = 2.609534e+05 FES = 742100 μ_F = 0.790 μ_{CR} = 0.039
 Diversity = 5.2592e+00 k_t = [2, 2, 2, 3]
 Generation 7430: Best = 2.608227e+05 FES = 743100 μ_F = 0.791 μ_{CR} = 0.042
 Diversity = 5.1426e+00 k_t = [1, 1, 1, 1]
 Generation 7440: Best = 2.607713e+05 FES = 744100 μ_F = 0.817 μ_{CR} = 0.037
 Diversity = 5.4905e+00 k_t = [2, 2, 3, 2]
 Generation 7450: Best = 2.606990e+05 FES = 745100 μ_F = 0.822 μ_{CR} = 0.036
 Diversity = 5.6859e+00 k_t = [1, 1, 1, 1]
 Generation 7460: Best = 2.604621e+05 FES = 746100 μ_F = 0.817 μ_{CR} = 0.044
 Diversity = 6.3822e+00 k_t = [3, 3, 4, 2]
 Generation 7470: Best = 2.603155e+05 FES = 747100 μ_F = 0.814 μ_{CR} = 0.038
 Diversity = 6.7029e+00 k_t = [1, 1, 1, 1]
 Generation 7480: Best = 2.602442e+05 FES = 748100 μ_F = 0.820 μ_{CR} = 0.032
 Diversity = 6.4118e+00 k_t = [2, 4, 3, 2]
 Generation 7490: Best = 2.599827e+05 FES = 749100 μ_F = 0.790 μ_{CR} = 0.035
 Diversity = 5.8355e+00 k_t = [1, 1, 1, 1]
 Generation 7500: Best = 2.598781e+05 FES = 750100 μ_F = 0.777 μ_{CR} = 0.034
 Diversity = 5.3280e+00 k_t = [2, 4, 3, 4]
 Generation 7510: Best = 2.598047e+05 FES = 751100 μ_F = 0.779 μ_{CR} = 0.038
 Diversity = 4.3426e+00 k_t = [1, 1, 1, 1]

Generation 7520: Best = 2.596778e+05 FES = 752100 μ_F = 0.785 μ_{CR} = 0.051
 Diversity = 4.2135e+00 k_t = [4, 2, 4, 3]
 Generation 7530: Best = 2.596222e+05 FES = 753100 μ_F = 0.797 μ_{CR} = 0.042
 Diversity = 4.1751e+00 k_t = [1, 1, 1, 1]
 Generation 7540: Best = 2.595146e+05 FES = 754100 μ_F = 0.798 μ_{CR} = 0.044
 Diversity = 4.1467e+00 k_t = [2, 3, 3, 2]
 Generation 7550: Best = 2.593836e+05 FES = 755100 μ_F = 0.790 μ_{CR} = 0.048
 Diversity = 4.2976e+00 k_t = [1, 1, 1, 1]
 Generation 7560: Best = 2.592256e+05 FES = 756100 μ_F = 0.809 μ_{CR} = 0.048
 Diversity = 4.6017e+00 k_t = [4, 3, 2, 4]
 Generation 7570: Best = 2.590858e+05 FES = 757100 μ_F = 0.807 μ_{CR} = 0.044
 Diversity = 4.5089e+00 k_t = [1, 1, 1, 1]
 Generation 7580: Best = 2.589175e+05 FES = 758100 μ_F = 0.785 μ_{CR} = 0.048
 Diversity = 4.3413e+00 k_t = [3, 3, 3, 3]
 Generation 7590: Best = 2.588062e+05 FES = 759100 μ_F = 0.793 μ_{CR} = 0.052
 Diversity = 4.2349e+00 k_t = [1, 1, 1, 1]
 Generation 7600: Best = 2.587097e+05 FES = 760100 μ_F = 0.796 μ_{CR} = 0.047
 Diversity = 4.0134e+00 k_t = [4, 2, 2, 2]
 Generation 7610: Best = 2.586171e+05 FES = 761100 μ_F = 0.797 μ_{CR} = 0.047
 Diversity = 4.1410e+00 k_t = [1, 1, 1, 1]
 Generation 7620: Best = 2.585576e+05 FES = 762100 μ_F = 0.773 μ_{CR} = 0.047
 Diversity = 3.9374e+00 k_t = [4, 3, 2, 2]
 Generation 7630: Best = 2.584751e+05 FES = 763100 μ_F = 0.770 μ_{CR} = 0.049
 Diversity = 3.8624e+00 k_t = [1, 1, 1, 1]
 Generation 7640: Best = 2.583453e+05 FES = 764100 μ_F = 0.782 μ_{CR} = 0.059
 Diversity = 3.9090e+00 k_t = [3, 3, 4, 3]
 Generation 7650: Best = 2.581965e+05 FES = 765100 μ_F = 0.795 μ_{CR} = 0.059
 Diversity = 4.2367e+00 k_t = [1, 1, 1, 1]
 Generation 7660: Best = 2.581402e+05 FES = 766100 μ_F = 0.792 μ_{CR} = 0.050
 Diversity = 3.8349e+00 k_t = [2, 2, 4, 4]
 Generation 7670: Best = 2.580072e+05 FES = 767100 μ_F = 0.808 μ_{CR} = 0.047
 Diversity = 3.5867e+00 k_t = [1, 1, 1, 1]
 Generation 7680: Best = 2.579633e+05 FES = 768100 μ_F = 0.821 μ_{CR} = 0.047
 Diversity = 3.9635e+00 k_t = [2, 3, 2, 2]
 Generation 7690: Best = 2.578535e+05 FES = 769100 μ_F = 0.822 μ_{CR} = 0.050
 Diversity = 4.0481e+00 k_t = [1, 1, 1, 1]
 Generation 7700: Best = 2.577395e+05 FES = 770100 μ_F = 0.806 μ_{CR} = 0.050
 Diversity = 4.0535e+00 k_t = [3, 3, 3, 2]
 Generation 7710: Best = 2.576194e+05 FES = 771100 μ_F = 0.810 μ_{CR} = 0.054
 Diversity = 4.1172e+00 k_t = [1, 1, 1, 1]
 Generation 7720: Best = 2.573366e+05 FES = 772100 μ_F = 0.814 μ_{CR} = 0.055
 Diversity = 4.5229e+00 k_t = [2, 3, 3, 4]
 Generation 7730: Best = 2.572604e+05 FES = 773100 μ_F = 0.834 μ_{CR} = 0.051
 Diversity = 4.5742e+00 k_t = [1, 1, 1, 1]
 Generation 7740: Best = 2.571410e+05 FES = 774100 μ_F = 0.829 μ_{CR} = 0.049
 Diversity = 5.0002e+00 k_t = [4, 2, 3, 4]
 Generation 7750: Best = 2.570334e+05 FES = 775100 μ_F = 0.824 μ_{CR} = 0.041
 Diversity = 4.7579e+00 k_t = [1, 1, 1, 1]
 Generation 7760: Best = 2.569845e+05 FES = 776100 μ_F = 0.821 μ_{CR} = 0.039
 Diversity = 4.6182e+00 k_t = [4, 3, 2, 3]
 Generation 7770: Best = 2.567698e+05 FES = 777100 μ_F = 0.810 μ_{CR} = 0.041
 Diversity = 4.6176e+00 k_t = [1, 1, 1, 1]
 Generation 7780: Best = 2.566831e+05 FES = 778100 μ_F = 0.805 μ_{CR} = 0.042
 Diversity = 5.3433e+00 k_t = [4, 2, 3, 2]

Generation 7790: Best = 2.565517e+05 FES = 779100 μ_F = 0.802 μ_{CR} = 0.044
 Diversity = 5.1824e+00 k_t = [1, 1, 1, 1]
 Generation 7800: Best = 2.564154e+05 FES = 780100 μ_F = 0.801 μ_{CR} = 0.037
 Diversity = 4.7971e+00 k_t = [3, 2, 3, 3]
 Generation 7810: Best = 2.563743e+05 FES = 781100 μ_F = 0.803 μ_{CR} = 0.050
 Diversity = 4.5745e+00 k_t = [1, 1, 1, 1]
 Generation 7820: Best = 2.563208e+05 FES = 782100 μ_F = 0.805 μ_{CR} = 0.053
 Diversity = 4.5400e+00 k_t = [4, 4, 3, 2]
 Generation 7830: Best = 2.561891e+05 FES = 783100 μ_F = 0.790 μ_{CR} = 0.049
 Diversity = 4.4605e+00 k_t = [1, 1, 1, 1]
 Generation 7840: Best = 2.560499e+05 FES = 784100 μ_F = 0.777 μ_{CR} = 0.048
 Diversity = 4.2280e+00 k_t = [4, 2, 3, 4]
 Generation 7850: Best = 2.559273e+05 FES = 785100 μ_F = 0.789 μ_{CR} = 0.053
 Diversity = 4.4530e+00 k_t = [1, 1, 1, 1]
 Generation 7860: Best = 2.558307e+05 FES = 786100 μ_F = 0.787 μ_{CR} = 0.053
 Diversity = 3.9454e+00 k_t = [2, 3, 2, 2]
 Generation 7870: Best = 2.557641e+05 FES = 787100 μ_F = 0.779 μ_{CR} = 0.056
 Diversity = 4.0020e+00 k_t = [1, 1, 1, 1]
 Generation 7880: Best = 2.556327e+05 FES = 788100 μ_F = 0.782 μ_{CR} = 0.050
 Diversity = 3.8834e+00 k_t = [4, 4, 4, 2]
 Generation 7890: Best = 2.556043e+05 FES = 789100 μ_F = 0.789 μ_{CR} = 0.048
 Diversity = 3.7329e+00 k_t = [1, 1, 1, 1]
 Generation 7900: Best = 2.555508e+05 FES = 790100 μ_F = 0.792 μ_{CR} = 0.053
 Diversity = 3.4203e+00 k_t = [4, 4, 2, 4]
 Generation 7910: Best = 2.554105e+05 FES = 791100 μ_F = 0.787 μ_{CR} = 0.057
 Diversity = 3.1952e+00 k_t = [1, 1, 1, 1]
 Generation 7920: Best = 2.553444e+05 FES = 792100 μ_F = 0.799 μ_{CR} = 0.051
 Diversity = 3.7615e+00 k_t = [4, 3, 3, 4]
 Generation 7930: Best = 2.552442e+05 FES = 793100 μ_F = 0.802 μ_{CR} = 0.053
 Diversity = 3.5268e+00 k_t = [1, 1, 1, 1]
 Generation 7940: Best = 2.551792e+05 FES = 794100 μ_F = 0.794 μ_{CR} = 0.053
 Diversity = 3.6415e+00 k_t = [4, 2, 4, 3]
 Generation 7950: Best = 2.549587e+05 FES = 795100 μ_F = 0.793 μ_{CR} = 0.060
 Diversity = 3.5970e+00 k_t = [1, 1, 1, 1]
 Generation 7960: Best = 2.549237e+05 FES = 796100 μ_F = 0.791 μ_{CR} = 0.058
 Diversity = 3.4070e+00 k_t = [4, 4, 4, 4]
 Generation 7970: Best = 2.548193e+05 FES = 797100 μ_F = 0.797 μ_{CR} = 0.055
 Diversity = 3.2601e+00 k_t = [1, 1, 1, 1]
 Generation 7980: Best = 2.546586e+05 FES = 798100 μ_F = 0.820 μ_{CR} = 0.055
 Diversity = 3.5611e+00 k_t = [4, 3, 4, 4]
 Generation 7990: Best = 2.545893e+05 FES = 799100 μ_F = 0.827 μ_{CR} = 0.053
 Diversity = 4.0072e+00 k_t = [1, 1, 1, 1]
 Generation 8000: Best = 2.544908e+05 FES = 800100 μ_F = 0.826 μ_{CR} = 0.047
 Diversity = 3.8815e+00 k_t = [2, 4, 2, 2]
 Generation 8010: Best = 2.543778e+05 FES = 801100 μ_F = 0.830 μ_{CR} = 0.048
 Diversity = 3.8216e+00 k_t = [1, 1, 1, 1]
 Generation 8020: Best = 2.542587e+05 FES = 802100 μ_F = 0.832 μ_{CR} = 0.052
 Diversity = 4.3368e+00 k_t = [2, 3, 2, 2]
 Generation 8030: Best = 2.540571e+05 FES = 803100 μ_F = 0.842 μ_{CR} = 0.049
 Diversity = 4.6868e+00 k_t = [1, 1, 1, 1]
 Generation 8040: Best = 2.539727e+05 FES = 804100 μ_F = 0.834 μ_{CR} = 0.044
 Diversity = 4.8951e+00 k_t = [4, 2, 3, 4]
 Generation 8050: Best = 2.538114e+05 FES = 805100 μ_F = 0.823 μ_{CR} = 0.043
 Diversity = 4.8727e+00 k_t = [1, 1, 1, 1]

Generation 8060: Best = 2.536948e+05 FES = 806100 μ_F = 0.823 μ_{CR} = 0.036
 Diversity = 4.6162e+00 k_t = [3, 2, 3, 3]
 Generation 8070: Best = 2.536361e+05 FES = 807100 μ_F = 0.805 μ_{CR} = 0.045
 Diversity = 4.4870e+00 k_t = [1, 1, 1, 1]
 Generation 8080: Best = 2.533879e+05 FES = 808100 μ_F = 0.806 μ_{CR} = 0.040
 Diversity = 4.3990e+00 k_t = [2, 3, 4, 4]
 Generation 8090: Best = 2.533038e+05 FES = 809100 μ_F = 0.817 μ_{CR} = 0.042
 Diversity = 4.3137e+00 k_t = [1, 1, 1, 1]
 Generation 8100: Best = 2.532284e+05 FES = 810100 μ_F = 0.797 μ_{CR} = 0.046
 Diversity = 4.1402e+00 k_t = [3, 2, 3, 3]
 Generation 8110: Best = 2.531244e+05 FES = 811100 μ_F = 0.796 μ_{CR} = 0.044
 Diversity = 3.7992e+00 k_t = [1, 1, 1, 1]
 Generation 8120: Best = 2.530254e+05 FES = 812100 μ_F = 0.772 μ_{CR} = 0.040
 Diversity = 3.4706e+00 k_t = [3, 4, 2, 3]
 Generation 8130: Best = 2.529119e+05 FES = 813100 μ_F = 0.766 μ_{CR} = 0.042
 Diversity = 3.2824e+00 k_t = [1, 1, 1, 1]
 Generation 8140: Best = 2.528350e+05 FES = 814100 μ_F = 0.765 μ_{CR} = 0.037
 Diversity = 3.1233e+00 k_t = [2, 4, 3, 3]
 Generation 8150: Best = 2.527314e+05 FES = 815100 μ_F = 0.758 μ_{CR} = 0.045
 Diversity = 3.0108e+00 k_t = [1, 1, 1, 1]
 Generation 8160: Best = 2.526796e+05 FES = 816100 μ_F = 0.739 μ_{CR} = 0.054
 Diversity = 3.0100e+00 k_t = [2, 3, 4, 2]
 Generation 8170: Best = 2.525657e+05 FES = 817100 μ_F = 0.753 μ_{CR} = 0.049
 Diversity = 2.8019e+00 k_t = [1, 1, 1, 1]
 Generation 8180: Best = 2.524942e+05 FES = 818100 μ_F = 0.761 μ_{CR} = 0.050
 Diversity = 2.8304e+00 k_t = [4, 2, 4, 4]
 Generation 8190: Best = 2.524325e+05 FES = 819100 μ_F = 0.774 μ_{CR} = 0.051
 Diversity = 2.3930e+00 k_t = [1, 1, 1, 1]
 Generation 8200: Best = 2.523331e+05 FES = 820100 μ_F = 0.786 μ_{CR} = 0.048
 Diversity = 2.6322e+00 k_t = [3, 3, 4, 2]
 Generation 8210: Best = 2.523009e+05 FES = 821100 μ_F = 0.784 μ_{CR} = 0.047
 Diversity = 2.3803e+00 k_t = [1, 1, 1, 1]
 Generation 8220: Best = 2.522423e+05 FES = 822100 μ_F = 0.802 μ_{CR} = 0.049
 Diversity = 2.4369e+00 k_t = [2, 3, 4, 4]
 Generation 8230: Best = 2.521034e+05 FES = 823100 μ_F = 0.796 μ_{CR} = 0.055
 Diversity = 2.6571e+00 k_t = [1, 1, 1, 1]
 Generation 8240: Best = 2.520325e+05 FES = 824100 μ_F = 0.781 μ_{CR} = 0.052
 Diversity = 2.8356e+00 k_t = [3, 4, 2, 2]
 Generation 8250: Best = 2.518869e+05 FES = 825100 μ_F = 0.779 μ_{CR} = 0.056
 Diversity = 2.8633e+00 k_t = [1, 1, 1, 1]
 Generation 8260: Best = 2.518076e+05 FES = 826100 μ_F = 0.777 μ_{CR} = 0.050
 Diversity = 2.7017e+00 k_t = [2, 2, 4, 4]
 Generation 8270: Best = 2.517252e+05 FES = 827100 μ_F = 0.785 μ_{CR} = 0.049
 Diversity = 2.3400e+00 k_t = [1, 1, 1, 1]
 Generation 8280: Best = 2.516006e+05 FES = 828100 μ_F = 0.791 μ_{CR} = 0.053
 Diversity = 2.4845e+00 k_t = [3, 2, 3, 2]
 Generation 8290: Best = 2.514750e+05 FES = 829100 μ_F = 0.780 μ_{CR} = 0.056
 Diversity = 2.8118e+00 k_t = [1, 1, 1, 1]
 Generation 8300: Best = 2.514410e+05 FES = 830100 μ_F = 0.784 μ_{CR} = 0.055
 Diversity = 2.8573e+00 k_t = [2, 3, 4, 2]
 Generation 8310: Best = 2.513853e+05 FES = 831100 μ_F = 0.779 μ_{CR} = 0.055
 Diversity = 3.1573e+00 k_t = [1, 1, 1, 1]
 Generation 8320: Best = 2.513021e+05 FES = 832100 μ_F = 0.799 μ_{CR} = 0.042
 Diversity = 3.3828e+00 k_t = [2, 3, 2, 4]

Generation 8330: Best = 2.511013e+05 FES = 833100 μ_F = 0.816 μ_{CR} = 0.042
 Diversity = 3.2298e+00 k_t = [1, 1, 1, 1]
 Generation 8340: Best = 2.509759e+05 FES = 834100 μ_F = 0.832 μ_{CR} = 0.040
 Diversity = 3.4040e+00 k_t = [3, 3, 2, 2]
 Generation 8350: Best = 2.507025e+05 FES = 835100 μ_F = 0.825 μ_{CR} = 0.044
 Diversity = 4.1044e+00 k_t = [1, 1, 1, 1]
 Generation 8360: Best = 2.505596e+05 FES = 836100 μ_F = 0.833 μ_{CR} = 0.045
 Diversity = 3.8262e+00 k_t = [3, 4, 4, 4]
 Generation 8370: Best = 2.504249e+05 FES = 837100 μ_F = 0.841 μ_{CR} = 0.048
 Diversity = 3.9239e+00 k_t = [1, 1, 1, 1]
 Generation 8380: Best = 2.502167e+05 FES = 838100 μ_F = 0.843 μ_{CR} = 0.045
 Diversity = 4.0367e+00 k_t = [2, 4, 3, 4]
 Generation 8390: Best = 2.501644e+05 FES = 839100 μ_F = 0.836 μ_{CR} = 0.046
 Diversity = 3.6684e+00 k_t = [1, 1, 1, 1]
 Generation 8400: Best = 2.500455e+05 FES = 840100 μ_F = 0.826 μ_{CR} = 0.046
 Diversity = 3.9957e+00 k_t = [2, 3, 4, 2]
 Generation 8410: Best = 2.498164e+05 FES = 841100 μ_F = 0.824 μ_{CR} = 0.043
 Diversity = 4.3262e+00 k_t = [1, 1, 1, 1]
 Generation 8420: Best = 2.496526e+05 FES = 842100 μ_F = 0.829 μ_{CR} = 0.039
 Diversity = 4.2038e+00 k_t = [4, 3, 2, 4]
 Generation 8430: Best = 2.496138e+05 FES = 843100 μ_F = 0.836 μ_{CR} = 0.045
 Diversity = 3.7412e+00 k_t = [1, 1, 1, 1]
 Generation 8440: Best = 2.493697e+05 FES = 844100 μ_F = 0.835 μ_{CR} = 0.050
 Diversity = 4.2263e+00 k_t = [3, 2, 3, 2]
 Generation 8450: Best = 2.492324e+05 FES = 845100 μ_F = 0.834 μ_{CR} = 0.041
 Diversity = 4.9051e+00 k_t = [1, 1, 1, 1]
 Generation 8460: Best = 2.490217e+05 FES = 846100 μ_F = 0.832 μ_{CR} = 0.042
 Diversity = 4.7513e+00 k_t = [3, 2, 3, 4]
 Generation 8470: Best = 2.488642e+05 FES = 847100 μ_F = 0.836 μ_{CR} = 0.046
 Diversity = 4.7109e+00 k_t = [1, 1, 1, 1]
 Generation 8480: Best = 2.486223e+05 FES = 848100 μ_F = 0.833 μ_{CR} = 0.044
 Diversity = 4.9364e+00 k_t = [2, 4, 4, 3]
 Generation 8490: Best = 2.484748e+05 FES = 849100 μ_F = 0.814 μ_{CR} = 0.039
 Diversity = 4.3966e+00 k_t = [1, 1, 1, 1]
 Generation 8500: Best = 2.483269e+05 FES = 850100 μ_F = 0.824 μ_{CR} = 0.036
 Diversity = 4.5524e+00 k_t = [4, 4, 2, 2]
 Generation 8510: Best = 2.481958e+05 FES = 851100 μ_F = 0.821 μ_{CR} = 0.038
 Diversity = 4.5474e+00 k_t = [1, 1, 1, 1]
 Generation 8520: Best = 2.480886e+05 FES = 852100 μ_F = 0.812 μ_{CR} = 0.037
 Diversity = 4.7488e+00 k_t = [3, 3, 4, 2]
 Generation 8530: Best = 2.479343e+05 FES = 853100 μ_F = 0.801 μ_{CR} = 0.040
 Diversity = 5.1009e+00 k_t = [1, 1, 1, 1]
 Generation 8540: Best = 2.477958e+05 FES = 854100 μ_F = 0.806 μ_{CR} = 0.040
 Diversity = 5.2127e+00 k_t = [3, 3, 3, 2]
 Generation 8550: Best = 2.476381e+05 FES = 855100 μ_F = 0.787 μ_{CR} = 0.048
 Diversity = 5.2615e+00 k_t = [1, 1, 1, 1]
 Generation 8560: Best = 2.474850e+05 FES = 856100 μ_F = 0.799 μ_{CR} = 0.044
 Diversity = 5.1614e+00 k_t = [3, 4, 4, 2]
 Generation 8570: Best = 2.473893e+05 FES = 857100 μ_F = 0.818 μ_{CR} = 0.046
 Diversity = 4.8839e+00 k_t = [1, 1, 1, 1]
 Generation 8580: Best = 2.472932e+05 FES = 858100 μ_F = 0.815 μ_{CR} = 0.048
 Diversity = 4.9943e+00 k_t = [3, 4, 3, 2]
 Generation 8590: Best = 2.471517e+05 FES = 859100 μ_F = 0.805 μ_{CR} = 0.047
 Diversity = 5.2722e+00 k_t = [1, 1, 1, 1]

Generation 8600: Best = 2.470209e+05 FES = 860100 μ_F = 0.804 μ_{CR} = 0.051
 Diversity = 4.7470e+00 k_t = [4, 2, 3, 3]
 Generation 8610: Best = 2.469603e+05 FES = 861100 μ_F = 0.804 μ_{CR} = 0.052
 Diversity = 4.6852e+00 k_t = [1, 1, 1, 1]
 Generation 8620: Best = 2.468301e+05 FES = 862100 μ_F = 0.803 μ_{CR} = 0.047
 Diversity = 4.8354e+00 k_t = [4, 2, 3, 2]
 Generation 8630: Best = 2.467104e+05 FES = 863100 μ_F = 0.806 μ_{CR} = 0.041
 Diversity = 4.7415e+00 k_t = [1, 1, 1, 1]
 Generation 8640: Best = 2.466116e+05 FES = 864100 μ_F = 0.799 μ_{CR} = 0.044
 Diversity = 4.6380e+00 k_t = [3, 2, 3, 4]
 Generation 8650: Best = 2.463527e+05 FES = 865100 μ_F = 0.813 μ_{CR} = 0.051
 Diversity = 4.6673e+00 k_t = [1, 1, 1, 1]
 Generation 8660: Best = 2.462994e+05 FES = 866100 μ_F = 0.804 μ_{CR} = 0.052
 Diversity = 4.9839e+00 k_t = [2, 4, 2, 2]
 Generation 8670: Best = 2.461284e+05 FES = 867100 μ_F = 0.816 μ_{CR} = 0.054
 Diversity = 4.9330e+00 k_t = [1, 1, 1, 1]
 Generation 8680: Best = 2.459505e+05 FES = 868100 μ_F = 0.816 μ_{CR} = 0.049
 Diversity = 4.8698e+00 k_t = [2, 3, 4, 3]
 Generation 8690: Best = 2.457276e+05 FES = 869100 μ_F = 0.804 μ_{CR} = 0.041
 Diversity = 4.7634e+00 k_t = [1, 1, 1, 1]
 Generation 8700: Best = 2.456240e+05 FES = 870100 μ_F = 0.804 μ_{CR} = 0.035
 Diversity = 4.5634e+00 k_t = [3, 2, 2, 3]
 Generation 8710: Best = 2.455419e+05 FES = 871100 μ_F = 0.804 μ_{CR} = 0.043
 Diversity = 4.1115e+00 k_t = [1, 1, 1, 1]
 Generation 8720: Best = 2.454606e+05 FES = 872100 μ_F = 0.803 μ_{CR} = 0.040
 Diversity = 4.0321e+00 k_t = [3, 4, 3, 4]
 Generation 8730: Best = 2.453271e+05 FES = 873100 μ_F = 0.790 μ_{CR} = 0.049
 Diversity = 3.6510e+00 k_t = [1, 1, 1, 1]
 Generation 8740: Best = 2.451613e+05 FES = 874100 μ_F = 0.788 μ_{CR} = 0.046
 Diversity = 3.5109e+00 k_t = [2, 4, 4, 3]
 Generation 8750: Best = 2.449954e+05 FES = 875100 μ_F = 0.796 μ_{CR} = 0.043
 Diversity = 3.4893e+00 k_t = [1, 1, 1, 1]
 Generation 8760: Best = 2.448705e+05 FES = 876100 μ_F = 0.794 μ_{CR} = 0.038
 Diversity = 3.9476e+00 k_t = [3, 4, 4, 2]
 Generation 8770: Best = 2.446091e+05 FES = 877100 μ_F = 0.807 μ_{CR} = 0.041
 Diversity = 4.1985e+00 k_t = [1, 1, 1, 1]
 Generation 8780: Best = 2.444497e+05 FES = 878100 μ_F = 0.816 μ_{CR} = 0.033
 Diversity = 4.3755e+00 k_t = [3, 2, 4, 3]
 Generation 8790: Best = 2.443145e+05 FES = 879100 μ_F = 0.820 μ_{CR} = 0.037
 Diversity = 4.4549e+00 k_t = [1, 1, 1, 1]
 Generation 8800: Best = 2.441014e+05 FES = 880100 μ_F = 0.806 μ_{CR} = 0.042
 Diversity = 4.1843e+00 k_t = [2, 3, 3, 4]
 Generation 8810: Best = 2.440105e+05 FES = 881100 μ_F = 0.804 μ_{CR} = 0.042
 Diversity = 3.7029e+00 k_t = [1, 1, 1, 1]
 Generation 8820: Best = 2.438779e+05 FES = 882100 μ_F = 0.818 μ_{CR} = 0.045
 Diversity = 3.5647e+00 k_t = [3, 2, 2, 2]
 Generation 8830: Best = 2.436658e+05 FES = 883100 μ_F = 0.818 μ_{CR} = 0.049
 Diversity = 3.7529e+00 k_t = [1, 1, 1, 1]
 Generation 8840: Best = 2.436289e+05 FES = 884100 μ_F = 0.818 μ_{CR} = 0.044
 Diversity = 3.7404e+00 k_t = [4, 2, 4, 3]
 Generation 8850: Best = 2.433173e+05 FES = 885100 μ_F = 0.814 μ_{CR} = 0.043
 Diversity = 3.7932e+00 k_t = [1, 1, 1, 1]
 Generation 8860: Best = 2.432215e+05 FES = 886100 μ_F = 0.814 μ_{CR} = 0.044
 Diversity = 3.8693e+00 k_t = [3, 3, 2, 2]

Generation 8870: Best = 2.430945e+05 FES = 887100 μ_F = 0.821 μ_{CR} = 0.041
 Diversity = 4.1398e+00 k_t = [1, 1, 1, 1]
 Generation 8880: Best = 2.429819e+05 FES = 888100 μ_F = 0.823 μ_{CR} = 0.040
 Diversity = 4.2318e+00 k_t = [4, 4, 2, 3]
 Generation 8890: Best = 2.427403e+05 FES = 889100 μ_F = 0.822 μ_{CR} = 0.044
 Diversity = 4.1911e+00 k_t = [1, 1, 1, 1]
 Generation 8900: Best = 2.423689e+05 FES = 890100 μ_F = 0.831 μ_{CR} = 0.048
 Diversity = 4.3903e+00 k_t = [3, 4, 3, 2]
 Generation 8910: Best = 2.422355e+05 FES = 891100 μ_F = 0.831 μ_{CR} = 0.040
 Diversity = 4.8501e+00 k_t = [1, 1, 1, 1]
 Generation 8920: Best = 2.420797e+05 FES = 892100 μ_F = 0.809 μ_{CR} = 0.040
 Diversity = 4.5219e+00 k_t = [3, 4, 4, 2]
 Generation 8930: Best = 2.419175e+05 FES = 893100 μ_F = 0.800 μ_{CR} = 0.039
 Diversity = 4.4013e+00 k_t = [1, 1, 1, 1]
 Generation 8940: Best = 2.417685e+05 FES = 894100 μ_F = 0.791 μ_{CR} = 0.041
 Diversity = 4.5099e+00 k_t = [3, 4, 4, 2]
 Generation 8950: Best = 2.416632e+05 FES = 895100 μ_F = 0.790 μ_{CR} = 0.042
 Diversity = 4.1540e+00 k_t = [1, 1, 1, 1]
 Generation 8960: Best = 2.415703e+05 FES = 896100 μ_F = 0.787 μ_{CR} = 0.039
 Diversity = 3.7897e+00 k_t = [4, 2, 4, 3]
 Generation 8970: Best = 2.413690e+05 FES = 897100 μ_F = 0.803 μ_{CR} = 0.041
 Diversity = 3.7022e+00 k_t = [1, 1, 1, 1]
 Generation 8980: Best = 2.412296e+05 FES = 898100 μ_F = 0.807 μ_{CR} = 0.039
 Diversity = 3.8711e+00 k_t = [2, 2, 2, 4]
 Generation 8990: Best = 2.411216e+05 FES = 899100 μ_F = 0.807 μ_{CR} = 0.034
 Diversity = 4.0035e+00 k_t = [1, 1, 1, 1]
 Generation 9000: Best = 2.410187e+05 FES = 900100 μ_F = 0.798 μ_{CR} = 0.042
 Diversity = 3.6045e+00 k_t = [4, 3, 3, 3]
 Generation 9010: Best = 2.408527e+05 FES = 901100 μ_F = 0.787 μ_{CR} = 0.038
 Diversity = 3.5513e+00 k_t = [1, 1, 1, 1]
 Generation 9020: Best = 2.407875e+05 FES = 902100 μ_F = 0.791 μ_{CR} = 0.037
 Diversity = 3.9153e+00 k_t = [4, 3, 2, 2]
 Generation 9030: Best = 2.406110e+05 FES = 903100 μ_F = 0.772 μ_{CR} = 0.036
 Diversity = 4.0290e+00 k_t = [1, 1, 1, 1]
 Generation 9040: Best = 2.405290e+05 FES = 904100 μ_F = 0.790 μ_{CR} = 0.046
 Diversity = 4.0895e+00 k_t = [4, 2, 2, 3]
 Generation 9050: Best = 2.404176e+05 FES = 905100 μ_F = 0.792 μ_{CR} = 0.044
 Diversity = 3.6850e+00 k_t = [1, 1, 1, 1]
 Generation 9060: Best = 2.402457e+05 FES = 906100 μ_F = 0.777 μ_{CR} = 0.038
 Diversity = 3.2466e+00 k_t = [2, 4, 2, 3]
 Generation 9070: Best = 2.402127e+05 FES = 907100 μ_F = 0.769 μ_{CR} = 0.038
 Diversity = 2.9955e+00 k_t = [1, 1, 1, 1]
 Generation 9080: Best = 2.401346e+05 FES = 908100 μ_F = 0.765 μ_{CR} = 0.043
 Diversity = 2.5785e+00 k_t = [2, 3, 4, 4]
 Generation 9090: Best = 2.399257e+05 FES = 909100 μ_F = 0.756 μ_{CR} = 0.049
 Diversity = 2.5295e+00 k_t = [1, 1, 1, 1]
 Generation 9100: Best = 2.399257e+05 FES = 910100 μ_F = 0.759 μ_{CR} = 0.046
 Diversity = 2.4326e+00 k_t = [3, 3, 2, 4]
 Generation 9110: Best = 2.398699e+05 FES = 911100 μ_F = 0.776 μ_{CR} = 0.044
 Diversity = 2.3688e+00 k_t = [1, 1, 1, 1]
 Generation 9120: Best = 2.397778e+05 FES = 912100 μ_F = 0.781 μ_{CR} = 0.056
 Diversity = 2.2284e+00 k_t = [3, 2, 4, 3]
 Generation 9130: Best = 2.396730e+05 FES = 913100 μ_F = 0.792 μ_{CR} = 0.046
 Diversity = 2.2066e+00 k_t = [1, 1, 1, 1]

Generation 9140: Best = 2.396032e+05 FES = 914100 μ_F = 0.801 μ_{CR} = 0.042
 Diversity = 2.5244e+00 k_t = [2, 4, 4, 2]
 Generation 9150: Best = 2.395563e+05 FES = 915100 μ_F = 0.789 μ_{CR} = 0.047
 Diversity = 2.5978e+00 k_t = [1, 1, 1, 1]
 Generation 9160: Best = 2.394015e+05 FES = 916100 μ_F = 0.799 μ_{CR} = 0.055
 Diversity = 2.7506e+00 k_t = [3, 3, 4, 3]
 Generation 9170: Best = 2.392792e+05 FES = 917100 μ_F = 0.805 μ_{CR} = 0.048
 Diversity = 2.9669e+00 k_t = [1, 1, 1, 1]
 Generation 9180: Best = 2.391012e+05 FES = 918100 μ_F = 0.806 μ_{CR} = 0.050
 Diversity = 3.1299e+00 k_t = [3, 2, 4, 2]
 Generation 9190: Best = 2.389845e+05 FES = 919100 μ_F = 0.814 μ_{CR} = 0.047
 Diversity = 3.4250e+00 k_t = [1, 1, 1, 1]
 Generation 9200: Best = 2.388777e+05 FES = 920100 μ_F = 0.815 μ_{CR} = 0.046
 Diversity = 3.4038e+00 k_t = [4, 4, 2, 3]
 Generation 9210: Best = 2.387308e+05 FES = 921100 μ_F = 0.810 μ_{CR} = 0.050
 Diversity = 3.2811e+00 k_t = [1, 1, 1, 1]
 Generation 9220: Best = 2.386175e+05 FES = 922100 μ_F = 0.812 μ_{CR} = 0.054
 Diversity = 3.5597e+00 k_t = [3, 2, 3, 2]
 Generation 9230: Best = 2.384469e+05 FES = 923100 μ_F = 0.816 μ_{CR} = 0.059
 Diversity = 3.7861e+00 k_t = [1, 1, 1, 1]
 Generation 9240: Best = 2.379574e+05 FES = 924100 μ_F = 0.816 μ_{CR} = 0.057
 Diversity = 3.8104e+00 k_t = [4, 4, 4, 2]
 Generation 9250: Best = 2.377890e+05 FES = 925100 μ_F = 0.823 μ_{CR} = 0.053
 Diversity = 4.0058e+00 k_t = [1, 1, 1, 1]
 Generation 9260: Best = 2.373915e+05 FES = 926100 μ_F = 0.835 μ_{CR} = 0.046
 Diversity = 3.6732e+00 k_t = [4, 4, 4, 2]
 Generation 9270: Best = 2.372211e+05 FES = 927100 μ_F = 0.840 μ_{CR} = 0.041
 Diversity = 3.8363e+00 k_t = [1, 1, 1, 1]
 Generation 9280: Best = 2.369220e+05 FES = 928100 μ_F = 0.827 μ_{CR} = 0.035
 Diversity = 4.0410e+00 k_t = [2, 4, 2, 4]
 Generation 9290: Best = 2.367779e+05 FES = 929100 μ_F = 0.829 μ_{CR} = 0.038
 Diversity = 3.6232e+00 k_t = [1, 1, 1, 1]
 Generation 9300: Best = 2.365933e+05 FES = 930100 μ_F = 0.808 μ_{CR} = 0.047
 Diversity = 3.3307e+00 k_t = [3, 4, 3, 3]
 Generation 9310: Best = 2.364816e+05 FES = 931100 μ_F = 0.797 μ_{CR} = 0.041
 Diversity = 3.1050e+00 k_t = [1, 1, 1, 1]
 Generation 9320: Best = 2.362399e+05 FES = 932100 μ_F = 0.800 μ_{CR} = 0.045
 Diversity = 2.7720e+00 k_t = [4, 4, 4, 4]
 Generation 9330: Best = 2.360538e+05 FES = 933100 μ_F = 0.805 μ_{CR} = 0.045
 Diversity = 2.9430e+00 k_t = [1, 1, 1, 1]
 Generation 9340: Best = 2.358670e+05 FES = 934100 μ_F = 0.797 μ_{CR} = 0.039
 Diversity = 3.0092e+00 k_t = [3, 2, 2, 2]
 Generation 9350: Best = 2.357508e+05 FES = 935100 μ_F = 0.804 μ_{CR} = 0.040
 Diversity = 3.0703e+00 k_t = [1, 1, 1, 1]
 Generation 9360: Best = 2.355799e+05 FES = 936100 μ_F = 0.823 μ_{CR} = 0.047
 Diversity = 3.0424e+00 k_t = [4, 2, 3, 3]
 Generation 9370: Best = 2.353853e+05 FES = 937100 μ_F = 0.818 μ_{CR} = 0.049
 Diversity = 3.2421e+00 k_t = [1, 1, 1, 1]
 Generation 9380: Best = 2.352666e+05 FES = 938100 μ_F = 0.809 μ_{CR} = 0.045
 Diversity = 3.4166e+00 k_t = [3, 3, 3, 3]
 Generation 9390: Best = 2.350943e+05 FES = 939100 μ_F = 0.815 μ_{CR} = 0.046
 Diversity = 3.3321e+00 k_t = [1, 1, 1, 1]
 Generation 9400: Best = 2.348516e+05 FES = 940100 μ_F = 0.809 μ_{CR} = 0.047
 Diversity = 3.5028e+00 k_t = [2, 3, 4, 3]

Generation 9410: Best = 2.347182e+05 FES = 941100 μ_F = 0.808 μ_{CR} = 0.054
 Diversity = 3.2393e+00 k_t = [1, 1, 1, 1]
 Generation 9420: Best = 2.346508e+05 FES = 942100 μ_F = 0.802 μ_{CR} = 0.042
 Diversity = 2.9727e+00 k_t = [2, 3, 2, 3]
 Generation 9430: Best = 2.345142e+05 FES = 943100 μ_F = 0.804 μ_{CR} = 0.042
 Diversity = 3.0143e+00 k_t = [1, 1, 1, 1]
 Generation 9440: Best = 2.344009e+05 FES = 944100 μ_F = 0.784 μ_{CR} = 0.053
 Diversity = 2.8724e+00 k_t = [4, 2, 3, 3]
 Generation 9450: Best = 2.342381e+05 FES = 945100 μ_F = 0.786 μ_{CR} = 0.049
 Diversity = 2.6909e+00 k_t = [1, 1, 1, 1]
 Generation 9460: Best = 2.340878e+05 FES = 946100 μ_F = 0.774 μ_{CR} = 0.045
 Diversity = 2.5011e+00 k_t = [4, 2, 3, 4]
 Generation 9470: Best = 2.340244e+05 FES = 947100 μ_F = 0.787 μ_{CR} = 0.045
 Diversity = 2.0828e+00 k_t = [1, 1, 1, 1]
 Generation 9480: Best = 2.339084e+05 FES = 948100 μ_F = 0.784 μ_{CR} = 0.049
 Diversity = 1.8718e+00 k_t = [4, 2, 3, 4]
 Generation 9490: Best = 2.338359e+05 FES = 949100 μ_F = 0.795 μ_{CR} = 0.052
 Diversity = 1.5550e+00 k_t = [1, 1, 1, 1]
 Generation 9500: Best = 2.337766e+05 FES = 950100 μ_F = 0.798 μ_{CR} = 0.048
 Diversity = 1.7972e+00 k_t = [2, 3, 4, 4]
 Generation 9510: Best = 2.336555e+05 FES = 951100 μ_F = 0.809 μ_{CR} = 0.045
 Diversity = 1.9675e+00 k_t = [1, 1, 1, 1]
 Generation 9520: Best = 2.335443e+05 FES = 952100 μ_F = 0.810 μ_{CR} = 0.050
 Diversity = 1.8338e+00 k_t = [2, 2, 2, 4]
 Generation 9530: Best = 2.334464e+05 FES = 953100 μ_F = 0.804 μ_{CR} = 0.056
 Diversity = 1.9980e+00 k_t = [1, 1, 1, 1]
 Generation 9540: Best = 2.332797e+05 FES = 954100 μ_F = 0.800 μ_{CR} = 0.061
 Diversity = 2.0859e+00 k_t = [3, 3, 4, 2]
 Generation 9550: Best = 2.332534e+05 FES = 955100 μ_F = 0.812 μ_{CR} = 0.048
 Diversity = 2.2279e+00 k_t = [1, 1, 1, 1]
 Generation 9560: Best = 2.331409e+05 FES = 956100 μ_F = 0.805 μ_{CR} = 0.055
 Diversity = 2.1633e+00 k_t = [3, 4, 4, 4]
 Generation 9570: Best = 2.330754e+05 FES = 957100 μ_F = 0.805 μ_{CR} = 0.056
 Diversity = 2.3033e+00 k_t = [1, 1, 1, 1]
 Generation 9580: Best = 2.329671e+05 FES = 958100 μ_F = 0.794 μ_{CR} = 0.055
 Diversity = 2.0125e+00 k_t = [3, 3, 3, 4]
 Generation 9590: Best = 2.329487e+05 FES = 959100 μ_F = 0.799 μ_{CR} = 0.046
 Diversity = 2.1010e+00 k_t = [1, 1, 1, 1]
 Generation 9600: Best = 2.328808e+05 FES = 960100 μ_F = 0.792 μ_{CR} = 0.048
 Diversity = 1.8921e+00 k_t = [4, 4, 4, 3]
 Generation 9610: Best = 2.327990e+05 FES = 961100 μ_F = 0.797 μ_{CR} = 0.056
 Diversity = 2.0173e+00 k_t = [1, 1, 1, 1]
 Generation 9620: Best = 2.327118e+05 FES = 962100 μ_F = 0.814 μ_{CR} = 0.052
 Diversity = 2.2208e+00 k_t = [4, 2, 2, 3]
 Generation 9630: Best = 2.325581e+05 FES = 963100 μ_F = 0.806 μ_{CR} = 0.053
 Diversity = 2.2814e+00 k_t = [1, 1, 1, 1]
 Generation 9640: Best = 2.325056e+05 FES = 964100 μ_F = 0.811 μ_{CR} = 0.058
 Diversity = 2.7088e+00 k_t = [2, 2, 2, 2]
 Generation 9650: Best = 2.324092e+05 FES = 965100 μ_F = 0.813 μ_{CR} = 0.045
 Diversity = 2.9868e+00 k_t = [1, 1, 1, 1]
 Generation 9660: Best = 2.323204e+05 FES = 966100 μ_F = 0.800 μ_{CR} = 0.052
 Diversity = 2.7639e+00 k_t = [4, 4, 2, 3]
 Generation 9670: Best = 2.322519e+05 FES = 967100 μ_F = 0.795 μ_{CR} = 0.053
 Diversity = 2.9005e+00 k_t = [1, 1, 1, 1]

Generation 9680: Best = 2.321247e+05 FES = 968100 μ_F = 0.805 μ_{CR} = 0.057
 Diversity = 2.7512e+00 k_t = [3, 3, 4, 2]
 Generation 9690: Best = 2.320917e+05 FES = 969100 μ_F = 0.788 μ_{CR} = 0.050
 Diversity = 3.0189e+00 k_t = [1, 1, 1, 1]
 Generation 9700: Best = 2.320055e+05 FES = 970100 μ_F = 0.779 μ_{CR} = 0.046
 Diversity = 2.8737e+00 k_t = [4, 4, 4, 3]
 Generation 9710: Best = 2.319573e+05 FES = 971100 μ_F = 0.772 μ_{CR} = 0.050
 Diversity = 2.5817e+00 k_t = [1, 1, 1, 1]
 Generation 9720: Best = 2.318570e+05 FES = 972100 μ_F = 0.773 μ_{CR} = 0.048
 Diversity = 2.4367e+00 k_t = [4, 2, 4, 3]
 Generation 9730: Best = 2.318140e+05 FES = 973100 μ_F = 0.798 μ_{CR} = 0.046
 Diversity = 2.6862e+00 k_t = [1, 1, 1, 1]
 Generation 9740: Best = 2.317366e+05 FES = 974100 μ_F = 0.797 μ_{CR} = 0.052
 Diversity = 2.5358e+00 k_t = [2, 3, 3, 3]
 Generation 9750: Best = 2.316908e+05 FES = 975100 μ_F = 0.792 μ_{CR} = 0.052
 Diversity = 2.5268e+00 k_t = [1, 1, 1, 1]
 Generation 9760: Best = 2.315955e+05 FES = 976100 μ_F = 0.798 μ_{CR} = 0.052
 Diversity = 2.4283e+00 k_t = [4, 4, 3, 2]
 Generation 9770: Best = 2.315424e+05 FES = 977100 μ_F = 0.797 μ_{CR} = 0.052
 Diversity = 2.6766e+00 k_t = [1, 1, 1, 1]
 Generation 9780: Best = 2.314407e+05 FES = 978100 μ_F = 0.804 μ_{CR} = 0.047
 Diversity = 2.7934e+00 k_t = [4, 4, 4, 4]
 Generation 9790: Best = 2.313735e+05 FES = 979100 μ_F = 0.813 μ_{CR} = 0.042
 Diversity = 2.8277e+00 k_t = [1, 1, 1, 1]
 Generation 9800: Best = 2.313356e+05 FES = 980100 μ_F = 0.802 μ_{CR} = 0.039
 Diversity = 2.6177e+00 k_t = [2, 2, 3, 4]
 Generation 9810: Best = 2.312411e+05 FES = 981100 μ_F = 0.795 μ_{CR} = 0.047
 Diversity = 2.6271e+00 k_t = [1, 1, 1, 1]
 Generation 9820: Best = 2.311456e+05 FES = 982100 μ_F = 0.785 μ_{CR} = 0.046
 Diversity = 2.6231e+00 k_t = [4, 2, 2, 3]
 Generation 9830: Best = 2.310257e+05 FES = 983100 μ_F = 0.797 μ_{CR} = 0.052
 Diversity = 2.9402e+00 k_t = [1, 1, 1, 1]
 Generation 9840: Best = 2.309196e+05 FES = 984100 μ_F = 0.818 μ_{CR} = 0.051
 Diversity = 2.9211e+00 k_t = [3, 4, 2, 2]
 Generation 9850: Best = 2.308122e+05 FES = 985100 μ_F = 0.816 μ_{CR} = 0.061
 Diversity = 3.3287e+00 k_t = [1, 1, 1, 1]
 Generation 9860: Best = 2.306915e+05 FES = 986100 μ_F = 0.806 μ_{CR} = 0.061
 Diversity = 3.4418e+00 k_t = [2, 2, 2, 4]
 Generation 9870: Best = 2.305867e+05 FES = 987100 μ_F = 0.805 μ_{CR} = 0.065
 Diversity = 3.2503e+00 k_t = [1, 1, 1, 1]
 Generation 9880: Best = 2.304762e+05 FES = 988100 μ_F = 0.801 μ_{CR} = 0.053
 Diversity = 3.3080e+00 k_t = [3, 3, 2, 4]
 Generation 9890: Best = 2.303589e+05 FES = 989100 μ_F = 0.812 μ_{CR} = 0.047
 Diversity = 3.6172e+00 k_t = [1, 1, 1, 1]
 Generation 9900: Best = 2.302614e+05 FES = 990100 μ_F = 0.816 μ_{CR} = 0.044
 Diversity = 3.6580e+00 k_t = [3, 2, 3, 3]
 Generation 9910: Best = 2.301702e+05 FES = 991100 μ_F = 0.800 μ_{CR} = 0.053
 Diversity = 3.9835e+00 k_t = [1, 1, 1, 1]
 Generation 9920: Best = 2.300194e+05 FES = 992100 μ_F = 0.803 μ_{CR} = 0.046
 Diversity = 3.9627e+00 k_t = [2, 4, 3, 2]
 Generation 9930: Best = 2.299133e+05 FES = 993100 μ_F = 0.809 μ_{CR} = 0.044
 Diversity = 3.9198e+00 k_t = [1, 1, 1, 1]
 Generation 9940: Best = 2.297736e+05 FES = 994100 μ_F = 0.811 μ_{CR} = 0.047
 Diversity = 3.4790e+00 k_t = [2, 2, 4, 4]

Generation 9950: Best = 2.296461e+05 FES = 995100 μ_F = 0.831 μ_{CR} = 0.043
 Diversity = 3.3807e+00 k_t = [1, 1, 1, 1]
 Generation 9960: Best = 2.295841e+05 FES = 996100 μ_F = 0.826 μ_{CR} = 0.042
 Diversity = 3.1692e+00 k_t = [2, 3, 2, 3]
 Generation 9970: Best = 2.294905e+05 FES = 997100 μ_F = 0.819 μ_{CR} = 0.042
 Diversity = 3.1395e+00 k_t = [1, 1, 1, 1]
 Generation 9980: Best = 2.294742e+05 FES = 998100 μ_F = 0.811 μ_{CR} = 0.044
 Diversity = 3.0154e+00 k_t = [3, 2, 4, 4]
 Generation 9990: Best = 2.293936e+05 FES = 999100 μ_F = 0.814 μ_{CR} = 0.046
 Diversity = 3.0184e+00 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 2.292570e+05

Total Function Evals = 1000000

Total Successful Updates = 224541

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Results:

Best Fitness : 2.292570e+05

Global Optimum : 1.800000e+03

Error (|best-optimum|) : 2.274570e+05

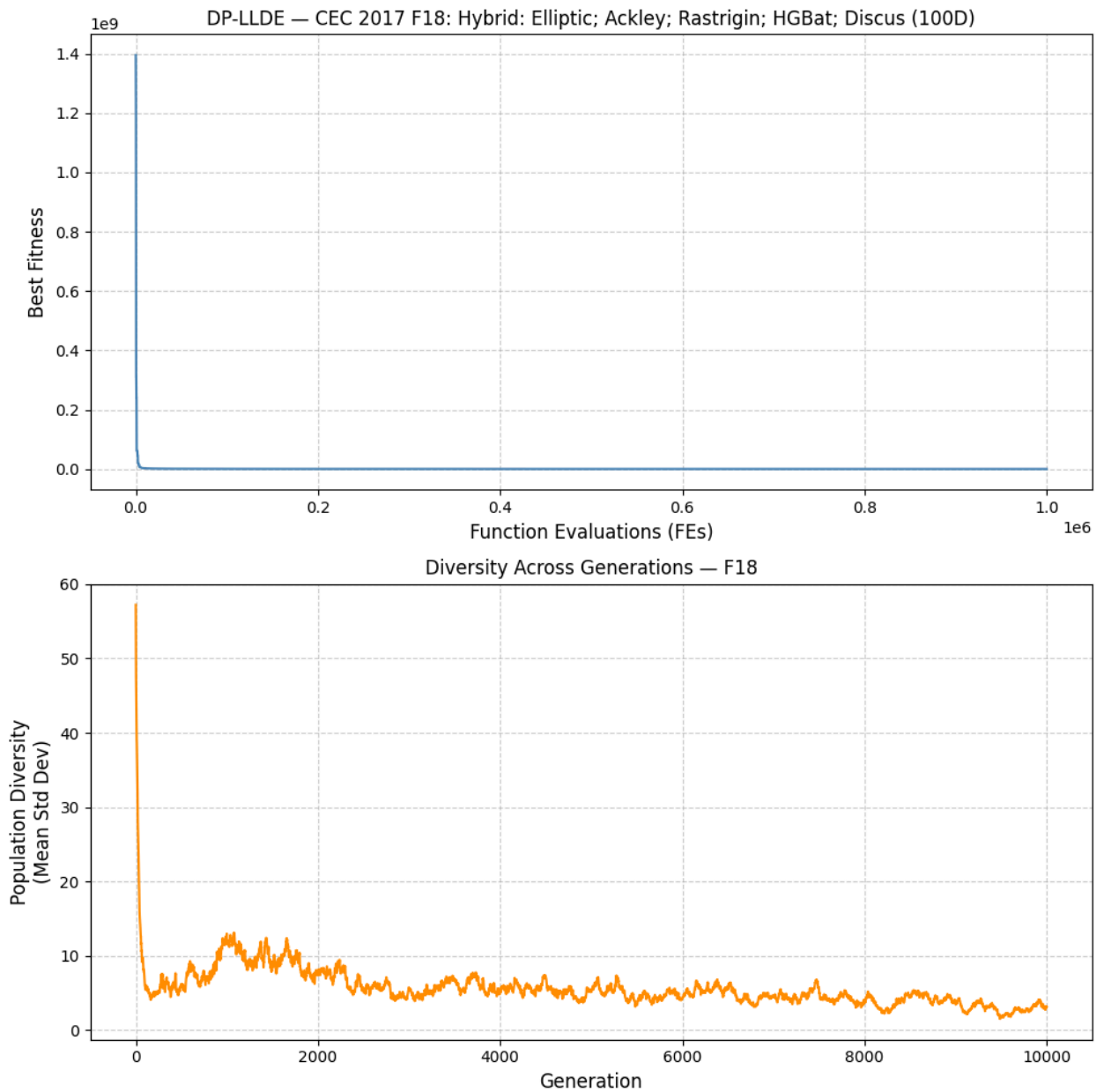
Best Solution (5 dim) : [-99.98230091 10.64699811 -8.753987 -19.47891621
4.75837102]

D^(0) (init. diversity): 5.7220e+01

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✓ Checkpoint saved to: /content/drive/My Drive/thesis_results/all_results_checkpoint.pkl



Graph saved as convergence_F18.png

✓ Plot saved to: /content/drive/My Drive/thesis_results/plots/convergence_F18.png

```
In [ ]: # =====
# F19 — Hybrid Function 9 (Bent Cigar; Rastrigin; Griewank+Rosenbrock; Weierst
# =====
run_and_plot('F19', make_wrapper(hybrid.f19), 1900.0, 'Hybrid: Bent Cigar; Ras
```

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=
DP-LLDE on CEC 2017 F19 – Single Run
=====
=
Function      : Hybrid: Bent Cigar; Rastrigin; Griewank+Rosenbrock; Weierstras
s; Schaffer (100D)
Bounds       : [-100, 100]
Global optimum : 1900.0
Initial D^(0) = 5.7220e+01 | Diversity threshold = 5.7220e+00
Generation 0: Best Fitness = 4.192174e+11
Generation  10: Best = 7.835822e+10 FES =  1100   $\mu_F$  = 0.574   $\mu_{CR}$  = 0.350
Diversity = 3.5668e+01 k_t = [1, 1, 1, 1]
Generation  20: Best = 1.561877e+10 FES =  2100   $\mu_F$  = 0.628   $\mu_{CR}$  = 0.349
Diversity = 2.4730e+01 k_t = [2, 3, 2, 3]
Generation  30: Best = 2.738769e+09 FES =  3100   $\mu_F$  = 0.644   $\mu_{CR}$  = 0.350
Diversity = 1.8793e+01 k_t = [1, 1, 1, 1]
Generation  40: Best = 3.643730e+08 FES =  4100   $\mu_F$  = 0.664   $\mu_{CR}$  = 0.354
Diversity = 1.4261e+01 k_t = [3, 2, 3, 3]
Generation  50: Best = 8.518182e+07 FES =  5100   $\mu_F$  = 0.666   $\mu_{CR}$  = 0.357
Diversity = 1.0982e+01 k_t = [1, 1, 1, 1]
Generation  60: Best = 2.343889e+07 FES =  6100   $\mu_F$  = 0.669   $\mu_{CR}$  = 0.359
Diversity = 7.9759e+00 k_t = [3, 4, 4, 3]
Generation  70: Best = 7.576746e+06 FES =  7100   $\mu_F$  = 0.679   $\mu_{CR}$  = 0.366
Diversity = 6.6838e+00 k_t = [1, 1, 1, 1]
Generation  80: Best = 2.915321e+06 FES =  8100   $\mu_F$  = 0.677   $\mu_{CR}$  = 0.375
Diversity = 5.0877e+00 k_t = [3, 4, 2, 4]
Generation  90: Best = 1.243892e+06 FES =  9100   $\mu_F$  = 0.696   $\mu_{CR}$  = 0.372
Diversity = 3.9975e+00 k_t = [1, 1, 1, 1]
Generation 100: Best = 7.101664e+05 FES = 10100   $\mu_F$  = 0.701   $\mu_{CR}$  = 0.381
Diversity = 3.2931e+00 k_t = [2, 2, 4, 3]
Generation 110: Best = 2.448208e+05 FES = 11100   $\mu_F$  = 0.725   $\mu_{CR}$  = 0.388
Diversity = 3.8735e+00 k_t = [1, 1, 1, 1]
Generation 120: Best = 1.302543e+05 FES = 12100   $\mu_F$  = 0.734   $\mu_{CR}$  = 0.390
Diversity = 4.0807e+00 k_t = [3, 3, 2, 4]
Generation 130: Best = 7.374631e+04 FES = 13100   $\mu_F$  = 0.745   $\mu_{CR}$  = 0.390
Diversity = 4.0372e+00 k_t = [1, 1, 1, 1]
Generation 140: Best = 3.422524e+04 FES = 14100   $\mu_F$  = 0.755   $\mu_{CR}$  = 0.392
Diversity = 4.3588e+00 k_t = [2, 4, 2, 3]
Generation 150: Best = 1.672300e+04 FES = 15100   $\mu_F$  = 0.762   $\mu_{CR}$  = 0.393
Diversity = 4.7432e+00 k_t = [1, 1, 1, 1]
Generation 160: Best = 1.031923e+04 FES = 16100   $\mu_F$  = 0.759   $\mu_{CR}$  = 0.403
Diversity = 5.0093e+00 k_t = [4, 3, 2, 2]
Generation 170: Best = 6.563558e+03 FES = 17100   $\mu_F$  = 0.758   $\mu_{CR}$  = 0.411
Diversity = 5.9700e+00 k_t = [1, 1, 1, 1]
Generation 180: Best = 5.165819e+03 FES = 18100   $\mu_F$  = 0.766   $\mu_{CR}$  = 0.406
Diversity = 5.9350e+00 k_t = [3, 4, 4, 3]
Generation 190: Best = 4.449466e+03 FES = 19100   $\mu_F$  = 0.768   $\mu_{CR}$  = 0.414
Diversity = 6.1600e+00 k_t = [1, 1, 1, 1]
Generation 200: Best = 4.023269e+03 FES = 20100   $\mu_F$  = 0.759   $\mu_{CR}$  = 0.414
Diversity = 6.3486e+00 k_t = [2, 2, 2, 4]
Generation 210: Best = 3.751813e+03 FES = 21100   $\mu_F$  = 0.771   $\mu_{CR}$  = 0.422
Diversity = 6.8397e+00 k_t = [1, 1, 1, 1]
Generation 220: Best = 3.615922e+03 FES = 22100   $\mu_F$  = 0.763   $\mu_{CR}$  = 0.413

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Diversity = 6.9021e+00	k_t = [4, 2, 2, 3]				
Generation 230: Best = 3.517087e+03	FES = 23100	$\mu_F = 0.751$	$\mu_{CR} = 0.417$		
Diversity = 6.9484e+00	k_t = [1, 1, 1, 1]				
Generation 240: Best = 3.484609e+03	FES = 24100	$\mu_F = 0.742$	$\mu_{CR} = 0.422$		
Diversity = 6.7960e+00	k_t = [4, 2, 4, 2]				
Generation 250: Best = 3.456246e+03	FES = 25100	$\mu_F = 0.749$	$\mu_{CR} = 0.406$		
Diversity = 6.8476e+00	k_t = [1, 1, 1, 1]				
Generation 260: Best = 3.424845e+03	FES = 26100	$\mu_F = 0.745$	$\mu_{CR} = 0.390$		
Diversity = 6.4526e+00	k_t = [2, 4, 4, 3]				
Generation 270: Best = 3.388521e+03	FES = 27100	$\mu_F = 0.693$	$\mu_{CR} = 0.357$		
Diversity = 6.0977e+00	k_t = [1, 1, 1, 1]				
Generation 280: Best = 3.388521e+03	FES = 28100	$\mu_F = 0.652$	$\mu_{CR} = 0.359$		
Diversity = 5.5915e+00	k_t = [4, 2, 2, 4]				
Generation 290: Best = 3.386160e+03	FES = 29100	$\mu_F = 0.645$	$\mu_{CR} = 0.333$		
Diversity = 5.3283e+00	k_t = [1, 1, 1, 1]				
Generation 300: Best = 3.373728e+03	FES = 30100	$\mu_F = 0.638$	$\mu_{CR} = 0.324$		
Diversity = 5.0002e+00	k_t = [3, 3, 2, 4]				
Generation 310: Best = 3.363715e+03	FES = 31100	$\mu_F = 0.643$	$\mu_{CR} = 0.294$		
Diversity = 4.6082e+00	k_t = [1, 1, 1, 1]				
Generation 320: Best = 3.362276e+03	FES = 32100	$\mu_F = 0.629$	$\mu_{CR} = 0.259$		
Diversity = 4.3498e+00	k_t = [4, 2, 4, 3]				
Generation 330: Best = 3.347017e+03	FES = 33100	$\mu_F = 0.629$	$\mu_{CR} = 0.218$		
Diversity = 4.1397e+00	k_t = [1, 1, 1, 1]				
Generation 340: Best = 3.346248e+03	FES = 34100	$\mu_F = 0.656$	$\mu_{CR} = 0.203$		
Diversity = 3.9777e+00	k_t = [3, 4, 4, 4]				
Generation 350: Best = 3.329387e+03	FES = 35100	$\mu_F = 0.697$	$\mu_{CR} = 0.165$		
Diversity = 3.7340e+00	k_t = [1, 1, 1, 1]				
Generation 360: Best = 3.314371e+03	FES = 36100	$\mu_F = 0.701$	$\mu_{CR} = 0.138$		
Diversity = 3.3356e+00	k_t = [3, 2, 2, 3]				
Generation 370: Best = 3.313454e+03	FES = 37100	$\mu_F = 0.718$	$\mu_{CR} = 0.120$		
Diversity = 3.0969e+00	k_t = [1, 1, 1, 1]				
Generation 380: Best = 3.303845e+03	FES = 38100	$\mu_F = 0.736$	$\mu_{CR} = 0.117$		
Diversity = 2.8212e+00	k_t = [3, 3, 3, 3]				
Generation 390: Best = 3.298150e+03	FES = 39100	$\mu_F = 0.744$	$\mu_{CR} = 0.113$		
Diversity = 2.8343e+00	k_t = [1, 1, 1, 1]				
Generation 400: Best = 3.279249e+03	FES = 40100	$\mu_F = 0.747$	$\mu_{CR} = 0.110$		
Diversity = 2.5308e+00	k_t = [3, 3, 3, 3]				
Generation 410: Best = 3.270805e+03	FES = 41100	$\mu_F = 0.758$	$\mu_{CR} = 0.113$		
Diversity = 2.5679e+00	k_t = [1, 1, 1, 1]				
Generation 420: Best = 3.264327e+03	FES = 42100	$\mu_F = 0.763$	$\mu_{CR} = 0.118$		
Diversity = 2.5461e+00	k_t = [3, 4, 3, 4]				
Generation 430: Best = 3.246026e+03	FES = 43100	$\mu_F = 0.749$	$\mu_{CR} = 0.125$		
Diversity = 2.0851e+00	k_t = [1, 1, 1, 1]				
Generation 440: Best = 3.244986e+03	FES = 44100	$\mu_F = 0.756$	$\mu_{CR} = 0.124$		
Diversity = 2.0701e+00	k_t = [2, 3, 4, 2]				
Generation 450: Best = 3.236758e+03	FES = 45100	$\mu_F = 0.754$	$\mu_{CR} = 0.127$		
Diversity = 2.0148e+00	k_t = [1, 1, 1, 1]				
Generation 460: Best = 3.236758e+03	FES = 46100	$\mu_F = 0.746$	$\mu_{CR} = 0.114$		
Diversity = 1.9925e+00	k_t = [4, 4, 2, 3]				
Generation 470: Best = 3.229240e+03	FES = 47100	$\mu_F = 0.732$	$\mu_{CR} = 0.112$		
Diversity = 1.8359e+00	k_t = [1, 1, 1, 1]				
Generation 480: Best = 3.224721e+03	FES = 48100	$\mu_F = 0.724$	$\mu_{CR} = 0.106$		
Diversity = 1.7190e+00	k_t = [4, 2, 3, 3]				
Generation 490: Best = 3.224575e+03	FES = 49100	$\mu_F = 0.726$	$\mu_{CR} = 0.103$		

Diversity = 1.6757e+00	k_t = [1, 1, 1, 1]					
Generation 500: Best = 3.220170e+03	FES = 50100	$\mu_F = 0.725$	$\mu_{CR} = 0.095$			
Diversity = 1.6410e+00	k_t = [3, 2, 3, 4]					
Generation 510: Best = 3.219759e+03	FES = 51100	$\mu_F = 0.751$	$\mu_{CR} = 0.081$			
Diversity = 1.5826e+00	k_t = [1, 1, 1, 1]					
Generation 520: Best = 3.218258e+03	FES = 52100	$\mu_F = 0.750$	$\mu_{CR} = 0.077$			
Diversity = 1.6070e+00	k_t = [4, 4, 4, 3]					
Generation 530: Best = 3.212919e+03	FES = 53100	$\mu_F = 0.746$	$\mu_{CR} = 0.076$			
Diversity = 1.6932e+00	k_t = [1, 1, 1, 1]					
Generation 540: Best = 3.211748e+03	FES = 54100	$\mu_F = 0.762$	$\mu_{CR} = 0.071$			
Diversity = 1.5923e+00	k_t = [3, 3, 3, 4]					
Generation 550: Best = 3.211748e+03	FES = 55100	$\mu_F = 0.761$	$\mu_{CR} = 0.075$			
Diversity = 1.5606e+00	k_t = [1, 1, 1, 1]					
Generation 560: Best = 3.209432e+03	FES = 56100	$\mu_F = 0.756$	$\mu_{CR} = 0.067$			
Diversity = 1.5058e+00	k_t = [2, 2, 3, 3]					
Generation 570: Best = 3.208054e+03	FES = 57100	$\mu_F = 0.766$	$\mu_{CR} = 0.068$			
Diversity = 1.4835e+00	k_t = [1, 1, 1, 1]					
Generation 580: Best = 3.207759e+03	FES = 58100	$\mu_F = 0.764$	$\mu_{CR} = 0.065$			
Diversity = 1.4908e+00	k_t = [4, 2, 4, 4]					
Generation 590: Best = 3.206226e+03	FES = 59100	$\mu_F = 0.793$	$\mu_{CR} = 0.065$			
Diversity = 1.5259e+00	k_t = [1, 1, 1, 1]					
Generation 600: Best = 3.202509e+03	FES = 60100	$\mu_F = 0.786$	$\mu_{CR} = 0.064$			
Diversity = 1.5211e+00	k_t = [4, 4, 3, 4]					
Generation 610: Best = 3.201420e+03	FES = 61100	$\mu_F = 0.815$	$\mu_{CR} = 0.058$			
Diversity = 1.4708e+00	k_t = [1, 1, 1, 1]					
Generation 620: Best = 3.201336e+03	FES = 62100	$\mu_F = 0.790$	$\mu_{CR} = 0.058$			
Diversity = 1.4035e+00	k_t = [3, 2, 4, 3]					
Generation 630: Best = 3.199831e+03	FES = 63100	$\mu_F = 0.814$	$\mu_{CR} = 0.068$			
Diversity = 1.3560e+00	k_t = [1, 1, 1, 1]					
Generation 640: Best = 3.199243e+03	FES = 64100	$\mu_F = 0.824$	$\mu_{CR} = 0.070$			
Diversity = 1.4170e+00	k_t = [4, 4, 2, 3]					
Generation 650: Best = 3.194876e+03	FES = 65100	$\mu_F = 0.824$	$\mu_{CR} = 0.072$			
Diversity = 1.4682e+00	k_t = [1, 1, 1, 1]					
Generation 660: Best = 3.194876e+03	FES = 66100	$\mu_F = 0.823$	$\mu_{CR} = 0.066$			
Diversity = 1.5481e+00	k_t = [2, 4, 3, 3]					
Generation 670: Best = 3.189192e+03	FES = 67100	$\mu_F = 0.821$	$\mu_{CR} = 0.067$			
Diversity = 1.6511e+00	k_t = [1, 1, 1, 1]					
Generation 680: Best = 3.188083e+03	FES = 68100	$\mu_F = 0.833$	$\mu_{CR} = 0.073$			
Diversity = 1.6593e+00	k_t = [2, 4, 4, 3]					
Generation 690: Best = 3.187558e+03	FES = 69100	$\mu_F = 0.825$	$\mu_{CR} = 0.076$			
Diversity = 1.7157e+00	k_t = [1, 1, 1, 1]					
Generation 700: Best = 3.183368e+03	FES = 70100	$\mu_F = 0.828$	$\mu_{CR} = 0.072$			
Diversity = 1.7209e+00	k_t = [4, 2, 4, 4]					
Generation 710: Best = 3.183236e+03	FES = 71100	$\mu_F = 0.820$	$\mu_{CR} = 0.071$			
Diversity = 1.7059e+00	k_t = [1, 1, 1, 1]					
Generation 720: Best = 3.182677e+03	FES = 72100	$\mu_F = 0.782$	$\mu_{CR} = 0.071$			
Diversity = 1.7007e+00	k_t = [3, 4, 4, 2]					
Generation 730: Best = 3.181145e+03	FES = 73100	$\mu_F = 0.780$	$\mu_{CR} = 0.069$			
Diversity = 1.7982e+00	k_t = [1, 1, 1, 1]					
Generation 740: Best = 3.179158e+03	FES = 74100	$\mu_F = 0.783$	$\mu_{CR} = 0.055$			
Diversity = 1.7815e+00	k_t = [3, 4, 2, 2]					
Generation 750: Best = 3.178593e+03	FES = 75100	$\mu_F = 0.776$	$\mu_{CR} = 0.057$			
Diversity = 1.8113e+00	k_t = [1, 1, 1, 1]					
Generation 760: Best = 3.178210e+03	FES = 76100	$\mu_F = 0.771$	$\mu_{CR} = 0.056$			

Diversity = 1.8975e+00	k_t = [3, 2, 2, 3]				
Generation 770: Best = 3.177929e+03	FES = 77100	$\mu_F = 0.779$	$\mu_{CR} = 0.054$		
Diversity = 1.9547e+00	k_t = [1, 1, 1, 1]				
Generation 780: Best = 3.176972e+03	FES = 78100	$\mu_F = 0.774$	$\mu_{CR} = 0.058$		
Diversity = 1.9938e+00	k_t = [4, 4, 2, 4]				
Generation 790: Best = 3.176846e+03	FES = 79100	$\mu_F = 0.754$	$\mu_{CR} = 0.057$		
Diversity = 1.9397e+00	k_t = [1, 1, 1, 1]				
Generation 800: Best = 3.175459e+03	FES = 80100	$\mu_F = 0.761$	$\mu_{CR} = 0.056$		
Diversity = 1.8820e+00	k_t = [3, 3, 2, 4]				
Generation 810: Best = 3.173013e+03	FES = 81100	$\mu_F = 0.759$	$\mu_{CR} = 0.059$		
Diversity = 1.8883e+00	k_t = [1, 1, 1, 1]				
Generation 820: Best = 3.171848e+03	FES = 82100	$\mu_F = 0.765$	$\mu_{CR} = 0.058$		
Diversity = 1.9021e+00	k_t = [2, 3, 2, 4]				
Generation 830: Best = 3.171210e+03	FES = 83100	$\mu_F = 0.736$	$\mu_{CR} = 0.065$		
Diversity = 1.8848e+00	k_t = [1, 1, 1, 1]				
Generation 840: Best = 3.170851e+03	FES = 84100	$\mu_F = 0.778$	$\mu_{CR} = 0.065$		
Diversity = 1.9297e+00	k_t = [3, 2, 3, 4]				
Generation 850: Best = 3.170250e+03	FES = 85100	$\mu_F = 0.770$	$\mu_{CR} = 0.061$		
Diversity = 1.9114e+00	k_t = [1, 1, 1, 1]				
Generation 860: Best = 3.169808e+03	FES = 86100	$\mu_F = 0.772$	$\mu_{CR} = 0.050$		
Diversity = 1.9300e+00	k_t = [3, 3, 4, 2]				
Generation 870: Best = 3.169505e+03	FES = 87100	$\mu_F = 0.801$	$\mu_{CR} = 0.056$		
Diversity = 1.9504e+00	k_t = [1, 1, 1, 1]				
Generation 880: Best = 3.169505e+03	FES = 88100	$\mu_F = 0.827$	$\mu_{CR} = 0.056$		
Diversity = 1.9229e+00	k_t = [2, 2, 4, 3]				
Generation 890: Best = 3.168540e+03	FES = 89100	$\mu_F = 0.784$	$\mu_{CR} = 0.059$		
Diversity = 1.9098e+00	k_t = [1, 1, 1, 1]				
Generation 900: Best = 3.168539e+03	FES = 90100	$\mu_F = 0.780$	$\mu_{CR} = 0.057$		
Diversity = 1.9351e+00	k_t = [3, 4, 3, 4]				
Generation 910: Best = 3.168282e+03	FES = 91100	$\mu_F = 0.766$	$\mu_{CR} = 0.054$		
Diversity = 1.9213e+00	k_t = [1, 1, 1, 1]				
Generation 920: Best = 3.167371e+03	FES = 92100	$\mu_F = 0.783$	$\mu_{CR} = 0.051$		
Diversity = 1.9038e+00	k_t = [3, 3, 3, 2]				
Generation 930: Best = 3.167292e+03	FES = 93100	$\mu_F = 0.759$	$\mu_{CR} = 0.046$		
Diversity = 1.9163e+00	k_t = [1, 1, 1, 1]				
Generation 940: Best = 3.166423e+03	FES = 94100	$\mu_F = 0.782$	$\mu_{CR} = 0.045$		
Diversity = 1.9413e+00	k_t = [2, 2, 2, 3]				
Generation 950: Best = 3.166090e+03	FES = 95100	$\mu_F = 0.766$	$\mu_{CR} = 0.047$		
Diversity = 1.9645e+00	k_t = [1, 1, 1, 1]				
Generation 960: Best = 3.166090e+03	FES = 96100	$\mu_F = 0.773$	$\mu_{CR} = 0.049$		
Diversity = 1.9421e+00	k_t = [3, 4, 2, 4]				
Generation 970: Best = 3.166090e+03	FES = 97100	$\mu_F = 0.761$	$\mu_{CR} = 0.046$		
Diversity = 1.9391e+00	k_t = [1, 1, 1, 1]				
Generation 980: Best = 3.166090e+03	FES = 98100	$\mu_F = 0.768$	$\mu_{CR} = 0.044$		
Diversity = 1.9819e+00	k_t = [2, 4, 4, 3]				
Generation 990: Best = 3.166001e+03	FES = 99100	$\mu_F = 0.746$	$\mu_{CR} = 0.039$		
Diversity = 1.9954e+00	k_t = [1, 1, 1, 1]				
Generation 1000: Best = 3.166001e+03	FES = 100100	$\mu_F = 0.727$	$\mu_{CR} = 0.047$		
Diversity = 1.9764e+00	k_t = [3, 4, 2, 4]				
Generation 1010: Best = 3.165753e+03	FES = 101100	$\mu_F = 0.707$	$\mu_{CR} = 0.050$		
Diversity = 1.9595e+00	k_t = [1, 1, 1, 1]				
Generation 1020: Best = 3.165196e+03	FES = 102100	$\mu_F = 0.724$	$\mu_{CR} = 0.043$		
Diversity = 1.9523e+00	k_t = [3, 3, 4, 2]				
Generation 1030: Best = 3.164766e+03	FES = 103100	$\mu_F = 0.724$	$\mu_{CR} = 0.044$		

Diversity = 1.9371e+00	k_t = [1, 1, 1, 1]			
Generation 1040: Best = 3.164291e+03	FES = 104100	$\mu_F = 0.713$	$\mu_{CR} = 0.048$	
Diversity = 1.9466e+00	k_t = [2, 3, 2, 3]			
Generation 1050: Best = 3.164226e+03	FES = 105100	$\mu_F = 0.717$	$\mu_{CR} = 0.052$	
Diversity = 1.9434e+00	k_t = [1, 1, 1, 1]			
Generation 1060: Best = 3.164226e+03	FES = 106100	$\mu_F = 0.742$	$\mu_{CR} = 0.054$	
Diversity = 1.9488e+00	k_t = [4, 4, 2, 2]			
Generation 1070: Best = 3.163776e+03	FES = 107100	$\mu_F = 0.717$	$\mu_{CR} = 0.052$	
Diversity = 1.9281e+00	k_t = [1, 1, 1, 1]			
Generation 1080: Best = 3.163618e+03	FES = 108100	$\mu_F = 0.751$	$\mu_{CR} = 0.047$	
Diversity = 1.9337e+00	k_t = [2, 3, 3, 2]			
Generation 1090: Best = 3.163618e+03	FES = 109100	$\mu_F = 0.812$	$\mu_{CR} = 0.051$	
Diversity = 1.9254e+00	k_t = [1, 1, 1, 1]			
Generation 1100: Best = 3.163618e+03	FES = 110100	$\mu_F = 0.772$	$\mu_{CR} = 0.048$	
Diversity = 1.9253e+00	k_t = [4, 3, 2, 2]			
Generation 1110: Best = 3.163598e+03	FES = 111100	$\mu_F = 0.756$	$\mu_{CR} = 0.041$	
Diversity = 1.9437e+00	k_t = [1, 1, 1, 1]			
Generation 1120: Best = 3.162679e+03	FES = 112100	$\mu_F = 0.744$	$\mu_{CR} = 0.042$	
Diversity = 1.9286e+00	k_t = [3, 2, 2, 3]			
Generation 1130: Best = 3.162679e+03	FES = 113100	$\mu_F = 0.752$	$\mu_{CR} = 0.049$	
Diversity = 1.9367e+00	k_t = [1, 1, 1, 1]			
Generation 1140: Best = 3.162679e+03	FES = 114100	$\mu_F = 0.752$	$\mu_{CR} = 0.046$	
Diversity = 1.9131e+00	k_t = [3, 4, 2, 3]			
Generation 1150: Best = 3.162392e+03	FES = 115100	$\mu_F = 0.756$	$\mu_{CR} = 0.047$	
Diversity = 1.8935e+00	k_t = [1, 1, 1, 1]			
Generation 1160: Best = 3.162146e+03	FES = 116100	$\mu_F = 0.768$	$\mu_{CR} = 0.042$	
Diversity = 1.8677e+00	k_t = [4, 3, 4, 4]			
Generation 1170: Best = 3.162098e+03	FES = 117100	$\mu_F = 0.779$	$\mu_{CR} = 0.033$	
Diversity = 1.8682e+00	k_t = [1, 1, 1, 1]			
Generation 1180: Best = 3.161750e+03	FES = 118100	$\mu_F = 0.812$	$\mu_{CR} = 0.038$	
Diversity = 1.8523e+00	k_t = [2, 4, 2, 4]			
Generation 1190: Best = 3.161750e+03	FES = 119100	$\mu_F = 0.788$	$\mu_{CR} = 0.038$	
Diversity = 1.8432e+00	k_t = [1, 1, 1, 1]			
Generation 1200: Best = 3.161644e+03	FES = 120100	$\mu_F = 0.726$	$\mu_{CR} = 0.044$	
Diversity = 1.8542e+00	k_t = [4, 3, 2, 4]			
Generation 1210: Best = 3.161531e+03	FES = 121100	$\mu_F = 0.739$	$\mu_{CR} = 0.044$	
Diversity = 1.8533e+00	k_t = [1, 1, 1, 1]			
Generation 1220: Best = 3.161531e+03	FES = 122100	$\mu_F = 0.754$	$\mu_{CR} = 0.038$	
Diversity = 1.8337e+00	k_t = [2, 3, 3, 4]			
Generation 1230: Best = 3.161281e+03	FES = 123100	$\mu_F = 0.748$	$\mu_{CR} = 0.044$	
Diversity = 1.8206e+00	k_t = [1, 1, 1, 1]			
Generation 1240: Best = 3.161281e+03	FES = 124100	$\mu_F = 0.777$	$\mu_{CR} = 0.040$	
Diversity = 1.8075e+00	k_t = [3, 3, 2, 4]			
Generation 1250: Best = 3.161281e+03	FES = 125100	$\mu_F = 0.741$	$\mu_{CR} = 0.039$	
Diversity = 1.7908e+00	k_t = [1, 1, 1, 1]			
Generation 1260: Best = 3.161281e+03	FES = 126100	$\mu_F = 0.756$	$\mu_{CR} = 0.037$	
Diversity = 1.7716e+00	k_t = [4, 4, 3, 4]			
Generation 1270: Best = 3.160166e+03	FES = 127100	$\mu_F = 0.739$	$\mu_{CR} = 0.038$	
Diversity = 1.7694e+00	k_t = [1, 1, 1, 1]			
Generation 1280: Best = 3.159897e+03	FES = 128100	$\mu_F = 0.768$	$\mu_{CR} = 0.038$	
Diversity = 1.7498e+00	k_t = [2, 3, 2, 4]			
Generation 1290: Best = 3.159313e+03	FES = 129100	$\mu_F = 0.750$	$\mu_{CR} = 0.042$	
Diversity = 1.7307e+00	k_t = [1, 1, 1, 1]			
Generation 1300: Best = 3.158592e+03	FES = 130100	$\mu_F = 0.751$	$\mu_{CR} = 0.043$	

Diversity = 1.7272e+00	k_t = [2, 3, 3, 2]			
Generation 1310: Best = 3.158547e+03	FES = 131100	$\mu_F = 0.729$	$\mu_{CR} = 0.035$	
Diversity = 1.7253e+00	k_t = [1, 1, 1, 1]			
Generation 1320: Best = 3.158517e+03	FES = 132100	$\mu_F = 0.708$	$\mu_{CR} = 0.029$	
Diversity = 1.7246e+00	k_t = [4, 4, 3, 2]			
Generation 1330: Best = 3.158517e+03	FES = 133100	$\mu_F = 0.696$	$\mu_{CR} = 0.030$	
Diversity = 1.7248e+00	k_t = [1, 1, 1, 1]			
Generation 1340: Best = 3.158501e+03	FES = 134100	$\mu_F = 0.700$	$\mu_{CR} = 0.030$	
Diversity = 1.7220e+00	k_t = [3, 3, 2, 4]			
Generation 1350: Best = 3.158477e+03	FES = 135100	$\mu_F = 0.698$	$\mu_{CR} = 0.032$	
Diversity = 1.7245e+00	k_t = [1, 1, 1, 1]			
Generation 1360: Best = 3.158477e+03	FES = 136100	$\mu_F = 0.677$	$\mu_{CR} = 0.038$	
Diversity = 1.7219e+00	k_t = [2, 3, 3, 3]			
Generation 1370: Best = 3.158477e+03	FES = 137100	$\mu_F = 0.701$	$\mu_{CR} = 0.037$	
Diversity = 1.7183e+00	k_t = [1, 1, 1, 1]			
Generation 1380: Best = 3.158477e+03	FES = 138100	$\mu_F = 0.692$	$\mu_{CR} = 0.034$	
Diversity = 1.7162e+00	k_t = [4, 2, 2, 2]			
Generation 1390: Best = 3.158477e+03	FES = 139100	$\mu_F = 0.647$	$\mu_{CR} = 0.037$	
Diversity = 1.7121e+00	k_t = [1, 1, 1, 1]			
Generation 1400: Best = 3.158463e+03	FES = 140100	$\mu_F = 0.656$	$\mu_{CR} = 0.041$	
Diversity = 1.6918e+00	k_t = [4, 3, 2, 4]			
Generation 1410: Best = 3.158408e+03	FES = 141100	$\mu_F = 0.642$	$\mu_{CR} = 0.036$	
Diversity = 1.6861e+00	k_t = [1, 1, 1, 1]			
Generation 1420: Best = 3.158361e+03	FES = 142100	$\mu_F = 0.618$	$\mu_{CR} = 0.032$	
Diversity = 1.6834e+00	k_t = [3, 2, 3, 2]			
Generation 1430: Best = 3.158361e+03	FES = 143100	$\mu_F = 0.646$	$\mu_{CR} = 0.027$	
Diversity = 1.6647e+00	k_t = [1, 1, 1, 1]			
Generation 1440: Best = 3.158327e+03	FES = 144100	$\mu_F = 0.704$	$\mu_{CR} = 0.041$	
Diversity = 1.6631e+00	k_t = [3, 3, 2, 2]			
Generation 1450: Best = 3.158327e+03	FES = 145100	$\mu_F = 0.744$	$\mu_{CR} = 0.042$	
Diversity = 1.6626e+00	k_t = [1, 1, 1, 1]			
Generation 1460: Best = 3.158327e+03	FES = 146100	$\mu_F = 0.778$	$\mu_{CR} = 0.037$	
Diversity = 1.6618e+00	k_t = [3, 3, 2, 3]			
Generation 1470: Best = 3.158122e+03	FES = 147100	$\mu_F = 0.766$	$\mu_{CR} = 0.038$	
Diversity = 1.6652e+00	k_t = [1, 1, 1, 1]			
Generation 1480: Best = 3.158122e+03	FES = 148100	$\mu_F = 0.799$	$\mu_{CR} = 0.038$	
Diversity = 1.6809e+00	k_t = [2, 2, 2, 3]			
Generation 1490: Best = 3.157810e+03	FES = 149100	$\mu_F = 0.784$	$\mu_{CR} = 0.032$	
Diversity = 1.6756e+00	k_t = [1, 1, 1, 1]			
Generation 1500: Best = 3.157810e+03	FES = 150100	$\mu_F = 0.790$	$\mu_{CR} = 0.033$	
Diversity = 1.6583e+00	k_t = [4, 4, 3, 4]			
Generation 1510: Best = 3.157810e+03	FES = 151100	$\mu_F = 0.774$	$\mu_{CR} = 0.039$	
Diversity = 1.6371e+00	k_t = [1, 1, 1, 1]			
Generation 1520: Best = 3.157735e+03	FES = 152100	$\mu_F = 0.742$	$\mu_{CR} = 0.033$	
Diversity = 1.6120e+00	k_t = [4, 4, 2, 4]			
Generation 1530: Best = 3.157735e+03	FES = 153100	$\mu_F = 0.779$	$\mu_{CR} = 0.037$	
Diversity = 1.6063e+00	k_t = [1, 1, 1, 1]			
Generation 1540: Best = 3.157735e+03	FES = 154100	$\mu_F = 0.789$	$\mu_{CR} = 0.032$	
Diversity = 1.5998e+00	k_t = [2, 2, 3, 4]			
Generation 1550: Best = 3.157611e+03	FES = 155100	$\mu_F = 0.744$	$\mu_{CR} = 0.030$	
Diversity = 1.5993e+00	k_t = [1, 1, 1, 1]			
Generation 1560: Best = 3.157611e+03	FES = 156100	$\mu_F = 0.783$	$\mu_{CR} = 0.029$	
Diversity = 1.5987e+00	k_t = [2, 4, 4, 2]			
Generation 1570: Best = 3.157589e+03	FES = 157100	$\mu_F = 0.804$	$\mu_{CR} = 0.026$	

Diversity = 1.6033e+00	k_t = [1, 1, 1, 1]			
Generation 1580: Best = 3.157589e+03	FES = 158100	$\mu_F = 0.815$	$\mu_{CR} = 0.028$	
Diversity = 1.6005e+00	k_t = [4, 2, 2, 2]			
Generation 1590: Best = 3.157406e+03	FES = 159100	$\mu_F = 0.842$	$\mu_{CR} = 0.032$	
Diversity = 1.6172e+00	k_t = [1, 1, 1, 1]			
Generation 1600: Best = 3.157406e+03	FES = 160100	$\mu_F = 0.863$	$\mu_{CR} = 0.030$	
Diversity = 1.5932e+00	k_t = [3, 2, 2, 4]			
Generation 1610: Best = 3.157406e+03	FES = 161100	$\mu_F = 0.834$	$\mu_{CR} = 0.036$	
Diversity = 1.5922e+00	k_t = [1, 1, 1, 1]			
Generation 1620: Best = 3.157406e+03	FES = 162100	$\mu_F = 0.829$	$\mu_{CR} = 0.029$	
Diversity = 1.5798e+00	k_t = [3, 3, 4, 2]			
Generation 1630: Best = 3.157406e+03	FES = 163100	$\mu_F = 0.833$	$\mu_{CR} = 0.033$	
Diversity = 1.5799e+00	k_t = [1, 1, 1, 1]			
Generation 1640: Best = 3.157406e+03	FES = 164100	$\mu_F = 0.815$	$\mu_{CR} = 0.039$	
Diversity = 1.5789e+00	k_t = [4, 4, 2, 2]			
Generation 1650: Best = 3.157406e+03	FES = 165100	$\mu_F = 0.788$	$\mu_{CR} = 0.038$	
Diversity = 1.5800e+00	k_t = [1, 1, 1, 1]			
Generation 1660: Best = 3.157351e+03	FES = 166100	$\mu_F = 0.822$	$\mu_{CR} = 0.033$	
Diversity = 1.5616e+00	k_t = [4, 4, 2, 2]			
Generation 1670: Best = 3.157351e+03	FES = 167100	$\mu_F = 0.844$	$\mu_{CR} = 0.036$	
Diversity = 1.5515e+00	k_t = [1, 1, 1, 1]			
Generation 1680: Best = 3.156771e+03	FES = 168100	$\mu_F = 0.821$	$\mu_{CR} = 0.041$	
Diversity = 1.5432e+00	k_t = [3, 3, 2, 3]			
Generation 1690: Best = 3.156771e+03	FES = 169100	$\mu_F = 0.790$	$\mu_{CR} = 0.037$	
Diversity = 1.5413e+00	k_t = [1, 1, 1, 1]			
Generation 1700: Best = 3.156771e+03	FES = 170100	$\mu_F = 0.816$	$\mu_{CR} = 0.036$	
Diversity = 1.5503e+00	k_t = [3, 3, 3, 3]			
Generation 1710: Best = 3.156766e+03	FES = 171100	$\mu_F = 0.871$	$\mu_{CR} = 0.026$	
Diversity = 1.5657e+00	k_t = [1, 1, 1, 1]			
Generation 1720: Best = 3.156766e+03	FES = 172100	$\mu_F = 0.856$	$\mu_{CR} = 0.038$	
Diversity = 1.5428e+00	k_t = [4, 2, 2, 2]			
Generation 1730: Best = 3.156728e+03	FES = 173100	$\mu_F = 0.857$	$\mu_{CR} = 0.028$	
Diversity = 1.5536e+00	k_t = [1, 1, 1, 1]			
Generation 1740: Best = 3.156728e+03	FES = 174100	$\mu_F = 0.841$	$\mu_{CR} = 0.028$	
Diversity = 1.5521e+00	k_t = [4, 3, 4, 2]			
Generation 1750: Best = 3.156676e+03	FES = 175100	$\mu_F = 0.836$	$\mu_{CR} = 0.030$	
Diversity = 1.5497e+00	k_t = [1, 1, 1, 1]			
Generation 1760: Best = 3.156658e+03	FES = 176100	$\mu_F = 0.855$	$\mu_{CR} = 0.031$	
Diversity = 1.5346e+00	k_t = [3, 4, 4, 3]			
Generation 1770: Best = 3.156658e+03	FES = 177100	$\mu_F = 0.866$	$\mu_{CR} = 0.040$	
Diversity = 1.5295e+00	k_t = [1, 1, 1, 1]			
Generation 1780: Best = 3.156658e+03	FES = 178100	$\mu_F = 0.793$	$\mu_{CR} = 0.044$	
Diversity = 1.4961e+00	k_t = [2, 2, 4, 4]			
Generation 1790: Best = 3.156658e+03	FES = 179100	$\mu_F = 0.793$	$\mu_{CR} = 0.036$	
Diversity = 1.5142e+00	k_t = [1, 1, 1, 1]			
Generation 1800: Best = 3.156658e+03	FES = 180100	$\mu_F = 0.758$	$\mu_{CR} = 0.042$	
Diversity = 1.5077e+00	k_t = [3, 3, 3, 2]			
Generation 1810: Best = 3.156658e+03	FES = 181100	$\mu_F = 0.765$	$\mu_{CR} = 0.044$	
Diversity = 1.5045e+00	k_t = [1, 1, 1, 1]			
Generation 1820: Best = 3.156373e+03	FES = 182100	$\mu_F = 0.765$	$\mu_{CR} = 0.030$	
Diversity = 1.5055e+00	k_t = [2, 4, 3, 2]			
Generation 1830: Best = 3.156373e+03	FES = 183100	$\mu_F = 0.797$	$\mu_{CR} = 0.033$	
Diversity = 1.5060e+00	k_t = [1, 1, 1, 1]			
Generation 1840: Best = 3.156373e+03	FES = 184100	$\mu_F = 0.699$	$\mu_{CR} = 0.036$	

Diversity = 1.5044e+00	k_t = [3, 4, 3, 3]			
Generation 1850: Best = 3.156373e+03	FES = 185100	$\mu_F = 0.683$	$\mu_{CR} = 0.030$	
Diversity = 1.4951e+00	k_t = [1, 1, 1, 1]			
Generation 1860: Best = 3.156370e+03	FES = 186100	$\mu_F = 0.750$	$\mu_{CR} = 0.029$	
Diversity = 1.5109e+00	k_t = [3, 3, 2, 2]			
Generation 1870: Best = 3.156370e+03	FES = 187100	$\mu_F = 0.741$	$\mu_{CR} = 0.043$	
Diversity = 1.5036e+00	k_t = [1, 1, 1, 1]			
Generation 1880: Best = 3.156370e+03	FES = 188100	$\mu_F = 0.771$	$\mu_{CR} = 0.035$	
Diversity = 1.5027e+00	k_t = [3, 4, 3, 4]			
Generation 1890: Best = 3.156075e+03	FES = 189100	$\mu_F = 0.770$	$\mu_{CR} = 0.028$	
Diversity = 1.5062e+00	k_t = [1, 1, 1, 1]			
Generation 1900: Best = 3.156075e+03	FES = 190100	$\mu_F = 0.727$	$\mu_{CR} = 0.031$	
Diversity = 1.5047e+00	k_t = [4, 4, 2, 3]			
Generation 1910: Best = 3.156075e+03	FES = 191100	$\mu_F = 0.650$	$\mu_{CR} = 0.036$	
Diversity = 1.4994e+00	k_t = [1, 1, 1, 1]			
Generation 1920: Best = 3.156075e+03	FES = 192100	$\mu_F = 0.662$	$\mu_{CR} = 0.034$	
Diversity = 1.5030e+00	k_t = [4, 3, 3, 3]			
Generation 1930: Best = 3.156075e+03	FES = 193100	$\mu_F = 0.668$	$\mu_{CR} = 0.034$	
Diversity = 1.4872e+00	k_t = [1, 1, 1, 1]			
Generation 1940: Best = 3.156075e+03	FES = 194100	$\mu_F = 0.642$	$\mu_{CR} = 0.030$	
Diversity = 1.4846e+00	k_t = [4, 4, 4, 2]			
Generation 1950: Best = 3.156075e+03	FES = 195100	$\mu_F = 0.700$	$\mu_{CR} = 0.027$	
Diversity = 1.4857e+00	k_t = [1, 1, 1, 1]			
Generation 1960: Best = 3.156075e+03	FES = 196100	$\mu_F = 0.743$	$\mu_{CR} = 0.035$	
Diversity = 1.4985e+00	k_t = [2, 2, 4, 3]			
Generation 1970: Best = 3.156075e+03	FES = 197100	$\mu_F = 0.748$	$\mu_{CR} = 0.026$	
Diversity = 1.4903e+00	k_t = [1, 1, 1, 1]			
Generation 1980: Best = 3.156073e+03	FES = 198100	$\mu_F = 0.786$	$\mu_{CR} = 0.029$	
Diversity = 1.4925e+00	k_t = [3, 3, 3, 3]			
Generation 1990: Best = 3.155882e+03	FES = 199100	$\mu_F = 0.740$	$\mu_{CR} = 0.030$	
Diversity = 1.4861e+00	k_t = [1, 1, 1, 1]			
Generation 2000: Best = 3.155882e+03	FES = 200100	$\mu_F = 0.759$	$\mu_{CR} = 0.037$	
Diversity = 1.4967e+00	k_t = [4, 4, 4, 4]			
Generation 2010: Best = 3.155882e+03	FES = 201100	$\mu_F = 0.760$	$\mu_{CR} = 0.034$	
Diversity = 1.4905e+00	k_t = [1, 1, 1, 1]			
Generation 2020: Best = 3.155803e+03	FES = 202100	$\mu_F = 0.755$	$\mu_{CR} = 0.036$	
Diversity = 1.4835e+00	k_t = [3, 3, 2, 3]			
Generation 2030: Best = 3.155803e+03	FES = 203100	$\mu_F = 0.741$	$\mu_{CR} = 0.033$	
Diversity = 1.4829e+00	k_t = [1, 1, 1, 1]			
Generation 2040: Best = 3.155803e+03	FES = 204100	$\mu_F = 0.779$	$\mu_{CR} = 0.037$	
Diversity = 1.4854e+00	k_t = [4, 3, 3, 2]			
Generation 2050: Best = 3.155803e+03	FES = 205100	$\mu_F = 0.712$	$\mu_{CR} = 0.037$	
Diversity = 1.5007e+00	k_t = [1, 1, 1, 1]			
Generation 2060: Best = 3.155625e+03	FES = 206100	$\mu_F = 0.752$	$\mu_{CR} = 0.039$	
Diversity = 1.4957e+00	k_t = [3, 2, 4, 4]			
Generation 2070: Best = 3.155518e+03	FES = 207100	$\mu_F = 0.765$	$\mu_{CR} = 0.029$	
Diversity = 1.4969e+00	k_t = [1, 1, 1, 1]			
Generation 2080: Best = 3.155518e+03	FES = 208100	$\mu_F = 0.762$	$\mu_{CR} = 0.039$	
Diversity = 1.4693e+00	k_t = [4, 2, 4, 4]			
Generation 2090: Best = 3.155346e+03	FES = 209100	$\mu_F = 0.811$	$\mu_{CR} = 0.032$	
Diversity = 1.4710e+00	k_t = [1, 1, 1, 1]			
Generation 2100: Best = 3.155346e+03	FES = 210100	$\mu_F = 0.796$	$\mu_{CR} = 0.031$	
Diversity = 1.4549e+00	k_t = [2, 3, 4, 2]			
Generation 2110: Best = 3.155346e+03	FES = 211100	$\mu_F = 0.786$	$\mu_{CR} = 0.032$	

Diversity = 1.4573e+00	k_t = [1, 1, 1, 1]			
Generation 2120: Best = 3.155344e+03	FES = 212100	$\mu_F = 0.801$	$\mu_{CR} = 0.042$	
Diversity = 1.4353e+00	k_t = [3, 2, 3, 4]			
Generation 2130: Best = 3.155323e+03	FES = 213100	$\mu_F = 0.802$	$\mu_{CR} = 0.034$	
Diversity = 1.4411e+00	k_t = [1, 1, 1, 1]			
Generation 2140: Best = 3.155286e+03	FES = 214100	$\mu_F = 0.847$	$\mu_{CR} = 0.030$	
Diversity = 1.4266e+00	k_t = [3, 2, 2, 2]			
Generation 2150: Best = 3.155242e+03	FES = 215100	$\mu_F = 0.822$	$\mu_{CR} = 0.035$	
Diversity = 1.3993e+00	k_t = [1, 1, 1, 1]			
Generation 2160: Best = 3.155242e+03	FES = 216100	$\mu_F = 0.815$	$\mu_{CR} = 0.033$	
Diversity = 1.4032e+00	k_t = [2, 3, 3, 3]			
Generation 2170: Best = 3.155242e+03	FES = 217100	$\mu_F = 0.830$	$\mu_{CR} = 0.030$	
Diversity = 1.4157e+00	k_t = [1, 1, 1, 1]			
Generation 2180: Best = 3.155242e+03	FES = 218100	$\mu_F = 0.852$	$\mu_{CR} = 0.028$	
Diversity = 1.4119e+00	k_t = [3, 3, 4, 2]			
Generation 2190: Best = 3.155242e+03	FES = 219100	$\mu_F = 0.827$	$\mu_{CR} = 0.033$	
Diversity = 1.3963e+00	k_t = [1, 1, 1, 1]			
Generation 2200: Best = 3.155232e+03	FES = 220100	$\mu_F = 0.828$	$\mu_{CR} = 0.033$	
Diversity = 1.3966e+00	k_t = [2, 2, 4, 2]			
Generation 2210: Best = 3.155131e+03	FES = 221100	$\mu_F = 0.820$	$\mu_{CR} = 0.031$	
Diversity = 1.3970e+00	k_t = [1, 1, 1, 1]			
Generation 2220: Best = 3.155105e+03	FES = 222100	$\mu_F = 0.805$	$\mu_{CR} = 0.036$	
Diversity = 1.3948e+00	k_t = [3, 4, 2, 3]			
Generation 2230: Best = 3.155105e+03	FES = 223100	$\mu_F = 0.737$	$\mu_{CR} = 0.040$	
Diversity = 1.3941e+00	k_t = [1, 1, 1, 1]			
Generation 2240: Best = 3.154991e+03	FES = 224100	$\mu_F = 0.751$	$\mu_{CR} = 0.035$	
Diversity = 1.3861e+00	k_t = [4, 4, 4, 2]			
Generation 2250: Best = 3.154924e+03	FES = 225100	$\mu_F = 0.784$	$\mu_{CR} = 0.039$	
Diversity = 1.3695e+00	k_t = [1, 1, 1, 1]			
Generation 2260: Best = 3.154916e+03	FES = 226100	$\mu_F = 0.784$	$\mu_{CR} = 0.036$	
Diversity = 1.3727e+00	k_t = [2, 4, 4, 3]			
Generation 2270: Best = 3.154905e+03	FES = 227100	$\mu_F = 0.771$	$\mu_{CR} = 0.038$	
Diversity = 1.3610e+00	k_t = [1, 1, 1, 1]			
Generation 2280: Best = 3.154905e+03	FES = 228100	$\mu_F = 0.805$	$\mu_{CR} = 0.039$	
Diversity = 1.3375e+00	k_t = [3, 2, 4, 3]			
Generation 2290: Best = 3.154885e+03	FES = 229100	$\mu_F = 0.807$	$\mu_{CR} = 0.039$	
Diversity = 1.3397e+00	k_t = [1, 1, 1, 1]			
Generation 2300: Best = 3.154560e+03	FES = 230100	$\mu_F = 0.822$	$\mu_{CR} = 0.043$	
Diversity = 1.3245e+00	k_t = [2, 4, 2, 4]			
Generation 2310: Best = 3.154396e+03	FES = 231100	$\mu_F = 0.809$	$\mu_{CR} = 0.043$	
Diversity = 1.3134e+00	k_t = [1, 1, 1, 1]			
Generation 2320: Best = 3.154396e+03	FES = 232100	$\mu_F = 0.802$	$\mu_{CR} = 0.050$	
Diversity = 1.3112e+00	k_t = [3, 4, 3, 2]			
Generation 2330: Best = 3.154340e+03	FES = 233100	$\mu_F = 0.809$	$\mu_{CR} = 0.051$	
Diversity = 1.3108e+00	k_t = [1, 1, 1, 1]			
Generation 2340: Best = 3.154201e+03	FES = 234100	$\mu_F = 0.745$	$\mu_{CR} = 0.039$	
Diversity = 1.2936e+00	k_t = [2, 3, 3, 2]			
Generation 2350: Best = 3.154201e+03	FES = 235100	$\mu_F = 0.764$	$\mu_{CR} = 0.041$	
Diversity = 1.2888e+00	k_t = [1, 1, 1, 1]			
Generation 2360: Best = 3.154178e+03	FES = 236100	$\mu_F = 0.735$	$\mu_{CR} = 0.053$	
Diversity = 1.2838e+00	k_t = [4, 3, 4, 2]			
Generation 2370: Best = 3.153980e+03	FES = 237100	$\mu_F = 0.764$	$\mu_{CR} = 0.047$	
Diversity = 1.2852e+00	k_t = [1, 1, 1, 1]			
Generation 2380: Best = 3.153966e+03	FES = 238100	$\mu_F = 0.733$	$\mu_{CR} = 0.040$	

Diversity = 1.2704e+00	k_t = [4, 3, 2, 2]			
Generation 2390: Best = 3.153966e+03	FES = 239100	$\mu_F = 0.730$	$\mu_{CR} = 0.042$	
Diversity = 1.2669e+00	k_t = [1, 1, 1, 1]			
Generation 2400: Best = 3.153859e+03	FES = 240100	$\mu_F = 0.729$	$\mu_{CR} = 0.054$	
Diversity = 1.2657e+00	k_t = [4, 4, 3, 2]			
Generation 2410: Best = 3.153464e+03	FES = 241100	$\mu_F = 0.701$	$\mu_{CR} = 0.066$	
Diversity = 1.2572e+00	k_t = [1, 1, 1, 1]			
Generation 2420: Best = 3.153253e+03	FES = 242100	$\mu_F = 0.741$	$\mu_{CR} = 0.060$	
Diversity = 1.2730e+00	k_t = [4, 3, 4, 2]			
Generation 2430: Best = 3.153160e+03	FES = 243100	$\mu_F = 0.744$	$\mu_{CR} = 0.062$	
Diversity = 1.2556e+00	k_t = [1, 1, 1, 1]			
Generation 2440: Best = 3.153119e+03	FES = 244100	$\mu_F = 0.745$	$\mu_{CR} = 0.054$	
Diversity = 1.2281e+00	k_t = [3, 3, 3, 4]			
Generation 2450: Best = 3.153119e+03	FES = 245100	$\mu_F = 0.762$	$\mu_{CR} = 0.045$	
Diversity = 1.2147e+00	k_t = [1, 1, 1, 1]			
Generation 2460: Best = 3.152995e+03	FES = 246100	$\mu_F = 0.791$	$\mu_{CR} = 0.054$	
Diversity = 1.1866e+00	k_t = [2, 4, 4, 4]			
Generation 2470: Best = 3.152966e+03	FES = 247100	$\mu_F = 0.808$	$\mu_{CR} = 0.056$	
Diversity = 1.1698e+00	k_t = [1, 1, 1, 1]			
Generation 2480: Best = 3.152958e+03	FES = 248100	$\mu_F = 0.807$	$\mu_{CR} = 0.052$	
Diversity = 1.1600e+00	k_t = [3, 4, 3, 4]			
Generation 2490: Best = 3.152903e+03	FES = 249100	$\mu_F = 0.798$	$\mu_{CR} = 0.054$	
Diversity = 1.1531e+00	k_t = [1, 1, 1, 1]			
Generation 2500: Best = 3.152622e+03	FES = 250100	$\mu_F = 0.794$	$\mu_{CR} = 0.060$	
Diversity = 1.1294e+00	k_t = [3, 2, 4, 4]			
Generation 2510: Best = 3.152262e+03	FES = 251100	$\mu_F = 0.810$	$\mu_{CR} = 0.055$	
Diversity = 1.1011e+00	k_t = [1, 1, 1, 1]			
Generation 2520: Best = 3.152208e+03	FES = 252100	$\mu_F = 0.810$	$\mu_{CR} = 0.061$	
Diversity = 1.0488e+00	k_t = [2, 4, 4, 4]			
Generation 2530: Best = 3.152196e+03	FES = 253100	$\mu_F = 0.796$	$\mu_{CR} = 0.067$	
Diversity = 9.8102e-01	k_t = [1, 1, 1, 1]			
Generation 2540: Best = 3.152173e+03	FES = 254100	$\mu_F = 0.812$	$\mu_{CR} = 0.056$	
Diversity = 9.5239e-01	k_t = [4, 3, 2, 3]			
Generation 2550: Best = 3.152173e+03	FES = 255100	$\mu_F = 0.804$	$\mu_{CR} = 0.052$	
Diversity = 9.2885e-01	k_t = [1, 1, 1, 1]			
Generation 2560: Best = 3.152151e+03	FES = 256100	$\mu_F = 0.774$	$\mu_{CR} = 0.057$	
Diversity = 9.2689e-01	k_t = [2, 2, 4, 2]			
Generation 2570: Best = 3.152151e+03	FES = 257100	$\mu_F = 0.752$	$\mu_{CR} = 0.057$	
Diversity = 8.9432e-01	k_t = [1, 1, 1, 1]			
Generation 2580: Best = 3.151785e+03	FES = 258100	$\mu_F = 0.773$	$\mu_{CR} = 0.058$	
Diversity = 8.3282e-01	k_t = [4, 4, 4, 4]			
Generation 2590: Best = 3.151785e+03	FES = 259100	$\mu_F = 0.779$	$\mu_{CR} = 0.054$	
Diversity = 7.9810e-01	k_t = [1, 1, 1, 1]			
Generation 2600: Best = 3.151785e+03	FES = 260100	$\mu_F = 0.766$	$\mu_{CR} = 0.057$	
Diversity = 7.6905e-01	k_t = [3, 4, 4, 4]			
Generation 2610: Best = 3.151785e+03	FES = 261100	$\mu_F = 0.770$	$\mu_{CR} = 0.059$	
Diversity = 7.7527e-01	k_t = [1, 1, 1, 1]			
Generation 2620: Best = 3.151782e+03	FES = 262100	$\mu_F = 0.769$	$\mu_{CR} = 0.055$	
Diversity = 7.7638e-01	k_t = [4, 3, 4, 2]			
Generation 2630: Best = 3.151782e+03	FES = 263100	$\mu_F = 0.751$	$\mu_{CR} = 0.060$	
Diversity = 7.6162e-01	k_t = [1, 1, 1, 1]			
Generation 2640: Best = 3.151685e+03	FES = 264100	$\mu_F = 0.745$	$\mu_{CR} = 0.062$	
Diversity = 7.5389e-01	k_t = [4, 4, 2, 3]			
Generation 2650: Best = 3.151641e+03	FES = 265100	$\mu_F = 0.705$	$\mu_{CR} = 0.064$	

Diversity = 7.4260e-01	k_t = [1, 1, 1, 1]			
Generation 2660: Best = 3.151641e+03	FES = 266100	$\mu_F = 0.700$	$\mu_{CR} = 0.060$	
Diversity = 7.1277e-01	k_t = [2, 4, 4, 4]			
Generation 2670: Best = 3.151600e+03	FES = 267100	$\mu_F = 0.739$	$\mu_{CR} = 0.057$	
Diversity = 7.0639e-01	k_t = [1, 1, 1, 1]			
Generation 2680: Best = 3.151491e+03	FES = 268100	$\mu_F = 0.770$	$\mu_{CR} = 0.060$	
Diversity = 6.8508e-01	k_t = [4, 4, 2, 4]			
Generation 2690: Best = 3.151426e+03	FES = 269100	$\mu_F = 0.772$	$\mu_{CR} = 0.051$	
Diversity = 6.8128e-01	k_t = [1, 1, 1, 1]			
Generation 2700: Best = 3.151326e+03	FES = 270100	$\mu_F = 0.767$	$\mu_{CR} = 0.045$	
Diversity = 6.7032e-01	k_t = [3, 2, 3, 2]			
Generation 2710: Best = 3.151244e+03	FES = 271100	$\mu_F = 0.774$	$\mu_{CR} = 0.053$	
Diversity = 6.7127e-01	k_t = [1, 1, 1, 1]			
Generation 2720: Best = 3.151188e+03	FES = 272100	$\mu_F = 0.809$	$\mu_{CR} = 0.051$	
Diversity = 6.5627e-01	k_t = [2, 2, 3, 3]			
Generation 2730: Best = 3.151173e+03	FES = 273100	$\mu_F = 0.775$	$\mu_{CR} = 0.058$	
Diversity = 6.4917e-01	k_t = [1, 1, 1, 1]			
Generation 2740: Best = 3.151173e+03	FES = 274100	$\mu_F = 0.770$	$\mu_{CR} = 0.061$	
Diversity = 6.2829e-01	k_t = [3, 2, 4, 4]			
Generation 2750: Best = 3.151173e+03	FES = 275100	$\mu_F = 0.779$	$\mu_{CR} = 0.055$	
Diversity = 6.2977e-01	k_t = [1, 1, 1, 1]			
Generation 2760: Best = 3.151145e+03	FES = 276100	$\mu_F = 0.791$	$\mu_{CR} = 0.052$	
Diversity = 6.2688e-01	k_t = [3, 2, 4, 3]			
Generation 2770: Best = 3.151053e+03	FES = 277100	$\mu_F = 0.787$	$\mu_{CR} = 0.060$	
Diversity = 6.2494e-01	k_t = [1, 1, 1, 1]			
Generation 2780: Best = 3.150974e+03	FES = 278100	$\mu_F = 0.774$	$\mu_{CR} = 0.047$	
Diversity = 6.2863e-01	k_t = [4, 4, 3, 3]			
Generation 2790: Best = 3.150883e+03	FES = 279100	$\mu_F = 0.768$	$\mu_{CR} = 0.050$	
Diversity = 6.2237e-01	k_t = [1, 1, 1, 1]			
Generation 2800: Best = 3.150883e+03	FES = 280100	$\mu_F = 0.807$	$\mu_{CR} = 0.056$	
Diversity = 6.2071e-01	k_t = [3, 4, 3, 2]			
Generation 2810: Best = 3.150883e+03	FES = 281100	$\mu_F = 0.772$	$\mu_{CR} = 0.058$	
Diversity = 6.1223e-01	k_t = [1, 1, 1, 1]			
Generation 2820: Best = 3.150883e+03	FES = 282100	$\mu_F = 0.715$	$\mu_{CR} = 0.057$	
Diversity = 5.9612e-01	k_t = [4, 3, 4, 4]			
Generation 2830: Best = 3.150817e+03	FES = 283100	$\mu_F = 0.722$	$\mu_{CR} = 0.049$	
Diversity = 5.9265e-01	k_t = [1, 1, 1, 1]			
Generation 2840: Best = 3.150803e+03	FES = 284100	$\mu_F = 0.708$	$\mu_{CR} = 0.052$	
Diversity = 5.7702e-01	k_t = [4, 2, 2, 4]			
Generation 2850: Best = 3.150790e+03	FES = 285100	$\mu_F = 0.709$	$\mu_{CR} = 0.045$	
Diversity = 5.7733e-01	k_t = [1, 1, 1, 1]			
Generation 2860: Best = 3.150719e+03	FES = 286100	$\mu_F = 0.745$	$\mu_{CR} = 0.048$	
Diversity = 5.7696e-01	k_t = [4, 4, 2, 2]			
Generation 2870: Best = 3.150700e+03	FES = 287100	$\mu_F = 0.740$	$\mu_{CR} = 0.045$	
Diversity = 5.7899e-01	k_t = [1, 1, 1, 1]			
Generation 2880: Best = 3.150672e+03	FES = 288100	$\mu_F = 0.744$	$\mu_{CR} = 0.047$	
Diversity = 5.7172e-01	k_t = [2, 2, 3, 2]			
Generation 2890: Best = 3.150611e+03	FES = 289100	$\mu_F = 0.702$	$\mu_{CR} = 0.056$	
Diversity = 5.7561e-01	k_t = [1, 1, 1, 1]			
Generation 2900: Best = 3.150611e+03	FES = 290100	$\mu_F = 0.717$	$\mu_{CR} = 0.045$	
Diversity = 5.7515e-01	k_t = [3, 2, 2, 4]			
Generation 2910: Best = 3.150559e+03	FES = 291100	$\mu_F = 0.757$	$\mu_{CR} = 0.045$	
Diversity = 5.6406e-01	k_t = [1, 1, 1, 1]			
Generation 2920: Best = 3.150519e+03	FES = 292100	$\mu_F = 0.770$	$\mu_{CR} = 0.049$	

Diversity = 5.6536e-01	k_t = [2, 4, 4, 2]			
Generation 2930: Best = 3.150519e+03	FES = 293100	$\mu_F = 0.736$	$\mu_{CR} = 0.044$	
Diversity = 5.6377e-01	k_t = [1, 1, 1, 1]			
Generation 2940: Best = 3.150519e+03	FES = 294100	$\mu_F = 0.721$	$\mu_{CR} = 0.041$	
Diversity = 5.6270e-01	k_t = [4, 2, 2, 3]			
Generation 2950: Best = 3.150462e+03	FES = 295100	$\mu_F = 0.758$	$\mu_{CR} = 0.043$	
Diversity = 5.6418e-01	k_t = [1, 1, 1, 1]			
Generation 2960: Best = 3.150417e+03	FES = 296100	$\mu_F = 0.741$	$\mu_{CR} = 0.060$	
Diversity = 5.4882e-01	k_t = [4, 3, 2, 2]			
Generation 2970: Best = 3.150417e+03	FES = 297100	$\mu_F = 0.740$	$\mu_{CR} = 0.055$	
Diversity = 5.4784e-01	k_t = [1, 1, 1, 1]			
Generation 2980: Best = 3.150417e+03	FES = 298100	$\mu_F = 0.732$	$\mu_{CR} = 0.046$	
Diversity = 5.3782e-01	k_t = [4, 4, 2, 4]			
Generation 2990: Best = 3.150413e+03	FES = 299100	$\mu_F = 0.758$	$\mu_{CR} = 0.049$	
Diversity = 5.3828e-01	k_t = [1, 1, 1, 1]			
Generation 3000: Best = 3.150390e+03	FES = 300100	$\mu_F = 0.770$	$\mu_{CR} = 0.043$	
Diversity = 5.4001e-01	k_t = [4, 3, 4, 4]			
Generation 3010: Best = 3.150390e+03	FES = 301100	$\mu_F = 0.744$	$\mu_{CR} = 0.049$	
Diversity = 5.4184e-01	k_t = [1, 1, 1, 1]			
Generation 3020: Best = 3.150359e+03	FES = 302100	$\mu_F = 0.737$	$\mu_{CR} = 0.055$	
Diversity = 5.4000e-01	k_t = [2, 3, 2, 4]			
Generation 3030: Best = 3.150334e+03	FES = 303100	$\mu_F = 0.739$	$\mu_{CR} = 0.044$	
Diversity = 5.3627e-01	k_t = [1, 1, 1, 1]			
Generation 3040: Best = 3.150283e+03	FES = 304100	$\mu_F = 0.749$	$\mu_{CR} = 0.046$	
Diversity = 5.1650e-01	k_t = [4, 3, 4, 4]			
Generation 3050: Best = 3.150283e+03	FES = 305100	$\mu_F = 0.728$	$\mu_{CR} = 0.039$	
Diversity = 5.1498e-01	k_t = [1, 1, 1, 1]			
Generation 3060: Best = 3.150283e+03	FES = 306100	$\mu_F = 0.744$	$\mu_{CR} = 0.044$	
Diversity = 5.2054e-01	k_t = [4, 4, 4, 2]			
Generation 3070: Best = 3.150275e+03	FES = 307100	$\mu_F = 0.691$	$\mu_{CR} = 0.045$	
Diversity = 5.1764e-01	k_t = [1, 1, 1, 1]			
Generation 3080: Best = 3.150225e+03	FES = 308100	$\mu_F = 0.724$	$\mu_{CR} = 0.051$	
Diversity = 5.1465e-01	k_t = [4, 2, 3, 3]			
Generation 3090: Best = 3.150225e+03	FES = 309100	$\mu_F = 0.729$	$\mu_{CR} = 0.049$	
Diversity = 5.1311e-01	k_t = [1, 1, 1, 1]			
Generation 3100: Best = 3.150213e+03	FES = 310100	$\mu_F = 0.751$	$\mu_{CR} = 0.039$	
Diversity = 5.1305e-01	k_t = [2, 2, 3, 2]			
Generation 3110: Best = 3.150213e+03	FES = 311100	$\mu_F = 0.762$	$\mu_{CR} = 0.032$	
Diversity = 5.1140e-01	k_t = [1, 1, 1, 1]			
Generation 3120: Best = 3.150167e+03	FES = 312100	$\mu_F = 0.787$	$\mu_{CR} = 0.035$	
Diversity = 5.0982e-01	k_t = [4, 4, 3, 3]			
Generation 3130: Best = 3.150155e+03	FES = 313100	$\mu_F = 0.788$	$\mu_{CR} = 0.034$	
Diversity = 5.1550e-01	k_t = [1, 1, 1, 1]			
Generation 3140: Best = 3.150140e+03	FES = 314100	$\mu_F = 0.810$	$\mu_{CR} = 0.037$	
Diversity = 5.1372e-01	k_t = [2, 3, 4, 2]			
Generation 3150: Best = 3.150140e+03	FES = 315100	$\mu_F = 0.787$	$\mu_{CR} = 0.043$	
Diversity = 5.1146e-01	k_t = [1, 1, 1, 1]			
Generation 3160: Best = 3.150140e+03	FES = 316100	$\mu_F = 0.779$	$\mu_{CR} = 0.039$	
Diversity = 5.1453e-01	k_t = [4, 4, 2, 4]			
Generation 3170: Best = 3.150104e+03	FES = 317100	$\mu_F = 0.786$	$\mu_{CR} = 0.035$	
Diversity = 5.1710e-01	k_t = [1, 1, 1, 1]			
Generation 3180: Best = 3.150043e+03	FES = 318100	$\mu_F = 0.779$	$\mu_{CR} = 0.034$	
Diversity = 5.0936e-01	k_t = [4, 2, 4, 2]			
Generation 3190: Best = 3.150003e+03	FES = 319100	$\mu_F = 0.783$	$\mu_{CR} = 0.040$	

Diversity = 5.1493e-01	k_t = [1, 1, 1, 1]			
Generation 3200: Best = 3.149995e+03	FES = 320100	$\mu_F = 0.778$	$\mu_{CR} = 0.038$	
Diversity = 5.0846e-01	k_t = [4, 3, 3, 3]			
Generation 3210: Best = 3.149964e+03	FES = 321100	$\mu_F = 0.802$	$\mu_{CR} = 0.036$	
Diversity = 5.0808e-01	k_t = [1, 1, 1, 1]			
Generation 3220: Best = 3.149909e+03	FES = 322100	$\mu_F = 0.805$	$\mu_{CR} = 0.036$	
Diversity = 5.0083e-01	k_t = [2, 3, 4, 4]			
Generation 3230: Best = 3.149909e+03	FES = 323100	$\mu_F = 0.769$	$\mu_{CR} = 0.042$	
Diversity = 5.0080e-01	k_t = [1, 1, 1, 1]			
Generation 3240: Best = 3.149909e+03	FES = 324100	$\mu_F = 0.780$	$\mu_{CR} = 0.037$	
Diversity = 4.9141e-01	k_t = [4, 3, 2, 4]			
Generation 3250: Best = 3.149909e+03	FES = 325100	$\mu_F = 0.775$	$\mu_{CR} = 0.036$	
Diversity = 4.8746e-01	k_t = [1, 1, 1, 1]			
Generation 3260: Best = 3.149908e+03	FES = 326100	$\mu_F = 0.761$	$\mu_{CR} = 0.039$	
Diversity = 4.8753e-01	k_t = [2, 4, 3, 2]			
Generation 3270: Best = 3.149873e+03	FES = 327100	$\mu_F = 0.724$	$\mu_{CR} = 0.047$	
Diversity = 4.8219e-01	k_t = [1, 1, 1, 1]			
Generation 3280: Best = 3.149873e+03	FES = 328100	$\mu_F = 0.737$	$\mu_{CR} = 0.046$	
Diversity = 4.7748e-01	k_t = [4, 2, 2, 2]			
Generation 3290: Best = 3.149873e+03	FES = 329100	$\mu_F = 0.736$	$\mu_{CR} = 0.041$	
Diversity = 4.7551e-01	k_t = [1, 1, 1, 1]			
Generation 3300: Best = 3.149866e+03	FES = 330100	$\mu_F = 0.738$	$\mu_{CR} = 0.037$	
Diversity = 4.7582e-01	k_t = [2, 4, 2, 4]			
Generation 3310: Best = 3.149856e+03	FES = 331100	$\mu_F = 0.760$	$\mu_{CR} = 0.040$	
Diversity = 4.7527e-01	k_t = [1, 1, 1, 1]			
Generation 3320: Best = 3.149856e+03	FES = 332100	$\mu_F = 0.772$	$\mu_{CR} = 0.042$	
Diversity = 4.7301e-01	k_t = [4, 2, 4, 4]			
Generation 3330: Best = 3.149829e+03	FES = 333100	$\mu_F = 0.766$	$\mu_{CR} = 0.039$	
Diversity = 4.6965e-01	k_t = [1, 1, 1, 1]			
Generation 3340: Best = 3.149824e+03	FES = 334100	$\mu_F = 0.729$	$\mu_{CR} = 0.039$	
Diversity = 4.6799e-01	k_t = [2, 3, 4, 3]			
Generation 3350: Best = 3.149809e+03	FES = 335100	$\mu_F = 0.721$	$\mu_{CR} = 0.038$	
Diversity = 4.7059e-01	k_t = [1, 1, 1, 1]			
Generation 3360: Best = 3.149809e+03	FES = 336100	$\mu_F = 0.748$	$\mu_{CR} = 0.053$	
Diversity = 4.7134e-01	k_t = [2, 4, 2, 4]			
Generation 3370: Best = 3.149791e+03	FES = 337100	$\mu_F = 0.745$	$\mu_{CR} = 0.043$	
Diversity = 4.6361e-01	k_t = [1, 1, 1, 1]			
Generation 3380: Best = 3.149791e+03	FES = 338100	$\mu_F = 0.755$	$\mu_{CR} = 0.042$	
Diversity = 4.6252e-01	k_t = [3, 2, 3, 4]			
Generation 3390: Best = 3.149791e+03	FES = 339100	$\mu_F = 0.775$	$\mu_{CR} = 0.033$	
Diversity = 4.6119e-01	k_t = [1, 1, 1, 1]			
Generation 3400: Best = 3.149791e+03	FES = 340100	$\mu_F = 0.788$	$\mu_{CR} = 0.040$	
Diversity = 4.4170e-01	k_t = [2, 3, 4, 4]			
Generation 3410: Best = 3.149791e+03	FES = 341100	$\mu_F = 0.769$	$\mu_{CR} = 0.039$	
Diversity = 4.4153e-01	k_t = [1, 1, 1, 1]			
Generation 3420: Best = 3.149791e+03	FES = 342100	$\mu_F = 0.754$	$\mu_{CR} = 0.039$	
Diversity = 4.4517e-01	k_t = [2, 4, 3, 3]			
Generation 3430: Best = 3.149791e+03	FES = 343100	$\mu_F = 0.745$	$\mu_{CR} = 0.038$	
Diversity = 4.4550e-01	k_t = [1, 1, 1, 1]			
Generation 3440: Best = 3.149791e+03	FES = 344100	$\mu_F = 0.753$	$\mu_{CR} = 0.036$	
Diversity = 4.4191e-01	k_t = [4, 2, 2, 3]			
Generation 3450: Best = 3.149781e+03	FES = 345100	$\mu_F = 0.755$	$\mu_{CR} = 0.035$	
Diversity = 4.3744e-01	k_t = [1, 1, 1, 1]			
Generation 3460: Best = 3.149781e+03	FES = 346100	$\mu_F = 0.753$	$\mu_{CR} = 0.037$	

Diversity = 4.3634e-01	k_t = [4, 3, 3, 2]			
Generation 3470: Best = 3.149781e+03	FES = 347100	$\mu_F = 0.738$	$\mu_{CR} = 0.043$	
Diversity = 4.3662e-01	k_t = [1, 1, 1, 1]			
Generation 3480: Best = 3.149776e+03	FES = 348100	$\mu_F = 0.755$	$\mu_{CR} = 0.036$	
Diversity = 4.2981e-01	k_t = [4, 4, 2, 4]			
Generation 3490: Best = 3.149776e+03	FES = 349100	$\mu_F = 0.669$	$\mu_{CR} = 0.040$	
Diversity = 4.2667e-01	k_t = [1, 1, 1, 1]			
Generation 3500: Best = 3.149776e+03	FES = 350100	$\mu_F = 0.663$	$\mu_{CR} = 0.036$	
Diversity = 4.2262e-01	k_t = [3, 2, 4, 3]			
Generation 3510: Best = 3.149776e+03	FES = 351100	$\mu_F = 0.659$	$\mu_{CR} = 0.044$	
Diversity = 4.2202e-01	k_t = [1, 1, 1, 1]			
Generation 3520: Best = 3.149775e+03	FES = 352100	$\mu_F = 0.676$	$\mu_{CR} = 0.042$	
Diversity = 4.2067e-01	k_t = [3, 4, 3, 4]			
Generation 3530: Best = 3.149775e+03	FES = 353100	$\mu_F = 0.699$	$\mu_{CR} = 0.044$	
Diversity = 4.1577e-01	k_t = [1, 1, 1, 1]			
Generation 3540: Best = 3.149715e+03	FES = 354100	$\mu_F = 0.706$	$\mu_{CR} = 0.050$	
Diversity = 4.0890e-01	k_t = [3, 4, 2, 3]			
Generation 3550: Best = 3.149706e+03	FES = 355100	$\mu_F = 0.718$	$\mu_{CR} = 0.041$	
Diversity = 4.0812e-01	k_t = [1, 1, 1, 1]			
Generation 3560: Best = 3.149684e+03	FES = 356100	$\mu_F = 0.720$	$\mu_{CR} = 0.041$	
Diversity = 4.0491e-01	k_t = [4, 2, 3, 3]			
Generation 3570: Best = 3.149681e+03	FES = 357100	$\mu_F = 0.749$	$\mu_{CR} = 0.034$	
Diversity = 4.0843e-01	k_t = [1, 1, 1, 1]			
Generation 3580: Best = 3.149681e+03	FES = 358100	$\mu_F = 0.754$	$\mu_{CR} = 0.035$	
Diversity = 4.0600e-01	k_t = [4, 4, 3, 2]			
Generation 3590: Best = 3.149681e+03	FES = 359100	$\mu_F = 0.753$	$\mu_{CR} = 0.040$	
Diversity = 4.1096e-01	k_t = [1, 1, 1, 1]			
Generation 3600: Best = 3.149675e+03	FES = 360100	$\mu_F = 0.781$	$\mu_{CR} = 0.040$	
Diversity = 4.0662e-01	k_t = [2, 3, 4, 4]			
Generation 3610: Best = 3.149675e+03	FES = 361100	$\mu_F = 0.779$	$\mu_{CR} = 0.036$	
Diversity = 3.9717e-01	k_t = [1, 1, 1, 1]			
Generation 3620: Best = 3.149654e+03	FES = 362100	$\mu_F = 0.752$	$\mu_{CR} = 0.043$	
Diversity = 3.9386e-01	k_t = [2, 3, 3, 3]			
Generation 3630: Best = 3.149622e+03	FES = 363100	$\mu_F = 0.704$	$\mu_{CR} = 0.035$	
Diversity = 3.9575e-01	k_t = [1, 1, 1, 1]			
Generation 3640: Best = 3.149589e+03	FES = 364100	$\mu_F = 0.704$	$\mu_{CR} = 0.042$	
Diversity = 3.9620e-01	k_t = [3, 4, 3, 2]			
Generation 3650: Best = 3.149588e+03	FES = 365100	$\mu_F = 0.657$	$\mu_{CR} = 0.044$	
Diversity = 3.9587e-01	k_t = [1, 1, 1, 1]			
Generation 3660: Best = 3.149579e+03	FES = 366100	$\mu_F = 0.675$	$\mu_{CR} = 0.041$	
Diversity = 3.8176e-01	k_t = [4, 4, 4, 4]			
Generation 3670: Best = 3.149579e+03	FES = 367100	$\mu_F = 0.705$	$\mu_{CR} = 0.038$	
Diversity = 3.7900e-01	k_t = [1, 1, 1, 1]			
Generation 3680: Best = 3.149579e+03	FES = 368100	$\mu_F = 0.677$	$\mu_{CR} = 0.035$	
Diversity = 3.8075e-01	k_t = [2, 2, 2, 3]			
Generation 3690: Best = 3.149569e+03	FES = 369100	$\mu_F = 0.680$	$\mu_{CR} = 0.036$	
Diversity = 3.8050e-01	k_t = [1, 1, 1, 1]			
Generation 3700: Best = 3.149568e+03	FES = 370100	$\mu_F = 0.657$	$\mu_{CR} = 0.035$	
Diversity = 3.6828e-01	k_t = [4, 2, 3, 4]			
Generation 3710: Best = 3.149565e+03	FES = 371100	$\mu_F = 0.669$	$\mu_{CR} = 0.039$	
Diversity = 3.6699e-01	k_t = [1, 1, 1, 1]			
Generation 3720: Best = 3.149549e+03	FES = 372100	$\mu_F = 0.711$	$\mu_{CR} = 0.046$	
Diversity = 3.5872e-01	k_t = [4, 3, 2, 4]			
Generation 3730: Best = 3.149549e+03	FES = 373100	$\mu_F = 0.718$	$\mu_{CR} = 0.039$	

Diversity = 3.5580e-01	k_t = [1, 1, 1, 1]			
Generation 3740: Best = 3.149451e+03	FES = 374100	$\mu_F = 0.738$	$\mu_{CR} = 0.046$	
Diversity = 3.5343e-01	k_t = [4, 3, 4, 3]			
Generation 3750: Best = 3.149451e+03	FES = 375100	$\mu_F = 0.754$	$\mu_{CR} = 0.040$	
Diversity = 3.5311e-01	k_t = [1, 1, 1, 1]			
Generation 3760: Best = 3.149451e+03	FES = 376100	$\mu_F = 0.789$	$\mu_{CR} = 0.042$	
Diversity = 3.5686e-01	k_t = [3, 2, 3, 4]			
Generation 3770: Best = 3.149437e+03	FES = 377100	$\mu_F = 0.795$	$\mu_{CR} = 0.040$	
Diversity = 3.5461e-01	k_t = [1, 1, 1, 1]			
Generation 3780: Best = 3.149437e+03	FES = 378100	$\mu_F = 0.812$	$\mu_{CR} = 0.043$	
Diversity = 3.4848e-01	k_t = [4, 2, 3, 3]			
Generation 3790: Best = 3.149437e+03	FES = 379100	$\mu_F = 0.795$	$\mu_{CR} = 0.037$	
Diversity = 3.4059e-01	k_t = [1, 1, 1, 1]			
Generation 3800: Best = 3.149437e+03	FES = 380100	$\mu_F = 0.810$	$\mu_{CR} = 0.036$	
Diversity = 3.4213e-01	k_t = [3, 4, 2, 2]			
Generation 3810: Best = 3.149437e+03	FES = 381100	$\mu_F = 0.767$	$\mu_{CR} = 0.030$	
Diversity = 3.4226e-01	k_t = [1, 1, 1, 1]			
Generation 3820: Best = 3.149431e+03	FES = 382100	$\mu_F = 0.797$	$\mu_{CR} = 0.029$	
Diversity = 3.3893e-01	k_t = [2, 4, 2, 4]			
Generation 3830: Best = 3.149431e+03	FES = 383100	$\mu_F = 0.764$	$\mu_{CR} = 0.036$	
Diversity = 3.3114e-01	k_t = [1, 1, 1, 1]			
Generation 3840: Best = 3.149431e+03	FES = 384100	$\mu_F = 0.783$	$\mu_{CR} = 0.034$	
Diversity = 3.1790e-01	k_t = [4, 2, 3, 2]			
Generation 3850: Best = 3.149427e+03	FES = 385100	$\mu_F = 0.761$	$\mu_{CR} = 0.034$	
Diversity = 3.1677e-01	k_t = [1, 1, 1, 1]			
Generation 3860: Best = 3.149427e+03	FES = 386100	$\mu_F = 0.779$	$\mu_{CR} = 0.032$	
Diversity = 3.1061e-01	k_t = [2, 4, 2, 3]			
Generation 3870: Best = 3.149426e+03	FES = 387100	$\mu_F = 0.763$	$\mu_{CR} = 0.035$	
Diversity = 3.1056e-01	k_t = [1, 1, 1, 1]			
Generation 3880: Best = 3.149426e+03	FES = 388100	$\mu_F = 0.788$	$\mu_{CR} = 0.035$	
Diversity = 3.0838e-01	k_t = [3, 4, 3, 3]			
Generation 3890: Best = 3.149426e+03	FES = 389100	$\mu_F = 0.718$	$\mu_{CR} = 0.043$	
Diversity = 3.0890e-01	k_t = [1, 1, 1, 1]			
Generation 3900: Best = 3.149425e+03	FES = 390100	$\mu_F = 0.690$	$\mu_{CR} = 0.041$	
Diversity = 3.0757e-01	k_t = [3, 2, 4, 4]			
Generation 3910: Best = 3.149425e+03	FES = 391100	$\mu_F = 0.681$	$\mu_{CR} = 0.036$	
Diversity = 3.0738e-01	k_t = [1, 1, 1, 1]			
Generation 3920: Best = 3.149418e+03	FES = 392100	$\mu_F = 0.715$	$\mu_{CR} = 0.036$	
Diversity = 2.9815e-01	k_t = [3, 2, 4, 4]			
Generation 3930: Best = 3.149418e+03	FES = 393100	$\mu_F = 0.714$	$\mu_{CR} = 0.044$	
Diversity = 2.9639e-01	k_t = [1, 1, 1, 1]			
Generation 3940: Best = 3.149418e+03	FES = 394100	$\mu_F = 0.764$	$\mu_{CR} = 0.038$	
Diversity = 2.9807e-01	k_t = [3, 4, 2, 2]			
Generation 3950: Best = 3.149418e+03	FES = 395100	$\mu_F = 0.766$	$\mu_{CR} = 0.037$	
Diversity = 3.0016e-01	k_t = [1, 1, 1, 1]			
Generation 3960: Best = 3.149413e+03	FES = 396100	$\mu_F = 0.777$	$\mu_{CR} = 0.033$	
Diversity = 2.9761e-01	k_t = [3, 3, 4, 2]			
Generation 3970: Best = 3.149413e+03	FES = 397100	$\mu_F = 0.777$	$\mu_{CR} = 0.041$	
Diversity = 2.9027e-01	k_t = [1, 1, 1, 1]			
Generation 3980: Best = 3.149403e+03	FES = 398100	$\mu_F = 0.752$	$\mu_{CR} = 0.043$	
Diversity = 2.8963e-01	k_t = [4, 2, 2, 3]			
Generation 3990: Best = 3.149403e+03	FES = 399100	$\mu_F = 0.733$	$\mu_{CR} = 0.043$	
Diversity = 2.8947e-01	k_t = [1, 1, 1, 1]			
Generation 4000: Best = 3.149403e+03	FES = 400100	$\mu_F = 0.756$	$\mu_{CR} = 0.040$	

Diversity = 2.8136e-01	k_t = [3, 4, 4, 4]			
Generation 4010: Best = 3.149400e+03	FES = 401100	$\mu_F = 0.732$	$\mu_{CR} = 0.042$	
Diversity = 2.8063e-01	k_t = [1, 1, 1, 1]			
Generation 4020: Best = 3.149399e+03	FES = 402100	$\mu_F = 0.787$	$\mu_{CR} = 0.034$	
Diversity = 2.6414e-01	k_t = [2, 3, 3, 3]			
Generation 4030: Best = 3.149394e+03	FES = 403100	$\mu_F = 0.747$	$\mu_{CR} = 0.033$	
Diversity = 2.6460e-01	k_t = [1, 1, 1, 1]			
Generation 4040: Best = 3.149394e+03	FES = 404100	$\mu_F = 0.708$	$\mu_{CR} = 0.037$	
Diversity = 2.5905e-01	k_t = [2, 2, 2, 2]			
Generation 4050: Best = 3.149393e+03	FES = 405100	$\mu_F = 0.704$	$\mu_{CR} = 0.043$	
Diversity = 2.5308e-01	k_t = [1, 1, 1, 1]			
Generation 4060: Best = 3.149391e+03	FES = 406100	$\mu_F = 0.708$	$\mu_{CR} = 0.035$	
Diversity = 2.4527e-01	k_t = [2, 4, 4, 4]			
Generation 4070: Best = 3.149391e+03	FES = 407100	$\mu_F = 0.697$	$\mu_{CR} = 0.044$	
Diversity = 2.4450e-01	k_t = [1, 1, 1, 1]			
Generation 4080: Best = 3.149386e+03	FES = 408100	$\mu_F = 0.716$	$\mu_{CR} = 0.038$	
Diversity = 2.4479e-01	k_t = [3, 4, 2, 3]			
Generation 4090: Best = 3.149383e+03	FES = 409100	$\mu_F = 0.761$	$\mu_{CR} = 0.032$	
Diversity = 2.4492e-01	k_t = [1, 1, 1, 1]			
Generation 4100: Best = 3.149378e+03	FES = 410100	$\mu_F = 0.751$	$\mu_{CR} = 0.039$	
Diversity = 2.4141e-01	k_t = [4, 4, 4, 2]			
Generation 4110: Best = 3.149378e+03	FES = 411100	$\mu_F = 0.754$	$\mu_{CR} = 0.041$	
Diversity = 2.4302e-01	k_t = [1, 1, 1, 1]			
Generation 4120: Best = 3.149377e+03	FES = 412100	$\mu_F = 0.758$	$\mu_{CR} = 0.037$	
Diversity = 2.4138e-01	k_t = [4, 3, 3, 2]			
Generation 4130: Best = 3.149372e+03	FES = 413100	$\mu_F = 0.766$	$\mu_{CR} = 0.038$	
Diversity = 2.4137e-01	k_t = [1, 1, 1, 1]			
Generation 4140: Best = 3.149360e+03	FES = 414100	$\mu_F = 0.774$	$\mu_{CR} = 0.038$	
Diversity = 2.4007e-01	k_t = [4, 2, 3, 4]			
Generation 4150: Best = 3.149311e+03	FES = 415100	$\mu_F = 0.802$	$\mu_{CR} = 0.038$	
Diversity = 2.3932e-01	k_t = [1, 1, 1, 1]			
Generation 4160: Best = 3.149302e+03	FES = 416100	$\mu_F = 0.798$	$\mu_{CR} = 0.038$	
Diversity = 2.4007e-01	k_t = [2, 2, 2, 4]			
Generation 4170: Best = 3.149302e+03	FES = 417100	$\mu_F = 0.769$	$\mu_{CR} = 0.039$	
Diversity = 2.3755e-01	k_t = [1, 1, 1, 1]			
Generation 4180: Best = 3.149300e+03	FES = 418100	$\mu_F = 0.776$	$\mu_{CR} = 0.041$	
Diversity = 2.3136e-01	k_t = [3, 3, 2, 2]			
Generation 4190: Best = 3.149300e+03	FES = 419100	$\mu_F = 0.690$	$\mu_{CR} = 0.038$	
Diversity = 2.2838e-01	k_t = [1, 1, 1, 1]			
Generation 4200: Best = 3.149300e+03	FES = 420100	$\mu_F = 0.694$	$\mu_{CR} = 0.042$	
Diversity = 2.1163e-01	k_t = [2, 3, 3, 4]			
Generation 4210: Best = 3.149286e+03	FES = 421100	$\mu_F = 0.722$	$\mu_{CR} = 0.041$	
Diversity = 2.2209e-01	k_t = [1, 1, 1, 1]			
Generation 4220: Best = 3.149240e+03	FES = 422100	$\mu_F = 0.735$	$\mu_{CR} = 0.051$	
Diversity = 2.2887e-01	k_t = [4, 2, 2, 2]			
Generation 4230: Best = 3.149240e+03	FES = 423100	$\mu_F = 0.709$	$\mu_{CR} = 0.049$	
Diversity = 2.3116e-01	k_t = [1, 1, 1, 1]			
Generation 4240: Best = 3.149240e+03	FES = 424100	$\mu_F = 0.726$	$\mu_{CR} = 0.050$	
Diversity = 2.2669e-01	k_t = [4, 3, 3, 3]			
Generation 4250: Best = 3.149238e+03	FES = 425100	$\mu_F = 0.687$	$\mu_{CR} = 0.047$	
Diversity = 2.3098e-01	k_t = [1, 1, 1, 1]			
Generation 4260: Best = 3.149217e+03	FES = 426100	$\mu_F = 0.706$	$\mu_{CR} = 0.042$	
Diversity = 2.3091e-01	k_t = [3, 4, 2, 3]			
Generation 4270: Best = 3.149183e+03	FES = 427100	$\mu_F = 0.684$	$\mu_{CR} = 0.049$	

Diversity = 2.4144e-01	k_t = [1, 1, 1, 1]			
Generation 4280: Best = 3.149182e+03	FES = 428100	$\mu_F = 0.672$	$\mu_{CR} = 0.047$	
Diversity = 2.4166e-01	k_t = [2, 4, 4, 2]			
Generation 4290: Best = 3.149182e+03	FES = 429100	$\mu_F = 0.695$	$\mu_{CR} = 0.053$	
Diversity = 2.4566e-01	k_t = [1, 1, 1, 1]			
Generation 4300: Best = 3.149181e+03	FES = 430100	$\mu_F = 0.728$	$\mu_{CR} = 0.049$	
Diversity = 2.4305e-01	k_t = [2, 4, 4, 4]			
Generation 4310: Best = 3.149165e+03	FES = 431100	$\mu_F = 0.691$	$\mu_{CR} = 0.058$	
Diversity = 2.3590e-01	k_t = [1, 1, 1, 1]			
Generation 4320: Best = 3.149165e+03	FES = 432100	$\mu_F = 0.680$	$\mu_{CR} = 0.056$	
Diversity = 2.3208e-01	k_t = [4, 4, 2, 4]			
Generation 4330: Best = 3.149164e+03	FES = 433100	$\mu_F = 0.682$	$\mu_{CR} = 0.053$	
Diversity = 2.3699e-01	k_t = [1, 1, 1, 1]			
Generation 4340: Best = 3.149164e+03	FES = 434100	$\mu_F = 0.685$	$\mu_{CR} = 0.049$	
Diversity = 2.3912e-01	k_t = [2, 3, 3, 2]			
Generation 4350: Best = 3.149150e+03	FES = 435100	$\mu_F = 0.631$	$\mu_{CR} = 0.054$	
Diversity = 2.2708e-01	k_t = [1, 1, 1, 1]			
Generation 4360: Best = 3.149145e+03	FES = 436100	$\mu_F = 0.657$	$\mu_{CR} = 0.043$	
Diversity = 2.1374e-01	k_t = [3, 2, 2, 4]			
Generation 4370: Best = 3.149095e+03	FES = 437100	$\mu_F = 0.633$	$\mu_{CR} = 0.051$	
Diversity = 1.3582e-01	k_t = [1, 1, 1, 1]			
Generation 4380: Best = 3.149084e+03	FES = 438100	$\mu_F = 0.655$	$\mu_{CR} = 0.059$	
Diversity = 1.1380e-01	k_t = [4, 2, 4, 2]			
Generation 4390: Best = 3.149083e+03	FES = 439100	$\mu_F = 0.635$	$\mu_{CR} = 0.051$	
Diversity = 1.1439e-01	k_t = [1, 1, 1, 1]			
Generation 4400: Best = 3.149079e+03	FES = 440100	$\mu_F = 0.657$	$\mu_{CR} = 0.057$	
Diversity = 1.0404e-01	k_t = [2, 4, 4, 2]			
Generation 4410: Best = 3.149077e+03	FES = 441100	$\mu_F = 0.695$	$\mu_{CR} = 0.056$	
Diversity = 1.0344e-01	k_t = [1, 1, 1, 1]			
Generation 4420: Best = 3.149074e+03	FES = 442100	$\mu_F = 0.714$	$\mu_{CR} = 0.057$	
Diversity = 1.0124e-01	k_t = [4, 3, 3, 4]			
Generation 4430: Best = 3.149069e+03	FES = 443100	$\mu_F = 0.718$	$\mu_{CR} = 0.062$	
Diversity = 9.1344e-02	k_t = [1, 1, 1, 1]			
Generation 4440: Best = 3.149063e+03	FES = 444100	$\mu_F = 0.733$	$\mu_{CR} = 0.073$	
Diversity = 5.5108e-02	k_t = [3, 2, 4, 4]			
Generation 4450: Best = 3.149045e+03	FES = 445100	$\mu_F = 0.696$	$\mu_{CR} = 0.071$	
Diversity = 5.2531e-02	k_t = [1, 1, 1, 1]			
Generation 4460: Best = 3.149045e+03	FES = 446100	$\mu_F = 0.683$	$\mu_{CR} = 0.069$	
Diversity = 4.6906e-02	k_t = [2, 4, 2, 3]			
Generation 4470: Best = 3.149045e+03	FES = 447100	$\mu_F = 0.685$	$\mu_{CR} = 0.066$	
Diversity = 5.0879e-02	k_t = [1, 1, 1, 1]			
Generation 4480: Best = 3.149035e+03	FES = 448100	$\mu_F = 0.672$	$\mu_{CR} = 0.078$	
Diversity = 4.6884e-02	k_t = [3, 3, 4, 2]			
Generation 4490: Best = 3.149022e+03	FES = 449100	$\mu_F = 0.626$	$\mu_{CR} = 0.072$	
Diversity = 4.1998e-02	k_t = [1, 1, 1, 1]			
Generation 4500: Best = 3.149019e+03	FES = 450100	$\mu_F = 0.624$	$\mu_{CR} = 0.068$	
Diversity = 4.0176e-02	k_t = [2, 4, 2, 2]			
Generation 4510: Best = 3.149006e+03	FES = 451100	$\mu_F = 0.647$	$\mu_{CR} = 0.078$	
Diversity = 3.9729e-02	k_t = [1, 1, 1, 1]			
Generation 4520: Best = 3.149006e+03	FES = 452100	$\mu_F = 0.650$	$\mu_{CR} = 0.070$	
Diversity = 3.7245e-02	k_t = [2, 3, 2, 2]			
Generation 4530: Best = 3.149004e+03	FES = 453100	$\mu_F = 0.662$	$\mu_{CR} = 0.068$	
Diversity = 3.7181e-02	k_t = [1, 1, 1, 1]			
Generation 4540: Best = 3.149004e+03	FES = 454100	$\mu_F = 0.666$	$\mu_{CR} = 0.073$	

Diversity = 3.4880e-02 k_t = [2, 4, 2, 3]
 Generation 4550: Best = 3.149001e+03 FES = 455100 μ_F = 0.667 μ_{CR} = 0.069
 Diversity = 3.4214e-02 k_t = [1, 1, 1, 1]
 Generation 4560: Best = 3.148992e+03 FES = 456100 μ_F = 0.645 μ_{CR} = 0.073
 Diversity = 2.9820e-02 k_t = [2, 3, 3, 3]
 Generation 4570: Best = 3.148983e+03 FES = 457100 μ_F = 0.628 μ_{CR} = 0.069
 Diversity = 2.7789e-02 k_t = [1, 1, 1, 1]
 Generation 4580: Best = 3.148983e+03 FES = 458100 μ_F = 0.660 μ_{CR} = 0.061
 Diversity = 2.5986e-02 k_t = [4, 2, 4, 2]
 Generation 4590: Best = 3.148983e+03 FES = 459100 μ_F = 0.669 μ_{CR} = 0.061
 Diversity = 2.6355e-02 k_t = [1, 1, 1, 1]
 Generation 4600: Best = 3.148982e+03 FES = 460100 μ_F = 0.672 μ_{CR} = 0.055
 Diversity = 2.6529e-02 k_t = [2, 4, 3, 3]
 Generation 4610: Best = 3.148979e+03 FES = 461100 μ_F = 0.698 μ_{CR} = 0.060
 Diversity = 2.5782e-02 k_t = [1, 1, 1, 1]
 Generation 4620: Best = 3.148977e+03 FES = 462100 μ_F = 0.743 μ_{CR} = 0.053
 Diversity = 2.6399e-02 k_t = [3, 4, 3, 3]
 Generation 4630: Best = 3.148967e+03 FES = 463100 μ_F = 0.745 μ_{CR} = 0.059
 Diversity = 2.9167e-02 k_t = [1, 1, 1, 1]
 Generation 4640: Best = 3.148961e+03 FES = 464100 μ_F = 0.735 μ_{CR} = 0.058
 Diversity = 2.5078e-02 k_t = [4, 3, 4, 4]
 Generation 4650: Best = 3.148961e+03 FES = 465100 μ_F = 0.720 μ_{CR} = 0.054
 Diversity = 2.3548e-02 k_t = [1, 1, 1, 1]
 Generation 4660: Best = 3.148958e+03 FES = 466100 μ_F = 0.717 μ_{CR} = 0.048
 Diversity = 2.2589e-02 k_t = [4, 3, 4, 3]
 Generation 4670: Best = 3.148955e+03 FES = 467100 μ_F = 0.706 μ_{CR} = 0.052
 Diversity = 2.0952e-02 k_t = [1, 1, 1, 1]
 Generation 4680: Best = 3.148953e+03 FES = 468100 μ_F = 0.722 μ_{CR} = 0.055
 Diversity = 2.0301e-02 k_t = [3, 2, 3, 2]
 Generation 4690: Best = 3.148953e+03 FES = 469100 μ_F = 0.740 μ_{CR} = 0.053
 Diversity = 2.0077e-02 k_t = [1, 1, 1, 1]
 Generation 4700: Best = 3.148952e+03 FES = 470100 μ_F = 0.744 μ_{CR} = 0.046
 Diversity = 2.2769e-02 k_t = [2, 3, 4, 4]
 Generation 4710: Best = 3.148951e+03 FES = 471100 μ_F = 0.709 μ_{CR} = 0.049
 Diversity = 1.9360e-02 k_t = [1, 1, 1, 1]
 Generation 4720: Best = 3.148949e+03 FES = 472100 μ_F = 0.720 μ_{CR} = 0.047
 Diversity = 1.9570e-02 k_t = [2, 4, 3, 2]
 Generation 4730: Best = 3.148947e+03 FES = 473100 μ_F = 0.762 μ_{CR} = 0.042
 Diversity = 1.9774e-02 k_t = [1, 1, 1, 1]
 Generation 4740: Best = 3.148945e+03 FES = 474100 μ_F = 0.741 μ_{CR} = 0.043
 Diversity = 1.9388e-02 k_t = [3, 2, 3, 2]
 Generation 4750: Best = 3.148945e+03 FES = 475100 μ_F = 0.728 μ_{CR} = 0.047
 Diversity = 1.8946e-02 k_t = [1, 1, 1, 1]
 Generation 4760: Best = 3.148944e+03 FES = 476100 μ_F = 0.725 μ_{CR} = 0.047
 Diversity = 1.8042e-02 k_t = [3, 2, 2, 4]
 Generation 4770: Best = 3.148944e+03 FES = 477100 μ_F = 0.698 μ_{CR} = 0.050
 Diversity = 1.5829e-02 k_t = [1, 1, 1, 1]
 Generation 4780: Best = 3.148940e+03 FES = 478100 μ_F = 0.727 μ_{CR} = 0.050
 Diversity = 1.5770e-02 k_t = [2, 4, 2, 4]
 Generation 4790: Best = 3.148940e+03 FES = 479100 μ_F = 0.728 μ_{CR} = 0.061
 Diversity = 1.2274e-02 k_t = [1, 1, 1, 1]
 Generation 4800: Best = 3.148938e+03 FES = 480100 μ_F = 0.751 μ_{CR} = 0.057
 Diversity = 9.8828e-03 k_t = [3, 2, 4, 4]
 Generation 4810: Best = 3.148937e+03 FES = 481100 μ_F = 0.757 μ_{CR} = 0.057

Diversity = 8.8555e-03	k_t = [1, 1, 1, 1]			
Generation 4820: Best = 3.148937e+03	FES = 482100	$\mu_F = 0.746$	$\mu_{CR} = 0.064$	
Diversity = 8.9267e-03	k_t = [3, 4, 2, 4]			
Generation 4830: Best = 3.148935e+03	FES = 483100	$\mu_F = 0.759$	$\mu_{CR} = 0.067$	
Diversity = 8.8553e-03	k_t = [1, 1, 1, 1]			
Generation 4840: Best = 3.148932e+03	FES = 484100	$\mu_F = 0.771$	$\mu_{CR} = 0.061$	
Diversity = 9.2047e-03	k_t = [3, 4, 3, 3]			
Generation 4850: Best = 3.148931e+03	FES = 485100	$\mu_F = 0.755$	$\mu_{CR} = 0.071$	
Diversity = 9.0647e-03	k_t = [1, 1, 1, 1]			
Generation 4860: Best = 3.148928e+03	FES = 486100	$\mu_F = 0.736$	$\mu_{CR} = 0.071$	
Diversity = 8.5870e-03	k_t = [3, 3, 2, 3]			
Generation 4870: Best = 3.148928e+03	FES = 487100	$\mu_F = 0.741$	$\mu_{CR} = 0.065$	
Diversity = 8.2320e-03	k_t = [1, 1, 1, 1]			
Generation 4880: Best = 3.148926e+03	FES = 488100	$\mu_F = 0.753$	$\mu_{CR} = 0.072$	
Diversity = 7.8472e-03	k_t = [2, 2, 2, 3]			
Generation 4890: Best = 3.148924e+03	FES = 489100	$\mu_F = 0.754$	$\mu_{CR} = 0.082$	
Diversity = 7.5304e-03	k_t = [1, 1, 1, 1]			
Generation 4900: Best = 3.148924e+03	FES = 490100	$\mu_F = 0.757$	$\mu_{CR} = 0.085$	
Diversity = 7.0744e-03	k_t = [2, 4, 3, 3]			
Generation 4910: Best = 3.148922e+03	FES = 491100	$\mu_F = 0.758$	$\mu_{CR} = 0.089$	
Diversity = 7.1557e-03	k_t = [1, 1, 1, 1]			
Generation 4920: Best = 3.148920e+03	FES = 492100	$\mu_F = 0.751$	$\mu_{CR} = 0.091$	
Diversity = 7.6180e-03	k_t = [3, 2, 2, 2]			
Generation 4930: Best = 3.148919e+03	FES = 493100	$\mu_F = 0.741$	$\mu_{CR} = 0.092$	
Diversity = 8.2040e-03	k_t = [1, 1, 1, 1]			
Generation 4940: Best = 3.148917e+03	FES = 494100	$\mu_F = 0.731$	$\mu_{CR} = 0.089$	
Diversity = 8.4420e-03	k_t = [3, 4, 3, 3]			
Generation 4950: Best = 3.148916e+03	FES = 495100	$\mu_F = 0.707$	$\mu_{CR} = 0.095$	
Diversity = 7.7197e-03	k_t = [1, 1, 1, 1]			
Generation 4960: Best = 3.148914e+03	FES = 496100	$\mu_F = 0.698$	$\mu_{CR} = 0.098$	
Diversity = 7.3143e-03	k_t = [4, 4, 3, 2]			
Generation 4970: Best = 3.148913e+03	FES = 497100	$\mu_F = 0.691$	$\mu_{CR} = 0.099$	
Diversity = 6.9264e-03	k_t = [1, 1, 1, 1]			
Generation 4980: Best = 3.148912e+03	FES = 498100	$\mu_F = 0.672$	$\mu_{CR} = 0.103$	
Diversity = 6.4459e-03	k_t = [3, 3, 2, 3]			
Generation 4990: Best = 3.148911e+03	FES = 499100	$\mu_F = 0.669$	$\mu_{CR} = 0.105$	
Diversity = 6.1722e-03	k_t = [1, 1, 1, 1]			
Generation 5000: Best = 3.148909e+03	FES = 500100	$\mu_F = 0.675$	$\mu_{CR} = 0.105$	
Diversity = 5.7319e-03	k_t = [3, 4, 2, 4]			
Generation 5010: Best = 3.148908e+03	FES = 501100	$\mu_F = 0.699$	$\mu_{CR} = 0.096$	
Diversity = 5.2539e-03	k_t = [1, 1, 1, 1]			
Generation 5020: Best = 3.148908e+03	FES = 502100	$\mu_F = 0.710$	$\mu_{CR} = 0.092$	
Diversity = 5.2680e-03	k_t = [3, 2, 4, 4]			
Generation 5030: Best = 3.148907e+03	FES = 503100	$\mu_F = 0.711$	$\mu_{CR} = 0.092$	
Diversity = 5.8255e-03	k_t = [1, 1, 1, 1]			
Generation 5040: Best = 3.148907e+03	FES = 504100	$\mu_F = 0.710$	$\mu_{CR} = 0.090$	
Diversity = 4.9527e-03	k_t = [3, 3, 4, 3]			
Generation 5050: Best = 3.148906e+03	FES = 505100	$\mu_F = 0.720$	$\mu_{CR} = 0.090$	
Diversity = 4.9384e-03	k_t = [1, 1, 1, 1]			
Generation 5060: Best = 3.148906e+03	FES = 506100	$\mu_F = 0.722$	$\mu_{CR} = 0.090$	
Diversity = 5.7133e-03	k_t = [4, 3, 3, 4]			
Generation 5070: Best = 3.148904e+03	FES = 507100	$\mu_F = 0.728$	$\mu_{CR} = 0.083$	
Diversity = 6.4231e-03	k_t = [1, 1, 1, 1]			
Generation 5080: Best = 3.148904e+03	FES = 508100	$\mu_F = 0.742$	$\mu_{CR} = 0.081$	

Diversity = 7.0771e-03	k_t = [3, 3, 4, 3]			
Generation 5090: Best = 3.148904e+03	FES = 509100	$\mu_F = 0.735$	$\mu_{CR} = 0.081$	
Diversity = 7.7716e-03	k_t = [1, 1, 1, 1]			
Generation 5100: Best = 3.148903e+03	FES = 510100	$\mu_F = 0.722$	$\mu_{CR} = 0.078$	
Diversity = 8.2521e-03	k_t = [3, 3, 2, 2]			
Generation 5110: Best = 3.148903e+03	FES = 511100	$\mu_F = 0.737$	$\mu_{CR} = 0.066$	
Diversity = 8.2923e-03	k_t = [1, 1, 1, 1]			
Generation 5120: Best = 3.148903e+03	FES = 512100	$\mu_F = 0.727$	$\mu_{CR} = 0.072$	
Diversity = 8.0301e-03	k_t = [4, 4, 3, 4]			
Generation 5130: Best = 3.148902e+03	FES = 513100	$\mu_F = 0.736$	$\mu_{CR} = 0.079$	
Diversity = 7.1279e-03	k_t = [1, 1, 1, 1]			
Generation 5140: Best = 3.148902e+03	FES = 514100	$\mu_F = 0.751$	$\mu_{CR} = 0.072$	
Diversity = 6.0484e-03	k_t = [2, 4, 4, 4]			
Generation 5150: Best = 3.148902e+03	FES = 515100	$\mu_F = 0.737$	$\mu_{CR} = 0.065$	
Diversity = 5.6955e-03	k_t = [1, 1, 1, 1]			
Generation 5160: Best = 3.148901e+03	FES = 516100	$\mu_F = 0.737$	$\mu_{CR} = 0.078$	
Diversity = 5.4819e-03	k_t = [3, 3, 4, 4]			
Generation 5170: Best = 3.148901e+03	FES = 517100	$\mu_F = 0.729$	$\mu_{CR} = 0.075$	
Diversity = 5.8104e-03	k_t = [1, 1, 1, 1]			
Generation 5180: Best = 3.148901e+03	FES = 518100	$\mu_F = 0.698$	$\mu_{CR} = 0.085$	
Diversity = 5.7572e-03	k_t = [4, 4, 2, 4]			
Generation 5190: Best = 3.148900e+03	FES = 519100	$\mu_F = 0.699$	$\mu_{CR} = 0.079$	
Diversity = 5.4432e-03	k_t = [1, 1, 1, 1]			
Generation 5200: Best = 3.148900e+03	FES = 520100	$\mu_F = 0.723$	$\mu_{CR} = 0.081$	
Diversity = 5.0468e-03	k_t = [3, 3, 3, 4]			
Generation 5210: Best = 3.148900e+03	FES = 521100	$\mu_F = 0.698$	$\mu_{CR} = 0.095$	
Diversity = 4.7310e-03	k_t = [1, 1, 1, 1]			
Generation 5220: Best = 3.148900e+03	FES = 522100	$\mu_F = 0.685$	$\mu_{CR} = 0.099$	
Diversity = 5.0987e-03	k_t = [3, 3, 4, 3]			
Generation 5230: Best = 3.148900e+03	FES = 523100	$\mu_F = 0.672$	$\mu_{CR} = 0.078$	
Diversity = 5.0526e-03	k_t = [1, 1, 1, 1]			
Generation 5240: Best = 3.148900e+03	FES = 524100	$\mu_F = 0.692$	$\mu_{CR} = 0.068$	
Diversity = 4.4602e-03	k_t = [4, 3, 4, 4]			
Generation 5250: Best = 3.148899e+03	FES = 525100	$\mu_F = 0.670$	$\mu_{CR} = 0.071$	
Diversity = 3.2909e-03	k_t = [1, 1, 1, 1]			
Generation 5260: Best = 3.148899e+03	FES = 526100	$\mu_F = 0.682$	$\mu_{CR} = 0.064$	
Diversity = 3.2380e-03	k_t = [2, 2, 4, 2]			
Generation 5270: Best = 3.148899e+03	FES = 527100	$\mu_F = 0.699$	$\mu_{CR} = 0.061$	
Diversity = 3.0146e-03	k_t = [1, 1, 1, 1]			
Generation 5280: Best = 3.148899e+03	FES = 528100	$\mu_F = 0.727$	$\mu_{CR} = 0.064$	
Diversity = 2.9947e-03	k_t = [2, 2, 3, 2]			
Generation 5290: Best = 3.148899e+03	FES = 529100	$\mu_F = 0.738$	$\mu_{CR} = 0.061$	
Diversity = 2.8190e-03	k_t = [1, 1, 1, 1]			
Generation 5300: Best = 3.148899e+03	FES = 530100	$\mu_F = 0.753$	$\mu_{CR} = 0.066$	
Diversity = 2.6969e-03	k_t = [4, 2, 2, 3]			
Generation 5310: Best = 3.148899e+03	FES = 531100	$\mu_F = 0.747$	$\mu_{CR} = 0.065$	
Diversity = 2.8022e-03	k_t = [1, 1, 1, 1]			
Generation 5320: Best = 3.148898e+03	FES = 532100	$\mu_F = 0.737$	$\mu_{CR} = 0.081$	
Diversity = 3.1775e-03	k_t = [4, 4, 4, 3]			
Generation 5330: Best = 3.148898e+03	FES = 533100	$\mu_F = 0.742$	$\mu_{CR} = 0.090$	
Diversity = 3.2596e-03	k_t = [1, 1, 1, 1]			
Generation 5340: Best = 3.148898e+03	FES = 534100	$\mu_F = 0.737$	$\mu_{CR} = 0.102$	
Diversity = 3.1258e-03	k_t = [3, 4, 3, 2]			
Generation 5350: Best = 3.148897e+03	FES = 535100	$\mu_F = 0.726$	$\mu_{CR} = 0.108$	

Diversity = 3.4257e-03	k_t = [1, 1, 1, 1]			
Generation 5360: Best = 3.148897e+03	FES = 536100	$\mu_F = 0.748$	$\mu_{CR} = 0.105$	
Diversity = 3.3602e-03	k_t = [4, 3, 2, 4]			
Generation 5370: Best = 3.148897e+03	FES = 537100	$\mu_F = 0.753$	$\mu_{CR} = 0.110$	
Diversity = 3.6244e-03	k_t = [1, 1, 1, 1]			
Generation 5380: Best = 3.148896e+03	FES = 538100	$\mu_F = 0.749$	$\mu_{CR} = 0.100$	
Diversity = 3.7724e-03	k_t = [3, 2, 2, 3]			
Generation 5390: Best = 3.148896e+03	FES = 539100	$\mu_F = 0.746$	$\mu_{CR} = 0.109$	
Diversity = 3.9524e-03	k_t = [1, 1, 1, 1]			
Generation 5400: Best = 3.148896e+03	FES = 540100	$\mu_F = 0.753$	$\mu_{CR} = 0.111$	
Diversity = 3.7394e-03	k_t = [3, 2, 3, 3]			
Generation 5410: Best = 3.148895e+03	FES = 541100	$\mu_F = 0.755$	$\mu_{CR} = 0.098$	
Diversity = 3.9308e-03	k_t = [1, 1, 1, 1]			
Generation 5420: Best = 3.148895e+03	FES = 542100	$\mu_F = 0.776$	$\mu_{CR} = 0.094$	
Diversity = 4.1109e-03	k_t = [2, 3, 3, 2]			
Generation 5430: Best = 3.148895e+03	FES = 543100	$\mu_F = 0.773$	$\mu_{CR} = 0.097$	
Diversity = 4.3148e-03	k_t = [1, 1, 1, 1]			
Generation 5440: Best = 3.148895e+03	FES = 544100	$\mu_F = 0.765$	$\mu_{CR} = 0.097$	
Diversity = 5.5467e-03	k_t = [4, 2, 2, 3]			
Generation 5450: Best = 3.148895e+03	FES = 545100	$\mu_F = 0.766$	$\mu_{CR} = 0.087$	
Diversity = 5.1711e-03	k_t = [1, 1, 1, 1]			
Generation 5460: Best = 3.148894e+03	FES = 546100	$\mu_F = 0.787$	$\mu_{CR} = 0.092$	
Diversity = 5.2608e-03	k_t = [2, 2, 4, 3]			
Generation 5470: Best = 3.148894e+03	FES = 547100	$\mu_F = 0.757$	$\mu_{CR} = 0.094$	
Diversity = 4.8381e-03	k_t = [1, 1, 1, 1]			
Generation 5480: Best = 3.148894e+03	FES = 548100	$\mu_F = 0.756$	$\mu_{CR} = 0.102$	
Diversity = 4.2715e-03	k_t = [3, 2, 3, 3]			
Generation 5490: Best = 3.148894e+03	FES = 549100	$\mu_F = 0.750$	$\mu_{CR} = 0.097$	
Diversity = 4.9528e-03	k_t = [1, 1, 1, 1]			
Generation 5500: Best = 3.148894e+03	FES = 550100	$\mu_F = 0.760$	$\mu_{CR} = 0.097$	
Diversity = 4.1799e-03	k_t = [4, 4, 2, 4]			
Generation 5510: Best = 3.148893e+03	FES = 551100	$\mu_F = 0.759$	$\mu_{CR} = 0.086$	
Diversity = 4.1135e-03	k_t = [1, 1, 1, 1]			
Generation 5520: Best = 3.148893e+03	FES = 552100	$\mu_F = 0.759$	$\mu_{CR} = 0.091$	
Diversity = 3.9374e-03	k_t = [4, 3, 4, 4]			
Generation 5530: Best = 3.148893e+03	FES = 553100	$\mu_F = 0.758$	$\mu_{CR} = 0.099$	
Diversity = 3.8160e-03	k_t = [1, 1, 1, 1]			
Generation 5540: Best = 3.148893e+03	FES = 554100	$\mu_F = 0.774$	$\mu_{CR} = 0.091$	
Diversity = 4.2728e-03	k_t = [2, 3, 2, 4]			
Generation 5550: Best = 3.148893e+03	FES = 555100	$\mu_F = 0.755$	$\mu_{CR} = 0.090$	
Diversity = 3.8945e-03	k_t = [1, 1, 1, 1]			
Generation 5560: Best = 3.148892e+03	FES = 556100	$\mu_F = 0.764$	$\mu_{CR} = 0.091$	
Diversity = 3.7432e-03	k_t = [2, 2, 4, 4]			
Generation 5570: Best = 3.148892e+03	FES = 557100	$\mu_F = 0.783$	$\mu_{CR} = 0.092$	
Diversity = 3.7620e-03	k_t = [1, 1, 1, 1]			
Generation 5580: Best = 3.148892e+03	FES = 558100	$\mu_F = 0.779$	$\mu_{CR} = 0.086$	
Diversity = 3.9892e-03	k_t = [2, 4, 2, 3]			
Generation 5590: Best = 3.148892e+03	FES = 559100	$\mu_F = 0.780$	$\mu_{CR} = 0.085$	
Diversity = 4.0561e-03	k_t = [1, 1, 1, 1]			
Generation 5600: Best = 3.148892e+03	FES = 560100	$\mu_F = 0.786$	$\mu_{CR} = 0.088$	
Diversity = 4.8321e-03	k_t = [3, 3, 4, 3]			
Generation 5610: Best = 3.148892e+03	FES = 561100	$\mu_F = 0.788$	$\mu_{CR} = 0.090$	
Diversity = 4.9462e-03	k_t = [1, 1, 1, 1]			
Generation 5620: Best = 3.148892e+03	FES = 562100	$\mu_F = 0.767$	$\mu_{CR} = 0.093$	

Diversity = 4.5713e-03 k_t = [3, 2, 3, 3]
 Generation 5630: Best = 3.148892e+03 FES = 563100 μ_F = 0.767 μ_{CR} = 0.083
 Diversity = 4.8884e-03 k_t = [1, 1, 1, 1]
 Generation 5640: Best = 3.148891e+03 FES = 564100 μ_F = 0.781 μ_{CR} = 0.092
 Diversity = 4.9612e-03 k_t = [3, 3, 4, 3]
 Generation 5650: Best = 3.148891e+03 FES = 565100 μ_F = 0.773 μ_{CR} = 0.097
 Diversity = 5.6456e-03 k_t = [1, 1, 1, 1]
 Generation 5660: Best = 3.148891e+03 FES = 566100 μ_F = 0.761 μ_{CR} = 0.089
 Diversity = 5.1212e-03 k_t = [2, 2, 3, 4]
 Generation 5670: Best = 3.148891e+03 FES = 567100 μ_F = 0.758 μ_{CR} = 0.092
 Diversity = 4.9746e-03 k_t = [1, 1, 1, 1]
 Generation 5680: Best = 3.148891e+03 FES = 568100 μ_F = 0.767 μ_{CR} = 0.090
 Diversity = 4.9109e-03 k_t = [2, 2, 2, 3]
 Generation 5690: Best = 3.148891e+03 FES = 569100 μ_F = 0.759 μ_{CR} = 0.087
 Diversity = 4.8240e-03 k_t = [1, 1, 1, 1]
 Generation 5700: Best = 3.148891e+03 FES = 570100 μ_F = 0.770 μ_{CR} = 0.078
 Diversity = 4.4941e-03 k_t = [2, 2, 4, 3]
 Generation 5710: Best = 3.148891e+03 FES = 571100 μ_F = 0.763 μ_{CR} = 0.081
 Diversity = 4.2818e-03 k_t = [1, 1, 1, 1]
 Generation 5720: Best = 3.148890e+03 FES = 572100 μ_F = 0.771 μ_{CR} = 0.084
 Diversity = 4.0499e-03 k_t = [2, 4, 4, 4]
 Generation 5730: Best = 3.148890e+03 FES = 573100 μ_F = 0.752 μ_{CR} = 0.079
 Diversity = 4.0034e-03 k_t = [1, 1, 1, 1]
 Generation 5740: Best = 3.148890e+03 FES = 574100 μ_F = 0.747 μ_{CR} = 0.082
 Diversity = 4.0515e-03 k_t = [3, 3, 4, 3]
 Generation 5750: Best = 3.148890e+03 FES = 575100 μ_F = 0.763 μ_{CR} = 0.089
 Diversity = 4.0910e-03 k_t = [1, 1, 1, 1]
 Generation 5760: Best = 3.148890e+03 FES = 576100 μ_F = 0.780 μ_{CR} = 0.087
 Diversity = 4.2624e-03 k_t = [3, 3, 3, 3]
 Generation 5770: Best = 3.148890e+03 FES = 577100 μ_F = 0.792 μ_{CR} = 0.087
 Diversity = 4.3405e-03 k_t = [1, 1, 1, 1]
 Generation 5780: Best = 3.148890e+03 FES = 578100 μ_F = 0.790 μ_{CR} = 0.083
 Diversity = 4.0214e-03 k_t = [3, 3, 3, 2]
 Generation 5790: Best = 3.148890e+03 FES = 579100 μ_F = 0.782 μ_{CR} = 0.076
 Diversity = 4.2944e-03 k_t = [1, 1, 1, 1]
 Generation 5800: Best = 3.148890e+03 FES = 580100 μ_F = 0.792 μ_{CR} = 0.074
 Diversity = 4.0580e-03 k_t = [3, 4, 3, 3]
 Generation 5810: Best = 3.148890e+03 FES = 581100 μ_F = 0.795 μ_{CR} = 0.081
 Diversity = 4.3025e-03 k_t = [1, 1, 1, 1]
 Generation 5820: Best = 3.148890e+03 FES = 582100 μ_F = 0.791 μ_{CR} = 0.083
 Diversity = 4.2081e-03 k_t = [3, 2, 3, 2]
 Generation 5830: Best = 3.148889e+03 FES = 583100 μ_F = 0.792 μ_{CR} = 0.075
 Diversity = 4.2138e-03 k_t = [1, 1, 1, 1]
 Generation 5840: Best = 3.148889e+03 FES = 584100 μ_F = 0.796 μ_{CR} = 0.077
 Diversity = 4.1154e-03 k_t = [4, 3, 2, 3]
 Generation 5850: Best = 3.148889e+03 FES = 585100 μ_F = 0.775 μ_{CR} = 0.070
 Diversity = 4.0347e-03 k_t = [1, 1, 1, 1]
 Generation 5860: Best = 3.148889e+03 FES = 586100 μ_F = 0.778 μ_{CR} = 0.070
 Diversity = 3.7788e-03 k_t = [3, 4, 2, 3]
 Generation 5870: Best = 3.148889e+03 FES = 587100 μ_F = 0.794 μ_{CR} = 0.072
 Diversity = 4.1413e-03 k_t = [1, 1, 1, 1]
 Generation 5880: Best = 3.148889e+03 FES = 588100 μ_F = 0.779 μ_{CR} = 0.085
 Diversity = 4.3342e-03 k_t = [4, 2, 3, 3]
 Generation 5890: Best = 3.148889e+03 FES = 589100 μ_F = 0.756 μ_{CR} = 0.087

Diversity = 4.1078e-03	k_t = [1, 1, 1, 1]			
Generation 5900: Best = 3.148889e+03	FES = 590100	$\mu_F = 0.762$	$\mu_{CR} = 0.086$	
Diversity = 3.8843e-03	k_t = [3, 3, 4, 3]			
Generation 5910: Best = 3.148889e+03	FES = 591100	$\mu_F = 0.771$	$\mu_{CR} = 0.081$	
Diversity = 3.9458e-03	k_t = [1, 1, 1, 1]			
Generation 5920: Best = 3.148889e+03	FES = 592100	$\mu_F = 0.793$	$\mu_{CR} = 0.087$	
Diversity = 4.4838e-03	k_t = [4, 3, 4, 4]			
Generation 5930: Best = 3.148889e+03	FES = 593100	$\mu_F = 0.789$	$\mu_{CR} = 0.083$	
Diversity = 4.6601e-03	k_t = [1, 1, 1, 1]			
Generation 5940: Best = 3.148889e+03	FES = 594100	$\mu_F = 0.784$	$\mu_{CR} = 0.090$	
Diversity = 4.1001e-03	k_t = [2, 3, 2, 4]			
Generation 5950: Best = 3.148889e+03	FES = 595100	$\mu_F = 0.776$	$\mu_{CR} = 0.080$	
Diversity = 3.8211e-03	k_t = [1, 1, 1, 1]			
Generation 5960: Best = 3.148889e+03	FES = 596100	$\mu_F = 0.758$	$\mu_{CR} = 0.083$	
Diversity = 4.1197e-03	k_t = [3, 4, 3, 4]			
Generation 5970: Best = 3.148889e+03	FES = 597100	$\mu_F = 0.757$	$\mu_{CR} = 0.072$	
Diversity = 4.4848e-03	k_t = [1, 1, 1, 1]			
Generation 5980: Best = 3.148888e+03	FES = 598100	$\mu_F = 0.760$	$\mu_{CR} = 0.077$	
Diversity = 4.2952e-03	k_t = [3, 3, 4, 2]			
Generation 5990: Best = 3.148888e+03	FES = 599100	$\mu_F = 0.755$	$\mu_{CR} = 0.081$	
Diversity = 4.9171e-03	k_t = [1, 1, 1, 1]			
Generation 6000: Best = 3.148888e+03	FES = 600100	$\mu_F = 0.772$	$\mu_{CR} = 0.077$	
Diversity = 5.3271e-03	k_t = [3, 3, 2, 2]			
Generation 6010: Best = 3.148888e+03	FES = 601100	$\mu_F = 0.765$	$\mu_{CR} = 0.077$	
Diversity = 4.6459e-03	k_t = [1, 1, 1, 1]			
Generation 6020: Best = 3.148888e+03	FES = 602100	$\mu_F = 0.772$	$\mu_{CR} = 0.079$	
Diversity = 4.3041e-03	k_t = [3, 4, 2, 4]			
Generation 6030: Best = 3.148888e+03	FES = 603100	$\mu_F = 0.775$	$\mu_{CR} = 0.084$	
Diversity = 4.2894e-03	k_t = [1, 1, 1, 1]			
Generation 6040: Best = 3.148888e+03	FES = 604100	$\mu_F = 0.773$	$\mu_{CR} = 0.085$	
Diversity = 4.6820e-03	k_t = [3, 3, 3, 3]			
Generation 6050: Best = 3.148888e+03	FES = 605100	$\mu_F = 0.770$	$\mu_{CR} = 0.094$	
Diversity = 4.6699e-03	k_t = [1, 1, 1, 1]			
Generation 6060: Best = 3.148888e+03	FES = 606100	$\mu_F = 0.773$	$\mu_{CR} = 0.087$	
Diversity = 4.6772e-03	k_t = [3, 4, 3, 3]			
Generation 6070: Best = 3.148888e+03	FES = 607100	$\mu_F = 0.769$	$\mu_{CR} = 0.086$	
Diversity = 4.6925e-03	k_t = [1, 1, 1, 1]			
Generation 6080: Best = 3.148887e+03	FES = 608100	$\mu_F = 0.775$	$\mu_{CR} = 0.088$	
Diversity = 5.3234e-03	k_t = [3, 3, 3, 3]			
Generation 6090: Best = 3.148887e+03	FES = 609100	$\mu_F = 0.781$	$\mu_{CR} = 0.076$	
Diversity = 5.1278e-03	k_t = [1, 1, 1, 1]			
Generation 6100: Best = 3.148887e+03	FES = 610100	$\mu_F = 0.793$	$\mu_{CR} = 0.082$	
Diversity = 5.1165e-03	k_t = [2, 3, 2, 2]			
Generation 6110: Best = 3.148887e+03	FES = 611100	$\mu_F = 0.806$	$\mu_{CR} = 0.075$	
Diversity = 4.7844e-03	k_t = [1, 1, 1, 1]			
Generation 6120: Best = 3.148887e+03	FES = 612100	$\mu_F = 0.813$	$\mu_{CR} = 0.082$	
Diversity = 5.1753e-03	k_t = [2, 2, 4, 3]			
Generation 6130: Best = 3.148887e+03	FES = 613100	$\mu_F = 0.804$	$\mu_{CR} = 0.082$	
Diversity = 5.7211e-03	k_t = [1, 1, 1, 1]			
Generation 6140: Best = 3.148887e+03	FES = 614100	$\mu_F = 0.795$	$\mu_{CR} = 0.078$	
Diversity = 5.2280e-03	k_t = [3, 4, 3, 4]			
Generation 6150: Best = 3.148887e+03	FES = 615100	$\mu_F = 0.793$	$\mu_{CR} = 0.079$	
Diversity = 4.7217e-03	k_t = [1, 1, 1, 1]			
Generation 6160: Best = 3.148887e+03	FES = 616100	$\mu_F = 0.792$	$\mu_{CR} = 0.083$	

Diversity = 4.2989e-03 k_t = [3, 4, 3, 3]
 Generation 6170: Best = 3.148887e+03 FES = 617100 μ_F = 0.775 μ_{CR} = 0.091
 Diversity = 4.4595e-03 k_t = [1, 1, 1, 1]
 Generation 6180: Best = 3.148887e+03 FES = 618100 μ_F = 0.767 μ_{CR} = 0.086
 Diversity = 4.5260e-03 k_t = [2, 3, 2, 2]
 Generation 6190: Best = 3.148886e+03 FES = 619100 μ_F = 0.778 μ_{CR} = 0.088
 Diversity = 5.5213e-03 k_t = [1, 1, 1, 1]
 Generation 6200: Best = 3.148886e+03 FES = 620100 μ_F = 0.792 μ_{CR} = 0.083
 Diversity = 5.3997e-03 k_t = [2, 4, 3, 4]
 Generation 6210: Best = 3.148886e+03 FES = 621100 μ_F = 0.804 μ_{CR} = 0.083
 Diversity = 5.3294e-03 k_t = [1, 1, 1, 1]
 Generation 6220: Best = 3.148886e+03 FES = 622100 μ_F = 0.796 μ_{CR} = 0.080
 Diversity = 5.1838e-03 k_t = [3, 2, 3, 4]
 Generation 6230: Best = 3.148886e+03 FES = 623100 μ_F = 0.793 μ_{CR} = 0.082
 Diversity = 6.1362e-03 k_t = [1, 1, 1, 1]
 Generation 6240: Best = 3.148886e+03 FES = 624100 μ_F = 0.795 μ_{CR} = 0.082
 Diversity = 5.8517e-03 k_t = [4, 2, 4, 3]
 Generation 6250: Best = 3.148886e+03 FES = 625100 μ_F = 0.788 μ_{CR} = 0.077
 Diversity = 5.9725e-03 k_t = [1, 1, 1, 1]
 Generation 6260: Best = 3.148886e+03 FES = 626100 μ_F = 0.794 μ_{CR} = 0.075
 Diversity = 5.6741e-03 k_t = [4, 4, 3, 4]
 Generation 6270: Best = 3.148885e+03 FES = 627100 μ_F = 0.782 μ_{CR} = 0.082
 Diversity = 5.2633e-03 k_t = [1, 1, 1, 1]
 Generation 6280: Best = 3.148885e+03 FES = 628100 μ_F = 0.793 μ_{CR} = 0.077
 Diversity = 4.3889e-03 k_t = [4, 2, 3, 4]
 Generation 6290: Best = 3.148885e+03 FES = 629100 μ_F = 0.791 μ_{CR} = 0.078
 Diversity = 4.2286e-03 k_t = [1, 1, 1, 1]
 Generation 6300: Best = 3.148885e+03 FES = 630100 μ_F = 0.774 μ_{CR} = 0.077
 Diversity = 4.0335e-03 k_t = [2, 2, 4, 2]
 Generation 6310: Best = 3.148885e+03 FES = 631100 μ_F = 0.775 μ_{CR} = 0.077
 Diversity = 4.7784e-03 k_t = [1, 1, 1, 1]
 Generation 6320: Best = 3.148885e+03 FES = 632100 μ_F = 0.786 μ_{CR} = 0.073
 Diversity = 5.2446e-03 k_t = [3, 4, 2, 4]
 Generation 6330: Best = 3.148885e+03 FES = 633100 μ_F = 0.776 μ_{CR} = 0.067
 Diversity = 4.8838e-03 k_t = [1, 1, 1, 1]
 Generation 6340: Best = 3.148885e+03 FES = 634100 μ_F = 0.760 μ_{CR} = 0.074
 Diversity = 5.0231e-03 k_t = [2, 3, 2, 4]
 Generation 6350: Best = 3.148885e+03 FES = 635100 μ_F = 0.758 μ_{CR} = 0.083
 Diversity = 4.5346e-03 k_t = [1, 1, 1, 1]
 Generation 6360: Best = 3.148885e+03 FES = 636100 μ_F = 0.751 μ_{CR} = 0.078
 Diversity = 4.9742e-03 k_t = [3, 2, 4, 2]
 Generation 6370: Best = 3.148885e+03 FES = 637100 μ_F = 0.774 μ_{CR} = 0.087
 Diversity = 5.7403e-03 k_t = [1, 1, 1, 1]
 Generation 6380: Best = 3.148885e+03 FES = 638100 μ_F = 0.773 μ_{CR} = 0.081
 Diversity = 6.2169e-03 k_t = [3, 2, 3, 2]
 Generation 6390: Best = 3.148884e+03 FES = 639100 μ_F = 0.772 μ_{CR} = 0.078
 Diversity = 7.1988e-03 k_t = [1, 1, 1, 1]
 Generation 6400: Best = 3.148884e+03 FES = 640100 μ_F = 0.778 μ_{CR} = 0.080
 Diversity = 6.9775e-03 k_t = [2, 4, 4, 3]
 Generation 6410: Best = 3.148884e+03 FES = 641100 μ_F = 0.779 μ_{CR} = 0.082
 Diversity = 6.6482e-03 k_t = [1, 1, 1, 1]
 Generation 6420: Best = 3.148884e+03 FES = 642100 μ_F = 0.789 μ_{CR} = 0.082
 Diversity = 6.5270e-03 k_t = [3, 3, 2, 4]
 Generation 6430: Best = 3.148884e+03 FES = 643100 μ_F = 0.811 μ_{CR} = 0.081

Diversity = 5.9026e-03	k_t = [1, 1, 1, 1]			
Generation 6440: Best = 3.148884e+03	FES = 644100	$\mu_F = 0.796$	$\mu_{CR} = 0.077$	
Diversity = 6.6718e-03	k_t = [3, 3, 4, 3]			
Generation 6450: Best = 3.148884e+03	FES = 645100	$\mu_F = 0.797$	$\mu_{CR} = 0.073$	
Diversity = 6.9843e-03	k_t = [1, 1, 1, 1]			
Generation 6460: Best = 3.148883e+03	FES = 646100	$\mu_F = 0.796$	$\mu_{CR} = 0.078$	
Diversity = 6.3952e-03	k_t = [4, 3, 2, 2]			
Generation 6470: Best = 3.148883e+03	FES = 647100	$\mu_F = 0.778$	$\mu_{CR} = 0.079$	
Diversity = 5.9693e-03	k_t = [1, 1, 1, 1]			
Generation 6480: Best = 3.148883e+03	FES = 648100	$\mu_F = 0.786$	$\mu_{CR} = 0.082$	
Diversity = 6.2957e-03	k_t = [3, 4, 4, 3]			
Generation 6490: Best = 3.148883e+03	FES = 649100	$\mu_F = 0.793$	$\mu_{CR} = 0.083$	
Diversity = 7.1869e-03	k_t = [1, 1, 1, 1]			
Generation 6500: Best = 3.148883e+03	FES = 650100	$\mu_F = 0.789$	$\mu_{CR} = 0.076$	
Diversity = 6.8651e-03	k_t = [4, 3, 2, 3]			
Generation 6510: Best = 3.148883e+03	FES = 651100	$\mu_F = 0.767$	$\mu_{CR} = 0.088$	
Diversity = 6.6995e-03	k_t = [1, 1, 1, 1]			
Generation 6520: Best = 3.148882e+03	FES = 652100	$\mu_F = 0.784$	$\mu_{CR} = 0.079$	
Diversity = 7.1298e-03	k_t = [2, 4, 2, 4]			
Generation 6530: Best = 3.148882e+03	FES = 653100	$\mu_F = 0.766$	$\mu_{CR} = 0.077$	
Diversity = 7.3479e-03	k_t = [1, 1, 1, 1]			
Generation 6540: Best = 3.148882e+03	FES = 654100	$\mu_F = 0.764$	$\mu_{CR} = 0.069$	
Diversity = 6.4231e-03	k_t = [3, 2, 4, 4]			
Generation 6550: Best = 3.148882e+03	FES = 655100	$\mu_F = 0.783$	$\mu_{CR} = 0.070$	
Diversity = 7.0003e-03	k_t = [1, 1, 1, 1]			
Generation 6560: Best = 3.148882e+03	FES = 656100	$\mu_F = 0.788$	$\mu_{CR} = 0.073$	
Diversity = 6.3690e-03	k_t = [4, 2, 2, 4]			
Generation 6570: Best = 3.148881e+03	FES = 657100	$\mu_F = 0.779$	$\mu_{CR} = 0.077$	
Diversity = 8.1332e-03	k_t = [1, 1, 1, 1]			
Generation 6580: Best = 3.148881e+03	FES = 658100	$\mu_F = 0.773$	$\mu_{CR} = 0.082$	
Diversity = 8.2504e-03	k_t = [4, 4, 4, 4]			
Generation 6590: Best = 3.148881e+03	FES = 659100	$\mu_F = 0.775$	$\mu_{CR} = 0.074$	
Diversity = 7.1714e-03	k_t = [1, 1, 1, 1]			
Generation 6600: Best = 3.148881e+03	FES = 660100	$\mu_F = 0.778$	$\mu_{CR} = 0.072$	
Diversity = 6.7336e-03	k_t = [4, 4, 4, 3]			
Generation 6610: Best = 3.148881e+03	FES = 661100	$\mu_F = 0.767$	$\mu_{CR} = 0.073$	
Diversity = 7.1549e-03	k_t = [1, 1, 1, 1]			
Generation 6620: Best = 3.148881e+03	FES = 662100	$\mu_F = 0.789$	$\mu_{CR} = 0.070$	
Diversity = 7.1927e-03	k_t = [3, 4, 3, 4]			
Generation 6630: Best = 3.148881e+03	FES = 663100	$\mu_F = 0.777$	$\mu_{CR} = 0.076$	
Diversity = 8.0049e-03	k_t = [1, 1, 1, 1]			
Generation 6640: Best = 3.148881e+03	FES = 664100	$\mu_F = 0.781$	$\mu_{CR} = 0.074$	
Diversity = 7.7385e-03	k_t = [4, 3, 3, 2]			
Generation 6650: Best = 3.148880e+03	FES = 665100	$\mu_F = 0.799$	$\mu_{CR} = 0.075$	
Diversity = 7.8681e-03	k_t = [1, 1, 1, 1]			
Generation 6660: Best = 3.148880e+03	FES = 666100	$\mu_F = 0.777$	$\mu_{CR} = 0.065$	
Diversity = 8.4126e-03	k_t = [4, 2, 4, 3]			
Generation 6670: Best = 3.148880e+03	FES = 667100	$\mu_F = 0.787$	$\mu_{CR} = 0.066$	
Diversity = 8.4639e-03	k_t = [1, 1, 1, 1]			
Generation 6680: Best = 3.148880e+03	FES = 668100	$\mu_F = 0.783$	$\mu_{CR} = 0.070$	
Diversity = 9.0522e-03	k_t = [4, 4, 3, 2]			
Generation 6690: Best = 3.148880e+03	FES = 669100	$\mu_F = 0.770$	$\mu_{CR} = 0.079$	
Diversity = 9.7358e-03	k_t = [1, 1, 1, 1]			
Generation 6700: Best = 3.148880e+03	FES = 670100	$\mu_F = 0.797$	$\mu_{CR} = 0.076$	

Diversity = 1.0650e-02	k_t = [2, 3, 4, 2]			
Generation 6710: Best = 3.148879e+03	FES = 671100	$\mu_F = 0.780$	$\mu_{CR} = 0.078$	
Diversity = 1.0029e-02	k_t = [1, 1, 1, 1]			
Generation 6720: Best = 3.148879e+03	FES = 672100	$\mu_F = 0.780$	$\mu_{CR} = 0.081$	
Diversity = 1.0053e-02	k_t = [2, 4, 2, 2]			
Generation 6730: Best = 3.148879e+03	FES = 673100	$\mu_F = 0.751$	$\mu_{CR} = 0.074$	
Diversity = 9.1156e-03	k_t = [1, 1, 1, 1]			
Generation 6740: Best = 3.148879e+03	FES = 674100	$\mu_F = 0.738$	$\mu_{CR} = 0.076$	
Diversity = 8.9107e-03	k_t = [4, 3, 2, 3]			
Generation 6750: Best = 3.148879e+03	FES = 675100	$\mu_F = 0.771$	$\mu_{CR} = 0.076$	
Diversity = 1.0273e-02	k_t = [1, 1, 1, 1]			
Generation 6760: Best = 3.148879e+03	FES = 676100	$\mu_F = 0.754$	$\mu_{CR} = 0.074$	
Diversity = 1.0488e-02	k_t = [4, 3, 3, 2]			
Generation 6770: Best = 3.148879e+03	FES = 677100	$\mu_F = 0.780$	$\mu_{CR} = 0.077$	
Diversity = 1.2229e-02	k_t = [1, 1, 1, 1]			
Generation 6780: Best = 3.148879e+03	FES = 678100	$\mu_F = 0.785$	$\mu_{CR} = 0.080$	
Diversity = 1.1116e-02	k_t = [3, 3, 4, 4]			
Generation 6790: Best = 3.148878e+03	FES = 679100	$\mu_F = 0.796$	$\mu_{CR} = 0.072$	
Diversity = 1.0617e-02	k_t = [1, 1, 1, 1]			
Generation 6800: Best = 3.148878e+03	FES = 680100	$\mu_F = 0.805$	$\mu_{CR} = 0.065$	
Diversity = 1.2676e-02	k_t = [4, 2, 2, 2]			
Generation 6810: Best = 3.148878e+03	FES = 681100	$\mu_F = 0.800$	$\mu_{CR} = 0.058$	
Diversity = 1.3470e-02	k_t = [1, 1, 1, 1]			
Generation 6820: Best = 3.148878e+03	FES = 682100	$\mu_F = 0.815$	$\mu_{CR} = 0.058$	
Diversity = 1.3835e-02	k_t = [2, 4, 4, 4]			
Generation 6830: Best = 3.148878e+03	FES = 683100	$\mu_F = 0.806$	$\mu_{CR} = 0.065$	
Diversity = 1.2985e-02	k_t = [1, 1, 1, 1]			
Generation 6840: Best = 3.148878e+03	FES = 684100	$\mu_F = 0.771$	$\mu_{CR} = 0.071$	
Diversity = 1.1626e-02	k_t = [3, 4, 4, 2]			
Generation 6850: Best = 3.148877e+03	FES = 685100	$\mu_F = 0.767$	$\mu_{CR} = 0.077$	
Diversity = 1.1431e-02	k_t = [1, 1, 1, 1]			
Generation 6860: Best = 3.148877e+03	FES = 686100	$\mu_F = 0.773$	$\mu_{CR} = 0.073$	
Diversity = 1.1156e-02	k_t = [3, 2, 2, 2]			
Generation 6870: Best = 3.148877e+03	FES = 687100	$\mu_F = 0.791$	$\mu_{CR} = 0.065$	
Diversity = 1.2887e-02	k_t = [1, 1, 1, 1]			
Generation 6880: Best = 3.148877e+03	FES = 688100	$\mu_F = 0.788$	$\mu_{CR} = 0.065$	
Diversity = 1.2230e-02	k_t = [4, 3, 3, 4]			
Generation 6890: Best = 3.148876e+03	FES = 689100	$\mu_F = 0.778$	$\mu_{CR} = 0.065$	
Diversity = 1.7985e-02	k_t = [1, 1, 1, 1]			
Generation 6900: Best = 3.148875e+03	FES = 690100	$\mu_F = 0.782$	$\mu_{CR} = 0.064$	
Diversity = 1.7392e-02	k_t = [4, 4, 4, 3]			
Generation 6910: Best = 3.148875e+03	FES = 691100	$\mu_F = 0.785$	$\mu_{CR} = 0.068$	
Diversity = 2.0376e-02	k_t = [1, 1, 1, 1]			
Generation 6920: Best = 3.148875e+03	FES = 692100	$\mu_F = 0.772$	$\mu_{CR} = 0.072$	
Diversity = 2.1299e-02	k_t = [2, 3, 4, 4]			
Generation 6930: Best = 3.148875e+03	FES = 693100	$\mu_F = 0.769$	$\mu_{CR} = 0.065$	
Diversity = 1.9759e-02	k_t = [1, 1, 1, 1]			
Generation 6940: Best = 3.148874e+03	FES = 694100	$\mu_F = 0.787$	$\mu_{CR} = 0.062$	
Diversity = 1.9966e-02	k_t = [4, 3, 2, 3]			
Generation 6950: Best = 3.148873e+03	FES = 695100	$\mu_F = 0.784$	$\mu_{CR} = 0.060$	
Diversity = 1.9703e-02	k_t = [1, 1, 1, 1]			
Generation 6960: Best = 3.148873e+03	FES = 696100	$\mu_F = 0.777$	$\mu_{CR} = 0.067$	
Diversity = 2.0247e-02	k_t = [3, 2, 2, 4]			
Generation 6970: Best = 3.148873e+03	FES = 697100	$\mu_F = 0.784$	$\mu_{CR} = 0.066$	

Diversity = 2.2310e-02 k_t = [1, 1, 1, 1]
 Generation 6980: Best = 3.148873e+03 FES = 698100 μ_F = 0.777 μ_{CR} = 0.059
 Diversity = 1.9753e-02 k_t = [4, 3, 3, 4]
 Generation 6990: Best = 3.148873e+03 FES = 699100 μ_F = 0.781 μ_{CR} = 0.054
 Diversity = 1.8385e-02 k_t = [1, 1, 1, 1]
 Generation 7000: Best = 3.148872e+03 FES = 700100 μ_F = 0.794 μ_{CR} = 0.051
 Diversity = 1.8335e-02 k_t = [4, 3, 4, 4]
 Generation 7010: Best = 3.148872e+03 FES = 701100 μ_F = 0.804 μ_{CR} = 0.051
 Diversity = 1.8796e-02 k_t = [1, 1, 1, 1]
 Generation 7020: Best = 3.148872e+03 FES = 702100 μ_F = 0.771 μ_{CR} = 0.063
 Diversity = 1.7053e-02 k_t = [4, 4, 2, 2]
 Generation 7030: Best = 3.148872e+03 FES = 703100 μ_F = 0.775 μ_{CR} = 0.061
 Diversity = 1.7232e-02 k_t = [1, 1, 1, 1]
 Generation 7040: Best = 3.148872e+03 FES = 704100 μ_F = 0.771 μ_{CR} = 0.060
 Diversity = 1.7308e-02 k_t = [3, 2, 2, 3]
 Generation 7050: Best = 3.148872e+03 FES = 705100 μ_F = 0.780 μ_{CR} = 0.059
 Diversity = 1.8508e-02 k_t = [1, 1, 1, 1]
 Generation 7060: Best = 3.148871e+03 FES = 706100 μ_F = 0.755 μ_{CR} = 0.067
 Diversity = 1.7198e-02 k_t = [4, 3, 4, 4]
 Generation 7070: Best = 3.148870e+03 FES = 707100 μ_F = 0.786 μ_{CR} = 0.064
 Diversity = 1.7211e-02 k_t = [1, 1, 1, 1]
 Generation 7080: Best = 3.148870e+03 FES = 708100 μ_F = 0.791 μ_{CR} = 0.070
 Diversity = 2.0205e-02 k_t = [4, 4, 3, 3]
 Generation 7090: Best = 3.148870e+03 FES = 709100 μ_F = 0.792 μ_{CR} = 0.062
 Diversity = 1.9737e-02 k_t = [1, 1, 1, 1]
 Generation 7100: Best = 3.148870e+03 FES = 710100 μ_F = 0.794 μ_{CR} = 0.058
 Diversity = 2.1641e-02 k_t = [3, 4, 3, 3]
 Generation 7110: Best = 3.148870e+03 FES = 711100 μ_F = 0.810 μ_{CR} = 0.063
 Diversity = 2.1409e-02 k_t = [1, 1, 1, 1]
 Generation 7120: Best = 3.148870e+03 FES = 712100 μ_F = 0.816 μ_{CR} = 0.059
 Diversity = 2.0463e-02 k_t = [4, 3, 2, 3]
 Generation 7130: Best = 3.148870e+03 FES = 713100 μ_F = 0.825 μ_{CR} = 0.069
 Diversity = 1.7447e-02 k_t = [1, 1, 1, 1]
 Generation 7140: Best = 3.148870e+03 FES = 714100 μ_F = 0.811 μ_{CR} = 0.063
 Diversity = 1.5034e-02 k_t = [3, 4, 3, 2]
 Generation 7150: Best = 3.148870e+03 FES = 715100 μ_F = 0.811 μ_{CR} = 0.065
 Diversity = 1.5895e-02 k_t = [1, 1, 1, 1]
 Generation 7160: Best = 3.148870e+03 FES = 716100 μ_F = 0.791 μ_{CR} = 0.061
 Diversity = 1.4551e-02 k_t = [2, 3, 4, 3]
 Generation 7170: Best = 3.148869e+03 FES = 717100 μ_F = 0.784 μ_{CR} = 0.063
 Diversity = 1.6060e-02 k_t = [1, 1, 1, 1]
 Generation 7180: Best = 3.148869e+03 FES = 718100 μ_F = 0.777 μ_{CR} = 0.065
 Diversity = 1.5295e-02 k_t = [3, 2, 3, 3]
 Generation 7190: Best = 3.148869e+03 FES = 719100 μ_F = 0.774 μ_{CR} = 0.062
 Diversity = 1.5868e-02 k_t = [1, 1, 1, 1]
 Generation 7200: Best = 3.148869e+03 FES = 720100 μ_F = 0.778 μ_{CR} = 0.059
 Diversity = 1.5522e-02 k_t = [4, 2, 3, 3]
 Generation 7210: Best = 3.148869e+03 FES = 721100 μ_F = 0.775 μ_{CR} = 0.060
 Diversity = 1.6966e-02 k_t = [1, 1, 1, 1]
 Generation 7220: Best = 3.148869e+03 FES = 722100 μ_F = 0.781 μ_{CR} = 0.063
 Diversity = 1.6663e-02 k_t = [4, 2, 4, 3]
 Generation 7230: Best = 3.148869e+03 FES = 723100 μ_F = 0.755 μ_{CR} = 0.053
 Diversity = 1.4659e-02 k_t = [1, 1, 1, 1]
 Generation 7240: Best = 3.148869e+03 FES = 724100 μ_F = 0.750 μ_{CR} = 0.063

Diversity = 1.3524e-02	k_t = [2, 3, 2, 4]			
Generation 7250: Best = 3.148868e+03	FES = 725100	$\mu_F = 0.762$	$\mu_{CR} = 0.062$	
Diversity = 1.1946e-02	k_t = [1, 1, 1, 1]			
Generation 7260: Best = 3.148868e+03	FES = 726100	$\mu_F = 0.791$	$\mu_{CR} = 0.062$	
Diversity = 1.5333e-02	k_t = [2, 2, 4, 3]			
Generation 7270: Best = 3.148868e+03	FES = 727100	$\mu_F = 0.798$	$\mu_{CR} = 0.063$	
Diversity = 1.8021e-02	k_t = [1, 1, 1, 1]			
Generation 7280: Best = 3.148868e+03	FES = 728100	$\mu_F = 0.822$	$\mu_{CR} = 0.064$	
Diversity = 1.7024e-02	k_t = [2, 4, 4, 4]			
Generation 7290: Best = 3.148868e+03	FES = 729100	$\mu_F = 0.812$	$\mu_{CR} = 0.063$	
Diversity = 1.6195e-02	k_t = [1, 1, 1, 1]			
Generation 7300: Best = 3.148867e+03	FES = 730100	$\mu_F = 0.815$	$\mu_{CR} = 0.070$	
Diversity = 1.6791e-02	k_t = [4, 4, 2, 3]			
Generation 7310: Best = 3.148867e+03	FES = 731100	$\mu_F = 0.822$	$\mu_{CR} = 0.073$	
Diversity = 2.2146e-02	k_t = [1, 1, 1, 1]			
Generation 7320: Best = 3.148867e+03	FES = 732100	$\mu_F = 0.831$	$\mu_{CR} = 0.068$	
Diversity = 2.1553e-02	k_t = [4, 2, 4, 3]			
Generation 7330: Best = 3.148867e+03	FES = 733100	$\mu_F = 0.816$	$\mu_{CR} = 0.073$	
Diversity = 1.9178e-02	k_t = [1, 1, 1, 1]			
Generation 7340: Best = 3.148867e+03	FES = 734100	$\mu_F = 0.803$	$\mu_{CR} = 0.069$	
Diversity = 1.6637e-02	k_t = [4, 4, 2, 4]			
Generation 7350: Best = 3.148867e+03	FES = 735100	$\mu_F = 0.808$	$\mu_{CR} = 0.056$	
Diversity = 1.8312e-02	k_t = [1, 1, 1, 1]			
Generation 7360: Best = 3.148867e+03	FES = 736100	$\mu_F = 0.795$	$\mu_{CR} = 0.053$	
Diversity = 1.8087e-02	k_t = [3, 2, 3, 3]			
Generation 7370: Best = 3.148866e+03	FES = 737100	$\mu_F = 0.784$	$\mu_{CR} = 0.063$	
Diversity = 1.7324e-02	k_t = [1, 1, 1, 1]			
Generation 7380: Best = 3.148866e+03	FES = 738100	$\mu_F = 0.794$	$\mu_{CR} = 0.063$	
Diversity = 1.5862e-02	k_t = [3, 2, 4, 4]			
Generation 7390: Best = 3.148866e+03	FES = 739100	$\mu_F = 0.792$	$\mu_{CR} = 0.069$	
Diversity = 1.6049e-02	k_t = [1, 1, 1, 1]			
Generation 7400: Best = 3.148866e+03	FES = 740100	$\mu_F = 0.785$	$\mu_{CR} = 0.059$	
Diversity = 1.8129e-02	k_t = [2, 3, 4, 4]			
Generation 7410: Best = 3.148865e+03	FES = 741100	$\mu_F = 0.784$	$\mu_{CR} = 0.064$	
Diversity = 1.9644e-02	k_t = [1, 1, 1, 1]			
Generation 7420: Best = 3.148865e+03	FES = 742100	$\mu_F = 0.798$	$\mu_{CR} = 0.063$	
Diversity = 2.7784e-02	k_t = [2, 3, 4, 3]			
Generation 7430: Best = 3.148864e+03	FES = 743100	$\mu_F = 0.787$	$\mu_{CR} = 0.059$	
Diversity = 3.2414e-02	k_t = [1, 1, 1, 1]			
Generation 7440: Best = 3.148864e+03	FES = 744100	$\mu_F = 0.796$	$\mu_{CR} = 0.059$	
Diversity = 3.2423e-02	k_t = [2, 3, 4, 3]			
Generation 7450: Best = 3.148863e+03	FES = 745100	$\mu_F = 0.795$	$\mu_{CR} = 0.059$	
Diversity = 3.4730e-02	k_t = [1, 1, 1, 1]			
Generation 7460: Best = 3.148862e+03	FES = 746100	$\mu_F = 0.799$	$\mu_{CR} = 0.064$	
Diversity = 3.9066e-02	k_t = [2, 3, 3, 4]			
Generation 7470: Best = 3.148862e+03	FES = 747100	$\mu_F = 0.764$	$\mu_{CR} = 0.061$	
Diversity = 4.1113e-02	k_t = [1, 1, 1, 1]			
Generation 7480: Best = 3.148862e+03	FES = 748100	$\mu_F = 0.769$	$\mu_{CR} = 0.061$	
Diversity = 3.4023e-02	k_t = [3, 3, 2, 4]			
Generation 7490: Best = 3.148862e+03	FES = 749100	$\mu_F = 0.784$	$\mu_{CR} = 0.061$	
Diversity = 2.9171e-02	k_t = [1, 1, 1, 1]			
Generation 7500: Best = 3.148862e+03	FES = 750100	$\mu_F = 0.789$	$\mu_{CR} = 0.060$	
Diversity = 2.4578e-02	k_t = [2, 4, 3, 3]			
Generation 7510: Best = 3.148862e+03	FES = 751100	$\mu_F = 0.781$	$\mu_{CR} = 0.073$	

Diversity = 2.3301e-02	k_t = [1, 1, 1, 1]			
Generation 7520: Best = 3.148862e+03	FES = 752100	$\mu_F = 0.786$	$\mu_{CR} = 0.065$	
Diversity = 2.1873e-02	k_t = [2, 3, 2, 4]			
Generation 7530: Best = 3.148862e+03	FES = 753100	$\mu_F = 0.790$	$\mu_{CR} = 0.067$	
Diversity = 1.9436e-02	k_t = [1, 1, 1, 1]			
Generation 7540: Best = 3.148862e+03	FES = 754100	$\mu_F = 0.792$	$\mu_{CR} = 0.064$	
Diversity = 1.7823e-02	k_t = [4, 2, 2, 4]			
Generation 7550: Best = 3.148862e+03	FES = 755100	$\mu_F = 0.808$	$\mu_{CR} = 0.069$	
Diversity = 1.5564e-02	k_t = [1, 1, 1, 1]			
Generation 7560: Best = 3.148861e+03	FES = 756100	$\mu_F = 0.795$	$\mu_{CR} = 0.073$	
Diversity = 1.4680e-02	k_t = [2, 3, 4, 2]			
Generation 7570: Best = 3.148861e+03	FES = 757100	$\mu_F = 0.781$	$\mu_{CR} = 0.069$	
Diversity = 1.4221e-02	k_t = [1, 1, 1, 1]			
Generation 7580: Best = 3.148861e+03	FES = 758100	$\mu_F = 0.787$	$\mu_{CR} = 0.064$	
Diversity = 1.2210e-02	k_t = [4, 2, 4, 3]			
Generation 7590: Best = 3.148861e+03	FES = 759100	$\mu_F = 0.793$	$\mu_{CR} = 0.061$	
Diversity = 1.2096e-02	k_t = [1, 1, 1, 1]			
Generation 7600: Best = 3.148861e+03	FES = 760100	$\mu_F = 0.795$	$\mu_{CR} = 0.069$	
Diversity = 1.2471e-02	k_t = [4, 3, 4, 3]			
Generation 7610: Best = 3.148861e+03	FES = 761100	$\mu_F = 0.789$	$\mu_{CR} = 0.063$	
Diversity = 1.2017e-02	k_t = [1, 1, 1, 1]			
Generation 7620: Best = 3.148861e+03	FES = 762100	$\mu_F = 0.779$	$\mu_{CR} = 0.073$	
Diversity = 1.0838e-02	k_t = [4, 4, 3, 2]			
Generation 7630: Best = 3.148861e+03	FES = 763100	$\mu_F = 0.772$	$\mu_{CR} = 0.069$	
Diversity = 1.1162e-02	k_t = [1, 1, 1, 1]			
Generation 7640: Best = 3.148861e+03	FES = 764100	$\mu_F = 0.789$	$\mu_{CR} = 0.071$	
Diversity = 1.0082e-02	k_t = [2, 2, 3, 4]			
Generation 7650: Best = 3.148861e+03	FES = 765100	$\mu_F = 0.784$	$\mu_{CR} = 0.072$	
Diversity = 1.0316e-02	k_t = [1, 1, 1, 1]			
Generation 7660: Best = 3.148861e+03	FES = 766100	$\mu_F = 0.764$	$\mu_{CR} = 0.070$	
Diversity = 9.1447e-03	k_t = [2, 3, 2, 3]			
Generation 7670: Best = 3.148861e+03	FES = 767100	$\mu_F = 0.771$	$\mu_{CR} = 0.070$	
Diversity = 1.0591e-02	k_t = [1, 1, 1, 1]			
Generation 7680: Best = 3.148861e+03	FES = 768100	$\mu_F = 0.760$	$\mu_{CR} = 0.074$	
Diversity = 9.7629e-03	k_t = [4, 4, 3, 4]			
Generation 7690: Best = 3.148860e+03	FES = 769100	$\mu_F = 0.768$	$\mu_{CR} = 0.067$	
Diversity = 1.0728e-02	k_t = [1, 1, 1, 1]			
Generation 7700: Best = 3.148860e+03	FES = 770100	$\mu_F = 0.785$	$\mu_{CR} = 0.070$	
Diversity = 1.1507e-02	k_t = [4, 2, 2, 3]			
Generation 7710: Best = 3.148860e+03	FES = 771100	$\mu_F = 0.794$	$\mu_{CR} = 0.068$	
Diversity = 1.1768e-02	k_t = [1, 1, 1, 1]			
Generation 7720: Best = 3.148860e+03	FES = 772100	$\mu_F = 0.805$	$\mu_{CR} = 0.065$	
Diversity = 1.1324e-02	k_t = [4, 2, 4, 2]			
Generation 7730: Best = 3.148860e+03	FES = 773100	$\mu_F = 0.807$	$\mu_{CR} = 0.068$	
Diversity = 1.3449e-02	k_t = [1, 1, 1, 1]			
Generation 7740: Best = 3.148860e+03	FES = 774100	$\mu_F = 0.813$	$\mu_{CR} = 0.068$	
Diversity = 1.4849e-02	k_t = [3, 4, 3, 2]			
Generation 7750: Best = 3.148860e+03	FES = 775100	$\mu_F = 0.811$	$\mu_{CR} = 0.058$	
Diversity = 1.4031e-02	k_t = [1, 1, 1, 1]			
Generation 7760: Best = 3.148860e+03	FES = 776100	$\mu_F = 0.817$	$\mu_{CR} = 0.059$	
Diversity = 1.3648e-02	k_t = [3, 3, 2, 2]			
Generation 7770: Best = 3.148860e+03	FES = 777100	$\mu_F = 0.814$	$\mu_{CR} = 0.058$	
Diversity = 1.4684e-02	k_t = [1, 1, 1, 1]			
Generation 7780: Best = 3.148859e+03	FES = 778100	$\mu_F = 0.811$	$\mu_{CR} = 0.060$	

Diversity = 1.3247e-02	k_t = [2, 3, 3, 3]			
Generation 7790: Best = 3.148859e+03	FES = 779100	$\mu_F = 0.790$	$\mu_{CR} = 0.067$	
Diversity = 1.4130e-02	k_t = [1, 1, 1, 1]			
Generation 7800: Best = 3.148859e+03	FES = 780100	$\mu_F = 0.782$	$\mu_{CR} = 0.079$	
Diversity = 1.4307e-02	k_t = [4, 3, 2, 3]			
Generation 7810: Best = 3.148859e+03	FES = 781100	$\mu_F = 0.764$	$\mu_{CR} = 0.079$	
Diversity = 1.2313e-02	k_t = [1, 1, 1, 1]			
Generation 7820: Best = 3.148859e+03	FES = 782100	$\mu_F = 0.765$	$\mu_{CR} = 0.068$	
Diversity = 1.2073e-02	k_t = [3, 3, 3, 2]			
Generation 7830: Best = 3.148859e+03	FES = 783100	$\mu_F = 0.774$	$\mu_{CR} = 0.059$	
Diversity = 1.3037e-02	k_t = [1, 1, 1, 1]			
Generation 7840: Best = 3.148859e+03	FES = 784100	$\mu_F = 0.773$	$\mu_{CR} = 0.058$	
Diversity = 1.2039e-02	k_t = [3, 4, 4, 4]			
Generation 7850: Best = 3.148859e+03	FES = 785100	$\mu_F = 0.778$	$\mu_{CR} = 0.056$	
Diversity = 1.2995e-02	k_t = [1, 1, 1, 1]			
Generation 7860: Best = 3.148858e+03	FES = 786100	$\mu_F = 0.803$	$\mu_{CR} = 0.060$	
Diversity = 1.3097e-02	k_t = [3, 2, 4, 3]			
Generation 7870: Best = 3.148858e+03	FES = 787100	$\mu_F = 0.783$	$\mu_{CR} = 0.056$	
Diversity = 1.2577e-02	k_t = [1, 1, 1, 1]			
Generation 7880: Best = 3.148858e+03	FES = 788100	$\mu_F = 0.784$	$\mu_{CR} = 0.065$	
Diversity = 1.1045e-02	k_t = [3, 3, 4, 4]			
Generation 7890: Best = 3.148858e+03	FES = 789100	$\mu_F = 0.791$	$\mu_{CR} = 0.066$	
Diversity = 1.0351e-02	k_t = [1, 1, 1, 1]			
Generation 7900: Best = 3.148858e+03	FES = 790100	$\mu_F = 0.792$	$\mu_{CR} = 0.071$	
Diversity = 1.2062e-02	k_t = [4, 4, 2, 2]			
Generation 7910: Best = 3.148858e+03	FES = 791100	$\mu_F = 0.786$	$\mu_{CR} = 0.065$	
Diversity = 1.1453e-02	k_t = [1, 1, 1, 1]			
Generation 7920: Best = 3.148858e+03	FES = 792100	$\mu_F = 0.780$	$\mu_{CR} = 0.067$	
Diversity = 1.1000e-02	k_t = [4, 3, 3, 2]			
Generation 7930: Best = 3.148858e+03	FES = 793100	$\mu_F = 0.780$	$\mu_{CR} = 0.059$	
Diversity = 1.0772e-02	k_t = [1, 1, 1, 1]			
Generation 7940: Best = 3.148858e+03	FES = 794100	$\mu_F = 0.790$	$\mu_{CR} = 0.066$	
Diversity = 1.0477e-02	k_t = [4, 3, 3, 3]			
Generation 7950: Best = 3.148858e+03	FES = 795100	$\mu_F = 0.779$	$\mu_{CR} = 0.069$	
Diversity = 1.1213e-02	k_t = [1, 1, 1, 1]			
Generation 7960: Best = 3.148858e+03	FES = 796100	$\mu_F = 0.796$	$\mu_{CR} = 0.068$	
Diversity = 1.1342e-02	k_t = [4, 2, 3, 2]			
Generation 7970: Best = 3.148858e+03	FES = 797100	$\mu_F = 0.798$	$\mu_{CR} = 0.066$	
Diversity = 1.3452e-02	k_t = [1, 1, 1, 1]			
Generation 7980: Best = 3.148858e+03	FES = 798100	$\mu_F = 0.784$	$\mu_{CR} = 0.066$	
Diversity = 1.4186e-02	k_t = [3, 4, 2, 2]			
Generation 7990: Best = 3.148858e+03	FES = 799100	$\mu_F = 0.760$	$\mu_{CR} = 0.070$	
Diversity = 1.3363e-02	k_t = [1, 1, 1, 1]			
Generation 8000: Best = 3.148857e+03	FES = 800100	$\mu_F = 0.762$	$\mu_{CR} = 0.064$	
Diversity = 1.3385e-02	k_t = [2, 2, 2, 2]			
Generation 8010: Best = 3.148857e+03	FES = 801100	$\mu_F = 0.764$	$\mu_{CR} = 0.067$	
Diversity = 1.3765e-02	k_t = [1, 1, 1, 1]			
Generation 8020: Best = 3.148857e+03	FES = 802100	$\mu_F = 0.769$	$\mu_{CR} = 0.066$	
Diversity = 1.2220e-02	k_t = [2, 3, 4, 4]			
Generation 8030: Best = 3.148857e+03	FES = 803100	$\mu_F = 0.758$	$\mu_{CR} = 0.073$	
Diversity = 1.3475e-02	k_t = [1, 1, 1, 1]			
Generation 8040: Best = 3.148857e+03	FES = 804100	$\mu_F = 0.763$	$\mu_{CR} = 0.064$	
Diversity = 1.4510e-02	k_t = [3, 4, 2, 2]			
Generation 8050: Best = 3.148857e+03	FES = 805100	$\mu_F = 0.772$	$\mu_{CR} = 0.068$	

Diversity = 1.6639e-02	k_t = [1, 1, 1, 1]			
Generation 8060: Best = 3.148857e+03	FES = 806100	$\mu_F = 0.779$	$\mu_{CR} = 0.067$	
Diversity = 1.6835e-02	k_t = [3, 4, 2, 2]			
Generation 8070: Best = 3.148857e+03	FES = 807100	$\mu_F = 0.788$	$\mu_{CR} = 0.064$	
Diversity = 1.6303e-02	k_t = [1, 1, 1, 1]			
Generation 8080: Best = 3.148857e+03	FES = 808100	$\mu_F = 0.783$	$\mu_{CR} = 0.065$	
Diversity = 1.5501e-02	k_t = [4, 2, 2, 3]			
Generation 8090: Best = 3.148857e+03	FES = 809100	$\mu_F = 0.776$	$\mu_{CR} = 0.067$	
Diversity = 1.4649e-02	k_t = [1, 1, 1, 1]			
Generation 8100: Best = 3.148857e+03	FES = 810100	$\mu_F = 0.772$	$\mu_{CR} = 0.062$	
Diversity = 1.5388e-02	k_t = [2, 2, 4, 4]			
Generation 8110: Best = 3.148856e+03	FES = 811100	$\mu_F = 0.776$	$\mu_{CR} = 0.053$	
Diversity = 1.5478e-02	k_t = [1, 1, 1, 1]			
Generation 8120: Best = 3.148856e+03	FES = 812100	$\mu_F = 0.774$	$\mu_{CR} = 0.057$	
Diversity = 1.2676e-02	k_t = [3, 4, 3, 4]			
Generation 8130: Best = 3.148856e+03	FES = 813100	$\mu_F = 0.748$	$\mu_{CR} = 0.066$	
Diversity = 1.2834e-02	k_t = [1, 1, 1, 1]			
Generation 8140: Best = 3.148856e+03	FES = 814100	$\mu_F = 0.741$	$\mu_{CR} = 0.066$	
Diversity = 1.3291e-02	k_t = [3, 4, 2, 4]			
Generation 8150: Best = 3.148856e+03	FES = 815100	$\mu_F = 0.755$	$\mu_{CR} = 0.065$	
Diversity = 1.4558e-02	k_t = [1, 1, 1, 1]			
Generation 8160: Best = 3.148856e+03	FES = 816100	$\mu_F = 0.738$	$\mu_{CR} = 0.065$	
Diversity = 1.6869e-02	k_t = [4, 3, 2, 2]			
Generation 8170: Best = 3.148856e+03	FES = 817100	$\mu_F = 0.760$	$\mu_{CR} = 0.057$	
Diversity = 1.7973e-02	k_t = [1, 1, 1, 1]			
Generation 8180: Best = 3.148856e+03	FES = 818100	$\mu_F = 0.768$	$\mu_{CR} = 0.064$	
Diversity = 2.0361e-02	k_t = [4, 3, 2, 2]			
Generation 8190: Best = 3.148856e+03	FES = 819100	$\mu_F = 0.758$	$\mu_{CR} = 0.056$	
Diversity = 2.0249e-02	k_t = [1, 1, 1, 1]			
Generation 8200: Best = 3.148856e+03	FES = 820100	$\mu_F = 0.774$	$\mu_{CR} = 0.056$	
Diversity = 2.2781e-02	k_t = [3, 2, 2, 3]			
Generation 8210: Best = 3.148856e+03	FES = 821100	$\mu_F = 0.785$	$\mu_{CR} = 0.054$	
Diversity = 2.2894e-02	k_t = [1, 1, 1, 1]			
Generation 8220: Best = 3.148856e+03	FES = 822100	$\mu_F = 0.783$	$\mu_{CR} = 0.059$	
Diversity = 2.3926e-02	k_t = [4, 3, 3, 2]			
Generation 8230: Best = 3.148856e+03	FES = 823100	$\mu_F = 0.777$	$\mu_{CR} = 0.060$	
Diversity = 2.5948e-02	k_t = [1, 1, 1, 1]			
Generation 8240: Best = 3.148856e+03	FES = 824100	$\mu_F = 0.777$	$\mu_{CR} = 0.061$	
Diversity = 2.3530e-02	k_t = [2, 4, 4, 4]			
Generation 8250: Best = 3.148855e+03	FES = 825100	$\mu_F = 0.788$	$\mu_{CR} = 0.051$	
Diversity = 1.8162e-02	k_t = [1, 1, 1, 1]			
Generation 8260: Best = 3.148855e+03	FES = 826100	$\mu_F = 0.774$	$\mu_{CR} = 0.057$	
Diversity = 1.6222e-02	k_t = [4, 4, 4, 3]			
Generation 8270: Best = 3.148855e+03	FES = 827100	$\mu_F = 0.757$	$\mu_{CR} = 0.057$	
Diversity = 1.4578e-02	k_t = [1, 1, 1, 1]			
Generation 8280: Best = 3.148855e+03	FES = 828100	$\mu_F = 0.756$	$\mu_{CR} = 0.057$	
Diversity = 1.4179e-02	k_t = [2, 4, 3, 2]			
Generation 8290: Best = 3.148855e+03	FES = 829100	$\mu_F = 0.774$	$\mu_{CR} = 0.058$	
Diversity = 1.4379e-02	k_t = [1, 1, 1, 1]			
Generation 8300: Best = 3.148855e+03	FES = 830100	$\mu_F = 0.767$	$\mu_{CR} = 0.061$	
Diversity = 1.2741e-02	k_t = [2, 3, 3, 3]			
Generation 8310: Best = 3.148855e+03	FES = 831100	$\mu_F = 0.763$	$\mu_{CR} = 0.065$	
Diversity = 1.1923e-02	k_t = [1, 1, 1, 1]			
Generation 8320: Best = 3.148855e+03	FES = 832100	$\mu_F = 0.767$	$\mu_{CR} = 0.067$	

Diversity = 1.1190e-02	k_t = [2, 3, 3, 4]			
Generation 8330: Best = 3.148855e+03	FES = 833100	$\mu_F = 0.766$	$\mu_{CR} = 0.067$	
Diversity = 1.1129e-02	k_t = [1, 1, 1, 1]			
Generation 8340: Best = 3.148855e+03	FES = 834100	$\mu_F = 0.765$	$\mu_{CR} = 0.068$	
Diversity = 1.2747e-02	k_t = [3, 4, 3, 2]			
Generation 8350: Best = 3.148855e+03	FES = 835100	$\mu_F = 0.772$	$\mu_{CR} = 0.066$	
Diversity = 1.3540e-02	k_t = [1, 1, 1, 1]			
Generation 8360: Best = 3.148854e+03	FES = 836100	$\mu_F = 0.773$	$\mu_{CR} = 0.067$	
Diversity = 1.3140e-02	k_t = [2, 2, 3, 4]			
Generation 8370: Best = 3.148854e+03	FES = 837100	$\mu_F = 0.792$	$\mu_{CR} = 0.065$	
Diversity = 1.3415e-02	k_t = [1, 1, 1, 1]			
Generation 8380: Best = 3.148854e+03	FES = 838100	$\mu_F = 0.798$	$\mu_{CR} = 0.062$	
Diversity = 1.3664e-02	k_t = [4, 3, 3, 2]			
Generation 8390: Best = 3.148854e+03	FES = 839100	$\mu_F = 0.788$	$\mu_{CR} = 0.054$	
Diversity = 1.2685e-02	k_t = [1, 1, 1, 1]			
Generation 8400: Best = 3.148854e+03	FES = 840100	$\mu_F = 0.800$	$\mu_{CR} = 0.051$	
Diversity = 1.2300e-02	k_t = [2, 3, 2, 4]			
Generation 8410: Best = 3.148854e+03	FES = 841100	$\mu_F = 0.830$	$\mu_{CR} = 0.057$	
Diversity = 1.0547e-02	k_t = [1, 1, 1, 1]			
Generation 8420: Best = 3.148854e+03	FES = 842100	$\mu_F = 0.832$	$\mu_{CR} = 0.058$	
Diversity = 1.0273e-02	k_t = [3, 4, 2, 2]			
Generation 8430: Best = 3.148854e+03	FES = 843100	$\mu_F = 0.838$	$\mu_{CR} = 0.065$	
Diversity = 9.4102e-03	k_t = [1, 1, 1, 1]			
Generation 8440: Best = 3.148854e+03	FES = 844100	$\mu_F = 0.825$	$\mu_{CR} = 0.068$	
Diversity = 9.0249e-03	k_t = [4, 2, 4, 4]			
Generation 8450: Best = 3.148854e+03	FES = 845100	$\mu_F = 0.816$	$\mu_{CR} = 0.065$	
Diversity = 1.0204e-02	k_t = [1, 1, 1, 1]			
Generation 8460: Best = 3.148854e+03	FES = 846100	$\mu_F = 0.824$	$\mu_{CR} = 0.062$	
Diversity = 1.1146e-02	k_t = [4, 4, 3, 3]			
Generation 8470: Best = 3.148854e+03	FES = 847100	$\mu_F = 0.834$	$\mu_{CR} = 0.063$	
Diversity = 1.0451e-02	k_t = [1, 1, 1, 1]			
Generation 8480: Best = 3.148854e+03	FES = 848100	$\mu_F = 0.815$	$\mu_{CR} = 0.064$	
Diversity = 9.6478e-03	k_t = [3, 4, 3, 4]			
Generation 8490: Best = 3.148854e+03	FES = 849100	$\mu_F = 0.824$	$\mu_{CR} = 0.053$	
Diversity = 8.6298e-03	k_t = [1, 1, 1, 1]			
Generation 8500: Best = 3.148854e+03	FES = 850100	$\mu_F = 0.791$	$\mu_{CR} = 0.062$	
Diversity = 7.4387e-03	k_t = [4, 3, 2, 4]			
Generation 8510: Best = 3.148854e+03	FES = 851100	$\mu_F = 0.773$	$\mu_{CR} = 0.068$	
Diversity = 7.2092e-03	k_t = [1, 1, 1, 1]			
Generation 8520: Best = 3.148854e+03	FES = 852100	$\mu_F = 0.777$	$\mu_{CR} = 0.068$	
Diversity = 8.1693e-03	k_t = [4, 2, 2, 2]			
Generation 8530: Best = 3.148854e+03	FES = 853100	$\mu_F = 0.757$	$\mu_{CR} = 0.066$	
Diversity = 7.7352e-03	k_t = [1, 1, 1, 1]			
Generation 8540: Best = 3.148854e+03	FES = 854100	$\mu_F = 0.740$	$\mu_{CR} = 0.073$	
Diversity = 8.2572e-03	k_t = [3, 3, 4, 2]			
Generation 8550: Best = 3.148854e+03	FES = 855100	$\mu_F = 0.744$	$\mu_{CR} = 0.068$	
Diversity = 7.3056e-03	k_t = [1, 1, 1, 1]			
Generation 8560: Best = 3.148853e+03	FES = 856100	$\mu_F = 0.753$	$\mu_{CR} = 0.061$	
Diversity = 6.5297e-03	k_t = [2, 3, 4, 4]			
Generation 8570: Best = 3.148853e+03	FES = 857100	$\mu_F = 0.755$	$\mu_{CR} = 0.064$	
Diversity = 6.2355e-03	k_t = [1, 1, 1, 1]			
Generation 8580: Best = 3.148853e+03	FES = 858100	$\mu_F = 0.775$	$\mu_{CR} = 0.063$	
Diversity = 6.8308e-03	k_t = [4, 4, 2, 3]			
Generation 8590: Best = 3.148853e+03	FES = 859100	$\mu_F = 0.790$	$\mu_{CR} = 0.059$	

Diversity = 7.3625e-03	k_t = [1, 1, 1, 1]			
Generation 8600: Best = 3.148853e+03	FES = 860100	$\mu_F = 0.779$	$\mu_{CR} = 0.068$	
Diversity = 7.6518e-03	k_t = [4, 3, 2, 3]			
Generation 8610: Best = 3.148853e+03	FES = 861100	$\mu_F = 0.785$	$\mu_{CR} = 0.072$	
Diversity = 8.1996e-03	k_t = [1, 1, 1, 1]			
Generation 8620: Best = 3.148853e+03	FES = 862100	$\mu_F = 0.815$	$\mu_{CR} = 0.069$	
Diversity = 7.1951e-03	k_t = [2, 2, 2, 3]			
Generation 8630: Best = 3.148853e+03	FES = 863100	$\mu_F = 0.818$	$\mu_{CR} = 0.071$	
Diversity = 8.2475e-03	k_t = [1, 1, 1, 1]			
Generation 8640: Best = 3.148853e+03	FES = 864100	$\mu_F = 0.811$	$\mu_{CR} = 0.074$	
Diversity = 8.7601e-03	k_t = [3, 4, 3, 4]			
Generation 8650: Best = 3.148853e+03	FES = 865100	$\mu_F = 0.823$	$\mu_{CR} = 0.072$	
Diversity = 9.9853e-03	k_t = [1, 1, 1, 1]			
Generation 8660: Best = 3.148853e+03	FES = 866100	$\mu_F = 0.821$	$\mu_{CR} = 0.072$	
Diversity = 1.0351e-02	k_t = [3, 2, 4, 2]			
Generation 8670: Best = 3.148853e+03	FES = 867100	$\mu_F = 0.817$	$\mu_{CR} = 0.072$	
Diversity = 1.0787e-02	k_t = [1, 1, 1, 1]			
Generation 8680: Best = 3.148853e+03	FES = 868100	$\mu_F = 0.808$	$\mu_{CR} = 0.070$	
Diversity = 1.1260e-02	k_t = [4, 3, 2, 3]			
Generation 8690: Best = 3.148852e+03	FES = 869100	$\mu_F = 0.800$	$\mu_{CR} = 0.063$	
Diversity = 1.1995e-02	k_t = [1, 1, 1, 1]			
Generation 8700: Best = 3.148852e+03	FES = 870100	$\mu_F = 0.803$	$\mu_{CR} = 0.076$	
Diversity = 1.0882e-02	k_t = [2, 3, 3, 4]			
Generation 8710: Best = 3.148852e+03	FES = 871100	$\mu_F = 0.798$	$\mu_{CR} = 0.078$	
Diversity = 1.0396e-02	k_t = [1, 1, 1, 1]			
Generation 8720: Best = 3.148852e+03	FES = 872100	$\mu_F = 0.799$	$\mu_{CR} = 0.067$	
Diversity = 1.0432e-02	k_t = [4, 3, 3, 3]			
Generation 8730: Best = 3.148852e+03	FES = 873100	$\mu_F = 0.785$	$\mu_{CR} = 0.070$	
Diversity = 1.0562e-02	k_t = [1, 1, 1, 1]			
Generation 8740: Best = 3.148852e+03	FES = 874100	$\mu_F = 0.782$	$\mu_{CR} = 0.067$	
Diversity = 1.0640e-02	k_t = [3, 4, 2, 4]			
Generation 8750: Best = 3.148852e+03	FES = 875100	$\mu_F = 0.783$	$\mu_{CR} = 0.068$	
Diversity = 1.0401e-02	k_t = [1, 1, 1, 1]			
Generation 8760: Best = 3.148852e+03	FES = 876100	$\mu_F = 0.790$	$\mu_{CR} = 0.065$	
Diversity = 1.0456e-02	k_t = [4, 4, 3, 3]			
Generation 8770: Best = 3.148852e+03	FES = 877100	$\mu_F = 0.775$	$\mu_{CR} = 0.070$	
Diversity = 9.7809e-03	k_t = [1, 1, 1, 1]			
Generation 8780: Best = 3.148852e+03	FES = 878100	$\mu_F = 0.768$	$\mu_{CR} = 0.071$	
Diversity = 9.5927e-03	k_t = [4, 3, 4, 3]			
Generation 8790: Best = 3.148852e+03	FES = 879100	$\mu_F = 0.765$	$\mu_{CR} = 0.059$	
Diversity = 8.8451e-03	k_t = [1, 1, 1, 1]			
Generation 8800: Best = 3.148852e+03	FES = 880100	$\mu_F = 0.744$	$\mu_{CR} = 0.063$	
Diversity = 8.4562e-03	k_t = [3, 4, 3, 3]			
Generation 8810: Best = 3.148852e+03	FES = 881100	$\mu_F = 0.747$	$\mu_{CR} = 0.059$	
Diversity = 8.6385e-03	k_t = [1, 1, 1, 1]			
Generation 8820: Best = 3.148852e+03	FES = 882100	$\mu_F = 0.756$	$\mu_{CR} = 0.054$	
Diversity = 7.8028e-03	k_t = [4, 3, 4, 4]			
Generation 8830: Best = 3.148852e+03	FES = 883100	$\mu_F = 0.774$	$\mu_{CR} = 0.062$	
Diversity = 7.9459e-03	k_t = [1, 1, 1, 1]			
Generation 8840: Best = 3.148851e+03	FES = 884100	$\mu_F = 0.788$	$\mu_{CR} = 0.067$	
Diversity = 8.7300e-03	k_t = [3, 2, 2, 4]			
Generation 8850: Best = 3.148851e+03	FES = 885100	$\mu_F = 0.807$	$\mu_{CR} = 0.066$	
Diversity = 8.5294e-03	k_t = [1, 1, 1, 1]			
Generation 8860: Best = 3.148851e+03	FES = 886100	$\mu_F = 0.805$	$\mu_{CR} = 0.067$	

Diversity = 8.6572e-03	k_t = [3, 4, 2, 4]			
Generation 8870: Best = 3.148851e+03	FES = 887100	$\mu_F = 0.801$	$\mu_{CR} = 0.066$	
Diversity = 8.8439e-03	k_t = [1, 1, 1, 1]			
Generation 8880: Best = 3.148851e+03	FES = 888100	$\mu_F = 0.803$	$\mu_{CR} = 0.072$	
Diversity = 8.0263e-03	k_t = [3, 4, 3, 3]			
Generation 8890: Best = 3.148851e+03	FES = 889100	$\mu_F = 0.803$	$\mu_{CR} = 0.076$	
Diversity = 8.0589e-03	k_t = [1, 1, 1, 1]			
Generation 8900: Best = 3.148851e+03	FES = 890100	$\mu_F = 0.791$	$\mu_{CR} = 0.064$	
Diversity = 8.7979e-03	k_t = [3, 3, 3, 3]			
Generation 8910: Best = 3.148851e+03	FES = 891100	$\mu_F = 0.789$	$\mu_{CR} = 0.064$	
Diversity = 8.6044e-03	k_t = [1, 1, 1, 1]			
Generation 8920: Best = 3.148851e+03	FES = 892100	$\mu_F = 0.786$	$\mu_{CR} = 0.069$	
Diversity = 8.3214e-03	k_t = [4, 3, 2, 4]			
Generation 8930: Best = 3.148851e+03	FES = 893100	$\mu_F = 0.774$	$\mu_{CR} = 0.074$	
Diversity = 7.0514e-03	k_t = [1, 1, 1, 1]			
Generation 8940: Best = 3.148851e+03	FES = 894100	$\mu_F = 0.765$	$\mu_{CR} = 0.077$	
Diversity = 8.2387e-03	k_t = [3, 4, 2, 2]			
Generation 8950: Best = 3.148851e+03	FES = 895100	$\mu_F = 0.761$	$\mu_{CR} = 0.073$	
Diversity = 8.8833e-03	k_t = [1, 1, 1, 1]			
Generation 8960: Best = 3.148851e+03	FES = 896100	$\mu_F = 0.766$	$\mu_{CR} = 0.078$	
Diversity = 9.4187e-03	k_t = [4, 2, 4, 2]			
Generation 8970: Best = 3.148850e+03	FES = 897100	$\mu_F = 0.754$	$\mu_{CR} = 0.070$	
Diversity = 9.7014e-03	k_t = [1, 1, 1, 1]			
Generation 8980: Best = 3.148850e+03	FES = 898100	$\mu_F = 0.763$	$\mu_{CR} = 0.064$	
Diversity = 9.3733e-03	k_t = [3, 2, 3, 4]			
Generation 8990: Best = 3.148850e+03	FES = 899100	$\mu_F = 0.799$	$\mu_{CR} = 0.063$	
Diversity = 1.0834e-02	k_t = [1, 1, 1, 1]			
Generation 9000: Best = 3.148850e+03	FES = 900100	$\mu_F = 0.787$	$\mu_{CR} = 0.059$	
Diversity = 1.0530e-02	k_t = [2, 3, 3, 4]			
Generation 9010: Best = 3.148850e+03	FES = 901100	$\mu_F = 0.801$	$\mu_{CR} = 0.063$	
Diversity = 1.1532e-02	k_t = [1, 1, 1, 1]			
Generation 9020: Best = 3.148850e+03	FES = 902100	$\mu_F = 0.793$	$\mu_{CR} = 0.066$	
Diversity = 1.1024e-02	k_t = [3, 3, 3, 2]			
Generation 9030: Best = 3.148850e+03	FES = 903100	$\mu_F = 0.789$	$\mu_{CR} = 0.056$	
Diversity = 1.0765e-02	k_t = [1, 1, 1, 1]			
Generation 9040: Best = 3.148850e+03	FES = 904100	$\mu_F = 0.767$	$\mu_{CR} = 0.060$	
Diversity = 1.1481e-02	k_t = [3, 4, 3, 4]			
Generation 9050: Best = 3.148849e+03	FES = 905100	$\mu_F = 0.794$	$\mu_{CR} = 0.064$	
Diversity = 1.1202e-02	k_t = [1, 1, 1, 1]			
Generation 9060: Best = 3.148849e+03	FES = 906100	$\mu_F = 0.786$	$\mu_{CR} = 0.071$	
Diversity = 1.1191e-02	k_t = [4, 2, 3, 4]			
Generation 9070: Best = 3.148849e+03	FES = 907100	$\mu_F = 0.789$	$\mu_{CR} = 0.076$	
Diversity = 1.0974e-02	k_t = [1, 1, 1, 1]			
Generation 9080: Best = 3.148849e+03	FES = 908100	$\mu_F = 0.793$	$\mu_{CR} = 0.074$	
Diversity = 9.7325e-03	k_t = [3, 4, 2, 4]			
Generation 9090: Best = 3.148849e+03	FES = 909100	$\mu_F = 0.779$	$\mu_{CR} = 0.076$	
Diversity = 1.0273e-02	k_t = [1, 1, 1, 1]			
Generation 9100: Best = 3.148849e+03	FES = 910100	$\mu_F = 0.771$	$\mu_{CR} = 0.071$	
Diversity = 1.0380e-02	k_t = [3, 2, 3, 2]			
Generation 9110: Best = 3.148849e+03	FES = 911100	$\mu_F = 0.765$	$\mu_{CR} = 0.067$	
Diversity = 1.0617e-02	k_t = [1, 1, 1, 1]			
Generation 9120: Best = 3.148848e+03	FES = 912100	$\mu_F = 0.765$	$\mu_{CR} = 0.067$	
Diversity = 1.0203e-02	k_t = [3, 4, 3, 4]			
Generation 9130: Best = 3.148848e+03	FES = 913100	$\mu_F = 0.777$	$\mu_{CR} = 0.073$	

Diversity = 1.2880e-02	k_t = [1, 1, 1, 1]			
Generation 9140: Best = 3.148848e+03	FES = 914100	$\mu_F = 0.780$	$\mu_{CR} = 0.071$	
Diversity = 1.2459e-02	k_t = [4, 3, 2, 2]			
Generation 9150: Best = 3.148848e+03	FES = 915100	$\mu_F = 0.771$	$\mu_{CR} = 0.075$	
Diversity = 1.1506e-02	k_t = [1, 1, 1, 1]			
Generation 9160: Best = 3.148848e+03	FES = 916100	$\mu_F = 0.792$	$\mu_{CR} = 0.064$	
Diversity = 9.7406e-03	k_t = [3, 4, 3, 2]			
Generation 9170: Best = 3.148848e+03	FES = 917100	$\mu_F = 0.783$	$\mu_{CR} = 0.069$	
Diversity = 9.5905e-03	k_t = [1, 1, 1, 1]			
Generation 9180: Best = 3.148848e+03	FES = 918100	$\mu_F = 0.780$	$\mu_{CR} = 0.071$	
Diversity = 8.8572e-03	k_t = [4, 2, 4, 2]			
Generation 9190: Best = 3.148848e+03	FES = 919100	$\mu_F = 0.773$	$\mu_{CR} = 0.070$	
Diversity = 9.2217e-03	k_t = [1, 1, 1, 1]			
Generation 9200: Best = 3.148848e+03	FES = 920100	$\mu_F = 0.783$	$\mu_{CR} = 0.061$	
Diversity = 7.9943e-03	k_t = [2, 4, 3, 4]			
Generation 9210: Best = 3.148848e+03	FES = 921100	$\mu_F = 0.787$	$\mu_{CR} = 0.066$	
Diversity = 7.5970e-03	k_t = [1, 1, 1, 1]			
Generation 9220: Best = 3.148848e+03	FES = 922100	$\mu_F = 0.776$	$\mu_{CR} = 0.077$	
Diversity = 7.6116e-03	k_t = [3, 2, 2, 2]			
Generation 9230: Best = 3.148848e+03	FES = 923100	$\mu_F = 0.789$	$\mu_{CR} = 0.075$	
Diversity = 8.1654e-03	k_t = [1, 1, 1, 1]			
Generation 9240: Best = 3.148848e+03	FES = 924100	$\mu_F = 0.795$	$\mu_{CR} = 0.069$	
Diversity = 7.6609e-03	k_t = [4, 3, 2, 3]			
Generation 9250: Best = 3.148848e+03	FES = 925100	$\mu_F = 0.778$	$\mu_{CR} = 0.074$	
Diversity = 8.9086e-03	k_t = [1, 1, 1, 1]			
Generation 9260: Best = 3.148848e+03	FES = 926100	$\mu_F = 0.776$	$\mu_{CR} = 0.066$	
Diversity = 7.9757e-03	k_t = [2, 2, 2, 3]			
Generation 9270: Best = 3.148847e+03	FES = 927100	$\mu_F = 0.797$	$\mu_{CR} = 0.061$	
Diversity = 8.8082e-03	k_t = [1, 1, 1, 1]			
Generation 9280: Best = 3.148847e+03	FES = 928100	$\mu_F = 0.807$	$\mu_{CR} = 0.072$	
Diversity = 8.6831e-03	k_t = [3, 2, 4, 4]			
Generation 9290: Best = 3.148847e+03	FES = 929100	$\mu_F = 0.789$	$\mu_{CR} = 0.069$	
Diversity = 8.4399e-03	k_t = [1, 1, 1, 1]			
Generation 9300: Best = 3.148847e+03	FES = 930100	$\mu_F = 0.785$	$\mu_{CR} = 0.069$	
Diversity = 8.2291e-03	k_t = [4, 3, 4, 4]			
Generation 9310: Best = 3.148847e+03	FES = 931100	$\mu_F = 0.780$	$\mu_{CR} = 0.068$	
Diversity = 7.5936e-03	k_t = [1, 1, 1, 1]			
Generation 9320: Best = 3.148847e+03	FES = 932100	$\mu_F = 0.758$	$\mu_{CR} = 0.072$	
Diversity = 9.0050e-03	k_t = [3, 2, 2, 3]			
Generation 9330: Best = 3.148847e+03	FES = 933100	$\mu_F = 0.767$	$\mu_{CR} = 0.074$	
Diversity = 8.8716e-03	k_t = [1, 1, 1, 1]			
Generation 9340: Best = 3.148847e+03	FES = 934100	$\mu_F = 0.777$	$\mu_{CR} = 0.067$	
Diversity = 8.5226e-03	k_t = [4, 2, 3, 3]			
Generation 9350: Best = 3.148847e+03	FES = 935100	$\mu_F = 0.762$	$\mu_{CR} = 0.067$	
Diversity = 9.6515e-03	k_t = [1, 1, 1, 1]			
Generation 9360: Best = 3.148847e+03	FES = 936100	$\mu_F = 0.777$	$\mu_{CR} = 0.066$	
Diversity = 8.5187e-03	k_t = [3, 2, 3, 4]			
Generation 9370: Best = 3.148847e+03	FES = 937100	$\mu_F = 0.778$	$\mu_{CR} = 0.071$	
Diversity = 8.1614e-03	k_t = [1, 1, 1, 1]			
Generation 9380: Best = 3.148847e+03	FES = 938100	$\mu_F = 0.778$	$\mu_{CR} = 0.067$	
Diversity = 8.2925e-03	k_t = [3, 2, 3, 2]			
Generation 9390: Best = 3.148847e+03	FES = 939100	$\mu_F = 0.789$	$\mu_{CR} = 0.065$	
Diversity = 7.9762e-03	k_t = [1, 1, 1, 1]			
Generation 9400: Best = 3.148847e+03	FES = 940100	$\mu_F = 0.765$	$\mu_{CR} = 0.066$	

Diversity = 7.7238e-03 k_t = [4, 2, 2, 2]
 Generation 9410: Best = 3.148847e+03 FES = 941100 μ_F = 0.774 μ_{CR} = 0.067
 Diversity = 7.5108e-03 k_t = [1, 1, 1, 1]
 Generation 9420: Best = 3.148847e+03 FES = 942100 μ_F = 0.778 μ_{CR} = 0.069
 Diversity = 7.7724e-03 k_t = [2, 4, 3, 4]
 Generation 9430: Best = 3.148847e+03 FES = 943100 μ_F = 0.777 μ_{CR} = 0.064
 Diversity = 7.9001e-03 k_t = [1, 1, 1, 1]
 Generation 9440: Best = 3.148846e+03 FES = 944100 μ_F = 0.780 μ_{CR} = 0.067
 Diversity = 7.6960e-03 k_t = [2, 2, 4, 2]
 Generation 9450: Best = 3.148846e+03 FES = 945100 μ_F = 0.776 μ_{CR} = 0.066
 Diversity = 8.3922e-03 k_t = [1, 1, 1, 1]
 Generation 9460: Best = 3.148846e+03 FES = 946100 μ_F = 0.757 μ_{CR} = 0.061
 Diversity = 7.3281e-03 k_t = [2, 2, 2, 4]
 Generation 9470: Best = 3.148846e+03 FES = 947100 μ_F = 0.767 μ_{CR} = 0.068
 Diversity = 7.9832e-03 k_t = [1, 1, 1, 1]
 Generation 9480: Best = 3.148846e+03 FES = 948100 μ_F = 0.767 μ_{CR} = 0.068
 Diversity = 7.5691e-03 k_t = [4, 2, 3, 3]
 Generation 9490: Best = 3.148846e+03 FES = 949100 μ_F = 0.786 μ_{CR} = 0.075
 Diversity = 7.9980e-03 k_t = [1, 1, 1, 1]
 Generation 9500: Best = 3.148846e+03 FES = 950100 μ_F = 0.789 μ_{CR} = 0.072
 Diversity = 8.0174e-03 k_t = [2, 2, 4, 2]
 Generation 9510: Best = 3.148846e+03 FES = 951100 μ_F = 0.773 μ_{CR} = 0.070
 Diversity = 8.2137e-03 k_t = [1, 1, 1, 1]
 Generation 9520: Best = 3.148846e+03 FES = 952100 μ_F = 0.781 μ_{CR} = 0.067
 Diversity = 7.8417e-03 k_t = [2, 4, 3, 4]
 Generation 9530: Best = 3.148846e+03 FES = 953100 μ_F = 0.779 μ_{CR} = 0.065
 Diversity = 8.2702e-03 k_t = [1, 1, 1, 1]
 Generation 9540: Best = 3.148846e+03 FES = 954100 μ_F = 0.774 μ_{CR} = 0.069
 Diversity = 7.2847e-03 k_t = [4, 4, 4, 2]
 Generation 9550: Best = 3.148846e+03 FES = 955100 μ_F = 0.761 μ_{CR} = 0.076
 Diversity = 7.4874e-03 k_t = [1, 1, 1, 1]
 Generation 9560: Best = 3.148846e+03 FES = 956100 μ_F = 0.765 μ_{CR} = 0.071
 Diversity = 6.3666e-03 k_t = [4, 3, 4, 4]
 Generation 9570: Best = 3.148846e+03 FES = 957100 μ_F = 0.766 μ_{CR} = 0.065
 Diversity = 6.1406e-03 k_t = [1, 1, 1, 1]
 Generation 9580: Best = 3.148846e+03 FES = 958100 μ_F = 0.779 μ_{CR} = 0.068
 Diversity = 6.0858e-03 k_t = [2, 3, 4, 3]
 Generation 9590: Best = 3.148846e+03 FES = 959100 μ_F = 0.774 μ_{CR} = 0.061
 Diversity = 6.3032e-03 k_t = [1, 1, 1, 1]
 Generation 9600: Best = 3.148846e+03 FES = 960100 μ_F = 0.784 μ_{CR} = 0.064
 Diversity = 6.6058e-03 k_t = [3, 2, 3, 3]
 Generation 9610: Best = 3.148846e+03 FES = 961100 μ_F = 0.793 μ_{CR} = 0.066
 Diversity = 7.4101e-03 k_t = [1, 1, 1, 1]
 Generation 9620: Best = 3.148846e+03 FES = 962100 μ_F = 0.803 μ_{CR} = 0.073
 Diversity = 6.0495e-03 k_t = [2, 3, 4, 4]
 Generation 9630: Best = 3.148846e+03 FES = 963100 μ_F = 0.800 μ_{CR} = 0.072
 Diversity = 5.6842e-03 k_t = [1, 1, 1, 1]
 Generation 9640: Best = 3.148846e+03 FES = 964100 μ_F = 0.773 μ_{CR} = 0.067
 Diversity = 6.0032e-03 k_t = [4, 2, 2, 3]
 Generation 9650: Best = 3.148846e+03 FES = 965100 μ_F = 0.788 μ_{CR} = 0.066
 Diversity = 6.5629e-03 k_t = [1, 1, 1, 1]
 Generation 9660: Best = 3.148846e+03 FES = 966100 μ_F = 0.774 μ_{CR} = 0.067
 Diversity = 6.0860e-03 k_t = [3, 2, 2, 2]
 Generation 9670: Best = 3.148846e+03 FES = 967100 μ_F = 0.785 μ_{CR} = 0.069

Diversity = 6.5408e-03	k_t = [1, 1, 1, 1]			
Generation 9680: Best = 3.148846e+03	FES = 968100	$\mu_F = 0.775$	$\mu_{CR} = 0.067$	
Diversity = 6.6226e-03	k_t = [4, 4, 4, 3]			
Generation 9690: Best = 3.148846e+03	FES = 969100	$\mu_F = 0.778$	$\mu_{CR} = 0.067$	
Diversity = 6.5413e-03	k_t = [1, 1, 1, 1]			
Generation 9700: Best = 3.148846e+03	FES = 970100	$\mu_F = 0.768$	$\mu_{CR} = 0.072$	
Diversity = 6.1481e-03	k_t = [3, 3, 2, 3]			
Generation 9710: Best = 3.148845e+03	FES = 971100	$\mu_F = 0.766$	$\mu_{CR} = 0.062$	
Diversity = 5.8405e-03	k_t = [1, 1, 1, 1]			
Generation 9720: Best = 3.148845e+03	FES = 972100	$\mu_F = 0.778$	$\mu_{CR} = 0.071$	
Diversity = 6.2296e-03	k_t = [4, 3, 4, 2]			
Generation 9730: Best = 3.148845e+03	FES = 973100	$\mu_F = 0.780$	$\mu_{CR} = 0.072$	
Diversity = 7.3586e-03	k_t = [1, 1, 1, 1]			
Generation 9740: Best = 3.148845e+03	FES = 974100	$\mu_F = 0.775$	$\mu_{CR} = 0.065$	
Diversity = 6.7811e-03	k_t = [4, 4, 3, 3]			
Generation 9750: Best = 3.148845e+03	FES = 975100	$\mu_F = 0.779$	$\mu_{CR} = 0.067$	
Diversity = 6.4029e-03	k_t = [1, 1, 1, 1]			
Generation 9760: Best = 3.148845e+03	FES = 976100	$\mu_F = 0.784$	$\mu_{CR} = 0.064$	
Diversity = 5.6855e-03	k_t = [3, 3, 3, 3]			
Generation 9770: Best = 3.148845e+03	FES = 977100	$\mu_F = 0.781$	$\mu_{CR} = 0.057$	
Diversity = 5.8478e-03	k_t = [1, 1, 1, 1]			
Generation 9780: Best = 3.148845e+03	FES = 978100	$\mu_F = 0.779$	$\mu_{CR} = 0.074$	
Diversity = 5.9528e-03	k_t = [4, 4, 2, 2]			
Generation 9790: Best = 3.148845e+03	FES = 979100	$\mu_F = 0.788$	$\mu_{CR} = 0.078$	
Diversity = 6.2322e-03	k_t = [1, 1, 1, 1]			
Generation 9800: Best = 3.148845e+03	FES = 980100	$\mu_F = 0.795$	$\mu_{CR} = 0.076$	
Diversity = 5.6484e-03	k_t = [4, 4, 2, 3]			
Generation 9810: Best = 3.148845e+03	FES = 981100	$\mu_F = 0.796$	$\mu_{CR} = 0.075$	
Diversity = 6.2560e-03	k_t = [1, 1, 1, 1]			
Generation 9820: Best = 3.148845e+03	FES = 982100	$\mu_F = 0.803$	$\mu_{CR} = 0.075$	
Diversity = 6.9299e-03	k_t = [2, 4, 3, 2]			
Generation 9830: Best = 3.148845e+03	FES = 983100	$\mu_F = 0.796$	$\mu_{CR} = 0.081$	
Diversity = 7.3483e-03	k_t = [1, 1, 1, 1]			
Generation 9840: Best = 3.148845e+03	FES = 984100	$\mu_F = 0.807$	$\mu_{CR} = 0.070$	
Diversity = 7.8443e-03	k_t = [3, 2, 3, 4]			
Generation 9850: Best = 3.148845e+03	FES = 985100	$\mu_F = 0.800$	$\mu_{CR} = 0.068$	
Diversity = 8.9645e-03	k_t = [1, 1, 1, 1]			
Generation 9860: Best = 3.148845e+03	FES = 986100	$\mu_F = 0.795$	$\mu_{CR} = 0.064$	
Diversity = 8.3717e-03	k_t = [3, 3, 4, 4]			
Generation 9870: Best = 3.148845e+03	FES = 987100	$\mu_F = 0.794$	$\mu_{CR} = 0.058$	
Diversity = 8.8679e-03	k_t = [1, 1, 1, 1]			
Generation 9880: Best = 3.148845e+03	FES = 988100	$\mu_F = 0.803$	$\mu_{CR} = 0.058$	
Diversity = 9.9486e-03	k_t = [3, 4, 2, 3]			
Generation 9890: Best = 3.148845e+03	FES = 989100	$\mu_F = 0.797$	$\mu_{CR} = 0.061$	
Diversity = 9.6314e-03	k_t = [1, 1, 1, 1]			
Generation 9900: Best = 3.148844e+03	FES = 990100	$\mu_F = 0.791$	$\mu_{CR} = 0.066$	
Diversity = 9.7162e-03	k_t = [4, 2, 3, 2]			
Generation 9910: Best = 3.148844e+03	FES = 991100	$\mu_F = 0.773$	$\mu_{CR} = 0.067$	
Diversity = 9.5676e-03	k_t = [1, 1, 1, 1]			
Generation 9920: Best = 3.148844e+03	FES = 992100	$\mu_F = 0.771$	$\mu_{CR} = 0.063$	
Diversity = 8.8966e-03	k_t = [4, 2, 2, 2]			
Generation 9930: Best = 3.148844e+03	FES = 993100	$\mu_F = 0.786$	$\mu_{CR} = 0.059$	
Diversity = 9.0026e-03	k_t = [1, 1, 1, 1]			
Generation 9940: Best = 3.148844e+03	FES = 994100	$\mu_F = 0.769$	$\mu_{CR} = 0.065$	

```

Diversity = 8.0389e-03 k_t = [4, 2, 4, 2]
Generation 9950: Best = 3.148844e+03 FES = 995100  $\mu_F$  = 0.770  $\mu_{CR}$  = 0.065
Diversity = 7.4930e-03 k_t = [1, 1, 1, 1]
Generation 9960: Best = 3.148844e+03 FES = 996100  $\mu_F$  = 0.779  $\mu_{CR}$  = 0.067
Diversity = 6.4181e-03 k_t = [3, 4, 2, 3]
Generation 9970: Best = 3.148844e+03 FES = 997100  $\mu_F$  = 0.768  $\mu_{CR}$  = 0.070
Diversity = 6.2644e-03 k_t = [1, 1, 1, 1]
Generation 9980: Best = 3.148844e+03 FES = 998100  $\mu_F$  = 0.787  $\mu_{CR}$  = 0.065
Diversity = 5.7890e-03 k_t = [3, 3, 4, 4]
Generation 9990: Best = 3.148844e+03 FES = 999100  $\mu_F$  = 0.769  $\mu_{CR}$  = 0.065
Diversity = 6.2841e-03 k_t = [1, 1, 1, 1]

```

Optimization complete.

```

Best Fitness          = 3.148844e+03
Total Function Evals  = 1000000
Total Successful Updates = 154639

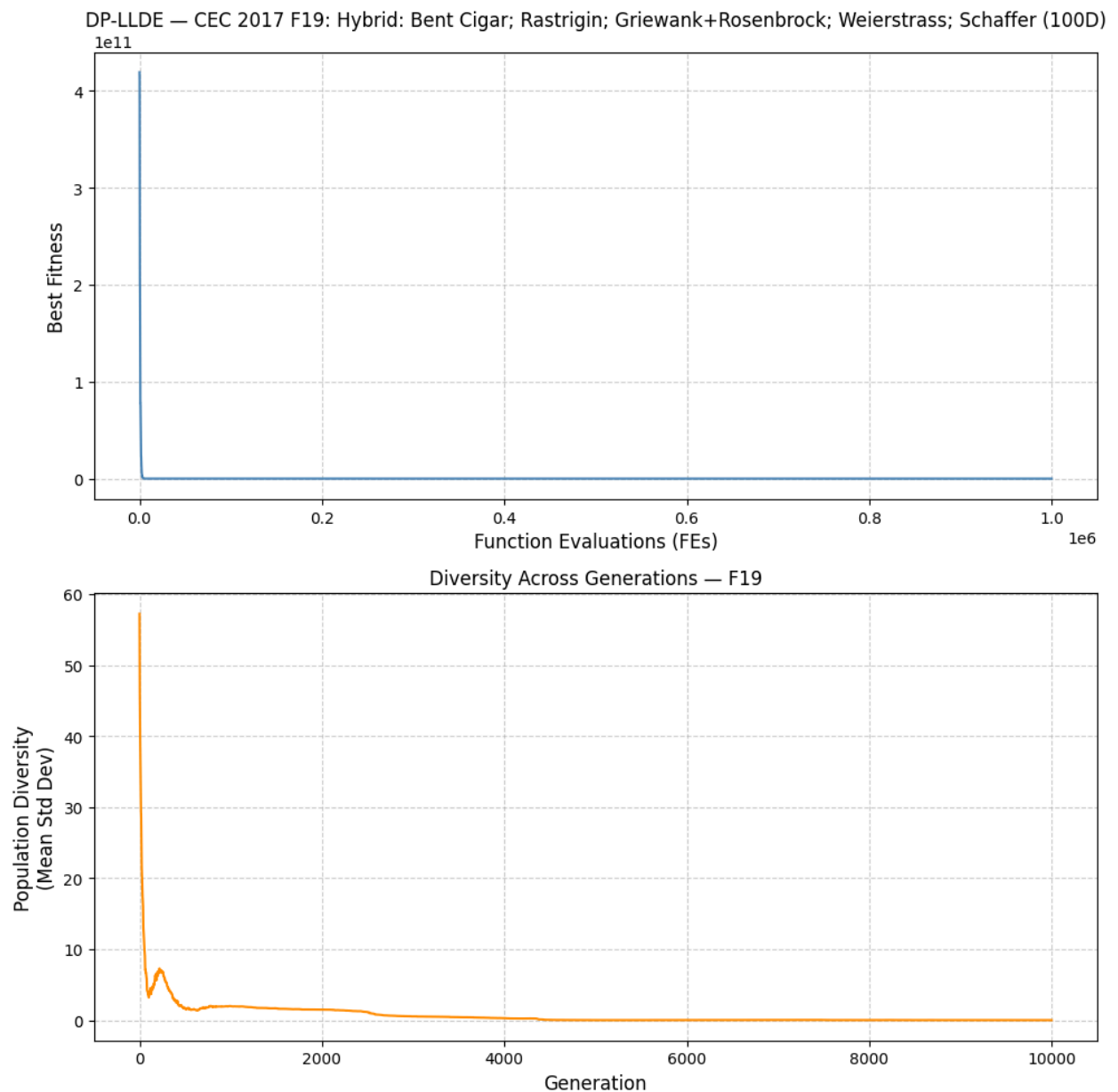
```

```

=====
=
Results:
  Best Fitness          : 3.148844e+03
  Global Optimum        : 1.900000e+03
  Error (|best-optimum|) : 1.248844e+03
  Best Solution (5 dim) : [-19.1604246  19.07836885  13.83585674  11.25780823
-37.94790683]
  D^(0) (init. diversity): 5.7220e+01
=====
=

```

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results/all_results_checkpoint.pkl



Graph saved as convergence_F19.png

✓ Plot saved to: /content/drive/My Drive/thesis_results/plots/convergence_F19.png

```
In [ ]: # =====
# F20 — Hybrid Function 10 (HappyCat; Katsuura; Ackley; Rastrigin; Schwefel; S
# =====
run_and_plot('F20', make_wrapper(hybrid.f20), 2000.0, 'Hybrid: HappyCat; Katsu
```


=====

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DP-LLDE on CEC 2017 F20 – Single Run

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=

Function : Hybrid: HappyCat; Katsuura; Ackley; Rastrigin; Schwefel; Schaffer (100D)

Bounds : [-100, 100]

Global optimum : 2000.0

Initial $D^*(0) = 5.7220e+01$ | Diversity threshold = 5.7220e+00

Generation 0: Best Fitness = 8.995901e+03

Generation 10: Best = 8.494499e+03 FES = 1100 $\mu_F = 0.570$ $\mu_{CR} = 0.356$

Diversity = 4.6988e+01 $k_t = [1, 1, 1, 1]$

Generation 20: Best = 8.319003e+03 FES = 2100 $\mu_F = 0.646$ $\mu_{CR} = 0.305$

Diversity = 4.4529e+01 $k_t = [2, 3, 2, 3]$

Generation 30: Best = 8.087676e+03 FES = 3100 $\mu_F = 0.688$ $\mu_{CR} = 0.265$

Diversity = 4.3886e+01 $k_t = [1, 1, 1, 1]$

Generation 40: Best = 8.087676e+03 FES = 4100 $\mu_F = 0.715$ $\mu_{CR} = 0.205$

Diversity = 4.2873e+01 $k_t = [3, 2, 3, 4]$

Generation 50: Best = 8.085169e+03 FES = 5100 $\mu_F = 0.735$ $\mu_{CR} = 0.116$

Diversity = 4.2582e+01 $k_t = [1, 1, 1, 1]$

Generation 60: Best = 8.085169e+03 FES = 6100 $\mu_F = 0.767$ $\mu_{CR} = 0.083$

Diversity = 4.2437e+01 $k_t = [3, 3, 2, 3]$

Generation 70: Best = 7.605076e+03 FES = 7100 $\mu_F = 0.798$ $\mu_{CR} = 0.059$

Diversity = 4.2983e+01 $k_t = [1, 1, 1, 1]$

Generation 80: Best = 7.572201e+03 FES = 8100 $\mu_F = 0.820$ $\mu_{CR} = 0.058$

Diversity = 4.3473e+01 $k_t = [4, 2, 4, 3]$

Generation 90: Best = 7.570379e+03 FES = 9100 $\mu_F = 0.834$ $\mu_{CR} = 0.052$

Diversity = 4.3395e+01 $k_t = [1, 1, 1, 1]$

Generation 100: Best = 7.339830e+03 FES = 10100 $\mu_F = 0.830$ $\mu_{CR} = 0.045$

Diversity = 4.3536e+01 $k_t = [4, 2, 3, 3]$

Generation 110: Best = 6.752093e+03 FES = 11100 $\mu_F = 0.855$ $\mu_{CR} = 0.045$

Diversity = 4.3698e+01 $k_t = [1, 1, 1, 1]$

Generation 120: Best = 6.752068e+03 FES = 12100 $\mu_F = 0.863$ $\mu_{CR} = 0.053$

Diversity = 4.3799e+01 $k_t = [4, 2, 2, 2]$

Generation 130: Best = 6.752068e+03 FES = 13100 $\mu_F = 0.868$ $\mu_{CR} = 0.047$

Diversity = 4.3755e+01 $k_t = [1, 1, 1, 1]$

Generation 140: Best = 6.752068e+03 FES = 14100 $\mu_F = 0.880$ $\mu_{CR} = 0.044$

Diversity = 4.3402e+01 $k_t = [3, 2, 3, 4]$

Generation 150: Best = 6.713250e+03 FES = 15100 $\mu_F = 0.895$ $\mu_{CR} = 0.048$

Diversity = 4.3876e+01 $k_t = [1, 1, 1, 1]$

Generation 160: Best = 6.713250e+03 FES = 16100 $\mu_F = 0.858$ $\mu_{CR} = 0.037$

Diversity = 4.3881e+01 $k_t = [2, 4, 3, 2]$

Generation 170: Best = 6.691404e+03 FES = 17100 $\mu_F = 0.850$ $\mu_{CR} = 0.040$

Diversity = 4.4028e+01 $k_t = [1, 1, 1, 1]$

Generation 180: Best = 6.691404e+03 FES = 18100 $\mu_F = 0.843$ $\mu_{CR} = 0.038$

Diversity = 4.3437e+01 $k_t = [2, 2, 4, 4]$

Generation 190: Best = 6.679357e+03 FES = 19100 $\mu_F = 0.860$ $\mu_{CR} = 0.033$

Diversity = 4.3553e+01 $k_t = [1, 1, 1, 1]$

Generation 200: Best = 6.679357e+03 FES = 20100 $\mu_F = 0.868$ $\mu_{CR} = 0.036$

Diversity = 4.3205e+01 $k_t = [4, 2, 3, 4]$

Generation 210: Best = 6.588242e+03 FES = 21100 $\mu_F = 0.873$ $\mu_{CR} = 0.031$

Diversity = 4.3313e+01 $k_t = [1, 1, 1, 1]$

Generation 220: Best = 6.588242e+03 FES = 22100 $\mu_F = 0.826$ $\mu_{CR} = 0.043$

Diversity = 4.2940e+01	k_t = [4, 4, 3, 4]				
Generation 230: Best = 6.588242e+03	FES = 23100	$\mu_F = 0.851$	$\mu_{CR} = 0.046$		
Diversity = 4.2898e+01	k_t = [1, 1, 1, 1]				
Generation 240: Best = 6.335788e+03	FES = 24100	$\mu_F = 0.852$	$\mu_{CR} = 0.039$		
Diversity = 4.2533e+01	k_t = [4, 2, 2, 4]				
Generation 250: Best = 6.335788e+03	FES = 25100	$\mu_F = 0.833$	$\mu_{CR} = 0.036$		
Diversity = 4.2537e+01	k_t = [1, 1, 1, 1]				
Generation 260: Best = 6.335788e+03	FES = 26100	$\mu_F = 0.795$	$\mu_{CR} = 0.040$		
Diversity = 4.2407e+01	k_t = [2, 4, 4, 3]				
Generation 270: Best = 6.335788e+03	FES = 27100	$\mu_F = 0.827$	$\mu_{CR} = 0.039$		
Diversity = 4.2283e+01	k_t = [1, 1, 1, 1]				
Generation 280: Best = 6.328507e+03	FES = 28100	$\mu_F = 0.856$	$\mu_{CR} = 0.041$		
Diversity = 4.1683e+01	k_t = [4, 4, 3, 4]				
Generation 290: Best = 6.328436e+03	FES = 29100	$\mu_F = 0.850$	$\mu_{CR} = 0.041$		
Diversity = 4.1620e+01	k_t = [1, 1, 1, 1]				
Generation 300: Best = 6.328258e+03	FES = 30100	$\mu_F = 0.835$	$\mu_{CR} = 0.045$		
Diversity = 4.1055e+01	k_t = [3, 2, 4, 4]				
Generation 310: Best = 6.328258e+03	FES = 31100	$\mu_F = 0.830$	$\mu_{CR} = 0.038$		
Diversity = 4.0732e+01	k_t = [1, 1, 1, 1]				
Generation 320: Best = 6.328258e+03	FES = 32100	$\mu_F = 0.824$	$\mu_{CR} = 0.058$		
Diversity = 4.0658e+01	k_t = [4, 2, 2, 2]				
Generation 330: Best = 6.286031e+03	FES = 33100	$\mu_F = 0.825$	$\mu_{CR} = 0.045$		
Diversity = 4.0671e+01	k_t = [1, 1, 1, 1]				
Generation 340: Best = 6.286031e+03	FES = 34100	$\mu_F = 0.853$	$\mu_{CR} = 0.040$		
Diversity = 4.0672e+01	k_t = [2, 3, 4, 3]				
Generation 350: Best = 6.286031e+03	FES = 35100	$\mu_F = 0.890$	$\mu_{CR} = 0.038$		
Diversity = 4.0783e+01	k_t = [1, 1, 1, 1]				
Generation 360: Best = 6.036103e+03	FES = 36100	$\mu_F = 0.897$	$\mu_{CR} = 0.037$		
Diversity = 4.0780e+01	k_t = [4, 2, 2, 3]				
Generation 370: Best = 6.024958e+03	FES = 37100	$\mu_F = 0.899$	$\mu_{CR} = 0.035$		
Diversity = 4.0673e+01	k_t = [1, 1, 1, 1]				
Generation 380: Best = 6.024958e+03	FES = 38100	$\mu_F = 0.882$	$\mu_{CR} = 0.040$		
Diversity = 4.0774e+01	k_t = [2, 3, 4, 2]				
Generation 390: Best = 6.024958e+03	FES = 39100	$\mu_F = 0.783$	$\mu_{CR} = 0.044$		
Diversity = 4.0651e+01	k_t = [1, 1, 1, 1]				
Generation 400: Best = 6.019920e+03	FES = 40100	$\mu_F = 0.831$	$\mu_{CR} = 0.045$		
Diversity = 4.0769e+01	k_t = [3, 4, 2, 4]				
Generation 410: Best = 6.019920e+03	FES = 41100	$\mu_F = 0.852$	$\mu_{CR} = 0.036$		
Diversity = 4.0732e+01	k_t = [1, 1, 1, 1]				
Generation 420: Best = 6.014732e+03	FES = 42100	$\mu_F = 0.856$	$\mu_{CR} = 0.035$		
Diversity = 4.0661e+01	k_t = [2, 2, 2, 3]				
Generation 430: Best = 6.014732e+03	FES = 43100	$\mu_F = 0.866$	$\mu_{CR} = 0.028$		
Diversity = 4.0624e+01	k_t = [1, 1, 1, 1]				
Generation 440: Best = 6.014732e+03	FES = 44100	$\mu_F = 0.864$	$\mu_{CR} = 0.026$		
Diversity = 4.0646e+01	k_t = [3, 3, 2, 2]				
Generation 450: Best = 6.013777e+03	FES = 45100	$\mu_F = 0.856$	$\mu_{CR} = 0.026$		
Diversity = 4.0609e+01	k_t = [1, 1, 1, 1]				
Generation 460: Best = 6.013777e+03	FES = 46100	$\mu_F = 0.830$	$\mu_{CR} = 0.044$		
Diversity = 4.0649e+01	k_t = [2, 2, 3, 2]				
Generation 470: Best = 6.013777e+03	FES = 47100	$\mu_F = 0.852$	$\mu_{CR} = 0.036$		
Diversity = 4.0689e+01	k_t = [1, 1, 1, 1]				
Generation 480: Best = 6.013777e+03	FES = 48100	$\mu_F = 0.864$	$\mu_{CR} = 0.043$		
Diversity = 4.0772e+01	k_t = [2, 3, 2, 3]				
Generation 490: Best = 5.983558e+03	FES = 49100	$\mu_F = 0.831$	$\mu_{CR} = 0.042$		

Diversity = 4.0860e+01	k_t = [1, 1, 1, 1]					
Generation 500: Best = 5.983558e+03	FES = 50100	$\mu_F = 0.854$	$\mu_{CR} = 0.047$			
Diversity = 4.0769e+01	k_t = [4, 2, 4, 4]					
Generation 510: Best = 5.983558e+03	FES = 51100	$\mu_F = 0.839$	$\mu_{CR} = 0.036$			
Diversity = 4.0709e+01	k_t = [1, 1, 1, 1]					
Generation 520: Best = 5.983558e+03	FES = 52100	$\mu_F = 0.821$	$\mu_{CR} = 0.042$			
Diversity = 4.0687e+01	k_t = [2, 2, 3, 2]					
Generation 530: Best = 5.983539e+03	FES = 53100	$\mu_F = 0.806$	$\mu_{CR} = 0.046$			
Diversity = 4.0672e+01	k_t = [1, 1, 1, 1]					
Generation 540: Best = 5.983539e+03	FES = 54100	$\mu_F = 0.819$	$\mu_{CR} = 0.040$			
Diversity = 4.0492e+01	k_t = [4, 3, 3, 4]					
Generation 550: Best = 5.983539e+03	FES = 55100	$\mu_F = 0.821$	$\mu_{CR} = 0.027$			
Diversity = 4.0478e+01	k_t = [1, 1, 1, 1]					
Generation 560: Best = 5.983539e+03	FES = 56100	$\mu_F = 0.841$	$\mu_{CR} = 0.023$			
Diversity = 4.0410e+01	k_t = [3, 3, 2, 3]					
Generation 570: Best = 5.983539e+03	FES = 57100	$\mu_F = 0.861$	$\mu_{CR} = 0.026$			
Diversity = 4.0525e+01	k_t = [1, 1, 1, 1]					
Generation 580: Best = 5.983539e+03	FES = 58100	$\mu_F = 0.881$	$\mu_{CR} = 0.022$			
Diversity = 4.0477e+01	k_t = [3, 2, 3, 4]					
Generation 590: Best = 5.983539e+03	FES = 59100	$\mu_F = 0.901$	$\mu_{CR} = 0.024$			
Diversity = 4.0753e+01	k_t = [1, 1, 1, 1]					
Generation 600: Best = 5.946472e+03	FES = 60100	$\mu_F = 0.897$	$\mu_{CR} = 0.025$			
Diversity = 4.0773e+01	k_t = [2, 4, 2, 2]					
Generation 610: Best = 5.946472e+03	FES = 61100	$\mu_F = 0.862$	$\mu_{CR} = 0.033$			
Diversity = 4.0797e+01	k_t = [1, 1, 1, 1]					
Generation 620: Best = 5.946472e+03	FES = 62100	$\mu_F = 0.807$	$\mu_{CR} = 0.029$			
Diversity = 4.0814e+01	k_t = [3, 4, 2, 2]					
Generation 630: Best = 5.902730e+03	FES = 63100	$\mu_F = 0.741$	$\mu_{CR} = 0.026$			
Diversity = 4.0779e+01	k_t = [1, 1, 1, 1]					
Generation 640: Best = 5.902730e+03	FES = 64100	$\mu_F = 0.768$	$\mu_{CR} = 0.035$			
Diversity = 4.0715e+01	k_t = [2, 3, 3, 4]					
Generation 650: Best = 5.902730e+03	FES = 65100	$\mu_F = 0.786$	$\mu_{CR} = 0.029$			
Diversity = 4.0670e+01	k_t = [1, 1, 1, 1]					
Generation 660: Best = 5.902730e+03	FES = 66100	$\mu_F = 0.718$	$\mu_{CR} = 0.034$			
Diversity = 4.0625e+01	k_t = [2, 4, 4, 3]					
Generation 670: Best = 5.902730e+03	FES = 67100	$\mu_F = 0.687$	$\mu_{CR} = 0.032$			
Diversity = 4.0615e+01	k_t = [1, 1, 1, 1]					
Generation 680: Best = 5.902636e+03	FES = 68100	$\mu_F = 0.708$	$\mu_{CR} = 0.043$			
Diversity = 4.0558e+01	k_t = [2, 2, 2, 3]					
Generation 690: Best = 5.902492e+03	FES = 69100	$\mu_F = 0.762$	$\mu_{CR} = 0.040$			
Diversity = 4.0527e+01	k_t = [1, 1, 1, 1]					
Generation 700: Best = 5.902492e+03	FES = 70100	$\mu_F = 0.743$	$\mu_{CR} = 0.034$			
Diversity = 4.0458e+01	k_t = [4, 2, 2, 4]					
Generation 710: Best = 5.884883e+03	FES = 71100	$\mu_F = 0.769$	$\mu_{CR} = 0.043$			
Diversity = 4.0436e+01	k_t = [1, 1, 1, 1]					
Generation 720: Best = 5.562758e+03	FES = 72100	$\mu_F = 0.826$	$\mu_{CR} = 0.031$			
Diversity = 4.0287e+01	k_t = [2, 2, 3, 4]					
Generation 730: Best = 5.562758e+03	FES = 73100	$\mu_F = 0.846$	$\mu_{CR} = 0.020$			
Diversity = 4.0227e+01	k_t = [1, 1, 1, 1]					
Generation 740: Best = 5.558150e+03	FES = 74100	$\mu_F = 0.852$	$\mu_{CR} = 0.024$			
Diversity = 4.0233e+01	k_t = [4, 2, 4, 2]					
Generation 750: Best = 5.557780e+03	FES = 75100	$\mu_F = 0.860$	$\mu_{CR} = 0.026$			
Diversity = 4.0436e+01	k_t = [1, 1, 1, 1]					
Generation 760: Best = 5.557780e+03	FES = 76100	$\mu_F = 0.866$	$\mu_{CR} = 0.036$			

Diversity = 4.0481e+01	k_t = [4, 3, 2, 3]				
Generation 770: Best = 5.557780e+03	FES = 77100	$\mu_F = 0.766$	$\mu_{CR} = 0.036$		
Diversity = 4.0446e+01	k_t = [1, 1, 1, 1]				
Generation 780: Best = 5.421248e+03	FES = 78100	$\mu_F = 0.791$	$\mu_{CR} = 0.044$		
Diversity = 4.0475e+01	k_t = [4, 3, 3, 2]				
Generation 790: Best = 5.421248e+03	FES = 79100	$\mu_F = 0.802$	$\mu_{CR} = 0.037$		
Diversity = 4.0462e+01	k_t = [1, 1, 1, 1]				
Generation 800: Best = 5.417219e+03	FES = 80100	$\mu_F = 0.852$	$\mu_{CR} = 0.034$		
Diversity = 4.0230e+01	k_t = [2, 4, 2, 4]				
Generation 810: Best = 5.417219e+03	FES = 81100	$\mu_F = 0.840$	$\mu_{CR} = 0.037$		
Diversity = 4.0204e+01	k_t = [1, 1, 1, 1]				
Generation 820: Best = 5.417219e+03	FES = 82100	$\mu_F = 0.809$	$\mu_{CR} = 0.030$		
Diversity = 3.9995e+01	k_t = [4, 4, 3, 4]				
Generation 830: Best = 5.405491e+03	FES = 83100	$\mu_F = 0.761$	$\mu_{CR} = 0.038$		
Diversity = 3.9969e+01	k_t = [1, 1, 1, 1]				
Generation 840: Best = 5.405491e+03	FES = 84100	$\mu_F = 0.812$	$\mu_{CR} = 0.029$		
Diversity = 3.9983e+01	k_t = [4, 2, 3, 3]				
Generation 850: Best = 5.405491e+03	FES = 85100	$\mu_F = 0.826$	$\mu_{CR} = 0.037$		
Diversity = 3.9958e+01	k_t = [1, 1, 1, 1]				
Generation 860: Best = 5.110552e+03	FES = 86100	$\mu_F = 0.700$	$\mu_{CR} = 0.039$		
Diversity = 3.9918e+01	k_t = [4, 2, 4, 4]				
Generation 870: Best = 4.950050e+03	FES = 87100	$\mu_F = 0.683$	$\mu_{CR} = 0.031$		
Diversity = 3.9832e+01	k_t = [1, 1, 1, 1]				
Generation 880: Best = 4.946319e+03	FES = 88100	$\mu_F = 0.736$	$\mu_{CR} = 0.038$		
Diversity = 3.9786e+01	k_t = [4, 3, 2, 4]				
Generation 890: Best = 4.861950e+03	FES = 89100	$\mu_F = 0.724$	$\mu_{CR} = 0.044$		
Diversity = 3.9689e+01	k_t = [1, 1, 1, 1]				
Generation 900: Best = 4.859749e+03	FES = 90100	$\mu_F = 0.758$	$\mu_{CR} = 0.041$		
Diversity = 3.9426e+01	k_t = [3, 3, 2, 4]				
Generation 910: Best = 4.859299e+03	FES = 91100	$\mu_F = 0.753$	$\mu_{CR} = 0.030$		
Diversity = 3.9250e+01	k_t = [1, 1, 1, 1]				
Generation 920: Best = 4.859299e+03	FES = 92100	$\mu_F = 0.735$	$\mu_{CR} = 0.043$		
Diversity = 3.9220e+01	k_t = [2, 4, 4, 2]				
Generation 930: Best = 4.859299e+03	FES = 93100	$\mu_F = 0.782$	$\mu_{CR} = 0.035$		
Diversity = 3.9239e+01	k_t = [1, 1, 1, 1]				
Generation 940: Best = 4.844274e+03	FES = 94100	$\mu_F = 0.776$	$\mu_{CR} = 0.031$		
Diversity = 3.9494e+01	k_t = [4, 3, 2, 4]				
Generation 950: Best = 4.844274e+03	FES = 95100	$\mu_F = 0.793$	$\mu_{CR} = 0.026$		
Diversity = 3.9474e+01	k_t = [1, 1, 1, 1]				
Generation 960: Best = 4.844274e+03	FES = 96100	$\mu_F = 0.787$	$\mu_{CR} = 0.032$		
Diversity = 3.9368e+01	k_t = [3, 3, 4, 3]				
Generation 970: Best = 4.844274e+03	FES = 97100	$\mu_F = 0.793$	$\mu_{CR} = 0.039$		
Diversity = 3.9247e+01	k_t = [1, 1, 1, 1]				
Generation 980: Best = 4.844274e+03	FES = 98100	$\mu_F = 0.759$	$\mu_{CR} = 0.056$		
Diversity = 3.9183e+01	k_t = [3, 4, 2, 3]				
Generation 990: Best = 4.844274e+03	FES = 99100	$\mu_F = 0.806$	$\mu_{CR} = 0.042$		
Diversity = 3.9186e+01	k_t = [1, 1, 1, 1]				
Generation 1000: Best = 4.844274e+03	FES = 100100	$\mu_F = 0.831$	$\mu_{CR} = 0.038$		
Diversity = 3.8900e+01	k_t = [4, 2, 3, 4]				
Generation 1010: Best = 4.844274e+03	FES = 101100	$\mu_F = 0.800$	$\mu_{CR} = 0.034$		
Diversity = 3.8895e+01	k_t = [1, 1, 1, 1]				
Generation 1020: Best = 4.805272e+03	FES = 102100	$\mu_F = 0.839$	$\mu_{CR} = 0.043$		
Diversity = 3.8806e+01	k_t = [3, 4, 2, 4]				
Generation 1030: Best = 4.798726e+03	FES = 103100	$\mu_F = 0.850$	$\mu_{CR} = 0.035$		

Diversity = 3.8825e+01	k_t = [1, 1, 1, 1]			
Generation 1040: Best = 4.715803e+03	FES = 104100	$\mu_F = 0.869$	$\mu_{CR} = 0.032$	
Diversity = 3.8898e+01	k_t = [3, 2, 2, 3]			
Generation 1050: Best = 4.712639e+03	FES = 105100	$\mu_F = 0.815$	$\mu_{CR} = 0.030$	
Diversity = 3.8926e+01	k_t = [1, 1, 1, 1]			
Generation 1060: Best = 4.712639e+03	FES = 106100	$\mu_F = 0.855$	$\mu_{CR} = 0.044$	
Diversity = 3.8862e+01	k_t = [4, 2, 4, 4]			
Generation 1070: Best = 4.712639e+03	FES = 107100	$\mu_F = 0.845$	$\mu_{CR} = 0.045$	
Diversity = 3.8762e+01	k_t = [1, 1, 1, 1]			
Generation 1080: Best = 4.579616e+03	FES = 108100	$\mu_F = 0.848$	$\mu_{CR} = 0.050$	
Diversity = 3.8728e+01	k_t = [3, 2, 4, 3]			
Generation 1090: Best = 4.513909e+03	FES = 109100	$\mu_F = 0.863$	$\mu_{CR} = 0.038$	
Diversity = 3.8696e+01	k_t = [1, 1, 1, 1]			
Generation 1100: Best = 4.510758e+03	FES = 110100	$\mu_F = 0.869$	$\mu_{CR} = 0.047$	
Diversity = 3.8707e+01	k_t = [3, 2, 2, 2]			
Generation 1110: Best = 4.510758e+03	FES = 111100	$\mu_F = 0.887$	$\mu_{CR} = 0.044$	
Diversity = 3.8715e+01	k_t = [1, 1, 1, 1]			
Generation 1120: Best = 4.491128e+03	FES = 112100	$\mu_F = 0.901$	$\mu_{CR} = 0.040$	
Diversity = 3.8139e+01	k_t = [2, 4, 3, 4]			
Generation 1130: Best = 4.484664e+03	FES = 113100	$\mu_F = 0.871$	$\mu_{CR} = 0.039$	
Diversity = 3.7898e+01	k_t = [1, 1, 1, 1]			
Generation 1140: Best = 4.467712e+03	FES = 114100	$\mu_F = 0.864$	$\mu_{CR} = 0.037$	
Diversity = 3.7657e+01	k_t = [3, 4, 4, 2]			
Generation 1150: Best = 4.467712e+03	FES = 115100	$\mu_F = 0.851$	$\mu_{CR} = 0.033$	
Diversity = 3.7724e+01	k_t = [1, 1, 1, 1]			
Generation 1160: Best = 4.467423e+03	FES = 116100	$\mu_F = 0.863$	$\mu_{CR} = 0.041$	
Diversity = 3.7714e+01	k_t = [4, 2, 4, 3]			
Generation 1170: Best = 4.425463e+03	FES = 117100	$\mu_F = 0.856$	$\mu_{CR} = 0.045$	
Diversity = 3.7660e+01	k_t = [1, 1, 1, 1]			
Generation 1180: Best = 4.389258e+03	FES = 118100	$\mu_F = 0.879$	$\mu_{CR} = 0.044$	
Diversity = 3.7480e+01	k_t = [3, 4, 4, 3]			
Generation 1190: Best = 4.377552e+03	FES = 119100	$\mu_F = 0.854$	$\mu_{CR} = 0.047$	
Diversity = 3.7374e+01	k_t = [1, 1, 1, 1]			
Generation 1200: Best = 4.274147e+03	FES = 120100	$\mu_F = 0.874$	$\mu_{CR} = 0.052$	
Diversity = 3.7148e+01	k_t = [4, 2, 2, 4]			
Generation 1210: Best = 4.272800e+03	FES = 121100	$\mu_F = 0.865$	$\mu_{CR} = 0.070$	
Diversity = 3.6827e+01	k_t = [1, 1, 1, 1]			
Generation 1220: Best = 4.203079e+03	FES = 122100	$\mu_F = 0.850$	$\mu_{CR} = 0.077$	
Diversity = 3.6827e+01	k_t = [3, 4, 3, 3]			
Generation 1230: Best = 4.195726e+03	FES = 123100	$\mu_F = 0.852$	$\mu_{CR} = 0.078$	
Diversity = 3.6777e+01	k_t = [1, 1, 1, 1]			
Generation 1240: Best = 4.072302e+03	FES = 124100	$\mu_F = 0.821$	$\mu_{CR} = 0.089$	
Diversity = 3.6624e+01	k_t = [4, 2, 3, 3]			
Generation 1250: Best = 4.007957e+03	FES = 125100	$\mu_F = 0.847$	$\mu_{CR} = 0.089$	
Diversity = 3.6598e+01	k_t = [1, 1, 1, 1]			
Generation 1260: Best = 3.960927e+03	FES = 126100	$\mu_F = 0.864$	$\mu_{CR} = 0.106$	
Diversity = 3.6700e+01	k_t = [3, 3, 2, 2]			
Generation 1270: Best = 3.935813e+03	FES = 127100	$\mu_F = 0.881$	$\mu_{CR} = 0.107$	
Diversity = 3.6260e+01	k_t = [1, 1, 1, 1]			
Generation 1280: Best = 3.836182e+03	FES = 128100	$\mu_F = 0.894$	$\mu_{CR} = 0.123$	
Diversity = 3.4849e+01	k_t = [2, 4, 3, 2]			
Generation 1290: Best = 3.836182e+03	FES = 129100	$\mu_F = 0.887$	$\mu_{CR} = 0.139$	
Diversity = 3.1179e+01	k_t = [1, 1, 1, 1]			
Generation 1300: Best = 3.762750e+03	FES = 130100	$\mu_F = 0.856$	$\mu_{CR} = 0.155$	

Diversity = 2.6505e+01	k_t = [3, 3, 2, 3]			
Generation 1310: Best = 3.759551e+03	FES = 131100	$\mu_F = 0.820$	$\mu_{CR} = 0.154$	
Diversity = 2.3239e+01	k_t = [1, 1, 1, 1]			
Generation 1320: Best = 3.708636e+03	FES = 132100	$\mu_F = 0.784$	$\mu_{CR} = 0.133$	
Diversity = 2.0223e+01	k_t = [2, 4, 4, 3]			
Generation 1330: Best = 3.683472e+03	FES = 133100	$\mu_F = 0.768$	$\mu_{CR} = 0.115$	
Diversity = 1.9197e+01	k_t = [1, 1, 1, 1]			
Generation 1340: Best = 3.652792e+03	FES = 134100	$\mu_F = 0.746$	$\mu_{CR} = 0.107$	
Diversity = 1.8018e+01	k_t = [3, 4, 4, 3]			
Generation 1350: Best = 3.640225e+03	FES = 135100	$\mu_F = 0.728$	$\mu_{CR} = 0.091$	
Diversity = 1.6881e+01	k_t = [1, 1, 1, 1]			
Generation 1360: Best = 3.617630e+03	FES = 136100	$\mu_F = 0.711$	$\mu_{CR} = 0.081$	
Diversity = 1.6516e+01	k_t = [4, 2, 2, 2]			
Generation 1370: Best = 3.617630e+03	FES = 137100	$\mu_F = 0.694$	$\mu_{CR} = 0.073$	
Diversity = 1.5654e+01	k_t = [1, 1, 1, 1]			
Generation 1380: Best = 3.597547e+03	FES = 138100	$\mu_F = 0.703$	$\mu_{CR} = 0.068$	
Diversity = 1.5374e+01	k_t = [4, 2, 2, 3]			
Generation 1390: Best = 3.595620e+03	FES = 139100	$\mu_F = 0.678$	$\mu_{CR} = 0.068$	
Diversity = 1.5278e+01	k_t = [1, 1, 1, 1]			
Generation 1400: Best = 3.589379e+03	FES = 140100	$\mu_F = 0.703$	$\mu_{CR} = 0.064$	
Diversity = 1.4981e+01	k_t = [3, 4, 4, 2]			
Generation 1410: Best = 3.589379e+03	FES = 141100	$\mu_F = 0.750$	$\mu_{CR} = 0.052$	
Diversity = 1.4732e+01	k_t = [1, 1, 1, 1]			
Generation 1420: Best = 3.587790e+03	FES = 142100	$\mu_F = 0.697$	$\mu_{CR} = 0.064$	
Diversity = 1.4711e+01	k_t = [3, 3, 4, 3]			
Generation 1430: Best = 3.584806e+03	FES = 143100	$\mu_F = 0.692$	$\mu_{CR} = 0.060$	
Diversity = 1.4617e+01	k_t = [1, 1, 1, 1]			
Generation 1440: Best = 3.562439e+03	FES = 144100	$\mu_F = 0.691$	$\mu_{CR} = 0.068$	
Diversity = 1.4262e+01	k_t = [4, 4, 3, 3]			
Generation 1450: Best = 3.561490e+03	FES = 145100	$\mu_F = 0.708$	$\mu_{CR} = 0.056$	
Diversity = 1.4138e+01	k_t = [1, 1, 1, 1]			
Generation 1460: Best = 3.560848e+03	FES = 146100	$\mu_F = 0.743$	$\mu_{CR} = 0.056$	
Diversity = 1.4024e+01	k_t = [3, 3, 3, 3]			
Generation 1470: Best = 3.560848e+03	FES = 147100	$\mu_F = 0.743$	$\mu_{CR} = 0.058$	
Diversity = 1.3898e+01	k_t = [1, 1, 1, 1]			
Generation 1480: Best = 3.559313e+03	FES = 148100	$\mu_F = 0.747$	$\mu_{CR} = 0.054$	
Diversity = 1.3770e+01	k_t = [3, 3, 4, 4]			
Generation 1490: Best = 3.559313e+03	FES = 149100	$\mu_F = 0.769$	$\mu_{CR} = 0.046$	
Diversity = 1.3684e+01	k_t = [1, 1, 1, 1]			
Generation 1500: Best = 3.559313e+03	FES = 150100	$\mu_F = 0.802$	$\mu_{CR} = 0.049$	
Diversity = 1.3661e+01	k_t = [4, 3, 2, 2]			
Generation 1510: Best = 3.559313e+03	FES = 151100	$\mu_F = 0.769$	$\mu_{CR} = 0.050$	
Diversity = 1.3635e+01	k_t = [1, 1, 1, 1]			
Generation 1520: Best = 3.552174e+03	FES = 152100	$\mu_F = 0.787$	$\mu_{CR} = 0.047$	
Diversity = 1.3629e+01	k_t = [4, 4, 3, 3]			
Generation 1530: Best = 3.552161e+03	FES = 153100	$\mu_F = 0.742$	$\mu_{CR} = 0.047$	
Diversity = 1.3625e+01	k_t = [1, 1, 1, 1]			
Generation 1540: Best = 3.544066e+03	FES = 154100	$\mu_F = 0.760$	$\mu_{CR} = 0.050$	
Diversity = 1.3248e+01	k_t = [2, 4, 3, 4]			
Generation 1550: Best = 3.536040e+03	FES = 155100	$\mu_F = 0.737$	$\mu_{CR} = 0.051$	
Diversity = 1.3185e+01	k_t = [1, 1, 1, 1]			
Generation 1560: Best = 3.528670e+03	FES = 156100	$\mu_F = 0.707$	$\mu_{CR} = 0.046$	
Diversity = 1.3156e+01	k_t = [3, 3, 2, 2]			
Generation 1570: Best = 3.527457e+03	FES = 157100	$\mu_F = 0.708$	$\mu_{CR} = 0.041$	

Diversity = 1.3209e+01	k_t = [1, 1, 1, 1]			
Generation 1580: Best = 3.527372e+03	FES = 158100	$\mu_F = 0.697$	$\mu_{CR} = 0.047$	
Diversity = 1.3196e+01	k_t = [4, 2, 3, 2]			
Generation 1590: Best = 3.527372e+03	FES = 159100	$\mu_F = 0.688$	$\mu_{CR} = 0.042$	
Diversity = 1.3230e+01	k_t = [1, 1, 1, 1]			
Generation 1600: Best = 3.525102e+03	FES = 160100	$\mu_F = 0.713$	$\mu_{CR} = 0.041$	
Diversity = 1.3264e+01	k_t = [2, 2, 4, 3]			
Generation 1610: Best = 3.525102e+03	FES = 161100	$\mu_F = 0.736$	$\mu_{CR} = 0.037$	
Diversity = 1.3263e+01	k_t = [1, 1, 1, 1]			
Generation 1620: Best = 3.524079e+03	FES = 162100	$\mu_F = 0.744$	$\mu_{CR} = 0.050$	
Diversity = 1.3189e+01	k_t = [4, 3, 2, 2]			
Generation 1630: Best = 3.523644e+03	FES = 163100	$\mu_F = 0.781$	$\mu_{CR} = 0.042$	
Diversity = 1.3246e+01	k_t = [1, 1, 1, 1]			
Generation 1640: Best = 3.522104e+03	FES = 164100	$\mu_F = 0.781$	$\mu_{CR} = 0.037$	
Diversity = 1.3263e+01	k_t = [2, 4, 4, 4]			
Generation 1650: Best = 3.514913e+03	FES = 165100	$\mu_F = 0.812$	$\mu_{CR} = 0.036$	
Diversity = 1.3234e+01	k_t = [1, 1, 1, 1]			
Generation 1660: Best = 3.514913e+03	FES = 166100	$\mu_F = 0.835$	$\mu_{CR} = 0.031$	
Diversity = 1.3044e+01	k_t = [4, 4, 4, 4]			
Generation 1670: Best = 3.514913e+03	FES = 167100	$\mu_F = 0.805$	$\mu_{CR} = 0.030$	
Diversity = 1.3038e+01	k_t = [1, 1, 1, 1]			
Generation 1680: Best = 3.507201e+03	FES = 168100	$\mu_F = 0.829$	$\mu_{CR} = 0.027$	
Diversity = 1.2989e+01	k_t = [2, 3, 2, 4]			
Generation 1690: Best = 3.507201e+03	FES = 169100	$\mu_F = 0.829$	$\mu_{CR} = 0.028$	
Diversity = 1.2991e+01	k_t = [1, 1, 1, 1]			
Generation 1700: Best = 3.497873e+03	FES = 170100	$\mu_F = 0.803$	$\mu_{CR} = 0.036$	
Diversity = 1.3023e+01	k_t = [2, 2, 4, 2]			
Generation 1710: Best = 3.497873e+03	FES = 171100	$\mu_F = 0.805$	$\mu_{CR} = 0.033$	
Diversity = 1.3063e+01	k_t = [1, 1, 1, 1]			
Generation 1720: Best = 3.491610e+03	FES = 172100	$\mu_F = 0.702$	$\mu_{CR} = 0.049$	
Diversity = 1.3042e+01	k_t = [3, 2, 3, 2]			
Generation 1730: Best = 3.491610e+03	FES = 173100	$\mu_F = 0.699$	$\mu_{CR} = 0.046$	
Diversity = 1.3016e+01	k_t = [1, 1, 1, 1]			
Generation 1740: Best = 3.490863e+03	FES = 174100	$\mu_F = 0.717$	$\mu_{CR} = 0.038$	
Diversity = 1.3017e+01	k_t = [3, 4, 3, 3]			
Generation 1750: Best = 3.490863e+03	FES = 175100	$\mu_F = 0.730$	$\mu_{CR} = 0.035$	
Diversity = 1.3031e+01	k_t = [1, 1, 1, 1]			
Generation 1760: Best = 3.490863e+03	FES = 176100	$\mu_F = 0.711$	$\mu_{CR} = 0.028$	
Diversity = 1.3111e+01	k_t = [4, 4, 4, 4]			
Generation 1770: Best = 3.490863e+03	FES = 177100	$\mu_F = 0.682$	$\mu_{CR} = 0.035$	
Diversity = 1.3145e+01	k_t = [1, 1, 1, 1]			
Generation 1780: Best = 3.490863e+03	FES = 178100	$\mu_F = 0.693$	$\mu_{CR} = 0.036$	
Diversity = 1.3100e+01	k_t = [2, 2, 2, 3]			
Generation 1790: Best = 3.490863e+03	FES = 179100	$\mu_F = 0.704$	$\mu_{CR} = 0.042$	
Diversity = 1.3112e+01	k_t = [1, 1, 1, 1]			
Generation 1800: Best = 3.487773e+03	FES = 180100	$\mu_F = 0.692$	$\mu_{CR} = 0.038$	
Diversity = 1.3022e+01	k_t = [3, 4, 4, 4]			
Generation 1810: Best = 3.487707e+03	FES = 181100	$\mu_F = 0.695$	$\mu_{CR} = 0.035$	
Diversity = 1.2927e+01	k_t = [1, 1, 1, 1]			
Generation 1820: Best = 3.487707e+03	FES = 182100	$\mu_F = 0.734$	$\mu_{CR} = 0.030$	
Diversity = 1.2956e+01	k_t = [2, 4, 2, 2]			
Generation 1830: Best = 3.483936e+03	FES = 183100	$\mu_F = 0.738$	$\mu_{CR} = 0.034$	
Diversity = 1.2924e+01	k_t = [1, 1, 1, 1]			
Generation 1840: Best = 3.480791e+03	FES = 184100	$\mu_F = 0.693$	$\mu_{CR} = 0.040$	

Diversity = 1.2897e+01	k_t = [4, 2, 3, 3]			
Generation 1850: Best = 3.480413e+03	FES = 185100	$\mu_F = 0.720$	$\mu_{CR} = 0.035$	
Diversity = 1.2890e+01	k_t = [1, 1, 1, 1]			
Generation 1860: Best = 3.480229e+03	FES = 186100	$\mu_F = 0.799$	$\mu_{CR} = 0.034$	
Diversity = 1.2959e+01	k_t = [2, 3, 3, 3]			
Generation 1870: Best = 3.480181e+03	FES = 187100	$\mu_F = 0.758$	$\mu_{CR} = 0.036$	
Diversity = 1.2941e+01	k_t = [1, 1, 1, 1]			
Generation 1880: Best = 3.480181e+03	FES = 188100	$\mu_F = 0.748$	$\mu_{CR} = 0.040$	
Diversity = 1.2937e+01	k_t = [2, 2, 2, 2]			
Generation 1890: Best = 3.480084e+03	FES = 189100	$\mu_F = 0.790$	$\mu_{CR} = 0.033$	
Diversity = 1.2937e+01	k_t = [1, 1, 1, 1]			
Generation 1900: Best = 3.479707e+03	FES = 190100	$\mu_F = 0.821$	$\mu_{CR} = 0.032$	
Diversity = 1.2905e+01	k_t = [3, 3, 3, 4]			
Generation 1910: Best = 3.479238e+03	FES = 191100	$\mu_F = 0.867$	$\mu_{CR} = 0.027$	
Diversity = 1.2946e+01	k_t = [1, 1, 1, 1]			
Generation 1920: Best = 3.479238e+03	FES = 192100	$\mu_F = 0.837$	$\mu_{CR} = 0.032$	
Diversity = 1.2911e+01	k_t = [4, 3, 3, 3]			
Generation 1930: Best = 3.479238e+03	FES = 193100	$\mu_F = 0.814$	$\mu_{CR} = 0.030$	
Diversity = 1.2960e+01	k_t = [1, 1, 1, 1]			
Generation 1940: Best = 3.479033e+03	FES = 194100	$\mu_F = 0.846$	$\mu_{CR} = 0.026$	
Diversity = 1.2862e+01	k_t = [4, 4, 3, 4]			
Generation 1950: Best = 3.478230e+03	FES = 195100	$\mu_F = 0.843$	$\mu_{CR} = 0.027$	
Diversity = 1.2882e+01	k_t = [1, 1, 1, 1]			
Generation 1960: Best = 3.477880e+03	FES = 196100	$\mu_F = 0.786$	$\mu_{CR} = 0.029$	
Diversity = 1.2896e+01	k_t = [2, 3, 2, 2]			
Generation 1970: Best = 3.477660e+03	FES = 197100	$\mu_F = 0.817$	$\mu_{CR} = 0.034$	
Diversity = 1.2920e+01	k_t = [1, 1, 1, 1]			
Generation 1980: Best = 3.476973e+03	FES = 198100	$\mu_F = 0.824$	$\mu_{CR} = 0.033$	
Diversity = 1.2699e+01	k_t = [3, 4, 3, 4]			
Generation 1990: Best = 3.476973e+03	FES = 199100	$\mu_F = 0.778$	$\mu_{CR} = 0.031$	
Diversity = 1.2676e+01	k_t = [1, 1, 1, 1]			
Generation 2000: Best = 3.476973e+03	FES = 200100	$\mu_F = 0.780$	$\mu_{CR} = 0.027$	
Diversity = 1.2717e+01	k_t = [2, 4, 3, 3]			
Generation 2010: Best = 3.476962e+03	FES = 201100	$\mu_F = 0.743$	$\mu_{CR} = 0.031$	
Diversity = 1.2692e+01	k_t = [1, 1, 1, 1]			
Generation 2020: Best = 3.476020e+03	FES = 202100	$\mu_F = 0.760$	$\mu_{CR} = 0.037$	
Diversity = 1.2687e+01	k_t = [3, 3, 3, 3]			
Generation 2030: Best = 3.472100e+03	FES = 203100	$\mu_F = 0.820$	$\mu_{CR} = 0.029$	
Diversity = 1.2687e+01	k_t = [1, 1, 1, 1]			
Generation 2040: Best = 3.470655e+03	FES = 204100	$\mu_F = 0.810$	$\mu_{CR} = 0.031$	
Diversity = 1.2701e+01	k_t = [3, 4, 4, 3]			
Generation 2050: Best = 3.470655e+03	FES = 205100	$\mu_F = 0.809$	$\mu_{CR} = 0.026$	
Diversity = 1.2673e+01	k_t = [1, 1, 1, 1]			
Generation 2060: Best = 3.470655e+03	FES = 206100	$\mu_F = 0.812$	$\mu_{CR} = 0.025$	
Diversity = 1.2693e+01	k_t = [4, 2, 3, 2]			
Generation 2070: Best = 3.470655e+03	FES = 207100	$\mu_F = 0.807$	$\mu_{CR} = 0.026$	
Diversity = 1.2714e+01	k_t = [1, 1, 1, 1]			
Generation 2080: Best = 3.470433e+03	FES = 208100	$\mu_F = 0.792$	$\mu_{CR} = 0.035$	
Diversity = 1.2707e+01	k_t = [2, 3, 4, 3]			
Generation 2090: Best = 3.470433e+03	FES = 209100	$\mu_F = 0.830$	$\mu_{CR} = 0.032$	
Diversity = 1.2756e+01	k_t = [1, 1, 1, 1]			
Generation 2100: Best = 3.470433e+03	FES = 210100	$\mu_F = 0.816$	$\mu_{CR} = 0.035$	
Diversity = 1.2716e+01	k_t = [4, 3, 4, 3]			
Generation 2110: Best = 3.470433e+03	FES = 211100	$\mu_F = 0.793$	$\mu_{CR} = 0.032$	

Diversity = 1.2727e+01	k_t = [1, 1, 1, 1]			
Generation 2120: Best = 3.470433e+03	FES = 212100	$\mu_F = 0.753$	$\mu_{CR} = 0.030$	
Diversity = 1.2703e+01	k_t = [3, 4, 3, 2]			
Generation 2130: Best = 3.470430e+03	FES = 213100	$\mu_F = 0.775$	$\mu_{CR} = 0.028$	
Diversity = 1.2619e+01	k_t = [1, 1, 1, 1]			
Generation 2140: Best = 3.469253e+03	FES = 214100	$\mu_F = 0.806$	$\mu_{CR} = 0.033$	
Diversity = 1.2612e+01	k_t = [3, 2, 4, 2]			
Generation 2150: Best = 3.469189e+03	FES = 215100	$\mu_F = 0.823$	$\mu_{CR} = 0.037$	
Diversity = 1.2589e+01	k_t = [1, 1, 1, 1]			
Generation 2160: Best = 3.468511e+03	FES = 216100	$\mu_F = 0.763$	$\mu_{CR} = 0.033$	
Diversity = 1.2564e+01	k_t = [2, 4, 4, 3]			
Generation 2170: Best = 3.468511e+03	FES = 217100	$\mu_F = 0.754$	$\mu_{CR} = 0.025$	
Diversity = 1.2596e+01	k_t = [1, 1, 1, 1]			
Generation 2180: Best = 3.468472e+03	FES = 218100	$\mu_F = 0.685$	$\mu_{CR} = 0.026$	
Diversity = 1.2567e+01	k_t = [2, 3, 4, 2]			
Generation 2190: Best = 3.468472e+03	FES = 219100	$\mu_F = 0.703$	$\mu_{CR} = 0.042$	
Diversity = 1.2572e+01	k_t = [1, 1, 1, 1]			
Generation 2200: Best = 3.465801e+03	FES = 220100	$\mu_F = 0.683$	$\mu_{CR} = 0.033$	
Diversity = 1.2605e+01	k_t = [4, 4, 2, 2]			
Generation 2210: Best = 3.465506e+03	FES = 221100	$\mu_F = 0.757$	$\mu_{CR} = 0.027$	
Diversity = 1.2581e+01	k_t = [1, 1, 1, 1]			
Generation 2220: Best = 3.465325e+03	FES = 222100	$\mu_F = 0.726$	$\mu_{CR} = 0.035$	
Diversity = 1.2562e+01	k_t = [4, 2, 4, 3]			
Generation 2230: Best = 3.465212e+03	FES = 223100	$\mu_F = 0.677$	$\mu_{CR} = 0.031$	
Diversity = 1.2577e+01	k_t = [1, 1, 1, 1]			
Generation 2240: Best = 3.465212e+03	FES = 224100	$\mu_F = 0.724$	$\mu_{CR} = 0.023$	
Diversity = 1.2549e+01	k_t = [2, 4, 3, 3]			
Generation 2250: Best = 3.465212e+03	FES = 225100	$\mu_F = 0.745$	$\mu_{CR} = 0.029$	
Diversity = 1.2567e+01	k_t = [1, 1, 1, 1]			
Generation 2260: Best = 3.465212e+03	FES = 226100	$\mu_F = 0.806$	$\mu_{CR} = 0.026$	
Diversity = 1.2376e+01	k_t = [3, 4, 2, 4]			
Generation 2270: Best = 3.465212e+03	FES = 227100	$\mu_F = 0.825$	$\mu_{CR} = 0.030$	
Diversity = 1.2320e+01	k_t = [1, 1, 1, 1]			
Generation 2280: Best = 3.464636e+03	FES = 228100	$\mu_F = 0.819$	$\mu_{CR} = 0.026$	
Diversity = 1.2259e+01	k_t = [4, 2, 3, 4]			
Generation 2290: Best = 3.464636e+03	FES = 229100	$\mu_F = 0.791$	$\mu_{CR} = 0.035$	
Diversity = 1.2241e+01	k_t = [1, 1, 1, 1]			
Generation 2300: Best = 3.464636e+03	FES = 230100	$\mu_F = 0.832$	$\mu_{CR} = 0.033$	
Diversity = 1.2163e+01	k_t = [4, 3, 3, 4]			
Generation 2310: Best = 3.464628e+03	FES = 231100	$\mu_F = 0.798$	$\mu_{CR} = 0.041$	
Diversity = 1.2195e+01	k_t = [1, 1, 1, 1]			
Generation 2320: Best = 3.463788e+03	FES = 232100	$\mu_F = 0.774$	$\mu_{CR} = 0.032$	
Diversity = 1.2162e+01	k_t = [2, 4, 4, 2]			
Generation 2330: Best = 3.463617e+03	FES = 233100	$\mu_F = 0.786$	$\mu_{CR} = 0.037$	
Diversity = 1.2108e+01	k_t = [1, 1, 1, 1]			
Generation 2340: Best = 3.463335e+03	FES = 234100	$\mu_F = 0.775$	$\mu_{CR} = 0.045$	
Diversity = 1.2070e+01	k_t = [2, 4, 4, 2]			
Generation 2350: Best = 3.461904e+03	FES = 235100	$\mu_F = 0.825$	$\mu_{CR} = 0.039$	
Diversity = 1.2084e+01	k_t = [1, 1, 1, 1]			
Generation 2360: Best = 3.461904e+03	FES = 236100	$\mu_F = 0.827$	$\mu_{CR} = 0.052$	
Diversity = 1.1733e+01	k_t = [3, 2, 3, 4]			
Generation 2370: Best = 3.461904e+03	FES = 237100	$\mu_F = 0.814$	$\mu_{CR} = 0.037$	
Diversity = 1.1634e+01	k_t = [1, 1, 1, 1]			
Generation 2380: Best = 3.461904e+03	FES = 238100	$\mu_F = 0.810$	$\mu_{CR} = 0.035$	

Diversity = 1.1522e+01	k_t = [2, 3, 4, 4]			
Generation 2390: Best = 3.461904e+03	FES = 239100	$\mu_F = 0.744$	$\mu_{CR} = 0.033$	
Diversity = 1.1478e+01	k_t = [1, 1, 1, 1]			
Generation 2400: Best = 3.459529e+03	FES = 240100	$\mu_F = 0.743$	$\mu_{CR} = 0.038$	
Diversity = 1.1341e+01	k_t = [4, 3, 2, 4]			
Generation 2410: Best = 3.459529e+03	FES = 241100	$\mu_F = 0.776$	$\mu_{CR} = 0.039$	
Diversity = 1.1144e+01	k_t = [1, 1, 1, 1]			
Generation 2420: Best = 3.459529e+03	FES = 242100	$\mu_F = 0.759$	$\mu_{CR} = 0.040$	
Diversity = 1.0974e+01	k_t = [3, 3, 2, 4]			
Generation 2430: Best = 3.459529e+03	FES = 243100	$\mu_F = 0.773$	$\mu_{CR} = 0.036$	
Diversity = 1.0895e+01	k_t = [1, 1, 1, 1]			
Generation 2440: Best = 3.459356e+03	FES = 244100	$\mu_F = 0.789$	$\mu_{CR} = 0.037$	
Diversity = 1.0918e+01	k_t = [2, 2, 2, 3]			
Generation 2450: Best = 3.458235e+03	FES = 245100	$\mu_F = 0.767$	$\mu_{CR} = 0.036$	
Diversity = 1.0902e+01	k_t = [1, 1, 1, 1]			
Generation 2460: Best = 3.458074e+03	FES = 246100	$\mu_F = 0.738$	$\mu_{CR} = 0.034$	
Diversity = 1.0751e+01	k_t = [2, 4, 4, 4]			
Generation 2470: Best = 3.458074e+03	FES = 247100	$\mu_F = 0.801$	$\mu_{CR} = 0.046$	
Diversity = 1.0652e+01	k_t = [1, 1, 1, 1]			
Generation 2480: Best = 3.458074e+03	FES = 248100	$\mu_F = 0.781$	$\mu_{CR} = 0.039$	
Diversity = 1.0604e+01	k_t = [3, 4, 3, 2]			
Generation 2490: Best = 3.458074e+03	FES = 249100	$\mu_F = 0.817$	$\mu_{CR} = 0.029$	
Diversity = 1.0636e+01	k_t = [1, 1, 1, 1]			
Generation 2500: Best = 3.457611e+03	FES = 250100	$\mu_F = 0.834$	$\mu_{CR} = 0.031$	
Diversity = 1.0536e+01	k_t = [2, 2, 4, 2]			
Generation 2510: Best = 3.457130e+03	FES = 251100	$\mu_F = 0.816$	$\mu_{CR} = 0.038$	
Diversity = 1.0545e+01	k_t = [1, 1, 1, 1]			
Generation 2520: Best = 3.456073e+03	FES = 252100	$\mu_F = 0.811$	$\mu_{CR} = 0.035$	
Diversity = 1.0092e+01	k_t = [4, 3, 2, 4]			
Generation 2530: Best = 3.456073e+03	FES = 253100	$\mu_F = 0.824$	$\mu_{CR} = 0.038$	
Diversity = 9.9777e+00	k_t = [1, 1, 1, 1]			
Generation 2540: Best = 3.456073e+03	FES = 254100	$\mu_F = 0.723$	$\mu_{CR} = 0.040$	
Diversity = 9.9711e+00	k_t = [2, 2, 4, 2]			
Generation 2550: Best = 3.455986e+03	FES = 255100	$\mu_F = 0.692$	$\mu_{CR} = 0.037$	
Diversity = 9.8621e+00	k_t = [1, 1, 1, 1]			
Generation 2560: Best = 3.455986e+03	FES = 256100	$\mu_F = 0.741$	$\mu_{CR} = 0.037$	
Diversity = 9.7846e+00	k_t = [4, 4, 4, 2]			
Generation 2570: Best = 3.455986e+03	FES = 257100	$\mu_F = 0.735$	$\mu_{CR} = 0.036$	
Diversity = 9.7735e+00	k_t = [1, 1, 1, 1]			
Generation 2580: Best = 3.455986e+03	FES = 258100	$\mu_F = 0.727$	$\mu_{CR} = 0.040$	
Diversity = 9.7163e+00	k_t = [4, 4, 2, 4]			
Generation 2590: Best = 3.455977e+03	FES = 259100	$\mu_F = 0.728$	$\mu_{CR} = 0.036$	
Diversity = 9.6049e+00	k_t = [1, 1, 1, 1]			
Generation 2600: Best = 3.455977e+03	FES = 260100	$\mu_F = 0.770$	$\mu_{CR} = 0.040$	
Diversity = 9.4518e+00	k_t = [2, 4, 3, 4]			
Generation 2610: Best = 3.455977e+03	FES = 261100	$\mu_F = 0.714$	$\mu_{CR} = 0.040$	
Diversity = 9.3420e+00	k_t = [1, 1, 1, 1]			
Generation 2620: Best = 3.455977e+03	FES = 262100	$\mu_F = 0.734$	$\mu_{CR} = 0.035$	
Diversity = 9.0916e+00	k_t = [2, 2, 2, 4]			
Generation 2630: Best = 3.451123e+03	FES = 263100	$\mu_F = 0.735$	$\mu_{CR} = 0.035$	
Diversity = 9.0256e+00	k_t = [1, 1, 1, 1]			
Generation 2640: Best = 3.450583e+03	FES = 264100	$\mu_F = 0.784$	$\mu_{CR} = 0.036$	
Diversity = 8.9113e+00	k_t = [2, 3, 2, 4]			
Generation 2650: Best = 3.450085e+03	FES = 265100	$\mu_F = 0.783$	$\mu_{CR} = 0.031$	

Diversity = 8.8797e+00	k_t = [1, 1, 1, 1]			
Generation 2660: Best = 3.450085e+03	FES = 266100	$\mu_F = 0.778$	$\mu_{CR} = 0.032$	
Diversity = 8.7688e+00	k_t = [4, 4, 2, 3]			
Generation 2670: Best = 3.449115e+03	FES = 267100	$\mu_F = 0.761$	$\mu_{CR} = 0.034$	
Diversity = 8.7643e+00	k_t = [1, 1, 1, 1]			
Generation 2680: Best = 3.448681e+03	FES = 268100	$\mu_F = 0.802$	$\mu_{CR} = 0.031$	
Diversity = 8.5691e+00	k_t = [3, 2, 2, 4]			
Generation 2690: Best = 3.448681e+03	FES = 269100	$\mu_F = 0.826$	$\mu_{CR} = 0.039$	
Diversity = 8.5046e+00	k_t = [1, 1, 1, 1]			
Generation 2700: Best = 3.447826e+03	FES = 270100	$\mu_F = 0.795$	$\mu_{CR} = 0.035$	
Diversity = 8.3565e+00	k_t = [3, 3, 2, 4]			
Generation 2710: Best = 3.447513e+03	FES = 271100	$\mu_F = 0.788$	$\mu_{CR} = 0.048$	
Diversity = 8.2911e+00	k_t = [1, 1, 1, 1]			
Generation 2720: Best = 3.447510e+03	FES = 272100	$\mu_F = 0.677$	$\mu_{CR} = 0.051$	
Diversity = 8.2413e+00	k_t = [3, 3, 3, 4]			
Generation 2730: Best = 3.445077e+03	FES = 273100	$\mu_F = 0.689$	$\mu_{CR} = 0.059$	
Diversity = 8.1737e+00	k_t = [1, 1, 1, 1]			
Generation 2740: Best = 3.445076e+03	FES = 274100	$\mu_F = 0.701$	$\mu_{CR} = 0.055$	
Diversity = 8.1678e+00	k_t = [4, 3, 2, 2]			
Generation 2750: Best = 3.443112e+03	FES = 275100	$\mu_F = 0.734$	$\mu_{CR} = 0.048$	
Diversity = 8.0778e+00	k_t = [1, 1, 1, 1]			
Generation 2760: Best = 3.443112e+03	FES = 276100	$\mu_F = 0.739$	$\mu_{CR} = 0.051$	
Diversity = 8.0715e+00	k_t = [2, 3, 4, 2]			
Generation 2770: Best = 3.442869e+03	FES = 277100	$\mu_F = 0.766$	$\mu_{CR} = 0.041$	
Diversity = 8.0471e+00	k_t = [1, 1, 1, 1]			
Generation 2780: Best = 3.440793e+03	FES = 278100	$\mu_F = 0.774$	$\mu_{CR} = 0.042$	
Diversity = 8.0273e+00	k_t = [2, 4, 2, 4]			
Generation 2790: Best = 3.440119e+03	FES = 279100	$\mu_F = 0.816$	$\mu_{CR} = 0.041$	
Diversity = 7.8924e+00	k_t = [1, 1, 1, 1]			
Generation 2800: Best = 3.438942e+03	FES = 280100	$\mu_F = 0.811$	$\mu_{CR} = 0.049$	
Diversity = 7.8598e+00	k_t = [3, 2, 4, 3]			
Generation 2810: Best = 3.438942e+03	FES = 281100	$\mu_F = 0.829$	$\mu_{CR} = 0.047$	
Diversity = 7.8951e+00	k_t = [1, 1, 1, 1]			
Generation 2820: Best = 3.438130e+03	FES = 282100	$\mu_F = 0.828$	$\mu_{CR} = 0.040$	
Diversity = 7.6081e+00	k_t = [2, 4, 3, 4]			
Generation 2830: Best = 3.438130e+03	FES = 283100	$\mu_F = 0.818$	$\mu_{CR} = 0.051$	
Diversity = 7.5145e+00	k_t = [1, 1, 1, 1]			
Generation 2840: Best = 3.437726e+03	FES = 284100	$\mu_F = 0.822$	$\mu_{CR} = 0.060$	
Diversity = 7.4384e+00	k_t = [2, 2, 3, 2]			
Generation 2850: Best = 3.436279e+03	FES = 285100	$\mu_F = 0.804$	$\mu_{CR} = 0.052$	
Diversity = 7.4062e+00	k_t = [1, 1, 1, 1]			
Generation 2860: Best = 3.434381e+03	FES = 286100	$\mu_F = 0.831$	$\mu_{CR} = 0.040$	
Diversity = 7.4186e+00	k_t = [2, 3, 4, 3]			
Generation 2870: Best = 3.434381e+03	FES = 287100	$\mu_F = 0.807$	$\mu_{CR} = 0.048$	
Diversity = 7.4168e+00	k_t = [1, 1, 1, 1]			
Generation 2880: Best = 3.433857e+03	FES = 288100	$\mu_F = 0.808$	$\mu_{CR} = 0.050$	
Diversity = 6.9036e+00	k_t = [3, 2, 2, 4]			
Generation 2890: Best = 3.433857e+03	FES = 289100	$\mu_F = 0.790$	$\mu_{CR} = 0.049$	
Diversity = 6.8928e+00	k_t = [1, 1, 1, 1]			
Generation 2900: Best = 3.433174e+03	FES = 290100	$\mu_F = 0.811$	$\mu_{CR} = 0.042$	
Diversity = 6.7204e+00	k_t = [4, 2, 2, 2]			
Generation 2910: Best = 3.433039e+03	FES = 291100	$\mu_F = 0.801$	$\mu_{CR} = 0.039$	
Diversity = 6.7128e+00	k_t = [1, 1, 1, 1]			
Generation 2920: Best = 3.432028e+03	FES = 292100	$\mu_F = 0.814$	$\mu_{CR} = 0.045$	

Diversity = 6.7476e+00	k_t = [3, 2, 4, 2]			
Generation 2930: Best = 3.431843e+03	FES = 293100	$\mu_F = 0.806$	$\mu_{CR} = 0.046$	
Diversity = 6.7307e+00	k_t = [1, 1, 1, 1]			
Generation 2940: Best = 3.431843e+03	FES = 294100	$\mu_F = 0.837$	$\mu_{CR} = 0.046$	
Diversity = 6.5866e+00	k_t = [2, 4, 2, 4]			
Generation 2950: Best = 3.431843e+03	FES = 295100	$\mu_F = 0.808$	$\mu_{CR} = 0.049$	
Diversity = 6.5584e+00	k_t = [1, 1, 1, 1]			
Generation 2960: Best = 3.431762e+03	FES = 296100	$\mu_F = 0.789$	$\mu_{CR} = 0.053$	
Diversity = 6.4992e+00	k_t = [2, 4, 3, 2]			
Generation 2970: Best = 3.431652e+03	FES = 297100	$\mu_F = 0.779$	$\mu_{CR} = 0.047$	
Diversity = 6.4650e+00	k_t = [1, 1, 1, 1]			
Generation 2980: Best = 3.431230e+03	FES = 298100	$\mu_F = 0.798$	$\mu_{CR} = 0.045$	
Diversity = 6.3840e+00	k_t = [4, 4, 4, 4]			
Generation 2990: Best = 3.430503e+03	FES = 299100	$\mu_F = 0.822$	$\mu_{CR} = 0.041$	
Diversity = 6.3047e+00	k_t = [1, 1, 1, 1]			
Generation 3000: Best = 3.429648e+03	FES = 300100	$\mu_F = 0.825$	$\mu_{CR} = 0.046$	
Diversity = 5.8939e+00	k_t = [2, 3, 4, 4]			
Generation 3010: Best = 3.429648e+03	FES = 301100	$\mu_F = 0.815$	$\mu_{CR} = 0.041$	
Diversity = 5.7786e+00	k_t = [1, 1, 1, 1]			
Generation 3020: Best = 3.429473e+03	FES = 302100	$\mu_F = 0.812$	$\mu_{CR} = 0.039$	
Diversity = 5.7369e+00	k_t = [2, 4, 3, 4]			
Generation 3030: Best = 3.428876e+03	FES = 303100	$\mu_F = 0.791$	$\mu_{CR} = 0.047$	
Diversity = 5.7476e+00	k_t = [1, 1, 1, 1]			
Generation 3040: Best = 3.428876e+03	FES = 304100	$\mu_F = 0.793$	$\mu_{CR} = 0.043$	
Diversity = 5.7618e+00	k_t = [4, 3, 2, 3]			
Generation 3050: Best = 3.428876e+03	FES = 305100	$\mu_F = 0.795$	$\mu_{CR} = 0.041$	
Diversity = 5.7518e+00	k_t = [1, 1, 1, 1]			
Generation 3060: Best = 3.428876e+03	FES = 306100	$\mu_F = 0.796$	$\mu_{CR} = 0.037$	
Diversity = 5.5780e+00	k_t = [3, 4, 2, 4]			
Generation 3070: Best = 3.428211e+03	FES = 307100	$\mu_F = 0.833$	$\mu_{CR} = 0.039$	
Diversity = 5.5677e+00	k_t = [1, 1, 1, 1]			
Generation 3080: Best = 3.427859e+03	FES = 308100	$\mu_F = 0.856$	$\mu_{CR} = 0.043$	
Diversity = 5.5963e+00	k_t = [2, 3, 4, 4]			
Generation 3090: Best = 3.426983e+03	FES = 309100	$\mu_F = 0.818$	$\mu_{CR} = 0.041$	
Diversity = 5.6152e+00	k_t = [1, 1, 1, 1]			
Generation 3100: Best = 3.426983e+03	FES = 310100	$\mu_F = 0.771$	$\mu_{CR} = 0.038$	
Diversity = 5.6383e+00	k_t = [3, 4, 2, 4]			
Generation 3110: Best = 3.426375e+03	FES = 311100	$\mu_F = 0.779$	$\mu_{CR} = 0.032$	
Diversity = 5.6527e+00	k_t = [1, 1, 1, 1]			
Generation 3120: Best = 3.426300e+03	FES = 312100	$\mu_F = 0.764$	$\mu_{CR} = 0.037$	
Diversity = 5.6858e+00	k_t = [4, 2, 3, 3]			
Generation 3130: Best = 3.426300e+03	FES = 313100	$\mu_F = 0.794$	$\mu_{CR} = 0.034$	
Diversity = 5.6950e+00	k_t = [1, 1, 1, 1]			
Generation 3140: Best = 3.426046e+03	FES = 314100	$\mu_F = 0.801$	$\mu_{CR} = 0.041$	
Diversity = 5.6751e+00	k_t = [2, 4, 4, 3]			
Generation 3150: Best = 3.426046e+03	FES = 315100	$\mu_F = 0.808$	$\mu_{CR} = 0.042$	
Diversity = 5.6571e+00	k_t = [1, 1, 1, 1]			
Generation 3160: Best = 3.426046e+03	FES = 316100	$\mu_F = 0.830$	$\mu_{CR} = 0.035$	
Diversity = 5.7495e+00	k_t = [2, 4, 4, 4]			
Generation 3170: Best = 3.426046e+03	FES = 317100	$\mu_F = 0.838$	$\mu_{CR} = 0.045$	
Diversity = 5.8001e+00	k_t = [1, 1, 1, 1]			
Generation 3180: Best = 3.426046e+03	FES = 318100	$\mu_F = 0.835$	$\mu_{CR} = 0.035$	
Diversity = 5.7635e+00	k_t = [3, 4, 2, 3]			
Generation 3190: Best = 3.426046e+03	FES = 319100	$\mu_F = 0.819$	$\mu_{CR} = 0.031$	

Diversity = 5.8477e+00	k_t = [1, 1, 1, 1]			
Generation 3200: Best = 3.426046e+03	FES = 320100	$\mu_F = 0.829$	$\mu_{CR} = 0.033$	
Diversity = 5.8369e+00	k_t = [4, 3, 2, 4]			
Generation 3210: Best = 3.425399e+03	FES = 321100	$\mu_F = 0.828$	$\mu_{CR} = 0.033$	
Diversity = 5.8292e+00	k_t = [1, 1, 1, 1]			
Generation 3220: Best = 3.425399e+03	FES = 322100	$\mu_F = 0.824$	$\mu_{CR} = 0.029$	
Diversity = 5.8421e+00	k_t = [3, 2, 4, 3]			
Generation 3230: Best = 3.425257e+03	FES = 323100	$\mu_F = 0.810$	$\mu_{CR} = 0.032$	
Diversity = 5.8856e+00	k_t = [1, 1, 1, 1]			
Generation 3240: Best = 3.424997e+03	FES = 324100	$\mu_F = 0.813$	$\mu_{CR} = 0.032$	
Diversity = 5.8960e+00	k_t = [3, 3, 4, 2]			
Generation 3250: Best = 3.424862e+03	FES = 325100	$\mu_F = 0.778$	$\mu_{CR} = 0.034$	
Diversity = 5.8867e+00	k_t = [1, 1, 1, 1]			
Generation 3260: Best = 3.424257e+03	FES = 326100	$\mu_F = 0.792$	$\mu_{CR} = 0.033$	
Diversity = 5.8476e+00	k_t = [2, 2, 3, 4]			
Generation 3270: Best = 3.424151e+03	FES = 327100	$\mu_F = 0.816$	$\mu_{CR} = 0.038$	
Diversity = 5.8553e+00	k_t = [1, 1, 1, 1]			
Generation 3280: Best = 3.424151e+03	FES = 328100	$\mu_F = 0.822$	$\mu_{CR} = 0.041$	
Diversity = 5.8555e+00	k_t = [2, 2, 2, 2]			
Generation 3290: Best = 3.424151e+03	FES = 329100	$\mu_F = 0.766$	$\mu_{CR} = 0.043$	
Diversity = 5.8415e+00	k_t = [1, 1, 1, 1]			
Generation 3300: Best = 3.424151e+03	FES = 330100	$\mu_F = 0.772$	$\mu_{CR} = 0.041$	
Diversity = 5.7994e+00	k_t = [2, 4, 3, 3]			
Generation 3310: Best = 3.423791e+03	FES = 331100	$\mu_F = 0.766$	$\mu_{CR} = 0.041$	
Diversity = 5.8467e+00	k_t = [1, 1, 1, 1]			
Generation 3320: Best = 3.423773e+03	FES = 332100	$\mu_F = 0.805$	$\mu_{CR} = 0.032$	
Diversity = 5.8469e+00	k_t = [4, 4, 3, 3]			
Generation 3330: Best = 3.423699e+03	FES = 333100	$\mu_F = 0.820$	$\mu_{CR} = 0.034$	
Diversity = 5.8904e+00	k_t = [1, 1, 1, 1]			
Generation 3340: Best = 3.423371e+03	FES = 334100	$\mu_F = 0.810$	$\mu_{CR} = 0.039$	
Diversity = 5.9154e+00	k_t = [3, 4, 3, 2]			
Generation 3350: Best = 3.423371e+03	FES = 335100	$\mu_F = 0.826$	$\mu_{CR} = 0.032$	
Diversity = 5.9281e+00	k_t = [1, 1, 1, 1]			
Generation 3360: Best = 3.423109e+03	FES = 336100	$\mu_F = 0.791$	$\mu_{CR} = 0.035$	
Diversity = 5.8769e+00	k_t = [3, 4, 3, 4]			
Generation 3370: Best = 3.423040e+03	FES = 337100	$\mu_F = 0.769$	$\mu_{CR} = 0.030$	
Diversity = 5.8147e+00	k_t = [1, 1, 1, 1]			
Generation 3380: Best = 3.423040e+03	FES = 338100	$\mu_F = 0.812$	$\mu_{CR} = 0.033$	
Diversity = 5.7863e+00	k_t = [3, 3, 4, 4]			
Generation 3390: Best = 3.423040e+03	FES = 339100	$\mu_F = 0.780$	$\mu_{CR} = 0.039$	
Diversity = 5.7826e+00	k_t = [1, 1, 1, 1]			
Generation 3400: Best = 3.422656e+03	FES = 340100	$\mu_F = 0.783$	$\mu_{CR} = 0.038$	
Diversity = 5.7841e+00	k_t = [4, 2, 4, 3]			
Generation 3410: Best = 3.422234e+03	FES = 341100	$\mu_F = 0.815$	$\mu_{CR} = 0.028$	
Diversity = 5.8119e+00	k_t = [1, 1, 1, 1]			
Generation 3420: Best = 3.421939e+03	FES = 342100	$\mu_F = 0.831$	$\mu_{CR} = 0.034$	
Diversity = 5.8259e+00	k_t = [2, 3, 3, 2]			
Generation 3430: Best = 3.421570e+03	FES = 343100	$\mu_F = 0.796$	$\mu_{CR} = 0.028$	
Diversity = 5.8127e+00	k_t = [1, 1, 1, 1]			
Generation 3440: Best = 3.421570e+03	FES = 344100	$\mu_F = 0.813$	$\mu_{CR} = 0.038$	
Diversity = 5.8174e+00	k_t = [3, 3, 3, 4]			
Generation 3450: Best = 3.420639e+03	FES = 345100	$\mu_F = 0.748$	$\mu_{CR} = 0.038$	
Diversity = 5.8318e+00	k_t = [1, 1, 1, 1]			
Generation 3460: Best = 3.420639e+03	FES = 346100	$\mu_F = 0.732$	$\mu_{CR} = 0.037$	

Diversity = 5.8557e+00	k_t = [3, 2, 2, 2]			
Generation 3470: Best = 3.419781e+03	FES = 347100	$\mu_F = 0.746$	$\mu_{CR} = 0.042$	
Diversity = 5.8777e+00	k_t = [1, 1, 1, 1]			
Generation 3480: Best = 3.419711e+03	FES = 348100	$\mu_F = 0.773$	$\mu_{CR} = 0.036$	
Diversity = 5.8947e+00	k_t = [4, 2, 2, 3]			
Generation 3490: Best = 3.419711e+03	FES = 349100	$\mu_F = 0.742$	$\mu_{CR} = 0.039$	
Diversity = 5.9153e+00	k_t = [1, 1, 1, 1]			
Generation 3500: Best = 3.419691e+03	FES = 350100	$\mu_F = 0.780$	$\mu_{CR} = 0.038$	
Diversity = 5.9467e+00	k_t = [2, 4, 4, 4]			
Generation 3510: Best = 3.419677e+03	FES = 351100	$\mu_F = 0.770$	$\mu_{CR} = 0.035$	
Diversity = 5.9673e+00	k_t = [1, 1, 1, 1]			
Generation 3520: Best = 3.419677e+03	FES = 352100	$\mu_F = 0.818$	$\mu_{CR} = 0.031$	
Diversity = 5.9502e+00	k_t = [3, 2, 3, 3]			
Generation 3530: Best = 3.419677e+03	FES = 353100	$\mu_F = 0.794$	$\mu_{CR} = 0.033$	
Diversity = 5.9606e+00	k_t = [1, 1, 1, 1]			
Generation 3540: Best = 3.419677e+03	FES = 354100	$\mu_F = 0.827$	$\mu_{CR} = 0.042$	
Diversity = 5.9799e+00	k_t = [4, 3, 2, 2]			
Generation 3550: Best = 3.419677e+03	FES = 355100	$\mu_F = 0.820$	$\mu_{CR} = 0.035$	
Diversity = 5.9933e+00	k_t = [1, 1, 1, 1]			
Generation 3560: Best = 3.419677e+03	FES = 356100	$\mu_F = 0.841$	$\mu_{CR} = 0.035$	
Diversity = 6.0309e+00	k_t = [4, 2, 4, 3]			
Generation 3570: Best = 3.419658e+03	FES = 357100	$\mu_F = 0.842$	$\mu_{CR} = 0.035$	
Diversity = 6.0333e+00	k_t = [1, 1, 1, 1]			
Generation 3580: Best = 3.419655e+03	FES = 358100	$\mu_F = 0.819$	$\mu_{CR} = 0.032$	
Diversity = 6.0641e+00	k_t = [4, 3, 3, 3]			
Generation 3590: Best = 3.419655e+03	FES = 359100	$\mu_F = 0.811$	$\mu_{CR} = 0.030$	
Diversity = 6.1045e+00	k_t = [1, 1, 1, 1]			
Generation 3600: Best = 3.419601e+03	FES = 360100	$\mu_F = 0.758$	$\mu_{CR} = 0.037$	
Diversity = 6.0980e+00	k_t = [3, 3, 4, 4]			
Generation 3610: Best = 3.419599e+03	FES = 361100	$\mu_F = 0.762$	$\mu_{CR} = 0.032$	
Diversity = 6.0751e+00	k_t = [1, 1, 1, 1]			
Generation 3620: Best = 3.419522e+03	FES = 362100	$\mu_F = 0.739$	$\mu_{CR} = 0.033$	
Diversity = 6.0784e+00	k_t = [2, 2, 2, 3]			
Generation 3630: Best = 3.419485e+03	FES = 363100	$\mu_F = 0.785$	$\mu_{CR} = 0.023$	
Diversity = 6.0886e+00	k_t = [1, 1, 1, 1]			
Generation 3640: Best = 3.418480e+03	FES = 364100	$\mu_F = 0.799$	$\mu_{CR} = 0.030$	
Diversity = 6.1153e+00	k_t = [4, 2, 2, 2]			
Generation 3650: Best = 3.418480e+03	FES = 365100	$\mu_F = 0.753$	$\mu_{CR} = 0.032$	
Diversity = 6.1247e+00	k_t = [1, 1, 1, 1]			
Generation 3660: Best = 3.418480e+03	FES = 366100	$\mu_F = 0.739$	$\mu_{CR} = 0.030$	
Diversity = 6.1132e+00	k_t = [2, 2, 2, 3]			
Generation 3670: Best = 3.418480e+03	FES = 367100	$\mu_F = 0.746$	$\mu_{CR} = 0.030$	
Diversity = 6.0984e+00	k_t = [1, 1, 1, 1]			
Generation 3680: Best = 3.418480e+03	FES = 368100	$\mu_F = 0.760$	$\mu_{CR} = 0.030$	
Diversity = 6.1063e+00	k_t = [2, 2, 3, 2]			
Generation 3690: Best = 3.418480e+03	FES = 369100	$\mu_F = 0.754$	$\mu_{CR} = 0.026$	
Diversity = 6.1198e+00	k_t = [1, 1, 1, 1]			
Generation 3700: Best = 3.418480e+03	FES = 370100	$\mu_F = 0.800$	$\mu_{CR} = 0.035$	
Diversity = 6.0794e+00	k_t = [2, 2, 3, 3]			
Generation 3710: Best = 3.418480e+03	FES = 371100	$\mu_F = 0.800$	$\mu_{CR} = 0.032$	
Diversity = 6.0624e+00	k_t = [1, 1, 1, 1]			
Generation 3720: Best = 3.418480e+03	FES = 372100	$\mu_F = 0.800$	$\mu_{CR} = 0.030$	
Diversity = 6.0653e+00	k_t = [3, 3, 4, 3]			
Generation 3730: Best = 3.418480e+03	FES = 373100	$\mu_F = 0.770$	$\mu_{CR} = 0.027$	

Diversity = 6.0754e+00	k_t = [1, 1, 1, 1]			
Generation 3740: Best = 3.418480e+03	FES = 374100	$\mu_F = 0.752$	$\mu_{CR} = 0.033$	
Diversity = 6.0663e+00	k_t = [2, 3, 3, 3]			
Generation 3750: Best = 3.418480e+03	FES = 375100	$\mu_F = 0.768$	$\mu_{CR} = 0.039$	
Diversity = 6.0695e+00	k_t = [1, 1, 1, 1]			
Generation 3760: Best = 3.418281e+03	FES = 376100	$\mu_F = 0.776$	$\mu_{CR} = 0.040$	
Diversity = 6.0699e+00	k_t = [4, 2, 2, 3]			
Generation 3770: Best = 3.418271e+03	FES = 377100	$\mu_F = 0.815$	$\mu_{CR} = 0.029$	
Diversity = 6.0858e+00	k_t = [1, 1, 1, 1]			
Generation 3780: Best = 3.418271e+03	FES = 378100	$\mu_F = 0.820$	$\mu_{CR} = 0.026$	
Diversity = 6.0985e+00	k_t = [4, 2, 3, 2]			
Generation 3790: Best = 3.418271e+03	FES = 379100	$\mu_F = 0.817$	$\mu_{CR} = 0.031$	
Diversity = 6.1042e+00	k_t = [1, 1, 1, 1]			
Generation 3800: Best = 3.418271e+03	FES = 380100	$\mu_F = 0.807$	$\mu_{CR} = 0.040$	
Diversity = 6.0921e+00	k_t = [3, 4, 3, 3]			
Generation 3810: Best = 3.418066e+03	FES = 381100	$\mu_F = 0.824$	$\mu_{CR} = 0.031$	
Diversity = 6.1027e+00	k_t = [1, 1, 1, 1]			
Generation 3820: Best = 3.418064e+03	FES = 382100	$\mu_F = 0.821$	$\mu_{CR} = 0.030$	
Diversity = 6.0908e+00	k_t = [2, 3, 3, 2]			
Generation 3830: Best = 3.418021e+03	FES = 383100	$\mu_F = 0.801$	$\mu_{CR} = 0.030$	
Diversity = 6.0939e+00	k_t = [1, 1, 1, 1]			
Generation 3840: Best = 3.417803e+03	FES = 384100	$\mu_F = 0.810$	$\mu_{CR} = 0.028$	
Diversity = 6.0934e+00	k_t = [2, 4, 2, 3]			
Generation 3850: Best = 3.417740e+03	FES = 385100	$\mu_F = 0.748$	$\mu_{CR} = 0.028$	
Diversity = 6.1072e+00	k_t = [1, 1, 1, 1]			
Generation 3860: Best = 3.417700e+03	FES = 386100	$\mu_F = 0.817$	$\mu_{CR} = 0.033$	
Diversity = 6.1159e+00	k_t = [2, 2, 4, 4]			
Generation 3870: Best = 3.417700e+03	FES = 387100	$\mu_F = 0.833$	$\mu_{CR} = 0.029$	
Diversity = 6.0598e+00	k_t = [1, 1, 1, 1]			
Generation 3880: Best = 3.417700e+03	FES = 388100	$\mu_F = 0.818$	$\mu_{CR} = 0.024$	
Diversity = 6.0506e+00	k_t = [3, 3, 2, 2]			
Generation 3890: Best = 3.417693e+03	FES = 389100	$\mu_F = 0.815$	$\mu_{CR} = 0.024$	
Diversity = 6.0629e+00	k_t = [1, 1, 1, 1]			
Generation 3900: Best = 3.417649e+03	FES = 390100	$\mu_F = 0.822$	$\mu_{CR} = 0.032$	
Diversity = 6.0422e+00	k_t = [4, 3, 3, 2]			
Generation 3910: Best = 3.417258e+03	FES = 391100	$\mu_F = 0.803$	$\mu_{CR} = 0.031$	
Diversity = 6.0377e+00	k_t = [1, 1, 1, 1]			
Generation 3920: Best = 3.417258e+03	FES = 392100	$\mu_F = 0.800$	$\mu_{CR} = 0.031$	
Diversity = 6.0464e+00	k_t = [4, 2, 4, 3]			
Generation 3930: Best = 3.416975e+03	FES = 393100	$\mu_F = 0.828$	$\mu_{CR} = 0.030$	
Diversity = 6.0477e+00	k_t = [1, 1, 1, 1]			
Generation 3940: Best = 3.416904e+03	FES = 394100	$\mu_F = 0.803$	$\mu_{CR} = 0.026$	
Diversity = 6.0408e+00	k_t = [3, 2, 4, 3]			
Generation 3950: Best = 3.416286e+03	FES = 395100	$\mu_F = 0.835$	$\mu_{CR} = 0.025$	
Diversity = 6.0559e+00	k_t = [1, 1, 1, 1]			
Generation 3960: Best = 3.416286e+03	FES = 396100	$\mu_F = 0.849$	$\mu_{CR} = 0.025$	
Diversity = 6.0581e+00	k_t = [3, 2, 4, 2]			
Generation 3970: Best = 3.416286e+03	FES = 397100	$\mu_F = 0.800$	$\mu_{CR} = 0.029$	
Diversity = 6.0906e+00	k_t = [1, 1, 1, 1]			
Generation 3980: Best = 3.416286e+03	FES = 398100	$\mu_F = 0.833$	$\mu_{CR} = 0.026$	
Diversity = 6.1240e+00	k_t = [2, 3, 3, 2]			
Generation 3990: Best = 3.416286e+03	FES = 399100	$\mu_F = 0.806$	$\mu_{CR} = 0.022$	
Diversity = 6.1158e+00	k_t = [1, 1, 1, 1]			
Generation 4000: Best = 3.416209e+03	FES = 400100	$\mu_F = 0.826$	$\mu_{CR} = 0.028$	

Diversity = 6.0131e+00	k_t = [2, 2, 2, 4]			
Generation 4010: Best = 3.416209e+03	FES = 401100	$\mu_F = 0.845$	$\mu_{CR} = 0.024$	
Diversity = 6.0265e+00	k_t = [1, 1, 1, 1]			
Generation 4020: Best = 3.416209e+03	FES = 402100	$\mu_F = 0.814$	$\mu_{CR} = 0.024$	
Diversity = 6.0308e+00	k_t = [3, 2, 2, 4]			
Generation 4030: Best = 3.416209e+03	FES = 403100	$\mu_F = 0.792$	$\mu_{CR} = 0.023$	
Diversity = 6.0447e+00	k_t = [1, 1, 1, 1]			
Generation 4040: Best = 3.416172e+03	FES = 404100	$\mu_F = 0.823$	$\mu_{CR} = 0.023$	
Diversity = 6.0489e+00	k_t = [3, 3, 4, 2]			
Generation 4050: Best = 3.416172e+03	FES = 405100	$\mu_F = 0.816$	$\mu_{CR} = 0.024$	
Diversity = 6.0632e+00	k_t = [1, 1, 1, 1]			
Generation 4060: Best = 3.416172e+03	FES = 406100	$\mu_F = 0.826$	$\mu_{CR} = 0.023$	
Diversity = 6.0884e+00	k_t = [2, 4, 3, 2]			
Generation 4070: Best = 3.416172e+03	FES = 407100	$\mu_F = 0.841$	$\mu_{CR} = 0.025$	
Diversity = 6.0738e+00	k_t = [1, 1, 1, 1]			
Generation 4080: Best = 3.416172e+03	FES = 408100	$\mu_F = 0.832$	$\mu_{CR} = 0.036$	
Diversity = 6.0126e+00	k_t = [4, 3, 4, 4]			
Generation 4090: Best = 3.416172e+03	FES = 409100	$\mu_F = 0.846$	$\mu_{CR} = 0.030$	
Diversity = 6.0155e+00	k_t = [1, 1, 1, 1]			
Generation 4100: Best = 3.416172e+03	FES = 410100	$\mu_F = 0.866$	$\mu_{CR} = 0.030$	
Diversity = 5.9880e+00	k_t = [4, 3, 4, 4]			
Generation 4110: Best = 3.416172e+03	FES = 411100	$\mu_F = 0.877$	$\mu_{CR} = 0.029$	
Diversity = 5.9866e+00	k_t = [1, 1, 1, 1]			
Generation 4120: Best = 3.416172e+03	FES = 412100	$\mu_F = 0.832$	$\mu_{CR} = 0.031$	
Diversity = 5.9753e+00	k_t = [3, 4, 2, 3]			
Generation 4130: Best = 3.416086e+03	FES = 413100	$\mu_F = 0.868$	$\mu_{CR} = 0.039$	
Diversity = 6.0351e+00	k_t = [1, 1, 1, 1]			
Generation 4140: Best = 3.416066e+03	FES = 414100	$\mu_F = 0.877$	$\mu_{CR} = 0.037$	
Diversity = 6.0396e+00	k_t = [3, 2, 4, 3]			
Generation 4150: Best = 3.416066e+03	FES = 415100	$\mu_F = 0.892$	$\mu_{CR} = 0.025$	
Diversity = 6.0421e+00	k_t = [1, 1, 1, 1]			
Generation 4160: Best = 3.415961e+03	FES = 416100	$\mu_F = 0.893$	$\mu_{CR} = 0.026$	
Diversity = 5.9884e+00	k_t = [3, 3, 2, 4]			
Generation 4170: Best = 3.415565e+03	FES = 417100	$\mu_F = 0.842$	$\mu_{CR} = 0.031$	
Diversity = 6.0170e+00	k_t = [1, 1, 1, 1]			
Generation 4180: Best = 3.415565e+03	FES = 418100	$\mu_F = 0.843$	$\mu_{CR} = 0.029$	
Diversity = 6.0610e+00	k_t = [3, 4, 3, 3]			
Generation 4190: Best = 3.415565e+03	FES = 419100	$\mu_F = 0.867$	$\mu_{CR} = 0.032$	
Diversity = 6.0169e+00	k_t = [1, 1, 1, 1]			
Generation 4200: Best = 3.415369e+03	FES = 420100	$\mu_F = 0.876$	$\mu_{CR} = 0.040$	
Diversity = 6.0120e+00	k_t = [2, 3, 3, 2]			
Generation 4210: Best = 3.415369e+03	FES = 421100	$\mu_F = 0.837$	$\mu_{CR} = 0.037$	
Diversity = 6.0127e+00	k_t = [1, 1, 1, 1]			
Generation 4220: Best = 3.415369e+03	FES = 422100	$\mu_F = 0.820$	$\mu_{CR} = 0.041$	
Diversity = 5.9351e+00	k_t = [3, 2, 4, 4]			
Generation 4230: Best = 3.415110e+03	FES = 423100	$\mu_F = 0.828$	$\mu_{CR} = 0.030$	
Diversity = 5.9404e+00	k_t = [1, 1, 1, 1]			
Generation 4240: Best = 3.414673e+03	FES = 424100	$\mu_F = 0.841$	$\mu_{CR} = 0.027$	
Diversity = 5.9470e+00	k_t = [2, 4, 4, 2]			
Generation 4250: Best = 3.414673e+03	FES = 425100	$\mu_F = 0.847$	$\mu_{CR} = 0.026$	
Diversity = 5.9549e+00	k_t = [1, 1, 1, 1]			
Generation 4260: Best = 3.414673e+03	FES = 426100	$\mu_F = 0.850$	$\mu_{CR} = 0.033$	
Diversity = 5.9370e+00	k_t = [4, 2, 2, 2]			
Generation 4270: Best = 3.414617e+03	FES = 427100	$\mu_F = 0.832$	$\mu_{CR} = 0.027$	

Diversity = 5.9484e+00	k_t = [1, 1, 1, 1]			
Generation 4280: Best = 3.414617e+03	FES = 428100	$\mu_F = 0.802$	$\mu_{CR} = 0.026$	
Diversity = 5.9420e+00	k_t = [4, 3, 3, 2]			
Generation 4290: Best = 3.414617e+03	FES = 429100	$\mu_F = 0.782$	$\mu_{CR} = 0.032$	
Diversity = 5.9504e+00	k_t = [1, 1, 1, 1]			
Generation 4300: Best = 3.414617e+03	FES = 430100	$\mu_F = 0.793$	$\mu_{CR} = 0.026$	
Diversity = 5.8878e+00	k_t = [4, 4, 4, 4]			
Generation 4310: Best = 3.414617e+03	FES = 431100	$\mu_F = 0.757$	$\mu_{CR} = 0.032$	
Diversity = 5.8771e+00	k_t = [1, 1, 1, 1]			
Generation 4320: Best = 3.414455e+03	FES = 432100	$\mu_F = 0.731$	$\mu_{CR} = 0.031$	
Diversity = 5.8704e+00	k_t = [2, 3, 4, 2]			
Generation 4330: Best = 3.414432e+03	FES = 433100	$\mu_F = 0.811$	$\mu_{CR} = 0.035$	
Diversity = 5.8722e+00	k_t = [1, 1, 1, 1]			
Generation 4340: Best = 3.414104e+03	FES = 434100	$\mu_F = 0.749$	$\mu_{CR} = 0.034$	
Diversity = 5.7966e+00	k_t = [4, 4, 4, 4]			
Generation 4350: Best = 3.414102e+03	FES = 435100	$\mu_F = 0.755$	$\mu_{CR} = 0.035$	
Diversity = 5.7865e+00	k_t = [1, 1, 1, 1]			
Generation 4360: Best = 3.413707e+03	FES = 436100	$\mu_F = 0.706$	$\mu_{CR} = 0.044$	
Diversity = 5.7869e+00	k_t = [4, 4, 2, 2]			
Generation 4370: Best = 3.413707e+03	FES = 437100	$\mu_F = 0.720$	$\mu_{CR} = 0.032$	
Diversity = 5.7772e+00	k_t = [1, 1, 1, 1]			
Generation 4380: Best = 3.413707e+03	FES = 438100	$\mu_F = 0.719$	$\mu_{CR} = 0.033$	
Diversity = 5.7654e+00	k_t = [4, 3, 4, 2]			
Generation 4390: Best = 3.413588e+03	FES = 439100	$\mu_F = 0.755$	$\mu_{CR} = 0.029$	
Diversity = 5.7618e+00	k_t = [1, 1, 1, 1]			
Generation 4400: Best = 3.413588e+03	FES = 440100	$\mu_F = 0.769$	$\mu_{CR} = 0.028$	
Diversity = 5.7605e+00	k_t = [2, 4, 4, 4]			
Generation 4410: Best = 3.413588e+03	FES = 441100	$\mu_F = 0.765$	$\mu_{CR} = 0.027$	
Diversity = 5.7559e+00	k_t = [1, 1, 1, 1]			
Generation 4420: Best = 3.413588e+03	FES = 442100	$\mu_F = 0.757$	$\mu_{CR} = 0.031$	
Diversity = 5.7697e+00	k_t = [3, 2, 2, 3]			
Generation 4430: Best = 3.413584e+03	FES = 443100	$\mu_F = 0.817$	$\mu_{CR} = 0.028$	
Diversity = 5.8216e+00	k_t = [1, 1, 1, 1]			
Generation 4440: Best = 3.413554e+03	FES = 444100	$\mu_F = 0.870$	$\mu_{CR} = 0.036$	
Diversity = 5.8386e+00	k_t = [2, 4, 3, 2]			
Generation 4450: Best = 3.413262e+03	FES = 445100	$\mu_F = 0.844$	$\mu_{CR} = 0.025$	
Diversity = 5.8377e+00	k_t = [1, 1, 1, 1]			
Generation 4460: Best = 3.413259e+03	FES = 446100	$\mu_F = 0.875$	$\mu_{CR} = 0.035$	
Diversity = 5.8634e+00	k_t = [2, 3, 2, 4]			
Generation 4470: Best = 3.413259e+03	FES = 447100	$\mu_F = 0.887$	$\mu_{CR} = 0.024$	
Diversity = 5.8788e+00	k_t = [1, 1, 1, 1]			
Generation 4480: Best = 3.413259e+03	FES = 448100	$\mu_F = 0.890$	$\mu_{CR} = 0.025$	
Diversity = 5.9270e+00	k_t = [4, 2, 2, 2]			
Generation 4490: Best = 3.413259e+03	FES = 449100	$\mu_F = 0.880$	$\mu_{CR} = 0.027$	
Diversity = 5.9290e+00	k_t = [1, 1, 1, 1]			
Generation 4500: Best = 3.413247e+03	FES = 450100	$\mu_F = 0.874$	$\mu_{CR} = 0.034$	
Diversity = 5.9415e+00	k_t = [3, 3, 3, 3]			
Generation 4510: Best = 3.413247e+03	FES = 451100	$\mu_F = 0.874$	$\mu_{CR} = 0.027$	
Diversity = 5.9568e+00	k_t = [1, 1, 1, 1]			
Generation 4520: Best = 3.413247e+03	FES = 452100	$\mu_F = 0.813$	$\mu_{CR} = 0.026$	
Diversity = 5.9441e+00	k_t = [3, 3, 4, 4]			
Generation 4530: Best = 3.413212e+03	FES = 453100	$\mu_F = 0.806$	$\mu_{CR} = 0.027$	
Diversity = 5.9715e+00	k_t = [1, 1, 1, 1]			
Generation 4540: Best = 3.413212e+03	FES = 454100	$\mu_F = 0.829$	$\mu_{CR} = 0.025$	

Diversity = 5.9881e+00	k_t = [2, 4, 4, 3]			
Generation 4550: Best = 3.413212e+03	FES = 455100	$\mu_F = 0.853$	$\mu_{CR} = 0.035$	
Diversity = 6.0007e+00	k_t = [1, 1, 1, 1]			
Generation 4560: Best = 3.413057e+03	FES = 456100	$\mu_F = 0.813$	$\mu_{CR} = 0.030$	
Diversity = 5.9756e+00	k_t = [2, 4, 2, 2]			
Generation 4570: Best = 3.413057e+03	FES = 457100	$\mu_F = 0.785$	$\mu_{CR} = 0.030$	
Diversity = 5.9685e+00	k_t = [1, 1, 1, 1]			
Generation 4580: Best = 3.413057e+03	FES = 458100	$\mu_F = 0.793$	$\mu_{CR} = 0.039$	
Diversity = 5.9285e+00	k_t = [4, 2, 2, 4]			
Generation 4590: Best = 3.413057e+03	FES = 459100	$\mu_F = 0.834$	$\mu_{CR} = 0.037$	
Diversity = 5.9261e+00	k_t = [1, 1, 1, 1]			
Generation 4600: Best = 3.413048e+03	FES = 460100	$\mu_F = 0.888$	$\mu_{CR} = 0.042$	
Diversity = 5.9278e+00	k_t = [2, 3, 3, 2]			
Generation 4610: Best = 3.413048e+03	FES = 461100	$\mu_F = 0.886$	$\mu_{CR} = 0.033$	
Diversity = 5.9409e+00	k_t = [1, 1, 1, 1]			
Generation 4620: Best = 3.413046e+03	FES = 462100	$\mu_F = 0.868$	$\mu_{CR} = 0.035$	
Diversity = 5.9357e+00	k_t = [3, 4, 3, 3]			
Generation 4630: Best = 3.413046e+03	FES = 463100	$\mu_F = 0.867$	$\mu_{CR} = 0.037$	
Diversity = 5.9828e+00	k_t = [1, 1, 1, 1]			
Generation 4640: Best = 3.413046e+03	FES = 464100	$\mu_F = 0.901$	$\mu_{CR} = 0.034$	
Diversity = 5.9940e+00	k_t = [4, 4, 4, 3]			
Generation 4650: Best = 3.412811e+03	FES = 465100	$\mu_F = 0.856$	$\mu_{CR} = 0.035$	
Diversity = 6.0031e+00	k_t = [1, 1, 1, 1]			
Generation 4660: Best = 3.412811e+03	FES = 466100	$\mu_F = 0.818$	$\mu_{CR} = 0.032$	
Diversity = 5.9981e+00	k_t = [3, 3, 2, 2]			
Generation 4670: Best = 3.412789e+03	FES = 467100	$\mu_F = 0.826$	$\mu_{CR} = 0.041$	
Diversity = 6.0081e+00	k_t = [1, 1, 1, 1]			
Generation 4680: Best = 3.412789e+03	FES = 468100	$\mu_F = 0.812$	$\mu_{CR} = 0.045$	
Diversity = 6.0090e+00	k_t = [3, 4, 3, 3]			
Generation 4690: Best = 3.412782e+03	FES = 469100	$\mu_F = 0.786$	$\mu_{CR} = 0.042$	
Diversity = 5.9784e+00	k_t = [1, 1, 1, 1]			
Generation 4700: Best = 3.412782e+03	FES = 470100	$\mu_F = 0.794$	$\mu_{CR} = 0.046$	
Diversity = 5.9968e+00	k_t = [2, 2, 2, 4]			
Generation 4710: Best = 3.412644e+03	FES = 471100	$\mu_F = 0.792$	$\mu_{CR} = 0.051$	
Diversity = 5.9699e+00	k_t = [1, 1, 1, 1]			
Generation 4720: Best = 3.412480e+03	FES = 472100	$\mu_F = 0.811$	$\mu_{CR} = 0.048$	
Diversity = 5.9713e+00	k_t = [4, 3, 2, 3]			
Generation 4730: Best = 3.412441e+03	FES = 473100	$\mu_F = 0.773$	$\mu_{CR} = 0.047$	
Diversity = 5.9570e+00	k_t = [1, 1, 1, 1]			
Generation 4740: Best = 3.412441e+03	FES = 474100	$\mu_F = 0.778$	$\mu_{CR} = 0.046$	
Diversity = 5.9370e+00	k_t = [3, 4, 3, 2]			
Generation 4750: Best = 3.412360e+03	FES = 475100	$\mu_F = 0.763$	$\mu_{CR} = 0.048$	
Diversity = 5.9135e+00	k_t = [1, 1, 1, 1]			
Generation 4760: Best = 3.412293e+03	FES = 476100	$\mu_F = 0.769$	$\mu_{CR} = 0.045$	
Diversity = 5.8558e+00	k_t = [4, 3, 4, 4]			
Generation 4770: Best = 3.412242e+03	FES = 477100	$\mu_F = 0.736$	$\mu_{CR} = 0.051$	
Diversity = 5.8166e+00	k_t = [1, 1, 1, 1]			
Generation 4780: Best = 3.412118e+03	FES = 478100	$\mu_F = 0.758$	$\mu_{CR} = 0.050$	
Diversity = 5.7965e+00	k_t = [2, 4, 3, 4]			
Generation 4790: Best = 3.411867e+03	FES = 479100	$\mu_F = 0.756$	$\mu_{CR} = 0.039$	
Diversity = 5.7892e+00	k_t = [1, 1, 1, 1]			
Generation 4800: Best = 3.411740e+03	FES = 480100	$\mu_F = 0.780$	$\mu_{CR} = 0.044$	
Diversity = 5.7926e+00	k_t = [2, 3, 4, 2]			
Generation 4810: Best = 3.411529e+03	FES = 481100	$\mu_F = 0.775$	$\mu_{CR} = 0.045$	

Diversity = 5.7645e+00	k_t = [1, 1, 1, 1]			
Generation 4820: Best = 3.411430e+03	FES = 482100	$\mu_F = 0.809$	$\mu_{CR} = 0.052$	
Diversity = 5.6490e+00	k_t = [2, 2, 4, 4]			
Generation 4830: Best = 3.411428e+03	FES = 483100	$\mu_F = 0.807$	$\mu_{CR} = 0.051$	
Diversity = 5.5775e+00	k_t = [1, 1, 1, 1]			
Generation 4840: Best = 3.411240e+03	FES = 484100	$\mu_F = 0.817$	$\mu_{CR} = 0.060$	
Diversity = 5.4165e+00	k_t = [3, 4, 4, 4]			
Generation 4850: Best = 3.411218e+03	FES = 485100	$\mu_F = 0.812$	$\mu_{CR} = 0.063$	
Diversity = 5.1932e+00	k_t = [1, 1, 1, 1]			
Generation 4860: Best = 3.411218e+03	FES = 486100	$\mu_F = 0.824$	$\mu_{CR} = 0.060$	
Diversity = 4.7345e+00	k_t = [4, 3, 3, 2]			
Generation 4870: Best = 3.410778e+03	FES = 487100	$\mu_F = 0.822$	$\mu_{CR} = 0.054$	
Diversity = 4.4860e+00	k_t = [1, 1, 1, 1]			
Generation 4880: Best = 3.410326e+03	FES = 488100	$\mu_F = 0.793$	$\mu_{CR} = 0.064$	
Diversity = 4.1280e+00	k_t = [3, 2, 3, 2]			
Generation 4890: Best = 3.410326e+03	FES = 489100	$\mu_F = 0.769$	$\mu_{CR} = 0.067$	
Diversity = 3.9369e+00	k_t = [1, 1, 1, 1]			
Generation 4900: Best = 3.410326e+03	FES = 490100	$\mu_F = 0.742$	$\mu_{CR} = 0.076$	
Diversity = 3.6262e+00	k_t = [2, 3, 2, 3]			
Generation 4910: Best = 3.410326e+03	FES = 491100	$\mu_F = 0.735$	$\mu_{CR} = 0.087$	
Diversity = 3.2994e+00	k_t = [1, 1, 1, 1]			
Generation 4920: Best = 3.410123e+03	FES = 492100	$\mu_F = 0.733$	$\mu_{CR} = 0.080$	
Diversity = 3.0367e+00	k_t = [3, 3, 4, 2]			
Generation 4930: Best = 3.409769e+03	FES = 493100	$\mu_F = 0.737$	$\mu_{CR} = 0.066$	
Diversity = 2.9139e+00	k_t = [1, 1, 1, 1]			
Generation 4940: Best = 3.409769e+03	FES = 494100	$\mu_F = 0.714$	$\mu_{CR} = 0.066$	
Diversity = 2.8315e+00	k_t = [2, 2, 3, 2]			
Generation 4950: Best = 3.409444e+03	FES = 495100	$\mu_F = 0.722$	$\mu_{CR} = 0.068$	
Diversity = 2.7500e+00	k_t = [1, 1, 1, 1]			
Generation 4960: Best = 3.409202e+03	FES = 496100	$\mu_F = 0.712$	$\mu_{CR} = 0.068$	
Diversity = 2.6062e+00	k_t = [3, 2, 3, 2]			
Generation 4970: Best = 3.409173e+03	FES = 497100	$\mu_F = 0.720$	$\mu_{CR} = 0.058$	
Diversity = 2.5843e+00	k_t = [1, 1, 1, 1]			
Generation 4980: Best = 3.408947e+03	FES = 498100	$\mu_F = 0.733$	$\mu_{CR} = 0.059$	
Diversity = 2.5067e+00	k_t = [3, 2, 4, 4]			
Generation 4990: Best = 3.408940e+03	FES = 499100	$\mu_F = 0.735$	$\mu_{CR} = 0.062$	
Diversity = 2.4356e+00	k_t = [1, 1, 1, 1]			
Generation 5000: Best = 3.408750e+03	FES = 500100	$\mu_F = 0.724$	$\mu_{CR} = 0.061$	
Diversity = 2.3535e+00	k_t = [2, 4, 4, 3]			
Generation 5010: Best = 3.408750e+03	FES = 501100	$\mu_F = 0.734$	$\mu_{CR} = 0.050$	
Diversity = 2.2155e+00	k_t = [1, 1, 1, 1]			
Generation 5020: Best = 3.408725e+03	FES = 502100	$\mu_F = 0.726$	$\mu_{CR} = 0.050$	
Diversity = 2.1432e+00	k_t = [2, 3, 4, 3]			
Generation 5030: Best = 3.408530e+03	FES = 503100	$\mu_F = 0.724$	$\mu_{CR} = 0.058$	
Diversity = 2.0877e+00	k_t = [1, 1, 1, 1]			
Generation 5040: Best = 3.408517e+03	FES = 504100	$\mu_F = 0.741$	$\mu_{CR} = 0.056$	
Diversity = 2.0570e+00	k_t = [4, 3, 2, 2]			
Generation 5050: Best = 3.408505e+03	FES = 505100	$\mu_F = 0.765$	$\mu_{CR} = 0.044$	
Diversity = 2.0460e+00	k_t = [1, 1, 1, 1]			
Generation 5060: Best = 3.408292e+03	FES = 506100	$\mu_F = 0.772$	$\mu_{CR} = 0.042$	
Diversity = 1.9945e+00	k_t = [4, 3, 3, 4]			
Generation 5070: Best = 3.408086e+03	FES = 507100	$\mu_F = 0.790$	$\mu_{CR} = 0.040$	
Diversity = 1.9393e+00	k_t = [1, 1, 1, 1]			
Generation 5080: Best = 3.408086e+03	FES = 508100	$\mu_F = 0.811$	$\mu_{CR} = 0.051$	

Diversity = 1.8910e+00	k_t = [4, 4, 2, 3]			
Generation 5090: Best = 3.408041e+03	FES = 509100	$\mu_F = 0.774$	$\mu_{CR} = 0.052$	
Diversity = 1.8572e+00	k_t = [1, 1, 1, 1]			
Generation 5100: Best = 3.408019e+03	FES = 510100	$\mu_F = 0.793$	$\mu_{CR} = 0.043$	
Diversity = 1.8488e+00	k_t = [3, 3, 2, 2]			
Generation 5110: Best = 3.408019e+03	FES = 511100	$\mu_F = 0.755$	$\mu_{CR} = 0.035$	
Diversity = 1.8208e+00	k_t = [1, 1, 1, 1]			
Generation 5120: Best = 3.407989e+03	FES = 512100	$\mu_F = 0.752$	$\mu_{CR} = 0.039$	
Diversity = 1.7907e+00	k_t = [4, 3, 4, 2]			
Generation 5130: Best = 3.407989e+03	FES = 513100	$\mu_F = 0.744$	$\mu_{CR} = 0.047$	
Diversity = 1.8237e+00	k_t = [1, 1, 1, 1]			
Generation 5140: Best = 3.407989e+03	FES = 514100	$\mu_F = 0.729$	$\mu_{CR} = 0.046$	
Diversity = 1.8008e+00	k_t = [2, 2, 4, 3]			
Generation 5150: Best = 3.407959e+03	FES = 515100	$\mu_F = 0.715$	$\mu_{CR} = 0.044$	
Diversity = 1.7966e+00	k_t = [1, 1, 1, 1]			
Generation 5160: Best = 3.407954e+03	FES = 516100	$\mu_F = 0.741$	$\mu_{CR} = 0.040$	
Diversity = 1.7869e+00	k_t = [4, 2, 4, 4]			
Generation 5170: Best = 3.407940e+03	FES = 517100	$\mu_F = 0.777$	$\mu_{CR} = 0.034$	
Diversity = 1.7737e+00	k_t = [1, 1, 1, 1]			
Generation 5180: Best = 3.407940e+03	FES = 518100	$\mu_F = 0.789$	$\mu_{CR} = 0.032$	
Diversity = 1.7454e+00	k_t = [2, 2, 2, 4]			
Generation 5190: Best = 3.407940e+03	FES = 519100	$\mu_F = 0.809$	$\mu_{CR} = 0.028$	
Diversity = 1.7435e+00	k_t = [1, 1, 1, 1]			
Generation 5200: Best = 3.407838e+03	FES = 520100	$\mu_F = 0.774$	$\mu_{CR} = 0.031$	
Diversity = 1.7381e+00	k_t = [4, 3, 2, 3]			
Generation 5210: Best = 3.407828e+03	FES = 521100	$\mu_F = 0.740$	$\mu_{CR} = 0.032$	
Diversity = 1.7680e+00	k_t = [1, 1, 1, 1]			
Generation 5220: Best = 3.407825e+03	FES = 522100	$\mu_F = 0.751$	$\mu_{CR} = 0.031$	
Diversity = 1.7786e+00	k_t = [2, 4, 2, 2]			
Generation 5230: Best = 3.407820e+03	FES = 523100	$\mu_F = 0.760$	$\mu_{CR} = 0.046$	
Diversity = 1.7937e+00	k_t = [1, 1, 1, 1]			
Generation 5240: Best = 3.407782e+03	FES = 524100	$\mu_F = 0.769$	$\mu_{CR} = 0.035$	
Diversity = 1.7945e+00	k_t = [2, 4, 2, 2]			
Generation 5250: Best = 3.407742e+03	FES = 525100	$\mu_F = 0.757$	$\mu_{CR} = 0.030$	
Diversity = 1.8114e+00	k_t = [1, 1, 1, 1]			
Generation 5260: Best = 3.407736e+03	FES = 526100	$\mu_F = 0.734$	$\mu_{CR} = 0.032$	
Diversity = 1.8024e+00	k_t = [2, 4, 2, 4]			
Generation 5270: Best = 3.407736e+03	FES = 527100	$\mu_F = 0.708$	$\mu_{CR} = 0.028$	
Diversity = 1.7931e+00	k_t = [1, 1, 1, 1]			
Generation 5280: Best = 3.407736e+03	FES = 528100	$\mu_F = 0.701$	$\mu_{CR} = 0.027$	
Diversity = 1.7928e+00	k_t = [2, 4, 3, 2]			
Generation 5290: Best = 3.407735e+03	FES = 529100	$\mu_F = 0.744$	$\mu_{CR} = 0.027$	
Diversity = 1.7964e+00	k_t = [1, 1, 1, 1]			
Generation 5300: Best = 3.407734e+03	FES = 530100	$\mu_F = 0.770$	$\mu_{CR} = 0.036$	
Diversity = 1.8079e+00	k_t = [2, 4, 4, 4]			
Generation 5310: Best = 3.407699e+03	FES = 531100	$\mu_F = 0.764$	$\mu_{CR} = 0.034$	
Diversity = 1.8170e+00	k_t = [1, 1, 1, 1]			
Generation 5320: Best = 3.407684e+03	FES = 532100	$\mu_F = 0.779$	$\mu_{CR} = 0.029$	
Diversity = 1.8171e+00	k_t = [4, 4, 2, 3]			
Generation 5330: Best = 3.407645e+03	FES = 533100	$\mu_F = 0.783$	$\mu_{CR} = 0.026$	
Diversity = 1.8133e+00	k_t = [1, 1, 1, 1]			
Generation 5340: Best = 3.407584e+03	FES = 534100	$\mu_F = 0.770$	$\mu_{CR} = 0.029$	
Diversity = 1.8082e+00	k_t = [2, 2, 2, 2]			
Generation 5350: Best = 3.407584e+03	FES = 535100	$\mu_F = 0.781$	$\mu_{CR} = 0.028$	

Diversity = 1.8155e+00	k_t = [1, 1, 1, 1]			
Generation 5360: Best = 3.407584e+03	FES = 536100	$\mu_F = 0.826$	$\mu_{CR} = 0.032$	
Diversity = 1.8069e+00	k_t = [4, 2, 3, 3]			
Generation 5370: Best = 3.407579e+03	FES = 537100	$\mu_F = 0.805$	$\mu_{CR} = 0.023$	
Diversity = 1.8124e+00	k_t = [1, 1, 1, 1]			
Generation 5380: Best = 3.407579e+03	FES = 538100	$\mu_F = 0.805$	$\mu_{CR} = 0.027$	
Diversity = 1.8125e+00	k_t = [3, 3, 2, 2]			
Generation 5390: Best = 3.407579e+03	FES = 539100	$\mu_F = 0.788$	$\mu_{CR} = 0.029$	
Diversity = 1.8092e+00	k_t = [1, 1, 1, 1]			
Generation 5400: Best = 3.407563e+03	FES = 540100	$\mu_F = 0.774$	$\mu_{CR} = 0.027$	
Diversity = 1.8158e+00	k_t = [3, 4, 4, 3]			
Generation 5410: Best = 3.407522e+03	FES = 541100	$\mu_F = 0.831$	$\mu_{CR} = 0.026$	
Diversity = 1.8148e+00	k_t = [1, 1, 1, 1]			
Generation 5420: Best = 3.407522e+03	FES = 542100	$\mu_F = 0.772$	$\mu_{CR} = 0.025$	
Diversity = 1.8136e+00	k_t = [2, 4, 3, 2]			
Generation 5430: Best = 3.407492e+03	FES = 543100	$\mu_F = 0.798$	$\mu_{CR} = 0.026$	
Diversity = 1.8139e+00	k_t = [1, 1, 1, 1]			
Generation 5440: Best = 3.407492e+03	FES = 544100	$\mu_F = 0.789$	$\mu_{CR} = 0.027$	
Diversity = 1.8037e+00	k_t = [4, 4, 2, 3]			
Generation 5450: Best = 3.407378e+03	FES = 545100	$\mu_F = 0.780$	$\mu_{CR} = 0.028$	
Diversity = 1.7961e+00	k_t = [1, 1, 1, 1]			
Generation 5460: Best = 3.407371e+03	FES = 546100	$\mu_F = 0.802$	$\mu_{CR} = 0.030$	
Diversity = 1.7759e+00	k_t = [4, 2, 3, 2]			
Generation 5470: Best = 3.407371e+03	FES = 547100	$\mu_F = 0.775$	$\mu_{CR} = 0.026$	
Diversity = 1.7753e+00	k_t = [1, 1, 1, 1]			
Generation 5480: Best = 3.407371e+03	FES = 548100	$\mu_F = 0.779$	$\mu_{CR} = 0.027$	
Diversity = 1.7744e+00	k_t = [4, 2, 3, 3]			
Generation 5490: Best = 3.407371e+03	FES = 549100	$\mu_F = 0.662$	$\mu_{CR} = 0.031$	
Diversity = 1.7805e+00	k_t = [1, 1, 1, 1]			
Generation 5500: Best = 3.407371e+03	FES = 550100	$\mu_F = 0.728$	$\mu_{CR} = 0.028$	
Diversity = 1.7485e+00	k_t = [2, 2, 4, 4]			
Generation 5510: Best = 3.407367e+03	FES = 551100	$\mu_F = 0.756$	$\mu_{CR} = 0.028$	
Diversity = 1.7289e+00	k_t = [1, 1, 1, 1]			
Generation 5520: Best = 3.407367e+03	FES = 552100	$\mu_F = 0.795$	$\mu_{CR} = 0.028$	
Diversity = 1.7344e+00	k_t = [4, 2, 3, 3]			
Generation 5530: Best = 3.407345e+03	FES = 553100	$\mu_F = 0.789$	$\mu_{CR} = 0.023$	
Diversity = 1.7317e+00	k_t = [1, 1, 1, 1]			
Generation 5540: Best = 3.407345e+03	FES = 554100	$\mu_F = 0.784$	$\mu_{CR} = 0.018$	
Diversity = 1.7289e+00	k_t = [2, 4, 4, 2]			
Generation 5550: Best = 3.407345e+03	FES = 555100	$\mu_F = 0.769$	$\mu_{CR} = 0.026$	
Diversity = 1.7368e+00	k_t = [1, 1, 1, 1]			
Generation 5560: Best = 3.407322e+03	FES = 556100	$\mu_F = 0.825$	$\mu_{CR} = 0.027$	
Diversity = 1.7356e+00	k_t = [3, 3, 4, 2]			
Generation 5570: Best = 3.407322e+03	FES = 557100	$\mu_F = 0.795$	$\mu_{CR} = 0.022$	
Diversity = 1.7436e+00	k_t = [1, 1, 1, 1]			
Generation 5580: Best = 3.407322e+03	FES = 558100	$\mu_F = 0.773$	$\mu_{CR} = 0.022$	
Diversity = 1.7296e+00	k_t = [3, 2, 3, 3]			
Generation 5590: Best = 3.407322e+03	FES = 559100	$\mu_F = 0.762$	$\mu_{CR} = 0.026$	
Diversity = 1.7567e+00	k_t = [1, 1, 1, 1]			
Generation 5600: Best = 3.407309e+03	FES = 560100	$\mu_F = 0.763$	$\mu_{CR} = 0.028$	
Diversity = 1.7578e+00	k_t = [3, 3, 3, 3]			
Generation 5610: Best = 3.407309e+03	FES = 561100	$\mu_F = 0.713$	$\mu_{CR} = 0.031$	
Diversity = 1.7641e+00	k_t = [1, 1, 1, 1]			
Generation 5620: Best = 3.407306e+03	FES = 562100	$\mu_F = 0.760$	$\mu_{CR} = 0.025$	

Diversity = 1.7656e+00	k_t = [3, 3, 4, 4]			
Generation 5630: Best = 3.407306e+03	FES = 563100	$\mu_F = 0.777$	$\mu_{CR} = 0.026$	
Diversity = 1.7589e+00	k_t = [1, 1, 1, 1]			
Generation 5640: Best = 3.407286e+03	FES = 564100	$\mu_F = 0.792$	$\mu_{CR} = 0.032$	
Diversity = 1.7561e+00	k_t = [3, 4, 3, 4]			
Generation 5650: Best = 3.407286e+03	FES = 565100	$\mu_F = 0.812$	$\mu_{CR} = 0.036$	
Diversity = 1.7594e+00	k_t = [1, 1, 1, 1]			
Generation 5660: Best = 3.407286e+03	FES = 566100	$\mu_F = 0.842$	$\mu_{CR} = 0.026$	
Diversity = 1.7700e+00	k_t = [2, 4, 3, 4]			
Generation 5670: Best = 3.407286e+03	FES = 567100	$\mu_F = 0.825$	$\mu_{CR} = 0.027$	
Diversity = 1.7668e+00	k_t = [1, 1, 1, 1]			
Generation 5680: Best = 3.407286e+03	FES = 568100	$\mu_F = 0.826$	$\mu_{CR} = 0.019$	
Diversity = 1.7663e+00	k_t = [2, 3, 4, 2]			
Generation 5690: Best = 3.407249e+03	FES = 569100	$\mu_F = 0.748$	$\mu_{CR} = 0.024$	
Diversity = 1.7647e+00	k_t = [1, 1, 1, 1]			
Generation 5700: Best = 3.407249e+03	FES = 570100	$\mu_F = 0.769$	$\mu_{CR} = 0.029$	
Diversity = 1.7647e+00	k_t = [4, 3, 3, 3]			
Generation 5710: Best = 3.407249e+03	FES = 571100	$\mu_F = 0.803$	$\mu_{CR} = 0.023$	
Diversity = 1.7647e+00	k_t = [1, 1, 1, 1]			
Generation 5720: Best = 3.407249e+03	FES = 572100	$\mu_F = 0.821$	$\mu_{CR} = 0.023$	
Diversity = 1.7306e+00	k_t = [4, 4, 3, 4]			
Generation 5730: Best = 3.407249e+03	FES = 573100	$\mu_F = 0.797$	$\mu_{CR} = 0.021$	
Diversity = 1.7306e+00	k_t = [1, 1, 1, 1]			
Generation 5740: Best = 3.407249e+03	FES = 574100	$\mu_F = 0.786$	$\mu_{CR} = 0.024$	
Diversity = 1.7333e+00	k_t = [2, 4, 4, 3]			
Generation 5750: Best = 3.407249e+03	FES = 575100	$\mu_F = 0.752$	$\mu_{CR} = 0.021$	
Diversity = 1.7333e+00	k_t = [1, 1, 1, 1]			
Generation 5760: Best = 3.407249e+03	FES = 576100	$\mu_F = 0.791$	$\mu_{CR} = 0.019$	
Diversity = 1.7436e+00	k_t = [2, 2, 4, 4]			
Generation 5770: Best = 3.407249e+03	FES = 577100	$\mu_F = 0.753$	$\mu_{CR} = 0.027$	
Diversity = 1.7366e+00	k_t = [1, 1, 1, 1]			
Generation 5780: Best = 3.407249e+03	FES = 578100	$\mu_F = 0.793$	$\mu_{CR} = 0.025$	
Diversity = 1.7415e+00	k_t = [4, 4, 4, 4]			
Generation 5790: Best = 3.407233e+03	FES = 579100	$\mu_F = 0.756$	$\mu_{CR} = 0.034$	
Diversity = 1.7454e+00	k_t = [1, 1, 1, 1]			
Generation 5800: Best = 3.407233e+03	FES = 580100	$\mu_F = 0.759$	$\mu_{CR} = 0.029$	
Diversity = 1.7554e+00	k_t = [4, 3, 3, 4]			
Generation 5810: Best = 3.407232e+03	FES = 581100	$\mu_F = 0.766$	$\mu_{CR} = 0.025$	
Diversity = 1.7546e+00	k_t = [1, 1, 1, 1]			
Generation 5820: Best = 3.407232e+03	FES = 582100	$\mu_F = 0.779$	$\mu_{CR} = 0.022$	
Diversity = 1.7570e+00	k_t = [3, 4, 2, 3]			
Generation 5830: Best = 3.407216e+03	FES = 583100	$\mu_F = 0.790$	$\mu_{CR} = 0.019$	
Diversity = 1.7569e+00	k_t = [1, 1, 1, 1]			
Generation 5840: Best = 3.407216e+03	FES = 584100	$\mu_F = 0.818$	$\mu_{CR} = 0.022$	
Diversity = 1.7376e+00	k_t = [3, 2, 4, 2]			
Generation 5850: Best = 3.407216e+03	FES = 585100	$\mu_F = 0.824$	$\mu_{CR} = 0.026$	
Diversity = 1.7457e+00	k_t = [1, 1, 1, 1]			
Generation 5860: Best = 3.407216e+03	FES = 586100	$\mu_F = 0.841$	$\mu_{CR} = 0.018$	
Diversity = 1.7457e+00	k_t = [4, 3, 3, 3]			
Generation 5870: Best = 3.407215e+03	FES = 587100	$\mu_F = 0.822$	$\mu_{CR} = 0.018$	
Diversity = 1.7457e+00	k_t = [1, 1, 1, 1]			
Generation 5880: Best = 3.407141e+03	FES = 588100	$\mu_F = 0.778$	$\mu_{CR} = 0.025$	
Diversity = 1.7456e+00	k_t = [3, 4, 4, 3]			
Generation 5890: Best = 3.407141e+03	FES = 589100	$\mu_F = 0.712$	$\mu_{CR} = 0.023$	

Diversity = 1.7526e+00	k_t = [1, 1, 1, 1]			
Generation 5900: Best = 3.407141e+03	FES = 590100	$\mu_F = 0.743$	$\mu_{CR} = 0.023$	
Diversity = 1.7514e+00	k_t = [3, 4, 2, 2]			
Generation 5910: Best = 3.407134e+03	FES = 591100	$\mu_F = 0.750$	$\mu_{CR} = 0.025$	
Diversity = 1.7556e+00	k_t = [1, 1, 1, 1]			
Generation 5920: Best = 3.407133e+03	FES = 592100	$\mu_F = 0.669$	$\mu_{CR} = 0.021$	
Diversity = 1.7579e+00	k_t = [2, 3, 4, 2]			
Generation 5930: Best = 3.407131e+03	FES = 593100	$\mu_F = 0.659$	$\mu_{CR} = 0.027$	
Diversity = 1.7617e+00	k_t = [1, 1, 1, 1]			
Generation 5940: Best = 3.407131e+03	FES = 594100	$\mu_F = 0.667$	$\mu_{CR} = 0.030$	
Diversity = 1.7617e+00	k_t = [3, 2, 4, 3]			
Generation 5950: Best = 3.407131e+03	FES = 595100	$\mu_F = 0.722$	$\mu_{CR} = 0.027$	
Diversity = 1.7635e+00	k_t = [1, 1, 1, 1]			
Generation 5960: Best = 3.407123e+03	FES = 596100	$\mu_F = 0.724$	$\mu_{CR} = 0.028$	
Diversity = 1.7867e+00	k_t = [4, 4, 3, 4]			
Generation 5970: Best = 3.407122e+03	FES = 597100	$\mu_F = 0.781$	$\mu_{CR} = 0.025$	
Diversity = 1.7887e+00	k_t = [1, 1, 1, 1]			
Generation 5980: Best = 3.407101e+03	FES = 598100	$\mu_F = 0.719$	$\mu_{CR} = 0.022$	
Diversity = 1.7887e+00	k_t = [3, 2, 4, 2]			
Generation 5990: Best = 3.407101e+03	FES = 599100	$\mu_F = 0.698$	$\mu_{CR} = 0.022$	
Diversity = 1.7887e+00	k_t = [1, 1, 1, 1]			
Generation 6000: Best = 3.407101e+03	FES = 600100	$\mu_F = 0.740$	$\mu_{CR} = 0.021$	
Diversity = 1.7925e+00	k_t = [3, 4, 3, 2]			
Generation 6010: Best = 3.407101e+03	FES = 601100	$\mu_F = 0.719$	$\mu_{CR} = 0.030$	
Diversity = 1.7924e+00	k_t = [1, 1, 1, 1]			
Generation 6020: Best = 3.407071e+03	FES = 602100	$\mu_F = 0.737$	$\mu_{CR} = 0.030$	
Diversity = 1.7936e+00	k_t = [2, 4, 2, 2]			
Generation 6030: Best = 3.407071e+03	FES = 603100	$\mu_F = 0.762$	$\mu_{CR} = 0.022$	
Diversity = 1.7936e+00	k_t = [1, 1, 1, 1]			
Generation 6040: Best = 3.407064e+03	FES = 604100	$\mu_F = 0.748$	$\mu_{CR} = 0.021$	
Diversity = 1.7891e+00	k_t = [2, 3, 3, 2]			
Generation 6050: Best = 3.407061e+03	FES = 605100	$\mu_F = 0.750$	$\mu_{CR} = 0.022$	
Diversity = 1.7881e+00	k_t = [1, 1, 1, 1]			
Generation 6060: Best = 3.407058e+03	FES = 606100	$\mu_F = 0.752$	$\mu_{CR} = 0.023$	
Diversity = 1.7880e+00	k_t = [4, 4, 2, 2]			
Generation 6070: Best = 3.407057e+03	FES = 607100	$\mu_F = 0.755$	$\mu_{CR} = 0.022$	
Diversity = 1.7881e+00	k_t = [1, 1, 1, 1]			
Generation 6080: Best = 3.407057e+03	FES = 608100	$\mu_F = 0.755$	$\mu_{CR} = 0.028$	
Diversity = 1.7912e+00	k_t = [2, 2, 4, 2]			
Generation 6090: Best = 3.407044e+03	FES = 609100	$\mu_F = 0.773$	$\mu_{CR} = 0.027$	
Diversity = 1.7912e+00	k_t = [1, 1, 1, 1]			
Generation 6100: Best = 3.407044e+03	FES = 610100	$\mu_F = 0.811$	$\mu_{CR} = 0.024$	
Diversity = 1.7945e+00	k_t = [4, 4, 2, 2]			
Generation 6110: Best = 3.407044e+03	FES = 611100	$\mu_F = 0.799$	$\mu_{CR} = 0.033$	
Diversity = 1.7998e+00	k_t = [1, 1, 1, 1]			
Generation 6120: Best = 3.407044e+03	FES = 612100	$\mu_F = 0.789$	$\mu_{CR} = 0.028$	
Diversity = 1.7997e+00	k_t = [3, 3, 2, 2]			
Generation 6130: Best = 3.407044e+03	FES = 613100	$\mu_F = 0.803$	$\mu_{CR} = 0.029$	
Diversity = 1.8011e+00	k_t = [1, 1, 1, 1]			
Generation 6140: Best = 3.407040e+03	FES = 614100	$\mu_F = 0.813$	$\mu_{CR} = 0.027$	
Diversity = 1.8032e+00	k_t = [3, 4, 2, 2]			
Generation 6150: Best = 3.407040e+03	FES = 615100	$\mu_F = 0.800$	$\mu_{CR} = 0.027$	
Diversity = 1.8071e+00	k_t = [1, 1, 1, 1]			
Generation 6160: Best = 3.407040e+03	FES = 616100	$\mu_F = 0.772$	$\mu_{CR} = 0.024$	

Diversity = 1.8071e+00	k_t = [4, 2, 4, 2]			
Generation 6170: Best = 3.407040e+03	FES = 617100	$\mu_F = 0.724$	$\mu_{CR} = 0.020$	
Diversity = 1.8056e+00	k_t = [1, 1, 1, 1]			
Generation 6180: Best = 3.407033e+03	FES = 618100	$\mu_F = 0.734$	$\mu_{CR} = 0.026$	
Diversity = 1.8021e+00	k_t = [3, 2, 2, 2]			
Generation 6190: Best = 3.407033e+03	FES = 619100	$\mu_F = 0.760$	$\mu_{CR} = 0.025$	
Diversity = 1.8034e+00	k_t = [1, 1, 1, 1]			
Generation 6200: Best = 3.407033e+03	FES = 620100	$\mu_F = 0.685$	$\mu_{CR} = 0.026$	
Diversity = 1.8034e+00	k_t = [2, 2, 2, 3]			
Generation 6210: Best = 3.407032e+03	FES = 621100	$\mu_F = 0.686$	$\mu_{CR} = 0.024$	
Diversity = 1.7925e+00	k_t = [1, 1, 1, 1]			
Generation 6220: Best = 3.407032e+03	FES = 622100	$\mu_F = 0.726$	$\mu_{CR} = 0.026$	
Diversity = 1.7925e+00	k_t = [4, 3, 4, 3]			
Generation 6230: Best = 3.407007e+03	FES = 623100	$\mu_F = 0.742$	$\mu_{CR} = 0.024$	
Diversity = 1.7913e+00	k_t = [1, 1, 1, 1]			
Generation 6240: Best = 3.407007e+03	FES = 624100	$\mu_F = 0.750$	$\mu_{CR} = 0.026$	
Diversity = 1.7936e+00	k_t = [2, 4, 3, 3]			
Generation 6250: Best = 3.407007e+03	FES = 625100	$\mu_F = 0.706$	$\mu_{CR} = 0.022$	
Diversity = 1.7928e+00	k_t = [1, 1, 1, 1]			
Generation 6260: Best = 3.407007e+03	FES = 626100	$\mu_F = 0.643$	$\mu_{CR} = 0.020$	
Diversity = 1.7952e+00	k_t = [2, 3, 2, 2]			
Generation 6270: Best = 3.407007e+03	FES = 627100	$\mu_F = 0.606$	$\mu_{CR} = 0.021$	
Diversity = 1.7935e+00	k_t = [1, 1, 1, 1]			
Generation 6280: Best = 3.407005e+03	FES = 628100	$\mu_F = 0.680$	$\mu_{CR} = 0.018$	
Diversity = 1.7935e+00	k_t = [3, 2, 2, 2]			
Generation 6290: Best = 3.407005e+03	FES = 629100	$\mu_F = 0.734$	$\mu_{CR} = 0.021$	
Diversity = 1.7935e+00	k_t = [1, 1, 1, 1]			
Generation 6300: Best = 3.407003e+03	FES = 630100	$\mu_F = 0.664$	$\mu_{CR} = 0.020$	
Diversity = 1.7934e+00	k_t = [4, 4, 3, 3]			
Generation 6310: Best = 3.407003e+03	FES = 631100	$\mu_F = 0.646$	$\mu_{CR} = 0.018$	
Diversity = 1.7941e+00	k_t = [1, 1, 1, 1]			
Generation 6320: Best = 3.407002e+03	FES = 632100	$\mu_F = 0.625$	$\mu_{CR} = 0.021$	
Diversity = 1.7827e+00	k_t = [2, 3, 4, 4]			
Generation 6330: Best = 3.407001e+03	FES = 633100	$\mu_F = 0.551$	$\mu_{CR} = 0.021$	
Diversity = 1.7805e+00	k_t = [1, 1, 1, 1]			
Generation 6340: Best = 3.407001e+03	FES = 634100	$\mu_F = 0.620$	$\mu_{CR} = 0.023$	
Diversity = 1.7804e+00	k_t = [4, 2, 2, 3]			
Generation 6350: Best = 3.406955e+03	FES = 635100	$\mu_F = 0.620$	$\mu_{CR} = 0.028$	
Diversity = 1.7839e+00	k_t = [1, 1, 1, 1]			
Generation 6360: Best = 3.406955e+03	FES = 636100	$\mu_F = 0.624$	$\mu_{CR} = 0.027$	
Diversity = 1.7839e+00	k_t = [3, 2, 3, 3]			
Generation 6370: Best = 3.406955e+03	FES = 637100	$\mu_F = 0.640$	$\mu_{CR} = 0.033$	
Diversity = 1.7839e+00	k_t = [1, 1, 1, 1]			
Generation 6380: Best = 3.406954e+03	FES = 638100	$\mu_F = 0.669$	$\mu_{CR} = 0.030$	
Diversity = 1.7860e+00	k_t = [3, 4, 4, 2]			
Generation 6390: Best = 3.406950e+03	FES = 639100	$\mu_F = 0.648$	$\mu_{CR} = 0.026$	
Diversity = 1.7861e+00	k_t = [1, 1, 1, 1]			
Generation 6400: Best = 3.406945e+03	FES = 640100	$\mu_F = 0.676$	$\mu_{CR} = 0.026$	
Diversity = 1.7861e+00	k_t = [2, 3, 3, 2]			
Generation 6410: Best = 3.406945e+03	FES = 641100	$\mu_F = 0.714$	$\mu_{CR} = 0.024$	
Diversity = 1.7876e+00	k_t = [1, 1, 1, 1]			
Generation 6420: Best = 3.406945e+03	FES = 642100	$\mu_F = 0.751$	$\mu_{CR} = 0.025$	
Diversity = 1.7863e+00	k_t = [3, 2, 4, 3]			
Generation 6430: Best = 3.406945e+03	FES = 643100	$\mu_F = 0.770$	$\mu_{CR} = 0.027$	

Diversity = 1.7803e+00	k_t = [1, 1, 1, 1]				
Generation 6440: Best = 3.406945e+03	FES = 644100	μ_F = 0.691	μ_{CR} = 0.025		
Diversity = 1.7803e+00	k_t = [3, 4, 3, 2]				
Generation 6450: Best = 3.406945e+03	FES = 645100	μ_F = 0.699	μ_{CR} = 0.024		
Diversity = 1.7803e+00	k_t = [1, 1, 1, 1]				
Generation 6460: Best = 3.406945e+03	FES = 646100	μ_F = 0.626	μ_{CR} = 0.024		
Diversity = 1.7721e+00	k_t = [4, 4, 2, 4]				
Generation 6470: Best = 3.406945e+03	FES = 647100	μ_F = 0.598	μ_{CR} = 0.029		
Diversity = 1.7721e+00	k_t = [1, 1, 1, 1]				
Generation 6480: Best = 3.406945e+03	FES = 648100	μ_F = 0.582	μ_{CR} = 0.028		
Diversity = 1.7730e+00	k_t = [2, 2, 2, 3]				
Generation 6490: Best = 3.406945e+03	FES = 649100	μ_F = 0.572	μ_{CR} = 0.033		
Diversity = 1.7661e+00	k_t = [1, 1, 1, 1]				
Generation 6500: Best = 3.406945e+03	FES = 650100	μ_F = 0.651	μ_{CR} = 0.026		
Diversity = 1.7661e+00	k_t = [2, 2, 2, 3]				
Generation 6510: Best = 3.406931e+03	FES = 651100	μ_F = 0.659	μ_{CR} = 0.033		
Diversity = 1.7661e+00	k_t = [1, 1, 1, 1]				
Generation 6520: Best = 3.406931e+03	FES = 652100	μ_F = 0.672	μ_{CR} = 0.023		
Diversity = 1.7661e+00	k_t = [3, 2, 4, 3]				
Generation 6530: Best = 3.406931e+03	FES = 653100	μ_F = 0.742	μ_{CR} = 0.022		
Diversity = 1.7661e+00	k_t = [1, 1, 1, 1]				
Generation 6540: Best = 3.406929e+03	FES = 654100	μ_F = 0.733	μ_{CR} = 0.023		
Diversity = 1.7615e+00	k_t = [4, 3, 4, 2]				
Generation 6550: Best = 3.406917e+03	FES = 655100	μ_F = 0.604	μ_{CR} = 0.026		
Diversity = 1.7613e+00	k_t = [1, 1, 1, 1]				
Generation 6560: Best = 3.406915e+03	FES = 656100	μ_F = 0.620	μ_{CR} = 0.025		
Diversity = 1.7591e+00	k_t = [2, 3, 4, 2]				
Generation 6570: Best = 3.406915e+03	FES = 657100	μ_F = 0.588	μ_{CR} = 0.023		
Diversity = 1.7585e+00	k_t = [1, 1, 1, 1]				
Generation 6580: Best = 3.406915e+03	FES = 658100	μ_F = 0.523	μ_{CR} = 0.029		
Diversity = 1.7585e+00	k_t = [4, 2, 3, 4]				
Generation 6590: Best = 3.406915e+03	FES = 659100	μ_F = 0.482	μ_{CR} = 0.026		
Diversity = 1.7577e+00	k_t = [1, 1, 1, 1]				
Generation 6600: Best = 3.406915e+03	FES = 660100	μ_F = 0.485	μ_{CR} = 0.026		
Diversity = 1.7560e+00	k_t = [4, 4, 4, 3]				
Generation 6610: Best = 3.406911e+03	FES = 661100	μ_F = 0.525	μ_{CR} = 0.024		
Diversity = 1.7560e+00	k_t = [1, 1, 1, 1]				
Generation 6620: Best = 3.406879e+03	FES = 662100	μ_F = 0.476	μ_{CR} = 0.022		
Diversity = 1.7557e+00	k_t = [3, 2, 4, 3]				
Generation 6630: Best = 3.406879e+03	FES = 663100	μ_F = 0.501	μ_{CR} = 0.025		
Diversity = 1.7556e+00	k_t = [1, 1, 1, 1]				
Generation 6640: Best = 3.406879e+03	FES = 664100	μ_F = 0.509	μ_{CR} = 0.022		
Diversity = 1.7553e+00	k_t = [3, 2, 2, 3]				
Generation 6650: Best = 3.406879e+03	FES = 665100	μ_F = 0.434	μ_{CR} = 0.023		
Diversity = 1.7578e+00	k_t = [1, 1, 1, 1]				
Generation 6660: Best = 3.406879e+03	FES = 666100	μ_F = 0.454	μ_{CR} = 0.022		
Diversity = 1.7578e+00	k_t = [3, 2, 2, 2]				
Generation 6670: Best = 3.406874e+03	FES = 667100	μ_F = 0.494	μ_{CR} = 0.023		
Diversity = 1.7577e+00	k_t = [1, 1, 1, 1]				
Generation 6680: Best = 3.406859e+03	FES = 668100	μ_F = 0.520	μ_{CR} = 0.022		
Diversity = 1.7576e+00	k_t = [2, 2, 3, 2]				
Generation 6690: Best = 3.406859e+03	FES = 669100	μ_F = 0.528	μ_{CR} = 0.024		
Diversity = 1.7575e+00	k_t = [1, 1, 1, 1]				
Generation 6700: Best = 3.406859e+03	FES = 670100	μ_F = 0.610	μ_{CR} = 0.025		

Diversity = 1.7708e+00	k_t = [4, 2, 4, 4]			
Generation 6710: Best = 3.406859e+03	FES = 671100	$\mu_F = 0.678$	$\mu_{CR} = 0.021$	
Diversity = 1.7696e+00	k_t = [1, 1, 1, 1]			
Generation 6720: Best = 3.406859e+03	FES = 672100	$\mu_F = 0.666$	$\mu_{CR} = 0.020$	
Diversity = 1.7696e+00	k_t = [2, 2, 4, 2]			
Generation 6730: Best = 3.406859e+03	FES = 673100	$\mu_F = 0.683$	$\mu_{CR} = 0.022$	
Diversity = 1.7744e+00	k_t = [1, 1, 1, 1]			
Generation 6740: Best = 3.406859e+03	FES = 674100	$\mu_F = 0.708$	$\mu_{CR} = 0.026$	
Diversity = 1.7745e+00	k_t = [4, 4, 4, 3]			
Generation 6750: Best = 3.406859e+03	FES = 675100	$\mu_F = 0.763$	$\mu_{CR} = 0.027$	
Diversity = 1.7745e+00	k_t = [1, 1, 1, 1]			
Generation 6760: Best = 3.406851e+03	FES = 676100	$\mu_F = 0.767$	$\mu_{CR} = 0.020$	
Diversity = 1.7766e+00	k_t = [2, 3, 4, 2]			
Generation 6770: Best = 3.406850e+03	FES = 677100	$\mu_F = 0.797$	$\mu_{CR} = 0.018$	
Diversity = 1.7766e+00	k_t = [1, 1, 1, 1]			
Generation 6780: Best = 3.406850e+03	FES = 678100	$\mu_F = 0.792$	$\mu_{CR} = 0.020$	
Diversity = 1.7766e+00	k_t = [4, 2, 2, 2]			
Generation 6790: Best = 3.406838e+03	FES = 679100	$\mu_F = 0.855$	$\mu_{CR} = 0.020$	
Diversity = 1.7766e+00	k_t = [1, 1, 1, 1]			
Generation 6800: Best = 3.406831e+03	FES = 680100	$\mu_F = 0.858$	$\mu_{CR} = 0.015$	
Diversity = 1.7766e+00	k_t = [2, 3, 3, 3]			
Generation 6810: Best = 3.406831e+03	FES = 681100	$\mu_F = 0.845$	$\mu_{CR} = 0.019$	
Diversity = 1.7766e+00	k_t = [1, 1, 1, 1]			
Generation 6820: Best = 3.406831e+03	FES = 682100	$\mu_F = 0.846$	$\mu_{CR} = 0.023$	
Diversity = 1.7766e+00	k_t = [4, 2, 3, 2]			
Generation 6830: Best = 3.406831e+03	FES = 683100	$\mu_F = 0.835$	$\mu_{CR} = 0.018$	
Diversity = 1.7766e+00	k_t = [1, 1, 1, 1]			
Generation 6840: Best = 3.406831e+03	FES = 684100	$\mu_F = 0.801$	$\mu_{CR} = 0.018$	
Diversity = 1.7766e+00	k_t = [2, 2, 2, 2]			
Generation 6850: Best = 3.406831e+03	FES = 685100	$\mu_F = 0.767$	$\mu_{CR} = 0.020$	
Diversity = 1.7770e+00	k_t = [1, 1, 1, 1]			
Generation 6860: Best = 3.406831e+03	FES = 686100	$\mu_F = 0.757$	$\mu_{CR} = 0.018$	
Diversity = 1.7770e+00	k_t = [4, 3, 4, 3]			
Generation 6870: Best = 3.406831e+03	FES = 687100	$\mu_F = 0.706$	$\mu_{CR} = 0.022$	
Diversity = 1.7925e+00	k_t = [1, 1, 1, 1]			
Generation 6880: Best = 3.406828e+03	FES = 688100	$\mu_F = 0.768$	$\mu_{CR} = 0.027$	
Diversity = 1.8167e+00	k_t = [3, 4, 3, 4]			
Generation 6890: Best = 3.406824e+03	FES = 689100	$\mu_F = 0.758$	$\mu_{CR} = 0.022$	
Diversity = 1.8167e+00	k_t = [1, 1, 1, 1]			
Generation 6900: Best = 3.406814e+03	FES = 690100	$\mu_F = 0.794$	$\mu_{CR} = 0.020$	
Diversity = 1.8167e+00	k_t = [3, 3, 3, 2]			
Generation 6910: Best = 3.406814e+03	FES = 691100	$\mu_F = 0.764$	$\mu_{CR} = 0.016$	
Diversity = 1.8167e+00	k_t = [1, 1, 1, 1]			
Generation 6920: Best = 3.406814e+03	FES = 692100	$\mu_F = 0.761$	$\mu_{CR} = 0.028$	
Diversity = 1.8100e+00	k_t = [4, 4, 4, 2]			
Generation 6930: Best = 3.406811e+03	FES = 693100	$\mu_F = 0.718$	$\mu_{CR} = 0.023$	
Diversity = 1.8100e+00	k_t = [1, 1, 1, 1]			
Generation 6940: Best = 3.406811e+03	FES = 694100	$\mu_F = 0.639$	$\mu_{CR} = 0.022$	
Diversity = 1.8100e+00	k_t = [4, 2, 2, 3]			
Generation 6950: Best = 3.406808e+03	FES = 695100	$\mu_F = 0.699$	$\mu_{CR} = 0.023$	
Diversity = 1.8099e+00	k_t = [1, 1, 1, 1]			
Generation 6960: Best = 3.406808e+03	FES = 696100	$\mu_F = 0.738$	$\mu_{CR} = 0.020$	
Diversity = 1.8087e+00	k_t = [3, 3, 4, 4]			
Generation 6970: Best = 3.406800e+03	FES = 697100	$\mu_F = 0.704$	$\mu_{CR} = 0.026$	

Diversity = 1.8086e+00	k_t = [1, 1, 1, 1]			
Generation 6980: Best = 3.406800e+03	FES = 698100	$\mu_F = 0.741$	$\mu_{CR} = 0.029$	
Diversity = 1.8075e+00	k_t = [4, 4, 2, 4]			
Generation 6990: Best = 3.406800e+03	FES = 699100	$\mu_F = 0.700$	$\mu_{CR} = 0.023$	
Diversity = 1.8040e+00	k_t = [1, 1, 1, 1]			
Generation 7000: Best = 3.406779e+03	FES = 700100	$\mu_F = 0.657$	$\mu_{CR} = 0.026$	
Diversity = 1.8039e+00	k_t = [3, 2, 4, 2]			
Generation 7010: Best = 3.406779e+03	FES = 701100	$\mu_F = 0.582$	$\mu_{CR} = 0.025$	
Diversity = 1.8039e+00	k_t = [1, 1, 1, 1]			
Generation 7020: Best = 3.406779e+03	FES = 702100	$\mu_F = 0.612$	$\mu_{CR} = 0.024$	
Diversity = 1.8039e+00	k_t = [2, 4, 4, 3]			
Generation 7030: Best = 3.406779e+03	FES = 703100	$\mu_F = 0.627$	$\mu_{CR} = 0.024$	
Diversity = 1.8097e+00	k_t = [1, 1, 1, 1]			
Generation 7040: Best = 3.406779e+03	FES = 704100	$\mu_F = 0.672$	$\mu_{CR} = 0.025$	
Diversity = 1.8135e+00	k_t = [3, 4, 4, 4]			
Generation 7050: Best = 3.406773e+03	FES = 705100	$\mu_F = 0.699$	$\mu_{CR} = 0.022$	
Diversity = 1.8181e+00	k_t = [1, 1, 1, 1]			
Generation 7060: Best = 3.406773e+03	FES = 706100	$\mu_F = 0.672$	$\mu_{CR} = 0.022$	
Diversity = 1.8181e+00	k_t = [2, 4, 3, 3]			
Generation 7070: Best = 3.406773e+03	FES = 707100	$\mu_F = 0.678$	$\mu_{CR} = 0.023$	
Diversity = 1.8183e+00	k_t = [1, 1, 1, 1]			
Generation 7080: Best = 3.406771e+03	FES = 708100	$\mu_F = 0.716$	$\mu_{CR} = 0.034$	
Diversity = 1.8232e+00	k_t = [4, 2, 3, 3]			
Generation 7090: Best = 3.406764e+03	FES = 709100	$\mu_F = 0.745$	$\mu_{CR} = 0.024$	
Diversity = 1.8251e+00	k_t = [1, 1, 1, 1]			
Generation 7100: Best = 3.406764e+03	FES = 710100	$\mu_F = 0.729$	$\mu_{CR} = 0.028$	
Diversity = 1.8267e+00	k_t = [2, 4, 2, 4]			
Generation 7110: Best = 3.406764e+03	FES = 711100	$\mu_F = 0.775$	$\mu_{CR} = 0.027$	
Diversity = 1.8267e+00	k_t = [1, 1, 1, 1]			
Generation 7120: Best = 3.406764e+03	FES = 712100	$\mu_F = 0.704$	$\mu_{CR} = 0.022$	
Diversity = 1.8359e+00	k_t = [3, 4, 4, 2]			
Generation 7130: Best = 3.406762e+03	FES = 713100	$\mu_F = 0.746$	$\mu_{CR} = 0.025$	
Diversity = 1.8357e+00	k_t = [1, 1, 1, 1]			
Generation 7140: Best = 3.406762e+03	FES = 714100	$\mu_F = 0.745$	$\mu_{CR} = 0.025$	
Diversity = 1.8128e+00	k_t = [3, 2, 4, 4]			
Generation 7150: Best = 3.406762e+03	FES = 715100	$\mu_F = 0.722$	$\mu_{CR} = 0.019$	
Diversity = 1.8303e+00	k_t = [1, 1, 1, 1]			
Generation 7160: Best = 3.406762e+03	FES = 716100	$\mu_F = 0.718$	$\mu_{CR} = 0.014$	
Diversity = 1.8303e+00	k_t = [2, 3, 4, 3]			
Generation 7170: Best = 3.406762e+03	FES = 717100	$\mu_F = 0.695$	$\mu_{CR} = 0.015$	
Diversity = 1.8310e+00	k_t = [1, 1, 1, 1]			
Generation 7180: Best = 3.406748e+03	FES = 718100	$\mu_F = 0.627$	$\mu_{CR} = 0.020$	
Diversity = 1.8310e+00	k_t = [2, 2, 4, 3]			
Generation 7190: Best = 3.406748e+03	FES = 719100	$\mu_F = 0.716$	$\mu_{CR} = 0.020$	
Diversity = 1.8310e+00	k_t = [1, 1, 1, 1]			
Generation 7200: Best = 3.406748e+03	FES = 720100	$\mu_F = 0.784$	$\mu_{CR} = 0.021$	
Diversity = 1.8255e+00	k_t = [4, 2, 4, 4]			
Generation 7210: Best = 3.406748e+03	FES = 721100	$\mu_F = 0.825$	$\mu_{CR} = 0.022$	
Diversity = 1.8255e+00	k_t = [1, 1, 1, 1]			
Generation 7220: Best = 3.406666e+03	FES = 722100	$\mu_F = 0.823$	$\mu_{CR} = 0.020$	
Diversity = 1.8254e+00	k_t = [4, 4, 4, 2]			
Generation 7230: Best = 3.406666e+03	FES = 723100	$\mu_F = 0.785$	$\mu_{CR} = 0.018$	
Diversity = 1.8065e+00	k_t = [1, 1, 1, 1]			
Generation 7240: Best = 3.406666e+03	FES = 724100	$\mu_F = 0.759$	$\mu_{CR} = 0.024$	

Diversity = 1.8132e+00	k_t = [3, 3, 3, 4]			
Generation 7250: Best = 3.406664e+03	FES = 725100	$\mu_F = 0.776$	$\mu_{CR} = 0.030$	
Diversity = 1.8125e+00	k_t = [1, 1, 1, 1]			
Generation 7260: Best = 3.406664e+03	FES = 726100	$\mu_F = 0.817$	$\mu_{CR} = 0.029$	
Diversity = 1.7994e+00	k_t = [4, 3, 4, 2]			
Generation 7270: Best = 3.406664e+03	FES = 727100	$\mu_F = 0.800$	$\mu_{CR} = 0.026$	
Diversity = 1.7994e+00	k_t = [1, 1, 1, 1]			
Generation 7280: Best = 3.406664e+03	FES = 728100	$\mu_F = 0.834$	$\mu_{CR} = 0.028$	
Diversity = 1.7994e+00	k_t = [4, 2, 4, 3]			
Generation 7290: Best = 3.406628e+03	FES = 729100	$\mu_F = 0.841$	$\mu_{CR} = 0.023$	
Diversity = 1.7994e+00	k_t = [1, 1, 1, 1]			
Generation 7300: Best = 3.406628e+03	FES = 730100	$\mu_F = 0.838$	$\mu_{CR} = 0.029$	
Diversity = 1.7994e+00	k_t = [4, 4, 3, 3]			
Generation 7310: Best = 3.406627e+03	FES = 731100	$\mu_F = 0.799$	$\mu_{CR} = 0.025$	
Diversity = 1.7994e+00	k_t = [1, 1, 1, 1]			
Generation 7320: Best = 3.406627e+03	FES = 732100	$\mu_F = 0.770$	$\mu_{CR} = 0.026$	
Diversity = 1.7958e+00	k_t = [4, 3, 3, 4]			
Generation 7330: Best = 3.406627e+03	FES = 733100	$\mu_F = 0.803$	$\mu_{CR} = 0.020$	
Diversity = 1.7990e+00	k_t = [1, 1, 1, 1]			
Generation 7340: Best = 3.406627e+03	FES = 734100	$\mu_F = 0.782$	$\mu_{CR} = 0.019$	
Diversity = 1.7726e+00	k_t = [4, 4, 4, 4]			
Generation 7350: Best = 3.406627e+03	FES = 735100	$\mu_F = 0.795$	$\mu_{CR} = 0.020$	
Diversity = 1.7741e+00	k_t = [1, 1, 1, 1]			
Generation 7360: Best = 3.406623e+03	FES = 736100	$\mu_F = 0.747$	$\mu_{CR} = 0.022$	
Diversity = 1.7741e+00	k_t = [2, 2, 3, 2]			
Generation 7370: Best = 3.406623e+03	FES = 737100	$\mu_F = 0.740$	$\mu_{CR} = 0.020$	
Diversity = 1.7741e+00	k_t = [1, 1, 1, 1]			
Generation 7380: Best = 3.406623e+03	FES = 738100	$\mu_F = 0.633$	$\mu_{CR} = 0.023$	
Diversity = 1.7820e+00	k_t = [4, 3, 3, 4]			
Generation 7390: Best = 3.406623e+03	FES = 739100	$\mu_F = 0.648$	$\mu_{CR} = 0.022$	
Diversity = 1.7820e+00	k_t = [1, 1, 1, 1]			
Generation 7400: Best = 3.406622e+03	FES = 740100	$\mu_F = 0.589$	$\mu_{CR} = 0.023$	
Diversity = 1.7800e+00	k_t = [4, 3, 3, 3]			
Generation 7410: Best = 3.406621e+03	FES = 741100	$\mu_F = 0.592$	$\mu_{CR} = 0.022$	
Diversity = 1.7799e+00	k_t = [1, 1, 1, 1]			
Generation 7420: Best = 3.406621e+03	FES = 742100	$\mu_F = 0.609$	$\mu_{CR} = 0.020$	
Diversity = 1.7774e+00	k_t = [4, 4, 3, 2]			
Generation 7430: Best = 3.406620e+03	FES = 743100	$\mu_F = 0.642$	$\mu_{CR} = 0.019$	
Diversity = 1.7773e+00	k_t = [1, 1, 1, 1]			
Generation 7440: Best = 3.406620e+03	FES = 744100	$\mu_F = 0.662$	$\mu_{CR} = 0.022$	
Diversity = 1.7769e+00	k_t = [4, 2, 2, 2]			
Generation 7450: Best = 3.406620e+03	FES = 745100	$\mu_F = 0.675$	$\mu_{CR} = 0.022$	
Diversity = 1.7768e+00	k_t = [1, 1, 1, 1]			
Generation 7460: Best = 3.406620e+03	FES = 746100	$\mu_F = 0.709$	$\mu_{CR} = 0.019$	
Diversity = 1.7738e+00	k_t = [2, 2, 4, 4]			
Generation 7470: Best = 3.406620e+03	FES = 747100	$\mu_F = 0.736$	$\mu_{CR} = 0.019$	
Diversity = 1.7738e+00	k_t = [1, 1, 1, 1]			
Generation 7480: Best = 3.406612e+03	FES = 748100	$\mu_F = 0.811$	$\mu_{CR} = 0.018$	
Diversity = 1.7589e+00	k_t = [2, 2, 4, 4]			
Generation 7490: Best = 3.406612e+03	FES = 749100	$\mu_F = 0.754$	$\mu_{CR} = 0.022$	
Diversity = 1.7589e+00	k_t = [1, 1, 1, 1]			
Generation 7500: Best = 3.406612e+03	FES = 750100	$\mu_F = 0.755$	$\mu_{CR} = 0.026$	
Diversity = 1.7572e+00	k_t = [4, 4, 3, 3]			
Generation 7510: Best = 3.406612e+03	FES = 751100	$\mu_F = 0.772$	$\mu_{CR} = 0.017$	

Diversity = 1.7572e+00	k_t = [1, 1, 1, 1]			
Generation 7520: Best = 3.406612e+03	FES = 752100	$\mu_F = 0.774$	$\mu_{CR} = 0.016$	
Diversity = 1.7556e+00	k_t = [2, 2, 2, 4]			
Generation 7530: Best = 3.406612e+03	FES = 753100	$\mu_F = 0.769$	$\mu_{CR} = 0.017$	
Diversity = 1.7557e+00	k_t = [1, 1, 1, 1]			
Generation 7540: Best = 3.406612e+03	FES = 754100	$\mu_F = 0.814$	$\mu_{CR} = 0.015$	
Diversity = 1.7557e+00	k_t = [3, 4, 3, 3]			
Generation 7550: Best = 3.406612e+03	FES = 755100	$\mu_F = 0.868$	$\mu_{CR} = 0.022$	
Diversity = 1.7557e+00	k_t = [1, 1, 1, 1]			
Generation 7560: Best = 3.406607e+03	FES = 756100	$\mu_F = 0.833$	$\mu_{CR} = 0.024$	
Diversity = 1.7557e+00	k_t = [3, 3, 2, 2]			
Generation 7570: Best = 3.406607e+03	FES = 757100	$\mu_F = 0.839$	$\mu_{CR} = 0.022$	
Diversity = 1.7557e+00	k_t = [1, 1, 1, 1]			
Generation 7580: Best = 3.406605e+03	FES = 758100	$\mu_F = 0.755$	$\mu_{CR} = 0.032$	
Diversity = 1.7557e+00	k_t = [4, 3, 3, 2]			
Generation 7590: Best = 3.406605e+03	FES = 759100	$\mu_F = 0.799$	$\mu_{CR} = 0.030$	
Diversity = 1.7557e+00	k_t = [1, 1, 1, 1]			
Generation 7600: Best = 3.406605e+03	FES = 760100	$\mu_F = 0.791$	$\mu_{CR} = 0.025$	
Diversity = 1.7557e+00	k_t = [3, 3, 3, 2]			
Generation 7610: Best = 3.406605e+03	FES = 761100	$\mu_F = 0.767$	$\mu_{CR} = 0.024$	
Diversity = 1.7557e+00	k_t = [1, 1, 1, 1]			
Generation 7620: Best = 3.406605e+03	FES = 762100	$\mu_F = 0.773$	$\mu_{CR} = 0.026$	
Diversity = 1.7557e+00	k_t = [3, 3, 2, 3]			
Generation 7630: Best = 3.406605e+03	FES = 763100	$\mu_F = 0.783$	$\mu_{CR} = 0.018$	
Diversity = 1.7555e+00	k_t = [1, 1, 1, 1]			
Generation 7640: Best = 3.406605e+03	FES = 764100	$\mu_F = 0.746$	$\mu_{CR} = 0.015$	
Diversity = 1.7618e+00	k_t = [3, 4, 4, 2]			
Generation 7650: Best = 3.406605e+03	FES = 765100	$\mu_F = 0.795$	$\mu_{CR} = 0.018$	
Diversity = 1.7617e+00	k_t = [1, 1, 1, 1]			
Generation 7660: Best = 3.406605e+03	FES = 766100	$\mu_F = 0.808$	$\mu_{CR} = 0.019$	
Diversity = 1.7617e+00	k_t = [3, 4, 4, 2]			
Generation 7670: Best = 3.406605e+03	FES = 767100	$\mu_F = 0.786$	$\mu_{CR} = 0.016$	
Diversity = 1.7576e+00	k_t = [1, 1, 1, 1]			
Generation 7680: Best = 3.406605e+03	FES = 768100	$\mu_F = 0.784$	$\mu_{CR} = 0.019$	
Diversity = 1.7496e+00	k_t = [4, 3, 4, 4]			
Generation 7690: Best = 3.406605e+03	FES = 769100	$\mu_F = 0.750$	$\mu_{CR} = 0.023$	
Diversity = 1.7496e+00	k_t = [1, 1, 1, 1]			
Generation 7700: Best = 3.406605e+03	FES = 770100	$\mu_F = 0.763$	$\mu_{CR} = 0.030$	
Diversity = 1.7496e+00	k_t = [4, 4, 2, 2]			
Generation 7710: Best = 3.406605e+03	FES = 771100	$\mu_F = 0.705$	$\mu_{CR} = 0.028$	
Diversity = 1.7496e+00	k_t = [1, 1, 1, 1]			
Generation 7720: Best = 3.406605e+03	FES = 772100	$\mu_F = 0.746$	$\mu_{CR} = 0.021$	
Diversity = 1.7496e+00	k_t = [2, 3, 3, 2]			
Generation 7730: Best = 3.406605e+03	FES = 773100	$\mu_F = 0.734$	$\mu_{CR} = 0.017$	
Diversity = 1.7496e+00	k_t = [1, 1, 1, 1]			
Generation 7740: Best = 3.406605e+03	FES = 774100	$\mu_F = 0.695$	$\mu_{CR} = 0.020$	
Diversity = 1.7496e+00	k_t = [4, 2, 4, 2]			
Generation 7750: Best = 3.406604e+03	FES = 775100	$\mu_F = 0.670$	$\mu_{CR} = 0.020$	
Diversity = 1.7496e+00	k_t = [1, 1, 1, 1]			
Generation 7760: Best = 3.406588e+03	FES = 776100	$\mu_F = 0.691$	$\mu_{CR} = 0.028$	
Diversity = 1.7492e+00	k_t = [2, 4, 4, 4]			
Generation 7770: Best = 3.406587e+03	FES = 777100	$\mu_F = 0.764$	$\mu_{CR} = 0.029$	
Diversity = 1.7491e+00	k_t = [1, 1, 1, 1]			
Generation 7780: Best = 3.406573e+03	FES = 778100	$\mu_F = 0.708$	$\mu_{CR} = 0.025$	

Diversity = 1.7491e+00	k_t = [2, 2, 3, 3]			
Generation 7790: Best = 3.406573e+03	FES = 779100	$\mu_F = 0.721$	$\mu_{CR} = 0.025$	
Diversity = 1.7491e+00	k_t = [1, 1, 1, 1]			
Generation 7800: Best = 3.406573e+03	FES = 780100	$\mu_F = 0.669$	$\mu_{CR} = 0.024$	
Diversity = 1.7448e+00	k_t = [3, 4, 2, 4]			
Generation 7810: Best = 3.406563e+03	FES = 781100	$\mu_F = 0.742$	$\mu_{CR} = 0.020$	
Diversity = 1.7447e+00	k_t = [1, 1, 1, 1]			
Generation 7820: Best = 3.406563e+03	FES = 782100	$\mu_F = 0.721$	$\mu_{CR} = 0.020$	
Diversity = 1.7413e+00	k_t = [3, 3, 4, 2]			
Generation 7830: Best = 3.406563e+03	FES = 783100	$\mu_F = 0.735$	$\mu_{CR} = 0.022$	
Diversity = 1.7413e+00	k_t = [1, 1, 1, 1]			
Generation 7840: Best = 3.406563e+03	FES = 784100	$\mu_F = 0.768$	$\mu_{CR} = 0.020$	
Diversity = 1.7313e+00	k_t = [4, 4, 4, 4]			
Generation 7850: Best = 3.406563e+03	FES = 785100	$\mu_F = 0.758$	$\mu_{CR} = 0.024$	
Diversity = 1.7313e+00	k_t = [1, 1, 1, 1]			
Generation 7860: Best = 3.406563e+03	FES = 786100	$\mu_F = 0.813$	$\mu_{CR} = 0.023$	
Diversity = 1.7433e+00	k_t = [3, 4, 2, 4]			
Generation 7870: Best = 3.406563e+03	FES = 787100	$\mu_F = 0.796$	$\mu_{CR} = 0.018$	
Diversity = 1.7433e+00	k_t = [1, 1, 1, 1]			
Generation 7880: Best = 3.406562e+03	FES = 788100	$\mu_F = 0.810$	$\mu_{CR} = 0.020$	
Diversity = 1.7433e+00	k_t = [3, 3, 3, 3]			
Generation 7890: Best = 3.406562e+03	FES = 789100	$\mu_F = 0.794$	$\mu_{CR} = 0.019$	
Diversity = 1.7433e+00	k_t = [1, 1, 1, 1]			
Generation 7900: Best = 3.406562e+03	FES = 790100	$\mu_F = 0.804$	$\mu_{CR} = 0.029$	
Diversity = 1.7426e+00	k_t = [2, 4, 3, 4]			
Generation 7910: Best = 3.406562e+03	FES = 791100	$\mu_F = 0.734$	$\mu_{CR} = 0.023$	
Diversity = 1.7426e+00	k_t = [1, 1, 1, 1]			
Generation 7920: Best = 3.406562e+03	FES = 792100	$\mu_F = 0.696$	$\mu_{CR} = 0.022$	
Diversity = 1.7422e+00	k_t = [2, 4, 3, 4]			
Generation 7930: Best = 3.406562e+03	FES = 793100	$\mu_F = 0.678$	$\mu_{CR} = 0.022$	
Diversity = 1.7422e+00	k_t = [1, 1, 1, 1]			
Generation 7940: Best = 3.406562e+03	FES = 794100	$\mu_F = 0.723$	$\mu_{CR} = 0.032$	
Diversity = 1.7422e+00	k_t = [2, 4, 4, 2]			
Generation 7950: Best = 3.406562e+03	FES = 795100	$\mu_F = 0.756$	$\mu_{CR} = 0.023$	
Diversity = 1.7421e+00	k_t = [1, 1, 1, 1]			
Generation 7960: Best = 3.406547e+03	FES = 796100	$\mu_F = 0.743$	$\mu_{CR} = 0.023$	
Diversity = 1.7421e+00	k_t = [4, 2, 3, 2]			
Generation 7970: Best = 3.406524e+03	FES = 797100	$\mu_F = 0.738$	$\mu_{CR} = 0.025$	
Diversity = 1.7421e+00	k_t = [1, 1, 1, 1]			
Generation 7980: Best = 3.406468e+03	FES = 798100	$\mu_F = 0.773$	$\mu_{CR} = 0.021$	
Diversity = 1.7421e+00	k_t = [3, 3, 3, 3]			
Generation 7990: Best = 3.406468e+03	FES = 799100	$\mu_F = 0.807$	$\mu_{CR} = 0.026$	
Diversity = 1.7421e+00	k_t = [1, 1, 1, 1]			
Generation 8000: Best = 3.406462e+03	FES = 800100	$\mu_F = 0.829$	$\mu_{CR} = 0.029$	
Diversity = 1.7421e+00	k_t = [3, 3, 4, 3]			
Generation 8010: Best = 3.406462e+03	FES = 801100	$\mu_F = 0.769$	$\mu_{CR} = 0.025$	
Diversity = 1.7421e+00	k_t = [1, 1, 1, 1]			
Generation 8020: Best = 3.406460e+03	FES = 802100	$\mu_F = 0.808$	$\mu_{CR} = 0.021$	
Diversity = 1.7418e+00	k_t = [2, 2, 3, 4]			
Generation 8030: Best = 3.406450e+03	FES = 803100	$\mu_F = 0.834$	$\mu_{CR} = 0.021$	
Diversity = 1.7418e+00	k_t = [1, 1, 1, 1]			
Generation 8040: Best = 3.406449e+03	FES = 804100	$\mu_F = 0.856$	$\mu_{CR} = 0.021$	
Diversity = 1.7418e+00	k_t = [2, 2, 4, 2]			
Generation 8050: Best = 3.406449e+03	FES = 805100	$\mu_F = 0.755$	$\mu_{CR} = 0.019$	

Diversity = 1.7418e+00	k_t = [1, 1, 1, 1]			
Generation 8060: Best = 3.406449e+03	FES = 806100	$\mu_F = 0.725$	$\mu_{CR} = 0.020$	
Diversity = 1.7418e+00	k_t = [3, 4, 4, 3]			
Generation 8070: Best = 3.406449e+03	FES = 807100	$\mu_F = 0.732$	$\mu_{CR} = 0.019$	
Diversity = 1.7418e+00	k_t = [1, 1, 1, 1]			
Generation 8080: Best = 3.406449e+03	FES = 808100	$\mu_F = 0.715$	$\mu_{CR} = 0.021$	
Diversity = 1.7418e+00	k_t = [2, 4, 2, 2]			
Generation 8090: Best = 3.406449e+03	FES = 809100	$\mu_F = 0.756$	$\mu_{CR} = 0.019$	
Diversity = 1.7418e+00	k_t = [1, 1, 1, 1]			
Generation 8100: Best = 3.406449e+03	FES = 810100	$\mu_F = 0.779$	$\mu_{CR} = 0.020$	
Diversity = 1.7460e+00	k_t = [4, 3, 4, 4]			
Generation 8110: Best = 3.406448e+03	FES = 811100	$\mu_F = 0.739$	$\mu_{CR} = 0.023$	
Diversity = 1.7460e+00	k_t = [1, 1, 1, 1]			
Generation 8120: Best = 3.406446e+03	FES = 812100	$\mu_F = 0.756$	$\mu_{CR} = 0.022$	
Diversity = 1.7460e+00	k_t = [2, 4, 4, 2]			
Generation 8130: Best = 3.406446e+03	FES = 813100	$\mu_F = 0.790$	$\mu_{CR} = 0.021$	
Diversity = 1.7460e+00	k_t = [1, 1, 1, 1]			
Generation 8140: Best = 3.406445e+03	FES = 814100	$\mu_F = 0.769$	$\mu_{CR} = 0.023$	
Diversity = 1.7460e+00	k_t = [4, 2, 2, 3]			
Generation 8150: Best = 3.406445e+03	FES = 815100	$\mu_F = 0.784$	$\mu_{CR} = 0.024$	
Diversity = 1.7460e+00	k_t = [1, 1, 1, 1]			
Generation 8160: Best = 3.406445e+03	FES = 816100	$\mu_F = 0.820$	$\mu_{CR} = 0.026$	
Diversity = 1.7474e+00	k_t = [2, 3, 3, 4]			
Generation 8170: Best = 3.406445e+03	FES = 817100	$\mu_F = 0.802$	$\mu_{CR} = 0.021$	
Diversity = 1.7491e+00	k_t = [1, 1, 1, 1]			
Generation 8180: Best = 3.406445e+03	FES = 818100	$\mu_F = 0.755$	$\mu_{CR} = 0.021$	
Diversity = 1.7491e+00	k_t = [3, 3, 2, 3]			
Generation 8190: Best = 3.406445e+03	FES = 819100	$\mu_F = 0.793$	$\mu_{CR} = 0.020$	
Diversity = 1.7491e+00	k_t = [1, 1, 1, 1]			
Generation 8200: Best = 3.406445e+03	FES = 820100	$\mu_F = 0.818$	$\mu_{CR} = 0.029$	
Diversity = 1.7491e+00	k_t = [4, 2, 3, 2]			
Generation 8210: Best = 3.406445e+03	FES = 821100	$\mu_F = 0.802$	$\mu_{CR} = 0.026$	
Diversity = 1.7492e+00	k_t = [1, 1, 1, 1]			
Generation 8220: Best = 3.406445e+03	FES = 822100	$\mu_F = 0.773$	$\mu_{CR} = 0.028$	
Diversity = 1.7492e+00	k_t = [2, 2, 2, 3]			
Generation 8230: Best = 3.406438e+03	FES = 823100	$\mu_F = 0.794$	$\mu_{CR} = 0.023$	
Diversity = 1.7467e+00	k_t = [1, 1, 1, 1]			
Generation 8240: Best = 3.406438e+03	FES = 824100	$\mu_F = 0.810$	$\mu_{CR} = 0.020$	
Diversity = 1.7467e+00	k_t = [2, 4, 3, 2]			
Generation 8250: Best = 3.406438e+03	FES = 825100	$\mu_F = 0.788$	$\mu_{CR} = 0.021$	
Diversity = 1.7467e+00	k_t = [1, 1, 1, 1]			
Generation 8260: Best = 3.406437e+03	FES = 826100	$\mu_F = 0.796$	$\mu_{CR} = 0.020$	
Diversity = 1.7427e+00	k_t = [2, 4, 2, 4]			
Generation 8270: Best = 3.406435e+03	FES = 827100	$\mu_F = 0.744$	$\mu_{CR} = 0.022$	
Diversity = 1.7427e+00	k_t = [1, 1, 1, 1]			
Generation 8280: Best = 3.406333e+03	FES = 828100	$\mu_F = 0.723$	$\mu_{CR} = 0.025$	
Diversity = 1.7427e+00	k_t = [4, 2, 4, 3]			
Generation 8290: Best = 3.406333e+03	FES = 829100	$\mu_F = 0.706$	$\mu_{CR} = 0.024$	
Diversity = 1.7427e+00	k_t = [1, 1, 1, 1]			
Generation 8300: Best = 3.406333e+03	FES = 830100	$\mu_F = 0.640$	$\mu_{CR} = 0.023$	
Diversity = 1.7427e+00	k_t = [3, 4, 3, 3]			
Generation 8310: Best = 3.406332e+03	FES = 831100	$\mu_F = 0.574$	$\mu_{CR} = 0.024$	
Diversity = 1.7427e+00	k_t = [1, 1, 1, 1]			
Generation 8320: Best = 3.406331e+03	FES = 832100	$\mu_F = 0.570$	$\mu_{CR} = 0.022$	

Diversity = 1.7427e+00	k_t = [3, 4, 2, 2]			
Generation 8330: Best = 3.406331e+03	FES = 833100	$\mu_F = 0.618$	$\mu_{CR} = 0.022$	
Diversity = 1.7427e+00	k_t = [1, 1, 1, 1]			
Generation 8340: Best = 3.406325e+03	FES = 834100	$\mu_F = 0.601$	$\mu_{CR} = 0.024$	
Diversity = 1.7427e+00	k_t = [4, 4, 4, 3]			
Generation 8350: Best = 3.406317e+03	FES = 835100	$\mu_F = 0.606$	$\mu_{CR} = 0.019$	
Diversity = 1.7427e+00	k_t = [1, 1, 1, 1]			
Generation 8360: Best = 3.406317e+03	FES = 836100	$\mu_F = 0.646$	$\mu_{CR} = 0.020$	
Diversity = 1.7396e+00	k_t = [4, 4, 4, 4]			
Generation 8370: Best = 3.406317e+03	FES = 837100	$\mu_F = 0.665$	$\mu_{CR} = 0.026$	
Diversity = 1.7397e+00	k_t = [1, 1, 1, 1]			
Generation 8380: Best = 3.406316e+03	FES = 838100	$\mu_F = 0.747$	$\mu_{CR} = 0.026$	
Diversity = 1.7396e+00	k_t = [2, 4, 4, 3]			
Generation 8390: Best = 3.406316e+03	FES = 839100	$\mu_F = 0.718$	$\mu_{CR} = 0.021$	
Diversity = 1.7396e+00	k_t = [1, 1, 1, 1]			
Generation 8400: Best = 3.406316e+03	FES = 840100	$\mu_F = 0.738$	$\mu_{CR} = 0.022$	
Diversity = 1.7331e+00	k_t = [2, 3, 4, 4]			
Generation 8410: Best = 3.406315e+03	FES = 841100	$\mu_F = 0.753$	$\mu_{CR} = 0.026$	
Diversity = 1.7331e+00	k_t = [1, 1, 1, 1]			
Generation 8420: Best = 3.406315e+03	FES = 842100	$\mu_F = 0.586$	$\mu_{CR} = 0.026$	
Diversity = 1.7331e+00	k_t = [3, 2, 4, 2]			
Generation 8430: Best = 3.406315e+03	FES = 843100	$\mu_F = 0.631$	$\mu_{CR} = 0.019$	
Diversity = 1.7331e+00	k_t = [1, 1, 1, 1]			
Generation 8440: Best = 3.406315e+03	FES = 844100	$\mu_F = 0.570$	$\mu_{CR} = 0.019$	
Diversity = 1.7331e+00	k_t = [2, 3, 2, 2]			
Generation 8450: Best = 3.406315e+03	FES = 845100	$\mu_F = 0.664$	$\mu_{CR} = 0.018$	
Diversity = 1.7331e+00	k_t = [1, 1, 1, 1]			
Generation 8460: Best = 3.406315e+03	FES = 846100	$\mu_F = 0.718$	$\mu_{CR} = 0.024$	
Diversity = 1.7331e+00	k_t = [4, 2, 3, 3]			
Generation 8470: Best = 3.406315e+03	FES = 847100	$\mu_F = 0.605$	$\mu_{CR} = 0.030$	
Diversity = 1.7330e+00	k_t = [1, 1, 1, 1]			
Generation 8480: Best = 3.406300e+03	FES = 848100	$\mu_F = 0.642$	$\mu_{CR} = 0.035$	
Diversity = 1.7330e+00	k_t = [4, 4, 4, 3]			
Generation 8490: Best = 3.406300e+03	FES = 849100	$\mu_F = 0.626$	$\mu_{CR} = 0.033$	
Diversity = 1.7331e+00	k_t = [1, 1, 1, 1]			
Generation 8500: Best = 3.406300e+03	FES = 850100	$\mu_F = 0.617$	$\mu_{CR} = 0.030$	
Diversity = 1.7330e+00	k_t = [3, 3, 3, 2]			
Generation 8510: Best = 3.406298e+03	FES = 851100	$\mu_F = 0.613$	$\mu_{CR} = 0.024$	
Diversity = 1.7331e+00	k_t = [1, 1, 1, 1]			
Generation 8520: Best = 3.406298e+03	FES = 852100	$\mu_F = 0.601$	$\mu_{CR} = 0.018$	
Diversity = 1.7330e+00	k_t = [3, 3, 2, 2]			
Generation 8530: Best = 3.406286e+03	FES = 853100	$\mu_F = 0.628$	$\mu_{CR} = 0.024$	
Diversity = 1.7330e+00	k_t = [1, 1, 1, 1]			
Generation 8540: Best = 3.406286e+03	FES = 854100	$\mu_F = 0.689$	$\mu_{CR} = 0.022$	
Diversity = 1.7359e+00	k_t = [2, 2, 4, 4]			
Generation 8550: Best = 3.406286e+03	FES = 855100	$\mu_F = 0.650$	$\mu_{CR} = 0.023$	
Diversity = 1.7359e+00	k_t = [1, 1, 1, 1]			
Generation 8560: Best = 3.406285e+03	FES = 856100	$\mu_F = 0.662$	$\mu_{CR} = 0.023$	
Diversity = 1.7359e+00	k_t = [2, 3, 3, 2]			
Generation 8570: Best = 3.406285e+03	FES = 857100	$\mu_F = 0.676$	$\mu_{CR} = 0.020$	
Diversity = 1.7359e+00	k_t = [1, 1, 1, 1]			
Generation 8580: Best = 3.406285e+03	FES = 858100	$\mu_F = 0.699$	$\mu_{CR} = 0.021$	
Diversity = 1.7359e+00	k_t = [2, 3, 4, 2]			
Generation 8590: Best = 3.406285e+03	FES = 859100	$\mu_F = 0.656$	$\mu_{CR} = 0.027$	

Diversity = 1.7359e+00	k_t = [1, 1, 1, 1]			
Generation 8600: Best = 3.406283e+03	FES = 860100	$\mu_F = 0.584$	$\mu_{CR} = 0.020$	
Diversity = 1.7359e+00	k_t = [3, 4, 3, 4]			
Generation 8610: Best = 3.406283e+03	FES = 861100	$\mu_F = 0.599$	$\mu_{CR} = 0.021$	
Diversity = 1.7359e+00	k_t = [1, 1, 1, 1]			
Generation 8620: Best = 3.406277e+03	FES = 862100	$\mu_F = 0.599$	$\mu_{CR} = 0.018$	
Diversity = 1.7359e+00	k_t = [2, 4, 3, 2]			
Generation 8630: Best = 3.406277e+03	FES = 863100	$\mu_F = 0.686$	$\mu_{CR} = 0.022$	
Diversity = 1.7359e+00	k_t = [1, 1, 1, 1]			
Generation 8640: Best = 3.406242e+03	FES = 864100	$\mu_F = 0.687$	$\mu_{CR} = 0.025$	
Diversity = 1.7173e+00	k_t = [3, 4, 2, 4]			
Generation 8650: Best = 3.406242e+03	FES = 865100	$\mu_F = 0.706$	$\mu_{CR} = 0.035$	
Diversity = 1.7172e+00	k_t = [1, 1, 1, 1]			
Generation 8660: Best = 3.406242e+03	FES = 866100	$\mu_F = 0.716$	$\mu_{CR} = 0.024$	
Diversity = 1.7172e+00	k_t = [4, 3, 3, 2]			
Generation 8670: Best = 3.406242e+03	FES = 867100	$\mu_F = 0.752$	$\mu_{CR} = 0.021$	
Diversity = 1.7172e+00	k_t = [1, 1, 1, 1]			
Generation 8680: Best = 3.406242e+03	FES = 868100	$\mu_F = 0.781$	$\mu_{CR} = 0.028$	
Diversity = 1.6968e+00	k_t = [4, 3, 3, 4]			
Generation 8690: Best = 3.406242e+03	FES = 869100	$\mu_F = 0.792$	$\mu_{CR} = 0.022$	
Diversity = 1.6968e+00	k_t = [1, 1, 1, 1]			
Generation 8700: Best = 3.406242e+03	FES = 870100	$\mu_F = 0.746$	$\mu_{CR} = 0.026$	
Diversity = 1.6968e+00	k_t = [3, 3, 2, 2]			
Generation 8710: Best = 3.406233e+03	FES = 871100	$\mu_F = 0.712$	$\mu_{CR} = 0.022$	
Diversity = 1.6968e+00	k_t = [1, 1, 1, 1]			
Generation 8720: Best = 3.406233e+03	FES = 872100	$\mu_F = 0.723$	$\mu_{CR} = 0.025$	
Diversity = 1.6968e+00	k_t = [4, 3, 4, 2]			
Generation 8730: Best = 3.406233e+03	FES = 873100	$\mu_F = 0.753$	$\mu_{CR} = 0.019$	
Diversity = 1.6968e+00	k_t = [1, 1, 1, 1]			
Generation 8740: Best = 3.406233e+03	FES = 874100	$\mu_F = 0.779$	$\mu_{CR} = 0.021$	
Diversity = 1.6968e+00	k_t = [3, 3, 2, 2]			
Generation 8750: Best = 3.406233e+03	FES = 875100	$\mu_F = 0.726$	$\mu_{CR} = 0.020$	
Diversity = 1.6968e+00	k_t = [1, 1, 1, 1]			
Generation 8760: Best = 3.406233e+03	FES = 876100	$\mu_F = 0.794$	$\mu_{CR} = 0.024$	
Diversity = 1.7010e+00	k_t = [4, 4, 2, 4]			
Generation 8770: Best = 3.406233e+03	FES = 877100	$\mu_F = 0.806$	$\mu_{CR} = 0.020$	
Diversity = 1.7010e+00	k_t = [1, 1, 1, 1]			
Generation 8780: Best = 3.406233e+03	FES = 878100	$\mu_F = 0.772$	$\mu_{CR} = 0.021$	
Diversity = 1.7010e+00	k_t = [2, 3, 2, 2]			
Generation 8790: Best = 3.406233e+03	FES = 879100	$\mu_F = 0.638$	$\mu_{CR} = 0.022$	
Diversity = 1.7010e+00	k_t = [1, 1, 1, 1]			
Generation 8800: Best = 3.406232e+03	FES = 880100	$\mu_F = 0.661$	$\mu_{CR} = 0.028$	
Diversity = 1.7018e+00	k_t = [3, 2, 2, 4]			
Generation 8810: Best = 3.406232e+03	FES = 881100	$\mu_F = 0.707$	$\mu_{CR} = 0.030$	
Diversity = 1.7018e+00	k_t = [1, 1, 1, 1]			
Generation 8820: Best = 3.406232e+03	FES = 882100	$\mu_F = 0.737$	$\mu_{CR} = 0.032$	
Diversity = 1.6952e+00	k_t = [4, 3, 4, 4]			
Generation 8830: Best = 3.406232e+03	FES = 883100	$\mu_F = 0.760$	$\mu_{CR} = 0.024$	
Diversity = 1.6953e+00	k_t = [1, 1, 1, 1]			
Generation 8840: Best = 3.406232e+03	FES = 884100	$\mu_F = 0.785$	$\mu_{CR} = 0.019$	
Diversity = 1.6953e+00	k_t = [4, 4, 3, 3]			
Generation 8850: Best = 3.406232e+03	FES = 885100	$\mu_F = 0.777$	$\mu_{CR} = 0.022$	
Diversity = 1.6952e+00	k_t = [1, 1, 1, 1]			
Generation 8860: Best = 3.406193e+03	FES = 886100	$\mu_F = 0.790$	$\mu_{CR} = 0.024$	

Diversity = 1.6913e+00	k_t = [4, 3, 4, 4]			
Generation 8870: Best = 3.406193e+03	FES = 887100	$\mu_F = 0.856$	$\mu_{CR} = 0.020$	
Diversity = 1.6914e+00	k_t = [1, 1, 1, 1]			
Generation 8880: Best = 3.406191e+03	FES = 888100	$\mu_F = 0.828$	$\mu_{CR} = 0.022$	
Diversity = 1.6759e+00	k_t = [3, 4, 2, 4]			
Generation 8890: Best = 3.406191e+03	FES = 889100	$\mu_F = 0.869$	$\mu_{CR} = 0.021$	
Diversity = 1.6759e+00	k_t = [1, 1, 1, 1]			
Generation 8900: Best = 3.406185e+03	FES = 890100	$\mu_F = 0.871$	$\mu_{CR} = 0.027$	
Diversity = 1.6759e+00	k_t = [4, 4, 3, 3]			
Generation 8910: Best = 3.406185e+03	FES = 891100	$\mu_F = 0.855$	$\mu_{CR} = 0.023$	
Diversity = 1.6759e+00	k_t = [1, 1, 1, 1]			
Generation 8920: Best = 3.406177e+03	FES = 892100	$\mu_F = 0.876$	$\mu_{CR} = 0.022$	
Diversity = 1.6759e+00	k_t = [3, 2, 3, 2]			
Generation 8930: Best = 3.406177e+03	FES = 893100	$\mu_F = 0.748$	$\mu_{CR} = 0.017$	
Diversity = 1.6759e+00	k_t = [1, 1, 1, 1]			
Generation 8940: Best = 3.406176e+03	FES = 894100	$\mu_F = 0.753$	$\mu_{CR} = 0.019$	
Diversity = 1.6759e+00	k_t = [3, 3, 2, 2]			
Generation 8950: Best = 3.406176e+03	FES = 895100	$\mu_F = 0.799$	$\mu_{CR} = 0.016$	
Diversity = 1.6759e+00	k_t = [1, 1, 1, 1]			
Generation 8960: Best = 3.406176e+03	FES = 896100	$\mu_F = 0.820$	$\mu_{CR} = 0.020$	
Diversity = 1.6850e+00	k_t = [3, 2, 2, 4]			
Generation 8970: Best = 3.406176e+03	FES = 897100	$\mu_F = 0.722$	$\mu_{CR} = 0.021$	
Diversity = 1.6851e+00	k_t = [1, 1, 1, 1]			
Generation 8980: Best = 3.406170e+03	FES = 898100	$\mu_F = 0.740$	$\mu_{CR} = 0.017$	
Diversity = 1.6850e+00	k_t = [3, 4, 3, 3]			
Generation 8990: Best = 3.406156e+03	FES = 899100	$\mu_F = 0.716$	$\mu_{CR} = 0.019$	
Diversity = 1.6850e+00	k_t = [1, 1, 1, 1]			
Generation 9000: Best = 3.406156e+03	FES = 900100	$\mu_F = 0.752$	$\mu_{CR} = 0.030$	
Diversity = 1.6850e+00	k_t = [4, 3, 2, 2]			
Generation 9010: Best = 3.406156e+03	FES = 901100	$\mu_F = 0.781$	$\mu_{CR} = 0.024$	
Diversity = 1.6850e+00	k_t = [1, 1, 1, 1]			
Generation 9020: Best = 3.406156e+03	FES = 902100	$\mu_F = 0.771$	$\mu_{CR} = 0.023$	
Diversity = 1.6850e+00	k_t = [3, 2, 3, 3]			
Generation 9030: Best = 3.406156e+03	FES = 903100	$\mu_F = 0.806$	$\mu_{CR} = 0.023$	
Diversity = 1.6851e+00	k_t = [1, 1, 1, 1]			
Generation 9040: Best = 3.406155e+03	FES = 904100	$\mu_F = 0.801$	$\mu_{CR} = 0.020$	
Diversity = 1.6850e+00	k_t = [2, 3, 2, 3]			
Generation 9050: Best = 3.406155e+03	FES = 905100	$\mu_F = 0.775$	$\mu_{CR} = 0.015$	
Diversity = 1.6850e+00	k_t = [1, 1, 1, 1]			
Generation 9060: Best = 3.406155e+03	FES = 906100	$\mu_F = 0.772$	$\mu_{CR} = 0.027$	
Diversity = 1.6851e+00	k_t = [4, 2, 4, 3]			
Generation 9070: Best = 3.406153e+03	FES = 907100	$\mu_F = 0.772$	$\mu_{CR} = 0.029$	
Diversity = 1.6850e+00	k_t = [1, 1, 1, 1]			
Generation 9080: Best = 3.406153e+03	FES = 908100	$\mu_F = 0.799$	$\mu_{CR} = 0.032$	
Diversity = 1.6851e+00	k_t = [4, 3, 4, 3]			
Generation 9090: Best = 3.406152e+03	FES = 909100	$\mu_F = 0.815$	$\mu_{CR} = 0.027$	
Diversity = 1.6851e+00	k_t = [1, 1, 1, 1]			
Generation 9100: Best = 3.406152e+03	FES = 910100	$\mu_F = 0.851$	$\mu_{CR} = 0.026$	
Diversity = 1.6633e+00	k_t = [3, 4, 4, 4]			
Generation 9110: Best = 3.406151e+03	FES = 911100	$\mu_F = 0.765$	$\mu_{CR} = 0.031$	
Diversity = 1.6633e+00	k_t = [1, 1, 1, 1]			
Generation 9120: Best = 3.406151e+03	FES = 912100	$\mu_F = 0.759$	$\mu_{CR} = 0.024$	
Diversity = 1.6633e+00	k_t = [2, 3, 4, 2]			
Generation 9130: Best = 3.406151e+03	FES = 913100	$\mu_F = 0.741$	$\mu_{CR} = 0.030$	

Diversity = 1.6633e+00	k_t = [1, 1, 1, 1]			
Generation 9140: Best = 3.406121e+03	FES = 914100	$\mu_F = 0.734$	$\mu_{CR} = 0.024$	
Diversity = 1.6633e+00	k_t = [3, 3, 3, 2]			
Generation 9150: Best = 3.406121e+03	FES = 915100	$\mu_F = 0.724$	$\mu_{CR} = 0.024$	
Diversity = 1.6634e+00	k_t = [1, 1, 1, 1]			
Generation 9160: Best = 3.406120e+03	FES = 916100	$\mu_F = 0.706$	$\mu_{CR} = 0.021$	
Diversity = 1.6634e+00	k_t = [3, 3, 3, 3]			
Generation 9170: Best = 3.406092e+03	FES = 917100	$\mu_F = 0.737$	$\mu_{CR} = 0.021$	
Diversity = 1.6634e+00	k_t = [1, 1, 1, 1]			
Generation 9180: Best = 3.406091e+03	FES = 918100	$\mu_F = 0.676$	$\mu_{CR} = 0.032$	
Diversity = 1.6660e+00	k_t = [3, 3, 4, 4]			
Generation 9190: Best = 3.406086e+03	FES = 919100	$\mu_F = 0.733$	$\mu_{CR} = 0.030$	
Diversity = 1.6660e+00	k_t = [1, 1, 1, 1]			
Generation 9200: Best = 3.406086e+03	FES = 920100	$\mu_F = 0.726$	$\mu_{CR} = 0.024$	
Diversity = 1.6660e+00	k_t = [2, 2, 3, 3]			
Generation 9210: Best = 3.406086e+03	FES = 921100	$\mu_F = 0.787$	$\mu_{CR} = 0.031$	
Diversity = 1.6660e+00	k_t = [1, 1, 1, 1]			
Generation 9220: Best = 3.406086e+03	FES = 922100	$\mu_F = 0.804$	$\mu_{CR} = 0.023$	
Diversity = 1.6370e+00	k_t = [2, 2, 3, 4]			
Generation 9230: Best = 3.406072e+03	FES = 923100	$\mu_F = 0.773$	$\mu_{CR} = 0.025$	
Diversity = 1.6370e+00	k_t = [1, 1, 1, 1]			
Generation 9240: Best = 3.406072e+03	FES = 924100	$\mu_F = 0.732$	$\mu_{CR} = 0.026$	
Diversity = 1.6370e+00	k_t = [2, 2, 4, 2]			
Generation 9250: Best = 3.406062e+03	FES = 925100	$\mu_F = 0.778$	$\mu_{CR} = 0.026$	
Diversity = 1.6370e+00	k_t = [1, 1, 1, 1]			
Generation 9260: Best = 3.406062e+03	FES = 926100	$\mu_F = 0.754$	$\mu_{CR} = 0.029$	
Diversity = 1.6370e+00	k_t = [4, 4, 3, 3]			
Generation 9270: Best = 3.406062e+03	FES = 927100	$\mu_F = 0.726$	$\mu_{CR} = 0.024$	
Diversity = 1.6370e+00	k_t = [1, 1, 1, 1]			
Generation 9280: Best = 3.406061e+03	FES = 928100	$\mu_F = 0.670$	$\mu_{CR} = 0.023$	
Diversity = 1.6298e+00	k_t = [2, 4, 3, 4]			
Generation 9290: Best = 3.406057e+03	FES = 929100	$\mu_F = 0.645$	$\mu_{CR} = 0.030$	
Diversity = 1.6298e+00	k_t = [1, 1, 1, 1]			
Generation 9300: Best = 3.406046e+03	FES = 930100	$\mu_F = 0.670$	$\mu_{CR} = 0.035$	
Diversity = 1.6298e+00	k_t = [2, 4, 4, 3]			
Generation 9310: Best = 3.406043e+03	FES = 931100	$\mu_F = 0.600$	$\mu_{CR} = 0.036$	
Diversity = 1.6299e+00	k_t = [1, 1, 1, 1]			
Generation 9320: Best = 3.406017e+03	FES = 932100	$\mu_F = 0.554$	$\mu_{CR} = 0.042$	
Diversity = 1.6299e+00	k_t = [4, 2, 2, 2]			
Generation 9330: Best = 3.406017e+03	FES = 933100	$\mu_F = 0.609$	$\mu_{CR} = 0.036$	
Diversity = 1.6298e+00	k_t = [1, 1, 1, 1]			
Generation 9340: Best = 3.406017e+03	FES = 934100	$\mu_F = 0.601$	$\mu_{CR} = 0.039$	
Diversity = 1.6282e+00	k_t = [3, 2, 2, 4]			
Generation 9350: Best = 3.406017e+03	FES = 935100	$\mu_F = 0.584$	$\mu_{CR} = 0.028$	
Diversity = 1.6282e+00	k_t = [1, 1, 1, 1]			
Generation 9360: Best = 3.406017e+03	FES = 936100	$\mu_F = 0.577$	$\mu_{CR} = 0.042$	
Diversity = 1.6282e+00	k_t = [4, 2, 4, 3]			
Generation 9370: Best = 3.406013e+03	FES = 937100	$\mu_F = 0.619$	$\mu_{CR} = 0.035$	
Diversity = 1.6282e+00	k_t = [1, 1, 1, 1]			
Generation 9380: Best = 3.406009e+03	FES = 938100	$\mu_F = 0.616$	$\mu_{CR} = 0.029$	
Diversity = 1.6282e+00	k_t = [4, 2, 2, 2]			
Generation 9390: Best = 3.406009e+03	FES = 939100	$\mu_F = 0.626$	$\mu_{CR} = 0.026$	
Diversity = 1.6282e+00	k_t = [1, 1, 1, 1]			
Generation 9400: Best = 3.406009e+03	FES = 940100	$\mu_F = 0.611$	$\mu_{CR} = 0.026$	

Diversity = 1.6282e+00	k_t = [4, 3, 2, 3]			
Generation 9410: Best = 3.406009e+03	FES = 941100	$\mu_F = 0.606$	$\mu_{CR} = 0.041$	
Diversity = 1.6282e+00	k_t = [1, 1, 1, 1]			
Generation 9420: Best = 3.406006e+03	FES = 942100	$\mu_F = 0.638$	$\mu_{CR} = 0.028$	
Diversity = 1.6051e+00	k_t = [2, 4, 2, 4]			
Generation 9430: Best = 3.406006e+03	FES = 943100	$\mu_F = 0.664$	$\mu_{CR} = 0.026$	
Diversity = 1.6051e+00	k_t = [1, 1, 1, 1]			
Generation 9440: Best = 3.406006e+03	FES = 944100	$\mu_F = 0.687$	$\mu_{CR} = 0.023$	
Diversity = 1.6051e+00	k_t = [2, 3, 3, 2]			
Generation 9450: Best = 3.406006e+03	FES = 945100	$\mu_F = 0.720$	$\mu_{CR} = 0.023$	
Diversity = 1.6051e+00	k_t = [1, 1, 1, 1]			
Generation 9460: Best = 3.406006e+03	FES = 946100	$\mu_F = 0.720$	$\mu_{CR} = 0.030$	
Diversity = 1.5820e+00	k_t = [2, 2, 2, 4]			
Generation 9470: Best = 3.406006e+03	FES = 947100	$\mu_F = 0.776$	$\mu_{CR} = 0.021$	
Diversity = 1.5821e+00	k_t = [1, 1, 1, 1]			
Generation 9480: Best = 3.406006e+03	FES = 948100	$\mu_F = 0.770$	$\mu_{CR} = 0.018$	
Diversity = 1.5821e+00	k_t = [3, 2, 4, 2]			
Generation 9490: Best = 3.406006e+03	FES = 949100	$\mu_F = 0.780$	$\mu_{CR} = 0.019$	
Diversity = 1.5821e+00	k_t = [1, 1, 1, 1]			
Generation 9500: Best = 3.405971e+03	FES = 950100	$\mu_F = 0.693$	$\mu_{CR} = 0.032$	
Diversity = 1.5820e+00	k_t = [2, 2, 4, 3]			
Generation 9510: Best = 3.405960e+03	FES = 951100	$\mu_F = 0.714$	$\mu_{CR} = 0.031$	
Diversity = 1.5820e+00	k_t = [1, 1, 1, 1]			
Generation 9520: Best = 3.405960e+03	FES = 952100	$\mu_F = 0.704$	$\mu_{CR} = 0.028$	
Diversity = 1.5820e+00	k_t = [2, 4, 2, 2]			
Generation 9530: Best = 3.405960e+03	FES = 953100	$\mu_F = 0.662$	$\mu_{CR} = 0.022$	
Diversity = 1.5820e+00	k_t = [1, 1, 1, 1]			
Generation 9540: Best = 3.405960e+03	FES = 954100	$\mu_F = 0.612$	$\mu_{CR} = 0.027$	
Diversity = 1.5820e+00	k_t = [2, 3, 3, 3]			
Generation 9550: Best = 3.405960e+03	FES = 955100	$\mu_F = 0.649$	$\mu_{CR} = 0.024$	
Diversity = 1.5821e+00	k_t = [1, 1, 1, 1]			
Generation 9560: Best = 3.405960e+03	FES = 956100	$\mu_F = 0.618$	$\mu_{CR} = 0.029$	
Diversity = 1.5850e+00	k_t = [3, 3, 2, 4]			
Generation 9570: Best = 3.405960e+03	FES = 957100	$\mu_F = 0.649$	$\mu_{CR} = 0.027$	
Diversity = 1.5850e+00	k_t = [1, 1, 1, 1]			
Generation 9580: Best = 3.405960e+03	FES = 958100	$\mu_F = 0.654$	$\mu_{CR} = 0.031$	
Diversity = 1.5789e+00	k_t = [3, 2, 2, 4]			
Generation 9590: Best = 3.405960e+03	FES = 959100	$\mu_F = 0.616$	$\mu_{CR} = 0.024$	
Diversity = 1.5789e+00	k_t = [1, 1, 1, 1]			
Generation 9600: Best = 3.405952e+03	FES = 960100	$\mu_F = 0.618$	$\mu_{CR} = 0.030$	
Diversity = 1.5789e+00	k_t = [3, 4, 3, 3]			
Generation 9610: Best = 3.405952e+03	FES = 961100	$\mu_F = 0.675$	$\mu_{CR} = 0.020$	
Diversity = 1.5789e+00	k_t = [1, 1, 1, 1]			
Generation 9620: Best = 3.405952e+03	FES = 962100	$\mu_F = 0.683$	$\mu_{CR} = 0.026$	
Diversity = 1.5789e+00	k_t = [2, 4, 3, 3]			
Generation 9630: Best = 3.405944e+03	FES = 963100	$\mu_F = 0.718$	$\mu_{CR} = 0.022$	
Diversity = 1.5789e+00	k_t = [1, 1, 1, 1]			
Generation 9640: Best = 3.405944e+03	FES = 964100	$\mu_F = 0.647$	$\mu_{CR} = 0.027$	
Diversity = 1.5789e+00	k_t = [2, 4, 4, 2]			
Generation 9650: Best = 3.405939e+03	FES = 965100	$\mu_F = 0.700$	$\mu_{CR} = 0.030$	
Diversity = 1.5789e+00	k_t = [1, 1, 1, 1]			
Generation 9660: Best = 3.405939e+03	FES = 966100	$\mu_F = 0.713$	$\mu_{CR} = 0.031$	
Diversity = 1.5790e+00	k_t = [4, 4, 4, 3]			
Generation 9670: Best = 3.405939e+03	FES = 967100	$\mu_F = 0.686$	$\mu_{CR} = 0.030$	

Diversity = 1.5790e+00	k_t = [1, 1, 1, 1]			
Generation 9680: Best = 3.405935e+03	FES = 968100	$\mu_F = 0.714$	$\mu_{CR} = 0.023$	
Diversity = 1.5790e+00	k_t = [2, 4, 3, 2]			
Generation 9690: Best = 3.405935e+03	FES = 969100	$\mu_F = 0.780$	$\mu_{CR} = 0.024$	
Diversity = 1.5789e+00	k_t = [1, 1, 1, 1]			
Generation 9700: Best = 3.405929e+03	FES = 970100	$\mu_F = 0.807$	$\mu_{CR} = 0.020$	
Diversity = 1.5789e+00	k_t = [3, 2, 2, 3]			
Generation 9710: Best = 3.405927e+03	FES = 971100	$\mu_F = 0.806$	$\mu_{CR} = 0.031$	
Diversity = 1.5790e+00	k_t = [1, 1, 1, 1]			
Generation 9720: Best = 3.405927e+03	FES = 972100	$\mu_F = 0.764$	$\mu_{CR} = 0.030$	
Diversity = 1.5789e+00	k_t = [4, 4, 3, 3]			
Generation 9730: Best = 3.405927e+03	FES = 973100	$\mu_F = 0.752$	$\mu_{CR} = 0.027$	
Diversity = 1.5789e+00	k_t = [1, 1, 1, 1]			
Generation 9740: Best = 3.405927e+03	FES = 974100	$\mu_F = 0.725$	$\mu_{CR} = 0.021$	
Diversity = 1.5789e+00	k_t = [2, 4, 2, 3]			
Generation 9750: Best = 3.405927e+03	FES = 975100	$\mu_F = 0.690$	$\mu_{CR} = 0.023$	
Diversity = 1.5789e+00	k_t = [1, 1, 1, 1]			
Generation 9760: Best = 3.405923e+03	FES = 976100	$\mu_F = 0.718$	$\mu_{CR} = 0.019$	
Diversity = 1.5789e+00	k_t = [3, 4, 4, 2]			
Generation 9770: Best = 3.405923e+03	FES = 977100	$\mu_F = 0.735$	$\mu_{CR} = 0.026$	
Diversity = 1.5789e+00	k_t = [1, 1, 1, 1]			
Generation 9780: Best = 3.405922e+03	FES = 978100	$\mu_F = 0.768$	$\mu_{CR} = 0.022$	
Diversity = 1.5754e+00	k_t = [3, 3, 2, 4]			
Generation 9790: Best = 3.405920e+03	FES = 979100	$\mu_F = 0.691$	$\mu_{CR} = 0.023$	
Diversity = 1.5754e+00	k_t = [1, 1, 1, 1]			
Generation 9800: Best = 3.405920e+03	FES = 980100	$\mu_F = 0.731$	$\mu_{CR} = 0.021$	
Diversity = 1.5754e+00	k_t = [3, 4, 3, 2]			
Generation 9810: Best = 3.405918e+03	FES = 981100	$\mu_F = 0.763$	$\mu_{CR} = 0.019$	
Diversity = 1.5754e+00	k_t = [1, 1, 1, 1]			
Generation 9820: Best = 3.405918e+03	FES = 982100	$\mu_F = 0.793$	$\mu_{CR} = 0.021$	
Diversity = 1.5514e+00	k_t = [3, 3, 3, 4]			
Generation 9830: Best = 3.405911e+03	FES = 983100	$\mu_F = 0.829$	$\mu_{CR} = 0.022$	
Diversity = 1.5514e+00	k_t = [1, 1, 1, 1]			
Generation 9840: Best = 3.405910e+03	FES = 984100	$\mu_F = 0.781$	$\mu_{CR} = 0.020$	
Diversity = 1.5473e+00	k_t = [2, 2, 2, 4]			
Generation 9850: Best = 3.405910e+03	FES = 985100	$\mu_F = 0.799$	$\mu_{CR} = 0.017$	
Diversity = 1.5473e+00	k_t = [1, 1, 1, 1]			
Generation 9860: Best = 3.405906e+03	FES = 986100	$\mu_F = 0.766$	$\mu_{CR} = 0.028$	
Diversity = 1.5473e+00	k_t = [3, 4, 2, 3]			
Generation 9870: Best = 3.405862e+03	FES = 987100	$\mu_F = 0.781$	$\mu_{CR} = 0.028$	
Diversity = 1.5473e+00	k_t = [1, 1, 1, 1]			
Generation 9880: Best = 3.405862e+03	FES = 988100	$\mu_F = 0.742$	$\mu_{CR} = 0.026$	
Diversity = 1.5389e+00	k_t = [4, 3, 2, 4]			
Generation 9890: Best = 3.405844e+03	FES = 989100	$\mu_F = 0.793$	$\mu_{CR} = 0.030$	
Diversity = 1.5377e+00	k_t = [1, 1, 1, 1]			
Generation 9900: Best = 3.405844e+03	FES = 990100	$\mu_F = 0.803$	$\mu_{CR} = 0.038$	
Diversity = 1.5377e+00	k_t = [4, 3, 4, 2]			
Generation 9910: Best = 3.405827e+03	FES = 991100	$\mu_F = 0.750$	$\mu_{CR} = 0.031$	
Diversity = 1.5378e+00	k_t = [1, 1, 1, 1]			
Generation 9920: Best = 3.405827e+03	FES = 992100	$\mu_F = 0.793$	$\mu_{CR} = 0.033$	
Diversity = 1.5344e+00	k_t = [4, 3, 4, 4]			
Generation 9930: Best = 3.405827e+03	FES = 993100	$\mu_F = 0.790$	$\mu_{CR} = 0.032$	
Diversity = 1.5344e+00	k_t = [1, 1, 1, 1]			
Generation 9940: Best = 3.405825e+03	FES = 994100	$\mu_F = 0.757$	$\mu_{CR} = 0.032$	

```

Diversity = 1.5344e+00 k_t = [3, 2, 3, 2]
Generation 9950: Best = 3.405824e+03 FES = 995100  $\mu_F$  = 0.701  $\mu_{CR}$  = 0.028
Diversity = 1.5292e+00 k_t = [1, 1, 1, 1]
Generation 9960: Best = 3.405824e+03 FES = 996100  $\mu_F$  = 0.677  $\mu_{CR}$  = 0.039
Diversity = 1.5292e+00 k_t = [3, 2, 4, 2]
Generation 9970: Best = 3.405817e+03 FES = 997100  $\mu_F$  = 0.723  $\mu_{CR}$  = 0.038
Diversity = 1.5292e+00 k_t = [1, 1, 1, 1]
Generation 9980: Best = 3.405810e+03 FES = 998100  $\mu_F$  = 0.779  $\mu_{CR}$  = 0.043
Diversity = 1.5292e+00 k_t = [4, 3, 4, 3]
Generation 9990: Best = 3.405810e+03 FES = 999100  $\mu_F$  = 0.726  $\mu_{CR}$  = 0.034
Diversity = 1.5292e+00 k_t = [1, 1, 1, 1]

```

Optimization complete.

```

Best Fitness          = 3.405808e+03
Total Function Evals  = 1000000
Total Successful Updates = 41442

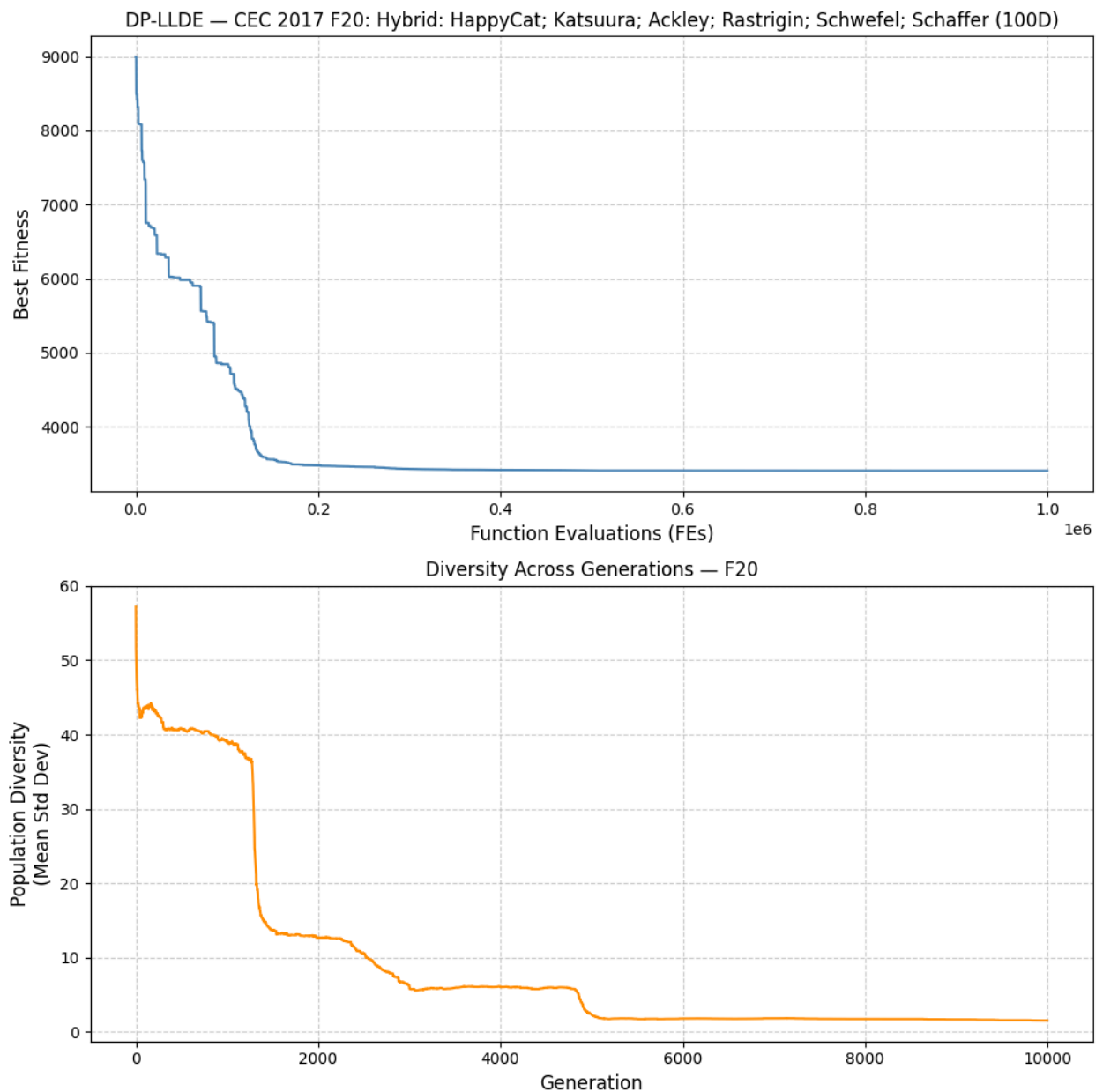
```

```

=====
=
Results:
  Best Fitness          : 3.405808e+03
  Global Optimum        : 2.000000e+03
  Error (|best-optimum|) : 1.405808e+03
  Best Solution (5 dim) : [-37.98335729  59.2646515  -9.7078163  48.64760262
96.68941792]
  D^(0) (init. diversity): 5.7220e+01
=====
=

```

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results/all_results_checkpoint.pkl



Graph saved as convergence_F20.png

✓ Plot saved to: /content/drive/My Drive/thesis_results/plots/convergence_F20.png

6.4 Composite Functions (F21-F30)

```
In [ ]: # =====
# F21 – Composition Function 1 (Rosenbrock; Elliptic; Rastrigin)
# =====
run_and_plot('F21', make_wrapper(composition.f21), 2100.0, 'Composition: Rosen
```

=====

=

DP-LLDE on CEC 2017 F21 – Single Run

=====

=

Function : Composition: Rosenbrock; Elliptic; Rastrigin (100D)

Bounds : [-100, 100]

Global optimum : 2100.0

Initial $D^*(0) = 5.7220e+01$ | Diversity threshold = 5.7220e+00

Generation 0: Best Fitness = 5.273291e+03

Generation 10:	Best = 4.257103e+03	FES = 1100	$\mu_F = 0.593$	$\mu_{CR} = 0.356$
Diversity = 4.0315e+01	$k_t = [1, 1, 1, 1]$			
Generation 20:	Best = 3.922576e+03	FES = 2100	$\mu_F = 0.638$	$\mu_{CR} = 0.366$
Diversity = 2.7873e+01	$k_t = [2, 3, 2, 3]$			
Generation 30:	Best = 3.652269e+03	FES = 3100	$\mu_F = 0.656$	$\mu_{CR} = 0.374$
Diversity = 2.1850e+01	$k_t = [1, 1, 1, 1]$			
Generation 40:	Best = 3.524308e+03	FES = 4100	$\mu_F = 0.657$	$\mu_{CR} = 0.359$
Diversity = 1.8193e+01	$k_t = [3, 2, 3, 4]$			
Generation 50:	Best = 3.457825e+03	FES = 5100	$\mu_F = 0.655$	$\mu_{CR} = 0.351$
Diversity = 1.6120e+01	$k_t = [1, 1, 1, 1]$			
Generation 60:	Best = 3.418651e+03	FES = 6100	$\mu_F = 0.651$	$\mu_{CR} = 0.350$
Diversity = 1.4850e+01	$k_t = [3, 4, 4, 3]$			
Generation 70:	Best = 3.352522e+03	FES = 7100	$\mu_F = 0.665$	$\mu_{CR} = 0.338$
Diversity = 1.3932e+01	$k_t = [1, 1, 1, 1]$			
Generation 80:	Best = 3.329635e+03	FES = 8100	$\mu_F = 0.675$	$\mu_{CR} = 0.336$
Diversity = 1.3184e+01	$k_t = [3, 4, 2, 4]$			
Generation 90:	Best = 3.329635e+03	FES = 9100	$\mu_F = 0.624$	$\mu_{CR} = 0.280$
Diversity = 1.2628e+01	$k_t = [1, 1, 1, 1]$			
Generation 100:	Best = 3.329635e+03	FES = 10100	$\mu_F = 0.665$	$\mu_{CR} = 0.201$
Diversity = 1.2448e+01	$k_t = [2, 4, 4, 2]$			
Generation 110:	Best = 3.312472e+03	FES = 11100	$\mu_F = 0.667$	$\mu_{CR} = 0.141$
Diversity = 1.2384e+01	$k_t = [1, 1, 1, 1]$			
Generation 120:	Best = 3.284002e+03	FES = 12100	$\mu_F = 0.713$	$\mu_{CR} = 0.099$
Diversity = 1.2073e+01	$k_t = [2, 2, 3, 4]$			
Generation 130:	Best = 3.283134e+03	FES = 13100	$\mu_F = 0.758$	$\mu_{CR} = 0.075$
Diversity = 1.1739e+01	$k_t = [1, 1, 1, 1]$			
Generation 140:	Best = 3.242629e+03	FES = 14100	$\mu_F = 0.758$	$\mu_{CR} = 0.062$
Diversity = 1.1615e+01	$k_t = [4, 4, 3, 4]$			
Generation 150:	Best = 3.242629e+03	FES = 15100	$\mu_F = 0.683$	$\mu_{CR} = 0.077$
Diversity = 1.1538e+01	$k_t = [1, 1, 1, 1]$			
Generation 160:	Best = 3.181494e+03	FES = 16100	$\mu_F = 0.699$	$\mu_{CR} = 0.062$
Diversity = 1.1416e+01	$k_t = [3, 2, 2, 4]$			
Generation 170:	Best = 3.157933e+03	FES = 17100	$\mu_F = 0.676$	$\mu_{CR} = 0.064$
Diversity = 1.1337e+01	$k_t = [1, 1, 1, 1]$			
Generation 180:	Best = 3.157933e+03	FES = 18100	$\mu_F = 0.639$	$\mu_{CR} = 0.059$
Diversity = 1.1212e+01	$k_t = [2, 2, 3, 3]$			
Generation 190:	Best = 3.144603e+03	FES = 19100	$\mu_F = 0.627$	$\mu_{CR} = 0.053$
Diversity = 1.1148e+01	$k_t = [1, 1, 1, 1]$			
Generation 200:	Best = 3.133383e+03	FES = 20100	$\mu_F = 0.658$	$\mu_{CR} = 0.056$
Diversity = 1.1060e+01	$k_t = [4, 4, 2, 3]$			
Generation 210:	Best = 3.133383e+03	FES = 21100	$\mu_F = 0.644$	$\mu_{CR} = 0.057$
Diversity = 1.1030e+01	$k_t = [1, 1, 1, 1]$			
Generation 220:	Best = 3.133383e+03	FES = 22100	$\mu_F = 0.640$	$\mu_{CR} = 0.062$
Diversity = 1.1006e+01	$k_t = [4, 4, 4, 3]$			

Generation 230: Best = 3.133383e+03	FES = 23100	$\mu_F = 0.637$	$\mu_{CR} = 0.048$
Diversity = 1.0967e+01	$k_t = [1, 1, 1, 1]$		
Generation 240: Best = 3.133383e+03	FES = 24100	$\mu_F = 0.661$	$\mu_{CR} = 0.045$
Diversity = 1.0949e+01	$k_t = [2, 3, 4, 2]$		
Generation 250: Best = 3.133383e+03	FES = 25100	$\mu_F = 0.590$	$\mu_{CR} = 0.038$
Diversity = 1.0892e+01	$k_t = [1, 1, 1, 1]$		
Generation 260: Best = 3.133383e+03	FES = 26100	$\mu_F = 0.565$	$\mu_{CR} = 0.044$
Diversity = 1.0837e+01	$k_t = [4, 4, 2, 3]$		
Generation 270: Best = 3.097964e+03	FES = 27100	$\mu_F = 0.562$	$\mu_{CR} = 0.044$
Diversity = 1.0769e+01	$k_t = [1, 1, 1, 1]$		
Generation 280: Best = 3.097964e+03	FES = 28100	$\mu_F = 0.576$	$\mu_{CR} = 0.042$
Diversity = 1.0764e+01	$k_t = [3, 4, 3, 3]$		
Generation 290: Best = 3.096086e+03	FES = 29100	$\mu_F = 0.608$	$\mu_{CR} = 0.039$
Diversity = 1.0618e+01	$k_t = [1, 1, 1, 1]$		
Generation 300: Best = 3.089338e+03	FES = 30100	$\mu_F = 0.597$	$\mu_{CR} = 0.034$
Diversity = 1.0579e+01	$k_t = [2, 4, 4, 3]$		
Generation 310: Best = 3.074502e+03	FES = 31100	$\mu_F = 0.586$	$\mu_{CR} = 0.036$
Diversity = 1.0579e+01	$k_t = [1, 1, 1, 1]$		
Generation 320: Best = 3.062451e+03	FES = 32100	$\mu_F = 0.536$	$\mu_{CR} = 0.041$
Diversity = 1.0531e+01	$k_t = [4, 3, 4, 3]$		
Generation 330: Best = 3.018953e+03	FES = 33100	$\mu_F = 0.454$	$\mu_{CR} = 0.043$
Diversity = 1.0495e+01	$k_t = [1, 1, 1, 1]$		
Generation 340: Best = 3.018953e+03	FES = 34100	$\mu_F = 0.414$	$\mu_{CR} = 0.050$
Diversity = 1.0315e+01	$k_t = [3, 3, 2, 4]$		
Generation 350: Best = 3.018953e+03	FES = 35100	$\mu_F = 0.422$	$\mu_{CR} = 0.051$
Diversity = 1.0323e+01	$k_t = [1, 1, 1, 1]$		
Generation 360: Best = 3.018083e+03	FES = 36100	$\mu_F = 0.487$	$\mu_{CR} = 0.042$
Diversity = 1.0324e+01	$k_t = [4, 3, 4, 4]$		
Generation 370: Best = 3.011615e+03	FES = 37100	$\mu_F = 0.468$	$\mu_{CR} = 0.052$
Diversity = 1.0296e+01	$k_t = [1, 1, 1, 1]$		
Generation 380: Best = 3.011615e+03	FES = 38100	$\mu_F = 0.433$	$\mu_{CR} = 0.052$
Diversity = 1.0256e+01	$k_t = [3, 3, 2, 2]$		
Generation 390: Best = 3.011615e+03	FES = 39100	$\mu_F = 0.468$	$\mu_{CR} = 0.060$
Diversity = 1.0230e+01	$k_t = [1, 1, 1, 1]$		
Generation 400: Best = 3.010286e+03	FES = 40100	$\mu_F = 0.459$	$\mu_{CR} = 0.057$
Diversity = 1.0226e+01	$k_t = [4, 4, 2, 3]$		
Generation 410: Best = 3.010286e+03	FES = 41100	$\mu_F = 0.452$	$\mu_{CR} = 0.059$
Diversity = 1.0237e+01	$k_t = [1, 1, 1, 1]$		
Generation 420: Best = 3.010286e+03	FES = 42100	$\mu_F = 0.468$	$\mu_{CR} = 0.051$
Diversity = 1.0224e+01	$k_t = [2, 3, 3, 2]$		
Generation 430: Best = 3.010286e+03	FES = 43100	$\mu_F = 0.375$	$\mu_{CR} = 0.069$
Diversity = 1.0200e+01	$k_t = [1, 1, 1, 1]$		
Generation 440: Best = 3.010286e+03	FES = 44100	$\mu_F = 0.402$	$\mu_{CR} = 0.054$
Diversity = 1.0228e+01	$k_t = [4, 2, 2, 2]$		
Generation 450: Best = 3.010286e+03	FES = 45100	$\mu_F = 0.407$	$\mu_{CR} = 0.053$
Diversity = 1.0222e+01	$k_t = [1, 1, 1, 1]$		
Generation 460: Best = 3.010286e+03	FES = 46100	$\mu_F = 0.472$	$\mu_{CR} = 0.050$
Diversity = 1.0244e+01	$k_t = [2, 3, 4, 2]$		
Generation 470: Best = 3.009732e+03	FES = 47100	$\mu_F = 0.421$	$\mu_{CR} = 0.045$
Diversity = 1.0220e+01	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 2.995529e+03	FES = 48100	$\mu_F = 0.453$	$\mu_{CR} = 0.049$
Diversity = 1.0213e+01	$k_t = [3, 4, 3, 4]$		
Generation 490: Best = 2.995529e+03	FES = 49100	$\mu_F = 0.460$	$\mu_{CR} = 0.047$
Diversity = 1.0196e+01	$k_t = [1, 1, 1, 1]$		

Generation 500: Best = 2.995529e+03	FES = 50100	$\mu_F = 0.461$	$\mu_{CR} = 0.041$
Diversity = 1.0178e+01	k_t = [2, 4, 3, 3]		
Generation 510: Best = 2.972123e+03	FES = 51100	$\mu_F = 0.518$	$\mu_{CR} = 0.041$
Diversity = 1.0161e+01	k_t = [1, 1, 1, 1]		
Generation 520: Best = 2.971214e+03	FES = 52100	$\mu_F = 0.472$	$\mu_{CR} = 0.047$
Diversity = 1.0154e+01	k_t = [3, 3, 3, 2]		
Generation 530: Best = 2.971214e+03	FES = 53100	$\mu_F = 0.470$	$\mu_{CR} = 0.031$
Diversity = 1.0168e+01	k_t = [1, 1, 1, 1]		
Generation 540: Best = 2.971030e+03	FES = 54100	$\mu_F = 0.519$	$\mu_{CR} = 0.037$
Diversity = 1.0156e+01	k_t = [4, 4, 2, 2]		
Generation 550: Best = 2.971030e+03	FES = 55100	$\mu_F = 0.515$	$\mu_{CR} = 0.036$
Diversity = 1.0152e+01	k_t = [1, 1, 1, 1]		
Generation 560: Best = 2.971030e+03	FES = 56100	$\mu_F = 0.570$	$\mu_{CR} = 0.039$
Diversity = 1.0083e+01	k_t = [2, 2, 4, 3]		
Generation 570: Best = 2.971030e+03	FES = 57100	$\mu_F = 0.515$	$\mu_{CR} = 0.045$
Diversity = 1.0072e+01	k_t = [1, 1, 1, 1]		
Generation 580: Best = 2.968877e+03	FES = 58100	$\mu_F = 0.574$	$\mu_{CR} = 0.043$
Diversity = 9.9560e+00	k_t = [4, 4, 4, 4]		
Generation 590: Best = 2.968877e+03	FES = 59100	$\mu_F = 0.526$	$\mu_{CR} = 0.031$
Diversity = 9.9464e+00	k_t = [1, 1, 1, 1]		
Generation 600: Best = 2.942178e+03	FES = 60100	$\mu_F = 0.472$	$\mu_{CR} = 0.039$
Diversity = 9.9308e+00	k_t = [4, 4, 2, 3]		
Generation 610: Best = 2.932392e+03	FES = 61100	$\mu_F = 0.440$	$\mu_{CR} = 0.037$
Diversity = 9.9152e+00	k_t = [1, 1, 1, 1]		
Generation 620: Best = 2.932392e+03	FES = 62100	$\mu_F = 0.519$	$\mu_{CR} = 0.030$
Diversity = 9.9460e+00	k_t = [2, 4, 4, 4]		
Generation 630: Best = 2.930490e+03	FES = 63100	$\mu_F = 0.472$	$\mu_{CR} = 0.049$
Diversity = 9.9661e+00	k_t = [1, 1, 1, 1]		
Generation 640: Best = 2.930490e+03	FES = 64100	$\mu_F = 0.415$	$\mu_{CR} = 0.048$
Diversity = 9.9411e+00	k_t = [2, 4, 3, 2]		
Generation 650: Best = 2.927748e+03	FES = 65100	$\mu_F = 0.405$	$\mu_{CR} = 0.038$
Diversity = 9.9326e+00	k_t = [1, 1, 1, 1]		
Generation 660: Best = 2.927591e+03	FES = 66100	$\mu_F = 0.427$	$\mu_{CR} = 0.047$
Diversity = 9.9483e+00	k_t = [4, 2, 3, 4]		
Generation 670: Best = 2.927591e+03	FES = 67100	$\mu_F = 0.410$	$\mu_{CR} = 0.041$
Diversity = 9.9379e+00	k_t = [1, 1, 1, 1]		
Generation 680: Best = 2.924842e+03	FES = 68100	$\mu_F = 0.385$	$\mu_{CR} = 0.051$
Diversity = 9.9260e+00	k_t = [2, 3, 2, 2]		
Generation 690: Best = 2.923188e+03	FES = 69100	$\mu_F = 0.370$	$\mu_{CR} = 0.048$
Diversity = 9.9100e+00	k_t = [1, 1, 1, 1]		
Generation 700: Best = 2.923188e+03	FES = 70100	$\mu_F = 0.354$	$\mu_{CR} = 0.045$
Diversity = 9.8670e+00	k_t = [2, 2, 3, 4]		
Generation 710: Best = 2.909775e+03	FES = 71100	$\mu_F = 0.320$	$\mu_{CR} = 0.044$
Diversity = 9.8369e+00	k_t = [1, 1, 1, 1]		
Generation 720: Best = 2.891278e+03	FES = 72100	$\mu_F = 0.327$	$\mu_{CR} = 0.035$
Diversity = 9.8085e+00	k_t = [4, 3, 3, 2]		
Generation 730: Best = 2.891278e+03	FES = 73100	$\mu_F = 0.315$	$\mu_{CR} = 0.039$
Diversity = 9.8021e+00	k_t = [1, 1, 1, 1]		
Generation 740: Best = 2.891171e+03	FES = 74100	$\mu_F = 0.274$	$\mu_{CR} = 0.055$
Diversity = 9.7721e+00	k_t = [2, 2, 4, 2]		
Generation 750: Best = 2.886534e+03	FES = 75100	$\mu_F = 0.258$	$\mu_{CR} = 0.050$
Diversity = 9.7600e+00	k_t = [1, 1, 1, 1]		
Generation 760: Best = 2.876842e+03	FES = 76100	$\mu_F = 0.307$	$\mu_{CR} = 0.044$
Diversity = 9.7142e+00	k_t = [3, 2, 4, 4]		

Generation 770: Best = 2.876842e+03	FES = 77100	$\mu_F = 0.316$	$\mu_{CR} = 0.040$
Diversity = 9.7024e+00	k_t = [1, 1, 1, 1]		
Generation 780: Best = 2.876842e+03	FES = 78100	$\mu_F = 0.270$	$\mu_{CR} = 0.043$
Diversity = 9.6933e+00	k_t = [3, 4, 3, 2]		
Generation 790: Best = 2.875572e+03	FES = 79100	$\mu_F = 0.367$	$\mu_{CR} = 0.038$
Diversity = 9.6873e+00	k_t = [1, 1, 1, 1]		
Generation 800: Best = 2.875425e+03	FES = 80100	$\mu_F = 0.332$	$\mu_{CR} = 0.056$
Diversity = 9.6821e+00	k_t = [3, 3, 2, 3]		
Generation 810: Best = 2.875425e+03	FES = 81100	$\mu_F = 0.382$	$\mu_{CR} = 0.049$
Diversity = 9.6732e+00	k_t = [1, 1, 1, 1]		
Generation 820: Best = 2.869305e+03	FES = 82100	$\mu_F = 0.408$	$\mu_{CR} = 0.040$
Diversity = 9.6170e+00	k_t = [2, 2, 3, 4]		
Generation 830: Best = 2.869305e+03	FES = 83100	$\mu_F = 0.402$	$\mu_{CR} = 0.042$
Diversity = 9.6094e+00	k_t = [1, 1, 1, 1]		
Generation 840: Best = 2.869305e+03	FES = 84100	$\mu_F = 0.387$	$\mu_{CR} = 0.041$
Diversity = 9.6156e+00	k_t = [4, 4, 2, 4]		
Generation 850: Best = 2.869305e+03	FES = 85100	$\mu_F = 0.415$	$\mu_{CR} = 0.038$
Diversity = 9.6114e+00	k_t = [1, 1, 1, 1]		
Generation 860: Best = 2.869248e+03	FES = 86100	$\mu_F = 0.422$	$\mu_{CR} = 0.045$
Diversity = 9.6147e+00	k_t = [4, 2, 4, 4]		
Generation 870: Best = 2.869248e+03	FES = 87100	$\mu_F = 0.424$	$\mu_{CR} = 0.045$
Diversity = 9.6130e+00	k_t = [1, 1, 1, 1]		
Generation 880: Best = 2.869248e+03	FES = 88100	$\mu_F = 0.375$	$\mu_{CR} = 0.043$
Diversity = 9.6048e+00	k_t = [4, 4, 3, 4]		
Generation 890: Best = 2.867930e+03	FES = 89100	$\mu_F = 0.369$	$\mu_{CR} = 0.039$
Diversity = 9.5987e+00	k_t = [1, 1, 1, 1]		
Generation 900: Best = 2.867930e+03	FES = 90100	$\mu_F = 0.414$	$\mu_{CR} = 0.041$
Diversity = 9.5320e+00	k_t = [4, 2, 2, 4]		
Generation 910: Best = 2.867930e+03	FES = 91100	$\mu_F = 0.422$	$\mu_{CR} = 0.039$
Diversity = 9.5249e+00	k_t = [1, 1, 1, 1]		
Generation 920: Best = 2.867930e+03	FES = 92100	$\mu_F = 0.394$	$\mu_{CR} = 0.038$
Diversity = 9.5251e+00	k_t = [4, 2, 4, 4]		
Generation 930: Best = 2.867930e+03	FES = 93100	$\mu_F = 0.341$	$\mu_{CR} = 0.039$
Diversity = 9.5220e+00	k_t = [1, 1, 1, 1]		
Generation 940: Best = 2.867930e+03	FES = 94100	$\mu_F = 0.326$	$\mu_{CR} = 0.046$
Diversity = 9.5168e+00	k_t = [3, 4, 4, 2]		
Generation 950: Best = 2.863629e+03	FES = 95100	$\mu_F = 0.353$	$\mu_{CR} = 0.042$
Diversity = 9.5089e+00	k_t = [1, 1, 1, 1]		
Generation 960: Best = 2.863217e+03	FES = 96100	$\mu_F = 0.356$	$\mu_{CR} = 0.039$
Diversity = 9.5480e+00	k_t = [3, 2, 2, 4]		
Generation 970: Best = 2.863217e+03	FES = 97100	$\mu_F = 0.359$	$\mu_{CR} = 0.048$
Diversity = 9.5409e+00	k_t = [1, 1, 1, 1]		
Generation 980: Best = 2.863217e+03	FES = 98100	$\mu_F = 0.416$	$\mu_{CR} = 0.054$
Diversity = 9.5679e+00	k_t = [2, 4, 4, 4]		
Generation 990: Best = 2.854163e+03	FES = 99100	$\mu_F = 0.371$	$\mu_{CR} = 0.039$
Diversity = 9.5666e+00	k_t = [1, 1, 1, 1]		
Generation 1000: Best = 2.854163e+03	FES = 100100	$\mu_F = 0.367$	$\mu_{CR} = 0.039$
Diversity = 9.5621e+00	k_t = [4, 3, 3, 2]		
Generation 1010: Best = 2.854163e+03	FES = 101100	$\mu_F = 0.436$	$\mu_{CR} = 0.037$
Diversity = 9.5606e+00	k_t = [1, 1, 1, 1]		
Generation 1020: Best = 2.854163e+03	FES = 102100	$\mu_F = 0.417$	$\mu_{CR} = 0.035$
Diversity = 9.5556e+00	k_t = [3, 3, 3, 3]		
Generation 1030: Best = 2.844590e+03	FES = 103100	$\mu_F = 0.436$	$\mu_{CR} = 0.026$
Diversity = 9.5635e+00	k_t = [1, 1, 1, 1]		

Generation 1040: Best = 2.844590e+03 FES = 104100 μ_F = 0.387 μ_{CR} = 0.033
 Diversity = 9.5593e+00 k_t = [4, 3, 3, 2]
 Generation 1050: Best = 2.844590e+03 FES = 105100 μ_F = 0.460 μ_{CR} = 0.030
 Diversity = 9.5741e+00 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 2.843207e+03 FES = 106100 μ_F = 0.495 μ_{CR} = 0.029
 Diversity = 9.5597e+00 k_t = [3, 3, 3, 4]
 Generation 1070: Best = 2.843207e+03 FES = 107100 μ_F = 0.464 μ_{CR} = 0.034
 Diversity = 9.5589e+00 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 2.843207e+03 FES = 108100 μ_F = 0.466 μ_{CR} = 0.028
 Diversity = 9.5011e+00 k_t = [4, 2, 2, 4]
 Generation 1090: Best = 2.833425e+03 FES = 109100 μ_F = 0.409 μ_{CR} = 0.046
 Diversity = 9.4927e+00 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 2.833425e+03 FES = 110100 μ_F = 0.423 μ_{CR} = 0.037
 Diversity = 9.4933e+00 k_t = [4, 4, 4, 3]
 Generation 1110: Best = 2.833425e+03 FES = 111100 μ_F = 0.477 μ_{CR} = 0.030
 Diversity = 9.4582e+00 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 2.833425e+03 FES = 112100 μ_F = 0.499 μ_{CR} = 0.028
 Diversity = 9.4575e+00 k_t = [4, 3, 2, 3]
 Generation 1130: Best = 2.833425e+03 FES = 113100 μ_F = 0.540 μ_{CR} = 0.028
 Diversity = 9.4539e+00 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 2.830390e+03 FES = 114100 μ_F = 0.512 μ_{CR} = 0.029
 Diversity = 9.4601e+00 k_t = [3, 3, 4, 3]
 Generation 1150: Best = 2.825545e+03 FES = 115100 μ_F = 0.511 μ_{CR} = 0.030
 Diversity = 9.4599e+00 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 2.825405e+03 FES = 116100 μ_F = 0.472 μ_{CR} = 0.036
 Diversity = 9.4888e+00 k_t = [4, 2, 4, 4]
 Generation 1170: Best = 2.825405e+03 FES = 117100 μ_F = 0.393 μ_{CR} = 0.035
 Diversity = 9.4889e+00 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 2.825405e+03 FES = 118100 μ_F = 0.453 μ_{CR} = 0.036
 Diversity = 9.4861e+00 k_t = [2, 3, 2, 2]
 Generation 1190: Best = 2.824779e+03 FES = 119100 μ_F = 0.458 μ_{CR} = 0.036
 Diversity = 9.4843e+00 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 2.824656e+03 FES = 120100 μ_F = 0.445 μ_{CR} = 0.038
 Diversity = 9.4810e+00 k_t = [3, 2, 3, 4]
 Generation 1210: Best = 2.823914e+03 FES = 121100 μ_F = 0.409 μ_{CR} = 0.041
 Diversity = 9.4743e+00 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 2.815281e+03 FES = 122100 μ_F = 0.367 μ_{CR} = 0.035
 Diversity = 9.4690e+00 k_t = [3, 4, 3, 3]
 Generation 1230: Best = 2.815281e+03 FES = 123100 μ_F = 0.291 μ_{CR} = 0.043
 Diversity = 9.4580e+00 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 2.815281e+03 FES = 124100 μ_F = 0.296 μ_{CR} = 0.045
 Diversity = 9.4597e+00 k_t = [3, 4, 2, 4]
 Generation 1250: Best = 2.812297e+03 FES = 125100 μ_F = 0.322 μ_{CR} = 0.045
 Diversity = 9.4595e+00 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 2.812297e+03 FES = 126100 μ_F = 0.370 μ_{CR} = 0.041
 Diversity = 9.4583e+00 k_t = [2, 4, 3, 2]
 Generation 1270: Best = 2.812297e+03 FES = 127100 μ_F = 0.337 μ_{CR} = 0.042
 Diversity = 9.4596e+00 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 2.812297e+03 FES = 128100 μ_F = 0.302 μ_{CR} = 0.045
 Diversity = 9.4564e+00 k_t = [2, 4, 4, 3]
 Generation 1290: Best = 2.812110e+03 FES = 129100 μ_F = 0.355 μ_{CR} = 0.039
 Diversity = 9.4545e+00 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 2.812110e+03 FES = 130100 μ_F = 0.382 μ_{CR} = 0.037
 Diversity = 9.3882e+00 k_t = [2, 3, 3, 4]

Generation 1310: Best = 2.808396e+03 FES = 131100 μ_F = 0.314 μ_{CR} = 0.041
 Diversity = 9.3899e+00 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 2.806201e+03 FES = 132100 μ_F = 0.322 μ_{CR} = 0.041
 Diversity = 9.3852e+00 k_t = [4, 2, 4, 3]
 Generation 1330: Best = 2.806201e+03 FES = 133100 μ_F = 0.272 μ_{CR} = 0.043
 Diversity = 9.3777e+00 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 2.800610e+03 FES = 134100 μ_F = 0.267 μ_{CR} = 0.052
 Diversity = 9.3733e+00 k_t = [2, 3, 4, 2]
 Generation 1350: Best = 2.800610e+03 FES = 135100 μ_F = 0.265 μ_{CR} = 0.054
 Diversity = 9.3702e+00 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 2.789426e+03 FES = 136100 μ_F = 0.282 μ_{CR} = 0.042
 Diversity = 9.3651e+00 k_t = [3, 2, 4, 2]
 Generation 1370: Best = 2.789426e+03 FES = 137100 μ_F = 0.232 μ_{CR} = 0.044
 Diversity = 9.3645e+00 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 2.783897e+03 FES = 138100 μ_F = 0.189 μ_{CR} = 0.053
 Diversity = 9.3557e+00 k_t = [2, 4, 3, 2]
 Generation 1390: Best = 2.783530e+03 FES = 139100 μ_F = 0.232 μ_{CR} = 0.054
 Diversity = 9.3506e+00 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 2.777943e+03 FES = 140100 μ_F = 0.238 μ_{CR} = 0.066
 Diversity = 9.3231e+00 k_t = [2, 2, 2, 2]
 Generation 1410: Best = 2.774556e+03 FES = 141100 μ_F = 0.272 μ_{CR} = 0.055
 Diversity = 9.3200e+00 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 2.765022e+03 FES = 142100 μ_F = 0.324 μ_{CR} = 0.044
 Diversity = 9.2795e+00 k_t = [2, 4, 4, 4]
 Generation 1430: Best = 2.765022e+03 FES = 143100 μ_F = 0.270 μ_{CR} = 0.039
 Diversity = 9.2771e+00 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 2.765022e+03 FES = 144100 μ_F = 0.261 μ_{CR} = 0.043
 Diversity = 9.2752e+00 k_t = [3, 3, 4, 3]
 Generation 1450: Best = 2.763674e+03 FES = 145100 μ_F = 0.317 μ_{CR} = 0.030
 Diversity = 9.2730e+00 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 2.763674e+03 FES = 146100 μ_F = 0.304 μ_{CR} = 0.033
 Diversity = 9.2683e+00 k_t = [4, 2, 2, 3]
 Generation 1470: Best = 2.762527e+03 FES = 147100 μ_F = 0.266 μ_{CR} = 0.040
 Diversity = 9.2626e+00 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 2.759208e+03 FES = 148100 μ_F = 0.302 μ_{CR} = 0.038
 Diversity = 9.2569e+00 k_t = [3, 3, 4, 4]
 Generation 1490: Best = 2.751502e+03 FES = 149100 μ_F = 0.321 μ_{CR} = 0.039
 Diversity = 9.2546e+00 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 2.751077e+03 FES = 150100 μ_F = 0.354 μ_{CR} = 0.030
 Diversity = 9.2852e+00 k_t = [3, 2, 4, 4]
 Generation 1510: Best = 2.745490e+03 FES = 151100 μ_F = 0.432 μ_{CR} = 0.032
 Diversity = 9.2862e+00 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 2.742636e+03 FES = 152100 μ_F = 0.456 μ_{CR} = 0.036
 Diversity = 9.3291e+00 k_t = [4, 4, 3, 4]
 Generation 1530: Best = 2.742636e+03 FES = 153100 μ_F = 0.416 μ_{CR} = 0.038
 Diversity = 9.3295e+00 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 2.742636e+03 FES = 154100 μ_F = 0.316 μ_{CR} = 0.037
 Diversity = 9.3297e+00 k_t = [3, 3, 2, 3]
 Generation 1550: Best = 2.738077e+03 FES = 155100 μ_F = 0.326 μ_{CR} = 0.029
 Diversity = 9.3203e+00 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 2.734149e+03 FES = 156100 μ_F = 0.312 μ_{CR} = 0.032
 Diversity = 9.3204e+00 k_t = [3, 4, 2, 2]
 Generation 1570: Best = 2.728929e+03 FES = 157100 μ_F = 0.269 μ_{CR} = 0.038
 Diversity = 9.3177e+00 k_t = [1, 1, 1, 1]

Generation 1580: Best = 2.728929e+03 FES = 158100 μ_F = 0.230 μ_{CR} = 0.045
 Diversity = 9.3148e+00 k_t = [2, 4, 4, 3]
 Generation 1590: Best = 2.727924e+03 FES = 159100 μ_F = 0.235 μ_{CR} = 0.042
 Diversity = 9.3094e+00 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 2.727924e+03 FES = 160100 μ_F = 0.238 μ_{CR} = 0.035
 Diversity = 9.3036e+00 k_t = [2, 4, 3, 4]
 Generation 1610: Best = 2.727924e+03 FES = 161100 μ_F = 0.276 μ_{CR} = 0.030
 Diversity = 9.2720e+00 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 2.727924e+03 FES = 162100 μ_F = 0.234 μ_{CR} = 0.033
 Diversity = 9.2708e+00 k_t = [4, 2, 2, 3]
 Generation 1630: Best = 2.727924e+03 FES = 163100 μ_F = 0.219 μ_{CR} = 0.042
 Diversity = 9.2706e+00 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 2.727574e+03 FES = 164100 μ_F = 0.182 μ_{CR} = 0.041
 Diversity = 9.2692e+00 k_t = [4, 3, 2, 2]
 Generation 1650: Best = 2.727574e+03 FES = 165100 μ_F = 0.229 μ_{CR} = 0.042
 Diversity = 9.2666e+00 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 2.722120e+03 FES = 166100 μ_F = 0.248 μ_{CR} = 0.036
 Diversity = 9.2633e+00 k_t = [3, 4, 2, 3]
 Generation 1670: Best = 2.722120e+03 FES = 167100 μ_F = 0.282 μ_{CR} = 0.048
 Diversity = 9.3084e+00 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 2.722120e+03 FES = 168100 μ_F = 0.320 μ_{CR} = 0.039
 Diversity = 9.3087e+00 k_t = [3, 3, 3, 3]
 Generation 1690: Best = 2.722120e+03 FES = 169100 μ_F = 0.310 μ_{CR} = 0.044
 Diversity = 9.3088e+00 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 2.715247e+03 FES = 170100 μ_F = 0.410 μ_{CR} = 0.045
 Diversity = 9.3029e+00 k_t = [3, 3, 4, 4]
 Generation 1710: Best = 2.713493e+03 FES = 171100 μ_F = 0.409 μ_{CR} = 0.044
 Diversity = 9.3022e+00 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 2.713493e+03 FES = 172100 μ_F = 0.397 μ_{CR} = 0.051
 Diversity = 9.3017e+00 k_t = [4, 2, 2, 3]
 Generation 1730: Best = 2.709233e+03 FES = 173100 μ_F = 0.380 μ_{CR} = 0.050
 Diversity = 9.3015e+00 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 2.708673e+03 FES = 174100 μ_F = 0.348 μ_{CR} = 0.048
 Diversity = 9.3012e+00 k_t = [4, 4, 3, 3]
 Generation 1750: Best = 2.702416e+03 FES = 175100 μ_F = 0.334 μ_{CR} = 0.054
 Diversity = 9.2996e+00 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 2.702416e+03 FES = 176100 μ_F = 0.321 μ_{CR} = 0.042
 Diversity = 9.2991e+00 k_t = [4, 3, 2, 3]
 Generation 1770: Best = 2.702318e+03 FES = 177100 μ_F = 0.316 μ_{CR} = 0.041
 Diversity = 9.2971e+00 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 2.699178e+03 FES = 178100 μ_F = 0.381 μ_{CR} = 0.031
 Diversity = 9.2944e+00 k_t = [2, 4, 3, 2]
 Generation 1790: Best = 2.699031e+03 FES = 179100 μ_F = 0.336 μ_{CR} = 0.034
 Diversity = 9.2934e+00 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 2.698976e+03 FES = 180100 μ_F = 0.325 μ_{CR} = 0.035
 Diversity = 9.3135e+00 k_t = [4, 2, 3, 4]
 Generation 1810: Best = 2.692472e+03 FES = 181100 μ_F = 0.297 μ_{CR} = 0.046
 Diversity = 9.3127e+00 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 2.691632e+03 FES = 182100 μ_F = 0.293 μ_{CR} = 0.047
 Diversity = 9.3094e+00 k_t = [2, 2, 3, 2]
 Generation 1830: Best = 2.689297e+03 FES = 183100 μ_F = 0.278 μ_{CR} = 0.045
 Diversity = 9.3098e+00 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 2.689297e+03 FES = 184100 μ_F = 0.398 μ_{CR} = 0.039
 Diversity = 9.2974e+00 k_t = [2, 2, 2, 4]

Generation 1850: Best = 2.689297e+03	FES = 185100	$\mu_F = 0.321$	$\mu_{CR} = 0.028$
Diversity = 9.2970e+00	$k_t = [1, 1, 1, 1]$		
Generation 1860: Best = 2.689297e+03	FES = 186100	$\mu_F = 0.363$	$\mu_{CR} = 0.038$
Diversity = 9.2929e+00	$k_t = [3, 2, 2, 2]$		
Generation 1870: Best = 2.689297e+03	FES = 187100	$\mu_F = 0.374$	$\mu_{CR} = 0.033$
Diversity = 9.2927e+00	$k_t = [1, 1, 1, 1]$		
Generation 1880: Best = 2.689297e+03	FES = 188100	$\mu_F = 0.345$	$\mu_{CR} = 0.036$
Diversity = 9.2918e+00	$k_t = [2, 4, 4, 4]$		
Generation 1890: Best = 2.685054e+03	FES = 189100	$\mu_F = 0.296$	$\mu_{CR} = 0.035$
Diversity = 9.2908e+00	$k_t = [1, 1, 1, 1]$		
Generation 1900: Best = 2.685038e+03	FES = 190100	$\mu_F = 0.330$	$\mu_{CR} = 0.038$
Diversity = 9.2902e+00	$k_t = [3, 2, 3, 4]$		
Generation 1910: Best = 2.683548e+03	FES = 191100	$\mu_F = 0.362$	$\mu_{CR} = 0.036$
Diversity = 9.2892e+00	$k_t = [1, 1, 1, 1]$		
Generation 1920: Best = 2.683548e+03	FES = 192100	$\mu_F = 0.377$	$\mu_{CR} = 0.038$
Diversity = 9.2812e+00	$k_t = [4, 2, 4, 2]$		
Generation 1930: Best = 2.683548e+03	FES = 193100	$\mu_F = 0.409$	$\mu_{CR} = 0.031$
Diversity = 9.2818e+00	$k_t = [1, 1, 1, 1]$		
Generation 1940: Best = 2.683548e+03	FES = 194100	$\mu_F = 0.350$	$\mu_{CR} = 0.036$
Diversity = 9.3007e+00	$k_t = [3, 3, 2, 2]$		
Generation 1950: Best = 2.683548e+03	FES = 195100	$\mu_F = 0.387$	$\mu_{CR} = 0.031$
Diversity = 9.3002e+00	$k_t = [1, 1, 1, 1]$		
Generation 1960: Best = 2.683422e+03	FES = 196100	$\mu_F = 0.400$	$\mu_{CR} = 0.024$
Diversity = 9.2985e+00	$k_t = [2, 2, 2, 3]$		
Generation 1970: Best = 2.682650e+03	FES = 197100	$\mu_F = 0.374$	$\mu_{CR} = 0.028$
Diversity = 9.2985e+00	$k_t = [1, 1, 1, 1]$		
Generation 1980: Best = 2.681047e+03	FES = 198100	$\mu_F = 0.370$	$\mu_{CR} = 0.025$
Diversity = 9.2976e+00	$k_t = [3, 3, 2, 2]$		
Generation 1990: Best = 2.681047e+03	FES = 199100	$\mu_F = 0.376$	$\mu_{CR} = 0.033$
Diversity = 9.2731e+00	$k_t = [1, 1, 1, 1]$		
Generation 2000: Best = 2.681047e+03	FES = 200100	$\mu_F = 0.457$	$\mu_{CR} = 0.029$
Diversity = 9.2590e+00	$k_t = [4, 4, 4, 2]$		
Generation 2010: Best = 2.680572e+03	FES = 201100	$\mu_F = 0.444$	$\mu_{CR} = 0.032$
Diversity = 9.2562e+00	$k_t = [1, 1, 1, 1]$		
Generation 2020: Best = 2.680572e+03	FES = 202100	$\mu_F = 0.407$	$\mu_{CR} = 0.041$
Diversity = 9.2547e+00	$k_t = [4, 2, 3, 3]$		
Generation 2030: Best = 2.680572e+03	FES = 203100	$\mu_F = 0.431$	$\mu_{CR} = 0.032$
Diversity = 9.2118e+00	$k_t = [1, 1, 1, 1]$		
Generation 2040: Best = 2.680572e+03	FES = 204100	$\mu_F = 0.435$	$\mu_{CR} = 0.027$
Diversity = 9.1299e+00	$k_t = [4, 3, 4, 4]$		
Generation 2050: Best = 2.680572e+03	FES = 205100	$\mu_F = 0.515$	$\mu_{CR} = 0.038$
Diversity = 9.0904e+00	$k_t = [1, 1, 1, 1]$		
Generation 2060: Best = 2.680572e+03	FES = 206100	$\mu_F = 0.559$	$\mu_{CR} = 0.039$
Diversity = 9.0763e+00	$k_t = [2, 3, 2, 4]$		
Generation 2070: Best = 2.680572e+03	FES = 207100	$\mu_F = 0.489$	$\mu_{CR} = 0.037$
Diversity = 9.0752e+00	$k_t = [1, 1, 1, 1]$		
Generation 2080: Best = 2.677661e+03	FES = 208100	$\mu_F = 0.523$	$\mu_{CR} = 0.030$
Diversity = 9.0755e+00	$k_t = [4, 4, 4, 2]$		
Generation 2090: Best = 2.677661e+03	FES = 209100	$\mu_F = 0.487$	$\mu_{CR} = 0.035$
Diversity = 9.0745e+00	$k_t = [1, 1, 1, 1]$		
Generation 2100: Best = 2.676366e+03	FES = 210100	$\mu_F = 0.434$	$\mu_{CR} = 0.035$
Diversity = 9.0741e+00	$k_t = [4, 3, 4, 3]$		
Generation 2110: Best = 2.674333e+03	FES = 211100	$\mu_F = 0.517$	$\mu_{CR} = 0.031$
Diversity = 9.0759e+00	$k_t = [1, 1, 1, 1]$		

Generation 2120: Best = 2.674333e+03 FES = 212100 μ_F = 0.520 μ_{CR} = 0.024
 Diversity = 9.0759e+00 k_t = [3, 3, 3, 3]
 Generation 2130: Best = 2.669935e+03 FES = 213100 μ_F = 0.519 μ_{CR} = 0.030
 Diversity = 9.0767e+00 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 2.661811e+03 FES = 214100 μ_F = 0.570 μ_{CR} = 0.046
 Diversity = 9.0771e+00 k_t = [2, 3, 2, 2]
 Generation 2150: Best = 2.661811e+03 FES = 215100 μ_F = 0.582 μ_{CR} = 0.045
 Diversity = 9.0772e+00 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 2.661811e+03 FES = 216100 μ_F = 0.620 μ_{CR} = 0.043
 Diversity = 9.0520e+00 k_t = [2, 4, 2, 4]
 Generation 2170: Best = 2.661811e+03 FES = 217100 μ_F = 0.597 μ_{CR} = 0.040
 Diversity = 8.9651e+00 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 2.661811e+03 FES = 218100 μ_F = 0.601 μ_{CR} = 0.031
 Diversity = 8.9649e+00 k_t = [2, 2, 3, 3]
 Generation 2190: Best = 2.660043e+03 FES = 219100 μ_F = 0.495 μ_{CR} = 0.041
 Diversity = 8.9631e+00 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 2.659341e+03 FES = 220100 μ_F = 0.560 μ_{CR} = 0.032
 Diversity = 8.9179e+00 k_t = [2, 2, 3, 2]
 Generation 2210: Best = 2.659341e+03 FES = 221100 μ_F = 0.476 μ_{CR} = 0.030
 Diversity = 8.9188e+00 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 2.648437e+03 FES = 222100 μ_F = 0.488 μ_{CR} = 0.045
 Diversity = 8.9190e+00 k_t = [2, 4, 3, 2]
 Generation 2230: Best = 2.648437e+03 FES = 223100 μ_F = 0.406 μ_{CR} = 0.037
 Diversity = 8.9194e+00 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 2.648437e+03 FES = 224100 μ_F = 0.351 μ_{CR} = 0.024
 Diversity = 8.9155e+00 k_t = [3, 4, 2, 3]
 Generation 2250: Best = 2.644288e+03 FES = 225100 μ_F = 0.386 μ_{CR} = 0.039
 Diversity = 8.9031e+00 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 2.640793e+03 FES = 226100 μ_F = 0.371 μ_{CR} = 0.033
 Diversity = 8.9005e+00 k_t = [3, 3, 2, 2]
 Generation 2270: Best = 2.635175e+03 FES = 227100 μ_F = 0.312 μ_{CR} = 0.045
 Diversity = 8.8391e+00 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 2.632857e+03 FES = 228100 μ_F = 0.396 μ_{CR} = 0.043
 Diversity = 8.8697e+00 k_t = [2, 2, 2, 4]
 Generation 2290: Best = 2.632857e+03 FES = 229100 μ_F = 0.453 μ_{CR} = 0.041
 Diversity = 8.8193e+00 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 2.632857e+03 FES = 230100 μ_F = 0.444 μ_{CR} = 0.035
 Diversity = 8.8189e+00 k_t = [4, 2, 2, 2]
 Generation 2310: Best = 2.632857e+03 FES = 231100 μ_F = 0.506 μ_{CR} = 0.026
 Diversity = 8.8187e+00 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 2.631488e+03 FES = 232100 μ_F = 0.584 μ_{CR} = 0.023
 Diversity = 8.7810e+00 k_t = [3, 4, 3, 4]
 Generation 2330: Best = 2.631488e+03 FES = 233100 μ_F = 0.586 μ_{CR} = 0.029
 Diversity = 8.7825e+00 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 2.631488e+03 FES = 234100 μ_F = 0.550 μ_{CR} = 0.031
 Diversity = 8.7834e+00 k_t = [3, 2, 2, 3]
 Generation 2350: Best = 2.631488e+03 FES = 235100 μ_F = 0.524 μ_{CR} = 0.031
 Diversity = 8.7840e+00 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 2.629865e+03 FES = 236100 μ_F = 0.597 μ_{CR} = 0.024
 Diversity = 8.7936e+00 k_t = [4, 4, 2, 4]
 Generation 2370: Best = 2.629865e+03 FES = 237100 μ_F = 0.523 μ_{CR} = 0.022
 Diversity = 8.7944e+00 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 2.624465e+03 FES = 238100 μ_F = 0.498 μ_{CR} = 0.043
 Diversity = 8.7956e+00 k_t = [4, 3, 4, 4]

Generation 2390: Best = 2.616956e+03 FES = 239100 μ_F = 0.447 μ_{CR} = 0.040
 Diversity = 8.7941e+00 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 2.616956e+03 FES = 240100 μ_F = 0.475 μ_{CR} = 0.040
 Diversity = 8.7618e+00 k_t = [4, 3, 4, 4]
 Generation 2410: Best = 2.616214e+03 FES = 241100 μ_F = 0.448 μ_{CR} = 0.050
 Diversity = 8.7604e+00 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 2.615771e+03 FES = 242100 μ_F = 0.436 μ_{CR} = 0.041
 Diversity = 8.7373e+00 k_t = [4, 4, 4, 4]
 Generation 2430: Best = 2.614483e+03 FES = 243100 μ_F = 0.459 μ_{CR} = 0.031
 Diversity = 8.7399e+00 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 2.611002e+03 FES = 244100 μ_F = 0.447 μ_{CR} = 0.057
 Diversity = 8.7401e+00 k_t = [4, 3, 2, 4]
 Generation 2450: Best = 2.610250e+03 FES = 245100 μ_F = 0.511 μ_{CR} = 0.041
 Diversity = 8.6648e+00 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 2.610250e+03 FES = 246100 μ_F = 0.533 μ_{CR} = 0.045
 Diversity = 8.6370e+00 k_t = [3, 2, 3, 4]
 Generation 2470: Best = 2.610250e+03 FES = 247100 μ_F = 0.474 μ_{CR} = 0.042
 Diversity = 8.6376e+00 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 2.608690e+03 FES = 248100 μ_F = 0.362 μ_{CR} = 0.058
 Diversity = 8.6367e+00 k_t = [3, 3, 2, 3]
 Generation 2490: Best = 2.608690e+03 FES = 249100 μ_F = 0.309 μ_{CR} = 0.062
 Diversity = 8.6356e+00 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 2.608690e+03 FES = 250100 μ_F = 0.325 μ_{CR} = 0.045
 Diversity = 8.6366e+00 k_t = [3, 2, 4, 3]
 Generation 2510: Best = 2.608690e+03 FES = 251100 μ_F = 0.370 μ_{CR} = 0.044
 Diversity = 8.6390e+00 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 2.608690e+03 FES = 252100 μ_F = 0.382 μ_{CR} = 0.038
 Diversity = 8.5865e+00 k_t = [3, 3, 3, 4]
 Generation 2530: Best = 2.608690e+03 FES = 253100 μ_F = 0.421 μ_{CR} = 0.040
 Diversity = 8.5994e+00 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 2.608636e+03 FES = 254100 μ_F = 0.414 μ_{CR} = 0.051
 Diversity = 8.5821e+00 k_t = [4, 2, 4, 4]
 Generation 2550: Best = 2.608636e+03 FES = 255100 μ_F = 0.387 μ_{CR} = 0.046
 Diversity = 8.5933e+00 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 2.605577e+03 FES = 256100 μ_F = 0.381 μ_{CR} = 0.047
 Diversity = 8.5813e+00 k_t = [3, 2, 2, 4]
 Generation 2570: Best = 2.605577e+03 FES = 257100 μ_F = 0.404 μ_{CR} = 0.042
 Diversity = 8.5828e+00 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 2.604520e+03 FES = 258100 μ_F = 0.351 μ_{CR} = 0.067
 Diversity = 8.5436e+00 k_t = [4, 2, 4, 3]
 Generation 2590: Best = 2.603775e+03 FES = 259100 μ_F = 0.391 μ_{CR} = 0.060
 Diversity = 8.5037e+00 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 2.603775e+03 FES = 260100 μ_F = 0.419 μ_{CR} = 0.059
 Diversity = 8.4454e+00 k_t = [4, 2, 4, 3]
 Generation 2610: Best = 2.602673e+03 FES = 261100 μ_F = 0.455 μ_{CR} = 0.081
 Diversity = 8.4482e+00 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 2.601540e+03 FES = 262100 μ_F = 0.439 μ_{CR} = 0.065
 Diversity = 8.4473e+00 k_t = [4, 2, 2, 3]
 Generation 2630: Best = 2.600899e+03 FES = 263100 μ_F = 0.475 μ_{CR} = 0.072
 Diversity = 8.4521e+00 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 2.598465e+03 FES = 264100 μ_F = 0.583 μ_{CR} = 0.074
 Diversity = 8.3318e+00 k_t = [4, 4, 2, 3]
 Generation 2650: Best = 2.597786e+03 FES = 265100 μ_F = 0.651 μ_{CR} = 0.088
 Diversity = 8.3095e+00 k_t = [1, 1, 1, 1]

Generation 2660: Best = 2.595192e+03 FES = 266100 μ_F = 0.708 μ_{CR} = 0.069
 Diversity = 8.2714e+00 k_t = [4, 2, 4, 2]
 Generation 2670: Best = 2.592811e+03 FES = 267100 μ_F = 0.651 μ_{CR} = 0.075
 Diversity = 8.2690e+00 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 2.592283e+03 FES = 268100 μ_F = 0.647 μ_{CR} = 0.077
 Diversity = 8.2560e+00 k_t = [4, 2, 3, 2]
 Generation 2690: Best = 2.587319e+03 FES = 269100 μ_F = 0.726 μ_{CR} = 0.105
 Diversity = 8.1973e+00 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 2.585914e+03 FES = 270100 μ_F = 0.707 μ_{CR} = 0.100
 Diversity = 8.1463e+00 k_t = [4, 2, 4, 3]
 Generation 2710: Best = 2.585914e+03 FES = 271100 μ_F = 0.700 μ_{CR} = 0.097
 Diversity = 8.0986e+00 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 2.583998e+03 FES = 272100 μ_F = 0.709 μ_{CR} = 0.118
 Diversity = 8.0406e+00 k_t = [2, 2, 4, 2]
 Generation 2730: Best = 2.583998e+03 FES = 273100 μ_F = 0.699 μ_{CR} = 0.129
 Diversity = 8.0297e+00 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 2.582635e+03 FES = 274100 μ_F = 0.708 μ_{CR} = 0.110
 Diversity = 8.0470e+00 k_t = [3, 4, 3, 2]
 Generation 2750: Best = 2.579427e+03 FES = 275100 μ_F = 0.737 μ_{CR} = 0.114
 Diversity = 7.9504e+00 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 2.578947e+03 FES = 276100 μ_F = 0.740 μ_{CR} = 0.107
 Diversity = 7.9482e+00 k_t = [4, 4, 3, 3]
 Generation 2770: Best = 2.575570e+03 FES = 277100 μ_F = 0.780 μ_{CR} = 0.144
 Diversity = 7.9318e+00 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 2.568127e+03 FES = 278100 μ_F = 0.773 μ_{CR} = 0.146
 Diversity = 7.8243e+00 k_t = [4, 3, 4, 3]
 Generation 2790: Best = 2.567754e+03 FES = 279100 μ_F = 0.795 μ_{CR} = 0.165
 Diversity = 7.6897e+00 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 2.563720e+03 FES = 280100 μ_F = 0.779 μ_{CR} = 0.180
 Diversity = 7.4323e+00 k_t = [2, 3, 3, 3]
 Generation 2810: Best = 2.558831e+03 FES = 281100 μ_F = 0.779 μ_{CR} = 0.194
 Diversity = 7.0558e+00 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 2.557055e+03 FES = 282100 μ_F = 0.799 μ_{CR} = 0.191
 Diversity = 5.6362e+00 k_t = [4, 4, 2, 4]
 Generation 2830: Best = 2.554098e+03 FES = 283100 μ_F = 0.802 μ_{CR} = 0.197
 Diversity = 4.1028e+00 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 2.551581e+03 FES = 284100 μ_F = 0.800 μ_{CR} = 0.193
 Diversity = 9.7883e-01 k_t = [2, 3, 4, 2]
 Generation 2850: Best = 2.548144e+03 FES = 285100 μ_F = 0.761 μ_{CR} = 0.192
 Diversity = 3.2623e-01 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 2.546761e+03 FES = 286100 μ_F = 0.721 μ_{CR} = 0.187
 Diversity = 2.6884e-01 k_t = [4, 2, 2, 2]
 Generation 2870: Best = 2.544327e+03 FES = 287100 μ_F = 0.690 μ_{CR} = 0.198
 Diversity = 2.1488e-01 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 2.543548e+03 FES = 288100 μ_F = 0.677 μ_{CR} = 0.197
 Diversity = 1.6918e-01 k_t = [2, 3, 2, 4]
 Generation 2890: Best = 2.542302e+03 FES = 289100 μ_F = 0.662 μ_{CR} = 0.199
 Diversity = 1.3198e-01 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 2.541904e+03 FES = 290100 μ_F = 0.665 μ_{CR} = 0.202
 Diversity = 1.0097e-01 k_t = [4, 4, 4, 4]
 Generation 2910: Best = 2.541265e+03 FES = 291100 μ_F = 0.659 μ_{CR} = 0.207
 Diversity = 7.6397e-02 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 2.540820e+03 FES = 292100 μ_F = 0.670 μ_{CR} = 0.201
 Diversity = 5.8731e-02 k_t = [4, 4, 3, 4]

Generation 2930: Best = 2.540536e+03 FES = 293100 μ_F = 0.678 μ_{CR} = 0.207
 Diversity = 5.0677e-02 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 2.540231e+03 FES = 294100 μ_F = 0.689 μ_{CR} = 0.213
 Diversity = 4.6561e-02 k_t = [2, 2, 3, 4]
 Generation 2950: Best = 2.540096e+03 FES = 295100 μ_F = 0.705 μ_{CR} = 0.205
 Diversity = 4.1685e-02 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 2.539902e+03 FES = 296100 μ_F = 0.716 μ_{CR} = 0.204
 Diversity = 3.8514e-02 k_t = [4, 3, 3, 4]
 Generation 2970: Best = 2.539732e+03 FES = 297100 μ_F = 0.718 μ_{CR} = 0.204
 Diversity = 3.8028e-02 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 2.539553e+03 FES = 298100 μ_F = 0.712 μ_{CR} = 0.208
 Diversity = 3.5161e-02 k_t = [2, 4, 4, 4]
 Generation 2990: Best = 2.539501e+03 FES = 299100 μ_F = 0.712 μ_{CR} = 0.203
 Diversity = 3.3376e-02 k_t = [1, 1, 1, 1]
 Generation 3000: Best = 2.539373e+03 FES = 300100 μ_F = 0.713 μ_{CR} = 0.204
 Diversity = 3.1462e-02 k_t = [2, 3, 3, 3]
 Generation 3010: Best = 2.539257e+03 FES = 301100 μ_F = 0.710 μ_{CR} = 0.204
 Diversity = 3.2420e-02 k_t = [1, 1, 1, 1]
 Generation 3020: Best = 2.539165e+03 FES = 302100 μ_F = 0.708 μ_{CR} = 0.204
 Diversity = 2.8570e-02 k_t = [4, 4, 3, 3]
 Generation 3030: Best = 2.539068e+03 FES = 303100 μ_F = 0.698 μ_{CR} = 0.204
 Diversity = 2.6966e-02 k_t = [1, 1, 1, 1]
 Generation 3040: Best = 2.539002e+03 FES = 304100 μ_F = 0.701 μ_{CR} = 0.212
 Diversity = 2.6068e-02 k_t = [4, 3, 4, 2]
 Generation 3050: Best = 2.538902e+03 FES = 305100 μ_F = 0.712 μ_{CR} = 0.215
 Diversity = 2.6138e-02 k_t = [1, 1, 1, 1]
 Generation 3060: Best = 2.538816e+03 FES = 306100 μ_F = 0.706 μ_{CR} = 0.223
 Diversity = 2.4540e-02 k_t = [4, 2, 4, 2]
 Generation 3070: Best = 2.538744e+03 FES = 307100 μ_F = 0.715 μ_{CR} = 0.223
 Diversity = 2.4590e-02 k_t = [1, 1, 1, 1]
 Generation 3080: Best = 2.538689e+03 FES = 308100 μ_F = 0.712 μ_{CR} = 0.232
 Diversity = 2.3019e-02 k_t = [2, 4, 2, 3]
 Generation 3090: Best = 2.538617e+03 FES = 309100 μ_F = 0.710 μ_{CR} = 0.241
 Diversity = 2.3004e-02 k_t = [1, 1, 1, 1]
 Generation 3100: Best = 2.538570e+03 FES = 310100 μ_F = 0.715 μ_{CR} = 0.235
 Diversity = 2.0511e-02 k_t = [2, 3, 4, 4]
 Generation 3110: Best = 2.538523e+03 FES = 311100 μ_F = 0.718 μ_{CR} = 0.237
 Diversity = 1.9415e-02 k_t = [1, 1, 1, 1]
 Generation 3120: Best = 2.538497e+03 FES = 312100 μ_F = 0.707 μ_{CR} = 0.230
 Diversity = 1.7152e-02 k_t = [2, 4, 4, 3]
 Generation 3130: Best = 2.538455e+03 FES = 313100 μ_F = 0.713 μ_{CR} = 0.225
 Diversity = 1.7379e-02 k_t = [1, 1, 1, 1]
 Generation 3140: Best = 2.538431e+03 FES = 314100 μ_F = 0.719 μ_{CR} = 0.222
 Diversity = 1.7385e-02 k_t = [2, 2, 4, 2]
 Generation 3150: Best = 2.538412e+03 FES = 315100 μ_F = 0.707 μ_{CR} = 0.211
 Diversity = 1.6689e-02 k_t = [1, 1, 1, 1]
 Generation 3160: Best = 2.538387e+03 FES = 316100 μ_F = 0.704 μ_{CR} = 0.215
 Diversity = 1.6310e-02 k_t = [2, 3, 3, 2]
 Generation 3170: Best = 2.538366e+03 FES = 317100 μ_F = 0.700 μ_{CR} = 0.213
 Diversity = 1.5285e-02 k_t = [1, 1, 1, 1]
 Generation 3180: Best = 2.538332e+03 FES = 318100 μ_F = 0.709 μ_{CR} = 0.219
 Diversity = 1.4754e-02 k_t = [4, 3, 3, 2]
 Generation 3190: Best = 2.538316e+03 FES = 319100 μ_F = 0.730 μ_{CR} = 0.208
 Diversity = 1.5346e-02 k_t = [1, 1, 1, 1]

Generation 3200: Best = 2.538292e+03 FES = 320100 μ_F = 0.736 μ_{CR} = 0.204
 Diversity = 1.3756e-02 k_t = [4, 4, 4, 3]
 Generation 3210: Best = 2.538271e+03 FES = 321100 μ_F = 0.746 μ_{CR} = 0.213
 Diversity = 1.3096e-02 k_t = [1, 1, 1, 1]
 Generation 3220: Best = 2.538248e+03 FES = 322100 μ_F = 0.743 μ_{CR} = 0.210
 Diversity = 1.2715e-02 k_t = [4, 2, 2, 3]
 Generation 3230: Best = 2.538219e+03 FES = 323100 μ_F = 0.746 μ_{CR} = 0.212
 Diversity = 1.2989e-02 k_t = [1, 1, 1, 1]
 Generation 3240: Best = 2.538213e+03 FES = 324100 μ_F = 0.742 μ_{CR} = 0.197
 Diversity = 1.2506e-02 k_t = [3, 3, 2, 3]
 Generation 3250: Best = 2.538201e+03 FES = 325100 μ_F = 0.734 μ_{CR} = 0.198
 Diversity = 1.2100e-02 k_t = [1, 1, 1, 1]
 Generation 3260: Best = 2.538178e+03 FES = 326100 μ_F = 0.731 μ_{CR} = 0.195
 Diversity = 1.1267e-02 k_t = [4, 4, 2, 3]
 Generation 3270: Best = 2.538165e+03 FES = 327100 μ_F = 0.729 μ_{CR} = 0.199
 Diversity = 1.0990e-02 k_t = [1, 1, 1, 1]
 Generation 3280: Best = 2.538153e+03 FES = 328100 μ_F = 0.726 μ_{CR} = 0.207
 Diversity = 1.0188e-02 k_t = [3, 3, 4, 4]
 Generation 3290: Best = 2.538142e+03 FES = 329100 μ_F = 0.732 μ_{CR} = 0.212
 Diversity = 9.6723e-03 k_t = [1, 1, 1, 1]
 Generation 3300: Best = 2.538137e+03 FES = 330100 μ_F = 0.715 μ_{CR} = 0.217
 Diversity = 9.1975e-03 k_t = [4, 2, 2, 2]
 Generation 3310: Best = 2.538130e+03 FES = 331100 μ_F = 0.710 μ_{CR} = 0.208
 Diversity = 8.8437e-03 k_t = [1, 1, 1, 1]
 Generation 3320: Best = 2.538121e+03 FES = 332100 μ_F = 0.715 μ_{CR} = 0.205
 Diversity = 8.3351e-03 k_t = [2, 3, 2, 3]
 Generation 3330: Best = 2.538109e+03 FES = 333100 μ_F = 0.723 μ_{CR} = 0.216
 Diversity = 9.0494e-03 k_t = [1, 1, 1, 1]
 Generation 3340: Best = 2.538101e+03 FES = 334100 μ_F = 0.729 μ_{CR} = 0.219
 Diversity = 8.3170e-03 k_t = [3, 4, 3, 3]
 Generation 3350: Best = 2.538090e+03 FES = 335100 μ_F = 0.722 μ_{CR} = 0.226
 Diversity = 8.0396e-03 k_t = [1, 1, 1, 1]
 Generation 3360: Best = 2.538085e+03 FES = 336100 μ_F = 0.715 μ_{CR} = 0.231
 Diversity = 8.1782e-03 k_t = [3, 4, 2, 2]
 Generation 3370: Best = 2.538077e+03 FES = 337100 μ_F = 0.716 μ_{CR} = 0.232
 Diversity = 8.5623e-03 k_t = [1, 1, 1, 1]
 Generation 3380: Best = 2.538070e+03 FES = 338100 μ_F = 0.710 μ_{CR} = 0.230
 Diversity = 7.5715e-03 k_t = [2, 3, 4, 2]
 Generation 3390: Best = 2.538064e+03 FES = 339100 μ_F = 0.711 μ_{CR} = 0.240
 Diversity = 7.0459e-03 k_t = [1, 1, 1, 1]
 Generation 3400: Best = 2.538059e+03 FES = 340100 μ_F = 0.708 μ_{CR} = 0.247
 Diversity = 6.4207e-03 k_t = [2, 4, 4, 4]
 Generation 3410: Best = 2.538055e+03 FES = 341100 μ_F = 0.702 μ_{CR} = 0.248
 Diversity = 6.1677e-03 k_t = [1, 1, 1, 1]
 Generation 3420: Best = 2.538050e+03 FES = 342100 μ_F = 0.693 μ_{CR} = 0.251
 Diversity = 5.8041e-03 k_t = [3, 4, 2, 2]
 Generation 3430: Best = 2.538046e+03 FES = 343100 μ_F = 0.697 μ_{CR} = 0.250
 Diversity = 5.4421e-03 k_t = [1, 1, 1, 1]
 Generation 3440: Best = 2.538040e+03 FES = 344100 μ_F = 0.717 μ_{CR} = 0.255
 Diversity = 5.5135e-03 k_t = [2, 4, 4, 2]
 Generation 3450: Best = 2.538036e+03 FES = 345100 μ_F = 0.717 μ_{CR} = 0.257
 Diversity = 5.6871e-03 k_t = [1, 1, 1, 1]
 Generation 3460: Best = 2.538031e+03 FES = 346100 μ_F = 0.726 μ_{CR} = 0.268
 Diversity = 5.4418e-03 k_t = [3, 3, 3, 3]

Generation 3470: Best = 2.538027e+03 FES = 347100 μ_F = 0.720 μ_{CR} = 0.268
 Diversity = 5.7936e-03 k_t = [1, 1, 1, 1]
 Generation 3480: Best = 2.538022e+03 FES = 348100 μ_F = 0.708 μ_{CR} = 0.265
 Diversity = 5.6833e-03 k_t = [3, 2, 2, 2]
 Generation 3490: Best = 2.538020e+03 FES = 349100 μ_F = 0.701 μ_{CR} = 0.257
 Diversity = 5.5829e-03 k_t = [1, 1, 1, 1]
 Generation 3500: Best = 2.538017e+03 FES = 350100 μ_F = 0.705 μ_{CR} = 0.250
 Diversity = 5.1158e-03 k_t = [4, 2, 3, 3]
 Generation 3510: Best = 2.538014e+03 FES = 351100 μ_F = 0.705 μ_{CR} = 0.238
 Diversity = 4.8004e-03 k_t = [1, 1, 1, 1]
 Generation 3520: Best = 2.538012e+03 FES = 352100 μ_F = 0.703 μ_{CR} = 0.240
 Diversity = 4.5690e-03 k_t = [3, 2, 2, 2]
 Generation 3530: Best = 2.538010e+03 FES = 353100 μ_F = 0.712 μ_{CR} = 0.227
 Diversity = 4.3743e-03 k_t = [1, 1, 1, 1]
 Generation 3540: Best = 2.538009e+03 FES = 354100 μ_F = 0.722 μ_{CR} = 0.226
 Diversity = 3.9696e-03 k_t = [4, 4, 3, 4]
 Generation 3550: Best = 2.538006e+03 FES = 355100 μ_F = 0.721 μ_{CR} = 0.219
 Diversity = 3.7252e-03 k_t = [1, 1, 1, 1]
 Generation 3560: Best = 2.538004e+03 FES = 356100 μ_F = 0.729 μ_{CR} = 0.215
 Diversity = 3.8485e-03 k_t = [2, 3, 3, 4]
 Generation 3570: Best = 2.538001e+03 FES = 357100 μ_F = 0.716 μ_{CR} = 0.217
 Diversity = 3.8111e-03 k_t = [1, 1, 1, 1]
 Generation 3580: Best = 2.538001e+03 FES = 358100 μ_F = 0.708 μ_{CR} = 0.213
 Diversity = 3.7630e-03 k_t = [2, 2, 3, 2]
 Generation 3590: Best = 2.537998e+03 FES = 359100 μ_F = 0.708 μ_{CR} = 0.209
 Diversity = 4.1464e-03 k_t = [1, 1, 1, 1]
 Generation 3600: Best = 2.537996e+03 FES = 360100 μ_F = 0.709 μ_{CR} = 0.202
 Diversity = 4.2249e-03 k_t = [3, 3, 2, 3]
 Generation 3610: Best = 2.537992e+03 FES = 361100 μ_F = 0.695 μ_{CR} = 0.209
 Diversity = 4.2215e-03 k_t = [1, 1, 1, 1]
 Generation 3620: Best = 2.537991e+03 FES = 362100 μ_F = 0.695 μ_{CR} = 0.217
 Diversity = 4.0046e-03 k_t = [2, 3, 4, 3]
 Generation 3630: Best = 2.537989e+03 FES = 363100 μ_F = 0.700 μ_{CR} = 0.220
 Diversity = 3.8587e-03 k_t = [1, 1, 1, 1]
 Generation 3640: Best = 2.537988e+03 FES = 364100 μ_F = 0.703 μ_{CR} = 0.226
 Diversity = 3.6859e-03 k_t = [2, 2, 3, 4]
 Generation 3650: Best = 2.537987e+03 FES = 365100 μ_F = 0.704 μ_{CR} = 0.209
 Diversity = 3.4815e-03 k_t = [1, 1, 1, 1]
 Generation 3660: Best = 2.537985e+03 FES = 366100 μ_F = 0.705 μ_{CR} = 0.208
 Diversity = 3.2701e-03 k_t = [3, 4, 4, 2]
 Generation 3670: Best = 2.537984e+03 FES = 367100 μ_F = 0.703 μ_{CR} = 0.208
 Diversity = 3.2576e-03 k_t = [1, 1, 1, 1]
 Generation 3680: Best = 2.537983e+03 FES = 368100 μ_F = 0.710 μ_{CR} = 0.205
 Diversity = 2.9456e-03 k_t = [4, 3, 3, 2]
 Generation 3690: Best = 2.537982e+03 FES = 369100 μ_F = 0.711 μ_{CR} = 0.205
 Diversity = 2.7467e-03 k_t = [1, 1, 1, 1]
 Generation 3700: Best = 2.537981e+03 FES = 370100 μ_F = 0.719 μ_{CR} = 0.207
 Diversity = 2.6083e-03 k_t = [4, 3, 2, 3]
 Generation 3710: Best = 2.537980e+03 FES = 371100 μ_F = 0.735 μ_{CR} = 0.209
 Diversity = 2.6642e-03 k_t = [1, 1, 1, 1]
 Generation 3720: Best = 2.537979e+03 FES = 372100 μ_F = 0.741 μ_{CR} = 0.204
 Diversity = 2.6571e-03 k_t = [2, 2, 4, 3]
 Generation 3730: Best = 2.537978e+03 FES = 373100 μ_F = 0.737 μ_{CR} = 0.209
 Diversity = 2.7027e-03 k_t = [1, 1, 1, 1]

Generation 3740: Best = 2.537977e+03 FES = 374100 μ_F = 0.739 μ_{CR} = 0.207
 Diversity = 2.6360e-03 k_t = [3, 4, 3, 2]
 Generation 3750: Best = 2.537976e+03 FES = 375100 μ_F = 0.732 μ_{CR} = 0.214
 Diversity = 2.5852e-03 k_t = [1, 1, 1, 1]
 Generation 3760: Best = 2.537975e+03 FES = 376100 μ_F = 0.721 μ_{CR} = 0.217
 Diversity = 2.4610e-03 k_t = [3, 2, 3, 2]
 Generation 3770: Best = 2.537975e+03 FES = 377100 μ_F = 0.713 μ_{CR} = 0.223
 Diversity = 2.5455e-03 k_t = [1, 1, 1, 1]
 Generation 3780: Best = 2.537974e+03 FES = 378100 μ_F = 0.714 μ_{CR} = 0.225
 Diversity = 2.5057e-03 k_t = [4, 4, 2, 2]
 Generation 3790: Best = 2.537973e+03 FES = 379100 μ_F = 0.716 μ_{CR} = 0.223
 Diversity = 2.3708e-03 k_t = [1, 1, 1, 1]
 Generation 3800: Best = 2.537973e+03 FES = 380100 μ_F = 0.703 μ_{CR} = 0.219
 Diversity = 2.1909e-03 k_t = [2, 2, 2, 2]
 Generation 3810: Best = 2.537972e+03 FES = 381100 μ_F = 0.709 μ_{CR} = 0.221
 Diversity = 2.2709e-03 k_t = [1, 1, 1, 1]
 Generation 3820: Best = 2.537971e+03 FES = 382100 μ_F = 0.727 μ_{CR} = 0.218
 Diversity = 2.2208e-03 k_t = [2, 2, 2, 3]
 Generation 3830: Best = 2.537970e+03 FES = 383100 μ_F = 0.708 μ_{CR} = 0.222
 Diversity = 2.2147e-03 k_t = [1, 1, 1, 1]
 Generation 3840: Best = 2.537970e+03 FES = 384100 μ_F = 0.707 μ_{CR} = 0.218
 Diversity = 2.1038e-03 k_t = [4, 2, 2, 4]
 Generation 3850: Best = 2.537969e+03 FES = 385100 μ_F = 0.710 μ_{CR} = 0.225
 Diversity = 2.0115e-03 k_t = [1, 1, 1, 1]
 Generation 3860: Best = 2.537969e+03 FES = 386100 μ_F = 0.714 μ_{CR} = 0.224
 Diversity = 1.7850e-03 k_t = [2, 2, 3, 4]
 Generation 3870: Best = 2.537969e+03 FES = 387100 μ_F = 0.706 μ_{CR} = 0.229
 Diversity = 1.7433e-03 k_t = [1, 1, 1, 1]
 Generation 3880: Best = 2.537968e+03 FES = 388100 μ_F = 0.711 μ_{CR} = 0.226
 Diversity = 1.6385e-03 k_t = [4, 2, 2, 3]
 Generation 3890: Best = 2.537968e+03 FES = 389100 μ_F = 0.712 μ_{CR} = 0.226
 Diversity = 1.5546e-03 k_t = [1, 1, 1, 1]
 Generation 3900: Best = 2.537968e+03 FES = 390100 μ_F = 0.711 μ_{CR} = 0.215
 Diversity = 1.5362e-03 k_t = [3, 2, 2, 2]
 Generation 3910: Best = 2.537967e+03 FES = 391100 μ_F = 0.722 μ_{CR} = 0.218
 Diversity = 1.5994e-03 k_t = [1, 1, 1, 1]
 Generation 3920: Best = 2.537967e+03 FES = 392100 μ_F = 0.724 μ_{CR} = 0.220
 Diversity = 1.5332e-03 k_t = [2, 3, 4, 3]
 Generation 3930: Best = 2.537967e+03 FES = 393100 μ_F = 0.716 μ_{CR} = 0.216
 Diversity = 1.5224e-03 k_t = [1, 1, 1, 1]
 Generation 3940: Best = 2.537966e+03 FES = 394100 μ_F = 0.723 μ_{CR} = 0.211
 Diversity = 1.4460e-03 k_t = [2, 3, 3, 3]
 Generation 3950: Best = 2.537966e+03 FES = 395100 μ_F = 0.732 μ_{CR} = 0.198
 Diversity = 1.4480e-03 k_t = [1, 1, 1, 1]
 Generation 3960: Best = 2.537966e+03 FES = 396100 μ_F = 0.733 μ_{CR} = 0.207
 Diversity = 1.5814e-03 k_t = [4, 3, 2, 2]
 Generation 3970: Best = 2.537965e+03 FES = 397100 μ_F = 0.736 μ_{CR} = 0.204
 Diversity = 1.6701e-03 k_t = [1, 1, 1, 1]
 Generation 3980: Best = 2.537965e+03 FES = 398100 μ_F = 0.735 μ_{CR} = 0.209
 Diversity = 1.6808e-03 k_t = [2, 2, 2, 2]
 Generation 3990: Best = 2.537965e+03 FES = 399100 μ_F = 0.717 μ_{CR} = 0.206
 Diversity = 1.7782e-03 k_t = [1, 1, 1, 1]
 Generation 4000: Best = 2.537964e+03 FES = 400100 μ_F = 0.719 μ_{CR} = 0.212
 Diversity = 1.6328e-03 k_t = [2, 3, 3, 4]

Generation 4010: Best = 2.537964e+03 FES = 401100 μ_F = 0.710 μ_{CR} = 0.213
 Diversity = 1.6001e-03 k_t = [1, 1, 1, 1]
 Generation 4020: Best = 2.537964e+03 FES = 402100 μ_F = 0.705 μ_{CR} = 0.212
 Diversity = 1.4710e-03 k_t = [2, 3, 3, 4]
 Generation 4030: Best = 2.537964e+03 FES = 403100 μ_F = 0.709 μ_{CR} = 0.214
 Diversity = 1.5240e-03 k_t = [1, 1, 1, 1]
 Generation 4040: Best = 2.537963e+03 FES = 404100 μ_F = 0.711 μ_{CR} = 0.220
 Diversity = 1.4703e-03 k_t = [2, 4, 4, 2]
 Generation 4050: Best = 2.537963e+03 FES = 405100 μ_F = 0.713 μ_{CR} = 0.223
 Diversity = 1.3911e-03 k_t = [1, 1, 1, 1]
 Generation 4060: Best = 2.537963e+03 FES = 406100 μ_F = 0.703 μ_{CR} = 0.223
 Diversity = 1.3267e-03 k_t = [2, 3, 4, 2]
 Generation 4070: Best = 2.537963e+03 FES = 407100 μ_F = 0.693 μ_{CR} = 0.216
 Diversity = 1.3006e-03 k_t = [1, 1, 1, 1]
 Generation 4080: Best = 2.537963e+03 FES = 408100 μ_F = 0.708 μ_{CR} = 0.212
 Diversity = 1.2244e-03 k_t = [2, 4, 4, 4]
 Generation 4090: Best = 2.537962e+03 FES = 409100 μ_F = 0.697 μ_{CR} = 0.205
 Diversity = 1.1470e-03 k_t = [1, 1, 1, 1]
 Generation 4100: Best = 2.537962e+03 FES = 410100 μ_F = 0.697 μ_{CR} = 0.207
 Diversity = 9.6201e-04 k_t = [4, 4, 2, 3]
 Generation 4110: Best = 2.537962e+03 FES = 411100 μ_F = 0.693 μ_{CR} = 0.205
 Diversity = 8.5540e-04 k_t = [1, 1, 1, 1]
 Generation 4120: Best = 2.537962e+03 FES = 412100 μ_F = 0.698 μ_{CR} = 0.211
 Diversity = 7.7985e-04 k_t = [4, 2, 2, 4]
 Generation 4130: Best = 2.537962e+03 FES = 413100 μ_F = 0.703 μ_{CR} = 0.218
 Diversity = 7.7694e-04 k_t = [1, 1, 1, 1]
 Generation 4140: Best = 2.537962e+03 FES = 414100 μ_F = 0.714 μ_{CR} = 0.222
 Diversity = 7.4946e-04 k_t = [3, 2, 4, 4]
 Generation 4150: Best = 2.537962e+03 FES = 415100 μ_F = 0.719 μ_{CR} = 0.227
 Diversity = 7.6843e-04 k_t = [1, 1, 1, 1]
 Generation 4160: Best = 2.537962e+03 FES = 416100 μ_F = 0.718 μ_{CR} = 0.234
 Diversity = 7.7564e-04 k_t = [3, 3, 3, 3]
 Generation 4170: Best = 2.537962e+03 FES = 417100 μ_F = 0.714 μ_{CR} = 0.231
 Diversity = 7.7860e-04 k_t = [1, 1, 1, 1]
 Generation 4180: Best = 2.537962e+03 FES = 418100 μ_F = 0.724 μ_{CR} = 0.231
 Diversity = 7.9345e-04 k_t = [4, 4, 3, 3]
 Generation 4190: Best = 2.537962e+03 FES = 419100 μ_F = 0.726 μ_{CR} = 0.227
 Diversity = 8.2662e-04 k_t = [1, 1, 1, 1]
 Generation 4200: Best = 2.537962e+03 FES = 420100 μ_F = 0.740 μ_{CR} = 0.237
 Diversity = 7.7036e-04 k_t = [2, 4, 4, 4]
 Generation 4210: Best = 2.537961e+03 FES = 421100 μ_F = 0.737 μ_{CR} = 0.235
 Diversity = 7.7614e-04 k_t = [1, 1, 1, 1]
 Generation 4220: Best = 2.537961e+03 FES = 422100 μ_F = 0.738 μ_{CR} = 0.243
 Diversity = 8.5431e-04 k_t = [2, 4, 3, 2]
 Generation 4230: Best = 2.537961e+03 FES = 423100 μ_F = 0.750 μ_{CR} = 0.245
 Diversity = 9.5163e-04 k_t = [1, 1, 1, 1]
 Generation 4240: Best = 2.537961e+03 FES = 424100 μ_F = 0.749 μ_{CR} = 0.241
 Diversity = 9.6217e-04 k_t = [3, 3, 3, 3]
 Generation 4250: Best = 2.537961e+03 FES = 425100 μ_F = 0.747 μ_{CR} = 0.251
 Diversity = 1.1108e-03 k_t = [1, 1, 1, 1]
 Generation 4260: Best = 2.537961e+03 FES = 426100 μ_F = 0.734 μ_{CR} = 0.246
 Diversity = 1.0405e-03 k_t = [2, 4, 2, 3]
 Generation 4270: Best = 2.537961e+03 FES = 427100 μ_F = 0.729 μ_{CR} = 0.240
 Diversity = 9.9816e-04 k_t = [1, 1, 1, 1]

Generation 4280: Best = 2.537961e+03 FES = 428100 μ_F = 0.701 μ_{CR} = 0.239
 Diversity = 8.6543e-04 k_t = [3, 2, 3, 3]
 Generation 4290: Best = 2.537961e+03 FES = 429100 μ_F = 0.705 μ_{CR} = 0.235
 Diversity = 8.8612e-04 k_t = [1, 1, 1, 1]
 Generation 4300: Best = 2.537960e+03 FES = 430100 μ_F = 0.714 μ_{CR} = 0.239
 Diversity = 8.2208e-04 k_t = [3, 4, 3, 4]
 Generation 4310: Best = 2.537960e+03 FES = 431100 μ_F = 0.705 μ_{CR} = 0.245
 Diversity = 8.0601e-04 k_t = [1, 1, 1, 1]
 Generation 4320: Best = 2.537960e+03 FES = 432100 μ_F = 0.702 μ_{CR} = 0.237
 Diversity = 7.2695e-04 k_t = [3, 4, 4, 3]
 Generation 4330: Best = 2.537960e+03 FES = 433100 μ_F = 0.701 μ_{CR} = 0.240
 Diversity = 6.6520e-04 k_t = [1, 1, 1, 1]
 Generation 4340: Best = 2.537960e+03 FES = 434100 μ_F = 0.709 μ_{CR} = 0.242
 Diversity = 6.6074e-04 k_t = [4, 2, 3, 3]
 Generation 4350: Best = 2.537960e+03 FES = 435100 μ_F = 0.691 μ_{CR} = 0.253
 Diversity = 6.2691e-04 k_t = [1, 1, 1, 1]
 Generation 4360: Best = 2.537960e+03 FES = 436100 μ_F = 0.691 μ_{CR} = 0.243
 Diversity = 6.7372e-04 k_t = [4, 3, 3, 2]
 Generation 4370: Best = 2.537960e+03 FES = 437100 μ_F = 0.694 μ_{CR} = 0.249
 Diversity = 6.8940e-04 k_t = [1, 1, 1, 1]
 Generation 4380: Best = 2.537960e+03 FES = 438100 μ_F = 0.700 μ_{CR} = 0.241
 Diversity = 6.5078e-04 k_t = [2, 4, 2, 2]
 Generation 4390: Best = 2.537960e+03 FES = 439100 μ_F = 0.718 μ_{CR} = 0.238
 Diversity = 6.5963e-04 k_t = [1, 1, 1, 1]
 Generation 4400: Best = 2.537960e+03 FES = 440100 μ_F = 0.717 μ_{CR} = 0.231
 Diversity = 6.2143e-04 k_t = [4, 3, 4, 4]
 Generation 4410: Best = 2.537960e+03 FES = 441100 μ_F = 0.716 μ_{CR} = 0.234
 Diversity = 5.7862e-04 k_t = [1, 1, 1, 1]
 Generation 4420: Best = 2.537960e+03 FES = 442100 μ_F = 0.714 μ_{CR} = 0.239
 Diversity = 5.4916e-04 k_t = [4, 4, 4, 3]
 Generation 4430: Best = 2.537960e+03 FES = 443100 μ_F = 0.702 μ_{CR} = 0.235
 Diversity = 5.2900e-04 k_t = [1, 1, 1, 1]
 Generation 4440: Best = 2.537960e+03 FES = 444100 μ_F = 0.696 μ_{CR} = 0.240
 Diversity = 4.8910e-04 k_t = [2, 2, 4, 3]
 Generation 4450: Best = 2.537960e+03 FES = 445100 μ_F = 0.703 μ_{CR} = 0.241
 Diversity = 5.0832e-04 k_t = [1, 1, 1, 1]
 Generation 4460: Best = 2.537959e+03 FES = 446100 μ_F = 0.694 μ_{CR} = 0.240
 Diversity = 4.8317e-04 k_t = [3, 2, 3, 2]
 Generation 4470: Best = 2.537959e+03 FES = 447100 μ_F = 0.694 μ_{CR} = 0.241
 Diversity = 4.6009e-04 k_t = [1, 1, 1, 1]
 Generation 4480: Best = 2.537959e+03 FES = 448100 μ_F = 0.712 μ_{CR} = 0.248
 Diversity = 4.5729e-04 k_t = [3, 3, 2, 4]
 Generation 4490: Best = 2.537959e+03 FES = 449100 μ_F = 0.720 μ_{CR} = 0.252
 Diversity = 5.1348e-04 k_t = [1, 1, 1, 1]
 Generation 4500: Best = 2.537959e+03 FES = 450100 μ_F = 0.728 μ_{CR} = 0.244
 Diversity = 5.3317e-04 k_t = [2, 2, 3, 2]
 Generation 4510: Best = 2.537959e+03 FES = 451100 μ_F = 0.738 μ_{CR} = 0.251
 Diversity = 5.3234e-04 k_t = [1, 1, 1, 1]
 Generation 4520: Best = 2.537959e+03 FES = 452100 μ_F = 0.734 μ_{CR} = 0.249
 Diversity = 4.9119e-04 k_t = [3, 3, 2, 4]
 Generation 4530: Best = 2.537959e+03 FES = 453100 μ_F = 0.700 μ_{CR} = 0.236
 Diversity = 4.6028e-04 k_t = [1, 1, 1, 1]
 Generation 4540: Best = 2.537959e+03 FES = 454100 μ_F = 0.698 μ_{CR} = 0.238
 Diversity = 4.3770e-04 k_t = [3, 2, 4, 3]

Generation 4550: Best = 2.537959e+03 FES = 455100 μ_F = 0.709 μ_{CR} = 0.240
 Diversity = 4.4397e-04 k_t = [1, 1, 1, 1]
 Generation 4560: Best = 2.537959e+03 FES = 456100 μ_F = 0.719 μ_{CR} = 0.244
 Diversity = 3.9583e-04 k_t = [4, 2, 2, 4]
 Generation 4570: Best = 2.537959e+03 FES = 457100 μ_F = 0.729 μ_{CR} = 0.247
 Diversity = 4.0554e-04 k_t = [1, 1, 1, 1]
 Generation 4580: Best = 2.537959e+03 FES = 458100 μ_F = 0.724 μ_{CR} = 0.247
 Diversity = 4.0577e-04 k_t = [2, 4, 2, 3]
 Generation 4590: Best = 2.537959e+03 FES = 459100 μ_F = 0.728 μ_{CR} = 0.240
 Diversity = 4.2424e-04 k_t = [1, 1, 1, 1]
 Generation 4600: Best = 2.537959e+03 FES = 460100 μ_F = 0.730 μ_{CR} = 0.238
 Diversity = 4.1700e-04 k_t = [4, 2, 4, 3]
 Generation 4610: Best = 2.537959e+03 FES = 461100 μ_F = 0.735 μ_{CR} = 0.238
 Diversity = 4.3295e-04 k_t = [1, 1, 1, 1]
 Generation 4620: Best = 2.537959e+03 FES = 462100 μ_F = 0.730 μ_{CR} = 0.240
 Diversity = 3.8569e-04 k_t = [2, 3, 2, 3]
 Generation 4630: Best = 2.537959e+03 FES = 463100 μ_F = 0.721 μ_{CR} = 0.243
 Diversity = 3.8872e-04 k_t = [1, 1, 1, 1]
 Generation 4640: Best = 2.537959e+03 FES = 464100 μ_F = 0.709 μ_{CR} = 0.246
 Diversity = 3.3906e-04 k_t = [3, 4, 4, 3]
 Generation 4650: Best = 2.537959e+03 FES = 465100 μ_F = 0.713 μ_{CR} = 0.250
 Diversity = 3.2079e-04 k_t = [1, 1, 1, 1]
 Generation 4660: Best = 2.537959e+03 FES = 466100 μ_F = 0.708 μ_{CR} = 0.254
 Diversity = 2.9636e-04 k_t = [2, 2, 4, 2]
 Generation 4670: Best = 2.537959e+03 FES = 467100 μ_F = 0.701 μ_{CR} = 0.254
 Diversity = 2.9765e-04 k_t = [1, 1, 1, 1]
 Generation 4680: Best = 2.537959e+03 FES = 468100 μ_F = 0.704 μ_{CR} = 0.250
 Diversity = 2.8192e-04 k_t = [2, 4, 2, 2]
 Generation 4690: Best = 2.537959e+03 FES = 469100 μ_F = 0.711 μ_{CR} = 0.240
 Diversity = 2.8406e-04 k_t = [1, 1, 1, 1]
 Generation 4700: Best = 2.537959e+03 FES = 470100 μ_F = 0.704 μ_{CR} = 0.243
 Diversity = 2.5857e-04 k_t = [2, 3, 4, 4]
 Generation 4710: Best = 2.537959e+03 FES = 471100 μ_F = 0.693 μ_{CR} = 0.243
 Diversity = 2.3048e-04 k_t = [1, 1, 1, 1]
 Generation 4720: Best = 2.537959e+03 FES = 472100 μ_F = 0.690 μ_{CR} = 0.246
 Diversity = 2.1557e-04 k_t = [3, 2, 4, 4]
 Generation 4730: Best = 2.537959e+03 FES = 473100 μ_F = 0.723 μ_{CR} = 0.246
 Diversity = 2.3101e-04 k_t = [1, 1, 1, 1]
 Generation 4740: Best = 2.537959e+03 FES = 474100 μ_F = 0.717 μ_{CR} = 0.244
 Diversity = 2.4180e-04 k_t = [4, 2, 4, 2]
 Generation 4750: Best = 2.537959e+03 FES = 475100 μ_F = 0.719 μ_{CR} = 0.244
 Diversity = 2.6873e-04 k_t = [1, 1, 1, 1]
 Generation 4760: Best = 2.537959e+03 FES = 476100 μ_F = 0.726 μ_{CR} = 0.248
 Diversity = 2.7442e-04 k_t = [3, 2, 3, 4]
 Generation 4770: Best = 2.537959e+03 FES = 477100 μ_F = 0.730 μ_{CR} = 0.242
 Diversity = 2.6063e-04 k_t = [1, 1, 1, 1]
 Generation 4780: Best = 2.537959e+03 FES = 478100 μ_F = 0.742 μ_{CR} = 0.241
 Diversity = 2.4460e-04 k_t = [4, 3, 2, 4]
 Generation 4790: Best = 2.537959e+03 FES = 479100 μ_F = 0.749 μ_{CR} = 0.239
 Diversity = 2.3024e-04 k_t = [1, 1, 1, 1]
 Generation 4800: Best = 2.537959e+03 FES = 480100 μ_F = 0.744 μ_{CR} = 0.235
 Diversity = 2.1011e-04 k_t = [2, 4, 3, 3]
 Generation 4810: Best = 2.537959e+03 FES = 481100 μ_F = 0.743 μ_{CR} = 0.234
 Diversity = 2.2536e-04 k_t = [1, 1, 1, 1]

Generation 4820: Best = 2.537959e+03 FES = 482100 μ_F = 0.730 μ_{CR} = 0.227
 Diversity = 2.0927e-04 k_t = [3, 3, 4, 4]
 Generation 4830: Best = 2.537959e+03 FES = 483100 μ_F = 0.721 μ_{CR} = 0.232
 Diversity = 2.0750e-04 k_t = [1, 1, 1, 1]
 Generation 4840: Best = 2.537959e+03 FES = 484100 μ_F = 0.719 μ_{CR} = 0.236
 Diversity = 2.0564e-04 k_t = [4, 2, 2, 4]
 Generation 4850: Best = 2.537959e+03 FES = 485100 μ_F = 0.712 μ_{CR} = 0.241
 Diversity = 2.1447e-04 k_t = [1, 1, 1, 1]
 Generation 4860: Best = 2.537959e+03 FES = 486100 μ_F = 0.710 μ_{CR} = 0.247
 Diversity = 2.0268e-04 k_t = [3, 2, 2, 3]
 Generation 4870: Best = 2.537959e+03 FES = 487100 μ_F = 0.711 μ_{CR} = 0.252
 Diversity = 2.1082e-04 k_t = [1, 1, 1, 1]
 Generation 4880: Best = 2.537959e+03 FES = 488100 μ_F = 0.730 μ_{CR} = 0.251
 Diversity = 2.0181e-04 k_t = [4, 2, 2, 3]
 Generation 4890: Best = 2.537959e+03 FES = 489100 μ_F = 0.719 μ_{CR} = 0.256
 Diversity = 2.0958e-04 k_t = [1, 1, 1, 1]
 Generation 4900: Best = 2.537959e+03 FES = 490100 μ_F = 0.720 μ_{CR} = 0.249
 Diversity = 1.9501e-04 k_t = [4, 4, 4, 2]
 Generation 4910: Best = 2.537959e+03 FES = 491100 μ_F = 0.726 μ_{CR} = 0.244
 Diversity = 1.8182e-04 k_t = [1, 1, 1, 1]
 Generation 4920: Best = 2.537959e+03 FES = 492100 μ_F = 0.713 μ_{CR} = 0.247
 Diversity = 1.6740e-04 k_t = [3, 4, 2, 2]
 Generation 4930: Best = 2.537959e+03 FES = 493100 μ_F = 0.719 μ_{CR} = 0.247
 Diversity = 1.7136e-04 k_t = [1, 1, 1, 1]
 Generation 4940: Best = 2.537959e+03 FES = 494100 μ_F = 0.714 μ_{CR} = 0.241
 Diversity = 1.5560e-04 k_t = [4, 2, 2, 4]
 Generation 4950: Best = 2.537959e+03 FES = 495100 μ_F = 0.727 μ_{CR} = 0.239
 Diversity = 1.5926e-04 k_t = [1, 1, 1, 1]
 Generation 4960: Best = 2.537959e+03 FES = 496100 μ_F = 0.729 μ_{CR} = 0.245
 Diversity = 1.6013e-04 k_t = [4, 3, 4, 3]
 Generation 4970: Best = 2.537959e+03 FES = 497100 μ_F = 0.731 μ_{CR} = 0.246
 Diversity = 1.7461e-04 k_t = [1, 1, 1, 1]
 Generation 4980: Best = 2.537959e+03 FES = 498100 μ_F = 0.729 μ_{CR} = 0.251
 Diversity = 1.7212e-04 k_t = [4, 2, 3, 3]
 Generation 4990: Best = 2.537959e+03 FES = 499100 μ_F = 0.721 μ_{CR} = 0.254
 Diversity = 1.7444e-04 k_t = [1, 1, 1, 1]
 Generation 5000: Best = 2.537959e+03 FES = 500100 μ_F = 0.713 μ_{CR} = 0.246
 Diversity = 1.6261e-04 k_t = [2, 2, 2, 2]
 Generation 5010: Best = 2.537959e+03 FES = 501100 μ_F = 0.697 μ_{CR} = 0.244
 Diversity = 1.5990e-04 k_t = [1, 1, 1, 1]
 Generation 5020: Best = 2.537959e+03 FES = 502100 μ_F = 0.714 μ_{CR} = 0.246
 Diversity = 1.6268e-04 k_t = [2, 4, 2, 2]
 Generation 5030: Best = 2.537959e+03 FES = 503100 μ_F = 0.714 μ_{CR} = 0.249
 Diversity = 1.7085e-04 k_t = [1, 1, 1, 1]
 Generation 5040: Best = 2.537959e+03 FES = 504100 μ_F = 0.712 μ_{CR} = 0.251
 Diversity = 1.6564e-04 k_t = [2, 2, 3, 2]
 Generation 5050: Best = 2.537959e+03 FES = 505100 μ_F = 0.720 μ_{CR} = 0.246
 Diversity = 1.6769e-04 k_t = [1, 1, 1, 1]
 Generation 5060: Best = 2.537959e+03 FES = 506100 μ_F = 0.724 μ_{CR} = 0.246
 Diversity = 1.6175e-04 k_t = [3, 2, 2, 4]
 Generation 5070: Best = 2.537959e+03 FES = 507100 μ_F = 0.726 μ_{CR} = 0.240
 Diversity = 1.6287e-04 k_t = [1, 1, 1, 1]
 Generation 5080: Best = 2.537959e+03 FES = 508100 μ_F = 0.726 μ_{CR} = 0.244
 Diversity = 1.5092e-04 k_t = [2, 3, 3, 3]

Generation 5090: Best = 2.537959e+03 FES = 509100 μ_F = 0.718 μ_{CR} = 0.241
 Diversity = 1.5708e-04 k_t = [1, 1, 1, 1]
 Generation 5100: Best = 2.537959e+03 FES = 510100 μ_F = 0.706 μ_{CR} = 0.234
 Diversity = 1.4070e-04 k_t = [3, 4, 4, 4]
 Generation 5110: Best = 2.537959e+03 FES = 511100 μ_F = 0.715 μ_{CR} = 0.236
 Diversity = 1.3597e-04 k_t = [1, 1, 1, 1]
 Generation 5120: Best = 2.537959e+03 FES = 512100 μ_F = 0.719 μ_{CR} = 0.239
 Diversity = 1.3127e-04 k_t = [2, 4, 4, 2]
 Generation 5130: Best = 2.537959e+03 FES = 513100 μ_F = 0.711 μ_{CR} = 0.234
 Diversity = 1.2971e-04 k_t = [1, 1, 1, 1]
 Generation 5140: Best = 2.537959e+03 FES = 514100 μ_F = 0.707 μ_{CR} = 0.245
 Diversity = 1.3443e-04 k_t = [4, 4, 2, 2]
 Generation 5150: Best = 2.537959e+03 FES = 515100 μ_F = 0.700 μ_{CR} = 0.241
 Diversity = 1.2591e-04 k_t = [1, 1, 1, 1]
 Generation 5160: Best = 2.537959e+03 FES = 516100 μ_F = 0.703 μ_{CR} = 0.233
 Diversity = 1.1365e-04 k_t = [3, 2, 4, 3]
 Generation 5170: Best = 2.537959e+03 FES = 517100 μ_F = 0.710 μ_{CR} = 0.243
 Diversity = 1.1993e-04 k_t = [1, 1, 1, 1]
 Generation 5180: Best = 2.537959e+03 FES = 518100 μ_F = 0.717 μ_{CR} = 0.252
 Diversity = 1.1750e-04 k_t = [4, 4, 3, 2]
 Generation 5190: Best = 2.537959e+03 FES = 519100 μ_F = 0.723 μ_{CR} = 0.258
 Diversity = 1.2485e-04 k_t = [1, 1, 1, 1]
 Generation 5200: Best = 2.537959e+03 FES = 520100 μ_F = 0.727 μ_{CR} = 0.250
 Diversity = 1.2248e-04 k_t = [4, 2, 4, 3]
 Generation 5210: Best = 2.537959e+03 FES = 521100 μ_F = 0.721 μ_{CR} = 0.257
 Diversity = 1.3430e-04 k_t = [1, 1, 1, 1]
 Generation 5220: Best = 2.537959e+03 FES = 522100 μ_F = 0.733 μ_{CR} = 0.257
 Diversity = 1.3911e-04 k_t = [4, 3, 2, 3]
 Generation 5230: Best = 2.537959e+03 FES = 523100 μ_F = 0.740 μ_{CR} = 0.259
 Diversity = 1.7033e-04 k_t = [1, 1, 1, 1]
 Generation 5240: Best = 2.537959e+03 FES = 524100 μ_F = 0.736 μ_{CR} = 0.253
 Diversity = 1.5909e-04 k_t = [4, 4, 3, 2]
 Generation 5250: Best = 2.537959e+03 FES = 525100 μ_F = 0.729 μ_{CR} = 0.253
 Diversity = 1.6634e-04 k_t = [1, 1, 1, 1]
 Generation 5260: Best = 2.537959e+03 FES = 526100 μ_F = 0.732 μ_{CR} = 0.256
 Diversity = 1.5110e-04 k_t = [2, 3, 2, 4]
 Generation 5270: Best = 2.537959e+03 FES = 527100 μ_F = 0.724 μ_{CR} = 0.257
 Diversity = 1.6211e-04 k_t = [1, 1, 1, 1]
 Generation 5280: Best = 2.537959e+03 FES = 528100 μ_F = 0.715 μ_{CR} = 0.256
 Diversity = 1.5012e-04 k_t = [2, 3, 3, 2]
 Generation 5290: Best = 2.537959e+03 FES = 529100 μ_F = 0.702 μ_{CR} = 0.264
 Diversity = 1.3516e-04 k_t = [1, 1, 1, 1]
 Generation 5300: Best = 2.537959e+03 FES = 530100 μ_F = 0.691 μ_{CR} = 0.257
 Diversity = 1.2191e-04 k_t = [3, 2, 4, 2]
 Generation 5310: Best = 2.537959e+03 FES = 531100 μ_F = 0.696 μ_{CR} = 0.258
 Diversity = 1.2129e-04 k_t = [1, 1, 1, 1]
 Generation 5320: Best = 2.537959e+03 FES = 532100 μ_F = 0.693 μ_{CR} = 0.262
 Diversity = 1.1259e-04 k_t = [2, 3, 4, 4]
 Generation 5330: Best = 2.537959e+03 FES = 533100 μ_F = 0.682 μ_{CR} = 0.262
 Diversity = 1.0828e-04 k_t = [1, 1, 1, 1]
 Generation 5340: Best = 2.537959e+03 FES = 534100 μ_F = 0.699 μ_{CR} = 0.272
 Diversity = 1.0265e-04 k_t = [3, 4, 4, 4]
 Generation 5350: Best = 2.537959e+03 FES = 535100 μ_F = 0.690 μ_{CR} = 0.271
 Diversity = 9.2045e-05 k_t = [1, 1, 1, 1]

Generation 5360: Best = 2.537959e+03 FES = 536100 μ_F = 0.702 μ_{CR} = 0.262
 Diversity = 8.7010e-05 k_t = [3, 3, 3, 3]
 Generation 5370: Best = 2.537959e+03 FES = 537100 μ_F = 0.706 μ_{CR} = 0.249
 Diversity = 8.6113e-05 k_t = [1, 1, 1, 1]
 Generation 5380: Best = 2.537959e+03 FES = 538100 μ_F = 0.720 μ_{CR} = 0.250
 Diversity = 8.4788e-05 k_t = [2, 2, 3, 2]
 Generation 5390: Best = 2.537959e+03 FES = 539100 μ_F = 0.714 μ_{CR} = 0.241
 Diversity = 8.3837e-05 k_t = [1, 1, 1, 1]
 Generation 5400: Best = 2.537959e+03 FES = 540100 μ_F = 0.723 μ_{CR} = 0.243
 Diversity = 7.7937e-05 k_t = [3, 3, 3, 2]
 Generation 5410: Best = 2.537959e+03 FES = 541100 μ_F = 0.719 μ_{CR} = 0.261
 Diversity = 7.9600e-05 k_t = [1, 1, 1, 1]
 Generation 5420: Best = 2.537959e+03 FES = 542100 μ_F = 0.703 μ_{CR} = 0.270
 Diversity = 7.8046e-05 k_t = [4, 4, 2, 3]
 Generation 5430: Best = 2.537959e+03 FES = 543100 μ_F = 0.714 μ_{CR} = 0.266
 Diversity = 8.5671e-05 k_t = [1, 1, 1, 1]
 Generation 5440: Best = 2.537959e+03 FES = 544100 μ_F = 0.716 μ_{CR} = 0.266
 Diversity = 8.4437e-05 k_t = [4, 3, 2, 3]
 Generation 5450: Best = 2.537959e+03 FES = 545100 μ_F = 0.727 μ_{CR} = 0.266
 Diversity = 9.1799e-05 k_t = [1, 1, 1, 1]
 Generation 5460: Best = 2.537959e+03 FES = 546100 μ_F = 0.729 μ_{CR} = 0.271
 Diversity = 1.0138e-04 k_t = [2, 2, 4, 2]
 Generation 5470: Best = 2.537959e+03 FES = 547100 μ_F = 0.722 μ_{CR} = 0.283
 Diversity = 1.2665e-04 k_t = [1, 1, 1, 1]
 Generation 5480: Best = 2.537959e+03 FES = 548100 μ_F = 0.716 μ_{CR} = 0.283
 Diversity = 1.2483e-04 k_t = [3, 3, 3, 2]
 Generation 5490: Best = 2.537959e+03 FES = 549100 μ_F = 0.712 μ_{CR} = 0.287
 Diversity = 1.3043e-04 k_t = [1, 1, 1, 1]
 Generation 5500: Best = 2.537959e+03 FES = 550100 μ_F = 0.721 μ_{CR} = 0.285
 Diversity = 1.1622e-04 k_t = [3, 3, 3, 4]
 Generation 5510: Best = 2.537959e+03 FES = 551100 μ_F = 0.733 μ_{CR} = 0.285
 Diversity = 1.1720e-04 k_t = [1, 1, 1, 1]
 Generation 5520: Best = 2.537959e+03 FES = 552100 μ_F = 0.727 μ_{CR} = 0.286
 Diversity = 1.0783e-04 k_t = [4, 4, 3, 4]
 Generation 5530: Best = 2.537959e+03 FES = 553100 μ_F = 0.736 μ_{CR} = 0.283
 Diversity = 9.8800e-05 k_t = [1, 1, 1, 1]
 Generation 5540: Best = 2.537959e+03 FES = 554100 μ_F = 0.724 μ_{CR} = 0.288
 Diversity = 9.2596e-05 k_t = [3, 2, 4, 4]
 Generation 5550: Best = 2.537959e+03 FES = 555100 μ_F = 0.738 μ_{CR} = 0.290
 Diversity = 1.0712e-04 k_t = [1, 1, 1, 1]
 Generation 5560: Best = 2.537959e+03 FES = 556100 μ_F = 0.749 μ_{CR} = 0.291
 Diversity = 1.1359e-04 k_t = [3, 2, 3, 4]
 Generation 5570: Best = 2.537959e+03 FES = 557100 μ_F = 0.740 μ_{CR} = 0.298
 Diversity = 1.1566e-04 k_t = [1, 1, 1, 1]
 Generation 5580: Best = 2.537959e+03 FES = 558100 μ_F = 0.727 μ_{CR} = 0.301
 Diversity = 1.0036e-04 k_t = [2, 2, 2, 4]
 Generation 5590: Best = 2.537959e+03 FES = 559100 μ_F = 0.724 μ_{CR} = 0.306
 Diversity = 1.0282e-04 k_t = [1, 1, 1, 1]
 Generation 5600: Best = 2.537959e+03 FES = 560100 μ_F = 0.726 μ_{CR} = 0.307
 Diversity = 1.0100e-04 k_t = [2, 3, 4, 3]
 Generation 5610: Best = 2.537959e+03 FES = 561100 μ_F = 0.717 μ_{CR} = 0.306
 Diversity = 9.9175e-05 k_t = [1, 1, 1, 1]
 Generation 5620: Best = 2.537959e+03 FES = 562100 μ_F = 0.725 μ_{CR} = 0.302
 Diversity = 9.4170e-05 k_t = [4, 4, 4, 2]

Generation 5630: Best = 2.537959e+03 FES = 563100 μ_F = 0.716 μ_{CR} = 0.295
 Diversity = 8.5927e-05 k_t = [1, 1, 1, 1]
 Generation 5640: Best = 2.537959e+03 FES = 564100 μ_F = 0.728 μ_{CR} = 0.284
 Diversity = 8.3557e-05 k_t = [4, 4, 3, 2]
 Generation 5650: Best = 2.537959e+03 FES = 565100 μ_F = 0.719 μ_{CR} = 0.288
 Diversity = 7.7702e-05 k_t = [1, 1, 1, 1]
 Generation 5660: Best = 2.537959e+03 FES = 566100 μ_F = 0.706 μ_{CR} = 0.283
 Diversity = 6.8872e-05 k_t = [4, 3, 4, 2]
 Generation 5670: Best = 2.537959e+03 FES = 567100 μ_F = 0.701 μ_{CR} = 0.286
 Diversity = 6.0263e-05 k_t = [1, 1, 1, 1]
 Generation 5680: Best = 2.537959e+03 FES = 568100 μ_F = 0.689 μ_{CR} = 0.271
 Diversity = 5.0484e-05 k_t = [3, 3, 2, 2]
 Generation 5690: Best = 2.537959e+03 FES = 569100 μ_F = 0.689 μ_{CR} = 0.273
 Diversity = 4.7062e-05 k_t = [1, 1, 1, 1]
 Generation 5700: Best = 2.537959e+03 FES = 570100 μ_F = 0.697 μ_{CR} = 0.272
 Diversity = 4.3462e-05 k_t = [4, 4, 2, 2]
 Generation 5710: Best = 2.537959e+03 FES = 571100 μ_F = 0.702 μ_{CR} = 0.269
 Diversity = 4.0805e-05 k_t = [1, 1, 1, 1]
 Generation 5720: Best = 2.537959e+03 FES = 572100 μ_F = 0.686 μ_{CR} = 0.278
 Diversity = 3.5964e-05 k_t = [3, 4, 2, 4]
 Generation 5730: Best = 2.537959e+03 FES = 573100 μ_F = 0.698 μ_{CR} = 0.273
 Diversity = 3.3251e-05 k_t = [1, 1, 1, 1]
 Generation 5740: Best = 2.537959e+03 FES = 574100 μ_F = 0.697 μ_{CR} = 0.267
 Diversity = 2.9316e-05 k_t = [4, 4, 3, 3]
 Generation 5750: Best = 2.537959e+03 FES = 575100 μ_F = 0.691 μ_{CR} = 0.274
 Diversity = 2.6430e-05 k_t = [1, 1, 1, 1]
 Generation 5760: Best = 2.537959e+03 FES = 576100 μ_F = 0.692 μ_{CR} = 0.268
 Diversity = 2.4178e-05 k_t = [4, 4, 2, 3]
 Generation 5770: Best = 2.537959e+03 FES = 577100 μ_F = 0.709 μ_{CR} = 0.269
 Diversity = 2.2665e-05 k_t = [1, 1, 1, 1]
 Generation 5780: Best = 2.537959e+03 FES = 578100 μ_F = 0.708 μ_{CR} = 0.265
 Diversity = 2.1381e-05 k_t = [3, 3, 3, 3]
 Generation 5790: Best = 2.537959e+03 FES = 579100 μ_F = 0.718 μ_{CR} = 0.264
 Diversity = 2.7870e-05 k_t = [1, 1, 1, 1]
 Generation 5800: Best = 2.537959e+03 FES = 580100 μ_F = 0.718 μ_{CR} = 0.258
 Diversity = 2.7594e-05 k_t = [4, 2, 2, 2]
 Generation 5810: Best = 2.537959e+03 FES = 581100 μ_F = 0.714 μ_{CR} = 0.251
 Diversity = 2.8705e-05 k_t = [1, 1, 1, 1]
 Generation 5820: Best = 2.537959e+03 FES = 582100 μ_F = 0.709 μ_{CR} = 0.239
 Diversity = 2.5441e-05 k_t = [2, 2, 4, 4]
 Generation 5830: Best = 2.537959e+03 FES = 583100 μ_F = 0.710 μ_{CR} = 0.235
 Diversity = 2.5767e-05 k_t = [1, 1, 1, 1]
 Generation 5840: Best = 2.537959e+03 FES = 584100 μ_F = 0.715 μ_{CR} = 0.232
 Diversity = 2.6694e-05 k_t = [4, 2, 4, 4]
 Generation 5850: Best = 2.537959e+03 FES = 585100 μ_F = 0.721 μ_{CR} = 0.239
 Diversity = 2.7500e-05 k_t = [1, 1, 1, 1]
 Generation 5860: Best = 2.537959e+03 FES = 586100 μ_F = 0.733 μ_{CR} = 0.240
 Diversity = 2.5628e-05 k_t = [2, 4, 3, 4]
 Generation 5870: Best = 2.537959e+03 FES = 587100 μ_F = 0.738 μ_{CR} = 0.249
 Diversity = 2.8561e-05 k_t = [1, 1, 1, 1]
 Generation 5880: Best = 2.537959e+03 FES = 588100 μ_F = 0.752 μ_{CR} = 0.252
 Diversity = 3.1970e-05 k_t = [3, 2, 2, 2]
 Generation 5890: Best = 2.537959e+03 FES = 589100 μ_F = 0.759 μ_{CR} = 0.251
 Diversity = 4.0690e-05 k_t = [1, 1, 1, 1]

Generation 5900: Best = 2.537959e+03	FES = 590100	$\mu_F = 0.763$	$\mu_{CR} = 0.251$
Diversity = 4.4132e-05	$k_t = [3, 4, 4, 4]$		
Generation 5910: Best = 2.537959e+03	FES = 591100	$\mu_F = 0.746$	$\mu_{CR} = 0.254$
Diversity = 4.2492e-05	$k_t = [1, 1, 1, 1]$		
Generation 5920: Best = 2.537959e+03	FES = 592100	$\mu_F = 0.725$	$\mu_{CR} = 0.261$
Diversity = 3.7671e-05	$k_t = [2, 3, 4, 2]$		
Generation 5930: Best = 2.537959e+03	FES = 593100	$\mu_F = 0.721$	$\mu_{CR} = 0.249$
Diversity = 3.7557e-05	$k_t = [1, 1, 1, 1]$		
Generation 5940: Best = 2.537959e+03	FES = 594100	$\mu_F = 0.717$	$\mu_{CR} = 0.249$
Diversity = 3.3971e-05	$k_t = [4, 4, 3, 2]$		
Generation 5950: Best = 2.537959e+03	FES = 595100	$\mu_F = 0.716$	$\mu_{CR} = 0.256$
Diversity = 3.2427e-05	$k_t = [1, 1, 1, 1]$		
Generation 5960: Best = 2.537959e+03	FES = 596100	$\mu_F = 0.711$	$\mu_{CR} = 0.257$
Diversity = 2.9014e-05	$k_t = [4, 4, 4, 2]$		
Generation 5970: Best = 2.537959e+03	FES = 597100	$\mu_F = 0.719$	$\mu_{CR} = 0.260$
Diversity = 2.8377e-05	$k_t = [1, 1, 1, 1]$		
Generation 5980: Best = 2.537959e+03	FES = 598100	$\mu_F = 0.716$	$\mu_{CR} = 0.264$
Diversity = 2.4863e-05	$k_t = [3, 3, 4, 3]$		
Generation 5990: Best = 2.537959e+03	FES = 599100	$\mu_F = 0.698$	$\mu_{CR} = 0.259$
Diversity = 2.2326e-05	$k_t = [1, 1, 1, 1]$		
Generation 6000: Best = 2.537959e+03	FES = 600100	$\mu_F = 0.691$	$\mu_{CR} = 0.258$
Diversity = 1.9559e-05	$k_t = [4, 2, 3, 2]$		
Generation 6010: Best = 2.537959e+03	FES = 601100	$\mu_F = 0.695$	$\mu_{CR} = 0.251$
Diversity = 1.8516e-05	$k_t = [1, 1, 1, 1]$		
Generation 6020: Best = 2.537959e+03	FES = 602100	$\mu_F = 0.704$	$\mu_{CR} = 0.252$
Diversity = 1.7279e-05	$k_t = [3, 2, 2, 3]$		
Generation 6030: Best = 2.537959e+03	FES = 603100	$\mu_F = 0.708$	$\mu_{CR} = 0.247$
Diversity = 1.7930e-05	$k_t = [1, 1, 1, 1]$		
Generation 6040: Best = 2.537959e+03	FES = 604100	$\mu_F = 0.714$	$\mu_{CR} = 0.248$
Diversity = 1.6900e-05	$k_t = [3, 4, 2, 4]$		
Generation 6050: Best = 2.537959e+03	FES = 605100	$\mu_F = 0.725$	$\mu_{CR} = 0.251$
Diversity = 1.7185e-05	$k_t = [1, 1, 1, 1]$		
Generation 6060: Best = 2.537959e+03	FES = 606100	$\mu_F = 0.722$	$\mu_{CR} = 0.252$
Diversity = 1.6925e-05	$k_t = [3, 4, 2, 2]$		
Generation 6070: Best = 2.537959e+03	FES = 607100	$\mu_F = 0.724$	$\mu_{CR} = 0.248$
Diversity = 1.7368e-05	$k_t = [1, 1, 1, 1]$		
Generation 6080: Best = 2.537959e+03	FES = 608100	$\mu_F = 0.720$	$\mu_{CR} = 0.254$
Diversity = 1.7163e-05	$k_t = [4, 3, 3, 3]$		
Generation 6090: Best = 2.537959e+03	FES = 609100	$\mu_F = 0.728$	$\mu_{CR} = 0.246$
Diversity = 1.7733e-05	$k_t = [1, 1, 1, 1]$		
Generation 6100: Best = 2.537959e+03	FES = 610100	$\mu_F = 0.722$	$\mu_{CR} = 0.247$
Diversity = 1.6210e-05	$k_t = [4, 3, 4, 3]$		
Generation 6110: Best = 2.537959e+03	FES = 611100	$\mu_F = 0.723$	$\mu_{CR} = 0.254$
Diversity = 1.6497e-05	$k_t = [1, 1, 1, 1]$		
Generation 6120: Best = 2.537959e+03	FES = 612100	$\mu_F = 0.718$	$\mu_{CR} = 0.246$
Diversity = 1.6950e-05	$k_t = [3, 3, 2, 2]$		
Generation 6130: Best = 2.537959e+03	FES = 613100	$\mu_F = 0.716$	$\mu_{CR} = 0.240$
Diversity = 1.8202e-05	$k_t = [1, 1, 1, 1]$		
Generation 6140: Best = 2.537959e+03	FES = 614100	$\mu_F = 0.715$	$\mu_{CR} = 0.243$
Diversity = 1.9332e-05	$k_t = [3, 4, 3, 2]$		
Generation 6150: Best = 2.537959e+03	FES = 615100	$\mu_F = 0.723$	$\mu_{CR} = 0.238$
Diversity = 2.1612e-05	$k_t = [1, 1, 1, 1]$		
Generation 6160: Best = 2.537959e+03	FES = 616100	$\mu_F = 0.721$	$\mu_{CR} = 0.236$
Diversity = 2.0515e-05	$k_t = [3, 4, 3, 3]$		

Generation 6170: Best = 2.537959e+03 FES = 617100 μ_F = 0.721 μ_{CR} = 0.235
 Diversity = 2.1323e-05 k_t = [1, 1, 1, 1]
 Generation 6180: Best = 2.537959e+03 FES = 618100 μ_F = 0.725 μ_{CR} = 0.234
 Diversity = 1.9665e-05 k_t = [4, 3, 3, 4]
 Generation 6190: Best = 2.537959e+03 FES = 619100 μ_F = 0.723 μ_{CR} = 0.237
 Diversity = 1.9168e-05 k_t = [1, 1, 1, 1]
 Generation 6200: Best = 2.537959e+03 FES = 620100 μ_F = 0.731 μ_{CR} = 0.235
 Diversity = 1.8418e-05 k_t = [3, 4, 4, 2]
 Generation 6210: Best = 2.537959e+03 FES = 621100 μ_F = 0.711 μ_{CR} = 0.229
 Diversity = 1.8011e-05 k_t = [1, 1, 1, 1]
 Generation 6220: Best = 2.537959e+03 FES = 622100 μ_F = 0.705 μ_{CR} = 0.222
 Diversity = 1.7172e-05 k_t = [3, 4, 2, 2]
 Generation 6230: Best = 2.537959e+03 FES = 623100 μ_F = 0.701 μ_{CR} = 0.221
 Diversity = 1.5947e-05 k_t = [1, 1, 1, 1]
 Generation 6240: Best = 2.537959e+03 FES = 624100 μ_F = 0.709 μ_{CR} = 0.232
 Diversity = 1.4925e-05 k_t = [2, 2, 4, 4]
 Generation 6250: Best = 2.537959e+03 FES = 625100 μ_F = 0.700 μ_{CR} = 0.237
 Diversity = 1.6457e-05 k_t = [1, 1, 1, 1]
 Generation 6260: Best = 2.537959e+03 FES = 626100 μ_F = 0.717 μ_{CR} = 0.245
 Diversity = 1.7459e-05 k_t = [3, 4, 3, 2]
 Generation 6270: Best = 2.537959e+03 FES = 627100 μ_F = 0.730 μ_{CR} = 0.253
 Diversity = 1.8339e-05 k_t = [1, 1, 1, 1]
 Generation 6280: Best = 2.537959e+03 FES = 628100 μ_F = 0.745 μ_{CR} = 0.250
 Diversity = 1.8964e-05 k_t = [2, 4, 4, 4]
 Generation 6290: Best = 2.537959e+03 FES = 629100 μ_F = 0.755 μ_{CR} = 0.260
 Diversity = 2.1143e-05 k_t = [1, 1, 1, 1]
 Generation 6300: Best = 2.537959e+03 FES = 630100 μ_F = 0.759 μ_{CR} = 0.257
 Diversity = 2.4050e-05 k_t = [3, 3, 4, 2]
 Generation 6310: Best = 2.537959e+03 FES = 631100 μ_F = 0.754 μ_{CR} = 0.251
 Diversity = 2.5363e-05 k_t = [1, 1, 1, 1]
 Generation 6320: Best = 2.537959e+03 FES = 632100 μ_F = 0.739 μ_{CR} = 0.248
 Diversity = 2.4204e-05 k_t = [3, 2, 2, 4]
 Generation 6330: Best = 2.537959e+03 FES = 633100 μ_F = 0.734 μ_{CR} = 0.241
 Diversity = 2.3838e-05 k_t = [1, 1, 1, 1]
 Generation 6340: Best = 2.537959e+03 FES = 634100 μ_F = 0.741 μ_{CR} = 0.246
 Diversity = 2.2652e-05 k_t = [4, 3, 3, 4]
 Generation 6350: Best = 2.537959e+03 FES = 635100 μ_F = 0.722 μ_{CR} = 0.236
 Diversity = 2.1584e-05 k_t = [1, 1, 1, 1]
 Generation 6360: Best = 2.537959e+03 FES = 636100 μ_F = 0.713 μ_{CR} = 0.236
 Diversity = 2.0418e-05 k_t = [4, 2, 4, 2]
 Generation 6370: Best = 2.537959e+03 FES = 637100 μ_F = 0.715 μ_{CR} = 0.242
 Diversity = 1.8652e-05 k_t = [1, 1, 1, 1]
 Generation 6380: Best = 2.537959e+03 FES = 638100 μ_F = 0.711 μ_{CR} = 0.246
 Diversity = 1.8450e-05 k_t = [2, 4, 3, 3]
 Generation 6390: Best = 2.537959e+03 FES = 639100 μ_F = 0.716 μ_{CR} = 0.254
 Diversity = 2.0700e-05 k_t = [1, 1, 1, 1]
 Generation 6400: Best = 2.537959e+03 FES = 640100 μ_F = 0.709 μ_{CR} = 0.257
 Diversity = 1.8262e-05 k_t = [3, 3, 2, 4]
 Generation 6410: Best = 2.537959e+03 FES = 641100 μ_F = 0.712 μ_{CR} = 0.259
 Diversity = 1.9469e-05 k_t = [1, 1, 1, 1]
 Generation 6420: Best = 2.537959e+03 FES = 642100 μ_F = 0.705 μ_{CR} = 0.255
 Diversity = 1.9197e-05 k_t = [3, 4, 4, 3]
 Generation 6430: Best = 2.537959e+03 FES = 643100 μ_F = 0.716 μ_{CR} = 0.256
 Diversity = 2.1325e-05 k_t = [1, 1, 1, 1]

Generation 6440: Best = 2.537959e+03 FES = 644100 μ_F = 0.728 μ_{CR} = 0.259
 Diversity = 2.0805e-05 k_t = [4, 3, 2, 4]
 Generation 6450: Best = 2.537959e+03 FES = 645100 μ_F = 0.720 μ_{CR} = 0.252
 Diversity = 2.0623e-05 k_t = [1, 1, 1, 1]
 Generation 6460: Best = 2.537959e+03 FES = 646100 μ_F = 0.708 μ_{CR} = 0.242
 Diversity = 1.8991e-05 k_t = [3, 2, 4, 2]
 Generation 6470: Best = 2.537959e+03 FES = 647100 μ_F = 0.703 μ_{CR} = 0.245
 Diversity = 1.8487e-05 k_t = [1, 1, 1, 1]
 Generation 6480: Best = 2.537959e+03 FES = 648100 μ_F = 0.707 μ_{CR} = 0.248
 Diversity = 1.7506e-05 k_t = [2, 4, 4, 4]
 Generation 6490: Best = 2.537959e+03 FES = 649100 μ_F = 0.698 μ_{CR} = 0.253
 Diversity = 1.6322e-05 k_t = [1, 1, 1, 1]
 Generation 6500: Best = 2.537959e+03 FES = 650100 μ_F = 0.684 μ_{CR} = 0.257
 Diversity = 1.3701e-05 k_t = [2, 3, 4, 4]
 Generation 6510: Best = 2.537959e+03 FES = 651100 μ_F = 0.710 μ_{CR} = 0.269
 Diversity = 1.3928e-05 k_t = [1, 1, 1, 1]
 Generation 6520: Best = 2.537959e+03 FES = 652100 μ_F = 0.709 μ_{CR} = 0.266
 Diversity = 1.3475e-05 k_t = [3, 2, 2, 4]
 Generation 6530: Best = 2.537959e+03 FES = 653100 μ_F = 0.709 μ_{CR} = 0.274
 Diversity = 1.3112e-05 k_t = [1, 1, 1, 1]
 Generation 6540: Best = 2.537959e+03 FES = 654100 μ_F = 0.718 μ_{CR} = 0.272
 Diversity = 1.2129e-05 k_t = [4, 3, 4, 2]
 Generation 6550: Best = 2.537959e+03 FES = 655100 μ_F = 0.721 μ_{CR} = 0.271
 Diversity = 1.1924e-05 k_t = [1, 1, 1, 1]
 Generation 6560: Best = 2.537959e+03 FES = 656100 μ_F = 0.727 μ_{CR} = 0.268
 Diversity = 1.2016e-05 k_t = [3, 3, 2, 4]
 Generation 6570: Best = 2.537959e+03 FES = 657100 μ_F = 0.726 μ_{CR} = 0.276
 Diversity = 1.2416e-05 k_t = [1, 1, 1, 1]
 Generation 6580: Best = 2.537959e+03 FES = 658100 μ_F = 0.730 μ_{CR} = 0.269
 Diversity = 1.1995e-05 k_t = [4, 3, 3, 2]
 Generation 6590: Best = 2.537959e+03 FES = 659100 μ_F = 0.725 μ_{CR} = 0.267
 Diversity = 1.1094e-05 k_t = [1, 1, 1, 1]
 Generation 6600: Best = 2.537959e+03 FES = 660100 μ_F = 0.726 μ_{CR} = 0.260
 Diversity = 9.2305e-06 k_t = [4, 3, 2, 4]
 Generation 6610: Best = 2.537959e+03 FES = 661100 μ_F = 0.730 μ_{CR} = 0.267
 Diversity = 9.4157e-06 k_t = [1, 1, 1, 1]
 Generation 6620: Best = 2.537959e+03 FES = 662100 μ_F = 0.726 μ_{CR} = 0.263
 Diversity = 8.9765e-06 k_t = [2, 2, 4, 2]
 Generation 6630: Best = 2.537959e+03 FES = 663100 μ_F = 0.736 μ_{CR} = 0.262
 Diversity = 9.1840e-06 k_t = [1, 1, 1, 1]
 Generation 6640: Best = 2.537959e+03 FES = 664100 μ_F = 0.721 μ_{CR} = 0.269
 Diversity = 8.3053e-06 k_t = [2, 4, 3, 3]
 Generation 6650: Best = 2.537959e+03 FES = 665100 μ_F = 0.717 μ_{CR} = 0.278
 Diversity = 8.1577e-06 k_t = [1, 1, 1, 1]
 Generation 6660: Best = 2.537959e+03 FES = 666100 μ_F = 0.706 μ_{CR} = 0.268
 Diversity = 7.6264e-06 k_t = [4, 3, 3, 4]
 Generation 6670: Best = 2.537959e+03 FES = 667100 μ_F = 0.713 μ_{CR} = 0.274
 Diversity = 7.4707e-06 k_t = [1, 1, 1, 1]
 Generation 6680: Best = 2.537959e+03 FES = 668100 μ_F = 0.715 μ_{CR} = 0.264
 Diversity = 7.1604e-06 k_t = [2, 4, 2, 2]
 Generation 6690: Best = 2.537959e+03 FES = 669100 μ_F = 0.717 μ_{CR} = 0.264
 Diversity = 7.5538e-06 k_t = [1, 1, 1, 1]
 Generation 6700: Best = 2.537959e+03 FES = 670100 μ_F = 0.720 μ_{CR} = 0.269
 Diversity = 7.5094e-06 k_t = [2, 4, 4, 3]

Generation 6710: Best = 2.537959e+03 FES = 671100 μ_F = 0.711 μ_{CR} = 0.264
 Diversity = 8.2554e-06 k_t = [1, 1, 1, 1]
 Generation 6720: Best = 2.537959e+03 FES = 672100 μ_F = 0.722 μ_{CR} = 0.258
 Diversity = 8.4030e-06 k_t = [2, 4, 2, 3]
 Generation 6730: Best = 2.537959e+03 FES = 673100 μ_F = 0.715 μ_{CR} = 0.262
 Diversity = 8.6168e-06 k_t = [1, 1, 1, 1]
 Generation 6740: Best = 2.537959e+03 FES = 674100 μ_F = 0.718 μ_{CR} = 0.266
 Diversity = 8.7767e-06 k_t = [2, 4, 3, 4]
 Generation 6750: Best = 2.537959e+03 FES = 675100 μ_F = 0.730 μ_{CR} = 0.265
 Diversity = 8.5538e-06 k_t = [1, 1, 1, 1]
 Generation 6760: Best = 2.537959e+03 FES = 676100 μ_F = 0.726 μ_{CR} = 0.266
 Diversity = 8.1311e-06 k_t = [4, 2, 3, 4]
 Generation 6770: Best = 2.537959e+03 FES = 677100 μ_F = 0.716 μ_{CR} = 0.266
 Diversity = 8.5242e-06 k_t = [1, 1, 1, 1]
 Generation 6780: Best = 2.537959e+03 FES = 678100 μ_F = 0.713 μ_{CR} = 0.264
 Diversity = 8.0907e-06 k_t = [2, 4, 4, 2]
 Generation 6790: Best = 2.537959e+03 FES = 679100 μ_F = 0.712 μ_{CR} = 0.255
 Diversity = 7.8852e-06 k_t = [1, 1, 1, 1]
 Generation 6800: Best = 2.537959e+03 FES = 680100 μ_F = 0.708 μ_{CR} = 0.261
 Diversity = 7.4468e-06 k_t = [2, 4, 3, 4]
 Generation 6810: Best = 2.537959e+03 FES = 681100 μ_F = 0.711 μ_{CR} = 0.253
 Diversity = 7.1776e-06 k_t = [1, 1, 1, 1]
 Generation 6820: Best = 2.537959e+03 FES = 682100 μ_F = 0.715 μ_{CR} = 0.249
 Diversity = 6.5865e-06 k_t = [2, 4, 4, 4]
 Generation 6830: Best = 2.537959e+03 FES = 683100 μ_F = 0.720 μ_{CR} = 0.239
 Diversity = 6.4857e-06 k_t = [1, 1, 1, 1]
 Generation 6840: Best = 2.537959e+03 FES = 684100 μ_F = 0.712 μ_{CR} = 0.232
 Diversity = 5.8931e-06 k_t = [4, 2, 2, 3]
 Generation 6850: Best = 2.537959e+03 FES = 685100 μ_F = 0.706 μ_{CR} = 0.235
 Diversity = 5.5703e-06 k_t = [1, 1, 1, 1]
 Generation 6860: Best = 2.537959e+03 FES = 686100 μ_F = 0.693 μ_{CR} = 0.238
 Diversity = 4.7547e-06 k_t = [2, 3, 4, 4]
 Generation 6870: Best = 2.537959e+03 FES = 687100 μ_F = 0.686 μ_{CR} = 0.240
 Diversity = 4.3601e-06 k_t = [1, 1, 1, 1]
 Generation 6880: Best = 2.537959e+03 FES = 688100 μ_F = 0.688 μ_{CR} = 0.241
 Diversity = 4.0493e-06 k_t = [3, 2, 2, 2]
 Generation 6890: Best = 2.537959e+03 FES = 689100 μ_F = 0.696 μ_{CR} = 0.243
 Diversity = 4.1232e-06 k_t = [1, 1, 1, 1]
 Generation 6900: Best = 2.537959e+03 FES = 690100 μ_F = 0.694 μ_{CR} = 0.244
 Diversity = 3.7156e-06 k_t = [3, 3, 2, 4]
 Generation 6910: Best = 2.537959e+03 FES = 691100 μ_F = 0.710 μ_{CR} = 0.250
 Diversity = 3.6708e-06 k_t = [1, 1, 1, 1]
 Generation 6920: Best = 2.537959e+03 FES = 692100 μ_F = 0.718 μ_{CR} = 0.254
 Diversity = 3.6656e-06 k_t = [3, 2, 2, 3]
 Generation 6930: Best = 2.537959e+03 FES = 693100 μ_F = 0.723 μ_{CR} = 0.251
 Diversity = 3.7407e-06 k_t = [1, 1, 1, 1]
 Generation 6940: Best = 2.537959e+03 FES = 694100 μ_F = 0.717 μ_{CR} = 0.255
 Diversity = 3.4291e-06 k_t = [2, 3, 3, 3]
 Generation 6950: Best = 2.537959e+03 FES = 695100 μ_F = 0.719 μ_{CR} = 0.261
 Diversity = 3.5867e-06 k_t = [1, 1, 1, 1]
 Generation 6960: Best = 2.537959e+03 FES = 696100 μ_F = 0.722 μ_{CR} = 0.263
 Diversity = 3.4655e-06 k_t = [4, 2, 3, 3]
 Generation 6970: Best = 2.537959e+03 FES = 697100 μ_F = 0.732 μ_{CR} = 0.263
 Diversity = 3.9967e-06 k_t = [1, 1, 1, 1]

Generation 6980: Best = 2.537959e+03 FES = 698100 μ_F = 0.743 μ_{CR} = 0.270
 Diversity = 3.7993e-06 k_t = [4, 4, 4, 4]
 Generation 6990: Best = 2.537959e+03 FES = 699100 μ_F = 0.734 μ_{CR} = 0.276
 Diversity = 4.0169e-06 k_t = [1, 1, 1, 1]
 Generation 7000: Best = 2.537959e+03 FES = 700100 μ_F = 0.747 μ_{CR} = 0.278
 Diversity = 4.5983e-06 k_t = [3, 3, 3, 4]
 Generation 7010: Best = 2.537959e+03 FES = 701100 μ_F = 0.749 μ_{CR} = 0.273
 Diversity = 4.9817e-06 k_t = [1, 1, 1, 1]
 Generation 7020: Best = 2.537959e+03 FES = 702100 μ_F = 0.733 μ_{CR} = 0.277
 Diversity = 4.4655e-06 k_t = [3, 4, 4, 4]
 Generation 7030: Best = 2.537959e+03 FES = 703100 μ_F = 0.737 μ_{CR} = 0.273
 Diversity = 4.6963e-06 k_t = [1, 1, 1, 1]
 Generation 7040: Best = 2.537959e+03 FES = 704100 μ_F = 0.723 μ_{CR} = 0.283
 Diversity = 4.6678e-06 k_t = [4, 2, 3, 2]
 Generation 7050: Best = 2.537959e+03 FES = 705100 μ_F = 0.723 μ_{CR} = 0.275
 Diversity = 4.5888e-06 k_t = [1, 1, 1, 1]
 Generation 7060: Best = 2.537959e+03 FES = 706100 μ_F = 0.707 μ_{CR} = 0.272
 Diversity = 4.2830e-06 k_t = [2, 2, 3, 3]
 Generation 7070: Best = 2.537959e+03 FES = 707100 μ_F = 0.709 μ_{CR} = 0.272
 Diversity = 4.5201e-06 k_t = [1, 1, 1, 1]
 Generation 7080: Best = 2.537959e+03 FES = 708100 μ_F = 0.720 μ_{CR} = 0.279
 Diversity = 4.8868e-06 k_t = [4, 3, 3, 3]
 Generation 7090: Best = 2.537959e+03 FES = 709100 μ_F = 0.731 μ_{CR} = 0.266
 Diversity = 5.0799e-06 k_t = [1, 1, 1, 1]
 Generation 7100: Best = 2.537959e+03 FES = 710100 μ_F = 0.726 μ_{CR} = 0.267
 Diversity = 4.9019e-06 k_t = [3, 2, 2, 3]
 Generation 7110: Best = 2.537959e+03 FES = 711100 μ_F = 0.729 μ_{CR} = 0.264
 Diversity = 4.8182e-06 k_t = [1, 1, 1, 1]
 Generation 7120: Best = 2.537959e+03 FES = 712100 μ_F = 0.724 μ_{CR} = 0.261
 Diversity = 4.6390e-06 k_t = [3, 3, 4, 2]
 Generation 7130: Best = 2.537959e+03 FES = 713100 μ_F = 0.720 μ_{CR} = 0.265
 Diversity = 4.6732e-06 k_t = [1, 1, 1, 1]
 Generation 7140: Best = 2.537959e+03 FES = 714100 μ_F = 0.718 μ_{CR} = 0.267
 Diversity = 4.1719e-06 k_t = [3, 3, 3, 4]
 Generation 7150: Best = 2.537959e+03 FES = 715100 μ_F = 0.717 μ_{CR} = 0.272
 Diversity = 3.8280e-06 k_t = [1, 1, 1, 1]
 Generation 7160: Best = 2.537959e+03 FES = 716100 μ_F = 0.725 μ_{CR} = 0.265
 Diversity = 4.3170e-06 k_t = [2, 4, 2, 2]
 Generation 7170: Best = 2.537959e+03 FES = 717100 μ_F = 0.716 μ_{CR} = 0.267
 Diversity = 4.4704e-06 k_t = [1, 1, 1, 1]
 Generation 7180: Best = 2.537959e+03 FES = 718100 μ_F = 0.710 μ_{CR} = 0.270
 Diversity = 4.2845e-06 k_t = [3, 4, 3, 3]
 Generation 7190: Best = 2.537959e+03 FES = 719100 μ_F = 0.706 μ_{CR} = 0.269
 Diversity = 3.6828e-06 k_t = [1, 1, 1, 1]
 Generation 7200: Best = 2.537959e+03 FES = 720100 μ_F = 0.706 μ_{CR} = 0.261
 Diversity = 3.3195e-06 k_t = [2, 2, 3, 4]
 Generation 7210: Best = 2.537959e+03 FES = 721100 μ_F = 0.709 μ_{CR} = 0.257
 Diversity = 3.0555e-06 k_t = [1, 1, 1, 1]
 Generation 7220: Best = 2.537959e+03 FES = 722100 μ_F = 0.709 μ_{CR} = 0.261
 Diversity = 2.7057e-06 k_t = [3, 2, 4, 3]
 Generation 7230: Best = 2.537959e+03 FES = 723100 μ_F = 0.725 μ_{CR} = 0.258
 Diversity = 2.5631e-06 k_t = [1, 1, 1, 1]
 Generation 7240: Best = 2.537959e+03 FES = 724100 μ_F = 0.733 μ_{CR} = 0.258
 Diversity = 2.4355e-06 k_t = [3, 4, 4, 4]

Generation 7250: Best = 2.537959e+03 FES = 725100 μ_F = 0.739 μ_{CR} = 0.257
 Diversity = 2.4479e-06 k_t = [1, 1, 1, 1]
 Generation 7260: Best = 2.537959e+03 FES = 726100 μ_F = 0.733 μ_{CR} = 0.272
 Diversity = 2.4937e-06 k_t = [3, 3, 3, 3]
 Generation 7270: Best = 2.537959e+03 FES = 727100 μ_F = 0.725 μ_{CR} = 0.271
 Diversity = 2.7822e-06 k_t = [1, 1, 1, 1]
 Generation 7280: Best = 2.537959e+03 FES = 728100 μ_F = 0.722 μ_{CR} = 0.262
 Diversity = 2.5229e-06 k_t = [3, 4, 4, 4]
 Generation 7290: Best = 2.537959e+03 FES = 729100 μ_F = 0.727 μ_{CR} = 0.266
 Diversity = 2.7296e-06 k_t = [1, 1, 1, 1]
 Generation 7300: Best = 2.537959e+03 FES = 730100 μ_F = 0.737 μ_{CR} = 0.274
 Diversity = 3.0821e-06 k_t = [3, 4, 3, 2]
 Generation 7310: Best = 2.537959e+03 FES = 731100 μ_F = 0.751 μ_{CR} = 0.280
 Diversity = 3.5994e-06 k_t = [1, 1, 1, 1]
 Generation 7320: Best = 2.537959e+03 FES = 732100 μ_F = 0.755 μ_{CR} = 0.287
 Diversity = 4.0533e-06 k_t = [3, 2, 2, 2]
 Generation 7330: Best = 2.537959e+03 FES = 733100 μ_F = 0.753 μ_{CR} = 0.294
 Diversity = 4.7245e-06 k_t = [1, 1, 1, 1]
 Generation 7340: Best = 2.537959e+03 FES = 734100 μ_F = 0.745 μ_{CR} = 0.297
 Diversity = 4.6272e-06 k_t = [4, 4, 2, 3]
 Generation 7350: Best = 2.537959e+03 FES = 735100 μ_F = 0.745 μ_{CR} = 0.304
 Diversity = 5.1335e-06 k_t = [1, 1, 1, 1]
 Generation 7360: Best = 2.537959e+03 FES = 736100 μ_F = 0.741 μ_{CR} = 0.298
 Diversity = 4.9960e-06 k_t = [4, 4, 4, 3]
 Generation 7370: Best = 2.537959e+03 FES = 737100 μ_F = 0.739 μ_{CR} = 0.303
 Diversity = 5.0138e-06 k_t = [1, 1, 1, 1]
 Generation 7380: Best = 2.537959e+03 FES = 738100 μ_F = 0.716 μ_{CR} = 0.291
 Diversity = 4.4503e-06 k_t = [2, 3, 2, 2]
 Generation 7390: Best = 2.537959e+03 FES = 739100 μ_F = 0.692 μ_{CR} = 0.294
 Diversity = 3.8621e-06 k_t = [1, 1, 1, 1]
 Generation 7400: Best = 2.537959e+03 FES = 740100 μ_F = 0.690 μ_{CR} = 0.298
 Diversity = 3.4630e-06 k_t = [4, 3, 3, 2]
 Generation 7410: Best = 2.537959e+03 FES = 741100 μ_F = 0.686 μ_{CR} = 0.310
 Diversity = 3.3311e-06 k_t = [1, 1, 1, 1]
 Generation 7420: Best = 2.537959e+03 FES = 742100 μ_F = 0.693 μ_{CR} = 0.302
 Diversity = 3.0373e-06 k_t = [4, 4, 2, 4]
 Generation 7430: Best = 2.537959e+03 FES = 743100 μ_F = 0.697 μ_{CR} = 0.307
 Diversity = 3.2272e-06 k_t = [1, 1, 1, 1]
 Generation 7440: Best = 2.537959e+03 FES = 744100 μ_F = 0.721 μ_{CR} = 0.316
 Diversity = 3.4753e-06 k_t = [2, 4, 2, 3]
 Generation 7450: Best = 2.537959e+03 FES = 745100 μ_F = 0.721 μ_{CR} = 0.320
 Diversity = 3.8698e-06 k_t = [1, 1, 1, 1]
 Generation 7460: Best = 2.537959e+03 FES = 746100 μ_F = 0.737 μ_{CR} = 0.317
 Diversity = 4.0571e-06 k_t = [2, 2, 4, 2]
 Generation 7470: Best = 2.537959e+03 FES = 747100 μ_F = 0.753 μ_{CR} = 0.313
 Diversity = 4.8668e-06 k_t = [1, 1, 1, 1]
 Generation 7480: Best = 2.537959e+03 FES = 748100 μ_F = 0.750 μ_{CR} = 0.311
 Diversity = 5.0053e-06 k_t = [3, 3, 4, 2]
 Generation 7490: Best = 2.537959e+03 FES = 749100 μ_F = 0.747 μ_{CR} = 0.302
 Diversity = 5.8184e-06 k_t = [1, 1, 1, 1]
 Generation 7500: Best = 2.537959e+03 FES = 750100 μ_F = 0.743 μ_{CR} = 0.293
 Diversity = 5.6674e-06 k_t = [4, 4, 2, 3]
 Generation 7510: Best = 2.537959e+03 FES = 751100 μ_F = 0.752 μ_{CR} = 0.296
 Diversity = 5.8337e-06 k_t = [1, 1, 1, 1]

Generation 7520: Best = 2.537959e+03 FES = 752100 μ_F = 0.736 μ_{CR} = 0.292
 Diversity = 5.1506e-06 k_t = [3, 3, 4, 3]
 Generation 7530: Best = 2.537959e+03 FES = 753100 μ_F = 0.723 μ_{CR} = 0.294
 Diversity = 4.9633e-06 k_t = [1, 1, 1, 1]
 Generation 7540: Best = 2.537959e+03 FES = 754100 μ_F = 0.722 μ_{CR} = 0.305
 Diversity = 4.9440e-06 k_t = [4, 2, 4, 2]
 Generation 7550: Best = 2.537959e+03 FES = 755100 μ_F = 0.717 μ_{CR} = 0.301
 Diversity = 5.1209e-06 k_t = [1, 1, 1, 1]
 Generation 7560: Best = 2.537959e+03 FES = 756100 μ_F = 0.720 μ_{CR} = 0.293
 Diversity = 4.9055e-06 k_t = [2, 2, 2, 4]
 Generation 7570: Best = 2.537959e+03 FES = 757100 μ_F = 0.718 μ_{CR} = 0.295
 Diversity = 5.2412e-06 k_t = [1, 1, 1, 1]
 Generation 7580: Best = 2.537959e+03 FES = 758100 μ_F = 0.716 μ_{CR} = 0.303
 Diversity = 5.1223e-06 k_t = [4, 3, 4, 2]
 Generation 7590: Best = 2.537959e+03 FES = 759100 μ_F = 0.717 μ_{CR} = 0.313
 Diversity = 5.1369e-06 k_t = [1, 1, 1, 1]
 Generation 7600: Best = 2.537959e+03 FES = 760100 μ_F = 0.726 μ_{CR} = 0.314
 Diversity = 5.3808e-06 k_t = [3, 4, 2, 4]
 Generation 7610: Best = 2.537959e+03 FES = 761100 μ_F = 0.726 μ_{CR} = 0.310
 Diversity = 5.0825e-06 k_t = [1, 1, 1, 1]
 Generation 7620: Best = 2.537959e+03 FES = 762100 μ_F = 0.723 μ_{CR} = 0.316
 Diversity = 4.7626e-06 k_t = [2, 3, 3, 4]
 Generation 7630: Best = 2.537959e+03 FES = 763100 μ_F = 0.710 μ_{CR} = 0.313
 Diversity = 5.3507e-06 k_t = [1, 1, 1, 1]
 Generation 7640: Best = 2.537959e+03 FES = 764100 μ_F = 0.703 μ_{CR} = 0.309
 Diversity = 4.9255e-06 k_t = [2, 3, 2, 4]
 Generation 7650: Best = 2.537959e+03 FES = 765100 μ_F = 0.707 μ_{CR} = 0.311
 Diversity = 4.7530e-06 k_t = [1, 1, 1, 1]
 Generation 7660: Best = 2.537959e+03 FES = 766100 μ_F = 0.699 μ_{CR} = 0.307
 Diversity = 4.4677e-06 k_t = [2, 2, 3, 3]
 Generation 7670: Best = 2.537959e+03 FES = 767100 μ_F = 0.703 μ_{CR} = 0.302
 Diversity = 4.5939e-06 k_t = [1, 1, 1, 1]
 Generation 7680: Best = 2.537959e+03 FES = 768100 μ_F = 0.711 μ_{CR} = 0.292
 Diversity = 4.0640e-06 k_t = [4, 3, 4, 4]
 Generation 7690: Best = 2.537959e+03 FES = 769100 μ_F = 0.719 μ_{CR} = 0.281
 Diversity = 4.1215e-06 k_t = [1, 1, 1, 1]
 Generation 7700: Best = 2.537959e+03 FES = 770100 μ_F = 0.709 μ_{CR} = 0.283
 Diversity = 3.5084e-06 k_t = [3, 3, 2, 4]
 Generation 7710: Best = 2.537959e+03 FES = 771100 μ_F = 0.711 μ_{CR} = 0.281
 Diversity = 3.4323e-06 k_t = [1, 1, 1, 1]
 Generation 7720: Best = 2.537959e+03 FES = 772100 μ_F = 0.705 μ_{CR} = 0.279
 Diversity = 3.2698e-06 k_t = [3, 3, 3, 3]
 Generation 7730: Best = 2.537959e+03 FES = 773100 μ_F = 0.708 μ_{CR} = 0.283
 Diversity = 3.2291e-06 k_t = [1, 1, 1, 1]
 Generation 7740: Best = 2.537959e+03 FES = 774100 μ_F = 0.712 μ_{CR} = 0.286
 Diversity = 3.1400e-06 k_t = [4, 2, 3, 3]
 Generation 7750: Best = 2.537959e+03 FES = 775100 μ_F = 0.720 μ_{CR} = 0.282
 Diversity = 3.5328e-06 k_t = [1, 1, 1, 1]
 Generation 7760: Best = 2.537959e+03 FES = 776100 μ_F = 0.719 μ_{CR} = 0.281
 Diversity = 3.3047e-06 k_t = [4, 2, 2, 3]
 Generation 7770: Best = 2.537959e+03 FES = 777100 μ_F = 0.719 μ_{CR} = 0.281
 Diversity = 3.3342e-06 k_t = [1, 1, 1, 1]
 Generation 7780: Best = 2.537959e+03 FES = 778100 μ_F = 0.719 μ_{CR} = 0.284
 Diversity = 3.2983e-06 k_t = [2, 3, 4, 3]

Generation 7790: Best = 2.537959e+03 FES = 779100 μ_F = 0.724 μ_{CR} = 0.290
 Diversity = 3.4984e-06 k_t = [1, 1, 1, 1]
 Generation 7800: Best = 2.537959e+03 FES = 780100 μ_F = 0.726 μ_{CR} = 0.301
 Diversity = 3.6860e-06 k_t = [3, 3, 2, 4]
 Generation 7810: Best = 2.537959e+03 FES = 781100 μ_F = 0.727 μ_{CR} = 0.296
 Diversity = 3.7388e-06 k_t = [1, 1, 1, 1]
 Generation 7820: Best = 2.537959e+03 FES = 782100 μ_F = 0.726 μ_{CR} = 0.293
 Diversity = 3.5230e-06 k_t = [4, 3, 3, 3]
 Generation 7830: Best = 2.537959e+03 FES = 783100 μ_F = 0.733 μ_{CR} = 0.292
 Diversity = 3.9295e-06 k_t = [1, 1, 1, 1]
 Generation 7840: Best = 2.537959e+03 FES = 784100 μ_F = 0.738 μ_{CR} = 0.298
 Diversity = 4.0850e-06 k_t = [4, 2, 3, 2]
 Generation 7850: Best = 2.537959e+03 FES = 785100 μ_F = 0.732 μ_{CR} = 0.292
 Diversity = 4.2695e-06 k_t = [1, 1, 1, 1]
 Generation 7860: Best = 2.537959e+03 FES = 786100 μ_F = 0.726 μ_{CR} = 0.289
 Diversity = 4.0340e-06 k_t = [4, 4, 3, 3]
 Generation 7870: Best = 2.537959e+03 FES = 787100 μ_F = 0.720 μ_{CR} = 0.286
 Diversity = 4.0460e-06 k_t = [1, 1, 1, 1]
 Generation 7880: Best = 2.537959e+03 FES = 788100 μ_F = 0.701 μ_{CR} = 0.278
 Diversity = 3.3744e-06 k_t = [3, 4, 3, 3]
 Generation 7890: Best = 2.537959e+03 FES = 789100 μ_F = 0.698 μ_{CR} = 0.272
 Diversity = 3.1036e-06 k_t = [1, 1, 1, 1]
 Generation 7900: Best = 2.537959e+03 FES = 790100 μ_F = 0.707 μ_{CR} = 0.267
 Diversity = 2.8082e-06 k_t = [2, 2, 3, 3]
 Generation 7910: Best = 2.537959e+03 FES = 791100 μ_F = 0.713 μ_{CR} = 0.270
 Diversity = 2.9634e-06 k_t = [1, 1, 1, 1]
 Generation 7920: Best = 2.537959e+03 FES = 792100 μ_F = 0.712 μ_{CR} = 0.274
 Diversity = 2.9013e-06 k_t = [3, 2, 3, 3]
 Generation 7930: Best = 2.537959e+03 FES = 793100 μ_F = 0.706 μ_{CR} = 0.268
 Diversity = 2.9543e-06 k_t = [1, 1, 1, 1]
 Generation 7940: Best = 2.537959e+03 FES = 794100 μ_F = 0.698 μ_{CR} = 0.266
 Diversity = 2.6862e-06 k_t = [3, 3, 4, 2]
 Generation 7950: Best = 2.537959e+03 FES = 795100 μ_F = 0.696 μ_{CR} = 0.269
 Diversity = 2.4483e-06 k_t = [1, 1, 1, 1]
 Generation 7960: Best = 2.537959e+03 FES = 796100 μ_F = 0.694 μ_{CR} = 0.267
 Diversity = 2.0995e-06 k_t = [3, 2, 3, 2]
 Generation 7970: Best = 2.537959e+03 FES = 797100 μ_F = 0.694 μ_{CR} = 0.265
 Diversity = 1.9909e-06 k_t = [1, 1, 1, 1]
 Generation 7980: Best = 2.537959e+03 FES = 798100 μ_F = 0.681 μ_{CR} = 0.263
 Diversity = 1.8138e-06 k_t = [2, 2, 3, 2]
 Generation 7990: Best = 2.537959e+03 FES = 799100 μ_F = 0.686 μ_{CR} = 0.262
 Diversity = 1.7859e-06 k_t = [1, 1, 1, 1]
 Generation 8000: Best = 2.537959e+03 FES = 800100 μ_F = 0.692 μ_{CR} = 0.263
 Diversity = 1.7099e-06 k_t = [2, 3, 4, 2]
 Generation 8010: Best = 2.537959e+03 FES = 801100 μ_F = 0.696 μ_{CR} = 0.259
 Diversity = 1.6263e-06 k_t = [1, 1, 1, 1]
 Generation 8020: Best = 2.537959e+03 FES = 802100 μ_F = 0.715 μ_{CR} = 0.260
 Diversity = 1.7016e-06 k_t = [4, 2, 3, 2]
 Generation 8030: Best = 2.537959e+03 FES = 803100 μ_F = 0.720 μ_{CR} = 0.263
 Diversity = 1.7827e-06 k_t = [1, 1, 1, 1]
 Generation 8040: Best = 2.537959e+03 FES = 804100 μ_F = 0.738 μ_{CR} = 0.260
 Diversity = 1.8902e-06 k_t = [4, 4, 3, 2]
 Generation 8050: Best = 2.537959e+03 FES = 805100 μ_F = 0.743 μ_{CR} = 0.264
 Diversity = 2.0417e-06 k_t = [1, 1, 1, 1]

Generation 8060: Best = 2.537959e+03 FES = 806100 μ_F = 0.743 μ_{CR} = 0.274
 Diversity = 2.0076e-06 k_t = [2, 3, 4, 2]
 Generation 8070: Best = 2.537959e+03 FES = 807100 μ_F = 0.747 μ_{CR} = 0.268
 Diversity = 2.2726e-06 k_t = [1, 1, 1, 1]
 Generation 8080: Best = 2.537959e+03 FES = 808100 μ_F = 0.741 μ_{CR} = 0.262
 Diversity = 2.2859e-06 k_t = [3, 2, 3, 2]
 Generation 8090: Best = 2.537959e+03 FES = 809100 μ_F = 0.744 μ_{CR} = 0.273
 Diversity = 2.6215e-06 k_t = [1, 1, 1, 1]
 Generation 8100: Best = 2.537959e+03 FES = 810100 μ_F = 0.734 μ_{CR} = 0.278
 Diversity = 2.3899e-06 k_t = [3, 2, 4, 4]
 Generation 8110: Best = 2.537959e+03 FES = 811100 μ_F = 0.735 μ_{CR} = 0.282
 Diversity = 2.7977e-06 k_t = [1, 1, 1, 1]
 Generation 8120: Best = 2.537959e+03 FES = 812100 μ_F = 0.746 μ_{CR} = 0.286
 Diversity = 3.0367e-06 k_t = [4, 4, 2, 4]
 Generation 8130: Best = 2.537959e+03 FES = 813100 μ_F = 0.759 μ_{CR} = 0.297
 Diversity = 3.6116e-06 k_t = [1, 1, 1, 1]
 Generation 8140: Best = 2.537959e+03 FES = 814100 μ_F = 0.757 μ_{CR} = 0.294
 Diversity = 3.5989e-06 k_t = [2, 3, 2, 3]
 Generation 8150: Best = 2.537959e+03 FES = 815100 μ_F = 0.751 μ_{CR} = 0.293
 Diversity = 4.4822e-06 k_t = [1, 1, 1, 1]
 Generation 8160: Best = 2.537959e+03 FES = 816100 μ_F = 0.735 μ_{CR} = 0.298
 Diversity = 4.4952e-06 k_t = [4, 2, 3, 2]
 Generation 8170: Best = 2.537959e+03 FES = 817100 μ_F = 0.742 μ_{CR} = 0.301
 Diversity = 5.4405e-06 k_t = [1, 1, 1, 1]
 Generation 8180: Best = 2.537959e+03 FES = 818100 μ_F = 0.735 μ_{CR} = 0.311
 Diversity = 5.7415e-06 k_t = [3, 3, 4, 3]
 Generation 8190: Best = 2.537959e+03 FES = 819100 μ_F = 0.739 μ_{CR} = 0.313
 Diversity = 6.7757e-06 k_t = [1, 1, 1, 1]
 Generation 8200: Best = 2.537959e+03 FES = 820100 μ_F = 0.745 μ_{CR} = 0.311
 Diversity = 7.6983e-06 k_t = [2, 2, 2, 2]
 Generation 8210: Best = 2.537959e+03 FES = 821100 μ_F = 0.729 μ_{CR} = 0.309
 Diversity = 8.9674e-06 k_t = [1, 1, 1, 1]
 Generation 8220: Best = 2.537959e+03 FES = 822100 μ_F = 0.716 μ_{CR} = 0.295
 Diversity = 7.8901e-06 k_t = [4, 4, 4, 3]
 Generation 8230: Best = 2.537959e+03 FES = 823100 μ_F = 0.712 μ_{CR} = 0.291
 Diversity = 7.4702e-06 k_t = [1, 1, 1, 1]
 Generation 8240: Best = 2.537959e+03 FES = 824100 μ_F = 0.696 μ_{CR} = 0.291
 Diversity = 6.9287e-06 k_t = [3, 4, 4, 3]
 Generation 8250: Best = 2.537959e+03 FES = 825100 μ_F = 0.699 μ_{CR} = 0.282
 Diversity = 7.0667e-06 k_t = [1, 1, 1, 1]
 Generation 8260: Best = 2.537959e+03 FES = 826100 μ_F = 0.700 μ_{CR} = 0.281
 Diversity = 6.7847e-06 k_t = [2, 3, 2, 4]
 Generation 8270: Best = 2.537959e+03 FES = 827100 μ_F = 0.697 μ_{CR} = 0.288
 Diversity = 6.4543e-06 k_t = [1, 1, 1, 1]
 Generation 8280: Best = 2.537959e+03 FES = 828100 μ_F = 0.698 μ_{CR} = 0.286
 Diversity = 5.8298e-06 k_t = [4, 4, 3, 2]
 Generation 8290: Best = 2.537959e+03 FES = 829100 μ_F = 0.703 μ_{CR} = 0.273
 Diversity = 5.3776e-06 k_t = [1, 1, 1, 1]
 Generation 8300: Best = 2.537959e+03 FES = 830100 μ_F = 0.708 μ_{CR} = 0.272
 Diversity = 4.9585e-06 k_t = [2, 3, 3, 4]
 Generation 8310: Best = 2.537959e+03 FES = 831100 μ_F = 0.707 μ_{CR} = 0.272
 Diversity = 5.2882e-06 k_t = [1, 1, 1, 1]
 Generation 8320: Best = 2.537959e+03 FES = 832100 μ_F = 0.694 μ_{CR} = 0.272
 Diversity = 5.1278e-06 k_t = [4, 3, 2, 3]

Generation 8330: Best = 2.537959e+03 FES = 833100 μ_F = 0.705 μ_{CR} = 0.266
 Diversity = 4.9998e-06 k_t = [1, 1, 1, 1]
 Generation 8340: Best = 2.537959e+03 FES = 834100 μ_F = 0.721 μ_{CR} = 0.261
 Diversity = 5.7257e-06 k_t = [3, 3, 2, 3]
 Generation 8350: Best = 2.537959e+03 FES = 835100 μ_F = 0.724 μ_{CR} = 0.260
 Diversity = 6.5061e-06 k_t = [1, 1, 1, 1]
 Generation 8360: Best = 2.537959e+03 FES = 836100 μ_F = 0.713 μ_{CR} = 0.259
 Diversity = 5.8956e-06 k_t = [2, 4, 3, 3]
 Generation 8370: Best = 2.537959e+03 FES = 837100 μ_F = 0.720 μ_{CR} = 0.250
 Diversity = 6.1843e-06 k_t = [1, 1, 1, 1]
 Generation 8380: Best = 2.537959e+03 FES = 838100 μ_F = 0.729 μ_{CR} = 0.251
 Diversity = 6.8268e-06 k_t = [3, 4, 4, 2]
 Generation 8390: Best = 2.537959e+03 FES = 839100 μ_F = 0.748 μ_{CR} = 0.257
 Diversity = 8.3733e-06 k_t = [1, 1, 1, 1]
 Generation 8400: Best = 2.537959e+03 FES = 840100 μ_F = 0.752 μ_{CR} = 0.256
 Diversity = 9.8854e-06 k_t = [3, 3, 4, 3]
 Generation 8410: Best = 2.537959e+03 FES = 841100 μ_F = 0.744 μ_{CR} = 0.253
 Diversity = 9.9365e-06 k_t = [1, 1, 1, 1]
 Generation 8420: Best = 2.537959e+03 FES = 842100 μ_F = 0.757 μ_{CR} = 0.258
 Diversity = 1.1732e-05 k_t = [3, 2, 3, 2]
 Generation 8430: Best = 2.537959e+03 FES = 843100 μ_F = 0.746 μ_{CR} = 0.264
 Diversity = 1.3744e-05 k_t = [1, 1, 1, 1]
 Generation 8440: Best = 2.537959e+03 FES = 844100 μ_F = 0.753 μ_{CR} = 0.261
 Diversity = 1.3914e-05 k_t = [2, 2, 3, 2]
 Generation 8450: Best = 2.537959e+03 FES = 845100 μ_F = 0.744 μ_{CR} = 0.261
 Diversity = 1.4422e-05 k_t = [1, 1, 1, 1]
 Generation 8460: Best = 2.537959e+03 FES = 846100 μ_F = 0.729 μ_{CR} = 0.261
 Diversity = 1.2267e-05 k_t = [3, 3, 4, 4]
 Generation 8470: Best = 2.537959e+03 FES = 847100 μ_F = 0.734 μ_{CR} = 0.266
 Diversity = 1.2503e-05 k_t = [1, 1, 1, 1]
 Generation 8480: Best = 2.537959e+03 FES = 848100 μ_F = 0.717 μ_{CR} = 0.269
 Diversity = 1.1141e-05 k_t = [3, 2, 2, 2]
 Generation 8490: Best = 2.537959e+03 FES = 849100 μ_F = 0.723 μ_{CR} = 0.251
 Diversity = 1.1139e-05 k_t = [1, 1, 1, 1]
 Generation 8500: Best = 2.537959e+03 FES = 850100 μ_F = 0.726 μ_{CR} = 0.260
 Diversity = 1.1127e-05 k_t = [3, 2, 3, 2]
 Generation 8510: Best = 2.537959e+03 FES = 851100 μ_F = 0.722 μ_{CR} = 0.254
 Diversity = 1.1045e-05 k_t = [1, 1, 1, 1]
 Generation 8520: Best = 2.537959e+03 FES = 852100 μ_F = 0.708 μ_{CR} = 0.252
 Diversity = 9.6777e-06 k_t = [2, 3, 2, 2]
 Generation 8530: Best = 2.537959e+03 FES = 853100 μ_F = 0.693 μ_{CR} = 0.258
 Diversity = 9.8952e-06 k_t = [1, 1, 1, 1]
 Generation 8540: Best = 2.537959e+03 FES = 854100 μ_F = 0.690 μ_{CR} = 0.274
 Diversity = 8.9112e-06 k_t = [2, 3, 3, 3]
 Generation 8550: Best = 2.537959e+03 FES = 855100 μ_F = 0.689 μ_{CR} = 0.274
 Diversity = 8.0903e-06 k_t = [1, 1, 1, 1]
 Generation 8560: Best = 2.537959e+03 FES = 856100 μ_F = 0.678 μ_{CR} = 0.272
 Diversity = 6.5792e-06 k_t = [2, 2, 3, 3]
 Generation 8570: Best = 2.537959e+03 FES = 857100 μ_F = 0.677 μ_{CR} = 0.276
 Diversity = 6.0232e-06 k_t = [1, 1, 1, 1]
 Generation 8580: Best = 2.537959e+03 FES = 858100 μ_F = 0.671 μ_{CR} = 0.273
 Diversity = 5.3826e-06 k_t = [2, 4, 4, 3]
 Generation 8590: Best = 2.537959e+03 FES = 859100 μ_F = 0.680 μ_{CR} = 0.275
 Diversity = 5.0314e-06 k_t = [1, 1, 1, 1]

Generation 8600: Best = 2.537959e+03 FES = 860100 μ_F = 0.697 μ_{CR} = 0.279
 Diversity = 4.5582e-06 k_t = [4, 2, 3, 4]
 Generation 8610: Best = 2.537959e+03 FES = 861100 μ_F = 0.698 μ_{CR} = 0.283
 Diversity = 4.4685e-06 k_t = [1, 1, 1, 1]
 Generation 8620: Best = 2.537959e+03 FES = 862100 μ_F = 0.703 μ_{CR} = 0.280
 Diversity = 4.0502e-06 k_t = [2, 2, 2, 4]
 Generation 8630: Best = 2.537959e+03 FES = 863100 μ_F = 0.705 μ_{CR} = 0.276
 Diversity = 3.6689e-06 k_t = [1, 1, 1, 1]
 Generation 8640: Best = 2.537959e+03 FES = 864100 μ_F = 0.712 μ_{CR} = 0.278
 Diversity = 3.3508e-06 k_t = [4, 4, 4, 4]
 Generation 8650: Best = 2.537959e+03 FES = 865100 μ_F = 0.725 μ_{CR} = 0.281
 Diversity = 3.2047e-06 k_t = [1, 1, 1, 1]
 Generation 8660: Best = 2.537959e+03 FES = 866100 μ_F = 0.738 μ_{CR} = 0.283
 Diversity = 3.0377e-06 k_t = [4, 4, 4, 4]
 Generation 8670: Best = 2.537959e+03 FES = 867100 μ_F = 0.728 μ_{CR} = 0.287
 Diversity = 3.0299e-06 k_t = [1, 1, 1, 1]
 Generation 8680: Best = 2.537959e+03 FES = 868100 μ_F = 0.734 μ_{CR} = 0.282
 Diversity = 3.1941e-06 k_t = [2, 3, 4, 4]
 Generation 8690: Best = 2.537959e+03 FES = 869100 μ_F = 0.734 μ_{CR} = 0.291
 Diversity = 3.4599e-06 k_t = [1, 1, 1, 1]
 Generation 8700: Best = 2.537959e+03 FES = 870100 μ_F = 0.737 μ_{CR} = 0.295
 Diversity = 3.7469e-06 k_t = [2, 4, 4, 3]
 Generation 8710: Best = 2.537959e+03 FES = 871100 μ_F = 0.743 μ_{CR} = 0.293
 Diversity = 4.3965e-06 k_t = [1, 1, 1, 1]
 Generation 8720: Best = 2.537959e+03 FES = 872100 μ_F = 0.745 μ_{CR} = 0.294
 Diversity = 5.1608e-06 k_t = [2, 4, 4, 4]
 Generation 8730: Best = 2.537959e+03 FES = 873100 μ_F = 0.748 μ_{CR} = 0.296
 Diversity = 5.4781e-06 k_t = [1, 1, 1, 1]
 Generation 8740: Best = 2.537959e+03 FES = 874100 μ_F = 0.749 μ_{CR} = 0.286
 Diversity = 6.0195e-06 k_t = [2, 4, 2, 2]
 Generation 8750: Best = 2.537959e+03 FES = 875100 μ_F = 0.737 μ_{CR} = 0.291
 Diversity = 6.5416e-06 k_t = [1, 1, 1, 1]
 Generation 8760: Best = 2.537959e+03 FES = 876100 μ_F = 0.736 μ_{CR} = 0.286
 Diversity = 6.7020e-06 k_t = [2, 2, 3, 2]
 Generation 8770: Best = 2.537959e+03 FES = 877100 μ_F = 0.729 μ_{CR} = 0.281
 Diversity = 6.7136e-06 k_t = [1, 1, 1, 1]
 Generation 8780: Best = 2.537959e+03 FES = 878100 μ_F = 0.724 μ_{CR} = 0.287
 Diversity = 6.4352e-06 k_t = [3, 4, 3, 4]
 Generation 8790: Best = 2.537959e+03 FES = 879100 μ_F = 0.738 μ_{CR} = 0.290
 Diversity = 7.2144e-06 k_t = [1, 1, 1, 1]
 Generation 8800: Best = 2.537959e+03 FES = 880100 μ_F = 0.741 μ_{CR} = 0.277
 Diversity = 7.1189e-06 k_t = [4, 3, 4, 3]
 Generation 8810: Best = 2.537959e+03 FES = 881100 μ_F = 0.736 μ_{CR} = 0.277
 Diversity = 6.9256e-06 k_t = [1, 1, 1, 1]
 Generation 8820: Best = 2.537959e+03 FES = 882100 μ_F = 0.727 μ_{CR} = 0.278
 Diversity = 7.0088e-06 k_t = [3, 4, 2, 2]
 Generation 8830: Best = 2.537959e+03 FES = 883100 μ_F = 0.720 μ_{CR} = 0.268
 Diversity = 7.4682e-06 k_t = [1, 1, 1, 1]
 Generation 8840: Best = 2.537959e+03 FES = 884100 μ_F = 0.718 μ_{CR} = 0.255
 Diversity = 6.9406e-06 k_t = [2, 2, 3, 2]
 Generation 8850: Best = 2.537959e+03 FES = 885100 μ_F = 0.706 μ_{CR} = 0.246
 Diversity = 6.5866e-06 k_t = [1, 1, 1, 1]
 Generation 8860: Best = 2.537959e+03 FES = 886100 μ_F = 0.707 μ_{CR} = 0.240
 Diversity = 5.9596e-06 k_t = [4, 3, 3, 2]

Generation 8870: Best = 2.537959e+03 FES = 887100 μ_F = 0.689 μ_{CR} = 0.238
 Diversity = 5.1267e-06 k_t = [1, 1, 1, 1]
 Generation 8880: Best = 2.537959e+03 FES = 888100 μ_F = 0.686 μ_{CR} = 0.233
 Diversity = 4.5049e-06 k_t = [2, 2, 4, 3]
 Generation 8890: Best = 2.537959e+03 FES = 889100 μ_F = 0.680 μ_{CR} = 0.236
 Diversity = 3.8869e-06 k_t = [1, 1, 1, 1]
 Generation 8900: Best = 2.537959e+03 FES = 890100 μ_F = 0.690 μ_{CR} = 0.241
 Diversity = 3.5885e-06 k_t = [4, 3, 2, 3]
 Generation 8910: Best = 2.537959e+03 FES = 891100 μ_F = 0.711 μ_{CR} = 0.235
 Diversity = 3.8171e-06 k_t = [1, 1, 1, 1]
 Generation 8920: Best = 2.537959e+03 FES = 892100 μ_F = 0.716 μ_{CR} = 0.243
 Diversity = 4.4226e-06 k_t = [3, 2, 2, 3]
 Generation 8930: Best = 2.537959e+03 FES = 893100 μ_F = 0.713 μ_{CR} = 0.244
 Diversity = 4.9400e-06 k_t = [1, 1, 1, 1]
 Generation 8940: Best = 2.537959e+03 FES = 894100 μ_F = 0.708 μ_{CR} = 0.246
 Diversity = 4.8437e-06 k_t = [2, 3, 3, 2]
 Generation 8950: Best = 2.537959e+03 FES = 895100 μ_F = 0.707 μ_{CR} = 0.256
 Diversity = 5.2296e-06 k_t = [1, 1, 1, 1]
 Generation 8960: Best = 2.537959e+03 FES = 896100 μ_F = 0.704 μ_{CR} = 0.253
 Diversity = 4.9147e-06 k_t = [4, 2, 2, 3]
 Generation 8970: Best = 2.537959e+03 FES = 897100 μ_F = 0.704 μ_{CR} = 0.245
 Diversity = 4.7897e-06 k_t = [1, 1, 1, 1]
 Generation 8980: Best = 2.537959e+03 FES = 898100 μ_F = 0.717 μ_{CR} = 0.248
 Diversity = 4.9074e-06 k_t = [2, 4, 3, 2]
 Generation 8990: Best = 2.537959e+03 FES = 899100 μ_F = 0.711 μ_{CR} = 0.245
 Diversity = 5.2547e-06 k_t = [1, 1, 1, 1]
 Generation 9000: Best = 2.537959e+03 FES = 900100 μ_F = 0.710 μ_{CR} = 0.246
 Diversity = 5.4663e-06 k_t = [3, 4, 2, 2]
 Generation 9010: Best = 2.537959e+03 FES = 901100 μ_F = 0.701 μ_{CR} = 0.254
 Diversity = 4.9304e-06 k_t = [1, 1, 1, 1]
 Generation 9020: Best = 2.537959e+03 FES = 902100 μ_F = 0.693 μ_{CR} = 0.253
 Diversity = 4.2415e-06 k_t = [4, 4, 2, 3]
 Generation 9030: Best = 2.537959e+03 FES = 903100 μ_F = 0.689 μ_{CR} = 0.258
 Diversity = 3.8135e-06 k_t = [1, 1, 1, 1]
 Generation 9040: Best = 2.537959e+03 FES = 904100 μ_F = 0.665 μ_{CR} = 0.255
 Diversity = 3.2891e-06 k_t = [3, 3, 2, 2]
 Generation 9050: Best = 2.537959e+03 FES = 905100 μ_F = 0.667 μ_{CR} = 0.257
 Diversity = 3.0710e-06 k_t = [1, 1, 1, 1]
 Generation 9060: Best = 2.537959e+03 FES = 906100 μ_F = 0.665 μ_{CR} = 0.256
 Diversity = 2.6413e-06 k_t = [3, 3, 2, 3]
 Generation 9070: Best = 2.537959e+03 FES = 907100 μ_F = 0.680 μ_{CR} = 0.256
 Diversity = 2.5498e-06 k_t = [1, 1, 1, 1]
 Generation 9080: Best = 2.537959e+03 FES = 908100 μ_F = 0.697 μ_{CR} = 0.252
 Diversity = 2.5463e-06 k_t = [2, 3, 4, 4]
 Generation 9090: Best = 2.537959e+03 FES = 909100 μ_F = 0.696 μ_{CR} = 0.252
 Diversity = 2.6123e-06 k_t = [1, 1, 1, 1]
 Generation 9100: Best = 2.537959e+03 FES = 910100 μ_F = 0.691 μ_{CR} = 0.256
 Diversity = 2.4367e-06 k_t = [4, 4, 3, 3]
 Generation 9110: Best = 2.537959e+03 FES = 911100 μ_F = 0.692 μ_{CR} = 0.266
 Diversity = 2.3642e-06 k_t = [1, 1, 1, 1]
 Generation 9120: Best = 2.537959e+03 FES = 912100 μ_F = 0.703 μ_{CR} = 0.260
 Diversity = 2.3849e-06 k_t = [3, 4, 3, 2]
 Generation 9130: Best = 2.537959e+03 FES = 913100 μ_F = 0.706 μ_{CR} = 0.266
 Diversity = 2.7551e-06 k_t = [1, 1, 1, 1]

Generation 9140: Best = 2.537959e+03 FES = 914100 μ_F = 0.701 μ_{CR} = 0.263
 Diversity = 2.5507e-06 k_t = [3, 2, 4, 2]
 Generation 9150: Best = 2.537959e+03 FES = 915100 μ_F = 0.693 μ_{CR} = 0.262
 Diversity = 2.5432e-06 k_t = [1, 1, 1, 1]
 Generation 9160: Best = 2.537959e+03 FES = 916100 μ_F = 0.708 μ_{CR} = 0.265
 Diversity = 2.4385e-06 k_t = [2, 2, 4, 3]
 Generation 9170: Best = 2.537959e+03 FES = 917100 μ_F = 0.703 μ_{CR} = 0.264
 Diversity = 2.3551e-06 k_t = [1, 1, 1, 1]
 Generation 9180: Best = 2.537959e+03 FES = 918100 μ_F = 0.705 μ_{CR} = 0.259
 Diversity = 2.2362e-06 k_t = [2, 3, 2, 3]
 Generation 9190: Best = 2.537959e+03 FES = 919100 μ_F = 0.707 μ_{CR} = 0.260
 Diversity = 2.2067e-06 k_t = [1, 1, 1, 1]
 Generation 9200: Best = 2.537959e+03 FES = 920100 μ_F = 0.712 μ_{CR} = 0.262
 Diversity = 1.9313e-06 k_t = [3, 2, 4, 2]
 Generation 9210: Best = 2.537959e+03 FES = 921100 μ_F = 0.706 μ_{CR} = 0.266
 Diversity = 1.9290e-06 k_t = [1, 1, 1, 1]
 Generation 9220: Best = 2.537959e+03 FES = 922100 μ_F = 0.702 μ_{CR} = 0.260
 Diversity = 1.8259e-06 k_t = [3, 2, 2, 4]
 Generation 9230: Best = 2.537959e+03 FES = 923100 μ_F = 0.699 μ_{CR} = 0.255
 Diversity = 1.8037e-06 k_t = [1, 1, 1, 1]
 Generation 9240: Best = 2.537959e+03 FES = 924100 μ_F = 0.693 μ_{CR} = 0.247
 Diversity = 1.6808e-06 k_t = [2, 2, 4, 2]
 Generation 9250: Best = 2.537959e+03 FES = 925100 μ_F = 0.677 μ_{CR} = 0.245
 Diversity = 1.6006e-06 k_t = [1, 1, 1, 1]
 Generation 9260: Best = 2.537959e+03 FES = 926100 μ_F = 0.683 μ_{CR} = 0.237
 Diversity = 1.3378e-06 k_t = [4, 4, 3, 3]
 Generation 9270: Best = 2.537959e+03 FES = 927100 μ_F = 0.682 μ_{CR} = 0.234
 Diversity = 1.2019e-06 k_t = [1, 1, 1, 1]
 Generation 9280: Best = 2.537959e+03 FES = 928100 μ_F = 0.693 μ_{CR} = 0.230
 Diversity = 1.0985e-06 k_t = [2, 4, 2, 3]
 Generation 9290: Best = 2.537959e+03 FES = 929100 μ_F = 0.678 μ_{CR} = 0.228
 Diversity = 1.0609e-06 k_t = [1, 1, 1, 1]
 Generation 9300: Best = 2.537959e+03 FES = 930100 μ_F = 0.681 μ_{CR} = 0.229
 Diversity = 9.7245e-07 k_t = [2, 3, 2, 2]
 Generation 9310: Best = 2.537959e+03 FES = 931100 μ_F = 0.675 μ_{CR} = 0.236
 Diversity = 9.9475e-07 k_t = [1, 1, 1, 1]
 Generation 9320: Best = 2.537959e+03 FES = 932100 μ_F = 0.697 μ_{CR} = 0.233
 Diversity = 9.3767e-07 k_t = [2, 3, 4, 2]
 Generation 9330: Best = 2.537959e+03 FES = 933100 μ_F = 0.701 μ_{CR} = 0.234
 Diversity = 9.9173e-07 k_t = [1, 1, 1, 1]
 Generation 9340: Best = 2.537959e+03 FES = 934100 μ_F = 0.720 μ_{CR} = 0.238
 Diversity = 9.4199e-07 k_t = [4, 4, 2, 4]
 Generation 9350: Best = 2.537959e+03 FES = 935100 μ_F = 0.726 μ_{CR} = 0.235
 Diversity = 8.8777e-07 k_t = [1, 1, 1, 1]
 Generation 9360: Best = 2.537959e+03 FES = 936100 μ_F = 0.732 μ_{CR} = 0.237
 Diversity = 8.9291e-07 k_t = [2, 3, 4, 3]
 Generation 9370: Best = 2.537959e+03 FES = 937100 μ_F = 0.729 μ_{CR} = 0.232
 Diversity = 1.0094e-06 k_t = [1, 1, 1, 1]
 Generation 9380: Best = 2.537959e+03 FES = 938100 μ_F = 0.738 μ_{CR} = 0.231
 Diversity = 9.7857e-07 k_t = [2, 4, 3, 4]
 Generation 9390: Best = 2.537959e+03 FES = 939100 μ_F = 0.746 μ_{CR} = 0.227
 Diversity = 1.0582e-06 k_t = [1, 1, 1, 1]
 Generation 9400: Best = 2.537959e+03 FES = 940100 μ_F = 0.750 μ_{CR} = 0.221
 Diversity = 1.1464e-06 k_t = [2, 3, 3, 2]

Generation 9410: Best = 2.537959e+03 FES = 941100 μ_F = 0.748 μ_{CR} = 0.217
 Diversity = 1.2959e-06 k_t = [1, 1, 1, 1]
 Generation 9420: Best = 2.537959e+03 FES = 942100 μ_F = 0.743 μ_{CR} = 0.228
 Diversity = 1.2506e-06 k_t = [2, 3, 4, 4]
 Generation 9430: Best = 2.537959e+03 FES = 943100 μ_F = 0.734 μ_{CR} = 0.222
 Diversity = 1.2179e-06 k_t = [1, 1, 1, 1]
 Generation 9440: Best = 2.537959e+03 FES = 944100 μ_F = 0.722 μ_{CR} = 0.223
 Diversity = 1.0564e-06 k_t = [3, 3, 3, 2]
 Generation 9450: Best = 2.537959e+03 FES = 945100 μ_F = 0.716 μ_{CR} = 0.225
 Diversity = 9.6841e-07 k_t = [1, 1, 1, 1]
 Generation 9460: Best = 2.537959e+03 FES = 946100 μ_F = 0.712 μ_{CR} = 0.229
 Diversity = 8.7587e-07 k_t = [3, 4, 2, 3]
 Generation 9470: Best = 2.537959e+03 FES = 947100 μ_F = 0.718 μ_{CR} = 0.228
 Diversity = 8.7102e-07 k_t = [1, 1, 1, 1]
 Generation 9480: Best = 2.537959e+03 FES = 948100 μ_F = 0.729 μ_{CR} = 0.225
 Diversity = 8.2654e-07 k_t = [3, 2, 2, 4]
 Generation 9490: Best = 2.537959e+03 FES = 949100 μ_F = 0.733 μ_{CR} = 0.228
 Diversity = 8.7210e-07 k_t = [1, 1, 1, 1]
 Generation 9500: Best = 2.537959e+03 FES = 950100 μ_F = 0.741 μ_{CR} = 0.225
 Diversity = 8.6983e-07 k_t = [2, 4, 4, 3]
 Generation 9510: Best = 2.537959e+03 FES = 951100 μ_F = 0.735 μ_{CR} = 0.223
 Diversity = 8.8456e-07 k_t = [1, 1, 1, 1]
 Generation 9520: Best = 2.537959e+03 FES = 952100 μ_F = 0.729 μ_{CR} = 0.212
 Diversity = 8.5498e-07 k_t = [3, 2, 4, 2]
 Generation 9530: Best = 2.537959e+03 FES = 953100 μ_F = 0.725 μ_{CR} = 0.211
 Diversity = 9.2715e-07 k_t = [1, 1, 1, 1]
 Generation 9540: Best = 2.537959e+03 FES = 954100 μ_F = 0.733 μ_{CR} = 0.213
 Diversity = 9.7599e-07 k_t = [2, 2, 2, 4]
 Generation 9550: Best = 2.537959e+03 FES = 955100 μ_F = 0.726 μ_{CR} = 0.214
 Diversity = 1.0965e-06 k_t = [1, 1, 1, 1]
 Generation 9560: Best = 2.537959e+03 FES = 956100 μ_F = 0.738 μ_{CR} = 0.222
 Diversity = 1.1380e-06 k_t = [4, 4, 2, 3]
 Generation 9570: Best = 2.537959e+03 FES = 957100 μ_F = 0.739 μ_{CR} = 0.238
 Diversity = 1.2969e-06 k_t = [1, 1, 1, 1]
 Generation 9580: Best = 2.537959e+03 FES = 958100 μ_F = 0.731 μ_{CR} = 0.233
 Diversity = 1.1776e-06 k_t = [3, 4, 2, 3]
 Generation 9590: Best = 2.537959e+03 FES = 959100 μ_F = 0.731 μ_{CR} = 0.233
 Diversity = 1.2789e-06 k_t = [1, 1, 1, 1]
 Generation 9600: Best = 2.537959e+03 FES = 960100 μ_F = 0.727 μ_{CR} = 0.234
 Diversity = 1.2499e-06 k_t = [4, 3, 3, 3]
 Generation 9610: Best = 2.537959e+03 FES = 961100 μ_F = 0.728 μ_{CR} = 0.243
 Diversity = 1.3105e-06 k_t = [1, 1, 1, 1]
 Generation 9620: Best = 2.537959e+03 FES = 962100 μ_F = 0.738 μ_{CR} = 0.241
 Diversity = 1.4847e-06 k_t = [4, 4, 2, 4]
 Generation 9630: Best = 2.537959e+03 FES = 963100 μ_F = 0.740 μ_{CR} = 0.236
 Diversity = 1.4591e-06 k_t = [1, 1, 1, 1]
 Generation 9640: Best = 2.537959e+03 FES = 964100 μ_F = 0.742 μ_{CR} = 0.237
 Diversity = 1.3275e-06 k_t = [2, 4, 3, 4]
 Generation 9650: Best = 2.537959e+03 FES = 965100 μ_F = 0.739 μ_{CR} = 0.227
 Diversity = 1.2985e-06 k_t = [1, 1, 1, 1]
 Generation 9660: Best = 2.537959e+03 FES = 966100 μ_F = 0.736 μ_{CR} = 0.224
 Diversity = 1.4123e-06 k_t = [2, 2, 2, 3]
 Generation 9670: Best = 2.537959e+03 FES = 967100 μ_F = 0.732 μ_{CR} = 0.223
 Diversity = 1.4561e-06 k_t = [1, 1, 1, 1]

Generation 9680: Best = 2.537959e+03 FES = 968100 μ_F = 0.726 μ_{CR} = 0.225
 Diversity = 1.3377e-06 k_t = [2, 2, 4, 4]
 Generation 9690: Best = 2.537959e+03 FES = 969100 μ_F = 0.738 μ_{CR} = 0.224
 Diversity = 1.4036e-06 k_t = [1, 1, 1, 1]
 Generation 9700: Best = 2.537959e+03 FES = 970100 μ_F = 0.738 μ_{CR} = 0.238
 Diversity = 1.3561e-06 k_t = [4, 4, 4, 3]
 Generation 9710: Best = 2.537959e+03 FES = 971100 μ_F = 0.729 μ_{CR} = 0.227
 Diversity = 1.4699e-06 k_t = [1, 1, 1, 1]
 Generation 9720: Best = 2.537959e+03 FES = 972100 μ_F = 0.734 μ_{CR} = 0.245
 Diversity = 1.8035e-06 k_t = [4, 4, 2, 4]
 Generation 9730: Best = 2.537959e+03 FES = 973100 μ_F = 0.729 μ_{CR} = 0.247
 Diversity = 1.8309e-06 k_t = [1, 1, 1, 1]
 Generation 9740: Best = 2.537959e+03 FES = 974100 μ_F = 0.744 μ_{CR} = 0.255
 Diversity = 1.9512e-06 k_t = [3, 3, 2, 3]
 Generation 9750: Best = 2.537959e+03 FES = 975100 μ_F = 0.743 μ_{CR} = 0.262
 Diversity = 2.1144e-06 k_t = [1, 1, 1, 1]
 Generation 9760: Best = 2.537959e+03 FES = 976100 μ_F = 0.749 μ_{CR} = 0.262
 Diversity = 2.1532e-06 k_t = [4, 4, 3, 2]
 Generation 9770: Best = 2.537959e+03 FES = 977100 μ_F = 0.746 μ_{CR} = 0.255
 Diversity = 2.1782e-06 k_t = [1, 1, 1, 1]
 Generation 9780: Best = 2.537959e+03 FES = 978100 μ_F = 0.731 μ_{CR} = 0.250
 Diversity = 2.3233e-06 k_t = [2, 3, 2, 2]
 Generation 9790: Best = 2.537959e+03 FES = 979100 μ_F = 0.735 μ_{CR} = 0.251
 Diversity = 2.5529e-06 k_t = [1, 1, 1, 1]
 Generation 9800: Best = 2.537959e+03 FES = 980100 μ_F = 0.728 μ_{CR} = 0.248
 Diversity = 2.3421e-06 k_t = [2, 2, 3, 2]
 Generation 9810: Best = 2.537959e+03 FES = 981100 μ_F = 0.709 μ_{CR} = 0.248
 Diversity = 2.1870e-06 k_t = [1, 1, 1, 1]
 Generation 9820: Best = 2.537959e+03 FES = 982100 μ_F = 0.707 μ_{CR} = 0.245
 Diversity = 2.0565e-06 k_t = [4, 2, 3, 4]
 Generation 9830: Best = 2.537959e+03 FES = 983100 μ_F = 0.723 μ_{CR} = 0.247
 Diversity = 2.2778e-06 k_t = [1, 1, 1, 1]
 Generation 9840: Best = 2.537959e+03 FES = 984100 μ_F = 0.732 μ_{CR} = 0.251
 Diversity = 2.5809e-06 k_t = [2, 3, 3, 2]
 Generation 9850: Best = 2.537959e+03 FES = 985100 μ_F = 0.734 μ_{CR} = 0.245
 Diversity = 2.6574e-06 k_t = [1, 1, 1, 1]
 Generation 9860: Best = 2.537959e+03 FES = 986100 μ_F = 0.735 μ_{CR} = 0.247
 Diversity = 2.5861e-06 k_t = [3, 3, 3, 3]
 Generation 9870: Best = 2.537959e+03 FES = 987100 μ_F = 0.713 μ_{CR} = 0.241
 Diversity = 2.4795e-06 k_t = [1, 1, 1, 1]
 Generation 9880: Best = 2.537959e+03 FES = 988100 μ_F = 0.714 μ_{CR} = 0.237
 Diversity = 2.3215e-06 k_t = [2, 3, 3, 3]
 Generation 9890: Best = 2.537959e+03 FES = 989100 μ_F = 0.704 μ_{CR} = 0.245
 Diversity = 2.2954e-06 k_t = [1, 1, 1, 1]
 Generation 9900: Best = 2.537959e+03 FES = 990100 μ_F = 0.702 μ_{CR} = 0.247
 Diversity = 2.2459e-06 k_t = [2, 2, 2, 3]
 Generation 9910: Best = 2.537959e+03 FES = 991100 μ_F = 0.694 μ_{CR} = 0.246
 Diversity = 2.2716e-06 k_t = [1, 1, 1, 1]
 Generation 9920: Best = 2.537959e+03 FES = 992100 μ_F = 0.706 μ_{CR} = 0.248
 Diversity = 2.2030e-06 k_t = [4, 2, 2, 2]
 Generation 9930: Best = 2.537959e+03 FES = 993100 μ_F = 0.712 μ_{CR} = 0.250
 Diversity = 2.2271e-06 k_t = [1, 1, 1, 1]
 Generation 9940: Best = 2.537959e+03 FES = 994100 μ_F = 0.707 μ_{CR} = 0.248
 Diversity = 2.1464e-06 k_t = [4, 4, 3, 3]

Generation 9950: Best = 2.537959e+03 FES = 995100 μ_F = 0.719 μ_{CR} = 0.249
 Diversity = 2.2000e-06 k_t = [1, 1, 1, 1]
 Generation 9960: Best = 2.537959e+03 FES = 996100 μ_F = 0.724 μ_{CR} = 0.254
 Diversity = 2.3936e-06 k_t = [2, 4, 2, 3]
 Generation 9970: Best = 2.537959e+03 FES = 997100 μ_F = 0.734 μ_{CR} = 0.261
 Diversity = 2.9130e-06 k_t = [1, 1, 1, 1]
 Generation 9980: Best = 2.537959e+03 FES = 998100 μ_F = 0.729 μ_{CR} = 0.271
 Diversity = 3.1032e-06 k_t = [4, 4, 3, 3]
 Generation 9990: Best = 2.537959e+03 FES = 999100 μ_F = 0.743 μ_{CR} = 0.268
 Diversity = 3.3361e-06 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 2.537959e+03

Total Function Evals = 1000000

Total Successful Updates = 327253

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Results:

Best Fitness : 2.537959e+03

Global Optimum : 2.100000e+03

Error (|best-optimum|) : 4.379585e+02

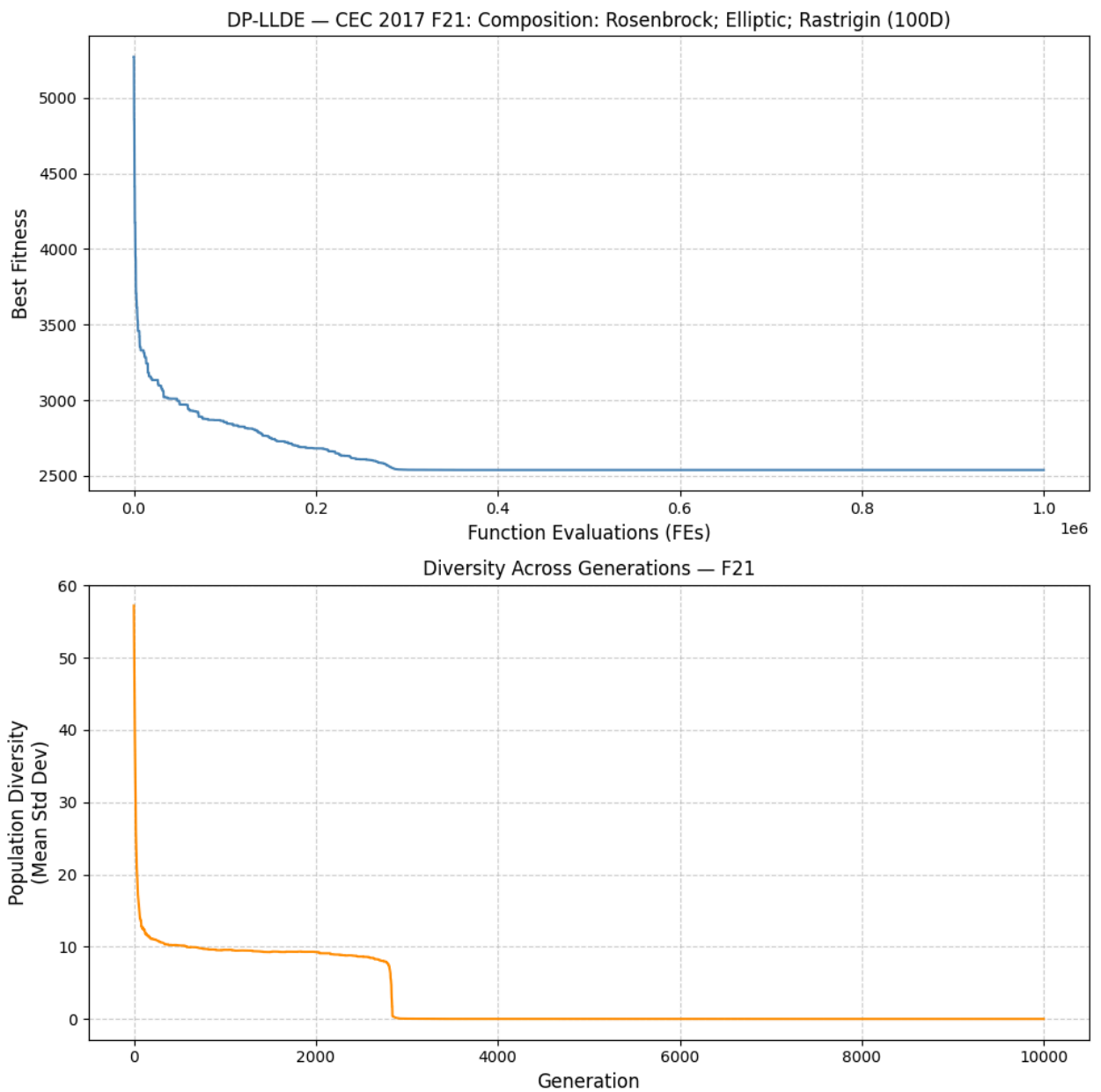
Best Solution (5 dim) : [2.43966028 3.85393567 60.20934927 47.12548754 49.28796278]

D^(0) (init. diversity): 5.7220e+01

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✓ Checkpoint saved to: /content/drive/My Drive/thesis_results/all_results_checkpoint.pkl



Graph saved as convergence_F21.png

✓ Plot saved to: /content/drive/My Drive/thesis_results/plots/convergence_F21.png

```
In [ ]: # =====
# F22 — Composition Function 2 (Rastrigin; Griewank; Schwefel)
# =====
run_and_plot('F22', make_wrapper(composition.f22), 2200.0, 'Composition: Rastr
```

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DP-LLDE on CEC 2017 F22 – Single Run

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Function : Composition: Rastrigin; Griewank; Schwefel (100D)
Bounds : [-100, 100]
Global optimum : 2200.0
Initial $D^*(0) = 5.7220e+01$ | Diversity threshold = 5.7220e+00
Generation 0: Best Fitness = 4.002781e+04
Generation 10: Best = 3.677280e+04 FES = 1100 $\mu_F = 0.613$ $\mu_{CR} = 0.333$
Diversity = 4.8452e+01 $k_t = [1, 1, 1, 1]$
Generation 20: Best = 3.597930e+04 FES = 2100 $\mu_F = 0.669$ $\mu_{CR} = 0.323$
Diversity = 4.7623e+01 $k_t = [2, 3, 2, 3]$
Generation 30: Best = 3.525111e+04 FES = 3100 $\mu_F = 0.631$ $\mu_{CR} = 0.321$
Diversity = 4.7422e+01 $k_t = [1, 1, 1, 1]$
Generation 40: Best = 3.499026e+04 FES = 4100 $\mu_F = 0.658$ $\mu_{CR} = 0.301$
Diversity = 4.6700e+01 $k_t = [3, 2, 3, 3]$
Generation 50: Best = 3.499026e+04 FES = 5100 $\mu_F = 0.671$ $\mu_{CR} = 0.265$
Diversity = 4.6588e+01 $k_t = [1, 1, 1, 1]$
Generation 60: Best = 3.499026e+04 FES = 6100 $\mu_F = 0.796$ $\mu_{CR} = 0.228$
Diversity = 4.6697e+01 $k_t = [4, 2, 2, 2]$
Generation 70: Best = 3.499026e+04 FES = 7100 $\mu_F = 0.822$ $\mu_{CR} = 0.171$
Diversity = 4.6735e+01 $k_t = [1, 1, 1, 1]$
Generation 80: Best = 3.498441e+04 FES = 8100 $\mu_F = 0.810$ $\mu_{CR} = 0.105$
Diversity = 4.6588e+01 $k_t = [2, 3, 4, 3]$
Generation 90: Best = 3.423293e+04 FES = 9100 $\mu_F = 0.818$ $\mu_{CR} = 0.076$
Diversity = 4.6568e+01 $k_t = [1, 1, 1, 1]$
Generation 100: Best = 3.322354e+04 FES = 10100 $\mu_F = 0.824$ $\mu_{CR} = 0.054$
Diversity = 4.7350e+01 $k_t = [3, 2, 4, 3]$
Generation 110: Best = 3.322354e+04 FES = 11100 $\mu_F = 0.820$ $\mu_{CR} = 0.050$
Diversity = 4.7449e+01 $k_t = [1, 1, 1, 1]$
Generation 120: Best = 3.212474e+04 FES = 12100 $\mu_F = 0.784$ $\mu_{CR} = 0.048$
Diversity = 4.7885e+01 $k_t = [4, 2, 4, 4]$
Generation 130: Best = 3.212474e+04 FES = 13100 $\mu_F = 0.788$ $\mu_{CR} = 0.045$
Diversity = 4.7796e+01 $k_t = [1, 1, 1, 1]$
Generation 140: Best = 3.212474e+04 FES = 14100 $\mu_F = 0.791$ $\mu_{CR} = 0.050$
Diversity = 4.7788e+01 $k_t = [3, 2, 4, 2]$
Generation 150: Best = 3.212474e+04 FES = 15100 $\mu_F = 0.773$ $\mu_{CR} = 0.042$
Diversity = 4.7505e+01 $k_t = [1, 1, 1, 1]$
Generation 160: Best = 3.193851e+04 FES = 16100 $\mu_F = 0.817$ $\mu_{CR} = 0.038$
Diversity = 4.7824e+01 $k_t = [4, 3, 3, 4]$
Generation 170: Best = 3.127454e+04 FES = 17100 $\mu_F = 0.788$ $\mu_{CR} = 0.036$
Diversity = 4.7871e+01 $k_t = [1, 1, 1, 1]$
Generation 180: Best = 3.127454e+04 FES = 18100 $\mu_F = 0.807$ $\mu_{CR} = 0.035$
Diversity = 4.7869e+01 $k_t = [2, 2, 4, 4]$
Generation 190: Best = 3.092027e+04 FES = 19100 $\mu_F = 0.842$ $\mu_{CR} = 0.023$
Diversity = 4.7733e+01 $k_t = [1, 1, 1, 1]$
Generation 200: Best = 3.066814e+04 FES = 20100 $\mu_F = 0.872$ $\mu_{CR} = 0.031$
Diversity = 4.7631e+01 $k_t = [4, 2, 4, 3]$
Generation 210: Best = 3.066814e+04 FES = 21100 $\mu_F = 0.853$ $\mu_{CR} = 0.028$
Diversity = 4.7557e+01 $k_t = [1, 1, 1, 1]$
Generation 220: Best = 3.037602e+04 FES = 22100 $\mu_F = 0.768$ $\mu_{CR} = 0.033$
Diversity = 4.7391e+01 $k_t = [3, 4, 3, 3]$

Generation 230: Best = 3.037602e+04	FES = 23100	$\mu_F = 0.722$	$\mu_{CR} = 0.032$
Diversity = 4.7161e+01	$k_t = [1, 1, 1, 1]$		
Generation 240: Best = 3.037602e+04	FES = 24100	$\mu_F = 0.766$	$\mu_{CR} = 0.030$
Diversity = 4.7157e+01	$k_t = [2, 3, 3, 2]$		
Generation 250: Best = 3.037602e+04	FES = 25100	$\mu_F = 0.798$	$\mu_{CR} = 0.024$
Diversity = 4.7196e+01	$k_t = [1, 1, 1, 1]$		
Generation 260: Best = 3.037602e+04	FES = 26100	$\mu_F = 0.773$	$\mu_{CR} = 0.021$
Diversity = 4.7241e+01	$k_t = [2, 3, 4, 4]$		
Generation 270: Best = 3.037602e+04	FES = 27100	$\mu_F = 0.831$	$\mu_{CR} = 0.019$
Diversity = 4.7379e+01	$k_t = [1, 1, 1, 1]$		
Generation 280: Best = 3.035642e+04	FES = 28100	$\mu_F = 0.782$	$\mu_{CR} = 0.021$
Diversity = 4.7407e+01	$k_t = [4, 4, 3, 3]$		
Generation 290: Best = 3.035642e+04	FES = 29100	$\mu_F = 0.729$	$\mu_{CR} = 0.028$
Diversity = 4.7371e+01	$k_t = [1, 1, 1, 1]$		
Generation 300: Best = 3.035642e+04	FES = 30100	$\mu_F = 0.717$	$\mu_{CR} = 0.024$
Diversity = 4.7300e+01	$k_t = [4, 2, 3, 3]$		
Generation 310: Best = 3.035642e+04	FES = 31100	$\mu_F = 0.738$	$\mu_{CR} = 0.029$
Diversity = 4.7078e+01	$k_t = [1, 1, 1, 1]$		
Generation 320: Best = 3.035642e+04	FES = 32100	$\mu_F = 0.695$	$\mu_{CR} = 0.030$
Diversity = 4.6965e+01	$k_t = [4, 3, 4, 2]$		
Generation 330: Best = 2.945173e+04	FES = 33100	$\mu_F = 0.759$	$\mu_{CR} = 0.026$
Diversity = 4.6951e+01	$k_t = [1, 1, 1, 1]$		
Generation 340: Best = 2.945173e+04	FES = 34100	$\mu_F = 0.752$	$\mu_{CR} = 0.027$
Diversity = 4.7152e+01	$k_t = [3, 3, 4, 3]$		
Generation 350: Best = 2.945173e+04	FES = 35100	$\mu_F = 0.740$	$\mu_{CR} = 0.026$
Diversity = 4.7151e+01	$k_t = [1, 1, 1, 1]$		
Generation 360: Best = 2.944890e+04	FES = 36100	$\mu_F = 0.771$	$\mu_{CR} = 0.024$
Diversity = 4.6740e+01	$k_t = [2, 3, 3, 2]$		
Generation 370: Best = 2.944890e+04	FES = 37100	$\mu_F = 0.791$	$\mu_{CR} = 0.024$
Diversity = 4.6496e+01	$k_t = [1, 1, 1, 1]$		
Generation 380: Best = 2.944890e+04	FES = 38100	$\mu_F = 0.786$	$\mu_{CR} = 0.025$
Diversity = 4.6541e+01	$k_t = [2, 2, 3, 2]$		
Generation 390: Best = 2.944890e+04	FES = 39100	$\mu_F = 0.772$	$\mu_{CR} = 0.022$
Diversity = 4.6534e+01	$k_t = [1, 1, 1, 1]$		
Generation 400: Best = 2.944890e+04	FES = 40100	$\mu_F = 0.541$	$\mu_{CR} = 0.034$
Diversity = 4.6522e+01	$k_t = [4, 4, 3, 3]$		
Generation 410: Best = 2.944890e+04	FES = 41100	$\mu_F = 0.449$	$\mu_{CR} = 0.045$
Diversity = 4.6517e+01	$k_t = [1, 1, 1, 1]$		
Generation 420: Best = 2.944890e+04	FES = 42100	$\mu_F = 0.511$	$\mu_{CR} = 0.036$
Diversity = 4.6497e+01	$k_t = [3, 4, 3, 3]$		
Generation 430: Best = 2.944890e+04	FES = 43100	$\mu_F = 0.604$	$\mu_{CR} = 0.024$
Diversity = 4.6480e+01	$k_t = [1, 1, 1, 1]$		
Generation 440: Best = 2.944890e+04	FES = 44100	$\mu_F = 0.565$	$\mu_{CR} = 0.026$
Diversity = 4.6392e+01	$k_t = [4, 4, 3, 3]$		
Generation 450: Best = 2.944890e+04	FES = 45100	$\mu_F = 0.545$	$\mu_{CR} = 0.029$
Diversity = 4.6357e+01	$k_t = [1, 1, 1, 1]$		
Generation 460: Best = 2.944890e+04	FES = 46100	$\mu_F = 0.502$	$\mu_{CR} = 0.034$
Diversity = 4.6022e+01	$k_t = [4, 3, 2, 4]$		
Generation 470: Best = 2.944890e+04	FES = 47100	$\mu_F = 0.531$	$\mu_{CR} = 0.025$
Diversity = 4.6008e+01	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 2.926485e+04	FES = 48100	$\mu_F = 0.597$	$\mu_{CR} = 0.019$
Diversity = 4.5861e+01	$k_t = [3, 4, 3, 4]$		
Generation 490: Best = 2.926485e+04	FES = 49100	$\mu_F = 0.573$	$\mu_{CR} = 0.021$
Diversity = 4.5662e+01	$k_t = [1, 1, 1, 1]$		

Generation 500: Best = 2.926485e+04	FES = 50100	$\mu_F = 0.597$	$\mu_{CR} = 0.025$
Diversity = 4.5633e+01	$k_t = [3, 3, 4, 2]$		
Generation 510: Best = 2.924857e+04	FES = 51100	$\mu_F = 0.621$	$\mu_{CR} = 0.028$
Diversity = 4.5544e+01	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 2.898571e+04	FES = 52100	$\mu_F = 0.550$	$\mu_{CR} = 0.029$
Diversity = 4.5532e+01	$k_t = [3, 3, 2, 3]$		
Generation 530: Best = 2.898571e+04	FES = 53100	$\mu_F = 0.495$	$\mu_{CR} = 0.036$
Diversity = 4.5526e+01	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 2.898571e+04	FES = 54100	$\mu_F = 0.504$	$\mu_{CR} = 0.043$
Diversity = 4.5507e+01	$k_t = [2, 3, 4, 2]$		
Generation 550: Best = 2.898571e+04	FES = 55100	$\mu_F = 0.382$	$\mu_{CR} = 0.036$
Diversity = 4.5439e+01	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 2.879676e+04	FES = 56100	$\mu_F = 0.438$	$\mu_{CR} = 0.034$
Diversity = 4.5508e+01	$k_t = [3, 4, 4, 4]$		
Generation 570: Best = 2.879676e+04	FES = 57100	$\mu_F = 0.318$	$\mu_{CR} = 0.056$
Diversity = 4.5343e+01	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 2.879676e+04	FES = 58100	$\mu_F = 0.240$	$\mu_{CR} = 0.054$
Diversity = 4.5315e+01	$k_t = [3, 4, 2, 2]$		
Generation 590: Best = 2.875884e+04	FES = 59100	$\mu_F = 0.242$	$\mu_{CR} = 0.049$
Diversity = 4.5265e+01	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 2.862309e+04	FES = 60100	$\mu_F = 0.283$	$\mu_{CR} = 0.039$
Diversity = 4.5251e+01	$k_t = [4, 3, 3, 4]$		
Generation 610: Best = 2.846454e+04	FES = 61100	$\mu_F = 0.258$	$\mu_{CR} = 0.055$
Diversity = 4.5217e+01	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 2.841412e+04	FES = 62100	$\mu_F = 0.307$	$\mu_{CR} = 0.045$
Diversity = 4.5194e+01	$k_t = [2, 4, 4, 3]$		
Generation 630: Best = 2.841412e+04	FES = 63100	$\mu_F = 0.267$	$\mu_{CR} = 0.042$
Diversity = 4.5176e+01	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 2.841412e+04	FES = 64100	$\mu_F = 0.253$	$\mu_{CR} = 0.045$
Diversity = 4.5238e+01	$k_t = [4, 3, 3, 4]$		
Generation 650: Best = 2.833874e+04	FES = 65100	$\mu_F = 0.169$	$\mu_{CR} = 0.057$
Diversity = 4.5165e+01	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 2.833874e+04	FES = 66100	$\mu_F = 0.241$	$\mu_{CR} = 0.040$
Diversity = 4.5149e+01	$k_t = [4, 2, 3, 3]$		
Generation 670: Best = 2.809638e+04	FES = 67100	$\mu_F = 0.291$	$\mu_{CR} = 0.040$
Diversity = 4.5133e+01	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 2.809638e+04	FES = 68100	$\mu_F = 0.295$	$\mu_{CR} = 0.030$
Diversity = 4.5121e+01	$k_t = [3, 4, 3, 3]$		
Generation 690: Best = 2.806716e+04	FES = 69100	$\mu_F = 0.315$	$\mu_{CR} = 0.048$
Diversity = 4.5105e+01	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 2.806716e+04	FES = 70100	$\mu_F = 0.329$	$\mu_{CR} = 0.046$
Diversity = 4.4919e+01	$k_t = [4, 2, 3, 4]$		
Generation 710: Best = 2.802260e+04	FES = 71100	$\mu_F = 0.290$	$\mu_{CR} = 0.042$
Diversity = 4.4902e+01	$k_t = [1, 1, 1, 1]$		
Generation 720: Best = 2.802260e+04	FES = 72100	$\mu_F = 0.216$	$\mu_{CR} = 0.043$
Diversity = 4.4889e+01	$k_t = [2, 3, 2, 3]$		
Generation 730: Best = 2.743606e+04	FES = 73100	$\mu_F = 0.316$	$\mu_{CR} = 0.027$
Diversity = 4.4887e+01	$k_t = [1, 1, 1, 1]$		
Generation 740: Best = 2.743606e+04	FES = 74100	$\mu_F = 0.456$	$\mu_{CR} = 0.028$
Diversity = 4.4886e+01	$k_t = [4, 4, 4, 3]$		
Generation 750: Best = 2.743606e+04	FES = 75100	$\mu_F = 0.434$	$\mu_{CR} = 0.034$
Diversity = 4.4782e+01	$k_t = [1, 1, 1, 1]$		
Generation 760: Best = 2.743606e+04	FES = 76100	$\mu_F = 0.539$	$\mu_{CR} = 0.026$
Diversity = 4.4640e+01	$k_t = [4, 2, 2, 4]$		

Generation 770: Best = 2.743606e+04	FES = 77100	$\mu_F = 0.532$	$\mu_{CR} = 0.029$
Diversity = 4.4637e+01	k_t = [1, 1, 1, 1]		
Generation 780: Best = 2.743606e+04	FES = 78100	$\mu_F = 0.513$	$\mu_{CR} = 0.033$
Diversity = 4.4686e+01	k_t = [2, 4, 2, 4]		
Generation 790: Best = 2.743606e+04	FES = 79100	$\mu_F = 0.515$	$\mu_{CR} = 0.027$
Diversity = 4.4683e+01	k_t = [1, 1, 1, 1]		
Generation 800: Best = 2.743606e+04	FES = 80100	$\mu_F = 0.543$	$\mu_{CR} = 0.022$
Diversity = 4.4681e+01	k_t = [4, 3, 4, 3]		
Generation 810: Best = 2.743606e+04	FES = 81100	$\mu_F = 0.502$	$\mu_{CR} = 0.018$
Diversity = 4.4676e+01	k_t = [1, 1, 1, 1]		
Generation 820: Best = 2.743606e+04	FES = 82100	$\mu_F = 0.525$	$\mu_{CR} = 0.024$
Diversity = 4.4675e+01	k_t = [2, 3, 2, 3]		
Generation 830: Best = 2.743606e+04	FES = 83100	$\mu_F = 0.441$	$\mu_{CR} = 0.027$
Diversity = 4.4671e+01	k_t = [1, 1, 1, 1]		
Generation 840: Best = 2.743606e+04	FES = 84100	$\mu_F = 0.370$	$\mu_{CR} = 0.022$
Diversity = 4.4662e+01	k_t = [2, 3, 4, 3]		
Generation 850: Best = 2.743606e+04	FES = 85100	$\mu_F = 0.351$	$\mu_{CR} = 0.020$
Diversity = 4.4651e+01	k_t = [1, 1, 1, 1]		
Generation 860: Best = 2.743606e+04	FES = 86100	$\mu_F = 0.348$	$\mu_{CR} = 0.023$
Diversity = 4.4634e+01	k_t = [4, 2, 3, 2]		
Generation 870: Best = 2.743606e+04	FES = 87100	$\mu_F = 0.352$	$\mu_{CR} = 0.022$
Diversity = 4.4621e+01	k_t = [1, 1, 1, 1]		
Generation 880: Best = 2.743606e+04	FES = 88100	$\mu_F = 0.326$	$\mu_{CR} = 0.034$
Diversity = 4.4614e+01	k_t = [3, 3, 3, 2]		
Generation 890: Best = 2.743388e+04	FES = 89100	$\mu_F = 0.323$	$\mu_{CR} = 0.043$
Diversity = 4.4608e+01	k_t = [1, 1, 1, 1]		
Generation 900: Best = 2.743388e+04	FES = 90100	$\mu_F = 0.340$	$\mu_{CR} = 0.030$
Diversity = 4.4592e+01	k_t = [4, 2, 2, 3]		
Generation 910: Best = 2.743388e+04	FES = 91100	$\mu_F = 0.362$	$\mu_{CR} = 0.019$
Diversity = 4.4533e+01	k_t = [1, 1, 1, 1]		
Generation 920: Best = 2.704878e+04	FES = 92100	$\mu_F = 0.274$	$\mu_{CR} = 0.025$
Diversity = 4.4529e+01	k_t = [4, 4, 2, 4]		
Generation 930: Best = 2.704878e+04	FES = 93100	$\mu_F = 0.305$	$\mu_{CR} = 0.028$
Diversity = 4.4512e+01	k_t = [1, 1, 1, 1]		
Generation 940: Best = 2.650245e+04	FES = 94100	$\mu_F = 0.208$	$\mu_{CR} = 0.037$
Diversity = 4.4450e+01	k_t = [2, 3, 4, 2]		
Generation 950: Best = 2.650245e+04	FES = 95100	$\mu_F = 0.174$	$\mu_{CR} = 0.041$
Diversity = 4.4444e+01	k_t = [1, 1, 1, 1]		
Generation 960: Best = 2.650245e+04	FES = 96100	$\mu_F = 0.209$	$\mu_{CR} = 0.050$
Diversity = 4.4316e+01	k_t = [3, 2, 4, 4]		
Generation 970: Best = 2.650245e+04	FES = 97100	$\mu_F = 0.210$	$\mu_{CR} = 0.037$
Diversity = 4.4302e+01	k_t = [1, 1, 1, 1]		
Generation 980: Best = 2.650245e+04	FES = 98100	$\mu_F = 0.268$	$\mu_{CR} = 0.024$
Diversity = 4.4293e+01	k_t = [2, 2, 4, 3]		
Generation 990: Best = 2.650245e+04	FES = 99100	$\mu_F = 0.283$	$\mu_{CR} = 0.021$
Diversity = 4.4278e+01	k_t = [1, 1, 1, 1]		
Generation 1000: Best = 2.650245e+04	FES = 100100	$\mu_F = 0.275$	$\mu_{CR} = 0.036$
Diversity = 4.4451e+01	k_t = [4, 2, 2, 4]		
Generation 1010: Best = 2.650245e+04	FES = 101100	$\mu_F = 0.231$	$\mu_{CR} = 0.029$
Diversity = 4.4445e+01	k_t = [1, 1, 1, 1]		
Generation 1020: Best = 2.650245e+04	FES = 102100	$\mu_F = 0.322$	$\mu_{CR} = 0.026$
Diversity = 4.4659e+01	k_t = [3, 3, 3, 4]		
Generation 1030: Best = 2.609699e+04	FES = 103100	$\mu_F = 0.393$	$\mu_{CR} = 0.027$
Diversity = 4.4650e+01	k_t = [1, 1, 1, 1]		

Generation 1040: Best = 2.609699e+04 FES = 104100 μ_F = 0.488 μ_{CR} = 0.019
 Diversity = 4.4452e+01 k_t = [2, 2, 4, 4]
 Generation 1050: Best = 2.609699e+04 FES = 105100 μ_F = 0.459 μ_{CR} = 0.019
 Diversity = 4.4454e+01 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 2.599515e+04 FES = 106100 μ_F = 0.408 μ_{CR} = 0.032
 Diversity = 4.4453e+01 k_t = [4, 3, 3, 3]
 Generation 1070: Best = 2.599515e+04 FES = 107100 μ_F = 0.278 μ_{CR} = 0.039
 Diversity = 4.4445e+01 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 2.599515e+04 FES = 108100 μ_F = 0.191 μ_{CR} = 0.041
 Diversity = 4.4439e+01 k_t = [3, 3, 2, 4]
 Generation 1090: Best = 2.599515e+04 FES = 109100 μ_F = 0.208 μ_{CR} = 0.060
 Diversity = 4.4431e+01 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 2.598425e+04 FES = 110100 μ_F = 0.237 μ_{CR} = 0.059
 Diversity = 4.4355e+01 k_t = [2, 3, 4, 4]
 Generation 1110: Best = 2.598425e+04 FES = 111100 μ_F = 0.191 μ_{CR} = 0.047
 Diversity = 4.4349e+01 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 2.598425e+04 FES = 112100 μ_F = 0.143 μ_{CR} = 0.050
 Diversity = 4.4338e+01 k_t = [3, 4, 2, 3]
 Generation 1130: Best = 2.598425e+04 FES = 113100 μ_F = 0.183 μ_{CR} = 0.046
 Diversity = 4.4338e+01 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 2.598425e+04 FES = 114100 μ_F = 0.230 μ_{CR} = 0.042
 Diversity = 4.4347e+01 k_t = [4, 3, 2, 4]
 Generation 1150: Best = 2.598425e+04 FES = 115100 μ_F = 0.199 μ_{CR} = 0.043
 Diversity = 4.4339e+01 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 2.598425e+04 FES = 116100 μ_F = 0.125 μ_{CR} = 0.048
 Diversity = 4.4317e+01 k_t = [2, 4, 3, 3]
 Generation 1170: Best = 2.598425e+04 FES = 117100 μ_F = 0.147 μ_{CR} = 0.043
 Diversity = 4.4292e+01 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 2.570980e+04 FES = 118100 μ_F = 0.217 μ_{CR} = 0.046
 Diversity = 4.4278e+01 k_t = [3, 2, 2, 3]
 Generation 1190: Best = 2.567376e+04 FES = 119100 μ_F = 0.169 μ_{CR} = 0.061
 Diversity = 4.4184e+01 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 2.567376e+04 FES = 120100 μ_F = 0.248 μ_{CR} = 0.059
 Diversity = 4.4155e+01 k_t = [2, 4, 3, 2]
 Generation 1210: Best = 2.567376e+04 FES = 121100 μ_F = 0.232 μ_{CR} = 0.035
 Diversity = 4.4159e+01 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 2.567376e+04 FES = 122100 μ_F = 0.226 μ_{CR} = 0.044
 Diversity = 4.4143e+01 k_t = [3, 4, 4, 2]
 Generation 1230: Best = 2.550993e+04 FES = 123100 μ_F = 0.144 μ_{CR} = 0.046
 Diversity = 4.4139e+01 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 2.550993e+04 FES = 124100 μ_F = 0.127 μ_{CR} = 0.043
 Diversity = 4.4119e+01 k_t = [2, 2, 4, 2]
 Generation 1250: Best = 2.550993e+04 FES = 125100 μ_F = 0.204 μ_{CR} = 0.052
 Diversity = 4.4113e+01 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 2.550993e+04 FES = 126100 μ_F = 0.256 μ_{CR} = 0.040
 Diversity = 4.4588e+01 k_t = [3, 2, 2, 4]
 Generation 1270: Best = 2.549777e+04 FES = 127100 μ_F = 0.286 μ_{CR} = 0.034
 Diversity = 4.4694e+01 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 2.521149e+04 FES = 128100 μ_F = 0.238 μ_{CR} = 0.033
 Diversity = 4.4683e+01 k_t = [3, 4, 2, 3]
 Generation 1290: Best = 2.521149e+04 FES = 129100 μ_F = 0.286 μ_{CR} = 0.036
 Diversity = 4.4674e+01 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 2.521149e+04 FES = 130100 μ_F = 0.307 μ_{CR} = 0.033
 Diversity = 4.4623e+01 k_t = [2, 2, 4, 4]

Generation 1310: Best = 2.521149e+04 FES = 131100 μ_F = 0.332 μ_{CR} = 0.029
 Diversity = 4.4627e+01 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 2.521149e+04 FES = 132100 μ_F = 0.300 μ_{CR} = 0.030
 Diversity = 4.4626e+01 k_t = [3, 3, 3, 2]
 Generation 1330: Best = 2.521149e+04 FES = 133100 μ_F = 0.227 μ_{CR} = 0.031
 Diversity = 4.4618e+01 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 2.516355e+04 FES = 134100 μ_F = 0.321 μ_{CR} = 0.030
 Diversity = 4.4614e+01 k_t = [4, 4, 3, 3]
 Generation 1350: Best = 2.516355e+04 FES = 135100 μ_F = 0.257 μ_{CR} = 0.026
 Diversity = 4.4614e+01 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 2.516355e+04 FES = 136100 μ_F = 0.252 μ_{CR} = 0.020
 Diversity = 4.4617e+01 k_t = [3, 3, 2, 3]
 Generation 1370: Best = 2.478845e+04 FES = 137100 μ_F = 0.191 μ_{CR} = 0.023
 Diversity = 4.4614e+01 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 2.476351e+04 FES = 138100 μ_F = 0.281 μ_{CR} = 0.040
 Diversity = 4.4644e+01 k_t = [4, 3, 3, 4]
 Generation 1390: Best = 2.476351e+04 FES = 139100 μ_F = 0.209 μ_{CR} = 0.049
 Diversity = 4.4644e+01 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 2.476351e+04 FES = 140100 μ_F = 0.232 μ_{CR} = 0.048
 Diversity = 4.4634e+01 k_t = [4, 3, 2, 3]
 Generation 1410: Best = 2.476351e+04 FES = 141100 μ_F = 0.160 μ_{CR} = 0.037
 Diversity = 4.4627e+01 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 2.474091e+04 FES = 142100 μ_F = 0.103 μ_{CR} = 0.037
 Diversity = 4.4622e+01 k_t = [3, 3, 2, 3]
 Generation 1430: Best = 2.464034e+04 FES = 143100 μ_F = 0.140 μ_{CR} = 0.048
 Diversity = 4.4610e+01 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 2.464034e+04 FES = 144100 μ_F = 0.128 μ_{CR} = 0.060
 Diversity = 4.4597e+01 k_t = [4, 3, 4, 3]
 Generation 1450: Best = 2.464034e+04 FES = 145100 μ_F = 0.167 μ_{CR} = 0.057
 Diversity = 4.4589e+01 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 2.464034e+04 FES = 146100 μ_F = 0.247 μ_{CR} = 0.068
 Diversity = 4.4531e+01 k_t = [3, 4, 2, 4]
 Generation 1470: Best = 2.464034e+04 FES = 147100 μ_F = 0.208 μ_{CR} = 0.043
 Diversity = 4.4528e+01 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 2.447726e+04 FES = 148100 μ_F = 0.129 μ_{CR} = 0.088
 Diversity = 4.4513e+01 k_t = [3, 4, 3, 3]
 Generation 1490: Best = 2.447726e+04 FES = 149100 μ_F = 0.078 μ_{CR} = 0.080
 Diversity = 4.4505e+01 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 2.447726e+04 FES = 150100 μ_F = 0.226 μ_{CR} = 0.051
 Diversity = 4.4539e+01 k_t = [2, 2, 2, 3]
 Generation 1510: Best = 2.447726e+04 FES = 151100 μ_F = 0.169 μ_{CR} = 0.048
 Diversity = 4.4530e+01 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 2.447726e+04 FES = 152100 μ_F = 0.140 μ_{CR} = 0.046
 Diversity = 4.4526e+01 k_t = [2, 3, 4, 2]
 Generation 1530: Best = 2.447587e+04 FES = 153100 μ_F = 0.102 μ_{CR} = 0.045
 Diversity = 4.4522e+01 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 2.447587e+04 FES = 154100 μ_F = 0.144 μ_{CR} = 0.063
 Diversity = 4.4594e+01 k_t = [4, 4, 3, 2]
 Generation 1550: Best = 2.403701e+04 FES = 155100 μ_F = 0.225 μ_{CR} = 0.058
 Diversity = 4.4502e+01 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 2.403701e+04 FES = 156100 μ_F = 0.207 μ_{CR} = 0.066
 Diversity = 4.4492e+01 k_t = [2, 2, 2, 2]
 Generation 1570: Best = 2.403701e+04 FES = 157100 μ_F = 0.190 μ_{CR} = 0.044
 Diversity = 4.4491e+01 k_t = [1, 1, 1, 1]

Generation 1580: Best = 2.403701e+04 FES = 158100 μ_F = 0.268 μ_{CR} = 0.035
 Diversity = 4.4259e+01 k_t = [3, 2, 3, 4]
 Generation 1590: Best = 2.403701e+04 FES = 159100 μ_F = 0.160 μ_{CR} = 0.043
 Diversity = 4.4254e+01 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 2.403701e+04 FES = 160100 μ_F = 0.134 μ_{CR} = 0.055
 Diversity = 4.4245e+01 k_t = [4, 4, 2, 4]
 Generation 1610: Best = 2.392476e+04 FES = 161100 μ_F = 0.093 μ_{CR} = 0.052
 Diversity = 4.4237e+01 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 2.357631e+04 FES = 162100 μ_F = 0.163 μ_{CR} = 0.057
 Diversity = 4.4168e+01 k_t = [2, 4, 2, 2]
 Generation 1630: Best = 2.357631e+04 FES = 163100 μ_F = 0.149 μ_{CR} = 0.044
 Diversity = 4.4151e+01 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 2.343598e+04 FES = 164100 μ_F = 0.111 μ_{CR} = 0.052
 Diversity = 4.4138e+01 k_t = [3, 2, 2, 4]
 Generation 1650: Best = 2.330313e+04 FES = 165100 μ_F = 0.149 μ_{CR} = 0.038
 Diversity = 4.4127e+01 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 2.323586e+04 FES = 166100 μ_F = 0.109 μ_{CR} = 0.048
 Diversity = 4.4127e+01 k_t = [3, 4, 2, 3]
 Generation 1670: Best = 2.323586e+04 FES = 167100 μ_F = 0.120 μ_{CR} = 0.048
 Diversity = 4.4123e+01 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 2.323586e+04 FES = 168100 μ_F = 0.084 μ_{CR} = 0.054
 Diversity = 4.4105e+01 k_t = [2, 4, 3, 2]
 Generation 1690: Best = 2.323586e+04 FES = 169100 μ_F = 0.071 μ_{CR} = 0.050
 Diversity = 4.4095e+01 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 2.323586e+04 FES = 170100 μ_F = 0.143 μ_{CR} = 0.069
 Diversity = 4.3956e+01 k_t = [4, 4, 2, 4]
 Generation 1710: Best = 2.318884e+04 FES = 171100 μ_F = 0.137 μ_{CR} = 0.060
 Diversity = 4.3951e+01 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 2.318884e+04 FES = 172100 μ_F = 0.102 μ_{CR} = 0.083
 Diversity = 4.3937e+01 k_t = [2, 4, 2, 3]
 Generation 1730: Best = 2.307696e+04 FES = 173100 μ_F = 0.089 μ_{CR} = 0.066
 Diversity = 4.3936e+01 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 2.307696e+04 FES = 174100 μ_F = 0.152 μ_{CR} = 0.072
 Diversity = 4.3938e+01 k_t = [4, 4, 4, 2]
 Generation 1750: Best = 2.301653e+04 FES = 175100 μ_F = 0.142 μ_{CR} = 0.053
 Diversity = 4.3935e+01 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 2.301653e+04 FES = 176100 μ_F = 0.194 μ_{CR} = 0.035
 Diversity = 4.3903e+01 k_t = [2, 2, 4, 2]
 Generation 1770: Best = 2.301653e+04 FES = 177100 μ_F = 0.205 μ_{CR} = 0.023
 Diversity = 4.3896e+01 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 2.301653e+04 FES = 178100 μ_F = 0.176 μ_{CR} = 0.025
 Diversity = 4.3886e+01 k_t = [2, 2, 4, 4]
 Generation 1790: Best = 2.300781e+04 FES = 179100 μ_F = 0.211 μ_{CR} = 0.025
 Diversity = 4.3883e+01 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 2.291358e+04 FES = 180100 μ_F = 0.304 μ_{CR} = 0.039
 Diversity = 4.3829e+01 k_t = [3, 2, 4, 4]
 Generation 1810: Best = 2.291358e+04 FES = 181100 μ_F = 0.247 μ_{CR} = 0.025
 Diversity = 4.3828e+01 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 2.291358e+04 FES = 182100 μ_F = 0.335 μ_{CR} = 0.024
 Diversity = 4.3826e+01 k_t = [2, 3, 2, 2]
 Generation 1830: Best = 2.252664e+04 FES = 183100 μ_F = 0.243 μ_{CR} = 0.030
 Diversity = 4.3822e+01 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 2.252664e+04 FES = 184100 μ_F = 0.202 μ_{CR} = 0.034
 Diversity = 4.3817e+01 k_t = [4, 4, 4, 3]

Generation 1850: Best = 2.252664e+04 FES = 185100 μ_F = 0.146 μ_{CR} = 0.041
 Diversity = 4.3814e+01 k_t = [1, 1, 1, 1]
 Generation 1860: Best = 2.252664e+04 FES = 186100 μ_F = 0.119 μ_{CR} = 0.046
 Diversity = 4.3810e+01 k_t = [2, 4, 2, 3]
 Generation 1870: Best = 2.240793e+04 FES = 187100 μ_F = 0.125 μ_{CR} = 0.047
 Diversity = 4.3801e+01 k_t = [1, 1, 1, 1]
 Generation 1880: Best = 2.240608e+04 FES = 188100 μ_F = 0.141 μ_{CR} = 0.042
 Diversity = 4.3800e+01 k_t = [2, 2, 2, 2]
 Generation 1890: Best = 2.240608e+04 FES = 189100 μ_F = 0.103 μ_{CR} = 0.053
 Diversity = 4.3787e+01 k_t = [1, 1, 1, 1]
 Generation 1900: Best = 2.239563e+04 FES = 190100 μ_F = 0.117 μ_{CR} = 0.046
 Diversity = 4.3783e+01 k_t = [2, 4, 2, 2]
 Generation 1910: Best = 2.230208e+04 FES = 191100 μ_F = 0.104 μ_{CR} = 0.047
 Diversity = 4.3775e+01 k_t = [1, 1, 1, 1]
 Generation 1920: Best = 2.230208e+04 FES = 192100 μ_F = 0.127 μ_{CR} = 0.039
 Diversity = 4.3762e+01 k_t = [3, 4, 4, 3]
 Generation 1930: Best = 2.230208e+04 FES = 193100 μ_F = 0.125 μ_{CR} = 0.060
 Diversity = 4.3752e+01 k_t = [1, 1, 1, 1]
 Generation 1940: Best = 2.226275e+04 FES = 194100 μ_F = 0.121 μ_{CR} = 0.070
 Diversity = 4.3749e+01 k_t = [4, 2, 2, 2]
 Generation 1950: Best = 2.226275e+04 FES = 195100 μ_F = 0.097 μ_{CR} = 0.062
 Diversity = 4.3745e+01 k_t = [1, 1, 1, 1]
 Generation 1960: Best = 2.226275e+04 FES = 196100 μ_F = 0.096 μ_{CR} = 0.050
 Diversity = 4.3743e+01 k_t = [3, 2, 4, 2]
 Generation 1970: Best = 2.222327e+04 FES = 197100 μ_F = 0.077 μ_{CR} = 0.045
 Diversity = 4.3740e+01 k_t = [1, 1, 1, 1]
 Generation 1980: Best = 2.222327e+04 FES = 198100 μ_F = 0.121 μ_{CR} = 0.057
 Diversity = 4.3733e+01 k_t = [4, 3, 3, 3]
 Generation 1990: Best = 2.222327e+04 FES = 199100 μ_F = 0.081 μ_{CR} = 0.048
 Diversity = 4.3725e+01 k_t = [1, 1, 1, 1]
 Generation 2000: Best = 2.222113e+04 FES = 200100 μ_F = 0.167 μ_{CR} = 0.048
 Diversity = 4.3829e+01 k_t = [3, 4, 3, 2]
 Generation 2010: Best = 2.211044e+04 FES = 201100 μ_F = 0.131 μ_{CR} = 0.063
 Diversity = 4.3826e+01 k_t = [1, 1, 1, 1]
 Generation 2020: Best = 2.210954e+04 FES = 202100 μ_F = 0.072 μ_{CR} = 0.073
 Diversity = 4.3816e+01 k_t = [3, 2, 4, 2]
 Generation 2030: Best = 2.207051e+04 FES = 203100 μ_F = 0.055 μ_{CR} = 0.066
 Diversity = 4.3803e+01 k_t = [1, 1, 1, 1]
 Generation 2040: Best = 2.207051e+04 FES = 204100 μ_F = 0.073 μ_{CR} = 0.073
 Diversity = 4.3804e+01 k_t = [2, 2, 2, 2]
 Generation 2050: Best = 2.207051e+04 FES = 205100 μ_F = 0.070 μ_{CR} = 0.055
 Diversity = 4.3797e+01 k_t = [1, 1, 1, 1]
 Generation 2060: Best = 2.207051e+04 FES = 206100 μ_F = 0.162 μ_{CR} = 0.047
 Diversity = 4.3595e+01 k_t = [3, 2, 2, 4]
 Generation 2070: Best = 2.207051e+04 FES = 207100 μ_F = 0.150 μ_{CR} = 0.042
 Diversity = 4.3593e+01 k_t = [1, 1, 1, 1]
 Generation 2080: Best = 2.193830e+04 FES = 208100 μ_F = 0.204 μ_{CR} = 0.046
 Diversity = 4.3590e+01 k_t = [3, 2, 2, 2]
 Generation 2090: Best = 2.193830e+04 FES = 209100 μ_F = 0.187 μ_{CR} = 0.037
 Diversity = 4.3588e+01 k_t = [1, 1, 1, 1]
 Generation 2100: Best = 2.193830e+04 FES = 210100 μ_F = 0.160 μ_{CR} = 0.030
 Diversity = 4.3586e+01 k_t = [3, 3, 3, 2]
 Generation 2110: Best = 2.193830e+04 FES = 211100 μ_F = 0.214 μ_{CR} = 0.041
 Diversity = 4.3509e+01 k_t = [1, 1, 1, 1]

Generation 2120: Best = 2.191109e+04 FES = 212100 μ_F = 0.157 μ_{CR} = 0.024
 Diversity = 4.3509e+01 k_t = [2, 3, 3, 3]
 Generation 2130: Best = 2.166412e+04 FES = 213100 μ_F = 0.110 μ_{CR} = 0.034
 Diversity = 4.3505e+01 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 2.164676e+04 FES = 214100 μ_F = 0.174 μ_{CR} = 0.045
 Diversity = 4.3485e+01 k_t = [2, 3, 3, 4]
 Generation 2150: Best = 2.164676e+04 FES = 215100 μ_F = 0.159 μ_{CR} = 0.042
 Diversity = 4.3483e+01 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 2.164676e+04 FES = 216100 μ_F = 0.157 μ_{CR} = 0.041
 Diversity = 4.3479e+01 k_t = [3, 2, 3, 2]
 Generation 2170: Best = 2.164407e+04 FES = 217100 μ_F = 0.115 μ_{CR} = 0.062
 Diversity = 4.3473e+01 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 2.164407e+04 FES = 218100 μ_F = 0.154 μ_{CR} = 0.053
 Diversity = 4.3474e+01 k_t = [3, 2, 2, 3]
 Generation 2190: Best = 2.164407e+04 FES = 219100 μ_F = 0.094 μ_{CR} = 0.077
 Diversity = 4.3468e+01 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 2.159918e+04 FES = 220100 μ_F = 0.085 μ_{CR} = 0.070
 Diversity = 4.3457e+01 k_t = [2, 4, 2, 4]
 Generation 2210: Best = 2.158711e+04 FES = 221100 μ_F = 0.057 μ_{CR} = 0.079
 Diversity = 4.3448e+01 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 2.140021e+04 FES = 222100 μ_F = 0.071 μ_{CR} = 0.069
 Diversity = 4.3446e+01 k_t = [2, 4, 2, 2]
 Generation 2230: Best = 2.140021e+04 FES = 223100 μ_F = 0.060 μ_{CR} = 0.061
 Diversity = 4.3441e+01 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 2.138798e+04 FES = 224100 μ_F = 0.054 μ_{CR} = 0.057
 Diversity = 4.3430e+01 k_t = [2, 3, 4, 4]
 Generation 2250: Best = 2.136089e+04 FES = 225100 μ_F = 0.065 μ_{CR} = 0.048
 Diversity = 4.3422e+01 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 2.136089e+04 FES = 226100 μ_F = 0.058 μ_{CR} = 0.053
 Diversity = 4.3417e+01 k_t = [2, 4, 2, 3]
 Generation 2270: Best = 2.136089e+04 FES = 227100 μ_F = 0.145 μ_{CR} = 0.039
 Diversity = 4.3410e+01 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 2.136089e+04 FES = 228100 μ_F = 0.100 μ_{CR} = 0.033
 Diversity = 4.3409e+01 k_t = [3, 2, 3, 2]
 Generation 2290: Best = 2.136089e+04 FES = 229100 μ_F = 0.280 μ_{CR} = 0.025
 Diversity = 4.3408e+01 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 2.136089e+04 FES = 230100 μ_F = 0.278 μ_{CR} = 0.021
 Diversity = 4.3408e+01 k_t = [4, 3, 3, 3]
 Generation 2310: Best = 2.136089e+04 FES = 231100 μ_F = 0.232 μ_{CR} = 0.022
 Diversity = 4.3406e+01 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 2.121164e+04 FES = 232100 μ_F = 0.235 μ_{CR} = 0.027
 Diversity = 4.3463e+01 k_t = [2, 4, 2, 4]
 Generation 2330: Best = 2.114117e+04 FES = 233100 μ_F = 0.210 μ_{CR} = 0.034
 Diversity = 4.3491e+01 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 2.114117e+04 FES = 234100 μ_F = 0.221 μ_{CR} = 0.026
 Diversity = 4.3491e+01 k_t = [4, 2, 3, 3]
 Generation 2350: Best = 2.113292e+04 FES = 235100 μ_F = 0.166 μ_{CR} = 0.032
 Diversity = 4.3491e+01 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 2.113007e+04 FES = 236100 μ_F = 0.103 μ_{CR} = 0.050
 Diversity = 4.3490e+01 k_t = [4, 4, 3, 2]
 Generation 2370: Best = 2.097326e+04 FES = 237100 μ_F = 0.115 μ_{CR} = 0.038
 Diversity = 4.3491e+01 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 2.094822e+04 FES = 238100 μ_F = 0.085 μ_{CR} = 0.038
 Diversity = 4.3491e+01 k_t = [2, 2, 4, 2]

Generation 2390: Best = 2.094806e+04 FES = 239100 μ_F = 0.086 μ_{CR} = 0.047
 Diversity = 4.3487e+01 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 2.093063e+04 FES = 240100 μ_F = 0.066 μ_{CR} = 0.041
 Diversity = 4.3481e+01 k_t = [2, 4, 2, 2]
 Generation 2410: Best = 2.093063e+04 FES = 241100 μ_F = 0.081 μ_{CR} = 0.045
 Diversity = 4.3474e+01 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 2.093063e+04 FES = 242100 μ_F = 0.160 μ_{CR} = 0.044
 Diversity = 4.3474e+01 k_t = [4, 4, 4, 3]
 Generation 2430: Best = 2.093063e+04 FES = 243100 μ_F = 0.142 μ_{CR} = 0.034
 Diversity = 4.3463e+01 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 2.082489e+04 FES = 244100 μ_F = 0.186 μ_{CR} = 0.039
 Diversity = 4.3539e+01 k_t = [4, 2, 2, 4]
 Generation 2450: Best = 2.070081e+04 FES = 245100 μ_F = 0.164 μ_{CR} = 0.034
 Diversity = 4.3537e+01 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 2.064316e+04 FES = 246100 μ_F = 0.204 μ_{CR} = 0.052
 Diversity = 4.3594e+01 k_t = [3, 2, 4, 4]
 Generation 2470: Best = 2.064316e+04 FES = 247100 μ_F = 0.131 μ_{CR} = 0.037
 Diversity = 4.3594e+01 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 2.060068e+04 FES = 248100 μ_F = 0.142 μ_{CR} = 0.033
 Diversity = 4.3594e+01 k_t = [4, 4, 4, 2]
 Generation 2490: Best = 2.059943e+04 FES = 249100 μ_F = 0.101 μ_{CR} = 0.034
 Diversity = 4.3589e+01 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 2.059943e+04 FES = 250100 μ_F = 0.089 μ_{CR} = 0.031
 Diversity = 4.3586e+01 k_t = [3, 2, 3, 3]
 Generation 2510: Best = 2.059721e+04 FES = 251100 μ_F = 0.113 μ_{CR} = 0.040
 Diversity = 4.3584e+01 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 2.059721e+04 FES = 252100 μ_F = 0.145 μ_{CR} = 0.050
 Diversity = 4.3421e+01 k_t = [2, 2, 2, 4]
 Generation 2530: Best = 2.059721e+04 FES = 253100 μ_F = 0.124 μ_{CR} = 0.052
 Diversity = 4.3421e+01 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 2.059721e+04 FES = 254100 μ_F = 0.133 μ_{CR} = 0.026
 Diversity = 4.3413e+01 k_t = [2, 2, 3, 3]
 Generation 2550: Best = 2.059721e+04 FES = 255100 μ_F = 0.075 μ_{CR} = 0.054
 Diversity = 4.3411e+01 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 2.059721e+04 FES = 256100 μ_F = 0.073 μ_{CR} = 0.062
 Diversity = 4.3408e+01 k_t = [2, 2, 3, 3]
 Generation 2570: Best = 2.059721e+04 FES = 257100 μ_F = 0.185 μ_{CR} = 0.060
 Diversity = 4.3314e+01 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 2.059721e+04 FES = 258100 μ_F = 0.133 μ_{CR} = 0.082
 Diversity = 4.3313e+01 k_t = [3, 2, 2, 2]
 Generation 2590: Best = 2.022729e+04 FES = 259100 μ_F = 0.187 μ_{CR} = 0.073
 Diversity = 4.3276e+01 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 2.022729e+04 FES = 260100 μ_F = 0.209 μ_{CR} = 0.053
 Diversity = 4.3276e+01 k_t = [3, 2, 2, 4]
 Generation 2610: Best = 2.016229e+04 FES = 261100 μ_F = 0.137 μ_{CR} = 0.047
 Diversity = 4.3274e+01 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 2.016229e+04 FES = 262100 μ_F = 0.094 μ_{CR} = 0.043
 Diversity = 4.3273e+01 k_t = [4, 2, 3, 2]
 Generation 2630: Best = 2.016229e+04 FES = 263100 μ_F = 0.154 μ_{CR} = 0.055
 Diversity = 4.3274e+01 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 2.016229e+04 FES = 264100 μ_F = 0.109 μ_{CR} = 0.044
 Diversity = 4.3272e+01 k_t = [2, 3, 2, 4]
 Generation 2650: Best = 2.016229e+04 FES = 265100 μ_F = 0.074 μ_{CR} = 0.046
 Diversity = 4.3269e+01 k_t = [1, 1, 1, 1]

Generation 2660: Best = 2.014614e+04 FES = 266100 μ_F = 0.135 μ_{CR} = 0.047
 Diversity = 4.3234e+01 k_t = [2, 2, 4, 4]
 Generation 2670: Best = 2.004664e+04 FES = 267100 μ_F = 0.134 μ_{CR} = 0.046
 Diversity = 4.3230e+01 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 2.004664e+04 FES = 268100 μ_F = 0.230 μ_{CR} = 0.039
 Diversity = 4.3232e+01 k_t = [3, 2, 3, 3]
 Generation 2690: Best = 2.004664e+04 FES = 269100 μ_F = 0.215 μ_{CR} = 0.024
 Diversity = 4.3233e+01 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 2.004664e+04 FES = 270100 μ_F = 0.130 μ_{CR} = 0.036
 Diversity = 4.3234e+01 k_t = [3, 2, 3, 2]
 Generation 2710: Best = 2.004664e+04 FES = 271100 μ_F = 0.128 μ_{CR} = 0.029
 Diversity = 4.3234e+01 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 1.992971e+04 FES = 272100 μ_F = 0.177 μ_{CR} = 0.039
 Diversity = 4.3192e+01 k_t = [2, 3, 2, 4]
 Generation 2730: Best = 1.992971e+04 FES = 273100 μ_F = 0.138 μ_{CR} = 0.035
 Diversity = 4.3191e+01 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 1.979196e+04 FES = 274100 μ_F = 0.139 μ_{CR} = 0.030
 Diversity = 4.3188e+01 k_t = [3, 4, 2, 3]
 Generation 2750: Best = 1.979196e+04 FES = 275100 μ_F = 0.126 μ_{CR} = 0.042
 Diversity = 4.3188e+01 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 1.979196e+04 FES = 276100 μ_F = 0.241 μ_{CR} = 0.031
 Diversity = 4.3171e+01 k_t = [3, 3, 4, 4]
 Generation 2770: Best = 1.976682e+04 FES = 277100 μ_F = 0.205 μ_{CR} = 0.027
 Diversity = 4.3170e+01 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 1.976682e+04 FES = 278100 μ_F = 0.188 μ_{CR} = 0.037
 Diversity = 4.3169e+01 k_t = [4, 2, 4, 3]
 Generation 2790: Best = 1.976682e+04 FES = 279100 μ_F = 0.144 μ_{CR} = 0.036
 Diversity = 4.3163e+01 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 1.972697e+04 FES = 280100 μ_F = 0.150 μ_{CR} = 0.026
 Diversity = 4.3164e+01 k_t = [3, 2, 2, 2]
 Generation 2810: Best = 1.972697e+04 FES = 281100 μ_F = 0.145 μ_{CR} = 0.028
 Diversity = 4.3163e+01 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 1.972610e+04 FES = 282100 μ_F = 0.133 μ_{CR} = 0.034
 Diversity = 4.3162e+01 k_t = [2, 4, 2, 2]
 Generation 2830: Best = 1.972446e+04 FES = 283100 μ_F = 0.098 μ_{CR} = 0.037
 Diversity = 4.3159e+01 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 1.972446e+04 FES = 284100 μ_F = 0.069 μ_{CR} = 0.051
 Diversity = 4.3155e+01 k_t = [3, 2, 2, 2]
 Generation 2850: Best = 1.972446e+04 FES = 285100 μ_F = 0.071 μ_{CR} = 0.036
 Diversity = 4.3152e+01 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 1.972446e+04 FES = 286100 μ_F = 0.161 μ_{CR} = 0.043
 Diversity = 4.3062e+01 k_t = [4, 3, 2, 4]
 Generation 2870: Best = 1.972446e+04 FES = 287100 μ_F = 0.122 μ_{CR} = 0.032
 Diversity = 4.3061e+01 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 1.963891e+04 FES = 288100 μ_F = 0.147 μ_{CR} = 0.034
 Diversity = 4.3060e+01 k_t = [3, 2, 2, 3]
 Generation 2890: Best = 1.963891e+04 FES = 289100 μ_F = 0.119 μ_{CR} = 0.035
 Diversity = 4.3058e+01 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 1.963891e+04 FES = 290100 μ_F = 0.157 μ_{CR} = 0.056
 Diversity = 4.3054e+01 k_t = [4, 4, 4, 2]
 Generation 2910: Best = 1.963891e+04 FES = 291100 μ_F = 0.143 μ_{CR} = 0.048
 Diversity = 4.3053e+01 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 1.963891e+04 FES = 292100 μ_F = 0.090 μ_{CR} = 0.034
 Diversity = 4.3052e+01 k_t = [2, 2, 4, 4]

Generation 2930: Best = 1.962085e+04	FES = 293100	μ_F = 0.123	μ_{CR} = 0.030
Diversity = 4.3044e+01	k_t = [1, 1, 1, 1]		
Generation 2940: Best = 1.962085e+04	FES = 294100	μ_F = 0.088	μ_{CR} = 0.037
Diversity = 4.3042e+01	k_t = [2, 4, 3, 2]		
Generation 2950: Best = 1.962085e+04	FES = 295100	μ_F = 0.081	μ_{CR} = 0.034
Diversity = 4.3040e+01	k_t = [1, 1, 1, 1]		
Generation 2960: Best = 1.961351e+04	FES = 296100	μ_F = 0.138	μ_{CR} = 0.061
Diversity = 4.3089e+01	k_t = [3, 3, 3, 2]		
Generation 2970: Best = 1.961351e+04	FES = 297100	μ_F = 0.114	μ_{CR} = 0.039
Diversity = 4.3084e+01	k_t = [1, 1, 1, 1]		
Generation 2980: Best = 1.960702e+04	FES = 298100	μ_F = 0.086	μ_{CR} = 0.049
Diversity = 4.3084e+01	k_t = [2, 4, 4, 2]		
Generation 2990: Best = 1.960088e+04	FES = 299100	μ_F = 0.068	μ_{CR} = 0.045
Diversity = 4.3080e+01	k_t = [1, 1, 1, 1]		
Generation 3000: Best = 1.956751e+04	FES = 300100	μ_F = 0.094	μ_{CR} = 0.051
Diversity = 4.3068e+01	k_t = [2, 2, 2, 4]		
Generation 3010: Best = 1.956751e+04	FES = 301100	μ_F = 0.074	μ_{CR} = 0.054
Diversity = 4.3066e+01	k_t = [1, 1, 1, 1]		
Generation 3020: Best = 1.956751e+04	FES = 302100	μ_F = 0.160	μ_{CR} = 0.055
Diversity = 4.2839e+01	k_t = [3, 3, 2, 4]		
Generation 3030: Best = 1.956751e+04	FES = 303100	μ_F = 0.124	μ_{CR} = 0.044
Diversity = 4.2838e+01	k_t = [1, 1, 1, 1]		
Generation 3040: Best = 1.955631e+04	FES = 304100	μ_F = 0.163	μ_{CR} = 0.047
Diversity = 4.2787e+01	k_t = [4, 2, 3, 4]		
Generation 3050: Best = 1.955631e+04	FES = 305100	μ_F = 0.170	μ_{CR} = 0.040
Diversity = 4.2787e+01	k_t = [1, 1, 1, 1]		
Generation 3060: Best = 1.955631e+04	FES = 306100	μ_F = 0.154	μ_{CR} = 0.069
Diversity = 4.2787e+01	k_t = [4, 4, 3, 2]		
Generation 3070: Best = 1.955631e+04	FES = 307100	μ_F = 0.114	μ_{CR} = 0.052
Diversity = 4.2786e+01	k_t = [1, 1, 1, 1]		
Generation 3080: Best = 1.955631e+04	FES = 308100	μ_F = 0.134	μ_{CR} = 0.042
Diversity = 4.2784e+01	k_t = [3, 4, 4, 2]		
Generation 3090: Best = 1.955631e+04	FES = 309100	μ_F = 0.076	μ_{CR} = 0.031
Diversity = 4.2782e+01	k_t = [1, 1, 1, 1]		
Generation 3100: Best = 1.955631e+04	FES = 310100	μ_F = 0.107	μ_{CR} = 0.034
Diversity = 4.2778e+01	k_t = [4, 2, 2, 2]		
Generation 3110: Best = 1.955631e+04	FES = 311100	μ_F = 0.086	μ_{CR} = 0.054
Diversity = 4.2777e+01	k_t = [1, 1, 1, 1]		
Generation 3120: Best = 1.955631e+04	FES = 312100	μ_F = 0.296	μ_{CR} = 0.042
Diversity = 4.2650e+01	k_t = [3, 3, 4, 4]		
Generation 3130: Best = 1.955631e+04	FES = 313100	μ_F = 0.322	μ_{CR} = 0.034
Diversity = 4.2650e+01	k_t = [1, 1, 1, 1]		
Generation 3140: Best = 1.952939e+04	FES = 314100	μ_F = 0.257	μ_{CR} = 0.032
Diversity = 4.2650e+01	k_t = [4, 2, 3, 3]		
Generation 3150: Best = 1.952939e+04	FES = 315100	μ_F = 0.232	μ_{CR} = 0.031
Diversity = 4.2650e+01	k_t = [1, 1, 1, 1]		
Generation 3160: Best = 1.952939e+04	FES = 316100	μ_F = 0.284	μ_{CR} = 0.039
Diversity = 4.2575e+01	k_t = [3, 3, 3, 4]		
Generation 3170: Best = 1.952939e+04	FES = 317100	μ_F = 0.297	μ_{CR} = 0.028
Diversity = 4.2426e+01	k_t = [1, 1, 1, 1]		
Generation 3180: Best = 1.952939e+04	FES = 318100	μ_F = 0.333	μ_{CR} = 0.019
Diversity = 4.2295e+01	k_t = [2, 2, 4, 4]		
Generation 3190: Best = 1.952939e+04	FES = 319100	μ_F = 0.242	μ_{CR} = 0.030
Diversity = 4.2297e+01	k_t = [1, 1, 1, 1]		

Generation 3200: Best = 1.952939e+04 FES = 320100 μ_F = 0.209 μ_{CR} = 0.033
 Diversity = 4.2294e+01 k_t = [2, 4, 3, 3]
 Generation 3210: Best = 1.952939e+04 FES = 321100 μ_F = 0.263 μ_{CR} = 0.027
 Diversity = 4.2223e+01 k_t = [1, 1, 1, 1]
 Generation 3220: Best = 1.947003e+04 FES = 322100 μ_F = 0.212 μ_{CR} = 0.041
 Diversity = 4.2223e+01 k_t = [4, 3, 4, 3]
 Generation 3230: Best = 1.947003e+04 FES = 323100 μ_F = 0.161 μ_{CR} = 0.030
 Diversity = 4.2222e+01 k_t = [1, 1, 1, 1]
 Generation 3240: Best = 1.947003e+04 FES = 324100 μ_F = 0.155 μ_{CR} = 0.032
 Diversity = 4.2221e+01 k_t = [4, 3, 2, 3]
 Generation 3250: Best = 1.947003e+04 FES = 325100 μ_F = 0.156 μ_{CR} = 0.031
 Diversity = 4.2221e+01 k_t = [1, 1, 1, 1]
 Generation 3260: Best = 1.947003e+04 FES = 326100 μ_F = 0.227 μ_{CR} = 0.030
 Diversity = 4.2220e+01 k_t = [4, 3, 2, 4]
 Generation 3270: Best = 1.947003e+04 FES = 327100 μ_F = 0.153 μ_{CR} = 0.030
 Diversity = 4.2224e+01 k_t = [1, 1, 1, 1]
 Generation 3280: Best = 1.943948e+04 FES = 328100 μ_F = 0.246 μ_{CR} = 0.038
 Diversity = 4.2170e+01 k_t = [4, 4, 2, 2]
 Generation 3290: Best = 1.943948e+04 FES = 329100 μ_F = 0.196 μ_{CR} = 0.036
 Diversity = 4.2170e+01 k_t = [1, 1, 1, 1]
 Generation 3300: Best = 1.943948e+04 FES = 330100 μ_F = 0.146 μ_{CR} = 0.037
 Diversity = 4.2169e+01 k_t = [2, 4, 2, 3]
 Generation 3310: Best = 1.943948e+04 FES = 331100 μ_F = 0.092 μ_{CR} = 0.045
 Diversity = 4.2167e+01 k_t = [1, 1, 1, 1]
 Generation 3320: Best = 1.933615e+04 FES = 332100 μ_F = 0.065 μ_{CR} = 0.050
 Diversity = 4.2164e+01 k_t = [2, 3, 4, 4]
 Generation 3330: Best = 1.933615e+04 FES = 333100 μ_F = 0.087 μ_{CR} = 0.033
 Diversity = 4.2155e+01 k_t = [1, 1, 1, 1]
 Generation 3340: Best = 1.933615e+04 FES = 334100 μ_F = 0.179 μ_{CR} = 0.038
 Diversity = 4.2151e+01 k_t = [4, 2, 2, 2]
 Generation 3350: Best = 1.933615e+04 FES = 335100 μ_F = 0.137 μ_{CR} = 0.040
 Diversity = 4.2147e+01 k_t = [1, 1, 1, 1]
 Generation 3360: Best = 1.932215e+04 FES = 336100 μ_F = 0.087 μ_{CR} = 0.037
 Diversity = 4.2145e+01 k_t = [3, 4, 2, 3]
 Generation 3370: Best = 1.932215e+04 FES = 337100 μ_F = 0.142 μ_{CR} = 0.045
 Diversity = 4.2144e+01 k_t = [1, 1, 1, 1]
 Generation 3380: Best = 1.932215e+04 FES = 338100 μ_F = 0.107 μ_{CR} = 0.043
 Diversity = 4.2143e+01 k_t = [4, 3, 3, 2]
 Generation 3390: Best = 1.930055e+04 FES = 339100 μ_F = 0.089 μ_{CR} = 0.047
 Diversity = 4.2141e+01 k_t = [1, 1, 1, 1]
 Generation 3400: Best = 1.930055e+04 FES = 340100 μ_F = 0.070 μ_{CR} = 0.040
 Diversity = 4.2141e+01 k_t = [4, 2, 3, 2]
 Generation 3410: Best = 1.913742e+04 FES = 341100 μ_F = 0.108 μ_{CR} = 0.038
 Diversity = 4.2141e+01 k_t = [1, 1, 1, 1]
 Generation 3420: Best = 1.913742e+04 FES = 342100 μ_F = 0.175 μ_{CR} = 0.049
 Diversity = 4.2076e+01 k_t = [2, 2, 3, 3]
 Generation 3430: Best = 1.913742e+04 FES = 343100 μ_F = 0.129 μ_{CR} = 0.042
 Diversity = 4.2074e+01 k_t = [1, 1, 1, 1]
 Generation 3440: Best = 1.913742e+04 FES = 344100 μ_F = 0.092 μ_{CR} = 0.043
 Diversity = 4.2073e+01 k_t = [2, 4, 3, 3]
 Generation 3450: Best = 1.913742e+04 FES = 345100 μ_F = 0.057 μ_{CR} = 0.052
 Diversity = 4.2070e+01 k_t = [1, 1, 1, 1]
 Generation 3460: Best = 1.913620e+04 FES = 346100 μ_F = 0.180 μ_{CR} = 0.047
 Diversity = 4.2072e+01 k_t = [4, 2, 4, 2]

Generation 3470: Best = 1.913620e+04 FES = 347100 μ_F = 0.131 μ_{CR} = 0.043
 Diversity = 4.2068e+01 k_t = [1, 1, 1, 1]
 Generation 3480: Best = 1.908290e+04 FES = 348100 μ_F = 0.160 μ_{CR} = 0.038
 Diversity = 4.1666e+01 k_t = [3, 2, 3, 4]
 Generation 3490: Best = 1.908290e+04 FES = 349100 μ_F = 0.179 μ_{CR} = 0.033
 Diversity = 4.1666e+01 k_t = [1, 1, 1, 1]
 Generation 3500: Best = 1.908290e+04 FES = 350100 μ_F = 0.164 μ_{CR} = 0.028
 Diversity = 4.1664e+01 k_t = [2, 3, 4, 2]
 Generation 3510: Best = 1.908290e+04 FES = 351100 μ_F = 0.113 μ_{CR} = 0.028
 Diversity = 4.1663e+01 k_t = [1, 1, 1, 1]
 Generation 3520: Best = 1.908054e+04 FES = 352100 μ_F = 0.169 μ_{CR} = 0.031
 Diversity = 4.1661e+01 k_t = [4, 3, 2, 4]
 Generation 3530: Best = 1.908054e+04 FES = 353100 μ_F = 0.214 μ_{CR} = 0.032
 Diversity = 4.1544e+01 k_t = [1, 1, 1, 1]
 Generation 3540: Best = 1.908054e+04 FES = 354100 μ_F = 0.185 μ_{CR} = 0.029
 Diversity = 4.1543e+01 k_t = [4, 4, 2, 2]
 Generation 3550: Best = 1.900656e+04 FES = 355100 μ_F = 0.149 μ_{CR} = 0.025
 Diversity = 4.1541e+01 k_t = [1, 1, 1, 1]
 Generation 3560: Best = 1.900656e+04 FES = 356100 μ_F = 0.236 μ_{CR} = 0.044
 Diversity = 4.1540e+01 k_t = [3, 2, 2, 4]
 Generation 3570: Best = 1.900656e+04 FES = 357100 μ_F = 0.210 μ_{CR} = 0.052
 Diversity = 4.1538e+01 k_t = [1, 1, 1, 1]
 Generation 3580: Best = 1.898372e+04 FES = 358100 μ_F = 0.290 μ_{CR} = 0.044
 Diversity = 4.1539e+01 k_t = [3, 2, 2, 2]
 Generation 3590: Best = 1.898372e+04 FES = 359100 μ_F = 0.257 μ_{CR} = 0.044
 Diversity = 4.1540e+01 k_t = [1, 1, 1, 1]
 Generation 3600: Best = 1.893126e+04 FES = 360100 μ_F = 0.203 μ_{CR} = 0.033
 Diversity = 4.1540e+01 k_t = [3, 4, 3, 2]
 Generation 3610: Best = 1.893126e+04 FES = 361100 μ_F = 0.143 μ_{CR} = 0.050
 Diversity = 4.1541e+01 k_t = [1, 1, 1, 1]
 Generation 3620: Best = 1.893126e+04 FES = 362100 μ_F = 0.106 μ_{CR} = 0.045
 Diversity = 4.1538e+01 k_t = [4, 2, 2, 2]
 Generation 3630: Best = 1.893126e+04 FES = 363100 μ_F = 0.075 μ_{CR} = 0.052
 Diversity = 4.1537e+01 k_t = [1, 1, 1, 1]
 Generation 3640: Best = 1.893126e+04 FES = 364100 μ_F = 0.139 μ_{CR} = 0.046
 Diversity = 4.1581e+01 k_t = [2, 4, 4, 4]
 Generation 3650: Best = 1.893126e+04 FES = 365100 μ_F = 0.126 μ_{CR} = 0.057
 Diversity = 4.1580e+01 k_t = [1, 1, 1, 1]
 Generation 3660: Best = 1.890243e+04 FES = 366100 μ_F = 0.249 μ_{CR} = 0.048
 Diversity = 4.1654e+01 k_t = [4, 2, 2, 4]
 Generation 3670: Best = 1.890243e+04 FES = 367100 μ_F = 0.218 μ_{CR} = 0.042
 Diversity = 4.1654e+01 k_t = [1, 1, 1, 1]
 Generation 3680: Best = 1.890243e+04 FES = 368100 μ_F = 0.208 μ_{CR} = 0.052
 Diversity = 4.1494e+01 k_t = [3, 4, 4, 4]
 Generation 3690: Best = 1.890243e+04 FES = 369100 μ_F = 0.203 μ_{CR} = 0.046
 Diversity = 4.1493e+01 k_t = [1, 1, 1, 1]
 Generation 3700: Best = 1.890243e+04 FES = 370100 μ_F = 0.201 μ_{CR} = 0.077
 Diversity = 4.1493e+01 k_t = [3, 4, 3, 2]
 Generation 3710: Best = 1.889836e+04 FES = 371100 μ_F = 0.137 μ_{CR} = 0.056
 Diversity = 4.1493e+01 k_t = [1, 1, 1, 1]
 Generation 3720: Best = 1.889836e+04 FES = 372100 μ_F = 0.105 μ_{CR} = 0.055
 Diversity = 4.1489e+01 k_t = [3, 3, 3, 2]
 Generation 3730: Best = 1.889752e+04 FES = 373100 μ_F = 0.112 μ_{CR} = 0.031
 Diversity = 4.1489e+01 k_t = [1, 1, 1, 1]

Generation 3740: Best = 1.889752e+04 FES = 374100 μ_F = 0.100 μ_{CR} = 0.042
 Diversity = 4.1489e+01 k_t = [4, 4, 4, 2]
 Generation 3750: Best = 1.889752e+04 FES = 375100 μ_F = 0.081 μ_{CR} = 0.053
 Diversity = 4.1486e+01 k_t = [1, 1, 1, 1]
 Generation 3760: Best = 1.889752e+04 FES = 376100 μ_F = 0.065 μ_{CR} = 0.047
 Diversity = 4.1481e+01 k_t = [4, 2, 3, 3]
 Generation 3770: Best = 1.889752e+04 FES = 377100 μ_F = 0.145 μ_{CR} = 0.039
 Diversity = 4.1489e+01 k_t = [1, 1, 1, 1]
 Generation 3780: Best = 1.889231e+04 FES = 378100 μ_F = 0.158 μ_{CR} = 0.029
 Diversity = 4.1381e+01 k_t = [2, 2, 2, 2]
 Generation 3790: Best = 1.885445e+04 FES = 379100 μ_F = 0.268 μ_{CR} = 0.038
 Diversity = 4.1057e+01 k_t = [1, 1, 1, 1]
 Generation 3800: Best = 1.883243e+04 FES = 380100 μ_F = 0.281 μ_{CR} = 0.038
 Diversity = 4.1029e+01 k_t = [3, 3, 3, 4]
 Generation 3810: Best = 1.883243e+04 FES = 381100 μ_F = 0.198 μ_{CR} = 0.037
 Diversity = 4.1030e+01 k_t = [1, 1, 1, 1]
 Generation 3820: Best = 1.883243e+04 FES = 382100 μ_F = 0.240 μ_{CR} = 0.034
 Diversity = 4.0772e+01 k_t = [2, 2, 2, 4]
 Generation 3830: Best = 1.883243e+04 FES = 383100 μ_F = 0.160 μ_{CR} = 0.030
 Diversity = 4.0772e+01 k_t = [1, 1, 1, 1]
 Generation 3840: Best = 1.883243e+04 FES = 384100 μ_F = 0.213 μ_{CR} = 0.053
 Diversity = 4.0771e+01 k_t = [4, 2, 2, 2]
 Generation 3850: Best = 1.881017e+04 FES = 385100 μ_F = 0.135 μ_{CR} = 0.053
 Diversity = 4.0771e+01 k_t = [1, 1, 1, 1]
 Generation 3860: Best = 1.872612e+04 FES = 386100 μ_F = 0.315 μ_{CR} = 0.049
 Diversity = 4.0770e+01 k_t = [4, 4, 4, 2]
 Generation 3870: Best = 1.868554e+04 FES = 387100 μ_F = 0.263 μ_{CR} = 0.053
 Diversity = 4.0770e+01 k_t = [1, 1, 1, 1]
 Generation 3880: Best = 1.868554e+04 FES = 388100 μ_F = 0.328 μ_{CR} = 0.057
 Diversity = 4.0770e+01 k_t = [4, 2, 4, 3]
 Generation 3890: Best = 1.868554e+04 FES = 389100 μ_F = 0.286 μ_{CR} = 0.041
 Diversity = 4.0769e+01 k_t = [1, 1, 1, 1]
 Generation 3900: Best = 1.868554e+04 FES = 390100 μ_F = 0.345 μ_{CR} = 0.050
 Diversity = 4.0740e+01 k_t = [3, 3, 4, 4]
 Generation 3910: Best = 1.868554e+04 FES = 391100 μ_F = 0.348 μ_{CR} = 0.041
 Diversity = 4.0740e+01 k_t = [1, 1, 1, 1]
 Generation 3920: Best = 1.864476e+04 FES = 392100 μ_F = 0.241 μ_{CR} = 0.036
 Diversity = 4.0740e+01 k_t = [4, 4, 2, 3]
 Generation 3930: Best = 1.864476e+04 FES = 393100 μ_F = 0.224 μ_{CR} = 0.033
 Diversity = 4.0740e+01 k_t = [1, 1, 1, 1]
 Generation 3940: Best = 1.864476e+04 FES = 394100 μ_F = 0.187 μ_{CR} = 0.038
 Diversity = 4.0739e+01 k_t = [4, 2, 3, 3]
 Generation 3950: Best = 1.855661e+04 FES = 395100 μ_F = 0.197 μ_{CR} = 0.033
 Diversity = 4.0741e+01 k_t = [1, 1, 1, 1]
 Generation 3960: Best = 1.855661e+04 FES = 396100 μ_F = 0.235 μ_{CR} = 0.027
 Diversity = 4.0742e+01 k_t = [4, 3, 2, 2]
 Generation 3970: Best = 1.855661e+04 FES = 397100 μ_F = 0.258 μ_{CR} = 0.023
 Diversity = 4.0742e+01 k_t = [1, 1, 1, 1]
 Generation 3980: Best = 1.855661e+04 FES = 398100 μ_F = 0.306 μ_{CR} = 0.034
 Diversity = 4.0871e+01 k_t = [2, 2, 2, 4]
 Generation 3990: Best = 1.855661e+04 FES = 399100 μ_F = 0.283 μ_{CR} = 0.048
 Diversity = 4.0871e+01 k_t = [1, 1, 1, 1]
 Generation 4000: Best = 1.855661e+04 FES = 400100 μ_F = 0.259 μ_{CR} = 0.037
 Diversity = 4.0875e+01 k_t = [3, 3, 2, 4]

Generation 4010: Best = 1.847475e+04 FES = 401100 μ_F = 0.209 μ_{CR} = 0.029
 Diversity = 4.0872e+01 k_t = [1, 1, 1, 1]
 Generation 4020: Best = 1.847475e+04 FES = 402100 μ_F = 0.160 μ_{CR} = 0.034
 Diversity = 4.0871e+01 k_t = [4, 4, 3, 2]
 Generation 4030: Best = 1.841308e+04 FES = 403100 μ_F = 0.159 μ_{CR} = 0.042
 Diversity = 4.0871e+01 k_t = [1, 1, 1, 1]
 Generation 4040: Best = 1.841308e+04 FES = 404100 μ_F = 0.106 μ_{CR} = 0.049
 Diversity = 4.0872e+01 k_t = [3, 3, 4, 3]
 Generation 4050: Best = 1.841308e+04 FES = 405100 μ_F = 0.081 μ_{CR} = 0.052
 Diversity = 4.0871e+01 k_t = [1, 1, 1, 1]
 Generation 4060: Best = 1.841308e+04 FES = 406100 μ_F = 0.176 μ_{CR} = 0.039
 Diversity = 4.0872e+01 k_t = [2, 2, 2, 2]
 Generation 4070: Best = 1.841308e+04 FES = 407100 μ_F = 0.147 μ_{CR} = 0.051
 Diversity = 4.0873e+01 k_t = [1, 1, 1, 1]
 Generation 4080: Best = 1.839979e+04 FES = 408100 μ_F = 0.107 μ_{CR} = 0.040
 Diversity = 4.0872e+01 k_t = [3, 2, 3, 2]
 Generation 4090: Best = 1.839979e+04 FES = 409100 μ_F = 0.241 μ_{CR} = 0.037
 Diversity = 4.0870e+01 k_t = [1, 1, 1, 1]
 Generation 4100: Best = 1.828767e+04 FES = 410100 μ_F = 0.143 μ_{CR} = 0.046
 Diversity = 4.0871e+01 k_t = [4, 3, 4, 2]
 Generation 4110: Best = 1.828767e+04 FES = 411100 μ_F = 0.115 μ_{CR} = 0.043
 Diversity = 4.0871e+01 k_t = [1, 1, 1, 1]
 Generation 4120: Best = 1.828767e+04 FES = 412100 μ_F = 0.167 μ_{CR} = 0.035
 Diversity = 4.0868e+01 k_t = [2, 3, 3, 3]
 Generation 4130: Best = 1.828767e+04 FES = 413100 μ_F = 0.145 μ_{CR} = 0.038
 Diversity = 4.0868e+01 k_t = [1, 1, 1, 1]
 Generation 4140: Best = 1.828767e+04 FES = 414100 μ_F = 0.122 μ_{CR} = 0.043
 Diversity = 4.0867e+01 k_t = [3, 3, 4, 3]
 Generation 4150: Best = 1.819654e+04 FES = 415100 μ_F = 0.086 μ_{CR} = 0.055
 Diversity = 4.0866e+01 k_t = [1, 1, 1, 1]
 Generation 4160: Best = 1.819654e+04 FES = 416100 μ_F = 0.072 μ_{CR} = 0.042
 Diversity = 4.0865e+01 k_t = [3, 3, 2, 3]
 Generation 4170: Best = 1.819654e+04 FES = 417100 μ_F = 0.061 μ_{CR} = 0.039
 Diversity = 4.0865e+01 k_t = [1, 1, 1, 1]
 Generation 4180: Best = 1.819654e+04 FES = 418100 μ_F = 0.151 μ_{CR} = 0.035
 Diversity = 4.0887e+01 k_t = [2, 4, 2, 4]
 Generation 4190: Best = 1.818222e+04 FES = 419100 μ_F = 0.211 μ_{CR} = 0.023
 Diversity = 4.0888e+01 k_t = [1, 1, 1, 1]
 Generation 4200: Best = 1.818222e+04 FES = 420100 μ_F = 0.193 μ_{CR} = 0.025
 Diversity = 4.0888e+01 k_t = [2, 4, 3, 2]
 Generation 4210: Best = 1.816807e+04 FES = 421100 μ_F = 0.125 μ_{CR} = 0.028
 Diversity = 4.0888e+01 k_t = [1, 1, 1, 1]
 Generation 4220: Best = 1.815177e+04 FES = 422100 μ_F = 0.122 μ_{CR} = 0.039
 Diversity = 4.0888e+01 k_t = [2, 4, 2, 4]
 Generation 4230: Best = 1.815177e+04 FES = 423100 μ_F = 0.090 μ_{CR} = 0.055
 Diversity = 4.0889e+01 k_t = [1, 1, 1, 1]
 Generation 4240: Best = 1.815177e+04 FES = 424100 μ_F = 0.083 μ_{CR} = 0.060
 Diversity = 4.0889e+01 k_t = [4, 4, 4, 4]
 Generation 4250: Best = 1.815013e+04 FES = 425100 μ_F = 0.089 μ_{CR} = 0.035
 Diversity = 4.0889e+01 k_t = [1, 1, 1, 1]
 Generation 4260: Best = 1.815013e+04 FES = 426100 μ_F = 0.159 μ_{CR} = 0.063
 Diversity = 4.1074e+01 k_t = [4, 2, 4, 4]
 Generation 4270: Best = 1.815013e+04 FES = 427100 μ_F = 0.140 μ_{CR} = 0.061
 Diversity = 4.1074e+01 k_t = [1, 1, 1, 1]

Generation 4280: Best = 1.815013e+04 FES = 428100 μ_F = 0.212 μ_{CR} = 0.046
 Diversity = 4.1063e+01 k_t = [4, 3, 4, 4]
 Generation 4290: Best = 1.812281e+04 FES = 429100 μ_F = 0.280 μ_{CR} = 0.047
 Diversity = 4.1132e+01 k_t = [1, 1, 1, 1]
 Generation 4300: Best = 1.812281e+04 FES = 430100 μ_F = 0.201 μ_{CR} = 0.041
 Diversity = 4.1131e+01 k_t = [4, 2, 4, 2]
 Generation 4310: Best = 1.812281e+04 FES = 431100 μ_F = 0.202 μ_{CR} = 0.045
 Diversity = 4.1130e+01 k_t = [1, 1, 1, 1]
 Generation 4320: Best = 1.812281e+04 FES = 432100 μ_F = 0.262 μ_{CR} = 0.052
 Diversity = 4.1259e+01 k_t = [4, 3, 2, 4]
 Generation 4330: Best = 1.808111e+04 FES = 433100 μ_F = 0.270 μ_{CR} = 0.043
 Diversity = 4.1259e+01 k_t = [1, 1, 1, 1]
 Generation 4340: Best = 1.808111e+04 FES = 434100 μ_F = 0.203 μ_{CR} = 0.062
 Diversity = 4.1259e+01 k_t = [3, 3, 4, 2]
 Generation 4350: Best = 1.808111e+04 FES = 435100 μ_F = 0.223 μ_{CR} = 0.047
 Diversity = 4.1259e+01 k_t = [1, 1, 1, 1]
 Generation 4360: Best = 1.807799e+04 FES = 436100 μ_F = 0.230 μ_{CR} = 0.053
 Diversity = 4.1259e+01 k_t = [2, 2, 4, 3]
 Generation 4370: Best = 1.801839e+04 FES = 437100 μ_F = 0.214 μ_{CR} = 0.036
 Diversity = 4.1259e+01 k_t = [1, 1, 1, 1]
 Generation 4380: Best = 1.801465e+04 FES = 438100 μ_F = 0.164 μ_{CR} = 0.032
 Diversity = 4.1258e+01 k_t = [3, 3, 2, 3]
 Generation 4390: Best = 1.795557e+04 FES = 439100 μ_F = 0.144 μ_{CR} = 0.024
 Diversity = 4.1260e+01 k_t = [1, 1, 1, 1]
 Generation 4400: Best = 1.795557e+04 FES = 440100 μ_F = 0.097 μ_{CR} = 0.044
 Diversity = 4.1258e+01 k_t = [2, 3, 3, 2]
 Generation 4410: Best = 1.795557e+04 FES = 441100 μ_F = 0.077 μ_{CR} = 0.044
 Diversity = 4.1257e+01 k_t = [1, 1, 1, 1]
 Generation 4420: Best = 1.795557e+04 FES = 442100 μ_F = 0.139 μ_{CR} = 0.039
 Diversity = 4.1258e+01 k_t = [3, 3, 3, 2]
 Generation 4430: Best = 1.788716e+04 FES = 443100 μ_F = 0.110 μ_{CR} = 0.064
 Diversity = 4.1255e+01 k_t = [1, 1, 1, 1]
 Generation 4440: Best = 1.788716e+04 FES = 444100 μ_F = 0.085 μ_{CR} = 0.063
 Diversity = 4.1256e+01 k_t = [3, 2, 4, 2]
 Generation 4450: Best = 1.788716e+04 FES = 445100 μ_F = 0.068 μ_{CR} = 0.055
 Diversity = 4.1256e+01 k_t = [1, 1, 1, 1]
 Generation 4460: Best = 1.788180e+04 FES = 446100 μ_F = 0.065 μ_{CR} = 0.047
 Diversity = 4.1257e+01 k_t = [3, 2, 2, 4]
 Generation 4470: Best = 1.788180e+04 FES = 447100 μ_F = 0.097 μ_{CR} = 0.056
 Diversity = 4.1259e+01 k_t = [1, 1, 1, 1]
 Generation 4480: Best = 1.788180e+04 FES = 448100 μ_F = 0.078 μ_{CR} = 0.055
 Diversity = 4.1258e+01 k_t = [3, 4, 4, 2]
 Generation 4490: Best = 1.788180e+04 FES = 449100 μ_F = 0.095 μ_{CR} = 0.079
 Diversity = 4.1258e+01 k_t = [1, 1, 1, 1]
 Generation 4500: Best = 1.788180e+04 FES = 450100 μ_F = 0.163 μ_{CR} = 0.068
 Diversity = 4.1258e+01 k_t = [3, 4, 2, 3]
 Generation 4510: Best = 1.788180e+04 FES = 451100 μ_F = 0.156 μ_{CR} = 0.060
 Diversity = 4.1254e+01 k_t = [1, 1, 1, 1]
 Generation 4520: Best = 1.760496e+04 FES = 452100 μ_F = 0.191 μ_{CR} = 0.055
 Diversity = 4.1255e+01 k_t = [4, 4, 3, 3]
 Generation 4530: Best = 1.760496e+04 FES = 453100 μ_F = 0.144 μ_{CR} = 0.040
 Diversity = 4.1255e+01 k_t = [1, 1, 1, 1]
 Generation 4540: Best = 1.758433e+04 FES = 454100 μ_F = 0.180 μ_{CR} = 0.033
 Diversity = 4.1254e+01 k_t = [2, 4, 3, 2]

Generation 4550: Best = 1.758433e+04 FES = 455100 μ_F = 0.142 μ_{CR} = 0.032
 Diversity = 4.1253e+01 k_t = [1, 1, 1, 1]
 Generation 4560: Best = 1.756023e+04 FES = 456100 μ_F = 0.104 μ_{CR} = 0.029
 Diversity = 4.1252e+01 k_t = [2, 4, 2, 2]
 Generation 4570: Best = 1.752417e+04 FES = 457100 μ_F = 0.075 μ_{CR} = 0.040
 Diversity = 4.1251e+01 k_t = [1, 1, 1, 1]
 Generation 4580: Best = 1.752417e+04 FES = 458100 μ_F = 0.147 μ_{CR} = 0.044
 Diversity = 4.1400e+01 k_t = [4, 2, 2, 4]
 Generation 4590: Best = 1.752417e+04 FES = 459100 μ_F = 0.131 μ_{CR} = 0.036
 Diversity = 4.1400e+01 k_t = [1, 1, 1, 1]
 Generation 4600: Best = 1.752417e+04 FES = 460100 μ_F = 0.089 μ_{CR} = 0.047
 Diversity = 4.1397e+01 k_t = [3, 4, 2, 3]
 Generation 4610: Best = 1.752417e+04 FES = 461100 μ_F = 0.143 μ_{CR} = 0.034
 Diversity = 4.1397e+01 k_t = [1, 1, 1, 1]
 Generation 4620: Best = 1.752417e+04 FES = 462100 μ_F = 0.193 μ_{CR} = 0.081
 Diversity = 4.1667e+01 k_t = [2, 4, 4, 4]
 Generation 4630: Best = 1.752417e+04 FES = 463100 μ_F = 0.212 μ_{CR} = 0.056
 Diversity = 4.1665e+01 k_t = [1, 1, 1, 1]
 Generation 4640: Best = 1.747952e+04 FES = 464100 μ_F = 0.155 μ_{CR} = 0.057
 Diversity = 4.1666e+01 k_t = [3, 4, 4, 3]
 Generation 4650: Best = 1.747952e+04 FES = 465100 μ_F = 0.115 μ_{CR} = 0.058
 Diversity = 4.1667e+01 k_t = [1, 1, 1, 1]
 Generation 4660: Best = 1.744296e+04 FES = 466100 μ_F = 0.085 μ_{CR} = 0.066
 Diversity = 4.1667e+01 k_t = [3, 2, 2, 3]
 Generation 4670: Best = 1.744296e+04 FES = 467100 μ_F = 0.105 μ_{CR} = 0.055
 Diversity = 4.1665e+01 k_t = [1, 1, 1, 1]
 Generation 4680: Best = 1.744296e+04 FES = 468100 μ_F = 0.104 μ_{CR} = 0.058
 Diversity = 4.1663e+01 k_t = [4, 2, 4, 2]
 Generation 4690: Best = 1.743458e+04 FES = 469100 μ_F = 0.149 μ_{CR} = 0.042
 Diversity = 4.1662e+01 k_t = [1, 1, 1, 1]
 Generation 4700: Best = 1.740709e+04 FES = 470100 μ_F = 0.116 μ_{CR} = 0.053
 Diversity = 4.1662e+01 k_t = [4, 2, 4, 4]
 Generation 4710: Best = 1.738126e+04 FES = 471100 μ_F = 0.104 μ_{CR} = 0.050
 Diversity = 4.1662e+01 k_t = [1, 1, 1, 1]
 Generation 4720: Best = 1.738126e+04 FES = 472100 μ_F = 0.106 μ_{CR} = 0.059
 Diversity = 4.1662e+01 k_t = [3, 4, 2, 2]
 Generation 4730: Best = 1.738079e+04 FES = 473100 μ_F = 0.085 μ_{CR} = 0.050
 Diversity = 4.1661e+01 k_t = [1, 1, 1, 1]
 Generation 4740: Best = 1.725301e+04 FES = 474100 μ_F = 0.144 μ_{CR} = 0.044
 Diversity = 4.1659e+01 k_t = [4, 2, 4, 3]
 Generation 4750: Best = 1.725301e+04 FES = 475100 μ_F = 0.163 μ_{CR} = 0.045
 Diversity = 4.1659e+01 k_t = [1, 1, 1, 1]
 Generation 4760: Best = 1.724443e+04 FES = 476100 μ_F = 0.185 μ_{CR} = 0.045
 Diversity = 4.1658e+01 k_t = [3, 2, 4, 3]
 Generation 4770: Best = 1.724443e+04 FES = 477100 μ_F = 0.198 μ_{CR} = 0.045
 Diversity = 4.1660e+01 k_t = [1, 1, 1, 1]
 Generation 4780: Best = 1.724443e+04 FES = 478100 μ_F = 0.187 μ_{CR} = 0.033
 Diversity = 4.1660e+01 k_t = [4, 3, 4, 2]
 Generation 4790: Best = 1.722549e+04 FES = 479100 μ_F = 0.112 μ_{CR} = 0.056
 Diversity = 4.1659e+01 k_t = [1, 1, 1, 1]
 Generation 4800: Best = 1.722549e+04 FES = 480100 μ_F = 0.119 μ_{CR} = 0.047
 Diversity = 4.1657e+01 k_t = [4, 4, 4, 4]
 Generation 4810: Best = 1.722549e+04 FES = 481100 μ_F = 0.102 μ_{CR} = 0.040
 Diversity = 4.1657e+01 k_t = [1, 1, 1, 1]

Generation 4820: Best = 1.717958e+04 FES = 482100 μ_F = 0.134 μ_{CR} = 0.040
 Diversity = 4.1657e+01 k_t = [4, 3, 4, 2]
 Generation 4830: Best = 1.717958e+04 FES = 483100 μ_F = 0.104 μ_{CR} = 0.062
 Diversity = 4.1658e+01 k_t = [1, 1, 1, 1]
 Generation 4840: Best = 1.716817e+04 FES = 484100 μ_F = 0.187 μ_{CR} = 0.048
 Diversity = 4.1861e+01 k_t = [3, 4, 4, 4]
 Generation 4850: Best = 1.716817e+04 FES = 485100 μ_F = 0.179 μ_{CR} = 0.042
 Diversity = 4.1862e+01 k_t = [1, 1, 1, 1]
 Generation 4860: Best = 1.715897e+04 FES = 486100 μ_F = 0.263 μ_{CR} = 0.033
 Diversity = 4.1861e+01 k_t = [3, 2, 4, 3]
 Generation 4870: Best = 1.715897e+04 FES = 487100 μ_F = 0.217 μ_{CR} = 0.036
 Diversity = 4.1861e+01 k_t = [1, 1, 1, 1]
 Generation 4880: Best = 1.715897e+04 FES = 488100 μ_F = 0.204 μ_{CR} = 0.051
 Diversity = 4.1863e+01 k_t = [3, 2, 3, 4]
 Generation 4890: Best = 1.715897e+04 FES = 489100 μ_F = 0.225 μ_{CR} = 0.058
 Diversity = 4.1863e+01 k_t = [1, 1, 1, 1]
 Generation 4900: Best = 1.715897e+04 FES = 490100 μ_F = 0.209 μ_{CR} = 0.074
 Diversity = 4.1864e+01 k_t = [2, 2, 3, 2]
 Generation 4910: Best = 1.715897e+04 FES = 491100 μ_F = 0.284 μ_{CR} = 0.068
 Diversity = 4.1778e+01 k_t = [1, 1, 1, 1]
 Generation 4920: Best = 1.715897e+04 FES = 492100 μ_F = 0.242 μ_{CR} = 0.074
 Diversity = 4.1779e+01 k_t = [2, 2, 2, 2]
 Generation 4930: Best = 1.715897e+04 FES = 493100 μ_F = 0.191 μ_{CR} = 0.055
 Diversity = 4.1779e+01 k_t = [1, 1, 1, 1]
 Generation 4940: Best = 1.715897e+04 FES = 494100 μ_F = 0.137 μ_{CR} = 0.049
 Diversity = 4.1779e+01 k_t = [3, 3, 3, 2]
 Generation 4950: Best = 1.715897e+04 FES = 495100 μ_F = 0.119 μ_{CR} = 0.042
 Diversity = 4.1780e+01 k_t = [1, 1, 1, 1]
 Generation 4960: Best = 1.704381e+04 FES = 496100 μ_F = 0.092 μ_{CR} = 0.052
 Diversity = 4.1777e+01 k_t = [2, 2, 4, 2]
 Generation 4970: Best = 1.703221e+04 FES = 497100 μ_F = 0.048 μ_{CR} = 0.042
 Diversity = 4.1777e+01 k_t = [1, 1, 1, 1]
 Generation 4980: Best = 1.703133e+04 FES = 498100 μ_F = 0.054 μ_{CR} = 0.040
 Diversity = 4.1774e+01 k_t = [2, 3, 3, 3]
 Generation 4990: Best = 1.703133e+04 FES = 499100 μ_F = 0.056 μ_{CR} = 0.059
 Diversity = 4.1772e+01 k_t = [1, 1, 1, 1]
 Generation 5000: Best = 1.703133e+04 FES = 500100 μ_F = 0.106 μ_{CR} = 0.055
 Diversity = 4.1770e+01 k_t = [3, 4, 3, 2]
 Generation 5010: Best = 1.703133e+04 FES = 501100 μ_F = 0.085 μ_{CR} = 0.041
 Diversity = 4.1770e+01 k_t = [1, 1, 1, 1]
 Generation 5020: Best = 1.703133e+04 FES = 502100 μ_F = 0.092 μ_{CR} = 0.037
 Diversity = 4.1770e+01 k_t = [3, 3, 3, 3]
 Generation 5030: Best = 1.701367e+04 FES = 503100 μ_F = 0.092 μ_{CR} = 0.049
 Diversity = 4.1767e+01 k_t = [1, 1, 1, 1]
 Generation 5040: Best = 1.700197e+04 FES = 504100 μ_F = 0.105 μ_{CR} = 0.036
 Diversity = 4.1766e+01 k_t = [4, 3, 2, 3]
 Generation 5050: Best = 1.700197e+04 FES = 505100 μ_F = 0.088 μ_{CR} = 0.040
 Diversity = 4.1764e+01 k_t = [1, 1, 1, 1]
 Generation 5060: Best = 1.699160e+04 FES = 506100 μ_F = 0.058 μ_{CR} = 0.047
 Diversity = 4.1762e+01 k_t = [3, 2, 2, 2]
 Generation 5070: Best = 1.699160e+04 FES = 507100 μ_F = 0.061 μ_{CR} = 0.043
 Diversity = 4.1762e+01 k_t = [1, 1, 1, 1]
 Generation 5080: Best = 1.698503e+04 FES = 508100 μ_F = 0.074 μ_{CR} = 0.048
 Diversity = 4.1761e+01 k_t = [4, 3, 2, 2]

Generation 5090: Best = 1.698503e+04 FES = 509100 μ_F = 0.077 μ_{CR} = 0.036
 Diversity = 4.1763e+01 k_t = [1, 1, 1, 1]
 Generation 5100: Best = 1.698454e+04 FES = 510100 μ_F = 0.096 μ_{CR} = 0.032
 Diversity = 4.1762e+01 k_t = [3, 4, 2, 2]
 Generation 5110: Best = 1.698454e+04 FES = 511100 μ_F = 0.098 μ_{CR} = 0.028
 Diversity = 4.1761e+01 k_t = [1, 1, 1, 1]
 Generation 5120: Best = 1.697306e+04 FES = 512100 μ_F = 0.086 μ_{CR} = 0.032
 Diversity = 4.1760e+01 k_t = [3, 3, 2, 2]
 Generation 5130: Best = 1.694582e+04 FES = 513100 μ_F = 0.128 μ_{CR} = 0.041
 Diversity = 4.1760e+01 k_t = [1, 1, 1, 1]
 Generation 5140: Best = 1.694419e+04 FES = 514100 μ_F = 0.243 μ_{CR} = 0.032
 Diversity = 4.1760e+01 k_t = [2, 4, 2, 2]
 Generation 5150: Best = 1.694419e+04 FES = 515100 μ_F = 0.167 μ_{CR} = 0.030
 Diversity = 4.1759e+01 k_t = [1, 1, 1, 1]
 Generation 5160: Best = 1.694419e+04 FES = 516100 μ_F = 0.279 μ_{CR} = 0.032
 Diversity = 4.1718e+01 k_t = [3, 2, 4, 4]
 Generation 5170: Best = 1.694419e+04 FES = 517100 μ_F = 0.356 μ_{CR} = 0.030
 Diversity = 4.1752e+01 k_t = [1, 1, 1, 1]
 Generation 5180: Best = 1.690704e+04 FES = 518100 μ_F = 0.355 μ_{CR} = 0.033
 Diversity = 4.1754e+01 k_t = [3, 4, 4, 4]
 Generation 5190: Best = 1.690704e+04 FES = 519100 μ_F = 0.343 μ_{CR} = 0.032
 Diversity = 4.1755e+01 k_t = [1, 1, 1, 1]
 Generation 5200: Best = 1.689957e+04 FES = 520100 μ_F = 0.372 μ_{CR} = 0.023
 Diversity = 4.1646e+01 k_t = [3, 2, 4, 4]
 Generation 5210: Best = 1.689957e+04 FES = 521100 μ_F = 0.363 μ_{CR} = 0.028
 Diversity = 4.1646e+01 k_t = [1, 1, 1, 1]
 Generation 5220: Best = 1.689957e+04 FES = 522100 μ_F = 0.305 μ_{CR} = 0.035
 Diversity = 4.1646e+01 k_t = [2, 3, 2, 3]
 Generation 5230: Best = 1.689957e+04 FES = 523100 μ_F = 0.323 μ_{CR} = 0.027
 Diversity = 4.1645e+01 k_t = [1, 1, 1, 1]
 Generation 5240: Best = 1.689957e+04 FES = 524100 μ_F = 0.309 μ_{CR} = 0.028
 Diversity = 4.1614e+01 k_t = [4, 4, 2, 4]
 Generation 5250: Best = 1.689957e+04 FES = 525100 μ_F = 0.343 μ_{CR} = 0.046
 Diversity = 4.1617e+01 k_t = [1, 1, 1, 1]
 Generation 5260: Best = 1.689957e+04 FES = 526100 μ_F = 0.370 μ_{CR} = 0.037
 Diversity = 4.1616e+01 k_t = [3, 3, 2, 2]
 Generation 5270: Best = 1.689957e+04 FES = 527100 μ_F = 0.279 μ_{CR} = 0.037
 Diversity = 4.1614e+01 k_t = [1, 1, 1, 1]
 Generation 5280: Best = 1.689957e+04 FES = 528100 μ_F = 0.207 μ_{CR} = 0.035
 Diversity = 4.1613e+01 k_t = [2, 4, 4, 3]
 Generation 5290: Best = 1.689817e+04 FES = 529100 μ_F = 0.231 μ_{CR} = 0.041
 Diversity = 4.1614e+01 k_t = [1, 1, 1, 1]
 Generation 5300: Best = 1.689817e+04 FES = 530100 μ_F = 0.223 μ_{CR} = 0.036
 Diversity = 4.1614e+01 k_t = [2, 2, 2, 2]
 Generation 5310: Best = 1.686621e+04 FES = 531100 μ_F = 0.236 μ_{CR} = 0.040
 Diversity = 4.1614e+01 k_t = [1, 1, 1, 1]
 Generation 5320: Best = 1.686621e+04 FES = 532100 μ_F = 0.182 μ_{CR} = 0.033
 Diversity = 4.1614e+01 k_t = [3, 3, 2, 3]
 Generation 5330: Best = 1.686621e+04 FES = 533100 μ_F = 0.165 μ_{CR} = 0.030
 Diversity = 4.1615e+01 k_t = [1, 1, 1, 1]
 Generation 5340: Best = 1.686120e+04 FES = 534100 μ_F = 0.149 μ_{CR} = 0.058
 Diversity = 4.1616e+01 k_t = [3, 2, 2, 3]
 Generation 5350: Best = 1.686120e+04 FES = 535100 μ_F = 0.144 μ_{CR} = 0.060
 Diversity = 4.1614e+01 k_t = [1, 1, 1, 1]

Generation 5360: Best = 1.681878e+04 FES = 536100 μ_F = 0.102 μ_{CR} = 0.059
 Diversity = 4.1612e+01 k_t = [3, 2, 3, 4]
 Generation 5370: Best = 1.681878e+04 FES = 537100 μ_F = 0.117 μ_{CR} = 0.048
 Diversity = 4.1613e+01 k_t = [1, 1, 1, 1]
 Generation 5380: Best = 1.681563e+04 FES = 538100 μ_F = 0.100 μ_{CR} = 0.050
 Diversity = 4.1613e+01 k_t = [4, 4, 2, 4]
 Generation 5390: Best = 1.681563e+04 FES = 539100 μ_F = 0.134 μ_{CR} = 0.059
 Diversity = 4.1611e+01 k_t = [1, 1, 1, 1]
 Generation 5400: Best = 1.681040e+04 FES = 540100 μ_F = 0.191 μ_{CR} = 0.050
 Diversity = 4.1611e+01 k_t = [4, 4, 4, 4]
 Generation 5410: Best = 1.681040e+04 FES = 541100 μ_F = 0.180 μ_{CR} = 0.038
 Diversity = 4.1609e+01 k_t = [1, 1, 1, 1]
 Generation 5420: Best = 1.681040e+04 FES = 542100 μ_F = 0.164 μ_{CR} = 0.025
 Diversity = 4.1609e+01 k_t = [3, 4, 3, 2]
 Generation 5430: Best = 1.681040e+04 FES = 543100 μ_F = 0.154 μ_{CR} = 0.031
 Diversity = 4.1608e+01 k_t = [1, 1, 1, 1]
 Generation 5440: Best = 1.680917e+04 FES = 544100 μ_F = 0.101 μ_{CR} = 0.044
 Diversity = 4.1606e+01 k_t = [4, 3, 4, 3]
 Generation 5450: Best = 1.680879e+04 FES = 545100 μ_F = 0.188 μ_{CR} = 0.053
 Diversity = 4.1607e+01 k_t = [1, 1, 1, 1]
 Generation 5460: Best = 1.680867e+04 FES = 546100 μ_F = 0.277 μ_{CR} = 0.067
 Diversity = 4.1412e+01 k_t = [4, 2, 2, 4]
 Generation 5470: Best = 1.680867e+04 FES = 547100 μ_F = 0.230 μ_{CR} = 0.074
 Diversity = 4.1411e+01 k_t = [1, 1, 1, 1]
 Generation 5480: Best = 1.680867e+04 FES = 548100 μ_F = 0.137 μ_{CR} = 0.091
 Diversity = 4.1409e+01 k_t = [3, 2, 3, 3]
 Generation 5490: Best = 1.680867e+04 FES = 549100 μ_F = 0.122 μ_{CR} = 0.114
 Diversity = 4.1412e+01 k_t = [1, 1, 1, 1]
 Generation 5500: Best = 1.674811e+04 FES = 550100 μ_F = 0.131 μ_{CR} = 0.110
 Diversity = 4.1412e+01 k_t = [4, 3, 4, 3]
 Generation 5510: Best = 1.674811e+04 FES = 551100 μ_F = 0.109 μ_{CR} = 0.104
 Diversity = 4.1412e+01 k_t = [1, 1, 1, 1]
 Generation 5520: Best = 1.672554e+04 FES = 552100 μ_F = 0.271 μ_{CR} = 0.083
 Diversity = 4.1308e+01 k_t = [2, 2, 3, 3]
 Generation 5530: Best = 1.672554e+04 FES = 553100 μ_F = 0.219 μ_{CR} = 0.070
 Diversity = 4.1308e+01 k_t = [1, 1, 1, 1]
 Generation 5540: Best = 1.672554e+04 FES = 554100 μ_F = 0.230 μ_{CR} = 0.065
 Diversity = 4.1308e+01 k_t = [2, 4, 3, 2]
 Generation 5550: Best = 1.671476e+04 FES = 555100 μ_F = 0.243 μ_{CR} = 0.048
 Diversity = 4.1309e+01 k_t = [1, 1, 1, 1]
 Generation 5560: Best = 1.671476e+04 FES = 556100 μ_F = 0.184 μ_{CR} = 0.057
 Diversity = 4.1310e+01 k_t = [3, 4, 3, 3]
 Generation 5570: Best = 1.671476e+04 FES = 557100 μ_F = 0.234 μ_{CR} = 0.048
 Diversity = 4.1307e+01 k_t = [1, 1, 1, 1]
 Generation 5580: Best = 1.671047e+04 FES = 558100 μ_F = 0.286 μ_{CR} = 0.056
 Diversity = 4.1307e+01 k_t = [2, 4, 2, 3]
 Generation 5590: Best = 1.671047e+04 FES = 559100 μ_F = 0.183 μ_{CR} = 0.054
 Diversity = 4.1305e+01 k_t = [1, 1, 1, 1]
 Generation 5600: Best = 1.671047e+04 FES = 560100 μ_F = 0.163 μ_{CR} = 0.049
 Diversity = 4.1304e+01 k_t = [2, 2, 2, 2]
 Generation 5610: Best = 1.665646e+04 FES = 561100 μ_F = 0.372 μ_{CR} = 0.056
 Diversity = 4.1231e+01 k_t = [1, 1, 1, 1]
 Generation 5620: Best = 1.665646e+04 FES = 562100 μ_F = 0.437 μ_{CR} = 0.045
 Diversity = 4.1164e+01 k_t = [4, 2, 2, 3]

Generation 5630: Best = 1.665646e+04 FES = 563100 μ_F = 0.402 μ_{CR} = 0.056
 Diversity = 4.1163e+01 k_t = [1, 1, 1, 1]
 Generation 5640: Best = 1.665646e+04 FES = 564100 μ_F = 0.343 μ_{CR} = 0.038
 Diversity = 4.1163e+01 k_t = [2, 3, 2, 3]
 Generation 5650: Best = 1.665476e+04 FES = 565100 μ_F = 0.307 μ_{CR} = 0.030
 Diversity = 4.1163e+01 k_t = [1, 1, 1, 1]
 Generation 5660: Best = 1.661828e+04 FES = 566100 μ_F = 0.308 μ_{CR} = 0.028
 Diversity = 4.1163e+01 k_t = [2, 2, 3, 2]
 Generation 5670: Best = 1.661828e+04 FES = 567100 μ_F = 0.302 μ_{CR} = 0.033
 Diversity = 4.1164e+01 k_t = [1, 1, 1, 1]
 Generation 5680: Best = 1.659377e+04 FES = 568100 μ_F = 0.295 μ_{CR} = 0.038
 Diversity = 4.1164e+01 k_t = [4, 2, 3, 2]
 Generation 5690: Best = 1.659377e+04 FES = 569100 μ_F = 0.296 μ_{CR} = 0.036
 Diversity = 4.1164e+01 k_t = [1, 1, 1, 1]
 Generation 5700: Best = 1.656941e+04 FES = 570100 μ_F = 0.348 μ_{CR} = 0.032
 Diversity = 4.1158e+01 k_t = [2, 2, 4, 4]
 Generation 5710: Best = 1.656941e+04 FES = 571100 μ_F = 0.313 μ_{CR} = 0.027
 Diversity = 4.1158e+01 k_t = [1, 1, 1, 1]
 Generation 5720: Best = 1.645693e+04 FES = 572100 μ_F = 0.318 μ_{CR} = 0.030
 Diversity = 4.1159e+01 k_t = [2, 3, 2, 3]
 Generation 5730: Best = 1.645693e+04 FES = 573100 μ_F = 0.253 μ_{CR} = 0.045
 Diversity = 4.1158e+01 k_t = [1, 1, 1, 1]
 Generation 5740: Best = 1.645693e+04 FES = 574100 μ_F = 0.178 μ_{CR} = 0.046
 Diversity = 4.1158e+01 k_t = [2, 2, 4, 2]
 Generation 5750: Best = 1.645693e+04 FES = 575100 μ_F = 0.177 μ_{CR} = 0.035
 Diversity = 4.1157e+01 k_t = [1, 1, 1, 1]
 Generation 5760: Best = 1.643467e+04 FES = 576100 μ_F = 0.208 μ_{CR} = 0.032
 Diversity = 4.0978e+01 k_t = [4, 3, 2, 2]
 Generation 5770: Best = 1.643467e+04 FES = 577100 μ_F = 0.202 μ_{CR} = 0.026
 Diversity = 4.0979e+01 k_t = [1, 1, 1, 1]
 Generation 5780: Best = 1.643239e+04 FES = 578100 μ_F = 0.185 μ_{CR} = 0.039
 Diversity = 4.0980e+01 k_t = [2, 4, 3, 3]
 Generation 5790: Best = 1.643239e+04 FES = 579100 μ_F = 0.156 μ_{CR} = 0.047
 Diversity = 4.0981e+01 k_t = [1, 1, 1, 1]
 Generation 5800: Best = 1.643239e+04 FES = 580100 μ_F = 0.200 μ_{CR} = 0.041
 Diversity = 4.0891e+01 k_t = [4, 3, 4, 4]
 Generation 5810: Best = 1.640441e+04 FES = 581100 μ_F = 0.190 μ_{CR} = 0.044
 Diversity = 4.0890e+01 k_t = [1, 1, 1, 1]
 Generation 5820: Best = 1.628780e+04 FES = 582100 μ_F = 0.242 μ_{CR} = 0.093
 Diversity = 4.0830e+01 k_t = [3, 2, 4, 4]
 Generation 5830: Best = 1.628780e+04 FES = 583100 μ_F = 0.293 μ_{CR} = 0.121
 Diversity = 4.0843e+01 k_t = [1, 1, 1, 1]
 Generation 5840: Best = 1.628780e+04 FES = 584100 μ_F = 0.331 μ_{CR} = 0.095
 Diversity = 4.0773e+01 k_t = [2, 4, 2, 3]
 Generation 5850: Best = 1.628780e+04 FES = 585100 μ_F = 0.263 μ_{CR} = 0.067
 Diversity = 4.0773e+01 k_t = [1, 1, 1, 1]
 Generation 5860: Best = 1.626171e+04 FES = 586100 μ_F = 0.285 μ_{CR} = 0.065
 Diversity = 4.0604e+01 k_t = [4, 2, 2, 4]
 Generation 5870: Best = 1.626171e+04 FES = 587100 μ_F = 0.304 μ_{CR} = 0.068
 Diversity = 4.0604e+01 k_t = [1, 1, 1, 1]
 Generation 5880: Best = 1.626171e+04 FES = 588100 μ_F = 0.278 μ_{CR} = 0.054
 Diversity = 4.0604e+01 k_t = [3, 3, 3, 2]
 Generation 5890: Best = 1.626171e+04 FES = 589100 μ_F = 0.232 μ_{CR} = 0.064
 Diversity = 4.0604e+01 k_t = [1, 1, 1, 1]

Generation 5900: Best = 1.621825e+04 FES = 590100 μ_F = 0.222 μ_{CR} = 0.056
 Diversity = 4.0737e+01 k_t = [2, 4, 4, 4]
 Generation 5910: Best = 1.621522e+04 FES = 591100 μ_F = 0.226 μ_{CR} = 0.071
 Diversity = 4.0737e+01 k_t = [1, 1, 1, 1]
 Generation 5920: Best = 1.620566e+04 FES = 592100 μ_F = 0.389 μ_{CR} = 0.084
 Diversity = 4.0617e+01 k_t = [3, 4, 2, 4]
 Generation 5930: Best = 1.620566e+04 FES = 593100 μ_F = 0.441 μ_{CR} = 0.072
 Diversity = 4.0530e+01 k_t = [1, 1, 1, 1]
 Generation 5940: Best = 1.619836e+04 FES = 594100 μ_F = 0.484 μ_{CR} = 0.070
 Diversity = 4.0529e+01 k_t = [2, 2, 3, 3]
 Generation 5950: Best = 1.614579e+04 FES = 595100 μ_F = 0.457 μ_{CR} = 0.064
 Diversity = 4.0528e+01 k_t = [1, 1, 1, 1]
 Generation 5960: Best = 1.614579e+04 FES = 596100 μ_F = 0.479 μ_{CR} = 0.081
 Diversity = 4.0530e+01 k_t = [2, 3, 3, 3]
 Generation 5970: Best = 1.614579e+04 FES = 597100 μ_F = 0.489 μ_{CR} = 0.079
 Diversity = 4.0506e+01 k_t = [1, 1, 1, 1]
 Generation 5980: Best = 1.614579e+04 FES = 598100 μ_F = 0.483 μ_{CR} = 0.078
 Diversity = 4.0506e+01 k_t = [2, 3, 2, 2]
 Generation 5990: Best = 1.614579e+04 FES = 599100 μ_F = 0.537 μ_{CR} = 0.064
 Diversity = 4.0506e+01 k_t = [1, 1, 1, 1]
 Generation 6000: Best = 1.614579e+04 FES = 600100 μ_F = 0.578 μ_{CR} = 0.066
 Diversity = 4.0452e+01 k_t = [3, 3, 3, 2]
 Generation 6010: Best = 1.612878e+04 FES = 601100 μ_F = 0.594 μ_{CR} = 0.080
 Diversity = 4.0452e+01 k_t = [1, 1, 1, 1]
 Generation 6020: Best = 1.612878e+04 FES = 602100 μ_F = 0.653 μ_{CR} = 0.088
 Diversity = 4.0350e+01 k_t = [2, 3, 2, 3]
 Generation 6030: Best = 1.612158e+04 FES = 603100 μ_F = 0.427 μ_{CR} = 0.064
 Diversity = 4.0351e+01 k_t = [1, 1, 1, 1]
 Generation 6040: Best = 1.612158e+04 FES = 604100 μ_F = 0.475 μ_{CR} = 0.076
 Diversity = 4.0246e+01 k_t = [3, 2, 3, 2]
 Generation 6050: Best = 1.612158e+04 FES = 605100 μ_F = 0.421 μ_{CR} = 0.088
 Diversity = 4.0246e+01 k_t = [1, 1, 1, 1]
 Generation 6060: Best = 1.602694e+04 FES = 606100 μ_F = 0.402 μ_{CR} = 0.091
 Diversity = 4.0247e+01 k_t = [3, 2, 4, 3]
 Generation 6070: Best = 1.602298e+04 FES = 607100 μ_F = 0.336 μ_{CR} = 0.087
 Diversity = 4.0246e+01 k_t = [1, 1, 1, 1]
 Generation 6080: Best = 1.602298e+04 FES = 608100 μ_F = 0.423 μ_{CR} = 0.077
 Diversity = 4.0399e+01 k_t = [2, 2, 2, 2]
 Generation 6090: Best = 1.602298e+04 FES = 609100 μ_F = 0.397 μ_{CR} = 0.074
 Diversity = 4.0399e+01 k_t = [1, 1, 1, 1]
 Generation 6100: Best = 1.594662e+04 FES = 610100 μ_F = 0.462 μ_{CR} = 0.075
 Diversity = 4.0414e+01 k_t = [2, 3, 2, 4]
 Generation 6110: Best = 1.592342e+04 FES = 611100 μ_F = 0.456 μ_{CR} = 0.088
 Diversity = 4.0415e+01 k_t = [1, 1, 1, 1]
 Generation 6120: Best = 1.592342e+04 FES = 612100 μ_F = 0.553 μ_{CR} = 0.080
 Diversity = 4.0535e+01 k_t = [2, 4, 3, 3]
 Generation 6130: Best = 1.592342e+04 FES = 613100 μ_F = 0.623 μ_{CR} = 0.085
 Diversity = 4.0505e+01 k_t = [1, 1, 1, 1]
 Generation 6140: Best = 1.592342e+04 FES = 614100 μ_F = 0.632 μ_{CR} = 0.091
 Diversity = 3.9951e+01 k_t = [4, 4, 2, 4]
 Generation 6150: Best = 1.592342e+04 FES = 615100 μ_F = 0.576 μ_{CR} = 0.067
 Diversity = 3.9951e+01 k_t = [1, 1, 1, 1]
 Generation 6160: Best = 1.592342e+04 FES = 616100 μ_F = 0.644 μ_{CR} = 0.063
 Diversity = 3.9966e+01 k_t = [4, 4, 2, 3]

Generation 6170: Best = 1.587927e+04 FES = 617100 μ_F = 0.627 μ_{CR} = 0.097
 Diversity = 3.9935e+01 k_t = [1, 1, 1, 1]
 Generation 6180: Best = 1.587927e+04 FES = 618100 μ_F = 0.633 μ_{CR} = 0.074
 Diversity = 3.9848e+01 k_t = [2, 3, 4, 2]
 Generation 6190: Best = 1.582718e+04 FES = 619100 μ_F = 0.667 μ_{CR} = 0.108
 Diversity = 3.9848e+01 k_t = [1, 1, 1, 1]
 Generation 6200: Best = 1.582718e+04 FES = 620100 μ_F = 0.728 μ_{CR} = 0.125
 Diversity = 3.9721e+01 k_t = [4, 3, 4, 4]
 Generation 6210: Best = 1.582718e+04 FES = 621100 μ_F = 0.710 μ_{CR} = 0.145
 Diversity = 3.9718e+01 k_t = [1, 1, 1, 1]
 Generation 6220: Best = 1.580605e+04 FES = 622100 μ_F = 0.688 μ_{CR} = 0.128
 Diversity = 3.9717e+01 k_t = [3, 4, 4, 3]
 Generation 6230: Best = 1.580308e+04 FES = 623100 μ_F = 0.697 μ_{CR} = 0.103
 Diversity = 3.9696e+01 k_t = [1, 1, 1, 1]
 Generation 6240: Best = 1.579224e+04 FES = 624100 μ_F = 0.739 μ_{CR} = 0.080
 Diversity = 3.9697e+01 k_t = [3, 4, 2, 2]
 Generation 6250: Best = 1.579224e+04 FES = 625100 μ_F = 0.752 μ_{CR} = 0.152
 Diversity = 3.9614e+01 k_t = [1, 1, 1, 1]
 Generation 6260: Best = 1.576013e+04 FES = 626100 μ_F = 0.795 μ_{CR} = 0.139
 Diversity = 3.9088e+01 k_t = [4, 4, 3, 4]
 Generation 6270: Best = 1.576013e+04 FES = 627100 μ_F = 0.841 μ_{CR} = 0.132
 Diversity = 3.8772e+01 k_t = [1, 1, 1, 1]
 Generation 6280: Best = 1.576013e+04 FES = 628100 μ_F = 0.720 μ_{CR} = 0.151
 Diversity = 3.8771e+01 k_t = [2, 3, 3, 2]
 Generation 6290: Best = 1.576013e+04 FES = 629100 μ_F = 0.681 μ_{CR} = 0.148
 Diversity = 3.8766e+01 k_t = [1, 1, 1, 1]
 Generation 6300: Best = 1.563921e+04 FES = 630100 μ_F = 0.698 μ_{CR} = 0.170
 Diversity = 3.8771e+01 k_t = [4, 2, 2, 3]
 Generation 6310: Best = 1.563092e+04 FES = 631100 μ_F = 0.697 μ_{CR} = 0.193
 Diversity = 3.8618e+01 k_t = [1, 1, 1, 1]
 Generation 6320: Best = 1.562697e+04 FES = 632100 μ_F = 0.703 μ_{CR} = 0.232
 Diversity = 3.8516e+01 k_t = [2, 4, 2, 3]
 Generation 6330: Best = 1.562697e+04 FES = 633100 μ_F = 0.660 μ_{CR} = 0.252
 Diversity = 3.8516e+01 k_t = [1, 1, 1, 1]
 Generation 6340: Best = 1.562697e+04 FES = 634100 μ_F = 0.617 μ_{CR} = 0.248
 Diversity = 3.8517e+01 k_t = [3, 3, 3, 3]
 Generation 6350: Best = 1.557188e+04 FES = 635100 μ_F = 0.577 μ_{CR} = 0.224
 Diversity = 3.8207e+01 k_t = [1, 1, 1, 1]
 Generation 6360: Best = 1.556343e+04 FES = 636100 μ_F = 0.615 μ_{CR} = 0.231
 Diversity = 3.8043e+01 k_t = [2, 4, 4, 3]
 Generation 6370: Best = 1.553308e+04 FES = 637100 μ_F = 0.608 μ_{CR} = 0.195
 Diversity = 3.8037e+01 k_t = [1, 1, 1, 1]
 Generation 6380: Best = 1.550825e+04 FES = 638100 μ_F = 0.636 μ_{CR} = 0.136
 Diversity = 3.8036e+01 k_t = [2, 2, 2, 3]
 Generation 6390: Best = 1.550825e+04 FES = 639100 μ_F = 0.677 μ_{CR} = 0.154
 Diversity = 3.7878e+01 k_t = [1, 1, 1, 1]
 Generation 6400: Best = 1.550825e+04 FES = 640100 μ_F = 0.643 μ_{CR} = 0.151
 Diversity = 3.7583e+01 k_t = [2, 3, 4, 4]
 Generation 6410: Best = 1.548611e+04 FES = 641100 μ_F = 0.703 μ_{CR} = 0.182
 Diversity = 3.7341e+01 k_t = [1, 1, 1, 1]
 Generation 6420: Best = 1.547125e+04 FES = 642100 μ_F = 0.704 μ_{CR} = 0.160
 Diversity = 3.7340e+01 k_t = [3, 2, 4, 3]
 Generation 6430: Best = 1.545238e+04 FES = 643100 μ_F = 0.700 μ_{CR} = 0.117
 Diversity = 3.7341e+01 k_t = [1, 1, 1, 1]

Generation 6440: Best = 1.545238e+04 FES = 644100 μ_F = 0.728 μ_{CR} = 0.121
 Diversity = 3.6965e+01 k_t = [3, 3, 2, 3]
 Generation 6450: Best = 1.544440e+04 FES = 645100 μ_F = 0.710 μ_{CR} = 0.140
 Diversity = 3.7004e+01 k_t = [1, 1, 1, 1]
 Generation 6460: Best = 1.537975e+04 FES = 646100 μ_F = 0.743 μ_{CR} = 0.125
 Diversity = 3.5977e+01 k_t = [2, 4, 2, 4]
 Generation 6470: Best = 1.537975e+04 FES = 647100 μ_F = 0.770 μ_{CR} = 0.153
 Diversity = 3.5747e+01 k_t = [1, 1, 1, 1]
 Generation 6480: Best = 1.533730e+04 FES = 648100 μ_F = 0.829 μ_{CR} = 0.133
 Diversity = 3.5394e+01 k_t = [2, 2, 4, 2]
 Generation 6490: Best = 1.532100e+04 FES = 649100 μ_F = 0.834 μ_{CR} = 0.142
 Diversity = 3.4869e+01 k_t = [1, 1, 1, 1]
 Generation 6500: Best = 1.531166e+04 FES = 650100 μ_F = 0.807 μ_{CR} = 0.136
 Diversity = 3.4389e+01 k_t = [2, 2, 3, 2]
 Generation 6510: Best = 1.525298e+04 FES = 651100 μ_F = 0.775 μ_{CR} = 0.145
 Diversity = 3.3665e+01 k_t = [1, 1, 1, 1]
 Generation 6520: Best = 1.517122e+04 FES = 652100 μ_F = 0.766 μ_{CR} = 0.150
 Diversity = 3.3371e+01 k_t = [3, 3, 2, 4]
 Generation 6530: Best = 1.512461e+04 FES = 653100 μ_F = 0.781 μ_{CR} = 0.150
 Diversity = 3.2592e+01 k_t = [1, 1, 1, 1]
 Generation 6540: Best = 1.509035e+04 FES = 654100 μ_F = 0.786 μ_{CR} = 0.158
 Diversity = 3.1593e+01 k_t = [4, 4, 4, 4]
 Generation 6550: Best = 1.501284e+04 FES = 655100 μ_F = 0.798 μ_{CR} = 0.158
 Diversity = 3.0869e+01 k_t = [1, 1, 1, 1]
 Generation 6560: Best = 1.498253e+04 FES = 656100 μ_F = 0.809 μ_{CR} = 0.165
 Diversity = 2.9494e+01 k_t = [4, 4, 2, 4]
 Generation 6570: Best = 1.494519e+04 FES = 657100 μ_F = 0.802 μ_{CR} = 0.168
 Diversity = 2.7795e+01 k_t = [1, 1, 1, 1]
 Generation 6580: Best = 1.492870e+04 FES = 658100 μ_F = 0.812 μ_{CR} = 0.169
 Diversity = 1.9336e+01 k_t = [3, 2, 4, 2]
 Generation 6590: Best = 1.484025e+04 FES = 659100 μ_F = 0.785 μ_{CR} = 0.171
 Diversity = 1.7456e+01 k_t = [1, 1, 1, 1]
 Generation 6600: Best = 1.484025e+04 FES = 660100 μ_F = 0.760 μ_{CR} = 0.172
 Diversity = 1.5918e+01 k_t = [4, 3, 2, 4]
 Generation 6610: Best = 1.481210e+04 FES = 661100 μ_F = 0.755 μ_{CR} = 0.163
 Diversity = 1.5920e+01 k_t = [1, 1, 1, 1]
 Generation 6620: Best = 1.480126e+04 FES = 662100 μ_F = 0.747 μ_{CR} = 0.162
 Diversity = 1.5921e+01 k_t = [3, 4, 2, 4]
 Generation 6630: Best = 1.476910e+04 FES = 663100 μ_F = 0.748 μ_{CR} = 0.162
 Diversity = 1.5922e+01 k_t = [1, 1, 1, 1]
 Generation 6640: Best = 1.474846e+04 FES = 664100 μ_F = 0.747 μ_{CR} = 0.166
 Diversity = 1.5923e+01 k_t = [3, 4, 3, 3]
 Generation 6650: Best = 1.471370e+04 FES = 665100 μ_F = 0.732 μ_{CR} = 0.168
 Diversity = 1.5177e+01 k_t = [1, 1, 1, 1]
 Generation 6660: Best = 1.469068e+04 FES = 666100 μ_F = 0.710 μ_{CR} = 0.165
 Diversity = 1.5177e+01 k_t = [4, 3, 4, 3]
 Generation 6670: Best = 1.466962e+04 FES = 667100 μ_F = 0.704 μ_{CR} = 0.171
 Diversity = 1.2407e+01 k_t = [1, 1, 1, 1]
 Generation 6680: Best = 1.464414e+04 FES = 668100 μ_F = 0.709 μ_{CR} = 0.174
 Diversity = 1.0370e+01 k_t = [4, 2, 3, 2]
 Generation 6690: Best = 1.464266e+04 FES = 669100 μ_F = 0.700 μ_{CR} = 0.168
 Diversity = 4.8877e+00 k_t = [1, 1, 1, 1]
 Generation 6700: Best = 1.463559e+04 FES = 670100 μ_F = 0.702 μ_{CR} = 0.165
 Diversity = 1.5318e-01 k_t = [2, 3, 3, 3]

Generation 6710: Best = 1.461571e+04 FES = 671100 μ_F = 0.676 μ_{CR} = 0.167
 Diversity = 1.2765e-01 k_t = [1, 1, 1, 1]
 Generation 6720: Best = 1.460044e+04 FES = 672100 μ_F = 0.654 μ_{CR} = 0.165
 Diversity = 1.0856e-01 k_t = [3, 2, 4, 2]
 Generation 6730: Best = 1.459551e+04 FES = 673100 μ_F = 0.649 μ_{CR} = 0.152
 Diversity = 9.9051e-02 k_t = [1, 1, 1, 1]
 Generation 6740: Best = 1.458901e+04 FES = 674100 μ_F = 0.629 μ_{CR} = 0.145
 Diversity = 7.7343e-02 k_t = [4, 3, 4, 4]
 Generation 6750: Best = 1.458390e+04 FES = 675100 μ_F = 0.616 μ_{CR} = 0.150
 Diversity = 6.1395e-02 k_t = [1, 1, 1, 1]
 Generation 6760: Best = 1.458161e+04 FES = 676100 μ_F = 0.626 μ_{CR} = 0.141
 Diversity = 5.2704e-02 k_t = [4, 3, 2, 4]
 Generation 6770: Best = 1.457702e+04 FES = 677100 μ_F = 0.632 μ_{CR} = 0.125
 Diversity = 4.5705e-02 k_t = [1, 1, 1, 1]
 Generation 6780: Best = 1.457574e+04 FES = 678100 μ_F = 0.630 μ_{CR} = 0.114
 Diversity = 4.1246e-02 k_t = [4, 2, 4, 3]
 Generation 6790: Best = 1.457360e+04 FES = 679100 μ_F = 0.615 μ_{CR} = 0.133
 Diversity = 3.3775e-02 k_t = [1, 1, 1, 1]
 Generation 6800: Best = 1.457214e+04 FES = 680100 μ_F = 0.616 μ_{CR} = 0.139
 Diversity = 3.0415e-02 k_t = [4, 4, 2, 2]
 Generation 6810: Best = 1.456914e+04 FES = 681100 μ_F = 0.654 μ_{CR} = 0.129
 Diversity = 2.9294e-02 k_t = [1, 1, 1, 1]
 Generation 6820: Best = 1.456801e+04 FES = 682100 μ_F = 0.657 μ_{CR} = 0.120
 Diversity = 2.7637e-02 k_t = [2, 2, 2, 3]
 Generation 6830: Best = 1.456738e+04 FES = 683100 μ_F = 0.658 μ_{CR} = 0.121
 Diversity = 2.6939e-02 k_t = [1, 1, 1, 1]
 Generation 6840: Best = 1.456633e+04 FES = 684100 μ_F = 0.654 μ_{CR} = 0.111
 Diversity = 2.3815e-02 k_t = [3, 2, 4, 4]
 Generation 6850: Best = 1.456472e+04 FES = 685100 μ_F = 0.652 μ_{CR} = 0.118
 Diversity = 2.2754e-02 k_t = [1, 1, 1, 1]
 Generation 6860: Best = 1.456111e+04 FES = 686100 μ_F = 0.660 μ_{CR} = 0.124
 Diversity = 2.3429e-02 k_t = [2, 2, 2, 2]
 Generation 6870: Best = 1.456088e+04 FES = 687100 μ_F = 0.678 μ_{CR} = 0.123
 Diversity = 2.5226e-02 k_t = [1, 1, 1, 1]
 Generation 6880: Best = 1.455995e+04 FES = 688100 μ_F = 0.689 μ_{CR} = 0.125
 Diversity = 2.4130e-02 k_t = [4, 3, 3, 4]
 Generation 6890: Best = 1.455831e+04 FES = 689100 μ_F = 0.698 μ_{CR} = 0.130
 Diversity = 2.3600e-02 k_t = [1, 1, 1, 1]
 Generation 6900: Best = 1.455615e+04 FES = 690100 μ_F = 0.723 μ_{CR} = 0.125
 Diversity = 2.4099e-02 k_t = [2, 4, 3, 4]
 Generation 6910: Best = 1.455519e+04 FES = 691100 μ_F = 0.717 μ_{CR} = 0.116
 Diversity = 2.3988e-02 k_t = [1, 1, 1, 1]
 Generation 6920: Best = 1.455427e+04 FES = 692100 μ_F = 0.712 μ_{CR} = 0.122
 Diversity = 2.2571e-02 k_t = [2, 3, 4, 2]
 Generation 6930: Best = 1.455292e+04 FES = 693100 μ_F = 0.710 μ_{CR} = 0.113
 Diversity = 2.2859e-02 k_t = [1, 1, 1, 1]
 Generation 6940: Best = 1.455052e+04 FES = 694100 μ_F = 0.707 μ_{CR} = 0.116
 Diversity = 2.2677e-02 k_t = [2, 4, 2, 2]
 Generation 6950: Best = 1.454980e+04 FES = 695100 μ_F = 0.724 μ_{CR} = 0.120
 Diversity = 2.3446e-02 k_t = [1, 1, 1, 1]
 Generation 6960: Best = 1.454898e+04 FES = 696100 μ_F = 0.721 μ_{CR} = 0.117
 Diversity = 2.3462e-02 k_t = [3, 3, 4, 2]
 Generation 6970: Best = 1.454765e+04 FES = 697100 μ_F = 0.729 μ_{CR} = 0.110
 Diversity = 2.2963e-02 k_t = [1, 1, 1, 1]

Generation 6980: Best = 1.454713e+04 FES = 698100 μ_F = 0.725 μ_{CR} = 0.101
 Diversity = 2.0496e-02 k_t = [2, 3, 3, 4]
 Generation 6990: Best = 1.454614e+04 FES = 699100 μ_F = 0.729 μ_{CR} = 0.103
 Diversity = 1.9621e-02 k_t = [1, 1, 1, 1]
 Generation 7000: Best = 1.454490e+04 FES = 700100 μ_F = 0.734 μ_{CR} = 0.117
 Diversity = 2.0494e-02 k_t = [2, 2, 2, 4]
 Generation 7010: Best = 1.454362e+04 FES = 701100 μ_F = 0.727 μ_{CR} = 0.114
 Diversity = 2.1118e-02 k_t = [1, 1, 1, 1]
 Generation 7020: Best = 1.454240e+04 FES = 702100 μ_F = 0.727 μ_{CR} = 0.125
 Diversity = 2.0576e-02 k_t = [4, 4, 2, 4]
 Generation 7030: Best = 1.454119e+04 FES = 703100 μ_F = 0.721 μ_{CR} = 0.124
 Diversity = 1.9403e-02 k_t = [1, 1, 1, 1]
 Generation 7040: Best = 1.454090e+04 FES = 704100 μ_F = 0.724 μ_{CR} = 0.117
 Diversity = 1.8615e-02 k_t = [3, 4, 2, 4]
 Generation 7050: Best = 1.454058e+04 FES = 705100 μ_F = 0.718 μ_{CR} = 0.112
 Diversity = 1.6772e-02 k_t = [1, 1, 1, 1]
 Generation 7060: Best = 1.453979e+04 FES = 706100 μ_F = 0.718 μ_{CR} = 0.096
 Diversity = 1.5298e-02 k_t = [2, 4, 4, 2]
 Generation 7070: Best = 1.453952e+04 FES = 707100 μ_F = 0.704 μ_{CR} = 0.095
 Diversity = 1.4439e-02 k_t = [1, 1, 1, 1]
 Generation 7080: Best = 1.453872e+04 FES = 708100 μ_F = 0.717 μ_{CR} = 0.094
 Diversity = 1.3987e-02 k_t = [2, 4, 2, 4]
 Generation 7090: Best = 1.453858e+04 FES = 709100 μ_F = 0.725 μ_{CR} = 0.098
 Diversity = 1.3929e-02 k_t = [1, 1, 1, 1]
 Generation 7100: Best = 1.453832e+04 FES = 710100 μ_F = 0.720 μ_{CR} = 0.092
 Diversity = 1.4002e-02 k_t = [4, 2, 4, 4]
 Generation 7110: Best = 1.453792e+04 FES = 711100 μ_F = 0.724 μ_{CR} = 0.091
 Diversity = 1.2873e-02 k_t = [1, 1, 1, 1]
 Generation 7120: Best = 1.453749e+04 FES = 712100 μ_F = 0.720 μ_{CR} = 0.090
 Diversity = 1.2157e-02 k_t = [3, 2, 3, 3]
 Generation 7130: Best = 1.453733e+04 FES = 713100 μ_F = 0.719 μ_{CR} = 0.087
 Diversity = 1.1617e-02 k_t = [1, 1, 1, 1]
 Generation 7140: Best = 1.453717e+04 FES = 714100 μ_F = 0.701 μ_{CR} = 0.087
 Diversity = 1.1425e-02 k_t = [2, 4, 3, 2]
 Generation 7150: Best = 1.453701e+04 FES = 715100 μ_F = 0.673 μ_{CR} = 0.101
 Diversity = 1.1399e-02 k_t = [1, 1, 1, 1]
 Generation 7160: Best = 1.453675e+04 FES = 716100 μ_F = 0.684 μ_{CR} = 0.089
 Diversity = 1.0972e-02 k_t = [4, 3, 2, 2]
 Generation 7170: Best = 1.453671e+04 FES = 717100 μ_F = 0.673 μ_{CR} = 0.088
 Diversity = 1.0347e-02 k_t = [1, 1, 1, 1]
 Generation 7180: Best = 1.453649e+04 FES = 718100 μ_F = 0.668 μ_{CR} = 0.085
 Diversity = 9.2935e-03 k_t = [3, 3, 2, 2]
 Generation 7190: Best = 1.453634e+04 FES = 719100 μ_F = 0.673 μ_{CR} = 0.082
 Diversity = 8.8024e-03 k_t = [1, 1, 1, 1]
 Generation 7200: Best = 1.453614e+04 FES = 720100 μ_F = 0.681 μ_{CR} = 0.074
 Diversity = 7.5747e-03 k_t = [2, 3, 3, 4]
 Generation 7210: Best = 1.453606e+04 FES = 721100 μ_F = 0.690 μ_{CR} = 0.075
 Diversity = 7.0694e-03 k_t = [1, 1, 1, 1]
 Generation 7220: Best = 1.453595e+04 FES = 722100 μ_F = 0.700 μ_{CR} = 0.070
 Diversity = 6.7634e-03 k_t = [3, 2, 4, 2]
 Generation 7230: Best = 1.453574e+04 FES = 723100 μ_F = 0.671 μ_{CR} = 0.082
 Diversity = 6.4934e-03 k_t = [1, 1, 1, 1]
 Generation 7240: Best = 1.453567e+04 FES = 724100 μ_F = 0.679 μ_{CR} = 0.076
 Diversity = 6.6611e-03 k_t = [2, 4, 4, 2]

Generation 7250: Best = 1.453558e+04 FES = 725100 μ_F = 0.689 μ_{CR} = 0.071
 Diversity = 6.7732e-03 k_t = [1, 1, 1, 1]
 Generation 7260: Best = 1.453554e+04 FES = 726100 μ_F = 0.698 μ_{CR} = 0.071
 Diversity = 6.5294e-03 k_t = [3, 3, 4, 3]
 Generation 7270: Best = 1.453546e+04 FES = 727100 μ_F = 0.706 μ_{CR} = 0.070
 Diversity = 6.5771e-03 k_t = [1, 1, 1, 1]
 Generation 7280: Best = 1.453541e+04 FES = 728100 μ_F = 0.728 μ_{CR} = 0.070
 Diversity = 6.0431e-03 k_t = [4, 4, 2, 4]
 Generation 7290: Best = 1.453534e+04 FES = 729100 μ_F = 0.753 μ_{CR} = 0.066
 Diversity = 5.9202e-03 k_t = [1, 1, 1, 1]
 Generation 7300: Best = 1.453507e+04 FES = 730100 μ_F = 0.750 μ_{CR} = 0.076
 Diversity = 6.4218e-03 k_t = [4, 4, 2, 2]
 Generation 7310: Best = 1.453505e+04 FES = 731100 μ_F = 0.752 μ_{CR} = 0.075
 Diversity = 6.5327e-03 k_t = [1, 1, 1, 1]
 Generation 7320: Best = 1.453483e+04 FES = 732100 μ_F = 0.745 μ_{CR} = 0.081
 Diversity = 6.6342e-03 k_t = [4, 2, 3, 3]
 Generation 7330: Best = 1.453473e+04 FES = 733100 μ_F = 0.748 μ_{CR} = 0.075
 Diversity = 6.6543e-03 k_t = [1, 1, 1, 1]
 Generation 7340: Best = 1.453472e+04 FES = 734100 μ_F = 0.756 μ_{CR} = 0.081
 Diversity = 6.2407e-03 k_t = [4, 4, 2, 3]
 Generation 7350: Best = 1.453468e+04 FES = 735100 μ_F = 0.748 μ_{CR} = 0.073
 Diversity = 6.3012e-03 k_t = [1, 1, 1, 1]
 Generation 7360: Best = 1.453461e+04 FES = 736100 μ_F = 0.751 μ_{CR} = 0.073
 Diversity = 5.7413e-03 k_t = [4, 3, 2, 3]
 Generation 7370: Best = 1.453446e+04 FES = 737100 μ_F = 0.745 μ_{CR} = 0.070
 Diversity = 5.8336e-03 k_t = [1, 1, 1, 1]
 Generation 7380: Best = 1.453437e+04 FES = 738100 μ_F = 0.756 μ_{CR} = 0.075
 Diversity = 6.0327e-03 k_t = [2, 3, 2, 3]
 Generation 7390: Best = 1.453436e+04 FES = 739100 μ_F = 0.742 μ_{CR} = 0.068
 Diversity = 5.7404e-03 k_t = [1, 1, 1, 1]
 Generation 7400: Best = 1.453412e+04 FES = 740100 μ_F = 0.747 μ_{CR} = 0.078
 Diversity = 6.0267e-03 k_t = [2, 2, 4, 4]
 Generation 7410: Best = 1.453409e+04 FES = 741100 μ_F = 0.753 μ_{CR} = 0.083
 Diversity = 6.3157e-03 k_t = [1, 1, 1, 1]
 Generation 7420: Best = 1.453374e+04 FES = 742100 μ_F = 0.738 μ_{CR} = 0.085
 Diversity = 6.7457e-03 k_t = [2, 3, 4, 2]
 Generation 7430: Best = 1.453362e+04 FES = 743100 μ_F = 0.724 μ_{CR} = 0.081
 Diversity = 7.0827e-03 k_t = [1, 1, 1, 1]
 Generation 7440: Best = 1.453337e+04 FES = 744100 μ_F = 0.741 μ_{CR} = 0.085
 Diversity = 7.2101e-03 k_t = [4, 3, 4, 2]
 Generation 7450: Best = 1.453317e+04 FES = 745100 μ_F = 0.751 μ_{CR} = 0.095
 Diversity = 7.7664e-03 k_t = [1, 1, 1, 1]
 Generation 7460: Best = 1.453279e+04 FES = 746100 μ_F = 0.747 μ_{CR} = 0.108
 Diversity = 7.9520e-03 k_t = [4, 4, 3, 3]
 Generation 7470: Best = 1.453256e+04 FES = 747100 μ_F = 0.749 μ_{CR} = 0.111
 Diversity = 8.5297e-03 k_t = [1, 1, 1, 1]
 Generation 7480: Best = 1.453253e+04 FES = 748100 μ_F = 0.729 μ_{CR} = 0.110
 Diversity = 7.9863e-03 k_t = [2, 4, 2, 4]
 Generation 7490: Best = 1.453182e+04 FES = 749100 μ_F = 0.734 μ_{CR} = 0.100
 Diversity = 7.9241e-03 k_t = [1, 1, 1, 1]
 Generation 7500: Best = 1.453177e+04 FES = 750100 μ_F = 0.728 μ_{CR} = 0.112
 Diversity = 7.5652e-03 k_t = [3, 2, 4, 4]
 Generation 7510: Best = 1.453168e+04 FES = 751100 μ_F = 0.725 μ_{CR} = 0.107
 Diversity = 6.6124e-03 k_t = [1, 1, 1, 1]

Generation 7520: Best = 1.453167e+04 FES = 752100 μ_F = 0.720 μ_{CR} = 0.100
 Diversity = 5.9358e-03 k_t = [2, 4, 3, 3]
 Generation 7530: Best = 1.453153e+04 FES = 753100 μ_F = 0.731 μ_{CR} = 0.088
 Diversity = 5.7616e-03 k_t = [1, 1, 1, 1]
 Generation 7540: Best = 1.453140e+04 FES = 754100 μ_F = 0.720 μ_{CR} = 0.088
 Diversity = 5.7572e-03 k_t = [4, 4, 2, 2]
 Generation 7550: Best = 1.453137e+04 FES = 755100 μ_F = 0.707 μ_{CR} = 0.084
 Diversity = 5.8057e-03 k_t = [1, 1, 1, 1]
 Generation 7560: Best = 1.453130e+04 FES = 756100 μ_F = 0.716 μ_{CR} = 0.078
 Diversity = 5.6390e-03 k_t = [3, 3, 4, 2]
 Generation 7570: Best = 1.453130e+04 FES = 757100 μ_F = 0.724 μ_{CR} = 0.076
 Diversity = 5.5991e-03 k_t = [1, 1, 1, 1]
 Generation 7580: Best = 1.453124e+04 FES = 758100 μ_F = 0.719 μ_{CR} = 0.090
 Diversity = 5.3992e-03 k_t = [3, 4, 4, 2]
 Generation 7590: Best = 1.453120e+04 FES = 759100 μ_F = 0.718 μ_{CR} = 0.079
 Diversity = 5.1708e-03 k_t = [1, 1, 1, 1]
 Generation 7600: Best = 1.453115e+04 FES = 760100 μ_F = 0.717 μ_{CR} = 0.073
 Diversity = 4.9510e-03 k_t = [3, 3, 4, 2]
 Generation 7610: Best = 1.453106e+04 FES = 761100 μ_F = 0.700 μ_{CR} = 0.072
 Diversity = 4.9036e-03 k_t = [1, 1, 1, 1]
 Generation 7620: Best = 1.453104e+04 FES = 762100 μ_F = 0.703 μ_{CR} = 0.074
 Diversity = 4.9201e-03 k_t = [2, 3, 2, 2]
 Generation 7630: Best = 1.453100e+04 FES = 763100 μ_F = 0.714 μ_{CR} = 0.078
 Diversity = 5.0721e-03 k_t = [1, 1, 1, 1]
 Generation 7640: Best = 1.453089e+04 FES = 764100 μ_F = 0.702 μ_{CR} = 0.079
 Diversity = 5.0366e-03 k_t = [3, 4, 4, 2]
 Generation 7650: Best = 1.453088e+04 FES = 765100 μ_F = 0.711 μ_{CR} = 0.081
 Diversity = 4.9106e-03 k_t = [1, 1, 1, 1]
 Generation 7660: Best = 1.453078e+04 FES = 766100 μ_F = 0.703 μ_{CR} = 0.081
 Diversity = 4.5072e-03 k_t = [2, 4, 3, 4]
 Generation 7670: Best = 1.453076e+04 FES = 767100 μ_F = 0.707 μ_{CR} = 0.079
 Diversity = 4.2707e-03 k_t = [1, 1, 1, 1]
 Generation 7680: Best = 1.453074e+04 FES = 768100 μ_F = 0.700 μ_{CR} = 0.084
 Diversity = 4.2871e-03 k_t = [3, 3, 4, 3]
 Generation 7690: Best = 1.453066e+04 FES = 769100 μ_F = 0.714 μ_{CR} = 0.080
 Diversity = 4.4292e-03 k_t = [1, 1, 1, 1]
 Generation 7700: Best = 1.453060e+04 FES = 770100 μ_F = 0.705 μ_{CR} = 0.083
 Diversity = 4.1780e-03 k_t = [4, 3, 2, 3]
 Generation 7710: Best = 1.453060e+04 FES = 771100 μ_F = 0.706 μ_{CR} = 0.084
 Diversity = 4.0288e-03 k_t = [1, 1, 1, 1]
 Generation 7720: Best = 1.453056e+04 FES = 772100 μ_F = 0.716 μ_{CR} = 0.081
 Diversity = 4.1465e-03 k_t = [4, 3, 3, 2]
 Generation 7730: Best = 1.453050e+04 FES = 773100 μ_F = 0.722 μ_{CR} = 0.076
 Diversity = 4.1736e-03 k_t = [1, 1, 1, 1]
 Generation 7740: Best = 1.453045e+04 FES = 774100 μ_F = 0.731 μ_{CR} = 0.078
 Diversity = 4.3552e-03 k_t = [2, 2, 4, 3]
 Generation 7750: Best = 1.453038e+04 FES = 775100 μ_F = 0.735 μ_{CR} = 0.078
 Diversity = 4.5414e-03 k_t = [1, 1, 1, 1]
 Generation 7760: Best = 1.453031e+04 FES = 776100 μ_F = 0.721 μ_{CR} = 0.079
 Diversity = 4.4243e-03 k_t = [3, 3, 4, 4]
 Generation 7770: Best = 1.453023e+04 FES = 777100 μ_F = 0.711 μ_{CR} = 0.075
 Diversity = 4.1967e-03 k_t = [1, 1, 1, 1]
 Generation 7780: Best = 1.453021e+04 FES = 778100 μ_F = 0.704 μ_{CR} = 0.081
 Diversity = 4.4962e-03 k_t = [3, 3, 3, 2]

Generation 7790: Best = 1.453018e+04 FES = 779100 μ_F = 0.708 μ_{CR} = 0.079
 Diversity = 4.5333e-03 k_t = [1, 1, 1, 1]
 Generation 7800: Best = 1.453016e+04 FES = 780100 μ_F = 0.694 μ_{CR} = 0.092
 Diversity = 4.2141e-03 k_t = [2, 2, 4, 4]
 Generation 7810: Best = 1.453010e+04 FES = 781100 μ_F = 0.713 μ_{CR} = 0.081
 Diversity = 3.7940e-03 k_t = [1, 1, 1, 1]
 Generation 7820: Best = 1.453005e+04 FES = 782100 μ_F = 0.737 μ_{CR} = 0.075
 Diversity = 3.7915e-03 k_t = [4, 3, 3, 2]
 Generation 7830: Best = 1.453001e+04 FES = 783100 μ_F = 0.735 μ_{CR} = 0.072
 Diversity = 3.9694e-03 k_t = [1, 1, 1, 1]
 Generation 7840: Best = 1.452997e+04 FES = 784100 μ_F = 0.733 μ_{CR} = 0.070
 Diversity = 3.9459e-03 k_t = [4, 3, 3, 2]
 Generation 7850: Best = 1.452994e+04 FES = 785100 μ_F = 0.722 μ_{CR} = 0.077
 Diversity = 4.1649e-03 k_t = [1, 1, 1, 1]
 Generation 7860: Best = 1.452989e+04 FES = 786100 μ_F = 0.739 μ_{CR} = 0.073
 Diversity = 4.2931e-03 k_t = [3, 2, 4, 2]
 Generation 7870: Best = 1.452979e+04 FES = 787100 μ_F = 0.747 μ_{CR} = 0.081
 Diversity = 4.5642e-03 k_t = [1, 1, 1, 1]
 Generation 7880: Best = 1.452976e+04 FES = 788100 μ_F = 0.746 μ_{CR} = 0.080
 Diversity = 4.5128e-03 k_t = [2, 3, 3, 4]
 Generation 7890: Best = 1.452967e+04 FES = 789100 μ_F = 0.745 μ_{CR} = 0.078
 Diversity = 4.5653e-03 k_t = [1, 1, 1, 1]
 Generation 7900: Best = 1.452953e+04 FES = 790100 μ_F = 0.741 μ_{CR} = 0.079
 Diversity = 4.5855e-03 k_t = [3, 3, 2, 2]
 Generation 7910: Best = 1.452946e+04 FES = 791100 μ_F = 0.745 μ_{CR} = 0.086
 Diversity = 4.8582e-03 k_t = [1, 1, 1, 1]
 Generation 7920: Best = 1.452937e+04 FES = 792100 μ_F = 0.750 μ_{CR} = 0.093
 Diversity = 4.9485e-03 k_t = [3, 4, 4, 2]
 Generation 7930: Best = 1.452931e+04 FES = 793100 μ_F = 0.747 μ_{CR} = 0.091
 Diversity = 5.1927e-03 k_t = [1, 1, 1, 1]
 Generation 7940: Best = 1.452927e+04 FES = 794100 μ_F = 0.761 μ_{CR} = 0.088
 Diversity = 4.9414e-03 k_t = [3, 4, 3, 2]
 Generation 7950: Best = 1.452925e+04 FES = 795100 μ_F = 0.762 μ_{CR} = 0.079
 Diversity = 4.9552e-03 k_t = [1, 1, 1, 1]
 Generation 7960: Best = 1.452918e+04 FES = 796100 μ_F = 0.754 μ_{CR} = 0.077
 Diversity = 4.7700e-03 k_t = [4, 4, 2, 2]
 Generation 7970: Best = 1.452909e+04 FES = 797100 μ_F = 0.745 μ_{CR} = 0.083
 Diversity = 4.7105e-03 k_t = [1, 1, 1, 1]
 Generation 7980: Best = 1.452907e+04 FES = 798100 μ_F = 0.742 μ_{CR} = 0.087
 Diversity = 4.4785e-03 k_t = [3, 3, 2, 4]
 Generation 7990: Best = 1.452898e+04 FES = 799100 μ_F = 0.733 μ_{CR} = 0.091
 Diversity = 4.3620e-03 k_t = [1, 1, 1, 1]
 Generation 8000: Best = 1.452891e+04 FES = 800100 μ_F = 0.746 μ_{CR} = 0.094
 Diversity = 4.4294e-03 k_t = [3, 4, 3, 4]
 Generation 8010: Best = 1.452884e+04 FES = 801100 μ_F = 0.757 μ_{CR} = 0.094
 Diversity = 4.6296e-03 k_t = [1, 1, 1, 1]
 Generation 8020: Best = 1.452878e+04 FES = 802100 μ_F = 0.736 μ_{CR} = 0.088
 Diversity = 4.5723e-03 k_t = [3, 4, 2, 2]
 Generation 8030: Best = 1.452873e+04 FES = 803100 μ_F = 0.704 μ_{CR} = 0.093
 Diversity = 4.4623e-03 k_t = [1, 1, 1, 1]
 Generation 8040: Best = 1.452871e+04 FES = 804100 μ_F = 0.716 μ_{CR} = 0.086
 Diversity = 3.9039e-03 k_t = [3, 4, 4, 3]
 Generation 8050: Best = 1.452868e+04 FES = 805100 μ_F = 0.724 μ_{CR} = 0.082
 Diversity = 3.7617e-03 k_t = [1, 1, 1, 1]

Generation 8060: Best = 1.452862e+04 FES = 806100 μ_F = 0.713 μ_{CR} = 0.084
 Diversity = 3.6559e-03 k_t = [3, 3, 2, 3]
 Generation 8070: Best = 1.452859e+04 FES = 807100 μ_F = 0.716 μ_{CR} = 0.086
 Diversity = 3.7628e-03 k_t = [1, 1, 1, 1]
 Generation 8080: Best = 1.452852e+04 FES = 808100 μ_F = 0.714 μ_{CR} = 0.088
 Diversity = 3.7857e-03 k_t = [2, 2, 3, 3]
 Generation 8090: Best = 1.452850e+04 FES = 809100 μ_F = 0.711 μ_{CR} = 0.083
 Diversity = 3.6734e-03 k_t = [1, 1, 1, 1]
 Generation 8100: Best = 1.452849e+04 FES = 810100 μ_F = 0.733 μ_{CR} = 0.080
 Diversity = 3.3328e-03 k_t = [3, 3, 3, 3]
 Generation 8110: Best = 1.452846e+04 FES = 811100 μ_F = 0.735 μ_{CR} = 0.081
 Diversity = 3.1965e-03 k_t = [1, 1, 1, 1]
 Generation 8120: Best = 1.452843e+04 FES = 812100 μ_F = 0.716 μ_{CR} = 0.077
 Diversity = 3.2484e-03 k_t = [4, 4, 4, 2]
 Generation 8130: Best = 1.452840e+04 FES = 813100 μ_F = 0.715 μ_{CR} = 0.082
 Diversity = 3.2036e-03 k_t = [1, 1, 1, 1]
 Generation 8140: Best = 1.452838e+04 FES = 814100 μ_F = 0.722 μ_{CR} = 0.082
 Diversity = 3.1626e-03 k_t = [2, 2, 3, 2]
 Generation 8150: Best = 1.452835e+04 FES = 815100 μ_F = 0.727 μ_{CR} = 0.079
 Diversity = 3.3021e-03 k_t = [1, 1, 1, 1]
 Generation 8160: Best = 1.452830e+04 FES = 816100 μ_F = 0.725 μ_{CR} = 0.080
 Diversity = 3.4047e-03 k_t = [3, 4, 4, 2]
 Generation 8170: Best = 1.452829e+04 FES = 817100 μ_F = 0.716 μ_{CR} = 0.075
 Diversity = 3.3858e-03 k_t = [1, 1, 1, 1]
 Generation 8180: Best = 1.452827e+04 FES = 818100 μ_F = 0.706 μ_{CR} = 0.084
 Diversity = 3.0816e-03 k_t = [2, 3, 3, 3]
 Generation 8190: Best = 1.452824e+04 FES = 819100 μ_F = 0.717 μ_{CR} = 0.087
 Diversity = 3.0536e-03 k_t = [1, 1, 1, 1]
 Generation 8200: Best = 1.452817e+04 FES = 820100 μ_F = 0.711 μ_{CR} = 0.097
 Diversity = 3.2481e-03 k_t = [3, 2, 2, 2]
 Generation 8210: Best = 1.452811e+04 FES = 821100 μ_F = 0.696 μ_{CR} = 0.094
 Diversity = 3.2938e-03 k_t = [1, 1, 1, 1]
 Generation 8220: Best = 1.452806e+04 FES = 822100 μ_F = 0.708 μ_{CR} = 0.098
 Diversity = 3.1939e-03 k_t = [4, 4, 2, 2]
 Generation 8230: Best = 1.452804e+04 FES = 823100 μ_F = 0.725 μ_{CR} = 0.098
 Diversity = 3.4013e-03 k_t = [1, 1, 1, 1]
 Generation 8240: Best = 1.452800e+04 FES = 824100 μ_F = 0.713 μ_{CR} = 0.108
 Diversity = 3.0874e-03 k_t = [2, 2, 4, 4]
 Generation 8250: Best = 1.452797e+04 FES = 825100 μ_F = 0.723 μ_{CR} = 0.107
 Diversity = 3.0030e-03 k_t = [1, 1, 1, 1]
 Generation 8260: Best = 1.452789e+04 FES = 826100 μ_F = 0.735 μ_{CR} = 0.100
 Diversity = 3.0214e-03 k_t = [3, 3, 4, 3]
 Generation 8270: Best = 1.452775e+04 FES = 827100 μ_F = 0.743 μ_{CR} = 0.099
 Diversity = 3.2695e-03 k_t = [1, 1, 1, 1]
 Generation 8280: Best = 1.452772e+04 FES = 828100 μ_F = 0.761 μ_{CR} = 0.102
 Diversity = 3.6949e-03 k_t = [2, 3, 4, 3]
 Generation 8290: Best = 1.452768e+04 FES = 829100 μ_F = 0.771 μ_{CR} = 0.105
 Diversity = 3.8197e-03 k_t = [1, 1, 1, 1]
 Generation 8300: Best = 1.452763e+04 FES = 830100 μ_F = 0.765 μ_{CR} = 0.110
 Diversity = 3.6521e-03 k_t = [2, 4, 3, 4]
 Generation 8310: Best = 1.452745e+04 FES = 831100 μ_F = 0.771 μ_{CR} = 0.120
 Diversity = 3.8349e-03 k_t = [1, 1, 1, 1]
 Generation 8320: Best = 1.452737e+04 FES = 832100 μ_F = 0.767 μ_{CR} = 0.119
 Diversity = 4.0843e-03 k_t = [4, 3, 4, 3]

Generation 8330: Best = 1.452731e+04 FES = 833100 μ_F = 0.768 μ_{CR} = 0.129
 Diversity = 4.6529e-03 k_t = [1, 1, 1, 1]
 Generation 8340: Best = 1.452719e+04 FES = 834100 μ_F = 0.778 μ_{CR} = 0.123
 Diversity = 4.8414e-03 k_t = [3, 2, 4, 3]
 Generation 8350: Best = 1.452697e+04 FES = 835100 μ_F = 0.786 μ_{CR} = 0.131
 Diversity = 5.0264e-03 k_t = [1, 1, 1, 1]
 Generation 8360: Best = 1.452669e+04 FES = 836100 μ_F = 0.778 μ_{CR} = 0.136
 Diversity = 5.4214e-03 k_t = [4, 4, 2, 4]
 Generation 8370: Best = 1.452664e+04 FES = 837100 μ_F = 0.777 μ_{CR} = 0.131
 Diversity = 5.6394e-03 k_t = [1, 1, 1, 1]
 Generation 8380: Best = 1.452653e+04 FES = 838100 μ_F = 0.757 μ_{CR} = 0.125
 Diversity = 5.0894e-03 k_t = [3, 4, 2, 4]
 Generation 8390: Best = 1.452645e+04 FES = 839100 μ_F = 0.743 μ_{CR} = 0.115
 Diversity = 4.8427e-03 k_t = [1, 1, 1, 1]
 Generation 8400: Best = 1.452641e+04 FES = 840100 μ_F = 0.744 μ_{CR} = 0.114
 Diversity = 4.6396e-03 k_t = [4, 2, 4, 3]
 Generation 8410: Best = 1.452632e+04 FES = 841100 μ_F = 0.762 μ_{CR} = 0.126
 Diversity = 4.8179e-03 k_t = [1, 1, 1, 1]
 Generation 8420: Best = 1.452617e+04 FES = 842100 μ_F = 0.751 μ_{CR} = 0.125
 Diversity = 5.1116e-03 k_t = [3, 2, 2, 3]
 Generation 8430: Best = 1.452610e+04 FES = 843100 μ_F = 0.730 μ_{CR} = 0.123
 Diversity = 5.2442e-03 k_t = [1, 1, 1, 1]
 Generation 8440: Best = 1.452603e+04 FES = 844100 μ_F = 0.726 μ_{CR} = 0.115
 Diversity = 5.3251e-03 k_t = [4, 4, 4, 2]
 Generation 8450: Best = 1.452595e+04 FES = 845100 μ_F = 0.710 μ_{CR} = 0.115
 Diversity = 5.1258e-03 k_t = [1, 1, 1, 1]
 Generation 8460: Best = 1.452586e+04 FES = 846100 μ_F = 0.703 μ_{CR} = 0.121
 Diversity = 4.8536e-03 k_t = [2, 4, 3, 3]
 Generation 8470: Best = 1.452574e+04 FES = 847100 μ_F = 0.710 μ_{CR} = 0.115
 Diversity = 4.8812e-03 k_t = [1, 1, 1, 1]
 Generation 8480: Best = 1.452568e+04 FES = 848100 μ_F = 0.692 μ_{CR} = 0.114
 Diversity = 4.7394e-03 k_t = [4, 4, 4, 3]
 Generation 8490: Best = 1.452557e+04 FES = 849100 μ_F = 0.711 μ_{CR} = 0.114
 Diversity = 4.7837e-03 k_t = [1, 1, 1, 1]
 Generation 8500: Best = 1.452549e+04 FES = 850100 μ_F = 0.708 μ_{CR} = 0.116
 Diversity = 4.5300e-03 k_t = [4, 2, 4, 2]
 Generation 8510: Best = 1.452544e+04 FES = 851100 μ_F = 0.716 μ_{CR} = 0.115
 Diversity = 4.7609e-03 k_t = [1, 1, 1, 1]
 Generation 8520: Best = 1.452538e+04 FES = 852100 μ_F = 0.721 μ_{CR} = 0.120
 Diversity = 4.6239e-03 k_t = [4, 4, 2, 3]
 Generation 8530: Best = 1.452535e+04 FES = 853100 μ_F = 0.717 μ_{CR} = 0.115
 Diversity = 4.6153e-03 k_t = [1, 1, 1, 1]
 Generation 8540: Best = 1.452532e+04 FES = 854100 μ_F = 0.716 μ_{CR} = 0.112
 Diversity = 4.3926e-03 k_t = [4, 4, 3, 3]
 Generation 8550: Best = 1.452523e+04 FES = 855100 μ_F = 0.721 μ_{CR} = 0.111
 Diversity = 4.2931e-03 k_t = [1, 1, 1, 1]
 Generation 8560: Best = 1.452518e+04 FES = 856100 μ_F = 0.738 μ_{CR} = 0.108
 Diversity = 4.3962e-03 k_t = [4, 4, 2, 3]
 Generation 8570: Best = 1.452509e+04 FES = 857100 μ_F = 0.750 μ_{CR} = 0.101
 Diversity = 4.4859e-03 k_t = [1, 1, 1, 1]
 Generation 8580: Best = 1.452507e+04 FES = 858100 μ_F = 0.756 μ_{CR} = 0.105
 Diversity = 4.2708e-03 k_t = [2, 3, 4, 4]
 Generation 8590: Best = 1.452499e+04 FES = 859100 μ_F = 0.749 μ_{CR} = 0.106
 Diversity = 4.0665e-03 k_t = [1, 1, 1, 1]

Generation 8600: Best = 1.452493e+04 FES = 860100 μ_F = 0.751 μ_{CR} = 0.107
 Diversity = 4.1712e-03 k_t = [3, 2, 4, 2]
 Generation 8610: Best = 1.452479e+04 FES = 861100 μ_F = 0.750 μ_{CR} = 0.113
 Diversity = 4.2587e-03 k_t = [1, 1, 1, 1]
 Generation 8620: Best = 1.452473e+04 FES = 862100 μ_F = 0.769 μ_{CR} = 0.108
 Diversity = 4.4530e-03 k_t = [3, 2, 4, 4]
 Generation 8630: Best = 1.452470e+04 FES = 863100 μ_F = 0.753 μ_{CR} = 0.107
 Diversity = 4.3806e-03 k_t = [1, 1, 1, 1]
 Generation 8640: Best = 1.452461e+04 FES = 864100 μ_F = 0.757 μ_{CR} = 0.111
 Diversity = 4.5442e-03 k_t = [2, 2, 3, 2]
 Generation 8650: Best = 1.452457e+04 FES = 865100 μ_F = 0.755 μ_{CR} = 0.106
 Diversity = 4.8722e-03 k_t = [1, 1, 1, 1]
 Generation 8660: Best = 1.452439e+04 FES = 866100 μ_F = 0.757 μ_{CR} = 0.109
 Diversity = 5.1884e-03 k_t = [4, 3, 3, 2]
 Generation 8670: Best = 1.452432e+04 FES = 867100 μ_F = 0.728 μ_{CR} = 0.109
 Diversity = 5.2020e-03 k_t = [1, 1, 1, 1]
 Generation 8680: Best = 1.452424e+04 FES = 868100 μ_F = 0.730 μ_{CR} = 0.118
 Diversity = 5.1605e-03 k_t = [3, 3, 3, 3]
 Generation 8690: Best = 1.452415e+04 FES = 869100 μ_F = 0.738 μ_{CR} = 0.131
 Diversity = 5.2426e-03 k_t = [1, 1, 1, 1]
 Generation 8700: Best = 1.452404e+04 FES = 870100 μ_F = 0.727 μ_{CR} = 0.127
 Diversity = 5.2158e-03 k_t = [3, 4, 2, 2]
 Generation 8710: Best = 1.452399e+04 FES = 871100 μ_F = 0.734 μ_{CR} = 0.125
 Diversity = 5.0478e-03 k_t = [1, 1, 1, 1]
 Generation 8720: Best = 1.452393e+04 FES = 872100 μ_F = 0.745 μ_{CR} = 0.129
 Diversity = 5.0377e-03 k_t = [3, 3, 2, 2]
 Generation 8730: Best = 1.452388e+04 FES = 873100 μ_F = 0.730 μ_{CR} = 0.134
 Diversity = 4.9087e-03 k_t = [1, 1, 1, 1]
 Generation 8740: Best = 1.452376e+04 FES = 874100 μ_F = 0.723 μ_{CR} = 0.133
 Diversity = 4.8896e-03 k_t = [2, 2, 3, 2]
 Generation 8750: Best = 1.452367e+04 FES = 875100 μ_F = 0.729 μ_{CR} = 0.130
 Diversity = 4.8845e-03 k_t = [1, 1, 1, 1]
 Generation 8760: Best = 1.452361e+04 FES = 876100 μ_F = 0.736 μ_{CR} = 0.126
 Diversity = 4.5172e-03 k_t = [4, 4, 2, 4]
 Generation 8770: Best = 1.452357e+04 FES = 877100 μ_F = 0.722 μ_{CR} = 0.119
 Diversity = 4.1216e-03 k_t = [1, 1, 1, 1]
 Generation 8780: Best = 1.452353e+04 FES = 878100 μ_F = 0.715 μ_{CR} = 0.125
 Diversity = 3.9946e-03 k_t = [3, 2, 3, 3]
 Generation 8790: Best = 1.452350e+04 FES = 879100 μ_F = 0.714 μ_{CR} = 0.115
 Diversity = 3.8601e-03 k_t = [1, 1, 1, 1]
 Generation 8800: Best = 1.452343e+04 FES = 880100 μ_F = 0.702 μ_{CR} = 0.123
 Diversity = 3.7104e-03 k_t = [4, 3, 2, 3]
 Generation 8810: Best = 1.452339e+04 FES = 881100 μ_F = 0.703 μ_{CR} = 0.118
 Diversity = 3.6885e-03 k_t = [1, 1, 1, 1]
 Generation 8820: Best = 1.452337e+04 FES = 882100 μ_F = 0.696 μ_{CR} = 0.124
 Diversity = 3.3724e-03 k_t = [4, 4, 3, 4]
 Generation 8830: Best = 1.452334e+04 FES = 883100 μ_F = 0.692 μ_{CR} = 0.122
 Diversity = 2.9936e-03 k_t = [1, 1, 1, 1]
 Generation 8840: Best = 1.452332e+04 FES = 884100 μ_F = 0.700 μ_{CR} = 0.117
 Diversity = 2.9210e-03 k_t = [4, 4, 2, 2]
 Generation 8850: Best = 1.452329e+04 FES = 885100 μ_F = 0.710 μ_{CR} = 0.117
 Diversity = 3.0541e-03 k_t = [1, 1, 1, 1]
 Generation 8860: Best = 1.452327e+04 FES = 886100 μ_F = 0.713 μ_{CR} = 0.114
 Diversity = 2.8492e-03 k_t = [4, 2, 4, 4]

Generation 8870: Best = 1.452323e+04 FES = 887100 μ_F = 0.724 μ_{CR} = 0.114
 Diversity = 2.6566e-03 k_t = [1, 1, 1, 1]
 Generation 8880: Best = 1.452323e+04 FES = 888100 μ_F = 0.728 μ_{CR} = 0.113
 Diversity = 2.4646e-03 k_t = [4, 2, 2, 4]
 Generation 8890: Best = 1.452319e+04 FES = 889100 μ_F = 0.728 μ_{CR} = 0.106
 Diversity = 2.3685e-03 k_t = [1, 1, 1, 1]
 Generation 8900: Best = 1.452316e+04 FES = 890100 μ_F = 0.740 μ_{CR} = 0.114
 Diversity = 2.2779e-03 k_t = [3, 3, 2, 4]
 Generation 8910: Best = 1.452312e+04 FES = 891100 μ_F = 0.751 μ_{CR} = 0.117
 Diversity = 2.3255e-03 k_t = [1, 1, 1, 1]
 Generation 8920: Best = 1.452311e+04 FES = 892100 μ_F = 0.750 μ_{CR} = 0.120
 Diversity = 2.2222e-03 k_t = [2, 3, 4, 4]
 Generation 8930: Best = 1.452311e+04 FES = 893100 μ_F = 0.741 μ_{CR} = 0.120
 Diversity = 2.1346e-03 k_t = [1, 1, 1, 1]
 Generation 8940: Best = 1.452308e+04 FES = 894100 μ_F = 0.736 μ_{CR} = 0.120
 Diversity = 1.9894e-03 k_t = [2, 2, 4, 4]
 Generation 8950: Best = 1.452307e+04 FES = 895100 μ_F = 0.737 μ_{CR} = 0.111
 Diversity = 2.1078e-03 k_t = [1, 1, 1, 1]
 Generation 8960: Best = 1.452306e+04 FES = 896100 μ_F = 0.759 μ_{CR} = 0.099
 Diversity = 2.1825e-03 k_t = [2, 4, 4, 2]
 Generation 8970: Best = 1.452303e+04 FES = 897100 μ_F = 0.762 μ_{CR} = 0.100
 Diversity = 2.2134e-03 k_t = [1, 1, 1, 1]
 Generation 8980: Best = 1.452301e+04 FES = 898100 μ_F = 0.765 μ_{CR} = 0.106
 Diversity = 2.2207e-03 k_t = [4, 2, 3, 3]
 Generation 8990: Best = 1.452300e+04 FES = 899100 μ_F = 0.753 μ_{CR} = 0.096
 Diversity = 2.3692e-03 k_t = [1, 1, 1, 1]
 Generation 9000: Best = 1.452297e+04 FES = 900100 μ_F = 0.746 μ_{CR} = 0.100
 Diversity = 2.3372e-03 k_t = [3, 4, 2, 3]
 Generation 9010: Best = 1.452296e+04 FES = 901100 μ_F = 0.730 μ_{CR} = 0.100
 Diversity = 2.3052e-03 k_t = [1, 1, 1, 1]
 Generation 9020: Best = 1.452294e+04 FES = 902100 μ_F = 0.726 μ_{CR} = 0.111
 Diversity = 2.1857e-03 k_t = [4, 4, 4, 2]
 Generation 9030: Best = 1.452292e+04 FES = 903100 μ_F = 0.714 μ_{CR} = 0.119
 Diversity = 2.2968e-03 k_t = [1, 1, 1, 1]
 Generation 9040: Best = 1.452290e+04 FES = 904100 μ_F = 0.718 μ_{CR} = 0.123
 Diversity = 2.1572e-03 k_t = [3, 4, 2, 3]
 Generation 9050: Best = 1.452286e+04 FES = 905100 μ_F = 0.709 μ_{CR} = 0.120
 Diversity = 2.1725e-03 k_t = [1, 1, 1, 1]
 Generation 9060: Best = 1.452284e+04 FES = 906100 μ_F = 0.706 μ_{CR} = 0.123
 Diversity = 2.1680e-03 k_t = [2, 2, 4, 3]
 Generation 9070: Best = 1.452282e+04 FES = 907100 μ_F = 0.718 μ_{CR} = 0.122
 Diversity = 2.2510e-03 k_t = [1, 1, 1, 1]
 Generation 9080: Best = 1.452275e+04 FES = 908100 μ_F = 0.731 μ_{CR} = 0.119
 Diversity = 2.4692e-03 k_t = [3, 3, 4, 4]
 Generation 9090: Best = 1.452263e+04 FES = 909100 μ_F = 0.741 μ_{CR} = 0.128
 Diversity = 2.8908e-03 k_t = [1, 1, 1, 1]
 Generation 9100: Best = 1.452256e+04 FES = 910100 μ_F = 0.761 μ_{CR} = 0.134
 Diversity = 3.4515e-03 k_t = [3, 3, 2, 2]
 Generation 9110: Best = 1.452250e+04 FES = 911100 μ_F = 0.750 μ_{CR} = 0.140
 Diversity = 3.5124e-03 k_t = [1, 1, 1, 1]
 Generation 9120: Best = 1.452244e+04 FES = 912100 μ_F = 0.755 μ_{CR} = 0.153
 Diversity = 3.5115e-03 k_t = [3, 3, 4, 2]
 Generation 9130: Best = 1.452235e+04 FES = 913100 μ_F = 0.740 μ_{CR} = 0.155
 Diversity = 3.5986e-03 k_t = [1, 1, 1, 1]

Generation 9140: Best = 1.452235e+04 FES = 914100 μ_F = 0.722 μ_{CR} = 0.154
 Diversity = 3.1985e-03 k_t = [2, 2, 2, 3]
 Generation 9150: Best = 1.452233e+04 FES = 915100 μ_F = 0.732 μ_{CR} = 0.135
 Diversity = 3.1014e-03 k_t = [1, 1, 1, 1]
 Generation 9160: Best = 1.452230e+04 FES = 916100 μ_F = 0.715 μ_{CR} = 0.138
 Diversity = 2.8776e-03 k_t = [3, 2, 4, 2]
 Generation 9170: Best = 1.452228e+04 FES = 917100 μ_F = 0.721 μ_{CR} = 0.131
 Diversity = 2.7852e-03 k_t = [1, 1, 1, 1]
 Generation 9180: Best = 1.452226e+04 FES = 918100 μ_F = 0.713 μ_{CR} = 0.123
 Diversity = 2.6203e-03 k_t = [2, 4, 4, 2]
 Generation 9190: Best = 1.452225e+04 FES = 919100 μ_F = 0.696 μ_{CR} = 0.109
 Diversity = 2.5958e-03 k_t = [1, 1, 1, 1]
 Generation 9200: Best = 1.452222e+04 FES = 920100 μ_F = 0.702 μ_{CR} = 0.114
 Diversity = 2.4327e-03 k_t = [2, 3, 3, 3]
 Generation 9210: Best = 1.452220e+04 FES = 921100 μ_F = 0.693 μ_{CR} = 0.124
 Diversity = 2.2866e-03 k_t = [1, 1, 1, 1]
 Generation 9220: Best = 1.452218e+04 FES = 922100 μ_F = 0.697 μ_{CR} = 0.127
 Diversity = 2.2093e-03 k_t = [3, 3, 3, 2]
 Generation 9230: Best = 1.452216e+04 FES = 923100 μ_F = 0.708 μ_{CR} = 0.131
 Diversity = 2.1637e-03 k_t = [1, 1, 1, 1]
 Generation 9240: Best = 1.452213e+04 FES = 924100 μ_F = 0.708 μ_{CR} = 0.126
 Diversity = 2.1121e-03 k_t = [3, 4, 4, 4]
 Generation 9250: Best = 1.452211e+04 FES = 925100 μ_F = 0.719 μ_{CR} = 0.127
 Diversity = 2.1245e-03 k_t = [1, 1, 1, 1]
 Generation 9260: Best = 1.452208e+04 FES = 926100 μ_F = 0.722 μ_{CR} = 0.125
 Diversity = 2.2205e-03 k_t = [4, 4, 4, 4]
 Generation 9270: Best = 1.452204e+04 FES = 927100 μ_F = 0.733 μ_{CR} = 0.134
 Diversity = 2.2861e-03 k_t = [1, 1, 1, 1]
 Generation 9280: Best = 1.452203e+04 FES = 928100 μ_F = 0.716 μ_{CR} = 0.137
 Diversity = 2.4459e-03 k_t = [4, 2, 4, 2]
 Generation 9290: Best = 1.452199e+04 FES = 929100 μ_F = 0.729 μ_{CR} = 0.137
 Diversity = 2.5573e-03 k_t = [1, 1, 1, 1]
 Generation 9300: Best = 1.452197e+04 FES = 930100 μ_F = 0.733 μ_{CR} = 0.136
 Diversity = 2.5140e-03 k_t = [3, 4, 2, 3]
 Generation 9310: Best = 1.452195e+04 FES = 931100 μ_F = 0.738 μ_{CR} = 0.130
 Diversity = 2.4279e-03 k_t = [1, 1, 1, 1]
 Generation 9320: Best = 1.452194e+04 FES = 932100 μ_F = 0.748 μ_{CR} = 0.135
 Diversity = 2.3073e-03 k_t = [2, 2, 3, 4]
 Generation 9330: Best = 1.452192e+04 FES = 933100 μ_F = 0.753 μ_{CR} = 0.125
 Diversity = 2.2393e-03 k_t = [1, 1, 1, 1]
 Generation 9340: Best = 1.452186e+04 FES = 934100 μ_F = 0.750 μ_{CR} = 0.131
 Diversity = 2.4018e-03 k_t = [4, 2, 3, 3]
 Generation 9350: Best = 1.452186e+04 FES = 935100 μ_F = 0.760 μ_{CR} = 0.134
 Diversity = 2.4474e-03 k_t = [1, 1, 1, 1]
 Generation 9360: Best = 1.452185e+04 FES = 936100 μ_F = 0.739 μ_{CR} = 0.127
 Diversity = 2.2924e-03 k_t = [3, 2, 4, 2]
 Generation 9370: Best = 1.452184e+04 FES = 937100 μ_F = 0.739 μ_{CR} = 0.126
 Diversity = 2.2068e-03 k_t = [1, 1, 1, 1]
 Generation 9380: Best = 1.452183e+04 FES = 938100 μ_F = 0.736 μ_{CR} = 0.123
 Diversity = 2.0365e-03 k_t = [4, 2, 2, 4]
 Generation 9390: Best = 1.452182e+04 FES = 939100 μ_F = 0.733 μ_{CR} = 0.114
 Diversity = 2.0316e-03 k_t = [1, 1, 1, 1]
 Generation 9400: Best = 1.452179e+04 FES = 940100 μ_F = 0.722 μ_{CR} = 0.114
 Diversity = 1.9731e-03 k_t = [2, 4, 2, 4]

Generation 9410: Best = 1.452177e+04 FES = 941100 μ_F = 0.731 μ_{CR} = 0.129
 Diversity = 1.9240e-03 k_t = [1, 1, 1, 1]
 Generation 9420: Best = 1.452176e+04 FES = 942100 μ_F = 0.731 μ_{CR} = 0.128
 Diversity = 1.9585e-03 k_t = [2, 3, 2, 3]
 Generation 9430: Best = 1.452174e+04 FES = 943100 μ_F = 0.735 μ_{CR} = 0.113
 Diversity = 2.0512e-03 k_t = [1, 1, 1, 1]
 Generation 9440: Best = 1.452173e+04 FES = 944100 μ_F = 0.729 μ_{CR} = 0.118
 Diversity = 1.9182e-03 k_t = [3, 3, 4, 3]
 Generation 9450: Best = 1.452173e+04 FES = 945100 μ_F = 0.734 μ_{CR} = 0.112
 Diversity = 1.8346e-03 k_t = [1, 1, 1, 1]
 Generation 9460: Best = 1.452171e+04 FES = 946100 μ_F = 0.736 μ_{CR} = 0.107
 Diversity = 1.7510e-03 k_t = [4, 3, 2, 2]
 Generation 9470: Best = 1.452170e+04 FES = 947100 μ_F = 0.727 μ_{CR} = 0.101
 Diversity = 1.7844e-03 k_t = [1, 1, 1, 1]
 Generation 9480: Best = 1.452169e+04 FES = 948100 μ_F = 0.729 μ_{CR} = 0.098
 Diversity = 1.7037e-03 k_t = [2, 4, 3, 3]
 Generation 9490: Best = 1.452169e+04 FES = 949100 μ_F = 0.745 μ_{CR} = 0.098
 Diversity = 1.6371e-03 k_t = [1, 1, 1, 1]
 Generation 9500: Best = 1.452168e+04 FES = 950100 μ_F = 0.743 μ_{CR} = 0.101
 Diversity = 1.5081e-03 k_t = [2, 4, 3, 4]
 Generation 9510: Best = 1.452166e+04 FES = 951100 μ_F = 0.740 μ_{CR} = 0.101
 Diversity = 1.3938e-03 k_t = [1, 1, 1, 1]
 Generation 9520: Best = 1.452166e+04 FES = 952100 μ_F = 0.732 μ_{CR} = 0.099
 Diversity = 1.3893e-03 k_t = [4, 3, 2, 4]
 Generation 9530: Best = 1.452164e+04 FES = 953100 μ_F = 0.725 μ_{CR} = 0.105
 Diversity = 1.3824e-03 k_t = [1, 1, 1, 1]
 Generation 9540: Best = 1.452163e+04 FES = 954100 μ_F = 0.724 μ_{CR} = 0.107
 Diversity = 1.3416e-03 k_t = [4, 2, 2, 4]
 Generation 9550: Best = 1.452163e+04 FES = 955100 μ_F = 0.749 μ_{CR} = 0.107
 Diversity = 1.3758e-03 k_t = [1, 1, 1, 1]
 Generation 9560: Best = 1.452163e+04 FES = 956100 μ_F = 0.738 μ_{CR} = 0.110
 Diversity = 1.3878e-03 k_t = [3, 3, 2, 2]
 Generation 9570: Best = 1.452162e+04 FES = 957100 μ_F = 0.734 μ_{CR} = 0.103
 Diversity = 1.3867e-03 k_t = [1, 1, 1, 1]
 Generation 9580: Best = 1.452162e+04 FES = 958100 μ_F = 0.737 μ_{CR} = 0.104
 Diversity = 1.2638e-03 k_t = [4, 2, 3, 3]
 Generation 9590: Best = 1.452161e+04 FES = 959100 μ_F = 0.742 μ_{CR} = 0.099
 Diversity = 1.2935e-03 k_t = [1, 1, 1, 1]
 Generation 9600: Best = 1.452160e+04 FES = 960100 μ_F = 0.744 μ_{CR} = 0.102
 Diversity = 1.2771e-03 k_t = [4, 4, 3, 4]
 Generation 9610: Best = 1.452160e+04 FES = 961100 μ_F = 0.750 μ_{CR} = 0.102
 Diversity = 1.2788e-03 k_t = [1, 1, 1, 1]
 Generation 9620: Best = 1.452159e+04 FES = 962100 μ_F = 0.736 μ_{CR} = 0.106
 Diversity = 1.2545e-03 k_t = [3, 3, 4, 3]
 Generation 9630: Best = 1.452158e+04 FES = 963100 μ_F = 0.730 μ_{CR} = 0.112
 Diversity = 1.2793e-03 k_t = [1, 1, 1, 1]
 Generation 9640: Best = 1.452156e+04 FES = 964100 μ_F = 0.736 μ_{CR} = 0.106
 Diversity = 1.2634e-03 k_t = [4, 2, 3, 4]
 Generation 9650: Best = 1.452156e+04 FES = 965100 μ_F = 0.759 μ_{CR} = 0.113
 Diversity = 1.4259e-03 k_t = [1, 1, 1, 1]
 Generation 9660: Best = 1.452154e+04 FES = 966100 μ_F = 0.767 μ_{CR} = 0.120
 Diversity = 1.3756e-03 k_t = [3, 3, 3, 3]
 Generation 9670: Best = 1.452154e+04 FES = 967100 μ_F = 0.776 μ_{CR} = 0.124
 Diversity = 1.4727e-03 k_t = [1, 1, 1, 1]

Generation 9680: Best = 1.452152e+04 FES = 968100 μ_F = 0.760 μ_{CR} = 0.131
 Diversity = 1.4888e-03 k_t = [2, 3, 4, 4]
 Generation 9690: Best = 1.452148e+04 FES = 969100 μ_F = 0.770 μ_{CR} = 0.138
 Diversity = 1.6145e-03 k_t = [1, 1, 1, 1]
 Generation 9700: Best = 1.452144e+04 FES = 970100 μ_F = 0.778 μ_{CR} = 0.148
 Diversity = 1.6437e-03 k_t = [3, 3, 2, 4]
 Generation 9710: Best = 1.452141e+04 FES = 971100 μ_F = 0.786 μ_{CR} = 0.163
 Diversity = 1.8587e-03 k_t = [1, 1, 1, 1]
 Generation 9720: Best = 1.452138e+04 FES = 972100 μ_F = 0.766 μ_{CR} = 0.169
 Diversity = 1.8490e-03 k_t = [4, 2, 4, 3]
 Generation 9730: Best = 1.452136e+04 FES = 973100 μ_F = 0.767 μ_{CR} = 0.172
 Diversity = 1.7914e-03 k_t = [1, 1, 1, 1]
 Generation 9740: Best = 1.452135e+04 FES = 974100 μ_F = 0.759 μ_{CR} = 0.171
 Diversity = 1.6190e-03 k_t = [4, 3, 3, 4]
 Generation 9750: Best = 1.452134e+04 FES = 975100 μ_F = 0.751 μ_{CR} = 0.157
 Diversity = 1.6813e-03 k_t = [1, 1, 1, 1]
 Generation 9760: Best = 1.452133e+04 FES = 976100 μ_F = 0.739 μ_{CR} = 0.147
 Diversity = 1.5938e-03 k_t = [2, 2, 3, 3]
 Generation 9770: Best = 1.452132e+04 FES = 977100 μ_F = 0.734 μ_{CR} = 0.136
 Diversity = 1.5430e-03 k_t = [1, 1, 1, 1]
 Generation 9780: Best = 1.452131e+04 FES = 978100 μ_F = 0.747 μ_{CR} = 0.135
 Diversity = 1.5099e-03 k_t = [4, 3, 3, 3]
 Generation 9790: Best = 1.452130e+04 FES = 979100 μ_F = 0.749 μ_{CR} = 0.136
 Diversity = 1.4871e-03 k_t = [1, 1, 1, 1]
 Generation 9800: Best = 1.452129e+04 FES = 980100 μ_F = 0.756 μ_{CR} = 0.132
 Diversity = 1.4455e-03 k_t = [4, 3, 3, 3]
 Generation 9810: Best = 1.452128e+04 FES = 981100 μ_F = 0.751 μ_{CR} = 0.126
 Diversity = 1.4563e-03 k_t = [1, 1, 1, 1]
 Generation 9820: Best = 1.452128e+04 FES = 982100 μ_F = 0.744 μ_{CR} = 0.110
 Diversity = 1.4293e-03 k_t = [2, 3, 3, 3]
 Generation 9830: Best = 1.452127e+04 FES = 983100 μ_F = 0.736 μ_{CR} = 0.112
 Diversity = 1.4611e-03 k_t = [1, 1, 1, 1]
 Generation 9840: Best = 1.452126e+04 FES = 984100 μ_F = 0.732 μ_{CR} = 0.112
 Diversity = 1.3407e-03 k_t = [2, 3, 2, 3]
 Generation 9850: Best = 1.452125e+04 FES = 985100 μ_F = 0.713 μ_{CR} = 0.123
 Diversity = 1.4179e-03 k_t = [1, 1, 1, 1]
 Generation 9860: Best = 1.452124e+04 FES = 986100 μ_F = 0.711 μ_{CR} = 0.130
 Diversity = 1.4420e-03 k_t = [4, 3, 3, 2]
 Generation 9870: Best = 1.452123e+04 FES = 987100 μ_F = 0.702 μ_{CR} = 0.124
 Diversity = 1.4470e-03 k_t = [1, 1, 1, 1]
 Generation 9880: Best = 1.452122e+04 FES = 988100 μ_F = 0.715 μ_{CR} = 0.127
 Diversity = 1.4430e-03 k_t = [4, 4, 2, 2]
 Generation 9890: Best = 1.452121e+04 FES = 989100 μ_F = 0.720 μ_{CR} = 0.127
 Diversity = 1.5679e-03 k_t = [1, 1, 1, 1]
 Generation 9900: Best = 1.452120e+04 FES = 990100 μ_F = 0.713 μ_{CR} = 0.132
 Diversity = 1.6728e-03 k_t = [3, 4, 4, 2]
 Generation 9910: Best = 1.452117e+04 FES = 991100 μ_F = 0.727 μ_{CR} = 0.129
 Diversity = 1.8489e-03 k_t = [1, 1, 1, 1]
 Generation 9920: Best = 1.452117e+04 FES = 992100 μ_F = 0.716 μ_{CR} = 0.132
 Diversity = 1.9266e-03 k_t = [3, 2, 2, 2]
 Generation 9930: Best = 1.452116e+04 FES = 993100 μ_F = 0.720 μ_{CR} = 0.138
 Diversity = 1.9795e-03 k_t = [1, 1, 1, 1]
 Generation 9940: Best = 1.452115e+04 FES = 994100 μ_F = 0.728 μ_{CR} = 0.128
 Diversity = 1.8283e-03 k_t = [3, 2, 4, 2]

Generation 9950: Best = 1.452114e+04 FES = 995100 μ_F = 0.719 μ_{CR} = 0.130
 Diversity = 1.8190e-03 k_t = [1, 1, 1, 1]
 Generation 9960: Best = 1.452113e+04 FES = 996100 μ_F = 0.719 μ_{CR} = 0.128
 Diversity = 1.6415e-03 k_t = [2, 3, 2, 4]
 Generation 9970: Best = 1.452111e+04 FES = 997100 μ_F = 0.723 μ_{CR} = 0.124
 Diversity = 1.5814e-03 k_t = [1, 1, 1, 1]
 Generation 9980: Best = 1.452109e+04 FES = 998100 μ_F = 0.736 μ_{CR} = 0.127
 Diversity = 1.6321e-03 k_t = [3, 4, 4, 3]
 Generation 9990: Best = 1.452108e+04 FES = 999100 μ_F = 0.730 μ_{CR} = 0.127
 Diversity = 1.6820e-03 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 1.452107e+04

Total Function Evals = 1000000

Total Successful Updates = 96295

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Results:

Best Fitness : 1.452107e+04

Global Optimum : 2.200000e+03

Error (|best-optimum|) : 1.232107e+04

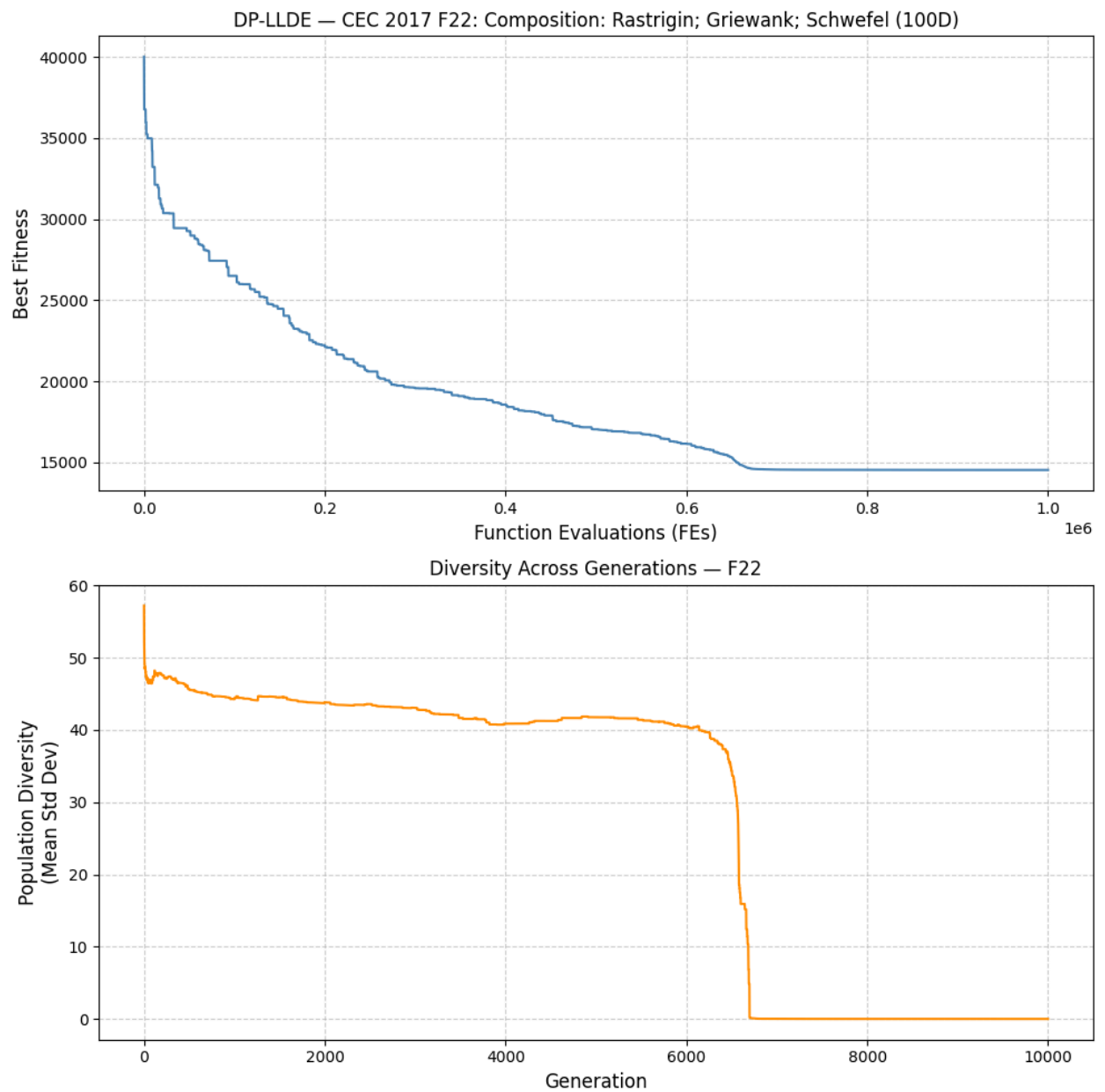
Best Solution (5 dim) : [41.80950743 67.7041659 -71.09101401 2.84010876
-68.82836186]

D^(0) (init. diversity): 5.7220e+01

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✓ Checkpoint saved to: /content/drive/My Drive/thesis_results/all_results_checkpoint.pkl



Graph saved as convergence_F22.png

✓ Plot saved to: /content/drive/My Drive/thesis_results/plots/convergence_F22.png

```
In [ ]: # =====
# F23 — Composition Function 3 (Rosenbrock; Ackley; Schwefel; Rastrigin)
# =====
run_and_plot('F23', make_wrapper(composition.f23), 2300.0, 'Composition: Rosen
```

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DP-LLDE on CEC 2017 F23 – Single Run

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Function : Composition: Rosenbrock; Ackley; Schwefel; Rastrigin (100D)
 Bounds : [-100, 100]
 Global optimum : 2300.0
 Initial $D^*(0) = 5.7220e+01$ | Diversity threshold = 5.7220e+00
 Generation 0: Best Fitness = 8.120533e+03

Generation 10:	Best = 5.975713e+03	FES = 1100	$\mu_F = 0.612$	$\mu_{CR} = 0.354$
Diversity = 4.9396e+01	$k_t = [1, 1, 1, 1]$			
Generation 20:	Best = 4.786974e+03	FES = 2100	$\mu_F = 0.680$	$\mu_{CR} = 0.350$
Diversity = 3.7472e+01	$k_t = [4, 3, 2, 2]$			
Generation 30:	Best = 4.417533e+03	FES = 3100	$\mu_F = 0.706$	$\mu_{CR} = 0.355$
Diversity = 2.9060e+01	$k_t = [1, 1, 1, 1]$			
Generation 40:	Best = 4.189001e+03	FES = 4100	$\mu_F = 0.726$	$\mu_{CR} = 0.339$
Diversity = 2.4739e+01	$k_t = [2, 4, 2, 2]$			
Generation 50:	Best = 4.137255e+03	FES = 5100	$\mu_F = 0.736$	$\mu_{CR} = 0.331$
Diversity = 2.3585e+01	$k_t = [1, 1, 1, 1]$			
Generation 60:	Best = 4.021661e+03	FES = 6100	$\mu_F = 0.725$	$\mu_{CR} = 0.315$
Diversity = 2.1719e+01	$k_t = [2, 3, 4, 2]$			
Generation 70:	Best = 3.976757e+03	FES = 7100	$\mu_F = 0.718$	$\mu_{CR} = 0.287$
Diversity = 2.0422e+01	$k_t = [1, 1, 1, 1]$			
Generation 80:	Best = 3.957849e+03	FES = 8100	$\mu_F = 0.720$	$\mu_{CR} = 0.264$
Diversity = 1.9544e+01	$k_t = [3, 4, 4, 2]$			
Generation 90:	Best = 3.942794e+03	FES = 9100	$\mu_F = 0.747$	$\mu_{CR} = 0.221$
Diversity = 1.9074e+01	$k_t = [1, 1, 1, 1]$			
Generation 100:	Best = 3.851351e+03	FES = 10100	$\mu_F = 0.751$	$\mu_{CR} = 0.179$
Diversity = 1.8424e+01	$k_t = [4, 2, 4, 4]$			
Generation 110:	Best = 3.845313e+03	FES = 11100	$\mu_F = 0.744$	$\mu_{CR} = 0.146$
Diversity = 1.7799e+01	$k_t = [1, 1, 1, 1]$			
Generation 120:	Best = 3.842785e+03	FES = 12100	$\mu_F = 0.738$	$\mu_{CR} = 0.113$
Diversity = 1.7351e+01	$k_t = [2, 4, 4, 2]$			
Generation 130:	Best = 3.833931e+03	FES = 13100	$\mu_F = 0.759$	$\mu_{CR} = 0.084$
Diversity = 1.7252e+01	$k_t = [1, 1, 1, 1]$			
Generation 140:	Best = 3.828402e+03	FES = 14100	$\mu_F = 0.764$	$\mu_{CR} = 0.058$
Diversity = 1.6700e+01	$k_t = [2, 4, 4, 3]$			
Generation 150:	Best = 3.806981e+03	FES = 15100	$\mu_F = 0.771$	$\mu_{CR} = 0.065$
Diversity = 1.6587e+01	$k_t = [1, 1, 1, 1]$			
Generation 160:	Best = 3.780315e+03	FES = 16100	$\mu_F = 0.765$	$\mu_{CR} = 0.066$
Diversity = 1.6488e+01	$k_t = [3, 3, 3, 3]$			
Generation 170:	Best = 3.772713e+03	FES = 17100	$\mu_F = 0.754$	$\mu_{CR} = 0.063$
Diversity = 1.6398e+01	$k_t = [1, 1, 1, 1]$			
Generation 180:	Best = 3.752011e+03	FES = 18100	$\mu_F = 0.784$	$\mu_{CR} = 0.062$
Diversity = 1.6506e+01	$k_t = [3, 4, 3, 4]$			
Generation 190:	Best = 3.752011e+03	FES = 19100	$\mu_F = 0.791$	$\mu_{CR} = 0.050$
Diversity = 1.6493e+01	$k_t = [1, 1, 1, 1]$			
Generation 200:	Best = 3.679298e+03	FES = 20100	$\mu_F = 0.795$	$\mu_{CR} = 0.050$
Diversity = 1.6355e+01	$k_t = [3, 2, 4, 2]$			
Generation 210:	Best = 3.679298e+03	FES = 21100	$\mu_F = 0.813$	$\mu_{CR} = 0.046$
Diversity = 1.6429e+01	$k_t = [1, 1, 1, 1]$			
Generation 220:	Best = 3.672745e+03	FES = 22100	$\mu_F = 0.818$	$\mu_{CR} = 0.045$
Diversity = 1.6427e+01	$k_t = [4, 3, 3, 2]$			

Generation 230: Best = 3.644670e+03	FES = 23100	$\mu_F = 0.835$	$\mu_{CR} = 0.044$
Diversity = 1.6382e+01	$k_t = [1, 1, 1, 1]$		
Generation 240: Best = 3.644670e+03	FES = 24100	$\mu_F = 0.802$	$\mu_{CR} = 0.042$
Diversity = 1.6365e+01	$k_t = [3, 3, 2, 2]$		
Generation 250: Best = 3.644670e+03	FES = 25100	$\mu_F = 0.850$	$\mu_{CR} = 0.044$
Diversity = 1.6363e+01	$k_t = [1, 1, 1, 1]$		
Generation 260: Best = 3.644670e+03	FES = 26100	$\mu_F = 0.874$	$\mu_{CR} = 0.046$
Diversity = 1.6344e+01	$k_t = [3, 4, 4, 4]$		
Generation 270: Best = 3.644670e+03	FES = 27100	$\mu_F = 0.845$	$\mu_{CR} = 0.043$
Diversity = 1.6295e+01	$k_t = [1, 1, 1, 1]$		
Generation 280: Best = 3.626516e+03	FES = 28100	$\mu_F = 0.811$	$\mu_{CR} = 0.044$
Diversity = 1.6360e+01	$k_t = [3, 4, 2, 2]$		
Generation 290: Best = 3.612475e+03	FES = 29100	$\mu_F = 0.827$	$\mu_{CR} = 0.039$
Diversity = 1.6380e+01	$k_t = [1, 1, 1, 1]$		
Generation 300: Best = 3.605708e+03	FES = 30100	$\mu_F = 0.812$	$\mu_{CR} = 0.039$
Diversity = 1.6032e+01	$k_t = [2, 3, 2, 4]$		
Generation 310: Best = 3.605708e+03	FES = 31100	$\mu_F = 0.860$	$\mu_{CR} = 0.032$
Diversity = 1.6034e+01	$k_t = [1, 1, 1, 1]$		
Generation 320: Best = 3.603081e+03	FES = 32100	$\mu_F = 0.841$	$\mu_{CR} = 0.027$
Diversity = 1.6071e+01	$k_t = [3, 3, 3, 3]$		
Generation 330: Best = 3.599449e+03	FES = 33100	$\mu_F = 0.849$	$\mu_{CR} = 0.030$
Diversity = 1.6058e+01	$k_t = [1, 1, 1, 1]$		
Generation 340: Best = 3.594490e+03	FES = 34100	$\mu_F = 0.820$	$\mu_{CR} = 0.038$
Diversity = 1.6067e+01	$k_t = [4, 4, 3, 3]$		
Generation 350: Best = 3.594490e+03	FES = 35100	$\mu_F = 0.807$	$\mu_{CR} = 0.035$
Diversity = 1.6053e+01	$k_t = [1, 1, 1, 1]$		
Generation 360: Best = 3.589259e+03	FES = 36100	$\mu_F = 0.788$	$\mu_{CR} = 0.026$
Diversity = 1.6052e+01	$k_t = [2, 3, 4, 3]$		
Generation 370: Best = 3.576180e+03	FES = 37100	$\mu_F = 0.751$	$\mu_{CR} = 0.030$
Diversity = 1.6041e+01	$k_t = [1, 1, 1, 1]$		
Generation 380: Best = 3.572918e+03	FES = 38100	$\mu_F = 0.755$	$\mu_{CR} = 0.032$
Diversity = 1.6025e+01	$k_t = [4, 2, 2, 2]$		
Generation 390: Best = 3.558004e+03	FES = 39100	$\mu_F = 0.737$	$\mu_{CR} = 0.046$
Diversity = 1.6007e+01	$k_t = [1, 1, 1, 1]$		
Generation 400: Best = 3.558004e+03	FES = 40100	$\mu_F = 0.745$	$\mu_{CR} = 0.036$
Diversity = 1.5981e+01	$k_t = [4, 2, 4, 3]$		
Generation 410: Best = 3.553529e+03	FES = 41100	$\mu_F = 0.717$	$\mu_{CR} = 0.040$
Diversity = 1.5961e+01	$k_t = [1, 1, 1, 1]$		
Generation 420: Best = 3.553153e+03	FES = 42100	$\mu_F = 0.739$	$\mu_{CR} = 0.029$
Diversity = 1.5952e+01	$k_t = [4, 2, 4, 3]$		
Generation 430: Best = 3.531565e+03	FES = 43100	$\mu_F = 0.786$	$\mu_{CR} = 0.029$
Diversity = 1.5961e+01	$k_t = [1, 1, 1, 1]$		
Generation 440: Best = 3.531565e+03	FES = 44100	$\mu_F = 0.785$	$\mu_{CR} = 0.032$
Diversity = 1.5965e+01	$k_t = [3, 2, 3, 2]$		
Generation 450: Best = 3.531565e+03	FES = 45100	$\mu_F = 0.810$	$\mu_{CR} = 0.035$
Diversity = 1.5974e+01	$k_t = [1, 1, 1, 1]$		
Generation 460: Best = 3.531565e+03	FES = 46100	$\mu_F = 0.775$	$\mu_{CR} = 0.033$
Diversity = 1.5970e+01	$k_t = [3, 3, 4, 3]$		
Generation 470: Best = 3.531565e+03	FES = 47100	$\mu_F = 0.761$	$\mu_{CR} = 0.032$
Diversity = 1.5977e+01	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 3.531565e+03	FES = 48100	$\mu_F = 0.771$	$\mu_{CR} = 0.030$
Diversity = 1.5972e+01	$k_t = [3, 4, 4, 3]$		
Generation 490: Best = 3.531565e+03	FES = 49100	$\mu_F = 0.818$	$\mu_{CR} = 0.022$
Diversity = 1.5957e+01	$k_t = [1, 1, 1, 1]$		

Generation 500: Best = 3.527906e+03	FES = 50100	$\mu_F = 0.794$	$\mu_{CR} = 0.030$
Diversity = 1.5931e+01	k_t = [3, 3, 4, 2]		
Generation 510: Best = 3.527906e+03	FES = 51100	$\mu_F = 0.809$	$\mu_{CR} = 0.025$
Diversity = 1.5923e+01	k_t = [1, 1, 1, 1]		
Generation 520: Best = 3.527906e+03	FES = 52100	$\mu_F = 0.851$	$\mu_{CR} = 0.027$
Diversity = 1.5930e+01	k_t = [3, 3, 3, 3]		
Generation 530: Best = 3.527906e+03	FES = 53100	$\mu_F = 0.814$	$\mu_{CR} = 0.027$
Diversity = 1.5882e+01	k_t = [1, 1, 1, 1]		
Generation 540: Best = 3.527906e+03	FES = 54100	$\mu_F = 0.703$	$\mu_{CR} = 0.027$
Diversity = 1.5798e+01	k_t = [3, 3, 4, 4]		
Generation 550: Best = 3.524651e+03	FES = 55100	$\mu_F = 0.717$	$\mu_{CR} = 0.039$
Diversity = 1.5779e+01	k_t = [1, 1, 1, 1]		
Generation 560: Best = 3.524651e+03	FES = 56100	$\mu_F = 0.741$	$\mu_{CR} = 0.031$
Diversity = 1.5759e+01	k_t = [4, 3, 4, 4]		
Generation 570: Best = 3.503475e+03	FES = 57100	$\mu_F = 0.794$	$\mu_{CR} = 0.032$
Diversity = 1.5755e+01	k_t = [1, 1, 1, 1]		
Generation 580: Best = 3.503475e+03	FES = 58100	$\mu_F = 0.708$	$\mu_{CR} = 0.036$
Diversity = 1.5680e+01	k_t = [2, 2, 4, 2]		
Generation 590: Best = 3.503475e+03	FES = 59100	$\mu_F = 0.649$	$\mu_{CR} = 0.036$
Diversity = 1.5673e+01	k_t = [1, 1, 1, 1]		
Generation 600: Best = 3.503475e+03	FES = 60100	$\mu_F = 0.609$	$\mu_{CR} = 0.026$
Diversity = 1.5634e+01	k_t = [3, 3, 3, 3]		
Generation 610: Best = 3.503475e+03	FES = 61100	$\mu_F = 0.641$	$\mu_{CR} = 0.023$
Diversity = 1.5623e+01	k_t = [1, 1, 1, 1]		
Generation 620: Best = 3.467418e+03	FES = 62100	$\mu_F = 0.583$	$\mu_{CR} = 0.037$
Diversity = 1.5474e+01	k_t = [2, 3, 2, 4]		
Generation 630: Best = 3.467418e+03	FES = 63100	$\mu_F = 0.657$	$\mu_{CR} = 0.032$
Diversity = 1.5452e+01	k_t = [1, 1, 1, 1]		
Generation 640: Best = 3.447265e+03	FES = 64100	$\mu_F = 0.658$	$\mu_{CR} = 0.027$
Diversity = 1.5459e+01	k_t = [2, 3, 2, 2]		
Generation 650: Best = 3.435013e+03	FES = 65100	$\mu_F = 0.620$	$\mu_{CR} = 0.027$
Diversity = 1.5413e+01	k_t = [1, 1, 1, 1]		
Generation 660: Best = 3.435013e+03	FES = 66100	$\mu_F = 0.679$	$\mu_{CR} = 0.027$
Diversity = 1.5351e+01	k_t = [3, 2, 2, 4]		
Generation 670: Best = 3.435013e+03	FES = 67100	$\mu_F = 0.615$	$\mu_{CR} = 0.034$
Diversity = 1.5317e+01	k_t = [1, 1, 1, 1]		
Generation 680: Best = 3.424002e+03	FES = 68100	$\mu_F = 0.630$	$\mu_{CR} = 0.031$
Diversity = 1.5314e+01	k_t = [2, 2, 2, 3]		
Generation 690: Best = 3.414096e+03	FES = 69100	$\mu_F = 0.615$	$\mu_{CR} = 0.038$
Diversity = 1.5282e+01	k_t = [1, 1, 1, 1]		
Generation 700: Best = 3.414096e+03	FES = 70100	$\mu_F = 0.535$	$\mu_{CR} = 0.030$
Diversity = 1.5264e+01	k_t = [2, 4, 4, 3]		
Generation 710: Best = 3.414096e+03	FES = 71100	$\mu_F = 0.520$	$\mu_{CR} = 0.029$
Diversity = 1.5250e+01	k_t = [1, 1, 1, 1]		
Generation 720: Best = 3.414096e+03	FES = 72100	$\mu_F = 0.509$	$\mu_{CR} = 0.046$
Diversity = 1.5212e+01	k_t = [4, 4, 2, 3]		
Generation 730: Best = 3.411132e+03	FES = 73100	$\mu_F = 0.518$	$\mu_{CR} = 0.037$
Diversity = 1.5194e+01	k_t = [1, 1, 1, 1]		
Generation 740: Best = 3.402219e+03	FES = 74100	$\mu_F = 0.440$	$\mu_{CR} = 0.049$
Diversity = 1.5177e+01	k_t = [4, 4, 3, 2]		
Generation 750: Best = 3.397952e+03	FES = 75100	$\mu_F = 0.469$	$\mu_{CR} = 0.039$
Diversity = 1.5172e+01	k_t = [1, 1, 1, 1]		
Generation 760: Best = 3.397952e+03	FES = 76100	$\mu_F = 0.473$	$\mu_{CR} = 0.038$
Diversity = 1.5167e+01	k_t = [4, 4, 3, 4]		

Generation 770:	Best = 3.397952e+03	FES = 77100	$\mu_F = 0.495$	$\mu_{CR} = 0.030$
Diversity = 1.5143e+01	$k_t = [1, 1, 1, 1]$			
Generation 780:	Best = 3.397952e+03	FES = 78100	$\mu_F = 0.544$	$\mu_{CR} = 0.034$
Diversity = 1.5140e+01	$k_t = [4, 2, 3, 3]$			
Generation 790:	Best = 3.397952e+03	FES = 79100	$\mu_F = 0.476$	$\mu_{CR} = 0.033$
Diversity = 1.5123e+01	$k_t = [1, 1, 1, 1]$			
Generation 800:	Best = 3.397952e+03	FES = 80100	$\mu_F = 0.452$	$\mu_{CR} = 0.027$
Diversity = 1.5120e+01	$k_t = [3, 3, 2, 3]$			
Generation 810:	Best = 3.393985e+03	FES = 81100	$\mu_F = 0.498$	$\mu_{CR} = 0.035$
Diversity = 1.5099e+01	$k_t = [1, 1, 1, 1]$			
Generation 820:	Best = 3.393985e+03	FES = 82100	$\mu_F = 0.604$	$\mu_{CR} = 0.039$
Diversity = 1.5116e+01	$k_t = [4, 4, 3, 4]$			
Generation 830:	Best = 3.393985e+03	FES = 83100	$\mu_F = 0.534$	$\mu_{CR} = 0.033$
Diversity = 1.5117e+01	$k_t = [1, 1, 1, 1]$			
Generation 840:	Best = 3.393985e+03	FES = 84100	$\mu_F = 0.489$	$\mu_{CR} = 0.032$
Diversity = 1.5094e+01	$k_t = [2, 4, 3, 2]$			
Generation 850:	Best = 3.393985e+03	FES = 85100	$\mu_F = 0.507$	$\mu_{CR} = 0.040$
Diversity = 1.5067e+01	$k_t = [1, 1, 1, 1]$			
Generation 860:	Best = 3.393985e+03	FES = 86100	$\mu_F = 0.524$	$\mu_{CR} = 0.042$
Diversity = 1.4986e+01	$k_t = [2, 2, 4, 4]$			
Generation 870:	Best = 3.389765e+03	FES = 87100	$\mu_F = 0.445$	$\mu_{CR} = 0.056$
Diversity = 1.4976e+01	$k_t = [1, 1, 1, 1]$			
Generation 880:	Best = 3.386340e+03	FES = 88100	$\mu_F = 0.456$	$\mu_{CR} = 0.051$
Diversity = 1.4953e+01	$k_t = [4, 2, 3, 3]$			
Generation 890:	Best = 3.386340e+03	FES = 89100	$\mu_F = 0.484$	$\mu_{CR} = 0.035$
Diversity = 1.4937e+01	$k_t = [1, 1, 1, 1]$			
Generation 900:	Best = 3.386340e+03	FES = 90100	$\mu_F = 0.568$	$\mu_{CR} = 0.029$
Diversity = 1.4907e+01	$k_t = [2, 4, 3, 4]$			
Generation 910:	Best = 3.379890e+03	FES = 91100	$\mu_F = 0.583$	$\mu_{CR} = 0.036$
Diversity = 1.4894e+01	$k_t = [1, 1, 1, 1]$			
Generation 920:	Best = 3.379890e+03	FES = 92100	$\mu_F = 0.567$	$\mu_{CR} = 0.042$
Diversity = 1.4898e+01	$k_t = [4, 4, 4, 2]$			
Generation 930:	Best = 3.379890e+03	FES = 93100	$\mu_F = 0.487$	$\mu_{CR} = 0.034$
Diversity = 1.4896e+01	$k_t = [1, 1, 1, 1]$			
Generation 940:	Best = 3.352230e+03	FES = 94100	$\mu_F = 0.447$	$\mu_{CR} = 0.033$
Diversity = 1.4878e+01	$k_t = [2, 2, 4, 2]$			
Generation 950:	Best = 3.352230e+03	FES = 95100	$\mu_F = 0.399$	$\mu_{CR} = 0.029$
Diversity = 1.4867e+01	$k_t = [1, 1, 1, 1]$			
Generation 960:	Best = 3.341709e+03	FES = 96100	$\mu_F = 0.460$	$\mu_{CR} = 0.028$
Diversity = 1.4867e+01	$k_t = [2, 4, 2, 2]$			
Generation 970:	Best = 3.326552e+03	FES = 97100	$\mu_F = 0.462$	$\mu_{CR} = 0.024$
Diversity = 1.4858e+01	$k_t = [1, 1, 1, 1]$			
Generation 980:	Best = 3.326552e+03	FES = 98100	$\mu_F = 0.515$	$\mu_{CR} = 0.032$
Diversity = 1.4844e+01	$k_t = [4, 4, 4, 4]$			
Generation 990:	Best = 3.326552e+03	FES = 99100	$\mu_F = 0.468$	$\mu_{CR} = 0.038$
Diversity = 1.4839e+01	$k_t = [1, 1, 1, 1]$			
Generation 1000:	Best = 3.326552e+03	FES = 100100	$\mu_F = 0.534$	$\mu_{CR} = 0.031$
Diversity = 1.4767e+01	$k_t = [3, 2, 2, 4]$			
Generation 1010:	Best = 3.326552e+03	FES = 101100	$\mu_F = 0.605$	$\mu_{CR} = 0.022$
Diversity = 1.4768e+01	$k_t = [1, 1, 1, 1]$			
Generation 1020:	Best = 3.326552e+03	FES = 102100	$\mu_F = 0.679$	$\mu_{CR} = 0.030$
Diversity = 1.4664e+01	$k_t = [4, 3, 2, 4]$			
Generation 1030:	Best = 3.326552e+03	FES = 103100	$\mu_F = 0.674$	$\mu_{CR} = 0.031$
Diversity = 1.4641e+01	$k_t = [1, 1, 1, 1]$			

Generation 1040: Best = 3.324488e+03 FES = 104100 μ_F = 0.725 μ_{CR} = 0.035
 Diversity = 1.4618e+01 k_t = [3, 4, 3, 3]
 Generation 1050: Best = 3.324488e+03 FES = 105100 μ_F = 0.673 μ_{CR} = 0.030
 Diversity = 1.4619e+01 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 3.324488e+03 FES = 106100 μ_F = 0.604 μ_{CR} = 0.037
 Diversity = 1.4608e+01 k_t = [4, 2, 4, 3]
 Generation 1070: Best = 3.321876e+03 FES = 107100 μ_F = 0.577 μ_{CR} = 0.030
 Diversity = 1.4610e+01 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 3.321876e+03 FES = 108100 μ_F = 0.605 μ_{CR} = 0.029
 Diversity = 1.4586e+01 k_t = [2, 4, 4, 4]
 Generation 1090: Best = 3.321876e+03 FES = 109100 μ_F = 0.582 μ_{CR} = 0.031
 Diversity = 1.4598e+01 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 3.321876e+03 FES = 110100 μ_F = 0.636 μ_{CR} = 0.026
 Diversity = 1.4587e+01 k_t = [2, 3, 3, 2]
 Generation 1110: Best = 3.311888e+03 FES = 111100 μ_F = 0.604 μ_{CR} = 0.026
 Diversity = 1.4586e+01 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 3.311888e+03 FES = 112100 μ_F = 0.645 μ_{CR} = 0.023
 Diversity = 1.4444e+01 k_t = [3, 3, 4, 4]
 Generation 1130: Best = 3.311888e+03 FES = 113100 μ_F = 0.512 μ_{CR} = 0.021
 Diversity = 1.4443e+01 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 3.311888e+03 FES = 114100 μ_F = 0.547 μ_{CR} = 0.027
 Diversity = 1.4435e+01 k_t = [4, 2, 2, 2]
 Generation 1150: Best = 3.311888e+03 FES = 115100 μ_F = 0.485 μ_{CR} = 0.031
 Diversity = 1.4432e+01 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 3.311888e+03 FES = 116100 μ_F = 0.562 μ_{CR} = 0.024
 Diversity = 1.4436e+01 k_t = [4, 2, 3, 2]
 Generation 1170: Best = 3.309305e+03 FES = 117100 μ_F = 0.559 μ_{CR} = 0.027
 Diversity = 1.4430e+01 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 3.309247e+03 FES = 118100 μ_F = 0.591 μ_{CR} = 0.028
 Diversity = 1.4449e+01 k_t = [2, 4, 2, 2]
 Generation 1190: Best = 3.309247e+03 FES = 119100 μ_F = 0.539 μ_{CR} = 0.026
 Diversity = 1.4446e+01 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 3.309247e+03 FES = 120100 μ_F = 0.509 μ_{CR} = 0.026
 Diversity = 1.4444e+01 k_t = [2, 4, 3, 4]
 Generation 1210: Best = 3.305234e+03 FES = 121100 μ_F = 0.533 μ_{CR} = 0.032
 Diversity = 1.4439e+01 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 3.305234e+03 FES = 122100 μ_F = 0.533 μ_{CR} = 0.032
 Diversity = 1.4433e+01 k_t = [3, 4, 4, 2]
 Generation 1230: Best = 3.303728e+03 FES = 123100 μ_F = 0.482 μ_{CR} = 0.032
 Diversity = 1.4398e+01 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 3.299584e+03 FES = 124100 μ_F = 0.526 μ_{CR} = 0.037
 Diversity = 1.4376e+01 k_t = [3, 3, 2, 2]
 Generation 1250: Best = 3.299061e+03 FES = 125100 μ_F = 0.440 μ_{CR} = 0.036
 Diversity = 1.4371e+01 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 3.299061e+03 FES = 126100 μ_F = 0.458 μ_{CR} = 0.031
 Diversity = 1.4375e+01 k_t = [4, 2, 4, 2]
 Generation 1270: Best = 3.299061e+03 FES = 127100 μ_F = 0.433 μ_{CR} = 0.032
 Diversity = 1.4364e+01 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 3.295404e+03 FES = 128100 μ_F = 0.420 μ_{CR} = 0.030
 Diversity = 1.4353e+01 k_t = [4, 2, 4, 2]
 Generation 1290: Best = 3.295404e+03 FES = 129100 μ_F = 0.482 μ_{CR} = 0.030
 Diversity = 1.4350e+01 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 3.295404e+03 FES = 130100 μ_F = 0.445 μ_{CR} = 0.021
 Diversity = 1.4341e+01 k_t = [2, 3, 3, 3]

Generation 1310: Best = 3.295404e+03 FES = 131100 μ_F = 0.518 μ_{CR} = 0.019
 Diversity = 1.4336e+01 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 3.295404e+03 FES = 132100 μ_F = 0.512 μ_{CR} = 0.021
 Diversity = 1.4331e+01 k_t = [2, 3, 3, 2]
 Generation 1330: Best = 3.295404e+03 FES = 133100 μ_F = 0.411 μ_{CR} = 0.036
 Diversity = 1.4322e+01 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 3.295404e+03 FES = 134100 μ_F = 0.386 μ_{CR} = 0.037
 Diversity = 1.4283e+01 k_t = [4, 3, 2, 4]
 Generation 1350: Best = 3.295404e+03 FES = 135100 μ_F = 0.317 μ_{CR} = 0.034
 Diversity = 1.4282e+01 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 3.295404e+03 FES = 136100 μ_F = 0.386 μ_{CR} = 0.048
 Diversity = 1.4274e+01 k_t = [4, 4, 3, 4]
 Generation 1370: Best = 3.294090e+03 FES = 137100 μ_F = 0.313 μ_{CR} = 0.035
 Diversity = 1.4265e+01 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 3.293659e+03 FES = 138100 μ_F = 0.402 μ_{CR} = 0.037
 Diversity = 1.4265e+01 k_t = [2, 3, 3, 4]
 Generation 1390: Best = 3.292704e+03 FES = 139100 μ_F = 0.433 μ_{CR} = 0.035
 Diversity = 1.4262e+01 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 3.292205e+03 FES = 140100 μ_F = 0.410 μ_{CR} = 0.048
 Diversity = 1.4209e+01 k_t = [4, 2, 4, 4]
 Generation 1410: Best = 3.292205e+03 FES = 141100 μ_F = 0.335 μ_{CR} = 0.052
 Diversity = 1.4197e+01 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 3.292205e+03 FES = 142100 μ_F = 0.238 μ_{CR} = 0.062
 Diversity = 1.4181e+01 k_t = [3, 3, 3, 3]
 Generation 1430: Best = 3.289846e+03 FES = 143100 μ_F = 0.244 μ_{CR} = 0.048
 Diversity = 1.4160e+01 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 3.284137e+03 FES = 144100 μ_F = 0.283 μ_{CR} = 0.043
 Diversity = 1.4172e+01 k_t = [2, 4, 2, 2]
 Generation 1450: Best = 3.284137e+03 FES = 145100 μ_F = 0.304 μ_{CR} = 0.038
 Diversity = 1.4160e+01 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 3.282754e+03 FES = 146100 μ_F = 0.293 μ_{CR} = 0.040
 Diversity = 1.4155e+01 k_t = [2, 3, 2, 4]
 Generation 1470: Best = 3.282754e+03 FES = 147100 μ_F = 0.467 μ_{CR} = 0.044
 Diversity = 1.4137e+01 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 3.279854e+03 FES = 148100 μ_F = 0.518 μ_{CR} = 0.037
 Diversity = 1.4136e+01 k_t = [2, 2, 2, 3]
 Generation 1490: Best = 3.279854e+03 FES = 149100 μ_F = 0.512 μ_{CR} = 0.043
 Diversity = 1.4134e+01 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 3.278270e+03 FES = 150100 μ_F = 0.549 μ_{CR} = 0.038
 Diversity = 1.4131e+01 k_t = [3, 4, 3, 2]
 Generation 1510: Best = 3.278270e+03 FES = 151100 μ_F = 0.537 μ_{CR} = 0.027
 Diversity = 1.4124e+01 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 3.275427e+03 FES = 152100 μ_F = 0.513 μ_{CR} = 0.033
 Diversity = 1.4116e+01 k_t = [2, 2, 2, 3]
 Generation 1530: Best = 3.268059e+03 FES = 153100 μ_F = 0.488 μ_{CR} = 0.033
 Diversity = 1.4115e+01 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 3.268059e+03 FES = 154100 μ_F = 0.442 μ_{CR} = 0.025
 Diversity = 1.4111e+01 k_t = [3, 4, 4, 2]
 Generation 1550: Best = 3.268059e+03 FES = 155100 μ_F = 0.460 μ_{CR} = 0.018
 Diversity = 1.4099e+01 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 3.260932e+03 FES = 156100 μ_F = 0.531 μ_{CR} = 0.019
 Diversity = 1.4079e+01 k_t = [2, 4, 2, 3]
 Generation 1570: Best = 3.257762e+03 FES = 157100 μ_F = 0.522 μ_{CR} = 0.021
 Diversity = 1.4070e+01 k_t = [1, 1, 1, 1]

Generation 1580: Best = 3.257762e+03 FES = 158100 μ_F = 0.519 μ_{CR} = 0.027
 Diversity = 1.4016e+01 k_t = [4, 3, 4, 4]
 Generation 1590: Best = 3.257762e+03 FES = 159100 μ_F = 0.520 μ_{CR} = 0.021
 Diversity = 1.4014e+01 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 3.257626e+03 FES = 160100 μ_F = 0.550 μ_{CR} = 0.023
 Diversity = 1.3990e+01 k_t = [2, 2, 2, 3]
 Generation 1610: Best = 3.253550e+03 FES = 161100 μ_F = 0.586 μ_{CR} = 0.024
 Diversity = 1.3979e+01 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 3.253550e+03 FES = 162100 μ_F = 0.589 μ_{CR} = 0.025
 Diversity = 1.3978e+01 k_t = [4, 2, 2, 3]
 Generation 1630: Best = 3.253550e+03 FES = 163100 μ_F = 0.668 μ_{CR} = 0.019
 Diversity = 1.3944e+01 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 3.253550e+03 FES = 164100 μ_F = 0.608 μ_{CR} = 0.032
 Diversity = 1.3945e+01 k_t = [4, 3, 4, 2]
 Generation 1650: Best = 3.253550e+03 FES = 165100 μ_F = 0.478 μ_{CR} = 0.041
 Diversity = 1.3940e+01 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 3.252022e+03 FES = 166100 μ_F = 0.570 μ_{CR} = 0.038
 Diversity = 1.3937e+01 k_t = [3, 3, 4, 2]
 Generation 1670: Best = 3.252022e+03 FES = 167100 μ_F = 0.537 μ_{CR} = 0.029
 Diversity = 1.3928e+01 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 3.252022e+03 FES = 168100 μ_F = 0.479 μ_{CR} = 0.041
 Diversity = 1.3925e+01 k_t = [2, 3, 4, 3]
 Generation 1690: Best = 3.243828e+03 FES = 169100 μ_F = 0.438 μ_{CR} = 0.040
 Diversity = 1.3923e+01 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 3.235875e+03 FES = 170100 μ_F = 0.450 μ_{CR} = 0.035
 Diversity = 1.3840e+01 k_t = [2, 3, 4, 4]
 Generation 1710: Best = 3.235875e+03 FES = 171100 μ_F = 0.422 μ_{CR} = 0.037
 Diversity = 1.3830e+01 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 3.235875e+03 FES = 172100 μ_F = 0.393 μ_{CR} = 0.036
 Diversity = 1.3818e+01 k_t = [4, 2, 4, 3]
 Generation 1730: Best = 3.235875e+03 FES = 173100 μ_F = 0.411 μ_{CR} = 0.034
 Diversity = 1.3822e+01 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 3.235875e+03 FES = 174100 μ_F = 0.363 μ_{CR} = 0.042
 Diversity = 1.3818e+01 k_t = [4, 3, 4, 2]
 Generation 1750: Best = 3.231477e+03 FES = 175100 μ_F = 0.409 μ_{CR} = 0.031
 Diversity = 1.3815e+01 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 3.223130e+03 FES = 176100 μ_F = 0.469 μ_{CR} = 0.034
 Diversity = 1.3800e+01 k_t = [4, 2, 2, 4]
 Generation 1770: Best = 3.223130e+03 FES = 177100 μ_F = 0.433 μ_{CR} = 0.029
 Diversity = 1.3788e+01 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 3.223130e+03 FES = 178100 μ_F = 0.384 μ_{CR} = 0.030
 Diversity = 1.3788e+01 k_t = [3, 2, 3, 3]
 Generation 1790: Best = 3.223130e+03 FES = 179100 μ_F = 0.303 μ_{CR} = 0.030
 Diversity = 1.3778e+01 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 3.223130e+03 FES = 180100 μ_F = 0.326 μ_{CR} = 0.042
 Diversity = 1.3773e+01 k_t = [2, 3, 2, 3]
 Generation 1810: Best = 3.223130e+03 FES = 181100 μ_F = 0.259 μ_{CR} = 0.071
 Diversity = 1.3765e+01 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 3.219065e+03 FES = 182100 μ_F = 0.215 μ_{CR} = 0.069
 Diversity = 1.3759e+01 k_t = [4, 4, 4, 2]
 Generation 1830: Best = 3.217586e+03 FES = 183100 μ_F = 0.211 μ_{CR} = 0.060
 Diversity = 1.3751e+01 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 3.217586e+03 FES = 184100 μ_F = 0.191 μ_{CR} = 0.051
 Diversity = 1.3750e+01 k_t = [2, 3, 3, 3]

Generation 1850: Best = 3.217586e+03 FES = 185100 μ_F = 0.280 μ_{CR} = 0.047
 Diversity = 1.3741e+01 k_t = [1, 1, 1, 1]
 Generation 1860: Best = 3.217586e+03 FES = 186100 μ_F = 0.214 μ_{CR} = 0.041
 Diversity = 1.3736e+01 k_t = [3, 3, 4, 3]
 Generation 1870: Best = 3.217321e+03 FES = 187100 μ_F = 0.209 μ_{CR} = 0.052
 Diversity = 1.3724e+01 k_t = [1, 1, 1, 1]
 Generation 1880: Best = 3.217321e+03 FES = 188100 μ_F = 0.199 μ_{CR} = 0.060
 Diversity = 1.3718e+01 k_t = [4, 3, 4, 2]
 Generation 1890: Best = 3.217321e+03 FES = 189100 μ_F = 0.194 μ_{CR} = 0.054
 Diversity = 1.3710e+01 k_t = [1, 1, 1, 1]
 Generation 1900: Best = 3.215322e+03 FES = 190100 μ_F = 0.217 μ_{CR} = 0.044
 Diversity = 1.3705e+01 k_t = [2, 4, 2, 2]
 Generation 1910: Best = 3.211407e+03 FES = 191100 μ_F = 0.251 μ_{CR} = 0.033
 Diversity = 1.3702e+01 k_t = [1, 1, 1, 1]
 Generation 1920: Best = 3.211407e+03 FES = 192100 μ_F = 0.221 μ_{CR} = 0.040
 Diversity = 1.3699e+01 k_t = [3, 2, 2, 3]
 Generation 1930: Best = 3.211407e+03 FES = 193100 μ_F = 0.276 μ_{CR} = 0.037
 Diversity = 1.3691e+01 k_t = [1, 1, 1, 1]
 Generation 1940: Best = 3.211407e+03 FES = 194100 μ_F = 0.471 μ_{CR} = 0.029
 Diversity = 1.3687e+01 k_t = [3, 4, 2, 3]
 Generation 1950: Best = 3.208696e+03 FES = 195100 μ_F = 0.503 μ_{CR} = 0.042
 Diversity = 1.3673e+01 k_t = [1, 1, 1, 1]
 Generation 1960: Best = 3.208696e+03 FES = 196100 μ_F = 0.529 μ_{CR} = 0.029
 Diversity = 1.3674e+01 k_t = [2, 3, 2, 3]
 Generation 1970: Best = 3.208696e+03 FES = 197100 μ_F = 0.511 μ_{CR} = 0.024
 Diversity = 1.3676e+01 k_t = [1, 1, 1, 1]
 Generation 1980: Best = 3.208200e+03 FES = 198100 μ_F = 0.511 μ_{CR} = 0.024
 Diversity = 1.3625e+01 k_t = [2, 3, 4, 3]
 Generation 1990: Best = 3.208200e+03 FES = 199100 μ_F = 0.473 μ_{CR} = 0.028
 Diversity = 1.3597e+01 k_t = [1, 1, 1, 1]
 Generation 2000: Best = 3.208200e+03 FES = 200100 μ_F = 0.470 μ_{CR} = 0.040
 Diversity = 1.3590e+01 k_t = [4, 4, 2, 3]
 Generation 2010: Best = 3.208200e+03 FES = 201100 μ_F = 0.408 μ_{CR} = 0.031
 Diversity = 1.3590e+01 k_t = [1, 1, 1, 1]
 Generation 2020: Best = 3.207042e+03 FES = 202100 μ_F = 0.343 μ_{CR} = 0.028
 Diversity = 1.3586e+01 k_t = [2, 4, 4, 3]
 Generation 2030: Best = 3.203777e+03 FES = 203100 μ_F = 0.332 μ_{CR} = 0.028
 Diversity = 1.3583e+01 k_t = [1, 1, 1, 1]
 Generation 2040: Best = 3.203777e+03 FES = 204100 μ_F = 0.374 μ_{CR} = 0.035
 Diversity = 1.3556e+01 k_t = [2, 2, 2, 4]
 Generation 2050: Best = 3.203777e+03 FES = 205100 μ_F = 0.436 μ_{CR} = 0.037
 Diversity = 1.3553e+01 k_t = [1, 1, 1, 1]
 Generation 2060: Best = 3.203777e+03 FES = 206100 μ_F = 0.286 μ_{CR} = 0.031
 Diversity = 1.3551e+01 k_t = [3, 4, 3, 2]
 Generation 2070: Best = 3.197649e+03 FES = 207100 μ_F = 0.241 μ_{CR} = 0.032
 Diversity = 1.3548e+01 k_t = [1, 1, 1, 1]
 Generation 2080: Best = 3.197649e+03 FES = 208100 μ_F = 0.199 μ_{CR} = 0.047
 Diversity = 1.3540e+01 k_t = [2, 2, 3, 2]
 Generation 2090: Best = 3.190906e+03 FES = 209100 μ_F = 0.190 μ_{CR} = 0.045
 Diversity = 1.3535e+01 k_t = [1, 1, 1, 1]
 Generation 2100: Best = 3.189323e+03 FES = 210100 μ_F = 0.197 μ_{CR} = 0.057
 Diversity = 1.3525e+01 k_t = [4, 2, 4, 4]
 Generation 2110: Best = 3.187455e+03 FES = 211100 μ_F = 0.214 μ_{CR} = 0.043
 Diversity = 1.3523e+01 k_t = [1, 1, 1, 1]

Generation 2120: Best = 3.184127e+03 FES = 212100 μ_F = 0.321 μ_{CR} = 0.056
 Diversity = 1.3492e+01 k_t = [4, 3, 4, 4]
 Generation 2130: Best = 3.175653e+03 FES = 213100 μ_F = 0.301 μ_{CR} = 0.045
 Diversity = 1.3488e+01 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 3.170376e+03 FES = 214100 μ_F = 0.349 μ_{CR} = 0.031
 Diversity = 1.3485e+01 k_t = [3, 2, 3, 2]
 Generation 2150: Best = 3.167354e+03 FES = 215100 μ_F = 0.303 μ_{CR} = 0.032
 Diversity = 1.3481e+01 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 3.165612e+03 FES = 216100 μ_F = 0.275 μ_{CR} = 0.040
 Diversity = 1.3474e+01 k_t = [2, 4, 4, 3]
 Generation 2170: Best = 3.165612e+03 FES = 217100 μ_F = 0.254 μ_{CR} = 0.039
 Diversity = 1.3474e+01 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 3.165612e+03 FES = 218100 μ_F = 0.253 μ_{CR} = 0.035
 Diversity = 1.3471e+01 k_t = [3, 2, 4, 3]
 Generation 2190: Best = 3.165612e+03 FES = 219100 μ_F = 0.238 μ_{CR} = 0.042
 Diversity = 1.3472e+01 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 3.165612e+03 FES = 220100 μ_F = 0.288 μ_{CR} = 0.039
 Diversity = 1.3423e+01 k_t = [2, 4, 4, 4]
 Generation 2210: Best = 3.163638e+03 FES = 221100 μ_F = 0.359 μ_{CR} = 0.034
 Diversity = 1.3426e+01 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 3.159595e+03 FES = 222100 μ_F = 0.365 μ_{CR} = 0.039
 Diversity = 1.3381e+01 k_t = [3, 2, 4, 2]
 Generation 2230: Best = 3.159595e+03 FES = 223100 μ_F = 0.438 μ_{CR} = 0.042
 Diversity = 1.3371e+01 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 3.159595e+03 FES = 224100 μ_F = 0.439 μ_{CR} = 0.037
 Diversity = 1.3331e+01 k_t = [4, 4, 2, 4]
 Generation 2250: Best = 3.159595e+03 FES = 225100 μ_F = 0.461 μ_{CR} = 0.041
 Diversity = 1.3327e+01 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 3.159595e+03 FES = 226100 μ_F = 0.535 μ_{CR} = 0.046
 Diversity = 1.3327e+01 k_t = [4, 3, 2, 2]
 Generation 2270: Best = 3.159595e+03 FES = 227100 μ_F = 0.532 μ_{CR} = 0.037
 Diversity = 1.3322e+01 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 3.159595e+03 FES = 228100 μ_F = 0.519 μ_{CR} = 0.036
 Diversity = 1.3317e+01 k_t = [2, 4, 2, 3]
 Generation 2290: Best = 3.159595e+03 FES = 229100 μ_F = 0.482 μ_{CR} = 0.023
 Diversity = 1.3316e+01 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 3.159595e+03 FES = 230100 μ_F = 0.574 μ_{CR} = 0.029
 Diversity = 1.3252e+01 k_t = [3, 3, 3, 4]
 Generation 2310: Best = 3.156364e+03 FES = 231100 μ_F = 0.642 μ_{CR} = 0.052
 Diversity = 1.3253e+01 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 3.156335e+03 FES = 232100 μ_F = 0.632 μ_{CR} = 0.046
 Diversity = 1.3221e+01 k_t = [3, 3, 2, 3]
 Generation 2330: Best = 3.156335e+03 FES = 233100 μ_F = 0.626 μ_{CR} = 0.043
 Diversity = 1.3233e+01 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 3.156335e+03 FES = 234100 μ_F = 0.517 μ_{CR} = 0.062
 Diversity = 1.3239e+01 k_t = [4, 4, 3, 4]
 Generation 2350: Best = 3.156335e+03 FES = 235100 μ_F = 0.421 μ_{CR} = 0.050
 Diversity = 1.3238e+01 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 3.156335e+03 FES = 236100 μ_F = 0.485 μ_{CR} = 0.042
 Diversity = 1.3237e+01 k_t = [3, 4, 3, 2]
 Generation 2370: Best = 3.156335e+03 FES = 237100 μ_F = 0.518 μ_{CR} = 0.026
 Diversity = 1.3238e+01 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 3.156335e+03 FES = 238100 μ_F = 0.599 μ_{CR} = 0.020
 Diversity = 1.3252e+01 k_t = [3, 4, 2, 4]

Generation 2390: Best = 3.156335e+03 FES = 239100 μ_F = 0.553 μ_{CR} = 0.025
 Diversity = 1.3253e+01 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 3.156335e+03 FES = 240100 μ_F = 0.547 μ_{CR} = 0.022
 Diversity = 1.3251e+01 k_t = [2, 4, 4, 2]
 Generation 2410: Best = 3.156335e+03 FES = 241100 μ_F = 0.544 μ_{CR} = 0.022
 Diversity = 1.3252e+01 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 3.156335e+03 FES = 242100 μ_F = 0.461 μ_{CR} = 0.026
 Diversity = 1.3254e+01 k_t = [2, 2, 3, 4]
 Generation 2430: Best = 3.156335e+03 FES = 243100 μ_F = 0.493 μ_{CR} = 0.029
 Diversity = 1.3251e+01 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 3.156335e+03 FES = 244100 μ_F = 0.599 μ_{CR} = 0.026
 Diversity = 1.3215e+01 k_t = [2, 4, 4, 4]
 Generation 2450: Best = 3.156335e+03 FES = 245100 μ_F = 0.657 μ_{CR} = 0.026
 Diversity = 1.3217e+01 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 3.155650e+03 FES = 246100 μ_F = 0.673 μ_{CR} = 0.042
 Diversity = 1.3217e+01 k_t = [4, 4, 4, 2]
 Generation 2470: Best = 3.153617e+03 FES = 247100 μ_F = 0.708 μ_{CR} = 0.033
 Diversity = 1.3217e+01 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 3.153617e+03 FES = 248100 μ_F = 0.776 μ_{CR} = 0.029
 Diversity = 1.3157e+01 k_t = [2, 2, 4, 3]
 Generation 2490: Best = 3.153327e+03 FES = 249100 μ_F = 0.733 μ_{CR} = 0.018
 Diversity = 1.3158e+01 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 3.150580e+03 FES = 250100 μ_F = 0.768 μ_{CR} = 0.029
 Diversity = 1.3154e+01 k_t = [3, 2, 3, 2]
 Generation 2510: Best = 3.148787e+03 FES = 251100 μ_F = 0.710 μ_{CR} = 0.028
 Diversity = 1.3152e+01 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 3.148787e+03 FES = 252100 μ_F = 0.690 μ_{CR} = 0.030
 Diversity = 1.3095e+01 k_t = [4, 4, 3, 4]
 Generation 2530: Best = 3.148787e+03 FES = 253100 μ_F = 0.615 μ_{CR} = 0.025
 Diversity = 1.3096e+01 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 3.148787e+03 FES = 254100 μ_F = 0.638 μ_{CR} = 0.027
 Diversity = 1.3092e+01 k_t = [4, 2, 3, 2]
 Generation 2550: Best = 3.148787e+03 FES = 255100 μ_F = 0.632 μ_{CR} = 0.023
 Diversity = 1.3094e+01 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 3.148787e+03 FES = 256100 μ_F = 0.626 μ_{CR} = 0.026
 Diversity = 1.3094e+01 k_t = [4, 2, 2, 3]
 Generation 2570: Best = 3.145837e+03 FES = 257100 μ_F = 0.564 μ_{CR} = 0.028
 Diversity = 1.3086e+01 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 3.145837e+03 FES = 258100 μ_F = 0.559 μ_{CR} = 0.028
 Diversity = 1.3088e+01 k_t = [2, 3, 2, 2]
 Generation 2590: Best = 3.145837e+03 FES = 259100 μ_F = 0.530 μ_{CR} = 0.026
 Diversity = 1.3044e+01 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 3.145837e+03 FES = 260100 μ_F = 0.470 μ_{CR} = 0.035
 Diversity = 1.3044e+01 k_t = [4, 4, 2, 3]
 Generation 2610: Best = 3.145837e+03 FES = 261100 μ_F = 0.525 μ_{CR} = 0.029
 Diversity = 1.3044e+01 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 3.142243e+03 FES = 262100 μ_F = 0.531 μ_{CR} = 0.037
 Diversity = 1.3031e+01 k_t = [4, 4, 4, 4]
 Generation 2630: Best = 3.142243e+03 FES = 263100 μ_F = 0.537 μ_{CR} = 0.053
 Diversity = 1.3031e+01 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 3.142243e+03 FES = 264100 μ_F = 0.627 μ_{CR} = 0.040
 Diversity = 1.3027e+01 k_t = [3, 3, 4, 4]
 Generation 2650: Best = 3.142243e+03 FES = 265100 μ_F = 0.656 μ_{CR} = 0.030
 Diversity = 1.3048e+01 k_t = [1, 1, 1, 1]

Generation 2660: Best = 3.140397e+03 FES = 266100 μ_F = 0.667 μ_{CR} = 0.042
 Diversity = 1.3050e+01 k_t = [4, 4, 3, 3]
 Generation 2670: Best = 3.140397e+03 FES = 267100 μ_F = 0.611 μ_{CR} = 0.028
 Diversity = 1.3032e+01 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 3.140397e+03 FES = 268100 μ_F = 0.634 μ_{CR} = 0.035
 Diversity = 1.3032e+01 k_t = [3, 2, 3, 2]
 Generation 2690: Best = 3.140375e+03 FES = 269100 μ_F = 0.626 μ_{CR} = 0.027
 Diversity = 1.3031e+01 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 3.138116e+03 FES = 270100 μ_F = 0.679 μ_{CR} = 0.033
 Diversity = 1.2974e+01 k_t = [4, 4, 3, 4]
 Generation 2710: Best = 3.137326e+03 FES = 271100 μ_F = 0.685 μ_{CR} = 0.044
 Diversity = 1.2970e+01 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 3.137326e+03 FES = 272100 μ_F = 0.649 μ_{CR} = 0.029
 Diversity = 1.2968e+01 k_t = [3, 4, 4, 2]
 Generation 2730: Best = 3.131044e+03 FES = 273100 μ_F = 0.700 μ_{CR} = 0.024
 Diversity = 1.2961e+01 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 3.130905e+03 FES = 274100 μ_F = 0.775 μ_{CR} = 0.025
 Diversity = 1.2954e+01 k_t = [4, 4, 4, 3]
 Generation 2750: Best = 3.130905e+03 FES = 275100 μ_F = 0.805 μ_{CR} = 0.032
 Diversity = 1.2959e+01 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 3.130905e+03 FES = 276100 μ_F = 0.791 μ_{CR} = 0.033
 Diversity = 1.2939e+01 k_t = [3, 3, 4, 2]
 Generation 2770: Best = 3.130905e+03 FES = 277100 μ_F = 0.648 μ_{CR} = 0.022
 Diversity = 1.2936e+01 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 3.130905e+03 FES = 278100 μ_F = 0.613 μ_{CR} = 0.021
 Diversity = 1.2925e+01 k_t = [2, 4, 4, 2]
 Generation 2790: Best = 3.129233e+03 FES = 279100 μ_F = 0.538 μ_{CR} = 0.032
 Diversity = 1.2925e+01 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 3.128922e+03 FES = 280100 μ_F = 0.523 μ_{CR} = 0.029
 Diversity = 1.2931e+01 k_t = [3, 3, 2, 4]
 Generation 2810: Best = 3.128922e+03 FES = 281100 μ_F = 0.543 μ_{CR} = 0.032
 Diversity = 1.2925e+01 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 3.122834e+03 FES = 282100 μ_F = 0.494 μ_{CR} = 0.032
 Diversity = 1.2922e+01 k_t = [2, 3, 2, 2]
 Generation 2830: Best = 3.121487e+03 FES = 283100 μ_F = 0.441 μ_{CR} = 0.041
 Diversity = 1.2920e+01 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 3.121464e+03 FES = 284100 μ_F = 0.521 μ_{CR} = 0.042
 Diversity = 1.2917e+01 k_t = [2, 3, 4, 2]
 Generation 2850: Best = 3.121464e+03 FES = 285100 μ_F = 0.466 μ_{CR} = 0.035
 Diversity = 1.2917e+01 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 3.121464e+03 FES = 286100 μ_F = 0.576 μ_{CR} = 0.056
 Diversity = 1.2823e+01 k_t = [4, 2, 2, 4]
 Generation 2870: Best = 3.121464e+03 FES = 287100 μ_F = 0.449 μ_{CR} = 0.051
 Diversity = 1.2821e+01 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 3.120288e+03 FES = 288100 μ_F = 0.534 μ_{CR} = 0.057
 Diversity = 1.2817e+01 k_t = [3, 4, 3, 2]
 Generation 2890: Best = 3.120288e+03 FES = 289100 μ_F = 0.440 μ_{CR} = 0.040
 Diversity = 1.2814e+01 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 3.109819e+03 FES = 290100 μ_F = 0.445 μ_{CR} = 0.044
 Diversity = 1.2812e+01 k_t = [4, 4, 3, 2]
 Generation 2910: Best = 3.109726e+03 FES = 291100 μ_F = 0.500 μ_{CR} = 0.036
 Diversity = 1.2807e+01 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 3.109726e+03 FES = 292100 μ_F = 0.517 μ_{CR} = 0.038
 Diversity = 1.2806e+01 k_t = [3, 4, 4, 2]

Generation 2930: Best = 3.109726e+03	FES = 293100	$\mu_F = 0.495$	$\mu_{CR} = 0.042$
Diversity = 1.2804e+01	$k_t = [1, 1, 1, 1]$		
Generation 2940: Best = 3.109726e+03	FES = 294100	$\mu_F = 0.585$	$\mu_{CR} = 0.036$
Diversity = 1.2771e+01	$k_t = [2, 3, 2, 4]$		
Generation 2950: Best = 3.109726e+03	FES = 295100	$\mu_F = 0.512$	$\mu_{CR} = 0.036$
Diversity = 1.2772e+01	$k_t = [1, 1, 1, 1]$		
Generation 2960: Best = 3.109726e+03	FES = 296100	$\mu_F = 0.545$	$\mu_{CR} = 0.033$
Diversity = 1.2771e+01	$k_t = [3, 4, 2, 4]$		
Generation 2970: Best = 3.109726e+03	FES = 297100	$\mu_F = 0.605$	$\mu_{CR} = 0.035$
Diversity = 1.2779e+01	$k_t = [1, 1, 1, 1]$		
Generation 2980: Best = 3.109726e+03	FES = 298100	$\mu_F = 0.573$	$\mu_{CR} = 0.042$
Diversity = 1.2781e+01	$k_t = [2, 2, 4, 2]$		
Generation 2990: Best = 3.109726e+03	FES = 299100	$\mu_F = 0.600$	$\mu_{CR} = 0.041$
Diversity = 1.2771e+01	$k_t = [1, 1, 1, 1]$		
Generation 3000: Best = 3.109726e+03	FES = 300100	$\mu_F = 0.612$	$\mu_{CR} = 0.035$
Diversity = 1.2769e+01	$k_t = [4, 3, 3, 2]$		
Generation 3010: Best = 3.109726e+03	FES = 301100	$\mu_F = 0.596$	$\mu_{CR} = 0.026$
Diversity = 1.2775e+01	$k_t = [1, 1, 1, 1]$		
Generation 3020: Best = 3.105899e+03	FES = 302100	$\mu_F = 0.568$	$\mu_{CR} = 0.041$
Diversity = 1.2770e+01	$k_t = [3, 4, 2, 3]$		
Generation 3030: Best = 3.099106e+03	FES = 303100	$\mu_F = 0.630$	$\mu_{CR} = 0.044$
Diversity = 1.2769e+01	$k_t = [1, 1, 1, 1]$		
Generation 3040: Best = 3.096736e+03	FES = 304100	$\mu_F = 0.587$	$\mu_{CR} = 0.049$
Diversity = 1.2768e+01	$k_t = [3, 3, 3, 3]$		
Generation 3050: Best = 3.096736e+03	FES = 305100	$\mu_F = 0.619$	$\mu_{CR} = 0.045$
Diversity = 1.2770e+01	$k_t = [1, 1, 1, 1]$		
Generation 3060: Best = 3.091716e+03	FES = 306100	$\mu_F = 0.692$	$\mu_{CR} = 0.058$
Diversity = 1.2774e+01	$k_t = [2, 4, 3, 3]$		
Generation 3070: Best = 3.091138e+03	FES = 307100	$\mu_F = 0.698$	$\mu_{CR} = 0.042$
Diversity = 1.2772e+01	$k_t = [1, 1, 1, 1]$		
Generation 3080: Best = 3.091138e+03	FES = 308100	$\mu_F = 0.742$	$\mu_{CR} = 0.057$
Diversity = 1.2753e+01	$k_t = [3, 2, 4, 4]$		
Generation 3090: Best = 3.090907e+03	FES = 309100	$\mu_F = 0.694$	$\mu_{CR} = 0.060$
Diversity = 1.2752e+01	$k_t = [1, 1, 1, 1]$		
Generation 3100: Best = 3.077418e+03	FES = 310100	$\mu_F = 0.734$	$\mu_{CR} = 0.070$
Diversity = 1.2726e+01	$k_t = [4, 4, 2, 2]$		
Generation 3110: Best = 3.077418e+03	FES = 311100	$\mu_F = 0.651$	$\mu_{CR} = 0.040$
Diversity = 1.2686e+01	$k_t = [1, 1, 1, 1]$		
Generation 3120: Best = 3.076971e+03	FES = 312100	$\mu_F = 0.732$	$\mu_{CR} = 0.044$
Diversity = 1.2687e+01	$k_t = [4, 3, 2, 4]$		
Generation 3130: Best = 3.073590e+03	FES = 313100	$\mu_F = 0.681$	$\mu_{CR} = 0.038$
Diversity = 1.2685e+01	$k_t = [1, 1, 1, 1]$		
Generation 3140: Best = 3.073078e+03	FES = 314100	$\mu_F = 0.611$	$\mu_{CR} = 0.068$
Diversity = 1.2682e+01	$k_t = [4, 4, 4, 2]$		
Generation 3150: Best = 3.070207e+03	FES = 315100	$\mu_F = 0.627$	$\mu_{CR} = 0.070$
Diversity = 1.2680e+01	$k_t = [1, 1, 1, 1]$		
Generation 3160: Best = 3.070207e+03	FES = 316100	$\mu_F = 0.649$	$\mu_{CR} = 0.054$
Diversity = 1.2623e+01	$k_t = [2, 2, 2, 4]$		
Generation 3170: Best = 3.069549e+03	FES = 317100	$\mu_F = 0.687$	$\mu_{CR} = 0.064$
Diversity = 1.2624e+01	$k_t = [1, 1, 1, 1]$		
Generation 3180: Best = 3.064171e+03	FES = 318100	$\mu_F = 0.669$	$\mu_{CR} = 0.069$
Diversity = 1.2595e+01	$k_t = [4, 2, 3, 3]$		
Generation 3190: Best = 3.061005e+03	FES = 319100	$\mu_F = 0.728$	$\mu_{CR} = 0.056$
Diversity = 1.2592e+01	$k_t = [1, 1, 1, 1]$		

Generation 3200: Best = 3.059685e+03 FES = 320100 μ_F = 0.709 μ_{CR} = 0.058
 Diversity = 1.2537e+01 k_t = [4, 3, 3, 3]
 Generation 3210: Best = 3.057267e+03 FES = 321100 μ_F = 0.617 μ_{CR} = 0.063
 Diversity = 1.2528e+01 k_t = [1, 1, 1, 1]
 Generation 3220: Best = 3.055413e+03 FES = 322100 μ_F = 0.621 μ_{CR} = 0.090
 Diversity = 1.2521e+01 k_t = [4, 2, 2, 2]
 Generation 3230: Best = 3.055413e+03 FES = 323100 μ_F = 0.575 μ_{CR} = 0.078
 Diversity = 1.2520e+01 k_t = [1, 1, 1, 1]
 Generation 3240: Best = 3.055395e+03 FES = 324100 μ_F = 0.572 μ_{CR} = 0.084
 Diversity = 1.2470e+01 k_t = [4, 2, 3, 4]
 Generation 3250: Best = 3.055339e+03 FES = 325100 μ_F = 0.658 μ_{CR} = 0.074
 Diversity = 1.2450e+01 k_t = [1, 1, 1, 1]
 Generation 3260: Best = 3.055218e+03 FES = 326100 μ_F = 0.691 μ_{CR} = 0.067
 Diversity = 1.2433e+01 k_t = [4, 3, 3, 3]
 Generation 3270: Best = 3.055218e+03 FES = 327100 μ_F = 0.651 μ_{CR} = 0.086
 Diversity = 1.2404e+01 k_t = [1, 1, 1, 1]
 Generation 3280: Best = 3.054602e+03 FES = 328100 μ_F = 0.600 μ_{CR} = 0.076
 Diversity = 1.2405e+01 k_t = [2, 2, 2, 2]
 Generation 3290: Best = 3.054602e+03 FES = 329100 μ_F = 0.672 μ_{CR} = 0.073
 Diversity = 1.2401e+01 k_t = [1, 1, 1, 1]
 Generation 3300: Best = 3.053844e+03 FES = 330100 μ_F = 0.685 μ_{CR} = 0.067
 Diversity = 1.2378e+01 k_t = [4, 4, 4, 2]
 Generation 3310: Best = 3.053844e+03 FES = 331100 μ_F = 0.621 μ_{CR} = 0.058
 Diversity = 1.2377e+01 k_t = [1, 1, 1, 1]
 Generation 3320: Best = 3.053702e+03 FES = 332100 μ_F = 0.614 μ_{CR} = 0.054
 Diversity = 1.2329e+01 k_t = [2, 4, 3, 4]
 Generation 3330: Best = 3.053702e+03 FES = 333100 μ_F = 0.615 μ_{CR} = 0.050
 Diversity = 1.2249e+01 k_t = [1, 1, 1, 1]
 Generation 3340: Best = 3.053315e+03 FES = 334100 μ_F = 0.580 μ_{CR} = 0.058
 Diversity = 1.2196e+01 k_t = [4, 3, 4, 3]
 Generation 3350: Best = 3.051504e+03 FES = 335100 μ_F = 0.583 μ_{CR} = 0.064
 Diversity = 1.2174e+01 k_t = [1, 1, 1, 1]
 Generation 3360: Best = 3.051504e+03 FES = 336100 μ_F = 0.634 μ_{CR} = 0.062
 Diversity = 1.2175e+01 k_t = [3, 2, 4, 2]
 Generation 3370: Best = 3.050824e+03 FES = 337100 μ_F = 0.635 μ_{CR} = 0.082
 Diversity = 1.2151e+01 k_t = [1, 1, 1, 1]
 Generation 3380: Best = 3.050340e+03 FES = 338100 μ_F = 0.672 μ_{CR} = 0.079
 Diversity = 1.2118e+01 k_t = [3, 3, 3, 3]
 Generation 3390: Best = 3.050340e+03 FES = 339100 μ_F = 0.756 μ_{CR} = 0.096
 Diversity = 1.2049e+01 k_t = [1, 1, 1, 1]
 Generation 3400: Best = 3.049957e+03 FES = 340100 μ_F = 0.805 μ_{CR} = 0.111
 Diversity = 1.1851e+01 k_t = [2, 3, 4, 4]
 Generation 3410: Best = 3.046020e+03 FES = 341100 μ_F = 0.785 μ_{CR} = 0.102
 Diversity = 1.1790e+01 k_t = [1, 1, 1, 1]
 Generation 3420: Best = 3.046020e+03 FES = 342100 μ_F = 0.790 μ_{CR} = 0.102
 Diversity = 1.1701e+01 k_t = [2, 2, 4, 4]
 Generation 3430: Best = 3.045868e+03 FES = 343100 μ_F = 0.816 μ_{CR} = 0.095
 Diversity = 1.1614e+01 k_t = [1, 1, 1, 1]
 Generation 3440: Best = 3.041636e+03 FES = 344100 μ_F = 0.832 μ_{CR} = 0.095
 Diversity = 1.1549e+01 k_t = [2, 3, 2, 2]
 Generation 3450: Best = 3.041185e+03 FES = 345100 μ_F = 0.846 μ_{CR} = 0.107
 Diversity = 1.1416e+01 k_t = [1, 1, 1, 1]
 Generation 3460: Best = 3.038640e+03 FES = 346100 μ_F = 0.869 μ_{CR} = 0.113
 Diversity = 1.0948e+01 k_t = [3, 2, 4, 4]

Generation 3470: Best = 3.037934e+03 FES = 347100 μ_F = 0.856 μ_{CR} = 0.113
 Diversity = 1.0735e+01 k_t = [1, 1, 1, 1]
 Generation 3480: Best = 3.034679e+03 FES = 348100 μ_F = 0.859 μ_{CR} = 0.121
 Diversity = 1.0529e+01 k_t = [4, 3, 2, 3]
 Generation 3490: Best = 3.029023e+03 FES = 349100 μ_F = 0.855 μ_{CR} = 0.135
 Diversity = 9.8774e+00 k_t = [1, 1, 1, 1]
 Generation 3500: Best = 3.028275e+03 FES = 350100 μ_F = 0.861 μ_{CR} = 0.143
 Diversity = 4.9404e+00 k_t = [3, 2, 2, 4]
 Generation 3510: Best = 3.027169e+03 FES = 351100 μ_F = 0.859 μ_{CR} = 0.142
 Diversity = 7.5400e-01 k_t = [1, 1, 1, 1]
 Generation 3520: Best = 3.026620e+03 FES = 352100 μ_F = 0.839 μ_{CR} = 0.153
 Diversity = 5.8275e-01 k_t = [2, 4, 2, 4]
 Generation 3530: Best = 3.024073e+03 FES = 353100 μ_F = 0.813 μ_{CR} = 0.152
 Diversity = 4.8102e-01 k_t = [1, 1, 1, 1]
 Generation 3540: Best = 3.021255e+03 FES = 354100 μ_F = 0.785 μ_{CR} = 0.151
 Diversity = 4.4213e-01 k_t = [4, 2, 4, 3]
 Generation 3550: Best = 3.020146e+03 FES = 355100 μ_F = 0.761 μ_{CR} = 0.162
 Diversity = 4.0011e-01 k_t = [1, 1, 1, 1]
 Generation 3560: Best = 3.018290e+03 FES = 356100 μ_F = 0.751 μ_{CR} = 0.169
 Diversity = 3.4270e-01 k_t = [4, 4, 4, 2]
 Generation 3570: Best = 3.016253e+03 FES = 357100 μ_F = 0.725 μ_{CR} = 0.180
 Diversity = 2.8975e-01 k_t = [1, 1, 1, 1]
 Generation 3580: Best = 3.014819e+03 FES = 358100 μ_F = 0.708 μ_{CR} = 0.193
 Diversity = 2.1832e-01 k_t = [2, 2, 3, 2]
 Generation 3590: Best = 3.013940e+03 FES = 359100 μ_F = 0.685 μ_{CR} = 0.191
 Diversity = 1.8919e-01 k_t = [1, 1, 1, 1]
 Generation 3600: Best = 3.013330e+03 FES = 360100 μ_F = 0.670 μ_{CR} = 0.195
 Diversity = 1.3973e-01 k_t = [3, 2, 3, 4]
 Generation 3610: Best = 3.012886e+03 FES = 361100 μ_F = 0.647 μ_{CR} = 0.194
 Diversity = 1.0418e-01 k_t = [1, 1, 1, 1]
 Generation 3620: Best = 3.012529e+03 FES = 362100 μ_F = 0.656 μ_{CR} = 0.203
 Diversity = 8.1854e-02 k_t = [2, 4, 2, 4]
 Generation 3630: Best = 3.012229e+03 FES = 363100 μ_F = 0.659 μ_{CR} = 0.200
 Diversity = 7.0249e-02 k_t = [1, 1, 1, 1]
 Generation 3640: Best = 3.012031e+03 FES = 364100 μ_F = 0.658 μ_{CR} = 0.205
 Diversity = 6.0528e-02 k_t = [4, 4, 2, 3]
 Generation 3650: Best = 3.011794e+03 FES = 365100 μ_F = 0.658 μ_{CR} = 0.206
 Diversity = 5.2461e-02 k_t = [1, 1, 1, 1]
 Generation 3660: Best = 3.011663e+03 FES = 366100 μ_F = 0.665 μ_{CR} = 0.204
 Diversity = 4.5610e-02 k_t = [3, 3, 2, 4]
 Generation 3670: Best = 3.011478e+03 FES = 367100 μ_F = 0.677 μ_{CR} = 0.201
 Diversity = 3.9721e-02 k_t = [1, 1, 1, 1]
 Generation 3680: Best = 3.011432e+03 FES = 368100 μ_F = 0.680 μ_{CR} = 0.198
 Diversity = 3.7253e-02 k_t = [4, 3, 2, 2]
 Generation 3690: Best = 3.011336e+03 FES = 369100 μ_F = 0.695 μ_{CR} = 0.200
 Diversity = 3.7195e-02 k_t = [1, 1, 1, 1]
 Generation 3700: Best = 3.011275e+03 FES = 370100 μ_F = 0.701 μ_{CR} = 0.194
 Diversity = 3.6401e-02 k_t = [4, 3, 3, 2]
 Generation 3710: Best = 3.011193e+03 FES = 371100 μ_F = 0.692 μ_{CR} = 0.206
 Diversity = 3.7039e-02 k_t = [1, 1, 1, 1]
 Generation 3720: Best = 3.011037e+03 FES = 372100 μ_F = 0.694 μ_{CR} = 0.219
 Diversity = 3.1198e-02 k_t = [4, 2, 4, 3]
 Generation 3730: Best = 3.010979e+03 FES = 373100 μ_F = 0.685 μ_{CR} = 0.219
 Diversity = 2.8616e-02 k_t = [1, 1, 1, 1]

Generation 3740: Best = 3.010946e+03 FES = 374100 μ_F = 0.702 μ_{CR} = 0.212
 Diversity = 2.5278e-02 k_t = [4, 4, 3, 3]
 Generation 3750: Best = 3.010880e+03 FES = 375100 μ_F = 0.715 μ_{CR} = 0.211
 Diversity = 2.5298e-02 k_t = [1, 1, 1, 1]
 Generation 3760: Best = 3.010818e+03 FES = 376100 μ_F = 0.716 μ_{CR} = 0.208
 Diversity = 2.4303e-02 k_t = [3, 2, 4, 2]
 Generation 3770: Best = 3.010782e+03 FES = 377100 μ_F = 0.711 μ_{CR} = 0.205
 Diversity = 2.3700e-02 k_t = [1, 1, 1, 1]
 Generation 3780: Best = 3.010763e+03 FES = 378100 μ_F = 0.692 μ_{CR} = 0.209
 Diversity = 2.1049e-02 k_t = [3, 3, 4, 2]
 Generation 3790: Best = 3.010727e+03 FES = 379100 μ_F = 0.689 μ_{CR} = 0.212
 Diversity = 1.9621e-02 k_t = [1, 1, 1, 1]
 Generation 3800: Best = 3.010708e+03 FES = 380100 μ_F = 0.692 μ_{CR} = 0.215
 Diversity = 1.6314e-02 k_t = [3, 3, 3, 4]
 Generation 3810: Best = 3.010692e+03 FES = 381100 μ_F = 0.686 μ_{CR} = 0.217
 Diversity = 1.4582e-02 k_t = [1, 1, 1, 1]
 Generation 3820: Best = 3.010682e+03 FES = 382100 μ_F = 0.679 μ_{CR} = 0.224
 Diversity = 1.2351e-02 k_t = [4, 2, 2, 4]
 Generation 3830: Best = 3.010666e+03 FES = 383100 μ_F = 0.688 μ_{CR} = 0.224
 Diversity = 1.1573e-02 k_t = [1, 1, 1, 1]
 Generation 3840: Best = 3.010657e+03 FES = 384100 μ_F = 0.695 μ_{CR} = 0.221
 Diversity = 1.1186e-02 k_t = [2, 2, 4, 2]
 Generation 3850: Best = 3.010644e+03 FES = 385100 μ_F = 0.692 μ_{CR} = 0.213
 Diversity = 1.0876e-02 k_t = [1, 1, 1, 1]
 Generation 3860: Best = 3.010636e+03 FES = 386100 μ_F = 0.693 μ_{CR} = 0.221
 Diversity = 1.0640e-02 k_t = [4, 2, 2, 2]
 Generation 3870: Best = 3.010629e+03 FES = 387100 μ_F = 0.701 μ_{CR} = 0.222
 Diversity = 1.0289e-02 k_t = [1, 1, 1, 1]
 Generation 3880: Best = 3.010621e+03 FES = 388100 μ_F = 0.705 μ_{CR} = 0.221
 Diversity = 9.0660e-03 k_t = [4, 3, 4, 4]
 Generation 3890: Best = 3.010617e+03 FES = 389100 μ_F = 0.701 μ_{CR} = 0.211
 Diversity = 8.0969e-03 k_t = [1, 1, 1, 1]
 Generation 3900: Best = 3.010611e+03 FES = 390100 μ_F = 0.702 μ_{CR} = 0.212
 Diversity = 7.2719e-03 k_t = [3, 3, 3, 4]
 Generation 3910: Best = 3.010608e+03 FES = 391100 μ_F = 0.704 μ_{CR} = 0.216
 Diversity = 6.7082e-03 k_t = [1, 1, 1, 1]
 Generation 3920: Best = 3.010606e+03 FES = 392100 μ_F = 0.710 μ_{CR} = 0.212
 Diversity = 6.2175e-03 k_t = [4, 3, 3, 2]
 Generation 3930: Best = 3.010602e+03 FES = 393100 μ_F = 0.704 μ_{CR} = 0.212
 Diversity = 6.1682e-03 k_t = [1, 1, 1, 1]
 Generation 3940: Best = 3.010599e+03 FES = 394100 μ_F = 0.710 μ_{CR} = 0.204
 Diversity = 5.6589e-03 k_t = [2, 4, 2, 3]
 Generation 3950: Best = 3.010596e+03 FES = 395100 μ_F = 0.714 μ_{CR} = 0.213
 Diversity = 5.7321e-03 k_t = [1, 1, 1, 1]
 Generation 3960: Best = 3.010594e+03 FES = 396100 μ_F = 0.718 μ_{CR} = 0.223
 Diversity = 5.4921e-03 k_t = [2, 2, 2, 3]
 Generation 3970: Best = 3.010589e+03 FES = 397100 μ_F = 0.728 μ_{CR} = 0.225
 Diversity = 6.0971e-03 k_t = [1, 1, 1, 1]
 Generation 3980: Best = 3.010585e+03 FES = 398100 μ_F = 0.729 μ_{CR} = 0.226
 Diversity = 5.9058e-03 k_t = [4, 4, 4, 3]
 Generation 3990: Best = 3.010581e+03 FES = 399100 μ_F = 0.741 μ_{CR} = 0.228
 Diversity = 6.0776e-03 k_t = [1, 1, 1, 1]
 Generation 4000: Best = 3.010576e+03 FES = 400100 μ_F = 0.730 μ_{CR} = 0.223
 Diversity = 6.2742e-03 k_t = [4, 2, 2, 4]

Generation 4010: Best = 3.010572e+03 FES = 401100 μ_F = 0.734 μ_{CR} = 0.225
 Diversity = 6.2516e-03 k_t = [1, 1, 1, 1]
 Generation 4020: Best = 3.010569e+03 FES = 402100 μ_F = 0.718 μ_{CR} = 0.228
 Diversity = 5.7733e-03 k_t = [3, 3, 3, 4]
 Generation 4030: Best = 3.010566e+03 FES = 403100 μ_F = 0.710 μ_{CR} = 0.228
 Diversity = 5.3160e-03 k_t = [1, 1, 1, 1]
 Generation 4040: Best = 3.010563e+03 FES = 404100 μ_F = 0.717 μ_{CR} = 0.232
 Diversity = 4.9662e-03 k_t = [3, 3, 3, 4]
 Generation 4050: Best = 3.010561e+03 FES = 405100 μ_F = 0.728 μ_{CR} = 0.228
 Diversity = 5.1119e-03 k_t = [1, 1, 1, 1]
 Generation 4060: Best = 3.010558e+03 FES = 406100 μ_F = 0.737 μ_{CR} = 0.219
 Diversity = 5.1985e-03 k_t = [4, 2, 4, 3]
 Generation 4070: Best = 3.010556e+03 FES = 407100 μ_F = 0.727 μ_{CR} = 0.226
 Diversity = 5.0349e-03 k_t = [1, 1, 1, 1]
 Generation 4080: Best = 3.010553e+03 FES = 408100 μ_F = 0.700 μ_{CR} = 0.229
 Diversity = 4.6596e-03 k_t = [3, 3, 3, 3]
 Generation 4090: Best = 3.010551e+03 FES = 409100 μ_F = 0.710 μ_{CR} = 0.232
 Diversity = 4.7412e-03 k_t = [1, 1, 1, 1]
 Generation 4100: Best = 3.010549e+03 FES = 410100 μ_F = 0.715 μ_{CR} = 0.236
 Diversity = 4.8089e-03 k_t = [2, 3, 2, 4]
 Generation 4110: Best = 3.010545e+03 FES = 411100 μ_F = 0.722 μ_{CR} = 0.219
 Diversity = 4.7800e-03 k_t = [1, 1, 1, 1]
 Generation 4120: Best = 3.010538e+03 FES = 412100 μ_F = 0.734 μ_{CR} = 0.219
 Diversity = 5.4142e-03 k_t = [2, 3, 3, 4]
 Generation 4130: Best = 3.010534e+03 FES = 413100 μ_F = 0.736 μ_{CR} = 0.227
 Diversity = 5.9761e-03 k_t = [1, 1, 1, 1]
 Generation 4140: Best = 3.010534e+03 FES = 414100 μ_F = 0.726 μ_{CR} = 0.231
 Diversity = 4.7679e-03 k_t = [3, 2, 3, 4]
 Generation 4150: Best = 3.010533e+03 FES = 415100 μ_F = 0.707 μ_{CR} = 0.223
 Diversity = 4.0186e-03 k_t = [1, 1, 1, 1]
 Generation 4160: Best = 3.010532e+03 FES = 416100 μ_F = 0.698 μ_{CR} = 0.222
 Diversity = 3.4528e-03 k_t = [4, 2, 3, 3]
 Generation 4170: Best = 3.010531e+03 FES = 417100 μ_F = 0.699 μ_{CR} = 0.227
 Diversity = 3.2395e-03 k_t = [1, 1, 1, 1]
 Generation 4180: Best = 3.010530e+03 FES = 418100 μ_F = 0.710 μ_{CR} = 0.234
 Diversity = 3.1054e-03 k_t = [2, 3, 4, 3]
 Generation 4190: Best = 3.010530e+03 FES = 419100 μ_F = 0.713 μ_{CR} = 0.230
 Diversity = 3.0316e-03 k_t = [1, 1, 1, 1]
 Generation 4200: Best = 3.010529e+03 FES = 420100 μ_F = 0.711 μ_{CR} = 0.221
 Diversity = 2.6345e-03 k_t = [4, 2, 4, 3]
 Generation 4210: Best = 3.010528e+03 FES = 421100 μ_F = 0.705 μ_{CR} = 0.232
 Diversity = 2.5657e-03 k_t = [1, 1, 1, 1]
 Generation 4220: Best = 3.010528e+03 FES = 422100 μ_F = 0.709 μ_{CR} = 0.237
 Diversity = 2.4278e-03 k_t = [4, 3, 4, 2]
 Generation 4230: Best = 3.010528e+03 FES = 423100 μ_F = 0.702 μ_{CR} = 0.237
 Diversity = 2.2542e-03 k_t = [1, 1, 1, 1]
 Generation 4240: Best = 3.010527e+03 FES = 424100 μ_F = 0.698 μ_{CR} = 0.238
 Diversity = 1.8632e-03 k_t = [3, 3, 2, 3]
 Generation 4250: Best = 3.010527e+03 FES = 425100 μ_F = 0.683 μ_{CR} = 0.242
 Diversity = 1.7965e-03 k_t = [1, 1, 1, 1]
 Generation 4260: Best = 3.010527e+03 FES = 426100 μ_F = 0.690 μ_{CR} = 0.231
 Diversity = 1.5200e-03 k_t = [3, 4, 4, 4]
 Generation 4270: Best = 3.010527e+03 FES = 427100 μ_F = 0.694 μ_{CR} = 0.228
 Diversity = 1.3488e-03 k_t = [1, 1, 1, 1]

Generation 4280: Best = 3.010527e+03 FES = 428100 μ_F = 0.694 μ_{CR} = 0.236
 Diversity = 1.1969e-03 k_t = [3, 4, 2, 4]
 Generation 4290: Best = 3.010527e+03 FES = 429100 μ_F = 0.702 μ_{CR} = 0.241
 Diversity = 1.2354e-03 k_t = [1, 1, 1, 1]
 Generation 4300: Best = 3.010526e+03 FES = 430100 μ_F = 0.703 μ_{CR} = 0.232
 Diversity = 1.2080e-03 k_t = [3, 3, 3, 3]
 Generation 4310: Best = 3.010526e+03 FES = 431100 μ_F = 0.699 μ_{CR} = 0.230
 Diversity = 1.1752e-03 k_t = [1, 1, 1, 1]
 Generation 4320: Best = 3.010526e+03 FES = 432100 μ_F = 0.691 μ_{CR} = 0.230
 Diversity = 1.0636e-03 k_t = [2, 3, 3, 2]
 Generation 4330: Best = 3.010526e+03 FES = 433100 μ_F = 0.703 μ_{CR} = 0.212
 Diversity = 1.0144e-03 k_t = [1, 1, 1, 1]
 Generation 4340: Best = 3.010526e+03 FES = 434100 μ_F = 0.708 μ_{CR} = 0.209
 Diversity = 9.1280e-04 k_t = [2, 2, 4, 3]
 Generation 4350: Best = 3.010526e+03 FES = 435100 μ_F = 0.705 μ_{CR} = 0.206
 Diversity = 9.5637e-04 k_t = [1, 1, 1, 1]
 Generation 4360: Best = 3.010526e+03 FES = 436100 μ_F = 0.712 μ_{CR} = 0.213
 Diversity = 9.5145e-04 k_t = [4, 4, 2, 2]
 Generation 4370: Best = 3.010526e+03 FES = 437100 μ_F = 0.714 μ_{CR} = 0.221
 Diversity = 9.2641e-04 k_t = [1, 1, 1, 1]
 Generation 4380: Best = 3.010526e+03 FES = 438100 μ_F = 0.719 μ_{CR} = 0.238
 Diversity = 9.0904e-04 k_t = [4, 2, 3, 4]
 Generation 4390: Best = 3.010526e+03 FES = 439100 μ_F = 0.717 μ_{CR} = 0.228
 Diversity = 9.6735e-04 k_t = [1, 1, 1, 1]
 Generation 4400: Best = 3.010525e+03 FES = 440100 μ_F = 0.725 μ_{CR} = 0.226
 Diversity = 9.2860e-04 k_t = [3, 2, 2, 4]
 Generation 4410: Best = 3.010525e+03 FES = 441100 μ_F = 0.712 μ_{CR} = 0.235
 Diversity = 9.3647e-04 k_t = [1, 1, 1, 1]
 Generation 4420: Best = 3.010525e+03 FES = 442100 μ_F = 0.707 μ_{CR} = 0.247
 Diversity = 8.9642e-04 k_t = [3, 4, 4, 4]
 Generation 4430: Best = 3.010525e+03 FES = 443100 μ_F = 0.709 μ_{CR} = 0.254
 Diversity = 8.5560e-04 k_t = [1, 1, 1, 1]
 Generation 4440: Best = 3.010525e+03 FES = 444100 μ_F = 0.709 μ_{CR} = 0.261
 Diversity = 8.2431e-04 k_t = [3, 4, 4, 2]
 Generation 4450: Best = 3.010525e+03 FES = 445100 μ_F = 0.711 μ_{CR} = 0.263
 Diversity = 7.7955e-04 k_t = [1, 1, 1, 1]
 Generation 4460: Best = 3.010525e+03 FES = 446100 μ_F = 0.690 μ_{CR} = 0.268
 Diversity = 6.8227e-04 k_t = [4, 3, 3, 4]
 Generation 4470: Best = 3.010525e+03 FES = 447100 μ_F = 0.697 μ_{CR} = 0.257
 Diversity = 6.3559e-04 k_t = [1, 1, 1, 1]
 Generation 4480: Best = 3.010525e+03 FES = 448100 μ_F = 0.700 μ_{CR} = 0.260
 Diversity = 6.2558e-04 k_t = [4, 2, 3, 2]
 Generation 4490: Best = 3.010525e+03 FES = 449100 μ_F = 0.713 μ_{CR} = 0.261
 Diversity = 6.2290e-04 k_t = [1, 1, 1, 1]
 Generation 4500: Best = 3.010525e+03 FES = 450100 μ_F = 0.715 μ_{CR} = 0.269
 Diversity = 6.0419e-04 k_t = [2, 4, 4, 2]
 Generation 4510: Best = 3.010525e+03 FES = 451100 μ_F = 0.712 μ_{CR} = 0.268
 Diversity = 5.4177e-04 k_t = [1, 1, 1, 1]
 Generation 4520: Best = 3.010525e+03 FES = 452100 μ_F = 0.711 μ_{CR} = 0.269
 Diversity = 5.6135e-04 k_t = [4, 2, 3, 2]
 Generation 4530: Best = 3.010525e+03 FES = 453100 μ_F = 0.711 μ_{CR} = 0.266
 Diversity = 6.0479e-04 k_t = [1, 1, 1, 1]
 Generation 4540: Best = 3.010525e+03 FES = 454100 μ_F = 0.709 μ_{CR} = 0.268
 Diversity = 5.7306e-04 k_t = [4, 4, 3, 3]

Generation 4550: Best = 3.010525e+03 FES = 455100 μ_F = 0.713 μ_{CR} = 0.274
 Diversity = 5.7664e-04 k_t = [1, 1, 1, 1]
 Generation 4560: Best = 3.010525e+03 FES = 456100 μ_F = 0.708 μ_{CR} = 0.276
 Diversity = 5.2150e-04 k_t = [3, 4, 3, 3]
 Generation 4570: Best = 3.010525e+03 FES = 457100 μ_F = 0.707 μ_{CR} = 0.267
 Diversity = 5.1718e-04 k_t = [1, 1, 1, 1]
 Generation 4580: Best = 3.010525e+03 FES = 458100 μ_F = 0.715 μ_{CR} = 0.272
 Diversity = 5.4087e-04 k_t = [2, 2, 4, 2]
 Generation 4590: Best = 3.010524e+03 FES = 459100 μ_F = 0.715 μ_{CR} = 0.273
 Diversity = 5.6836e-04 k_t = [1, 1, 1, 1]
 Generation 4600: Best = 3.010524e+03 FES = 460100 μ_F = 0.709 μ_{CR} = 0.276
 Diversity = 5.7242e-04 k_t = [3, 3, 3, 2]
 Generation 4610: Best = 3.010524e+03 FES = 461100 μ_F = 0.713 μ_{CR} = 0.280
 Diversity = 6.2442e-04 k_t = [1, 1, 1, 1]
 Generation 4620: Best = 3.010524e+03 FES = 462100 μ_F = 0.712 μ_{CR} = 0.281
 Diversity = 6.2089e-04 k_t = [4, 4, 3, 3]
 Generation 4630: Best = 3.010524e+03 FES = 463100 μ_F = 0.723 μ_{CR} = 0.289
 Diversity = 6.4846e-04 k_t = [1, 1, 1, 1]
 Generation 4640: Best = 3.010524e+03 FES = 464100 μ_F = 0.719 μ_{CR} = 0.288
 Diversity = 6.6405e-04 k_t = [4, 2, 2, 2]
 Generation 4650: Best = 3.010524e+03 FES = 465100 μ_F = 0.713 μ_{CR} = 0.279
 Diversity = 6.6319e-04 k_t = [1, 1, 1, 1]
 Generation 4660: Best = 3.010524e+03 FES = 466100 μ_F = 0.705 μ_{CR} = 0.274
 Diversity = 6.2382e-04 k_t = [4, 3, 3, 2]
 Generation 4670: Best = 3.010524e+03 FES = 467100 μ_F = 0.698 μ_{CR} = 0.281
 Diversity = 5.9482e-04 k_t = [1, 1, 1, 1]
 Generation 4680: Best = 3.010524e+03 FES = 468100 μ_F = 0.677 μ_{CR} = 0.283
 Diversity = 4.8493e-04 k_t = [4, 3, 2, 3]
 Generation 4690: Best = 3.010524e+03 FES = 469100 μ_F = 0.677 μ_{CR} = 0.279
 Diversity = 4.0112e-04 k_t = [1, 1, 1, 1]
 Generation 4700: Best = 3.010524e+03 FES = 470100 μ_F = 0.683 μ_{CR} = 0.282
 Diversity = 3.4977e-04 k_t = [3, 4, 4, 2]
 Generation 4710: Best = 3.010524e+03 FES = 471100 μ_F = 0.688 μ_{CR} = 0.282
 Diversity = 3.3560e-04 k_t = [1, 1, 1, 1]
 Generation 4720: Best = 3.010524e+03 FES = 472100 μ_F = 0.682 μ_{CR} = 0.279
 Diversity = 3.1815e-04 k_t = [3, 4, 2, 3]
 Generation 4730: Best = 3.010524e+03 FES = 473100 μ_F = 0.670 μ_{CR} = 0.258
 Diversity = 3.0585e-04 k_t = [1, 1, 1, 1]
 Generation 4740: Best = 3.010524e+03 FES = 474100 μ_F = 0.675 μ_{CR} = 0.263
 Diversity = 2.6763e-04 k_t = [4, 4, 3, 4]
 Generation 4750: Best = 3.010524e+03 FES = 475100 μ_F = 0.674 μ_{CR} = 0.269
 Diversity = 2.2915e-04 k_t = [1, 1, 1, 1]
 Generation 4760: Best = 3.010524e+03 FES = 476100 μ_F = 0.674 μ_{CR} = 0.272
 Diversity = 2.1958e-04 k_t = [2, 2, 4, 3]
 Generation 4770: Best = 3.010524e+03 FES = 477100 μ_F = 0.687 μ_{CR} = 0.284
 Diversity = 2.4267e-04 k_t = [1, 1, 1, 1]
 Generation 4780: Best = 3.010524e+03 FES = 478100 μ_F = 0.691 μ_{CR} = 0.278
 Diversity = 2.0504e-04 k_t = [3, 3, 3, 3]
 Generation 4790: Best = 3.010524e+03 FES = 479100 μ_F = 0.692 μ_{CR} = 0.269
 Diversity = 1.9497e-04 k_t = [1, 1, 1, 1]
 Generation 4800: Best = 3.010524e+03 FES = 480100 μ_F = 0.697 μ_{CR} = 0.271
 Diversity = 1.9351e-04 k_t = [4, 3, 3, 3]
 Generation 4810: Best = 3.010524e+03 FES = 481100 μ_F = 0.706 μ_{CR} = 0.265
 Diversity = 1.7258e-04 k_t = [1, 1, 1, 1]

Generation 4820: Best = 3.010524e+03 FES = 482100 μ_F = 0.703 μ_{CR} = 0.265
 Diversity = 1.4595e-04 k_t = [4, 3, 2, 4]
 Generation 4830: Best = 3.010524e+03 FES = 483100 μ_F = 0.698 μ_{CR} = 0.268
 Diversity = 1.3101e-04 k_t = [1, 1, 1, 1]
 Generation 4840: Best = 3.010524e+03 FES = 484100 μ_F = 0.706 μ_{CR} = 0.272
 Diversity = 1.2190e-04 k_t = [2, 3, 3, 4]
 Generation 4850: Best = 3.010524e+03 FES = 485100 μ_F = 0.699 μ_{CR} = 0.271
 Diversity = 1.2271e-04 k_t = [1, 1, 1, 1]
 Generation 4860: Best = 3.010524e+03 FES = 486100 μ_F = 0.705 μ_{CR} = 0.277
 Diversity = 1.1757e-04 k_t = [3, 3, 3, 2]
 Generation 4870: Best = 3.010524e+03 FES = 487100 μ_F = 0.704 μ_{CR} = 0.275
 Diversity = 1.1766e-04 k_t = [1, 1, 1, 1]
 Generation 4880: Best = 3.010524e+03 FES = 488100 μ_F = 0.697 μ_{CR} = 0.270
 Diversity = 1.0635e-04 k_t = [2, 4, 2, 2]
 Generation 4890: Best = 3.010524e+03 FES = 489100 μ_F = 0.686 μ_{CR} = 0.272
 Diversity = 1.0209e-04 k_t = [1, 1, 1, 1]
 Generation 4900: Best = 3.010524e+03 FES = 490100 μ_F = 0.683 μ_{CR} = 0.273
 Diversity = 9.2104e-05 k_t = [3, 4, 2, 2]
 Generation 4910: Best = 3.010524e+03 FES = 491100 μ_F = 0.690 μ_{CR} = 0.274
 Diversity = 8.7100e-05 k_t = [1, 1, 1, 1]
 Generation 4920: Best = 3.010524e+03 FES = 492100 μ_F = 0.688 μ_{CR} = 0.272
 Diversity = 7.3490e-05 k_t = [2, 3, 4, 4]
 Generation 4930: Best = 3.010524e+03 FES = 493100 μ_F = 0.683 μ_{CR} = 0.281
 Diversity = 7.2399e-05 k_t = [1, 1, 1, 1]
 Generation 4940: Best = 3.010524e+03 FES = 494100 μ_F = 0.696 μ_{CR} = 0.283
 Diversity = 7.1598e-05 k_t = [4, 4, 4, 3]
 Generation 4950: Best = 3.010524e+03 FES = 495100 μ_F = 0.700 μ_{CR} = 0.287
 Diversity = 7.5543e-05 k_t = [1, 1, 1, 1]
 Generation 4960: Best = 3.010524e+03 FES = 496100 μ_F = 0.710 μ_{CR} = 0.279
 Diversity = 7.5085e-05 k_t = [2, 3, 3, 3]
 Generation 4970: Best = 3.010524e+03 FES = 497100 μ_F = 0.707 μ_{CR} = 0.284
 Diversity = 8.0972e-05 k_t = [1, 1, 1, 1]
 Generation 4980: Best = 3.010524e+03 FES = 498100 μ_F = 0.724 μ_{CR} = 0.285
 Diversity = 9.0651e-05 k_t = [4, 4, 4, 2]
 Generation 4990: Best = 3.010524e+03 FES = 499100 μ_F = 0.742 μ_{CR} = 0.280
 Diversity = 1.0450e-04 k_t = [1, 1, 1, 1]
 Generation 5000: Best = 3.010524e+03 FES = 500100 μ_F = 0.737 μ_{CR} = 0.284
 Diversity = 1.0887e-04 k_t = [4, 3, 4, 3]
 Generation 5010: Best = 3.010524e+03 FES = 501100 μ_F = 0.743 μ_{CR} = 0.275
 Diversity = 1.0815e-04 k_t = [1, 1, 1, 1]
 Generation 5020: Best = 3.010524e+03 FES = 502100 μ_F = 0.737 μ_{CR} = 0.273
 Diversity = 1.0224e-04 k_t = [2, 3, 3, 4]
 Generation 5030: Best = 3.010524e+03 FES = 503100 μ_F = 0.727 μ_{CR} = 0.269
 Diversity = 1.0137e-04 k_t = [1, 1, 1, 1]
 Generation 5040: Best = 3.010524e+03 FES = 504100 μ_F = 0.713 μ_{CR} = 0.270
 Diversity = 9.3138e-05 k_t = [4, 3, 3, 2]
 Generation 5050: Best = 3.010524e+03 FES = 505100 μ_F = 0.706 μ_{CR} = 0.281
 Diversity = 9.0521e-05 k_t = [1, 1, 1, 1]
 Generation 5060: Best = 3.010524e+03 FES = 506100 μ_F = 0.704 μ_{CR} = 0.285
 Diversity = 8.6564e-05 k_t = [3, 2, 2, 2]
 Generation 5070: Best = 3.010524e+03 FES = 507100 μ_F = 0.713 μ_{CR} = 0.292
 Diversity = 8.8452e-05 k_t = [1, 1, 1, 1]
 Generation 5080: Best = 3.010524e+03 FES = 508100 μ_F = 0.712 μ_{CR} = 0.292
 Diversity = 8.8446e-05 k_t = [4, 3, 4, 3]

Generation 5090: Best = 3.010524e+03 FES = 509100 μ_F = 0.711 μ_{CR} = 0.295
 Diversity = 9.4694e-05 k_t = [1, 1, 1, 1]
 Generation 5100: Best = 3.010524e+03 FES = 510100 μ_F = 0.714 μ_{CR} = 0.294
 Diversity = 8.8168e-05 k_t = [2, 4, 3, 3]
 Generation 5110: Best = 3.010524e+03 FES = 511100 μ_F = 0.714 μ_{CR} = 0.285
 Diversity = 1.0334e-04 k_t = [1, 1, 1, 1]
 Generation 5120: Best = 3.010524e+03 FES = 512100 μ_F = 0.720 μ_{CR} = 0.281
 Diversity = 9.8530e-05 k_t = [2, 4, 2, 4]
 Generation 5130: Best = 3.010524e+03 FES = 513100 μ_F = 0.732 μ_{CR} = 0.277
 Diversity = 9.5825e-05 k_t = [1, 1, 1, 1]
 Generation 5140: Best = 3.010524e+03 FES = 514100 μ_F = 0.740 μ_{CR} = 0.274
 Diversity = 9.7688e-05 k_t = [4, 3, 3, 4]
 Generation 5150: Best = 3.010524e+03 FES = 515100 μ_F = 0.735 μ_{CR} = 0.263
 Diversity = 1.0047e-04 k_t = [1, 1, 1, 1]
 Generation 5160: Best = 3.010524e+03 FES = 516100 μ_F = 0.741 μ_{CR} = 0.272
 Diversity = 1.0169e-04 k_t = [2, 2, 3, 4]
 Generation 5170: Best = 3.010524e+03 FES = 517100 μ_F = 0.731 μ_{CR} = 0.269
 Diversity = 1.0988e-04 k_t = [1, 1, 1, 1]
 Generation 5180: Best = 3.010524e+03 FES = 518100 μ_F = 0.733 μ_{CR} = 0.274
 Diversity = 1.1037e-04 k_t = [3, 3, 4, 4]
 Generation 5190: Best = 3.010524e+03 FES = 519100 μ_F = 0.735 μ_{CR} = 0.271
 Diversity = 1.2508e-04 k_t = [1, 1, 1, 1]
 Generation 5200: Best = 3.010524e+03 FES = 520100 μ_F = 0.734 μ_{CR} = 0.270
 Diversity = 1.2072e-04 k_t = [4, 3, 4, 3]
 Generation 5210: Best = 3.010524e+03 FES = 521100 μ_F = 0.731 μ_{CR} = 0.273
 Diversity = 1.2020e-04 k_t = [1, 1, 1, 1]
 Generation 5220: Best = 3.010524e+03 FES = 522100 μ_F = 0.729 μ_{CR} = 0.271
 Diversity = 1.2374e-04 k_t = [3, 3, 4, 4]
 Generation 5230: Best = 3.010524e+03 FES = 523100 μ_F = 0.731 μ_{CR} = 0.269
 Diversity = 1.2473e-04 k_t = [1, 1, 1, 1]
 Generation 5240: Best = 3.010524e+03 FES = 524100 μ_F = 0.732 μ_{CR} = 0.264
 Diversity = 1.2172e-04 k_t = [4, 2, 3, 4]
 Generation 5250: Best = 3.010524e+03 FES = 525100 μ_F = 0.741 μ_{CR} = 0.263
 Diversity = 1.3783e-04 k_t = [1, 1, 1, 1]
 Generation 5260: Best = 3.010524e+03 FES = 526100 μ_F = 0.742 μ_{CR} = 0.271
 Diversity = 1.3628e-04 k_t = [3, 4, 3, 3]
 Generation 5270: Best = 3.010524e+03 FES = 527100 μ_F = 0.754 μ_{CR} = 0.266
 Diversity = 1.5430e-04 k_t = [1, 1, 1, 1]
 Generation 5280: Best = 3.010524e+03 FES = 528100 μ_F = 0.746 μ_{CR} = 0.248
 Diversity = 1.5607e-04 k_t = [2, 4, 3, 2]
 Generation 5290: Best = 3.010524e+03 FES = 529100 μ_F = 0.738 μ_{CR} = 0.242
 Diversity = 1.6776e-04 k_t = [1, 1, 1, 1]
 Generation 5300: Best = 3.010524e+03 FES = 530100 μ_F = 0.742 μ_{CR} = 0.239
 Diversity = 1.7280e-04 k_t = [2, 2, 4, 4]
 Generation 5310: Best = 3.010524e+03 FES = 531100 μ_F = 0.735 μ_{CR} = 0.242
 Diversity = 1.8709e-04 k_t = [1, 1, 1, 1]
 Generation 5320: Best = 3.010524e+03 FES = 532100 μ_F = 0.724 μ_{CR} = 0.248
 Diversity = 2.0660e-04 k_t = [3, 2, 2, 2]
 Generation 5330: Best = 3.010524e+03 FES = 533100 μ_F = 0.725 μ_{CR} = 0.248
 Diversity = 2.2195e-04 k_t = [1, 1, 1, 1]
 Generation 5340: Best = 3.010524e+03 FES = 534100 μ_F = 0.714 μ_{CR} = 0.247
 Diversity = 2.2355e-04 k_t = [4, 4, 4, 3]
 Generation 5350: Best = 3.010524e+03 FES = 535100 μ_F = 0.720 μ_{CR} = 0.242
 Diversity = 2.2401e-04 k_t = [1, 1, 1, 1]

Generation 5360: Best = 3.010524e+03 FES = 536100 μ_F = 0.723 μ_{CR} = 0.233
 Diversity = 2.1458e-04 k_t = [3, 4, 2, 4]
 Generation 5370: Best = 3.010524e+03 FES = 537100 μ_F = 0.715 μ_{CR} = 0.232
 Diversity = 2.2517e-04 k_t = [1, 1, 1, 1]
 Generation 5380: Best = 3.010524e+03 FES = 538100 μ_F = 0.701 μ_{CR} = 0.236
 Diversity = 1.9287e-04 k_t = [4, 3, 4, 3]
 Generation 5390: Best = 3.010524e+03 FES = 539100 μ_F = 0.699 μ_{CR} = 0.237
 Diversity = 1.8071e-04 k_t = [1, 1, 1, 1]
 Generation 5400: Best = 3.010524e+03 FES = 540100 μ_F = 0.690 μ_{CR} = 0.235
 Diversity = 1.6654e-04 k_t = [4, 2, 4, 2]
 Generation 5410: Best = 3.010524e+03 FES = 541100 μ_F = 0.699 μ_{CR} = 0.233
 Diversity = 1.6029e-04 k_t = [1, 1, 1, 1]
 Generation 5420: Best = 3.010524e+03 FES = 542100 μ_F = 0.703 μ_{CR} = 0.234
 Diversity = 1.4049e-04 k_t = [3, 4, 4, 4]
 Generation 5430: Best = 3.010524e+03 FES = 543100 μ_F = 0.703 μ_{CR} = 0.233
 Diversity = 1.4282e-04 k_t = [1, 1, 1, 1]
 Generation 5440: Best = 3.010524e+03 FES = 544100 μ_F = 0.717 μ_{CR} = 0.238
 Diversity = 1.3297e-04 k_t = [4, 3, 4, 4]
 Generation 5450: Best = 3.010524e+03 FES = 545100 μ_F = 0.731 μ_{CR} = 0.241
 Diversity = 1.3641e-04 k_t = [1, 1, 1, 1]
 Generation 5460: Best = 3.010524e+03 FES = 546100 μ_F = 0.733 μ_{CR} = 0.245
 Diversity = 1.6941e-04 k_t = [3, 2, 3, 2]
 Generation 5470: Best = 3.010524e+03 FES = 547100 μ_F = 0.728 μ_{CR} = 0.244
 Diversity = 1.7885e-04 k_t = [1, 1, 1, 1]
 Generation 5480: Best = 3.010524e+03 FES = 548100 μ_F = 0.726 μ_{CR} = 0.255
 Diversity = 1.7302e-04 k_t = [3, 4, 4, 3]
 Generation 5490: Best = 3.010524e+03 FES = 549100 μ_F = 0.737 μ_{CR} = 0.257
 Diversity = 1.8266e-04 k_t = [1, 1, 1, 1]
 Generation 5500: Best = 3.010524e+03 FES = 550100 μ_F = 0.722 μ_{CR} = 0.262
 Diversity = 2.0018e-04 k_t = [2, 2, 2, 2]
 Generation 5510: Best = 3.010524e+03 FES = 551100 μ_F = 0.729 μ_{CR} = 0.271
 Diversity = 2.2273e-04 k_t = [1, 1, 1, 1]
 Generation 5520: Best = 3.010524e+03 FES = 552100 μ_F = 0.727 μ_{CR} = 0.272
 Diversity = 2.3013e-04 k_t = [3, 2, 4, 4]
 Generation 5530: Best = 3.010524e+03 FES = 553100 μ_F = 0.726 μ_{CR} = 0.277
 Diversity = 2.2793e-04 k_t = [1, 1, 1, 1]
 Generation 5540: Best = 3.010524e+03 FES = 554100 μ_F = 0.731 μ_{CR} = 0.280
 Diversity = 2.5456e-04 k_t = [4, 4, 3, 2]
 Generation 5550: Best = 3.010524e+03 FES = 555100 μ_F = 0.719 μ_{CR} = 0.280
 Diversity = 2.6418e-04 k_t = [1, 1, 1, 1]
 Generation 5560: Best = 3.010524e+03 FES = 556100 μ_F = 0.728 μ_{CR} = 0.285
 Diversity = 2.7068e-04 k_t = [4, 3, 4, 3]
 Generation 5570: Best = 3.010524e+03 FES = 557100 μ_F = 0.728 μ_{CR} = 0.293
 Diversity = 2.7852e-04 k_t = [1, 1, 1, 1]
 Generation 5580: Best = 3.010524e+03 FES = 558100 μ_F = 0.719 μ_{CR} = 0.282
 Diversity = 2.7038e-04 k_t = [3, 4, 3, 2]
 Generation 5590: Best = 3.010524e+03 FES = 559100 μ_F = 0.713 μ_{CR} = 0.284
 Diversity = 2.8543e-04 k_t = [1, 1, 1, 1]
 Generation 5600: Best = 3.010524e+03 FES = 560100 μ_F = 0.710 μ_{CR} = 0.279
 Diversity = 2.8615e-04 k_t = [2, 2, 2, 2]
 Generation 5610: Best = 3.010524e+03 FES = 561100 μ_F = 0.702 μ_{CR} = 0.276
 Diversity = 2.8777e-04 k_t = [1, 1, 1, 1]
 Generation 5620: Best = 3.010524e+03 FES = 562100 μ_F = 0.700 μ_{CR} = 0.277
 Diversity = 2.5627e-04 k_t = [3, 4, 2, 4]

Generation 5630: Best = 3.010524e+03 FES = 563100 μ_F = 0.695 μ_{CR} = 0.285
 Diversity = 2.3051e-04 k_t = [1, 1, 1, 1]
 Generation 5640: Best = 3.010524e+03 FES = 564100 μ_F = 0.692 μ_{CR} = 0.287
 Diversity = 2.1801e-04 k_t = [4, 4, 3, 2]
 Generation 5650: Best = 3.010524e+03 FES = 565100 μ_F = 0.692 μ_{CR} = 0.285
 Diversity = 2.0478e-04 k_t = [1, 1, 1, 1]
 Generation 5660: Best = 3.010524e+03 FES = 566100 μ_F = 0.674 μ_{CR} = 0.284
 Diversity = 1.8639e-04 k_t = [4, 2, 3, 2]
 Generation 5670: Best = 3.010524e+03 FES = 567100 μ_F = 0.675 μ_{CR} = 0.280
 Diversity = 1.7145e-04 k_t = [1, 1, 1, 1]
 Generation 5680: Best = 3.010524e+03 FES = 568100 μ_F = 0.683 μ_{CR} = 0.280
 Diversity = 1.5799e-04 k_t = [2, 3, 2, 4]
 Generation 5690: Best = 3.010524e+03 FES = 569100 μ_F = 0.679 μ_{CR} = 0.286
 Diversity = 1.4818e-04 k_t = [1, 1, 1, 1]
 Generation 5700: Best = 3.010524e+03 FES = 570100 μ_F = 0.668 μ_{CR} = 0.274
 Diversity = 1.2808e-04 k_t = [3, 2, 4, 4]
 Generation 5710: Best = 3.010524e+03 FES = 571100 μ_F = 0.689 μ_{CR} = 0.280
 Diversity = 1.1898e-04 k_t = [1, 1, 1, 1]
 Generation 5720: Best = 3.010524e+03 FES = 572100 μ_F = 0.690 μ_{CR} = 0.271
 Diversity = 9.9916e-05 k_t = [3, 3, 4, 3]
 Generation 5730: Best = 3.010524e+03 FES = 573100 μ_F = 0.686 μ_{CR} = 0.285
 Diversity = 9.2664e-05 k_t = [1, 1, 1, 1]
 Generation 5740: Best = 3.010524e+03 FES = 574100 μ_F = 0.704 μ_{CR} = 0.285
 Diversity = 1.0517e-04 k_t = [4, 2, 3, 2]
 Generation 5750: Best = 3.010524e+03 FES = 575100 μ_F = 0.691 μ_{CR} = 0.280
 Diversity = 9.9487e-05 k_t = [1, 1, 1, 1]
 Generation 5760: Best = 3.010524e+03 FES = 576100 μ_F = 0.690 μ_{CR} = 0.279
 Diversity = 9.5910e-05 k_t = [3, 3, 2, 2]
 Generation 5770: Best = 3.010524e+03 FES = 577100 μ_F = 0.695 μ_{CR} = 0.278
 Diversity = 8.6445e-05 k_t = [1, 1, 1, 1]
 Generation 5780: Best = 3.010524e+03 FES = 578100 μ_F = 0.689 μ_{CR} = 0.278
 Diversity = 8.1209e-05 k_t = [4, 2, 2, 2]
 Generation 5790: Best = 3.010524e+03 FES = 579100 μ_F = 0.679 μ_{CR} = 0.278
 Diversity = 7.6836e-05 k_t = [1, 1, 1, 1]
 Generation 5800: Best = 3.010524e+03 FES = 580100 μ_F = 0.688 μ_{CR} = 0.286
 Diversity = 7.0760e-05 k_t = [4, 3, 4, 2]
 Generation 5810: Best = 3.010524e+03 FES = 581100 μ_F = 0.692 μ_{CR} = 0.286
 Diversity = 6.1659e-05 k_t = [1, 1, 1, 1]
 Generation 5820: Best = 3.010524e+03 FES = 582100 μ_F = 0.693 μ_{CR} = 0.298
 Diversity = 4.8881e-05 k_t = [4, 4, 4, 2]
 Generation 5830: Best = 3.010524e+03 FES = 583100 μ_F = 0.693 μ_{CR} = 0.304
 Diversity = 3.9746e-05 k_t = [1, 1, 1, 1]
 Generation 5840: Best = 3.010524e+03 FES = 584100 μ_F = 0.687 μ_{CR} = 0.299
 Diversity = 3.2145e-05 k_t = [4, 3, 4, 3]
 Generation 5850: Best = 3.010524e+03 FES = 585100 μ_F = 0.692 μ_{CR} = 0.294
 Diversity = 2.8863e-05 k_t = [1, 1, 1, 1]
 Generation 5860: Best = 3.010524e+03 FES = 586100 μ_F = 0.682 μ_{CR} = 0.287
 Diversity = 2.5657e-05 k_t = [4, 2, 3, 3]
 Generation 5870: Best = 3.010524e+03 FES = 587100 μ_F = 0.680 μ_{CR} = 0.285
 Diversity = 2.3542e-05 k_t = [1, 1, 1, 1]
 Generation 5880: Best = 3.010524e+03 FES = 588100 μ_F = 0.692 μ_{CR} = 0.280
 Diversity = 2.1174e-05 k_t = [4, 4, 2, 4]
 Generation 5890: Best = 3.010524e+03 FES = 589100 μ_F = 0.685 μ_{CR} = 0.284
 Diversity = 1.7823e-05 k_t = [1, 1, 1, 1]

Generation 5900: Best = 3.010524e+03 FES = 590100 μ_F = 0.697 μ_{CR} = 0.285
 Diversity = 1.6190e-05 k_t = [4, 4, 4, 4]
 Generation 5910: Best = 3.010524e+03 FES = 591100 μ_F = 0.710 μ_{CR} = 0.279
 Diversity = 1.5496e-05 k_t = [1, 1, 1, 1]
 Generation 5920: Best = 3.010524e+03 FES = 592100 μ_F = 0.713 μ_{CR} = 0.281
 Diversity = 1.4175e-05 k_t = [3, 4, 4, 3]
 Generation 5930: Best = 3.010524e+03 FES = 593100 μ_F = 0.707 μ_{CR} = 0.281
 Diversity = 1.3311e-05 k_t = [1, 1, 1, 1]
 Generation 5940: Best = 3.010524e+03 FES = 594100 μ_F = 0.697 μ_{CR} = 0.282
 Diversity = 1.2165e-05 k_t = [4, 3, 2, 2]
 Generation 5950: Best = 3.010524e+03 FES = 595100 μ_F = 0.691 μ_{CR} = 0.280
 Diversity = 1.2613e-05 k_t = [1, 1, 1, 1]
 Generation 5960: Best = 3.010524e+03 FES = 596100 μ_F = 0.688 μ_{CR} = 0.280
 Diversity = 1.1214e-05 k_t = [3, 4, 4, 4]
 Generation 5970: Best = 3.010524e+03 FES = 597100 μ_F = 0.688 μ_{CR} = 0.279
 Diversity = 1.1117e-05 k_t = [1, 1, 1, 1]
 Generation 5980: Best = 3.010524e+03 FES = 598100 μ_F = 0.693 μ_{CR} = 0.269
 Diversity = 1.0374e-05 k_t = [2, 2, 2, 4]
 Generation 5990: Best = 3.010524e+03 FES = 599100 μ_F = 0.695 μ_{CR} = 0.262
 Diversity = 1.0026e-05 k_t = [1, 1, 1, 1]
 Generation 6000: Best = 3.010524e+03 FES = 600100 μ_F = 0.701 μ_{CR} = 0.255
 Diversity = 9.8488e-06 k_t = [2, 4, 2, 4]
 Generation 6010: Best = 3.010524e+03 FES = 601100 μ_F = 0.718 μ_{CR} = 0.263
 Diversity = 1.0982e-05 k_t = [1, 1, 1, 1]
 Generation 6020: Best = 3.010524e+03 FES = 602100 μ_F = 0.731 μ_{CR} = 0.265
 Diversity = 1.1799e-05 k_t = [2, 3, 3, 3]
 Generation 6030: Best = 3.010524e+03 FES = 603100 μ_F = 0.723 μ_{CR} = 0.266
 Diversity = 1.2782e-05 k_t = [1, 1, 1, 1]
 Generation 6040: Best = 3.010524e+03 FES = 604100 μ_F = 0.722 μ_{CR} = 0.262
 Diversity = 1.2492e-05 k_t = [4, 4, 2, 2]
 Generation 6050: Best = 3.010524e+03 FES = 605100 μ_F = 0.714 μ_{CR} = 0.273
 Diversity = 1.2599e-05 k_t = [1, 1, 1, 1]
 Generation 6060: Best = 3.010524e+03 FES = 606100 μ_F = 0.719 μ_{CR} = 0.277
 Diversity = 1.2146e-05 k_t = [4, 2, 4, 3]
 Generation 6070: Best = 3.010524e+03 FES = 607100 μ_F = 0.714 μ_{CR} = 0.286
 Diversity = 1.1517e-05 k_t = [1, 1, 1, 1]
 Generation 6080: Best = 3.010524e+03 FES = 608100 μ_F = 0.708 μ_{CR} = 0.285
 Diversity = 1.0428e-05 k_t = [4, 3, 2, 3]
 Generation 6090: Best = 3.010524e+03 FES = 609100 μ_F = 0.705 μ_{CR} = 0.283
 Diversity = 9.8116e-06 k_t = [1, 1, 1, 1]
 Generation 6100: Best = 3.010524e+03 FES = 610100 μ_F = 0.701 μ_{CR} = 0.277
 Diversity = 9.2068e-06 k_t = [4, 2, 4, 3]
 Generation 6110: Best = 3.010524e+03 FES = 611100 μ_F = 0.692 μ_{CR} = 0.261
 Diversity = 9.2144e-06 k_t = [1, 1, 1, 1]
 Generation 6120: Best = 3.010524e+03 FES = 612100 μ_F = 0.696 μ_{CR} = 0.261
 Diversity = 8.5627e-06 k_t = [2, 2, 2, 4]
 Generation 6130: Best = 3.010524e+03 FES = 613100 μ_F = 0.701 μ_{CR} = 0.257
 Diversity = 8.6442e-06 k_t = [1, 1, 1, 1]
 Generation 6140: Best = 3.010524e+03 FES = 614100 μ_F = 0.707 μ_{CR} = 0.253
 Diversity = 8.7976e-06 k_t = [2, 2, 2, 2]
 Generation 6150: Best = 3.010524e+03 FES = 615100 μ_F = 0.710 μ_{CR} = 0.251
 Diversity = 9.1668e-06 k_t = [1, 1, 1, 1]
 Generation 6160: Best = 3.010524e+03 FES = 616100 μ_F = 0.712 μ_{CR} = 0.245
 Diversity = 8.2709e-06 k_t = [3, 3, 2, 3]

Generation 6170: Best = 3.010524e+03 FES = 617100 μ_F = 0.714 μ_{CR} = 0.254
 Diversity = 8.0691e-06 k_t = [1, 1, 1, 1]
 Generation 6180: Best = 3.010524e+03 FES = 618100 μ_F = 0.717 μ_{CR} = 0.258
 Diversity = 8.0367e-06 k_t = [3, 4, 4, 2]
 Generation 6190: Best = 3.010524e+03 FES = 619100 μ_F = 0.721 μ_{CR} = 0.254
 Diversity = 8.1874e-06 k_t = [1, 1, 1, 1]
 Generation 6200: Best = 3.010524e+03 FES = 620100 μ_F = 0.718 μ_{CR} = 0.265
 Diversity = 7.8588e-06 k_t = [2, 3, 4, 2]
 Generation 6210: Best = 3.010524e+03 FES = 621100 μ_F = 0.716 μ_{CR} = 0.261
 Diversity = 8.3899e-06 k_t = [1, 1, 1, 1]
 Generation 6220: Best = 3.010524e+03 FES = 622100 μ_F = 0.706 μ_{CR} = 0.261
 Diversity = 8.0103e-06 k_t = [3, 2, 2, 3]
 Generation 6230: Best = 3.010524e+03 FES = 623100 μ_F = 0.707 μ_{CR} = 0.258
 Diversity = 8.1442e-06 k_t = [1, 1, 1, 1]
 Generation 6240: Best = 3.010524e+03 FES = 624100 μ_F = 0.711 μ_{CR} = 0.257
 Diversity = 9.2350e-06 k_t = [3, 3, 2, 2]
 Generation 6250: Best = 3.010524e+03 FES = 625100 μ_F = 0.715 μ_{CR} = 0.258
 Diversity = 1.0221e-05 k_t = [1, 1, 1, 1]
 Generation 6260: Best = 3.010524e+03 FES = 626100 μ_F = 0.715 μ_{CR} = 0.265
 Diversity = 1.0363e-05 k_t = [2, 4, 3, 3]
 Generation 6270: Best = 3.010524e+03 FES = 627100 μ_F = 0.716 μ_{CR} = 0.269
 Diversity = 1.0303e-05 k_t = [1, 1, 1, 1]
 Generation 6280: Best = 3.010524e+03 FES = 628100 μ_F = 0.710 μ_{CR} = 0.256
 Diversity = 8.9032e-06 k_t = [4, 3, 4, 3]
 Generation 6290: Best = 3.010524e+03 FES = 629100 μ_F = 0.710 μ_{CR} = 0.249
 Diversity = 8.8814e-06 k_t = [1, 1, 1, 1]
 Generation 6300: Best = 3.010524e+03 FES = 630100 μ_F = 0.736 μ_{CR} = 0.240
 Diversity = 9.1145e-06 k_t = [2, 3, 4, 4]
 Generation 6310: Best = 3.010524e+03 FES = 631100 μ_F = 0.744 μ_{CR} = 0.239
 Diversity = 9.5968e-06 k_t = [1, 1, 1, 1]
 Generation 6320: Best = 3.010524e+03 FES = 632100 μ_F = 0.735 μ_{CR} = 0.245
 Diversity = 9.1540e-06 k_t = [4, 4, 4, 3]
 Generation 6330: Best = 3.010524e+03 FES = 633100 μ_F = 0.725 μ_{CR} = 0.247
 Diversity = 9.4614e-06 k_t = [1, 1, 1, 1]
 Generation 6340: Best = 3.010524e+03 FES = 634100 μ_F = 0.717 μ_{CR} = 0.246
 Diversity = 8.7003e-06 k_t = [2, 2, 3, 4]
 Generation 6350: Best = 3.010524e+03 FES = 635100 μ_F = 0.710 μ_{CR} = 0.246
 Diversity = 7.8446e-06 k_t = [1, 1, 1, 1]
 Generation 6360: Best = 3.010524e+03 FES = 636100 μ_F = 0.714 μ_{CR} = 0.248
 Diversity = 6.8426e-06 k_t = [4, 3, 4, 3]
 Generation 6370: Best = 3.010524e+03 FES = 637100 μ_F = 0.718 μ_{CR} = 0.248
 Diversity = 6.5003e-06 k_t = [1, 1, 1, 1]
 Generation 6380: Best = 3.010524e+03 FES = 638100 μ_F = 0.725 μ_{CR} = 0.258
 Diversity = 5.7968e-06 k_t = [2, 4, 3, 3]
 Generation 6390: Best = 3.010524e+03 FES = 639100 μ_F = 0.721 μ_{CR} = 0.264
 Diversity = 5.6353e-06 k_t = [1, 1, 1, 1]
 Generation 6400: Best = 3.010524e+03 FES = 640100 μ_F = 0.717 μ_{CR} = 0.265
 Diversity = 5.2862e-06 k_t = [3, 4, 2, 4]
 Generation 6410: Best = 3.010524e+03 FES = 641100 μ_F = 0.733 μ_{CR} = 0.263
 Diversity = 5.2402e-06 k_t = [1, 1, 1, 1]
 Generation 6420: Best = 3.010524e+03 FES = 642100 μ_F = 0.722 μ_{CR} = 0.260
 Diversity = 5.0155e-06 k_t = [4, 2, 4, 4]
 Generation 6430: Best = 3.010524e+03 FES = 643100 μ_F = 0.720 μ_{CR} = 0.264
 Diversity = 4.6798e-06 k_t = [1, 1, 1, 1]

Generation 6440: Best = 3.010524e+03 FES = 644100 μ_F = 0.702 μ_{CR} = 0.275
 Diversity = 4.5017e-06 k_t = [2, 3, 2, 4]
 Generation 6450: Best = 3.010524e+03 FES = 645100 μ_F = 0.708 μ_{CR} = 0.275
 Diversity = 4.3041e-06 k_t = [1, 1, 1, 1]
 Generation 6460: Best = 3.010524e+03 FES = 646100 μ_F = 0.723 μ_{CR} = 0.281
 Diversity = 4.2909e-06 k_t = [4, 4, 2, 4]
 Generation 6470: Best = 3.010524e+03 FES = 647100 μ_F = 0.723 μ_{CR} = 0.286
 Diversity = 4.7653e-06 k_t = [1, 1, 1, 1]
 Generation 6480: Best = 3.010524e+03 FES = 648100 μ_F = 0.734 μ_{CR} = 0.282
 Diversity = 5.2916e-06 k_t = [3, 3, 3, 2]
 Generation 6490: Best = 3.010524e+03 FES = 649100 μ_F = 0.723 μ_{CR} = 0.280
 Diversity = 5.4332e-06 k_t = [1, 1, 1, 1]
 Generation 6500: Best = 3.010524e+03 FES = 650100 μ_F = 0.722 μ_{CR} = 0.276
 Diversity = 5.2227e-06 k_t = [3, 4, 3, 3]
 Generation 6510: Best = 3.010524e+03 FES = 651100 μ_F = 0.727 μ_{CR} = 0.277
 Diversity = 5.1886e-06 k_t = [1, 1, 1, 1]
 Generation 6520: Best = 3.010524e+03 FES = 652100 μ_F = 0.724 μ_{CR} = 0.275
 Diversity = 4.4669e-06 k_t = [4, 3, 3, 4]
 Generation 6530: Best = 3.010524e+03 FES = 653100 μ_F = 0.721 μ_{CR} = 0.268
 Diversity = 4.4294e-06 k_t = [1, 1, 1, 1]
 Generation 6540: Best = 3.010524e+03 FES = 654100 μ_F = 0.712 μ_{CR} = 0.278
 Diversity = 4.0663e-06 k_t = [2, 3, 2, 3]
 Generation 6550: Best = 3.010524e+03 FES = 655100 μ_F = 0.706 μ_{CR} = 0.281
 Diversity = 4.0127e-06 k_t = [1, 1, 1, 1]
 Generation 6560: Best = 3.010524e+03 FES = 656100 μ_F = 0.701 μ_{CR} = 0.275
 Diversity = 3.7563e-06 k_t = [3, 4, 3, 2]
 Generation 6570: Best = 3.010524e+03 FES = 657100 μ_F = 0.704 μ_{CR} = 0.272
 Diversity = 3.6991e-06 k_t = [1, 1, 1, 1]
 Generation 6580: Best = 3.010524e+03 FES = 658100 μ_F = 0.706 μ_{CR} = 0.278
 Diversity = 3.7903e-06 k_t = [2, 3, 4, 2]
 Generation 6590: Best = 3.010524e+03 FES = 659100 μ_F = 0.709 μ_{CR} = 0.277
 Diversity = 3.9993e-06 k_t = [1, 1, 1, 1]
 Generation 6600: Best = 3.010524e+03 FES = 660100 μ_F = 0.699 μ_{CR} = 0.275
 Diversity = 3.6678e-06 k_t = [4, 3, 3, 4]
 Generation 6610: Best = 3.010524e+03 FES = 661100 μ_F = 0.681 μ_{CR} = 0.282
 Diversity = 3.2576e-06 k_t = [1, 1, 1, 1]
 Generation 6620: Best = 3.010524e+03 FES = 662100 μ_F = 0.689 μ_{CR} = 0.276
 Diversity = 2.8368e-06 k_t = [2, 3, 2, 4]
 Generation 6630: Best = 3.010524e+03 FES = 663100 μ_F = 0.691 μ_{CR} = 0.266
 Diversity = 2.6773e-06 k_t = [1, 1, 1, 1]
 Generation 6640: Best = 3.010524e+03 FES = 664100 μ_F = 0.700 μ_{CR} = 0.266
 Diversity = 2.8770e-06 k_t = [2, 3, 3, 2]
 Generation 6650: Best = 3.010524e+03 FES = 665100 μ_F = 0.705 μ_{CR} = 0.267
 Diversity = 2.8995e-06 k_t = [1, 1, 1, 1]
 Generation 6660: Best = 3.010524e+03 FES = 666100 μ_F = 0.700 μ_{CR} = 0.272
 Diversity = 2.6931e-06 k_t = [2, 2, 4, 2]
 Generation 6670: Best = 3.010524e+03 FES = 667100 μ_F = 0.696 μ_{CR} = 0.276
 Diversity = 2.5996e-06 k_t = [1, 1, 1, 1]
 Generation 6680: Best = 3.010524e+03 FES = 668100 μ_F = 0.685 μ_{CR} = 0.264
 Diversity = 2.2178e-06 k_t = [2, 4, 4, 4]
 Generation 6690: Best = 3.010524e+03 FES = 669100 μ_F = 0.693 μ_{CR} = 0.266
 Diversity = 2.2036e-06 k_t = [1, 1, 1, 1]
 Generation 6700: Best = 3.010524e+03 FES = 670100 μ_F = 0.672 μ_{CR} = 0.258
 Diversity = 1.9506e-06 k_t = [2, 4, 2, 2]

Generation 6710: Best = 3.010524e+03 FES = 671100 μ_F = 0.680 μ_{CR} = 0.257
 Diversity = 2.0444e-06 k_t = [1, 1, 1, 1]
 Generation 6720: Best = 3.010524e+03 FES = 672100 μ_F = 0.685 μ_{CR} = 0.269
 Diversity = 2.0675e-06 k_t = [2, 2, 2, 2]
 Generation 6730: Best = 3.010524e+03 FES = 673100 μ_F = 0.707 μ_{CR} = 0.280
 Diversity = 2.1428e-06 k_t = [1, 1, 1, 1]
 Generation 6740: Best = 3.010524e+03 FES = 674100 μ_F = 0.719 μ_{CR} = 0.278
 Diversity = 2.3175e-06 k_t = [4, 3, 4, 2]
 Generation 6750: Best = 3.010524e+03 FES = 675100 μ_F = 0.725 μ_{CR} = 0.267
 Diversity = 2.7308e-06 k_t = [1, 1, 1, 1]
 Generation 6760: Best = 3.010524e+03 FES = 676100 μ_F = 0.720 μ_{CR} = 0.264
 Diversity = 2.7709e-06 k_t = [3, 4, 2, 2]
 Generation 6770: Best = 3.010524e+03 FES = 677100 μ_F = 0.718 μ_{CR} = 0.268
 Diversity = 2.7852e-06 k_t = [1, 1, 1, 1]
 Generation 6780: Best = 3.010524e+03 FES = 678100 μ_F = 0.716 μ_{CR} = 0.270
 Diversity = 2.5661e-06 k_t = [3, 2, 2, 3]
 Generation 6790: Best = 3.010524e+03 FES = 679100 μ_F = 0.724 μ_{CR} = 0.259
 Diversity = 2.5730e-06 k_t = [1, 1, 1, 1]
 Generation 6800: Best = 3.010524e+03 FES = 680100 μ_F = 0.726 μ_{CR} = 0.252
 Diversity = 2.6799e-06 k_t = [4, 3, 3, 3]
 Generation 6810: Best = 3.010524e+03 FES = 681100 μ_F = 0.723 μ_{CR} = 0.246
 Diversity = 2.5278e-06 k_t = [1, 1, 1, 1]
 Generation 6820: Best = 3.010524e+03 FES = 682100 μ_F = 0.709 μ_{CR} = 0.246
 Diversity = 2.1781e-06 k_t = [4, 2, 4, 3]
 Generation 6830: Best = 3.010524e+03 FES = 683100 μ_F = 0.704 μ_{CR} = 0.243
 Diversity = 2.0982e-06 k_t = [1, 1, 1, 1]
 Generation 6840: Best = 3.010524e+03 FES = 684100 μ_F = 0.706 μ_{CR} = 0.252
 Diversity = 1.9990e-06 k_t = [3, 4, 4, 4]
 Generation 6850: Best = 3.010524e+03 FES = 685100 μ_F = 0.715 μ_{CR} = 0.238
 Diversity = 1.9782e-06 k_t = [1, 1, 1, 1]
 Generation 6860: Best = 3.010524e+03 FES = 686100 μ_F = 0.713 μ_{CR} = 0.243
 Diversity = 2.0358e-06 k_t = [4, 4, 4, 3]
 Generation 6870: Best = 3.010524e+03 FES = 687100 μ_F = 0.728 μ_{CR} = 0.239
 Diversity = 2.6133e-06 k_t = [1, 1, 1, 1]
 Generation 6880: Best = 3.010524e+03 FES = 688100 μ_F = 0.747 μ_{CR} = 0.238
 Diversity = 3.7343e-06 k_t = [2, 3, 3, 2]
 Generation 6890: Best = 3.010524e+03 FES = 689100 μ_F = 0.749 μ_{CR} = 0.239
 Diversity = 4.7494e-06 k_t = [1, 1, 1, 1]
 Generation 6900: Best = 3.010524e+03 FES = 690100 μ_F = 0.745 μ_{CR} = 0.239
 Diversity = 5.3735e-06 k_t = [2, 4, 3, 2]
 Generation 6910: Best = 3.010524e+03 FES = 691100 μ_F = 0.761 μ_{CR} = 0.249
 Diversity = 6.3264e-06 k_t = [1, 1, 1, 1]
 Generation 6920: Best = 3.010524e+03 FES = 692100 μ_F = 0.747 μ_{CR} = 0.248
 Diversity = 6.5097e-06 k_t = [2, 2, 3, 2]
 Generation 6930: Best = 3.010524e+03 FES = 693100 μ_F = 0.752 μ_{CR} = 0.248
 Diversity = 6.8505e-06 k_t = [1, 1, 1, 1]
 Generation 6940: Best = 3.010524e+03 FES = 694100 μ_F = 0.747 μ_{CR} = 0.247
 Diversity = 7.0370e-06 k_t = [2, 4, 4, 2]
 Generation 6950: Best = 3.010524e+03 FES = 695100 μ_F = 0.746 μ_{CR} = 0.247
 Diversity = 7.0115e-06 k_t = [1, 1, 1, 1]
 Generation 6960: Best = 3.010524e+03 FES = 696100 μ_F = 0.738 μ_{CR} = 0.256
 Diversity = 6.9917e-06 k_t = [2, 2, 2, 4]
 Generation 6970: Best = 3.010524e+03 FES = 697100 μ_F = 0.729 μ_{CR} = 0.256
 Diversity = 6.5376e-06 k_t = [1, 1, 1, 1]

Generation 6980: Best = 3.010524e+03 FES = 698100 μ_F = 0.712 μ_{CR} = 0.248
 Diversity = 5.5399e-06 k_t = [3, 3, 4, 4]
 Generation 6990: Best = 3.010524e+03 FES = 699100 μ_F = 0.704 μ_{CR} = 0.248
 Diversity = 4.8205e-06 k_t = [1, 1, 1, 1]
 Generation 7000: Best = 3.010524e+03 FES = 700100 μ_F = 0.700 μ_{CR} = 0.242
 Diversity = 4.4092e-06 k_t = [4, 4, 2, 4]
 Generation 7010: Best = 3.010524e+03 FES = 701100 μ_F = 0.701 μ_{CR} = 0.242
 Diversity = 4.3868e-06 k_t = [1, 1, 1, 1]
 Generation 7020: Best = 3.010524e+03 FES = 702100 μ_F = 0.713 μ_{CR} = 0.239
 Diversity = 4.0901e-06 k_t = [3, 4, 2, 4]
 Generation 7030: Best = 3.010524e+03 FES = 703100 μ_F = 0.694 μ_{CR} = 0.241
 Diversity = 3.5364e-06 k_t = [1, 1, 1, 1]
 Generation 7040: Best = 3.010524e+03 FES = 704100 μ_F = 0.678 μ_{CR} = 0.246
 Diversity = 2.8442e-06 k_t = [4, 3, 4, 3]
 Generation 7050: Best = 3.010524e+03 FES = 705100 μ_F = 0.673 μ_{CR} = 0.235
 Diversity = 2.4604e-06 k_t = [1, 1, 1, 1]
 Generation 7060: Best = 3.010524e+03 FES = 706100 μ_F = 0.670 μ_{CR} = 0.240
 Diversity = 2.0074e-06 k_t = [3, 3, 2, 3]
 Generation 7070: Best = 3.010524e+03 FES = 707100 μ_F = 0.668 μ_{CR} = 0.243
 Diversity = 1.9367e-06 k_t = [1, 1, 1, 1]
 Generation 7080: Best = 3.010524e+03 FES = 708100 μ_F = 0.670 μ_{CR} = 0.249
 Diversity = 1.6781e-06 k_t = [2, 3, 2, 4]
 Generation 7090: Best = 3.010524e+03 FES = 709100 μ_F = 0.686 μ_{CR} = 0.252
 Diversity = 1.6080e-06 k_t = [1, 1, 1, 1]
 Generation 7100: Best = 3.010524e+03 FES = 710100 μ_F = 0.701 μ_{CR} = 0.247
 Diversity = 1.6410e-06 k_t = [2, 2, 3, 2]
 Generation 7110: Best = 3.010524e+03 FES = 711100 μ_F = 0.709 μ_{CR} = 0.238
 Diversity = 1.6618e-06 k_t = [1, 1, 1, 1]
 Generation 7120: Best = 3.010524e+03 FES = 712100 μ_F = 0.696 μ_{CR} = 0.234
 Diversity = 1.4755e-06 k_t = [2, 3, 2, 4]
 Generation 7130: Best = 3.010524e+03 FES = 713100 μ_F = 0.712 μ_{CR} = 0.224
 Diversity = 1.4094e-06 k_t = [1, 1, 1, 1]
 Generation 7140: Best = 3.010524e+03 FES = 714100 μ_F = 0.730 μ_{CR} = 0.217
 Diversity = 1.3872e-06 k_t = [3, 4, 4, 3]
 Generation 7150: Best = 3.010524e+03 FES = 715100 μ_F = 0.736 μ_{CR} = 0.210
 Diversity = 1.5970e-06 k_t = [1, 1, 1, 1]
 Generation 7160: Best = 3.010524e+03 FES = 716100 μ_F = 0.752 μ_{CR} = 0.214
 Diversity = 1.9235e-06 k_t = [3, 2, 4, 3]
 Generation 7170: Best = 3.010524e+03 FES = 717100 μ_F = 0.735 μ_{CR} = 0.217
 Diversity = 2.0731e-06 k_t = [1, 1, 1, 1]
 Generation 7180: Best = 3.010524e+03 FES = 718100 μ_F = 0.734 μ_{CR} = 0.220
 Diversity = 2.0874e-06 k_t = [3, 2, 2, 4]
 Generation 7190: Best = 3.010524e+03 FES = 719100 μ_F = 0.732 μ_{CR} = 0.224
 Diversity = 2.4226e-06 k_t = [1, 1, 1, 1]
 Generation 7200: Best = 3.010524e+03 FES = 720100 μ_F = 0.734 μ_{CR} = 0.228
 Diversity = 2.4466e-06 k_t = [2, 4, 2, 4]
 Generation 7210: Best = 3.010524e+03 FES = 721100 μ_F = 0.720 μ_{CR} = 0.231
 Diversity = 2.5148e-06 k_t = [1, 1, 1, 1]
 Generation 7220: Best = 3.010524e+03 FES = 722100 μ_F = 0.712 μ_{CR} = 0.237
 Diversity = 2.2387e-06 k_t = [3, 4, 2, 3]
 Generation 7230: Best = 3.010524e+03 FES = 723100 μ_F = 0.719 μ_{CR} = 0.240
 Diversity = 2.2916e-06 k_t = [1, 1, 1, 1]
 Generation 7240: Best = 3.010524e+03 FES = 724100 μ_F = 0.722 μ_{CR} = 0.239
 Diversity = 2.1836e-06 k_t = [4, 3, 2, 3]

Generation 7250: Best = 3.010524e+03 FES = 725100 μ_F = 0.730 μ_{CR} = 0.241
 Diversity = 2.3661e-06 k_t = [1, 1, 1, 1]
 Generation 7260: Best = 3.010524e+03 FES = 726100 μ_F = 0.727 μ_{CR} = 0.236
 Diversity = 2.4211e-06 k_t = [2, 3, 4, 2]
 Generation 7270: Best = 3.010524e+03 FES = 727100 μ_F = 0.745 μ_{CR} = 0.231
 Diversity = 2.6196e-06 k_t = [1, 1, 1, 1]
 Generation 7280: Best = 3.010524e+03 FES = 728100 μ_F = 0.757 μ_{CR} = 0.228
 Diversity = 3.1132e-06 k_t = [4, 4, 3, 3]
 Generation 7290: Best = 3.010524e+03 FES = 729100 μ_F = 0.756 μ_{CR} = 0.229
 Diversity = 3.3789e-06 k_t = [1, 1, 1, 1]
 Generation 7300: Best = 3.010524e+03 FES = 730100 μ_F = 0.746 μ_{CR} = 0.229
 Diversity = 3.2694e-06 k_t = [3, 2, 3, 4]
 Generation 7310: Best = 3.010524e+03 FES = 731100 μ_F = 0.741 μ_{CR} = 0.231
 Diversity = 3.2709e-06 k_t = [1, 1, 1, 1]
 Generation 7320: Best = 3.010524e+03 FES = 732100 μ_F = 0.744 μ_{CR} = 0.230
 Diversity = 3.2639e-06 k_t = [2, 2, 4, 2]
 Generation 7330: Best = 3.010524e+03 FES = 733100 μ_F = 0.752 μ_{CR} = 0.237
 Diversity = 3.4655e-06 k_t = [1, 1, 1, 1]
 Generation 7340: Best = 3.010524e+03 FES = 734100 μ_F = 0.722 μ_{CR} = 0.239
 Diversity = 3.0189e-06 k_t = [3, 3, 2, 4]
 Generation 7350: Best = 3.010524e+03 FES = 735100 μ_F = 0.710 μ_{CR} = 0.230
 Diversity = 2.8791e-06 k_t = [1, 1, 1, 1]
 Generation 7360: Best = 3.010524e+03 FES = 736100 μ_F = 0.714 μ_{CR} = 0.241
 Diversity = 2.7182e-06 k_t = [4, 4, 4, 4]
 Generation 7370: Best = 3.010524e+03 FES = 737100 μ_F = 0.713 μ_{CR} = 0.240
 Diversity = 2.6804e-06 k_t = [1, 1, 1, 1]
 Generation 7380: Best = 3.010524e+03 FES = 738100 μ_F = 0.725 μ_{CR} = 0.248
 Diversity = 2.6776e-06 k_t = [3, 3, 3, 3]
 Generation 7390: Best = 3.010524e+03 FES = 739100 μ_F = 0.727 μ_{CR} = 0.250
 Diversity = 3.0399e-06 k_t = [1, 1, 1, 1]
 Generation 7400: Best = 3.010524e+03 FES = 740100 μ_F = 0.723 μ_{CR} = 0.251
 Diversity = 2.9587e-06 k_t = [3, 3, 3, 4]
 Generation 7410: Best = 3.010524e+03 FES = 741100 μ_F = 0.718 μ_{CR} = 0.261
 Diversity = 3.1644e-06 k_t = [1, 1, 1, 1]
 Generation 7420: Best = 3.010524e+03 FES = 742100 μ_F = 0.717 μ_{CR} = 0.266
 Diversity = 3.3341e-06 k_t = [2, 3, 2, 2]
 Generation 7430: Best = 3.010524e+03 FES = 743100 μ_F = 0.703 μ_{CR} = 0.257
 Diversity = 3.5982e-06 k_t = [1, 1, 1, 1]
 Generation 7440: Best = 3.010524e+03 FES = 744100 μ_F = 0.713 μ_{CR} = 0.256
 Diversity = 3.6234e-06 k_t = [2, 2, 2, 4]
 Generation 7450: Best = 3.010524e+03 FES = 745100 μ_F = 0.712 μ_{CR} = 0.267
 Diversity = 3.4842e-06 k_t = [1, 1, 1, 1]
 Generation 7460: Best = 3.010524e+03 FES = 746100 μ_F = 0.721 μ_{CR} = 0.277
 Diversity = 3.1933e-06 k_t = [3, 4, 3, 4]
 Generation 7470: Best = 3.010524e+03 FES = 747100 μ_F = 0.722 μ_{CR} = 0.282
 Diversity = 3.0919e-06 k_t = [1, 1, 1, 1]
 Generation 7480: Best = 3.010524e+03 FES = 748100 μ_F = 0.728 μ_{CR} = 0.285
 Diversity = 3.2565e-06 k_t = [4, 3, 2, 2]
 Generation 7490: Best = 3.010524e+03 FES = 749100 μ_F = 0.720 μ_{CR} = 0.293
 Diversity = 3.3174e-06 k_t = [1, 1, 1, 1]
 Generation 7500: Best = 3.010524e+03 FES = 750100 μ_F = 0.708 μ_{CR} = 0.296
 Diversity = 3.0683e-06 k_t = [4, 3, 2, 4]
 Generation 7510: Best = 3.010524e+03 FES = 751100 μ_F = 0.718 μ_{CR} = 0.291
 Diversity = 2.8339e-06 k_t = [1, 1, 1, 1]

Generation 7520: Best = 3.010524e+03 FES = 752100 μ_F = 0.734 μ_{CR} = 0.295
 Diversity = 2.9339e-06 k_t = [4, 4, 4, 4]
 Generation 7530: Best = 3.010524e+03 FES = 753100 μ_F = 0.746 μ_{CR} = 0.293
 Diversity = 3.5384e-06 k_t = [1, 1, 1, 1]
 Generation 7540: Best = 3.010524e+03 FES = 754100 μ_F = 0.738 μ_{CR} = 0.293
 Diversity = 3.0501e-06 k_t = [4, 3, 3, 4]
 Generation 7550: Best = 3.010524e+03 FES = 755100 μ_F = 0.738 μ_{CR} = 0.287
 Diversity = 3.0867e-06 k_t = [1, 1, 1, 1]
 Generation 7560: Best = 3.010524e+03 FES = 756100 μ_F = 0.729 μ_{CR} = 0.292
 Diversity = 3.4040e-06 k_t = [4, 2, 2, 2]
 Generation 7570: Best = 3.010524e+03 FES = 757100 μ_F = 0.711 μ_{CR} = 0.296
 Diversity = 3.2817e-06 k_t = [1, 1, 1, 1]
 Generation 7580: Best = 3.010524e+03 FES = 758100 μ_F = 0.704 μ_{CR} = 0.300
 Diversity = 2.6764e-06 k_t = [2, 2, 4, 4]
 Generation 7590: Best = 3.010524e+03 FES = 759100 μ_F = 0.703 μ_{CR} = 0.295
 Diversity = 2.4627e-06 k_t = [1, 1, 1, 1]
 Generation 7600: Best = 3.010524e+03 FES = 760100 μ_F = 0.695 μ_{CR} = 0.287
 Diversity = 2.4665e-06 k_t = [3, 4, 4, 3]
 Generation 7610: Best = 3.010524e+03 FES = 761100 μ_F = 0.695 μ_{CR} = 0.297
 Diversity = 2.4240e-06 k_t = [1, 1, 1, 1]
 Generation 7620: Best = 3.010524e+03 FES = 762100 μ_F = 0.707 μ_{CR} = 0.292
 Diversity = 2.2486e-06 k_t = [4, 4, 3, 3]
 Generation 7630: Best = 3.010524e+03 FES = 763100 μ_F = 0.699 μ_{CR} = 0.301
 Diversity = 2.1129e-06 k_t = [1, 1, 1, 1]
 Generation 7640: Best = 3.010524e+03 FES = 764100 μ_F = 0.695 μ_{CR} = 0.310
 Diversity = 2.1783e-06 k_t = [2, 2, 3, 2]
 Generation 7650: Best = 3.010524e+03 FES = 765100 μ_F = 0.695 μ_{CR} = 0.294
 Diversity = 2.1422e-06 k_t = [1, 1, 1, 1]
 Generation 7660: Best = 3.010524e+03 FES = 766100 μ_F = 0.700 μ_{CR} = 0.293
 Diversity = 2.1980e-06 k_t = [4, 2, 4, 3]
 Generation 7670: Best = 3.010524e+03 FES = 767100 μ_F = 0.694 μ_{CR} = 0.289
 Diversity = 2.2353e-06 k_t = [1, 1, 1, 1]
 Generation 7680: Best = 3.010524e+03 FES = 768100 μ_F = 0.709 μ_{CR} = 0.282
 Diversity = 2.3036e-06 k_t = [4, 4, 4, 3]
 Generation 7690: Best = 3.010524e+03 FES = 769100 μ_F = 0.713 μ_{CR} = 0.284
 Diversity = 2.4249e-06 k_t = [1, 1, 1, 1]
 Generation 7700: Best = 3.010524e+03 FES = 770100 μ_F = 0.719 μ_{CR} = 0.287
 Diversity = 2.4361e-06 k_t = [4, 2, 4, 3]
 Generation 7710: Best = 3.010524e+03 FES = 771100 μ_F = 0.715 μ_{CR} = 0.297
 Diversity = 2.4333e-06 k_t = [1, 1, 1, 1]
 Generation 7720: Best = 3.010524e+03 FES = 772100 μ_F = 0.712 μ_{CR} = 0.295
 Diversity = 2.3351e-06 k_t = [2, 4, 2, 3]
 Generation 7730: Best = 3.010524e+03 FES = 773100 μ_F = 0.697 μ_{CR} = 0.293
 Diversity = 2.1251e-06 k_t = [1, 1, 1, 1]
 Generation 7740: Best = 3.010524e+03 FES = 774100 μ_F = 0.701 μ_{CR} = 0.295
 Diversity = 1.9655e-06 k_t = [4, 4, 2, 3]
 Generation 7750: Best = 3.010524e+03 FES = 775100 μ_F = 0.701 μ_{CR} = 0.290
 Diversity = 1.8784e-06 k_t = [1, 1, 1, 1]
 Generation 7760: Best = 3.010524e+03 FES = 776100 μ_F = 0.698 μ_{CR} = 0.285
 Diversity = 1.8673e-06 k_t = [2, 2, 4, 2]
 Generation 7770: Best = 3.010524e+03 FES = 777100 μ_F = 0.707 μ_{CR} = 0.283
 Diversity = 2.0022e-06 k_t = [1, 1, 1, 1]
 Generation 7780: Best = 3.010524e+03 FES = 778100 μ_F = 0.724 μ_{CR} = 0.284
 Diversity = 2.0736e-06 k_t = [2, 4, 3, 4]

Generation 7790: Best = 3.010524e+03 FES = 779100 μ_F = 0.733 μ_{CR} = 0.275
 Diversity = 2.1526e-06 k_t = [1, 1, 1, 1]
 Generation 7800: Best = 3.010524e+03 FES = 780100 μ_F = 0.744 μ_{CR} = 0.277
 Diversity = 2.2945e-06 k_t = [4, 4, 2, 3]
 Generation 7810: Best = 3.010524e+03 FES = 781100 μ_F = 0.751 μ_{CR} = 0.280
 Diversity = 2.9131e-06 k_t = [1, 1, 1, 1]
 Generation 7820: Best = 3.010524e+03 FES = 782100 μ_F = 0.750 μ_{CR} = 0.290
 Diversity = 3.5834e-06 k_t = [2, 3, 2, 3]
 Generation 7830: Best = 3.010524e+03 FES = 783100 μ_F = 0.753 μ_{CR} = 0.286
 Diversity = 4.2406e-06 k_t = [1, 1, 1, 1]
 Generation 7840: Best = 3.010524e+03 FES = 784100 μ_F = 0.757 μ_{CR} = 0.299
 Diversity = 4.9038e-06 k_t = [3, 3, 2, 2]
 Generation 7850: Best = 3.010524e+03 FES = 785100 μ_F = 0.760 μ_{CR} = 0.292
 Diversity = 5.7690e-06 k_t = [1, 1, 1, 1]
 Generation 7860: Best = 3.010524e+03 FES = 786100 μ_F = 0.750 μ_{CR} = 0.295
 Diversity = 6.1928e-06 k_t = [4, 2, 2, 2]
 Generation 7870: Best = 3.010524e+03 FES = 787100 μ_F = 0.725 μ_{CR} = 0.292
 Diversity = 6.5649e-06 k_t = [1, 1, 1, 1]
 Generation 7880: Best = 3.010524e+03 FES = 788100 μ_F = 0.722 μ_{CR} = 0.295
 Diversity = 6.6136e-06 k_t = [2, 2, 3, 3]
 Generation 7890: Best = 3.010524e+03 FES = 789100 μ_F = 0.728 μ_{CR} = 0.297
 Diversity = 7.6901e-06 k_t = [1, 1, 1, 1]
 Generation 7900: Best = 3.010524e+03 FES = 790100 μ_F = 0.732 μ_{CR} = 0.302
 Diversity = 7.7953e-06 k_t = [4, 2, 4, 2]
 Generation 7910: Best = 3.010524e+03 FES = 791100 μ_F = 0.706 μ_{CR} = 0.308
 Diversity = 7.1294e-06 k_t = [1, 1, 1, 1]
 Generation 7920: Best = 3.010524e+03 FES = 792100 μ_F = 0.684 μ_{CR} = 0.315
 Diversity = 5.9286e-06 k_t = [3, 4, 3, 4]
 Generation 7930: Best = 3.010524e+03 FES = 793100 μ_F = 0.689 μ_{CR} = 0.308
 Diversity = 5.3320e-06 k_t = [1, 1, 1, 1]
 Generation 7940: Best = 3.010524e+03 FES = 794100 μ_F = 0.703 μ_{CR} = 0.322
 Diversity = 5.9615e-06 k_t = [2, 3, 4, 2]
 Generation 7950: Best = 3.010524e+03 FES = 795100 μ_F = 0.697 μ_{CR} = 0.311
 Diversity = 6.2397e-06 k_t = [1, 1, 1, 1]
 Generation 7960: Best = 3.010524e+03 FES = 796100 μ_F = 0.702 μ_{CR} = 0.315
 Diversity = 6.3667e-06 k_t = [3, 4, 4, 3]
 Generation 7970: Best = 3.010524e+03 FES = 797100 μ_F = 0.703 μ_{CR} = 0.315
 Diversity = 6.0237e-06 k_t = [1, 1, 1, 1]
 Generation 7980: Best = 3.010524e+03 FES = 798100 μ_F = 0.690 μ_{CR} = 0.310
 Diversity = 5.2309e-06 k_t = [2, 2, 2, 3]
 Generation 7990: Best = 3.010524e+03 FES = 799100 μ_F = 0.690 μ_{CR} = 0.305
 Diversity = 4.9710e-06 k_t = [1, 1, 1, 1]
 Generation 8000: Best = 3.010524e+03 FES = 800100 μ_F = 0.685 μ_{CR} = 0.306
 Diversity = 4.3959e-06 k_t = [4, 4, 4, 4]
 Generation 8010: Best = 3.010524e+03 FES = 801100 μ_F = 0.687 μ_{CR} = 0.314
 Diversity = 3.5538e-06 k_t = [1, 1, 1, 1]
 Generation 8020: Best = 3.010524e+03 FES = 802100 μ_F = 0.686 μ_{CR} = 0.310
 Diversity = 3.2303e-06 k_t = [2, 4, 3, 2]
 Generation 8030: Best = 3.010524e+03 FES = 803100 μ_F = 0.696 μ_{CR} = 0.314
 Diversity = 3.5211e-06 k_t = [1, 1, 1, 1]
 Generation 8040: Best = 3.010524e+03 FES = 804100 μ_F = 0.705 μ_{CR} = 0.308
 Diversity = 3.3613e-06 k_t = [3, 3, 3, 4]
 Generation 8050: Best = 3.010524e+03 FES = 805100 μ_F = 0.702 μ_{CR} = 0.307
 Diversity = 3.4501e-06 k_t = [1, 1, 1, 1]

Generation 8060: Best = 3.010524e+03 FES = 806100 μ_F = 0.710 μ_{CR} = 0.305
 Diversity = 3.3789e-06 k_t = [4, 3, 4, 3]
 Generation 8070: Best = 3.010524e+03 FES = 807100 μ_F = 0.713 μ_{CR} = 0.300
 Diversity = 3.1771e-06 k_t = [1, 1, 1, 1]
 Generation 8080: Best = 3.010524e+03 FES = 808100 μ_F = 0.702 μ_{CR} = 0.299
 Diversity = 3.0505e-06 k_t = [4, 4, 2, 2]
 Generation 8090: Best = 3.010524e+03 FES = 809100 μ_F = 0.700 μ_{CR} = 0.308
 Diversity = 3.2385e-06 k_t = [1, 1, 1, 1]
 Generation 8100: Best = 3.010524e+03 FES = 810100 μ_F = 0.701 μ_{CR} = 0.309
 Diversity = 2.8725e-06 k_t = [2, 4, 4, 4]
 Generation 8110: Best = 3.010524e+03 FES = 811100 μ_F = 0.711 μ_{CR} = 0.305
 Diversity = 2.8222e-06 k_t = [1, 1, 1, 1]
 Generation 8120: Best = 3.010524e+03 FES = 812100 μ_F = 0.717 μ_{CR} = 0.307
 Diversity = 2.9076e-06 k_t = [4, 4, 3, 3]
 Generation 8130: Best = 3.010524e+03 FES = 813100 μ_F = 0.718 μ_{CR} = 0.308
 Diversity = 2.8602e-06 k_t = [1, 1, 1, 1]
 Generation 8140: Best = 3.010524e+03 FES = 814100 μ_F = 0.703 μ_{CR} = 0.312
 Diversity = 2.5763e-06 k_t = [4, 2, 2, 4]
 Generation 8150: Best = 3.010524e+03 FES = 815100 μ_F = 0.713 μ_{CR} = 0.310
 Diversity = 2.5120e-06 k_t = [1, 1, 1, 1]
 Generation 8160: Best = 3.010524e+03 FES = 816100 μ_F = 0.714 μ_{CR} = 0.304
 Diversity = 2.5854e-06 k_t = [3, 3, 4, 2]
 Generation 8170: Best = 3.010524e+03 FES = 817100 μ_F = 0.709 μ_{CR} = 0.300
 Diversity = 2.5695e-06 k_t = [1, 1, 1, 1]
 Generation 8180: Best = 3.010524e+03 FES = 818100 μ_F = 0.713 μ_{CR} = 0.301
 Diversity = 2.4031e-06 k_t = [4, 3, 4, 2]
 Generation 8190: Best = 3.010524e+03 FES = 819100 μ_F = 0.711 μ_{CR} = 0.292
 Diversity = 2.0893e-06 k_t = [1, 1, 1, 1]
 Generation 8200: Best = 3.010524e+03 FES = 820100 μ_F = 0.695 μ_{CR} = 0.295
 Diversity = 1.7539e-06 k_t = [2, 3, 3, 4]
 Generation 8210: Best = 3.010524e+03 FES = 821100 μ_F = 0.707 μ_{CR} = 0.297
 Diversity = 1.7493e-06 k_t = [1, 1, 1, 1]
 Generation 8220: Best = 3.010524e+03 FES = 822100 μ_F = 0.717 μ_{CR} = 0.292
 Diversity = 1.8472e-06 k_t = [4, 3, 4, 3]
 Generation 8230: Best = 3.010524e+03 FES = 823100 μ_F = 0.714 μ_{CR} = 0.282
 Diversity = 1.9847e-06 k_t = [1, 1, 1, 1]
 Generation 8240: Best = 3.010524e+03 FES = 824100 μ_F = 0.708 μ_{CR} = 0.276
 Diversity = 1.8990e-06 k_t = [2, 3, 3, 3]
 Generation 8250: Best = 3.010524e+03 FES = 825100 μ_F = 0.709 μ_{CR} = 0.273
 Diversity = 1.8614e-06 k_t = [1, 1, 1, 1]
 Generation 8260: Best = 3.010524e+03 FES = 826100 μ_F = 0.707 μ_{CR} = 0.261
 Diversity = 1.5666e-06 k_t = [4, 3, 4, 4]
 Generation 8270: Best = 3.010524e+03 FES = 827100 μ_F = 0.708 μ_{CR} = 0.255
 Diversity = 1.5508e-06 k_t = [1, 1, 1, 1]
 Generation 8280: Best = 3.010524e+03 FES = 828100 μ_F = 0.703 μ_{CR} = 0.247
 Diversity = 1.3752e-06 k_t = [4, 2, 2, 4]
 Generation 8290: Best = 3.010524e+03 FES = 829100 μ_F = 0.714 μ_{CR} = 0.254
 Diversity = 1.3926e-06 k_t = [1, 1, 1, 1]
 Generation 8300: Best = 3.010524e+03 FES = 830100 μ_F = 0.712 μ_{CR} = 0.264
 Diversity = 1.2272e-06 k_t = [4, 2, 4, 3]
 Generation 8310: Best = 3.010524e+03 FES = 831100 μ_F = 0.717 μ_{CR} = 0.262
 Diversity = 1.1532e-06 k_t = [1, 1, 1, 1]
 Generation 8320: Best = 3.010524e+03 FES = 832100 μ_F = 0.715 μ_{CR} = 0.250
 Diversity = 1.1234e-06 k_t = [2, 4, 4, 3]

Generation 8330: Best = 3.010524e+03 FES = 833100 μ_F = 0.718 μ_{CR} = 0.251
 Diversity = 1.2263e-06 k_t = [1, 1, 1, 1]
 Generation 8340: Best = 3.010524e+03 FES = 834100 μ_F = 0.727 μ_{CR} = 0.241
 Diversity = 1.2619e-06 k_t = [4, 4, 3, 3]
 Generation 8350: Best = 3.010524e+03 FES = 835100 μ_F = 0.737 μ_{CR} = 0.237
 Diversity = 1.3313e-06 k_t = [1, 1, 1, 1]
 Generation 8360: Best = 3.010524e+03 FES = 836100 μ_F = 0.724 μ_{CR} = 0.235
 Diversity = 1.2526e-06 k_t = [3, 3, 3, 4]
 Generation 8370: Best = 3.010524e+03 FES = 837100 μ_F = 0.732 μ_{CR} = 0.240
 Diversity = 1.3553e-06 k_t = [1, 1, 1, 1]
 Generation 8380: Best = 3.010524e+03 FES = 838100 μ_F = 0.732 μ_{CR} = 0.237
 Diversity = 1.2639e-06 k_t = [3, 2, 4, 2]
 Generation 8390: Best = 3.010524e+03 FES = 839100 μ_F = 0.742 μ_{CR} = 0.245
 Diversity = 1.2667e-06 k_t = [1, 1, 1, 1]
 Generation 8400: Best = 3.010524e+03 FES = 840100 μ_F = 0.731 μ_{CR} = 0.247
 Diversity = 1.3412e-06 k_t = [3, 3, 3, 2]
 Generation 8410: Best = 3.010524e+03 FES = 841100 μ_F = 0.726 μ_{CR} = 0.255
 Diversity = 1.3248e-06 k_t = [1, 1, 1, 1]
 Generation 8420: Best = 3.010524e+03 FES = 842100 μ_F = 0.724 μ_{CR} = 0.263
 Diversity = 1.2475e-06 k_t = [3, 4, 3, 2]
 Generation 8430: Best = 3.010524e+03 FES = 843100 μ_F = 0.727 μ_{CR} = 0.261
 Diversity = 1.2463e-06 k_t = [1, 1, 1, 1]
 Generation 8440: Best = 3.010524e+03 FES = 844100 μ_F = 0.718 μ_{CR} = 0.246
 Diversity = 1.1402e-06 k_t = [2, 4, 2, 3]
 Generation 8450: Best = 3.010524e+03 FES = 845100 μ_F = 0.717 μ_{CR} = 0.252
 Diversity = 1.1417e-06 k_t = [1, 1, 1, 1]
 Generation 8460: Best = 3.010524e+03 FES = 846100 μ_F = 0.710 μ_{CR} = 0.246
 Diversity = 1.0456e-06 k_t = [3, 3, 3, 3]
 Generation 8470: Best = 3.010524e+03 FES = 847100 μ_F = 0.695 μ_{CR} = 0.238
 Diversity = 9.9938e-07 k_t = [1, 1, 1, 1]
 Generation 8480: Best = 3.010524e+03 FES = 848100 μ_F = 0.689 μ_{CR} = 0.241
 Diversity = 9.3840e-07 k_t = [2, 2, 2, 2]
 Generation 8490: Best = 3.010524e+03 FES = 849100 μ_F = 0.674 μ_{CR} = 0.241
 Diversity = 8.8345e-07 k_t = [1, 1, 1, 1]
 Generation 8500: Best = 3.010524e+03 FES = 850100 μ_F = 0.678 μ_{CR} = 0.238
 Diversity = 7.4359e-07 k_t = [3, 3, 4, 4]
 Generation 8510: Best = 3.010524e+03 FES = 851100 μ_F = 0.699 μ_{CR} = 0.238
 Diversity = 7.0003e-07 k_t = [1, 1, 1, 1]
 Generation 8520: Best = 3.010524e+03 FES = 852100 μ_F = 0.707 μ_{CR} = 0.240
 Diversity = 7.3917e-07 k_t = [3, 2, 4, 2]
 Generation 8530: Best = 3.010524e+03 FES = 853100 μ_F = 0.715 μ_{CR} = 0.238
 Diversity = 8.3201e-07 k_t = [1, 1, 1, 1]
 Generation 8540: Best = 3.010524e+03 FES = 854100 μ_F = 0.718 μ_{CR} = 0.237
 Diversity = 7.6266e-07 k_t = [4, 4, 2, 4]
 Generation 8550: Best = 3.010524e+03 FES = 855100 μ_F = 0.728 μ_{CR} = 0.240
 Diversity = 7.4309e-07 k_t = [1, 1, 1, 1]
 Generation 8560: Best = 3.010524e+03 FES = 856100 μ_F = 0.714 μ_{CR} = 0.232
 Diversity = 6.7597e-07 k_t = [3, 2, 3, 3]
 Generation 8570: Best = 3.010524e+03 FES = 857100 μ_F = 0.706 μ_{CR} = 0.233
 Diversity = 6.5333e-07 k_t = [1, 1, 1, 1]
 Generation 8580: Best = 3.010524e+03 FES = 858100 μ_F = 0.708 μ_{CR} = 0.234
 Diversity = 6.0565e-07 k_t = [4, 4, 4, 4]
 Generation 8590: Best = 3.010524e+03 FES = 859100 μ_F = 0.710 μ_{CR} = 0.222
 Diversity = 5.3269e-07 k_t = [1, 1, 1, 1]

Generation 8600: Best = 3.010524e+03 FES = 860100 μ_F = 0.703 μ_{CR} = 0.214
 Diversity = 5.0392e-07 k_t = [3, 4, 2, 2]
 Generation 8610: Best = 3.010524e+03 FES = 861100 μ_F = 0.693 μ_{CR} = 0.212
 Diversity = 4.7243e-07 k_t = [1, 1, 1, 1]
 Generation 8620: Best = 3.010524e+03 FES = 862100 μ_F = 0.697 μ_{CR} = 0.208
 Diversity = 4.3907e-07 k_t = [3, 4, 3, 3]
 Generation 8630: Best = 3.010524e+03 FES = 863100 μ_F = 0.699 μ_{CR} = 0.200
 Diversity = 4.5059e-07 k_t = [1, 1, 1, 1]
 Generation 8640: Best = 3.010524e+03 FES = 864100 μ_F = 0.703 μ_{CR} = 0.196
 Diversity = 4.2672e-07 k_t = [4, 3, 2, 3]
 Generation 8650: Best = 3.010524e+03 FES = 865100 μ_F = 0.703 μ_{CR} = 0.213
 Diversity = 4.2155e-07 k_t = [1, 1, 1, 1]
 Generation 8660: Best = 3.010524e+03 FES = 866100 μ_F = 0.711 μ_{CR} = 0.221
 Diversity = 3.6917e-07 k_t = [3, 4, 3, 4]
 Generation 8670: Best = 3.010524e+03 FES = 867100 μ_F = 0.710 μ_{CR} = 0.214
 Diversity = 3.4572e-07 k_t = [1, 1, 1, 1]
 Generation 8680: Best = 3.010524e+03 FES = 868100 μ_F = 0.708 μ_{CR} = 0.209
 Diversity = 3.3663e-07 k_t = [4, 4, 4, 4]
 Generation 8690: Best = 3.010524e+03 FES = 869100 μ_F = 0.722 μ_{CR} = 0.206
 Diversity = 3.2033e-07 k_t = [1, 1, 1, 1]
 Generation 8700: Best = 3.010524e+03 FES = 870100 μ_F = 0.730 μ_{CR} = 0.200
 Diversity = 3.1539e-07 k_t = [2, 4, 2, 4]
 Generation 8710: Best = 3.010524e+03 FES = 871100 μ_F = 0.728 μ_{CR} = 0.200
 Diversity = 3.0522e-07 k_t = [1, 1, 1, 1]
 Generation 8720: Best = 3.010524e+03 FES = 872100 μ_F = 0.726 μ_{CR} = 0.188
 Diversity = 3.1799e-07 k_t = [4, 3, 2, 4]
 Generation 8730: Best = 3.010524e+03 FES = 873100 μ_F = 0.732 μ_{CR} = 0.195
 Diversity = 3.2117e-07 k_t = [1, 1, 1, 1]
 Generation 8740: Best = 3.010524e+03 FES = 874100 μ_F = 0.748 μ_{CR} = 0.194
 Diversity = 3.0171e-07 k_t = [3, 2, 4, 4]
 Generation 8750: Best = 3.010524e+03 FES = 875100 μ_F = 0.746 μ_{CR} = 0.189
 Diversity = 3.1546e-07 k_t = [1, 1, 1, 1]
 Generation 8760: Best = 3.010524e+03 FES = 876100 μ_F = 0.740 μ_{CR} = 0.195
 Diversity = 2.9986e-07 k_t = [2, 3, 2, 3]
 Generation 8770: Best = 3.010524e+03 FES = 877100 μ_F = 0.730 μ_{CR} = 0.202
 Diversity = 3.1259e-07 k_t = [1, 1, 1, 1]
 Generation 8780: Best = 3.010524e+03 FES = 878100 μ_F = 0.738 μ_{CR} = 0.197
 Diversity = 3.2418e-07 k_t = [2, 3, 3, 2]
 Generation 8790: Best = 3.010524e+03 FES = 879100 μ_F = 0.724 μ_{CR} = 0.190
 Diversity = 3.2791e-07 k_t = [1, 1, 1, 1]
 Generation 8800: Best = 3.010524e+03 FES = 880100 μ_F = 0.730 μ_{CR} = 0.187
 Diversity = 3.0354e-07 k_t = [3, 2, 4, 4]
 Generation 8810: Best = 3.010524e+03 FES = 881100 μ_F = 0.735 μ_{CR} = 0.187
 Diversity = 3.2579e-07 k_t = [1, 1, 1, 1]
 Generation 8820: Best = 3.010524e+03 FES = 882100 μ_F = 0.736 μ_{CR} = 0.193
 Diversity = 3.2478e-07 k_t = [3, 2, 4, 2]
 Generation 8830: Best = 3.010524e+03 FES = 883100 μ_F = 0.742 μ_{CR} = 0.203
 Diversity = 3.5116e-07 k_t = [1, 1, 1, 1]
 Generation 8840: Best = 3.010524e+03 FES = 884100 μ_F = 0.732 μ_{CR} = 0.201
 Diversity = 3.0217e-07 k_t = [3, 4, 3, 3]
 Generation 8850: Best = 3.010524e+03 FES = 885100 μ_F = 0.725 μ_{CR} = 0.185
 Diversity = 2.7939e-07 k_t = [1, 1, 1, 1]
 Generation 8860: Best = 3.010524e+03 FES = 886100 μ_F = 0.726 μ_{CR} = 0.180
 Diversity = 2.5453e-07 k_t = [3, 4, 3, 2]

Generation 8870: Best = 3.010524e+03 FES = 887100 μ_F = 0.727 μ_{CR} = 0.175
 Diversity = 2.6192e-07 k_t = [1, 1, 1, 1]
 Generation 8880: Best = 3.010524e+03 FES = 888100 μ_F = 0.732 μ_{CR} = 0.174
 Diversity = 2.7208e-07 k_t = [3, 2, 4, 2]
 Generation 8890: Best = 3.010524e+03 FES = 889100 μ_F = 0.734 μ_{CR} = 0.180
 Diversity = 2.6317e-07 k_t = [1, 1, 1, 1]
 Generation 8900: Best = 3.010524e+03 FES = 890100 μ_F = 0.742 μ_{CR} = 0.181
 Diversity = 2.6301e-07 k_t = [2, 2, 2, 3]
 Generation 8910: Best = 3.010524e+03 FES = 891100 μ_F = 0.740 μ_{CR} = 0.175
 Diversity = 2.7413e-07 k_t = [1, 1, 1, 1]
 Generation 8920: Best = 3.010524e+03 FES = 892100 μ_F = 0.735 μ_{CR} = 0.177
 Diversity = 2.6166e-07 k_t = [2, 2, 3, 2]
 Generation 8930: Best = 3.010524e+03 FES = 893100 μ_F = 0.741 μ_{CR} = 0.173
 Diversity = 2.7098e-07 k_t = [1, 1, 1, 1]
 Generation 8940: Best = 3.010524e+03 FES = 894100 μ_F = 0.747 μ_{CR} = 0.183
 Diversity = 2.6976e-07 k_t = [2, 3, 2, 4]
 Generation 8950: Best = 3.010524e+03 FES = 895100 μ_F = 0.742 μ_{CR} = 0.178
 Diversity = 3.0266e-07 k_t = [1, 1, 1, 1]
 Generation 8960: Best = 3.010524e+03 FES = 896100 μ_F = 0.749 μ_{CR} = 0.171
 Diversity = 3.0931e-07 k_t = [3, 4, 3, 4]
 Generation 8970: Best = 3.010524e+03 FES = 897100 μ_F = 0.748 μ_{CR} = 0.171
 Diversity = 3.2101e-07 k_t = [1, 1, 1, 1]
 Generation 8980: Best = 3.010524e+03 FES = 898100 μ_F = 0.735 μ_{CR} = 0.172
 Diversity = 2.9880e-07 k_t = [4, 2, 2, 4]
 Generation 8990: Best = 3.010524e+03 FES = 899100 μ_F = 0.737 μ_{CR} = 0.167
 Diversity = 3.0898e-07 k_t = [1, 1, 1, 1]
 Generation 9000: Best = 3.010524e+03 FES = 900100 μ_F = 0.734 μ_{CR} = 0.166
 Diversity = 2.9803e-07 k_t = [3, 3, 2, 3]
 Generation 9010: Best = 3.010524e+03 FES = 901100 μ_F = 0.723 μ_{CR} = 0.161
 Diversity = 2.8658e-07 k_t = [1, 1, 1, 1]
 Generation 9020: Best = 3.010524e+03 FES = 902100 μ_F = 0.722 μ_{CR} = 0.163
 Diversity = 2.5812e-07 k_t = [2, 4, 2, 3]
 Generation 9030: Best = 3.010524e+03 FES = 903100 μ_F = 0.723 μ_{CR} = 0.166
 Diversity = 2.7404e-07 k_t = [1, 1, 1, 1]
 Generation 9040: Best = 3.010524e+03 FES = 904100 μ_F = 0.730 μ_{CR} = 0.166
 Diversity = 2.8955e-07 k_t = [2, 3, 2, 4]
 Generation 9050: Best = 3.010524e+03 FES = 905100 μ_F = 0.733 μ_{CR} = 0.162
 Diversity = 3.4034e-07 k_t = [1, 1, 1, 1]
 Generation 9060: Best = 3.010524e+03 FES = 906100 μ_F = 0.735 μ_{CR} = 0.166
 Diversity = 3.7849e-07 k_t = [3, 2, 2, 2]
 Generation 9070: Best = 3.010524e+03 FES = 907100 μ_F = 0.733 μ_{CR} = 0.170
 Diversity = 4.3114e-07 k_t = [1, 1, 1, 1]
 Generation 9080: Best = 3.010524e+03 FES = 908100 μ_F = 0.737 μ_{CR} = 0.171
 Diversity = 4.7778e-07 k_t = [3, 2, 2, 2]
 Generation 9090: Best = 3.010524e+03 FES = 909100 μ_F = 0.730 μ_{CR} = 0.173
 Diversity = 5.0037e-07 k_t = [1, 1, 1, 1]
 Generation 9100: Best = 3.010524e+03 FES = 910100 μ_F = 0.732 μ_{CR} = 0.166
 Diversity = 5.1489e-07 k_t = [2, 4, 4, 3]
 Generation 9110: Best = 3.010524e+03 FES = 911100 μ_F = 0.727 μ_{CR} = 0.166
 Diversity = 5.9478e-07 k_t = [1, 1, 1, 1]
 Generation 9120: Best = 3.010524e+03 FES = 912100 μ_F = 0.735 μ_{CR} = 0.173
 Diversity = 6.5589e-07 k_t = [3, 4, 2, 3]
 Generation 9130: Best = 3.010524e+03 FES = 913100 μ_F = 0.747 μ_{CR} = 0.179
 Diversity = 7.1594e-07 k_t = [1, 1, 1, 1]

Generation 9140: Best = 3.010524e+03 FES = 914100 μ_F = 0.752 μ_{CR} = 0.174
 Diversity = 7.1454e-07 k_t = [3, 4, 2, 3]
 Generation 9150: Best = 3.010524e+03 FES = 915100 μ_F = 0.750 μ_{CR} = 0.177
 Diversity = 8.1026e-07 k_t = [1, 1, 1, 1]
 Generation 9160: Best = 3.010524e+03 FES = 916100 μ_F = 0.745 μ_{CR} = 0.188
 Diversity = 8.8506e-07 k_t = [3, 4, 2, 4]
 Generation 9170: Best = 3.010524e+03 FES = 917100 μ_F = 0.748 μ_{CR} = 0.201
 Diversity = 9.8595e-07 k_t = [1, 1, 1, 1]
 Generation 9180: Best = 3.010524e+03 FES = 918100 μ_F = 0.748 μ_{CR} = 0.213
 Diversity = 1.1266e-06 k_t = [4, 4, 2, 4]
 Generation 9190: Best = 3.010524e+03 FES = 919100 μ_F = 0.753 μ_{CR} = 0.219
 Diversity = 1.2293e-06 k_t = [1, 1, 1, 1]
 Generation 9200: Best = 3.010524e+03 FES = 920100 μ_F = 0.745 μ_{CR} = 0.216
 Diversity = 1.2149e-06 k_t = [2, 4, 3, 4]
 Generation 9210: Best = 3.010524e+03 FES = 921100 μ_F = 0.762 μ_{CR} = 0.212
 Diversity = 1.3650e-06 k_t = [1, 1, 1, 1]
 Generation 9220: Best = 3.010524e+03 FES = 922100 μ_F = 0.745 μ_{CR} = 0.206
 Diversity = 1.4323e-06 k_t = [3, 3, 3, 2]
 Generation 9230: Best = 3.010524e+03 FES = 923100 μ_F = 0.735 μ_{CR} = 0.218
 Diversity = 1.4543e-06 k_t = [1, 1, 1, 1]
 Generation 9240: Best = 3.010524e+03 FES = 924100 μ_F = 0.732 μ_{CR} = 0.223
 Diversity = 1.3602e-06 k_t = [4, 4, 3, 2]
 Generation 9250: Best = 3.010524e+03 FES = 925100 μ_F = 0.727 μ_{CR} = 0.222
 Diversity = 1.3478e-06 k_t = [1, 1, 1, 1]
 Generation 9260: Best = 3.010524e+03 FES = 926100 μ_F = 0.728 μ_{CR} = 0.221
 Diversity = 1.2983e-06 k_t = [4, 2, 2, 3]
 Generation 9270: Best = 3.010524e+03 FES = 927100 μ_F = 0.732 μ_{CR} = 0.229
 Diversity = 1.3855e-06 k_t = [1, 1, 1, 1]
 Generation 9280: Best = 3.010524e+03 FES = 928100 μ_F = 0.731 μ_{CR} = 0.226
 Diversity = 1.3992e-06 k_t = [2, 2, 4, 2]
 Generation 9290: Best = 3.010524e+03 FES = 929100 μ_F = 0.733 μ_{CR} = 0.226
 Diversity = 1.4551e-06 k_t = [1, 1, 1, 1]
 Generation 9300: Best = 3.010524e+03 FES = 930100 μ_F = 0.723 μ_{CR} = 0.223
 Diversity = 1.4002e-06 k_t = [2, 3, 3, 4]
 Generation 9310: Best = 3.010524e+03 FES = 931100 μ_F = 0.719 μ_{CR} = 0.223
 Diversity = 1.4707e-06 k_t = [1, 1, 1, 1]
 Generation 9320: Best = 3.010524e+03 FES = 932100 μ_F = 0.717 μ_{CR} = 0.218
 Diversity = 1.5622e-06 k_t = [3, 4, 3, 2]
 Generation 9330: Best = 3.010524e+03 FES = 933100 μ_F = 0.710 μ_{CR} = 0.226
 Diversity = 1.7263e-06 k_t = [1, 1, 1, 1]
 Generation 9340: Best = 3.010524e+03 FES = 934100 μ_F = 0.729 μ_{CR} = 0.236
 Diversity = 1.9521e-06 k_t = [4, 2, 3, 3]
 Generation 9350: Best = 3.010524e+03 FES = 935100 μ_F = 0.726 μ_{CR} = 0.239
 Diversity = 2.0662e-06 k_t = [1, 1, 1, 1]
 Generation 9360: Best = 3.010524e+03 FES = 936100 μ_F = 0.724 μ_{CR} = 0.240
 Diversity = 1.8755e-06 k_t = [4, 2, 3, 3]
 Generation 9370: Best = 3.010524e+03 FES = 937100 μ_F = 0.735 μ_{CR} = 0.237
 Diversity = 1.9286e-06 k_t = [1, 1, 1, 1]
 Generation 9380: Best = 3.010524e+03 FES = 938100 μ_F = 0.727 μ_{CR} = 0.245
 Diversity = 1.8808e-06 k_t = [4, 3, 4, 2]
 Generation 9390: Best = 3.010524e+03 FES = 939100 μ_F = 0.732 μ_{CR} = 0.247
 Diversity = 2.1195e-06 k_t = [1, 1, 1, 1]
 Generation 9400: Best = 3.010524e+03 FES = 940100 μ_F = 0.735 μ_{CR} = 0.254
 Diversity = 2.2444e-06 k_t = [4, 3, 2, 2]

Generation 9410: Best = 3.010524e+03 FES = 941100 μ_F = 0.736 μ_{CR} = 0.242
 Diversity = 2.2230e-06 k_t = [1, 1, 1, 1]
 Generation 9420: Best = 3.010524e+03 FES = 942100 μ_F = 0.738 μ_{CR} = 0.236
 Diversity = 2.1748e-06 k_t = [2, 3, 2, 2]
 Generation 9430: Best = 3.010524e+03 FES = 943100 μ_F = 0.737 μ_{CR} = 0.234
 Diversity = 2.3730e-06 k_t = [1, 1, 1, 1]
 Generation 9440: Best = 3.010524e+03 FES = 944100 μ_F = 0.744 μ_{CR} = 0.235
 Diversity = 2.4173e-06 k_t = [3, 3, 2, 3]
 Generation 9450: Best = 3.010524e+03 FES = 945100 μ_F = 0.741 μ_{CR} = 0.246
 Diversity = 2.8466e-06 k_t = [1, 1, 1, 1]
 Generation 9460: Best = 3.010524e+03 FES = 946100 μ_F = 0.732 μ_{CR} = 0.245
 Diversity = 2.9299e-06 k_t = [4, 2, 2, 2]
 Generation 9470: Best = 3.010524e+03 FES = 947100 μ_F = 0.723 μ_{CR} = 0.251
 Diversity = 3.3043e-06 k_t = [1, 1, 1, 1]
 Generation 9480: Best = 3.010524e+03 FES = 948100 μ_F = 0.717 μ_{CR} = 0.249
 Diversity = 2.9963e-06 k_t = [3, 2, 2, 3]
 Generation 9490: Best = 3.010524e+03 FES = 949100 μ_F = 0.710 μ_{CR} = 0.248
 Diversity = 3.0888e-06 k_t = [1, 1, 1, 1]
 Generation 9500: Best = 3.010524e+03 FES = 950100 μ_F = 0.716 μ_{CR} = 0.256
 Diversity = 3.0208e-06 k_t = [4, 4, 3, 2]
 Generation 9510: Best = 3.010524e+03 FES = 951100 μ_F = 0.712 μ_{CR} = 0.257
 Diversity = 2.9171e-06 k_t = [1, 1, 1, 1]
 Generation 9520: Best = 3.010524e+03 FES = 952100 μ_F = 0.723 μ_{CR} = 0.263
 Diversity = 2.6620e-06 k_t = [2, 2, 2, 4]
 Generation 9530: Best = 3.010524e+03 FES = 953100 μ_F = 0.732 μ_{CR} = 0.257
 Diversity = 3.1054e-06 k_t = [1, 1, 1, 1]
 Generation 9540: Best = 3.010524e+03 FES = 954100 μ_F = 0.740 μ_{CR} = 0.250
 Diversity = 3.5026e-06 k_t = [3, 4, 4, 2]
 Generation 9550: Best = 3.010524e+03 FES = 955100 μ_F = 0.730 μ_{CR} = 0.253
 Diversity = 3.3688e-06 k_t = [1, 1, 1, 1]
 Generation 9560: Best = 3.010524e+03 FES = 956100 μ_F = 0.729 μ_{CR} = 0.251
 Diversity = 3.2526e-06 k_t = [4, 2, 4, 3]
 Generation 9570: Best = 3.010524e+03 FES = 957100 μ_F = 0.734 μ_{CR} = 0.260
 Diversity = 3.4440e-06 k_t = [1, 1, 1, 1]
 Generation 9580: Best = 3.010524e+03 FES = 958100 μ_F = 0.739 μ_{CR} = 0.261
 Diversity = 3.3542e-06 k_t = [4, 3, 2, 2]
 Generation 9590: Best = 3.010524e+03 FES = 959100 μ_F = 0.734 μ_{CR} = 0.259
 Diversity = 3.2882e-06 k_t = [1, 1, 1, 1]
 Generation 9600: Best = 3.010524e+03 FES = 960100 μ_F = 0.732 μ_{CR} = 0.265
 Diversity = 3.0638e-06 k_t = [3, 2, 4, 4]
 Generation 9610: Best = 3.010524e+03 FES = 961100 μ_F = 0.736 μ_{CR} = 0.265
 Diversity = 3.1538e-06 k_t = [1, 1, 1, 1]
 Generation 9620: Best = 3.010524e+03 FES = 962100 μ_F = 0.726 μ_{CR} = 0.261
 Diversity = 3.3600e-06 k_t = [2, 2, 2, 3]
 Generation 9630: Best = 3.010524e+03 FES = 963100 μ_F = 0.714 μ_{CR} = 0.257
 Diversity = 3.5789e-06 k_t = [1, 1, 1, 1]
 Generation 9640: Best = 3.010524e+03 FES = 964100 μ_F = 0.719 μ_{CR} = 0.257
 Diversity = 3.8077e-06 k_t = [4, 2, 3, 2]
 Generation 9650: Best = 3.010524e+03 FES = 965100 μ_F = 0.695 μ_{CR} = 0.266
 Diversity = 3.7232e-06 k_t = [1, 1, 1, 1]
 Generation 9660: Best = 3.010524e+03 FES = 966100 μ_F = 0.708 μ_{CR} = 0.258
 Diversity = 3.4170e-06 k_t = [4, 3, 4, 3]
 Generation 9670: Best = 3.010524e+03 FES = 967100 μ_F = 0.711 μ_{CR} = 0.261
 Diversity = 3.4478e-06 k_t = [1, 1, 1, 1]

Generation 9680: Best = 3.010524e+03	FES = 968100	$\mu_F = 0.718$	$\mu_{CR} = 0.260$
Diversity = 4.1545e-06	$k_t = [4, 3, 2, 2]$		
Generation 9690: Best = 3.010524e+03	FES = 969100	$\mu_F = 0.729$	$\mu_{CR} = 0.264$
Diversity = 4.6328e-06	$k_t = [1, 1, 1, 1]$		
Generation 9700: Best = 3.010524e+03	FES = 970100	$\mu_F = 0.740$	$\mu_{CR} = 0.263$
Diversity = 5.5940e-06	$k_t = [4, 2, 4, 2]$		
Generation 9710: Best = 3.010524e+03	FES = 971100	$\mu_F = 0.737$	$\mu_{CR} = 0.264$
Diversity = 6.3005e-06	$k_t = [1, 1, 1, 1]$		
Generation 9720: Best = 3.010524e+03	FES = 972100	$\mu_F = 0.754$	$\mu_{CR} = 0.269$
Diversity = 8.0346e-06	$k_t = [4, 4, 2, 2]$		
Generation 9730: Best = 3.010524e+03	FES = 973100	$\mu_F = 0.773$	$\mu_{CR} = 0.264$
Diversity = 1.1440e-05	$k_t = [1, 1, 1, 1]$		
Generation 9740: Best = 3.010524e+03	FES = 974100	$\mu_F = 0.777$	$\mu_{CR} = 0.265$
Diversity = 1.4192e-05	$k_t = [3, 2, 4, 3]$		
Generation 9750: Best = 3.010524e+03	FES = 975100	$\mu_F = 0.766$	$\mu_{CR} = 0.260$
Diversity = 1.8156e-05	$k_t = [1, 1, 1, 1]$		
Generation 9760: Best = 3.010524e+03	FES = 976100	$\mu_F = 0.771$	$\mu_{CR} = 0.258$
Diversity = 1.9603e-05	$k_t = [2, 4, 4, 4]$		
Generation 9770: Best = 3.010524e+03	FES = 977100	$\mu_F = 0.766$	$\mu_{CR} = 0.270$
Diversity = 2.0454e-05	$k_t = [1, 1, 1, 1]$		
Generation 9780: Best = 3.010524e+03	FES = 978100	$\mu_F = 0.749$	$\mu_{CR} = 0.268$
Diversity = 1.8925e-05	$k_t = [2, 4, 3, 3]$		
Generation 9790: Best = 3.010524e+03	FES = 979100	$\mu_F = 0.738$	$\mu_{CR} = 0.273$
Diversity = 1.9088e-05	$k_t = [1, 1, 1, 1]$		
Generation 9800: Best = 3.010524e+03	FES = 980100	$\mu_F = 0.730$	$\mu_{CR} = 0.275$
Diversity = 1.8961e-05	$k_t = [2, 4, 3, 3]$		
Generation 9810: Best = 3.010524e+03	FES = 981100	$\mu_F = 0.727$	$\mu_{CR} = 0.278$
Diversity = 1.8544e-05	$k_t = [1, 1, 1, 1]$		
Generation 9820: Best = 3.010524e+03	FES = 982100	$\mu_F = 0.731$	$\mu_{CR} = 0.291$
Diversity = 2.0133e-05	$k_t = [4, 2, 2, 3]$		
Generation 9830: Best = 3.010524e+03	FES = 983100	$\mu_F = 0.725$	$\mu_{CR} = 0.292$
Diversity = 2.1151e-05	$k_t = [1, 1, 1, 1]$		
Generation 9840: Best = 3.010524e+03	FES = 984100	$\mu_F = 0.722$	$\mu_{CR} = 0.274$
Diversity = 2.1153e-05	$k_t = [3, 4, 4, 2]$		
Generation 9850: Best = 3.010524e+03	FES = 985100	$\mu_F = 0.716$	$\mu_{CR} = 0.276$
Diversity = 2.0210e-05	$k_t = [1, 1, 1, 1]$		
Generation 9860: Best = 3.010524e+03	FES = 986100	$\mu_F = 0.710$	$\mu_{CR} = 0.270$
Diversity = 1.9239e-05	$k_t = [2, 3, 4, 3]$		
Generation 9870: Best = 3.010524e+03	FES = 987100	$\mu_F = 0.724$	$\mu_{CR} = 0.270$
Diversity = 2.2365e-05	$k_t = [1, 1, 1, 1]$		
Generation 9880: Best = 3.010524e+03	FES = 988100	$\mu_F = 0.732$	$\mu_{CR} = 0.274$
Diversity = 2.2860e-05	$k_t = [4, 4, 3, 4]$		
Generation 9890: Best = 3.010524e+03	FES = 989100	$\mu_F = 0.735$	$\mu_{CR} = 0.275$
Diversity = 2.1609e-05	$k_t = [1, 1, 1, 1]$		
Generation 9900: Best = 3.010524e+03	FES = 990100	$\mu_F = 0.730$	$\mu_{CR} = 0.280$
Diversity = 2.0881e-05	$k_t = [2, 4, 4, 3]$		
Generation 9910: Best = 3.010524e+03	FES = 991100	$\mu_F = 0.721$	$\mu_{CR} = 0.270$
Diversity = 2.1127e-05	$k_t = [1, 1, 1, 1]$		
Generation 9920: Best = 3.010524e+03	FES = 992100	$\mu_F = 0.706$	$\mu_{CR} = 0.271$
Diversity = 1.8035e-05	$k_t = [3, 4, 2, 4]$		
Generation 9930: Best = 3.010524e+03	FES = 993100	$\mu_F = 0.715$	$\mu_{CR} = 0.274$
Diversity = 1.9197e-05	$k_t = [1, 1, 1, 1]$		
Generation 9940: Best = 3.010524e+03	FES = 994100	$\mu_F = 0.725$	$\mu_{CR} = 0.287$
Diversity = 2.6203e-05	$k_t = [4, 2, 3, 2]$		

Generation 9950: Best = 3.010524e+03 FES = 995100 μ_F = 0.739 μ_{CR} = 0.295
 Diversity = 3.3801e-05 k_t = [1, 1, 1, 1]
 Generation 9960: Best = 3.010524e+03 FES = 996100 μ_F = 0.750 μ_{CR} = 0.296
 Diversity = 3.6334e-05 k_t = [2, 4, 2, 2]
 Generation 9970: Best = 3.010524e+03 FES = 997100 μ_F = 0.740 μ_{CR} = 0.295
 Diversity = 3.7551e-05 k_t = [1, 1, 1, 1]
 Generation 9980: Best = 3.010524e+03 FES = 998100 μ_F = 0.740 μ_{CR} = 0.294
 Diversity = 3.6259e-05 k_t = [4, 3, 2, 4]
 Generation 9990: Best = 3.010524e+03 FES = 999100 μ_F = 0.730 μ_{CR} = 0.294
 Diversity = 3.4190e-05 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 3.010524e+03

Total Function Evals = 1000000

Total Successful Updates = 299229

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Results:

Best Fitness : 3.010524e+03

Global Optimum : 2.300000e+03

Error (|best-optimum|) : 7.105236e+02

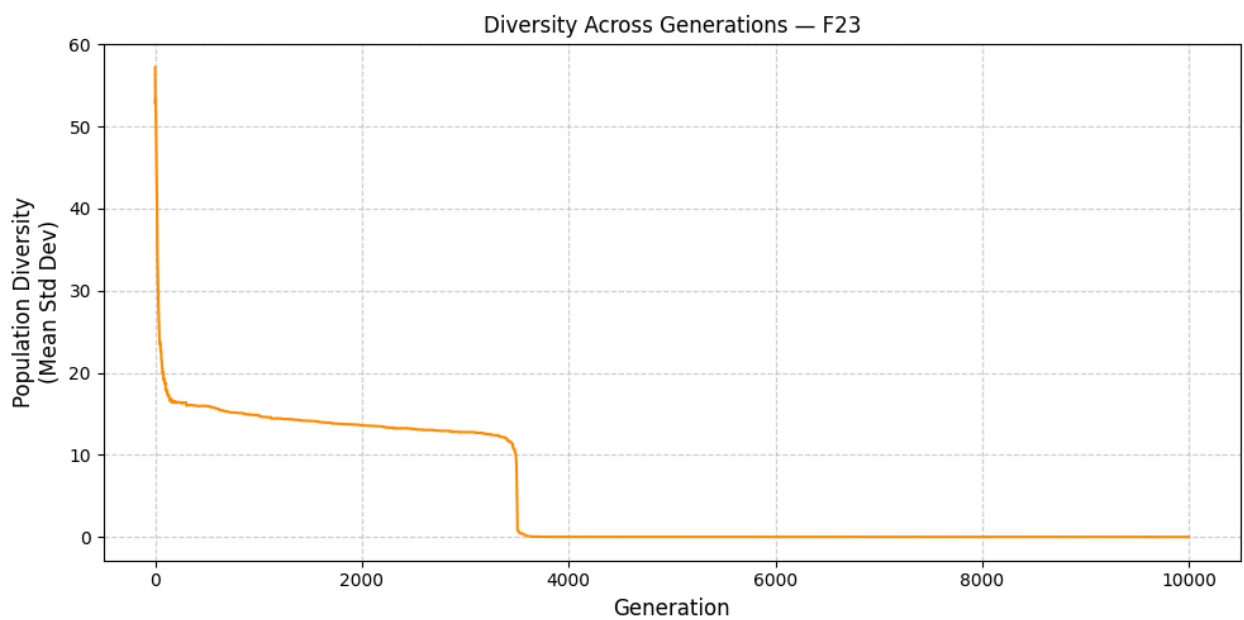
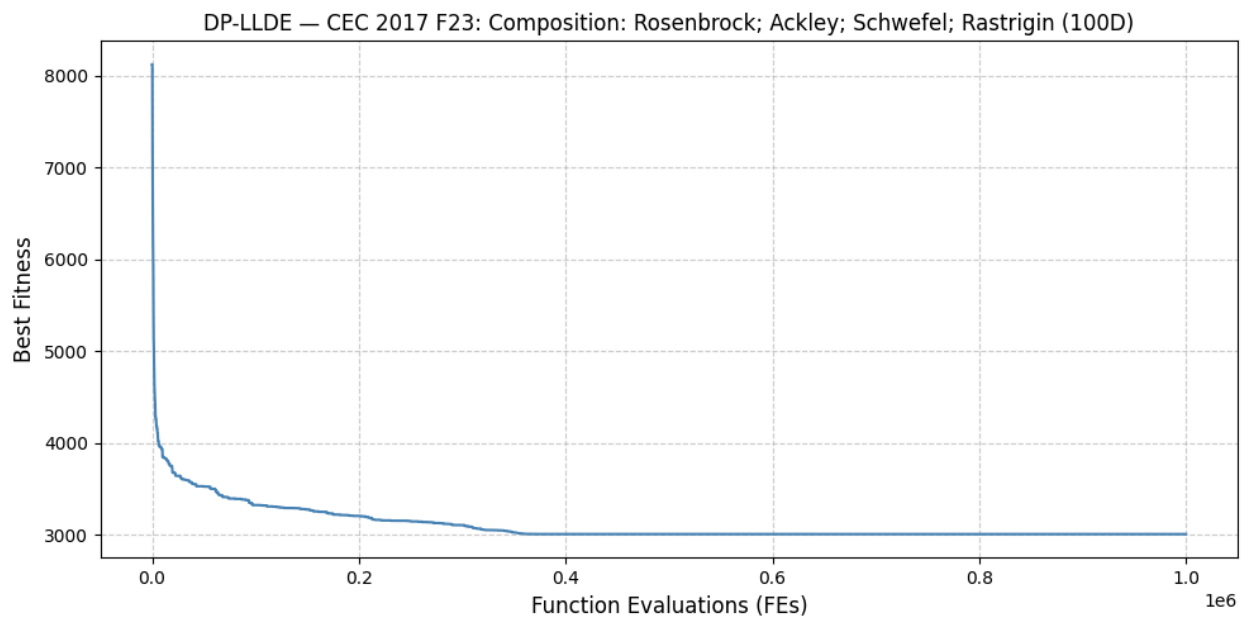
Best Solution (5 dim) : [-86.74013727 -95.36331553 -73.20681524 90.80787614
-66.56105762]

D^(0) (init. diversity): 5.7220e+01

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✓ Checkpoint saved to: /content/drive/My Drive/thesis_results/all_results_checkpoint.pkl



Graph saved as convergence_F23.png

✓ Plot saved to: /content/drive/My Drive/thesis_results/plots/convergence_F23.png

```
In [ ]: # =====
# F24 — Composition Function 4 (Ackley; Elliptic; Griewank; Rastrigin)
# =====
run_and_plot('F24', make_wrapper(composition.f24), 2400.0, 'Composition: Ackle
```

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DP-LLDE on CEC 2017 F24 – Single Run

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Function : Composition: Ackley; Elliptic; Griewank; Rastrigin (100D)
Bounds : [-100, 100]
Global optimum : 2400.0
Initial $D^*(0) = 5.7220e+01$ | Diversity threshold = 5.7220e+00
Generation 0: Best Fitness = 1.419682e+04
Generation 10: Best = 9.154106e+03 FES = 1100 $\mu_F = 0.598$ $\mu_{CR} = 0.351$
Diversity = 4.4085e+01 $k_t = [1, 1, 1, 1]$
Generation 20: Best = 6.833074e+03 FES = 2100 $\mu_F = 0.663$ $\mu_{CR} = 0.349$
Diversity = 3.4072e+01 $k_t = [2, 3, 2, 3]$
Generation 30: Best = 5.509125e+03 FES = 3100 $\mu_F = 0.696$ $\mu_{CR} = 0.355$
Diversity = 2.9091e+01 $k_t = [1, 1, 1, 1]$
Generation 40: Best = 5.011203e+03 FES = 4100 $\mu_F = 0.713$ $\mu_{CR} = 0.356$
Diversity = 2.3288e+01 $k_t = [3, 2, 2, 2]$
Generation 50: Best = 4.790614e+03 FES = 5100 $\mu_F = 0.719$ $\mu_{CR} = 0.354$
Diversity = 1.9629e+01 $k_t = [1, 1, 1, 1]$
Generation 60: Best = 4.543490e+03 FES = 6100 $\mu_F = 0.718$ $\mu_{CR} = 0.341$
Diversity = 1.6323e+01 $k_t = [4, 2, 2, 2]$
Generation 70: Best = 4.492522e+03 FES = 7100 $\mu_F = 0.715$ $\mu_{CR} = 0.328$
Diversity = 1.5113e+01 $k_t = [1, 1, 1, 1]$
Generation 80: Best = 4.425276e+03 FES = 8100 $\mu_F = 0.712$ $\mu_{CR} = 0.307$
Diversity = 1.3532e+01 $k_t = [4, 4, 2, 4]$
Generation 90: Best = 4.353423e+03 FES = 9100 $\mu_F = 0.702$ $\mu_{CR} = 0.293$
Diversity = 1.2520e+01 $k_t = [1, 1, 1, 1]$
Generation 100: Best = 4.335052e+03 FES = 10100 $\mu_F = 0.709$ $\mu_{CR} = 0.280$
Diversity = 1.1595e+01 $k_t = [2, 3, 2, 4]$
Generation 110: Best = 4.333507e+03 FES = 11100 $\mu_F = 0.692$ $\mu_{CR} = 0.242$
Diversity = 1.1245e+01 $k_t = [1, 1, 1, 1]$
Generation 120: Best = 4.290348e+03 FES = 12100 $\mu_F = 0.692$ $\mu_{CR} = 0.211$
Diversity = 1.0507e+01 $k_t = [2, 4, 2, 2]$
Generation 130: Best = 4.290348e+03 FES = 13100 $\mu_F = 0.660$ $\mu_{CR} = 0.175$
Diversity = 1.0214e+01 $k_t = [1, 1, 1, 1]$
Generation 140: Best = 4.236916e+03 FES = 14100 $\mu_F = 0.680$ $\mu_{CR} = 0.133$
Diversity = 9.8846e+00 $k_t = [3, 4, 4, 3]$
Generation 150: Best = 4.230979e+03 FES = 15100 $\mu_F = 0.709$ $\mu_{CR} = 0.101$
Diversity = 9.6586e+00 $k_t = [1, 1, 1, 1]$
Generation 160: Best = 4.187703e+03 FES = 16100 $\mu_F = 0.713$ $\mu_{CR} = 0.081$
Diversity = 9.5753e+00 $k_t = [4, 4, 2, 2]$
Generation 170: Best = 4.171550e+03 FES = 17100 $\mu_F = 0.717$ $\mu_{CR} = 0.079$
Diversity = 9.4720e+00 $k_t = [1, 1, 1, 1]$
Generation 180: Best = 4.166122e+03 FES = 18100 $\mu_F = 0.705$ $\mu_{CR} = 0.071$
Diversity = 9.2381e+00 $k_t = [3, 2, 3, 2]$
Generation 190: Best = 4.166122e+03 FES = 19100 $\mu_F = 0.715$ $\mu_{CR} = 0.061$
Diversity = 9.1753e+00 $k_t = [1, 1, 1, 1]$
Generation 200: Best = 4.165289e+03 FES = 20100 $\mu_F = 0.687$ $\mu_{CR} = 0.053$
Diversity = 9.1110e+00 $k_t = [4, 2, 3, 4]$
Generation 210: Best = 4.159848e+03 FES = 21100 $\mu_F = 0.679$ $\mu_{CR} = 0.056$
Diversity = 9.0434e+00 $k_t = [1, 1, 1, 1]$
Generation 220: Best = 4.153520e+03 FES = 22100 $\mu_F = 0.703$ $\mu_{CR} = 0.061$
Diversity = 8.8499e+00 $k_t = [4, 4, 4, 4]$

Generation 230:	Best = 4.104536e+03	FES = 23100	$\mu_F = 0.716$	$\mu_{CR} = 0.058$
Diversity = 8.7469e+00	$k_t = [1, 1, 1, 1]$			
Generation 240:	Best = 4.104355e+03	FES = 24100	$\mu_F = 0.749$	$\mu_{CR} = 0.046$
Diversity = 8.7442e+00	$k_t = [3, 2, 3, 3]$			
Generation 250:	Best = 4.104355e+03	FES = 25100	$\mu_F = 0.719$	$\mu_{CR} = 0.055$
Diversity = 8.7124e+00	$k_t = [1, 1, 1, 1]$			
Generation 260:	Best = 4.104355e+03	FES = 26100	$\mu_F = 0.758$	$\mu_{CR} = 0.054$
Diversity = 8.7562e+00	$k_t = [3, 2, 3, 4]$			
Generation 270:	Best = 4.104355e+03	FES = 27100	$\mu_F = 0.663$	$\mu_{CR} = 0.065$
Diversity = 8.7868e+00	$k_t = [1, 1, 1, 1]$			
Generation 280:	Best = 4.104355e+03	FES = 28100	$\mu_F = 0.674$	$\mu_{CR} = 0.045$
Diversity = 8.7671e+00	$k_t = [2, 4, 4, 4]$			
Generation 290:	Best = 4.097081e+03	FES = 29100	$\mu_F = 0.662$	$\mu_{CR} = 0.043$
Diversity = 8.7884e+00	$k_t = [1, 1, 1, 1]$			
Generation 300:	Best = 4.097081e+03	FES = 30100	$\mu_F = 0.620$	$\mu_{CR} = 0.048$
Diversity = 8.7731e+00	$k_t = [4, 4, 3, 2]$			
Generation 310:	Best = 4.097081e+03	FES = 31100	$\mu_F = 0.646$	$\mu_{CR} = 0.050$
Diversity = 8.7591e+00	$k_t = [1, 1, 1, 1]$			
Generation 320:	Best = 4.087691e+03	FES = 32100	$\mu_F = 0.675$	$\mu_{CR} = 0.047$
Diversity = 8.6991e+00	$k_t = [3, 3, 2, 2]$			
Generation 330:	Best = 4.066977e+03	FES = 33100	$\mu_F = 0.642$	$\mu_{CR} = 0.047$
Diversity = 8.6898e+00	$k_t = [1, 1, 1, 1]$			
Generation 340:	Best = 4.065951e+03	FES = 34100	$\mu_F = 0.617$	$\mu_{CR} = 0.047$
Diversity = 8.6727e+00	$k_t = [4, 4, 4, 2]$			
Generation 350:	Best = 4.065951e+03	FES = 35100	$\mu_F = 0.560$	$\mu_{CR} = 0.039$
Diversity = 8.6278e+00	$k_t = [1, 1, 1, 1]$			
Generation 360:	Best = 4.065951e+03	FES = 36100	$\mu_F = 0.594$	$\mu_{CR} = 0.049$
Diversity = 8.5481e+00	$k_t = [4, 2, 4, 4]$			
Generation 370:	Best = 4.061996e+03	FES = 37100	$\mu_F = 0.563$	$\mu_{CR} = 0.044$
Diversity = 8.4851e+00	$k_t = [1, 1, 1, 1]$			
Generation 380:	Best = 4.015000e+03	FES = 38100	$\mu_F = 0.553$	$\mu_{CR} = 0.054$
Diversity = 8.5021e+00	$k_t = [4, 2, 2, 3]$			
Generation 390:	Best = 4.012940e+03	FES = 39100	$\mu_F = 0.555$	$\mu_{CR} = 0.047$
Diversity = 8.4976e+00	$k_t = [1, 1, 1, 1]$			
Generation 400:	Best = 4.012940e+03	FES = 40100	$\mu_F = 0.573$	$\mu_{CR} = 0.040$
Diversity = 8.4490e+00	$k_t = [2, 2, 3, 3]$			
Generation 410:	Best = 4.012940e+03	FES = 41100	$\mu_F = 0.575$	$\mu_{CR} = 0.038$
Diversity = 8.4397e+00	$k_t = [1, 1, 1, 1]$			
Generation 420:	Best = 4.009015e+03	FES = 42100	$\mu_F = 0.576$	$\mu_{CR} = 0.038$
Diversity = 8.3162e+00	$k_t = [4, 3, 3, 4]$			
Generation 430:	Best = 4.008503e+03	FES = 43100	$\mu_F = 0.551$	$\mu_{CR} = 0.050$
Diversity = 8.2895e+00	$k_t = [1, 1, 1, 1]$			
Generation 440:	Best = 3.992527e+03	FES = 44100	$\mu_F = 0.565$	$\mu_{CR} = 0.037$
Diversity = 8.2773e+00	$k_t = [4, 2, 2, 3]$			
Generation 450:	Best = 3.974250e+03	FES = 45100	$\mu_F = 0.572$	$\mu_{CR} = 0.038$
Diversity = 8.2427e+00	$k_t = [1, 1, 1, 1]$			
Generation 460:	Best = 3.974250e+03	FES = 46100	$\mu_F = 0.548$	$\mu_{CR} = 0.044$
Diversity = 8.1730e+00	$k_t = [4, 3, 4, 4]$			
Generation 470:	Best = 3.958756e+03	FES = 47100	$\mu_F = 0.573$	$\mu_{CR} = 0.041$
Diversity = 8.1636e+00	$k_t = [1, 1, 1, 1]$			
Generation 480:	Best = 3.948437e+03	FES = 48100	$\mu_F = 0.580$	$\mu_{CR} = 0.039$
Diversity = 8.1924e+00	$k_t = [4, 2, 4, 3]$			
Generation 490:	Best = 3.948437e+03	FES = 49100	$\mu_F = 0.552$	$\mu_{CR} = 0.043$
Diversity = 8.1854e+00	$k_t = [1, 1, 1, 1]$			

Generation 500: Best = 3.948437e+03	FES = 50100	$\mu_F = 0.511$	$\mu_{CR} = 0.032$
Diversity = 8.1733e+00	$k_t = [2, 2, 3, 2]$		
Generation 510: Best = 3.948437e+03	FES = 51100	$\mu_F = 0.552$	$\mu_{CR} = 0.039$
Diversity = 8.1158e+00	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 3.948437e+03	FES = 52100	$\mu_F = 0.558$	$\mu_{CR} = 0.048$
Diversity = 8.0969e+00	$k_t = [4, 3, 4, 2]$		
Generation 530: Best = 3.948437e+03	FES = 53100	$\mu_F = 0.538$	$\mu_{CR} = 0.037$
Diversity = 8.0783e+00	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 3.948437e+03	FES = 54100	$\mu_F = 0.542$	$\mu_{CR} = 0.050$
Diversity = 8.0691e+00	$k_t = [4, 4, 3, 2]$		
Generation 550: Best = 3.946452e+03	FES = 55100	$\mu_F = 0.512$	$\mu_{CR} = 0.039$
Diversity = 8.0527e+00	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 3.944619e+03	FES = 56100	$\mu_F = 0.542$	$\mu_{CR} = 0.027$
Diversity = 8.0937e+00	$k_t = [2, 2, 2, 2]$		
Generation 570: Best = 3.944619e+03	FES = 57100	$\mu_F = 0.596$	$\mu_{CR} = 0.025$
Diversity = 8.0811e+00	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 3.941241e+03	FES = 58100	$\mu_F = 0.611$	$\mu_{CR} = 0.047$
Diversity = 7.9754e+00	$k_t = [2, 4, 3, 4]$		
Generation 590: Best = 3.941241e+03	FES = 59100	$\mu_F = 0.597$	$\mu_{CR} = 0.041$
Diversity = 7.9903e+00	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 3.941241e+03	FES = 60100	$\mu_F = 0.610$	$\mu_{CR} = 0.032$
Diversity = 7.9252e+00	$k_t = [2, 4, 3, 4]$		
Generation 610: Best = 3.938673e+03	FES = 61100	$\mu_F = 0.664$	$\mu_{CR} = 0.031$
Diversity = 7.9415e+00	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 3.935315e+03	FES = 62100	$\mu_F = 0.678$	$\mu_{CR} = 0.031$
Diversity = 7.8900e+00	$k_t = [2, 3, 3, 3]$		
Generation 630: Best = 3.935315e+03	FES = 63100	$\mu_F = 0.726$	$\mu_{CR} = 0.029$
Diversity = 7.8797e+00	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 3.933440e+03	FES = 64100	$\mu_F = 0.672$	$\mu_{CR} = 0.030$
Diversity = 7.8680e+00	$k_t = [3, 2, 3, 3]$		
Generation 650: Best = 3.930500e+03	FES = 65100	$\mu_F = 0.610$	$\mu_{CR} = 0.026$
Diversity = 7.8679e+00	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 3.930500e+03	FES = 66100	$\mu_F = 0.544$	$\mu_{CR} = 0.047$
Diversity = 7.8550e+00	$k_t = [3, 4, 3, 3]$		
Generation 670: Best = 3.909424e+03	FES = 67100	$\mu_F = 0.519$	$\mu_{CR} = 0.040$
Diversity = 7.8464e+00	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 3.909337e+03	FES = 68100	$\mu_F = 0.448$	$\mu_{CR} = 0.036$
Diversity = 7.8335e+00	$k_t = [3, 2, 4, 2]$		
Generation 690: Best = 3.906636e+03	FES = 69100	$\mu_F = 0.399$	$\mu_{CR} = 0.045$
Diversity = 7.8252e+00	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 3.883677e+03	FES = 70100	$\mu_F = 0.492$	$\mu_{CR} = 0.043$
Diversity = 7.8607e+00	$k_t = [2, 2, 3, 4]$		
Generation 710: Best = 3.883677e+03	FES = 71100	$\mu_F = 0.504$	$\mu_{CR} = 0.044$
Diversity = 7.8470e+00	$k_t = [1, 1, 1, 1]$		
Generation 720: Best = 3.882529e+03	FES = 72100	$\mu_F = 0.491$	$\mu_{CR} = 0.045$
Diversity = 7.8437e+00	$k_t = [4, 3, 2, 4]$		
Generation 730: Best = 3.879143e+03	FES = 73100	$\mu_F = 0.477$	$\mu_{CR} = 0.068$
Diversity = 7.8437e+00	$k_t = [1, 1, 1, 1]$		
Generation 740: Best = 3.879143e+03	FES = 74100	$\mu_F = 0.475$	$\mu_{CR} = 0.055$
Diversity = 7.8639e+00	$k_t = [2, 3, 2, 2]$		
Generation 750: Best = 3.879143e+03	FES = 75100	$\mu_F = 0.433$	$\mu_{CR} = 0.050$
Diversity = 7.8616e+00	$k_t = [1, 1, 1, 1]$		
Generation 760: Best = 3.879143e+03	FES = 76100	$\mu_F = 0.363$	$\mu_{CR} = 0.062$
Diversity = 7.8604e+00	$k_t = [2, 2, 4, 4]$		

Generation 770: Best = 3.877122e+03	FES = 77100	$\mu_F = 0.406$	$\mu_{CR} = 0.062$
Diversity = 7.8298e+00	$k_t = [1, 1, 1, 1]$		
Generation 780: Best = 3.877122e+03	FES = 78100	$\mu_F = 0.388$	$\mu_{CR} = 0.066$
Diversity = 7.7961e+00	$k_t = [3, 4, 3, 4]$		
Generation 790: Best = 3.877122e+03	FES = 79100	$\mu_F = 0.366$	$\mu_{CR} = 0.049$
Diversity = 7.7815e+00	$k_t = [1, 1, 1, 1]$		
Generation 800: Best = 3.876488e+03	FES = 80100	$\mu_F = 0.418$	$\mu_{CR} = 0.050$
Diversity = 7.7667e+00	$k_t = [3, 2, 4, 4]$		
Generation 810: Best = 3.868520e+03	FES = 81100	$\mu_F = 0.355$	$\mu_{CR} = 0.057$
Diversity = 7.7202e+00	$k_t = [1, 1, 1, 1]$		
Generation 820: Best = 3.868520e+03	FES = 82100	$\mu_F = 0.320$	$\mu_{CR} = 0.054$
Diversity = 7.7051e+00	$k_t = [3, 3, 3, 2]$		
Generation 830: Best = 3.863726e+03	FES = 83100	$\mu_F = 0.329$	$\mu_{CR} = 0.050$
Diversity = 7.6172e+00	$k_t = [1, 1, 1, 1]$		
Generation 840: Best = 3.857951e+03	FES = 84100	$\mu_F = 0.381$	$\mu_{CR} = 0.046$
Diversity = 7.5707e+00	$k_t = [2, 4, 4, 4]$		
Generation 850: Best = 3.857216e+03	FES = 85100	$\mu_F = 0.385$	$\mu_{CR} = 0.042$
Diversity = 7.5563e+00	$k_t = [1, 1, 1, 1]$		
Generation 860: Best = 3.857216e+03	FES = 86100	$\mu_F = 0.460$	$\mu_{CR} = 0.039$
Diversity = 7.5516e+00	$k_t = [4, 4, 3, 2]$		
Generation 870: Best = 3.857075e+03	FES = 87100	$\mu_F = 0.459$	$\mu_{CR} = 0.033$
Diversity = 7.5337e+00	$k_t = [1, 1, 1, 1]$		
Generation 880: Best = 3.857075e+03	FES = 88100	$\mu_F = 0.490$	$\mu_{CR} = 0.030$
Diversity = 7.5307e+00	$k_t = [2, 4, 4, 3]$		
Generation 890: Best = 3.852038e+03	FES = 89100	$\mu_F = 0.434$	$\mu_{CR} = 0.036$
Diversity = 7.5279e+00	$k_t = [1, 1, 1, 1]$		
Generation 900: Best = 3.852038e+03	FES = 90100	$\mu_F = 0.454$	$\mu_{CR} = 0.047$
Diversity = 7.5185e+00	$k_t = [4, 2, 4, 4]$		
Generation 910: Best = 3.849555e+03	FES = 91100	$\mu_F = 0.424$	$\mu_{CR} = 0.043$
Diversity = 7.5121e+00	$k_t = [1, 1, 1, 1]$		
Generation 920: Best = 3.849555e+03	FES = 92100	$\mu_F = 0.455$	$\mu_{CR} = 0.034$
Diversity = 7.4315e+00	$k_t = [4, 3, 3, 4]$		
Generation 930: Best = 3.835466e+03	FES = 93100	$\mu_F = 0.402$	$\mu_{CR} = 0.040$
Diversity = 7.4198e+00	$k_t = [1, 1, 1, 1]$		
Generation 940: Best = 3.834164e+03	FES = 94100	$\mu_F = 0.354$	$\mu_{CR} = 0.048$
Diversity = 7.4143e+00	$k_t = [2, 3, 2, 2]$		
Generation 950: Best = 3.829755e+03	FES = 95100	$\mu_F = 0.395$	$\mu_{CR} = 0.045$
Diversity = 7.4026e+00	$k_t = [1, 1, 1, 1]$		
Generation 960: Best = 3.829755e+03	FES = 96100	$\mu_F = 0.442$	$\mu_{CR} = 0.045$
Diversity = 7.3771e+00	$k_t = [2, 2, 3, 4]$		
Generation 970: Best = 3.829755e+03	FES = 97100	$\mu_F = 0.403$	$\mu_{CR} = 0.037$
Diversity = 7.3766e+00	$k_t = [1, 1, 1, 1]$		
Generation 980: Best = 3.829755e+03	FES = 98100	$\mu_F = 0.404$	$\mu_{CR} = 0.040$
Diversity = 7.3634e+00	$k_t = [3, 2, 2, 4]$		
Generation 990: Best = 3.822224e+03	FES = 99100	$\mu_F = 0.363$	$\mu_{CR} = 0.040$
Diversity = 7.3571e+00	$k_t = [1, 1, 1, 1]$		
Generation 1000: Best = 3.818784e+03	FES = 100100	$\mu_F = 0.462$	$\mu_{CR} = 0.039$
Diversity = 7.3398e+00	$k_t = [4, 4, 4, 4]$		
Generation 1010: Best = 3.812147e+03	FES = 101100	$\mu_F = 0.466$	$\mu_{CR} = 0.037$
Diversity = 7.3013e+00	$k_t = [1, 1, 1, 1]$		
Generation 1020: Best = 3.812147e+03	FES = 102100	$\mu_F = 0.529$	$\mu_{CR} = 0.037$
Diversity = 7.2740e+00	$k_t = [3, 3, 2, 4]$		
Generation 1030: Best = 3.812125e+03	FES = 103100	$\mu_F = 0.480$	$\mu_{CR} = 0.034$
Diversity = 7.2722e+00	$k_t = [1, 1, 1, 1]$		

Generation 1040: Best = 3.812125e+03 FES = 104100 μ_F = 0.507 μ_{CR} = 0.030
 Diversity = 7.2472e+00 k_t = [4, 2, 4, 4]
 Generation 1050: Best = 3.811222e+03 FES = 105100 μ_F = 0.502 μ_{CR} = 0.031
 Diversity = 7.2455e+00 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 3.811222e+03 FES = 106100 μ_F = 0.608 μ_{CR} = 0.029
 Diversity = 7.2593e+00 k_t = [2, 4, 3, 2]
 Generation 1070: Best = 3.805499e+03 FES = 107100 μ_F = 0.570 μ_{CR} = 0.030
 Diversity = 7.2608e+00 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 3.794427e+03 FES = 108100 μ_F = 0.442 μ_{CR} = 0.049
 Diversity = 7.2571e+00 k_t = [4, 3, 3, 3]
 Generation 1090: Best = 3.794427e+03 FES = 109100 μ_F = 0.400 μ_{CR} = 0.045
 Diversity = 7.2534e+00 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 3.793224e+03 FES = 110100 μ_F = 0.457 μ_{CR} = 0.037
 Diversity = 7.2484e+00 k_t = [3, 3, 3, 3]
 Generation 1110: Best = 3.792660e+03 FES = 111100 μ_F = 0.411 μ_{CR} = 0.038
 Diversity = 7.2416e+00 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 3.792660e+03 FES = 112100 μ_F = 0.422 μ_{CR} = 0.041
 Diversity = 7.2393e+00 k_t = [4, 3, 3, 2]
 Generation 1130: Best = 3.779809e+03 FES = 113100 μ_F = 0.409 μ_{CR} = 0.038
 Diversity = 7.2377e+00 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 3.779809e+03 FES = 114100 μ_F = 0.467 μ_{CR} = 0.041
 Diversity = 7.2309e+00 k_t = [2, 4, 3, 2]
 Generation 1150: Best = 3.779809e+03 FES = 115100 μ_F = 0.427 μ_{CR} = 0.052
 Diversity = 7.2341e+00 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 3.779809e+03 FES = 116100 μ_F = 0.521 μ_{CR} = 0.030
 Diversity = 7.2335e+00 k_t = [3, 3, 2, 3]
 Generation 1170: Best = 3.769477e+03 FES = 117100 μ_F = 0.562 μ_{CR} = 0.036
 Diversity = 7.2379e+00 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 3.765505e+03 FES = 118100 μ_F = 0.519 μ_{CR} = 0.037
 Diversity = 7.2387e+00 k_t = [4, 3, 3, 3]
 Generation 1190: Best = 3.765505e+03 FES = 119100 μ_F = 0.502 μ_{CR} = 0.030
 Diversity = 7.2472e+00 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 3.761542e+03 FES = 120100 μ_F = 0.510 μ_{CR} = 0.037
 Diversity = 7.2254e+00 k_t = [3, 2, 2, 2]
 Generation 1210: Best = 3.761286e+03 FES = 121100 μ_F = 0.554 μ_{CR} = 0.028
 Diversity = 7.2254e+00 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 3.761286e+03 FES = 122100 μ_F = 0.583 μ_{CR} = 0.029
 Diversity = 7.2064e+00 k_t = [4, 2, 2, 4]
 Generation 1230: Best = 3.761286e+03 FES = 123100 μ_F = 0.568 μ_{CR} = 0.030
 Diversity = 7.2153e+00 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 3.761286e+03 FES = 124100 μ_F = 0.515 μ_{CR} = 0.025
 Diversity = 7.2118e+00 k_t = [4, 3, 3, 2]
 Generation 1250: Best = 3.761286e+03 FES = 125100 μ_F = 0.624 μ_{CR} = 0.020
 Diversity = 7.2322e+00 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 3.759346e+03 FES = 126100 μ_F = 0.633 μ_{CR} = 0.020
 Diversity = 7.2005e+00 k_t = [3, 3, 3, 4]
 Generation 1270: Best = 3.754688e+03 FES = 127100 μ_F = 0.642 μ_{CR} = 0.022
 Diversity = 7.2002e+00 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 3.754466e+03 FES = 128100 μ_F = 0.529 μ_{CR} = 0.034
 Diversity = 7.2084e+00 k_t = [2, 2, 3, 4]
 Generation 1290: Best = 3.754466e+03 FES = 129100 μ_F = 0.504 μ_{CR} = 0.031
 Diversity = 7.2058e+00 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 3.754466e+03 FES = 130100 μ_F = 0.482 μ_{CR} = 0.031
 Diversity = 7.2017e+00 k_t = [2, 3, 4, 2]

Generation 1310: Best = 3.745416e+03 FES = 131100 μ_F = 0.463 μ_{CR} = 0.029
 Diversity = 7.2003e+00 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 3.745416e+03 FES = 132100 μ_F = 0.532 μ_{CR} = 0.030
 Diversity = 7.1967e+00 k_t = [3, 3, 2, 3]
 Generation 1330: Best = 3.745403e+03 FES = 133100 μ_F = 0.511 μ_{CR} = 0.045
 Diversity = 7.1962e+00 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 3.738872e+03 FES = 134100 μ_F = 0.460 μ_{CR} = 0.049
 Diversity = 7.1937e+00 k_t = [3, 3, 3, 3]
 Generation 1350: Best = 3.738872e+03 FES = 135100 μ_F = 0.469 μ_{CR} = 0.051
 Diversity = 7.1801e+00 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 3.738872e+03 FES = 136100 μ_F = 0.424 μ_{CR} = 0.047
 Diversity = 7.1797e+00 k_t = [3, 4, 2, 2]
 Generation 1370: Best = 3.738872e+03 FES = 137100 μ_F = 0.454 μ_{CR} = 0.037
 Diversity = 7.1780e+00 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 3.737798e+03 FES = 138100 μ_F = 0.433 μ_{CR} = 0.034
 Diversity = 7.1733e+00 k_t = [2, 2, 2, 2]
 Generation 1390: Best = 3.737798e+03 FES = 139100 μ_F = 0.444 μ_{CR} = 0.031
 Diversity = 7.1729e+00 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 3.737798e+03 FES = 140100 μ_F = 0.478 μ_{CR} = 0.031
 Diversity = 7.2001e+00 k_t = [2, 4, 3, 3]
 Generation 1410: Best = 3.735907e+03 FES = 141100 μ_F = 0.500 μ_{CR} = 0.029
 Diversity = 7.2004e+00 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 3.735907e+03 FES = 142100 μ_F = 0.436 μ_{CR} = 0.023
 Diversity = 7.1996e+00 k_t = [3, 2, 2, 2]
 Generation 1430: Best = 3.722696e+03 FES = 143100 μ_F = 0.440 μ_{CR} = 0.030
 Diversity = 7.1989e+00 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 3.721483e+03 FES = 144100 μ_F = 0.453 μ_{CR} = 0.039
 Diversity = 7.1644e+00 k_t = [4, 3, 3, 4]
 Generation 1450: Best = 3.721483e+03 FES = 145100 μ_F = 0.469 μ_{CR} = 0.035
 Diversity = 7.1616e+00 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 3.721483e+03 FES = 146100 μ_F = 0.454 μ_{CR} = 0.038
 Diversity = 7.1627e+00 k_t = [2, 3, 4, 3]
 Generation 1470: Best = 3.714760e+03 FES = 147100 μ_F = 0.397 μ_{CR} = 0.032
 Diversity = 7.1574e+00 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 3.714510e+03 FES = 148100 μ_F = 0.358 μ_{CR} = 0.039
 Diversity = 7.1531e+00 k_t = [4, 4, 4, 3]
 Generation 1490: Best = 3.703304e+03 FES = 149100 μ_F = 0.342 μ_{CR} = 0.041
 Diversity = 7.1477e+00 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 3.703283e+03 FES = 150100 μ_F = 0.378 μ_{CR} = 0.041
 Diversity = 7.1058e+00 k_t = [2, 2, 3, 4]
 Generation 1510: Best = 3.703283e+03 FES = 151100 μ_F = 0.390 μ_{CR} = 0.040
 Diversity = 7.1062e+00 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 3.702100e+03 FES = 152100 μ_F = 0.345 μ_{CR} = 0.040
 Diversity = 7.1073e+00 k_t = [3, 3, 2, 2]
 Generation 1530: Best = 3.702100e+03 FES = 153100 μ_F = 0.351 μ_{CR} = 0.042
 Diversity = 7.1042e+00 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 3.702100e+03 FES = 154100 μ_F = 0.302 μ_{CR} = 0.048
 Diversity = 7.0740e+00 k_t = [3, 3, 3, 3]
 Generation 1550: Best = 3.699154e+03 FES = 155100 μ_F = 0.370 μ_{CR} = 0.034
 Diversity = 7.0619e+00 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 3.699154e+03 FES = 156100 μ_F = 0.378 μ_{CR} = 0.035
 Diversity = 7.0435e+00 k_t = [4, 3, 3, 4]
 Generation 1570: Best = 3.699013e+03 FES = 157100 μ_F = 0.384 μ_{CR} = 0.040
 Diversity = 7.0419e+00 k_t = [1, 1, 1, 1]

Generation 1580: Best = 3.699013e+03 FES = 158100 μ_F = 0.402 μ_{CR} = 0.036
 Diversity = 6.9709e+00 k_t = [2, 4, 4, 4]
 Generation 1590: Best = 3.699013e+03 FES = 159100 μ_F = 0.327 μ_{CR} = 0.040
 Diversity = 6.9635e+00 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 3.696534e+03 FES = 160100 μ_F = 0.407 μ_{CR} = 0.035
 Diversity = 6.9461e+00 k_t = [3, 4, 2, 4]
 Generation 1610: Best = 3.695724e+03 FES = 161100 μ_F = 0.423 μ_{CR} = 0.044
 Diversity = 6.9448e+00 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 3.695724e+03 FES = 162100 μ_F = 0.489 μ_{CR} = 0.036
 Diversity = 6.9487e+00 k_t = [3, 3, 3, 2]
 Generation 1630: Best = 3.693366e+03 FES = 163100 μ_F = 0.512 μ_{CR} = 0.027
 Diversity = 6.9328e+00 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 3.693366e+03 FES = 164100 μ_F = 0.492 μ_{CR} = 0.048
 Diversity = 6.9385e+00 k_t = [4, 3, 4, 4]
 Generation 1650: Best = 3.683612e+03 FES = 165100 μ_F = 0.464 μ_{CR} = 0.037
 Diversity = 6.9424e+00 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 3.683100e+03 FES = 166100 μ_F = 0.446 μ_{CR} = 0.035
 Diversity = 6.9419e+00 k_t = [4, 3, 2, 4]
 Generation 1670: Best = 3.680544e+03 FES = 167100 μ_F = 0.480 μ_{CR} = 0.033
 Diversity = 6.9266e+00 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 3.680544e+03 FES = 168100 μ_F = 0.560 μ_{CR} = 0.031
 Diversity = 6.9689e+00 k_t = [2, 2, 2, 4]
 Generation 1690: Best = 3.680076e+03 FES = 169100 μ_F = 0.602 μ_{CR} = 0.022
 Diversity = 6.9710e+00 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 3.680076e+03 FES = 170100 μ_F = 0.556 μ_{CR} = 0.034
 Diversity = 6.9700e+00 k_t = [4, 3, 4, 3]
 Generation 1710: Best = 3.680076e+03 FES = 171100 μ_F = 0.523 μ_{CR} = 0.033
 Diversity = 6.9692e+00 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 3.680076e+03 FES = 172100 μ_F = 0.456 μ_{CR} = 0.032
 Diversity = 6.9701e+00 k_t = [3, 4, 3, 3]
 Generation 1730: Best = 3.680076e+03 FES = 173100 μ_F = 0.522 μ_{CR} = 0.029
 Diversity = 6.9552e+00 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 3.680073e+03 FES = 174100 μ_F = 0.460 μ_{CR} = 0.030
 Diversity = 6.9513e+00 k_t = [4, 4, 4, 2]
 Generation 1750: Best = 3.679456e+03 FES = 175100 μ_F = 0.441 μ_{CR} = 0.030
 Diversity = 6.9511e+00 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 3.679456e+03 FES = 176100 μ_F = 0.457 μ_{CR} = 0.029
 Diversity = 6.8704e+00 k_t = [4, 2, 3, 4]
 Generation 1770: Best = 3.678983e+03 FES = 177100 μ_F = 0.405 μ_{CR} = 0.030
 Diversity = 6.8706e+00 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 3.678653e+03 FES = 178100 μ_F = 0.401 μ_{CR} = 0.024
 Diversity = 6.8680e+00 k_t = [2, 2, 2, 4]
 Generation 1790: Best = 3.677607e+03 FES = 179100 μ_F = 0.372 μ_{CR} = 0.041
 Diversity = 6.8446e+00 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 3.674351e+03 FES = 180100 μ_F = 0.396 μ_{CR} = 0.030
 Diversity = 6.8303e+00 k_t = [3, 3, 2, 4]
 Generation 1810: Best = 3.674351e+03 FES = 181100 μ_F = 0.396 μ_{CR} = 0.041
 Diversity = 6.8321e+00 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 3.674348e+03 FES = 182100 μ_F = 0.407 μ_{CR} = 0.047
 Diversity = 6.8216e+00 k_t = [4, 2, 4, 4]
 Generation 1830: Best = 3.674348e+03 FES = 183100 μ_F = 0.338 μ_{CR} = 0.055
 Diversity = 6.8181e+00 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 3.672054e+03 FES = 184100 μ_F = 0.363 μ_{CR} = 0.072
 Diversity = 6.7958e+00 k_t = [4, 3, 3, 4]

Generation 1850: Best = 3.670523e+03	FES = 185100	$\mu_F = 0.382$	$\mu_{CR} = 0.054$
Diversity = 6.7973e+00	$k_t = [1, 1, 1, 1]$		
Generation 1860: Best = 3.670523e+03	FES = 186100	$\mu_F = 0.412$	$\mu_{CR} = 0.042$
Diversity = 6.7255e+00	$k_t = [4, 2, 4, 4]$		
Generation 1870: Best = 3.668381e+03	FES = 187100	$\mu_F = 0.453$	$\mu_{CR} = 0.043$
Diversity = 6.6979e+00	$k_t = [1, 1, 1, 1]$		
Generation 1880: Best = 3.668381e+03	FES = 188100	$\mu_F = 0.451$	$\mu_{CR} = 0.034$
Diversity = 6.6775e+00	$k_t = [4, 4, 4, 3]$		
Generation 1890: Best = 3.664753e+03	FES = 189100	$\mu_F = 0.384$	$\mu_{CR} = 0.046$
Diversity = 6.6799e+00	$k_t = [1, 1, 1, 1]$		
Generation 1900: Best = 3.661701e+03	FES = 190100	$\mu_F = 0.435$	$\mu_{CR} = 0.036$
Diversity = 6.6839e+00	$k_t = [4, 2, 2, 2]$		
Generation 1910: Best = 3.661490e+03	FES = 191100	$\mu_F = 0.408$	$\mu_{CR} = 0.027$
Diversity = 6.6763e+00	$k_t = [1, 1, 1, 1]$		
Generation 1920: Best = 3.661490e+03	FES = 192100	$\mu_F = 0.418$	$\mu_{CR} = 0.030$
Diversity = 6.6538e+00	$k_t = [2, 3, 3, 3]$		
Generation 1930: Best = 3.660386e+03	FES = 193100	$\mu_F = 0.348$	$\mu_{CR} = 0.030$
Diversity = 6.6558e+00	$k_t = [1, 1, 1, 1]$		
Generation 1940: Best = 3.660386e+03	FES = 194100	$\mu_F = 0.352$	$\mu_{CR} = 0.038$
Diversity = 6.6564e+00	$k_t = [2, 2, 2, 3]$		
Generation 1950: Best = 3.659659e+03	FES = 195100	$\mu_F = 0.358$	$\mu_{CR} = 0.030$
Diversity = 6.6559e+00	$k_t = [1, 1, 1, 1]$		
Generation 1960: Best = 3.659017e+03	FES = 196100	$\mu_F = 0.338$	$\mu_{CR} = 0.034$
Diversity = 6.6559e+00	$k_t = [2, 4, 3, 2]$		
Generation 1970: Best = 3.658630e+03	FES = 197100	$\mu_F = 0.292$	$\mu_{CR} = 0.037$
Diversity = 6.6524e+00	$k_t = [1, 1, 1, 1]$		
Generation 1980: Best = 3.656528e+03	FES = 198100	$\mu_F = 0.309$	$\mu_{CR} = 0.047$
Diversity = 6.5969e+00	$k_t = [4, 4, 4, 4]$		
Generation 1990: Best = 3.656335e+03	FES = 199100	$\mu_F = 0.271$	$\mu_{CR} = 0.046$
Diversity = 6.6057e+00	$k_t = [1, 1, 1, 1]$		
Generation 2000: Best = 3.654608e+03	FES = 200100	$\mu_F = 0.259$	$\mu_{CR} = 0.057$
Diversity = 6.6038e+00	$k_t = [2, 2, 3, 3]$		
Generation 2010: Best = 3.654312e+03	FES = 201100	$\mu_F = 0.269$	$\mu_{CR} = 0.041$
Diversity = 6.6012e+00	$k_t = [1, 1, 1, 1]$		
Generation 2020: Best = 3.653694e+03	FES = 202100	$\mu_F = 0.271$	$\mu_{CR} = 0.042$
Diversity = 6.5993e+00	$k_t = [2, 4, 4, 3]$		
Generation 2030: Best = 3.650802e+03	FES = 203100	$\mu_F = 0.258$	$\mu_{CR} = 0.057$
Diversity = 6.5996e+00	$k_t = [1, 1, 1, 1]$		
Generation 2040: Best = 3.650802e+03	FES = 204100	$\mu_F = 0.192$	$\mu_{CR} = 0.066$
Diversity = 6.5920e+00	$k_t = [2, 2, 2, 2]$		
Generation 2050: Best = 3.650193e+03	FES = 205100	$\mu_F = 0.248$	$\mu_{CR} = 0.055$
Diversity = 6.5907e+00	$k_t = [1, 1, 1, 1]$		
Generation 2060: Best = 3.646019e+03	FES = 206100	$\mu_F = 0.231$	$\mu_{CR} = 0.048$
Diversity = 6.5926e+00	$k_t = [2, 3, 4, 2]$		
Generation 2070: Best = 3.645651e+03	FES = 207100	$\mu_F = 0.259$	$\mu_{CR} = 0.051$
Diversity = 6.5905e+00	$k_t = [1, 1, 1, 1]$		
Generation 2080: Best = 3.630912e+03	FES = 208100	$\mu_F = 0.261$	$\mu_{CR} = 0.044$
Diversity = 6.5923e+00	$k_t = [3, 3, 2, 2]$		
Generation 2090: Best = 3.630912e+03	FES = 209100	$\mu_F = 0.302$	$\mu_{CR} = 0.049$
Diversity = 6.5740e+00	$k_t = [1, 1, 1, 1]$		
Generation 2100: Best = 3.630630e+03	FES = 210100	$\mu_F = 0.300$	$\mu_{CR} = 0.047$
Diversity = 6.5754e+00	$k_t = [3, 3, 2, 2]$		
Generation 2110: Best = 3.630098e+03	FES = 211100	$\mu_F = 0.330$	$\mu_{CR} = 0.044$
Diversity = 6.5747e+00	$k_t = [1, 1, 1, 1]$		

Generation 2120: Best = 3.628674e+03 FES = 212100 μ_F = 0.408 μ_{CR} = 0.045
 Diversity = 6.5561e+00 k_t = [3, 4, 3, 2]
 Generation 2130: Best = 3.628674e+03 FES = 213100 μ_F = 0.470 μ_{CR} = 0.040
 Diversity = 6.5549e+00 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 3.623460e+03 FES = 214100 μ_F = 0.512 μ_{CR} = 0.040
 Diversity = 6.4841e+00 k_t = [3, 4, 2, 4]
 Generation 2150: Best = 3.615878e+03 FES = 215100 μ_F = 0.510 μ_{CR} = 0.045
 Diversity = 6.4877e+00 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 3.613177e+03 FES = 216100 μ_F = 0.478 μ_{CR} = 0.049
 Diversity = 6.4876e+00 k_t = [2, 4, 4, 3]
 Generation 2170: Best = 3.611848e+03 FES = 217100 μ_F = 0.480 μ_{CR} = 0.033
 Diversity = 6.4912e+00 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 3.611848e+03 FES = 218100 μ_F = 0.415 μ_{CR} = 0.034
 Diversity = 6.4923e+00 k_t = [2, 2, 3, 2]
 Generation 2190: Best = 3.604799e+03 FES = 219100 μ_F = 0.398 μ_{CR} = 0.040
 Diversity = 6.4736e+00 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 3.603919e+03 FES = 220100 μ_F = 0.369 μ_{CR} = 0.066
 Diversity = 6.4179e+00 k_t = [2, 3, 3, 4]
 Generation 2210: Best = 3.596423e+03 FES = 221100 μ_F = 0.406 μ_{CR} = 0.079
 Diversity = 6.4150e+00 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 3.596423e+03 FES = 222100 μ_F = 0.471 μ_{CR} = 0.081
 Diversity = 6.4115e+00 k_t = [3, 4, 2, 3]
 Generation 2230: Best = 3.593141e+03 FES = 223100 μ_F = 0.452 μ_{CR} = 0.068
 Diversity = 6.4102e+00 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 3.593076e+03 FES = 224100 μ_F = 0.392 μ_{CR} = 0.059
 Diversity = 6.4066e+00 k_t = [2, 3, 2, 3]
 Generation 2250: Best = 3.589987e+03 FES = 225100 μ_F = 0.411 μ_{CR} = 0.047
 Diversity = 6.4073e+00 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 3.589530e+03 FES = 226100 μ_F = 0.419 μ_{CR} = 0.044
 Diversity = 6.4063e+00 k_t = [3, 3, 3, 2]
 Generation 2270: Best = 3.585283e+03 FES = 227100 μ_F = 0.472 μ_{CR} = 0.044
 Diversity = 6.4074e+00 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 3.585086e+03 FES = 228100 μ_F = 0.412 μ_{CR} = 0.048
 Diversity = 6.4087e+00 k_t = [2, 2, 3, 2]
 Generation 2290: Best = 3.585086e+03 FES = 229100 μ_F = 0.439 μ_{CR} = 0.068
 Diversity = 6.4058e+00 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 3.582586e+03 FES = 230100 μ_F = 0.532 μ_{CR} = 0.073
 Diversity = 6.3973e+00 k_t = [4, 4, 4, 2]
 Generation 2310: Best = 3.580526e+03 FES = 231100 μ_F = 0.631 μ_{CR} = 0.041
 Diversity = 6.3457e+00 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 3.579656e+03 FES = 232100 μ_F = 0.615 μ_{CR} = 0.051
 Diversity = 6.3460e+00 k_t = [4, 4, 2, 3]
 Generation 2330: Best = 3.578940e+03 FES = 233100 μ_F = 0.578 μ_{CR} = 0.040
 Diversity = 6.3599e+00 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 3.576564e+03 FES = 234100 μ_F = 0.614 μ_{CR} = 0.043
 Diversity = 6.3270e+00 k_t = [2, 3, 3, 2]
 Generation 2350: Best = 3.576514e+03 FES = 235100 μ_F = 0.565 μ_{CR} = 0.047
 Diversity = 6.3270e+00 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 3.576514e+03 FES = 236100 μ_F = 0.609 μ_{CR} = 0.050
 Diversity = 6.3249e+00 k_t = [4, 2, 3, 4]
 Generation 2370: Best = 3.575884e+03 FES = 237100 μ_F = 0.643 μ_{CR} = 0.038
 Diversity = 6.3140e+00 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 3.574169e+03 FES = 238100 μ_F = 0.678 μ_{CR} = 0.033
 Diversity = 6.3038e+00 k_t = [3, 4, 2, 4]

Generation 2390: Best = 3.574150e+03 FES = 239100 μ_F = 0.550 μ_{CR} = 0.046
 Diversity = 6.3020e+00 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 3.570270e+03 FES = 240100 μ_F = 0.520 μ_{CR} = 0.055
 Diversity = 6.3044e+00 k_t = [2, 2, 4, 2]
 Generation 2410: Best = 3.569183e+03 FES = 241100 μ_F = 0.543 μ_{CR} = 0.068
 Diversity = 6.3031e+00 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 3.567497e+03 FES = 242100 μ_F = 0.516 μ_{CR} = 0.068
 Diversity = 6.3140e+00 k_t = [3, 4, 2, 3]
 Generation 2430: Best = 3.562628e+03 FES = 243100 μ_F = 0.527 μ_{CR} = 0.065
 Diversity = 6.2856e+00 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 3.562628e+03 FES = 244100 μ_F = 0.634 μ_{CR} = 0.072
 Diversity = 6.2697e+00 k_t = [4, 3, 3, 4]
 Generation 2450: Best = 3.562221e+03 FES = 245100 μ_F = 0.604 μ_{CR} = 0.073
 Diversity = 6.2689e+00 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 3.561470e+03 FES = 246100 μ_F = 0.609 μ_{CR} = 0.067
 Diversity = 6.2466e+00 k_t = [2, 4, 4, 2]
 Generation 2470: Best = 3.561374e+03 FES = 247100 μ_F = 0.683 μ_{CR} = 0.057
 Diversity = 6.2184e+00 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 3.561084e+03 FES = 248100 μ_F = 0.709 μ_{CR} = 0.081
 Diversity = 6.1195e+00 k_t = [3, 4, 3, 4]
 Generation 2490: Best = 3.560379e+03 FES = 249100 μ_F = 0.639 μ_{CR} = 0.088
 Diversity = 6.0994e+00 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 3.560076e+03 FES = 250100 μ_F = 0.679 μ_{CR} = 0.084
 Diversity = 6.1025e+00 k_t = [2, 2, 2, 3]
 Generation 2510: Best = 3.557493e+03 FES = 251100 μ_F = 0.703 μ_{CR} = 0.060
 Diversity = 6.1048e+00 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 3.553470e+03 FES = 252100 μ_F = 0.731 μ_{CR} = 0.079
 Diversity = 6.0904e+00 k_t = [4, 3, 2, 2]
 Generation 2530: Best = 3.553470e+03 FES = 253100 μ_F = 0.753 μ_{CR} = 0.073
 Diversity = 6.0914e+00 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 3.552327e+03 FES = 254100 μ_F = 0.709 μ_{CR} = 0.082
 Diversity = 6.0516e+00 k_t = [3, 4, 3, 4]
 Generation 2550: Best = 3.551237e+03 FES = 255100 μ_F = 0.732 μ_{CR} = 0.110
 Diversity = 6.0088e+00 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 3.548921e+03 FES = 256100 μ_F = 0.771 μ_{CR} = 0.109
 Diversity = 6.0094e+00 k_t = [2, 3, 4, 3]
 Generation 2570: Best = 3.545120e+03 FES = 257100 μ_F = 0.770 μ_{CR} = 0.116
 Diversity = 5.9928e+00 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 3.542122e+03 FES = 258100 μ_F = 0.805 μ_{CR} = 0.107
 Diversity = 5.9627e+00 k_t = [3, 3, 2, 3]
 Generation 2590: Best = 3.538619e+03 FES = 259100 μ_F = 0.806 μ_{CR} = 0.116
 Diversity = 5.9217e+00 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 3.537899e+03 FES = 260100 μ_F = 0.827 μ_{CR} = 0.117
 Diversity = 5.7728e+00 k_t = [3, 2, 4, 4]
 Generation 2610: Best = 3.535512e+03 FES = 261100 μ_F = 0.823 μ_{CR} = 0.121
 Diversity = 5.7274e+00 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 3.532303e+03 FES = 262100 μ_F = 0.810 μ_{CR} = 0.122
 Diversity = 5.4496e+00 k_t = [3, 4, 4, 3]
 Generation 2630: Best = 3.528172e+03 FES = 263100 μ_F = 0.824 μ_{CR} = 0.129
 Diversity = 4.6398e+00 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 3.522662e+03 FES = 264100 μ_F = 0.827 μ_{CR} = 0.140
 Diversity = 1.3612e+00 k_t = [4, 2, 2, 4]
 Generation 2650: Best = 3.519830e+03 FES = 265100 μ_F = 0.814 μ_{CR} = 0.151
 Diversity = 5.0132e-01 k_t = [1, 1, 1, 1]

Generation 2660: Best = 3.511167e+03 FES = 266100 μ_F = 0.799 μ_{CR} = 0.154
 Diversity = 3.6163e-01 k_t = [2, 3, 3, 4]
 Generation 2670: Best = 3.510095e+03 FES = 267100 μ_F = 0.766 μ_{CR} = 0.146
 Diversity = 3.1471e-01 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 3.508227e+03 FES = 268100 μ_F = 0.733 μ_{CR} = 0.148
 Diversity = 2.8161e-01 k_t = [3, 2, 2, 2]
 Generation 2690: Best = 3.505149e+03 FES = 269100 μ_F = 0.721 μ_{CR} = 0.157
 Diversity = 2.5046e-01 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 3.503300e+03 FES = 270100 μ_F = 0.709 μ_{CR} = 0.169
 Diversity = 2.0698e-01 k_t = [3, 4, 2, 2]
 Generation 2710: Best = 3.501116e+03 FES = 271100 μ_F = 0.692 μ_{CR} = 0.180
 Diversity = 1.7197e-01 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 3.500705e+03 FES = 272100 μ_F = 0.682 μ_{CR} = 0.188
 Diversity = 1.4645e-01 k_t = [2, 3, 3, 2]
 Generation 2730: Best = 3.499688e+03 FES = 273100 μ_F = 0.674 μ_{CR} = 0.195
 Diversity = 1.3065e-01 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 3.498947e+03 FES = 274100 μ_F = 0.662 μ_{CR} = 0.195
 Diversity = 1.0685e-01 k_t = [2, 3, 4, 2]
 Generation 2750: Best = 3.498160e+03 FES = 275100 μ_F = 0.661 μ_{CR} = 0.190
 Diversity = 9.2073e-02 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 3.497882e+03 FES = 276100 μ_F = 0.670 μ_{CR} = 0.198
 Diversity = 7.5839e-02 k_t = [2, 4, 4, 4]
 Generation 2770: Best = 3.497701e+03 FES = 277100 μ_F = 0.680 μ_{CR} = 0.201
 Diversity = 6.7784e-02 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 3.497395e+03 FES = 278100 μ_F = 0.667 μ_{CR} = 0.203
 Diversity = 5.6429e-02 k_t = [3, 4, 2, 4]
 Generation 2790: Best = 3.497146e+03 FES = 279100 μ_F = 0.672 μ_{CR} = 0.197
 Diversity = 4.9220e-02 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 3.496957e+03 FES = 280100 μ_F = 0.675 μ_{CR} = 0.207
 Diversity = 4.4193e-02 k_t = [3, 4, 3, 2]
 Generation 2810: Best = 3.496758e+03 FES = 281100 μ_F = 0.687 μ_{CR} = 0.217
 Diversity = 4.0809e-02 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 3.496606e+03 FES = 282100 μ_F = 0.697 μ_{CR} = 0.211
 Diversity = 3.6667e-02 k_t = [4, 3, 4, 4]
 Generation 2830: Best = 3.496467e+03 FES = 283100 μ_F = 0.710 μ_{CR} = 0.214
 Diversity = 3.5401e-02 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 3.496333e+03 FES = 284100 μ_F = 0.708 μ_{CR} = 0.208
 Diversity = 3.2556e-02 k_t = [2, 2, 4, 4]
 Generation 2850: Best = 3.496189e+03 FES = 285100 μ_F = 0.721 μ_{CR} = 0.211
 Diversity = 3.3304e-02 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 3.496102e+03 FES = 286100 μ_F = 0.725 μ_{CR} = 0.210
 Diversity = 3.1900e-02 k_t = [3, 4, 4, 3]
 Generation 2870: Best = 3.495999e+03 FES = 287100 μ_F = 0.714 μ_{CR} = 0.216
 Diversity = 2.9947e-02 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 3.495896e+03 FES = 288100 μ_F = 0.714 μ_{CR} = 0.226
 Diversity = 2.9109e-02 k_t = [4, 3, 3, 2]
 Generation 2890: Best = 3.495775e+03 FES = 289100 μ_F = 0.721 μ_{CR} = 0.228
 Diversity = 2.8976e-02 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 3.495694e+03 FES = 290100 μ_F = 0.722 μ_{CR} = 0.224
 Diversity = 2.7164e-02 k_t = [3, 3, 3, 4]
 Generation 2910: Best = 3.495595e+03 FES = 291100 μ_F = 0.722 μ_{CR} = 0.228
 Diversity = 2.8361e-02 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 3.495496e+03 FES = 292100 μ_F = 0.726 μ_{CR} = 0.231
 Diversity = 2.7911e-02 k_t = [3, 4, 3, 2]

Generation 2930: Best = 3.495415e+03 FES = 293100 μ_F = 0.727 μ_{CR} = 0.237
 Diversity = 2.9523e-02 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 3.495326e+03 FES = 294100 μ_F = 0.717 μ_{CR} = 0.224
 Diversity = 2.8539e-02 k_t = [2, 3, 3, 2]
 Generation 2950: Best = 3.495251e+03 FES = 295100 μ_F = 0.731 μ_{CR} = 0.227
 Diversity = 2.8759e-02 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 3.495141e+03 FES = 296100 μ_F = 0.731 μ_{CR} = 0.228
 Diversity = 2.7277e-02 k_t = [4, 3, 2, 4]
 Generation 2970: Best = 3.495100e+03 FES = 297100 μ_F = 0.739 μ_{CR} = 0.221
 Diversity = 2.4865e-02 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 3.495060e+03 FES = 298100 μ_F = 0.745 μ_{CR} = 0.222
 Diversity = 2.2435e-02 k_t = [2, 2, 4, 4]
 Generation 2990: Best = 3.495025e+03 FES = 299100 μ_F = 0.728 μ_{CR} = 0.213
 Diversity = 2.1415e-02 k_t = [1, 1, 1, 1]
 Generation 3000: Best = 3.494983e+03 FES = 300100 μ_F = 0.717 μ_{CR} = 0.213
 Diversity = 1.9905e-02 k_t = [2, 2, 3, 3]
 Generation 3010: Best = 3.494953e+03 FES = 301100 μ_F = 0.703 μ_{CR} = 0.221
 Diversity = 1.9399e-02 k_t = [1, 1, 1, 1]
 Generation 3020: Best = 3.494924e+03 FES = 302100 μ_F = 0.705 μ_{CR} = 0.226
 Diversity = 1.6862e-02 k_t = [4, 4, 3, 3]
 Generation 3030: Best = 3.494898e+03 FES = 303100 μ_F = 0.697 μ_{CR} = 0.215
 Diversity = 1.5096e-02 k_t = [1, 1, 1, 1]
 Generation 3040: Best = 3.494881e+03 FES = 304100 μ_F = 0.695 μ_{CR} = 0.223
 Diversity = 1.3713e-02 k_t = [3, 4, 2, 3]
 Generation 3050: Best = 3.494864e+03 FES = 305100 μ_F = 0.698 μ_{CR} = 0.221
 Diversity = 1.2879e-02 k_t = [1, 1, 1, 1]
 Generation 3060: Best = 3.494850e+03 FES = 306100 μ_F = 0.708 μ_{CR} = 0.225
 Diversity = 1.1911e-02 k_t = [2, 3, 2, 3]
 Generation 3070: Best = 3.494836e+03 FES = 307100 μ_F = 0.708 μ_{CR} = 0.226
 Diversity = 1.1642e-02 k_t = [1, 1, 1, 1]
 Generation 3080: Best = 3.494824e+03 FES = 308100 μ_F = 0.698 μ_{CR} = 0.234
 Diversity = 1.0515e-02 k_t = [3, 4, 4, 2]
 Generation 3090: Best = 3.494810e+03 FES = 309100 μ_F = 0.704 μ_{CR} = 0.241
 Diversity = 1.0370e-02 k_t = [1, 1, 1, 1]
 Generation 3100: Best = 3.494798e+03 FES = 310100 μ_F = 0.711 μ_{CR} = 0.243
 Diversity = 9.6091e-03 k_t = [4, 3, 4, 3]
 Generation 3110: Best = 3.494788e+03 FES = 311100 μ_F = 0.711 μ_{CR} = 0.244
 Diversity = 9.1827e-03 k_t = [1, 1, 1, 1]
 Generation 3120: Best = 3.494777e+03 FES = 312100 μ_F = 0.704 μ_{CR} = 0.245
 Diversity = 8.4784e-03 k_t = [4, 2, 3, 2]
 Generation 3130: Best = 3.494775e+03 FES = 313100 μ_F = 0.708 μ_{CR} = 0.237
 Diversity = 8.4126e-03 k_t = [1, 1, 1, 1]
 Generation 3140: Best = 3.494764e+03 FES = 314100 μ_F = 0.704 μ_{CR} = 0.235
 Diversity = 7.8866e-03 k_t = [2, 4, 3, 2]
 Generation 3150: Best = 3.494755e+03 FES = 315100 μ_F = 0.716 μ_{CR} = 0.238
 Diversity = 8.2926e-03 k_t = [1, 1, 1, 1]
 Generation 3160: Best = 3.494743e+03 FES = 316100 μ_F = 0.712 μ_{CR} = 0.236
 Diversity = 8.5498e-03 k_t = [2, 4, 2, 2]
 Generation 3170: Best = 3.494741e+03 FES = 317100 μ_F = 0.727 μ_{CR} = 0.228
 Diversity = 9.3295e-03 k_t = [1, 1, 1, 1]
 Generation 3180: Best = 3.494731e+03 FES = 318100 μ_F = 0.727 μ_{CR} = 0.226
 Diversity = 8.7568e-03 k_t = [4, 4, 3, 2]
 Generation 3190: Best = 3.494720e+03 FES = 319100 μ_F = 0.735 μ_{CR} = 0.226
 Diversity = 9.1115e-03 k_t = [1, 1, 1, 1]

Generation 3200: Best = 3.494714e+03 FES = 320100 μ_F = 0.734 μ_{CR} = 0.235
 Diversity = 8.3132e-03 k_t = [3, 4, 4, 3]
 Generation 3210: Best = 3.494707e+03 FES = 321100 μ_F = 0.731 μ_{CR} = 0.229
 Diversity = 8.1616e-03 k_t = [1, 1, 1, 1]
 Generation 3220: Best = 3.494699e+03 FES = 322100 μ_F = 0.733 μ_{CR} = 0.231
 Diversity = 8.1436e-03 k_t = [4, 4, 4, 3]
 Generation 3230: Best = 3.494692e+03 FES = 323100 μ_F = 0.735 μ_{CR} = 0.224
 Diversity = 7.5574e-03 k_t = [1, 1, 1, 1]
 Generation 3240: Best = 3.494687e+03 FES = 324100 μ_F = 0.729 μ_{CR} = 0.230
 Diversity = 7.1955e-03 k_t = [2, 2, 4, 2]
 Generation 3250: Best = 3.494680e+03 FES = 325100 μ_F = 0.729 μ_{CR} = 0.231
 Diversity = 7.5312e-03 k_t = [1, 1, 1, 1]
 Generation 3260: Best = 3.494672e+03 FES = 326100 μ_F = 0.718 μ_{CR} = 0.237
 Diversity = 7.0438e-03 k_t = [3, 3, 4, 3]
 Generation 3270: Best = 3.494669e+03 FES = 327100 μ_F = 0.695 μ_{CR} = 0.247
 Diversity = 6.1756e-03 k_t = [1, 1, 1, 1]
 Generation 3280: Best = 3.494666e+03 FES = 328100 μ_F = 0.706 μ_{CR} = 0.236
 Diversity = 5.4261e-03 k_t = [4, 2, 3, 4]
 Generation 3290: Best = 3.494662e+03 FES = 329100 μ_F = 0.705 μ_{CR} = 0.230
 Diversity = 4.8547e-03 k_t = [1, 1, 1, 1]
 Generation 3300: Best = 3.494659e+03 FES = 330100 μ_F = 0.716 μ_{CR} = 0.237
 Diversity = 4.4913e-03 k_t = [4, 3, 3, 3]
 Generation 3310: Best = 3.494656e+03 FES = 331100 μ_F = 0.720 μ_{CR} = 0.241
 Diversity = 4.6620e-03 k_t = [1, 1, 1, 1]
 Generation 3320: Best = 3.494653e+03 FES = 332100 μ_F = 0.710 μ_{CR} = 0.241
 Diversity = 4.4394e-03 k_t = [2, 4, 4, 4]
 Generation 3330: Best = 3.494650e+03 FES = 333100 μ_F = 0.714 μ_{CR} = 0.245
 Diversity = 4.5073e-03 k_t = [1, 1, 1, 1]
 Generation 3340: Best = 3.494646e+03 FES = 334100 μ_F = 0.713 μ_{CR} = 0.256
 Diversity = 4.3320e-03 k_t = [4, 2, 4, 3]
 Generation 3350: Best = 3.494643e+03 FES = 335100 μ_F = 0.712 μ_{CR} = 0.258
 Diversity = 4.3597e-03 k_t = [1, 1, 1, 1]
 Generation 3360: Best = 3.494641e+03 FES = 336100 μ_F = 0.712 μ_{CR} = 0.263
 Diversity = 3.8429e-03 k_t = [4, 2, 2, 3]
 Generation 3370: Best = 3.494640e+03 FES = 337100 μ_F = 0.710 μ_{CR} = 0.265
 Diversity = 3.7857e-03 k_t = [1, 1, 1, 1]
 Generation 3380: Best = 3.494638e+03 FES = 338100 μ_F = 0.714 μ_{CR} = 0.266
 Diversity = 3.6831e-03 k_t = [3, 2, 2, 3]
 Generation 3390: Best = 3.494636e+03 FES = 339100 μ_F = 0.721 μ_{CR} = 0.262
 Diversity = 3.5296e-03 k_t = [1, 1, 1, 1]
 Generation 3400: Best = 3.494635e+03 FES = 340100 μ_F = 0.716 μ_{CR} = 0.263
 Diversity = 3.3739e-03 k_t = [3, 3, 2, 2]
 Generation 3410: Best = 3.494634e+03 FES = 341100 μ_F = 0.697 μ_{CR} = 0.253
 Diversity = 3.2665e-03 k_t = [1, 1, 1, 1]
 Generation 3420: Best = 3.494632e+03 FES = 342100 μ_F = 0.706 μ_{CR} = 0.258
 Diversity = 3.0347e-03 k_t = [4, 4, 4, 4]
 Generation 3430: Best = 3.494631e+03 FES = 343100 μ_F = 0.709 μ_{CR} = 0.259
 Diversity = 2.9409e-03 k_t = [1, 1, 1, 1]
 Generation 3440: Best = 3.494630e+03 FES = 344100 μ_F = 0.719 μ_{CR} = 0.250
 Diversity = 2.9458e-03 k_t = [4, 4, 4, 2]
 Generation 3450: Best = 3.494628e+03 FES = 345100 μ_F = 0.730 μ_{CR} = 0.239
 Diversity = 2.9957e-03 k_t = [1, 1, 1, 1]
 Generation 3460: Best = 3.494627e+03 FES = 346100 μ_F = 0.725 μ_{CR} = 0.249
 Diversity = 2.8928e-03 k_t = [2, 4, 2, 4]

Generation 3470: Best = 3.494626e+03 FES = 347100 μ_F = 0.721 μ_{CR} = 0.240
 Diversity = 2.9186e-03 k_t = [1, 1, 1, 1]
 Generation 3480: Best = 3.494624e+03 FES = 348100 μ_F = 0.730 μ_{CR} = 0.244
 Diversity = 2.9175e-03 k_t = [2, 2, 4, 3]
 Generation 3490: Best = 3.494622e+03 FES = 349100 μ_F = 0.722 μ_{CR} = 0.245
 Diversity = 3.0992e-03 k_t = [1, 1, 1, 1]
 Generation 3500: Best = 3.494620e+03 FES = 350100 μ_F = 0.726 μ_{CR} = 0.242
 Diversity = 3.2150e-03 k_t = [4, 3, 2, 2]
 Generation 3510: Best = 3.494618e+03 FES = 351100 μ_F = 0.725 μ_{CR} = 0.242
 Diversity = 3.1512e-03 k_t = [1, 1, 1, 1]
 Generation 3520: Best = 3.494617e+03 FES = 352100 μ_F = 0.736 μ_{CR} = 0.241
 Diversity = 3.1874e-03 k_t = [4, 4, 3, 2]
 Generation 3530: Best = 3.494616e+03 FES = 353100 μ_F = 0.733 μ_{CR} = 0.233
 Diversity = 2.9931e-03 k_t = [1, 1, 1, 1]
 Generation 3540: Best = 3.494615e+03 FES = 354100 μ_F = 0.730 μ_{CR} = 0.236
 Diversity = 2.5429e-03 k_t = [2, 3, 3, 3]
 Generation 3550: Best = 3.494614e+03 FES = 355100 μ_F = 0.719 μ_{CR} = 0.243
 Diversity = 2.5423e-03 k_t = [1, 1, 1, 1]
 Generation 3560: Best = 3.494613e+03 FES = 356100 μ_F = 0.720 μ_{CR} = 0.242
 Diversity = 2.3662e-03 k_t = [2, 4, 4, 4]
 Generation 3570: Best = 3.494613e+03 FES = 357100 μ_F = 0.717 μ_{CR} = 0.243
 Diversity = 2.2622e-03 k_t = [1, 1, 1, 1]
 Generation 3580: Best = 3.494612e+03 FES = 358100 μ_F = 0.732 μ_{CR} = 0.240
 Diversity = 2.2254e-03 k_t = [2, 4, 4, 3]
 Generation 3590: Best = 3.494611e+03 FES = 359100 μ_F = 0.729 μ_{CR} = 0.243
 Diversity = 2.2785e-03 k_t = [1, 1, 1, 1]
 Generation 3600: Best = 3.494610e+03 FES = 360100 μ_F = 0.728 μ_{CR} = 0.243
 Diversity = 2.2334e-03 k_t = [3, 2, 3, 2]
 Generation 3610: Best = 3.494609e+03 FES = 361100 μ_F = 0.718 μ_{CR} = 0.241
 Diversity = 2.4419e-03 k_t = [1, 1, 1, 1]
 Generation 3620: Best = 3.494608e+03 FES = 362100 μ_F = 0.705 μ_{CR} = 0.237
 Diversity = 2.3715e-03 k_t = [3, 3, 2, 2]
 Generation 3630: Best = 3.494608e+03 FES = 363100 μ_F = 0.706 μ_{CR} = 0.227
 Diversity = 2.4335e-03 k_t = [1, 1, 1, 1]
 Generation 3640: Best = 3.494607e+03 FES = 364100 μ_F = 0.714 μ_{CR} = 0.232
 Diversity = 2.3551e-03 k_t = [2, 4, 4, 3]
 Generation 3650: Best = 3.494606e+03 FES = 365100 μ_F = 0.716 μ_{CR} = 0.227
 Diversity = 2.4599e-03 k_t = [1, 1, 1, 1]
 Generation 3660: Best = 3.494605e+03 FES = 366100 μ_F = 0.712 μ_{CR} = 0.233
 Diversity = 2.4339e-03 k_t = [3, 4, 4, 2]
 Generation 3670: Best = 3.494604e+03 FES = 367100 μ_F = 0.725 μ_{CR} = 0.235
 Diversity = 2.6785e-03 k_t = [1, 1, 1, 1]
 Generation 3680: Best = 3.494602e+03 FES = 368100 μ_F = 0.731 μ_{CR} = 0.235
 Diversity = 2.6886e-03 k_t = [2, 2, 2, 3]
 Generation 3690: Best = 3.494602e+03 FES = 369100 μ_F = 0.724 μ_{CR} = 0.241
 Diversity = 2.8954e-03 k_t = [1, 1, 1, 1]
 Generation 3700: Best = 3.494601e+03 FES = 370100 μ_F = 0.718 μ_{CR} = 0.246
 Diversity = 2.8765e-03 k_t = [2, 3, 4, 2]
 Generation 3710: Best = 3.494600e+03 FES = 371100 μ_F = 0.699 μ_{CR} = 0.240
 Diversity = 2.7564e-03 k_t = [1, 1, 1, 1]
 Generation 3720: Best = 3.494599e+03 FES = 372100 μ_F = 0.705 μ_{CR} = 0.236
 Diversity = 2.7257e-03 k_t = [3, 2, 2, 3]
 Generation 3730: Best = 3.494597e+03 FES = 373100 μ_F = 0.710 μ_{CR} = 0.239
 Diversity = 2.9341e-03 k_t = [1, 1, 1, 1]

Generation 3740: Best = 3.494596e+03 FES = 374100 μ_F = 0.712 μ_{CR} = 0.231
 Diversity = 2.8865e-03 k_t = [3, 2, 3, 2]
 Generation 3750: Best = 3.494595e+03 FES = 375100 μ_F = 0.713 μ_{CR} = 0.233
 Diversity = 2.8171e-03 k_t = [1, 1, 1, 1]
 Generation 3760: Best = 3.494595e+03 FES = 376100 μ_F = 0.710 μ_{CR} = 0.234
 Diversity = 2.6244e-03 k_t = [2, 2, 2, 2]
 Generation 3770: Best = 3.494594e+03 FES = 377100 μ_F = 0.699 μ_{CR} = 0.224
 Diversity = 2.5056e-03 k_t = [1, 1, 1, 1]
 Generation 3780: Best = 3.494594e+03 FES = 378100 μ_F = 0.707 μ_{CR} = 0.219
 Diversity = 2.2757e-03 k_t = [4, 3, 2, 4]
 Generation 3790: Best = 3.494593e+03 FES = 379100 μ_F = 0.708 μ_{CR} = 0.221
 Diversity = 2.1492e-03 k_t = [1, 1, 1, 1]
 Generation 3800: Best = 3.494593e+03 FES = 380100 μ_F = 0.710 μ_{CR} = 0.216
 Diversity = 1.8932e-03 k_t = [3, 2, 3, 4]
 Generation 3810: Best = 3.494593e+03 FES = 381100 μ_F = 0.714 μ_{CR} = 0.215
 Diversity = 1.8087e-03 k_t = [1, 1, 1, 1]
 Generation 3820: Best = 3.494592e+03 FES = 382100 μ_F = 0.728 μ_{CR} = 0.209
 Diversity = 1.7523e-03 k_t = [4, 3, 4, 4]
 Generation 3830: Best = 3.494592e+03 FES = 383100 μ_F = 0.733 μ_{CR} = 0.206
 Diversity = 1.6765e-03 k_t = [1, 1, 1, 1]
 Generation 3840: Best = 3.494591e+03 FES = 384100 μ_F = 0.732 μ_{CR} = 0.214
 Diversity = 1.6047e-03 k_t = [3, 3, 4, 3]
 Generation 3850: Best = 3.494591e+03 FES = 385100 μ_F = 0.723 μ_{CR} = 0.220
 Diversity = 1.5541e-03 k_t = [1, 1, 1, 1]
 Generation 3860: Best = 3.494591e+03 FES = 386100 μ_F = 0.723 μ_{CR} = 0.220
 Diversity = 1.4638e-03 k_t = [3, 2, 2, 3]
 Generation 3870: Best = 3.494590e+03 FES = 387100 μ_F = 0.711 μ_{CR} = 0.217
 Diversity = 1.4833e-03 k_t = [1, 1, 1, 1]
 Generation 3880: Best = 3.494590e+03 FES = 388100 μ_F = 0.722 μ_{CR} = 0.227
 Diversity = 1.3407e-03 k_t = [4, 3, 3, 4]
 Generation 3890: Best = 3.494590e+03 FES = 389100 μ_F = 0.733 μ_{CR} = 0.228
 Diversity = 1.3315e-03 k_t = [1, 1, 1, 1]
 Generation 3900: Best = 3.494590e+03 FES = 390100 μ_F = 0.733 μ_{CR} = 0.233
 Diversity = 1.3398e-03 k_t = [2, 3, 3, 4]
 Generation 3910: Best = 3.494589e+03 FES = 391100 μ_F = 0.733 μ_{CR} = 0.232
 Diversity = 1.3748e-03 k_t = [1, 1, 1, 1]
 Generation 3920: Best = 3.494589e+03 FES = 392100 μ_F = 0.730 μ_{CR} = 0.236
 Diversity = 1.3020e-03 k_t = [3, 3, 2, 4]
 Generation 3930: Best = 3.494589e+03 FES = 393100 μ_F = 0.726 μ_{CR} = 0.250
 Diversity = 1.2127e-03 k_t = [1, 1, 1, 1]
 Generation 3940: Best = 3.494589e+03 FES = 394100 μ_F = 0.712 μ_{CR} = 0.242
 Diversity = 1.1269e-03 k_t = [4, 2, 2, 3]
 Generation 3950: Best = 3.494589e+03 FES = 395100 μ_F = 0.707 μ_{CR} = 0.242
 Diversity = 1.0924e-03 k_t = [1, 1, 1, 1]
 Generation 3960: Best = 3.494588e+03 FES = 396100 μ_F = 0.716 μ_{CR} = 0.239
 Diversity = 1.0775e-03 k_t = [4, 4, 2, 3]
 Generation 3970: Best = 3.494588e+03 FES = 397100 μ_F = 0.716 μ_{CR} = 0.242
 Diversity = 1.0411e-03 k_t = [1, 1, 1, 1]
 Generation 3980: Best = 3.494588e+03 FES = 398100 μ_F = 0.707 μ_{CR} = 0.246
 Diversity = 9.7078e-04 k_t = [2, 2, 3, 4]
 Generation 3990: Best = 3.494588e+03 FES = 399100 μ_F = 0.721 μ_{CR} = 0.246
 Diversity = 1.0080e-03 k_t = [1, 1, 1, 1]
 Generation 4000: Best = 3.494588e+03 FES = 400100 μ_F = 0.702 μ_{CR} = 0.242
 Diversity = 9.3277e-04 k_t = [2, 2, 2, 2]

Generation 4010: Best = 3.494588e+03 FES = 401100 μ_F = 0.696 μ_{CR} = 0.246
 Diversity = 9.1181e-04 k_t = [1, 1, 1, 1]
 Generation 4020: Best = 3.494588e+03 FES = 402100 μ_F = 0.688 μ_{CR} = 0.245
 Diversity = 8.3566e-04 k_t = [2, 2, 3, 2]
 Generation 4030: Best = 3.494588e+03 FES = 403100 μ_F = 0.685 μ_{CR} = 0.242
 Diversity = 7.6006e-04 k_t = [1, 1, 1, 1]
 Generation 4040: Best = 3.494588e+03 FES = 404100 μ_F = 0.682 μ_{CR} = 0.238
 Diversity = 7.4635e-04 k_t = [2, 2, 2, 2]
 Generation 4050: Best = 3.494588e+03 FES = 405100 μ_F = 0.680 μ_{CR} = 0.228
 Diversity = 6.5133e-04 k_t = [1, 1, 1, 1]
 Generation 4060: Best = 3.494588e+03 FES = 406100 μ_F = 0.695 μ_{CR} = 0.222
 Diversity = 6.2625e-04 k_t = [3, 4, 2, 3]
 Generation 4070: Best = 3.494588e+03 FES = 407100 μ_F = 0.694 μ_{CR} = 0.227
 Diversity = 6.2730e-04 k_t = [1, 1, 1, 1]
 Generation 4080: Best = 3.494587e+03 FES = 408100 μ_F = 0.696 μ_{CR} = 0.227
 Diversity = 5.6634e-04 k_t = [4, 4, 2, 3]
 Generation 4090: Best = 3.494587e+03 FES = 409100 μ_F = 0.704 μ_{CR} = 0.230
 Diversity = 5.8502e-04 k_t = [1, 1, 1, 1]
 Generation 4100: Best = 3.494587e+03 FES = 410100 μ_F = 0.720 μ_{CR} = 0.216
 Diversity = 5.7731e-04 k_t = [4, 4, 4, 3]
 Generation 4110: Best = 3.494587e+03 FES = 411100 μ_F = 0.724 μ_{CR} = 0.220
 Diversity = 6.1376e-04 k_t = [1, 1, 1, 1]
 Generation 4120: Best = 3.494587e+03 FES = 412100 μ_F = 0.728 μ_{CR} = 0.210
 Diversity = 6.0184e-04 k_t = [4, 2, 3, 2]
 Generation 4130: Best = 3.494587e+03 FES = 413100 μ_F = 0.725 μ_{CR} = 0.218
 Diversity = 5.8347e-04 k_t = [1, 1, 1, 1]
 Generation 4140: Best = 3.494587e+03 FES = 414100 μ_F = 0.716 μ_{CR} = 0.217
 Diversity = 4.8909e-04 k_t = [2, 4, 2, 4]
 Generation 4150: Best = 3.494587e+03 FES = 415100 μ_F = 0.710 μ_{CR} = 0.210
 Diversity = 4.3266e-04 k_t = [1, 1, 1, 1]
 Generation 4160: Best = 3.494587e+03 FES = 416100 μ_F = 0.694 μ_{CR} = 0.212
 Diversity = 4.0351e-04 k_t = [2, 4, 4, 2]
 Generation 4170: Best = 3.494587e+03 FES = 417100 μ_F = 0.697 μ_{CR} = 0.215
 Diversity = 3.7291e-04 k_t = [1, 1, 1, 1]
 Generation 4180: Best = 3.494587e+03 FES = 418100 μ_F = 0.703 μ_{CR} = 0.213
 Diversity = 3.3280e-04 k_t = [3, 2, 2, 4]
 Generation 4190: Best = 3.494587e+03 FES = 419100 μ_F = 0.718 μ_{CR} = 0.212
 Diversity = 3.1732e-04 k_t = [1, 1, 1, 1]
 Generation 4200: Best = 3.494587e+03 FES = 420100 μ_F = 0.726 μ_{CR} = 0.214
 Diversity = 3.0432e-04 k_t = [4, 4, 2, 2]
 Generation 4210: Best = 3.494587e+03 FES = 421100 μ_F = 0.724 μ_{CR} = 0.226
 Diversity = 3.1828e-04 k_t = [1, 1, 1, 1]
 Generation 4220: Best = 3.494587e+03 FES = 422100 μ_F = 0.722 μ_{CR} = 0.221
 Diversity = 3.0926e-04 k_t = [4, 3, 4, 4]
 Generation 4230: Best = 3.494587e+03 FES = 423100 μ_F = 0.728 μ_{CR} = 0.224
 Diversity = 3.2625e-04 k_t = [1, 1, 1, 1]
 Generation 4240: Best = 3.494587e+03 FES = 424100 μ_F = 0.738 μ_{CR} = 0.223
 Diversity = 3.2762e-04 k_t = [4, 3, 4, 4]
 Generation 4250: Best = 3.494587e+03 FES = 425100 μ_F = 0.738 μ_{CR} = 0.225
 Diversity = 3.1007e-04 k_t = [1, 1, 1, 1]
 Generation 4260: Best = 3.494587e+03 FES = 426100 μ_F = 0.734 μ_{CR} = 0.213
 Diversity = 2.9252e-04 k_t = [4, 3, 4, 2]
 Generation 4270: Best = 3.494587e+03 FES = 427100 μ_F = 0.737 μ_{CR} = 0.225
 Diversity = 3.0434e-04 k_t = [1, 1, 1, 1]

Generation 4280: Best = 3.494587e+03 FES = 428100 μ_F = 0.721 μ_{CR} = 0.224
 Diversity = 2.6312e-04 k_t = [2, 3, 2, 4]
 Generation 4290: Best = 3.494587e+03 FES = 429100 μ_F = 0.721 μ_{CR} = 0.211
 Diversity = 2.5587e-04 k_t = [1, 1, 1, 1]
 Generation 4300: Best = 3.494587e+03 FES = 430100 μ_F = 0.719 μ_{CR} = 0.211
 Diversity = 2.5585e-04 k_t = [2, 2, 3, 2]
 Generation 4310: Best = 3.494587e+03 FES = 431100 μ_F = 0.710 μ_{CR} = 0.205
 Diversity = 2.3743e-04 k_t = [1, 1, 1, 1]
 Generation 4320: Best = 3.494587e+03 FES = 432100 μ_F = 0.721 μ_{CR} = 0.213
 Diversity = 2.2219e-04 k_t = [3, 4, 3, 3]
 Generation 4330: Best = 3.494587e+03 FES = 433100 μ_F = 0.725 μ_{CR} = 0.207
 Diversity = 2.3870e-04 k_t = [1, 1, 1, 1]
 Generation 4340: Best = 3.494587e+03 FES = 434100 μ_F = 0.717 μ_{CR} = 0.210
 Diversity = 2.4185e-04 k_t = [2, 4, 4, 2]
 Generation 4350: Best = 3.494587e+03 FES = 435100 μ_F = 0.727 μ_{CR} = 0.213
 Diversity = 2.3910e-04 k_t = [1, 1, 1, 1]
 Generation 4360: Best = 3.494587e+03 FES = 436100 μ_F = 0.725 μ_{CR} = 0.212
 Diversity = 2.3628e-04 k_t = [3, 3, 3, 3]
 Generation 4370: Best = 3.494587e+03 FES = 437100 μ_F = 0.717 μ_{CR} = 0.214
 Diversity = 2.4199e-04 k_t = [1, 1, 1, 1]
 Generation 4380: Best = 3.494587e+03 FES = 438100 μ_F = 0.736 μ_{CR} = 0.219
 Diversity = 2.7177e-04 k_t = [3, 3, 2, 3]
 Generation 4390: Best = 3.494587e+03 FES = 439100 μ_F = 0.736 μ_{CR} = 0.225
 Diversity = 2.8482e-04 k_t = [1, 1, 1, 1]
 Generation 4400: Best = 3.494587e+03 FES = 440100 μ_F = 0.749 μ_{CR} = 0.231
 Diversity = 3.1736e-04 k_t = [4, 3, 2, 3]
 Generation 4410: Best = 3.494587e+03 FES = 441100 μ_F = 0.751 μ_{CR} = 0.231
 Diversity = 3.5448e-04 k_t = [1, 1, 1, 1]
 Generation 4420: Best = 3.494587e+03 FES = 442100 μ_F = 0.751 μ_{CR} = 0.226
 Diversity = 3.4595e-04 k_t = [2, 3, 3, 3]
 Generation 4430: Best = 3.494587e+03 FES = 443100 μ_F = 0.738 μ_{CR} = 0.233
 Diversity = 3.5572e-04 k_t = [1, 1, 1, 1]
 Generation 4440: Best = 3.494587e+03 FES = 444100 μ_F = 0.726 μ_{CR} = 0.228
 Diversity = 3.2452e-04 k_t = [2, 3, 3, 2]
 Generation 4450: Best = 3.494587e+03 FES = 445100 μ_F = 0.723 μ_{CR} = 0.227
 Diversity = 3.2630e-04 k_t = [1, 1, 1, 1]
 Generation 4460: Best = 3.494587e+03 FES = 446100 μ_F = 0.721 μ_{CR} = 0.223
 Diversity = 3.0869e-04 k_t = [2, 2, 4, 4]
 Generation 4470: Best = 3.494587e+03 FES = 447100 μ_F = 0.705 μ_{CR} = 0.223
 Diversity = 2.9611e-04 k_t = [1, 1, 1, 1]
 Generation 4480: Best = 3.494587e+03 FES = 448100 μ_F = 0.709 μ_{CR} = 0.231
 Diversity = 3.0051e-04 k_t = [3, 4, 2, 2]
 Generation 4490: Best = 3.494587e+03 FES = 449100 μ_F = 0.722 μ_{CR} = 0.238
 Diversity = 3.4604e-04 k_t = [1, 1, 1, 1]
 Generation 4500: Best = 3.494587e+03 FES = 450100 μ_F = 0.714 μ_{CR} = 0.232
 Diversity = 3.2536e-04 k_t = [2, 3, 2, 2]
 Generation 4510: Best = 3.494587e+03 FES = 451100 μ_F = 0.709 μ_{CR} = 0.228
 Diversity = 3.1231e-04 k_t = [1, 1, 1, 1]
 Generation 4520: Best = 3.494587e+03 FES = 452100 μ_F = 0.702 μ_{CR} = 0.230
 Diversity = 2.8561e-04 k_t = [4, 4, 3, 3]
 Generation 4530: Best = 3.494587e+03 FES = 453100 μ_F = 0.704 μ_{CR} = 0.231
 Diversity = 2.7166e-04 k_t = [1, 1, 1, 1]
 Generation 4540: Best = 3.494587e+03 FES = 454100 μ_F = 0.706 μ_{CR} = 0.240
 Diversity = 2.3410e-04 k_t = [4, 4, 3, 4]

Generation 4550: Best = 3.494587e+03 FES = 455100 μ_F = 0.721 μ_{CR} = 0.244
 Diversity = 2.2363e-04 k_t = [1, 1, 1, 1]
 Generation 4560: Best = 3.494587e+03 FES = 456100 μ_F = 0.722 μ_{CR} = 0.242
 Diversity = 2.2055e-04 k_t = [2, 2, 3, 3]
 Generation 4570: Best = 3.494587e+03 FES = 457100 μ_F = 0.706 μ_{CR} = 0.251
 Diversity = 2.2488e-04 k_t = [1, 1, 1, 1]
 Generation 4580: Best = 3.494587e+03 FES = 458100 μ_F = 0.721 μ_{CR} = 0.243
 Diversity = 2.1951e-04 k_t = [2, 2, 3, 4]
 Generation 4590: Best = 3.494587e+03 FES = 459100 μ_F = 0.735 μ_{CR} = 0.249
 Diversity = 2.5224e-04 k_t = [1, 1, 1, 1]
 Generation 4600: Best = 3.494587e+03 FES = 460100 μ_F = 0.730 μ_{CR} = 0.248
 Diversity = 2.4413e-04 k_t = [4, 4, 3, 2]
 Generation 4610: Best = 3.494587e+03 FES = 461100 μ_F = 0.732 μ_{CR} = 0.252
 Diversity = 2.5012e-04 k_t = [1, 1, 1, 1]
 Generation 4620: Best = 3.494587e+03 FES = 462100 μ_F = 0.738 μ_{CR} = 0.251
 Diversity = 2.6285e-04 k_t = [2, 2, 4, 3]
 Generation 4630: Best = 3.494587e+03 FES = 463100 μ_F = 0.745 μ_{CR} = 0.247
 Diversity = 3.0476e-04 k_t = [1, 1, 1, 1]
 Generation 4640: Best = 3.494587e+03 FES = 464100 μ_F = 0.740 μ_{CR} = 0.247
 Diversity = 3.1468e-04 k_t = [4, 4, 3, 2]
 Generation 4650: Best = 3.494587e+03 FES = 465100 μ_F = 0.735 μ_{CR} = 0.246
 Diversity = 3.0352e-04 k_t = [1, 1, 1, 1]
 Generation 4660: Best = 3.494587e+03 FES = 466100 μ_F = 0.725 μ_{CR} = 0.235
 Diversity = 2.7364e-04 k_t = [3, 2, 3, 4]
 Generation 4670: Best = 3.494587e+03 FES = 467100 μ_F = 0.720 μ_{CR} = 0.238
 Diversity = 2.8111e-04 k_t = [1, 1, 1, 1]
 Generation 4680: Best = 3.494587e+03 FES = 468100 μ_F = 0.728 μ_{CR} = 0.243
 Diversity = 2.9501e-04 k_t = [3, 4, 4, 3]
 Generation 4690: Best = 3.494586e+03 FES = 469100 μ_F = 0.723 μ_{CR} = 0.238
 Diversity = 3.0910e-04 k_t = [1, 1, 1, 1]
 Generation 4700: Best = 3.494586e+03 FES = 470100 μ_F = 0.729 μ_{CR} = 0.247
 Diversity = 3.2189e-04 k_t = [2, 3, 3, 3]
 Generation 4710: Best = 3.494586e+03 FES = 471100 μ_F = 0.732 μ_{CR} = 0.259
 Diversity = 3.8072e-04 k_t = [1, 1, 1, 1]
 Generation 4720: Best = 3.494586e+03 FES = 472100 μ_F = 0.736 μ_{CR} = 0.252
 Diversity = 3.9933e-04 k_t = [2, 3, 3, 4]
 Generation 4730: Best = 3.494586e+03 FES = 473100 μ_F = 0.726 μ_{CR} = 0.244
 Diversity = 3.7753e-04 k_t = [1, 1, 1, 1]
 Generation 4740: Best = 3.494586e+03 FES = 474100 μ_F = 0.729 μ_{CR} = 0.246
 Diversity = 3.7575e-04 k_t = [2, 2, 4, 2]
 Generation 4750: Best = 3.494586e+03 FES = 475100 μ_F = 0.722 μ_{CR} = 0.242
 Diversity = 3.6427e-04 k_t = [1, 1, 1, 1]
 Generation 4760: Best = 3.494586e+03 FES = 476100 μ_F = 0.712 μ_{CR} = 0.247
 Diversity = 3.0248e-04 k_t = [2, 2, 3, 4]
 Generation 4770: Best = 3.494586e+03 FES = 477100 μ_F = 0.710 μ_{CR} = 0.235
 Diversity = 2.8787e-04 k_t = [1, 1, 1, 1]
 Generation 4780: Best = 3.494586e+03 FES = 478100 μ_F = 0.732 μ_{CR} = 0.237
 Diversity = 2.8587e-04 k_t = [3, 2, 2, 4]
 Generation 4790: Best = 3.494586e+03 FES = 479100 μ_F = 0.720 μ_{CR} = 0.237
 Diversity = 2.6737e-04 k_t = [1, 1, 1, 1]
 Generation 4800: Best = 3.494586e+03 FES = 480100 μ_F = 0.719 μ_{CR} = 0.234
 Diversity = 2.4796e-04 k_t = [3, 2, 3, 2]
 Generation 4810: Best = 3.494586e+03 FES = 481100 μ_F = 0.713 μ_{CR} = 0.233
 Diversity = 2.4226e-04 k_t = [1, 1, 1, 1]

Generation 4820: Best = 3.494586e+03 FES = 482100 μ_F = 0.709 μ_{CR} = 0.225
 Diversity = 2.1615e-04 k_t = [3, 2, 3, 3]
 Generation 4830: Best = 3.494586e+03 FES = 483100 μ_F = 0.712 μ_{CR} = 0.220
 Diversity = 2.0514e-04 k_t = [1, 1, 1, 1]
 Generation 4840: Best = 3.494586e+03 FES = 484100 μ_F = 0.706 μ_{CR} = 0.215
 Diversity = 1.9016e-04 k_t = [3, 4, 4, 2]
 Generation 4850: Best = 3.494586e+03 FES = 485100 μ_F = 0.705 μ_{CR} = 0.212
 Diversity = 1.8035e-04 k_t = [1, 1, 1, 1]
 Generation 4860: Best = 3.494586e+03 FES = 486100 μ_F = 0.711 μ_{CR} = 0.205
 Diversity = 1.7128e-04 k_t = [4, 3, 3, 2]
 Generation 4870: Best = 3.494586e+03 FES = 487100 μ_F = 0.706 μ_{CR} = 0.203
 Diversity = 1.7738e-04 k_t = [1, 1, 1, 1]
 Generation 4880: Best = 3.494586e+03 FES = 488100 μ_F = 0.703 μ_{CR} = 0.204
 Diversity = 1.5223e-04 k_t = [2, 4, 2, 4]
 Generation 4890: Best = 3.494586e+03 FES = 489100 μ_F = 0.699 μ_{CR} = 0.212
 Diversity = 1.3795e-04 k_t = [1, 1, 1, 1]
 Generation 4900: Best = 3.494586e+03 FES = 490100 μ_F = 0.697 μ_{CR} = 0.210
 Diversity = 1.3074e-04 k_t = [3, 3, 3, 3]
 Generation 4910: Best = 3.494586e+03 FES = 491100 μ_F = 0.702 μ_{CR} = 0.217
 Diversity = 1.3227e-04 k_t = [1, 1, 1, 1]
 Generation 4920: Best = 3.494586e+03 FES = 492100 μ_F = 0.719 μ_{CR} = 0.217
 Diversity = 1.4051e-04 k_t = [2, 2, 4, 2]
 Generation 4930: Best = 3.494586e+03 FES = 493100 μ_F = 0.734 μ_{CR} = 0.221
 Diversity = 1.5212e-04 k_t = [1, 1, 1, 1]
 Generation 4940: Best = 3.494586e+03 FES = 494100 μ_F = 0.727 μ_{CR} = 0.223
 Diversity = 1.5151e-04 k_t = [3, 2, 3, 2]
 Generation 4950: Best = 3.494586e+03 FES = 495100 μ_F = 0.713 μ_{CR} = 0.225
 Diversity = 1.4332e-04 k_t = [1, 1, 1, 1]
 Generation 4960: Best = 3.494586e+03 FES = 496100 μ_F = 0.724 μ_{CR} = 0.232
 Diversity = 1.3494e-04 k_t = [2, 2, 2, 3]
 Generation 4970: Best = 3.494586e+03 FES = 497100 μ_F = 0.732 μ_{CR} = 0.238
 Diversity = 1.3722e-04 k_t = [1, 1, 1, 1]
 Generation 4980: Best = 3.494586e+03 FES = 498100 μ_F = 0.737 μ_{CR} = 0.227
 Diversity = 1.3180e-04 k_t = [2, 4, 3, 3]
 Generation 4990: Best = 3.494586e+03 FES = 499100 μ_F = 0.745 μ_{CR} = 0.227
 Diversity = 1.4770e-04 k_t = [1, 1, 1, 1]
 Generation 5000: Best = 3.494586e+03 FES = 500100 μ_F = 0.739 μ_{CR} = 0.223
 Diversity = 1.4108e-04 k_t = [2, 3, 4, 4]
 Generation 5010: Best = 3.494586e+03 FES = 501100 μ_F = 0.735 μ_{CR} = 0.226
 Diversity = 1.3574e-04 k_t = [1, 1, 1, 1]
 Generation 5020: Best = 3.494586e+03 FES = 502100 μ_F = 0.714 μ_{CR} = 0.230
 Diversity = 1.1305e-04 k_t = [3, 4, 4, 3]
 Generation 5030: Best = 3.494586e+03 FES = 503100 μ_F = 0.717 μ_{CR} = 0.232
 Diversity = 1.0942e-04 k_t = [1, 1, 1, 1]
 Generation 5040: Best = 3.494586e+03 FES = 504100 μ_F = 0.719 μ_{CR} = 0.240
 Diversity = 9.9588e-05 k_t = [3, 4, 4, 4]
 Generation 5050: Best = 3.494586e+03 FES = 505100 μ_F = 0.730 μ_{CR} = 0.239
 Diversity = 1.0202e-04 k_t = [1, 1, 1, 1]
 Generation 5060: Best = 3.494586e+03 FES = 506100 μ_F = 0.732 μ_{CR} = 0.230
 Diversity = 1.0524e-04 k_t = [3, 4, 3, 2]
 Generation 5070: Best = 3.494586e+03 FES = 507100 μ_F = 0.736 μ_{CR} = 0.214
 Diversity = 1.0511e-04 k_t = [1, 1, 1, 1]
 Generation 5080: Best = 3.494586e+03 FES = 508100 μ_F = 0.732 μ_{CR} = 0.212
 Diversity = 1.0321e-04 k_t = [4, 2, 3, 3]

Generation 5090: Best = 3.494586e+03 FES = 509100 μ_F = 0.726 μ_{CR} = 0.213
 Diversity = 1.0541e-04 k_t = [1, 1, 1, 1]
 Generation 5100: Best = 3.494586e+03 FES = 510100 μ_F = 0.737 μ_{CR} = 0.205
 Diversity = 9.8556e-05 k_t = [4, 4, 3, 3]
 Generation 5110: Best = 3.494586e+03 FES = 511100 μ_F = 0.729 μ_{CR} = 0.208
 Diversity = 9.4373e-05 k_t = [1, 1, 1, 1]
 Generation 5120: Best = 3.494586e+03 FES = 512100 μ_F = 0.716 μ_{CR} = 0.215
 Diversity = 8.6603e-05 k_t = [4, 3, 4, 2]
 Generation 5130: Best = 3.494586e+03 FES = 513100 μ_F = 0.728 μ_{CR} = 0.222
 Diversity = 9.0031e-05 k_t = [1, 1, 1, 1]
 Generation 5140: Best = 3.494586e+03 FES = 514100 μ_F = 0.733 μ_{CR} = 0.223
 Diversity = 9.3802e-05 k_t = [2, 2, 4, 2]
 Generation 5150: Best = 3.494586e+03 FES = 515100 μ_F = 0.742 μ_{CR} = 0.223
 Diversity = 9.9020e-05 k_t = [1, 1, 1, 1]
 Generation 5160: Best = 3.494586e+03 FES = 516100 μ_F = 0.741 μ_{CR} = 0.227
 Diversity = 9.1930e-05 k_t = [2, 4, 3, 2]
 Generation 5170: Best = 3.494586e+03 FES = 517100 μ_F = 0.734 μ_{CR} = 0.232
 Diversity = 8.9821e-05 k_t = [1, 1, 1, 1]
 Generation 5180: Best = 3.494586e+03 FES = 518100 μ_F = 0.725 μ_{CR} = 0.225
 Diversity = 8.3319e-05 k_t = [3, 4, 2, 3]
 Generation 5190: Best = 3.494586e+03 FES = 519100 μ_F = 0.719 μ_{CR} = 0.228
 Diversity = 8.3811e-05 k_t = [1, 1, 1, 1]
 Generation 5200: Best = 3.494586e+03 FES = 520100 μ_F = 0.724 μ_{CR} = 0.224
 Diversity = 7.8427e-05 k_t = [4, 4, 4, 2]
 Generation 5210: Best = 3.494586e+03 FES = 521100 μ_F = 0.721 μ_{CR} = 0.211
 Diversity = 7.1783e-05 k_t = [1, 1, 1, 1]
 Generation 5220: Best = 3.494586e+03 FES = 522100 μ_F = 0.724 μ_{CR} = 0.213
 Diversity = 6.3692e-05 k_t = [3, 4, 4, 4]
 Generation 5230: Best = 3.494586e+03 FES = 523100 μ_F = 0.725 μ_{CR} = 0.224
 Diversity = 5.8491e-05 k_t = [1, 1, 1, 1]
 Generation 5240: Best = 3.494586e+03 FES = 524100 μ_F = 0.721 μ_{CR} = 0.228
 Diversity = 5.5719e-05 k_t = [4, 4, 2, 3]
 Generation 5250: Best = 3.494586e+03 FES = 525100 μ_F = 0.722 μ_{CR} = 0.224
 Diversity = 5.4140e-05 k_t = [1, 1, 1, 1]
 Generation 5260: Best = 3.494586e+03 FES = 526100 μ_F = 0.735 μ_{CR} = 0.236
 Diversity = 5.3973e-05 k_t = [3, 3, 3, 3]
 Generation 5270: Best = 3.494586e+03 FES = 527100 μ_F = 0.719 μ_{CR} = 0.246
 Diversity = 5.3760e-05 k_t = [1, 1, 1, 1]
 Generation 5280: Best = 3.494586e+03 FES = 528100 μ_F = 0.708 μ_{CR} = 0.252
 Diversity = 4.8162e-05 k_t = [2, 4, 3, 3]
 Generation 5290: Best = 3.494586e+03 FES = 529100 μ_F = 0.714 μ_{CR} = 0.254
 Diversity = 4.7541e-05 k_t = [1, 1, 1, 1]
 Generation 5300: Best = 3.494586e+03 FES = 530100 μ_F = 0.712 μ_{CR} = 0.252
 Diversity = 4.1468e-05 k_t = [4, 4, 4, 4]
 Generation 5310: Best = 3.494586e+03 FES = 531100 μ_F = 0.714 μ_{CR} = 0.250
 Diversity = 4.1181e-05 k_t = [1, 1, 1, 1]
 Generation 5320: Best = 3.494586e+03 FES = 532100 μ_F = 0.702 μ_{CR} = 0.251
 Diversity = 3.8968e-05 k_t = [4, 2, 2, 2]
 Generation 5330: Best = 3.494586e+03 FES = 533100 μ_F = 0.704 μ_{CR} = 0.251
 Diversity = 3.6264e-05 k_t = [1, 1, 1, 1]
 Generation 5340: Best = 3.494586e+03 FES = 534100 μ_F = 0.707 μ_{CR} = 0.256
 Diversity = 3.2290e-05 k_t = [4, 3, 3, 3]
 Generation 5350: Best = 3.494586e+03 FES = 535100 μ_F = 0.717 μ_{CR} = 0.260
 Diversity = 3.0469e-05 k_t = [1, 1, 1, 1]

Generation 5360: Best = 3.494586e+03 FES = 536100 μ_F = 0.721 μ_{CR} = 0.257
 Diversity = 3.0529e-05 k_t = [2, 3, 3, 2]
 Generation 5370: Best = 3.494586e+03 FES = 537100 μ_F = 0.709 μ_{CR} = 0.251
 Diversity = 2.9679e-05 k_t = [1, 1, 1, 1]
 Generation 5380: Best = 3.494586e+03 FES = 538100 μ_F = 0.711 μ_{CR} = 0.245
 Diversity = 3.2148e-05 k_t = [2, 4, 2, 2]
 Generation 5390: Best = 3.494586e+03 FES = 539100 μ_F = 0.721 μ_{CR} = 0.239
 Diversity = 3.3240e-05 k_t = [1, 1, 1, 1]
 Generation 5400: Best = 3.494586e+03 FES = 540100 μ_F = 0.726 μ_{CR} = 0.244
 Diversity = 3.1918e-05 k_t = [3, 4, 4, 4]
 Generation 5410: Best = 3.494586e+03 FES = 541100 μ_F = 0.747 μ_{CR} = 0.245
 Diversity = 3.1943e-05 k_t = [1, 1, 1, 1]
 Generation 5420: Best = 3.494586e+03 FES = 542100 μ_F = 0.735 μ_{CR} = 0.255
 Diversity = 2.6585e-05 k_t = [4, 2, 3, 4]
 Generation 5430: Best = 3.494586e+03 FES = 543100 μ_F = 0.739 μ_{CR} = 0.245
 Diversity = 2.5684e-05 k_t = [1, 1, 1, 1]
 Generation 5440: Best = 3.494586e+03 FES = 544100 μ_F = 0.745 μ_{CR} = 0.244
 Diversity = 2.6052e-05 k_t = [2, 3, 2, 3]
 Generation 5450: Best = 3.494586e+03 FES = 545100 μ_F = 0.741 μ_{CR} = 0.247
 Diversity = 2.8856e-05 k_t = [1, 1, 1, 1]
 Generation 5460: Best = 3.494586e+03 FES = 546100 μ_F = 0.728 μ_{CR} = 0.245
 Diversity = 2.9368e-05 k_t = [3, 2, 3, 4]
 Generation 5470: Best = 3.494586e+03 FES = 547100 μ_F = 0.730 μ_{CR} = 0.245
 Diversity = 2.8944e-05 k_t = [1, 1, 1, 1]
 Generation 5480: Best = 3.494586e+03 FES = 548100 μ_F = 0.730 μ_{CR} = 0.245
 Diversity = 2.6548e-05 k_t = [3, 3, 4, 3]
 Generation 5490: Best = 3.494586e+03 FES = 549100 μ_F = 0.718 μ_{CR} = 0.237
 Diversity = 2.6679e-05 k_t = [1, 1, 1, 1]
 Generation 5500: Best = 3.494586e+03 FES = 550100 μ_F = 0.718 μ_{CR} = 0.228
 Diversity = 2.5352e-05 k_t = [3, 4, 2, 2]
 Generation 5510: Best = 3.494586e+03 FES = 551100 μ_F = 0.721 μ_{CR} = 0.226
 Diversity = 2.4716e-05 k_t = [1, 1, 1, 1]
 Generation 5520: Best = 3.494586e+03 FES = 552100 μ_F = 0.733 μ_{CR} = 0.230
 Diversity = 2.5110e-05 k_t = [4, 4, 4, 4]
 Generation 5530: Best = 3.494586e+03 FES = 553100 μ_F = 0.746 μ_{CR} = 0.212
 Diversity = 2.5824e-05 k_t = [1, 1, 1, 1]
 Generation 5540: Best = 3.494586e+03 FES = 554100 μ_F = 0.739 μ_{CR} = 0.218
 Diversity = 2.7115e-05 k_t = [3, 4, 3, 3]
 Generation 5550: Best = 3.494586e+03 FES = 555100 μ_F = 0.731 μ_{CR} = 0.214
 Diversity = 2.7039e-05 k_t = [1, 1, 1, 1]
 Generation 5560: Best = 3.494586e+03 FES = 556100 μ_F = 0.735 μ_{CR} = 0.220
 Diversity = 2.6508e-05 k_t = [4, 3, 3, 2]
 Generation 5570: Best = 3.494586e+03 FES = 557100 μ_F = 0.727 μ_{CR} = 0.222
 Diversity = 2.6165e-05 k_t = [1, 1, 1, 1]
 Generation 5580: Best = 3.494586e+03 FES = 558100 μ_F = 0.727 μ_{CR} = 0.221
 Diversity = 2.5801e-05 k_t = [3, 2, 2, 2]
 Generation 5590: Best = 3.494586e+03 FES = 559100 μ_F = 0.723 μ_{CR} = 0.215
 Diversity = 2.4696e-05 k_t = [1, 1, 1, 1]
 Generation 5600: Best = 3.494586e+03 FES = 560100 μ_F = 0.727 μ_{CR} = 0.208
 Diversity = 2.2498e-05 k_t = [4, 4, 2, 2]
 Generation 5610: Best = 3.494586e+03 FES = 561100 μ_F = 0.716 μ_{CR} = 0.212
 Diversity = 2.1324e-05 k_t = [1, 1, 1, 1]
 Generation 5620: Best = 3.494586e+03 FES = 562100 μ_F = 0.706 μ_{CR} = 0.216
 Diversity = 1.9046e-05 k_t = [4, 2, 4, 3]

Generation 5630: Best = 3.494586e+03 FES = 563100 μ_F = 0.713 μ_{CR} = 0.224
 Diversity = 1.7651e-05 k_t = [1, 1, 1, 1]
 Generation 5640: Best = 3.494586e+03 FES = 564100 μ_F = 0.698 μ_{CR} = 0.227
 Diversity = 1.5732e-05 k_t = [4, 3, 4, 3]
 Generation 5650: Best = 3.494586e+03 FES = 565100 μ_F = 0.694 μ_{CR} = 0.223
 Diversity = 1.5218e-05 k_t = [1, 1, 1, 1]
 Generation 5660: Best = 3.494586e+03 FES = 566100 μ_F = 0.710 μ_{CR} = 0.225
 Diversity = 1.5189e-05 k_t = [4, 2, 4, 3]
 Generation 5670: Best = 3.494586e+03 FES = 567100 μ_F = 0.722 μ_{CR} = 0.222
 Diversity = 1.6096e-05 k_t = [1, 1, 1, 1]
 Generation 5680: Best = 3.494586e+03 FES = 568100 μ_F = 0.726 μ_{CR} = 0.225
 Diversity = 1.6355e-05 k_t = [3, 2, 2, 2]
 Generation 5690: Best = 3.494586e+03 FES = 569100 μ_F = 0.736 μ_{CR} = 0.223
 Diversity = 1.7763e-05 k_t = [1, 1, 1, 1]
 Generation 5700: Best = 3.494586e+03 FES = 570100 μ_F = 0.745 μ_{CR} = 0.229
 Diversity = 1.7037e-05 k_t = [2, 2, 2, 3]
 Generation 5710: Best = 3.494586e+03 FES = 571100 μ_F = 0.735 μ_{CR} = 0.223
 Diversity = 1.6650e-05 k_t = [1, 1, 1, 1]
 Generation 5720: Best = 3.494586e+03 FES = 572100 μ_F = 0.730 μ_{CR} = 0.231
 Diversity = 1.6526e-05 k_t = [3, 4, 4, 2]
 Generation 5730: Best = 3.494586e+03 FES = 573100 μ_F = 0.725 μ_{CR} = 0.233
 Diversity = 1.6749e-05 k_t = [1, 1, 1, 1]
 Generation 5740: Best = 3.494586e+03 FES = 574100 μ_F = 0.726 μ_{CR} = 0.238
 Diversity = 1.6178e-05 k_t = [3, 3, 3, 4]
 Generation 5750: Best = 3.494586e+03 FES = 575100 μ_F = 0.745 μ_{CR} = 0.234
 Diversity = 1.7295e-05 k_t = [1, 1, 1, 1]
 Generation 5760: Best = 3.494586e+03 FES = 576100 μ_F = 0.747 μ_{CR} = 0.235
 Diversity = 2.1652e-05 k_t = [3, 3, 2, 2]
 Generation 5770: Best = 3.494586e+03 FES = 577100 μ_F = 0.744 μ_{CR} = 0.243
 Diversity = 2.6351e-05 k_t = [1, 1, 1, 1]
 Generation 5780: Best = 3.494586e+03 FES = 578100 μ_F = 0.743 μ_{CR} = 0.244
 Diversity = 2.5629e-05 k_t = [3, 4, 2, 4]
 Generation 5790: Best = 3.494586e+03 FES = 579100 μ_F = 0.737 μ_{CR} = 0.237
 Diversity = 2.3383e-05 k_t = [1, 1, 1, 1]
 Generation 5800: Best = 3.494586e+03 FES = 580100 μ_F = 0.718 μ_{CR} = 0.239
 Diversity = 2.0458e-05 k_t = [4, 2, 2, 4]
 Generation 5810: Best = 3.494586e+03 FES = 581100 μ_F = 0.704 μ_{CR} = 0.221
 Diversity = 1.7642e-05 k_t = [1, 1, 1, 1]
 Generation 5820: Best = 3.494586e+03 FES = 582100 μ_F = 0.714 μ_{CR} = 0.225
 Diversity = 1.6332e-05 k_t = [2, 4, 2, 4]
 Generation 5830: Best = 3.494586e+03 FES = 583100 μ_F = 0.705 μ_{CR} = 0.224
 Diversity = 1.5438e-05 k_t = [1, 1, 1, 1]
 Generation 5840: Best = 3.494586e+03 FES = 584100 μ_F = 0.710 μ_{CR} = 0.224
 Diversity = 1.4348e-05 k_t = [2, 2, 4, 3]
 Generation 5850: Best = 3.494586e+03 FES = 585100 μ_F = 0.701 μ_{CR} = 0.232
 Diversity = 1.4300e-05 k_t = [1, 1, 1, 1]
 Generation 5860: Best = 3.494586e+03 FES = 586100 μ_F = 0.709 μ_{CR} = 0.236
 Diversity = 1.4855e-05 k_t = [2, 3, 3, 2]
 Generation 5870: Best = 3.494586e+03 FES = 587100 μ_F = 0.727 μ_{CR} = 0.235
 Diversity = 1.5880e-05 k_t = [1, 1, 1, 1]
 Generation 5880: Best = 3.494586e+03 FES = 588100 μ_F = 0.718 μ_{CR} = 0.236
 Diversity = 1.3471e-05 k_t = [2, 2, 3, 4]
 Generation 5890: Best = 3.494586e+03 FES = 589100 μ_F = 0.732 μ_{CR} = 0.238
 Diversity = 1.3927e-05 k_t = [1, 1, 1, 1]

Generation 5900: Best = 3.494586e+03 FES = 590100 μ_F = 0.743 μ_{CR} = 0.247
 Diversity = 1.3827e-05 k_t = [2, 3, 2, 4]
 Generation 5910: Best = 3.494586e+03 FES = 591100 μ_F = 0.743 μ_{CR} = 0.255
 Diversity = 1.4412e-05 k_t = [1, 1, 1, 1]
 Generation 5920: Best = 3.494586e+03 FES = 592100 μ_F = 0.734 μ_{CR} = 0.244
 Diversity = 1.3329e-05 k_t = [4, 3, 4, 3]
 Generation 5930: Best = 3.494586e+03 FES = 593100 μ_F = 0.747 μ_{CR} = 0.249
 Diversity = 1.3545e-05 k_t = [1, 1, 1, 1]
 Generation 5940: Best = 3.494586e+03 FES = 594100 μ_F = 0.748 μ_{CR} = 0.260
 Diversity = 1.2409e-05 k_t = [3, 3, 3, 4]
 Generation 5950: Best = 3.494586e+03 FES = 595100 μ_F = 0.745 μ_{CR} = 0.257
 Diversity = 1.2742e-05 k_t = [1, 1, 1, 1]
 Generation 5960: Best = 3.494586e+03 FES = 596100 μ_F = 0.729 μ_{CR} = 0.255
 Diversity = 1.2735e-05 k_t = [2, 3, 4, 2]
 Generation 5970: Best = 3.494586e+03 FES = 597100 μ_F = 0.727 μ_{CR} = 0.250
 Diversity = 1.3026e-05 k_t = [1, 1, 1, 1]
 Generation 5980: Best = 3.494586e+03 FES = 598100 μ_F = 0.707 μ_{CR} = 0.249
 Diversity = 1.1011e-05 k_t = [4, 4, 3, 4]
 Generation 5990: Best = 3.494586e+03 FES = 599100 μ_F = 0.728 μ_{CR} = 0.245
 Diversity = 1.1849e-05 k_t = [1, 1, 1, 1]
 Generation 6000: Best = 3.494586e+03 FES = 600100 μ_F = 0.725 μ_{CR} = 0.249
 Diversity = 1.2538e-05 k_t = [4, 2, 3, 2]
 Generation 6010: Best = 3.494586e+03 FES = 601100 μ_F = 0.711 μ_{CR} = 0.253
 Diversity = 1.2807e-05 k_t = [1, 1, 1, 1]
 Generation 6020: Best = 3.494586e+03 FES = 602100 μ_F = 0.721 μ_{CR} = 0.255
 Diversity = 1.2696e-05 k_t = [3, 2, 2, 4]
 Generation 6030: Best = 3.494586e+03 FES = 603100 μ_F = 0.711 μ_{CR} = 0.258
 Diversity = 1.2068e-05 k_t = [1, 1, 1, 1]
 Generation 6040: Best = 3.494586e+03 FES = 604100 μ_F = 0.716 μ_{CR} = 0.254
 Diversity = 1.2512e-05 k_t = [4, 3, 2, 3]
 Generation 6050: Best = 3.494586e+03 FES = 605100 μ_F = 0.722 μ_{CR} = 0.254
 Diversity = 1.3316e-05 k_t = [1, 1, 1, 1]
 Generation 6060: Best = 3.494586e+03 FES = 606100 μ_F = 0.732 μ_{CR} = 0.247
 Diversity = 1.3308e-05 k_t = [3, 3, 2, 4]
 Generation 6070: Best = 3.494586e+03 FES = 607100 μ_F = 0.722 μ_{CR} = 0.235
 Diversity = 1.3222e-05 k_t = [1, 1, 1, 1]
 Generation 6080: Best = 3.494586e+03 FES = 608100 μ_F = 0.736 μ_{CR} = 0.232
 Diversity = 1.2725e-05 k_t = [3, 2, 4, 4]
 Generation 6090: Best = 3.494586e+03 FES = 609100 μ_F = 0.733 μ_{CR} = 0.232
 Diversity = 1.2566e-05 k_t = [1, 1, 1, 1]
 Generation 6100: Best = 3.494586e+03 FES = 610100 μ_F = 0.735 μ_{CR} = 0.235
 Diversity = 1.2668e-05 k_t = [4, 4, 3, 3]
 Generation 6110: Best = 3.494586e+03 FES = 611100 μ_F = 0.728 μ_{CR} = 0.234
 Diversity = 1.2495e-05 k_t = [1, 1, 1, 1]
 Generation 6120: Best = 3.494586e+03 FES = 612100 μ_F = 0.730 μ_{CR} = 0.240
 Diversity = 1.2275e-05 k_t = [2, 2, 2, 4]
 Generation 6130: Best = 3.494586e+03 FES = 613100 μ_F = 0.716 μ_{CR} = 0.244
 Diversity = 1.3451e-05 k_t = [1, 1, 1, 1]
 Generation 6140: Best = 3.494586e+03 FES = 614100 μ_F = 0.712 μ_{CR} = 0.245
 Diversity = 1.2812e-05 k_t = [2, 4, 2, 4]
 Generation 6150: Best = 3.494586e+03 FES = 615100 μ_F = 0.703 μ_{CR} = 0.248
 Diversity = 1.2251e-05 k_t = [1, 1, 1, 1]
 Generation 6160: Best = 3.494586e+03 FES = 616100 μ_F = 0.688 μ_{CR} = 0.250
 Diversity = 1.1262e-05 k_t = [3, 2, 4, 2]

Generation 6170: Best = 3.494586e+03	FES = 617100	$\mu_F = 0.675$	$\mu_{CR} = 0.251$
Diversity = 1.0471e-05	$k_t = [1, 1, 1, 1]$		
Generation 6180: Best = 3.494586e+03	FES = 618100	$\mu_F = 0.682$	$\mu_{CR} = 0.246$
Diversity = 9.0349e-06	$k_t = [2, 4, 2, 2]$		
Generation 6190: Best = 3.494586e+03	FES = 619100	$\mu_F = 0.663$	$\mu_{CR} = 0.246$
Diversity = 7.9030e-06	$k_t = [1, 1, 1, 1]$		
Generation 6200: Best = 3.494586e+03	FES = 620100	$\mu_F = 0.677$	$\mu_{CR} = 0.244$
Diversity = 6.5637e-06	$k_t = [3, 3, 2, 3]$		
Generation 6210: Best = 3.494586e+03	FES = 621100	$\mu_F = 0.684$	$\mu_{CR} = 0.242$
Diversity = 6.0216e-06	$k_t = [1, 1, 1, 1]$		
Generation 6220: Best = 3.494586e+03	FES = 622100	$\mu_F = 0.695$	$\mu_{CR} = 0.243$
Diversity = 5.5108e-06	$k_t = [3, 3, 2, 3]$		
Generation 6230: Best = 3.494586e+03	FES = 623100	$\mu_F = 0.715$	$\mu_{CR} = 0.238$
Diversity = 5.6006e-06	$k_t = [1, 1, 1, 1]$		
Generation 6240: Best = 3.494586e+03	FES = 624100	$\mu_F = 0.719$	$\mu_{CR} = 0.232$
Diversity = 5.3962e-06	$k_t = [3, 3, 4, 3]$		
Generation 6250: Best = 3.494586e+03	FES = 625100	$\mu_F = 0.723$	$\mu_{CR} = 0.232$
Diversity = 5.4603e-06	$k_t = [1, 1, 1, 1]$		
Generation 6260: Best = 3.494586e+03	FES = 626100	$\mu_F = 0.727$	$\mu_{CR} = 0.217$
Diversity = 5.2502e-06	$k_t = [4, 2, 4, 4]$		
Generation 6270: Best = 3.494586e+03	FES = 627100	$\mu_F = 0.742$	$\mu_{CR} = 0.221$
Diversity = 5.7170e-06	$k_t = [1, 1, 1, 1]$		
Generation 6280: Best = 3.494586e+03	FES = 628100	$\mu_F = 0.736$	$\mu_{CR} = 0.222$
Diversity = 5.3203e-06	$k_t = [2, 4, 4, 3]$		
Generation 6290: Best = 3.494586e+03	FES = 629100	$\mu_F = 0.735$	$\mu_{CR} = 0.227$
Diversity = 5.7982e-06	$k_t = [1, 1, 1, 1]$		
Generation 6300: Best = 3.494586e+03	FES = 630100	$\mu_F = 0.733$	$\mu_{CR} = 0.230$
Diversity = 6.4525e-06	$k_t = [2, 4, 3, 2]$		
Generation 6310: Best = 3.494586e+03	FES = 631100	$\mu_F = 0.732$	$\mu_{CR} = 0.224$
Diversity = 6.9648e-06	$k_t = [1, 1, 1, 1]$		
Generation 6320: Best = 3.494586e+03	FES = 632100	$\mu_F = 0.723$	$\mu_{CR} = 0.218$
Diversity = 6.0107e-06	$k_t = [4, 2, 4, 2]$		
Generation 6330: Best = 3.494586e+03	FES = 633100	$\mu_F = 0.730$	$\mu_{CR} = 0.228$
Diversity = 5.5230e-06	$k_t = [1, 1, 1, 1]$		
Generation 6340: Best = 3.494586e+03	FES = 634100	$\mu_F = 0.721$	$\mu_{CR} = 0.223$
Diversity = 4.8093e-06	$k_t = [4, 4, 3, 3]$		
Generation 6350: Best = 3.494586e+03	FES = 635100	$\mu_F = 0.712$	$\mu_{CR} = 0.223$
Diversity = 4.5732e-06	$k_t = [1, 1, 1, 1]$		
Generation 6360: Best = 3.494586e+03	FES = 636100	$\mu_F = 0.710$	$\mu_{CR} = 0.236$
Diversity = 4.1615e-06	$k_t = [4, 2, 3, 3]$		
Generation 6370: Best = 3.494586e+03	FES = 637100	$\mu_F = 0.701$	$\mu_{CR} = 0.239$
Diversity = 3.8774e-06	$k_t = [1, 1, 1, 1]$		
Generation 6380: Best = 3.494586e+03	FES = 638100	$\mu_F = 0.701$	$\mu_{CR} = 0.242$
Diversity = 3.5680e-06	$k_t = [4, 3, 3, 3]$		
Generation 6390: Best = 3.494586e+03	FES = 639100	$\mu_F = 0.691$	$\mu_{CR} = 0.236$
Diversity = 3.2149e-06	$k_t = [1, 1, 1, 1]$		
Generation 6400: Best = 3.494586e+03	FES = 640100	$\mu_F = 0.691$	$\mu_{CR} = 0.231$
Diversity = 2.7898e-06	$k_t = [4, 4, 4, 4]$		
Generation 6410: Best = 3.494586e+03	FES = 641100	$\mu_F = 0.703$	$\mu_{CR} = 0.232$
Diversity = 2.5748e-06	$k_t = [1, 1, 1, 1]$		
Generation 6420: Best = 3.494586e+03	FES = 642100	$\mu_F = 0.710$	$\mu_{CR} = 0.229$
Diversity = 2.5456e-06	$k_t = [2, 4, 4, 3]$		
Generation 6430: Best = 3.494586e+03	FES = 643100	$\mu_F = 0.708$	$\mu_{CR} = 0.232$
Diversity = 2.4894e-06	$k_t = [1, 1, 1, 1]$		

Generation 6440: Best = 3.494586e+03 FES = 644100 μ_F = 0.713 μ_{CR} = 0.226
 Diversity = 2.3550e-06 k_t = [2, 4, 2, 2]
 Generation 6450: Best = 3.494586e+03 FES = 645100 μ_F = 0.708 μ_{CR} = 0.225
 Diversity = 2.2739e-06 k_t = [1, 1, 1, 1]
 Generation 6460: Best = 3.494586e+03 FES = 646100 μ_F = 0.703 μ_{CR} = 0.215
 Diversity = 2.0740e-06 k_t = [2, 2, 2, 2]
 Generation 6470: Best = 3.494586e+03 FES = 647100 μ_F = 0.702 μ_{CR} = 0.214
 Diversity = 2.0801e-06 k_t = [1, 1, 1, 1]
 Generation 6480: Best = 3.494586e+03 FES = 648100 μ_F = 0.695 μ_{CR} = 0.214
 Diversity = 1.8575e-06 k_t = [3, 2, 2, 4]
 Generation 6490: Best = 3.494586e+03 FES = 649100 μ_F = 0.709 μ_{CR} = 0.215
 Diversity = 1.9118e-06 k_t = [1, 1, 1, 1]
 Generation 6500: Best = 3.494586e+03 FES = 650100 μ_F = 0.715 μ_{CR} = 0.212
 Diversity = 1.8558e-06 k_t = [4, 2, 4, 4]
 Generation 6510: Best = 3.494586e+03 FES = 651100 μ_F = 0.732 μ_{CR} = 0.211
 Diversity = 1.9916e-06 k_t = [1, 1, 1, 1]
 Generation 6520: Best = 3.494586e+03 FES = 652100 μ_F = 0.752 μ_{CR} = 0.212
 Diversity = 2.1377e-06 k_t = [4, 4, 3, 3]
 Generation 6530: Best = 3.494586e+03 FES = 653100 μ_F = 0.742 μ_{CR} = 0.219
 Diversity = 2.3048e-06 k_t = [1, 1, 1, 1]
 Generation 6540: Best = 3.494586e+03 FES = 654100 μ_F = 0.739 μ_{CR} = 0.208
 Diversity = 2.1970e-06 k_t = [3, 4, 4, 3]
 Generation 6550: Best = 3.494586e+03 FES = 655100 μ_F = 0.735 μ_{CR} = 0.215
 Diversity = 2.3735e-06 k_t = [1, 1, 1, 1]
 Generation 6560: Best = 3.494586e+03 FES = 656100 μ_F = 0.729 μ_{CR} = 0.217
 Diversity = 2.2653e-06 k_t = [4, 4, 4, 4]
 Generation 6570: Best = 3.494586e+03 FES = 657100 μ_F = 0.719 μ_{CR} = 0.222
 Diversity = 2.0515e-06 k_t = [1, 1, 1, 1]
 Generation 6580: Best = 3.494586e+03 FES = 658100 μ_F = 0.730 μ_{CR} = 0.217
 Diversity = 1.9900e-06 k_t = [3, 4, 3, 3]
 Generation 6590: Best = 3.494586e+03 FES = 659100 μ_F = 0.730 μ_{CR} = 0.217
 Diversity = 1.9906e-06 k_t = [1, 1, 1, 1]
 Generation 6600: Best = 3.494586e+03 FES = 660100 μ_F = 0.737 μ_{CR} = 0.222
 Diversity = 1.8251e-06 k_t = [2, 3, 2, 4]
 Generation 6610: Best = 3.494586e+03 FES = 661100 μ_F = 0.738 μ_{CR} = 0.216
 Diversity = 1.7430e-06 k_t = [1, 1, 1, 1]
 Generation 6620: Best = 3.494586e+03 FES = 662100 μ_F = 0.733 μ_{CR} = 0.219
 Diversity = 1.5679e-06 k_t = [2, 4, 3, 3]
 Generation 6630: Best = 3.494586e+03 FES = 663100 μ_F = 0.726 μ_{CR} = 0.216
 Diversity = 1.6079e-06 k_t = [1, 1, 1, 1]
 Generation 6640: Best = 3.494586e+03 FES = 664100 μ_F = 0.724 μ_{CR} = 0.215
 Diversity = 1.5737e-06 k_t = [3, 4, 2, 3]
 Generation 6650: Best = 3.494586e+03 FES = 665100 μ_F = 0.721 μ_{CR} = 0.213
 Diversity = 1.6353e-06 k_t = [1, 1, 1, 1]
 Generation 6660: Best = 3.494586e+03 FES = 666100 μ_F = 0.729 μ_{CR} = 0.213
 Diversity = 1.7779e-06 k_t = [2, 4, 2, 2]
 Generation 6670: Best = 3.494586e+03 FES = 667100 μ_F = 0.730 μ_{CR} = 0.218
 Diversity = 1.9047e-06 k_t = [1, 1, 1, 1]
 Generation 6680: Best = 3.494586e+03 FES = 668100 μ_F = 0.742 μ_{CR} = 0.229
 Diversity = 1.9131e-06 k_t = [3, 3, 2, 3]
 Generation 6690: Best = 3.494586e+03 FES = 669100 μ_F = 0.750 μ_{CR} = 0.232
 Diversity = 2.1779e-06 k_t = [1, 1, 1, 1]
 Generation 6700: Best = 3.494586e+03 FES = 670100 μ_F = 0.735 μ_{CR} = 0.219
 Diversity = 2.0845e-06 k_t = [3, 4, 3, 2]

Generation 6710: Best = 3.494586e+03 FES = 671100 μ_F = 0.722 μ_{CR} = 0.210
 Diversity = 2.1232e-06 k_t = [1, 1, 1, 1]
 Generation 6720: Best = 3.494586e+03 FES = 672100 μ_F = 0.719 μ_{CR} = 0.207
 Diversity = 1.9578e-06 k_t = [3, 4, 2, 4]
 Generation 6730: Best = 3.494586e+03 FES = 673100 μ_F = 0.729 μ_{CR} = 0.210
 Diversity = 1.9102e-06 k_t = [1, 1, 1, 1]
 Generation 6740: Best = 3.494586e+03 FES = 674100 μ_F = 0.726 μ_{CR} = 0.219
 Diversity = 1.7235e-06 k_t = [3, 2, 4, 3]
 Generation 6750: Best = 3.494586e+03 FES = 675100 μ_F = 0.718 μ_{CR} = 0.223
 Diversity = 1.8412e-06 k_t = [1, 1, 1, 1]
 Generation 6760: Best = 3.494586e+03 FES = 676100 μ_F = 0.712 μ_{CR} = 0.217
 Diversity = 1.6459e-06 k_t = [3, 3, 3, 4]
 Generation 6770: Best = 3.494586e+03 FES = 677100 μ_F = 0.710 μ_{CR} = 0.226
 Diversity = 1.5063e-06 k_t = [1, 1, 1, 1]
 Generation 6780: Best = 3.494586e+03 FES = 678100 μ_F = 0.717 μ_{CR} = 0.228
 Diversity = 1.4029e-06 k_t = [2, 2, 2, 2]
 Generation 6790: Best = 3.494586e+03 FES = 679100 μ_F = 0.710 μ_{CR} = 0.231
 Diversity = 1.3219e-06 k_t = [1, 1, 1, 1]
 Generation 6800: Best = 3.494586e+03 FES = 680100 μ_F = 0.706 μ_{CR} = 0.226
 Diversity = 1.2924e-06 k_t = [4, 2, 3, 2]
 Generation 6810: Best = 3.494586e+03 FES = 681100 μ_F = 0.707 μ_{CR} = 0.227
 Diversity = 1.2465e-06 k_t = [1, 1, 1, 1]
 Generation 6820: Best = 3.494586e+03 FES = 682100 μ_F = 0.712 μ_{CR} = 0.233
 Diversity = 1.1603e-06 k_t = [3, 4, 2, 3]
 Generation 6830: Best = 3.494586e+03 FES = 683100 μ_F = 0.709 μ_{CR} = 0.239
 Diversity = 1.1987e-06 k_t = [1, 1, 1, 1]
 Generation 6840: Best = 3.494586e+03 FES = 684100 μ_F = 0.709 μ_{CR} = 0.236
 Diversity = 1.1751e-06 k_t = [3, 4, 4, 4]
 Generation 6850: Best = 3.494586e+03 FES = 685100 μ_F = 0.714 μ_{CR} = 0.226
 Diversity = 1.1530e-06 k_t = [1, 1, 1, 1]
 Generation 6860: Best = 3.494586e+03 FES = 686100 μ_F = 0.718 μ_{CR} = 0.222
 Diversity = 1.0472e-06 k_t = [3, 4, 3, 4]
 Generation 6870: Best = 3.494586e+03 FES = 687100 μ_F = 0.727 μ_{CR} = 0.216
 Diversity = 9.9890e-07 k_t = [1, 1, 1, 1]
 Generation 6880: Best = 3.494586e+03 FES = 688100 μ_F = 0.729 μ_{CR} = 0.217
 Diversity = 9.7700e-07 k_t = [3, 2, 4, 2]
 Generation 6890: Best = 3.494586e+03 FES = 689100 μ_F = 0.742 μ_{CR} = 0.220
 Diversity = 1.1031e-06 k_t = [1, 1, 1, 1]
 Generation 6900: Best = 3.494586e+03 FES = 690100 μ_F = 0.750 μ_{CR} = 0.223
 Diversity = 1.2857e-06 k_t = [3, 2, 4, 2]
 Generation 6910: Best = 3.494586e+03 FES = 691100 μ_F = 0.750 μ_{CR} = 0.221
 Diversity = 1.4502e-06 k_t = [1, 1, 1, 1]
 Generation 6920: Best = 3.494586e+03 FES = 692100 μ_F = 0.752 μ_{CR} = 0.229
 Diversity = 1.3511e-06 k_t = [3, 2, 3, 3]
 Generation 6930: Best = 3.494586e+03 FES = 693100 μ_F = 0.752 μ_{CR} = 0.230
 Diversity = 1.4925e-06 k_t = [1, 1, 1, 1]
 Generation 6940: Best = 3.494586e+03 FES = 694100 μ_F = 0.758 μ_{CR} = 0.224
 Diversity = 1.5557e-06 k_t = [2, 2, 2, 4]
 Generation 6950: Best = 3.494586e+03 FES = 695100 μ_F = 0.761 μ_{CR} = 0.221
 Diversity = 1.8844e-06 k_t = [1, 1, 1, 1]
 Generation 6960: Best = 3.494586e+03 FES = 696100 μ_F = 0.742 μ_{CR} = 0.223
 Diversity = 1.7864e-06 k_t = [4, 3, 3, 4]
 Generation 6970: Best = 3.494586e+03 FES = 697100 μ_F = 0.740 μ_{CR} = 0.223
 Diversity = 1.8901e-06 k_t = [1, 1, 1, 1]

Generation 6980: Best = 3.494586e+03 FES = 698100 μ_F = 0.750 μ_{CR} = 0.224
 Diversity = 2.1150e-06 k_t = [3, 3, 3, 4]
 Generation 6990: Best = 3.494586e+03 FES = 699100 μ_F = 0.768 μ_{CR} = 0.226
 Diversity = 2.6163e-06 k_t = [1, 1, 1, 1]
 Generation 7000: Best = 3.494586e+03 FES = 700100 μ_F = 0.764 μ_{CR} = 0.232
 Diversity = 2.9494e-06 k_t = [4, 3, 2, 3]
 Generation 7010: Best = 3.494586e+03 FES = 701100 μ_F = 0.757 μ_{CR} = 0.227
 Diversity = 2.9604e-06 k_t = [1, 1, 1, 1]
 Generation 7020: Best = 3.494586e+03 FES = 702100 μ_F = 0.737 μ_{CR} = 0.230
 Diversity = 2.6092e-06 k_t = [3, 2, 4, 4]
 Generation 7030: Best = 3.494586e+03 FES = 703100 μ_F = 0.736 μ_{CR} = 0.237
 Diversity = 2.2742e-06 k_t = [1, 1, 1, 1]
 Generation 7040: Best = 3.494586e+03 FES = 704100 μ_F = 0.732 μ_{CR} = 0.241
 Diversity = 2.2015e-06 k_t = [2, 2, 3, 2]
 Generation 7050: Best = 3.494586e+03 FES = 705100 μ_F = 0.713 μ_{CR} = 0.246
 Diversity = 2.2223e-06 k_t = [1, 1, 1, 1]
 Generation 7060: Best = 3.494586e+03 FES = 706100 μ_F = 0.703 μ_{CR} = 0.244
 Diversity = 1.9476e-06 k_t = [4, 3, 2, 2]
 Generation 7070: Best = 3.494586e+03 FES = 707100 μ_F = 0.681 μ_{CR} = 0.241
 Diversity = 1.6706e-06 k_t = [1, 1, 1, 1]
 Generation 7080: Best = 3.494586e+03 FES = 708100 μ_F = 0.679 μ_{CR} = 0.240
 Diversity = 1.3153e-06 k_t = [3, 4, 3, 3]
 Generation 7090: Best = 3.494586e+03 FES = 709100 μ_F = 0.668 μ_{CR} = 0.232
 Diversity = 1.1347e-06 k_t = [1, 1, 1, 1]
 Generation 7100: Best = 3.494586e+03 FES = 710100 μ_F = 0.665 μ_{CR} = 0.224
 Diversity = 9.9297e-07 k_t = [4, 2, 2, 2]
 Generation 7110: Best = 3.494586e+03 FES = 711100 μ_F = 0.670 μ_{CR} = 0.224
 Diversity = 8.9439e-07 k_t = [1, 1, 1, 1]
 Generation 7120: Best = 3.494586e+03 FES = 712100 μ_F = 0.668 μ_{CR} = 0.230
 Diversity = 7.3190e-07 k_t = [4, 4, 2, 4]
 Generation 7130: Best = 3.494586e+03 FES = 713100 μ_F = 0.669 μ_{CR} = 0.232
 Diversity = 6.0632e-07 k_t = [1, 1, 1, 1]
 Generation 7140: Best = 3.494586e+03 FES = 714100 μ_F = 0.676 μ_{CR} = 0.230
 Diversity = 5.4396e-07 k_t = [2, 4, 2, 3]
 Generation 7150: Best = 3.494586e+03 FES = 715100 μ_F = 0.691 μ_{CR} = 0.223
 Diversity = 5.3519e-07 k_t = [1, 1, 1, 1]
 Generation 7160: Best = 3.494586e+03 FES = 716100 μ_F = 0.711 μ_{CR} = 0.227
 Diversity = 4.9984e-07 k_t = [3, 3, 2, 4]
 Generation 7170: Best = 3.494586e+03 FES = 717100 μ_F = 0.716 μ_{CR} = 0.233
 Diversity = 4.7963e-07 k_t = [1, 1, 1, 1]
 Generation 7180: Best = 3.494586e+03 FES = 718100 μ_F = 0.715 μ_{CR} = 0.228
 Diversity = 4.5274e-07 k_t = [2, 4, 3, 4]
 Generation 7190: Best = 3.494586e+03 FES = 719100 μ_F = 0.716 μ_{CR} = 0.223
 Diversity = 4.6002e-07 k_t = [1, 1, 1, 1]
 Generation 7200: Best = 3.494586e+03 FES = 720100 μ_F = 0.720 μ_{CR} = 0.222
 Diversity = 4.2076e-07 k_t = [3, 4, 3, 4]
 Generation 7210: Best = 3.494586e+03 FES = 721100 μ_F = 0.727 μ_{CR} = 0.224
 Diversity = 4.4399e-07 k_t = [1, 1, 1, 1]
 Generation 7220: Best = 3.494586e+03 FES = 722100 μ_F = 0.738 μ_{CR} = 0.226
 Diversity = 4.7867e-07 k_t = [4, 2, 2, 4]
 Generation 7230: Best = 3.494586e+03 FES = 723100 μ_F = 0.748 μ_{CR} = 0.224
 Diversity = 5.0611e-07 k_t = [1, 1, 1, 1]
 Generation 7240: Best = 3.494586e+03 FES = 724100 μ_F = 0.733 μ_{CR} = 0.227
 Diversity = 4.9047e-07 k_t = [2, 4, 2, 3]

Generation 7250: Best = 3.494586e+03 FES = 725100 μ_F = 0.746 μ_{CR} = 0.234
 Diversity = 5.3786e-07 k_t = [1, 1, 1, 1]
 Generation 7260: Best = 3.494586e+03 FES = 726100 μ_F = 0.745 μ_{CR} = 0.231
 Diversity = 5.0396e-07 k_t = [2, 3, 3, 3]
 Generation 7270: Best = 3.494586e+03 FES = 727100 μ_F = 0.727 μ_{CR} = 0.230
 Diversity = 4.9804e-07 k_t = [1, 1, 1, 1]
 Generation 7280: Best = 3.494586e+03 FES = 728100 μ_F = 0.733 μ_{CR} = 0.214
 Diversity = 5.3357e-07 k_t = [3, 4, 2, 2]
 Generation 7290: Best = 3.494586e+03 FES = 729100 μ_F = 0.722 μ_{CR} = 0.216
 Diversity = 5.2933e-07 k_t = [1, 1, 1, 1]
 Generation 7300: Best = 3.494586e+03 FES = 730100 μ_F = 0.721 μ_{CR} = 0.221
 Diversity = 5.0958e-07 k_t = [3, 4, 4, 2]
 Generation 7310: Best = 3.494586e+03 FES = 731100 μ_F = 0.723 μ_{CR} = 0.222
 Diversity = 5.1125e-07 k_t = [1, 1, 1, 1]
 Generation 7320: Best = 3.494586e+03 FES = 732100 μ_F = 0.733 μ_{CR} = 0.216
 Diversity = 5.0503e-07 k_t = [3, 2, 2, 3]
 Generation 7330: Best = 3.494586e+03 FES = 733100 μ_F = 0.727 μ_{CR} = 0.205
 Diversity = 5.1977e-07 k_t = [1, 1, 1, 1]
 Generation 7340: Best = 3.494586e+03 FES = 734100 μ_F = 0.732 μ_{CR} = 0.201
 Diversity = 4.7398e-07 k_t = [3, 2, 2, 4]
 Generation 7350: Best = 3.494586e+03 FES = 735100 μ_F = 0.734 μ_{CR} = 0.197
 Diversity = 4.6292e-07 k_t = [1, 1, 1, 1]
 Generation 7360: Best = 3.494586e+03 FES = 736100 μ_F = 0.720 μ_{CR} = 0.209
 Diversity = 4.2239e-07 k_t = [4, 2, 3, 3]
 Generation 7370: Best = 3.494586e+03 FES = 737100 μ_F = 0.726 μ_{CR} = 0.200
 Diversity = 4.4400e-07 k_t = [1, 1, 1, 1]
 Generation 7380: Best = 3.494586e+03 FES = 738100 μ_F = 0.734 μ_{CR} = 0.198
 Diversity = 4.3465e-07 k_t = [2, 3, 4, 3]
 Generation 7390: Best = 3.494586e+03 FES = 739100 μ_F = 0.742 μ_{CR} = 0.198
 Diversity = 4.8898e-07 k_t = [1, 1, 1, 1]
 Generation 7400: Best = 3.494586e+03 FES = 740100 μ_F = 0.739 μ_{CR} = 0.204
 Diversity = 4.9860e-07 k_t = [4, 4, 2, 2]
 Generation 7410: Best = 3.494586e+03 FES = 741100 μ_F = 0.746 μ_{CR} = 0.210
 Diversity = 5.4335e-07 k_t = [1, 1, 1, 1]
 Generation 7420: Best = 3.494586e+03 FES = 742100 μ_F = 0.735 μ_{CR} = 0.213
 Diversity = 5.5667e-07 k_t = [4, 4, 3, 3]
 Generation 7430: Best = 3.494586e+03 FES = 743100 μ_F = 0.721 μ_{CR} = 0.217
 Diversity = 5.7632e-07 k_t = [1, 1, 1, 1]
 Generation 7440: Best = 3.494586e+03 FES = 744100 μ_F = 0.730 μ_{CR} = 0.200
 Diversity = 5.5214e-07 k_t = [4, 2, 4, 4]
 Generation 7450: Best = 3.494586e+03 FES = 745100 μ_F = 0.747 μ_{CR} = 0.196
 Diversity = 5.9447e-07 k_t = [1, 1, 1, 1]
 Generation 7460: Best = 3.494586e+03 FES = 746100 μ_F = 0.752 μ_{CR} = 0.195
 Diversity = 5.9090e-07 k_t = [4, 3, 4, 2]
 Generation 7470: Best = 3.494586e+03 FES = 747100 μ_F = 0.767 μ_{CR} = 0.200
 Diversity = 6.0312e-07 k_t = [1, 1, 1, 1]
 Generation 7480: Best = 3.494586e+03 FES = 748100 μ_F = 0.749 μ_{CR} = 0.195
 Diversity = 5.9824e-07 k_t = [2, 2, 2, 3]
 Generation 7490: Best = 3.494586e+03 FES = 749100 μ_F = 0.731 μ_{CR} = 0.188
 Diversity = 6.3831e-07 k_t = [1, 1, 1, 1]
 Generation 7500: Best = 3.494586e+03 FES = 750100 μ_F = 0.741 μ_{CR} = 0.188
 Diversity = 6.1257e-07 k_t = [2, 3, 2, 3]
 Generation 7510: Best = 3.494586e+03 FES = 751100 μ_F = 0.738 μ_{CR} = 0.194
 Diversity = 6.1691e-07 k_t = [1, 1, 1, 1]

Generation 7520: Best = 3.494586e+03 FES = 752100 μ_F = 0.742 μ_{CR} = 0.197
 Diversity = 6.2295e-07 k_t = [3, 4, 4, 2]
 Generation 7530: Best = 3.494586e+03 FES = 753100 μ_F = 0.724 μ_{CR} = 0.197
 Diversity = 6.1350e-07 k_t = [1, 1, 1, 1]
 Generation 7540: Best = 3.494586e+03 FES = 754100 μ_F = 0.709 μ_{CR} = 0.208
 Diversity = 5.4280e-07 k_t = [2, 4, 2, 3]
 Generation 7550: Best = 3.494586e+03 FES = 755100 μ_F = 0.716 μ_{CR} = 0.202
 Diversity = 5.3278e-07 k_t = [1, 1, 1, 1]
 Generation 7560: Best = 3.494586e+03 FES = 756100 μ_F = 0.708 μ_{CR} = 0.199
 Diversity = 4.8462e-07 k_t = [3, 4, 3, 3]
 Generation 7570: Best = 3.494586e+03 FES = 757100 μ_F = 0.696 μ_{CR} = 0.194
 Diversity = 4.6977e-07 k_t = [1, 1, 1, 1]
 Generation 7580: Best = 3.494586e+03 FES = 758100 μ_F = 0.697 μ_{CR} = 0.198
 Diversity = 4.4151e-07 k_t = [4, 2, 2, 2]
 Generation 7590: Best = 3.494586e+03 FES = 759100 μ_F = 0.702 μ_{CR} = 0.210
 Diversity = 4.0480e-07 k_t = [1, 1, 1, 1]
 Generation 7600: Best = 3.494586e+03 FES = 760100 μ_F = 0.717 μ_{CR} = 0.214
 Diversity = 3.6507e-07 k_t = [4, 2, 4, 4]
 Generation 7610: Best = 3.494586e+03 FES = 761100 μ_F = 0.726 μ_{CR} = 0.206
 Diversity = 3.4217e-07 k_t = [1, 1, 1, 1]
 Generation 7620: Best = 3.494586e+03 FES = 762100 μ_F = 0.745 μ_{CR} = 0.214
 Diversity = 3.6229e-07 k_t = [4, 4, 3, 4]
 Generation 7630: Best = 3.494586e+03 FES = 763100 μ_F = 0.759 μ_{CR} = 0.212
 Diversity = 3.8456e-07 k_t = [1, 1, 1, 1]
 Generation 7640: Best = 3.494586e+03 FES = 764100 μ_F = 0.755 μ_{CR} = 0.211
 Diversity = 4.0819e-07 k_t = [2, 3, 3, 2]
 Generation 7650: Best = 3.494586e+03 FES = 765100 μ_F = 0.762 μ_{CR} = 0.214
 Diversity = 4.6994e-07 k_t = [1, 1, 1, 1]
 Generation 7660: Best = 3.494586e+03 FES = 766100 μ_F = 0.758 μ_{CR} = 0.208
 Diversity = 4.3668e-07 k_t = [2, 2, 3, 4]
 Generation 7670: Best = 3.494586e+03 FES = 767100 μ_F = 0.749 μ_{CR} = 0.209
 Diversity = 4.4661e-07 k_t = [1, 1, 1, 1]
 Generation 7680: Best = 3.494586e+03 FES = 768100 μ_F = 0.747 μ_{CR} = 0.209
 Diversity = 4.4369e-07 k_t = [3, 2, 4, 2]
 Generation 7690: Best = 3.494586e+03 FES = 769100 μ_F = 0.744 μ_{CR} = 0.211
 Diversity = 4.3944e-07 k_t = [1, 1, 1, 1]
 Generation 7700: Best = 3.494586e+03 FES = 770100 μ_F = 0.728 μ_{CR} = 0.200
 Diversity = 4.0848e-07 k_t = [3, 4, 2, 2]
 Generation 7710: Best = 3.494586e+03 FES = 771100 μ_F = 0.715 μ_{CR} = 0.196
 Diversity = 4.1950e-07 k_t = [1, 1, 1, 1]
 Generation 7720: Best = 3.494586e+03 FES = 772100 μ_F = 0.727 μ_{CR} = 0.200
 Diversity = 4.2130e-07 k_t = [2, 4, 2, 2]
 Generation 7730: Best = 3.494586e+03 FES = 773100 μ_F = 0.739 μ_{CR} = 0.199
 Diversity = 4.5318e-07 k_t = [1, 1, 1, 1]
 Generation 7740: Best = 3.494586e+03 FES = 774100 μ_F = 0.729 μ_{CR} = 0.196
 Diversity = 4.6011e-07 k_t = [3, 4, 2, 2]
 Generation 7750: Best = 3.494586e+03 FES = 775100 μ_F = 0.734 μ_{CR} = 0.199
 Diversity = 4.8810e-07 k_t = [1, 1, 1, 1]
 Generation 7760: Best = 3.494586e+03 FES = 776100 μ_F = 0.741 μ_{CR} = 0.209
 Diversity = 5.5274e-07 k_t = [2, 3, 4, 3]
 Generation 7770: Best = 3.494586e+03 FES = 777100 μ_F = 0.744 μ_{CR} = 0.215
 Diversity = 6.6073e-07 k_t = [1, 1, 1, 1]
 Generation 7780: Best = 3.494586e+03 FES = 778100 μ_F = 0.730 μ_{CR} = 0.209
 Diversity = 6.1962e-07 k_t = [3, 2, 4, 3]

Generation 7790: Best = 3.494586e+03 FES = 779100 μ_F = 0.730 μ_{CR} = 0.210
 Diversity = 6.4421e-07 k_t = [1, 1, 1, 1]
 Generation 7800: Best = 3.494586e+03 FES = 780100 μ_F = 0.735 μ_{CR} = 0.218
 Diversity = 6.6154e-07 k_t = [3, 4, 4, 3]
 Generation 7810: Best = 3.494586e+03 FES = 781100 μ_F = 0.723 μ_{CR} = 0.219
 Diversity = 6.6577e-07 k_t = [1, 1, 1, 1]
 Generation 7820: Best = 3.494586e+03 FES = 782100 μ_F = 0.721 μ_{CR} = 0.218
 Diversity = 5.9938e-07 k_t = [2, 3, 3, 3]
 Generation 7830: Best = 3.494586e+03 FES = 783100 μ_F = 0.724 μ_{CR} = 0.211
 Diversity = 6.1717e-07 k_t = [1, 1, 1, 1]
 Generation 7840: Best = 3.494586e+03 FES = 784100 μ_F = 0.715 μ_{CR} = 0.204
 Diversity = 6.0780e-07 k_t = [4, 3, 2, 3]
 Generation 7850: Best = 3.494586e+03 FES = 785100 μ_F = 0.706 μ_{CR} = 0.214
 Diversity = 5.9495e-07 k_t = [1, 1, 1, 1]
 Generation 7860: Best = 3.494586e+03 FES = 786100 μ_F = 0.719 μ_{CR} = 0.203
 Diversity = 6.2017e-07 k_t = [3, 2, 4, 3]
 Generation 7870: Best = 3.494586e+03 FES = 787100 μ_F = 0.724 μ_{CR} = 0.199
 Diversity = 6.4382e-07 k_t = [1, 1, 1, 1]
 Generation 7880: Best = 3.494586e+03 FES = 788100 μ_F = 0.732 μ_{CR} = 0.200
 Diversity = 6.6156e-07 k_t = [4, 2, 2, 3]
 Generation 7890: Best = 3.494586e+03 FES = 789100 μ_F = 0.731 μ_{CR} = 0.212
 Diversity = 7.3917e-07 k_t = [1, 1, 1, 1]
 Generation 7900: Best = 3.494586e+03 FES = 790100 μ_F = 0.718 μ_{CR} = 0.206
 Diversity = 6.9276e-07 k_t = [4, 2, 3, 3]
 Generation 7910: Best = 3.494586e+03 FES = 791100 μ_F = 0.716 μ_{CR} = 0.207
 Diversity = 7.3097e-07 k_t = [1, 1, 1, 1]
 Generation 7920: Best = 3.494586e+03 FES = 792100 μ_F = 0.712 μ_{CR} = 0.213
 Diversity = 6.9732e-07 k_t = [2, 2, 4, 3]
 Generation 7930: Best = 3.494586e+03 FES = 793100 μ_F = 0.722 μ_{CR} = 0.212
 Diversity = 7.2630e-07 k_t = [1, 1, 1, 1]
 Generation 7940: Best = 3.494586e+03 FES = 794100 μ_F = 0.714 μ_{CR} = 0.212
 Diversity = 6.5636e-07 k_t = [4, 4, 2, 2]
 Generation 7950: Best = 3.494586e+03 FES = 795100 μ_F = 0.710 μ_{CR} = 0.214
 Diversity = 5.9883e-07 k_t = [1, 1, 1, 1]
 Generation 7960: Best = 3.494586e+03 FES = 796100 μ_F = 0.713 μ_{CR} = 0.208
 Diversity = 4.9326e-07 k_t = [2, 2, 4, 4]
 Generation 7970: Best = 3.494586e+03 FES = 797100 μ_F = 0.709 μ_{CR} = 0.215
 Diversity = 4.2911e-07 k_t = [1, 1, 1, 1]
 Generation 7980: Best = 3.494586e+03 FES = 798100 μ_F = 0.712 μ_{CR} = 0.220
 Diversity = 4.0551e-07 k_t = [3, 3, 3, 2]
 Generation 7990: Best = 3.494586e+03 FES = 799100 μ_F = 0.717 μ_{CR} = 0.218
 Diversity = 4.2157e-07 k_t = [1, 1, 1, 1]
 Generation 8000: Best = 3.494586e+03 FES = 800100 μ_F = 0.711 μ_{CR} = 0.208
 Diversity = 3.8690e-07 k_t = [2, 4, 4, 3]
 Generation 8010: Best = 3.494586e+03 FES = 801100 μ_F = 0.692 μ_{CR} = 0.206
 Diversity = 3.5138e-07 k_t = [1, 1, 1, 1]
 Generation 8020: Best = 3.494586e+03 FES = 802100 μ_F = 0.691 μ_{CR} = 0.205
 Diversity = 2.8825e-07 k_t = [2, 2, 3, 4]
 Generation 8030: Best = 3.494586e+03 FES = 803100 μ_F = 0.701 μ_{CR} = 0.205
 Diversity = 2.5081e-07 k_t = [1, 1, 1, 1]
 Generation 8040: Best = 3.494586e+03 FES = 804100 μ_F = 0.707 μ_{CR} = 0.209
 Diversity = 2.2544e-07 k_t = [3, 4, 2, 4]
 Generation 8050: Best = 3.494586e+03 FES = 805100 μ_F = 0.726 μ_{CR} = 0.209
 Diversity = 2.1481e-07 k_t = [1, 1, 1, 1]

Generation 8060: Best = 3.494586e+03 FES = 806100 μ_F = 0.720 μ_{CR} = 0.203
 Diversity = 1.9833e-07 k_t = [2, 3, 3, 4]
 Generation 8070: Best = 3.494586e+03 FES = 807100 μ_F = 0.715 μ_{CR} = 0.189
 Diversity = 1.8071e-07 k_t = [1, 1, 1, 1]
 Generation 8080: Best = 3.494586e+03 FES = 808100 μ_F = 0.709 μ_{CR} = 0.197
 Diversity = 1.6399e-07 k_t = [2, 4, 3, 3]
 Generation 8090: Best = 3.494586e+03 FES = 809100 μ_F = 0.718 μ_{CR} = 0.199
 Diversity = 1.5595e-07 k_t = [1, 1, 1, 1]
 Generation 8100: Best = 3.494586e+03 FES = 810100 μ_F = 0.742 μ_{CR} = 0.206
 Diversity = 1.5728e-07 k_t = [3, 3, 4, 2]
 Generation 8110: Best = 3.494586e+03 FES = 811100 μ_F = 0.743 μ_{CR} = 0.204
 Diversity = 1.6593e-07 k_t = [1, 1, 1, 1]
 Generation 8120: Best = 3.494586e+03 FES = 812100 μ_F = 0.746 μ_{CR} = 0.201
 Diversity = 1.5792e-07 k_t = [2, 3, 2, 3]
 Generation 8130: Best = 3.494586e+03 FES = 813100 μ_F = 0.740 μ_{CR} = 0.197
 Diversity = 1.6689e-07 k_t = [1, 1, 1, 1]
 Generation 8140: Best = 3.494586e+03 FES = 814100 μ_F = 0.732 μ_{CR} = 0.195
 Diversity = 1.6579e-07 k_t = [3, 3, 3, 2]
 Generation 8150: Best = 3.494586e+03 FES = 815100 μ_F = 0.736 μ_{CR} = 0.193
 Diversity = 1.7457e-07 k_t = [1, 1, 1, 1]
 Generation 8160: Best = 3.494586e+03 FES = 816100 μ_F = 0.736 μ_{CR} = 0.193
 Diversity = 1.7013e-07 k_t = [2, 4, 4, 2]
 Generation 8170: Best = 3.494586e+03 FES = 817100 μ_F = 0.740 μ_{CR} = 0.197
 Diversity = 1.6617e-07 k_t = [1, 1, 1, 1]
 Generation 8180: Best = 3.494586e+03 FES = 818100 μ_F = 0.733 μ_{CR} = 0.200
 Diversity = 1.4202e-07 k_t = [2, 4, 2, 3]
 Generation 8190: Best = 3.494586e+03 FES = 819100 μ_F = 0.738 μ_{CR} = 0.198
 Diversity = 1.3809e-07 k_t = [1, 1, 1, 1]
 Generation 8200: Best = 3.494586e+03 FES = 820100 μ_F = 0.740 μ_{CR} = 0.194
 Diversity = 1.3392e-07 k_t = [2, 4, 2, 2]
 Generation 8210: Best = 3.494586e+03 FES = 821100 μ_F = 0.732 μ_{CR} = 0.187
 Diversity = 1.4012e-07 k_t = [1, 1, 1, 1]
 Generation 8220: Best = 3.494586e+03 FES = 822100 μ_F = 0.731 μ_{CR} = 0.191
 Diversity = 1.4150e-07 k_t = [3, 2, 4, 2]
 Generation 8230: Best = 3.494586e+03 FES = 823100 μ_F = 0.716 μ_{CR} = 0.187
 Diversity = 1.3751e-07 k_t = [1, 1, 1, 1]
 Generation 8240: Best = 3.494586e+03 FES = 824100 μ_F = 0.727 μ_{CR} = 0.183
 Diversity = 1.3227e-07 k_t = [4, 4, 4, 2]
 Generation 8250: Best = 3.494586e+03 FES = 825100 μ_F = 0.743 μ_{CR} = 0.182
 Diversity = 1.2929e-07 k_t = [1, 1, 1, 1]
 Generation 8260: Best = 3.494586e+03 FES = 826100 μ_F = 0.750 μ_{CR} = 0.179
 Diversity = 1.2940e-07 k_t = [4, 3, 2, 2]
 Generation 8270: Best = 3.494586e+03 FES = 827100 μ_F = 0.738 μ_{CR} = 0.183
 Diversity = 1.2996e-07 k_t = [1, 1, 1, 1]
 Generation 8280: Best = 3.494586e+03 FES = 828100 μ_F = 0.744 μ_{CR} = 0.177
 Diversity = 1.2398e-07 k_t = [3, 3, 2, 2]
 Generation 8290: Best = 3.494586e+03 FES = 829100 μ_F = 0.738 μ_{CR} = 0.176
 Diversity = 1.2158e-07 k_t = [1, 1, 1, 1]
 Generation 8300: Best = 3.494586e+03 FES = 830100 μ_F = 0.743 μ_{CR} = 0.177
 Diversity = 1.1613e-07 k_t = [4, 2, 2, 3]
 Generation 8310: Best = 3.494586e+03 FES = 831100 μ_F = 0.756 μ_{CR} = 0.183
 Diversity = 1.1831e-07 k_t = [1, 1, 1, 1]
 Generation 8320: Best = 3.494586e+03 FES = 832100 μ_F = 0.747 μ_{CR} = 0.176
 Diversity = 1.0949e-07 k_t = [3, 4, 4, 4]

Generation 8330: Best = 3.494586e+03 FES = 833100 μ_F = 0.763 μ_{CR} = 0.179
 Diversity = 1.0523e-07 k_t = [1, 1, 1, 1]
 Generation 8340: Best = 3.494586e+03 FES = 834100 μ_F = 0.758 μ_{CR} = 0.169
 Diversity = 1.0313e-07 k_t = [2, 4, 2, 2]
 Generation 8350: Best = 3.494586e+03 FES = 835100 μ_F = 0.767 μ_{CR} = 0.173
 Diversity = 1.0582e-07 k_t = [1, 1, 1, 1]
 Generation 8360: Best = 3.494586e+03 FES = 836100 μ_F = 0.768 μ_{CR} = 0.172
 Diversity = 1.0461e-07 k_t = [4, 2, 3, 2]
 Generation 8370: Best = 3.494586e+03 FES = 837100 μ_F = 0.766 μ_{CR} = 0.180
 Diversity = 1.0324e-07 k_t = [1, 1, 1, 1]
 Generation 8380: Best = 3.494586e+03 FES = 838100 μ_F = 0.756 μ_{CR} = 0.173
 Diversity = 9.7247e-08 k_t = [4, 4, 2, 3]
 Generation 8390: Best = 3.494586e+03 FES = 839100 μ_F = 0.743 μ_{CR} = 0.160
 Diversity = 9.7001e-08 k_t = [1, 1, 1, 1]
 Generation 8400: Best = 3.494586e+03 FES = 840100 μ_F = 0.754 μ_{CR} = 0.166
 Diversity = 9.6152e-08 k_t = [2, 2, 3, 2]
 Generation 8410: Best = 3.494586e+03 FES = 841100 μ_F = 0.754 μ_{CR} = 0.174
 Diversity = 9.6028e-08 k_t = [1, 1, 1, 1]
 Generation 8420: Best = 3.494586e+03 FES = 842100 μ_F = 0.736 μ_{CR} = 0.170
 Diversity = 9.1801e-08 k_t = [4, 2, 4, 4]
 Generation 8430: Best = 3.494586e+03 FES = 843100 μ_F = 0.728 μ_{CR} = 0.170
 Diversity = 9.1748e-08 k_t = [1, 1, 1, 1]
 Generation 8440: Best = 3.494586e+03 FES = 844100 μ_F = 0.698 μ_{CR} = 0.160
 Diversity = 8.4715e-08 k_t = [2, 2, 2, 4]
 Generation 8450: Best = 3.494586e+03 FES = 845100 μ_F = 0.717 μ_{CR} = 0.164
 Diversity = 8.5724e-08 k_t = [1, 1, 1, 1]
 Generation 8460: Best = 3.494586e+03 FES = 846100 μ_F = 0.737 μ_{CR} = 0.165
 Diversity = 8.3238e-08 k_t = [2, 4, 3, 3]
 Generation 8470: Best = 3.494586e+03 FES = 847100 μ_F = 0.741 μ_{CR} = 0.160
 Diversity = 8.4315e-08 k_t = [1, 1, 1, 1]
 Generation 8480: Best = 3.494586e+03 FES = 848100 μ_F = 0.744 μ_{CR} = 0.172
 Diversity = 8.6030e-08 k_t = [2, 4, 2, 3]
 Generation 8490: Best = 3.494586e+03 FES = 849100 μ_F = 0.737 μ_{CR} = 0.174
 Diversity = 8.6839e-08 k_t = [1, 1, 1, 1]
 Generation 8500: Best = 3.494586e+03 FES = 850100 μ_F = 0.752 μ_{CR} = 0.165
 Diversity = 8.9822e-08 k_t = [4, 4, 4, 3]
 Generation 8510: Best = 3.494586e+03 FES = 851100 μ_F = 0.770 μ_{CR} = 0.174
 Diversity = 1.0183e-07 k_t = [1, 1, 1, 1]
 Generation 8520: Best = 3.494586e+03 FES = 852100 μ_F = 0.788 μ_{CR} = 0.175
 Diversity = 1.1682e-07 k_t = [4, 4, 2, 4]
 Generation 8530: Best = 3.494586e+03 FES = 853100 μ_F = 0.793 μ_{CR} = 0.174
 Diversity = 1.3788e-07 k_t = [1, 1, 1, 1]
 Generation 8540: Best = 3.494586e+03 FES = 854100 μ_F = 0.781 μ_{CR} = 0.180
 Diversity = 1.3607e-07 k_t = [3, 2, 3, 4]
 Generation 8550: Best = 3.494586e+03 FES = 855100 μ_F = 0.770 μ_{CR} = 0.174
 Diversity = 1.5671e-07 k_t = [1, 1, 1, 1]
 Generation 8560: Best = 3.494586e+03 FES = 856100 μ_F = 0.762 μ_{CR} = 0.170
 Diversity = 1.6253e-07 k_t = [4, 3, 2, 3]
 Generation 8570: Best = 3.494586e+03 FES = 857100 μ_F = 0.765 μ_{CR} = 0.167
 Diversity = 1.6362e-07 k_t = [1, 1, 1, 1]
 Generation 8580: Best = 3.494586e+03 FES = 858100 μ_F = 0.762 μ_{CR} = 0.165
 Diversity = 1.4250e-07 k_t = [4, 3, 4, 4]
 Generation 8590: Best = 3.494586e+03 FES = 859100 μ_F = 0.764 μ_{CR} = 0.165
 Diversity = 1.3246e-07 k_t = [1, 1, 1, 1]

Generation 8600: Best = 3.494586e+03 FES = 860100 μ_F = 0.765 μ_{CR} = 0.170
 Diversity = 1.2159e-07 k_t = [4, 4, 3, 3]
 Generation 8610: Best = 3.494586e+03 FES = 861100 μ_F = 0.760 μ_{CR} = 0.169
 Diversity = 1.2389e-07 k_t = [1, 1, 1, 1]
 Generation 8620: Best = 3.494586e+03 FES = 862100 μ_F = 0.760 μ_{CR} = 0.176
 Diversity = 1.2094e-07 k_t = [4, 2, 4, 4]
 Generation 8630: Best = 3.494586e+03 FES = 863100 μ_F = 0.750 μ_{CR} = 0.174
 Diversity = 1.1914e-07 k_t = [1, 1, 1, 1]
 Generation 8640: Best = 3.494586e+03 FES = 864100 μ_F = 0.772 μ_{CR} = 0.179
 Diversity = 1.1980e-07 k_t = [4, 3, 4, 3]
 Generation 8650: Best = 3.494586e+03 FES = 865100 μ_F = 0.781 μ_{CR} = 0.176
 Diversity = 1.3061e-07 k_t = [1, 1, 1, 1]
 Generation 8660: Best = 3.494586e+03 FES = 866100 μ_F = 0.769 μ_{CR} = 0.174
 Diversity = 1.4059e-07 k_t = [4, 2, 2, 2]
 Generation 8670: Best = 3.494586e+03 FES = 867100 μ_F = 0.761 μ_{CR} = 0.170
 Diversity = 1.3773e-07 k_t = [1, 1, 1, 1]
 Generation 8680: Best = 3.494586e+03 FES = 868100 μ_F = 0.742 μ_{CR} = 0.164
 Diversity = 1.2411e-07 k_t = [3, 4, 3, 3]
 Generation 8690: Best = 3.494586e+03 FES = 869100 μ_F = 0.726 μ_{CR} = 0.165
 Diversity = 1.2446e-07 k_t = [1, 1, 1, 1]
 Generation 8700: Best = 3.494586e+03 FES = 870100 μ_F = 0.726 μ_{CR} = 0.164
 Diversity = 1.2360e-07 k_t = [3, 3, 3, 2]
 Generation 8710: Best = 3.494586e+03 FES = 871100 μ_F = 0.735 μ_{CR} = 0.169
 Diversity = 1.2303e-07 k_t = [1, 1, 1, 1]
 Generation 8720: Best = 3.494586e+03 FES = 872100 μ_F = 0.730 μ_{CR} = 0.163
 Diversity = 1.1468e-07 k_t = [3, 4, 2, 4]
 Generation 8730: Best = 3.494586e+03 FES = 873100 μ_F = 0.744 μ_{CR} = 0.159
 Diversity = 1.1306e-07 k_t = [1, 1, 1, 1]
 Generation 8740: Best = 3.494586e+03 FES = 874100 μ_F = 0.761 μ_{CR} = 0.167
 Diversity = 1.1850e-07 k_t = [2, 3, 3, 3]
 Generation 8750: Best = 3.494586e+03 FES = 875100 μ_F = 0.759 μ_{CR} = 0.171
 Diversity = 1.2995e-07 k_t = [1, 1, 1, 1]
 Generation 8760: Best = 3.494586e+03 FES = 876100 μ_F = 0.756 μ_{CR} = 0.172
 Diversity = 1.3321e-07 k_t = [3, 2, 3, 2]
 Generation 8770: Best = 3.494586e+03 FES = 877100 μ_F = 0.738 μ_{CR} = 0.166
 Diversity = 1.3330e-07 k_t = [1, 1, 1, 1]
 Generation 8780: Best = 3.494586e+03 FES = 878100 μ_F = 0.728 μ_{CR} = 0.165
 Diversity = 1.1722e-07 k_t = [3, 3, 4, 4]
 Generation 8790: Best = 3.494586e+03 FES = 879100 μ_F = 0.745 μ_{CR} = 0.171
 Diversity = 1.2208e-07 k_t = [1, 1, 1, 1]
 Generation 8800: Best = 3.494586e+03 FES = 880100 μ_F = 0.744 μ_{CR} = 0.189
 Diversity = 1.1617e-07 k_t = [4, 4, 4, 4]
 Generation 8810: Best = 3.494586e+03 FES = 881100 μ_F = 0.750 μ_{CR} = 0.182
 Diversity = 1.1624e-07 k_t = [1, 1, 1, 1]
 Generation 8820: Best = 3.494586e+03 FES = 882100 μ_F = 0.764 μ_{CR} = 0.178
 Diversity = 1.1810e-07 k_t = [3, 3, 3, 2]
 Generation 8830: Best = 3.494586e+03 FES = 883100 μ_F = 0.759 μ_{CR} = 0.183
 Diversity = 1.2588e-07 k_t = [1, 1, 1, 1]
 Generation 8840: Best = 3.494586e+03 FES = 884100 μ_F = 0.748 μ_{CR} = 0.191
 Diversity = 1.1801e-07 k_t = [3, 4, 3, 3]
 Generation 8850: Best = 3.494586e+03 FES = 885100 μ_F = 0.754 μ_{CR} = 0.193
 Diversity = 1.3395e-07 k_t = [1, 1, 1, 1]
 Generation 8860: Best = 3.494586e+03 FES = 886100 μ_F = 0.751 μ_{CR} = 0.203
 Diversity = 1.4776e-07 k_t = [3, 3, 2, 2]

Generation 8870: Best = 3.494586e+03 FES = 887100 μ_F = 0.755 μ_{CR} = 0.189
 Diversity = 1.5338e-07 k_t = [1, 1, 1, 1]
 Generation 8880: Best = 3.494586e+03 FES = 888100 μ_F = 0.749 μ_{CR} = 0.185
 Diversity = 1.4820e-07 k_t = [4, 4, 2, 2]
 Generation 8890: Best = 3.494586e+03 FES = 889100 μ_F = 0.750 μ_{CR} = 0.187
 Diversity = 1.4448e-07 k_t = [1, 1, 1, 1]
 Generation 8900: Best = 3.494586e+03 FES = 890100 μ_F = 0.751 μ_{CR} = 0.191
 Diversity = 1.3895e-07 k_t = [2, 3, 2, 2]
 Generation 8910: Best = 3.494586e+03 FES = 891100 μ_F = 0.753 μ_{CR} = 0.187
 Diversity = 1.4587e-07 k_t = [1, 1, 1, 1]
 Generation 8920: Best = 3.494586e+03 FES = 892100 μ_F = 0.740 μ_{CR} = 0.174
 Diversity = 1.3523e-07 k_t = [2, 3, 3, 3]
 Generation 8930: Best = 3.494586e+03 FES = 893100 μ_F = 0.742 μ_{CR} = 0.174
 Diversity = 1.3715e-07 k_t = [1, 1, 1, 1]
 Generation 8940: Best = 3.494586e+03 FES = 894100 μ_F = 0.746 μ_{CR} = 0.173
 Diversity = 1.2703e-07 k_t = [4, 2, 4, 3]
 Generation 8950: Best = 3.494586e+03 FES = 895100 μ_F = 0.749 μ_{CR} = 0.173
 Diversity = 1.3047e-07 k_t = [1, 1, 1, 1]
 Generation 8960: Best = 3.494586e+03 FES = 896100 μ_F = 0.740 μ_{CR} = 0.171
 Diversity = 1.2622e-07 k_t = [3, 2, 4, 2]
 Generation 8970: Best = 3.494586e+03 FES = 897100 μ_F = 0.753 μ_{CR} = 0.178
 Diversity = 1.3120e-07 k_t = [1, 1, 1, 1]
 Generation 8980: Best = 3.494586e+03 FES = 898100 μ_F = 0.754 μ_{CR} = 0.176
 Diversity = 1.3542e-07 k_t = [3, 3, 4, 2]
 Generation 8990: Best = 3.494586e+03 FES = 899100 μ_F = 0.750 μ_{CR} = 0.176
 Diversity = 1.3587e-07 k_t = [1, 1, 1, 1]
 Generation 9000: Best = 3.494586e+03 FES = 900100 μ_F = 0.759 μ_{CR} = 0.171
 Diversity = 1.1931e-07 k_t = [4, 3, 4, 4]
 Generation 9010: Best = 3.494586e+03 FES = 901100 μ_F = 0.757 μ_{CR} = 0.165
 Diversity = 1.1441e-07 k_t = [1, 1, 1, 1]
 Generation 9020: Best = 3.494586e+03 FES = 902100 μ_F = 0.769 μ_{CR} = 0.177
 Diversity = 1.1489e-07 k_t = [4, 3, 2, 4]
 Generation 9030: Best = 3.494586e+03 FES = 903100 μ_F = 0.767 μ_{CR} = 0.178
 Diversity = 1.2181e-07 k_t = [1, 1, 1, 1]
 Generation 9040: Best = 3.494586e+03 FES = 904100 μ_F = 0.758 μ_{CR} = 0.172
 Diversity = 1.1281e-07 k_t = [2, 4, 2, 4]
 Generation 9050: Best = 3.494586e+03 FES = 905100 μ_F = 0.760 μ_{CR} = 0.167
 Diversity = 1.1199e-07 k_t = [1, 1, 1, 1]
 Generation 9060: Best = 3.494586e+03 FES = 906100 μ_F = 0.748 μ_{CR} = 0.174
 Diversity = 1.0370e-07 k_t = [2, 3, 2, 4]
 Generation 9070: Best = 3.494586e+03 FES = 907100 μ_F = 0.750 μ_{CR} = 0.176
 Diversity = 1.0617e-07 k_t = [1, 1, 1, 1]
 Generation 9080: Best = 3.494586e+03 FES = 908100 μ_F = 0.742 μ_{CR} = 0.166
 Diversity = 1.0117e-07 k_t = [4, 4, 3, 3]
 Generation 9090: Best = 3.494586e+03 FES = 909100 μ_F = 0.751 μ_{CR} = 0.163
 Diversity = 1.0371e-07 k_t = [1, 1, 1, 1]
 Generation 9100: Best = 3.494586e+03 FES = 910100 μ_F = 0.739 μ_{CR} = 0.169
 Diversity = 9.2296e-08 k_t = [2, 2, 3, 4]
 Generation 9110: Best = 3.494586e+03 FES = 911100 μ_F = 0.732 μ_{CR} = 0.162
 Diversity = 9.3361e-08 k_t = [1, 1, 1, 1]
 Generation 9120: Best = 3.494586e+03 FES = 912100 μ_F = 0.735 μ_{CR} = 0.158
 Diversity = 8.6179e-08 k_t = [3, 3, 4, 4]
 Generation 9130: Best = 3.494586e+03 FES = 913100 μ_F = 0.748 μ_{CR} = 0.162
 Diversity = 8.9326e-08 k_t = [1, 1, 1, 1]

Generation 9140: Best = 3.494586e+03 FES = 914100 μ_F = 0.763 μ_{CR} = 0.166
 Diversity = 9.0821e-08 k_t = [4, 3, 2, 4]
 Generation 9150: Best = 3.494586e+03 FES = 915100 μ_F = 0.769 μ_{CR} = 0.166
 Diversity = 9.8312e-08 k_t = [1, 1, 1, 1]
 Generation 9160: Best = 3.494586e+03 FES = 916100 μ_F = 0.771 μ_{CR} = 0.174
 Diversity = 1.0153e-07 k_t = [3, 4, 4, 4]
 Generation 9170: Best = 3.494586e+03 FES = 917100 μ_F = 0.769 μ_{CR} = 0.176
 Diversity = 1.0327e-07 k_t = [1, 1, 1, 1]
 Generation 9180: Best = 3.494586e+03 FES = 918100 μ_F = 0.764 μ_{CR} = 0.172
 Diversity = 1.0313e-07 k_t = [2, 2, 2, 3]
 Generation 9190: Best = 3.494586e+03 FES = 919100 μ_F = 0.760 μ_{CR} = 0.167
 Diversity = 1.0277e-07 k_t = [1, 1, 1, 1]
 Generation 9200: Best = 3.494586e+03 FES = 920100 μ_F = 0.754 μ_{CR} = 0.173
 Diversity = 9.6571e-08 k_t = [3, 2, 3, 3]
 Generation 9210: Best = 3.494586e+03 FES = 921100 μ_F = 0.743 μ_{CR} = 0.170
 Diversity = 9.9177e-08 k_t = [1, 1, 1, 1]
 Generation 9220: Best = 3.494586e+03 FES = 922100 μ_F = 0.750 μ_{CR} = 0.168
 Diversity = 9.1549e-08 k_t = [2, 2, 2, 4]
 Generation 9230: Best = 3.494586e+03 FES = 923100 μ_F = 0.743 μ_{CR} = 0.154
 Diversity = 9.2005e-08 k_t = [1, 1, 1, 1]
 Generation 9240: Best = 3.494586e+03 FES = 924100 μ_F = 0.749 μ_{CR} = 0.159
 Diversity = 9.2329e-08 k_t = [3, 4, 4, 2]
 Generation 9250: Best = 3.494586e+03 FES = 925100 μ_F = 0.753 μ_{CR} = 0.168
 Diversity = 9.6241e-08 k_t = [1, 1, 1, 1]
 Generation 9260: Best = 3.494586e+03 FES = 926100 μ_F = 0.764 μ_{CR} = 0.170
 Diversity = 9.6675e-08 k_t = [3, 3, 4, 4]
 Generation 9270: Best = 3.494586e+03 FES = 927100 μ_F = 0.762 μ_{CR} = 0.178
 Diversity = 1.0763e-07 k_t = [1, 1, 1, 1]
 Generation 9280: Best = 3.494586e+03 FES = 928100 μ_F = 0.762 μ_{CR} = 0.175
 Diversity = 1.0669e-07 k_t = [4, 3, 4, 3]
 Generation 9290: Best = 3.494586e+03 FES = 929100 μ_F = 0.769 μ_{CR} = 0.175
 Diversity = 1.0493e-07 k_t = [1, 1, 1, 1]
 Generation 9300: Best = 3.494586e+03 FES = 930100 μ_F = 0.754 μ_{CR} = 0.172
 Diversity = 9.4582e-08 k_t = [2, 4, 4, 4]
 Generation 9310: Best = 3.494586e+03 FES = 931100 μ_F = 0.758 μ_{CR} = 0.174
 Diversity = 9.3129e-08 k_t = [1, 1, 1, 1]
 Generation 9320: Best = 3.494586e+03 FES = 932100 μ_F = 0.774 μ_{CR} = 0.169
 Diversity = 9.1181e-08 k_t = [3, 4, 2, 4]
 Generation 9330: Best = 3.494586e+03 FES = 933100 μ_F = 0.775 μ_{CR} = 0.170
 Diversity = 9.2578e-08 k_t = [1, 1, 1, 1]
 Generation 9340: Best = 3.494586e+03 FES = 934100 μ_F = 0.789 μ_{CR} = 0.169
 Diversity = 9.5889e-08 k_t = [2, 4, 2, 3]
 Generation 9350: Best = 3.494586e+03 FES = 935100 μ_F = 0.788 μ_{CR} = 0.172
 Diversity = 1.0752e-07 k_t = [1, 1, 1, 1]
 Generation 9360: Best = 3.494586e+03 FES = 936100 μ_F = 0.783 μ_{CR} = 0.175
 Diversity = 1.1316e-07 k_t = [4, 4, 3, 4]
 Generation 9370: Best = 3.494586e+03 FES = 937100 μ_F = 0.796 μ_{CR} = 0.174
 Diversity = 1.1889e-07 k_t = [1, 1, 1, 1]
 Generation 9380: Best = 3.494586e+03 FES = 938100 μ_F = 0.778 μ_{CR} = 0.167
 Diversity = 1.0935e-07 k_t = [2, 3, 4, 3]
 Generation 9390: Best = 3.494586e+03 FES = 939100 μ_F = 0.767 μ_{CR} = 0.168
 Diversity = 1.1041e-07 k_t = [1, 1, 1, 1]
 Generation 9400: Best = 3.494586e+03 FES = 940100 μ_F = 0.758 μ_{CR} = 0.175
 Diversity = 1.0454e-07 k_t = [2, 2, 4, 3]

Generation 9410: Best = 3.494586e+03 FES = 941100 μ_F = 0.760 μ_{CR} = 0.176
 Diversity = 1.0810e-07 k_t = [1, 1, 1, 1]
 Generation 9420: Best = 3.494586e+03 FES = 942100 μ_F = 0.763 μ_{CR} = 0.179
 Diversity = 1.0038e-07 k_t = [2, 2, 4, 3]
 Generation 9430: Best = 3.494586e+03 FES = 943100 μ_F = 0.769 μ_{CR} = 0.171
 Diversity = 1.1283e-07 k_t = [1, 1, 1, 1]
 Generation 9440: Best = 3.494586e+03 FES = 944100 μ_F = 0.757 μ_{CR} = 0.171
 Diversity = 1.1427e-07 k_t = [2, 3, 3, 2]
 Generation 9450: Best = 3.494586e+03 FES = 945100 μ_F = 0.752 μ_{CR} = 0.163
 Diversity = 1.1477e-07 k_t = [1, 1, 1, 1]
 Generation 9460: Best = 3.494586e+03 FES = 946100 μ_F = 0.757 μ_{CR} = 0.163
 Diversity = 1.1194e-07 k_t = [2, 3, 4, 3]
 Generation 9470: Best = 3.494586e+03 FES = 947100 μ_F = 0.762 μ_{CR} = 0.162
 Diversity = 1.3349e-07 k_t = [1, 1, 1, 1]
 Generation 9480: Best = 3.494586e+03 FES = 948100 μ_F = 0.773 μ_{CR} = 0.172
 Diversity = 1.3231e-07 k_t = [4, 2, 2, 4]
 Generation 9490: Best = 3.494586e+03 FES = 949100 μ_F = 0.770 μ_{CR} = 0.181
 Diversity = 1.2988e-07 k_t = [1, 1, 1, 1]
 Generation 9500: Best = 3.494586e+03 FES = 950100 μ_F = 0.749 μ_{CR} = 0.174
 Diversity = 1.2035e-07 k_t = [3, 3, 4, 4]
 Generation 9510: Best = 3.494586e+03 FES = 951100 μ_F = 0.762 μ_{CR} = 0.185
 Diversity = 1.2303e-07 k_t = [1, 1, 1, 1]
 Generation 9520: Best = 3.494586e+03 FES = 952100 μ_F = 0.764 μ_{CR} = 0.185
 Diversity = 1.2035e-07 k_t = [3, 2, 4, 4]
 Generation 9530: Best = 3.494586e+03 FES = 953100 μ_F = 0.774 μ_{CR} = 0.192
 Diversity = 1.1948e-07 k_t = [1, 1, 1, 1]
 Generation 9540: Best = 3.494586e+03 FES = 954100 μ_F = 0.759 μ_{CR} = 0.189
 Diversity = 1.1257e-07 k_t = [4, 2, 2, 4]
 Generation 9550: Best = 3.494586e+03 FES = 955100 μ_F = 0.763 μ_{CR} = 0.181
 Diversity = 1.1577e-07 k_t = [1, 1, 1, 1]
 Generation 9560: Best = 3.494586e+03 FES = 956100 μ_F = 0.743 μ_{CR} = 0.182
 Diversity = 1.0882e-07 k_t = [3, 4, 3, 3]
 Generation 9570: Best = 3.494586e+03 FES = 957100 μ_F = 0.739 μ_{CR} = 0.181
 Diversity = 1.0747e-07 k_t = [1, 1, 1, 1]
 Generation 9580: Best = 3.494586e+03 FES = 958100 μ_F = 0.746 μ_{CR} = 0.180
 Diversity = 1.0294e-07 k_t = [3, 4, 2, 4]
 Generation 9590: Best = 3.494586e+03 FES = 959100 μ_F = 0.760 μ_{CR} = 0.182
 Diversity = 1.0449e-07 k_t = [1, 1, 1, 1]
 Generation 9600: Best = 3.494586e+03 FES = 960100 μ_F = 0.759 μ_{CR} = 0.187
 Diversity = 9.6614e-08 k_t = [3, 2, 4, 4]
 Generation 9610: Best = 3.494586e+03 FES = 961100 μ_F = 0.755 μ_{CR} = 0.184
 Diversity = 1.0319e-07 k_t = [1, 1, 1, 1]
 Generation 9620: Best = 3.494586e+03 FES = 962100 μ_F = 0.770 μ_{CR} = 0.179
 Diversity = 1.0855e-07 k_t = [3, 4, 2, 2]
 Generation 9630: Best = 3.494586e+03 FES = 963100 μ_F = 0.772 μ_{CR} = 0.189
 Diversity = 1.2117e-07 k_t = [1, 1, 1, 1]
 Generation 9640: Best = 3.494586e+03 FES = 964100 μ_F = 0.770 μ_{CR} = 0.190
 Diversity = 1.4216e-07 k_t = [2, 3, 3, 2]
 Generation 9650: Best = 3.494586e+03 FES = 965100 μ_F = 0.770 μ_{CR} = 0.183
 Diversity = 1.5207e-07 k_t = [1, 1, 1, 1]
 Generation 9660: Best = 3.494586e+03 FES = 966100 μ_F = 0.766 μ_{CR} = 0.190
 Diversity = 1.4179e-07 k_t = [4, 2, 3, 2]
 Generation 9670: Best = 3.494586e+03 FES = 967100 μ_F = 0.758 μ_{CR} = 0.185
 Diversity = 1.3816e-07 k_t = [1, 1, 1, 1]

Generation 9680: Best = 3.494586e+03 FES = 968100 μ_F = 0.753 μ_{CR} = 0.172
 Diversity = 1.3029e-07 k_t = [2, 2, 4, 3]
 Generation 9690: Best = 3.494586e+03 FES = 969100 μ_F = 0.742 μ_{CR} = 0.163
 Diversity = 1.2722e-07 k_t = [1, 1, 1, 1]
 Generation 9700: Best = 3.494586e+03 FES = 970100 μ_F = 0.740 μ_{CR} = 0.168
 Diversity = 1.1865e-07 k_t = [3, 3, 3, 4]
 Generation 9710: Best = 3.494586e+03 FES = 971100 μ_F = 0.747 μ_{CR} = 0.176
 Diversity = 1.2137e-07 k_t = [1, 1, 1, 1]
 Generation 9720: Best = 3.494586e+03 FES = 972100 μ_F = 0.760 μ_{CR} = 0.170
 Diversity = 1.1701e-07 k_t = [4, 2, 3, 4]
 Generation 9730: Best = 3.494586e+03 FES = 973100 μ_F = 0.763 μ_{CR} = 0.170
 Diversity = 1.1726e-07 k_t = [1, 1, 1, 1]
 Generation 9740: Best = 3.494586e+03 FES = 974100 μ_F = 0.763 μ_{CR} = 0.171
 Diversity = 1.1664e-07 k_t = [4, 4, 4, 3]
 Generation 9750: Best = 3.494586e+03 FES = 975100 μ_F = 0.772 μ_{CR} = 0.170
 Diversity = 1.1488e-07 k_t = [1, 1, 1, 1]
 Generation 9760: Best = 3.494586e+03 FES = 976100 μ_F = 0.770 μ_{CR} = 0.166
 Diversity = 1.1138e-07 k_t = [2, 2, 4, 2]
 Generation 9770: Best = 3.494586e+03 FES = 977100 μ_F = 0.747 μ_{CR} = 0.162
 Diversity = 1.0891e-07 k_t = [1, 1, 1, 1]
 Generation 9780: Best = 3.494586e+03 FES = 978100 μ_F = 0.751 μ_{CR} = 0.163
 Diversity = 1.0481e-07 k_t = [3, 3, 3, 3]
 Generation 9790: Best = 3.494586e+03 FES = 979100 μ_F = 0.748 μ_{CR} = 0.163
 Diversity = 1.0381e-07 k_t = [1, 1, 1, 1]
 Generation 9800: Best = 3.494586e+03 FES = 980100 μ_F = 0.747 μ_{CR} = 0.161
 Diversity = 1.0107e-07 k_t = [4, 4, 4, 2]
 Generation 9810: Best = 3.494586e+03 FES = 981100 μ_F = 0.741 μ_{CR} = 0.170
 Diversity = 1.0136e-07 k_t = [1, 1, 1, 1]
 Generation 9820: Best = 3.494586e+03 FES = 982100 μ_F = 0.757 μ_{CR} = 0.175
 Diversity = 9.6833e-08 k_t = [3, 3, 4, 3]
 Generation 9830: Best = 3.494586e+03 FES = 983100 μ_F = 0.762 μ_{CR} = 0.173
 Diversity = 9.8198e-08 k_t = [1, 1, 1, 1]
 Generation 9840: Best = 3.494586e+03 FES = 984100 μ_F = 0.773 μ_{CR} = 0.172
 Diversity = 9.9828e-08 k_t = [2, 4, 4, 2]
 Generation 9850: Best = 3.494586e+03 FES = 985100 μ_F = 0.765 μ_{CR} = 0.164
 Diversity = 1.0153e-07 k_t = [1, 1, 1, 1]
 Generation 9860: Best = 3.494586e+03 FES = 986100 μ_F = 0.745 μ_{CR} = 0.156
 Diversity = 9.2151e-08 k_t = [4, 3, 3, 4]
 Generation 9870: Best = 3.494586e+03 FES = 987100 μ_F = 0.745 μ_{CR} = 0.147
 Diversity = 9.4540e-08 k_t = [1, 1, 1, 1]
 Generation 9880: Best = 3.494586e+03 FES = 988100 μ_F = 0.745 μ_{CR} = 0.143
 Diversity = 9.2907e-08 k_t = [3, 2, 3, 3]
 Generation 9890: Best = 3.494586e+03 FES = 989100 μ_F = 0.744 μ_{CR} = 0.139
 Diversity = 9.4760e-08 k_t = [1, 1, 1, 1]
 Generation 9900: Best = 3.494586e+03 FES = 990100 μ_F = 0.751 μ_{CR} = 0.136
 Diversity = 9.3325e-08 k_t = [3, 2, 2, 3]
 Generation 9910: Best = 3.494586e+03 FES = 991100 μ_F = 0.772 μ_{CR} = 0.145
 Diversity = 9.6341e-08 k_t = [1, 1, 1, 1]
 Generation 9920: Best = 3.494586e+03 FES = 992100 μ_F = 0.784 μ_{CR} = 0.150
 Diversity = 9.7191e-08 k_t = [2, 2, 4, 3]
 Generation 9930: Best = 3.494586e+03 FES = 993100 μ_F = 0.759 μ_{CR} = 0.157
 Diversity = 9.5217e-08 k_t = [1, 1, 1, 1]
 Generation 9940: Best = 3.494586e+03 FES = 994100 μ_F = 0.763 μ_{CR} = 0.148
 Diversity = 9.6015e-08 k_t = [3, 3, 2, 3]

Generation 9950: Best = 3.494586e+03 FES = 995100 μ_F = 0.752 μ_{CR} = 0.152
 Diversity = 9.5206e-08 k_t = [1, 1, 1, 1]
 Generation 9960: Best = 3.494586e+03 FES = 996100 μ_F = 0.707 μ_{CR} = 0.134
 Diversity = 9.1545e-08 k_t = [4, 3, 3, 3]
 Generation 9970: Best = 3.494586e+03 FES = 997100 μ_F = 0.686 μ_{CR} = 0.146
 Diversity = 9.1289e-08 k_t = [1, 1, 1, 1]
 Generation 9980: Best = 3.494586e+03 FES = 998100 μ_F = 0.667 μ_{CR} = 0.156
 Diversity = 8.5218e-08 k_t = [4, 2, 3, 4]
 Generation 9990: Best = 3.494586e+03 FES = 999100 μ_F = 0.659 μ_{CR} = 0.136
 Diversity = 8.6088e-08 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 3.494586e+03

Total Function Evals = 1000000

Total Successful Updates = 317307

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Results:

Best Fitness : 3.494586e+03

Global Optimum : 2.400000e+03

Error (|best-optimum|) : 1.094586e+03

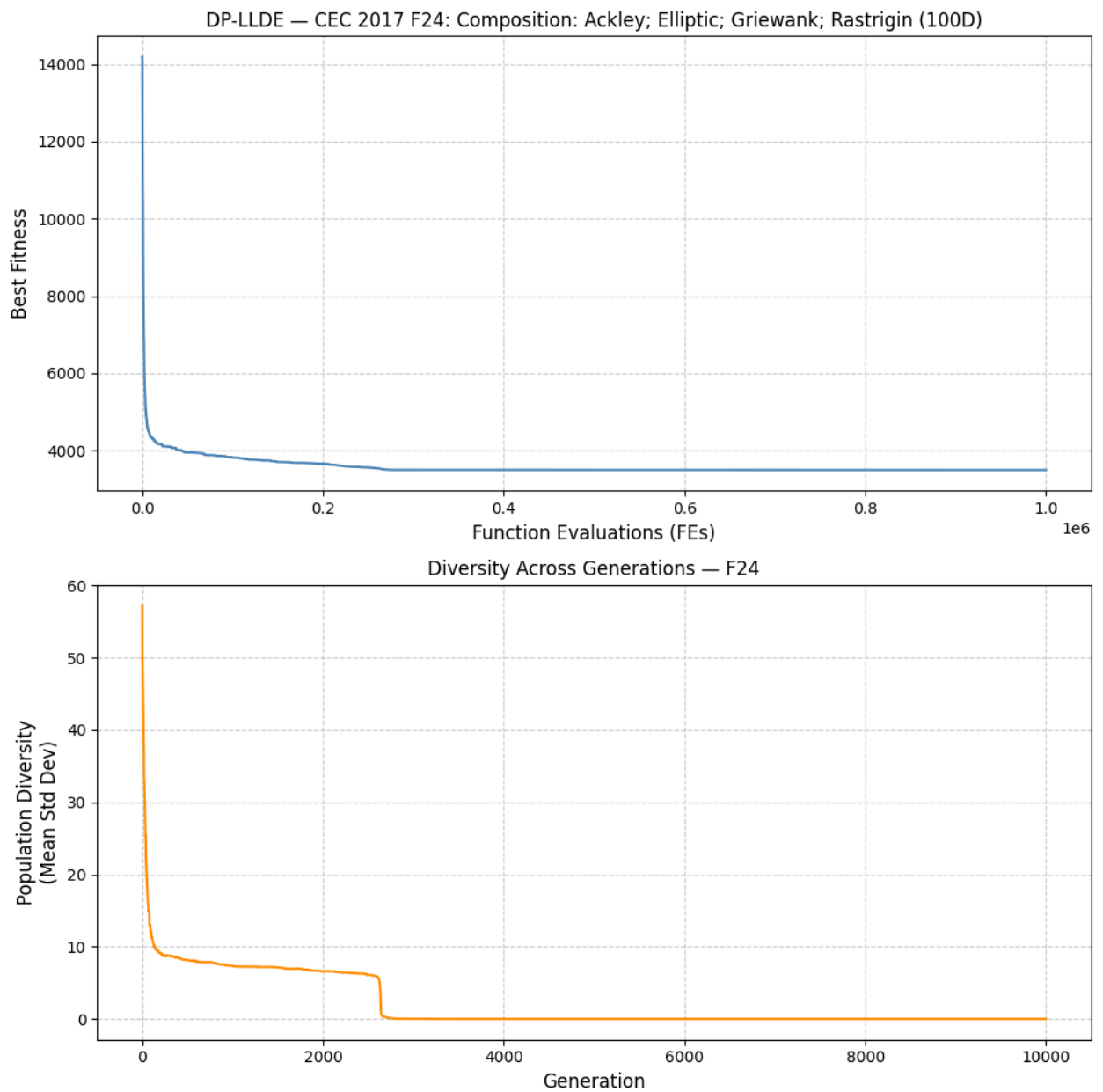
Best Solution (5 dim) : [78.16039782 55.3903569 -58.19901251 -77.98148422
70.98137932]

D^(0) (init. diversity): 5.7220e+01

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✓ Checkpoint saved to: /content/drive/My Drive/thesis_results/all_results_checkpoint.pkl



Graph saved as convergence_F24.png

✓ Plot saved to: /content/drive/My Drive/thesis_results/plots/convergence_F24.png

```
In [ ]: # =====
# F25 — Composition Function 5 (Rastrigin; HappyCat; Ackley; Discus; Rosenbrock)
# =====
run_and_plot('F25', make_wrapper(composition.f25), 2500.0, 'Composition: Rastrigin')
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DP-LLDE on CEC 2017 F25 – Single Run

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Function : Composition: Rastrigin; HappyCat; Ackley; Discus; Rosenbrock (100D)

Bounds : [-100, 100]

Global optimum : 2500.0

Initial $D^*(0) = 5.7220e+01$ | Diversity threshold = 5.7220e+00

Generation 0: Best Fitness = 1.233610e+05

Generation 10:	Best = 2.288207e+04	FES = 1100	$\mu_F = 0.543$	$\mu_{CR} = 0.352$
Diversity = 2.7939e+01	$k_t = [1, 1, 1, 1]$			
Generation 20:	Best = 1.280790e+04	FES = 2100	$\mu_F = 0.570$	$\mu_{CR} = 0.352$
Diversity = 1.4491e+01	$k_t = [2, 3, 2, 3]$			
Generation 30:	Best = 9.979159e+03	FES = 3100	$\mu_F = 0.594$	$\mu_{CR} = 0.356$
Diversity = 8.6587e+00	$k_t = [1, 1, 1, 1]$			
Generation 40:	Best = 8.063548e+03	FES = 4100	$\mu_F = 0.630$	$\mu_{CR} = 0.364$
Diversity = 6.0029e+00	$k_t = [3, 2, 3, 3]$			
Generation 50:	Best = 7.101019e+03	FES = 5100	$\mu_F = 0.657$	$\mu_{CR} = 0.366$
Diversity = 4.8073e+00	$k_t = [1, 1, 1, 1]$			
Generation 60:	Best = 6.461711e+03	FES = 6100	$\mu_F = 0.676$	$\mu_{CR} = 0.365$
Diversity = 3.5764e+00	$k_t = [3, 4, 4, 3]$			
Generation 70:	Best = 6.078052e+03	FES = 7100	$\mu_F = 0.708$	$\mu_{CR} = 0.374$
Diversity = 3.1864e+00	$k_t = [1, 1, 1, 1]$			
Generation 80:	Best = 5.709295e+03	FES = 8100	$\mu_F = 0.718$	$\mu_{CR} = 0.379$
Diversity = 2.8517e+00	$k_t = [2, 4, 4, 3]$			
Generation 90:	Best = 5.405729e+03	FES = 9100	$\mu_F = 0.731$	$\mu_{CR} = 0.378$
Diversity = 2.6952e+00	$k_t = [1, 1, 1, 1]$			
Generation 100:	Best = 5.138845e+03	FES = 10100	$\mu_F = 0.737$	$\mu_{CR} = 0.381$
Diversity = 2.4736e+00	$k_t = [4, 4, 2, 4]$			
Generation 110:	Best = 4.978455e+03	FES = 11100	$\mu_F = 0.746$	$\mu_{CR} = 0.389$
Diversity = 2.2214e+00	$k_t = [1, 1, 1, 1]$			
Generation 120:	Best = 4.852518e+03	FES = 12100	$\mu_F = 0.749$	$\mu_{CR} = 0.398$
Diversity = 1.9371e+00	$k_t = [3, 3, 2, 4]$			
Generation 130:	Best = 4.741293e+03	FES = 13100	$\mu_F = 0.760$	$\mu_{CR} = 0.402$
Diversity = 1.9131e+00	$k_t = [1, 1, 1, 1]$			
Generation 140:	Best = 4.608997e+03	FES = 14100	$\mu_F = 0.763$	$\mu_{CR} = 0.403$
Diversity = 1.9145e+00	$k_t = [2, 4, 2, 3]$			
Generation 150:	Best = 4.482643e+03	FES = 15100	$\mu_F = 0.763$	$\mu_{CR} = 0.406$
Diversity = 1.8858e+00	$k_t = [1, 1, 1, 1]$			
Generation 160:	Best = 4.398569e+03	FES = 16100	$\mu_F = 0.751$	$\mu_{CR} = 0.411$
Diversity = 1.7387e+00	$k_t = [4, 3, 2, 2]$			
Generation 170:	Best = 4.295290e+03	FES = 17100	$\mu_F = 0.749$	$\mu_{CR} = 0.418$
Diversity = 1.6565e+00	$k_t = [1, 1, 1, 1]$			
Generation 180:	Best = 4.207736e+03	FES = 18100	$\mu_F = 0.746$	$\mu_{CR} = 0.422$
Diversity = 1.4301e+00	$k_t = [3, 4, 2, 3]$			
Generation 190:	Best = 4.142017e+03	FES = 19100	$\mu_F = 0.750$	$\mu_{CR} = 0.432$
Diversity = 1.4076e+00	$k_t = [1, 1, 1, 1]$			
Generation 200:	Best = 4.075663e+03	FES = 20100	$\mu_F = 0.761$	$\mu_{CR} = 0.434$
Diversity = 1.3842e+00	$k_t = [2, 3, 3, 3]$			
Generation 210:	Best = 4.027562e+03	FES = 21100	$\mu_F = 0.759$	$\mu_{CR} = 0.445$
Diversity = 1.2997e+00	$k_t = [1, 1, 1, 1]$			
Generation 220:	Best = 3.986695e+03	FES = 22100	$\mu_F = 0.759$	$\mu_{CR} = 0.449$

Diversity = 1.0740e+00	k_t = [4, 2, 2, 3]				
Generation 230: Best = 3.942443e+03	FES = 23100	$\mu_F = 0.760$	$\mu_{CR} = 0.449$		
Diversity = 1.1026e+00	k_t = [1, 1, 1, 1]				
Generation 240: Best = 3.916367e+03	FES = 24100	$\mu_F = 0.760$	$\mu_{CR} = 0.456$		
Diversity = 9.3103e-01	k_t = [4, 2, 4, 2]				
Generation 250: Best = 3.884604e+03	FES = 25100	$\mu_F = 0.755$	$\mu_{CR} = 0.449$		
Diversity = 8.9418e-01	k_t = [1, 1, 1, 1]				
Generation 260: Best = 3.858081e+03	FES = 26100	$\mu_F = 0.757$	$\mu_{CR} = 0.456$		
Diversity = 8.6945e-01	k_t = [2, 4, 4, 3]				
Generation 270: Best = 3.829417e+03	FES = 27100	$\mu_F = 0.762$	$\mu_{CR} = 0.460$		
Diversity = 8.9969e-01	k_t = [1, 1, 1, 1]				
Generation 280: Best = 3.800662e+03	FES = 28100	$\mu_F = 0.761$	$\mu_{CR} = 0.460$		
Diversity = 8.2581e-01	k_t = [4, 2, 2, 4]				
Generation 290: Best = 3.778937e+03	FES = 29100	$\mu_F = 0.765$	$\mu_{CR} = 0.467$		
Diversity = 8.2069e-01	k_t = [1, 1, 1, 1]				
Generation 300: Best = 3.757942e+03	FES = 30100	$\mu_F = 0.758$	$\mu_{CR} = 0.463$		
Diversity = 7.4086e-01	k_t = [3, 3, 2, 4]				
Generation 310: Best = 3.737387e+03	FES = 31100	$\mu_F = 0.779$	$\mu_{CR} = 0.457$		
Diversity = 7.5499e-01	k_t = [1, 1, 1, 1]				
Generation 320: Best = 3.718994e+03	FES = 32100	$\mu_F = 0.774$	$\mu_{CR} = 0.455$		
Diversity = 6.5284e-01	k_t = [4, 2, 4, 3]				
Generation 330: Best = 3.705002e+03	FES = 33100	$\mu_F = 0.773$	$\mu_{CR} = 0.455$		
Diversity = 6.5904e-01	k_t = [1, 1, 1, 1]				
Generation 340: Best = 3.690523e+03	FES = 34100	$\mu_F = 0.770$	$\mu_{CR} = 0.450$		
Diversity = 6.4616e-01	k_t = [2, 3, 3, 3]				
Generation 350: Best = 3.676105e+03	FES = 35100	$\mu_F = 0.766$	$\mu_{CR} = 0.445$		
Diversity = 6.4077e-01	k_t = [1, 1, 1, 1]				
Generation 360: Best = 3.666354e+03	FES = 36100	$\mu_F = 0.766$	$\mu_{CR} = 0.451$		
Diversity = 6.2229e-01	k_t = [2, 4, 3, 3]				
Generation 370: Best = 3.656033e+03	FES = 37100	$\mu_F = 0.761$	$\mu_{CR} = 0.452$		
Diversity = 5.7197e-01	k_t = [1, 1, 1, 1]				
Generation 380: Best = 3.647364e+03	FES = 38100	$\mu_F = 0.755$	$\mu_{CR} = 0.454$		
Diversity = 5.2440e-01	k_t = [2, 3, 4, 2]				
Generation 390: Best = 3.637794e+03	FES = 39100	$\mu_F = 0.756$	$\mu_{CR} = 0.457$		
Diversity = 5.2719e-01	k_t = [1, 1, 1, 1]				
Generation 400: Best = 3.628292e+03	FES = 40100	$\mu_F = 0.765$	$\mu_{CR} = 0.457$		
Diversity = 5.2752e-01	k_t = [3, 4, 2, 3]				
Generation 410: Best = 3.621301e+03	FES = 41100	$\mu_F = 0.779$	$\mu_{CR} = 0.457$		
Diversity = 5.5477e-01	k_t = [1, 1, 1, 1]				
Generation 420: Best = 3.611786e+03	FES = 42100	$\mu_F = 0.770$	$\mu_{CR} = 0.453$		
Diversity = 5.4298e-01	k_t = [3, 2, 4, 3]				
Generation 430: Best = 3.604334e+03	FES = 43100	$\mu_F = 0.761$	$\mu_{CR} = 0.450$		
Diversity = 5.1964e-01	k_t = [1, 1, 1, 1]				
Generation 440: Best = 3.596131e+03	FES = 44100	$\mu_F = 0.768$	$\mu_{CR} = 0.453$		
Diversity = 5.1672e-01	k_t = [4, 2, 4, 2]				
Generation 450: Best = 3.588276e+03	FES = 45100	$\mu_F = 0.762$	$\mu_{CR} = 0.446$		
Diversity = 4.9614e-01	k_t = [1, 1, 1, 1]				
Generation 460: Best = 3.582346e+03	FES = 46100	$\mu_F = 0.766$	$\mu_{CR} = 0.440$		
Diversity = 5.0410e-01	k_t = [2, 2, 3, 2]				
Generation 470: Best = 3.574822e+03	FES = 47100	$\mu_F = 0.763$	$\mu_{CR} = 0.447$		
Diversity = 5.1864e-01	k_t = [1, 1, 1, 1]				
Generation 480: Best = 3.569032e+03	FES = 48100	$\mu_F = 0.757$	$\mu_{CR} = 0.453$		
Diversity = 4.8121e-01	k_t = [2, 2, 3, 4]				
Generation 490: Best = 3.563796e+03	FES = 49100	$\mu_F = 0.757$	$\mu_{CR} = 0.455$		

Diversity = 4.8223e-01	k_t = [1, 1, 1, 1]				
Generation 500: Best = 3.558803e+03	FES = 50100	$\mu_F = 0.746$	$\mu_{CR} = 0.450$		
Diversity = 4.3991e-01	k_t = [2, 3, 3, 2]				
Generation 510: Best = 3.553623e+03	FES = 51100	$\mu_F = 0.746$	$\mu_{CR} = 0.457$		
Diversity = 4.4920e-01	k_t = [1, 1, 1, 1]				
Generation 520: Best = 3.546793e+03	FES = 52100	$\mu_F = 0.735$	$\mu_{CR} = 0.459$		
Diversity = 4.1997e-01	k_t = [3, 2, 2, 2]				
Generation 530: Best = 3.543352e+03	FES = 53100	$\mu_F = 0.738$	$\mu_{CR} = 0.453$		
Diversity = 4.0061e-01	k_t = [1, 1, 1, 1]				
Generation 540: Best = 3.538275e+03	FES = 54100	$\mu_F = 0.750$	$\mu_{CR} = 0.445$		
Diversity = 3.9604e-01	k_t = [3, 3, 3, 2]				
Generation 550: Best = 3.531602e+03	FES = 55100	$\mu_F = 0.750$	$\mu_{CR} = 0.454$		
Diversity = 4.5797e-01	k_t = [1, 1, 1, 1]				
Generation 560: Best = 3.523587e+03	FES = 56100	$\mu_F = 0.762$	$\mu_{CR} = 0.452$		
Diversity = 4.6280e-01	k_t = [2, 3, 2, 2]				
Generation 570: Best = 3.516047e+03	FES = 57100	$\mu_F = 0.749$	$\mu_{CR} = 0.457$		
Diversity = 4.6418e-01	k_t = [1, 1, 1, 1]				
Generation 580: Best = 3.508486e+03	FES = 58100	$\mu_F = 0.750$	$\mu_{CR} = 0.449$		
Diversity = 4.5249e-01	k_t = [2, 3, 3, 3]				
Generation 590: Best = 3.502339e+03	FES = 59100	$\mu_F = 0.753$	$\mu_{CR} = 0.447$		
Diversity = 5.0359e-01	k_t = [1, 1, 1, 1]				
Generation 600: Best = 3.495191e+03	FES = 60100	$\mu_F = 0.754$	$\mu_{CR} = 0.450$		
Diversity = 4.9474e-01	k_t = [3, 4, 3, 4]				
Generation 610: Best = 3.487441e+03	FES = 61100	$\mu_F = 0.744$	$\mu_{CR} = 0.446$		
Diversity = 4.6966e-01	k_t = [1, 1, 1, 1]				
Generation 620: Best = 3.480789e+03	FES = 62100	$\mu_F = 0.748$	$\mu_{CR} = 0.459$		
Diversity = 4.4275e-01	k_t = [2, 2, 4, 3]				
Generation 630: Best = 3.471689e+03	FES = 63100	$\mu_F = 0.751$	$\mu_{CR} = 0.464$		
Diversity = 5.0214e-01	k_t = [1, 1, 1, 1]				
Generation 640: Best = 3.465398e+03	FES = 64100	$\mu_F = 0.751$	$\mu_{CR} = 0.465$		
Diversity = 4.4648e-01	k_t = [2, 4, 4, 4]				
Generation 650: Best = 3.459262e+03	FES = 65100	$\mu_F = 0.751$	$\mu_{CR} = 0.463$		
Diversity = 4.4393e-01	k_t = [1, 1, 1, 1]				
Generation 660: Best = 3.453160e+03	FES = 66100	$\mu_F = 0.762$	$\mu_{CR} = 0.465$		
Diversity = 4.2665e-01	k_t = [3, 2, 2, 4]				
Generation 670: Best = 3.447450e+03	FES = 67100	$\mu_F = 0.762$	$\mu_{CR} = 0.466$		
Diversity = 4.4566e-01	k_t = [1, 1, 1, 1]				
Generation 680: Best = 3.440954e+03	FES = 68100	$\mu_F = 0.757$	$\mu_{CR} = 0.462$		
Diversity = 3.9047e-01	k_t = [3, 3, 4, 3]				
Generation 690: Best = 3.437512e+03	FES = 69100	$\mu_F = 0.758$	$\mu_{CR} = 0.465$		
Diversity = 4.1099e-01	k_t = [1, 1, 1, 1]				
Generation 700: Best = 3.432725e+03	FES = 70100	$\mu_F = 0.748$	$\mu_{CR} = 0.463$		
Diversity = 3.8021e-01	k_t = [4, 2, 3, 3]				
Generation 710: Best = 3.429289e+03	FES = 71100	$\mu_F = 0.748$	$\mu_{CR} = 0.459$		
Diversity = 3.7263e-01	k_t = [1, 1, 1, 1]				
Generation 720: Best = 3.426503e+03	FES = 72100	$\mu_F = 0.734$	$\mu_{CR} = 0.462$		
Diversity = 2.9910e-01	k_t = [4, 4, 3, 4]				
Generation 730: Best = 3.425023e+03	FES = 73100	$\mu_F = 0.715$	$\mu_{CR} = 0.464$		
Diversity = 2.5361e-01	k_t = [1, 1, 1, 1]				
Generation 740: Best = 3.423474e+03	FES = 74100	$\mu_F = 0.698$	$\mu_{CR} = 0.465$		
Diversity = 2.2922e-01	k_t = [3, 4, 4, 4]				
Generation 750: Best = 3.422264e+03	FES = 75100	$\mu_F = 0.698$	$\mu_{CR} = 0.453$		
Diversity = 2.0227e-01	k_t = [1, 1, 1, 1]				
Generation 760: Best = 3.420681e+03	FES = 76100	$\mu_F = 0.698$	$\mu_{CR} = 0.459$		

Diversity = 1.9454e-01	k_t = [4, 4, 2, 2]				
Generation 770: Best = 3.419813e+03	FES = 77100	$\mu_F = 0.700$	$\mu_{CR} = 0.462$		
Diversity = 1.5872e-01	k_t = [1, 1, 1, 1]				
Generation 780: Best = 3.418816e+03	FES = 78100	$\mu_F = 0.711$	$\mu_{CR} = 0.451$		
Diversity = 1.4590e-01	k_t = [4, 2, 4, 2]				
Generation 790: Best = 3.418076e+03	FES = 79100	$\mu_F = 0.728$	$\mu_{CR} = 0.436$		
Diversity = 1.4910e-01	k_t = [1, 1, 1, 1]				
Generation 800: Best = 3.417268e+03	FES = 80100	$\mu_F = 0.739$	$\mu_{CR} = 0.432$		
Diversity = 1.3974e-01	k_t = [3, 4, 4, 4]				
Generation 810: Best = 3.416072e+03	FES = 81100	$\mu_F = 0.742$	$\mu_{CR} = 0.430$		
Diversity = 1.5402e-01	k_t = [1, 1, 1, 1]				
Generation 820: Best = 3.414587e+03	FES = 82100	$\mu_F = 0.741$	$\mu_{CR} = 0.429$		
Diversity = 1.4922e-01	k_t = [4, 4, 4, 4]				
Generation 830: Best = 3.413582e+03	FES = 83100	$\mu_F = 0.746$	$\mu_{CR} = 0.429$		
Diversity = 1.6588e-01	k_t = [1, 1, 1, 1]				
Generation 840: Best = 3.412102e+03	FES = 84100	$\mu_F = 0.739$	$\mu_{CR} = 0.424$		
Diversity = 1.7446e-01	k_t = [4, 3, 4, 2]				
Generation 850: Best = 3.411328e+03	FES = 85100	$\mu_F = 0.744$	$\mu_{CR} = 0.430$		
Diversity = 1.6693e-01	k_t = [1, 1, 1, 1]				
Generation 860: Best = 3.410321e+03	FES = 86100	$\mu_F = 0.743$	$\mu_{CR} = 0.433$		
Diversity = 1.6281e-01	k_t = [2, 2, 2, 4]				
Generation 870: Best = 3.409600e+03	FES = 87100	$\mu_F = 0.751$	$\mu_{CR} = 0.419$		
Diversity = 1.7024e-01	k_t = [1, 1, 1, 1]				
Generation 880: Best = 3.408951e+03	FES = 88100	$\mu_F = 0.740$	$\mu_{CR} = 0.401$		
Diversity = 1.6018e-01	k_t = [2, 3, 3, 2]				
Generation 890: Best = 3.408407e+03	FES = 89100	$\mu_F = 0.721$	$\mu_{CR} = 0.400$		
Diversity = 1.4937e-01	k_t = [1, 1, 1, 1]				
Generation 900: Best = 3.407569e+03	FES = 90100	$\mu_F = 0.718$	$\mu_{CR} = 0.392$		
Diversity = 1.4184e-01	k_t = [4, 3, 3, 2]				
Generation 910: Best = 3.406593e+03	FES = 91100	$\mu_F = 0.716$	$\mu_{CR} = 0.388$		
Diversity = 1.4167e-01	k_t = [1, 1, 1, 1]				
Generation 920: Best = 3.405537e+03	FES = 92100	$\mu_F = 0.721$	$\mu_{CR} = 0.391$		
Diversity = 1.4423e-01	k_t = [3, 2, 2, 3]				
Generation 930: Best = 3.404338e+03	FES = 93100	$\mu_F = 0.728$	$\mu_{CR} = 0.389$		
Diversity = 1.6508e-01	k_t = [1, 1, 1, 1]				
Generation 940: Best = 3.403644e+03	FES = 94100	$\mu_F = 0.734$	$\mu_{CR} = 0.385$		
Diversity = 1.5681e-01	k_t = [2, 4, 3, 3]				
Generation 950: Best = 3.402993e+03	FES = 95100	$\mu_F = 0.734$	$\mu_{CR} = 0.377$		
Diversity = 1.5755e-01	k_t = [1, 1, 1, 1]				
Generation 960: Best = 3.402361e+03	FES = 96100	$\mu_F = 0.729$	$\mu_{CR} = 0.362$		
Diversity = 1.4538e-01	k_t = [3, 3, 3, 3]				
Generation 970: Best = 3.401625e+03	FES = 97100	$\mu_F = 0.721$	$\mu_{CR} = 0.345$		
Diversity = 1.4535e-01	k_t = [1, 1, 1, 1]				
Generation 980: Best = 3.400887e+03	FES = 98100	$\mu_F = 0.725$	$\mu_{CR} = 0.346$		
Diversity = 1.3349e-01	k_t = [4, 4, 4, 2]				
Generation 990: Best = 3.400408e+03	FES = 99100	$\mu_F = 0.723$	$\mu_{CR} = 0.336$		
Diversity = 1.3064e-01	k_t = [1, 1, 1, 1]				
Generation 1000: Best = 3.399637e+03	FES = 100100	$\mu_F = 0.724$	$\mu_{CR} = 0.341$		
Diversity = 1.1195e-01	k_t = [2, 3, 3, 4]				
Generation 1010: Best = 3.399176e+03	FES = 101100	$\mu_F = 0.717$	$\mu_{CR} = 0.340$		
Diversity = 1.1277e-01	k_t = [1, 1, 1, 1]				
Generation 1020: Best = 3.398854e+03	FES = 102100	$\mu_F = 0.715$	$\mu_{CR} = 0.336$		
Diversity = 1.0576e-01	k_t = [3, 4, 2, 3]				
Generation 1030: Best = 3.398571e+03	FES = 103100	$\mu_F = 0.717$	$\mu_{CR} = 0.339$		

Diversity = 1.1025e-01	k_t = [1, 1, 1, 1]			
Generation 1040: Best = 3.397966e+03	FES = 104100	$\mu_F = 0.719$	$\mu_{CR} = 0.333$	
Diversity = 1.1283e-01	k_t = [4, 4, 2, 3]			
Generation 1050: Best = 3.397588e+03	FES = 105100	$\mu_F = 0.700$	$\mu_{CR} = 0.334$	
Diversity = 1.0728e-01	k_t = [1, 1, 1, 1]			
Generation 1060: Best = 3.397146e+03	FES = 106100	$\mu_F = 0.702$	$\mu_{CR} = 0.331$	
Diversity = 1.0195e-01	k_t = [4, 3, 2, 4]			
Generation 1070: Best = 3.396727e+03	FES = 107100	$\mu_F = 0.710$	$\mu_{CR} = 0.336$	
Diversity = 8.7219e-02	k_t = [1, 1, 1, 1]			
Generation 1080: Best = 3.396508e+03	FES = 108100	$\mu_F = 0.718$	$\mu_{CR} = 0.338$	
Diversity = 8.9364e-02	k_t = [3, 3, 2, 2]			
Generation 1090: Best = 3.396240e+03	FES = 109100	$\mu_F = 0.717$	$\mu_{CR} = 0.342$	
Diversity = 8.8598e-02	k_t = [1, 1, 1, 1]			
Generation 1100: Best = 3.395881e+03	FES = 110100	$\mu_F = 0.717$	$\mu_{CR} = 0.343$	
Diversity = 8.3920e-02	k_t = [3, 2, 4, 4]			
Generation 1110: Best = 3.395734e+03	FES = 111100	$\mu_F = 0.713$	$\mu_{CR} = 0.343$	
Diversity = 7.6665e-02	k_t = [1, 1, 1, 1]			
Generation 1120: Best = 3.395434e+03	FES = 112100	$\mu_F = 0.701$	$\mu_{CR} = 0.342$	
Diversity = 6.8454e-02	k_t = [4, 3, 4, 3]			
Generation 1130: Best = 3.395272e+03	FES = 113100	$\mu_F = 0.708$	$\mu_{CR} = 0.341$	
Diversity = 6.6746e-02	k_t = [1, 1, 1, 1]			
Generation 1140: Best = 3.395039e+03	FES = 114100	$\mu_F = 0.707$	$\mu_{CR} = 0.341$	
Diversity = 6.1389e-02	k_t = [4, 4, 4, 3]			
Generation 1150: Best = 3.394913e+03	FES = 115100	$\mu_F = 0.716$	$\mu_{CR} = 0.340$	
Diversity = 5.6146e-02	k_t = [1, 1, 1, 1]			
Generation 1160: Best = 3.394770e+03	FES = 116100	$\mu_F = 0.715$	$\mu_{CR} = 0.342$	
Diversity = 5.2826e-02	k_t = [4, 3, 4, 4]			
Generation 1170: Best = 3.394589e+03	FES = 117100	$\mu_F = 0.723$	$\mu_{CR} = 0.347$	
Diversity = 5.7868e-02	k_t = [1, 1, 1, 1]			
Generation 1180: Best = 3.394476e+03	FES = 118100	$\mu_F = 0.716$	$\mu_{CR} = 0.333$	
Diversity = 5.7613e-02	k_t = [4, 4, 4, 4]			
Generation 1190: Best = 3.394286e+03	FES = 119100	$\mu_F = 0.709$	$\mu_{CR} = 0.334$	
Diversity = 5.9095e-02	k_t = [1, 1, 1, 1]			
Generation 1200: Best = 3.394182e+03	FES = 120100	$\mu_F = 0.699$	$\mu_{CR} = 0.331$	
Diversity = 5.3973e-02	k_t = [4, 3, 3, 3]			
Generation 1210: Best = 3.393952e+03	FES = 121100	$\mu_F = 0.720$	$\mu_{CR} = 0.327$	
Diversity = 5.8013e-02	k_t = [1, 1, 1, 1]			
Generation 1220: Best = 3.393750e+03	FES = 122100	$\mu_F = 0.739$	$\mu_{CR} = 0.320$	
Diversity = 6.4194e-02	k_t = [2, 4, 2, 4]			
Generation 1230: Best = 3.393578e+03	FES = 123100	$\mu_F = 0.727$	$\mu_{CR} = 0.311$	
Diversity = 6.3662e-02	k_t = [1, 1, 1, 1]			
Generation 1240: Best = 3.393423e+03	FES = 124100	$\mu_F = 0.737$	$\mu_{CR} = 0.302$	
Diversity = 5.9307e-02	k_t = [3, 4, 4, 4]			
Generation 1250: Best = 3.393275e+03	FES = 125100	$\mu_F = 0.745$	$\mu_{CR} = 0.297$	
Diversity = 6.2488e-02	k_t = [1, 1, 1, 1]			
Generation 1260: Best = 3.393029e+03	FES = 126100	$\mu_F = 0.739$	$\mu_{CR} = 0.304$	
Diversity = 6.6098e-02	k_t = [4, 4, 2, 2]			
Generation 1270: Best = 3.392697e+03	FES = 127100	$\mu_F = 0.744$	$\mu_{CR} = 0.307$	
Diversity = 7.1175e-02	k_t = [1, 1, 1, 1]			
Generation 1280: Best = 3.392368e+03	FES = 128100	$\mu_F = 0.741$	$\mu_{CR} = 0.306$	
Diversity = 6.8902e-02	k_t = [2, 4, 3, 3]			
Generation 1290: Best = 3.392149e+03	FES = 129100	$\mu_F = 0.753$	$\mu_{CR} = 0.303$	
Diversity = 7.2641e-02	k_t = [1, 1, 1, 1]			
Generation 1300: Best = 3.391861e+03	FES = 130100	$\mu_F = 0.767$	$\mu_{CR} = 0.306$	

Diversity = 7.5768e-02	k_t = [2, 4, 2, 2]			
Generation 1310: Best = 3.391541e+03	FES = 131100	$\mu_F = 0.753$	$\mu_{CR} = 0.309$	
Diversity = 7.9515e-02	k_t = [1, 1, 1, 1]			
Generation 1320: Best = 3.391397e+03	FES = 132100	$\mu_F = 0.750$	$\mu_{CR} = 0.308$	
Diversity = 8.0808e-02	k_t = [3, 3, 4, 2]			
Generation 1330: Best = 3.391117e+03	FES = 133100	$\mu_F = 0.751$	$\mu_{CR} = 0.302$	
Diversity = 7.6992e-02	k_t = [1, 1, 1, 1]			
Generation 1340: Best = 3.390636e+03	FES = 134100	$\mu_F = 0.763$	$\mu_{CR} = 0.306$	
Diversity = 7.5749e-02	k_t = [4, 2, 3, 4]			
Generation 1350: Best = 3.390107e+03	FES = 135100	$\mu_F = 0.774$	$\mu_{CR} = 0.293$	
Diversity = 8.6296e-02	k_t = [1, 1, 1, 1]			
Generation 1360: Best = 3.388980e+03	FES = 136100	$\mu_F = 0.771$	$\mu_{CR} = 0.297$	
Diversity = 1.0201e-01	k_t = [3, 3, 4, 3]			
Generation 1370: Best = 3.388506e+03	FES = 137100	$\mu_F = 0.769$	$\mu_{CR} = 0.301$	
Diversity = 1.2442e-01	k_t = [1, 1, 1, 1]			
Generation 1380: Best = 3.388239e+03	FES = 138100	$\mu_F = 0.767$	$\mu_{CR} = 0.297$	
Diversity = 1.1086e-01	k_t = [2, 3, 2, 4]			
Generation 1390: Best = 3.387814e+03	FES = 139100	$\mu_F = 0.750$	$\mu_{CR} = 0.288$	
Diversity = 1.1065e-01	k_t = [1, 1, 1, 1]			
Generation 1400: Best = 3.387102e+03	FES = 140100	$\mu_F = 0.739$	$\mu_{CR} = 0.292$	
Diversity = 1.1762e-01	k_t = [4, 4, 4, 2]			
Generation 1410: Best = 3.386727e+03	FES = 141100	$\mu_F = 0.747$	$\mu_{CR} = 0.299$	
Diversity = 1.2210e-01	k_t = [1, 1, 1, 1]			
Generation 1420: Best = 3.386146e+03	FES = 142100	$\mu_F = 0.730$	$\mu_{CR} = 0.307$	
Diversity = 1.3126e-01	k_t = [4, 2, 2, 2]			
Generation 1430: Best = 3.385722e+03	FES = 143100	$\mu_F = 0.728$	$\mu_{CR} = 0.301$	
Diversity = 1.2718e-01	k_t = [1, 1, 1, 1]			
Generation 1440: Best = 3.385319e+03	FES = 144100	$\mu_F = 0.729$	$\mu_{CR} = 0.295$	
Diversity = 1.0970e-01	k_t = [4, 4, 4, 3]			
Generation 1450: Best = 3.384825e+03	FES = 145100	$\mu_F = 0.734$	$\mu_{CR} = 0.292$	
Diversity = 1.0932e-01	k_t = [1, 1, 1, 1]			
Generation 1460: Best = 3.384339e+03	FES = 146100	$\mu_F = 0.730$	$\mu_{CR} = 0.295$	
Diversity = 1.0141e-01	k_t = [4, 3, 3, 3]			
Generation 1470: Best = 3.384070e+03	FES = 147100	$\mu_F = 0.727$	$\mu_{CR} = 0.289$	
Diversity = 1.0033e-01	k_t = [1, 1, 1, 1]			
Generation 1480: Best = 3.383646e+03	FES = 148100	$\mu_F = 0.746$	$\mu_{CR} = 0.288$	
Diversity = 1.0283e-01	k_t = [4, 3, 4, 2]			
Generation 1490: Best = 3.383001e+03	FES = 149100	$\mu_F = 0.750$	$\mu_{CR} = 0.296$	
Diversity = 1.1256e-01	k_t = [1, 1, 1, 1]			
Generation 1500: Best = 3.382682e+03	FES = 150100	$\mu_F = 0.757$	$\mu_{CR} = 0.299$	
Diversity = 1.1679e-01	k_t = [4, 2, 3, 3]			
Generation 1510: Best = 3.382358e+03	FES = 151100	$\mu_F = 0.757$	$\mu_{CR} = 0.298$	
Diversity = 1.2062e-01	k_t = [1, 1, 1, 1]			
Generation 1520: Best = 3.381901e+03	FES = 152100	$\mu_F = 0.742$	$\mu_{CR} = 0.291$	
Diversity = 1.1639e-01	k_t = [2, 2, 3, 2]			
Generation 1530: Best = 3.381593e+03	FES = 153100	$\mu_F = 0.729$	$\mu_{CR} = 0.293$	
Diversity = 1.1738e-01	k_t = [1, 1, 1, 1]			
Generation 1540: Best = 3.381163e+03	FES = 154100	$\mu_F = 0.725$	$\mu_{CR} = 0.282$	
Diversity = 1.0584e-01	k_t = [3, 2, 2, 4]			
Generation 1550: Best = 3.380645e+03	FES = 155100	$\mu_F = 0.715$	$\mu_{CR} = 0.275$	
Diversity = 1.0637e-01	k_t = [1, 1, 1, 1]			
Generation 1560: Best = 3.380288e+03	FES = 156100	$\mu_F = 0.722$	$\mu_{CR} = 0.270$	
Diversity = 1.1024e-01	k_t = [2, 4, 3, 2]			
Generation 1570: Best = 3.379966e+03	FES = 157100	$\mu_F = 0.727$	$\mu_{CR} = 0.270$	

Diversity = 1.1200e-01	k_t = [1, 1, 1, 1]			
Generation 1580: Best = 3.379531e+03	FES = 158100	$\mu_F = 0.729$	$\mu_{CR} = 0.263$	
Diversity = 1.0943e-01	k_t = [4, 2, 2, 3]			
Generation 1590: Best = 3.379168e+03	FES = 159100	$\mu_F = 0.712$	$\mu_{CR} = 0.249$	
Diversity = 1.0815e-01	k_t = [1, 1, 1, 1]			
Generation 1600: Best = 3.378741e+03	FES = 160100	$\mu_F = 0.716$	$\mu_{CR} = 0.256$	
Diversity = 1.1471e-01	k_t = [3, 4, 3, 2]			
Generation 1610: Best = 3.378400e+03	FES = 161100	$\mu_F = 0.719$	$\mu_{CR} = 0.257$	
Diversity = 1.1286e-01	k_t = [1, 1, 1, 1]			
Generation 1620: Best = 3.378094e+03	FES = 162100	$\mu_F = 0.730$	$\mu_{CR} = 0.240$	
Diversity = 1.1228e-01	k_t = [4, 3, 4, 3]			
Generation 1630: Best = 3.377757e+03	FES = 163100	$\mu_F = 0.734$	$\mu_{CR} = 0.240$	
Diversity = 1.0853e-01	k_t = [1, 1, 1, 1]			
Generation 1640: Best = 3.377424e+03	FES = 164100	$\mu_F = 0.736$	$\mu_{CR} = 0.234$	
Diversity = 1.0777e-01	k_t = [4, 2, 4, 3]			
Generation 1650: Best = 3.377184e+03	FES = 165100	$\mu_F = 0.736$	$\mu_{CR} = 0.236$	
Diversity = 1.1007e-01	k_t = [1, 1, 1, 1]			
Generation 1660: Best = 3.376916e+03	FES = 166100	$\mu_F = 0.730$	$\mu_{CR} = 0.218$	
Diversity = 1.0210e-01	k_t = [4, 2, 4, 4]			
Generation 1670: Best = 3.376657e+03	FES = 167100	$\mu_F = 0.723$	$\mu_{CR} = 0.226$	
Diversity = 9.9929e-02	k_t = [1, 1, 1, 1]			
Generation 1680: Best = 3.376505e+03	FES = 168100	$\mu_F = 0.720$	$\mu_{CR} = 0.219$	
Diversity = 9.1355e-02	k_t = [2, 2, 3, 3]			
Generation 1690: Best = 3.376329e+03	FES = 169100	$\mu_F = 0.715$	$\mu_{CR} = 0.219$	
Diversity = 9.2122e-02	k_t = [1, 1, 1, 1]			
Generation 1700: Best = 3.376059e+03	FES = 170100	$\mu_F = 0.723$	$\mu_{CR} = 0.230$	
Diversity = 9.5154e-02	k_t = [2, 3, 4, 2]			
Generation 1710: Best = 3.375858e+03	FES = 171100	$\mu_F = 0.719$	$\mu_{CR} = 0.218$	
Diversity = 9.5592e-02	k_t = [1, 1, 1, 1]			
Generation 1720: Best = 3.375665e+03	FES = 172100	$\mu_F = 0.714$	$\mu_{CR} = 0.219$	
Diversity = 8.9842e-02	k_t = [4, 3, 2, 4]			
Generation 1730: Best = 3.375509e+03	FES = 173100	$\mu_F = 0.714$	$\mu_{CR} = 0.214$	
Diversity = 8.6891e-02	k_t = [1, 1, 1, 1]			
Generation 1740: Best = 3.375320e+03	FES = 174100	$\mu_F = 0.721$	$\mu_{CR} = 0.215$	
Diversity = 8.9340e-02	k_t = [3, 3, 3, 2]			
Generation 1750: Best = 3.375063e+03	FES = 175100	$\mu_F = 0.730$	$\mu_{CR} = 0.220$	
Diversity = 9.2400e-02	k_t = [1, 1, 1, 1]			
Generation 1760: Best = 3.374881e+03	FES = 176100	$\mu_F = 0.726$	$\mu_{CR} = 0.220$	
Diversity = 8.8079e-02	k_t = [2, 3, 4, 2]			
Generation 1770: Best = 3.374770e+03	FES = 177100	$\mu_F = 0.718$	$\mu_{CR} = 0.215$	
Diversity = 8.5425e-02	k_t = [1, 1, 1, 1]			
Generation 1780: Best = 3.374590e+03	FES = 178100	$\mu_F = 0.728$	$\mu_{CR} = 0.212$	
Diversity = 8.3506e-02	k_t = [3, 4, 4, 3]			
Generation 1790: Best = 3.374407e+03	FES = 179100	$\mu_F = 0.733$	$\mu_{CR} = 0.213$	
Diversity = 8.5177e-02	k_t = [1, 1, 1, 1]			
Generation 1800: Best = 3.374183e+03	FES = 180100	$\mu_F = 0.739$	$\mu_{CR} = 0.214$	
Diversity = 9.2666e-02	k_t = [4, 2, 2, 2]			
Generation 1810: Best = 3.374001e+03	FES = 181100	$\mu_F = 0.742$	$\mu_{CR} = 0.209$	
Diversity = 8.6635e-02	k_t = [1, 1, 1, 1]			
Generation 1820: Best = 3.373887e+03	FES = 182100	$\mu_F = 0.748$	$\mu_{CR} = 0.215$	
Diversity = 7.7832e-02	k_t = [3, 3, 3, 4]			
Generation 1830: Best = 3.373671e+03	FES = 183100	$\mu_F = 0.735$	$\mu_{CR} = 0.211$	
Diversity = 7.4418e-02	k_t = [1, 1, 1, 1]			
Generation 1840: Best = 3.373583e+03	FES = 184100	$\mu_F = 0.729$	$\mu_{CR} = 0.205$	

Diversity = 6.9939e-02	k_t = [2, 3, 4, 4]			
Generation 1850: Best = 3.373487e+03	FES = 185100	$\mu_F = 0.733$	$\mu_{CR} = 0.208$	
Diversity = 6.8242e-02	k_t = [1, 1, 1, 1]			
Generation 1860: Best = 3.373352e+03	FES = 186100	$\mu_F = 0.723$	$\mu_{CR} = 0.199$	
Diversity = 6.9128e-02	k_t = [4, 3, 2, 3]			
Generation 1870: Best = 3.373247e+03	FES = 187100	$\mu_F = 0.736$	$\mu_{CR} = 0.201$	
Diversity = 6.5366e-02	k_t = [1, 1, 1, 1]			
Generation 1880: Best = 3.373145e+03	FES = 188100	$\mu_F = 0.737$	$\mu_{CR} = 0.190$	
Diversity = 6.0584e-02	k_t = [2, 4, 3, 4]			
Generation 1890: Best = 3.373025e+03	FES = 189100	$\mu_F = 0.732$	$\mu_{CR} = 0.186$	
Diversity = 6.0612e-02	k_t = [1, 1, 1, 1]			
Generation 1900: Best = 3.372890e+03	FES = 190100	$\mu_F = 0.750$	$\mu_{CR} = 0.195$	
Diversity = 6.3999e-02	k_t = [2, 3, 3, 3]			
Generation 1910: Best = 3.372751e+03	FES = 191100	$\mu_F = 0.750$	$\mu_{CR} = 0.198$	
Diversity = 7.1507e-02	k_t = [1, 1, 1, 1]			
Generation 1920: Best = 3.372575e+03	FES = 192100	$\mu_F = 0.756$	$\mu_{CR} = 0.204$	
Diversity = 7.3830e-02	k_t = [3, 2, 4, 3]			
Generation 1930: Best = 3.372231e+03	FES = 193100	$\mu_F = 0.751$	$\mu_{CR} = 0.203$	
Diversity = 8.0974e-02	k_t = [1, 1, 1, 1]			
Generation 1940: Best = 3.371961e+03	FES = 194100	$\mu_F = 0.765$	$\mu_{CR} = 0.204$	
Diversity = 8.9694e-02	k_t = [3, 2, 3, 3]			
Generation 1950: Best = 3.371856e+03	FES = 195100	$\mu_F = 0.750$	$\mu_{CR} = 0.201$	
Diversity = 9.1735e-02	k_t = [1, 1, 1, 1]			
Generation 1960: Best = 3.371771e+03	FES = 196100	$\mu_F = 0.741$	$\mu_{CR} = 0.187$	
Diversity = 8.3662e-02	k_t = [2, 2, 2, 3]			
Generation 1970: Best = 3.371547e+03	FES = 197100	$\mu_F = 0.744$	$\mu_{CR} = 0.183$	
Diversity = 8.0219e-02	k_t = [1, 1, 1, 1]			
Generation 1980: Best = 3.371443e+03	FES = 198100	$\mu_F = 0.750$	$\mu_{CR} = 0.181$	
Diversity = 7.5023e-02	k_t = [4, 4, 3, 4]			
Generation 1990: Best = 3.371330e+03	FES = 199100	$\mu_F = 0.740$	$\mu_{CR} = 0.181$	
Diversity = 6.4595e-02	k_t = [1, 1, 1, 1]			
Generation 2000: Best = 3.371210e+03	FES = 200100	$\mu_F = 0.746$	$\mu_{CR} = 0.173$	
Diversity = 6.0710e-02	k_t = [4, 2, 3, 4]			
Generation 2010: Best = 3.371063e+03	FES = 201100	$\mu_F = 0.758$	$\mu_{CR} = 0.182$	
Diversity = 6.1387e-02	k_t = [1, 1, 1, 1]			
Generation 2020: Best = 3.370968e+03	FES = 202100	$\mu_F = 0.755$	$\mu_{CR} = 0.184$	
Diversity = 6.0990e-02	k_t = [4, 3, 3, 2]			
Generation 2030: Best = 3.370910e+03	FES = 203100	$\mu_F = 0.762$	$\mu_{CR} = 0.186$	
Diversity = 6.1070e-02	k_t = [1, 1, 1, 1]			
Generation 2040: Best = 3.370819e+03	FES = 204100	$\mu_F = 0.750$	$\mu_{CR} = 0.186$	
Diversity = 5.8053e-02	k_t = [4, 4, 4, 2]			
Generation 2050: Best = 3.370747e+03	FES = 205100	$\mu_F = 0.745$	$\mu_{CR} = 0.184$	
Diversity = 5.6385e-02	k_t = [1, 1, 1, 1]			
Generation 2060: Best = 3.370667e+03	FES = 206100	$\mu_F = 0.748$	$\mu_{CR} = 0.183$	
Diversity = 5.3865e-02	k_t = [3, 4, 3, 2]			
Generation 2070: Best = 3.370567e+03	FES = 207100	$\mu_F = 0.747$	$\mu_{CR} = 0.183$	
Diversity = 5.0221e-02	k_t = [1, 1, 1, 1]			
Generation 2080: Best = 3.370491e+03	FES = 208100	$\mu_F = 0.746$	$\mu_{CR} = 0.183$	
Diversity = 5.1719e-02	k_t = [2, 4, 3, 2]			
Generation 2090: Best = 3.370404e+03	FES = 209100	$\mu_F = 0.751$	$\mu_{CR} = 0.178$	
Diversity = 5.0682e-02	k_t = [1, 1, 1, 1]			
Generation 2100: Best = 3.370309e+03	FES = 210100	$\mu_F = 0.751$	$\mu_{CR} = 0.169$	
Diversity = 5.1512e-02	k_t = [4, 3, 4, 2]			
Generation 2110: Best = 3.370234e+03	FES = 211100	$\mu_F = 0.741$	$\mu_{CR} = 0.179$	

Diversity = 5.2285e-02	k_t = [1, 1, 1, 1]			
Generation 2120: Best = 3.370163e+03	FES = 212100	$\mu_F = 0.735$	$\mu_{CR} = 0.177$	
Diversity = 4.9528e-02	k_t = [2, 3, 3, 2]			
Generation 2130: Best = 3.370084e+03	FES = 213100	$\mu_F = 0.732$	$\mu_{CR} = 0.167$	
Diversity = 5.0211e-02	k_t = [1, 1, 1, 1]			
Generation 2140: Best = 3.370036e+03	FES = 214100	$\mu_F = 0.732$	$\mu_{CR} = 0.161$	
Diversity = 4.6782e-02	k_t = [3, 3, 4, 3]			
Generation 2150: Best = 3.369977e+03	FES = 215100	$\mu_F = 0.723$	$\mu_{CR} = 0.164$	
Diversity = 4.6948e-02	k_t = [1, 1, 1, 1]			
Generation 2160: Best = 3.369895e+03	FES = 216100	$\mu_F = 0.722$	$\mu_{CR} = 0.165$	
Diversity = 4.3268e-02	k_t = [3, 2, 4, 2]			
Generation 2170: Best = 3.369862e+03	FES = 217100	$\mu_F = 0.722$	$\mu_{CR} = 0.166$	
Diversity = 4.1590e-02	k_t = [1, 1, 1, 1]			
Generation 2180: Best = 3.369808e+03	FES = 218100	$\mu_F = 0.735$	$\mu_{CR} = 0.164$	
Diversity = 3.7751e-02	k_t = [3, 4, 3, 4]			
Generation 2190: Best = 3.369725e+03	FES = 219100	$\mu_F = 0.734$	$\mu_{CR} = 0.174$	
Diversity = 3.6070e-02	k_t = [1, 1, 1, 1]			
Generation 2200: Best = 3.369696e+03	FES = 220100	$\mu_F = 0.746$	$\mu_{CR} = 0.175$	
Diversity = 3.6094e-02	k_t = [4, 3, 2, 2]			
Generation 2210: Best = 3.369670e+03	FES = 221100	$\mu_F = 0.738$	$\mu_{CR} = 0.180$	
Diversity = 3.5452e-02	k_t = [1, 1, 1, 1]			
Generation 2220: Best = 3.369613e+03	FES = 222100	$\mu_F = 0.733$	$\mu_{CR} = 0.189$	
Diversity = 3.1011e-02	k_t = [4, 2, 2, 4]			
Generation 2230: Best = 3.369565e+03	FES = 223100	$\mu_F = 0.731$	$\mu_{CR} = 0.195$	
Diversity = 3.1330e-02	k_t = [1, 1, 1, 1]			
Generation 2240: Best = 3.369518e+03	FES = 224100	$\mu_F = 0.725$	$\mu_{CR} = 0.194$	
Diversity = 3.4009e-02	k_t = [3, 4, 2, 3]			
Generation 2250: Best = 3.369461e+03	FES = 225100	$\mu_F = 0.722$	$\mu_{CR} = 0.184$	
Diversity = 3.6996e-02	k_t = [1, 1, 1, 1]			
Generation 2260: Best = 3.369357e+03	FES = 226100	$\mu_F = 0.717$	$\mu_{CR} = 0.185$	
Diversity = 4.0830e-02	k_t = [4, 4, 4, 2]			
Generation 2270: Best = 3.369299e+03	FES = 227100	$\mu_F = 0.726$	$\mu_{CR} = 0.187$	
Diversity = 4.4532e-02	k_t = [1, 1, 1, 1]			
Generation 2280: Best = 3.369226e+03	FES = 228100	$\mu_F = 0.714$	$\mu_{CR} = 0.196$	
Diversity = 4.0877e-02	k_t = [3, 2, 3, 4]			
Generation 2290: Best = 3.369156e+03	FES = 229100	$\mu_F = 0.724$	$\mu_{CR} = 0.197$	
Diversity = 3.7676e-02	k_t = [1, 1, 1, 1]			
Generation 2300: Best = 3.369109e+03	FES = 230100	$\mu_F = 0.716$	$\mu_{CR} = 0.192$	
Diversity = 3.5832e-02	k_t = [2, 2, 2, 2]			
Generation 2310: Best = 3.369039e+03	FES = 231100	$\mu_F = 0.699$	$\mu_{CR} = 0.187$	
Diversity = 3.3105e-02	k_t = [1, 1, 1, 1]			
Generation 2320: Best = 3.368991e+03	FES = 232100	$\mu_F = 0.710$	$\mu_{CR} = 0.188$	
Diversity = 3.0533e-02	k_t = [3, 4, 2, 3]			
Generation 2330: Best = 3.368926e+03	FES = 233100	$\mu_F = 0.728$	$\mu_{CR} = 0.195$	
Diversity = 3.3318e-02	k_t = [1, 1, 1, 1]			
Generation 2340: Best = 3.368855e+03	FES = 234100	$\mu_F = 0.747$	$\mu_{CR} = 0.200$	
Diversity = 3.3367e-02	k_t = [2, 2, 4, 3]			
Generation 2350: Best = 3.368706e+03	FES = 235100	$\mu_F = 0.753$	$\mu_{CR} = 0.206$	
Diversity = 3.4351e-02	k_t = [1, 1, 1, 1]			
Generation 2360: Best = 3.368619e+03	FES = 236100	$\mu_F = 0.762$	$\mu_{CR} = 0.199$	
Diversity = 3.5247e-02	k_t = [2, 3, 3, 3]			
Generation 2370: Best = 3.368470e+03	FES = 237100	$\mu_F = 0.765$	$\mu_{CR} = 0.200$	
Diversity = 3.9297e-02	k_t = [1, 1, 1, 1]			
Generation 2380: Best = 3.368371e+03	FES = 238100	$\mu_F = 0.763$	$\mu_{CR} = 0.194$	

Diversity = 4.1788e-02	k_t = [4, 4, 4, 2]			
Generation 2390: Best = 3.368279e+03	FES = 239100	$\mu_F = 0.765$	$\mu_{CR} = 0.200$	
Diversity = 4.5432e-02	k_t = [1, 1, 1, 1]			
Generation 2400: Best = 3.368122e+03	FES = 240100	$\mu_F = 0.767$	$\mu_{CR} = 0.206$	
Diversity = 4.6000e-02	k_t = [4, 2, 3, 3]			
Generation 2410: Best = 3.368011e+03	FES = 241100	$\mu_F = 0.765$	$\mu_{CR} = 0.210$	
Diversity = 4.9735e-02	k_t = [1, 1, 1, 1]			
Generation 2420: Best = 3.367902e+03	FES = 242100	$\mu_F = 0.768$	$\mu_{CR} = 0.206$	
Diversity = 5.0269e-02	k_t = [2, 2, 3, 3]			
Generation 2430: Best = 3.367753e+03	FES = 243100	$\mu_F = 0.768$	$\mu_{CR} = 0.205$	
Diversity = 5.5523e-02	k_t = [1, 1, 1, 1]			
Generation 2440: Best = 3.367643e+03	FES = 244100	$\mu_F = 0.774$	$\mu_{CR} = 0.203$	
Diversity = 5.8387e-02	k_t = [2, 2, 2, 3]			
Generation 2450: Best = 3.367524e+03	FES = 245100	$\mu_F = 0.772$	$\mu_{CR} = 0.194$	
Diversity = 6.0969e-02	k_t = [1, 1, 1, 1]			
Generation 2460: Best = 3.367370e+03	FES = 246100	$\mu_F = 0.779$	$\mu_{CR} = 0.198$	
Diversity = 6.1632e-02	k_t = [4, 3, 3, 3]			
Generation 2470: Best = 3.367256e+03	FES = 247100	$\mu_F = 0.773$	$\mu_{CR} = 0.200$	
Diversity = 6.4013e-02	k_t = [1, 1, 1, 1]			
Generation 2480: Best = 3.367179e+03	FES = 248100	$\mu_F = 0.758$	$\mu_{CR} = 0.194$	
Diversity = 6.1836e-02	k_t = [3, 2, 3, 2]			
Generation 2490: Best = 3.367086e+03	FES = 249100	$\mu_F = 0.738$	$\mu_{CR} = 0.189$	
Diversity = 6.4250e-02	k_t = [1, 1, 1, 1]			
Generation 2500: Best = 3.366953e+03	FES = 250100	$\mu_F = 0.739$	$\mu_{CR} = 0.183$	
Diversity = 5.9164e-02	k_t = [4, 3, 3, 3]			
Generation 2510: Best = 3.366901e+03	FES = 251100	$\mu_F = 0.739$	$\mu_{CR} = 0.176$	
Diversity = 6.4734e-02	k_t = [1, 1, 1, 1]			
Generation 2520: Best = 3.366768e+03	FES = 252100	$\mu_F = 0.741$	$\mu_{CR} = 0.175$	
Diversity = 6.3697e-02	k_t = [2, 3, 2, 3]			
Generation 2530: Best = 3.366661e+03	FES = 253100	$\mu_F = 0.728$	$\mu_{CR} = 0.176$	
Diversity = 6.3635e-02	k_t = [1, 1, 1, 1]			
Generation 2540: Best = 3.366560e+03	FES = 254100	$\mu_F = 0.738$	$\mu_{CR} = 0.165$	
Diversity = 6.0561e-02	k_t = [2, 4, 3, 4]			
Generation 2550: Best = 3.366457e+03	FES = 255100	$\mu_F = 0.744$	$\mu_{CR} = 0.170$	
Diversity = 6.0608e-02	k_t = [1, 1, 1, 1]			
Generation 2560: Best = 3.366280e+03	FES = 256100	$\mu_F = 0.734$	$\mu_{CR} = 0.172$	
Diversity = 6.4996e-02	k_t = [4, 2, 4, 2]			
Generation 2570: Best = 3.366162e+03	FES = 257100	$\mu_F = 0.732$	$\mu_{CR} = 0.166$	
Diversity = 7.1323e-02	k_t = [1, 1, 1, 1]			
Generation 2580: Best = 3.366034e+03	FES = 258100	$\mu_F = 0.725$	$\mu_{CR} = 0.168$	
Diversity = 7.2026e-02	k_t = [2, 3, 4, 3]			
Generation 2590: Best = 3.365889e+03	FES = 259100	$\mu_F = 0.717$	$\mu_{CR} = 0.172$	
Diversity = 7.1806e-02	k_t = [1, 1, 1, 1]			
Generation 2600: Best = 3.365752e+03	FES = 260100	$\mu_F = 0.735$	$\mu_{CR} = 0.170$	
Diversity = 7.0428e-02	k_t = [3, 2, 2, 3]			
Generation 2610: Best = 3.365579e+03	FES = 261100	$\mu_F = 0.747$	$\mu_{CR} = 0.173$	
Diversity = 6.9072e-02	k_t = [1, 1, 1, 1]			
Generation 2620: Best = 3.365456e+03	FES = 262100	$\mu_F = 0.751$	$\mu_{CR} = 0.176$	
Diversity = 6.2553e-02	k_t = [3, 4, 3, 3]			
Generation 2630: Best = 3.365337e+03	FES = 263100	$\mu_F = 0.760$	$\mu_{CR} = 0.181$	
Diversity = 6.6513e-02	k_t = [1, 1, 1, 1]			
Generation 2640: Best = 3.365194e+03	FES = 264100	$\mu_F = 0.755$	$\mu_{CR} = 0.178$	
Diversity = 6.2986e-02	k_t = [4, 4, 2, 4]			
Generation 2650: Best = 3.365039e+03	FES = 265100	$\mu_F = 0.759$	$\mu_{CR} = 0.185$	

Diversity = 6.4054e-02	k_t = [1, 1, 1, 1]			
Generation 2660: Best = 3.364837e+03	FES = 266100	$\mu_F = 0.769$	$\mu_{CR} = 0.190$	
Diversity = 6.4440e-02	k_t = [2, 2, 4, 4]			
Generation 2670: Best = 3.364623e+03	FES = 267100	$\mu_F = 0.777$	$\mu_{CR} = 0.200$	
Diversity = 7.3947e-02	k_t = [1, 1, 1, 1]			
Generation 2680: Best = 3.364526e+03	FES = 268100	$\mu_F = 0.756$	$\mu_{CR} = 0.194$	
Diversity = 7.0407e-02	k_t = [4, 3, 3, 4]			
Generation 2690: Best = 3.364310e+03	FES = 269100	$\mu_F = 0.760$	$\mu_{CR} = 0.195$	
Diversity = 7.0429e-02	k_t = [1, 1, 1, 1]			
Generation 2700: Best = 3.364233e+03	FES = 270100	$\mu_F = 0.735$	$\mu_{CR} = 0.198$	
Diversity = 6.6388e-02	k_t = [2, 2, 3, 2]			
Generation 2710: Best = 3.364128e+03	FES = 271100	$\mu_F = 0.717$	$\mu_{CR} = 0.189$	
Diversity = 6.8416e-02	k_t = [1, 1, 1, 1]			
Generation 2720: Best = 3.364078e+03	FES = 272100	$\mu_F = 0.704$	$\mu_{CR} = 0.188$	
Diversity = 6.0357e-02	k_t = [2, 3, 2, 4]			
Generation 2730: Best = 3.363979e+03	FES = 273100	$\mu_F = 0.693$	$\mu_{CR} = 0.172$	
Diversity = 5.3321e-02	k_t = [1, 1, 1, 1]			
Generation 2740: Best = 3.363952e+03	FES = 274100	$\mu_F = 0.694$	$\mu_{CR} = 0.177$	
Diversity = 5.0464e-02	k_t = [4, 4, 4, 2]			
Generation 2750: Best = 3.363884e+03	FES = 275100	$\mu_F = 0.701$	$\mu_{CR} = 0.176$	
Diversity = 4.9530e-02	k_t = [1, 1, 1, 1]			
Generation 2760: Best = 3.363777e+03	FES = 276100	$\mu_F = 0.715$	$\mu_{CR} = 0.178$	
Diversity = 4.5555e-02	k_t = [2, 3, 3, 3]			
Generation 2770: Best = 3.363758e+03	FES = 277100	$\mu_F = 0.710$	$\mu_{CR} = 0.176$	
Diversity = 4.1386e-02	k_t = [1, 1, 1, 1]			
Generation 2780: Best = 3.363688e+03	FES = 278100	$\mu_F = 0.721$	$\mu_{CR} = 0.176$	
Diversity = 3.8263e-02	k_t = [2, 3, 4, 2]			
Generation 2790: Best = 3.363620e+03	FES = 279100	$\mu_F = 0.724$	$\mu_{CR} = 0.184$	
Diversity = 3.6713e-02	k_t = [1, 1, 1, 1]			
Generation 2800: Best = 3.363557e+03	FES = 280100	$\mu_F = 0.726$	$\mu_{CR} = 0.180$	
Diversity = 3.6484e-02	k_t = [4, 3, 4, 2]			
Generation 2810: Best = 3.363494e+03	FES = 281100	$\mu_F = 0.725$	$\mu_{CR} = 0.179$	
Diversity = 3.6870e-02	k_t = [1, 1, 1, 1]			
Generation 2820: Best = 3.363472e+03	FES = 282100	$\mu_F = 0.728$	$\mu_{CR} = 0.172$	
Diversity = 3.4069e-02	k_t = [4, 3, 2, 4]			
Generation 2830: Best = 3.363445e+03	FES = 283100	$\mu_F = 0.737$	$\mu_{CR} = 0.164$	
Diversity = 3.2682e-02	k_t = [1, 1, 1, 1]			
Generation 2840: Best = 3.363398e+03	FES = 284100	$\mu_F = 0.741$	$\mu_{CR} = 0.167$	
Diversity = 3.1770e-02	k_t = [2, 4, 3, 4]			
Generation 2850: Best = 3.363348e+03	FES = 285100	$\mu_F = 0.757$	$\mu_{CR} = 0.172$	
Diversity = 3.1242e-02	k_t = [1, 1, 1, 1]			
Generation 2860: Best = 3.363309e+03	FES = 286100	$\mu_F = 0.767$	$\mu_{CR} = 0.164$	
Diversity = 3.2862e-02	k_t = [4, 4, 4, 3]			
Generation 2870: Best = 3.363229e+03	FES = 287100	$\mu_F = 0.786$	$\mu_{CR} = 0.163$	
Diversity = 3.6299e-02	k_t = [1, 1, 1, 1]			
Generation 2880: Best = 3.363184e+03	FES = 288100	$\mu_F = 0.783$	$\mu_{CR} = 0.172$	
Diversity = 3.7236e-02	k_t = [3, 2, 2, 4]			
Generation 2890: Best = 3.363103e+03	FES = 289100	$\mu_F = 0.778$	$\mu_{CR} = 0.166$	
Diversity = 3.7437e-02	k_t = [1, 1, 1, 1]			
Generation 2900: Best = 3.362967e+03	FES = 290100	$\mu_F = 0.782$	$\mu_{CR} = 0.174$	
Diversity = 4.1675e-02	k_t = [4, 2, 4, 2]			
Generation 2910: Best = 3.362902e+03	FES = 291100	$\mu_F = 0.791$	$\mu_{CR} = 0.171$	
Diversity = 4.6424e-02	k_t = [1, 1, 1, 1]			
Generation 2920: Best = 3.362769e+03	FES = 292100	$\mu_F = 0.785$	$\mu_{CR} = 0.179$	

Diversity = 4.9121e-02	k_t = [3, 2, 3, 4]			
Generation 2930: Best = 3.362615e+03	FES = 293100	$\mu_F = 0.778$	$\mu_{CR} = 0.194$	
Diversity = 5.0791e-02	k_t = [1, 1, 1, 1]			
Generation 2940: Best = 3.362604e+03	FES = 294100	$\mu_F = 0.764$	$\mu_{CR} = 0.197$	
Diversity = 4.7544e-02	k_t = [4, 4, 3, 3]			
Generation 2950: Best = 3.362495e+03	FES = 295100	$\mu_F = 0.747$	$\mu_{CR} = 0.201$	
Diversity = 4.3280e-02	k_t = [1, 1, 1, 1]			
Generation 2960: Best = 3.362409e+03	FES = 296100	$\mu_F = 0.750$	$\mu_{CR} = 0.193$	
Diversity = 4.4030e-02	k_t = [2, 4, 3, 3]			
Generation 2970: Best = 3.362306e+03	FES = 297100	$\mu_F = 0.745$	$\mu_{CR} = 0.192$	
Diversity = 4.5832e-02	k_t = [1, 1, 1, 1]			
Generation 2980: Best = 3.362292e+03	FES = 298100	$\mu_F = 0.743$	$\mu_{CR} = 0.184$	
Diversity = 4.0809e-02	k_t = [2, 3, 3, 4]			
Generation 2990: Best = 3.362237e+03	FES = 299100	$\mu_F = 0.720$	$\mu_{CR} = 0.178$	
Diversity = 3.8665e-02	k_t = [1, 1, 1, 1]			
Generation 3000: Best = 3.362196e+03	FES = 300100	$\mu_F = 0.738$	$\mu_{CR} = 0.176$	
Diversity = 4.2649e-02	k_t = [4, 4, 3, 2]			
Generation 3010: Best = 3.362145e+03	FES = 301100	$\mu_F = 0.719$	$\mu_{CR} = 0.168$	
Diversity = 4.0386e-02	k_t = [1, 1, 1, 1]			
Generation 3020: Best = 3.362088e+03	FES = 302100	$\mu_F = 0.722$	$\mu_{CR} = 0.171$	
Diversity = 3.8804e-02	k_t = [3, 2, 3, 4]			
Generation 3030: Best = 3.362015e+03	FES = 303100	$\mu_F = 0.731$	$\mu_{CR} = 0.173$	
Diversity = 3.5782e-02	k_t = [1, 1, 1, 1]			
Generation 3040: Best = 3.361978e+03	FES = 304100	$\mu_F = 0.741$	$\mu_{CR} = 0.179$	
Diversity = 3.5851e-02	k_t = [2, 2, 4, 3]			
Generation 3050: Best = 3.361921e+03	FES = 305100	$\mu_F = 0.733$	$\mu_{CR} = 0.176$	
Diversity = 3.3794e-02	k_t = [1, 1, 1, 1]			
Generation 3060: Best = 3.361892e+03	FES = 306100	$\mu_F = 0.736$	$\mu_{CR} = 0.179$	
Diversity = 3.1569e-02	k_t = [3, 4, 3, 2]			
Generation 3070: Best = 3.361861e+03	FES = 307100	$\mu_F = 0.730$	$\mu_{CR} = 0.173$	
Diversity = 2.9947e-02	k_t = [1, 1, 1, 1]			
Generation 3080: Best = 3.361827e+03	FES = 308100	$\mu_F = 0.745$	$\mu_{CR} = 0.175$	
Diversity = 2.8574e-02	k_t = [3, 3, 3, 2]			
Generation 3090: Best = 3.361781e+03	FES = 309100	$\mu_F = 0.737$	$\mu_{CR} = 0.176$	
Diversity = 2.7745e-02	k_t = [1, 1, 1, 1]			
Generation 3100: Best = 3.361725e+03	FES = 310100	$\mu_F = 0.742$	$\mu_{CR} = 0.182$	
Diversity = 2.8430e-02	k_t = [4, 3, 2, 3]			
Generation 3110: Best = 3.361668e+03	FES = 311100	$\mu_F = 0.749$	$\mu_{CR} = 0.177$	
Diversity = 2.9001e-02	k_t = [1, 1, 1, 1]			
Generation 3120: Best = 3.361634e+03	FES = 312100	$\mu_F = 0.745$	$\mu_{CR} = 0.186$	
Diversity = 2.9223e-02	k_t = [4, 4, 2, 2]			
Generation 3130: Best = 3.361602e+03	FES = 313100	$\mu_F = 0.747$	$\mu_{CR} = 0.188$	
Diversity = 3.0959e-02	k_t = [1, 1, 1, 1]			
Generation 3140: Best = 3.361543e+03	FES = 314100	$\mu_F = 0.754$	$\mu_{CR} = 0.186$	
Diversity = 3.1195e-02	k_t = [2, 4, 2, 2]			
Generation 3150: Best = 3.361486e+03	FES = 315100	$\mu_F = 0.750$	$\mu_{CR} = 0.191$	
Diversity = 3.4943e-02	k_t = [1, 1, 1, 1]			
Generation 3160: Best = 3.361441e+03	FES = 316100	$\mu_F = 0.743$	$\mu_{CR} = 0.202$	
Diversity = 3.3674e-02	k_t = [2, 4, 4, 3]			
Generation 3170: Best = 3.361376e+03	FES = 317100	$\mu_F = 0.743$	$\mu_{CR} = 0.204$	
Diversity = 3.5050e-02	k_t = [1, 1, 1, 1]			
Generation 3180: Best = 3.361309e+03	FES = 318100	$\mu_F = 0.757$	$\mu_{CR} = 0.200$	
Diversity = 3.6867e-02	k_t = [3, 2, 4, 2]			
Generation 3190: Best = 3.361234e+03	FES = 319100	$\mu_F = 0.757$	$\mu_{CR} = 0.207$	

Diversity = 3.9985e-02	k_t = [1, 1, 1, 1]			
Generation 3200: Best = 3.361191e+03	FES = 320100	$\mu_F = 0.753$	$\mu_{CR} = 0.195$	
Diversity = 3.6652e-02	k_t = [3, 2, 4, 4]			
Generation 3210: Best = 3.361101e+03	FES = 321100	$\mu_F = 0.755$	$\mu_{CR} = 0.212$	
Diversity = 3.4667e-02	k_t = [1, 1, 1, 1]			
Generation 3220: Best = 3.361014e+03	FES = 322100	$\mu_F = 0.762$	$\mu_{CR} = 0.216$	
Diversity = 3.5436e-02	k_t = [3, 2, 4, 3]			
Generation 3230: Best = 3.360949e+03	FES = 323100	$\mu_F = 0.766$	$\mu_{CR} = 0.212$	
Diversity = 3.9667e-02	k_t = [1, 1, 1, 1]			
Generation 3240: Best = 3.360847e+03	FES = 324100	$\mu_F = 0.778$	$\mu_{CR} = 0.220$	
Diversity = 4.2271e-02	k_t = [4, 4, 4, 3]			
Generation 3250: Best = 3.360766e+03	FES = 325100	$\mu_F = 0.771$	$\mu_{CR} = 0.219$	
Diversity = 4.5798e-02	k_t = [1, 1, 1, 1]			
Generation 3260: Best = 3.360648e+03	FES = 326100	$\mu_F = 0.767$	$\mu_{CR} = 0.217$	
Diversity = 4.6075e-02	k_t = [4, 3, 2, 4]			
Generation 3270: Best = 3.360555e+03	FES = 327100	$\mu_F = 0.749$	$\mu_{CR} = 0.217$	
Diversity = 4.7646e-02	k_t = [1, 1, 1, 1]			
Generation 3280: Best = 3.360483e+03	FES = 328100	$\mu_F = 0.738$	$\mu_{CR} = 0.215$	
Diversity = 4.7161e-02	k_t = [3, 3, 2, 2]			
Generation 3290: Best = 3.360420e+03	FES = 329100	$\mu_F = 0.726$	$\mu_{CR} = 0.209$	
Diversity = 4.5533e-02	k_t = [1, 1, 1, 1]			
Generation 3300: Best = 3.360327e+03	FES = 330100	$\mu_F = 0.727$	$\mu_{CR} = 0.220$	
Diversity = 3.9322e-02	k_t = [4, 4, 3, 4]			
Generation 3310: Best = 3.360292e+03	FES = 331100	$\mu_F = 0.740$	$\mu_{CR} = 0.219$	
Diversity = 3.6872e-02	k_t = [1, 1, 1, 1]			
Generation 3320: Best = 3.360216e+03	FES = 332100	$\mu_F = 0.736$	$\mu_{CR} = 0.208$	
Diversity = 3.2670e-02	k_t = [3, 4, 2, 2]			
Generation 3330: Best = 3.360193e+03	FES = 333100	$\mu_F = 0.734$	$\mu_{CR} = 0.201$	
Diversity = 3.2000e-02	k_t = [1, 1, 1, 1]			
Generation 3340: Best = 3.360146e+03	FES = 334100	$\mu_F = 0.746$	$\mu_{CR} = 0.193$	
Diversity = 3.0092e-02	k_t = [3, 3, 4, 4]			
Generation 3350: Best = 3.360100e+03	FES = 335100	$\mu_F = 0.753$	$\mu_{CR} = 0.196$	
Diversity = 2.9167e-02	k_t = [1, 1, 1, 1]			
Generation 3360: Best = 3.360034e+03	FES = 336100	$\mu_F = 0.758$	$\mu_{CR} = 0.195$	
Diversity = 2.9010e-02	k_t = [4, 2, 3, 4]			
Generation 3370: Best = 3.359989e+03	FES = 337100	$\mu_F = 0.766$	$\mu_{CR} = 0.202$	
Diversity = 2.9043e-02	k_t = [1, 1, 1, 1]			
Generation 3380: Best = 3.359940e+03	FES = 338100	$\mu_F = 0.762$	$\mu_{CR} = 0.204$	
Diversity = 3.1264e-02	k_t = [2, 4, 4, 2]			
Generation 3390: Best = 3.359901e+03	FES = 339100	$\mu_F = 0.769$	$\mu_{CR} = 0.208$	
Diversity = 3.3166e-02	k_t = [1, 1, 1, 1]			
Generation 3400: Best = 3.359846e+03	FES = 340100	$\mu_F = 0.776$	$\mu_{CR} = 0.203$	
Diversity = 3.5380e-02	k_t = [2, 4, 3, 2]			
Generation 3410: Best = 3.359766e+03	FES = 341100	$\mu_F = 0.783$	$\mu_{CR} = 0.207$	
Diversity = 3.7255e-02	k_t = [1, 1, 1, 1]			
Generation 3420: Best = 3.359685e+03	FES = 342100	$\mu_F = 0.772$	$\mu_{CR} = 0.214$	
Diversity = 3.7180e-02	k_t = [3, 3, 4, 4]			
Generation 3430: Best = 3.359623e+03	FES = 343100	$\mu_F = 0.770$	$\mu_{CR} = 0.205$	
Diversity = 3.7974e-02	k_t = [1, 1, 1, 1]			
Generation 3440: Best = 3.359573e+03	FES = 344100	$\mu_F = 0.769$	$\mu_{CR} = 0.202$	
Diversity = 3.9222e-02	k_t = [4, 2, 3, 3]			
Generation 3450: Best = 3.359517e+03	FES = 345100	$\mu_F = 0.770$	$\mu_{CR} = 0.196$	
Diversity = 4.5033e-02	k_t = [1, 1, 1, 1]			
Generation 3460: Best = 3.359451e+03	FES = 346100	$\mu_F = 0.762$	$\mu_{CR} = 0.203$	

Diversity = 4.4545e-02	k_t = [2, 3, 2, 3]			
Generation 3470: Best = 3.359401e+03	FES = 347100	$\mu_F = 0.754$	$\mu_{CR} = 0.194$	
Diversity = 4.3306e-02	k_t = [1, 1, 1, 1]			
Generation 3480: Best = 3.359370e+03	FES = 348100	$\mu_F = 0.735$	$\mu_{CR} = 0.175$	
Diversity = 4.0905e-02	k_t = [3, 3, 4, 3]			
Generation 3490: Best = 3.359293e+03	FES = 349100	$\mu_F = 0.738$	$\mu_{CR} = 0.181$	
Diversity = 3.9185e-02	k_t = [1, 1, 1, 1]			
Generation 3500: Best = 3.359255e+03	FES = 350100	$\mu_F = 0.737$	$\mu_{CR} = 0.184$	
Diversity = 3.5603e-02	k_t = [2, 3, 4, 3]			
Generation 3510: Best = 3.359201e+03	FES = 351100	$\mu_F = 0.752$	$\mu_{CR} = 0.185$	
Diversity = 3.4399e-02	k_t = [1, 1, 1, 1]			
Generation 3520: Best = 3.359140e+03	FES = 352100	$\mu_F = 0.758$	$\mu_{CR} = 0.182$	
Diversity = 3.4300e-02	k_t = [2, 3, 3, 2]			
Generation 3530: Best = 3.359061e+03	FES = 353100	$\mu_F = 0.768$	$\mu_{CR} = 0.184$	
Diversity = 3.8789e-02	k_t = [1, 1, 1, 1]			
Generation 3540: Best = 3.358994e+03	FES = 354100	$\mu_F = 0.783$	$\mu_{CR} = 0.188$	
Diversity = 4.2457e-02	k_t = [2, 2, 4, 4]			
Generation 3550: Best = 3.358913e+03	FES = 355100	$\mu_F = 0.781$	$\mu_{CR} = 0.188$	
Diversity = 4.4135e-02	k_t = [1, 1, 1, 1]			
Generation 3560: Best = 3.358845e+03	FES = 356100	$\mu_F = 0.774$	$\mu_{CR} = 0.192$	
Diversity = 4.3932e-02	k_t = [2, 4, 3, 2]			
Generation 3570: Best = 3.358756e+03	FES = 357100	$\mu_F = 0.779$	$\mu_{CR} = 0.190$	
Diversity = 4.6099e-02	k_t = [1, 1, 1, 1]			
Generation 3580: Best = 3.358703e+03	FES = 358100	$\mu_F = 0.770$	$\mu_{CR} = 0.194$	
Diversity = 4.5790e-02	k_t = [2, 3, 4, 2]			
Generation 3590: Best = 3.358643e+03	FES = 359100	$\mu_F = 0.759$	$\mu_{CR} = 0.195$	
Diversity = 4.5818e-02	k_t = [1, 1, 1, 1]			
Generation 3600: Best = 3.358532e+03	FES = 360100	$\mu_F = 0.754$	$\mu_{CR} = 0.198$	
Diversity = 4.4729e-02	k_t = [3, 3, 4, 4]			
Generation 3610: Best = 3.358429e+03	FES = 361100	$\mu_F = 0.752$	$\mu_{CR} = 0.204$	
Diversity = 4.6420e-02	k_t = [1, 1, 1, 1]			
Generation 3620: Best = 3.358311e+03	FES = 362100	$\mu_F = 0.760$	$\mu_{CR} = 0.206$	
Diversity = 4.6626e-02	k_t = [2, 4, 2, 4]			
Generation 3630: Best = 3.358233e+03	FES = 363100	$\mu_F = 0.756$	$\mu_{CR} = 0.215$	
Diversity = 4.5430e-02	k_t = [1, 1, 1, 1]			
Generation 3640: Best = 3.358155e+03	FES = 364100	$\mu_F = 0.756$	$\mu_{CR} = 0.220$	
Diversity = 4.5782e-02	k_t = [4, 3, 2, 2]			
Generation 3650: Best = 3.358056e+03	FES = 365100	$\mu_F = 0.756$	$\mu_{CR} = 0.225$	
Diversity = 4.4447e-02	k_t = [1, 1, 1, 1]			
Generation 3660: Best = 3.357953e+03	FES = 366100	$\mu_F = 0.753$	$\mu_{CR} = 0.229$	
Diversity = 4.5162e-02	k_t = [3, 3, 4, 2]			
Generation 3670: Best = 3.357811e+03	FES = 367100	$\mu_F = 0.754$	$\mu_{CR} = 0.224$	
Diversity = 4.6264e-02	k_t = [1, 1, 1, 1]			
Generation 3680: Best = 3.357748e+03	FES = 368100	$\mu_F = 0.751$	$\mu_{CR} = 0.231$	
Diversity = 4.2292e-02	k_t = [2, 3, 3, 4]			
Generation 3690: Best = 3.357672e+03	FES = 369100	$\mu_F = 0.750$	$\mu_{CR} = 0.220$	
Diversity = 4.8418e-02	k_t = [1, 1, 1, 1]			
Generation 3700: Best = 3.357559e+03	FES = 370100	$\mu_F = 0.742$	$\mu_{CR} = 0.220$	
Diversity = 4.5132e-02	k_t = [2, 4, 4, 4]			
Generation 3710: Best = 3.357502e+03	FES = 371100	$\mu_F = 0.741$	$\mu_{CR} = 0.218$	
Diversity = 4.1743e-02	k_t = [1, 1, 1, 1]			
Generation 3720: Best = 3.357403e+03	FES = 372100	$\mu_F = 0.744$	$\mu_{CR} = 0.213$	
Diversity = 3.7851e-02	k_t = [3, 3, 2, 4]			
Generation 3730: Best = 3.357382e+03	FES = 373100	$\mu_F = 0.734$	$\mu_{CR} = 0.205$	

Diversity = 3.5792e-02	k_t = [1, 1, 1, 1]			
Generation 3740: Best = 3.357342e+03	FES = 374100	$\mu_F = 0.728$	$\mu_{CR} = 0.211$	
Diversity = 3.2075e-02	k_t = [2, 2, 2, 4]			
Generation 3750: Best = 3.357273e+03	FES = 375100	$\mu_F = 0.747$	$\mu_{CR} = 0.209$	
Diversity = 3.2385e-02	k_t = [1, 1, 1, 1]			
Generation 3760: Best = 3.357227e+03	FES = 376100	$\mu_F = 0.754$	$\mu_{CR} = 0.195$	
Diversity = 3.3674e-02	k_t = [4, 2, 4, 3]			
Generation 3770: Best = 3.357132e+03	FES = 377100	$\mu_F = 0.757$	$\mu_{CR} = 0.193$	
Diversity = 3.6611e-02	k_t = [1, 1, 1, 1]			
Generation 3780: Best = 3.357052e+03	FES = 378100	$\mu_F = 0.776$	$\mu_{CR} = 0.203$	
Diversity = 3.7728e-02	k_t = [2, 2, 3, 2]			
Generation 3790: Best = 3.357019e+03	FES = 379100	$\mu_F = 0.777$	$\mu_{CR} = 0.193$	
Diversity = 4.0402e-02	k_t = [1, 1, 1, 1]			
Generation 3800: Best = 3.356946e+03	FES = 380100	$\mu_F = 0.767$	$\mu_{CR} = 0.186$	
Diversity = 4.0569e-02	k_t = [2, 2, 4, 2]			
Generation 3810: Best = 3.356795e+03	FES = 381100	$\mu_F = 0.770$	$\mu_{CR} = 0.200$	
Diversity = 4.3896e-02	k_t = [1, 1, 1, 1]			
Generation 3820: Best = 3.356700e+03	FES = 382100	$\mu_F = 0.767$	$\mu_{CR} = 0.201$	
Diversity = 4.9997e-02	k_t = [3, 2, 2, 3]			
Generation 3830: Best = 3.356608e+03	FES = 383100	$\mu_F = 0.775$	$\mu_{CR} = 0.191$	
Diversity = 5.9728e-02	k_t = [1, 1, 1, 1]			
Generation 3840: Best = 3.356473e+03	FES = 384100	$\mu_F = 0.781$	$\mu_{CR} = 0.195$	
Diversity = 6.3710e-02	k_t = [4, 4, 2, 3]			
Generation 3850: Best = 3.356265e+03	FES = 385100	$\mu_F = 0.774$	$\mu_{CR} = 0.188$	
Diversity = 6.5292e-02	k_t = [1, 1, 1, 1]			
Generation 3860: Best = 3.356160e+03	FES = 386100	$\mu_F = 0.777$	$\mu_{CR} = 0.186$	
Diversity = 6.5491e-02	k_t = [3, 3, 4, 4]			
Generation 3870: Best = 3.355967e+03	FES = 387100	$\mu_F = 0.764$	$\mu_{CR} = 0.180$	
Diversity = 6.6805e-02	k_t = [1, 1, 1, 1]			
Generation 3880: Best = 3.355841e+03	FES = 388100	$\mu_F = 0.763$	$\mu_{CR} = 0.181$	
Diversity = 6.5204e-02	k_t = [3, 2, 4, 4]			
Generation 3890: Best = 3.355735e+03	FES = 389100	$\mu_F = 0.758$	$\mu_{CR} = 0.187$	
Diversity = 6.5090e-02	k_t = [1, 1, 1, 1]			
Generation 3900: Best = 3.355578e+03	FES = 390100	$\mu_F = 0.754$	$\mu_{CR} = 0.184$	
Diversity = 6.0029e-02	k_t = [3, 4, 3, 4]			
Generation 3910: Best = 3.355444e+03	FES = 391100	$\mu_F = 0.749$	$\mu_{CR} = 0.181$	
Diversity = 6.1269e-02	k_t = [1, 1, 1, 1]			
Generation 3920: Best = 3.355294e+03	FES = 392100	$\mu_F = 0.762$	$\mu_{CR} = 0.183$	
Diversity = 6.2295e-02	k_t = [2, 3, 4, 4]			
Generation 3930: Best = 3.355118e+03	FES = 393100	$\mu_F = 0.765$	$\mu_{CR} = 0.187$	
Diversity = 7.0413e-02	k_t = [1, 1, 1, 1]			
Generation 3940: Best = 3.354950e+03	FES = 394100	$\mu_F = 0.762$	$\mu_{CR} = 0.178$	
Diversity = 7.2470e-02	k_t = [4, 3, 2, 4]			
Generation 3950: Best = 3.354830e+03	FES = 395100	$\mu_F = 0.759$	$\mu_{CR} = 0.166$	
Diversity = 6.9633e-02	k_t = [1, 1, 1, 1]			
Generation 3960: Best = 3.354476e+03	FES = 396100	$\mu_F = 0.753$	$\mu_{CR} = 0.178$	
Diversity = 8.3556e-02	k_t = [2, 2, 2, 2]			
Generation 3970: Best = 3.354333e+03	FES = 397100	$\mu_F = 0.757$	$\mu_{CR} = 0.185$	
Diversity = 9.5388e-02	k_t = [1, 1, 1, 1]			
Generation 3980: Best = 3.354211e+03	FES = 398100	$\mu_F = 0.755$	$\mu_{CR} = 0.179$	
Diversity = 9.1080e-02	k_t = [4, 3, 3, 2]			
Generation 3990: Best = 3.354077e+03	FES = 399100	$\mu_F = 0.735$	$\mu_{CR} = 0.179$	
Diversity = 9.1687e-02	k_t = [1, 1, 1, 1]			
Generation 4000: Best = 3.353772e+03	FES = 400100	$\mu_F = 0.747$	$\mu_{CR} = 0.176$	

Diversity = 1.0635e-01	k_t = [2, 4, 2, 4]			
Generation 4010: Best = 3.353596e+03	FES = 401100	$\mu_F = 0.745$	$\mu_{CR} = 0.174$	
Diversity = 1.0165e-01	k_t = [1, 1, 1, 1]			
Generation 4020: Best = 3.353306e+03	FES = 402100	$\mu_F = 0.737$	$\mu_{CR} = 0.174$	
Diversity = 9.4704e-02	k_t = [3, 4, 2, 2]			
Generation 4030: Best = 3.353196e+03	FES = 403100	$\mu_F = 0.742$	$\mu_{CR} = 0.177$	
Diversity = 8.9917e-02	k_t = [1, 1, 1, 1]			
Generation 4040: Best = 3.352976e+03	FES = 404100	$\mu_F = 0.733$	$\mu_{CR} = 0.182$	
Diversity = 8.1931e-02	k_t = [4, 2, 2, 4]			
Generation 4050: Best = 3.352807e+03	FES = 405100	$\mu_F = 0.739$	$\mu_{CR} = 0.183$	
Diversity = 8.0611e-02	k_t = [1, 1, 1, 1]			
Generation 4060: Best = 3.352689e+03	FES = 406100	$\mu_F = 0.739$	$\mu_{CR} = 0.172$	
Diversity = 8.1722e-02	k_t = [2, 2, 4, 2]			
Generation 4070: Best = 3.352582e+03	FES = 407100	$\mu_F = 0.721$	$\mu_{CR} = 0.165$	
Diversity = 7.9687e-02	k_t = [1, 1, 1, 1]			
Generation 4080: Best = 3.352402e+03	FES = 408100	$\mu_F = 0.736$	$\mu_{CR} = 0.168$	
Diversity = 8.4312e-02	k_t = [2, 3, 3, 4]			
Generation 4090: Best = 3.352197e+03	FES = 409100	$\mu_F = 0.736$	$\mu_{CR} = 0.169$	
Diversity = 8.2711e-02	k_t = [1, 1, 1, 1]			
Generation 4100: Best = 3.352060e+03	FES = 410100	$\mu_F = 0.738$	$\mu_{CR} = 0.167$	
Diversity = 8.1597e-02	k_t = [2, 3, 3, 4]			
Generation 4110: Best = 3.351773e+03	FES = 411100	$\mu_F = 0.749$	$\mu_{CR} = 0.168$	
Diversity = 8.6356e-02	k_t = [1, 1, 1, 1]			
Generation 4120: Best = 3.351512e+03	FES = 412100	$\mu_F = 0.739$	$\mu_{CR} = 0.170$	
Diversity = 1.0259e-01	k_t = [2, 3, 4, 2]			
Generation 4130: Best = 3.351303e+03	FES = 413100	$\mu_F = 0.738$	$\mu_{CR} = 0.184$	
Diversity = 1.0246e-01	k_t = [1, 1, 1, 1]			
Generation 4140: Best = 3.350962e+03	FES = 414100	$\mu_F = 0.747$	$\mu_{CR} = 0.185$	
Diversity = 1.0192e-01	k_t = [4, 4, 4, 4]			
Generation 4150: Best = 3.350657e+03	FES = 415100	$\mu_F = 0.749$	$\mu_{CR} = 0.177$	
Diversity = 9.3199e-02	k_t = [1, 1, 1, 1]			
Generation 4160: Best = 3.350436e+03	FES = 416100	$\mu_F = 0.747$	$\mu_{CR} = 0.188$	
Diversity = 8.8804e-02	k_t = [4, 4, 2, 4]			
Generation 4170: Best = 3.350272e+03	FES = 417100	$\mu_F = 0.760$	$\mu_{CR} = 0.183$	
Diversity = 8.8116e-02	k_t = [1, 1, 1, 1]			
Generation 4180: Best = 3.350006e+03	FES = 418100	$\mu_F = 0.762$	$\mu_{CR} = 0.182$	
Diversity = 8.5401e-02	k_t = [4, 4, 3, 2]			
Generation 4190: Best = 3.349825e+03	FES = 419100	$\mu_F = 0.763$	$\mu_{CR} = 0.182$	
Diversity = 9.0790e-02	k_t = [1, 1, 1, 1]			
Generation 4200: Best = 3.349650e+03	FES = 420100	$\mu_F = 0.765$	$\mu_{CR} = 0.183$	
Diversity = 9.1092e-02	k_t = [3, 4, 3, 2]			
Generation 4210: Best = 3.349477e+03	FES = 421100	$\mu_F = 0.762$	$\mu_{CR} = 0.184$	
Diversity = 8.9352e-02	k_t = [1, 1, 1, 1]			
Generation 4220: Best = 3.349269e+03	FES = 422100	$\mu_F = 0.745$	$\mu_{CR} = 0.177$	
Diversity = 8.4459e-02	k_t = [3, 4, 4, 3]			
Generation 4230: Best = 3.349037e+03	FES = 423100	$\mu_F = 0.746$	$\mu_{CR} = 0.177$	
Diversity = 8.9332e-02	k_t = [1, 1, 1, 1]			
Generation 4240: Best = 3.348739e+03	FES = 424100	$\mu_F = 0.743$	$\mu_{CR} = 0.191$	
Diversity = 8.6639e-02	k_t = [4, 3, 3, 4]			
Generation 4250: Best = 3.348643e+03	FES = 425100	$\mu_F = 0.739$	$\mu_{CR} = 0.190$	
Diversity = 8.6949e-02	k_t = [1, 1, 1, 1]			
Generation 4260: Best = 3.348448e+03	FES = 426100	$\mu_F = 0.743$	$\mu_{CR} = 0.187$	
Diversity = 8.1906e-02	k_t = [3, 4, 2, 3]			
Generation 4270: Best = 3.348323e+03	FES = 427100	$\mu_F = 0.747$	$\mu_{CR} = 0.191$	

Diversity = 8.9010e-02	k_t = [1, 1, 1, 1]			
Generation 4280: Best = 3.348109e+03	FES = 428100	$\mu_F = 0.744$	$\mu_{CR} = 0.193$	
Diversity = 8.7299e-02	k_t = [4, 4, 2, 4]			
Generation 4290: Best = 3.347944e+03	FES = 429100	$\mu_F = 0.754$	$\mu_{CR} = 0.181$	
Diversity = 8.7589e-02	k_t = [1, 1, 1, 1]			
Generation 4300: Best = 3.347852e+03	FES = 430100	$\mu_F = 0.728$	$\mu_{CR} = 0.180$	
Diversity = 8.4981e-02	k_t = [3, 3, 2, 4]			
Generation 4310: Best = 3.347677e+03	FES = 431100	$\mu_F = 0.723$	$\mu_{CR} = 0.175$	
Diversity = 7.9796e-02	k_t = [1, 1, 1, 1]			
Generation 4320: Best = 3.347545e+03	FES = 432100	$\mu_F = 0.725$	$\mu_{CR} = 0.169$	
Diversity = 8.6221e-02	k_t = [3, 4, 2, 3]			
Generation 4330: Best = 3.347430e+03	FES = 433100	$\mu_F = 0.729$	$\mu_{CR} = 0.166$	
Diversity = 8.8996e-02	k_t = [1, 1, 1, 1]			
Generation 4340: Best = 3.347183e+03	FES = 434100	$\mu_F = 0.725$	$\mu_{CR} = 0.167$	
Diversity = 8.9832e-02	k_t = [3, 3, 2, 2]			
Generation 4350: Best = 3.347045e+03	FES = 435100	$\mu_F = 0.732$	$\mu_{CR} = 0.169$	
Diversity = 9.6274e-02	k_t = [1, 1, 1, 1]			
Generation 4360: Best = 3.346998e+03	FES = 436100	$\mu_F = 0.709$	$\mu_{CR} = 0.154$	
Diversity = 8.7200e-02	k_t = [2, 2, 4, 4]			
Generation 4370: Best = 3.346916e+03	FES = 437100	$\mu_F = 0.702$	$\mu_{CR} = 0.149$	
Diversity = 8.0779e-02	k_t = [1, 1, 1, 1]			
Generation 4380: Best = 3.346788e+03	FES = 438100	$\mu_F = 0.718$	$\mu_{CR} = 0.150$	
Diversity = 7.3687e-02	k_t = [2, 3, 4, 4]			
Generation 4390: Best = 3.346736e+03	FES = 439100	$\mu_F = 0.719$	$\mu_{CR} = 0.147$	
Diversity = 6.5861e-02	k_t = [1, 1, 1, 1]			
Generation 4400: Best = 3.346629e+03	FES = 440100	$\mu_F = 0.719$	$\mu_{CR} = 0.151$	
Diversity = 5.9880e-02	k_t = [4, 2, 3, 4]			
Generation 4410: Best = 3.346552e+03	FES = 441100	$\mu_F = 0.723$	$\mu_{CR} = 0.151$	
Diversity = 5.3869e-02	k_t = [1, 1, 1, 1]			
Generation 4420: Best = 3.346472e+03	FES = 442100	$\mu_F = 0.726$	$\mu_{CR} = 0.154$	
Diversity = 5.1987e-02	k_t = [4, 4, 4, 3]			
Generation 4430: Best = 3.346383e+03	FES = 443100	$\mu_F = 0.724$	$\mu_{CR} = 0.164$	
Diversity = 5.0096e-02	k_t = [1, 1, 1, 1]			
Generation 4440: Best = 3.346274e+03	FES = 444100	$\mu_F = 0.725$	$\mu_{CR} = 0.175$	
Diversity = 5.1842e-02	k_t = [3, 4, 2, 2]			
Generation 4450: Best = 3.346217e+03	FES = 445100	$\mu_F = 0.727$	$\mu_{CR} = 0.185$	
Diversity = 5.0510e-02	k_t = [1, 1, 1, 1]			
Generation 4460: Best = 3.346106e+03	FES = 446100	$\mu_F = 0.724$	$\mu_{CR} = 0.190$	
Diversity = 4.6103e-02	k_t = [3, 2, 4, 3]			
Generation 4470: Best = 3.346059e+03	FES = 447100	$\mu_F = 0.735$	$\mu_{CR} = 0.200$	
Diversity = 5.1031e-02	k_t = [1, 1, 1, 1]			
Generation 4480: Best = 3.345954e+03	FES = 448100	$\mu_F = 0.747$	$\mu_{CR} = 0.192$	
Diversity = 5.3253e-02	k_t = [2, 2, 2, 2]			
Generation 4490: Best = 3.345873e+03	FES = 449100	$\mu_F = 0.730$	$\mu_{CR} = 0.194$	
Diversity = 5.2836e-02	k_t = [1, 1, 1, 1]			
Generation 4500: Best = 3.345800e+03	FES = 450100	$\mu_F = 0.736$	$\mu_{CR} = 0.191$	
Diversity = 4.6658e-02	k_t = [2, 4, 3, 2]			
Generation 4510: Best = 3.345716e+03	FES = 451100	$\mu_F = 0.735$	$\mu_{CR} = 0.196$	
Diversity = 4.6978e-02	k_t = [1, 1, 1, 1]			
Generation 4520: Best = 3.345659e+03	FES = 452100	$\mu_F = 0.727$	$\mu_{CR} = 0.195$	
Diversity = 4.3240e-02	k_t = [3, 3, 2, 3]			
Generation 4530: Best = 3.345603e+03	FES = 453100	$\mu_F = 0.725$	$\mu_{CR} = 0.193$	
Diversity = 4.0778e-02	k_t = [1, 1, 1, 1]			
Generation 4540: Best = 3.345551e+03	FES = 454100	$\mu_F = 0.736$	$\mu_{CR} = 0.196$	

Diversity = 3.8582e-02	k_t = [2, 2, 3, 4]			
Generation 4550: Best = 3.345475e+03	FES = 455100	$\mu_F = 0.748$	$\mu_{CR} = 0.196$	
Diversity = 3.9656e-02	k_t = [1, 1, 1, 1]			
Generation 4560: Best = 3.345396e+03	FES = 456100	$\mu_F = 0.747$	$\mu_{CR} = 0.210$	
Diversity = 3.6623e-02	k_t = [4, 3, 2, 4]			
Generation 4570: Best = 3.345338e+03	FES = 457100	$\mu_F = 0.741$	$\mu_{CR} = 0.205$	
Diversity = 3.8001e-02	k_t = [1, 1, 1, 1]			
Generation 4580: Best = 3.345265e+03	FES = 458100	$\mu_F = 0.741$	$\mu_{CR} = 0.203$	
Diversity = 3.5341e-02	k_t = [4, 4, 2, 4]			
Generation 4590: Best = 3.345225e+03	FES = 459100	$\mu_F = 0.740$	$\mu_{CR} = 0.201$	
Diversity = 3.6130e-02	k_t = [1, 1, 1, 1]			
Generation 4600: Best = 3.345158e+03	FES = 460100	$\mu_F = 0.740$	$\mu_{CR} = 0.196$	
Diversity = 3.3412e-02	k_t = [4, 3, 4, 4]			
Generation 4610: Best = 3.345103e+03	FES = 461100	$\mu_F = 0.744$	$\mu_{CR} = 0.210$	
Diversity = 3.0703e-02	k_t = [1, 1, 1, 1]			
Generation 4620: Best = 3.345059e+03	FES = 462100	$\mu_F = 0.744$	$\mu_{CR} = 0.204$	
Diversity = 3.0537e-02	k_t = [2, 3, 2, 2]			
Generation 4630: Best = 3.345000e+03	FES = 463100	$\mu_F = 0.733$	$\mu_{CR} = 0.203$	
Diversity = 3.1498e-02	k_t = [1, 1, 1, 1]			
Generation 4640: Best = 3.344942e+03	FES = 464100	$\mu_F = 0.735$	$\mu_{CR} = 0.203$	
Diversity = 3.1765e-02	k_t = [2, 3, 2, 3]			
Generation 4650: Best = 3.344864e+03	FES = 465100	$\mu_F = 0.746$	$\mu_{CR} = 0.204$	
Diversity = 3.9439e-02	k_t = [1, 1, 1, 1]			
Generation 4660: Best = 3.344744e+03	FES = 466100	$\mu_F = 0.761$	$\mu_{CR} = 0.200$	
Diversity = 4.7448e-02	k_t = [3, 3, 4, 2]			
Generation 4670: Best = 3.344661e+03	FES = 467100	$\mu_F = 0.761$	$\mu_{CR} = 0.202$	
Diversity = 4.5616e-02	k_t = [1, 1, 1, 1]			
Generation 4680: Best = 3.344443e+03	FES = 468100	$\mu_F = 0.773$	$\mu_{CR} = 0.209$	
Diversity = 4.7577e-02	k_t = [4, 2, 2, 4]			
Generation 4690: Best = 3.344224e+03	FES = 469100	$\mu_F = 0.774$	$\mu_{CR} = 0.211$	
Diversity = 5.0337e-02	k_t = [1, 1, 1, 1]			
Generation 4700: Best = 3.344183e+03	FES = 470100	$\mu_F = 0.778$	$\mu_{CR} = 0.209$	
Diversity = 4.8804e-02	k_t = [4, 2, 2, 3]			
Generation 4710: Best = 3.344059e+03	FES = 471100	$\mu_F = 0.768$	$\mu_{CR} = 0.205$	
Diversity = 5.0455e-02	k_t = [1, 1, 1, 1]			
Generation 4720: Best = 3.343775e+03	FES = 472100	$\mu_F = 0.774$	$\mu_{CR} = 0.206$	
Diversity = 5.4992e-02	k_t = [3, 2, 3, 2]			
Generation 4730: Best = 3.343683e+03	FES = 473100	$\mu_F = 0.783$	$\mu_{CR} = 0.206$	
Diversity = 5.8610e-02	k_t = [1, 1, 1, 1]			
Generation 4740: Best = 3.343586e+03	FES = 474100	$\mu_F = 0.787$	$\mu_{CR} = 0.206$	
Diversity = 5.9395e-02	k_t = [2, 3, 2, 2]			
Generation 4750: Best = 3.343335e+03	FES = 475100	$\mu_F = 0.765$	$\mu_{CR} = 0.200$	
Diversity = 6.3457e-02	k_t = [1, 1, 1, 1]			
Generation 4760: Best = 3.343263e+03	FES = 476100	$\mu_F = 0.756$	$\mu_{CR} = 0.211$	
Diversity = 5.7693e-02	k_t = [3, 2, 3, 4]			
Generation 4770: Best = 3.343120e+03	FES = 477100	$\mu_F = 0.749$	$\mu_{CR} = 0.221$	
Diversity = 5.7538e-02	k_t = [1, 1, 1, 1]			
Generation 4780: Best = 3.342961e+03	FES = 478100	$\mu_F = 0.747$	$\mu_{CR} = 0.213$	
Diversity = 5.6536e-02	k_t = [2, 2, 3, 4]			
Generation 4790: Best = 3.342869e+03	FES = 479100	$\mu_F = 0.741$	$\mu_{CR} = 0.205$	
Diversity = 6.0771e-02	k_t = [1, 1, 1, 1]			
Generation 4800: Best = 3.342733e+03	FES = 480100	$\mu_F = 0.749$	$\mu_{CR} = 0.198$	
Diversity = 6.2722e-02	k_t = [2, 2, 3, 2]			
Generation 4810: Best = 3.342589e+03	FES = 481100	$\mu_F = 0.744$	$\mu_{CR} = 0.197$	

Diversity = 6.8577e-02	k_t = [1, 1, 1, 1]			
Generation 4820: Best = 3.342449e+03	FES = 482100	$\mu_F = 0.737$	$\mu_{CR} = 0.198$	
Diversity = 6.6292e-02	k_t = [2, 2, 2, 3]			
Generation 4830: Best = 3.342264e+03	FES = 483100	$\mu_F = 0.739$	$\mu_{CR} = 0.202$	
Diversity = 7.1415e-02	k_t = [1, 1, 1, 1]			
Generation 4840: Best = 3.342159e+03	FES = 484100	$\mu_F = 0.746$	$\mu_{CR} = 0.193$	
Diversity = 7.0498e-02	k_t = [2, 3, 2, 2]			
Generation 4850: Best = 3.341972e+03	FES = 485100	$\mu_F = 0.748$	$\mu_{CR} = 0.196$	
Diversity = 7.1794e-02	k_t = [1, 1, 1, 1]			
Generation 4860: Best = 3.341861e+03	FES = 486100	$\mu_F = 0.748$	$\mu_{CR} = 0.195$	
Diversity = 6.8524e-02	k_t = [2, 4, 3, 3]			
Generation 4870: Best = 3.341739e+03	FES = 487100	$\mu_F = 0.724$	$\mu_{CR} = 0.196$	
Diversity = 6.7634e-02	k_t = [1, 1, 1, 1]			
Generation 4880: Best = 3.341602e+03	FES = 488100	$\mu_F = 0.733$	$\mu_{CR} = 0.201$	
Diversity = 6.2742e-02	k_t = [3, 4, 4, 2]			
Generation 4890: Best = 3.341496e+03	FES = 489100	$\mu_F = 0.733$	$\mu_{CR} = 0.198$	
Diversity = 6.4264e-02	k_t = [1, 1, 1, 1]			
Generation 4900: Best = 3.341436e+03	FES = 490100	$\mu_F = 0.720$	$\mu_{CR} = 0.191$	
Diversity = 6.2306e-02	k_t = [4, 3, 3, 3]			
Generation 4910: Best = 3.341323e+03	FES = 491100	$\mu_F = 0.728$	$\mu_{CR} = 0.188$	
Diversity = 6.0384e-02	k_t = [1, 1, 1, 1]			
Generation 4920: Best = 3.341228e+03	FES = 492100	$\mu_F = 0.731$	$\mu_{CR} = 0.192$	
Diversity = 5.6856e-02	k_t = [3, 2, 3, 3]			
Generation 4930: Best = 3.341163e+03	FES = 493100	$\mu_F = 0.720$	$\mu_{CR} = 0.190$	
Diversity = 5.4703e-02	k_t = [1, 1, 1, 1]			
Generation 4940: Best = 3.341071e+03	FES = 494100	$\mu_F = 0.716$	$\mu_{CR} = 0.194$	
Diversity = 5.3142e-02	k_t = [4, 4, 2, 3]			
Generation 4950: Best = 3.340922e+03	FES = 495100	$\mu_F = 0.724$	$\mu_{CR} = 0.201$	
Diversity = 6.1208e-02	k_t = [1, 1, 1, 1]			
Generation 4960: Best = 3.340809e+03	FES = 496100	$\mu_F = 0.741$	$\mu_{CR} = 0.204$	
Diversity = 6.1944e-02	k_t = [4, 4, 2, 3]			
Generation 4970: Best = 3.340721e+03	FES = 497100	$\mu_F = 0.748$	$\mu_{CR} = 0.209$	
Diversity = 5.9219e-02	k_t = [1, 1, 1, 1]			
Generation 4980: Best = 3.340618e+03	FES = 498100	$\mu_F = 0.739$	$\mu_{CR} = 0.202$	
Diversity = 5.1380e-02	k_t = [2, 2, 3, 4]			
Generation 4990: Best = 3.340519e+03	FES = 499100	$\mu_F = 0.732$	$\mu_{CR} = 0.190$	
Diversity = 4.9604e-02	k_t = [1, 1, 1, 1]			
Generation 5000: Best = 3.340470e+03	FES = 500100	$\mu_F = 0.723$	$\mu_{CR} = 0.190$	
Diversity = 4.5603e-02	k_t = [2, 3, 2, 4]			
Generation 5010: Best = 3.340397e+03	FES = 501100	$\mu_F = 0.718$	$\mu_{CR} = 0.187$	
Diversity = 4.1265e-02	k_t = [1, 1, 1, 1]			
Generation 5020: Best = 3.340338e+03	FES = 502100	$\mu_F = 0.718$	$\mu_{CR} = 0.188$	
Diversity = 3.7766e-02	k_t = [3, 3, 3, 3]			
Generation 5030: Best = 3.340273e+03	FES = 503100	$\mu_F = 0.731$	$\mu_{CR} = 0.191$	
Diversity = 3.8273e-02	k_t = [1, 1, 1, 1]			
Generation 5040: Best = 3.340175e+03	FES = 504100	$\mu_F = 0.725$	$\mu_{CR} = 0.202$	
Diversity = 3.6299e-02	k_t = [4, 2, 2, 3]			
Generation 5050: Best = 3.340136e+03	FES = 505100	$\mu_F = 0.733$	$\mu_{CR} = 0.211$	
Diversity = 3.6737e-02	k_t = [1, 1, 1, 1]			
Generation 5060: Best = 3.340072e+03	FES = 506100	$\mu_F = 0.726$	$\mu_{CR} = 0.210$	
Diversity = 3.5447e-02	k_t = [4, 2, 4, 3]			
Generation 5070: Best = 3.340012e+03	FES = 507100	$\mu_F = 0.729$	$\mu_{CR} = 0.206$	
Diversity = 3.4988e-02	k_t = [1, 1, 1, 1]			
Generation 5080: Best = 3.339959e+03	FES = 508100	$\mu_F = 0.741$	$\mu_{CR} = 0.205$	

Diversity = 2.9419e-02	k_t = [3, 2, 3, 4]			
Generation 5090: Best = 3.339936e+03	FES = 509100	$\mu_F = 0.742$	$\mu_{CR} = 0.204$	
Diversity = 2.8894e-02	k_t = [1, 1, 1, 1]			
Generation 5100: Best = 3.339883e+03	FES = 510100	$\mu_F = 0.752$	$\mu_{CR} = 0.211$	
Diversity = 2.8262e-02	k_t = [3, 2, 3, 4]			
Generation 5110: Best = 3.339840e+03	FES = 511100	$\mu_F = 0.757$	$\mu_{CR} = 0.206$	
Diversity = 2.7873e-02	k_t = [1, 1, 1, 1]			
Generation 5120: Best = 3.339813e+03	FES = 512100	$\mu_F = 0.769$	$\mu_{CR} = 0.210$	
Diversity = 2.7490e-02	k_t = [2, 4, 3, 3]			
Generation 5130: Best = 3.339760e+03	FES = 513100	$\mu_F = 0.768$	$\mu_{CR} = 0.217$	
Diversity = 3.0766e-02	k_t = [1, 1, 1, 1]			
Generation 5140: Best = 3.339720e+03	FES = 514100	$\mu_F = 0.763$	$\mu_{CR} = 0.213$	
Diversity = 3.2769e-02	k_t = [3, 3, 4, 3]			
Generation 5150: Best = 3.339680e+03	FES = 515100	$\mu_F = 0.753$	$\mu_{CR} = 0.210$	
Diversity = 3.4615e-02	k_t = [1, 1, 1, 1]			
Generation 5160: Best = 3.339614e+03	FES = 516100	$\mu_F = 0.752$	$\mu_{CR} = 0.207$	
Diversity = 3.6129e-02	k_t = [3, 2, 4, 3]			
Generation 5170: Best = 3.339559e+03	FES = 517100	$\mu_F = 0.753$	$\mu_{CR} = 0.215$	
Diversity = 4.1585e-02	k_t = [1, 1, 1, 1]			
Generation 5180: Best = 3.339510e+03	FES = 518100	$\mu_F = 0.753$	$\mu_{CR} = 0.214$	
Diversity = 3.8871e-02	k_t = [2, 2, 3, 2]			
Generation 5190: Best = 3.339449e+03	FES = 519100	$\mu_F = 0.732$	$\mu_{CR} = 0.204$	
Diversity = 3.8280e-02	k_t = [1, 1, 1, 1]			
Generation 5200: Best = 3.339429e+03	FES = 520100	$\mu_F = 0.740$	$\mu_{CR} = 0.201$	
Diversity = 3.5587e-02	k_t = [3, 2, 3, 4]			
Generation 5210: Best = 3.339385e+03	FES = 521100	$\mu_F = 0.742$	$\mu_{CR} = 0.197$	
Diversity = 3.5046e-02	k_t = [1, 1, 1, 1]			
Generation 5220: Best = 3.339343e+03	FES = 522100	$\mu_F = 0.744$	$\mu_{CR} = 0.203$	
Diversity = 3.6816e-02	k_t = [3, 4, 3, 2]			
Generation 5230: Best = 3.339303e+03	FES = 523100	$\mu_F = 0.743$	$\mu_{CR} = 0.199$	
Diversity = 3.8586e-02	k_t = [1, 1, 1, 1]			
Generation 5240: Best = 3.339263e+03	FES = 524100	$\mu_F = 0.735$	$\mu_{CR} = 0.199$	
Diversity = 3.4653e-02	k_t = [3, 4, 3, 2]			
Generation 5250: Best = 3.339190e+03	FES = 525100	$\mu_F = 0.740$	$\mu_{CR} = 0.196$	
Diversity = 3.5872e-02	k_t = [1, 1, 1, 1]			
Generation 5260: Best = 3.339137e+03	FES = 526100	$\mu_F = 0.730$	$\mu_{CR} = 0.209$	
Diversity = 3.7853e-02	k_t = [3, 4, 3, 3]			
Generation 5270: Best = 3.339062e+03	FES = 527100	$\mu_F = 0.733$	$\mu_{CR} = 0.203$	
Diversity = 4.2082e-02	k_t = [1, 1, 1, 1]			
Generation 5280: Best = 3.338993e+03	FES = 528100	$\mu_F = 0.731$	$\mu_{CR} = 0.202$	
Diversity = 4.0266e-02	k_t = [2, 3, 3, 4]			
Generation 5290: Best = 3.338867e+03	FES = 529100	$\mu_F = 0.738$	$\mu_{CR} = 0.208$	
Diversity = 4.4374e-02	k_t = [1, 1, 1, 1]			
Generation 5300: Best = 3.338784e+03	FES = 530100	$\mu_F = 0.748$	$\mu_{CR} = 0.208$	
Diversity = 3.9801e-02	k_t = [4, 2, 3, 4]			
Generation 5310: Best = 3.338738e+03	FES = 531100	$\mu_F = 0.749$	$\mu_{CR} = 0.212$	
Diversity = 3.6930e-02	k_t = [1, 1, 1, 1]			
Generation 5320: Best = 3.338690e+03	FES = 532100	$\mu_F = 0.750$	$\mu_{CR} = 0.223$	
Diversity = 3.6883e-02	k_t = [4, 2, 3, 3]			
Generation 5330: Best = 3.338612e+03	FES = 533100	$\mu_F = 0.749$	$\mu_{CR} = 0.221$	
Diversity = 3.9123e-02	k_t = [1, 1, 1, 1]			
Generation 5340: Best = 3.338566e+03	FES = 534100	$\mu_F = 0.742$	$\mu_{CR} = 0.226$	
Diversity = 3.4776e-02	k_t = [2, 4, 3, 4]			
Generation 5350: Best = 3.338480e+03	FES = 535100	$\mu_F = 0.760$	$\mu_{CR} = 0.229$	

Diversity = 3.8664e-02	k_t = [1, 1, 1, 1]			
Generation 5360: Best = 3.338374e+03	FES = 536100	$\mu_F = 0.755$	$\mu_{CR} = 0.221$	
Diversity = 3.9774e-02	k_t = [3, 2, 4, 4]			
Generation 5370: Best = 3.338276e+03	FES = 537100	$\mu_F = 0.768$	$\mu_{CR} = 0.230$	
Diversity = 4.5856e-02	k_t = [1, 1, 1, 1]			
Generation 5380: Best = 3.338184e+03	FES = 538100	$\mu_F = 0.740$	$\mu_{CR} = 0.222$	
Diversity = 4.5222e-02	k_t = [4, 3, 4, 3]			
Generation 5390: Best = 3.338115e+03	FES = 539100	$\mu_F = 0.755$	$\mu_{CR} = 0.225$	
Diversity = 4.6944e-02	k_t = [1, 1, 1, 1]			
Generation 5400: Best = 3.338022e+03	FES = 540100	$\mu_F = 0.748$	$\mu_{CR} = 0.224$	
Diversity = 4.3598e-02	k_t = [2, 3, 2, 4]			
Generation 5410: Best = 3.337945e+03	FES = 541100	$\mu_F = 0.747$	$\mu_{CR} = 0.217$	
Diversity = 4.5714e-02	k_t = [1, 1, 1, 1]			
Generation 5420: Best = 3.337854e+03	FES = 542100	$\mu_F = 0.748$	$\mu_{CR} = 0.212$	
Diversity = 4.2706e-02	k_t = [4, 3, 4, 4]			
Generation 5430: Best = 3.337740e+03	FES = 543100	$\mu_F = 0.752$	$\mu_{CR} = 0.209$	
Diversity = 4.5859e-02	k_t = [1, 1, 1, 1]			
Generation 5440: Best = 3.337644e+03	FES = 544100	$\mu_F = 0.755$	$\mu_{CR} = 0.218$	
Diversity = 4.5965e-02	k_t = [2, 4, 4, 4]			
Generation 5450: Best = 3.337551e+03	FES = 545100	$\mu_F = 0.760$	$\mu_{CR} = 0.208$	
Diversity = 4.6130e-02	k_t = [1, 1, 1, 1]			
Generation 5460: Best = 3.337459e+03	FES = 546100	$\mu_F = 0.749$	$\mu_{CR} = 0.211$	
Diversity = 4.7185e-02	k_t = [2, 2, 3, 3]			
Generation 5470: Best = 3.337354e+03	FES = 547100	$\mu_F = 0.741$	$\mu_{CR} = 0.216$	
Diversity = 5.4068e-02	k_t = [1, 1, 1, 1]			
Generation 5480: Best = 3.337278e+03	FES = 548100	$\mu_F = 0.739$	$\mu_{CR} = 0.217$	
Diversity = 5.3651e-02	k_t = [2, 4, 4, 3]			
Generation 5490: Best = 3.337146e+03	FES = 549100	$\mu_F = 0.729$	$\mu_{CR} = 0.227$	
Diversity = 5.2837e-02	k_t = [1, 1, 1, 1]			
Generation 5500: Best = 3.337050e+03	FES = 550100	$\mu_F = 0.720$	$\mu_{CR} = 0.223$	
Diversity = 5.3736e-02	k_t = [4, 2, 2, 3]			
Generation 5510: Best = 3.336934e+03	FES = 551100	$\mu_F = 0.720$	$\mu_{CR} = 0.221$	
Diversity = 5.3648e-02	k_t = [1, 1, 1, 1]			
Generation 5520: Best = 3.336786e+03	FES = 552100	$\mu_F = 0.737$	$\mu_{CR} = 0.227$	
Diversity = 5.4917e-02	k_t = [2, 2, 4, 3]			
Generation 5530: Best = 3.336680e+03	FES = 553100	$\mu_F = 0.727$	$\mu_{CR} = 0.235$	
Diversity = 6.1219e-02	k_t = [1, 1, 1, 1]			
Generation 5540: Best = 3.336533e+03	FES = 554100	$\mu_F = 0.726$	$\mu_{CR} = 0.240$	
Diversity = 6.0798e-02	k_t = [2, 2, 2, 2]			
Generation 5550: Best = 3.336431e+03	FES = 555100	$\mu_F = 0.721$	$\mu_{CR} = 0.229$	
Diversity = 6.3730e-02	k_t = [1, 1, 1, 1]			
Generation 5560: Best = 3.336314e+03	FES = 556100	$\mu_F = 0.731$	$\mu_{CR} = 0.220$	
Diversity = 5.8779e-02	k_t = [4, 4, 2, 4]			
Generation 5570: Best = 3.336211e+03	FES = 557100	$\mu_F = 0.722$	$\mu_{CR} = 0.223$	
Diversity = 5.5220e-02	k_t = [1, 1, 1, 1]			
Generation 5580: Best = 3.336179e+03	FES = 558100	$\mu_F = 0.715$	$\mu_{CR} = 0.208$	
Diversity = 4.8309e-02	k_t = [4, 2, 2, 4]			
Generation 5590: Best = 3.336052e+03	FES = 559100	$\mu_F = 0.720$	$\mu_{CR} = 0.209$	
Diversity = 4.5584e-02	k_t = [1, 1, 1, 1]			
Generation 5600: Best = 3.335863e+03	FES = 560100	$\mu_F = 0.729$	$\mu_{CR} = 0.212$	
Diversity = 5.0471e-02	k_t = [2, 2, 2, 4]			
Generation 5610: Best = 3.335724e+03	FES = 561100	$\mu_F = 0.743$	$\mu_{CR} = 0.210$	
Diversity = 5.8213e-02	k_t = [1, 1, 1, 1]			
Generation 5620: Best = 3.335686e+03	FES = 562100	$\mu_F = 0.735$	$\mu_{CR} = 0.209$	

Diversity = 5.2443e-02 k_t = [2, 2, 2, 2]
 Generation 5630: Best = 3.335626e+03 FES = 563100 μ_F = 0.719 μ_{CR} = 0.213
 Diversity = 4.7819e-02 k_t = [1, 1, 1, 1]
 Generation 5640: Best = 3.335551e+03 FES = 564100 μ_F = 0.705 μ_{CR} = 0.217
 Diversity = 4.2474e-02 k_t = [3, 2, 3, 3]
 Generation 5650: Best = 3.335511e+03 FES = 565100 μ_F = 0.703 μ_{CR} = 0.214
 Diversity = 4.1105e-02 k_t = [1, 1, 1, 1]
 Generation 5660: Best = 3.335459e+03 FES = 566100 μ_F = 0.706 μ_{CR} = 0.217
 Diversity = 3.7980e-02 k_t = [2, 3, 2, 2]
 Generation 5670: Best = 3.335394e+03 FES = 567100 μ_F = 0.713 μ_{CR} = 0.204
 Diversity = 3.7841e-02 k_t = [1, 1, 1, 1]
 Generation 5680: Best = 3.335281e+03 FES = 568100 μ_F = 0.720 μ_{CR} = 0.199
 Diversity = 3.8777e-02 k_t = [2, 4, 4, 2]
 Generation 5690: Best = 3.335195e+03 FES = 569100 μ_F = 0.730 μ_{CR} = 0.209
 Diversity = 4.6434e-02 k_t = [1, 1, 1, 1]
 Generation 5700: Best = 3.335117e+03 FES = 570100 μ_F = 0.746 μ_{CR} = 0.210
 Diversity = 4.6407e-02 k_t = [3, 3, 3, 2]
 Generation 5710: Best = 3.334985e+03 FES = 571100 μ_F = 0.755 μ_{CR} = 0.214
 Diversity = 4.7887e-02 k_t = [1, 1, 1, 1]
 Generation 5720: Best = 3.334947e+03 FES = 572100 μ_F = 0.757 μ_{CR} = 0.212
 Diversity = 4.7743e-02 k_t = [2, 4, 3, 3]
 Generation 5730: Best = 3.334854e+03 FES = 573100 μ_F = 0.750 μ_{CR} = 0.222
 Diversity = 4.7779e-02 k_t = [1, 1, 1, 1]
 Generation 5740: Best = 3.334744e+03 FES = 574100 μ_F = 0.747 μ_{CR} = 0.217
 Diversity = 5.4512e-02 k_t = [4, 4, 4, 4]
 Generation 5750: Best = 3.334664e+03 FES = 575100 μ_F = 0.748 μ_{CR} = 0.215
 Diversity = 4.6725e-02 k_t = [1, 1, 1, 1]
 Generation 5760: Best = 3.334596e+03 FES = 576100 μ_F = 0.737 μ_{CR} = 0.209
 Diversity = 4.1452e-02 k_t = [4, 2, 4, 3]
 Generation 5770: Best = 3.334569e+03 FES = 577100 μ_F = 0.742 μ_{CR} = 0.199
 Diversity = 4.0330e-02 k_t = [1, 1, 1, 1]
 Generation 5780: Best = 3.334519e+03 FES = 578100 μ_F = 0.746 μ_{CR} = 0.201
 Diversity = 3.7657e-02 k_t = [3, 3, 2, 2]
 Generation 5790: Best = 3.334454e+03 FES = 579100 μ_F = 0.750 μ_{CR} = 0.196
 Diversity = 3.8740e-02 k_t = [1, 1, 1, 1]
 Generation 5800: Best = 3.334407e+03 FES = 580100 μ_F = 0.745 μ_{CR} = 0.191
 Diversity = 3.6472e-02 k_t = [4, 4, 2, 2]
 Generation 5810: Best = 3.334368e+03 FES = 581100 μ_F = 0.733 μ_{CR} = 0.195
 Diversity = 3.5473e-02 k_t = [1, 1, 1, 1]
 Generation 5820: Best = 3.334324e+03 FES = 582100 μ_F = 0.738 μ_{CR} = 0.199
 Diversity = 3.6369e-02 k_t = [2, 3, 3, 2]
 Generation 5830: Best = 3.334184e+03 FES = 583100 μ_F = 0.756 μ_{CR} = 0.192
 Diversity = 3.9846e-02 k_t = [1, 1, 1, 1]
 Generation 5840: Best = 3.334125e+03 FES = 584100 μ_F = 0.768 μ_{CR} = 0.192
 Diversity = 3.9226e-02 k_t = [2, 4, 4, 3]
 Generation 5850: Best = 3.334069e+03 FES = 585100 μ_F = 0.765 μ_{CR} = 0.187
 Diversity = 3.8716e-02 k_t = [1, 1, 1, 1]
 Generation 5860: Best = 3.333993e+03 FES = 586100 μ_F = 0.762 μ_{CR} = 0.196
 Diversity = 3.5151e-02 k_t = [4, 4, 2, 4]
 Generation 5870: Best = 3.333940e+03 FES = 587100 μ_F = 0.762 μ_{CR} = 0.199
 Diversity = 3.6176e-02 k_t = [1, 1, 1, 1]
 Generation 5880: Best = 3.333886e+03 FES = 588100 μ_F = 0.755 μ_{CR} = 0.192
 Diversity = 3.3610e-02 k_t = [3, 3, 4, 3]
 Generation 5890: Best = 3.333819e+03 FES = 589100 μ_F = 0.737 μ_{CR} = 0.188

Diversity = 3.4562e-02	k_t = [1, 1, 1, 1]			
Generation 5900: Best = 3.333749e+03	FES = 590100	$\mu_F = 0.744$	$\mu_{CR} = 0.191$	
Diversity = 3.6526e-02	k_t = [3, 3, 3, 2]			
Generation 5910: Best = 3.333695e+03	FES = 591100	$\mu_F = 0.740$	$\mu_{CR} = 0.190$	
Diversity = 3.2368e-02	k_t = [1, 1, 1, 1]			
Generation 5920: Best = 3.333667e+03	FES = 592100	$\mu_F = 0.730$	$\mu_{CR} = 0.196$	
Diversity = 2.8815e-02	k_t = [4, 4, 3, 3]			
Generation 5930: Best = 3.333631e+03	FES = 593100	$\mu_F = 0.732$	$\mu_{CR} = 0.196$	
Diversity = 2.5643e-02	k_t = [1, 1, 1, 1]			
Generation 5940: Best = 3.333605e+03	FES = 594100	$\mu_F = 0.727$	$\mu_{CR} = 0.191$	
Diversity = 2.3548e-02	k_t = [2, 4, 3, 3]			
Generation 5950: Best = 3.333572e+03	FES = 595100	$\mu_F = 0.718$	$\mu_{CR} = 0.192$	
Diversity = 2.4055e-02	k_t = [1, 1, 1, 1]			
Generation 5960: Best = 3.333557e+03	FES = 596100	$\mu_F = 0.723$	$\mu_{CR} = 0.205$	
Diversity = 2.2880e-02	k_t = [3, 4, 4, 2]			
Generation 5970: Best = 3.333522e+03	FES = 597100	$\mu_F = 0.722$	$\mu_{CR} = 0.211$	
Diversity = 2.2223e-02	k_t = [1, 1, 1, 1]			
Generation 5980: Best = 3.333505e+03	FES = 598100	$\mu_F = 0.716$	$\mu_{CR} = 0.206$	
Diversity = 2.0333e-02	k_t = [3, 2, 2, 4]			
Generation 5990: Best = 3.333490e+03	FES = 599100	$\mu_F = 0.711$	$\mu_{CR} = 0.209$	
Diversity = 1.9450e-02	k_t = [1, 1, 1, 1]			
Generation 6000: Best = 3.333455e+03	FES = 600100	$\mu_F = 0.718$	$\mu_{CR} = 0.207$	
Diversity = 1.9271e-02	k_t = [2, 3, 3, 2]			
Generation 6010: Best = 3.333441e+03	FES = 601100	$\mu_F = 0.711$	$\mu_{CR} = 0.219$	
Diversity = 1.9893e-02	k_t = [1, 1, 1, 1]			
Generation 6020: Best = 3.333425e+03	FES = 602100	$\mu_F = 0.710$	$\mu_{CR} = 0.216$	
Diversity = 1.6946e-02	k_t = [2, 4, 4, 3]			
Generation 6030: Best = 3.333411e+03	FES = 603100	$\mu_F = 0.710$	$\mu_{CR} = 0.221$	
Diversity = 1.6019e-02	k_t = [1, 1, 1, 1]			
Generation 6040: Best = 3.333397e+03	FES = 604100	$\mu_F = 0.702$	$\mu_{CR} = 0.222$	
Diversity = 1.5466e-02	k_t = [4, 3, 2, 2]			
Generation 6050: Best = 3.333381e+03	FES = 605100	$\mu_F = 0.697$	$\mu_{CR} = 0.216$	
Diversity = 1.4712e-02	k_t = [1, 1, 1, 1]			
Generation 6060: Best = 3.333346e+03	FES = 606100	$\mu_F = 0.718$	$\mu_{CR} = 0.224$	
Diversity = 1.6245e-02	k_t = [3, 2, 3, 4]			
Generation 6070: Best = 3.333304e+03	FES = 607100	$\mu_F = 0.742$	$\mu_{CR} = 0.230$	
Diversity = 1.8936e-02	k_t = [1, 1, 1, 1]			
Generation 6080: Best = 3.333287e+03	FES = 608100	$\mu_F = 0.747$	$\mu_{CR} = 0.220$	
Diversity = 1.7921e-02	k_t = [2, 4, 3, 2]			
Generation 6090: Best = 3.333259e+03	FES = 609100	$\mu_F = 0.739$	$\mu_{CR} = 0.233$	
Diversity = 1.7479e-02	k_t = [1, 1, 1, 1]			
Generation 6100: Best = 3.333237e+03	FES = 610100	$\mu_F = 0.740$	$\mu_{CR} = 0.233$	
Diversity = 1.6772e-02	k_t = [2, 3, 2, 2]			
Generation 6110: Best = 3.333223e+03	FES = 611100	$\mu_F = 0.733$	$\mu_{CR} = 0.240$	
Diversity = 1.6849e-02	k_t = [1, 1, 1, 1]			
Generation 6120: Best = 3.333205e+03	FES = 612100	$\mu_F = 0.723$	$\mu_{CR} = 0.243$	
Diversity = 1.6300e-02	k_t = [2, 3, 3, 2]			
Generation 6130: Best = 3.333187e+03	FES = 613100	$\mu_F = 0.715$	$\mu_{CR} = 0.236$	
Diversity = 1.5401e-02	k_t = [1, 1, 1, 1]			
Generation 6140: Best = 3.333175e+03	FES = 614100	$\mu_F = 0.713$	$\mu_{CR} = 0.234$	
Diversity = 1.3710e-02	k_t = [4, 4, 4, 3]			
Generation 6150: Best = 3.333161e+03	FES = 615100	$\mu_F = 0.718$	$\mu_{CR} = 0.240$	
Diversity = 1.3849e-02	k_t = [1, 1, 1, 1]			
Generation 6160: Best = 3.333150e+03	FES = 616100	$\mu_F = 0.727$	$\mu_{CR} = 0.236$	

Diversity = 1.3821e-02	k_t = [4, 3, 2, 2]			
Generation 6170: Best = 3.333143e+03	FES = 617100	$\mu_F = 0.707$	$\mu_{CR} = 0.221$	
Diversity = 1.3887e-02	k_t = [1, 1, 1, 1]			
Generation 6180: Best = 3.333132e+03	FES = 618100	$\mu_F = 0.712$	$\mu_{CR} = 0.222$	
Diversity = 1.3298e-02	k_t = [2, 3, 2, 4]			
Generation 6190: Best = 3.333121e+03	FES = 619100	$\mu_F = 0.713$	$\mu_{CR} = 0.215$	
Diversity = 1.2022e-02	k_t = [1, 1, 1, 1]			
Generation 6200: Best = 3.333112e+03	FES = 620100	$\mu_F = 0.722$	$\mu_{CR} = 0.208$	
Diversity = 1.2091e-02	k_t = [2, 2, 2, 2]			
Generation 6210: Best = 3.333106e+03	FES = 621100	$\mu_F = 0.733$	$\mu_{CR} = 0.205$	
Diversity = 1.3150e-02	k_t = [1, 1, 1, 1]			
Generation 6220: Best = 3.333098e+03	FES = 622100	$\mu_F = 0.745$	$\mu_{CR} = 0.200$	
Diversity = 1.3182e-02	k_t = [4, 4, 2, 2]			
Generation 6230: Best = 3.333090e+03	FES = 623100	$\mu_F = 0.767$	$\mu_{CR} = 0.188$	
Diversity = 1.4233e-02	k_t = [1, 1, 1, 1]			
Generation 6240: Best = 3.333081e+03	FES = 624100	$\mu_F = 0.766$	$\mu_{CR} = 0.181$	
Diversity = 1.4168e-02	k_t = [2, 2, 2, 2]			
Generation 6250: Best = 3.333075e+03	FES = 625100	$\mu_F = 0.773$	$\mu_{CR} = 0.168$	
Diversity = 1.4180e-02	k_t = [1, 1, 1, 1]			
Generation 6260: Best = 3.333067e+03	FES = 626100	$\mu_F = 0.757$	$\mu_{CR} = 0.163$	
Diversity = 1.3460e-02	k_t = [4, 4, 4, 2]			
Generation 6270: Best = 3.333057e+03	FES = 627100	$\mu_F = 0.755$	$\mu_{CR} = 0.164$	
Diversity = 1.3934e-02	k_t = [1, 1, 1, 1]			
Generation 6280: Best = 3.333042e+03	FES = 628100	$\mu_F = 0.761$	$\mu_{CR} = 0.158$	
Diversity = 1.3387e-02	k_t = [4, 2, 4, 2]			
Generation 6290: Best = 3.333035e+03	FES = 629100	$\mu_F = 0.769$	$\mu_{CR} = 0.164$	
Diversity = 1.4540e-02	k_t = [1, 1, 1, 1]			
Generation 6300: Best = 3.333019e+03	FES = 630100	$\mu_F = 0.762$	$\mu_{CR} = 0.160$	
Diversity = 1.4355e-02	k_t = [2, 4, 4, 2]			
Generation 6310: Best = 3.333013e+03	FES = 631100	$\mu_F = 0.760$	$\mu_{CR} = 0.163$	
Diversity = 1.3916e-02	k_t = [1, 1, 1, 1]			
Generation 6320: Best = 3.333012e+03	FES = 632100	$\mu_F = 0.762$	$\mu_{CR} = 0.162$	
Diversity = 1.4808e-02	k_t = [3, 4, 3, 2]			
Generation 6330: Best = 3.332999e+03	FES = 633100	$\mu_F = 0.760$	$\mu_{CR} = 0.161$	
Diversity = 1.4471e-02	k_t = [1, 1, 1, 1]			
Generation 6340: Best = 3.332990e+03	FES = 634100	$\mu_F = 0.769$	$\mu_{CR} = 0.159$	
Diversity = 1.3460e-02	k_t = [2, 3, 4, 4]			
Generation 6350: Best = 3.332965e+03	FES = 635100	$\mu_F = 0.774$	$\mu_{CR} = 0.159$	
Diversity = 1.3042e-02	k_t = [1, 1, 1, 1]			
Generation 6360: Best = 3.332950e+03	FES = 636100	$\mu_F = 0.773$	$\mu_{CR} = 0.165$	
Diversity = 1.3743e-02	k_t = [4, 2, 2, 4]			
Generation 6370: Best = 3.332923e+03	FES = 637100	$\mu_F = 0.791$	$\mu_{CR} = 0.182$	
Diversity = 1.5939e-02	k_t = [1, 1, 1, 1]			
Generation 6380: Best = 3.332876e+03	FES = 638100	$\mu_F = 0.797$	$\mu_{CR} = 0.189$	
Diversity = 1.7755e-02	k_t = [2, 4, 4, 2]			
Generation 6390: Best = 3.332837e+03	FES = 639100	$\mu_F = 0.803$	$\mu_{CR} = 0.186$	
Diversity = 1.9725e-02	k_t = [1, 1, 1, 1]			
Generation 6400: Best = 3.332802e+03	FES = 640100	$\mu_F = 0.795$	$\mu_{CR} = 0.193$	
Diversity = 2.1836e-02	k_t = [4, 4, 2, 2]			
Generation 6410: Best = 3.332777e+03	FES = 641100	$\mu_F = 0.783$	$\mu_{CR} = 0.191$	
Diversity = 2.3854e-02	k_t = [1, 1, 1, 1]			
Generation 6420: Best = 3.332756e+03	FES = 642100	$\mu_F = 0.766$	$\mu_{CR} = 0.189$	
Diversity = 2.2453e-02	k_t = [4, 4, 3, 4]			
Generation 6430: Best = 3.332722e+03	FES = 643100	$\mu_F = 0.750$	$\mu_{CR} = 0.184$	

Diversity = 2.2981e-02	k_t = [1, 1, 1, 1]			
Generation 6440: Best = 3.332662e+03	FES = 644100	$\mu_F = 0.758$	$\mu_{CR} = 0.188$	
Diversity = 2.4398e-02	k_t = [2, 4, 4, 4]			
Generation 6450: Best = 3.332633e+03	FES = 645100	$\mu_F = 0.763$	$\mu_{CR} = 0.184$	
Diversity = 2.6695e-02	k_t = [1, 1, 1, 1]			
Generation 6460: Best = 3.332606e+03	FES = 646100	$\mu_F = 0.768$	$\mu_{CR} = 0.182$	
Diversity = 2.6941e-02	k_t = [4, 4, 4, 3]			
Generation 6470: Best = 3.332572e+03	FES = 647100	$\mu_F = 0.765$	$\mu_{CR} = 0.178$	
Diversity = 2.7555e-02	k_t = [1, 1, 1, 1]			
Generation 6480: Best = 3.332515e+03	FES = 648100	$\mu_F = 0.759$	$\mu_{CR} = 0.173$	
Diversity = 2.7024e-02	k_t = [3, 2, 2, 2]			
Generation 6490: Best = 3.332510e+03	FES = 649100	$\mu_F = 0.762$	$\mu_{CR} = 0.180$	
Diversity = 2.7044e-02	k_t = [1, 1, 1, 1]			
Generation 6500: Best = 3.332475e+03	FES = 650100	$\mu_F = 0.742$	$\mu_{CR} = 0.184$	
Diversity = 2.4736e-02	k_t = [3, 3, 2, 4]			
Generation 6510: Best = 3.332453e+03	FES = 651100	$\mu_F = 0.728$	$\mu_{CR} = 0.184$	
Diversity = 2.5948e-02	k_t = [1, 1, 1, 1]			
Generation 6520: Best = 3.332434e+03	FES = 652100	$\mu_F = 0.728$	$\mu_{CR} = 0.190$	
Diversity = 2.4519e-02	k_t = [2, 4, 4, 3]			
Generation 6530: Best = 3.332407e+03	FES = 653100	$\mu_F = 0.716$	$\mu_{CR} = 0.183$	
Diversity = 2.2785e-02	k_t = [1, 1, 1, 1]			
Generation 6540: Best = 3.332375e+03	FES = 654100	$\mu_F = 0.722$	$\mu_{CR} = 0.185$	
Diversity = 2.0757e-02	k_t = [4, 3, 4, 3]			
Generation 6550: Best = 3.332364e+03	FES = 655100	$\mu_F = 0.726$	$\mu_{CR} = 0.183$	
Diversity = 1.8957e-02	k_t = [1, 1, 1, 1]			
Generation 6560: Best = 3.332346e+03	FES = 656100	$\mu_F = 0.723$	$\mu_{CR} = 0.182$	
Diversity = 1.7289e-02	k_t = [4, 2, 4, 4]			
Generation 6570: Best = 3.332330e+03	FES = 657100	$\mu_F = 0.724$	$\mu_{CR} = 0.192$	
Diversity = 1.6001e-02	k_t = [1, 1, 1, 1]			
Generation 6580: Best = 3.332319e+03	FES = 658100	$\mu_F = 0.723$	$\mu_{CR} = 0.198$	
Diversity = 1.5315e-02	k_t = [3, 3, 4, 2]			
Generation 6590: Best = 3.332300e+03	FES = 659100	$\mu_F = 0.725$	$\mu_{CR} = 0.204$	
Diversity = 1.6123e-02	k_t = [1, 1, 1, 1]			
Generation 6600: Best = 3.332273e+03	FES = 660100	$\mu_F = 0.731$	$\mu_{CR} = 0.199$	
Diversity = 1.6692e-02	k_t = [2, 2, 3, 3]			
Generation 6610: Best = 3.332250e+03	FES = 661100	$\mu_F = 0.748$	$\mu_{CR} = 0.202$	
Diversity = 1.8842e-02	k_t = [1, 1, 1, 1]			
Generation 6620: Best = 3.332217e+03	FES = 662100	$\mu_F = 0.747$	$\mu_{CR} = 0.196$	
Diversity = 2.0319e-02	k_t = [4, 3, 4, 4]			
Generation 6630: Best = 3.332121e+03	FES = 663100	$\mu_F = 0.757$	$\mu_{CR} = 0.194$	
Diversity = 2.3298e-02	k_t = [1, 1, 1, 1]			
Generation 6640: Best = 3.332085e+03	FES = 664100	$\mu_F = 0.759$	$\mu_{CR} = 0.188$	
Diversity = 2.3923e-02	k_t = [3, 4, 2, 3]			
Generation 6650: Best = 3.332048e+03	FES = 665100	$\mu_F = 0.762$	$\mu_{CR} = 0.195$	
Diversity = 2.3226e-02	k_t = [1, 1, 1, 1]			
Generation 6660: Best = 3.332009e+03	FES = 666100	$\mu_F = 0.764$	$\mu_{CR} = 0.200$	
Diversity = 2.1347e-02	k_t = [3, 4, 2, 4]			
Generation 6670: Best = 3.331968e+03	FES = 667100	$\mu_F = 0.785$	$\mu_{CR} = 0.195$	
Diversity = 2.4114e-02	k_t = [1, 1, 1, 1]			
Generation 6680: Best = 3.331900e+03	FES = 668100	$\mu_F = 0.787$	$\mu_{CR} = 0.205$	
Diversity = 2.8736e-02	k_t = [3, 4, 4, 4]			
Generation 6690: Best = 3.331855e+03	FES = 669100	$\mu_F = 0.772$	$\mu_{CR} = 0.192$	
Diversity = 2.8107e-02	k_t = [1, 1, 1, 1]			
Generation 6700: Best = 3.331819e+03	FES = 670100	$\mu_F = 0.776$	$\mu_{CR} = 0.191$	

Diversity = 2.6829e-02	k_t = [3, 3, 3, 4]			
Generation 6710: Best = 3.331769e+03	FES = 671100	$\mu_F = 0.774$	$\mu_{CR} = 0.193$	
Diversity = 2.4689e-02	k_t = [1, 1, 1, 1]			
Generation 6720: Best = 3.331747e+03	FES = 672100	$\mu_F = 0.775$	$\mu_{CR} = 0.197$	
Diversity = 2.5664e-02	k_t = [4, 2, 4, 2]			
Generation 6730: Best = 3.331703e+03	FES = 673100	$\mu_F = 0.772$	$\mu_{CR} = 0.196$	
Diversity = 2.5581e-02	k_t = [1, 1, 1, 1]			
Generation 6740: Best = 3.331668e+03	FES = 674100	$\mu_F = 0.778$	$\mu_{CR} = 0.190$	
Diversity = 2.6375e-02	k_t = [2, 3, 2, 3]			
Generation 6750: Best = 3.331635e+03	FES = 675100	$\mu_F = 0.768$	$\mu_{CR} = 0.189$	
Diversity = 2.8269e-02	k_t = [1, 1, 1, 1]			
Generation 6760: Best = 3.331609e+03	FES = 676100	$\mu_F = 0.756$	$\mu_{CR} = 0.187$	
Diversity = 2.8132e-02	k_t = [2, 4, 4, 2]			
Generation 6770: Best = 3.331586e+03	FES = 677100	$\mu_F = 0.723$	$\mu_{CR} = 0.185$	
Diversity = 2.5548e-02	k_t = [1, 1, 1, 1]			
Generation 6780: Best = 3.331553e+03	FES = 678100	$\mu_F = 0.708$	$\mu_{CR} = 0.191$	
Diversity = 2.3570e-02	k_t = [2, 2, 2, 2]			
Generation 6790: Best = 3.331531e+03	FES = 679100	$\mu_F = 0.715$	$\mu_{CR} = 0.181$	
Diversity = 2.2837e-02	k_t = [1, 1, 1, 1]			
Generation 6800: Best = 3.331516e+03	FES = 680100	$\mu_F = 0.717$	$\mu_{CR} = 0.185$	
Diversity = 1.9171e-02	k_t = [3, 2, 3, 4]			
Generation 6810: Best = 3.331499e+03	FES = 681100	$\mu_F = 0.721$	$\mu_{CR} = 0.176$	
Diversity = 1.7768e-02	k_t = [1, 1, 1, 1]			
Generation 6820: Best = 3.331481e+03	FES = 682100	$\mu_F = 0.735$	$\mu_{CR} = 0.183$	
Diversity = 1.7141e-02	k_t = [2, 4, 4, 2]			
Generation 6830: Best = 3.331470e+03	FES = 683100	$\mu_F = 0.734$	$\mu_{CR} = 0.187$	
Diversity = 1.7348e-02	k_t = [1, 1, 1, 1]			
Generation 6840: Best = 3.331459e+03	FES = 684100	$\mu_F = 0.731$	$\mu_{CR} = 0.177$	
Diversity = 1.6066e-02	k_t = [4, 3, 3, 4]			
Generation 6850: Best = 3.331448e+03	FES = 685100	$\mu_F = 0.732$	$\mu_{CR} = 0.175$	
Diversity = 1.5331e-02	k_t = [1, 1, 1, 1]			
Generation 6860: Best = 3.331439e+03	FES = 686100	$\mu_F = 0.712$	$\mu_{CR} = 0.186$	
Diversity = 1.4174e-02	k_t = [2, 4, 3, 3]			
Generation 6870: Best = 3.331427e+03	FES = 687100	$\mu_F = 0.723$	$\mu_{CR} = 0.197$	
Diversity = 1.4675e-02	k_t = [1, 1, 1, 1]			
Generation 6880: Best = 3.331418e+03	FES = 688100	$\mu_F = 0.728$	$\mu_{CR} = 0.201$	
Diversity = 1.4254e-02	k_t = [3, 4, 3, 2]			
Generation 6890: Best = 3.331407e+03	FES = 689100	$\mu_F = 0.722$	$\mu_{CR} = 0.197$	
Diversity = 1.3776e-02	k_t = [1, 1, 1, 1]			
Generation 6900: Best = 3.331397e+03	FES = 690100	$\mu_F = 0.723$	$\mu_{CR} = 0.196$	
Diversity = 1.2929e-02	k_t = [3, 3, 4, 3]			
Generation 6910: Best = 3.331390e+03	FES = 691100	$\mu_F = 0.728$	$\mu_{CR} = 0.197$	
Diversity = 1.1818e-02	k_t = [1, 1, 1, 1]			
Generation 6920: Best = 3.331381e+03	FES = 692100	$\mu_F = 0.735$	$\mu_{CR} = 0.206$	
Diversity = 1.2088e-02	k_t = [2, 2, 3, 3]			
Generation 6930: Best = 3.331372e+03	FES = 693100	$\mu_F = 0.743$	$\mu_{CR} = 0.204$	
Diversity = 1.3000e-02	k_t = [1, 1, 1, 1]			
Generation 6940: Best = 3.331365e+03	FES = 694100	$\mu_F = 0.734$	$\mu_{CR} = 0.208$	
Diversity = 1.2810e-02	k_t = [3, 3, 4, 2]			
Generation 6950: Best = 3.331359e+03	FES = 695100	$\mu_F = 0.725$	$\mu_{CR} = 0.205$	
Diversity = 1.2801e-02	k_t = [1, 1, 1, 1]			
Generation 6960: Best = 3.331350e+03	FES = 696100	$\mu_F = 0.734$	$\mu_{CR} = 0.200$	
Diversity = 1.1926e-02	k_t = [2, 2, 4, 3]			
Generation 6970: Best = 3.331340e+03	FES = 697100	$\mu_F = 0.733$	$\mu_{CR} = 0.202$	

Diversity = 1.3241e-02	k_t = [1, 1, 1, 1]			
Generation 6980: Best = 3.331321e+03	FES = 698100	$\mu_F = 0.740$	$\mu_{CR} = 0.204$	
Diversity = 1.6683e-02	k_t = [4, 2, 3, 3]			
Generation 6990: Best = 3.331303e+03	FES = 699100	$\mu_F = 0.741$	$\mu_{CR} = 0.205$	
Diversity = 1.7550e-02	k_t = [1, 1, 1, 1]			
Generation 7000: Best = 3.331287e+03	FES = 700100	$\mu_F = 0.754$	$\mu_{CR} = 0.210$	
Diversity = 1.5456e-02	k_t = [2, 3, 3, 4]			
Generation 7010: Best = 3.331265e+03	FES = 701100	$\mu_F = 0.756$	$\mu_{CR} = 0.207$	
Diversity = 1.7071e-02	k_t = [1, 1, 1, 1]			
Generation 7020: Best = 3.331218e+03	FES = 702100	$\mu_F = 0.767$	$\mu_{CR} = 0.207$	
Diversity = 2.0819e-02	k_t = [3, 3, 2, 4]			
Generation 7030: Best = 3.331193e+03	FES = 703100	$\mu_F = 0.762$	$\mu_{CR} = 0.201$	
Diversity = 2.2825e-02	k_t = [1, 1, 1, 1]			
Generation 7040: Best = 3.331156e+03	FES = 704100	$\mu_F = 0.761$	$\mu_{CR} = 0.207$	
Diversity = 2.4855e-02	k_t = [2, 4, 3, 2]			
Generation 7050: Best = 3.331125e+03	FES = 705100	$\mu_F = 0.750$	$\mu_{CR} = 0.194$	
Diversity = 2.5781e-02	k_t = [1, 1, 1, 1]			
Generation 7060: Best = 3.331085e+03	FES = 706100	$\mu_F = 0.736$	$\mu_{CR} = 0.196$	
Diversity = 2.2778e-02	k_t = [3, 3, 3, 4]			
Generation 7070: Best = 3.331074e+03	FES = 707100	$\mu_F = 0.746$	$\mu_{CR} = 0.192$	
Diversity = 2.3284e-02	k_t = [1, 1, 1, 1]			
Generation 7080: Best = 3.331034e+03	FES = 708100	$\mu_F = 0.753$	$\mu_{CR} = 0.188$	
Diversity = 2.3079e-02	k_t = [2, 3, 3, 3]			
Generation 7090: Best = 3.330981e+03	FES = 709100	$\mu_F = 0.750$	$\mu_{CR} = 0.197$	
Diversity = 2.9464e-02	k_t = [1, 1, 1, 1]			
Generation 7100: Best = 3.330938e+03	FES = 710100	$\mu_F = 0.743$	$\mu_{CR} = 0.192$	
Diversity = 2.7094e-02	k_t = [3, 4, 3, 4]			
Generation 7110: Best = 3.330897e+03	FES = 711100	$\mu_F = 0.746$	$\mu_{CR} = 0.190$	
Diversity = 2.5830e-02	k_t = [1, 1, 1, 1]			
Generation 7120: Best = 3.330861e+03	FES = 712100	$\mu_F = 0.737$	$\mu_{CR} = 0.191$	
Diversity = 2.4158e-02	k_t = [4, 4, 4, 3]			
Generation 7130: Best = 3.330840e+03	FES = 713100	$\mu_F = 0.744$	$\mu_{CR} = 0.191$	
Diversity = 2.6100e-02	k_t = [1, 1, 1, 1]			
Generation 7140: Best = 3.330808e+03	FES = 714100	$\mu_F = 0.733$	$\mu_{CR} = 0.188$	
Diversity = 2.3009e-02	k_t = [4, 3, 4, 4]			
Generation 7150: Best = 3.330800e+03	FES = 715100	$\mu_F = 0.733$	$\mu_{CR} = 0.192$	
Diversity = 2.0492e-02	k_t = [1, 1, 1, 1]			
Generation 7160: Best = 3.330768e+03	FES = 716100	$\mu_F = 0.738$	$\mu_{CR} = 0.193$	
Diversity = 2.1692e-02	k_t = [2, 4, 3, 4]			
Generation 7170: Best = 3.330739e+03	FES = 717100	$\mu_F = 0.746$	$\mu_{CR} = 0.194$	
Diversity = 2.7475e-02	k_t = [1, 1, 1, 1]			
Generation 7180: Best = 3.330707e+03	FES = 718100	$\mu_F = 0.744$	$\mu_{CR} = 0.192$	
Diversity = 3.1506e-02	k_t = [2, 2, 2, 2]			
Generation 7190: Best = 3.330654e+03	FES = 719100	$\mu_F = 0.733$	$\mu_{CR} = 0.194$	
Diversity = 3.1545e-02	k_t = [1, 1, 1, 1]			
Generation 7200: Best = 3.330635e+03	FES = 720100	$\mu_F = 0.725$	$\mu_{CR} = 0.195$	
Diversity = 2.6833e-02	k_t = [3, 3, 2, 3]			
Generation 7210: Best = 3.330594e+03	FES = 721100	$\mu_F = 0.730$	$\mu_{CR} = 0.196$	
Diversity = 2.5041e-02	k_t = [1, 1, 1, 1]			
Generation 7220: Best = 3.330570e+03	FES = 722100	$\mu_F = 0.738$	$\mu_{CR} = 0.196$	
Diversity = 2.2935e-02	k_t = [3, 2, 4, 2]			
Generation 7230: Best = 3.330546e+03	FES = 723100	$\mu_F = 0.742$	$\mu_{CR} = 0.196$	
Diversity = 2.2169e-02	k_t = [1, 1, 1, 1]			
Generation 7240: Best = 3.330517e+03	FES = 724100	$\mu_F = 0.745$	$\mu_{CR} = 0.194$	

Diversity = 2.0715e-02	k_t = [4, 4, 4, 2]			
Generation 7250: Best = 3.330486e+03	FES = 725100	$\mu_F = 0.748$	$\mu_{CR} = 0.212$	
Diversity = 1.9810e-02	k_t = [1, 1, 1, 1]			
Generation 7260: Best = 3.330471e+03	FES = 726100	$\mu_F = 0.756$	$\mu_{CR} = 0.206$	
Diversity = 2.0640e-02	k_t = [3, 2, 2, 3]			
Generation 7270: Best = 3.330451e+03	FES = 727100	$\mu_F = 0.749$	$\mu_{CR} = 0.212$	
Diversity = 2.0062e-02	k_t = [1, 1, 1, 1]			
Generation 7280: Best = 3.330426e+03	FES = 728100	$\mu_F = 0.747$	$\mu_{CR} = 0.213$	
Diversity = 1.8871e-02	k_t = [4, 2, 3, 3]			
Generation 7290: Best = 3.330414e+03	FES = 729100	$\mu_F = 0.725$	$\mu_{CR} = 0.212$	
Diversity = 1.8497e-02	k_t = [1, 1, 1, 1]			
Generation 7300: Best = 3.330385e+03	FES = 730100	$\mu_F = 0.730$	$\mu_{CR} = 0.207$	
Diversity = 1.8675e-02	k_t = [3, 2, 4, 3]			
Generation 7310: Best = 3.330363e+03	FES = 731100	$\mu_F = 0.733$	$\mu_{CR} = 0.208$	
Diversity = 1.8413e-02	k_t = [1, 1, 1, 1]			
Generation 7320: Best = 3.330341e+03	FES = 732100	$\mu_F = 0.733$	$\mu_{CR} = 0.199$	
Diversity = 1.6886e-02	k_t = [3, 2, 2, 3]			
Generation 7330: Best = 3.330324e+03	FES = 733100	$\mu_F = 0.728$	$\mu_{CR} = 0.199$	
Diversity = 1.7194e-02	k_t = [1, 1, 1, 1]			
Generation 7340: Best = 3.330300e+03	FES = 734100	$\mu_F = 0.731$	$\mu_{CR} = 0.201$	
Diversity = 1.8735e-02	k_t = [4, 3, 3, 2]			
Generation 7350: Best = 3.330262e+03	FES = 735100	$\mu_F = 0.740$	$\mu_{CR} = 0.191$	
Diversity = 2.2774e-02	k_t = [1, 1, 1, 1]			
Generation 7360: Best = 3.330243e+03	FES = 736100	$\mu_F = 0.744$	$\mu_{CR} = 0.193$	
Diversity = 1.8230e-02	k_t = [4, 3, 4, 3]			
Generation 7370: Best = 3.330226e+03	FES = 737100	$\mu_F = 0.746$	$\mu_{CR} = 0.191$	
Diversity = 1.7988e-02	k_t = [1, 1, 1, 1]			
Generation 7380: Best = 3.330209e+03	FES = 738100	$\mu_F = 0.732$	$\mu_{CR} = 0.199$	
Diversity = 1.6478e-02	k_t = [2, 3, 4, 4]			
Generation 7390: Best = 3.330196e+03	FES = 739100	$\mu_F = 0.741$	$\mu_{CR} = 0.199$	
Diversity = 1.6081e-02	k_t = [1, 1, 1, 1]			
Generation 7400: Best = 3.330189e+03	FES = 740100	$\mu_F = 0.732$	$\mu_{CR} = 0.184$	
Diversity = 1.5765e-02	k_t = [2, 4, 2, 2]			
Generation 7410: Best = 3.330174e+03	FES = 741100	$\mu_F = 0.726$	$\mu_{CR} = 0.188$	
Diversity = 1.5592e-02	k_t = [1, 1, 1, 1]			
Generation 7420: Best = 3.330162e+03	FES = 742100	$\mu_F = 0.732$	$\mu_{CR} = 0.195$	
Diversity = 1.5252e-02	k_t = [2, 4, 3, 3]			
Generation 7430: Best = 3.330148e+03	FES = 743100	$\mu_F = 0.732$	$\mu_{CR} = 0.198$	
Diversity = 1.6576e-02	k_t = [1, 1, 1, 1]			
Generation 7440: Best = 3.330141e+03	FES = 744100	$\mu_F = 0.718$	$\mu_{CR} = 0.193$	
Diversity = 1.6578e-02	k_t = [3, 4, 3, 2]			
Generation 7450: Best = 3.330129e+03	FES = 745100	$\mu_F = 0.720$	$\mu_{CR} = 0.194$	
Diversity = 1.6341e-02	k_t = [1, 1, 1, 1]			
Generation 7460: Best = 3.330116e+03	FES = 746100	$\mu_F = 0.724$	$\mu_{CR} = 0.195$	
Diversity = 1.4737e-02	k_t = [2, 3, 2, 4]			
Generation 7470: Best = 3.330102e+03	FES = 747100	$\mu_F = 0.739$	$\mu_{CR} = 0.196$	
Diversity = 1.6228e-02	k_t = [1, 1, 1, 1]			
Generation 7480: Best = 3.330077e+03	FES = 748100	$\mu_F = 0.743$	$\mu_{CR} = 0.191$	
Diversity = 1.6194e-02	k_t = [4, 4, 3, 4]			
Generation 7490: Best = 3.330064e+03	FES = 749100	$\mu_F = 0.746$	$\mu_{CR} = 0.197$	
Diversity = 1.6381e-02	k_t = [1, 1, 1, 1]			
Generation 7500: Best = 3.330013e+03	FES = 750100	$\mu_F = 0.762$	$\mu_{CR} = 0.203$	
Diversity = 1.9196e-02	k_t = [2, 4, 4, 3]			
Generation 7510: Best = 3.329912e+03	FES = 751100	$\mu_F = 0.775$	$\mu_{CR} = 0.200$	

Diversity = 2.8149e-02	k_t = [1, 1, 1, 1]			
Generation 7520: Best = 3.329879e+03	FES = 752100	$\mu_F = 0.755$	$\mu_{CR} = 0.200$	
Diversity = 2.6839e-02	k_t = [2, 4, 2, 2]			
Generation 7530: Best = 3.329832e+03	FES = 753100	$\mu_F = 0.757$	$\mu_{CR} = 0.206$	
Diversity = 2.7216e-02	k_t = [1, 1, 1, 1]			
Generation 7540: Best = 3.329811e+03	FES = 754100	$\mu_F = 0.749$	$\mu_{CR} = 0.209$	
Diversity = 2.7225e-02	k_t = [3, 2, 2, 2]			
Generation 7550: Best = 3.329749e+03	FES = 755100	$\mu_F = 0.742$	$\mu_{CR} = 0.214$	
Diversity = 2.6302e-02	k_t = [1, 1, 1, 1]			
Generation 7560: Best = 3.329737e+03	FES = 756100	$\mu_F = 0.739$	$\mu_{CR} = 0.204$	
Diversity = 2.3081e-02	k_t = [3, 4, 4, 2]			
Generation 7570: Best = 3.329710e+03	FES = 757100	$\mu_F = 0.755$	$\mu_{CR} = 0.206$	
Diversity = 2.2906e-02	k_t = [1, 1, 1, 1]			
Generation 7580: Best = 3.329692e+03	FES = 758100	$\mu_F = 0.745$	$\mu_{CR} = 0.194$	
Diversity = 2.3201e-02	k_t = [3, 4, 3, 2]			
Generation 7590: Best = 3.329616e+03	FES = 759100	$\mu_F = 0.753$	$\mu_{CR} = 0.206$	
Diversity = 3.1022e-02	k_t = [1, 1, 1, 1]			
Generation 7600: Best = 3.329578e+03	FES = 760100	$\mu_F = 0.749$	$\mu_{CR} = 0.209$	
Diversity = 2.9991e-02	k_t = [4, 2, 2, 2]			
Generation 7610: Best = 3.329550e+03	FES = 761100	$\mu_F = 0.753$	$\mu_{CR} = 0.206$	
Diversity = 2.8544e-02	k_t = [1, 1, 1, 1]			
Generation 7620: Best = 3.329506e+03	FES = 762100	$\mu_F = 0.754$	$\mu_{CR} = 0.195$	
Diversity = 2.8266e-02	k_t = [2, 3, 4, 3]			
Generation 7630: Best = 3.329477e+03	FES = 763100	$\mu_F = 0.761$	$\mu_{CR} = 0.200$	
Diversity = 2.7098e-02	k_t = [1, 1, 1, 1]			
Generation 7640: Best = 3.329436e+03	FES = 764100	$\mu_F = 0.750$	$\mu_{CR} = 0.194$	
Diversity = 2.5637e-02	k_t = [4, 4, 2, 4]			
Generation 7650: Best = 3.329421e+03	FES = 765100	$\mu_F = 0.733$	$\mu_{CR} = 0.196$	
Diversity = 2.3715e-02	k_t = [1, 1, 1, 1]			
Generation 7660: Best = 3.329393e+03	FES = 766100	$\mu_F = 0.739$	$\mu_{CR} = 0.195$	
Diversity = 2.3157e-02	k_t = [3, 3, 4, 3]			
Generation 7670: Best = 3.329369e+03	FES = 767100	$\mu_F = 0.736$	$\mu_{CR} = 0.204$	
Diversity = 2.3759e-02	k_t = [1, 1, 1, 1]			
Generation 7680: Best = 3.329345e+03	FES = 768100	$\mu_F = 0.725$	$\mu_{CR} = 0.197$	
Diversity = 2.5310e-02	k_t = [4, 2, 4, 3]			
Generation 7690: Best = 3.329315e+03	FES = 769100	$\mu_F = 0.723$	$\mu_{CR} = 0.194$	
Diversity = 2.5599e-02	k_t = [1, 1, 1, 1]			
Generation 7700: Best = 3.329303e+03	FES = 770100	$\mu_F = 0.729$	$\mu_{CR} = 0.183$	
Diversity = 2.3898e-02	k_t = [4, 2, 4, 4]			
Generation 7710: Best = 3.329247e+03	FES = 771100	$\mu_F = 0.744$	$\mu_{CR} = 0.173$	
Diversity = 2.3820e-02	k_t = [1, 1, 1, 1]			
Generation 7720: Best = 3.329210e+03	FES = 772100	$\mu_F = 0.749$	$\mu_{CR} = 0.173$	
Diversity = 2.5222e-02	k_t = [2, 2, 4, 4]			
Generation 7730: Best = 3.329183e+03	FES = 773100	$\mu_F = 0.751$	$\mu_{CR} = 0.176$	
Diversity = 2.2941e-02	k_t = [1, 1, 1, 1]			
Generation 7740: Best = 3.329156e+03	FES = 774100	$\mu_F = 0.751$	$\mu_{CR} = 0.185$	
Diversity = 2.2683e-02	k_t = [4, 3, 3, 2]			
Generation 7750: Best = 3.329136e+03	FES = 775100	$\mu_F = 0.747$	$\mu_{CR} = 0.190$	
Diversity = 2.2508e-02	k_t = [1, 1, 1, 1]			
Generation 7760: Best = 3.329110e+03	FES = 776100	$\mu_F = 0.739$	$\mu_{CR} = 0.195$	
Diversity = 2.0129e-02	k_t = [3, 4, 4, 4]			
Generation 7770: Best = 3.329095e+03	FES = 777100	$\mu_F = 0.726$	$\mu_{CR} = 0.191$	
Diversity = 1.7961e-02	k_t = [1, 1, 1, 1]			
Generation 7780: Best = 3.329077e+03	FES = 778100	$\mu_F = 0.719$	$\mu_{CR} = 0.195$	

Diversity = 1.7445e-02	k_t = [4, 4, 3, 3]			
Generation 7790: Best = 3.329061e+03	FES = 779100	$\mu_F = 0.726$	$\mu_{CR} = 0.191$	
Diversity = 1.8493e-02	k_t = [1, 1, 1, 1]			
Generation 7800: Best = 3.329038e+03	FES = 780100	$\mu_F = 0.711$	$\mu_{CR} = 0.204$	
Diversity = 1.7919e-02	k_t = [3, 3, 3, 4]			
Generation 7810: Best = 3.329016e+03	FES = 781100	$\mu_F = 0.713$	$\mu_{CR} = 0.207$	
Diversity = 1.8736e-02	k_t = [1, 1, 1, 1]			
Generation 7820: Best = 3.329003e+03	FES = 782100	$\mu_F = 0.712$	$\mu_{CR} = 0.195$	
Diversity = 1.7814e-02	k_t = [4, 4, 4, 2]			
Generation 7830: Best = 3.328986e+03	FES = 783100	$\mu_F = 0.702$	$\mu_{CR} = 0.201$	
Diversity = 1.8087e-02	k_t = [1, 1, 1, 1]			
Generation 7840: Best = 3.328969e+03	FES = 784100	$\mu_F = 0.700$	$\mu_{CR} = 0.205$	
Diversity = 1.6328e-02	k_t = [2, 4, 4, 4]			
Generation 7850: Best = 3.328952e+03	FES = 785100	$\mu_F = 0.715$	$\mu_{CR} = 0.209$	
Diversity = 1.6414e-02	k_t = [1, 1, 1, 1]			
Generation 7860: Best = 3.328909e+03	FES = 786100	$\mu_F = 0.721$	$\mu_{CR} = 0.217$	
Diversity = 1.7903e-02	k_t = [3, 3, 2, 3]			
Generation 7870: Best = 3.328861e+03	FES = 787100	$\mu_F = 0.726$	$\mu_{CR} = 0.226$	
Diversity = 1.7750e-02	k_t = [1, 1, 1, 1]			
Generation 7880: Best = 3.328813e+03	FES = 788100	$\mu_F = 0.742$	$\mu_{CR} = 0.233$	
Diversity = 2.0368e-02	k_t = [2, 3, 3, 4]			
Generation 7890: Best = 3.328794e+03	FES = 789100	$\mu_F = 0.739$	$\mu_{CR} = 0.232$	
Diversity = 1.9203e-02	k_t = [1, 1, 1, 1]			
Generation 7900: Best = 3.328775e+03	FES = 790100	$\mu_F = 0.738$	$\mu_{CR} = 0.226$	
Diversity = 1.8494e-02	k_t = [4, 3, 4, 2]			
Generation 7910: Best = 3.328754e+03	FES = 791100	$\mu_F = 0.738$	$\mu_{CR} = 0.227$	
Diversity = 1.6475e-02	k_t = [1, 1, 1, 1]			
Generation 7920: Best = 3.328746e+03	FES = 792100	$\mu_F = 0.742$	$\mu_{CR} = 0.218$	
Diversity = 1.5827e-02	k_t = [2, 4, 3, 3]			
Generation 7930: Best = 3.328733e+03	FES = 793100	$\mu_F = 0.726$	$\mu_{CR} = 0.217$	
Diversity = 1.5270e-02	k_t = [1, 1, 1, 1]			
Generation 7940: Best = 3.328721e+03	FES = 794100	$\mu_F = 0.731$	$\mu_{CR} = 0.203$	
Diversity = 1.4579e-02	k_t = [3, 4, 3, 4]			
Generation 7950: Best = 3.328705e+03	FES = 795100	$\mu_F = 0.744$	$\mu_{CR} = 0.204$	
Diversity = 1.4453e-02	k_t = [1, 1, 1, 1]			
Generation 7960: Best = 3.328691e+03	FES = 796100	$\mu_F = 0.747$	$\mu_{CR} = 0.198$	
Diversity = 1.3978e-02	k_t = [3, 2, 2, 4]			
Generation 7970: Best = 3.328676e+03	FES = 797100	$\mu_F = 0.733$	$\mu_{CR} = 0.196$	
Diversity = 1.4177e-02	k_t = [1, 1, 1, 1]			
Generation 7980: Best = 3.328654e+03	FES = 798100	$\mu_F = 0.744$	$\mu_{CR} = 0.205$	
Diversity = 1.3433e-02	k_t = [3, 2, 3, 3]			
Generation 7990: Best = 3.328643e+03	FES = 799100	$\mu_F = 0.738$	$\mu_{CR} = 0.207$	
Diversity = 1.3448e-02	k_t = [1, 1, 1, 1]			
Generation 8000: Best = 3.328632e+03	FES = 800100	$\mu_F = 0.731$	$\mu_{CR} = 0.215$	
Diversity = 1.3062e-02	k_t = [2, 4, 2, 2]			
Generation 8010: Best = 3.328621e+03	FES = 801100	$\mu_F = 0.727$	$\mu_{CR} = 0.209$	
Diversity = 1.2131e-02	k_t = [1, 1, 1, 1]			
Generation 8020: Best = 3.328609e+03	FES = 802100	$\mu_F = 0.734$	$\mu_{CR} = 0.214$	
Diversity = 1.1022e-02	k_t = [4, 4, 4, 3]			
Generation 8030: Best = 3.328601e+03	FES = 803100	$\mu_F = 0.740$	$\mu_{CR} = 0.216$	
Diversity = 1.0956e-02	k_t = [1, 1, 1, 1]			
Generation 8040: Best = 3.328591e+03	FES = 804100	$\mu_F = 0.746$	$\mu_{CR} = 0.219$	
Diversity = 1.0546e-02	k_t = [4, 3, 2, 3]			
Generation 8050: Best = 3.328576e+03	FES = 805100	$\mu_F = 0.757$	$\mu_{CR} = 0.223$	

Diversity = 1.1669e-02	k_t = [1, 1, 1, 1]			
Generation 8060: Best = 3.328561e+03	FES = 806100	$\mu_F = 0.749$	$\mu_{CR} = 0.228$	
Diversity = 1.2741e-02	k_t = [2, 4, 3, 3]			
Generation 8070: Best = 3.328544e+03	FES = 807100	$\mu_F = 0.753$	$\mu_{CR} = 0.232$	
Diversity = 1.3618e-02	k_t = [1, 1, 1, 1]			
Generation 8080: Best = 3.328534e+03	FES = 808100	$\mu_F = 0.743$	$\mu_{CR} = 0.222$	
Diversity = 1.2269e-02	k_t = [3, 4, 2, 3]			
Generation 8090: Best = 3.328530e+03	FES = 809100	$\mu_F = 0.731$	$\mu_{CR} = 0.228$	
Diversity = 1.1705e-02	k_t = [1, 1, 1, 1]			
Generation 8100: Best = 3.328519e+03	FES = 810100	$\mu_F = 0.739$	$\mu_{CR} = 0.236$	
Diversity = 1.1499e-02	k_t = [2, 2, 3, 2]			
Generation 8110: Best = 3.328511e+03	FES = 811100	$\mu_F = 0.749$	$\mu_{CR} = 0.241$	
Diversity = 1.2864e-02	k_t = [1, 1, 1, 1]			
Generation 8120: Best = 3.328502e+03	FES = 812100	$\mu_F = 0.747$	$\mu_{CR} = 0.239$	
Diversity = 1.2781e-02	k_t = [3, 3, 2, 2]			
Generation 8130: Best = 3.328486e+03	FES = 813100	$\mu_F = 0.743$	$\mu_{CR} = 0.228$	
Diversity = 1.3154e-02	k_t = [1, 1, 1, 1]			
Generation 8140: Best = 3.328481e+03	FES = 814100	$\mu_F = 0.739$	$\mu_{CR} = 0.233$	
Diversity = 1.1811e-02	k_t = [2, 3, 3, 3]			
Generation 8150: Best = 3.328471e+03	FES = 815100	$\mu_F = 0.745$	$\mu_{CR} = 0.229$	
Diversity = 1.2458e-02	k_t = [1, 1, 1, 1]			
Generation 8160: Best = 3.328465e+03	FES = 816100	$\mu_F = 0.742$	$\mu_{CR} = 0.229$	
Diversity = 1.1560e-02	k_t = [4, 3, 3, 4]			
Generation 8170: Best = 3.328456e+03	FES = 817100	$\mu_F = 0.737$	$\mu_{CR} = 0.226$	
Diversity = 1.0892e-02	k_t = [1, 1, 1, 1]			
Generation 8180: Best = 3.328447e+03	FES = 818100	$\mu_F = 0.735$	$\mu_{CR} = 0.223$	
Diversity = 1.0236e-02	k_t = [2, 3, 4, 2]			
Generation 8190: Best = 3.328441e+03	FES = 819100	$\mu_F = 0.734$	$\mu_{CR} = 0.218$	
Diversity = 1.0272e-02	k_t = [1, 1, 1, 1]			
Generation 8200: Best = 3.328435e+03	FES = 820100	$\mu_F = 0.729$	$\mu_{CR} = 0.212$	
Diversity = 9.9296e-03	k_t = [2, 2, 3, 2]			
Generation 8210: Best = 3.328428e+03	FES = 821100	$\mu_F = 0.738$	$\mu_{CR} = 0.207$	
Diversity = 1.0677e-02	k_t = [1, 1, 1, 1]			
Generation 8220: Best = 3.328414e+03	FES = 822100	$\mu_F = 0.735$	$\mu_{CR} = 0.209$	
Diversity = 1.1027e-02	k_t = [4, 2, 2, 3]			
Generation 8230: Best = 3.328407e+03	FES = 823100	$\mu_F = 0.748$	$\mu_{CR} = 0.204$	
Diversity = 1.1302e-02	k_t = [1, 1, 1, 1]			
Generation 8240: Best = 3.328394e+03	FES = 824100	$\mu_F = 0.753$	$\mu_{CR} = 0.198$	
Diversity = 1.1517e-02	k_t = [4, 3, 3, 2]			
Generation 8250: Best = 3.328378e+03	FES = 825100	$\mu_F = 0.765$	$\mu_{CR} = 0.200$	
Diversity = 1.2950e-02	k_t = [1, 1, 1, 1]			
Generation 8260: Best = 3.328356e+03	FES = 826100	$\mu_F = 0.771$	$\mu_{CR} = 0.204$	
Diversity = 1.3985e-02	k_t = [3, 3, 3, 4]			
Generation 8270: Best = 3.328340e+03	FES = 827100	$\mu_F = 0.775$	$\mu_{CR} = 0.192$	
Diversity = 1.5697e-02	k_t = [1, 1, 1, 1]			
Generation 8280: Best = 3.328324e+03	FES = 828100	$\mu_F = 0.761$	$\mu_{CR} = 0.195$	
Diversity = 1.5633e-02	k_t = [2, 3, 4, 2]			
Generation 8290: Best = 3.328312e+03	FES = 829100	$\mu_F = 0.751$	$\mu_{CR} = 0.196$	
Diversity = 1.6424e-02	k_t = [1, 1, 1, 1]			
Generation 8300: Best = 3.328301e+03	FES = 830100	$\mu_F = 0.749$	$\mu_{CR} = 0.193$	
Diversity = 1.5191e-02	k_t = [2, 4, 4, 3]			
Generation 8310: Best = 3.328280e+03	FES = 831100	$\mu_F = 0.753$	$\mu_{CR} = 0.197$	
Diversity = 1.5820e-02	k_t = [1, 1, 1, 1]			
Generation 8320: Best = 3.328266e+03	FES = 832100	$\mu_F = 0.764$	$\mu_{CR} = 0.200$	

Diversity = 1.7094e-02	k_t = [4, 2, 4, 4]			
Generation 8330: Best = 3.328250e+03	FES = 833100	$\mu_F = 0.751$	$\mu_{CR} = 0.199$	
Diversity = 1.6558e-02	k_t = [1, 1, 1, 1]			
Generation 8340: Best = 3.328231e+03	FES = 834100	$\mu_F = 0.738$	$\mu_{CR} = 0.195$	
Diversity = 1.6369e-02	k_t = [2, 3, 4, 3]			
Generation 8350: Best = 3.328214e+03	FES = 835100	$\mu_F = 0.740$	$\mu_{CR} = 0.184$	
Diversity = 1.7249e-02	k_t = [1, 1, 1, 1]			
Generation 8360: Best = 3.328201e+03	FES = 836100	$\mu_F = 0.731$	$\mu_{CR} = 0.183$	
Diversity = 1.7398e-02	k_t = [3, 2, 4, 2]			
Generation 8370: Best = 3.328169e+03	FES = 837100	$\mu_F = 0.732$	$\mu_{CR} = 0.188$	
Diversity = 1.9930e-02	k_t = [1, 1, 1, 1]			
Generation 8380: Best = 3.328156e+03	FES = 838100	$\mu_F = 0.738$	$\mu_{CR} = 0.185$	
Diversity = 1.5383e-02	k_t = [2, 4, 3, 4]			
Generation 8390: Best = 3.328147e+03	FES = 839100	$\mu_F = 0.731$	$\mu_{CR} = 0.179$	
Diversity = 1.3667e-02	k_t = [1, 1, 1, 1]			
Generation 8400: Best = 3.328135e+03	FES = 840100	$\mu_F = 0.742$	$\mu_{CR} = 0.180$	
Diversity = 1.3406e-02	k_t = [3, 4, 4, 2]			
Generation 8410: Best = 3.328122e+03	FES = 841100	$\mu_F = 0.747$	$\mu_{CR} = 0.190$	
Diversity = 1.3383e-02	k_t = [1, 1, 1, 1]			
Generation 8420: Best = 3.328113e+03	FES = 842100	$\mu_F = 0.750$	$\mu_{CR} = 0.193$	
Diversity = 1.3519e-02	k_t = [4, 3, 3, 4]			
Generation 8430: Best = 3.328102e+03	FES = 843100	$\mu_F = 0.749$	$\mu_{CR} = 0.189$	
Diversity = 1.3571e-02	k_t = [1, 1, 1, 1]			
Generation 8440: Best = 3.328091e+03	FES = 844100	$\mu_F = 0.749$	$\mu_{CR} = 0.194$	
Diversity = 1.4014e-02	k_t = [3, 2, 4, 4]			
Generation 8450: Best = 3.328079e+03	FES = 845100	$\mu_F = 0.739$	$\mu_{CR} = 0.190$	
Diversity = 1.4612e-02	k_t = [1, 1, 1, 1]			
Generation 8460: Best = 3.328070e+03	FES = 846100	$\mu_F = 0.723$	$\mu_{CR} = 0.188$	
Diversity = 1.5760e-02	k_t = [4, 3, 2, 2]			
Generation 8470: Best = 3.328052e+03	FES = 847100	$\mu_F = 0.728$	$\mu_{CR} = 0.185$	
Diversity = 1.6409e-02	k_t = [1, 1, 1, 1]			
Generation 8480: Best = 3.328037e+03	FES = 848100	$\mu_F = 0.740$	$\mu_{CR} = 0.183$	
Diversity = 1.3228e-02	k_t = [2, 4, 4, 4]			
Generation 8490: Best = 3.328029e+03	FES = 849100	$\mu_F = 0.730$	$\mu_{CR} = 0.187$	
Diversity = 1.2747e-02	k_t = [1, 1, 1, 1]			
Generation 8500: Best = 3.328020e+03	FES = 850100	$\mu_F = 0.740$	$\mu_{CR} = 0.182$	
Diversity = 1.2597e-02	k_t = [4, 4, 3, 2]			
Generation 8510: Best = 3.328012e+03	FES = 851100	$\mu_F = 0.736$	$\mu_{CR} = 0.182$	
Diversity = 1.2333e-02	k_t = [1, 1, 1, 1]			
Generation 8520: Best = 3.328006e+03	FES = 852100	$\mu_F = 0.732$	$\mu_{CR} = 0.187$	
Diversity = 1.2070e-02	k_t = [3, 4, 2, 2]			
Generation 8530: Best = 3.327995e+03	FES = 853100	$\mu_F = 0.721$	$\mu_{CR} = 0.182$	
Diversity = 1.2428e-02	k_t = [1, 1, 1, 1]			
Generation 8540: Best = 3.327987e+03	FES = 854100	$\mu_F = 0.733$	$\mu_{CR} = 0.181$	
Diversity = 1.2535e-02	k_t = [2, 2, 2, 2]			
Generation 8550: Best = 3.327976e+03	FES = 855100	$\mu_F = 0.728$	$\mu_{CR} = 0.191$	
Diversity = 1.3112e-02	k_t = [1, 1, 1, 1]			
Generation 8560: Best = 3.327965e+03	FES = 856100	$\mu_F = 0.736$	$\mu_{CR} = 0.188$	
Diversity = 1.2778e-02	k_t = [2, 4, 4, 2]			
Generation 8570: Best = 3.327957e+03	FES = 857100	$\mu_F = 0.747$	$\mu_{CR} = 0.195$	
Diversity = 1.2591e-02	k_t = [1, 1, 1, 1]			
Generation 8580: Best = 3.327949e+03	FES = 858100	$\mu_F = 0.737$	$\mu_{CR} = 0.196$	
Diversity = 1.1762e-02	k_t = [3, 4, 2, 3]			
Generation 8590: Best = 3.327941e+03	FES = 859100	$\mu_F = 0.730$	$\mu_{CR} = 0.185$	

Diversity = 1.1544e-02	k_t = [1, 1, 1, 1]			
Generation 8600: Best = 3.327935e+03	FES = 860100	$\mu_F = 0.737$	$\mu_{CR} = 0.179$	
Diversity = 1.0833e-02	k_t = [3, 4, 2, 3]			
Generation 8610: Best = 3.327927e+03	FES = 861100	$\mu_F = 0.730$	$\mu_{CR} = 0.184$	
Diversity = 1.0729e-02	k_t = [1, 1, 1, 1]			
Generation 8620: Best = 3.327918e+03	FES = 862100	$\mu_F = 0.728$	$\mu_{CR} = 0.195$	
Diversity = 1.1545e-02	k_t = [3, 3, 2, 2]			
Generation 8630: Best = 3.327912e+03	FES = 863100	$\mu_F = 0.732$	$\mu_{CR} = 0.195$	
Diversity = 1.2027e-02	k_t = [1, 1, 1, 1]			
Generation 8640: Best = 3.327903e+03	FES = 864100	$\mu_F = 0.729$	$\mu_{CR} = 0.199$	
Diversity = 1.1988e-02	k_t = [4, 2, 3, 3]			
Generation 8650: Best = 3.327892e+03	FES = 865100	$\mu_F = 0.729$	$\mu_{CR} = 0.201$	
Diversity = 1.2062e-02	k_t = [1, 1, 1, 1]			
Generation 8660: Best = 3.327886e+03	FES = 866100	$\mu_F = 0.727$	$\mu_{CR} = 0.191$	
Diversity = 1.0836e-02	k_t = [4, 3, 3, 3]			
Generation 8670: Best = 3.327881e+03	FES = 867100	$\mu_F = 0.724$	$\mu_{CR} = 0.193$	
Diversity = 9.7930e-03	k_t = [1, 1, 1, 1]			
Generation 8680: Best = 3.327874e+03	FES = 868100	$\mu_F = 0.729$	$\mu_{CR} = 0.208$	
Diversity = 8.5744e-03	k_t = [4, 4, 2, 4]			
Generation 8690: Best = 3.327869e+03	FES = 869100	$\mu_F = 0.736$	$\mu_{CR} = 0.205$	
Diversity = 8.4589e-03	k_t = [1, 1, 1, 1]			
Generation 8700: Best = 3.327863e+03	FES = 870100	$\mu_F = 0.743$	$\mu_{CR} = 0.213$	
Diversity = 7.9545e-03	k_t = [2, 3, 3, 4]			
Generation 8710: Best = 3.327858e+03	FES = 871100	$\mu_F = 0.734$	$\mu_{CR} = 0.213$	
Diversity = 8.2564e-03	k_t = [1, 1, 1, 1]			
Generation 8720: Best = 3.327853e+03	FES = 872100	$\mu_F = 0.729$	$\mu_{CR} = 0.226$	
Diversity = 7.8370e-03	k_t = [3, 4, 3, 3]			
Generation 8730: Best = 3.327849e+03	FES = 873100	$\mu_F = 0.736$	$\mu_{CR} = 0.224$	
Diversity = 8.1718e-03	k_t = [1, 1, 1, 1]			
Generation 8740: Best = 3.327846e+03	FES = 874100	$\mu_F = 0.739$	$\mu_{CR} = 0.228$	
Diversity = 7.8809e-03	k_t = [2, 4, 4, 4]			
Generation 8750: Best = 3.327842e+03	FES = 875100	$\mu_F = 0.737$	$\mu_{CR} = 0.225$	
Diversity = 7.5448e-03	k_t = [1, 1, 1, 1]			
Generation 8760: Best = 3.327839e+03	FES = 876100	$\mu_F = 0.732$	$\mu_{CR} = 0.213$	
Diversity = 7.0094e-03	k_t = [2, 3, 4, 2]			
Generation 8770: Best = 3.327837e+03	FES = 877100	$\mu_F = 0.734$	$\mu_{CR} = 0.207$	
Diversity = 7.0389e-03	k_t = [1, 1, 1, 1]			
Generation 8780: Best = 3.327833e+03	FES = 878100	$\mu_F = 0.723$	$\mu_{CR} = 0.212$	
Diversity = 7.1538e-03	k_t = [2, 2, 3, 2]			
Generation 8790: Best = 3.327831e+03	FES = 879100	$\mu_F = 0.710$	$\mu_{CR} = 0.213$	
Diversity = 6.9940e-03	k_t = [1, 1, 1, 1]			
Generation 8800: Best = 3.327829e+03	FES = 880100	$\mu_F = 0.710$	$\mu_{CR} = 0.218$	
Diversity = 6.2153e-03	k_t = [2, 4, 4, 4]			
Generation 8810: Best = 3.327825e+03	FES = 881100	$\mu_F = 0.715$	$\mu_{CR} = 0.221$	
Diversity = 5.7864e-03	k_t = [1, 1, 1, 1]			
Generation 8820: Best = 3.327823e+03	FES = 882100	$\mu_F = 0.732$	$\mu_{CR} = 0.218$	
Diversity = 5.8671e-03	k_t = [2, 4, 4, 3]			
Generation 8830: Best = 3.327820e+03	FES = 883100	$\mu_F = 0.731$	$\mu_{CR} = 0.209$	
Diversity = 6.0020e-03	k_t = [1, 1, 1, 1]			
Generation 8840: Best = 3.327817e+03	FES = 884100	$\mu_F = 0.741$	$\mu_{CR} = 0.217$	
Diversity = 5.7964e-03	k_t = [4, 3, 4, 3]			
Generation 8850: Best = 3.327814e+03	FES = 885100	$\mu_F = 0.755$	$\mu_{CR} = 0.215$	
Diversity = 6.3777e-03	k_t = [1, 1, 1, 1]			
Generation 8860: Best = 3.327806e+03	FES = 886100	$\mu_F = 0.769$	$\mu_{CR} = 0.210$	

Diversity = 7.5994e-03	k_t = [3, 2, 3, 4]			
Generation 8870: Best = 3.327793e+03	FES = 887100	$\mu_F = 0.782$	$\mu_{CR} = 0.211$	
Diversity = 8.6009e-03	k_t = [1, 1, 1, 1]			
Generation 8880: Best = 3.327783e+03	FES = 888100	$\mu_F = 0.768$	$\mu_{CR} = 0.207$	
Diversity = 8.3586e-03	k_t = [2, 2, 3, 3]			
Generation 8890: Best = 3.327772e+03	FES = 889100	$\mu_F = 0.773$	$\mu_{CR} = 0.215$	
Diversity = 9.6222e-03	k_t = [1, 1, 1, 1]			
Generation 8900: Best = 3.327757e+03	FES = 890100	$\mu_F = 0.774$	$\mu_{CR} = 0.215$	
Diversity = 9.7090e-03	k_t = [2, 4, 4, 3]			
Generation 8910: Best = 3.327750e+03	FES = 891100	$\mu_F = 0.765$	$\mu_{CR} = 0.212$	
Diversity = 1.0324e-02	k_t = [1, 1, 1, 1]			
Generation 8920: Best = 3.327747e+03	FES = 892100	$\mu_F = 0.758$	$\mu_{CR} = 0.215$	
Diversity = 9.8917e-03	k_t = [2, 4, 2, 3]			
Generation 8930: Best = 3.327739e+03	FES = 893100	$\mu_F = 0.753$	$\mu_{CR} = 0.223$	
Diversity = 1.0212e-02	k_t = [1, 1, 1, 1]			
Generation 8940: Best = 3.327736e+03	FES = 894100	$\mu_F = 0.735$	$\mu_{CR} = 0.225$	
Diversity = 9.4567e-03	k_t = [2, 3, 3, 3]			
Generation 8950: Best = 3.327729e+03	FES = 895100	$\mu_F = 0.729$	$\mu_{CR} = 0.226$	
Diversity = 9.7855e-03	k_t = [1, 1, 1, 1]			
Generation 8960: Best = 3.327727e+03	FES = 896100	$\mu_F = 0.696$	$\mu_{CR} = 0.230$	
Diversity = 8.4973e-03	k_t = [2, 2, 3, 3]			
Generation 8970: Best = 3.327724e+03	FES = 897100	$\mu_F = 0.687$	$\mu_{CR} = 0.218$	
Diversity = 7.6555e-03	k_t = [1, 1, 1, 1]			
Generation 8980: Best = 3.327721e+03	FES = 898100	$\mu_F = 0.686$	$\mu_{CR} = 0.234$	
Diversity = 6.3650e-03	k_t = [2, 4, 4, 4]			
Generation 8990: Best = 3.327719e+03	FES = 899100	$\mu_F = 0.686$	$\mu_{CR} = 0.239$	
Diversity = 5.9228e-03	k_t = [1, 1, 1, 1]			
Generation 9000: Best = 3.327717e+03	FES = 900100	$\mu_F = 0.690$	$\mu_{CR} = 0.240$	
Diversity = 5.0279e-03	k_t = [4, 4, 4, 4]			
Generation 9010: Best = 3.327715e+03	FES = 901100	$\mu_F = 0.698$	$\mu_{CR} = 0.239$	
Diversity = 4.8096e-03	k_t = [1, 1, 1, 1]			
Generation 9020: Best = 3.327713e+03	FES = 902100	$\mu_F = 0.714$	$\mu_{CR} = 0.247$	
Diversity = 4.3037e-03	k_t = [4, 2, 4, 4]			
Generation 9030: Best = 3.327712e+03	FES = 903100	$\mu_F = 0.731$	$\mu_{CR} = 0.238$	
Diversity = 4.0485e-03	k_t = [1, 1, 1, 1]			
Generation 9040: Best = 3.327708e+03	FES = 904100	$\mu_F = 0.731$	$\mu_{CR} = 0.243$	
Diversity = 4.2219e-03	k_t = [3, 2, 3, 4]			
Generation 9050: Best = 3.327707e+03	FES = 905100	$\mu_F = 0.723$	$\mu_{CR} = 0.242$	
Diversity = 4.5461e-03	k_t = [1, 1, 1, 1]			
Generation 9060: Best = 3.327706e+03	FES = 906100	$\mu_F = 0.721$	$\mu_{CR} = 0.237$	
Diversity = 4.3540e-03	k_t = [3, 3, 2, 2]			
Generation 9070: Best = 3.327704e+03	FES = 907100	$\mu_F = 0.715$	$\mu_{CR} = 0.237$	
Diversity = 4.3010e-03	k_t = [1, 1, 1, 1]			
Generation 9080: Best = 3.327703e+03	FES = 908100	$\mu_F = 0.705$	$\mu_{CR} = 0.238$	
Diversity = 3.9308e-03	k_t = [4, 4, 4, 3]			
Generation 9090: Best = 3.327702e+03	FES = 909100	$\mu_F = 0.706$	$\mu_{CR} = 0.243$	
Diversity = 3.7067e-03	k_t = [1, 1, 1, 1]			
Generation 9100: Best = 3.327700e+03	FES = 910100	$\mu_F = 0.710$	$\mu_{CR} = 0.233$	
Diversity = 3.3815e-03	k_t = [4, 4, 4, 4]			
Generation 9110: Best = 3.327699e+03	FES = 911100	$\mu_F = 0.728$	$\mu_{CR} = 0.238$	
Diversity = 3.2133e-03	k_t = [1, 1, 1, 1]			
Generation 9120: Best = 3.327698e+03	FES = 912100	$\mu_F = 0.727$	$\mu_{CR} = 0.237$	
Diversity = 3.2607e-03	k_t = [4, 3, 4, 4]			
Generation 9130: Best = 3.327696e+03	FES = 913100	$\mu_F = 0.742$	$\mu_{CR} = 0.229$	

Diversity = 3.4229e-03	k_t = [1, 1, 1, 1]			
Generation 9140: Best = 3.327695e+03	FES = 914100	$\mu_F = 0.745$	$\mu_{CR} = 0.229$	
Diversity = 3.6353e-03	k_t = [3, 2, 3, 2]			
Generation 9150: Best = 3.327693e+03	FES = 915100	$\mu_F = 0.746$	$\mu_{CR} = 0.239$	
Diversity = 3.9074e-03	k_t = [1, 1, 1, 1]			
Generation 9160: Best = 3.327691e+03	FES = 916100	$\mu_F = 0.746$	$\mu_{CR} = 0.243$	
Diversity = 4.0958e-03	k_t = [3, 3, 2, 2]			
Generation 9170: Best = 3.327690e+03	FES = 917100	$\mu_F = 0.772$	$\mu_{CR} = 0.240$	
Diversity = 4.3899e-03	k_t = [1, 1, 1, 1]			
Generation 9180: Best = 3.327689e+03	FES = 918100	$\mu_F = 0.765$	$\mu_{CR} = 0.249$	
Diversity = 4.6175e-03	k_t = [2, 3, 4, 2]			
Generation 9190: Best = 3.327687e+03	FES = 919100	$\mu_F = 0.754$	$\mu_{CR} = 0.233$	
Diversity = 4.5031e-03	k_t = [1, 1, 1, 1]			
Generation 9200: Best = 3.327687e+03	FES = 920100	$\mu_F = 0.745$	$\mu_{CR} = 0.235$	
Diversity = 4.0257e-03	k_t = [4, 4, 3, 4]			
Generation 9210: Best = 3.327686e+03	FES = 921100	$\mu_F = 0.725$	$\mu_{CR} = 0.227$	
Diversity = 3.3541e-03	k_t = [1, 1, 1, 1]			
Generation 9220: Best = 3.327685e+03	FES = 922100	$\mu_F = 0.710$	$\mu_{CR} = 0.223$	
Diversity = 2.9941e-03	k_t = [2, 2, 4, 3]			
Generation 9230: Best = 3.327684e+03	FES = 923100	$\mu_F = 0.722$	$\mu_{CR} = 0.212$	
Diversity = 3.0911e-03	k_t = [1, 1, 1, 1]			
Generation 9240: Best = 3.327684e+03	FES = 924100	$\mu_F = 0.731$	$\mu_{CR} = 0.216$	
Diversity = 2.8768e-03	k_t = [2, 3, 3, 4]			
Generation 9250: Best = 3.327684e+03	FES = 925100	$\mu_F = 0.726$	$\mu_{CR} = 0.215$	
Diversity = 2.7274e-03	k_t = [1, 1, 1, 1]			
Generation 9260: Best = 3.327683e+03	FES = 926100	$\mu_F = 0.734$	$\mu_{CR} = 0.217$	
Diversity = 2.6376e-03	k_t = [4, 4, 2, 2]			
Generation 9270: Best = 3.327683e+03	FES = 927100	$\mu_F = 0.727$	$\mu_{CR} = 0.218$	
Diversity = 2.5456e-03	k_t = [1, 1, 1, 1]			
Generation 9280: Best = 3.327682e+03	FES = 928100	$\mu_F = 0.736$	$\mu_{CR} = 0.212$	
Diversity = 2.5295e-03	k_t = [2, 3, 2, 3]			
Generation 9290: Best = 3.327681e+03	FES = 929100	$\mu_F = 0.750$	$\mu_{CR} = 0.212$	
Diversity = 3.4697e-03	k_t = [1, 1, 1, 1]			
Generation 9300: Best = 3.327674e+03	FES = 930100	$\mu_F = 0.761$	$\mu_{CR} = 0.211$	
Diversity = 4.8166e-03	k_t = [3, 4, 4, 2]			
Generation 9310: Best = 3.327672e+03	FES = 931100	$\mu_F = 0.763$	$\mu_{CR} = 0.207$	
Diversity = 5.0203e-03	k_t = [1, 1, 1, 1]			
Generation 9320: Best = 3.327669e+03	FES = 932100	$\mu_F = 0.767$	$\mu_{CR} = 0.200$	
Diversity = 5.2898e-03	k_t = [2, 3, 4, 2]			
Generation 9330: Best = 3.327666e+03	FES = 933100	$\mu_F = 0.767$	$\mu_{CR} = 0.205$	
Diversity = 5.7655e-03	k_t = [1, 1, 1, 1]			
Generation 9340: Best = 3.327664e+03	FES = 934100	$\mu_F = 0.756$	$\mu_{CR} = 0.196$	
Diversity = 5.3771e-03	k_t = [2, 3, 2, 4]			
Generation 9350: Best = 3.327663e+03	FES = 935100	$\mu_F = 0.746$	$\mu_{CR} = 0.196$	
Diversity = 4.5585e-03	k_t = [1, 1, 1, 1]			
Generation 9360: Best = 3.327661e+03	FES = 936100	$\mu_F = 0.749$	$\mu_{CR} = 0.198$	
Diversity = 4.2340e-03	k_t = [2, 2, 2, 2]			
Generation 9370: Best = 3.327660e+03	FES = 937100	$\mu_F = 0.736$	$\mu_{CR} = 0.195$	
Diversity = 4.2693e-03	k_t = [1, 1, 1, 1]			
Generation 9380: Best = 3.327659e+03	FES = 938100	$\mu_F = 0.731$	$\mu_{CR} = 0.194$	
Diversity = 4.2486e-03	k_t = [2, 3, 3, 2]			
Generation 9390: Best = 3.327659e+03	FES = 939100	$\mu_F = 0.723$	$\mu_{CR} = 0.189$	
Diversity = 4.1186e-03	k_t = [1, 1, 1, 1]			
Generation 9400: Best = 3.327657e+03	FES = 940100	$\mu_F = 0.717$	$\mu_{CR} = 0.190$	

Diversity = 3.6808e-03	k_t = [2, 4, 2, 3]			
Generation 9410: Best = 3.327656e+03	FES = 941100	$\mu_F = 0.716$	$\mu_{CR} = 0.195$	
Diversity = 3.5487e-03	k_t = [1, 1, 1, 1]			
Generation 9420: Best = 3.327655e+03	FES = 942100	$\mu_F = 0.731$	$\mu_{CR} = 0.206$	
Diversity = 3.4870e-03	k_t = [4, 4, 3, 2]			
Generation 9430: Best = 3.327654e+03	FES = 943100	$\mu_F = 0.736$	$\mu_{CR} = 0.215$	
Diversity = 3.5772e-03	k_t = [1, 1, 1, 1]			
Generation 9440: Best = 3.327652e+03	FES = 944100	$\mu_F = 0.736$	$\mu_{CR} = 0.212$	
Diversity = 3.7146e-03	k_t = [4, 2, 3, 2]			
Generation 9450: Best = 3.327650e+03	FES = 945100	$\mu_F = 0.743$	$\mu_{CR} = 0.221$	
Diversity = 3.7965e-03	k_t = [1, 1, 1, 1]			
Generation 9460: Best = 3.327649e+03	FES = 946100	$\mu_F = 0.752$	$\mu_{CR} = 0.214$	
Diversity = 3.8204e-03	k_t = [4, 3, 3, 4]			
Generation 9470: Best = 3.327647e+03	FES = 947100	$\mu_F = 0.763$	$\mu_{CR} = 0.218$	
Diversity = 4.0835e-03	k_t = [1, 1, 1, 1]			
Generation 9480: Best = 3.327645e+03	FES = 948100	$\mu_F = 0.756$	$\mu_{CR} = 0.213$	
Diversity = 4.3127e-03	k_t = [2, 3, 2, 2]			
Generation 9490: Best = 3.327643e+03	FES = 949100	$\mu_F = 0.756$	$\mu_{CR} = 0.214$	
Diversity = 4.6522e-03	k_t = [1, 1, 1, 1]			
Generation 9500: Best = 3.327642e+03	FES = 950100	$\mu_F = 0.745$	$\mu_{CR} = 0.218$	
Diversity = 4.7576e-03	k_t = [4, 4, 4, 2]			
Generation 9510: Best = 3.327641e+03	FES = 951100	$\mu_F = 0.745$	$\mu_{CR} = 0.219$	
Diversity = 4.5377e-03	k_t = [1, 1, 1, 1]			
Generation 9520: Best = 3.327640e+03	FES = 952100	$\mu_F = 0.724$	$\mu_{CR} = 0.224$	
Diversity = 3.7353e-03	k_t = [2, 4, 2, 4]			
Generation 9530: Best = 3.327639e+03	FES = 953100	$\mu_F = 0.708$	$\mu_{CR} = 0.235$	
Diversity = 3.2298e-03	k_t = [1, 1, 1, 1]			
Generation 9540: Best = 3.327638e+03	FES = 954100	$\mu_F = 0.714$	$\mu_{CR} = 0.229$	
Diversity = 3.1762e-03	k_t = [4, 2, 2, 3]			
Generation 9550: Best = 3.327637e+03	FES = 955100	$\mu_F = 0.725$	$\mu_{CR} = 0.228$	
Diversity = 3.1594e-03	k_t = [1, 1, 1, 1]			
Generation 9560: Best = 3.327636e+03	FES = 956100	$\mu_F = 0.731$	$\mu_{CR} = 0.224$	
Diversity = 2.9310e-03	k_t = [4, 3, 2, 4]			
Generation 9570: Best = 3.327635e+03	FES = 957100	$\mu_F = 0.719$	$\mu_{CR} = 0.226$	
Diversity = 2.6665e-03	k_t = [1, 1, 1, 1]			
Generation 9580: Best = 3.327635e+03	FES = 958100	$\mu_F = 0.735$	$\mu_{CR} = 0.230$	
Diversity = 2.6708e-03	k_t = [4, 4, 3, 3]			
Generation 9590: Best = 3.327634e+03	FES = 959100	$\mu_F = 0.735$	$\mu_{CR} = 0.222$	
Diversity = 2.5049e-03	k_t = [1, 1, 1, 1]			
Generation 9600: Best = 3.327634e+03	FES = 960100	$\mu_F = 0.736$	$\mu_{CR} = 0.221$	
Diversity = 2.3221e-03	k_t = [2, 3, 4, 4]			
Generation 9610: Best = 3.327633e+03	FES = 961100	$\mu_F = 0.727$	$\mu_{CR} = 0.218$	
Diversity = 2.2506e-03	k_t = [1, 1, 1, 1]			
Generation 9620: Best = 3.327632e+03	FES = 962100	$\mu_F = 0.745$	$\mu_{CR} = 0.218$	
Diversity = 2.2292e-03	k_t = [3, 3, 4, 4]			
Generation 9630: Best = 3.327632e+03	FES = 963100	$\mu_F = 0.754$	$\mu_{CR} = 0.229$	
Diversity = 2.4205e-03	k_t = [1, 1, 1, 1]			
Generation 9640: Best = 3.327631e+03	FES = 964100	$\mu_F = 0.750$	$\mu_{CR} = 0.224$	
Diversity = 2.4566e-03	k_t = [4, 3, 4, 3]			
Generation 9650: Best = 3.327630e+03	FES = 965100	$\mu_F = 0.760$	$\mu_{CR} = 0.219$	
Diversity = 2.5146e-03	k_t = [1, 1, 1, 1]			
Generation 9660: Best = 3.327629e+03	FES = 966100	$\mu_F = 0.760$	$\mu_{CR} = 0.208$	
Diversity = 2.5626e-03	k_t = [2, 3, 2, 4]			
Generation 9670: Best = 3.327628e+03	FES = 967100	$\mu_F = 0.768$	$\mu_{CR} = 0.211$	

Diversity = 2.7541e-03	k_t = [1, 1, 1, 1]			
Generation 9680: Best = 3.327628e+03	FES = 968100	$\mu_F = 0.782$	$\mu_{CR} = 0.211$	
Diversity = 2.7491e-03	k_t = [3, 2, 2, 3]			
Generation 9690: Best = 3.327627e+03	FES = 969100	$\mu_F = 0.765$	$\mu_{CR} = 0.222$	
Diversity = 2.8003e-03	k_t = [1, 1, 1, 1]			
Generation 9700: Best = 3.327626e+03	FES = 970100	$\mu_F = 0.759$	$\mu_{CR} = 0.228$	
Diversity = 2.7490e-03	k_t = [3, 3, 3, 2]			
Generation 9710: Best = 3.327626e+03	FES = 971100	$\mu_F = 0.748$	$\mu_{CR} = 0.230$	
Diversity = 2.8538e-03	k_t = [1, 1, 1, 1]			
Generation 9720: Best = 3.327625e+03	FES = 972100	$\mu_F = 0.749$	$\mu_{CR} = 0.229$	
Diversity = 2.6285e-03	k_t = [3, 3, 4, 4]			
Generation 9730: Best = 3.327624e+03	FES = 973100	$\mu_F = 0.750$	$\mu_{CR} = 0.239$	
Diversity = 2.5465e-03	k_t = [1, 1, 1, 1]			
Generation 9740: Best = 3.327623e+03	FES = 974100	$\mu_F = 0.750$	$\mu_{CR} = 0.242$	
Diversity = 2.6392e-03	k_t = [4, 2, 3, 2]			
Generation 9750: Best = 3.327623e+03	FES = 975100	$\mu_F = 0.739$	$\mu_{CR} = 0.244$	
Diversity = 2.5490e-03	k_t = [1, 1, 1, 1]			
Generation 9760: Best = 3.327622e+03	FES = 976100	$\mu_F = 0.735$	$\mu_{CR} = 0.247$	
Diversity = 2.3376e-03	k_t = [3, 2, 3, 2]			
Generation 9770: Best = 3.327622e+03	FES = 977100	$\mu_F = 0.717$	$\mu_{CR} = 0.240$	
Diversity = 2.3476e-03	k_t = [1, 1, 1, 1]			
Generation 9780: Best = 3.327622e+03	FES = 978100	$\mu_F = 0.709$	$\mu_{CR} = 0.246$	
Diversity = 2.0549e-03	k_t = [4, 4, 2, 4]			
Generation 9790: Best = 3.327621e+03	FES = 979100	$\mu_F = 0.700$	$\mu_{CR} = 0.238$	
Diversity = 1.7308e-03	k_t = [1, 1, 1, 1]			
Generation 9800: Best = 3.327621e+03	FES = 980100	$\mu_F = 0.700$	$\mu_{CR} = 0.234$	
Diversity = 1.3825e-03	k_t = [4, 3, 2, 4]			
Generation 9810: Best = 3.327621e+03	FES = 981100	$\mu_F = 0.701$	$\mu_{CR} = 0.228$	
Diversity = 1.2442e-03	k_t = [1, 1, 1, 1]			
Generation 9820: Best = 3.327621e+03	FES = 982100	$\mu_F = 0.712$	$\mu_{CR} = 0.225$	
Diversity = 1.1067e-03	k_t = [3, 4, 2, 3]			
Generation 9830: Best = 3.327621e+03	FES = 983100	$\mu_F = 0.726$	$\mu_{CR} = 0.229$	
Diversity = 1.1141e-03	k_t = [1, 1, 1, 1]			
Generation 9840: Best = 3.327621e+03	FES = 984100	$\mu_F = 0.740$	$\mu_{CR} = 0.235$	
Diversity = 1.1567e-03	k_t = [3, 2, 4, 4]			
Generation 9850: Best = 3.327620e+03	FES = 985100	$\mu_F = 0.750$	$\mu_{CR} = 0.233$	
Diversity = 1.2099e-03	k_t = [1, 1, 1, 1]			
Generation 9860: Best = 3.327620e+03	FES = 986100	$\mu_F = 0.738$	$\mu_{CR} = 0.231$	
Diversity = 1.1886e-03	k_t = [4, 4, 3, 4]			
Generation 9870: Best = 3.327620e+03	FES = 987100	$\mu_F = 0.736$	$\mu_{CR} = 0.230$	
Diversity = 1.3214e-03	k_t = [1, 1, 1, 1]			
Generation 9880: Best = 3.327620e+03	FES = 988100	$\mu_F = 0.740$	$\mu_{CR} = 0.240$	
Diversity = 1.2665e-03	k_t = [3, 4, 4, 4]			
Generation 9890: Best = 3.327620e+03	FES = 989100	$\mu_F = 0.735$	$\mu_{CR} = 0.237$	
Diversity = 1.2308e-03	k_t = [1, 1, 1, 1]			
Generation 9900: Best = 3.327620e+03	FES = 990100	$\mu_F = 0.731$	$\mu_{CR} = 0.229$	
Diversity = 1.1511e-03	k_t = [2, 2, 3, 3]			
Generation 9910: Best = 3.327619e+03	FES = 991100	$\mu_F = 0.711$	$\mu_{CR} = 0.218$	
Diversity = 1.0510e-03	k_t = [1, 1, 1, 1]			
Generation 9920: Best = 3.327619e+03	FES = 992100	$\mu_F = 0.705$	$\mu_{CR} = 0.229$	
Diversity = 9.3095e-04	k_t = [3, 2, 4, 2]			
Generation 9930: Best = 3.327619e+03	FES = 993100	$\mu_F = 0.705$	$\mu_{CR} = 0.236$	
Diversity = 8.3262e-04	k_t = [1, 1, 1, 1]			
Generation 9940: Best = 3.327619e+03	FES = 994100	$\mu_F = 0.698$	$\mu_{CR} = 0.240$	

Diversity = 7.1891e-04 k_t = [3, 2, 2, 3]
 Generation 9950: Best = 3.327619e+03 FES = 995100 μ_F = 0.682 μ_{CR} = 0.212
 Diversity = 6.4471e-04 k_t = [1, 1, 1, 1]
 Generation 9960: Best = 3.327619e+03 FES = 996100 μ_F = 0.688 μ_{CR} = 0.207
 Diversity = 5.6531e-04 k_t = [2, 4, 3, 4]
 Generation 9970: Best = 3.327619e+03 FES = 997100 μ_F = 0.700 μ_{CR} = 0.205
 Diversity = 5.8455e-04 k_t = [1, 1, 1, 1]
 Generation 9980: Best = 3.327619e+03 FES = 998100 μ_F = 0.705 μ_{CR} = 0.207
 Diversity = 5.2202e-04 k_t = [3, 4, 3, 4]
 Generation 9990: Best = 3.327619e+03 FES = 999100 μ_F = 0.710 μ_{CR} = 0.211
 Diversity = 4.5984e-04 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 3.327619e+03
 Total Function Evals = 1000000
 Total Successful Updates = 419210

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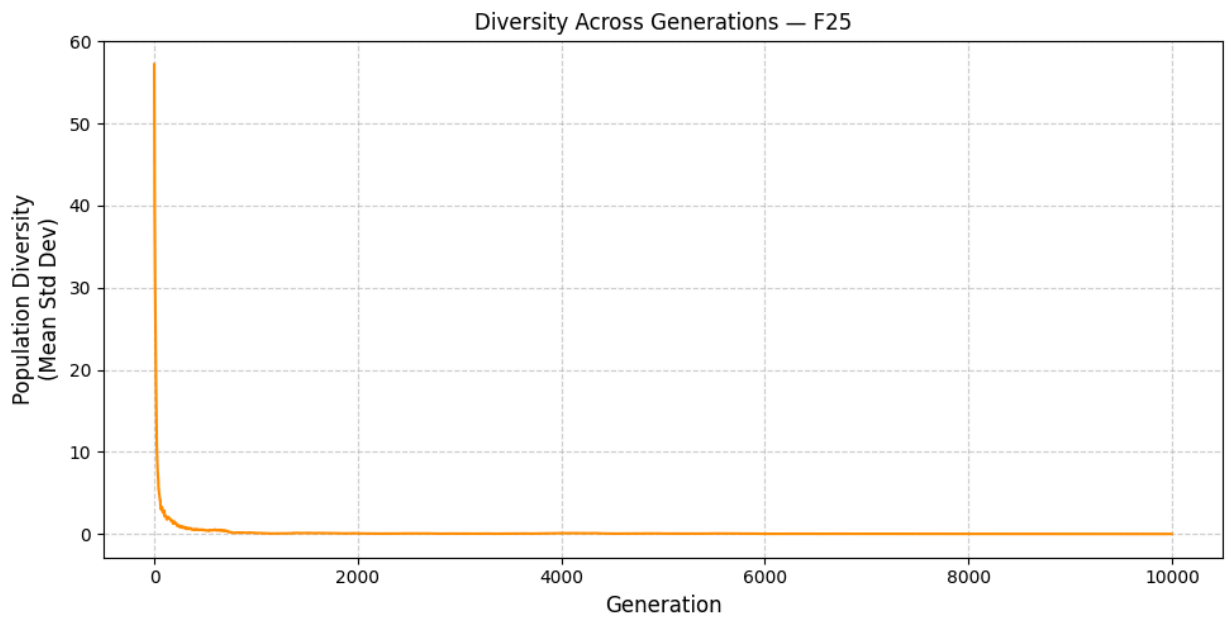
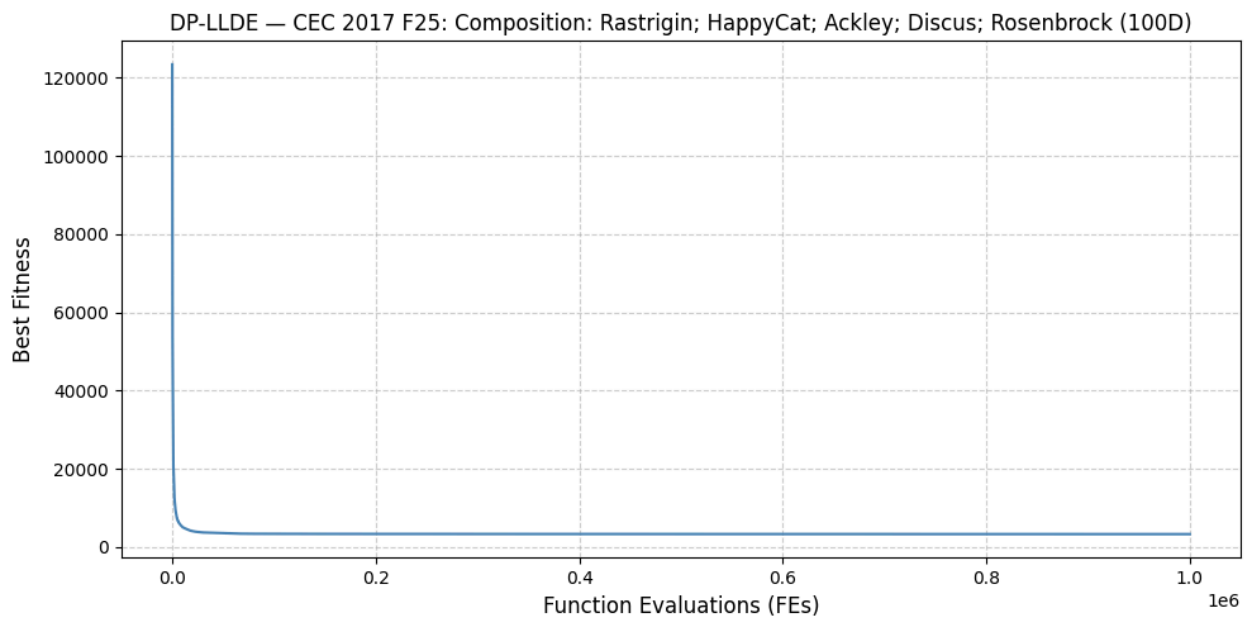
Results:

Best Fitness : 3.327619e+03
 Global Optimum : 2.500000e+03
 Error (|best-optimum|) : 8.276192e+02
 Best Solution (5 dim) : [-11.33952142 -7.40114744 45.35301177 3.54951648
 -15.97697859]
 D^(0) (init. diversity): 5.7220e+01

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✓ Checkpoint saved to: /content/drive/My Drive/thesis_results/all_results_checkpoint.pkl



Graph saved as convergence_F25.png

✓ Plot saved to: /content/drive/My Drive/thesis_results/plots/convergence_F25.png

```
In [ ]: # =====
# F26 — Composition Function 6 (Schaffer; Schwefel; Griewank; Rosenbrock; Rastrigin)
# =====
run_and_plot('F26', make_wrapper(composition.f26), 2600.0, 'Composition: Schaffer; Schwefel; Griewank; Rosenbrock; Rastrigin')
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DP-LLDE on CEC 2017 F26 – Single Run

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Function : Composition: Schaffer; Schwefel; Griewank; Rosenbrock; Rastrig
in (100D)

Bounds : [-100, 100]

Global optimum : 2600.0

Initial $D^*(0) = 5.7220e+01$ | Diversity threshold = 5.7220e+00

Generation 0: Best Fitness = 1.023547e+05

Generation 10: Best = 4.815651e+04 FES = 1100 $\mu_F = 0.556$ $\mu_{CR} = 0.351$

Diversity = 3.2739e+01 $k_t = [1, 1, 1, 1]$

Generation 20: Best = 3.565654e+04 FES = 2100 $\mu_F = 0.604$ $\mu_{CR} = 0.352$

Diversity = 2.1129e+01 $k_t = [2, 3, 2, 3]$

Generation 30: Best = 2.882728e+04 FES = 3100 $\mu_F = 0.632$ $\mu_{CR} = 0.358$

Diversity = 1.4504e+01 $k_t = [1, 1, 1, 1]$

Generation 40: Best = 2.607398e+04 FES = 4100 $\mu_F = 0.670$ $\mu_{CR} = 0.362$

Diversity = 1.0651e+01 $k_t = [3, 2, 3, 4]$

Generation 50: Best = 2.432062e+04 FES = 5100 $\mu_F = 0.684$ $\mu_{CR} = 0.351$

Diversity = 9.4756e+00 $k_t = [1, 1, 1, 1]$

Generation 60: Best = 2.277802e+04 FES = 6100 $\mu_F = 0.724$ $\mu_{CR} = 0.332$

Diversity = 9.5374e+00 $k_t = [4, 3, 2, 2]$

Generation 70: Best = 2.234477e+04 FES = 7100 $\mu_F = 0.710$ $\mu_{CR} = 0.306$

Diversity = 9.1698e+00 $k_t = [1, 1, 1, 1]$

Generation 80: Best = 2.167798e+04 FES = 8100 $\mu_F = 0.720$ $\mu_{CR} = 0.272$

Diversity = 8.7815e+00 $k_t = [3, 2, 3, 4]$

Generation 90: Best = 2.083404e+04 FES = 9100 $\mu_F = 0.723$ $\mu_{CR} = 0.235$

Diversity = 8.3561e+00 $k_t = [1, 1, 1, 1]$

Generation 100: Best = 2.083404e+04 FES = 10100 $\mu_F = 0.701$ $\mu_{CR} = 0.194$

Diversity = 7.6487e+00 $k_t = [2, 3, 2, 4]$

Generation 110: Best = 2.062084e+04 FES = 11100 $\mu_F = 0.648$ $\mu_{CR} = 0.178$

Diversity = 7.3907e+00 $k_t = [1, 1, 1, 1]$

Generation 120: Best = 2.009191e+04 FES = 12100 $\mu_F = 0.677$ $\mu_{CR} = 0.140$

Diversity = 7.1760e+00 $k_t = [4, 2, 3, 4]$

Generation 130: Best = 1.954648e+04 FES = 13100 $\mu_F = 0.714$ $\mu_{CR} = 0.107$

Diversity = 7.0174e+00 $k_t = [1, 1, 1, 1]$

Generation 140: Best = 1.945493e+04 FES = 14100 $\mu_F = 0.731$ $\mu_{CR} = 0.093$

Diversity = 6.8991e+00 $k_t = [2, 4, 3, 2]$

Generation 150: Best = 1.915200e+04 FES = 15100 $\mu_F = 0.739$ $\mu_{CR} = 0.083$

Diversity = 6.8428e+00 $k_t = [1, 1, 1, 1]$

Generation 160: Best = 1.906555e+04 FES = 16100 $\mu_F = 0.724$ $\mu_{CR} = 0.078$

Diversity = 6.7140e+00 $k_t = [3, 4, 3, 2]$

Generation 170: Best = 1.895356e+04 FES = 17100 $\mu_F = 0.713$ $\mu_{CR} = 0.077$

Diversity = 6.6916e+00 $k_t = [1, 1, 1, 1]$

Generation 180: Best = 1.876098e+04 FES = 18100 $\mu_F = 0.721$ $\mu_{CR} = 0.071$

Diversity = 6.6708e+00 $k_t = [2, 3, 4, 2]$

Generation 190: Best = 1.876098e+04 FES = 19100 $\mu_F = 0.749$ $\mu_{CR} = 0.077$

Diversity = 6.5047e+00 $k_t = [1, 1, 1, 1]$

Generation 200: Best = 1.850886e+04 FES = 20100 $\mu_F = 0.750$ $\mu_{CR} = 0.065$

Diversity = 6.5347e+00 $k_t = [2, 2, 4, 3]$

Generation 210: Best = 1.841976e+04 FES = 21100 $\mu_F = 0.729$ $\mu_{CR} = 0.054$

Diversity = 6.5316e+00 $k_t = [1, 1, 1, 1]$

Generation 220: Best = 1.841976e+04 FES = 22100 $\mu_F = 0.689$ $\mu_{CR} = 0.060$

Diversity = 6.4599e+00	k_t = [3, 3, 4, 2]				
Generation 230: Best = 1.830692e+04	FES = 23100	$\mu_F = 0.676$	$\mu_{CR} = 0.055$		
Diversity = 6.3611e+00	k_t = [1, 1, 1, 1]				
Generation 240: Best = 1.789334e+04	FES = 24100	$\mu_F = 0.681$	$\mu_{CR} = 0.055$		
Diversity = 6.2356e+00	k_t = [3, 2, 4, 4]				
Generation 250: Best = 1.789334e+04	FES = 25100	$\mu_F = 0.702$	$\mu_{CR} = 0.050$		
Diversity = 6.2722e+00	k_t = [1, 1, 1, 1]				
Generation 260: Best = 1.789025e+04	FES = 26100	$\mu_F = 0.657$	$\mu_{CR} = 0.052$		
Diversity = 6.1777e+00	k_t = [3, 2, 4, 4]				
Generation 270: Best = 1.789025e+04	FES = 27100	$\mu_F = 0.660$	$\mu_{CR} = 0.053$		
Diversity = 6.1632e+00	k_t = [1, 1, 1, 1]				
Generation 280: Best = 1.771279e+04	FES = 28100	$\mu_F = 0.693$	$\mu_{CR} = 0.056$		
Diversity = 6.1582e+00	k_t = [2, 2, 2, 4]				
Generation 290: Best = 1.771279e+04	FES = 29100	$\mu_F = 0.715$	$\mu_{CR} = 0.046$		
Diversity = 6.1310e+00	k_t = [1, 1, 1, 1]				
Generation 300: Best = 1.755056e+04	FES = 30100	$\mu_F = 0.694$	$\mu_{CR} = 0.059$		
Diversity = 6.0292e+00	k_t = [4, 4, 3, 4]				
Generation 310: Best = 1.716068e+04	FES = 31100	$\mu_F = 0.707$	$\mu_{CR} = 0.050$		
Diversity = 6.0418e+00	k_t = [1, 1, 1, 1]				
Generation 320: Best = 1.716068e+04	FES = 32100	$\mu_F = 0.679$	$\mu_{CR} = 0.058$		
Diversity = 5.8105e+00	k_t = [4, 2, 2, 4]				
Generation 330: Best = 1.716068e+04	FES = 33100	$\mu_F = 0.681$	$\mu_{CR} = 0.062$		
Diversity = 5.7500e+00	k_t = [1, 1, 1, 1]				
Generation 340: Best = 1.699956e+04	FES = 34100	$\mu_F = 0.666$	$\mu_{CR} = 0.064$		
Diversity = 5.7166e+00	k_t = [4, 3, 4, 2]				
Generation 350: Best = 1.699956e+04	FES = 35100	$\mu_F = 0.678$	$\mu_{CR} = 0.060$		
Diversity = 5.6887e+00	k_t = [1, 1, 1, 1]				
Generation 360: Best = 1.674582e+04	FES = 36100	$\mu_F = 0.709$	$\mu_{CR} = 0.059$		
Diversity = 5.6505e+00	k_t = [4, 2, 2, 2]				
Generation 370: Best = 1.668553e+04	FES = 37100	$\mu_F = 0.702$	$\mu_{CR} = 0.060$		
Diversity = 5.6414e+00	k_t = [1, 1, 1, 1]				
Generation 380: Best = 1.630715e+04	FES = 38100	$\mu_F = 0.699$	$\mu_{CR} = 0.060$		
Diversity = 5.4778e+00	k_t = [2, 3, 4, 3]				
Generation 390: Best = 1.629146e+04	FES = 39100	$\mu_F = 0.709$	$\mu_{CR} = 0.052$		
Diversity = 5.4467e+00	k_t = [1, 1, 1, 1]				
Generation 400: Best = 1.612123e+04	FES = 40100	$\mu_F = 0.750$	$\mu_{CR} = 0.046$		
Diversity = 5.3863e+00	k_t = [4, 4, 3, 2]				
Generation 410: Best = 1.605533e+04	FES = 41100	$\mu_F = 0.776$	$\mu_{CR} = 0.047$		
Diversity = 5.3659e+00	k_t = [1, 1, 1, 1]				
Generation 420: Best = 1.604991e+04	FES = 42100	$\mu_F = 0.767$	$\mu_{CR} = 0.055$		
Diversity = 5.2221e+00	k_t = [4, 2, 2, 4]				
Generation 430: Best = 1.579658e+04	FES = 43100	$\mu_F = 0.724$	$\mu_{CR} = 0.059$		
Diversity = 5.0508e+00	k_t = [1, 1, 1, 1]				
Generation 440: Best = 1.579658e+04	FES = 44100	$\mu_F = 0.723$	$\mu_{CR} = 0.053$		
Diversity = 5.0481e+00	k_t = [3, 4, 2, 2]				
Generation 450: Best = 1.579658e+04	FES = 45100	$\mu_F = 0.722$	$\mu_{CR} = 0.067$		
Diversity = 5.0297e+00	k_t = [1, 1, 1, 1]				
Generation 460: Best = 1.579302e+04	FES = 46100	$\mu_F = 0.697$	$\mu_{CR} = 0.069$		
Diversity = 4.8333e+00	k_t = [4, 3, 3, 3]				
Generation 470: Best = 1.570913e+04	FES = 47100	$\mu_F = 0.689$	$\mu_{CR} = 0.067$		
Diversity = 4.7826e+00	k_t = [1, 1, 1, 1]				
Generation 480: Best = 1.566847e+04	FES = 48100	$\mu_F = 0.697$	$\mu_{CR} = 0.072$		
Diversity = 4.5347e+00	k_t = [4, 4, 4, 4]				
Generation 490: Best = 1.565945e+04	FES = 49100	$\mu_F = 0.671$	$\mu_{CR} = 0.071$		

Diversity = 4.2681e+00	k_t = [1, 1, 1, 1]					
Generation 500: Best = 1.540067e+04	FES = 50100	$\mu_F = 0.668$	$\mu_{CR} = 0.071$			
Diversity = 4.1539e+00	k_t = [2, 4, 4, 2]					
Generation 510: Best = 1.540067e+04	FES = 51100	$\mu_F = 0.666$	$\mu_{CR} = 0.074$			
Diversity = 3.9929e+00	k_t = [1, 1, 1, 1]					
Generation 520: Best = 1.538988e+04	FES = 52100	$\mu_F = 0.668$	$\mu_{CR} = 0.074$			
Diversity = 3.5404e+00	k_t = [3, 3, 4, 3]					
Generation 530: Best = 1.510768e+04	FES = 53100	$\mu_F = 0.652$	$\mu_{CR} = 0.089$			
Diversity = 3.3069e+00	k_t = [1, 1, 1, 1]					
Generation 540: Best = 1.496407e+04	FES = 54100	$\mu_F = 0.649$	$\mu_{CR} = 0.101$			
Diversity = 2.9624e+00	k_t = [3, 3, 3, 4]					
Generation 550: Best = 1.474380e+04	FES = 55100	$\mu_F = 0.646$	$\mu_{CR} = 0.099$			
Diversity = 2.6734e+00	k_t = [1, 1, 1, 1]					
Generation 560: Best = 1.460693e+04	FES = 56100	$\mu_F = 0.622$	$\mu_{CR} = 0.105$			
Diversity = 2.3668e+00	k_t = [2, 4, 4, 2]					
Generation 570: Best = 1.452620e+04	FES = 57100	$\mu_F = 0.631$	$\mu_{CR} = 0.113$			
Diversity = 2.0416e+00	k_t = [1, 1, 1, 1]					
Generation 580: Best = 1.411573e+04	FES = 58100	$\mu_F = 0.638$	$\mu_{CR} = 0.124$			
Diversity = 1.7660e+00	k_t = [4, 2, 3, 3]					
Generation 590: Best = 1.376459e+04	FES = 59100	$\mu_F = 0.663$	$\mu_{CR} = 0.139$			
Diversity = 1.5421e+00	k_t = [1, 1, 1, 1]					
Generation 600: Best = 1.325986e+04	FES = 60100	$\mu_F = 0.681$	$\mu_{CR} = 0.147$			
Diversity = 8.8546e-01	k_t = [2, 2, 4, 4]					
Generation 610: Best = 1.290189e+04	FES = 61100	$\mu_F = 0.687$	$\mu_{CR} = 0.160$			
Diversity = 6.1829e-01	k_t = [1, 1, 1, 1]					
Generation 620: Best = 1.266893e+04	FES = 62100	$\mu_F = 0.693$	$\mu_{CR} = 0.169$			
Diversity = 4.5785e-01	k_t = [2, 3, 3, 4]					
Generation 630: Best = 1.255667e+04	FES = 63100	$\mu_F = 0.700$	$\mu_{CR} = 0.175$			
Diversity = 3.6545e-01	k_t = [1, 1, 1, 1]					
Generation 640: Best = 1.247579e+04	FES = 64100	$\mu_F = 0.706$	$\mu_{CR} = 0.175$			
Diversity = 3.1790e-01	k_t = [2, 2, 2, 4]					
Generation 650: Best = 1.241843e+04	FES = 65100	$\mu_F = 0.700$	$\mu_{CR} = 0.187$			
Diversity = 2.8242e-01	k_t = [1, 1, 1, 1]					
Generation 660: Best = 1.235404e+04	FES = 66100	$\mu_F = 0.680$	$\mu_{CR} = 0.204$			
Diversity = 2.3817e-01	k_t = [4, 3, 4, 4]					
Generation 670: Best = 1.230870e+04	FES = 67100	$\mu_F = 0.689$	$\mu_{CR} = 0.205$			
Diversity = 2.2623e-01	k_t = [1, 1, 1, 1]					
Generation 680: Best = 1.226401e+04	FES = 68100	$\mu_F = 0.697$	$\mu_{CR} = 0.215$			
Diversity = 1.9986e-01	k_t = [3, 3, 2, 4]					
Generation 690: Best = 1.223238e+04	FES = 69100	$\mu_F = 0.691$	$\mu_{CR} = 0.221$			
Diversity = 1.8639e-01	k_t = [1, 1, 1, 1]					
Generation 700: Best = 1.220064e+04	FES = 70100	$\mu_F = 0.698$	$\mu_{CR} = 0.228$			
Diversity = 1.6651e-01	k_t = [3, 2, 2, 4]					
Generation 710: Best = 1.217205e+04	FES = 71100	$\mu_F = 0.702$	$\mu_{CR} = 0.233$			
Diversity = 1.5257e-01	k_t = [1, 1, 1, 1]					
Generation 720: Best = 1.215565e+04	FES = 72100	$\mu_F = 0.698$	$\mu_{CR} = 0.230$			
Diversity = 1.3468e-01	k_t = [3, 2, 3, 4]					
Generation 730: Best = 1.213556e+04	FES = 73100	$\mu_F = 0.706$	$\mu_{CR} = 0.237$			
Diversity = 1.3031e-01	k_t = [1, 1, 1, 1]					
Generation 740: Best = 1.211308e+04	FES = 74100	$\mu_F = 0.711$	$\mu_{CR} = 0.227$			
Diversity = 1.2697e-01	k_t = [4, 3, 4, 2]					
Generation 750: Best = 1.209998e+04	FES = 75100	$\mu_F = 0.717$	$\mu_{CR} = 0.226$			
Diversity = 1.3254e-01	k_t = [1, 1, 1, 1]					
Generation 760: Best = 1.208545e+04	FES = 76100	$\mu_F = 0.704$	$\mu_{CR} = 0.223$			

Diversity = 1.2617e-01	k_t = [2, 3, 2, 2]				
Generation 770: Best = 1.207522e+04	FES = 77100	$\mu_F = 0.698$	$\mu_{CR} = 0.218$		
Diversity = 1.2287e-01	k_t = [1, 1, 1, 1]				
Generation 780: Best = 1.206665e+04	FES = 78100	$\mu_F = 0.691$	$\mu_{CR} = 0.223$		
Diversity = 1.0550e-01	k_t = [4, 3, 3, 2]				
Generation 790: Best = 1.205453e+04	FES = 79100	$\mu_F = 0.688$	$\mu_{CR} = 0.225$		
Diversity = 9.5231e-02	k_t = [1, 1, 1, 1]				
Generation 800: Best = 1.204429e+04	FES = 80100	$\mu_F = 0.689$	$\mu_{CR} = 0.232$		
Diversity = 8.3510e-02	k_t = [4, 4, 2, 2]				
Generation 810: Best = 1.203727e+04	FES = 81100	$\mu_F = 0.696$	$\mu_{CR} = 0.237$		
Diversity = 8.0818e-02	k_t = [1, 1, 1, 1]				
Generation 820: Best = 1.202827e+04	FES = 82100	$\mu_F = 0.692$	$\mu_{CR} = 0.238$		
Diversity = 7.5520e-02	k_t = [4, 3, 4, 3]				
Generation 830: Best = 1.202240e+04	FES = 83100	$\mu_F = 0.698$	$\mu_{CR} = 0.238$		
Diversity = 7.0153e-02	k_t = [1, 1, 1, 1]				
Generation 840: Best = 1.201623e+04	FES = 84100	$\mu_F = 0.699$	$\mu_{CR} = 0.238$		
Diversity = 6.5638e-02	k_t = [2, 4, 2, 4]				
Generation 850: Best = 1.201093e+04	FES = 85100	$\mu_F = 0.717$	$\mu_{CR} = 0.225$		
Diversity = 6.4538e-02	k_t = [1, 1, 1, 1]				
Generation 860: Best = 1.200654e+04	FES = 86100	$\mu_F = 0.712$	$\mu_{CR} = 0.225$		
Diversity = 5.6877e-02	k_t = [4, 4, 3, 3]				
Generation 870: Best = 1.200366e+04	FES = 87100	$\mu_F = 0.711$	$\mu_{CR} = 0.222$		
Diversity = 5.2803e-02	k_t = [1, 1, 1, 1]				
Generation 880: Best = 1.199976e+04	FES = 88100	$\mu_F = 0.721$	$\mu_{CR} = 0.227$		
Diversity = 5.1981e-02	k_t = [2, 2, 4, 3]				
Generation 890: Best = 1.199496e+04	FES = 89100	$\mu_F = 0.712$	$\mu_{CR} = 0.227$		
Diversity = 5.4151e-02	k_t = [1, 1, 1, 1]				
Generation 900: Best = 1.199127e+04	FES = 90100	$\mu_F = 0.708$	$\mu_{CR} = 0.229$		
Diversity = 5.4407e-02	k_t = [2, 3, 4, 3]				
Generation 910: Best = 1.198767e+04	FES = 91100	$\mu_F = 0.723$	$\mu_{CR} = 0.233$		
Diversity = 5.6118e-02	k_t = [1, 1, 1, 1]				
Generation 920: Best = 1.198411e+04	FES = 92100	$\mu_F = 0.723$	$\mu_{CR} = 0.235$		
Diversity = 5.4498e-02	k_t = [2, 4, 3, 3]				
Generation 930: Best = 1.198231e+04	FES = 93100	$\mu_F = 0.719$	$\mu_{CR} = 0.230$		
Diversity = 5.2740e-02	k_t = [1, 1, 1, 1]				
Generation 940: Best = 1.198014e+04	FES = 94100	$\mu_F = 0.719$	$\mu_{CR} = 0.222$		
Diversity = 4.7516e-02	k_t = [2, 2, 4, 4]				
Generation 950: Best = 1.197735e+04	FES = 95100	$\mu_F = 0.724$	$\mu_{CR} = 0.222$		
Diversity = 4.6698e-02	k_t = [1, 1, 1, 1]				
Generation 960: Best = 1.197537e+04	FES = 96100	$\mu_F = 0.721$	$\mu_{CR} = 0.221$		
Diversity = 4.5930e-02	k_t = [2, 3, 4, 2]				
Generation 970: Best = 1.197310e+04	FES = 97100	$\mu_F = 0.698$	$\mu_{CR} = 0.226$		
Diversity = 4.6360e-02	k_t = [1, 1, 1, 1]				
Generation 980: Best = 1.197107e+04	FES = 98100	$\mu_F = 0.713$	$\mu_{CR} = 0.228$		
Diversity = 4.3625e-02	k_t = [2, 3, 2, 2]				
Generation 990: Best = 1.196927e+04	FES = 99100	$\mu_F = 0.703$	$\mu_{CR} = 0.230$		
Diversity = 4.2833e-02	k_t = [1, 1, 1, 1]				
Generation 1000: Best = 1.196799e+04	FES = 100100	$\mu_F = 0.710$	$\mu_{CR} = 0.230$		
Diversity = 4.0670e-02	k_t = [4, 3, 4, 2]				
Generation 1010: Best = 1.196664e+04	FES = 101100	$\mu_F = 0.710$	$\mu_{CR} = 0.227$		
Diversity = 3.8226e-02	k_t = [1, 1, 1, 1]				
Generation 1020: Best = 1.196518e+04	FES = 102100	$\mu_F = 0.709$	$\mu_{CR} = 0.231$		
Diversity = 3.4037e-02	k_t = [2, 2, 2, 4]				
Generation 1030: Best = 1.196363e+04	FES = 103100	$\mu_F = 0.697$	$\mu_{CR} = 0.229$		

Diversity = 3.3036e-02	k_t = [1, 1, 1, 1]			
Generation 1040: Best = 1.196211e+04	FES = 104100	$\mu_F = 0.695$	$\mu_{CR} = 0.224$	
Diversity = 3.0132e-02	k_t = [4, 4, 3, 3]			
Generation 1050: Best = 1.196084e+04	FES = 105100	$\mu_F = 0.713$	$\mu_{CR} = 0.218$	
Diversity = 2.9347e-02	k_t = [1, 1, 1, 1]			
Generation 1060: Best = 1.195979e+04	FES = 106100	$\mu_F = 0.726$	$\mu_{CR} = 0.220$	
Diversity = 2.8124e-02	k_t = [2, 2, 4, 4]			
Generation 1070: Best = 1.195840e+04	FES = 107100	$\mu_F = 0.726$	$\mu_{CR} = 0.228$	
Diversity = 3.0083e-02	k_t = [1, 1, 1, 1]			
Generation 1080: Best = 1.195484e+04	FES = 108100	$\mu_F = 0.733$	$\mu_{CR} = 0.240$	
Diversity = 3.1447e-02	k_t = [2, 3, 2, 3]			
Generation 1090: Best = 1.195366e+04	FES = 109100	$\mu_F = 0.734$	$\mu_{CR} = 0.238$	
Diversity = 3.6392e-02	k_t = [1, 1, 1, 1]			
Generation 1100: Best = 1.195130e+04	FES = 110100	$\mu_F = 0.737$	$\mu_{CR} = 0.241$	
Diversity = 3.4140e-02	k_t = [3, 3, 3, 3]			
Generation 1110: Best = 1.194921e+04	FES = 111100	$\mu_F = 0.731$	$\mu_{CR} = 0.247$	
Diversity = 3.8195e-02	k_t = [1, 1, 1, 1]			
Generation 1120: Best = 1.194567e+04	FES = 112100	$\mu_F = 0.740$	$\mu_{CR} = 0.251$	
Diversity = 4.0476e-02	k_t = [4, 4, 3, 4]			
Generation 1130: Best = 1.194002e+04	FES = 113100	$\mu_F = 0.757$	$\mu_{CR} = 0.251$	
Diversity = 4.2299e-02	k_t = [1, 1, 1, 1]			
Generation 1140: Best = 1.193509e+04	FES = 114100	$\mu_F = 0.751$	$\mu_{CR} = 0.251$	
Diversity = 5.1778e-02	k_t = [4, 2, 4, 2]			
Generation 1150: Best = 1.193139e+04	FES = 115100	$\mu_F = 0.744$	$\mu_{CR} = 0.263$	
Diversity = 5.1347e-02	k_t = [1, 1, 1, 1]			
Generation 1160: Best = 1.192789e+04	FES = 116100	$\mu_F = 0.724$	$\mu_{CR} = 0.267$	
Diversity = 4.5839e-02	k_t = [3, 4, 2, 4]			
Generation 1170: Best = 1.192691e+04	FES = 117100	$\mu_F = 0.713$	$\mu_{CR} = 0.264$	
Diversity = 4.4468e-02	k_t = [1, 1, 1, 1]			
Generation 1180: Best = 1.192502e+04	FES = 118100	$\mu_F = 0.709$	$\mu_{CR} = 0.264$	
Diversity = 4.0333e-02	k_t = [2, 4, 2, 4]			
Generation 1190: Best = 1.192391e+04	FES = 119100	$\mu_F = 0.713$	$\mu_{CR} = 0.256$	
Diversity = 3.6981e-02	k_t = [1, 1, 1, 1]			
Generation 1200: Best = 1.192249e+04	FES = 120100	$\mu_F = 0.696$	$\mu_{CR} = 0.250$	
Diversity = 3.2560e-02	k_t = [4, 4, 4, 3]			
Generation 1210: Best = 1.192186e+04	FES = 121100	$\mu_F = 0.700$	$\mu_{CR} = 0.245$	
Diversity = 2.8700e-02	k_t = [1, 1, 1, 1]			
Generation 1220: Best = 1.192096e+04	FES = 122100	$\mu_F = 0.698$	$\mu_{CR} = 0.245$	
Diversity = 2.5051e-02	k_t = [4, 4, 2, 3]			
Generation 1230: Best = 1.192035e+04	FES = 123100	$\mu_F = 0.695$	$\mu_{CR} = 0.235$	
Diversity = 2.3030e-02	k_t = [1, 1, 1, 1]			
Generation 1240: Best = 1.191957e+04	FES = 124100	$\mu_F = 0.702$	$\mu_{CR} = 0.226$	
Diversity = 2.2556e-02	k_t = [2, 2, 2, 2]			
Generation 1250: Best = 1.191897e+04	FES = 125100	$\mu_F = 0.699$	$\mu_{CR} = 0.231$	
Diversity = 2.1740e-02	k_t = [1, 1, 1, 1]			
Generation 1260: Best = 1.191837e+04	FES = 126100	$\mu_F = 0.707$	$\mu_{CR} = 0.236$	
Diversity = 1.9769e-02	k_t = [2, 4, 3, 4]			
Generation 1270: Best = 1.191789e+04	FES = 127100	$\mu_F = 0.701$	$\mu_{CR} = 0.229$	
Diversity = 2.0205e-02	k_t = [1, 1, 1, 1]			
Generation 1280: Best = 1.191722e+04	FES = 128100	$\mu_F = 0.694$	$\mu_{CR} = 0.223$	
Diversity = 1.8396e-02	k_t = [4, 4, 2, 4]			
Generation 1290: Best = 1.191688e+04	FES = 129100	$\mu_F = 0.710$	$\mu_{CR} = 0.219$	
Diversity = 1.7552e-02	k_t = [1, 1, 1, 1]			
Generation 1300: Best = 1.191613e+04	FES = 130100	$\mu_F = 0.713$	$\mu_{CR} = 0.225$	

Diversity = 1.7658e-02	k_t = [4, 3, 2, 4]			
Generation 1310: Best = 1.191538e+04	FES = 131100	$\mu_F = 0.716$	$\mu_{CR} = 0.233$	
Diversity = 1.7854e-02	k_t = [1, 1, 1, 1]			
Generation 1320: Best = 1.191437e+04	FES = 132100	$\mu_F = 0.730$	$\mu_{CR} = 0.238$	
Diversity = 2.0473e-02	k_t = [3, 2, 4, 2]			
Generation 1330: Best = 1.191350e+04	FES = 133100	$\mu_F = 0.734$	$\mu_{CR} = 0.244$	
Diversity = 2.1993e-02	k_t = [1, 1, 1, 1]			
Generation 1340: Best = 1.191207e+04	FES = 134100	$\mu_F = 0.740$	$\mu_{CR} = 0.243$	
Diversity = 2.2371e-02	k_t = [3, 3, 4, 3]			
Generation 1350: Best = 1.191094e+04	FES = 135100	$\mu_F = 0.742$	$\mu_{CR} = 0.242$	
Diversity = 2.4970e-02	k_t = [1, 1, 1, 1]			
Generation 1360: Best = 1.190955e+04	FES = 136100	$\mu_F = 0.753$	$\mu_{CR} = 0.236$	
Diversity = 2.5611e-02	k_t = [3, 3, 4, 2]			
Generation 1370: Best = 1.190899e+04	FES = 137100	$\mu_F = 0.750$	$\mu_{CR} = 0.234$	
Diversity = 2.5075e-02	k_t = [1, 1, 1, 1]			
Generation 1380: Best = 1.190845e+04	FES = 138100	$\mu_F = 0.743$	$\mu_{CR} = 0.233$	
Diversity = 2.2056e-02	k_t = [4, 3, 3, 4]			
Generation 1390: Best = 1.190787e+04	FES = 139100	$\mu_F = 0.727$	$\mu_{CR} = 0.230$	
Diversity = 1.9136e-02	k_t = [1, 1, 1, 1]			
Generation 1400: Best = 1.190755e+04	FES = 140100	$\mu_F = 0.726$	$\mu_{CR} = 0.229$	
Diversity = 1.7280e-02	k_t = [3, 3, 2, 4]			
Generation 1410: Best = 1.190723e+04	FES = 141100	$\mu_F = 0.727$	$\mu_{CR} = 0.226$	
Diversity = 1.6818e-02	k_t = [1, 1, 1, 1]			
Generation 1420: Best = 1.190653e+04	FES = 142100	$\mu_F = 0.734$	$\mu_{CR} = 0.225$	
Diversity = 1.7010e-02	k_t = [3, 2, 2, 4]			
Generation 1430: Best = 1.190575e+04	FES = 143100	$\mu_F = 0.734$	$\mu_{CR} = 0.233$	
Diversity = 1.9365e-02	k_t = [1, 1, 1, 1]			
Generation 1440: Best = 1.190497e+04	FES = 144100	$\mu_F = 0.736$	$\mu_{CR} = 0.244$	
Diversity = 1.8927e-02	k_t = [2, 2, 2, 4]			
Generation 1450: Best = 1.190441e+04	FES = 145100	$\mu_F = 0.728$	$\mu_{CR} = 0.241$	
Diversity = 1.8843e-02	k_t = [1, 1, 1, 1]			
Generation 1460: Best = 1.190390e+04	FES = 146100	$\mu_F = 0.727$	$\mu_{CR} = 0.248$	
Diversity = 1.7685e-02	k_t = [3, 2, 4, 4]			
Generation 1470: Best = 1.190357e+04	FES = 147100	$\mu_F = 0.732$	$\mu_{CR} = 0.246$	
Diversity = 1.7989e-02	k_t = [1, 1, 1, 1]			
Generation 1480: Best = 1.190305e+04	FES = 148100	$\mu_F = 0.726$	$\mu_{CR} = 0.244$	
Diversity = 1.6573e-02	k_t = [2, 3, 4, 3]			
Generation 1490: Best = 1.190269e+04	FES = 149100	$\mu_F = 0.721$	$\mu_{CR} = 0.241$	
Diversity = 1.7305e-02	k_t = [1, 1, 1, 1]			
Generation 1500: Best = 1.190231e+04	FES = 150100	$\mu_F = 0.714$	$\mu_{CR} = 0.248$	
Diversity = 1.8613e-02	k_t = [3, 2, 3, 2]			
Generation 1510: Best = 1.190197e+04	FES = 151100	$\mu_F = 0.712$	$\mu_{CR} = 0.234$	
Diversity = 1.8196e-02	k_t = [1, 1, 1, 1]			
Generation 1520: Best = 1.190170e+04	FES = 152100	$\mu_F = 0.707$	$\mu_{CR} = 0.241$	
Diversity = 1.6776e-02	k_t = [3, 2, 2, 2]			
Generation 1530: Best = 1.190136e+04	FES = 153100	$\mu_F = 0.707$	$\mu_{CR} = 0.235$	
Diversity = 1.6719e-02	k_t = [1, 1, 1, 1]			
Generation 1540: Best = 1.190103e+04	FES = 154100	$\mu_F = 0.715$	$\mu_{CR} = 0.236$	
Diversity = 1.5207e-02	k_t = [2, 4, 3, 3]			
Generation 1550: Best = 1.190087e+04	FES = 155100	$\mu_F = 0.708$	$\mu_{CR} = 0.239$	
Diversity = 1.5605e-02	k_t = [1, 1, 1, 1]			
Generation 1560: Best = 1.190060e+04	FES = 156100	$\mu_F = 0.715$	$\mu_{CR} = 0.236$	
Diversity = 1.4983e-02	k_t = [2, 3, 4, 3]			
Generation 1570: Best = 1.190039e+04	FES = 157100	$\mu_F = 0.713$	$\mu_{CR} = 0.239$	

Diversity = 1.4793e-02	k_t = [1, 1, 1, 1]			
Generation 1580: Best = 1.190022e+04	FES = 158100	$\mu_F = 0.704$	$\mu_{CR} = 0.244$	
Diversity = 1.3498e-02	k_t = [4, 2, 4, 2]			
Generation 1590: Best = 1.189997e+04	FES = 159100	$\mu_F = 0.709$	$\mu_{CR} = 0.249$	
Diversity = 1.3377e-02	k_t = [1, 1, 1, 1]			
Generation 1600: Best = 1.189979e+04	FES = 160100	$\mu_F = 0.704$	$\mu_{CR} = 0.252$	
Diversity = 1.2629e-02	k_t = [4, 2, 2, 4]			
Generation 1610: Best = 1.189966e+04	FES = 161100	$\mu_F = 0.706$	$\mu_{CR} = 0.252$	
Diversity = 1.1159e-02	k_t = [1, 1, 1, 1]			
Generation 1620: Best = 1.189955e+04	FES = 162100	$\mu_F = 0.700$	$\mu_{CR} = 0.249$	
Diversity = 9.8542e-03	k_t = [2, 3, 2, 3]			
Generation 1630: Best = 1.189944e+04	FES = 163100	$\mu_F = 0.683$	$\mu_{CR} = 0.246$	
Diversity = 9.1127e-03	k_t = [1, 1, 1, 1]			
Generation 1640: Best = 1.189934e+04	FES = 164100	$\mu_F = 0.676$	$\mu_{CR} = 0.245$	
Diversity = 8.2175e-03	k_t = [3, 2, 3, 3]			
Generation 1650: Best = 1.189929e+04	FES = 165100	$\mu_F = 0.677$	$\mu_{CR} = 0.256$	
Diversity = 7.6690e-03	k_t = [1, 1, 1, 1]			
Generation 1660: Best = 1.189920e+04	FES = 166100	$\mu_F = 0.690$	$\mu_{CR} = 0.252$	
Diversity = 7.5941e-03	k_t = [2, 3, 3, 2]			
Generation 1670: Best = 1.189911e+04	FES = 167100	$\mu_F = 0.696$	$\mu_{CR} = 0.254$	
Diversity = 7.7459e-03	k_t = [1, 1, 1, 1]			
Generation 1680: Best = 1.189906e+04	FES = 168100	$\mu_F = 0.695$	$\mu_{CR} = 0.252$	
Diversity = 6.7883e-03	k_t = [3, 4, 4, 4]			
Generation 1690: Best = 1.189899e+04	FES = 169100	$\mu_F = 0.688$	$\mu_{CR} = 0.258$	
Diversity = 6.0276e-03	k_t = [1, 1, 1, 1]			
Generation 1700: Best = 1.189894e+04	FES = 170100	$\mu_F = 0.691$	$\mu_{CR} = 0.261$	
Diversity = 5.5686e-03	k_t = [4, 2, 2, 3]			
Generation 1710: Best = 1.189889e+04	FES = 171100	$\mu_F = 0.685$	$\mu_{CR} = 0.264$	
Diversity = 5.5240e-03	k_t = [1, 1, 1, 1]			
Generation 1720: Best = 1.189885e+04	FES = 172100	$\mu_F = 0.706$	$\mu_{CR} = 0.271$	
Diversity = 5.0967e-03	k_t = [3, 3, 4, 4]			
Generation 1730: Best = 1.189879e+04	FES = 173100	$\mu_F = 0.710$	$\mu_{CR} = 0.277$	
Diversity = 5.3305e-03	k_t = [1, 1, 1, 1]			
Generation 1740: Best = 1.189874e+04	FES = 174100	$\mu_F = 0.727$	$\mu_{CR} = 0.275$	
Diversity = 5.0448e-03	k_t = [4, 4, 3, 4]			
Generation 1750: Best = 1.189869e+04	FES = 175100	$\mu_F = 0.729$	$\mu_{CR} = 0.277$	
Diversity = 5.0567e-03	k_t = [1, 1, 1, 1]			
Generation 1760: Best = 1.189867e+04	FES = 176100	$\mu_F = 0.727$	$\mu_{CR} = 0.270$	
Diversity = 4.6616e-03	k_t = [3, 3, 2, 4]			
Generation 1770: Best = 1.189862e+04	FES = 177100	$\mu_F = 0.708$	$\mu_{CR} = 0.261$	
Diversity = 4.4927e-03	k_t = [1, 1, 1, 1]			
Generation 1780: Best = 1.189859e+04	FES = 178100	$\mu_F = 0.720$	$\mu_{CR} = 0.263$	
Diversity = 4.6933e-03	k_t = [4, 4, 3, 2]			
Generation 1790: Best = 1.189858e+04	FES = 179100	$\mu_F = 0.726$	$\mu_{CR} = 0.270$	
Diversity = 4.4922e-03	k_t = [1, 1, 1, 1]			
Generation 1800: Best = 1.189854e+04	FES = 180100	$\mu_F = 0.715$	$\mu_{CR} = 0.272$	
Diversity = 4.4860e-03	k_t = [2, 4, 3, 2]			
Generation 1810: Best = 1.189852e+04	FES = 181100	$\mu_F = 0.717$	$\mu_{CR} = 0.271$	
Diversity = 4.5690e-03	k_t = [1, 1, 1, 1]			
Generation 1820: Best = 1.189849e+04	FES = 182100	$\mu_F = 0.711$	$\mu_{CR} = 0.263$	
Diversity = 4.3252e-03	k_t = [4, 3, 4, 3]			
Generation 1830: Best = 1.189846e+04	FES = 183100	$\mu_F = 0.703$	$\mu_{CR} = 0.269$	
Diversity = 4.1586e-03	k_t = [1, 1, 1, 1]			
Generation 1840: Best = 1.189844e+04	FES = 184100	$\mu_F = 0.707$	$\mu_{CR} = 0.268$	

Diversity = 3.7340e-03	k_t = [3, 2, 3, 3]			
Generation 1850: Best = 1.189842e+04	FES = 185100	$\mu_F = 0.712$	$\mu_{CR} = 0.273$	
Diversity = 3.8851e-03	k_t = [1, 1, 1, 1]			
Generation 1860: Best = 1.189839e+04	FES = 186100	$\mu_F = 0.726$	$\mu_{CR} = 0.276$	
Diversity = 4.0077e-03	k_t = [3, 3, 2, 4]			
Generation 1870: Best = 1.189835e+04	FES = 187100	$\mu_F = 0.738$	$\mu_{CR} = 0.270$	
Diversity = 4.2858e-03	k_t = [1, 1, 1, 1]			
Generation 1880: Best = 1.189833e+04	FES = 188100	$\mu_F = 0.726$	$\mu_{CR} = 0.278$	
Diversity = 4.1516e-03	k_t = [3, 4, 4, 4]			
Generation 1890: Best = 1.189830e+04	FES = 189100	$\mu_F = 0.723$	$\mu_{CR} = 0.266$	
Diversity = 4.0955e-03	k_t = [1, 1, 1, 1]			
Generation 1900: Best = 1.189825e+04	FES = 190100	$\mu_F = 0.724$	$\mu_{CR} = 0.269$	
Diversity = 4.0482e-03	k_t = [4, 2, 4, 2]			
Generation 1910: Best = 1.189823e+04	FES = 191100	$\mu_F = 0.719$	$\mu_{CR} = 0.270$	
Diversity = 4.0696e-03	k_t = [1, 1, 1, 1]			
Generation 1920: Best = 1.189819e+04	FES = 192100	$\mu_F = 0.705$	$\mu_{CR} = 0.274$	
Diversity = 4.0001e-03	k_t = [4, 2, 4, 2]			
Generation 1930: Best = 1.189816e+04	FES = 193100	$\mu_F = 0.713$	$\mu_{CR} = 0.278$	
Diversity = 4.2023e-03	k_t = [1, 1, 1, 1]			
Generation 1940: Best = 1.189811e+04	FES = 194100	$\mu_F = 0.725$	$\mu_{CR} = 0.272$	
Diversity = 4.2051e-03	k_t = [2, 4, 2, 4]			
Generation 1950: Best = 1.189810e+04	FES = 195100	$\mu_F = 0.719$	$\mu_{CR} = 0.273$	
Diversity = 3.9222e-03	k_t = [1, 1, 1, 1]			
Generation 1960: Best = 1.189807e+04	FES = 196100	$\mu_F = 0.714$	$\mu_{CR} = 0.275$	
Diversity = 3.5980e-03	k_t = [4, 3, 4, 3]			
Generation 1970: Best = 1.189804e+04	FES = 197100	$\mu_F = 0.732$	$\mu_{CR} = 0.275$	
Diversity = 3.8226e-03	k_t = [1, 1, 1, 1]			
Generation 1980: Best = 1.189802e+04	FES = 198100	$\mu_F = 0.735$	$\mu_{CR} = 0.269$	
Diversity = 3.8619e-03	k_t = [3, 2, 3, 4]			
Generation 1990: Best = 1.189800e+04	FES = 199100	$\mu_F = 0.736$	$\mu_{CR} = 0.262$	
Diversity = 3.9280e-03	k_t = [1, 1, 1, 1]			
Generation 2000: Best = 1.189798e+04	FES = 200100	$\mu_F = 0.726$	$\mu_{CR} = 0.262$	
Diversity = 3.7626e-03	k_t = [3, 4, 4, 2]			
Generation 2010: Best = 1.189797e+04	FES = 201100	$\mu_F = 0.735$	$\mu_{CR} = 0.268$	
Diversity = 3.8094e-03	k_t = [1, 1, 1, 1]			
Generation 2020: Best = 1.189795e+04	FES = 202100	$\mu_F = 0.708$	$\mu_{CR} = 0.266$	
Diversity = 3.5608e-03	k_t = [2, 3, 3, 2]			
Generation 2030: Best = 1.189792e+04	FES = 203100	$\mu_F = 0.722$	$\mu_{CR} = 0.263$	
Diversity = 3.7285e-03	k_t = [1, 1, 1, 1]			
Generation 2040: Best = 1.189789e+04	FES = 204100	$\mu_F = 0.724$	$\mu_{CR} = 0.261$	
Diversity = 3.5424e-03	k_t = [3, 3, 4, 4]			
Generation 2050: Best = 1.189787e+04	FES = 205100	$\mu_F = 0.722$	$\mu_{CR} = 0.257$	
Diversity = 3.6808e-03	k_t = [1, 1, 1, 1]			
Generation 2060: Best = 1.189785e+04	FES = 206100	$\mu_F = 0.723$	$\mu_{CR} = 0.252$	
Diversity = 3.7184e-03	k_t = [3, 3, 3, 2]			
Generation 2070: Best = 1.189784e+04	FES = 207100	$\mu_F = 0.719$	$\mu_{CR} = 0.244$	
Diversity = 3.7482e-03	k_t = [1, 1, 1, 1]			
Generation 2080: Best = 1.189781e+04	FES = 208100	$\mu_F = 0.732$	$\mu_{CR} = 0.225$	
Diversity = 3.5522e-03	k_t = [2, 2, 2, 4]			
Generation 2090: Best = 1.189779e+04	FES = 209100	$\mu_F = 0.733$	$\mu_{CR} = 0.225$	
Diversity = 3.6949e-03	k_t = [1, 1, 1, 1]			
Generation 2100: Best = 1.189776e+04	FES = 210100	$\mu_F = 0.736$	$\mu_{CR} = 0.218$	
Diversity = 3.8796e-03	k_t = [4, 2, 3, 4]			
Generation 2110: Best = 1.189772e+04	FES = 211100	$\mu_F = 0.732$	$\mu_{CR} = 0.219$	

Diversity = 3.7835e-03	k_t = [1, 1, 1, 1]			
Generation 2120: Best = 1.189768e+04	FES = 212100	$\mu_F = 0.737$	$\mu_{CR} = 0.218$	
Diversity = 3.9998e-03	k_t = [4, 3, 3, 3]			
Generation 2130: Best = 1.189763e+04	FES = 213100	$\mu_F = 0.746$	$\mu_{CR} = 0.212$	
Diversity = 4.3622e-03	k_t = [1, 1, 1, 1]			
Generation 2140: Best = 1.189761e+04	FES = 214100	$\mu_F = 0.763$	$\mu_{CR} = 0.204$	
Diversity = 4.6650e-03	k_t = [2, 2, 2, 4]			
Generation 2150: Best = 1.189755e+04	FES = 215100	$\mu_F = 0.766$	$\mu_{CR} = 0.211$	
Diversity = 5.1825e-03	k_t = [1, 1, 1, 1]			
Generation 2160: Best = 1.189744e+04	FES = 216100	$\mu_F = 0.771$	$\mu_{CR} = 0.207$	
Diversity = 5.7859e-03	k_t = [3, 4, 3, 3]			
Generation 2170: Best = 1.189740e+04	FES = 217100	$\mu_F = 0.757$	$\mu_{CR} = 0.203$	
Diversity = 6.3030e-03	k_t = [1, 1, 1, 1]			
Generation 2180: Best = 1.189732e+04	FES = 218100	$\mu_F = 0.747$	$\mu_{CR} = 0.206$	
Diversity = 6.0887e-03	k_t = [4, 4, 3, 3]			
Generation 2190: Best = 1.189729e+04	FES = 219100	$\mu_F = 0.749$	$\mu_{CR} = 0.211$	
Diversity = 6.3195e-03	k_t = [1, 1, 1, 1]			
Generation 2200: Best = 1.189725e+04	FES = 220100	$\mu_F = 0.736$	$\mu_{CR} = 0.197$	
Diversity = 5.8950e-03	k_t = [3, 2, 2, 4]			
Generation 2210: Best = 1.189719e+04	FES = 221100	$\mu_F = 0.731$	$\mu_{CR} = 0.196$	
Diversity = 5.5387e-03	k_t = [1, 1, 1, 1]			
Generation 2220: Best = 1.189716e+04	FES = 222100	$\mu_F = 0.737$	$\mu_{CR} = 0.193$	
Diversity = 5.3017e-03	k_t = [2, 4, 4, 4]			
Generation 2230: Best = 1.189708e+04	FES = 223100	$\mu_F = 0.736$	$\mu_{CR} = 0.190$	
Diversity = 5.4256e-03	k_t = [1, 1, 1, 1]			
Generation 2240: Best = 1.189703e+04	FES = 224100	$\mu_F = 0.743$	$\mu_{CR} = 0.191$	
Diversity = 5.9575e-03	k_t = [2, 3, 2, 3]			
Generation 2250: Best = 1.189701e+04	FES = 225100	$\mu_F = 0.730$	$\mu_{CR} = 0.189$	
Diversity = 5.9601e-03	k_t = [1, 1, 1, 1]			
Generation 2260: Best = 1.189697e+04	FES = 226100	$\mu_F = 0.725$	$\mu_{CR} = 0.190$	
Diversity = 5.7827e-03	k_t = [4, 3, 4, 4]			
Generation 2270: Best = 1.189695e+04	FES = 227100	$\mu_F = 0.721$	$\mu_{CR} = 0.193$	
Diversity = 5.4634e-03	k_t = [1, 1, 1, 1]			
Generation 2280: Best = 1.189692e+04	FES = 228100	$\mu_F = 0.707$	$\mu_{CR} = 0.197$	
Diversity = 4.9853e-03	k_t = [3, 4, 2, 2]			
Generation 2290: Best = 1.189690e+04	FES = 229100	$\mu_F = 0.702$	$\mu_{CR} = 0.191$	
Diversity = 4.8350e-03	k_t = [1, 1, 1, 1]			
Generation 2300: Best = 1.189688e+04	FES = 230100	$\mu_F = 0.702$	$\mu_{CR} = 0.194$	
Diversity = 4.5197e-03	k_t = [4, 2, 4, 2]			
Generation 2310: Best = 1.189686e+04	FES = 231100	$\mu_F = 0.695$	$\mu_{CR} = 0.184$	
Diversity = 4.0348e-03	k_t = [1, 1, 1, 1]			
Generation 2320: Best = 1.189684e+04	FES = 232100	$\mu_F = 0.689$	$\mu_{CR} = 0.184$	
Diversity = 3.4911e-03	k_t = [4, 4, 2, 4]			
Generation 2330: Best = 1.189683e+04	FES = 233100	$\mu_F = 0.689$	$\mu_{CR} = 0.191$	
Diversity = 3.4442e-03	k_t = [1, 1, 1, 1]			
Generation 2340: Best = 1.189682e+04	FES = 234100	$\mu_F = 0.687$	$\mu_{CR} = 0.189$	
Diversity = 3.1977e-03	k_t = [4, 2, 3, 4]			
Generation 2350: Best = 1.189680e+04	FES = 235100	$\mu_F = 0.700$	$\mu_{CR} = 0.191$	
Diversity = 3.0258e-03	k_t = [1, 1, 1, 1]			
Generation 2360: Best = 1.189679e+04	FES = 236100	$\mu_F = 0.700$	$\mu_{CR} = 0.199$	
Diversity = 2.9407e-03	k_t = [3, 3, 4, 2]			
Generation 2370: Best = 1.189678e+04	FES = 237100	$\mu_F = 0.697$	$\mu_{CR} = 0.192$	
Diversity = 2.8470e-03	k_t = [1, 1, 1, 1]			
Generation 2380: Best = 1.189676e+04	FES = 238100	$\mu_F = 0.720$	$\mu_{CR} = 0.196$	

Diversity = 2.7621e-03	k_t = [3, 3, 4, 3]			
Generation 2390: Best = 1.189675e+04	FES = 239100	$\mu_F = 0.727$	$\mu_{CR} = 0.195$	
Diversity = 3.0031e-03	k_t = [1, 1, 1, 1]			
Generation 2400: Best = 1.189674e+04	FES = 240100	$\mu_F = 0.728$	$\mu_{CR} = 0.191$	
Diversity = 2.8806e-03	k_t = [3, 2, 4, 4]			
Generation 2410: Best = 1.189672e+04	FES = 241100	$\mu_F = 0.734$	$\mu_{CR} = 0.196$	
Diversity = 2.6377e-03	k_t = [1, 1, 1, 1]			
Generation 2420: Best = 1.189671e+04	FES = 242100	$\mu_F = 0.747$	$\mu_{CR} = 0.181$	
Diversity = 2.6391e-03	k_t = [2, 2, 4, 4]			
Generation 2430: Best = 1.189670e+04	FES = 243100	$\mu_F = 0.766$	$\mu_{CR} = 0.176$	
Diversity = 2.8509e-03	k_t = [1, 1, 1, 1]			
Generation 2440: Best = 1.189668e+04	FES = 244100	$\mu_F = 0.768$	$\mu_{CR} = 0.176$	
Diversity = 3.1059e-03	k_t = [2, 4, 4, 3]			
Generation 2450: Best = 1.189666e+04	FES = 245100	$\mu_F = 0.761$	$\mu_{CR} = 0.186$	
Diversity = 3.2787e-03	k_t = [1, 1, 1, 1]			
Generation 2460: Best = 1.189664e+04	FES = 246100	$\mu_F = 0.753$	$\mu_{CR} = 0.189$	
Diversity = 3.5773e-03	k_t = [3, 4, 3, 2]			
Generation 2470: Best = 1.189663e+04	FES = 247100	$\mu_F = 0.745$	$\mu_{CR} = 0.189$	
Diversity = 3.6286e-03	k_t = [1, 1, 1, 1]			
Generation 2480: Best = 1.189659e+04	FES = 248100	$\mu_F = 0.744$	$\mu_{CR} = 0.182$	
Diversity = 3.3893e-03	k_t = [4, 2, 3, 4]			
Generation 2490: Best = 1.189659e+04	FES = 249100	$\mu_F = 0.724$	$\mu_{CR} = 0.181$	
Diversity = 3.4447e-03	k_t = [1, 1, 1, 1]			
Generation 2500: Best = 1.189657e+04	FES = 250100	$\mu_F = 0.724$	$\mu_{CR} = 0.186$	
Diversity = 3.3003e-03	k_t = [4, 4, 2, 2]			
Generation 2510: Best = 1.189655e+04	FES = 251100	$\mu_F = 0.728$	$\mu_{CR} = 0.184$	
Diversity = 3.4046e-03	k_t = [1, 1, 1, 1]			
Generation 2520: Best = 1.189653e+04	FES = 252100	$\mu_F = 0.725$	$\mu_{CR} = 0.189$	
Diversity = 3.1285e-03	k_t = [2, 4, 2, 4]			
Generation 2530: Best = 1.189652e+04	FES = 253100	$\mu_F = 0.734$	$\mu_{CR} = 0.187$	
Diversity = 3.1321e-03	k_t = [1, 1, 1, 1]			
Generation 2540: Best = 1.189650e+04	FES = 254100	$\mu_F = 0.724$	$\mu_{CR} = 0.200$	
Diversity = 3.0255e-03	k_t = [4, 3, 2, 4]			
Generation 2550: Best = 1.189648e+04	FES = 255100	$\mu_F = 0.731$	$\mu_{CR} = 0.205$	
Diversity = 3.1033e-03	k_t = [1, 1, 1, 1]			
Generation 2560: Best = 1.189647e+04	FES = 256100	$\mu_F = 0.739$	$\mu_{CR} = 0.205$	
Diversity = 2.9822e-03	k_t = [2, 4, 3, 2]			
Generation 2570: Best = 1.189646e+04	FES = 257100	$\mu_F = 0.731$	$\mu_{CR} = 0.206$	
Diversity = 2.9571e-03	k_t = [1, 1, 1, 1]			
Generation 2580: Best = 1.189645e+04	FES = 258100	$\mu_F = 0.719$	$\mu_{CR} = 0.205$	
Diversity = 2.8069e-03	k_t = [2, 3, 3, 3]			
Generation 2590: Best = 1.189645e+04	FES = 259100	$\mu_F = 0.699$	$\mu_{CR} = 0.206$	
Diversity = 2.7048e-03	k_t = [1, 1, 1, 1]			
Generation 2600: Best = 1.189644e+04	FES = 260100	$\mu_F = 0.697$	$\mu_{CR} = 0.201$	
Diversity = 2.5052e-03	k_t = [2, 2, 2, 2]			
Generation 2610: Best = 1.189643e+04	FES = 261100	$\mu_F = 0.686$	$\mu_{CR} = 0.204$	
Diversity = 2.3998e-03	k_t = [1, 1, 1, 1]			
Generation 2620: Best = 1.189643e+04	FES = 262100	$\mu_F = 0.698$	$\mu_{CR} = 0.206$	
Diversity = 2.1964e-03	k_t = [3, 4, 4, 4]			
Generation 2630: Best = 1.189642e+04	FES = 263100	$\mu_F = 0.705$	$\mu_{CR} = 0.200$	
Diversity = 2.0025e-03	k_t = [1, 1, 1, 1]			
Generation 2640: Best = 1.189642e+04	FES = 264100	$\mu_F = 0.692$	$\mu_{CR} = 0.200$	
Diversity = 1.8486e-03	k_t = [4, 4, 4, 3]			
Generation 2650: Best = 1.189641e+04	FES = 265100	$\mu_F = 0.696$	$\mu_{CR} = 0.201$	

Diversity = 1.7540e-03	k_t = [1, 1, 1, 1]			
Generation 2660: Best = 1.189641e+04	FES = 266100	$\mu_F = 0.689$	$\mu_{CR} = 0.201$	
Diversity = 1.6505e-03	k_t = [4, 3, 2, 2]			
Generation 2670: Best = 1.189641e+04	FES = 267100	$\mu_F = 0.687$	$\mu_{CR} = 0.201$	
Diversity = 1.5857e-03	k_t = [1, 1, 1, 1]			
Generation 2680: Best = 1.189641e+04	FES = 268100	$\mu_F = 0.676$	$\mu_{CR} = 0.193$	
Diversity = 1.4174e-03	k_t = [2, 2, 4, 4]			
Generation 2690: Best = 1.189640e+04	FES = 269100	$\mu_F = 0.682$	$\mu_{CR} = 0.193$	
Diversity = 1.3011e-03	k_t = [1, 1, 1, 1]			
Generation 2700: Best = 1.189640e+04	FES = 270100	$\mu_F = 0.689$	$\mu_{CR} = 0.194$	
Diversity = 1.2423e-03	k_t = [2, 3, 3, 3]			
Generation 2710: Best = 1.189640e+04	FES = 271100	$\mu_F = 0.693$	$\mu_{CR} = 0.185$	
Diversity = 1.2017e-03	k_t = [1, 1, 1, 1]			
Generation 2720: Best = 1.189640e+04	FES = 272100	$\mu_F = 0.703$	$\mu_{CR} = 0.176$	
Diversity = 1.1876e-03	k_t = [2, 3, 2, 2]			
Generation 2730: Best = 1.189640e+04	FES = 273100	$\mu_F = 0.692$	$\mu_{CR} = 0.176$	
Diversity = 1.1587e-03	k_t = [1, 1, 1, 1]			
Generation 2740: Best = 1.189640e+04	FES = 274100	$\mu_F = 0.691$	$\mu_{CR} = 0.180$	
Diversity = 1.0843e-03	k_t = [4, 4, 2, 3]			
Generation 2750: Best = 1.189639e+04	FES = 275100	$\mu_F = 0.667$	$\mu_{CR} = 0.191$	
Diversity = 1.0946e-03	k_t = [1, 1, 1, 1]			
Generation 2760: Best = 1.189639e+04	FES = 276100	$\mu_F = 0.667$	$\mu_{CR} = 0.187$	
Diversity = 1.0157e-03	k_t = [4, 4, 3, 4]			
Generation 2770: Best = 1.189639e+04	FES = 277100	$\mu_F = 0.669$	$\mu_{CR} = 0.182$	
Diversity = 9.1767e-04	k_t = [1, 1, 1, 1]			
Generation 2780: Best = 1.189639e+04	FES = 278100	$\mu_F = 0.676$	$\mu_{CR} = 0.174$	
Diversity = 9.3599e-04	k_t = [4, 2, 2, 2]			
Generation 2790: Best = 1.189639e+04	FES = 279100	$\mu_F = 0.687$	$\mu_{CR} = 0.172$	
Diversity = 9.3736e-04	k_t = [1, 1, 1, 1]			
Generation 2800: Best = 1.189639e+04	FES = 280100	$\mu_F = 0.689$	$\mu_{CR} = 0.164$	
Diversity = 8.7499e-04	k_t = [3, 2, 4, 3]			
Generation 2810: Best = 1.189639e+04	FES = 281100	$\mu_F = 0.700$	$\mu_{CR} = 0.169$	
Diversity = 9.0960e-04	k_t = [1, 1, 1, 1]			
Generation 2820: Best = 1.189638e+04	FES = 282100	$\mu_F = 0.701$	$\mu_{CR} = 0.170$	
Diversity = 9.4744e-04	k_t = [2, 2, 2, 4]			
Generation 2830: Best = 1.189638e+04	FES = 283100	$\mu_F = 0.700$	$\mu_{CR} = 0.170$	
Diversity = 9.7107e-04	k_t = [1, 1, 1, 1]			
Generation 2840: Best = 1.189638e+04	FES = 284100	$\mu_F = 0.712$	$\mu_{CR} = 0.164$	
Diversity = 1.0625e-03	k_t = [3, 2, 2, 2]			
Generation 2850: Best = 1.189638e+04	FES = 285100	$\mu_F = 0.717$	$\mu_{CR} = 0.161$	
Diversity = 1.0767e-03	k_t = [1, 1, 1, 1]			
Generation 2860: Best = 1.189638e+04	FES = 286100	$\mu_F = 0.717$	$\mu_{CR} = 0.158$	
Diversity = 1.0154e-03	k_t = [2, 4, 3, 3]			
Generation 2870: Best = 1.189638e+04	FES = 287100	$\mu_F = 0.727$	$\mu_{CR} = 0.161$	
Diversity = 1.0145e-03	k_t = [1, 1, 1, 1]			
Generation 2880: Best = 1.189638e+04	FES = 288100	$\mu_F = 0.733$	$\mu_{CR} = 0.158$	
Diversity = 9.9476e-04	k_t = [4, 3, 4, 3]			
Generation 2890: Best = 1.189637e+04	FES = 289100	$\mu_F = 0.730$	$\mu_{CR} = 0.154$	
Diversity = 1.0231e-03	k_t = [1, 1, 1, 1]			
Generation 2900: Best = 1.189637e+04	FES = 290100	$\mu_F = 0.745$	$\mu_{CR} = 0.154$	
Diversity = 1.1107e-03	k_t = [4, 2, 2, 2]			
Generation 2910: Best = 1.189637e+04	FES = 291100	$\mu_F = 0.739$	$\mu_{CR} = 0.154$	
Diversity = 1.1528e-03	k_t = [1, 1, 1, 1]			
Generation 2920: Best = 1.189637e+04	FES = 292100	$\mu_F = 0.733$	$\mu_{CR} = 0.152$	

Diversity = 1.0466e-03	k_t = [3, 4, 2, 4]			
Generation 2930: Best = 1.189637e+04	FES = 293100	$\mu_F = 0.734$	$\mu_{CR} = 0.154$	
Diversity = 9.9001e-04	k_t = [1, 1, 1, 1]			
Generation 2940: Best = 1.189637e+04	FES = 294100	$\mu_F = 0.738$	$\mu_{CR} = 0.153$	
Diversity = 9.6922e-04	k_t = [4, 2, 3, 4]			
Generation 2950: Best = 1.189637e+04	FES = 295100	$\mu_F = 0.730$	$\mu_{CR} = 0.148$	
Diversity = 9.5680e-04	k_t = [1, 1, 1, 1]			
Generation 2960: Best = 1.189636e+04	FES = 296100	$\mu_F = 0.727$	$\mu_{CR} = 0.153$	
Diversity = 9.1049e-04	k_t = [2, 3, 3, 4]			
Generation 2970: Best = 1.189636e+04	FES = 297100	$\mu_F = 0.740$	$\mu_{CR} = 0.156$	
Diversity = 8.9702e-04	k_t = [1, 1, 1, 1]			
Generation 2980: Best = 1.189636e+04	FES = 298100	$\mu_F = 0.740$	$\mu_{CR} = 0.159$	
Diversity = 9.2056e-04	k_t = [4, 3, 3, 4]			
Generation 2990: Best = 1.189636e+04	FES = 299100	$\mu_F = 0.733$	$\mu_{CR} = 0.162$	
Diversity = 9.3868e-04	k_t = [1, 1, 1, 1]			
Generation 3000: Best = 1.189636e+04	FES = 300100	$\mu_F = 0.728$	$\mu_{CR} = 0.161$	
Diversity = 9.5926e-04	k_t = [2, 4, 2, 2]			
Generation 3010: Best = 1.189636e+04	FES = 301100	$\mu_F = 0.726$	$\mu_{CR} = 0.154$	
Diversity = 9.8906e-04	k_t = [1, 1, 1, 1]			
Generation 3020: Best = 1.189636e+04	FES = 302100	$\mu_F = 0.706$	$\mu_{CR} = 0.161$	
Diversity = 8.5664e-04	k_t = [4, 3, 4, 4]			
Generation 3030: Best = 1.189636e+04	FES = 303100	$\mu_F = 0.729$	$\mu_{CR} = 0.150$	
Diversity = 8.2481e-04	k_t = [1, 1, 1, 1]			
Generation 3040: Best = 1.189636e+04	FES = 304100	$\mu_F = 0.740$	$\mu_{CR} = 0.151$	
Diversity = 8.4177e-04	k_t = [4, 3, 4, 2]			
Generation 3050: Best = 1.189635e+04	FES = 305100	$\mu_F = 0.729$	$\mu_{CR} = 0.141$	
Diversity = 8.8128e-04	k_t = [1, 1, 1, 1]			
Generation 3060: Best = 1.189635e+04	FES = 306100	$\mu_F = 0.735$	$\mu_{CR} = 0.141$	
Diversity = 8.4734e-04	k_t = [3, 3, 4, 3]			
Generation 3070: Best = 1.189635e+04	FES = 307100	$\mu_F = 0.729$	$\mu_{CR} = 0.142$	
Diversity = 8.7350e-04	k_t = [1, 1, 1, 1]			
Generation 3080: Best = 1.189635e+04	FES = 308100	$\mu_F = 0.733$	$\mu_{CR} = 0.134$	
Diversity = 8.7615e-04	k_t = [2, 3, 4, 2]			
Generation 3090: Best = 1.189635e+04	FES = 309100	$\mu_F = 0.727$	$\mu_{CR} = 0.134$	
Diversity = 9.1290e-04	k_t = [1, 1, 1, 1]			
Generation 3100: Best = 1.189635e+04	FES = 310100	$\mu_F = 0.727$	$\mu_{CR} = 0.149$	
Diversity = 8.9474e-04	k_t = [4, 3, 4, 2]			
Generation 3110: Best = 1.189635e+04	FES = 311100	$\mu_F = 0.730$	$\mu_{CR} = 0.146$	
Diversity = 8.9041e-04	k_t = [1, 1, 1, 1]			
Generation 3120: Best = 1.189635e+04	FES = 312100	$\mu_F = 0.716$	$\mu_{CR} = 0.151$	
Diversity = 8.0241e-04	k_t = [3, 4, 2, 2]			
Generation 3130: Best = 1.189635e+04	FES = 313100	$\mu_F = 0.729$	$\mu_{CR} = 0.143$	
Diversity = 8.0194e-04	k_t = [1, 1, 1, 1]			
Generation 3140: Best = 1.189635e+04	FES = 314100	$\mu_F = 0.721$	$\mu_{CR} = 0.133$	
Diversity = 7.6634e-04	k_t = [3, 2, 4, 2]			
Generation 3150: Best = 1.189635e+04	FES = 315100	$\mu_F = 0.705$	$\mu_{CR} = 0.136$	
Diversity = 7.6171e-04	k_t = [1, 1, 1, 1]			
Generation 3160: Best = 1.189635e+04	FES = 316100	$\mu_F = 0.720$	$\mu_{CR} = 0.143$	
Diversity = 7.2689e-04	k_t = [2, 3, 2, 4]			
Generation 3170: Best = 1.189634e+04	FES = 317100	$\mu_F = 0.725$	$\mu_{CR} = 0.139$	
Diversity = 7.3648e-04	k_t = [1, 1, 1, 1]			
Generation 3180: Best = 1.189634e+04	FES = 318100	$\mu_F = 0.720$	$\mu_{CR} = 0.145$	
Diversity = 7.5273e-04	k_t = [3, 2, 2, 2]			
Generation 3190: Best = 1.189634e+04	FES = 319100	$\mu_F = 0.733$	$\mu_{CR} = 0.145$	

Diversity = 8.1460e-04	k_t = [1, 1, 1, 1]			
Generation 3200: Best = 1.189634e+04	FES = 320100	$\mu_F = 0.723$	$\mu_{CR} = 0.146$	
Diversity = 8.3497e-04	k_t = [4, 2, 2, 2]			
Generation 3210: Best = 1.189634e+04	FES = 321100	$\mu_F = 0.744$	$\mu_{CR} = 0.144$	
Diversity = 8.6819e-04	k_t = [1, 1, 1, 1]			
Generation 3220: Best = 1.189634e+04	FES = 322100	$\mu_F = 0.726$	$\mu_{CR} = 0.157$	
Diversity = 8.2435e-04	k_t = [3, 3, 3, 3]			
Generation 3230: Best = 1.189634e+04	FES = 323100	$\mu_F = 0.730$	$\mu_{CR} = 0.159$	
Diversity = 8.8791e-04	k_t = [1, 1, 1, 1]			
Generation 3240: Best = 1.189634e+04	FES = 324100	$\mu_F = 0.718$	$\mu_{CR} = 0.152$	
Diversity = 8.2364e-04	k_t = [3, 3, 3, 4]			
Generation 3250: Best = 1.189634e+04	FES = 325100	$\mu_F = 0.728$	$\mu_{CR} = 0.149$	
Diversity = 7.4636e-04	k_t = [1, 1, 1, 1]			
Generation 3260: Best = 1.189634e+04	FES = 326100	$\mu_F = 0.740$	$\mu_{CR} = 0.161$	
Diversity = 7.7897e-04	k_t = [4, 3, 3, 2]			
Generation 3270: Best = 1.189634e+04	FES = 327100	$\mu_F = 0.730$	$\mu_{CR} = 0.159$	
Diversity = 7.9919e-04	k_t = [1, 1, 1, 1]			
Generation 3280: Best = 1.189633e+04	FES = 328100	$\mu_F = 0.733$	$\mu_{CR} = 0.159$	
Diversity = 7.4221e-04	k_t = [4, 4, 4, 4]			
Generation 3290: Best = 1.189633e+04	FES = 329100	$\mu_F = 0.740$	$\mu_{CR} = 0.159$	
Diversity = 7.1201e-04	k_t = [1, 1, 1, 1]			
Generation 3300: Best = 1.189633e+04	FES = 330100	$\mu_F = 0.740$	$\mu_{CR} = 0.162$	
Diversity = 7.2555e-04	k_t = [2, 4, 2, 3]			
Generation 3310: Best = 1.189633e+04	FES = 331100	$\mu_F = 0.729$	$\mu_{CR} = 0.162$	
Diversity = 7.2556e-04	k_t = [1, 1, 1, 1]			
Generation 3320: Best = 1.189633e+04	FES = 332100	$\mu_F = 0.730$	$\mu_{CR} = 0.162$	
Diversity = 7.0078e-04	k_t = [4, 3, 2, 4]			
Generation 3330: Best = 1.189633e+04	FES = 333100	$\mu_F = 0.748$	$\mu_{CR} = 0.163$	
Diversity = 6.8511e-04	k_t = [1, 1, 1, 1]			
Generation 3340: Best = 1.189633e+04	FES = 334100	$\mu_F = 0.732$	$\mu_{CR} = 0.172$	
Diversity = 6.7173e-04	k_t = [4, 2, 2, 3]			
Generation 3350: Best = 1.189633e+04	FES = 335100	$\mu_F = 0.727$	$\mu_{CR} = 0.173$	
Diversity = 7.0206e-04	k_t = [1, 1, 1, 1]			
Generation 3360: Best = 1.189633e+04	FES = 336100	$\mu_F = 0.726$	$\mu_{CR} = 0.179$	
Diversity = 6.7567e-04	k_t = [3, 2, 2, 4]			
Generation 3370: Best = 1.189633e+04	FES = 337100	$\mu_F = 0.735$	$\mu_{CR} = 0.179$	
Diversity = 7.0739e-04	k_t = [1, 1, 1, 1]			
Generation 3380: Best = 1.189633e+04	FES = 338100	$\mu_F = 0.723$	$\mu_{CR} = 0.181$	
Diversity = 6.8526e-04	k_t = [2, 4, 2, 3]			
Generation 3390: Best = 1.189633e+04	FES = 339100	$\mu_F = 0.732$	$\mu_{CR} = 0.182$	
Diversity = 7.2765e-04	k_t = [1, 1, 1, 1]			
Generation 3400: Best = 1.189633e+04	FES = 340100	$\mu_F = 0.726$	$\mu_{CR} = 0.184$	
Diversity = 7.1826e-04	k_t = [2, 2, 4, 4]			
Generation 3410: Best = 1.189633e+04	FES = 341100	$\mu_F = 0.718$	$\mu_{CR} = 0.187$	
Diversity = 7.5546e-04	k_t = [1, 1, 1, 1]			
Generation 3420: Best = 1.189632e+04	FES = 342100	$\mu_F = 0.714$	$\mu_{CR} = 0.188$	
Diversity = 7.3299e-04	k_t = [2, 4, 2, 4]			
Generation 3430: Best = 1.189632e+04	FES = 343100	$\mu_F = 0.727$	$\mu_{CR} = 0.191$	
Diversity = 8.0464e-04	k_t = [1, 1, 1, 1]			
Generation 3440: Best = 1.189632e+04	FES = 344100	$\mu_F = 0.717$	$\mu_{CR} = 0.189$	
Diversity = 7.8670e-04	k_t = [3, 2, 3, 3]			
Generation 3450: Best = 1.189632e+04	FES = 345100	$\mu_F = 0.715$	$\mu_{CR} = 0.181$	
Diversity = 7.6290e-04	k_t = [1, 1, 1, 1]			
Generation 3460: Best = 1.189632e+04	FES = 346100	$\mu_F = 0.701$	$\mu_{CR} = 0.161$	

Diversity = 7.2466e-04 k_t = [4, 3, 4, 2]
 Generation 3470: Best = 1.189632e+04 FES = 347100 μ_F = 0.705 μ_{CR} = 0.163
 Diversity = 6.9117e-04 k_t = [1, 1, 1, 1]
 Generation 3480: Best = 1.189632e+04 FES = 348100 μ_F = 0.717 μ_{CR} = 0.159
 Diversity = 6.9593e-04 k_t = [3, 3, 3, 2]
 Generation 3490: Best = 1.189632e+04 FES = 349100 μ_F = 0.714 μ_{CR} = 0.171
 Diversity = 7.1881e-04 k_t = [1, 1, 1, 1]
 Generation 3500: Best = 1.189632e+04 FES = 350100 μ_F = 0.714 μ_{CR} = 0.169
 Diversity = 7.3054e-04 k_t = [4, 2, 4, 3]
 Generation 3510: Best = 1.189632e+04 FES = 351100 μ_F = 0.719 μ_{CR} = 0.166
 Diversity = 7.5085e-04 k_t = [1, 1, 1, 1]
 Generation 3520: Best = 1.189632e+04 FES = 352100 μ_F = 0.729 μ_{CR} = 0.160
 Diversity = 7.3507e-04 k_t = [2, 3, 3, 3]
 Generation 3530: Best = 1.189632e+04 FES = 353100 μ_F = 0.720 μ_{CR} = 0.157
 Diversity = 7.2387e-04 k_t = [1, 1, 1, 1]
 Generation 3540: Best = 1.189632e+04 FES = 354100 μ_F = 0.724 μ_{CR} = 0.152
 Diversity = 7.1658e-04 k_t = [2, 2, 3, 3]
 Generation 3550: Best = 1.189631e+04 FES = 355100 μ_F = 0.733 μ_{CR} = 0.151
 Diversity = 7.4698e-04 k_t = [1, 1, 1, 1]
 Generation 3560: Best = 1.189631e+04 FES = 356100 μ_F = 0.719 μ_{CR} = 0.149
 Diversity = 7.3021e-04 k_t = [2, 2, 2, 4]
 Generation 3570: Best = 1.189631e+04 FES = 357100 μ_F = 0.729 μ_{CR} = 0.144
 Diversity = 7.2559e-04 k_t = [1, 1, 1, 1]
 Generation 3580: Best = 1.189631e+04 FES = 358100 μ_F = 0.723 μ_{CR} = 0.158
 Diversity = 7.0010e-04 k_t = [4, 3, 4, 4]
 Generation 3590: Best = 1.189631e+04 FES = 359100 μ_F = 0.730 μ_{CR} = 0.167
 Diversity = 7.5729e-04 k_t = [1, 1, 1, 1]
 Generation 3600: Best = 1.189631e+04 FES = 360100 μ_F = 0.724 μ_{CR} = 0.166
 Diversity = 7.3818e-04 k_t = [4, 4, 2, 2]
 Generation 3610: Best = 1.189631e+04 FES = 361100 μ_F = 0.722 μ_{CR} = 0.158
 Diversity = 7.5465e-04 k_t = [1, 1, 1, 1]
 Generation 3620: Best = 1.189631e+04 FES = 362100 μ_F = 0.715 μ_{CR} = 0.149
 Diversity = 7.4552e-04 k_t = [4, 3, 2, 4]
 Generation 3630: Best = 1.189631e+04 FES = 363100 μ_F = 0.700 μ_{CR} = 0.151
 Diversity = 6.8742e-04 k_t = [1, 1, 1, 1]
 Generation 3640: Best = 1.189631e+04 FES = 364100 μ_F = 0.700 μ_{CR} = 0.147
 Diversity = 6.1276e-04 k_t = [4, 4, 3, 3]
 Generation 3650: Best = 1.189631e+04 FES = 365100 μ_F = 0.701 μ_{CR} = 0.145
 Diversity = 6.1046e-04 k_t = [1, 1, 1, 1]
 Generation 3660: Best = 1.189631e+04 FES = 366100 μ_F = 0.728 μ_{CR} = 0.138
 Diversity = 5.8535e-04 k_t = [4, 4, 2, 3]
 Generation 3670: Best = 1.189631e+04 FES = 367100 μ_F = 0.723 μ_{CR} = 0.142
 Diversity = 5.8730e-04 k_t = [1, 1, 1, 1]
 Generation 3680: Best = 1.189631e+04 FES = 368100 μ_F = 0.711 μ_{CR} = 0.140
 Diversity = 5.5157e-04 k_t = [2, 4, 2, 4]
 Generation 3690: Best = 1.189631e+04 FES = 369100 μ_F = 0.720 μ_{CR} = 0.147
 Diversity = 5.2258e-04 k_t = [1, 1, 1, 1]
 Generation 3700: Best = 1.189631e+04 FES = 370100 μ_F = 0.719 μ_{CR} = 0.155
 Diversity = 4.9640e-04 k_t = [3, 4, 4, 4]
 Generation 3710: Best = 1.189631e+04 FES = 371100 μ_F = 0.729 μ_{CR} = 0.137
 Diversity = 4.8793e-04 k_t = [1, 1, 1, 1]
 Generation 3720: Best = 1.189631e+04 FES = 372100 μ_F = 0.726 μ_{CR} = 0.147
 Diversity = 4.7915e-04 k_t = [4, 2, 3, 3]
 Generation 3730: Best = 1.189631e+04 FES = 373100 μ_F = 0.749 μ_{CR} = 0.146

Diversity = 4.9467e-04	k_t = [1, 1, 1, 1]			
Generation 3740: Best = 1.189631e+04	FES = 374100	$\mu_F = 0.746$	$\mu_{CR} = 0.158$	
Diversity = 5.1083e-04	k_t = [3, 2, 3, 3]			
Generation 3750: Best = 1.189631e+04	FES = 375100	$\mu_F = 0.738$	$\mu_{CR} = 0.160$	
Diversity = 5.1351e-04	k_t = [1, 1, 1, 1]			
Generation 3760: Best = 1.189631e+04	FES = 376100	$\mu_F = 0.727$	$\mu_{CR} = 0.153$	
Diversity = 4.8445e-04	k_t = [3, 2, 4, 4]			
Generation 3770: Best = 1.189630e+04	FES = 377100	$\mu_F = 0.724$	$\mu_{CR} = 0.157$	
Diversity = 4.9527e-04	k_t = [1, 1, 1, 1]			
Generation 3780: Best = 1.189630e+04	FES = 378100	$\mu_F = 0.736$	$\mu_{CR} = 0.146$	
Diversity = 5.0885e-04	k_t = [3, 3, 4, 4]			
Generation 3790: Best = 1.189630e+04	FES = 379100	$\mu_F = 0.733$	$\mu_{CR} = 0.164$	
Diversity = 5.6028e-04	k_t = [1, 1, 1, 1]			
Generation 3800: Best = 1.189630e+04	FES = 380100	$\mu_F = 0.742$	$\mu_{CR} = 0.166$	
Diversity = 6.6319e-04	k_t = [3, 2, 2, 2]			
Generation 3810: Best = 1.189630e+04	FES = 381100	$\mu_F = 0.736$	$\mu_{CR} = 0.163$	
Diversity = 7.1306e-04	k_t = [1, 1, 1, 1]			
Generation 3820: Best = 1.189630e+04	FES = 382100	$\mu_F = 0.730$	$\mu_{CR} = 0.166$	
Diversity = 7.4159e-04	k_t = [3, 3, 3, 2]			
Generation 3830: Best = 1.189630e+04	FES = 383100	$\mu_F = 0.737$	$\mu_{CR} = 0.169$	
Diversity = 8.1461e-04	k_t = [1, 1, 1, 1]			
Generation 3840: Best = 1.189630e+04	FES = 384100	$\mu_F = 0.733$	$\mu_{CR} = 0.173$	
Diversity = 7.9380e-04	k_t = [2, 4, 4, 2]			
Generation 3850: Best = 1.189630e+04	FES = 385100	$\mu_F = 0.732$	$\mu_{CR} = 0.182$	
Diversity = 7.6951e-04	k_t = [1, 1, 1, 1]			
Generation 3860: Best = 1.189630e+04	FES = 386100	$\mu_F = 0.720$	$\mu_{CR} = 0.186$	
Diversity = 8.0324e-04	k_t = [2, 2, 3, 2]			
Generation 3870: Best = 1.189629e+04	FES = 387100	$\mu_F = 0.719$	$\mu_{CR} = 0.179$	
Diversity = 8.1916e-04	k_t = [1, 1, 1, 1]			
Generation 3880: Best = 1.189629e+04	FES = 388100	$\mu_F = 0.722$	$\mu_{CR} = 0.180$	
Diversity = 8.0116e-04	k_t = [4, 3, 3, 2]			
Generation 3890: Best = 1.189629e+04	FES = 389100	$\mu_F = 0.722$	$\mu_{CR} = 0.173$	
Diversity = 7.7256e-04	k_t = [1, 1, 1, 1]			
Generation 3900: Best = 1.189629e+04	FES = 390100	$\mu_F = 0.699$	$\mu_{CR} = 0.174$	
Diversity = 6.5471e-04	k_t = [2, 4, 3, 4]			
Generation 3910: Best = 1.189629e+04	FES = 391100	$\mu_F = 0.704$	$\mu_{CR} = 0.179$	
Diversity = 6.3478e-04	k_t = [1, 1, 1, 1]			
Generation 3920: Best = 1.189629e+04	FES = 392100	$\mu_F = 0.696$	$\mu_{CR} = 0.175$	
Diversity = 5.9856e-04	k_t = [4, 4, 3, 4]			
Generation 3930: Best = 1.189629e+04	FES = 393100	$\mu_F = 0.689$	$\mu_{CR} = 0.177$	
Diversity = 5.5070e-04	k_t = [1, 1, 1, 1]			
Generation 3940: Best = 1.189629e+04	FES = 394100	$\mu_F = 0.687$	$\mu_{CR} = 0.173$	
Diversity = 5.0952e-04	k_t = [4, 4, 2, 3]			
Generation 3950: Best = 1.189629e+04	FES = 395100	$\mu_F = 0.693$	$\mu_{CR} = 0.168$	
Diversity = 5.4717e-04	k_t = [1, 1, 1, 1]			
Generation 3960: Best = 1.189629e+04	FES = 396100	$\mu_F = 0.690$	$\mu_{CR} = 0.171$	
Diversity = 4.9042e-04	k_t = [2, 4, 2, 4]			
Generation 3970: Best = 1.189629e+04	FES = 397100	$\mu_F = 0.696$	$\mu_{CR} = 0.175$	
Diversity = 4.7510e-04	k_t = [1, 1, 1, 1]			
Generation 3980: Best = 1.189629e+04	FES = 398100	$\mu_F = 0.696$	$\mu_{CR} = 0.167$	
Diversity = 4.8817e-04	k_t = [4, 2, 3, 3]			
Generation 3990: Best = 1.189629e+04	FES = 399100	$\mu_F = 0.701$	$\mu_{CR} = 0.169$	
Diversity = 4.8747e-04	k_t = [1, 1, 1, 1]			
Generation 4000: Best = 1.189629e+04	FES = 400100	$\mu_F = 0.701$	$\mu_{CR} = 0.168$	

Diversity = 4.3657e-04	k_t = [2, 3, 4, 3]			
Generation 4010: Best = 1.189629e+04	FES = 401100	$\mu_F = 0.696$	$\mu_{CR} = 0.170$	
Diversity = 4.2407e-04	k_t = [1, 1, 1, 1]			
Generation 4020: Best = 1.189629e+04	FES = 402100	$\mu_F = 0.693$	$\mu_{CR} = 0.174$	
Diversity = 3.7529e-04	k_t = [3, 4, 3, 3]			
Generation 4030: Best = 1.189629e+04	FES = 403100	$\mu_F = 0.690$	$\mu_{CR} = 0.171$	
Diversity = 3.8700e-04	k_t = [1, 1, 1, 1]			
Generation 4040: Best = 1.189629e+04	FES = 404100	$\mu_F = 0.685$	$\mu_{CR} = 0.164$	
Diversity = 3.5738e-04	k_t = [4, 4, 2, 3]			
Generation 4050: Best = 1.189629e+04	FES = 405100	$\mu_F = 0.679$	$\mu_{CR} = 0.155$	
Diversity = 3.1403e-04	k_t = [1, 1, 1, 1]			
Generation 4060: Best = 1.189629e+04	FES = 406100	$\mu_F = 0.668$	$\mu_{CR} = 0.155$	
Diversity = 2.9668e-04	k_t = [3, 2, 3, 2]			
Generation 4070: Best = 1.189629e+04	FES = 407100	$\mu_F = 0.676$	$\mu_{CR} = 0.161$	
Diversity = 3.0380e-04	k_t = [1, 1, 1, 1]			
Generation 4080: Best = 1.189629e+04	FES = 408100	$\mu_F = 0.674$	$\mu_{CR} = 0.160$	
Diversity = 2.6994e-04	k_t = [2, 3, 3, 4]			
Generation 4090: Best = 1.189629e+04	FES = 409100	$\mu_F = 0.672$	$\mu_{CR} = 0.145$	
Diversity = 2.5452e-04	k_t = [1, 1, 1, 1]			
Generation 4100: Best = 1.189629e+04	FES = 410100	$\mu_F = 0.688$	$\mu_{CR} = 0.148$	
Diversity = 2.5214e-04	k_t = [3, 4, 2, 4]			
Generation 4110: Best = 1.189629e+04	FES = 411100	$\mu_F = 0.706$	$\mu_{CR} = 0.143$	
Diversity = 2.4934e-04	k_t = [1, 1, 1, 1]			
Generation 4120: Best = 1.189629e+04	FES = 412100	$\mu_F = 0.716$	$\mu_{CR} = 0.134$	
Diversity = 2.5450e-04	k_t = [4, 3, 3, 2]			
Generation 4130: Best = 1.189629e+04	FES = 413100	$\mu_F = 0.710$	$\mu_{CR} = 0.134$	
Diversity = 2.5637e-04	k_t = [1, 1, 1, 1]			
Generation 4140: Best = 1.189629e+04	FES = 414100	$\mu_F = 0.724$	$\mu_{CR} = 0.133$	
Diversity = 2.5024e-04	k_t = [3, 4, 3, 4]			
Generation 4150: Best = 1.189629e+04	FES = 415100	$\mu_F = 0.729$	$\mu_{CR} = 0.128$	
Diversity = 2.3500e-04	k_t = [1, 1, 1, 1]			
Generation 4160: Best = 1.189629e+04	FES = 416100	$\mu_F = 0.721$	$\mu_{CR} = 0.127$	
Diversity = 2.2833e-04	k_t = [3, 2, 4, 4]			
Generation 4170: Best = 1.189629e+04	FES = 417100	$\mu_F = 0.720$	$\mu_{CR} = 0.137$	
Diversity = 2.3206e-04	k_t = [1, 1, 1, 1]			
Generation 4180: Best = 1.189629e+04	FES = 418100	$\mu_F = 0.728$	$\mu_{CR} = 0.139$	
Diversity = 2.4308e-04	k_t = [4, 2, 2, 2]			
Generation 4190: Best = 1.189629e+04	FES = 419100	$\mu_F = 0.724$	$\mu_{CR} = 0.147$	
Diversity = 2.4266e-04	k_t = [1, 1, 1, 1]			
Generation 4200: Best = 1.189629e+04	FES = 420100	$\mu_F = 0.735$	$\mu_{CR} = 0.151$	
Diversity = 2.5730e-04	k_t = [3, 2, 3, 2]			
Generation 4210: Best = 1.189629e+04	FES = 421100	$\mu_F = 0.734$	$\mu_{CR} = 0.147$	
Diversity = 2.9094e-04	k_t = [1, 1, 1, 1]			
Generation 4220: Best = 1.189629e+04	FES = 422100	$\mu_F = 0.743$	$\mu_{CR} = 0.149$	
Diversity = 2.8812e-04	k_t = [2, 4, 3, 4]			
Generation 4230: Best = 1.189629e+04	FES = 423100	$\mu_F = 0.733$	$\mu_{CR} = 0.152$	
Diversity = 2.9895e-04	k_t = [1, 1, 1, 1]			
Generation 4240: Best = 1.189629e+04	FES = 424100	$\mu_F = 0.734$	$\mu_{CR} = 0.144$	
Diversity = 2.8042e-04	k_t = [3, 3, 4, 3]			
Generation 4250: Best = 1.189629e+04	FES = 425100	$\mu_F = 0.720$	$\mu_{CR} = 0.142$	
Diversity = 2.8566e-04	k_t = [1, 1, 1, 1]			
Generation 4260: Best = 1.189629e+04	FES = 426100	$\mu_F = 0.719$	$\mu_{CR} = 0.138$	
Diversity = 2.7503e-04	k_t = [3, 4, 4, 2]			
Generation 4270: Best = 1.189629e+04	FES = 427100	$\mu_F = 0.733$	$\mu_{CR} = 0.129$	

Diversity = 2.7018e-04 k_t = [1, 1, 1, 1]
 Generation 4280: Best = 1.189629e+04 FES = 428100 μ_F = 0.724 μ_{CR} = 0.121
 Diversity = 2.3587e-04 k_t = [2, 2, 4, 4]
 Generation 4290: Best = 1.189629e+04 FES = 429100 μ_F = 0.739 μ_{CR} = 0.124
 Diversity = 2.3498e-04 k_t = [1, 1, 1, 1]
 Generation 4300: Best = 1.189629e+04 FES = 430100 μ_F = 0.738 μ_{CR} = 0.129
 Diversity = 2.3145e-04 k_t = [4, 2, 3, 3]
 Generation 4310: Best = 1.189629e+04 FES = 431100 μ_F = 0.754 μ_{CR} = 0.137
 Diversity = 2.4615e-04 k_t = [1, 1, 1, 1]
 Generation 4320: Best = 1.189629e+04 FES = 432100 μ_F = 0.753 μ_{CR} = 0.144
 Diversity = 2.6044e-04 k_t = [3, 2, 3, 3]
 Generation 4330: Best = 1.189629e+04 FES = 433100 μ_F = 0.742 μ_{CR} = 0.149
 Diversity = 2.6750e-04 k_t = [1, 1, 1, 1]
 Generation 4340: Best = 1.189629e+04 FES = 434100 μ_F = 0.742 μ_{CR} = 0.147
 Diversity = 2.8411e-04 k_t = [2, 2, 3, 3]
 Generation 4350: Best = 1.189629e+04 FES = 435100 μ_F = 0.731 μ_{CR} = 0.147
 Diversity = 2.9802e-04 k_t = [1, 1, 1, 1]
 Generation 4360: Best = 1.189629e+04 FES = 436100 μ_F = 0.732 μ_{CR} = 0.138
 Diversity = 2.7821e-04 k_t = [2, 2, 2, 3]
 Generation 4370: Best = 1.189628e+04 FES = 437100 μ_F = 0.727 μ_{CR} = 0.137
 Diversity = 2.7459e-04 k_t = [1, 1, 1, 1]
 Generation 4380: Best = 1.189628e+04 FES = 438100 μ_F = 0.728 μ_{CR} = 0.141
 Diversity = 2.6591e-04 k_t = [3, 3, 2, 4]
 Generation 4390: Best = 1.189628e+04 FES = 439100 μ_F = 0.725 μ_{CR} = 0.141
 Diversity = 2.7397e-04 k_t = [1, 1, 1, 1]
 Generation 4400: Best = 1.189628e+04 FES = 440100 μ_F = 0.735 μ_{CR} = 0.140
 Diversity = 2.7983e-04 k_t = [4, 4, 3, 4]
 Generation 4410: Best = 1.189628e+04 FES = 441100 μ_F = 0.738 μ_{CR} = 0.135
 Diversity = 2.6893e-04 k_t = [1, 1, 1, 1]
 Generation 4420: Best = 1.189628e+04 FES = 442100 μ_F = 0.738 μ_{CR} = 0.137
 Diversity = 2.6248e-04 k_t = [3, 2, 2, 3]
 Generation 4430: Best = 1.189628e+04 FES = 443100 μ_F = 0.720 μ_{CR} = 0.145
 Diversity = 2.5289e-04 k_t = [1, 1, 1, 1]
 Generation 4440: Best = 1.189628e+04 FES = 444100 μ_F = 0.739 μ_{CR} = 0.146
 Diversity = 2.4672e-04 k_t = [3, 3, 2, 4]
 Generation 4450: Best = 1.189628e+04 FES = 445100 μ_F = 0.714 μ_{CR} = 0.144
 Diversity = 2.3286e-04 k_t = [1, 1, 1, 1]
 Generation 4460: Best = 1.189628e+04 FES = 446100 μ_F = 0.708 μ_{CR} = 0.143
 Diversity = 2.1764e-04 k_t = [3, 3, 3, 4]
 Generation 4470: Best = 1.189628e+04 FES = 447100 μ_F = 0.716 μ_{CR} = 0.145
 Diversity = 2.0827e-04 k_t = [1, 1, 1, 1]
 Generation 4480: Best = 1.189628e+04 FES = 448100 μ_F = 0.711 μ_{CR} = 0.145
 Diversity = 1.9127e-04 k_t = [4, 3, 3, 3]
 Generation 4490: Best = 1.189628e+04 FES = 449100 μ_F = 0.699 μ_{CR} = 0.134
 Diversity = 1.8844e-04 k_t = [1, 1, 1, 1]
 Generation 4500: Best = 1.189628e+04 FES = 450100 μ_F = 0.710 μ_{CR} = 0.134
 Diversity = 1.7603e-04 k_t = [3, 2, 2, 3]
 Generation 4510: Best = 1.189628e+04 FES = 451100 μ_F = 0.707 μ_{CR} = 0.139
 Diversity = 1.7339e-04 k_t = [1, 1, 1, 1]
 Generation 4520: Best = 1.189628e+04 FES = 452100 μ_F = 0.717 μ_{CR} = 0.139
 Diversity = 1.6400e-04 k_t = [3, 4, 2, 3]
 Generation 4530: Best = 1.189628e+04 FES = 453100 μ_F = 0.722 μ_{CR} = 0.143
 Diversity = 1.7501e-04 k_t = [1, 1, 1, 1]
 Generation 4540: Best = 1.189628e+04 FES = 454100 μ_F = 0.732 μ_{CR} = 0.146

Diversity = 1.7417e-04	k_t = [4, 4, 4, 4]			
Generation 4550: Best = 1.189628e+04	FES = 455100	$\mu_F = 0.736$	$\mu_{CR} = 0.142$	
Diversity = 1.7030e-04	k_t = [1, 1, 1, 1]			
Generation 4560: Best = 1.189628e+04	FES = 456100	$\mu_F = 0.747$	$\mu_{CR} = 0.131$	
Diversity = 1.5988e-04	k_t = [4, 4, 3, 3]			
Generation 4570: Best = 1.189628e+04	FES = 457100	$\mu_F = 0.737$	$\mu_{CR} = 0.119$	
Diversity = 1.5945e-04	k_t = [1, 1, 1, 1]			
Generation 4580: Best = 1.189628e+04	FES = 458100	$\mu_F = 0.734$	$\mu_{CR} = 0.111$	
Diversity = 1.5129e-04	k_t = [2, 2, 2, 4]			
Generation 4590: Best = 1.189628e+04	FES = 459100	$\mu_F = 0.726$	$\mu_{CR} = 0.110$	
Diversity = 1.4461e-04	k_t = [1, 1, 1, 1]			
Generation 4600: Best = 1.189628e+04	FES = 460100	$\mu_F = 0.735$	$\mu_{CR} = 0.113$	
Diversity = 1.4487e-04	k_t = [2, 2, 2, 3]			
Generation 4610: Best = 1.189628e+04	FES = 461100	$\mu_F = 0.738$	$\mu_{CR} = 0.117$	
Diversity = 1.5815e-04	k_t = [1, 1, 1, 1]			
Generation 4620: Best = 1.189628e+04	FES = 462100	$\mu_F = 0.729$	$\mu_{CR} = 0.117$	
Diversity = 1.5645e-04	k_t = [4, 4, 4, 3]			
Generation 4630: Best = 1.189628e+04	FES = 463100	$\mu_F = 0.716$	$\mu_{CR} = 0.118$	
Diversity = 1.6074e-04	k_t = [1, 1, 1, 1]			
Generation 4640: Best = 1.189628e+04	FES = 464100	$\mu_F = 0.716$	$\mu_{CR} = 0.127$	
Diversity = 1.4967e-04	k_t = [3, 3, 3, 4]			
Generation 4650: Best = 1.189628e+04	FES = 465100	$\mu_F = 0.725$	$\mu_{CR} = 0.127$	
Diversity = 1.6980e-04	k_t = [1, 1, 1, 1]			
Generation 4660: Best = 1.189628e+04	FES = 466100	$\mu_F = 0.751$	$\mu_{CR} = 0.121$	
Diversity = 1.9879e-04	k_t = [4, 4, 2, 2]			
Generation 4670: Best = 1.189628e+04	FES = 467100	$\mu_F = 0.752$	$\mu_{CR} = 0.112$	
Diversity = 2.1571e-04	k_t = [1, 1, 1, 1]			
Generation 4680: Best = 1.189628e+04	FES = 468100	$\mu_F = 0.756$	$\mu_{CR} = 0.124$	
Diversity = 2.0505e-04	k_t = [3, 2, 2, 4]			
Generation 4690: Best = 1.189628e+04	FES = 469100	$\mu_F = 0.755$	$\mu_{CR} = 0.124$	
Diversity = 1.9520e-04	k_t = [1, 1, 1, 1]			
Generation 4700: Best = 1.189628e+04	FES = 470100	$\mu_F = 0.766$	$\mu_{CR} = 0.120$	
Diversity = 2.1121e-04	k_t = [2, 2, 3, 4]			
Generation 4710: Best = 1.189628e+04	FES = 471100	$\mu_F = 0.774$	$\mu_{CR} = 0.119$	
Diversity = 2.1926e-04	k_t = [1, 1, 1, 1]			
Generation 4720: Best = 1.189628e+04	FES = 472100	$\mu_F = 0.779$	$\mu_{CR} = 0.119$	
Diversity = 2.4505e-04	k_t = [4, 3, 4, 4]			
Generation 4730: Best = 1.189628e+04	FES = 473100	$\mu_F = 0.776$	$\mu_{CR} = 0.123$	
Diversity = 2.5071e-04	k_t = [1, 1, 1, 1]			
Generation 4740: Best = 1.189628e+04	FES = 474100	$\mu_F = 0.763$	$\mu_{CR} = 0.126$	
Diversity = 2.5332e-04	k_t = [4, 3, 2, 4]			
Generation 4750: Best = 1.189628e+04	FES = 475100	$\mu_F = 0.769$	$\mu_{CR} = 0.119$	
Diversity = 2.5587e-04	k_t = [1, 1, 1, 1]			
Generation 4760: Best = 1.189628e+04	FES = 476100	$\mu_F = 0.762$	$\mu_{CR} = 0.112$	
Diversity = 2.4179e-04	k_t = [2, 4, 3, 4]			
Generation 4770: Best = 1.189628e+04	FES = 477100	$\mu_F = 0.751$	$\mu_{CR} = 0.104$	
Diversity = 2.2866e-04	k_t = [1, 1, 1, 1]			
Generation 4780: Best = 1.189628e+04	FES = 478100	$\mu_F = 0.749$	$\mu_{CR} = 0.099$	
Diversity = 2.3032e-04	k_t = [4, 2, 4, 2]			
Generation 4790: Best = 1.189628e+04	FES = 479100	$\mu_F = 0.746$	$\mu_{CR} = 0.100$	
Diversity = 2.4606e-04	k_t = [1, 1, 1, 1]			
Generation 4800: Best = 1.189628e+04	FES = 480100	$\mu_F = 0.748$	$\mu_{CR} = 0.103$	
Diversity = 2.3939e-04	k_t = [3, 3, 4, 3]			
Generation 4810: Best = 1.189628e+04	FES = 481100	$\mu_F = 0.741$	$\mu_{CR} = 0.110$	

Diversity = 2.4838e-04	k_t = [1, 1, 1, 1]			
Generation 4820: Best = 1.189628e+04	FES = 482100	$\mu_F = 0.734$	$\mu_{CR} = 0.113$	
Diversity = 2.4921e-04	k_t = [4, 3, 2, 4]			
Generation 4830: Best = 1.189628e+04	FES = 483100	$\mu_F = 0.739$	$\mu_{CR} = 0.112$	
Diversity = 2.4685e-04	k_t = [1, 1, 1, 1]			
Generation 4840: Best = 1.189628e+04	FES = 484100	$\mu_F = 0.733$	$\mu_{CR} = 0.116$	
Diversity = 2.3459e-04	k_t = [3, 3, 2, 3]			
Generation 4850: Best = 1.189628e+04	FES = 485100	$\mu_F = 0.736$	$\mu_{CR} = 0.119$	
Diversity = 2.3382e-04	k_t = [1, 1, 1, 1]			
Generation 4860: Best = 1.189628e+04	FES = 486100	$\mu_F = 0.754$	$\mu_{CR} = 0.112$	
Diversity = 2.4187e-04	k_t = [4, 2, 2, 2]			
Generation 4870: Best = 1.189628e+04	FES = 487100	$\mu_F = 0.765$	$\mu_{CR} = 0.109$	
Diversity = 2.5690e-04	k_t = [1, 1, 1, 1]			
Generation 4880: Best = 1.189628e+04	FES = 488100	$\mu_F = 0.759$	$\mu_{CR} = 0.120$	
Diversity = 2.6036e-04	k_t = [3, 4, 4, 4]			
Generation 4890: Best = 1.189628e+04	FES = 489100	$\mu_F = 0.749$	$\mu_{CR} = 0.121$	
Diversity = 2.9647e-04	k_t = [1, 1, 1, 1]			
Generation 4900: Best = 1.189628e+04	FES = 490100	$\mu_F = 0.770$	$\mu_{CR} = 0.114$	
Diversity = 3.1869e-04	k_t = [2, 4, 2, 3]			
Generation 4910: Best = 1.189628e+04	FES = 491100	$\mu_F = 0.751$	$\mu_{CR} = 0.122$	
Diversity = 3.3498e-04	k_t = [1, 1, 1, 1]			
Generation 4920: Best = 1.189628e+04	FES = 492100	$\mu_F = 0.738$	$\mu_{CR} = 0.108$	
Diversity = 3.1606e-04	k_t = [4, 3, 2, 2]			
Generation 4930: Best = 1.189628e+04	FES = 493100	$\mu_F = 0.732$	$\mu_{CR} = 0.102$	
Diversity = 3.1318e-04	k_t = [1, 1, 1, 1]			
Generation 4940: Best = 1.189628e+04	FES = 494100	$\mu_F = 0.719$	$\mu_{CR} = 0.116$	
Diversity = 2.8024e-04	k_t = [2, 2, 3, 4]			
Generation 4950: Best = 1.189628e+04	FES = 495100	$\mu_F = 0.708$	$\mu_{CR} = 0.121$	
Diversity = 2.8307e-04	k_t = [1, 1, 1, 1]			
Generation 4960: Best = 1.189628e+04	FES = 496100	$\mu_F = 0.708$	$\mu_{CR} = 0.115$	
Diversity = 2.8478e-04	k_t = [2, 4, 2, 2]			
Generation 4970: Best = 1.189628e+04	FES = 497100	$\mu_F = 0.722$	$\mu_{CR} = 0.109$	
Diversity = 2.8564e-04	k_t = [1, 1, 1, 1]			
Generation 4980: Best = 1.189628e+04	FES = 498100	$\mu_F = 0.717$	$\mu_{CR} = 0.102$	
Diversity = 2.8263e-04	k_t = [4, 2, 4, 3]			
Generation 4990: Best = 1.189628e+04	FES = 499100	$\mu_F = 0.730$	$\mu_{CR} = 0.103$	
Diversity = 3.0084e-04	k_t = [1, 1, 1, 1]			
Generation 5000: Best = 1.189628e+04	FES = 500100	$\mu_F = 0.725$	$\mu_{CR} = 0.095$	
Diversity = 2.9778e-04	k_t = [3, 2, 2, 2]			
Generation 5010: Best = 1.189628e+04	FES = 501100	$\mu_F = 0.729$	$\mu_{CR} = 0.095$	
Diversity = 2.9017e-04	k_t = [1, 1, 1, 1]			
Generation 5020: Best = 1.189628e+04	FES = 502100	$\mu_F = 0.734$	$\mu_{CR} = 0.106$	
Diversity = 2.8797e-04	k_t = [4, 2, 4, 3]			
Generation 5030: Best = 1.189628e+04	FES = 503100	$\mu_F = 0.721$	$\mu_{CR} = 0.114$	
Diversity = 3.0373e-04	k_t = [1, 1, 1, 1]			
Generation 5040: Best = 1.189628e+04	FES = 504100	$\mu_F = 0.709$	$\mu_{CR} = 0.112$	
Diversity = 2.8461e-04	k_t = [2, 3, 3, 4]			
Generation 5050: Best = 1.189628e+04	FES = 505100	$\mu_F = 0.717$	$\mu_{CR} = 0.118$	
Diversity = 2.9404e-04	k_t = [1, 1, 1, 1]			
Generation 5060: Best = 1.189628e+04	FES = 506100	$\mu_F = 0.723$	$\mu_{CR} = 0.131$	
Diversity = 3.2378e-04	k_t = [4, 2, 4, 2]			
Generation 5070: Best = 1.189628e+04	FES = 507100	$\mu_F = 0.727$	$\mu_{CR} = 0.136$	
Diversity = 3.4669e-04	k_t = [1, 1, 1, 1]			
Generation 5080: Best = 1.189628e+04	FES = 508100	$\mu_F = 0.737$	$\mu_{CR} = 0.137$	

Diversity = 3.5094e-04	k_t = [3, 2, 2, 2]			
Generation 5090: Best = 1.189628e+04	FES = 509100	$\mu_F = 0.748$	$\mu_{CR} = 0.145$	
Diversity = 3.8131e-04	k_t = [1, 1, 1, 1]			
Generation 5100: Best = 1.189628e+04	FES = 510100	$\mu_F = 0.742$	$\mu_{CR} = 0.144$	
Diversity = 3.9472e-04	k_t = [2, 2, 3, 2]			
Generation 5110: Best = 1.189628e+04	FES = 511100	$\mu_F = 0.754$	$\mu_{CR} = 0.142$	
Diversity = 3.9709e-04	k_t = [1, 1, 1, 1]			
Generation 5120: Best = 1.189628e+04	FES = 512100	$\mu_F = 0.751$	$\mu_{CR} = 0.128$	
Diversity = 3.6303e-04	k_t = [4, 2, 2, 4]			
Generation 5130: Best = 1.189628e+04	FES = 513100	$\mu_F = 0.762$	$\mu_{CR} = 0.121$	
Diversity = 3.6920e-04	k_t = [1, 1, 1, 1]			
Generation 5140: Best = 1.189628e+04	FES = 514100	$\mu_F = 0.758$	$\mu_{CR} = 0.117$	
Diversity = 3.4652e-04	k_t = [3, 2, 4, 4]			
Generation 5150: Best = 1.189628e+04	FES = 515100	$\mu_F = 0.751$	$\mu_{CR} = 0.121$	
Diversity = 3.6838e-04	k_t = [1, 1, 1, 1]			
Generation 5160: Best = 1.189628e+04	FES = 516100	$\mu_F = 0.751$	$\mu_{CR} = 0.121$	
Diversity = 3.8582e-04	k_t = [4, 4, 4, 2]			
Generation 5170: Best = 1.189628e+04	FES = 517100	$\mu_F = 0.744$	$\mu_{CR} = 0.120$	
Diversity = 4.2836e-04	k_t = [1, 1, 1, 1]			
Generation 5180: Best = 1.189628e+04	FES = 518100	$\mu_F = 0.729$	$\mu_{CR} = 0.119$	
Diversity = 4.0162e-04	k_t = [2, 2, 2, 2]			
Generation 5190: Best = 1.189627e+04	FES = 519100	$\mu_F = 0.737$	$\mu_{CR} = 0.122$	
Diversity = 4.0002e-04	k_t = [1, 1, 1, 1]			
Generation 5200: Best = 1.189627e+04	FES = 520100	$\mu_F = 0.737$	$\mu_{CR} = 0.128$	
Diversity = 3.8284e-04	k_t = [4, 2, 4, 3]			
Generation 5210: Best = 1.189627e+04	FES = 521100	$\mu_F = 0.737$	$\mu_{CR} = 0.128$	
Diversity = 3.6631e-04	k_t = [1, 1, 1, 1]			
Generation 5220: Best = 1.189627e+04	FES = 522100	$\mu_F = 0.755$	$\mu_{CR} = 0.125$	
Diversity = 3.7319e-04	k_t = [4, 2, 2, 2]			
Generation 5230: Best = 1.189627e+04	FES = 523100	$\mu_F = 0.741$	$\mu_{CR} = 0.110$	
Diversity = 3.7337e-04	k_t = [1, 1, 1, 1]			
Generation 5240: Best = 1.189627e+04	FES = 524100	$\mu_F = 0.722$	$\mu_{CR} = 0.103$	
Diversity = 3.4815e-04	k_t = [3, 3, 3, 3]			
Generation 5250: Best = 1.189627e+04	FES = 525100	$\mu_F = 0.708$	$\mu_{CR} = 0.103$	
Diversity = 3.3466e-04	k_t = [1, 1, 1, 1]			
Generation 5260: Best = 1.189627e+04	FES = 526100	$\mu_F = 0.712$	$\mu_{CR} = 0.106$	
Diversity = 3.0436e-04	k_t = [3, 4, 3, 3]			
Generation 5270: Best = 1.189627e+04	FES = 527100	$\mu_F = 0.703$	$\mu_{CR} = 0.104$	
Diversity = 2.7505e-04	k_t = [1, 1, 1, 1]			
Generation 5280: Best = 1.189627e+04	FES = 528100	$\mu_F = 0.689$	$\mu_{CR} = 0.111$	
Diversity = 2.3859e-04	k_t = [2, 2, 4, 4]			
Generation 5290: Best = 1.189627e+04	FES = 529100	$\mu_F = 0.704$	$\mu_{CR} = 0.110$	
Diversity = 2.3437e-04	k_t = [1, 1, 1, 1]			
Generation 5300: Best = 1.189627e+04	FES = 530100	$\mu_F = 0.710$	$\mu_{CR} = 0.100$	
Diversity = 2.1721e-04	k_t = [4, 2, 3, 4]			
Generation 5310: Best = 1.189627e+04	FES = 531100	$\mu_F = 0.727$	$\mu_{CR} = 0.103$	
Diversity = 2.0140e-04	k_t = [1, 1, 1, 1]			
Generation 5320: Best = 1.189627e+04	FES = 532100	$\mu_F = 0.713$	$\mu_{CR} = 0.101$	
Diversity = 1.9014e-04	k_t = [4, 3, 4, 4]			
Generation 5330: Best = 1.189627e+04	FES = 533100	$\mu_F = 0.716$	$\mu_{CR} = 0.099$	
Diversity = 1.9758e-04	k_t = [1, 1, 1, 1]			
Generation 5340: Best = 1.189627e+04	FES = 534100	$\mu_F = 0.726$	$\mu_{CR} = 0.098$	
Diversity = 1.8632e-04	k_t = [2, 2, 2, 4]			
Generation 5350: Best = 1.189627e+04	FES = 535100	$\mu_F = 0.724$	$\mu_{CR} = 0.101$	

Diversity = 1.9802e-04	k_t = [1, 1, 1, 1]			
Generation 5360: Best = 1.189627e+04	FES = 536100	$\mu_F = 0.735$	$\mu_{CR} = 0.103$	
Diversity = 2.0543e-04	k_t = [2, 4, 3, 2]			
Generation 5370: Best = 1.189627e+04	FES = 537100	$\mu_F = 0.730$	$\mu_{CR} = 0.096$	
Diversity = 2.0819e-04	k_t = [1, 1, 1, 1]			
Generation 5380: Best = 1.189627e+04	FES = 538100	$\mu_F = 0.730$	$\mu_{CR} = 0.094$	
Diversity = 1.9528e-04	k_t = [3, 2, 4, 3]			
Generation 5390: Best = 1.189627e+04	FES = 539100	$\mu_F = 0.735$	$\mu_{CR} = 0.096$	
Diversity = 1.9864e-04	k_t = [1, 1, 1, 1]			
Generation 5400: Best = 1.189627e+04	FES = 540100	$\mu_F = 0.747$	$\mu_{CR} = 0.092$	
Diversity = 1.9498e-04	k_t = [2, 3, 3, 4]			
Generation 5410: Best = 1.189627e+04	FES = 541100	$\mu_F = 0.755$	$\mu_{CR} = 0.097$	
Diversity = 1.9489e-04	k_t = [1, 1, 1, 1]			
Generation 5420: Best = 1.189627e+04	FES = 542100	$\mu_F = 0.742$	$\mu_{CR} = 0.092$	
Diversity = 1.9858e-04	k_t = [2, 2, 4, 3]			
Generation 5430: Best = 1.189627e+04	FES = 543100	$\mu_F = 0.726$	$\mu_{CR} = 0.085$	
Diversity = 2.0143e-04	k_t = [1, 1, 1, 1]			
Generation 5440: Best = 1.189627e+04	FES = 544100	$\mu_F = 0.712$	$\mu_{CR} = 0.092$	
Diversity = 1.9434e-04	k_t = [3, 2, 2, 2]			
Generation 5450: Best = 1.189627e+04	FES = 545100	$\mu_F = 0.703$	$\mu_{CR} = 0.090$	
Diversity = 1.8785e-04	k_t = [1, 1, 1, 1]			
Generation 5460: Best = 1.189627e+04	FES = 546100	$\mu_F = 0.704$	$\mu_{CR} = 0.080$	
Diversity = 1.7950e-04	k_t = [4, 3, 3, 3]			
Generation 5470: Best = 1.189627e+04	FES = 547100	$\mu_F = 0.718$	$\mu_{CR} = 0.086$	
Diversity = 1.8137e-04	k_t = [1, 1, 1, 1]			
Generation 5480: Best = 1.189627e+04	FES = 548100	$\mu_F = 0.718$	$\mu_{CR} = 0.087$	
Diversity = 1.7619e-04	k_t = [2, 3, 2, 4]			
Generation 5490: Best = 1.189627e+04	FES = 549100	$\mu_F = 0.703$	$\mu_{CR} = 0.092$	
Diversity = 1.6093e-04	k_t = [1, 1, 1, 1]			
Generation 5500: Best = 1.189627e+04	FES = 550100	$\mu_F = 0.717$	$\mu_{CR} = 0.087$	
Diversity = 1.5852e-04	k_t = [2, 2, 3, 4]			
Generation 5510: Best = 1.189627e+04	FES = 551100	$\mu_F = 0.719$	$\mu_{CR} = 0.079$	
Diversity = 1.5259e-04	k_t = [1, 1, 1, 1]			
Generation 5520: Best = 1.189627e+04	FES = 552100	$\mu_F = 0.721$	$\mu_{CR} = 0.078$	
Diversity = 1.4361e-04	k_t = [2, 2, 2, 3]			
Generation 5530: Best = 1.189627e+04	FES = 553100	$\mu_F = 0.718$	$\mu_{CR} = 0.075$	
Diversity = 1.3947e-04	k_t = [1, 1, 1, 1]			
Generation 5540: Best = 1.189627e+04	FES = 554100	$\mu_F = 0.702$	$\mu_{CR} = 0.073$	
Diversity = 1.2822e-04	k_t = [4, 3, 3, 3]			
Generation 5550: Best = 1.189627e+04	FES = 555100	$\mu_F = 0.705$	$\mu_{CR} = 0.076$	
Diversity = 1.2523e-04	k_t = [1, 1, 1, 1]			
Generation 5560: Best = 1.189627e+04	FES = 556100	$\mu_F = 0.727$	$\mu_{CR} = 0.072$	
Diversity = 1.3002e-04	k_t = [2, 2, 4, 2]			
Generation 5570: Best = 1.189627e+04	FES = 557100	$\mu_F = 0.736$	$\mu_{CR} = 0.069$	
Diversity = 1.3111e-04	k_t = [1, 1, 1, 1]			
Generation 5580: Best = 1.189627e+04	FES = 558100	$\mu_F = 0.738$	$\mu_{CR} = 0.071$	
Diversity = 1.2509e-04	k_t = [2, 2, 4, 2]			
Generation 5590: Best = 1.189627e+04	FES = 559100	$\mu_F = 0.739$	$\mu_{CR} = 0.069$	
Diversity = 1.3169e-04	k_t = [1, 1, 1, 1]			
Generation 5600: Best = 1.189627e+04	FES = 560100	$\mu_F = 0.730$	$\mu_{CR} = 0.080$	
Diversity = 1.3305e-04	k_t = [2, 3, 3, 2]			
Generation 5610: Best = 1.189627e+04	FES = 561100	$\mu_F = 0.736$	$\mu_{CR} = 0.086$	
Diversity = 1.3114e-04	k_t = [1, 1, 1, 1]			
Generation 5620: Best = 1.189627e+04	FES = 562100	$\mu_F = 0.728$	$\mu_{CR} = 0.086$	

Diversity = 1.2686e-04	k_t = [3, 4, 4, 4]			
Generation 5630: Best = 1.189627e+04	FES = 563100	$\mu_F = 0.750$	$\mu_{CR} = 0.081$	
Diversity = 1.1997e-04	k_t = [1, 1, 1, 1]			
Generation 5640: Best = 1.189627e+04	FES = 564100	$\mu_F = 0.759$	$\mu_{CR} = 0.083$	
Diversity = 1.1957e-04	k_t = [4, 3, 4, 2]			
Generation 5650: Best = 1.189627e+04	FES = 565100	$\mu_F = 0.747$	$\mu_{CR} = 0.082$	
Diversity = 1.2178e-04	k_t = [1, 1, 1, 1]			
Generation 5660: Best = 1.189627e+04	FES = 566100	$\mu_F = 0.757$	$\mu_{CR} = 0.075$	
Diversity = 1.1873e-04	k_t = [2, 2, 3, 4]			
Generation 5670: Best = 1.189627e+04	FES = 567100	$\mu_F = 0.756$	$\mu_{CR} = 0.078$	
Diversity = 1.1408e-04	k_t = [1, 1, 1, 1]			
Generation 5680: Best = 1.189627e+04	FES = 568100	$\mu_F = 0.728$	$\mu_{CR} = 0.072$	
Diversity = 1.0717e-04	k_t = [2, 4, 2, 4]			
Generation 5690: Best = 1.189627e+04	FES = 569100	$\mu_F = 0.728$	$\mu_{CR} = 0.071$	
Diversity = 1.0155e-04	k_t = [1, 1, 1, 1]			
Generation 5700: Best = 1.189627e+04	FES = 570100	$\mu_F = 0.725$	$\mu_{CR} = 0.077$	
Diversity = 8.9435e-05	k_t = [3, 4, 4, 4]			
Generation 5710: Best = 1.189627e+04	FES = 571100	$\mu_F = 0.722$	$\mu_{CR} = 0.082$	
Diversity = 8.5430e-05	k_t = [1, 1, 1, 1]			
Generation 5720: Best = 1.189627e+04	FES = 572100	$\mu_F = 0.729$	$\mu_{CR} = 0.081$	
Diversity = 8.6144e-05	k_t = [3, 2, 4, 4]			
Generation 5730: Best = 1.189627e+04	FES = 573100	$\mu_F = 0.719$	$\mu_{CR} = 0.073$	
Diversity = 8.9612e-05	k_t = [1, 1, 1, 1]			
Generation 5740: Best = 1.189627e+04	FES = 574100	$\mu_F = 0.733$	$\mu_{CR} = 0.090$	
Diversity = 8.7917e-05	k_t = [3, 2, 4, 2]			
Generation 5750: Best = 1.189627e+04	FES = 575100	$\mu_F = 0.737$	$\mu_{CR} = 0.094$	
Diversity = 9.5402e-05	k_t = [1, 1, 1, 1]			
Generation 5760: Best = 1.189627e+04	FES = 576100	$\mu_F = 0.744$	$\mu_{CR} = 0.088$	
Diversity = 9.3673e-05	k_t = [4, 4, 3, 3]			
Generation 5770: Best = 1.189627e+04	FES = 577100	$\mu_F = 0.751$	$\mu_{CR} = 0.088$	
Diversity = 9.7173e-05	k_t = [1, 1, 1, 1]			
Generation 5780: Best = 1.189627e+04	FES = 578100	$\mu_F = 0.743$	$\mu_{CR} = 0.089$	
Diversity = 1.0062e-04	k_t = [2, 3, 3, 4]			
Generation 5790: Best = 1.189627e+04	FES = 579100	$\mu_F = 0.761$	$\mu_{CR} = 0.095$	
Diversity = 9.8009e-05	k_t = [1, 1, 1, 1]			
Generation 5800: Best = 1.189627e+04	FES = 580100	$\mu_F = 0.762$	$\mu_{CR} = 0.098$	
Diversity = 1.0381e-04	k_t = [2, 4, 4, 2]			
Generation 5810: Best = 1.189627e+04	FES = 581100	$\mu_F = 0.784$	$\mu_{CR} = 0.107$	
Diversity = 1.1701e-04	k_t = [1, 1, 1, 1]			
Generation 5820: Best = 1.189627e+04	FES = 582100	$\mu_F = 0.798$	$\mu_{CR} = 0.105$	
Diversity = 1.1658e-04	k_t = [3, 4, 4, 3]			
Generation 5830: Best = 1.189627e+04	FES = 583100	$\mu_F = 0.771$	$\mu_{CR} = 0.110$	
Diversity = 1.1603e-04	k_t = [1, 1, 1, 1]			
Generation 5840: Best = 1.189627e+04	FES = 584100	$\mu_F = 0.744$	$\mu_{CR} = 0.110$	
Diversity = 1.0856e-04	k_t = [3, 2, 2, 4]			
Generation 5850: Best = 1.189627e+04	FES = 585100	$\mu_F = 0.732$	$\mu_{CR} = 0.114$	
Diversity = 1.1030e-04	k_t = [1, 1, 1, 1]			
Generation 5860: Best = 1.189627e+04	FES = 586100	$\mu_F = 0.731$	$\mu_{CR} = 0.110$	
Diversity = 1.0212e-04	k_t = [4, 2, 4, 3]			
Generation 5870: Best = 1.189627e+04	FES = 587100	$\mu_F = 0.714$	$\mu_{CR} = 0.121$	
Diversity = 1.0209e-04	k_t = [1, 1, 1, 1]			
Generation 5880: Best = 1.189627e+04	FES = 588100	$\mu_F = 0.720$	$\mu_{CR} = 0.113$	
Diversity = 1.0430e-04	k_t = [3, 3, 4, 3]			
Generation 5890: Best = 1.189627e+04	FES = 589100	$\mu_F = 0.734$	$\mu_{CR} = 0.113$	

Diversity = 1.1071e-04	k_t = [1, 1, 1, 1]			
Generation 5900: Best = 1.189627e+04	FES = 590100	$\mu_F = 0.747$	$\mu_{CR} = 0.116$	
Diversity = 1.1343e-04	k_t = [3, 4, 2, 3]			
Generation 5910: Best = 1.189627e+04	FES = 591100	$\mu_F = 0.743$	$\mu_{CR} = 0.111$	
Diversity = 1.2028e-04	k_t = [1, 1, 1, 1]			
Generation 5920: Best = 1.189627e+04	FES = 592100	$\mu_F = 0.730$	$\mu_{CR} = 0.106$	
Diversity = 1.1917e-04	k_t = [4, 2, 2, 2]			
Generation 5930: Best = 1.189627e+04	FES = 593100	$\mu_F = 0.720$	$\mu_{CR} = 0.111$	
Diversity = 1.2781e-04	k_t = [1, 1, 1, 1]			
Generation 5940: Best = 1.189627e+04	FES = 594100	$\mu_F = 0.717$	$\mu_{CR} = 0.120$	
Diversity = 1.2945e-04	k_t = [2, 3, 2, 2]			
Generation 5950: Best = 1.189627e+04	FES = 595100	$\mu_F = 0.706$	$\mu_{CR} = 0.122$	
Diversity = 1.2452e-04	k_t = [1, 1, 1, 1]			
Generation 5960: Best = 1.189627e+04	FES = 596100	$\mu_F = 0.717$	$\mu_{CR} = 0.117$	
Diversity = 1.1668e-04	k_t = [2, 3, 2, 3]			
Generation 5970: Best = 1.189627e+04	FES = 597100	$\mu_F = 0.708$	$\mu_{CR} = 0.111$	
Diversity = 1.1637e-04	k_t = [1, 1, 1, 1]			
Generation 5980: Best = 1.189627e+04	FES = 598100	$\mu_F = 0.684$	$\mu_{CR} = 0.109$	
Diversity = 1.1007e-04	k_t = [3, 2, 2, 2]			
Generation 5990: Best = 1.189627e+04	FES = 599100	$\mu_F = 0.676$	$\mu_{CR} = 0.108$	
Diversity = 1.1008e-04	k_t = [1, 1, 1, 1]			
Generation 6000: Best = 1.189627e+04	FES = 600100	$\mu_F = 0.683$	$\mu_{CR} = 0.103$	
Diversity = 9.9969e-05	k_t = [2, 3, 3, 3]			
Generation 6010: Best = 1.189627e+04	FES = 601100	$\mu_F = 0.687$	$\mu_{CR} = 0.109$	
Diversity = 9.5761e-05	k_t = [1, 1, 1, 1]			
Generation 6020: Best = 1.189627e+04	FES = 602100	$\mu_F = 0.696$	$\mu_{CR} = 0.120$	
Diversity = 9.5161e-05	k_t = [2, 4, 3, 2]			
Generation 6030: Best = 1.189627e+04	FES = 603100	$\mu_F = 0.702$	$\mu_{CR} = 0.115$	
Diversity = 9.5138e-05	k_t = [1, 1, 1, 1]			
Generation 6040: Best = 1.189627e+04	FES = 604100	$\mu_F = 0.711$	$\mu_{CR} = 0.117$	
Diversity = 8.8975e-05	k_t = [3, 4, 3, 4]			
Generation 6050: Best = 1.189627e+04	FES = 605100	$\mu_F = 0.715$	$\mu_{CR} = 0.115$	
Diversity = 8.7882e-05	k_t = [1, 1, 1, 1]			
Generation 6060: Best = 1.189627e+04	FES = 606100	$\mu_F = 0.720$	$\mu_{CR} = 0.114$	
Diversity = 9.2019e-05	k_t = [4, 4, 2, 3]			
Generation 6070: Best = 1.189627e+04	FES = 607100	$\mu_F = 0.709$	$\mu_{CR} = 0.110$	
Diversity = 9.4769e-05	k_t = [1, 1, 1, 1]			
Generation 6080: Best = 1.189627e+04	FES = 608100	$\mu_F = 0.710$	$\mu_{CR} = 0.105$	
Diversity = 8.9491e-05	k_t = [4, 4, 3, 3]			
Generation 6090: Best = 1.189627e+04	FES = 609100	$\mu_F = 0.691$	$\mu_{CR} = 0.107$	
Diversity = 9.1723e-05	k_t = [1, 1, 1, 1]			
Generation 6100: Best = 1.189627e+04	FES = 610100	$\mu_F = 0.720$	$\mu_{CR} = 0.101$	
Diversity = 8.5961e-05	k_t = [3, 4, 4, 4]			
Generation 6110: Best = 1.189627e+04	FES = 611100	$\mu_F = 0.728$	$\mu_{CR} = 0.100$	
Diversity = 7.6878e-05	k_t = [1, 1, 1, 1]			
Generation 6120: Best = 1.189627e+04	FES = 612100	$\mu_F = 0.715$	$\mu_{CR} = 0.107$	
Diversity = 7.6818e-05	k_t = [3, 2, 3, 2]			
Generation 6130: Best = 1.189627e+04	FES = 613100	$\mu_F = 0.718$	$\mu_{CR} = 0.105$	
Diversity = 7.8920e-05	k_t = [1, 1, 1, 1]			
Generation 6140: Best = 1.189627e+04	FES = 614100	$\mu_F = 0.716$	$\mu_{CR} = 0.112$	
Diversity = 7.9299e-05	k_t = [3, 3, 3, 2]			
Generation 6150: Best = 1.189627e+04	FES = 615100	$\mu_F = 0.733$	$\mu_{CR} = 0.098$	
Diversity = 8.6427e-05	k_t = [1, 1, 1, 1]			
Generation 6160: Best = 1.189627e+04	FES = 616100	$\mu_F = 0.745$	$\mu_{CR} = 0.095$	

Diversity = 8.8308e-05	k_t = [4, 2, 3, 2]			
Generation 6170: Best = 1.189627e+04	FES = 617100	$\mu_F = 0.748$	$\mu_{CR} = 0.094$	
Diversity = 9.0981e-05	k_t = [1, 1, 1, 1]			
Generation 6180: Best = 1.189627e+04	FES = 618100	$\mu_F = 0.752$	$\mu_{CR} = 0.099$	
Diversity = 8.8745e-05	k_t = [4, 4, 2, 4]			
Generation 6190: Best = 1.189627e+04	FES = 619100	$\mu_F = 0.766$	$\mu_{CR} = 0.100$	
Diversity = 9.0452e-05	k_t = [1, 1, 1, 1]			
Generation 6200: Best = 1.189627e+04	FES = 620100	$\mu_F = 0.763$	$\mu_{CR} = 0.108$	
Diversity = 9.5464e-05	k_t = [4, 2, 4, 2]			
Generation 6210: Best = 1.189627e+04	FES = 621100	$\mu_F = 0.753$	$\mu_{CR} = 0.107$	
Diversity = 1.0087e-04	k_t = [1, 1, 1, 1]			
Generation 6220: Best = 1.189627e+04	FES = 622100	$\mu_F = 0.749$	$\mu_{CR} = 0.104$	
Diversity = 1.1222e-04	k_t = [3, 4, 2, 2]			
Generation 6230: Best = 1.189627e+04	FES = 623100	$\mu_F = 0.745$	$\mu_{CR} = 0.098$	
Diversity = 1.2070e-04	k_t = [1, 1, 1, 1]			
Generation 6240: Best = 1.189627e+04	FES = 624100	$\mu_F = 0.736$	$\mu_{CR} = 0.103$	
Diversity = 1.2380e-04	k_t = [3, 4, 2, 2]			
Generation 6250: Best = 1.189627e+04	FES = 625100	$\mu_F = 0.725$	$\mu_{CR} = 0.105$	
Diversity = 1.3035e-04	k_t = [1, 1, 1, 1]			
Generation 6260: Best = 1.189627e+04	FES = 626100	$\mu_F = 0.730$	$\mu_{CR} = 0.111$	
Diversity = 1.3057e-04	k_t = [4, 3, 3, 3]			
Generation 6270: Best = 1.189627e+04	FES = 627100	$\mu_F = 0.729$	$\mu_{CR} = 0.125$	
Diversity = 1.3563e-04	k_t = [1, 1, 1, 1]			
Generation 6280: Best = 1.189627e+04	FES = 628100	$\mu_F = 0.720$	$\mu_{CR} = 0.125$	
Diversity = 1.3473e-04	k_t = [4, 3, 2, 3]			
Generation 6290: Best = 1.189627e+04	FES = 629100	$\mu_F = 0.705$	$\mu_{CR} = 0.122$	
Diversity = 1.4166e-04	k_t = [1, 1, 1, 1]			
Generation 6300: Best = 1.189627e+04	FES = 630100	$\mu_F = 0.702$	$\mu_{CR} = 0.121$	
Diversity = 1.3011e-04	k_t = [4, 2, 2, 4]			
Generation 6310: Best = 1.189627e+04	FES = 631100	$\mu_F = 0.706$	$\mu_{CR} = 0.106$	
Diversity = 1.2945e-04	k_t = [1, 1, 1, 1]			
Generation 6320: Best = 1.189627e+04	FES = 632100	$\mu_F = 0.725$	$\mu_{CR} = 0.096$	
Diversity = 1.3593e-04	k_t = [2, 4, 3, 4]			
Generation 6330: Best = 1.189627e+04	FES = 633100	$\mu_F = 0.719$	$\mu_{CR} = 0.111$	
Diversity = 1.4360e-04	k_t = [1, 1, 1, 1]			
Generation 6340: Best = 1.189627e+04	FES = 634100	$\mu_F = 0.717$	$\mu_{CR} = 0.111$	
Diversity = 1.6117e-04	k_t = [3, 2, 3, 2]			
Generation 6350: Best = 1.189627e+04	FES = 635100	$\mu_F = 0.719$	$\mu_{CR} = 0.122$	
Diversity = 1.7771e-04	k_t = [1, 1, 1, 1]			
Generation 6360: Best = 1.189627e+04	FES = 636100	$\mu_F = 0.734$	$\mu_{CR} = 0.122$	
Diversity = 1.8709e-04	k_t = [3, 3, 3, 2]			
Generation 6370: Best = 1.189627e+04	FES = 637100	$\mu_F = 0.714$	$\mu_{CR} = 0.122$	
Diversity = 1.8691e-04	k_t = [1, 1, 1, 1]			
Generation 6380: Best = 1.189627e+04	FES = 638100	$\mu_F = 0.700$	$\mu_{CR} = 0.126$	
Diversity = 1.7543e-04	k_t = [3, 2, 2, 3]			
Generation 6390: Best = 1.189627e+04	FES = 639100	$\mu_F = 0.673$	$\mu_{CR} = 0.133$	
Diversity = 1.8355e-04	k_t = [1, 1, 1, 1]			
Generation 6400: Best = 1.189627e+04	FES = 640100	$\mu_F = 0.685$	$\mu_{CR} = 0.137$	
Diversity = 1.7434e-04	k_t = [4, 2, 2, 4]			
Generation 6410: Best = 1.189627e+04	FES = 641100	$\mu_F = 0.685$	$\mu_{CR} = 0.145$	
Diversity = 1.7649e-04	k_t = [1, 1, 1, 1]			
Generation 6420: Best = 1.189627e+04	FES = 642100	$\mu_F = 0.696$	$\mu_{CR} = 0.146$	
Diversity = 1.7824e-04	k_t = [2, 4, 4, 3]			
Generation 6430: Best = 1.189627e+04	FES = 643100	$\mu_F = 0.691$	$\mu_{CR} = 0.137$	

Diversity = 1.7463e-04	k_t = [1, 1, 1, 1]			
Generation 6440: Best = 1.189627e+04	FES = 644100	$\mu_F = 0.707$	$\mu_{CR} = 0.126$	
Diversity = 1.7369e-04	k_t = [3, 4, 3, 4]			
Generation 6450: Best = 1.189627e+04	FES = 645100	$\mu_F = 0.733$	$\mu_{CR} = 0.125$	
Diversity = 1.6649e-04	k_t = [1, 1, 1, 1]			
Generation 6460: Best = 1.189627e+04	FES = 646100	$\mu_F = 0.730$	$\mu_{CR} = 0.127$	
Diversity = 1.5654e-04	k_t = [3, 2, 2, 4]			
Generation 6470: Best = 1.189627e+04	FES = 647100	$\mu_F = 0.728$	$\mu_{CR} = 0.126$	
Diversity = 1.5776e-04	k_t = [1, 1, 1, 1]			
Generation 6480: Best = 1.189627e+04	FES = 648100	$\mu_F = 0.716$	$\mu_{CR} = 0.119$	
Diversity = 1.5796e-04	k_t = [2, 3, 2, 2]			
Generation 6490: Best = 1.189627e+04	FES = 649100	$\mu_F = 0.709$	$\mu_{CR} = 0.126$	
Diversity = 1.6344e-04	k_t = [1, 1, 1, 1]			
Generation 6500: Best = 1.189627e+04	FES = 650100	$\mu_F = 0.725$	$\mu_{CR} = 0.128$	
Diversity = 1.6871e-04	k_t = [4, 3, 3, 3]			
Generation 6510: Best = 1.189627e+04	FES = 651100	$\mu_F = 0.731$	$\mu_{CR} = 0.136$	
Diversity = 1.8685e-04	k_t = [1, 1, 1, 1]			
Generation 6520: Best = 1.189627e+04	FES = 652100	$\mu_F = 0.744$	$\mu_{CR} = 0.125$	
Diversity = 2.1763e-04	k_t = [2, 2, 2, 2]			
Generation 6530: Best = 1.189627e+04	FES = 653100	$\mu_F = 0.742$	$\mu_{CR} = 0.129$	
Diversity = 2.4088e-04	k_t = [1, 1, 1, 1]			
Generation 6540: Best = 1.189627e+04	FES = 654100	$\mu_F = 0.720$	$\mu_{CR} = 0.136$	
Diversity = 2.2576e-04	k_t = [3, 4, 2, 3]			
Generation 6550: Best = 1.189627e+04	FES = 655100	$\mu_F = 0.714$	$\mu_{CR} = 0.140$	
Diversity = 2.2707e-04	k_t = [1, 1, 1, 1]			
Generation 6560: Best = 1.189627e+04	FES = 656100	$\mu_F = 0.721$	$\mu_{CR} = 0.135$	
Diversity = 2.4303e-04	k_t = [4, 4, 4, 4]			
Generation 6570: Best = 1.189627e+04	FES = 657100	$\mu_F = 0.739$	$\mu_{CR} = 0.119$	
Diversity = 2.2381e-04	k_t = [1, 1, 1, 1]			
Generation 6580: Best = 1.189627e+04	FES = 658100	$\mu_F = 0.729$	$\mu_{CR} = 0.112$	
Diversity = 2.0762e-04	k_t = [2, 2, 3, 3]			
Generation 6590: Best = 1.189627e+04	FES = 659100	$\mu_F = 0.739$	$\mu_{CR} = 0.099$	
Diversity = 2.1055e-04	k_t = [1, 1, 1, 1]			
Generation 6600: Best = 1.189627e+04	FES = 660100	$\mu_F = 0.762$	$\mu_{CR} = 0.091$	
Diversity = 2.0292e-04	k_t = [2, 3, 2, 3]			
Generation 6610: Best = 1.189627e+04	FES = 661100	$\mu_F = 0.760$	$\mu_{CR} = 0.091$	
Diversity = 1.9614e-04	k_t = [1, 1, 1, 1]			
Generation 6620: Best = 1.189627e+04	FES = 662100	$\mu_F = 0.762$	$\mu_{CR} = 0.095$	
Diversity = 1.9238e-04	k_t = [4, 4, 3, 2]			
Generation 6630: Best = 1.189627e+04	FES = 663100	$\mu_F = 0.767$	$\mu_{CR} = 0.101$	
Diversity = 1.9675e-04	k_t = [1, 1, 1, 1]			
Generation 6640: Best = 1.189627e+04	FES = 664100	$\mu_F = 0.766$	$\mu_{CR} = 0.105$	
Diversity = 2.2188e-04	k_t = [4, 2, 4, 2]			
Generation 6650: Best = 1.189627e+04	FES = 665100	$\mu_F = 0.740$	$\mu_{CR} = 0.108$	
Diversity = 2.2020e-04	k_t = [1, 1, 1, 1]			
Generation 6660: Best = 1.189627e+04	FES = 666100	$\mu_F = 0.745$	$\mu_{CR} = 0.105$	
Diversity = 2.2030e-04	k_t = [3, 4, 4, 3]			
Generation 6670: Best = 1.189627e+04	FES = 667100	$\mu_F = 0.732$	$\mu_{CR} = 0.104$	
Diversity = 2.2303e-04	k_t = [1, 1, 1, 1]			
Generation 6680: Best = 1.189627e+04	FES = 668100	$\mu_F = 0.709$	$\mu_{CR} = 0.097$	
Diversity = 2.1996e-04	k_t = [4, 2, 2, 2]			
Generation 6690: Best = 1.189627e+04	FES = 669100	$\mu_F = 0.704$	$\mu_{CR} = 0.101$	
Diversity = 2.1494e-04	k_t = [1, 1, 1, 1]			
Generation 6700: Best = 1.189627e+04	FES = 670100	$\mu_F = 0.714$	$\mu_{CR} = 0.097$	

Diversity = 2.1670e-04	k_t = [2, 4, 3, 4]			
Generation 6710: Best = 1.189627e+04	FES = 671100	$\mu_F = 0.712$	$\mu_{CR} = 0.095$	
Diversity = 2.0161e-04	k_t = [1, 1, 1, 1]			
Generation 6720: Best = 1.189627e+04	FES = 672100	$\mu_F = 0.701$	$\mu_{CR} = 0.096$	
Diversity = 1.8620e-04	k_t = [4, 2, 2, 4]			
Generation 6730: Best = 1.189627e+04	FES = 673100	$\mu_F = 0.692$	$\mu_{CR} = 0.106$	
Diversity = 1.7363e-04	k_t = [1, 1, 1, 1]			
Generation 6740: Best = 1.189627e+04	FES = 674100	$\mu_F = 0.687$	$\mu_{CR} = 0.102$	
Diversity = 1.7374e-04	k_t = [2, 3, 3, 2]			
Generation 6750: Best = 1.189627e+04	FES = 675100	$\mu_F = 0.689$	$\mu_{CR} = 0.100$	
Diversity = 1.7970e-04	k_t = [1, 1, 1, 1]			
Generation 6760: Best = 1.189627e+04	FES = 676100	$\mu_F = 0.696$	$\mu_{CR} = 0.110$	
Diversity = 1.7306e-04	k_t = [2, 3, 3, 2]			
Generation 6770: Best = 1.189627e+04	FES = 677100	$\mu_F = 0.683$	$\mu_{CR} = 0.115$	
Diversity = 1.7797e-04	k_t = [1, 1, 1, 1]			
Generation 6780: Best = 1.189627e+04	FES = 678100	$\mu_F = 0.681$	$\mu_{CR} = 0.118$	
Diversity = 1.7920e-04	k_t = [2, 3, 3, 2]			
Generation 6790: Best = 1.189627e+04	FES = 679100	$\mu_F = 0.684$	$\mu_{CR} = 0.113$	
Diversity = 1.8880e-04	k_t = [1, 1, 1, 1]			
Generation 6800: Best = 1.189627e+04	FES = 680100	$\mu_F = 0.671$	$\mu_{CR} = 0.113$	
Diversity = 1.7938e-04	k_t = [2, 3, 2, 3]			
Generation 6810: Best = 1.189627e+04	FES = 681100	$\mu_F = 0.676$	$\mu_{CR} = 0.107$	
Diversity = 1.6636e-04	k_t = [1, 1, 1, 1]			
Generation 6820: Best = 1.189627e+04	FES = 682100	$\mu_F = 0.670$	$\mu_{CR} = 0.098$	
Diversity = 1.4515e-04	k_t = [4, 3, 2, 3]			
Generation 6830: Best = 1.189627e+04	FES = 683100	$\mu_F = 0.674$	$\mu_{CR} = 0.094$	
Diversity = 1.4373e-04	k_t = [1, 1, 1, 1]			
Generation 6840: Best = 1.189627e+04	FES = 684100	$\mu_F = 0.694$	$\mu_{CR} = 0.098$	
Diversity = 1.4276e-04	k_t = [2, 2, 3, 4]			
Generation 6850: Best = 1.189627e+04	FES = 685100	$\mu_F = 0.697$	$\mu_{CR} = 0.104$	
Diversity = 1.4324e-04	k_t = [1, 1, 1, 1]			
Generation 6860: Best = 1.189627e+04	FES = 686100	$\mu_F = 0.709$	$\mu_{CR} = 0.106$	
Diversity = 1.4868e-04	k_t = [2, 2, 2, 4]			
Generation 6870: Best = 1.189627e+04	FES = 687100	$\mu_F = 0.721$	$\mu_{CR} = 0.104$	
Diversity = 1.6343e-04	k_t = [1, 1, 1, 1]			
Generation 6880: Best = 1.189627e+04	FES = 688100	$\mu_F = 0.727$	$\mu_{CR} = 0.104$	
Diversity = 1.6655e-04	k_t = [3, 3, 2, 3]			
Generation 6890: Best = 1.189627e+04	FES = 689100	$\mu_F = 0.724$	$\mu_{CR} = 0.107$	
Diversity = 1.6513e-04	k_t = [1, 1, 1, 1]			
Generation 6900: Best = 1.189627e+04	FES = 690100	$\mu_F = 0.722$	$\mu_{CR} = 0.110$	
Diversity = 1.5369e-04	k_t = [4, 3, 3, 3]			
Generation 6910: Best = 1.189627e+04	FES = 691100	$\mu_F = 0.720$	$\mu_{CR} = 0.114$	
Diversity = 1.6391e-04	k_t = [1, 1, 1, 1]			
Generation 6920: Best = 1.189627e+04	FES = 692100	$\mu_F = 0.721$	$\mu_{CR} = 0.107$	
Diversity = 1.5705e-04	k_t = [2, 2, 2, 4]			
Generation 6930: Best = 1.189627e+04	FES = 693100	$\mu_F = 0.714$	$\mu_{CR} = 0.098$	
Diversity = 1.5229e-04	k_t = [1, 1, 1, 1]			
Generation 6940: Best = 1.189627e+04	FES = 694100	$\mu_F = 0.721$	$\mu_{CR} = 0.091$	
Diversity = 1.5615e-04	k_t = [3, 3, 3, 2]			
Generation 6950: Best = 1.189627e+04	FES = 695100	$\mu_F = 0.724$	$\mu_{CR} = 0.092$	
Diversity = 1.6693e-04	k_t = [1, 1, 1, 1]			
Generation 6960: Best = 1.189627e+04	FES = 696100	$\mu_F = 0.725$	$\mu_{CR} = 0.093$	
Diversity = 1.6618e-04	k_t = [2, 3, 3, 2]			
Generation 6970: Best = 1.189627e+04	FES = 697100	$\mu_F = 0.722$	$\mu_{CR} = 0.089$	

Diversity = 1.6643e-04	k_t = [1, 1, 1, 1]			
Generation 6980: Best = 1.189627e+04	FES = 698100	$\mu_F = 0.721$	$\mu_{CR} = 0.097$	
Diversity = 1.5993e-04	k_t = [4, 4, 2, 3]			
Generation 6990: Best = 1.189627e+04	FES = 699100	$\mu_F = 0.720$	$\mu_{CR} = 0.090$	
Diversity = 1.5946e-04	k_t = [1, 1, 1, 1]			
Generation 7000: Best = 1.189627e+04	FES = 700100	$\mu_F = 0.734$	$\mu_{CR} = 0.091$	
Diversity = 1.6224e-04	k_t = [4, 2, 2, 2]			
Generation 7010: Best = 1.189627e+04	FES = 701100	$\mu_F = 0.761$	$\mu_{CR} = 0.089$	
Diversity = 1.6942e-04	k_t = [1, 1, 1, 1]			
Generation 7020: Best = 1.189627e+04	FES = 702100	$\mu_F = 0.760$	$\mu_{CR} = 0.093$	
Diversity = 1.6409e-04	k_t = [2, 4, 3, 2]			
Generation 7030: Best = 1.189627e+04	FES = 703100	$\mu_F = 0.773$	$\mu_{CR} = 0.088$	
Diversity = 1.6430e-04	k_t = [1, 1, 1, 1]			
Generation 7040: Best = 1.189627e+04	FES = 704100	$\mu_F = 0.772$	$\mu_{CR} = 0.095$	
Diversity = 1.5210e-04	k_t = [4, 3, 4, 4]			
Generation 7050: Best = 1.189627e+04	FES = 705100	$\mu_F = 0.765$	$\mu_{CR} = 0.095$	
Diversity = 1.4880e-04	k_t = [1, 1, 1, 1]			
Generation 7060: Best = 1.189627e+04	FES = 706100	$\mu_F = 0.759$	$\mu_{CR} = 0.093$	
Diversity = 1.5812e-04	k_t = [2, 4, 2, 3]			
Generation 7070: Best = 1.189627e+04	FES = 707100	$\mu_F = 0.764$	$\mu_{CR} = 0.096$	
Diversity = 1.7265e-04	k_t = [1, 1, 1, 1]			
Generation 7080: Best = 1.189627e+04	FES = 708100	$\mu_F = 0.751$	$\mu_{CR} = 0.085$	
Diversity = 1.6750e-04	k_t = [3, 2, 3, 4]			
Generation 7090: Best = 1.189627e+04	FES = 709100	$\mu_F = 0.749$	$\mu_{CR} = 0.090$	
Diversity = 1.8604e-04	k_t = [1, 1, 1, 1]			
Generation 7100: Best = 1.189627e+04	FES = 710100	$\mu_F = 0.746$	$\mu_{CR} = 0.094$	
Diversity = 1.8727e-04	k_t = [2, 2, 4, 2]			
Generation 7110: Best = 1.189627e+04	FES = 711100	$\mu_F = 0.746$	$\mu_{CR} = 0.088$	
Diversity = 1.8039e-04	k_t = [1, 1, 1, 1]			
Generation 7120: Best = 1.189627e+04	FES = 712100	$\mu_F = 0.744$	$\mu_{CR} = 0.082$	
Diversity = 1.8242e-04	k_t = [3, 4, 2, 2]			
Generation 7130: Best = 1.189627e+04	FES = 713100	$\mu_F = 0.744$	$\mu_{CR} = 0.082$	
Diversity = 1.9821e-04	k_t = [1, 1, 1, 1]			
Generation 7140: Best = 1.189627e+04	FES = 714100	$\mu_F = 0.752$	$\mu_{CR} = 0.076$	
Diversity = 1.9109e-04	k_t = [2, 3, 2, 4]			
Generation 7150: Best = 1.189627e+04	FES = 715100	$\mu_F = 0.736$	$\mu_{CR} = 0.089$	
Diversity = 1.9303e-04	k_t = [1, 1, 1, 1]			
Generation 7160: Best = 1.189627e+04	FES = 716100	$\mu_F = 0.727$	$\mu_{CR} = 0.082$	
Diversity = 1.7900e-04	k_t = [3, 4, 2, 4]			
Generation 7170: Best = 1.189627e+04	FES = 717100	$\mu_F = 0.717$	$\mu_{CR} = 0.085$	
Diversity = 1.8303e-04	k_t = [1, 1, 1, 1]			
Generation 7180: Best = 1.189627e+04	FES = 718100	$\mu_F = 0.730$	$\mu_{CR} = 0.086$	
Diversity = 1.9266e-04	k_t = [4, 3, 3, 4]			
Generation 7190: Best = 1.189627e+04	FES = 719100	$\mu_F = 0.733$	$\mu_{CR} = 0.091$	
Diversity = 1.9750e-04	k_t = [1, 1, 1, 1]			
Generation 7200: Best = 1.189627e+04	FES = 720100	$\mu_F = 0.728$	$\mu_{CR} = 0.095$	
Diversity = 2.1012e-04	k_t = [2, 3, 4, 3]			
Generation 7210: Best = 1.189627e+04	FES = 721100	$\mu_F = 0.743$	$\mu_{CR} = 0.093$	
Diversity = 2.2374e-04	k_t = [1, 1, 1, 1]			
Generation 7220: Best = 1.189627e+04	FES = 722100	$\mu_F = 0.749$	$\mu_{CR} = 0.087$	
Diversity = 2.3619e-04	k_t = [3, 2, 2, 3]			
Generation 7230: Best = 1.189627e+04	FES = 723100	$\mu_F = 0.725$	$\mu_{CR} = 0.079$	
Diversity = 2.4781e-04	k_t = [1, 1, 1, 1]			
Generation 7240: Best = 1.189627e+04	FES = 724100	$\mu_F = 0.753$	$\mu_{CR} = 0.080$	

Diversity = 2.4083e-04	k_t = [4, 2, 4, 2]			
Generation 7250: Best = 1.189627e+04	FES = 725100	$\mu_F = 0.743$	$\mu_{CR} = 0.084$	
Diversity = 2.3183e-04	k_t = [1, 1, 1, 1]			
Generation 7260: Best = 1.189627e+04	FES = 726100	$\mu_F = 0.740$	$\mu_{CR} = 0.086$	
Diversity = 2.1091e-04	k_t = [3, 3, 4, 3]			
Generation 7270: Best = 1.189627e+04	FES = 727100	$\mu_F = 0.763$	$\mu_{CR} = 0.076$	
Diversity = 2.0048e-04	k_t = [1, 1, 1, 1]			
Generation 7280: Best = 1.189627e+04	FES = 728100	$\mu_F = 0.777$	$\mu_{CR} = 0.079$	
Diversity = 1.9620e-04	k_t = [3, 3, 3, 4]			
Generation 7290: Best = 1.189627e+04	FES = 729100	$\mu_F = 0.777$	$\mu_{CR} = 0.083$	
Diversity = 1.8837e-04	k_t = [1, 1, 1, 1]			
Generation 7300: Best = 1.189627e+04	FES = 730100	$\mu_F = 0.778$	$\mu_{CR} = 0.084$	
Diversity = 1.8623e-04	k_t = [3, 4, 2, 4]			
Generation 7310: Best = 1.189627e+04	FES = 731100	$\mu_F = 0.767$	$\mu_{CR} = 0.093$	
Diversity = 1.8305e-04	k_t = [1, 1, 1, 1]			
Generation 7320: Best = 1.189627e+04	FES = 732100	$\mu_F = 0.752$	$\mu_{CR} = 0.094$	
Diversity = 1.7746e-04	k_t = [4, 2, 3, 4]			
Generation 7330: Best = 1.189627e+04	FES = 733100	$\mu_F = 0.737$	$\mu_{CR} = 0.094$	
Diversity = 1.6106e-04	k_t = [1, 1, 1, 1]			
Generation 7340: Best = 1.189627e+04	FES = 734100	$\mu_F = 0.758$	$\mu_{CR} = 0.096$	
Diversity = 1.6448e-04	k_t = [4, 4, 2, 3]			
Generation 7350: Best = 1.189627e+04	FES = 735100	$\mu_F = 0.752$	$\mu_{CR} = 0.096$	
Diversity = 1.6881e-04	k_t = [1, 1, 1, 1]			
Generation 7360: Best = 1.189627e+04	FES = 736100	$\mu_F = 0.743$	$\mu_{CR} = 0.098$	
Diversity = 1.6057e-04	k_t = [3, 4, 4, 4]			
Generation 7370: Best = 1.189627e+04	FES = 737100	$\mu_F = 0.751$	$\mu_{CR} = 0.098$	
Diversity = 1.6848e-04	k_t = [1, 1, 1, 1]			
Generation 7380: Best = 1.189627e+04	FES = 738100	$\mu_F = 0.744$	$\mu_{CR} = 0.096$	
Diversity = 1.9141e-04	k_t = [4, 4, 4, 2]			
Generation 7390: Best = 1.189627e+04	FES = 739100	$\mu_F = 0.761$	$\mu_{CR} = 0.087$	
Diversity = 2.1064e-04	k_t = [1, 1, 1, 1]			
Generation 7400: Best = 1.189627e+04	FES = 740100	$\mu_F = 0.756$	$\mu_{CR} = 0.090$	
Diversity = 1.9785e-04	k_t = [4, 4, 3, 4]			
Generation 7410: Best = 1.189627e+04	FES = 741100	$\mu_F = 0.767$	$\mu_{CR} = 0.092$	
Diversity = 1.9355e-04	k_t = [1, 1, 1, 1]			
Generation 7420: Best = 1.189627e+04	FES = 742100	$\mu_F = 0.788$	$\mu_{CR} = 0.081$	
Diversity = 1.9037e-04	k_t = [4, 4, 3, 4]			
Generation 7430: Best = 1.189627e+04	FES = 743100	$\mu_F = 0.794$	$\mu_{CR} = 0.088$	
Diversity = 1.8567e-04	k_t = [1, 1, 1, 1]			
Generation 7440: Best = 1.189627e+04	FES = 744100	$\mu_F = 0.781$	$\mu_{CR} = 0.091$	
Diversity = 1.8469e-04	k_t = [3, 4, 2, 3]			
Generation 7450: Best = 1.189627e+04	FES = 745100	$\mu_F = 0.773$	$\mu_{CR} = 0.086$	
Diversity = 1.8317e-04	k_t = [1, 1, 1, 1]			
Generation 7460: Best = 1.189627e+04	FES = 746100	$\mu_F = 0.771$	$\mu_{CR} = 0.088$	
Diversity = 1.9556e-04	k_t = [3, 2, 4, 2]			
Generation 7470: Best = 1.189627e+04	FES = 747100	$\mu_F = 0.773$	$\mu_{CR} = 0.087$	
Diversity = 1.9610e-04	k_t = [1, 1, 1, 1]			
Generation 7480: Best = 1.189627e+04	FES = 748100	$\mu_F = 0.772$	$\mu_{CR} = 0.080$	
Diversity = 2.0326e-04	k_t = [3, 2, 2, 3]			
Generation 7490: Best = 1.189627e+04	FES = 749100	$\mu_F = 0.755$	$\mu_{CR} = 0.084$	
Diversity = 1.9635e-04	k_t = [1, 1, 1, 1]			
Generation 7500: Best = 1.189627e+04	FES = 750100	$\mu_F = 0.758$	$\mu_{CR} = 0.082$	
Diversity = 1.9222e-04	k_t = [2, 3, 3, 2]			
Generation 7510: Best = 1.189627e+04	FES = 751100	$\mu_F = 0.742$	$\mu_{CR} = 0.080$	

Diversity = 1.9138e-04	k_t = [1, 1, 1, 1]			
Generation 7520: Best = 1.189627e+04	FES = 752100	$\mu_F = 0.737$	$\mu_{CR} = 0.078$	
Diversity = 1.8282e-04	k_t = [4, 4, 2, 3]			
Generation 7530: Best = 1.189627e+04	FES = 753100	$\mu_F = 0.703$	$\mu_{CR} = 0.078$	
Diversity = 1.7339e-04	k_t = [1, 1, 1, 1]			
Generation 7540: Best = 1.189627e+04	FES = 754100	$\mu_F = 0.698$	$\mu_{CR} = 0.082$	
Diversity = 1.6788e-04	k_t = [3, 4, 2, 3]			
Generation 7550: Best = 1.189627e+04	FES = 755100	$\mu_F = 0.715$	$\mu_{CR} = 0.082$	
Diversity = 1.6223e-04	k_t = [1, 1, 1, 1]			
Generation 7560: Best = 1.189627e+04	FES = 756100	$\mu_F = 0.736$	$\mu_{CR} = 0.093$	
Diversity = 1.6486e-04	k_t = [2, 4, 4, 2]			
Generation 7570: Best = 1.189627e+04	FES = 757100	$\mu_F = 0.734$	$\mu_{CR} = 0.093$	
Diversity = 1.7127e-04	k_t = [1, 1, 1, 1]			
Generation 7580: Best = 1.189627e+04	FES = 758100	$\mu_F = 0.746$	$\mu_{CR} = 0.082$	
Diversity = 1.7208e-04	k_t = [2, 3, 2, 2]			
Generation 7590: Best = 1.189627e+04	FES = 759100	$\mu_F = 0.769$	$\mu_{CR} = 0.083$	
Diversity = 1.6857e-04	k_t = [1, 1, 1, 1]			
Generation 7600: Best = 1.189627e+04	FES = 760100	$\mu_F = 0.773$	$\mu_{CR} = 0.082$	
Diversity = 1.6420e-04	k_t = [4, 2, 2, 3]			
Generation 7610: Best = 1.189627e+04	FES = 761100	$\mu_F = 0.774$	$\mu_{CR} = 0.075$	
Diversity = 1.6096e-04	k_t = [1, 1, 1, 1]			
Generation 7620: Best = 1.189627e+04	FES = 762100	$\mu_F = 0.785$	$\mu_{CR} = 0.078$	
Diversity = 1.5067e-04	k_t = [3, 2, 4, 4]			
Generation 7630: Best = 1.189627e+04	FES = 763100	$\mu_F = 0.782$	$\mu_{CR} = 0.078$	
Diversity = 1.4552e-04	k_t = [1, 1, 1, 1]			
Generation 7640: Best = 1.189627e+04	FES = 764100	$\mu_F = 0.782$	$\mu_{CR} = 0.081$	
Diversity = 1.4066e-04	k_t = [4, 4, 2, 4]			
Generation 7650: Best = 1.189627e+04	FES = 765100	$\mu_F = 0.785$	$\mu_{CR} = 0.089$	
Diversity = 1.3571e-04	k_t = [1, 1, 1, 1]			
Generation 7660: Best = 1.189627e+04	FES = 766100	$\mu_F = 0.792$	$\mu_{CR} = 0.081$	
Diversity = 1.3647e-04	k_t = [2, 4, 3, 3]			
Generation 7670: Best = 1.189627e+04	FES = 767100	$\mu_F = 0.783$	$\mu_{CR} = 0.085$	
Diversity = 1.3429e-04	k_t = [1, 1, 1, 1]			
Generation 7680: Best = 1.189627e+04	FES = 768100	$\mu_F = 0.770$	$\mu_{CR} = 0.082$	
Diversity = 1.3931e-04	k_t = [3, 3, 2, 2]			
Generation 7690: Best = 1.189627e+04	FES = 769100	$\mu_F = 0.786$	$\mu_{CR} = 0.079$	
Diversity = 1.4241e-04	k_t = [1, 1, 1, 1]			
Generation 7700: Best = 1.189627e+04	FES = 770100	$\mu_F = 0.788$	$\mu_{CR} = 0.082$	
Diversity = 1.3840e-04	k_t = [4, 4, 3, 4]			
Generation 7710: Best = 1.189627e+04	FES = 771100	$\mu_F = 0.805$	$\mu_{CR} = 0.084$	
Diversity = 1.3297e-04	k_t = [1, 1, 1, 1]			
Generation 7720: Best = 1.189627e+04	FES = 772100	$\mu_F = 0.799$	$\mu_{CR} = 0.082$	
Diversity = 1.2136e-04	k_t = [4, 4, 3, 4]			
Generation 7730: Best = 1.189627e+04	FES = 773100	$\mu_F = 0.791$	$\mu_{CR} = 0.080$	
Diversity = 1.2473e-04	k_t = [1, 1, 1, 1]			
Generation 7740: Best = 1.189627e+04	FES = 774100	$\mu_F = 0.776$	$\mu_{CR} = 0.075$	
Diversity = 1.2514e-04	k_t = [4, 3, 4, 3]			
Generation 7750: Best = 1.189627e+04	FES = 775100	$\mu_F = 0.770$	$\mu_{CR} = 0.075$	
Diversity = 1.3262e-04	k_t = [1, 1, 1, 1]			
Generation 7760: Best = 1.189627e+04	FES = 776100	$\mu_F = 0.777$	$\mu_{CR} = 0.080$	
Diversity = 1.4835e-04	k_t = [4, 3, 3, 2]			
Generation 7770: Best = 1.189627e+04	FES = 777100	$\mu_F = 0.765$	$\mu_{CR} = 0.088$	
Diversity = 1.6837e-04	k_t = [1, 1, 1, 1]			
Generation 7780: Best = 1.189627e+04	FES = 778100	$\mu_F = 0.769$	$\mu_{CR} = 0.090$	

Diversity = 1.6788e-04	k_t = [2, 4, 4, 3]			
Generation 7790: Best = 1.189627e+04	FES = 779100	$\mu_F = 0.751$	$\mu_{CR} = 0.076$	
Diversity = 1.7232e-04	k_t = [1, 1, 1, 1]			
Generation 7800: Best = 1.189627e+04	FES = 780100	$\mu_F = 0.742$	$\mu_{CR} = 0.075$	
Diversity = 1.5495e-04	k_t = [3, 4, 4, 4]			
Generation 7810: Best = 1.189627e+04	FES = 781100	$\mu_F = 0.725$	$\mu_{CR} = 0.076$	
Diversity = 1.4816e-04	k_t = [1, 1, 1, 1]			
Generation 7820: Best = 1.189627e+04	FES = 782100	$\mu_F = 0.716$	$\mu_{CR} = 0.081$	
Diversity = 1.4483e-04	k_t = [4, 2, 3, 3]			
Generation 7830: Best = 1.189627e+04	FES = 783100	$\mu_F = 0.717$	$\mu_{CR} = 0.081$	
Diversity = 1.3671e-04	k_t = [1, 1, 1, 1]			
Generation 7840: Best = 1.189627e+04	FES = 784100	$\mu_F = 0.704$	$\mu_{CR} = 0.079$	
Diversity = 1.3119e-04	k_t = [2, 3, 3, 2]			
Generation 7850: Best = 1.189627e+04	FES = 785100	$\mu_F = 0.693$	$\mu_{CR} = 0.076$	
Diversity = 1.3032e-04	k_t = [1, 1, 1, 1]			
Generation 7860: Best = 1.189627e+04	FES = 786100	$\mu_F = 0.697$	$\mu_{CR} = 0.076$	
Diversity = 1.1812e-04	k_t = [2, 2, 4, 4]			
Generation 7870: Best = 1.189627e+04	FES = 787100	$\mu_F = 0.697$	$\mu_{CR} = 0.076$	
Diversity = 1.1234e-04	k_t = [1, 1, 1, 1]			
Generation 7880: Best = 1.189627e+04	FES = 788100	$\mu_F = 0.695$	$\mu_{CR} = 0.082$	
Diversity = 1.0831e-04	k_t = [3, 2, 3, 2]			
Generation 7890: Best = 1.189627e+04	FES = 789100	$\mu_F = 0.697$	$\mu_{CR} = 0.089$	
Diversity = 1.0780e-04	k_t = [1, 1, 1, 1]			
Generation 7900: Best = 1.189627e+04	FES = 790100	$\mu_F = 0.702$	$\mu_{CR} = 0.094$	
Diversity = 1.0489e-04	k_t = [4, 2, 4, 3]			
Generation 7910: Best = 1.189627e+04	FES = 791100	$\mu_F = 0.696$	$\mu_{CR} = 0.089$	
Diversity = 1.0259e-04	k_t = [1, 1, 1, 1]			
Generation 7920: Best = 1.189627e+04	FES = 792100	$\mu_F = 0.708$	$\mu_{CR} = 0.088$	
Diversity = 1.0640e-04	k_t = [2, 4, 2, 3]			
Generation 7930: Best = 1.189627e+04	FES = 793100	$\mu_F = 0.729$	$\mu_{CR} = 0.083$	
Diversity = 1.1016e-04	k_t = [1, 1, 1, 1]			
Generation 7940: Best = 1.189627e+04	FES = 794100	$\mu_F = 0.751$	$\mu_{CR} = 0.082$	
Diversity = 1.1619e-04	k_t = [4, 3, 2, 3]			
Generation 7950: Best = 1.189627e+04	FES = 795100	$\mu_F = 0.751$	$\mu_{CR} = 0.088$	
Diversity = 1.2915e-04	k_t = [1, 1, 1, 1]			
Generation 7960: Best = 1.189627e+04	FES = 796100	$\mu_F = 0.746$	$\mu_{CR} = 0.094$	
Diversity = 1.3039e-04	k_t = [3, 3, 3, 2]			
Generation 7970: Best = 1.189627e+04	FES = 797100	$\mu_F = 0.747$	$\mu_{CR} = 0.093$	
Diversity = 1.2753e-04	k_t = [1, 1, 1, 1]			
Generation 7980: Best = 1.189627e+04	FES = 798100	$\mu_F = 0.756$	$\mu_{CR} = 0.096$	
Diversity = 1.4487e-04	k_t = [2, 3, 3, 4]			
Generation 7990: Best = 1.189627e+04	FES = 799100	$\mu_F = 0.778$	$\mu_{CR} = 0.093$	
Diversity = 1.5051e-04	k_t = [1, 1, 1, 1]			
Generation 8000: Best = 1.189627e+04	FES = 800100	$\mu_F = 0.779$	$\mu_{CR} = 0.097$	
Diversity = 1.5310e-04	k_t = [4, 4, 4, 4]			
Generation 8010: Best = 1.189627e+04	FES = 801100	$\mu_F = 0.776$	$\mu_{CR} = 0.109$	
Diversity = 1.5403e-04	k_t = [1, 1, 1, 1]			
Generation 8020: Best = 1.189627e+04	FES = 802100	$\mu_F = 0.766$	$\mu_{CR} = 0.105$	
Diversity = 1.4418e-04	k_t = [2, 2, 3, 4]			
Generation 8030: Best = 1.189627e+04	FES = 803100	$\mu_F = 0.749$	$\mu_{CR} = 0.104$	
Diversity = 1.2837e-04	k_t = [1, 1, 1, 1]			
Generation 8040: Best = 1.189627e+04	FES = 804100	$\mu_F = 0.753$	$\mu_{CR} = 0.096$	
Diversity = 1.2164e-04	k_t = [2, 4, 4, 4]			
Generation 8050: Best = 1.189627e+04	FES = 805100	$\mu_F = 0.731$	$\mu_{CR} = 0.103$	

Diversity = 1.1621e-04	k_t = [1, 1, 1, 1]			
Generation 8060: Best = 1.189627e+04	FES = 806100	$\mu_F = 0.745$	$\mu_{CR} = 0.103$	
Diversity = 1.1942e-04	k_t = [2, 4, 4, 3]			
Generation 8070: Best = 1.189627e+04	FES = 807100	$\mu_F = 0.731$	$\mu_{CR} = 0.096$	
Diversity = 1.2697e-04	k_t = [1, 1, 1, 1]			
Generation 8080: Best = 1.189627e+04	FES = 808100	$\mu_F = 0.742$	$\mu_{CR} = 0.093$	
Diversity = 1.3327e-04	k_t = [4, 3, 4, 3]			
Generation 8090: Best = 1.189627e+04	FES = 809100	$\mu_F = 0.745$	$\mu_{CR} = 0.086$	
Diversity = 1.3709e-04	k_t = [1, 1, 1, 1]			
Generation 8100: Best = 1.189627e+04	FES = 810100	$\mu_F = 0.726$	$\mu_{CR} = 0.085$	
Diversity = 1.3804e-04	k_t = [2, 4, 2, 3]			
Generation 8110: Best = 1.189627e+04	FES = 811100	$\mu_F = 0.734$	$\mu_{CR} = 0.094$	
Diversity = 1.4151e-04	k_t = [1, 1, 1, 1]			
Generation 8120: Best = 1.189627e+04	FES = 812100	$\mu_F = 0.773$	$\mu_{CR} = 0.081$	
Diversity = 1.4403e-04	k_t = [2, 2, 4, 4]			
Generation 8130: Best = 1.189627e+04	FES = 813100	$\mu_F = 0.769$	$\mu_{CR} = 0.083$	
Diversity = 1.4495e-04	k_t = [1, 1, 1, 1]			
Generation 8140: Best = 1.189627e+04	FES = 814100	$\mu_F = 0.773$	$\mu_{CR} = 0.089$	
Diversity = 1.4889e-04	k_t = [2, 2, 2, 2]			
Generation 8150: Best = 1.189627e+04	FES = 815100	$\mu_F = 0.768$	$\mu_{CR} = 0.085$	
Diversity = 1.5707e-04	k_t = [1, 1, 1, 1]			
Generation 8160: Best = 1.189627e+04	FES = 816100	$\mu_F = 0.760$	$\mu_{CR} = 0.085$	
Diversity = 1.6526e-04	k_t = [2, 4, 3, 2]			
Generation 8170: Best = 1.189627e+04	FES = 817100	$\mu_F = 0.775$	$\mu_{CR} = 0.081$	
Diversity = 1.7796e-04	k_t = [1, 1, 1, 1]			
Generation 8180: Best = 1.189627e+04	FES = 818100	$\mu_F = 0.757$	$\mu_{CR} = 0.088$	
Diversity = 1.7593e-04	k_t = [2, 2, 2, 2]			
Generation 8190: Best = 1.189627e+04	FES = 819100	$\mu_F = 0.756$	$\mu_{CR} = 0.085$	
Diversity = 1.9052e-04	k_t = [1, 1, 1, 1]			
Generation 8200: Best = 1.189627e+04	FES = 820100	$\mu_F = 0.750$	$\mu_{CR} = 0.086$	
Diversity = 1.8698e-04	k_t = [2, 4, 4, 2]			
Generation 8210: Best = 1.189627e+04	FES = 821100	$\mu_F = 0.732$	$\mu_{CR} = 0.089$	
Diversity = 1.8313e-04	k_t = [1, 1, 1, 1]			
Generation 8220: Best = 1.189627e+04	FES = 822100	$\mu_F = 0.731$	$\mu_{CR} = 0.099$	
Diversity = 1.8209e-04	k_t = [2, 3, 4, 3]			
Generation 8230: Best = 1.189627e+04	FES = 823100	$\mu_F = 0.720$	$\mu_{CR} = 0.099$	
Diversity = 1.7959e-04	k_t = [1, 1, 1, 1]			
Generation 8240: Best = 1.189627e+04	FES = 824100	$\mu_F = 0.693$	$\mu_{CR} = 0.085$	
Diversity = 1.7028e-04	k_t = [2, 2, 2, 2]			
Generation 8250: Best = 1.189627e+04	FES = 825100	$\mu_F = 0.706$	$\mu_{CR} = 0.089$	
Diversity = 1.6287e-04	k_t = [1, 1, 1, 1]			
Generation 8260: Best = 1.189627e+04	FES = 826100	$\mu_F = 0.712$	$\mu_{CR} = 0.083$	
Diversity = 1.5977e-04	k_t = [3, 4, 3, 2]			
Generation 8270: Best = 1.189627e+04	FES = 827100	$\mu_F = 0.711$	$\mu_{CR} = 0.078$	
Diversity = 1.5915e-04	k_t = [1, 1, 1, 1]			
Generation 8280: Best = 1.189627e+04	FES = 828100	$\mu_F = 0.728$	$\mu_{CR} = 0.075$	
Diversity = 1.4688e-04	k_t = [4, 4, 4, 3]			
Generation 8290: Best = 1.189627e+04	FES = 829100	$\mu_F = 0.717$	$\mu_{CR} = 0.076$	
Diversity = 1.4136e-04	k_t = [1, 1, 1, 1]			
Generation 8300: Best = 1.189627e+04	FES = 830100	$\mu_F = 0.721$	$\mu_{CR} = 0.084$	
Diversity = 1.2651e-04	k_t = [4, 2, 2, 3]			
Generation 8310: Best = 1.189627e+04	FES = 831100	$\mu_F = 0.724$	$\mu_{CR} = 0.079$	
Diversity = 1.2183e-04	k_t = [1, 1, 1, 1]			
Generation 8320: Best = 1.189627e+04	FES = 832100	$\mu_F = 0.740$	$\mu_{CR} = 0.085$	

Diversity = 1.1734e-04 k_t = [2, 3, 3, 2]
 Generation 8330: Best = 1.189627e+04 FES = 833100 μ_F = 0.744 μ_{CR} = 0.089
 Diversity = 1.1486e-04 k_t = [1, 1, 1, 1]
 Generation 8340: Best = 1.189627e+04 FES = 834100 μ_F = 0.740 μ_{CR} = 0.087
 Diversity = 1.0995e-04 k_t = [3, 3, 3, 4]
 Generation 8350: Best = 1.189627e+04 FES = 835100 μ_F = 0.740 μ_{CR} = 0.077
 Diversity = 1.0839e-04 k_t = [1, 1, 1, 1]
 Generation 8360: Best = 1.189627e+04 FES = 836100 μ_F = 0.746 μ_{CR} = 0.088
 Diversity = 1.0385e-04 k_t = [3, 3, 2, 2]
 Generation 8370: Best = 1.189627e+04 FES = 837100 μ_F = 0.761 μ_{CR} = 0.090
 Diversity = 1.0920e-04 k_t = [1, 1, 1, 1]
 Generation 8380: Best = 1.189627e+04 FES = 838100 μ_F = 0.774 μ_{CR} = 0.080
 Diversity = 1.0287e-04 k_t = [3, 4, 3, 3]
 Generation 8390: Best = 1.189627e+04 FES = 839100 μ_F = 0.776 μ_{CR} = 0.079
 Diversity = 1.0361e-04 k_t = [1, 1, 1, 1]
 Generation 8400: Best = 1.189627e+04 FES = 840100 μ_F = 0.761 μ_{CR} = 0.075
 Diversity = 1.0336e-04 k_t = [3, 2, 2, 2]
 Generation 8410: Best = 1.189627e+04 FES = 841100 μ_F = 0.766 μ_{CR} = 0.077
 Diversity = 1.0636e-04 k_t = [1, 1, 1, 1]
 Generation 8420: Best = 1.189627e+04 FES = 842100 μ_F = 0.747 μ_{CR} = 0.083
 Diversity = 1.0210e-04 k_t = [3, 3, 3, 2]
 Generation 8430: Best = 1.189627e+04 FES = 843100 μ_F = 0.744 μ_{CR} = 0.082
 Diversity = 1.0580e-04 k_t = [1, 1, 1, 1]
 Generation 8440: Best = 1.189627e+04 FES = 844100 μ_F = 0.749 μ_{CR} = 0.075
 Diversity = 1.0087e-04 k_t = [3, 2, 2, 3]
 Generation 8450: Best = 1.189627e+04 FES = 845100 μ_F = 0.754 μ_{CR} = 0.075
 Diversity = 1.0489e-04 k_t = [1, 1, 1, 1]
 Generation 8460: Best = 1.189627e+04 FES = 846100 μ_F = 0.779 μ_{CR} = 0.077
 Diversity = 1.0355e-04 k_t = [2, 3, 3, 4]
 Generation 8470: Best = 1.189627e+04 FES = 847100 μ_F = 0.759 μ_{CR} = 0.079
 Diversity = 1.0102e-04 k_t = [1, 1, 1, 1]
 Generation 8480: Best = 1.189627e+04 FES = 848100 μ_F = 0.760 μ_{CR} = 0.075
 Diversity = 9.7389e-05 k_t = [2, 3, 4, 4]
 Generation 8490: Best = 1.189627e+04 FES = 849100 μ_F = 0.753 μ_{CR} = 0.077
 Diversity = 9.8379e-05 k_t = [1, 1, 1, 1]
 Generation 8500: Best = 1.189627e+04 FES = 850100 μ_F = 0.742 μ_{CR} = 0.072
 Diversity = 1.0217e-04 k_t = [3, 4, 4, 3]
 Generation 8510: Best = 1.189627e+04 FES = 851100 μ_F = 0.763 μ_{CR} = 0.074
 Diversity = 1.0674e-04 k_t = [1, 1, 1, 1]
 Generation 8520: Best = 1.189627e+04 FES = 852100 μ_F = 0.749 μ_{CR} = 0.074
 Diversity = 1.0386e-04 k_t = [2, 2, 3, 2]
 Generation 8530: Best = 1.189627e+04 FES = 853100 μ_F = 0.739 μ_{CR} = 0.081
 Diversity = 1.0093e-04 k_t = [1, 1, 1, 1]
 Generation 8540: Best = 1.189627e+04 FES = 854100 μ_F = 0.743 μ_{CR} = 0.086
 Diversity = 9.3212e-05 k_t = [4, 2, 4, 4]
 Generation 8550: Best = 1.189627e+04 FES = 855100 μ_F = 0.757 μ_{CR} = 0.076
 Diversity = 9.2206e-05 k_t = [1, 1, 1, 1]
 Generation 8560: Best = 1.189627e+04 FES = 856100 μ_F = 0.765 μ_{CR} = 0.068
 Diversity = 9.0700e-05 k_t = [4, 4, 2, 3]
 Generation 8570: Best = 1.189627e+04 FES = 857100 μ_F = 0.755 μ_{CR} = 0.070
 Diversity = 8.6202e-05 k_t = [1, 1, 1, 1]
 Generation 8580: Best = 1.189627e+04 FES = 858100 μ_F = 0.751 μ_{CR} = 0.076
 Diversity = 7.9779e-05 k_t = [3, 2, 2, 4]
 Generation 8590: Best = 1.189627e+04 FES = 859100 μ_F = 0.745 μ_{CR} = 0.067

Diversity = 7.3311e-05	k_t = [1, 1, 1, 1]			
Generation 8600: Best = 1.189627e+04	FES = 860100	$\mu_F = 0.763$	$\mu_{CR} = 0.069$	
Diversity = 7.2306e-05	k_t = [2, 3, 3, 2]			
Generation 8610: Best = 1.189627e+04	FES = 861100	$\mu_F = 0.768$	$\mu_{CR} = 0.078$	
Diversity = 7.6580e-05	k_t = [1, 1, 1, 1]			
Generation 8620: Best = 1.189627e+04	FES = 862100	$\mu_F = 0.751$	$\mu_{CR} = 0.078$	
Diversity = 7.8355e-05	k_t = [4, 3, 3, 3]			
Generation 8630: Best = 1.189627e+04	FES = 863100	$\mu_F = 0.761$	$\mu_{CR} = 0.087$	
Diversity = 7.9761e-05	k_t = [1, 1, 1, 1]			
Generation 8640: Best = 1.189627e+04	FES = 864100	$\mu_F = 0.753$	$\mu_{CR} = 0.080$	
Diversity = 8.1096e-05	k_t = [3, 3, 2, 4]			
Generation 8650: Best = 1.189627e+04	FES = 865100	$\mu_F = 0.759$	$\mu_{CR} = 0.077$	
Diversity = 8.4685e-05	k_t = [1, 1, 1, 1]			
Generation 8660: Best = 1.189627e+04	FES = 866100	$\mu_F = 0.753$	$\mu_{CR} = 0.078$	
Diversity = 8.4655e-05	k_t = [2, 3, 2, 3]			
Generation 8670: Best = 1.189627e+04	FES = 867100	$\mu_F = 0.756$	$\mu_{CR} = 0.083$	
Diversity = 8.2420e-05	k_t = [1, 1, 1, 1]			
Generation 8680: Best = 1.189627e+04	FES = 868100	$\mu_F = 0.764$	$\mu_{CR} = 0.077$	
Diversity = 7.9394e-05	k_t = [2, 4, 2, 4]			
Generation 8690: Best = 1.189626e+04	FES = 869100	$\mu_F = 0.763$	$\mu_{CR} = 0.081$	
Diversity = 7.9814e-05	k_t = [1, 1, 1, 1]			
Generation 8700: Best = 1.189626e+04	FES = 870100	$\mu_F = 0.768$	$\mu_{CR} = 0.086$	
Diversity = 7.8890e-05	k_t = [3, 3, 2, 3]			
Generation 8710: Best = 1.189626e+04	FES = 871100	$\mu_F = 0.756$	$\mu_{CR} = 0.080$	
Diversity = 8.1775e-05	k_t = [1, 1, 1, 1]			
Generation 8720: Best = 1.189626e+04	FES = 872100	$\mu_F = 0.740$	$\mu_{CR} = 0.082$	
Diversity = 8.2991e-05	k_t = [3, 4, 2, 2]			
Generation 8730: Best = 1.189626e+04	FES = 873100	$\mu_F = 0.743$	$\mu_{CR} = 0.088$	
Diversity = 8.6057e-05	k_t = [1, 1, 1, 1]			
Generation 8740: Best = 1.189626e+04	FES = 874100	$\mu_F = 0.723$	$\mu_{CR} = 0.085$	
Diversity = 8.1465e-05	k_t = [3, 4, 3, 2]			
Generation 8750: Best = 1.189626e+04	FES = 875100	$\mu_F = 0.742$	$\mu_{CR} = 0.088$	
Diversity = 8.5170e-05	k_t = [1, 1, 1, 1]			
Generation 8760: Best = 1.189626e+04	FES = 876100	$\mu_F = 0.738$	$\mu_{CR} = 0.089$	
Diversity = 8.8947e-05	k_t = [2, 3, 4, 2]			
Generation 8770: Best = 1.189626e+04	FES = 877100	$\mu_F = 0.744$	$\mu_{CR} = 0.090$	
Diversity = 9.2487e-05	k_t = [1, 1, 1, 1]			
Generation 8780: Best = 1.189626e+04	FES = 878100	$\mu_F = 0.757$	$\mu_{CR} = 0.089$	
Diversity = 9.5433e-05	k_t = [2, 4, 4, 4]			
Generation 8790: Best = 1.189626e+04	FES = 879100	$\mu_F = 0.737$	$\mu_{CR} = 0.087$	
Diversity = 9.5425e-05	k_t = [1, 1, 1, 1]			
Generation 8800: Best = 1.189626e+04	FES = 880100	$\mu_F = 0.746$	$\mu_{CR} = 0.092$	
Diversity = 1.0848e-04	k_t = [4, 2, 3, 3]			
Generation 8810: Best = 1.189626e+04	FES = 881100	$\mu_F = 0.755$	$\mu_{CR} = 0.095$	
Diversity = 1.2149e-04	k_t = [1, 1, 1, 1]			
Generation 8820: Best = 1.189626e+04	FES = 882100	$\mu_F = 0.762$	$\mu_{CR} = 0.098$	
Diversity = 1.3271e-04	k_t = [4, 4, 3, 2]			
Generation 8830: Best = 1.189626e+04	FES = 883100	$\mu_F = 0.768$	$\mu_{CR} = 0.089$	
Diversity = 1.4235e-04	k_t = [1, 1, 1, 1]			
Generation 8840: Best = 1.189626e+04	FES = 884100	$\mu_F = 0.767$	$\mu_{CR} = 0.099$	
Diversity = 1.4856e-04	k_t = [3, 2, 4, 4]			
Generation 8850: Best = 1.189626e+04	FES = 885100	$\mu_F = 0.769$	$\mu_{CR} = 0.094$	
Diversity = 1.6656e-04	k_t = [1, 1, 1, 1]			
Generation 8860: Best = 1.189626e+04	FES = 886100	$\mu_F = 0.771$	$\mu_{CR} = 0.095$	

Diversity = 1.7692e-04	k_t = [4, 4, 3, 2]			
Generation 8870: Best = 1.189626e+04	FES = 887100	$\mu_F = 0.768$	$\mu_{CR} = 0.102$	
Diversity = 1.9502e-04	k_t = [1, 1, 1, 1]			
Generation 8880: Best = 1.189626e+04	FES = 888100	$\mu_F = 0.754$	$\mu_{CR} = 0.107$	
Diversity = 1.9200e-04	k_t = [4, 2, 4, 3]			
Generation 8890: Best = 1.189626e+04	FES = 889100	$\mu_F = 0.741$	$\mu_{CR} = 0.109$	
Diversity = 1.8051e-04	k_t = [1, 1, 1, 1]			
Generation 8900: Best = 1.189626e+04	FES = 890100	$\mu_F = 0.731$	$\mu_{CR} = 0.098$	
Diversity = 1.5712e-04	k_t = [3, 4, 2, 4]			
Generation 8910: Best = 1.189626e+04	FES = 891100	$\mu_F = 0.741$	$\mu_{CR} = 0.091$	
Diversity = 1.4293e-04	k_t = [1, 1, 1, 1]			
Generation 8920: Best = 1.189626e+04	FES = 892100	$\mu_F = 0.746$	$\mu_{CR} = 0.095$	
Diversity = 1.4889e-04	k_t = [3, 2, 3, 2]			
Generation 8930: Best = 1.189626e+04	FES = 893100	$\mu_F = 0.750$	$\mu_{CR} = 0.095$	
Diversity = 1.4907e-04	k_t = [1, 1, 1, 1]			
Generation 8940: Best = 1.189626e+04	FES = 894100	$\mu_F = 0.737$	$\mu_{CR} = 0.087$	
Diversity = 1.4092e-04	k_t = [2, 2, 4, 4]			
Generation 8950: Best = 1.189626e+04	FES = 895100	$\mu_F = 0.734$	$\mu_{CR} = 0.086$	
Diversity = 1.3480e-04	k_t = [1, 1, 1, 1]			
Generation 8960: Best = 1.189626e+04	FES = 896100	$\mu_F = 0.757$	$\mu_{CR} = 0.086$	
Diversity = 1.3510e-04	k_t = [3, 2, 2, 3]			
Generation 8970: Best = 1.189626e+04	FES = 897100	$\mu_F = 0.747$	$\mu_{CR} = 0.087$	
Diversity = 1.5000e-04	k_t = [1, 1, 1, 1]			
Generation 8980: Best = 1.189626e+04	FES = 898100	$\mu_F = 0.737$	$\mu_{CR} = 0.088$	
Diversity = 1.6131e-04	k_t = [3, 2, 4, 2]			
Generation 8990: Best = 1.189626e+04	FES = 899100	$\mu_F = 0.744$	$\mu_{CR} = 0.090$	
Diversity = 1.6856e-04	k_t = [1, 1, 1, 1]			
Generation 9000: Best = 1.189626e+04	FES = 900100	$\mu_F = 0.731$	$\mu_{CR} = 0.093$	
Diversity = 1.6348e-04	k_t = [2, 2, 2, 4]			
Generation 9010: Best = 1.189626e+04	FES = 901100	$\mu_F = 0.723$	$\mu_{CR} = 0.086$	
Diversity = 1.4570e-04	k_t = [1, 1, 1, 1]			
Generation 9020: Best = 1.189626e+04	FES = 902100	$\mu_F = 0.730$	$\mu_{CR} = 0.089$	
Diversity = 1.3010e-04	k_t = [3, 3, 2, 3]			
Generation 9030: Best = 1.189626e+04	FES = 903100	$\mu_F = 0.739$	$\mu_{CR} = 0.080$	
Diversity = 1.3126e-04	k_t = [1, 1, 1, 1]			
Generation 9040: Best = 1.189626e+04	FES = 904100	$\mu_F = 0.747$	$\mu_{CR} = 0.081$	
Diversity = 1.2915e-04	k_t = [3, 2, 4, 2]			
Generation 9050: Best = 1.189626e+04	FES = 905100	$\mu_F = 0.748$	$\mu_{CR} = 0.080$	
Diversity = 1.3491e-04	k_t = [1, 1, 1, 1]			
Generation 9060: Best = 1.189626e+04	FES = 906100	$\mu_F = 0.759$	$\mu_{CR} = 0.084$	
Diversity = 1.2802e-04	k_t = [2, 3, 2, 2]			
Generation 9070: Best = 1.189626e+04	FES = 907100	$\mu_F = 0.765$	$\mu_{CR} = 0.077$	
Diversity = 1.3404e-04	k_t = [1, 1, 1, 1]			
Generation 9080: Best = 1.189626e+04	FES = 908100	$\mu_F = 0.782$	$\mu_{CR} = 0.082$	
Diversity = 1.3589e-04	k_t = [4, 4, 2, 3]			
Generation 9090: Best = 1.189626e+04	FES = 909100	$\mu_F = 0.762$	$\mu_{CR} = 0.076$	
Diversity = 1.3005e-04	k_t = [1, 1, 1, 1]			
Generation 9100: Best = 1.189626e+04	FES = 910100	$\mu_F = 0.764$	$\mu_{CR} = 0.081$	
Diversity = 1.3073e-04	k_t = [4, 3, 2, 2]			
Generation 9110: Best = 1.189626e+04	FES = 911100	$\mu_F = 0.765$	$\mu_{CR} = 0.080$	
Diversity = 1.3258e-04	k_t = [1, 1, 1, 1]			
Generation 9120: Best = 1.189626e+04	FES = 912100	$\mu_F = 0.758$	$\mu_{CR} = 0.075$	
Diversity = 1.2013e-04	k_t = [2, 3, 2, 4]			
Generation 9130: Best = 1.189626e+04	FES = 913100	$\mu_F = 0.750$	$\mu_{CR} = 0.078$	

Diversity = 1.1803e-04	k_t = [1, 1, 1, 1]			
Generation 9140: Best = 1.189626e+04	FES = 914100	$\mu_F = 0.768$	$\mu_{CR} = 0.077$	
Diversity = 1.1521e-04	k_t = [3, 3, 4, 4]			
Generation 9150: Best = 1.189626e+04	FES = 915100	$\mu_F = 0.775$	$\mu_{CR} = 0.076$	
Diversity = 1.1468e-04	k_t = [1, 1, 1, 1]			
Generation 9160: Best = 1.189626e+04	FES = 916100	$\mu_F = 0.774$	$\mu_{CR} = 0.079$	
Diversity = 1.1257e-04	k_t = [3, 4, 4, 4]			
Generation 9170: Best = 1.189626e+04	FES = 917100	$\mu_F = 0.789$	$\mu_{CR} = 0.073$	
Diversity = 1.1860e-04	k_t = [1, 1, 1, 1]			
Generation 9180: Best = 1.189626e+04	FES = 918100	$\mu_F = 0.773$	$\mu_{CR} = 0.073$	
Diversity = 1.2907e-04	k_t = [4, 4, 2, 3]			
Generation 9190: Best = 1.189626e+04	FES = 919100	$\mu_F = 0.782$	$\mu_{CR} = 0.079$	
Diversity = 1.4000e-04	k_t = [1, 1, 1, 1]			
Generation 9200: Best = 1.189626e+04	FES = 920100	$\mu_F = 0.787$	$\mu_{CR} = 0.086$	
Diversity = 1.4318e-04	k_t = [4, 3, 2, 2]			
Generation 9210: Best = 1.189626e+04	FES = 921100	$\mu_F = 0.791$	$\mu_{CR} = 0.091$	
Diversity = 1.5472e-04	k_t = [1, 1, 1, 1]			
Generation 9220: Best = 1.189626e+04	FES = 922100	$\mu_F = 0.780$	$\mu_{CR} = 0.094$	
Diversity = 1.7163e-04	k_t = [2, 3, 4, 2]			
Generation 9230: Best = 1.189626e+04	FES = 923100	$\mu_F = 0.775$	$\mu_{CR} = 0.087$	
Diversity = 1.8014e-04	k_t = [1, 1, 1, 1]			
Generation 9240: Best = 1.189626e+04	FES = 924100	$\mu_F = 0.787$	$\mu_{CR} = 0.086$	
Diversity = 1.7799e-04	k_t = [2, 2, 3, 2]			
Generation 9250: Best = 1.189626e+04	FES = 925100	$\mu_F = 0.774$	$\mu_{CR} = 0.079$	
Diversity = 1.8181e-04	k_t = [1, 1, 1, 1]			
Generation 9260: Best = 1.189626e+04	FES = 926100	$\mu_F = 0.769$	$\mu_{CR} = 0.077$	
Diversity = 1.6406e-04	k_t = [2, 3, 3, 4]			
Generation 9270: Best = 1.189626e+04	FES = 927100	$\mu_F = 0.742$	$\mu_{CR} = 0.079$	
Diversity = 1.4943e-04	k_t = [1, 1, 1, 1]			
Generation 9280: Best = 1.189626e+04	FES = 928100	$\mu_F = 0.745$	$\mu_{CR} = 0.086$	
Diversity = 1.3884e-04	k_t = [3, 2, 3, 4]			
Generation 9290: Best = 1.189626e+04	FES = 929100	$\mu_F = 0.740$	$\mu_{CR} = 0.094$	
Diversity = 1.3548e-04	k_t = [1, 1, 1, 1]			
Generation 9300: Best = 1.189626e+04	FES = 930100	$\mu_F = 0.728$	$\mu_{CR} = 0.091$	
Diversity = 1.3310e-04	k_t = [4, 2, 3, 2]			
Generation 9310: Best = 1.189626e+04	FES = 931100	$\mu_F = 0.732$	$\mu_{CR} = 0.089$	
Diversity = 1.3636e-04	k_t = [1, 1, 1, 1]			
Generation 9320: Best = 1.189626e+04	FES = 932100	$\mu_F = 0.730$	$\mu_{CR} = 0.082$	
Diversity = 1.3075e-04	k_t = [3, 2, 4, 4]			
Generation 9330: Best = 1.189626e+04	FES = 933100	$\mu_F = 0.722$	$\mu_{CR} = 0.082$	
Diversity = 1.2131e-04	k_t = [1, 1, 1, 1]			
Generation 9340: Best = 1.189626e+04	FES = 934100	$\mu_F = 0.748$	$\mu_{CR} = 0.086$	
Diversity = 1.1804e-04	k_t = [3, 4, 2, 3]			
Generation 9350: Best = 1.189626e+04	FES = 935100	$\mu_F = 0.724$	$\mu_{CR} = 0.087$	
Diversity = 1.2153e-04	k_t = [1, 1, 1, 1]			
Generation 9360: Best = 1.189626e+04	FES = 936100	$\mu_F = 0.727$	$\mu_{CR} = 0.079$	
Diversity = 1.2255e-04	k_t = [3, 4, 2, 4]			
Generation 9370: Best = 1.189626e+04	FES = 937100	$\mu_F = 0.728$	$\mu_{CR} = 0.083$	
Diversity = 1.1546e-04	k_t = [1, 1, 1, 1]			
Generation 9380: Best = 1.189626e+04	FES = 938100	$\mu_F = 0.726$	$\mu_{CR} = 0.084$	
Diversity = 1.1343e-04	k_t = [3, 3, 3, 4]			
Generation 9390: Best = 1.189626e+04	FES = 939100	$\mu_F = 0.754$	$\mu_{CR} = 0.082$	
Diversity = 1.1681e-04	k_t = [1, 1, 1, 1]			
Generation 9400: Best = 1.189626e+04	FES = 940100	$\mu_F = 0.774$	$\mu_{CR} = 0.078$	

Diversity = 1.3465e-04	k_t = [2, 4, 4, 2]			
Generation 9410: Best = 1.189626e+04	FES = 941100	$\mu_F = 0.781$	$\mu_{CR} = 0.079$	
Diversity = 1.4739e-04	k_t = [1, 1, 1, 1]			
Generation 9420: Best = 1.189626e+04	FES = 942100	$\mu_F = 0.776$	$\mu_{CR} = 0.091$	
Diversity = 1.4834e-04	k_t = [3, 4, 3, 2]			
Generation 9430: Best = 1.189626e+04	FES = 943100	$\mu_F = 0.761$	$\mu_{CR} = 0.088$	
Diversity = 1.4991e-04	k_t = [1, 1, 1, 1]			
Generation 9440: Best = 1.189626e+04	FES = 944100	$\mu_F = 0.760$	$\mu_{CR} = 0.083$	
Diversity = 1.5737e-04	k_t = [3, 3, 2, 2]			
Generation 9450: Best = 1.189626e+04	FES = 945100	$\mu_F = 0.756$	$\mu_{CR} = 0.082$	
Diversity = 1.6328e-04	k_t = [1, 1, 1, 1]			
Generation 9460: Best = 1.189626e+04	FES = 946100	$\mu_F = 0.761$	$\mu_{CR} = 0.082$	
Diversity = 1.5730e-04	k_t = [3, 3, 2, 2]			
Generation 9470: Best = 1.189626e+04	FES = 947100	$\mu_F = 0.754$	$\mu_{CR} = 0.078$	
Diversity = 1.4907e-04	k_t = [1, 1, 1, 1]			
Generation 9480: Best = 1.189626e+04	FES = 948100	$\mu_F = 0.744$	$\mu_{CR} = 0.090$	
Diversity = 1.4061e-04	k_t = [4, 4, 3, 2]			
Generation 9490: Best = 1.189626e+04	FES = 949100	$\mu_F = 0.731$	$\mu_{CR} = 0.086$	
Diversity = 1.4086e-04	k_t = [1, 1, 1, 1]			
Generation 9500: Best = 1.189626e+04	FES = 950100	$\mu_F = 0.718$	$\mu_{CR} = 0.082$	
Diversity = 1.3329e-04	k_t = [3, 4, 2, 3]			
Generation 9510: Best = 1.189626e+04	FES = 951100	$\mu_F = 0.734$	$\mu_{CR} = 0.081$	
Diversity = 1.2871e-04	k_t = [1, 1, 1, 1]			
Generation 9520: Best = 1.189626e+04	FES = 952100	$\mu_F = 0.743$	$\mu_{CR} = 0.081$	
Diversity = 1.2915e-04	k_t = [4, 2, 4, 2]			
Generation 9530: Best = 1.189626e+04	FES = 953100	$\mu_F = 0.749$	$\mu_{CR} = 0.083$	
Diversity = 1.3128e-04	k_t = [1, 1, 1, 1]			
Generation 9540: Best = 1.189626e+04	FES = 954100	$\mu_F = 0.737$	$\mu_{CR} = 0.078$	
Diversity = 1.1878e-04	k_t = [4, 3, 4, 3]			
Generation 9550: Best = 1.189626e+04	FES = 955100	$\mu_F = 0.733$	$\mu_{CR} = 0.079$	
Diversity = 1.1442e-04	k_t = [1, 1, 1, 1]			
Generation 9560: Best = 1.189626e+04	FES = 956100	$\mu_F = 0.707$	$\mu_{CR} = 0.077$	
Diversity = 1.0653e-04	k_t = [4, 2, 3, 2]			
Generation 9570: Best = 1.189626e+04	FES = 957100	$\mu_F = 0.744$	$\mu_{CR} = 0.075$	
Diversity = 1.0474e-04	k_t = [1, 1, 1, 1]			
Generation 9580: Best = 1.189626e+04	FES = 958100	$\mu_F = 0.730$	$\mu_{CR} = 0.079$	
Diversity = 9.7085e-05	k_t = [2, 3, 3, 2]			
Generation 9590: Best = 1.189626e+04	FES = 959100	$\mu_F = 0.721$	$\mu_{CR} = 0.080$	
Diversity = 9.5600e-05	k_t = [1, 1, 1, 1]			
Generation 9600: Best = 1.189626e+04	FES = 960100	$\mu_F = 0.716$	$\mu_{CR} = 0.076$	
Diversity = 8.7432e-05	k_t = [4, 4, 2, 4]			
Generation 9610: Best = 1.189626e+04	FES = 961100	$\mu_F = 0.703$	$\mu_{CR} = 0.077$	
Diversity = 7.9332e-05	k_t = [1, 1, 1, 1]			
Generation 9620: Best = 1.189626e+04	FES = 962100	$\mu_F = 0.703$	$\mu_{CR} = 0.080$	
Diversity = 7.3765e-05	k_t = [4, 4, 4, 3]			
Generation 9630: Best = 1.189626e+04	FES = 963100	$\mu_F = 0.708$	$\mu_{CR} = 0.082$	
Diversity = 7.3311e-05	k_t = [1, 1, 1, 1]			
Generation 9640: Best = 1.189626e+04	FES = 964100	$\mu_F = 0.729$	$\mu_{CR} = 0.076$	
Diversity = 7.7083e-05	k_t = [3, 2, 2, 3]			
Generation 9650: Best = 1.189626e+04	FES = 965100	$\mu_F = 0.720$	$\mu_{CR} = 0.066$	
Diversity = 7.9712e-05	k_t = [1, 1, 1, 1]			
Generation 9660: Best = 1.189626e+04	FES = 966100	$\mu_F = 0.689$	$\mu_{CR} = 0.066$	
Diversity = 8.1296e-05	k_t = [2, 2, 2, 3]			
Generation 9670: Best = 1.189626e+04	FES = 967100	$\mu_F = 0.695$	$\mu_{CR} = 0.076$	

Diversity = 8.3435e-05	k_t = [1, 1, 1, 1]			
Generation 9680: Best = 1.189626e+04	FES = 968100	$\mu_F = 0.710$	$\mu_{CR} = 0.079$	
Diversity = 8.4877e-05	k_t = [2, 2, 3, 2]			
Generation 9690: Best = 1.189626e+04	FES = 969100	$\mu_F = 0.711$	$\mu_{CR} = 0.082$	
Diversity = 8.4954e-05	k_t = [1, 1, 1, 1]			
Generation 9700: Best = 1.189626e+04	FES = 970100	$\mu_F = 0.712$	$\mu_{CR} = 0.072$	
Diversity = 8.4724e-05	k_t = [4, 2, 2, 2]			
Generation 9710: Best = 1.189626e+04	FES = 971100	$\mu_F = 0.709$	$\mu_{CR} = 0.067$	
Diversity = 8.5603e-05	k_t = [1, 1, 1, 1]			
Generation 9720: Best = 1.189626e+04	FES = 972100	$\mu_F = 0.709$	$\mu_{CR} = 0.067$	
Diversity = 8.4194e-05	k_t = [3, 4, 4, 2]			
Generation 9730: Best = 1.189626e+04	FES = 973100	$\mu_F = 0.714$	$\mu_{CR} = 0.063$	
Diversity = 8.5216e-05	k_t = [1, 1, 1, 1]			
Generation 9740: Best = 1.189626e+04	FES = 974100	$\mu_F = 0.710$	$\mu_{CR} = 0.069$	
Diversity = 7.9843e-05	k_t = [4, 3, 2, 4]			
Generation 9750: Best = 1.189626e+04	FES = 975100	$\mu_F = 0.729$	$\mu_{CR} = 0.074$	
Diversity = 7.6256e-05	k_t = [1, 1, 1, 1]			
Generation 9760: Best = 1.189626e+04	FES = 976100	$\mu_F = 0.751$	$\mu_{CR} = 0.079$	
Diversity = 7.8406e-05	k_t = [4, 3, 3, 4]			
Generation 9770: Best = 1.189626e+04	FES = 977100	$\mu_F = 0.762$	$\mu_{CR} = 0.074$	
Diversity = 8.4286e-05	k_t = [1, 1, 1, 1]			
Generation 9780: Best = 1.189626e+04	FES = 978100	$\mu_F = 0.750$	$\mu_{CR} = 0.080$	
Diversity = 8.1633e-05	k_t = [2, 2, 3, 4]			
Generation 9790: Best = 1.189626e+04	FES = 979100	$\mu_F = 0.756$	$\mu_{CR} = 0.089$	
Diversity = 8.2133e-05	k_t = [1, 1, 1, 1]			
Generation 9800: Best = 1.189626e+04	FES = 980100	$\mu_F = 0.760$	$\mu_{CR} = 0.084$	
Diversity = 7.8844e-05	k_t = [3, 2, 3, 4]			
Generation 9810: Best = 1.189626e+04	FES = 981100	$\mu_F = 0.763$	$\mu_{CR} = 0.081$	
Diversity = 7.7198e-05	k_t = [1, 1, 1, 1]			
Generation 9820: Best = 1.189626e+04	FES = 982100	$\mu_F = 0.769$	$\mu_{CR} = 0.073$	
Diversity = 7.8016e-05	k_t = [4, 3, 3, 2]			
Generation 9830: Best = 1.189626e+04	FES = 983100	$\mu_F = 0.791$	$\mu_{CR} = 0.075$	
Diversity = 8.2726e-05	k_t = [1, 1, 1, 1]			
Generation 9840: Best = 1.189626e+04	FES = 984100	$\mu_F = 0.783$	$\mu_{CR} = 0.071$	
Diversity = 9.1062e-05	k_t = [3, 2, 3, 4]			
Generation 9850: Best = 1.189626e+04	FES = 985100	$\mu_F = 0.787$	$\mu_{CR} = 0.074$	
Diversity = 9.1121e-05	k_t = [1, 1, 1, 1]			
Generation 9860: Best = 1.189626e+04	FES = 986100	$\mu_F = 0.773$	$\mu_{CR} = 0.084$	
Diversity = 8.8844e-05	k_t = [3, 4, 2, 2]			
Generation 9870: Best = 1.189626e+04	FES = 987100	$\mu_F = 0.765$	$\mu_{CR} = 0.078$	
Diversity = 8.8922e-05	k_t = [1, 1, 1, 1]			
Generation 9880: Best = 1.189626e+04	FES = 988100	$\mu_F = 0.774$	$\mu_{CR} = 0.076$	
Diversity = 8.2223e-05	k_t = [2, 2, 2, 3]			
Generation 9890: Best = 1.189626e+04	FES = 989100	$\mu_F = 0.758$	$\mu_{CR} = 0.072$	
Diversity = 8.2267e-05	k_t = [1, 1, 1, 1]			
Generation 9900: Best = 1.189626e+04	FES = 990100	$\mu_F = 0.777$	$\mu_{CR} = 0.072$	
Diversity = 8.1586e-05	k_t = [4, 4, 3, 2]			
Generation 9910: Best = 1.189626e+04	FES = 991100	$\mu_F = 0.761$	$\mu_{CR} = 0.074$	
Diversity = 7.9467e-05	k_t = [1, 1, 1, 1]			
Generation 9920: Best = 1.189626e+04	FES = 992100	$\mu_F = 0.741$	$\mu_{CR} = 0.076$	
Diversity = 7.8177e-05	k_t = [2, 2, 2, 3]			
Generation 9930: Best = 1.189626e+04	FES = 993100	$\mu_F = 0.745$	$\mu_{CR} = 0.066$	
Diversity = 7.5991e-05	k_t = [1, 1, 1, 1]			
Generation 9940: Best = 1.189626e+04	FES = 994100	$\mu_F = 0.739$	$\mu_{CR} = 0.067$	

```

Diversity = 7.3497e-05 k_t = [3, 2, 3, 4]
Generation 9950: Best = 1.189626e+04 FES = 995100  $\mu_F$  = 0.755  $\mu_{CR}$  = 0.071
Diversity = 7.1608e-05 k_t = [1, 1, 1, 1]
Generation 9960: Best = 1.189626e+04 FES = 996100  $\mu_F$  = 0.760  $\mu_{CR}$  = 0.080
Diversity = 7.4322e-05 k_t = [2, 2, 2, 3]
Generation 9970: Best = 1.189626e+04 FES = 997100  $\mu_F$  = 0.747  $\mu_{CR}$  = 0.077
Diversity = 7.4059e-05 k_t = [1, 1, 1, 1]
Generation 9980: Best = 1.189626e+04 FES = 998100  $\mu_F$  = 0.756  $\mu_{CR}$  = 0.083
Diversity = 7.5222e-05 k_t = [2, 2, 2, 3]
Generation 9990: Best = 1.189626e+04 FES = 999100  $\mu_F$  = 0.740  $\mu_{CR}$  = 0.078
Diversity = 7.4065e-05 k_t = [1, 1, 1, 1]

```

Optimization complete.

```

Best Fitness          = 1.189626e+04
Total Function Evals  = 1000000
Total Successful Updates = 277506

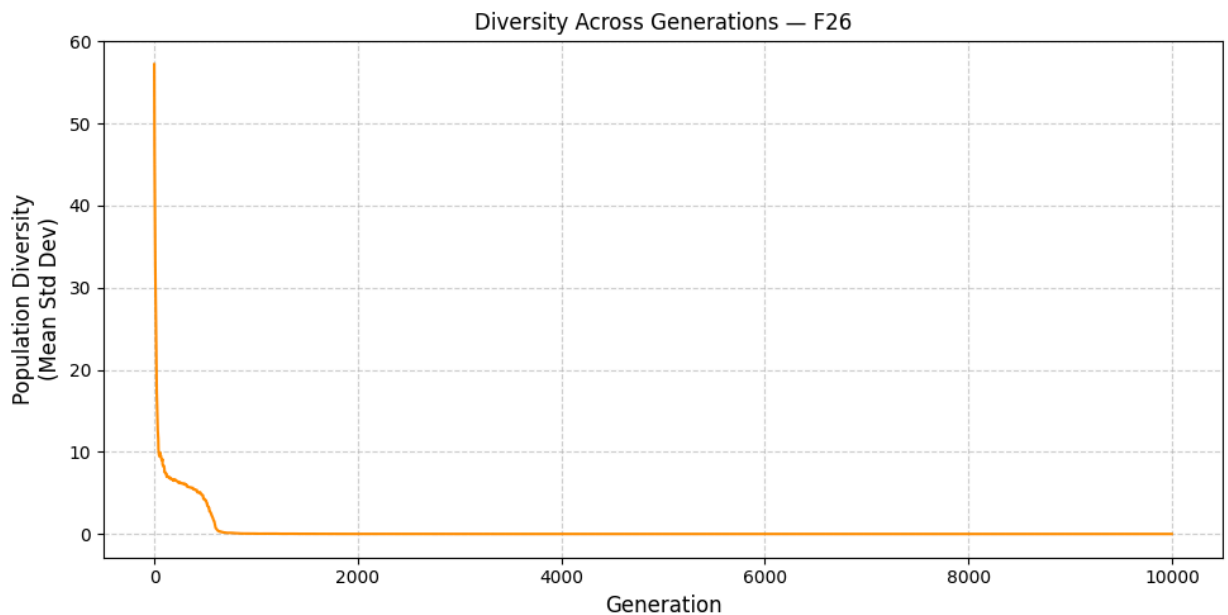
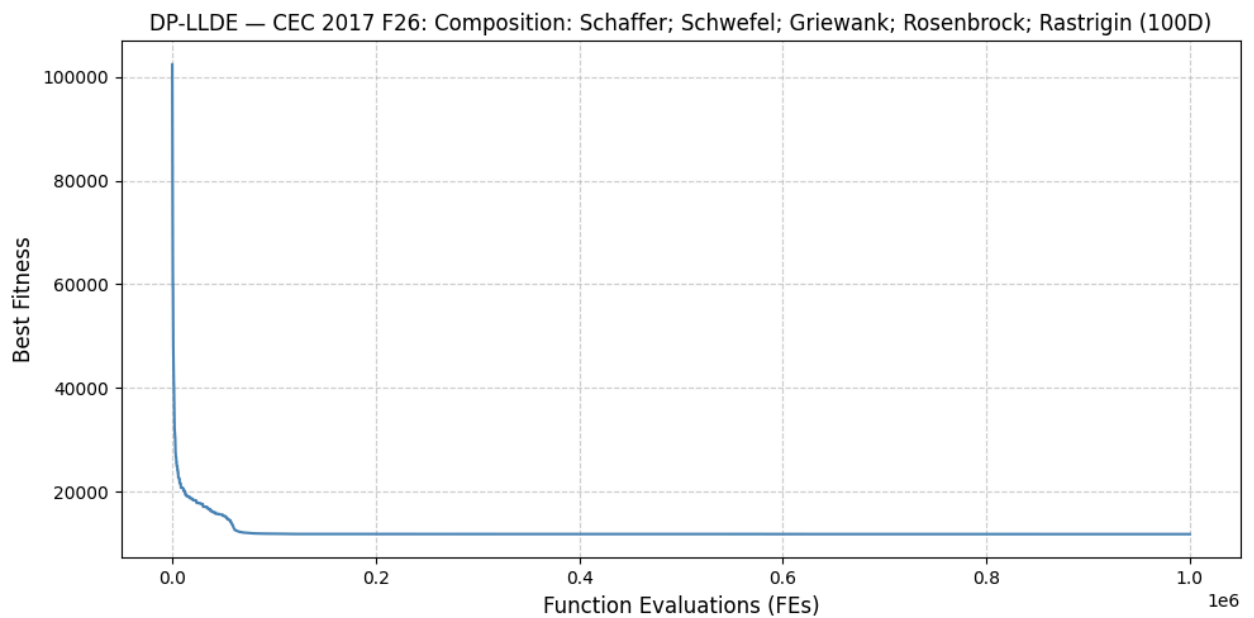
```

```

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=
Results:
  Best Fitness          : 1.189626e+04
  Global Optimum        : 2.600000e+03
  Error (|best-optimum|) : 9.296263e+03
  Best Solution (5 dim) : [ -9.02659396 -71.22210562 -37.57989094 -76.30484344
-15.0318107 ]
  D^(0) (init. diversity): 5.7220e+01
=====
=

```

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results/all_results_checkpoint.pkl



Graph saved as convergence_F26.png

✓ Plot saved to: /content/drive/My Drive/thesis_results/plots/convergence_F26.png

```
In [ ]: # =====
# F27 — Composition Function 7 (HGBat; Rastrigin; Schwefel; Bent Cigar; Ellipt
# =====
run_and_plot('F27', make_wrapper(composition.f27), 2700.0, 'Composition: HGBat
```

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DP-LLDE on CEC 2017 F27 – Single Run

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Function : Composition: HGBat; Rastrigin; Schwefel; Bent Cigar; Elliptic; Schaffer (100D)

Bounds : [-100, 100]

Global optimum : 2700.0

Initial $D^*(0) = 5.7220e+01$ | Diversity threshold = 5.7220e+00

Generation 0: Best Fitness = 1.447050e+04

Generation 10:	Best = 1.034820e+04	FES = 1100	$\mu_F = 0.594$	$\mu_{CR} = 0.347$
Diversity = 4.2631e+01	$k_t = [1, 1, 1, 1]$			
Generation 20:	Best = 7.347098e+03	FES = 2100	$\mu_F = 0.642$	$\mu_{CR} = 0.343$
Diversity = 3.2973e+01	$k_t = [2, 3, 2, 3]$			
Generation 30:	Best = 6.005856e+03	FES = 3100	$\mu_F = 0.654$	$\mu_{CR} = 0.340$
Diversity = 2.9046e+01	$k_t = [1, 1, 1, 1]$			
Generation 40:	Best = 5.511751e+03	FES = 4100	$\mu_F = 0.668$	$\mu_{CR} = 0.325$
Diversity = 2.4718e+01	$k_t = [3, 2, 2, 2]$			
Generation 50:	Best = 5.008384e+03	FES = 5100	$\mu_F = 0.668$	$\mu_{CR} = 0.327$
Diversity = 1.9572e+01	$k_t = [1, 1, 1, 1]$			
Generation 60:	Best = 4.665164e+03	FES = 6100	$\mu_F = 0.676$	$\mu_{CR} = 0.310$
Diversity = 1.6233e+01	$k_t = [3, 4, 4, 3]$			
Generation 70:	Best = 4.468939e+03	FES = 7100	$\mu_F = 0.683$	$\mu_{CR} = 0.302$
Diversity = 1.3696e+01	$k_t = [1, 1, 1, 1]$			
Generation 80:	Best = 4.306193e+03	FES = 8100	$\mu_F = 0.679$	$\mu_{CR} = 0.302$
Diversity = 1.1475e+01	$k_t = [4, 3, 3, 3]$			
Generation 90:	Best = 4.198666e+03	FES = 9100	$\mu_F = 0.682$	$\mu_{CR} = 0.297$
Diversity = 9.6678e+00	$k_t = [1, 1, 1, 1]$			
Generation 100:	Best = 4.146439e+03	FES = 10100	$\mu_F = 0.689$	$\mu_{CR} = 0.286$
Diversity = 7.9716e+00	$k_t = [2, 4, 4, 2]$			
Generation 110:	Best = 4.057598e+03	FES = 11100	$\mu_F = 0.696$	$\mu_{CR} = 0.274$
Diversity = 6.7523e+00	$k_t = [1, 1, 1, 1]$			
Generation 120:	Best = 4.024099e+03	FES = 12100	$\mu_F = 0.694$	$\mu_{CR} = 0.252$
Diversity = 6.0161e+00	$k_t = [3, 3, 3, 2]$			
Generation 130:	Best = 3.973854e+03	FES = 13100	$\mu_F = 0.683$	$\mu_{CR} = 0.241$
Diversity = 5.2448e+00	$k_t = [1, 1, 1, 1]$			
Generation 140:	Best = 3.908062e+03	FES = 14100	$\mu_F = 0.690$	$\mu_{CR} = 0.225$
Diversity = 4.7276e+00	$k_t = [2, 3, 4, 2]$			
Generation 150:	Best = 3.891007e+03	FES = 15100	$\mu_F = 0.672$	$\mu_{CR} = 0.213$
Diversity = 3.8809e+00	$k_t = [1, 1, 1, 1]$			
Generation 160:	Best = 3.862415e+03	FES = 16100	$\mu_F = 0.658$	$\mu_{CR} = 0.208$
Diversity = 2.9969e+00	$k_t = [2, 3, 4, 3]$			
Generation 170:	Best = 3.836254e+03	FES = 17100	$\mu_F = 0.659$	$\mu_{CR} = 0.198$
Diversity = 2.4050e+00	$k_t = [1, 1, 1, 1]$			
Generation 180:	Best = 3.811675e+03	FES = 18100	$\mu_F = 0.660$	$\mu_{CR} = 0.187$
Diversity = 1.9173e+00	$k_t = [4, 4, 2, 4]$			
Generation 190:	Best = 3.788321e+03	FES = 19100	$\mu_F = 0.677$	$\mu_{CR} = 0.184$
Diversity = 1.4514e+00	$k_t = [1, 1, 1, 1]$			
Generation 200:	Best = 3.774209e+03	FES = 20100	$\mu_F = 0.677$	$\mu_{CR} = 0.180$
Diversity = 1.1164e+00	$k_t = [4, 2, 4, 4]$			
Generation 210:	Best = 3.756699e+03	FES = 21100	$\mu_F = 0.685$	$\mu_{CR} = 0.173$
Diversity = 8.3526e-01	$k_t = [1, 1, 1, 1]$			
Generation 220:	Best = 3.749833e+03	FES = 22100	$\mu_F = 0.695$	$\mu_{CR} = 0.174$

Diversity = 7.0161e-01	k_t = [2, 4, 4, 3]				
Generation 230: Best = 3.739945e+03	FES = 23100	$\mu_F = 0.698$	$\mu_{CR} = 0.174$		
Diversity = 5.7933e-01	k_t = [1, 1, 1, 1]				
Generation 240: Best = 3.729490e+03	FES = 24100	$\mu_F = 0.710$	$\mu_{CR} = 0.185$		
Diversity = 4.8082e-01	k_t = [2, 4, 3, 3]				
Generation 250: Best = 3.723586e+03	FES = 25100	$\mu_F = 0.706$	$\mu_{CR} = 0.184$		
Diversity = 4.4799e-01	k_t = [1, 1, 1, 1]				
Generation 260: Best = 3.719646e+03	FES = 26100	$\mu_F = 0.711$	$\mu_{CR} = 0.183$		
Diversity = 3.8549e-01	k_t = [2, 2, 4, 3]				
Generation 270: Best = 3.715760e+03	FES = 27100	$\mu_F = 0.706$	$\mu_{CR} = 0.180$		
Diversity = 3.5565e-01	k_t = [1, 1, 1, 1]				
Generation 280: Best = 3.712911e+03	FES = 28100	$\mu_F = 0.705$	$\mu_{CR} = 0.182$		
Diversity = 3.3083e-01	k_t = [4, 3, 3, 4]				
Generation 290: Best = 3.709333e+03	FES = 29100	$\mu_F = 0.712$	$\mu_{CR} = 0.178$		
Diversity = 3.1445e-01	k_t = [1, 1, 1, 1]				
Generation 300: Best = 3.707379e+03	FES = 30100	$\mu_F = 0.701$	$\mu_{CR} = 0.174$		
Diversity = 2.8936e-01	k_t = [4, 3, 3, 4]				
Generation 310: Best = 3.704792e+03	FES = 31100	$\mu_F = 0.691$	$\mu_{CR} = 0.167$		
Diversity = 2.7860e-01	k_t = [1, 1, 1, 1]				
Generation 320: Best = 3.703941e+03	FES = 32100	$\mu_F = 0.688$	$\mu_{CR} = 0.163$		
Diversity = 2.6127e-01	k_t = [4, 2, 2, 4]				
Generation 330: Best = 3.702062e+03	FES = 33100	$\mu_F = 0.681$	$\mu_{CR} = 0.165$		
Diversity = 2.1999e-01	k_t = [1, 1, 1, 1]				
Generation 340: Best = 3.700592e+03	FES = 34100	$\mu_F = 0.694$	$\mu_{CR} = 0.166$		
Diversity = 2.0065e-01	k_t = [2, 3, 2, 4]				
Generation 350: Best = 3.699154e+03	FES = 35100	$\mu_F = 0.692$	$\mu_{CR} = 0.161$		
Diversity = 1.8585e-01	k_t = [1, 1, 1, 1]				
Generation 360: Best = 3.698262e+03	FES = 36100	$\mu_F = 0.671$	$\mu_{CR} = 0.158$		
Diversity = 1.7345e-01	k_t = [2, 3, 4, 2]				
Generation 370: Best = 3.697632e+03	FES = 37100	$\mu_F = 0.678$	$\mu_{CR} = 0.155$		
Diversity = 1.6159e-01	k_t = [1, 1, 1, 1]				
Generation 380: Best = 3.696864e+03	FES = 38100	$\mu_F = 0.684$	$\mu_{CR} = 0.156$		
Diversity = 1.4972e-01	k_t = [2, 2, 3, 2]				
Generation 390: Best = 3.696096e+03	FES = 39100	$\mu_F = 0.686$	$\mu_{CR} = 0.150$		
Diversity = 1.4687e-01	k_t = [1, 1, 1, 1]				
Generation 400: Best = 3.695430e+03	FES = 40100	$\mu_F = 0.685$	$\mu_{CR} = 0.153$		
Diversity = 1.3343e-01	k_t = [4, 2, 4, 3]				
Generation 410: Best = 3.694995e+03	FES = 41100	$\mu_F = 0.688$	$\mu_{CR} = 0.136$		
Diversity = 1.2776e-01	k_t = [1, 1, 1, 1]				
Generation 420: Best = 3.694465e+03	FES = 42100	$\mu_F = 0.688$	$\mu_{CR} = 0.128$		
Diversity = 1.1472e-01	k_t = [4, 3, 3, 4]				
Generation 430: Best = 3.693998e+03	FES = 43100	$\mu_F = 0.698$	$\mu_{CR} = 0.133$		
Diversity = 1.0873e-01	k_t = [1, 1, 1, 1]				
Generation 440: Best = 3.693530e+03	FES = 44100	$\mu_F = 0.702$	$\mu_{CR} = 0.139$		
Diversity = 1.0508e-01	k_t = [3, 2, 4, 3]				
Generation 450: Best = 3.692974e+03	FES = 45100	$\mu_F = 0.686$	$\mu_{CR} = 0.144$		
Diversity = 1.0063e-01	k_t = [1, 1, 1, 1]				
Generation 460: Best = 3.692408e+03	FES = 46100	$\mu_F = 0.707$	$\mu_{CR} = 0.144$		
Diversity = 9.7529e-02	k_t = [3, 3, 4, 3]				
Generation 470: Best = 3.692044e+03	FES = 47100	$\mu_F = 0.694$	$\mu_{CR} = 0.143$		
Diversity = 9.1757e-02	k_t = [1, 1, 1, 1]				
Generation 480: Best = 3.691495e+03	FES = 48100	$\mu_F = 0.694$	$\mu_{CR} = 0.147$		
Diversity = 9.0537e-02	k_t = [3, 3, 4, 2]				
Generation 490: Best = 3.691142e+03	FES = 49100	$\mu_F = 0.685$	$\mu_{CR} = 0.150$		

Diversity = 8.7478e-02	k_t = [1, 1, 1, 1]					
Generation 500: Best = 3.690900e+03	FES = 50100	$\mu_F = 0.683$	$\mu_{CR} = 0.147$			
Diversity = 8.4208e-02	k_t = [3, 3, 3, 2]					
Generation 510: Best = 3.690329e+03	FES = 51100	$\mu_F = 0.682$	$\mu_{CR} = 0.150$			
Diversity = 8.1130e-02	k_t = [1, 1, 1, 1]					
Generation 520: Best = 3.690200e+03	FES = 52100	$\mu_F = 0.684$	$\mu_{CR} = 0.149$			
Diversity = 7.3364e-02	k_t = [2, 2, 4, 3]					
Generation 530: Best = 3.689894e+03	FES = 53100	$\mu_F = 0.695$	$\mu_{CR} = 0.149$			
Diversity = 7.2346e-02	k_t = [1, 1, 1, 1]					
Generation 540: Best = 3.689551e+03	FES = 54100	$\mu_F = 0.703$	$\mu_{CR} = 0.143$			
Diversity = 7.3303e-02	k_t = [2, 2, 4, 2]					
Generation 550: Best = 3.689394e+03	FES = 55100	$\mu_F = 0.700$	$\mu_{CR} = 0.138$			
Diversity = 7.6214e-02	k_t = [1, 1, 1, 1]					
Generation 560: Best = 3.689151e+03	FES = 56100	$\mu_F = 0.700$	$\mu_{CR} = 0.129$			
Diversity = 7.3935e-02	k_t = [4, 4, 4, 3]					
Generation 570: Best = 3.688954e+03	FES = 57100	$\mu_F = 0.704$	$\mu_{CR} = 0.129$			
Diversity = 8.1381e-02	k_t = [1, 1, 1, 1]					
Generation 580: Best = 3.688705e+03	FES = 58100	$\mu_F = 0.704$	$\mu_{CR} = 0.131$			
Diversity = 8.1422e-02	k_t = [2, 2, 4, 2]					
Generation 590: Best = 3.688442e+03	FES = 59100	$\mu_F = 0.690$	$\mu_{CR} = 0.125$			
Diversity = 7.5697e-02	k_t = [1, 1, 1, 1]					
Generation 600: Best = 3.688329e+03	FES = 60100	$\mu_F = 0.694$	$\mu_{CR} = 0.110$			
Diversity = 7.8171e-02	k_t = [4, 2, 2, 3]					
Generation 610: Best = 3.688071e+03	FES = 61100	$\mu_F = 0.687$	$\mu_{CR} = 0.108$			
Diversity = 8.3228e-02	k_t = [1, 1, 1, 1]					
Generation 620: Best = 3.687784e+03	FES = 62100	$\mu_F = 0.684$	$\mu_{CR} = 0.113$			
Diversity = 8.2118e-02	k_t = [2, 2, 2, 2]					
Generation 630: Best = 3.687140e+03	FES = 63100	$\mu_F = 0.691$	$\mu_{CR} = 0.101$			
Diversity = 8.2488e-02	k_t = [1, 1, 1, 1]					
Generation 640: Best = 3.686908e+03	FES = 64100	$\mu_F = 0.708$	$\mu_{CR} = 0.110$			
Diversity = 9.0602e-02	k_t = [4, 2, 3, 4]					
Generation 650: Best = 3.686776e+03	FES = 65100	$\mu_F = 0.717$	$\mu_{CR} = 0.109$			
Diversity = 9.4753e-02	k_t = [1, 1, 1, 1]					
Generation 660: Best = 3.686439e+03	FES = 66100	$\mu_F = 0.709$	$\mu_{CR} = 0.103$			
Diversity = 1.1322e-01	k_t = [4, 2, 4, 2]					
Generation 670: Best = 3.685320e+03	FES = 67100	$\mu_F = 0.713$	$\mu_{CR} = 0.108$			
Diversity = 1.2249e-01	k_t = [1, 1, 1, 1]					
Generation 680: Best = 3.684012e+03	FES = 68100	$\mu_F = 0.724$	$\mu_{CR} = 0.101$			
Diversity = 1.6103e-01	k_t = [2, 2, 4, 2]					
Generation 690: Best = 3.683088e+03	FES = 69100	$\mu_F = 0.728$	$\mu_{CR} = 0.109$			
Diversity = 1.2051e-01	k_t = [1, 1, 1, 1]					
Generation 700: Best = 3.681859e+03	FES = 70100	$\mu_F = 0.732$	$\mu_{CR} = 0.123$			
Diversity = 1.0714e-01	k_t = [4, 2, 2, 2]					
Generation 710: Best = 3.681143e+03	FES = 71100	$\mu_F = 0.740$	$\mu_{CR} = 0.128$			
Diversity = 9.8268e-02	k_t = [1, 1, 1, 1]					
Generation 720: Best = 3.680596e+03	FES = 72100	$\mu_F = 0.750$	$\mu_{CR} = 0.135$			
Diversity = 9.5020e-02	k_t = [4, 3, 3, 4]					
Generation 730: Best = 3.680010e+03	FES = 73100	$\mu_F = 0.754$	$\mu_{CR} = 0.127$			
Diversity = 9.2478e-02	k_t = [1, 1, 1, 1]					
Generation 740: Best = 3.679794e+03	FES = 74100	$\mu_F = 0.730$	$\mu_{CR} = 0.123$			
Diversity = 8.7517e-02	k_t = [4, 2, 3, 3]					
Generation 750: Best = 3.679471e+03	FES = 75100	$\mu_F = 0.727$	$\mu_{CR} = 0.120$			
Diversity = 8.0893e-02	k_t = [1, 1, 1, 1]					
Generation 760: Best = 3.679409e+03	FES = 76100	$\mu_F = 0.721$	$\mu_{CR} = 0.130$			

Diversity = 7.1179e-02	k_t = [4, 4, 4, 3]				
Generation 770: Best = 3.679036e+03	FES = 77100	$\mu_F = 0.692$	$\mu_{CR} = 0.123$		
Diversity = 6.4595e-02	k_t = [1, 1, 1, 1]				
Generation 780: Best = 3.678817e+03	FES = 78100	$\mu_F = 0.697$	$\mu_{CR} = 0.121$		
Diversity = 5.9234e-02	k_t = [2, 3, 2, 3]				
Generation 790: Best = 3.678568e+03	FES = 79100	$\mu_F = 0.711$	$\mu_{CR} = 0.124$		
Diversity = 6.0582e-02	k_t = [1, 1, 1, 1]				
Generation 800: Best = 3.678303e+03	FES = 80100	$\mu_F = 0.716$	$\mu_{CR} = 0.125$		
Diversity = 6.2403e-02	k_t = [4, 3, 4, 2]				
Generation 810: Best = 3.677963e+03	FES = 81100	$\mu_F = 0.722$	$\mu_{CR} = 0.120$		
Diversity = 6.4564e-02	k_t = [1, 1, 1, 1]				
Generation 820: Best = 3.677945e+03	FES = 82100	$\mu_F = 0.738$	$\mu_{CR} = 0.125$		
Diversity = 6.1514e-02	k_t = [4, 4, 4, 4]				
Generation 830: Best = 3.677700e+03	FES = 83100	$\mu_F = 0.743$	$\mu_{CR} = 0.119$		
Diversity = 5.9630e-02	k_t = [1, 1, 1, 1]				
Generation 840: Best = 3.677375e+03	FES = 84100	$\mu_F = 0.753$	$\mu_{CR} = 0.120$		
Diversity = 5.8356e-02	k_t = [3, 4, 3, 4]				
Generation 850: Best = 3.677047e+03	FES = 85100	$\mu_F = 0.746$	$\mu_{CR} = 0.116$		
Diversity = 5.8132e-02	k_t = [1, 1, 1, 1]				
Generation 860: Best = 3.676842e+03	FES = 86100	$\mu_F = 0.746$	$\mu_{CR} = 0.119$		
Diversity = 6.2023e-02	k_t = [2, 3, 4, 2]				
Generation 870: Best = 3.676748e+03	FES = 87100	$\mu_F = 0.732$	$\mu_{CR} = 0.116$		
Diversity = 6.6192e-02	k_t = [1, 1, 1, 1]				
Generation 880: Best = 3.676575e+03	FES = 88100	$\mu_F = 0.723$	$\mu_{CR} = 0.118$		
Diversity = 5.9847e-02	k_t = [2, 2, 3, 4]				
Generation 890: Best = 3.676361e+03	FES = 89100	$\mu_F = 0.735$	$\mu_{CR} = 0.118$		
Diversity = 5.3739e-02	k_t = [1, 1, 1, 1]				
Generation 900: Best = 3.676270e+03	FES = 90100	$\mu_F = 0.733$	$\mu_{CR} = 0.125$		
Diversity = 5.1786e-02	k_t = [4, 2, 4, 3]				
Generation 910: Best = 3.676096e+03	FES = 91100	$\mu_F = 0.731$	$\mu_{CR} = 0.125$		
Diversity = 5.0346e-02	k_t = [1, 1, 1, 1]				
Generation 920: Best = 3.675958e+03	FES = 92100	$\mu_F = 0.713$	$\mu_{CR} = 0.122$		
Diversity = 4.7452e-02	k_t = [4, 4, 2, 3]				
Generation 930: Best = 3.675906e+03	FES = 93100	$\mu_F = 0.714$	$\mu_{CR} = 0.118$		
Diversity = 4.7103e-02	k_t = [1, 1, 1, 1]				
Generation 940: Best = 3.675801e+03	FES = 94100	$\mu_F = 0.710$	$\mu_{CR} = 0.113$		
Diversity = 4.5426e-02	k_t = [2, 3, 2, 3]				
Generation 950: Best = 3.675755e+03	FES = 95100	$\mu_F = 0.708$	$\mu_{CR} = 0.101$		
Diversity = 4.3269e-02	k_t = [1, 1, 1, 1]				
Generation 960: Best = 3.675628e+03	FES = 96100	$\mu_F = 0.712$	$\mu_{CR} = 0.098$		
Diversity = 4.2495e-02	k_t = [3, 2, 2, 3]				
Generation 970: Best = 3.675557e+03	FES = 97100	$\mu_F = 0.716$	$\mu_{CR} = 0.101$		
Diversity = 4.2187e-02	k_t = [1, 1, 1, 1]				
Generation 980: Best = 3.675479e+03	FES = 98100	$\mu_F = 0.718$	$\mu_{CR} = 0.102$		
Diversity = 3.9167e-02	k_t = [3, 2, 3, 4]				
Generation 990: Best = 3.675298e+03	FES = 99100	$\mu_F = 0.721$	$\mu_{CR} = 0.106$		
Diversity = 3.7059e-02	k_t = [1, 1, 1, 1]				
Generation 1000: Best = 3.675213e+03	FES = 100100	$\mu_F = 0.722$	$\mu_{CR} = 0.118$		
Diversity = 3.6947e-02	k_t = [4, 3, 4, 2]				
Generation 1010: Best = 3.675033e+03	FES = 101100	$\mu_F = 0.724$	$\mu_{CR} = 0.120$		
Diversity = 4.0304e-02	k_t = [1, 1, 1, 1]				
Generation 1020: Best = 3.674883e+03	FES = 102100	$\mu_F = 0.722$	$\mu_{CR} = 0.120$		
Diversity = 4.0570e-02	k_t = [2, 2, 4, 4]				
Generation 1030: Best = 3.674701e+03	FES = 103100	$\mu_F = 0.720$	$\mu_{CR} = 0.116$		

Diversity = 3.9682e-02	k_t = [1, 1, 1, 1]			
Generation 1040: Best = 3.674546e+03	FES = 104100	$\mu_F = 0.735$	$\mu_{CR} = 0.117$	
Diversity = 3.9301e-02	k_t = [3, 4, 2, 3]			
Generation 1050: Best = 3.674465e+03	FES = 105100	$\mu_F = 0.727$	$\mu_{CR} = 0.123$	
Diversity = 3.8125e-02	k_t = [1, 1, 1, 1]			
Generation 1060: Best = 3.674312e+03	FES = 106100	$\mu_F = 0.731$	$\mu_{CR} = 0.132$	
Diversity = 3.7653e-02	k_t = [3, 2, 4, 4]			
Generation 1070: Best = 3.673944e+03	FES = 107100	$\mu_F = 0.741$	$\mu_{CR} = 0.134$	
Diversity = 4.0033e-02	k_t = [1, 1, 1, 1]			
Generation 1080: Best = 3.673742e+03	FES = 108100	$\mu_F = 0.750$	$\mu_{CR} = 0.132$	
Diversity = 4.1214e-02	k_t = [2, 3, 2, 3]			
Generation 1090: Best = 3.673494e+03	FES = 109100	$\mu_F = 0.759$	$\mu_{CR} = 0.131$	
Diversity = 4.3163e-02	k_t = [1, 1, 1, 1]			
Generation 1100: Best = 3.673338e+03	FES = 110100	$\mu_F = 0.752$	$\mu_{CR} = 0.122$	
Diversity = 4.6614e-02	k_t = [4, 3, 2, 2]			
Generation 1110: Best = 3.673135e+03	FES = 111100	$\mu_F = 0.746$	$\mu_{CR} = 0.127$	
Diversity = 4.7989e-02	k_t = [1, 1, 1, 1]			
Generation 1120: Best = 3.672866e+03	FES = 112100	$\mu_F = 0.754$	$\mu_{CR} = 0.123$	
Diversity = 4.9828e-02	k_t = [2, 3, 3, 2]			
Generation 1130: Best = 3.672634e+03	FES = 113100	$\mu_F = 0.753$	$\mu_{CR} = 0.127$	
Diversity = 5.0354e-02	k_t = [1, 1, 1, 1]			
Generation 1140: Best = 3.672504e+03	FES = 114100	$\mu_F = 0.739$	$\mu_{CR} = 0.123$	
Diversity = 4.9727e-02	k_t = [4, 2, 3, 4]			
Generation 1150: Best = 3.672137e+03	FES = 115100	$\mu_F = 0.726$	$\mu_{CR} = 0.141$	
Diversity = 4.9140e-02	k_t = [1, 1, 1, 1]			
Generation 1160: Best = 3.671742e+03	FES = 116100	$\mu_F = 0.731$	$\mu_{CR} = 0.145$	
Diversity = 5.2213e-02	k_t = [2, 3, 3, 2]			
Generation 1170: Best = 3.671566e+03	FES = 117100	$\mu_F = 0.722$	$\mu_{CR} = 0.142$	
Diversity = 5.2554e-02	k_t = [1, 1, 1, 1]			
Generation 1180: Best = 3.671411e+03	FES = 118100	$\mu_F = 0.725$	$\mu_{CR} = 0.147$	
Diversity = 5.0794e-02	k_t = [4, 3, 3, 3]			
Generation 1190: Best = 3.670906e+03	FES = 119100	$\mu_F = 0.732$	$\mu_{CR} = 0.163$	
Diversity = 5.1375e-02	k_t = [1, 1, 1, 1]			
Generation 1200: Best = 3.670806e+03	FES = 120100	$\mu_F = 0.737$	$\mu_{CR} = 0.165$	
Diversity = 5.2182e-02	k_t = [3, 4, 3, 2]			
Generation 1210: Best = 3.670220e+03	FES = 121100	$\mu_F = 0.743$	$\mu_{CR} = 0.165$	
Diversity = 5.1506e-02	k_t = [1, 1, 1, 1]			
Generation 1220: Best = 3.669965e+03	FES = 122100	$\mu_F = 0.739$	$\mu_{CR} = 0.171$	
Diversity = 5.3175e-02	k_t = [4, 4, 2, 4]			
Generation 1230: Best = 3.669583e+03	FES = 123100	$\mu_F = 0.746$	$\mu_{CR} = 0.178$	
Diversity = 5.1348e-02	k_t = [1, 1, 1, 1]			
Generation 1240: Best = 3.669149e+03	FES = 124100	$\mu_F = 0.736$	$\mu_{CR} = 0.168$	
Diversity = 4.9972e-02	k_t = [3, 3, 2, 2]			
Generation 1250: Best = 3.669039e+03	FES = 125100	$\mu_F = 0.713$	$\mu_{CR} = 0.155$	
Diversity = 4.8323e-02	k_t = [1, 1, 1, 1]			
Generation 1260: Best = 3.668751e+03	FES = 126100	$\mu_F = 0.724$	$\mu_{CR} = 0.148$	
Diversity = 4.9773e-02	k_t = [2, 4, 2, 2]			
Generation 1270: Best = 3.668521e+03	FES = 127100	$\mu_F = 0.716$	$\mu_{CR} = 0.147$	
Diversity = 5.0955e-02	k_t = [1, 1, 1, 1]			
Generation 1280: Best = 3.668381e+03	FES = 128100	$\mu_F = 0.712$	$\mu_{CR} = 0.151$	
Diversity = 4.8081e-02	k_t = [2, 2, 2, 2]			
Generation 1290: Best = 3.668326e+03	FES = 129100	$\mu_F = 0.706$	$\mu_{CR} = 0.144$	
Diversity = 4.5746e-02	k_t = [1, 1, 1, 1]			
Generation 1300: Best = 3.668067e+03	FES = 130100	$\mu_F = 0.707$	$\mu_{CR} = 0.131$	

Diversity = 4.1714e-02	k_t = [4, 3, 2, 3]			
Generation 1310: Best = 3.667888e+03	FES = 131100	$\mu_F = 0.701$	$\mu_{CR} = 0.128$	
Diversity = 4.1680e-02	k_t = [1, 1, 1, 1]			
Generation 1320: Best = 3.667733e+03	FES = 132100	$\mu_F = 0.707$	$\mu_{CR} = 0.130$	
Diversity = 3.8463e-02	k_t = [4, 2, 3, 4]			
Generation 1330: Best = 3.667619e+03	FES = 133100	$\mu_F = 0.685$	$\mu_{CR} = 0.137$	
Diversity = 3.8820e-02	k_t = [1, 1, 1, 1]			
Generation 1340: Best = 3.667512e+03	FES = 134100	$\mu_F = 0.684$	$\mu_{CR} = 0.130$	
Diversity = 3.6390e-02	k_t = [3, 4, 2, 2]			
Generation 1350: Best = 3.667429e+03	FES = 135100	$\mu_F = 0.685$	$\mu_{CR} = 0.120$	
Diversity = 3.4963e-02	k_t = [1, 1, 1, 1]			
Generation 1360: Best = 3.667398e+03	FES = 136100	$\mu_F = 0.668$	$\mu_{CR} = 0.116$	
Diversity = 3.2461e-02	k_t = [4, 3, 3, 3]			
Generation 1370: Best = 3.667322e+03	FES = 137100	$\mu_F = 0.662$	$\mu_{CR} = 0.120$	
Diversity = 3.1237e-02	k_t = [1, 1, 1, 1]			
Generation 1380: Best = 3.667204e+03	FES = 138100	$\mu_F = 0.669$	$\mu_{CR} = 0.134$	
Diversity = 3.0293e-02	k_t = [4, 4, 3, 2]			
Generation 1390: Best = 3.667182e+03	FES = 139100	$\mu_F = 0.657$	$\mu_{CR} = 0.131$	
Diversity = 2.9662e-02	k_t = [1, 1, 1, 1]			
Generation 1400: Best = 3.667159e+03	FES = 140100	$\mu_F = 0.663$	$\mu_{CR} = 0.122$	
Diversity = 2.8074e-02	k_t = [3, 3, 4, 2]			
Generation 1410: Best = 3.667117e+03	FES = 141100	$\mu_F = 0.670$	$\mu_{CR} = 0.116$	
Diversity = 2.6551e-02	k_t = [1, 1, 1, 1]			
Generation 1420: Best = 3.667057e+03	FES = 142100	$\mu_F = 0.680$	$\mu_{CR} = 0.125$	
Diversity = 2.4067e-02	k_t = [3, 3, 4, 3]			
Generation 1430: Best = 3.667039e+03	FES = 143100	$\mu_F = 0.686$	$\mu_{CR} = 0.123$	
Diversity = 2.2666e-02	k_t = [1, 1, 1, 1]			
Generation 1440: Best = 3.667010e+03	FES = 144100	$\mu_F = 0.674$	$\mu_{CR} = 0.123$	
Diversity = 2.1328e-02	k_t = [2, 3, 4, 3]			
Generation 1450: Best = 3.666959e+03	FES = 145100	$\mu_F = 0.702$	$\mu_{CR} = 0.120$	
Diversity = 2.1986e-02	k_t = [1, 1, 1, 1]			
Generation 1460: Best = 3.666921e+03	FES = 146100	$\mu_F = 0.689$	$\mu_{CR} = 0.113$	
Diversity = 2.0183e-02	k_t = [4, 4, 4, 2]			
Generation 1470: Best = 3.666871e+03	FES = 147100	$\mu_F = 0.697$	$\mu_{CR} = 0.105$	
Diversity = 2.1291e-02	k_t = [1, 1, 1, 1]			
Generation 1480: Best = 3.666773e+03	FES = 148100	$\mu_F = 0.706$	$\mu_{CR} = 0.111$	
Diversity = 2.0728e-02	k_t = [2, 3, 2, 2]			
Generation 1490: Best = 3.666756e+03	FES = 149100	$\mu_F = 0.708$	$\mu_{CR} = 0.111$	
Diversity = 2.0786e-02	k_t = [1, 1, 1, 1]			
Generation 1500: Best = 3.666733e+03	FES = 150100	$\mu_F = 0.719$	$\mu_{CR} = 0.124$	
Diversity = 2.0781e-02	k_t = [3, 3, 4, 3]			
Generation 1510: Best = 3.666660e+03	FES = 151100	$\mu_F = 0.720$	$\mu_{CR} = 0.123$	
Diversity = 2.1024e-02	k_t = [1, 1, 1, 1]			
Generation 1520: Best = 3.666535e+03	FES = 152100	$\mu_F = 0.727$	$\mu_{CR} = 0.124$	
Diversity = 2.0210e-02	k_t = [4, 4, 4, 4]			
Generation 1530: Best = 3.666450e+03	FES = 153100	$\mu_F = 0.737$	$\mu_{CR} = 0.132$	
Diversity = 2.0757e-02	k_t = [1, 1, 1, 1]			
Generation 1540: Best = 3.666376e+03	FES = 154100	$\mu_F = 0.739$	$\mu_{CR} = 0.136$	
Diversity = 2.1266e-02	k_t = [4, 3, 3, 4]			
Generation 1550: Best = 3.666310e+03	FES = 155100	$\mu_F = 0.746$	$\mu_{CR} = 0.129$	
Diversity = 2.0913e-02	k_t = [1, 1, 1, 1]			
Generation 1560: Best = 3.666273e+03	FES = 156100	$\mu_F = 0.738$	$\mu_{CR} = 0.132$	
Diversity = 2.0070e-02	k_t = [3, 2, 4, 3]			
Generation 1570: Best = 3.666179e+03	FES = 157100	$\mu_F = 0.736$	$\mu_{CR} = 0.137$	

Diversity = 2.0276e-02	k_t = [1, 1, 1, 1]			
Generation 1580: Best = 3.666153e+03	FES = 158100	$\mu_F = 0.741$	$\mu_{CR} = 0.136$	
Diversity = 2.0145e-02	k_t = [3, 2, 4, 2]			
Generation 1590: Best = 3.666087e+03	FES = 159100	$\mu_F = 0.744$	$\mu_{CR} = 0.145$	
Diversity = 2.0786e-02	k_t = [1, 1, 1, 1]			
Generation 1600: Best = 3.666025e+03	FES = 160100	$\mu_F = 0.756$	$\mu_{CR} = 0.131$	
Diversity = 1.9553e-02	k_t = [2, 3, 3, 4]			
Generation 1610: Best = 3.665986e+03	FES = 161100	$\mu_F = 0.753$	$\mu_{CR} = 0.125$	
Diversity = 1.9325e-02	k_t = [1, 1, 1, 1]			
Generation 1620: Best = 3.665923e+03	FES = 162100	$\mu_F = 0.742$	$\mu_{CR} = 0.120$	
Diversity = 1.7473e-02	k_t = [2, 3, 2, 4]			
Generation 1630: Best = 3.665908e+03	FES = 163100	$\mu_F = 0.736$	$\mu_{CR} = 0.115$	
Diversity = 1.7453e-02	k_t = [1, 1, 1, 1]			
Generation 1640: Best = 3.665871e+03	FES = 164100	$\mu_F = 0.729$	$\mu_{CR} = 0.114$	
Diversity = 1.7216e-02	k_t = [3, 4, 4, 4]			
Generation 1650: Best = 3.665822e+03	FES = 165100	$\mu_F = 0.727$	$\mu_{CR} = 0.122$	
Diversity = 1.6067e-02	k_t = [1, 1, 1, 1]			
Generation 1660: Best = 3.665790e+03	FES = 166100	$\mu_F = 0.733$	$\mu_{CR} = 0.123$	
Diversity = 1.6000e-02	k_t = [2, 2, 3, 4]			
Generation 1670: Best = 3.665774e+03	FES = 167100	$\mu_F = 0.748$	$\mu_{CR} = 0.118$	
Diversity = 1.5024e-02	k_t = [1, 1, 1, 1]			
Generation 1680: Best = 3.665730e+03	FES = 168100	$\mu_F = 0.767$	$\mu_{CR} = 0.110$	
Diversity = 1.4657e-02	k_t = [4, 2, 3, 4]			
Generation 1690: Best = 3.665687e+03	FES = 169100	$\mu_F = 0.775$	$\mu_{CR} = 0.106$	
Diversity = 1.5339e-02	k_t = [1, 1, 1, 1]			
Generation 1700: Best = 3.665666e+03	FES = 170100	$\mu_F = 0.791$	$\mu_{CR} = 0.105$	
Diversity = 1.6133e-02	k_t = [2, 4, 3, 4]			
Generation 1710: Best = 3.665627e+03	FES = 171100	$\mu_F = 0.781$	$\mu_{CR} = 0.091$	
Diversity = 1.5468e-02	k_t = [1, 1, 1, 1]			
Generation 1720: Best = 3.665604e+03	FES = 172100	$\mu_F = 0.773$	$\mu_{CR} = 0.096$	
Diversity = 1.6474e-02	k_t = [2, 4, 4, 2]			
Generation 1730: Best = 3.665565e+03	FES = 173100	$\mu_F = 0.775$	$\mu_{CR} = 0.101$	
Diversity = 1.6988e-02	k_t = [1, 1, 1, 1]			
Generation 1740: Best = 3.665537e+03	FES = 174100	$\mu_F = 0.779$	$\mu_{CR} = 0.102$	
Diversity = 1.7342e-02	k_t = [3, 4, 4, 4]			
Generation 1750: Best = 3.665490e+03	FES = 175100	$\mu_F = 0.756$	$\mu_{CR} = 0.103$	
Diversity = 1.6690e-02	k_t = [1, 1, 1, 1]			
Generation 1760: Best = 3.665450e+03	FES = 176100	$\mu_F = 0.756$	$\mu_{CR} = 0.098$	
Diversity = 1.6132e-02	k_t = [4, 3, 4, 4]			
Generation 1770: Best = 3.665388e+03	FES = 177100	$\mu_F = 0.763$	$\mu_{CR} = 0.099$	
Diversity = 1.6311e-02	k_t = [1, 1, 1, 1]			
Generation 1780: Best = 3.665375e+03	FES = 178100	$\mu_F = 0.757$	$\mu_{CR} = 0.112$	
Diversity = 1.7118e-02	k_t = [4, 2, 4, 3]			
Generation 1790: Best = 3.665358e+03	FES = 179100	$\mu_F = 0.744$	$\mu_{CR} = 0.108$	
Diversity = 1.6981e-02	k_t = [1, 1, 1, 1]			
Generation 1800: Best = 3.665328e+03	FES = 180100	$\mu_F = 0.750$	$\mu_{CR} = 0.104$	
Diversity = 1.6028e-02	k_t = [3, 2, 2, 4]			
Generation 1810: Best = 3.665324e+03	FES = 181100	$\mu_F = 0.757$	$\mu_{CR} = 0.103$	
Diversity = 1.6268e-02	k_t = [1, 1, 1, 1]			
Generation 1820: Best = 3.665252e+03	FES = 182100	$\mu_F = 0.762$	$\mu_{CR} = 0.099$	
Diversity = 1.6296e-02	k_t = [3, 4, 3, 2]			
Generation 1830: Best = 3.665231e+03	FES = 183100	$\mu_F = 0.764$	$\mu_{CR} = 0.102$	
Diversity = 1.8579e-02	k_t = [1, 1, 1, 1]			
Generation 1840: Best = 3.665183e+03	FES = 184100	$\mu_F = 0.744$	$\mu_{CR} = 0.106$	

Diversity = 1.9667e-02	k_t = [2, 3, 4, 2]			
Generation 1850: Best = 3.665138e+03	FES = 185100	$\mu_F = 0.733$	$\mu_{CR} = 0.112$	
Diversity = 2.0640e-02	k_t = [1, 1, 1, 1]			
Generation 1860: Best = 3.665113e+03	FES = 186100	$\mu_F = 0.738$	$\mu_{CR} = 0.103$	
Diversity = 2.0271e-02	k_t = [3, 4, 2, 4]			
Generation 1870: Best = 3.665015e+03	FES = 187100	$\mu_F = 0.737$	$\mu_{CR} = 0.110$	
Diversity = 1.9299e-02	k_t = [1, 1, 1, 1]			
Generation 1880: Best = 3.665009e+03	FES = 188100	$\mu_F = 0.725$	$\mu_{CR} = 0.124$	
Diversity = 1.9629e-02	k_t = [3, 3, 3, 2]			
Generation 1890: Best = 3.664898e+03	FES = 189100	$\mu_F = 0.732$	$\mu_{CR} = 0.126$	
Diversity = 2.0764e-02	k_t = [1, 1, 1, 1]			
Generation 1900: Best = 3.664793e+03	FES = 190100	$\mu_F = 0.741$	$\mu_{CR} = 0.114$	
Diversity = 1.9630e-02	k_t = [2, 4, 3, 4]			
Generation 1910: Best = 3.664725e+03	FES = 191100	$\mu_F = 0.728$	$\mu_{CR} = 0.123$	
Diversity = 1.9039e-02	k_t = [1, 1, 1, 1]			
Generation 1920: Best = 3.664715e+03	FES = 192100	$\mu_F = 0.729$	$\mu_{CR} = 0.126$	
Diversity = 1.9908e-02	k_t = [3, 3, 4, 2]			
Generation 1930: Best = 3.664675e+03	FES = 193100	$\mu_F = 0.717$	$\mu_{CR} = 0.119$	
Diversity = 2.0083e-02	k_t = [1, 1, 1, 1]			
Generation 1940: Best = 3.664635e+03	FES = 194100	$\mu_F = 0.726$	$\mu_{CR} = 0.120$	
Diversity = 1.7963e-02	k_t = [3, 2, 3, 3]			
Generation 1950: Best = 3.664540e+03	FES = 195100	$\mu_F = 0.737$	$\mu_{CR} = 0.118$	
Diversity = 1.9854e-02	k_t = [1, 1, 1, 1]			
Generation 1960: Best = 3.664509e+03	FES = 196100	$\mu_F = 0.743$	$\mu_{CR} = 0.107$	
Diversity = 2.0335e-02	k_t = [3, 2, 4, 4]			
Generation 1970: Best = 3.664489e+03	FES = 197100	$\mu_F = 0.753$	$\mu_{CR} = 0.110$	
Diversity = 2.0368e-02	k_t = [1, 1, 1, 1]			
Generation 1980: Best = 3.664449e+03	FES = 198100	$\mu_F = 0.744$	$\mu_{CR} = 0.107$	
Diversity = 2.0347e-02	k_t = [4, 4, 3, 2]			
Generation 1990: Best = 3.664406e+03	FES = 199100	$\mu_F = 0.740$	$\mu_{CR} = 0.111$	
Diversity = 2.0211e-02	k_t = [1, 1, 1, 1]			
Generation 2000: Best = 3.664355e+03	FES = 200100	$\mu_F = 0.743$	$\mu_{CR} = 0.108$	
Diversity = 1.9406e-02	k_t = [4, 4, 3, 4]			
Generation 2010: Best = 3.664240e+03	FES = 201100	$\mu_F = 0.743$	$\mu_{CR} = 0.114$	
Diversity = 1.8999e-02	k_t = [1, 1, 1, 1]			
Generation 2020: Best = 3.664240e+03	FES = 202100	$\mu_F = 0.737$	$\mu_{CR} = 0.116$	
Diversity = 1.8818e-02	k_t = [3, 4, 2, 3]			
Generation 2030: Best = 3.664202e+03	FES = 203100	$\mu_F = 0.757$	$\mu_{CR} = 0.113$	
Diversity = 1.8795e-02	k_t = [1, 1, 1, 1]			
Generation 2040: Best = 3.664129e+03	FES = 204100	$\mu_F = 0.768$	$\mu_{CR} = 0.119$	
Diversity = 1.8228e-02	k_t = [2, 4, 2, 4]			
Generation 2050: Best = 3.664082e+03	FES = 205100	$\mu_F = 0.766$	$\mu_{CR} = 0.122$	
Diversity = 1.8128e-02	k_t = [1, 1, 1, 1]			
Generation 2060: Best = 3.664040e+03	FES = 206100	$\mu_F = 0.748$	$\mu_{CR} = 0.120$	
Diversity = 1.8534e-02	k_t = [3, 2, 4, 3]			
Generation 2070: Best = 3.664010e+03	FES = 207100	$\mu_F = 0.752$	$\mu_{CR} = 0.111$	
Diversity = 1.8720e-02	k_t = [1, 1, 1, 1]			
Generation 2080: Best = 3.663980e+03	FES = 208100	$\mu_F = 0.739$	$\mu_{CR} = 0.111$	
Diversity = 1.8552e-02	k_t = [4, 2, 3, 3]			
Generation 2090: Best = 3.663933e+03	FES = 209100	$\mu_F = 0.741$	$\mu_{CR} = 0.101$	
Diversity = 1.7516e-02	k_t = [1, 1, 1, 1]			
Generation 2100: Best = 3.663909e+03	FES = 210100	$\mu_F = 0.734$	$\mu_{CR} = 0.108$	
Diversity = 1.6431e-02	k_t = [4, 4, 4, 4]			
Generation 2110: Best = 3.663854e+03	FES = 211100	$\mu_F = 0.738$	$\mu_{CR} = 0.106$	

Diversity = 1.5605e-02	k_t = [1, 1, 1, 1]			
Generation 2120: Best = 3.663794e+03	FES = 212100	$\mu_F = 0.734$	$\mu_{CR} = 0.111$	
Diversity = 1.5351e-02	k_t = [4, 2, 3, 4]			
Generation 2130: Best = 3.663728e+03	FES = 213100	$\mu_F = 0.734$	$\mu_{CR} = 0.112$	
Diversity = 1.6696e-02	k_t = [1, 1, 1, 1]			
Generation 2140: Best = 3.663635e+03	FES = 214100	$\mu_F = 0.749$	$\mu_{CR} = 0.126$	
Diversity = 1.6351e-02	k_t = [2, 3, 3, 3]			
Generation 2150: Best = 3.663511e+03	FES = 215100	$\mu_F = 0.756$	$\mu_{CR} = 0.131$	
Diversity = 1.8435e-02	k_t = [1, 1, 1, 1]			
Generation 2160: Best = 3.663414e+03	FES = 216100	$\mu_F = 0.776$	$\mu_{CR} = 0.132$	
Diversity = 1.9340e-02	k_t = [4, 4, 4, 2]			
Generation 2170: Best = 3.663362e+03	FES = 217100	$\mu_F = 0.778$	$\mu_{CR} = 0.124$	
Diversity = 1.9962e-02	k_t = [1, 1, 1, 1]			
Generation 2180: Best = 3.663325e+03	FES = 218100	$\mu_F = 0.765$	$\mu_{CR} = 0.131$	
Diversity = 1.7329e-02	k_t = [4, 3, 3, 4]			
Generation 2190: Best = 3.663253e+03	FES = 219100	$\mu_F = 0.751$	$\mu_{CR} = 0.124$	
Diversity = 1.5737e-02	k_t = [1, 1, 1, 1]			
Generation 2200: Best = 3.663214e+03	FES = 220100	$\mu_F = 0.755$	$\mu_{CR} = 0.126$	
Diversity = 1.6226e-02	k_t = [3, 3, 2, 3]			
Generation 2210: Best = 3.663175e+03	FES = 221100	$\mu_F = 0.734$	$\mu_{CR} = 0.132$	
Diversity = 1.7537e-02	k_t = [1, 1, 1, 1]			
Generation 2220: Best = 3.663133e+03	FES = 222100	$\mu_F = 0.738$	$\mu_{CR} = 0.126$	
Diversity = 1.6871e-02	k_t = [2, 4, 3, 3]			
Generation 2230: Best = 3.663090e+03	FES = 223100	$\mu_F = 0.747$	$\mu_{CR} = 0.121$	
Diversity = 1.7324e-02	k_t = [1, 1, 1, 1]			
Generation 2240: Best = 3.663005e+03	FES = 224100	$\mu_F = 0.745$	$\mu_{CR} = 0.122$	
Diversity = 1.7634e-02	k_t = [4, 2, 2, 3]			
Generation 2250: Best = 3.662964e+03	FES = 225100	$\mu_F = 0.723$	$\mu_{CR} = 0.133$	
Diversity = 1.8026e-02	k_t = [1, 1, 1, 1]			
Generation 2260: Best = 3.662941e+03	FES = 226100	$\mu_F = 0.727$	$\mu_{CR} = 0.125$	
Diversity = 1.7426e-02	k_t = [3, 2, 2, 4]			
Generation 2270: Best = 3.662905e+03	FES = 227100	$\mu_F = 0.725$	$\mu_{CR} = 0.116$	
Diversity = 1.7195e-02	k_t = [1, 1, 1, 1]			
Generation 2280: Best = 3.662873e+03	FES = 228100	$\mu_F = 0.722$	$\mu_{CR} = 0.113$	
Diversity = 1.7145e-02	k_t = [2, 4, 3, 4]			
Generation 2290: Best = 3.662772e+03	FES = 229100	$\mu_F = 0.743$	$\mu_{CR} = 0.118$	
Diversity = 1.7214e-02	k_t = [1, 1, 1, 1]			
Generation 2300: Best = 3.662702e+03	FES = 230100	$\mu_F = 0.754$	$\mu_{CR} = 0.118$	
Diversity = 1.9409e-02	k_t = [3, 2, 3, 2]			
Generation 2310: Best = 3.662583e+03	FES = 231100	$\mu_F = 0.761$	$\mu_{CR} = 0.121$	
Diversity = 2.4079e-02	k_t = [1, 1, 1, 1]			
Generation 2320: Best = 3.662394e+03	FES = 232100	$\mu_F = 0.761$	$\mu_{CR} = 0.123$	
Diversity = 2.6552e-02	k_t = [3, 4, 4, 2]			
Generation 2330: Best = 3.662334e+03	FES = 233100	$\mu_F = 0.757$	$\mu_{CR} = 0.123$	
Diversity = 2.7900e-02	k_t = [1, 1, 1, 1]			
Generation 2340: Best = 3.662258e+03	FES = 234100	$\mu_F = 0.739$	$\mu_{CR} = 0.126$	
Diversity = 2.7985e-02	k_t = [4, 3, 3, 2]			
Generation 2350: Best = 3.662148e+03	FES = 235100	$\mu_F = 0.743$	$\mu_{CR} = 0.134$	
Diversity = 2.8587e-02	k_t = [1, 1, 1, 1]			
Generation 2360: Best = 3.662069e+03	FES = 236100	$\mu_F = 0.739$	$\mu_{CR} = 0.120$	
Diversity = 2.6487e-02	k_t = [2, 3, 4, 4]			
Generation 2370: Best = 3.662069e+03	FES = 237100	$\mu_F = 0.714$	$\mu_{CR} = 0.121$	
Diversity = 2.3699e-02	k_t = [1, 1, 1, 1]			
Generation 2380: Best = 3.662061e+03	FES = 238100	$\mu_F = 0.687$	$\mu_{CR} = 0.122$	

Diversity = 2.1311e-02	k_t = [2, 4, 2, 3]			
Generation 2390: Best = 3.662045e+03	FES = 239100	$\mu_F = 0.694$	$\mu_{CR} = 0.109$	
Diversity = 1.9496e-02	k_t = [1, 1, 1, 1]			
Generation 2400: Best = 3.662020e+03	FES = 240100	$\mu_F = 0.710$	$\mu_{CR} = 0.114$	
Diversity = 1.8240e-02	k_t = [2, 3, 2, 4]			
Generation 2410: Best = 3.661995e+03	FES = 241100	$\mu_F = 0.703$	$\mu_{CR} = 0.107$	
Diversity = 1.6480e-02	k_t = [1, 1, 1, 1]			
Generation 2420: Best = 3.661962e+03	FES = 242100	$\mu_F = 0.701$	$\mu_{CR} = 0.107$	
Diversity = 1.5864e-02	k_t = [3, 4, 4, 3]			
Generation 2430: Best = 3.661956e+03	FES = 243100	$\mu_F = 0.688$	$\mu_{CR} = 0.110$	
Diversity = 1.5706e-02	k_t = [1, 1, 1, 1]			
Generation 2440: Best = 3.661926e+03	FES = 244100	$\mu_F = 0.671$	$\mu_{CR} = 0.109$	
Diversity = 1.4673e-02	k_t = [4, 2, 2, 3]			
Generation 2450: Best = 3.661924e+03	FES = 245100	$\mu_F = 0.682$	$\mu_{CR} = 0.116$	
Diversity = 1.3778e-02	k_t = [1, 1, 1, 1]			
Generation 2460: Best = 3.661915e+03	FES = 246100	$\mu_F = 0.651$	$\mu_{CR} = 0.112$	
Diversity = 1.2205e-02	k_t = [2, 3, 3, 3]			
Generation 2470: Best = 3.661898e+03	FES = 247100	$\mu_F = 0.674$	$\mu_{CR} = 0.099$	
Diversity = 1.2245e-02	k_t = [1, 1, 1, 1]			
Generation 2480: Best = 3.661887e+03	FES = 248100	$\mu_F = 0.666$	$\mu_{CR} = 0.094$	
Diversity = 1.1730e-02	k_t = [4, 3, 2, 2]			
Generation 2490: Best = 3.661873e+03	FES = 249100	$\mu_F = 0.662$	$\mu_{CR} = 0.093$	
Diversity = 1.0635e-02	k_t = [1, 1, 1, 1]			
Generation 2500: Best = 3.661863e+03	FES = 250100	$\mu_F = 0.683$	$\mu_{CR} = 0.103$	
Diversity = 9.8708e-03	k_t = [4, 2, 4, 3]			
Generation 2510: Best = 3.661859e+03	FES = 251100	$\mu_F = 0.684$	$\mu_{CR} = 0.107$	
Diversity = 9.5315e-03	k_t = [1, 1, 1, 1]			
Generation 2520: Best = 3.661854e+03	FES = 252100	$\mu_F = 0.691$	$\mu_{CR} = 0.105$	
Diversity = 8.9604e-03	k_t = [3, 3, 3, 2]			
Generation 2530: Best = 3.661843e+03	FES = 253100	$\mu_F = 0.700$	$\mu_{CR} = 0.098$	
Diversity = 8.6675e-03	k_t = [1, 1, 1, 1]			
Generation 2540: Best = 3.661835e+03	FES = 254100	$\mu_F = 0.708$	$\mu_{CR} = 0.100$	
Diversity = 7.8179e-03	k_t = [2, 3, 4, 4]			
Generation 2550: Best = 3.661827e+03	FES = 255100	$\mu_F = 0.741$	$\mu_{CR} = 0.091$	
Diversity = 7.6051e-03	k_t = [1, 1, 1, 1]			
Generation 2560: Best = 3.661822e+03	FES = 256100	$\mu_F = 0.737$	$\mu_{CR} = 0.087$	
Diversity = 7.5696e-03	k_t = [2, 3, 3, 4]			
Generation 2570: Best = 3.661817e+03	FES = 257100	$\mu_F = 0.700$	$\mu_{CR} = 0.094$	
Diversity = 7.3842e-03	k_t = [1, 1, 1, 1]			
Generation 2580: Best = 3.661816e+03	FES = 258100	$\mu_F = 0.684$	$\mu_{CR} = 0.092$	
Diversity = 6.8996e-03	k_t = [3, 3, 2, 2]			
Generation 2590: Best = 3.661811e+03	FES = 259100	$\mu_F = 0.684$	$\mu_{CR} = 0.093$	
Diversity = 7.1546e-03	k_t = [1, 1, 1, 1]			
Generation 2600: Best = 3.661800e+03	FES = 260100	$\mu_F = 0.688$	$\mu_{CR} = 0.083$	
Diversity = 6.8001e-03	k_t = [3, 2, 3, 2]			
Generation 2610: Best = 3.661800e+03	FES = 261100	$\mu_F = 0.691$	$\mu_{CR} = 0.080$	
Diversity = 6.9824e-03	k_t = [1, 1, 1, 1]			
Generation 2620: Best = 3.661796e+03	FES = 262100	$\mu_F = 0.700$	$\mu_{CR} = 0.080$	
Diversity = 6.9236e-03	k_t = [3, 3, 4, 4]			
Generation 2630: Best = 3.661796e+03	FES = 263100	$\mu_F = 0.696$	$\mu_{CR} = 0.083$	
Diversity = 6.8104e-03	k_t = [1, 1, 1, 1]			
Generation 2640: Best = 3.661794e+03	FES = 264100	$\mu_F = 0.692$	$\mu_{CR} = 0.088$	
Diversity = 6.5792e-03	k_t = [2, 3, 3, 3]			
Generation 2650: Best = 3.661789e+03	FES = 265100	$\mu_F = 0.698$	$\mu_{CR} = 0.084$	

Diversity = 6.2943e-03	k_t = [1, 1, 1, 1]			
Generation 2660: Best = 3.661784e+03	FES = 266100	$\mu_F = 0.682$	$\mu_{CR} = 0.086$	
Diversity = 6.3628e-03	k_t = [4, 3, 2, 2]			
Generation 2670: Best = 3.661776e+03	FES = 267100	$\mu_F = 0.675$	$\mu_{CR} = 0.086$	
Diversity = 6.2259e-03	k_t = [1, 1, 1, 1]			
Generation 2680: Best = 3.661774e+03	FES = 268100	$\mu_F = 0.691$	$\mu_{CR} = 0.083$	
Diversity = 6.6334e-03	k_t = [3, 3, 3, 2]			
Generation 2690: Best = 3.661772e+03	FES = 269100	$\mu_F = 0.677$	$\mu_{CR} = 0.080$	
Diversity = 6.6789e-03	k_t = [1, 1, 1, 1]			
Generation 2700: Best = 3.661766e+03	FES = 270100	$\mu_F = 0.673$	$\mu_{CR} = 0.085$	
Diversity = 6.7724e-03	k_t = [3, 3, 3, 2]			
Generation 2710: Best = 3.661762e+03	FES = 271100	$\mu_F = 0.674$	$\mu_{CR} = 0.076$	
Diversity = 6.6926e-03	k_t = [1, 1, 1, 1]			
Generation 2720: Best = 3.661758e+03	FES = 272100	$\mu_F = 0.664$	$\mu_{CR} = 0.082$	
Diversity = 6.0484e-03	k_t = [4, 3, 4, 3]			
Generation 2730: Best = 3.661754e+03	FES = 273100	$\mu_F = 0.672$	$\mu_{CR} = 0.091$	
Diversity = 6.1059e-03	k_t = [1, 1, 1, 1]			
Generation 2740: Best = 3.661751e+03	FES = 274100	$\mu_F = 0.672$	$\mu_{CR} = 0.085$	
Diversity = 6.3399e-03	k_t = [4, 2, 3, 2]			
Generation 2750: Best = 3.661747e+03	FES = 275100	$\mu_F = 0.663$	$\mu_{CR} = 0.090$	
Diversity = 6.1318e-03	k_t = [1, 1, 1, 1]			
Generation 2760: Best = 3.661746e+03	FES = 276100	$\mu_F = 0.680$	$\mu_{CR} = 0.096$	
Diversity = 5.6564e-03	k_t = [4, 3, 4, 4]			
Generation 2770: Best = 3.661743e+03	FES = 277100	$\mu_F = 0.682$	$\mu_{CR} = 0.090$	
Diversity = 5.2996e-03	k_t = [1, 1, 1, 1]			
Generation 2780: Best = 3.661737e+03	FES = 278100	$\mu_F = 0.697$	$\mu_{CR} = 0.086$	
Diversity = 4.9059e-03	k_t = [3, 2, 2, 3]			
Generation 2790: Best = 3.661730e+03	FES = 279100	$\mu_F = 0.690$	$\mu_{CR} = 0.080$	
Diversity = 4.8718e-03	k_t = [1, 1, 1, 1]			
Generation 2800: Best = 3.661722e+03	FES = 280100	$\mu_F = 0.673$	$\mu_{CR} = 0.083$	
Diversity = 4.7071e-03	k_t = [2, 3, 2, 3]			
Generation 2810: Best = 3.661719e+03	FES = 281100	$\mu_F = 0.687$	$\mu_{CR} = 0.082$	
Diversity = 4.8574e-03	k_t = [1, 1, 1, 1]			
Generation 2820: Best = 3.661717e+03	FES = 282100	$\mu_F = 0.692$	$\mu_{CR} = 0.087$	
Diversity = 4.5867e-03	k_t = [2, 2, 4, 4]			
Generation 2830: Best = 3.661717e+03	FES = 283100	$\mu_F = 0.672$	$\mu_{CR} = 0.086$	
Diversity = 4.4781e-03	k_t = [1, 1, 1, 1]			
Generation 2840: Best = 3.661713e+03	FES = 284100	$\mu_F = 0.680$	$\mu_{CR} = 0.082$	
Diversity = 4.5052e-03	k_t = [4, 3, 4, 2]			
Generation 2850: Best = 3.661710e+03	FES = 285100	$\mu_F = 0.700$	$\mu_{CR} = 0.086$	
Diversity = 4.6033e-03	k_t = [1, 1, 1, 1]			
Generation 2860: Best = 3.661707e+03	FES = 286100	$\mu_F = 0.702$	$\mu_{CR} = 0.088$	
Diversity = 4.3923e-03	k_t = [3, 4, 3, 4]			
Generation 2870: Best = 3.661704e+03	FES = 287100	$\mu_F = 0.696$	$\mu_{CR} = 0.075$	
Diversity = 4.3557e-03	k_t = [1, 1, 1, 1]			
Generation 2880: Best = 3.661702e+03	FES = 288100	$\mu_F = 0.684$	$\mu_{CR} = 0.077$	
Diversity = 4.2579e-03	k_t = [2, 2, 4, 2]			
Generation 2890: Best = 3.661701e+03	FES = 289100	$\mu_F = 0.677$	$\mu_{CR} = 0.078$	
Diversity = 4.1446e-03	k_t = [1, 1, 1, 1]			
Generation 2900: Best = 3.661696e+03	FES = 290100	$\mu_F = 0.704$	$\mu_{CR} = 0.083$	
Diversity = 4.1167e-03	k_t = [4, 2, 2, 2]			
Generation 2910: Best = 3.661691e+03	FES = 291100	$\mu_F = 0.706$	$\mu_{CR} = 0.075$	
Diversity = 4.2539e-03	k_t = [1, 1, 1, 1]			
Generation 2920: Best = 3.661691e+03	FES = 292100	$\mu_F = 0.701$	$\mu_{CR} = 0.078$	

Diversity = 4.3290e-03	k_t = [3, 2, 2, 2]			
Generation 2930: Best = 3.661687e+03	FES = 293100	$\mu_F = 0.673$	$\mu_{CR} = 0.089$	
Diversity = 4.2748e-03	k_t = [1, 1, 1, 1]			
Generation 2940: Best = 3.661685e+03	FES = 294100	$\mu_F = 0.689$	$\mu_{CR} = 0.090$	
Diversity = 4.3424e-03	k_t = [3, 4, 4, 3]			
Generation 2950: Best = 3.661683e+03	FES = 295100	$\mu_F = 0.690$	$\mu_{CR} = 0.089$	
Diversity = 4.2087e-03	k_t = [1, 1, 1, 1]			
Generation 2960: Best = 3.661681e+03	FES = 296100	$\mu_F = 0.679$	$\mu_{CR} = 0.089$	
Diversity = 4.3958e-03	k_t = [2, 4, 4, 2]			
Generation 2970: Best = 3.661677e+03	FES = 297100	$\mu_F = 0.681$	$\mu_{CR} = 0.084$	
Diversity = 4.6201e-03	k_t = [1, 1, 1, 1]			
Generation 2980: Best = 3.661674e+03	FES = 298100	$\mu_F = 0.687$	$\mu_{CR} = 0.090$	
Diversity = 4.2295e-03	k_t = [3, 3, 4, 2]			
Generation 2990: Best = 3.661667e+03	FES = 299100	$\mu_F = 0.701$	$\mu_{CR} = 0.098$	
Diversity = 4.3102e-03	k_t = [1, 1, 1, 1]			
Generation 3000: Best = 3.661665e+03	FES = 300100	$\mu_F = 0.697$	$\mu_{CR} = 0.095$	
Diversity = 4.1165e-03	k_t = [4, 4, 2, 2]			
Generation 3010: Best = 3.661662e+03	FES = 301100	$\mu_F = 0.708$	$\mu_{CR} = 0.088$	
Diversity = 4.3250e-03	k_t = [1, 1, 1, 1]			
Generation 3020: Best = 3.661661e+03	FES = 302100	$\mu_F = 0.700$	$\mu_{CR} = 0.094$	
Diversity = 4.2244e-03	k_t = [3, 4, 2, 4]			
Generation 3030: Best = 3.661656e+03	FES = 303100	$\mu_F = 0.697$	$\mu_{CR} = 0.088$	
Diversity = 3.8859e-03	k_t = [1, 1, 1, 1]			
Generation 3040: Best = 3.661654e+03	FES = 304100	$\mu_F = 0.665$	$\mu_{CR} = 0.101$	
Diversity = 3.5921e-03	k_t = [3, 2, 3, 2]			
Generation 3050: Best = 3.661649e+03	FES = 305100	$\mu_F = 0.665$	$\mu_{CR} = 0.090$	
Diversity = 3.6404e-03	k_t = [1, 1, 1, 1]			
Generation 3060: Best = 3.661648e+03	FES = 306100	$\mu_F = 0.671$	$\mu_{CR} = 0.090$	
Diversity = 3.2407e-03	k_t = [4, 4, 4, 3]			
Generation 3070: Best = 3.661646e+03	FES = 307100	$\mu_F = 0.694$	$\mu_{CR} = 0.084$	
Diversity = 3.2756e-03	k_t = [1, 1, 1, 1]			
Generation 3080: Best = 3.661645e+03	FES = 308100	$\mu_F = 0.690$	$\mu_{CR} = 0.088$	
Diversity = 3.3677e-03	k_t = [3, 4, 2, 3]			
Generation 3090: Best = 3.661641e+03	FES = 309100	$\mu_F = 0.681$	$\mu_{CR} = 0.092$	
Diversity = 3.3104e-03	k_t = [1, 1, 1, 1]			
Generation 3100: Best = 3.661640e+03	FES = 310100	$\mu_F = 0.706$	$\mu_{CR} = 0.089$	
Diversity = 3.0606e-03	k_t = [2, 2, 2, 4]			
Generation 3110: Best = 3.661635e+03	FES = 311100	$\mu_F = 0.712$	$\mu_{CR} = 0.095$	
Diversity = 2.9737e-03	k_t = [1, 1, 1, 1]			
Generation 3120: Best = 3.661635e+03	FES = 312100	$\mu_F = 0.721$	$\mu_{CR} = 0.090$	
Diversity = 2.8589e-03	k_t = [3, 4, 4, 3]			
Generation 3130: Best = 3.661631e+03	FES = 313100	$\mu_F = 0.732$	$\mu_{CR} = 0.090$	
Diversity = 2.9765e-03	k_t = [1, 1, 1, 1]			
Generation 3140: Best = 3.661629e+03	FES = 314100	$\mu_F = 0.734$	$\mu_{CR} = 0.082$	
Diversity = 2.9949e-03	k_t = [3, 2, 2, 2]			
Generation 3150: Best = 3.661628e+03	FES = 315100	$\mu_F = 0.723$	$\mu_{CR} = 0.089$	
Diversity = 3.1028e-03	k_t = [1, 1, 1, 1]			
Generation 3160: Best = 3.661627e+03	FES = 316100	$\mu_F = 0.728$	$\mu_{CR} = 0.095$	
Diversity = 2.8979e-03	k_t = [3, 4, 2, 3]			
Generation 3170: Best = 3.661623e+03	FES = 317100	$\mu_F = 0.720$	$\mu_{CR} = 0.089$	
Diversity = 2.8205e-03	k_t = [1, 1, 1, 1]			
Generation 3180: Best = 3.661623e+03	FES = 318100	$\mu_F = 0.703$	$\mu_{CR} = 0.088$	
Diversity = 2.7849e-03	k_t = [2, 4, 3, 3]			
Generation 3190: Best = 3.661621e+03	FES = 319100	$\mu_F = 0.700$	$\mu_{CR} = 0.086$	

Diversity = 2.5830e-03	k_t = [1, 1, 1, 1]			
Generation 3200: Best = 3.661617e+03	FES = 320100	$\mu_F = 0.711$	$\mu_{CR} = 0.084$	
Diversity = 2.6238e-03	k_t = [2, 2, 3, 3]			
Generation 3210: Best = 3.661617e+03	FES = 321100	$\mu_F = 0.730$	$\mu_{CR} = 0.085$	
Diversity = 2.6915e-03	k_t = [1, 1, 1, 1]			
Generation 3220: Best = 3.661611e+03	FES = 322100	$\mu_F = 0.718$	$\mu_{CR} = 0.097$	
Diversity = 2.6106e-03	k_t = [2, 2, 2, 3]			
Generation 3230: Best = 3.661611e+03	FES = 323100	$\mu_F = 0.712$	$\mu_{CR} = 0.098$	
Diversity = 2.6215e-03	k_t = [1, 1, 1, 1]			
Generation 3240: Best = 3.661609e+03	FES = 324100	$\mu_F = 0.723$	$\mu_{CR} = 0.100$	
Diversity = 2.5331e-03	k_t = [4, 2, 3, 3]			
Generation 3250: Best = 3.661605e+03	FES = 325100	$\mu_F = 0.733$	$\mu_{CR} = 0.094$	
Diversity = 2.6196e-03	k_t = [1, 1, 1, 1]			
Generation 3260: Best = 3.661604e+03	FES = 326100	$\mu_F = 0.748$	$\mu_{CR} = 0.093$	
Diversity = 2.5718e-03	k_t = [3, 3, 3, 4]			
Generation 3270: Best = 3.661602e+03	FES = 327100	$\mu_F = 0.748$	$\mu_{CR} = 0.099$	
Diversity = 2.3528e-03	k_t = [1, 1, 1, 1]			
Generation 3280: Best = 3.661602e+03	FES = 328100	$\mu_F = 0.737$	$\mu_{CR} = 0.095$	
Diversity = 2.4488e-03	k_t = [2, 4, 3, 2]			
Generation 3290: Best = 3.661601e+03	FES = 329100	$\mu_F = 0.727$	$\mu_{CR} = 0.093$	
Diversity = 2.3267e-03	k_t = [1, 1, 1, 1]			
Generation 3300: Best = 3.661600e+03	FES = 330100	$\mu_F = 0.716$	$\mu_{CR} = 0.093$	
Diversity = 2.1255e-03	k_t = [4, 3, 2, 2]			
Generation 3310: Best = 3.661599e+03	FES = 331100	$\mu_F = 0.724$	$\mu_{CR} = 0.091$	
Diversity = 2.0688e-03	k_t = [1, 1, 1, 1]			
Generation 3320: Best = 3.661596e+03	FES = 332100	$\mu_F = 0.744$	$\mu_{CR} = 0.085$	
Diversity = 2.0455e-03	k_t = [3, 3, 2, 4]			
Generation 3330: Best = 3.661595e+03	FES = 333100	$\mu_F = 0.763$	$\mu_{CR} = 0.096$	
Diversity = 1.9424e-03	k_t = [1, 1, 1, 1]			
Generation 3340: Best = 3.661594e+03	FES = 334100	$\mu_F = 0.745$	$\mu_{CR} = 0.084$	
Diversity = 1.9566e-03	k_t = [2, 4, 4, 3]			
Generation 3350: Best = 3.661592e+03	FES = 335100	$\mu_F = 0.738$	$\mu_{CR} = 0.093$	
Diversity = 1.8953e-03	k_t = [1, 1, 1, 1]			
Generation 3360: Best = 3.661592e+03	FES = 336100	$\mu_F = 0.744$	$\mu_{CR} = 0.084$	
Diversity = 1.7884e-03	k_t = [3, 2, 3, 2]			
Generation 3370: Best = 3.661591e+03	FES = 337100	$\mu_F = 0.726$	$\mu_{CR} = 0.088$	
Diversity = 1.8321e-03	k_t = [1, 1, 1, 1]			
Generation 3380: Best = 3.661590e+03	FES = 338100	$\mu_F = 0.716$	$\mu_{CR} = 0.093$	
Diversity = 1.6454e-03	k_t = [3, 4, 4, 2]			
Generation 3390: Best = 3.661588e+03	FES = 339100	$\mu_F = 0.716$	$\mu_{CR} = 0.096$	
Diversity = 1.7525e-03	k_t = [1, 1, 1, 1]			
Generation 3400: Best = 3.661587e+03	FES = 340100	$\mu_F = 0.721$	$\mu_{CR} = 0.093$	
Diversity = 1.7301e-03	k_t = [2, 4, 3, 3]			
Generation 3410: Best = 3.661586e+03	FES = 341100	$\mu_F = 0.715$	$\mu_{CR} = 0.100$	
Diversity = 1.7608e-03	k_t = [1, 1, 1, 1]			
Generation 3420: Best = 3.661584e+03	FES = 342100	$\mu_F = 0.714$	$\mu_{CR} = 0.093$	
Diversity = 1.7816e-03	k_t = [4, 3, 3, 2]			
Generation 3430: Best = 3.661583e+03	FES = 343100	$\mu_F = 0.717$	$\mu_{CR} = 0.089$	
Diversity = 1.8783e-03	k_t = [1, 1, 1, 1]			
Generation 3440: Best = 3.661581e+03	FES = 344100	$\mu_F = 0.746$	$\mu_{CR} = 0.086$	
Diversity = 2.0245e-03	k_t = [4, 2, 3, 3]			
Generation 3450: Best = 3.661580e+03	FES = 345100	$\mu_F = 0.719$	$\mu_{CR} = 0.078$	
Diversity = 2.0596e-03	k_t = [1, 1, 1, 1]			
Generation 3460: Best = 3.661579e+03	FES = 346100	$\mu_F = 0.729$	$\mu_{CR} = 0.082$	

Diversity = 2.1415e-03	k_t = [4, 2, 4, 3]			
Generation 3470: Best = 3.661577e+03	FES = 347100	$\mu_F = 0.701$	$\mu_{CR} = 0.089$	
Diversity = 2.0237e-03	k_t = [1, 1, 1, 1]			
Generation 3480: Best = 3.661575e+03	FES = 348100	$\mu_F = 0.706$	$\mu_{CR} = 0.086$	
Diversity = 1.9106e-03	k_t = [2, 3, 3, 4]			
Generation 3490: Best = 3.661575e+03	FES = 349100	$\mu_F = 0.701$	$\mu_{CR} = 0.080$	
Diversity = 1.8083e-03	k_t = [1, 1, 1, 1]			
Generation 3500: Best = 3.661574e+03	FES = 350100	$\mu_F = 0.721$	$\mu_{CR} = 0.076$	
Diversity = 1.8856e-03	k_t = [3, 3, 4, 2]			
Generation 3510: Best = 3.661573e+03	FES = 351100	$\mu_F = 0.732$	$\mu_{CR} = 0.081$	
Diversity = 1.9484e-03	k_t = [1, 1, 1, 1]			
Generation 3520: Best = 3.661572e+03	FES = 352100	$\mu_F = 0.730$	$\mu_{CR} = 0.090$	
Diversity = 1.9009e-03	k_t = [3, 4, 4, 4]			
Generation 3530: Best = 3.661571e+03	FES = 353100	$\mu_F = 0.738$	$\mu_{CR} = 0.087$	
Diversity = 1.9997e-03	k_t = [1, 1, 1, 1]			
Generation 3540: Best = 3.661570e+03	FES = 354100	$\mu_F = 0.741$	$\mu_{CR} = 0.083$	
Diversity = 2.0044e-03	k_t = [4, 2, 2, 3]			
Generation 3550: Best = 3.661569e+03	FES = 355100	$\mu_F = 0.726$	$\mu_{CR} = 0.085$	
Diversity = 1.9522e-03	k_t = [1, 1, 1, 1]			
Generation 3560: Best = 3.661567e+03	FES = 356100	$\mu_F = 0.714$	$\mu_{CR} = 0.100$	
Diversity = 1.9457e-03	k_t = [4, 3, 3, 2]			
Generation 3570: Best = 3.661566e+03	FES = 357100	$\mu_F = 0.680$	$\mu_{CR} = 0.099$	
Diversity = 1.8591e-03	k_t = [1, 1, 1, 1]			
Generation 3580: Best = 3.661564e+03	FES = 358100	$\mu_F = 0.679$	$\mu_{CR} = 0.103$	
Diversity = 1.7620e-03	k_t = [3, 3, 3, 3]			
Generation 3590: Best = 3.661564e+03	FES = 359100	$\mu_F = 0.682$	$\mu_{CR} = 0.106$	
Diversity = 1.8212e-03	k_t = [1, 1, 1, 1]			
Generation 3600: Best = 3.661563e+03	FES = 360100	$\mu_F = 0.714$	$\mu_{CR} = 0.093$	
Diversity = 1.7940e-03	k_t = [3, 4, 2, 4]			
Generation 3610: Best = 3.661562e+03	FES = 361100	$\mu_F = 0.720$	$\mu_{CR} = 0.085$	
Diversity = 1.9752e-03	k_t = [1, 1, 1, 1]			
Generation 3620: Best = 3.661562e+03	FES = 362100	$\mu_F = 0.723$	$\mu_{CR} = 0.080$	
Diversity = 1.9441e-03	k_t = [3, 2, 3, 2]			
Generation 3630: Best = 3.661559e+03	FES = 363100	$\mu_F = 0.725$	$\mu_{CR} = 0.080$	
Diversity = 1.9480e-03	k_t = [1, 1, 1, 1]			
Generation 3640: Best = 3.661558e+03	FES = 364100	$\mu_F = 0.730$	$\mu_{CR} = 0.089$	
Diversity = 1.7919e-03	k_t = [3, 2, 2, 3]			
Generation 3650: Best = 3.661557e+03	FES = 365100	$\mu_F = 0.737$	$\mu_{CR} = 0.082$	
Diversity = 1.9586e-03	k_t = [1, 1, 1, 1]			
Generation 3660: Best = 3.661553e+03	FES = 366100	$\mu_F = 0.727$	$\mu_{CR} = 0.086$	
Diversity = 1.9404e-03	k_t = [4, 4, 3, 2]			
Generation 3670: Best = 3.661553e+03	FES = 367100	$\mu_F = 0.722$	$\mu_{CR} = 0.095$	
Diversity = 2.0568e-03	k_t = [1, 1, 1, 1]			
Generation 3680: Best = 3.661552e+03	FES = 368100	$\mu_F = 0.713$	$\mu_{CR} = 0.086$	
Diversity = 2.0082e-03	k_t = [4, 2, 3, 3]			
Generation 3690: Best = 3.661551e+03	FES = 369100	$\mu_F = 0.722$	$\mu_{CR} = 0.091$	
Diversity = 1.9574e-03	k_t = [1, 1, 1, 1]			
Generation 3700: Best = 3.661550e+03	FES = 370100	$\mu_F = 0.706$	$\mu_{CR} = 0.093$	
Diversity = 1.8356e-03	k_t = [4, 2, 2, 2]			
Generation 3710: Best = 3.661546e+03	FES = 371100	$\mu_F = 0.710$	$\mu_{CR} = 0.088$	
Diversity = 1.8911e-03	k_t = [1, 1, 1, 1]			
Generation 3720: Best = 3.661545e+03	FES = 372100	$\mu_F = 0.717$	$\mu_{CR} = 0.097$	
Diversity = 1.8363e-03	k_t = [4, 2, 2, 4]			
Generation 3730: Best = 3.661540e+03	FES = 373100	$\mu_F = 0.737$	$\mu_{CR} = 0.096$	

Diversity = 1.9815e-03	k_t = [1, 1, 1, 1]			
Generation 3740: Best = 3.661540e+03	FES = 374100	$\mu_F = 0.733$	$\mu_{CR} = 0.088$	
Diversity = 1.8470e-03	k_t = [2, 3, 2, 3]			
Generation 3750: Best = 3.661540e+03	FES = 375100	$\mu_F = 0.743$	$\mu_{CR} = 0.094$	
Diversity = 1.8222e-03	k_t = [1, 1, 1, 1]			
Generation 3760: Best = 3.661539e+03	FES = 376100	$\mu_F = 0.740$	$\mu_{CR} = 0.093$	
Diversity = 1.8034e-03	k_t = [2, 4, 3, 4]			
Generation 3770: Best = 3.661537e+03	FES = 377100	$\mu_F = 0.756$	$\mu_{CR} = 0.103$	
Diversity = 1.7978e-03	k_t = [1, 1, 1, 1]			
Generation 3780: Best = 3.661535e+03	FES = 378100	$\mu_F = 0.753$	$\mu_{CR} = 0.113$	
Diversity = 1.7883e-03	k_t = [2, 2, 2, 3]			
Generation 3790: Best = 3.661532e+03	FES = 379100	$\mu_F = 0.738$	$\mu_{CR} = 0.119$	
Diversity = 1.8468e-03	k_t = [1, 1, 1, 1]			
Generation 3800: Best = 3.661528e+03	FES = 380100	$\mu_F = 0.744$	$\mu_{CR} = 0.112$	
Diversity = 1.9216e-03	k_t = [3, 2, 2, 2]			
Generation 3810: Best = 3.661528e+03	FES = 381100	$\mu_F = 0.755$	$\mu_{CR} = 0.102$	
Diversity = 1.8630e-03	k_t = [1, 1, 1, 1]			
Generation 3820: Best = 3.661526e+03	FES = 382100	$\mu_F = 0.750$	$\mu_{CR} = 0.096$	
Diversity = 1.8451e-03	k_t = [2, 4, 4, 4]			
Generation 3830: Best = 3.661525e+03	FES = 383100	$\mu_F = 0.769$	$\mu_{CR} = 0.099$	
Diversity = 2.0140e-03	k_t = [1, 1, 1, 1]			
Generation 3840: Best = 3.661524e+03	FES = 384100	$\mu_F = 0.766$	$\mu_{CR} = 0.114$	
Diversity = 1.9613e-03	k_t = [2, 4, 4, 3]			
Generation 3850: Best = 3.661523e+03	FES = 385100	$\mu_F = 0.775$	$\mu_{CR} = 0.106$	
Diversity = 2.0456e-03	k_t = [1, 1, 1, 1]			
Generation 3860: Best = 3.661521e+03	FES = 386100	$\mu_F = 0.785$	$\mu_{CR} = 0.098$	
Diversity = 2.1197e-03	k_t = [2, 2, 4, 2]			
Generation 3870: Best = 3.661521e+03	FES = 387100	$\mu_F = 0.777$	$\mu_{CR} = 0.090$	
Diversity = 2.1830e-03	k_t = [1, 1, 1, 1]			
Generation 3880: Best = 3.661519e+03	FES = 388100	$\mu_F = 0.793$	$\mu_{CR} = 0.096$	
Diversity = 1.9559e-03	k_t = [3, 3, 3, 3]			
Generation 3890: Best = 3.661515e+03	FES = 389100	$\mu_F = 0.768$	$\mu_{CR} = 0.113$	
Diversity = 1.9615e-03	k_t = [1, 1, 1, 1]			
Generation 3900: Best = 3.661515e+03	FES = 390100	$\mu_F = 0.754$	$\mu_{CR} = 0.116$	
Diversity = 2.0009e-03	k_t = [2, 3, 4, 2]			
Generation 3910: Best = 3.661514e+03	FES = 391100	$\mu_F = 0.758$	$\mu_{CR} = 0.112$	
Diversity = 2.0724e-03	k_t = [1, 1, 1, 1]			
Generation 3920: Best = 3.661511e+03	FES = 392100	$\mu_F = 0.744$	$\mu_{CR} = 0.110$	
Diversity = 1.8741e-03	k_t = [3, 4, 3, 3]			
Generation 3930: Best = 3.661510e+03	FES = 393100	$\mu_F = 0.736$	$\mu_{CR} = 0.109$	
Diversity = 1.8717e-03	k_t = [1, 1, 1, 1]			
Generation 3940: Best = 3.661507e+03	FES = 394100	$\mu_F = 0.737$	$\mu_{CR} = 0.109$	
Diversity = 1.8831e-03	k_t = [4, 4, 3, 3]			
Generation 3950: Best = 3.661505e+03	FES = 395100	$\mu_F = 0.740$	$\mu_{CR} = 0.108$	
Diversity = 1.9277e-03	k_t = [1, 1, 1, 1]			
Generation 3960: Best = 3.661503e+03	FES = 396100	$\mu_F = 0.749$	$\mu_{CR} = 0.112$	
Diversity = 1.9523e-03	k_t = [4, 4, 2, 3]			
Generation 3970: Best = 3.661503e+03	FES = 397100	$\mu_F = 0.732$	$\mu_{CR} = 0.111$	
Diversity = 1.9985e-03	k_t = [1, 1, 1, 1]			
Generation 3980: Best = 3.661502e+03	FES = 398100	$\mu_F = 0.721$	$\mu_{CR} = 0.111$	
Diversity = 1.8453e-03	k_t = [4, 2, 2, 2]			
Generation 3990: Best = 3.661501e+03	FES = 399100	$\mu_F = 0.726$	$\mu_{CR} = 0.105$	
Diversity = 1.9451e-03	k_t = [1, 1, 1, 1]			
Generation 4000: Best = 3.661501e+03	FES = 400100	$\mu_F = 0.713$	$\mu_{CR} = 0.108$	

Diversity = 1.8134e-03	k_t = [4, 4, 2, 4]			
Generation 4010: Best = 3.661500e+03	FES = 401100	$\mu_F = 0.724$	$\mu_{CR} = 0.104$	
Diversity = 1.7126e-03	k_t = [1, 1, 1, 1]			
Generation 4020: Best = 3.661498e+03	FES = 402100	$\mu_F = 0.728$	$\mu_{CR} = 0.105$	
Diversity = 1.7048e-03	k_t = [4, 2, 4, 3]			
Generation 4030: Best = 3.661496e+03	FES = 403100	$\mu_F = 0.733$	$\mu_{CR} = 0.109$	
Diversity = 1.6717e-03	k_t = [1, 1, 1, 1]			
Generation 4040: Best = 3.661496e+03	FES = 404100	$\mu_F = 0.726$	$\mu_{CR} = 0.106$	
Diversity = 1.6478e-03	k_t = [4, 2, 2, 3]			
Generation 4050: Best = 3.661495e+03	FES = 405100	$\mu_F = 0.703$	$\mu_{CR} = 0.105$	
Diversity = 1.7125e-03	k_t = [1, 1, 1, 1]			
Generation 4060: Best = 3.661495e+03	FES = 406100	$\mu_F = 0.695$	$\mu_{CR} = 0.099$	
Diversity = 1.6874e-03	k_t = [2, 4, 4, 3]			
Generation 4070: Best = 3.661494e+03	FES = 407100	$\mu_F = 0.690$	$\mu_{CR} = 0.098$	
Diversity = 1.6769e-03	k_t = [1, 1, 1, 1]			
Generation 4080: Best = 3.661493e+03	FES = 408100	$\mu_F = 0.696$	$\mu_{CR} = 0.103$	
Diversity = 1.6566e-03	k_t = [3, 4, 4, 3]			
Generation 4090: Best = 3.661491e+03	FES = 409100	$\mu_F = 0.695$	$\mu_{CR} = 0.097$	
Diversity = 1.5821e-03	k_t = [1, 1, 1, 1]			
Generation 4100: Best = 3.661491e+03	FES = 410100	$\mu_F = 0.694$	$\mu_{CR} = 0.096$	
Diversity = 1.5365e-03	k_t = [3, 2, 4, 4]			
Generation 4110: Best = 3.661490e+03	FES = 411100	$\mu_F = 0.711$	$\mu_{CR} = 0.100$	
Diversity = 1.3753e-03	k_t = [1, 1, 1, 1]			
Generation 4120: Best = 3.661489e+03	FES = 412100	$\mu_F = 0.695$	$\mu_{CR} = 0.109$	
Diversity = 1.2481e-03	k_t = [4, 2, 4, 3]			
Generation 4130: Best = 3.661487e+03	FES = 413100	$\mu_F = 0.712$	$\mu_{CR} = 0.115$	
Diversity = 1.3001e-03	k_t = [1, 1, 1, 1]			
Generation 4140: Best = 3.661487e+03	FES = 414100	$\mu_F = 0.717$	$\mu_{CR} = 0.106$	
Diversity = 1.3273e-03	k_t = [4, 3, 2, 3]			
Generation 4150: Best = 3.661485e+03	FES = 415100	$\mu_F = 0.704$	$\mu_{CR} = 0.101$	
Diversity = 1.3605e-03	k_t = [1, 1, 1, 1]			
Generation 4160: Best = 3.661485e+03	FES = 416100	$\mu_F = 0.713$	$\mu_{CR} = 0.108$	
Diversity = 1.3899e-03	k_t = [3, 2, 2, 4]			
Generation 4170: Best = 3.661484e+03	FES = 417100	$\mu_F = 0.721$	$\mu_{CR} = 0.109$	
Diversity = 1.3777e-03	k_t = [1, 1, 1, 1]			
Generation 4180: Best = 3.661483e+03	FES = 418100	$\mu_F = 0.724$	$\mu_{CR} = 0.110$	
Diversity = 1.4641e-03	k_t = [4, 4, 2, 2]			
Generation 4190: Best = 3.661482e+03	FES = 419100	$\mu_F = 0.731$	$\mu_{CR} = 0.107$	
Diversity = 1.5077e-03	k_t = [1, 1, 1, 1]			
Generation 4200: Best = 3.661481e+03	FES = 420100	$\mu_F = 0.721$	$\mu_{CR} = 0.112$	
Diversity = 1.4742e-03	k_t = [4, 2, 2, 4]			
Generation 4210: Best = 3.661480e+03	FES = 421100	$\mu_F = 0.730$	$\mu_{CR} = 0.097$	
Diversity = 1.4205e-03	k_t = [1, 1, 1, 1]			
Generation 4220: Best = 3.661479e+03	FES = 422100	$\mu_F = 0.728$	$\mu_{CR} = 0.098$	
Diversity = 1.3268e-03	k_t = [3, 2, 4, 3]			
Generation 4230: Best = 3.661478e+03	FES = 423100	$\mu_F = 0.751$	$\mu_{CR} = 0.095$	
Diversity = 1.4280e-03	k_t = [1, 1, 1, 1]			
Generation 4240: Best = 3.661477e+03	FES = 424100	$\mu_F = 0.763$	$\mu_{CR} = 0.098$	
Diversity = 1.4187e-03	k_t = [2, 4, 2, 4]			
Generation 4250: Best = 3.661476e+03	FES = 425100	$\mu_F = 0.771$	$\mu_{CR} = 0.100$	
Diversity = 1.3388e-03	k_t = [1, 1, 1, 1]			
Generation 4260: Best = 3.661475e+03	FES = 426100	$\mu_F = 0.756$	$\mu_{CR} = 0.100$	
Diversity = 1.4455e-03	k_t = [4, 2, 4, 2]			
Generation 4270: Best = 3.661475e+03	FES = 427100	$\mu_F = 0.734$	$\mu_{CR} = 0.096$	

Diversity = 1.4052e-03	k_t = [1, 1, 1, 1]			
Generation 4280: Best = 3.661474e+03	FES = 428100	$\mu_F = 0.722$	$\mu_{CR} = 0.100$	
Diversity = 1.2336e-03	k_t = [2, 4, 2, 3]			
Generation 4290: Best = 3.661474e+03	FES = 429100	$\mu_F = 0.743$	$\mu_{CR} = 0.103$	
Diversity = 1.2826e-03	k_t = [1, 1, 1, 1]			
Generation 4300: Best = 3.661473e+03	FES = 430100	$\mu_F = 0.746$	$\mu_{CR} = 0.096$	
Diversity = 1.2418e-03	k_t = [3, 3, 3, 4]			
Generation 4310: Best = 3.661472e+03	FES = 431100	$\mu_F = 0.739$	$\mu_{CR} = 0.101$	
Diversity = 1.2601e-03	k_t = [1, 1, 1, 1]			
Generation 4320: Best = 3.661472e+03	FES = 432100	$\mu_F = 0.731$	$\mu_{CR} = 0.108$	
Diversity = 1.2179e-03	k_t = [4, 2, 3, 3]			
Generation 4330: Best = 3.661472e+03	FES = 433100	$\mu_F = 0.727$	$\mu_{CR} = 0.089$	
Diversity = 1.2158e-03	k_t = [1, 1, 1, 1]			
Generation 4340: Best = 3.661472e+03	FES = 434100	$\mu_F = 0.705$	$\mu_{CR} = 0.083$	
Diversity = 1.1895e-03	k_t = [4, 2, 3, 2]			
Generation 4350: Best = 3.661471e+03	FES = 435100	$\mu_F = 0.708$	$\mu_{CR} = 0.083$	
Diversity = 1.2148e-03	k_t = [1, 1, 1, 1]			
Generation 4360: Best = 3.661471e+03	FES = 436100	$\mu_F = 0.720$	$\mu_{CR} = 0.081$	
Diversity = 1.2755e-03	k_t = [3, 3, 2, 3]			
Generation 4370: Best = 3.661470e+03	FES = 437100	$\mu_F = 0.721$	$\mu_{CR} = 0.072$	
Diversity = 1.2774e-03	k_t = [1, 1, 1, 1]			
Generation 4380: Best = 3.661470e+03	FES = 438100	$\mu_F = 0.719$	$\mu_{CR} = 0.081$	
Diversity = 1.2416e-03	k_t = [2, 4, 2, 3]			
Generation 4390: Best = 3.661469e+03	FES = 439100	$\mu_F = 0.710$	$\mu_{CR} = 0.085$	
Diversity = 1.2150e-03	k_t = [1, 1, 1, 1]			
Generation 4400: Best = 3.661469e+03	FES = 440100	$\mu_F = 0.707$	$\mu_{CR} = 0.091$	
Diversity = 1.2001e-03	k_t = [2, 4, 2, 2]			
Generation 4410: Best = 3.661468e+03	FES = 441100	$\mu_F = 0.696$	$\mu_{CR} = 0.094$	
Diversity = 1.2153e-03	k_t = [1, 1, 1, 1]			
Generation 4420: Best = 3.661468e+03	FES = 442100	$\mu_F = 0.702$	$\mu_{CR} = 0.098$	
Diversity = 1.2031e-03	k_t = [2, 3, 3, 2]			
Generation 4430: Best = 3.661467e+03	FES = 443100	$\mu_F = 0.717$	$\mu_{CR} = 0.104$	
Diversity = 1.1875e-03	k_t = [1, 1, 1, 1]			
Generation 4440: Best = 3.661466e+03	FES = 444100	$\mu_F = 0.711$	$\mu_{CR} = 0.109$	
Diversity = 1.1880e-03	k_t = [2, 2, 2, 2]			
Generation 4450: Best = 3.661466e+03	FES = 445100	$\mu_F = 0.717$	$\mu_{CR} = 0.118$	
Diversity = 1.1490e-03	k_t = [1, 1, 1, 1]			
Generation 4460: Best = 3.661465e+03	FES = 446100	$\mu_F = 0.700$	$\mu_{CR} = 0.116$	
Diversity = 1.1471e-03	k_t = [3, 2, 3, 2]			
Generation 4470: Best = 3.661464e+03	FES = 447100	$\mu_F = 0.693$	$\mu_{CR} = 0.101$	
Diversity = 1.1636e-03	k_t = [1, 1, 1, 1]			
Generation 4480: Best = 3.661463e+03	FES = 448100	$\mu_F = 0.708$	$\mu_{CR} = 0.097$	
Diversity = 1.2518e-03	k_t = [3, 2, 2, 2]			
Generation 4490: Best = 3.661463e+03	FES = 449100	$\mu_F = 0.718$	$\mu_{CR} = 0.095$	
Diversity = 1.2370e-03	k_t = [1, 1, 1, 1]			
Generation 4500: Best = 3.661462e+03	FES = 450100	$\mu_F = 0.711$	$\mu_{CR} = 0.107$	
Diversity = 1.2045e-03	k_t = [2, 3, 4, 3]			
Generation 4510: Best = 3.661462e+03	FES = 451100	$\mu_F = 0.727$	$\mu_{CR} = 0.102$	
Diversity = 1.2493e-03	k_t = [1, 1, 1, 1]			
Generation 4520: Best = 3.661461e+03	FES = 452100	$\mu_F = 0.748$	$\mu_{CR} = 0.106$	
Diversity = 1.3910e-03	k_t = [2, 2, 2, 4]			
Generation 4530: Best = 3.661459e+03	FES = 453100	$\mu_F = 0.751$	$\mu_{CR} = 0.106$	
Diversity = 1.4325e-03	k_t = [1, 1, 1, 1]			
Generation 4540: Best = 3.661459e+03	FES = 454100	$\mu_F = 0.753$	$\mu_{CR} = 0.091$	

Diversity = 1.5435e-03 k_t = [2, 3, 2, 2]
 Generation 4550: Best = 3.661458e+03 FES = 455100 μ_F = 0.762 μ_{CR} = 0.101
 Diversity = 1.5264e-03 k_t = [1, 1, 1, 1]
 Generation 4560: Best = 3.661457e+03 FES = 456100 μ_F = 0.756 μ_{CR} = 0.110
 Diversity = 1.4950e-03 k_t = [4, 2, 2, 4]
 Generation 4570: Best = 3.661455e+03 FES = 457100 μ_F = 0.772 μ_{CR} = 0.119
 Diversity = 1.5654e-03 k_t = [1, 1, 1, 1]
 Generation 4580: Best = 3.661454e+03 FES = 458100 μ_F = 0.778 μ_{CR} = 0.112
 Diversity = 1.5802e-03 k_t = [3, 4, 4, 4]
 Generation 4590: Best = 3.661453e+03 FES = 459100 μ_F = 0.782 μ_{CR} = 0.114
 Diversity = 1.5904e-03 k_t = [1, 1, 1, 1]
 Generation 4600: Best = 3.661452e+03 FES = 460100 μ_F = 0.784 μ_{CR} = 0.111
 Diversity = 1.5552e-03 k_t = [2, 3, 4, 3]
 Generation 4610: Best = 3.661451e+03 FES = 461100 μ_F = 0.788 μ_{CR} = 0.115
 Diversity = 1.7791e-03 k_t = [1, 1, 1, 1]
 Generation 4620: Best = 3.661450e+03 FES = 462100 μ_F = 0.777 μ_{CR} = 0.106
 Diversity = 1.6265e-03 k_t = [3, 2, 2, 3]
 Generation 4630: Best = 3.661449e+03 FES = 463100 μ_F = 0.773 μ_{CR} = 0.113
 Diversity = 1.8271e-03 k_t = [1, 1, 1, 1]
 Generation 4640: Best = 3.661446e+03 FES = 464100 μ_F = 0.778 μ_{CR} = 0.114
 Diversity = 1.9262e-03 k_t = [4, 4, 4, 2]
 Generation 4650: Best = 3.661444e+03 FES = 465100 μ_F = 0.770 μ_{CR} = 0.117
 Diversity = 2.1488e-03 k_t = [1, 1, 1, 1]
 Generation 4660: Best = 3.661441e+03 FES = 466100 μ_F = 0.774 μ_{CR} = 0.123
 Diversity = 2.0000e-03 k_t = [3, 2, 3, 4]
 Generation 4670: Best = 3.661441e+03 FES = 467100 μ_F = 0.763 μ_{CR} = 0.119
 Diversity = 1.8751e-03 k_t = [1, 1, 1, 1]
 Generation 4680: Best = 3.661439e+03 FES = 468100 μ_F = 0.766 μ_{CR} = 0.117
 Diversity = 2.0167e-03 k_t = [4, 3, 2, 2]
 Generation 4690: Best = 3.661438e+03 FES = 469100 μ_F = 0.764 μ_{CR} = 0.114
 Diversity = 2.0733e-03 k_t = [1, 1, 1, 1]
 Generation 4700: Best = 3.661437e+03 FES = 470100 μ_F = 0.770 μ_{CR} = 0.100
 Diversity = 2.1538e-03 k_t = [4, 4, 3, 3]
 Generation 4710: Best = 3.661434e+03 FES = 471100 μ_F = 0.779 μ_{CR} = 0.102
 Diversity = 2.2340e-03 k_t = [1, 1, 1, 1]
 Generation 4720: Best = 3.661433e+03 FES = 472100 μ_F = 0.772 μ_{CR} = 0.096
 Diversity = 2.1392e-03 k_t = [2, 3, 3, 3]
 Generation 4730: Best = 3.661432e+03 FES = 473100 μ_F = 0.764 μ_{CR} = 0.093
 Diversity = 2.3919e-03 k_t = [1, 1, 1, 1]
 Generation 4740: Best = 3.661430e+03 FES = 474100 μ_F = 0.754 μ_{CR} = 0.101
 Diversity = 2.3448e-03 k_t = [2, 2, 2, 3]
 Generation 4750: Best = 3.661426e+03 FES = 475100 μ_F = 0.756 μ_{CR} = 0.105
 Diversity = 2.6424e-03 k_t = [1, 1, 1, 1]
 Generation 4760: Best = 3.661422e+03 FES = 476100 μ_F = 0.748 μ_{CR} = 0.108
 Diversity = 2.5349e-03 k_t = [2, 2, 4, 4]
 Generation 4770: Best = 3.661418e+03 FES = 477100 μ_F = 0.760 μ_{CR} = 0.120
 Diversity = 2.7897e-03 k_t = [1, 1, 1, 1]
 Generation 4780: Best = 3.661417e+03 FES = 478100 μ_F = 0.753 μ_{CR} = 0.120
 Diversity = 2.7491e-03 k_t = [4, 3, 4, 2]
 Generation 4790: Best = 3.661413e+03 FES = 479100 μ_F = 0.755 μ_{CR} = 0.121
 Diversity = 2.6844e-03 k_t = [1, 1, 1, 1]
 Generation 4800: Best = 3.661409e+03 FES = 480100 μ_F = 0.751 μ_{CR} = 0.126
 Diversity = 2.5584e-03 k_t = [4, 2, 3, 2]
 Generation 4810: Best = 3.661409e+03 FES = 481100 μ_F = 0.760 μ_{CR} = 0.126

Diversity = 2.8807e-03	k_t = [1, 1, 1, 1]			
Generation 4820: Best = 3.661408e+03	FES = 482100	$\mu_F = 0.755$	$\mu_{CR} = 0.125$	
Diversity = 2.8682e-03	k_t = [2, 2, 2, 3]			
Generation 4830: Best = 3.661404e+03	FES = 483100	$\mu_F = 0.757$	$\mu_{CR} = 0.123$	
Diversity = 2.7079e-03	k_t = [1, 1, 1, 1]			
Generation 4840: Best = 3.661402e+03	FES = 484100	$\mu_F = 0.768$	$\mu_{CR} = 0.125$	
Diversity = 2.7423e-03	k_t = [2, 2, 2, 3]			
Generation 4850: Best = 3.661397e+03	FES = 485100	$\mu_F = 0.756$	$\mu_{CR} = 0.128$	
Diversity = 2.7418e-03	k_t = [1, 1, 1, 1]			
Generation 4860: Best = 3.661396e+03	FES = 486100	$\mu_F = 0.750$	$\mu_{CR} = 0.130$	
Diversity = 3.1046e-03	k_t = [2, 3, 4, 2]			
Generation 4870: Best = 3.661394e+03	FES = 487100	$\mu_F = 0.734$	$\mu_{CR} = 0.124$	
Diversity = 3.1399e-03	k_t = [1, 1, 1, 1]			
Generation 4880: Best = 3.661391e+03	FES = 488100	$\mu_F = 0.740$	$\mu_{CR} = 0.114$	
Diversity = 2.8616e-03	k_t = [3, 3, 2, 4]			
Generation 4890: Best = 3.661388e+03	FES = 489100	$\mu_F = 0.761$	$\mu_{CR} = 0.114$	
Diversity = 2.8520e-03	k_t = [1, 1, 1, 1]			
Generation 4900: Best = 3.661385e+03	FES = 490100	$\mu_F = 0.760$	$\mu_{CR} = 0.111$	
Diversity = 2.6570e-03	k_t = [4, 2, 4, 4]			
Generation 4910: Best = 3.661383e+03	FES = 491100	$\mu_F = 0.747$	$\mu_{CR} = 0.099$	
Diversity = 2.8027e-03	k_t = [1, 1, 1, 1]			
Generation 4920: Best = 3.661380e+03	FES = 492100	$\mu_F = 0.755$	$\mu_{CR} = 0.102$	
Diversity = 2.7419e-03	k_t = [4, 2, 2, 4]			
Generation 4930: Best = 3.661379e+03	FES = 493100	$\mu_F = 0.738$	$\mu_{CR} = 0.102$	
Diversity = 2.6139e-03	k_t = [1, 1, 1, 1]			
Generation 4940: Best = 3.661378e+03	FES = 494100	$\mu_F = 0.750$	$\mu_{CR} = 0.112$	
Diversity = 2.5099e-03	k_t = [4, 2, 4, 4]			
Generation 4950: Best = 3.661376e+03	FES = 495100	$\mu_F = 0.747$	$\mu_{CR} = 0.109$	
Diversity = 2.4465e-03	k_t = [1, 1, 1, 1]			
Generation 4960: Best = 3.661372e+03	FES = 496100	$\mu_F = 0.762$	$\mu_{CR} = 0.107$	
Diversity = 2.3652e-03	k_t = [2, 3, 4, 3]			
Generation 4970: Best = 3.661372e+03	FES = 497100	$\mu_F = 0.772$	$\mu_{CR} = 0.114$	
Diversity = 2.4604e-03	k_t = [1, 1, 1, 1]			
Generation 4980: Best = 3.661370e+03	FES = 498100	$\mu_F = 0.764$	$\mu_{CR} = 0.114$	
Diversity = 2.5956e-03	k_t = [4, 4, 3, 3]			
Generation 4990: Best = 3.661368e+03	FES = 499100	$\mu_F = 0.769$	$\mu_{CR} = 0.116$	
Diversity = 2.6994e-03	k_t = [1, 1, 1, 1]			
Generation 5000: Best = 3.661366e+03	FES = 500100	$\mu_F = 0.761$	$\mu_{CR} = 0.115$	
Diversity = 2.4385e-03	k_t = [3, 3, 2, 4]			
Generation 5010: Best = 3.661363e+03	FES = 501100	$\mu_F = 0.752$	$\mu_{CR} = 0.117$	
Diversity = 2.3835e-03	k_t = [1, 1, 1, 1]			
Generation 5020: Best = 3.661359e+03	FES = 502100	$\mu_F = 0.753$	$\mu_{CR} = 0.123$	
Diversity = 2.4045e-03	k_t = [2, 4, 3, 3]			
Generation 5030: Best = 3.661357e+03	FES = 503100	$\mu_F = 0.765$	$\mu_{CR} = 0.112$	
Diversity = 2.6584e-03	k_t = [1, 1, 1, 1]			
Generation 5040: Best = 3.661355e+03	FES = 504100	$\mu_F = 0.763$	$\mu_{CR} = 0.108$	
Diversity = 2.5201e-03	k_t = [4, 3, 2, 2]			
Generation 5050: Best = 3.661353e+03	FES = 505100	$\mu_F = 0.760$	$\mu_{CR} = 0.121$	
Diversity = 2.6583e-03	k_t = [1, 1, 1, 1]			
Generation 5060: Best = 3.661352e+03	FES = 506100	$\mu_F = 0.756$	$\mu_{CR} = 0.113$	
Diversity = 2.5802e-03	k_t = [4, 2, 4, 4]			
Generation 5070: Best = 3.661347e+03	FES = 507100	$\mu_F = 0.755$	$\mu_{CR} = 0.122$	
Diversity = 2.6427e-03	k_t = [1, 1, 1, 1]			
Generation 5080: Best = 3.661345e+03	FES = 508100	$\mu_F = 0.747$	$\mu_{CR} = 0.112$	

Diversity = 2.7736e-03	k_t = [3, 4, 4, 2]			
Generation 5090: Best = 3.661343e+03	FES = 509100	$\mu_F = 0.753$	$\mu_{CR} = 0.109$	
Diversity = 2.9486e-03	k_t = [1, 1, 1, 1]			
Generation 5100: Best = 3.661340e+03	FES = 510100	$\mu_F = 0.734$	$\mu_{CR} = 0.123$	
Diversity = 2.9530e-03	k_t = [4, 2, 2, 3]			
Generation 5110: Best = 3.661337e+03	FES = 511100	$\mu_F = 0.739$	$\mu_{CR} = 0.121$	
Diversity = 3.0513e-03	k_t = [1, 1, 1, 1]			
Generation 5120: Best = 3.661336e+03	FES = 512100	$\mu_F = 0.740$	$\mu_{CR} = 0.115$	
Diversity = 3.0001e-03	k_t = [2, 3, 3, 3]			
Generation 5130: Best = 3.661335e+03	FES = 513100	$\mu_F = 0.740$	$\mu_{CR} = 0.113$	
Diversity = 3.1679e-03	k_t = [1, 1, 1, 1]			
Generation 5140: Best = 3.661333e+03	FES = 514100	$\mu_F = 0.740$	$\mu_{CR} = 0.102$	
Diversity = 3.1101e-03	k_t = [2, 2, 4, 4]			
Generation 5150: Best = 3.661332e+03	FES = 515100	$\mu_F = 0.724$	$\mu_{CR} = 0.109$	
Diversity = 3.0745e-03	k_t = [1, 1, 1, 1]			
Generation 5160: Best = 3.661329e+03	FES = 516100	$\mu_F = 0.740$	$\mu_{CR} = 0.106$	
Diversity = 3.0671e-03	k_t = [3, 3, 3, 4]			
Generation 5170: Best = 3.661323e+03	FES = 517100	$\mu_F = 0.735$	$\mu_{CR} = 0.100$	
Diversity = 3.0678e-03	k_t = [1, 1, 1, 1]			
Generation 5180: Best = 3.661323e+03	FES = 518100	$\mu_F = 0.733$	$\mu_{CR} = 0.102$	
Diversity = 2.8265e-03	k_t = [2, 3, 3, 4]			
Generation 5190: Best = 3.661320e+03	FES = 519100	$\mu_F = 0.750$	$\mu_{CR} = 0.101$	
Diversity = 3.0095e-03	k_t = [1, 1, 1, 1]			
Generation 5200: Best = 3.661317e+03	FES = 520100	$\mu_F = 0.758$	$\mu_{CR} = 0.112$	
Diversity = 3.1105e-03	k_t = [4, 3, 4, 3]			
Generation 5210: Best = 3.661312e+03	FES = 521100	$\mu_F = 0.764$	$\mu_{CR} = 0.105$	
Diversity = 3.3930e-03	k_t = [1, 1, 1, 1]			
Generation 5220: Best = 3.661310e+03	FES = 522100	$\mu_F = 0.760$	$\mu_{CR} = 0.121$	
Diversity = 3.3278e-03	k_t = [2, 4, 3, 2]			
Generation 5230: Best = 3.661304e+03	FES = 523100	$\mu_F = 0.766$	$\mu_{CR} = 0.113$	
Diversity = 3.3406e-03	k_t = [1, 1, 1, 1]			
Generation 5240: Best = 3.661303e+03	FES = 524100	$\mu_F = 0.765$	$\mu_{CR} = 0.118$	
Diversity = 3.2203e-03	k_t = [4, 2, 2, 2]			
Generation 5250: Best = 3.661300e+03	FES = 525100	$\mu_F = 0.747$	$\mu_{CR} = 0.119$	
Diversity = 3.4858e-03	k_t = [1, 1, 1, 1]			
Generation 5260: Best = 3.661299e+03	FES = 526100	$\mu_F = 0.759$	$\mu_{CR} = 0.109$	
Diversity = 3.3653e-03	k_t = [4, 3, 3, 2]			
Generation 5270: Best = 3.661295e+03	FES = 527100	$\mu_F = 0.745$	$\mu_{CR} = 0.114$	
Diversity = 3.3426e-03	k_t = [1, 1, 1, 1]			
Generation 5280: Best = 3.661294e+03	FES = 528100	$\mu_F = 0.742$	$\mu_{CR} = 0.106$	
Diversity = 3.3492e-03	k_t = [2, 4, 4, 2]			
Generation 5290: Best = 3.661290e+03	FES = 529100	$\mu_F = 0.749$	$\mu_{CR} = 0.101$	
Diversity = 3.3150e-03	k_t = [1, 1, 1, 1]			
Generation 5300: Best = 3.661289e+03	FES = 530100	$\mu_F = 0.744$	$\mu_{CR} = 0.102$	
Diversity = 3.0531e-03	k_t = [4, 2, 2, 4]			
Generation 5310: Best = 3.661286e+03	FES = 531100	$\mu_F = 0.737$	$\mu_{CR} = 0.108$	
Diversity = 3.0018e-03	k_t = [1, 1, 1, 1]			
Generation 5320: Best = 3.661286e+03	FES = 532100	$\mu_F = 0.733$	$\mu_{CR} = 0.114$	
Diversity = 2.7799e-03	k_t = [3, 2, 2, 4]			
Generation 5330: Best = 3.661285e+03	FES = 533100	$\mu_F = 0.740$	$\mu_{CR} = 0.119$	
Diversity = 2.7877e-03	k_t = [1, 1, 1, 1]			
Generation 5340: Best = 3.661283e+03	FES = 534100	$\mu_F = 0.735$	$\mu_{CR} = 0.113$	
Diversity = 2.6714e-03	k_t = [3, 2, 4, 3]			
Generation 5350: Best = 3.661281e+03	FES = 535100	$\mu_F = 0.723$	$\mu_{CR} = 0.105$	

Diversity = 2.6791e-03	k_t = [1, 1, 1, 1]			
Generation 5360: Best = 3.661280e+03	FES = 536100	$\mu_F = 0.732$	$\mu_{CR} = 0.107$	
Diversity = 2.7305e-03	k_t = [4, 4, 3, 2]			
Generation 5370: Best = 3.661278e+03	FES = 537100	$\mu_F = 0.700$	$\mu_{CR} = 0.108$	
Diversity = 2.6966e-03	k_t = [1, 1, 1, 1]			
Generation 5380: Best = 3.661277e+03	FES = 538100	$\mu_F = 0.702$	$\mu_{CR} = 0.105$	
Diversity = 2.7465e-03	k_t = [4, 4, 4, 2]			
Generation 5390: Best = 3.661276e+03	FES = 539100	$\mu_F = 0.694$	$\mu_{CR} = 0.101$	
Diversity = 2.6347e-03	k_t = [1, 1, 1, 1]			
Generation 5400: Best = 3.661275e+03	FES = 540100	$\mu_F = 0.704$	$\mu_{CR} = 0.098$	
Diversity = 2.5740e-03	k_t = [3, 4, 3, 2]			
Generation 5410: Best = 3.661273e+03	FES = 541100	$\mu_F = 0.705$	$\mu_{CR} = 0.103$	
Diversity = 2.5427e-03	k_t = [1, 1, 1, 1]			
Generation 5420: Best = 3.661273e+03	FES = 542100	$\mu_F = 0.709$	$\mu_{CR} = 0.101$	
Diversity = 2.4051e-03	k_t = [2, 3, 4, 2]			
Generation 5430: Best = 3.661271e+03	FES = 543100	$\mu_F = 0.712$	$\mu_{CR} = 0.105$	
Diversity = 2.5167e-03	k_t = [1, 1, 1, 1]			
Generation 5440: Best = 3.661270e+03	FES = 544100	$\mu_F = 0.711$	$\mu_{CR} = 0.102$	
Diversity = 2.5857e-03	k_t = [3, 2, 3, 2]			
Generation 5450: Best = 3.661270e+03	FES = 545100	$\mu_F = 0.704$	$\mu_{CR} = 0.103$	
Diversity = 2.6635e-03	k_t = [1, 1, 1, 1]			
Generation 5460: Best = 3.661269e+03	FES = 546100	$\mu_F = 0.698$	$\mu_{CR} = 0.101$	
Diversity = 2.5459e-03	k_t = [4, 2, 2, 3]			
Generation 5470: Best = 3.661268e+03	FES = 547100	$\mu_F = 0.700$	$\mu_{CR} = 0.099$	
Diversity = 2.4636e-03	k_t = [1, 1, 1, 1]			
Generation 5480: Best = 3.661267e+03	FES = 548100	$\mu_F = 0.685$	$\mu_{CR} = 0.086$	
Diversity = 2.2968e-03	k_t = [4, 4, 4, 4]			
Generation 5490: Best = 3.661266e+03	FES = 549100	$\mu_F = 0.697$	$\mu_{CR} = 0.084$	
Diversity = 2.3341e-03	k_t = [1, 1, 1, 1]			
Generation 5500: Best = 3.661264e+03	FES = 550100	$\mu_F = 0.703$	$\mu_{CR} = 0.086$	
Diversity = 2.2705e-03	k_t = [4, 3, 2, 4]			
Generation 5510: Best = 3.661260e+03	FES = 551100	$\mu_F = 0.707$	$\mu_{CR} = 0.092$	
Diversity = 2.1927e-03	k_t = [1, 1, 1, 1]			
Generation 5520: Best = 3.661260e+03	FES = 552100	$\mu_F = 0.729$	$\mu_{CR} = 0.097$	
Diversity = 2.1940e-03	k_t = [2, 4, 3, 2]			
Generation 5530: Best = 3.661259e+03	FES = 553100	$\mu_F = 0.737$	$\mu_{CR} = 0.102$	
Diversity = 2.4856e-03	k_t = [1, 1, 1, 1]			
Generation 5540: Best = 3.661259e+03	FES = 554100	$\mu_F = 0.715$	$\mu_{CR} = 0.106$	
Diversity = 2.5811e-03	k_t = [2, 2, 3, 3]			
Generation 5550: Best = 3.661257e+03	FES = 555100	$\mu_F = 0.708$	$\mu_{CR} = 0.101$	
Diversity = 2.5584e-03	k_t = [1, 1, 1, 1]			
Generation 5560: Best = 3.661256e+03	FES = 556100	$\mu_F = 0.731$	$\mu_{CR} = 0.102$	
Diversity = 2.5152e-03	k_t = [2, 4, 3, 4]			
Generation 5570: Best = 3.661253e+03	FES = 557100	$\mu_F = 0.716$	$\mu_{CR} = 0.110$	
Diversity = 2.5155e-03	k_t = [1, 1, 1, 1]			
Generation 5580: Best = 3.661253e+03	FES = 558100	$\mu_F = 0.738$	$\mu_{CR} = 0.108$	
Diversity = 2.3756e-03	k_t = [2, 4, 3, 4]			
Generation 5590: Best = 3.661253e+03	FES = 559100	$\mu_F = 0.726$	$\mu_{CR} = 0.100$	
Diversity = 2.2718e-03	k_t = [1, 1, 1, 1]			
Generation 5600: Best = 3.661252e+03	FES = 560100	$\mu_F = 0.733$	$\mu_{CR} = 0.096$	
Diversity = 2.4087e-03	k_t = [2, 4, 2, 2]			
Generation 5610: Best = 3.661252e+03	FES = 561100	$\mu_F = 0.710$	$\mu_{CR} = 0.095$	
Diversity = 2.3763e-03	k_t = [1, 1, 1, 1]			
Generation 5620: Best = 3.661251e+03	FES = 562100	$\mu_F = 0.705$	$\mu_{CR} = 0.098$	

Diversity = 2.1605e-03	k_t = [2, 2, 3, 3]			
Generation 5630: Best = 3.661250e+03	FES = 563100	$\mu_F = 0.702$	$\mu_{CR} = 0.098$	
Diversity = 2.0868e-03	k_t = [1, 1, 1, 1]			
Generation 5640: Best = 3.661249e+03	FES = 564100	$\mu_F = 0.725$	$\mu_{CR} = 0.088$	
Diversity = 1.9129e-03	k_t = [2, 4, 2, 2]			
Generation 5650: Best = 3.661249e+03	FES = 565100	$\mu_F = 0.702$	$\mu_{CR} = 0.082$	
Diversity = 1.8115e-03	k_t = [1, 1, 1, 1]			
Generation 5660: Best = 3.661248e+03	FES = 566100	$\mu_F = 0.691$	$\mu_{CR} = 0.078$	
Diversity = 1.6599e-03	k_t = [3, 3, 3, 4]			
Generation 5670: Best = 3.661248e+03	FES = 567100	$\mu_F = 0.697$	$\mu_{CR} = 0.081$	
Diversity = 1.6519e-03	k_t = [1, 1, 1, 1]			
Generation 5680: Best = 3.661247e+03	FES = 568100	$\mu_F = 0.700$	$\mu_{CR} = 0.095$	
Diversity = 1.6831e-03	k_t = [2, 4, 3, 3]			
Generation 5690: Best = 3.661247e+03	FES = 569100	$\mu_F = 0.698$	$\mu_{CR} = 0.089$	
Diversity = 1.5756e-03	k_t = [1, 1, 1, 1]			
Generation 5700: Best = 3.661247e+03	FES = 570100	$\mu_F = 0.677$	$\mu_{CR} = 0.093$	
Diversity = 1.3529e-03	k_t = [3, 3, 3, 4]			
Generation 5710: Best = 3.661247e+03	FES = 571100	$\mu_F = 0.686$	$\mu_{CR} = 0.085$	
Diversity = 1.2207e-03	k_t = [1, 1, 1, 1]			
Generation 5720: Best = 3.661247e+03	FES = 572100	$\mu_F = 0.673$	$\mu_{CR} = 0.083$	
Diversity = 1.1162e-03	k_t = [4, 2, 3, 2]			
Generation 5730: Best = 3.661246e+03	FES = 573100	$\mu_F = 0.692$	$\mu_{CR} = 0.085$	
Diversity = 1.0548e-03	k_t = [1, 1, 1, 1]			
Generation 5740: Best = 3.661246e+03	FES = 574100	$\mu_F = 0.687$	$\mu_{CR} = 0.076$	
Diversity = 1.0240e-03	k_t = [3, 4, 3, 3]			
Generation 5750: Best = 3.661245e+03	FES = 575100	$\mu_F = 0.720$	$\mu_{CR} = 0.076$	
Diversity = 1.0149e-03	k_t = [1, 1, 1, 1]			
Generation 5760: Best = 3.661245e+03	FES = 576100	$\mu_F = 0.730$	$\mu_{CR} = 0.084$	
Diversity = 9.5880e-04	k_t = [3, 4, 3, 4]			
Generation 5770: Best = 3.661245e+03	FES = 577100	$\mu_F = 0.728$	$\mu_{CR} = 0.078$	
Diversity = 9.9669e-04	k_t = [1, 1, 1, 1]			
Generation 5780: Best = 3.661245e+03	FES = 578100	$\mu_F = 0.717$	$\mu_{CR} = 0.086$	
Diversity = 9.7966e-04	k_t = [4, 4, 2, 3]			
Generation 5790: Best = 3.661245e+03	FES = 579100	$\mu_F = 0.715$	$\mu_{CR} = 0.090$	
Diversity = 1.0721e-03	k_t = [1, 1, 1, 1]			
Generation 5800: Best = 3.661245e+03	FES = 580100	$\mu_F = 0.717$	$\mu_{CR} = 0.090$	
Diversity = 9.9566e-04	k_t = [2, 2, 4, 2]			
Generation 5810: Best = 3.661244e+03	FES = 581100	$\mu_F = 0.734$	$\mu_{CR} = 0.090$	
Diversity = 1.0217e-03	k_t = [1, 1, 1, 1]			
Generation 5820: Best = 3.661244e+03	FES = 582100	$\mu_F = 0.741$	$\mu_{CR} = 0.085$	
Diversity = 1.0240e-03	k_t = [4, 2, 4, 4]			
Generation 5830: Best = 3.661244e+03	FES = 583100	$\mu_F = 0.751$	$\mu_{CR} = 0.075$	
Diversity = 1.0480e-03	k_t = [1, 1, 1, 1]			
Generation 5840: Best = 3.661244e+03	FES = 584100	$\mu_F = 0.761$	$\mu_{CR} = 0.080$	
Diversity = 9.8439e-04	k_t = [4, 3, 4, 4]			
Generation 5850: Best = 3.661243e+03	FES = 585100	$\mu_F = 0.735$	$\mu_{CR} = 0.088$	
Diversity = 9.2399e-04	k_t = [1, 1, 1, 1]			
Generation 5860: Best = 3.661243e+03	FES = 586100	$\mu_F = 0.734$	$\mu_{CR} = 0.089$	
Diversity = 9.3378e-04	k_t = [3, 2, 3, 2]			
Generation 5870: Best = 3.661243e+03	FES = 587100	$\mu_F = 0.728$	$\mu_{CR} = 0.082$	
Diversity = 9.2038e-04	k_t = [1, 1, 1, 1]			
Generation 5880: Best = 3.661243e+03	FES = 588100	$\mu_F = 0.747$	$\mu_{CR} = 0.086$	
Diversity = 9.0388e-04	k_t = [4, 4, 4, 4]			
Generation 5890: Best = 3.661243e+03	FES = 589100	$\mu_F = 0.740$	$\mu_{CR} = 0.086$	

Diversity = 9.4102e-04	k_t = [1, 1, 1, 1]			
Generation 5900: Best = 3.661243e+03	FES = 590100	$\mu_F = 0.717$	$\mu_{CR} = 0.082$	
Diversity = 9.3327e-04	k_t = [4, 4, 3, 2]			
Generation 5910: Best = 3.661242e+03	FES = 591100	$\mu_F = 0.702$	$\mu_{CR} = 0.078$	
Diversity = 9.6704e-04	k_t = [1, 1, 1, 1]			
Generation 5920: Best = 3.661242e+03	FES = 592100	$\mu_F = 0.681$	$\mu_{CR} = 0.070$	
Diversity = 8.9006e-04	k_t = [4, 2, 4, 4]			
Generation 5930: Best = 3.661242e+03	FES = 593100	$\mu_F = 0.678$	$\mu_{CR} = 0.075$	
Diversity = 8.2474e-04	k_t = [1, 1, 1, 1]			
Generation 5940: Best = 3.661242e+03	FES = 594100	$\mu_F = 0.691$	$\mu_{CR} = 0.071$	
Diversity = 7.9116e-04	k_t = [2, 2, 4, 3]			
Generation 5950: Best = 3.661242e+03	FES = 595100	$\mu_F = 0.702$	$\mu_{CR} = 0.071$	
Diversity = 8.0484e-04	k_t = [1, 1, 1, 1]			
Generation 5960: Best = 3.661242e+03	FES = 596100	$\mu_F = 0.704$	$\mu_{CR} = 0.079$	
Diversity = 7.6739e-04	k_t = [4, 2, 3, 2]			
Generation 5970: Best = 3.661242e+03	FES = 597100	$\mu_F = 0.687$	$\mu_{CR} = 0.079$	
Diversity = 7.0981e-04	k_t = [1, 1, 1, 1]			
Generation 5980: Best = 3.661242e+03	FES = 598100	$\mu_F = 0.704$	$\mu_{CR} = 0.083$	
Diversity = 6.2562e-04	k_t = [2, 4, 4, 3]			
Generation 5990: Best = 3.661242e+03	FES = 599100	$\mu_F = 0.702$	$\mu_{CR} = 0.080$	
Diversity = 6.4521e-04	k_t = [1, 1, 1, 1]			
Generation 6000: Best = 3.661242e+03	FES = 600100	$\mu_F = 0.697$	$\mu_{CR} = 0.083$	
Diversity = 6.0075e-04	k_t = [3, 3, 3, 3]			
Generation 6010: Best = 3.661241e+03	FES = 601100	$\mu_F = 0.704$	$\mu_{CR} = 0.088$	
Diversity = 5.8848e-04	k_t = [1, 1, 1, 1]			
Generation 6020: Best = 3.661241e+03	FES = 602100	$\mu_F = 0.732$	$\mu_{CR} = 0.085$	
Diversity = 5.8387e-04	k_t = [3, 2, 3, 3]			
Generation 6030: Best = 3.661241e+03	FES = 603100	$\mu_F = 0.743$	$\mu_{CR} = 0.080$	
Diversity = 6.0783e-04	k_t = [1, 1, 1, 1]			
Generation 6040: Best = 3.661241e+03	FES = 604100	$\mu_F = 0.729$	$\mu_{CR} = 0.081$	
Diversity = 6.0939e-04	k_t = [3, 3, 2, 3]			
Generation 6050: Best = 3.661241e+03	FES = 605100	$\mu_F = 0.722$	$\mu_{CR} = 0.078$	
Diversity = 5.4998e-04	k_t = [1, 1, 1, 1]			
Generation 6060: Best = 3.661241e+03	FES = 606100	$\mu_F = 0.723$	$\mu_{CR} = 0.077$	
Diversity = 5.7982e-04	k_t = [4, 2, 2, 4]			
Generation 6070: Best = 3.661241e+03	FES = 607100	$\mu_F = 0.735$	$\mu_{CR} = 0.076$	
Diversity = 5.7154e-04	k_t = [1, 1, 1, 1]			
Generation 6080: Best = 3.661241e+03	FES = 608100	$\mu_F = 0.741$	$\mu_{CR} = 0.074$	
Diversity = 5.4067e-04	k_t = [2, 4, 4, 2]			
Generation 6090: Best = 3.661241e+03	FES = 609100	$\mu_F = 0.750$	$\mu_{CR} = 0.077$	
Diversity = 5.8508e-04	k_t = [1, 1, 1, 1]			
Generation 6100: Best = 3.661241e+03	FES = 610100	$\mu_F = 0.744$	$\mu_{CR} = 0.079$	
Diversity = 5.6684e-04	k_t = [4, 2, 4, 2]			
Generation 6110: Best = 3.661241e+03	FES = 611100	$\mu_F = 0.744$	$\mu_{CR} = 0.078$	
Diversity = 5.7772e-04	k_t = [1, 1, 1, 1]			
Generation 6120: Best = 3.661240e+03	FES = 612100	$\mu_F = 0.721$	$\mu_{CR} = 0.072$	
Diversity = 5.1368e-04	k_t = [3, 2, 2, 4]			
Generation 6130: Best = 3.661240e+03	FES = 613100	$\mu_F = 0.742$	$\mu_{CR} = 0.078$	
Diversity = 5.2553e-04	k_t = [1, 1, 1, 1]			
Generation 6140: Best = 3.661240e+03	FES = 614100	$\mu_F = 0.738$	$\mu_{CR} = 0.077$	
Diversity = 5.0799e-04	k_t = [3, 3, 3, 3]			
Generation 6150: Best = 3.661240e+03	FES = 615100	$\mu_F = 0.736$	$\mu_{CR} = 0.082$	
Diversity = 5.3658e-04	k_t = [1, 1, 1, 1]			
Generation 6160: Best = 3.661240e+03	FES = 616100	$\mu_F = 0.727$	$\mu_{CR} = 0.085$	

Diversity = 5.5617e-04	k_t = [2, 2, 4, 2]			
Generation 6170: Best = 3.661240e+03	FES = 617100	$\mu_F = 0.740$	$\mu_{CR} = 0.084$	
Diversity = 5.3801e-04	k_t = [1, 1, 1, 1]			
Generation 6180: Best = 3.661240e+03	FES = 618100	$\mu_F = 0.732$	$\mu_{CR} = 0.090$	
Diversity = 5.6984e-04	k_t = [4, 2, 4, 2]			
Generation 6190: Best = 3.661240e+03	FES = 619100	$\mu_F = 0.733$	$\mu_{CR} = 0.090$	
Diversity = 5.8082e-04	k_t = [1, 1, 1, 1]			
Generation 6200: Best = 3.661239e+03	FES = 620100	$\mu_F = 0.746$	$\mu_{CR} = 0.091$	
Diversity = 6.0734e-04	k_t = [2, 3, 4, 4]			
Generation 6210: Best = 3.661239e+03	FES = 621100	$\mu_F = 0.763$	$\mu_{CR} = 0.102$	
Diversity = 6.3322e-04	k_t = [1, 1, 1, 1]			
Generation 6220: Best = 3.661239e+03	FES = 622100	$\mu_F = 0.747$	$\mu_{CR} = 0.099$	
Diversity = 6.0543e-04	k_t = [4, 3, 4, 3]			
Generation 6230: Best = 3.661239e+03	FES = 623100	$\mu_F = 0.751$	$\mu_{CR} = 0.095$	
Diversity = 5.9303e-04	k_t = [1, 1, 1, 1]			
Generation 6240: Best = 3.661239e+03	FES = 624100	$\mu_F = 0.744$	$\mu_{CR} = 0.097$	
Diversity = 5.7251e-04	k_t = [3, 3, 3, 4]			
Generation 6250: Best = 3.661239e+03	FES = 625100	$\mu_F = 0.729$	$\mu_{CR} = 0.092$	
Diversity = 5.2692e-04	k_t = [1, 1, 1, 1]			
Generation 6260: Best = 3.661239e+03	FES = 626100	$\mu_F = 0.735$	$\mu_{CR} = 0.104$	
Diversity = 4.6321e-04	k_t = [3, 4, 3, 4]			
Generation 6270: Best = 3.661239e+03	FES = 627100	$\mu_F = 0.736$	$\mu_{CR} = 0.098$	
Diversity = 4.6885e-04	k_t = [1, 1, 1, 1]			
Generation 6280: Best = 3.661239e+03	FES = 628100	$\mu_F = 0.734$	$\mu_{CR} = 0.091$	
Diversity = 4.7947e-04	k_t = [3, 4, 4, 4]			
Generation 6290: Best = 3.661238e+03	FES = 629100	$\mu_F = 0.746$	$\mu_{CR} = 0.100$	
Diversity = 4.8538e-04	k_t = [1, 1, 1, 1]			
Generation 6300: Best = 3.661238e+03	FES = 630100	$\mu_F = 0.731$	$\mu_{CR} = 0.100$	
Diversity = 5.1468e-04	k_t = [2, 3, 3, 2]			
Generation 6310: Best = 3.661238e+03	FES = 631100	$\mu_F = 0.715$	$\mu_{CR} = 0.102$	
Diversity = 5.1556e-04	k_t = [1, 1, 1, 1]			
Generation 6320: Best = 3.661238e+03	FES = 632100	$\mu_F = 0.718$	$\mu_{CR} = 0.105$	
Diversity = 4.6734e-04	k_t = [2, 3, 4, 3]			
Generation 6330: Best = 3.661238e+03	FES = 633100	$\mu_F = 0.706$	$\mu_{CR} = 0.106$	
Diversity = 4.6883e-04	k_t = [1, 1, 1, 1]			
Generation 6340: Best = 3.661238e+03	FES = 634100	$\mu_F = 0.712$	$\mu_{CR} = 0.103$	
Diversity = 4.9858e-04	k_t = [2, 3, 2, 2]			
Generation 6350: Best = 3.661238e+03	FES = 635100	$\mu_F = 0.705$	$\mu_{CR} = 0.097$	
Diversity = 5.0024e-04	k_t = [1, 1, 1, 1]			
Generation 6360: Best = 3.661238e+03	FES = 636100	$\mu_F = 0.706$	$\mu_{CR} = 0.097$	
Diversity = 4.6033e-04	k_t = [2, 3, 3, 2]			
Generation 6370: Best = 3.661238e+03	FES = 637100	$\mu_F = 0.701$	$\mu_{CR} = 0.090$	
Diversity = 4.5267e-04	k_t = [1, 1, 1, 1]			
Generation 6380: Best = 3.661238e+03	FES = 638100	$\mu_F = 0.704$	$\mu_{CR} = 0.089$	
Diversity = 4.3835e-04	k_t = [2, 4, 3, 2]			
Generation 6390: Best = 3.661238e+03	FES = 639100	$\mu_F = 0.710$	$\mu_{CR} = 0.086$	
Diversity = 4.6042e-04	k_t = [1, 1, 1, 1]			
Generation 6400: Best = 3.661237e+03	FES = 640100	$\mu_F = 0.717$	$\mu_{CR} = 0.088$	
Diversity = 4.7808e-04	k_t = [3, 2, 3, 3]			
Generation 6410: Best = 3.661237e+03	FES = 641100	$\mu_F = 0.733$	$\mu_{CR} = 0.097$	
Diversity = 4.9908e-04	k_t = [1, 1, 1, 1]			
Generation 6420: Best = 3.661237e+03	FES = 642100	$\mu_F = 0.723$	$\mu_{CR} = 0.102$	
Diversity = 5.0025e-04	k_t = [3, 3, 4, 2]			
Generation 6430: Best = 3.661237e+03	FES = 643100	$\mu_F = 0.736$	$\mu_{CR} = 0.106$	

Diversity = 5.5658e-04	k_t = [1, 1, 1, 1]			
Generation 6440: Best = 3.661237e+03	FES = 644100	$\mu_F = 0.736$	$\mu_{CR} = 0.115$	
Diversity = 5.7831e-04	k_t = [2, 3, 2, 3]			
Generation 6450: Best = 3.661237e+03	FES = 645100	$\mu_F = 0.737$	$\mu_{CR} = 0.124$	
Diversity = 6.0331e-04	k_t = [1, 1, 1, 1]			
Generation 6460: Best = 3.661236e+03	FES = 646100	$\mu_F = 0.742$	$\mu_{CR} = 0.123$	
Diversity = 5.6809e-04	k_t = [4, 4, 4, 3]			
Generation 6470: Best = 3.661236e+03	FES = 647100	$\mu_F = 0.738$	$\mu_{CR} = 0.121$	
Diversity = 4.8950e-04	k_t = [1, 1, 1, 1]			
Generation 6480: Best = 3.661236e+03	FES = 648100	$\mu_F = 0.734$	$\mu_{CR} = 0.123$	
Diversity = 5.0664e-04	k_t = [3, 3, 4, 3]			
Generation 6490: Best = 3.661236e+03	FES = 649100	$\mu_F = 0.735$	$\mu_{CR} = 0.125$	
Diversity = 5.2448e-04	k_t = [1, 1, 1, 1]			
Generation 6500: Best = 3.661236e+03	FES = 650100	$\mu_F = 0.725$	$\mu_{CR} = 0.124$	
Diversity = 5.2381e-04	k_t = [3, 4, 2, 2]			
Generation 6510: Best = 3.661236e+03	FES = 651100	$\mu_F = 0.731$	$\mu_{CR} = 0.120$	
Diversity = 5.7907e-04	k_t = [1, 1, 1, 1]			
Generation 6520: Best = 3.661235e+03	FES = 652100	$\mu_F = 0.721$	$\mu_{CR} = 0.115$	
Diversity = 5.1938e-04	k_t = [4, 4, 3, 4]			
Generation 6530: Best = 3.661235e+03	FES = 653100	$\mu_F = 0.735$	$\mu_{CR} = 0.107$	
Diversity = 5.4250e-04	k_t = [1, 1, 1, 1]			
Generation 6540: Best = 3.661235e+03	FES = 654100	$\mu_F = 0.758$	$\mu_{CR} = 0.119$	
Diversity = 5.2281e-04	k_t = [4, 2, 2, 3]			
Generation 6550: Best = 3.661235e+03	FES = 655100	$\mu_F = 0.747$	$\mu_{CR} = 0.121$	
Diversity = 6.0716e-04	k_t = [1, 1, 1, 1]			
Generation 6560: Best = 3.661235e+03	FES = 656100	$\mu_F = 0.731$	$\mu_{CR} = 0.121$	
Diversity = 5.5489e-04	k_t = [2, 2, 3, 4]			
Generation 6570: Best = 3.661234e+03	FES = 657100	$\mu_F = 0.738$	$\mu_{CR} = 0.125$	
Diversity = 5.9409e-04	k_t = [1, 1, 1, 1]			
Generation 6580: Best = 3.661234e+03	FES = 658100	$\mu_F = 0.744$	$\mu_{CR} = 0.134$	
Diversity = 6.0901e-04	k_t = [2, 4, 3, 2]			
Generation 6590: Best = 3.661234e+03	FES = 659100	$\mu_F = 0.760$	$\mu_{CR} = 0.148$	
Diversity = 6.4585e-04	k_t = [1, 1, 1, 1]			
Generation 6600: Best = 3.661233e+03	FES = 660100	$\mu_F = 0.765$	$\mu_{CR} = 0.140$	
Diversity = 5.9038e-04	k_t = [4, 3, 4, 3]			
Generation 6610: Best = 3.661233e+03	FES = 661100	$\mu_F = 0.771$	$\mu_{CR} = 0.143$	
Diversity = 6.3761e-04	k_t = [1, 1, 1, 1]			
Generation 6620: Best = 3.661233e+03	FES = 662100	$\mu_F = 0.751$	$\mu_{CR} = 0.158$	
Diversity = 6.8747e-04	k_t = [3, 3, 4, 2]			
Generation 6630: Best = 3.661232e+03	FES = 663100	$\mu_F = 0.757$	$\mu_{CR} = 0.154$	
Diversity = 7.3764e-04	k_t = [1, 1, 1, 1]			
Generation 6640: Best = 3.661232e+03	FES = 664100	$\mu_F = 0.756$	$\mu_{CR} = 0.152$	
Diversity = 7.0929e-04	k_t = [2, 2, 3, 4]			
Generation 6650: Best = 3.661232e+03	FES = 665100	$\mu_F = 0.753$	$\mu_{CR} = 0.144$	
Diversity = 7.2462e-04	k_t = [1, 1, 1, 1]			
Generation 6660: Best = 3.661232e+03	FES = 666100	$\mu_F = 0.766$	$\mu_{CR} = 0.135$	
Diversity = 7.7960e-04	k_t = [3, 4, 4, 2]			
Generation 6670: Best = 3.661231e+03	FES = 667100	$\mu_F = 0.767$	$\mu_{CR} = 0.135$	
Diversity = 8.5305e-04	k_t = [1, 1, 1, 1]			
Generation 6680: Best = 3.661231e+03	FES = 668100	$\mu_F = 0.781$	$\mu_{CR} = 0.132$	
Diversity = 8.1875e-04	k_t = [3, 2, 4, 4]			
Generation 6690: Best = 3.661230e+03	FES = 669100	$\mu_F = 0.798$	$\mu_{CR} = 0.132$	
Diversity = 8.8371e-04	k_t = [1, 1, 1, 1]			
Generation 6700: Best = 3.661230e+03	FES = 670100	$\mu_F = 0.783$	$\mu_{CR} = 0.138$	

Diversity = 9.3360e-04	k_t = [4, 3, 3, 4]			
Generation 6710: Best = 3.661230e+03	FES = 671100	$\mu_F = 0.781$	$\mu_{CR} = 0.126$	
Diversity = 9.7592e-04	k_t = [1, 1, 1, 1]			
Generation 6720: Best = 3.661229e+03	FES = 672100	$\mu_F = 0.765$	$\mu_{CR} = 0.122$	
Diversity = 1.0336e-03	k_t = [4, 2, 3, 2]			
Generation 6730: Best = 3.661229e+03	FES = 673100	$\mu_F = 0.753$	$\mu_{CR} = 0.124$	
Diversity = 1.0883e-03	k_t = [1, 1, 1, 1]			
Generation 6740: Best = 3.661228e+03	FES = 674100	$\mu_F = 0.750$	$\mu_{CR} = 0.134$	
Diversity = 1.0352e-03	k_t = [4, 2, 2, 3]			
Generation 6750: Best = 3.661228e+03	FES = 675100	$\mu_F = 0.767$	$\mu_{CR} = 0.144$	
Diversity = 1.0791e-03	k_t = [1, 1, 1, 1]			
Generation 6760: Best = 3.661228e+03	FES = 676100	$\mu_F = 0.775$	$\mu_{CR} = 0.144$	
Diversity = 1.0846e-03	k_t = [4, 2, 2, 2]			
Generation 6770: Best = 3.661228e+03	FES = 677100	$\mu_F = 0.770$	$\mu_{CR} = 0.142$	
Diversity = 1.1027e-03	k_t = [1, 1, 1, 1]			
Generation 6780: Best = 3.661227e+03	FES = 678100	$\mu_F = 0.749$	$\mu_{CR} = 0.136$	
Diversity = 9.9456e-04	k_t = [3, 3, 3, 2]			
Generation 6790: Best = 3.661227e+03	FES = 679100	$\mu_F = 0.743$	$\mu_{CR} = 0.130$	
Diversity = 1.0679e-03	k_t = [1, 1, 1, 1]			
Generation 6800: Best = 3.661226e+03	FES = 680100	$\mu_F = 0.741$	$\mu_{CR} = 0.128$	
Diversity = 1.0485e-03	k_t = [4, 2, 2, 4]			
Generation 6810: Best = 3.661226e+03	FES = 681100	$\mu_F = 0.743$	$\mu_{CR} = 0.134$	
Diversity = 1.0975e-03	k_t = [1, 1, 1, 1]			
Generation 6820: Best = 3.661226e+03	FES = 682100	$\mu_F = 0.748$	$\mu_{CR} = 0.133$	
Diversity = 1.0288e-03	k_t = [4, 4, 3, 4]			
Generation 6830: Best = 3.661225e+03	FES = 683100	$\mu_F = 0.762$	$\mu_{CR} = 0.135$	
Diversity = 1.1700e-03	k_t = [1, 1, 1, 1]			
Generation 6840: Best = 3.661225e+03	FES = 684100	$\mu_F = 0.741$	$\mu_{CR} = 0.128$	
Diversity = 1.1376e-03	k_t = [3, 4, 4, 2]			
Generation 6850: Best = 3.661225e+03	FES = 685100	$\mu_F = 0.744$	$\mu_{CR} = 0.124$	
Diversity = 1.1301e-03	k_t = [1, 1, 1, 1]			
Generation 6860: Best = 3.661225e+03	FES = 686100	$\mu_F = 0.761$	$\mu_{CR} = 0.110$	
Diversity = 1.0457e-03	k_t = [3, 2, 4, 3]			
Generation 6870: Best = 3.661224e+03	FES = 687100	$\mu_F = 0.736$	$\mu_{CR} = 0.108$	
Diversity = 1.0456e-03	k_t = [1, 1, 1, 1]			
Generation 6880: Best = 3.661224e+03	FES = 688100	$\mu_F = 0.732$	$\mu_{CR} = 0.119$	
Diversity = 1.0421e-03	k_t = [2, 4, 3, 2]			
Generation 6890: Best = 3.661224e+03	FES = 689100	$\mu_F = 0.728$	$\mu_{CR} = 0.110$	
Diversity = 1.0126e-03	k_t = [1, 1, 1, 1]			
Generation 6900: Best = 3.661223e+03	FES = 690100	$\mu_F = 0.718$	$\mu_{CR} = 0.119$	
Diversity = 9.8940e-04	k_t = [4, 4, 4, 4]			
Generation 6910: Best = 3.661223e+03	FES = 691100	$\mu_F = 0.736$	$\mu_{CR} = 0.113$	
Diversity = 9.8832e-04	k_t = [1, 1, 1, 1]			
Generation 6920: Best = 3.661223e+03	FES = 692100	$\mu_F = 0.735$	$\mu_{CR} = 0.112$	
Diversity = 9.3171e-04	k_t = [2, 2, 2, 4]			
Generation 6930: Best = 3.661223e+03	FES = 693100	$\mu_F = 0.733$	$\mu_{CR} = 0.114$	
Diversity = 8.9547e-04	k_t = [1, 1, 1, 1]			
Generation 6940: Best = 3.661223e+03	FES = 694100	$\mu_F = 0.724$	$\mu_{CR} = 0.113$	
Diversity = 8.4381e-04	k_t = [3, 2, 4, 4]			
Generation 6950: Best = 3.661222e+03	FES = 695100	$\mu_F = 0.710$	$\mu_{CR} = 0.126$	
Diversity = 8.9746e-04	k_t = [1, 1, 1, 1]			
Generation 6960: Best = 3.661222e+03	FES = 696100	$\mu_F = 0.715$	$\mu_{CR} = 0.114$	
Diversity = 8.1622e-04	k_t = [3, 4, 2, 3]			
Generation 6970: Best = 3.661222e+03	FES = 697100	$\mu_F = 0.743$	$\mu_{CR} = 0.111$	

Diversity = 8.8109e-04	k_t = [1, 1, 1, 1]			
Generation 6980: Best = 3.661222e+03	FES = 698100	$\mu_F = 0.741$	$\mu_{CR} = 0.107$	
Diversity = 7.8538e-04	k_t = [3, 2, 3, 3]			
Generation 6990: Best = 3.661222e+03	FES = 699100	$\mu_F = 0.748$	$\mu_{CR} = 0.113$	
Diversity = 8.3747e-04	k_t = [1, 1, 1, 1]			
Generation 7000: Best = 3.661222e+03	FES = 700100	$\mu_F = 0.739$	$\mu_{CR} = 0.109$	
Diversity = 8.0537e-04	k_t = [2, 3, 4, 2]			
Generation 7010: Best = 3.661222e+03	FES = 701100	$\mu_F = 0.744$	$\mu_{CR} = 0.100$	
Diversity = 7.9147e-04	k_t = [1, 1, 1, 1]			
Generation 7020: Best = 3.661222e+03	FES = 702100	$\mu_F = 0.750$	$\mu_{CR} = 0.084$	
Diversity = 7.8011e-04	k_t = [3, 3, 2, 2]			
Generation 7030: Best = 3.661222e+03	FES = 703100	$\mu_F = 0.763$	$\mu_{CR} = 0.085$	
Diversity = 7.4669e-04	k_t = [1, 1, 1, 1]			
Generation 7040: Best = 3.661221e+03	FES = 704100	$\mu_F = 0.763$	$\mu_{CR} = 0.092$	
Diversity = 7.5175e-04	k_t = [2, 2, 2, 3]			
Generation 7050: Best = 3.661221e+03	FES = 705100	$\mu_F = 0.752$	$\mu_{CR} = 0.094$	
Diversity = 7.4938e-04	k_t = [1, 1, 1, 1]			
Generation 7060: Best = 3.661221e+03	FES = 706100	$\mu_F = 0.752$	$\mu_{CR} = 0.101$	
Diversity = 7.1993e-04	k_t = [2, 3, 3, 2]			
Generation 7070: Best = 3.661221e+03	FES = 707100	$\mu_F = 0.740$	$\mu_{CR} = 0.102$	
Diversity = 6.9730e-04	k_t = [1, 1, 1, 1]			
Generation 7080: Best = 3.661221e+03	FES = 708100	$\mu_F = 0.735$	$\mu_{CR} = 0.104$	
Diversity = 6.7246e-04	k_t = [2, 3, 4, 4]			
Generation 7090: Best = 3.661221e+03	FES = 709100	$\mu_F = 0.728$	$\mu_{CR} = 0.099$	
Diversity = 6.6942e-04	k_t = [1, 1, 1, 1]			
Generation 7100: Best = 3.661221e+03	FES = 710100	$\mu_F = 0.723$	$\mu_{CR} = 0.092$	
Diversity = 6.3070e-04	k_t = [2, 4, 2, 3]			
Generation 7110: Best = 3.661220e+03	FES = 711100	$\mu_F = 0.732$	$\mu_{CR} = 0.089$	
Diversity = 6.5459e-04	k_t = [1, 1, 1, 1]			
Generation 7120: Best = 3.661220e+03	FES = 712100	$\mu_F = 0.738$	$\mu_{CR} = 0.105$	
Diversity = 7.1177e-04	k_t = [4, 2, 2, 3]			
Generation 7130: Best = 3.661220e+03	FES = 713100	$\mu_F = 0.709$	$\mu_{CR} = 0.111$	
Diversity = 6.8600e-04	k_t = [1, 1, 1, 1]			
Generation 7140: Best = 3.661220e+03	FES = 714100	$\mu_F = 0.720$	$\mu_{CR} = 0.109$	
Diversity = 6.5938e-04	k_t = [2, 3, 4, 3]			
Generation 7150: Best = 3.661220e+03	FES = 715100	$\mu_F = 0.707$	$\mu_{CR} = 0.112$	
Diversity = 6.4132e-04	k_t = [1, 1, 1, 1]			
Generation 7160: Best = 3.661219e+03	FES = 716100	$\mu_F = 0.704$	$\mu_{CR} = 0.111$	
Diversity = 5.9908e-04	k_t = [3, 3, 3, 2]			
Generation 7170: Best = 3.661219e+03	FES = 717100	$\mu_F = 0.711$	$\mu_{CR} = 0.108$	
Diversity = 6.4678e-04	k_t = [1, 1, 1, 1]			
Generation 7180: Best = 3.661219e+03	FES = 718100	$\mu_F = 0.707$	$\mu_{CR} = 0.114$	
Diversity = 6.0128e-04	k_t = [3, 3, 2, 3]			
Generation 7190: Best = 3.661219e+03	FES = 719100	$\mu_F = 0.717$	$\mu_{CR} = 0.113$	
Diversity = 6.2700e-04	k_t = [1, 1, 1, 1]			
Generation 7200: Best = 3.661219e+03	FES = 720100	$\mu_F = 0.730$	$\mu_{CR} = 0.108$	
Diversity = 6.5859e-04	k_t = [2, 2, 4, 3]			
Generation 7210: Best = 3.661219e+03	FES = 721100	$\mu_F = 0.742$	$\mu_{CR} = 0.116$	
Diversity = 6.7535e-04	k_t = [1, 1, 1, 1]			
Generation 7220: Best = 3.661218e+03	FES = 722100	$\mu_F = 0.732$	$\mu_{CR} = 0.116$	
Diversity = 6.9382e-04	k_t = [2, 3, 2, 3]			
Generation 7230: Best = 3.661218e+03	FES = 723100	$\mu_F = 0.720$	$\mu_{CR} = 0.111$	
Diversity = 7.3932e-04	k_t = [1, 1, 1, 1]			
Generation 7240: Best = 3.661218e+03	FES = 724100	$\mu_F = 0.719$	$\mu_{CR} = 0.120$	

Diversity = 7.0952e-04	k_t = [4, 2, 2, 3]			
Generation 7250: Best = 3.661218e+03	FES = 725100	$\mu_F = 0.725$	$\mu_{CR} = 0.118$	
Diversity = 7.7650e-04	k_t = [1, 1, 1, 1]			
Generation 7260: Best = 3.661218e+03	FES = 726100	$\mu_F = 0.733$	$\mu_{CR} = 0.100$	
Diversity = 7.2255e-04	k_t = [2, 4, 3, 2]			
Generation 7270: Best = 3.661216e+03	FES = 727100	$\mu_F = 0.754$	$\mu_{CR} = 0.107$	
Diversity = 7.6575e-04	k_t = [1, 1, 1, 1]			
Generation 7280: Best = 3.661216e+03	FES = 728100	$\mu_F = 0.758$	$\mu_{CR} = 0.105$	
Diversity = 8.2505e-04	k_t = [3, 4, 3, 4]			
Generation 7290: Best = 3.661216e+03	FES = 729100	$\mu_F = 0.757$	$\mu_{CR} = 0.112$	
Diversity = 9.1732e-04	k_t = [1, 1, 1, 1]			
Generation 7300: Best = 3.661215e+03	FES = 730100	$\mu_F = 0.764$	$\mu_{CR} = 0.111$	
Diversity = 8.8416e-04	k_t = [3, 3, 3, 2]			
Generation 7310: Best = 3.661215e+03	FES = 731100	$\mu_F = 0.773$	$\mu_{CR} = 0.111$	
Diversity = 1.0172e-03	k_t = [1, 1, 1, 1]			
Generation 7320: Best = 3.661214e+03	FES = 732100	$\mu_F = 0.773$	$\mu_{CR} = 0.113$	
Diversity = 1.1731e-03	k_t = [2, 2, 3, 2]			
Generation 7330: Best = 3.661214e+03	FES = 733100	$\mu_F = 0.782$	$\mu_{CR} = 0.110$	
Diversity = 1.2227e-03	k_t = [1, 1, 1, 1]			
Generation 7340: Best = 3.661214e+03	FES = 734100	$\mu_F = 0.780$	$\mu_{CR} = 0.109$	
Diversity = 1.2215e-03	k_t = [4, 4, 2, 2]			
Generation 7350: Best = 3.661213e+03	FES = 735100	$\mu_F = 0.765$	$\mu_{CR} = 0.109$	
Diversity = 1.2415e-03	k_t = [1, 1, 1, 1]			
Generation 7360: Best = 3.661213e+03	FES = 736100	$\mu_F = 0.757$	$\mu_{CR} = 0.106$	
Diversity = 1.1796e-03	k_t = [3, 3, 2, 4]			
Generation 7370: Best = 3.661213e+03	FES = 737100	$\mu_F = 0.754$	$\mu_{CR} = 0.104$	
Diversity = 1.2318e-03	k_t = [1, 1, 1, 1]			
Generation 7380: Best = 3.661212e+03	FES = 738100	$\mu_F = 0.762$	$\mu_{CR} = 0.102$	
Diversity = 1.3611e-03	k_t = [3, 4, 2, 4]			
Generation 7390: Best = 3.661211e+03	FES = 739100	$\mu_F = 0.772$	$\mu_{CR} = 0.115$	
Diversity = 1.4326e-03	k_t = [1, 1, 1, 1]			
Generation 7400: Best = 3.661211e+03	FES = 740100	$\mu_F = 0.763$	$\mu_{CR} = 0.133$	
Diversity = 1.5687e-03	k_t = [3, 3, 2, 2]			
Generation 7410: Best = 3.661210e+03	FES = 741100	$\mu_F = 0.733$	$\mu_{CR} = 0.132$	
Diversity = 1.4979e-03	k_t = [1, 1, 1, 1]			
Generation 7420: Best = 3.661210e+03	FES = 742100	$\mu_F = 0.732$	$\mu_{CR} = 0.119$	
Diversity = 1.3766e-03	k_t = [4, 3, 3, 3]			
Generation 7430: Best = 3.661210e+03	FES = 743100	$\mu_F = 0.730$	$\mu_{CR} = 0.115$	
Diversity = 1.3564e-03	k_t = [1, 1, 1, 1]			
Generation 7440: Best = 3.661209e+03	FES = 744100	$\mu_F = 0.733$	$\mu_{CR} = 0.107$	
Diversity = 1.2775e-03	k_t = [2, 4, 4, 2]			
Generation 7450: Best = 3.661209e+03	FES = 745100	$\mu_F = 0.729$	$\mu_{CR} = 0.102$	
Diversity = 1.1289e-03	k_t = [1, 1, 1, 1]			
Generation 7460: Best = 3.661209e+03	FES = 746100	$\mu_F = 0.740$	$\mu_{CR} = 0.096$	
Diversity = 1.1372e-03	k_t = [4, 2, 2, 4]			
Generation 7470: Best = 3.661209e+03	FES = 747100	$\mu_F = 0.735$	$\mu_{CR} = 0.105$	
Diversity = 1.0968e-03	k_t = [1, 1, 1, 1]			
Generation 7480: Best = 3.661208e+03	FES = 748100	$\mu_F = 0.733$	$\mu_{CR} = 0.116$	
Diversity = 1.0510e-03	k_t = [3, 2, 2, 4]			
Generation 7490: Best = 3.661208e+03	FES = 749100	$\mu_F = 0.754$	$\mu_{CR} = 0.118$	
Diversity = 1.0302e-03	k_t = [1, 1, 1, 1]			
Generation 7500: Best = 3.661208e+03	FES = 750100	$\mu_F = 0.771$	$\mu_{CR} = 0.120$	
Diversity = 1.0516e-03	k_t = [2, 2, 4, 2]			
Generation 7510: Best = 3.661207e+03	FES = 751100	$\mu_F = 0.756$	$\mu_{CR} = 0.117$	

Diversity = 1.0179e-03	k_t = [1, 1, 1, 1]			
Generation 7520: Best = 3.661207e+03	FES = 752100	$\mu_F = 0.754$	$\mu_{CR} = 0.110$	
Diversity = 9.0406e-04	k_t = [2, 4, 4, 4]			
Generation 7530: Best = 3.661207e+03	FES = 753100	$\mu_F = 0.765$	$\mu_{CR} = 0.098$	
Diversity = 8.7777e-04	k_t = [1, 1, 1, 1]			
Generation 7540: Best = 3.661207e+03	FES = 754100	$\mu_F = 0.764$	$\mu_{CR} = 0.097$	
Diversity = 8.6224e-04	k_t = [2, 3, 2, 4]			
Generation 7550: Best = 3.661207e+03	FES = 755100	$\mu_F = 0.791$	$\mu_{CR} = 0.095$	
Diversity = 8.6016e-04	k_t = [1, 1, 1, 1]			
Generation 7560: Best = 3.661206e+03	FES = 756100	$\mu_F = 0.794$	$\mu_{CR} = 0.097$	
Diversity = 9.1153e-04	k_t = [2, 3, 3, 3]			
Generation 7570: Best = 3.661206e+03	FES = 757100	$\mu_F = 0.784$	$\mu_{CR} = 0.109$	
Diversity = 9.9559e-04	k_t = [1, 1, 1, 1]			
Generation 7580: Best = 3.661206e+03	FES = 758100	$\mu_F = 0.755$	$\mu_{CR} = 0.105$	
Diversity = 9.2152e-04	k_t = [3, 4, 4, 2]			
Generation 7590: Best = 3.661206e+03	FES = 759100	$\mu_F = 0.755$	$\mu_{CR} = 0.096$	
Diversity = 1.0367e-03	k_t = [1, 1, 1, 1]			
Generation 7600: Best = 3.661206e+03	FES = 760100	$\mu_F = 0.755$	$\mu_{CR} = 0.093$	
Diversity = 9.5842e-04	k_t = [2, 4, 4, 4]			
Generation 7610: Best = 3.661205e+03	FES = 761100	$\mu_F = 0.750$	$\mu_{CR} = 0.093$	
Diversity = 9.8135e-04	k_t = [1, 1, 1, 1]			
Generation 7620: Best = 3.661205e+03	FES = 762100	$\mu_F = 0.750$	$\mu_{CR} = 0.092$	
Diversity = 1.0085e-03	k_t = [4, 3, 2, 2]			
Generation 7630: Best = 3.661205e+03	FES = 763100	$\mu_F = 0.745$	$\mu_{CR} = 0.099$	
Diversity = 1.0767e-03	k_t = [1, 1, 1, 1]			
Generation 7640: Best = 3.661204e+03	FES = 764100	$\mu_F = 0.742$	$\mu_{CR} = 0.100$	
Diversity = 1.0067e-03	k_t = [4, 2, 4, 4]			
Generation 7650: Best = 3.661204e+03	FES = 765100	$\mu_F = 0.752$	$\mu_{CR} = 0.097$	
Diversity = 9.1568e-04	k_t = [1, 1, 1, 1]			
Generation 7660: Best = 3.661204e+03	FES = 766100	$\mu_F = 0.752$	$\mu_{CR} = 0.104$	
Diversity = 8.6778e-04	k_t = [3, 4, 4, 4]			
Generation 7670: Best = 3.661204e+03	FES = 767100	$\mu_F = 0.750$	$\mu_{CR} = 0.107$	
Diversity = 8.5926e-04	k_t = [1, 1, 1, 1]			
Generation 7680: Best = 3.661204e+03	FES = 768100	$\mu_F = 0.765$	$\mu_{CR} = 0.102$	
Diversity = 8.5126e-04	k_t = [4, 4, 3, 3]			
Generation 7690: Best = 3.661203e+03	FES = 769100	$\mu_F = 0.767$	$\mu_{CR} = 0.110$	
Diversity = 9.4519e-04	k_t = [1, 1, 1, 1]			
Generation 7700: Best = 3.661203e+03	FES = 770100	$\mu_F = 0.766$	$\mu_{CR} = 0.108$	
Diversity = 9.4676e-04	k_t = [2, 4, 3, 2]			
Generation 7710: Best = 3.661203e+03	FES = 771100	$\mu_F = 0.764$	$\mu_{CR} = 0.105$	
Diversity = 9.8960e-04	k_t = [1, 1, 1, 1]			
Generation 7720: Best = 3.661203e+03	FES = 772100	$\mu_F = 0.745$	$\mu_{CR} = 0.116$	
Diversity = 8.9577e-04	k_t = [4, 3, 3, 4]			
Generation 7730: Best = 3.661202e+03	FES = 773100	$\mu_F = 0.745$	$\mu_{CR} = 0.119$	
Diversity = 9.0488e-04	k_t = [1, 1, 1, 1]			
Generation 7740: Best = 3.661202e+03	FES = 774100	$\mu_F = 0.736$	$\mu_{CR} = 0.119$	
Diversity = 8.6047e-04	k_t = [4, 4, 4, 3]			
Generation 7750: Best = 3.661202e+03	FES = 775100	$\mu_F = 0.743$	$\mu_{CR} = 0.116$	
Diversity = 8.7236e-04	k_t = [1, 1, 1, 1]			
Generation 7760: Best = 3.661201e+03	FES = 776100	$\mu_F = 0.746$	$\mu_{CR} = 0.115$	
Diversity = 8.9265e-04	k_t = [2, 3, 2, 4]			
Generation 7770: Best = 3.661201e+03	FES = 777100	$\mu_F = 0.759$	$\mu_{CR} = 0.105$	
Diversity = 8.8134e-04	k_t = [1, 1, 1, 1]			
Generation 7780: Best = 3.661201e+03	FES = 778100	$\mu_F = 0.747$	$\mu_{CR} = 0.100$	

Diversity = 9.1339e-04	k_t = [3, 4, 3, 2]			
Generation 7790: Best = 3.661200e+03	FES = 779100	$\mu_F = 0.735$	$\mu_{CR} = 0.115$	
Diversity = 1.0831e-03	k_t = [1, 1, 1, 1]			
Generation 7800: Best = 3.661199e+03	FES = 780100	$\mu_F = 0.736$	$\mu_{CR} = 0.113$	
Diversity = 1.1084e-03	k_t = [3, 4, 2, 2]			
Generation 7810: Best = 3.661199e+03	FES = 781100	$\mu_F = 0.728$	$\mu_{CR} = 0.134$	
Diversity = 1.1163e-03	k_t = [1, 1, 1, 1]			
Generation 7820: Best = 3.661198e+03	FES = 782100	$\mu_F = 0.732$	$\mu_{CR} = 0.133$	
Diversity = 1.1388e-03	k_t = [3, 2, 4, 2]			
Generation 7830: Best = 3.661197e+03	FES = 783100	$\mu_F = 0.744$	$\mu_{CR} = 0.128$	
Diversity = 1.3486e-03	k_t = [1, 1, 1, 1]			
Generation 7840: Best = 3.661196e+03	FES = 784100	$\mu_F = 0.744$	$\mu_{CR} = 0.127$	
Diversity = 1.4284e-03	k_t = [3, 2, 3, 3]			
Generation 7850: Best = 3.661195e+03	FES = 785100	$\mu_F = 0.759$	$\mu_{CR} = 0.128$	
Diversity = 1.5061e-03	k_t = [1, 1, 1, 1]			
Generation 7860: Best = 3.661194e+03	FES = 786100	$\mu_F = 0.760$	$\mu_{CR} = 0.123$	
Diversity = 1.3991e-03	k_t = [3, 3, 4, 4]			
Generation 7870: Best = 3.661194e+03	FES = 787100	$\mu_F = 0.753$	$\mu_{CR} = 0.128$	
Diversity = 1.3878e-03	k_t = [1, 1, 1, 1]			
Generation 7880: Best = 3.661193e+03	FES = 788100	$\mu_F = 0.743$	$\mu_{CR} = 0.133$	
Diversity = 1.2886e-03	k_t = [3, 3, 3, 2]			
Generation 7890: Best = 3.661193e+03	FES = 789100	$\mu_F = 0.753$	$\mu_{CR} = 0.120$	
Diversity = 1.2870e-03	k_t = [1, 1, 1, 1]			
Generation 7900: Best = 3.661193e+03	FES = 790100	$\mu_F = 0.739$	$\mu_{CR} = 0.130$	
Diversity = 1.0795e-03	k_t = [4, 2, 4, 4]			
Generation 7910: Best = 3.661193e+03	FES = 791100	$\mu_F = 0.732$	$\mu_{CR} = 0.130$	
Diversity = 1.1157e-03	k_t = [1, 1, 1, 1]			
Generation 7920: Best = 3.661192e+03	FES = 792100	$\mu_F = 0.729$	$\mu_{CR} = 0.114$	
Diversity = 1.0610e-03	k_t = [4, 2, 2, 4]			
Generation 7930: Best = 3.661192e+03	FES = 793100	$\mu_F = 0.705$	$\mu_{CR} = 0.110$	
Diversity = 1.0366e-03	k_t = [1, 1, 1, 1]			
Generation 7940: Best = 3.661192e+03	FES = 794100	$\mu_F = 0.695$	$\mu_{CR} = 0.107$	
Diversity = 9.7010e-04	k_t = [4, 3, 2, 2]			
Generation 7950: Best = 3.661191e+03	FES = 795100	$\mu_F = 0.696$	$\mu_{CR} = 0.097$	
Diversity = 9.5603e-04	k_t = [1, 1, 1, 1]			
Generation 7960: Best = 3.661191e+03	FES = 796100	$\mu_F = 0.708$	$\mu_{CR} = 0.100$	
Diversity = 9.9992e-04	k_t = [3, 3, 2, 3]			
Generation 7970: Best = 3.661191e+03	FES = 797100	$\mu_F = 0.721$	$\mu_{CR} = 0.104$	
Diversity = 1.0232e-03	k_t = [1, 1, 1, 1]			
Generation 7980: Best = 3.661190e+03	FES = 798100	$\mu_F = 0.725$	$\mu_{CR} = 0.109$	
Diversity = 1.0929e-03	k_t = [2, 2, 3, 3]			
Generation 7990: Best = 3.661190e+03	FES = 799100	$\mu_F = 0.722$	$\mu_{CR} = 0.109$	
Diversity = 1.1174e-03	k_t = [1, 1, 1, 1]			
Generation 8000: Best = 3.661189e+03	FES = 800100	$\mu_F = 0.731$	$\mu_{CR} = 0.105$	
Diversity = 1.0747e-03	k_t = [2, 2, 3, 4]			
Generation 8010: Best = 3.661188e+03	FES = 801100	$\mu_F = 0.753$	$\mu_{CR} = 0.103$	
Diversity = 1.1527e-03	k_t = [1, 1, 1, 1]			
Generation 8020: Best = 3.661188e+03	FES = 802100	$\mu_F = 0.742$	$\mu_{CR} = 0.102$	
Diversity = 1.1082e-03	k_t = [4, 2, 3, 2]			
Generation 8030: Best = 3.661188e+03	FES = 803100	$\mu_F = 0.743$	$\mu_{CR} = 0.107$	
Diversity = 1.1554e-03	k_t = [1, 1, 1, 1]			
Generation 8040: Best = 3.661188e+03	FES = 804100	$\mu_F = 0.747$	$\mu_{CR} = 0.110$	
Diversity = 1.2465e-03	k_t = [2, 2, 2, 2]			
Generation 8050: Best = 3.661188e+03	FES = 805100	$\mu_F = 0.751$	$\mu_{CR} = 0.108$	

Diversity = 1.1487e-03	k_t = [1, 1, 1, 1]			
Generation 8060: Best = 3.661187e+03	FES = 806100	$\mu_F = 0.765$	$\mu_{CR} = 0.110$	
Diversity = 1.0158e-03	k_t = [4, 3, 4, 4]			
Generation 8070: Best = 3.661187e+03	FES = 807100	$\mu_F = 0.777$	$\mu_{CR} = 0.110$	
Diversity = 1.0128e-03	k_t = [1, 1, 1, 1]			
Generation 8080: Best = 3.661186e+03	FES = 808100	$\mu_F = 0.774$	$\mu_{CR} = 0.110$	
Diversity = 1.0812e-03	k_t = [3, 2, 2, 2]			
Generation 8090: Best = 3.661186e+03	FES = 809100	$\mu_F = 0.775$	$\mu_{CR} = 0.117$	
Diversity = 1.2757e-03	k_t = [1, 1, 1, 1]			
Generation 8100: Best = 3.661185e+03	FES = 810100	$\mu_F = 0.772$	$\mu_{CR} = 0.120$	
Diversity = 1.3761e-03	k_t = [3, 2, 2, 4]			
Generation 8110: Best = 3.661184e+03	FES = 811100	$\mu_F = 0.771$	$\mu_{CR} = 0.117$	
Diversity = 1.3836e-03	k_t = [1, 1, 1, 1]			
Generation 8120: Best = 3.661184e+03	FES = 812100	$\mu_F = 0.779$	$\mu_{CR} = 0.120$	
Diversity = 1.4003e-03	k_t = [4, 2, 3, 4]			
Generation 8130: Best = 3.661183e+03	FES = 813100	$\mu_F = 0.769$	$\mu_{CR} = 0.122$	
Diversity = 1.4014e-03	k_t = [1, 1, 1, 1]			
Generation 8140: Best = 3.661182e+03	FES = 814100	$\mu_F = 0.777$	$\mu_{CR} = 0.119$	
Diversity = 1.3463e-03	k_t = [2, 4, 4, 4]			
Generation 8150: Best = 3.661182e+03	FES = 815100	$\mu_F = 0.782$	$\mu_{CR} = 0.123$	
Diversity = 1.4159e-03	k_t = [1, 1, 1, 1]			
Generation 8160: Best = 3.661181e+03	FES = 816100	$\mu_F = 0.789$	$\mu_{CR} = 0.123$	
Diversity = 1.5366e-03	k_t = [3, 2, 4, 2]			
Generation 8170: Best = 3.661181e+03	FES = 817100	$\mu_F = 0.775$	$\mu_{CR} = 0.127$	
Diversity = 1.6199e-03	k_t = [1, 1, 1, 1]			
Generation 8180: Best = 3.661180e+03	FES = 818100	$\mu_F = 0.760$	$\mu_{CR} = 0.123$	
Diversity = 1.5996e-03	k_t = [4, 4, 2, 3]			
Generation 8190: Best = 3.661179e+03	FES = 819100	$\mu_F = 0.752$	$\mu_{CR} = 0.116$	
Diversity = 1.5886e-03	k_t = [1, 1, 1, 1]			
Generation 8200: Best = 3.661179e+03	FES = 820100	$\mu_F = 0.735$	$\mu_{CR} = 0.112$	
Diversity = 1.4985e-03	k_t = [3, 3, 4, 3]			
Generation 8210: Best = 3.661178e+03	FES = 821100	$\mu_F = 0.738$	$\mu_{CR} = 0.112$	
Diversity = 1.3454e-03	k_t = [1, 1, 1, 1]			
Generation 8220: Best = 3.661178e+03	FES = 822100	$\mu_F = 0.745$	$\mu_{CR} = 0.113$	
Diversity = 1.3702e-03	k_t = [3, 2, 3, 3]			
Generation 8230: Best = 3.661177e+03	FES = 823100	$\mu_F = 0.748$	$\mu_{CR} = 0.116$	
Diversity = 1.4244e-03	k_t = [1, 1, 1, 1]			
Generation 8240: Best = 3.661177e+03	FES = 824100	$\mu_F = 0.741$	$\mu_{CR} = 0.116$	
Diversity = 1.3797e-03	k_t = [3, 3, 4, 3]			
Generation 8250: Best = 3.661176e+03	FES = 825100	$\mu_F = 0.748$	$\mu_{CR} = 0.114$	
Diversity = 1.4397e-03	k_t = [1, 1, 1, 1]			
Generation 8260: Best = 3.661174e+03	FES = 826100	$\mu_F = 0.741$	$\mu_{CR} = 0.118$	
Diversity = 1.5111e-03	k_t = [4, 3, 2, 3]			
Generation 8270: Best = 3.661173e+03	FES = 827100	$\mu_F = 0.751$	$\mu_{CR} = 0.122$	
Diversity = 1.7472e-03	k_t = [1, 1, 1, 1]			
Generation 8280: Best = 3.661173e+03	FES = 828100	$\mu_F = 0.745$	$\mu_{CR} = 0.124$	
Diversity = 1.6699e-03	k_t = [2, 2, 4, 3]			
Generation 8290: Best = 3.661172e+03	FES = 829100	$\mu_F = 0.729$	$\mu_{CR} = 0.114$	
Diversity = 1.4946e-03	k_t = [1, 1, 1, 1]			
Generation 8300: Best = 3.661172e+03	FES = 830100	$\mu_F = 0.729$	$\mu_{CR} = 0.108$	
Diversity = 1.3755e-03	k_t = [2, 2, 2, 2]			
Generation 8310: Best = 3.661172e+03	FES = 831100	$\mu_F = 0.702$	$\mu_{CR} = 0.107$	
Diversity = 1.3899e-03	k_t = [1, 1, 1, 1]			
Generation 8320: Best = 3.661172e+03	FES = 832100	$\mu_F = 0.696$	$\mu_{CR} = 0.097$	

Diversity = 1.2367e-03	k_t = [2, 4, 3, 4]			
Generation 8330: Best = 3.661171e+03	FES = 833100	$\mu_F = 0.717$	$\mu_{CR} = 0.099$	
Diversity = 1.1412e-03	k_t = [1, 1, 1, 1]			
Generation 8340: Best = 3.661171e+03	FES = 834100	$\mu_F = 0.723$	$\mu_{CR} = 0.094$	
Diversity = 1.0624e-03	k_t = [4, 4, 4, 4]			
Generation 8350: Best = 3.661171e+03	FES = 835100	$\mu_F = 0.736$	$\mu_{CR} = 0.092$	
Diversity = 1.0760e-03	k_t = [1, 1, 1, 1]			
Generation 8360: Best = 3.661170e+03	FES = 836100	$\mu_F = 0.748$	$\mu_{CR} = 0.101$	
Diversity = 1.0901e-03	k_t = [2, 4, 3, 3]			
Generation 8370: Best = 3.661170e+03	FES = 837100	$\mu_F = 0.762$	$\mu_{CR} = 0.095$	
Diversity = 1.1925e-03	k_t = [1, 1, 1, 1]			
Generation 8380: Best = 3.661170e+03	FES = 838100	$\mu_F = 0.764$	$\mu_{CR} = 0.100$	
Diversity = 1.2296e-03	k_t = [4, 3, 4, 4]			
Generation 8390: Best = 3.661169e+03	FES = 839100	$\mu_F = 0.746$	$\mu_{CR} = 0.107$	
Diversity = 1.2960e-03	k_t = [1, 1, 1, 1]			
Generation 8400: Best = 3.661169e+03	FES = 840100	$\mu_F = 0.742$	$\mu_{CR} = 0.111$	
Diversity = 1.2931e-03	k_t = [2, 2, 2, 3]			
Generation 8410: Best = 3.661169e+03	FES = 841100	$\mu_F = 0.736$	$\mu_{CR} = 0.102$	
Diversity = 1.3484e-03	k_t = [1, 1, 1, 1]			
Generation 8420: Best = 3.661169e+03	FES = 842100	$\mu_F = 0.718$	$\mu_{CR} = 0.096$	
Diversity = 1.3047e-03	k_t = [4, 3, 2, 3]			
Generation 8430: Best = 3.661169e+03	FES = 843100	$\mu_F = 0.717$	$\mu_{CR} = 0.092$	
Diversity = 1.2409e-03	k_t = [1, 1, 1, 1]			
Generation 8440: Best = 3.661168e+03	FES = 844100	$\mu_F = 0.729$	$\mu_{CR} = 0.099$	
Diversity = 1.2844e-03	k_t = [4, 4, 4, 2]			
Generation 8450: Best = 3.661168e+03	FES = 845100	$\mu_F = 0.723$	$\mu_{CR} = 0.100$	
Diversity = 1.2074e-03	k_t = [1, 1, 1, 1]			
Generation 8460: Best = 3.661168e+03	FES = 846100	$\mu_F = 0.726$	$\mu_{CR} = 0.099$	
Diversity = 1.1628e-03	k_t = [4, 2, 4, 2]			
Generation 8470: Best = 3.661167e+03	FES = 847100	$\mu_F = 0.730$	$\mu_{CR} = 0.094$	
Diversity = 1.1681e-03	k_t = [1, 1, 1, 1]			
Generation 8480: Best = 3.661166e+03	FES = 848100	$\mu_F = 0.748$	$\mu_{CR} = 0.086$	
Diversity = 1.1560e-03	k_t = [4, 2, 4, 4]			
Generation 8490: Best = 3.661166e+03	FES = 849100	$\mu_F = 0.771$	$\mu_{CR} = 0.086$	
Diversity = 1.1924e-03	k_t = [1, 1, 1, 1]			
Generation 8500: Best = 3.661165e+03	FES = 850100	$\mu_F = 0.781$	$\mu_{CR} = 0.082$	
Diversity = 1.1951e-03	k_t = [2, 2, 3, 4]			
Generation 8510: Best = 3.661165e+03	FES = 851100	$\mu_F = 0.777$	$\mu_{CR} = 0.089$	
Diversity = 1.2099e-03	k_t = [1, 1, 1, 1]			
Generation 8520: Best = 3.661164e+03	FES = 852100	$\mu_F = 0.779$	$\mu_{CR} = 0.095$	
Diversity = 1.2401e-03	k_t = [3, 2, 3, 3]			
Generation 8530: Best = 3.661163e+03	FES = 853100	$\mu_F = 0.786$	$\mu_{CR} = 0.102$	
Diversity = 1.3305e-03	k_t = [1, 1, 1, 1]			
Generation 8540: Best = 3.661162e+03	FES = 854100	$\mu_F = 0.775$	$\mu_{CR} = 0.100$	
Diversity = 1.4691e-03	k_t = [3, 2, 2, 2]			
Generation 8550: Best = 3.661161e+03	FES = 855100	$\mu_F = 0.782$	$\mu_{CR} = 0.111$	
Diversity = 1.6313e-03	k_t = [1, 1, 1, 1]			
Generation 8560: Best = 3.661159e+03	FES = 856100	$\mu_F = 0.775$	$\mu_{CR} = 0.119$	
Diversity = 1.7851e-03	k_t = [2, 2, 2, 4]			
Generation 8570: Best = 3.661158e+03	FES = 857100	$\mu_F = 0.760$	$\mu_{CR} = 0.121$	
Diversity = 1.8268e-03	k_t = [1, 1, 1, 1]			
Generation 8580: Best = 3.661156e+03	FES = 858100	$\mu_F = 0.756$	$\mu_{CR} = 0.134$	
Diversity = 1.9432e-03	k_t = [4, 3, 3, 4]			
Generation 8590: Best = 3.661155e+03	FES = 859100	$\mu_F = 0.753$	$\mu_{CR} = 0.134$	

Diversity = 1.8376e-03	k_t = [1, 1, 1, 1]			
Generation 8600: Best = 3.661154e+03	FES = 860100	$\mu_F = 0.747$	$\mu_{CR} = 0.122$	
Diversity = 1.7288e-03	k_t = [2, 2, 4, 2]			
Generation 8610: Best = 3.661154e+03	FES = 861100	$\mu_F = 0.748$	$\mu_{CR} = 0.123$	
Diversity = 1.6380e-03	k_t = [1, 1, 1, 1]			
Generation 8620: Best = 3.661152e+03	FES = 862100	$\mu_F = 0.765$	$\mu_{CR} = 0.121$	
Diversity = 1.5980e-03	k_t = [4, 3, 3, 3]			
Generation 8630: Best = 3.661151e+03	FES = 863100	$\mu_F = 0.765$	$\mu_{CR} = 0.125$	
Diversity = 1.7979e-03	k_t = [1, 1, 1, 1]			
Generation 8640: Best = 3.661150e+03	FES = 864100	$\mu_F = 0.769$	$\mu_{CR} = 0.118$	
Diversity = 1.6873e-03	k_t = [4, 2, 2, 4]			
Generation 8650: Best = 3.661149e+03	FES = 865100	$\mu_F = 0.762$	$\mu_{CR} = 0.125$	
Diversity = 1.6552e-03	k_t = [1, 1, 1, 1]			
Generation 8660: Best = 3.661148e+03	FES = 866100	$\mu_F = 0.745$	$\mu_{CR} = 0.126$	
Diversity = 1.6303e-03	k_t = [3, 3, 4, 2]			
Generation 8670: Best = 3.661147e+03	FES = 867100	$\mu_F = 0.723$	$\mu_{CR} = 0.115$	
Diversity = 1.5890e-03	k_t = [1, 1, 1, 1]			
Generation 8680: Best = 3.661147e+03	FES = 868100	$\mu_F = 0.730$	$\mu_{CR} = 0.122$	
Diversity = 1.5215e-03	k_t = [4, 4, 3, 2]			
Generation 8690: Best = 3.661147e+03	FES = 869100	$\mu_F = 0.731$	$\mu_{CR} = 0.130$	
Diversity = 1.5757e-03	k_t = [1, 1, 1, 1]			
Generation 8700: Best = 3.661146e+03	FES = 870100	$\mu_F = 0.746$	$\mu_{CR} = 0.130$	
Diversity = 1.5453e-03	k_t = [4, 2, 4, 3]			
Generation 8710: Best = 3.661145e+03	FES = 871100	$\mu_F = 0.745$	$\mu_{CR} = 0.123$	
Diversity = 1.6036e-03	k_t = [1, 1, 1, 1]			
Generation 8720: Best = 3.661145e+03	FES = 872100	$\mu_F = 0.755$	$\mu_{CR} = 0.122$	
Diversity = 1.7567e-03	k_t = [3, 3, 2, 3]			
Generation 8730: Best = 3.661143e+03	FES = 873100	$\mu_F = 0.769$	$\mu_{CR} = 0.117$	
Diversity = 1.7239e-03	k_t = [1, 1, 1, 1]			
Generation 8740: Best = 3.661143e+03	FES = 874100	$\mu_F = 0.771$	$\mu_{CR} = 0.109$	
Diversity = 1.7643e-03	k_t = [2, 2, 3, 3]			
Generation 8750: Best = 3.661142e+03	FES = 875100	$\mu_F = 0.767$	$\mu_{CR} = 0.100$	
Diversity = 1.7313e-03	k_t = [1, 1, 1, 1]			
Generation 8760: Best = 3.661141e+03	FES = 876100	$\mu_F = 0.751$	$\mu_{CR} = 0.100$	
Diversity = 1.8770e-03	k_t = [2, 2, 4, 3]			
Generation 8770: Best = 3.661140e+03	FES = 877100	$\mu_F = 0.743$	$\mu_{CR} = 0.101$	
Diversity = 2.0722e-03	k_t = [1, 1, 1, 1]			
Generation 8780: Best = 3.661140e+03	FES = 878100	$\mu_F = 0.732$	$\mu_{CR} = 0.101$	
Diversity = 2.0188e-03	k_t = [4, 3, 4, 3]			
Generation 8790: Best = 3.661139e+03	FES = 879100	$\mu_F = 0.712$	$\mu_{CR} = 0.107$	
Diversity = 1.9443e-03	k_t = [1, 1, 1, 1]			
Generation 8800: Best = 3.661139e+03	FES = 880100	$\mu_F = 0.718$	$\mu_{CR} = 0.108$	
Diversity = 1.6388e-03	k_t = [4, 4, 2, 3]			
Generation 8810: Best = 3.661138e+03	FES = 881100	$\mu_F = 0.727$	$\mu_{CR} = 0.107$	
Diversity = 1.6706e-03	k_t = [1, 1, 1, 1]			
Generation 8820: Best = 3.661138e+03	FES = 882100	$\mu_F = 0.727$	$\mu_{CR} = 0.101$	
Diversity = 1.5909e-03	k_t = [2, 2, 4, 2]			
Generation 8830: Best = 3.661137e+03	FES = 883100	$\mu_F = 0.722$	$\mu_{CR} = 0.114$	
Diversity = 1.6215e-03	k_t = [1, 1, 1, 1]			
Generation 8840: Best = 3.661137e+03	FES = 884100	$\mu_F = 0.714$	$\mu_{CR} = 0.114$	
Diversity = 1.5637e-03	k_t = [2, 3, 3, 2]			
Generation 8850: Best = 3.661136e+03	FES = 885100	$\mu_F = 0.724$	$\mu_{CR} = 0.094$	
Diversity = 1.5588e-03	k_t = [1, 1, 1, 1]			
Generation 8860: Best = 3.661136e+03	FES = 886100	$\mu_F = 0.729$	$\mu_{CR} = 0.091$	

Diversity = 1.5249e-03	k_t = [3, 2, 2, 4]			
Generation 8870: Best = 3.661136e+03	FES = 887100	$\mu_F = 0.733$	$\mu_{CR} = 0.094$	
Diversity = 1.5579e-03	k_t = [1, 1, 1, 1]			
Generation 8880: Best = 3.661135e+03	FES = 888100	$\mu_F = 0.736$	$\mu_{CR} = 0.105$	
Diversity = 1.5129e-03	k_t = [4, 3, 2, 3]			
Generation 8890: Best = 3.661135e+03	FES = 889100	$\mu_F = 0.719$	$\mu_{CR} = 0.106$	
Diversity = 1.4270e-03	k_t = [1, 1, 1, 1]			
Generation 8900: Best = 3.661134e+03	FES = 890100	$\mu_F = 0.716$	$\mu_{CR} = 0.110$	
Diversity = 1.3618e-03	k_t = [3, 4, 3, 3]			
Generation 8910: Best = 3.661134e+03	FES = 891100	$\mu_F = 0.720$	$\mu_{CR} = 0.102$	
Diversity = 1.3865e-03	k_t = [1, 1, 1, 1]			
Generation 8920: Best = 3.661134e+03	FES = 892100	$\mu_F = 0.728$	$\mu_{CR} = 0.102$	
Diversity = 1.2863e-03	k_t = [2, 4, 2, 4]			
Generation 8930: Best = 3.661134e+03	FES = 893100	$\mu_F = 0.723$	$\mu_{CR} = 0.108$	
Diversity = 1.3126e-03	k_t = [1, 1, 1, 1]			
Generation 8940: Best = 3.661133e+03	FES = 894100	$\mu_F = 0.726$	$\mu_{CR} = 0.109$	
Diversity = 1.1688e-03	k_t = [4, 4, 4, 4]			
Generation 8950: Best = 3.661133e+03	FES = 895100	$\mu_F = 0.745$	$\mu_{CR} = 0.112$	
Diversity = 1.2184e-03	k_t = [1, 1, 1, 1]			
Generation 8960: Best = 3.661133e+03	FES = 896100	$\mu_F = 0.754$	$\mu_{CR} = 0.106$	
Diversity = 1.3319e-03	k_t = [3, 3, 3, 3]			
Generation 8970: Best = 3.661132e+03	FES = 897100	$\mu_F = 0.749$	$\mu_{CR} = 0.104$	
Diversity = 1.3588e-03	k_t = [1, 1, 1, 1]			
Generation 8980: Best = 3.661132e+03	FES = 898100	$\mu_F = 0.738$	$\mu_{CR} = 0.105$	
Diversity = 1.3068e-03	k_t = [4, 3, 4, 3]			
Generation 8990: Best = 3.661132e+03	FES = 899100	$\mu_F = 0.749$	$\mu_{CR} = 0.103$	
Diversity = 1.2492e-03	k_t = [1, 1, 1, 1]			
Generation 9000: Best = 3.661131e+03	FES = 900100	$\mu_F = 0.745$	$\mu_{CR} = 0.111$	
Diversity = 1.2169e-03	k_t = [2, 4, 3, 3]			
Generation 9010: Best = 3.661131e+03	FES = 901100	$\mu_F = 0.743$	$\mu_{CR} = 0.106$	
Diversity = 1.2501e-03	k_t = [1, 1, 1, 1]			
Generation 9020: Best = 3.661130e+03	FES = 902100	$\mu_F = 0.728$	$\mu_{CR} = 0.099$	
Diversity = 1.2709e-03	k_t = [4, 2, 4, 2]			
Generation 9030: Best = 3.661130e+03	FES = 903100	$\mu_F = 0.703$	$\mu_{CR} = 0.110$	
Diversity = 1.3441e-03	k_t = [1, 1, 1, 1]			
Generation 9040: Best = 3.661130e+03	FES = 904100	$\mu_F = 0.706$	$\mu_{CR} = 0.113$	
Diversity = 1.3517e-03	k_t = [4, 2, 2, 2]			
Generation 9050: Best = 3.661130e+03	FES = 905100	$\mu_F = 0.697$	$\mu_{CR} = 0.125$	
Diversity = 1.2621e-03	k_t = [1, 1, 1, 1]			
Generation 9060: Best = 3.661129e+03	FES = 906100	$\mu_F = 0.714$	$\mu_{CR} = 0.114$	
Diversity = 1.2044e-03	k_t = [3, 2, 4, 2]			
Generation 9070: Best = 3.661129e+03	FES = 907100	$\mu_F = 0.719$	$\mu_{CR} = 0.109$	
Diversity = 1.3021e-03	k_t = [1, 1, 1, 1]			
Generation 9080: Best = 3.661129e+03	FES = 908100	$\mu_F = 0.731$	$\mu_{CR} = 0.111$	
Diversity = 1.3211e-03	k_t = [3, 3, 4, 3]			
Generation 9090: Best = 3.661128e+03	FES = 909100	$\mu_F = 0.744$	$\mu_{CR} = 0.114$	
Diversity = 1.3402e-03	k_t = [1, 1, 1, 1]			
Generation 9100: Best = 3.661128e+03	FES = 910100	$\mu_F = 0.736$	$\mu_{CR} = 0.115$	
Diversity = 1.2895e-03	k_t = [4, 3, 3, 3]			
Generation 9110: Best = 3.661128e+03	FES = 911100	$\mu_F = 0.737$	$\mu_{CR} = 0.110$	
Diversity = 1.3039e-03	k_t = [1, 1, 1, 1]			
Generation 9120: Best = 3.661127e+03	FES = 912100	$\mu_F = 0.728$	$\mu_{CR} = 0.107$	
Diversity = 1.2984e-03	k_t = [2, 4, 3, 4]			
Generation 9130: Best = 3.661127e+03	FES = 913100	$\mu_F = 0.703$	$\mu_{CR} = 0.109$	

Diversity = 1.2818e-03	k_t = [1, 1, 1, 1]			
Generation 9140: Best = 3.661127e+03	FES = 914100	$\mu_F = 0.704$	$\mu_{CR} = 0.107$	
Diversity = 1.1095e-03	k_t = [2, 2, 3, 3]			
Generation 9150: Best = 3.661127e+03	FES = 915100	$\mu_F = 0.701$	$\mu_{CR} = 0.110$	
Diversity = 1.0412e-03	k_t = [1, 1, 1, 1]			
Generation 9160: Best = 3.661126e+03	FES = 916100	$\mu_F = 0.706$	$\mu_{CR} = 0.102$	
Diversity = 1.0495e-03	k_t = [3, 3, 3, 3]			
Generation 9170: Best = 3.661126e+03	FES = 917100	$\mu_F = 0.710$	$\mu_{CR} = 0.100$	
Diversity = 9.8738e-04	k_t = [1, 1, 1, 1]			
Generation 9180: Best = 3.661126e+03	FES = 918100	$\mu_F = 0.704$	$\mu_{CR} = 0.094$	
Diversity = 9.3865e-04	k_t = [2, 2, 2, 4]			
Generation 9190: Best = 3.661126e+03	FES = 919100	$\mu_F = 0.713$	$\mu_{CR} = 0.091$	
Diversity = 9.2649e-04	k_t = [1, 1, 1, 1]			
Generation 9200: Best = 3.661126e+03	FES = 920100	$\mu_F = 0.723$	$\mu_{CR} = 0.094$	
Diversity = 9.7117e-04	k_t = [4, 2, 4, 2]			
Generation 9210: Best = 3.661126e+03	FES = 921100	$\mu_F = 0.718$	$\mu_{CR} = 0.094$	
Diversity = 9.7307e-04	k_t = [1, 1, 1, 1]			
Generation 9220: Best = 3.661125e+03	FES = 922100	$\mu_F = 0.726$	$\mu_{CR} = 0.089$	
Diversity = 9.5494e-04	k_t = [4, 2, 4, 2]			
Generation 9230: Best = 3.661125e+03	FES = 923100	$\mu_F = 0.729$	$\mu_{CR} = 0.093$	
Diversity = 9.3814e-04	k_t = [1, 1, 1, 1]			
Generation 9240: Best = 3.661125e+03	FES = 924100	$\mu_F = 0.715$	$\mu_{CR} = 0.099$	
Diversity = 8.7481e-04	k_t = [3, 2, 3, 4]			
Generation 9250: Best = 3.661125e+03	FES = 925100	$\mu_F = 0.719$	$\mu_{CR} = 0.104$	
Diversity = 8.9347e-04	k_t = [1, 1, 1, 1]			
Generation 9260: Best = 3.661125e+03	FES = 926100	$\mu_F = 0.709$	$\mu_{CR} = 0.096$	
Diversity = 9.1284e-04	k_t = [2, 4, 2, 3]			
Generation 9270: Best = 3.661124e+03	FES = 927100	$\mu_F = 0.716$	$\mu_{CR} = 0.093$	
Diversity = 9.3936e-04	k_t = [1, 1, 1, 1]			
Generation 9280: Best = 3.661124e+03	FES = 928100	$\mu_F = 0.713$	$\mu_{CR} = 0.101$	
Diversity = 9.0301e-04	k_t = [2, 3, 3, 3]			
Generation 9290: Best = 3.661124e+03	FES = 929100	$\mu_F = 0.714$	$\mu_{CR} = 0.098$	
Diversity = 9.1688e-04	k_t = [1, 1, 1, 1]			
Generation 9300: Best = 3.661124e+03	FES = 930100	$\mu_F = 0.739$	$\mu_{CR} = 0.103$	
Diversity = 9.2306e-04	k_t = [4, 2, 3, 2]			
Generation 9310: Best = 3.661123e+03	FES = 931100	$\mu_F = 0.737$	$\mu_{CR} = 0.095$	
Diversity = 9.6033e-04	k_t = [1, 1, 1, 1]			
Generation 9320: Best = 3.661123e+03	FES = 932100	$\mu_F = 0.740$	$\mu_{CR} = 0.099$	
Diversity = 8.8290e-04	k_t = [2, 2, 3, 4]			
Generation 9330: Best = 3.661123e+03	FES = 933100	$\mu_F = 0.748$	$\mu_{CR} = 0.097$	
Diversity = 8.9497e-04	k_t = [1, 1, 1, 1]			
Generation 9340: Best = 3.661123e+03	FES = 934100	$\mu_F = 0.736$	$\mu_{CR} = 0.103$	
Diversity = 8.1673e-04	k_t = [2, 2, 3, 2]			
Generation 9350: Best = 3.661123e+03	FES = 935100	$\mu_F = 0.743$	$\mu_{CR} = 0.104$	
Diversity = 8.3975e-04	k_t = [1, 1, 1, 1]			
Generation 9360: Best = 3.661123e+03	FES = 936100	$\mu_F = 0.733$	$\mu_{CR} = 0.108$	
Diversity = 9.0202e-04	k_t = [3, 4, 3, 3]			
Generation 9370: Best = 3.661122e+03	FES = 937100	$\mu_F = 0.730$	$\mu_{CR} = 0.110$	
Diversity = 9.1798e-04	k_t = [1, 1, 1, 1]			
Generation 9380: Best = 3.661122e+03	FES = 938100	$\mu_F = 0.732$	$\mu_{CR} = 0.118$	
Diversity = 9.2325e-04	k_t = [3, 2, 2, 3]			
Generation 9390: Best = 3.661122e+03	FES = 939100	$\mu_F = 0.744$	$\mu_{CR} = 0.119$	
Diversity = 9.3899e-04	k_t = [1, 1, 1, 1]			
Generation 9400: Best = 3.661121e+03	FES = 940100	$\mu_F = 0.740$	$\mu_{CR} = 0.113$	

Diversity = 1.0392e-03 k_t = [4, 4, 2, 2]
 Generation 9410: Best = 3.661121e+03 FES = 941100 μ_F = 0.736 μ_{CR} = 0.119
 Diversity = 1.1333e-03 k_t = [1, 1, 1, 1]
 Generation 9420: Best = 3.661121e+03 FES = 942100 μ_F = 0.751 μ_{CR} = 0.116
 Diversity = 1.1447e-03 k_t = [3, 3, 3, 4]
 Generation 9430: Best = 3.661120e+03 FES = 943100 μ_F = 0.755 μ_{CR} = 0.110
 Diversity = 1.1274e-03 k_t = [1, 1, 1, 1]
 Generation 9440: Best = 3.661120e+03 FES = 944100 μ_F = 0.747 μ_{CR} = 0.118
 Diversity = 1.1535e-03 k_t = [3, 3, 2, 3]
 Generation 9450: Best = 3.661120e+03 FES = 945100 μ_F = 0.743 μ_{CR} = 0.117
 Diversity = 1.2197e-03 k_t = [1, 1, 1, 1]
 Generation 9460: Best = 3.661119e+03 FES = 946100 μ_F = 0.744 μ_{CR} = 0.111
 Diversity = 1.1248e-03 k_t = [4, 3, 3, 4]
 Generation 9470: Best = 3.661119e+03 FES = 947100 μ_F = 0.747 μ_{CR} = 0.108
 Diversity = 1.1170e-03 k_t = [1, 1, 1, 1]
 Generation 9480: Best = 3.661119e+03 FES = 948100 μ_F = 0.744 μ_{CR} = 0.118
 Diversity = 1.0262e-03 k_t = [2, 4, 2, 4]
 Generation 9490: Best = 3.661118e+03 FES = 949100 μ_F = 0.751 μ_{CR} = 0.113
 Diversity = 1.1000e-03 k_t = [1, 1, 1, 1]
 Generation 9500: Best = 3.661118e+03 FES = 950100 μ_F = 0.732 μ_{CR} = 0.106
 Diversity = 1.0762e-03 k_t = [2, 4, 4, 2]
 Generation 9510: Best = 3.661118e+03 FES = 951100 μ_F = 0.726 μ_{CR} = 0.107
 Diversity = 1.1041e-03 k_t = [1, 1, 1, 1]
 Generation 9520: Best = 3.661118e+03 FES = 952100 μ_F = 0.715 μ_{CR} = 0.101
 Diversity = 1.0684e-03 k_t = [4, 3, 4, 4]
 Generation 9530: Best = 3.661118e+03 FES = 953100 μ_F = 0.734 μ_{CR} = 0.114
 Diversity = 1.0547e-03 k_t = [1, 1, 1, 1]
 Generation 9540: Best = 3.661117e+03 FES = 954100 μ_F = 0.730 μ_{CR} = 0.106
 Diversity = 1.0467e-03 k_t = [3, 2, 3, 2]
 Generation 9550: Best = 3.661117e+03 FES = 955100 μ_F = 0.722 μ_{CR} = 0.094
 Diversity = 1.0729e-03 k_t = [1, 1, 1, 1]
 Generation 9560: Best = 3.661117e+03 FES = 956100 μ_F = 0.729 μ_{CR} = 0.100
 Diversity = 9.5010e-04 k_t = [3, 3, 4, 4]
 Generation 9570: Best = 3.661117e+03 FES = 957100 μ_F = 0.709 μ_{CR} = 0.106
 Diversity = 9.5765e-04 k_t = [1, 1, 1, 1]
 Generation 9580: Best = 3.661117e+03 FES = 958100 μ_F = 0.699 μ_{CR} = 0.104
 Diversity = 8.7839e-04 k_t = [2, 3, 4, 4]
 Generation 9590: Best = 3.661116e+03 FES = 959100 μ_F = 0.692 μ_{CR} = 0.108
 Diversity = 8.9768e-04 k_t = [1, 1, 1, 1]
 Generation 9600: Best = 3.661116e+03 FES = 960100 μ_F = 0.697 μ_{CR} = 0.116
 Diversity = 8.8466e-04 k_t = [2, 4, 3, 2]
 Generation 9610: Best = 3.661116e+03 FES = 961100 μ_F = 0.706 μ_{CR} = 0.117
 Diversity = 9.8135e-04 k_t = [1, 1, 1, 1]
 Generation 9620: Best = 3.661116e+03 FES = 962100 μ_F = 0.706 μ_{CR} = 0.110
 Diversity = 9.7142e-04 k_t = [2, 4, 4, 3]
 Generation 9630: Best = 3.661115e+03 FES = 963100 μ_F = 0.713 μ_{CR} = 0.116
 Diversity = 9.6099e-04 k_t = [1, 1, 1, 1]
 Generation 9640: Best = 3.661115e+03 FES = 964100 μ_F = 0.717 μ_{CR} = 0.126
 Diversity = 9.9334e-04 k_t = [2, 2, 2, 3]
 Generation 9650: Best = 3.661114e+03 FES = 965100 μ_F = 0.726 μ_{CR} = 0.124
 Diversity = 1.0374e-03 k_t = [1, 1, 1, 1]
 Generation 9660: Best = 3.661114e+03 FES = 966100 μ_F = 0.740 μ_{CR} = 0.120
 Diversity = 1.0984e-03 k_t = [2, 2, 2, 3]
 Generation 9670: Best = 3.661114e+03 FES = 967100 μ_F = 0.732 μ_{CR} = 0.115

Diversity = 1.0115e-03	k_t = [1, 1, 1, 1]			
Generation 9680: Best = 3.661114e+03	FES = 968100	$\mu_F = 0.723$	$\mu_{CR} = 0.111$	
Diversity = 9.2697e-04	k_t = [3, 4, 2, 4]			
Generation 9690: Best = 3.661114e+03	FES = 969100	$\mu_F = 0.723$	$\mu_{CR} = 0.110$	
Diversity = 8.2807e-04	k_t = [1, 1, 1, 1]			
Generation 9700: Best = 3.661113e+03	FES = 970100	$\mu_F = 0.713$	$\mu_{CR} = 0.106$	
Diversity = 7.6156e-04	k_t = [3, 4, 4, 4]			
Generation 9710: Best = 3.661113e+03	FES = 971100	$\mu_F = 0.737$	$\mu_{CR} = 0.098$	
Diversity = 6.6976e-04	k_t = [1, 1, 1, 1]			
Generation 9720: Best = 3.661113e+03	FES = 972100	$\mu_F = 0.747$	$\mu_{CR} = 0.103$	
Diversity = 6.8367e-04	k_t = [3, 3, 4, 4]			
Generation 9730: Best = 3.661113e+03	FES = 973100	$\mu_F = 0.758$	$\mu_{CR} = 0.111$	
Diversity = 7.5049e-04	k_t = [1, 1, 1, 1]			
Generation 9740: Best = 3.661113e+03	FES = 974100	$\mu_F = 0.748$	$\mu_{CR} = 0.110$	
Diversity = 8.8928e-04	k_t = [2, 4, 2, 2]			
Generation 9750: Best = 3.661112e+03	FES = 975100	$\mu_F = 0.760$	$\mu_{CR} = 0.116$	
Diversity = 9.8438e-04	k_t = [1, 1, 1, 1]			
Generation 9760: Best = 3.661112e+03	FES = 976100	$\mu_F = 0.740$	$\mu_{CR} = 0.111$	
Diversity = 1.0092e-03	k_t = [3, 4, 2, 2]			
Generation 9770: Best = 3.661112e+03	FES = 977100	$\mu_F = 0.755$	$\mu_{CR} = 0.106$	
Diversity = 1.1064e-03	k_t = [1, 1, 1, 1]			
Generation 9780: Best = 3.661111e+03	FES = 978100	$\mu_F = 0.754$	$\mu_{CR} = 0.110$	
Diversity = 1.0133e-03	k_t = [3, 2, 2, 3]			
Generation 9790: Best = 3.661111e+03	FES = 979100	$\mu_F = 0.758$	$\mu_{CR} = 0.104$	
Diversity = 1.0893e-03	k_t = [1, 1, 1, 1]			
Generation 9800: Best = 3.661110e+03	FES = 980100	$\mu_F = 0.763$	$\mu_{CR} = 0.110$	
Diversity = 1.1258e-03	k_t = [3, 2, 3, 3]			
Generation 9810: Best = 3.661110e+03	FES = 981100	$\mu_F = 0.763$	$\mu_{CR} = 0.111$	
Diversity = 1.1983e-03	k_t = [1, 1, 1, 1]			
Generation 9820: Best = 3.661109e+03	FES = 982100	$\mu_F = 0.767$	$\mu_{CR} = 0.107$	
Diversity = 1.2235e-03	k_t = [4, 2, 2, 2]			
Generation 9830: Best = 3.661108e+03	FES = 983100	$\mu_F = 0.762$	$\mu_{CR} = 0.103$	
Diversity = 1.3455e-03	k_t = [1, 1, 1, 1]			
Generation 9840: Best = 3.661107e+03	FES = 984100	$\mu_F = 0.768$	$\mu_{CR} = 0.102$	
Diversity = 1.3857e-03	k_t = [4, 4, 3, 4]			
Generation 9850: Best = 3.661107e+03	FES = 985100	$\mu_F = 0.767$	$\mu_{CR} = 0.100$	
Diversity = 1.3709e-03	k_t = [1, 1, 1, 1]			
Generation 9860: Best = 3.661107e+03	FES = 986100	$\mu_F = 0.774$	$\mu_{CR} = 0.106$	
Diversity = 1.4366e-03	k_t = [2, 3, 2, 4]			
Generation 9870: Best = 3.661106e+03	FES = 987100	$\mu_F = 0.751$	$\mu_{CR} = 0.114$	
Diversity = 1.5581e-03	k_t = [1, 1, 1, 1]			
Generation 9880: Best = 3.661105e+03	FES = 988100	$\mu_F = 0.743$	$\mu_{CR} = 0.114$	
Diversity = 1.5691e-03	k_t = [2, 3, 2, 2]			
Generation 9890: Best = 3.661104e+03	FES = 989100	$\mu_F = 0.752$	$\mu_{CR} = 0.105$	
Diversity = 1.5406e-03	k_t = [1, 1, 1, 1]			
Generation 9900: Best = 3.661103e+03	FES = 990100	$\mu_F = 0.760$	$\mu_{CR} = 0.108$	
Diversity = 1.5924e-03	k_t = [2, 2, 2, 4]			
Generation 9910: Best = 3.661103e+03	FES = 991100	$\mu_F = 0.765$	$\mu_{CR} = 0.116$	
Diversity = 1.6535e-03	k_t = [1, 1, 1, 1]			
Generation 9920: Best = 3.661102e+03	FES = 992100	$\mu_F = 0.769$	$\mu_{CR} = 0.110$	
Diversity = 1.6007e-03	k_t = [2, 2, 4, 3]			
Generation 9930: Best = 3.661101e+03	FES = 993100	$\mu_F = 0.753$	$\mu_{CR} = 0.124$	
Diversity = 1.6000e-03	k_t = [1, 1, 1, 1]			
Generation 9940: Best = 3.661101e+03	FES = 994100	$\mu_F = 0.765$	$\mu_{CR} = 0.110$	

```

Diversity = 1.5887e-03 k_t = [4, 3, 4, 4]
Generation 9950: Best = 3.661100e+03 FES = 995100  $\mu_F$  = 0.745  $\mu_{CR}$  = 0.112
Diversity = 1.5461e-03 k_t = [1, 1, 1, 1]
Generation 9960: Best = 3.661100e+03 FES = 996100  $\mu_F$  = 0.741  $\mu_{CR}$  = 0.106
Diversity = 1.4723e-03 k_t = [3, 4, 3, 3]
Generation 9970: Best = 3.661098e+03 FES = 997100  $\mu_F$  = 0.734  $\mu_{CR}$  = 0.112
Diversity = 1.5144e-03 k_t = [1, 1, 1, 1]
Generation 9980: Best = 3.661098e+03 FES = 998100  $\mu_F$  = 0.720  $\mu_{CR}$  = 0.116
Diversity = 1.4556e-03 k_t = [2, 4, 3, 3]
Generation 9990: Best = 3.661097e+03 FES = 999100  $\mu_F$  = 0.720  $\mu_{CR}$  = 0.120
Diversity = 1.4578e-03 k_t = [1, 1, 1, 1]

```

Optimization complete.

```

Best Fitness          = 3.661097e+03
Total Function Evals  = 1000000
Total Successful Updates = 239286

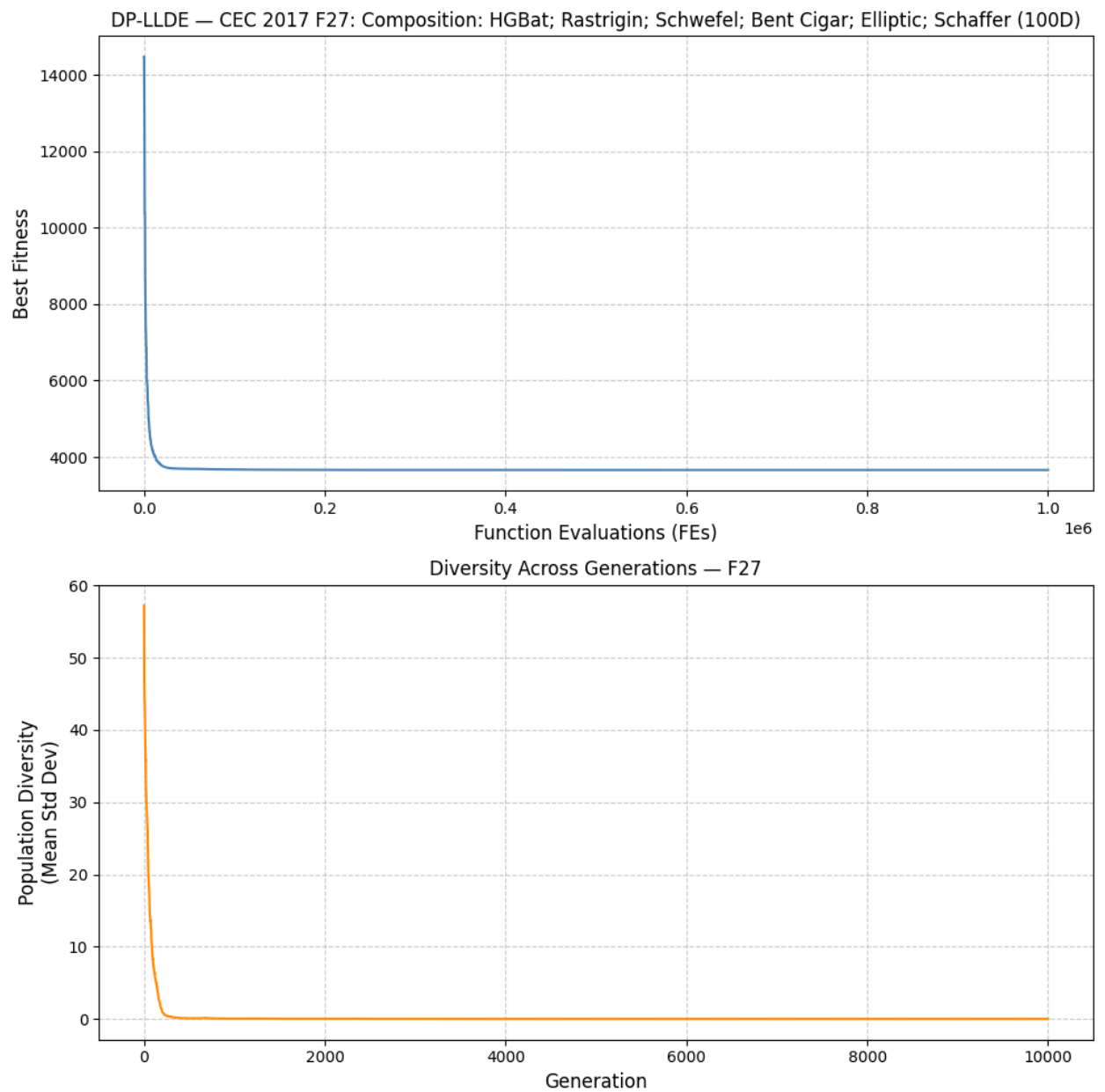
```

```

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=
Results:
  Best Fitness          : 3.661097e+03
  Global Optimum        : 2.700000e+03
  Error (|best-optimum|) : 9.610973e+02
  Best Solution (5 dim) : [-37.32379431  99.99999847 -81.37126277  44.73213314
32.31521893]
  D^(0) (init. diversity): 5.7220e+01
=====
=

```

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results/all_results_checkpoint.pkl



Graph saved as convergence_F27.png

✓ Plot saved to: /content/drive/My Drive/thesis_results/plots/convergence_F27.png

```
In [16]: # =====
# F28 — Composition Function 8 (Ackley; Griewank; Discus; Rosenbrock; HappyCat)
# =====
run_and_plot('F28', make_wrapper(composition.f28), 2800.0, 'Composition: Ackle
```


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DP-LLDE on CEC 2017 F28 – Single Run

=====

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Function : Composition: Ackley; Griewank; Discus; Rosenbrock; HappyCat; S
chaffer (100D)

Bounds : [-100, 100]

Global optimum : 2800.0

Initial $D^*(0) = 5.7220e+01$ | Diversity threshold = 5.7220e+00

Generation 0: Best Fitness = 6.506878e+04

Generation 10: Best = 2.914383e+04 FES = 1100 $\mu_F = 0.556$ $\mu_{CR} = 0.354$

Diversity = 3.0840e+01 $k_t = [1, 1, 1, 1]$

Generation 20: Best = 2.051206e+04 FES = 2100 $\mu_F = 0.582$ $\mu_{CR} = 0.359$

Diversity = 1.7451e+01 $k_t = [2, 3, 2, 3]$

Generation 30: Best = 1.579697e+04 FES = 3100 $\mu_F = 0.613$ $\mu_{CR} = 0.363$

Diversity = 1.0759e+01 $k_t = [1, 1, 1, 1]$

Generation 40: Best = 1.313852e+04 FES = 4100 $\mu_F = 0.633$ $\mu_{CR} = 0.365$

Diversity = 6.8770e+00 $k_t = [3, 2, 3, 4]$

Generation 50: Best = 1.166303e+04 FES = 5100 $\mu_F = 0.667$ $\mu_{CR} = 0.369$

Diversity = 5.7205e+00 $k_t = [1, 1, 1, 1]$

Generation 60: Best = 1.073381e+04 FES = 6100 $\mu_F = 0.690$ $\mu_{CR} = 0.370$

Diversity = 4.4115e+00 $k_t = [4, 2, 2, 2]$

Generation 70: Best = 9.763463e+03 FES = 7100 $\mu_F = 0.718$ $\mu_{CR} = 0.378$

Diversity = 4.0893e+00 $k_t = [1, 1, 1, 1]$

Generation 80: Best = 9.098197e+03 FES = 8100 $\mu_F = 0.728$ $\mu_{CR} = 0.383$

Diversity = 3.7670e+00 $k_t = [4, 3, 3, 3]$

Generation 90: Best = 8.405667e+03 FES = 9100 $\mu_F = 0.731$ $\mu_{CR} = 0.382$

Diversity = 3.3491e+00 $k_t = [1, 1, 1, 1]$

Generation 100: Best = 7.930209e+03 FES = 10100 $\mu_F = 0.747$ $\mu_{CR} = 0.384$

Diversity = 3.2461e+00 $k_t = [2, 4, 4, 2]$

Generation 110: Best = 7.539863e+03 FES = 11100 $\mu_F = 0.755$ $\mu_{CR} = 0.385$

Diversity = 3.6067e+00 $k_t = [1, 1, 1, 1]$

Generation 120: Best = 7.090328e+03 FES = 12100 $\mu_F = 0.751$ $\mu_{CR} = 0.389$

Diversity = 3.2108e+00 $k_t = [3, 3, 2, 4]$

Generation 130: Best = 6.762272e+03 FES = 13100 $\mu_F = 0.752$ $\mu_{CR} = 0.396$

Diversity = 3.0367e+00 $k_t = [1, 1, 1, 1]$

Generation 140: Best = 6.302406e+03 FES = 14100 $\mu_F = 0.758$ $\mu_{CR} = 0.406$

Diversity = 2.9897e+00 $k_t = [3, 2, 4, 2]$

Generation 150: Best = 6.001465e+03 FES = 15100 $\mu_F = 0.763$ $\mu_{CR} = 0.409$

Diversity = 3.1393e+00 $k_t = [1, 1, 1, 1]$

Generation 160: Best = 5.613213e+03 FES = 16100 $\mu_F = 0.755$ $\mu_{CR} = 0.410$

Diversity = 2.9008e+00 $k_t = [4, 3, 4, 3]$

Generation 170: Best = 5.332535e+03 FES = 17100 $\mu_F = 0.753$ $\mu_{CR} = 0.417$

Diversity = 2.6224e+00 $k_t = [1, 1, 1, 1]$

Generation 180: Best = 5.131565e+03 FES = 18100 $\mu_F = 0.755$ $\mu_{CR} = 0.416$

Diversity = 2.2081e+00 $k_t = [3, 4, 4, 3]$

Generation 190: Best = 4.931789e+03 FES = 19100 $\mu_F = 0.759$ $\mu_{CR} = 0.420$

Diversity = 2.0891e+00 $k_t = [1, 1, 1, 1]$

Generation 200: Best = 4.818155e+03 FES = 20100 $\mu_F = 0.752$ $\mu_{CR} = 0.416$

Diversity = 1.8160e+00 $k_t = [4, 4, 3, 3]$

Generation 210: Best = 4.694416e+03 FES = 21100 $\mu_F = 0.769$ $\mu_{CR} = 0.428$

Diversity = 1.7354e+00 $k_t = [1, 1, 1, 1]$

Generation 220: Best = 4.586948e+03 FES = 22100 $\mu_F = 0.763$ $\mu_{CR} = 0.427$

Diversity = 1.5891e+00	k_t = [4, 2, 2, 3]				
Generation 230: Best = 4.508493e+03	FES = 23100	$\mu_F = 0.764$	$\mu_{CR} = 0.431$		
Diversity = 1.5462e+00	k_t = [1, 1, 1, 1]				
Generation 240: Best = 4.428727e+03	FES = 24100	$\mu_F = 0.764$	$\mu_{CR} = 0.431$		
Diversity = 1.4681e+00	k_t = [4, 2, 4, 2]				
Generation 250: Best = 4.352732e+03	FES = 25100	$\mu_F = 0.770$	$\mu_{CR} = 0.428$		
Diversity = 1.4276e+00	k_t = [1, 1, 1, 1]				
Generation 260: Best = 4.275809e+03	FES = 26100	$\mu_F = 0.775$	$\mu_{CR} = 0.427$		
Diversity = 1.3572e+00	k_t = [2, 4, 4, 3]				
Generation 270: Best = 4.222797e+03	FES = 27100	$\mu_F = 0.769$	$\mu_{CR} = 0.431$		
Diversity = 1.3674e+00	k_t = [1, 1, 1, 1]				
Generation 280: Best = 4.163163e+03	FES = 28100	$\mu_F = 0.763$	$\mu_{CR} = 0.431$		
Diversity = 1.3583e+00	k_t = [4, 2, 2, 4]				
Generation 290: Best = 4.106083e+03	FES = 29100	$\mu_F = 0.771$	$\mu_{CR} = 0.441$		
Diversity = 1.2662e+00	k_t = [1, 1, 1, 1]				
Generation 300: Best = 4.061997e+03	FES = 30100	$\mu_F = 0.764$	$\mu_{CR} = 0.444$		
Diversity = 1.1703e+00	k_t = [3, 3, 2, 4]				
Generation 310: Best = 4.029013e+03	FES = 31100	$\mu_F = 0.772$	$\mu_{CR} = 0.444$		
Diversity = 1.1017e+00	k_t = [1, 1, 1, 1]				
Generation 320: Best = 3.985741e+03	FES = 32100	$\mu_F = 0.762$	$\mu_{CR} = 0.446$		
Diversity = 1.0489e+00	k_t = [4, 2, 4, 3]				
Generation 330: Best = 3.952963e+03	FES = 33100	$\mu_F = 0.767$	$\mu_{CR} = 0.452$		
Diversity = 9.5659e-01	k_t = [1, 1, 1, 1]				
Generation 340: Best = 3.922399e+03	FES = 34100	$\mu_F = 0.773$	$\mu_{CR} = 0.455$		
Diversity = 9.4105e-01	k_t = [2, 3, 3, 3]				
Generation 350: Best = 3.898191e+03	FES = 35100	$\mu_F = 0.772$	$\mu_{CR} = 0.454$		
Diversity = 9.8967e-01	k_t = [1, 1, 1, 1]				
Generation 360: Best = 3.865497e+03	FES = 36100	$\mu_F = 0.770$	$\mu_{CR} = 0.452$		
Diversity = 9.4304e-01	k_t = [2, 4, 3, 3]				
Generation 370: Best = 3.840744e+03	FES = 37100	$\mu_F = 0.767$	$\mu_{CR} = 0.455$		
Diversity = 9.4700e-01	k_t = [1, 1, 1, 1]				
Generation 380: Best = 3.818166e+03	FES = 38100	$\mu_F = 0.752$	$\mu_{CR} = 0.457$		
Diversity = 8.2188e-01	k_t = [2, 3, 4, 2]				
Generation 390: Best = 3.796058e+03	FES = 39100	$\mu_F = 0.755$	$\mu_{CR} = 0.459$		
Diversity = 7.8233e-01	k_t = [1, 1, 1, 1]				
Generation 400: Best = 3.777036e+03	FES = 40100	$\mu_F = 0.766$	$\mu_{CR} = 0.462$		
Diversity = 7.4305e-01	k_t = [3, 4, 2, 3]				
Generation 410: Best = 3.760441e+03	FES = 41100	$\mu_F = 0.779$	$\mu_{CR} = 0.471$		
Diversity = 7.5340e-01	k_t = [1, 1, 1, 1]				
Generation 420: Best = 3.744129e+03	FES = 42100	$\mu_F = 0.777$	$\mu_{CR} = 0.472$		
Diversity = 7.4470e-01	k_t = [3, 2, 4, 3]				
Generation 430: Best = 3.725470e+03	FES = 43100	$\mu_F = 0.768$	$\mu_{CR} = 0.470$		
Diversity = 7.7848e-01	k_t = [1, 1, 1, 1]				
Generation 440: Best = 3.709986e+03	FES = 44100	$\mu_F = 0.769$	$\mu_{CR} = 0.481$		
Diversity = 7.0953e-01	k_t = [4, 2, 4, 2]				
Generation 450: Best = 3.697614e+03	FES = 45100	$\mu_F = 0.765$	$\mu_{CR} = 0.477$		
Diversity = 6.6886e-01	k_t = [1, 1, 1, 1]				
Generation 460: Best = 3.685274e+03	FES = 46100	$\mu_F = 0.764$	$\mu_{CR} = 0.478$		
Diversity = 6.3777e-01	k_t = [2, 2, 3, 2]				
Generation 470: Best = 3.675384e+03	FES = 47100	$\mu_F = 0.760$	$\mu_{CR} = 0.484$		
Diversity = 6.4046e-01	k_t = [1, 1, 1, 1]				
Generation 480: Best = 3.661963e+03	FES = 48100	$\mu_F = 0.760$	$\mu_{CR} = 0.491$		
Diversity = 5.4618e-01	k_t = [2, 2, 3, 4]				
Generation 490: Best = 3.652614e+03	FES = 49100	$\mu_F = 0.772$	$\mu_{CR} = 0.493$		

Diversity = 6.1760e-01	k_t = [1, 1, 1, 1]				
Generation 500: Best = 3.642648e+03	FES = 50100	$\mu_F = 0.774$	$\mu_{CR} = 0.495$		
Diversity = 6.3352e-01	k_t = [2, 3, 3, 2]				
Generation 510: Best = 3.632147e+03	FES = 51100	$\mu_F = 0.766$	$\mu_{CR} = 0.500$		
Diversity = 6.3998e-01	k_t = [1, 1, 1, 1]				
Generation 520: Best = 3.622528e+03	FES = 52100	$\mu_F = 0.755$	$\mu_{CR} = 0.506$		
Diversity = 5.4386e-01	k_t = [3, 2, 2, 2]				
Generation 530: Best = 3.614061e+03	FES = 53100	$\mu_F = 0.750$	$\mu_{CR} = 0.506$		
Diversity = 5.0172e-01	k_t = [1, 1, 1, 1]				
Generation 540: Best = 3.608515e+03	FES = 54100	$\mu_F = 0.740$	$\mu_{CR} = 0.501$		
Diversity = 4.2534e-01	k_t = [3, 3, 3, 2]				
Generation 550: Best = 3.600612e+03	FES = 55100	$\mu_F = 0.735$	$\mu_{CR} = 0.506$		
Diversity = 4.3103e-01	k_t = [1, 1, 1, 1]				
Generation 560: Best = 3.593080e+03	FES = 56100	$\mu_F = 0.756$	$\mu_{CR} = 0.506$		
Diversity = 4.7490e-01	k_t = [2, 3, 2, 2]				
Generation 570: Best = 3.585329e+03	FES = 57100	$\mu_F = 0.747$	$\mu_{CR} = 0.519$		
Diversity = 5.1064e-01	k_t = [1, 1, 1, 1]				
Generation 580: Best = 3.578199e+03	FES = 58100	$\mu_F = 0.747$	$\mu_{CR} = 0.509$		
Diversity = 4.4216e-01	k_t = [2, 3, 3, 3]				
Generation 590: Best = 3.571816e+03	FES = 59100	$\mu_F = 0.751$	$\mu_{CR} = 0.514$		
Diversity = 5.0737e-01	k_t = [1, 1, 1, 1]				
Generation 600: Best = 3.564312e+03	FES = 60100	$\mu_F = 0.758$	$\mu_{CR} = 0.512$		
Diversity = 4.6499e-01	k_t = [2, 3, 3, 2]				
Generation 610: Best = 3.558569e+03	FES = 61100	$\mu_F = 0.751$	$\mu_{CR} = 0.513$		
Diversity = 4.6577e-01	k_t = [1, 1, 1, 1]				
Generation 620: Best = 3.553178e+03	FES = 62100	$\mu_F = 0.744$	$\mu_{CR} = 0.516$		
Diversity = 4.0801e-01	k_t = [3, 2, 4, 4]				
Generation 630: Best = 3.549265e+03	FES = 63100	$\mu_F = 0.750$	$\mu_{CR} = 0.513$		
Diversity = 4.3946e-01	k_t = [1, 1, 1, 1]				
Generation 640: Best = 3.542692e+03	FES = 64100	$\mu_F = 0.747$	$\mu_{CR} = 0.515$		
Diversity = 4.1723e-01	k_t = [3, 3, 4, 3]				
Generation 650: Best = 3.536001e+03	FES = 65100	$\mu_F = 0.746$	$\mu_{CR} = 0.524$		
Diversity = 4.6016e-01	k_t = [1, 1, 1, 1]				
Generation 660: Best = 3.531411e+03	FES = 66100	$\mu_F = 0.746$	$\mu_{CR} = 0.527$		
Diversity = 4.1402e-01	k_t = [2, 4, 4, 2]				
Generation 670: Best = 3.528031e+03	FES = 67100	$\mu_F = 0.748$	$\mu_{CR} = 0.528$		
Diversity = 3.9418e-01	k_t = [1, 1, 1, 1]				
Generation 680: Best = 3.523046e+03	FES = 68100	$\mu_F = 0.746$	$\mu_{CR} = 0.529$		
Diversity = 3.8376e-01	k_t = [4, 2, 4, 4]				
Generation 690: Best = 3.519482e+03	FES = 69100	$\mu_F = 0.748$	$\mu_{CR} = 0.531$		
Diversity = 4.1459e-01	k_t = [1, 1, 1, 1]				
Generation 700: Best = 3.514600e+03	FES = 70100	$\mu_F = 0.750$	$\mu_{CR} = 0.532$		
Diversity = 4.1018e-01	k_t = [2, 4, 4, 2]				
Generation 710: Best = 3.511142e+03	FES = 71100	$\mu_F = 0.748$	$\mu_{CR} = 0.534$		
Diversity = 3.8837e-01	k_t = [1, 1, 1, 1]				
Generation 720: Best = 3.506933e+03	FES = 72100	$\mu_F = 0.750$	$\mu_{CR} = 0.533$		
Diversity = 3.8120e-01	k_t = [2, 2, 2, 4]				
Generation 730: Best = 3.503894e+03	FES = 73100	$\mu_F = 0.743$	$\mu_{CR} = 0.535$		
Diversity = 3.8057e-01	k_t = [1, 1, 1, 1]				
Generation 740: Best = 3.500847e+03	FES = 74100	$\mu_F = 0.723$	$\mu_{CR} = 0.541$		
Diversity = 3.3037e-01	k_t = [3, 2, 2, 4]				
Generation 750: Best = 3.498882e+03	FES = 75100	$\mu_F = 0.715$	$\mu_{CR} = 0.532$		
Diversity = 3.3227e-01	k_t = [1, 1, 1, 1]				
Generation 760: Best = 3.497385e+03	FES = 76100	$\mu_F = 0.704$	$\mu_{CR} = 0.533$		

Diversity = 2.8871e-01	k_t = [2, 3, 2, 4]				
Generation 770: Best = 3.494988e+03	FES = 77100	$\mu_F = 0.701$	$\mu_{CR} = 0.540$		
Diversity = 2.7183e-01	k_t = [1, 1, 1, 1]				
Generation 780: Best = 3.493745e+03	FES = 78100	$\mu_F = 0.706$	$\mu_{CR} = 0.540$		
Diversity = 2.4127e-01	k_t = [4, 4, 3, 2]				
Generation 790: Best = 3.492611e+03	FES = 79100	$\mu_F = 0.696$	$\mu_{CR} = 0.527$		
Diversity = 2.0430e-01	k_t = [1, 1, 1, 1]				
Generation 800: Best = 3.491234e+03	FES = 80100	$\mu_F = 0.709$	$\mu_{CR} = 0.528$		
Diversity = 1.9924e-01	k_t = [3, 2, 3, 2]				
Generation 810: Best = 3.490423e+03	FES = 81100	$\mu_F = 0.707$	$\mu_{CR} = 0.524$		
Diversity = 1.7618e-01	k_t = [1, 1, 1, 1]				
Generation 820: Best = 3.489395e+03	FES = 82100	$\mu_F = 0.715$	$\mu_{CR} = 0.516$		
Diversity = 1.7088e-01	k_t = [2, 2, 4, 3]				
Generation 830: Best = 3.488786e+03	FES = 83100	$\mu_F = 0.718$	$\mu_{CR} = 0.524$		
Diversity = 1.6879e-01	k_t = [1, 1, 1, 1]				
Generation 840: Best = 3.487286e+03	FES = 84100	$\mu_F = 0.722$	$\mu_{CR} = 0.533$		
Diversity = 1.6995e-01	k_t = [4, 2, 2, 3]				
Generation 850: Best = 3.486754e+03	FES = 85100	$\mu_F = 0.723$	$\mu_{CR} = 0.533$		
Diversity = 1.6394e-01	k_t = [1, 1, 1, 1]				
Generation 860: Best = 3.486001e+03	FES = 86100	$\mu_F = 0.724$	$\mu_{CR} = 0.535$		
Diversity = 1.5334e-01	k_t = [2, 3, 4, 2]				
Generation 870: Best = 3.485056e+03	FES = 87100	$\mu_F = 0.721$	$\mu_{CR} = 0.525$		
Diversity = 1.5276e-01	k_t = [1, 1, 1, 1]				
Generation 880: Best = 3.484786e+03	FES = 88100	$\mu_F = 0.710$	$\mu_{CR} = 0.522$		
Diversity = 1.2970e-01	k_t = [4, 4, 3, 3]				
Generation 890: Best = 3.484148e+03	FES = 89100	$\mu_F = 0.718$	$\mu_{CR} = 0.516$		
Diversity = 1.3053e-01	k_t = [1, 1, 1, 1]				
Generation 900: Best = 3.483560e+03	FES = 90100	$\mu_F = 0.743$	$\mu_{CR} = 0.512$		
Diversity = 1.2914e-01	k_t = [4, 3, 3, 3]				
Generation 910: Best = 3.483208e+03	FES = 91100	$\mu_F = 0.744$	$\mu_{CR} = 0.507$		
Diversity = 1.3613e-01	k_t = [1, 1, 1, 1]				
Generation 920: Best = 3.482417e+03	FES = 92100	$\mu_F = 0.744$	$\mu_{CR} = 0.517$		
Diversity = 1.3459e-01	k_t = [2, 2, 4, 3]				
Generation 930: Best = 3.481954e+03	FES = 93100	$\mu_F = 0.728$	$\mu_{CR} = 0.518$		
Diversity = 1.3953e-01	k_t = [1, 1, 1, 1]				
Generation 940: Best = 3.481490e+03	FES = 94100	$\mu_F = 0.729$	$\mu_{CR} = 0.529$		
Diversity = 1.2412e-01	k_t = [3, 2, 2, 3]				
Generation 950: Best = 3.480837e+03	FES = 95100	$\mu_F = 0.727$	$\mu_{CR} = 0.517$		
Diversity = 1.2626e-01	k_t = [1, 1, 1, 1]				
Generation 960: Best = 3.480396e+03	FES = 96100	$\mu_F = 0.728$	$\mu_{CR} = 0.521$		
Diversity = 1.2297e-01	k_t = [2, 4, 2, 4]				
Generation 970: Best = 3.480021e+03	FES = 97100	$\mu_F = 0.736$	$\mu_{CR} = 0.519$		
Diversity = 1.2747e-01	k_t = [1, 1, 1, 1]				
Generation 980: Best = 3.479509e+03	FES = 98100	$\mu_F = 0.728$	$\mu_{CR} = 0.521$		
Diversity = 1.2779e-01	k_t = [2, 2, 2, 2]				
Generation 990: Best = 3.478912e+03	FES = 99100	$\mu_F = 0.716$	$\mu_{CR} = 0.537$		
Diversity = 1.3102e-01	k_t = [1, 1, 1, 1]				
Generation 1000: Best = 3.478519e+03	FES = 100100	$\mu_F = 0.713$	$\mu_{CR} = 0.521$		
Diversity = 1.1194e-01	k_t = [3, 4, 3, 4]				
Generation 1010: Best = 3.477907e+03	FES = 101100	$\mu_F = 0.721$	$\mu_{CR} = 0.523$		
Diversity = 1.1402e-01	k_t = [1, 1, 1, 1]				
Generation 1020: Best = 3.477499e+03	FES = 102100	$\mu_F = 0.731$	$\mu_{CR} = 0.519$		
Diversity = 1.1505e-01	k_t = [3, 4, 2, 3]				
Generation 1030: Best = 3.477103e+03	FES = 103100	$\mu_F = 0.737$	$\mu_{CR} = 0.530$		

Diversity = 1.0953e-01	k_t = [1, 1, 1, 1]			
Generation 1040: Best = 3.476924e+03	FES = 104100	$\mu_F = 0.719$	$\mu_{CR} = 0.515$	
Diversity = 8.4330e-02	k_t = [4, 4, 2, 3]			
Generation 1050: Best = 3.476634e+03	FES = 105100	$\mu_F = 0.734$	$\mu_{CR} = 0.518$	
Diversity = 7.6642e-02	k_t = [1, 1, 1, 1]			
Generation 1060: Best = 3.476168e+03	FES = 106100	$\mu_F = 0.743$	$\mu_{CR} = 0.521$	
Diversity = 7.6762e-02	k_t = [4, 3, 2, 4]			
Generation 1070: Best = 3.475960e+03	FES = 107100	$\mu_F = 0.743$	$\mu_{CR} = 0.520$	
Diversity = 6.9499e-02	k_t = [1, 1, 1, 1]			
Generation 1080: Best = 3.475768e+03	FES = 108100	$\mu_F = 0.730$	$\mu_{CR} = 0.513$	
Diversity = 6.0393e-02	k_t = [3, 3, 3, 2]			
Generation 1090: Best = 3.475539e+03	FES = 109100	$\mu_F = 0.740$	$\mu_{CR} = 0.506$	
Diversity = 6.3292e-02	k_t = [1, 1, 1, 1]			
Generation 1100: Best = 3.475373e+03	FES = 110100	$\mu_F = 0.729$	$\mu_{CR} = 0.501$	
Diversity = 5.9712e-02	k_t = [3, 3, 3, 3]			
Generation 1110: Best = 3.475163e+03	FES = 111100	$\mu_F = 0.746$	$\mu_{CR} = 0.494$	
Diversity = 6.4964e-02	k_t = [1, 1, 1, 1]			
Generation 1120: Best = 3.474874e+03	FES = 112100	$\mu_F = 0.740$	$\mu_{CR} = 0.499$	
Diversity = 5.8752e-02	k_t = [3, 4, 2, 4]			
Generation 1130: Best = 3.474656e+03	FES = 113100	$\mu_F = 0.744$	$\mu_{CR} = 0.503$	
Diversity = 6.6995e-02	k_t = [1, 1, 1, 1]			
Generation 1140: Best = 3.474401e+03	FES = 114100	$\mu_F = 0.747$	$\mu_{CR} = 0.503$	
Diversity = 6.6873e-02	k_t = [4, 3, 4, 4]			
Generation 1150: Best = 3.474224e+03	FES = 115100	$\mu_F = 0.739$	$\mu_{CR} = 0.507$	
Diversity = 6.0706e-02	k_t = [1, 1, 1, 1]			
Generation 1160: Best = 3.474002e+03	FES = 116100	$\mu_F = 0.761$	$\mu_{CR} = 0.508$	
Diversity = 6.0877e-02	k_t = [3, 3, 3, 4]			
Generation 1170: Best = 3.473760e+03	FES = 117100	$\mu_F = 0.761$	$\mu_{CR} = 0.488$	
Diversity = 7.1738e-02	k_t = [1, 1, 1, 1]			
Generation 1180: Best = 3.473530e+03	FES = 118100	$\mu_F = 0.749$	$\mu_{CR} = 0.482$	
Diversity = 7.2158e-02	k_t = [3, 2, 4, 4]			
Generation 1190: Best = 3.473307e+03	FES = 119100	$\mu_F = 0.750$	$\mu_{CR} = 0.472$	
Diversity = 7.9855e-02	k_t = [1, 1, 1, 1]			
Generation 1200: Best = 3.473085e+03	FES = 120100	$\mu_F = 0.747$	$\mu_{CR} = 0.457$	
Diversity = 7.7962e-02	k_t = [4, 4, 2, 3]			
Generation 1210: Best = 3.472828e+03	FES = 121100	$\mu_F = 0.737$	$\mu_{CR} = 0.459$	
Diversity = 7.5163e-02	k_t = [1, 1, 1, 1]			
Generation 1220: Best = 3.472644e+03	FES = 122100	$\mu_F = 0.741$	$\mu_{CR} = 0.451$	
Diversity = 7.9777e-02	k_t = [2, 4, 2, 4]			
Generation 1230: Best = 3.472333e+03	FES = 123100	$\mu_F = 0.746$	$\mu_{CR} = 0.442$	
Diversity = 8.9465e-02	k_t = [1, 1, 1, 1]			
Generation 1240: Best = 3.471922e+03	FES = 124100	$\mu_F = 0.751$	$\mu_{CR} = 0.433$	
Diversity = 9.8296e-02	k_t = [2, 2, 3, 2]			
Generation 1250: Best = 3.471522e+03	FES = 125100	$\mu_F = 0.736$	$\mu_{CR} = 0.441$	
Diversity = 1.1647e-01	k_t = [1, 1, 1, 1]			
Generation 1260: Best = 3.471257e+03	FES = 126100	$\mu_F = 0.731$	$\mu_{CR} = 0.444$	
Diversity = 1.0512e-01	k_t = [3, 2, 4, 3]			
Generation 1270: Best = 3.470892e+03	FES = 127100	$\mu_F = 0.730$	$\mu_{CR} = 0.441$	
Diversity = 1.0909e-01	k_t = [1, 1, 1, 1]			
Generation 1280: Best = 3.470490e+03	FES = 128100	$\mu_F = 0.717$	$\mu_{CR} = 0.445$	
Diversity = 1.0687e-01	k_t = [4, 2, 4, 2]			
Generation 1290: Best = 3.470206e+03	FES = 129100	$\mu_F = 0.722$	$\mu_{CR} = 0.451$	
Diversity = 1.0155e-01	k_t = [1, 1, 1, 1]			
Generation 1300: Best = 3.469691e+03	FES = 130100	$\mu_F = 0.708$	$\mu_{CR} = 0.453$	

Diversity = 9.0079e-02	k_t = [4, 2, 4, 2]			
Generation 1310: Best = 3.469382e+03	FES = 131100	$\mu_F = 0.714$	$\mu_{CR} = 0.453$	
Diversity = 9.3005e-02	k_t = [1, 1, 1, 1]			
Generation 1320: Best = 3.469092e+03	FES = 132100	$\mu_F = 0.715$	$\mu_{CR} = 0.451$	
Diversity = 8.3614e-02	k_t = [3, 2, 3, 4]			
Generation 1330: Best = 3.468728e+03	FES = 133100	$\mu_F = 0.724$	$\mu_{CR} = 0.455$	
Diversity = 9.2723e-02	k_t = [1, 1, 1, 1]			
Generation 1340: Best = 3.468350e+03	FES = 134100	$\mu_F = 0.727$	$\mu_{CR} = 0.451$	
Diversity = 8.4229e-02	k_t = [4, 3, 4, 4]			
Generation 1350: Best = 3.468052e+03	FES = 135100	$\mu_F = 0.731$	$\mu_{CR} = 0.443$	
Diversity = 8.0507e-02	k_t = [1, 1, 1, 1]			
Generation 1360: Best = 3.467784e+03	FES = 136100	$\mu_F = 0.734$	$\mu_{CR} = 0.441$	
Diversity = 7.9757e-02	k_t = [2, 2, 4, 2]			
Generation 1370: Best = 3.467589e+03	FES = 137100	$\mu_F = 0.719$	$\mu_{CR} = 0.451$	
Diversity = 8.2346e-02	k_t = [1, 1, 1, 1]			
Generation 1380: Best = 3.467329e+03	FES = 138100	$\mu_F = 0.723$	$\mu_{CR} = 0.447$	
Diversity = 7.3331e-02	k_t = [2, 4, 2, 4]			
Generation 1390: Best = 3.467089e+03	FES = 139100	$\mu_F = 0.738$	$\mu_{CR} = 0.448$	
Diversity = 8.5237e-02	k_t = [1, 1, 1, 1]			
Generation 1400: Best = 3.466821e+03	FES = 140100	$\mu_F = 0.741$	$\mu_{CR} = 0.444$	
Diversity = 9.2817e-02	k_t = [2, 4, 2, 3]			
Generation 1410: Best = 3.466468e+03	FES = 141100	$\mu_F = 0.730$	$\mu_{CR} = 0.437$	
Diversity = 1.0090e-01	k_t = [1, 1, 1, 1]			
Generation 1420: Best = 3.466101e+03	FES = 142100	$\mu_F = 0.725$	$\mu_{CR} = 0.437$	
Diversity = 1.0088e-01	k_t = [2, 3, 2, 3]			
Generation 1430: Best = 3.465713e+03	FES = 143100	$\mu_F = 0.726$	$\mu_{CR} = 0.430$	
Diversity = 1.1151e-01	k_t = [1, 1, 1, 1]			
Generation 1440: Best = 3.465357e+03	FES = 144100	$\mu_F = 0.716$	$\mu_{CR} = 0.438$	
Diversity = 9.6409e-02	k_t = [2, 2, 4, 4]			
Generation 1450: Best = 3.464846e+03	FES = 145100	$\mu_F = 0.728$	$\mu_{CR} = 0.431$	
Diversity = 1.0490e-01	k_t = [1, 1, 1, 1]			
Generation 1460: Best = 3.464491e+03	FES = 146100	$\mu_F = 0.720$	$\mu_{CR} = 0.426$	
Diversity = 9.6572e-02	k_t = [3, 3, 3, 3]			
Generation 1470: Best = 3.464130e+03	FES = 147100	$\mu_F = 0.734$	$\mu_{CR} = 0.421$	
Diversity = 1.0856e-01	k_t = [1, 1, 1, 1]			
Generation 1480: Best = 3.463643e+03	FES = 148100	$\mu_F = 0.739$	$\mu_{CR} = 0.416$	
Diversity = 1.1167e-01	k_t = [3, 3, 4, 4]			
Generation 1490: Best = 3.463307e+03	FES = 149100	$\mu_F = 0.733$	$\mu_{CR} = 0.413$	
Diversity = 1.1065e-01	k_t = [1, 1, 1, 1]			
Generation 1500: Best = 3.463093e+03	FES = 150100	$\mu_F = 0.726$	$\mu_{CR} = 0.410$	
Diversity = 1.0254e-01	k_t = [3, 2, 4, 4]			
Generation 1510: Best = 3.462861e+03	FES = 151100	$\mu_F = 0.716$	$\mu_{CR} = 0.405$	
Diversity = 8.7368e-02	k_t = [1, 1, 1, 1]			
Generation 1520: Best = 3.462706e+03	FES = 152100	$\mu_F = 0.705$	$\mu_{CR} = 0.401$	
Diversity = 7.8097e-02	k_t = [3, 3, 4, 2]			
Generation 1530: Best = 3.462505e+03	FES = 153100	$\mu_F = 0.706$	$\mu_{CR} = 0.398$	
Diversity = 7.6252e-02	k_t = [1, 1, 1, 1]			
Generation 1540: Best = 3.462280e+03	FES = 154100	$\mu_F = 0.716$	$\mu_{CR} = 0.401$	
Diversity = 7.8172e-02	k_t = [3, 2, 3, 2]			
Generation 1550: Best = 3.462062e+03	FES = 155100	$\mu_F = 0.721$	$\mu_{CR} = 0.395$	
Diversity = 8.2108e-02	k_t = [1, 1, 1, 1]			
Generation 1560: Best = 3.461750e+03	FES = 156100	$\mu_F = 0.736$	$\mu_{CR} = 0.395$	
Diversity = 9.6227e-02	k_t = [2, 3, 4, 2]			
Generation 1570: Best = 3.461460e+03	FES = 157100	$\mu_F = 0.740$	$\mu_{CR} = 0.398$	

Diversity = 1.0417e-01	k_t = [1, 1, 1, 1]			
Generation 1580: Best = 3.461158e+03	FES = 158100	$\mu_F = 0.744$	$\mu_{CR} = 0.395$	
Diversity = 9.0383e-02	k_t = [4, 3, 3, 3]			
Generation 1590: Best = 3.460926e+03	FES = 159100	$\mu_F = 0.733$	$\mu_{CR} = 0.395$	
Diversity = 8.8401e-02	k_t = [1, 1, 1, 1]			
Generation 1600: Best = 3.460667e+03	FES = 160100	$\mu_F = 0.734$	$\mu_{CR} = 0.402$	
Diversity = 8.9693e-02	k_t = [4, 3, 2, 3]			
Generation 1610: Best = 3.460439e+03	FES = 161100	$\mu_F = 0.740$	$\mu_{CR} = 0.401$	
Diversity = 9.2520e-02	k_t = [1, 1, 1, 1]			
Generation 1620: Best = 3.460130e+03	FES = 162100	$\mu_F = 0.726$	$\mu_{CR} = 0.398$	
Diversity = 9.1986e-02	k_t = [3, 4, 2, 2]			
Generation 1630: Best = 3.459948e+03	FES = 163100	$\mu_F = 0.735$	$\mu_{CR} = 0.395$	
Diversity = 9.3217e-02	k_t = [1, 1, 1, 1]			
Generation 1640: Best = 3.459682e+03	FES = 164100	$\mu_F = 0.720$	$\mu_{CR} = 0.387$	
Diversity = 8.9433e-02	k_t = [3, 2, 2, 4]			
Generation 1650: Best = 3.459382e+03	FES = 165100	$\mu_F = 0.740$	$\mu_{CR} = 0.379$	
Diversity = 9.9381e-02	k_t = [1, 1, 1, 1]			
Generation 1660: Best = 3.459038e+03	FES = 166100	$\mu_F = 0.738$	$\mu_{CR} = 0.376$	
Diversity = 1.0709e-01	k_t = [4, 4, 3, 2]			
Generation 1670: Best = 3.458704e+03	FES = 167100	$\mu_F = 0.745$	$\mu_{CR} = 0.370$	
Diversity = 1.0388e-01	k_t = [1, 1, 1, 1]			
Generation 1680: Best = 3.458494e+03	FES = 168100	$\mu_F = 0.736$	$\mu_{CR} = 0.370$	
Diversity = 9.6610e-02	k_t = [4, 3, 2, 3]			
Generation 1690: Best = 3.458167e+03	FES = 169100	$\mu_F = 0.743$	$\mu_{CR} = 0.366$	
Diversity = 9.8639e-02	k_t = [1, 1, 1, 1]			
Generation 1700: Best = 3.458008e+03	FES = 170100	$\mu_F = 0.741$	$\mu_{CR} = 0.368$	
Diversity = 9.5949e-02	k_t = [3, 2, 2, 2]			
Generation 1710: Best = 3.457748e+03	FES = 171100	$\mu_F = 0.749$	$\mu_{CR} = 0.363$	
Diversity = 9.3668e-02	k_t = [1, 1, 1, 1]			
Generation 1720: Best = 3.457540e+03	FES = 172100	$\mu_F = 0.752$	$\mu_{CR} = 0.367$	
Diversity = 9.1913e-02	k_t = [4, 3, 4, 2]			
Generation 1730: Best = 3.457292e+03	FES = 173100	$\mu_F = 0.744$	$\mu_{CR} = 0.360$	
Diversity = 9.6980e-02	k_t = [1, 1, 1, 1]			
Generation 1740: Best = 3.457110e+03	FES = 174100	$\mu_F = 0.741$	$\mu_{CR} = 0.352$	
Diversity = 9.3038e-02	k_t = [4, 4, 3, 2]			
Generation 1750: Best = 3.456892e+03	FES = 175100	$\mu_F = 0.741$	$\mu_{CR} = 0.345$	
Diversity = 9.6880e-02	k_t = [1, 1, 1, 1]			
Generation 1760: Best = 3.456642e+03	FES = 176100	$\mu_F = 0.741$	$\mu_{CR} = 0.342$	
Diversity = 1.0511e-01	k_t = [4, 2, 4, 2]			
Generation 1770: Best = 3.456416e+03	FES = 177100	$\mu_F = 0.728$	$\mu_{CR} = 0.334$	
Diversity = 1.0894e-01	k_t = [1, 1, 1, 1]			
Generation 1780: Best = 3.456174e+03	FES = 178100	$\mu_F = 0.719$	$\mu_{CR} = 0.323$	
Diversity = 1.0890e-01	k_t = [4, 2, 2, 3]			
Generation 1790: Best = 3.455954e+03	FES = 179100	$\mu_F = 0.734$	$\mu_{CR} = 0.314$	
Diversity = 1.0999e-01	k_t = [1, 1, 1, 1]			
Generation 1800: Best = 3.455719e+03	FES = 180100	$\mu_F = 0.730$	$\mu_{CR} = 0.306$	
Diversity = 1.1757e-01	k_t = [3, 2, 4, 2]			
Generation 1810: Best = 3.455549e+03	FES = 181100	$\mu_F = 0.723$	$\mu_{CR} = 0.296$	
Diversity = 1.2383e-01	k_t = [1, 1, 1, 1]			
Generation 1820: Best = 3.455261e+03	FES = 182100	$\mu_F = 0.727$	$\mu_{CR} = 0.292$	
Diversity = 1.1616e-01	k_t = [2, 3, 4, 3]			
Generation 1830: Best = 3.455096e+03	FES = 183100	$\mu_F = 0.726$	$\mu_{CR} = 0.294$	
Diversity = 1.2134e-01	k_t = [1, 1, 1, 1]			
Generation 1840: Best = 3.454804e+03	FES = 184100	$\mu_F = 0.725$	$\mu_{CR} = 0.286$	

Diversity = 1.1240e-01	k_t = [4, 4, 2, 4]			
Generation 1850: Best = 3.454559e+03	FES = 185100	$\mu_F = 0.727$	$\mu_{CR} = 0.273$	
Diversity = 1.0949e-01	k_t = [1, 1, 1, 1]			
Generation 1860: Best = 3.454321e+03	FES = 186100	$\mu_F = 0.735$	$\mu_{CR} = 0.265$	
Diversity = 1.0952e-01	k_t = [2, 2, 4, 4]			
Generation 1870: Best = 3.454025e+03	FES = 187100	$\mu_F = 0.734$	$\mu_{CR} = 0.260$	
Diversity = 1.1450e-01	k_t = [1, 1, 1, 1]			
Generation 1880: Best = 3.453807e+03	FES = 188100	$\mu_F = 0.721$	$\mu_{CR} = 0.258$	
Diversity = 1.1234e-01	k_t = [2, 2, 2, 2]			
Generation 1890: Best = 3.453529e+03	FES = 189100	$\mu_F = 0.714$	$\mu_{CR} = 0.256$	
Diversity = 1.1708e-01	k_t = [1, 1, 1, 1]			
Generation 1900: Best = 3.453191e+03	FES = 190100	$\mu_F = 0.704$	$\mu_{CR} = 0.251$	
Diversity = 1.0462e-01	k_t = [4, 4, 2, 2]			
Generation 1910: Best = 3.453006e+03	FES = 191100	$\mu_F = 0.719$	$\mu_{CR} = 0.249$	
Diversity = 9.9082e-02	k_t = [1, 1, 1, 1]			
Generation 1920: Best = 3.452564e+03	FES = 192100	$\mu_F = 0.723$	$\mu_{CR} = 0.246$	
Diversity = 1.0410e-01	k_t = [3, 4, 2, 2]			
Generation 1930: Best = 3.452308e+03	FES = 193100	$\mu_F = 0.747$	$\mu_{CR} = 0.260$	
Diversity = 1.2130e-01	k_t = [1, 1, 1, 1]			
Generation 1940: Best = 3.452037e+03	FES = 194100	$\mu_F = 0.751$	$\mu_{CR} = 0.248$	
Diversity = 1.2025e-01	k_t = [4, 3, 4, 2]			
Generation 1950: Best = 3.451740e+03	FES = 195100	$\mu_F = 0.758$	$\mu_{CR} = 0.249$	
Diversity = 1.2407e-01	k_t = [1, 1, 1, 1]			
Generation 1960: Best = 3.451498e+03	FES = 196100	$\mu_F = 0.747$	$\mu_{CR} = 0.244$	
Diversity = 1.1611e-01	k_t = [4, 4, 2, 4]			
Generation 1970: Best = 3.451083e+03	FES = 197100	$\mu_F = 0.746$	$\mu_{CR} = 0.238$	
Diversity = 1.1789e-01	k_t = [1, 1, 1, 1]			
Generation 1980: Best = 3.450885e+03	FES = 198100	$\mu_F = 0.750$	$\mu_{CR} = 0.243$	
Diversity = 1.2336e-01	k_t = [3, 4, 4, 2]			
Generation 1990: Best = 3.450668e+03	FES = 199100	$\mu_F = 0.742$	$\mu_{CR} = 0.233$	
Diversity = 1.2243e-01	k_t = [1, 1, 1, 1]			
Generation 2000: Best = 3.450341e+03	FES = 200100	$\mu_F = 0.726$	$\mu_{CR} = 0.225$	
Diversity = 1.2139e-01	k_t = [3, 3, 3, 2]			
Generation 2010: Best = 3.449926e+03	FES = 201100	$\mu_F = 0.735$	$\mu_{CR} = 0.219$	
Diversity = 1.3799e-01	k_t = [1, 1, 1, 1]			
Generation 2020: Best = 3.449758e+03	FES = 202100	$\mu_F = 0.744$	$\mu_{CR} = 0.218$	
Diversity = 1.4315e-01	k_t = [3, 4, 2, 3]			
Generation 2030: Best = 3.449249e+03	FES = 203100	$\mu_F = 0.737$	$\mu_{CR} = 0.217$	
Diversity = 1.4342e-01	k_t = [1, 1, 1, 1]			
Generation 2040: Best = 3.449089e+03	FES = 204100	$\mu_F = 0.752$	$\mu_{CR} = 0.206$	
Diversity = 1.4932e-01	k_t = [4, 4, 2, 2]			
Generation 2050: Best = 3.448818e+03	FES = 205100	$\mu_F = 0.751$	$\mu_{CR} = 0.209$	
Diversity = 1.5056e-01	k_t = [1, 1, 1, 1]			
Generation 2060: Best = 3.448576e+03	FES = 206100	$\mu_F = 0.740$	$\mu_{CR} = 0.199$	
Diversity = 1.3394e-01	k_t = [4, 4, 4, 2]			
Generation 2070: Best = 3.448301e+03	FES = 207100	$\mu_F = 0.742$	$\mu_{CR} = 0.210$	
Diversity = 1.2929e-01	k_t = [1, 1, 1, 1]			
Generation 2080: Best = 3.448017e+03	FES = 208100	$\mu_F = 0.738$	$\mu_{CR} = 0.209$	
Diversity = 1.2968e-01	k_t = [2, 2, 3, 4]			
Generation 2090: Best = 3.447770e+03	FES = 209100	$\mu_F = 0.735$	$\mu_{CR} = 0.198$	
Diversity = 1.3852e-01	k_t = [1, 1, 1, 1]			
Generation 2100: Best = 3.447520e+03	FES = 210100	$\mu_F = 0.728$	$\mu_{CR} = 0.196$	
Diversity = 1.2711e-01	k_t = [3, 2, 2, 4]			
Generation 2110: Best = 3.447251e+03	FES = 211100	$\mu_F = 0.725$	$\mu_{CR} = 0.195$	

Diversity = 1.3057e-01	k_t = [1, 1, 1, 1]			
Generation 2120: Best = 3.446986e+03	FES = 212100	$\mu_F = 0.734$	$\mu_{CR} = 0.196$	
Diversity = 1.3012e-01	k_t = [3, 2, 3, 4]			
Generation 2130: Best = 3.446778e+03	FES = 213100	$\mu_F = 0.742$	$\mu_{CR} = 0.188$	
Diversity = 1.3326e-01	k_t = [1, 1, 1, 1]			
Generation 2140: Best = 3.446507e+03	FES = 214100	$\mu_F = 0.732$	$\mu_{CR} = 0.186$	
Diversity = 1.2701e-01	k_t = [4, 4, 3, 4]			
Generation 2150: Best = 3.446347e+03	FES = 215100	$\mu_F = 0.733$	$\mu_{CR} = 0.187$	
Diversity = 1.2262e-01	k_t = [1, 1, 1, 1]			
Generation 2160: Best = 3.446102e+03	FES = 216100	$\mu_F = 0.725$	$\mu_{CR} = 0.178$	
Diversity = 1.2243e-01	k_t = [2, 3, 2, 4]			
Generation 2170: Best = 3.445887e+03	FES = 217100	$\mu_F = 0.737$	$\mu_{CR} = 0.171$	
Diversity = 1.2238e-01	k_t = [1, 1, 1, 1]			
Generation 2180: Best = 3.445634e+03	FES = 218100	$\mu_F = 0.734$	$\mu_{CR} = 0.179$	
Diversity = 1.2014e-01	k_t = [3, 4, 2, 4]			
Generation 2190: Best = 3.445452e+03	FES = 219100	$\mu_F = 0.737$	$\mu_{CR} = 0.183$	
Diversity = 1.1649e-01	k_t = [1, 1, 1, 1]			
Generation 2200: Best = 3.445256e+03	FES = 220100	$\mu_F = 0.743$	$\mu_{CR} = 0.186$	
Diversity = 1.0924e-01	k_t = [4, 4, 3, 3]			
Generation 2210: Best = 3.445124e+03	FES = 221100	$\mu_F = 0.746$	$\mu_{CR} = 0.180$	
Diversity = 1.0662e-01	k_t = [1, 1, 1, 1]			
Generation 2220: Best = 3.444964e+03	FES = 222100	$\mu_F = 0.754$	$\mu_{CR} = 0.178$	
Diversity = 1.0037e-01	k_t = [3, 2, 2, 4]			
Generation 2230: Best = 3.444814e+03	FES = 223100	$\mu_F = 0.753$	$\mu_{CR} = 0.170$	
Diversity = 8.8889e-02	k_t = [1, 1, 1, 1]			
Generation 2240: Best = 3.444647e+03	FES = 224100	$\mu_F = 0.765$	$\mu_{CR} = 0.166$	
Diversity = 8.5587e-02	k_t = [3, 3, 3, 2]			
Generation 2250: Best = 3.444524e+03	FES = 225100	$\mu_F = 0.775$	$\mu_{CR} = 0.177$	
Diversity = 9.2579e-02	k_t = [1, 1, 1, 1]			
Generation 2260: Best = 3.444353e+03	FES = 226100	$\mu_F = 0.778$	$\mu_{CR} = 0.172$	
Diversity = 9.2376e-02	k_t = [4, 2, 2, 3]			
Generation 2270: Best = 3.444239e+03	FES = 227100	$\mu_F = 0.792$	$\mu_{CR} = 0.167$	
Diversity = 9.5377e-02	k_t = [1, 1, 1, 1]			
Generation 2280: Best = 3.444065e+03	FES = 228100	$\mu_F = 0.787$	$\mu_{CR} = 0.165$	
Diversity = 8.8703e-02	k_t = [4, 3, 2, 4]			
Generation 2290: Best = 3.443869e+03	FES = 229100	$\mu_F = 0.791$	$\mu_{CR} = 0.163$	
Diversity = 9.3917e-02	k_t = [1, 1, 1, 1]			
Generation 2300: Best = 3.443646e+03	FES = 230100	$\mu_F = 0.784$	$\mu_{CR} = 0.176$	
Diversity = 1.0049e-01	k_t = [4, 2, 4, 2]			
Generation 2310: Best = 3.443420e+03	FES = 231100	$\mu_F = 0.783$	$\mu_{CR} = 0.181$	
Diversity = 1.0862e-01	k_t = [1, 1, 1, 1]			
Generation 2320: Best = 3.443123e+03	FES = 232100	$\mu_F = 0.791$	$\mu_{CR} = 0.173$	
Diversity = 1.2448e-01	k_t = [2, 2, 2, 4]			
Generation 2330: Best = 3.442811e+03	FES = 233100	$\mu_F = 0.777$	$\mu_{CR} = 0.175$	
Diversity = 1.3704e-01	k_t = [1, 1, 1, 1]			
Generation 2340: Best = 3.442372e+03	FES = 234100	$\mu_F = 0.766$	$\mu_{CR} = 0.181$	
Diversity = 1.3726e-01	k_t = [2, 4, 3, 4]			
Generation 2350: Best = 3.442019e+03	FES = 235100	$\mu_F = 0.775$	$\mu_{CR} = 0.183$	
Diversity = 1.5100e-01	k_t = [1, 1, 1, 1]			
Generation 2360: Best = 3.441547e+03	FES = 236100	$\mu_F = 0.784$	$\mu_{CR} = 0.186$	
Diversity = 1.5911e-01	k_t = [3, 2, 4, 4]			
Generation 2370: Best = 3.441296e+03	FES = 237100	$\mu_F = 0.772$	$\mu_{CR} = 0.178$	
Diversity = 1.7205e-01	k_t = [1, 1, 1, 1]			
Generation 2380: Best = 3.440951e+03	FES = 238100	$\mu_F = 0.775$	$\mu_{CR} = 0.165$	

Diversity = 1.8053e-01	k_t = [2, 4, 3, 4]			
Generation 2390: Best = 3.440498e+03	FES = 239100	$\mu_F = 0.766$	$\mu_{CR} = 0.164$	
Diversity = 1.7924e-01	k_t = [1, 1, 1, 1]			
Generation 2400: Best = 3.440210e+03	FES = 240100	$\mu_F = 0.758$	$\mu_{CR} = 0.168$	
Diversity = 1.8845e-01	k_t = [3, 4, 3, 3]			
Generation 2410: Best = 3.439890e+03	FES = 241100	$\mu_F = 0.745$	$\mu_{CR} = 0.171$	
Diversity = 1.9300e-01	k_t = [1, 1, 1, 1]			
Generation 2420: Best = 3.439438e+03	FES = 242100	$\mu_F = 0.731$	$\mu_{CR} = 0.164$	
Diversity = 1.9237e-01	k_t = [4, 3, 2, 2]			
Generation 2430: Best = 3.438989e+03	FES = 243100	$\mu_F = 0.754$	$\mu_{CR} = 0.150$	
Diversity = 2.4259e-01	k_t = [1, 1, 1, 1]			
Generation 2440: Best = 3.438479e+03	FES = 244100	$\mu_F = 0.763$	$\mu_{CR} = 0.139$	
Diversity = 2.3606e-01	k_t = [2, 4, 2, 4]			
Generation 2450: Best = 3.438172e+03	FES = 245100	$\mu_F = 0.762$	$\mu_{CR} = 0.132$	
Diversity = 2.1713e-01	k_t = [1, 1, 1, 1]			
Generation 2460: Best = 3.437835e+03	FES = 246100	$\mu_F = 0.766$	$\mu_{CR} = 0.133$	
Diversity = 1.9147e-01	k_t = [3, 4, 3, 4]			
Generation 2470: Best = 3.437377e+03	FES = 247100	$\mu_F = 0.772$	$\mu_{CR} = 0.128$	
Diversity = 1.7620e-01	k_t = [1, 1, 1, 1]			
Generation 2480: Best = 3.437126e+03	FES = 248100	$\mu_F = 0.790$	$\mu_{CR} = 0.131$	
Diversity = 1.8617e-01	k_t = [2, 2, 3, 2]			
Generation 2490: Best = 3.436808e+03	FES = 249100	$\mu_F = 0.789$	$\mu_{CR} = 0.132$	
Diversity = 1.8770e-01	k_t = [1, 1, 1, 1]			
Generation 2500: Best = 3.436520e+03	FES = 250100	$\mu_F = 0.779$	$\mu_{CR} = 0.142$	
Diversity = 1.9121e-01	k_t = [3, 2, 2, 2]			
Generation 2510: Best = 3.436180e+03	FES = 251100	$\mu_F = 0.767$	$\mu_{CR} = 0.142$	
Diversity = 1.8701e-01	k_t = [1, 1, 1, 1]			
Generation 2520: Best = 3.435866e+03	FES = 252100	$\mu_F = 0.761$	$\mu_{CR} = 0.141$	
Diversity = 1.7909e-01	k_t = [4, 4, 4, 3]			
Generation 2530: Best = 3.435381e+03	FES = 253100	$\mu_F = 0.763$	$\mu_{CR} = 0.141$	
Diversity = 1.8488e-01	k_t = [1, 1, 1, 1]			
Generation 2540: Best = 3.435246e+03	FES = 254100	$\mu_F = 0.764$	$\mu_{CR} = 0.145$	
Diversity = 1.7429e-01	k_t = [3, 3, 4, 3]			
Generation 2550: Best = 3.434979e+03	FES = 255100	$\mu_F = 0.748$	$\mu_{CR} = 0.149$	
Diversity = 1.8030e-01	k_t = [1, 1, 1, 1]			
Generation 2560: Best = 3.434716e+03	FES = 256100	$\mu_F = 0.747$	$\mu_{CR} = 0.147$	
Diversity = 1.7187e-01	k_t = [3, 2, 4, 2]			
Generation 2570: Best = 3.434546e+03	FES = 257100	$\mu_F = 0.738$	$\mu_{CR} = 0.156$	
Diversity = 1.6931e-01	k_t = [1, 1, 1, 1]			
Generation 2580: Best = 3.434118e+03	FES = 258100	$\mu_F = 0.754$	$\mu_{CR} = 0.157$	
Diversity = 1.8029e-01	k_t = [3, 3, 2, 4]			
Generation 2590: Best = 3.433786e+03	FES = 259100	$\mu_F = 0.749$	$\mu_{CR} = 0.160$	
Diversity = 1.6863e-01	k_t = [1, 1, 1, 1]			
Generation 2600: Best = 3.433425e+03	FES = 260100	$\mu_F = 0.762$	$\mu_{CR} = 0.157$	
Diversity = 1.7297e-01	k_t = [4, 2, 3, 2]			
Generation 2610: Best = 3.433171e+03	FES = 261100	$\mu_F = 0.772$	$\mu_{CR} = 0.152$	
Diversity = 1.8080e-01	k_t = [1, 1, 1, 1]			
Generation 2620: Best = 3.432824e+03	FES = 262100	$\mu_F = 0.773$	$\mu_{CR} = 0.149$	
Diversity = 1.6286e-01	k_t = [4, 2, 2, 4]			
Generation 2630: Best = 3.432248e+03	FES = 263100	$\mu_F = 0.758$	$\mu_{CR} = 0.147$	
Diversity = 1.5469e-01	k_t = [1, 1, 1, 1]			
Generation 2640: Best = 3.432190e+03	FES = 264100	$\mu_F = 0.771$	$\mu_{CR} = 0.146$	
Diversity = 1.6393e-01	k_t = [4, 2, 3, 2]			
Generation 2650: Best = 3.432055e+03	FES = 265100	$\mu_F = 0.764$	$\mu_{CR} = 0.138$	

Diversity = 1.6119e-01	k_t = [1, 1, 1, 1]			
Generation 2660: Best = 3.431787e+03	FES = 266100	$\mu_F = 0.762$	$\mu_{CR} = 0.147$	
Diversity = 1.5604e-01	k_t = [3, 4, 2, 2]			
Generation 2670: Best = 3.431582e+03	FES = 267100	$\mu_F = 0.766$	$\mu_{CR} = 0.143$	
Diversity = 1.5034e-01	k_t = [1, 1, 1, 1]			
Generation 2680: Best = 3.431352e+03	FES = 268100	$\mu_F = 0.755$	$\mu_{CR} = 0.145$	
Diversity = 1.3441e-01	k_t = [2, 3, 4, 4]			
Generation 2690: Best = 3.431137e+03	FES = 269100	$\mu_F = 0.753$	$\mu_{CR} = 0.137$	
Diversity = 1.3421e-01	k_t = [1, 1, 1, 1]			
Generation 2700: Best = 3.430846e+03	FES = 270100	$\mu_F = 0.758$	$\mu_{CR} = 0.147$	
Diversity = 1.2575e-01	k_t = [3, 2, 4, 4]			
Generation 2710: Best = 3.430658e+03	FES = 271100	$\mu_F = 0.773$	$\mu_{CR} = 0.146$	
Diversity = 1.3345e-01	k_t = [1, 1, 1, 1]			
Generation 2720: Best = 3.430492e+03	FES = 272100	$\mu_F = 0.773$	$\mu_{CR} = 0.146$	
Diversity = 1.3072e-01	k_t = [2, 2, 4, 2]			
Generation 2730: Best = 3.430287e+03	FES = 273100	$\mu_F = 0.771$	$\mu_{CR} = 0.136$	
Diversity = 1.3108e-01	k_t = [1, 1, 1, 1]			
Generation 2740: Best = 3.430172e+03	FES = 274100	$\mu_F = 0.767$	$\mu_{CR} = 0.143$	
Diversity = 1.2683e-01	k_t = [2, 4, 4, 4]			
Generation 2750: Best = 3.429957e+03	FES = 275100	$\mu_F = 0.758$	$\mu_{CR} = 0.140$	
Diversity = 1.2416e-01	k_t = [1, 1, 1, 1]			
Generation 2760: Best = 3.429745e+03	FES = 276100	$\mu_F = 0.759$	$\mu_{CR} = 0.141$	
Diversity = 1.2203e-01	k_t = [4, 2, 2, 4]			
Generation 2770: Best = 3.429550e+03	FES = 277100	$\mu_F = 0.757$	$\mu_{CR} = 0.146$	
Diversity = 1.2377e-01	k_t = [1, 1, 1, 1]			
Generation 2780: Best = 3.429326e+03	FES = 278100	$\mu_F = 0.751$	$\mu_{CR} = 0.149$	
Diversity = 1.2507e-01	k_t = [2, 4, 4, 2]			
Generation 2790: Best = 3.428897e+03	FES = 279100	$\mu_F = 0.762$	$\mu_{CR} = 0.154$	
Diversity = 1.2811e-01	k_t = [1, 1, 1, 1]			
Generation 2800: Best = 3.428525e+03	FES = 280100	$\mu_F = 0.774$	$\mu_{CR} = 0.151$	
Diversity = 1.5216e-01	k_t = [3, 4, 4, 2]			
Generation 2810: Best = 3.428323e+03	FES = 281100	$\mu_F = 0.779$	$\mu_{CR} = 0.151$	
Diversity = 1.5783e-01	k_t = [1, 1, 1, 1]			
Generation 2820: Best = 3.428095e+03	FES = 282100	$\mu_F = 0.779$	$\mu_{CR} = 0.142$	
Diversity = 1.4670e-01	k_t = [4, 2, 2, 3]			
Generation 2830: Best = 3.427836e+03	FES = 283100	$\mu_F = 0.768$	$\mu_{CR} = 0.151$	
Diversity = 1.5474e-01	k_t = [1, 1, 1, 1]			
Generation 2840: Best = 3.427441e+03	FES = 284100	$\mu_F = 0.776$	$\mu_{CR} = 0.151$	
Diversity = 1.5182e-01	k_t = [4, 4, 3, 4]			
Generation 2850: Best = 3.427165e+03	FES = 285100	$\mu_F = 0.772$	$\mu_{CR} = 0.155$	
Diversity = 1.5876e-01	k_t = [1, 1, 1, 1]			
Generation 2860: Best = 3.426888e+03	FES = 286100	$\mu_F = 0.767$	$\mu_{CR} = 0.157$	
Diversity = 1.5615e-01	k_t = [3, 4, 4, 4]			
Generation 2870: Best = 3.426620e+03	FES = 287100	$\mu_F = 0.758$	$\mu_{CR} = 0.161$	
Diversity = 1.3971e-01	k_t = [1, 1, 1, 1]			
Generation 2880: Best = 3.426267e+03	FES = 288100	$\mu_F = 0.776$	$\mu_{CR} = 0.162$	
Diversity = 1.4784e-01	k_t = [3, 2, 2, 4]			
Generation 2890: Best = 3.426078e+03	FES = 289100	$\mu_F = 0.782$	$\mu_{CR} = 0.155$	
Diversity = 1.5304e-01	k_t = [1, 1, 1, 1]			
Generation 2900: Best = 3.425715e+03	FES = 290100	$\mu_F = 0.790$	$\mu_{CR} = 0.157$	
Diversity = 1.5444e-01	k_t = [4, 4, 3, 4]			
Generation 2910: Best = 3.425374e+03	FES = 291100	$\mu_F = 0.798$	$\mu_{CR} = 0.152$	
Diversity = 1.5207e-01	k_t = [1, 1, 1, 1]			
Generation 2920: Best = 3.425210e+03	FES = 292100	$\mu_F = 0.795$	$\mu_{CR} = 0.147$	

Diversity = 1.4414e-01	k_t = [3, 4, 3, 4]			
Generation 2930: Best = 3.424885e+03	FES = 293100	$\mu_F = 0.778$	$\mu_{CR} = 0.149$	
Diversity = 1.4139e-01	k_t = [1, 1, 1, 1]			
Generation 2940: Best = 3.424625e+03	FES = 294100	$\mu_F = 0.774$	$\mu_{CR} = 0.153$	
Diversity = 1.4007e-01	k_t = [2, 3, 4, 4]			
Generation 2950: Best = 3.424331e+03	FES = 295100	$\mu_F = 0.764$	$\mu_{CR} = 0.160$	
Diversity = 1.4076e-01	k_t = [1, 1, 1, 1]			
Generation 2960: Best = 3.423979e+03	FES = 296100	$\mu_F = 0.774$	$\mu_{CR} = 0.157$	
Diversity = 1.5067e-01	k_t = [3, 4, 4, 2]			
Generation 2970: Best = 3.423490e+03	FES = 297100	$\mu_F = 0.777$	$\mu_{CR} = 0.151$	
Diversity = 1.5731e-01	k_t = [1, 1, 1, 1]			
Generation 2980: Best = 3.423324e+03	FES = 298100	$\mu_F = 0.777$	$\mu_{CR} = 0.148$	
Diversity = 1.5975e-01	k_t = [3, 4, 2, 3]			
Generation 2990: Best = 3.423131e+03	FES = 299100	$\mu_F = 0.778$	$\mu_{CR} = 0.152$	
Diversity = 1.6143e-01	k_t = [1, 1, 1, 1]			
Generation 3000: Best = 3.422627e+03	FES = 300100	$\mu_F = 0.767$	$\mu_{CR} = 0.157$	
Diversity = 1.5867e-01	k_t = [3, 2, 4, 3]			
Generation 3010: Best = 3.422349e+03	FES = 301100	$\mu_F = 0.756$	$\mu_{CR} = 0.161$	
Diversity = 1.6100e-01	k_t = [1, 1, 1, 1]			
Generation 3020: Best = 3.421894e+03	FES = 302100	$\mu_F = 0.749$	$\mu_{CR} = 0.164$	
Diversity = 1.4428e-01	k_t = [4, 3, 4, 3]			
Generation 3030: Best = 3.421569e+03	FES = 303100	$\mu_F = 0.761$	$\mu_{CR} = 0.165$	
Diversity = 1.5511e-01	k_t = [1, 1, 1, 1]			
Generation 3040: Best = 3.421108e+03	FES = 304100	$\mu_F = 0.759$	$\mu_{CR} = 0.181$	
Diversity = 1.5435e-01	k_t = [2, 4, 3, 3]			
Generation 3050: Best = 3.420712e+03	FES = 305100	$\mu_F = 0.758$	$\mu_{CR} = 0.177$	
Diversity = 1.7240e-01	k_t = [1, 1, 1, 1]			
Generation 3060: Best = 3.420382e+03	FES = 306100	$\mu_F = 0.749$	$\mu_{CR} = 0.177$	
Diversity = 1.6643e-01	k_t = [4, 4, 3, 3]			
Generation 3070: Best = 3.419961e+03	FES = 307100	$\mu_F = 0.734$	$\mu_{CR} = 0.180$	
Diversity = 1.6989e-01	k_t = [1, 1, 1, 1]			
Generation 3080: Best = 3.419644e+03	FES = 308100	$\mu_F = 0.731$	$\mu_{CR} = 0.176$	
Diversity = 1.5730e-01	k_t = [2, 2, 3, 4]			
Generation 3090: Best = 3.419313e+03	FES = 309100	$\mu_F = 0.744$	$\mu_{CR} = 0.171$	
Diversity = 1.5738e-01	k_t = [1, 1, 1, 1]			
Generation 3100: Best = 3.418968e+03	FES = 310100	$\mu_F = 0.728$	$\mu_{CR} = 0.184$	
Diversity = 1.6270e-01	k_t = [2, 4, 2, 2]			
Generation 3110: Best = 3.418560e+03	FES = 311100	$\mu_F = 0.736$	$\mu_{CR} = 0.178$	
Diversity = 1.7209e-01	k_t = [1, 1, 1, 1]			
Generation 3120: Best = 3.418329e+03	FES = 312100	$\mu_F = 0.717$	$\mu_{CR} = 0.177$	
Diversity = 1.5873e-01	k_t = [2, 4, 2, 3]			
Generation 3130: Best = 3.417988e+03	FES = 313100	$\mu_F = 0.728$	$\mu_{CR} = 0.173$	
Diversity = 1.5340e-01	k_t = [1, 1, 1, 1]			
Generation 3140: Best = 3.417351e+03	FES = 314100	$\mu_F = 0.740$	$\mu_{CR} = 0.177$	
Diversity = 1.5827e-01	k_t = [3, 4, 3, 2]			
Generation 3150: Best = 3.417080e+03	FES = 315100	$\mu_F = 0.739$	$\mu_{CR} = 0.180$	
Diversity = 1.6802e-01	k_t = [1, 1, 1, 1]			
Generation 3160: Best = 3.416367e+03	FES = 316100	$\mu_F = 0.734$	$\mu_{CR} = 0.175$	
Diversity = 1.5844e-01	k_t = [3, 3, 3, 3]			
Generation 3170: Best = 3.415909e+03	FES = 317100	$\mu_F = 0.741$	$\mu_{CR} = 0.176$	
Diversity = 1.6904e-01	k_t = [1, 1, 1, 1]			
Generation 3180: Best = 3.415732e+03	FES = 318100	$\mu_F = 0.746$	$\mu_{CR} = 0.184$	
Diversity = 1.5383e-01	k_t = [2, 4, 4, 4]			
Generation 3190: Best = 3.415272e+03	FES = 319100	$\mu_F = 0.752$	$\mu_{CR} = 0.190$	

Diversity = 1.4568e-01	k_t = [1, 1, 1, 1]			
Generation 3200: Best = 3.414889e+03	FES = 320100	$\mu_F = 0.729$	$\mu_{CR} = 0.191$	
Diversity = 1.3006e-01	k_t = [2, 2, 3, 4]			
Generation 3210: Best = 3.414399e+03	FES = 321100	$\mu_F = 0.733$	$\mu_{CR} = 0.193$	
Diversity = 1.3081e-01	k_t = [1, 1, 1, 1]			
Generation 3220: Best = 3.414134e+03	FES = 322100	$\mu_F = 0.750$	$\mu_{CR} = 0.194$	
Diversity = 1.3271e-01	k_t = [2, 4, 3, 4]			
Generation 3230: Best = 3.413643e+03	FES = 323100	$\mu_F = 0.766$	$\mu_{CR} = 0.192$	
Diversity = 1.4664e-01	k_t = [1, 1, 1, 1]			
Generation 3240: Best = 3.413337e+03	FES = 324100	$\mu_F = 0.767$	$\mu_{CR} = 0.178$	
Diversity = 1.5236e-01	k_t = [3, 3, 4, 2]			
Generation 3250: Best = 3.412772e+03	FES = 325100	$\mu_F = 0.779$	$\mu_{CR} = 0.179$	
Diversity = 1.6386e-01	k_t = [1, 1, 1, 1]			
Generation 3260: Best = 3.412312e+03	FES = 326100	$\mu_F = 0.779$	$\mu_{CR} = 0.170$	
Diversity = 1.7398e-01	k_t = [2, 4, 4, 2]			
Generation 3270: Best = 3.411736e+03	FES = 327100	$\mu_F = 0.771$	$\mu_{CR} = 0.164$	
Diversity = 1.8018e-01	k_t = [1, 1, 1, 1]			
Generation 3280: Best = 3.411475e+03	FES = 328100	$\mu_F = 0.760$	$\mu_{CR} = 0.171$	
Diversity = 1.8504e-01	k_t = [4, 2, 4, 2]			
Generation 3290: Best = 3.411069e+03	FES = 329100	$\mu_F = 0.760$	$\mu_{CR} = 0.185$	
Diversity = 1.7465e-01	k_t = [1, 1, 1, 1]			
Generation 3300: Best = 3.410755e+03	FES = 330100	$\mu_F = 0.765$	$\mu_{CR} = 0.187$	
Diversity = 1.6081e-01	k_t = [3, 2, 4, 4]			
Generation 3310: Best = 3.410299e+03	FES = 331100	$\mu_F = 0.770$	$\mu_{CR} = 0.185$	
Diversity = 1.6447e-01	k_t = [1, 1, 1, 1]			
Generation 3320: Best = 3.409360e+03	FES = 332100	$\mu_F = 0.774$	$\mu_{CR} = 0.178$	
Diversity = 1.7271e-01	k_t = [2, 3, 3, 4]			
Generation 3330: Best = 3.409135e+03	FES = 333100	$\mu_F = 0.778$	$\mu_{CR} = 0.172$	
Diversity = 1.6694e-01	k_t = [1, 1, 1, 1]			
Generation 3340: Best = 3.408585e+03	FES = 334100	$\mu_F = 0.777$	$\mu_{CR} = 0.162$	
Diversity = 1.7740e-01	k_t = [2, 2, 3, 2]			
Generation 3350: Best = 3.408328e+03	FES = 335100	$\mu_F = 0.767$	$\mu_{CR} = 0.164$	
Diversity = 1.8170e-01	k_t = [1, 1, 1, 1]			
Generation 3360: Best = 3.408053e+03	FES = 336100	$\mu_F = 0.775$	$\mu_{CR} = 0.162$	
Diversity = 1.5963e-01	k_t = [3, 3, 4, 2]			
Generation 3370: Best = 3.407663e+03	FES = 337100	$\mu_F = 0.766$	$\mu_{CR} = 0.159$	
Diversity = 1.5519e-01	k_t = [1, 1, 1, 1]			
Generation 3380: Best = 3.407396e+03	FES = 338100	$\mu_F = 0.745$	$\mu_{CR} = 0.151$	
Diversity = 1.4565e-01	k_t = [4, 3, 3, 2]			
Generation 3390: Best = 3.407173e+03	FES = 339100	$\mu_F = 0.740$	$\mu_{CR} = 0.165$	
Diversity = 1.3803e-01	k_t = [1, 1, 1, 1]			
Generation 3400: Best = 3.406835e+03	FES = 340100	$\mu_F = 0.744$	$\mu_{CR} = 0.164$	
Diversity = 1.3685e-01	k_t = [2, 4, 3, 4]			
Generation 3410: Best = 3.406557e+03	FES = 341100	$\mu_F = 0.757$	$\mu_{CR} = 0.174$	
Diversity = 1.3662e-01	k_t = [1, 1, 1, 1]			
Generation 3420: Best = 3.406109e+03	FES = 342100	$\mu_F = 0.755$	$\mu_{CR} = 0.174$	
Diversity = 1.3834e-01	k_t = [4, 4, 4, 2]			
Generation 3430: Best = 3.405779e+03	FES = 343100	$\mu_F = 0.759$	$\mu_{CR} = 0.170$	
Diversity = 1.4983e-01	k_t = [1, 1, 1, 1]			
Generation 3440: Best = 3.405297e+03	FES = 344100	$\mu_F = 0.770$	$\mu_{CR} = 0.170$	
Diversity = 1.5488e-01	k_t = [2, 3, 4, 3]			
Generation 3450: Best = 3.404777e+03	FES = 345100	$\mu_F = 0.765$	$\mu_{CR} = 0.173$	
Diversity = 1.6893e-01	k_t = [1, 1, 1, 1]			
Generation 3460: Best = 3.404198e+03	FES = 346100	$\mu_F = 0.767$	$\mu_{CR} = 0.171$	

Diversity = 1.8561e-01	k_t = [4, 2, 2, 2]			
Generation 3470: Best = 3.403843e+03	FES = 347100	$\mu_F = 0.764$	$\mu_{CR} = 0.164$	
Diversity = 1.9430e-01	k_t = [1, 1, 1, 1]			
Generation 3480: Best = 3.403264e+03	FES = 348100	$\mu_F = 0.747$	$\mu_{CR} = 0.171$	
Diversity = 1.7916e-01	k_t = [3, 2, 3, 3]			
Generation 3490: Best = 3.402859e+03	FES = 349100	$\mu_F = 0.734$	$\mu_{CR} = 0.178$	
Diversity = 1.7779e-01	k_t = [1, 1, 1, 1]			
Generation 3500: Best = 3.402431e+03	FES = 350100	$\mu_F = 0.737$	$\mu_{CR} = 0.181$	
Diversity = 1.6538e-01	k_t = [2, 3, 4, 4]			
Generation 3510: Best = 3.402126e+03	FES = 351100	$\mu_F = 0.744$	$\mu_{CR} = 0.173$	
Diversity = 1.6250e-01	k_t = [1, 1, 1, 1]			
Generation 3520: Best = 3.401691e+03	FES = 352100	$\mu_F = 0.742$	$\mu_{CR} = 0.170$	
Diversity = 1.5133e-01	k_t = [3, 2, 3, 4]			
Generation 3530: Best = 3.401338e+03	FES = 353100	$\mu_F = 0.752$	$\mu_{CR} = 0.179$	
Diversity = 1.5693e-01	k_t = [1, 1, 1, 1]			
Generation 3540: Best = 3.401005e+03	FES = 354100	$\mu_F = 0.756$	$\mu_{CR} = 0.180$	
Diversity = 1.5449e-01	k_t = [2, 3, 3, 4]			
Generation 3550: Best = 3.400712e+03	FES = 355100	$\mu_F = 0.751$	$\mu_{CR} = 0.176$	
Diversity = 1.5164e-01	k_t = [1, 1, 1, 1]			
Generation 3560: Best = 3.400410e+03	FES = 356100	$\mu_F = 0.746$	$\mu_{CR} = 0.167$	
Diversity = 1.4908e-01	k_t = [4, 3, 4, 3]			
Generation 3570: Best = 3.399966e+03	FES = 357100	$\mu_F = 0.755$	$\mu_{CR} = 0.176$	
Diversity = 1.5316e-01	k_t = [1, 1, 1, 1]			
Generation 3580: Best = 3.399687e+03	FES = 358100	$\mu_F = 0.778$	$\mu_{CR} = 0.181$	
Diversity = 1.5039e-01	k_t = [3, 2, 3, 3]			
Generation 3590: Best = 3.399344e+03	FES = 359100	$\mu_F = 0.775$	$\mu_{CR} = 0.180$	
Diversity = 1.5030e-01	k_t = [1, 1, 1, 1]			
Generation 3600: Best = 3.399133e+03	FES = 360100	$\mu_F = 0.756$	$\mu_{CR} = 0.176$	
Diversity = 1.3830e-01	k_t = [4, 2, 3, 4]			
Generation 3610: Best = 3.398797e+03	FES = 361100	$\mu_F = 0.749$	$\mu_{CR} = 0.178$	
Diversity = 1.3826e-01	k_t = [1, 1, 1, 1]			
Generation 3620: Best = 3.398652e+03	FES = 362100	$\mu_F = 0.748$	$\mu_{CR} = 0.177$	
Diversity = 1.3434e-01	k_t = [4, 2, 4, 4]			
Generation 3630: Best = 3.398309e+03	FES = 363100	$\mu_F = 0.742$	$\mu_{CR} = 0.170$	
Diversity = 1.3460e-01	k_t = [1, 1, 1, 1]			
Generation 3640: Best = 3.398100e+03	FES = 364100	$\mu_F = 0.751$	$\mu_{CR} = 0.175$	
Diversity = 1.2569e-01	k_t = [2, 4, 4, 4]			
Generation 3650: Best = 3.397745e+03	FES = 365100	$\mu_F = 0.751$	$\mu_{CR} = 0.185$	
Diversity = 1.3280e-01	k_t = [1, 1, 1, 1]			
Generation 3660: Best = 3.397587e+03	FES = 366100	$\mu_F = 0.744$	$\mu_{CR} = 0.183$	
Diversity = 1.2583e-01	k_t = [4, 3, 4, 3]			
Generation 3670: Best = 3.397346e+03	FES = 367100	$\mu_F = 0.738$	$\mu_{CR} = 0.184$	
Diversity = 1.2341e-01	k_t = [1, 1, 1, 1]			
Generation 3680: Best = 3.397179e+03	FES = 368100	$\mu_F = 0.730$	$\mu_{CR} = 0.182$	
Diversity = 1.1201e-01	k_t = [3, 3, 3, 4]			
Generation 3690: Best = 3.396998e+03	FES = 369100	$\mu_F = 0.729$	$\mu_{CR} = 0.183$	
Diversity = 1.0523e-01	k_t = [1, 1, 1, 1]			
Generation 3700: Best = 3.396772e+03	FES = 370100	$\mu_F = 0.747$	$\mu_{CR} = 0.181$	
Diversity = 1.0469e-01	k_t = [3, 4, 4, 3]			
Generation 3710: Best = 3.396605e+03	FES = 371100	$\mu_F = 0.753$	$\mu_{CR} = 0.183$	
Diversity = 1.0225e-01	k_t = [1, 1, 1, 1]			
Generation 3720: Best = 3.396288e+03	FES = 372100	$\mu_F = 0.756$	$\mu_{CR} = 0.181$	
Diversity = 1.0605e-01	k_t = [2, 4, 2, 2]			
Generation 3730: Best = 3.396166e+03	FES = 373100	$\mu_F = 0.745$	$\mu_{CR} = 0.182$	

Diversity = 1.0877e-01	k_t = [1, 1, 1, 1]			
Generation 3740: Best = 3.395893e+03	FES = 374100	$\mu_F = 0.760$	$\mu_{CR} = 0.177$	
Diversity = 1.0421e-01	k_t = [4, 4, 4, 4]			
Generation 3750: Best = 3.395744e+03	FES = 375100	$\mu_F = 0.766$	$\mu_{CR} = 0.183$	
Diversity = 9.9274e-02	k_t = [1, 1, 1, 1]			
Generation 3760: Best = 3.395584e+03	FES = 376100	$\mu_F = 0.762$	$\mu_{CR} = 0.182$	
Diversity = 1.0145e-01	k_t = [4, 2, 2, 2]			
Generation 3770: Best = 3.395306e+03	FES = 377100	$\mu_F = 0.765$	$\mu_{CR} = 0.174$	
Diversity = 1.1008e-01	k_t = [1, 1, 1, 1]			
Generation 3780: Best = 3.395033e+03	FES = 378100	$\mu_F = 0.763$	$\mu_{CR} = 0.169$	
Diversity = 1.1658e-01	k_t = [3, 4, 3, 3]			
Generation 3790: Best = 3.394807e+03	FES = 379100	$\mu_F = 0.767$	$\mu_{CR} = 0.174$	
Diversity = 1.2146e-01	k_t = [1, 1, 1, 1]			
Generation 3800: Best = 3.394402e+03	FES = 380100	$\mu_F = 0.777$	$\mu_{CR} = 0.164$	
Diversity = 1.2433e-01	k_t = [3, 2, 2, 4]			
Generation 3810: Best = 3.394158e+03	FES = 381100	$\mu_F = 0.772$	$\mu_{CR} = 0.170$	
Diversity = 1.2667e-01	k_t = [1, 1, 1, 1]			
Generation 3820: Best = 3.394070e+03	FES = 382100	$\mu_F = 0.768$	$\mu_{CR} = 0.163$	
Diversity = 1.3093e-01	k_t = [4, 4, 3, 2]			
Generation 3830: Best = 3.393652e+03	FES = 383100	$\mu_F = 0.786$	$\mu_{CR} = 0.156$	
Diversity = 1.3721e-01	k_t = [1, 1, 1, 1]			
Generation 3840: Best = 3.393383e+03	FES = 384100	$\mu_F = 0.777$	$\mu_{CR} = 0.159$	
Diversity = 1.5035e-01	k_t = [4, 2, 2, 2]			
Generation 3850: Best = 3.392984e+03	FES = 385100	$\mu_F = 0.763$	$\mu_{CR} = 0.161$	
Diversity = 1.6148e-01	k_t = [1, 1, 1, 1]			
Generation 3860: Best = 3.392439e+03	FES = 386100	$\mu_F = 0.769$	$\mu_{CR} = 0.164$	
Diversity = 1.5314e-01	k_t = [4, 2, 2, 4]			
Generation 3870: Best = 3.391936e+03	FES = 387100	$\mu_F = 0.772$	$\mu_{CR} = 0.166$	
Diversity = 1.5709e-01	k_t = [1, 1, 1, 1]			
Generation 3880: Best = 3.391672e+03	FES = 388100	$\mu_F = 0.774$	$\mu_{CR} = 0.167$	
Diversity = 1.4832e-01	k_t = [2, 4, 4, 4]			
Generation 3890: Best = 3.391448e+03	FES = 389100	$\mu_F = 0.774$	$\mu_{CR} = 0.165$	
Diversity = 1.4624e-01	k_t = [1, 1, 1, 1]			
Generation 3900: Best = 3.391163e+03	FES = 390100	$\mu_F = 0.762$	$\mu_{CR} = 0.159$	
Diversity = 1.3480e-01	k_t = [4, 3, 2, 3]			
Generation 3910: Best = 3.390903e+03	FES = 391100	$\mu_F = 0.776$	$\mu_{CR} = 0.158$	
Diversity = 1.4145e-01	k_t = [1, 1, 1, 1]			
Generation 3920: Best = 3.390415e+03	FES = 392100	$\mu_F = 0.775$	$\mu_{CR} = 0.168$	
Diversity = 1.4179e-01	k_t = [2, 4, 4, 3]			
Generation 3930: Best = 3.389967e+03	FES = 393100	$\mu_F = 0.763$	$\mu_{CR} = 0.177$	
Diversity = 1.6074e-01	k_t = [1, 1, 1, 1]			
Generation 3940: Best = 3.389840e+03	FES = 394100	$\mu_F = 0.770$	$\mu_{CR} = 0.172$	
Diversity = 1.4310e-01	k_t = [4, 3, 2, 3]			
Generation 3950: Best = 3.389641e+03	FES = 395100	$\mu_F = 0.760$	$\mu_{CR} = 0.164$	
Diversity = 1.3847e-01	k_t = [1, 1, 1, 1]			
Generation 3960: Best = 3.389374e+03	FES = 396100	$\mu_F = 0.747$	$\mu_{CR} = 0.163$	
Diversity = 1.3261e-01	k_t = [2, 3, 4, 3]			
Generation 3970: Best = 3.389086e+03	FES = 397100	$\mu_F = 0.739$	$\mu_{CR} = 0.167$	
Diversity = 1.3065e-01	k_t = [1, 1, 1, 1]			
Generation 3980: Best = 3.388843e+03	FES = 398100	$\mu_F = 0.729$	$\mu_{CR} = 0.179$	
Diversity = 1.2711e-01	k_t = [2, 3, 2, 3]			
Generation 3990: Best = 3.388536e+03	FES = 399100	$\mu_F = 0.743$	$\mu_{CR} = 0.175$	
Diversity = 1.2787e-01	k_t = [1, 1, 1, 1]			
Generation 4000: Best = 3.388388e+03	FES = 400100	$\mu_F = 0.740$	$\mu_{CR} = 0.163$	

Diversity = 1.2472e-01 k_t = [2, 4, 2, 3]
 Generation 4010: Best = 3.388252e+03 FES = 401100 μ_F = 0.744 μ_{CR} = 0.155
 Diversity = 1.2380e-01 k_t = [1, 1, 1, 1]
 Generation 4020: Best = 3.387993e+03 FES = 402100 μ_F = 0.739 μ_{CR} = 0.153
 Diversity = 1.2277e-01 k_t = [3, 2, 2, 2]
 Generation 4030: Best = 3.387822e+03 FES = 403100 μ_F = 0.735 μ_{CR} = 0.157
 Diversity = 1.1997e-01 k_t = [1, 1, 1, 1]
 Generation 4040: Best = 3.387700e+03 FES = 404100 μ_F = 0.738 μ_{CR} = 0.164
 Diversity = 1.0534e-01 k_t = [2, 3, 4, 4]
 Generation 4050: Best = 3.387472e+03 FES = 405100 μ_F = 0.749 μ_{CR} = 0.157
 Diversity = 9.9694e-02 k_t = [1, 1, 1, 1]
 Generation 4060: Best = 3.387408e+03 FES = 406100 μ_F = 0.738 μ_{CR} = 0.163
 Diversity = 9.7461e-02 k_t = [2, 2, 4, 3]
 Generation 4070: Best = 3.387300e+03 FES = 407100 μ_F = 0.717 μ_{CR} = 0.164
 Diversity = 9.8734e-02 k_t = [1, 1, 1, 1]
 Generation 4080: Best = 3.387069e+03 FES = 408100 μ_F = 0.724 μ_{CR} = 0.165
 Diversity = 1.0092e-01 k_t = [2, 4, 3, 2]
 Generation 4090: Best = 3.386964e+03 FES = 409100 μ_F = 0.724 μ_{CR} = 0.164
 Diversity = 1.0104e-01 k_t = [1, 1, 1, 1]
 Generation 4100: Best = 3.386828e+03 FES = 410100 μ_F = 0.724 μ_{CR} = 0.163
 Diversity = 9.3068e-02 k_t = [3, 4, 4, 3]
 Generation 4110: Best = 3.386720e+03 FES = 411100 μ_F = 0.737 μ_{CR} = 0.167
 Diversity = 9.3086e-02 k_t = [1, 1, 1, 1]
 Generation 4120: Best = 3.386566e+03 FES = 412100 μ_F = 0.738 μ_{CR} = 0.176
 Diversity = 8.9823e-02 k_t = [3, 3, 2, 4]
 Generation 4130: Best = 3.386401e+03 FES = 413100 μ_F = 0.764 μ_{CR} = 0.175
 Diversity = 8.8948e-02 k_t = [1, 1, 1, 1]
 Generation 4140: Best = 3.386283e+03 FES = 414100 μ_F = 0.778 μ_{CR} = 0.180
 Diversity = 9.1323e-02 k_t = [3, 2, 2, 3]
 Generation 4150: Best = 3.386127e+03 FES = 415100 μ_F = 0.772 μ_{CR} = 0.184
 Diversity = 9.4839e-02 k_t = [1, 1, 1, 1]
 Generation 4160: Best = 3.385921e+03 FES = 416100 μ_F = 0.780 μ_{CR} = 0.179
 Diversity = 1.0011e-01 k_t = [2, 2, 3, 4]
 Generation 4170: Best = 3.385659e+03 FES = 417100 μ_F = 0.788 μ_{CR} = 0.167
 Diversity = 1.0247e-01 k_t = [1, 1, 1, 1]
 Generation 4180: Best = 3.385623e+03 FES = 418100 μ_F = 0.765 μ_{CR} = 0.165
 Diversity = 1.0125e-01 k_t = [3, 3, 3, 2]
 Generation 4190: Best = 3.385491e+03 FES = 419100 μ_F = 0.761 μ_{CR} = 0.158
 Diversity = 1.0040e-01 k_t = [1, 1, 1, 1]
 Generation 4200: Best = 3.385373e+03 FES = 420100 μ_F = 0.773 μ_{CR} = 0.165
 Diversity = 1.0250e-01 k_t = [2, 4, 2, 2]
 Generation 4210: Best = 3.385137e+03 FES = 421100 μ_F = 0.766 μ_{CR} = 0.171
 Diversity = 1.0507e-01 k_t = [1, 1, 1, 1]
 Generation 4220: Best = 3.385021e+03 FES = 422100 μ_F = 0.757 μ_{CR} = 0.173
 Diversity = 1.0119e-01 k_t = [4, 2, 4, 3]
 Generation 4230: Best = 3.384844e+03 FES = 423100 μ_F = 0.754 μ_{CR} = 0.177
 Diversity = 1.0504e-01 k_t = [1, 1, 1, 1]
 Generation 4240: Best = 3.384572e+03 FES = 424100 μ_F = 0.754 μ_{CR} = 0.168
 Diversity = 1.0123e-01 k_t = [4, 2, 3, 3]
 Generation 4250: Best = 3.384452e+03 FES = 425100 μ_F = 0.763 μ_{CR} = 0.166
 Diversity = 1.1041e-01 k_t = [1, 1, 1, 1]
 Generation 4260: Best = 3.384242e+03 FES = 426100 μ_F = 0.764 μ_{CR} = 0.165
 Diversity = 1.1124e-01 k_t = [4, 2, 4, 4]
 Generation 4270: Best = 3.384111e+03 FES = 427100 μ_F = 0.761 μ_{CR} = 0.157

Diversity = 1.1367e-01	k_t = [1, 1, 1, 1]			
Generation 4280: Best = 3.383987e+03	FES = 428100	$\mu_F = 0.764$	$\mu_{CR} = 0.158$	
Diversity = 1.2104e-01	k_t = [2, 2, 3, 2]			
Generation 4290: Best = 3.383764e+03	FES = 429100	$\mu_F = 0.748$	$\mu_{CR} = 0.171$	
Diversity = 1.2753e-01	k_t = [1, 1, 1, 1]			
Generation 4300: Best = 3.383542e+03	FES = 430100	$\mu_F = 0.749$	$\mu_{CR} = 0.167$	
Diversity = 1.2650e-01	k_t = [3, 3, 2, 2]			
Generation 4310: Best = 3.383439e+03	FES = 431100	$\mu_F = 0.745$	$\mu_{CR} = 0.153$	
Diversity = 1.2198e-01	k_t = [1, 1, 1, 1]			
Generation 4320: Best = 3.383286e+03	FES = 432100	$\mu_F = 0.748$	$\mu_{CR} = 0.149$	
Diversity = 1.1434e-01	k_t = [4, 4, 4, 4]			
Generation 4330: Best = 3.383119e+03	FES = 433100	$\mu_F = 0.754$	$\mu_{CR} = 0.152$	
Diversity = 1.1151e-01	k_t = [1, 1, 1, 1]			
Generation 4340: Best = 3.382908e+03	FES = 434100	$\mu_F = 0.749$	$\mu_{CR} = 0.152$	
Diversity = 1.0742e-01	k_t = [2, 3, 4, 3]			
Generation 4350: Best = 3.382678e+03	FES = 435100	$\mu_F = 0.748$	$\mu_{CR} = 0.164$	
Diversity = 9.9568e-02	k_t = [1, 1, 1, 1]			
Generation 4360: Best = 3.382587e+03	FES = 436100	$\mu_F = 0.758$	$\mu_{CR} = 0.172$	
Diversity = 9.7075e-02	k_t = [2, 4, 2, 4]			
Generation 4370: Best = 3.382469e+03	FES = 437100	$\mu_F = 0.763$	$\mu_{CR} = 0.172$	
Diversity = 8.4963e-02	k_t = [1, 1, 1, 1]			
Generation 4380: Best = 3.382377e+03	FES = 438100	$\mu_F = 0.766$	$\mu_{CR} = 0.176$	
Diversity = 8.4336e-02	k_t = [3, 4, 2, 3]			
Generation 4390: Best = 3.382284e+03	FES = 439100	$\mu_F = 0.752$	$\mu_{CR} = 0.184$	
Diversity = 8.1447e-02	k_t = [1, 1, 1, 1]			
Generation 4400: Best = 3.382142e+03	FES = 440100	$\mu_F = 0.755$	$\mu_{CR} = 0.186$	
Diversity = 7.6849e-02	k_t = [3, 2, 3, 4]			
Generation 4410: Best = 3.382006e+03	FES = 441100	$\mu_F = 0.752$	$\mu_{CR} = 0.187$	
Diversity = 8.1389e-02	k_t = [1, 1, 1, 1]			
Generation 4420: Best = 3.381941e+03	FES = 442100	$\mu_F = 0.760$	$\mu_{CR} = 0.185$	
Diversity = 7.9451e-02	k_t = [4, 3, 3, 4]			
Generation 4430: Best = 3.381819e+03	FES = 443100	$\mu_F = 0.762$	$\mu_{CR} = 0.181$	
Diversity = 8.1850e-02	k_t = [1, 1, 1, 1]			
Generation 4440: Best = 3.381730e+03	FES = 444100	$\mu_F = 0.744$	$\mu_{CR} = 0.194$	
Diversity = 7.6426e-02	k_t = [4, 4, 4, 3]			
Generation 4450: Best = 3.381602e+03	FES = 445100	$\mu_F = 0.744$	$\mu_{CR} = 0.193$	
Diversity = 7.9654e-02	k_t = [1, 1, 1, 1]			
Generation 4460: Best = 3.381486e+03	FES = 446100	$\mu_F = 0.745$	$\mu_{CR} = 0.196$	
Diversity = 7.7140e-02	k_t = [4, 3, 2, 4]			
Generation 4470: Best = 3.381324e+03	FES = 447100	$\mu_F = 0.754$	$\mu_{CR} = 0.196$	
Diversity = 8.4278e-02	k_t = [1, 1, 1, 1]			
Generation 4480: Best = 3.381172e+03	FES = 448100	$\mu_F = 0.756$	$\mu_{CR} = 0.191$	
Diversity = 9.0807e-02	k_t = [2, 2, 3, 2]			
Generation 4490: Best = 3.381001e+03	FES = 449100	$\mu_F = 0.748$	$\mu_{CR} = 0.196$	
Diversity = 9.7788e-02	k_t = [1, 1, 1, 1]			
Generation 4500: Best = 3.380811e+03	FES = 450100	$\mu_F = 0.750$	$\mu_{CR} = 0.205$	
Diversity = 9.3789e-02	k_t = [2, 3, 3, 3]			
Generation 4510: Best = 3.380637e+03	FES = 451100	$\mu_F = 0.760$	$\mu_{CR} = 0.201$	
Diversity = 1.0227e-01	k_t = [1, 1, 1, 1]			
Generation 4520: Best = 3.380517e+03	FES = 452100	$\mu_F = 0.744$	$\mu_{CR} = 0.202$	
Diversity = 9.2533e-02	k_t = [3, 4, 3, 4]			
Generation 4530: Best = 3.380337e+03	FES = 453100	$\mu_F = 0.741$	$\mu_{CR} = 0.197$	
Diversity = 8.5527e-02	k_t = [1, 1, 1, 1]			
Generation 4540: Best = 3.380240e+03	FES = 454100	$\mu_F = 0.743$	$\mu_{CR} = 0.198$	

Diversity = 8.4439e-02 k_t = [3, 2, 4, 3]
 Generation 4550: Best = 3.380000e+03 FES = 455100 μ_F = 0.745 μ_{CR} = 0.187
 Diversity = 9.0617e-02 k_t = [1, 1, 1, 1]
 Generation 4560: Best = 3.379901e+03 FES = 456100 μ_F = 0.742 μ_{CR} = 0.195
 Diversity = 9.0921e-02 k_t = [2, 3, 4, 2]
 Generation 4570: Best = 3.379816e+03 FES = 457100 μ_F = 0.722 μ_{CR} = 0.200
 Diversity = 8.3794e-02 k_t = [1, 1, 1, 1]
 Generation 4580: Best = 3.379710e+03 FES = 458100 μ_F = 0.743 μ_{CR} = 0.189
 Diversity = 7.9333e-02 k_t = [2, 3, 4, 4]
 Generation 4590: Best = 3.379578e+03 FES = 459100 μ_F = 0.740 μ_{CR} = 0.191
 Diversity = 8.2852e-02 k_t = [1, 1, 1, 1]
 Generation 4600: Best = 3.379482e+03 FES = 460100 μ_F = 0.734 μ_{CR} = 0.196
 Diversity = 8.0612e-02 k_t = [2, 2, 4, 4]
 Generation 4610: Best = 3.379376e+03 FES = 461100 μ_F = 0.733 μ_{CR} = 0.192
 Diversity = 7.4689e-02 k_t = [1, 1, 1, 1]
 Generation 4620: Best = 3.379320e+03 FES = 462100 μ_F = 0.733 μ_{CR} = 0.195
 Diversity = 7.5107e-02 k_t = [2, 3, 4, 2]
 Generation 4630: Best = 3.379224e+03 FES = 463100 μ_F = 0.744 μ_{CR} = 0.192
 Diversity = 8.2580e-02 k_t = [1, 1, 1, 1]
 Generation 4640: Best = 3.379099e+03 FES = 464100 μ_F = 0.737 μ_{CR} = 0.192
 Diversity = 7.5889e-02 k_t = [2, 3, 2, 4]
 Generation 4650: Best = 3.379066e+03 FES = 465100 μ_F = 0.757 μ_{CR} = 0.189
 Diversity = 8.1462e-02 k_t = [1, 1, 1, 1]
 Generation 4660: Best = 3.378917e+03 FES = 466100 μ_F = 0.753 μ_{CR} = 0.182
 Diversity = 8.5805e-02 k_t = [4, 4, 2, 2]
 Generation 4670: Best = 3.378843e+03 FES = 467100 μ_F = 0.740 μ_{CR} = 0.184
 Diversity = 8.5994e-02 k_t = [1, 1, 1, 1]
 Generation 4680: Best = 3.378767e+03 FES = 468100 μ_F = 0.740 μ_{CR} = 0.183
 Diversity = 7.8511e-02 k_t = [4, 2, 3, 2]
 Generation 4690: Best = 3.378647e+03 FES = 469100 μ_F = 0.735 μ_{CR} = 0.182
 Diversity = 8.1198e-02 k_t = [1, 1, 1, 1]
 Generation 4700: Best = 3.378557e+03 FES = 470100 μ_F = 0.737 μ_{CR} = 0.190
 Diversity = 7.3739e-02 k_t = [2, 2, 4, 3]
 Generation 4710: Best = 3.378478e+03 FES = 471100 μ_F = 0.729 μ_{CR} = 0.195
 Diversity = 7.6396e-02 k_t = [1, 1, 1, 1]
 Generation 4720: Best = 3.378374e+03 FES = 472100 μ_F = 0.732 μ_{CR} = 0.186
 Diversity = 7.2100e-02 k_t = [2, 4, 4, 3]
 Generation 4730: Best = 3.378295e+03 FES = 473100 μ_F = 0.747 μ_{CR} = 0.182
 Diversity = 7.3303e-02 k_t = [1, 1, 1, 1]
 Generation 4740: Best = 3.378217e+03 FES = 474100 μ_F = 0.748 μ_{CR} = 0.173
 Diversity = 7.0320e-02 k_t = [2, 4, 2, 4]
 Generation 4750: Best = 3.378104e+03 FES = 475100 μ_F = 0.734 μ_{CR} = 0.182
 Diversity = 6.8086e-02 k_t = [1, 1, 1, 1]
 Generation 4760: Best = 3.377989e+03 FES = 476100 μ_F = 0.742 μ_{CR} = 0.185
 Diversity = 7.2373e-02 k_t = [3, 3, 4, 3]
 Generation 4770: Best = 3.377904e+03 FES = 477100 μ_F = 0.756 μ_{CR} = 0.178
 Diversity = 7.7976e-02 k_t = [1, 1, 1, 1]
 Generation 4780: Best = 3.377817e+03 FES = 478100 μ_F = 0.752 μ_{CR} = 0.185
 Diversity = 8.2623e-02 k_t = [4, 4, 4, 2]
 Generation 4790: Best = 3.377698e+03 FES = 479100 μ_F = 0.737 μ_{CR} = 0.185
 Diversity = 8.4820e-02 k_t = [1, 1, 1, 1]
 Generation 4800: Best = 3.377601e+03 FES = 480100 μ_F = 0.727 μ_{CR} = 0.182
 Diversity = 8.3265e-02 k_t = [2, 3, 2, 2]
 Generation 4810: Best = 3.377514e+03 FES = 481100 μ_F = 0.729 μ_{CR} = 0.175

Diversity = 8.2607e-02	k_t = [1, 1, 1, 1]			
Generation 4820: Best = 3.377437e+03	FES = 482100	$\mu_F = 0.725$	$\mu_{CR} = 0.175$	
Diversity = 8.0788e-02	k_t = [3, 4, 3, 2]			
Generation 4830: Best = 3.377326e+03	FES = 483100	$\mu_F = 0.711$	$\mu_{CR} = 0.168$	
Diversity = 7.4774e-02	k_t = [1, 1, 1, 1]			
Generation 4840: Best = 3.377302e+03	FES = 484100	$\mu_F = 0.705$	$\mu_{CR} = 0.164$	
Diversity = 6.7535e-02	k_t = [2, 3, 3, 3]			
Generation 4850: Best = 3.377222e+03	FES = 485100	$\mu_F = 0.714$	$\mu_{CR} = 0.167$	
Diversity = 6.6536e-02	k_t = [1, 1, 1, 1]			
Generation 4860: Best = 3.377116e+03	FES = 486100	$\mu_F = 0.737$	$\mu_{CR} = 0.172$	
Diversity = 7.3828e-02	k_t = [3, 3, 4, 2]			
Generation 4870: Best = 3.377007e+03	FES = 487100	$\mu_F = 0.746$	$\mu_{CR} = 0.171$	
Diversity = 7.5480e-02	k_t = [1, 1, 1, 1]			
Generation 4880: Best = 3.376910e+03	FES = 488100	$\mu_F = 0.747$	$\mu_{CR} = 0.174$	
Diversity = 7.1566e-02	k_t = [2, 4, 3, 4]			
Generation 4890: Best = 3.376802e+03	FES = 489100	$\mu_F = 0.748$	$\mu_{CR} = 0.175$	
Diversity = 7.5699e-02	k_t = [1, 1, 1, 1]			
Generation 4900: Best = 3.376693e+03	FES = 490100	$\mu_F = 0.735$	$\mu_{CR} = 0.168$	
Diversity = 7.5732e-02	k_t = [4, 3, 4, 2]			
Generation 4910: Best = 3.376609e+03	FES = 491100	$\mu_F = 0.752$	$\mu_{CR} = 0.169$	
Diversity = 7.8716e-02	k_t = [1, 1, 1, 1]			
Generation 4920: Best = 3.376496e+03	FES = 492100	$\mu_F = 0.740$	$\mu_{CR} = 0.172$	
Diversity = 7.5331e-02	k_t = [3, 3, 2, 4]			
Generation 4930: Best = 3.376434e+03	FES = 493100	$\mu_F = 0.744$	$\mu_{CR} = 0.159$	
Diversity = 7.3529e-02	k_t = [1, 1, 1, 1]			
Generation 4940: Best = 3.376333e+03	FES = 494100	$\mu_F = 0.727$	$\mu_{CR} = 0.171$	
Diversity = 6.6616e-02	k_t = [3, 3, 4, 3]			
Generation 4950: Best = 3.376232e+03	FES = 495100	$\mu_F = 0.740$	$\mu_{CR} = 0.176$	
Diversity = 6.9647e-02	k_t = [1, 1, 1, 1]			
Generation 4960: Best = 3.376180e+03	FES = 496100	$\mu_F = 0.745$	$\mu_{CR} = 0.178$	
Diversity = 6.7437e-02	k_t = [4, 2, 4, 4]			
Generation 4970: Best = 3.376101e+03	FES = 497100	$\mu_F = 0.750$	$\mu_{CR} = 0.183$	
Diversity = 6.4775e-02	k_t = [1, 1, 1, 1]			
Generation 4980: Best = 3.376044e+03	FES = 498100	$\mu_F = 0.759$	$\mu_{CR} = 0.186$	
Diversity = 5.7489e-02	k_t = [4, 4, 3, 4]			
Generation 4990: Best = 3.375949e+03	FES = 499100	$\mu_F = 0.766$	$\mu_{CR} = 0.181$	
Diversity = 5.5820e-02	k_t = [1, 1, 1, 1]			
Generation 5000: Best = 3.375880e+03	FES = 500100	$\mu_F = 0.773$	$\mu_{CR} = 0.183$	
Diversity = 6.0310e-02	k_t = [4, 4, 3, 3]			
Generation 5010: Best = 3.375828e+03	FES = 501100	$\mu_F = 0.762$	$\mu_{CR} = 0.184$	
Diversity = 5.9447e-02	k_t = [1, 1, 1, 1]			
Generation 5020: Best = 3.375777e+03	FES = 502100	$\mu_F = 0.773$	$\mu_{CR} = 0.177$	
Diversity = 5.8583e-02	k_t = [3, 4, 4, 2]			
Generation 5030: Best = 3.375711e+03	FES = 503100	$\mu_F = 0.770$	$\mu_{CR} = 0.186$	
Diversity = 5.9007e-02	k_t = [1, 1, 1, 1]			
Generation 5040: Best = 3.375619e+03	FES = 504100	$\mu_F = 0.770$	$\mu_{CR} = 0.186$	
Diversity = 6.5125e-02	k_t = [3, 4, 4, 2]			
Generation 5050: Best = 3.375539e+03	FES = 505100	$\mu_F = 0.769$	$\mu_{CR} = 0.192$	
Diversity = 7.2355e-02	k_t = [1, 1, 1, 1]			
Generation 5060: Best = 3.375464e+03	FES = 506100	$\mu_F = 0.754$	$\mu_{CR} = 0.188$	
Diversity = 7.2756e-02	k_t = [2, 4, 4, 3]			
Generation 5070: Best = 3.375399e+03	FES = 507100	$\mu_F = 0.737$	$\mu_{CR} = 0.185$	
Diversity = 7.3827e-02	k_t = [1, 1, 1, 1]			
Generation 5080: Best = 3.375353e+03	FES = 508100	$\mu_F = 0.713$	$\mu_{CR} = 0.186$	

Diversity = 6.7913e-02	k_t = [2, 4, 4, 3]			
Generation 5090: Best = 3.375293e+03	FES = 509100	$\mu_F = 0.711$	$\mu_{CR} = 0.185$	
Diversity = 6.3023e-02	k_t = [1, 1, 1, 1]			
Generation 5100: Best = 3.375233e+03	FES = 510100	$\mu_F = 0.719$	$\mu_{CR} = 0.181$	
Diversity = 5.5327e-02	k_t = [4, 2, 2, 4]			
Generation 5110: Best = 3.375181e+03	FES = 511100	$\mu_F = 0.727$	$\mu_{CR} = 0.174$	
Diversity = 5.2860e-02	k_t = [1, 1, 1, 1]			
Generation 5120: Best = 3.375144e+03	FES = 512100	$\mu_F = 0.730$	$\mu_{CR} = 0.180$	
Diversity = 5.1515e-02	k_t = [2, 3, 2, 3]			
Generation 5130: Best = 3.375088e+03	FES = 513100	$\mu_F = 0.748$	$\mu_{CR} = 0.185$	
Diversity = 5.2959e-02	k_t = [1, 1, 1, 1]			
Generation 5140: Best = 3.375018e+03	FES = 514100	$\mu_F = 0.754$	$\mu_{CR} = 0.181$	
Diversity = 5.5118e-02	k_t = [2, 3, 3, 2]			
Generation 5150: Best = 3.374954e+03	FES = 515100	$\mu_F = 0.746$	$\mu_{CR} = 0.193$	
Diversity = 5.9635e-02	k_t = [1, 1, 1, 1]			
Generation 5160: Best = 3.374879e+03	FES = 516100	$\mu_F = 0.749$	$\mu_{CR} = 0.195$	
Diversity = 6.1217e-02	k_t = [3, 4, 4, 2]			
Generation 5170: Best = 3.374834e+03	FES = 517100	$\mu_F = 0.755$	$\mu_{CR} = 0.194$	
Diversity = 5.9206e-02	k_t = [1, 1, 1, 1]			
Generation 5180: Best = 3.374753e+03	FES = 518100	$\mu_F = 0.759$	$\mu_{CR} = 0.196$	
Diversity = 5.9758e-02	k_t = [2, 3, 3, 2]			
Generation 5190: Best = 3.374675e+03	FES = 519100	$\mu_F = 0.758$	$\mu_{CR} = 0.195$	
Diversity = 6.2137e-02	k_t = [1, 1, 1, 1]			
Generation 5200: Best = 3.374610e+03	FES = 520100	$\mu_F = 0.750$	$\mu_{CR} = 0.198$	
Diversity = 5.6930e-02	k_t = [2, 4, 3, 4]			
Generation 5210: Best = 3.374562e+03	FES = 521100	$\mu_F = 0.745$	$\mu_{CR} = 0.199$	
Diversity = 5.2800e-02	k_t = [1, 1, 1, 1]			
Generation 5220: Best = 3.374497e+03	FES = 522100	$\mu_F = 0.737$	$\mu_{CR} = 0.193$	
Diversity = 5.3012e-02	k_t = [4, 3, 2, 4]			
Generation 5230: Best = 3.374441e+03	FES = 523100	$\mu_F = 0.742$	$\mu_{CR} = 0.190$	
Diversity = 5.3455e-02	k_t = [1, 1, 1, 1]			
Generation 5240: Best = 3.374368e+03	FES = 524100	$\mu_F = 0.750$	$\mu_{CR} = 0.193$	
Diversity = 5.4566e-02	k_t = [3, 3, 2, 3]			
Generation 5250: Best = 3.374298e+03	FES = 525100	$\mu_F = 0.755$	$\mu_{CR} = 0.195$	
Diversity = 5.9654e-02	k_t = [1, 1, 1, 1]			
Generation 5260: Best = 3.374218e+03	FES = 526100	$\mu_F = 0.751$	$\mu_{CR} = 0.191$	
Diversity = 6.1668e-02	k_t = [4, 4, 3, 2]			
Generation 5270: Best = 3.374133e+03	FES = 527100	$\mu_F = 0.756$	$\mu_{CR} = 0.193$	
Diversity = 6.5371e-02	k_t = [1, 1, 1, 1]			
Generation 5280: Best = 3.374061e+03	FES = 528100	$\mu_F = 0.748$	$\mu_{CR} = 0.195$	
Diversity = 6.2566e-02	k_t = [2, 3, 4, 2]			
Generation 5290: Best = 3.374023e+03	FES = 529100	$\mu_F = 0.741$	$\mu_{CR} = 0.193$	
Diversity = 6.4213e-02	k_t = [1, 1, 1, 1]			
Generation 5300: Best = 3.373933e+03	FES = 530100	$\mu_F = 0.746$	$\mu_{CR} = 0.184$	
Diversity = 5.9155e-02	k_t = [3, 3, 4, 4]			
Generation 5310: Best = 3.373890e+03	FES = 531100	$\mu_F = 0.752$	$\mu_{CR} = 0.185$	
Diversity = 6.0475e-02	k_t = [1, 1, 1, 1]			
Generation 5320: Best = 3.373750e+03	FES = 532100	$\mu_F = 0.750$	$\mu_{CR} = 0.181$	
Diversity = 5.9310e-02	k_t = [2, 2, 4, 3]			
Generation 5330: Best = 3.373717e+03	FES = 533100	$\mu_F = 0.740$	$\mu_{CR} = 0.185$	
Diversity = 6.2770e-02	k_t = [1, 1, 1, 1]			
Generation 5340: Best = 3.373462e+03	FES = 534100	$\mu_F = 0.741$	$\mu_{CR} = 0.192$	
Diversity = 6.2269e-02	k_t = [3, 4, 3, 4]			
Generation 5350: Best = 3.373372e+03	FES = 535100	$\mu_F = 0.751$	$\mu_{CR} = 0.202$	

Diversity = 6.8199e-02	k_t = [1, 1, 1, 1]			
Generation 5360: Best = 3.373107e+03	FES = 536100	$\mu_F = 0.756$	$\mu_{CR} = 0.211$	
Diversity = 7.8806e-02	k_t = [2, 2, 2, 2]			
Generation 5370: Best = 3.372938e+03	FES = 537100	$\mu_F = 0.766$	$\mu_{CR} = 0.216$	
Diversity = 8.9824e-02	k_t = [1, 1, 1, 1]			
Generation 5380: Best = 3.372772e+03	FES = 538100	$\mu_F = 0.749$	$\mu_{CR} = 0.213$	
Diversity = 8.5039e-02	k_t = [3, 3, 4, 4]			
Generation 5390: Best = 3.372612e+03	FES = 539100	$\mu_F = 0.741$	$\mu_{CR} = 0.218$	
Diversity = 8.7700e-02	k_t = [1, 1, 1, 1]			
Generation 5400: Best = 3.372443e+03	FES = 540100	$\mu_F = 0.745$	$\mu_{CR} = 0.230$	
Diversity = 8.1796e-02	k_t = [2, 3, 2, 4]			
Generation 5410: Best = 3.372335e+03	FES = 541100	$\mu_F = 0.742$	$\mu_{CR} = 0.228$	
Diversity = 8.3344e-02	k_t = [1, 1, 1, 1]			
Generation 5420: Best = 3.372191e+03	FES = 542100	$\mu_F = 0.737$	$\mu_{CR} = 0.230$	
Diversity = 8.2207e-02	k_t = [2, 2, 4, 4]			
Generation 5430: Best = 3.372076e+03	FES = 543100	$\mu_F = 0.729$	$\mu_{CR} = 0.231$	
Diversity = 7.9563e-02	k_t = [1, 1, 1, 1]			
Generation 5440: Best = 3.371939e+03	FES = 544100	$\mu_F = 0.739$	$\mu_{CR} = 0.227$	
Diversity = 7.7154e-02	k_t = [4, 2, 2, 2]			
Generation 5450: Best = 3.371837e+03	FES = 545100	$\mu_F = 0.734$	$\mu_{CR} = 0.229$	
Diversity = 7.5938e-02	k_t = [1, 1, 1, 1]			
Generation 5460: Best = 3.371769e+03	FES = 546100	$\mu_F = 0.741$	$\mu_{CR} = 0.237$	
Diversity = 7.5786e-02	k_t = [3, 2, 4, 2]			
Generation 5470: Best = 3.371681e+03	FES = 547100	$\mu_F = 0.718$	$\mu_{CR} = 0.236$	
Diversity = 7.3747e-02	k_t = [1, 1, 1, 1]			
Generation 5480: Best = 3.371536e+03	FES = 548100	$\mu_F = 0.714$	$\mu_{CR} = 0.244$	
Diversity = 7.4036e-02	k_t = [3, 2, 4, 3]			
Generation 5490: Best = 3.371397e+03	FES = 549100	$\mu_F = 0.715$	$\mu_{CR} = 0.238$	
Diversity = 7.9331e-02	k_t = [1, 1, 1, 1]			
Generation 5500: Best = 3.371231e+03	FES = 550100	$\mu_F = 0.725$	$\mu_{CR} = 0.247$	
Diversity = 7.5963e-02	k_t = [3, 2, 2, 4]			
Generation 5510: Best = 3.371094e+03	FES = 551100	$\mu_F = 0.726$	$\mu_{CR} = 0.246$	
Diversity = 7.6449e-02	k_t = [1, 1, 1, 1]			
Generation 5520: Best = 3.371002e+03	FES = 552100	$\mu_F = 0.729$	$\mu_{CR} = 0.251$	
Diversity = 7.5898e-02	k_t = [2, 2, 2, 4]			
Generation 5530: Best = 3.370874e+03	FES = 553100	$\mu_F = 0.719$	$\mu_{CR} = 0.255$	
Diversity = 8.0215e-02	k_t = [1, 1, 1, 1]			
Generation 5540: Best = 3.370793e+03	FES = 554100	$\mu_F = 0.728$	$\mu_{CR} = 0.241$	
Diversity = 7.3984e-02	k_t = [2, 2, 2, 4]			
Generation 5550: Best = 3.370735e+03	FES = 555100	$\mu_F = 0.717$	$\mu_{CR} = 0.237$	
Diversity = 6.7312e-02	k_t = [1, 1, 1, 1]			
Generation 5560: Best = 3.370644e+03	FES = 556100	$\mu_F = 0.719$	$\mu_{CR} = 0.228$	
Diversity = 6.0486e-02	k_t = [4, 2, 3, 4]			
Generation 5570: Best = 3.370529e+03	FES = 557100	$\mu_F = 0.718$	$\mu_{CR} = 0.233$	
Diversity = 6.5075e-02	k_t = [1, 1, 1, 1]			
Generation 5580: Best = 3.370360e+03	FES = 558100	$\mu_F = 0.724$	$\mu_{CR} = 0.238$	
Diversity = 7.0445e-02	k_t = [4, 2, 3, 2]			
Generation 5590: Best = 3.370267e+03	FES = 559100	$\mu_F = 0.741$	$\mu_{CR} = 0.236$	
Diversity = 7.3975e-02	k_t = [1, 1, 1, 1]			
Generation 5600: Best = 3.370091e+03	FES = 560100	$\mu_F = 0.743$	$\mu_{CR} = 0.235$	
Diversity = 7.2053e-02	k_t = [4, 2, 3, 4]			
Generation 5610: Best = 3.369934e+03	FES = 561100	$\mu_F = 0.753$	$\mu_{CR} = 0.236$	
Diversity = 6.7604e-02	k_t = [1, 1, 1, 1]			
Generation 5620: Best = 3.369759e+03	FES = 562100	$\mu_F = 0.767$	$\mu_{CR} = 0.236$	

Diversity = 7.3696e-02	k_t = [2, 3, 2, 2]			
Generation 5630: Best = 3.369607e+03	FES = 563100	$\mu_F = 0.777$	$\mu_{CR} = 0.230$	
Diversity = 8.0242e-02	k_t = [1, 1, 1, 1]			
Generation 5640: Best = 3.369488e+03	FES = 564100	$\mu_F = 0.770$	$\mu_{CR} = 0.233$	
Diversity = 7.9196e-02	k_t = [4, 2, 4, 4]			
Generation 5650: Best = 3.369291e+03	FES = 565100	$\mu_F = 0.760$	$\mu_{CR} = 0.225$	
Diversity = 7.9218e-02	k_t = [1, 1, 1, 1]			
Generation 5660: Best = 3.369186e+03	FES = 566100	$\mu_F = 0.766$	$\mu_{CR} = 0.224$	
Diversity = 8.2914e-02	k_t = [4, 4, 4, 2]			
Generation 5670: Best = 3.369047e+03	FES = 567100	$\mu_F = 0.774$	$\mu_{CR} = 0.230$	
Diversity = 8.8861e-02	k_t = [1, 1, 1, 1]			
Generation 5680: Best = 3.368913e+03	FES = 568100	$\mu_F = 0.771$	$\mu_{CR} = 0.217$	
Diversity = 8.2897e-02	k_t = [4, 2, 2, 4]			
Generation 5690: Best = 3.368747e+03	FES = 569100	$\mu_F = 0.756$	$\mu_{CR} = 0.209$	
Diversity = 7.8275e-02	k_t = [1, 1, 1, 1]			
Generation 5700: Best = 3.368646e+03	FES = 570100	$\mu_F = 0.749$	$\mu_{CR} = 0.210$	
Diversity = 7.7058e-02	k_t = [3, 4, 2, 2]			
Generation 5710: Best = 3.368520e+03	FES = 571100	$\mu_F = 0.749$	$\mu_{CR} = 0.218$	
Diversity = 8.1992e-02	k_t = [1, 1, 1, 1]			
Generation 5720: Best = 3.368351e+03	FES = 572100	$\mu_F = 0.755$	$\mu_{CR} = 0.223$	
Diversity = 8.7567e-02	k_t = [3, 4, 2, 2]			
Generation 5730: Best = 3.368200e+03	FES = 573100	$\mu_F = 0.754$	$\mu_{CR} = 0.225$	
Diversity = 9.3761e-02	k_t = [1, 1, 1, 1]			
Generation 5740: Best = 3.368068e+03	FES = 574100	$\mu_F = 0.755$	$\mu_{CR} = 0.224$	
Diversity = 9.7166e-02	k_t = [2, 3, 2, 2]			
Generation 5750: Best = 3.367832e+03	FES = 575100	$\mu_F = 0.743$	$\mu_{CR} = 0.221$	
Diversity = 1.0658e-01	k_t = [1, 1, 1, 1]			
Generation 5760: Best = 3.367655e+03	FES = 576100	$\mu_F = 0.731$	$\mu_{CR} = 0.224$	
Diversity = 1.0843e-01	k_t = [3, 3, 2, 2]			
Generation 5770: Best = 3.367547e+03	FES = 577100	$\mu_F = 0.723$	$\mu_{CR} = 0.218$	
Diversity = 1.0691e-01	k_t = [1, 1, 1, 1]			
Generation 5780: Best = 3.367328e+03	FES = 578100	$\mu_F = 0.711$	$\mu_{CR} = 0.218$	
Diversity = 9.7568e-02	k_t = [3, 2, 4, 2]			
Generation 5790: Best = 3.367252e+03	FES = 579100	$\mu_F = 0.710$	$\mu_{CR} = 0.213$	
Diversity = 9.2242e-02	k_t = [1, 1, 1, 1]			
Generation 5800: Best = 3.367132e+03	FES = 580100	$\mu_F = 0.706$	$\mu_{CR} = 0.208$	
Diversity = 8.5564e-02	k_t = [3, 2, 2, 2]			
Generation 5810: Best = 3.367015e+03	FES = 581100	$\mu_F = 0.701$	$\mu_{CR} = 0.210$	
Diversity = 8.4717e-02	k_t = [1, 1, 1, 1]			
Generation 5820: Best = 3.366935e+03	FES = 582100	$\mu_F = 0.706$	$\mu_{CR} = 0.207$	
Diversity = 7.7436e-02	k_t = [2, 2, 3, 4]			
Generation 5830: Best = 3.366823e+03	FES = 583100	$\mu_F = 0.713$	$\mu_{CR} = 0.205$	
Diversity = 7.3215e-02	k_t = [1, 1, 1, 1]			
Generation 5840: Best = 3.366745e+03	FES = 584100	$\mu_F = 0.714$	$\mu_{CR} = 0.204$	
Diversity = 7.1801e-02	k_t = [4, 3, 3, 2]			
Generation 5850: Best = 3.366645e+03	FES = 585100	$\mu_F = 0.717$	$\mu_{CR} = 0.201$	
Diversity = 6.8618e-02	k_t = [1, 1, 1, 1]			
Generation 5860: Best = 3.366544e+03	FES = 586100	$\mu_F = 0.724$	$\mu_{CR} = 0.214$	
Diversity = 6.3158e-02	k_t = [2, 4, 4, 3]			
Generation 5870: Best = 3.366455e+03	FES = 587100	$\mu_F = 0.727$	$\mu_{CR} = 0.222$	
Diversity = 6.5457e-02	k_t = [1, 1, 1, 1]			
Generation 5880: Best = 3.366389e+03	FES = 588100	$\mu_F = 0.737$	$\mu_{CR} = 0.220$	
Diversity = 5.7744e-02	k_t = [4, 3, 3, 4]			
Generation 5890: Best = 3.366294e+03	FES = 589100	$\mu_F = 0.760$	$\mu_{CR} = 0.223$	

Diversity = 6.2177e-02	k_t = [1, 1, 1, 1]			
Generation 5900: Best = 3.366188e+03	FES = 590100	$\mu_F = 0.750$	$\mu_{CR} = 0.222$	
Diversity = 6.7454e-02	k_t = [2, 3, 4, 4]			
Generation 5910: Best = 3.366001e+03	FES = 591100	$\mu_F = 0.748$	$\mu_{CR} = 0.220$	
Diversity = 7.3999e-02	k_t = [1, 1, 1, 1]			
Generation 5920: Best = 3.365903e+03	FES = 592100	$\mu_F = 0.754$	$\mu_{CR} = 0.221$	
Diversity = 7.8113e-02	k_t = [4, 3, 3, 4]			
Generation 5930: Best = 3.365816e+03	FES = 593100	$\mu_F = 0.756$	$\mu_{CR} = 0.215$	
Diversity = 8.0120e-02	k_t = [1, 1, 1, 1]			
Generation 5940: Best = 3.365694e+03	FES = 594100	$\mu_F = 0.746$	$\mu_{CR} = 0.211$	
Diversity = 8.1092e-02	k_t = [3, 4, 2, 3]			
Generation 5950: Best = 3.365590e+03	FES = 595100	$\mu_F = 0.749$	$\mu_{CR} = 0.206$	
Diversity = 8.4323e-02	k_t = [1, 1, 1, 1]			
Generation 5960: Best = 3.365435e+03	FES = 596100	$\mu_F = 0.743$	$\mu_{CR} = 0.208$	
Diversity = 8.2302e-02	k_t = [2, 4, 4, 4]			
Generation 5970: Best = 3.365325e+03	FES = 597100	$\mu_F = 0.750$	$\mu_{CR} = 0.203$	
Diversity = 8.0956e-02	k_t = [1, 1, 1, 1]			
Generation 5980: Best = 3.365202e+03	FES = 598100	$\mu_F = 0.749$	$\mu_{CR} = 0.204$	
Diversity = 8.4986e-02	k_t = [2, 2, 2, 2]			
Generation 5990: Best = 3.365084e+03	FES = 599100	$\mu_F = 0.743$	$\mu_{CR} = 0.200$	
Diversity = 8.3640e-02	k_t = [1, 1, 1, 1]			
Generation 6000: Best = 3.364941e+03	FES = 600100	$\mu_F = 0.742$	$\mu_{CR} = 0.197$	
Diversity = 7.9341e-02	k_t = [2, 4, 2, 3]			
Generation 6010: Best = 3.364862e+03	FES = 601100	$\mu_F = 0.749$	$\mu_{CR} = 0.201$	
Diversity = 8.2681e-02	k_t = [1, 1, 1, 1]			
Generation 6020: Best = 3.364717e+03	FES = 602100	$\mu_F = 0.752$	$\mu_{CR} = 0.210$	
Diversity = 8.0116e-02	k_t = [4, 3, 3, 4]			
Generation 6030: Best = 3.364530e+03	FES = 603100	$\mu_F = 0.751$	$\mu_{CR} = 0.227$	
Diversity = 8.4876e-02	k_t = [1, 1, 1, 1]			
Generation 6040: Best = 3.364429e+03	FES = 604100	$\mu_F = 0.752$	$\mu_{CR} = 0.226$	
Diversity = 8.7440e-02	k_t = [3, 3, 3, 3]			
Generation 6050: Best = 3.364271e+03	FES = 605100	$\mu_F = 0.737$	$\mu_{CR} = 0.241$	
Diversity = 8.7834e-02	k_t = [1, 1, 1, 1]			
Generation 6060: Best = 3.364119e+03	FES = 606100	$\mu_F = 0.741$	$\mu_{CR} = 0.241$	
Diversity = 8.1646e-02	k_t = [3, 4, 3, 3]			
Generation 6070: Best = 3.363954e+03	FES = 607100	$\mu_F = 0.739$	$\mu_{CR} = 0.239$	
Diversity = 8.1028e-02	k_t = [1, 1, 1, 1]			
Generation 6080: Best = 3.363865e+03	FES = 608100	$\mu_F = 0.738$	$\mu_{CR} = 0.229$	
Diversity = 8.0655e-02	k_t = [2, 3, 2, 2]			
Generation 6090: Best = 3.363751e+03	FES = 609100	$\mu_F = 0.742$	$\mu_{CR} = 0.227$	
Diversity = 8.1068e-02	k_t = [1, 1, 1, 1]			
Generation 6100: Best = 3.363647e+03	FES = 610100	$\mu_F = 0.746$	$\mu_{CR} = 0.225$	
Diversity = 7.6902e-02	k_t = [2, 3, 2, 2]			
Generation 6110: Best = 3.363507e+03	FES = 611100	$\mu_F = 0.741$	$\mu_{CR} = 0.228$	
Diversity = 8.2419e-02	k_t = [1, 1, 1, 1]			
Generation 6120: Best = 3.363346e+03	FES = 612100	$\mu_F = 0.752$	$\mu_{CR} = 0.223$	
Diversity = 7.8408e-02	k_t = [4, 3, 4, 3]			
Generation 6130: Best = 3.363222e+03	FES = 613100	$\mu_F = 0.749$	$\mu_{CR} = 0.219$	
Diversity = 8.7404e-02	k_t = [1, 1, 1, 1]			
Generation 6140: Best = 3.363054e+03	FES = 614100	$\mu_F = 0.737$	$\mu_{CR} = 0.217$	
Diversity = 8.2561e-02	k_t = [2, 2, 4, 2]			
Generation 6150: Best = 3.362977e+03	FES = 615100	$\mu_F = 0.745$	$\mu_{CR} = 0.211$	
Diversity = 8.3898e-02	k_t = [1, 1, 1, 1]			
Generation 6160: Best = 3.362904e+03	FES = 616100	$\mu_F = 0.742$	$\mu_{CR} = 0.205$	

Diversity = 7.9439e-02	k_t = [3, 3, 2, 2]			
Generation 6170: Best = 3.362816e+03	FES = 617100	$\mu_F = 0.724$	$\mu_{CR} = 0.205$	
Diversity = 7.7032e-02	k_t = [1, 1, 1, 1]			
Generation 6180: Best = 3.362699e+03	FES = 618100	$\mu_F = 0.722$	$\mu_{CR} = 0.204$	
Diversity = 7.1601e-02	k_t = [4, 4, 2, 4]			
Generation 6190: Best = 3.362636e+03	FES = 619100	$\mu_F = 0.729$	$\mu_{CR} = 0.207$	
Diversity = 6.7932e-02	k_t = [1, 1, 1, 1]			
Generation 6200: Best = 3.362493e+03	FES = 620100	$\mu_F = 0.725$	$\mu_{CR} = 0.208$	
Diversity = 6.7045e-02	k_t = [4, 3, 2, 3]			
Generation 6210: Best = 3.362442e+03	FES = 621100	$\mu_F = 0.719$	$\mu_{CR} = 0.213$	
Diversity = 6.6816e-02	k_t = [1, 1, 1, 1]			
Generation 6220: Best = 3.362353e+03	FES = 622100	$\mu_F = 0.731$	$\mu_{CR} = 0.202$	
Diversity = 6.4676e-02	k_t = [4, 4, 2, 4]			
Generation 6230: Best = 3.362246e+03	FES = 623100	$\mu_F = 0.742$	$\mu_{CR} = 0.208$	
Diversity = 6.9574e-02	k_t = [1, 1, 1, 1]			
Generation 6240: Best = 3.362165e+03	FES = 624100	$\mu_F = 0.739$	$\mu_{CR} = 0.207$	
Diversity = 6.2836e-02	k_t = [4, 3, 4, 3]			
Generation 6250: Best = 3.362075e+03	FES = 625100	$\mu_F = 0.744$	$\mu_{CR} = 0.201$	
Diversity = 6.3336e-02	k_t = [1, 1, 1, 1]			
Generation 6260: Best = 3.361983e+03	FES = 626100	$\mu_F = 0.747$	$\mu_{CR} = 0.205$	
Diversity = 6.5578e-02	k_t = [3, 2, 2, 2]			
Generation 6270: Best = 3.361909e+03	FES = 627100	$\mu_F = 0.751$	$\mu_{CR} = 0.208$	
Diversity = 7.3738e-02	k_t = [1, 1, 1, 1]			
Generation 6280: Best = 3.361811e+03	FES = 628100	$\mu_F = 0.750$	$\mu_{CR} = 0.214$	
Diversity = 7.2561e-02	k_t = [4, 3, 2, 4]			
Generation 6290: Best = 3.361699e+03	FES = 629100	$\mu_F = 0.741$	$\mu_{CR} = 0.220$	
Diversity = 6.9481e-02	k_t = [1, 1, 1, 1]			
Generation 6300: Best = 3.361596e+03	FES = 630100	$\mu_F = 0.750$	$\mu_{CR} = 0.217$	
Diversity = 7.4386e-02	k_t = [3, 2, 3, 2]			
Generation 6310: Best = 3.361477e+03	FES = 631100	$\mu_F = 0.748$	$\mu_{CR} = 0.218$	
Diversity = 8.0703e-02	k_t = [1, 1, 1, 1]			
Generation 6320: Best = 3.361338e+03	FES = 632100	$\mu_F = 0.750$	$\mu_{CR} = 0.214$	
Diversity = 8.5948e-02	k_t = [2, 3, 2, 2]			
Generation 6330: Best = 3.361234e+03	FES = 633100	$\mu_F = 0.746$	$\mu_{CR} = 0.218$	
Diversity = 9.2202e-02	k_t = [1, 1, 1, 1]			
Generation 6340: Best = 3.361129e+03	FES = 634100	$\mu_F = 0.733$	$\mu_{CR} = 0.216$	
Diversity = 8.2166e-02	k_t = [3, 4, 4, 4]			
Generation 6350: Best = 3.361012e+03	FES = 635100	$\mu_F = 0.734$	$\mu_{CR} = 0.212$	
Diversity = 7.5998e-02	k_t = [1, 1, 1, 1]			
Generation 6360: Best = 3.360952e+03	FES = 636100	$\mu_F = 0.723$	$\mu_{CR} = 0.208$	
Diversity = 7.0943e-02	k_t = [2, 2, 2, 4]			
Generation 6370: Best = 3.360841e+03	FES = 637100	$\mu_F = 0.722$	$\mu_{CR} = 0.202$	
Diversity = 6.9287e-02	k_t = [1, 1, 1, 1]			
Generation 6380: Best = 3.360754e+03	FES = 638100	$\mu_F = 0.735$	$\mu_{CR} = 0.205$	
Diversity = 6.5025e-02	k_t = [4, 3, 3, 3]			
Generation 6390: Best = 3.360683e+03	FES = 639100	$\mu_F = 0.732$	$\mu_{CR} = 0.207$	
Diversity = 6.4212e-02	k_t = [1, 1, 1, 1]			
Generation 6400: Best = 3.360618e+03	FES = 640100	$\mu_F = 0.725$	$\mu_{CR} = 0.210$	
Diversity = 6.3511e-02	k_t = [2, 3, 2, 3]			
Generation 6410: Best = 3.360525e+03	FES = 641100	$\mu_F = 0.728$	$\mu_{CR} = 0.214$	
Diversity = 6.9562e-02	k_t = [1, 1, 1, 1]			
Generation 6420: Best = 3.360451e+03	FES = 642100	$\mu_F = 0.729$	$\mu_{CR} = 0.209$	
Diversity = 6.4822e-02	k_t = [4, 4, 3, 3]			
Generation 6430: Best = 3.360331e+03	FES = 643100	$\mu_F = 0.739$	$\mu_{CR} = 0.225$	

Diversity = 6.6099e-02 k_t = [1, 1, 1, 1]
 Generation 6440: Best = 3.360294e+03 FES = 644100 μ_F = 0.746 μ_{CR} = 0.223
 Diversity = 6.6414e-02 k_t = [2, 4, 2, 4]
 Generation 6450: Best = 3.360200e+03 FES = 645100 μ_F = 0.754 μ_{CR} = 0.210
 Diversity = 6.2831e-02 k_t = [1, 1, 1, 1]
 Generation 6460: Best = 3.360122e+03 FES = 646100 μ_F = 0.759 μ_{CR} = 0.209
 Diversity = 6.4955e-02 k_t = [2, 4, 3, 2]
 Generation 6470: Best = 3.360023e+03 FES = 647100 μ_F = 0.753 μ_{CR} = 0.204
 Diversity = 6.7217e-02 k_t = [1, 1, 1, 1]
 Generation 6480: Best = 3.359949e+03 FES = 648100 μ_F = 0.749 μ_{CR} = 0.203
 Diversity = 6.5571e-02 k_t = [4, 4, 3, 4]
 Generation 6490: Best = 3.359866e+03 FES = 649100 μ_F = 0.747 μ_{CR} = 0.192
 Diversity = 6.6520e-02 k_t = [1, 1, 1, 1]
 Generation 6500: Best = 3.359813e+03 FES = 650100 μ_F = 0.747 μ_{CR} = 0.192
 Diversity = 6.3943e-02 k_t = [4, 2, 2, 4]
 Generation 6510: Best = 3.359733e+03 FES = 651100 μ_F = 0.757 μ_{CR} = 0.190
 Diversity = 6.2425e-02 k_t = [1, 1, 1, 1]
 Generation 6520: Best = 3.359657e+03 FES = 652100 μ_F = 0.757 μ_{CR} = 0.189
 Diversity = 6.3401e-02 k_t = [3, 2, 4, 3]
 Generation 6530: Best = 3.359592e+03 FES = 653100 μ_F = 0.756 μ_{CR} = 0.188
 Diversity = 6.7353e-02 k_t = [1, 1, 1, 1]
 Generation 6540: Best = 3.359531e+03 FES = 654100 μ_F = 0.749 μ_{CR} = 0.193
 Diversity = 6.7387e-02 k_t = [2, 3, 2, 2]
 Generation 6550: Best = 3.359453e+03 FES = 655100 μ_F = 0.755 μ_{CR} = 0.193
 Diversity = 7.2574e-02 k_t = [1, 1, 1, 1]
 Generation 6560: Best = 3.359356e+03 FES = 656100 μ_F = 0.761 μ_{CR} = 0.201
 Diversity = 6.5694e-02 k_t = [3, 2, 4, 4]
 Generation 6570: Best = 3.359296e+03 FES = 657100 μ_F = 0.776 μ_{CR} = 0.198
 Diversity = 6.5916e-02 k_t = [1, 1, 1, 1]
 Generation 6580: Best = 3.359211e+03 FES = 658100 μ_F = 0.762 μ_{CR} = 0.203
 Diversity = 6.3071e-02 k_t = [3, 3, 2, 3]
 Generation 6590: Best = 3.359162e+03 FES = 659100 μ_F = 0.752 μ_{CR} = 0.196
 Diversity = 6.2130e-02 k_t = [1, 1, 1, 1]
 Generation 6600: Best = 3.359099e+03 FES = 660100 μ_F = 0.746 μ_{CR} = 0.190
 Diversity = 6.0509e-02 k_t = [2, 3, 4, 2]
 Generation 6610: Best = 3.359023e+03 FES = 661100 μ_F = 0.753 μ_{CR} = 0.198
 Diversity = 6.5646e-02 k_t = [1, 1, 1, 1]
 Generation 6620: Best = 3.358967e+03 FES = 662100 μ_F = 0.741 μ_{CR} = 0.198
 Diversity = 6.0278e-02 k_t = [4, 4, 3, 4]
 Generation 6630: Best = 3.358895e+03 FES = 663100 μ_F = 0.738 μ_{CR} = 0.193
 Diversity = 6.0344e-02 k_t = [1, 1, 1, 1]
 Generation 6640: Best = 3.358842e+03 FES = 664100 μ_F = 0.745 μ_{CR} = 0.189
 Diversity = 5.8540e-02 k_t = [2, 3, 2, 3]
 Generation 6650: Best = 3.358715e+03 FES = 665100 μ_F = 0.753 μ_{CR} = 0.192
 Diversity = 6.3075e-02 k_t = [1, 1, 1, 1]
 Generation 6660: Best = 3.358611e+03 FES = 666100 μ_F = 0.752 μ_{CR} = 0.203
 Diversity = 6.9065e-02 k_t = [2, 4, 3, 2]
 Generation 6670: Best = 3.358530e+03 FES = 667100 μ_F = 0.763 μ_{CR} = 0.209
 Diversity = 7.2974e-02 k_t = [1, 1, 1, 1]
 Generation 6680: Best = 3.358410e+03 FES = 668100 μ_F = 0.757 μ_{CR} = 0.213
 Diversity = 7.5723e-02 k_t = [3, 2, 3, 3]
 Generation 6690: Best = 3.358342e+03 FES = 669100 μ_F = 0.742 μ_{CR} = 0.208
 Diversity = 7.6462e-02 k_t = [1, 1, 1, 1]
 Generation 6700: Best = 3.358216e+03 FES = 670100 μ_F = 0.751 μ_{CR} = 0.217

Diversity = 6.9592e-02	k_t = [4, 4, 4, 3]			
Generation 6710: Best = 3.358169e+03	FES = 671100	$\mu_F = 0.747$	$\mu_{CR} = 0.215$	
Diversity = 6.9043e-02	k_t = [1, 1, 1, 1]			
Generation 6720: Best = 3.358090e+03	FES = 672100	$\mu_F = 0.745$	$\mu_{CR} = 0.222$	
Diversity = 6.4951e-02	k_t = [4, 2, 2, 4]			
Generation 6730: Best = 3.358022e+03	FES = 673100	$\mu_F = 0.746$	$\mu_{CR} = 0.223$	
Diversity = 6.2601e-02	k_t = [1, 1, 1, 1]			
Generation 6740: Best = 3.357961e+03	FES = 674100	$\mu_F = 0.750$	$\mu_{CR} = 0.220$	
Diversity = 6.7345e-02	k_t = [3, 3, 4, 2]			
Generation 6750: Best = 3.357902e+03	FES = 675100	$\mu_F = 0.747$	$\mu_{CR} = 0.218$	
Diversity = 6.4913e-02	k_t = [1, 1, 1, 1]			
Generation 6760: Best = 3.357844e+03	FES = 676100	$\mu_F = 0.736$	$\mu_{CR} = 0.227$	
Diversity = 6.5188e-02	k_t = [4, 2, 2, 2]			
Generation 6770: Best = 3.357782e+03	FES = 677100	$\mu_F = 0.710$	$\mu_{CR} = 0.224$	
Diversity = 6.4750e-02	k_t = [1, 1, 1, 1]			
Generation 6780: Best = 3.357726e+03	FES = 678100	$\mu_F = 0.704$	$\mu_{CR} = 0.225$	
Diversity = 5.6477e-02	k_t = [4, 3, 3, 3]			
Generation 6790: Best = 3.357693e+03	FES = 679100	$\mu_F = 0.703$	$\mu_{CR} = 0.236$	
Diversity = 5.0349e-02	k_t = [1, 1, 1, 1]			
Generation 6800: Best = 3.357660e+03	FES = 680100	$\mu_F = 0.713$	$\mu_{CR} = 0.232$	
Diversity = 4.7521e-02	k_t = [4, 2, 3, 3]			
Generation 6810: Best = 3.357621e+03	FES = 681100	$\mu_F = 0.713$	$\mu_{CR} = 0.232$	
Diversity = 4.8375e-02	k_t = [1, 1, 1, 1]			
Generation 6820: Best = 3.357577e+03	FES = 682100	$\mu_F = 0.713$	$\mu_{CR} = 0.231$	
Diversity = 4.9239e-02	k_t = [3, 2, 2, 2]			
Generation 6830: Best = 3.357519e+03	FES = 683100	$\mu_F = 0.712$	$\mu_{CR} = 0.227$	
Diversity = 4.8938e-02	k_t = [1, 1, 1, 1]			
Generation 6840: Best = 3.357475e+03	FES = 684100	$\mu_F = 0.703$	$\mu_{CR} = 0.227$	
Diversity = 4.3923e-02	k_t = [3, 2, 2, 4]			
Generation 6850: Best = 3.357453e+03	FES = 685100	$\mu_F = 0.707$	$\mu_{CR} = 0.225$	
Diversity = 4.1487e-02	k_t = [1, 1, 1, 1]			
Generation 6860: Best = 3.357425e+03	FES = 686100	$\mu_F = 0.714$	$\mu_{CR} = 0.226$	
Diversity = 3.9437e-02	k_t = [2, 4, 3, 4]			
Generation 6870: Best = 3.357371e+03	FES = 687100	$\mu_F = 0.722$	$\mu_{CR} = 0.217$	
Diversity = 3.9922e-02	k_t = [1, 1, 1, 1]			
Generation 6880: Best = 3.357337e+03	FES = 688100	$\mu_F = 0.728$	$\mu_{CR} = 0.214$	
Diversity = 4.1863e-02	k_t = [3, 2, 4, 2]			
Generation 6890: Best = 3.357297e+03	FES = 689100	$\mu_F = 0.722$	$\mu_{CR} = 0.217$	
Diversity = 4.2805e-02	k_t = [1, 1, 1, 1]			
Generation 6900: Best = 3.357280e+03	FES = 690100	$\mu_F = 0.721$	$\mu_{CR} = 0.217$	
Diversity = 4.0671e-02	k_t = [2, 4, 4, 3]			
Generation 6910: Best = 3.357237e+03	FES = 691100	$\mu_F = 0.710$	$\mu_{CR} = 0.213$	
Diversity = 4.0374e-02	k_t = [1, 1, 1, 1]			
Generation 6920: Best = 3.357209e+03	FES = 692100	$\mu_F = 0.725$	$\mu_{CR} = 0.213$	
Diversity = 3.9895e-02	k_t = [4, 3, 3, 4]			
Generation 6930: Best = 3.357173e+03	FES = 693100	$\mu_F = 0.718$	$\mu_{CR} = 0.215$	
Diversity = 3.9137e-02	k_t = [1, 1, 1, 1]			
Generation 6940: Best = 3.357149e+03	FES = 694100	$\mu_F = 0.728$	$\mu_{CR} = 0.218$	
Diversity = 3.8059e-02	k_t = [2, 3, 4, 3]			
Generation 6950: Best = 3.357121e+03	FES = 695100	$\mu_F = 0.727$	$\mu_{CR} = 0.214$	
Diversity = 3.8574e-02	k_t = [1, 1, 1, 1]			
Generation 6960: Best = 3.357080e+03	FES = 696100	$\mu_F = 0.738$	$\mu_{CR} = 0.223$	
Diversity = 4.1606e-02	k_t = [3, 4, 4, 2]			
Generation 6970: Best = 3.357029e+03	FES = 697100	$\mu_F = 0.737$	$\mu_{CR} = 0.216$	

Diversity = 4.6459e-02	k_t = [1, 1, 1, 1]			
Generation 6980: Best = 3.356981e+03	FES = 698100	$\mu_F = 0.743$	$\mu_{CR} = 0.208$	
Diversity = 4.7227e-02	k_t = [4, 4, 2, 2]			
Generation 6990: Best = 3.356939e+03	FES = 699100	$\mu_F = 0.745$	$\mu_{CR} = 0.205$	
Diversity = 4.9117e-02	k_t = [1, 1, 1, 1]			
Generation 7000: Best = 3.356908e+03	FES = 700100	$\mu_F = 0.743$	$\mu_{CR} = 0.215$	
Diversity = 4.3652e-02	k_t = [2, 2, 2, 4]			
Generation 7010: Best = 3.356861e+03	FES = 701100	$\mu_F = 0.752$	$\mu_{CR} = 0.208$	
Diversity = 4.7495e-02	k_t = [1, 1, 1, 1]			
Generation 7020: Best = 3.356829e+03	FES = 702100	$\mu_F = 0.749$	$\mu_{CR} = 0.216$	
Diversity = 4.6579e-02	k_t = [4, 3, 2, 3]			
Generation 7030: Best = 3.356793e+03	FES = 703100	$\mu_F = 0.739$	$\mu_{CR} = 0.217$	
Diversity = 4.5416e-02	k_t = [1, 1, 1, 1]			
Generation 7040: Best = 3.356753e+03	FES = 704100	$\mu_F = 0.735$	$\mu_{CR} = 0.215$	
Diversity = 4.1907e-02	k_t = [3, 2, 4, 2]			
Generation 7050: Best = 3.356722e+03	FES = 705100	$\mu_F = 0.744$	$\mu_{CR} = 0.219$	
Diversity = 4.4377e-02	k_t = [1, 1, 1, 1]			
Generation 7060: Best = 3.356676e+03	FES = 706100	$\mu_F = 0.746$	$\mu_{CR} = 0.206$	
Diversity = 4.4431e-02	k_t = [3, 3, 2, 2]			
Generation 7070: Best = 3.356643e+03	FES = 707100	$\mu_F = 0.737$	$\mu_{CR} = 0.210$	
Diversity = 4.4580e-02	k_t = [1, 1, 1, 1]			
Generation 7080: Best = 3.356619e+03	FES = 708100	$\mu_F = 0.742$	$\mu_{CR} = 0.204$	
Diversity = 4.3676e-02	k_t = [4, 3, 2, 3]			
Generation 7090: Best = 3.356574e+03	FES = 709100	$\mu_F = 0.740$	$\mu_{CR} = 0.204$	
Diversity = 4.5511e-02	k_t = [1, 1, 1, 1]			
Generation 7100: Best = 3.356546e+03	FES = 710100	$\mu_F = 0.738$	$\mu_{CR} = 0.207$	
Diversity = 4.3946e-02	k_t = [2, 4, 2, 3]			
Generation 7110: Best = 3.356510e+03	FES = 711100	$\mu_F = 0.731$	$\mu_{CR} = 0.207$	
Diversity = 4.5252e-02	k_t = [1, 1, 1, 1]			
Generation 7120: Best = 3.356477e+03	FES = 712100	$\mu_F = 0.725$	$\mu_{CR} = 0.213$	
Diversity = 4.0547e-02	k_t = [4, 3, 3, 4]			
Generation 7130: Best = 3.356447e+03	FES = 713100	$\mu_F = 0.741$	$\mu_{CR} = 0.212$	
Diversity = 4.2841e-02	k_t = [1, 1, 1, 1]			
Generation 7140: Best = 3.356410e+03	FES = 714100	$\mu_F = 0.744$	$\mu_{CR} = 0.219$	
Diversity = 4.2875e-02	k_t = [2, 2, 4, 2]			
Generation 7150: Best = 3.356372e+03	FES = 715100	$\mu_F = 0.734$	$\mu_{CR} = 0.220$	
Diversity = 4.3552e-02	k_t = [1, 1, 1, 1]			
Generation 7160: Best = 3.356334e+03	FES = 716100	$\mu_F = 0.741$	$\mu_{CR} = 0.214$	
Diversity = 4.4198e-02	k_t = [3, 2, 4, 2]			
Generation 7170: Best = 3.356301e+03	FES = 717100	$\mu_F = 0.733$	$\mu_{CR} = 0.213$	
Diversity = 4.3365e-02	k_t = [1, 1, 1, 1]			
Generation 7180: Best = 3.356272e+03	FES = 718100	$\mu_F = 0.746$	$\mu_{CR} = 0.202$	
Diversity = 4.4384e-02	k_t = [4, 4, 4, 2]			
Generation 7190: Best = 3.356253e+03	FES = 719100	$\mu_F = 0.742$	$\mu_{CR} = 0.199$	
Diversity = 4.3713e-02	k_t = [1, 1, 1, 1]			
Generation 7200: Best = 3.356222e+03	FES = 720100	$\mu_F = 0.740$	$\mu_{CR} = 0.192$	
Diversity = 4.2132e-02	k_t = [2, 2, 2, 2]			
Generation 7210: Best = 3.356208e+03	FES = 721100	$\mu_F = 0.741$	$\mu_{CR} = 0.193$	
Diversity = 4.4062e-02	k_t = [1, 1, 1, 1]			
Generation 7220: Best = 3.356151e+03	FES = 722100	$\mu_F = 0.748$	$\mu_{CR} = 0.207$	
Diversity = 4.3459e-02	k_t = [3, 2, 4, 4]			
Generation 7230: Best = 3.356124e+03	FES = 723100	$\mu_F = 0.742$	$\mu_{CR} = 0.200$	
Diversity = 4.2867e-02	k_t = [1, 1, 1, 1]			
Generation 7240: Best = 3.356089e+03	FES = 724100	$\mu_F = 0.751$	$\mu_{CR} = 0.195$	

Diversity = 4.6201e-02	k_t = [3, 4, 3, 3]			
Generation 7250: Best = 3.356053e+03	FES = 725100	$\mu_F = 0.748$	$\mu_{CR} = 0.196$	
Diversity = 4.9032e-02	k_t = [1, 1, 1, 1]			
Generation 7260: Best = 3.356004e+03	FES = 726100	$\mu_F = 0.757$	$\mu_{CR} = 0.184$	
Diversity = 4.5721e-02	k_t = [4, 4, 3, 3]			
Generation 7270: Best = 3.355989e+03	FES = 727100	$\mu_F = 0.752$	$\mu_{CR} = 0.185$	
Diversity = 4.4968e-02	k_t = [1, 1, 1, 1]			
Generation 7280: Best = 3.355964e+03	FES = 728100	$\mu_F = 0.745$	$\mu_{CR} = 0.185$	
Diversity = 4.0671e-02	k_t = [4, 3, 4, 4]			
Generation 7290: Best = 3.355927e+03	FES = 729100	$\mu_F = 0.750$	$\mu_{CR} = 0.190$	
Diversity = 4.0764e-02	k_t = [1, 1, 1, 1]			
Generation 7300: Best = 3.355893e+03	FES = 730100	$\mu_F = 0.748$	$\mu_{CR} = 0.191$	
Diversity = 4.2847e-02	k_t = [2, 2, 4, 2]			
Generation 7310: Best = 3.355860e+03	FES = 731100	$\mu_F = 0.744$	$\mu_{CR} = 0.191$	
Diversity = 4.3401e-02	k_t = [1, 1, 1, 1]			
Generation 7320: Best = 3.355826e+03	FES = 732100	$\mu_F = 0.745$	$\mu_{CR} = 0.191$	
Diversity = 4.3036e-02	k_t = [2, 3, 2, 2]			
Generation 7330: Best = 3.355809e+03	FES = 733100	$\mu_F = 0.759$	$\mu_{CR} = 0.188$	
Diversity = 4.3652e-02	k_t = [1, 1, 1, 1]			
Generation 7340: Best = 3.355787e+03	FES = 734100	$\mu_F = 0.743$	$\mu_{CR} = 0.185$	
Diversity = 4.5721e-02	k_t = [2, 2, 3, 2]			
Generation 7350: Best = 3.355752e+03	FES = 735100	$\mu_F = 0.722$	$\mu_{CR} = 0.187$	
Diversity = 4.6375e-02	k_t = [1, 1, 1, 1]			
Generation 7360: Best = 3.355721e+03	FES = 736100	$\mu_F = 0.707$	$\mu_{CR} = 0.187$	
Diversity = 4.6208e-02	k_t = [2, 2, 3, 2]			
Generation 7370: Best = 3.355696e+03	FES = 737100	$\mu_F = 0.694$	$\mu_{CR} = 0.195$	
Diversity = 4.5470e-02	k_t = [1, 1, 1, 1]			
Generation 7380: Best = 3.355682e+03	FES = 738100	$\mu_F = 0.689$	$\mu_{CR} = 0.197$	
Diversity = 3.9347e-02	k_t = [4, 2, 4, 2]			
Generation 7390: Best = 3.355668e+03	FES = 739100	$\mu_F = 0.689$	$\mu_{CR} = 0.204$	
Diversity = 3.4222e-02	k_t = [1, 1, 1, 1]			
Generation 7400: Best = 3.355646e+03	FES = 740100	$\mu_F = 0.699$	$\mu_{CR} = 0.200$	
Diversity = 3.0458e-02	k_t = [4, 2, 4, 3]			
Generation 7410: Best = 3.355627e+03	FES = 741100	$\mu_F = 0.710$	$\mu_{CR} = 0.199$	
Diversity = 2.9032e-02	k_t = [1, 1, 1, 1]			
Generation 7420: Best = 3.355611e+03	FES = 742100	$\mu_F = 0.707$	$\mu_{CR} = 0.201$	
Diversity = 2.7959e-02	k_t = [3, 2, 2, 3]			
Generation 7430: Best = 3.355593e+03	FES = 743100	$\mu_F = 0.723$	$\mu_{CR} = 0.195$	
Diversity = 2.9277e-02	k_t = [1, 1, 1, 1]			
Generation 7440: Best = 3.355582e+03	FES = 744100	$\mu_F = 0.728$	$\mu_{CR} = 0.200$	
Diversity = 2.9053e-02	k_t = [4, 2, 3, 3]			
Generation 7450: Best = 3.355558e+03	FES = 745100	$\mu_F = 0.727$	$\mu_{CR} = 0.211$	
Diversity = 2.8133e-02	k_t = [1, 1, 1, 1]			
Generation 7460: Best = 3.355549e+03	FES = 746100	$\mu_F = 0.725$	$\mu_{CR} = 0.218$	
Diversity = 2.5475e-02	k_t = [3, 3, 4, 4]			
Generation 7470: Best = 3.355534e+03	FES = 747100	$\mu_F = 0.713$	$\mu_{CR} = 0.221$	
Diversity = 2.4689e-02	k_t = [1, 1, 1, 1]			
Generation 7480: Best = 3.355526e+03	FES = 748100	$\mu_F = 0.726$	$\mu_{CR} = 0.221$	
Diversity = 2.4705e-02	k_t = [4, 4, 2, 3]			
Generation 7490: Best = 3.355516e+03	FES = 749100	$\mu_F = 0.733$	$\mu_{CR} = 0.226$	
Diversity = 2.3274e-02	k_t = [1, 1, 1, 1]			
Generation 7500: Best = 3.355496e+03	FES = 750100	$\mu_F = 0.737$	$\mu_{CR} = 0.228$	
Diversity = 2.2672e-02	k_t = [4, 4, 3, 2]			
Generation 7510: Best = 3.355482e+03	FES = 751100	$\mu_F = 0.740$	$\mu_{CR} = 0.225$	

Diversity = 2.4082e-02	k_t = [1, 1, 1, 1]			
Generation 7520: Best = 3.355468e+03	FES = 752100	$\mu_F = 0.746$	$\mu_{CR} = 0.230$	
Diversity = 2.4740e-02	k_t = [2, 2, 4, 3]			
Generation 7530: Best = 3.355450e+03	FES = 753100	$\mu_F = 0.749$	$\mu_{CR} = 0.222$	
Diversity = 2.6078e-02	k_t = [1, 1, 1, 1]			
Generation 7540: Best = 3.355438e+03	FES = 754100	$\mu_F = 0.759$	$\mu_{CR} = 0.223$	
Diversity = 2.6835e-02	k_t = [2, 3, 3, 4]			
Generation 7550: Best = 3.355416e+03	FES = 755100	$\mu_F = 0.747$	$\mu_{CR} = 0.226$	
Diversity = 2.7207e-02	k_t = [1, 1, 1, 1]			
Generation 7560: Best = 3.355398e+03	FES = 756100	$\mu_F = 0.749$	$\mu_{CR} = 0.228$	
Diversity = 2.9945e-02	k_t = [3, 2, 2, 3]			
Generation 7570: Best = 3.355377e+03	FES = 757100	$\mu_F = 0.749$	$\mu_{CR} = 0.225$	
Diversity = 3.2407e-02	k_t = [1, 1, 1, 1]			
Generation 7580: Best = 3.355360e+03	FES = 758100	$\mu_F = 0.740$	$\mu_{CR} = 0.215$	
Diversity = 3.4879e-02	k_t = [3, 2, 4, 2]			
Generation 7590: Best = 3.355346e+03	FES = 759100	$\mu_F = 0.729$	$\mu_{CR} = 0.213$	
Diversity = 3.4951e-02	k_t = [1, 1, 1, 1]			
Generation 7600: Best = 3.355325e+03	FES = 760100	$\mu_F = 0.718$	$\mu_{CR} = 0.214$	
Diversity = 3.0591e-02	k_t = [4, 4, 3, 3]			
Generation 7610: Best = 3.355303e+03	FES = 761100	$\mu_F = 0.723$	$\mu_{CR} = 0.221$	
Diversity = 3.1770e-02	k_t = [1, 1, 1, 1]			
Generation 7620: Best = 3.355285e+03	FES = 762100	$\mu_F = 0.731$	$\mu_{CR} = 0.222$	
Diversity = 3.3448e-02	k_t = [2, 4, 3, 2]			
Generation 7630: Best = 3.355278e+03	FES = 763100	$\mu_F = 0.723$	$\mu_{CR} = 0.218$	
Diversity = 3.1562e-02	k_t = [1, 1, 1, 1]			
Generation 7640: Best = 3.355264e+03	FES = 764100	$\mu_F = 0.716$	$\mu_{CR} = 0.214$	
Diversity = 2.8402e-02	k_t = [4, 2, 3, 2]			
Generation 7650: Best = 3.355247e+03	FES = 765100	$\mu_F = 0.717$	$\mu_{CR} = 0.225$	
Diversity = 2.8554e-02	k_t = [1, 1, 1, 1]			
Generation 7660: Best = 3.355235e+03	FES = 766100	$\mu_F = 0.715$	$\mu_{CR} = 0.223$	
Diversity = 2.5177e-02	k_t = [3, 3, 3, 4]			
Generation 7670: Best = 3.355224e+03	FES = 767100	$\mu_F = 0.727$	$\mu_{CR} = 0.223$	
Diversity = 2.4872e-02	k_t = [1, 1, 1, 1]			
Generation 7680: Best = 3.355213e+03	FES = 768100	$\mu_F = 0.745$	$\mu_{CR} = 0.223$	
Diversity = 2.4435e-02	k_t = [4, 3, 2, 4]			
Generation 7690: Best = 3.355194e+03	FES = 769100	$\mu_F = 0.740$	$\mu_{CR} = 0.224$	
Diversity = 2.3351e-02	k_t = [1, 1, 1, 1]			
Generation 7700: Best = 3.355184e+03	FES = 770100	$\mu_F = 0.744$	$\mu_{CR} = 0.226$	
Diversity = 2.2957e-02	k_t = [4, 3, 4, 4]			
Generation 7710: Best = 3.355169e+03	FES = 771100	$\mu_F = 0.748$	$\mu_{CR} = 0.218$	
Diversity = 2.2731e-02	k_t = [1, 1, 1, 1]			
Generation 7720: Best = 3.355160e+03	FES = 772100	$\mu_F = 0.757$	$\mu_{CR} = 0.221$	
Diversity = 2.3027e-02	k_t = [2, 2, 4, 3]			
Generation 7730: Best = 3.355149e+03	FES = 773100	$\mu_F = 0.750$	$\mu_{CR} = 0.224$	
Diversity = 2.5905e-02	k_t = [1, 1, 1, 1]			
Generation 7740: Best = 3.355136e+03	FES = 774100	$\mu_F = 0.741$	$\mu_{CR} = 0.223$	
Diversity = 2.4005e-02	k_t = [4, 2, 2, 2]			
Generation 7750: Best = 3.355126e+03	FES = 775100	$\mu_F = 0.743$	$\mu_{CR} = 0.224$	
Diversity = 2.5842e-02	k_t = [1, 1, 1, 1]			
Generation 7760: Best = 3.355112e+03	FES = 776100	$\mu_F = 0.753$	$\mu_{CR} = 0.235$	
Diversity = 2.5929e-02	k_t = [4, 3, 3, 2]			
Generation 7770: Best = 3.355100e+03	FES = 777100	$\mu_F = 0.747$	$\mu_{CR} = 0.227$	
Diversity = 2.7361e-02	k_t = [1, 1, 1, 1]			
Generation 7780: Best = 3.355085e+03	FES = 778100	$\mu_F = 0.750$	$\mu_{CR} = 0.233$	

Diversity = 2.7125e-02	k_t = [2, 2, 2, 3]			
Generation 7790: Best = 3.355074e+03	FES = 779100	$\mu_F = 0.757$	$\mu_{CR} = 0.235$	
Diversity = 3.0036e-02	k_t = [1, 1, 1, 1]			
Generation 7800: Best = 3.355052e+03	FES = 780100	$\mu_F = 0.758$	$\mu_{CR} = 0.228$	
Diversity = 3.2161e-02	k_t = [4, 3, 3, 2]			
Generation 7810: Best = 3.355037e+03	FES = 781100	$\mu_F = 0.755$	$\mu_{CR} = 0.222$	
Diversity = 3.2369e-02	k_t = [1, 1, 1, 1]			
Generation 7820: Best = 3.355014e+03	FES = 782100	$\mu_F = 0.745$	$\mu_{CR} = 0.210$	
Diversity = 3.1180e-02	k_t = [3, 2, 4, 2]			
Generation 7830: Best = 3.354996e+03	FES = 783100	$\mu_F = 0.738$	$\mu_{CR} = 0.209$	
Diversity = 3.2712e-02	k_t = [1, 1, 1, 1]			
Generation 7840: Best = 3.354976e+03	FES = 784100	$\mu_F = 0.743$	$\mu_{CR} = 0.205$	
Diversity = 2.9734e-02	k_t = [3, 3, 4, 4]			
Generation 7850: Best = 3.354952e+03	FES = 785100	$\mu_F = 0.753$	$\mu_{CR} = 0.207$	
Diversity = 3.0439e-02	k_t = [1, 1, 1, 1]			
Generation 7860: Best = 3.354925e+03	FES = 786100	$\mu_F = 0.755$	$\mu_{CR} = 0.199$	
Diversity = 3.4081e-02	k_t = [3, 3, 4, 2]			
Generation 7870: Best = 3.354905e+03	FES = 787100	$\mu_F = 0.753$	$\mu_{CR} = 0.205$	
Diversity = 3.2528e-02	k_t = [1, 1, 1, 1]			
Generation 7880: Best = 3.354880e+03	FES = 788100	$\mu_F = 0.753$	$\mu_{CR} = 0.212$	
Diversity = 3.0683e-02	k_t = [3, 4, 3, 4]			
Generation 7890: Best = 3.354872e+03	FES = 789100	$\mu_F = 0.754$	$\mu_{CR} = 0.222$	
Diversity = 3.1494e-02	k_t = [1, 1, 1, 1]			
Generation 7900: Best = 3.354847e+03	FES = 790100	$\mu_F = 0.753$	$\mu_{CR} = 0.221$	
Diversity = 3.1747e-02	k_t = [2, 2, 4, 3]			
Generation 7910: Best = 3.354821e+03	FES = 791100	$\mu_F = 0.761$	$\mu_{CR} = 0.221$	
Diversity = 3.2521e-02	k_t = [1, 1, 1, 1]			
Generation 7920: Best = 3.354802e+03	FES = 792100	$\mu_F = 0.752$	$\mu_{CR} = 0.228$	
Diversity = 3.1961e-02	k_t = [3, 2, 3, 2]			
Generation 7930: Best = 3.354784e+03	FES = 793100	$\mu_F = 0.755$	$\mu_{CR} = 0.224$	
Diversity = 3.2371e-02	k_t = [1, 1, 1, 1]			
Generation 7940: Best = 3.354766e+03	FES = 794100	$\mu_F = 0.748$	$\mu_{CR} = 0.227$	
Diversity = 2.9462e-02	k_t = [4, 4, 2, 4]			
Generation 7950: Best = 3.354751e+03	FES = 795100	$\mu_F = 0.748$	$\mu_{CR} = 0.225$	
Diversity = 3.0274e-02	k_t = [1, 1, 1, 1]			
Generation 7960: Best = 3.354726e+03	FES = 796100	$\mu_F = 0.752$	$\mu_{CR} = 0.231$	
Diversity = 2.9789e-02	k_t = [4, 2, 3, 3]			
Generation 7970: Best = 3.354711e+03	FES = 797100	$\mu_F = 0.754$	$\mu_{CR} = 0.238$	
Diversity = 3.1135e-02	k_t = [1, 1, 1, 1]			
Generation 7980: Best = 3.354695e+03	FES = 798100	$\mu_F = 0.746$	$\mu_{CR} = 0.250$	
Diversity = 3.0345e-02	k_t = [3, 3, 2, 2]			
Generation 7990: Best = 3.354680e+03	FES = 799100	$\mu_F = 0.734$	$\mu_{CR} = 0.243$	
Diversity = 3.1265e-02	k_t = [1, 1, 1, 1]			
Generation 8000: Best = 3.354662e+03	FES = 800100	$\mu_F = 0.731$	$\mu_{CR} = 0.247$	
Diversity = 3.0584e-02	k_t = [2, 4, 4, 2]			
Generation 8010: Best = 3.354646e+03	FES = 801100	$\mu_F = 0.718$	$\mu_{CR} = 0.238$	
Diversity = 2.9948e-02	k_t = [1, 1, 1, 1]			
Generation 8020: Best = 3.354630e+03	FES = 802100	$\mu_F = 0.727$	$\mu_{CR} = 0.239$	
Diversity = 2.9252e-02	k_t = [3, 3, 2, 3]			
Generation 8030: Best = 3.354619e+03	FES = 803100	$\mu_F = 0.718$	$\mu_{CR} = 0.241$	
Diversity = 2.6733e-02	k_t = [1, 1, 1, 1]			
Generation 8040: Best = 3.354605e+03	FES = 804100	$\mu_F = 0.728$	$\mu_{CR} = 0.238$	
Diversity = 2.5312e-02	k_t = [4, 4, 4, 3]			
Generation 8050: Best = 3.354593e+03	FES = 805100	$\mu_F = 0.737$	$\mu_{CR} = 0.237$	

Diversity = 2.4103e-02	k_t = [1, 1, 1, 1]			
Generation 8060: Best = 3.354583e+03	FES = 806100	$\mu_F = 0.746$	$\mu_{CR} = 0.237$	
Diversity = 2.4159e-02	k_t = [4, 4, 2, 3]			
Generation 8070: Best = 3.354561e+03	FES = 807100	$\mu_F = 0.741$	$\mu_{CR} = 0.245$	
Diversity = 2.5626e-02	k_t = [1, 1, 1, 1]			
Generation 8080: Best = 3.354550e+03	FES = 808100	$\mu_F = 0.741$	$\mu_{CR} = 0.251$	
Diversity = 2.6590e-02	k_t = [3, 3, 2, 2]			
Generation 8090: Best = 3.354534e+03	FES = 809100	$\mu_F = 0.744$	$\mu_{CR} = 0.253$	
Diversity = 2.8742e-02	k_t = [1, 1, 1, 1]			
Generation 8100: Best = 3.354523e+03	FES = 810100	$\mu_F = 0.750$	$\mu_{CR} = 0.245$	
Diversity = 2.7634e-02	k_t = [2, 4, 4, 4]			
Generation 8110: Best = 3.354510e+03	FES = 811100	$\mu_F = 0.744$	$\mu_{CR} = 0.235$	
Diversity = 2.7438e-02	k_t = [1, 1, 1, 1]			
Generation 8120: Best = 3.354498e+03	FES = 812100	$\mu_F = 0.747$	$\mu_{CR} = 0.226$	
Diversity = 2.7499e-02	k_t = [2, 4, 3, 2]			
Generation 8130: Best = 3.354481e+03	FES = 813100	$\mu_F = 0.746$	$\mu_{CR} = 0.228$	
Diversity = 2.7857e-02	k_t = [1, 1, 1, 1]			
Generation 8140: Best = 3.354475e+03	FES = 814100	$\mu_F = 0.735$	$\mu_{CR} = 0.228$	
Diversity = 2.5615e-02	k_t = [3, 4, 4, 4]			
Generation 8150: Best = 3.354465e+03	FES = 815100	$\mu_F = 0.726$	$\mu_{CR} = 0.228$	
Diversity = 2.2938e-02	k_t = [1, 1, 1, 1]			
Generation 8160: Best = 3.354452e+03	FES = 816100	$\mu_F = 0.726$	$\mu_{CR} = 0.220$	
Diversity = 2.1031e-02	k_t = [2, 2, 2, 4]			
Generation 8170: Best = 3.354446e+03	FES = 817100	$\mu_F = 0.730$	$\mu_{CR} = 0.217$	
Diversity = 2.1337e-02	k_t = [1, 1, 1, 1]			
Generation 8180: Best = 3.354436e+03	FES = 818100	$\mu_F = 0.738$	$\mu_{CR} = 0.217$	
Diversity = 2.0961e-02	k_t = [3, 2, 2, 4]			
Generation 8190: Best = 3.354432e+03	FES = 819100	$\mu_F = 0.738$	$\mu_{CR} = 0.215$	
Diversity = 2.0602e-02	k_t = [1, 1, 1, 1]			
Generation 8200: Best = 3.354418e+03	FES = 820100	$\mu_F = 0.747$	$\mu_{CR} = 0.219$	
Diversity = 2.1443e-02	k_t = [3, 3, 4, 3]			
Generation 8210: Best = 3.354410e+03	FES = 821100	$\mu_F = 0.747$	$\mu_{CR} = 0.210$	
Diversity = 2.2638e-02	k_t = [1, 1, 1, 1]			
Generation 8220: Best = 3.354405e+03	FES = 822100	$\mu_F = 0.733$	$\mu_{CR} = 0.204$	
Diversity = 1.9378e-02	k_t = [3, 2, 4, 4]			
Generation 8230: Best = 3.354398e+03	FES = 823100	$\mu_F = 0.745$	$\mu_{CR} = 0.196$	
Diversity = 1.8905e-02	k_t = [1, 1, 1, 1]			
Generation 8240: Best = 3.354390e+03	FES = 824100	$\mu_F = 0.759$	$\mu_{CR} = 0.192$	
Diversity = 2.0878e-02	k_t = [2, 3, 2, 2]			
Generation 8250: Best = 3.354377e+03	FES = 825100	$\mu_F = 0.754$	$\mu_{CR} = 0.198$	
Diversity = 2.1841e-02	k_t = [1, 1, 1, 1]			
Generation 8260: Best = 3.354370e+03	FES = 826100	$\mu_F = 0.739$	$\mu_{CR} = 0.207$	
Diversity = 2.0376e-02	k_t = [2, 2, 2, 3]			
Generation 8270: Best = 3.354365e+03	FES = 827100	$\mu_F = 0.749$	$\mu_{CR} = 0.211$	
Diversity = 2.1383e-02	k_t = [1, 1, 1, 1]			
Generation 8280: Best = 3.354350e+03	FES = 828100	$\mu_F = 0.740$	$\mu_{CR} = 0.211$	
Diversity = 2.2230e-02	k_t = [2, 2, 3, 2]			
Generation 8290: Best = 3.354346e+03	FES = 829100	$\mu_F = 0.732$	$\mu_{CR} = 0.212$	
Diversity = 2.1935e-02	k_t = [1, 1, 1, 1]			
Generation 8300: Best = 3.354334e+03	FES = 830100	$\mu_F = 0.731$	$\mu_{CR} = 0.214$	
Diversity = 2.1351e-02	k_t = [3, 4, 3, 2]			
Generation 8310: Best = 3.354322e+03	FES = 831100	$\mu_F = 0.740$	$\mu_{CR} = 0.210$	
Diversity = 2.1690e-02	k_t = [1, 1, 1, 1]			
Generation 8320: Best = 3.354313e+03	FES = 832100	$\mu_F = 0.742$	$\mu_{CR} = 0.210$	

Diversity = 2.0023e-02	k_t = [2, 3, 3, 4]			
Generation 8330: Best = 3.354307e+03	FES = 833100	$\mu_F = 0.740$	$\mu_{CR} = 0.205$	
Diversity = 2.1531e-02	k_t = [1, 1, 1, 1]			
Generation 8340: Best = 3.354295e+03	FES = 834100	$\mu_F = 0.747$	$\mu_{CR} = 0.207$	
Diversity = 2.3173e-02	k_t = [4, 2, 4, 2]			
Generation 8350: Best = 3.354284e+03	FES = 835100	$\mu_F = 0.751$	$\mu_{CR} = 0.205$	
Diversity = 2.3951e-02	k_t = [1, 1, 1, 1]			
Generation 8360: Best = 3.354272e+03	FES = 836100	$\mu_F = 0.752$	$\mu_{CR} = 0.202$	
Diversity = 2.3183e-02	k_t = [4, 4, 3, 4]			
Generation 8370: Best = 3.354257e+03	FES = 837100	$\mu_F = 0.752$	$\mu_{CR} = 0.214$	
Diversity = 2.3673e-02	k_t = [1, 1, 1, 1]			
Generation 8380: Best = 3.354240e+03	FES = 838100	$\mu_F = 0.765$	$\mu_{CR} = 0.212$	
Diversity = 2.4138e-02	k_t = [3, 4, 4, 2]			
Generation 8390: Best = 3.354226e+03	FES = 839100	$\mu_F = 0.772$	$\mu_{CR} = 0.213$	
Diversity = 2.6157e-02	k_t = [1, 1, 1, 1]			
Generation 8400: Best = 3.354212e+03	FES = 840100	$\mu_F = 0.767$	$\mu_{CR} = 0.217$	
Diversity = 2.8087e-02	k_t = [4, 4, 2, 2]			
Generation 8410: Best = 3.354200e+03	FES = 841100	$\mu_F = 0.749$	$\mu_{CR} = 0.227$	
Diversity = 2.8180e-02	k_t = [1, 1, 1, 1]			
Generation 8420: Best = 3.354186e+03	FES = 842100	$\mu_F = 0.750$	$\mu_{CR} = 0.222$	
Diversity = 2.7989e-02	k_t = [4, 2, 3, 2]			
Generation 8430: Best = 3.354166e+03	FES = 843100	$\mu_F = 0.748$	$\mu_{CR} = 0.227$	
Diversity = 3.0981e-02	k_t = [1, 1, 1, 1]			
Generation 8440: Best = 3.354133e+03	FES = 844100	$\mu_F = 0.741$	$\mu_{CR} = 0.230$	
Diversity = 3.1330e-02	k_t = [2, 4, 3, 3]			
Generation 8450: Best = 3.354114e+03	FES = 845100	$\mu_F = 0.739$	$\mu_{CR} = 0.231$	
Diversity = 3.1922e-02	k_t = [1, 1, 1, 1]			
Generation 8460: Best = 3.354091e+03	FES = 846100	$\mu_F = 0.753$	$\mu_{CR} = 0.237$	
Diversity = 3.1926e-02	k_t = [2, 2, 3, 4]			
Generation 8470: Best = 3.354073e+03	FES = 847100	$\mu_F = 0.757$	$\mu_{CR} = 0.239$	
Diversity = 3.2666e-02	k_t = [1, 1, 1, 1]			
Generation 8480: Best = 3.354057e+03	FES = 848100	$\mu_F = 0.751$	$\mu_{CR} = 0.234$	
Diversity = 3.0451e-02	k_t = [4, 4, 4, 4]			
Generation 8490: Best = 3.354037e+03	FES = 849100	$\mu_F = 0.738$	$\mu_{CR} = 0.245$	
Diversity = 3.0427e-02	k_t = [1, 1, 1, 1]			
Generation 8500: Best = 3.354018e+03	FES = 850100	$\mu_F = 0.742$	$\mu_{CR} = 0.242$	
Diversity = 2.7984e-02	k_t = [4, 3, 2, 3]			
Generation 8510: Best = 3.354008e+03	FES = 851100	$\mu_F = 0.727$	$\mu_{CR} = 0.242$	
Diversity = 2.8892e-02	k_t = [1, 1, 1, 1]			
Generation 8520: Best = 3.353992e+03	FES = 852100	$\mu_F = 0.725$	$\mu_{CR} = 0.244$	
Diversity = 2.7832e-02	k_t = [4, 3, 4, 4]			
Generation 8530: Best = 3.353974e+03	FES = 853100	$\mu_F = 0.733$	$\mu_{CR} = 0.239$	
Diversity = 2.7533e-02	k_t = [1, 1, 1, 1]			
Generation 8540: Best = 3.353962e+03	FES = 854100	$\mu_F = 0.732$	$\mu_{CR} = 0.242$	
Diversity = 2.5708e-02	k_t = [4, 4, 4, 3]			
Generation 8550: Best = 3.353951e+03	FES = 855100	$\mu_F = 0.746$	$\mu_{CR} = 0.245$	
Diversity = 2.4695e-02	k_t = [1, 1, 1, 1]			
Generation 8560: Best = 3.353934e+03	FES = 856100	$\mu_F = 0.753$	$\mu_{CR} = 0.248$	
Diversity = 2.4754e-02	k_t = [3, 4, 2, 4]			
Generation 8570: Best = 3.353920e+03	FES = 857100	$\mu_F = 0.747$	$\mu_{CR} = 0.244$	
Diversity = 2.5582e-02	k_t = [1, 1, 1, 1]			
Generation 8580: Best = 3.353914e+03	FES = 858100	$\mu_F = 0.746$	$\mu_{CR} = 0.249$	
Diversity = 2.4890e-02	k_t = [4, 2, 3, 3]			
Generation 8590: Best = 3.353901e+03	FES = 859100	$\mu_F = 0.747$	$\mu_{CR} = 0.235$	

Diversity = 2.4693e-02	k_t = [1, 1, 1, 1]			
Generation 8600: Best = 3.353893e+03	FES = 860100	$\mu_F = 0.753$	$\mu_{CR} = 0.228$	
Diversity = 2.3931e-02	k_t = [2, 3, 3, 3]			
Generation 8610: Best = 3.353886e+03	FES = 861100	$\mu_F = 0.743$	$\mu_{CR} = 0.231$	
Diversity = 2.3242e-02	k_t = [1, 1, 1, 1]			
Generation 8620: Best = 3.353874e+03	FES = 862100	$\mu_F = 0.734$	$\mu_{CR} = 0.235$	
Diversity = 2.2954e-02	k_t = [4, 3, 3, 2]			
Generation 8630: Best = 3.353863e+03	FES = 863100	$\mu_F = 0.742$	$\mu_{CR} = 0.231$	
Diversity = 2.2888e-02	k_t = [1, 1, 1, 1]			
Generation 8640: Best = 3.353854e+03	FES = 864100	$\mu_F = 0.743$	$\mu_{CR} = 0.227$	
Diversity = 2.3246e-02	k_t = [4, 2, 2, 3]			
Generation 8650: Best = 3.353843e+03	FES = 865100	$\mu_F = 0.736$	$\mu_{CR} = 0.232$	
Diversity = 2.4470e-02	k_t = [1, 1, 1, 1]			
Generation 8660: Best = 3.353828e+03	FES = 866100	$\mu_F = 0.750$	$\mu_{CR} = 0.236$	
Diversity = 2.5564e-02	k_t = [4, 2, 2, 2]			
Generation 8670: Best = 3.353805e+03	FES = 867100	$\mu_F = 0.743$	$\mu_{CR} = 0.233$	
Diversity = 2.7025e-02	k_t = [1, 1, 1, 1]			
Generation 8680: Best = 3.353780e+03	FES = 868100	$\mu_F = 0.751$	$\mu_{CR} = 0.234$	
Diversity = 2.7402e-02	k_t = [4, 2, 4, 4]			
Generation 8690: Best = 3.353769e+03	FES = 869100	$\mu_F = 0.735$	$\mu_{CR} = 0.230$	
Diversity = 2.6489e-02	k_t = [1, 1, 1, 1]			
Generation 8700: Best = 3.353752e+03	FES = 870100	$\mu_F = 0.746$	$\mu_{CR} = 0.235$	
Diversity = 2.5974e-02	k_t = [4, 4, 2, 2]			
Generation 8710: Best = 3.353738e+03	FES = 871100	$\mu_F = 0.743$	$\mu_{CR} = 0.236$	
Diversity = 2.5474e-02	k_t = [1, 1, 1, 1]			
Generation 8720: Best = 3.353728e+03	FES = 872100	$\mu_F = 0.743$	$\mu_{CR} = 0.238$	
Diversity = 2.5477e-02	k_t = [4, 3, 4, 2]			
Generation 8730: Best = 3.353714e+03	FES = 873100	$\mu_F = 0.739$	$\mu_{CR} = 0.227$	
Diversity = 2.6451e-02	k_t = [1, 1, 1, 1]			
Generation 8740: Best = 3.353685e+03	FES = 874100	$\mu_F = 0.741$	$\mu_{CR} = 0.225$	
Diversity = 2.7030e-02	k_t = [3, 4, 2, 2]			
Generation 8750: Best = 3.353677e+03	FES = 875100	$\mu_F = 0.742$	$\mu_{CR} = 0.219$	
Diversity = 2.7187e-02	k_t = [1, 1, 1, 1]			
Generation 8760: Best = 3.353657e+03	FES = 876100	$\mu_F = 0.751$	$\mu_{CR} = 0.217$	
Diversity = 2.6890e-02	k_t = [3, 2, 3, 3]			
Generation 8770: Best = 3.353647e+03	FES = 877100	$\mu_F = 0.744$	$\mu_{CR} = 0.215$	
Diversity = 2.9145e-02	k_t = [1, 1, 1, 1]			
Generation 8780: Best = 3.353634e+03	FES = 878100	$\mu_F = 0.735$	$\mu_{CR} = 0.209$	
Diversity = 2.5508e-02	k_t = [3, 4, 2, 2]			
Generation 8790: Best = 3.353624e+03	FES = 879100	$\mu_F = 0.719$	$\mu_{CR} = 0.209$	
Diversity = 2.3502e-02	k_t = [1, 1, 1, 1]			
Generation 8800: Best = 3.353616e+03	FES = 880100	$\mu_F = 0.718$	$\mu_{CR} = 0.218$	
Diversity = 2.2402e-02	k_t = [2, 3, 4, 2]			
Generation 8810: Best = 3.353610e+03	FES = 881100	$\mu_F = 0.721$	$\mu_{CR} = 0.214$	
Diversity = 2.3123e-02	k_t = [1, 1, 1, 1]			
Generation 8820: Best = 3.353592e+03	FES = 882100	$\mu_F = 0.716$	$\mu_{CR} = 0.215$	
Diversity = 2.1750e-02	k_t = [3, 2, 4, 3]			
Generation 8830: Best = 3.353585e+03	FES = 883100	$\mu_F = 0.717$	$\mu_{CR} = 0.214$	
Diversity = 2.3034e-02	k_t = [1, 1, 1, 1]			
Generation 8840: Best = 3.353575e+03	FES = 884100	$\mu_F = 0.711$	$\mu_{CR} = 0.214$	
Diversity = 2.2285e-02	k_t = [4, 3, 2, 2]			
Generation 8850: Best = 3.353570e+03	FES = 885100	$\mu_F = 0.708$	$\mu_{CR} = 0.209$	
Diversity = 2.2086e-02	k_t = [1, 1, 1, 1]			
Generation 8860: Best = 3.353560e+03	FES = 886100	$\mu_F = 0.712$	$\mu_{CR} = 0.213$	

Diversity = 2.0427e-02	k_t = [2, 3, 3, 3]			
Generation 8870: Best = 3.353553e+03	FES = 887100	$\mu_F = 0.713$	$\mu_{CR} = 0.216$	
Diversity = 2.0391e-02	k_t = [1, 1, 1, 1]			
Generation 8880: Best = 3.353541e+03	FES = 888100	$\mu_F = 0.720$	$\mu_{CR} = 0.214$	
Diversity = 1.9852e-02	k_t = [3, 2, 3, 4]			
Generation 8890: Best = 3.353535e+03	FES = 889100	$\mu_F = 0.727$	$\mu_{CR} = 0.214$	
Diversity = 1.9649e-02	k_t = [1, 1, 1, 1]			
Generation 8900: Best = 3.353526e+03	FES = 890100	$\mu_F = 0.721$	$\mu_{CR} = 0.214$	
Diversity = 2.0096e-02	k_t = [3, 4, 3, 2]			
Generation 8910: Best = 3.353517e+03	FES = 891100	$\mu_F = 0.724$	$\mu_{CR} = 0.224$	
Diversity = 1.9527e-02	k_t = [1, 1, 1, 1]			
Generation 8920: Best = 3.353509e+03	FES = 892100	$\mu_F = 0.727$	$\mu_{CR} = 0.217$	
Diversity = 1.7935e-02	k_t = [3, 3, 3, 3]			
Generation 8930: Best = 3.353502e+03	FES = 893100	$\mu_F = 0.732$	$\mu_{CR} = 0.218$	
Diversity = 1.8315e-02	k_t = [1, 1, 1, 1]			
Generation 8940: Best = 3.353492e+03	FES = 894100	$\mu_F = 0.737$	$\mu_{CR} = 0.228$	
Diversity = 1.8721e-02	k_t = [4, 2, 3, 2]			
Generation 8950: Best = 3.353482e+03	FES = 895100	$\mu_F = 0.750$	$\mu_{CR} = 0.227$	
Diversity = 2.0981e-02	k_t = [1, 1, 1, 1]			
Generation 8960: Best = 3.353468e+03	FES = 896100	$\mu_F = 0.761$	$\mu_{CR} = 0.229$	
Diversity = 2.1636e-02	k_t = [4, 4, 2, 3]			
Generation 8970: Best = 3.353454e+03	FES = 897100	$\mu_F = 0.765$	$\mu_{CR} = 0.233$	
Diversity = 2.1718e-02	k_t = [1, 1, 1, 1]			
Generation 8980: Best = 3.353445e+03	FES = 898100	$\mu_F = 0.753$	$\mu_{CR} = 0.236$	
Diversity = 2.2629e-02	k_t = [4, 4, 2, 2]			
Generation 8990: Best = 3.353437e+03	FES = 899100	$\mu_F = 0.749$	$\mu_{CR} = 0.234$	
Diversity = 2.3573e-02	k_t = [1, 1, 1, 1]			
Generation 9000: Best = 3.353423e+03	FES = 900100	$\mu_F = 0.757$	$\mu_{CR} = 0.230$	
Diversity = 2.3009e-02	k_t = [4, 4, 2, 4]			
Generation 9010: Best = 3.353410e+03	FES = 901100	$\mu_F = 0.752$	$\mu_{CR} = 0.238$	
Diversity = 2.1553e-02	k_t = [1, 1, 1, 1]			
Generation 9020: Best = 3.353388e+03	FES = 902100	$\mu_F = 0.742$	$\mu_{CR} = 0.232$	
Diversity = 2.4074e-02	k_t = [3, 2, 3, 2]			
Generation 9030: Best = 3.353382e+03	FES = 903100	$\mu_F = 0.744$	$\mu_{CR} = 0.231$	
Diversity = 2.6019e-02	k_t = [1, 1, 1, 1]			
Generation 9040: Best = 3.353365e+03	FES = 904100	$\mu_F = 0.738$	$\mu_{CR} = 0.228$	
Diversity = 2.6240e-02	k_t = [2, 2, 2, 3]			
Generation 9050: Best = 3.353352e+03	FES = 905100	$\mu_F = 0.743$	$\mu_{CR} = 0.221$	
Diversity = 2.7705e-02	k_t = [1, 1, 1, 1]			
Generation 9060: Best = 3.353341e+03	FES = 906100	$\mu_F = 0.751$	$\mu_{CR} = 0.205$	
Diversity = 2.5466e-02	k_t = [2, 4, 3, 4]			
Generation 9070: Best = 3.353331e+03	FES = 907100	$\mu_F = 0.758$	$\mu_{CR} = 0.206$	
Diversity = 2.6011e-02	k_t = [1, 1, 1, 1]			
Generation 9080: Best = 3.353323e+03	FES = 908100	$\mu_F = 0.768$	$\mu_{CR} = 0.204$	
Diversity = 2.6288e-02	k_t = [4, 3, 2, 3]			
Generation 9090: Best = 3.353306e+03	FES = 909100	$\mu_F = 0.755$	$\mu_{CR} = 0.210$	
Diversity = 2.6584e-02	k_t = [1, 1, 1, 1]			
Generation 9100: Best = 3.353300e+03	FES = 910100	$\mu_F = 0.749$	$\mu_{CR} = 0.208$	
Diversity = 2.5697e-02	k_t = [4, 3, 4, 2]			
Generation 9110: Best = 3.353290e+03	FES = 911100	$\mu_F = 0.746$	$\mu_{CR} = 0.211$	
Diversity = 2.5309e-02	k_t = [1, 1, 1, 1]			
Generation 9120: Best = 3.353278e+03	FES = 912100	$\mu_F = 0.747$	$\mu_{CR} = 0.207$	
Diversity = 2.3054e-02	k_t = [4, 4, 4, 4]			
Generation 9130: Best = 3.353271e+03	FES = 913100	$\mu_F = 0.746$	$\mu_{CR} = 0.217$	

Diversity = 2.2056e-02	k_t = [1, 1, 1, 1]			
Generation 9140: Best = 3.353261e+03	FES = 914100	$\mu_F = 0.740$	$\mu_{CR} = 0.223$	
Diversity = 2.1951e-02	k_t = [3, 3, 3, 2]			
Generation 9150: Best = 3.353248e+03	FES = 915100	$\mu_F = 0.744$	$\mu_{CR} = 0.229$	
Diversity = 2.2190e-02	k_t = [1, 1, 1, 1]			
Generation 9160: Best = 3.353240e+03	FES = 916100	$\mu_F = 0.730$	$\mu_{CR} = 0.227$	
Diversity = 2.2326e-02	k_t = [4, 2, 2, 2]			
Generation 9170: Best = 3.353228e+03	FES = 917100	$\mu_F = 0.739$	$\mu_{CR} = 0.226$	
Diversity = 2.2947e-02	k_t = [1, 1, 1, 1]			
Generation 9180: Best = 3.353216e+03	FES = 918100	$\mu_F = 0.743$	$\mu_{CR} = 0.227$	
Diversity = 2.2589e-02	k_t = [2, 3, 2, 2]			
Generation 9190: Best = 3.353185e+03	FES = 919100	$\mu_F = 0.753$	$\mu_{CR} = 0.230$	
Diversity = 2.7551e-02	k_t = [1, 1, 1, 1]			
Generation 9200: Best = 3.353156e+03	FES = 920100	$\mu_F = 0.766$	$\mu_{CR} = 0.241$	
Diversity = 2.9932e-02	k_t = [2, 2, 3, 4]			
Generation 9210: Best = 3.353127e+03	FES = 921100	$\mu_F = 0.771$	$\mu_{CR} = 0.239$	
Diversity = 3.2191e-02	k_t = [1, 1, 1, 1]			
Generation 9220: Best = 3.353107e+03	FES = 922100	$\mu_F = 0.762$	$\mu_{CR} = 0.227$	
Diversity = 3.1472e-02	k_t = [2, 2, 3, 4]			
Generation 9230: Best = 3.353098e+03	FES = 923100	$\mu_F = 0.756$	$\mu_{CR} = 0.226$	
Diversity = 3.1653e-02	k_t = [1, 1, 1, 1]			
Generation 9240: Best = 3.353082e+03	FES = 924100	$\mu_F = 0.756$	$\mu_{CR} = 0.228$	
Diversity = 3.1675e-02	k_t = [2, 4, 3, 2]			
Generation 9250: Best = 3.353059e+03	FES = 925100	$\mu_F = 0.739$	$\mu_{CR} = 0.219$	
Diversity = 2.9658e-02	k_t = [1, 1, 1, 1]			
Generation 9260: Best = 3.353042e+03	FES = 926100	$\mu_F = 0.743$	$\mu_{CR} = 0.226$	
Diversity = 2.8877e-02	k_t = [3, 3, 3, 2]			
Generation 9270: Best = 3.353025e+03	FES = 927100	$\mu_F = 0.734$	$\mu_{CR} = 0.229$	
Diversity = 2.9364e-02	k_t = [1, 1, 1, 1]			
Generation 9280: Best = 3.353004e+03	FES = 928100	$\mu_F = 0.744$	$\mu_{CR} = 0.232$	
Diversity = 2.9303e-02	k_t = [4, 3, 3, 4]			
Generation 9290: Best = 3.352989e+03	FES = 929100	$\mu_F = 0.740$	$\mu_{CR} = 0.235$	
Diversity = 2.6385e-02	k_t = [1, 1, 1, 1]			
Generation 9300: Best = 3.352978e+03	FES = 930100	$\mu_F = 0.740$	$\mu_{CR} = 0.242$	
Diversity = 2.5494e-02	k_t = [3, 3, 4, 2]			
Generation 9310: Best = 3.352969e+03	FES = 931100	$\mu_F = 0.751$	$\mu_{CR} = 0.234$	
Diversity = 2.6069e-02	k_t = [1, 1, 1, 1]			
Generation 9320: Best = 3.352956e+03	FES = 932100	$\mu_F = 0.749$	$\mu_{CR} = 0.233$	
Diversity = 2.4554e-02	k_t = [4, 2, 3, 3]			
Generation 9330: Best = 3.352943e+03	FES = 933100	$\mu_F = 0.750$	$\mu_{CR} = 0.233$	
Diversity = 2.4969e-02	k_t = [1, 1, 1, 1]			
Generation 9340: Best = 3.352932e+03	FES = 934100	$\mu_F = 0.735$	$\mu_{CR} = 0.230$	
Diversity = 2.4223e-02	k_t = [3, 3, 2, 3]			
Generation 9350: Best = 3.352914e+03	FES = 935100	$\mu_F = 0.740$	$\mu_{CR} = 0.231$	
Diversity = 2.6124e-02	k_t = [1, 1, 1, 1]			
Generation 9360: Best = 3.352908e+03	FES = 936100	$\mu_F = 0.736$	$\mu_{CR} = 0.220$	
Diversity = 2.4939e-02	k_t = [4, 3, 3, 3]			
Generation 9370: Best = 3.352895e+03	FES = 937100	$\mu_F = 0.741$	$\mu_{CR} = 0.216$	
Diversity = 2.3958e-02	k_t = [1, 1, 1, 1]			
Generation 9380: Best = 3.352881e+03	FES = 938100	$\mu_F = 0.744$	$\mu_{CR} = 0.219$	
Diversity = 2.4722e-02	k_t = [2, 4, 4, 2]			
Generation 9390: Best = 3.352873e+03	FES = 939100	$\mu_F = 0.739$	$\mu_{CR} = 0.224$	
Diversity = 2.4009e-02	k_t = [1, 1, 1, 1]			
Generation 9400: Best = 3.352860e+03	FES = 940100	$\mu_F = 0.755$	$\mu_{CR} = 0.226$	

Diversity = 2.4735e-02	k_t = [2, 3, 3, 2]			
Generation 9410: Best = 3.352852e+03	FES = 941100	$\mu_F = 0.738$	$\mu_{CR} = 0.224$	
Diversity = 2.5751e-02	k_t = [1, 1, 1, 1]			
Generation 9420: Best = 3.352830e+03	FES = 942100	$\mu_F = 0.740$	$\mu_{CR} = 0.224$	
Diversity = 2.3862e-02	k_t = [3, 2, 2, 3]			
Generation 9430: Best = 3.352808e+03	FES = 943100	$\mu_F = 0.755$	$\mu_{CR} = 0.227$	
Diversity = 2.7523e-02	k_t = [1, 1, 1, 1]			
Generation 9440: Best = 3.352795e+03	FES = 944100	$\mu_F = 0.767$	$\mu_{CR} = 0.212$	
Diversity = 2.9796e-02	k_t = [2, 2, 4, 3]			
Generation 9450: Best = 3.352768e+03	FES = 945100	$\mu_F = 0.764$	$\mu_{CR} = 0.211$	
Diversity = 3.3636e-02	k_t = [1, 1, 1, 1]			
Generation 9460: Best = 3.352741e+03	FES = 946100	$\mu_F = 0.745$	$\mu_{CR} = 0.220$	
Diversity = 3.3910e-02	k_t = [2, 4, 4, 2]			
Generation 9470: Best = 3.352724e+03	FES = 947100	$\mu_F = 0.754$	$\mu_{CR} = 0.224$	
Diversity = 3.3352e-02	k_t = [1, 1, 1, 1]			
Generation 9480: Best = 3.352698e+03	FES = 948100	$\mu_F = 0.749$	$\mu_{CR} = 0.216$	
Diversity = 3.2566e-02	k_t = [3, 4, 4, 2]			
Generation 9490: Best = 3.352689e+03	FES = 949100	$\mu_F = 0.751$	$\mu_{CR} = 0.216$	
Diversity = 3.3914e-02	k_t = [1, 1, 1, 1]			
Generation 9500: Best = 3.352664e+03	FES = 950100	$\mu_F = 0.744$	$\mu_{CR} = 0.212$	
Diversity = 3.2785e-02	k_t = [2, 4, 4, 3]			
Generation 9510: Best = 3.352642e+03	FES = 951100	$\mu_F = 0.736$	$\mu_{CR} = 0.207$	
Diversity = 3.2128e-02	k_t = [1, 1, 1, 1]			
Generation 9520: Best = 3.352629e+03	FES = 952100	$\mu_F = 0.726$	$\mu_{CR} = 0.205$	
Diversity = 3.1277e-02	k_t = [3, 3, 2, 2]			
Generation 9530: Best = 3.352622e+03	FES = 953100	$\mu_F = 0.733$	$\mu_{CR} = 0.207$	
Diversity = 2.9743e-02	k_t = [1, 1, 1, 1]			
Generation 9540: Best = 3.352607e+03	FES = 954100	$\mu_F = 0.728$	$\mu_{CR} = 0.206$	
Diversity = 2.7286e-02	k_t = [4, 2, 3, 3]			
Generation 9550: Best = 3.352595e+03	FES = 955100	$\mu_F = 0.727$	$\mu_{CR} = 0.200$	
Diversity = 2.6189e-02	k_t = [1, 1, 1, 1]			
Generation 9560: Best = 3.352581e+03	FES = 956100	$\mu_F = 0.734$	$\mu_{CR} = 0.194$	
Diversity = 2.5872e-02	k_t = [3, 4, 3, 4]			
Generation 9570: Best = 3.352565e+03	FES = 957100	$\mu_F = 0.749$	$\mu_{CR} = 0.200$	
Diversity = 2.7296e-02	k_t = [1, 1, 1, 1]			
Generation 9580: Best = 3.352551e+03	FES = 958100	$\mu_F = 0.731$	$\mu_{CR} = 0.205$	
Diversity = 2.6893e-02	k_t = [2, 3, 4, 3]			
Generation 9590: Best = 3.352532e+03	FES = 959100	$\mu_F = 0.730$	$\mu_{CR} = 0.205$	
Diversity = 2.7206e-02	k_t = [1, 1, 1, 1]			
Generation 9600: Best = 3.352521e+03	FES = 960100	$\mu_F = 0.726$	$\mu_{CR} = 0.211$	
Diversity = 2.6339e-02	k_t = [4, 2, 4, 3]			
Generation 9610: Best = 3.352507e+03	FES = 961100	$\mu_F = 0.739$	$\mu_{CR} = 0.210$	
Diversity = 2.7287e-02	k_t = [1, 1, 1, 1]			
Generation 9620: Best = 3.352498e+03	FES = 962100	$\mu_F = 0.738$	$\mu_{CR} = 0.213$	
Diversity = 2.7801e-02	k_t = [4, 2, 2, 3]			
Generation 9630: Best = 3.352485e+03	FES = 963100	$\mu_F = 0.749$	$\mu_{CR} = 0.215$	
Diversity = 2.6547e-02	k_t = [1, 1, 1, 1]			
Generation 9640: Best = 3.352471e+03	FES = 964100	$\mu_F = 0.755$	$\mu_{CR} = 0.211$	
Diversity = 2.7037e-02	k_t = [4, 4, 2, 2]			
Generation 9650: Best = 3.352461e+03	FES = 965100	$\mu_F = 0.751$	$\mu_{CR} = 0.211$	
Diversity = 2.7507e-02	k_t = [1, 1, 1, 1]			
Generation 9660: Best = 3.352448e+03	FES = 966100	$\mu_F = 0.743$	$\mu_{CR} = 0.207$	
Diversity = 2.6395e-02	k_t = [4, 2, 4, 3]			
Generation 9670: Best = 3.352436e+03	FES = 967100	$\mu_F = 0.738$	$\mu_{CR} = 0.216$	

Diversity = 2.7714e-02	k_t = [1, 1, 1, 1]			
Generation 9680: Best = 3.352423e+03	FES = 968100	$\mu_F = 0.732$	$\mu_{CR} = 0.217$	
Diversity = 2.6012e-02	k_t = [2, 4, 4, 3]			
Generation 9690: Best = 3.352416e+03	FES = 969100	$\mu_F = 0.735$	$\mu_{CR} = 0.213$	
Diversity = 2.5599e-02	k_t = [1, 1, 1, 1]			
Generation 9700: Best = 3.352405e+03	FES = 970100	$\mu_F = 0.735$	$\mu_{CR} = 0.212$	
Diversity = 2.4143e-02	k_t = [4, 4, 2, 2]			
Generation 9710: Best = 3.352397e+03	FES = 971100	$\mu_F = 0.735$	$\mu_{CR} = 0.210$	
Diversity = 2.3590e-02	k_t = [1, 1, 1, 1]			
Generation 9720: Best = 3.352388e+03	FES = 972100	$\mu_F = 0.729$	$\mu_{CR} = 0.213$	
Diversity = 2.1548e-02	k_t = [4, 3, 2, 4]			
Generation 9730: Best = 3.352382e+03	FES = 973100	$\mu_F = 0.744$	$\mu_{CR} = 0.216$	
Diversity = 2.0665e-02	k_t = [1, 1, 1, 1]			
Generation 9740: Best = 3.352366e+03	FES = 974100	$\mu_F = 0.724$	$\mu_{CR} = 0.209$	
Diversity = 2.0822e-02	k_t = [3, 3, 2, 2]			
Generation 9750: Best = 3.352353e+03	FES = 975100	$\mu_F = 0.738$	$\mu_{CR} = 0.216$	
Diversity = 2.1465e-02	k_t = [1, 1, 1, 1]			
Generation 9760: Best = 3.352350e+03	FES = 976100	$\mu_F = 0.742$	$\mu_{CR} = 0.220$	
Diversity = 1.9402e-02	k_t = [3, 3, 3, 3]			
Generation 9770: Best = 3.352342e+03	FES = 977100	$\mu_F = 0.728$	$\mu_{CR} = 0.223$	
Diversity = 1.8652e-02	k_t = [1, 1, 1, 1]			
Generation 9780: Best = 3.352333e+03	FES = 978100	$\mu_F = 0.727$	$\mu_{CR} = 0.220$	
Diversity = 1.7342e-02	k_t = [2, 3, 3, 4]			
Generation 9790: Best = 3.352324e+03	FES = 979100	$\mu_F = 0.730$	$\mu_{CR} = 0.220$	
Diversity = 1.7400e-02	k_t = [1, 1, 1, 1]			
Generation 9800: Best = 3.352320e+03	FES = 980100	$\mu_F = 0.739$	$\mu_{CR} = 0.219$	
Diversity = 1.5764e-02	k_t = [4, 4, 4, 3]			
Generation 9810: Best = 3.352310e+03	FES = 981100	$\mu_F = 0.756$	$\mu_{CR} = 0.211$	
Diversity = 1.6501e-02	k_t = [1, 1, 1, 1]			
Generation 9820: Best = 3.352306e+03	FES = 982100	$\mu_F = 0.765$	$\mu_{CR} = 0.215$	
Diversity = 1.7385e-02	k_t = [2, 2, 2, 3]			
Generation 9830: Best = 3.352297e+03	FES = 983100	$\mu_F = 0.753$	$\mu_{CR} = 0.210$	
Diversity = 1.8494e-02	k_t = [1, 1, 1, 1]			
Generation 9840: Best = 3.352292e+03	FES = 984100	$\mu_F = 0.742$	$\mu_{CR} = 0.209$	
Diversity = 1.8248e-02	k_t = [2, 2, 2, 2]			
Generation 9850: Best = 3.352286e+03	FES = 985100	$\mu_F = 0.738$	$\mu_{CR} = 0.205$	
Diversity = 1.7372e-02	k_t = [1, 1, 1, 1]			
Generation 9860: Best = 3.352281e+03	FES = 986100	$\mu_F = 0.733$	$\mu_{CR} = 0.197$	
Diversity = 1.6601e-02	k_t = [2, 3, 2, 3]			
Generation 9870: Best = 3.352273e+03	FES = 987100	$\mu_F = 0.746$	$\mu_{CR} = 0.207$	
Diversity = 1.8411e-02	k_t = [1, 1, 1, 1]			
Generation 9880: Best = 3.352268e+03	FES = 988100	$\mu_F = 0.737$	$\mu_{CR} = 0.201$	
Diversity = 1.8467e-02	k_t = [4, 2, 2, 4]			
Generation 9890: Best = 3.352261e+03	FES = 989100	$\mu_F = 0.731$	$\mu_{CR} = 0.195$	
Diversity = 1.7795e-02	k_t = [1, 1, 1, 1]			
Generation 9900: Best = 3.352255e+03	FES = 990100	$\mu_F = 0.735$	$\mu_{CR} = 0.200$	
Diversity = 1.8787e-02	k_t = [2, 2, 4, 2]			
Generation 9910: Best = 3.352245e+03	FES = 991100	$\mu_F = 0.746$	$\mu_{CR} = 0.203$	
Diversity = 2.0924e-02	k_t = [1, 1, 1, 1]			
Generation 9920: Best = 3.352225e+03	FES = 992100	$\mu_F = 0.752$	$\mu_{CR} = 0.206$	
Diversity = 2.3596e-02	k_t = [4, 2, 3, 2]			
Generation 9930: Best = 3.352202e+03	FES = 993100	$\mu_F = 0.755$	$\mu_{CR} = 0.219$	
Diversity = 2.6344e-02	k_t = [1, 1, 1, 1]			
Generation 9940: Best = 3.352188e+03	FES = 994100	$\mu_F = 0.754$	$\mu_{CR} = 0.224$	

```

Diversity = 2.7932e-02 k_t = [4, 4, 3, 2]
Generation 9950: Best = 3.352176e+03 FES = 995100  $\mu_F$  = 0.758  $\mu_{CR}$  = 0.225
Diversity = 2.7043e-02 k_t = [1, 1, 1, 1]
Generation 9960: Best = 3.352166e+03 FES = 996100  $\mu_F$  = 0.750  $\mu_{CR}$  = 0.216
Diversity = 2.5954e-02 k_t = [4, 3, 4, 3]
Generation 9970: Best = 3.352147e+03 FES = 997100  $\mu_F$  = 0.743  $\mu_{CR}$  = 0.216
Diversity = 2.6872e-02 k_t = [1, 1, 1, 1]
Generation 9980: Best = 3.352116e+03 FES = 998100  $\mu_F$  = 0.738  $\mu_{CR}$  = 0.221
Diversity = 2.7155e-02 k_t = [3, 2, 4, 3]
Generation 9990: Best = 3.352105e+03 FES = 999100  $\mu_F$  = 0.749  $\mu_{CR}$  = 0.225
Diversity = 2.8935e-02 k_t = [1, 1, 1, 1]

```

Optimization complete.

```

Best Fitness          = 3.352089e+03
Total Function Evals  = 1000000
Total Successful Updates = 421473

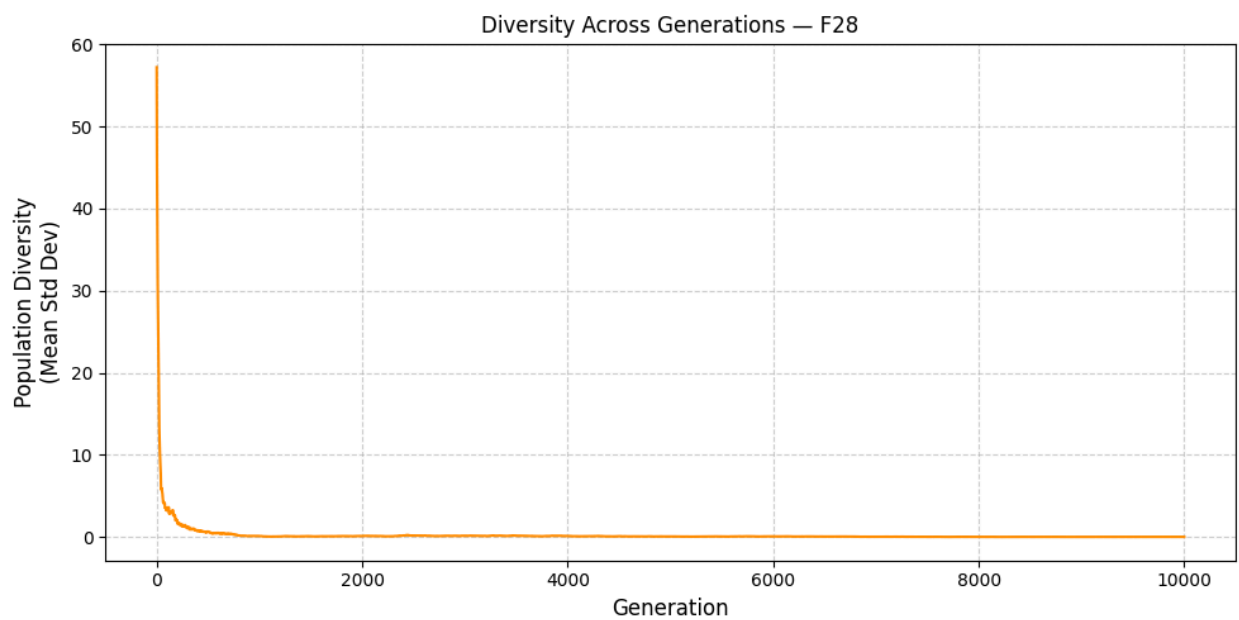
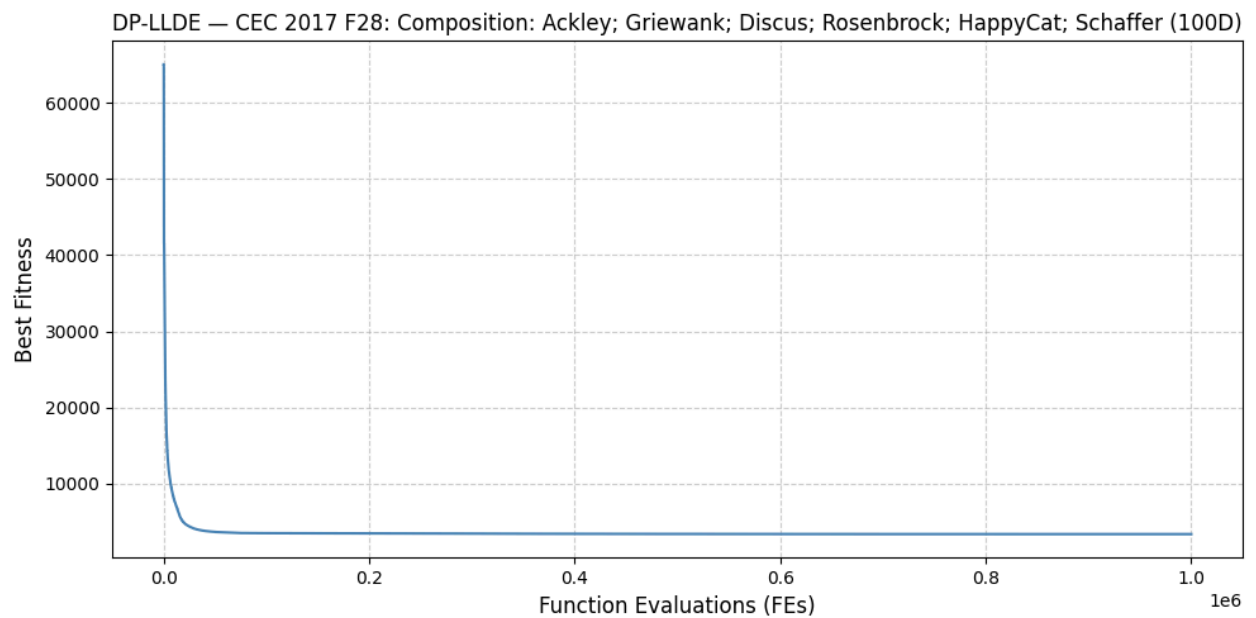
```

```

=====
=
Results:
  Best Fitness          : 3.352089e+03
  Global Optimum        : 2.800000e+03
  Error (|best-optimum|) : 5.520893e+02
  Best Solution (5 dim) : [-27.80321243 -19.18985798 -16.09512084  58.56948637
-24.50913418]
  D^(0) (init. diversity): 5.7220e+01
=====
=

```

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results/all_results_checkpoint.pkl



Graph saved as convergence_F28.png

✓ Plot saved to: /content/drive/My Drive/thesis_results/plots/convergence_F28.png

```
In [17]: # =====
# F29 — Composition Function 9 (F15; F16; F17)
# =====
run_and_plot('F29', make_wrapper(composition.f29), 2900.0, 'Composition: F15;
```

```

=====
=
DP-LLDE on CEC 2017 F29 – Single Run
=====
=
Function      : Composition: F15; F16; F17 (100D)
Bounds       : [-100, 100]
Global optimum : 2900.0
Initial D^(0) = 5.7220e+01 | Diversity threshold = 5.7220e+00
Generation 0: Best Fitness = 1.114421e+07
Generation 10: Best = 3.415933e+04 FES = 1100  $\mu_F$  = 0.564  $\mu_{CR}$  = 0.351
Diversity = 3.6822e+01 k_t = [1, 1, 1, 1]
Generation 20: Best = 2.064804e+04 FES = 2100  $\mu_F$  = 0.607  $\mu_{CR}$  = 0.343
Diversity = 2.5840e+01 k_t = [2, 3, 2, 3]
Generation 30: Best = 1.379743e+04 FES = 3100  $\mu_F$  = 0.631  $\mu_{CR}$  = 0.355
Diversity = 2.1814e+01 k_t = [1, 1, 1, 1]
Generation 40: Best = 1.263778e+04 FES = 4100  $\mu_F$  = 0.671  $\mu_{CR}$  = 0.350
Diversity = 2.0017e+01 k_t = [3, 2, 2, 2]
Generation 50: Best = 1.100340e+04 FES = 5100  $\mu_F$  = 0.685  $\mu_{CR}$  = 0.343
Diversity = 1.9616e+01 k_t = [1, 1, 1, 1]
Generation 60: Best = 1.094659e+04 FES = 6100  $\mu_F$  = 0.692  $\mu_{CR}$  = 0.286
Diversity = 1.9557e+01 k_t = [3, 4, 4, 3]
Generation 70: Best = 1.077689e+04 FES = 7100  $\mu_F$  = 0.649  $\mu_{CR}$  = 0.270
Diversity = 1.8196e+01 k_t = [1, 1, 1, 1]
Generation 80: Best = 1.052950e+04 FES = 8100  $\mu_F$  = 0.648  $\mu_{CR}$  = 0.235
Diversity = 1.7592e+01 k_t = [4, 3, 3, 3]
Generation 90: Best = 1.031513e+04 FES = 9100  $\mu_F$  = 0.687  $\mu_{CR}$  = 0.186
Diversity = 1.7914e+01 k_t = [1, 1, 1, 1]
Generation 100: Best = 1.017003e+04 FES = 10100  $\mu_F$  = 0.682  $\mu_{CR}$  = 0.133
Diversity = 1.7746e+01 k_t = [4, 4, 4, 4]
Generation 110: Best = 1.017003e+04 FES = 11100  $\mu_F$  = 0.708  $\mu_{CR}$  = 0.099
Diversity = 1.7535e+01 k_t = [1, 1, 1, 1]
Generation 120: Best = 1.000383e+04 FES = 12100  $\mu_F$  = 0.709  $\mu_{CR}$  = 0.078
Diversity = 1.7516e+01 k_t = [3, 2, 4, 2]
Generation 130: Best = 9.899939e+03 FES = 13100  $\mu_F$  = 0.731  $\mu_{CR}$  = 0.063
Diversity = 1.7311e+01 k_t = [1, 1, 1, 1]
Generation 140: Best = 9.631534e+03 FES = 14100  $\mu_F$  = 0.739  $\mu_{CR}$  = 0.064
Diversity = 1.7223e+01 k_t = [3, 3, 4, 2]
Generation 150: Best = 9.155603e+03 FES = 15100  $\mu_F$  = 0.756  $\mu_{CR}$  = 0.060
Diversity = 1.7202e+01 k_t = [1, 1, 1, 1]
Generation 160: Best = 9.090317e+03 FES = 16100  $\mu_F$  = 0.768  $\mu_{CR}$  = 0.054
Diversity = 1.7144e+01 k_t = [4, 3, 4, 2]
Generation 170: Best = 8.968738e+03 FES = 17100  $\mu_F$  = 0.760  $\mu_{CR}$  = 0.054
Diversity = 1.7089e+01 k_t = [1, 1, 1, 1]
Generation 180: Best = 8.641990e+03 FES = 18100  $\mu_F$  = 0.758  $\mu_{CR}$  = 0.052
Diversity = 1.7008e+01 k_t = [3, 4, 4, 2]
Generation 190: Best = 8.613761e+03 FES = 19100  $\mu_F$  = 0.772  $\mu_{CR}$  = 0.047
Diversity = 1.6995e+01 k_t = [1, 1, 1, 1]
Generation 200: Best = 8.366947e+03 FES = 20100  $\mu_F$  = 0.762  $\mu_{CR}$  = 0.046
Diversity = 1.6982e+01 k_t = [4, 4, 2, 3]
Generation 210: Best = 8.333606e+03 FES = 21100  $\mu_F$  = 0.785  $\mu_{CR}$  = 0.046
Diversity = 1.6932e+01 k_t = [1, 1, 1, 1]
Generation 220: Best = 8.270361e+03 FES = 22100  $\mu_F$  = 0.758  $\mu_{CR}$  = 0.048
Diversity = 1.6682e+01 k_t = [3, 3, 2, 4]

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Generation 230: Best = 8.270361e+03	FES = 23100	$\mu_F = 0.729$	$\mu_{CR} = 0.056$
Diversity = 1.6539e+01	k_t = [1, 1, 1, 1]		
Generation 240: Best = 8.270361e+03	FES = 24100	$\mu_F = 0.737$	$\mu_{CR} = 0.057$
Diversity = 1.6653e+01	k_t = [3, 3, 2, 3]		
Generation 250: Best = 8.249268e+03	FES = 25100	$\mu_F = 0.761$	$\mu_{CR} = 0.051$
Diversity = 1.6708e+01	k_t = [1, 1, 1, 1]		
Generation 260: Best = 8.232752e+03	FES = 26100	$\mu_F = 0.800$	$\mu_{CR} = 0.048$
Diversity = 1.6563e+01	k_t = [2, 4, 3, 4]		
Generation 270: Best = 8.232752e+03	FES = 27100	$\mu_F = 0.794$	$\mu_{CR} = 0.040$
Diversity = 1.6574e+01	k_t = [1, 1, 1, 1]		
Generation 280: Best = 8.232752e+03	FES = 28100	$\mu_F = 0.827$	$\mu_{CR} = 0.037$
Diversity = 1.6548e+01	k_t = [2, 4, 3, 2]		
Generation 290: Best = 8.232752e+03	FES = 29100	$\mu_F = 0.829$	$\mu_{CR} = 0.044$
Diversity = 1.6588e+01	k_t = [1, 1, 1, 1]		
Generation 300: Best = 8.232752e+03	FES = 30100	$\mu_F = 0.841$	$\mu_{CR} = 0.045$
Diversity = 1.6728e+01	k_t = [2, 4, 4, 3]		
Generation 310: Best = 8.212866e+03	FES = 31100	$\mu_F = 0.851$	$\mu_{CR} = 0.048$
Diversity = 1.6760e+01	k_t = [1, 1, 1, 1]		
Generation 320: Best = 8.126872e+03	FES = 32100	$\mu_F = 0.850$	$\mu_{CR} = 0.039$
Diversity = 1.6786e+01	k_t = [2, 4, 4, 3]		
Generation 330: Best = 8.115680e+03	FES = 33100	$\mu_F = 0.822$	$\mu_{CR} = 0.038$
Diversity = 1.6780e+01	k_t = [1, 1, 1, 1]		
Generation 340: Best = 8.071630e+03	FES = 34100	$\mu_F = 0.800$	$\mu_{CR} = 0.040$
Diversity = 1.6810e+01	k_t = [3, 3, 4, 2]		
Generation 350: Best = 7.959840e+03	FES = 35100	$\mu_F = 0.826$	$\mu_{CR} = 0.039$
Diversity = 1.6761e+01	k_t = [1, 1, 1, 1]		
Generation 360: Best = 7.712706e+03	FES = 36100	$\mu_F = 0.818$	$\mu_{CR} = 0.040$
Diversity = 1.6866e+01	k_t = [3, 2, 2, 2]		
Generation 370: Best = 7.712706e+03	FES = 37100	$\mu_F = 0.771$	$\mu_{CR} = 0.040$
Diversity = 1.6848e+01	k_t = [1, 1, 1, 1]		
Generation 380: Best = 7.712706e+03	FES = 38100	$\mu_F = 0.750$	$\mu_{CR} = 0.047$
Diversity = 1.6674e+01	k_t = [4, 4, 3, 4]		
Generation 390: Best = 7.637773e+03	FES = 39100	$\mu_F = 0.807$	$\mu_{CR} = 0.038$
Diversity = 1.6682e+01	k_t = [1, 1, 1, 1]		
Generation 400: Best = 7.637773e+03	FES = 40100	$\mu_F = 0.777$	$\mu_{CR} = 0.037$
Diversity = 1.6649e+01	k_t = [2, 2, 2, 3]		
Generation 410: Best = 7.621092e+03	FES = 41100	$\mu_F = 0.755$	$\mu_{CR} = 0.038$
Diversity = 1.6733e+01	k_t = [1, 1, 1, 1]		
Generation 420: Best = 7.621092e+03	FES = 42100	$\mu_F = 0.771$	$\mu_{CR} = 0.042$
Diversity = 1.6726e+01	k_t = [2, 4, 2, 3]		
Generation 430: Best = 7.521104e+03	FES = 43100	$\mu_F = 0.768$	$\mu_{CR} = 0.037$
Diversity = 1.6773e+01	k_t = [1, 1, 1, 1]		
Generation 440: Best = 7.521104e+03	FES = 44100	$\mu_F = 0.766$	$\mu_{CR} = 0.045$
Diversity = 1.6657e+01	k_t = [4, 2, 4, 4]		
Generation 450: Best = 7.521104e+03	FES = 45100	$\mu_F = 0.793$	$\mu_{CR} = 0.042$
Diversity = 1.6632e+01	k_t = [1, 1, 1, 1]		
Generation 460: Best = 7.521104e+03	FES = 46100	$\mu_F = 0.779$	$\mu_{CR} = 0.039$
Diversity = 1.6652e+01	k_t = [2, 3, 3, 3]		
Generation 470: Best = 7.511230e+03	FES = 47100	$\mu_F = 0.804$	$\mu_{CR} = 0.039$
Diversity = 1.6607e+01	k_t = [1, 1, 1, 1]		
Generation 480: Best = 7.508726e+03	FES = 48100	$\mu_F = 0.812$	$\mu_{CR} = 0.036$
Diversity = 1.6706e+01	k_t = [4, 4, 3, 4]		
Generation 490: Best = 7.501628e+03	FES = 49100	$\mu_F = 0.778$	$\mu_{CR} = 0.043$
Diversity = 1.6708e+01	k_t = [1, 1, 1, 1]		

Generation 500: Best = 7.430079e+03	FES = 50100	$\mu_F = 0.752$	$\mu_{CR} = 0.034$
Diversity = 1.6728e+01	$k_t = [2, 2, 2, 2]$		
Generation 510: Best = 7.430079e+03	FES = 51100	$\mu_F = 0.730$	$\mu_{CR} = 0.037$
Diversity = 1.6709e+01	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 7.370776e+03	FES = 52100	$\mu_F = 0.715$	$\mu_{CR} = 0.050$
Diversity = 1.6575e+01	$k_t = [2, 3, 3, 4]$		
Generation 530: Best = 7.219624e+03	FES = 53100	$\mu_F = 0.688$	$\mu_{CR} = 0.051$
Diversity = 1.6643e+01	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 7.210886e+03	FES = 54100	$\mu_F = 0.700$	$\mu_{CR} = 0.041$
Diversity = 1.6527e+01	$k_t = [3, 2, 3, 3]$		
Generation 550: Best = 7.210886e+03	FES = 55100	$\mu_F = 0.755$	$\mu_{CR} = 0.038$
Diversity = 1.6523e+01	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 7.042295e+03	FES = 56100	$\mu_F = 0.774$	$\mu_{CR} = 0.036$
Diversity = 1.6503e+01	$k_t = [3, 4, 4, 4]$		
Generation 570: Best = 7.029462e+03	FES = 57100	$\mu_F = 0.794$	$\mu_{CR} = 0.037$
Diversity = 1.6502e+01	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 6.874958e+03	FES = 58100	$\mu_F = 0.797$	$\mu_{CR} = 0.044$
Diversity = 1.6498e+01	$k_t = [4, 4, 2, 3]$		
Generation 590: Best = 6.874958e+03	FES = 59100	$\mu_F = 0.793$	$\mu_{CR} = 0.033$
Diversity = 1.6522e+01	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 6.801312e+03	FES = 60100	$\mu_F = 0.741$	$\mu_{CR} = 0.039$
Diversity = 1.6501e+01	$k_t = [3, 3, 3, 2]$		
Generation 610: Best = 6.801312e+03	FES = 61100	$\mu_F = 0.723$	$\mu_{CR} = 0.042$
Diversity = 1.6435e+01	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 6.801312e+03	FES = 62100	$\mu_F = 0.711$	$\mu_{CR} = 0.045$
Diversity = 1.6403e+01	$k_t = [4, 4, 4, 2]$		
Generation 630: Best = 6.801312e+03	FES = 63100	$\mu_F = 0.699$	$\mu_{CR} = 0.043$
Diversity = 1.6366e+01	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 6.801312e+03	FES = 64100	$\mu_F = 0.755$	$\mu_{CR} = 0.040$
Diversity = 1.6357e+01	$k_t = [4, 3, 3, 4]$		
Generation 650: Best = 6.801312e+03	FES = 65100	$\mu_F = 0.760$	$\mu_{CR} = 0.042$
Diversity = 1.6347e+01	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 6.801312e+03	FES = 66100	$\mu_F = 0.792$	$\mu_{CR} = 0.039$
Diversity = 1.6250e+01	$k_t = [3, 2, 3, 3]$		
Generation 670: Best = 6.801312e+03	FES = 67100	$\mu_F = 0.817$	$\mu_{CR} = 0.041$
Diversity = 1.6240e+01	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 6.801312e+03	FES = 68100	$\mu_F = 0.822$	$\mu_{CR} = 0.041$
Diversity = 1.6262e+01	$k_t = [2, 2, 4, 2]$		
Generation 690: Best = 6.775948e+03	FES = 69100	$\mu_F = 0.834$	$\mu_{CR} = 0.040$
Diversity = 1.6288e+01	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 6.735573e+03	FES = 70100	$\mu_F = 0.798$	$\mu_{CR} = 0.040$
Diversity = 1.6305e+01	$k_t = [3, 2, 2, 3]$		
Generation 710: Best = 6.735573e+03	FES = 71100	$\mu_F = 0.801$	$\mu_{CR} = 0.035$
Diversity = 1.6303e+01	$k_t = [1, 1, 1, 1]$		
Generation 720: Best = 6.732816e+03	FES = 72100	$\mu_F = 0.791$	$\mu_{CR} = 0.034$
Diversity = 1.6314e+01	$k_t = [4, 3, 3, 2]$		
Generation 730: Best = 6.732752e+03	FES = 73100	$\mu_F = 0.830$	$\mu_{CR} = 0.036$
Diversity = 1.6273e+01	$k_t = [1, 1, 1, 1]$		
Generation 740: Best = 6.732752e+03	FES = 74100	$\mu_F = 0.810$	$\mu_{CR} = 0.035$
Diversity = 1.6294e+01	$k_t = [3, 4, 4, 2]$		
Generation 750: Best = 6.732752e+03	FES = 75100	$\mu_F = 0.810$	$\mu_{CR} = 0.034$
Diversity = 1.6358e+01	$k_t = [1, 1, 1, 1]$		
Generation 760: Best = 6.699505e+03	FES = 76100	$\mu_F = 0.810$	$\mu_{CR} = 0.034$
Diversity = 1.6333e+01	$k_t = [4, 4, 4, 3]$		

Generation 770: Best = 6.699505e+03	FES = 77100	$\mu_F = 0.796$	$\mu_{CR} = 0.028$
Diversity = 1.6308e+01	k_t = [1, 1, 1, 1]		
Generation 780: Best = 6.699505e+03	FES = 78100	$\mu_F = 0.827$	$\mu_{CR} = 0.038$
Diversity = 1.6155e+01	k_t = [3, 3, 2, 4]		
Generation 790: Best = 6.699505e+03	FES = 79100	$\mu_F = 0.862$	$\mu_{CR} = 0.044$
Diversity = 1.6139e+01	k_t = [1, 1, 1, 1]		
Generation 800: Best = 6.699505e+03	FES = 80100	$\mu_F = 0.816$	$\mu_{CR} = 0.050$
Diversity = 1.6076e+01	k_t = [2, 3, 2, 3]		
Generation 810: Best = 6.651580e+03	FES = 81100	$\mu_F = 0.818$	$\mu_{CR} = 0.038$
Diversity = 1.5994e+01	k_t = [1, 1, 1, 1]		
Generation 820: Best = 6.651580e+03	FES = 82100	$\mu_F = 0.792$	$\mu_{CR} = 0.028$
Diversity = 1.5937e+01	k_t = [3, 3, 4, 3]		
Generation 830: Best = 6.618420e+03	FES = 83100	$\mu_F = 0.775$	$\mu_{CR} = 0.028$
Diversity = 1.5957e+01	k_t = [1, 1, 1, 1]		
Generation 840: Best = 6.513394e+03	FES = 84100	$\mu_F = 0.716$	$\mu_{CR} = 0.033$
Diversity = 1.6012e+01	k_t = [3, 2, 3, 3]		
Generation 850: Best = 6.513394e+03	FES = 85100	$\mu_F = 0.749$	$\mu_{CR} = 0.029$
Diversity = 1.6035e+01	k_t = [1, 1, 1, 1]		
Generation 860: Best = 6.513394e+03	FES = 86100	$\mu_F = 0.792$	$\mu_{CR} = 0.042$
Diversity = 1.6083e+01	k_t = [3, 3, 3, 2]		
Generation 870: Best = 6.513018e+03	FES = 87100	$\mu_F = 0.790$	$\mu_{CR} = 0.032$
Diversity = 1.6089e+01	k_t = [1, 1, 1, 1]		
Generation 880: Best = 6.512086e+03	FES = 88100	$\mu_F = 0.778$	$\mu_{CR} = 0.045$
Diversity = 1.6108e+01	k_t = [3, 2, 4, 3]		
Generation 890: Best = 6.512086e+03	FES = 89100	$\mu_F = 0.782$	$\mu_{CR} = 0.031$
Diversity = 1.6021e+01	k_t = [1, 1, 1, 1]		
Generation 900: Best = 6.509189e+03	FES = 90100	$\mu_F = 0.758$	$\mu_{CR} = 0.031$
Diversity = 1.6007e+01	k_t = [4, 3, 3, 2]		
Generation 910: Best = 6.509189e+03	FES = 91100	$\mu_F = 0.775$	$\mu_{CR} = 0.028$
Diversity = 1.5940e+01	k_t = [1, 1, 1, 1]		
Generation 920: Best = 6.509189e+03	FES = 92100	$\mu_F = 0.721$	$\mu_{CR} = 0.023$
Diversity = 1.5983e+01	k_t = [4, 4, 2, 2]		
Generation 930: Best = 6.497728e+03	FES = 93100	$\mu_F = 0.715$	$\mu_{CR} = 0.025$
Diversity = 1.5847e+01	k_t = [1, 1, 1, 1]		
Generation 940: Best = 6.469148e+03	FES = 94100	$\mu_F = 0.788$	$\mu_{CR} = 0.030$
Diversity = 1.5749e+01	k_t = [2, 4, 2, 4]		
Generation 950: Best = 6.408407e+03	FES = 95100	$\mu_F = 0.817$	$\mu_{CR} = 0.033$
Diversity = 1.5641e+01	k_t = [1, 1, 1, 1]		
Generation 960: Best = 6.408407e+03	FES = 96100	$\mu_F = 0.841$	$\mu_{CR} = 0.028$
Diversity = 1.5676e+01	k_t = [2, 3, 3, 3]		
Generation 970: Best = 6.398637e+03	FES = 97100	$\mu_F = 0.820$	$\mu_{CR} = 0.031$
Diversity = 1.5647e+01	k_t = [1, 1, 1, 1]		
Generation 980: Best = 6.398637e+03	FES = 98100	$\mu_F = 0.820$	$\mu_{CR} = 0.029$
Diversity = 1.5648e+01	k_t = [4, 2, 3, 2]		
Generation 990: Best = 6.398637e+03	FES = 99100	$\mu_F = 0.842$	$\mu_{CR} = 0.032$
Diversity = 1.5628e+01	k_t = [1, 1, 1, 1]		
Generation 1000: Best = 6.398637e+03	FES = 100100	$\mu_F = 0.887$	$\mu_{CR} = 0.036$
Diversity = 1.5424e+01	k_t = [2, 4, 2, 4]		
Generation 1010: Best = 6.397276e+03	FES = 101100	$\mu_F = 0.776$	$\mu_{CR} = 0.043$
Diversity = 1.5419e+01	k_t = [1, 1, 1, 1]		
Generation 1020: Best = 6.396775e+03	FES = 102100	$\mu_F = 0.796$	$\mu_{CR} = 0.038$
Diversity = 1.5318e+01	k_t = [4, 3, 4, 4]		
Generation 1030: Best = 6.394444e+03	FES = 103100	$\mu_F = 0.778$	$\mu_{CR} = 0.029$
Diversity = 1.5306e+01	k_t = [1, 1, 1, 1]		

Generation 1040: Best = 6.390631e+03 FES = 104100 μ_F = 0.819 μ_{CR} = 0.043
 Diversity = 1.5298e+01 k_t = [3, 2, 4, 2]
 Generation 1050: Best = 6.390631e+03 FES = 105100 μ_F = 0.769 μ_{CR} = 0.038
 Diversity = 1.5251e+01 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 6.389323e+03 FES = 106100 μ_F = 0.787 μ_{CR} = 0.038
 Diversity = 1.5224e+01 k_t = [4, 2, 4, 2]
 Generation 1070: Best = 6.389323e+03 FES = 107100 μ_F = 0.812 μ_{CR} = 0.038
 Diversity = 1.5278e+01 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 6.366365e+03 FES = 108100 μ_F = 0.813 μ_{CR} = 0.039
 Diversity = 1.5326e+01 k_t = [3, 4, 3, 3]
 Generation 1090: Best = 6.366365e+03 FES = 109100 μ_F = 0.855 μ_{CR} = 0.040
 Diversity = 1.5336e+01 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 6.340817e+03 FES = 110100 μ_F = 0.792 μ_{CR} = 0.033
 Diversity = 1.5349e+01 k_t = [2, 4, 2, 2]
 Generation 1110: Best = 6.314839e+03 FES = 111100 μ_F = 0.780 μ_{CR} = 0.027
 Diversity = 1.5263e+01 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 6.314839e+03 FES = 112100 μ_F = 0.719 μ_{CR} = 0.035
 Diversity = 1.5239e+01 k_t = [3, 2, 4, 3]
 Generation 1130: Best = 6.314839e+03 FES = 113100 μ_F = 0.722 μ_{CR} = 0.043
 Diversity = 1.5182e+01 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 6.314839e+03 FES = 114100 μ_F = 0.772 μ_{CR} = 0.039
 Diversity = 1.5123e+01 k_t = [4, 4, 3, 4]
 Generation 1150: Best = 6.314839e+03 FES = 115100 μ_F = 0.777 μ_{CR} = 0.038
 Diversity = 1.5113e+01 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 6.314839e+03 FES = 116100 μ_F = 0.767 μ_{CR} = 0.040
 Diversity = 1.4999e+01 k_t = [4, 3, 3, 4]
 Generation 1170: Best = 6.314839e+03 FES = 117100 μ_F = 0.779 μ_{CR} = 0.033
 Diversity = 1.4950e+01 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 6.314839e+03 FES = 118100 μ_F = 0.824 μ_{CR} = 0.027
 Diversity = 1.4823e+01 k_t = [4, 4, 4, 2]
 Generation 1190: Best = 6.312450e+03 FES = 119100 μ_F = 0.842 μ_{CR} = 0.033
 Diversity = 1.4792e+01 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 6.312450e+03 FES = 120100 μ_F = 0.862 μ_{CR} = 0.030
 Diversity = 1.4462e+01 k_t = [3, 4, 3, 4]
 Generation 1210: Best = 6.295679e+03 FES = 121100 μ_F = 0.856 μ_{CR} = 0.029
 Diversity = 1.4305e+01 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 6.295679e+03 FES = 122100 μ_F = 0.849 μ_{CR} = 0.034
 Diversity = 1.4102e+01 k_t = [3, 4, 4, 4]
 Generation 1230: Best = 6.253870e+03 FES = 123100 μ_F = 0.870 μ_{CR} = 0.040
 Diversity = 1.4065e+01 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 6.253870e+03 FES = 124100 μ_F = 0.851 μ_{CR} = 0.039
 Diversity = 1.3963e+01 k_t = [2, 4, 2, 3]
 Generation 1250: Best = 6.248569e+03 FES = 125100 μ_F = 0.862 μ_{CR} = 0.030
 Diversity = 1.3889e+01 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 6.248569e+03 FES = 126100 μ_F = 0.831 μ_{CR} = 0.032
 Diversity = 1.3603e+01 k_t = [3, 2, 4, 4]
 Generation 1270: Best = 6.218609e+03 FES = 127100 μ_F = 0.831 μ_{CR} = 0.032
 Diversity = 1.3578e+01 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 6.218609e+03 FES = 128100 μ_F = 0.822 μ_{CR} = 0.033
 Diversity = 1.3597e+01 k_t = [3, 4, 2, 2]
 Generation 1290: Best = 6.218609e+03 FES = 129100 μ_F = 0.834 μ_{CR} = 0.040
 Diversity = 1.3581e+01 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 6.206680e+03 FES = 130100 μ_F = 0.815 μ_{CR} = 0.040
 Diversity = 1.3559e+01 k_t = [3, 4, 3, 2]

Generation 1310: Best = 6.198289e+03 FES = 131100 μ_F = 0.846 μ_{CR} = 0.040
 Diversity = 1.3427e+01 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 6.198289e+03 FES = 132100 μ_F = 0.828 μ_{CR} = 0.037
 Diversity = 1.3424e+01 k_t = [2, 3, 2, 2]
 Generation 1330: Best = 6.195841e+03 FES = 133100 μ_F = 0.825 μ_{CR} = 0.040
 Diversity = 1.3421e+01 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 6.195841e+03 FES = 134100 μ_F = 0.846 μ_{CR} = 0.040
 Diversity = 1.3366e+01 k_t = [4, 3, 3, 2]
 Generation 1350: Best = 6.194961e+03 FES = 135100 μ_F = 0.832 μ_{CR} = 0.045
 Diversity = 1.3348e+01 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 6.194961e+03 FES = 136100 μ_F = 0.862 μ_{CR} = 0.039
 Diversity = 1.3196e+01 k_t = [3, 3, 3, 3]
 Generation 1370: Best = 6.191760e+03 FES = 137100 μ_F = 0.836 μ_{CR} = 0.044
 Diversity = 1.3221e+01 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 6.187127e+03 FES = 138100 μ_F = 0.854 μ_{CR} = 0.047
 Diversity = 1.3238e+01 k_t = [2, 3, 4, 2]
 Generation 1390: Best = 6.181282e+03 FES = 139100 μ_F = 0.875 μ_{CR} = 0.043
 Diversity = 1.3340e+01 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 6.181282e+03 FES = 140100 μ_F = 0.889 μ_{CR} = 0.043
 Diversity = 1.3130e+01 k_t = [3, 2, 4, 2]
 Generation 1410: Best = 6.056191e+03 FES = 141100 μ_F = 0.849 μ_{CR} = 0.058
 Diversity = 1.3036e+01 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 6.056191e+03 FES = 142100 μ_F = 0.816 μ_{CR} = 0.054
 Diversity = 1.2879e+01 k_t = [4, 4, 2, 3]
 Generation 1430: Best = 6.048057e+03 FES = 143100 μ_F = 0.811 μ_{CR} = 0.048
 Diversity = 1.2829e+01 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 6.048057e+03 FES = 144100 μ_F = 0.821 μ_{CR} = 0.075
 Diversity = 1.2696e+01 k_t = [2, 2, 3, 2]
 Generation 1450: Best = 6.021893e+03 FES = 145100 μ_F = 0.819 μ_{CR} = 0.056
 Diversity = 1.2495e+01 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 6.021421e+03 FES = 146100 μ_F = 0.825 μ_{CR} = 0.044
 Diversity = 1.2415e+01 k_t = [4, 2, 3, 2]
 Generation 1470: Best = 5.977285e+03 FES = 147100 μ_F = 0.827 μ_{CR} = 0.039
 Diversity = 1.2445e+01 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 5.977285e+03 FES = 148100 μ_F = 0.839 μ_{CR} = 0.058
 Diversity = 1.2122e+01 k_t = [4, 3, 4, 4]
 Generation 1490: Best = 5.976381e+03 FES = 149100 μ_F = 0.839 μ_{CR} = 0.062
 Diversity = 1.1959e+01 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 5.976381e+03 FES = 150100 μ_F = 0.802 μ_{CR} = 0.057
 Diversity = 1.1961e+01 k_t = [4, 4, 2, 2]
 Generation 1510: Best = 5.976381e+03 FES = 151100 μ_F = 0.788 μ_{CR} = 0.065
 Diversity = 1.1650e+01 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 5.976381e+03 FES = 152100 μ_F = 0.794 μ_{CR} = 0.069
 Diversity = 1.1037e+01 k_t = [3, 3, 4, 4]
 Generation 1530: Best = 5.961372e+03 FES = 153100 μ_F = 0.792 μ_{CR} = 0.066
 Diversity = 1.0830e+01 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 5.920272e+03 FES = 154100 μ_F = 0.803 μ_{CR} = 0.070
 Diversity = 1.0549e+01 k_t = [4, 2, 2, 2]
 Generation 1550: Best = 5.898705e+03 FES = 155100 μ_F = 0.790 μ_{CR} = 0.072
 Diversity = 1.0351e+01 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 5.849506e+03 FES = 156100 μ_F = 0.776 μ_{CR} = 0.080
 Diversity = 9.8521e+00 k_t = [4, 2, 3, 3]
 Generation 1570: Best = 5.848177e+03 FES = 157100 μ_F = 0.774 μ_{CR} = 0.079
 Diversity = 9.3958e+00 k_t = [1, 1, 1, 1]

Generation 1580: Best = 5.799386e+03 FES = 158100 μ_F = 0.767 μ_{CR} = 0.074
 Diversity = 8.7978e+00 k_t = [3, 2, 4, 2]
 Generation 1590: Best = 5.794189e+03 FES = 159100 μ_F = 0.760 μ_{CR} = 0.072
 Diversity = 8.1825e+00 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 5.755824e+03 FES = 160100 μ_F = 0.754 μ_{CR} = 0.077
 Diversity = 7.8654e+00 k_t = [2, 4, 3, 3]
 Generation 1610: Best = 5.733183e+03 FES = 161100 μ_F = 0.759 μ_{CR} = 0.078
 Diversity = 7.3192e+00 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 5.701347e+03 FES = 162100 μ_F = 0.746 μ_{CR} = 0.081
 Diversity = 7.0007e+00 k_t = [2, 4, 3, 3]
 Generation 1630: Best = 5.701347e+03 FES = 163100 μ_F = 0.734 μ_{CR} = 0.077
 Diversity = 6.8083e+00 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 5.688975e+03 FES = 164100 μ_F = 0.750 μ_{CR} = 0.068
 Diversity = 6.3811e+00 k_t = [2, 2, 4, 2]
 Generation 1650: Best = 5.670521e+03 FES = 165100 μ_F = 0.741 μ_{CR} = 0.069
 Diversity = 6.1546e+00 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 5.650006e+03 FES = 166100 μ_F = 0.713 μ_{CR} = 0.075
 Diversity = 6.0137e+00 k_t = [2, 3, 2, 4]
 Generation 1670: Best = 5.628025e+03 FES = 167100 μ_F = 0.708 μ_{CR} = 0.066
 Diversity = 5.9877e+00 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 5.620231e+03 FES = 168100 μ_F = 0.721 μ_{CR} = 0.066
 Diversity = 5.8797e+00 k_t = [3, 2, 4, 4]
 Generation 1690: Best = 5.615297e+03 FES = 169100 μ_F = 0.728 μ_{CR} = 0.060
 Diversity = 5.7569e+00 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 5.608063e+03 FES = 170100 μ_F = 0.746 μ_{CR} = 0.065
 Diversity = 5.5361e+00 k_t = [3, 3, 2, 4]
 Generation 1710: Best = 5.592174e+03 FES = 171100 μ_F = 0.726 μ_{CR} = 0.059
 Diversity = 5.3978e+00 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 5.580789e+03 FES = 172100 μ_F = 0.719 μ_{CR} = 0.058
 Diversity = 5.2810e+00 k_t = [4, 4, 3, 4]
 Generation 1730: Best = 5.575744e+03 FES = 173100 μ_F = 0.696 μ_{CR} = 0.066
 Diversity = 5.2749e+00 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 5.561612e+03 FES = 174100 μ_F = 0.697 μ_{CR} = 0.063
 Diversity = 5.1933e+00 k_t = [4, 2, 3, 2]
 Generation 1750: Best = 5.561612e+03 FES = 175100 μ_F = 0.704 μ_{CR} = 0.057
 Diversity = 5.1329e+00 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 5.551565e+03 FES = 176100 μ_F = 0.721 μ_{CR} = 0.057
 Diversity = 5.1047e+00 k_t = [4, 2, 3, 2]
 Generation 1770: Best = 5.551093e+03 FES = 177100 μ_F = 0.715 μ_{CR} = 0.053
 Diversity = 5.0351e+00 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 5.539274e+03 FES = 178100 μ_F = 0.694 μ_{CR} = 0.056
 Diversity = 5.0212e+00 k_t = [4, 2, 3, 3]
 Generation 1790: Best = 5.539274e+03 FES = 179100 μ_F = 0.699 μ_{CR} = 0.051
 Diversity = 4.9568e+00 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 5.523441e+03 FES = 180100 μ_F = 0.729 μ_{CR} = 0.059
 Diversity = 4.8362e+00 k_t = [4, 3, 3, 4]
 Generation 1810: Best = 5.498680e+03 FES = 181100 μ_F = 0.720 μ_{CR} = 0.051
 Diversity = 4.7240e+00 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 5.498680e+03 FES = 182100 μ_F = 0.706 μ_{CR} = 0.058
 Diversity = 4.6793e+00 k_t = [3, 3, 3, 4]
 Generation 1830: Best = 5.495937e+03 FES = 183100 μ_F = 0.699 μ_{CR} = 0.058
 Diversity = 4.6234e+00 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 5.495503e+03 FES = 184100 μ_F = 0.722 μ_{CR} = 0.054
 Diversity = 4.5095e+00 k_t = [4, 4, 2, 4]

Generation 1850: Best = 5.473616e+03 FES = 185100 μ_F = 0.725 μ_{CR} = 0.064
 Diversity = 4.4428e+00 k_t = [1, 1, 1, 1]
 Generation 1860: Best = 5.465660e+03 FES = 186100 μ_F = 0.707 μ_{CR} = 0.069
 Diversity = 4.2930e+00 k_t = [2, 4, 4, 4]
 Generation 1870: Best = 5.447047e+03 FES = 187100 μ_F = 0.710 μ_{CR} = 0.065
 Diversity = 4.2105e+00 k_t = [1, 1, 1, 1]
 Generation 1880: Best = 5.447047e+03 FES = 188100 μ_F = 0.734 μ_{CR} = 0.062
 Diversity = 4.0147e+00 k_t = [4, 2, 2, 4]
 Generation 1890: Best = 5.441123e+03 FES = 189100 μ_F = 0.736 μ_{CR} = 0.061
 Diversity = 3.7664e+00 k_t = [1, 1, 1, 1]
 Generation 1900: Best = 5.441123e+03 FES = 190100 μ_F = 0.755 μ_{CR} = 0.063
 Diversity = 3.7384e+00 k_t = [3, 3, 3, 3]
 Generation 1910: Best = 5.441123e+03 FES = 191100 μ_F = 0.752 μ_{CR} = 0.062
 Diversity = 3.6635e+00 k_t = [1, 1, 1, 1]
 Generation 1920: Best = 5.439893e+03 FES = 192100 μ_F = 0.743 μ_{CR} = 0.067
 Diversity = 3.3113e+00 k_t = [4, 3, 3, 2]
 Generation 1930: Best = 5.430901e+03 FES = 193100 μ_F = 0.721 μ_{CR} = 0.068
 Diversity = 3.2150e+00 k_t = [1, 1, 1, 1]
 Generation 1940: Best = 5.406518e+03 FES = 194100 μ_F = 0.731 μ_{CR} = 0.078
 Diversity = 2.9522e+00 k_t = [2, 4, 3, 2]
 Generation 1950: Best = 5.399346e+03 FES = 195100 μ_F = 0.743 μ_{CR} = 0.086
 Diversity = 2.6713e+00 k_t = [1, 1, 1, 1]
 Generation 1960: Best = 5.392158e+03 FES = 196100 μ_F = 0.762 μ_{CR} = 0.092
 Diversity = 2.3611e+00 k_t = [3, 4, 3, 3]
 Generation 1970: Best = 5.373469e+03 FES = 197100 μ_F = 0.762 μ_{CR} = 0.104
 Diversity = 2.1018e+00 k_t = [1, 1, 1, 1]
 Generation 1980: Best = 5.356209e+03 FES = 198100 μ_F = 0.766 μ_{CR} = 0.109
 Diversity = 1.6763e+00 k_t = [4, 4, 2, 4]
 Generation 1990: Best = 5.349208e+03 FES = 199100 μ_F = 0.772 μ_{CR} = 0.109
 Diversity = 1.3900e+00 k_t = [1, 1, 1, 1]
 Generation 2000: Best = 5.332753e+03 FES = 200100 μ_F = 0.761 μ_{CR} = 0.107
 Diversity = 1.3475e+00 k_t = [2, 4, 3, 2]
 Generation 2010: Best = 5.320666e+03 FES = 201100 μ_F = 0.745 μ_{CR} = 0.115
 Diversity = 1.2714e+00 k_t = [1, 1, 1, 1]
 Generation 2020: Best = 5.317050e+03 FES = 202100 μ_F = 0.739 μ_{CR} = 0.110
 Diversity = 1.1559e+00 k_t = [4, 4, 4, 2]
 Generation 2030: Best = 5.310314e+03 FES = 203100 μ_F = 0.750 μ_{CR} = 0.108
 Diversity = 1.0853e+00 k_t = [1, 1, 1, 1]
 Generation 2040: Best = 5.308085e+03 FES = 204100 μ_F = 0.753 μ_{CR} = 0.104
 Diversity = 9.8384e-01 k_t = [2, 2, 2, 3]
 Generation 2050: Best = 5.300023e+03 FES = 205100 μ_F = 0.732 μ_{CR} = 0.107
 Diversity = 8.7300e-01 k_t = [1, 1, 1, 1]
 Generation 2060: Best = 5.296096e+03 FES = 206100 μ_F = 0.720 μ_{CR} = 0.105
 Diversity = 7.6335e-01 k_t = [2, 4, 4, 2]
 Generation 2070: Best = 5.287989e+03 FES = 207100 μ_F = 0.745 μ_{CR} = 0.101
 Diversity = 7.1332e-01 k_t = [1, 1, 1, 1]
 Generation 2080: Best = 5.287238e+03 FES = 208100 μ_F = 0.742 μ_{CR} = 0.100
 Diversity = 6.1896e-01 k_t = [3, 3, 4, 3]
 Generation 2090: Best = 5.282870e+03 FES = 209100 μ_F = 0.744 μ_{CR} = 0.103
 Diversity = 5.6393e-01 k_t = [1, 1, 1, 1]
 Generation 2100: Best = 5.280634e+03 FES = 210100 μ_F = 0.734 μ_{CR} = 0.098
 Diversity = 4.8057e-01 k_t = [4, 3, 4, 4]
 Generation 2110: Best = 5.276655e+03 FES = 211100 μ_F = 0.733 μ_{CR} = 0.101
 Diversity = 4.4883e-01 k_t = [1, 1, 1, 1]

Generation 2120: Best = 5.275232e+03 FES = 212100 μ_F = 0.739 μ_{CR} = 0.098
 Diversity = 4.1284e-01 k_t = [3, 2, 4, 2]
 Generation 2130: Best = 5.272463e+03 FES = 213100 μ_F = 0.736 μ_{CR} = 0.094
 Diversity = 4.0453e-01 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 5.271784e+03 FES = 214100 μ_F = 0.741 μ_{CR} = 0.090
 Diversity = 3.9066e-01 k_t = [4, 4, 2, 2]
 Generation 2150: Best = 5.269958e+03 FES = 215100 μ_F = 0.735 μ_{CR} = 0.094
 Diversity = 3.7561e-01 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 5.266665e+03 FES = 216100 μ_F = 0.725 μ_{CR} = 0.096
 Diversity = 3.5572e-01 k_t = [3, 2, 3, 2]
 Generation 2170: Best = 5.264634e+03 FES = 217100 μ_F = 0.737 μ_{CR} = 0.101
 Diversity = 3.5294e-01 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 5.263522e+03 FES = 218100 μ_F = 0.743 μ_{CR} = 0.096
 Diversity = 3.2176e-01 k_t = [3, 2, 4, 3]
 Generation 2190: Best = 5.261365e+03 FES = 219100 μ_F = 0.735 μ_{CR} = 0.100
 Diversity = 3.0542e-01 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 5.260392e+03 FES = 220100 μ_F = 0.735 μ_{CR} = 0.092
 Diversity = 2.7383e-01 k_t = [2, 3, 4, 4]
 Generation 2210: Best = 5.260036e+03 FES = 221100 μ_F = 0.724 μ_{CR} = 0.086
 Diversity = 2.6129e-01 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 5.257642e+03 FES = 222100 μ_F = 0.714 μ_{CR} = 0.093
 Diversity = 2.4344e-01 k_t = [4, 2, 2, 3]
 Generation 2230: Best = 5.257003e+03 FES = 223100 μ_F = 0.721 μ_{CR} = 0.094
 Diversity = 2.3567e-01 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 5.255818e+03 FES = 224100 μ_F = 0.725 μ_{CR} = 0.096
 Diversity = 2.2801e-01 k_t = [4, 2, 4, 3]
 Generation 2250: Best = 5.254851e+03 FES = 225100 μ_F = 0.734 μ_{CR} = 0.109
 Diversity = 2.2144e-01 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 5.253259e+03 FES = 226100 μ_F = 0.734 μ_{CR} = 0.120
 Diversity = 2.0959e-01 k_t = [4, 4, 2, 4]
 Generation 2270: Best = 5.251871e+03 FES = 227100 μ_F = 0.737 μ_{CR} = 0.134
 Diversity = 1.9369e-01 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 5.250528e+03 FES = 228100 μ_F = 0.746 μ_{CR} = 0.142
 Diversity = 1.7869e-01 k_t = [2, 4, 3, 3]
 Generation 2290: Best = 5.249438e+03 FES = 229100 μ_F = 0.752 μ_{CR} = 0.143
 Diversity = 1.7889e-01 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 5.244524e+03 FES = 230100 μ_F = 0.773 μ_{CR} = 0.155
 Diversity = 2.3887e-01 k_t = [3, 4, 2, 4]
 Generation 2310: Best = 5.236658e+03 FES = 231100 μ_F = 0.789 μ_{CR} = 0.163
 Diversity = 2.8961e-01 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 5.234931e+03 FES = 232100 μ_F = 0.786 μ_{CR} = 0.176
 Diversity = 1.5463e-01 k_t = [2, 3, 2, 4]
 Generation 2330: Best = 5.232940e+03 FES = 233100 μ_F = 0.772 μ_{CR} = 0.176
 Diversity = 1.7113e-01 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 5.225819e+03 FES = 234100 μ_F = 0.779 μ_{CR} = 0.175
 Diversity = 2.3598e-01 k_t = [4, 4, 2, 3]
 Generation 2350: Best = 5.221015e+03 FES = 235100 μ_F = 0.794 μ_{CR} = 0.183
 Diversity = 2.3828e-01 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 5.219948e+03 FES = 236100 μ_F = 0.782 μ_{CR} = 0.186
 Diversity = 1.7986e-01 k_t = [3, 3, 2, 2]
 Generation 2370: Best = 5.218856e+03 FES = 237100 μ_F = 0.746 μ_{CR} = 0.179
 Diversity = 1.5800e-01 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 5.218669e+03 FES = 238100 μ_F = 0.723 μ_{CR} = 0.172
 Diversity = 1.2996e-01 k_t = [4, 2, 3, 4]

Generation 2390: Best = 5.218138e+03 FES = 239100 μ_F = 0.704 μ_{CR} = 0.166
 Diversity = 1.1821e-01 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 5.217600e+03 FES = 240100 μ_F = 0.699 μ_{CR} = 0.169
 Diversity = 9.6303e-02 k_t = [4, 2, 4, 4]
 Generation 2410: Best = 5.216934e+03 FES = 241100 μ_F = 0.715 μ_{CR} = 0.168
 Diversity = 9.0121e-02 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 5.216828e+03 FES = 242100 μ_F = 0.743 μ_{CR} = 0.164
 Diversity = 8.0493e-02 k_t = [2, 4, 2, 4]
 Generation 2430: Best = 5.216669e+03 FES = 243100 μ_F = 0.738 μ_{CR} = 0.161
 Diversity = 7.6565e-02 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 5.216493e+03 FES = 244100 μ_F = 0.721 μ_{CR} = 0.145
 Diversity = 7.1644e-02 k_t = [3, 3, 3, 2]
 Generation 2450: Best = 5.216375e+03 FES = 245100 μ_F = 0.725 μ_{CR} = 0.146
 Diversity = 6.9728e-02 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 5.216139e+03 FES = 246100 μ_F = 0.725 μ_{CR} = 0.146
 Diversity = 6.7080e-02 k_t = [3, 3, 2, 3]
 Generation 2470: Best = 5.215889e+03 FES = 247100 μ_F = 0.715 μ_{CR} = 0.146
 Diversity = 6.2935e-02 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 5.215833e+03 FES = 248100 μ_F = 0.713 μ_{CR} = 0.144
 Diversity = 5.2748e-02 k_t = [4, 3, 4, 4]
 Generation 2490: Best = 5.215722e+03 FES = 249100 μ_F = 0.728 μ_{CR} = 0.141
 Diversity = 5.4075e-02 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 5.215587e+03 FES = 250100 μ_F = 0.732 μ_{CR} = 0.150
 Diversity = 4.9871e-02 k_t = [4, 2, 2, 2]
 Generation 2510: Best = 5.215482e+03 FES = 251100 μ_F = 0.745 μ_{CR} = 0.140
 Diversity = 4.7312e-02 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 5.215427e+03 FES = 252100 μ_F = 0.747 μ_{CR} = 0.141
 Diversity = 4.4740e-02 k_t = [2, 4, 4, 4]
 Generation 2530: Best = 5.215350e+03 FES = 253100 μ_F = 0.754 μ_{CR} = 0.141
 Diversity = 4.4872e-02 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 5.215262e+03 FES = 254100 μ_F = 0.761 μ_{CR} = 0.138
 Diversity = 4.5027e-02 k_t = [2, 2, 3, 3]
 Generation 2550: Best = 5.215134e+03 FES = 255100 μ_F = 0.742 μ_{CR} = 0.131
 Diversity = 4.5746e-02 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 5.215022e+03 FES = 256100 μ_F = 0.753 μ_{CR} = 0.134
 Diversity = 4.4889e-02 k_t = [2, 3, 4, 4]
 Generation 2570: Best = 5.215008e+03 FES = 257100 μ_F = 0.769 μ_{CR} = 0.123
 Diversity = 4.5268e-02 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 5.214970e+03 FES = 258100 μ_F = 0.778 μ_{CR} = 0.123
 Diversity = 4.4419e-02 k_t = [4, 2, 2, 2]
 Generation 2590: Best = 5.214949e+03 FES = 259100 μ_F = 0.781 μ_{CR} = 0.113
 Diversity = 4.5338e-02 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 5.214871e+03 FES = 260100 μ_F = 0.793 μ_{CR} = 0.112
 Diversity = 4.4798e-02 k_t = [2, 4, 4, 4]
 Generation 2610: Best = 5.214871e+03 FES = 261100 μ_F = 0.780 μ_{CR} = 0.105
 Diversity = 4.4706e-02 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 5.214815e+03 FES = 262100 μ_F = 0.758 μ_{CR} = 0.109
 Diversity = 4.4157e-02 k_t = [4, 4, 2, 4]
 Generation 2630: Best = 5.214806e+03 FES = 263100 μ_F = 0.753 μ_{CR} = 0.104
 Diversity = 4.3251e-02 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 5.214679e+03 FES = 264100 μ_F = 0.750 μ_{CR} = 0.107
 Diversity = 4.3383e-02 k_t = [4, 4, 2, 4]
 Generation 2650: Best = 5.214674e+03 FES = 265100 μ_F = 0.747 μ_{CR} = 0.098
 Diversity = 4.3660e-02 k_t = [1, 1, 1, 1]

Generation 2660: Best = 5.214651e+03 FES = 266100 μ_F = 0.744 μ_{CR} = 0.093
 Diversity = 4.3528e-02 k_t = [3, 4, 2, 3]
 Generation 2670: Best = 5.214651e+03 FES = 267100 μ_F = 0.730 μ_{CR} = 0.102
 Diversity = 4.3495e-02 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 5.214596e+03 FES = 268100 μ_F = 0.733 μ_{CR} = 0.093
 Diversity = 4.3281e-02 k_t = [4, 4, 4, 2]
 Generation 2690: Best = 5.214583e+03 FES = 269100 μ_F = 0.745 μ_{CR} = 0.084
 Diversity = 4.2926e-02 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 5.214581e+03 FES = 270100 μ_F = 0.739 μ_{CR} = 0.078
 Diversity = 4.2969e-02 k_t = [2, 4, 2, 4]
 Generation 2710: Best = 5.214581e+03 FES = 271100 μ_F = 0.750 μ_{CR} = 0.076
 Diversity = 4.2898e-02 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 5.214567e+03 FES = 272100 μ_F = 0.770 μ_{CR} = 0.075
 Diversity = 4.2617e-02 k_t = [3, 2, 2, 4]
 Generation 2730: Best = 5.214498e+03 FES = 273100 μ_F = 0.780 μ_{CR} = 0.078
 Diversity = 4.2157e-02 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 5.214498e+03 FES = 274100 μ_F = 0.783 μ_{CR} = 0.080
 Diversity = 4.1875e-02 k_t = [4, 2, 3, 4]
 Generation 2750: Best = 5.214485e+03 FES = 275100 μ_F = 0.785 μ_{CR} = 0.080
 Diversity = 4.1995e-02 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 5.214452e+03 FES = 276100 μ_F = 0.804 μ_{CR} = 0.089
 Diversity = 4.2138e-02 k_t = [3, 4, 2, 2]
 Generation 2770: Best = 5.214413e+03 FES = 277100 μ_F = 0.803 μ_{CR} = 0.087
 Diversity = 4.2336e-02 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 5.214381e+03 FES = 278100 μ_F = 0.811 μ_{CR} = 0.087
 Diversity = 4.1892e-02 k_t = [3, 2, 3, 4]
 Generation 2790: Best = 5.214300e+03 FES = 279100 μ_F = 0.802 μ_{CR} = 0.096
 Diversity = 4.2265e-02 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 5.214294e+03 FES = 280100 μ_F = 0.788 μ_{CR} = 0.105
 Diversity = 4.2018e-02 k_t = [4, 3, 4, 4]
 Generation 2810: Best = 5.214291e+03 FES = 281100 μ_F = 0.776 μ_{CR} = 0.112
 Diversity = 4.0744e-02 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 5.214266e+03 FES = 282100 μ_F = 0.774 μ_{CR} = 0.120
 Diversity = 3.8289e-02 k_t = [3, 3, 3, 3]
 Generation 2830: Best = 5.214241e+03 FES = 283100 μ_F = 0.763 μ_{CR} = 0.116
 Diversity = 3.6126e-02 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 5.214215e+03 FES = 284100 μ_F = 0.762 μ_{CR} = 0.126
 Diversity = 3.2858e-02 k_t = [4, 3, 3, 4]
 Generation 2850: Best = 5.214181e+03 FES = 285100 μ_F = 0.737 μ_{CR} = 0.142
 Diversity = 2.5627e-02 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 5.214124e+03 FES = 286100 μ_F = 0.750 μ_{CR} = 0.148
 Diversity = 2.3254e-02 k_t = [2, 2, 3, 2]
 Generation 2870: Best = 5.214095e+03 FES = 287100 μ_F = 0.754 μ_{CR} = 0.153
 Diversity = 2.1932e-02 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 5.214031e+03 FES = 288100 μ_F = 0.755 μ_{CR} = 0.157
 Diversity = 1.7487e-02 k_t = [3, 2, 3, 4]
 Generation 2890: Best = 5.214001e+03 FES = 289100 μ_F = 0.761 μ_{CR} = 0.163
 Diversity = 1.6604e-02 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 5.213965e+03 FES = 290100 μ_F = 0.761 μ_{CR} = 0.169
 Diversity = 1.3965e-02 k_t = [2, 3, 2, 2]
 Generation 2910: Best = 5.213935e+03 FES = 291100 μ_F = 0.740 μ_{CR} = 0.165
 Diversity = 1.3137e-02 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 5.213903e+03 FES = 292100 μ_F = 0.727 μ_{CR} = 0.164
 Diversity = 1.1112e-02 k_t = [3, 4, 3, 3]

Generation 2930: Best = 5.213853e+03 FES = 293100 μ_F = 0.732 μ_{CR} = 0.163
 Diversity = 1.1154e-02 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 5.213823e+03 FES = 294100 μ_F = 0.722 μ_{CR} = 0.163
 Diversity = 1.0347e-02 k_t = [2, 3, 3, 3]
 Generation 2950: Best = 5.213778e+03 FES = 295100 μ_F = 0.701 μ_{CR} = 0.173
 Diversity = 9.4060e-03 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 5.213734e+03 FES = 296100 μ_F = 0.702 μ_{CR} = 0.178
 Diversity = 9.8365e-03 k_t = [3, 4, 2, 2]
 Generation 2970: Best = 5.213709e+03 FES = 297100 μ_F = 0.700 μ_{CR} = 0.170
 Diversity = 9.8287e-03 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 5.213676e+03 FES = 298100 μ_F = 0.692 μ_{CR} = 0.165
 Diversity = 8.4940e-03 k_t = [4, 2, 3, 2]
 Generation 2990: Best = 5.213662e+03 FES = 299100 μ_F = 0.694 μ_{CR} = 0.165
 Diversity = 7.9300e-03 k_t = [1, 1, 1, 1]
 Generation 3000: Best = 5.213635e+03 FES = 300100 μ_F = 0.700 μ_{CR} = 0.167
 Diversity = 7.1721e-03 k_t = [2, 2, 2, 4]
 Generation 3010: Best = 5.213599e+03 FES = 301100 μ_F = 0.707 μ_{CR} = 0.169
 Diversity = 7.0918e-03 k_t = [1, 1, 1, 1]
 Generation 3020: Best = 5.213568e+03 FES = 302100 μ_F = 0.716 μ_{CR} = 0.173
 Diversity = 6.7276e-03 k_t = [2, 2, 4, 3]
 Generation 3030: Best = 5.213545e+03 FES = 303100 μ_F = 0.732 μ_{CR} = 0.180
 Diversity = 6.6405e-03 k_t = [1, 1, 1, 1]
 Generation 3040: Best = 5.213526e+03 FES = 304100 μ_F = 0.722 μ_{CR} = 0.178
 Diversity = 6.9381e-03 k_t = [3, 3, 3, 2]
 Generation 3050: Best = 5.213498e+03 FES = 305100 μ_F = 0.729 μ_{CR} = 0.180
 Diversity = 6.8700e-03 k_t = [1, 1, 1, 1]
 Generation 3060: Best = 5.213476e+03 FES = 306100 μ_F = 0.725 μ_{CR} = 0.178
 Diversity = 6.9353e-03 k_t = [3, 3, 3, 2]
 Generation 3070: Best = 5.213455e+03 FES = 307100 μ_F = 0.727 μ_{CR} = 0.171
 Diversity = 6.8428e-03 k_t = [1, 1, 1, 1]
 Generation 3080: Best = 5.213436e+03 FES = 308100 μ_F = 0.733 μ_{CR} = 0.158
 Diversity = 5.9501e-03 k_t = [4, 4, 2, 3]
 Generation 3090: Best = 5.213424e+03 FES = 309100 μ_F = 0.743 μ_{CR} = 0.161
 Diversity = 5.6742e-03 k_t = [1, 1, 1, 1]
 Generation 3100: Best = 5.213405e+03 FES = 310100 μ_F = 0.746 μ_{CR} = 0.157
 Diversity = 5.3087e-03 k_t = [2, 2, 3, 4]
 Generation 3110: Best = 5.213393e+03 FES = 311100 μ_F = 0.742 μ_{CR} = 0.158
 Diversity = 5.5930e-03 k_t = [1, 1, 1, 1]
 Generation 3120: Best = 5.213373e+03 FES = 312100 μ_F = 0.743 μ_{CR} = 0.157
 Diversity = 5.8406e-03 k_t = [2, 3, 3, 2]
 Generation 3130: Best = 5.213357e+03 FES = 313100 μ_F = 0.734 μ_{CR} = 0.150
 Diversity = 6.2918e-03 k_t = [1, 1, 1, 1]
 Generation 3140: Best = 5.213340e+03 FES = 314100 μ_F = 0.744 μ_{CR} = 0.156
 Diversity = 6.4640e-03 k_t = [2, 4, 4, 4]
 Generation 3150: Best = 5.213322e+03 FES = 315100 μ_F = 0.735 μ_{CR} = 0.163
 Diversity = 6.3223e-03 k_t = [1, 1, 1, 1]
 Generation 3160: Best = 5.213301e+03 FES = 316100 μ_F = 0.741 μ_{CR} = 0.171
 Diversity = 5.7697e-03 k_t = [3, 2, 3, 4]
 Generation 3170: Best = 5.213281e+03 FES = 317100 μ_F = 0.752 μ_{CR} = 0.172
 Diversity = 5.6378e-03 k_t = [1, 1, 1, 1]
 Generation 3180: Best = 5.213276e+03 FES = 318100 μ_F = 0.745 μ_{CR} = 0.173
 Diversity = 5.3675e-03 k_t = [4, 4, 4, 3]
 Generation 3190: Best = 5.213265e+03 FES = 319100 μ_F = 0.744 μ_{CR} = 0.170
 Diversity = 5.2263e-03 k_t = [1, 1, 1, 1]

Generation 3200: Best = 5.213254e+03 FES = 320100 μ_F = 0.726 μ_{CR} = 0.156
 Diversity = 4.9150e-03 k_t = [2, 3, 2, 2]
 Generation 3210: Best = 5.213250e+03 FES = 321100 μ_F = 0.718 μ_{CR} = 0.152
 Diversity = 4.5308e-03 k_t = [1, 1, 1, 1]
 Generation 3220: Best = 5.213230e+03 FES = 322100 μ_F = 0.724 μ_{CR} = 0.156
 Diversity = 4.2639e-03 k_t = [2, 3, 3, 2]
 Generation 3230: Best = 5.213223e+03 FES = 323100 μ_F = 0.731 μ_{CR} = 0.162
 Diversity = 4.6300e-03 k_t = [1, 1, 1, 1]
 Generation 3240: Best = 5.213219e+03 FES = 324100 μ_F = 0.721 μ_{CR} = 0.161
 Diversity = 4.3928e-03 k_t = [2, 3, 2, 3]
 Generation 3250: Best = 5.213208e+03 FES = 325100 μ_F = 0.726 μ_{CR} = 0.156
 Diversity = 3.9301e-03 k_t = [1, 1, 1, 1]
 Generation 3260: Best = 5.213197e+03 FES = 326100 μ_F = 0.734 μ_{CR} = 0.162
 Diversity = 4.0118e-03 k_t = [4, 2, 3, 3]
 Generation 3270: Best = 5.213186e+03 FES = 327100 μ_F = 0.745 μ_{CR} = 0.166
 Diversity = 4.0209e-03 k_t = [1, 1, 1, 1]
 Generation 3280: Best = 5.213179e+03 FES = 328100 μ_F = 0.746 μ_{CR} = 0.166
 Diversity = 4.1501e-03 k_t = [4, 2, 2, 2]
 Generation 3290: Best = 5.213170e+03 FES = 329100 μ_F = 0.751 μ_{CR} = 0.162
 Diversity = 4.0754e-03 k_t = [1, 1, 1, 1]
 Generation 3300: Best = 5.213164e+03 FES = 330100 μ_F = 0.747 μ_{CR} = 0.157
 Diversity = 3.9205e-03 k_t = [2, 3, 2, 3]
 Generation 3310: Best = 5.213156e+03 FES = 331100 μ_F = 0.745 μ_{CR} = 0.163
 Diversity = 4.1152e-03 k_t = [1, 1, 1, 1]
 Generation 3320: Best = 5.213143e+03 FES = 332100 μ_F = 0.737 μ_{CR} = 0.158
 Diversity = 3.9957e-03 k_t = [2, 4, 2, 2]
 Generation 3330: Best = 5.213139e+03 FES = 333100 μ_F = 0.740 μ_{CR} = 0.160
 Diversity = 4.0862e-03 k_t = [1, 1, 1, 1]
 Generation 3340: Best = 5.213134e+03 FES = 334100 μ_F = 0.754 μ_{CR} = 0.156
 Diversity = 4.2674e-03 k_t = [2, 2, 4, 2]
 Generation 3350: Best = 5.213126e+03 FES = 335100 μ_F = 0.757 μ_{CR} = 0.167
 Diversity = 4.1608e-03 k_t = [1, 1, 1, 1]
 Generation 3360: Best = 5.213118e+03 FES = 336100 μ_F = 0.765 μ_{CR} = 0.169
 Diversity = 4.0915e-03 k_t = [4, 2, 3, 3]
 Generation 3370: Best = 5.213114e+03 FES = 337100 μ_F = 0.742 μ_{CR} = 0.165
 Diversity = 4.0675e-03 k_t = [1, 1, 1, 1]
 Generation 3380: Best = 5.213106e+03 FES = 338100 μ_F = 0.732 μ_{CR} = 0.156
 Diversity = 4.0324e-03 k_t = [3, 4, 3, 2]
 Generation 3390: Best = 5.213096e+03 FES = 339100 μ_F = 0.740 μ_{CR} = 0.163
 Diversity = 4.1909e-03 k_t = [1, 1, 1, 1]
 Generation 3400: Best = 5.213091e+03 FES = 340100 μ_F = 0.753 μ_{CR} = 0.173
 Diversity = 4.0401e-03 k_t = [4, 3, 4, 3]
 Generation 3410: Best = 5.213085e+03 FES = 341100 μ_F = 0.739 μ_{CR} = 0.171
 Diversity = 3.9363e-03 k_t = [1, 1, 1, 1]
 Generation 3420: Best = 5.213078e+03 FES = 342100 μ_F = 0.735 μ_{CR} = 0.174
 Diversity = 3.8331e-03 k_t = [3, 3, 3, 3]
 Generation 3430: Best = 5.213072e+03 FES = 343100 μ_F = 0.729 μ_{CR} = 0.170
 Diversity = 3.6959e-03 k_t = [1, 1, 1, 1]
 Generation 3440: Best = 5.213066e+03 FES = 344100 μ_F = 0.730 μ_{CR} = 0.159
 Diversity = 3.4722e-03 k_t = [2, 3, 2, 3]
 Generation 3450: Best = 5.213060e+03 FES = 345100 μ_F = 0.723 μ_{CR} = 0.156
 Diversity = 3.5222e-03 k_t = [1, 1, 1, 1]
 Generation 3460: Best = 5.213053e+03 FES = 346100 μ_F = 0.729 μ_{CR} = 0.154
 Diversity = 3.2920e-03 k_t = [4, 4, 4, 4]

Generation 3470: Best = 5.213048e+03 FES = 347100 μ_F = 0.729 μ_{CR} = 0.154
 Diversity = 3.2341e-03 k_t = [1, 1, 1, 1]
 Generation 3480: Best = 5.213039e+03 FES = 348100 μ_F = 0.736 μ_{CR} = 0.164
 Diversity = 3.0587e-03 k_t = [2, 3, 4, 3]
 Generation 3490: Best = 5.213036e+03 FES = 349100 μ_F = 0.741 μ_{CR} = 0.166
 Diversity = 3.0961e-03 k_t = [1, 1, 1, 1]
 Generation 3500: Best = 5.213031e+03 FES = 350100 μ_F = 0.742 μ_{CR} = 0.157
 Diversity = 2.9053e-03 k_t = [2, 2, 3, 4]
 Generation 3510: Best = 5.213028e+03 FES = 351100 μ_F = 0.735 μ_{CR} = 0.143
 Diversity = 2.7623e-03 k_t = [1, 1, 1, 1]
 Generation 3520: Best = 5.213020e+03 FES = 352100 μ_F = 0.743 μ_{CR} = 0.149
 Diversity = 2.6511e-03 k_t = [3, 4, 4, 2]
 Generation 3530: Best = 5.213011e+03 FES = 353100 μ_F = 0.753 μ_{CR} = 0.156
 Diversity = 2.9231e-03 k_t = [1, 1, 1, 1]
 Generation 3540: Best = 5.213003e+03 FES = 354100 μ_F = 0.762 μ_{CR} = 0.163
 Diversity = 3.0049e-03 k_t = [2, 2, 2, 3]
 Generation 3550: Best = 5.212999e+03 FES = 355100 μ_F = 0.741 μ_{CR} = 0.151
 Diversity = 3.0563e-03 k_t = [1, 1, 1, 1]
 Generation 3560: Best = 5.212996e+03 FES = 356100 μ_F = 0.755 μ_{CR} = 0.142
 Diversity = 2.9360e-03 k_t = [4, 3, 4, 4]
 Generation 3570: Best = 5.212993e+03 FES = 357100 μ_F = 0.761 μ_{CR} = 0.136
 Diversity = 2.6714e-03 k_t = [1, 1, 1, 1]
 Generation 3580: Best = 5.212988e+03 FES = 358100 μ_F = 0.758 μ_{CR} = 0.136
 Diversity = 2.6989e-03 k_t = [2, 2, 3, 2]
 Generation 3590: Best = 5.212985e+03 FES = 359100 μ_F = 0.743 μ_{CR} = 0.138
 Diversity = 2.8102e-03 k_t = [1, 1, 1, 1]
 Generation 3600: Best = 5.212981e+03 FES = 360100 μ_F = 0.742 μ_{CR} = 0.145
 Diversity = 2.8494e-03 k_t = [3, 2, 2, 2]
 Generation 3610: Best = 5.212977e+03 FES = 361100 μ_F = 0.731 μ_{CR} = 0.143
 Diversity = 2.8156e-03 k_t = [1, 1, 1, 1]
 Generation 3620: Best = 5.212974e+03 FES = 362100 μ_F = 0.738 μ_{CR} = 0.139
 Diversity = 2.7995e-03 k_t = [2, 3, 3, 3]
 Generation 3630: Best = 5.212967e+03 FES = 363100 μ_F = 0.741 μ_{CR} = 0.135
 Diversity = 2.8239e-03 k_t = [1, 1, 1, 1]
 Generation 3640: Best = 5.212961e+03 FES = 364100 μ_F = 0.731 μ_{CR} = 0.140
 Diversity = 2.9560e-03 k_t = [2, 2, 4, 2]
 Generation 3650: Best = 5.212955e+03 FES = 365100 μ_F = 0.733 μ_{CR} = 0.148
 Diversity = 3.0016e-03 k_t = [1, 1, 1, 1]
 Generation 3660: Best = 5.212947e+03 FES = 366100 μ_F = 0.751 μ_{CR} = 0.157
 Diversity = 2.9577e-03 k_t = [2, 3, 2, 3]
 Generation 3670: Best = 5.212941e+03 FES = 367100 μ_F = 0.750 μ_{CR} = 0.152
 Diversity = 3.1371e-03 k_t = [1, 1, 1, 1]
 Generation 3680: Best = 5.212937e+03 FES = 368100 μ_F = 0.744 μ_{CR} = 0.148
 Diversity = 2.9794e-03 k_t = [2, 2, 3, 2]
 Generation 3690: Best = 5.212935e+03 FES = 369100 μ_F = 0.756 μ_{CR} = 0.148
 Diversity = 3.0972e-03 k_t = [1, 1, 1, 1]
 Generation 3700: Best = 5.212928e+03 FES = 370100 μ_F = 0.755 μ_{CR} = 0.151
 Diversity = 2.9011e-03 k_t = [2, 2, 4, 2]
 Generation 3710: Best = 5.212922e+03 FES = 371100 μ_F = 0.746 μ_{CR} = 0.149
 Diversity = 3.0413e-03 k_t = [1, 1, 1, 1]
 Generation 3720: Best = 5.212916e+03 FES = 372100 μ_F = 0.741 μ_{CR} = 0.157
 Diversity = 2.9431e-03 k_t = [2, 4, 2, 4]
 Generation 3730: Best = 5.212910e+03 FES = 373100 μ_F = 0.737 μ_{CR} = 0.152
 Diversity = 2.8835e-03 k_t = [1, 1, 1, 1]

Generation 3740: Best = 5.212907e+03 FES = 374100 μ_F = 0.736 μ_{CR} = 0.157
 Diversity = 2.8090e-03 k_t = [4, 2, 2, 3]
 Generation 3750: Best = 5.212904e+03 FES = 375100 μ_F = 0.752 μ_{CR} = 0.158
 Diversity = 2.9594e-03 k_t = [1, 1, 1, 1]
 Generation 3760: Best = 5.212898e+03 FES = 376100 μ_F = 0.743 μ_{CR} = 0.177
 Diversity = 2.9419e-03 k_t = [2, 3, 2, 2]
 Generation 3770: Best = 5.212896e+03 FES = 377100 μ_F = 0.731 μ_{CR} = 0.174
 Diversity = 3.0491e-03 k_t = [1, 1, 1, 1]
 Generation 3780: Best = 5.212888e+03 FES = 378100 μ_F = 0.734 μ_{CR} = 0.173
 Diversity = 2.7406e-03 k_t = [3, 4, 3, 4]
 Generation 3790: Best = 5.212884e+03 FES = 379100 μ_F = 0.740 μ_{CR} = 0.173
 Diversity = 2.9374e-03 k_t = [1, 1, 1, 1]
 Generation 3800: Best = 5.212881e+03 FES = 380100 μ_F = 0.723 μ_{CR} = 0.169
 Diversity = 2.8034e-03 k_t = [4, 2, 3, 3]
 Generation 3810: Best = 5.212877e+03 FES = 381100 μ_F = 0.721 μ_{CR} = 0.165
 Diversity = 2.6467e-03 k_t = [1, 1, 1, 1]
 Generation 3820: Best = 5.212874e+03 FES = 382100 μ_F = 0.752 μ_{CR} = 0.163
 Diversity = 2.5579e-03 k_t = [4, 3, 3, 3]
 Generation 3830: Best = 5.212871e+03 FES = 383100 μ_F = 0.746 μ_{CR} = 0.155
 Diversity = 2.7163e-03 k_t = [1, 1, 1, 1]
 Generation 3840: Best = 5.212867e+03 FES = 384100 μ_F = 0.753 μ_{CR} = 0.149
 Diversity = 2.5696e-03 k_t = [2, 4, 4, 4]
 Generation 3850: Best = 5.212864e+03 FES = 385100 μ_F = 0.756 μ_{CR} = 0.145
 Diversity = 2.5309e-03 k_t = [1, 1, 1, 1]
 Generation 3860: Best = 5.212861e+03 FES = 386100 μ_F = 0.744 μ_{CR} = 0.141
 Diversity = 2.5898e-03 k_t = [4, 4, 4, 3]
 Generation 3870: Best = 5.212859e+03 FES = 387100 μ_F = 0.741 μ_{CR} = 0.141
 Diversity = 2.4949e-03 k_t = [1, 1, 1, 1]
 Generation 3880: Best = 5.212855e+03 FES = 388100 μ_F = 0.734 μ_{CR} = 0.139
 Diversity = 2.5058e-03 k_t = [3, 4, 4, 2]
 Generation 3890: Best = 5.212852e+03 FES = 389100 μ_F = 0.726 μ_{CR} = 0.138
 Diversity = 2.5417e-03 k_t = [1, 1, 1, 1]
 Generation 3900: Best = 5.212849e+03 FES = 390100 μ_F = 0.736 μ_{CR} = 0.131
 Diversity = 2.4187e-03 k_t = [2, 4, 3, 4]
 Generation 3910: Best = 5.212847e+03 FES = 391100 μ_F = 0.730 μ_{CR} = 0.135
 Diversity = 2.4201e-03 k_t = [1, 1, 1, 1]
 Generation 3920: Best = 5.212846e+03 FES = 392100 μ_F = 0.742 μ_{CR} = 0.127
 Diversity = 2.4460e-03 k_t = [3, 3, 4, 2]
 Generation 3930: Best = 5.212845e+03 FES = 393100 μ_F = 0.719 μ_{CR} = 0.130
 Diversity = 2.4582e-03 k_t = [1, 1, 1, 1]
 Generation 3940: Best = 5.212843e+03 FES = 394100 μ_F = 0.733 μ_{CR} = 0.123
 Diversity = 2.2849e-03 k_t = [3, 3, 2, 2]
 Generation 3950: Best = 5.212840e+03 FES = 395100 μ_F = 0.738 μ_{CR} = 0.125
 Diversity = 2.3517e-03 k_t = [1, 1, 1, 1]
 Generation 3960: Best = 5.212837e+03 FES = 396100 μ_F = 0.723 μ_{CR} = 0.119
 Diversity = 2.1370e-03 k_t = [3, 4, 2, 4]
 Generation 3970: Best = 5.212836e+03 FES = 397100 μ_F = 0.710 μ_{CR} = 0.128
 Diversity = 2.0263e-03 k_t = [1, 1, 1, 1]
 Generation 3980: Best = 5.212835e+03 FES = 398100 μ_F = 0.726 μ_{CR} = 0.126
 Diversity = 2.0576e-03 k_t = [4, 4, 4, 2]
 Generation 3990: Best = 5.212832e+03 FES = 399100 μ_F = 0.723 μ_{CR} = 0.132
 Diversity = 2.0681e-03 k_t = [1, 1, 1, 1]
 Generation 4000: Best = 5.212831e+03 FES = 400100 μ_F = 0.723 μ_{CR} = 0.129
 Diversity = 2.0975e-03 k_t = [4, 2, 2, 2]

Generation 4010: Best = 5.212828e+03 FES = 401100 μ_F = 0.721 μ_{CR} = 0.127
 Diversity = 2.0112e-03 k_t = [1, 1, 1, 1]
 Generation 4020: Best = 5.212827e+03 FES = 402100 μ_F = 0.719 μ_{CR} = 0.133
 Diversity = 1.9355e-03 k_t = [3, 2, 2, 4]
 Generation 4030: Best = 5.212824e+03 FES = 403100 μ_F = 0.709 μ_{CR} = 0.134
 Diversity = 1.8770e-03 k_t = [1, 1, 1, 1]
 Generation 4040: Best = 5.212823e+03 FES = 404100 μ_F = 0.722 μ_{CR} = 0.133
 Diversity = 1.8614e-03 k_t = [2, 4, 2, 2]
 Generation 4050: Best = 5.212821e+03 FES = 405100 μ_F = 0.738 μ_{CR} = 0.139
 Diversity = 1.9745e-03 k_t = [1, 1, 1, 1]
 Generation 4060: Best = 5.212817e+03 FES = 406100 μ_F = 0.737 μ_{CR} = 0.140
 Diversity = 1.9806e-03 k_t = [4, 2, 4, 3]
 Generation 4070: Best = 5.212813e+03 FES = 407100 μ_F = 0.751 μ_{CR} = 0.145
 Diversity = 2.1067e-03 k_t = [1, 1, 1, 1]
 Generation 4080: Best = 5.212812e+03 FES = 408100 μ_F = 0.754 μ_{CR} = 0.138
 Diversity = 2.0672e-03 k_t = [4, 2, 2, 4]
 Generation 4090: Best = 5.212811e+03 FES = 409100 μ_F = 0.759 μ_{CR} = 0.130
 Diversity = 2.1073e-03 k_t = [1, 1, 1, 1]
 Generation 4100: Best = 5.212809e+03 FES = 410100 μ_F = 0.756 μ_{CR} = 0.126
 Diversity = 2.1426e-03 k_t = [2, 3, 4, 3]
 Generation 4110: Best = 5.212806e+03 FES = 411100 μ_F = 0.753 μ_{CR} = 0.116
 Diversity = 2.2327e-03 k_t = [1, 1, 1, 1]
 Generation 4120: Best = 5.212805e+03 FES = 412100 μ_F = 0.758 μ_{CR} = 0.119
 Diversity = 2.1052e-03 k_t = [2, 2, 4, 2]
 Generation 4130: Best = 5.212804e+03 FES = 413100 μ_F = 0.757 μ_{CR} = 0.122
 Diversity = 2.1550e-03 k_t = [1, 1, 1, 1]
 Generation 4140: Best = 5.212802e+03 FES = 414100 μ_F = 0.747 μ_{CR} = 0.125
 Diversity = 2.1593e-03 k_t = [4, 3, 4, 2]
 Generation 4150: Best = 5.212800e+03 FES = 415100 μ_F = 0.734 μ_{CR} = 0.136
 Diversity = 2.1998e-03 k_t = [1, 1, 1, 1]
 Generation 4160: Best = 5.212798e+03 FES = 416100 μ_F = 0.734 μ_{CR} = 0.123
 Diversity = 2.0017e-03 k_t = [2, 3, 4, 3]
 Generation 4170: Best = 5.212796e+03 FES = 417100 μ_F = 0.731 μ_{CR} = 0.125
 Diversity = 2.0788e-03 k_t = [1, 1, 1, 1]
 Generation 4180: Best = 5.212794e+03 FES = 418100 μ_F = 0.753 μ_{CR} = 0.128
 Diversity = 2.0298e-03 k_t = [4, 4, 2, 4]
 Generation 4190: Best = 5.212790e+03 FES = 419100 μ_F = 0.757 μ_{CR} = 0.133
 Diversity = 2.0589e-03 k_t = [1, 1, 1, 1]
 Generation 4200: Best = 5.212789e+03 FES = 420100 μ_F = 0.755 μ_{CR} = 0.143
 Diversity = 2.1588e-03 k_t = [2, 3, 4, 2]
 Generation 4210: Best = 5.212787e+03 FES = 421100 μ_F = 0.742 μ_{CR} = 0.154
 Diversity = 2.2775e-03 k_t = [1, 1, 1, 1]
 Generation 4220: Best = 5.212785e+03 FES = 422100 μ_F = 0.734 μ_{CR} = 0.141
 Diversity = 2.1083e-03 k_t = [4, 2, 2, 4]
 Generation 4230: Best = 5.212782e+03 FES = 423100 μ_F = 0.739 μ_{CR} = 0.142
 Diversity = 1.9987e-03 k_t = [1, 1, 1, 1]
 Generation 4240: Best = 5.212779e+03 FES = 424100 μ_F = 0.750 μ_{CR} = 0.124
 Diversity = 2.0673e-03 k_t = [2, 3, 2, 2]
 Generation 4250: Best = 5.212776e+03 FES = 425100 μ_F = 0.769 μ_{CR} = 0.130
 Diversity = 2.3537e-03 k_t = [1, 1, 1, 1]
 Generation 4260: Best = 5.212773e+03 FES = 426100 μ_F = 0.774 μ_{CR} = 0.138
 Diversity = 2.5950e-03 k_t = [3, 3, 2, 2]
 Generation 4270: Best = 5.212770e+03 FES = 427100 μ_F = 0.769 μ_{CR} = 0.141
 Diversity = 2.6490e-03 k_t = [1, 1, 1, 1]

Generation 4280: Best = 5.212767e+03 FES = 428100 μ_F = 0.774 μ_{CR} = 0.132
 Diversity = 2.5077e-03 k_t = [2, 2, 2, 3]
 Generation 4290: Best = 5.212765e+03 FES = 429100 μ_F = 0.767 μ_{CR} = 0.136
 Diversity = 2.5881e-03 k_t = [1, 1, 1, 1]
 Generation 4300: Best = 5.212758e+03 FES = 430100 μ_F = 0.764 μ_{CR} = 0.155
 Diversity = 2.7067e-03 k_t = [2, 2, 2, 4]
 Generation 4310: Best = 5.212754e+03 FES = 431100 μ_F = 0.762 μ_{CR} = 0.158
 Diversity = 2.8522e-03 k_t = [1, 1, 1, 1]
 Generation 4320: Best = 5.212752e+03 FES = 432100 μ_F = 0.751 μ_{CR} = 0.149
 Diversity = 2.7438e-03 k_t = [2, 3, 2, 3]
 Generation 4330: Best = 5.212746e+03 FES = 433100 μ_F = 0.752 μ_{CR} = 0.152
 Diversity = 2.8160e-03 k_t = [1, 1, 1, 1]
 Generation 4340: Best = 5.212742e+03 FES = 434100 μ_F = 0.741 μ_{CR} = 0.150
 Diversity = 2.7203e-03 k_t = [2, 4, 3, 2]
 Generation 4350: Best = 5.212740e+03 FES = 435100 μ_F = 0.731 μ_{CR} = 0.150
 Diversity = 2.5910e-03 k_t = [1, 1, 1, 1]
 Generation 4360: Best = 5.212737e+03 FES = 436100 μ_F = 0.728 μ_{CR} = 0.140
 Diversity = 2.4569e-03 k_t = [2, 2, 4, 3]
 Generation 4370: Best = 5.212735e+03 FES = 437100 μ_F = 0.724 μ_{CR} = 0.132
 Diversity = 2.3806e-03 k_t = [1, 1, 1, 1]
 Generation 4380: Best = 5.212734e+03 FES = 438100 μ_F = 0.716 μ_{CR} = 0.131
 Diversity = 2.1795e-03 k_t = [2, 2, 4, 4]
 Generation 4390: Best = 5.212732e+03 FES = 439100 μ_F = 0.711 μ_{CR} = 0.144
 Diversity = 1.9177e-03 k_t = [1, 1, 1, 1]
 Generation 4400: Best = 5.212729e+03 FES = 440100 μ_F = 0.707 μ_{CR} = 0.148
 Diversity = 1.8695e-03 k_t = [3, 2, 3, 4]
 Generation 4410: Best = 5.212728e+03 FES = 441100 μ_F = 0.719 μ_{CR} = 0.139
 Diversity = 1.6930e-03 k_t = [1, 1, 1, 1]
 Generation 4420: Best = 5.212727e+03 FES = 442100 μ_F = 0.726 μ_{CR} = 0.135
 Diversity = 1.5805e-03 k_t = [2, 3, 4, 4]
 Generation 4430: Best = 5.212726e+03 FES = 443100 μ_F = 0.719 μ_{CR} = 0.129
 Diversity = 1.5618e-03 k_t = [1, 1, 1, 1]
 Generation 4440: Best = 5.212724e+03 FES = 444100 μ_F = 0.722 μ_{CR} = 0.124
 Diversity = 1.5178e-03 k_t = [2, 4, 2, 3]
 Generation 4450: Best = 5.212723e+03 FES = 445100 μ_F = 0.731 μ_{CR} = 0.123
 Diversity = 1.4762e-03 k_t = [1, 1, 1, 1]
 Generation 4460: Best = 5.212723e+03 FES = 446100 μ_F = 0.736 μ_{CR} = 0.122
 Diversity = 1.3788e-03 k_t = [2, 2, 3, 3]
 Generation 4470: Best = 5.212722e+03 FES = 447100 μ_F = 0.746 μ_{CR} = 0.121
 Diversity = 1.3548e-03 k_t = [1, 1, 1, 1]
 Generation 4480: Best = 5.212721e+03 FES = 448100 μ_F = 0.738 μ_{CR} = 0.116
 Diversity = 1.3736e-03 k_t = [2, 4, 3, 2]
 Generation 4490: Best = 5.212720e+03 FES = 449100 μ_F = 0.747 μ_{CR} = 0.122
 Diversity = 1.4245e-03 k_t = [1, 1, 1, 1]
 Generation 4500: Best = 5.212719e+03 FES = 450100 μ_F = 0.754 μ_{CR} = 0.123
 Diversity = 1.4478e-03 k_t = [3, 3, 3, 3]
 Generation 4510: Best = 5.212718e+03 FES = 451100 μ_F = 0.748 μ_{CR} = 0.121
 Diversity = 1.5479e-03 k_t = [1, 1, 1, 1]
 Generation 4520: Best = 5.212717e+03 FES = 452100 μ_F = 0.756 μ_{CR} = 0.119
 Diversity = 1.5751e-03 k_t = [2, 4, 2, 3]
 Generation 4530: Best = 5.212715e+03 FES = 453100 μ_F = 0.757 μ_{CR} = 0.122
 Diversity = 1.6292e-03 k_t = [1, 1, 1, 1]
 Generation 4540: Best = 5.212715e+03 FES = 454100 μ_F = 0.761 μ_{CR} = 0.122
 Diversity = 1.5658e-03 k_t = [4, 3, 2, 3]

Generation 4550: Best = 5.212713e+03 FES = 455100 μ_F = 0.754 μ_{CR} = 0.128
 Diversity = 1.5220e-03 k_t = [1, 1, 1, 1]
 Generation 4560: Best = 5.212712e+03 FES = 456100 μ_F = 0.752 μ_{CR} = 0.125
 Diversity = 1.5475e-03 k_t = [2, 3, 2, 3]
 Generation 4570: Best = 5.212712e+03 FES = 457100 μ_F = 0.751 μ_{CR} = 0.125
 Diversity = 1.6443e-03 k_t = [1, 1, 1, 1]
 Generation 4580: Best = 5.212711e+03 FES = 458100 μ_F = 0.762 μ_{CR} = 0.125
 Diversity = 1.5684e-03 k_t = [4, 2, 2, 4]
 Generation 4590: Best = 5.212710e+03 FES = 459100 μ_F = 0.743 μ_{CR} = 0.135
 Diversity = 1.4766e-03 k_t = [1, 1, 1, 1]
 Generation 4600: Best = 5.212709e+03 FES = 460100 μ_F = 0.738 μ_{CR} = 0.142
 Diversity = 1.4303e-03 k_t = [2, 3, 3, 4]
 Generation 4610: Best = 5.212708e+03 FES = 461100 μ_F = 0.747 μ_{CR} = 0.135
 Diversity = 1.4526e-03 k_t = [1, 1, 1, 1]
 Generation 4620: Best = 5.212707e+03 FES = 462100 μ_F = 0.748 μ_{CR} = 0.138
 Diversity = 1.3731e-03 k_t = [4, 3, 3, 3]
 Generation 4630: Best = 5.212706e+03 FES = 463100 μ_F = 0.758 μ_{CR} = 0.152
 Diversity = 1.4352e-03 k_t = [1, 1, 1, 1]
 Generation 4640: Best = 5.212705e+03 FES = 464100 μ_F = 0.751 μ_{CR} = 0.145
 Diversity = 1.4269e-03 k_t = [3, 2, 3, 4]
 Generation 4650: Best = 5.212704e+03 FES = 465100 μ_F = 0.737 μ_{CR} = 0.140
 Diversity = 1.3049e-03 k_t = [1, 1, 1, 1]
 Generation 4660: Best = 5.212704e+03 FES = 466100 μ_F = 0.734 μ_{CR} = 0.142
 Diversity = 1.2759e-03 k_t = [2, 4, 4, 2]
 Generation 4670: Best = 5.212703e+03 FES = 467100 μ_F = 0.743 μ_{CR} = 0.137
 Diversity = 1.2336e-03 k_t = [1, 1, 1, 1]
 Generation 4680: Best = 5.212702e+03 FES = 468100 μ_F = 0.725 μ_{CR} = 0.131
 Diversity = 1.3041e-03 k_t = [3, 4, 3, 2]
 Generation 4690: Best = 5.212701e+03 FES = 469100 μ_F = 0.728 μ_{CR} = 0.123
 Diversity = 1.3320e-03 k_t = [1, 1, 1, 1]
 Generation 4700: Best = 5.212701e+03 FES = 470100 μ_F = 0.722 μ_{CR} = 0.128
 Diversity = 1.2584e-03 k_t = [2, 2, 2, 4]
 Generation 4710: Best = 5.212699e+03 FES = 471100 μ_F = 0.714 μ_{CR} = 0.131
 Diversity = 1.2538e-03 k_t = [1, 1, 1, 1]
 Generation 4720: Best = 5.212698e+03 FES = 472100 μ_F = 0.719 μ_{CR} = 0.136
 Diversity = 1.2983e-03 k_t = [3, 4, 3, 2]
 Generation 4730: Best = 5.212697e+03 FES = 473100 μ_F = 0.732 μ_{CR} = 0.139
 Diversity = 1.4036e-03 k_t = [1, 1, 1, 1]
 Generation 4740: Best = 5.212695e+03 FES = 474100 μ_F = 0.735 μ_{CR} = 0.147
 Diversity = 1.3370e-03 k_t = [3, 2, 2, 3]
 Generation 4750: Best = 5.212694e+03 FES = 475100 μ_F = 0.731 μ_{CR} = 0.144
 Diversity = 1.3588e-03 k_t = [1, 1, 1, 1]
 Generation 4760: Best = 5.212693e+03 FES = 476100 μ_F = 0.750 μ_{CR} = 0.144
 Diversity = 1.3067e-03 k_t = [3, 3, 2, 3]
 Generation 4770: Best = 5.212692e+03 FES = 477100 μ_F = 0.743 μ_{CR} = 0.145
 Diversity = 1.3549e-03 k_t = [1, 1, 1, 1]
 Generation 4780: Best = 5.212692e+03 FES = 478100 μ_F = 0.753 μ_{CR} = 0.146
 Diversity = 1.2765e-03 k_t = [3, 2, 3, 4]
 Generation 4790: Best = 5.212691e+03 FES = 479100 μ_F = 0.750 μ_{CR} = 0.143
 Diversity = 1.2937e-03 k_t = [1, 1, 1, 1]
 Generation 4800: Best = 5.212690e+03 FES = 480100 μ_F = 0.757 μ_{CR} = 0.144
 Diversity = 1.2625e-03 k_t = [3, 4, 4, 4]
 Generation 4810: Best = 5.212689e+03 FES = 481100 μ_F = 0.765 μ_{CR} = 0.145
 Diversity = 1.3082e-03 k_t = [1, 1, 1, 1]

Generation 4820: Best = 5.212688e+03 FES = 482100 μ_F = 0.758 μ_{CR} = 0.152
 Diversity = 1.3074e-03 k_t = [4, 4, 2, 2]
 Generation 4830: Best = 5.212687e+03 FES = 483100 μ_F = 0.752 μ_{CR} = 0.156
 Diversity = 1.3071e-03 k_t = [1, 1, 1, 1]
 Generation 4840: Best = 5.212686e+03 FES = 484100 μ_F = 0.744 μ_{CR} = 0.153
 Diversity = 1.3371e-03 k_t = [4, 2, 3, 2]
 Generation 4850: Best = 5.212685e+03 FES = 485100 μ_F = 0.748 μ_{CR} = 0.151
 Diversity = 1.3653e-03 k_t = [1, 1, 1, 1]
 Generation 4860: Best = 5.212684e+03 FES = 486100 μ_F = 0.742 μ_{CR} = 0.152
 Diversity = 1.3148e-03 k_t = [3, 3, 2, 2]
 Generation 4870: Best = 5.212683e+03 FES = 487100 μ_F = 0.751 μ_{CR} = 0.151
 Diversity = 1.3696e-03 k_t = [1, 1, 1, 1]
 Generation 4880: Best = 5.212682e+03 FES = 488100 μ_F = 0.752 μ_{CR} = 0.152
 Diversity = 1.3555e-03 k_t = [4, 4, 4, 2]
 Generation 4890: Best = 5.212681e+03 FES = 489100 μ_F = 0.749 μ_{CR} = 0.151
 Diversity = 1.3571e-03 k_t = [1, 1, 1, 1]
 Generation 4900: Best = 5.212680e+03 FES = 490100 μ_F = 0.744 μ_{CR} = 0.156
 Diversity = 1.3108e-03 k_t = [2, 2, 4, 3]
 Generation 4910: Best = 5.212679e+03 FES = 491100 μ_F = 0.742 μ_{CR} = 0.150
 Diversity = 1.4033e-03 k_t = [1, 1, 1, 1]
 Generation 4920: Best = 5.212678e+03 FES = 492100 μ_F = 0.742 μ_{CR} = 0.150
 Diversity = 1.3408e-03 k_t = [2, 2, 2, 3]
 Generation 4930: Best = 5.212677e+03 FES = 493100 μ_F = 0.725 μ_{CR} = 0.148
 Diversity = 1.3574e-03 k_t = [1, 1, 1, 1]
 Generation 4940: Best = 5.212675e+03 FES = 494100 μ_F = 0.737 μ_{CR} = 0.160
 Diversity = 1.4056e-03 k_t = [2, 4, 3, 2]
 Generation 4950: Best = 5.212674e+03 FES = 495100 μ_F = 0.741 μ_{CR} = 0.161
 Diversity = 1.4770e-03 k_t = [1, 1, 1, 1]
 Generation 4960: Best = 5.212672e+03 FES = 496100 μ_F = 0.729 μ_{CR} = 0.161
 Diversity = 1.3842e-03 k_t = [2, 2, 4, 3]
 Generation 4970: Best = 5.212671e+03 FES = 497100 μ_F = 0.732 μ_{CR} = 0.162
 Diversity = 1.4555e-03 k_t = [1, 1, 1, 1]
 Generation 4980: Best = 5.212669e+03 FES = 498100 μ_F = 0.732 μ_{CR} = 0.163
 Diversity = 1.3376e-03 k_t = [3, 4, 2, 4]
 Generation 4990: Best = 5.212666e+03 FES = 499100 μ_F = 0.739 μ_{CR} = 0.160
 Diversity = 1.5156e-03 k_t = [1, 1, 1, 1]
 Generation 5000: Best = 5.212664e+03 FES = 500100 μ_F = 0.740 μ_{CR} = 0.155
 Diversity = 1.4999e-03 k_t = [3, 2, 2, 2]
 Generation 5010: Best = 5.212664e+03 FES = 501100 μ_F = 0.750 μ_{CR} = 0.157
 Diversity = 1.5543e-03 k_t = [1, 1, 1, 1]
 Generation 5020: Best = 5.212662e+03 FES = 502100 μ_F = 0.750 μ_{CR} = 0.162
 Diversity = 1.5591e-03 k_t = [3, 3, 4, 3]
 Generation 5030: Best = 5.212661e+03 FES = 503100 μ_F = 0.735 μ_{CR} = 0.154
 Diversity = 1.5639e-03 k_t = [1, 1, 1, 1]
 Generation 5040: Best = 5.212660e+03 FES = 504100 μ_F = 0.729 μ_{CR} = 0.160
 Diversity = 1.4875e-03 k_t = [3, 3, 3, 3]
 Generation 5050: Best = 5.212659e+03 FES = 505100 μ_F = 0.737 μ_{CR} = 0.165
 Diversity = 1.5833e-03 k_t = [1, 1, 1, 1]
 Generation 5060: Best = 5.212657e+03 FES = 506100 μ_F = 0.741 μ_{CR} = 0.163
 Diversity = 1.5453e-03 k_t = [4, 2, 2, 2]
 Generation 5070: Best = 5.212656e+03 FES = 507100 μ_F = 0.739 μ_{CR} = 0.159
 Diversity = 1.6002e-03 k_t = [1, 1, 1, 1]
 Generation 5080: Best = 5.212655e+03 FES = 508100 μ_F = 0.731 μ_{CR} = 0.154
 Diversity = 1.4573e-03 k_t = [4, 4, 2, 3]

Generation 5090: Best = 5.212653e+03 FES = 509100 μ_F = 0.737 μ_{CR} = 0.140
 Diversity = 1.4676e-03 k_t = [1, 1, 1, 1]
 Generation 5100: Best = 5.212652e+03 FES = 510100 μ_F = 0.731 μ_{CR} = 0.139
 Diversity = 1.4620e-03 k_t = [4, 2, 4, 3]
 Generation 5110: Best = 5.212650e+03 FES = 511100 μ_F = 0.724 μ_{CR} = 0.144
 Diversity = 1.5154e-03 k_t = [1, 1, 1, 1]
 Generation 5120: Best = 5.212649e+03 FES = 512100 μ_F = 0.735 μ_{CR} = 0.138
 Diversity = 1.5343e-03 k_t = [3, 4, 2, 3]
 Generation 5130: Best = 5.212647e+03 FES = 513100 μ_F = 0.738 μ_{CR} = 0.142
 Diversity = 1.4956e-03 k_t = [1, 1, 1, 1]
 Generation 5140: Best = 5.212646e+03 FES = 514100 μ_F = 0.738 μ_{CR} = 0.145
 Diversity = 1.4373e-03 k_t = [2, 3, 2, 3]
 Generation 5150: Best = 5.212645e+03 FES = 515100 μ_F = 0.738 μ_{CR} = 0.153
 Diversity = 1.5634e-03 k_t = [1, 1, 1, 1]
 Generation 5160: Best = 5.212644e+03 FES = 516100 μ_F = 0.738 μ_{CR} = 0.146
 Diversity = 1.5481e-03 k_t = [3, 2, 3, 2]
 Generation 5170: Best = 5.212643e+03 FES = 517100 μ_F = 0.736 μ_{CR} = 0.149
 Diversity = 1.5151e-03 k_t = [1, 1, 1, 1]
 Generation 5180: Best = 5.212642e+03 FES = 518100 μ_F = 0.736 μ_{CR} = 0.131
 Diversity = 1.3368e-03 k_t = [2, 3, 3, 3]
 Generation 5190: Best = 5.212641e+03 FES = 519100 μ_F = 0.731 μ_{CR} = 0.122
 Diversity = 1.4378e-03 k_t = [1, 1, 1, 1]
 Generation 5200: Best = 5.212639e+03 FES = 520100 μ_F = 0.739 μ_{CR} = 0.129
 Diversity = 1.3628e-03 k_t = [4, 4, 2, 2]
 Generation 5210: Best = 5.212638e+03 FES = 521100 μ_F = 0.741 μ_{CR} = 0.136
 Diversity = 1.4619e-03 k_t = [1, 1, 1, 1]
 Generation 5220: Best = 5.212638e+03 FES = 522100 μ_F = 0.751 μ_{CR} = 0.141
 Diversity = 1.3912e-03 k_t = [3, 2, 3, 2]
 Generation 5230: Best = 5.212637e+03 FES = 523100 μ_F = 0.742 μ_{CR} = 0.141
 Diversity = 1.3949e-03 k_t = [1, 1, 1, 1]
 Generation 5240: Best = 5.212635e+03 FES = 524100 μ_F = 0.759 μ_{CR} = 0.149
 Diversity = 1.3327e-03 k_t = [2, 2, 3, 3]
 Generation 5250: Best = 5.212634e+03 FES = 525100 μ_F = 0.778 μ_{CR} = 0.161
 Diversity = 1.4353e-03 k_t = [1, 1, 1, 1]
 Generation 5260: Best = 5.212633e+03 FES = 526100 μ_F = 0.784 μ_{CR} = 0.148
 Diversity = 1.3558e-03 k_t = [2, 4, 4, 4]
 Generation 5270: Best = 5.212629e+03 FES = 527100 μ_F = 0.789 μ_{CR} = 0.156
 Diversity = 1.4929e-03 k_t = [1, 1, 1, 1]
 Generation 5280: Best = 5.212627e+03 FES = 528100 μ_F = 0.795 μ_{CR} = 0.152
 Diversity = 1.6569e-03 k_t = [2, 3, 3, 3]
 Generation 5290: Best = 5.212625e+03 FES = 529100 μ_F = 0.802 μ_{CR} = 0.148
 Diversity = 1.8897e-03 k_t = [1, 1, 1, 1]
 Generation 5300: Best = 5.212623e+03 FES = 530100 μ_F = 0.795 μ_{CR} = 0.150
 Diversity = 1.9617e-03 k_t = [4, 2, 3, 2]
 Generation 5310: Best = 5.212622e+03 FES = 531100 μ_F = 0.776 μ_{CR} = 0.147
 Diversity = 2.0089e-03 k_t = [1, 1, 1, 1]
 Generation 5320: Best = 5.212620e+03 FES = 532100 μ_F = 0.753 μ_{CR} = 0.147
 Diversity = 1.9964e-03 k_t = [3, 3, 2, 2]
 Generation 5330: Best = 5.212619e+03 FES = 533100 μ_F = 0.747 μ_{CR} = 0.151
 Diversity = 1.9894e-03 k_t = [1, 1, 1, 1]
 Generation 5340: Best = 5.212618e+03 FES = 534100 μ_F = 0.744 μ_{CR} = 0.137
 Diversity = 1.8468e-03 k_t = [4, 2, 4, 4]
 Generation 5350: Best = 5.212616e+03 FES = 535100 μ_F = 0.737 μ_{CR} = 0.139
 Diversity = 1.7926e-03 k_t = [1, 1, 1, 1]

Generation 5360: Best = 5.212615e+03 FES = 536100 μ_F = 0.738 μ_{CR} = 0.131
 Diversity = 1.8224e-03 k_t = [2, 3, 4, 2]
 Generation 5370: Best = 5.212613e+03 FES = 537100 μ_F = 0.739 μ_{CR} = 0.128
 Diversity = 1.8415e-03 k_t = [1, 1, 1, 1]
 Generation 5380: Best = 5.212611e+03 FES = 538100 μ_F = 0.731 μ_{CR} = 0.133
 Diversity = 1.7490e-03 k_t = [2, 3, 3, 3]
 Generation 5390: Best = 5.212610e+03 FES = 539100 μ_F = 0.748 μ_{CR} = 0.126
 Diversity = 1.7759e-03 k_t = [1, 1, 1, 1]
 Generation 5400: Best = 5.212608e+03 FES = 540100 μ_F = 0.750 μ_{CR} = 0.133
 Diversity = 1.7502e-03 k_t = [2, 2, 2, 4]
 Generation 5410: Best = 5.212607e+03 FES = 541100 μ_F = 0.751 μ_{CR} = 0.131
 Diversity = 1.8135e-03 k_t = [1, 1, 1, 1]
 Generation 5420: Best = 5.212605e+03 FES = 542100 μ_F = 0.761 μ_{CR} = 0.127
 Diversity = 1.7297e-03 k_t = [2, 4, 2, 4]
 Generation 5430: Best = 5.212603e+03 FES = 543100 μ_F = 0.759 μ_{CR} = 0.120
 Diversity = 1.8504e-03 k_t = [1, 1, 1, 1]
 Generation 5440: Best = 5.212602e+03 FES = 544100 μ_F = 0.770 μ_{CR} = 0.120
 Diversity = 1.8606e-03 k_t = [4, 2, 3, 4]
 Generation 5450: Best = 5.212599e+03 FES = 545100 μ_F = 0.758 μ_{CR} = 0.133
 Diversity = 1.8864e-03 k_t = [1, 1, 1, 1]
 Generation 5460: Best = 5.212598e+03 FES = 546100 μ_F = 0.760 μ_{CR} = 0.126
 Diversity = 1.8166e-03 k_t = [3, 4, 3, 3]
 Generation 5470: Best = 5.212596e+03 FES = 547100 μ_F = 0.748 μ_{CR} = 0.125
 Diversity = 1.9098e-03 k_t = [1, 1, 1, 1]
 Generation 5480: Best = 5.212594e+03 FES = 548100 μ_F = 0.758 μ_{CR} = 0.118
 Diversity = 1.8401e-03 k_t = [2, 3, 4, 4]
 Generation 5490: Best = 5.212593e+03 FES = 549100 μ_F = 0.754 μ_{CR} = 0.123
 Diversity = 1.9583e-03 k_t = [1, 1, 1, 1]
 Generation 5500: Best = 5.212592e+03 FES = 550100 μ_F = 0.745 μ_{CR} = 0.122
 Diversity = 1.9447e-03 k_t = [4, 3, 4, 3]
 Generation 5510: Best = 5.212587e+03 FES = 551100 μ_F = 0.746 μ_{CR} = 0.124
 Diversity = 2.0283e-03 k_t = [1, 1, 1, 1]
 Generation 5520: Best = 5.212586e+03 FES = 552100 μ_F = 0.746 μ_{CR} = 0.133
 Diversity = 2.1156e-03 k_t = [2, 3, 2, 2]
 Generation 5530: Best = 5.212586e+03 FES = 553100 μ_F = 0.738 μ_{CR} = 0.128
 Diversity = 2.1085e-03 k_t = [1, 1, 1, 1]
 Generation 5540: Best = 5.212584e+03 FES = 554100 μ_F = 0.733 μ_{CR} = 0.129
 Diversity = 1.9969e-03 k_t = [3, 4, 3, 4]
 Generation 5550: Best = 5.212583e+03 FES = 555100 μ_F = 0.740 μ_{CR} = 0.123
 Diversity = 1.9662e-03 k_t = [1, 1, 1, 1]
 Generation 5560: Best = 5.212578e+03 FES = 556100 μ_F = 0.732 μ_{CR} = 0.127
 Diversity = 2.1219e-03 k_t = [4, 4, 3, 2]
 Generation 5570: Best = 5.212576e+03 FES = 557100 μ_F = 0.751 μ_{CR} = 0.127
 Diversity = 2.3022e-03 k_t = [1, 1, 1, 1]
 Generation 5580: Best = 5.212575e+03 FES = 558100 μ_F = 0.759 μ_{CR} = 0.119
 Diversity = 2.3461e-03 k_t = [3, 2, 4, 2]
 Generation 5590: Best = 5.212574e+03 FES = 559100 μ_F = 0.765 μ_{CR} = 0.125
 Diversity = 2.3769e-03 k_t = [1, 1, 1, 1]
 Generation 5600: Best = 5.212573e+03 FES = 560100 μ_F = 0.755 μ_{CR} = 0.115
 Diversity = 2.2170e-03 k_t = [3, 2, 2, 4]
 Generation 5610: Best = 5.212571e+03 FES = 561100 μ_F = 0.756 μ_{CR} = 0.111
 Diversity = 2.1023e-03 k_t = [1, 1, 1, 1]
 Generation 5620: Best = 5.212569e+03 FES = 562100 μ_F = 0.772 μ_{CR} = 0.104
 Diversity = 2.2457e-03 k_t = [2, 3, 3, 2]

Generation 5630: Best = 5.212568e+03 FES = 563100 μ_F = 0.772 μ_{CR} = 0.106
 Diversity = 2.2895e-03 k_t = [1, 1, 1, 1]
 Generation 5640: Best = 5.212567e+03 FES = 564100 μ_F = 0.745 μ_{CR} = 0.106
 Diversity = 2.0018e-03 k_t = [2, 4, 4, 4]
 Generation 5650: Best = 5.212565e+03 FES = 565100 μ_F = 0.742 μ_{CR} = 0.116
 Diversity = 1.9248e-03 k_t = [1, 1, 1, 1]
 Generation 5660: Best = 5.212564e+03 FES = 566100 μ_F = 0.754 μ_{CR} = 0.112
 Diversity = 1.9064e-03 k_t = [2, 4, 2, 3]
 Generation 5670: Best = 5.212563e+03 FES = 567100 μ_F = 0.752 μ_{CR} = 0.119
 Diversity = 1.9986e-03 k_t = [1, 1, 1, 1]
 Generation 5680: Best = 5.212560e+03 FES = 568100 μ_F = 0.750 μ_{CR} = 0.113
 Diversity = 1.8545e-03 k_t = [3, 3, 4, 4]
 Generation 5690: Best = 5.212559e+03 FES = 569100 μ_F = 0.756 μ_{CR} = 0.110
 Diversity = 1.8969e-03 k_t = [1, 1, 1, 1]
 Generation 5700: Best = 5.212558e+03 FES = 570100 μ_F = 0.762 μ_{CR} = 0.124
 Diversity = 1.8118e-03 k_t = [3, 3, 2, 4]
 Generation 5710: Best = 5.212556e+03 FES = 571100 μ_F = 0.757 μ_{CR} = 0.117
 Diversity = 1.7143e-03 k_t = [1, 1, 1, 1]
 Generation 5720: Best = 5.212555e+03 FES = 572100 μ_F = 0.748 μ_{CR} = 0.124
 Diversity = 1.6553e-03 k_t = [4, 3, 3, 3]
 Generation 5730: Best = 5.212553e+03 FES = 573100 μ_F = 0.744 μ_{CR} = 0.123
 Diversity = 1.6126e-03 k_t = [1, 1, 1, 1]
 Generation 5740: Best = 5.212552e+03 FES = 574100 μ_F = 0.742 μ_{CR} = 0.124
 Diversity = 1.6114e-03 k_t = [3, 3, 3, 2]
 Generation 5750: Best = 5.212551e+03 FES = 575100 μ_F = 0.738 μ_{CR} = 0.130
 Diversity = 1.6559e-03 k_t = [1, 1, 1, 1]
 Generation 5760: Best = 5.212550e+03 FES = 576100 μ_F = 0.741 μ_{CR} = 0.122
 Diversity = 1.5894e-03 k_t = [3, 2, 4, 2]
 Generation 5770: Best = 5.212549e+03 FES = 577100 μ_F = 0.731 μ_{CR} = 0.123
 Diversity = 1.5879e-03 k_t = [1, 1, 1, 1]
 Generation 5780: Best = 5.212548e+03 FES = 578100 μ_F = 0.724 μ_{CR} = 0.122
 Diversity = 1.5310e-03 k_t = [2, 4, 4, 3]
 Generation 5790: Best = 5.212547e+03 FES = 579100 μ_F = 0.721 μ_{CR} = 0.117
 Diversity = 1.5205e-03 k_t = [1, 1, 1, 1]
 Generation 5800: Best = 5.212545e+03 FES = 580100 μ_F = 0.729 μ_{CR} = 0.112
 Diversity = 1.4211e-03 k_t = [3, 2, 2, 3]
 Generation 5810: Best = 5.212545e+03 FES = 581100 μ_F = 0.730 μ_{CR} = 0.105
 Diversity = 1.4807e-03 k_t = [1, 1, 1, 1]
 Generation 5820: Best = 5.212544e+03 FES = 582100 μ_F = 0.743 μ_{CR} = 0.111
 Diversity = 1.3958e-03 k_t = [3, 4, 3, 4]
 Generation 5830: Best = 5.212543e+03 FES = 583100 μ_F = 0.746 μ_{CR} = 0.117
 Diversity = 1.2525e-03 k_t = [1, 1, 1, 1]
 Generation 5840: Best = 5.212543e+03 FES = 584100 μ_F = 0.746 μ_{CR} = 0.109
 Diversity = 1.1913e-03 k_t = [4, 3, 3, 3]
 Generation 5850: Best = 5.212542e+03 FES = 585100 μ_F = 0.750 μ_{CR} = 0.108
 Diversity = 1.2325e-03 k_t = [1, 1, 1, 1]
 Generation 5860: Best = 5.212541e+03 FES = 586100 μ_F = 0.762 μ_{CR} = 0.109
 Diversity = 1.2094e-03 k_t = [4, 4, 3, 2]
 Generation 5870: Best = 5.212541e+03 FES = 587100 μ_F = 0.751 μ_{CR} = 0.109
 Diversity = 1.2197e-03 k_t = [1, 1, 1, 1]
 Generation 5880: Best = 5.212540e+03 FES = 588100 μ_F = 0.751 μ_{CR} = 0.118
 Diversity = 1.2141e-03 k_t = [3, 4, 4, 3]
 Generation 5890: Best = 5.212538e+03 FES = 589100 μ_F = 0.745 μ_{CR} = 0.114
 Diversity = 1.2360e-03 k_t = [1, 1, 1, 1]

Generation 5900: Best = 5.212537e+03 FES = 590100 μ_F = 0.769 μ_{CR} = 0.116
 Diversity = 1.3352e-03 k_t = [3, 3, 4, 2]
 Generation 5910: Best = 5.212536e+03 FES = 591100 μ_F = 0.782 μ_{CR} = 0.123
 Diversity = 1.3940e-03 k_t = [1, 1, 1, 1]
 Generation 5920: Best = 5.212534e+03 FES = 592100 μ_F = 0.782 μ_{CR} = 0.130
 Diversity = 1.4667e-03 k_t = [3, 4, 3, 2]
 Generation 5930: Best = 5.212532e+03 FES = 593100 μ_F = 0.781 μ_{CR} = 0.127
 Diversity = 1.6789e-03 k_t = [1, 1, 1, 1]
 Generation 5940: Best = 5.212531e+03 FES = 594100 μ_F = 0.779 μ_{CR} = 0.122
 Diversity = 1.7349e-03 k_t = [4, 3, 3, 2]
 Generation 5950: Best = 5.212530e+03 FES = 595100 μ_F = 0.768 μ_{CR} = 0.121
 Diversity = 1.8626e-03 k_t = [1, 1, 1, 1]
 Generation 5960: Best = 5.212529e+03 FES = 596100 μ_F = 0.750 μ_{CR} = 0.120
 Diversity = 1.8406e-03 k_t = [2, 2, 3, 4]
 Generation 5970: Best = 5.212527e+03 FES = 597100 μ_F = 0.742 μ_{CR} = 0.127
 Diversity = 1.7836e-03 k_t = [1, 1, 1, 1]
 Generation 5980: Best = 5.212525e+03 FES = 598100 μ_F = 0.732 μ_{CR} = 0.133
 Diversity = 1.6853e-03 k_t = [3, 3, 3, 2]
 Generation 5990: Best = 5.212522e+03 FES = 599100 μ_F = 0.730 μ_{CR} = 0.125
 Diversity = 1.7932e-03 k_t = [1, 1, 1, 1]
 Generation 6000: Best = 5.212521e+03 FES = 600100 μ_F = 0.729 μ_{CR} = 0.130
 Diversity = 1.7441e-03 k_t = [4, 3, 3, 2]
 Generation 6010: Best = 5.212519e+03 FES = 601100 μ_F = 0.722 μ_{CR} = 0.138
 Diversity = 1.7015e-03 k_t = [1, 1, 1, 1]
 Generation 6020: Best = 5.212518e+03 FES = 602100 μ_F = 0.726 μ_{CR} = 0.142
 Diversity = 1.6019e-03 k_t = [4, 4, 3, 4]
 Generation 6030: Best = 5.212517e+03 FES = 603100 μ_F = 0.716 μ_{CR} = 0.138
 Diversity = 1.4834e-03 k_t = [1, 1, 1, 1]
 Generation 6040: Best = 5.212515e+03 FES = 604100 μ_F = 0.724 μ_{CR} = 0.136
 Diversity = 1.3608e-03 k_t = [2, 4, 3, 4]
 Generation 6050: Best = 5.212514e+03 FES = 605100 μ_F = 0.728 μ_{CR} = 0.129
 Diversity = 1.3182e-03 k_t = [1, 1, 1, 1]
 Generation 6060: Best = 5.212513e+03 FES = 606100 μ_F = 0.747 μ_{CR} = 0.131
 Diversity = 1.3004e-03 k_t = [4, 3, 2, 3]
 Generation 6070: Best = 5.212512e+03 FES = 607100 μ_F = 0.749 μ_{CR} = 0.127
 Diversity = 1.3603e-03 k_t = [1, 1, 1, 1]
 Generation 6080: Best = 5.212510e+03 FES = 608100 μ_F = 0.732 μ_{CR} = 0.127
 Diversity = 1.3561e-03 k_t = [4, 2, 3, 2]
 Generation 6090: Best = 5.212510e+03 FES = 609100 μ_F = 0.720 μ_{CR} = 0.125
 Diversity = 1.3352e-03 k_t = [1, 1, 1, 1]
 Generation 6100: Best = 5.212508e+03 FES = 610100 μ_F = 0.720 μ_{CR} = 0.131
 Diversity = 1.2880e-03 k_t = [2, 3, 3, 4]
 Generation 6110: Best = 5.212507e+03 FES = 611100 μ_F = 0.729 μ_{CR} = 0.141
 Diversity = 1.1950e-03 k_t = [1, 1, 1, 1]
 Generation 6120: Best = 5.212505e+03 FES = 612100 μ_F = 0.737 μ_{CR} = 0.139
 Diversity = 1.2374e-03 k_t = [3, 4, 4, 2]
 Generation 6130: Best = 5.212503e+03 FES = 613100 μ_F = 0.752 μ_{CR} = 0.137
 Diversity = 1.3407e-03 k_t = [1, 1, 1, 1]
 Generation 6140: Best = 5.212503e+03 FES = 614100 μ_F = 0.749 μ_{CR} = 0.131
 Diversity = 1.4842e-03 k_t = [2, 4, 3, 2]
 Generation 6150: Best = 5.212502e+03 FES = 615100 μ_F = 0.740 μ_{CR} = 0.132
 Diversity = 1.5802e-03 k_t = [1, 1, 1, 1]
 Generation 6160: Best = 5.212500e+03 FES = 616100 μ_F = 0.742 μ_{CR} = 0.137
 Diversity = 1.4214e-03 k_t = [3, 3, 3, 3]

Generation 6170: Best = 5.212499e+03 FES = 617100 μ_F = 0.749 μ_{CR} = 0.128
 Diversity = 1.3613e-03 k_t = [1, 1, 1, 1]
 Generation 6180: Best = 5.212497e+03 FES = 618100 μ_F = 0.755 μ_{CR} = 0.135
 Diversity = 1.4244e-03 k_t = [3, 2, 4, 2]
 Generation 6190: Best = 5.212495e+03 FES = 619100 μ_F = 0.747 μ_{CR} = 0.132
 Diversity = 1.4763e-03 k_t = [1, 1, 1, 1]
 Generation 6200: Best = 5.212494e+03 FES = 620100 μ_F = 0.745 μ_{CR} = 0.124
 Diversity = 1.4231e-03 k_t = [4, 4, 3, 3]
 Generation 6210: Best = 5.212493e+03 FES = 621100 μ_F = 0.734 μ_{CR} = 0.122
 Diversity = 1.4245e-03 k_t = [1, 1, 1, 1]
 Generation 6220: Best = 5.212491e+03 FES = 622100 μ_F = 0.744 μ_{CR} = 0.121
 Diversity = 1.3734e-03 k_t = [2, 4, 4, 2]
 Generation 6230: Best = 5.212490e+03 FES = 623100 μ_F = 0.767 μ_{CR} = 0.138
 Diversity = 1.4625e-03 k_t = [1, 1, 1, 1]
 Generation 6240: Best = 5.212488e+03 FES = 624100 μ_F = 0.765 μ_{CR} = 0.156
 Diversity = 1.5459e-03 k_t = [4, 3, 4, 4]
 Generation 6250: Best = 5.212487e+03 FES = 625100 μ_F = 0.757 μ_{CR} = 0.149
 Diversity = 1.5109e-03 k_t = [1, 1, 1, 1]
 Generation 6260: Best = 5.212485e+03 FES = 626100 μ_F = 0.750 μ_{CR} = 0.153
 Diversity = 1.5874e-03 k_t = [3, 4, 3, 2]
 Generation 6270: Best = 5.212484e+03 FES = 627100 μ_F = 0.756 μ_{CR} = 0.153
 Diversity = 1.5964e-03 k_t = [1, 1, 1, 1]
 Generation 6280: Best = 5.212483e+03 FES = 628100 μ_F = 0.753 μ_{CR} = 0.160
 Diversity = 1.5277e-03 k_t = [3, 3, 4, 2]
 Generation 6290: Best = 5.212481e+03 FES = 629100 μ_F = 0.739 μ_{CR} = 0.168
 Diversity = 1.5273e-03 k_t = [1, 1, 1, 1]
 Generation 6300: Best = 5.212479e+03 FES = 630100 μ_F = 0.738 μ_{CR} = 0.156
 Diversity = 1.5495e-03 k_t = [3, 4, 2, 4]
 Generation 6310: Best = 5.212477e+03 FES = 631100 μ_F = 0.731 μ_{CR} = 0.150
 Diversity = 1.6606e-03 k_t = [1, 1, 1, 1]
 Generation 6320: Best = 5.212477e+03 FES = 632100 μ_F = 0.740 μ_{CR} = 0.145
 Diversity = 1.6443e-03 k_t = [2, 4, 2, 3]
 Generation 6330: Best = 5.212475e+03 FES = 633100 μ_F = 0.730 μ_{CR} = 0.143
 Diversity = 1.6256e-03 k_t = [1, 1, 1, 1]
 Generation 6340: Best = 5.212473e+03 FES = 634100 μ_F = 0.741 μ_{CR} = 0.146
 Diversity = 1.5930e-03 k_t = [3, 4, 4, 3]
 Generation 6350: Best = 5.212473e+03 FES = 635100 μ_F = 0.745 μ_{CR} = 0.144
 Diversity = 1.6376e-03 k_t = [1, 1, 1, 1]
 Generation 6360: Best = 5.212471e+03 FES = 636100 μ_F = 0.741 μ_{CR} = 0.141
 Diversity = 1.5652e-03 k_t = [2, 2, 4, 3]
 Generation 6370: Best = 5.212470e+03 FES = 637100 μ_F = 0.746 μ_{CR} = 0.146
 Diversity = 1.6098e-03 k_t = [1, 1, 1, 1]
 Generation 6380: Best = 5.212469e+03 FES = 638100 μ_F = 0.768 μ_{CR} = 0.149
 Diversity = 1.6985e-03 k_t = [4, 2, 4, 3]
 Generation 6390: Best = 5.212468e+03 FES = 639100 μ_F = 0.779 μ_{CR} = 0.138
 Diversity = 1.6636e-03 k_t = [1, 1, 1, 1]
 Generation 6400: Best = 5.212466e+03 FES = 640100 μ_F = 0.783 μ_{CR} = 0.128
 Diversity = 1.7653e-03 k_t = [4, 2, 4, 2]
 Generation 6410: Best = 5.212465e+03 FES = 641100 μ_F = 0.765 μ_{CR} = 0.122
 Diversity = 1.8132e-03 k_t = [1, 1, 1, 1]
 Generation 6420: Best = 5.212464e+03 FES = 642100 μ_F = 0.746 μ_{CR} = 0.122
 Diversity = 1.6877e-03 k_t = [4, 4, 3, 4]
 Generation 6430: Best = 5.212463e+03 FES = 643100 μ_F = 0.744 μ_{CR} = 0.126
 Diversity = 1.5382e-03 k_t = [1, 1, 1, 1]

Generation 6440: Best = 5.212463e+03 FES = 644100 μ_F = 0.751 μ_{CR} = 0.129
 Diversity = 1.4318e-03 k_t = [4, 3, 4, 4]
 Generation 6450: Best = 5.212462e+03 FES = 645100 μ_F = 0.752 μ_{CR} = 0.123
 Diversity = 1.4234e-03 k_t = [1, 1, 1, 1]
 Generation 6460: Best = 5.212462e+03 FES = 646100 μ_F = 0.763 μ_{CR} = 0.125
 Diversity = 1.3047e-03 k_t = [4, 4, 2, 4]
 Generation 6470: Best = 5.212461e+03 FES = 647100 μ_F = 0.765 μ_{CR} = 0.124
 Diversity = 1.2818e-03 k_t = [1, 1, 1, 1]
 Generation 6480: Best = 5.212460e+03 FES = 648100 μ_F = 0.756 μ_{CR} = 0.117
 Diversity = 1.2629e-03 k_t = [4, 3, 4, 3]
 Generation 6490: Best = 5.212459e+03 FES = 649100 μ_F = 0.764 μ_{CR} = 0.125
 Diversity = 1.2638e-03 k_t = [1, 1, 1, 1]
 Generation 6500: Best = 5.212458e+03 FES = 650100 μ_F = 0.759 μ_{CR} = 0.126
 Diversity = 1.2412e-03 k_t = [3, 2, 2, 3]
 Generation 6510: Best = 5.212457e+03 FES = 651100 μ_F = 0.754 μ_{CR} = 0.128
 Diversity = 1.3363e-03 k_t = [1, 1, 1, 1]
 Generation 6520: Best = 5.212456e+03 FES = 652100 μ_F = 0.744 μ_{CR} = 0.120
 Diversity = 1.2906e-03 k_t = [4, 2, 4, 4]
 Generation 6530: Best = 5.212456e+03 FES = 653100 μ_F = 0.746 μ_{CR} = 0.124
 Diversity = 1.2326e-03 k_t = [1, 1, 1, 1]
 Generation 6540: Best = 5.212454e+03 FES = 654100 μ_F = 0.761 μ_{CR} = 0.121
 Diversity = 1.2583e-03 k_t = [4, 2, 3, 4]
 Generation 6550: Best = 5.212454e+03 FES = 655100 μ_F = 0.750 μ_{CR} = 0.119
 Diversity = 1.2570e-03 k_t = [1, 1, 1, 1]
 Generation 6560: Best = 5.212453e+03 FES = 656100 μ_F = 0.744 μ_{CR} = 0.112
 Diversity = 1.2273e-03 k_t = [3, 4, 4, 3]
 Generation 6570: Best = 5.212452e+03 FES = 657100 μ_F = 0.736 μ_{CR} = 0.115
 Diversity = 1.1852e-03 k_t = [1, 1, 1, 1]
 Generation 6580: Best = 5.212452e+03 FES = 658100 μ_F = 0.735 μ_{CR} = 0.117
 Diversity = 1.1571e-03 k_t = [3, 4, 4, 4]
 Generation 6590: Best = 5.212451e+03 FES = 659100 μ_F = 0.736 μ_{CR} = 0.121
 Diversity = 1.1109e-03 k_t = [1, 1, 1, 1]
 Generation 6600: Best = 5.212451e+03 FES = 660100 μ_F = 0.752 μ_{CR} = 0.123
 Diversity = 1.1855e-03 k_t = [2, 4, 3, 2]
 Generation 6610: Best = 5.212450e+03 FES = 661100 μ_F = 0.750 μ_{CR} = 0.121
 Diversity = 1.1686e-03 k_t = [1, 1, 1, 1]
 Generation 6620: Best = 5.212449e+03 FES = 662100 μ_F = 0.748 μ_{CR} = 0.126
 Diversity = 1.1307e-03 k_t = [3, 3, 2, 4]
 Generation 6630: Best = 5.212448e+03 FES = 663100 μ_F = 0.756 μ_{CR} = 0.125
 Diversity = 1.0765e-03 k_t = [1, 1, 1, 1]
 Generation 6640: Best = 5.212447e+03 FES = 664100 μ_F = 0.746 μ_{CR} = 0.131
 Diversity = 9.9418e-04 k_t = [3, 4, 4, 4]
 Generation 6650: Best = 5.212447e+03 FES = 665100 μ_F = 0.759 μ_{CR} = 0.123
 Diversity = 1.0469e-03 k_t = [1, 1, 1, 1]
 Generation 6660: Best = 5.212447e+03 FES = 666100 μ_F = 0.756 μ_{CR} = 0.117
 Diversity = 1.0654e-03 k_t = [2, 2, 4, 2]
 Generation 6670: Best = 5.212446e+03 FES = 667100 μ_F = 0.762 μ_{CR} = 0.115
 Diversity = 1.0534e-03 k_t = [1, 1, 1, 1]
 Generation 6680: Best = 5.212445e+03 FES = 668100 μ_F = 0.750 μ_{CR} = 0.125
 Diversity = 1.0194e-03 k_t = [3, 2, 4, 2]
 Generation 6690: Best = 5.212445e+03 FES = 669100 μ_F = 0.746 μ_{CR} = 0.121
 Diversity = 1.0911e-03 k_t = [1, 1, 1, 1]
 Generation 6700: Best = 5.212444e+03 FES = 670100 μ_F = 0.749 μ_{CR} = 0.123
 Diversity = 1.0361e-03 k_t = [2, 4, 4, 4]

Generation 6710: Best = 5.212443e+03 FES = 671100 μ_F = 0.755 μ_{CR} = 0.139
 Diversity = 1.1024e-03 k_t = [1, 1, 1, 1]
 Generation 6720: Best = 5.212441e+03 FES = 672100 μ_F = 0.744 μ_{CR} = 0.148
 Diversity = 1.0937e-03 k_t = [2, 4, 4, 2]
 Generation 6730: Best = 5.212441e+03 FES = 673100 μ_F = 0.744 μ_{CR} = 0.145
 Diversity = 1.1242e-03 k_t = [1, 1, 1, 1]
 Generation 6740: Best = 5.212440e+03 FES = 674100 μ_F = 0.751 μ_{CR} = 0.150
 Diversity = 1.1870e-03 k_t = [4, 3, 2, 2]
 Generation 6750: Best = 5.212439e+03 FES = 675100 μ_F = 0.745 μ_{CR} = 0.152
 Diversity = 1.2772e-03 k_t = [1, 1, 1, 1]
 Generation 6760: Best = 5.212438e+03 FES = 676100 μ_F = 0.755 μ_{CR} = 0.147
 Diversity = 1.1460e-03 k_t = [3, 2, 3, 4]
 Generation 6770: Best = 5.212437e+03 FES = 677100 μ_F = 0.754 μ_{CR} = 0.146
 Diversity = 1.1010e-03 k_t = [1, 1, 1, 1]
 Generation 6780: Best = 5.212437e+03 FES = 678100 μ_F = 0.753 μ_{CR} = 0.148
 Diversity = 1.1035e-03 k_t = [4, 3, 4, 3]
 Generation 6790: Best = 5.212436e+03 FES = 679100 μ_F = 0.732 μ_{CR} = 0.151
 Diversity = 1.0933e-03 k_t = [1, 1, 1, 1]
 Generation 6800: Best = 5.212434e+03 FES = 680100 μ_F = 0.742 μ_{CR} = 0.152
 Diversity = 1.1211e-03 k_t = [4, 3, 3, 2]
 Generation 6810: Best = 5.212434e+03 FES = 681100 μ_F = 0.741 μ_{CR} = 0.150
 Diversity = 1.2057e-03 k_t = [1, 1, 1, 1]
 Generation 6820: Best = 5.212433e+03 FES = 682100 μ_F = 0.757 μ_{CR} = 0.154
 Diversity = 1.2886e-03 k_t = [4, 2, 2, 2]
 Generation 6830: Best = 5.212432e+03 FES = 683100 μ_F = 0.756 μ_{CR} = 0.163
 Diversity = 1.3622e-03 k_t = [1, 1, 1, 1]
 Generation 6840: Best = 5.212431e+03 FES = 684100 μ_F = 0.759 μ_{CR} = 0.158
 Diversity = 1.3486e-03 k_t = [2, 4, 2, 4]
 Generation 6850: Best = 5.212430e+03 FES = 685100 μ_F = 0.760 μ_{CR} = 0.148
 Diversity = 1.3162e-03 k_t = [1, 1, 1, 1]
 Generation 6860: Best = 5.212430e+03 FES = 686100 μ_F = 0.753 μ_{CR} = 0.159
 Diversity = 1.3240e-03 k_t = [4, 4, 4, 3]
 Generation 6870: Best = 5.212429e+03 FES = 687100 μ_F = 0.758 μ_{CR} = 0.157
 Diversity = 1.3417e-03 k_t = [1, 1, 1, 1]
 Generation 6880: Best = 5.212428e+03 FES = 688100 μ_F = 0.753 μ_{CR} = 0.157
 Diversity = 1.2816e-03 k_t = [4, 4, 4, 4]
 Generation 6890: Best = 5.212427e+03 FES = 689100 μ_F = 0.742 μ_{CR} = 0.155
 Diversity = 1.1805e-03 k_t = [1, 1, 1, 1]
 Generation 6900: Best = 5.212426e+03 FES = 690100 μ_F = 0.738 μ_{CR} = 0.154
 Diversity = 1.1483e-03 k_t = [4, 3, 3, 4]
 Generation 6910: Best = 5.212426e+03 FES = 691100 μ_F = 0.742 μ_{CR} = 0.159
 Diversity = 1.1750e-03 k_t = [1, 1, 1, 1]
 Generation 6920: Best = 5.212425e+03 FES = 692100 μ_F = 0.738 μ_{CR} = 0.163
 Diversity = 1.1234e-03 k_t = [2, 4, 3, 4]
 Generation 6930: Best = 5.212425e+03 FES = 693100 μ_F = 0.749 μ_{CR} = 0.159
 Diversity = 1.0738e-03 k_t = [1, 1, 1, 1]
 Generation 6940: Best = 5.212424e+03 FES = 694100 μ_F = 0.755 μ_{CR} = 0.158
 Diversity = 1.0503e-03 k_t = [2, 4, 3, 4]
 Generation 6950: Best = 5.212423e+03 FES = 695100 μ_F = 0.745 μ_{CR} = 0.164
 Diversity = 1.0030e-03 k_t = [1, 1, 1, 1]
 Generation 6960: Best = 5.212423e+03 FES = 696100 μ_F = 0.744 μ_{CR} = 0.167
 Diversity = 9.9339e-04 k_t = [2, 2, 3, 4]
 Generation 6970: Best = 5.212422e+03 FES = 697100 μ_F = 0.754 μ_{CR} = 0.162
 Diversity = 9.9086e-04 k_t = [1, 1, 1, 1]

Generation 6980: Best = 5.212422e+03 FES = 698100 μ_F = 0.749 μ_{CR} = 0.158
 Diversity = 1.0107e-03 k_t = [2, 2, 2, 3]
 Generation 6990: Best = 5.212421e+03 FES = 699100 μ_F = 0.739 μ_{CR} = 0.152
 Diversity = 1.0870e-03 k_t = [1, 1, 1, 1]
 Generation 7000: Best = 5.212421e+03 FES = 700100 μ_F = 0.737 μ_{CR} = 0.153
 Diversity = 1.0903e-03 k_t = [3, 2, 3, 4]
 Generation 7010: Best = 5.212420e+03 FES = 701100 μ_F = 0.733 μ_{CR} = 0.145
 Diversity = 1.0354e-03 k_t = [1, 1, 1, 1]
 Generation 7020: Best = 5.212420e+03 FES = 702100 μ_F = 0.735 μ_{CR} = 0.138
 Diversity = 1.0673e-03 k_t = [4, 3, 3, 2]
 Generation 7030: Best = 5.212419e+03 FES = 703100 μ_F = 0.740 μ_{CR} = 0.137
 Diversity = 1.0407e-03 k_t = [1, 1, 1, 1]
 Generation 7040: Best = 5.212419e+03 FES = 704100 μ_F = 0.742 μ_{CR} = 0.138
 Diversity = 9.0141e-04 k_t = [2, 2, 2, 3]
 Generation 7050: Best = 5.212419e+03 FES = 705100 μ_F = 0.738 μ_{CR} = 0.133
 Diversity = 9.4864e-04 k_t = [1, 1, 1, 1]
 Generation 7060: Best = 5.212418e+03 FES = 706100 μ_F = 0.722 μ_{CR} = 0.137
 Diversity = 8.6936e-04 k_t = [2, 3, 4, 4]
 Generation 7070: Best = 5.212418e+03 FES = 707100 μ_F = 0.715 μ_{CR} = 0.146
 Diversity = 8.5096e-04 k_t = [1, 1, 1, 1]
 Generation 7080: Best = 5.212417e+03 FES = 708100 μ_F = 0.711 μ_{CR} = 0.150
 Diversity = 7.8515e-04 k_t = [3, 2, 3, 4]
 Generation 7090: Best = 5.212417e+03 FES = 709100 μ_F = 0.720 μ_{CR} = 0.156
 Diversity = 7.9448e-04 k_t = [1, 1, 1, 1]
 Generation 7100: Best = 5.212417e+03 FES = 710100 μ_F = 0.729 μ_{CR} = 0.143
 Diversity = 7.3903e-04 k_t = [4, 2, 3, 4]
 Generation 7110: Best = 5.212416e+03 FES = 711100 μ_F = 0.729 μ_{CR} = 0.134
 Diversity = 7.3761e-04 k_t = [1, 1, 1, 1]
 Generation 7120: Best = 5.212416e+03 FES = 712100 μ_F = 0.723 μ_{CR} = 0.144
 Diversity = 6.9535e-04 k_t = [3, 4, 4, 4]
 Generation 7130: Best = 5.212416e+03 FES = 713100 μ_F = 0.736 μ_{CR} = 0.148
 Diversity = 6.5927e-04 k_t = [1, 1, 1, 1]
 Generation 7140: Best = 5.212415e+03 FES = 714100 μ_F = 0.742 μ_{CR} = 0.143
 Diversity = 6.5151e-04 k_t = [3, 2, 3, 4]
 Generation 7150: Best = 5.212415e+03 FES = 715100 μ_F = 0.753 μ_{CR} = 0.144
 Diversity = 6.9342e-04 k_t = [1, 1, 1, 1]
 Generation 7160: Best = 5.212414e+03 FES = 716100 μ_F = 0.772 μ_{CR} = 0.151
 Diversity = 7.0322e-04 k_t = [4, 2, 4, 4]
 Generation 7170: Best = 5.212414e+03 FES = 717100 μ_F = 0.767 μ_{CR} = 0.159
 Diversity = 7.7163e-04 k_t = [1, 1, 1, 1]
 Generation 7180: Best = 5.212414e+03 FES = 718100 μ_F = 0.764 μ_{CR} = 0.158
 Diversity = 7.7718e-04 k_t = [3, 3, 4, 3]
 Generation 7190: Best = 5.212413e+03 FES = 719100 μ_F = 0.760 μ_{CR} = 0.159
 Diversity = 7.5743e-04 k_t = [1, 1, 1, 1]
 Generation 7200: Best = 5.212412e+03 FES = 720100 μ_F = 0.761 μ_{CR} = 0.162
 Diversity = 8.1520e-04 k_t = [2, 3, 2, 3]
 Generation 7210: Best = 5.212412e+03 FES = 721100 μ_F = 0.759 μ_{CR} = 0.164
 Diversity = 9.0194e-04 k_t = [1, 1, 1, 1]
 Generation 7220: Best = 5.212411e+03 FES = 722100 μ_F = 0.769 μ_{CR} = 0.175
 Diversity = 9.8564e-04 k_t = [4, 3, 4, 2]
 Generation 7230: Best = 5.212411e+03 FES = 723100 μ_F = 0.764 μ_{CR} = 0.178
 Diversity = 1.0145e-03 k_t = [1, 1, 1, 1]
 Generation 7240: Best = 5.212410e+03 FES = 724100 μ_F = 0.766 μ_{CR} = 0.168
 Diversity = 9.9376e-04 k_t = [4, 2, 3, 2]

Generation 7250: Best = 5.212410e+03 FES = 725100 μ_F = 0.760 μ_{CR} = 0.174
 Diversity = 1.0566e-03 k_t = [1, 1, 1, 1]
 Generation 7260: Best = 5.212409e+03 FES = 726100 μ_F = 0.747 μ_{CR} = 0.171
 Diversity = 9.7526e-04 k_t = [2, 4, 2, 4]
 Generation 7270: Best = 5.212408e+03 FES = 727100 μ_F = 0.735 μ_{CR} = 0.163
 Diversity = 9.5062e-04 k_t = [1, 1, 1, 1]
 Generation 7280: Best = 5.212408e+03 FES = 728100 μ_F = 0.749 μ_{CR} = 0.161
 Diversity = 8.9597e-04 k_t = [3, 2, 2, 4]
 Generation 7290: Best = 5.212408e+03 FES = 729100 μ_F = 0.746 μ_{CR} = 0.167
 Diversity = 8.1804e-04 k_t = [1, 1, 1, 1]
 Generation 7300: Best = 5.212407e+03 FES = 730100 μ_F = 0.747 μ_{CR} = 0.165
 Diversity = 7.7157e-04 k_t = [3, 2, 2, 3]
 Generation 7310: Best = 5.212407e+03 FES = 731100 μ_F = 0.756 μ_{CR} = 0.160
 Diversity = 8.4344e-04 k_t = [1, 1, 1, 1]
 Generation 7320: Best = 5.212406e+03 FES = 732100 μ_F = 0.747 μ_{CR} = 0.170
 Diversity = 8.5503e-04 k_t = [4, 2, 4, 3]
 Generation 7330: Best = 5.212406e+03 FES = 733100 μ_F = 0.748 μ_{CR} = 0.183
 Diversity = 9.0941e-04 k_t = [1, 1, 1, 1]
 Generation 7340: Best = 5.212406e+03 FES = 734100 μ_F = 0.737 μ_{CR} = 0.188
 Diversity = 7.8398e-04 k_t = [3, 3, 4, 3]
 Generation 7350: Best = 5.212405e+03 FES = 735100 μ_F = 0.744 μ_{CR} = 0.182
 Diversity = 7.6482e-04 k_t = [1, 1, 1, 1]
 Generation 7360: Best = 5.212405e+03 FES = 736100 μ_F = 0.744 μ_{CR} = 0.177
 Diversity = 7.5375e-04 k_t = [3, 4, 2, 4]
 Generation 7370: Best = 5.212404e+03 FES = 737100 μ_F = 0.737 μ_{CR} = 0.175
 Diversity = 7.4741e-04 k_t = [1, 1, 1, 1]
 Generation 7380: Best = 5.212404e+03 FES = 738100 μ_F = 0.749 μ_{CR} = 0.177
 Diversity = 7.4219e-04 k_t = [4, 2, 3, 4]
 Generation 7390: Best = 5.212403e+03 FES = 739100 μ_F = 0.754 μ_{CR} = 0.180
 Diversity = 8.1268e-04 k_t = [1, 1, 1, 1]
 Generation 7400: Best = 5.212403e+03 FES = 740100 μ_F = 0.761 μ_{CR} = 0.176
 Diversity = 7.8492e-04 k_t = [4, 4, 3, 4]
 Generation 7410: Best = 5.212403e+03 FES = 741100 μ_F = 0.788 μ_{CR} = 0.172
 Diversity = 7.6921e-04 k_t = [1, 1, 1, 1]
 Generation 7420: Best = 5.212403e+03 FES = 742100 μ_F = 0.772 μ_{CR} = 0.168
 Diversity = 7.7151e-04 k_t = [4, 3, 4, 3]
 Generation 7430: Best = 5.212402e+03 FES = 743100 μ_F = 0.763 μ_{CR} = 0.158
 Diversity = 8.2195e-04 k_t = [1, 1, 1, 1]
 Generation 7440: Best = 5.212402e+03 FES = 744100 μ_F = 0.756 μ_{CR} = 0.159
 Diversity = 8.3182e-04 k_t = [4, 3, 2, 4]
 Generation 7450: Best = 5.212401e+03 FES = 745100 μ_F = 0.748 μ_{CR} = 0.163
 Diversity = 8.3199e-04 k_t = [1, 1, 1, 1]
 Generation 7460: Best = 5.212401e+03 FES = 746100 μ_F = 0.759 μ_{CR} = 0.175
 Diversity = 8.6334e-04 k_t = [2, 3, 3, 3]
 Generation 7470: Best = 5.212400e+03 FES = 747100 μ_F = 0.760 μ_{CR} = 0.176
 Diversity = 9.8304e-04 k_t = [1, 1, 1, 1]
 Generation 7480: Best = 5.212400e+03 FES = 748100 μ_F = 0.751 μ_{CR} = 0.167
 Diversity = 1.0046e-03 k_t = [4, 2, 2, 3]
 Generation 7490: Best = 5.212400e+03 FES = 749100 μ_F = 0.745 μ_{CR} = 0.162
 Diversity = 1.0248e-03 k_t = [1, 1, 1, 1]
 Generation 7500: Best = 5.212399e+03 FES = 750100 μ_F = 0.730 μ_{CR} = 0.166
 Diversity = 1.0384e-03 k_t = [3, 3, 4, 2]
 Generation 7510: Best = 5.212399e+03 FES = 751100 μ_F = 0.737 μ_{CR} = 0.157
 Diversity = 1.0161e-03 k_t = [1, 1, 1, 1]

Generation 7520: Best = 5.212398e+03 FES = 752100 μ_F = 0.742 μ_{CR} = 0.153
 Diversity = 1.0482e-03 k_t = [2, 2, 3, 2]
 Generation 7530: Best = 5.212397e+03 FES = 753100 μ_F = 0.724 μ_{CR} = 0.153
 Diversity = 1.1319e-03 k_t = [1, 1, 1, 1]
 Generation 7540: Best = 5.212396e+03 FES = 754100 μ_F = 0.727 μ_{CR} = 0.160
 Diversity = 1.0869e-03 k_t = [4, 2, 3, 4]
 Generation 7550: Best = 5.212396e+03 FES = 755100 μ_F = 0.725 μ_{CR} = 0.149
 Diversity = 1.0021e-03 k_t = [1, 1, 1, 1]
 Generation 7560: Best = 5.212396e+03 FES = 756100 μ_F = 0.728 μ_{CR} = 0.142
 Diversity = 9.4216e-04 k_t = [4, 4, 3, 4]
 Generation 7570: Best = 5.212396e+03 FES = 757100 μ_F = 0.742 μ_{CR} = 0.146
 Diversity = 8.7391e-04 k_t = [1, 1, 1, 1]
 Generation 7580: Best = 5.212396e+03 FES = 758100 μ_F = 0.744 μ_{CR} = 0.145
 Diversity = 8.5138e-04 k_t = [2, 3, 4, 3]
 Generation 7590: Best = 5.212395e+03 FES = 759100 μ_F = 0.761 μ_{CR} = 0.153
 Diversity = 8.6378e-04 k_t = [1, 1, 1, 1]
 Generation 7600: Best = 5.212395e+03 FES = 760100 μ_F = 0.762 μ_{CR} = 0.150
 Diversity = 9.0713e-04 k_t = [4, 2, 3, 2]
 Generation 7610: Best = 5.212395e+03 FES = 761100 μ_F = 0.764 μ_{CR} = 0.143
 Diversity = 9.3534e-04 k_t = [1, 1, 1, 1]
 Generation 7620: Best = 5.212394e+03 FES = 762100 μ_F = 0.775 μ_{CR} = 0.144
 Diversity = 9.9120e-04 k_t = [2, 2, 4, 2]
 Generation 7630: Best = 5.212393e+03 FES = 763100 μ_F = 0.769 μ_{CR} = 0.159
 Diversity = 1.1609e-03 k_t = [1, 1, 1, 1]
 Generation 7640: Best = 5.212393e+03 FES = 764100 μ_F = 0.768 μ_{CR} = 0.153
 Diversity = 1.1663e-03 k_t = [2, 2, 3, 2]
 Generation 7650: Best = 5.212392e+03 FES = 765100 μ_F = 0.761 μ_{CR} = 0.148
 Diversity = 1.1588e-03 k_t = [1, 1, 1, 1]
 Generation 7660: Best = 5.212392e+03 FES = 766100 μ_F = 0.756 μ_{CR} = 0.152
 Diversity = 1.0826e-03 k_t = [2, 3, 3, 3]
 Generation 7670: Best = 5.212391e+03 FES = 767100 μ_F = 0.746 μ_{CR} = 0.142
 Diversity = 1.0814e-03 k_t = [1, 1, 1, 1]
 Generation 7680: Best = 5.212391e+03 FES = 768100 μ_F = 0.748 μ_{CR} = 0.143
 Diversity = 1.0469e-03 k_t = [3, 2, 3, 4]
 Generation 7690: Best = 5.212390e+03 FES = 769100 μ_F = 0.739 μ_{CR} = 0.139
 Diversity = 1.0489e-03 k_t = [1, 1, 1, 1]
 Generation 7700: Best = 5.212390e+03 FES = 770100 μ_F = 0.744 μ_{CR} = 0.140
 Diversity = 9.8938e-04 k_t = [2, 2, 3, 4]
 Generation 7710: Best = 5.212390e+03 FES = 771100 μ_F = 0.738 μ_{CR} = 0.135
 Diversity = 9.6610e-04 k_t = [1, 1, 1, 1]
 Generation 7720: Best = 5.212389e+03 FES = 772100 μ_F = 0.730 μ_{CR} = 0.133
 Diversity = 9.4017e-04 k_t = [2, 2, 2, 2]
 Generation 7730: Best = 5.212389e+03 FES = 773100 μ_F = 0.710 μ_{CR} = 0.132
 Diversity = 1.0324e-03 k_t = [1, 1, 1, 1]
 Generation 7740: Best = 5.212388e+03 FES = 774100 μ_F = 0.739 μ_{CR} = 0.133
 Diversity = 1.0572e-03 k_t = [4, 2, 3, 4]
 Generation 7750: Best = 5.212388e+03 FES = 775100 μ_F = 0.743 μ_{CR} = 0.127
 Diversity = 1.0871e-03 k_t = [1, 1, 1, 1]
 Generation 7760: Best = 5.212387e+03 FES = 776100 μ_F = 0.742 μ_{CR} = 0.128
 Diversity = 1.0947e-03 k_t = [2, 4, 4, 2]
 Generation 7770: Best = 5.212386e+03 FES = 777100 μ_F = 0.749 μ_{CR} = 0.126
 Diversity = 1.1766e-03 k_t = [1, 1, 1, 1]
 Generation 7780: Best = 5.212386e+03 FES = 778100 μ_F = 0.748 μ_{CR} = 0.129
 Diversity = 1.1474e-03 k_t = [4, 2, 3, 2]

Generation 7790: Best = 5.212385e+03 FES = 779100 μ_F = 0.750 μ_{CR} = 0.124
 Diversity = 1.2038e-03 k_t = [1, 1, 1, 1]
 Generation 7800: Best = 5.212385e+03 FES = 780100 μ_F = 0.746 μ_{CR} = 0.132
 Diversity = 1.1133e-03 k_t = [2, 3, 2, 4]
 Generation 7810: Best = 5.212384e+03 FES = 781100 μ_F = 0.742 μ_{CR} = 0.133
 Diversity = 1.0739e-03 k_t = [1, 1, 1, 1]
 Generation 7820: Best = 5.212383e+03 FES = 782100 μ_F = 0.730 μ_{CR} = 0.130
 Diversity = 1.1111e-03 k_t = [4, 4, 2, 2]
 Generation 7830: Best = 5.212383e+03 FES = 783100 μ_F = 0.747 μ_{CR} = 0.131
 Diversity = 1.0875e-03 k_t = [1, 1, 1, 1]
 Generation 7840: Best = 5.212382e+03 FES = 784100 μ_F = 0.750 μ_{CR} = 0.126
 Diversity = 1.0063e-03 k_t = [4, 4, 2, 3]
 Generation 7850: Best = 5.212382e+03 FES = 785100 μ_F = 0.736 μ_{CR} = 0.122
 Diversity = 9.8126e-04 k_t = [1, 1, 1, 1]
 Generation 7860: Best = 5.212382e+03 FES = 786100 μ_F = 0.738 μ_{CR} = 0.118
 Diversity = 9.7323e-04 k_t = [2, 2, 4, 2]
 Generation 7870: Best = 5.212382e+03 FES = 787100 μ_F = 0.739 μ_{CR} = 0.114
 Diversity = 9.3147e-04 k_t = [1, 1, 1, 1]
 Generation 7880: Best = 5.212381e+03 FES = 788100 μ_F = 0.734 μ_{CR} = 0.118
 Diversity = 8.3803e-04 k_t = [4, 4, 4, 4]
 Generation 7890: Best = 5.212381e+03 FES = 789100 μ_F = 0.750 μ_{CR} = 0.124
 Diversity = 8.3725e-04 k_t = [1, 1, 1, 1]
 Generation 7900: Best = 5.212380e+03 FES = 790100 μ_F = 0.764 μ_{CR} = 0.124
 Diversity = 8.9034e-04 k_t = [4, 3, 3, 4]
 Generation 7910: Best = 5.212380e+03 FES = 791100 μ_F = 0.750 μ_{CR} = 0.132
 Diversity = 8.9691e-04 k_t = [1, 1, 1, 1]
 Generation 7920: Best = 5.212380e+03 FES = 792100 μ_F = 0.761 μ_{CR} = 0.131
 Diversity = 9.0401e-04 k_t = [3, 4, 2, 2]
 Generation 7930: Best = 5.212379e+03 FES = 793100 μ_F = 0.756 μ_{CR} = 0.140
 Diversity = 9.0611e-04 k_t = [1, 1, 1, 1]
 Generation 7940: Best = 5.212379e+03 FES = 794100 μ_F = 0.745 μ_{CR} = 0.141
 Diversity = 8.7080e-04 k_t = [2, 3, 2, 4]
 Generation 7950: Best = 5.212379e+03 FES = 795100 μ_F = 0.750 μ_{CR} = 0.134
 Diversity = 9.1266e-04 k_t = [1, 1, 1, 1]
 Generation 7960: Best = 5.212378e+03 FES = 796100 μ_F = 0.747 μ_{CR} = 0.131
 Diversity = 9.0296e-04 k_t = [4, 4, 4, 2]
 Generation 7970: Best = 5.212378e+03 FES = 797100 μ_F = 0.748 μ_{CR} = 0.132
 Diversity = 8.9404e-04 k_t = [1, 1, 1, 1]
 Generation 7980: Best = 5.212378e+03 FES = 798100 μ_F = 0.752 μ_{CR} = 0.131
 Diversity = 9.3226e-04 k_t = [2, 4, 2, 2]
 Generation 7990: Best = 5.212378e+03 FES = 799100 μ_F = 0.739 μ_{CR} = 0.137
 Diversity = 9.4153e-04 k_t = [1, 1, 1, 1]
 Generation 8000: Best = 5.212377e+03 FES = 800100 μ_F = 0.728 μ_{CR} = 0.140
 Diversity = 9.0937e-04 k_t = [2, 4, 2, 3]
 Generation 8010: Best = 5.212377e+03 FES = 801100 μ_F = 0.719 μ_{CR} = 0.137
 Diversity = 8.8597e-04 k_t = [1, 1, 1, 1]
 Generation 8020: Best = 5.212376e+03 FES = 802100 μ_F = 0.726 μ_{CR} = 0.128
 Diversity = 8.3496e-04 k_t = [3, 2, 3, 4]
 Generation 8030: Best = 5.212376e+03 FES = 803100 μ_F = 0.741 μ_{CR} = 0.125
 Diversity = 8.0040e-04 k_t = [1, 1, 1, 1]
 Generation 8040: Best = 5.212376e+03 FES = 804100 μ_F = 0.722 μ_{CR} = 0.131
 Diversity = 7.3793e-04 k_t = [2, 3, 2, 4]
 Generation 8050: Best = 5.212376e+03 FES = 805100 μ_F = 0.724 μ_{CR} = 0.124
 Diversity = 7.6756e-04 k_t = [1, 1, 1, 1]

Generation 8060: Best = 5.212375e+03 FES = 806100 μ_F = 0.716 μ_{CR} = 0.125
 Diversity = 7.0665e-04 k_t = [3, 4, 3, 3]
 Generation 8070: Best = 5.212375e+03 FES = 807100 μ_F = 0.704 μ_{CR} = 0.134
 Diversity = 6.8473e-04 k_t = [1, 1, 1, 1]
 Generation 8080: Best = 5.212375e+03 FES = 808100 μ_F = 0.703 μ_{CR} = 0.129
 Diversity = 6.3255e-04 k_t = [2, 2, 3, 4]
 Generation 8090: Best = 5.212375e+03 FES = 809100 μ_F = 0.715 μ_{CR} = 0.121
 Diversity = 5.9422e-04 k_t = [1, 1, 1, 1]
 Generation 8100: Best = 5.212375e+03 FES = 810100 μ_F = 0.718 μ_{CR} = 0.122
 Diversity = 5.8475e-04 k_t = [4, 2, 4, 3]
 Generation 8110: Best = 5.212375e+03 FES = 811100 μ_F = 0.731 μ_{CR} = 0.121
 Diversity = 6.0269e-04 k_t = [1, 1, 1, 1]
 Generation 8120: Best = 5.212374e+03 FES = 812100 μ_F = 0.721 μ_{CR} = 0.119
 Diversity = 5.8598e-04 k_t = [2, 3, 3, 4]
 Generation 8130: Best = 5.212374e+03 FES = 813100 μ_F = 0.736 μ_{CR} = 0.124
 Diversity = 6.0355e-04 k_t = [1, 1, 1, 1]
 Generation 8140: Best = 5.212374e+03 FES = 814100 μ_F = 0.738 μ_{CR} = 0.134
 Diversity = 6.5158e-04 k_t = [2, 2, 4, 2]
 Generation 8150: Best = 5.212373e+03 FES = 815100 μ_F = 0.753 μ_{CR} = 0.140
 Diversity = 7.6848e-04 k_t = [1, 1, 1, 1]
 Generation 8160: Best = 5.212373e+03 FES = 816100 μ_F = 0.743 μ_{CR} = 0.141
 Diversity = 7.7246e-04 k_t = [2, 3, 4, 4]
 Generation 8170: Best = 5.212372e+03 FES = 817100 μ_F = 0.738 μ_{CR} = 0.129
 Diversity = 7.1281e-04 k_t = [1, 1, 1, 1]
 Generation 8180: Best = 5.212372e+03 FES = 818100 μ_F = 0.742 μ_{CR} = 0.137
 Diversity = 7.2539e-04 k_t = [4, 4, 2, 2]
 Generation 8190: Best = 5.212372e+03 FES = 819100 μ_F = 0.756 μ_{CR} = 0.150
 Diversity = 8.0498e-04 k_t = [1, 1, 1, 1]
 Generation 8200: Best = 5.212371e+03 FES = 820100 μ_F = 0.758 μ_{CR} = 0.158
 Diversity = 7.4904e-04 k_t = [2, 3, 3, 4]
 Generation 8210: Best = 5.212371e+03 FES = 821100 μ_F = 0.763 μ_{CR} = 0.146
 Diversity = 7.6754e-04 k_t = [1, 1, 1, 1]
 Generation 8220: Best = 5.212370e+03 FES = 822100 μ_F = 0.747 μ_{CR} = 0.143
 Diversity = 7.4816e-04 k_t = [3, 3, 2, 4]
 Generation 8230: Best = 5.212370e+03 FES = 823100 μ_F = 0.746 μ_{CR} = 0.140
 Diversity = 7.7638e-04 k_t = [1, 1, 1, 1]
 Generation 8240: Best = 5.212370e+03 FES = 824100 μ_F = 0.755 μ_{CR} = 0.134
 Diversity = 7.8283e-04 k_t = [2, 4, 4, 4]
 Generation 8250: Best = 5.212369e+03 FES = 825100 μ_F = 0.738 μ_{CR} = 0.136
 Diversity = 7.4946e-04 k_t = [1, 1, 1, 1]
 Generation 8260: Best = 5.212369e+03 FES = 826100 μ_F = 0.746 μ_{CR} = 0.144
 Diversity = 7.7355e-04 k_t = [4, 2, 4, 2]
 Generation 8270: Best = 5.212369e+03 FES = 827100 μ_F = 0.742 μ_{CR} = 0.145
 Diversity = 7.8910e-04 k_t = [1, 1, 1, 1]
 Generation 8280: Best = 5.212368e+03 FES = 828100 μ_F = 0.743 μ_{CR} = 0.143
 Diversity = 8.2049e-04 k_t = [4, 3, 4, 2]
 Generation 8290: Best = 5.212368e+03 FES = 829100 μ_F = 0.741 μ_{CR} = 0.149
 Diversity = 8.3924e-04 k_t = [1, 1, 1, 1]
 Generation 8300: Best = 5.212367e+03 FES = 830100 μ_F = 0.742 μ_{CR} = 0.155
 Diversity = 7.7211e-04 k_t = [4, 3, 2, 3]
 Generation 8310: Best = 5.212367e+03 FES = 831100 μ_F = 0.752 μ_{CR} = 0.150
 Diversity = 8.5037e-04 k_t = [1, 1, 1, 1]
 Generation 8320: Best = 5.212366e+03 FES = 832100 μ_F = 0.743 μ_{CR} = 0.147
 Diversity = 8.0662e-04 k_t = [4, 3, 4, 2]

Generation 8330: Best = 5.212366e+03 FES = 833100 μ_F = 0.754 μ_{CR} = 0.142
 Diversity = 7.6429e-04 k_t = [1, 1, 1, 1]
 Generation 8340: Best = 5.212366e+03 FES = 834100 μ_F = 0.780 μ_{CR} = 0.150
 Diversity = 7.8767e-04 k_t = [4, 3, 2, 3]
 Generation 8350: Best = 5.212365e+03 FES = 835100 μ_F = 0.757 μ_{CR} = 0.152
 Diversity = 7.8973e-04 k_t = [1, 1, 1, 1]
 Generation 8360: Best = 5.212364e+03 FES = 836100 μ_F = 0.744 μ_{CR} = 0.146
 Diversity = 7.6937e-04 k_t = [3, 3, 3, 3]
 Generation 8370: Best = 5.212364e+03 FES = 837100 μ_F = 0.751 μ_{CR} = 0.142
 Diversity = 8.4427e-04 k_t = [1, 1, 1, 1]
 Generation 8380: Best = 5.212364e+03 FES = 838100 μ_F = 0.767 μ_{CR} = 0.141
 Diversity = 8.4769e-04 k_t = [4, 4, 3, 4]
 Generation 8390: Best = 5.212363e+03 FES = 839100 μ_F = 0.772 μ_{CR} = 0.145
 Diversity = 8.7482e-04 k_t = [1, 1, 1, 1]
 Generation 8400: Best = 5.212362e+03 FES = 840100 μ_F = 0.778 μ_{CR} = 0.149
 Diversity = 9.3382e-04 k_t = [2, 4, 2, 4]
 Generation 8410: Best = 5.212362e+03 FES = 841100 μ_F = 0.774 μ_{CR} = 0.155
 Diversity = 9.5042e-04 k_t = [1, 1, 1, 1]
 Generation 8420: Best = 5.212361e+03 FES = 842100 μ_F = 0.768 μ_{CR} = 0.158
 Diversity = 9.4510e-04 k_t = [4, 2, 4, 4]
 Generation 8430: Best = 5.212361e+03 FES = 843100 μ_F = 0.760 μ_{CR} = 0.169
 Diversity = 9.4741e-04 k_t = [1, 1, 1, 1]
 Generation 8440: Best = 5.212361e+03 FES = 844100 μ_F = 0.746 μ_{CR} = 0.165
 Diversity = 8.9461e-04 k_t = [3, 2, 2, 3]
 Generation 8450: Best = 5.212360e+03 FES = 845100 μ_F = 0.739 μ_{CR} = 0.155
 Diversity = 9.3063e-04 k_t = [1, 1, 1, 1]
 Generation 8460: Best = 5.212359e+03 FES = 846100 μ_F = 0.740 μ_{CR} = 0.154
 Diversity = 9.5959e-04 k_t = [4, 2, 4, 2]
 Generation 8470: Best = 5.212359e+03 FES = 847100 μ_F = 0.739 μ_{CR} = 0.163
 Diversity = 9.6156e-04 k_t = [1, 1, 1, 1]
 Generation 8480: Best = 5.212358e+03 FES = 848100 μ_F = 0.749 μ_{CR} = 0.169
 Diversity = 9.9912e-04 k_t = [2, 3, 2, 3]
 Generation 8490: Best = 5.212358e+03 FES = 849100 μ_F = 0.752 μ_{CR} = 0.164
 Diversity = 1.0713e-03 k_t = [1, 1, 1, 1]
 Generation 8500: Best = 5.212357e+03 FES = 850100 μ_F = 0.741 μ_{CR} = 0.170
 Diversity = 1.0067e-03 k_t = [2, 2, 4, 3]
 Generation 8510: Best = 5.212357e+03 FES = 851100 μ_F = 0.734 μ_{CR} = 0.174
 Diversity = 9.8954e-04 k_t = [1, 1, 1, 1]
 Generation 8520: Best = 5.212356e+03 FES = 852100 μ_F = 0.731 μ_{CR} = 0.180
 Diversity = 1.0197e-03 k_t = [4, 4, 3, 4]
 Generation 8530: Best = 5.212355e+03 FES = 853100 μ_F = 0.736 μ_{CR} = 0.174
 Diversity = 1.0610e-03 k_t = [1, 1, 1, 1]
 Generation 8540: Best = 5.212355e+03 FES = 854100 μ_F = 0.720 μ_{CR} = 0.181
 Diversity = 1.0283e-03 k_t = [2, 4, 2, 4]
 Generation 8550: Best = 5.212355e+03 FES = 855100 μ_F = 0.727 μ_{CR} = 0.176
 Diversity = 1.0442e-03 k_t = [1, 1, 1, 1]
 Generation 8560: Best = 5.212354e+03 FES = 856100 μ_F = 0.717 μ_{CR} = 0.175
 Diversity = 1.0646e-03 k_t = [2, 3, 4, 2]
 Generation 8570: Best = 5.212354e+03 FES = 857100 μ_F = 0.723 μ_{CR} = 0.174
 Diversity = 1.0161e-03 k_t = [1, 1, 1, 1]
 Generation 8580: Best = 5.212353e+03 FES = 858100 μ_F = 0.720 μ_{CR} = 0.177
 Diversity = 9.5625e-04 k_t = [3, 3, 4, 2]
 Generation 8590: Best = 5.212353e+03 FES = 859100 μ_F = 0.711 μ_{CR} = 0.173
 Diversity = 8.7850e-04 k_t = [1, 1, 1, 1]

Generation 8600: Best = 5.212353e+03 FES = 860100 μ_F = 0.694 μ_{CR} = 0.172
 Diversity = 7.6955e-04 k_t = [2, 2, 3, 4]
 Generation 8610: Best = 5.212353e+03 FES = 861100 μ_F = 0.694 μ_{CR} = 0.167
 Diversity = 6.8076e-04 k_t = [1, 1, 1, 1]
 Generation 8620: Best = 5.212352e+03 FES = 862100 μ_F = 0.711 μ_{CR} = 0.166
 Diversity = 6.1898e-04 k_t = [2, 2, 4, 4]
 Generation 8630: Best = 5.212352e+03 FES = 863100 μ_F = 0.703 μ_{CR} = 0.159
 Diversity = 6.0984e-04 k_t = [1, 1, 1, 1]
 Generation 8640: Best = 5.212352e+03 FES = 864100 μ_F = 0.708 μ_{CR} = 0.168
 Diversity = 6.1323e-04 k_t = [2, 4, 4, 3]
 Generation 8650: Best = 5.212352e+03 FES = 865100 μ_F = 0.702 μ_{CR} = 0.170
 Diversity = 6.0616e-04 k_t = [1, 1, 1, 1]
 Generation 8660: Best = 5.212352e+03 FES = 866100 μ_F = 0.715 μ_{CR} = 0.162
 Diversity = 6.0838e-04 k_t = [2, 4, 4, 2]
 Generation 8670: Best = 5.212352e+03 FES = 867100 μ_F = 0.716 μ_{CR} = 0.158
 Diversity = 5.8483e-04 k_t = [1, 1, 1, 1]
 Generation 8680: Best = 5.212351e+03 FES = 868100 μ_F = 0.716 μ_{CR} = 0.141
 Diversity = 5.3965e-04 k_t = [3, 2, 4, 2]
 Generation 8690: Best = 5.212351e+03 FES = 869100 μ_F = 0.712 μ_{CR} = 0.124
 Diversity = 4.9591e-04 k_t = [1, 1, 1, 1]
 Generation 8700: Best = 5.212351e+03 FES = 870100 μ_F = 0.719 μ_{CR} = 0.122
 Diversity = 4.5985e-04 k_t = [2, 2, 3, 4]
 Generation 8710: Best = 5.212351e+03 FES = 871100 μ_F = 0.736 μ_{CR} = 0.121
 Diversity = 4.7158e-04 k_t = [1, 1, 1, 1]
 Generation 8720: Best = 5.212351e+03 FES = 872100 μ_F = 0.755 μ_{CR} = 0.130
 Diversity = 4.6286e-04 k_t = [2, 3, 2, 4]
 Generation 8730: Best = 5.212351e+03 FES = 873100 μ_F = 0.742 μ_{CR} = 0.127
 Diversity = 4.6531e-04 k_t = [1, 1, 1, 1]
 Generation 8740: Best = 5.212351e+03 FES = 874100 μ_F = 0.745 μ_{CR} = 0.130
 Diversity = 4.3184e-04 k_t = [4, 3, 4, 4]
 Generation 8750: Best = 5.212351e+03 FES = 875100 μ_F = 0.754 μ_{CR} = 0.138
 Diversity = 4.2534e-04 k_t = [1, 1, 1, 1]
 Generation 8760: Best = 5.212351e+03 FES = 876100 μ_F = 0.751 μ_{CR} = 0.143
 Diversity = 4.3604e-04 k_t = [2, 4, 2, 4]
 Generation 8770: Best = 5.212351e+03 FES = 877100 μ_F = 0.734 μ_{CR} = 0.141
 Diversity = 4.2882e-04 k_t = [1, 1, 1, 1]
 Generation 8780: Best = 5.212350e+03 FES = 878100 μ_F = 0.738 μ_{CR} = 0.135
 Diversity = 4.3850e-04 k_t = [2, 2, 2, 2]
 Generation 8790: Best = 5.212350e+03 FES = 879100 μ_F = 0.741 μ_{CR} = 0.131
 Diversity = 4.3777e-04 k_t = [1, 1, 1, 1]
 Generation 8800: Best = 5.212350e+03 FES = 880100 μ_F = 0.747 μ_{CR} = 0.126
 Diversity = 4.3444e-04 k_t = [3, 3, 2, 2]
 Generation 8810: Best = 5.212350e+03 FES = 881100 μ_F = 0.754 μ_{CR} = 0.122
 Diversity = 4.3265e-04 k_t = [1, 1, 1, 1]
 Generation 8820: Best = 5.212350e+03 FES = 882100 μ_F = 0.775 μ_{CR} = 0.117
 Diversity = 4.6119e-04 k_t = [3, 2, 3, 2]
 Generation 8830: Best = 5.212350e+03 FES = 883100 μ_F = 0.766 μ_{CR} = 0.120
 Diversity = 4.7562e-04 k_t = [1, 1, 1, 1]
 Generation 8840: Best = 5.212350e+03 FES = 884100 μ_F = 0.776 μ_{CR} = 0.119
 Diversity = 4.5965e-04 k_t = [3, 4, 3, 4]
 Generation 8850: Best = 5.212350e+03 FES = 885100 μ_F = 0.778 μ_{CR} = 0.122
 Diversity = 4.7703e-04 k_t = [1, 1, 1, 1]
 Generation 8860: Best = 5.212350e+03 FES = 886100 μ_F = 0.775 μ_{CR} = 0.130
 Diversity = 4.9608e-04 k_t = [3, 2, 3, 4]

Generation 8870: Best = 5.212350e+03 FES = 887100 μ_F = 0.777 μ_{CR} = 0.136
 Diversity = 5.1703e-04 k_t = [1, 1, 1, 1]
 Generation 8880: Best = 5.212349e+03 FES = 888100 μ_F = 0.770 μ_{CR} = 0.144
 Diversity = 5.2464e-04 k_t = [4, 3, 2, 3]
 Generation 8890: Best = 5.212349e+03 FES = 889100 μ_F = 0.769 μ_{CR} = 0.141
 Diversity = 5.4149e-04 k_t = [1, 1, 1, 1]
 Generation 8900: Best = 5.212349e+03 FES = 890100 μ_F = 0.766 μ_{CR} = 0.132
 Diversity = 5.6216e-04 k_t = [3, 4, 2, 2]
 Generation 8910: Best = 5.212349e+03 FES = 891100 μ_F = 0.771 μ_{CR} = 0.127
 Diversity = 6.0744e-04 k_t = [1, 1, 1, 1]
 Generation 8920: Best = 5.212349e+03 FES = 892100 μ_F = 0.757 μ_{CR} = 0.132
 Diversity = 5.8257e-04 k_t = [3, 3, 2, 4]
 Generation 8930: Best = 5.212349e+03 FES = 893100 μ_F = 0.766 μ_{CR} = 0.132
 Diversity = 5.9415e-04 k_t = [1, 1, 1, 1]
 Generation 8940: Best = 5.212349e+03 FES = 894100 μ_F = 0.767 μ_{CR} = 0.121
 Diversity = 6.0658e-04 k_t = [4, 3, 4, 4]
 Generation 8950: Best = 5.212348e+03 FES = 895100 μ_F = 0.767 μ_{CR} = 0.132
 Diversity = 6.0365e-04 k_t = [1, 1, 1, 1]
 Generation 8960: Best = 5.212348e+03 FES = 896100 μ_F = 0.758 μ_{CR} = 0.130
 Diversity = 6.0029e-04 k_t = [4, 4, 3, 2]
 Generation 8970: Best = 5.212348e+03 FES = 897100 μ_F = 0.740 μ_{CR} = 0.135
 Diversity = 5.9115e-04 k_t = [1, 1, 1, 1]
 Generation 8980: Best = 5.212348e+03 FES = 898100 μ_F = 0.736 μ_{CR} = 0.128
 Diversity = 6.0921e-04 k_t = [3, 2, 4, 2]
 Generation 8990: Best = 5.212348e+03 FES = 899100 μ_F = 0.722 μ_{CR} = 0.129
 Diversity = 6.1450e-04 k_t = [1, 1, 1, 1]
 Generation 9000: Best = 5.212348e+03 FES = 900100 μ_F = 0.723 μ_{CR} = 0.136
 Diversity = 6.0691e-04 k_t = [3, 2, 2, 3]
 Generation 9010: Best = 5.212348e+03 FES = 901100 μ_F = 0.728 μ_{CR} = 0.133
 Diversity = 6.2867e-04 k_t = [1, 1, 1, 1]
 Generation 9020: Best = 5.212348e+03 FES = 902100 μ_F = 0.716 μ_{CR} = 0.129
 Diversity = 5.9297e-04 k_t = [3, 3, 4, 3]
 Generation 9030: Best = 5.212347e+03 FES = 903100 μ_F = 0.715 μ_{CR} = 0.121
 Diversity = 6.0385e-04 k_t = [1, 1, 1, 1]
 Generation 9040: Best = 5.212347e+03 FES = 904100 μ_F = 0.729 μ_{CR} = 0.120
 Diversity = 6.0243e-04 k_t = [2, 3, 2, 4]
 Generation 9050: Best = 5.212347e+03 FES = 905100 μ_F = 0.725 μ_{CR} = 0.123
 Diversity = 6.1636e-04 k_t = [1, 1, 1, 1]
 Generation 9060: Best = 5.212347e+03 FES = 906100 μ_F = 0.730 μ_{CR} = 0.127
 Diversity = 6.0602e-04 k_t = [4, 2, 4, 4]
 Generation 9070: Best = 5.212347e+03 FES = 907100 μ_F = 0.742 μ_{CR} = 0.123
 Diversity = 6.2754e-04 k_t = [1, 1, 1, 1]
 Generation 9080: Best = 5.212347e+03 FES = 908100 μ_F = 0.738 μ_{CR} = 0.117
 Diversity = 6.2996e-04 k_t = [3, 3, 2, 3]
 Generation 9090: Best = 5.212346e+03 FES = 909100 μ_F = 0.766 μ_{CR} = 0.113
 Diversity = 6.8613e-04 k_t = [1, 1, 1, 1]
 Generation 9100: Best = 5.212346e+03 FES = 910100 μ_F = 0.767 μ_{CR} = 0.118
 Diversity = 6.7727e-04 k_t = [4, 4, 4, 4]
 Generation 9110: Best = 5.212346e+03 FES = 911100 μ_F = 0.762 μ_{CR} = 0.120
 Diversity = 6.8609e-04 k_t = [1, 1, 1, 1]
 Generation 9120: Best = 5.212346e+03 FES = 912100 μ_F = 0.760 μ_{CR} = 0.124
 Diversity = 7.0289e-04 k_t = [2, 4, 4, 4]
 Generation 9130: Best = 5.212345e+03 FES = 913100 μ_F = 0.746 μ_{CR} = 0.134
 Diversity = 6.9175e-04 k_t = [1, 1, 1, 1]

Generation 9140: Best = 5.212345e+03 FES = 914100 μ_F = 0.751 μ_{CR} = 0.130
 Diversity = 7.4850e-04 k_t = [3, 3, 3, 2]
 Generation 9150: Best = 5.212345e+03 FES = 915100 μ_F = 0.746 μ_{CR} = 0.128
 Diversity = 7.9818e-04 k_t = [1, 1, 1, 1]
 Generation 9160: Best = 5.212344e+03 FES = 916100 μ_F = 0.748 μ_{CR} = 0.121
 Diversity = 8.0017e-04 k_t = [4, 4, 2, 2]
 Generation 9170: Best = 5.212344e+03 FES = 917100 μ_F = 0.767 μ_{CR} = 0.133
 Diversity = 8.9211e-04 k_t = [1, 1, 1, 1]
 Generation 9180: Best = 5.212343e+03 FES = 918100 μ_F = 0.773 μ_{CR} = 0.128
 Diversity = 9.2449e-04 k_t = [4, 4, 2, 2]
 Generation 9190: Best = 5.212343e+03 FES = 919100 μ_F = 0.764 μ_{CR} = 0.120
 Diversity = 9.9553e-04 k_t = [1, 1, 1, 1]
 Generation 9200: Best = 5.212343e+03 FES = 920100 μ_F = 0.760 μ_{CR} = 0.122
 Diversity = 1.0269e-03 k_t = [2, 4, 4, 3]
 Generation 9210: Best = 5.212343e+03 FES = 921100 μ_F = 0.746 μ_{CR} = 0.112
 Diversity = 9.9501e-04 k_t = [1, 1, 1, 1]
 Generation 9220: Best = 5.212343e+03 FES = 922100 μ_F = 0.737 μ_{CR} = 0.112
 Diversity = 9.3587e-04 k_t = [4, 4, 2, 4]
 Generation 9230: Best = 5.212342e+03 FES = 923100 μ_F = 0.731 μ_{CR} = 0.121
 Diversity = 8.6672e-04 k_t = [1, 1, 1, 1]
 Generation 9240: Best = 5.212342e+03 FES = 924100 μ_F = 0.747 μ_{CR} = 0.119
 Diversity = 8.6609e-04 k_t = [2, 4, 2, 3]
 Generation 9250: Best = 5.212342e+03 FES = 925100 μ_F = 0.740 μ_{CR} = 0.112
 Diversity = 9.0344e-04 k_t = [1, 1, 1, 1]
 Generation 9260: Best = 5.212341e+03 FES = 926100 μ_F = 0.753 μ_{CR} = 0.117
 Diversity = 8.7477e-04 k_t = [2, 4, 4, 4]
 Generation 9270: Best = 5.212340e+03 FES = 927100 μ_F = 0.778 μ_{CR} = 0.124
 Diversity = 8.7307e-04 k_t = [1, 1, 1, 1]
 Generation 9280: Best = 5.212340e+03 FES = 928100 μ_F = 0.772 μ_{CR} = 0.124
 Diversity = 9.3048e-04 k_t = [4, 2, 4, 2]
 Generation 9290: Best = 5.212340e+03 FES = 929100 μ_F = 0.783 μ_{CR} = 0.121
 Diversity = 9.8812e-04 k_t = [1, 1, 1, 1]
 Generation 9300: Best = 5.212339e+03 FES = 930100 μ_F = 0.775 μ_{CR} = 0.128
 Diversity = 9.8297e-04 k_t = [3, 2, 3, 2]
 Generation 9310: Best = 5.212339e+03 FES = 931100 μ_F = 0.766 μ_{CR} = 0.129
 Diversity = 9.8495e-04 k_t = [1, 1, 1, 1]
 Generation 9320: Best = 5.212339e+03 FES = 932100 μ_F = 0.761 μ_{CR} = 0.140
 Diversity = 9.3498e-04 k_t = [3, 3, 4, 4]
 Generation 9330: Best = 5.212338e+03 FES = 933100 μ_F = 0.765 μ_{CR} = 0.139
 Diversity = 8.9900e-04 k_t = [1, 1, 1, 1]
 Generation 9340: Best = 5.212338e+03 FES = 934100 μ_F = 0.760 μ_{CR} = 0.137
 Diversity = 9.0753e-04 k_t = [4, 2, 4, 3]
 Generation 9350: Best = 5.212337e+03 FES = 935100 μ_F = 0.757 μ_{CR} = 0.136
 Diversity = 9.2582e-04 k_t = [1, 1, 1, 1]
 Generation 9360: Best = 5.212337e+03 FES = 936100 μ_F = 0.758 μ_{CR} = 0.121
 Diversity = 8.5511e-04 k_t = [3, 2, 2, 3]
 Generation 9370: Best = 5.212337e+03 FES = 937100 μ_F = 0.752 μ_{CR} = 0.118
 Diversity = 8.7794e-04 k_t = [1, 1, 1, 1]
 Generation 9380: Best = 5.212337e+03 FES = 938100 μ_F = 0.757 μ_{CR} = 0.114
 Diversity = 9.1045e-04 k_t = [4, 4, 4, 2]
 Generation 9390: Best = 5.212336e+03 FES = 939100 μ_F = 0.757 μ_{CR} = 0.113
 Diversity = 9.1630e-04 k_t = [1, 1, 1, 1]
 Generation 9400: Best = 5.212336e+03 FES = 940100 μ_F = 0.750 μ_{CR} = 0.104
 Diversity = 8.5664e-04 k_t = [3, 4, 2, 3]

Generation 9410: Best = 5.212335e+03 FES = 941100 μ_F = 0.751 μ_{CR} = 0.119
 Diversity = 8.6180e-04 k_t = [1, 1, 1, 1]
 Generation 9420: Best = 5.212335e+03 FES = 942100 μ_F = 0.739 μ_{CR} = 0.119
 Diversity = 9.0088e-04 k_t = [4, 3, 4, 2]
 Generation 9430: Best = 5.212335e+03 FES = 943100 μ_F = 0.739 μ_{CR} = 0.118
 Diversity = 9.3895e-04 k_t = [1, 1, 1, 1]
 Generation 9440: Best = 5.212335e+03 FES = 944100 μ_F = 0.732 μ_{CR} = 0.109
 Diversity = 8.0895e-04 k_t = [2, 4, 2, 4]
 Generation 9450: Best = 5.212334e+03 FES = 945100 μ_F = 0.734 μ_{CR} = 0.100
 Diversity = 7.6210e-04 k_t = [1, 1, 1, 1]
 Generation 9460: Best = 5.212334e+03 FES = 946100 μ_F = 0.729 μ_{CR} = 0.105
 Diversity = 7.5754e-04 k_t = [2, 4, 2, 2]
 Generation 9470: Best = 5.212334e+03 FES = 947100 μ_F = 0.742 μ_{CR} = 0.107
 Diversity = 7.7445e-04 k_t = [1, 1, 1, 1]
 Generation 9480: Best = 5.212334e+03 FES = 948100 μ_F = 0.750 μ_{CR} = 0.113
 Diversity = 7.7932e-04 k_t = [3, 4, 2, 3]
 Generation 9490: Best = 5.212334e+03 FES = 949100 μ_F = 0.746 μ_{CR} = 0.124
 Diversity = 7.7523e-04 k_t = [1, 1, 1, 1]
 Generation 9500: Best = 5.212333e+03 FES = 950100 μ_F = 0.725 μ_{CR} = 0.126
 Diversity = 7.0784e-04 k_t = [3, 4, 3, 3]
 Generation 9510: Best = 5.212333e+03 FES = 951100 μ_F = 0.724 μ_{CR} = 0.129
 Diversity = 7.6048e-04 k_t = [1, 1, 1, 1]
 Generation 9520: Best = 5.212333e+03 FES = 952100 μ_F = 0.731 μ_{CR} = 0.136
 Diversity = 7.4390e-04 k_t = [2, 4, 2, 3]
 Generation 9530: Best = 5.212333e+03 FES = 953100 μ_F = 0.722 μ_{CR} = 0.120
 Diversity = 7.5663e-04 k_t = [1, 1, 1, 1]
 Generation 9540: Best = 5.212333e+03 FES = 954100 μ_F = 0.726 μ_{CR} = 0.111
 Diversity = 7.7044e-04 k_t = [4, 4, 3, 3]
 Generation 9550: Best = 5.212332e+03 FES = 955100 μ_F = 0.717 μ_{CR} = 0.108
 Diversity = 7.5396e-04 k_t = [1, 1, 1, 1]
 Generation 9560: Best = 5.212332e+03 FES = 956100 μ_F = 0.721 μ_{CR} = 0.112
 Diversity = 7.1254e-04 k_t = [2, 2, 2, 2]
 Generation 9570: Best = 5.212332e+03 FES = 957100 μ_F = 0.723 μ_{CR} = 0.117
 Diversity = 7.5453e-04 k_t = [1, 1, 1, 1]
 Generation 9580: Best = 5.212332e+03 FES = 958100 μ_F = 0.738 μ_{CR} = 0.108
 Diversity = 7.3701e-04 k_t = [2, 3, 3, 4]
 Generation 9590: Best = 5.212331e+03 FES = 959100 μ_F = 0.737 μ_{CR} = 0.117
 Diversity = 7.2377e-04 k_t = [1, 1, 1, 1]
 Generation 9600: Best = 5.212331e+03 FES = 960100 μ_F = 0.730 μ_{CR} = 0.128
 Diversity = 7.2639e-04 k_t = [4, 2, 3, 3]
 Generation 9610: Best = 5.212330e+03 FES = 961100 μ_F = 0.732 μ_{CR} = 0.130
 Diversity = 8.2082e-04 k_t = [1, 1, 1, 1]
 Generation 9620: Best = 5.212329e+03 FES = 962100 μ_F = 0.742 μ_{CR} = 0.133
 Diversity = 8.9077e-04 k_t = [2, 2, 3, 2]
 Generation 9630: Best = 5.212329e+03 FES = 963100 μ_F = 0.767 μ_{CR} = 0.139
 Diversity = 1.0031e-03 k_t = [1, 1, 1, 1]
 Generation 9640: Best = 5.212329e+03 FES = 964100 μ_F = 0.752 μ_{CR} = 0.133
 Diversity = 9.9905e-04 k_t = [2, 3, 3, 2]
 Generation 9650: Best = 5.212328e+03 FES = 965100 μ_F = 0.746 μ_{CR} = 0.131
 Diversity = 9.6057e-04 k_t = [1, 1, 1, 1]
 Generation 9660: Best = 5.212328e+03 FES = 966100 μ_F = 0.758 μ_{CR} = 0.131
 Diversity = 9.1589e-04 k_t = [4, 4, 2, 4]
 Generation 9670: Best = 5.212327e+03 FES = 967100 μ_F = 0.764 μ_{CR} = 0.131
 Diversity = 9.2790e-04 k_t = [1, 1, 1, 1]

Generation 9680: Best = 5.212327e+03 FES = 968100 μ_F = 0.773 μ_{CR} = 0.135
 Diversity = 9.1387e-04 k_t = [2, 2, 3, 4]
 Generation 9690: Best = 5.212326e+03 FES = 969100 μ_F = 0.770 μ_{CR} = 0.136
 Diversity = 9.1560e-04 k_t = [1, 1, 1, 1]
 Generation 9700: Best = 5.212326e+03 FES = 970100 μ_F = 0.773 μ_{CR} = 0.150
 Diversity = 9.2706e-04 k_t = [3, 2, 2, 4]
 Generation 9710: Best = 5.212325e+03 FES = 971100 μ_F = 0.771 μ_{CR} = 0.154
 Diversity = 9.5572e-04 k_t = [1, 1, 1, 1]
 Generation 9720: Best = 5.212325e+03 FES = 972100 μ_F = 0.761 μ_{CR} = 0.146
 Diversity = 9.2450e-04 k_t = [2, 4, 4, 3]
 Generation 9730: Best = 5.212324e+03 FES = 973100 μ_F = 0.751 μ_{CR} = 0.150
 Diversity = 1.0004e-03 k_t = [1, 1, 1, 1]
 Generation 9740: Best = 5.212323e+03 FES = 974100 μ_F = 0.763 μ_{CR} = 0.149
 Diversity = 9.6454e-04 k_t = [4, 2, 3, 4]
 Generation 9750: Best = 5.212323e+03 FES = 975100 μ_F = 0.741 μ_{CR} = 0.147
 Diversity = 1.0197e-03 k_t = [1, 1, 1, 1]
 Generation 9760: Best = 5.212322e+03 FES = 976100 μ_F = 0.757 μ_{CR} = 0.152
 Diversity = 1.0721e-03 k_t = [4, 2, 2, 2]
 Generation 9770: Best = 5.212322e+03 FES = 977100 μ_F = 0.761 μ_{CR} = 0.143
 Diversity = 1.0698e-03 k_t = [1, 1, 1, 1]
 Generation 9780: Best = 5.212322e+03 FES = 978100 μ_F = 0.757 μ_{CR} = 0.144
 Diversity = 9.8073e-04 k_t = [4, 3, 2, 3]
 Generation 9790: Best = 5.212319e+03 FES = 979100 μ_F = 0.752 μ_{CR} = 0.145
 Diversity = 1.0048e-03 k_t = [1, 1, 1, 1]
 Generation 9800: Best = 5.212318e+03 FES = 980100 μ_F = 0.762 μ_{CR} = 0.141
 Diversity = 1.2043e-03 k_t = [4, 4, 4, 3]
 Generation 9810: Best = 5.212317e+03 FES = 981100 μ_F = 0.779 μ_{CR} = 0.147
 Diversity = 1.3611e-03 k_t = [1, 1, 1, 1]
 Generation 9820: Best = 5.212314e+03 FES = 982100 μ_F = 0.777 μ_{CR} = 0.146
 Diversity = 1.4111e-03 k_t = [4, 3, 4, 3]
 Generation 9830: Best = 5.212313e+03 FES = 983100 μ_F = 0.767 μ_{CR} = 0.152
 Diversity = 1.6047e-03 k_t = [1, 1, 1, 1]
 Generation 9840: Best = 5.212312e+03 FES = 984100 μ_F = 0.746 μ_{CR} = 0.164
 Diversity = 1.6049e-03 k_t = [3, 4, 3, 2]
 Generation 9850: Best = 5.212311e+03 FES = 985100 μ_F = 0.750 μ_{CR} = 0.161
 Diversity = 1.5554e-03 k_t = [1, 1, 1, 1]
 Generation 9860: Best = 5.212310e+03 FES = 986100 μ_F = 0.759 μ_{CR} = 0.170
 Diversity = 1.4404e-03 k_t = [4, 3, 4, 3]
 Generation 9870: Best = 5.212308e+03 FES = 987100 μ_F = 0.753 μ_{CR} = 0.175
 Diversity = 1.4516e-03 k_t = [1, 1, 1, 1]
 Generation 9880: Best = 5.212307e+03 FES = 988100 μ_F = 0.751 μ_{CR} = 0.175
 Diversity = 1.5239e-03 k_t = [4, 4, 4, 3]
 Generation 9890: Best = 5.212305e+03 FES = 989100 μ_F = 0.735 μ_{CR} = 0.169
 Diversity = 1.4472e-03 k_t = [1, 1, 1, 1]
 Generation 9900: Best = 5.212304e+03 FES = 990100 μ_F = 0.740 μ_{CR} = 0.169
 Diversity = 1.3413e-03 k_t = [2, 4, 2, 4]
 Generation 9910: Best = 5.212303e+03 FES = 991100 μ_F = 0.751 μ_{CR} = 0.173
 Diversity = 1.3329e-03 k_t = [1, 1, 1, 1]
 Generation 9920: Best = 5.212302e+03 FES = 992100 μ_F = 0.749 μ_{CR} = 0.179
 Diversity = 1.2345e-03 k_t = [4, 3, 2, 4]
 Generation 9930: Best = 5.212301e+03 FES = 993100 μ_F = 0.744 μ_{CR} = 0.172
 Diversity = 1.1494e-03 k_t = [1, 1, 1, 1]
 Generation 9940: Best = 5.212300e+03 FES = 994100 μ_F = 0.751 μ_{CR} = 0.185
 Diversity = 1.1482e-03 k_t = [2, 2, 2, 4]

Generation 9950: Best = 5.212298e+03 FES = 995100 μ_F = 0.760 μ_{CR} = 0.182
 Diversity = 1.2655e-03 k_t = [1, 1, 1, 1]
 Generation 9960: Best = 5.212297e+03 FES = 996100 μ_F = 0.767 μ_{CR} = 0.177
 Diversity = 1.2717e-03 k_t = [2, 3, 4, 4]
 Generation 9970: Best = 5.212295e+03 FES = 997100 μ_F = 0.779 μ_{CR} = 0.169
 Diversity = 1.3983e-03 k_t = [1, 1, 1, 1]
 Generation 9980: Best = 5.212295e+03 FES = 998100 μ_F = 0.777 μ_{CR} = 0.170
 Diversity = 1.3369e-03 k_t = [3, 4, 4, 4]
 Generation 9990: Best = 5.212293e+03 FES = 999100 μ_F = 0.758 μ_{CR} = 0.177
 Diversity = 1.3719e-03 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 5.212292e+03

Total Function Evals = 1000000

Total Successful Updates = 251695

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Results:

Best Fitness : 5.212292e+03

Global Optimum : 2.900000e+03

Error (|best-optimum|) : 2.312292e+03

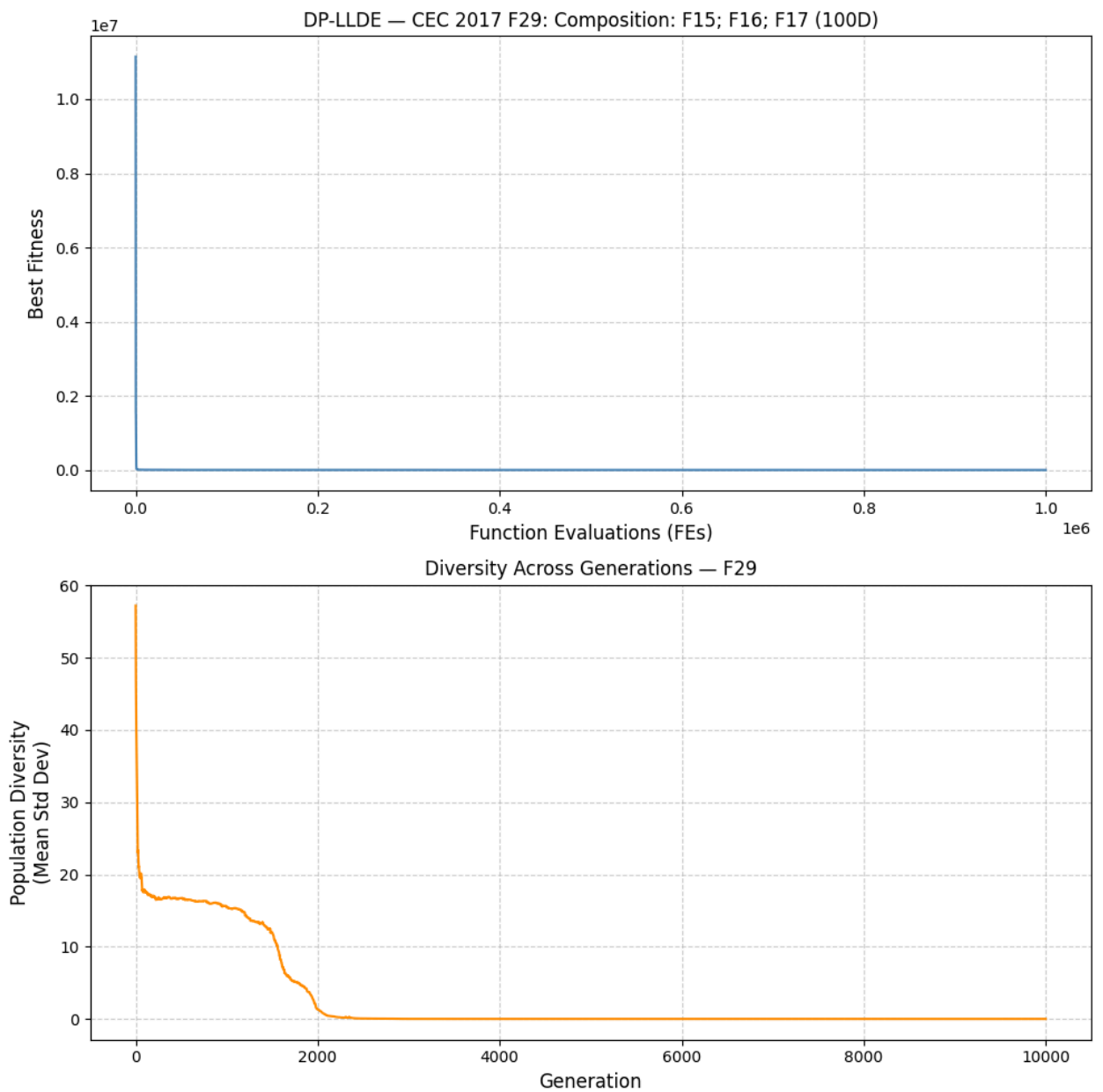
Best Solution (5 dim) : [3.1187364 -48.16019824 -26.9684865 42.13331888
95.84905067]

$D^{(0)}$ (init. diversity): 5.7220e+01

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✓ Checkpoint saved to: /content/drive/My Drive/thesis_results/all_results_checkpoint.pkl



Graph saved as convergence_F29.png

✓ Plot saved to: /content/drive/My Drive/thesis_results/plots/convergence_F29.png

```
In [18]: # =====
# F30 — Composition Function 10 (F15; F18; F19)
# =====
run_and_plot('F30', make_wrapper(composition.f30), 3000.0, 'Composition: F15;
```

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DP-LLDE on CEC 2017 F30 – Single Run

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Function : Composition: F15; F18; F19 (100D)

Bounds : [-100, 100]

Global optimum : 3000.0

Initial $D^*(0) = 5.7220e+01$ | Diversity threshold = 5.7220e+00

Generation 0: Best Fitness = 7.713945e+11

Generation 10: Best = 1.712127e+11 FES = 1100 $\mu_F = 0.572$ $\mu_{CR} = 0.350$

Diversity = 3.3579e+01 $k_t = [1, 1, 1, 1]$

Generation 20: Best = 5.922253e+10 FES = 2100 $\mu_F = 0.621$ $\mu_{CR} = 0.351$

Diversity = 2.5152e+01 $k_t = [2, 3, 2, 3]$

Generation 30: Best = 2.085903e+10 FES = 3100 $\mu_F = 0.656$ $\mu_{CR} = 0.358$

Diversity = 2.1562e+01 $k_t = [1, 1, 1, 1]$

Generation 40: Best = 8.961671e+09 FES = 4100 $\mu_F = 0.683$ $\mu_{CR} = 0.359$

Diversity = 1.7399e+01 $k_t = [3, 2, 3, 4]$

Generation 50: Best = 3.285298e+09 FES = 5100 $\mu_F = 0.705$ $\mu_{CR} = 0.357$

Diversity = 1.5877e+01 $k_t = [1, 1, 1, 1]$

Generation 60: Best = 1.182371e+09 FES = 6100 $\mu_F = 0.711$ $\mu_{CR} = 0.348$

Diversity = 1.3267e+01 $k_t = [2, 2, 3, 3]$

Generation 70: Best = 5.412320e+08 FES = 7100 $\mu_F = 0.711$ $\mu_{CR} = 0.352$

Diversity = 1.1704e+01 $k_t = [1, 1, 1, 1]$

Generation 80: Best = 2.275678e+08 FES = 8100 $\mu_F = 0.718$ $\mu_{CR} = 0.356$

Diversity = 1.0046e+01 $k_t = [4, 3, 4, 2]$

Generation 90: Best = 9.981642e+07 FES = 9100 $\mu_F = 0.724$ $\mu_{CR} = 0.372$

Diversity = 8.5789e+00 $k_t = [1, 1, 1, 1]$

Generation 100: Best = 6.567815e+07 FES = 10100 $\mu_F = 0.728$ $\mu_{CR} = 0.368$

Diversity = 7.5381e+00 $k_t = [2, 2, 4, 3]$

Generation 110: Best = 5.029056e+07 FES = 11100 $\mu_F = 0.708$ $\mu_{CR} = 0.366$

Diversity = 6.2809e+00 $k_t = [1, 1, 1, 1]$

Generation 120: Best = 3.700865e+07 FES = 12100 $\mu_F = 0.714$ $\mu_{CR} = 0.368$

Diversity = 6.1847e+00 $k_t = [2, 4, 2, 2]$

Generation 130: Best = 2.733174e+07 FES = 13100 $\mu_F = 0.719$ $\mu_{CR} = 0.362$

Diversity = 5.7667e+00 $k_t = [1, 1, 1, 1]$

Generation 140: Best = 2.104114e+07 FES = 14100 $\mu_F = 0.723$ $\mu_{CR} = 0.376$

Diversity = 5.2481e+00 $k_t = [2, 3, 4, 2]$

Generation 150: Best = 1.606005e+07 FES = 15100 $\mu_F = 0.723$ $\mu_{CR} = 0.382$

Diversity = 5.2497e+00 $k_t = [1, 1, 1, 1]$

Generation 160: Best = 1.315087e+07 FES = 16100 $\mu_F = 0.721$ $\mu_{CR} = 0.374$

Diversity = 5.1126e+00 $k_t = [4, 2, 3, 2]$

Generation 170: Best = 9.824815e+06 FES = 17100 $\mu_F = 0.729$ $\mu_{CR} = 0.367$

Diversity = 4.8218e+00 $k_t = [1, 1, 1, 1]$

Generation 180: Best = 8.850775e+06 FES = 18100 $\mu_F = 0.736$ $\mu_{CR} = 0.368$

Diversity = 4.7877e+00 $k_t = [2, 3, 2, 2]$

Generation 190: Best = 7.004387e+06 FES = 19100 $\mu_F = 0.735$ $\mu_{CR} = 0.362$

Diversity = 5.1958e+00 $k_t = [1, 1, 1, 1]$

Generation 200: Best = 5.606256e+06 FES = 20100 $\mu_F = 0.732$ $\mu_{CR} = 0.370$

Diversity = 5.0158e+00 $k_t = [4, 4, 2, 4]$

Generation 210: Best = 5.109428e+06 FES = 21100 $\mu_F = 0.743$ $\mu_{CR} = 0.364$

Diversity = 4.1818e+00 $k_t = [1, 1, 1, 1]$

Generation 220: Best = 4.609774e+06 FES = 22100 $\mu_F = 0.745$ $\mu_{CR} = 0.356$

Diversity = 3.9620e+00 $k_t = [4, 4, 2, 3]$

Generation 230: Best = 4.269347e+06	FES = 23100	$\mu_F = 0.725$	$\mu_{CR} = 0.361$
Diversity = 3.5916e+00	$k_t = [1, 1, 1, 1]$		
Generation 240: Best = 3.978976e+06	FES = 24100	$\mu_F = 0.725$	$\mu_{CR} = 0.355$
Diversity = 3.2071e+00	$k_t = [2, 2, 3, 4]$		
Generation 250: Best = 3.768836e+06	FES = 25100	$\mu_F = 0.732$	$\mu_{CR} = 0.346$
Diversity = 3.2130e+00	$k_t = [1, 1, 1, 1]$		
Generation 260: Best = 3.488899e+06	FES = 26100	$\mu_F = 0.729$	$\mu_{CR} = 0.350$
Diversity = 3.0860e+00	$k_t = [3, 4, 3, 2]$		
Generation 270: Best = 3.205407e+06	FES = 27100	$\mu_F = 0.742$	$\mu_{CR} = 0.345$
Diversity = 3.3392e+00	$k_t = [1, 1, 1, 1]$		
Generation 280: Best = 2.976987e+06	FES = 28100	$\mu_F = 0.746$	$\mu_{CR} = 0.326$
Diversity = 3.2249e+00	$k_t = [2, 4, 4, 3]$		
Generation 290: Best = 2.877029e+06	FES = 29100	$\mu_F = 0.752$	$\mu_{CR} = 0.309$
Diversity = 3.4442e+00	$k_t = [1, 1, 1, 1]$		
Generation 300: Best = 2.766819e+06	FES = 30100	$\mu_F = 0.748$	$\mu_{CR} = 0.289$
Diversity = 3.3575e+00	$k_t = [2, 3, 4, 4]$		
Generation 310: Best = 2.613103e+06	FES = 31100	$\mu_F = 0.755$	$\mu_{CR} = 0.286$
Diversity = 3.1897e+00	$k_t = [1, 1, 1, 1]$		
Generation 320: Best = 2.403739e+06	FES = 32100	$\mu_F = 0.769$	$\mu_{CR} = 0.282$
Diversity = 3.4892e+00	$k_t = [2, 3, 3, 4]$		
Generation 330: Best = 2.229807e+06	FES = 33100	$\mu_F = 0.774$	$\mu_{CR} = 0.273$
Diversity = 3.8331e+00	$k_t = [1, 1, 1, 1]$		
Generation 340: Best = 2.067926e+06	FES = 34100	$\mu_F = 0.785$	$\mu_{CR} = 0.255$
Diversity = 4.0737e+00	$k_t = [2, 2, 3, 3]$		
Generation 350: Best = 1.992786e+06	FES = 35100	$\mu_F = 0.770$	$\mu_{CR} = 0.241$
Diversity = 4.2761e+00	$k_t = [1, 1, 1, 1]$		
Generation 360: Best = 1.854592e+06	FES = 36100	$\mu_F = 0.771$	$\mu_{CR} = 0.242$
Diversity = 4.3044e+00	$k_t = [2, 4, 3, 3]$		
Generation 370: Best = 1.767333e+06	FES = 37100	$\mu_F = 0.766$	$\mu_{CR} = 0.231$
Diversity = 4.3071e+00	$k_t = [1, 1, 1, 1]$		
Generation 380: Best = 1.687319e+06	FES = 38100	$\mu_F = 0.773$	$\mu_{CR} = 0.229$
Diversity = 4.1420e+00	$k_t = [3, 4, 2, 3]$		
Generation 390: Best = 1.607921e+06	FES = 39100	$\mu_F = 0.771$	$\mu_{CR} = 0.203$
Diversity = 4.5239e+00	$k_t = [1, 1, 1, 1]$		
Generation 400: Best = 1.550002e+06	FES = 40100	$\mu_F = 0.773$	$\mu_{CR} = 0.182$
Diversity = 4.5363e+00	$k_t = [3, 4, 2, 3]$		
Generation 410: Best = 1.460398e+06	FES = 41100	$\mu_F = 0.786$	$\mu_{CR} = 0.176$
Diversity = 4.9617e+00	$k_t = [1, 1, 1, 1]$		
Generation 420: Best = 1.325900e+06	FES = 42100	$\mu_F = 0.796$	$\mu_{CR} = 0.180$
Diversity = 5.0745e+00	$k_t = [4, 3, 3, 3]$		
Generation 430: Best = 1.244243e+06	FES = 43100	$\mu_F = 0.790$	$\mu_{CR} = 0.178$
Diversity = 5.7208e+00	$k_t = [1, 1, 1, 1]$		
Generation 440: Best = 1.168962e+06	FES = 44100	$\mu_F = 0.788$	$\mu_{CR} = 0.156$
Diversity = 5.7882e+00	$k_t = [2, 4, 2, 2]$		
Generation 450: Best = 1.084275e+06	FES = 45100	$\mu_F = 0.779$	$\mu_{CR} = 0.141$
Diversity = 6.2368e+00	$k_t = [1, 1, 1, 1]$		
Generation 460: Best = 9.437746e+05	FES = 46100	$\mu_F = 0.786$	$\mu_{CR} = 0.148$
Diversity = 6.9354e+00	$k_t = [4, 4, 4, 3]$		
Generation 470: Best = 8.975530e+05	FES = 47100	$\mu_F = 0.764$	$\mu_{CR} = 0.143$
Diversity = 7.0301e+00	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 8.213122e+05	FES = 48100	$\mu_F = 0.771$	$\mu_{CR} = 0.129$
Diversity = 6.9028e+00	$k_t = [2, 4, 4, 2]$		
Generation 490: Best = 7.796345e+05	FES = 49100	$\mu_F = 0.782$	$\mu_{CR} = 0.120$
Diversity = 6.6479e+00	$k_t = [1, 1, 1, 1]$		

Generation 500:	Best = 7.236320e+05	FES = 50100	$\mu_F = 0.787$	$\mu_{CR} = 0.117$
Diversity = 6.7373e+00	$k_t = [3, 2, 2, 3]$			
Generation 510:	Best = 6.809523e+05	FES = 51100	$\mu_F = 0.782$	$\mu_{CR} = 0.109$
Diversity = 6.7668e+00	$k_t = [1, 1, 1, 1]$			
Generation 520:	Best = 6.412626e+05	FES = 52100	$\mu_F = 0.781$	$\mu_{CR} = 0.120$
Diversity = 6.5364e+00	$k_t = [4, 2, 4, 3]$			
Generation 530:	Best = 5.833725e+05	FES = 53100	$\mu_F = 0.790$	$\mu_{CR} = 0.114$
Diversity = 6.9898e+00	$k_t = [1, 1, 1, 1]$			
Generation 540:	Best = 5.227965e+05	FES = 54100	$\mu_F = 0.795$	$\mu_{CR} = 0.109$
Diversity = 7.3661e+00	$k_t = [3, 3, 3, 4]$			
Generation 550:	Best = 4.828970e+05	FES = 55100	$\mu_F = 0.796$	$\mu_{CR} = 0.111$
Diversity = 7.4831e+00	$k_t = [1, 1, 1, 1]$			
Generation 560:	Best = 4.325974e+05	FES = 56100	$\mu_F = 0.785$	$\mu_{CR} = 0.111$
Diversity = 7.9575e+00	$k_t = [3, 2, 4, 2]$			
Generation 570:	Best = 3.957394e+05	FES = 57100	$\mu_F = 0.797$	$\mu_{CR} = 0.117$
Diversity = 8.1347e+00	$k_t = [1, 1, 1, 1]$			
Generation 580:	Best = 3.497939e+05	FES = 58100	$\mu_F = 0.787$	$\mu_{CR} = 0.107$
Diversity = 7.4133e+00	$k_t = [3, 2, 3, 4]$			
Generation 590:	Best = 3.159896e+05	FES = 59100	$\mu_F = 0.788$	$\mu_{CR} = 0.104$
Diversity = 7.4203e+00	$k_t = [1, 1, 1, 1]$			
Generation 600:	Best = 2.988483e+05	FES = 60100	$\mu_F = 0.792$	$\mu_{CR} = 0.098$
Diversity = 7.6911e+00	$k_t = [4, 3, 4, 2]$			
Generation 610:	Best = 2.553335e+05	FES = 61100	$\mu_F = 0.797$	$\mu_{CR} = 0.105$
Diversity = 8.1549e+00	$k_t = [1, 1, 1, 1]$			
Generation 620:	Best = 2.249548e+05	FES = 62100	$\mu_F = 0.808$	$\mu_{CR} = 0.094$
Diversity = 8.6539e+00	$k_t = [3, 4, 2, 3]$			
Generation 630:	Best = 2.095612e+05	FES = 63100	$\mu_F = 0.808$	$\mu_{CR} = 0.090$
Diversity = 9.1232e+00	$k_t = [1, 1, 1, 1]$			
Generation 640:	Best = 1.876862e+05	FES = 64100	$\mu_F = 0.808$	$\mu_{CR} = 0.095$
Diversity = 9.3240e+00	$k_t = [4, 3, 4, 3]$			
Generation 650:	Best = 1.730718e+05	FES = 65100	$\mu_F = 0.807$	$\mu_{CR} = 0.092$
Diversity = 9.9427e+00	$k_t = [1, 1, 1, 1]$			
Generation 660:	Best = 1.522130e+05	FES = 66100	$\mu_F = 0.794$	$\mu_{CR} = 0.103$
Diversity = 8.8347e+00	$k_t = [2, 4, 2, 3]$			
Generation 670:	Best = 1.427485e+05	FES = 67100	$\mu_F = 0.792$	$\mu_{CR} = 0.095$
Diversity = 8.1379e+00	$k_t = [1, 1, 1, 1]$			
Generation 680:	Best = 1.260543e+05	FES = 68100	$\mu_F = 0.777$	$\mu_{CR} = 0.104$
Diversity = 7.6345e+00	$k_t = [2, 2, 3, 2]$			
Generation 690:	Best = 1.197895e+05	FES = 69100	$\mu_F = 0.787$	$\mu_{CR} = 0.101$
Diversity = 7.9501e+00	$k_t = [1, 1, 1, 1]$			
Generation 700:	Best = 1.099173e+05	FES = 70100	$\mu_F = 0.783$	$\mu_{CR} = 0.099$
Diversity = 7.6263e+00	$k_t = [2, 3, 4, 2]$			
Generation 710:	Best = 1.045906e+05	FES = 71100	$\mu_F = 0.784$	$\mu_{CR} = 0.101$
Diversity = 7.6130e+00	$k_t = [1, 1, 1, 1]$			
Generation 720:	Best = 9.351589e+04	FES = 72100	$\mu_F = 0.792$	$\mu_{CR} = 0.103$
Diversity = 7.0310e+00	$k_t = [2, 3, 4, 4]$			
Generation 730:	Best = 8.351889e+04	FES = 73100	$\mu_F = 0.796$	$\mu_{CR} = 0.104$
Diversity = 6.5922e+00	$k_t = [1, 1, 1, 1]$			
Generation 740:	Best = 7.983577e+04	FES = 74100	$\mu_F = 0.800$	$\mu_{CR} = 0.102$
Diversity = 6.8275e+00	$k_t = [4, 2, 2, 3]$			
Generation 750:	Best = 7.439208e+04	FES = 75100	$\mu_F = 0.815$	$\mu_{CR} = 0.101$
Diversity = 7.4511e+00	$k_t = [1, 1, 1, 1]$			
Generation 760:	Best = 7.021597e+04	FES = 76100	$\mu_F = 0.804$	$\mu_{CR} = 0.095$
Diversity = 7.3506e+00	$k_t = [2, 2, 2, 3]$			

Generation 770: Best = 6.194613e+04	FES = 77100	$\mu_F = 0.798$	$\mu_{CR} = 0.097$
Diversity = 7.2437e+00	$k_t = [1, 1, 1, 1]$		
Generation 780: Best = 5.659223e+04	FES = 78100	$\mu_F = 0.794$	$\mu_{CR} = 0.098$
Diversity = 7.5122e+00	$k_t = [4, 4, 4, 2]$		
Generation 790: Best = 5.283755e+04	FES = 79100	$\mu_F = 0.782$	$\mu_{CR} = 0.102$
Diversity = 7.4939e+00	$k_t = [1, 1, 1, 1]$		
Generation 800: Best = 5.012501e+04	FES = 80100	$\mu_F = 0.788$	$\mu_{CR} = 0.094$
Diversity = 6.9912e+00	$k_t = [3, 2, 2, 3]$		
Generation 810: Best = 4.764731e+04	FES = 81100	$\mu_F = 0.782$	$\mu_{CR} = 0.086$
Diversity = 6.3491e+00	$k_t = [1, 1, 1, 1]$		
Generation 820: Best = 4.517302e+04	FES = 82100	$\mu_F = 0.775$	$\mu_{CR} = 0.091$
Diversity = 6.0312e+00	$k_t = [3, 4, 2, 3]$		
Generation 830: Best = 4.225600e+04	FES = 83100	$\mu_F = 0.780$	$\mu_{CR} = 0.089$
Diversity = 6.1360e+00	$k_t = [1, 1, 1, 1]$		
Generation 840: Best = 3.969704e+04	FES = 84100	$\mu_F = 0.784$	$\mu_{CR} = 0.090$
Diversity = 5.9107e+00	$k_t = [4, 2, 3, 4]$		
Generation 850: Best = 3.778619e+04	FES = 85100	$\mu_F = 0.769$	$\mu_{CR} = 0.090$
Diversity = 4.8696e+00	$k_t = [1, 1, 1, 1]$		
Generation 860: Best = 3.666595e+04	FES = 86100	$\mu_F = 0.781$	$\mu_{CR} = 0.093$
Diversity = 4.4108e+00	$k_t = [4, 3, 2, 2]$		
Generation 870: Best = 3.510017e+04	FES = 87100	$\mu_F = 0.779$	$\mu_{CR} = 0.090$
Diversity = 4.0592e+00	$k_t = [1, 1, 1, 1]$		
Generation 880: Best = 3.345268e+04	FES = 88100	$\mu_F = 0.787$	$\mu_{CR} = 0.096$
Diversity = 4.0064e+00	$k_t = [3, 2, 2, 3]$		
Generation 890: Best = 3.208430e+04	FES = 89100	$\mu_F = 0.791$	$\mu_{CR} = 0.099$
Diversity = 3.7806e+00	$k_t = [1, 1, 1, 1]$		
Generation 900: Best = 3.078456e+04	FES = 90100	$\mu_F = 0.792$	$\mu_{CR} = 0.105$
Diversity = 3.8308e+00	$k_t = [4, 3, 2, 3]$		
Generation 910: Best = 2.969750e+04	FES = 91100	$\mu_F = 0.786$	$\mu_{CR} = 0.102$
Diversity = 3.6420e+00	$k_t = [1, 1, 1, 1]$		
Generation 920: Best = 2.823289e+04	FES = 92100	$\mu_F = 0.796$	$\mu_{CR} = 0.103$
Diversity = 4.0696e+00	$k_t = [3, 2, 4, 2]$		
Generation 930: Best = 2.733088e+04	FES = 93100	$\mu_F = 0.798$	$\mu_{CR} = 0.105$
Diversity = 3.8353e+00	$k_t = [1, 1, 1, 1]$		
Generation 940: Best = 2.615950e+04	FES = 94100	$\mu_F = 0.787$	$\mu_{CR} = 0.093$
Diversity = 3.6233e+00	$k_t = [2, 4, 2, 3]$		
Generation 950: Best = 2.543485e+04	FES = 95100	$\mu_F = 0.795$	$\mu_{CR} = 0.088$
Diversity = 3.4405e+00	$k_t = [1, 1, 1, 1]$		
Generation 960: Best = 2.453425e+04	FES = 96100	$\mu_F = 0.792$	$\mu_{CR} = 0.097$
Diversity = 3.0280e+00	$k_t = [2, 2, 4, 3]$		
Generation 970: Best = 2.338097e+04	FES = 97100	$\mu_F = 0.791$	$\mu_{CR} = 0.096$
Diversity = 2.8928e+00	$k_t = [1, 1, 1, 1]$		
Generation 980: Best = 2.275982e+04	FES = 98100	$\mu_F = 0.798$	$\mu_{CR} = 0.093$
Diversity = 3.2062e+00	$k_t = [2, 2, 4, 2]$		
Generation 990: Best = 2.209595e+04	FES = 99100	$\mu_F = 0.787$	$\mu_{CR} = 0.091$
Diversity = 2.9543e+00	$k_t = [1, 1, 1, 1]$		
Generation 1000: Best = 2.107052e+04	FES = 100100	$\mu_F = 0.781$	$\mu_{CR} = 0.089$
Diversity = 2.8740e+00	$k_t = [2, 4, 2, 3]$		
Generation 1010: Best = 2.079478e+04	FES = 101100	$\mu_F = 0.779$	$\mu_{CR} = 0.089$
Diversity = 2.7135e+00	$k_t = [1, 1, 1, 1]$		
Generation 1020: Best = 2.020530e+04	FES = 102100	$\mu_F = 0.764$	$\mu_{CR} = 0.094$
Diversity = 2.4915e+00	$k_t = [4, 3, 4, 3]$		
Generation 1030: Best = 1.986348e+04	FES = 103100	$\mu_F = 0.773$	$\mu_{CR} = 0.096$
Diversity = 2.3130e+00	$k_t = [1, 1, 1, 1]$		

Generation 1040: Best = 1.972232e+04 FES = 104100 μ_F = 0.755 μ_{CR} = 0.087
 Diversity = 2.0204e+00 k_t = [4, 4, 2, 2]
 Generation 1050: Best = 1.934635e+04 FES = 105100 μ_F = 0.766 μ_{CR} = 0.089
 Diversity = 1.8745e+00 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 1.896208e+04 FES = 106100 μ_F = 0.769 μ_{CR} = 0.092
 Diversity = 1.7705e+00 k_t = [2, 2, 4, 4]
 Generation 1070: Best = 1.883815e+04 FES = 107100 μ_F = 0.780 μ_{CR} = 0.087
 Diversity = 1.6836e+00 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 1.854523e+04 FES = 108100 μ_F = 0.784 μ_{CR} = 0.097
 Diversity = 1.6269e+00 k_t = [3, 4, 3, 3]
 Generation 1090: Best = 1.845294e+04 FES = 109100 μ_F = 0.785 μ_{CR} = 0.098
 Diversity = 1.6180e+00 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 1.828448e+04 FES = 110100 μ_F = 0.788 μ_{CR} = 0.101
 Diversity = 1.5806e+00 k_t = [2, 4, 2, 3]
 Generation 1110: Best = 1.803418e+04 FES = 111100 μ_F = 0.794 μ_{CR} = 0.102
 Diversity = 1.6942e+00 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 1.790754e+04 FES = 112100 μ_F = 0.798 μ_{CR} = 0.101
 Diversity = 1.6670e+00 k_t = [4, 3, 2, 2]
 Generation 1130: Best = 1.776002e+04 FES = 113100 μ_F = 0.787 μ_{CR} = 0.095
 Diversity = 1.6461e+00 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 1.758427e+04 FES = 114100 μ_F = 0.795 μ_{CR} = 0.091
 Diversity = 1.6093e+00 k_t = [4, 2, 3, 4]
 Generation 1150: Best = 1.736909e+04 FES = 115100 μ_F = 0.808 μ_{CR} = 0.088
 Diversity = 1.7153e+00 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 1.711266e+04 FES = 116100 μ_F = 0.816 μ_{CR} = 0.085
 Diversity = 1.7756e+00 k_t = [2, 3, 2, 3]
 Generation 1170: Best = 1.685311e+04 FES = 117100 μ_F = 0.823 μ_{CR} = 0.095
 Diversity = 2.0834e+00 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 1.657267e+04 FES = 118100 μ_F = 0.829 μ_{CR} = 0.089
 Diversity = 2.1538e+00 k_t = [4, 3, 2, 3]
 Generation 1190: Best = 1.649671e+04 FES = 119100 μ_F = 0.815 μ_{CR} = 0.087
 Diversity = 1.8505e+00 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 1.636295e+04 FES = 120100 μ_F = 0.793 μ_{CR} = 0.087
 Diversity = 1.6367e+00 k_t = [4, 4, 3, 2]
 Generation 1210: Best = 1.624004e+04 FES = 121100 μ_F = 0.783 μ_{CR} = 0.090
 Diversity = 1.6345e+00 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 1.613246e+04 FES = 122100 μ_F = 0.781 μ_{CR} = 0.091
 Diversity = 1.5061e+00 k_t = [4, 2, 4, 4]
 Generation 1230: Best = 1.600760e+04 FES = 123100 μ_F = 0.770 μ_{CR} = 0.087
 Diversity = 1.3490e+00 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 1.582237e+04 FES = 124100 μ_F = 0.784 μ_{CR} = 0.087
 Diversity = 1.3758e+00 k_t = [3, 4, 3, 2]
 Generation 1250: Best = 1.577290e+04 FES = 125100 μ_F = 0.780 μ_{CR} = 0.093
 Diversity = 1.4721e+00 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 1.550133e+04 FES = 126100 μ_F = 0.803 μ_{CR} = 0.082
 Diversity = 1.8821e+00 k_t = [4, 4, 4, 3]
 Generation 1270: Best = 1.540931e+04 FES = 127100 μ_F = 0.792 μ_{CR} = 0.093
 Diversity = 1.3670e+00 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 1.537306e+04 FES = 128100 μ_F = 0.792 μ_{CR} = 0.093
 Diversity = 1.1916e+00 k_t = [4, 4, 4, 3]
 Generation 1290: Best = 1.526190e+04 FES = 129100 μ_F = 0.788 μ_{CR} = 0.090
 Diversity = 1.1494e+00 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 1.515113e+04 FES = 130100 μ_F = 0.788 μ_{CR} = 0.086
 Diversity = 1.1597e+00 k_t = [4, 3, 2, 2]

Generation 1310: Best = 1.506496e+04 FES = 131100 μ_F = 0.788 μ_{CR} = 0.094
 Diversity = 1.1761e+00 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 1.499368e+04 FES = 132100 μ_F = 0.778 μ_{CR} = 0.098
 Diversity = 1.2014e+00 k_t = [4, 4, 4, 2]
 Generation 1330: Best = 1.487054e+04 FES = 133100 μ_F = 0.777 μ_{CR} = 0.095
 Diversity = 1.3014e+00 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 1.460064e+04 FES = 134100 μ_F = 0.786 μ_{CR} = 0.095
 Diversity = 1.3793e+00 k_t = [2, 2, 4, 2]
 Generation 1350: Best = 1.448440e+04 FES = 135100 μ_F = 0.788 μ_{CR} = 0.092
 Diversity = 1.2396e+00 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 1.442966e+04 FES = 136100 μ_F = 0.772 μ_{CR} = 0.093
 Diversity = 1.0647e+00 k_t = [3, 3, 4, 4]
 Generation 1370: Best = 1.437586e+04 FES = 137100 μ_F = 0.772 μ_{CR} = 0.088
 Diversity = 9.4257e-01 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 1.431596e+04 FES = 138100 μ_F = 0.764 μ_{CR} = 0.087
 Diversity = 8.3411e-01 k_t = [2, 4, 3, 3]
 Generation 1390: Best = 1.428084e+04 FES = 139100 μ_F = 0.760 μ_{CR} = 0.084
 Diversity = 7.7477e-01 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 1.424421e+04 FES = 140100 μ_F = 0.762 μ_{CR} = 0.087
 Diversity = 7.2391e-01 k_t = [3, 4, 3, 3]
 Generation 1410: Best = 1.417849e+04 FES = 141100 μ_F = 0.771 μ_{CR} = 0.081
 Diversity = 7.0672e-01 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 1.416298e+04 FES = 142100 μ_F = 0.766 μ_{CR} = 0.087
 Diversity = 6.2017e-01 k_t = [4, 3, 4, 3]
 Generation 1430: Best = 1.413158e+04 FES = 143100 μ_F = 0.769 μ_{CR} = 0.085
 Diversity = 5.4428e-01 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 1.411728e+04 FES = 144100 μ_F = 0.757 μ_{CR} = 0.084
 Diversity = 4.9951e-01 k_t = [3, 2, 4, 3]
 Generation 1450: Best = 1.406059e+04 FES = 145100 μ_F = 0.784 μ_{CR} = 0.078
 Diversity = 4.9409e-01 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 1.403352e+04 FES = 146100 μ_F = 0.802 μ_{CR} = 0.086
 Diversity = 4.9733e-01 k_t = [2, 3, 2, 4]
 Generation 1470: Best = 1.401316e+04 FES = 147100 μ_F = 0.807 μ_{CR} = 0.082
 Diversity = 4.9026e-01 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 1.395267e+04 FES = 148100 μ_F = 0.814 μ_{CR} = 0.075
 Diversity = 4.7975e-01 k_t = [2, 3, 4, 4]
 Generation 1490: Best = 1.394314e+04 FES = 149100 μ_F = 0.804 μ_{CR} = 0.086
 Diversity = 4.3844e-01 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 1.392199e+04 FES = 150100 μ_F = 0.812 μ_{CR} = 0.080
 Diversity = 4.2402e-01 k_t = [4, 4, 2, 4]
 Generation 1510: Best = 1.387675e+04 FES = 151100 μ_F = 0.816 μ_{CR} = 0.081
 Diversity = 4.2093e-01 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 1.386323e+04 FES = 152100 μ_F = 0.815 μ_{CR} = 0.085
 Diversity = 4.2895e-01 k_t = [2, 4, 2, 4]
 Generation 1530: Best = 1.384558e+04 FES = 153100 μ_F = 0.809 μ_{CR} = 0.075
 Diversity = 4.2302e-01 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 1.381235e+04 FES = 154100 μ_F = 0.801 μ_{CR} = 0.073
 Diversity = 4.4064e-01 k_t = [3, 3, 4, 2]
 Generation 1550: Best = 1.379207e+04 FES = 155100 μ_F = 0.809 μ_{CR} = 0.079
 Diversity = 4.8313e-01 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 1.377920e+04 FES = 156100 μ_F = 0.794 μ_{CR} = 0.074
 Diversity = 4.8402e-01 k_t = [3, 4, 4, 4]
 Generation 1570: Best = 1.374353e+04 FES = 157100 μ_F = 0.798 μ_{CR} = 0.079
 Diversity = 4.8826e-01 k_t = [1, 1, 1, 1]

Generation 1580: Best = 1.372412e+04 FES = 158100 μ_F = 0.788 μ_{CR} = 0.083
 Diversity = 4.5529e-01 k_t = [2, 3, 2, 3]
 Generation 1590: Best = 1.371286e+04 FES = 159100 μ_F = 0.777 μ_{CR} = 0.085
 Diversity = 4.4268e-01 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 1.368554e+04 FES = 160100 μ_F = 0.787 μ_{CR} = 0.084
 Diversity = 4.3291e-01 k_t = [3, 2, 3, 4]
 Generation 1610: Best = 1.366738e+04 FES = 161100 μ_F = 0.795 μ_{CR} = 0.085
 Diversity = 4.4485e-01 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 1.365879e+04 FES = 162100 μ_F = 0.797 μ_{CR} = 0.079
 Diversity = 4.5643e-01 k_t = [4, 3, 2, 2]
 Generation 1630: Best = 1.363145e+04 FES = 163100 μ_F = 0.799 μ_{CR} = 0.078
 Diversity = 4.7250e-01 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 1.363033e+04 FES = 164100 μ_F = 0.801 μ_{CR} = 0.072
 Diversity = 4.7712e-01 k_t = [4, 4, 4, 2]
 Generation 1650: Best = 1.361089e+04 FES = 165100 μ_F = 0.789 μ_{CR} = 0.068
 Diversity = 4.7510e-01 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 1.360061e+04 FES = 166100 μ_F = 0.791 μ_{CR} = 0.080
 Diversity = 4.7379e-01 k_t = [2, 3, 4, 3]
 Generation 1670: Best = 1.357813e+04 FES = 167100 μ_F = 0.794 μ_{CR} = 0.076
 Diversity = 4.6708e-01 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 1.356446e+04 FES = 168100 μ_F = 0.802 μ_{CR} = 0.078
 Diversity = 4.4361e-01 k_t = [3, 3, 3, 4]
 Generation 1690: Best = 1.354021e+04 FES = 169100 μ_F = 0.792 μ_{CR} = 0.079
 Diversity = 4.3998e-01 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 1.353532e+04 FES = 170100 μ_F = 0.795 μ_{CR} = 0.070
 Diversity = 4.2588e-01 k_t = [3, 3, 2, 2]
 Generation 1710: Best = 1.349656e+04 FES = 171100 μ_F = 0.808 μ_{CR} = 0.075
 Diversity = 4.8602e-01 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 1.346957e+04 FES = 172100 μ_F = 0.811 μ_{CR} = 0.073
 Diversity = 4.9434e-01 k_t = [2, 2, 2, 4]
 Generation 1730: Best = 1.345773e+04 FES = 173100 μ_F = 0.807 μ_{CR} = 0.080
 Diversity = 4.5544e-01 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 1.342167e+04 FES = 174100 μ_F = 0.806 μ_{CR} = 0.082
 Diversity = 4.4427e-01 k_t = [3, 2, 4, 4]
 Generation 1750: Best = 1.338876e+04 FES = 175100 μ_F = 0.797 μ_{CR} = 0.082
 Diversity = 4.3685e-01 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 1.336853e+04 FES = 176100 μ_F = 0.805 μ_{CR} = 0.089
 Diversity = 4.3870e-01 k_t = [3, 4, 3, 3]
 Generation 1770: Best = 1.336652e+04 FES = 177100 μ_F = 0.816 μ_{CR} = 0.094
 Diversity = 4.2177e-01 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 1.334750e+04 FES = 178100 μ_F = 0.809 μ_{CR} = 0.090
 Diversity = 4.3059e-01 k_t = [2, 3, 2, 2]
 Generation 1790: Best = 1.334309e+04 FES = 179100 μ_F = 0.814 μ_{CR} = 0.084
 Diversity = 4.4386e-01 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 1.332934e+04 FES = 180100 μ_F = 0.792 μ_{CR} = 0.098
 Diversity = 4.3881e-01 k_t = [4, 3, 4, 2]
 Generation 1810: Best = 1.331268e+04 FES = 181100 μ_F = 0.796 μ_{CR} = 0.099
 Diversity = 4.2583e-01 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 1.330030e+04 FES = 182100 μ_F = 0.800 μ_{CR} = 0.083
 Diversity = 3.9042e-01 k_t = [3, 3, 2, 2]
 Generation 1830: Best = 1.327960e+04 FES = 183100 μ_F = 0.797 μ_{CR} = 0.081
 Diversity = 3.9508e-01 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 1.327178e+04 FES = 184100 μ_F = 0.803 μ_{CR} = 0.082
 Diversity = 3.9458e-01 k_t = [3, 2, 4, 2]

Generation 1850: Best = 1.326666e+04	FES = 185100	$\mu_F = 0.789$	$\mu_{CR} = 0.077$
Diversity = 4.0130e-01	k_t = [1, 1, 1, 1]		
Generation 1860: Best = 1.325582e+04	FES = 186100	$\mu_F = 0.777$	$\mu_{CR} = 0.074$
Diversity = 4.0818e-01	k_t = [2, 3, 4, 2]		
Generation 1870: Best = 1.324307e+04	FES = 187100	$\mu_F = 0.782$	$\mu_{CR} = 0.066$
Diversity = 3.9538e-01	k_t = [1, 1, 1, 1]		
Generation 1880: Best = 1.322439e+04	FES = 188100	$\mu_F = 0.786$	$\mu_{CR} = 0.066$
Diversity = 3.6702e-01	k_t = [3, 3, 3, 4]		
Generation 1890: Best = 1.321624e+04	FES = 189100	$\mu_F = 0.791$	$\mu_{CR} = 0.073$
Diversity = 3.6145e-01	k_t = [1, 1, 1, 1]		
Generation 1900: Best = 1.320533e+04	FES = 190100	$\mu_F = 0.774$	$\mu_{CR} = 0.074$
Diversity = 3.4213e-01	k_t = [2, 3, 4, 2]		
Generation 1910: Best = 1.319851e+04	FES = 191100	$\mu_F = 0.789$	$\mu_{CR} = 0.071$
Diversity = 3.5208e-01	k_t = [1, 1, 1, 1]		
Generation 1920: Best = 1.319645e+04	FES = 192100	$\mu_F = 0.801$	$\mu_{CR} = 0.068$
Diversity = 3.3807e-01	k_t = [2, 3, 2, 3]		
Generation 1930: Best = 1.318229e+04	FES = 193100	$\mu_F = 0.777$	$\mu_{CR} = 0.072$
Diversity = 3.3317e-01	k_t = [1, 1, 1, 1]		
Generation 1940: Best = 1.317609e+04	FES = 194100	$\mu_F = 0.776$	$\mu_{CR} = 0.069$
Diversity = 3.3789e-01	k_t = [2, 3, 4, 2]		
Generation 1950: Best = 1.316698e+04	FES = 195100	$\mu_F = 0.774$	$\mu_{CR} = 0.075$
Diversity = 3.2529e-01	k_t = [1, 1, 1, 1]		
Generation 1960: Best = 1.316002e+04	FES = 196100	$\mu_F = 0.791$	$\mu_{CR} = 0.077$
Diversity = 3.2854e-01	k_t = [3, 2, 4, 3]		
Generation 1970: Best = 1.315178e+04	FES = 197100	$\mu_F = 0.802$	$\mu_{CR} = 0.074$
Diversity = 3.3563e-01	k_t = [1, 1, 1, 1]		
Generation 1980: Best = 1.314154e+04	FES = 198100	$\mu_F = 0.788$	$\mu_{CR} = 0.071$
Diversity = 3.6586e-01	k_t = [2, 3, 3, 2]		
Generation 1990: Best = 1.313461e+04	FES = 199100	$\mu_F = 0.795$	$\mu_{CR} = 0.060$
Diversity = 3.8607e-01	k_t = [1, 1, 1, 1]		
Generation 2000: Best = 1.312794e+04	FES = 200100	$\mu_F = 0.781$	$\mu_{CR} = 0.071$
Diversity = 3.8141e-01	k_t = [2, 3, 2, 2]		
Generation 2010: Best = 1.310748e+04	FES = 201100	$\mu_F = 0.797$	$\mu_{CR} = 0.066$
Diversity = 3.9103e-01	k_t = [1, 1, 1, 1]		
Generation 2020: Best = 1.309366e+04	FES = 202100	$\mu_F = 0.791$	$\mu_{CR} = 0.073$
Diversity = 3.8619e-01	k_t = [3, 3, 2, 4]		
Generation 2030: Best = 1.308205e+04	FES = 203100	$\mu_F = 0.793$	$\mu_{CR} = 0.072$
Diversity = 3.9170e-01	k_t = [1, 1, 1, 1]		
Generation 2040: Best = 1.304015e+04	FES = 204100	$\mu_F = 0.796$	$\mu_{CR} = 0.083$
Diversity = 4.0121e-01	k_t = [2, 3, 2, 3]		
Generation 2050: Best = 1.296928e+04	FES = 205100	$\mu_F = 0.802$	$\mu_{CR} = 0.083$
Diversity = 5.0197e-01	k_t = [1, 1, 1, 1]		
Generation 2060: Best = 1.275948e+04	FES = 206100	$\mu_F = 0.817$	$\mu_{CR} = 0.093$
Diversity = 6.2283e-01	k_t = [4, 3, 4, 4]		
Generation 2070: Best = 1.268990e+04	FES = 207100	$\mu_F = 0.825$	$\mu_{CR} = 0.111$
Diversity = 5.7614e-01	k_t = [1, 1, 1, 1]		
Generation 2080: Best = 1.267376e+04	FES = 208100	$\mu_F = 0.823$	$\mu_{CR} = 0.112$
Diversity = 5.0348e-01	k_t = [2, 4, 4, 4]		
Generation 2090: Best = 1.264836e+04	FES = 209100	$\mu_F = 0.817$	$\mu_{CR} = 0.101$
Diversity = 4.6648e-01	k_t = [1, 1, 1, 1]		
Generation 2100: Best = 1.264552e+04	FES = 210100	$\mu_F = 0.816$	$\mu_{CR} = 0.096$
Diversity = 4.3417e-01	k_t = [2, 2, 4, 3]		
Generation 2110: Best = 1.263499e+04	FES = 211100	$\mu_F = 0.805$	$\mu_{CR} = 0.097$
Diversity = 3.8754e-01	k_t = [1, 1, 1, 1]		

Generation 2120: Best = 1.262678e+04 FES = 212100 μ_F = 0.787 μ_{CR} = 0.088
 Diversity = 3.8077e-01 k_t = [2, 4, 4, 2]
 Generation 2130: Best = 1.262403e+04 FES = 213100 μ_F = 0.790 μ_{CR} = 0.078
 Diversity = 3.6764e-01 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 1.260662e+04 FES = 214100 μ_F = 0.794 μ_{CR} = 0.072
 Diversity = 3.4993e-01 k_t = [3, 2, 3, 4]
 Generation 2150: Best = 1.259860e+04 FES = 215100 μ_F = 0.796 μ_{CR} = 0.074
 Diversity = 3.2501e-01 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 1.259807e+04 FES = 216100 μ_F = 0.790 μ_{CR} = 0.075
 Diversity = 2.9887e-01 k_t = [2, 4, 4, 4]
 Generation 2170: Best = 1.259127e+04 FES = 217100 μ_F = 0.791 μ_{CR} = 0.069
 Diversity = 2.9796e-01 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 1.258348e+04 FES = 218100 μ_F = 0.793 μ_{CR} = 0.068
 Diversity = 2.9020e-01 k_t = [4, 4, 3, 3]
 Generation 2190: Best = 1.257851e+04 FES = 219100 μ_F = 0.787 μ_{CR} = 0.071
 Diversity = 2.9698e-01 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 1.257851e+04 FES = 220100 μ_F = 0.792 μ_{CR} = 0.069
 Diversity = 2.8799e-01 k_t = [2, 4, 2, 3]
 Generation 2210: Best = 1.257771e+04 FES = 221100 μ_F = 0.794 μ_{CR} = 0.064
 Diversity = 2.7616e-01 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 1.257234e+04 FES = 222100 μ_F = 0.791 μ_{CR} = 0.066
 Diversity = 2.6841e-01 k_t = [2, 2, 2, 3]
 Generation 2230: Best = 1.257082e+04 FES = 223100 μ_F = 0.803 μ_{CR} = 0.069
 Diversity = 2.6628e-01 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 1.256896e+04 FES = 224100 μ_F = 0.812 μ_{CR} = 0.056
 Diversity = 2.5991e-01 k_t = [4, 2, 3, 4]
 Generation 2250: Best = 1.256366e+04 FES = 225100 μ_F = 0.787 μ_{CR} = 0.054
 Diversity = 2.6388e-01 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 1.256242e+04 FES = 226100 μ_F = 0.798 μ_{CR} = 0.064
 Diversity = 2.5898e-01 k_t = [3, 4, 4, 2]
 Generation 2270: Best = 1.256108e+04 FES = 227100 μ_F = 0.777 μ_{CR} = 0.068
 Diversity = 2.5911e-01 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 1.255884e+04 FES = 228100 μ_F = 0.782 μ_{CR} = 0.061
 Diversity = 2.5391e-01 k_t = [2, 4, 2, 4]
 Generation 2290: Best = 1.255757e+04 FES = 229100 μ_F = 0.797 μ_{CR} = 0.053
 Diversity = 2.5019e-01 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 1.255509e+04 FES = 230100 μ_F = 0.789 μ_{CR} = 0.053
 Diversity = 2.5086e-01 k_t = [3, 3, 4, 4]
 Generation 2310: Best = 1.255471e+04 FES = 231100 μ_F = 0.821 μ_{CR} = 0.054
 Diversity = 2.5231e-01 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 1.255294e+04 FES = 232100 μ_F = 0.839 μ_{CR} = 0.049
 Diversity = 2.4564e-01 k_t = [3, 3, 3, 4]
 Generation 2330: Best = 1.255149e+04 FES = 233100 μ_F = 0.844 μ_{CR} = 0.054
 Diversity = 2.5084e-01 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 1.254852e+04 FES = 234100 μ_F = 0.843 μ_{CR} = 0.044
 Diversity = 2.4990e-01 k_t = [2, 4, 4, 4]
 Generation 2350: Best = 1.254777e+04 FES = 235100 μ_F = 0.842 μ_{CR} = 0.054
 Diversity = 2.4638e-01 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 1.254323e+04 FES = 236100 μ_F = 0.833 μ_{CR} = 0.052
 Diversity = 2.5207e-01 k_t = [4, 4, 2, 3]
 Generation 2370: Best = 1.254235e+04 FES = 237100 μ_F = 0.835 μ_{CR} = 0.056
 Diversity = 2.5516e-01 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 1.254068e+04 FES = 238100 μ_F = 0.823 μ_{CR} = 0.055
 Diversity = 2.5125e-01 k_t = [2, 3, 4, 2]

Generation 2390: Best = 1.253706e+04 FES = 239100 μ_F = 0.811 μ_{CR} = 0.062
 Diversity = 2.5155e-01 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 1.253618e+04 FES = 240100 μ_F = 0.778 μ_{CR} = 0.058
 Diversity = 2.5679e-01 k_t = [2, 2, 4, 2]
 Generation 2410: Best = 1.253191e+04 FES = 241100 μ_F = 0.773 μ_{CR} = 0.057
 Diversity = 2.5848e-01 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 1.253185e+04 FES = 242100 μ_F = 0.737 μ_{CR} = 0.061
 Diversity = 2.4446e-01 k_t = [3, 4, 2, 3]
 Generation 2430: Best = 1.252708e+04 FES = 243100 μ_F = 0.754 μ_{CR} = 0.060
 Diversity = 2.4488e-01 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 1.252600e+04 FES = 244100 μ_F = 0.778 μ_{CR} = 0.059
 Diversity = 2.4377e-01 k_t = [2, 4, 2, 4]
 Generation 2450: Best = 1.252235e+04 FES = 245100 μ_F = 0.783 μ_{CR} = 0.061
 Diversity = 2.4576e-01 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 1.251580e+04 FES = 246100 μ_F = 0.780 μ_{CR} = 0.071
 Diversity = 2.5702e-01 k_t = [4, 2, 3, 3]
 Generation 2470: Best = 1.251569e+04 FES = 247100 μ_F = 0.776 μ_{CR} = 0.059
 Diversity = 2.5945e-01 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 1.250874e+04 FES = 248100 μ_F = 0.782 μ_{CR} = 0.061
 Diversity = 2.6366e-01 k_t = [4, 2, 4, 2]
 Generation 2490: Best = 1.250767e+04 FES = 249100 μ_F = 0.782 μ_{CR} = 0.055
 Diversity = 2.6610e-01 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 1.250767e+04 FES = 250100 μ_F = 0.774 μ_{CR} = 0.058
 Diversity = 2.6949e-01 k_t = [2, 2, 3, 2]
 Generation 2510: Best = 1.250554e+04 FES = 251100 μ_F = 0.778 μ_{CR} = 0.058
 Diversity = 2.6168e-01 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 1.250473e+04 FES = 252100 μ_F = 0.780 μ_{CR} = 0.055
 Diversity = 2.5674e-01 k_t = [2, 4, 4, 4]
 Generation 2530: Best = 1.250213e+04 FES = 253100 μ_F = 0.788 μ_{CR} = 0.051
 Diversity = 2.4677e-01 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 1.249577e+04 FES = 254100 μ_F = 0.803 μ_{CR} = 0.055
 Diversity = 2.4395e-01 k_t = [4, 2, 4, 4]
 Generation 2550: Best = 1.249553e+04 FES = 255100 μ_F = 0.794 μ_{CR} = 0.054
 Diversity = 2.5417e-01 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 1.249197e+04 FES = 256100 μ_F = 0.804 μ_{CR} = 0.059
 Diversity = 2.3886e-01 k_t = [4, 2, 2, 4]
 Generation 2570: Best = 1.248808e+04 FES = 257100 μ_F = 0.801 μ_{CR} = 0.053
 Diversity = 2.3609e-01 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 1.248222e+04 FES = 258100 μ_F = 0.785 μ_{CR} = 0.061
 Diversity = 2.2472e-01 k_t = [2, 3, 3, 4]
 Generation 2590: Best = 1.248114e+04 FES = 259100 μ_F = 0.812 μ_{CR} = 0.057
 Diversity = 2.3157e-01 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 1.247979e+04 FES = 260100 μ_F = 0.821 μ_{CR} = 0.058
 Diversity = 2.2890e-01 k_t = [4, 2, 3, 4]
 Generation 2610: Best = 1.247307e+04 FES = 261100 μ_F = 0.832 μ_{CR} = 0.059
 Diversity = 2.3664e-01 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 1.246571e+04 FES = 262100 μ_F = 0.831 μ_{CR} = 0.061
 Diversity = 2.5354e-01 k_t = [2, 2, 4, 3]
 Generation 2630: Best = 1.246346e+04 FES = 263100 μ_F = 0.819 μ_{CR} = 0.059
 Diversity = 2.6318e-01 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 1.245333e+04 FES = 264100 μ_F = 0.817 μ_{CR} = 0.060
 Diversity = 2.7276e-01 k_t = [3, 4, 2, 2]
 Generation 2650: Best = 1.243371e+04 FES = 265100 μ_F = 0.809 μ_{CR} = 0.065
 Diversity = 2.9644e-01 k_t = [1, 1, 1, 1]

Generation 2660: Best = 1.243069e+04 FES = 266100 μ_F = 0.801 μ_{CR} = 0.069
 Diversity = 3.0651e-01 k_t = [2, 4, 3, 3]
 Generation 2670: Best = 1.241760e+04 FES = 267100 μ_F = 0.811 μ_{CR} = 0.064
 Diversity = 3.1588e-01 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 1.241327e+04 FES = 268100 μ_F = 0.801 μ_{CR} = 0.073
 Diversity = 2.9785e-01 k_t = [4, 3, 2, 3]
 Generation 2690: Best = 1.239953e+04 FES = 269100 μ_F = 0.797 μ_{CR} = 0.082
 Diversity = 2.9511e-01 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 1.239802e+04 FES = 270100 μ_F = 0.797 μ_{CR} = 0.083
 Diversity = 2.9694e-01 k_t = [3, 3, 3, 2]
 Generation 2710: Best = 1.238551e+04 FES = 271100 μ_F = 0.784 μ_{CR} = 0.078
 Diversity = 2.9469e-01 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 1.237006e+04 FES = 272100 μ_F = 0.807 μ_{CR} = 0.080
 Diversity = 3.0555e-01 k_t = [4, 3, 4, 3]
 Generation 2730: Best = 1.236527e+04 FES = 273100 μ_F = 0.823 μ_{CR} = 0.072
 Diversity = 3.1350e-01 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 1.236527e+04 FES = 274100 μ_F = 0.815 μ_{CR} = 0.074
 Diversity = 2.9937e-01 k_t = [3, 2, 3, 4]
 Generation 2750: Best = 1.235827e+04 FES = 275100 μ_F = 0.824 μ_{CR} = 0.069
 Diversity = 2.9423e-01 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 1.234747e+04 FES = 276100 μ_F = 0.823 μ_{CR} = 0.071
 Diversity = 2.8982e-01 k_t = [2, 4, 3, 4]
 Generation 2770: Best = 1.234605e+04 FES = 277100 μ_F = 0.824 μ_{CR} = 0.071
 Diversity = 2.7325e-01 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 1.234181e+04 FES = 278100 μ_F = 0.828 μ_{CR} = 0.071
 Diversity = 2.8003e-01 k_t = [4, 3, 2, 2]
 Generation 2790: Best = 1.232978e+04 FES = 279100 μ_F = 0.826 μ_{CR} = 0.069
 Diversity = 2.8422e-01 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 1.231830e+04 FES = 280100 μ_F = 0.814 μ_{CR} = 0.065
 Diversity = 2.6373e-01 k_t = [4, 3, 3, 4]
 Generation 2810: Best = 1.231354e+04 FES = 281100 μ_F = 0.804 μ_{CR} = 0.063
 Diversity = 2.6949e-01 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 1.230880e+04 FES = 282100 μ_F = 0.810 μ_{CR} = 0.059
 Diversity = 2.8212e-01 k_t = [3, 3, 2, 2]
 Generation 2830: Best = 1.230680e+04 FES = 283100 μ_F = 0.784 μ_{CR} = 0.071
 Diversity = 2.9035e-01 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 1.229481e+04 FES = 284100 μ_F = 0.792 μ_{CR} = 0.071
 Diversity = 2.9485e-01 k_t = [2, 3, 4, 4]
 Generation 2850: Best = 1.228603e+04 FES = 285100 μ_F = 0.818 μ_{CR} = 0.061
 Diversity = 3.0386e-01 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 1.228533e+04 FES = 286100 μ_F = 0.780 μ_{CR} = 0.065
 Diversity = 3.1248e-01 k_t = [2, 4, 3, 2]
 Generation 2870: Best = 1.226846e+04 FES = 287100 μ_F = 0.792 μ_{CR} = 0.069
 Diversity = 3.0804e-01 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 1.226621e+04 FES = 288100 μ_F = 0.778 μ_{CR} = 0.068
 Diversity = 3.0080e-01 k_t = [4, 3, 4, 2]
 Generation 2890: Best = 1.225852e+04 FES = 289100 μ_F = 0.792 μ_{CR} = 0.071
 Diversity = 3.0618e-01 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 1.225178e+04 FES = 290100 μ_F = 0.798 μ_{CR} = 0.067
 Diversity = 3.1016e-01 k_t = [4, 2, 4, 3]
 Generation 2910: Best = 1.225178e+04 FES = 291100 μ_F = 0.795 μ_{CR} = 0.071
 Diversity = 3.2646e-01 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 1.224321e+04 FES = 292100 μ_F = 0.792 μ_{CR} = 0.062
 Diversity = 3.2971e-01 k_t = [2, 3, 2, 3]

Generation 2930: Best = 1.223933e+04 FES = 293100 μ_F = 0.788 μ_{CR} = 0.060
 Diversity = 3.2373e-01 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 1.223424e+04 FES = 294100 μ_F = 0.800 μ_{CR} = 0.068
 Diversity = 3.2200e-01 k_t = [3, 4, 2, 2]
 Generation 2950: Best = 1.223258e+04 FES = 295100 μ_F = 0.789 μ_{CR} = 0.061
 Diversity = 3.2207e-01 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 1.222945e+04 FES = 296100 μ_F = 0.768 μ_{CR} = 0.058
 Diversity = 3.0894e-01 k_t = [3, 4, 3, 3]
 Generation 2970: Best = 1.222701e+04 FES = 297100 μ_F = 0.800 μ_{CR} = 0.062
 Diversity = 3.1824e-01 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 1.222082e+04 FES = 298100 μ_F = 0.765 μ_{CR} = 0.067
 Diversity = 3.1999e-01 k_t = [3, 2, 3, 2]
 Generation 2990: Best = 1.221850e+04 FES = 299100 μ_F = 0.776 μ_{CR} = 0.062
 Diversity = 3.1034e-01 k_t = [1, 1, 1, 1]
 Generation 3000: Best = 1.221576e+04 FES = 300100 μ_F = 0.770 μ_{CR} = 0.067
 Diversity = 3.0562e-01 k_t = [4, 3, 2, 4]
 Generation 3010: Best = 1.220885e+04 FES = 301100 μ_F = 0.771 μ_{CR} = 0.067
 Diversity = 3.0238e-01 k_t = [1, 1, 1, 1]
 Generation 3020: Best = 1.220579e+04 FES = 302100 μ_F = 0.770 μ_{CR} = 0.063
 Diversity = 2.7868e-01 k_t = [4, 3, 3, 4]
 Generation 3030: Best = 1.220329e+04 FES = 303100 μ_F = 0.771 μ_{CR} = 0.063
 Diversity = 2.8120e-01 k_t = [1, 1, 1, 1]
 Generation 3040: Best = 1.220021e+04 FES = 304100 μ_F = 0.775 μ_{CR} = 0.060
 Diversity = 2.7539e-01 k_t = [3, 4, 3, 3]
 Generation 3050: Best = 1.219750e+04 FES = 305100 μ_F = 0.793 μ_{CR} = 0.065
 Diversity = 2.7852e-01 k_t = [1, 1, 1, 1]
 Generation 3060: Best = 1.219282e+04 FES = 306100 μ_F = 0.832 μ_{CR} = 0.066
 Diversity = 2.7369e-01 k_t = [2, 4, 3, 4]
 Generation 3070: Best = 1.219149e+04 FES = 307100 μ_F = 0.815 μ_{CR} = 0.063
 Diversity = 2.7079e-01 k_t = [1, 1, 1, 1]
 Generation 3080: Best = 1.218313e+04 FES = 308100 μ_F = 0.798 μ_{CR} = 0.062
 Diversity = 2.6165e-01 k_t = [4, 2, 2, 4]
 Generation 3090: Best = 1.218305e+04 FES = 309100 μ_F = 0.810 μ_{CR} = 0.061
 Diversity = 2.6620e-01 k_t = [1, 1, 1, 1]
 Generation 3100: Best = 1.217857e+04 FES = 310100 μ_F = 0.830 μ_{CR} = 0.059
 Diversity = 2.6635e-01 k_t = [4, 4, 3, 3]
 Generation 3110: Best = 1.217411e+04 FES = 311100 μ_F = 0.827 μ_{CR} = 0.057
 Diversity = 2.8605e-01 k_t = [1, 1, 1, 1]
 Generation 3120: Best = 1.217264e+04 FES = 312100 μ_F = 0.837 μ_{CR} = 0.069
 Diversity = 2.8941e-01 k_t = [3, 2, 3, 3]
 Generation 3130: Best = 1.216410e+04 FES = 313100 μ_F = 0.818 μ_{CR} = 0.074
 Diversity = 2.8464e-01 k_t = [1, 1, 1, 1]
 Generation 3140: Best = 1.216403e+04 FES = 314100 μ_F = 0.821 μ_{CR} = 0.063
 Diversity = 2.7924e-01 k_t = [2, 2, 4, 4]
 Generation 3150: Best = 1.216162e+04 FES = 315100 μ_F = 0.816 μ_{CR} = 0.072
 Diversity = 2.8885e-01 k_t = [1, 1, 1, 1]
 Generation 3160: Best = 1.215675e+04 FES = 316100 μ_F = 0.802 μ_{CR} = 0.064
 Diversity = 2.8543e-01 k_t = [2, 4, 4, 2]
 Generation 3170: Best = 1.215191e+04 FES = 317100 μ_F = 0.804 μ_{CR} = 0.060
 Diversity = 2.7258e-01 k_t = [1, 1, 1, 1]
 Generation 3180: Best = 1.215191e+04 FES = 318100 μ_F = 0.770 μ_{CR} = 0.058
 Diversity = 2.6591e-01 k_t = [4, 4, 4, 3]
 Generation 3190: Best = 1.214661e+04 FES = 319100 μ_F = 0.785 μ_{CR} = 0.060
 Diversity = 2.7242e-01 k_t = [1, 1, 1, 1]

Generation 3200: Best = 1.214326e+04 FES = 320100 μ_F = 0.797 μ_{CR} = 0.054
 Diversity = 2.6705e-01 k_t = [2, 4, 4, 3]
 Generation 3210: Best = 1.214278e+04 FES = 321100 μ_F = 0.790 μ_{CR} = 0.051
 Diversity = 2.6765e-01 k_t = [1, 1, 1, 1]
 Generation 3220: Best = 1.213857e+04 FES = 322100 μ_F = 0.795 μ_{CR} = 0.075
 Diversity = 2.4955e-01 k_t = [4, 2, 4, 2]
 Generation 3230: Best = 1.213778e+04 FES = 323100 μ_F = 0.792 μ_{CR} = 0.064
 Diversity = 2.3810e-01 k_t = [1, 1, 1, 1]
 Generation 3240: Best = 1.213770e+04 FES = 324100 μ_F = 0.798 μ_{CR} = 0.058
 Diversity = 2.3208e-01 k_t = [3, 3, 3, 4]
 Generation 3250: Best = 1.213574e+04 FES = 325100 μ_F = 0.781 μ_{CR} = 0.058
 Diversity = 2.3607e-01 k_t = [1, 1, 1, 1]
 Generation 3260: Best = 1.213423e+04 FES = 326100 μ_F = 0.774 μ_{CR} = 0.050
 Diversity = 2.3775e-01 k_t = [3, 2, 4, 2]
 Generation 3270: Best = 1.213423e+04 FES = 327100 μ_F = 0.790 μ_{CR} = 0.050
 Diversity = 2.3876e-01 k_t = [1, 1, 1, 1]
 Generation 3280: Best = 1.213255e+04 FES = 328100 μ_F = 0.778 μ_{CR} = 0.046
 Diversity = 2.3842e-01 k_t = [4, 4, 3, 3]
 Generation 3290: Best = 1.213097e+04 FES = 329100 μ_F = 0.804 μ_{CR} = 0.049
 Diversity = 2.3064e-01 k_t = [1, 1, 1, 1]
 Generation 3300: Best = 1.212947e+04 FES = 330100 μ_F = 0.824 μ_{CR} = 0.050
 Diversity = 2.1678e-01 k_t = [3, 2, 4, 4]
 Generation 3310: Best = 1.212821e+04 FES = 331100 μ_F = 0.842 μ_{CR} = 0.053
 Diversity = 2.1885e-01 k_t = [1, 1, 1, 1]
 Generation 3320: Best = 1.212678e+04 FES = 332100 μ_F = 0.832 μ_{CR} = 0.058
 Diversity = 2.1387e-01 k_t = [2, 3, 4, 4]
 Generation 3330: Best = 1.212550e+04 FES = 333100 μ_F = 0.820 μ_{CR} = 0.054
 Diversity = 2.1543e-01 k_t = [1, 1, 1, 1]
 Generation 3340: Best = 1.212270e+04 FES = 334100 μ_F = 0.827 μ_{CR} = 0.052
 Diversity = 2.1434e-01 k_t = [3, 4, 4, 3]
 Generation 3350: Best = 1.212114e+04 FES = 335100 μ_F = 0.849 μ_{CR} = 0.040
 Diversity = 2.1996e-01 k_t = [1, 1, 1, 1]
 Generation 3360: Best = 1.211993e+04 FES = 336100 μ_F = 0.856 μ_{CR} = 0.042
 Diversity = 2.2066e-01 k_t = [2, 3, 3, 3]
 Generation 3370: Best = 1.211823e+04 FES = 337100 μ_F = 0.835 μ_{CR} = 0.040
 Diversity = 2.2482e-01 k_t = [1, 1, 1, 1]
 Generation 3380: Best = 1.211749e+04 FES = 338100 μ_F = 0.824 μ_{CR} = 0.055
 Diversity = 2.1927e-01 k_t = [2, 2, 4, 2]
 Generation 3390: Best = 1.211656e+04 FES = 339100 μ_F = 0.806 μ_{CR} = 0.048
 Diversity = 2.1468e-01 k_t = [1, 1, 1, 1]
 Generation 3400: Best = 1.211417e+04 FES = 340100 μ_F = 0.821 μ_{CR} = 0.052
 Diversity = 2.0733e-01 k_t = [2, 4, 2, 3]
 Generation 3410: Best = 1.210646e+04 FES = 341100 μ_F = 0.805 μ_{CR} = 0.053
 Diversity = 2.1522e-01 k_t = [1, 1, 1, 1]
 Generation 3420: Best = 1.210549e+04 FES = 342100 μ_F = 0.779 μ_{CR} = 0.055
 Diversity = 2.1040e-01 k_t = [3, 2, 2, 3]
 Generation 3430: Best = 1.210494e+04 FES = 343100 μ_F = 0.787 μ_{CR} = 0.054
 Diversity = 2.0695e-01 k_t = [1, 1, 1, 1]
 Generation 3440: Best = 1.210171e+04 FES = 344100 μ_F = 0.765 μ_{CR} = 0.054
 Diversity = 1.9896e-01 k_t = [2, 2, 2, 3]
 Generation 3450: Best = 1.210171e+04 FES = 345100 μ_F = 0.748 μ_{CR} = 0.057
 Diversity = 1.9742e-01 k_t = [1, 1, 1, 1]
 Generation 3460: Best = 1.210043e+04 FES = 346100 μ_F = 0.758 μ_{CR} = 0.060
 Diversity = 1.9007e-01 k_t = [4, 2, 3, 4]

Generation 3470: Best = 1.209801e+04 FES = 347100 μ_F = 0.761 μ_{CR} = 0.064
 Diversity = 1.7726e-01 k_t = [1, 1, 1, 1]
 Generation 3480: Best = 1.209285e+04 FES = 348100 μ_F = 0.754 μ_{CR} = 0.063
 Diversity = 1.7321e-01 k_t = [4, 3, 4, 3]
 Generation 3490: Best = 1.209234e+04 FES = 349100 μ_F = 0.766 μ_{CR} = 0.060
 Diversity = 1.7156e-01 k_t = [1, 1, 1, 1]
 Generation 3500: Best = 1.209092e+04 FES = 350100 μ_F = 0.768 μ_{CR} = 0.061
 Diversity = 1.7073e-01 k_t = [2, 2, 4, 2]
 Generation 3510: Best = 1.208665e+04 FES = 351100 μ_F = 0.795 μ_{CR} = 0.054
 Diversity = 1.7468e-01 k_t = [1, 1, 1, 1]
 Generation 3520: Best = 1.208252e+04 FES = 352100 μ_F = 0.767 μ_{CR} = 0.066
 Diversity = 1.7666e-01 k_t = [4, 3, 4, 2]
 Generation 3530: Best = 1.207835e+04 FES = 353100 μ_F = 0.773 μ_{CR} = 0.071
 Diversity = 1.8868e-01 k_t = [1, 1, 1, 1]
 Generation 3540: Best = 1.207543e+04 FES = 354100 μ_F = 0.772 μ_{CR} = 0.066
 Diversity = 1.8184e-01 k_t = [3, 3, 3, 3]
 Generation 3550: Best = 1.207340e+04 FES = 355100 μ_F = 0.773 μ_{CR} = 0.063
 Diversity = 1.7243e-01 k_t = [1, 1, 1, 1]
 Generation 3560: Best = 1.207217e+04 FES = 356100 μ_F = 0.769 μ_{CR} = 0.062
 Diversity = 1.7474e-01 k_t = [4, 4, 3, 2]
 Generation 3570: Best = 1.206815e+04 FES = 357100 μ_F = 0.768 μ_{CR} = 0.060
 Diversity = 1.7726e-01 k_t = [1, 1, 1, 1]
 Generation 3580: Best = 1.206768e+04 FES = 358100 μ_F = 0.763 μ_{CR} = 0.061
 Diversity = 1.7721e-01 k_t = [4, 2, 3, 3]
 Generation 3590: Best = 1.206627e+04 FES = 359100 μ_F = 0.767 μ_{CR} = 0.065
 Diversity = 1.8193e-01 k_t = [1, 1, 1, 1]
 Generation 3600: Best = 1.206111e+04 FES = 360100 μ_F = 0.790 μ_{CR} = 0.055
 Diversity = 1.8138e-01 k_t = [2, 3, 2, 3]
 Generation 3610: Best = 1.205885e+04 FES = 361100 μ_F = 0.779 μ_{CR} = 0.070
 Diversity = 1.8702e-01 k_t = [1, 1, 1, 1]
 Generation 3620: Best = 1.205819e+04 FES = 362100 μ_F = 0.763 μ_{CR} = 0.065
 Diversity = 1.7972e-01 k_t = [2, 2, 4, 3]
 Generation 3630: Best = 1.205818e+04 FES = 363100 μ_F = 0.785 μ_{CR} = 0.061
 Diversity = 1.8463e-01 k_t = [1, 1, 1, 1]
 Generation 3640: Best = 1.205554e+04 FES = 364100 μ_F = 0.797 μ_{CR} = 0.067
 Diversity = 1.6779e-01 k_t = [2, 3, 3, 4]
 Generation 3650: Best = 1.205281e+04 FES = 365100 μ_F = 0.804 μ_{CR} = 0.056
 Diversity = 1.6587e-01 k_t = [1, 1, 1, 1]
 Generation 3660: Best = 1.205009e+04 FES = 366100 μ_F = 0.781 μ_{CR} = 0.059
 Diversity = 1.6193e-01 k_t = [4, 2, 3, 3]
 Generation 3670: Best = 1.204972e+04 FES = 367100 μ_F = 0.776 μ_{CR} = 0.058
 Diversity = 1.5590e-01 k_t = [1, 1, 1, 1]
 Generation 3680: Best = 1.204682e+04 FES = 368100 μ_F = 0.784 μ_{CR} = 0.061
 Diversity = 1.5698e-01 k_t = [2, 3, 4, 4]
 Generation 3690: Best = 1.204653e+04 FES = 369100 μ_F = 0.776 μ_{CR} = 0.064
 Diversity = 1.5638e-01 k_t = [1, 1, 1, 1]
 Generation 3700: Best = 1.204634e+04 FES = 370100 μ_F = 0.788 μ_{CR} = 0.060
 Diversity = 1.5602e-01 k_t = [4, 3, 3, 3]
 Generation 3710: Best = 1.204505e+04 FES = 371100 μ_F = 0.788 μ_{CR} = 0.059
 Diversity = 1.5268e-01 k_t = [1, 1, 1, 1]
 Generation 3720: Best = 1.204355e+04 FES = 372100 μ_F = 0.777 μ_{CR} = 0.061
 Diversity = 1.4998e-01 k_t = [4, 4, 3, 4]
 Generation 3730: Best = 1.204224e+04 FES = 373100 μ_F = 0.759 μ_{CR} = 0.072
 Diversity = 1.4985e-01 k_t = [1, 1, 1, 1]

Generation 3740: Best = 1.203898e+04 FES = 374100 μ_F = 0.764 μ_{CR} = 0.070
 Diversity = 1.4290e-01 k_t = [4, 2, 3, 3]
 Generation 3750: Best = 1.203825e+04 FES = 375100 μ_F = 0.753 μ_{CR} = 0.069
 Diversity = 1.4268e-01 k_t = [1, 1, 1, 1]
 Generation 3760: Best = 1.203757e+04 FES = 376100 μ_F = 0.746 μ_{CR} = 0.067
 Diversity = 1.4248e-01 k_t = [4, 4, 4, 3]
 Generation 3770: Best = 1.203428e+04 FES = 377100 μ_F = 0.757 μ_{CR} = 0.066
 Diversity = 1.4165e-01 k_t = [1, 1, 1, 1]
 Generation 3780: Best = 1.203403e+04 FES = 378100 μ_F = 0.756 μ_{CR} = 0.062
 Diversity = 1.3468e-01 k_t = [3, 3, 3, 4]
 Generation 3790: Best = 1.203370e+04 FES = 379100 μ_F = 0.738 μ_{CR} = 0.070
 Diversity = 1.3126e-01 k_t = [1, 1, 1, 1]
 Generation 3800: Best = 1.203275e+04 FES = 380100 μ_F = 0.770 μ_{CR} = 0.072
 Diversity = 1.2798e-01 k_t = [3, 3, 3, 3]
 Generation 3810: Best = 1.203157e+04 FES = 381100 μ_F = 0.765 μ_{CR} = 0.072
 Diversity = 1.2065e-01 k_t = [1, 1, 1, 1]
 Generation 3820: Best = 1.203118e+04 FES = 382100 μ_F = 0.769 μ_{CR} = 0.072
 Diversity = 1.2062e-01 k_t = [3, 4, 2, 4]
 Generation 3830: Best = 1.202993e+04 FES = 383100 μ_F = 0.756 μ_{CR} = 0.074
 Diversity = 1.1273e-01 k_t = [1, 1, 1, 1]
 Generation 3840: Best = 1.202860e+04 FES = 384100 μ_F = 0.754 μ_{CR} = 0.086
 Diversity = 1.0996e-01 k_t = [3, 4, 4, 4]
 Generation 3850: Best = 1.202829e+04 FES = 385100 μ_F = 0.746 μ_{CR} = 0.078
 Diversity = 1.0710e-01 k_t = [1, 1, 1, 1]
 Generation 3860: Best = 1.202695e+04 FES = 386100 μ_F = 0.742 μ_{CR} = 0.070
 Diversity = 1.0498e-01 k_t = [4, 3, 2, 3]
 Generation 3870: Best = 1.202633e+04 FES = 387100 μ_F = 0.703 μ_{CR} = 0.071
 Diversity = 9.4209e-02 k_t = [1, 1, 1, 1]
 Generation 3880: Best = 1.202554e+04 FES = 388100 μ_F = 0.700 μ_{CR} = 0.079
 Diversity = 9.0065e-02 k_t = [3, 4, 2, 4]
 Generation 3890: Best = 1.202458e+04 FES = 389100 μ_F = 0.689 μ_{CR} = 0.084
 Diversity = 8.9332e-02 k_t = [1, 1, 1, 1]
 Generation 3900: Best = 1.202384e+04 FES = 390100 μ_F = 0.702 μ_{CR} = 0.086
 Diversity = 8.4523e-02 k_t = [4, 2, 3, 4]
 Generation 3910: Best = 1.202375e+04 FES = 391100 μ_F = 0.716 μ_{CR} = 0.085
 Diversity = 8.0129e-02 k_t = [1, 1, 1, 1]
 Generation 3920: Best = 1.202248e+04 FES = 392100 μ_F = 0.750 μ_{CR} = 0.081
 Diversity = 8.3054e-02 k_t = [3, 3, 4, 3]
 Generation 3930: Best = 1.202238e+04 FES = 393100 μ_F = 0.759 μ_{CR} = 0.069
 Diversity = 7.8548e-02 k_t = [1, 1, 1, 1]
 Generation 3940: Best = 1.202193e+04 FES = 394100 μ_F = 0.774 μ_{CR} = 0.073
 Diversity = 7.6054e-02 k_t = [4, 2, 3, 4]
 Generation 3950: Best = 1.202144e+04 FES = 395100 μ_F = 0.780 μ_{CR} = 0.071
 Diversity = 7.5777e-02 k_t = [1, 1, 1, 1]
 Generation 3960: Best = 1.202111e+04 FES = 396100 μ_F = 0.791 μ_{CR} = 0.062
 Diversity = 7.3875e-02 k_t = [3, 4, 2, 2]
 Generation 3970: Best = 1.202054e+04 FES = 397100 μ_F = 0.789 μ_{CR} = 0.067
 Diversity = 7.6192e-02 k_t = [1, 1, 1, 1]
 Generation 3980: Best = 1.202001e+04 FES = 398100 μ_F = 0.782 μ_{CR} = 0.065
 Diversity = 7.0086e-02 k_t = [3, 4, 2, 3]
 Generation 3990: Best = 1.202001e+04 FES = 399100 μ_F = 0.792 μ_{CR} = 0.065
 Diversity = 7.3124e-02 k_t = [1, 1, 1, 1]
 Generation 4000: Best = 1.201969e+04 FES = 400100 μ_F = 0.788 μ_{CR} = 0.068
 Diversity = 6.9513e-02 k_t = [4, 4, 4, 2]

Generation 4010: Best = 1.201901e+04 FES = 401100 μ_F = 0.799 μ_{CR} = 0.073
 Diversity = 7.1851e-02 k_t = [1, 1, 1, 1]
 Generation 4020: Best = 1.201876e+04 FES = 402100 μ_F = 0.796 μ_{CR} = 0.075
 Diversity = 7.1677e-02 k_t = [2, 4, 4, 3]
 Generation 4030: Best = 1.201836e+04 FES = 403100 μ_F = 0.787 μ_{CR} = 0.081
 Diversity = 6.4709e-02 k_t = [1, 1, 1, 1]
 Generation 4040: Best = 1.201809e+04 FES = 404100 μ_F = 0.780 μ_{CR} = 0.080
 Diversity = 6.5249e-02 k_t = [3, 2, 2, 4]
 Generation 4050: Best = 1.201739e+04 FES = 405100 μ_F = 0.803 μ_{CR} = 0.084
 Diversity = 6.2422e-02 k_t = [1, 1, 1, 1]
 Generation 4060: Best = 1.201700e+04 FES = 406100 μ_F = 0.812 μ_{CR} = 0.097
 Diversity = 6.0306e-02 k_t = [4, 3, 4, 2]
 Generation 4070: Best = 1.201630e+04 FES = 407100 μ_F = 0.800 μ_{CR} = 0.098
 Diversity = 5.9263e-02 k_t = [1, 1, 1, 1]
 Generation 4080: Best = 1.201482e+04 FES = 408100 μ_F = 0.798 μ_{CR} = 0.102
 Diversity = 5.9403e-02 k_t = [4, 2, 2, 2]
 Generation 4090: Best = 1.201439e+04 FES = 409100 μ_F = 0.783 μ_{CR} = 0.098
 Diversity = 6.2611e-02 k_t = [1, 1, 1, 1]
 Generation 4100: Best = 1.201339e+04 FES = 410100 μ_F = 0.780 μ_{CR} = 0.103
 Diversity = 6.0433e-02 k_t = [3, 3, 2, 2]
 Generation 4110: Best = 1.201228e+04 FES = 411100 μ_F = 0.773 μ_{CR} = 0.113
 Diversity = 5.9953e-02 k_t = [1, 1, 1, 1]
 Generation 4120: Best = 1.201139e+04 FES = 412100 μ_F = 0.772 μ_{CR} = 0.115
 Diversity = 5.6205e-02 k_t = [2, 3, 2, 3]
 Generation 4130: Best = 1.201109e+04 FES = 413100 μ_F = 0.773 μ_{CR} = 0.127
 Diversity = 5.0918e-02 k_t = [1, 1, 1, 1]
 Generation 4140: Best = 1.201048e+04 FES = 414100 μ_F = 0.762 μ_{CR} = 0.121
 Diversity = 4.4090e-02 k_t = [3, 3, 3, 3]
 Generation 4150: Best = 1.201014e+04 FES = 415100 μ_F = 0.766 μ_{CR} = 0.124
 Diversity = 4.6886e-02 k_t = [1, 1, 1, 1]
 Generation 4160: Best = 1.200974e+04 FES = 416100 μ_F = 0.774 μ_{CR} = 0.115
 Diversity = 4.0887e-02 k_t = [2, 3, 2, 4]
 Generation 4170: Best = 1.200925e+04 FES = 417100 μ_F = 0.753 μ_{CR} = 0.106
 Diversity = 3.6799e-02 k_t = [1, 1, 1, 1]
 Generation 4180: Best = 1.200892e+04 FES = 418100 μ_F = 0.754 μ_{CR} = 0.111
 Diversity = 3.2652e-02 k_t = [2, 4, 3, 4]
 Generation 4190: Best = 1.200841e+04 FES = 419100 μ_F = 0.755 μ_{CR} = 0.130
 Diversity = 3.0467e-02 k_t = [1, 1, 1, 1]
 Generation 4200: Best = 1.200734e+04 FES = 420100 μ_F = 0.756 μ_{CR} = 0.130
 Diversity = 3.3135e-02 k_t = [4, 3, 4, 3]
 Generation 4210: Best = 1.200699e+04 FES = 421100 μ_F = 0.768 μ_{CR} = 0.139
 Diversity = 3.5968e-02 k_t = [1, 1, 1, 1]
 Generation 4220: Best = 1.200624e+04 FES = 422100 μ_F = 0.768 μ_{CR} = 0.136
 Diversity = 3.8358e-02 k_t = [4, 2, 3, 3]
 Generation 4230: Best = 1.200536e+04 FES = 423100 μ_F = 0.777 μ_{CR} = 0.141
 Diversity = 4.4682e-02 k_t = [1, 1, 1, 1]
 Generation 4240: Best = 1.200507e+04 FES = 424100 μ_F = 0.771 μ_{CR} = 0.138
 Diversity = 3.6477e-02 k_t = [2, 2, 4, 3]
 Generation 4250: Best = 1.200442e+04 FES = 425100 μ_F = 0.781 μ_{CR} = 0.137
 Diversity = 4.2800e-02 k_t = [1, 1, 1, 1]
 Generation 4260: Best = 1.200421e+04 FES = 426100 μ_F = 0.769 μ_{CR} = 0.138
 Diversity = 3.4332e-02 k_t = [4, 4, 3, 3]
 Generation 4270: Best = 1.200366e+04 FES = 427100 μ_F = 0.758 μ_{CR} = 0.137
 Diversity = 3.4141e-02 k_t = [1, 1, 1, 1]

Generation 4280: Best = 1.200335e+04 FES = 428100 μ_F = 0.748 μ_{CR} = 0.146
 Diversity = 3.0610e-02 k_t = [3, 3, 3, 3]
 Generation 4290: Best = 1.200283e+04 FES = 429100 μ_F = 0.736 μ_{CR} = 0.137
 Diversity = 2.7666e-02 k_t = [1, 1, 1, 1]
 Generation 4300: Best = 1.200252e+04 FES = 430100 μ_F = 0.747 μ_{CR} = 0.136
 Diversity = 2.3537e-02 k_t = [2, 4, 2, 4]
 Generation 4310: Best = 1.200240e+04 FES = 431100 μ_F = 0.738 μ_{CR} = 0.132
 Diversity = 2.0012e-02 k_t = [1, 1, 1, 1]
 Generation 4320: Best = 1.200210e+04 FES = 432100 μ_F = 0.729 μ_{CR} = 0.128
 Diversity = 2.1064e-02 k_t = [2, 2, 2, 3]
 Generation 4330: Best = 1.200198e+04 FES = 433100 μ_F = 0.748 μ_{CR} = 0.125
 Diversity = 2.2627e-02 k_t = [1, 1, 1, 1]
 Generation 4340: Best = 1.200173e+04 FES = 434100 μ_F = 0.782 μ_{CR} = 0.124
 Diversity = 1.7818e-02 k_t = [3, 2, 4, 4]
 Generation 4350: Best = 1.200153e+04 FES = 435100 μ_F = 0.807 μ_{CR} = 0.121
 Diversity = 2.0993e-02 k_t = [1, 1, 1, 1]
 Generation 4360: Best = 1.200144e+04 FES = 436100 μ_F = 0.794 μ_{CR} = 0.125
 Diversity = 2.1891e-02 k_t = [3, 2, 2, 3]
 Generation 4370: Best = 1.200115e+04 FES = 437100 μ_F = 0.803 μ_{CR} = 0.125
 Diversity = 2.2215e-02 k_t = [1, 1, 1, 1]
 Generation 4380: Best = 1.200101e+04 FES = 438100 μ_F = 0.801 μ_{CR} = 0.130
 Diversity = 1.9993e-02 k_t = [4, 3, 3, 3]
 Generation 4390: Best = 1.200092e+04 FES = 439100 μ_F = 0.779 μ_{CR} = 0.123
 Diversity = 2.0163e-02 k_t = [1, 1, 1, 1]
 Generation 4400: Best = 1.200069e+04 FES = 440100 μ_F = 0.778 μ_{CR} = 0.132
 Diversity = 1.7533e-02 k_t = [3, 4, 4, 2]
 Generation 4410: Best = 1.200056e+04 FES = 441100 μ_F = 0.795 μ_{CR} = 0.140
 Diversity = 1.9680e-02 k_t = [1, 1, 1, 1]
 Generation 4420: Best = 1.200031e+04 FES = 442100 μ_F = 0.789 μ_{CR} = 0.142
 Diversity = 1.8081e-02 k_t = [3, 2, 3, 2]
 Generation 4430: Best = 1.200013e+04 FES = 443100 μ_F = 0.775 μ_{CR} = 0.141
 Diversity = 1.9343e-02 k_t = [1, 1, 1, 1]
 Generation 4440: Best = 1.199994e+04 FES = 444100 μ_F = 0.759 μ_{CR} = 0.133
 Diversity = 1.8682e-02 k_t = [3, 4, 3, 3]
 Generation 4450: Best = 1.199982e+04 FES = 445100 μ_F = 0.756 μ_{CR} = 0.137
 Diversity = 1.4936e-02 k_t = [1, 1, 1, 1]
 Generation 4460: Best = 1.199973e+04 FES = 446100 μ_F = 0.746 μ_{CR} = 0.139
 Diversity = 1.2280e-02 k_t = [4, 4, 2, 3]
 Generation 4470: Best = 1.199957e+04 FES = 447100 μ_F = 0.760 μ_{CR} = 0.148
 Diversity = 1.1927e-02 k_t = [1, 1, 1, 1]
 Generation 4480: Best = 1.199945e+04 FES = 448100 μ_F = 0.757 μ_{CR} = 0.144
 Diversity = 1.0907e-02 k_t = [3, 2, 4, 4]
 Generation 4490: Best = 1.199934e+04 FES = 449100 μ_F = 0.758 μ_{CR} = 0.136
 Diversity = 1.1286e-02 k_t = [1, 1, 1, 1]
 Generation 4500: Best = 1.199899e+04 FES = 450100 μ_F = 0.772 μ_{CR} = 0.147
 Diversity = 1.1689e-02 k_t = [2, 3, 3, 4]
 Generation 4510: Best = 1.199890e+04 FES = 451100 μ_F = 0.773 μ_{CR} = 0.144
 Diversity = 1.1497e-02 k_t = [1, 1, 1, 1]
 Generation 4520: Best = 1.199879e+04 FES = 452100 μ_F = 0.771 μ_{CR} = 0.144
 Diversity = 1.1066e-02 k_t = [4, 4, 4, 4]
 Generation 4530: Best = 1.199866e+04 FES = 453100 μ_F = 0.765 μ_{CR} = 0.150
 Diversity = 1.0371e-02 k_t = [1, 1, 1, 1]
 Generation 4540: Best = 1.199851e+04 FES = 454100 μ_F = 0.759 μ_{CR} = 0.152
 Diversity = 1.0806e-02 k_t = [4, 2, 3, 3]

Generation 4550: Best = 1.199844e+04 FES = 455100 μ_F = 0.768 μ_{CR} = 0.154
 Diversity = 1.0857e-02 k_t = [1, 1, 1, 1]
 Generation 4560: Best = 1.199824e+04 FES = 456100 μ_F = 0.778 μ_{CR} = 0.159
 Diversity = 1.2190e-02 k_t = [3, 2, 4, 3]
 Generation 4570: Best = 1.199807e+04 FES = 457100 μ_F = 0.761 μ_{CR} = 0.156
 Diversity = 1.5376e-02 k_t = [1, 1, 1, 1]
 Generation 4580: Best = 1.199787e+04 FES = 458100 μ_F = 0.768 μ_{CR} = 0.154
 Diversity = 1.4999e-02 k_t = [4, 4, 3, 2]
 Generation 4590: Best = 1.199779e+04 FES = 459100 μ_F = 0.756 μ_{CR} = 0.145
 Diversity = 1.4894e-02 k_t = [1, 1, 1, 1]
 Generation 4600: Best = 1.199766e+04 FES = 460100 μ_F = 0.753 μ_{CR} = 0.155
 Diversity = 1.4425e-02 k_t = [4, 2, 3, 3]
 Generation 4610: Best = 1.199751e+04 FES = 461100 μ_F = 0.757 μ_{CR} = 0.160
 Diversity = 1.3262e-02 k_t = [1, 1, 1, 1]
 Generation 4620: Best = 1.199749e+04 FES = 462100 μ_F = 0.760 μ_{CR} = 0.152
 Diversity = 1.2297e-02 k_t = [4, 3, 3, 4]
 Generation 4630: Best = 1.199741e+04 FES = 463100 μ_F = 0.759 μ_{CR} = 0.158
 Diversity = 1.2188e-02 k_t = [1, 1, 1, 1]
 Generation 4640: Best = 1.199733e+04 FES = 464100 μ_F = 0.759 μ_{CR} = 0.168
 Diversity = 1.0078e-02 k_t = [3, 4, 4, 3]
 Generation 4650: Best = 1.199724e+04 FES = 465100 μ_F = 0.759 μ_{CR} = 0.167
 Diversity = 1.0359e-02 k_t = [1, 1, 1, 1]
 Generation 4660: Best = 1.199711e+04 FES = 466100 μ_F = 0.757 μ_{CR} = 0.167
 Diversity = 1.0136e-02 k_t = [4, 2, 3, 3]
 Generation 4670: Best = 1.199700e+04 FES = 467100 μ_F = 0.750 μ_{CR} = 0.168
 Diversity = 1.1972e-02 k_t = [1, 1, 1, 1]
 Generation 4680: Best = 1.199690e+04 FES = 468100 μ_F = 0.755 μ_{CR} = 0.178
 Diversity = 1.2338e-02 k_t = [4, 4, 3, 2]
 Generation 4690: Best = 1.199684e+04 FES = 469100 μ_F = 0.770 μ_{CR} = 0.181
 Diversity = 1.1620e-02 k_t = [1, 1, 1, 1]
 Generation 4700: Best = 1.199676e+04 FES = 470100 μ_F = 0.767 μ_{CR} = 0.173
 Diversity = 1.0660e-02 k_t = [3, 3, 2, 2]
 Generation 4710: Best = 1.199670e+04 FES = 471100 μ_F = 0.753 μ_{CR} = 0.183
 Diversity = 1.1102e-02 k_t = [1, 1, 1, 1]
 Generation 4720: Best = 1.199667e+04 FES = 472100 μ_F = 0.752 μ_{CR} = 0.174
 Diversity = 8.6335e-03 k_t = [4, 3, 3, 4]
 Generation 4730: Best = 1.199661e+04 FES = 473100 μ_F = 0.753 μ_{CR} = 0.167
 Diversity = 8.9893e-03 k_t = [1, 1, 1, 1]
 Generation 4740: Best = 1.199654e+04 FES = 474100 μ_F = 0.767 μ_{CR} = 0.160
 Diversity = 7.9370e-03 k_t = [3, 2, 3, 4]
 Generation 4750: Best = 1.199649e+04 FES = 475100 μ_F = 0.755 μ_{CR} = 0.163
 Diversity = 8.3176e-03 k_t = [1, 1, 1, 1]
 Generation 4760: Best = 1.199637e+04 FES = 476100 μ_F = 0.758 μ_{CR} = 0.157
 Diversity = 9.9927e-03 k_t = [4, 3, 3, 2]
 Generation 4770: Best = 1.199628e+04 FES = 477100 μ_F = 0.765 μ_{CR} = 0.163
 Diversity = 1.0969e-02 k_t = [1, 1, 1, 1]
 Generation 4780: Best = 1.199624e+04 FES = 478100 μ_F = 0.770 μ_{CR} = 0.159
 Diversity = 1.0332e-02 k_t = [2, 2, 4, 2]
 Generation 4790: Best = 1.199617e+04 FES = 479100 μ_F = 0.760 μ_{CR} = 0.154
 Diversity = 9.9516e-03 k_t = [1, 1, 1, 1]
 Generation 4800: Best = 1.199612e+04 FES = 480100 μ_F = 0.762 μ_{CR} = 0.161
 Diversity = 8.8179e-03 k_t = [2, 3, 2, 4]
 Generation 4810: Best = 1.199605e+04 FES = 481100 μ_F = 0.766 μ_{CR} = 0.149
 Diversity = 8.5221e-03 k_t = [1, 1, 1, 1]

Generation 4820: Best = 1.199600e+04 FES = 482100 μ_F = 0.771 μ_{CR} = 0.148
 Diversity = 8.3352e-03 k_t = [4, 3, 4, 3]
 Generation 4830: Best = 1.199594e+04 FES = 483100 μ_F = 0.757 μ_{CR} = 0.149
 Diversity = 9.4003e-03 k_t = [1, 1, 1, 1]
 Generation 4840: Best = 1.199590e+04 FES = 484100 μ_F = 0.763 μ_{CR} = 0.140
 Diversity = 8.7976e-03 k_t = [2, 3, 3, 3]
 Generation 4850: Best = 1.199586e+04 FES = 485100 μ_F = 0.767 μ_{CR} = 0.142
 Diversity = 9.8521e-03 k_t = [1, 1, 1, 1]
 Generation 4860: Best = 1.199579e+04 FES = 486100 μ_F = 0.763 μ_{CR} = 0.152
 Diversity = 9.1538e-03 k_t = [2, 3, 4, 2]
 Generation 4870: Best = 1.199570e+04 FES = 487100 μ_F = 0.764 μ_{CR} = 0.165
 Diversity = 9.6441e-03 k_t = [1, 1, 1, 1]
 Generation 4880: Best = 1.199563e+04 FES = 488100 μ_F = 0.769 μ_{CR} = 0.174
 Diversity = 9.9466e-03 k_t = [4, 3, 2, 3]
 Generation 4890: Best = 1.199556e+04 FES = 489100 μ_F = 0.758 μ_{CR} = 0.181
 Diversity = 9.1532e-03 k_t = [1, 1, 1, 1]
 Generation 4900: Best = 1.199546e+04 FES = 490100 μ_F = 0.756 μ_{CR} = 0.181
 Diversity = 9.0421e-03 k_t = [2, 3, 4, 4]
 Generation 4910: Best = 1.199540e+04 FES = 491100 μ_F = 0.749 μ_{CR} = 0.186
 Diversity = 9.0499e-03 k_t = [1, 1, 1, 1]
 Generation 4920: Best = 1.199534e+04 FES = 492100 μ_F = 0.756 μ_{CR} = 0.187
 Diversity = 7.9709e-03 k_t = [2, 2, 2, 4]
 Generation 4930: Best = 1.199526e+04 FES = 493100 μ_F = 0.764 μ_{CR} = 0.177
 Diversity = 7.9762e-03 k_t = [1, 1, 1, 1]
 Generation 4940: Best = 1.199520e+04 FES = 494100 μ_F = 0.760 μ_{CR} = 0.173
 Diversity = 9.4585e-03 k_t = [3, 4, 2, 2]
 Generation 4950: Best = 1.199516e+04 FES = 495100 μ_F = 0.761 μ_{CR} = 0.165
 Diversity = 8.2259e-03 k_t = [1, 1, 1, 1]
 Generation 4960: Best = 1.199511e+04 FES = 496100 μ_F = 0.747 μ_{CR} = 0.155
 Diversity = 8.1016e-03 k_t = [2, 3, 2, 4]
 Generation 4970: Best = 1.199506e+04 FES = 497100 μ_F = 0.764 μ_{CR} = 0.160
 Diversity = 7.5459e-03 k_t = [1, 1, 1, 1]
 Generation 4980: Best = 1.199501e+04 FES = 498100 μ_F = 0.759 μ_{CR} = 0.169
 Diversity = 7.9237e-03 k_t = [2, 2, 3, 2]
 Generation 4990: Best = 1.199498e+04 FES = 499100 μ_F = 0.756 μ_{CR} = 0.166
 Diversity = 8.4572e-03 k_t = [1, 1, 1, 1]
 Generation 5000: Best = 1.199495e+04 FES = 500100 μ_F = 0.756 μ_{CR} = 0.160
 Diversity = 8.0472e-03 k_t = [3, 2, 2, 2]
 Generation 5010: Best = 1.199490e+04 FES = 501100 μ_F = 0.752 μ_{CR} = 0.160
 Diversity = 7.8957e-03 k_t = [1, 1, 1, 1]
 Generation 5020: Best = 1.199485e+04 FES = 502100 μ_F = 0.758 μ_{CR} = 0.161
 Diversity = 6.7842e-03 k_t = [4, 4, 3, 4]
 Generation 5030: Best = 1.199483e+04 FES = 503100 μ_F = 0.767 μ_{CR} = 0.157
 Diversity = 6.6571e-03 k_t = [1, 1, 1, 1]
 Generation 5040: Best = 1.199481e+04 FES = 504100 μ_F = 0.777 μ_{CR} = 0.159
 Diversity = 6.7401e-03 k_t = [4, 3, 3, 2]
 Generation 5050: Best = 1.199477e+04 FES = 505100 μ_F = 0.774 μ_{CR} = 0.162
 Diversity = 7.1684e-03 k_t = [1, 1, 1, 1]
 Generation 5060: Best = 1.199472e+04 FES = 506100 μ_F = 0.773 μ_{CR} = 0.165
 Diversity = 7.2408e-03 k_t = [4, 3, 3, 2]
 Generation 5070: Best = 1.199469e+04 FES = 507100 μ_F = 0.785 μ_{CR} = 0.175
 Diversity = 7.5248e-03 k_t = [1, 1, 1, 1]
 Generation 5080: Best = 1.199460e+04 FES = 508100 μ_F = 0.785 μ_{CR} = 0.189
 Diversity = 7.1332e-03 k_t = [3, 2, 3, 3]

Generation 5090: Best = 1.199459e+04 FES = 509100 μ_F = 0.789 μ_{CR} = 0.196
 Diversity = 7.4341e-03 k_t = [1, 1, 1, 1]
 Generation 5100: Best = 1.199455e+04 FES = 510100 μ_F = 0.787 μ_{CR} = 0.192
 Diversity = 7.1264e-03 k_t = [4, 2, 3, 4]
 Generation 5110: Best = 1.199452e+04 FES = 511100 μ_F = 0.790 μ_{CR} = 0.176
 Diversity = 6.6658e-03 k_t = [1, 1, 1, 1]
 Generation 5120: Best = 1.199449e+04 FES = 512100 μ_F = 0.787 μ_{CR} = 0.180
 Diversity = 5.8116e-03 k_t = [3, 3, 4, 4]
 Generation 5130: Best = 1.199445e+04 FES = 513100 μ_F = 0.785 μ_{CR} = 0.185
 Diversity = 5.8145e-03 k_t = [1, 1, 1, 1]
 Generation 5140: Best = 1.199441e+04 FES = 514100 μ_F = 0.789 μ_{CR} = 0.183
 Diversity = 6.2957e-03 k_t = [3, 4, 2, 2]
 Generation 5150: Best = 1.199438e+04 FES = 515100 μ_F = 0.770 μ_{CR} = 0.185
 Diversity = 6.8627e-03 k_t = [1, 1, 1, 1]
 Generation 5160: Best = 1.199434e+04 FES = 516100 μ_F = 0.778 μ_{CR} = 0.188
 Diversity = 6.6556e-03 k_t = [3, 3, 4, 4]
 Generation 5170: Best = 1.199428e+04 FES = 517100 μ_F = 0.759 μ_{CR} = 0.193
 Diversity = 6.1184e-03 k_t = [1, 1, 1, 1]
 Generation 5180: Best = 1.199424e+04 FES = 518100 μ_F = 0.761 μ_{CR} = 0.197
 Diversity = 6.3757e-03 k_t = [4, 3, 3, 4]
 Generation 5190: Best = 1.199422e+04 FES = 519100 μ_F = 0.754 μ_{CR} = 0.186
 Diversity = 6.2029e-03 k_t = [1, 1, 1, 1]
 Generation 5200: Best = 1.199419e+04 FES = 520100 μ_F = 0.765 μ_{CR} = 0.178
 Diversity = 6.0181e-03 k_t = [2, 3, 4, 2]
 Generation 5210: Best = 1.199417e+04 FES = 521100 μ_F = 0.769 μ_{CR} = 0.178
 Diversity = 6.2258e-03 k_t = [1, 1, 1, 1]
 Generation 5220: Best = 1.199412e+04 FES = 522100 μ_F = 0.763 μ_{CR} = 0.187
 Diversity = 5.2845e-03 k_t = [2, 4, 4, 4]
 Generation 5230: Best = 1.199407e+04 FES = 523100 μ_F = 0.765 μ_{CR} = 0.182
 Diversity = 5.8023e-03 k_t = [1, 1, 1, 1]
 Generation 5240: Best = 1.199405e+04 FES = 524100 μ_F = 0.769 μ_{CR} = 0.187
 Diversity = 6.5177e-03 k_t = [2, 2, 3, 4]
 Generation 5250: Best = 1.199400e+04 FES = 525100 μ_F = 0.754 μ_{CR} = 0.187
 Diversity = 6.5473e-03 k_t = [1, 1, 1, 1]
 Generation 5260: Best = 1.199397e+04 FES = 526100 μ_F = 0.744 μ_{CR} = 0.189
 Diversity = 6.3401e-03 k_t = [4, 2, 3, 4]
 Generation 5270: Best = 1.199394e+04 FES = 527100 μ_F = 0.737 μ_{CR} = 0.187
 Diversity = 6.0498e-03 k_t = [1, 1, 1, 1]
 Generation 5280: Best = 1.199389e+04 FES = 528100 μ_F = 0.757 μ_{CR} = 0.194
 Diversity = 5.5773e-03 k_t = [2, 3, 2, 3]
 Generation 5290: Best = 1.199385e+04 FES = 529100 μ_F = 0.759 μ_{CR} = 0.206
 Diversity = 6.1135e-03 k_t = [1, 1, 1, 1]
 Generation 5300: Best = 1.199382e+04 FES = 530100 μ_F = 0.754 μ_{CR} = 0.209
 Diversity = 4.7224e-03 k_t = [3, 3, 2, 4]
 Generation 5310: Best = 1.199377e+04 FES = 531100 μ_F = 0.761 μ_{CR} = 0.207
 Diversity = 6.4018e-03 k_t = [1, 1, 1, 1]
 Generation 5320: Best = 1.199366e+04 FES = 532100 μ_F = 0.770 μ_{CR} = 0.213
 Diversity = 6.8844e-03 k_t = [2, 3, 2, 3]
 Generation 5330: Best = 1.199352e+04 FES = 533100 μ_F = 0.793 μ_{CR} = 0.217
 Diversity = 9.9253e-03 k_t = [1, 1, 1, 1]
 Generation 5340: Best = 1.199346e+04 FES = 534100 μ_F = 0.782 μ_{CR} = 0.229
 Diversity = 8.8984e-03 k_t = [4, 4, 3, 4]
 Generation 5350: Best = 1.199344e+04 FES = 535100 μ_F = 0.771 μ_{CR} = 0.229
 Diversity = 9.4140e-03 k_t = [1, 1, 1, 1]

Generation 5360: Best = 1.199340e+04 FES = 536100 μ_F = 0.776 μ_{CR} = 0.228
 Diversity = 7.1842e-03 k_t = [2, 2, 2, 4]
 Generation 5370: Best = 1.199337e+04 FES = 537100 μ_F = 0.761 μ_{CR} = 0.207
 Diversity = 6.3500e-03 k_t = [1, 1, 1, 1]
 Generation 5380: Best = 1.199334e+04 FES = 538100 μ_F = 0.769 μ_{CR} = 0.201
 Diversity = 6.4634e-03 k_t = [3, 2, 4, 3]
 Generation 5390: Best = 1.199333e+04 FES = 539100 μ_F = 0.767 μ_{CR} = 0.196
 Diversity = 6.3287e-03 k_t = [1, 1, 1, 1]
 Generation 5400: Best = 1.199330e+04 FES = 540100 μ_F = 0.776 μ_{CR} = 0.209
 Diversity = 6.1192e-03 k_t = [3, 4, 4, 4]
 Generation 5410: Best = 1.199328e+04 FES = 541100 μ_F = 0.768 μ_{CR} = 0.200
 Diversity = 5.7175e-03 k_t = [1, 1, 1, 1]
 Generation 5420: Best = 1.199326e+04 FES = 542100 μ_F = 0.756 μ_{CR} = 0.205
 Diversity = 5.2903e-03 k_t = [4, 3, 4, 2]
 Generation 5430: Best = 1.199323e+04 FES = 543100 μ_F = 0.756 μ_{CR} = 0.210
 Diversity = 5.5536e-03 k_t = [1, 1, 1, 1]
 Generation 5440: Best = 1.199320e+04 FES = 544100 μ_F = 0.764 μ_{CR} = 0.201
 Diversity = 5.0763e-03 k_t = [3, 2, 2, 3]
 Generation 5450: Best = 1.199318e+04 FES = 545100 μ_F = 0.771 μ_{CR} = 0.206
 Diversity = 5.4151e-03 k_t = [1, 1, 1, 1]
 Generation 5460: Best = 1.199314e+04 FES = 546100 μ_F = 0.761 μ_{CR} = 0.211
 Diversity = 4.8987e-03 k_t = [4, 3, 4, 4]
 Generation 5470: Best = 1.199311e+04 FES = 547100 μ_F = 0.767 μ_{CR} = 0.209
 Diversity = 5.1786e-03 k_t = [1, 1, 1, 1]
 Generation 5480: Best = 1.199309e+04 FES = 548100 μ_F = 0.768 μ_{CR} = 0.214
 Diversity = 5.0148e-03 k_t = [3, 2, 3, 3]
 Generation 5490: Best = 1.199307e+04 FES = 549100 μ_F = 0.768 μ_{CR} = 0.220
 Diversity = 5.3279e-03 k_t = [1, 1, 1, 1]
 Generation 5500: Best = 1.199306e+04 FES = 550100 μ_F = 0.763 μ_{CR} = 0.212
 Diversity = 4.8824e-03 k_t = [3, 3, 4, 2]
 Generation 5510: Best = 1.199304e+04 FES = 551100 μ_F = 0.739 μ_{CR} = 0.214
 Diversity = 4.8525e-03 k_t = [1, 1, 1, 1]
 Generation 5520: Best = 1.199303e+04 FES = 552100 μ_F = 0.738 μ_{CR} = 0.204
 Diversity = 4.2703e-03 k_t = [4, 4, 2, 2]
 Generation 5530: Best = 1.199302e+04 FES = 553100 μ_F = 0.765 μ_{CR} = 0.199
 Diversity = 4.1674e-03 k_t = [1, 1, 1, 1]
 Generation 5540: Best = 1.199301e+04 FES = 554100 μ_F = 0.775 μ_{CR} = 0.189
 Diversity = 3.8000e-03 k_t = [2, 4, 2, 4]
 Generation 5550: Best = 1.199299e+04 FES = 555100 μ_F = 0.768 μ_{CR} = 0.192
 Diversity = 3.7293e-03 k_t = [1, 1, 1, 1]
 Generation 5560: Best = 1.199298e+04 FES = 556100 μ_F = 0.764 μ_{CR} = 0.194
 Diversity = 3.8432e-03 k_t = [4, 4, 2, 2]
 Generation 5570: Best = 1.199296e+04 FES = 557100 μ_F = 0.770 μ_{CR} = 0.195
 Diversity = 4.4133e-03 k_t = [1, 1, 1, 1]
 Generation 5580: Best = 1.199295e+04 FES = 558100 μ_F = 0.773 μ_{CR} = 0.199
 Diversity = 3.8055e-03 k_t = [3, 2, 2, 3]
 Generation 5590: Best = 1.199294e+04 FES = 559100 μ_F = 0.773 μ_{CR} = 0.198
 Diversity = 4.2569e-03 k_t = [1, 1, 1, 1]
 Generation 5600: Best = 1.199293e+04 FES = 560100 μ_F = 0.764 μ_{CR} = 0.201
 Diversity = 3.7374e-03 k_t = [3, 2, 2, 2]
 Generation 5610: Best = 1.199292e+04 FES = 561100 μ_F = 0.754 μ_{CR} = 0.194
 Diversity = 3.6349e-03 k_t = [1, 1, 1, 1]
 Generation 5620: Best = 1.199290e+04 FES = 562100 μ_F = 0.753 μ_{CR} = 0.191
 Diversity = 3.7026e-03 k_t = [2, 3, 3, 2]

Generation 5630: Best = 1.199290e+04 FES = 563100 μ_F = 0.758 μ_{CR} = 0.192
 Diversity = 3.5752e-03 k_t = [1, 1, 1, 1]
 Generation 5640: Best = 1.199289e+04 FES = 564100 μ_F = 0.763 μ_{CR} = 0.191
 Diversity = 2.9285e-03 k_t = [2, 3, 4, 4]
 Generation 5650: Best = 1.199288e+04 FES = 565100 μ_F = 0.755 μ_{CR} = 0.188
 Diversity = 2.8645e-03 k_t = [1, 1, 1, 1]
 Generation 5660: Best = 1.199287e+04 FES = 566100 μ_F = 0.757 μ_{CR} = 0.193
 Diversity = 2.8984e-03 k_t = [3, 3, 4, 2]
 Generation 5670: Best = 1.199286e+04 FES = 567100 μ_F = 0.757 μ_{CR} = 0.196
 Diversity = 3.1791e-03 k_t = [1, 1, 1, 1]
 Generation 5680: Best = 1.199284e+04 FES = 568100 μ_F = 0.767 μ_{CR} = 0.199
 Diversity = 2.8773e-03 k_t = [4, 2, 3, 4]
 Generation 5690: Best = 1.199283e+04 FES = 569100 μ_F = 0.765 μ_{CR} = 0.207
 Diversity = 2.7477e-03 k_t = [1, 1, 1, 1]
 Generation 5700: Best = 1.199281e+04 FES = 570100 μ_F = 0.779 μ_{CR} = 0.223
 Diversity = 3.1359e-03 k_t = [3, 2, 2, 3]
 Generation 5710: Best = 1.199280e+04 FES = 571100 μ_F = 0.784 μ_{CR} = 0.226
 Diversity = 4.0542e-03 k_t = [1, 1, 1, 1]
 Generation 5720: Best = 1.199279e+04 FES = 572100 μ_F = 0.781 μ_{CR} = 0.224
 Diversity = 3.4658e-03 k_t = [3, 4, 3, 4]
 Generation 5730: Best = 1.199278e+04 FES = 573100 μ_F = 0.778 μ_{CR} = 0.228
 Diversity = 3.3051e-03 k_t = [1, 1, 1, 1]
 Generation 5740: Best = 1.199276e+04 FES = 574100 μ_F = 0.767 μ_{CR} = 0.227
 Diversity = 3.5573e-03 k_t = [2, 3, 2, 2]
 Generation 5750: Best = 1.199275e+04 FES = 575100 μ_F = 0.772 μ_{CR} = 0.230
 Diversity = 3.2597e-03 k_t = [1, 1, 1, 1]
 Generation 5760: Best = 1.199274e+04 FES = 576100 μ_F = 0.768 μ_{CR} = 0.234
 Diversity = 2.8732e-03 k_t = [4, 4, 4, 3]
 Generation 5770: Best = 1.199273e+04 FES = 577100 μ_F = 0.773 μ_{CR} = 0.236
 Diversity = 3.1054e-03 k_t = [1, 1, 1, 1]
 Generation 5780: Best = 1.199272e+04 FES = 578100 μ_F = 0.782 μ_{CR} = 0.235
 Diversity = 3.0949e-03 k_t = [4, 2, 4, 4]
 Generation 5790: Best = 1.199271e+04 FES = 579100 μ_F = 0.780 μ_{CR} = 0.248
 Diversity = 3.1045e-03 k_t = [1, 1, 1, 1]
 Generation 5800: Best = 1.199270e+04 FES = 580100 μ_F = 0.776 μ_{CR} = 0.252
 Diversity = 2.8954e-03 k_t = [4, 4, 4, 2]
 Generation 5810: Best = 1.199269e+04 FES = 581100 μ_F = 0.779 μ_{CR} = 0.250
 Diversity = 3.4558e-03 k_t = [1, 1, 1, 1]
 Generation 5820: Best = 1.199267e+04 FES = 582100 μ_F = 0.788 μ_{CR} = 0.260
 Diversity = 3.6757e-03 k_t = [3, 3, 4, 3]
 Generation 5830: Best = 1.199265e+04 FES = 583100 μ_F = 0.800 μ_{CR} = 0.260
 Diversity = 4.1798e-03 k_t = [1, 1, 1, 1]
 Generation 5840: Best = 1.199265e+04 FES = 584100 μ_F = 0.778 μ_{CR} = 0.246
 Diversity = 3.1828e-03 k_t = [4, 4, 3, 3]
 Generation 5850: Best = 1.199264e+04 FES = 585100 μ_F = 0.782 μ_{CR} = 0.242
 Diversity = 3.5719e-03 k_t = [1, 1, 1, 1]
 Generation 5860: Best = 1.199263e+04 FES = 586100 μ_F = 0.765 μ_{CR} = 0.231
 Diversity = 2.9988e-03 k_t = [2, 3, 3, 3]
 Generation 5870: Best = 1.199262e+04 FES = 587100 μ_F = 0.764 μ_{CR} = 0.237
 Diversity = 3.3279e-03 k_t = [1, 1, 1, 1]
 Generation 5880: Best = 1.199262e+04 FES = 588100 μ_F = 0.754 μ_{CR} = 0.226
 Diversity = 2.6719e-03 k_t = [2, 4, 2, 4]
 Generation 5890: Best = 1.199261e+04 FES = 589100 μ_F = 0.743 μ_{CR} = 0.223
 Diversity = 2.5632e-03 k_t = [1, 1, 1, 1]

Generation 5900: Best = 1.199260e+04 FES = 590100 μ_F = 0.748 μ_{CR} = 0.217
 Diversity = 2.5593e-03 k_t = [2, 2, 3, 3]
 Generation 5910: Best = 1.199260e+04 FES = 591100 μ_F = 0.741 μ_{CR} = 0.219
 Diversity = 2.5382e-03 k_t = [1, 1, 1, 1]
 Generation 5920: Best = 1.199259e+04 FES = 592100 μ_F = 0.743 μ_{CR} = 0.214
 Diversity = 2.3030e-03 k_t = [4, 2, 2, 4]
 Generation 5930: Best = 1.199259e+04 FES = 593100 μ_F = 0.747 μ_{CR} = 0.208
 Diversity = 2.4734e-03 k_t = [1, 1, 1, 1]
 Generation 5940: Best = 1.199259e+04 FES = 594100 μ_F = 0.759 μ_{CR} = 0.200
 Diversity = 2.2218e-03 k_t = [3, 2, 4, 3]
 Generation 5950: Best = 1.199258e+04 FES = 595100 μ_F = 0.764 μ_{CR} = 0.210
 Diversity = 2.4133e-03 k_t = [1, 1, 1, 1]
 Generation 5960: Best = 1.199258e+04 FES = 596100 μ_F = 0.757 μ_{CR} = 0.209
 Diversity = 2.2931e-03 k_t = [4, 3, 4, 2]
 Generation 5970: Best = 1.199257e+04 FES = 597100 μ_F = 0.763 μ_{CR} = 0.215
 Diversity = 2.1483e-03 k_t = [1, 1, 1, 1]
 Generation 5980: Best = 1.199257e+04 FES = 598100 μ_F = 0.764 μ_{CR} = 0.210
 Diversity = 1.7301e-03 k_t = [4, 2, 3, 3]
 Generation 5990: Best = 1.199256e+04 FES = 599100 μ_F = 0.772 μ_{CR} = 0.210
 Diversity = 1.8943e-03 k_t = [1, 1, 1, 1]
 Generation 6000: Best = 1.199255e+04 FES = 600100 μ_F = 0.768 μ_{CR} = 0.209
 Diversity = 2.2578e-03 k_t = [3, 4, 2, 2]
 Generation 6010: Best = 1.199255e+04 FES = 601100 μ_F = 0.777 μ_{CR} = 0.207
 Diversity = 2.3634e-03 k_t = [1, 1, 1, 1]
 Generation 6020: Best = 1.199254e+04 FES = 602100 μ_F = 0.774 μ_{CR} = 0.215
 Diversity = 2.5522e-03 k_t = [4, 4, 3, 2]
 Generation 6030: Best = 1.199254e+04 FES = 603100 μ_F = 0.779 μ_{CR} = 0.208
 Diversity = 2.5559e-03 k_t = [1, 1, 1, 1]
 Generation 6040: Best = 1.199253e+04 FES = 604100 μ_F = 0.768 μ_{CR} = 0.220
 Diversity = 2.0671e-03 k_t = [4, 3, 4, 4]
 Generation 6050: Best = 1.199253e+04 FES = 605100 μ_F = 0.783 μ_{CR} = 0.215
 Diversity = 1.8639e-03 k_t = [1, 1, 1, 1]
 Generation 6060: Best = 1.199252e+04 FES = 606100 μ_F = 0.789 μ_{CR} = 0.219
 Diversity = 2.0578e-03 k_t = [3, 2, 2, 3]
 Generation 6070: Best = 1.199252e+04 FES = 607100 μ_F = 0.794 μ_{CR} = 0.227
 Diversity = 2.2248e-03 k_t = [1, 1, 1, 1]
 Generation 6080: Best = 1.199251e+04 FES = 608100 μ_F = 0.792 μ_{CR} = 0.237
 Diversity = 2.0252e-03 k_t = [2, 2, 2, 3]
 Generation 6090: Best = 1.199251e+04 FES = 609100 μ_F = 0.799 μ_{CR} = 0.216
 Diversity = 2.1930e-03 k_t = [1, 1, 1, 1]
 Generation 6100: Best = 1.199250e+04 FES = 610100 μ_F = 0.798 μ_{CR} = 0.212
 Diversity = 2.0457e-03 k_t = [3, 3, 2, 4]
 Generation 6110: Best = 1.199250e+04 FES = 611100 μ_F = 0.792 μ_{CR} = 0.205
 Diversity = 2.0628e-03 k_t = [1, 1, 1, 1]
 Generation 6120: Best = 1.199249e+04 FES = 612100 μ_F = 0.786 μ_{CR} = 0.213
 Diversity = 2.0174e-03 k_t = [3, 3, 2, 4]
 Generation 6130: Best = 1.199249e+04 FES = 613100 μ_F = 0.797 μ_{CR} = 0.217
 Diversity = 2.3856e-03 k_t = [1, 1, 1, 1]
 Generation 6140: Best = 1.199248e+04 FES = 614100 μ_F = 0.785 μ_{CR} = 0.211
 Diversity = 2.0378e-03 k_t = [4, 3, 2, 4]
 Generation 6150: Best = 1.199248e+04 FES = 615100 μ_F = 0.788 μ_{CR} = 0.214
 Diversity = 2.0867e-03 k_t = [1, 1, 1, 1]
 Generation 6160: Best = 1.199247e+04 FES = 616100 μ_F = 0.788 μ_{CR} = 0.210
 Diversity = 2.1414e-03 k_t = [2, 2, 4, 3]

Generation 6170: Best = 1.199247e+04 FES = 617100 μ_F = 0.780 μ_{CR} = 0.207
 Diversity = 2.2997e-03 k_t = [1, 1, 1, 1]
 Generation 6180: Best = 1.199245e+04 FES = 618100 μ_F = 0.786 μ_{CR} = 0.212
 Diversity = 2.3860e-03 k_t = [4, 4, 4, 3]
 Generation 6190: Best = 1.199244e+04 FES = 619100 μ_F = 0.787 μ_{CR} = 0.216
 Diversity = 2.8417e-03 k_t = [1, 1, 1, 1]
 Generation 6200: Best = 1.199243e+04 FES = 620100 μ_F = 0.782 μ_{CR} = 0.222
 Diversity = 2.4084e-03 k_t = [2, 4, 4, 3]
 Generation 6210: Best = 1.199242e+04 FES = 621100 μ_F = 0.781 μ_{CR} = 0.220
 Diversity = 2.5572e-03 k_t = [1, 1, 1, 1]
 Generation 6220: Best = 1.199241e+04 FES = 622100 μ_F = 0.790 μ_{CR} = 0.214
 Diversity = 2.7662e-03 k_t = [4, 2, 4, 4]
 Generation 6230: Best = 1.199240e+04 FES = 623100 μ_F = 0.782 μ_{CR} = 0.211
 Diversity = 2.4795e-03 k_t = [1, 1, 1, 1]
 Generation 6240: Best = 1.199239e+04 FES = 624100 μ_F = 0.783 μ_{CR} = 0.210
 Diversity = 2.2572e-03 k_t = [3, 2, 4, 3]
 Generation 6250: Best = 1.199239e+04 FES = 625100 μ_F = 0.785 μ_{CR} = 0.210
 Diversity = 2.4645e-03 k_t = [1, 1, 1, 1]
 Generation 6260: Best = 1.199238e+04 FES = 626100 μ_F = 0.774 μ_{CR} = 0.202
 Diversity = 2.4100e-03 k_t = [3, 4, 4, 3]
 Generation 6270: Best = 1.199238e+04 FES = 627100 μ_F = 0.785 μ_{CR} = 0.204
 Diversity = 2.3488e-03 k_t = [1, 1, 1, 1]
 Generation 6280: Best = 1.199237e+04 FES = 628100 μ_F = 0.783 μ_{CR} = 0.209
 Diversity = 2.0157e-03 k_t = [2, 2, 2, 2]
 Generation 6290: Best = 1.199237e+04 FES = 629100 μ_F = 0.775 μ_{CR} = 0.210
 Diversity = 2.3361e-03 k_t = [1, 1, 1, 1]
 Generation 6300: Best = 1.199236e+04 FES = 630100 μ_F = 0.775 μ_{CR} = 0.208
 Diversity = 2.4641e-03 k_t = [3, 4, 2, 3]
 Generation 6310: Best = 1.199235e+04 FES = 631100 μ_F = 0.767 μ_{CR} = 0.207
 Diversity = 2.6380e-03 k_t = [1, 1, 1, 1]
 Generation 6320: Best = 1.199234e+04 FES = 632100 μ_F = 0.770 μ_{CR} = 0.213
 Diversity = 2.7100e-03 k_t = [3, 3, 4, 3]
 Generation 6330: Best = 1.199234e+04 FES = 633100 μ_F = 0.769 μ_{CR} = 0.214
 Diversity = 2.6287e-03 k_t = [1, 1, 1, 1]
 Generation 6340: Best = 1.199233e+04 FES = 634100 μ_F = 0.760 μ_{CR} = 0.204
 Diversity = 2.3692e-03 k_t = [3, 3, 3, 4]
 Generation 6350: Best = 1.199233e+04 FES = 635100 μ_F = 0.763 μ_{CR} = 0.203
 Diversity = 2.2081e-03 k_t = [1, 1, 1, 1]
 Generation 6360: Best = 1.199232e+04 FES = 636100 μ_F = 0.773 μ_{CR} = 0.206
 Diversity = 2.4122e-03 k_t = [3, 3, 4, 2]
 Generation 6370: Best = 1.199232e+04 FES = 637100 μ_F = 0.769 μ_{CR} = 0.212
 Diversity = 2.3696e-03 k_t = [1, 1, 1, 1]
 Generation 6380: Best = 1.199231e+04 FES = 638100 μ_F = 0.771 μ_{CR} = 0.210
 Diversity = 2.0495e-03 k_t = [4, 2, 4, 4]
 Generation 6390: Best = 1.199231e+04 FES = 639100 μ_F = 0.778 μ_{CR} = 0.215
 Diversity = 2.3416e-03 k_t = [1, 1, 1, 1]
 Generation 6400: Best = 1.199230e+04 FES = 640100 μ_F = 0.785 μ_{CR} = 0.211
 Diversity = 2.1344e-03 k_t = [4, 4, 4, 4]
 Generation 6410: Best = 1.199229e+04 FES = 641100 μ_F = 0.783 μ_{CR} = 0.212
 Diversity = 2.3547e-03 k_t = [1, 1, 1, 1]
 Generation 6420: Best = 1.199229e+04 FES = 642100 μ_F = 0.790 μ_{CR} = 0.218
 Diversity = 2.8030e-03 k_t = [4, 2, 4, 2]
 Generation 6430: Best = 1.199228e+04 FES = 643100 μ_F = 0.782 μ_{CR} = 0.226
 Diversity = 2.8613e-03 k_t = [1, 1, 1, 1]

Generation 6440: Best = 1.199227e+04 FES = 644100 μ_F = 0.793 μ_{CR} = 0.238
 Diversity = 2.3708e-03 k_t = [2, 4, 4, 4]
 Generation 6450: Best = 1.199226e+04 FES = 645100 μ_F = 0.796 μ_{CR} = 0.242
 Diversity = 2.8312e-03 k_t = [1, 1, 1, 1]
 Generation 6460: Best = 1.199225e+04 FES = 646100 μ_F = 0.793 μ_{CR} = 0.240
 Diversity = 2.3041e-03 k_t = [2, 4, 4, 4]
 Generation 6470: Best = 1.199224e+04 FES = 647100 μ_F = 0.788 μ_{CR} = 0.238
 Diversity = 2.6553e-03 k_t = [1, 1, 1, 1]
 Generation 6480: Best = 1.199223e+04 FES = 648100 μ_F = 0.791 μ_{CR} = 0.234
 Diversity = 2.6101e-03 k_t = [3, 2, 4, 4]
 Generation 6490: Best = 1.199222e+04 FES = 649100 μ_F = 0.788 μ_{CR} = 0.232
 Diversity = 2.8498e-03 k_t = [1, 1, 1, 1]
 Generation 6500: Best = 1.199222e+04 FES = 650100 μ_F = 0.785 μ_{CR} = 0.230
 Diversity = 2.3415e-03 k_t = [3, 4, 3, 4]
 Generation 6510: Best = 1.199221e+04 FES = 651100 μ_F = 0.771 μ_{CR} = 0.241
 Diversity = 2.4706e-03 k_t = [1, 1, 1, 1]
 Generation 6520: Best = 1.199220e+04 FES = 652100 μ_F = 0.772 μ_{CR} = 0.241
 Diversity = 2.3473e-03 k_t = [3, 2, 2, 3]
 Generation 6530: Best = 1.199219e+04 FES = 653100 μ_F = 0.768 μ_{CR} = 0.248
 Diversity = 2.9452e-03 k_t = [1, 1, 1, 1]
 Generation 6540: Best = 1.199219e+04 FES = 654100 μ_F = 0.765 μ_{CR} = 0.258
 Diversity = 2.6345e-03 k_t = [2, 4, 2, 4]
 Generation 6550: Best = 1.199218e+04 FES = 655100 μ_F = 0.752 μ_{CR} = 0.258
 Diversity = 2.4465e-03 k_t = [1, 1, 1, 1]
 Generation 6560: Best = 1.199218e+04 FES = 656100 μ_F = 0.749 μ_{CR} = 0.255
 Diversity = 2.1753e-03 k_t = [2, 2, 3, 3]
 Generation 6570: Best = 1.199217e+04 FES = 657100 μ_F = 0.749 μ_{CR} = 0.254
 Diversity = 2.3451e-03 k_t = [1, 1, 1, 1]
 Generation 6580: Best = 1.199217e+04 FES = 658100 μ_F = 0.751 μ_{CR} = 0.253
 Diversity = 2.2200e-03 k_t = [3, 4, 4, 3]
 Generation 6590: Best = 1.199216e+04 FES = 659100 μ_F = 0.742 μ_{CR} = 0.248
 Diversity = 2.0110e-03 k_t = [1, 1, 1, 1]
 Generation 6600: Best = 1.199216e+04 FES = 660100 μ_F = 0.743 μ_{CR} = 0.246
 Diversity = 2.0156e-03 k_t = [3, 3, 4, 3]
 Generation 6610: Best = 1.199216e+04 FES = 661100 μ_F = 0.739 μ_{CR} = 0.245
 Diversity = 2.1522e-03 k_t = [1, 1, 1, 1]
 Generation 6620: Best = 1.199215e+04 FES = 662100 μ_F = 0.745 μ_{CR} = 0.240
 Diversity = 1.7894e-03 k_t = [2, 4, 3, 4]
 Generation 6630: Best = 1.199215e+04 FES = 663100 μ_F = 0.759 μ_{CR} = 0.229
 Diversity = 1.8869e-03 k_t = [1, 1, 1, 1]
 Generation 6640: Best = 1.199215e+04 FES = 664100 μ_F = 0.769 μ_{CR} = 0.228
 Diversity = 2.2493e-03 k_t = [4, 2, 2, 2]
 Generation 6650: Best = 1.199214e+04 FES = 665100 μ_F = 0.758 μ_{CR} = 0.223
 Diversity = 2.1870e-03 k_t = [1, 1, 1, 1]
 Generation 6660: Best = 1.199214e+04 FES = 666100 μ_F = 0.764 μ_{CR} = 0.214
 Diversity = 1.7745e-03 k_t = [3, 4, 2, 4]
 Generation 6670: Best = 1.199214e+04 FES = 667100 μ_F = 0.771 μ_{CR} = 0.225
 Diversity = 1.9821e-03 k_t = [1, 1, 1, 1]
 Generation 6680: Best = 1.199213e+04 FES = 668100 μ_F = 0.777 μ_{CR} = 0.228
 Diversity = 2.1273e-03 k_t = [2, 2, 2, 3]
 Generation 6690: Best = 1.199213e+04 FES = 669100 μ_F = 0.779 μ_{CR} = 0.219
 Diversity = 2.2419e-03 k_t = [1, 1, 1, 1]
 Generation 6700: Best = 1.199212e+04 FES = 670100 μ_F = 0.781 μ_{CR} = 0.227
 Diversity = 1.9948e-03 k_t = [4, 3, 2, 4]

Generation 6710: Best = 1.199212e+04 FES = 671100 μ_F = 0.786 μ_{CR} = 0.235
 Diversity = 1.9354e-03 k_t = [1, 1, 1, 1]
 Generation 6720: Best = 1.199211e+04 FES = 672100 μ_F = 0.796 μ_{CR} = 0.233
 Diversity = 2.1917e-03 k_t = [4, 4, 4, 2]
 Generation 6730: Best = 1.199211e+04 FES = 673100 μ_F = 0.789 μ_{CR} = 0.230
 Diversity = 2.5856e-03 k_t = [1, 1, 1, 1]
 Generation 6740: Best = 1.199210e+04 FES = 674100 μ_F = 0.780 μ_{CR} = 0.231
 Diversity = 2.2283e-03 k_t = [2, 3, 4, 2]
 Generation 6750: Best = 1.199210e+04 FES = 675100 μ_F = 0.773 μ_{CR} = 0.229
 Diversity = 2.3308e-03 k_t = [1, 1, 1, 1]
 Generation 6760: Best = 1.199209e+04 FES = 676100 μ_F = 0.756 μ_{CR} = 0.225
 Diversity = 2.4864e-03 k_t = [2, 4, 3, 2]
 Generation 6770: Best = 1.199209e+04 FES = 677100 μ_F = 0.761 μ_{CR} = 0.221
 Diversity = 2.3255e-03 k_t = [1, 1, 1, 1]
 Generation 6780: Best = 1.199209e+04 FES = 678100 μ_F = 0.757 μ_{CR} = 0.220
 Diversity = 2.1836e-03 k_t = [4, 2, 4, 2]
 Generation 6790: Best = 1.199208e+04 FES = 679100 μ_F = 0.753 μ_{CR} = 0.215
 Diversity = 2.1364e-03 k_t = [1, 1, 1, 1]
 Generation 6800: Best = 1.199208e+04 FES = 680100 μ_F = 0.756 μ_{CR} = 0.221
 Diversity = 1.9943e-03 k_t = [2, 3, 3, 3]
 Generation 6810: Best = 1.199208e+04 FES = 681100 μ_F = 0.759 μ_{CR} = 0.216
 Diversity = 2.3790e-03 k_t = [1, 1, 1, 1]
 Generation 6820: Best = 1.199207e+04 FES = 682100 μ_F = 0.759 μ_{CR} = 0.203
 Diversity = 1.9459e-03 k_t = [4, 3, 2, 3]
 Generation 6830: Best = 1.199207e+04 FES = 683100 μ_F = 0.742 μ_{CR} = 0.204
 Diversity = 1.7699e-03 k_t = [1, 1, 1, 1]
 Generation 6840: Best = 1.199207e+04 FES = 684100 μ_F = 0.744 μ_{CR} = 0.195
 Diversity = 1.5431e-03 k_t = [3, 3, 3, 4]
 Generation 6850: Best = 1.199207e+04 FES = 685100 μ_F = 0.760 μ_{CR} = 0.194
 Diversity = 1.7019e-03 k_t = [1, 1, 1, 1]
 Generation 6860: Best = 1.199206e+04 FES = 686100 μ_F = 0.770 μ_{CR} = 0.193
 Diversity = 1.7089e-03 k_t = [4, 2, 4, 3]
 Generation 6870: Best = 1.199206e+04 FES = 687100 μ_F = 0.775 μ_{CR} = 0.197
 Diversity = 1.7199e-03 k_t = [1, 1, 1, 1]
 Generation 6880: Best = 1.199205e+04 FES = 688100 μ_F = 0.787 μ_{CR} = 0.205
 Diversity = 1.9157e-03 k_t = [3, 3, 3, 3]
 Generation 6890: Best = 1.199205e+04 FES = 689100 μ_F = 0.780 μ_{CR} = 0.208
 Diversity = 2.0464e-03 k_t = [1, 1, 1, 1]
 Generation 6900: Best = 1.199204e+04 FES = 690100 μ_F = 0.783 μ_{CR} = 0.207
 Diversity = 2.0003e-03 k_t = [3, 2, 4, 2]
 Generation 6910: Best = 1.199204e+04 FES = 691100 μ_F = 0.780 μ_{CR} = 0.199
 Diversity = 2.1172e-03 k_t = [1, 1, 1, 1]
 Generation 6920: Best = 1.199204e+04 FES = 692100 μ_F = 0.793 μ_{CR} = 0.202
 Diversity = 2.0152e-03 k_t = [2, 3, 4, 2]
 Generation 6930: Best = 1.199204e+04 FES = 693100 μ_F = 0.790 μ_{CR} = 0.197
 Diversity = 1.8517e-03 k_t = [1, 1, 1, 1]
 Generation 6940: Best = 1.199204e+04 FES = 694100 μ_F = 0.792 μ_{CR} = 0.195
 Diversity = 1.8630e-03 k_t = [3, 4, 4, 2]
 Generation 6950: Best = 1.199203e+04 FES = 695100 μ_F = 0.773 μ_{CR} = 0.207
 Diversity = 1.8493e-03 k_t = [1, 1, 1, 1]
 Generation 6960: Best = 1.199203e+04 FES = 696100 μ_F = 0.773 μ_{CR} = 0.205
 Diversity = 1.6382e-03 k_t = [3, 3, 4, 3]
 Generation 6970: Best = 1.199202e+04 FES = 697100 μ_F = 0.774 μ_{CR} = 0.208
 Diversity = 2.2275e-03 k_t = [1, 1, 1, 1]

Generation 6980: Best = 1.199202e+04 FES = 698100 μ_F = 0.779 μ_{CR} = 0.214
 Diversity = 2.1549e-03 k_t = [4, 4, 4, 2]
 Generation 6990: Best = 1.199201e+04 FES = 699100 μ_F = 0.780 μ_{CR} = 0.209
 Diversity = 2.1419e-03 k_t = [1, 1, 1, 1]
 Generation 7000: Best = 1.199201e+04 FES = 700100 μ_F = 0.794 μ_{CR} = 0.218
 Diversity = 2.1404e-03 k_t = [4, 3, 2, 4]
 Generation 7010: Best = 1.199201e+04 FES = 701100 μ_F = 0.777 μ_{CR} = 0.216
 Diversity = 2.1416e-03 k_t = [1, 1, 1, 1]
 Generation 7020: Best = 1.199200e+04 FES = 702100 μ_F = 0.770 μ_{CR} = 0.218
 Diversity = 2.3402e-03 k_t = [3, 2, 2, 2]
 Generation 7030: Best = 1.199199e+04 FES = 703100 μ_F = 0.767 μ_{CR} = 0.225
 Diversity = 2.8070e-03 k_t = [1, 1, 1, 1]
 Generation 7040: Best = 1.199199e+04 FES = 704100 μ_F = 0.772 μ_{CR} = 0.224
 Diversity = 2.3162e-03 k_t = [2, 2, 4, 3]
 Generation 7050: Best = 1.199198e+04 FES = 705100 μ_F = 0.774 μ_{CR} = 0.229
 Diversity = 2.2021e-03 k_t = [1, 1, 1, 1]
 Generation 7060: Best = 1.199198e+04 FES = 706100 μ_F = 0.783 μ_{CR} = 0.227
 Diversity = 1.9415e-03 k_t = [4, 4, 2, 4]
 Generation 7070: Best = 1.199197e+04 FES = 707100 μ_F = 0.767 μ_{CR} = 0.232
 Diversity = 2.0888e-03 k_t = [1, 1, 1, 1]
 Generation 7080: Best = 1.199197e+04 FES = 708100 μ_F = 0.774 μ_{CR} = 0.237
 Diversity = 2.1173e-03 k_t = [2, 4, 2, 2]
 Generation 7090: Best = 1.199196e+04 FES = 709100 μ_F = 0.779 μ_{CR} = 0.238
 Diversity = 2.3812e-03 k_t = [1, 1, 1, 1]
 Generation 7100: Best = 1.199196e+04 FES = 710100 μ_F = 0.773 μ_{CR} = 0.241
 Diversity = 2.2223e-03 k_t = [4, 4, 3, 4]
 Generation 7110: Best = 1.199195e+04 FES = 711100 μ_F = 0.778 μ_{CR} = 0.237
 Diversity = 2.5069e-03 k_t = [1, 1, 1, 1]
 Generation 7120: Best = 1.199194e+04 FES = 712100 μ_F = 0.769 μ_{CR} = 0.241
 Diversity = 2.0655e-03 k_t = [4, 2, 4, 4]
 Generation 7130: Best = 1.199194e+04 FES = 713100 μ_F = 0.764 μ_{CR} = 0.247
 Diversity = 2.1521e-03 k_t = [1, 1, 1, 1]
 Generation 7140: Best = 1.199193e+04 FES = 714100 μ_F = 0.771 μ_{CR} = 0.251
 Diversity = 2.1443e-03 k_t = [4, 4, 2, 3]
 Generation 7150: Best = 1.199193e+04 FES = 715100 μ_F = 0.775 μ_{CR} = 0.249
 Diversity = 2.2282e-03 k_t = [1, 1, 1, 1]
 Generation 7160: Best = 1.199192e+04 FES = 716100 μ_F = 0.765 μ_{CR} = 0.249
 Diversity = 1.8847e-03 k_t = [3, 2, 3, 2]
 Generation 7170: Best = 1.199192e+04 FES = 717100 μ_F = 0.742 μ_{CR} = 0.250
 Diversity = 1.9228e-03 k_t = [1, 1, 1, 1]
 Generation 7180: Best = 1.199192e+04 FES = 718100 μ_F = 0.739 μ_{CR} = 0.253
 Diversity = 1.4997e-03 k_t = [4, 2, 2, 3]
 Generation 7190: Best = 1.199191e+04 FES = 719100 μ_F = 0.737 μ_{CR} = 0.245
 Diversity = 1.7083e-03 k_t = [1, 1, 1, 1]
 Generation 7200: Best = 1.199191e+04 FES = 720100 μ_F = 0.747 μ_{CR} = 0.243
 Diversity = 1.2018e-03 k_t = [2, 2, 4, 3]
 Generation 7210: Best = 1.199191e+04 FES = 721100 μ_F = 0.748 μ_{CR} = 0.257
 Diversity = 1.3072e-03 k_t = [1, 1, 1, 1]
 Generation 7220: Best = 1.199190e+04 FES = 722100 μ_F = 0.744 μ_{CR} = 0.254
 Diversity = 1.4732e-03 k_t = [2, 4, 2, 4]
 Generation 7230: Best = 1.199190e+04 FES = 723100 μ_F = 0.748 μ_{CR} = 0.254
 Diversity = 2.0821e-03 k_t = [1, 1, 1, 1]
 Generation 7240: Best = 1.199190e+04 FES = 724100 μ_F = 0.754 μ_{CR} = 0.258
 Diversity = 1.6249e-03 k_t = [3, 2, 4, 4]

Generation 7250: Best = 1.199190e+04 FES = 725100 μ_F = 0.762 μ_{CR} = 0.253
 Diversity = 1.4990e-03 k_t = [1, 1, 1, 1]
 Generation 7260: Best = 1.199189e+04 FES = 726100 μ_F = 0.770 μ_{CR} = 0.255
 Diversity = 1.5450e-03 k_t = [3, 2, 3, 4]
 Generation 7270: Best = 1.199189e+04 FES = 727100 μ_F = 0.783 μ_{CR} = 0.250
 Diversity = 1.7643e-03 k_t = [1, 1, 1, 1]
 Generation 7280: Best = 1.199188e+04 FES = 728100 μ_F = 0.786 μ_{CR} = 0.239
 Diversity = 1.9128e-03 k_t = [2, 3, 2, 3]
 Generation 7290: Best = 1.199188e+04 FES = 729100 μ_F = 0.781 μ_{CR} = 0.248
 Diversity = 1.8875e-03 k_t = [1, 1, 1, 1]
 Generation 7300: Best = 1.199188e+04 FES = 730100 μ_F = 0.766 μ_{CR} = 0.243
 Diversity = 1.9997e-03 k_t = [4, 3, 3, 2]
 Generation 7310: Best = 1.199187e+04 FES = 731100 μ_F = 0.773 μ_{CR} = 0.240
 Diversity = 2.1628e-03 k_t = [1, 1, 1, 1]
 Generation 7320: Best = 1.199187e+04 FES = 732100 μ_F = 0.771 μ_{CR} = 0.238
 Diversity = 1.8811e-03 k_t = [3, 2, 2, 3]
 Generation 7330: Best = 1.199186e+04 FES = 733100 μ_F = 0.781 μ_{CR} = 0.232
 Diversity = 2.1922e-03 k_t = [1, 1, 1, 1]
 Generation 7340: Best = 1.199186e+04 FES = 734100 μ_F = 0.777 μ_{CR} = 0.230
 Diversity = 1.9454e-03 k_t = [3, 4, 4, 3]
 Generation 7350: Best = 1.199186e+04 FES = 735100 μ_F = 0.776 μ_{CR} = 0.233
 Diversity = 2.0924e-03 k_t = [1, 1, 1, 1]
 Generation 7360: Best = 1.199185e+04 FES = 736100 μ_F = 0.770 μ_{CR} = 0.236
 Diversity = 2.0665e-03 k_t = [2, 2, 2, 3]
 Generation 7370: Best = 1.199185e+04 FES = 737100 μ_F = 0.755 μ_{CR} = 0.225
 Diversity = 2.2391e-03 k_t = [1, 1, 1, 1]
 Generation 7380: Best = 1.199184e+04 FES = 738100 μ_F = 0.758 μ_{CR} = 0.221
 Diversity = 1.6984e-03 k_t = [4, 4, 2, 4]
 Generation 7390: Best = 1.199184e+04 FES = 739100 μ_F = 0.762 μ_{CR} = 0.224
 Diversity = 1.8685e-03 k_t = [1, 1, 1, 1]
 Generation 7400: Best = 1.199183e+04 FES = 740100 μ_F = 0.767 μ_{CR} = 0.238
 Diversity = 2.3350e-03 k_t = [4, 4, 4, 2]
 Generation 7410: Best = 1.199182e+04 FES = 741100 μ_F = 0.767 μ_{CR} = 0.237
 Diversity = 2.6570e-03 k_t = [1, 1, 1, 1]
 Generation 7420: Best = 1.199182e+04 FES = 742100 μ_F = 0.762 μ_{CR} = 0.238
 Diversity = 2.2543e-03 k_t = [4, 4, 2, 2]
 Generation 7430: Best = 1.199182e+04 FES = 743100 μ_F = 0.774 μ_{CR} = 0.245
 Diversity = 2.3584e-03 k_t = [1, 1, 1, 1]
 Generation 7440: Best = 1.199181e+04 FES = 744100 μ_F = 0.760 μ_{CR} = 0.231
 Diversity = 2.2957e-03 k_t = [2, 3, 2, 2]
 Generation 7450: Best = 1.199181e+04 FES = 745100 μ_F = 0.756 μ_{CR} = 0.222
 Diversity = 2.2617e-03 k_t = [1, 1, 1, 1]
 Generation 7460: Best = 1.199180e+04 FES = 746100 μ_F = 0.768 μ_{CR} = 0.228
 Diversity = 2.0280e-03 k_t = [4, 3, 3, 4]
 Generation 7470: Best = 1.199180e+04 FES = 747100 μ_F = 0.768 μ_{CR} = 0.227
 Diversity = 1.8329e-03 k_t = [1, 1, 1, 1]
 Generation 7480: Best = 1.199180e+04 FES = 748100 μ_F = 0.762 μ_{CR} = 0.229
 Diversity = 1.9192e-03 k_t = [3, 3, 3, 2]
 Generation 7490: Best = 1.199179e+04 FES = 749100 μ_F = 0.752 μ_{CR} = 0.228
 Diversity = 1.9229e-03 k_t = [1, 1, 1, 1]
 Generation 7500: Best = 1.199179e+04 FES = 750100 μ_F = 0.746 μ_{CR} = 0.226
 Diversity = 1.7057e-03 k_t = [3, 4, 2, 2]
 Generation 7510: Best = 1.199179e+04 FES = 751100 μ_F = 0.754 μ_{CR} = 0.216
 Diversity = 1.6351e-03 k_t = [1, 1, 1, 1]

Generation 7520: Best = 1.199178e+04 FES = 752100 μ_F = 0.761 μ_{CR} = 0.219
 Diversity = 1.6176e-03 k_t = [3, 2, 2, 3]
 Generation 7530: Best = 1.199178e+04 FES = 753100 μ_F = 0.761 μ_{CR} = 0.221
 Diversity = 1.8571e-03 k_t = [1, 1, 1, 1]
 Generation 7540: Best = 1.199178e+04 FES = 754100 μ_F = 0.767 μ_{CR} = 0.221
 Diversity = 1.6065e-03 k_t = [2, 2, 4, 3]
 Generation 7550: Best = 1.199177e+04 FES = 755100 μ_F = 0.775 μ_{CR} = 0.216
 Diversity = 1.6219e-03 k_t = [1, 1, 1, 1]
 Generation 7560: Best = 1.199177e+04 FES = 756100 μ_F = 0.785 μ_{CR} = 0.214
 Diversity = 1.5922e-03 k_t = [3, 2, 2, 2]
 Generation 7570: Best = 1.199177e+04 FES = 757100 μ_F = 0.781 μ_{CR} = 0.210
 Diversity = 1.5834e-03 k_t = [1, 1, 1, 1]
 Generation 7580: Best = 1.199177e+04 FES = 758100 μ_F = 0.776 μ_{CR} = 0.212
 Diversity = 1.7509e-03 k_t = [2, 4, 4, 2]
 Generation 7590: Best = 1.199176e+04 FES = 759100 μ_F = 0.777 μ_{CR} = 0.208
 Diversity = 1.9776e-03 k_t = [1, 1, 1, 1]
 Generation 7600: Best = 1.199175e+04 FES = 760100 μ_F = 0.777 μ_{CR} = 0.212
 Diversity = 2.1315e-03 k_t = [2, 4, 4, 4]
 Generation 7610: Best = 1.199174e+04 FES = 761100 μ_F = 0.780 μ_{CR} = 0.217
 Diversity = 2.7855e-03 k_t = [1, 1, 1, 1]
 Generation 7620: Best = 1.199173e+04 FES = 762100 μ_F = 0.790 μ_{CR} = 0.218
 Diversity = 2.4708e-03 k_t = [4, 4, 3, 3]
 Generation 7630: Best = 1.199173e+04 FES = 763100 μ_F = 0.792 μ_{CR} = 0.216
 Diversity = 2.4167e-03 k_t = [1, 1, 1, 1]
 Generation 7640: Best = 1.199172e+04 FES = 764100 μ_F = 0.793 μ_{CR} = 0.215
 Diversity = 2.5155e-03 k_t = [2, 4, 2, 3]
 Generation 7650: Best = 1.199172e+04 FES = 765100 μ_F = 0.782 μ_{CR} = 0.210
 Diversity = 2.6008e-03 k_t = [1, 1, 1, 1]
 Generation 7660: Best = 1.199171e+04 FES = 766100 μ_F = 0.783 μ_{CR} = 0.216
 Diversity = 2.2689e-03 k_t = [3, 2, 3, 3]
 Generation 7670: Best = 1.199171e+04 FES = 767100 μ_F = 0.781 μ_{CR} = 0.220
 Diversity = 2.4074e-03 k_t = [1, 1, 1, 1]
 Generation 7680: Best = 1.199170e+04 FES = 768100 μ_F = 0.778 μ_{CR} = 0.221
 Diversity = 2.2381e-03 k_t = [2, 2, 2, 3]
 Generation 7690: Best = 1.199170e+04 FES = 769100 μ_F = 0.772 μ_{CR} = 0.212
 Diversity = 2.4121e-03 k_t = [1, 1, 1, 1]
 Generation 7700: Best = 1.199169e+04 FES = 770100 μ_F = 0.764 μ_{CR} = 0.203
 Diversity = 2.1748e-03 k_t = [4, 3, 2, 3]
 Generation 7710: Best = 1.199168e+04 FES = 771100 μ_F = 0.764 μ_{CR} = 0.207
 Diversity = 2.3823e-03 k_t = [1, 1, 1, 1]
 Generation 7720: Best = 1.199168e+04 FES = 772100 μ_F = 0.772 μ_{CR} = 0.209
 Diversity = 2.2158e-03 k_t = [4, 4, 2, 4]
 Generation 7730: Best = 1.199167e+04 FES = 773100 μ_F = 0.771 μ_{CR} = 0.198
 Diversity = 2.0708e-03 k_t = [1, 1, 1, 1]
 Generation 7740: Best = 1.199167e+04 FES = 774100 μ_F = 0.762 μ_{CR} = 0.199
 Diversity = 1.8250e-03 k_t = [4, 2, 2, 4]
 Generation 7750: Best = 1.199167e+04 FES = 775100 μ_F = 0.766 μ_{CR} = 0.207
 Diversity = 2.0150e-03 k_t = [1, 1, 1, 1]
 Generation 7760: Best = 1.199166e+04 FES = 776100 μ_F = 0.759 μ_{CR} = 0.212
 Diversity = 1.8397e-03 k_t = [4, 2, 2, 2]
 Generation 7770: Best = 1.199166e+04 FES = 777100 μ_F = 0.757 μ_{CR} = 0.225
 Diversity = 2.0505e-03 k_t = [1, 1, 1, 1]
 Generation 7780: Best = 1.199166e+04 FES = 778100 μ_F = 0.758 μ_{CR} = 0.218
 Diversity = 1.8125e-03 k_t = [2, 3, 4, 4]

Generation 7790: Best = 1.199166e+04 FES = 779100 μ_F = 0.760 μ_{CR} = 0.224
 Diversity = 1.7388e-03 k_t = [1, 1, 1, 1]
 Generation 7800: Best = 1.199165e+04 FES = 780100 μ_F = 0.752 μ_{CR} = 0.224
 Diversity = 1.6250e-03 k_t = [2, 3, 3, 4]
 Generation 7810: Best = 1.199165e+04 FES = 781100 μ_F = 0.764 μ_{CR} = 0.228
 Diversity = 1.9174e-03 k_t = [1, 1, 1, 1]
 Generation 7820: Best = 1.199165e+04 FES = 782100 μ_F = 0.768 μ_{CR} = 0.236
 Diversity = 1.9118e-03 k_t = [3, 3, 4, 2]
 Generation 7830: Best = 1.199164e+04 FES = 783100 μ_F = 0.762 μ_{CR} = 0.236
 Diversity = 1.7085e-03 k_t = [1, 1, 1, 1]
 Generation 7840: Best = 1.199164e+04 FES = 784100 μ_F = 0.749 μ_{CR} = 0.238
 Diversity = 1.5802e-03 k_t = [4, 3, 3, 4]
 Generation 7850: Best = 1.199164e+04 FES = 785100 μ_F = 0.748 μ_{CR} = 0.228
 Diversity = 1.5701e-03 k_t = [1, 1, 1, 1]
 Generation 7860: Best = 1.199163e+04 FES = 786100 μ_F = 0.759 μ_{CR} = 0.225
 Diversity = 1.5280e-03 k_t = [2, 3, 4, 4]
 Generation 7870: Best = 1.199163e+04 FES = 787100 μ_F = 0.772 μ_{CR} = 0.220
 Diversity = 1.8639e-03 k_t = [1, 1, 1, 1]
 Generation 7880: Best = 1.199162e+04 FES = 788100 μ_F = 0.794 μ_{CR} = 0.226
 Diversity = 1.6311e-03 k_t = [3, 2, 4, 3]
 Generation 7890: Best = 1.199162e+04 FES = 789100 μ_F = 0.791 μ_{CR} = 0.235
 Diversity = 1.8235e-03 k_t = [1, 1, 1, 1]
 Generation 7900: Best = 1.199162e+04 FES = 790100 μ_F = 0.779 μ_{CR} = 0.233
 Diversity = 1.7598e-03 k_t = [2, 4, 2, 4]
 Generation 7910: Best = 1.199162e+04 FES = 791100 μ_F = 0.775 μ_{CR} = 0.231
 Diversity = 1.7746e-03 k_t = [1, 1, 1, 1]
 Generation 7920: Best = 1.199162e+04 FES = 792100 μ_F = 0.764 μ_{CR} = 0.238
 Diversity = 1.6692e-03 k_t = [3, 3, 3, 2]
 Generation 7930: Best = 1.199161e+04 FES = 793100 μ_F = 0.764 μ_{CR} = 0.240
 Diversity = 1.7907e-03 k_t = [1, 1, 1, 1]
 Generation 7940: Best = 1.199161e+04 FES = 794100 μ_F = 0.777 μ_{CR} = 0.243
 Diversity = 1.6887e-03 k_t = [4, 3, 4, 2]
 Generation 7950: Best = 1.199160e+04 FES = 795100 μ_F = 0.782 μ_{CR} = 0.239
 Diversity = 2.1886e-03 k_t = [1, 1, 1, 1]
 Generation 7960: Best = 1.199160e+04 FES = 796100 μ_F = 0.776 μ_{CR} = 0.231
 Diversity = 1.9277e-03 k_t = [2, 2, 4, 2]
 Generation 7970: Best = 1.199160e+04 FES = 797100 μ_F = 0.776 μ_{CR} = 0.228
 Diversity = 1.7271e-03 k_t = [1, 1, 1, 1]
 Generation 7980: Best = 1.199160e+04 FES = 798100 μ_F = 0.767 μ_{CR} = 0.228
 Diversity = 1.4748e-03 k_t = [3, 3, 4, 4]
 Generation 7990: Best = 1.199159e+04 FES = 799100 μ_F = 0.772 μ_{CR} = 0.227
 Diversity = 1.6733e-03 k_t = [1, 1, 1, 1]
 Generation 8000: Best = 1.199159e+04 FES = 800100 μ_F = 0.772 μ_{CR} = 0.222
 Diversity = 1.6998e-03 k_t = [2, 2, 4, 3]
 Generation 8010: Best = 1.199159e+04 FES = 801100 μ_F = 0.757 μ_{CR} = 0.219
 Diversity = 1.6544e-03 k_t = [1, 1, 1, 1]
 Generation 8020: Best = 1.199159e+04 FES = 802100 μ_F = 0.761 μ_{CR} = 0.213
 Diversity = 1.4764e-03 k_t = [3, 4, 3, 4]
 Generation 8030: Best = 1.199158e+04 FES = 803100 μ_F = 0.764 μ_{CR} = 0.203
 Diversity = 1.5487e-03 k_t = [1, 1, 1, 1]
 Generation 8040: Best = 1.199158e+04 FES = 804100 μ_F = 0.766 μ_{CR} = 0.194
 Diversity = 1.5893e-03 k_t = [2, 2, 4, 3]
 Generation 8050: Best = 1.199158e+04 FES = 805100 μ_F = 0.777 μ_{CR} = 0.197
 Diversity = 1.6758e-03 k_t = [1, 1, 1, 1]

Generation 8060: Best = 1.199158e+04 FES = 806100 μ_F = 0.761 μ_{CR} = 0.206
 Diversity = 1.5369e-03 k_t = [3, 3, 2, 4]
 Generation 8070: Best = 1.199158e+04 FES = 807100 μ_F = 0.754 μ_{CR} = 0.206
 Diversity = 1.6367e-03 k_t = [1, 1, 1, 1]
 Generation 8080: Best = 1.199157e+04 FES = 808100 μ_F = 0.743 μ_{CR} = 0.199
 Diversity = 1.4720e-03 k_t = [2, 3, 4, 2]
 Generation 8090: Best = 1.199157e+04 FES = 809100 μ_F = 0.749 μ_{CR} = 0.194
 Diversity = 1.5539e-03 k_t = [1, 1, 1, 1]
 Generation 8100: Best = 1.199157e+04 FES = 810100 μ_F = 0.740 μ_{CR} = 0.203
 Diversity = 1.3959e-03 k_t = [3, 4, 4, 4]
 Generation 8110: Best = 1.199157e+04 FES = 811100 μ_F = 0.749 μ_{CR} = 0.202
 Diversity = 1.4083e-03 k_t = [1, 1, 1, 1]
 Generation 8120: Best = 1.199157e+04 FES = 812100 μ_F = 0.756 μ_{CR} = 0.197
 Diversity = 1.1584e-03 k_t = [4, 2, 2, 4]
 Generation 8130: Best = 1.199156e+04 FES = 813100 μ_F = 0.750 μ_{CR} = 0.196
 Diversity = 1.1302e-03 k_t = [1, 1, 1, 1]
 Generation 8140: Best = 1.199156e+04 FES = 814100 μ_F = 0.761 μ_{CR} = 0.200
 Diversity = 1.2050e-03 k_t = [4, 2, 3, 3]
 Generation 8150: Best = 1.199156e+04 FES = 815100 μ_F = 0.758 μ_{CR} = 0.210
 Diversity = 1.3708e-03 k_t = [1, 1, 1, 1]
 Generation 8160: Best = 1.199156e+04 FES = 816100 μ_F = 0.758 μ_{CR} = 0.207
 Diversity = 1.2232e-03 k_t = [3, 2, 4, 4]
 Generation 8170: Best = 1.199156e+04 FES = 817100 μ_F = 0.763 μ_{CR} = 0.211
 Diversity = 1.1014e-03 k_t = [1, 1, 1, 1]
 Generation 8180: Best = 1.199156e+04 FES = 818100 μ_F = 0.764 μ_{CR} = 0.202
 Diversity = 9.6043e-04 k_t = [3, 2, 4, 4]
 Generation 8190: Best = 1.199156e+04 FES = 819100 μ_F = 0.774 μ_{CR} = 0.201
 Diversity = 1.1825e-03 k_t = [1, 1, 1, 1]
 Generation 8200: Best = 1.199155e+04 FES = 820100 μ_F = 0.785 μ_{CR} = 0.201
 Diversity = 1.2222e-03 k_t = [3, 4, 4, 3]
 Generation 8210: Best = 1.199155e+04 FES = 821100 μ_F = 0.793 μ_{CR} = 0.200
 Diversity = 1.6041e-03 k_t = [1, 1, 1, 1]
 Generation 8220: Best = 1.199155e+04 FES = 822100 μ_F = 0.793 μ_{CR} = 0.205
 Diversity = 1.5095e-03 k_t = [4, 3, 3, 4]
 Generation 8230: Best = 1.199154e+04 FES = 823100 μ_F = 0.805 μ_{CR} = 0.212
 Diversity = 1.8720e-03 k_t = [1, 1, 1, 1]
 Generation 8240: Best = 1.199154e+04 FES = 824100 μ_F = 0.813 μ_{CR} = 0.203
 Diversity = 1.9794e-03 k_t = [4, 4, 2, 4]
 Generation 8250: Best = 1.199153e+04 FES = 825100 μ_F = 0.818 μ_{CR} = 0.201
 Diversity = 2.0088e-03 k_t = [1, 1, 1, 1]
 Generation 8260: Best = 1.199153e+04 FES = 826100 μ_F = 0.813 μ_{CR} = 0.203
 Diversity = 1.7565e-03 k_t = [4, 4, 3, 2]
 Generation 8270: Best = 1.199153e+04 FES = 827100 μ_F = 0.806 μ_{CR} = 0.207
 Diversity = 1.7924e-03 k_t = [1, 1, 1, 1]
 Generation 8280: Best = 1.199152e+04 FES = 828100 μ_F = 0.801 μ_{CR} = 0.198
 Diversity = 1.8817e-03 k_t = [3, 2, 3, 4]
 Generation 8290: Best = 1.199152e+04 FES = 829100 μ_F = 0.803 μ_{CR} = 0.202
 Diversity = 1.9982e-03 k_t = [1, 1, 1, 1]
 Generation 8300: Best = 1.199152e+04 FES = 830100 μ_F = 0.799 μ_{CR} = 0.196
 Diversity = 1.9339e-03 k_t = [2, 2, 2, 2]
 Generation 8310: Best = 1.199151e+04 FES = 831100 μ_F = 0.789 μ_{CR} = 0.205
 Diversity = 2.0767e-03 k_t = [1, 1, 1, 1]
 Generation 8320: Best = 1.199150e+04 FES = 832100 μ_F = 0.790 μ_{CR} = 0.208
 Diversity = 2.5651e-03 k_t = [3, 4, 2, 3]

Generation 8330: Best = 1.199150e+04 FES = 833100 μ_F = 0.793 μ_{CR} = 0.212
 Diversity = 2.3595e-03 k_t = [1, 1, 1, 1]
 Generation 8340: Best = 1.199150e+04 FES = 834100 μ_F = 0.785 μ_{CR} = 0.226
 Diversity = 2.1610e-03 k_t = [3, 2, 4, 2]
 Generation 8350: Best = 1.199149e+04 FES = 835100 μ_F = 0.781 μ_{CR} = 0.225
 Diversity = 2.3485e-03 k_t = [1, 1, 1, 1]
 Generation 8360: Best = 1.199149e+04 FES = 836100 μ_F = 0.779 μ_{CR} = 0.232
 Diversity = 2.1942e-03 k_t = [3, 4, 4, 4]
 Generation 8370: Best = 1.199148e+04 FES = 837100 μ_F = 0.775 μ_{CR} = 0.226
 Diversity = 2.0238e-03 k_t = [1, 1, 1, 1]
 Generation 8380: Best = 1.199148e+04 FES = 838100 μ_F = 0.782 μ_{CR} = 0.220
 Diversity = 1.8688e-03 k_t = [2, 3, 4, 4]
 Generation 8390: Best = 1.199147e+04 FES = 839100 μ_F = 0.788 μ_{CR} = 0.223
 Diversity = 2.5726e-03 k_t = [1, 1, 1, 1]
 Generation 8400: Best = 1.199146e+04 FES = 840100 μ_F = 0.808 μ_{CR} = 0.225
 Diversity = 3.0463e-03 k_t = [3, 4, 3, 3]
 Generation 8410: Best = 1.199145e+04 FES = 841100 μ_F = 0.797 μ_{CR} = 0.226
 Diversity = 3.4532e-03 k_t = [1, 1, 1, 1]
 Generation 8420: Best = 1.199144e+04 FES = 842100 μ_F = 0.788 μ_{CR} = 0.223
 Diversity = 2.8908e-03 k_t = [4, 3, 2, 4]
 Generation 8430: Best = 1.199144e+04 FES = 843100 μ_F = 0.788 μ_{CR} = 0.215
 Diversity = 2.8527e-03 k_t = [1, 1, 1, 1]
 Generation 8440: Best = 1.199143e+04 FES = 844100 μ_F = 0.781 μ_{CR} = 0.220
 Diversity = 1.9722e-03 k_t = [3, 4, 4, 3]
 Generation 8450: Best = 1.199143e+04 FES = 845100 μ_F = 0.789 μ_{CR} = 0.233
 Diversity = 2.1495e-03 k_t = [1, 1, 1, 1]
 Generation 8460: Best = 1.199143e+04 FES = 846100 μ_F = 0.783 μ_{CR} = 0.224
 Diversity = 1.8666e-03 k_t = [3, 2, 3, 3]
 Generation 8470: Best = 1.199143e+04 FES = 847100 μ_F = 0.765 μ_{CR} = 0.223
 Diversity = 1.7452e-03 k_t = [1, 1, 1, 1]
 Generation 8480: Best = 1.199142e+04 FES = 848100 μ_F = 0.758 μ_{CR} = 0.220
 Diversity = 1.4793e-03 k_t = [3, 2, 4, 4]
 Generation 8490: Best = 1.199142e+04 FES = 849100 μ_F = 0.763 μ_{CR} = 0.218
 Diversity = 1.4655e-03 k_t = [1, 1, 1, 1]
 Generation 8500: Best = 1.199142e+04 FES = 850100 μ_F = 0.763 μ_{CR} = 0.221
 Diversity = 1.3964e-03 k_t = [4, 4, 3, 2]
 Generation 8510: Best = 1.199142e+04 FES = 851100 μ_F = 0.755 μ_{CR} = 0.217
 Diversity = 1.3873e-03 k_t = [1, 1, 1, 1]
 Generation 8520: Best = 1.199142e+04 FES = 852100 μ_F = 0.751 μ_{CR} = 0.223
 Diversity = 1.1764e-03 k_t = [2, 4, 2, 3]
 Generation 8530: Best = 1.199141e+04 FES = 853100 μ_F = 0.754 μ_{CR} = 0.227
 Diversity = 1.2401e-03 k_t = [1, 1, 1, 1]
 Generation 8540: Best = 1.199141e+04 FES = 854100 μ_F = 0.763 μ_{CR} = 0.228
 Diversity = 1.0909e-03 k_t = [4, 4, 4, 4]
 Generation 8550: Best = 1.199141e+04 FES = 855100 μ_F = 0.772 μ_{CR} = 0.224
 Diversity = 1.1625e-03 k_t = [1, 1, 1, 1]
 Generation 8560: Best = 1.199141e+04 FES = 856100 μ_F = 0.781 μ_{CR} = 0.237
 Diversity = 1.1796e-03 k_t = [2, 2, 2, 4]
 Generation 8570: Best = 1.199141e+04 FES = 857100 μ_F = 0.764 μ_{CR} = 0.244
 Diversity = 1.2688e-03 k_t = [1, 1, 1, 1]
 Generation 8580: Best = 1.199140e+04 FES = 858100 μ_F = 0.768 μ_{CR} = 0.235
 Diversity = 1.2219e-03 k_t = [3, 4, 2, 4]
 Generation 8590: Best = 1.199140e+04 FES = 859100 μ_F = 0.771 μ_{CR} = 0.232
 Diversity = 1.3328e-03 k_t = [1, 1, 1, 1]

Generation 8600: Best = 1.199140e+04 FES = 860100 μ_F = 0.770 μ_{CR} = 0.222
 Diversity = 1.0953e-03 k_t = [4, 4, 3, 4]
 Generation 8610: Best = 1.199140e+04 FES = 861100 μ_F = 0.769 μ_{CR} = 0.230
 Diversity = 1.2141e-03 k_t = [1, 1, 1, 1]
 Generation 8620: Best = 1.199140e+04 FES = 862100 μ_F = 0.766 μ_{CR} = 0.221
 Diversity = 1.0422e-03 k_t = [2, 2, 4, 4]
 Generation 8630: Best = 1.199140e+04 FES = 863100 μ_F = 0.758 μ_{CR} = 0.216
 Diversity = 1.0000e-03 k_t = [1, 1, 1, 1]
 Generation 8640: Best = 1.199140e+04 FES = 864100 μ_F = 0.759 μ_{CR} = 0.213
 Diversity = 9.5572e-04 k_t = [3, 4, 4, 4]
 Generation 8650: Best = 1.199139e+04 FES = 865100 μ_F = 0.776 μ_{CR} = 0.210
 Diversity = 1.0051e-03 k_t = [1, 1, 1, 1]
 Generation 8660: Best = 1.199139e+04 FES = 866100 μ_F = 0.789 μ_{CR} = 0.211
 Diversity = 1.2531e-03 k_t = [3, 2, 4, 2]
 Generation 8670: Best = 1.199139e+04 FES = 867100 μ_F = 0.798 μ_{CR} = 0.212
 Diversity = 1.3804e-03 k_t = [1, 1, 1, 1]
 Generation 8680: Best = 1.199139e+04 FES = 868100 μ_F = 0.795 μ_{CR} = 0.217
 Diversity = 1.1624e-03 k_t = [4, 4, 2, 3]
 Generation 8690: Best = 1.199139e+04 FES = 869100 μ_F = 0.800 μ_{CR} = 0.219
 Diversity = 1.3108e-03 k_t = [1, 1, 1, 1]
 Generation 8700: Best = 1.199139e+04 FES = 870100 μ_F = 0.798 μ_{CR} = 0.213
 Diversity = 1.2347e-03 k_t = [2, 3, 2, 3]
 Generation 8710: Best = 1.199138e+04 FES = 871100 μ_F = 0.796 μ_{CR} = 0.221
 Diversity = 1.3794e-03 k_t = [1, 1, 1, 1]
 Generation 8720: Best = 1.199138e+04 FES = 872100 μ_F = 0.810 μ_{CR} = 0.223
 Diversity = 1.2864e-03 k_t = [4, 4, 3, 4]
 Generation 8730: Best = 1.199138e+04 FES = 873100 μ_F = 0.807 μ_{CR} = 0.232
 Diversity = 1.6286e-03 k_t = [1, 1, 1, 1]
 Generation 8740: Best = 1.199137e+04 FES = 874100 μ_F = 0.799 μ_{CR} = 0.239
 Diversity = 1.6200e-03 k_t = [3, 4, 3, 2]
 Generation 8750: Best = 1.199137e+04 FES = 875100 μ_F = 0.780 μ_{CR} = 0.240
 Diversity = 1.8508e-03 k_t = [1, 1, 1, 1]
 Generation 8760: Best = 1.199137e+04 FES = 876100 μ_F = 0.779 μ_{CR} = 0.234
 Diversity = 1.7037e-03 k_t = [2, 2, 2, 4]
 Generation 8770: Best = 1.199136e+04 FES = 877100 μ_F = 0.780 μ_{CR} = 0.231
 Diversity = 1.7077e-03 k_t = [1, 1, 1, 1]
 Generation 8780: Best = 1.199136e+04 FES = 878100 μ_F = 0.780 μ_{CR} = 0.232
 Diversity = 1.3274e-03 k_t = [4, 4, 4, 4]
 Generation 8790: Best = 1.199136e+04 FES = 879100 μ_F = 0.763 μ_{CR} = 0.233
 Diversity = 1.3708e-03 k_t = [1, 1, 1, 1]
 Generation 8800: Best = 1.199135e+04 FES = 880100 μ_F = 0.768 μ_{CR} = 0.236
 Diversity = 1.5874e-03 k_t = [2, 4, 4, 3]
 Generation 8810: Best = 1.199135e+04 FES = 881100 μ_F = 0.780 μ_{CR} = 0.252
 Diversity = 2.0754e-03 k_t = [1, 1, 1, 1]
 Generation 8820: Best = 1.199134e+04 FES = 882100 μ_F = 0.791 μ_{CR} = 0.249
 Diversity = 2.0065e-03 k_t = [4, 2, 2, 3]
 Generation 8830: Best = 1.199134e+04 FES = 883100 μ_F = 0.785 μ_{CR} = 0.247
 Diversity = 1.7153e-03 k_t = [1, 1, 1, 1]
 Generation 8840: Best = 1.199133e+04 FES = 884100 μ_F = 0.794 μ_{CR} = 0.250
 Diversity = 1.8321e-03 k_t = [4, 2, 2, 3]
 Generation 8850: Best = 1.199133e+04 FES = 885100 μ_F = 0.782 μ_{CR} = 0.244
 Diversity = 1.9042e-03 k_t = [1, 1, 1, 1]
 Generation 8860: Best = 1.199132e+04 FES = 886100 μ_F = 0.781 μ_{CR} = 0.241
 Diversity = 2.1469e-03 k_t = [2, 2, 4, 2]

Generation 8870: Best = 1.199131e+04 FES = 887100 μ_F = 0.774 μ_{CR} = 0.249
 Diversity = 2.5718e-03 k_t = [1, 1, 1, 1]
 Generation 8880: Best = 1.199131e+04 FES = 888100 μ_F = 0.779 μ_{CR} = 0.249
 Diversity = 2.4332e-03 k_t = [3, 2, 4, 2]
 Generation 8890: Best = 1.199130e+04 FES = 889100 μ_F = 0.783 μ_{CR} = 0.248
 Diversity = 2.5192e-03 k_t = [1, 1, 1, 1]
 Generation 8900: Best = 1.199130e+04 FES = 890100 μ_F = 0.778 μ_{CR} = 0.263
 Diversity = 2.0445e-03 k_t = [3, 3, 2, 3]
 Generation 8910: Best = 1.199128e+04 FES = 891100 μ_F = 0.779 μ_{CR} = 0.273
 Diversity = 2.5838e-03 k_t = [1, 1, 1, 1]
 Generation 8920: Best = 1.199128e+04 FES = 892100 μ_F = 0.768 μ_{CR} = 0.276
 Diversity = 2.7450e-03 k_t = [2, 3, 3, 2]
 Generation 8930: Best = 1.199127e+04 FES = 893100 μ_F = 0.761 μ_{CR} = 0.280
 Diversity = 2.9522e-03 k_t = [1, 1, 1, 1]
 Generation 8940: Best = 1.199126e+04 FES = 894100 μ_F = 0.755 μ_{CR} = 0.274
 Diversity = 2.4743e-03 k_t = [3, 2, 3, 4]
 Generation 8950: Best = 1.199126e+04 FES = 895100 μ_F = 0.745 μ_{CR} = 0.271
 Diversity = 2.5681e-03 k_t = [1, 1, 1, 1]
 Generation 8960: Best = 1.199125e+04 FES = 896100 μ_F = 0.754 μ_{CR} = 0.275
 Diversity = 2.3421e-03 k_t = [2, 4, 2, 3]
 Generation 8970: Best = 1.199124e+04 FES = 897100 μ_F = 0.756 μ_{CR} = 0.273
 Diversity = 2.3853e-03 k_t = [1, 1, 1, 1]
 Generation 8980: Best = 1.199123e+04 FES = 898100 μ_F = 0.768 μ_{CR} = 0.265
 Diversity = 2.1148e-03 k_t = [3, 4, 4, 3]
 Generation 8990: Best = 1.199123e+04 FES = 899100 μ_F = 0.774 μ_{CR} = 0.268
 Diversity = 2.3133e-03 k_t = [1, 1, 1, 1]
 Generation 9000: Best = 1.199122e+04 FES = 900100 μ_F = 0.781 μ_{CR} = 0.261
 Diversity = 2.1034e-03 k_t = [4, 4, 2, 2]
 Generation 9010: Best = 1.199122e+04 FES = 901100 μ_F = 0.765 μ_{CR} = 0.255
 Diversity = 1.9806e-03 k_t = [1, 1, 1, 1]
 Generation 9020: Best = 1.199122e+04 FES = 902100 μ_F = 0.756 μ_{CR} = 0.244
 Diversity = 1.6913e-03 k_t = [4, 2, 4, 4]
 Generation 9030: Best = 1.199121e+04 FES = 903100 μ_F = 0.749 μ_{CR} = 0.252
 Diversity = 1.8368e-03 k_t = [1, 1, 1, 1]
 Generation 9040: Best = 1.199121e+04 FES = 904100 μ_F = 0.747 μ_{CR} = 0.247
 Diversity = 1.6072e-03 k_t = [3, 2, 3, 4]
 Generation 9050: Best = 1.199121e+04 FES = 905100 μ_F = 0.745 μ_{CR} = 0.242
 Diversity = 1.5520e-03 k_t = [1, 1, 1, 1]
 Generation 9060: Best = 1.199121e+04 FES = 906100 μ_F = 0.747 μ_{CR} = 0.239
 Diversity = 1.3945e-03 k_t = [3, 4, 3, 4]
 Generation 9070: Best = 1.199121e+04 FES = 907100 μ_F = 0.746 μ_{CR} = 0.250
 Diversity = 1.3030e-03 k_t = [1, 1, 1, 1]
 Generation 9080: Best = 1.199120e+04 FES = 908100 μ_F = 0.750 μ_{CR} = 0.252
 Diversity = 1.3100e-03 k_t = [3, 3, 3, 4]
 Generation 9090: Best = 1.199120e+04 FES = 909100 μ_F = 0.766 μ_{CR} = 0.255
 Diversity = 1.3479e-03 k_t = [1, 1, 1, 1]
 Generation 9100: Best = 1.199120e+04 FES = 910100 μ_F = 0.770 μ_{CR} = 0.249
 Diversity = 1.5488e-03 k_t = [3, 3, 3, 4]
 Generation 9110: Best = 1.199119e+04 FES = 911100 μ_F = 0.771 μ_{CR} = 0.264
 Diversity = 1.7190e-03 k_t = [1, 1, 1, 1]
 Generation 9120: Best = 1.199119e+04 FES = 912100 μ_F = 0.772 μ_{CR} = 0.261
 Diversity = 1.4748e-03 k_t = [4, 4, 3, 4]
 Generation 9130: Best = 1.199119e+04 FES = 913100 μ_F = 0.794 μ_{CR} = 0.264
 Diversity = 1.6300e-03 k_t = [1, 1, 1, 1]

Generation 9140: Best = 1.199119e+04 FES = 914100 $\mu_F = 0.777$ $\mu_{CR} = 0.257$
 Diversity = 1.4772e-03 $k_t = [4, 2, 3, 4]$
 Generation 9150: Best = 1.199118e+04 FES = 915100 $\mu_F = 0.783$ $\mu_{CR} = 0.264$
 Diversity = 1.5971e-03 $k_t = [1, 1, 1, 1]$
 Generation 9160: Best = 1.199118e+04 FES = 916100 $\mu_F = 0.777$ $\mu_{CR} = 0.264$
 Diversity = 1.4622e-03 $k_t = [2, 3, 2, 2]$
 Generation 9170: Best = 1.199118e+04 FES = 917100 $\mu_F = 0.779$ $\mu_{CR} = 0.260$
 Diversity = 1.4862e-03 $k_t = [1, 1, 1, 1]$
 Generation 9180: Best = 1.199118e+04 FES = 918100 $\mu_F = 0.776$ $\mu_{CR} = 0.250$
 Diversity = 1.1950e-03 $k_t = [4, 2, 2, 4]$
 Generation 9190: Best = 1.199117e+04 FES = 919100 $\mu_F = 0.778$ $\mu_{CR} = 0.251$
 Diversity = 1.2637e-03 $k_t = [1, 1, 1, 1]$
 Generation 9200: Best = 1.199117e+04 FES = 920100 $\mu_F = 0.770$ $\mu_{CR} = 0.249$
 Diversity = 1.3861e-03 $k_t = [2, 3, 4, 2]$
 Generation 9210: Best = 1.199117e+04 FES = 921100 $\mu_F = 0.766$ $\mu_{CR} = 0.260$
 Diversity = 1.6921e-03 $k_t = [1, 1, 1, 1]$
 Generation 9220: Best = 1.199116e+04 FES = 922100 $\mu_F = 0.779$ $\mu_{CR} = 0.255$
 Diversity = 1.7851e-03 $k_t = [3, 4, 4, 4]$
 Generation 9230: Best = 1.199116e+04 FES = 923100 $\mu_F = 0.763$ $\mu_{CR} = 0.242$
 Diversity = 1.7084e-03 $k_t = [1, 1, 1, 1]$
 Generation 9240: Best = 1.199116e+04 FES = 924100 $\mu_F = 0.769$ $\mu_{CR} = 0.245$
 Diversity = 1.5399e-03 $k_t = [4, 3, 4, 4]$
 Generation 9250: Best = 1.199116e+04 FES = 925100 $\mu_F = 0.775$ $\mu_{CR} = 0.244$
 Diversity = 1.6009e-03 $k_t = [1, 1, 1, 1]$
 Generation 9260: Best = 1.199115e+04 FES = 926100 $\mu_F = 0.761$ $\mu_{CR} = 0.247$
 Diversity = 1.4520e-03 $k_t = [3, 3, 2, 4]$
 Generation 9270: Best = 1.199115e+04 FES = 927100 $\mu_F = 0.758$ $\mu_{CR} = 0.238$
 Diversity = 1.3507e-03 $k_t = [1, 1, 1, 1]$
 Generation 9280: Best = 1.199115e+04 FES = 928100 $\mu_F = 0.765$ $\mu_{CR} = 0.224$
 Diversity = 1.3668e-03 $k_t = [2, 3, 4, 2]$
 Generation 9290: Best = 1.199115e+04 FES = 929100 $\mu_F = 0.767$ $\mu_{CR} = 0.228$
 Diversity = 1.6597e-03 $k_t = [1, 1, 1, 1]$
 Generation 9300: Best = 1.199115e+04 FES = 930100 $\mu_F = 0.757$ $\mu_{CR} = 0.236$
 Diversity = 1.2465e-03 $k_t = [3, 4, 4, 4]$
 Generation 9310: Best = 1.199114e+04 FES = 931100 $\mu_F = 0.757$ $\mu_{CR} = 0.226$
 Diversity = 1.3618e-03 $k_t = [1, 1, 1, 1]$
 Generation 9320: Best = 1.199114e+04 FES = 932100 $\mu_F = 0.763$ $\mu_{CR} = 0.223$
 Diversity = 1.4470e-03 $k_t = [3, 2, 4, 2]$
 Generation 9330: Best = 1.199114e+04 FES = 933100 $\mu_F = 0.763$ $\mu_{CR} = 0.222$
 Diversity = 1.8541e-03 $k_t = [1, 1, 1, 1]$
 Generation 9340: Best = 1.199113e+04 FES = 934100 $\mu_F = 0.762$ $\mu_{CR} = 0.225$
 Diversity = 1.5507e-03 $k_t = [2, 4, 4, 4]$
 Generation 9350: Best = 1.199113e+04 FES = 935100 $\mu_F = 0.775$ $\mu_{CR} = 0.228$
 Diversity = 1.6547e-03 $k_t = [1, 1, 1, 1]$
 Generation 9360: Best = 1.199113e+04 FES = 936100 $\mu_F = 0.774$ $\mu_{CR} = 0.224$
 Diversity = 1.3166e-03 $k_t = [3, 3, 3, 4]$
 Generation 9370: Best = 1.199113e+04 FES = 937100 $\mu_F = 0.785$ $\mu_{CR} = 0.223$
 Diversity = 1.6783e-03 $k_t = [1, 1, 1, 1]$
 Generation 9380: Best = 1.199112e+04 FES = 938100 $\mu_F = 0.789$ $\mu_{CR} = 0.219$
 Diversity = 1.9521e-03 $k_t = [3, 4, 4, 2]$
 Generation 9390: Best = 1.199112e+04 FES = 939100 $\mu_F = 0.792$ $\mu_{CR} = 0.221$
 Diversity = 1.9800e-03 $k_t = [1, 1, 1, 1]$
 Generation 9400: Best = 1.199111e+04 FES = 940100 $\mu_F = 0.792$ $\mu_{CR} = 0.227$
 Diversity = 1.7760e-03 $k_t = [3, 4, 4, 3]$

Generation 9410: Best = 1.199111e+04 FES = 941100 μ_F = 0.797 μ_{CR} = 0.231
 Diversity = 2.1113e-03 k_t = [1, 1, 1, 1]
 Generation 9420: Best = 1.199110e+04 FES = 942100 μ_F = 0.795 μ_{CR} = 0.226
 Diversity = 1.8801e-03 k_t = [2, 2, 3, 3]
 Generation 9430: Best = 1.199110e+04 FES = 943100 μ_F = 0.800 μ_{CR} = 0.222
 Diversity = 1.9314e-03 k_t = [1, 1, 1, 1]
 Generation 9440: Best = 1.199110e+04 FES = 944100 μ_F = 0.788 μ_{CR} = 0.227
 Diversity = 1.8284e-03 k_t = [4, 3, 4, 3]
 Generation 9450: Best = 1.199109e+04 FES = 945100 μ_F = 0.793 μ_{CR} = 0.236
 Diversity = 1.8686e-03 k_t = [1, 1, 1, 1]
 Generation 9460: Best = 1.199109e+04 FES = 946100 μ_F = 0.789 μ_{CR} = 0.239
 Diversity = 1.5795e-03 k_t = [4, 2, 3, 3]
 Generation 9470: Best = 1.199109e+04 FES = 947100 μ_F = 0.766 μ_{CR} = 0.239
 Diversity = 1.4883e-03 k_t = [1, 1, 1, 1]
 Generation 9480: Best = 1.199109e+04 FES = 948100 μ_F = 0.763 μ_{CR} = 0.224
 Diversity = 1.3956e-03 k_t = [2, 2, 2, 4]
 Generation 9490: Best = 1.199108e+04 FES = 949100 μ_F = 0.767 μ_{CR} = 0.229
 Diversity = 1.7130e-03 k_t = [1, 1, 1, 1]
 Generation 9500: Best = 1.199108e+04 FES = 950100 μ_F = 0.778 μ_{CR} = 0.224
 Diversity = 1.8187e-03 k_t = [3, 3, 4, 2]
 Generation 9510: Best = 1.199108e+04 FES = 951100 μ_F = 0.761 μ_{CR} = 0.219
 Diversity = 1.9182e-03 k_t = [1, 1, 1, 1]
 Generation 9520: Best = 1.199107e+04 FES = 952100 μ_F = 0.761 μ_{CR} = 0.219
 Diversity = 1.5510e-03 k_t = [4, 3, 3, 4]
 Generation 9530: Best = 1.199107e+04 FES = 953100 μ_F = 0.764 μ_{CR} = 0.226
 Diversity = 1.5925e-03 k_t = [1, 1, 1, 1]
 Generation 9540: Best = 1.199107e+04 FES = 954100 μ_F = 0.758 μ_{CR} = 0.232
 Diversity = 1.4355e-03 k_t = [3, 2, 3, 3]
 Generation 9550: Best = 1.199107e+04 FES = 955100 μ_F = 0.760 μ_{CR} = 0.229
 Diversity = 1.5364e-03 k_t = [1, 1, 1, 1]
 Generation 9560: Best = 1.199106e+04 FES = 956100 μ_F = 0.751 μ_{CR} = 0.240
 Diversity = 1.2276e-03 k_t = [4, 2, 2, 3]
 Generation 9570: Best = 1.199106e+04 FES = 957100 μ_F = 0.748 μ_{CR} = 0.243
 Diversity = 1.5358e-03 k_t = [1, 1, 1, 1]
 Generation 9580: Best = 1.199106e+04 FES = 958100 μ_F = 0.756 μ_{CR} = 0.252
 Diversity = 1.3122e-03 k_t = [3, 4, 4, 2]
 Generation 9590: Best = 1.199106e+04 FES = 959100 μ_F = 0.768 μ_{CR} = 0.247
 Diversity = 1.3089e-03 k_t = [1, 1, 1, 1]
 Generation 9600: Best = 1.199106e+04 FES = 960100 μ_F = 0.758 μ_{CR} = 0.234
 Diversity = 1.2261e-03 k_t = [3, 4, 3, 2]
 Generation 9610: Best = 1.199105e+04 FES = 961100 μ_F = 0.757 μ_{CR} = 0.229
 Diversity = 1.2740e-03 k_t = [1, 1, 1, 1]
 Generation 9620: Best = 1.199105e+04 FES = 962100 μ_F = 0.762 μ_{CR} = 0.230
 Diversity = 1.3343e-03 k_t = [4, 2, 4, 2]
 Generation 9630: Best = 1.199105e+04 FES = 963100 μ_F = 0.763 μ_{CR} = 0.237
 Diversity = 1.2501e-03 k_t = [1, 1, 1, 1]
 Generation 9640: Best = 1.199105e+04 FES = 964100 μ_F = 0.766 μ_{CR} = 0.236
 Diversity = 1.0899e-03 k_t = [2, 2, 4, 4]
 Generation 9650: Best = 1.199105e+04 FES = 965100 μ_F = 0.767 μ_{CR} = 0.238
 Diversity = 1.3136e-03 k_t = [1, 1, 1, 1]
 Generation 9660: Best = 1.199104e+04 FES = 966100 μ_F = 0.749 μ_{CR} = 0.241
 Diversity = 1.3149e-03 k_t = [2, 3, 2, 4]
 Generation 9670: Best = 1.199104e+04 FES = 967100 μ_F = 0.760 μ_{CR} = 0.237
 Diversity = 1.4854e-03 k_t = [1, 1, 1, 1]

Generation 9680: Best = 1.199104e+04 FES = 968100 μ_F = 0.760 μ_{CR} = 0.237
 Diversity = 1.3261e-03 k_t = [3, 2, 3, 4]
 Generation 9690: Best = 1.199104e+04 FES = 969100 μ_F = 0.778 μ_{CR} = 0.239
 Diversity = 1.6122e-03 k_t = [1, 1, 1, 1]
 Generation 9700: Best = 1.199103e+04 FES = 970100 μ_F = 0.775 μ_{CR} = 0.231
 Diversity = 1.3786e-03 k_t = [3, 2, 3, 4]
 Generation 9710: Best = 1.199103e+04 FES = 971100 μ_F = 0.765 μ_{CR} = 0.239
 Diversity = 1.4143e-03 k_t = [1, 1, 1, 1]
 Generation 9720: Best = 1.199103e+04 FES = 972100 μ_F = 0.760 μ_{CR} = 0.235
 Diversity = 1.1979e-03 k_t = [2, 4, 3, 3]
 Generation 9730: Best = 1.199103e+04 FES = 973100 μ_F = 0.765 μ_{CR} = 0.236
 Diversity = 1.1595e-03 k_t = [1, 1, 1, 1]
 Generation 9740: Best = 1.199103e+04 FES = 974100 μ_F = 0.765 μ_{CR} = 0.230
 Diversity = 1.3304e-03 k_t = [4, 2, 3, 3]
 Generation 9750: Best = 1.199103e+04 FES = 975100 μ_F = 0.784 μ_{CR} = 0.230
 Diversity = 1.3152e-03 k_t = [1, 1, 1, 1]
 Generation 9760: Best = 1.199102e+04 FES = 976100 μ_F = 0.789 μ_{CR} = 0.231
 Diversity = 1.3240e-03 k_t = [3, 2, 2, 4]
 Generation 9770: Best = 1.199102e+04 FES = 977100 μ_F = 0.787 μ_{CR} = 0.230
 Diversity = 1.4228e-03 k_t = [1, 1, 1, 1]
 Generation 9780: Best = 1.199102e+04 FES = 978100 μ_F = 0.786 μ_{CR} = 0.237
 Diversity = 1.3917e-03 k_t = [2, 2, 4, 4]
 Generation 9790: Best = 1.199102e+04 FES = 979100 μ_F = 0.793 μ_{CR} = 0.241
 Diversity = 1.5296e-03 k_t = [1, 1, 1, 1]
 Generation 9800: Best = 1.199101e+04 FES = 980100 μ_F = 0.786 μ_{CR} = 0.231
 Diversity = 1.8703e-03 k_t = [2, 3, 3, 2]
 Generation 9810: Best = 1.199101e+04 FES = 981100 μ_F = 0.775 μ_{CR} = 0.231
 Diversity = 1.8919e-03 k_t = [1, 1, 1, 1]
 Generation 9820: Best = 1.199101e+04 FES = 982100 μ_F = 0.774 μ_{CR} = 0.225
 Diversity = 1.6451e-03 k_t = [4, 4, 4, 3]
 Generation 9830: Best = 1.199100e+04 FES = 983100 μ_F = 0.776 μ_{CR} = 0.230
 Diversity = 1.6510e-03 k_t = [1, 1, 1, 1]
 Generation 9840: Best = 1.199100e+04 FES = 984100 μ_F = 0.771 μ_{CR} = 0.237
 Diversity = 1.4378e-03 k_t = [3, 3, 3, 2]
 Generation 9850: Best = 1.199100e+04 FES = 985100 μ_F = 0.781 μ_{CR} = 0.240
 Diversity = 1.8460e-03 k_t = [1, 1, 1, 1]
 Generation 9860: Best = 1.199099e+04 FES = 986100 μ_F = 0.784 μ_{CR} = 0.240
 Diversity = 1.8560e-03 k_t = [4, 3, 2, 4]
 Generation 9870: Best = 1.199099e+04 FES = 987100 μ_F = 0.794 μ_{CR} = 0.233
 Diversity = 2.1115e-03 k_t = [1, 1, 1, 1]
 Generation 9880: Best = 1.199098e+04 FES = 988100 μ_F = 0.787 μ_{CR} = 0.240
 Diversity = 1.9159e-03 k_t = [2, 4, 3, 4]
 Generation 9890: Best = 1.199098e+04 FES = 989100 μ_F = 0.788 μ_{CR} = 0.249
 Diversity = 2.3483e-03 k_t = [1, 1, 1, 1]
 Generation 9900: Best = 1.199097e+04 FES = 990100 μ_F = 0.784 μ_{CR} = 0.242
 Diversity = 2.4054e-03 k_t = [2, 4, 3, 3]
 Generation 9910: Best = 1.199096e+04 FES = 991100 μ_F = 0.770 μ_{CR} = 0.233
 Diversity = 2.6805e-03 k_t = [1, 1, 1, 1]
 Generation 9920: Best = 1.199095e+04 FES = 992100 μ_F = 0.782 μ_{CR} = 0.227
 Diversity = 2.4596e-03 k_t = [3, 4, 2, 3]
 Generation 9930: Best = 1.199095e+04 FES = 993100 μ_F = 0.793 μ_{CR} = 0.225
 Diversity = 2.4788e-03 k_t = [1, 1, 1, 1]
 Generation 9940: Best = 1.199094e+04 FES = 994100 μ_F = 0.780 μ_{CR} = 0.223
 Diversity = 2.6660e-03 k_t = [3, 2, 3, 4]

Generation 9950: Best = 1.199093e+04 FES = 995100 μ_F = 0.794 μ_{CR} = 0.222
 Diversity = 3.1881e-03 k_t = [1, 1, 1, 1]
 Generation 9960: Best = 1.199092e+04 FES = 996100 μ_F = 0.792 μ_{CR} = 0.208
 Diversity = 3.0578e-03 k_t = [3, 2, 4, 3]
 Generation 9970: Best = 1.199092e+04 FES = 997100 μ_F = 0.796 μ_{CR} = 0.198
 Diversity = 3.3947e-03 k_t = [1, 1, 1, 1]
 Generation 9980: Best = 1.199091e+04 FES = 998100 μ_F = 0.790 μ_{CR} = 0.206
 Diversity = 3.0976e-03 k_t = [2, 3, 4, 2]
 Generation 9990: Best = 1.199091e+04 FES = 999100 μ_F = 0.777 μ_{CR} = 0.203
 Diversity = 2.9301e-03 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 1.199091e+04

Total Function Evals = 1000000

Total Successful Updates = 331689

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Results:

Best Fitness : 1.199091e+04

Global Optimum : 3.000000e+03

Error (|best-optimum|) : 8.990907e+03

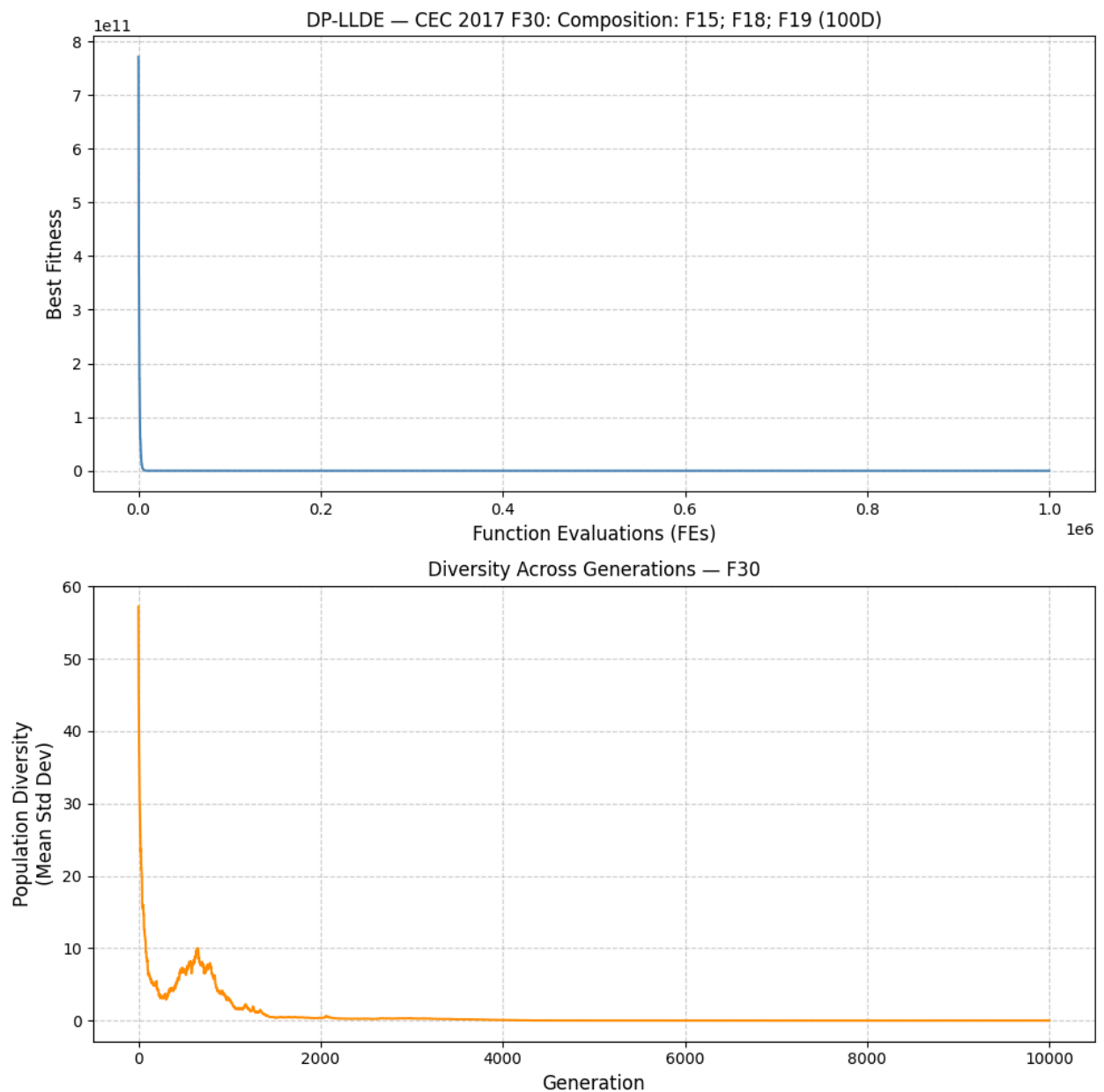
Best Solution (5 dim) : [-99.9999945 -31.3970843 50.79102909 5.29880908
100.]

D^(0) (init. diversity): 5.7220e+01

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✓ Checkpoint saved to: /content/drive/My Drive/thesis_results/all_results_checkpoint.pkl



Graph saved as convergence_F30.png

✓ Plot saved to: /content/drive/My Drive/thesis_results/plots/convergence_F30.png

Section 7. Compiled Results

Print out a summary of all results in a tabular format

```
In [19]: print('=' * 80)
print('SUMMARY — DP-LLDE on CEC 2017 (100D), Single Run, Seed=42')
print('=' * 80)
print(f'{"Function":<10} {"Best Fitness":>18} {"Global Optimum":>18} {"Error |
```

```

print('-' * 70)
for fn in [f'F{i}' for i in range(1, 31)]:
    if fn not in all_results:
        continue
    r = all_results[fn]
    if r['best_fit'] is None:
        print(f'{fn:<10} {"N/A (deprecated)":>18} {r["f_global"]:>18.2f} {"N/A (deprecated)":>18.2f} {r["error"]:>18.2f}')
    else:
        print(f'{fn:<10} {r["best_fit"]:>18.6e} {r["f_global"]:>18.2f} {r["error"]:>18.2f}')
print('=' * 80)

```

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SUMMARY – DP-LLDE on CEC 2017 (100D), Single Run, Seed=42				
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Function	Best Fitness	Global Optimum	Error	best-opt

F1	3.427554e+08	100.00	3.427553e+08	
F2	2.018874e+39	200.00	2.018874e+39	
F3	4.666286e+04	300.00	4.636286e+04	
F4	6.151044e+02	400.00	2.151044e+02	
F5	8.392794e+02	500.00	3.392794e+02	
F6	6.196271e+02	600.00	1.962708e+01	
F7	1.176144e+03	700.00	4.761444e+02	
F8	1.158187e+03	800.00	3.581867e+02	
F9	2.258795e+03	900.00	1.358795e+03	
F10	1.180711e+04	1000.00	1.080711e+04	
F11	1.390262e+03	1100.00	2.902615e+02	
F12	3.401814e+06	1200.00	3.400614e+06	
F13	4.970705e+03	1300.00	3.670705e+03	
F14	1.034320e+05	1400.00	1.020320e+05	
F15	1.956554e+03	1500.00	4.565540e+02	
F16	3.993394e+03	1600.00	2.393394e+03	
F17	3.843024e+03	1700.00	2.143024e+03	
F18	2.292570e+05	1800.00	2.274570e+05	
F19	3.148844e+03	1900.00	1.248844e+03	
F20	3.405808e+03	2000.00	1.405808e+03	
F21	2.537959e+03	2100.00	4.379585e+02	
F22	1.452107e+04	2200.00	1.232107e+04	
F23	3.010524e+03	2300.00	7.105236e+02	
F24	3.494586e+03	2400.00	1.094586e+03	
F25	3.327619e+03	2500.00	8.276192e+02	
F26	1.189626e+04	2600.00	9.296263e+03	
F27	3.661097e+03	2700.00	9.610973e+02	
F28	3.352089e+03	2800.00	5.520893e+02	
F29	5.212292e+03	2900.00	2.312292e+03	
F30	1.199091e+04	3000.00	8.990907e+03	
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